

TREATISE ON THE CIRCLE

Volume III — Geometry, Analysis and Signals on the Circle

From curvature to transforms: differential geometry, sphere, topology, DFT–FFT, signals and telecommunications



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Preface

Volume II is the introduction: parametrisation, integral calculus, complex numbers and the first rudiments of harmonic analysis on the circle, together with the elementary applications to electrical engineering, directional statistics and the discrete circle. This volume is *the rest*: everything the circle generates once it is granted more than one dimension, more than one structure and more than one domain. It presupposes Volume II and, where needed, Volume I, recalled by explicit cross-reference and never summarised.

How it is organised. The volume is divided into six Parts.

- **Part A — Differential and global geometry** (Chapters 1–14): curvature and osculating circle, space curves, global theorems on plane curves, isometries, the complex circle, the sphere and spherical trigonometry, geodesics, surfaces, arbitrary dimension, the topology of the circle, the circle as a group, tangencies and packings.
- **Part B — Complex and harmonic analysis on the disc** (Chapters 15–18): Cauchy and residues, harmonic functions and the Poisson kernel, convergence of Fourier series, partial differential equations on the circle and in the disc.
- **Part C — Transforms** (Chapters 19–23): Fourier, Laplace, Z and DTFT, the discrete transform and the FFT algorithm.
- **Part D — Signals, control, telecommunications** (Chapters 24–27): sampling and linear systems, the circle in control and transmission lines, complex baseband and modulations, OFDM.
- **Part E — Applications to automation** (Chapters 28–30): $SO(3)$ and quaternions, Lie groups and robot kinematics, interpolation and trajectories.
- **Part F — Numerics and research** (Chapters 31–33): circle fitting, open problems and research literacy, epilogue.

Reading levels. Each chapter declares its level: **A** (upper secondary school), **B** (first year of a STEM degree), **C** (undergraduate), **D** (graduate). A reader at level A–B must be able to skip the sections marked C–D without losing the thread: forward references are always explicit and the advanced sections are preceded by a prerequisites box.

Prerequisites. Differential and integral calculus in several variables, linear algebra, complex numbers and elements of Fourier series: everything that is developed in Volume II. Appendix A recalls in condensed form the results of Volumes I and II used in the text; Appendix C collects the tables of transforms and identities; Appendix D the index of symbols.

Note on the September 2026 supplement The volume now also includes Part G on state, estimation, dynamics and digital control. Chapters 1–33 and Parts A–F retain their numbering; the new Part adds Chapters 34–37. The Riemann section supplements Chapter 15 and the Gauss section Chapter 32. The application chain connects the kinematics of Chapters 28–30 to dynamics, feedback and estimation; the reference Z-transform chapter is Chapter 21.

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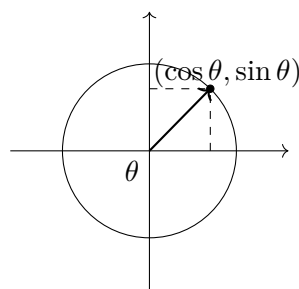
Part A

Differential and global geometry

CHAPTER 1

Prologue: the circle as guiding thread

This volume has a single protagonist: the circle. Not as an isolated figure, but as a *seed* from which curvature, isometries, the sphere, geometry in any dimension and even topology sprout — and, on the applied side, robot kinematics and machine motion control. Before entering the mathematics, it is worth recalling where the circle comes from and why it has traversed three thousand years of history remaining at the center of geometric thought.



Unit circle

Figure 1.1. The circle generates the trigonometric functions: a point on the unit circle has coordinates $(\cos \theta, \sin \theta)$; the two projections trace sine and cosine as θ varies — the first, fundamental link between the circle and analysis.

1.1. Three thousand years around a circle

The circle is among the first mathematical objects studied with rigor. Already the Babylonians and Egyptians estimated its area; but the first *scientific* treatment is by Archimedes of Syracuse (287–212 BC), who in *On the Measurement of the Circle* proved, by contradiction and with the method of exhaustion, that the area of a circle equals that of a right triangle with legs the radius and the circumference, i.e. $A = \frac{1}{2}rC$. Using inscribed and circumscribed polygons up to 96 sides, Archimedes enclosed π between $\frac{223}{71} \approx 3.1408$ and $\frac{22}{7} \approx 3.1429$: the first algorithm in history to approximate a constant with controlled precision.

Remark 1.1: The method of exhaustion as ancestor of calculus. Enclosing a curve between polygons that approximate it ever better is, in embryo, the idea of limit and of integral. The same scheme — approximating the curved with the rectilinear and passing to the limit — returns throughout the volume: in arc length (Chapter 1), in the calculation of the area of the sphere (Chapter 9) and in the volume of the n -dimensional ball (Chapter 12).

1.2. The quadrature of the circle

For over two millennia geometers asked whether, with ruler and compass, one could construct a square of area equal to that of a given circle: the celebrated *quadrature of the circle*. The question turned out to be deeper than expected: constructing with ruler and compass is equivalent to

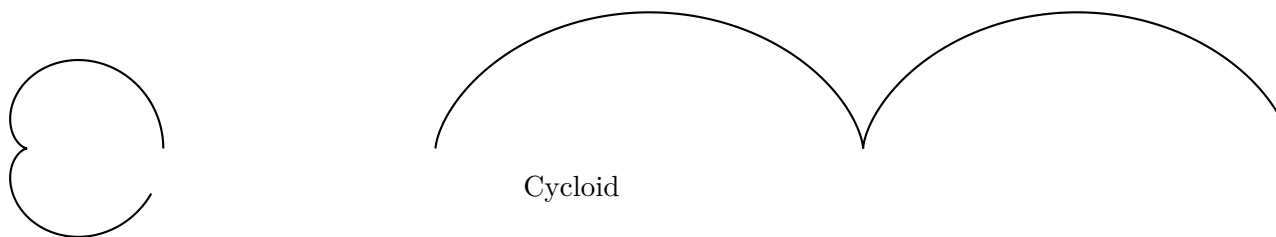
admitting only numbers obtainable from rational operations and square roots, and squaring the circle would require constructing $\sqrt{\pi}$. The breakthrough came in 1882, when Ferdinand von Lindemann proved that π is **transcendental** (it is not the root of any polynomial with rational coefficients): since constructible numbers are algebraic, π is not constructible and the quadrature of the circle is *impossible*.

***Remark 1.2: An anecdote for students: the “law on π ”.** In 1897 an amateur physician convinced the Indiana House to pass a bill (the famous Indiana Pi Bill) which, attempting to “square the circle”, implied wrong values of π (including 3.2). The State Senate, warned by a mathematician passing through, rejected it. The moral: a mathematical constant is not established by decree. The transcendence of π is not a detail, but the deep reason why a circular arc is not a polynomial curve and must be represented, in modern CNC controllers, via rational curves (NURBS, Chapter 15).*

1.3. The circle as generator

The guiding idea of the volume is that almost every geometric object we will encounter is “the circle in a richer context”.

- **Curvature** (Chapter 2): the circle is the only curve with constant curvature, and every curve is locally approximated by its osculating circle.
- **Isometries** (Chapter 4): the circle is the only conic invariant under the whole group $O(2)$; its symmetries are $SO(2) \cong \mathbb{T}$.
- **Sphere** (Chapters 8 and 9): the sphere is the circle “raised” to three dimensions, with great circles in place of lines.
- **Dimension n** (Chapter 12): S^{n-1} generalizes the circle to every dimension, with surprising phenomena.
- **Topology** (Chapter 13): $\pi_1(S^1) = \mathbb{Z}$ makes the circle topologically unique and lies at the root of the Hopf fibration.
- **Robotics and automation**: the constant-curvature arc is the “brick” of circular interpolation; rotations ($SO(3)$, quaternions) preserve the sphere; the topology of the cable governs cable twisting.



Cardioid $r = 1 + \cos \theta$

Figure 1.1: Two curves generated by the motion of a circle: the *cardioid* (trace of a point of a circle rolling on an equal circle) and the *cycloid* (point of a circle rolling on a line). The circle is not only a static figure: in motion it generates an entire family of curves, many of which recur in the volume (the cycloid as brachistochrone and as evolute of itself).

Remark 1.3: Why an automation engineer cares. Every rotary motion of a rotating axis, every G02/G03 interpolation of a machine tool, every orientation of a tool around an axis is, mathematically, an element of the theory of the circle and its symmetries. Understanding the circle deeply — its curvature, its isometries, its extension to the sphere and to $SO(3)$ — means understanding the language with which PLCs and robots are programmed. This volume holds together the two souls: mathematical rigor and industrial application.

Self-assessment quiz

1. Archimedes proved that the area of the circle is: (a) πr (b) $\frac{1}{2}rC$ (c) $2\pi r$ (d) r^2
2. The quadrature of the circle is impossible because π is: (a) irrational (b) transcendental (c) negative (d) very large
3. The method of exhaustion is the ancestor of: (a) linear algebra (b) integral calculus (c) number theory (d) topology
4. A circular arc in CNC is represented exactly with: (a) polynomials (b) NURBS (rational) (c) lines (d) natural splines
5. The group of rotations of the circle in the plane is: (a) $\mathbb{Z}$ (b) $SO(2)$ (c) $SO(3)$ (d) $\mathbb{R}$

Solutions: Q1 (b); Q2 (b) Lindemann 1882; Q3 (b); Q4 (b); Q5 (b) $SO(2) \cong \mathbb{T}$.

Chapter sources

Sources and bibliography

- [1] Archimedes, *On the Measurement of the Circle*.
- [2] *Squaring the Circle*, MacTutor History of Mathematics (University of St Andrews).
- [3] *Squaring the Circle*, AMS Feature Column.
- [4] *Pi* and *Squaring the circle*, reference encyclopedic entries.
- [5] Lindemann (1882), transcendence of π .
- [6] History of the *Indiana Pi Bill* (1897).

CHAPTER 2

Plane curves: curvature, osculating circle, evolutes

In the previous volumes the circle appeared as a set of equidistant points (Volume I), as the multiplicative group of complex numbers of modulus one (Volume II, Chapter 3) and as the domain of harmonic analysis (Volume II, Chapter 4). One last point of view remains, historically the latest and conceptually the deepest: the circle as the *unit of measurement of curvature*. This is the viewpoint of differential geometry, and it grows out of an elementary question — how curved is a curve? — whose answer is: as curved as the circle that resembles it most.

INFO

The idea in one line. To each point of a regular curve one attaches the circle that best approximates it, the *osculating* circle; the reciprocal of its radius is the curvature. The circle is then the curve of constant curvature, and curvature, by the fundamental theorem, determines the curve up to a rigid motion: prescribing it means designing the curve. The *global* theorems that follow from this theory — rotation index, isoperimetric inequality, four vertices — are collected in Chapter 4.

Warning. Conventions of this chapter.

- Curves are *plane*, of class C^2 at least, parametrised by an interval of $\mathbb{R}$; we write $\gamma(t) = (x(t), y(t))$, or $\gamma(s)$ when the parameter is arc length.
- A curve is **regular** if $\gamma'(t) \neq 0$ for every t ; **closed** if it is defined on $[a, b]$ with γ and all derivatives agreeing at the endpoints; **simple** if it is injective on $[a, b]$.
- The positive orientation is counterclockwise; with this choice the curvature of a convex curve traversed counterclockwise is positive.
- We systematically identify $\mathbb{R}^2$ with $\mathbb{C}$, as in Chapter 3, whenever this shortens the computations.

INFO

Level and prerequisites. Level B–C. Sections 2.1 and 2.2 are level B and require only one-variable differential calculus and the dot product in the plane; Sections 2.3 and 2.4 are level B–C and additionally use differentiation of vector-valued functions and the identification $\mathbb{R}^2 \simeq \mathbb{C}$. Prerequisites inside the treatise: circle, tangents and normals (Volume I, Chapters 2 and 4); complex numbers in polar form (Volume II, Chapter 3); the definite integral and Fourier series (Volume II, Chapter 4), the latter used only in Chapter 4. No prior knowledge of differential geometry is assumed.

2.1. Arc length and the unit tangent

Definition 2.1: Arc length and the arc-length parameter. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be regular. The **length** of γ is

$$L = \int_a^b \|\gamma'(t)\| dt, \quad \|\gamma'(t)\| = \sqrt{x'(t)^2 + y'(t)^2},$$

and the **arc-length parameter** is the function

$$s(t) = \int_a^t \|\gamma'(u)\| du.$$

Since $s'(t) = \|\gamma'(t)\| > 0$, the function s is strictly increasing and invertible: one may reparametrise γ by s , obtaining a curve with $\|\gamma'(s)\| = 1$ for every s , said to be **parametrised by arc length**.

Example 2.1: The circle in arc length. For the circle of radius r centred at the origin, $\gamma(t) = (r \cos t, r \sin t)$ with $t \in [0, 2\pi]$:

$$\gamma'(t) = (-r \sin t, r \cos t), \quad \|\gamma'(t)\| = r\sqrt{\sin^2 t + \cos^2 t} = r,$$

a constant. Hence $s(t) = rt$, and the total length is $L = 2\pi r$ — the circumference formula, here *derived* rather than assumed. Reparametrising with $t = s/r$:

$$\gamma(s) = \left(r \cos \frac{s}{r}, r \sin \frac{s}{r} \right), \quad s \in [0, 2\pi r],$$

and indeed $\|\gamma'(s)\| = 1$. ✓

Definition 2.2: Unit tangent and unit normal. For a curve parametrised by arc length one sets

$$T(s) = \gamma'(s) \quad (\text{unit tangent}),$$

and defines the **unit normal** $N(s)$ as the vector obtained by rotating $T(s)$ by $+90^\circ$:

$$T = (T_1, T_2) \implies N = (-T_2, T_1).$$

Under the identification $\mathbb{R}^2 \cong \mathbb{C}$ this is simply $N = iT$ (Theorem (Volume II, Chapter 3) of Chapter 3).

Lemma 2.1: The derivative of a unit vector is orthogonal to it. Let $v : I \rightarrow \mathbb{R}^2$ be differentiable with $\|v(s)\| = c$ constant. Then $\langle v(s), v'(s) \rangle = 0$ for every s .

Proof. From $\|v\|^2 = \langle v, v \rangle = c^2$, differentiating and using bilinearity and symmetry of the scalar product,

$$0 = \frac{d}{ds} \langle v, v \rangle = 2\langle v, v' \rangle,$$

which is the claim. Geometrically: a vector of fixed length can only rotate, and the velocity of rotation is perpendicular to the vector itself. □

2.2. Curvature and the osculating circle

Definition 2.3: Curvature. Let γ be a plane curve of class C^2 parametrised by arc length. By Lemma 2.1, $T'(s)$ is orthogonal to $T(s)$, hence parallel to $N(s)$: there is a unique scalar function $k(s)$ with

$$T'(s) = k(s) N(s).$$

The function k is the **signed curvature** of γ . Its absolute value $|k|$ is the **geometric curvature**; where $k(s) \neq 0$, the number

$$\rho(s) = \frac{1}{|k(s)|}$$

is the **radius of curvature**.

Theorem 2.1: Frenet formulas in the plane. For a plane C^2 curve parametrised by arc length,

$$T'(s) = k(s) N(s), \quad N'(s) = -k(s) T(s).$$

Proof. **Step 1 — the first formula.** This is Definition 2.3.

Step 2 — the second. Since $\|N\| = 1$, Lemma 2.1 gives $N' \perp N$, hence N' is parallel to T : there is $\lambda(s)$ with $N' = \lambda T$.

Step 3 — determining λ . From $\langle T, N \rangle = 0$, differentiating:

$$0 = \langle T', N \rangle + \langle T, N' \rangle = k \langle N, N \rangle + \lambda \langle T, T \rangle = k + \lambda,$$

whence $\lambda = -k$.

Step 4 — the complex reading. Identifying $\mathbb{R}^2$ with $\mathbb{C}$ and writing $T(s) = e^{i\vartheta(s)}$ (legitimate because $|T| = 1$, by Theorem (Volume II, Chapter 3) of Chapter 3),

$$T'(s) = i\vartheta'(s)e^{i\vartheta(s)} = \vartheta'(s) N(s),$$

hence $k = \vartheta'$: **curvature is the angular velocity of the tangent**. This formulation, which we shall use repeatedly, shows that the whole Frenet calculus in the plane reduces to differentiating a complex exponential. $\square$

Theorem 2.2: Curvature of the circle. The circle of radius r traversed counterclockwise has constant curvature $k = 1/r$; traversed clockwise, $k = -1/r$. A straight line has zero curvature.

Proof. **Step 1 — direct computation.** From Example 2.1, $\gamma(s) = (r \cos(s/r), r \sin(s/r))$, hence

$$T(s) = \left(-\sin \frac{s}{r}, \cos \frac{s}{r} \right), \quad T'(s) = \left(-\frac{1}{r} \cos \frac{s}{r}, -\frac{1}{r} \sin \frac{s}{r} \right).$$

Step 2 — comparison with N . Rotating T by $+90^\circ$ as in Definition 2.2:

$$N(s) = \left(-\cos \frac{s}{r}, -\sin \frac{s}{r} \right),$$

the unit vector pointing *towards the centre*. Comparing, $T' = \frac{1}{r}N$, i.e. $k = 1/r$.

Step 3 — clockwise. Substituting $s \mapsto -s$ changes the sign of T but not that of T' with respect to s ; repeating the computation gives $k = -1/r$.

Step 4 — the line. For $\gamma(s) = p + sv$ with $\|v\| = 1$ we have $T = v$ constant, hence $T' = 0$ and $k = 0$.

Step 5 — dimensional check. k has the dimensions of an inverse length: m^{-1} . A large circle is barely curved, a small one very much so: the curvature of a circle of radius 1 km is 10^{-3} m^{-1} , that of a circle of radius 1 mm is 10^3 m^{-1} . ✓ □

Theorem 2.3: Curvature formula for arbitrary parametrisations. Let $\gamma(t) = (x(t), y(t))$ be regular of class C^2 , not necessarily parametrised by arc length. Then

$$k(t) = \frac{x'(t)y''(t) - y'(t)x''(t)}{(x'(t)^2 + y'(t)^2)^{3/2}}.$$

In particular, for the graph of a function $y = f(x)$,

$$k(x) = \frac{f''(x)}{(1 + f'(x)^2)^{3/2}}.$$

Proof. Step 1 — change of parameter. Let $v(t) = \|\gamma'(t)\|$, so $ds/dt = v$. By the chain rule,

$$T = \frac{d\gamma}{ds} = \frac{\gamma'(t)}{v}, \quad \frac{dT}{ds} = \frac{1}{v} \frac{d}{dt} \left(\frac{\gamma'}{v} \right) = \frac{\gamma''v - \gamma'v'}{v^3}.$$

Step 2 — projection onto N . The curvature is the component of dT/ds along $N = \frac{1}{v}(-y', x')$:

$$k = \left\langle \frac{dT}{ds}, N \right\rangle = \frac{1}{v^4} \left\langle \gamma''v - \gamma'v', (-y', x') \right\rangle.$$

The term with γ' gives $\langle (x', y'), (-y', x') \rangle = 0$, so only

$$k = \frac{v}{v^4} \left\langle (x'', y''), (-y', x') \right\rangle = \frac{x'y'' - y'x''}{v^3}$$

survives, which is the stated formula since $v^3 = (x'^2 + y'^2)^{3/2}$.

Step 3 — the graph case. With $x = t$, $y = f(t)$: $x' = 1$, $x'' = 0$, $y' = f'$, $y'' = f''$, whence $k = f''/(1 + f'^2)^{3/2}$.

Step 4 — check on the circle. With $\gamma(t) = (r \cos t, r \sin t)$:

$$x' = -r \sin t, \quad y' = r \cos t, \quad x'' = -r \cos t, \quad y'' = -r \sin t,$$

$$x'y'' - y'x'' = r^2 \sin^2 t + r^2 \cos^2 t = r^2, \quad (x'^2 + y'^2)^{3/2} = r^3,$$

hence $k = r^2/r^3 = 1/r$, in agreement with Theorem 2.2. ✓ □

Definition 2.4: Osculating circle. Let γ be of class C^2 with $k(s_0) \neq 0$. The **osculating circle** of γ at $\gamma(s_0)$ is the circle of radius $\rho(s_0) = 1/|k(s_0)|$ and centre

$$c(s_0) = \gamma(s_0) + \frac{1}{k(s_0)} N(s_0),$$

called the **centre of curvature**. The centre lies on the side towards which the curve bends.

Theorem 2.4: Characterisation of the osculating circle. Among all circles tangent to γ at $P = \gamma(s_0)$, the osculating circle is the only one having second-order contact with γ : that is, the only one for which the distance between curve and circle is $o((s - s_0)^2)$ as $s \rightarrow s_0$. Equivalently, it is the limit of the circles through three points of the curve as the

three points tend to P .

Proof. Step 1 — Taylor expansion of the curve. Set $s_0 = 0$ and $P = \gamma(0)$. By Taylor to second order, using $\gamma' = T$ and $\gamma'' = T' = kN$:

$$\gamma(s) = P + sT(0) + \frac{s^2}{2}k(0)N(0) + o(s^2).$$

Step 2 — expansion of the osculating circle. The circle with centre $c = P + \frac{1}{k(0)}N(0)$ and radius $\rho = 1/|k(0)|$, parametrised by its own arc length from P and with the same tangent, has by Theorem 2.2 constant curvature $k(0)$; its Taylor expansion at P is therefore, by the same computation as in Step 1,

$$\sigma(s) = P + sT(0) + \frac{s^2}{2}k(0)N(0) + o(s^2).$$

Step 3 — comparison. The two expansions agree up to and including second order: $\gamma(s) - \sigma(s) = o(s^2)$.

Step 4 — uniqueness. A circle tangent at P with radius $\rho' \neq \rho$ has curvature $k' \neq k(0)$; its expansion differs from that of γ already in the degree-2 term, by $\frac{s^2}{2}(k' - k(0))N(0) \neq o(s^2)$. So the contact is only of first order.

Step 5 — the limit of circles through three points. Let A, B, C be three distinct points of γ near P . By the circumscribed circle theorem (Volume I) there is a unique circle through them. As A, B, C tend to P , the perpendicular bisectors of AB and BC tend to lines perpendicular to the tangent at P , and their intersection tends to the point at distance $\lim \frac{|AB||BC||CA|}{4 \text{Area}(ABC)}$ from P : applying the circumradius formula and expanding as in Step 1 gives exactly $1/|k(0)|$. The details are Exercise S2.

Step 6 — why “osculating”. The name, introduced by Leibniz, comes from the Latin *osculari*, “to kiss”: the circle does not merely touch the curve (first-order contact, i.e. tangency), but adheres to it to second order. $\square$

Remark 2.1: Curvature as a “local circle”. Theorem 2.4 justifies the definition of curvature: to measure how curved a curve is means to find the circle that resembles it most, and read off its radius. In this sense the circle is not one curve among others, but the unit of measurement of the curvature of all the others — just as the straight line is the unit of measurement of direction. That is why this chapter belongs in a treatise on the circle.

Example 2.2: Curvature of a parabola and of an ellipse. Parabola $y = x^2$. By Theorem 2.3, with $f' = 2x$ and $f'' = 2$:

$$k(x) = \frac{2}{(1 + 4x^2)^{3/2}}.$$

At the vertex $x = 0$: $k = 2$, hence $\rho = 0.5$: the osculating circle has radius $1/2$ and centre $(0, 0.5)$. For $x = 1$: $k = 2/5^{3/2} = 2/11.1803 = 0.178885$, hence $\rho = 5.5902$: the parabola “straightens out” quickly as one moves from the vertex.

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. With $\gamma(t) = (a \cos t, b \sin t)$:

$$x' = -a \sin t, \quad y' = b \cos t, \quad x'' = -a \cos t, \quad y'' = -b \sin t,$$

$$x'y'' - y'x'' = ab \sin^2 t + ab \cos^2 t = ab,$$

$$k(t) = \frac{ab}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}}.$$

At the ends of the major axis ($t = 0$, point $(a, 0)$): $k = ab/b^3 = a/b^2$, hence $\rho = b^2/a$. At the ends of the minor axis ($t = \pi/2$): $k = b/a^2$, $\rho = a^2/b$.

Numerical check. With $a = 3$, $b = 2$: $\rho(0) = 4/3 = 1.\overline{33}$ and $\rho(\pi/2) = 9/2 = 4.5$. The ratio of the two extreme radii of curvature is $(a/b)^3 = (1.5)^3 = 3.375$, and indeed $4.5/1.\overline{33} = 3.375$. ✓

Degeneracy check. For $a = b = r$ both formulas give $\rho = r$, the circle: the curvature is constant and the ellipse degenerates into the circle, as it must. ✓

Theorem 2.5: Characterisation of the circle by curvature. Let γ be a connected plane curve of class C^2 with constant curvature k . Then:

- (i) if $k = 0$, γ is contained in a straight line;
- (ii) if $k \neq 0$, γ is contained in a circle of radius $1/|k|$.

Proof. (i) If $k \equiv 0$ then $T' = 0$, hence T is a constant v and $\gamma(s) = \gamma(0) + sv$: a line.

(ii) Let $k \neq 0$ be constant. Consider

$$c(s) = \gamma(s) + \frac{1}{k} N(s),$$

the centre of curvature of Definition 2.4. Differentiating and using the Frenet formulas (Theorem 2.1),

$$c'(s) = T(s) + \frac{1}{k} N'(s) = T(s) + \frac{1}{k} (-kT(s)) = 0.$$

So c is constant, say c_0 . But then

$$\|\gamma(s) - c_0\| = \left\| -\frac{1}{k} N(s) \right\| = \frac{1}{|k|}$$

for every s : all points of γ are at distance $1/|k|$ from the fixed point c_0 , so by the very definition of a circle (Volume I), γ lies on the circle with centre c_0 and radius $1/|k|$.

Step 3 — the role of the C^2 hypothesis. The proof uses only the existence and differentiability of N , guaranteed by $\gamma \in C^2$. The curve need not be closed: an open curve of constant curvature is an *arc* of a circle.

Step 4 — consistency check. The computation also displays the geometric meaning: when the curvature is constant, the centre of curvature does not move. When it varies, the centre of curvature describes a curve — the evolute — the subject of the next section. □

Best practice. To check that a curve is a circle, the computation of Theorem 2.5(ii) — showing that the centre of curvature is constant — is almost always quicker than manipulating the equation of the curve directly. It also works in reverse: if in some problem the centre of curvature turns out to be constant, one has already proved that the curve is an arc of a circle, with no further computation.

Proposition 2.1: Tait–Kneser theorem. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be of class C^3 , parametrised by arc length, with curvature $k > 0$ and free of vertices, that is $k'(s) \neq 0$ for every s . Then for $a \leq s_1 < s_2 \leq b$ the osculating disc at $\gamma(s_1)$ is contained in the interior of the one at $\gamma(s_2)$, or vice versa, according to the direction in which $\rho = 1/k$ is monotone. In particular the osculating circles at two distinct points of the arc are disjoint and nested.

Proof. Step 1 — velocity of the evolute. Let $e(s) = \gamma(s) + \rho(s)N(s)$ be the centre of curvature, that is the evolute. By Theorem 2.6(i),

$$e'(s) = \rho'(s) N(s).$$

Since $k > 0$ and $k' \neq 0$, the derivative $\rho' = -k'/k^2$ never vanishes either and, being continuous, it has constant sign on $[a, b]$: ρ is strictly monotone.

Step 2 — estimate of the distance between the centres. Integrating,

$$e(s_2) - e(s_1) = \int_{s_1}^{s_2} \rho'(\sigma) N(\sigma) d\sigma,$$

and by the triangle inequality for integrals of vector-valued functions,

$$\|e(s_2) - e(s_1)\| \leq \int_{s_1}^{s_2} |\rho'(\sigma)| d\sigma = |\rho(s_2) - \rho(s_1)|,$$

where the last equality uses that ρ' has constant sign (Step 1).

Step 3 — the inequality is strict. Suppose by contradiction that equality holds and let u be the unit vector of $e(s_2) - e(s_1)$ (non-zero, otherwise $\rho(s_2) = \rho(s_1)$, against strict monotonicity). Then

$$\int_{s_1}^{s_2} (|\rho'(\sigma)| - \rho'(\sigma) \langle N(\sigma), u \rangle) d\sigma = 0,$$

and the integrand is continuous and non-negative, because $|\langle N, u \rangle| \leq 1$; hence it vanishes identically, which together with $\rho'(\sigma) \neq 0$ forces $\langle N(\sigma), u \rangle = \pm 1$, that is $N(\sigma) = \pm u$ constant on $[s_1, s_2]$. But then $N' \equiv 0$, whereas the Frenet formulae (Theorem 2.1) give $N' = -kT$ with $\|N'\| = k > 0$: a contradiction. Therefore

$$\|e(s_2) - e(s_1)\| < |\rho(s_2) - \rho(s_1)|.$$

Step 4 — from the inequality to the nesting. Two closed discs $D(c_1, r_1)$ and $D(c_2, r_2)$ satisfy $D(c_1, r_1) \subset \text{int } D(c_2, r_2)$ if and only if $\|c_2 - c_1\| < r_2 - r_1$: indeed for every $x \in D(c_1, r_1)$ one has $\|x - c_2\| \leq \|x - c_1\| + \|c_1 - c_2\| < r_1 + (r_2 - r_1) = r_2$, and conversely the point $c_1 + r_1 \frac{c_1 - c_2}{\|c_1 - c_2\|}$ (or any boundary point if $c_1 = c_2$) realises the distance $\|c_1 - c_2\| + r_1$, which must stay $< r_2$. If ρ is increasing one applies Step 3 with $c_1 = e(s_1)$, $r_1 = \rho(s_1)$, $c_2 = e(s_2)$, $r_2 = \rho(s_2)$; if it is decreasing the roles are exchanged.

Step 5 — independent numerical check. On the logarithmic spiral $\gamma(t) = e^t(\cos t, \sin t)$ one has $\rho(t) = \sqrt{2}e^t$ and $e(t) = e^t(-\sin t, \cos t)$ (Exercise 2.18 and a direct computation of the centre of curvature). For $t = 0$ and $t = 1$:

$$\|e(1) - e(0)\| = 2.334881 < \rho(1) - \rho(0) = 3.844231 - 1.414214 = 2.430017. \checkmark$$

The two values are recomputed independently in the script `verifica_cap02_cap04.py` (block [C2-7]), where the centre of curvature is obtained from the general formula $\gamma + \frac{1}{k}N$ and not from the closed form. $\square$

Remark 2.2: Why it looks impossible. The osculating circles all «kiss» the same curve, and yet they never meet: the curve is contained in none of them, it crosses each of them passing from the inside to the outside. The apparent paradox dissolves once one observes that the contact is of second order but not of third: where $k' \neq 0$, the osculating circle cuts the curve instead of staying on one side of it (Theorem 2.4). The family of osculating circles of a vertex-free arc therefore fills an annular region of the plane in layers, like the skins of an onion.

Example 2.3: Curvature of the catenary. For the catenary $y = a \cosh(x/a)$, with $a > 0$, one has $y' = \sinh(x/a)$, $y'' = \frac{1}{a} \cosh(x/a)$ and $1 + y'^2 = \cosh^2(x/a)$; the formula for graphs (Theorem 2.3) gives

$$k(x) = \frac{y''}{(1 + y'^2)^{3/2}} = \frac{\frac{1}{a} \cosh(x/a)}{\cosh^3(x/a)} = \frac{1}{a \cosh^2(x/a)}, \quad \rho(x) = a \cosh^2 \frac{x}{a} = \frac{y^2}{a}.$$

The radius of curvature therefore equals the square of the ordinate divided by a ; at the vertex $x = 0$ it equals $\rho(0) = a$.

Independent check. With $a = 1$ and $x = 1$ the parametrisation $\gamma(t) = (t, \cosh t)$ gives $x' = 1$, $y' = \sinh t$, $x'' = 0$, $y'' = \cosh t$, hence $k = \cosh t / \cosh^3 t = 1 / \cosh^2 t$. Numerically $\cosh 1 = 1.543081$, $\cosh^2 1 = 2.381098$ and $k(1) = 0.419974$, while the closed formula gives $\rho(1) = 2.381098$ and $1/\rho(1) = 0.419974$. ✓

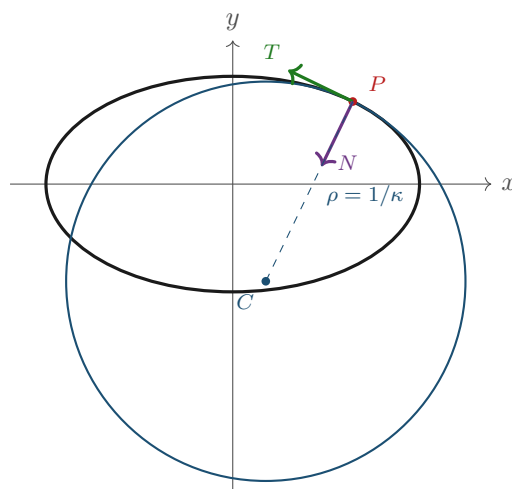


Figure 2.1: The Frenet frame and the osculating circle at a point P of the ellipse $x^2/a^2 + y^2/b^2 = 1$ with $a = 2.6$, $b = 1.5$. The unit tangent T and unit normal N form a moving orthonormal basis; the osculating circle has centre $C = P + \rho N$ on the inner normal and radius $\rho = 1/\kappa$. Here $\kappa = 0.360$ and $\rho = 2.779$. The osculating circle is the unique circle having second-order contact with the curve: among all circles tangent at P , it is the one that also reproduces the curvature.

2.3. Evolutes and involutes

Definition 2.5: Evolute. Let γ be of class C^3 with $k(s) \neq 0$ for every s . The **evolute** of γ is the curve described by the centre of curvature:

$$e(s) = \gamma(s) + \frac{1}{k(s)} N(s).$$

Theorem 2.6: Properties of the evolute. Let γ be as above and $\rho = 1/k$ (with sign). Then:

- (i) $e'(s) = \rho'(s) N(s)$; in particular the evolute is tangent to the normal of γ , i.e. **the evolute is the envelope of the normals** of γ ;
- (ii) e is regular exactly where $k'(s) \neq 0$; the points with $k'(s) = 0$ (**vertices** of γ) correspond to singular points — **cusps** — of the evolute;

- (iii) the length of the arc of evolute between s_1 and s_2 , if ρ is monotone on $[s_1, s_2]$, equals $|\rho(s_2) - \rho(s_1)|$;
 (iv) γ is a circle if and only if its evolute reduces to a point.

Proof. (i) Differentiating $e = \gamma + \rho N$ and using the Frenet formulas (Theorem 2.1) with $\rho = 1/k$:

$$e' = \gamma' + \rho' N + \rho N' = T + \rho' N - \rho k T = T + \rho' N - T = \rho' N,$$

having used $\rho k = 1$. So e' is parallel to N , i.e. to the normal of γ at the corresponding point: the normal line to γ at $\gamma(s)$ is tangent to e at $e(s)$, which is precisely the definition of an envelope.

(ii) By (i), $e' = 0$ iff $\rho' = 0$, and since $\rho = 1/k$ with $k \neq 0$, this is equivalent to $k' = 0$.

(iii) The length is

$$\int_{s_1}^{s_2} \|e'(s)\| ds = \int_{s_1}^{s_2} |\rho'(s)| ds = \left| \int_{s_1}^{s_2} \rho'(s) ds \right| = |\rho(s_2) - \rho(s_1)|,$$

the last but one step using that ρ' has constant sign (ρ monotone).

(iv) If γ is a circle, ρ is constant and by (i) $e' = 0$: the evolute is the centre. Conversely, if e is constant then $\rho' = 0$, hence k is constant and by Theorem 2.5 γ is a circle. $\square$

Example 2.4: Evolute of the ellipse: the astroid. For the ellipse $\gamma(t) = (a \cos t, b \sin t)$ one obtains, by the computation of Example 2.2 and Definition 2.5,

$$e(t) = \left(\frac{a^2 - b^2}{a} \cos^3 t, \frac{b^2 - a^2}{b} \sin^3 t \right),$$

a stretched **astroid**, with Cartesian equation

$$(aX)^{2/3} + (bY)^{2/3} = (a^2 - b^2)^{2/3}.$$

Check at the vertices. For $t = 0$ (the point $(a, 0)$) one finds $e(0) = (\frac{a^2 - b^2}{a}, 0)$, whose distance from $(a, 0)$ is

$$a - \frac{a^2 - b^2}{a} = \frac{b^2}{a} = \rho(0),$$

in agreement with Example 2.2. $\checkmark$

The four cusps. The astroid has four cusps, at $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$: these are the four vertices of the ellipse, the ends of the axes, where $k' = 0$. This is the first encounter with Theorem 4.6, proved later: *every* simple closed curve has at least four vertices, and the ellipse has exactly four.

Limiting case. For $a = b$ the evolute collapses to the origin, consistently with Theorem 2.6(iv). $\checkmark$

Definition 2.6: Involute. Let γ be regular, parametrised by arc length, and $a \in \mathbb{R}$. The **involute** of γ with parameter a is

$$\iota_a(s) = \gamma(s) + (a - s) T(s).$$

Intuitively: it is the trajectory of the end of a taut, inextensible thread unwinding from the curve.

Theorem 2.7: Evolute and involute are inverse operations. With the previous notation, at points where $k(s) \neq 0$ and $a \neq s$:

- (i) $\iota'_a(s) = (a - s)k(s)N(s)$: the involute is regular where $k \neq 0$ and $s \neq a$, and its tangent is orthogonal to that of γ ;
- (ii) the evolute of the involute ι_a is γ itself.

Proof. (i) Differentiating and using $\gamma' = T$, $T' = kN$:

$$\iota'_a = T - T + (a - s)T' = (a - s)kN.$$

(ii) Suppose, to fix ideas, $a - s > 0$ and $k > 0$; the other cases are handled the same way with the signs changed.

Unit tangent. By (i), $\iota'_a = (a - s)kN$ with $(a - s)k > 0$, hence $\|\iota'_a\| = (a - s)k$ and the unit tangent of the involute is $\tilde{T} = N$. Its unit normal, obtained by rotating $\tilde{T}$ by $+90^\circ$, is $\tilde{N} = iN = -T$ (Definition 2.2).

Arc length. Writing σ for the arc length of ι_a , we have $d\sigma/ds = (a - s)k$.

Curvature. Differentiating $\tilde{T} = N$ with respect to σ and using $N' = -kT$:

$$\frac{d\tilde{T}}{d\sigma} = \frac{1}{(a - s)k} \frac{dN}{ds} = \frac{-kT}{(a - s)k} = \frac{-T}{a - s} = \frac{1}{a - s} \tilde{N},$$

so the curvature of the involute is $\tilde{k} = \frac{1}{a - s}$ and its radius of curvature is $\tilde{\rho} = a - s$.

Centre of curvature. By Definition 2.4,

$$\iota_a(s) + \frac{1}{\tilde{k}} \tilde{N} = \gamma(s) + (a - s)T(s) + (a - s)(-T(s)) = \gamma(s).$$

The centre of curvature of the involute is therefore the corresponding point of γ : the evolute of ι_a is γ .

Step 3 — check via length. The verification can also be made without signs, using Theorem 2.6(iii): the length of the arc of γ between s_1 and s_2 is $s_2 - s_1$, which is exactly the variation of the radius of curvature $\tilde{\rho} = a - s$ of the involute, in absolute value. This is the precise formulation of the unwinding-thread picture: the unwound thread is as long as the arc traversed. $\square$

Example 2.5: The involute of the circle and the profile of gear teeth. Let γ be the circle of radius r , parametrised by $\gamma(t) = (r \cos t, r \sin t)$ with arc length $s = rt$ (Example 2.1) and $T(t) = (-\sin t, \cos t)$. The involute with $a = 0$ is

$$\iota(t) = (r \cos t, r \sin t) - rt(-\sin t, \cos t) = r(\cos t + t \sin t, \sin t - t \cos t).$$

Step 1 — speed and arc length. By Theorem 2.7(i), with $k = 1/r$ and $s = rt$:

$$\|\iota'(t)\| = \left| \frac{d\iota}{ds} \right| \cdot r = |-s|k \cdot r = rt.$$

So the length of the involute from 0 to t is

$$\Lambda(t) = \int_0^t ru \, du = \frac{rt^2}{2}.$$

Step 2 — radius of curvature. By Theorem 2.7(ii), the centre of curvature of ι at t is $\gamma(t)$, hence

$$\tilde{\rho}(t) = \|\iota(t) - \gamma(t)\| = rt,$$

growing linearly: the involute of the circle “opens out” progressively.

Step 3 — why gears. In a gear drive one wants the ratio of angular velocities to be *constant* throughout the meshing, otherwise the transmission generates vibration and noise. The corresponding geometric condition (the *fundamental law of gearing*) is that the common normal to the two profiles at the point of contact should always pass through one fixed point of the line of centres, the *pitch point*. The involute profile satisfies it: by Theorem 2.7(i) the normal to the involute at $\iota(t)$ is the tangent to the base circle at $\gamma(t)$, and the common tangents to two fixed base circles form a fixed line — the *line of action*. The intersection of the line of action with the line of centres is therefore fixed, and the gear ratio is constant.

Step 4 — practical consequence. Since the geometry depends only on the base circles, the gear ratio stays constant *even if the centre distance varies slightly* — through manufacturing tolerances, bearing wear or thermal expansion. This is why the involute profile has almost everywhere supplanted the cycloidal profile, which is instead sensitive to centre distance. International standardisation (ISO 53, which defines the standard basic rack tooth profile for involute cylindrical gears) codifies this choice.

Step 5 — numerical check. For $r = 30$ mm and $t = 0.5$ rad:

$$\begin{aligned}\iota(0.5) &= 30(\cos 0.5 + 0.5 \sin 0.5, \sin 0.5 - 0.5 \cos 0.5) = 30(0.877583 + 0.239713, 0.479426 - 0.438791), \\ &= (33.519, 1.219) \text{ mm}.\end{aligned}$$

Distance from the point $\gamma(0.5) = (26.327; 14.383)$ mm:

$$\sqrt{(33.519 - 26.327)^2 + (1.219 - 14.383)^2} = \sqrt{51.72 + 173.29} = \sqrt{225.01} = 15.000 \text{ mm},$$

against $rt = 30 \times 0.5 = 15$ mm predicted in Step 2. ✓

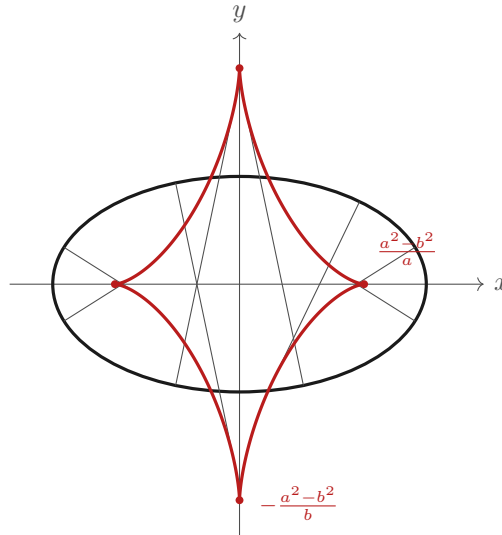


Figure 2.2: The evolute of the ellipse, that is the locus of the centres of the osculating circles, with parametric equations $X = \frac{a^2-b^2}{a} \cos^3 t$, $Y = -\frac{a^2-b^2}{b} \sin^3 t$: a stretched astroid. The thin lines are the normals to the ellipse, and the evolute is their envelope: each normal touches it at the centre of curvature of the point it starts from. The four corner points (cusps) correspond to the four vertices of the ellipse, i.e. the points of stationary curvature: they are exactly the four predicted by the four-vertex theorem. Note that the evolute leaves the ellipse along the minor axis, because the radius of curvature at the vertices $(0, \pm b)$ equals $a^2/b = 4.507 > b$.

2.4. Curves generated by rolling circles

Definition 2.7: Cycloid, epicycloid, hypocycloid.

- The **cycloid** is the curve traced by a point on the circumference of a circle of radius r rolling without slipping along a straight line:

$$\gamma(t) = r(t - \sin t, 1 - \cos t).$$

- The **epicycloid** is traced by a point of a circle of radius r rolling externally on a fixed circle of radius R :

$$\gamma(t) = \left((R+r) \cos t - r \cos \frac{R+r}{r} t, (R+r) \sin t - r \sin \frac{R+r}{r} t \right).$$

- The **hypocycloid** is the analogous curve for internal rolling:

$$\gamma(t) = \left((R-r) \cos t + r \cos \frac{R-r}{r} t, (R-r) \sin t - r \sin \frac{R-r}{r} t \right).$$

Theorem 2.8: The two-circle theorem of Cardano (al-Tusi couple). *If a circle of radius r rolls without slipping inside a circle of radius $R = 2r$, every point of the circumference of the moving circle describes a diameter of the fixed circle, traversed back and forth with simple harmonic motion.*

Proof. **Step 1 — substitution in the formula.** In the hypocycloid of Definition 2.7 put $R = 2r$, so $R - r = r$ and $\frac{R-r}{r} = 1$:

$$\gamma(t) = (r \cos t + r \cos t, r \sin t - r \sin t) = (2r \cos t, 0).$$

Step 2 — reading. The point moves along the x -axis between $-2r$ and $+2r$: it describes exactly the diameter of the fixed circle, with law of motion $x(t) = 2r \cos t$, i.e. simple harmonic motion of amplitude $2r$.

Step 3 — independent synthetic proof. Let O be the centre of the fixed circle (radius $2r$), C the centre of the moving circle (radius r), Q the contact point and P the tracing point. Since the rolling is internal, $|OC| = 2r - r = r$: hence O lies on the moving circle, being at distance r from its centre C .

Let Q_0 be the initial position of the contact point, chosen so that initially P coincides with Q_0 . Rolling without slipping means that the arc Q_0Q of the fixed circle and the arc QP of the moving circle have the same length. Writing α for the central angle $\widehat{Q_0OQ}$ in the fixed circle and β for the central angle $\widehat{QCP}$ in the moving circle, equality of arcs gives

$$2r \alpha = r \beta \implies \beta = 2\alpha.$$

Now Q , P and O all lie on the moving circle: by the inscribed angle theorem (Volume I), the angle $\widehat{QOP}$, inscribed in the moving circle and subtending the arc QP , equals half the corresponding central angle β , that is α .

But $\widehat{Q_0OQ} = \alpha$ by construction: hence $\widehat{QOP} = \widehat{Q_0OQ}$, and the rays OP and OQ_0 make equal and opposite angles with OQ . It follows that P always lies on the fixed line OQ_0 . Since moreover $|OP| \leq |OC| + |CP| = 2r$, the point P traverses the diameter of the fixed circle along OQ_0 . ✓

Step 4 — historical and technical interest. The construction appears in Nasir al-Din al-Tusi (13th century) as a device for generating rectilinear alternating motion out of circular motions, within reformed Ptolemaic astronomy; it is known in the West as “Cardano’s circles”,

from *De subtilitate* (1550). Mechanically, it is the principle of double-crank couplings and of certain straight-line mechanisms without guides.

Step 5 — consistency with Chapter 4. The projection of a uniform circular motion onto an axis is a harmonic motion (Chapter 4, and Chapter 6 for phasors). Cardano's theorem says that the same harmonic motion is obtained, without projecting, as the composition of two rotations of opposite sense and frequencies 1 and -2 — exactly the decomposition $\cos t = \frac{e^{it} + e^{-it}}{2}$ of Theorem (Volume II, Chapter 3) of Chapter 3, read kinematically. $\square$

Example 2.6: Notable curves as special cases. Cardioid ($R = r$, one-cusped epicycloid):

$$\gamma(t) = r(2 \cos t - \cos 2t, 2 \sin t - \sin 2t).$$

Nephroid ($R = 2r$, two-cusped epicycloid): it is the reflection caustic of a circle illuminated by parallel rays.

Deltoid ($R = 3r$, three-cusped hypocycloid).

Astroid ($R = 4r$, four-cusped hypocycloid):

$$\gamma(t) = R(\cos^3 t, \sin^3 t),$$

the same curve met as the evolute of the ellipse in Example 2.4.

General rule. If $R/r = n$ is an integer, the hypocycloid has n cusps and closes after one revolution; if R/r is a rational p/q in lowest terms, it closes after q revolutions with p cusps; if it is irrational it never closes and is dense in the annulus. This last fact is, once again, the non-latticeness criterion of Theorem (Volume II, Chapter 6) of Chapter 8, and the same phenomenon as the density of the orbits of an irrational rotation.

Numerical check on the astroid. With $R = 4$, $r = 1$, $t = \pi/6$:

$$(R - r) \cos t + r \cos 3t = 3 \times 0.866025 + \cos \frac{\pi}{2} = 2.598076 + 0 = 2.598076,$$

$$(R - r) \sin t - r \sin 3t = 3 \times 0.5 - 1 = 0.5.$$

Compare with $R(\cos^3 t, \sin^3 t) = 4(0.649519; 0.125) = (2.598076; 0.5)$. $\checkmark$

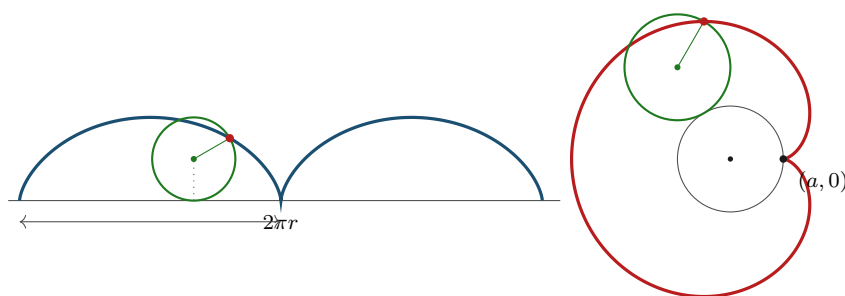


Figure 2.3: Two curves generated by a rolling circle. Left, the cycloid: a point rigidly attached to a circle of radius r rolling without slipping along a line traces $x = r(t - \sin t)$, $y = r(1 - \cos t)$; each arch has base $2\pi r$, height $2r$, length $8r$ and area $3\pi r^2$. Right, the cardioid, obtained by rolling a circle of radius a outside a fixed circle of the same radius: $x = 2a \cos t - a \cos 2t$, $y = 2a \sin t - a \sin 2t$. In green the generating circle in the position shown, in red the tracing point. It is the same mechanism as Ptolemy's circles and Fourier series: composing two circular motions.

2.5. The fundamental theorem of plane curves

The previous sections have extracted a number from the curve, its curvature. It remains to understand how much information that number carries: the answer is *all* of it. Given the function $k(s)$, the curve exists, is unique up to a rigid motion, and is built by integrating twice. This is the passage from *analysis* (measuring a given curve) to *synthesis* (designing a curve by prescribing its curvature), and it is the reason why this chapter has direct applications to trajectory planning.

Theorem 2.9: Fundamental theorem of plane curves. Let $k : [0, L] \rightarrow \mathbb{R}$ be continuous, let $p_0 \in \mathbb{R}^2$ and let T_0 be a unit vector of $\mathbb{R}^2$. Then:

- (i) existence: there is a curve $\gamma \in C^2([0, L]; \mathbb{R}^2)$ parametrised by arc length with $\gamma(0) = p_0$, $\gamma'(0) = T_0$ and signed curvature equal to k ;
- (ii) uniqueness: such a curve is unique;
- (iii) rigidity: two C^2 curves parametrised by arc length with the same curvature function differ by a direct rigid motion of the plane (a rotation followed by a translation).

Proof. **Step 1 — construction.** Let ϑ_0 be the angle of T_0 , that is $T_0 = (\cos \vartheta_0, \sin \vartheta_0)$, and define

$$\vartheta(s) = \vartheta_0 + \int_0^s k(\sigma) d\sigma, \quad \gamma(s) = p_0 + \int_0^s (\cos \vartheta(\sigma), \sin \vartheta(\sigma)) d\sigma.$$

Both integrals exist because the integrands are continuous.

Step 2 — checking the required properties. By the fundamental theorem of calculus, $\gamma'(s) = (\cos \vartheta(s), \sin \vartheta(s))$, which has norm 1: hence γ is parametrised by arc length and $T = (\cos \vartheta, \sin \vartheta)$. Moreover $\gamma(0) = p_0$ and $\gamma'(0) = T_0$. Differentiating once more,

$$T'(s) = \vartheta'(s) (-\sin \vartheta(s), \cos \vartheta(s)) = k(s) N(s),$$

having recognised in $N = (-\sin \vartheta, \cos \vartheta)$ the rotation of T by $+90^\circ$, that is the normal unit vector of Definition 2.3. Comparison with the Frenet formulae (Theorem 2.1) shows that the curvature of γ is exactly k . Since k is continuous, $\vartheta \in C^1$ and $\gamma \in C^2$. This proves (i).

Step 3 — uniqueness for fixed initial data. Let $\tilde{\gamma}$ be another curve with the same data. Its tangent angle $\tilde{\vartheta}$ is a continuous function with $\tilde{\vartheta}' = k$ and $\tilde{\vartheta}(0) \equiv \vartheta_0 \pmod{2\pi}$; replacing $\tilde{\vartheta}$ by $\tilde{\vartheta} - 2m\pi$ for a suitable integer m one may assume $\tilde{\vartheta}(0) = \vartheta_0$. Then $(\tilde{\vartheta} - \vartheta)' = 0$ and $(\tilde{\vartheta} - \vartheta)(0) = 0$, hence $\tilde{\vartheta} \equiv \vartheta$; therefore $\tilde{\gamma}' = \gamma'$ and, integrating with $\tilde{\gamma}(0) = \gamma(0) = p_0$, $\tilde{\gamma} \equiv \gamma$. This proves (ii).

Step 4 — reduction of rigidity to Step 3. Let γ_1, γ_2 be two C^2 curves parametrised by arc length with the same curvature k . Let R be the rotation taking $T_1(0)$ to $T_2(0)$ — it exists and is unique, because a rotation of the plane is determined by the image of one unit vector — and let $b = \gamma_2(0) - R\gamma_1(0)$. Put $M(x) = Rx + b$, a direct rigid motion, and $\tilde{\gamma}_1 = M \circ \gamma_1$.

Step 5 — curvature is invariant under direct rigid motions. If $\tilde{\gamma} = R\gamma + b$ with R a rotation, then $\tilde{\gamma}' = R\gamma'$ still has norm 1, $\tilde{T} = RT$ and $\tilde{N} = RN$ (because R commutes with the rotation by $+90^\circ$, both being rotations). Hence

$$\tilde{T}' = R T' = R(kN) = k RN = k \tilde{N},$$

that is, $\tilde{\gamma}$ has the same curvature k . (The detailed computation, with the behaviour under reflections and under reversal of orientation, is Exercise 2.15.)

Step 6 — conclusion. By Step 5, $\tilde{\gamma}_1$ has curvature k ; by construction $\tilde{\gamma}_1(0) = \gamma_2(0)$ and $\tilde{T}_1(0) = RT_1(0) = T_2(0)$. The two curves $\tilde{\gamma}_1$ and γ_2 therefore solve the same problem as in Step 3, and by uniqueness they coincide: $\gamma_2 = M \circ \gamma_1$. This proves (iii).

Step 7 — reading. Curvature is a *complete system of invariants* for plane curves: it contains all the geometric information and nothing else. Applied to constant k , the theorem returns Theorem 2.5: up to rigid motions there is only one curve of constant curvature $c \neq 0$, and the circle of radius $1/|c|$ realises it. $\square$

Example 2.7: Reconstruction of the curve of constant curvature. Apply the construction of Step 1 to $k(s) \equiv c > 0$, with $p_0 = (0, 0)$ and $T_0 = (1, 0)$, that is $\vartheta_0 = 0$.

Step 1. $\vartheta(s) = \int_0^s c \, d\sigma = cs$.

Step 2. $T(s) = (\cos cs, \sin cs)$.

Step 3. Integrating,

$$\gamma(s) = \left(\frac{1}{c} \sin cs, \frac{1}{c} (1 - \cos cs) \right).$$

Step 4 — recognition. $x^2 + (y - \frac{1}{c})^2 = \frac{1}{c^2} (\sin^2 cs + \cos^2 cs) = \frac{1}{c^2}$: it is the circle with centre $(0, 1/c)$ and radius $1/c$, traversed counterclockwise starting from the origin. ✓

Remark 2.3: The circle as a prototype, and the synthesis of trajectories. Example 2.7 is the simplest case of a general procedure: one prescribes the law $k(s)$ and integrates twice. The three elementary laws are

$$k(s) = 0 \Rightarrow \text{segment}; \quad k(s) = \frac{1}{r} \Rightarrow \text{circular arc}; \quad k(s) = k_0 + cs \Rightarrow \text{clothoid},$$

and the third is the subject of the next section. The point is not aesthetic: the curvature of a trajectory travelled by a wheeled vehicle is proportional to the steering angle, which cannot change instantaneously; prescribing a continuous k means prescribing a physically executable manoeuvre.

2.6. Clothoids and curvature-continuous planning

A direct transition between a straight stretch ($k = 0$) and a circular arc ($k = 1/r$) has *discontinuous* curvature: at the junction point the steering angle would have to jump from 0 to a finite value in an instant, that is with infinite angular velocity. The cure is to interpose a stretch in which the curvature grows linearly along the arc: the clothoid.

Definition 2.8: Clothoid (Euler spiral). A **clothoid** (or *Euler spiral*, or *Cornu spiral*) is a plane curve parametrised by arc length whose curvature is an affine function of s :

$$k(s) = k_0 + cs, \quad c \neq 0.$$

The number c , of dimension $[\text{length}]^{-2}$, is called the **sharpness parameter** of the transition.

Proposition 2.2: Explicit form of the clothoid and Fresnel integrals. Let $c > 0$ and consider the clothoid with $k_0 = 0$ leaving the origin with horizontal tangent, that is the one given by Theorem 2.9 with $k(s) = cs$, $p_0 = (0, 0)$, $T_0 = (1, 0)$. Then:

(i) the tangent angle is $\varphi(s) = \frac{1}{2}cs^2$ and

$$\gamma(s) = \left(\int_0^s \cos \frac{c\sigma^2}{2} d\sigma, \int_0^s \sin \frac{c\sigma^2}{2} d\sigma \right);$$

(ii) setting $a = \sqrt{\pi/c}$ and $\tau = s/a$, one has

$$\gamma(s) = a (C(\tau), S(\tau)), \quad C(\tau) = \int_0^\tau \cos \frac{\pi u^2}{2} du, \quad S(\tau) = \int_0^\tau \sin \frac{\pi u^2}{2} du,$$

where C and S are the Fresnel integrals: every clothoid is therefore similar to the normalised clothoid $\tau \mapsto (C(\tau), S(\tau))$, with similarity ratio a ;

(iii) the identity $\rho(s)s = 1/c$ holds for $s > 0$: the product of the radius of curvature and the arc length is constant along the curve. This is the property that characterises the clothoid among all plane curves.

Proof. Step 1 — the tangent angle. By Step 1 of the proof of Theorem 2.9, with $\vartheta_0 = 0$,

$$\varphi(s) = \int_0^s c\sigma \, d\sigma = \frac{cs^2}{2},$$

and $\gamma(s) = \int_0^s (\cos \varphi, \sin \varphi)$: this is (i).

Step 2 — normalisation. In (i) set $\sigma = au$ with $a = \sqrt{\pi/c}$, that is $ca^2 = \pi$. Then $\frac{c\sigma^2}{2} = \frac{ca^2u^2}{2} = \frac{\pi u^2}{2}$ and $d\sigma = a \, du$, while $\sigma = s$ corresponds to $u = s/a = \tau$. Substituting,

$$\int_0^s \cos \frac{c\sigma^2}{2} \, d\sigma = a \int_0^\tau \cos \frac{\pi u^2}{2} \, du = a C(\tau),$$

and likewise for the second component: this is (ii).

Step 3 — the identity $\rho s = 1/c$. By construction $k(s) = cs$, hence $\rho(s) = 1/|k(s)| = 1/(cs)$ for $s > 0$ and $\rho(s)s = 1/c$.

Step 4 — the characterisation. Conversely, if a curve parametrised by arc length satisfies $\rho(s)s = \text{const.} = 1/c$ on an interval with $s > 0$, then $k(s) = \pm cs$, and by Theorem 2.9 the curve is, up to a rigid motion, the clothoid of part (i) (or its mirror image).

Step 5 — dimensional check. $[c] = \text{m}^{-2}$, hence $[a] = \sqrt{\text{m}^2} = \text{m}$ and τ is dimensionless, as it must be for the argument of C and S . $\checkmark$ $\square$

Remark 2.4: What we import: the asymptotic value of the Fresnel integrals. As $\tau \rightarrow +\infty$ the normalised clothoid converges to the point $(\frac{1}{2}, \frac{1}{2})$, and as $\tau \rightarrow -\infty$ to its opposite: the spiral winds around two «eyes». The ingredient is the value

$$\lim_{\tau \rightarrow +\infty} C(\tau) = \lim_{\tau \rightarrow +\infty} S(\tau) = \frac{1}{2},$$

which is a classical result on the Fresnel integrals and which we **import** here: it is not proved in this treatise, because it requires the value of the Gauss integral $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$ and a contour argument in the complex plane (or, equivalently, the Fourier transform of the Gaussian, Volume II, Chapter 4). The source is the NIST Digital Library of Mathematical Functions, §7.2: formulae (7.2.7) and (7.2.8) are the definitions of C and S used here, and (7.2.9) gives precisely $\lim_{x \rightarrow \infty} C(x) = \lim_{x \rightarrow \infty} S(x) = \frac{1}{2}$. Nothing that follows depends on this limit: the transition formulae use only values of C and S at finite points, computed by quadrature.

Proposition 2.3: The curvature-continuous transition between a straight stretch and an arc. Suppose one wants to join a straight stretch to a circular arc of radius $r > 0$ by a clothoid stretch $k(s) = cs$, $s \in [0, L]$, travelled at constant speed $v > 0$. Then:

(i) the curvature is continuous at the two junction points if and only if $cL = 1/r$, that is

$$L = \frac{1}{rc};$$

(ii) in that case the angular deviation of the stretch is $\Delta\varphi = \frac{cL^2}{2} = \frac{L}{2r}$;

(iii) the final point is $\gamma(L) = a(C(\tau_L), S(\tau_L))$ with $a = \sqrt{\pi/c}$ and $\tau_L = L/a$;

(iv) if the steering rate imposes $|\dot{k}| \leq \dot{k}_{\max}$, then $L \geq \frac{v}{r \dot{k}_{\max}}$.

Proof. Step 1 — continuity at the entrance. At the beginning of the stretch $k(0) = 0$, which is the curvature of the straight line: the junction is continuous for every choice of c and L .

Step 2 — continuity at the exit. At the end of the stretch $k(L) = cL$; the curvature of the arc is $1/r$. Continuity is therefore equivalent to $cL = 1/r$, that is $L = 1/(rc)$: this is (i).

Step 3 — angular deviation. By Step 1 of the proof of Proposition 2.2, $\varphi(L) = cL^2/2$; substituting $c = 1/(rL)$ one obtains $\varphi(L) = L/(2r)$: this is (ii).

Step 4 — final point. This is Proposition 2.2(ii) evaluated at $s = L$: this is (iii).

Step 5 — kinematic constraint. Along the trajectory $s = vt$, hence

$$\dot{k} = \frac{dk}{dt} = \frac{dk}{ds} \frac{ds}{dt} = cv.$$

The constraint $cv \leq \dot{k}_{\max}$ gives $c \leq \dot{k}_{\max}/v$ and, substituting into (i),

$$L = \frac{1}{rc} \geq \frac{v}{r\dot{k}_{\max}},$$

which is (iv). The case of equality corresponds to steering at the maximum admissible rate throughout the stretch. $\square$

Proposition 2.4: The shift of the transition. With the notation of Proposition 2.3, the **shift** of the transition is the quantity

$$p = y(L) - r(1 - \cos \Delta\varphi),$$

that is, how far the circular arc must be moved inwards with respect to the straight stretch so that the clothoid joins them. Then

$$p = \frac{L^2}{24r} + O\left(\frac{L^4}{r^3}\right) \quad (L/r \rightarrow 0).$$

Proof. Step 1 — expansion of $y(L)$. From Proposition 2.2(i) and the expansion $\sin x = x - \frac{x^3}{6} + O(x^5)$ with $x = c\sigma^2/2$,

$$y(L) = \int_0^L \sin \frac{c\sigma^2}{2} d\sigma = \int_0^L \left(\frac{c\sigma^2}{2} - \frac{c^3\sigma^6}{48} + O(c^5\sigma^{10}) \right) d\sigma = \frac{cL^3}{6} - \frac{c^3L^7}{336} + O(c^5L^{11}).$$

With $cL = 1/r$ the leading term equals $\frac{L^2}{6r}$ and the next one $\frac{L^4}{336r^3}$.

Step 2 — expansion of the comparison term. From $\Delta\varphi = L/(2r)$ and $1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + O(x^6)$,

$$r(1 - \cos \Delta\varphi) = r \left(\frac{L^2}{8r^2} - \frac{L^4}{384r^4} + O\left(\frac{L^6}{r^6}\right) \right) = \frac{L^2}{8r} - \frac{L^4}{384r^3} + O\left(\frac{L^6}{r^5}\right).$$

Step 3 — difference. Subtracting,

$$p = \left(\frac{1}{6} - \frac{1}{8} \right) \frac{L^2}{r} + O\left(\frac{L^4}{r^3}\right) = \frac{L^2}{24r} + O\left(\frac{L^4}{r^3}\right),$$

because $\frac{1}{6} - \frac{1}{8} = \frac{4-3}{24} = \frac{1}{24}$.

Step 4 — practical reading. The formula $p \simeq L^2/(24r)$ is the one used in railway and road practice to size the displacement of the track or of the lane; Step 3 shows that the error is of fourth order in L/r , hence negligible as long as the transition is long compared with the deviation it produces. $\square$

Example 2.8: A clothoid transition, with numerical check. A vehicle passes from a straight stretch to an arc of radius $r = 5$ m through a clothoid stretch of parameter $c = 0.04$ m⁻².

Step 1 — length of the stretch. By Proposition 2.3(i), $L = 1/(rc) = 1/(5 \times 0.04) = 5$ m.

Step 2 — angular deviation. By (ii), $\Delta\varphi = cL^2/2 = 0.04 \times 25/2 = 0.5$ rad = 28.647890°.

Step 3 — scale factor and Fresnel argument. $a = \sqrt{\pi/c} = \sqrt{\pi/0.04} = 8.862269$ m and $\tau_L = L/a = 5/8.862269 = 0.564190$.

Step 4 — final point. By composite Simpson quadrature one obtains $C(\tau_L) = 0.550247$ and $S(\tau_L) = 0.092366$, hence

$$\gamma(L) = a(C(\tau_L), S(\tau_L)) = (4.876438; 0.818570) \text{ m},$$

with chord $\|\gamma(L)\| = 4.944665$ m.

Step 5 — shift. The exact value is $p = y(L) - r(1 - \cos \Delta\varphi) = 0.818570 - 5 \times (1 - \cos 0.5) = 0.206483$ m, while the estimate of Proposition 2.4 gives $L^2/(24r) = 25/120 = 0.208333$ m: a discrepancy of 0.001850 m, that is 0.90%. Consistent with the error $O(L^4/r^3)$: here $L/r = 1$ and the estimate remains accurate to within 1%. ✓

Independent check. All the values are recomputed in the script `verifica_cap02_cap04.py` (block [C2-3]) in two distinct ways: direct integration of $\int_0^L \cos(c\sigma^2/2) d\sigma$ and reduction to the Fresnel integrals via Proposition 2.2(ii); the two ways agree to within 10^{-8} .

2.6.1 Application to automation

The kinematic constraint of a mobile robot or of a wheeled vehicle is the continuity of the curvature: k is proportional to the steering angle, which cannot change instantaneously. A segment $\rightarrow$ arc transition has discontinuous curvature (a jump from 0 to $1/r$) and would require steering at infinite rate; an interposed clothoid stretch makes k continuous — hence the position C^2 — with steering at constant angular rate, as Step 5 of the proof of Proposition 2.3 shows. Clothoid splines are the standard for the transition curves of roads and railways and, today, for the trajectory planning of autonomous vehicles: they reduce lateral *jerk*, actuator wear and the risk of rollover. A stretch with constant k remains a circular arc, the building block of the circular motions of the tool centre point in industrial robotics.

2.6.2 Circular interpolation in PLC and CNC controllers

The case $k \equiv 1/r$ (circular arc) is programmed directly in motion controllers. In the ISO language (G-code) of **Siemens SINUMERIK** CNCs the arc is commanded with **G02** (clockwise) and **G03** (counterclockwise), after selecting the interpolation plane with **G17/G18/G19** (xy , zx , yz) and indicating the centre through the incremental coordinates **I**, **J**; helical interpolation adds the feed along the orthogonal axis. In the PLC world the standard **PLCopen Motion Control** defines the function block **MC_MoveCircularAbsolute**, available in equivalent form on **Siemens** (TIA Portal, Technology Objects), **Schneider Electric** (PacDrive/Lexium), **Rockwell** (Studio 5000, MCCM instructions on a Coordinate System), **Delta** and **Elco**: one specifies the end point, the centre or the radius, and the direction. The mathematics is that of this chapter: an arc is the locus of constant curvature $1/r$ (Theorem 2.5), and a continuous junction to a straight stretch requires a transition with variable curvature — a clothoid or a spline — which advanced controllers realise with *look-ahead* and smoothing functions in order to limit the jerk.

2.6.3 Computing with Python

The formulae of this chapter translate directly into code. The following fragment computes the signed curvature and the osculating circle of a parametric curve, and the final point of a clothoid transition; it can be used to redo the exercises.

Listing 2.1: Curvature, osculating circle and clothoid transition

```
import numpy as np
from scipy.integrate import quad          # or a hand-written Simpson
rule

def curvatura(gp, gpp):
    """Signed curvature from gamma'(t)=gp and gamma''(t)=gpp in the
        plane."""
    xp, yp = gp; xpp, ypp = gpp
    return (xp*ypp - yp*xpp) / (xp**2 + yp**2)**1.5

def cerchio_oscultatore(P, gp, gpp):
    k = curvatura(gp, gpp)
    T = np.array(gp, float); T /= np.linalg.norm(T)
    N = np.array([-T[1], T[0]])          # normal rotated by +90
    degrees
    centro = np.array(P, float) + (1.0/k)*N
    return centro, 1.0/abs(k)           # (centre, radius rho)

def clotoide(c, L):
    """End point and angular deviation of the transition k(s)=c*s on [0,
        L]."""
    x = quad(lambda s: np.cos(0.5*c*s*s), 0.0, L)[0]
    y = quad(lambda s: np.sin(0.5*c*s*s), 0.0, L)[0]
    return (x, y), 0.5*c*L*L

# Parabola y=x^2 at the point (1,1): gamma=(t, t^2), gamma'=(1,2t),
# gamma''=(0,2)
print(cerchio_oscultatore((1, 1), (1, 2), (0, 2))) # -> ([-4. 3.5],
5.5902...)
# Transition with c=0.04 m^-2 and r=5 m, hence L=1/(r c)=5 m
print(clotoide(0.04, 5.0)) # -> ((4.876438,
0.818570), 0.5)
```

The results agree with Exercise 2.19 (centre $(-4, 7/2)$, radius $5\sqrt{5}/2 \simeq 5.590170$) and with Example 2.8. ✓

2.7. Assigned exercises

The answers are in the Teacher's key (Section 2.11).

Exercise 2.1: Osculating circle and evolute of the hyperbola. *Aim:* to apply the general curvature formula, the centre of curvature and the definition of evolute to a non-symmetric curve, and to recognise a vertex. *Estimated time:* 40 minutes. *Type:* symbolic with numerical check.

Consider the branch $\gamma(t) = (t, 1/t)$, $t > 0$, of the hyperbola $xy = 1$.

- (a) Compute $k(t)$ with Theorem 2.3 and check that $k(1) = 1/\sqrt{2}$.
- (b) Determine the centre and radius of the osculating circle at $(1, 1)$.

- (c) Derive the evolute in parametric form and show that $e(t) = \left(\frac{3t^4 + 1}{2t^3}, \frac{t^4 + 3}{2t} \right)$.
- (d) Prove that $t = 1$ is a vertex ($\rho'(1) = 0$) and that it is the only one on the branch.
- (e) Check (c) for $t = 0.7$ and $t = 1.6$ to two decimal places.

Exercise 2.2: The evolute of the cycloid is a cycloid. *Aim:* to prove Huygens's result underlying the isochronous clock, and to connect it with the properties of the evolute. *Estimated time:* 45 minutes. *Type:* symbolic/proof-based.

Let $\gamma(t) = r(t - \sin t, 1 - \cos t)$, $t \in (0, 2\pi)$.

- (a) Check that $\|\gamma'(t)\| = 2r \sin \frac{t}{2}$ and that $\rho(t) = 4r \sin \frac{t}{2}$.
- (b) Compute the centre of curvature and show that $e(t) = r(t + \sin t, \cos t - 1)$.
- (c) Setting $u = t + \pi$, prove that $e(t) = r(u - \sin u, 1 - \cos u) - (\pi r, 2r)$: the evolute is the same cycloid translated by $(-\pi r, -2r)$.
- (d) Deduce, from Theorem 2.6(iii), the length of the arc of the evolute between $t = \pi/2$ and $t = \pi$, and compare it with $8r$, the length of one arch.
- (e) Explain in one line why this result makes the cycloidal pendulum isochronous.

Exercise 2.3: Sizing a clothoid transition. *Aim:* to use Proposition 2.3 in a design problem, with a kinematic constraint and a check of the shift approximation. *Estimated time:* 50 minutes. *Type:* numerical/computational.

A self-driving vehicle travels a straight stretch at $v = 8$ m/s and must enter a curve of radius $r = 40$ m. The maximum rate of change of curvature allowed by the steering system is $\dot{k}_{\max} = 0.02 \text{ m}^{-1}\text{s}^{-1}$.

- (a) Determine the minimum length L of the transition and the corresponding c .
- (b) Compute $\Delta\varphi$ in radians and in degrees.
- (c) Compute $a = \sqrt{\pi/c}$, τ_L and the final point $\gamma(L)$ to four decimal places, by numerical quadrature.
- (d) Compute the exact shift and compare it with $L^2/(24r)$; comment on the relative discrepancy in the light of the term $O(L^4/r^3)$ and of the ratio L/r obtained.
- (e) Repeat (a)–(d) halving $\dot{k}_{\max}$ and discuss what changes for passenger comfort.

2.8. Worked exercises

Exercise 2.4: The trochoid: inflections, cusps and curvature at the vertex. *Type:* symbolic with numerical check. Consider the *trochoid*

$$\gamma(t) = (rt - d \sin t, r - d \cos t), \quad r > 0, d > 0,$$

described by a point rigidly attached to a circle of radius r rolling on a line, placed at distance d from the centre ($d < r$ curtate cycloid, $d = r$ cycloid, $d > r$ prolate cycloid). Compute $k(t)$, determine the inflection points and the value $k(\pi)$, and discuss the three cases.

Solution. Strategy. One applies the formula of Theorem 2.3; the sign of the numerator governs the inflections, and the case $d = r$ must be treated separately because the curve has cusps.

Step 1 — derivatives. $x' = r - d \cos t$, $y' = d \sin t$, $x'' = d \sin t$, $y'' = d \cos t$.

Step 2 — numerator.

$$x'y'' - y'x'' = (r - d \cos t) d \cos t - d \sin t \cdot d \sin t = dr \cos t - d^2 \cos^2 t - d^2 \sin^2 t = d(r \cos t - d).$$

Step 3 — denominator.

$$x'^2 + y'^2 = (r - d \cos t)^2 + d^2 \sin^2 t = r^2 - 2rd \cos t + d^2,$$

which is $\geq (r - d)^2$ and vanishes only if $d = r$ and $\cos t = 1$: these are the cusps of the ordinary cycloid, where the curve is not regular.

Step 4 — curvature.

$$k(t) = \frac{d(r \cos t - d)}{(r^2 - 2rd \cos t + d^2)^{3/2}}.$$

Step 5 — inflections. $k(t) = 0$ if and only if $\cos t = d/r$, which has a solution if and only if $d \leq r$. Hence the curtate cycloid ($d < r$) has two inflections per period, at $t = \pm \arccos(d/r)$; the ordinary cycloid ($d = r$) has $\cos t = 1$, which however is a cusp and not an inflection; the prolate cycloid ($d > r$) has no inflections and the curvature has constant (negative) sign, because $r \cos t - d \leq r - d < 0$.

Step 6 — the highest point. At $t = \pi$ (maximum ordinate $r + d$):

$$k(\pi) = \frac{d(-r - d)}{(r^2 + 2rd + d^2)^{3/2}} = \frac{-d(r + d)}{(r + d)^3} = -\frac{d}{(r + d)^2}.$$

Independent check. With $r = 1$ the script `verifica_cap02_cap04.py` (block [C2-2]) recomputes $k(\pi)$ from the general formula and finds -0.222222 for $d = 0.5$, -0.250000 for $d = 1$ and -0.240000 for $d = 1.5$, against the values $-d/(r + d)^2$ equal to $-2/9$, $-1/4$ and $-1.5/6.25$. ✓ In the case $d = r = 1$ one recovers $\rho(\pi) = 1/|k(\pi)| = 4r$, in agreement with Exercise 2.8.

Interpretation. The sign of k tells which way the curve bends: the curtate cycloid, which never touches the rolling line, alternates concave and convex stretches and therefore has inflections; the prolate one forms loops and stays curved on the same side throughout. The ordinary cycloid is the transitional case, where the two inflections degenerate into a cusp.

Typical mistakes. (a) Forgetting that at $t = 0$ with $d = r$ the curve is *not* regular and applying the formula there anyway: the denominator vanishes. (b) Concluding that the prolate cycloid has inflections because it «changes direction»: it is the tangent that changes direction, not the sign of the curvature. (c) Using $\|\gamma'\| = 2r \sin(t/2)$, valid only for $d = r$.

Extension. Show that for $d < r$ the minimum radius of curvature is attained at the lowest point $t = 0$ and equals

$$\rho_{\min} = \frac{((r - d)^2)^{3/2}}{d(r - d)} = \frac{(r - d)^2}{d},$$

and check that $\rho_{\min} \rightarrow 0$ as $d \rightarrow r^-$: this is how the cusp is formed. (With $r = 1$ and $d = 0.5$; 0.8 ; 0.95 ; 0.99 one obtains respectively 0.500000 ; 0.050000 ; 0.002632 ; 0.000101 ; the script `verifica_cap02_cap04.py`, block [C2-2], recomputes them by minimising $\rho(t)$ by sampling. ✓)

Exercise 2.5: Nested osculating circles on the logarithmic spiral. *Type:* mixed. For the logarithmic spiral $\gamma(t) = e^t(\cos t, \sin t)$ compute the radius of curvature and the evolute in closed form, check the hypothesis of Proposition 2.1 and verify the nesting numerically between $t = 0$ and $t = 1$.

Solution. Strategy. One computes γ' and γ'' , applies the curvature formula, builds the centre of curvature $e = \gamma + \rho N$ and compares $\|e(1) - e(0)\|$ with $\rho(1) - \rho(0)$.

Step 1 — derivatives. $\gamma'(t) = e^t(\cos t - \sin t, \sin t + \cos t)$ and $\gamma''(t) = e^t(-2 \sin t, 2 \cos t)$, as one checks by differentiating again.

Step 2 — speed. $\|\gamma'(t)\|^2 = e^{2t}[(\cos t - \sin t)^2 + (\sin t + \cos t)^2] = 2e^{2t}$, hence $\|\gamma'\| = \sqrt{2}e^t$ and the arc length from 0 to T equals $\sqrt{2}(e^T - 1)$ (Exercise 2.17).

Step 3 — curvature.

$$x'y'' - y'x'' = e^{2t}[(\cos t - \sin t)2 \cos t + (\sin t + \cos t)2 \sin t] = 2e^{2t},$$

and $(x'^2 + y'^2)^{3/2} = (2e^{2t})^{3/2} = 2\sqrt{2}e^{3t}$, hence

$$k(t) = \frac{2e^{2t}}{2\sqrt{2}e^{3t}} = \frac{1}{\sqrt{2}e^t}, \quad \rho(t) = \sqrt{2}e^t.$$

Step 4 — hypothesis of Tait–Kneser. $k > 0$ everywhere and $k'(t) = -\frac{1}{\sqrt{2}}e^{-t} \neq 0$: the spiral is free of vertices and ρ is strictly increasing. The hypotheses of Proposition 2.1 are satisfied.

Step 5 — evolute. The unit tangent is $T = \frac{1}{\sqrt{2}}(\cos t - \sin t, \sin t + \cos t)$ and its rotation by $+90^\circ$ is $N = \frac{1}{\sqrt{2}}(-\sin t - \cos t, \cos t - \sin t)$. Hence

$$e(t) = \gamma(t) + \rho(t)N(t) = e^t(\cos t, \sin t) + e^t(-\sin t - \cos t, \cos t - \sin t) = e^t(-\sin t, \cos t).$$

The evolute is the same spiral rotated by $+90^\circ$: the logarithmic spiral is, up to a rigid motion, its own evolute.

Independent check. The script `verifica_cap02_cap04.py` (block [C2-7]) computes the centre of curvature from the general formula $\gamma + \frac{1}{k}N$, without using the closed form of Step 5, and finds $e(0) = (0; 1)$, $e(1) = (-2.287355; 1.468694)$, hence

$$\|e(1) - e(0)\| = 2.334881 < \rho(1) - \rho(0) = 3.844231 - 1.414214 = 2.430017. \quad \checkmark$$

The osculating disc at $t = 0$ is therefore contained in the interior of the one at $t = 1$, as Step 4 of the proof of Proposition 2.1 predicts.

Interpretation. The logarithmic spiral is *self-similar*: the transformation $t \mapsto t + \alpha$ maps it into itself scaled by e^α . All geometric quantities follow the same exponential law — arc length, radius of curvature, distance from the pole — and this is why the nesting of the osculating circles can be seen with the naked eye: each circle is the image of the previous one under a similarity of ratio e .

Typical mistakes. (a) Using t as arc length: it is not, and the formula $k = \vartheta'$ must be corrected by dividing by $\|\gamma'\|$. (b) Taking N as the *outward* normal instead of the rotation of T by $+90^\circ$: one gets the centre on the wrong side. (c) Confusing the nesting of the discs with the inclusion of the curve in one of them: the curve crosses every osculating circle.

Extension. Check that for the general logarithmic spiral $\gamma(t) = e^{bt}(\cos t, \sin t)$ one has $\rho(t) = \frac{\sqrt{1+b^2}}{b}e^{bt}$ and that the evolute is again a logarithmic spiral with the same b ; discuss the limit $b \rightarrow 0^+$, in which the spiral degenerates into the unit circle and the evolute into its centre (Theorem 2.6(iv)).

Exercise 2.6: Minimum length of a transition for a mobile robot. *Type:* numerical/applied. A mobile robot travelling at $v = 1$ m/s must join a straight stretch to an arc of radius $r = 2$ m with a clothoid $k(s) = cs$. The maximum rate of change of the curvature is

$\dot{k}_{\max} = 0.5 \text{ m}^{-1}\text{s}^{-1}$. Determine the minimum length L of the stretch, the angular deviation and the shift, and check the estimate $L^2/(24r)$.

Solution. Strategy. One uses Proposition 2.3(iv) for the kinematic constraint and (i) for the length, then Proposition 2.4 for the shift.

Step 1 — constraint on the sharpness. From Step 5 of the proof of Proposition 2.3, $\dot{k} = cv$; imposing $\dot{k} \leq \dot{k}_{\max}$ one obtains $c \leq \dot{k}_{\max}/v = 0.5 \text{ m}^{-2}$.

Step 2 — minimum length. Continuity requires $cL = 1/r = 0.5 \text{ m}^{-1}$, hence $L = 1/(rc)$; since L decreases as c grows, the minimum length occurs for $c = 0.5 \text{ m}^{-2}$:

$$L_{\min} = \frac{1}{rc} = \frac{1}{2 \times 0.5} = 1 \text{ m}.$$

Step 3 — angular deviation. $\Delta\varphi = cL^2/2 = 0.5 \times 1/2 = 0.25 \text{ rad} = 14.323945^\circ$; alternatively $\Delta\varphi = L/(2r) = 1/4 = 0.25 \text{ rad}$. ✓ (The two formulae of Proposition 2.3(ii) agree, as they must.)

Step 4 — final point. $a = \sqrt{\pi/c} = \sqrt{2\pi} = 2.506628 \text{ m}$ and $\tau_L = L/a = 0.398942$. By composite Simpson quadrature $C(\tau_L) = 0.396456$ and $S(\tau_L) = 0.033097$, hence

$$\gamma(L) = a(C(\tau_L), S(\tau_L)) = (0.993768; 0.082962) \text{ m},$$

with chord $\|\gamma(L)\| = 0.997225 \text{ m}$, smaller than the length $L = 1 \text{ m}$ of the stretch, as it must be.

Step 5 — shift. The exact value is $p = y(L) - r(1 - \cos \Delta\varphi) = 0.082962 - 2 \times (1 - \cos 0.25) = 0.082962 - 0.062175 = 0.020787 \text{ m}$, against the estimate $L^2/(24r) = 1/48 = 0.020833 \text{ m}$: a discrepancy of 0.000046 m , that is 0.22% . Here $L/r = 0.5$, half of the case of Example 2.8, and the relative error drops from 0.90% to 0.22% , that is by a factor 4: consistent with the term $O(L^4/r^3)$ of Proposition 2.4, whose relative weight with respect to $L^2/(24r)$ is $O((L/r)^2)$. ✓

Interpretation. One metre of clothoid and two centimetres of displacement of the arc suffice to make the manoeuvre executable with steering at constant rate. If the transition were omitted, the steering would have to reach the angle corresponding to $r = 2 \text{ m}$ instantaneously: physically impossible, and in practice it translates into a trajectory error and an impulsive load on the actuators.

Typical mistakes. (a) Confusing $\dot{k}$ (derivative with respect to *time*) with $k' = c$ (derivative with respect to *arc length*): the conversion factor is the speed v . (b) Imposing $c \geq \dot{k}_{\max}/v$ instead of $c \leq$: one would obtain a maximum length instead of a minimum one. (c) Using $\Delta\varphi = cL$ instead of $cL^2/2$.

Extension. Build the complete curvature profile straight–clothoid–arc–clothoid–straight of Exercise 2.23 with $r = 2 \text{ m}$ and $L = 1 \text{ m}$ for both transitions, and compute the total length of the path needed for an overall deviation of 90° .

2.9. Numerical and symbolic exercises with solutions

The exercises below come from the treatment of plane curves in Volume II and are reproduced in full, together with their solutions.

2.9.1 Numerical exercises

Exercise 2.7: N1 — Osculating circle of a logarithm. For the curve $y = \ln x$ compute the curvature, the radius of curvature and the centre of curvature at the point $(1, 0)$, and check that the distance between point and centre equals the radius.

Exercise 2.8: N2 — Curvature of the cycloid. For the cycloid $\gamma(t) = r(t - \sin t, 1 - \cos t)$ compute the curvature as a function of t , the radius of curvature at $t = \pi$ and at $t = \pi/2$, and the length of one arch.

Exercise 2.9: N3 — Evolute of the parabola. Determine the evolute of the parabola $y = x^2$ and check, for $x = 1$, that the distance between point and centre of curvature equals the radius of curvature computed in Example 2.2.

Exercise 2.10: N4 — Circle involute for a gear. A gear has a base circle of radius $r = 50$ mm. Compute the coordinates of the involute point at $t = 1$ rad, the length of the involute from $t = 0$ to $t = 1$, and the radius of curvature there.

Exercise 2.11: N5 — A Cardano mechanism. A circle of radius $r = 50$ mm rolls internally in a fixed circle of radius $R = 100$ mm. Compute the stroke of the tracing point, its position at $t = 40^\circ$, and its maximum speed if the centre of the moving circle makes 30 revolutions per minute.

Exercise 2.12: N6 — Hypocycloids that close and hypocycloids that do not. Determine the number of cusps and the number of revolutions needed for closure for $R/r = 4$, $R/r = 5/2$ and $R/r = \pi$. For the first case compute the total length of the curve with $R = 40$ mm and compare it with the perimeter of the circumscribed circle.

2.9.2 Symbolic exercises

Exercise 2.13: S1 — The circle through three points. Complete Step 5 of the proof of Theorem 2.4: prove that the circle through $\gamma(s_0 - \varepsilon)$, $\gamma(s_0)$, $\gamma(s_0 + \varepsilon)$ tends, as $\varepsilon \rightarrow 0$, to the osculating circle.

Exercise 2.14: S2 — The hypocycloid in complex form. Write the hypocycloid of Definition 2.7 as a sum of two complex exponentials, and deduce Theorem 2.8 purely algebraically.

Exercise 2.15: S3 — Invariance of curvature. Prove that the signed curvature is invariant under rotations and translations, changes sign under reflections, and changes sign under reversal of orientation. Deduce that the geometric curvature $|k|$ is invariant under all isometries of the plane.

Exercise 2.16: S4 — The evolute of the ellipse. Verify by direct computation the formula for the evolute of the ellipse given in Example 2.4.

2.9.3 Solutions to the numerical exercises

Solution. N1. *Step 1 — derivatives.* $f(x) = \ln x$, $f'(x) = 1/x$, $f''(x) = -1/x^2$; at $x = 1$: $f = 0$, $f' = 1$, $f'' = -1$.

Step 2 — curvature. By Theorem 2.3,

$$k = \frac{f''}{(1 + f'^2)^{3/2}} = \frac{-1}{2^{3/2}} = \frac{-1}{2.828427} = -0.353553.$$

The negative sign means the curve is concave downwards (with x increasing).

Step 3 — radius of curvature. $\rho = 1/|k| = 2.828427 = 2\sqrt{2}$.

Step 4 — unit vectors. $\gamma'(x) = (1, 1)$, hence $T = \frac{1}{\sqrt{2}}(1, 1) = (0.707107; 0.707107)$ and, rotating by $+90^\circ$, $N = (-0.707107; 0.707107)$.

Step 5 — centre of curvature. By Definition 2.4,

$$c = \gamma + \frac{1}{k} N = (1, 0) + (-2.828427) (-0.707107; 0.707107) = (1 + 2, 0 - 2) = (3, -2).$$

Step 6 — check.

$$\|c - \gamma\| = \sqrt{(3 - 1)^2 + (-2 - 0)^2} = \sqrt{8} = 2.828427 = \rho. \checkmark$$

Step 7 — plausibility check. The centre lies *below* the curve, on the concave side, as it must for $f'' < 0$. $\checkmark$

Solution. N2. *Step 1 — derivatives.*

$$x' = r(1 - \cos t), \quad y' = r \sin t, \quad x'' = r \sin t, \quad y'' = r \cos t.$$

Step 2 — numerator.

$$x'y'' - y'x'' = r^2[(1 - \cos t) \cos t - \sin^2 t] = r^2[\cos t - \cos^2 t - \sin^2 t] = r^2(\cos t - 1).$$

Step 3 — denominator.

$$x'^2 + y'^2 = r^2[(1 - \cos t)^2 + \sin^2 t] = r^2(2 - 2 \cos t),$$

hence $(x'^2 + y'^2)^{3/2} = r^3(2 - 2 \cos t)^{3/2}$.

Step 4 — curvature. By Theorem 2.3, using $1 - \cos t = 2 \sin^2(t/2)$:

$$k(t) = \frac{-r^2(1 - \cos t)}{r^3(2(1 - \cos t))^{3/2}} = \frac{-1}{r 2^{3/2} \sqrt{1 - \cos t}} = \frac{-1}{r 2^{3/2} \cdot \sqrt{2} |\sin(t/2)|} = \frac{-1}{4r |\sin(t/2)|}.$$

Hence

$$\rho(t) = 4r |\sin(t/2)|.$$

Step 5 — required values.

$$\rho(\pi) = 4r \sin \frac{\pi}{2} = 4r, \quad \rho\left(\frac{\pi}{4}\right) = 4r \sin \frac{\pi}{4} = 4r \times 0.707107 = 2.828427 r.$$

Step 6 — length of one arch.

$$L = \int_0^{2\pi} \sqrt{x'^2 + y'^2} dt = \int_0^{2\pi} r \sqrt{2 - 2 \cos t} dt = \int_0^{2\pi} 2r \sin \frac{t}{2} dt = 2r \left[-2 \cos \frac{t}{2} \right]_0^{2\pi} = 8r.$$

Step 7 — reading. At the cusps ($t = 0, 2\pi$) we have $\rho = 0$: the curvature diverges, consistently with the fact that there the cycloid has a singular point ($\gamma' = 0$). At the highest point ($t = \pi$) the radius of curvature is $4r$, i.e. *four times* the radius of the generating circle: a notoriously counter-intuitive result, but confirmed by the computation.

Consistency check. The arch length $8r$ is four times the diameter of the generating circle; this is the result obtained by Christopher Wren in 1658 by pre-analytic methods.

Solution. N3. *Step 1 — formulas for a graph.* For $y = f(x)$, from Definition 2.5 and Theorem 2.3 one obtains

$$X = x - \frac{f'(1 + f'^2)}{f''}, \quad Y = f + \frac{1 + f'^2}{f''}.$$

Step 2 — application to $y = x^2$. With $f' = 2x$, $f'' = 2$:

$$X = x - \frac{2x(1 + 4x^2)}{2} = x - x - 4x^3 = -4x^3,$$

$$Y = x^2 + \frac{1 + 4x^2}{2} = \frac{1}{2} + 3x^2.$$

Step 3 — Cartesian form. From $x = (-X/4)^{1/3}$ and $x^2 = (Y - \frac{1}{2})/3$,

$$\left(\frac{X}{4}\right)^2 = \left(\frac{Y - \frac{1}{2}}{3}\right)^3,$$

a *semicubical parabola* (Neile's parabola), with cusp at $(0, \frac{1}{2})$, the centre of curvature of the vertex of the parabola.

Step 4 — check for $x = 1$. Point of the parabola: $(1, 1)$. Centre of curvature: $X = -4$, $Y = 0.5 + 3 = 3.5$. Distance:

$$\sqrt{(-4 - 1)^2 + (3.5 - 1)^2} = \sqrt{25 + 6.25} = \sqrt{31.25} = 5.590170.$$

In Example 2.2 we found $\rho(1) = 5.5902$. ✓

Step 5 — only one cusp. The evolute of the parabola has a single cusp, not four: there is no contradiction with Theorem 4.6, which concerns *closed* curves. The parabola is open, and has a single vertex (in the sense of Definition 4.4: $k' = 0$ only at $x = 0$).

Solution. N4. *Step 1 — coordinates.* From Example 2.5 with $r = 50$ mm and $t = 1$ rad:

$$\iota(1) = 50(\cos 1 + \sin 1, \sin 1 - \cos 1).$$

With $\cos 1 = 0.540302$ and $\sin 1 = 0.841471$:

$$\iota(1) = 50(1.381773, 0.301169) = (69.0887; 15.0584) \text{ mm}.$$

Step 2 — length of the involute. By Step 1 of Example 2.5,

$$\Lambda(1) = \frac{rt^2}{2} = \frac{50 \times 1}{2} = 25.00 \text{ mm}.$$

Step 3 — radius of curvature. By Step 2 of the same Example, $\tilde{\rho} = rt = 50.00$ mm.

Step 4 — check. The corresponding base-circle point is $\gamma(1) = 50(\cos 1, \sin 1) = (27.0151; 42.0735)$ mm; its distance from $\iota(1)$ is

$$\sqrt{(69.0887 - 27.0151)^2 + (15.0584 - 42.0735)^2} = \sqrt{1770.18 + 729.82} = \sqrt{2500.00} = 50.00 \text{ mm}. \checkmark$$

Step 5 — distance from the centre. $\|\iota(1)\| = \sqrt{4773.25 + 226.75} = \sqrt{5000.00} = 70.711$ mm, i.e. $r\sqrt{1+t^2} = 50\sqrt{2}$: a general formula following from Pythagoras applied to the right triangle with legs r and rt (the base-circle radius and the unwound thread, mutually orthogonal). ✓

Solution. N5. *Step 1 — checking the conditions.* $R = 2r$ with $r = 50$ mm: we are in the case of Theorem 2.8.

Step 2 — stroke. The point describes the diameter of the fixed circle, of length $2R = 200$ mm; the stroke is therefore 200 mm, i.e. ± 100 mm about the centre.

Step 3 — position at $t = 40^\circ$.

$$x(t) = 2r \cos t = 100 \cos 40^\circ = 100 \times 0.766044 = 76.604 \text{ mm}, \quad y = 0.$$

Step 4 — speed. At 30 rpm the angular velocity of the centre is

$$\omega = \frac{30 \times 2\pi}{60} = \pi \text{ rad/s} = 3.14159 \text{ rad/s}.$$

Since $x(t) = 100 \cos(\omega t)$ mm,

$$\dot{x} = -100 \omega \sin(\omega t), \quad |\dot{x}|_{\max} = 100 \times 3.14159 = 314.16 \text{ mm/s}.$$

Step 5 — reading. The motion is harmonic with amplitude 100 mm and angular frequency π rad/s, i.e. period 2 s: two seconds out and back. The speed is greatest at mid-stroke and zero at the ends, where the point reverses. It is the same velocity profile as an infinitely long connecting rod — and indeed the Cardano mechanism is used precisely to obtain a rigorously harmonic reciprocating motion, without the second-order error introduced by the ordinary crank-and-rod linkage.

Step 6 — acceleration. $\ddot{x} = -100\omega^2 \cos \omega t$, with maximum $100 \times 9.8696 = 986.96 \text{ mm/s}^2 = 0.987 \text{ m/s}^2$: modest, but proportional to the *square* of the rotational speed. At 300 rpm it would become 98.7 m/s^2 , about $10g$.

Solution. N6. *Step 1 — closure criterion.* The hypocycloid closes after q revolutions if $R/r = p/q$ with p, q coprime positive integers, and then has p cusps; it never closes if R/r is irrational.

Step 2 — the three cases.

R/r	cusps	revolutions to close	name
4	4	1	astroid
$5/2$	5	2	five-cusped hypocycloid
π	—	never	curve dense in the annulus

Step 3 — the irrational case. If $R/r \notin \mathbb{Q}$, the cusp points are $\{2\pi n R/r \bmod 2\pi\}$, the orbit of an irrational rotation of the circle: it is dense, so the curve never closes and densely fills the annulus with radii $R - 2r$ and R . It is the same phenomenon — and the same non-latticeness criterion — as in Theorem (Volume II, Chapter 6) of Chapter 8.

Step 4 — length of the astroid. With $R = 40$ mm and $r = 10$ mm the astroid is $\gamma(t) = R(\cos^3 t, \sin^3 t)$, hence

$$\gamma'(t) = 3R(-\cos^2 t \sin t, \sin^2 t \cos t), \quad \|\gamma'\| = 3R|\sin t \cos t| = \frac{3R}{2}|\sin 2t|.$$

Integrating over the period:

$$L = \frac{3R}{2} \int_0^{2\pi} |\sin 2t| dt = \frac{3R}{2} \times 4 = 6R = 240 \text{ mm},$$

using $\int_0^{2\pi} |\sin 2t| dt = 4$.

Step 5 — comparison. The circumscribed circle has radius $R = 40$ mm and perimeter $2\pi R = 251.33$ mm. The astroid, though touching it at four points, is shorter: $240/251.33 = 0.9549$, i.e. 95.5%. There is no contradiction with the isoperimetric inequality, which compares area with perimeter and not two perimeters: the astroid has area $\frac{3\pi R^2}{8} = 1885.0 \text{ mm}^2$ against $\pi R^2 = 5026.5 \text{ mm}^2$ for the circle, and its isoperimetric ratio is

$$\frac{4\pi A}{L^2} = \frac{4\pi \times 1885.0}{57600} = 0.4112,$$

far from 1: the astroid is a very “inefficient” curve. ✓

2.9.4 Solutions to the symbolic exercises

Solution. S1. *Step 1 — the three points.* Set $s_0 = 0$ and consider $P_- = \gamma(-\varepsilon)$, $P_0 = \gamma(0)$, $P_+ = \gamma(\varepsilon)$. By Taylor, with $T = T(0)$, $N = N(0)$, $k = k(0)$:

$$\gamma(\pm\varepsilon) - \gamma(0) = \pm\varepsilon T + \frac{\varepsilon^2}{2}k N + O(\varepsilon^3).$$

Step 2 — sides of the triangle.

$$|P_+P_0| = \varepsilon + O(\varepsilon^3), \quad |P_0P_-| = \varepsilon + O(\varepsilon^3), \quad |P_+P_-| = 2\varepsilon + O(\varepsilon^3).$$

Step 3 — area of the triangle.

$$2 \text{ Area} = \left| \det \left(\gamma(\varepsilon) - \gamma(0), \gamma(-\varepsilon) - \gamma(0) \right) \right|.$$

Substituting the expansions and using $\det(T, T) = \det(N, N) = 0$, $\det(T, N) = 1$, $\det(N, T) = -1$:

$$\det = \varepsilon \cdot \frac{\varepsilon^2 k}{2} \det(T, N) + \frac{\varepsilon^2 k}{2} \cdot (-\varepsilon) \det(N, T) + O(\varepsilon^4) = \varepsilon^3 k + O(\varepsilon^4).$$

Hence $\text{Area} = \frac{1}{2}|k|\varepsilon^3 + O(\varepsilon^4)$.

Step 4 — circumradius. By the classical formula (Volume I) $R_{\text{circ}} = \frac{abc}{4 \text{ Area}}$ with a, b, c the sides:

$$R_{\text{circ}} = \frac{\varepsilon \cdot \varepsilon \cdot 2\varepsilon + O(\varepsilon^4)}{4 \cdot \frac{1}{2}|k|\varepsilon^3 + O(\varepsilon^4)} = \frac{2\varepsilon^3(1 + O(\varepsilon))}{2|k|\varepsilon^3(1 + O(\varepsilon))} \xrightarrow{\varepsilon \rightarrow 0} \frac{1}{|k|} = \rho.$$

Step 5 — the centre. The circumcentre lies on the perpendicular bisector of P_-P_+ , which in the limit tends to the normal line of γ at P_0 ; since its distance from P_0 tends to ρ and it lies on the concave side (sign of k), the limit is exactly $\gamma(0) + \frac{1}{k}N(0)$, the centre of curvature of Definition 2.4.

Step 6 — the case $k = 0$. If $k(0) = 0$ the area is $o(\varepsilon^3)$ and the radius diverges: the circle through three points degenerates into the tangent line, consistently with the absence of an osculating circle at inflection points.

Solution. S2. *Step 1 — complex form.* Identifying $\mathbb{R}^2$ with $\mathbb{C}$, the hypocycloid of Definition 2.7 reads

$$z(t) = (R - r)e^{it} + r e^{-i\frac{R-r}{r}t}.$$

Indeed, taking real and imaginary parts and using Euler's formula (Theorem (Volume II, Chapter 3) of Chapter 3):

$$\operatorname{Re} z = (R - r) \cos t + r \cos \frac{R-r}{r}t, \quad \operatorname{Im} z = (R - r) \sin t - r \sin \frac{R-r}{r}t,$$

exactly the two given components.

Step 2 — kinematic reading. The formula exhibits the hypocycloid as the sum of two uniform circular motions: one of radius $R - r$ and angular velocity $+1$ (the centre of the moving circle about O), one of radius r and angular velocity $-\frac{R-r}{r}$ (the point about the moving centre, in the opposite sense). It is the same scheme as the phasors of Chapter 6 and as the Ptolemaic epicycles.

Step 3 — the case $R = 2r$. Substituting, $R - r = r$ and $\frac{R-r}{r} = 1$:

$$z(t) = r e^{it} + r e^{-it} = r(e^{it} + e^{-it}) = 2r \cos t,$$

by the inverse Euler formula (Theorem (Volume II, Chapter 3) of Chapter 3).

Step 4 — conclusion. $z(t)$ is a real number for every t : the point stays on the real axis, i.e. on the diameter of the fixed circle, moving along it with law $2r \cos t$, simple harmonic motion. This is Theorem 2.8, proved in four lines of algebra.

Step 5 — generalisation. The same computation shows that the hypocycloid is closed if and only if the two frequencies 1 and $-\frac{R-r}{r}$ are commensurable, i.e. if $R/r \in \mathbb{Q}$: the criterion of Exercise N6 is recovered with no geometry at all. In the language of Chapter 4 the curve is a *two-term trigonometric polynomial*: its closure is the periodicity of a sum of two exponentials.

Solution. S3. *Step 1 — translations.* If $\tilde{\gamma} = \gamma + b$ with b constant, then $\tilde{\gamma}' = \gamma'$: the unit tangent, the normal and all derivatives are unchanged, hence $\tilde{k} = k$.

Step 2 — rotations. Let A be a rotation, i.e. (in the complex language of Chapter 3) multiplication by $e^{i\alpha}$. Then $\tilde{T} = e^{i\alpha}T$ and, since rotating by $+90^\circ$ is multiplication by i , which commutes with $e^{i\alpha}$,

$$\tilde{N} = i\tilde{T} = ie^{i\alpha}T = e^{i\alpha}iT = e^{i\alpha}N.$$

Hence $\tilde{T}' = e^{i\alpha}T' = e^{i\alpha}kN = k\tilde{N}$, i.e. $\tilde{k} = k$. Arc length is unchanged because $|e^{i\alpha}| = 1$.

Step 3 — reflections. Let A be reflection in the real axis, i.e. conjugation $z \mapsto \bar{z}$. Then $\tilde{T} = \bar{T}$, but

$$\tilde{N} = i\tilde{T} = i\bar{T} = -i\bar{T} = -\bar{N},$$

because $\overline{iw} = -i\bar{w}$ for every w . Hence $\tilde{T}' = \overline{T'} = \bar{kN} = k\bar{N} = -k\tilde{N}$, i.e. $\tilde{k} = -k$. Every reflection is a composition of this one with a rotation, so the result holds for all of them.

Step 4 — reversal of orientation. With $\tilde{\gamma}(s) = \gamma(-s)$: $\tilde{T}(s) = -T(-s)$ and $\tilde{N}(s) = -N(-s)$, while

$$\frac{d\tilde{T}}{ds}(s) = T'(-s) = k(-s)N(-s) = -k(-s)\tilde{N}(s),$$

hence $\tilde{k}(s) = -k(-s)$: the curvature changes sign.

Step 5 — conclusion. The *signed* curvature is invariant under direct rigid motions (translations and rotations) and changes sign under the indirect ones and under reversal of the direction of travel. The *geometric* curvature $|k|$ is therefore invariant under all isometries of

the plane and under every reparametrisation: it is a quantity intrinsic to the figure, not to its description. ✓

Step 6 — check on the circle. The counterclockwise circle has $k = 1/r$, the clockwise one $k = -1/r$ (Theorem 2.2); the absolute value is $1/r$ in both cases, as it must be for what is the same figure. ✓

Solution. S4. *Step 1 — data.* $\gamma(t) = (a \cos t, b \sin t)$, with $\gamma'(t) = (-a \sin t, b \cos t)$ and

$$D(t) = \|\gamma'(t)\| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}.$$

Step 2 — unit vectors.

$$T = \frac{1}{D}(-a \sin t, b \cos t), \quad N = \frac{1}{D}(-b \cos t, -a \sin t),$$

having rotated T by $+90^\circ$ as in Definition 2.2.

Step 3 — curvature. From Example 2.2, $k = ab/D^3$, hence $1/k = D^3/(ab)$.

Step 4 — centre of curvature.

$$e = \gamma + \frac{1}{k} N = (a \cos t, b \sin t) + \frac{D^3}{ab} \cdot \frac{1}{D}(-b \cos t, -a \sin t),$$

that is, componentwise,

$$X = a \cos t - \frac{D^2}{a} \cos t = \frac{\cos t}{a}(a^2 - D^2), \quad Y = b \sin t - \frac{D^2}{b} \sin t = \frac{\sin t}{b}(b^2 - D^2).$$

Step 5 — simplification. With $D^2 = a^2 \sin^2 t + b^2 \cos^2 t$:

$$a^2 - D^2 = a^2(1 - \sin^2 t) - b^2 \cos^2 t = (a^2 - b^2) \cos^2 t,$$

$$b^2 - D^2 = b^2(1 - \cos^2 t) - a^2 \sin^2 t = (b^2 - a^2) \sin^2 t.$$

Substituting,

$$X = \frac{a^2 - b^2}{a} \cos^3 t, \quad Y = \frac{b^2 - a^2}{b} \sin^3 t,$$

which is the formula of Example 2.4. ✓

Step 6 — Cartesian form. From $\cos t = (\frac{aX}{a^2 - b^2})^{1/3}$ and $\sin t = (\frac{bY}{b^2 - a^2})^{1/3}$, squaring and adding:

$$(aX)^{2/3} + (bY)^{2/3} = (a^2 - b^2)^{2/3},$$

the equation of an astroid. ✓

Step 7 — limiting cases. For $a = b$ one gets $X = Y = 0$: the evolute is the centre, consistently with Theorem 2.6(iv). For $b \rightarrow 0$ (ellipse degenerating into a segment) the cusps go to infinity along Y , consistently with the curvature tending to zero along the long sides. ✓

2.9.5 Exercises from Chapter 2 of Volume III

The exercises below come from the plane-curves chapter of the previous edition of Volume III and are reproduced with their solutions.

Exercise 2.17. Compute the arc length of the logarithmic spiral $\gamma(t) = (e^t \cos t, e^t \sin t)$ from $t = 0$ to $t = T$.

Step-by-step proof. Passo 1. $\gamma'(t) = e^t(\cos t - \sin t, \sin t + \cos t)$. **Passo 2.** $|\gamma'(t)|^2 = e^{2t}[(\cos t - \sin t)^2 + (\sin t + \cos t)^2] = e^{2t} \cdot 2$, so $|\gamma'(t)| = \sqrt{2} e^t$. **Passo 3.** $s = \int_0^T \sqrt{2} e^t dt = \sqrt{2}(e^T - 1)$. $\square$

Exercise 2.18. Compute the curvature of the spiral $\gamma(t) = (e^t \cos t, e^t \sin t)$.

Solution. Passo 1. $x' = e^t(\cos t - \sin t)$, $y' = e^t(\sin t + \cos t)$; $x'' = -2e^t \sin t$, $y'' = 2e^t \cos t$. **Passo 2.** $x'y'' - y'x'' = e^{2t}[(\cos t - \sin t)2 \cos t - (\sin t + \cos t)(-2 \sin t)] = 2e^{2t}$. **Passo 3.** $(x'^2 + y'^2)^{3/2} = (2e^{2t})^{3/2} = 2\sqrt{2} e^{3t}$. **Passo 4.** $\kappa = \frac{2e^{2t}}{2\sqrt{2} e^{3t}} = \frac{1}{\sqrt{2} e^t}$: the curvature decays exponentially along the spiral.

Exercise 2.19. Determine the center and radius of the osculating circle of the parabola $y = x^2$ at the point $(1, 1)$.

Solution. With $\gamma(t) = (t, t^2)$ at $t = 1$: $\gamma' = (1, 2)$, $\gamma'' = (0, 2)$, $|\gamma'|^2 = 5$. Curvature $\kappa = \frac{1 \cdot 2 - 2 \cdot 0}{5^{3/2}} = \frac{2}{5\sqrt{5}}$, hence $\rho = \frac{5\sqrt{5}}{2}$. The unit tangent is $T = \frac{1}{\sqrt{5}}(1, 2)$, the normal $+90^\circ$ is $N = \frac{1}{\sqrt{5}}(-2, 1)$. Center:

$$c = (1, 1) + \frac{1}{\kappa}N = (1, 1) + \frac{5\sqrt{5}}{2} \cdot \frac{1}{\sqrt{5}}(-2, 1) = (1, 1) + \frac{5}{2}(-2, 1) = \left(-4, \frac{7}{2}\right).$$

Osculating circle: center $(-4, 7/2)$, radius $5\sqrt{5}/2 \approx 5.59$.

Exercise 2.20. Verify that, for the ellipse with $a > b$, the radius of curvature is maximum on the minor axis and minimum on the major axis.

Solution. $\rho(t) = \frac{1}{ab}g(t)^{3/2}$ with $g(t) = a^2 \sin^2 t + b^2 \cos^2 t = b^2 + (a^2 - b^2) \sin^2 t$. Since $a > b$, g is minimum for $\sin t = 0$ (major axis, $\rho = b^2/a$) and maximum for $\sin^2 t = 1$ (minor axis, $\rho = a^2/b$). Since $a^2/b > b^2/a$, the claim follows.

Exercise 2.21. Compute the radius of curvature of the catenary $y = a \cosh(x/a)$ at the vertex $x = 0$ and interpret its value.

Solution. From Example 2.14, $\rho = a \cosh^2(x/a)$. At $x = 0$: $\cosh 0 = 1$, hence $\rho(0) = a$. At the vertex the radius of curvature equals the parameter a of the catenary (the minimum height of the curve above the directrix): the osculating circle at the vertex has center at $(0, 2a)$ and radius a .

Exercise 2.22. For the cardioid in polar form $r(\theta) = 1 + \cos \theta$, compute the curvature using the polar formula $\kappa = \frac{r^2 + 2r'^2 - rr''}{(r^2 + r'^2)^{3/2}}$.

Solution. Passo 1. $r = 1 + \cos \theta$, $r' = -\sin \theta$, $r'' = -\cos \theta$. **Passo 2.** Numerator: $r^2 + 2r'^2 - rr'' = (1 + \cos \theta)^2 + 2 \sin^2 \theta + (1 + \cos \theta) \cos \theta$. Expanding: $1 + 2 \cos \theta + \cos^2 \theta +$

$2\sin^2\theta + \cos\theta + \cos^2\theta = 3 + 3\cos\theta = 3(1 + \cos\theta)$. **Passo 3.** Denominator: $r^2 + r'^2 = (1 + \cos\theta)^2 + \sin^2\theta = 2 + 2\cos\theta = 2(1 + \cos\theta)$, raised to $3/2$: $[2(1 + \cos\theta)]^{3/2}$. **Passo 4.**

$$\kappa = \frac{3(1 + \cos\theta)}{[2(1 + \cos\theta)]^{3/2}} = \frac{3}{2^{3/2}\sqrt{1 + \cos\theta}} :$$

the curvature diverges at the cusp ($\theta = \pi$, $r = 0$), consistent with the geometry of the cardioid.

Exercise 2.23. A robotic path consists of: straight line, clothoid, circle arc of radius r , clothoid, straight line. Write the piecewise curvature law $\kappa(s)$ and verify that it is continuous.

Solution. With lengths L_1 (entry clothoid) and arc starting at $s = s_1$:

$$\kappa(s) = \begin{cases} 0 & \text{initial straight line} \\ c(s - s_0) & \text{clothoid, } s \in [s_0, s_0 + L_1], \ c = \frac{1}{rL_1} \\ 1/r & \text{arc} \\ 1/r - c(s - s_2) & \text{exit clothoid} \\ 0 & \text{final straight line} \end{cases}$$

At the ends of each segment the values match ($0 \rightarrow 0$, then $cL_1 = 1/r$, then $1/r \rightarrow 1/r$, then $1/r$ decreases to 0): κ is continuous and piecewise linear. It is the typical curvature profile of an industrial trajectory with limited jerk.

2.10. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher's key (Section 2.11).

- (Definition) At the point $\gamma(s_0)$ of a curve with $k(s_0) \neq 0$, the *osculating circle* is the circle with
 - centre $\gamma(s_0) + k(s_0)N(s_0)$ and radius $|k(s_0)|$
 - centre $\gamma(s_0) + \frac{1}{k(s_0)}N(s_0)$ and radius $1/|k(s_0)|$
 - centre $\gamma(s_0) + \frac{1}{k(s_0)}T(s_0)$ and radius $1/|k(s_0)|$
 - centre $\gamma(s_0)$ and radius $1/|k(s_0)|$
- (Hypothesis) In the Fundamental theorem of plane curves, the hypothesis that guarantees $\gamma \in C^2$ is
 - k bounded
 - k continuous
 - k differentiable
 - k of constant sign
- (Symbolic computation) The radius of curvature of the catenary $y = a \cosh(x/a)$ equals
 - $a \cosh(x/a)$
 - $a \cosh^2(x/a)$
 - $a \sinh^2(x/a)$
 - $a^2 \cosh(x/a)$
- (Numerical computation) For a clothoid transition $k(s) = cs$ with $c = 0.04 \text{ m}^{-2}$ towards an arc of radius $r = 5 \text{ m}$, the length of the stretch and the angular deviation are
 - $L = 5 \text{ m}$, $\Delta\varphi = 0.5 \text{ rad}$
 - $L = 5 \text{ m}$, $\Delta\varphi = 0.25 \text{ rad}$
 - $L = 0.2 \text{ m}$, $\Delta\varphi = 0.5 \text{ rad}$
 - $L = 25 \text{ m}$, $\Delta\varphi = 12.5 \text{ rad}$
- (Proof step) In the proof of the Tait–Kneser theorem, the inequality $\|e(s_2) - e(s_1)\| < |\rho(s_2) - \rho(s_1)|$ is *strict* because

-
- A) ρ is strictly monotone B) equality would force N to be constant, whereas $\|N'\| = k > 0$ C) the evolute is regular D) the two discs are disjoint
6. (Error diagnosis) A student computes the curvature of the spiral $\gamma(t) = e^t(\cos t, \sin t)$ as $k = \vartheta'(t)$, where ϑ is the tangent angle, and obtains $k \equiv 1$. The mistake is
- A) the formula $k = \vartheta'$ holds only for convex curves B) t is not the arc length: one needs $k = \vartheta'(t)/\|\gamma'(t)\|$ C) the tangent angle of the spiral is not differentiable D) the absolute value of ϑ' must be used
7. (Transfer) A wheeled vehicle cannot travel a direct straight-line $\rightarrow$ arc transition because
- A) the trajectory would not be continuous B) the curvature would jump from 0 to $1/r$ and the steering angle would have to change instantaneously C) the arc length would not be defined D) the circular arc cannot be parametrised by arc length
8. (Source and metacognition) The reduction of the clothoid to the Fresnel integrals and the limit values $C(\infty) = S(\infty) = \frac{1}{2}$ are found in
- A) do Carmo, *Differential Geometry of Curves and Surfaces*, §1-5 B) NIST Digital Library of Mathematical Functions, §7.2, formulae (7.2.7)–(7.2.9) C) Hopf, *Über die Drehung der Tangenten...* D) Struik, *Lectures on Classical Differential Geometry*, ch. 4

2.10.1 Supplementary questionnaire

- The curvature of a circle of radius r traversed counterclockwise is

A) r B) $1/r$ C) r^2 D) $2\pi r$
- The osculating circle has contact with the curve of order

A) zero B) first C) second D) third
- A plane curve with constant non-zero curvature is

A) a line B) an arc of a circle C) a parabola D) an ellipse
- The evolute of a curve is

A) the parallel curve B) the locus of the centres of curvature C) the reflected curve D) the tangent
- The evolute is the envelope of

A) the tangents B) the normals C) the chords D) the secants
- The evolute reduces to a point if and only if the curve is

A) a line B) a circle C) a parabola D) an astroid
- The circle-involute profile is used in gears because

A) it is easier to cut B) it keeps the gear ratio constant even when the centre distance varies C) it reduces weight D) it is stronger
- The radius of curvature of the cycloid at its highest point is

A) r B) $2r$ C) $4r$ D) $8r$
- A circle of radius r rolling inside one of radius $2r$ generates

A) a cardioid B) a diameter C) an astroid D) a circle

10. The Fundamental theorem of plane curves states that $k(s)$ determines the curve
 A) uniquely B) up to a direct rigid motion C) up to a homothety D) only if $k > 0$
11. A clothoid is characterised by curvature
 A) constant B) zero C) affine in the arc length D) quadratic in time
12. In SINUMERIK G-code a clockwise circular arc is programmed with
 A) G01 B) G02 C) G03 D) G00

BEST PRACTICE

Answer key.

1-B: Theorem 2.2. **2-C:** Theorem 2.4. **3-B:** Theorem 2.5(ii). **4-B:** Definition 2.5.
5-B: Theorem 2.6(i). **6-B:** Theorem 2.6(iv). **7-B:** Example 2.5, Steps 3–4. **8-C:**
 Exercise N2: $\rho(\pi) = 4r$. **9-B:** Theorem 2.8. **10-B:** Theorem 2.9(iii). **11-C:**
 Definition 2.8, $k(s) = k_0 + cs$. **12-B:** Section 2.6.2; G03 is the counterclockwise one.

2.11. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnosis.

- 1) **B.** *Rationale:* this is Definition 2.4: the centre lies on the normal at distance $\rho = 1/|k|$, on the side towards which the curve bends, and the radius is ρ . *Distractors:* A swaps radius and curvature; C puts the centre on the tangent instead of the normal; D leaves the centre on the curve. *Diagnosed error:* confusion between k and $\rho = 1/|k|$. *Source:* Theorem 2.4. *Prerequisite:* the unit vectors T and N . *Remedial work:* redo Exercise 2.19 drawing the circle.
- 2) **B.** *Rationale:* in Step 2 of the proof of Theorem 2.9, the continuity of k gives $\vartheta \in C^1$ and hence $\gamma \in C^2$; nothing more is needed. *Distractors:* A is not enough (a bounded but discontinuous k makes ϑ only Lipschitz); C is stronger than necessary and would give $\gamma \in C^3$; D is irrelevant. *Diagnosed error:* asking of the hypothesis more than the proof uses. *Source:* Theorem 2.9, Step 2. *Prerequisite:* the fundamental theorem of integral calculus. *Remedial work:* reread Step 2 marking where continuity enters.
- 3) **B.** *Rationale:* from Example 2.3, $k = 1/(a \cosh^2(x/a))$ and $\rho = a \cosh^2(x/a) = y^2/a$. *Distractors:* A is y , that is the ordinate, not the radius; C confuses $\cosh$ with $\sinh$; D gets the power of a wrong and is not even homogeneous. *Diagnosed error:* forgetting the exponent $3/2$ in the denominator of the formula for graphs. *Source:* Theorem 2.3. *Prerequisite:* hyperbolic functions. *Remedial work:* a dimensional check, $[\rho] = \text{length}$.
- 4) **A.** *Rationale:* by Proposition 2.3(i), $L = 1/(rc) = 1/(5 \times 0.04) = 5$ m, and by (ii) $\Delta\varphi = cL^2/2 = 0.04 \times 25/2 = 0.5$ rad (Example 2.8). *Distractors:* B uses $\Delta\varphi = L/(4r)$; C swaps the roles of r and c ; D forgets to divide by r . *Diagnosed error:* using $\Delta\varphi = cL$ instead of $cL^2/2$. *Source:* Proposition 2.3. *Prerequisite:* integration of an affine function. *Remedial work:* redo Steps 1–2 of Example 2.8.
- 5) **B.** *Rationale:* this is exactly Step 3 of the proof of Proposition 2.1: equality in the integral triangle inequality would force N to be constant, but $N' = -kT$ has norm $k > 0$. *Distractors:* A gives only the *weak* inequality (Step 2); C and D are consequences, not causes. *Diagnosed error:* confusing the weak inequality with the strict one. *Source:* Proposition 2.1, Steps 2–3. *Prerequisite:* the triangle inequality for vector-valued integrals. *Remedial work:* rewrite Step 3 isolating the case of equality.
- 6) **B.** *Rationale:* the formula $k = d\vartheta/ds$ is with respect to arc length; in an arbitrary

parameter one has $k = \vartheta'(t)/\|\gamma'(t)\|$. For the spiral $\vartheta' = 1$ and $\|\gamma'\| = \sqrt{2}e^t$, whence $k = 1/(\sqrt{2}e^t)$ (Exercise 2.5, Step 3). *Distractors*: A invents a non-existent hypothesis; C is false ($\vartheta(t) = t + \pi/4$ is smooth); D does not change the size of the error. *Diagnosed error*: using a non-natural parameter as if it were the natural one. *Source*: Theorem 2.3. *Prerequisite*: the chain rule. *Remedial work*: recompute k for the spiral with the general formula and compare.

7) B. *Rationale*: this is Remark 2.3 and Section 2.6.1: k is proportional to the steering angle, which is a continuous physical variable. *Distractors*: A is false, the trajectory is continuous (indeed C^1); C and D are meaningless statements, the arc is regular. *Diagnosed error*: confusing the regularity of the trajectory with that of the curvature. *Source*: Proposition 2.3, Step 5. *Prerequisite*: the notion of continuity. *Remedial work*: draw $k(s)$ for the path of Exercise 2.23.

8) B. *Rationale*: definitions (7.2.7)–(7.2.8) and limit (7.2.9) of the DLMF are exactly what Proposition 2.2(ii) and Remark 2.4 use. *Distractors*: A deals with plane curves but not with the Fresnel integrals; C is about the rotation index; D does not contain the limit values. *Diagnosed error*: citing a differential-geometry textbook for a result of special-function analysis. *Source*: NIST DLMF, §7.2. *Prerequisite*: being able to tell which kind of source fits which claim. *Remedial work*: check formula (7.2.9) online and compare it with Remark 2.4.

BEST PRACTICE

Assigned exercises — answers.

2.1 (hyperbola $xy = 1$). (a) $k(t) = \frac{2t^3}{(t^4 + 1)^{3/2}}$, hence $k(1) = 2/2^{3/2} = 1/\sqrt{2} = 0.707107$.

(b) $\rho(1) = \sqrt{2} = 1.414214$; $T(1) = \frac{1}{\sqrt{2}}(1, -1)$, $N(1) = \frac{1}{\sqrt{2}}(1, 1)$, centre $(1, 1) + \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1) = (2, 2)$. (c) $e(t) = \left(\frac{3t^4 + 1}{2t^3}, \frac{t^4 + 3}{2t} \right)$; for $t = 1$ this gives $(2, 2)$, consistent with (b). (d)

$\rho(t) = \frac{(t^4 + 1)^{3/2}}{2t^3}$ and $\rho'(t) = \frac{3(t^4 + 1)^{1/2}(t^4 - 1)}{2t^4}$, which for $t > 0$ vanishes only at $t = 1$: a single vertex, a minimum (ρ decreases for $t < 1$ and increases for $t > 1$). (e) $e(0.7) = (2.507726; 2.314357)$ and $e(1.6) = (2.522070; 2.985500)$; `verifica_cap02_cap04.py`, block [C2-1]. ✓

2.2 (evolute of the cycloid). (a) $\gamma' = r(1 - \cos t, \sin t)$, $\|\gamma'\|^2 = r^2(2 - 2\cos t) = 4r^2 \sin^2 \frac{t}{2}$, hence $\|\gamma'\| = 2r \sin \frac{t}{2}$ on $(0, 2\pi)$; with $k = 1/(4r \sin \frac{t}{2})$ one gets $\rho = 4r \sin \frac{t}{2}$. (b) $e(t) = r(t + \sin t, \cos t - 1)$. (c) with $u = t + \pi$: $\sin u = -\sin t$ and $\cos u = -\cos t$, hence $r(u - \sin u) - \pi r = r(t + \pi + \sin t) - \pi r = r(t + \sin t)$ and $r(1 - \cos u) - 2r = r(1 + \cos t) - 2r = r(\cos t - 1)$: this coincides with (b). (d) By Theorem 2.6(iii) the length of the arc of the evolute is $|\rho(\pi) - \rho(\pi/2)| = 4r - 4r \sin \frac{\pi}{4} = 4r - 2\sqrt{2}r = 1.171573r$, that is 14.64% of the length $8r$ of one arch. (e) Because a pendulum whose suspension is constrained between two cycloidal arcs describes the involute of the cycloid, which is again a cycloid (Huygens 1673); on a cycloid the period of the oscillations does not depend on the amplitude. Numerical checks in `verifica_cap02_cap04.py`, block [C2-4]. ✓

2.3 (clothoid transition). (a) $c \leq \dot{k}_{\max}/v = 0.02/8 = 0.0025 \text{ m}^{-2}$; with equality, $L = 1/(rc) = 1/(40 \times 0.0025) = 10 \text{ m}$. (b) $\Delta\varphi = cL^2/2 = 0.125 \text{ rad} = 7.161972^\circ$ (and also $L/(2r) = 10/80$). (c) $a = \sqrt{\pi/0.0025} = 35.449077 \text{ m}$, $\tau_L = 0.282095$, $C(\tau_L) = 0.281654$, $S(\tau_L) = 0.011741$, hence $\gamma(L) = (9.984386; 0.416202) \text{ m}$ and chord 9.993057 m . (d) $p = 0.416202 - 40(1 - \cos 0.125) = 0.104109 \text{ m}$ against $L^2/(24r) = 100/960 = 0.104167 \text{ m}$: relative discrepancy 0.0558%. The term $O(L^4/r^3)$ of Proposition 2.4 weighs, relative to the leading term $L^2/(24r)$, as $O((L/r)^2)$: passing from $L/r = 1$ (Example 2.8, discrepancy

0.90%) to $L/r = 0.25$ the relative discrepancy must be divided by 16, and indeed $0.90/16 = 0.056$. ✓(e) With $\dot{k}_{\max} = 0.01 \text{ m}^{-1}\text{s}^{-1}$: $c = 0.00125 \text{ m}^{-2}$, $L = 20 \text{ m}$, $\Delta\varphi = 0.25 \text{ rad} = 14.323945^\circ$, $a = 50.132565 \text{ m}$, $\tau_L = 0.398942$, $\gamma(L) = (19.875361; 1.659241) \text{ m}$, $p = 0.415738 \text{ m}$ against $L^2/(24r) = 0.416667 \text{ m}$ (discrepancy 0.22%). The transition doubles in length and the offset quadruples: one gains in comfort (the lateral acceleration grows more slowly) and loses in available space. All the values are recomputed in `verifica_cap02_cap04.py`, block [C2-3]. ✓

2.12. Chapter formulary

Length and arc length

$$L = \int_a^b \|\gamma'(t)\| dt, \quad \frac{ds}{dt} = \|\gamma'(t)\|$$

Frenet formulas in the plane

$$T' = k N, \quad N' = -k T, \quad k = \frac{d\vartheta}{ds}, \quad N = iT \quad (\text{in } \mathbb{C})$$

Curvature

$$k = \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}}, \quad k_{\text{graph}} = \frac{f''}{(1 + f'^2)^{3/2}}, \quad \rho = \frac{1}{|k|}$$

$$\text{circle of radius } r : k = \frac{1}{r}; \quad \text{line} : k = 0$$

Osculating circle and evolute

$$c(s) = \gamma(s) + \frac{1}{k(s)}N(s), \quad e'(s) = \rho'(s)N(s), \quad L_{\text{evolute}} = |\Delta\rho|$$

Involute

$$\iota_a(s) = \gamma(s) + (a - s)T(s), \quad \iota'_a = (a - s)k N$$

$$\text{circle involute} : \iota(t) = r(\cos t + t \sin t, \sin t - t \cos t), \quad \tilde{\rho} = rt, \quad \Lambda = \frac{rt^2}{2}$$

Rolling curves

$$\text{cycloid} : \gamma = r(t - \sin t, 1 - \cos t), \quad \rho = 4r|\sin \frac{t}{2}|, \quad L_{\text{arch}} = 8r$$

$$\text{hypocycloid (in } \mathbb{C}) : z(t) = (R - r)e^{it} + re^{-i\frac{R-r}{r}t}; \quad R = 2r \Rightarrow z = 2r \cos t$$

Evolute of the ellipse

$$e(t) = \left(\frac{a^2 - b^2}{a} \cos^3 t, \frac{b^2 - a^2}{b} \sin^3 t \right), \quad \rho_{\max} = \frac{a^2}{b}, \quad \rho_{\min} = \frac{b^2}{a}$$

Fundamental theorem (synthesis of a curve)

$$\vartheta(s) = \vartheta_0 + \int_0^s k, \quad \gamma(s) = p_0 + \int_0^s (\cos \vartheta, \sin \vartheta),$$

$$k \equiv 0 \Rightarrow \text{straight line}; \quad k \equiv \frac{1}{r} \Rightarrow \text{circle}; \quad k = k_0 + cs \Rightarrow \text{clothoid}$$

Nested osculating circles (Tait–Kneser)

$$k > 0, \quad k' \neq 0 \implies \|e(s_2) - e(s_1)\| < |\rho(s_2) - \rho(s_1)|$$

Clothoid and Fresnel integrals

$$\varphi(s) = \frac{cs^2}{2}, \quad \gamma(s) = a(C(\tau), S(\tau)), \quad a = \sqrt{\frac{\pi}{c}}, \quad \tau = \frac{s}{a}, \quad \rho(s)s = \frac{1}{c}$$

$$C(\tau) = \int_0^\tau \cos \frac{\pi u^2}{2} du, \quad S(\tau) = \int_0^\tau \sin \frac{\pi u^2}{2} du$$

Straight-line-arc transition

$$L = \frac{1}{rc}, \quad \Delta\varphi = \frac{cL^2}{2} = \frac{L}{2r}, \quad \dot{k} = cv \leq \dot{k}_{\max} \Rightarrow L \geq \frac{v}{r \dot{k}_{\max}}$$

$$p = y(L) - r(1 - \cos \Delta\varphi) = \frac{L^2}{24r} + O\left(\frac{L^4}{r^3}\right)$$

Catenary

$$y = a \cosh \frac{x}{a} \Rightarrow k = \frac{1}{a \cosh^2(x/a)}, \quad \rho = a \cosh^2 \frac{x}{a} = \frac{y^2}{a}$$

Sources and bibliography

- [1] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, Englewood Cliffs, 1976 (the author was at IMPA, Rio de Janeiro). Reference text for the Frenet formulas, the rotation index, the four-vertex theorem and the isoperimetric inequality; main source of Sections 2.2 and 4.1. South American source.
- [2] D. J. Struik, *Lectures on Classical Differential Geometry*, Addison-Wesley, Reading, 1950 (2nd ed. Dover, 1988). Evolutes, involutes, cycloidal curves; a classical treatment with extensive historical framing.
- [3] A. Gray, E. Abbena, S. Salamon, *Modern Differential Geometry of Curves and Surfaces with Mathematica*, 3rd ed., Chapman & Hall/CRC, 2006. Explicit formulas for evolutes, involutes, cycloids and their numerical checks.
- [4] Nasir al-Din al-Tusi, *al-Tadhkira fi 'ilm al-hay'a* (Memoir on Astronomy), c. 1261. First description of the “al-Tusi couple”, the mechanism of Theorem 2.8. Middle Eastern source.
- [5] G. Cardano, *De subtilitate libri XXI*, Nuremberg, 1550. Description of “Cardano’s circles”, whence the Western name of the same mechanism.
- [6] J. Wallis, *Tractatus duo. Prior, de cycloide...*, Oxford, 1659. Publication of Christopher Wren’s 1658 rectification of the cycloid ($L = 8r$, checked in Exercise N2).
- [7] ISO 53:1998, *Cylindrical gears for general and heavy engineering — Standard basic rack tooth profile*. The standard codifying the circle-involute profile discussed in Example 2.5.
- [8] MIT OpenCourseWare, *18.950 Differential Geometry* — <https://ocw.mit.edu/>. Teaching material on plane curves, curvature and global theorems.
- [9] NPTEL, Indian Institute of Technology, *Differential Geometry of Curves and Surfaces* — <https://nptel.ac.in/>. Asian university course on the same topics.
- [10] UNAM, Facultad de Ciencias, *Geometría Diferencial*, university syllabus. Latin American reference.

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- [11] C. A. Barbieri, D. Miller-Cotto, S. N. Clerjuste, K. Chawla, *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*, Educational Psychology Review 35, art. 11, 2023 — <https://doi.org/10.1007/s10648-023-09745-1>.
- [12] H. L. Roediger III, J. D. Karpicke, *Test-Enhanced Learning*, Psychological Science 17(3), 2006 — <https://doi.org/10.1111/j.1467-9280.2006.01693.x>.
- [13] E. Ghys, S. Tabachnikov, V. Timorin, *Osculating curves: around the Tait–Kneser theorem*, The Mathematical Intelligencer **35**(1), 2013, pp. 61–66. The nesting of osculating circles along a vertex-free arc, recalled in Remark 4.7.
- [14] A. Pressley, *Elementary Differential Geometry*, 2nd ed., Springer Undergraduate Mathematics Series, Springer, London, 2010, XII+474 pp., ISBN 978-1-84882-890-2, chs. 1–2 (plane curves, signed curvature, fundamental theorem). Verified on 5 September 2026 at <https://link.springer.com/book/10.1007/978-1-84882-891-9>.
- [15] NIST, *Digital Library of Mathematical Functions*, ch. 7 «Error Function, Dawson’s and Fresnel Integrals», §7.2 «Definitions», formulae (7.2.7), (7.2.8) and (7.2.9). Definition of the Fresnel integrals C and S used in Proposition 2.2 and limit values $C(\infty) = S(\infty) = \frac{1}{2}$ declared as imported in Remark 2.4. Verified on 5 September 2026 at <https://dlmf.nist.gov/7.2>.
- [16] A. Shaju, S. Southward, M. Ahmadian, *Enhancing Autonomous Vehicle Navigation with a Clothoid-Based Lateral Controller*, Applied Sciences **14** (2024), no. 5, art. 1817, 21 pp., doi:10.3390/app14051817 (published on 22 February 2024; a version is also available as arXiv:2608.07740). Industrial use of clothoid transitions with adaptive *lookahead*, referred to in Section 2.6.1. Verified on 5 September 2026 at <https://doi.org/10.3390/app14051817> and <https://arxiv.org/abs/2608.07740>.
- [17] S. Oh, Q. Chen, H. E. Tseng, G. Pandey, G. Orosz, *Sharable Clothoid-based Continuous Motion Planning for Connected Automated Vehicles*, arXiv:2312.10880 (18 December 2023). Real-time planning with three consecutive clothoids; it confirms the role of the constraint $\dot{k} \leq \dot{k}_{\max}$ of Proposition 2.3(iv). Verified on 5 September 2026 at <https://arxiv.org/abs/2312.10880>.

CHAPTER 3

Space curves: Frenet–Serret and torsion

A plane curve never leaves its own plane: one number describes it, the signed curvature, and the fundamental theorem of plane curves (Chapter 2) says that this number determines it up to a rigid motion. In space a second degree of freedom appears: besides bending, a curve may *twist*, that is, leave the plane that best approximates it. The quantity measuring this departure is the **torsion**, and the fundamental theorem of space curves states that the pair (κ, τ) determines the curve up to a direct rigid motion: two functions of one variable contain the whole geometry of a thread in space.

The chapter follows a single line. First the moving Frenet frame $\{T, N, B\}$ is built and its derivatives are shown to be expressible in terms of κ and τ alone (Section 3.1); then the fundamental theorem, existence and uniqueness, is proved (Section 3.2) by reducing it to a linear Cauchy problem whose solvability is proved in full in the chapter appendix (Section 3.11); the theorem is then used to recognise the **circular helix** as the only curve with constant non-zero curvature and torsion, and generalised helices are characterised by Lancret's theorem (Section 3.3); the **osculating circle** and the **osculating sphere** are introduced (Section 3.4); the chapter closes with the necessary and sufficient condition for a curve to lie on a sphere (Section 3.5).

The circle appears at every step: it is the osculating circle that defines curvature; it is a circle that a curve with $\tau \equiv 0$ and constant κ must be; it is the circular helix — the circle «drawn out» along its own axis — that is the only curve with constant κ and τ ; and it is again a circle that the osculating sphere cuts on the osculating plane.

INFO

Level and prerequisites. Level **B–C**; Section 3.5 and the appendix are of level **C**.

Internal prerequisites. Arc length, reparametrisation by arc length, curvature and osculating circle of plane curves, fundamental theorem of plane curves (Chapter 2); the orthogonal group, $SO(3)$ and direct rigid motions (Chapter 5); scalar product, cross product and triple product in $\mathbb{R}^3$ (Volume I, Chapter 2).

External prerequisites, assumed and not reproved. Exactly one, stated in full in Remark 3.6: the completeness of the space $C([\alpha, \beta], \mathbb{R}^n)$ of continuous functions with the maximum norm. Everything else — including existence and uniqueness for the Frenet system and the contraction mapping theorem — is proved in this chapter.

What you will learn. To build the Frenet frame of a space curve and derive the Frenet–Serret formulas from it; to compute κ and τ in an arbitrary parameter and to recognise when a curve is a straight line, when it is planar, when it is a circle, when it is a helix; to use the fundamental theorem as a *tool* (to identify a curve from κ and τ , not merely to contemplate its statement); to determine osculating circles and spheres; to decide with a single computation whether a curve lies on a sphere, and to know why the condition « $\rho^2 + (\rho'\sigma)^2$ constant», which one often reads, is *not* enough.

Warning. Chapter conventions.

- A curve is a map $\gamma : I \rightarrow \mathbb{R}^3$ defined on an open interval $I \subseteq \mathbb{R}$; the prime always denotes the derivative with respect to the parameter written, and s is reserved for arc length (*unit-speed parametrisation*: $|\gamma'(s)| \equiv 1$).
- $\langle \cdot, \cdot \rangle$ is the standard scalar product, $u \times v$ the cross product, $\det(u, v, w) = \langle u \times v, w \rangle$ the triple product.
- **Sign of the torsion.** We adopt

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N,$$

that is $\tau = -\langle B', N \rangle$; with this choice the right-handed helix has $\tau > 0$ and $\tau = \langle \gamma' \times \gamma'', \gamma''' \rangle / |\gamma' \times \gamma''|^2$. **do Carmo, Gallier and Powell use the opposite convention** ($B' = +\tau N$): in their texts the torsion of the right-handed helix is negative and the formula in an arbitrary parameter carries a minus sign. Whenever a source is cited, the change of sign is flagged if it matters.

- $\rho = 1/\kappa$ is the **radius of curvature** and $\sigma = 1/\tau$ the **radius of torsion**; ρ' always means $d\rho/ds$.
- Decimals with the point; all digits shown are exact (rounded at the last digit written).

3.1. The Frenet frame and the torsion

Definition 3.1: Regular, biregular, unit-speed curve. Let $I \subseteq \mathbb{R}$ be an open interval. A curve $\gamma : I \rightarrow \mathbb{R}^3$ of class C^k ($k \geq 1$) is called **regular** if $\gamma'(t) \neq 0$ for every $t \in I$. It is **parametrised by arc length** if $|\gamma'(s)| = 1$ for every $s \in I$; in that case one writes s instead of t . A curve parametrised by arc length, of class C^k with $k \geq 2$, is called **biregular** if $\gamma''(s) \neq 0$ for every $s \in I$. An arbitrary regular curve is called biregular if its reparametrisation by arc length is.

Every regular curve of class C^k admits a reparametrisation by arc length of the same class (Chapter 2): the proof given there for plane curves does not use the dimension of the target space and holds verbatim in $\mathbb{R}^3$. From now on, unless otherwise stated, curves are of class at least C^3 and parametrised by arc length.

Definition 3.2: Frenet trihedron. Let $\gamma : I \rightarrow \mathbb{R}^3$ be biregular, parametrised by arc length. One sets

$$T = \gamma', \quad \kappa = |T'| = |\gamma''| > 0, \quad N = \frac{T'}{\kappa}, \quad B = T \times N.$$

T is the unit **tangent**, N the **principal normal**, B the **binormal**, κ the **curvature**. The triple $\{T, N, B\}$ is the **Frenet frame**. The plane through $\gamma(s)$ spanned by $T(s), N(s)$ is the **osculating plane**, the one spanned by $N(s), B(s)$ the **normal plane**, the one spanned by $T(s), B(s)$ the **rectifying plane**.

Lemma 3.1: *The Frenet trihedron is a positive orthonormal basis. For every $s \in I$ the triple $\{T(s), N(s), B(s)\}$ is an orthonormal basis of $\mathbb{R}^3$ with $\det(T, N, B) = 1$; moreover*

$$N = B \times T, \quad T = N \times B.$$

Proof. By hypothesis $|T| = 1$. Differentiating the identity $\langle T, T \rangle = 1$ gives $2\langle T', T \rangle = 0$, hence $T' \perp T$ and so $N \perp T$; moreover $|N| = |T'|/\kappa = 1$ by definition of κ . The cross product of two orthogonal unit vectors is a unit vector orthogonal to both: $|B| = |T| |N| \sin \frac{\pi}{2} = 1$ and $B \perp T$, $B \perp N$. Hence $\{T, N, B\}$ is orthonormal, and $\det(T, N, B) = \langle T \times N, B \rangle = \langle B, B \rangle = 1 > 0$: the basis is positive.

For the last part one uses the double cross product identity $u \times (v \times w) = v\langle u, w \rangle - w\langle u, v \rangle$:

$$B \times T = (T \times N) \times T = -T \times (T \times N) = -(T\langle T, N \rangle - N\langle T, T \rangle) = N,$$

because $\langle T, N \rangle = 0$ and $\langle T, T \rangle = 1$; similarly $N \times B = N \times (T \times N) = T\langle N, N \rangle - N\langle N, T \rangle = T$.
□

Lemma 3.2: *The derivative of the binormal is parallel to the normal. Let γ be biregular of class C^3 , parametrised by arc length. Then $B'(s)$ is parallel to $N(s)$ for every $s \in I$.*

Proof. From $|B| = 1$, that is $\langle B, B \rangle = 1$, differentiation gives $\langle B', B \rangle = 0$: hence $B' \perp B$. Moreover, differentiating $B = T \times N$ and using $T' = \kappa N$,

$$B' = T' \times N + T \times N' = \kappa(N \times N) + T \times N' = T \times N',$$

because $N \times N = 0$. A cross product $T \times N'$ is orthogonal to T , hence $\langle B', T \rangle = 0$: so $B' \perp T$. Since $\{T, N, B\}$ is an orthonormal basis (Lemma 3.1), a vector orthogonal to both T and B lies on the line spanned by N .
□

Definition 3.3: Torsion. By Lemma 3.2 there exists a unique real number $\tau(s)$ such that

$$B'(s) = -\tau(s) N(s);$$

$\tau(s) = -\langle B'(s), N(s) \rangle$ is called the **torsion** of γ at s . The number $\sigma = 1/\tau$, defined where $\tau \neq 0$, is the **radius of torsion**; $\rho = 1/\kappa$ is the **radius of curvature**.

Theorem 3.1: Frenet–Serret formulas. Let $\gamma : I \rightarrow \mathbb{R}^3$ be biregular of class C^3 , parametrised by arc length, with curvature $\kappa > 0$ and torsion τ . Then

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N,$$

that is, in matrix form,

$$\frac{d}{ds} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix},$$

where the coefficient matrix is **skew-symmetric**.

Proof. The first formula is the very definition of N and κ : $T' = \kappa N$ because $N = T'/\kappa$ and $\kappa = |T'|$. The third is Definition 3.3, made legitimate by Lemma 3.2.

The second remains. By Lemma 3.1 one has $N = B \times T$; differentiating,

$$N' = B' \times T + B \times T' = (-\tau N) \times T + B \times (\kappa N) = -\tau(N \times T) + \kappa(B \times N).$$

From Lemma 3.1, $N \times T = -(T \times N) = -B$ and $B \times N = -(N \times B) = -T$. Substituting,

$$N' = -\tau(-B) + \kappa(-T) = -\kappa T + \tau B.$$

The skew-symmetry of the matrix is no accident: it is the infinitesimal translation of the fact

that $\{T, N, B\}$ stays orthonormal. Indeed, if $X_1 = T$, $X_2 = N$, $X_3 = B$ and $X'_i = \sum_k a_{ik} X_k$, differentiating $\langle X_i, X_j \rangle = \delta_{ij}$ gives $0 = \langle X'_i, X_j \rangle + \langle X_i, X'_j \rangle = a_{ij} + a_{ji}$. $\square$

Proposition 3.1: The Darboux vector. Setting $\omega = \tau T + \kappa B$, for every $X \in \{T, N, B\}$ one has

$$X' = \omega \times X,$$

and $|\omega| = \sqrt{\kappa^2 + \tau^2}$. The vector ω is called the **Darboux vector**: the Frenet frame rotates instantaneously about ω with angular velocity $|\omega|$.

Proof. One computes the three cross products using $T \times T = N \times N = B \times B = 0$ and the relations of Lemma 3.1, namely $T \times N = B$, $N \times B = T$, $B \times T = N$:

$$\omega \times T = \tau(T \times T) + \kappa(B \times T) = \kappa N = T';$$

$$\omega \times N = \tau(T \times N) + \kappa(B \times N) = \tau B - \kappa T = N';$$

$$\omega \times B = \tau(T \times B) + \kappa(B \times B) = -\tau N = B',$$

where $T \times B = -(B \times T) = -N$ was used. Finally, since $T \perp B$ and $|T| = |B| = 1$, Pythagoras' theorem gives $|\omega|^2 = \tau^2 + \kappa^2$. $\square$

Remark 3.1: Why the Darboux vector matters in robotics. The relation $X' = \omega \times X$ is formally identical to the velocity distribution law of a rotating rigid body: the Frenet frame behaves like a frame attached to a body rotating with angular velocity ω . The component of ω along T equals τ : it is the rotation of the tool about the direction of travel. It is exactly what the rotation minimizing frames used in trajectory generation cancel, when one wants to carry a tool along a curve without parasitic rotation; the link with Lie groups is taken up in Chapter 29 and the applications in Chapter 30.

Proposition 3.2: Curvature and torsion in an arbitrary parameter. Let $\gamma : J \rightarrow \mathbb{R}^3$ be regular of class C^3 with $\gamma' \times \gamma'' \neq 0$. Then γ is biregular and, writing $v = |\gamma'|$,

$$\kappa = \frac{|\gamma' \times \gamma''|}{|\gamma'|^3}, \quad \tau = \frac{\langle \gamma' \times \gamma'', \gamma''' \rangle}{|\gamma' \times \gamma''|^2} = \frac{\det(\gamma', \gamma'', \gamma''')}{|\gamma' \times \gamma''|^2}.$$

Moreover $\gamma' \times \gamma'' = v^3 \kappa B$: the binormal is the unit vector of $\gamma' \times \gamma''$.

Proof. Let $s(t)$ be the arc length, $s'(t) = v(t) > 0$. By the chain rule $\gamma' = vT$; differentiating again, and using $T' = \kappa N$ with respect to s (hence $dT/dt = v\kappa N$),

$$\gamma'' = v'T + v^2 \kappa N.$$

Therefore

$$\gamma' \times \gamma'' = vT \times (v'T + v^2 \kappa N) = vv'(T \times T) + v^3 \kappa (T \times N) = v^3 \kappa B,$$

and since $|B| = 1$, $v > 0$, $\kappa > 0$ we get $|\gamma' \times \gamma''| = v^3 \kappa$, that is the first formula. (That $\gamma' \times \gamma'' \neq 0$ is equivalent to $\kappa > 0$, i.e. to biregularity, is read off the same identity.)

Differentiating a third time and using $dN/dt = v(-\kappa T + \tau B)$,

$$\gamma''' = (v'' - v^3 \kappa^2)T + (v'v\kappa + (v^2 \kappa)')N + v^3 \kappa \tau B.$$

Only the component along B is needed: since $\gamma' \times \gamma'' = v^3 \kappa B$ and $\{T, N, B\}$ is orthonormal,

$$\langle \gamma' \times \gamma'', \gamma''' \rangle = v^3 \kappa \langle B, \gamma''' \rangle = v^3 \kappa \cdot v^3 \kappa \tau = v^6 \kappa^2 \tau = |\gamma' \times \gamma''|^2 \tau,$$

whence the second formula. The equality between triple product and determinant is the very definition of the triple product. $\square$

Proposition 3.3: Straight lines and plane curves. Let $\gamma : I \rightarrow \mathbb{R}^3$ be parametrised by arc length.

(i) If γ is of class C^2 and $\kappa \equiv 0$, then $\gamma(I)$ is contained in a straight line.

(ii) If γ is biregular of class C^3 , then $\gamma(I)$ is contained in a plane if and only if $\tau \equiv 0$.

Proof. (i) $\kappa = |\gamma''| \equiv 0$ gives $\gamma'' \equiv 0$, hence $\gamma' \equiv a$ constant with $|a| = 1$ and $\gamma(s) = \gamma(s_0) + (s - s_0)a$: this is the parametrisation of a line.

(ii) If $\tau \equiv 0$. Then $B' = -\tau N \equiv 0$, that is B is a constant vector, call it B_0 ($|B_0| = 1$). Fixing $s_0 \in I$ and setting $h(s) = \langle \gamma(s) - \gamma(s_0), B_0 \rangle$, one has $h'(s) = \langle \gamma'(s), B_0 \rangle = \langle T(s), B(s) \rangle = 0$ and $h(s_0) = 0$, hence $h \equiv 0$: the curve lies in the plane through $\gamma(s_0)$ orthogonal to B_0 .

Only if. Suppose $\gamma(I) \subseteq \Pi = \{x : \langle x - p, n \rangle = 0\}$ with $|n| = 1$. Differentiating $\langle \gamma - p, n \rangle \equiv 0$ gives $\langle T, n \rangle \equiv 0$; differentiating again, $\langle T', n \rangle = \kappa \langle N, n \rangle \equiv 0$ and, since $\kappa > 0$, $\langle N, n \rangle \equiv 0$. So n is orthogonal to T and to N , that is, it lies on the line spanned by B : both being unit vectors, $\langle B, n \rangle$ equals $+1$ or -1 at every point. The function $s \mapsto \langle B(s), n \rangle$ is continuous with values in $\{-1, +1\}$ on an interval, hence constant; therefore $B = \varepsilon n$ with ε a fixed sign, B is constant, $B' \equiv 0$ and $\tau = -\langle B', N \rangle \equiv 0$. $\square$

Proposition 3.4: Invariance under direct rigid motions. Let $F(x) = Rx + c$ with $R \in SO(3)$ and $c \in \mathbb{R}^3$ (**direct rigid motion**), and let γ be biregular parametrised by arc length. Then $\tilde{\gamma} = F \circ \gamma$ is again parametrised by arc length, biregular, and

$$\tilde{T} = RT, \quad \tilde{N} = RN, \quad \tilde{B} = RB, \quad \tilde{\kappa} = \kappa, \quad \tilde{\tau} = \tau.$$

If instead $\det R = -1$ one still has $\tilde{\kappa} = \kappa$, but $\tilde{\tau} = -\tau$.

Proof. Step 1: the cross product under $R \in SO(3)$. For all $u, v, w \in \mathbb{R}^3$,

$$\langle Ru \times Rv, w \rangle = \det(Ru, Rv, w) = \det(Ru, Rv, R(R^\top w)) = \det(R) \det(u, v, R^\top w),$$

using $RR^\top = I$ and multiplicativity of the determinant. Now $\det(u, v, R^\top w) = \langle u \times v, R^\top w \rangle = \langle R(u \times v), w \rangle$ because $R^\top$ is the adjoint of R . Since w is arbitrary,

$$Ru \times Rv = \det(R) R(u \times v).$$

For $R \in SO(3)$, $\det R = 1$ and hence $Ru \times Rv = R(u \times v)$.

Step 2. $\tilde{\gamma}' = R\gamma' = RT$ and $|RT| = |T| = 1$ (orthogonal matrices preserve the norm): $\tilde{\gamma}$ is parametrised by arc length and $\tilde{T} = RT$. Next $\tilde{T}' = RT' = \kappa RN$, hence $\tilde{\kappa} = |\tilde{T}'| = \kappa|RN| = \kappa > 0$ (biregularity is preserved) and $\tilde{N} = \tilde{T}'/\tilde{\kappa} = RN$. By Step 1, $\tilde{B} = \tilde{T} \times \tilde{N} = RT \times RN = R(T \times N) = RB$. Finally $\tilde{B}' = RB' = -\tau RN = -\tau \tilde{N}$, that is $\tilde{\tau} = \tau$.

Case $\det R = -1$. The steps concerning T, κ, N do not use the determinant and remain valid; Step 1 instead gives $\tilde{B} = RT \times RN = -R(T \times N) = -RB$, whence $\tilde{B}' = -RB' = \tau RN = \tau \tilde{N} = -(-\tau)\tilde{N}$ and therefore $\tilde{\tau} = -\tau$. A reflection exchanges right-handed and left-handed helices: torsion is an *oriented* invariant, curvature is not. $\square$

Example 3.1: The twisted cubic. Let $\gamma(t) = (t, t^2, t^3)$. Then $\gamma' = (1, 2t, 3t^2)$, $\gamma'' = (0, 2, 6t)$, $\gamma''' = (0, 0, 6)$ and

$$\gamma' \times \gamma'' = (2t \cdot 6t - 3t^2 \cdot 2, \quad 3t^2 \cdot 0 - 1 \cdot 6t, \quad 1 \cdot 2 - 2t \cdot 0) = (6t^2, -6t, 2).$$

Hence $|\gamma' \times \gamma''|^2 = 36t^4 + 36t^2 + 4 = 4(9t^4 + 9t^2 + 1)$, $|\gamma'|^2 = 1 + 4t^2 + 9t^4$ and $\langle \gamma' \times \gamma'', \gamma''' \rangle = 12$.

By Proposition 3.2,

$$\kappa(t) = \frac{2\sqrt{9t^4 + 9t^2 + 1}}{(1 + 4t^2 + 9t^4)^{3/2}}, \quad \tau(t) = \frac{12}{4(9t^4 + 9t^2 + 1)} = \frac{3}{9t^4 + 9t^2 + 1}.$$

At $t = 0$: $\kappa(0) = 2/1 = 2$ and $\tau(0) = 3$. *Independent check at $t = 0$* : $|\gamma'(0)| = 1$ (the parameter is already arc length at $t = 0$), $\gamma''(0) = (0, 2, 0)$ and hence $\kappa(0) = |\gamma''(0)| = 2$ ✓; moreover $T = (1, 0, 0)$, $N = (0, 1, 0)$, $B = (0, 0, 1)$ and the component of $\gamma'''(0) = (0, 0, 6)$ along B equals $6 = v^3 \kappa \tau = 1 \cdot 2 \cdot 3$ ✓.

At $t = 1$: $9 + 9 + 1 = 19$ and $1 + 4 + 9 = 14$, hence $\kappa(1) = 2\sqrt{19}/14^{3/2} = 8.7177979/52.3832034 = 0.1664235$ and $\tau(1) = 3/19 = 0.1578947$. *Independent check*: the direct numerical computation of $|\gamma' \times \gamma''|/|\gamma'|^3$ and of $\langle \gamma' \times \gamma'', \gamma''' \rangle / |\gamma' \times \gamma''|^2$ with high-precision numerical derivatives gives 0.166423535003 and 0.157894736842 , that is the same digits. ✓

Since $\tau(t) > 0$ for every t , the twisted cubic is planar on no arc (Proposition 3.3(ii)); and since $\kappa(t) \rightarrow 0$ and $\tau(t) \rightarrow 0$ as $t \rightarrow \pm\infty$, far from the origin the curve is more and more «almost straight».

3.2. The fundamental theorem of space curves

The Frenet–Serret formulas say that the derivatives of the frame are expressed by means of κ and τ alone. Read backwards, this means that κ and τ *prescribe* the motion of the frame, and hence — by integrating the tangent — the curve. The following theorem makes this idea exact. Its proof rests on a single analytic fact, existence and uniqueness of the solution of a *linear* Cauchy problem: that fact is proved in full in the chapter appendix (Lemma 3.4), and is not imported.

Theorem 3.2: Fundamental theorem of space curves. *Let $I \subseteq \mathbb{R}$ be an interval, $s_0 \in I$, $\kappa : I \rightarrow \mathbb{R}$ of class C^1 with $\kappa(s) > 0$ for every s , and $\tau : I \rightarrow \mathbb{R}$ of class C^1 .*

- (i) (Existence.) *Given a point $p_0 \in \mathbb{R}^3$ and a positive orthonormal basis $\{T_0, N_0, B_0\}$ of $\mathbb{R}^3$, there exists a curve $\gamma : I \rightarrow \mathbb{R}^3$ of class C^3 , parametrised by arc length, biregular, with curvature κ , torsion τ , $\gamma(s_0) = p_0$ and Frenet frame at s_0 equal to $\{T_0, N_0, B_0\}$.*
- (ii) (Uniqueness with initial data.) *Such a curve is unique.*
- (iii) (Uniqueness up to isometries.) *If γ and $\tilde{\gamma}$ are two biregular curves on I , parametrised by arc length, with the same curvature κ and the same torsion τ , then there exists exactly one direct rigid motion $F(x) = Rx + c$, $R \in SO(3)$, such that $\tilde{\gamma} = F \circ \gamma$.*

Proof. Step 1 — the Frenet system as a linear Cauchy problem. We look for three functions $T, N, B : I \rightarrow \mathbb{R}^3$ satisfying the Frenet–Serret formulas. Collecting them in the vector $X = (T, N, B) \in \mathbb{R}^9$ and writing I_3 for the 3×3 identity matrix, the three formulas amount to the single equation

$$X'(s) = \mathcal{A}(s)X(s), \quad \mathcal{A}(s) = \begin{pmatrix} 0 & \kappa(s)I_3 & 0 \\ -\kappa(s)I_3 & 0 & \tau(s)I_3 \\ 0 & -\tau(s)I_3 & 0 \end{pmatrix} \in M_9(\mathbb{R}).$$

$\mathcal{A}$ is continuous on I (indeed of class C^1), because κ and τ are. By Lemma 3.4 of the appendix there exists exactly one $X \in C^1(I, \mathbb{R}^9)$ with $X(s_0) = (T_0, N_0, B_0)$; and since $\mathcal{A}$ is C^1 , the equation itself gives $X' \in C^1$, that is $X \in C^2$.

Step 2 — orthonormality is preserved. Let $X_1 = T$, $X_2 = N$, $X_3 = B$ be the three vector components of the solution and let $g_{ij}(s) = \langle X_i(s), X_j(s) \rangle$. Writing $X'_i = \sum_k a_{ik} X_k$

with

$$A = (a_{ik}) = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}, \quad A^\top = -A,$$

one has

$$g'_{ij} = \langle X'_i, X_j \rangle + \langle X_i, X'_j \rangle = \sum_k a_{ik} g_{kj} + \sum_k a_{jk} g_{ik},$$

that is, in matrix form, $G' = AG + GA^\top$ for $G = (g_{ij})$. This is again a *linear* system (in the nine unknowns g_{ij}) with continuous coefficients. The constant function $G \equiv I_3$ solves it, because $AI_3 + I_3A^\top = A + A^\top = 0$, and satisfies the initial datum $G(s_0) = I_3$ (the triple $\{T_0, N_0, B_0\}$ is orthonormal). By the uniqueness part of Lemma 3.4 applied to this system, $G \equiv I_3$: $\{T(s), N(s), B(s)\}$ is orthonormal for every $s \in I$.

Step 3 — the orientation is preserved. The function $s \mapsto \det(T(s), N(s), B(s))$ is continuous and, the triple being orthonormal, takes only the values ± 1 ; on an interval a continuous function with values in $\{-1, +1\}$ is constant, and at s_0 it equals $+1$. Hence $\det(T, N, B) \equiv 1$ and, by Lemma 3.1 applied to the positive orthonormal basis $\{T, N, B\}$, $B = T \times N$.

Step 4 — the curve. Set

$$\gamma(s) = p_0 + \int_{s_0}^s T(u) du.$$

Then $\gamma \in C^3$ (because $T \in C^2$), $\gamma' = T$ with $|T| = 1$: γ is parametrised by arc length and $\gamma(s_0) = p_0$. Moreover $\gamma'' = T' = \kappa N$ with $|N| = 1$ and $\kappa > 0$, hence $|\gamma''| = \kappa > 0$: γ is biregular, its curvature is κ and its principal normal is $\gamma''/|\gamma''| = N$. Its binormal is then $T \times N = B$ (Step 3) and its torsion is, by definition, the unique number $\tilde{\tau}$ with $B' = -\tilde{\tau}N$: the third equation of the system says $B' = -\tau N$, hence $\tilde{\tau} = \tau$. The Frenet frame of γ at s_0 is $\{T_0, N_0, B_0\}$. This proves (i).

Step 5 — uniqueness with initial data (ii). If γ and $\hat{\gamma}$ both satisfy (i), their Frenet frames solve the same linear system of Step 1 with the same initial datum (T_0, N_0, B_0) ; by the uniqueness part of Lemma 3.4 they coincide, in particular $\gamma' = T = \hat{\gamma}'$. Then $(\gamma - \hat{\gamma})' \equiv 0$ and $\gamma(s_0) = p_0 = \hat{\gamma}(s_0)$, hence $\gamma = \hat{\gamma}$.

Step 6 — uniqueness up to isometries (iii): construction of F . Let $\{T, N, B\}$ and $\{\tilde{T}, \tilde{N}, \tilde{B}\}$ be the frames of γ and $\tilde{\gamma}$. At s_0 both triples are positive orthonormal bases of $\mathbb{R}^3$ (Lemma 3.1); hence there is exactly one matrix R carrying the first to the second, namely $R = [\tilde{T}(s_0) \tilde{N}(s_0) \tilde{B}(s_0)] [T(s_0) N(s_0) B(s_0)]^{-1}$, a product of two orthogonal matrices of determinant 1, hence $R \in SO(3)$. Set $c = \tilde{\gamma}(s_0) - R\gamma(s_0)$ and $F(x) = Rx + c$, and $\beta = F \circ \gamma$. By Proposition 3.4, β is parametrised by arc length, biregular, has curvature κ and torsion τ , and its Frenet frame is $\{RT, RN, RB\}$; by construction $\beta(s_0) = \tilde{\gamma}(s_0)$ and the frame of β at s_0 coincides with that of $\tilde{\gamma}$.

Step 7 — $\beta = \tilde{\gamma}$. Set, for $s \in I$,

$$f(s) = |T_\beta - \tilde{T}|^2 + |N_\beta - \tilde{N}|^2 + |B_\beta - \tilde{B}|^2 = 6 - 2(\langle T_\beta, \tilde{T} \rangle + \langle N_\beta, \tilde{N} \rangle + \langle B_\beta, \tilde{B} \rangle),$$

where the last equality uses that all six vectors are unit vectors. The two frames satisfy the same Frenet–Serret formulas, with the same κ and τ . Differentiating the three scalar products:

$$\langle T_\beta, \tilde{T} \rangle' = \kappa \langle N_\beta, \tilde{T} \rangle + \kappa \langle T_\beta, \tilde{N} \rangle,$$

$$\langle N_\beta, \tilde{N} \rangle' = -\kappa \langle T_\beta, \tilde{N} \rangle + \tau \langle B_\beta, \tilde{N} \rangle - \kappa \langle N_\beta, \tilde{T} \rangle + \tau \langle N_\beta, \tilde{B} \rangle,$$

$$\langle B_\beta, \tilde{B} \rangle' = -\tau \langle N_\beta, \tilde{B} \rangle - \tau \langle B_\beta, \tilde{N} \rangle.$$

Adding, the six terms cancel in pairs: $\kappa \langle N_\beta, \tilde{T} \rangle$ with $-\kappa \langle N_\beta, \tilde{T} \rangle$, $\kappa \langle T_\beta, \tilde{N} \rangle$ with $-\kappa \langle T_\beta, \tilde{N} \rangle$, $\tau \langle B_\beta, \tilde{N} \rangle$ with $-\tau \langle B_\beta, \tilde{N} \rangle$, $\tau \langle N_\beta, \tilde{B} \rangle$ with $-\tau \langle N_\beta, \tilde{B} \rangle$. Hence $f' \equiv 0$ and f is constant. Since

$f(s_0) = 0$ (Step 6), $f \equiv 0$, that is $T_\beta = \tilde{T}$ on all of I . Then $(\beta - \tilde{\gamma})' \equiv 0$ and $\beta(s_0) = \tilde{\gamma}(s_0)$, hence $\beta = \tilde{\gamma}$, i.e. $\tilde{\gamma} = F \circ \gamma$.

Step 8 — uniqueness of F . If $F_1(x) = R_1x + c_1$ and $F_2(x) = R_2x + c_2$ are two direct rigid motions with $F_1 \circ \gamma = F_2 \circ \gamma$, then $(R_1 - R_2)\gamma(s) = c_2 - c_1$ for every s ; differentiating, $(R_1 - R_2)T(s) = 0$ for every s , and differentiating again $(R_1 - R_2)N(s) = 0$ and $(R_1 - R_2)B(s) = 0$. Since $\{T, N, B\}$ is a basis of $\mathbb{R}^3$, $R_1 = R_2$ and hence $c_1 = c_2$. $\square$

Remark 3.2: What the fundamental theorem does not say. Three warnings, all necessary.

- (a) The hypothesis $\kappa > 0$ everywhere cannot be dropped. If κ vanishes at a point, N and B are not defined there and the system of Step 1 becomes meaningless; there exist non-planar C^∞ curves with τ defined and zero wherever it exists, obtained by gluing two planar arcs lying in different planes through a stretch of zero curvature. Proposition 3.3(ii) does presuppose biregularity.
- (b) Uniqueness is up to direct rigid motions, not up to arbitrary isometries: by Proposition 3.4 a reflection changes the sign of τ . The mirror image of a right-handed helix has the same curvature and opposite torsion, and cannot be superimposed on the original by any rigid motion.
- (c) The theorem is local in the sense that it concerns the parametrisation, not the image: nothing prevents the constructed curve from self-intersecting. Global properties (rotation number, knots, Fenchel's theorem) belong to Chapter 4 and its sequels.

3.3. The circular helix and generalised helices

Definition 3.4: Circular helix. Given $a > 0$ and $b \in \mathbb{R}$, the **circular helix** of radius a and parameter b is the curve

$$\gamma(t) = (a \cos t, a \sin t, bt), \quad t \in \mathbb{R}.$$

The number $p = 2\pi b$ is its **pitch** (advance along the axis per full turn); the helix is **right-handed** if $b > 0$, **left-handed** if $b < 0$; for $b = 0$ it degenerates into the circle of radius a in the plane $z = 0$.

Theorem 3.3: Curvature and torsion of the circular helix. The circular helix of radius $a > 0$ and parameter $b \neq 0$ is biregular, has $|\gamma'| = c$ constant with $c = \sqrt{a^2 + b^2}$, and

$$\kappa = \frac{a}{a^2 + b^2}, \quad \tau = \frac{b}{a^2 + b^2},$$

both **constant**. Its parametrisation by arc length is $\gamma(s) = (a \cos \frac{s}{c}, a \sin \frac{s}{c}, b \frac{s}{c})$, and its principal normal $N = (-\cos \frac{s}{c}, -\sin \frac{s}{c}, 0)$ points at every instant towards the z axis.

Proof. One computes

$$\gamma' = (-a \sin t, a \cos t, b), \quad \gamma'' = (-a \cos t, -a \sin t, 0), \quad \gamma''' = (a \sin t, -a \cos t, 0).$$

At once $|\gamma'|^2 = a^2 \sin^2 t + a^2 \cos^2 t + b^2 = a^2 + b^2 = c^2$, constant and non-zero: the curve is regular and the reparametrisation by arc length is $s = ct$, that is $t = s/c$, which gives the stated formula.

The cross product, component by component:

$$\gamma' \times \gamma'' = (a \cos t \cdot 0 - b \cdot (-a \sin t), \quad b \cdot (-a \cos t) - (-a \sin t) \cdot 0, \quad (-a \sin t)(-a \sin t) - (a \cos t)(-a \cos t)),$$

that is $\gamma' \times \gamma'' = (ab \sin t, -ab \cos t, a^2)$, which is never zero because $a > 0$: the curve is biregular. Its modulus is

$$|\gamma' \times \gamma''| = \sqrt{a^2 b^2 \sin^2 t + a^2 b^2 \cos^2 t + a^4} = \sqrt{a^2 b^2 + a^4} = a\sqrt{a^2 + b^2} = ac.$$

By Proposition 3.2,

$$\kappa = \frac{ac}{c^3} = \frac{a}{c^2} = \frac{a}{a^2 + b^2}.$$

For the torsion,

$$\langle \gamma' \times \gamma'', \gamma''' \rangle = ab \sin t \cdot a \sin t + (-ab \cos t)(-a \cos t) + a^2 \cdot 0 = a^2 b (\sin^2 t + \cos^2 t) = a^2 b,$$

while $|\gamma' \times \gamma''|^2 = a^2 c^2$; hence $\tau = a^2 b / (a^2 c^2) = b / c^2 = b / (a^2 + b^2)$.

Independent check of the normal. From the arc-length parametrisation, $T = \gamma'(s) = \frac{1}{c}(-a \sin \frac{s}{c}, a \cos \frac{s}{c}, b)$ and $T' = \frac{a}{c^2}(-\cos \frac{s}{c}, -\sin \frac{s}{c}, 0)$: its modulus is $a/c^2 = \kappa$ ✓, and its unit vector is $N = (-\cos \frac{s}{c}, -\sin \frac{s}{c}, 0)$, which is the unit vector pointing from $(a \cos \frac{s}{c}, a \sin \frac{s}{c})$ towards the origin of the horizontal plane, that is towards the axis. ✓ $\square$

Corollary 3.1: Curves with constant curvature and torsion. Let $\gamma : I \rightarrow \mathbb{R}^3$ be biregular of class C^3 , parametrised by arc length, with $\kappa \equiv \kappa_0 > 0$ and $\tau \equiv \tau_0$ constant.

(i) If $\tau_0 = 0$, then $\gamma(I)$ is contained in a circle of radius $1/\kappa_0$.

(ii) If $\tau_0 \neq 0$, then there is a direct rigid motion F such that $\gamma = F \circ \eta$, where η is the circular helix, parametrised by arc length, of radius and parameter

$$a = \frac{\kappa_0}{\kappa_0^2 + \tau_0^2}, \quad b = \frac{\tau_0}{\kappa_0^2 + \tau_0^2}.$$

In particular the circular helix is, up to direct rigid motions, the only curve with constant curvature and torsion both non-zero.

Proof. (i) If $\tau_0 = 0$ the curve is planar (Proposition 3.3(ii)) and lies in the osculating plane, which is fixed. Consider $q(s) = \gamma(s) + \frac{1}{\kappa_0} N(s)$. By the Frenet–Serret formulas,

$$q' = T + \frac{1}{\kappa_0}(-\kappa_0 T + \tau_0 B) = T - T + 0 = 0,$$

hence q is a fixed point q_0 and $|\gamma(s) - q_0| = \frac{1}{\kappa_0} |N(s)| = \frac{1}{\kappa_0}$ for every s : the curve lies on the circle of the osculating plane with centre q_0 and radius $1/\kappa_0$.

(ii) We look for $a > 0$ and $b \neq 0$ with $\frac{a}{a^2 + b^2} = \kappa_0$ and $\frac{b}{a^2 + b^2} = \tau_0$. Setting $c^2 = a^2 + b^2$, the two conditions read $a = \kappa_0 c^2$ and $b = \tau_0 c^2$; substituting into $c^2 = a^2 + b^2$ gives

$$c^2 = c^4(\kappa_0^2 + \tau_0^2) \implies c^2 = \frac{1}{\kappa_0^2 + \tau_0^2}$$

(because $c > 0$), and hence $a = \kappa_0 / (\kappa_0^2 + \tau_0^2) > 0$, $b = \tau_0 / (\kappa_0^2 + \tau_0^2) \neq 0$: the solution exists, is unique, and has $a > 0$ as required. By Theorem 3.3 the circular helix η with those parameters, parametrised by arc length, has curvature κ_0 and torsion τ_0 . The two curves γ and η thus have the same curvature and the same torsion on I (restricting η to I), and Theorem 3.2(iii) gives a direct rigid motion F with $\gamma = F \circ \eta$.

Check. For $\kappa_0 = \tau_0 = \frac{1}{2}$ one gets $\kappa_0^2 + \tau_0^2 = \frac{1}{2}$, hence $a = b = 1$: the helix $\eta(t) = (\cos t, \sin t, t)$, which has $|\eta'| = \sqrt{2}$ and, by Theorem 3.3, $\kappa = \tau = 1/(1+1) = \frac{1}{2}$. ✓ $\square$

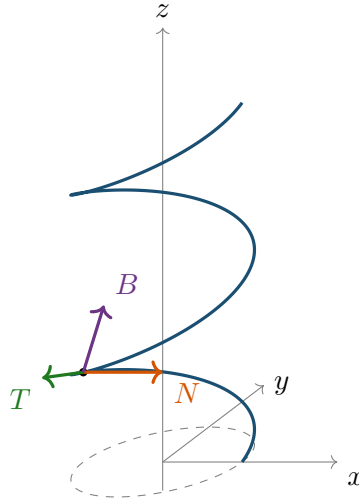


Figure 3.1: The circular helix $\gamma(t) = (\cos t, \sin t, \frac{1}{2}t)$ ($a = 1$, $b = \frac{1}{2}$, two turns) with the Frenet frame, drawn with unit vectors, at the point $t = \pi$. Coordinates computed by hand: $\gamma(\pi) = (-1, 0, \frac{\pi}{2}) = (-1, 0, 1.5707963)$; $\gamma'(\pi) = (0, -1, \frac{1}{2})$ of modulus $c = \sqrt{5}/2 = 1.1180340$, hence $T = (0, -0.8944272, 0.4472136)$; $N = (1, 0, 0)$, pointing towards the z axis; $B = T \times N = (0, 0.4472136, 0.8944272)$. Curvature and torsion, by Theorem 3.3: $\kappa = 1/(1 + \frac{1}{4}) = 0.8$ and $\tau = \frac{1}{2}/(1 + \frac{1}{4}) = 0.4$; pitch $2\pi b = \pi = 3.1415927$. The normal always points towards the axis: it is the dashed base circle that commands the curvature, and the vertical advance that produces the torsion.

Definition 3.5: Generalised helix. A biregular curve γ is a **generalised helix** (or curve of *constant slope*) if there exists a fixed unit vector $u \in \mathbb{R}^3$ such that $\langle T(s), u \rangle$ is constant, that is, if the tangent makes a constant angle with a fixed direction. The unit vector u is called the **axis** of the helix.

Theorem 3.4: Lancret's theorem. A biregular curve $\gamma : I \rightarrow \mathbb{R}^3$ of class C^3 , parametrised by arc length, is a generalised helix if and only if the ratio τ/κ is constant on I . If $\theta \in (0, \pi)$ is the angle between the tangent and the axis, then

$$\frac{\tau}{\kappa} = \cot \theta,$$

and the axis is $u = \cos \theta T + \sin \theta B$.

Proof. Necessity. Let $\langle T, u \rangle = \cos \theta$ be constant, with $|u| = 1$. If $\cos \theta = \pm 1$ then $T = \pm u$ would be constant and hence $\kappa = |T'| = 0$, against biregularity: so $\theta \in (0, \pi)$ and $\sin \theta > 0$. Differentiating $\langle T, u \rangle = \cos \theta$ gives $\langle T', u \rangle = \kappa \langle N, u \rangle = 0$, and since $\kappa > 0$,

$$\langle N, u \rangle \equiv 0.$$

Writing u in the orthonormal basis $\{T, N, B\}$, $u = \langle u, T \rangle T + \langle u, N \rangle N + \langle u, B \rangle B = \cos \theta T + \lambda B$ with $\lambda = \langle u, B \rangle$. The coefficient λ is constant, because $\lambda' = \langle u, B' \rangle = -\tau \langle u, N \rangle = 0$; and from $|u| = 1$ it follows that $\cos^2 \theta + \lambda^2 = 1$, i.e. $\lambda = \pm \sin \theta$. It cannot be $\lambda = 0$ (otherwise $u = \pm T$ and again $\kappa = 0$). Differentiating now u , which is constant,

$$0 = u' = \cos \theta T' + \lambda B' = \cos \theta \kappa N - \lambda \tau N = (\kappa \cos \theta - \lambda \tau) N,$$

and since $|N| = 1$ we get $\kappa \cos \theta = \lambda \tau$, that is

$$\frac{\tau}{\kappa} = \frac{\cos \theta}{\lambda} = \pm \frac{\cos \theta}{\sin \theta} = \pm \cot \theta,$$

constant. Replacing u by $-u$ if necessary (which changes θ into $\pi - \theta$ and λ into $-\lambda$) one reduces to the case $\lambda = +\sin \theta$, which gives the stated form.

Sufficiency. Let $\tau/\kappa \equiv m$ be constant. Since the cotangent is a bijection of $(0, \pi)$ onto $\mathbb{R}$, there is a unique $\theta \in (0, \pi)$ with $\cot \theta = m$; set

$$u(s) = \cos \theta T(s) + \sin \theta B(s).$$

It is a unit vector, because $T \perp B$ and $\cos^2 \theta + \sin^2 \theta = 1$. Differentiating and using Frenet–Serret,

$$u' = \cos \theta \kappa N + \sin \theta (-\tau N) = (\kappa \cos \theta - \tau \sin \theta) N = \kappa \sin \theta \left(\cot \theta - \frac{\tau}{\kappa} \right) N = 0,$$

because $\cot \theta = m = \tau/\kappa$. Hence u is a fixed unit vector and $\langle T, u \rangle = \cos \theta$ is constant: γ is a generalised helix with axis u . $\square$

Remark 3.3: Lancret and the circular helix. Theorem 3.4 is strictly weaker than Corollary 3.1: it demands that τ/κ be constant, not that κ and τ be constant separately. The circular helix satisfies both conditions, with $\tau/\kappa = b/a = \cot \theta$: indeed the angle θ between the tangent and the z axis satisfies $\cos \theta = \langle T, e_3 \rangle = b/c$ and $\sin \theta = a/c$, whence $\cot \theta = b/a$. $\checkmark$ But there exist generalised helices that are not circular: for instance $\gamma(t) = (3t - t^3, 3t^2, 3t + t^3)$ has $\kappa(t) = \tau(t) = \frac{1}{3(1+t^2)^2}$, hence $\tau/\kappa \equiv 1$ and $\theta = 45^\circ$, but neither κ nor τ is constant (see Assigned exercise 3.2).

Example 3.2: The helical weld bead. A helical weld bead on a pipe of radius $a = 50$ mm must have pitch (advance per turn) $p = 30$ mm. From the pitch one gets

$$b = \frac{p}{2\pi} = \frac{30}{2\pi} = 4.7746483 \text{ mm},$$

hence $a^2 + b^2 = 2500 + 22.7972663 = 2522.7972663 \text{ mm}^2$ and, by Theorem 3.3,

$$\kappa = \frac{50}{2522.7972663} = 1.9819270 \cdot 10^{-2} \text{ mm}^{-1}, \quad \tau = \frac{4.7746483}{2522.7972663} = 1.8926009 \cdot 10^{-3} \text{ mm}^{-1}.$$

Independent check. The ratio κ/τ must equal a/b (the denominator cancels in the two formulas): $1.9819270 \cdot 10^{-2} / 1.8926009 \cdot 10^{-3} = 10.4719755$ and $a/b = 50/4.7746483 = 10.4719755 \checkmark$; moreover $10.4719755 = 2\pi \cdot 50/30 = 100\pi/30$, that is exactly $\frac{10\pi}{3}$. The radius of curvature is $\rho = 1/\kappa = (a^2 + b^2)/a = 2522.7972663/50 = 50.4559453 \text{ mm}$, just larger than the radius of the pipe: the trajectory «opens up» by half a millimetre with respect to the circle.

Reading. The curvature dominates the torsion by an order of magnitude ($\kappa/\tau \approx 10.5$): the path is almost circular. By Theorem 3.4 the helix angle with respect to the pipe axis satisfies $\cot \theta = \tau/\kappa = b/a = 0.0954930$, that is $\theta = 84.5451966^\circ$: the tangent is almost orthogonal to the axis, as it must be for a pitch small compared with the circumference. The modulus of the Darboux vector (Proposition 3.1) is $|\omega| = \sqrt{\kappa^2 + \tau^2} = 1/c = 1/\sqrt{2522.7972663} = 1/50.2274553 = 1.9909430 \cdot 10^{-2} \text{ mm}^{-1}$: the welding head, advancing one millimetre along the bead, rotates by $1.99 \cdot 10^{-2}$ radians, of which only $1.89 \cdot 10^{-3}$ about the direction of travel.

3.4. Osculating circle and osculating sphere

In the plane the osculating circle is defined as the limit of the circles through three points of the curve as they come together (Chapter 2). In space it is more convenient to use an equivalent and more manageable definition, the one by *order of contact*, which has the advantage of telling at once *in which plane* the circle lies.

Definition 3.6: Contact of order k . Two curves γ, δ parametrised by arc length have **contact of order $\geq k$** at the point $\gamma(s_0) = \delta(u_0)$ if

$$\gamma^{(j)}(s_0) = \delta^{(j)}(u_0) \quad \text{for } j = 0, 1, \dots, k.$$

Definition 3.7: Osculating circle in space. Let γ be biregular and $s_0 \in I$. The **osculating circle** of γ at $\gamma(s_0)$ is the circle of the osculating plane at $\gamma(s_0)$ with centre

$$q = \gamma(s_0) + \rho(s_0) N(s_0) \quad \text{and radius} \quad \rho(s_0) = \frac{1}{\kappa(s_0)}.$$

The point q is called the **centre of curvature**, and the line through q orthogonal to the osculating plane is called the **polar axis** at s_0 .

Theorem 3.5: The osculating circle is the unique circle with second-order contact. Let γ be biregular, $s_0 \in I$, $p = \gamma(s_0)$. Among all circles of $\mathbb{R}^3$ through p there is exactly one having contact of order ≥ 2 with γ at p , and it is the osculating circle of Definition 3.7.

Proof. Step 1 — what $c(0), c'(0), c''(0)$ are for a circle. Let C be the circle with centre q and radius $r > 0$ contained in the plane Π through q orthogonal to the unit vector m . A parametrisation by arc length of it is $c(u) = q + r \cos \frac{u}{r} e_1 + r \sin \frac{u}{r} e_2$ with $\{e_1, e_2, m\}$ a positive orthonormal basis; choosing the origin of the parameter at the point $p \in C$ one may assume $c(0) = p$, that is $e_1 = (p - q)/r$. Then

$$c'(0) = e_2, \quad c''(0) = -\frac{1}{r} e_1 = -\frac{p - q}{r^2} = \frac{q - p}{r^2}.$$

In particular the curvature of the circle is $|c''| = 1/r$ and its principal normal at p is $(q - p)/r$: it points at the centre.

Step 2 — imposing the contact. Contact of order ≥ 2 requires $c(0) = \gamma(s_0) = p$ (already imposed), $c'(0) = \gamma'(s_0) = T(s_0)$ and $c''(0) = \gamma''(s_0) = \kappa(s_0)N(s_0)$. From the second one gets $e_2 = T(s_0)$. From the third,

$$\frac{q - p}{r^2} = \kappa(s_0)N(s_0) \implies q - p = r^2 \kappa(s_0)N(s_0).$$

Taking moduli and recalling that $|q - p| = r$ (the centre is at distance r from a point of the circle) and $|N| = 1$:

$$r = r^2 \kappa(s_0) \implies r = \frac{1}{\kappa(s_0)} = \rho(s_0),$$

because $r > 0$. Substituting, $q = p + \rho(s_0)^2 \kappa(s_0)N(s_0) = p + \rho(s_0)N(s_0)$.

Step 3 — the plane. The plane Π of the circle contains q and p , hence the direction $q - p$, parallel to $N(s_0)$, and contains $e_2 = T(s_0)$: it is therefore the plane through p spanned by $T(s_0)$ and $N(s_0)$, that is the osculating plane.

Centre, radius and plane are thus uniquely determined: the circle with contact of order ≥ 2 exists and is unique, and it is the one of Definition 3.7. Conversely, that circle does have contact of order ≥ 2 , because the three computations of Step 1 give exactly p , $T(s_0)$ and $\kappa(s_0)N(s_0)$ once one sets $r = \rho(s_0)$, $e_1 = -N(s_0)$, $e_2 = T(s_0)$. $\square$

Definition 3.8: Osculating sphere. Let γ be biregular of class C^4 and $s_0 \in I$. A sphere $S(q, r)$ with centre q and radius r is called **osculating** to γ at s_0 if the function

$$h(s) = |\gamma(s) - q|^2 - r^2$$

satisfies $h(s_0) = h'(s_0) = h''(s_0) = h'''(s_0) = 0$ (contact of order ≥ 3 with the sphere).

Theorem 3.6: Centre and radius of the osculating sphere. Let $\gamma : I \rightarrow \mathbb{R}^3$ be biregular of class C^4 , parametrised by arc length, and let $s_0 \in I$ with $\tau(s_0) \neq 0$. Then there is exactly one osculating sphere at s_0 , and its centre and radius are

$$q = \gamma + \rho N + \rho' \sigma B, \quad r = \sqrt{\rho^2 + (\rho' \sigma)^2},$$

everything evaluated at s_0 , where $\rho = 1/\kappa$, $\sigma = 1/\tau$ and $\rho' = d\rho/ds$ (de Saint Venant's formula). In particular the centre lies on the polar axis, and $r \geq \rho$: **the osculating sphere is never smaller than the osculating circle.**

Proof. Set $e(s) = \gamma(s) - q$, so that $h = \langle e, e \rangle - r^2$ and $e' = T$. Differentiating three times with the Frenet-Serret formulas:

$$h' = 2\langle e, T \rangle;$$

$$h'' = 2(\langle T, T \rangle + \langle e, T' \rangle) = 2(1 + \kappa \langle e, N \rangle);$$

$$h''' = 2(\kappa' \langle e, N \rangle + \kappa \langle e, N' \rangle), \quad \langle e, N' \rangle = \langle T, N \rangle + \langle e, N' \rangle = -\kappa \langle e, T \rangle + \tau \langle e, B \rangle.$$

The contact conditions then give, in order:

- from $h'(s_0) = 0$: $\langle e, T \rangle = 0$;
- from $h''(s_0) = 0$: $1 + \kappa \langle e, N \rangle = 0$, that is $\langle e, N \rangle = -1/\kappa = -\rho$;
- from $h'''(s_0) = 0$, using $\langle e, T \rangle = 0$: $\kappa' \langle e, N \rangle + \kappa \tau \langle e, B \rangle = 0$, that is $-\kappa' \rho + \kappa \tau \langle e, B \rangle = 0$ and hence

$$\langle e, B \rangle = \frac{\kappa' \rho}{\kappa \tau} = \frac{\kappa'}{\kappa^2 \tau} = -\frac{\rho'}{\tau} = -\rho' \sigma,$$

because $\rho = 1/\kappa$ gives $\rho' = -\kappa'/\kappa^2$.

Since $\{T, N, B\}$ is an orthonormal basis,

$$e = \langle e, T \rangle T + \langle e, N \rangle N + \langle e, B \rangle B = -\rho N - \rho' \sigma B,$$

whence $q = \gamma - e = \gamma + \rho N + \rho' \sigma B$: the centre is uniquely determined. Finally $h(s_0) = 0$ forces $r^2 = |e|^2 = \rho^2 + (\rho' \sigma)^2$, which determines $r > 0$ (it is positive because $\rho > 0$). Conversely, with these choices the four conditions hold by construction: the osculating sphere exists and is unique.

The centre q differs from the centre of curvature $\gamma + \rho N$ by the vector $\rho' \sigma B$, parallel to B : it therefore lies on the polar axis of Definition 3.7. And $r^2 = \rho^2 + (\rho' \sigma)^2 \geq \rho^2$ gives $r \geq \rho$, with equality if and only if $\rho'(s_0) = 0$. $\square$

Remark 3.4: The osculating circle as a section of the osculating sphere. The osculating sphere meets the osculating plane in a circle, and that circle is the osculating circle. Indeed the distance of the centre $q = \gamma + \rho N + \rho' \sigma B$ from the osculating plane (which passes through γ and is orthogonal to B) equals $|\langle q - \gamma, B \rangle| = |\rho' \sigma|$, and the section circle has as centre the projection of q onto the plane, that is $\gamma + \rho N$, and radius $\sqrt{r^2 - (\rho' \sigma)^2} = \rho$: exactly the centre and radius of the osculating circle. The circle is the trace of the sphere

on the plane that best approximates the curve — one more occurrence of the same object in a chapter that talks about everything else.

3.5. Spherical curves

When does a curve lie on a sphere? The answer is a single equation involving κ , τ and their derivatives, and it can be read off the osculating sphere: a curve is spherical exactly when its osculating sphere *does not change*.

Theorem 3.7: Condition for lying on a sphere. Let $\gamma : I \rightarrow \mathbb{R}^3$ be biregular of class C^4 , parametrised by arc length, with $\tau(s) \neq 0$ for every $s \in I$. Setting $\rho = 1/\kappa$ and $\sigma = 1/\tau$, the following are equivalent:

- (i) $\gamma(I)$ is contained in a sphere;
 - (ii) $\varsigma := \tau\rho + (\rho'\sigma)' \equiv 0$ on I ;
 - (iii) the osculating sphere is the same at every point of I .
- In that case the sphere is unique, with centre $q = \gamma + \rho N + \rho'\sigma B$ (constant) and radius $r = \sqrt{\rho^2 + (\rho'\sigma)^2}$ (constant).

Proof. Step 1 — the derivative of the osculating centre. Let $q(s) = \gamma(s) + \rho(s)N(s) + \rho'(s)\sigma(s)B(s)$ be the centre of the osculating sphere (Theorem 3.6), defined on all of I because τ never vanishes. Differentiating with the Frenet–Serret formulas:

$$\gamma' = T, \quad (\rho N)' = \rho'N + \rho(-\kappa T + \tau B) = \rho'N - T + \rho\tau B,$$

$$(\rho'\sigma B)' = (\rho'\sigma)'B + \rho'\sigma(-\tau N) = (\rho'\sigma)'B - \rho'N,$$

where $\rho\kappa = 1$ was used in the second line and $\sigma\tau = 1$ in the third. Adding, the terms in T cancel ($T - T = 0$) and so do those in N ($\rho'N - \rho'N = 0$); what remains is

$$q' = (\tau\rho + (\rho'\sigma)')B = \varsigma B$$

Step 2 — (ii) $\Rightarrow$ (i) and (iii). If $\varsigma \equiv 0$ then $q' \equiv 0$, that is q is a fixed point q_0 . Moreover

$$\frac{d}{ds}(\rho^2 + (\rho'\sigma)^2) = 2\rho\rho' + 2(\rho'\sigma)(\rho'\sigma)' = 2\rho\rho' + 2(\rho'\sigma)(-\tau\rho) = 2\rho\rho' - 2\rho\rho'\sigma\tau = 0,$$

having substituted $(\rho'\sigma)' = -\tau\rho$ (which is (ii)) and used $\sigma\tau = 1$. Hence the radius r is constant as well: the osculating sphere does not change, which is (iii). And since $|\gamma(s) - q_0|^2 = |e(s)|^2 = \rho^2 + (\rho'\sigma)^2 = r^2$ (see the proof of Theorem 3.6, where $e = -\rho N - \rho'\sigma B$), the whole curve lies on the sphere $S(q_0, r)$: this is (i).

Step 3 — (i) $\Rightarrow$ (ii). Let $|\gamma(s) - q_0| = r_0$ for every s , with q_0 and $r_0 > 0$ fixed, and set $e = \gamma - q_0$. From $\langle e, e \rangle = r_0^2$ constant one obtains, by differentiating three times exactly as in the proof of Theorem 3.6,

$$\langle e, T \rangle = 0, \quad \langle e, N \rangle = -\rho, \quad \langle e, B \rangle = -\rho'\sigma,$$

(the first by differentiating once, the second twice, the third three times). Therefore $e = -\rho N - \rho'\sigma B$ and $q_0 = \gamma - e = \gamma + \rho N + \rho'\sigma B = q(s)$ for every s : the osculating centre coincides with q_0 , hence it is constant, hence $q' \equiv 0$ and by Step 1 $\varsigma B \equiv 0$; since $|B| = 1$, $\varsigma \equiv 0$.

Step 4 — (iii) $\Rightarrow$ (ii). If the osculating sphere is the same at every point, its centre $q(s)$ is constant, and one concludes as above.

Uniqueness of the sphere. If $\gamma(I)$ lay on two distinct spheres, their intersection would be empty, a point or a circle; but a biregular curve with $\tau \neq 0$ is not planar (Proposition 3.3(ii)) and hence cannot be contained in a circle nor reduce to a point. So the sphere is unique, and coincides with the osculating sphere. $\square$

Corollary 3.2: Minimum curvature of a spherical curve. Let $\gamma : I \rightarrow \mathbb{R}^3$ be a biregular curve of class C^4 , parametrised by arc length, contained in the sphere of radius r , with τ never zero. Then

$$\kappa(s) \geq \frac{1}{r} \quad \text{for every } s \in I,$$

and equality holds at one point of I at most. In particular $\kappa > 1/r$ everywhere except possibly at one point: a curve drawn on a sphere of radius r is always more curved than a great circle, except at most at one instant.

Proof. By Theorem 3.7 the curve satisfies condition (ii) and, at every point,

$$r^2 = \rho(s)^2 + (\rho'(s)\sigma(s))^2 \geq \rho(s)^2,$$

hence $0 < \rho(s) \leq r$ and $\kappa(s) = 1/\rho(s) \geq 1/r$: this is the first part.

The equality $\kappa(s_1) = 1/r$ is equivalent to $\rho(s_1) = r$, that is, by the same identity, to $f(s_1) = 0$ where $f = \rho'\sigma$. Now condition (ii) says exactly

$$f'(s) = (\rho'\sigma)'(s) = -\tau(s)\rho(s) \quad \text{for every } s \in I.$$

By hypothesis τ never vanishes on I and is continuous, hence it has constant sign on the interval I ; and $\rho > 0$. Therefore f' never vanishes and has constant sign: f is **strictly monotone** on I . A strictly monotone function vanishes at most once, hence $f(s_1) = 0$ — that is $\kappa(s_1) = 1/r$ — can happen at one point of I at most. $\square$

Remark 3.5: « $\rho^2 + (\rho'\sigma)^2$ constant» is not enough. One often reads that a curve is spherical if and only if $\rho^2 + (\rho'\sigma)^2$ is constant. The condition is **necessary** — Theorem 3.7 says so, that quantity being r^2 — but it is **not sufficient**. Counterexample: the circular helix with $a = b = 1$ has $\kappa = \tau = \frac{1}{2}$ constant (Theorem 3.3), hence $\rho = 2$, $\rho' = 0$ and

$$\rho^2 + (\rho'\sigma)^2 = 4 \quad \text{constant,}$$

and yet $\varsigma = \tau\rho + (\rho'\sigma)' = \frac{1}{2} \cdot 2 + 0 = 1 \neq 0$: by Theorem 3.7 the helix is not spherical — as is obvious anyway, since it is unbounded. $\checkmark$

The two conditions are equivalent under the additional hypothesis that ρ' never vanishes. Indeed, differentiating $\rho^2 + (\rho'\sigma)^2 = \text{const}$ one gets $\rho\rho' + (\rho'\sigma)(\rho'\sigma)' = 0$; dividing by $\rho' \neq 0$,

$$\rho + \sigma(\rho'\sigma)' = 0 \quad \Longleftrightarrow \quad \tau\rho + (\rho'\sigma)' = 0,$$

where the last equivalence is obtained by multiplying by $\tau = 1/\sigma$. It is exactly the case $\rho' \equiv 0$ — constant curvature, that is the helix — that escapes, and that provides the only counterexample within the class of curves with constant κ and τ .

Example 3.3: Viviani's curve. Consider, for $t \in (-\pi, \pi)$,

$$\gamma(t) = (1 + \cos t, \sin t, 2 \sin \frac{t}{2}).$$

It is spherical, and one sees it at once:

$$|\gamma(t)|^2 = (1 + \cos t)^2 + \sin^2 t + 4 \sin^2 \frac{t}{2} = 1 + 2 \cos t + \cos^2 t + \sin^2 t + 2(1 - \cos t) = 4,$$

using $2 \sin^2 \frac{t}{2} = 1 - \cos t$. Hence γ lies on the sphere with centre the origin and radius 2. (It is the intersection of that sphere with the cylinder $(x - 1)^2 + y^2 = 1$: the classical «Viviani window».)

Curvature and torsion. With $c = \cos \frac{t}{2}$ and $s_* = \sin \frac{t}{2}$ one finds $\gamma' = (-\sin t, \cos t, c)$, $\gamma'' = (-\cos t, -\sin t, -\frac{1}{2}s_*)$, $\gamma''' = (\sin t, -\cos t, -\frac{1}{4}c)$, hence $|\gamma'|^2 = 1 + c^2 = \frac{3+\cos t}{2}$ and, after simplification,

$$\gamma' \times \gamma'' = \left(s_*(c^2 + \frac{1}{2}), -c^3, 1\right), \quad |\gamma' \times \gamma''|^2 = \frac{13 + 3 \cos t}{8}, \quad \langle \gamma' \times \gamma'', \gamma''' \rangle = \frac{3}{4}c.$$

By Proposition 3.2,

$$\kappa(t) = \frac{\sqrt{13 + 3 \cos t}}{(3 + \cos t)^{3/2}}, \quad \tau(t) = \frac{6 \cos \frac{t}{2}}{13 + 3 \cos t}.$$

Independent check: the direct numerical computation of κ and τ from the derivatives (high-precision finite differences) gives, at $t = 0.9$, the values 0.559406433527 and 0.363454048833, while the closed formulas give the same fourteen digits. ✓ At $t = \frac{\pi}{2}$ the formulas give $\kappa = \sqrt{13}/3^{3/2} = 0.6938887$ and $\tau = 3\sqrt{2}/13 = 0.3263570$, confirmed by the direct computation. ✓

Centre and radius of the osculating sphere at $t = 0$. Here $\cos 0 = 1$, hence $\kappa(0) = \sqrt{16}/4^{3/2} = 4/8 = \frac{1}{2}$, $\tau(0) = 6/16 = \frac{3}{8}$, $\rho(0) = 2$. Since κ is an even function of t and the reparametrisation by arc length is odd, $\rho'(0) = 0$. Moreover $\gamma(0) = (2, 0, 0)$, $\gamma'(0) = (0, 1, 1)$ with $|\gamma'(0)| = \sqrt{2}$ ✓ ($= \sqrt{(3+1)/2}$), $\gamma' \times \gamma''(0) = (0, -1, 1)$ of modulus $\sqrt{2}$ ✓ ($= \sqrt{16/8}$); hence $T = \frac{1}{\sqrt{2}}(0, 1, 1)$, $B = \frac{1}{\sqrt{2}}(0, -1, 1)$ and

$$N = B \times T = \frac{1}{2}[(0, -1, 1) \times (0, 1, 1)] = \frac{1}{2}(-1 \cdot 1 - 1 \cdot 1, 0 - 0, 0 - 0) = (-1, 0, 0).$$

The formula of Theorem 3.6 then gives

$$q = \gamma(0) + \rho(0)N + 0 \cdot B = (2, 0, 0) + 2(-1, 0, 0) = (0, 0, 0), \quad r = \sqrt{\rho^2 + 0} = 2,$$

that is exactly the sphere the curve lies on. ✓

Verification of condition (ii). At $t = 0.9$ one has $\rho = 1.7876090$, $\rho' = -0.3259869$, $\sigma = 1/\tau = 2.7513794$, $\rho'\sigma = -0.8969135$; hence $\rho^2 + (\rho'\sigma)^2 = 3.1955461 + 0.8044539 = 4.0000000 = r^2$ ✓, and $\tau\rho = 0.6497137$ while the numerical computation of $(\rho'\sigma)'$ gives -0.6497137 : the sum ς vanishes to forty digits. ✓ Finally $\kappa(0.9) = 0.5594064 > 1/r = 0.5$, as Corollary 3.2 prescribes; and the point $t = 0$, where $\kappa = \frac{1}{2} = 1/r$ exactly, is *the only* point of equality — as the corollary allows, $\rho'\sigma$ being strictly monotone. ✓

3.6. Assigned exercises

The answers are in the Teacher's key (Section 7.15).

Exercise 3.1: The twisted cubic: curvature, torsion, osculating circle. *Aim:* to use the formulas in an arbitrary parameter and to construct the osculating circle at a point.

Estimated time: 40 minutes. *Type:* symbolic with numerical check.

Let $\gamma(t) = (t, t^2, t^3)$.

- (a) Derive $\kappa(t)$ and $\tau(t)$ as in Example 3.1 and check that $\tau(t) > 0$ for every t : the curve is planar on no arc.
- (b) Compute κ and τ at $t = 2$ (closed form and numerical value to seven decimal places).
- (c) Determine the Frenet frame at $t = 0$ and the osculating circle at $\gamma(0)$: centre, radius, plane. Check that the centre is at exactly the radius from the point.
- (d) Is the twisted cubic a generalised helix? Answer with Theorem 3.4, computing τ/κ at $t = 0$ and at $t = 1$.

Exercise 3.2: A generalised helix that is not circular. *Aim:* to distinguish Lancret's theorem from Corollary 3.1. *Estimated time:* 45 minutes. *Type:* symbolic/proof-based.

Let $\gamma(t) = (3t - t^3, 3t^2, 3t + t^3)$.

- (a) Check that $\gamma'(t) = 3(1 - t^2, 2t, 1 + t^2)$ and that $|\gamma'(t)| = 3\sqrt{2}(1 + t^2)$.
- (b) Compute $\gamma' \times \gamma''$ and show that $\kappa(t) = \tau(t) = \frac{1}{3(1 + t^2)^2}$.
- (c) Deduce from Theorem 3.4 that γ is a generalised helix, determine the angle θ and write the axis u explicitly at $t = 0$.
- (d) Explain why γ is *not* congruent to any circular helix, although τ/κ is constant: which hypothesis of Corollary 3.1 fails?

Exercise 3.3: Spherical or not? Three curves examined. *Aim:* to apply the criterion of Theorem 3.7 and to recognise its hypotheses. *Estimated time:* 45 minutes. *Type:* numerical/computational.

For each of the following curves, say whether it is contained in a sphere and, if so, determine centre and radius.

- (a) The circular helix with $a = 3$, $b = 4$.
- (b) The circle $\gamma(t) = (3 \cos t, 3 \sin t, 4)$.
- (c) Viviani's curve of Example 3.3, read at $t = \frac{\pi}{2}$: compute κ , τ , ρ and check Corollary 3.2.
- (d) For (b), which hypothesis of Theorem 3.7 fails? Why is the answer nevertheless «yes, it is spherical», and why in that case is the sphere *not* unique?

3.7. Worked exercises

Exercise 3.4: Designing a helix from κ and τ . *Type:* symbolic with numerical check. A welding head must follow a trajectory with constant curvature $\kappa_0 = 0.12 \text{ mm}^{-1}$ and constant torsion $\tau_0 = 0.16 \text{ mm}^{-1}$.

- (a) Determine radius, parameter and pitch of the corresponding circular helix.
- (b) Compute the angular velocity of the Frenet frame and the helix angle.
- (c) Check the result by recomputing κ and τ from the helix found.

Solution. Strategy. Corollary 3.1 says that the trajectory exists, is unique up to direct rigid motions, and is the circular helix with $a = \kappa_0/(\kappa_0^2 + \tau_0^2)$ and $b = \tau_0/(\kappa_0^2 + \tau_0^2)$. One applies the formula and checks backwards with Theorem 3.3.

Step 1 — the parameters. $\kappa_0^2 + \tau_0^2 = 0.0144 + 0.0256 = 0.04 \text{ mm}^{-2}$. Hence

$$a = \frac{0.12}{0.04} = 3 \text{ mm}, \quad b = \frac{0.16}{0.04} = 4 \text{ mm}.$$

The pitch is $p = 2\pi b = 8\pi = 25.1327412 \text{ mm}$, and $c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5 \text{ mm}$.

Step 2 — direct check. By Theorem 3.3 the helix with $a = 3$, $b = 4$ has

$$\kappa = \frac{3}{9+16} = \frac{3}{25} = 0.12, \quad \tau = \frac{4}{9+16} = \frac{4}{25} = 0.16,$$

that is the initial data. ✓

Step 3 — angular velocity and angle. By Proposition 3.1,

$$|\omega| = \sqrt{\kappa_0^2 + \tau_0^2} = \sqrt{0.04} = 0.2 \text{ rad/mm},$$

which coincides with $1/c = 1/5 = 0.2$ ✓ (for the helix $\kappa^2 + \tau^2 = (a^2 + b^2)/(a^2 + b^2)^2 = 1/c^2$). The helix angle with respect to the axis satisfies, by Theorem 3.4, $\cot \theta = \tau_0/\kappa_0 = b/a = 4/3$, hence $\theta = \operatorname{arccot} \frac{4}{3} = 36.8698976^\circ$; equivalently $\cos \theta = b/c = 4/5 = 0.8$ and $\sin \theta = a/c = 3/5 = 0.6$, the classical (3, 4, 5) triple. ✓

Interpretation. At equal advance speed, the head rotates by 0.2 rad for every millimetre travelled; of this rotation the component about the direction of travel is $\tau_0 = 0.16$ rad/mm, that is 80%. It is the opposite of Example 3.2, where torsion was one tenth of curvature: here the pitch is large compared with the radius and the trajectory is «taut», almost a straight climb.

Typical errors. (1) Confusing the parameter b with the pitch: the pitch is $2\pi b$, not b . (2) Swapping the two formulas and getting $a = 4$, $b = 3$: the mistake is caught by recomputing κ , which would come out 0.16 instead of 0.12. (3) Forgetting that the helix is determined only up to a direct rigid motion: the position of the axis and the handedness are not given by κ and τ .

Extension. What happens if one asks for $\tau_0 = 0$? Corollary 3.1(i) answers: the trajectory is the circle of radius $1/\kappa_0 = 1/0.12 = 8.3333333$ mm, and the formula for b correctly gives $b = 0$, that is the degenerate helix.

Exercise 3.5: Viviani's curve is spherical: checking with the criterion. *Type:* proof-based with double numerical check. Consider $\gamma(t) = (1 + \cos t, \sin t, 2 \sin \frac{t}{2})$ on $(-\pi, \pi)$.

- (a) Check directly that the curve lies on a sphere and identify it.
- (b) Verify condition (ii) of Theorem 3.7 at $t = 0$, exactly.
- (c) Verify the same condition at $t = 0.9$ numerically and compare centre and radius of the osculating sphere with those found in (a).

Solution. Strategy. Point (a) is a three-line algebraic computation; (b) and (c) are the verification that the *criterion* returns what (a) has already established — this is how one checks a criterion: by applying it where the answer is known.

Step 1 — lying on the sphere. Using $2 \sin^2 \frac{t}{2} = 1 - \cos t$,

$$|\gamma|^2 = (1 + \cos t)^2 + \sin^2 t + 4 \sin^2 \frac{t}{2} = 1 + 2 \cos t + 1 + 2 - 2 \cos t = 4.$$

The curve lies on the sphere with centre $O = (0, 0, 0)$ and radius $r = 2$.

Step 2 — hypotheses of the theorem. $|\gamma'|^2 = 1 + \cos^2 \frac{t}{2} = \frac{3+\cos t}{2} \geq 1 > 0$: regular. $|\gamma' \times \gamma''|^2 = \frac{13+3 \cos t}{8} \geq \frac{10}{8} > 0$: biregular. $\tau(t) = \frac{6 \cos(t/2)}{13+3 \cos t}$ vanishes only for $\cos \frac{t}{2} = 0$, that is $t = \pm\pi$, excluded from the interval: $\tau \neq 0$ on $(-\pi, \pi)$. ✓ All the hypotheses of Theorem 3.7 hold.

Step 3 — exact verification at $t = 0$. As in Example 3.3: $\kappa(0) = \frac{1}{2}$, $\tau(0) = \frac{3}{8}$, $\rho(0) = 2$, and $\rho'(0) = 0$ because κ is even in t and the reparametrisation $s(t)$ is odd, so that $\rho(s)$ is even in s and its derivative vanishes at $s = 0$. Hence $\rho'\sigma$ vanishes at $t = 0$, and the centre

of the osculating sphere is

$$q = \gamma(0) + \rho(0)N(0) = (2, 0, 0) + 2(-1, 0, 0) = (0, 0, 0) = O,$$

with $r = \sqrt{\rho(0)^2 + 0} = 2$: it coincides with the sphere of Step 1. ✓ As for condition (ii), at $t = 0$ it forces $(\rho'\sigma)'(0) = -\tau(0)\rho(0) = -\frac{3}{8} \cdot 2 = -0.75$: the numerical computation of $(\rho'\sigma)'$ at $t = 0$ gives exactly -0.75 . ✓

Step 4 — numerical verification at $t = 0.9$. From the closed forms:

$$\kappa = 0.5594064, \quad \tau = 0.3634540, \quad \rho = 1.7876090, \quad \sigma = 2.7513794, \quad |\gamma'| = 1.3456615.$$

The derivative with respect to s is obtained by dividing the one with respect to t by $|\gamma'|$: $\rho' = -0.3259869$, hence $\rho'\sigma = -0.8969135$. Then

$$\rho^2 + (\rho'\sigma)^2 = 3.1955461 + 0.8044539 = 4.0000000 = r^2 \quad \checkmark$$

and

$$\tau\rho = 0.6497137, \quad (\rho'\sigma)' = -0.6497137, \quad \varsigma = 0 \quad \checkmark$$

(the sum vanishes to forty decimal places in the computation performed). The centre of the osculating sphere, computed with $q = \gamma + \rho N + \rho'\sigma B$, comes out $(0, 0, 0)$ to forty digits, and the radius 2: the osculating sphere *does not change*, as condition (iii) of Theorem 3.7 requires. ✓

Interpretation. Viviani's curve is the intersection of the sphere $x^2 + y^2 + z^2 = 4$ with the cylinder $(x - 1)^2 + y^2 = 1$, that is with the cylinder whose diameter is the radius of the sphere. It is the first historical example of a non-planar spherical curve studied analytically, and it shows concretely that «lying on a sphere» is a *differential* condition on κ and τ : one does not need the equation of the sphere to notice it.

Typical errors. (1) Differentiating ρ with respect to t instead of s : condition (ii) is written in arc length, and one is off by a factor $|\gamma'|$. (2) Using the form « $\rho^2 + (\rho'\sigma)^2$ constant» as a criterion: that would be a mistake, as Remark 3.5 shows. (3) Extending the curve to $t = \pm\pi$, where $\tau = 0$ and the hypotheses fail (it is the double point of the curve).

Extension. Check that at the point $t = 0$ the radius of curvature attains its maximum $\rho = 2 = r$: it is the only point at which, by Corollary 3.2, the equality $\kappa = 1/r$ can hold. Geometrically it is the point where the curve is as «straight» as possible on that sphere: there it osculates a great circle.

3.8. Self-assessment quiz

- (Definition) With the convention of this chapter, the torsion of a biregular curve is defined by
A) $T' = \tau N$ B) $B' = -\tau N$ C) $N' = \tau B$ only D) $\tau = |B'|$
- (Hypothesis) In Theorem 3.2 the hypothesis $\kappa(s) > 0$ for *every* s serves to
A) make the curve bounded B) guarantee that N , B and hence the Frenet system are defined at every point C) make τ positive D) ensure that the curve is simple
- (Symbolic computation) For the circular helix of radius a and parameter b the ratio τ/κ equals
A) a/b B) b/a C) ab D) $b/(a^2 + b^2)$

-
4. (Numerical computation) The circular helix with $a = 3$ and $b = 4$ has Darboux vector of modulus
 A) 0.12 B) 0.16 C) 0.20 D) 0.28
 5. (Proof step) In the proof of Theorem 3.2 orthonormality of the frame is preserved because
 A) the coefficient matrix is skew-symmetric, so that $G \equiv I_3$ solves $G' = AG + GA^\top$ with the same initial datum B) the solution is bounded C) $\kappa > 0$ D) the integral of a unit vector is a unit vector
 6. (Error diagnosis) A student writes: «The curve has $\rho^2 + (\rho'\sigma)^2$ constant, hence it lies on a sphere.» The mistake is
 A) the right quantity is $\rho^2 - (\rho'\sigma)^2$ B) that condition is only necessary: one needs $\tau\rho + (\rho'\sigma)' = 0$, and the circular helix is a counterexample C) the hypothesis $\kappa > 0$ is missing D) the formula holds only for plane curves
 7. (Transfer) A biregular curve of class C^3 with τ/κ constant and κ non-constant is
 A) necessarily a circular helix B) necessarily planar C) a generalised helix but not a circular helix D) necessarily spherical
 8. (Source and metacognition) The complete proof of existence in the fundamental theorem of space curves, reduced to the existence and uniqueness theorem for ordinary differential equations, is found in
 A) do Carmo, *Differential Geometry of Curves and Surfaces*, Section 1-5 for the statement and the appendix to Chapter 4 for existence B) do Carmo, Section 2-4 C) do Carmo, Chapter 5 D) do Carmo, appendix to Chapter 1

3.9. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnosis.

- 1) **B.** *Rationale:* this is Definition 3.3, made legitimate by Lemma 3.2 ($B' \parallel N$). *Distractors:* A is the first Frenet formula with κ replaced by τ ; C is true but incomplete (it is part of the second formula, and by itself it does not define τ because it does not say that N' has no other components); D is false because $|B'| = |\tau|$: the modulus loses the sign, which is precisely the information distinguishing right-handed from left-handed helices (Proposition 3.4). *Error diagnosed:* confusing an oriented quantity with its modulus. *Source:* Powell, M435, Definition 2.32 (with the opposite sign convention). *Prerequisite:* derivative of a unit vector. *Remedial work:* recompute B' for the right-handed helix and for its mirror image.
- 2) **B.** *Rationale:* if $\kappa(s_1) = 0$ the normal $N = T'/\kappa$ is not defined at s_1 , and with it neither are B , τ and the system of Step 1. *Distractors:* A is false (the helix has $\kappa > 0$ and is unbounded); C is false, τ has arbitrary sign; D is false, the theorem says nothing about self-intersections (Remark 3.2(c)). *Error diagnosed:* attributing to a hypothesis a conclusion that does not belong to it. *Source:* Gallier, ch. 19, §19.8, warning on the role of $c(s) > 0$. *Prerequisite:* definition of N . *Remedial work:* reread Remark 3.2(a).
- 3) **B.** *Rationale:* from Theorem 3.3, $\tau/\kappa = \frac{b/(a^2+b^2)}{a/(a^2+b^2)} = b/a$; the common denominator cancels. *Distractors:* A inverts the ratio; C is dimensionally wrong (τ/κ is dimensionless, ab is an area); D is τ , not the ratio. *Error diagnosed:* failing to cancel the common factor, or swapping numerator and denominator. *Source:* Theorem 3.3; Gallier, Problem 19.20. *Prerequisite:* algebra of fractions. *Remedial work:* check that $\tau/\kappa = \cot \theta$ is the reciprocal

of the slope, as Theorem 3.4 requires.

4) **C.** *Rationale:* $\kappa = 3/25 = 0.12$, $\tau = 4/25 = 0.16$ and $|\omega| = \sqrt{\kappa^2 + \tau^2} = \sqrt{0.0144 + 0.0256} = \sqrt{0.04} = 0.2$; equivalently $1/\sqrt{a^2 + b^2} = 1/5 = 0.2$. *Distractors:* A is κ ; B is τ ; D is the sum $\kappa + \tau$, that is the mistake of adding moduli instead of squares. *Error diagnosed:* adding orthogonal components without Pythagoras' theorem. *Source:* Proposition 3.1. *Prerequisite:* norm of a combination of orthogonal unit vectors. *Remedial work:* redo the computation with $a = b = 1$, where $|\omega| = \sqrt{2}/2 = 0.7071068$.

5) **A.** *Rationale:* this is Step 2 of the proof: the matrix A of the system satisfies $A + A^\top = 0$, hence the constant matrix $G \equiv I_3$ solves $G' = AG + GA^\top$; uniqueness for linear systems (Lemma 3.4) does the rest. *Distractors:* B is irrelevant (a bounded solution is not thereby orthonormal); C is needed elsewhere, to guarantee that the system exists; D is false (the integral of a unit vector is in general not a unit vector: $\int_0^1 T$ has modulus ≤ 1 , not $= 1$). *Error diagnosed:* accepting orthonormality as «obvious» instead of deducing it from skew-symmetry. *Source:* Wang, *Fundamental Theorem of the Local Theory of Curves*, §5; do Carmo, appendix to Chapter 4. *Prerequisite:* uniqueness for linear systems. *Remedial work:* write out the six equations for g_{ij} and check that $G \equiv I_3$ solves them.

6) **B.** *Rationale:* this is exactly Remark 3.5: the circular helix with $a = b = 1$ has $\rho = 2$ constant, hence $\rho^2 + (\rho'\sigma)^2 = 4$ constant, but $\varsigma = \tau\rho = 1 \neq 0$ and the curve is unbounded. *Distractors:* A invents a sign; C is false, $\kappa > 0$ is already among the hypotheses; D is false, for plane curves $\tau = 0$ and σ is not even defined. *Error diagnosed:* mistaking a necessary condition for a sufficient one. *Source:* Theorem 3.7; da Silva, Theorem 2.3.1. *Prerequisite:* distinguishing «only if» from «if». *Remedial work:* redo the computation on the helix with $a = b = 1$ and then on Viviani's curve.

7) **C.** *Rationale:* τ/κ constant gives a generalised helix by Theorem 3.4; but Corollary 3.1 requires κ and τ constant separately, and here κ is not. The example $\gamma(t) = (3t - t^3, 3t^2, 3t + t^3)$ of Remark 3.3 is precisely of this kind. *Distractors:* A confuses Lancret with the corollary; B is false (τ/κ constant and non-zero excludes $\tau \equiv 0$); D is unrelated: nothing links constant slope to lying on a sphere (and indeed helices are never spherical). *Error diagnosed:* confusing «constant ratio» with «both constant». *Source:* Theorem 3.4. *Prerequisite:* reading the hypotheses. *Remedial work:* compute κ and τ of the example at $t = 0$ and $t = 1$ and check that the ratio stays 1 while the values change.

8) **A.** *Rationale:* do Carmo states the theorem in Section 1-5 and defers the proof of existence to the appendix to Chapter 4, where the existence and uniqueness theorem for ordinary differential equations is proved. *Distractors:* B, C and D point to sections dealing with surfaces (Chapter 2), intrinsic geometry (Chapter 4) or the appendix to Chapter 1, which concerns continuity and differentiability. *Error diagnosed:* citing a text «from memory» without checking where the proof is. *Source:* do Carmo, Section 1-5; Powell, M435, Theorem 2.39, which refers explicitly to the pages of do Carmo. *Prerequisite:* none. *Remedial work:* compare the structure of that proof with the one of Theorem 3.2 and of the appendix to this chapter, which instead carries it out in full.

BEST PRACTICE

Assigned exercise 3.1 — answers. (a) As in Example 3.1: $\gamma' \times \gamma'' = (6t^2, -6t, 2)$, $\langle \gamma' \times \gamma'', \gamma''' \rangle = 12$ and

$$\kappa(t) = \frac{2\sqrt{9t^4 + 9t^2 + 1}}{(1 + 4t^2 + 9t^4)^{3/2}}, \quad \tau(t) = \frac{3}{9t^4 + 9t^2 + 1}.$$

The denominator of τ is $9t^4 + 9t^2 + 1 \geq 1 > 0$, hence $\tau(t) = 3/(9t^4 + 9t^2 + 1) > 0$ for every t : by Proposition 3.3(ii) no arc is planar.

(b) At $t = 2$: $9 \cdot 16 + 9 \cdot 4 + 1 = 144 + 36 + 1 = 181$ and $1 + 16 + 144 = 161$, hence

$$\kappa(2) = \frac{2\sqrt{181}}{161^{3/2}} = 0.0131714, \quad \tau(2) = \frac{3}{181} = 0.0165746.$$

Direct numerical check (high-precision derivatives): 0.013171355 and 0.016574586. ✓

(c) At $t = 0$: $\gamma'(0) = (1, 0, 0)$ of modulus 1 (the parameter is arc length at that point), $\gamma''(0) = (0, 2, 0)$, hence $T = (1, 0, 0)$, $\kappa(0) = 2$, $N = (0, 1, 0)$, $B = T \times N = (0, 0, 1)$. The osculating circle (Definition 3.7) has radius $\rho = 1/2 = 0.5$ and centre

$$q = \gamma(0) + \rho N = (0, 0, 0) + \frac{1}{2}(0, 1, 0) = (0, \frac{1}{2}, 0),$$

and lies in the osculating plane, spanned by $T = (1, 0, 0)$ and $N = (0, 1, 0)$, that is the plane $z = 0$. Check: $|q - \gamma(0)| = |(0, \frac{1}{2}, 0)| = \frac{1}{2} = \rho$. ✓

(d) τ/κ at $t = 0$ equals $3/2 = 1.5$; at $t = 1$ it equals $0.1578947/0.1664235 = 0.9487525$. It is not constant, hence by Theorem 3.4 the twisted cubic is *not* a generalised helix.

BEST PRACTICE

Assigned exercise 3.2 — answers. (a) $\gamma' = (3 - 3t^2, 6t, 3 + 3t^2) = 3(1 - t^2, 2t, 1 + t^2)$ and

$$|\gamma'|^2 = 9[(1 - t^2)^2 + 4t^2 + (1 + t^2)^2] = 9[2 + 4t^2 + 2t^4] = 18(1 + t^2)^2,$$

hence $|\gamma'| = 3\sqrt{2}(1 + t^2)$.

(b) $\gamma'' = 6(-t, 1, t)$ and $\gamma''' = 6(-1, 0, 1)$. Then

$$\gamma' \times \gamma'' = 18[(1 - t^2, 2t, 1 + t^2) \times (-t, 1, t)] = 18(t^2 - 1, -2t, 1 + t^2),$$

of modulus $18\sqrt{(t^2 - 1)^2 + 4t^2 + (1 + t^2)^2} = 18\sqrt{2}(1 + t^2)$. Hence

$$\kappa = \frac{18\sqrt{2}(1 + t^2)}{(3\sqrt{2}(1 + t^2))^3} = \frac{18\sqrt{2}(1 + t^2)}{54\sqrt{2}(1 + t^2)^3} = \frac{1}{3(1 + t^2)^2}.$$

Moreover $\langle \gamma' \times \gamma'', \gamma''' \rangle = 18 \cdot 6[-(t^2 - 1) + 0 + (1 + t^2)] = 108 \cdot 2 = 216$ and $|\gamma' \times \gamma''|^2 = 648(1 + t^2)^2$, hence $\tau = 216/(648(1 + t^2)^2) = 1/(3(1 + t^2)^2) = \kappa$. Numerical check at $t = 0.7$: both equal 0.150143386935. ✓

(c) $\tau/\kappa \equiv 1 = \cot \theta$ gives $\theta = \frac{\pi}{4} = 45^\circ$. At $t = 0$: $\gamma'(0) = 3(1, 0, 1)$, $|\gamma'(0)| = 3\sqrt{2}$, hence $T(0) = \frac{1}{\sqrt{2}}(1, 0, 1)$; $\gamma' \times \gamma''(0) = 18(-1, 0, 1)$, hence $B(0) = \frac{1}{\sqrt{2}}(-1, 0, 1)$. The axis is

$$u = \cos \frac{\pi}{4} T + \sin \frac{\pi}{4} B = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}[(1, 0, 1) + (-1, 0, 1)] = \frac{1}{2}(0, 0, 2) = (0, 0, 1),$$

that is the z axis. Check: $\langle T(t), (0, 0, 1) \rangle = \frac{3(1+t^2)}{3\sqrt{2}(1+t^2)} = \frac{1}{\sqrt{2}} = \cos 45^\circ$ for every t . ✓

(d) The hypothesis that κ and τ be *constant* fails: here $\kappa(0) = \frac{1}{3} = 0.3333333$ and $\kappa(1) = \frac{1}{12} = 0.0833333$. Theorem 3.4 requires only the ratio to be constant, Corollary 3.1 requires both functions to be constant; and by Theorem 3.2 two congruent curves have the *same* κ and τ , which for the circular helix are constant: hence no circular helix can be congruent to this curve.

BEST PRACTICE

Assigned exercise 3.3 — answers. (a) *No*. For the helix with $a = 3$, $b = 4$ one has

$\kappa = 0.12$, $\tau = 0.16$, $\rho = 1/0.12 = 8.3333333$ constant, hence $\rho' = 0$ and

$$\varsigma = \tau\rho + (\rho'\sigma)' = 0.16 \cdot 8.3333333 + 0 = 1.3333333 = \frac{4}{3} \neq 0 :$$

by Theorem 3.7 the curve is not spherical. (Note that $\varsigma = \tau\rho = b/a = 4/3$, in agreement with Remark 3.5.) Besides, the helix is unbounded.

(b) *Yes*. The circle $\gamma(t) = (3 \cos t, 3 \sin t, 4)$ has $|\gamma(t)|^2 = 9 + 16 = 25$: it lies on the sphere with centre the origin and radius 5. But it also lies on every sphere with centre $(0, 0, 4 + h)$ and radius $\sqrt{9 + h^2}$, for every $h \in \mathbb{R}$.

(c) At $t = \frac{\pi}{2}$: $\cos t = 0$ and $\cos \frac{t}{2} = \frac{\sqrt{2}}{2}$, hence

$$\kappa = \frac{\sqrt{13}}{3^{3/2}} = 0.6938887, \quad \tau = \frac{6 \cdot \frac{\sqrt{2}}{2}}{13} = \frac{3\sqrt{2}}{13} = 0.3263570, \quad \rho = 1.4411534.$$

Since $\rho = 1.4411534 < 2 = r$, one has $\kappa = 0.6938887 > 1/r = 0.5$: Corollary 3.2 is verified. ✓

(d) In case (b) the hypothesis $\tau \neq 0$ fails: the circle is planar, $\tau \equiv 0$, and $\sigma = 1/\tau$ is not even defined, so the criterion of Theorem 3.7 does not apply. The answer is still «yes» because lying on a sphere is checked directly; and the sphere is not unique precisely because the curve is planar: the spheres containing the circle are exactly those whose centre lies on the *axis* of the circle — the line through the centre $(0, 0, 4)$ orthogonal to the plane $z = 4$ — with radius fixed by Pythagoras' theorem, that is the one-parameter family described in (b). This is exactly the case excluded at the end of the proof of Theorem 3.7, where uniqueness is obtained by observing that a curve with $\tau \neq 0$ cannot lie on a circle.

3.10. Chapter formulary

Frenet frame and Frenet–Serret formulas (arc length, $\kappa > 0$).

$$T = \gamma', \quad \kappa = |\gamma''|, \quad N = \frac{\gamma''}{\kappa}, \quad B = T \times N, \quad N = B \times T, \quad T = N \times B.$$

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N; \quad \frac{d}{ds} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}.$$

Darboux vector: $\omega = \tau T + \kappa B$, $X' = \omega \times X$, $|\omega| = \sqrt{\kappa^2 + \tau^2}$.

Arbitrary parameter ($v = |\gamma'|$).

$$\gamma' = vT, \quad \gamma'' = v'T + v^2\kappa N, \quad \gamma' \times \gamma'' = v^3\kappa B,$$

$$\kappa = \frac{|\gamma' \times \gamma''|}{|\gamma'|^3}, \quad \tau = \frac{\det(\gamma', \gamma'', \gamma''')}{|\gamma' \times \gamma''|^2}, \quad B = \frac{\gamma' \times \gamma''}{|\gamma' \times \gamma''|}.$$

Mind the sign: do Carmo, Gallier and Powell define $B' = +\tau N$ and hence have the formula with a minus sign.

Notable cases.

$$\kappa \equiv 0 \iff \text{straight line}; \quad \tau \equiv 0 \iff \text{plane curve};$$

$\kappa \equiv \kappa_0 > 0$ and $\tau \equiv 0 \iff$ circle of radius $1/\kappa_0$.

Circular helix $\gamma(t) = (a \cos t, a \sin t, bt)$, $c = \sqrt{a^2 + b^2}$, pitch $p = 2\pi b$:

$$|\gamma'| = c, \quad \kappa = \frac{a}{a^2 + b^2}, \quad \tau = \frac{b}{a^2 + b^2}, \quad |\omega| = \frac{1}{c};$$

$$\text{from constant } (\kappa_0, \tau_0): \quad a = \frac{\kappa_0}{\kappa_0^2 + \tau_0^2}, \quad b = \frac{\tau_0}{\kappa_0^2 + \tau_0^2}.$$

Lancret: generalised helix $\iff \frac{\tau}{\kappa} = \cot \theta$ constant; axis $u = \cos \theta T + \sin \theta B$.

Osculating circle and sphere ($\rho = 1/\kappa$, $\sigma = 1/\tau$, $\rho' = d\rho/ds$).

osculating circle: centre $\gamma + \rho N$, radius ρ , in the osculating plane;

osculating sphere: centre $q = \gamma + \rho N + \rho' \sigma B$, radius $r = \sqrt{\rho^2 + (\rho' \sigma)^2} \geq \rho$.

Spherical curves: with $\kappa > 0$ and τ never zero,

$$\gamma \text{ lies on a sphere } \iff \varsigma := \tau \rho + (\rho' \sigma)' \equiv 0 \iff q' \equiv 0,$$

and then centre q and radius r are the ones written above, both constant. The condition $\rho^2 + (\rho' \sigma)^2 = \text{const}$ alone is **necessary and not sufficient** (counterexample: the circular helix); it becomes sufficient if ρ' never vanishes. On a sphere of radius r : $\kappa \geq 1/r$, with equality at one point at most.

3.11. Appendix: existence and uniqueness for linear systems

Theorem 3.2 rests on a single analytic fact: the Cauchy problem $X' = \mathcal{A}(s)X$, $X(s_0) = X_0$, with $\mathcal{A}$ continuous, has exactly one solution on the whole interval. It is the *linear* case of the Picard–Lindelöf theorem, and in this form the proof closes without needing anything that is not elementary: we give it in full, so that no link is missing in the chapter. The strategy is the classical one: the problem is rewritten as the search for a fixed point of an integral operator, and the contraction mapping theorem is applied.

Lemma 3.3: Contraction mapping theorem (Banach–Caccioppoli). *Let (X, d) be a non-empty complete metric space and let $\Phi : X \rightarrow X$ satisfy $d(\Phi x, \Phi y) \leq \lambda d(x, y)$ for all $x, y \in X$, with a fixed $\lambda \in (0, 1)$. Then Φ has exactly one fixed point. Moreover, if for some $m \geq 1$ the iterate Φ^m is a contraction (without Φ being one), Φ still has exactly one fixed point.*

Proof. Existence. Choose $x_0 \in X$ and set $x_{n+1} = \Phi x_n$. By induction $d(x_{n+1}, x_n) \leq \lambda^n d(x_1, x_0)$. For $m > n$, the triangle inequality and the sum of the geometric series give

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq d(x_1, x_0) \sum_{k=n}^{\infty} \lambda^k = \frac{\lambda^n}{1-\lambda} d(x_1, x_0) \xrightarrow{n \rightarrow \infty} 0.$$

The sequence is Cauchy, and by completeness it converges to some $x_* \in X$. Since Φ is Lipschitz, hence continuous, $\Phi x_* = \lim_n \Phi x_n = \lim_n x_{n+1} = x_*$.

Uniqueness. If $\Phi x = x$ and $\Phi y = y$, then $d(x, y) = d(\Phi x, \Phi y) \leq \lambda d(x, y)$, that is $(1-\lambda)d(x, y) \leq 0$; since $1-\lambda > 0$ and $d \geq 0$, we get $d(x, y) = 0$ and $x = y$.

Case Φ^m a contraction. By what has already been proved Φ^m has a unique fixed point x_* . Then $\Phi^m(\Phi x_*) = \Phi(\Phi^m x_*) = \Phi x_*$, that is Φx_* is a fixed point of Φ^m ; by uniqueness, $\Phi x_* = x_*$.

Conversely every fixed point of Φ is a fixed point of Φ^m , hence equal to x_* . $\square$

Lemma 3.4: Linear Cauchy problem. Let $I \subseteq \mathbb{R}$ be an interval, $\mathcal{A} : I \rightarrow M_n(\mathbb{R})$ continuous, $s_0 \in I$ and $X_0 \in \mathbb{R}^n$. Then there exists exactly one function $X \in C^1(I, \mathbb{R}^n)$ such that

$$X'(s) = \mathcal{A}(s)X(s) \quad \text{for every } s \in I, \quad X(s_0) = X_0.$$

Proof. Step 1 — reduction to a compact interval. Let $[\alpha, \beta] \subseteq I$ be a compact interval with $s_0 \in [\alpha, \beta]$. Since $\mathcal{A}$ is continuous and $[\alpha, \beta]$ compact, the function $s \mapsto \|\mathcal{A}(s)\|$ (operator norm associated with the Euclidean norm) is continuous and bounded: let $L = \max_{s \in [\alpha, \beta]} \|\mathcal{A}(s)\| < +\infty$, so that $|\mathcal{A}(s)v| \leq L|v|$ for every $v \in \mathbb{R}^n$ and every $s \in [\alpha, \beta]$.

Step 2 — the Picard operator. Let $E = C([\alpha, \beta], \mathbb{R}^n)$ with the norm $\|X\|_\infty = \max_{s \in [\alpha, \beta]} |X(s)|$; $(E, \|\cdot\|_\infty)$ is a Banach space (Remark 3.6). Define $\Phi : E \rightarrow E$ by

$$(\Phi X)(s) = X_0 + \int_{s_0}^s \mathcal{A}(u)X(u) du.$$

The integrand is continuous, hence ΦX is well defined, of class C^1 , and in particular continuous: Φ maps E into itself. By the fundamental theorem of calculus, $X \in E$ satisfies $\Phi X = X$ if and only if X is of class C^1 , satisfies $X' = \mathcal{A}X$ on $[\alpha, \beta]$ and $X(s_0) = X_0$: solving the Cauchy problem is the same as finding a fixed point of Φ .

Step 3 — the iterated estimate. We prove by induction on $m \geq 1$ that for all $X, Y \in E$ and every $s \in [\alpha, \beta]$

$$|(\Phi^m X)(s) - (\Phi^m Y)(s)| \leq \frac{L^m |s - s_0|^m}{m!} \|X - Y\|_\infty.$$

For $m = 1$:

$$|(\Phi X)(s) - (\Phi Y)(s)| = \left| \int_{s_0}^s \mathcal{A}(u)(X(u) - Y(u)) du \right| \leq \left| \int_{s_0}^s L |X(u) - Y(u)| du \right|,$$

and the last quantity is $\leq L |s - s_0| \|X - Y\|_\infty$. Assuming the estimate for m , one has

$$\begin{aligned} |(\Phi^{m+1} X)(s) - (\Phi^{m+1} Y)(s)| &\leq \left| \int_{s_0}^s L |(\Phi^m X)(u) - (\Phi^m Y)(u)| du \right| \\ &\leq \left| \int_{s_0}^s \frac{L^{m+1} |u - s_0|^m}{m!} du \right| \cdot \|X - Y\|_\infty = \frac{L^{m+1} |s - s_0|^{m+1}}{(m+1)!} \|X - Y\|_\infty, \end{aligned}$$

as required.

Step 4 — conclusion on $[\alpha, \beta]$. Setting $\ell = \beta - \alpha$, Step 3 gives

$$\|\Phi^m X - \Phi^m Y\|_\infty \leq \frac{(L\ell)^m}{m!} \|X - Y\|_\infty.$$

Since $\sum_m (L\ell)^m / m! = e^{L\ell}$ converges, the general term tends to zero: there is an m with $(L\ell)^m / m! < 1$, and for that m the iterate Φ^m is a contraction. By Lemma 3.3, Φ has exactly one fixed point in E : the Cauchy problem has exactly one solution on $[\alpha, \beta]$.

Step 5 — from $[\alpha, \beta]$ to I . Write I as the union of an increasing sequence of compact intervals $[\alpha_k, \beta_k]$ containing s_0 (for instance $[\alpha_k, \beta_k] = [a_k, b_k]$ with $a_k \downarrow \inf I$, $b_k \uparrow \sup I$ inside I). By Step 4 on each of them there is a unique solution X_k ; by uniqueness, X_{k+1} restricted to $[\alpha_k, \beta_k]$ coincides with X_k . The function X defined on I by $X(s) = X_k(s)$ for $s \in [\alpha_k, \beta_k]$ is therefore well posed, is of class C^1 , solves the problem, and is unique: two solutions on I agree on every compact set and hence on I . $\square$

Remark 3.6: What we import. In the whole chapter exactly one fact is **imported**, that is, used without proof: the completeness of the space $(C([\alpha, \beta], \mathbb{R}^n), \|\cdot\|_\infty)$, namely the fact that a sequence of continuous functions on a compact set, Cauchy for the maximum norm, converges uniformly to a continuous function. It is the classical theorem on uniform convergence from a first analysis course; a complete proof, in exactly the form used here, is in Wang, *Fundamental Theorem of the Local Theory of Curves*, Theorem 3.4 (University of Chicago REU 2018), verified and read in full.

What is missing, exactly: that step alone. The contraction mapping theorem is proved (Lemma 3.3), the iterated estimate is proved (Lemma 3.4, Step 3), the extension to the whole interval is proved (Step 5), and the fundamental theorem of curves is deduced from these (Theorem 3.2). No other result of the chapter is imported: in particular the non-linear case of the Picard–Lindelöf theorem is neither needed nor invoked.

Sources and bibliography

- [1] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice–Hall, Englewood Cliffs, 1976 (revised and updated edition, Dover, Mineola, 2016). Chapter 1, Section 1-5 «The Local Theory of Curves Parametrized by Arc Length»: Frenet trihedron, curvature, torsion, Frenet–Serret formulas and the statement of the FUNDAMENTAL THEOREM OF THE LOCAL THEORY OF CURVES; the proof of existence is deferred to the appendix to Chapter 4. *Warning:* do Carmo defines the torsion by $B' = +\tau N$, the opposite sign to the one adopted here. The precise pages (20–21 for uniqueness, 309–311 for existence, 22 for the exercises of Section 1-5) are those indicated by Powell, M435 (source no. 2), §2.7 and Exercise 2.33; it was not possible to check them against a copy of the book, and they are reported here with this explicit reservation — verified as above on 5 September 2026.
- [2] M. Powell, *M435: Introduction to Differential Geometry*, course notes, §2 «Geometry of Curves». Text read in full: Definition 2.28 (principal normal), Definition 2.29 (osculating plane), Definition 2.30 (binormal), Proposition 2.31 ($|B| = 1$, $B \perp B'$, $B' \perp T$), Definition 2.32 (torsion, with the explicit warning «many authors use a minus sign in this definition. We are following Do Carmo»), Theorem 2.34 (Serret–Frenet formulas, with the proof via $N = B \wedge T$ and the matrix variant based on skew-symmetry), Proposition 2.37 ($\kappa \equiv 0 \Rightarrow$ line, $\tau \equiv 0 \Rightarrow$ planar), Theorem 2.39 (fundamental theorem, with the reference to the pages of do Carmo), §2.8 «Spherical curves» (curves on the sphere, geodesic curvature, Theorem 2.48 and Corollary 2.49: the geodesics of S^2 are arcs of great circles). Available at maths.gla.ac.uk/~mpowell/M435-intro-chapter-1-geometry-of-curves.pdf — verified on 5 September 2026.
- [3] J. Gallier, *Geometric Methods and Applications for Computer Science and Engineering*, 2nd edition, Springer, *Texts in Applied Mathematics* 38, Chapter 19 «Basics of the Differential Geometry of Curves», pp. 529 ff. Text read in full: §19.6 «The Frenet Frame», §19.7 «Torsion», §19.8 «The Frenet Equations» with Lemma 19.7 (Frenet–Serret formulas and formulas for κ and τ in an arbitrary parameter), the computation of curvature and torsion of the circular helix $f(t) = (a \cos t, a \sin t, kt)$ — taken up in Problem 19.20 —, and §19.9 «Osculating Spheres» with Definition 19.15 and Lemma 19.8, giving de Saint Venant’s formula (1845) $\Omega - M = R\mathbf{n} - \mathcal{T}R'\mathbf{b}$ for the centre of the osculating sphere, attributed by Gallier to Kreyszig (osculating spheres go back to Fuss, 1806). *Warning:* Gallier too uses $b' = \tau n$, so his τ and his $\mathcal{T} = 1/\tau$ have the opposite sign to ours, and de Saint Venant’s formula becomes, in our convention, $q - \gamma = \rho N + \rho'\sigma B$. Available at cis.upenn.edu/~cis6100/gma-v2-chap19.pdf — verified on 5 September 2026.

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- [4] K. Wang, *Fundamental Theorem of the Local Theory of Curves*, University of Chicago REU 2018. Text read in full: §3 «The Space of Continuous Functions» (Theorem 3.4: completeness of $C_b(V, \mathbb{R}^d)$ with the sup norm — the only result imported in this chapter; Theorem 3.7: contraction mapping theorem, reproved here in Lemma 3.3); §4 «Local Existence and Uniqueness of Solutions to ODEs» (Theorem 4.3 and Corollary 4.4: existence and uniqueness with the iterated estimate $\|\Phi^m X - \Phi^m Y\| \leq \frac{(\int L)^m}{m!} \|X - Y\|$, reproved here in Lemma 3.4); §5 (Theorem 5.6: invariance of arc length, curvature and torsion under rigid motions; Theorem 5.7: fundamental theorem, with the uniqueness proof via the function $|t - \bar{t}|^2 + |n - \bar{n}|^2 + |b - \bar{b}|^2$, taken up in Step 7 of our proof). *Warning:* Wang follows do Carmo's convention, $b' = \tau n$. Available at math.uchicago.edu/~may/REU2018/REUPapers/Wang,Kaixin.pdf — verified on 5 September 2026.
- [5] L. C. B. da Silva, *Differential Geometry of Rotation Minimizing Frames, Spherical Curves, and Quantum Mechanics of a Constrained Particle*, thesis, arXiv:1806.08830 (22 June 2018), doi.org/10.48550/arXiv.1806.08830. Text read for the parts used: §2.2 «Frenet frame and Frenet equations for a space curve», which adopts *our* convention $b' = -\tau n$ (equation 2.17); §2.3 «Spherical curves and osculating spheres», where formula (2.24) gives centre and radius of the osculating sphere, $P_S = \alpha + \rho n + \frac{1}{\tau} \frac{d\rho}{ds} b$ and $R_S = \sqrt{\rho^2 + \frac{1}{\tau^2} (\rho')^2}$, and Theorem 2.3.1 — attributed to Kreyszig (1991) and Kühnel (2010) — states that a regular C^4 curve with non-zero torsion lies on a sphere of radius r if and only if $\varsigma(s) = \tau \rho + \frac{d}{ds}(\frac{\rho'}{\tau}) = 0$: this is Theorem 3.7 of this chapter, proved here in both directions. §3.2 also reports the equivalent characterisation of Bishop (1975) by means of *rotation minimizing frames* (Theorem 3.2.1) — verified on 5 September 2026.

CHAPTER 4

Global theorems on plane curves

Chapter 2 studied curvature *locally*, point by point. Here we collect the *global* theorems, which concern the curve as a whole and in which the circle systematically emerges as the extremal object: the rotation index, the support function and curves of constant width with the theorems of Barbier and Blaschke–Lebesgue, the isoperimetric inequality, the four-vertex theorem, and finally the global form of Gauss–Bonnet, which carries the same pattern of reasoning from curves to closed surfaces.

INFO

The idea in one line. Every theorem of this chapter has the form *an integral quantity of the curve is constrained*, and the circle is the equality case: the total curvature equals 2π , the perimeter of a convex body of constant width equals πw , the area at fixed perimeter is maximal, the points of stationary curvature are at least four.

Warning. Conventions of the chapter. The conventions of Chapter 2 apply: plane curves of class C^2 at least, regular ($\gamma' \neq 0$), *closed* when γ and all its derivatives agree at the endpoints, *simple* when γ is injective on $[a, b]$; positive orientation is counterclockwise; s is arc length, T and N are the unit tangent and normal, k is the signed curvature and $\rho = 1/|k|$ the radius of curvature. Decimal separators are points: 0.5.

INFO

Level and prerequisites. Level C–D. The whole chapter rests on Chapter 2: arc length and the Frenet formulas (Theorem 2.1), the curvature of the circle (Theorem 2.2), the characterisation of curves of constant curvature (Theorem 2.5), the evolute and its properties (Theorem 2.6). One also needs Fourier series with Parseval’s identity (Volume II, Chapter 4), used in Barbier’s and the isoperimetric theorem; the Jordan curve theorem (Appendix A), used only in the general case of the four-vertex theorem; and, for Section 4.6, the local form of Gauss–Bonnet on a geodesic triangle, proved for the sphere in Chapter 7.

4.1. The rotation index theorem

Definition 4.1: Rotation index. Let $\gamma : [0, L] \rightarrow \mathbb{R}^2$ be a closed regular curve parametrised by arc length, and let $\vartheta : [0, L] \rightarrow \mathbb{R}$ be a continuous determination of the angle of the unit tangent, i.e. a continuous function with $T(s) = (\cos \vartheta(s), \sin \vartheta(s))$. Since $T(L) = T(0)$, the difference $\vartheta(L) - \vartheta(0)$ is an integer multiple of 2π ; the number

$$\text{ind}(\gamma) = \frac{\vartheta(L) - \vartheta(0)}{2\pi} \in \mathbb{Z}$$

is the **rotation index** (or *turning number*) of γ .

Theorem 4.1: Rotation index theorem (Umlaufsatz, Hopf 1935). Let γ be a closed, regular and simple plane curve. Then $\text{ind}(\gamma) = \pm 1$, the sign being $+1$ if the curve is traversed counterclockwise. Consequently

$$\int_0^L k(s) ds = 2\pi \quad (\text{total curvature}).$$

Proof. Step 1 — reduction to the integral of the curvature. By Step 4 of the proof of Theorem 2.1, $k = \vartheta'$; hence

$$\int_0^L k(s) ds = \vartheta(L) - \vartheta(0) = 2\pi \text{ind}(\gamma),$$

and the claim about the integral follows from the one about the index.

Step 2 — choice of the starting point. Since γ is closed and continuous, its image is compact; let $p = \gamma(0)$ be a point of minimal ordinate (it exists by Weierstrass). Choose coordinates so that p is the origin and $T(0) = (1, 0)$: the whole curve then lies in the half-plane $y \geq 0$, and the line $y = 0$ is a *supporting line*.

Step 3 — the secant map. Let

$$\Delta = \{(s_1, s_2) : 0 \leq s_1 \leq s_2 \leq L\},$$

a closed triangle in the plane, and define $\psi : \Delta \rightarrow S^1$ by

$$\psi(s_1, s_2) = \frac{\gamma(s_2) - \gamma(s_1)}{\|\gamma(s_2) - \gamma(s_1)\|} \quad \text{if } s_1 < s_2 \text{ and } (s_1, s_2) \neq (0, L),$$

$$\psi(s, s) = T(s), \quad \psi(0, L) = -T(0).$$

This is well defined because γ is *simple*: the denominator vanishes only on the diagonal and at the vertex $(0, L)$. Continuity on the diagonal follows from the expansion $\gamma(s_2) - \gamma(s_1) = (s_2 - s_1)T(s_1) + o(s_2 - s_1)$; at the vertex $(0, L)$ one uses the same expansion with $\gamma(L) = \gamma(0)$ and $T(L) = T(0)$, which gives the unit vector $-T(0)$.

Step 4 — lifting. The triangle Δ is convex, hence simply connected; a continuous map into S^1 defined on a simply connected domain admits a continuous *lift* $\Psi : \Delta \rightarrow \mathbb{R}$ with $\psi = (\cos \Psi, \sin \Psi)$, unique up to an additive constant multiple of 2π (lifting theorem; see the bibliography). Fix $\Psi(0, 0) = 0$.

Step 5 — the index as the variation along the hypotenuse. On the diagonal $\psi(s, s) = T(s)$, so $\Psi(s, s)$ is a continuous determination of ϑ , and

$$2\pi \text{ind}(\gamma) = \Psi(L, L) - \Psi(0, 0) = \Psi(L, L).$$

Step 6 — computation along the two legs. By continuity of the lift, $\Psi(L, L)$ can be computed by travelling around the boundary of Δ along the legs instead of the hypotenuse.

First leg $s_1 = 0$, s_2 from 0 to L : here $\psi(0, s_2)$ is the unit vector from $p = \gamma(0)$ to $\gamma(s_2)$. Since the whole curve lies in the half-plane $y \geq 0$ and p has $y = 0$, this vector always has non-negative second component: it never enters the lower half-plane. Starting from $\Psi(0, 0) = 0$ (the vector $T(0) = (1, 0)$) and ending at $\psi(0, L) = -T(0) = (-1, 0)$, the continuous determination must equal π at the end (the only alternative, $-\pi$, would require crossing the lower half-plane). Hence $\Psi(0, L) = \pi$.

Second leg $s_2 = L$, s_1 from 0 to L : now $\psi(s_1, L)$ is the unit vector from $\gamma(s_1)$ to $\gamma(L) = p$, the opposite kind of vector; it always has non-positive second component, and goes from $-T(0)$ to $\psi(L, L) = T(L) = T(0)$. Passing through the lower half-plane, the continuous determination goes from π to 2π .

Step 7 — conclusion. $\Psi(L, L) = 2\pi$, hence $\text{ind}(\gamma) = 1$. If the curve is traversed clockwise one gets -1 by symmetry (apply the result to the reflected curve).

Step 8 — check on the circle. For the counterclockwise circle of radius r , $k = 1/r$ is constant and $L = 2\pi r$, so

$$\int_0^L k \, ds = \frac{2\pi r}{r} = 2\pi. \quad \checkmark$$

And for an ellipse, for any convex curve, for a non-convex but simple curve, the result is the same: the total curvature does not depend on the shape. It is a *topological invariant*.

Step 9 — necessity of simplicity. The lemniscate traversed once is closed and regular but not simple, and has index 0; the curve traversing a circle twice has index 2. Simplicity is therefore essential, and it is exactly the hypothesis that makes ψ well defined in Step 3. $\square$

Corollary 4.1: Total curvature and mean radius. For a simple closed curve of length L , the mean curvature is $\bar{k} = 2\pi/L$, the reciprocal of the radius of the circle of equal perimeter. In particular there is at least one point where $k(s) \geq 2\pi/L$ and at least one where $k(s) \leq 2\pi/L$.

Proof. The integral mean of k is $\frac{1}{L} \int_0^L k \, ds = 2\pi/L$ by Theorem 4.1. A circle of perimeter L has radius $L/2\pi$ and curvature $2\pi/L$: the mean therefore coincides with the value for the isoperimetric circle. The last assertion is the elementary property that a continuous function attains values both $\geq$ and $\leq$ its own mean. $\square$

Exercise 4.1. Verify the theorem for the circle of radius r traversed counterclockwise.

Solution. On the circle $\kappa \equiv 1/r$ and the length is $L = 2\pi r$, hence $\oint \kappa \, ds = \frac{1}{r} \cdot 2\pi r = 2\pi$: index $i = +1$. The tangent makes exactly one full turn, consistently with the counterclockwise orientation.

4.2. The support function

The previous sections described a curve through its points. For *convex* curves there is a dual description, in which the primary object is not the point but the *tangent line*: this is the support function, and with it the theorems of this and the next sections are proved by turning geometric problems into problems about Fourier series — that is, by referring them back to Chapter 4.

Definition 4.2: Support function. Let $K \subset \mathbb{R}^2$ be a compact convex set containing the origin in its interior. The **support function** of K is

$$h(\theta) = \max_{x \in K} \langle x, u(\theta) \rangle, \quad u(\theta) = (\cos \theta, \sin \theta).$$

Geometrically, $h(\theta)$ is the signed distance from the origin to the supporting line of K orthogonal to $u(\theta)$.

Theorem 4.2: Reconstruction of the curve from the support function. Let K be convex with boundary ∂K of class C^2 and everywhere positive curvature. Then h is of class C^2 and:

(i) the point of ∂K whose outer normal is $u(\theta)$ is

$$\gamma(\theta) = h(\theta) u(\theta) + h'(\theta) u'(\theta), \quad u'(\theta) = (-\sin \theta, \cos \theta);$$

(ii) the radius of curvature there is

$$\rho(\theta) = h(\theta) + h''(\theta) > 0;$$

(iii) (Cauchy's formula) the perimeter of K is

$$L = \int_0^{2\pi} h(\theta) d\theta;$$

(iv) the area of K is

$$A = \frac{1}{2} \int_0^{2\pi} (h(\theta)^2 - h'(\theta)^2) d\theta.$$

Proof. (i) The point $\gamma(\theta)$ is characterised by two conditions: it lies on the supporting line, i.e. $\langle \gamma(\theta), u(\theta) \rangle = h(\theta)$; and its tangent is orthogonal to $u(\theta)$, i.e. $\gamma'(\theta) \parallel u'(\theta)$.

Write $\gamma = \alpha u + \beta u'$ in the moving orthonormal basis $\{u, u'\}$ (orthonormal since u and u' are orthogonal unit vectors). The first condition gives $\alpha = h$. Differentiating, and using $u'' = -u$:

$$\gamma' = h'u + hu' + \beta'u' - \beta u = (h' - \beta)u + (h + \beta')u'.$$

The second condition forces the component along u to vanish: $\beta = h'$. Substituting gives the formula.

(ii) With $\beta = h'$, the previous computation gives

$$\gamma'(\theta) = (h + h'') u'(\theta).$$

On the other hand $u'(\theta)$ is the unit tangent (up to sign) and θ is the angle of the normal, hence also the angle of the tangent up to $\pi/2$: by Step 4 of the proof of Theorem 2.1, $k = d\vartheta/ds = d\theta/ds$, so

$$\rho = \frac{1}{k} = \frac{ds}{d\theta} = \|\gamma'(\theta)\| = h + h'',$$

positive by the hypothesis of positive curvature.

(iii) Integrating $ds = \rho d\theta$ over a full turn,

$$L = \int_0^{2\pi} (h + h'') d\theta = \int_0^{2\pi} h d\theta + [h']_0^{2\pi} = \int_0^{2\pi} h d\theta,$$

since h' is 2π -periodic and the boundary term vanishes.

(iv) The area enclosed by a simple closed curve is, by Green's theorem,

$$A = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} \det(\gamma(\theta), \gamma'(\theta)) d\theta.$$

With $\gamma = hu + h'u'$ and $\gamma' = (h + h'')u'$, and using $\det(u, u') = 1$, $\det(u', u') = 0$:

$$\det(\gamma, \gamma') = h(h + h'').$$

Hence

$$A = \frac{1}{2} \int_0^{2\pi} (h^2 + hh'') d\theta = \frac{1}{2} \int_0^{2\pi} (h^2 - h'^2) d\theta,$$

having integrated by parts $\int hh'' = [hh'] - \int h'^2 = -\int h'^2$ (boundary term zero by periodicity).

Step 5 — check on the circle. For the circle of radius r centred at the origin, $h \equiv r$: then $\gamma(\theta) = r u(\theta)$ (the circle itself), $\rho = r + 0 = r \checkmark$, $L = 2\pi r \checkmark$ and $A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \pi r^2 \checkmark$.

Step 6 — check on a translated circle. For the circle of radius r centred at $c = (c_1, c_2)$ one has $h(\theta) = r + c_1 \cos \theta + c_2 \sin \theta$. Then $h'' = -c_1 \cos \theta - c_2 \sin \theta$ and $\rho = h + h'' = r \checkmark$; moreover $L = 2\pi r$ (the cosine and sine terms integrate to zero) $\checkmark$. For the area, $\int h^2 = 2\pi r^2 + \pi(c_1^2 + c_2^2)$ and $\int h'^2 = \pi(c_1^2 + c_2^2)$, hence $A = \pi r^2 \checkmark$, independent of the translation, as it must be. $\square$

Remark 4.1: Why the support function simplifies everything. The formula $\rho = h + h''$ is the heart of the method. It turns a geometric quantity — the radius of curvature — into a linear differential operator applied to h ; and since h is 2π -periodic, it can be expanded in a Fourier series (Chapter 4), where the operator $1 + \frac{d^2}{d\theta^2}$ becomes multiplication by $1 - n^2$. All the theorems that follow are, in essence, the computation of that multiplication. Note the special value $n = 1$, for which $1 - n^2 = 0$: the first harmonics contribute nothing to the curvature — indeed they correspond to mere translations of the convex body, as Step 6 shows.

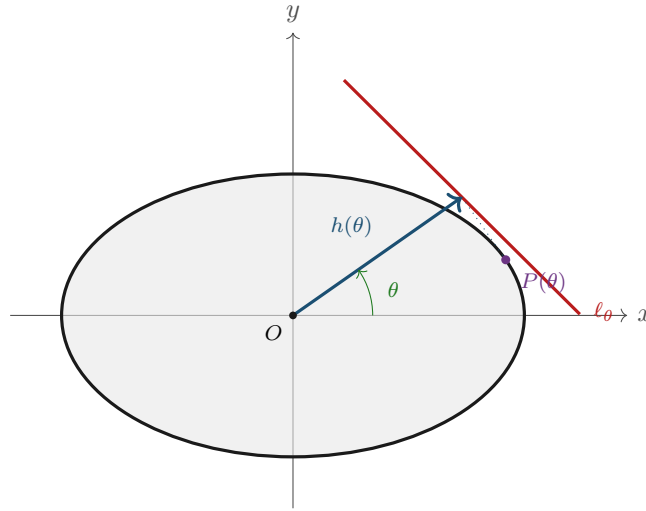


Figure 4.1: The support function of a convex body. For every direction θ one takes the supporting line ℓ_θ orthogonal to $(\cos \theta, \sin \theta)$ that leaves the body entirely on one side; $h(\theta)$ is its signed distance from the origin, and $P(\theta)$ the contact point. The function h determines the convex body completely, and turns geometric problems into Fourier analysis: the perimeter is $L = \int_0^{2\pi} h$ (Cauchy's formula), the area is $A = \frac{1}{2} \int_0^{2\pi} (h^2 - h'^2)$, the width in direction θ is $w(\theta) = h(\theta) + h(\theta + \pi)$ and the radius of curvature is $\rho = h + h''$. For the ellipse drawn here $h(\theta) = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}$.

4.3. Curves of constant width and Barbier's theorem

Definition 4.3: Width. The **width** of a convex body K in direction θ is the distance between the two supporting lines orthogonal to $u(\theta)$:

$$w(\theta) = h(\theta) + h(\theta + \pi).$$

K has **constant width** w if $w(\theta) = w$ for every θ .

Example 4.1: The Reuleaux triangle. Take an equilateral triangle of side w and, for each vertex, draw the arc of circle of radius w centred at that vertex and lying between the other two. The figure bounded by the three arcs is the **Reuleaux triangle**.

Constant width. Every supporting line touches the figure either at a vertex or along an arc. If it touches the arc opposite the vertex V , the parallel supporting line on the other side passes through V , and the distance between the two is the radius of the arc, namely w . Since every direction falls into one of these cases, the width is constantly w .

Perimeter. Each arc subtends an angle of $60^\circ = \pi/3$ at its centre and has radius w , hence length $w\pi/3$; the three arcs give

$$L = 3 \cdot \frac{\pi w}{3} = \pi w,$$

exactly the perimeter of the circle of diameter w . This is no coincidence: it is Theorem 4.3.

Area. The area is that of the equilateral triangle plus three circular segments:

$$A = \frac{\sqrt{3}}{4}w^2 + 3 \left(\frac{\pi w^2}{6} - \frac{\sqrt{3}}{4}w^2 \right) = \frac{\pi w^2}{2} - \frac{\sqrt{3}}{2}w^2 = \frac{\pi - \sqrt{3}}{2}w^2.$$

Numerically, $\frac{3.141593-1.732051}{2} = 0.704771$, against $\pi w^2/4 = 0.785398 w^2$ for the circle of diameter w : the Reuleaux triangle has 89.73% of the area of the circle of the same width.

Applications. Curves of constant width appear in the design of polygonal coins (which must pass the selectors of vending machines in every orientation: the British 20p and 50p coins are Reuleaux heptagons), in cam mechanisms, and in dimensional inspection: *measuring a constant-width body with a calliper always gives the same value, so it is not enough to guarantee that a part is cylindrical*. This last is a recurrent error in quality control, and is why geometrical tolerancing standards prescribe measuring *roundness* (ISO 1101) and not merely width.

Theorem 4.3: Barbier's theorem. *Every convex curve of constant width w has perimeter*

$$L = \pi w,$$

independently of its shape. In particular all curves of constant width w have the same perimeter as the circle of diameter w .

Proof. First proof (Cauchy's formula). By Theorem 4.2(iii),

$$L = \int_0^{2\pi} h(\theta) d\theta.$$

Substituting $\theta \rightarrow \theta + \pi$ in the integral (a change of variable which, by periodicity, does not alter the value),

$$L = \int_0^{2\pi} h(\theta + \pi) d\theta.$$

Adding the two expressions and dividing by two,

$$L = \frac{1}{2} \int_0^{2\pi} (h(\theta) + h(\theta + \pi)) d\theta = \frac{1}{2} \int_0^{2\pi} w(\theta) d\theta = \frac{1}{2} \cdot 2\pi w = \pi w.$$

Second proof (Fourier series). Expand h in a Fourier series (Chapter 4):

$$h(\theta) = a_0 + \sum_{n \geq 1} (a_n \cos n\theta + b_n \sin n\theta).$$

Then

$$w(\theta) = h(\theta) + h(\theta + \pi) = 2a_0 + 2 \sum_{n \text{ even}, n \geq 2} (a_n \cos n\theta + b_n \sin n\theta),$$

because $\cos n(\theta + \pi) = (-1)^n \cos n\theta$ and similarly for the sine: the odd harmonics cancel, the even ones double. The width is therefore constant if and only if $a_n = b_n = 0$ for every even $n \geq 2$, and then $w = 2a_0$. By Cauchy's formula, $L = 2\pi a_0 = \pi w$.

Third proof (via curvature). From Theorem 4.2(ii), $\rho(\theta) = h(\theta) + h''(\theta)$. Then

$$\rho(\theta) + \rho(\theta + \pi) = (h(\theta) + h(\theta + \pi)) + (h''(\theta) + h''(\theta + \pi)) = w(\theta) + w''(\theta) = w,$$

w being constant. Integrating over half a turn,

$$L = \int_0^{2\pi} \rho \, d\theta = \int_0^\pi (\rho(\theta) + \rho(\theta + \pi)) \, d\theta = \int_0^\pi w \, d\theta = \pi w.$$

Step 4 — check on the Reuleaux triangle. Example 4.1 gave $L = \pi w$ by an independent computation. ✓

Step 5 — reading. Barbier's theorem is the first of a series of results in which the circle is *not* characterised: the perimeter does not distinguish the circle from the other curves of constant width. What does distinguish them is the area, as the next theorem says. □

Theorem 4.4: Extremes of area at constant width. Among all convex bodies of constant width w :

- (i) the maximum area is attained by the circle of diameter w , with $A = \pi w^2/4$;
- (ii) the minimum is attained by the Reuleaux triangle, with $A = \frac{\pi - \sqrt{3}}{2} w^2$ (Blaschke–Lebesgue theorem).

Proof. [of part (i) only] By Theorem 4.3 all convex bodies of constant width w have the same perimeter $L = \pi w$. By the isoperimetric inequality (Theorem 4.5, proved in the next section), among all closed curves of given perimeter the area is maximal for the circle and equals $L^2/4\pi$. Here

$$A \leq \frac{L^2}{4\pi} = \frac{\pi^2 w^2}{4\pi} = \frac{\pi w^2}{4},$$

with equality only for the circle, which indeed has constant width w (its diameter).

On part (ii). The Blaschke–Lebesgue theorem was proved independently by Wilhelm Blaschke (1915) and Henri Lebesgue (1914). The original proof goes through compactness of the space of convex bodies in the Hausdorff metric and an analysis of the minimisers; Section 4.3.1 gives a shorter and complete one, using only Minkowski's mixed areas and Pál's hexagon theorem. The minimum is 89.73% of the maximum, as checked in Example 4.1. □

4.3.1 Part (ii): the Blaschke–Lebesgue theorem, proved

The original proof of Blaschke and Lebesgue goes through compactness of the space of convex bodies and an analysis of the minimisers. There is, however, a markedly shorter route, due essentially to Chakerian, which uses only two tools: Minkowski's *mixed areas* and a theorem of Pál from 1920 on the existence of a circumscribed regular hexagon. We prove Pál's theorem in full, because it is elementary and instructive; of the mixed areas we import only two standard properties, which we state explicitly.

Lemma 4.1: Pál's hexagon theorem (1920). Let K be a convex body of constant width w . Then there exists a regular hexagon of width w (that is, of inradius $w/2$) containing K .

Proof. Let h be the support function of K (Definition 4.2), which is continuous, and let $u(\theta) = (\cos \theta, \sin \theta)$.

Step 1 — the circumscribed equiangular hexagon. Fix θ and consider the six directions $u_k = u(\theta + k\pi/3)$, $k = 0, \dots, 5$, and the six supporting half-planes

$$\Pi_k = \{x \in \mathbb{R}^2 : \langle x, u_k \rangle \leq h_k\}, \quad h_k = h\left(\theta + \frac{k\pi}{3}\right).$$

Since each Π_k contains K , so does $H(\theta) = \bigcap_{k=0}^5 \Pi_k$; it is a hexagon with all angles equal to 120° (the side normals are equally spaced), whose opposite sides are at distance

$$h_k + h_{k+3} = h\left(\theta + \frac{k\pi}{3}\right) + h\left(\theta + \frac{k\pi}{3} + \pi\right) = w$$

by constant width. It remains to choose θ so that $H(\theta)$ is *regular*.

Step 2 — when the hexagon is regular. The hexagon $H(\theta)$ is regular if and only if some translation brings all six side distances from the origin to the common value $w/2$: that is, if and only if there exists $t \in \mathbb{R}^2$ with

$$h_k - \langle t, u_k \rangle = \frac{w}{2} \quad \text{for every } k = 0, \dots, 5.$$

The equations with $k = 3, 4, 5$ follow from those with $k = 0, 1, 2$: indeed $u_{k+3} = -u_k$ and $h_{k+3} = w - h_k$, hence

$$h_{k+3} - \langle t, u_{k+3} \rangle = w - h_k + \langle t, u_k \rangle = w - (h_k - \langle t, u_k \rangle) = w - \frac{w}{2} = \frac{w}{2}.$$

Three equations in two unknowns remain. The three vectors u_0, u_1, u_2 satisfy the single linear relation

$$u_0 - u_1 + u_2 = 0,$$

as one checks at once by setting $\theta = 0$: $u_0 = (1, 0)$, $u_1 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, $u_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, and $u_0 + u_2 = u_1$ (for general θ one merely rotates). Since u_0 and u_1 are independent, the system of the first two equations has exactly one solution t ; it satisfies the third as well if and only if the right-hand sides obey the same linear relation, that is if and only if

$$\left(h_0 - \frac{w}{2}\right) - \left(h_1 - \frac{w}{2}\right) + \left(h_2 - \frac{w}{2}\right) = 0, \quad \text{i.e.} \quad h_0 - h_1 + h_2 = \frac{w}{2}.$$

Step 3 — existence of θ . Put

$$F(\theta) = h(\theta) - h\left(\theta + \frac{\pi}{3}\right) + h\left(\theta + \frac{2\pi}{3}\right) - \frac{w}{2},$$

a continuous function on $\mathbb{R}$. Compute F one sixth of a turn later, using $h(\theta + \pi) = w - h(\theta)$:

$$F\left(\theta + \frac{\pi}{3}\right) = h\left(\theta + \frac{\pi}{3}\right) - h\left(\theta + \frac{2\pi}{3}\right) + h(\theta + \pi) - \frac{w}{2} = h_1 - h_2 + w - h_0 - \frac{w}{2} = -F(\theta).$$

So F changes sign (weakly) between θ and $\theta + \pi/3$: if $F(0) > 0$ then $F(\pi/3) < 0$, and by the intermediate value theorem there is $\theta^* \in (0, \pi/3)$ with $F(\theta^*) = 0$; if $F(0) < 0$ one argues the same way; if $F(0) = 0$ take $\theta^* = 0$.

Step 4 — conclusion. For $\theta = \theta^*$ the condition of Step 2 holds, hence there is t such that the hexagon $H(\theta^*)$ translated by $-t$ has all six sides at distance $w/2$ from the origin: it is a regular hexagon of inradius $w/2$, that is of width w , and it contains K . $\square$

*Remark 4.2: What we import from the theory of convex bodies. The final step needs two standard facts about Minkowski's **mixed area**, which we do not prove and cite explicitly (Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, §5.1):*

(M1) (Minkowski) *for two convex bodies K, L in the plane and $\lambda \geq 0$ the area of the Minkowski sum is a quadratic polynomial*

$$A(K + \lambda L) = A(K) + 2\lambda A(K, L) + \lambda^2 A(L),$$

which defines the mixed area $A(K, L)$; it is symmetric, invariant under translations

of either argument, and satisfies $A(K, K) = A(K)$;
(M2) (monotonicity) if $K_1 \subseteq K_2$ then $A(K_1, L) \leq A(K_2, L)$, for every L .
Both are elementary results within the theory, but proving them requires the machinery of convex bodies which we have not built here. Everything else in the proof that follows is complete.

Lemma 4.2: *The sum $K + (-K)$ is a disc. Let K be a convex body of constant width w . Then*

$$K + (-K) = D(0, w),$$

the disc of centre the origin and radius w .

Proof. The support function of a Minkowski sum is the sum of the support functions, $h_{K+L} = h_K + h_L$: indeed

$$h_{K+L}(u) = \max_{x \in K, y \in L} \langle x + y, u \rangle = \max_{x \in K} \langle x, u \rangle + \max_{y \in L} \langle y, u \rangle.$$

Moreover $h_{-K}(u) = \max_{x \in K} \langle -x, u \rangle = h_K(-u)$. Hence, for every direction u ,

$$h_{K+(-K)}(u) = h_K(u) + h_K(-u) = w$$

by constant width. But the constant w is exactly the support function of the disc $D(0, w)$. Since a convex body is determined by its support function (Theorem 4.2), the two bodies coincide. $\square$

Proof. Proof of part (ii) of Theorem 4.4.

Step 1 — the starting identity. Applying (M1) with $L = -K$ and $\lambda = 1$, and using $A(-K) = A(K)$ (central symmetry preserves area),

$$A(K + (-K)) = A(K) + 2A(K, -K) + A(-K) = 2A(K) + 2A(K, -K).$$

By Lemma 4.2 the left-hand side equals πw^2 , hence

$$\pi w^2 = 2A(K) + 2A(K, -K).$$

Step 2 — bounding the mixed area. By Lemma 4.1 there is a regular hexagon H of width w with $K \subseteq H$. Passing to opposites, $-K \subseteq -H$. A regular hexagon is centrally symmetric, so $-H$ is a translate of H ; by translation invariance and monotonicity (M2), applied twice,

$$A(K, -K) \leq A(H, -K) \leq A(H, -H) = A(H, H) = A(H).$$

Step 3 — the area of the hexagon. A regular hexagon of inradius ρ has perimeter $6 \cdot 2\rho/\sqrt{3}$ and area $\frac{1}{2}p\rho = 2\sqrt{3}\rho^2$. With $\rho = w/2$,

$$A(H) = 2\sqrt{3} \cdot \frac{w^2}{4} = \frac{\sqrt{3}}{2} w^2.$$

Step 4 — conclusion. Substituting into Step 1,

$$\pi w^2 = 2A(K) + 2A(K, -K) \leq 2A(K) + \sqrt{3} w^2,$$

whence

$$A(K) \geq \frac{\pi - \sqrt{3}}{2} w^2.$$

Step 5 — the minimum is attained. The Reuleaux triangle has constant width w (Example 4.1) and area exactly $\frac{\pi - \sqrt{3}}{2} w^2 = 0.704771 w^2$: the inequality is therefore an equality for it, and the minimum is attained. Numerically the ratio to the maximum is

$$\frac{(\pi - \sqrt{3})/2}{\pi/4} = \frac{0.704771}{0.785398} = 0.897342,$$

that is 89.73%, in agreement with the figure recorded earlier. $\square$

Remark 4.3: What is left out: uniqueness. The proof above establishes the inequality and the fact that the Reuleaux triangle attains it. It does not establish that this is the only minimiser up to isometry: that requires a further analysis of the equality case, which means tracing back the conditions under which (M2) is an equality. Uniqueness is true and is proved in the sources cited; here we record it as a statement, keeping it clearly distinct from what we have proved.

It is worth noticing how little was needed: constant width enters exactly twice, in Lemma 4.1 and in Lemma 4.2, and in both cases only through the identity $h(\theta) + h(\theta + \pi) = w$. All the rest is algebra of mixed areas.

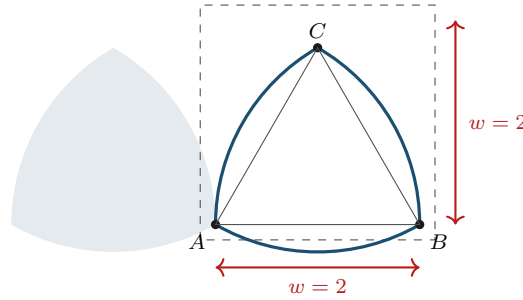


Figure 4.2: The Reuleaux triangle: the simplest curve of constant width after the circle. It is made of three circular arcs of radius w , each centred at the opposite vertex. In every direction the distance between the two supporting lines is always w , because one passes through a vertex and the other is tangent to the arc centred at that vertex: the shape therefore rotates inside a square of side w while always touching all four sides. By Barbier's theorem the perimeter is πw , exactly as for the circle of diameter w ; but the area is $\frac{1}{2}(\pi - \sqrt{3})w^2 \simeq 0.7048 w^2$, smaller than the circle's, and by the Blaschke–Lebesgue theorem it is the smallest possible.

4.4. The isoperimetric inequality

Theorem 4.5: Isoperimetric inequality. Let γ be a closed, simple plane curve of class C^1 , of length L , enclosing an area A . Then

$$4\pi A \leq L^2,$$

with equality if and only if γ is a circle.

Proof. [for convex curves, via the support function (Hurwitz)] **Step 1 — series expansion.** Let h be the support function, expanded in a Fourier series as in the second proof of Theorem 4.3:

$$h(\theta) = a_0 + \sum_{n \geq 1} (a_n \cos n\theta + b_n \sin n\theta).$$

Step 2 — perimeter. By Theorem 4.2(iii) and orthogonality of the trigonometric functions (Theorem (Volume II, Chapter 4) of Chapter 4),

$$L = \int_0^{2\pi} h \, d\theta = 2\pi a_0.$$

Step 3 — area. By Parseval's identity (Theorem (Volume II, Chapter 4) of Chapter 4) in real form,

$$\int_0^{2\pi} h^2 \, d\theta = 2\pi a_0^2 + \pi \sum_{n \geq 1} (a_n^2 + b_n^2),$$

and since $h'(\theta) = \sum_{n \geq 1} n(-a_n \sin n\theta + b_n \cos n\theta)$,

$$\int_0^{2\pi} h'^2 \, d\theta = \pi \sum_{n \geq 1} n^2 (a_n^2 + b_n^2).$$

Hence, by Theorem 4.2(iv),

$$A = \frac{1}{2} \left[2\pi a_0^2 + \pi \sum_{n \geq 1} (1 - n^2)(a_n^2 + b_n^2) \right] = \pi a_0^2 - \frac{\pi}{2} \sum_{n \geq 2} (n^2 - 1)(a_n^2 + b_n^2),$$

the $n = 1$ term having disappeared because $1 - n^2 = 0$.

Step 4 — the inequality. From $L = 2\pi a_0$ we get $L^2/(4\pi) = \pi a_0^2$, hence

$$\frac{L^2}{4\pi} - A = \frac{\pi}{2} \sum_{n \geq 2} (n^2 - 1)(a_n^2 + b_n^2) \geq 0,$$

since $n^2 - 1 > 0$ for $n \geq 2$. This is the claim.

Step 5 — equality case. Equality holds iff $a_n = b_n = 0$ for every $n \geq 2$, i.e.

$$h(\theta) = a_0 + a_1 \cos \theta + b_1 \sin \theta,$$

which by Step 6 of the proof of Theorem 4.2 is the support function of the circle of radius a_0 centred at (a_1, b_1) . So equality characterises the circle.

Step 6 — the non-convex case. If γ is not convex, consider its *convex hull*: it has greater or equal area and smaller or equal perimeter (replacing a concave arc by its chord shortens the boundary and adds area). Hence $4\pi A \leq 4\pi A_{\text{conv}} \leq L_{\text{conv}}^2 \leq L^2$, and the inequality extends. The equalities require γ to equal its convex hull, i.e. to be convex, and then the circle by Step 5.

Step 7 — the isoperimetric deficit. The quantity $\frac{L^2}{4\pi} - A$ measures “how far the curve is from being a circle”, and Step 4 expresses it as a sum of squares of the higher harmonics of the support function. It is the same pattern as Wirtinger's inequality, and the same pattern by which, in Chapter 4, Parseval's identity quantified the approximation error: the energy of the neglected harmonics.

Step 8 — check on the ellipse. For the ellipse with semi-axes $a = 3$, $b = 2$, $A = \pi ab = 18.8496$ and the perimeter (an elliptic integral, numerically $L = 15.8654$) give

$$\frac{L^2}{4\pi} = \frac{251.711}{12.5664} = 20.0305 > 18.8496 = A. \quad \checkmark$$

The isoperimetric ratio $4\pi A/L^2 = 0.94104 < 1$: the ellipse encloses 94.1% of the area a circle of the same perimeter would enclose. $\square$

Hurwitz: Fourier series of the curve itself. Up to scaling, let $L = 2\pi$ and parametrize by arc length $s \in [0, 2\pi]$, with $x'^2 + y'^2 = 1$; translating, assume $\int_0^{2\pi} x \, ds = 0$. Expand in Fourier series $x = \sum_n (a_n \cos ns + b_n \sin ns)$ and $y = \sum_n (c_n \cos ns + d_n \sin ns)$. By Parseval's identity,

$$\int_0^{2\pi} (x'^2 + y'^2) \, ds = \pi \sum_{n \geq 1} n^2 (a_n^2 + b_n^2 + c_n^2 + d_n^2) = 2\pi,$$

while the area (Green's formula) is $A = \int_0^{2\pi} xy' \, ds = \pi \sum_{n \geq 1} n(a_n d_n - b_n c_n)$. Hence

$$\frac{L^2}{4\pi} - A = \pi - A = \frac{\pi}{2} \sum_{n \geq 1} [n^2 (a_n^2 + b_n^2 + c_n^2 + d_n^2) - 2n(a_n d_n - b_n c_n)].$$

Each term is bounded below using $n \geq 1$: $n^2(a_n^2 + d_n^2) - 2na_n d_n \geq n(a_n^2 + d_n^2) - 2na_n d_n = n(a_n - d_n)^2 \geq 0$ (and similarly for b_n, c_n), so the sum is ≥ 0 : $L^2 \geq 4\pi A$. Equality. Requires every term to be zero: possible only if $a_n = b_n = c_n = d_n = 0$ for $n \geq 2$ and, for $n = 1$, $a_1 = d_1$, $b_1 = -c_1$. Then $x = a_1 \cos s + b_1 \sin s$, $y = -b_1 \cos s + a_1 \sin s$, which satisfies $x^2 + y^2 = a_1^2 + b_1^2$: a circle. $\square$

Remark 4.4: History and scope. The isoperimetric problem is very old — the legend of Dido, who obtained as much land as she could enclose with an ox hide, is its narrative statement — and the answer “the circle” was already held to be obvious by Zenodorus (2nd century BC). The first correct modern proof is due to Karl Weierstrass, by the calculus of variations; Jakob Steiner's proof (1838), by symmetrisation, showed that if a maximum exists then it is the circle, but not its existence — a gap filled precisely by Weierstrass. The proof given here is due to Adolf Hurwitz (1901) and is the shortest: it replaces all the geometry by Parseval's identity.

Remark 4.5: The circle as maximum enclosure. Equivalently: among all curves of given length, the circle encloses the maximum area. It is the reason why soap bubbles and pressurized pipe sections tend to the circle, and a guiding principle in shape optimization (minimize perimeter for given area).

Exercise 4.2. How much “less efficient” is the square than the circle at enclosing area?

Solution. For the circle $L^2/(4\pi A) = 1$. For a square of side ℓ : $L = 4\ell$, $A = \ell^2$, hence $L^2/(4\pi A) = 16\ell^2/(4\pi\ell^2) = 4/\pi \approx 1.273$. The isoperimetric ratio exceeds 1 by 27%: for the same perimeter the square encloses $\pi/4 \approx 78.5\%$ of the circle's area.

4.5. The four-vertex theorem

Definition 4.4: Vertex of a curve. A **vertex** of a plane C^3 curve is a point where the curvature is stationary, i.e. $k'(s) = 0$.

Theorem 4.6: Four-vertex theorem. Every closed, simple plane curve of class C^3 that is convex has at least four vertices.

Proof. Step 1 — at least two vertices always exist. The function k is continuous on the compact $[0, L]$ and periodic, hence attains a maximum and a minimum; at those points $k' = 0$. If maximum and minimum coincide, k is constant and by Theorem 2.5 the curve is a circle, which has $k' \equiv 0$: every point is a vertex and the claim is trivially true. So assume k is not constant and let A and B be the points of maximum and minimum.

Step 2 — hypothesis for contradiction. Suppose A and B are the *only* vertices. Then k' vanishes only at A and B , hence has constant sign on each of the two arcs into which A and B divide the curve: positive on the arc along which k increases from B to A , negative on the other.

Step 3 — choice of coordinates. Let r be the line through A and B . Since the curve is convex, a line meets it in at most two points (or in a segment, excluded here by the strict convexity implied by $k > 0$; for general convex curves one argues on the hull). So r meets the curve only at A and B , and the two arcs lie in the two opposite half-planes determined by r . Choose coordinates with r the x -axis: then $y(s)$ has constant sign on each arc, opposite on the two, and vanishes only at A and B .

Step 4 — the product has constant sign. By construction $k'(s)$ and $y(s)$ have constant sign on each arc. Reversing the y -axis if necessary, we may assume they have the *same* sign on both arcs, so

$$k'(s)y(s) \geq 0 \quad \text{for every } s,$$

with equality only at A and B . Hence

$$\int_0^L k'(s)y(s) ds > 0.$$

Step 5 — but the integral vanishes. Integrate by parts; the boundary term vanishes because the curve is closed and k, y are periodic:

$$\int_0^L k'(s)y(s) ds = - \int_0^L k(s)y'(s) ds.$$

Now $y' = T_2$, the second component of the unit tangent; and by the Frenet formulas (Theorem 2.1), with $T = (T_1, T_2)$ and $N = (-T_2, T_1)$,

$$T' = kN \implies T'_1 = -kT_2 \implies ky' = kT_2 = -T'_1.$$

Hence

$$- \int_0^L ky' ds = \int_0^L T'_1(s) ds = [T_1(s)]_0^L = 0,$$

because $T(L) = T(0)$.

Step 6 — contradiction and conclusion. Steps 4 and 5 are incompatible: the integral is simultaneously positive and zero. So the hypothesis of Step 2 is false and there are more than two vertices. But the vertices of a non-constant periodic function come in even number, because between two consecutive maxima there is a minimum: hence there are at least four.

Step 7 — the general case. Convexity was used only in Step 3, to guarantee that the line AB separates the two arcs. The theorem in fact holds for *every* simple closed curve, convex or not: it was stated for convex curves by Syamadas Mukhopadhyaya (1909) and proved in general by Adolf Kneser (1912); an elementary proof of the general case is due to Robert Osserman (1985). Section 4.5.1 proves the general case in full, importing only the Jordan curve theorem.

Step 8 — check on the ellipse. For a non-circular ellipse, from Example 2.2,

$$k(t) = \frac{ab}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}},$$

and the derivative vanishes where that of the denominator does, i.e. where $(a^2 - b^2) \sin t \cos t = 0$: at $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. Exactly four vertices, the ends of the axes — and, by Theorem 2.6(ii), the four cusps of the astroid evolute of Example 2.4. ✓

Step 9 — why it is a theorem about the circle. The equality case — infinitely many vertices — occurs only for the circle, as Step 1 shows. The theorem therefore says that *every* closed curve is “at least four times like a circle”: at four points at least its curvature is stationary, i.e. the curve is locally indistinguishable from a circle to third order. □

4.5.1 The general case: non-convex curves

Step 7 of the previous proof declared the convexity hypothesis superfluous, referring to the literature. We close that reference here with a complete proof, along Osserman’s minimal enclosing circle route (1985): the only imported statement is the Jordan curve theorem, which does not concern the circle. First two local tools on the osculating circle, useful elsewhere as well; then the proof.

Lemma 4.3: The osculating circle crosses the curve where $k' \neq 0$. Let γ be a plane curve of class C^3 parametrised by arc length, with $k(s_0) > 0$, and let σ be its osculating circle at $p = \gamma(s_0)$ (Definition 2.4). If $k'(s_0) \neq 0$, then γ crosses σ at p : every neighbourhood of s_0 contains points of the curve inside the osculating disc and points outside it.

Proof. Set $s_0 = 0$, $\rho = 1/k(0)$, and let $C = \gamma(0) + \rho N(0)$ be the centre of the osculating circle. Consider the C^3 function

$$f(s) = |\gamma(s) - C|^2 - \rho^2,$$

which is negative where the curve is inside the osculating disc, positive where it is outside, zero where it meets it. We compute the first three derivatives using the Frenet formulas (Theorem 2.1), $T' = kN$ and $N' = -kT$.

Order 0. $f(0) = |\gamma(0) - C|^2 - \rho^2 = |-\rho N(0)|^2 - \rho^2 = 0$.

Order 1. $f'(s) = 2\langle T, \gamma - C \rangle$, and at 0

$$f'(0) = 2\langle T(0), -\rho N(0) \rangle = 0,$$

because $T \perp N$: the osculating circle is tangent to the curve.

Order 2. $f''(s) = 2\langle T', \gamma - C \rangle + 2\langle T, T \rangle = 2k\langle N, \gamma - C \rangle + 2$, and at 0

$$f''(0) = 2k(0)\langle N(0), -\rho N(0) \rangle + 2 = -2k(0)\rho + 2 = -2 + 2 = 0,$$

because $\rho = 1/k(0)$: this is the second-order contact condition that *defines* the osculating circle (Theorem 2.4).

Order 3. Differentiating once more,

$$f'''(s) = 2k'\langle N, \gamma - C \rangle + 2k\langle N', \gamma - C \rangle + 2k\langle N, T \rangle = 2k'\langle N, \gamma - C \rangle - 2k^2\langle T, \gamma - C \rangle,$$

and at 0 the second term vanishes (as at first order), while the first gives

$$f'''(0) = 2k'(0)\langle N(0), -\rho N(0) \rangle = -2\rho k'(0) = -\frac{2k'(0)}{k(0)}.$$

Conclusion. By Taylor’s formula with Peano remainder,

$$f(s) = -\frac{k'(0)}{3k(0)} s^3 + o(s^3) \quad (s \rightarrow 0).$$

If $k'(0) \neq 0$ the coefficient of s^3 is nonzero, hence f has the same sign as $-k'(0)s^3/(3k(0))$ in a punctured neighbourhood of 0: opposite signs on the left and on the right. The curve therefore passes from one side of the osculating circle to the other. □

Corollary 4.2: *Where the osculating circle supports, there is a vertex.* Let $p = \gamma(s_0)$ be a point at which the osculating circle σ supports the curve, that is, at which there is a neighbourhood of s_0 whose points all lie on the same (weak) side of σ : all in the closed osculating disc, or all in its closed complement. Then p is a vertex.

Proof. If $k'(s_0) \neq 0$, then by Lemma 4.3 every neighbourhood of s_0 would contain points on both sides of σ , against the support hypothesis. Hence $k'(s_0) = 0$, that is p is a vertex by Definition 4.4. (If $k(s_0) = 0$ the osculating circle degenerates into a line, and the same computation applies to the signed distance from the tangent line, whose expansion starts with the term $\frac{1}{6}k'(s_0)s^3$ as soon as $k(s_0) = 0$.) $\square$

Definition 4.5: Vertex as a local extremum. For a regular curve of class C^2 , parametrised by arc length with signed curvature k , a parameter s_0 (or the point $\gamma(s_0)$) is a **vertex** if k has a local maximum or a local minimum at s_0 , that is, if there is $\eta > 0$ such that $k(s_0) \geq k(s)$ for every $|s - s_0| < \eta$, or $k(s_0) \leq k(s)$ for every $|s - s_0| < \eta$. If the curve is of class C^3 , then k is C^1 and $k'(s_0) = 0$ at every vertex (Fermat's theorem): one recovers Definition 4.4. For C^3 curves the two notions are used interchangeably in what follows.

What we import. A single statement, which does not concern the circle and is a prerequisite from the topology of the plane.

INFO

(J) Jordan curve theorem. *If $J \subset \mathbb{R}^2$ is a Jordan curve (a homeomorphic image of a circle), then $\mathbb{R}^2 \setminus J$ has exactly two connected components, one bounded, $\text{int } J$, and one unbounded, $\text{ext } J$, and J is the boundary of each of the two: every point of J is a limit of points of both.* Sources: J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000, §63; C. Thomassen, *The Jordan–Schönflies theorem and the classification of surfaces*, American Mathematical Monthly 99 (1992), no. 2, pp. 116–130. For the curve $\alpha = \gamma(\mathbb{R})$ we set $F = \alpha \cup \text{int } \alpha$, the closed region enclosed. Everything else in this subsection is proved.

Local structure and orientation.

Lemma 4.4: *Two sides of a Jordan curve at a smooth point.* Let J be a Jordan curve with a parametrisation $\varphi : \mathbb{R}/\ell\mathbb{Z} \rightarrow J$ (a homeomorphism), let $p = \varphi(u_0)$, and suppose φ is of class C^1 with $\varphi' \neq 0$ on an open interval containing u_0 . Fix $\eta > 0$ such that $\delta(u) = \varphi(u_0 + u)$ is C^1 and regular on $[-\eta, \eta]$; the rest of J , that is $J \setminus \delta((-\eta, \eta))$, is compact and does not contain p , hence it has positive distance from p , and the same holds for $\delta([- \eta, \eta] \setminus (-u_1, u_1))$ for every $0 < u_1 < \eta$. Let n be a unit vector normal to $\delta'(0)$. Then there is an open rectangle $\mathcal{R}$ centred at p such that:

- (i) $\mathcal{R} \setminus J$ has exactly two connected components, $\mathcal{R}^+$ and $\mathcal{R}^-$; $\mathcal{R}^+$ contains the points $p + tn$ and $\mathcal{R}^-$ the points $p - tn$, for small $t > 0$;
- (ii) one of the two components is contained in $\text{int } J$ and the other in $\text{ext } J$;
- (iii) (continuity of the side) if J is a regular C^1 arc also near the points $\delta(u)$, $|u|$ small, and $n(u)$ is the normal that varies continuously with $n(0) = n$, then the component of $\mathbb{R}^2 \setminus J$ that contains $\delta(u) + tn(u)$ (small $t > 0$) is the same for all u near 0.

Proof. (i) Coordinates with $p = 0$, $\delta'(0)$ parallel to e_1 and $n = e_2$. Since $x'(0) \neq 0$, by the inverse function theorem the first coordinate $x = x(u)$ is a C^1 diffeomorphism of a neighbourhood of 0 onto an interval $(-a, a)$, and the arc is the graph $y = \varphi_0(x)$ of some $\varphi_0 \in C^1(-a, a)$ with $\varphi_0(0) = 0$, $\varphi_0'(0) = 0$. Taking $0 < a' < a$ and $b > c_0 := \max_{|x| \leq a'} |\varphi_0(x)|$, let $\mathcal{R} =$

$(-a', a') \times (-b, b)$; shrinking a' and b we may assume that $\mathcal{R}$ contains no other point of J (the rest of J and the points $\delta(u)$ with $|u|$ not small are at positive distance from p). Then $\mathcal{R} \setminus J = \mathcal{R}^+ \cup \mathcal{R}^-$ with $\mathcal{R}^\pm = \{(x, y) \in \mathcal{R} : \pm(y - \varphi_0(x)) > 0\}$. Each of them is connected: the strip $\{c_0 < y < b\} \cap \mathcal{R}$ is a rectangle contained in $\mathcal{R}^+$, and every point of $\mathcal{R}^+$ is joined to it by a vertical segment that stays above the graph; similarly for $\mathcal{R}^-$. The two sets are open and disjoint, hence they are the components; $(0, t) \in \mathcal{R}^+$ for $0 < t < b$.

(ii) $\mathcal{R}^+$ is connected and does not meet J , hence it lies in $\text{int } J$ or in $\text{ext } J$; the same for $\mathcal{R}^-$. By (J), p is a limit both of points of $\text{int } J$ and of points of $\text{ext } J$, and those close enough to p lie in $\mathcal{R}^+ \cup \mathcal{R}^-$: if both components lay in $\text{int } J$, no point of $\text{ext } J$ would be near p , absurd; symmetrically for $\text{ext } J$.

(iii) In the coordinates of (i), for small $|u|$ we have $\delta(u) = (x(u), \varphi_0(x(u)))$ and $n(u) \rightarrow (0, 1)$. Since $|\varphi'_0| \leq \frac{1}{2}$ near 0, for small u we have $n_2(u) > \frac{1}{2}|n_1(u)|$ and, for small $t > 0$, the point $\delta(u) + tn(u)$ has ordinate $\varphi_0(x(u)) + tn_2(u) > \varphi_0(x(u)) + \frac{1}{2}t|n_1(u)| \geq \varphi_0(x(u) + tn_1(u))$: it lies above the graph, that is in $\mathcal{R}^+$. The local component “on the side of $n(u)$ ” at $\delta(u)$ and $\mathcal{R}^+$ therefore have a point in common, and lie in the same component of $\mathbb{R}^2 \setminus J$. $\square$

Corollary 4.3: Positive orientation. *For the curve $\alpha = \gamma(\mathbb{R})$ exactly one of the two alternatives holds: $N(s)$ points towards $\text{int } \alpha$ for every s , or $-N(s)$ points towards $\text{int } \alpha$ for every s . Replacing $\gamma(s)$ by $\gamma(-s)$ if necessary, one obtains that N is the **inward normal** at every point: $\mathcal{R}^+ \subset \text{int } \alpha$ on the side of N and $\mathcal{R}^- \subset \text{ext } \alpha$ on the side of $-N$. This is the positive orientation, for which a circle traversed counterclockwise has $k \equiv 1/r > 0$.*

Proof. Let A be the set of parameters $s \in \mathbb{R}/L\mathbb{Z}$ such that $\gamma(s) + tN(s) \in \text{int } \alpha$ for small $t > 0$. By Lemma 4.4(ii) every parameter lies in A or in the analogous set A' with $\text{ext } \alpha$; by Lemma 4.4(iii) both sets are open. A and A' are complementary and open in the connected space $\mathbb{R}/L\mathbb{Z}$: one of the two is empty. $\square$

Lemma 4.5: Cusp. *Let J be a Jordan curve with parametrisation $\varphi : \mathbb{R}/\ell\mathbb{Z} \rightarrow J$, $p = \varphi(u_0)$, and suppose φ is of class C^1 and regular, with one-sided derivatives at u_0 , on $[u_0, u_0 + \eta]$ and on $[u_0 - \eta, u_0]$. Put $\delta_1(x) = \varphi(u_0 + x)$ and $\delta_2(x) = \varphi(u_0 - x)$ for $x \in [0, \eta]$: two regular injective C^1 arcs issuing from p . Suppose they have the same initial direction $\delta'_1(0) \parallel \delta'_2(0) \parallel e_1$ (same sense) and that in the local graphs $y = \varphi_1(x)$, $y = \varphi_2(x)$ ($0 \leq x < a$) one has $\varphi_1(x) > \varphi_2(x)$ for $0 < x < a$. Then there is a rectangle $\mathcal{R}$ centred at p such that $\mathcal{R} \setminus J$ has exactly two components: the **cusp** $\{(x, y) \in \mathcal{R} : 0 < x, \varphi_2(x) < y < \varphi_1(x)\}$ and the **rest**, which contains $\{x < 0\} \cap \mathcal{R}$, $\{y > \varphi_1(x), x \geq 0\}$ and $\{y < \varphi_2(x), x \geq 0\}$.*

Proof. The rest of J , that is $J \setminus \varphi((u_0 - \eta, u_0 + \eta))$, is compact and does not contain p : it has positive distance from p , even when it passes close to p . As in Lemma 4.4 one constructs $\mathcal{R} = (-a', a') \times (-b, b)$, smaller than that distance, with $\mathcal{R} \cap J$ equal to the two graphs over $[0, a']$. The cusp is connected (vertical segments up to the curve $y = \frac{1}{2}(\varphi_1 + \varphi_2)$, which is connected). The rest is connected: the part $\{x < 0\}$ is a rectangle; every point with $x \geq 0$ above φ_1 is joined vertically to a high horizontal strip and from there to the part $x < 0$; similarly below φ_2 . The two parts are open and disjoint. $\square$

The enclosing circle and the contact points. From now on α is positively oriented, Γ is the circle of Lemma 4.6, with centre O and radius R , D is the open disc and $\bar{D}$ the closed one, $K = \alpha \cap \Gamma$ is the *contact set*. Arcs of Γ are traversed counterclockwise; for $p \in \Gamma$, τ_p is the counterclockwise unit tangent to Γ at p , and $i\tau_p = (O - p)/R$. A *gap* is a connected component of $\Gamma \setminus K$: a maximal open arc of Γ free of contact points, with endpoints in K .

Lemma 4.6: Minimal enclosing circle. *There is a closed disc $\bar{D}(O, R)$ of minimal radius among those containing α , with $R > 0$; and it is unique.*

Proof. The function $f(c) = \max_{x \in \alpha} |x - c|$ is well defined (α is compact), continuous (it is 1-Lipschitz) and $f(c) \geq |c| - \max_{x \in \alpha} |x| \rightarrow +\infty$ as $|c| \rightarrow \infty$: it has a global minimum at a point O ; put $R = f(O)$. By definition $\alpha \subset \bar{D}(O, R)$, and no closed disc of smaller radius contains α . $R > 0$ because α has more than one point. Uniqueness: if $O_1 \neq O_2$ were both minimal, with $O_m = \frac{1}{2}(O_1 + O_2)$ and $\delta = \frac{1}{2}|O_1 - O_2| > 0$, the parallelogram identity would give $|x - O_m|^2 = \frac{1}{2}(|x - O_1|^2 + |x - O_2|^2) - \delta^2 \leq R^2 - \delta^2$ for every $x \in \alpha$, and $f(O_m) < R$: absurd. $\square$

Lemma 4.7: The contact set does not lie in an open semicircle. *$K \neq \emptyset$ and K is not contained in any open semicircle of Γ . Consequently: (a) K has at least two points, and if it has exactly two they are antipodal; (b) every gap has angular width $\leq \pi$; (c) if $K \neq \Gamma$ then gaps exist, and if K contains no arc of Γ then every gap has distinct endpoints and there are at least two distinct gaps.*

Proof. If $K = \emptyset$, then $\max_{x \in \alpha} |x - O| < R$ and a smaller disc would contain α . Let u be a unit vector and suppose $K \subset \{p \in \Gamma : \langle p - O, u \rangle > 0\}$. Since K is compact there is $\delta > 0$ with $\langle p - O, u \rangle \geq \delta$ on K . The set $\alpha_{\text{far}} = \{x \in \alpha : \langle x - O, u \rangle \leq \delta/2\}$ is compact and does not meet Γ , hence, if it is non-empty, $\max_{\alpha_{\text{far}}} |x - O| = R - \eta$ with $\eta > 0$. Take $0 < \varepsilon < \min(\delta, \eta)$ and $O' = O + \varepsilon u$. For $x \in \alpha$ with $\langle x - O, u \rangle \geq \delta/2$: $|x - O'|^2 = |x - O|^2 - 2\varepsilon \langle x - O, u \rangle + \varepsilon^2 \leq R^2 - \varepsilon\delta + \varepsilon^2 < R^2$; for $x \in \alpha_{\text{far}}$: $|x - O'| \leq R - \eta + \varepsilon < R$. Hence $f(O') < R$: absurd. (a) A single point, or two non-antipodal points, lie in an open semicircle. (b) If a gap had width $> \pi$, its complement would be a closed arc of width $< \pi$ containing K , contained in an open semicircle. (c) $\Gamma \setminus K$ is open in Γ ; its components are open arcs with endpoints in K (the case “ Γ minus one point” is excluded by (a)). If K contains no arc, every gap (b, a) has $b \neq a$, and the complementary closed arc $[a, b]$ is not contained in K : it contains a point of $\Gamma \setminus K$ and with it a gap $G_2 \subset (a, b)$, distinct from $G_1 = (b, a)$. $\square$

Lemma 4.8: Comparison with a circle. *Let σ be the circle with centre c and radius $r > 0$, let $\gamma(s_0) \in \sigma$ and let $n = (c - \gamma(s_0))/r$ be the unit vector towards the centre. If in a neighbourhood of s_0 the curve lies in the closed disc $\bar{D}(c, r)$, then $\langle T(s_0), n \rangle = 0$ and $\langle \gamma''(s_0), n \rangle \geq 1/r$; if it lies in the closed complement of the open disc, then $\langle T(s_0), n \rangle = 0$ and $\langle \gamma''(s_0), n \rangle \leq 1/r$.*

Proof. $g(s) = |\gamma(s) - c|^2 - r^2$ is C^2 with $g(s_0) = 0$. In the first case $g \leq 0$ near s_0 : interior maximum, $g'(s_0) = 2\langle T, \gamma - c \rangle(s_0) = 0$ and $g''(s_0) = 2 + 2\langle \gamma''(s_0), -rn \rangle \leq 0$, that is $\langle \gamma''(s_0), n \rangle \geq 1/r$. In the second case $g \geq 0$, interior minimum, with the inequalities reversed. $\square$

Lemma 4.9: At the contact points. *Let $p = \gamma(s_p) \in K$. Then the curve is tangent to Γ at p , the inward normal points towards the centre, $N(s_p) = (O - p)/R$, hence $T(s_p) = \tau_p$, and $k(s_p) \geq 1/R$.*

Proof. Since $\alpha \subset \bar{D}$, Lemma 4.8 with $\sigma = \Gamma$ gives $T(s_p) \perp (O - p)$, that is $T(s_p) = \pm \tau_p$, and $\langle \gamma''(s_p), n \rangle \geq 1/R$ with $n = (O - p)/R$. It remains to show that $N(s_p) = n$. In the coordinates of Lemma 4.4 at p : $p = 0$, $T(s_p) = e_1$, $N(s_p) = e_2$, $O = \pm Re_2$. If $O = -Re_2$, the point $(0, t)$ with small $t > 0$, which lies in $\mathcal{R}^+ \subset \text{int } \alpha \subset \bar{D}$ (Corollary 4.3), would be at distance $R + t > R$ from O : absurd. Hence $O = Re_2$, $N(s_p) = n$, $k(s_p) = \langle \gamma''(s_p), N(s_p) \rangle \geq 1/R$, and from $i\tau_p = n = iT(s_p)$ it follows that $T(s_p) = \tau_p$. $\square$

The gaps. Let $G = (b, a)$ be a gap, the open counterclockwise arc from b to a , of width $\theta \leq \pi$, with $b \neq a$; let $s_b < s_a < s_b + L$ be the parameters of b and a . Let ℓ be the line ba , let ν be the unit normal to ℓ pointing towards G (the arc, of width $\leq \pi$, lies in a closed half-plane of ℓ) and $H_+ = \{x : \langle x - b, \nu \rangle > 0\}$.

Lemma 4.10: *From an endpoint of a gap the curve sets off counterclockwise and reaches the other endpoint without touching Γ . (a) $T(s_b) = \tau_b$ and $\langle \tau_b, \nu \rangle > 0$. (b) There is $\varepsilon > 0$ such that $\gamma(s) \in D \cap H_+$ for every $s \in (s_b, s_b + \varepsilon)$. (c) The first parameter $s > s_b$ with $\gamma(s) \in K$ is s_a : the arc $\alpha' = \gamma([s_b, s_a])$ satisfies $\alpha' \cap \Gamma = \{b, a\}$.*

Proof. (a) The first part is Lemma 4.9. The line ℓ is a secant through b , not the tangent: $\langle \tau_b, \nu \rangle \neq 0$. With $b = O + Re^{i\phi_b}$, the function $h(\phi) = \langle O + Re^{i(\phi_b + \phi)} - b, \nu \rangle$ vanishes at 0, is positive for $0 < \phi < \theta$, and $h'(0) = R\langle \tau_b, \nu \rangle$: were it negative, h would be negative just after 0. (b) Near s_b one has $\gamma(s) \neq O$; let $\psi(s)$ be the argument of $\gamma(s) - O$, a C^1 function with $\psi(s_b) = \phi_b$ and, by the formula $\frac{d}{ds} \arg z = \langle z', iz \rangle / |z|^2$ and the identity $i(b - O) = R\tau_b$, $\psi'(s_b) = \langle \tau_b, R\tau_b \rangle / R^2 = 1/R > 0$. Moreover $h_\gamma(s) = \langle \gamma(s) - b, \nu \rangle$ has $h_\gamma(s_b) = 0$ and $h'_\gamma(s_b) = \langle \tau_b, \nu \rangle > 0$. There is therefore $\varepsilon > 0$ such that for $s \in (s_b, s_b + \varepsilon)$: $\phi_b < \psi(s) < \phi_b + \theta$ and $\gamma(s) \in H_+$. If for such an s we had $\gamma(s) \in \Gamma$, then $\gamma(s) = O + Re^{i\psi(s)} \in G$; but $G \cap K = \emptyset$ and $\gamma(s) \in K$: absurd. Hence $\gamma(s) \in D$.

(c) The set $\Sigma = \{s \in [s_b + \varepsilon, s_b + L] : \gamma(s) \in K\}$ is closed and contains s_a : it has a minimum s' , which by (b) is the first parameter $> s_b$ with $\gamma(s') \in K$. Let $P' = \gamma(s')$ and suppose, for contradiction, that $P' \neq a$. Let $\beta_1 = \gamma([s_b, s'])$, $\beta_2 = \gamma([s', s_b + L])$ and let Γ_1 be the closed counterclockwise arc of Γ from b to P' . Since β_1 has no interior contacts, $\beta_1 \cap \Gamma_1 = \{b, P'\}$ and $J = \beta_1 \cup \Gamma_1$ is a Jordan curve, parametrised by running along β_1 from b to P' and then along Γ_1 from P' to b clockwise; at b the two arcs issuing from it are β_1 and Γ_1 traversed counterclockwise, both with initial direction τ_b . The arc Γ_1 contains G and, after G , the point a , interior to Γ_1 (because $P' \neq a$ and $P' \notin G$); and a is interior to β_2 .

Fact 1: the exterior of $\bar{D}$ lies in $\text{ext } J$, being connected, unbounded and disjoint from $J \subset \bar{D}$.

Fact 2: at every $x \in \beta_2^\circ \cap J$ the local side interior to D is $\text{int } J$. Such an x is a contact point on Γ_1 , distinct from b, P' , at positive distance from β_1 (the curve is simple): near x , J coincides with Γ . Lemma 4.4 at x gives two sides, one in $\text{int } J$ and one in $\text{ext } J$; the side exterior to D contains points outside $\bar{D}$, which lie in $\text{ext } J$ (Fact 1); hence the side interior to D is $\text{int } J$.

Fact 3: $\beta_2^\circ \setminus J$ lies entirely in $\text{int } J$ or entirely in $\text{ext } J$. Define $g(s)$, for $s \in (s', s_b + L)$, as the component of $\mathbb{R}^2 \setminus J$ containing $\gamma(s)$ if $\gamma(s) \notin J$, and $g(s) = \text{int}$ if $\gamma(s) \in J$. g is locally constant: at a parameter with $\gamma(s) \notin J$ the nearby points lie in the same open component; at a parameter s_x with $x = \gamma(s_x) \in J$ the nearby points $\gamma(s)$ lie in the rectangle of Lemma 4.4 at x and in $\bar{D}$: either on $\Gamma = J$ (and $g = \text{int}$ by definition) or on the side interior to D , which is $\text{int } J$ by Fact 2. A locally constant function on an interval is constant.

Fact 4: $g(s_a) = \text{int}$, because $a \in J$.

Fact 5: $g(s) = \text{ext}$ for $s < s_b + L$ near $s_b + L$. Near $b = \gamma(s_b + L)$ the curve J consists of the initial piece of β_1 and that of Γ_1 , both issuing from b with direction τ_b ; the rest of J is at positive distance from b . In the coordinates with $b = 0$, $\tau_b = e_1$, $O = Re_2$: Γ_1 is the graph of $\varphi_2(x) = R - \sqrt{R^2 - x^2}$ and β_1 the graph of some $\varphi_1 \geq \varphi_2$ (because $\beta_1 \subset \bar{D}$) with $\varphi_1 > \varphi_2$ for small $x > 0$ (because $\beta_1^\circ \cap \Gamma = \emptyset$). Lemma 4.5 applies: the rest contains points with $x > 0$, $y < \varphi_2(x)$, that is outside $\bar{D}$, hence $\text{rest} \subset \text{ext } J$ (Fact 1 and connectedness). The points $\gamma(s)$ with $s < s_b + L$ near $s_b + L$ have abscissa $x(s) = (s - s_b - L) + o(s - s_b - L) < 0$: they lie in the rest, that is in $\text{ext } J$.

Facts 3, 4, 5 are incompatible. Hence $P' = a$. □

Lemma 4.11: *Between two consecutive contacts the curvature drops below $1/R$. With the previous notation, let $\alpha' = \gamma([s_b, s_a])$, let $\bar{G}$ be the closed counterclockwise arc of Γ from b to a , let m be the midpoint of $\bar{ba}$ and $d = \frac{1}{2}|a - b| > 0$. There exists $s_1 \in (s_b, s_a)$ with $k(s_1) < 1/R$.*

Proof. *Step 1 — the pencil through b and a .* Every circle through b and a has centre $m + t\nu$ and radius $\sqrt{d^2 + t^2}$. For $x \in H_+$ put $t(x) = \frac{|x - m|^2 - d^2}{2\langle x - m, \nu \rangle}$: it is the parameter of the unique circle of the pencil through x , and for a circle of the pencil of parameter t_1

$$x \text{ is interior to it} \iff |x - m|^2 - 2t_1\langle x - m, \nu \rangle < d^2 \iff t(x) < t_1. \quad (*)$$

Γ belongs to the pencil: $O = m + t_0\nu$ with $t_0 \leq 0$, because $\bar{G}$ has width $\leq \pi$ and lies in $\bar{H}_+$, hence $O \notin H_+$. For the points $y \in G \subset H_+$ one has $t(y) = t_0$.

Step 2 — the point Q and the circle C' . By Lemma 4.10(b) the point $Q = \gamma(s_b + \varepsilon/2)$ lies in $D \cap H_+$ and is interior to α' . By (*) with $t_1 = t_0$: $t_Q := t(Q) < t_0 \leq 0$. Let C' be the circle of the pencil of parameter t_Q , centre $c' = m + t_Q\nu$, radius $R' = \sqrt{d^2 + t_Q^2}$: from $t_Q < t_0 \leq 0$ it follows that $t_Q^2 > t_0^2$ and $R' > R$. Let D' be its open disc.

Step 3 — translations towards $-\nu$. For $t \geq 0$ let $D'_t = D' - t\nu$ (centre $c' - t\nu$, radius R') and let C'_t be its boundary. (i) $D'_t \cap \bar{H}_+ \subset D'$: if $x + t\nu \in D'$ and $\langle x - m, \nu \rangle \geq 0$, then $\langle x - c', \nu \rangle = \langle x - m, \nu \rangle - t_Q > 0$ and $|x - c'|^2 = |(x + t\nu) - c'|^2 - 2t\langle x - c', \nu \rangle - t^2 \leq |(x + t\nu) - c'|^2 < R'^2$. (ii) $D'_t \cap \bar{G} = \emptyset$ for every $t \geq 0$: a $y \in G$ has $t(y) = t_0 > t_Q$, hence $y \notin D'$ by (*) and $y \notin D'_t$ by (i); and $|b - c'| = |a - c'| = R'$. (iii) $b, a \notin \bar{D}'_t$ for $t > 0$: $|b - c' + t\nu|^2 = R'^2 + 2t\langle b - c', \nu \rangle + t^2 = R'^2 - 2t t_Q + t^2 > R'^2$ (we used $\langle b - m, \nu \rangle = 0$); likewise for a .

Step 4 — the last contact. Let $\mathcal{T} = \{t \geq 0 : \bar{D}'_t \cap \alpha' \neq \emptyset\}$. We have $0 \in \mathcal{T}$ ($Q \in C'$); $\mathcal{T}$ is bounded ($\alpha' \subset \bar{D}(O, R)$ and $\bar{D}'_t$ moves away from it) and closed (by compactness of α' and continuity). Let $t^* = \max \mathcal{T}$: $\bar{D}'_{t^*} \cap \alpha' \neq \emptyset$ and $D'_{t^*} \cap \alpha' = \emptyset$ (if some $x \in \alpha'$ had $|x - c' + t^*\nu| < R'$, by continuity the same would hold for t slightly larger than t^*). Let $Q_1 = \gamma(s_1) \in \alpha' \cap C'_{t^*}$.

Step 5 — Q_1 is interior to α' . If $t^* > 0$, by (iii) $Q_1 \neq b, a$; if $t^* = 0$ take $Q_1 = Q$. In either case $s_1 \in (s_b, s_a)$ and $\gamma(s_1) \notin \Gamma$ (Lemma 4.10(c)).

Step 6 — the disc lies on the side of the inward normal. Let $J_1 = \alpha' \cup \bar{G}$, a Jordan curve ($\alpha' \cap \bar{G} = \{b, a\}$), parametrised by running along α' from b to a and then along $\bar{G}$ from a to b clockwise; let $\mathcal{L} = \text{int } J_1$. (α) $\mathcal{L} \subset \bar{D}$: the exterior of $\bar{D}$ is connected, unbounded and disjoint from J_1 , hence contained in $\text{ext } J_1$. (β) *The cusp at b lies in $\mathcal{L}$.* In the coordinates of Fact 5 (with α' and $\bar{G}$ in place of β_1 and Γ_1) Lemma 4.5 gives the cusp $\{0 < x, \varphi_2 < y < \varphi_1\}$, connected and disjoint from J_1 . Let $y_0 \in G$ be near b : near y_0 , J_1 coincides with Γ (the arc α' is at positive distance from y_0); Lemma 4.4 at y_0 gives two sides, and the one exterior to D contains points outside $\bar{D}$, that is of $\text{ext } J_1$ by (α): the side interior to D at y_0 is $\mathcal{L}$. The cusp contains points of that side, hence cusp $\subset \mathcal{L}$. (γ) *At every $s \in (s_b, s_a)$ the local side of α' on the side of $-N(s)$ is $\mathcal{L}$.* By Lemma 4.4 applied to J_1 at $\gamma(s)$ (where J_1 coincides with the C^2 arc α' , and $\bar{G}$ is far away because $\gamma(s) \notin \Gamma$) the two sides are $\mathcal{L}$ and $\text{ext } J_1$. For s slightly larger than s_b the point $\gamma(s) - tN(s)$ (small $t > 0$) lies in the cusp (in the coordinates of Fact 5, $\gamma(s) = (x(s), \varphi_1(x(s)))$ and $N(s) \rightarrow e_2$): the side of $-N(s)$ is $\mathcal{L}$ near s_b , and by the continuity of the side (Lemma 4.4(iii)) it is so on the whole of (s_b, s_a) . (δ) $D'_{t^*} \cap \mathcal{L} = \emptyset$. D'_{t^*} is connected and does not meet J_1 (Step 4 for α' , Step 3(ii) for $\bar{G}$): if it met $\mathcal{L}$ it would be contained in it; but it has radius $R' > R$ and does not lie in $\bar{D} \supset \mathcal{L}$. (ε) Let n_c be the unit vector from Q_1 towards the centre $c'' = c' - t^*\nu$ of C'_{t^*} . The points $Q_1 + tn_c$, $0 < t < R'$, lie in D'_{t^*} , hence neither in J_1 nor in $\mathcal{L}$; for small t they lie in the rectangle of Lemma 4.4 at Q_1 , and therefore in the side $\text{ext } J_1$, which by (γ) is the side of $+N(s_1)$. Since α' lies in the closed

complement of D'_{t*} and touches C'_{t*} at Q_1 , Lemma 4.8 gives $T(s_1) \perp n_c$, that is $n_c = \pm N(s_1)$; the position just found gives $n_c = N(s_1)$.

Step 7 — the inequality. Again by Lemma 4.8, $\langle \gamma''(s_1), n_c \rangle \leq 1/R'$, that is $k(s_1) = \langle \gamma''(s_1), N(s_1) \rangle \leq 1/R' < 1/R$. $\square$

Theorem 4.7: Four-vertex theorem, general case. *Every closed, simple, regular plane curve of class C^2 — convex or not — has at least four vertices, that is at least four distinct points at which the signed curvature has a local extremum. If the curve is of class C^3 , the curvature has at least four distinct stationary points.*

Proof. Case I: K contains an arc of Γ . On such an arc the curve coincides with Γ , traversed counterclockwise (Lemma 4.9): $k \equiv 1/R$ on an open interval of parameters, and every point of it is a vertex (a local maximum and a local minimum at once). There are infinitely many vertices.

Case II: K contains no arc. By Lemma 4.7(c) there are two distinct gaps $G_1 = (b_1, a_2)$ and $G_2 = (b_2, a_3)$, with distinct endpoints and width $\leq \pi$. By Lemma 4.10(c) the arcs $\alpha'_1 = \gamma([s_{b_1}, s_{a_2}])$ and $\alpha'_2 = \gamma([s_{b_2}, s_{a_3}])$ (with $s_{b_i} < s_{a_{i+1}} < s_{b_i} + L$) contain no interior contacts; by Lemma 4.11 there are $Q_1 = \gamma(q_1)$ interior to α'_1 and $Q_2 = \gamma(q_2)$ interior to α'_2 with $k(q_1), k(q_2) < 1/R$, whereas by Lemma 4.9 at the four contact points b_1, a_2, b_2, a_3 one has $k \geq 1/R$.

The arcs have disjoint interiors. The open parameter intervals $I_1 = (s_{b_1}, s_{a_2})$ and $I_2 = (s_{b_2}, s_{a_3})$, in $\mathbb{R}/L\mathbb{Z}$, are free of contact parameters and have endpoints that are contact parameters: they are connected components of the complement of the closed set of contact parameters, hence they coincide or are disjoint; they coincide only if $b_1 = b_2$, but then G_1 and G_2 would both be the gap starting at b_1 , that is equal. Hence $I_1 \cap I_2 = \emptyset$.

Cyclic order. Running along the curve from q_1 counterclockwise one stays in α'_1 up to a_2 and then leaves it; since $q_2 \notin I_1$ and $s_{a_2} \notin I_2$, the closed arc $[q_1, q_2]$ contains s_{a_2} , and symmetrically $[q_2, q_1 + L]$ contains s_{a_3} .

Two local maxima. On $[q_1, q_2]$ the maximum of k is $\geq k(s_{a_2}) \geq 1/R > \max(k(q_1), k(q_2))$: it is attained at an interior point M_1 , a local maximum of k . In the same way there is $M_2 \in (q_2, q_1 + L)$ with $k(M_2) \geq 1/R$.

Two local minima. On $[s_{b_1}, s_{a_2}]$ the minimum is $\leq k(q_1) < 1/R \leq \min(k(s_{b_1}), k(s_{a_2}))$, attained at an interior point $m_1 \in I_1$; in the same way $m_2 \in I_2$.

They are distinct. M_1 and M_2 lie in disjoint open arcs; $m_1 \in I_1$ and $m_2 \in I_2$, disjoint; an M_i ($k \geq 1/R$) and an m_j ($k < 1/R$) do not coincide. The curve has at least four vertices; if it is C^3 , at each of them $k' = 0$. $\square$

Remark 4.6: Strong form, the support theorem, and why “simple” is needed.

Strong form (Osseman, 1985). *If K contains no arc and the distinct gaps are at least $n \geq 2$, the same scheme gives $2n$ vertices: the intervals I_i are distinct components, hence disjoint, and each of them contains a local minimum m_i ; ordering the q_i cyclically, every open arc between two consecutive ones leaves its own I_i through a contact parameter, where $k \geq 1/R$, and therefore contains a local maximum M_i ; the n arcs are disjoint and $M_i \neq m_j$ because of the value of k . In the figure, the curve $r = 1 + 0.35 \cos 3\theta$ has $n = 3$ contacts and exactly $6 = 2n$ vertices.*

The osculating support theorem. Petrunin and Zamora Barrera (2022) prove, by means of the “moon in a puddle” lemma, that a closed simple regular curve is supported by its own osculating circles at at least four points, at two of them from inside and at two from outside; from Corollary 4.2 four vertices follow once more. In earlier editions that theorem was imported; the proof given above, by Osseman’s route, no longer needs it, and it is quoted here only as a second road.

Why “simple” is needed. The limaçon $r = 1 + 2\cos\theta$ is closed, regular and C^∞ , but it self-intersects. With $c = \cos\theta$ its curvature is $k = (9 + 6c)/(5 + 4c)^{3/2}$ and $\frac{dk}{dc} = \frac{6(5 + 4c) - 6(9 + 6c)}{(5 + 4c)^{5/2}} = \frac{-24 - 12c}{(5 + 4c)^{5/2}} < 0$ for $|c| \leq 1$: $k'(\theta) = \frac{dk}{dc} \cdot (-\sin\theta)$ vanishes only at $\theta = 0, \pi$. Exactly two vertices, $k_{\min} = 15/27 = 5/9$ and $k_{\max} = 3$ (values confirmed by DeTurck and others, 2007, for the same limaçon rotated). In the proof, simplicity enters in Lemma 4.4 (only two sides, one inner and one outer) and in Lemma 4.10 (β_2 does not cross β_1).

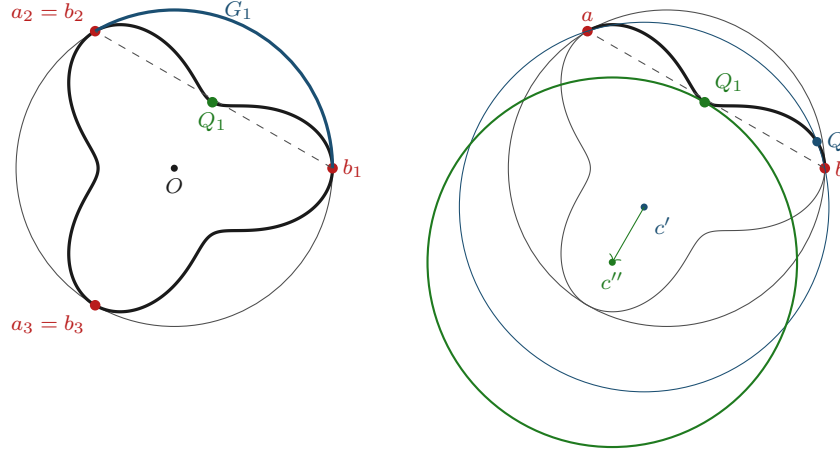


Figure 4.3: The proof on the curve $r = 1 + 0.35\cos 3\theta$ (non-convex: k drops to -5.92 in the indentations). Left: the minimal enclosing circle has radius $R = 1.35$ and three contact points, at $0^\circ, 120^\circ, 240^\circ$, where $k = 2.469 \geq 1/R = 0.741$; the three gaps have width $120^\circ \leq 180^\circ$. Right, the gap G_1 from b to a : the circle C' through b, a and the point Q of the curve at 10° has radius $R' = 1.576 > R$ and centre $c' = (-0.191, -0.331)$; translated in the direction $-\nu$ by $t^* = 0.544$ it touches the arc α' for the last time at the point Q_1 at 60° , where $k(Q_1) = -5.92 \leq 1/R' = 0.634 < 1/R$, and its centre c'' lies on the side of the inward normal ($\langle n_c, N \rangle = 1$). The curve has exactly $6 = 2 \cdot 3$ vertices, at $0^\circ, 60^\circ, \dots, 300^\circ$. Values computed by hand and with an independent script.

INFO

Independent numerical check. A second reader wrote a script (`verifica_quattro_vertici.py`, attached to the sources) that generates 796 simple closed non-convex curves in polar form, computes the minimal enclosing circle by an exact method and puts every lemma to the test: contacts never inside an open semicircle; $k \geq 1/R$ at the contacts (minimum margin 0.08); in 10 400 configurations of Lemma 4.11, including 750 in which the arc crosses the chord again, the disc of the last contact always lies on the side of the inward normal and $k(Q_1) \leq 1/R' < 1/R$; at least 4 local extrema, and at least $2n$ with n gaps. No counterexample. A numerical check is not a proof: here it serves to rule out slips in the computations.

Remark 4.7: Two from inside and two from outside: maxima and minima. The proof of Theorem 4.7 says more than the mere existence of four vertices: it distinguishes the two types. At the contact points with the enclosing circle the curvature is $\geq 1/R$ and the local maxima fall there; under every gap the curvature drops below $1/R$ and the local minima fall there. There are therefore at least two local maxima and at least two local minima of

the curvature, which is the strong form of the four-vertex theorem and coincides with what Step 6 of the proof of Theorem 4.6 derives, in the convex case, from the parity of the number of extrema. The same is said, in other words, by the osculating support theorem of Petrunin and Zamora Barrera (Remark 4.6): where the osculating circle supports from inside — the curve lies locally outside the osculating disc — the radius of curvature is a local minimum, hence the curvature is a local maximum; where it supports from outside the opposite happens. The link with nested osculating circles is the Tait–Kneser theorem: along an arc free of vertices the osculating circles are pairwise disjoint and nested one inside the other. It is precisely the vertices that break the nesting, and that is why their existence is not a detail but the condition for a closed curve to be able to close up on itself.

Corollary 4.4: *The evolute of a closed convex curve has at least four cusps. If γ is closed, simple, convex and C^3 with $k > 0$ and k non-constant, its evolute has at least four cusps.*

Proof. By Theorem 2.6(ii) the singularities of the evolute are exactly the points with $k' = 0$, the vertices of γ ; by Theorem 4.6 there are at least four. $\square$

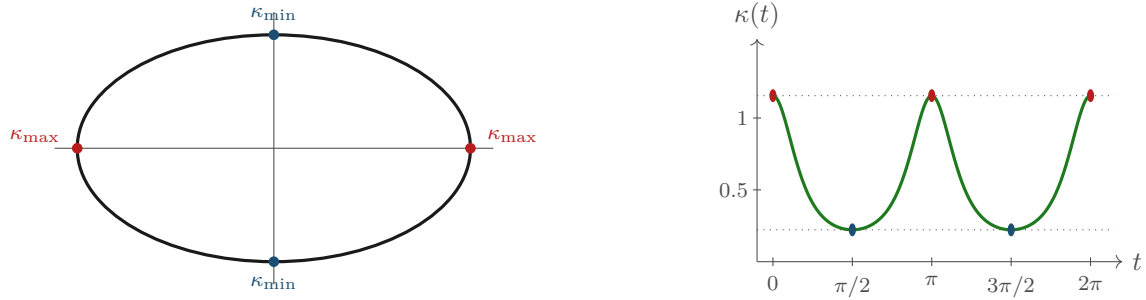


Figure 4.4: The four-vertex theorem on the ellipse. Left, the four vertices, i.e. the points of stationary curvature: two maxima at the ends of the major axis ($\kappa = a/b^2 = 1.156$) and two minima at the ends of the minor axis ($\kappa = b/a^2 = 0.222$). Right, the graph of $\kappa(t) = ab/(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}$ over a full turn: it has exactly two maxima and two minima. The theorem states that this is the smallest possible number: every simple closed plane curve has at least four vertices. The circle is the degenerate case in which the curvature is constant and every point is a vertex.

Remark 4.8: *The ellipse has exactly four vertices. The ellipse (Chapter 1... recte Chapter 2) achieves the minimum: four vertices, at the four endpoints of the axes, where κ is maximum (major axis) or minimum (minor axis). The circle is the degenerate case where κ is constant and every point is a “vertex”: the theorem is trivially satisfied but with no isolated vertices.*

Exercise 4.3. Identify the vertices of the ellipse $x^2/a^2 + y^2/b^2 = 1$ with $a > b$.

Solution. From the curvature $\kappa(t) = ab(a^2 \sin^2 t + b^2 \cos^2 t)^{-3/2}$ (Chapter 1... recte Chapter 2), the extrema occur where $\frac{d}{dt}(a^2 \sin^2 t + b^2 \cos^2 t) = 0$, i.e., $\sin t \cos t = 0$: at the four points $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, i.e., $(\pm a, 0)$ (maximum curvature) and $(0, \pm b)$ (minimum curvature). Exactly four vertices.

4.6. The global Gauss–Bonnet theorem

The preceding theorems tied an integral quantity of a plane curve — the total curvature, the perimeter, the area — to a discrete datum: an integer index, the number of vertices, the case of equality. The global Gauss–Bonnet theorem carries the same pattern into dimension two: the integral of the Gaussian curvature over a closed surface *does not depend on the metric*, but only on the topology, and equals 2π times the Euler characteristic. Deforming the surface without tearing it, the curvature is redistributed but its integral does not change by an iota.

Remark 4.9: What we import for the global Gauss–Bonnet theorem. The proof that follows is complete starting from three statements that this volume does **not** prove and that we declare here, with the source.

(P1) Local form of Gauss–Bonnet on a geodesic triangle. If T is a geodesic triangle of an oriented regular surface, contained in a normal neighbourhood, with interior angles α, β, γ , then

$$\iint_T K \, dA = \alpha + \beta + \gamma - \pi.$$

For the sphere this is proved in full in Chapter 7 (Theorem 7.11), where $K = 1/r^2$ and the statement coincides with Girard’s theorem on the spherical excess. For an arbitrary regular surface we import it from do Carmo, §4-5. What is missing, in order to close it, is: the covariant derivative along a curve and parallel transport, which belong to Chapter 10 and to the theory of surfaces, not to that of plane curves.

(P2) Existence of a geodesic triangulation. Every compact regular surface admits a finite triangulation whose triangles are geodesic and each contained in a normal neighbourhood. do Carmo, §4-5, states and uses it without proving it, referring to T. Radó (Acta Szeged 2, 1926), who proved the triangulability of every topological surface.

(P3) Topological invariance of the Euler characteristic. The number $\chi = V - E + F$ does not depend on the chosen triangulation, but only on the surface: this is again Radó (1926), taken up in do Carmo, §4-5. For closed orientable surfaces one has moreover $\chi = 2 - 2g$, with g the genus.

Everything else — the combinatorial count, the sum of the angles, the passage from $2\pi V - \pi F$ to $2\pi\chi$ — is proved below, and the numerical checks are independent.

Definition 4.6: Triangulation and Euler characteristic. A **triangulation** of a compact surface M without boundary is a finite family of closed subsets $T_1, \dots, T_F$, each homeomorphic to a closed triangle of the plane, such that

- (a) $\bigcup_i T_i = M$;
- (b) two distinct triangles are disjoint, or have exactly one vertex in common, or exactly one whole edge.

Denoting by V , E and F the numbers of vertices, edges and faces, the **Euler characteristic** of the triangulation is $\chi = V - E + F$.

Lemma 4.12: Every edge belongs to two faces. In a triangulation of a compact surface without boundary one has

$$3F = 2E.$$

Proof. Step 1 — the set to be counted. Consider the set

$$\mathcal{I} = \{(T, \ell) : T \text{ a face, } \ell \text{ an edge of } T\},$$

and compute its cardinality in two ways.

Step 2 — counting by faces. Every face is homeomorphic to a triangle and therefore has exactly three edges: $|\mathcal{I}| = 3F$.

Step 3 — counting by edges. Let ℓ be an edge and let p be an interior point of ℓ . Since M is a surface without boundary, p has a neighbourhood homeomorphic to an open disc, and inside that neighbourhood ℓ divides it into two half-discs; by condition (b) of Definition 4.6 each half-disc is covered by the points of a single face. Hence every edge belongs to exactly two faces and $|\mathcal{I}| = 2E$.

Step 4 — conclusion. $3F = |\mathcal{I}| = 2E$. □

***Lemma 4.13: Sum of all the angles of a triangulation.** Let $T_1, \dots, T_F$ be a triangulation of a compact surface without boundary M and let $\alpha_i, \beta_i, \gamma_i$ be the interior angles of T_i . Then*

$$\sum_{i=1}^F (\alpha_i + \beta_i + \gamma_i) = 2\pi V.$$

Proof. **Step 1 — reorganising the sum.** Every angle in the sum is the angle of a face at one of its vertices. Grouping by vertex,

$$\sum_{i=1}^F (\alpha_i + \beta_i + \gamma_i) = \sum_{p \in \{\text{vertices}\}} \sum_{T \ni p} \vartheta_T(p),$$

where $\vartheta_T(p)$ is the angle of the face T at the vertex p .

Step 2 — the angles around a vertex add up to 2π . Let p be a vertex. Since M is a surface without boundary, p has a neighbourhood U homeomorphic to a disc; the faces containing p meet U in angular sectors which, by condition (b) of Definition 4.6, follow one another around p pairwise adjacent along an edge, covering U exactly once and without overlaps. The faces around p therefore form a cycle and their angles add up to a full angle:

$$\sum_{T \ni p} \vartheta_T(p) = 2\pi.$$

Step 3 — conclusion. Summing over all V vertices one obtains $2\pi V$. □

***Theorem 4.8: Global Gauss–Bonnet.** Let M be a compact, orientable regular surface without boundary, with Gaussian curvature K . Then*

$$\iint_M K \, dA = 2\pi \chi(M),$$

where $\chi(M) = V - E + F$ is the Euler characteristic of any triangulation of M . In particular the integral of the curvature does not depend on the metric.

Proof. **Step 1 — triangulation.** By (P2) of Remark 4.9 choose a geodesic triangulation $T_1, \dots, T_F$ of M , with interior angles $\alpha_i, \beta_i, \gamma_i$.

Step 2 — additivity of the integral. The triangles cover M and meet only along edges and vertices, which are finite unions of curves and of points, hence sets of measure zero for the area measure of M . Therefore

$$\iint_M K \, dA = \sum_{i=1}^F \iint_{T_i} K \, dA.$$

Step 3 — local form on each triangle. By (P1),

$$\iint_{T_i} K \, dA = \alpha_i + \beta_i + \gamma_i - \pi \quad (i = 1, \dots, F).$$

Step 4 — summation. Summing over i and using Lemma 4.13,

$$\iint_M K dA = \sum_{i=1}^F (\alpha_i + \beta_i + \gamma_i) - \pi F = 2\pi V - \pi F.$$

Step 5 — the combinatorics. By Lemma 4.12, $E = \frac{3}{2}F$, hence

$$2\pi(V - E + F) = 2\pi V - 2\pi \cdot \frac{3F}{2} + 2\pi F = 2\pi V - 3\pi F + 2\pi F = 2\pi V - \pi F.$$

The right-hand side coincides with that of Step 4.

Step 6 — conclusion. Combining Steps 4 and 5,

$$\iint_M K dA = 2\pi(V - E + F) = 2\pi\chi(M),$$

and by (P3) the value $V - E + F$ does not depend on the chosen triangulation: the left-hand side, which does not depend on it by construction, is therefore a topological invariant. Since the left-hand side depends on the metric only through K and dA , while the right-hand side does not depend on it at all, the final assertion follows as well.

Step 7 — consistency check on the sign. If $K > 0$ everywhere, the left-hand side is positive, hence $\chi(M) > 0$; if $K \equiv 0$, then $\chi(M) = 0$. These are the two cases checked in Examples 4.2 and 4.3. $\square$

Example 4.2: The sphere with the octahedral triangulation. Take on S^2 (radius 1) the inscribed regular octahedron, with vertices $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, $(0, 0, \pm 1)$, and project its edges onto the great circles: one obtains a geodesic triangulation with

$$V = 6, \quad E = 12, \quad F = 8.$$

Step 1 — combinatorial check. $3F = 24 = 2E$ (Lemma 4.12) $\checkmark$, and $\chi = 6 - 12 + 8 = 2$.

Step 2 — the eight triangles. Every face is a *triangular* triangle: its three vertices are pairwise orthogonal as unit vectors of $\mathbb{R}^3$, hence $\alpha = \beta = \gamma = \pi/2$. The excess equals $\alpha + \beta + \gamma - \pi = \frac{3\pi}{2} - \pi = \frac{\pi}{2}$, and the area, since the sphere is divided into eight congruent parts, equals $4\pi/8 = \pi/2$: the two quantities coincide, as (P1) predicts with $K \equiv 1$. $\checkmark$

Step 3 — sum of the angles. $\sum_i (\alpha_i + \beta_i + \gamma_i) = 8 \times \frac{3\pi}{2} = 12\pi = 2\pi V = 2\pi \times 6$ (Lemma 4.13) $\checkmark$.

Step 4 — the two expressions of Step 5 of the proof. $2\pi V - \pi F = 12\pi - 8\pi = 4\pi$ and $2\pi\chi = 2\pi \times 2 = 4\pi$. $\checkmark$

Step 5 — independence of the radius. For $S^2(O, r)$ one has $K = 1/r^2$ and area $4\pi r^2$, hence

$$\iint_{S^2(O, r)} K dA = \frac{1}{r^2} \cdot 4\pi r^2 = 4\pi$$

for *every* r : the total curvature does not see the geometry, only the topology. $\checkmark$ (Checks in `verifica_cap02_cap04.py`, blocks [C4-4] and [C4-5].)

Example 4.3: The torus: zero total curvature. Consider the torus of revolution with radii $R > r > 0$,

$$\sigma(u, v) = ((R + r \cos u) \cos v, (R + r \cos u) \sin v, r \sin u), \quad u, v \in [0, 2\pi],$$

for which $K(u) = \frac{\cos u}{r(R + r \cos u)}$ and $dA = r(R + r \cos u) du dv$.

Step 1 — the integral. The product $K dA$ simplifies:

$$\iint_{T^2} K dA = \int_0^{2\pi} \int_0^{2\pi} \cos u du dv = 2\pi \int_0^{2\pi} \cos u du = 0.$$

Step 2 — comparison with the theorem. $\chi(T^2) = 0$, hence $2\pi\chi = 0$: consistent. ✓

Step 3 — check on two triangulations. The 3×3 grid of the fundamental square, with each small square divided into two triangles, gives $V = 9$, $E = 27$, $F = 18$: $3F = 54 = 2E$ ✓ and $\chi = 9 - 27 + 18 = 0$ ✓. The *minimal* triangulation of the torus has $V = 7$, $E = 21$, $F = 14$ — the combinatorics is that of the complete graph K_7 embedded in the torus, described by Möbius and realised as a polyhedron embedded in $\mathbb{R}^3$ by Á. Császár in 1949 — and gives $3F = 42 = 2E$ ✓, $\chi = 7 - 21 + 14 = 0$ ✓.

Step 4 — the flat torus. On the flat torus $\mathbb{R}^2/\mathbb{Z}^2$ the triangles of the grid are Euclidean, with angle sum π each: the total sum equals $\pi F = 18\pi$, which is also $2\pi V = 2\pi \times 9 = 18\pi$ ✓ (Lemma 4.13), and $2\pi V - \pi F = 0$, as it must be for $K \equiv 0$.

Step 5 — where the curvature is. On the torus of revolution with $R = 3$ and $r = 1$, the curvature equals $K(0) = \frac{1}{1 \times 4} = 0.25$ on the outermost circle and $K(\pi) = \frac{-1}{1 \times 2} = -0.5$ on the innermost one: the outer part is convex, the inner part is saddle-shaped, and the two contributions cancel exactly. ✓

Corollary 4.5: No everywhere-positive curvature outside the sphere. Let M be a compact, orientable regular surface without boundary with $K > 0$ at every point. Then $\chi(M) = 2$, that is M has genus 0. In particular no metric on the torus — nor on a closed orientable surface of genus ≥ 1 — has everywhere-positive Gaussian curvature.

Proof. **Step 1.** If $K > 0$ on a compact set and $dA > 0$, then $\iint_M K dA > 0$; by Theorem 4.8, $2\pi\chi(M) > 0$, hence $\chi(M) \geq 1$.

Step 2. By (P3) of Remark 4.9, a closed orientable surface has $\chi = 2 - 2g$ with $g \geq 0$ an integer: the possible values are $2, 0, -2, -4, \dots$, and the only one ≥ 1 is $\chi = 2$, that is $g = 0$.

Step 3. The torus has $g = 1$ and $\chi = 0$: it cannot carry a metric with everywhere-positive curvature. The computation of Example 4.3 shows this concretely: whatever the metric, the positive curvature of the outer part must be exactly compensated by negative curvature elsewhere. □

Remark 4.10: Why it is the same theorem as the Umlaufsatz. Section 4.1 established $\int_0^L k ds = 2\pi \operatorname{ind}(\gamma)$, with $\operatorname{ind} = \pm 1$ for simple curves. It is the same statement in dimension one: the integral of a curvature equals 2π times an integer depending only on the topology of the object, not on its shape. The link is more than an analogy: the Gauss–Bonnet form for a region with boundary contains both terms, $\iint_R K dA + \oint_{\partial R} k_g ds + \sum(\text{exterior angles}) = 2\pi\chi(R)$, and it reduces to the Umlaufsatz when R is a simply connected plane region ($K \equiv 0$, $\chi = 1$).

Exercise 4.4: Total curvature of a genus-two surface. What is the value of $\iint_M K dA$ for a closed orientable surface of genus 2 (the «double torus»)? And what is the average sign of the curvature?

Solution. By (P3) of Remark 4.9, $\chi = 2 - 2g = 2 - 4 = -2$, hence by Theorem 4.8

$$\iint_M K dA = 2\pi \times (-2) = -4\pi.$$

The integral is negative: on average the curvature is negative, that is, the surface is predominantly saddle-shaped. By Corollary 4.5 it cannot be everywhere-positive; a fortiori it cannot be everywhere zero either, because that would give $\chi = 0$. A smooth double torus has positive curvature on the «outer» parts of the two handles and negative curvature in the throats, with balance -4π independently of how one deforms it. ✓

4.7. Assigned exercises

The answers are in the Teacher's key (Section 4.11).

Exercise 4.5: The n -sided Reuleaux polygons. *Aim:* to construct an infinite family of curves of constant width, to check Barbier's theorem on it and to place it between the two extremes of the Blaschke–Lebesgue theorem. *Estimated time:* 45 minutes. *Type:* symbolic with numerical check.

For odd $n \geq 3$, the **Reuleaux polygon** of width w is obtained from a regular n -gon by replacing each side with the circular arc of radius w centred at the opposite vertex.

- Prove that it is a convex body of constant width w and deduce its perimeter from Theorem 4.3.
- Prove that its area equals $A_n = \frac{1}{2} \left(\pi - n \tan \frac{\pi}{2n} \right) w^2$.
- Compute A_3 , A_5 and A_7 for $w = 1$ and their ratios to the area of the circle of diameter 1.
- Check that A_n increases with n and that $A_n \rightarrow \pi w^2/4$ as $n \rightarrow +\infty$, tabulating $n = 9, 51, 501, 5001$.
- Interpret (c) and (d) in the light of Theorem 4.4.

Exercise 4.6: A three-lobed polar curve: index, area, vertices. *Aim:* to apply, in a single example, the Umlaufsatz, the isoperimetric inequality and the four-vertex theorem to a simple but *non-convex* curve. *Estimated time:* 50 minutes. *Type:* numerical/computational. Consider the polar curve $r(\theta) = 1 + a \cos 3\theta$ with $a = 0.3$.

- Check that the curve is closed, regular and simple, and that it has no intersection with the origin.
- Compute the enclosed area in closed form and verify it numerically.
- Compute the length L by quadrature and the isoperimetric ratio $L^2/(4\pi A)$; comment on its value in the light of Theorem 4.5.
- Compute the total curvature $\int k \, ds$ and verify Theorem 4.1.
- Determine the number and the positions of the vertices and compare with Theorem 4.6; check that the curve is not convex by computing $k(0)$ and $k(\pi/3)$.

Exercise 4.7: Gauss–Bonnet on two triangulations. *Aim:* to carry out in full the count of the proof of Theorem 4.8 on explicit triangulations. *Estimated time:* 35 minutes. *Type:* combinatorial/numerical.

- For the regular icosahedron inscribed in the sphere, projected onto the great circles, determine V , E , F ; check Lemma 4.12 and compute χ , $2\pi V - \pi F$ and $\iint K \, dA$.
- Deduce the average angular excess of a triangle of the triangulation and compare it with that of the octahedron of Example 4.2.
- For the torus, check $3F = 2E$ and $\chi = 0$ on the diagonalised 3×3 grid and on the minimal triangulation with 7 vertices.
- Explain why no triangulation of the torus can have $\chi \neq 0$, even though it may have

arbitrarily large V , E , F .

- (e) Construct a triangulation of the torus with exactly $F = 100$ faces (or prove that none exists) and compute its V and E compatible with $\chi = 0$.

4.8. Worked exercises

Exercise 4.8: Convex bodies prescribed by a single harmonic. *Type:* symbolic with numerical check. For an integer $n \geq 2$ and $\varepsilon \in \mathbb{R}$ consider the 2π -periodic function

$$h(\theta) = 1 + \varepsilon \cos n\theta.$$

Determine for which ε the function h is the support function of a convex body of class C^2 , and compute its perimeter, area and isoperimetric deficit.

Solution. Strategy. One uses the convexity criterion of Theorem 4.2: h is the support function of a smooth convex body if and only if $h + h'' > 0$; then one applies Cauchy's formulae for L and for A .

Step 1 — radius of curvature. $h'' = -n^2\varepsilon \cos n\theta$, hence

$$\rho(\theta) = h(\theta) + h''(\theta) = 1 - (n^2 - 1)\varepsilon \cos n\theta.$$

Step 2 — convexity threshold. The minimum of ρ equals $1 - (n^2 - 1)|\varepsilon|$, hence $\rho > 0$ everywhere if and only if

$$|\varepsilon| < \frac{1}{n^2 - 1}.$$

For $n = 3$ the threshold is $1/8 = 0.125$; for $n = 4$ it is $1/15 = 0.066667$; for $n = 5$ it is $1/24 = 0.041667$: the higher the harmonic, the smaller its amplitude must be for the body to remain convex.

Step 3 — perimeter. By Cauchy's formula (Theorem 4.2(iii)),

$$L = \int_0^{2\pi} h \, d\theta = 2\pi + \varepsilon \int_0^{2\pi} \cos n\theta \, d\theta = 2\pi,$$

because $\int_0^{2\pi} \cos n\theta \, d\theta = 0$ for $n \geq 1$: all these convex bodies have the same perimeter as the unit circle, for every admissible ε .

Step 4 — area. From $h' = -n\varepsilon \sin n\theta$ and $\int_0^{2\pi} \cos^2 = \int_0^{2\pi} \sin^2 = \pi$,

$$\int_0^{2\pi} h^2 = 2\pi + \pi\varepsilon^2, \quad \int_0^{2\pi} h'^2 = \pi n^2 \varepsilon^2,$$

hence, by Theorem 4.2(iv),

$$A = \frac{1}{2} \left(2\pi + \pi\varepsilon^2 - \pi n^2 \varepsilon^2 \right) = \pi - \frac{(n^2 - 1)\pi\varepsilon^2}{2}.$$

Step 5 — isoperimetric deficit. Since $L^2/(4\pi) = 4\pi^2/(4\pi) = \pi$,

$$\frac{L^2}{4\pi} - A = \frac{(n^2 - 1)\pi\varepsilon^2}{2} \geq 0,$$

with equality if and only if $\varepsilon = 0$, that is for the circle: this is exactly Step 4 of the proof of Theorem 4.5, restricted to a single harmonic.

Independent check. The script `verifica_cap02_cap04.py` (block [C4-1]) computes $\min(h + h'')$ by sampling, L and A by Simpson quadrature, and compares them with the formulae of Steps 2–4. For $n = 3$ and $\varepsilon = 0.05; 0.1; 0.124$ it finds respectively $\min \rho = 0.600000; 0.200000; 0.008000$ (that is $1 - 8\varepsilon$), $L = 6.283185 = 2\pi$ and deficit $0.031416; 0.125664; 0.193221$, against $4\pi\varepsilon^2$. For $n = 2, 4, 5$ with $\varepsilon = 0.02$ it verifies the general formula. ✓

Interpretation. The perimeter «sees» only the mean of h (the zeroth harmonic), the area sees all the harmonics with weight $n^2 - 1$: it is the same hierarchy that Parseval's identity imposes on the energy of a signal. The isoperimetric deficit is the energy of the higher harmonics of the support function.

Typical mistakes. (a) Using $A = \frac{1}{2} \int h^2$, forgetting the term $-h'^2$: one would obtain an area *larger* than that of the circle, contradicting Theorem 4.5. (b) Believing that $|\varepsilon| < 1$ suffices for convexity: for $n = 5$ the threshold is already 0.042. (c) Confusing $\rho = h + h''$ with h itself.

Extension. Consider $h(\theta) = 1 + \varepsilon \cos n\theta$ with n odd and check that the width $w(\theta) = h(\theta) + h(\theta + \pi)$ is constant $= 2$: these are smooth non-circular convex bodies of constant width, for which Theorem 4.3 predicts $L = 2\pi = \pi w$, in agreement with Step 3.

Exercise 4.9: Descartes's theorem on the total angular defect. *Type:* proof-based/combinatorial. Let P be a convex polyhedron whose faces are plane triangles. For every vertex p define the *angular defect*

$$\delta_p = 2\pi - \sum_{T \ni p} \vartheta_T(p).$$

Prove that $\sum_p \delta_p = 4\pi$, and check it on the Platonic solids (for which polygonal faces are also allowed, counted with the angle sum of the face).

Solution. Strategy. One counts the angles in two ways, exactly as in Lemmas 4.12 and 4.13, the only difference being that the triangles are now *Euclidean* and not geodesic: the sum of their angles is π , not π plus an excess.

Step 1 — sum of the defects. By definition,

$$\sum_p \delta_p = 2\pi V - \sum_p \sum_{T \ni p} \vartheta_T(p) = 2\pi V - \sum_T (\alpha_T + \beta_T + \gamma_T),$$

where in the second equality the sum has been reorganised by faces instead of by vertices (it is Step 1 of Lemma 4.13, read backwards).

Step 2 — the triangles are plane. Every face is a Euclidean triangle, hence $\alpha_T + \beta_T + \gamma_T = \pi$ and the sum equals πF :

$$\sum_p \delta_p = 2\pi V - \pi F.$$

Step 3 — the combinatorics. The boundary of P is a compact surface without boundary, hence Lemma 4.12 applies, $3F = 2E$, and by Step 5 of the proof of Theorem 4.8 $2\pi V - \pi F = 2\pi(V - E + F) = 2\pi\chi$.

Step 4 — the topology. The boundary of a convex polyhedron is homeomorphic to S^2 , hence $\chi = 2$ and

$$\sum_p \delta_p = 2\pi \times 2 = 4\pi.$$

Step 5 — polygonal faces. If the faces are polygons with m_T sides, Step 2 becomes $\sum_T (m_T - 2)\pi$ and Step 3 is adapted by counting $\sum_T m_T = 2E$; one obtains again $2\pi V - 2\pi E + 2\pi F = 2\pi\chi = 4\pi$.

Independent check. For the five Platonic solids, with ϑ the angle of a face at one of its vertices:

solid	V	faces per vertex	ϑ	δ_p
tetrahedron	4	3	$\pi/3$	π
cube	8	3	$\pi/2$	$\pi/2$
octahedron	6	4	$\pi/3$	$2\pi/3$
dodecahedron	20	3	$3\pi/5$	$\pi/5$
icosahedron	12	5	$\pi/3$	$\pi/3$

and in all five cases $V \delta_p = 4\pi$: 4π ; $8 \cdot \pi/2 = 4\pi$; $6 \cdot 2\pi/3 = 4\pi$; $20 \cdot \pi/5 = 4\pi$; $12 \cdot \pi/3 = 4\pi$. ✓ (Block [C4-4] of the script.)

Interpretation. The angular defect is «curvature concentrated at the vertices»: a polyhedron is flat everywhere except at the vertices, where all the curvature accumulates. Descartes's theorem (around 1630, rediscovered and published much later) is therefore the exact discrete version of Theorem 4.8, with $\iint K dA$ replaced by $\sum_p \delta_p$.

Typical mistakes. (a) Summing the angles of the faces instead of the defects of the vertices. (b) Forgetting that the result holds for the boundary of a convex polyhedron because it is a topological sphere: a toroidal polyhedron (for instance Császár's) has total defect 0. (c) Using $3F = 2E$ for polyhedra with non-triangular faces.

Extension. Check that for the Császár polyhedron ($V = 7$, $E = 21$, $F = 14$, topologically a torus) the sum of the angular defects equals $2\pi \cdot 7 - \pi \cdot 14 = 0$, consistently with $\chi = 0$ (Example 4.3).

Exercise 4.10: Why a calliper does not recognise a Reuleaux profile. *Type:* applied with numerical check. A turned workpiece has a Reuleaux-triangle cross section of width w . Show that a vernier calliper always measures w in every direction, compute the radii of the circumscribed and inscribed circles, and determine the circularity error that an inspection according to ISO 1101 would detect.

Solution. Strategy. The calliper measures the distance between two parallel supporting lines, that is the width $w(\theta)$; by Theorem 4.3 and the definition of constant width this does not depend on θ . The circularity error, on the other hand, is the difference between the radii of the minimal circumscribed circle and the maximal inscribed circle, and it is not zero.

Step 1 — the calliper measures the width. The two jaws of a calliper are plane and parallel: the reading is $w(\theta) = h(\theta) + h(\theta + \pi)$, that is the width in the direction θ (Definition 4.3). For the Reuleaux triangle this equals w for every θ (Example 4.1): the calliper always gives the same reading, exactly as for a cylinder of diameter w .

Step 2 — circumscribed circle. By symmetry the centre of the two circles is the centroid O of the equilateral triangle of side w from which the profile is built. The distance from O to a vertex is the circumradius of the equilateral triangle,

$$R = \frac{w}{\sqrt{3}} = 0.577350 w,$$

and the three vertices are the points of the profile farthest from O : R is therefore the radius of the minimal circumscribed circle.

Step 3 — inscribed circle. The points nearest to O lie on the arcs, at distance $w - R$ from the centre: indeed the arc opposite a vertex V has centre at V and radius w , and the point of that arc nearest to O lies on the line VO , at distance $w - |VO| = w - R$ from O .

Hence

$$r = w - R = w \left(1 - \frac{1}{\sqrt{3}} \right) = 0.422650 w.$$

Note that $R + r = w$: this is the relation between the two supporting lines orthogonal to the vertex–arc direction, that is a case of constant width.

Step 4 — circularity error. According to ISO 1101 circularity is the difference between the radii of two concentric circles enclosing the profile:

$$R - r = w \left(\frac{2}{\sqrt{3}} - 1 \right) = 0.154701 w.$$

For $w = 40$ mm one obtains $R = 23.094011$ mm, $r = 16.905989$ mm and circularity error 6.188022 mm: enormous compared with any machining tolerance.

Independent check. The script `verifica_cap02_cap04.py` (block [C4-2]) recomputes R , r , $R + r$ and $R - r$ for $w = 1$ and finds 0.577350; 0.422650; 1.000000; 0.154701. ✓

Interpretation. Constant width and circularity are different properties: the first concerns the *supporting lines*, the second the concentric *circles*. An inspection with a calliper (or with two parallel rollers) therefore cannot certify the roundness of a shaft: one needs a dial gauge on a V-block with a correctly chosen angle, or a three-point measurement with non-equispaced angles, or a coordinate measuring machine. It is a classical and costly metrological mistake.

Typical mistakes. (a) Concluding from $L = \pi w$ (Theorem 4.3) that the profile is a circle: the perimeter does not characterise the shape. (b) Taking $w/2$ as the inscribed radius. (c) Confusing circularity (ISO 1101) with the dimensional tolerance on the diameter.

Extension. Compute R , r and the circularity error for the Reuleaux polygons with 5 and 7 sides and check that they tend respectively to $w/2$, $w/2$ and 0 as $n \rightarrow \infty$, consistently with Exercise 4.5(d).

4.9. Numerical and symbolic exercises with solutions

The exercises below come from the treatment of plane curves in Volume II and are reproduced in full, together with their solutions.

4.9.1 Numerical exercises

Exercise 4.11: N1 — A convex body given by its support function. Let $h(\theta) = 10 + \cos 3\theta$ (in mm). Check that h defines a convex body; compute its perimeter, area, minimum and maximum radii of curvature, and the isoperimetric deficit $L^2/4\pi - A$; compare the latter with the formula of Step 4 of the proof of Theorem 4.5.

Exercise 4.12: N2 — Reuleaux triangle versus circle. For a Reuleaux triangle of width $w = 40$ mm compute the perimeter and area, and compare them with those of the circle of diameter 40 mm. Comment in the light of Theorems 4.3 and 4.4.

Exercise 4.13: N3 — The four vertices of an ellipse. For the ellipse with semi-axes $a = 5$ and $b = 3$ determine the four vertices, the corresponding curvatures and radii of curvature, and the four cusps of the evolute.

Exercise 4.14: N4 — Total curvature of an ellipse. Verify analytically that the total curvature of the ellipse with semi-axes a and b equals 2π , as predicted by Theorem 4.1. (Hint: use $\int_0^{2\pi} \frac{dt}{a^2 \sin^2 t + b^2 \cos^2 t} = \frac{2\pi}{ab}$.)

4.9.2 Symbolic exercises

Exercise 4.15: S1 — An enclosed curve cannot be too little curved. Let γ be a closed curve contained in a circle of radius R . Prove that there is a point where $|k| \geq 1/R$.

Exercise 4.16: S2 — Wirtinger's inequality. Let f be 2π -periodic of class C^1 with $\int_0^{2\pi} f = 0$. Prove that

$$\int_0^{2\pi} f'(\theta)^2 d\theta \geq \int_0^{2\pi} f(\theta)^2 d\theta,$$

with equality if and only if $f(\theta) = a \cos \theta + b \sin \theta$.

Exercise 4.17: S3 — Double normals of a constant-width curve. Let K be convex of constant width w and class C^2 . Prove that $\rho(\theta) + \rho(\theta + \pi) = w$, that the diameter of K equals w , and that the normal at a point is also the normal at the opposite point (a *double normal*).

Exercise 4.18: S4 — Steiner's formula for parallel curves. Let K be convex with perimeter L and area A , and let K_d be the convex body of points at distance $\leq d$ from K . Prove, using the support function, that

$$L_d = L + 2\pi d, \quad A_d = A + Ld + \pi d^2.$$

Interpret the term πd^2 .

Exercise 4.19: S5 — Constant width and central symmetry. Prove that a convex body of constant width possessing a centre of symmetry is a circle. Deduce that the Reuleaux triangle has no centre of symmetry.

4.9.3 Solutions to the numerical exercises

Solution. N1. Step 1 — convexity. By Theorem 4.2(ii),

$$\rho(\theta) = h + h'' = (10 + \cos 3\theta) + (-9 \cos 3\theta) = 10 - 8 \cos 3\theta.$$

Since $\cos 3\theta \in [-1, 1]$, $\rho \in [2, 18]$, always positive: h is indeed the support function of a convex body with smooth boundary.

Step 2 — extreme radii.

$$\rho_{\min} = 10 - 8 = 2 \text{ mm} \quad (\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}), \quad \rho_{\max} = 10 + 8 = 18 \text{ mm} \quad (\theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3}).$$

There are therefore six vertices, in agreement with Theorem 4.6 (at least four).

Step 3 — perimeter. By Cauchy's formula,

$$L = \int_0^{2\pi} (10 + \cos 3\theta) d\theta = 20\pi = 62.8319 \text{ mm},$$

the integral of the cosine over a whole number of periods being zero.

Step 4 — area. With $h' = -3 \sin 3\theta$:

$$\int_0^{2\pi} h^2 d\theta = \int_0^{2\pi} (100 + 20 \cos 3\theta + \cos^2 3\theta) d\theta = 200\pi + 0 + \pi = 201\pi,$$

$$\int_0^{2\pi} h'^2 d\theta = 9 \int_0^{2\pi} \sin^2 3\theta d\theta = 9\pi.$$

Hence

$$A = \frac{1}{2}(201\pi - 9\pi) = 96\pi = 301.593 \text{ mm}^2.$$

Step 5 — isoperimetric deficit.

$$\frac{L^2}{4\pi} = \frac{(20\pi)^2}{4\pi} = 100\pi = 314.159 \text{ mm}^2, \quad \frac{L^2}{4\pi} - A = 4\pi = 12.566 \text{ mm}^2.$$

Step 6 — comparison with the theoretical formula. Step 4 of the proof of Theorem 4.5 predicts

$$\frac{L^2}{4\pi} - A = \frac{\pi}{2} \sum_{n \geq 2} (n^2 - 1)(a_n^2 + b_n^2) = \frac{\pi}{2} (3^2 - 1) \cdot 1^2 = 4\pi. \checkmark$$

The only harmonic present is $n = 3$ with $a_3 = 1$.

Step 7 — isoperimetric ratio. $4\pi A/L^2 = 96\pi/(100\pi) = 0.96$: the body encloses 96% of the area of the circle of equal perimeter.

Solution. N2. *Step 1 — perimeter of the Reuleaux triangle.* By Theorem 4.3, $L = \pi w = \pi \times 40 = 125.664 \text{ mm}$.

Step 2 — area of the Reuleaux triangle. From Example 4.1,

$$A = \frac{\pi - \sqrt{3}}{2} w^2 = 0.704771 \times 1600 = 1127.63 \text{ mm}^2.$$

Step 3 — circle of diameter 40 mm.

$$L_o = \pi \times 40 = 125.664 \text{ mm}, \quad A_o = \frac{\pi \times 40^2}{4} = 400\pi = 1256.64 \text{ mm}^2.$$

Step 4 — comparison. The perimeters are *identical* (Theorem 4.3); the areas are not:

$$\frac{A}{A_o} = \frac{1127.63}{1256.64} = 0.89734,$$

i.e. 89.73%. By Theorem 4.4 these are the two extremes: no figure of width 40 mm has area greater than 1256.64 mm^2 nor smaller than 1127.63 mm^2 .

Step 5 — practical reading. A machine part measured with a vernier calliper always reads 40.00 mm in every orientation, whether it is cylindrical or has a Reuleaux-triangle cross-section. The difference — over 10% of area, and a completely different shape — is undetectable with that instrument. One needs a dial gauge on a rotary table, or a roundness measurement per ISO 1101.

Solution. N3. *Step 1 — position of the vertices.* From Example 2.2,

$$k(t) = \frac{ab}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}}.$$

The derivative vanishes when that of the denominator does, i.e. when $2(a^2 - b^2) \sin t \cos t = 0$: for $a \neq b$, at $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. The four vertices are therefore $(\pm 5, 0)$ and $(0, \pm 3)$.

Step 2 — curvatures. With $a = 5, b = 3$:

$$k(0) = \frac{ab}{b^3} = \frac{a}{b^2} = \frac{5}{9} = 0.555556, \quad \rho = \frac{b^2}{a} = \frac{9}{5} = 1.8,$$

$$k\left(\frac{\pi}{2}\right) = \frac{b}{a^2} = \frac{3}{25} = 0.12, \quad \rho = \frac{a^2}{b} = \frac{25}{3} = 8.3\bar{3}.$$

Step 3 — cusps of the evolute. From Example 2.4,

$$e(t) = \left(\frac{a^2 - b^2}{a} \cos^3 t, \frac{b^2 - a^2}{b} \sin^3 t \right) = (3.2 \cos^3 t, -5.3\bar{3} \sin^3 t),$$

since $a^2 - b^2 = 16$. The four cusps are

$$(\pm 3.2; 0) \quad \text{and} \quad (0; \mp 5.3\bar{3}).$$

Step 4 — check. From the vertex $(5, 0)$ to the centre of curvature $(3.2; 0)$ the distance is $5 - 3.2 = 1.8 = \rho(0) \checkmark$. From the vertex $(0, 3)$ to the centre $(0; -5.3\bar{3})$ the distance is $3 + 5.3\bar{3} = 8.3\bar{3} = \rho(\pi/2) \checkmark$.

Step 5 — reading. The radius of curvature varies by a factor $8.3\bar{3}/1.8 = 4.63 = (a/b)^3$: the ellipse is far “flatter” at the ends of the minor axis. This is why an approximate four-arc ellipse template uses such different radii.

Solution. N4. *Step 1 — line element.*

$$\|\gamma'(t)\| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}.$$

Step 2 — the product $k ds$. From Example 2.2,

$$k(t) \|\gamma'(t)\| = \frac{ab}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}} \cdot (a^2 \sin^2 t + b^2 \cos^2 t)^{1/2} = \frac{ab}{a^2 \sin^2 t + b^2 \cos^2 t}.$$

Step 3 — integration. Using the suggested integral,

$$\int_0^L k ds = \int_0^{2\pi} \frac{ab dt}{a^2 \sin^2 t + b^2 \cos^2 t} = ab \cdot \frac{2\pi}{ab} = 2\pi. \checkmark$$

Step 4 — check of the suggested integral. Dividing numerator and denominator by $\cos^2 t$ and setting $u = \tan t$ on $(-\pi/2, \pi/2)$:

$$\int_{-\pi/2}^{\pi/2} \frac{dt}{a^2 \sin^2 t + b^2 \cos^2 t} = \int_{-\infty}^{+\infty} \frac{du}{a^2 u^2 + b^2} = \frac{1}{ab} \left[\arctan \frac{au}{b} \right]_{-\infty}^{+\infty} = \frac{\pi}{ab},$$

and over the full period this doubles: $2\pi/(ab)$. $\checkmark$

Step 5 — reading. The integral does not depend on a and b : the total curvature is 2π for every ellipse, for the circle ($a = b$) as for a very elongated ellipse. This is exactly the topological invariance asserted by Theorem 4.1: flattening the ellipse concentrates the curvature at the two ends without changing its integral.

4.9.4 Solutions to the symbolic exercises

Solution. S1. *Step 1 — the distance function.* Let c be the centre of the containing circle and put

$$f(s) = \|\gamma(s) - c\|^2.$$

f is continuous and periodic, hence attains a maximum at some s_0 ; let $d = \sqrt{f(s_0)} \leq R$ be the maximal distance.

Step 2 — first-order condition.

$$f'(s) = 2\langle \gamma(s) - c, T(s) \rangle,$$

which vanishes at s_0 : the vector $\gamma(s_0) - c$ is orthogonal to $T(s_0)$, hence parallel to $N(s_0)$: $\gamma(s_0) - c = \lambda N(s_0)$ with $|\lambda| = d$.

Step 3 — second-order condition. Differentiating again and using $T' = kN$:

$$f''(s) = 2\left(\langle T, T \rangle + \langle \gamma - c, kN \rangle\right) = 2\left(1 + k\langle \gamma - c, N \rangle\right).$$

At a maximum $f''(s_0) \leq 0$, hence

$$1 + k(s_0) \lambda \leq 0 \implies k(s_0) \lambda \leq -1.$$

Step 4 — conclusion. Taking absolute values, $|k(s_0)| d \geq 1$, i.e.

$$|k(s_0)| \geq \frac{1}{d} \geq \frac{1}{R}.$$

Step 5 — equality case and reading. Equality $|k| = 1/R$ everywhere forces, by Theorem 2.5, the curve to be the circle itself. The result therefore reads: *no closed curve can lie inside a circle while being everywhere less curved than it.* It is the geometric analogue of the maximum principle, and the proof — a second-order comparison at a contact point — is its elementary form.

Solution. S2. *Step 1 — series expansion.* Since f is 2π -periodic and C^1 , write its real Fourier series (Chapter 4); the hypothesis $\int_0^{2\pi} f = 0$ means the constant coefficient vanishes:

$$f(\theta) = \sum_{n \geq 1} (a_n \cos n\theta + b_n \sin n\theta).$$

Step 2 — the two energies. By Parseval's identity (Theorem (Volume II, Chapter 4) of Chapter 4),

$$\int_0^{2\pi} f^2 = \pi \sum_{n \geq 1} (a_n^2 + b_n^2),$$

and since $f'(\theta) = \sum_{n \geq 1} n(-a_n \sin n\theta + b_n \cos n\theta)$ has coefficients na_n, nb_n (up to sign and swap, irrelevant for squares),

$$\int_0^{2\pi} f'^2 = \pi \sum_{n \geq 1} n^2 (a_n^2 + b_n^2).$$

Step 3 — difference.

$$\int_0^{2\pi} f'^2 - \int_0^{2\pi} f^2 = \pi \sum_{n \geq 1} (n^2 - 1)(a_n^2 + b_n^2) \geq 0,$$

since $n^2 - 1 \geq 0$ for $n \geq 1$.

Step 4 — equality case. The sum vanishes iff $a_n = b_n = 0$ for every $n \geq 2$, i.e. $f(\theta) = a_1 \cos \theta + b_1 \sin \theta$.

Step 5 — why the zero-mean hypothesis is needed. Without it the inequality is false: for $f \equiv 1$ we have $\int f'^2 = 0 < 2\pi = \int f^2$. Indeed the $n = 0$ term gives $n^2 - 1 = -1 < 0$, the only negative term of the sum. It plays the same role as the constant term in the proof of the isoperimetric inequality, where it is not cancelled but balanced by the perimeter.

Step 6 — connection. With $f = h - a_0$ (support function with its mean removed), Wirtinger's inequality gives a bound on $\int h'^2$; the full computation, carried out in Step 3 of the proof of Theorem 4.5, is a refined version of this inequality in which the $n = 1$ term is also tracked.

Solution. S3. *Step 1 — sum of the radii of curvature.* By Theorem 4.2(ii),

$$\rho(\theta) + \rho(\theta + \pi) = (h(\theta) + h(\theta + \pi)) + (h''(\theta) + h''(\theta + \pi)) = w(\theta) + w''(\theta) = w + 0 = w,$$

because w is constant and the second derivative of a constant is zero.

Step 2 — the chord between opposite points. From $u(\theta + \pi) = -u(\theta)$ and $u'(\theta + \pi) = -u'(\theta)$, Theorem 4.2(i) gives

$$\gamma(\theta + \pi) = -h(\theta + \pi)u(\theta) - h'(\theta + \pi)u'(\theta),$$

hence

$$\gamma(\theta) - \gamma(\theta + \pi) = (h(\theta) + h(\theta + \pi))u(\theta) + (h'(\theta) + h'(\theta + \pi))u'(\theta).$$

Step 3 — the tangential term vanishes. Differentiating the identity $h(\theta) + h(\theta + \pi) = w$ (constant) gives $h'(\theta) + h'(\theta + \pi) = 0$. Hence

$$\gamma(\theta) - \gamma(\theta + \pi) = w u(\theta).$$

Step 4 — double normal. The vector $u(\theta)$ is the normal to γ at $\gamma(\theta)$ and is also, up to sign, the normal at $\gamma(\theta + \pi)$: the chord joining the two points is therefore *orthogonal to the curve at both*. It is called a *double normal*; its length is w .

Step 5 — the diameter. For any pair of points $P, Q \in K$, the distance $|PQ|$ does not exceed the width of K in the direction of $P - Q$, which is w . Hence $\text{diam}(K) \leq w$; and Step 3 exhibits pairs at distance exactly w . So $\text{diam}(K) = w$.

Step 6 — mechanical consequence. Step 1 says the two opposite radii of curvature are complementary to w : where the curve is very curved (ρ small) the opposite point is very flat. That is why a body of constant width rolls under a plank keeping its height constant — but its *centre* oscillates, and for that reason it cannot replace a wheel on an axle.

Solution. S4. *Step 1 — support function of the parallel body.* K_d is the set of points at distance $\leq d$ from K , i.e. the Minkowski sum $K + D_d$ with D_d the disc of radius d . Since the support function of a Minkowski sum is the sum of the support functions (immediate from Definition 4.2: the maximum of a sum of independent terms is the sum of the maxima), and since the disc of radius d has constant support function d :

$$h_d(\theta) = h(\theta) + d.$$

Step 2 — perimeter. By Cauchy's formula (Theorem 4.2(iii)),

$$L_d = \int_0^{2\pi} (h + d) d\theta = L + 2\pi d.$$

Step 3 — area. Since $h'_d = h'$,

$$A_d = \frac{1}{2} \int_0^{2\pi} [(h+d)^2 - h'^2] d\theta = \frac{1}{2} \int_0^{2\pi} (h^2 - h'^2) d\theta + d \int_0^{2\pi} h d\theta + \frac{d^2}{2} \int_0^{2\pi} d\theta,$$

that is

$$A_d = A + Ld + \pi d^2.$$

Step 4 — interpretation. Ld is the area of the “strip” of width d built along the boundary; πd^2 is the area of the circular sectors filling the exterior angles. By Theorem 4.1, the exterior angles of a simple closed curve total 2π : the sectors therefore reassemble into a *full disc* of radius d , of area πd^2 . The formula thus also holds for polygons, where the sectors are concentrated at the vertices.

Step 5 — check on the circle. With K the circle of radius r : $L = 2\pi r$, $A = \pi r^2$, and K_d is the circle of radius $r + d$:

$$L_d = 2\pi(r + d) = 2\pi r + 2\pi d \checkmark, \quad A_d = \pi(r + d)^2 = \pi r^2 + 2\pi r d + \pi d^2 = A + Ld + \pi d^2 \checkmark.$$

Step 6 — check on the degenerate segment. For K a segment of length ℓ (degenerate convex body, $L = 2\ell$, $A = 0$), K_d is a “stadium”: two half-discs of radius d and a rectangle $\ell \times 2d$. $\text{Area} = 2\ell d + \pi d^2 = A + Ld + \pi d^2 \checkmark$; $\text{perimeter} = 2\ell + 2\pi d = L + 2\pi d \checkmark$.

Solution. S5. *Step 1 — translating central symmetry.* If K has a centre of symmetry, choose the origin there: then $K = -K$. By Definition 4.2,

$$h_{-K}(\theta) = \max_{x \in K} \langle -x, u(\theta) \rangle = \max_{x \in K} \langle x, -u(\theta) \rangle = h(\theta + \pi),$$

so $K = -K$ is equivalent to $h(\theta) = h(\theta + \pi)$ for every θ .

Step 2 — combining with constant width. The constant-width hypothesis is $h(\theta) + h(\theta + \pi) = w$. Substituting Step 1,

$$2h(\theta) = w \implies h(\theta) = \frac{w}{2} \quad \text{for every } \theta.$$

Step 3 — conclusion. A constant support function $= w/2$ is, by Step 5 of the proof of Theorem 4.2, that of the circle of radius $w/2$ centred at the origin. So K is a circle.

Step 4 — the Reuleaux triangle. It has constant width (Example 4.1) but is not a circle — its area is 89.73% of that of the circle of equal width — hence, by Steps 1–3, it has no centre of symmetry. $\checkmark$

Step 5 — reading. The result singles out the circle at the intersection of two properties neither of which suffices on its own: constant width (shared with infinitely many other figures, by Theorem 4.3) and central symmetry (shared with ellipses, rectangles, parallelograms). It is another instance of the chapter’s recurrent motif: the circle as an equality case, or as the intersection of families.

Step 6 — a technical consequence. No constant-width profile other than the circle can rotate about a fixed axis through a centre: there is no centre. That is why bodies of constant width can *roll* under a plank but cannot serve as a wheel on a hub.

4.10. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 4.11).

-
1. (Definition) A *vertex* of a regular plane curve is a point where
 - A) the curvature vanishes, $k(s) = 0$ B) the curvature is stationary, $k'(s) = 0$ C) the curve meets its own evolute D) the tangent is parallel to the axis of ordinates
 2. (Hypothesis) In the Umlaufsatz, the hypothesis that brings the rotation index to the value ± 1 , rather than to an arbitrary integer, is that the curve be
 - A) convex B) simple, that is free of self-intersections C) of class C^∞ D) of constant curvature sign
 3. (Symbolic computation) If $h(\theta) = 1 + \varepsilon \cos n\theta$ is a support function, the radius of curvature $\rho = h + h''$ equals
 - A) $1 + (n^2 - 1)\varepsilon \cos n\theta$ B) $1 - (n^2 - 1)\varepsilon \cos n\theta$ C) $1 - n^2\varepsilon \cos n\theta$ D) $1 + \varepsilon \cos n\theta$
 4. (Numerical computation) The Reuleaux triangle of width $w = 1$ has area and ratio to the circle of diameter 1 equal to
 - A) $A = \frac{\pi - \sqrt{3}}{2} = 0.704771$, that is 89.73% B) $A = \frac{\pi}{4} = 0.785398$, that is 100% C) $A = \pi - \sqrt{3} = 1.409542$, that is 179.47% D) $A = \frac{\sqrt{3}}{2} = 0.866025$, that is 110.27%
 5. (Proof step) In the proof of the global Gauss–Bonnet theorem the passage from the expression $2\pi V - \pi F$ to $2\pi\chi = 2\pi(V - E + F)$ is obtained
 - A) from the additivity of the integral over the triangles B) from the relation $3F = 2E$, that is from the fact that every edge belongs to two faces C) from the existence of a geodesic triangulation D) from the local form of Gauss–Bonnet on a single triangle
 6. (Error diagnosis) A student computes the area of a convex body from its support function as $A = \frac{1}{2} \int_0^{2\pi} h^2 d\theta$; for $h = 1 + 0.1 \cos 3\theta$ he obtains $A = 3.157301$ and concludes that this convex body has a larger area than the circle of equal perimeter $L = 2\pi$. The mistake is
 - A) having omitted the term $-h'^2$ in Cauchy's formula for the area, which instead gives $A = 3.015929 < \pi$ B) having miscomputed the perimeter, which is not 2π C) having integrated over $[0, 2\pi]$ instead of over $[0, \pi]$ D) having used an h which for $\varepsilon = 0.1$ is not the support function of any convex body
 7. (Transfer) A ground shaft is inspected by measuring its diameter with a vernier calliper in many directions: all the readings give 40.00 mm. Can one conclude that the cross section is circular?
 - A) yes, because constant width characterises the circle B) no: every profile of constant width gives the same reading, and for a Reuleaux triangle the circularity error would be 6.188 mm C) yes, provided the calliper is calibrated according to ISO 1101 D) no, but only because the calliper has finite resolution
 8. (Source and metacognition) The elementary proof of the four-vertex theorem in the *general* case (simple, not necessarily convex curves) by means of the minimal circumscribed circle, followed in Section 4.5.1, is found in
 - A) S. Mukhopadhyaya, Bulletin of the Calcutta Mathematical Society **1** (1909) B) R. Osserman, *The Four-or-More Vertex Theorem*, The American Mathematical Monthly **92**(5), 1985, pp. 332–337 C) H. Hopf, Compositio Mathematica **2** (1935) D) A. Hurwitz, Annales de l'École Normale Supérieure **19** (1902)
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4.10.1 Supplementary questionnaire

1. The total curvature of a simple closed curve is
A) π B) 2π C) shape-dependent D) 4π
2. The formula $\rho = h + h''$ relates the radius of curvature to
A) arc length B) the support function C) the area D) the width
3. Cauchy's formula for the perimeter of a convex body is
A) $L = \int h \, d\theta$ B) $L = \int h^2 d\theta$ C) $L = 2\pi h$ D) $L = \int h' \, d\theta$
4. Barbier's theorem states that a curve of constant width w has perimeter
A) $2\pi w$ B) πw C) $\pi w/2$ D) shape-dependent
5. Among convex bodies of constant width w , the minimum area is attained by
A) the circle B) the Reuleaux triangle C) the square D) none
6. Does a vernier calliper distinguish a cylinder from a Reuleaux profile?
A) yes B) no, it measures the same width in every direction C) only if calibrated
D) only for large diameters
7. The isoperimetric inequality states
A) $4\pi A \leq L^2$ B) $4\pi A \geq L^2$ C) $A \leq L$ D) $A = L^2/4\pi$
8. In the isoperimetric equality case the curve is
A) an ellipse B) a circle C) a regular polygon D) a constant-width curve
9. The four-vertex theorem states that every simple closed curve has at least
A) two points of stationary curvature B) four C) three D) none
10. The vertices of a curve correspond, on the evolute, to
A) regular points B) cusps C) inflections D) asymptotes
11. A convex body of constant width with a centre of symmetry is
A) a square B) a circle C) an ellipse D) a Reuleaux triangle
12. The global Gauss–Bonnet theorem states that, for a compact orientable surface without boundary, $\iint_M K \, dA$ equals
A) $2\pi\chi(M)$ B) 4π for every M C) 0 for every M D) the area of M
13. In a triangulation of a compact surface without boundary one always has
A) $2F = 3E$ B) $3F = 2E$ C) $F = E$ D) $V = E - F$
14. The Euler characteristic of a triangulation of the torus equals
A) 2 B) 0 C) -2 D) it depends on the chosen triangulation

BEST PRACTICE

Answer key.

1-B: Theorem 4.1; a topological invariant. **2-B:** Theorem 4.2(ii). **3-A:** Theorem 4.2(iii).
4-B: Theorem 4.3. **5-B:** Theorem 4.4(ii). **6-B:** Example 4.1 and Exercise N2. **7-A:**

Theorem 4.5. **8-B**: same theorem, Step 5. **9-B**: Theorem 4.6. **10-B**: Theorem 2.6(ii) and Corollary 4.4. **11-B**: Exercise S5. **12-A**: Theorem 4.8. **13-B**: Lemma 4.12. **14-B**: Example 4.3; by (P3) of Remark 4.9 the value does not depend on the triangulation.

4.11. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnosis.

1) B. *Rationale*: this is Definition 4.4: a vertex is a critical point of the curvature, $k'(s) = 0$, not a point where the curvature vanishes. *Distractors*: A confuses the vertex with the inflection ($k = 0$); C describes a property that vertices in general do not have; D depends on the frame of reference, whereas the notion of vertex is invariant under rigid motions. *Diagnosed error*: confusion between the vanishing of k and the stationarity of k . *Source*: Definition 4.4 and Theorem 4.6. *Prerequisite*: critical point of a differentiable function. *Remedial work*: compute k and k' for the ellipse (Remark 4.8) and mark the four vertices, checking that $k \neq 0$ at each of them.

2) B. *Rationale*: Theorem 4.1 holds for every closed regular curve with $\text{ind} \in \mathbb{Z}$ (Definition 4.1); it is the hypothesis of *simplicity* that forces $\text{ind} = \pm 1$, and the sign depends only on the orientation. *Distractors*: A is stronger than necessary (convex curves are a subset of the simple ones); C is not needed, C^2 suffices; D is false as a hypothesis and false as a conclusion: the curve of Exercise 4.6 is simple with k of varying sign and still has index 1. *Diagnosed error*: attributing to convexity what depends on the absence of self-intersections. *Source*: Theorem 4.1 (Hopf 1935). *Prerequisite*: continuous lifting of the tangent angle. *Remedial work*: compute the index of a lemniscate (which is closed and regular but not simple) and compare it with ± 1 .

3) B. *Rationale*: $h'' = -n^2 \varepsilon \cos n\theta$, hence $\rho = h + h'' = 1 - (n^2 - 1)\varepsilon \cos n\theta$: this is Step 1 of Exercise 4.8. *Distractors*: A gets the sign wrong and would suggest that the admissible amplitude grows with n ; C forgets the contribution of h and gives $\rho < 0$ already for $\varepsilon > 1/n^2$; D confuses ρ with h . *Diagnosed error*: applying $\rho = h + h''$ without differentiating twice, or getting the sign of $\cos''$ wrong. *Source*: Theorem 4.2(ii). *Prerequisite*: derivatives of $\cos n\theta$. *Remedial work*: check that for $\varepsilon = 0$ one recovers the unit circle, $\rho \equiv 1$.

4) A. *Rationale*: for $n = 3$ the formula $A_n = \frac{1}{2}(\pi - n \tan \frac{\pi}{2n})w^2$ of Exercise 4.5(b) gives $A_3 = \frac{1}{2}(\pi - 3 \tan \frac{\pi}{6}) = \frac{1}{2}(\pi - \sqrt{3}) = 0.704771$, against $\pi/4 = 0.785398$ for the circle: the ratio is 0.897342, that is 89.73%. *Distractors*: B is the area of the circle and contradicts Theorem 4.4(ii), which gives the Reuleaux triangle as the *strict* minimum; C forgets the factor $\frac{1}{2}$ and exceeds the area of the circle, contradicting Theorem 4.4(i); D is the area of the equilateral triangle of side 1, that is of the figure *before* replacing the sides with the arcs. *Diagnosed error*: using the formula without the plausibility check $A_{\text{Reuleaux}} < A_{\text{circle}}$. *Source*: Theorem 4.4 and Example 4.1; values recomputed in `verifica_cap02_cap04.py`, block [C4-2]. ✓ *Prerequisite*: $\tan(\pi/6) = 1/\sqrt{3}$. *Remedial work*: tabulate A_n/A_c for $n = 3, 5, 7$ and observe the monotonicity.

5) B. *Rationale*: Step 4 of the proof of Theorem 4.8 reaches $2\pi V - \pi F$; Step 5 uses $E = \frac{3}{2}F$, that is Lemma 4.12, to rewrite $2\pi(V - E + F)$ in the same form. *Distractors*: A is needed at Step 2, to split the integral over the triangles; C is (P2), the hypothesis that makes Step 1 possible; D is (P1), used at Step 3. *Diagnosed error*: failing to distinguish which of the four ingredients acts at which step. *Source*: Theorem 4.8, Steps 4–5, and Lemma 4.12. *Prerequisite*: double counting of a finite set. *Remedial work*: redo Steps 4–5 on the octahedron (Example 4.2) substituting the numbers line by line.

6) A. *Rationale*: Cauchy's formula for the area is $A = \frac{1}{2} \int_0^{2\pi} (h^2 - h'^2) d\theta$ (Theorem 4.2(iv)):

the term $-h'^2$ is not a detail, it is what makes the isoperimetric inequality true. With $h = 1 + 0.1 \cos 3\theta$ one has $\frac{1}{2} \int h^2 = 3.157301$ but $\frac{1}{2} \int (h^2 - h'^2) = 3.015929 < \pi = 3.141593$, in agreement with Step 5 of Exercise 4.8, which predicts $A = \pi - 4\pi\varepsilon^2 = \pi - 0.125664$. *Distractors*: B is false, Step 3 of the same exercise gives $L = 2\pi$ exactly; C would change the domain of a 2π -periodic function for no reason; D is false, the convexity threshold for $n = 3$ is $|\varepsilon| < 1/8 = 0.125$ and 0.1 respects it. *Diagnosed error*: memorising the area formula by analogy with the polar one $\frac{1}{2} \int r^2$, which holds for a different parametrisation. *Source*: Theorem 4.2(iv); numerical check in `verifica_cap02_cap04.py`, block [C4-1]. ✓ *Prerequisite*: $\int_0^{2\pi} \cos^2 = \int_0^{2\pi} \sin^2 = \pi$. *Remedial work*: compute the two expressions for $h \equiv 1$ (where they coincide) and for $h = 1 + \varepsilon \cos 3\theta$ (where they do not), and explain the difference.

7) **B. Rationale**: the calliper measures the width $w(\theta) = h(\theta) + h(\theta + \pi)$ (Definition 4.3), which is constant for *all* curves of constant width, not only for the circle; worked Exercise 4.10 computes for the Reuleaux triangle with $w = 40$ mm the radii $R = 23.094011$ mm and $r = 16.905989$ mm, that is a circularity error $R - r = 6.188022$ mm. *Distractors*: A reverses the implication: the circle has constant width, but it is not the only one; C attributes to calibration a problem of principle, and ISO 1101 defines circularity precisely as $R - r$, which the calliper does not measure; D mistakes a systematic error of method for an error of resolution. *Diagnosed error*: mistaking a necessary condition for a sufficient one. *Source*: Theorem 4.3, Example 4.1, Exercise 4.10; ISO 1101. *Prerequisite*: the distinction between width and diameter. *Remedial work*: draw the Reuleaux triangle between two parallel lines and rotate it, then between two concentric circles.

8) **B. Rationale**: the proof followed in Section 4.5.1 is Osserman's, based on the minimal circumscribed circle and on the contact set (Lemmas 4.6 and 4.7). *Distractors*: A is the first statement, but for convex curves *only*; C is the Umlaufsatz; D is the isoperimetric proof by Fourier series. *Diagnosed error*: citing the paper that gave the theorem its name instead of the one containing the proof used. *Source*: R. Osserman, The American Mathematical Monthly **92**(5), 1985, pp. 332–337, doi:10.2307/2323126. *Prerequisite*: being able to tell historical priority from the source of the proof. *Remedial work*: compare the years (1909, 1912, 1935, 1985, 2021) with the results listed in the *Sources* section.

BEST PRACTICE

Assigned exercises — answers.

4.5 (Reuleaux polygons). (a) The supporting lines orthogonal to any direction touch the profile at a vertex and on the opposite arc, of radius w centred at that vertex: the distance is w for every direction, hence the width is constant and by Theorem 4.3 $L = \pi w$, the same perimeter as the circle of diameter w . (b) The profile is the union of the regular n -gon and of n circular segments; adding the areas and simplifying one obtains $A_n = \frac{1}{2}(\pi - n \tan \frac{\pi}{2n})w^2$. (c) For $w = 1$: $A_3 = 0.704771$, $A_5 = 0.758497$, $A_7 = 0.771944$, against $A_{\text{circle}} = \pi/4 = 0.785398$; the ratios are 89.7342%, 96.5748% and 98.2870%. (d) $A_9 = 0.777325$ (98.9721%), $A_{51} = 0.785150$ (99.9684%), $A_{501} = 0.785396$ (99.9997%), $A_{5001} = 0.785398$: the sequence increases and tends to $\pi w^2/4$, because $n \tan \frac{\pi}{2n} \rightarrow \pi/2$. (e) All these convex bodies have the same perimeter πw but different areas, between the minimum $A_3 = \frac{\pi - \sqrt{3}}{2}w^2$ and the maximum $\pi w^2/4$: they are an explicit family filling almost the whole interval predicted by Theorem 4.4, and they show that both extremes are attained only at the ends of the family. Values recomputed in `verifica_cap02_cap04.py`, block [C4-2]. ✓

4.6 (three-lobed polar curve, $a = 0.3$). (a) $r(\theta) = 1 + 0.3 \cos 3\theta \in [0.7; 1.3]$ is always positive, so the curve does not pass through the origin; $\theta \mapsto \theta$ is injective on $[0, 2\pi)$ and different radii give different points, so the curve is simple; it is closed and regular because $r^2 + r'^2 > 0$.

(b) $A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \pi(1 + \frac{a^2}{2}) = 1.045\pi = 3.282964$, confirmed by Simpson quadrature with the same value to nine decimal places. (c) $L = \int_0^{2\pi} \sqrt{r^2 + r'^2} d\theta = 7.426640$, hence $L^2/(4\pi A) = 1.336930 > 1$ and $L^2 - 4\pi A = 13.900034 > 0$: the inequality of Theorem 4.5 holds and is widely strict, as it must be for a curve that is not a circle. (d) $\int k ds = 2\pi$ to seven decimal places, that is $\text{ind}(\gamma) = 1$ (Theorem 4.1): the curve is simple and traversed counterclockwise. (e) k' changes sign at six points, $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$, that is the three maxima and the three minima of r : six vertices, at least four as Theorem 4.6 requires. The curve is *not* convex: $k(0) = 2.366864 > 0$ but $k(\pi/3) = -4.081633 < 0$; the general form of Theorem 4.7 therefore applies. Checks in `verifica_cap02_cap04.py`, block [C4-3]. ✓

4.7 (Gauss–Bonnet on two triangulations). (a) Icosahedron: $V = 12, E = 30, F = 20$; $3F = 60 = 2E$ (Lemma 4.12) ✓, $\chi = 12 - 30 + 20 = 2$, $2\pi V - \pi F = 24\pi - 20\pi = 4\pi = 2\pi\chi$ and $\iint K dA = 4\pi$, the same value as for the octahedron. (b) The total angular excess is $\iint K dA = 4\pi$ shared among F triangles: for the icosahedron $4\pi/20 = \pi/5 = 0.628319$ rad $= 36^\circ$, for the octahedron $4\pi/8 = \pi/2 = 1.570796$ rad $= 90^\circ$. The finer the triangulation, the more the triangles are «almost Euclidean»: the average excess tends to zero like $1/F$, but its sum stays 4π . (c) Diagonalised 3×3 grid: $V = 9, E = 27, F = 18, 3F = 54 = 2E$ ✓, $\chi = 0$ ✓. Minimal triangulation (Császár polyhedron): $V = 7, E = 21, F = 14, 3F = 42 = 2E$ ✓, $\chi = 0$ ✓. (d) Because by (P3) of Remark 4.9 χ does not depend on the triangulation: from $3F = 2E$ it follows that $E = \frac{3}{2}F$ and hence $\chi = V - \frac{1}{2}F$, that is V and F are tied by $V = \chi + \frac{1}{2}F$; as F grows, V grows in the same proportion and the difference stays 0. (e) It exists: take the 5×10 grid of the fundamental square with each cell divided by a diagonal. Then $F = 2 \times 5 \times 10 = 100, E = 3 \times 50 = 150, V = 50$, and $\chi = 50 - 150 + 100 = 0$ ✓; the general constraints $E = \frac{3}{2}F$ and $V = \frac{1}{2}F$ are satisfied, and $F = 100$ is even, as it must be for $E = \frac{3}{2}F$ to be an integer. All the counts are repeated in `verifica_cap02_cap04.py`, block [C4-4]. ✓

4.12. Chapter formulary

Rotation index

$$\int_0^L k ds = 2\pi \text{ind}(\gamma); \quad \gamma \text{ closed simple} \Rightarrow \text{ind} = \pm 1$$

Support function

$$h(\theta) = \max_{x \in K} \langle x, u(\theta) \rangle, \quad \gamma(\theta) = hu + h'u', \quad \rho = h + h''$$

$$L = \int_0^{2\pi} h d\theta, \quad A = \frac{1}{2} \int_0^{2\pi} (h^2 - h'^2) d\theta$$

Width and Barbier's theorem

$$w(\theta) = h(\theta) + h(\theta + \pi); \quad w \text{ constant} \Rightarrow L = \pi w, \quad \rho(\theta) + \rho(\theta + \pi) = w$$

$$\text{Reuleaux : } A = \frac{\pi - \sqrt{3}}{2} w^2 \approx 0.70477 w^2; \quad \text{circle : } A = \frac{\pi w^2}{4} \approx 0.78540 w^2$$

Isoperimetric inequality

$$4\pi A \leq L^2, \quad \frac{L^2}{4\pi} - A = \frac{\pi}{2} \sum_{n \geq 2} (n^2 - 1)(a_n^2 + b_n^2)$$

Steiner's formulas (parallel curves)

$$L_d = L + 2\pi d, \quad A_d = A + Ld + \pi d^2$$

Convex bodies prescribed by a single harmonic

$$h(\theta) = 1 + \varepsilon \cos n\theta \implies \rho = 1 - (n^2 - 1)\varepsilon \cos n\theta, \quad |\varepsilon| < \frac{1}{n^2 - 1},$$

$$L = 2\pi, \quad A = \pi - \frac{(n^2 - 1)\pi\varepsilon^2}{2}, \quad \frac{L^2}{4\pi} - A = \frac{(n^2 - 1)\pi\varepsilon^2}{2}$$

Reuleaux polygons (n odd, width w)

$$L = \pi w, \quad A_n = \frac{1}{2} \left(\pi - n \tan \frac{\pi}{2n} \right) w^2, \quad A_3 = \frac{\pi - \sqrt{3}}{2} w^2, \quad A_n \nearrow \frac{\pi w^2}{4}$$

$$\text{Reuleaux triangle: } R = \frac{w}{\sqrt{3}}, \quad r = w - R, \quad R + r = w, \quad R - r = w \left(\frac{2}{\sqrt{3}} - 1 \right)$$

Triangulations and Euler characteristic

$$3F = 2E, \quad \chi = V - E + F = V - \frac{F}{2}, \quad \sum_{i=1}^F (\alpha_i + \beta_i + \gamma_i) = 2\pi V$$

Global Gauss–Bonnet

$$\iint_M K \, dA = 2\pi V - \pi F = 2\pi \chi(M); \quad \chi = 2 - 2g \quad (\text{closed orientable})$$

$$\chi(S^2) = 2 \implies \iint K \, dA = 4\pi; \quad \chi(T^2) = 0 \implies \iint K \, dA = 0$$

Angular defect (Descartes, discrete version)

$$\delta_p = 2\pi - \sum_{T \ni p} \vartheta_T(p), \quad \sum_p \delta_p = 2\pi V - \pi F = 2\pi \chi = 4\pi \quad (\text{convex polyhedron})$$

Sources and bibliography

- [1] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, Englewood Cliffs, 1976 (the author was at IMPA, Rio de Janeiro). Reference text for the Frenet formulas, the rotation index, the four-vertex theorem and the isoperimetric inequality; main source of Sections 2.2 and 4.1. South American source.
- [2] A. Gray, E. Abbena, S. Salamon, *Modern Differential Geometry of Curves and Surfaces with Mathematica*, 3rd ed., Chapman & Hall/CRC, 2006. Explicit formulas for evolutes, involutes, cycloids and their numerical checks.
- [3] H. Hopf, *Über die Drehung der Tangenten und Sehnen ebener Kurven*, *Compositio Mathematica*, vol. 2, 1935, pp. 50–62. Original proof of the *Umlaufsatz*, followed in Theorem 4.1.
- [4] A. Hurwitz, *Sur quelques applications géométriques des séries de Fourier*, *Annales scientifiques de l'École Normale Supérieure*, series 3, vol. 19, 1902, pp. 357–408. Proof of the isoperimetric inequality by Fourier series, followed in Theorem 4.5.
- [5] E. Barbier, *Note sur le problème de l'aiguille et le jeu du joint couvert*, *Journal de*

mathématiques pures et appliquées, series 2, vol. 5, 1860, pp. 273–286. Original statement of Theorem 4.3.

- [6] W. Blaschke, *Konvexe Bereiche gegebener konstanter Breite und kleinsten Inhalts*, Mathematische Annalen, vol. 76, 1915, pp. 504–513.
- [7] H. Lebesgue, *Sur le problème des isopérimètres et sur les domaines de largeur constante*, Bulletin de la Société Mathématique de France, Comptes Rendus des Séances, vol. 7, 1914, pp. 72–76. With the previous item, Theorem 4.4(ii).
- [8] S. Mukhopadhyaya, *New methods in the geometry of a plane arc*, Bulletin of the Calcutta Mathematical Society, vol. 1, 1909, pp. 31–37. First statement of the four-vertex theorem, for convex curves. Asian source.
- [9] A. Kneser, *Bemerkungen über die Anzahl der Extreme der Krümmung auf geschlossenen Kurven und über verwandte Fragen in einer nicht euklidischen Geometrie*, in *Festschrift Heinrich Weber*, Teubner, Leipzig, 1912, pp. 170–180. Extension of the four-vertex theorem to all simple closed curves.
- [10] R. Osserman, *The four-or-more vertex theorem*, The American Mathematical Monthly, vol. 92, no. 5, 1985, pp. 332–337. Elementary proof of the general case.
- [11] T. Bonnesen, W. Fenchel, *Theorie der konvexen Körper*, Springer, Berlin, 1934. Support function, Cauchy and Steiner formulas, constant width.
- [12] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge University Press, 2014. Modern and complete treatment of the theory of convex bodies.
- [13] I. M. Yaglom, V. G. Boltyanskii, *Convex Figures*, Holt, Rinehart and Winston, New York, 1961 (translated from the Russian edition, 1951). Curves of constant width and the Reuleaux triangle, treated elementarily.
- [14] H. Rademacher, O. Toeplitz, *The Enjoyment of Mathematics*, Princeton University Press, 1957. Expository chapters on isoperimetry and constant width.
- [15] ISO 1101, *Geometrical product specifications (GPS) — Geometrical tolerancing*, International Organization for Standardization, Geneva. Definition of the roundness tolerance cited in Example 4.1 and Exercise N2.
- [16] AIMS South Africa, *African Institute for Mathematical Sciences — teaching materials* — <https://aims.ac.za/>. African institutional source on geometry and analysis.
- [17] C. A. Barbieri, D. Miller-Cotto, S. N. Clerjuste, K. Chawla, *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*, Educational Psychology Review 35, art. 11, 2023 — <https://doi.org/10.1007/s10648-023-09745-1>.
- [18] H. L. Roediger III, J. D. Karpicke, *Test-Enhanced Learning*, Psychological Science 17(3), 2006 — <https://doi.org/10.1111/j.1467-9280.2006.01693.x>.
- [19] J. Pál, *Ein Minimumproblem für Ovale*, Mathematische Annalen **83** (1921), pp. 311–319. Contains the theorem on the regular hexagon circumscribed about a body of constant width, proved in full here in Lemma 4.1.
- [20] G. D. Chakerian, *Sets of Constant Width*, Pacific Journal of Mathematics **19**(1), 1966, pp. 13–21. The mixed-area proof of the Blaschke–Lebesgue theorem, followed in Section 4.3.1.

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- [21] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge University Press, 2014, §5.1. Minkowski mixed areas: existence, symmetry and monotonicity, that is the only two facts (M1) and (M2) imported in Remark 4.2.
- [22] W. Blaschke, *Konvexe Bereiche gegebener konstanter Breite und kleinsten Inhalts*, Mathematische Annalen **76** (1915), pp. 504–513; H. Lebesgue, *Sur le problème des isopérimètres et sur les domaines de largeur constante*, Bulletin de la Société Mathématique de France, Comptes Rendus des Séances **7** (1914), pp. 72–76. The two original, independent proofs, including uniqueness of the minimiser.
- [23] E. Harrell, *A direct proof of a theorem of Blaschke and Lebesgue*, Journal of Geometric Analysis **12**(1), 2002, pp. 81–88. An alternative variational proof, with a discussion of the equality case.
- [24] **A. Petrunin, S. Zamora Barrera**, *Moon in a puddle and the four-vertex theorem*, arXiv:2107.08455 [math.DG], 2021; published in The American Mathematical Monthly **129**(5), 2022, pp. 475–479, doi:10.1080/00029890.2022.2041355. It contains the *moon in a puddle* lemma and the osculating support theorem, recalled in Remark 4.6 as a second road to the general case, which Section 4.5.1 proves along Osserman’s route.
- [25] D. DeTurck, H. Gluck, D. Pomerleano, D. S. Vela-Vick, *The four vertex theorem and its converse*, Notices of the American Mathematical Society **54** (2007), no. 2, pp. 192–207 (arXiv:math/0609268). An exposition of Osserman’s steps and check values for the limaçon; the points that this exposition leaves implicit (the order of the contacts, the side of the disc at the last contact) are proved in Section 4.5.1.
- [26] J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, Upper Saddle River, 2000, §63; C. Thomassen, *The Jordan–Schönflies theorem and the classification of surfaces*, The American Mathematical Monthly **99** (1992), no. 2, pp. 116–130. The Jordan curve theorem, the only prerequisite imported in Section 4.5.1. Sources verified on 4 September 2026.
- [27] **R. Osserman**, *The Four-or-More Vertex Theorem*, The American Mathematical Monthly **92**(5), 1985, pp. 332–337, doi:10.2307/2323126. An elementary proof by means of the smallest enclosing circle, with a discussion of the exact number of vertices.
- [28] S. Mukhopadhyaya, *New methods in the geometry of a plane arc*, Bulletin of the Calcutta Mathematical Society **1**, 1909, pp. 31–37 (statement for convex curves); A. Kneser, *Bemerkungen über die Anzahl der Extreme der Krümmung auf geschlossenen Kurven und über verwandte Fragen in einer nicht euklidischen Geometrie*, in *Festschrift Heinrich Weber*, Teubner, Leipzig, 1912, pp. 170–180 (general case).
- [29] do Carmo, *Differential Geometry of Curves and Surfaces*, Dover 2016 (global theorems).
- [30] Hurwitz (1901), isoperimetric proof via Fourier series.
- [31] Hopf, *Umlaufsatz*.
- [32] Four-vertex theorem (Mukhopadhyaya 1909, Kneser).
- [33] Gray, *Modern Differential Geometry of Curves and Surfaces*.
- [34] Chern, *A Simple Intrinsic Proof of the Gauss–Bonnet Formula*.
- [35] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, revised and updated edition, Dover Publications, Mineola (New York), 2016, 512 pp., ISBN 978-0-486-80699-0; chapter 4 «The Intrinsic Geometry of Surfaces», §4-5 «The Gauss–Bonnet Theo-
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rem and Its Applications». Source of the local form of Gauss–Bonnet on a geodesic triangle (P1), of the existence of a geodesic triangulation (P2) and of the invariance of χ (P3), declared as imported in Remark 4.9. Verified on 5 September 2026 at <https://books.google.com/books?id=uXF6DQAAQBAJ>.

- [36] S.-S. Chern, *A Simple Intrinsic Proof of the Gauss–Bonnet Formula for Closed Riemannian Manifolds*, *Annals of Mathematics* **45** (1944), no. 4, pp. 747–752, doi:10.2307/1969302. Intrinsic proof of the Gauss–Bonnet formula, historically the first one independent of the embedding in $\mathbb{R}^3$; cited here as the source of Theorem 4.8, of which the text gives the proof by triangulation. Verified on 5 September 2026 at <https://doi.org/10.2307/1969302> and <https://www.jstor.org/stable/1969302>.
- [37] T. Radó, *Über den Begriff der Riemannschen Fläche*, *Acta Litterarum ac Scientiarum Regiae Universitatis Hungaricae Francisco-Josephinae, Sectio Scientiarum Mathematicarum*, vol. 2 (1926), pp. 101–121. Triangulability of every topological surface: it is the source of (P2) and (P3) of Remark 4.9. Verified on 5 September 2026 at <http://acta.bibl.u-szeged.hu> where volume 2 is dated 1926 (in the literature the work is often cited as «Radó 1925»).
- [38] Á. Császár, *A polyhedron without diagonals*, *Acta Scientiarum Mathematicarum* (Szeged) **13** (1949–1950), pp. 140–142. Realisation in $\mathbb{R}^3$ of the minimal triangulation of the torus with $V = 7$, $E = 21$, $F = 14$, used in Example 4.3 and in Exercise 4.7(c). Verified on 5 September 2026 at <https://mathworld.wolfram.com/CsaszarPolyhedron.html>.
- [39] P. J. Federico, *Descartes on Polyhedra: A Study of the «De Solidorum Elementis»*, *Sources in the History of Mathematics and Physical Sciences*, vol. 4, Springer-Verlag, New York, 1982, 144 pp., ISBN 0-387-90760-2, doi:10.1007/978-1-4612-5759-2. Edition and study of Descartes’s manuscript in which the total angular defect $= 4\pi$ appears, proved in Exercise 4.9. Verified on 5 September 2026 at <https://link.springer.com/book/10.1007/978-1-4612-5759-2>.

Isometries and the orthogonal group

A circle does not change if one rotates it about its own centre, and it does not change if one reflects it in any one of its diametral lines. This visual triviality conceals a precise and surprisingly strong statement: the transformations of the plane that preserve distances and fix a point form a group, the **orthogonal group** $O(2)$, and the circle is not merely a figure invariant under that group — it is, among all centred conics, the *only* invariant figure. The circle, in other words, is not defined only by its equation: it is defined by its symmetries.

The chapter follows this idea in four steps. First we prove that every isometry of the plane has the form $x \mapsto Ax + b$ with A orthogonal (and hence that isometries map circles to circles, with the same radius: result R1.5 of the Plan). Then we classify $O(2)$: only two families, rotations and reflections, and $SO(2)$ is the circle itself, both as a group and as a topological space. Then we prove, by a computation with quadratic forms, that the circle is the only centred conic invariant under all of $O(2)$. Finally we go up one dimension: Euler's rotation theorem says that every element of $SO(3)$ is a rotation about an axis, and that axis exists because $+1$ is always an eigenvalue. It is the bridge towards orientation in space, quaternions and robot kinematics.

INFO

Level and prerequisites. Level **A–C** (secondary school for the geometric part, first-year STEM for the classification of $O(2)$, undergraduate for the final section on $SO(3)$).

Internal prerequisites. Metric definition of the circle and Euclidean distance (Volume I, Chapter 2); scalar product, norm and orthogonality in the Cartesian plane (Volume I, Chapter 7); polar form of complex numbers and the exponential $e^{i\theta}$ (Volume II, Chapter 3); conics in canonical form (Volume II, Chapter 5, and Chapter 2 of this volume for plane curves).

External prerequisites, assumed and not reproved (a first course in linear algebra):

- the determinant of a square matrix, with $\det(AB) = \det A \cdot \det B$ and $\det A^\top = \det A$;
- the characteristic polynomial $p_A(\lambda) = \det(A - \lambda I)$; the eigenvalues are its roots;
- the fundamental theorem of algebra (a polynomial of degree n has n complex roots counted with multiplicity) and the fact that the complex roots of a polynomial with real coefficients come in conjugate pairs;
- a square matrix with linearly independent columns is invertible.

Everything else — the classification of $O(2)$, the spectrum, the uniqueness of the invariant conic, Euler's theorem — is proved here.

What one learns. How to recognise an isometry from its matrix; how to classify the elements of $O(2)$ and compose them without multiplying matrices; how to prove that the circle is characterised by its symmetries; how to find the axis and the angle of a rotation of space starting from its matrix.

Warning. Conventions of this chapter.

- Vectors of $\mathbb{R}^n$ are *columns*; $x^\top y = \langle x, y \rangle$ is the standard scalar product and $|x| = \sqrt{x^\top x}$ the Euclidean norm.
- I is the identity matrix (of order clear from the context) and $A^\top$ the transpose.

- Angles are in radians; rotations are *anticlockwise* (positive in the sense from $\mathbf{e}_1$ to $\mathbf{e}_2$).
- $C(P, r) = \{x : |x - P| = r\}$ is the circle of centre P and radius $r > 0$.
- $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ is the unit circle of the complex plane, identified with $S^1 \subset \mathbb{R}^2$ via $x + iy \leftrightarrow (x, y)$.
- “Centred conic” means: a level set $\{x \in \mathbb{R}^2 : x^\top Q x = 1\}$ with Q a real symmetric 2×2 matrix. The centre is therefore the origin by construction.

5.1. Isometries of the plane and orthogonal matrices

Definition 5.1: Isometry. An **isometry** of $\mathbb{R}^n$ is a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that preserves distances:

$$|f(x) - f(y)| = |x - y| \quad \text{for all } x, y \in \mathbb{R}^n.$$

Neither linearity nor continuity nor surjectivity is required: they are consequences, as we shall see.

Definition 5.2: Orthogonal matrices. A real matrix A of order n is **orthogonal** if $A^\top A = I$. The set

$$O(n) = \{A \in \mathbb{R}^{n \times n} : A^\top A = I\}$$

is the **orthogonal group**; the subset $SO(n) = \{A \in O(n) : \det A = +1\}$ is the **special orthogonal group**, or group of proper rotations.

Proposition 5.1: Three equivalent characterisations. For a real matrix A of order n the following are equivalent:

- (i) $A^\top A = I$ (that is, $A \in O(n)$);
- (ii) $|Ax| = |x|$ for every $x \in \mathbb{R}^n$;
- (iii) $(Ax)^\top (Ay) = x^\top y$ for all $x, y \in \mathbb{R}^n$;
- (iv) the columns of A form an orthonormal basis of $\mathbb{R}^n$.

Proof. **(i) $\Rightarrow$ (ii).** $|Ax|^2 = (Ax)^\top (Ax) = x^\top A^\top A x = x^\top I x = x^\top x = |x|^2$, and norms are ≥ 0 : hence $|Ax| = |x|$.

(ii) $\Rightarrow$ (iii). Apply (ii) to $x + y$:

$$|A(x + y)|^2 = |Ax + Ay|^2 = |Ax|^2 + 2(Ax)^\top (Ay) + |Ay|^2 = |x|^2 + 2(Ax)^\top (Ay) + |y|^2,$$

whereas $|A(x + y)|^2 = |x + y|^2 = |x|^2 + 2x^\top y + |y|^2$. Equating and cancelling $|x|^2 + |y|^2$ we obtain $2(Ax)^\top (Ay) = 2x^\top y$, that is, (iii).

(iii) $\Rightarrow$ (i). Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the vectors of the canonical basis. The (i, j) entry of $A^\top A$ is $(A\mathbf{e}_i)^\top (A\mathbf{e}_j)$, which by (iii) equals $\mathbf{e}_i^\top \mathbf{e}_j = \delta_{ij}$: hence $A^\top A = I$.

(i) $\Leftrightarrow$ (iv). The j -th column of A is $A_j = Ae_j$, and the (i, j) entry of $A^\top A$ is $A_i^\top A_j$. To say $A^\top A = I$ is therefore exactly to say $A_i^\top A_j = \delta_{ij}$, that is, that the n columns are pairwise orthogonal and of norm 1. An orthonormal system of n vectors in $\mathbb{R}^n$ is linearly independent (if $\sum_j c_j A_j = 0$, taking the scalar product with A_i gives $c_i = 0$), hence it is a basis. $\square$

Corollary 5.1: $O(n)$ is a group and $\det A = \pm 1$. $O(n)$ is a group under matrix multiplication, with $A^{-1} = A^\top$; for every $A \in O(n)$ one has $\det A = \pm 1$; and $SO(n)$ is a subgroup of $O(n)$.

Proof. From $A^\top A = I$ it follows that A is invertible (the columns are a basis by Proposition 5.1(iv)) and that $A^{-1} = A^\top$; then also $AA^\top = I$, that is, $(A^\top)^\top A^\top = I$: the transpose $A^{-1} = A^\top$ is orthogonal. If $A, B \in O(n)$ then $(AB)^\top(AB) = B^\top A^\top AB = B^\top B = I$, hence $AB \in O(n)$; and $I \in O(n)$. The determinant: $1 = \det I = \det(A^\top A) = \det A^\top \cdot \det A = (\det A)^2$, hence $\det A = \pm 1$. Finally, $\det(AB) = \det A \det B = 1$ and $\det(A^{-1}) = (\det A)^{-1} = 1$ for $A, B \in SO(n)$: $SO(n)$ is a subgroup. $\square$

Theorem 5.1: Structure of isometries: $f(x) = Ax + b$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an isometry. Then there exist a unique $A \in O(n)$ and a unique $b \in \mathbb{R}^n$ such that

$$f(x) = Ax + b \quad \text{for every } x \in \mathbb{R}^n.$$

In particular every isometry is affine, bijective, and its inverse is again an isometry.

Proof. **Step 1 — reduction to the case $f(0) = 0$.** Set $b = f(0)$ and $g(x) = f(x) - b$. Then $g(0) = 0$ and g is still an isometry, because $|g(x) - g(y)| = |f(x) - f(y)| = |x - y|$. It suffices to prove that g is linear with orthogonal matrix.

Step 2 — g preserves the norm. From $g(0) = 0$: $|g(x)| = |g(x) - g(0)| = |x - 0| = |x|$.

Step 3 — g preserves the scalar product. For all x, y ,

$$|x - y|^2 = |g(x) - g(y)|^2 = |g(x)|^2 - 2g(x)^\top g(y) + |g(y)|^2 = |x|^2 - 2g(x)^\top g(y) + |y|^2,$$

using Step 2; but $|x - y|^2 = |x|^2 - 2x^\top y + |y|^2$. Comparing, $g(x)^\top g(y) = x^\top y$ for all x, y .

Step 4 — g is linear. Let $u_i = g(e_i)$, $i = 1, \dots, n$. By Step 3, $u_i^\top u_j = e_i^\top e_j = \delta_{ij}$: the u_i form an orthonormal basis of $\mathbb{R}^n$. Fix $x \in \mathbb{R}^n$ and expand $g(x)$ in this basis: since the basis is orthonormal, $g(x) = \sum_i (u_i^\top g(x)) u_i$. But by Step 3

$$u_i^\top g(x) = g(e_i)^\top g(x) = e_i^\top x = x_i,$$

hence $g(x) = \sum_i x_i u_i = Ax$, where A is the matrix with columns $u_1, \dots, u_n$. The map $x \mapsto Ax$ is linear, and $A \in O(n)$ by Proposition 5.1(iv).

Step 5 — uniqueness and invertibility. If $Ax + b = A'x + b'$ for every x , setting $x = 0$ gives $b = b'$, and then $Ax = A'x$ for every x , that is, $A = A'$ (apply it to the vectors of the canonical basis). Since A is invertible, f is bijective with $f^{-1}(y) = A^{-1}(y - b) = A^\top y - A^\top b$, which is again of the form “orthogonal plus translation”, hence an isometry. $\square$

Theorem 5.2: Isometries map circles to circles (R1.5). Let f be an isometry of $\mathbb{R}^2$ and let $C(P, r)$ be a circle. Then

$$f(C(P, r)) = C(f(P), r) :$$

the image is a circle of equal radius, with centre the image of the centre. Moreover f preserves incidences: it maps lines to lines, chords to chords, tangents to tangents, and preserves the number of common points of two figures.

Proof. **Inclusion $\subseteq$.** If $|x - P| = r$, then $|f(x) - f(P)| = |x - P| = r$, hence $f(x) \in C(f(P), r)$.

Inclusion $\supseteq$. By Theorem 5.1 f is bijective and f^{-1} is an isometry. Let $y \in C(f(P), r)$ and $x = f^{-1}(y)$; then

$$|x - P| = |f^{-1}(y) - f^{-1}(f(P))| = |y - f(P)| = r,$$

hence $x \in C(P, r)$ and $y = f(x) \in f(C(P, r))$.

Incidences. By Theorem 5.1, $f(x) = Ax + b$ is affine and invertible: it maps the line $\{p + tv : t \in \mathbb{R}\}$ to the line $\{f(p) + tAv : t \in \mathbb{R}\}$ (and $Av \neq 0$ because A is invertible). Being

bijection, $f(X \cap Y) = f(X) \cap f(Y)$ for every pair of subsets, hence the number of common points is preserved; in particular a tangent line (a single common point with the circle) goes to a tangent line, a secant to a secant, an external line to an external one. The lengths of chords are preserved because f preserves distances. $\square$

Theorem 5.3: *$O(2)$ preserves every centred circle. For every $A \in O(2)$ and every $r > 0$, the map $x \mapsto Ax$ is a bijection of the circle $C(O, r) = \{x : |x| = r\}$ onto itself.*

Proof. This is the case $P = O$, $b = 0$ of Theorem 5.2: from $|Ax| = |x|$ (Proposition 5.1(ii)) we get $A(C(O, r)) \subseteq C(O, r)$, and since $A^{-1} = A^\top \in O(2)$ (Corollary 5.1) the same inclusion holds for A^{-1} , that is, $C(O, r) \subseteq A(C(O, r))$. $\square$

Remark 5.1: *The symmetry group of the circle is all of $O(2)$. Theorem 5.3 says that $O(2)$ is contained in the symmetry group of $C(O, r)$. The converse holds as well: an isometry f that maps $C(O, r)$ onto itself fixes the centre, because the centre is the only point equidistant from all the points of the circle (Volume I, Chapter 2), and therefore $f(0) = 0$: by Theorem 5.1 f has the form $x \mapsto Ax$ with $A \in O(2)$. The symmetry group of the circle is therefore exactly $O(2)$; it is an infinite, non-discrete group, unlike the finite dihedral group of a regular polygon. The fact is recorded, with the same statement, in the MIT algebra notes (RES.18-011, Lecture 14, Example 14.4), cited among the sources.*

Example 5.1: A concrete isometry, recognised from its matrix. Let

$$f(x) = Ax + b, \quad A = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

The columns of A are $\frac{1}{5}(3, 4)^\top$ and $\frac{1}{5}(-4, 3)^\top$: they have norm $\frac{1}{5}\sqrt{9+16} = \frac{5}{5} = 1$ and scalar product $\frac{1}{25}(3 \cdot (-4) + 4 \cdot 3) = 0$. Hence $A \in O(2)$ by Proposition 5.1(iv), and $\det A = \frac{1}{25}(9 + 16) = 1$: it is a rotation, by the angle $\theta = \arccos \frac{3}{5}$. *Independent check:* $\cos \theta = 0.6$ and $\sin \theta = 0.8$ give $\theta = 0.927295 \text{ rad} = 53.130102^\circ$, and $\cos^2 \theta + \sin^2 \theta = 0.36 + 0.64 = 1$. $\checkmark$ The image of the circle $C((2, 1)^\top, 3)$ is, by Theorem 5.2, the circle of radius 3 and centre

$$f\left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}\right) = \frac{1}{5} \begin{pmatrix} 3 \cdot 2 - 4 \cdot 1 \\ 4 \cdot 2 + 3 \cdot 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 \\ 11 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 0.2 \end{pmatrix}.$$

Independent check: the point $(5, 1)^\top$ lies on $C((2, 1)^\top, 3)$; its image is $\frac{1}{5}(15 - 4, 20 + 3)^\top + (1, -2)^\top = (2.2, 4.6)^\top - (0, 0)^\top = (2.2, 4.6)^\top$, and $|(3.2, 2.6)^\top - (1.4, 0.2)^\top| = |(1.8, 2.4)^\top| = \sqrt{3.24 + 5.76} = \sqrt{9} = 3$. $\checkmark$

5.2. The classification of $O(2)$

Theorem 5.4: *Classification of $O(2)$. Every $A \in O(2)$ is, for a unique $\theta \in [0, 2\pi)$, a rotation or a reflection:*

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (\det = +1), \quad S_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \quad (\det = -1).$$

Hence $O(2) = SO(2) \cup \{S_\theta\}$, with $SO(2) = \{R_\theta\}$.

Proof. Step 1 — the first column. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in O(2)$. By Proposition 5.1(iv) the columns are orthonormal:

$$a^2 + c^2 = 1, \quad b^2 + d^2 = 1, \quad ab + cd = 0.$$

From $a^2 + c^2 = 1$ the point (a, c) lies on the unit circle, hence there exists exactly one $\theta \in [0, 2\pi)$ with $(a, c) = (\cos \theta, \sin \theta)$: this is the parametrisation of the circle (Volume II, Chapter 1), surjective and injective on $[0, 2\pi)$.

Step 2 — the second column. The condition $ab + cd = 0$ says that the vector (b, d) is orthogonal to $(a, c) = (\cos \theta, \sin \theta)$. In the plane the orthogonal complement of a non-zero vector is a line: indeed $\{(b, d) : ab + cd = 0\}$ is the kernel of the non-zero 1×2 matrix $(a \ c)$, which has rank 1, hence a kernel of dimension 1. A basis of this kernel is $(-\sin \theta, \cos \theta)$, because $\cos \theta \cdot (-\sin \theta) + \sin \theta \cdot \cos \theta = 0$ and the vector is non-zero. Hence $(b, d) = \lambda(-\sin \theta, \cos \theta)$ for some $\lambda \in \mathbb{R}$; the condition $b^2 + d^2 = 1$ gives $\lambda^2(\sin^2 \theta + \cos^2 \theta) = \lambda^2 = 1$, that is, $\lambda = \pm 1$.

Step 3 — the two cases. For $\lambda = +1$, $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = R_\theta$, with $\det A = \cos^2 \theta + \sin^2 \theta = +1$. For $\lambda = -1$, $A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = S_\theta$, with $\det A = -\cos^2 \theta - \sin^2 \theta = -1$.

The two cases are mutually exclusive because $\det$ equals $+1$ in the first and -1 in the second, and θ is unique by Step 1.

Step 4 — every R_θ and every S_θ lies in $O(2)$. The columns of R_θ and of S_θ are orthonormal by construction, so the two families are indeed contained in $O(2)$: the classification is exact, not merely necessary. $\square$

Proposition 5.2: S_θ is the reflection in the line of inclination $\theta/2$. For every θ , the matrix S_θ fixes pointwise the line

$$\ell_\theta = \left\{ t \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix}^\top : t \in \mathbb{R} \right\}$$

and changes the sign of every vector orthogonal to ℓ_θ . Hence S_θ is the orthogonal reflection in the line through the origin making the angle $\theta/2$ with the x -axis. In particular $S_\theta^2 = I$.

Proof. Set $u = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2})^\top$ and $v = (-\sin \frac{\theta}{2}, \cos \frac{\theta}{2})^\top$; u, v is an orthonormal basis of $\mathbb{R}^2$. By the subtraction formulae,

$$\begin{aligned} S_\theta u &= \begin{pmatrix} \cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2} \\ \sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} \cos(\theta - \frac{\theta}{2}) \\ \sin(\theta - \frac{\theta}{2}) \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix} = u, \\ S_\theta v &= \begin{pmatrix} -\cos \theta \sin \frac{\theta}{2} + \sin \theta \cos \frac{\theta}{2} \\ -\sin \theta \sin \frac{\theta}{2} - \cos \theta \cos \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} \sin(\theta - \frac{\theta}{2}) \\ -\cos(\theta - \frac{\theta}{2}) \end{pmatrix} = \begin{pmatrix} \sin \frac{\theta}{2} \\ -\cos \frac{\theta}{2} \end{pmatrix} = -v. \end{aligned}$$

On a basis S_θ therefore acts as the identity on u and as $-\text{id}$ on v : it is the orthogonal reflection in the line $\mathbb{R}u = \ell_\theta$. Consequently $S_\theta^2 u = u$ and $S_\theta^2 v = v$, that is, $S_\theta^2 = I$. *Independent numerical check:* for $\theta = 1.3$ one obtains $|S_\theta u - u| = 1.1 \cdot 10^{-16}$ and $|S_\theta v + v| = 1.1 \cdot 10^{-16}$, within machine error. $\checkmark$ $\square$

Proposition 5.3: Multiplication table of $O(2)$. For all α, β (with the indices understood modulo 2π):

$$R_\alpha R_\beta = R_{\alpha+\beta}, \quad R_\alpha S_\beta = S_{\alpha+\beta}, \quad S_\alpha R_\beta = S_{\alpha-\beta}, \quad S_\alpha S_\beta = R_{\alpha-\beta}.$$

Consequently: $SO(2)$ is abelian; $O(2)$ is not; $R_\alpha^{-1} = R_{-\alpha}$ and $S_\alpha^{-1} = S_\alpha$; the product of two reflections is the rotation by an angle equal to twice the angle between the two axes, and every rotation is a product of two reflections.

Proof. All four identities are obtained by multiplying the matrices and using the addition formulae $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$ and $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ (Volume I, Chapter 8). In full, the fourth case:

$$\begin{aligned} S_\alpha S_\beta &= \begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & \sin \beta \\ \sin \beta & -\cos \beta \end{pmatrix} = \begin{pmatrix} \cos \alpha \cos \beta + \sin \alpha \sin \beta & \cos \alpha \sin \beta - \sin \alpha \cos \beta \\ \sin \alpha \cos \beta - \cos \alpha \sin \beta & \sin \alpha \sin \beta + \cos \alpha \cos \beta \end{pmatrix} \\ &= \begin{pmatrix} \cos(\alpha - \beta) & -\sin(\alpha - \beta) \\ \sin(\alpha - \beta) & \cos(\alpha - \beta) \end{pmatrix} = R_{\alpha - \beta}. \end{aligned}$$

The other three are carried out in the same way. From the first identity, $R_\alpha R_\beta = R_{\alpha + \beta} = R_{\beta + \alpha} = R_\beta R_\alpha$: $SO(2)$ is abelian. From the second and the third, $R_\alpha S_0 = S_\alpha$ whereas $S_0 R_\alpha = S_{-\alpha}$: the two products differ for $\alpha \notin \{0, \pi\}$, hence $O(2)$ is not abelian. From $R_\alpha R_{-\alpha} = R_0 = I$ it follows that $R_\alpha^{-1} = R_{-\alpha}$; from $S_\alpha S_\alpha = R_0 = I$ it follows that $S_\alpha^{-1} = S_\alpha$ (in agreement with Proposition 5.2).

The factor two. By Proposition 5.2 the axes of S_α and S_β have inclinations $\alpha/2$ and $\beta/2$, hence they make with each other the angle $\varphi = \frac{\alpha - \beta}{2}$; and $S_\alpha S_\beta = R_{\alpha - \beta} = R_{2\varphi}$. Conversely, every R_γ is written as $S_\gamma S_0$: every rotation is a product of two reflections. *Independent numerical check:* with $\alpha = 1.1$, $\beta = 0.4$ one obtains $\max |S_\alpha S_\beta - R_{\alpha - \beta}| = 1.1 \cdot 10^{-16}$, $\max |R_\alpha S_\beta - S_{\alpha + \beta}| = 1.4 \cdot 10^{-16}$ and $\max |S_\alpha R_\beta - S_{\alpha - \beta}| = 1.1 \cdot 10^{-16}$. $\checkmark$ $\square$

Corollary 5.2: $SO(2)$ has index two in $O(2)$. $SO(2)$ is a normal subgroup of $O(2)$ of index 2, and $O(2)/SO(2) \cong \{\pm 1\}$. The two cosets are $SO(2)$ (the rotations) and $S_0 SO(2)$ (the reflections).

Proof. The map $\det : O(2) \rightarrow \{\pm 1\}$ is a group homomorphism ($\det(AB) = \det A \det B$), surjective because $\det R_0 = +1$ and $\det S_0 = -1$ (Theorem 5.4), with kernel $SO(2)$ by definition. The kernel of a homomorphism is normal, and by the first isomorphism theorem $O(2)/SO(2) \cong \{\pm 1\}$, a group of order 2: the index is 2. The non-trivial coset consists of the elements of determinant -1 , that is, of the reflections; and $S_0 R_\beta = S_{-\beta}$ (Proposition 5.3) shows that these are exactly those of the form $S_0 A$ with $A \in SO(2)$. $\square$

Theorem 5.5: Spectrum of R_θ and of S_θ . For $\theta \in (0, 2\pi)$, $\theta \neq \pi$, the rotation R_θ has complex conjugate eigenvalues $e^{\pm i\theta}$ and no real eigenvector: no direction of the plane is fixed. For $\theta = 0$ one has $R_0 = I$ (eigenvalue 1, double) and for $\theta = \pi$ one has $R_\pi = -I$ (eigenvalue -1 , double). Every reflection S_θ , on the contrary, has real eigenvalues $+1$ and -1 , simple, with orthogonal eigenspaces.

Proof. Rotations. $\text{tr } R_\theta = 2 \cos \theta$ and $\det R_\theta = 1$, hence the characteristic polynomial is

$$p(\lambda) = \lambda^2 - 2 \cos \theta \lambda + 1,$$

with discriminant $\Delta = 4 \cos^2 \theta - 4 = -4 \sin^2 \theta \leq 0$. If $\sin \theta \neq 0$ (that is, $\theta \neq 0, \pi$) then $\Delta < 0$: the roots are not real and equal $\cos \theta \pm i \sin \theta = e^{\pm i\theta}$ (one checks directly: $(e^{i\theta})^2 - 2 \cos \theta e^{i\theta} + 1 = e^{i\theta}(e^{i\theta} + e^{-i\theta} - 2 \cos \theta) = 0$). A real eigenvector would correspond to a real eigenvalue, which does not exist: hence there are no fixed directions. If $\theta = 0$, $R_0 = I$ and every vector is an eigenvector of eigenvalue 1; if $\theta = \pi$, $R_\pi = -I$ and every vector is an eigenvector of eigenvalue -1 (in both cases $e^{\pm i\theta}$ is real and coincides with that value: the formula remains true).

Reflections. $\operatorname{tr} S_\theta = \cos \theta - \cos \theta = 0$ and $\det S_\theta = -1$, hence $p(\lambda) = \lambda^2 - 1$, with roots ± 1 , simple. The eigenvectors have already been exhibited in Proposition 5.2: u for $+1$ and v for -1 , orthogonal to each other. This also proves that S_θ is diagonalisable over $\mathbb{R}$: in the basis $\{u, v\}$ one has $S_\theta = \operatorname{diag}(1, -1)$. $\square$

Remark 5.2: Geometric interpretation of the spectrum. The absence of real eigenvectors of R_θ ($\theta \neq 0, \pi$) is the algebraic translation of an elementary fact: a non-trivial rotation of the plane fixes no direction. The reflection S_θ , on the contrary, fixes the axis (eigenvalue $+1$) and reverses the orthogonal direction (eigenvalue -1): its determinant -1 is precisely the product $(+1) \cdot (-1)$ and measures the reversal of orientation. In Chapter 28 the same computation in dimension three will distinguish proper and improper rotations.

Theorem 5.6: $SO(2) \cong \mathbb{T}$ as groups and as topological spaces. The map

$$\Phi : \mathbb{T} \longrightarrow SO(2), \quad \Phi(e^{i\theta}) = R_\theta,$$

is well defined, bijective, and is a group isomorphism between $(\mathbb{T}, \cdot)$ and $(SO(2), \cdot)$. Moreover Φ is a homeomorphism, if one endows $SO(2) \subset \mathbb{R}^{2 \times 2} \cong \mathbb{R}^4$ with the induced topology: $SO(2)$ and $\mathbb{T}$ are the same topological group. In particular $SO(2)$ is abelian, compact and connected.

Proof. Well defined. If $e^{i\theta} = e^{i\theta'}$, then $\theta - \theta' \in 2\pi\mathbb{Z}$, and since $\cos$ and $\sin$ are 2π -periodic, $R_\theta = R_{\theta'}$: the value does not depend on the chosen representative.

Homomorphism. By Proposition 5.3, $\Phi(e^{i\theta}e^{i\varphi}) = \Phi(e^{i(\theta+\varphi)}) = R_{\theta+\varphi} = R_\theta R_\varphi = \Phi(e^{i\theta})\Phi(e^{i\varphi})$.

Bijectivity. Surjectivity is Theorem 5.4 (every element of $SO(2)$ is an R_θ). Injectivity: if $R_\theta = I$ then $\cos \theta = 1$ and $\sin \theta = 0$, that is, $\theta \in 2\pi\mathbb{Z}$, that is, $e^{i\theta} = 1$; the kernel is trivial and a homomorphism with trivial kernel is injective.

Homeomorphism. Φ is continuous because the four entries of R_θ are continuous functions of $(\cos \theta, \sin \theta)$: indeed, writing $z = x + iy \in \mathbb{T}$,

$$\Phi(z) = \begin{pmatrix} x & -y \\ y & x \end{pmatrix},$$

which is even the restriction to $\mathbb{T}$ of a linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^4$. Its inverse $\Phi^{-1}\left(\begin{smallmatrix} a & -c \\ c & a \end{smallmatrix}\right) = a + ic$ is the restriction of a linear projection, hence continuous. Two continuous, mutually inverse maps form a homeomorphism.

Consequences. $\mathbb{T}$ is abelian (complex multiplication is commutative), compact and connected (Chapter 12, Theorem 12.1); these properties are transported through the homeomorphism-isomorphism Φ . $\square$

Remark 5.3: Where this identification continues. The isomorphism Φ of Theorem 5.6 is the doorway to two later chapters, and we do not go further here. The topology of $\mathbb{T}$ — compactness, connectedness, the covering $t \mapsto e^{it}$, the winding number and $\pi_1(\mathbb{T}) \cong \mathbb{Z}$ — is the content of Chapter 12. The structure of topological group and of Lie group of $\mathbb{T}$ — the invariant measure $d\theta/2\pi$, the continuous characters $z \mapsto z^n$, orthogonality and duality with $\mathbb{Z}$ — is the content of Chapter 13. Here only the algebraic fact matters: the group of rotations of the plane is the circle, and hence every statement about the circle as a group is automatically a statement about rotations, and vice versa.

5.3. Quadratic forms and the uniqueness of the circle

Definition 5.3: Quadratic form and centred conic. Let $Q = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ be a real symmetric matrix. The associated **quadratic form** is

$$q(x) = x^\top Q x = a x_1^2 + 2b x_1 x_2 + c x_2^2,$$

and the **centred conic** of matrix Q is the level set $\mathcal{C}_Q = \{x \in \mathbb{R}^2 : q(x) = 1\}$. A linear map A leaves $\mathcal{C}_Q$ **invariant** if $A(\mathcal{C}_Q) = \mathcal{C}_Q$.

Theorem 5.7: Invariance of the form of the circle. The equation $x^\top x = r^2$ is invariant under every $A \in O(2)$: if $w = Av$ and $v^\top v = r^2$, then $w^\top w = r^2$. Equivalently, $A^\top I A = I$ for every $A \in O(2)$.

Proof. $w^\top w = (Av)^\top (Av) = v^\top A^\top A v = v^\top I v = v^\top v = r^2$, using $A^\top A = I$. The equivalent form is the same identity read as an equality of matrices. $\square$

Lemma 5.1: $SO(2)$ acts transitively on every centred circle. Let $x, y \in \mathbb{R}^2$ with $|x| = |y| = r > 0$. Then there exists θ with $R_\theta x = y$. In particular, the orbit of a point $x \neq 0$ under $SO(2)$ is the whole circle $C(O, |x|)$.

Proof. Since $|x/r| = 1$, there exists α with $x = r(\cos \alpha, \sin \alpha)^\top$; similarly $y = r(\cos \beta, \sin \beta)^\top$. Then, by the addition formulae,

$$R_{\beta-\alpha} x = r \begin{pmatrix} \cos(\beta - \alpha) \cos \alpha - \sin(\beta - \alpha) \sin \alpha \\ \sin(\beta - \alpha) \cos \alpha + \cos(\beta - \alpha) \sin \alpha \end{pmatrix} = r \begin{pmatrix} \cos \beta \\ \sin \beta \end{pmatrix} = y.$$

Conversely, every $R_\theta x$ has norm $|x|$ (Theorem 5.3): the orbit is contained in the circle, and by what has just been proved it exhausts it. $\square$

Theorem 5.8: The circle is the only $O(2)$ -invariant centred conic. Let Q be a real symmetric 2×2 matrix and let $\mathcal{C}_Q = \{x : x^\top Q x = 1\}$ be non-empty. If $A(\mathcal{C}_Q) = \mathcal{C}_Q$ for every $A \in O(2)$, then

$$Q = \lambda I \quad \text{with } \lambda > 0, \quad \mathcal{C}_Q = C(O, 1/\sqrt{\lambda}) :$$

$\mathcal{C}_Q$ is a circle centred at the origin. Conversely, every centred circle is $O(2)$ -invariant (Theorem 5.7).

Proof. Step 1 — invariance produces a whole circle inside $\mathcal{C}_Q$. Let $x_0 \in \mathcal{C}_Q$ and $r = |x_0|$. We have $r > 0$, because $0^\top Q 0 = 0 \neq 1$. By hypothesis $R_\theta x_0 \in \mathcal{C}_Q$ for every θ , and by Lemma 5.1 the points $R_\theta x_0$ describe the whole circle $C(O, r)$. Hence

$$q(x) = x^\top Q x = 1 \quad \text{for every } x \text{ with } |x| = r. \quad (*)$$

Step 2 — what (*) imposes. Write $Q = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ and parametrise $x = r(\cos t, \sin t)^\top$. Then

$$q(x) = r^2(a \cos^2 t + 2b \cos t \sin t + c \sin^2 t).$$

Using $\cos^2 t = \frac{1+\cos 2t}{2}$, $\sin^2 t = \frac{1-\cos 2t}{2}$ and $2 \cos t \sin t = \sin 2t$ (Volume I, Chapter 8):

$$q(x) = r^2 \left(\underbrace{\frac{a+c}{2}}_{=: \mu} + \underbrace{\frac{a-c}{2}}_{=: \nu} \cos 2t + \underbrace{b}_{=: \rho} \sin 2t \right).$$

Independent check: for $a = 2$, $b = -0.7$, $c = 1.3$, $r = 1.9$ and seven equispaced values of t in $[0, 2\pi]$, the two expressions agree with maximum error $1.8 \cdot 10^{-15}$. ✓

Condition (*) says $r^2(\mu + \nu \cos 2t + \rho \sin 2t) = 1$ for *every* t . We evaluate it at three points:

$$t = 0 : r^2(\mu + \nu) = 1; \quad t = \frac{\pi}{2} : r^2(\mu - \nu) = 1; \quad t = \frac{\pi}{4} : r^2(\mu + \rho) = 1.$$

Subtracting the first two: $2r^2\nu = 0$, and since $r > 0$ we get $\nu = 0$, hence $a = c$ and $r^2\mu = 1$. Substituting in the third: $r^2\rho = 0$, hence $\rho = b = 0$.

Step 3 — conclusion. From $a = c$ and $b = 0$ it follows that $Q = aI$ with $a = \mu = 1/r^2 > 0$. Then

$$\mathcal{C}_Q = \{x : a|x|^2 = 1\} = \{x : |x| = 1/\sqrt{a}\} = C(O, r),$$

which is the circle of centre the origin and radius $r = 1/\sqrt{a}$. The converse is Theorem 5.7. □

Remark 5.4: Why the proof does not use the spectral theorem. The short route would be: “invariance is equivalent to $A^\top QA = Q$ for every A , hence Q commutes with all rotations, hence every direction consists of eigenvectors and Q is a multiple of the identity”. That route requires two extra things: that the equality of the sets $\mathcal{C}_Q = \mathcal{C}_{A^\top QA}$ imply the equality of the matrices (true for non-degenerate conics, but to be proved) and the spectral theorem for symmetric matrices. The proof given above avoids both: it uses only the parametrisation of the circle and the independence of the functions $1, \cos 2t, \sin 2t$, verified by hand at three values of t . It is the simplest case of Fourier analysis: a function that is constant on the circle has no harmonics of order 2.

Corollary 5.3: The symmetry group of a non-circular ellipse is finite. Let $E = \{x : x_1^2/\alpha^2 + x_2^2/\beta^2 = 1\}$ with $\alpha > \beta > 0$. Then $\{A \in O(2) : A(E) = E\}$ is the group of order 4 $\{I, R_\pi, S_0, S_\pi\}$, isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (the Klein group): the identity, the half-turn rotation and the reflections in the two axes.

Proof. Here $Q = \text{diag}(1/\alpha^2, 1/\beta^2)$ is not a multiple of the identity because $\alpha \neq \beta$; by Theorem 5.8, E cannot be invariant under *all* of $O(2)$. The four matrices listed leave E invariant: I and $R_\pi = -I$ because $q(-x) = q(x)$; $S_0 = \text{diag}(1, -1)$ and $S_\pi = \text{diag}(-1, 1)$ because q depends only on x_1^2 and x_2^2 . Conversely, let $A \in O(2)$ with $A(E) = E$. The points of E of maximal norm are $(\pm\alpha, 0)^\top$ and those of minimal norm $(0, \pm\beta)^\top$ (because on E one has $|x|^2 = x_1^2 + x_2^2$ with $x_1^2/\alpha^2 + x_2^2/\beta^2 = 1$, that is, $|x|^2 = \beta^2 + x_1^2(1 - \beta^2/\alpha^2)$, increasing in $x_1^2 \in [0, \alpha^2]$). Since A preserves the norm, it permutes these two sets: $A(\alpha, 0)^\top = (\pm\alpha, 0)^\top$ and $A(0, \beta)^\top = (0, \pm\beta)^\top$, that is, $Ae_1 = \pm e_1$ and $Ae_2 = \pm e_2$. The four combinations of signs give exactly $I, S_0, S_\pi, -I = R_\pi$. Each has square I , hence the group is the Klein group. □

Example 5.2: The ellipse rotated by a quarter turn. Let $\alpha = 2$, $\beta = 1$, that is, $Q = \text{diag}(1/4, 1)$. Under $R_{\pi/2}$ the matrix of the conic becomes $R_{\pi/2}^\top QR_{\pi/2}$. The computation by hand:

$$R_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_{\pi/2}^\top QR_{\pi/2} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1/4 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1/4 \end{pmatrix},$$

that is, the ellipse $x_1^2 + x_2^2/4 = 1$: the axes have been exchanged and the conic is *different*

from the previous one. *Independent numerical check:* the computation carried out in double precision returns $\text{diag}(1.0; 0.25)$ with vanishing off-diagonal part. ✓ In accordance with Theorem 5.8, only for $\alpha = \beta$ do the two matrices coincide: and then the conic is a circle.

5.4. From $O(2)$ to $SO(3)$: Euler's rotation theorem

Definition 5.4: $O(3)$ and $SO(3)$. $O(3) = \{A \in \mathbb{R}^{3 \times 3} : A^\top A = I\}$ is the group of linear isometries of space, and $SO(3) = \{A \in O(3) : \det A = +1\}$ the subgroup of **proper rotations**. The elements of $O(3) \setminus SO(3)$ (determinant -1) are the **improper** isometries: they reverse orientation.

Lemma 5.2: Every $A \in SO(3)$ has $+1$ among its eigenvalues. For every $A \in SO(3)$ one has $\det(A - I) = 0$; hence there exists $u \in \mathbb{R}^3$, $u \neq 0$, with $Au = u$.

Proof. We use only $A^\top A = I$ and $\det A = 1$:

$$\det(A - I) \stackrel{(1)}{=} \det A \cdot \det(A - I) \stackrel{(2)}{=} \det(A^\top(A - I))^\top$$

— let us work it out calmly. Since $\det A = 1$,

$$\det(A - I) = \det(A - I) \cdot \det A^\top = \det((A - I)A^\top) = \det(AA^\top - A^\top) = \det(I - A^\top).$$

On the other hand $\det(I - A^\top) = \det((I - A)^\top) = \det(I - A)$, and $\det(I - A) = \det(-(A - I)) = (-1)^3 \det(A - I) = -\det(A - I)$, because the matrix has order 3. Putting things together:

$$\det(A - I) = -\det(A - I) \implies 2 \det(A - I) = 0 \implies \det(A - I) = 0.$$

A vanishing determinant means that the matrix $A - I$ is singular, that is, that the homogeneous system $(A - I)u = 0$ has a solution $u \neq 0$: that solution satisfies $Au = u$. *Independent numerical check:* on 2000 randomly generated matrices of $SO(3)$ (QR factorisation of Gaussian matrices, with a sign correction to impose $\det = +1$; seed 20260905) the maximum of $|\det(A - I)|$ is $2.6 \cdot 10^{-15}$, that is, zero within machine error. ✓ $\square$

Theorem 5.9: Euler's rotation theorem. Let $A \in SO(3)$. Then there exists a unit vector $u \in \mathbb{R}^3$ with $Au = u$ (the **axis**), the plane $u^\perp$ is invariant under A , and the restriction of A to $u^\perp$ is a rotation by an angle θ : in a suitable orthonormal basis $\{u, w_1, w_2\}$,

$$A \longleftrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}.$$

Every rotation of space is therefore a rotation about an axis.

Proof. Step 1 — the axis. By Lemma 5.2 there exists $u_0 \neq 0$ with $Au_0 = u_0$; setting $u = u_0/|u_0|$, one still has $Au = u$ by linearity, and $|u| = 1$.

Step 2 — the orthogonal plane is invariant. Let $v \perp u$, that is, $u^\top v = 0$. Then, using Proposition 5.1(iii) and $Au = u$,

$$u^\top(Av) = (Au)^\top(Av) = u^\top v = 0,$$

hence $Av \perp u$: $A(u^\perp) \subseteq u^\perp$. Since A is invertible and $u^\perp$ is finite dimensional, $A(u^\perp) = u^\perp$.

Step 3 — block form. Complete u to an orthonormal basis $\{u, w_1, w_2\}$ of $\mathbb{R}^3$ (the vectors w_1, w_2 are an orthonormal basis of the plane $u^\perp$). Let P be the matrix with columns u, w_1, w_2 : $P \in O(3)$ by Proposition 5.1(iv). Consider $M = P^\top A P$. Its first column is $P^\top A u = P^\top u = (1, 0, 0)^\top$, because the rows of $P^\top$ are $u^\top, w_1^\top, w_2^\top$ and u is orthonormal to w_1, w_2 . Its first row has entries $u^\top A w_j$ for $j = 1, 2$: since $A^{-1}u = A^\top u = u$ (from $Au = u$ it follows that $u = A^\top A u = A^\top u$),

$$u^\top A w_j = (A^\top u)^\top w_j = u^\top w_j = 0.$$

Hence

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & B & \end{pmatrix}, \quad B \in \mathbb{R}^{2 \times 2}.$$

Step 4 — the 2×2 block is a plane rotation. $M = P^\top A P$ is orthogonal (a product of orthogonal matrices, Corollary 5.1) and $\det M = \det P^\top \det A \det P = \det A = 1$ (because $\det P^\top \det P = \det(P^\top P) = 1$). From $M^\top M = I$, read blockwise, it follows that $B^\top B = I$, that is, $B \in O(2)$; and expanding $\det M$ along the first row one obtains $\det M = 1 \cdot \det B$, hence $\det B = 1$, that is, $B \in SO(2)$. By Theorem 5.4, $B = R_\theta$ for some $\theta \in [0, 2\pi)$. This is exactly the statement: in the basis $\{u, w_1, w_2\}$ the matrix of A is the one written above. $\square$

Corollary 5.4: Rotation angle from the trace. For every $A \in SO(3)$ with axis u and angle θ one has

$$\operatorname{tr} A = 1 + 2 \cos \theta, \quad \text{that is,} \quad \cos \theta = \frac{\operatorname{tr} A - 1}{2}.$$

In particular $\operatorname{tr} A \in [-1, 3]$, with $\operatorname{tr} A = 3$ if and only if $A = I$.

Proof. The trace is invariant under conjugation: $\operatorname{tr}(P^\top A P) = \operatorname{tr}(A P P^\top) = \operatorname{tr} A$ (one uses $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ and $P P^\top = I$). From the form of Theorem 5.9, $\operatorname{tr} A = \operatorname{tr} M = 1 + \cos \theta + \cos \theta = 1 + 2 \cos \theta$. Since $\cos \theta \in [-1, 1]$, $\operatorname{tr} A \in [-1, 3]$; and $\operatorname{tr} A = 3$ forces $\cos \theta = 1$, that is, $\theta = 0$, that is, $M = I$ and hence $A = P M P^\top = I$. $\square$

Corollary 5.5: The product of two rotations is a rotation. If $A, B \in SO(3)$, then $AB \in SO(3)$: by Theorem 5.9 AB too is a rotation about an axis, in general different from those of A and of B .

Proof. $SO(3)$ is a group (Corollary 5.1), hence $AB \in SO(3)$; the rest is Theorem 5.9 applied to AB . That the axis is in general different is shown by Remark 5.5. $\square$

Remark 5.5: Non-commutativity of $SO(3)$, with an exact counterexample. Unlike $SO(2)$ (Proposition 5.3), the group $SO(3)$ is not commutative. Take the two quarter-turn rotations about the z - and x -axes:

$$A = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The direct computation gives

$$AB = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix},$$

which are different. Both have trace 0, hence by Corollary 5.4 they are rotations by $\theta = 120^\circ$; but the axes are different: AB fixes $(1, 1, 1)^\top/\sqrt{3}$, whereas BA fixes $(1, -1, 1)^\top/\sqrt{3}$ (one checks directly: $AB(1, 1, 1)^\top = (1, 1, 1)^\top$ and $BA(1, -1, 1)^\top = (1, -1, 1)^\top$). Independent numerical check: the computation in double precision confirms $\text{tr}(AB) = \text{tr}(BA) = 1.2 \cdot 10^{-16}$ (that is, 0), $\theta = 120.000000^\circ$ for both, and axes $\mp(0.577350; 0.577350; 0.577350)$ and $\mp(0.577350; -0.577350; 0.577350)$, with $1/\sqrt{3} = 0.577350$. ✓

It is the fact that makes programming the orientation of a six-axis robot delicate: the order in which two rotations are applied changes the result. Euler angles, gimbal lock and quaternions — the tools with which industry lives with this non-commutativity — are the subject of Chapter 28.

5.4.1 Planar orientation of the TCP

For a robot operating in the plane (vertical axis of rotation, as in many palletising stations) the orientation of the tool — the *TCP*, *tool centre point* — is an element of $SO(2)$: it can be encoded as a 2×2 matrix, as a unit complex number or as a single angle θ . The three encodings are equivalent by Theorem 5.6, but they do not have the same cost: composing two rotations costs one complex multiplication (four real multiplications and two real additions) against the eight multiplications and four additions of the 2×2 matrix product; and in the angular encoding a single addition suffices, with the sole care of the reduction modulo 2π . It is a computational advantage exploited in controllers.

The jump to a free three-dimensional orientation, on the other hand, requires $SO(3)$, which is not commutative (Remark 5.5), with the complications that will be seen: Euler angles, *gimbal lock*, quaternions.

5.4.2 The rotation matrix inside the controller

Whatever the representation exposed to the user (angles or quaternions), *all* robot controllers — **KUKA**, **ABB**, **FANUC**, **Yaskawa** — work internally with 4×4 homogeneous matrices containing a 3×3 rotation matrix belonging to $SO(3)$, that is, by Theorem 5.9, a rotation about an axis. For a vertical axis of rotation (palletising, rotary tables driven by **Siemens/Rockwell** PLCs), the orientation reduces to an element of $SO(2)$: a single angle, with the commutative arithmetic of Proposition 5.3 and of Theorem 5.6. The difference between the two cases is not one of convenience but of algebraic structure: it is the difference between an abelian group and one that is not.

5.5. Assigned exercises

Answers are in the Teacher's key (Section 5.8).

Exercise 5.1: The ellipse is not $O(2)$ -invariant. *Aim:* to use Theorem 5.8 and Corollary 5.3 on a concrete case. *Estimated time:* 25 minutes. *Type:* symbolic/geometric.

Consider the ellipse $x^2/\alpha^2 + y^2/\beta^2 = 1$ with $\alpha \neq \beta$.

- Write down the matrix Q of the conic and explain, citing Theorem 5.8, why the ellipse cannot be $O(2)$ -invariant.
- Compute $R_{\pi/2}^\top Q R_{\pi/2}$ and describe the image conic.
- Determine for which θ one has $R_\theta^\top Q R_\theta = Q$.
- List all the elements of $O(2)$ that leave the ellipse invariant and recognise the structure of the group they form.

Exercise 5.2: A rotation of space with zero trace. *Aim:* to pass from the matrix to axis and angle by Euler's theorem. *Estimated time:* 30 minutes. *Type:* numerical/symbolic. Let $A \in SO(3)$ with $\text{tr } A = 0$.

- Determine the rotation angle θ .
- Check that the cyclic permutation of the axes $\mathbf{e}_1 \mapsto \mathbf{e}_2 \mapsto \mathbf{e}_3 \mapsto \mathbf{e}_1$ is such a rotation: write down its matrix, check that it belongs to $SO(3)$ and determine its axis.
- Verify the axis by explicitly solving the system $(A - I)u = 0$.
- Compute A^3 and explain the result in terms of θ .

Exercise 5.3: Recognising and composing elements of $O(2)$. *Aim:* to apply the classification and the multiplication table without multiplying matrices. *Estimated time:* 35 minutes. *Type:* symbolic with a numerical check.

Consider

$$M_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad M_2 = \frac{1}{13} \begin{pmatrix} 5 & 12 \\ 12 & -5 \end{pmatrix}, \quad M_3 = \frac{1}{2} \begin{pmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{pmatrix}.$$

- Check that the three matrices belong to $O(2)$ and decide, for each, whether it is a rotation or a reflection, determining the parameter θ of Theorem 5.4.
- For the reflections, give the equation of the axis (Proposition 5.2).
- Compute M_1M_2 , M_2M_1 and $M_2M_3M_2$ using *only* Proposition 5.3, and check the results with the matrix product.
- Decide whether the subgroup of $O(2)$ generated by M_1 is finite, and if so what its order is. The same question for the subgroup generated by M_2 and for the one generated by M_1 and M_2 together.

5.6. Worked exercises

Exercise 5.4: The product of two reflections is a rotation. *Type:* symbolic/proof-based. Prove that the product of two plane reflections $S_\alpha S_\beta$ is a rotation, determine its angle, and interpret the result geometrically. Deduce that $O(2)$ is generated by reflections alone, and that every rotation can be written as a product of two reflections in infinitely many ways.

Solution. Strategy. Multiply the two matrices and recognise the subtraction formulae; then translate the result in terms of axes, using Proposition 5.2.

Step 1 — the computation. This is the fourth identity of Proposition 5.3, carried out there in full:

$$S_\alpha S_\beta = R_{\alpha-\beta}.$$

That the result is a rotation can also be seen without computations, from the determinant: $\det(S_\alpha S_\beta) = (-1)(-1) = +1$, hence the product lies in $SO(2)$, which by Theorem 5.4 consists of rotations only.

Step 2 — the angle, in terms of axes. By Proposition 5.2, the axis of S_α has inclination $\alpha/2$ and that of S_β inclination $\beta/2$; the angle between the two axes is $\varphi = \frac{\alpha-\beta}{2}$. Hence

$$S_\alpha S_\beta = R_{2\varphi} :$$

the product of two reflections is the rotation by twice the angle between the two axes, in the sense that goes from the second axis to the first.

Step 3 — generation. Every rotation R_γ is written $R_\gamma = S_\gamma S_0$ (the case $\alpha = \gamma, \beta = 0$), and every reflection is already a reflection: hence the reflections generate all of $O(2)$. The expression is not unique: for every β one has $R_\gamma = S_{\gamma+\beta} S_\beta$, that is, one may choose the first axis *freely* and the other is then determined. Geometrically: only the angle *between* the two axes matters, not their absolute position.

Independent check. (i) Numerical: with $\alpha = 1.1$ and $\beta = 0.4$, $\max |S_\alpha S_\beta - R_{0.7}| = 1.1 \cdot 10^{-16}$. ✓ (ii) On an exact case: $S_\pi = \text{diag}(-1, 1)$ (reflection in the y -axis, inclination $\pi/2$) and $S_0 = \text{diag}(1, -1)$ (reflection in the x -axis, inclination 0); the product is $\text{diag}(-1, -1) = -I = R_\pi$, and indeed $\alpha - \beta = \pi$, with angle between the axes $\pi/2$ and rotation angle twice that, π . ✓ (iii) Consistency with Corollary 5.2: the product of two elements of the non-trivial coset lies in the subgroup, as it must in a quotient of order 2.

Interpretation. The “factor two” is the same one that governs the central angle and the inscribed angle (Volume I, Chapter 4) and will reappear, in dimension three, in the double covering $SU(2) \rightarrow SO(3)$ (Chapter 28): a unit quaternion that rotates by θ is built with the angle $\theta/2$. It is the algebraic trace of the fact that reflections are “square roots” of rotations.

Typical errors. (1) Concluding that the rotation angle is $\alpha - \beta$ *between the axes*: no, $\alpha - \beta$ is the difference of the *parameters*, and the angle between the axes is half of it. (2) Forgetting that $S_\alpha S_\beta \neq S_\beta S_\alpha$ in general (one gets $R_{\beta-\alpha}$, the inverse rotation). (3) Believing that the product of three reflections is a rotation: it has determinant -1 , hence it is a reflection.

Extension. The same statement holds in $\mathbb{R}^n$: by the n -reflections theorem (D. Klain, notes cited among the sources, Theorem 3), every matrix of $O(n)$ is a product of at most n reflections, and $\det A = (-1)^k$ if A is a product of k reflections. In $\mathbb{R}^3$ this gives: the rotations are the products of 0 or 2 reflections, the improper isometries the products of 1 or 3.

Exercise 5.5: The eigenvectors of $S_{\pi/4}$. *Type:* numerical/symbolic. Find the eigenvalues and eigenvectors of $S_{\pi/4}$, identify its axis of reflection and check consistency with Proposition 5.2 and with Theorem 5.5.

Solution. Strategy. Write down the matrix, compute the characteristic polynomial, solve the two linear systems, and compare the axis found with the theoretical prediction ($\theta/2 = \pi/8$).

Step 1 — the matrix. With $\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$,

$$S_{\pi/4} = \begin{pmatrix} \cos \frac{\pi}{4} & \sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & -\cos \frac{\pi}{4} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Step 2 — the spectrum. $\text{tr} = 0$ and $\det = \frac{1}{2}(-1 - 1) = -1$, hence $p(\lambda) = \lambda^2 - 1$ and the eigenvalues are ± 1 , as predicted by Theorem 5.5.

Step 3 — the eigenvectors. For $\lambda = +1$ one solves $(S_{\pi/4} - I)v = 0$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 - \sqrt{2} & 1 \\ 1 & -1 - \sqrt{2} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies (1 - \sqrt{2})v_1 + v_2 = 0 \implies v_2 = (\sqrt{2} - 1)v_1.$$

An eigenvector is therefore $v^{(+)} = (1, \sqrt{2} - 1)^\top$. For $\lambda = -1$, the same reduction gives $(1 + \sqrt{2})v_1 + v_2 = 0$, that is, $v^{(-)} = (1, -1 - \sqrt{2})^\top$. The two are orthogonal: $1 \cdot 1 + (\sqrt{2} - 1)(-1 - \sqrt{2}) = 1 - (\sqrt{2} - 1)(\sqrt{2} + 1) = 1 - (2 - 1) = 0$. ✓

Step 4 — the axis. The axis is the line $\mathbb{R}v^{(+)}$, of slope $\sqrt{2} - 1 = 0.414214$, that is, of inclination $\arctan(\sqrt{2} - 1) = 0.392699 \text{ rad} = 22.5^\circ$. This is exactly $\theta/2 = \frac{\pi/4}{2} = \frac{\pi}{8} = 22.5^\circ$, as Proposition 5.2 predicts. ✓

Independent check. (i) The proposition gives directly the eigenvector $u = (\cos \frac{\pi}{8}, \sin \frac{\pi}{8})^\top = (0.923880; 0.382683)^\top$; the ratio of the components is $0.382683/0.923880 = 0.414214 = \sqrt{2} - 1$: the same line. ✓ (ii) Numerical: the diagonalisation in double precision of $S_{\pi/4}$ returns eigenvalues $\{-1, +1\}$ and axis angle 22.5° . ✓

Interpretation. The reflection $S_{\pi/4}$ exchanges the coordinate axes (it maps $(1, 0)$ to $(1, 1)/\sqrt{2}$ and vice versa) and its axis is the bisector of that exchange: it is the case $\theta = \pi/4$ of the general remark “the parameter θ is twice the inclination of the axis”. The factor two, again.

Typical errors. (1) Writing that the axis has inclination $\pi/4$ (that is, θ instead of $\theta/2$): it is the most frequent error, and it is unmasked at once by checking that $S_{\pi/4}(1, 1)^\top = \frac{1}{\sqrt{2}}(2, 0)^\top \neq (1, 1)^\top$: the bisector of the first quadrant is *not* fixed. (2) Taking $(1, 1)^\top$ as an eigenvector of $S_{\pi/4}$, confusing it with $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is $S_{\pi/2}$ and really does have its axis at 45° . (3) Forgetting the normalisation and concluding that $|v^{(+)}| = 1$.

Extension. For a general θ the eigenvectors are $(\cos \frac{\theta}{2}, \sin \frac{\theta}{2})^\top$ and $(-\sin \frac{\theta}{2}, \cos \frac{\theta}{2})^\top$: the change-of-basis matrix is $R_{\theta/2}$, and indeed $R_{\theta/2}^\top S_\theta R_{\theta/2} = S_0 = \text{diag}(1, -1)$. All plane reflections are therefore conjugate to one another in $O(2)$: up to a change of orthonormal basis, there is only one reflection.

Exercise 5.6: Axis and angle of a rotation of space. *Type:* numerical/computational with a proof-based part. Consider

$$A = \frac{1}{3} \begin{pmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{pmatrix}.$$

- (a) Check that $A \in SO(3)$.
- (b) Determine the rotation angle.
- (c) Determine the axis.
- (d) Check the result with an independent verification.

Solution. Strategy. Check $A^\top A = I$ and $\det A = 1$ using the columns alone; obtain the angle from Corollary 5.4; find the axis by solving $(A - I)u = 0$, guaranteed solvable by Lemma 5.2; finally check by reconstructing the action of A on a vector orthogonal to the axis.

Step 1 — $A \in SO(3)$. The columns are $\frac{1}{3}(2, 2, -1)^\top$, $\frac{1}{3}(-1, 2, 2)^\top$, $\frac{1}{3}(2, -1, 2)^\top$. Each has norm

$$\frac{1}{3}\sqrt{4 + 4 + 1} = \frac{1}{3}\sqrt{9} = 1,$$

and the pairwise scalar products equal $\frac{1}{9}(2 \cdot (-1) + 2 \cdot 2 + (-1) \cdot 2) = \frac{1}{9}(-2 + 4 - 2) = 0$, $\frac{1}{9}(2 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 2) = \frac{1}{9}(4 - 2 - 2) = 0$, $\frac{1}{9}((-1) \cdot 2 + 2 \cdot (-1) + 2 \cdot 2) = \frac{1}{9}(-2 - 2 + 4) = 0$. By Proposition 5.1(iv), $A \in O(3)$. The determinant, expanded along the first row:

$$27 \det A = 2(2 \cdot 2 - (-1) \cdot 2) - (-1)(2 \cdot 2 - (-1)(-1)) + 2(2 \cdot 2 - 2 \cdot (-1)) = 2 \cdot 6 + 1 \cdot 3 + 2 \cdot 6 = 27,$$

hence $\det A = +1$ and $A \in SO(3)$.

Step 2 — **the angle.** $\text{tr } A = \frac{1}{3}(2 + 2 + 2) = 2$, hence by Corollary 5.4

$$\cos \theta = \frac{2 - 1}{2} = \frac{1}{2} \implies \theta = 60^\circ.$$

Step 3 — the axis. Solve $(A - I)u = 0$, that is, $3(A - I)u = 0$:

$$\begin{pmatrix} -1 & -1 & 2 \\ 2 & -1 & -1 \\ -1 & 2 & -1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = 0.$$

Adding the first and the second row: $u_1 - 2u_2 + u_3 = 0$, which is the opposite of the third (the three rows are therefore dependent, as they must be). From the first row $u_3 = \frac{u_1 + u_2}{2}$; substituting in the second, $2u_1 - u_2 - \frac{u_1 + u_2}{2} = 0$, that is, $4u_1 - 2u_2 - u_1 - u_2 = 0$, that is, $3u_1 = 3u_2$, that is, $u_1 = u_2$. Then $u_3 = u_1$: the axis is the line generated by $(1, 1, 1)^\top$, and the unit vector is

$$u = \frac{1}{\sqrt{3}}(1, 1, 1)^\top = (0.577350; 0.577350; 0.577350)^\top.$$

Step 4 — independent check. (i) One verifies directly that $Au = u$: the first component of $3A(1, 1, 1)^\top$ is $2 - 1 + 2 = 3$; the second $2 + 2 - 1 = 3$; the third $-1 + 2 + 2 = 3$. Hence $A(1, 1, 1)^\top = (1, 1, 1)^\top$. ✓ (ii) We check the angle on a vector orthogonal to the axis. Let $w = (1, -1, 0)^\top$, with $u^\top w = 0$ and $|w|^2 = 2$. Then $3Aw = (2 + 1, 2 - 2, -1 - 2)^\top = (3, 0, -3)^\top$, that is, $Aw = (1, 0, -1)^\top$, of norm $\sqrt{2} = |w|$. ✓ The angle between w and Aw satisfies

$$\cos \angle(w, Aw) = \frac{w^\top Aw}{|w| |Aw|} = \frac{1 \cdot 1 + (-1) \cdot 0 + 0 \cdot (-1)}{2} = \frac{1}{2},$$

that is, 60° : it agrees with the value of Step 2. ✓ (iii) Since $\theta = 60^\circ$, one must have $A^6 = I$; and indeed $\text{tr}(A^2) = 1 + 2 \cos 120^\circ = 0$ and $\text{tr}(A^3) = 1 + 2 \cos 180^\circ = -1$, values confirmed by the double-precision computation. ✓

Interpretation. The matrix A rotates by 60° about the main diagonal of the cube. The powers of A generate a cyclic subgroup of order 6 of $SO(3)$; it is the copy inside $SO(3)$ of the group $\mathbb{Z}/6\mathbb{Z}$, and it shows concretely how discrete symmetries nest inside the continuous group of rotations.

Typical errors. (1) Using $\cos \theta = \text{tr } A/3$ (the average of the diagonal entries): the correct formula is $\cos \theta = (\text{tr } A - 1)/2$, because one of the three directions is fixed and contributes 1, not $\cos \theta$. (2) Looking for the axis among the columns of A : the axis is an eigenvector, not a column. (3) Forgetting to normalise u and then using it in formulae that require $|u| = 1$. (4) Deducing the *sense* of the rotation from the trace alone: the trace gives $\cos \theta$, hence θ up to sign; the sense requires the orientation of the axis (see Chapter 28).

Extension. The same procedure — the trace for the angle, the kernel of $A - I$ for the axis — is the standard algorithm in robot controllers for passing from the rotation matrix to the “axis-angle” representation. The skew-symmetric part $\frac{1}{2}(A - A^\top)$ moreover provides the

oriented axis: here $\frac{1}{2}(A - A^\top) = \frac{1}{6} \begin{pmatrix} 0 & -3 & 3 \\ 3 & 0 & -3 \\ -3 & 3 & 0 \end{pmatrix}$, whose associated vector is $\frac{1}{2}(1, 1, 1)^\top$,

parallel to u and with the same sense, of norm $\sin \theta = \sin 60^\circ = \frac{\sqrt{3}}{2}$: indeed $|\frac{1}{2}(1, 1, 1)^\top| = \frac{\sqrt{3}}{2}$. ✓

5.7. Self-assessment quiz

1. (Definition) A real matrix A of order n is *orthogonal* when

A) $\det A = 1$ B) $A^\top A = I$ C) A is symmetric D) A is invertible

2. (Hypothesis) In Theorem 5.8 the hypothesis that makes the conclusion “ Q is a multiple of

- the identity” true is
- A) that Q be invertible B) that $\mathcal{C}_Q$ be invariant under *all* $A \in O(2)$ C) that Q be positive definite D) that $\mathcal{C}_Q$ be bounded
3. (Symbolic computation) The product $S_\alpha S_\beta$ of two plane reflections equals
- A) $S_{\alpha+\beta}$ B) $R_{\alpha-\beta}$ C) $R_{(\alpha-\beta)/2}$ D) I
4. (Numerical computation) If $A \in SO(3)$ and $\text{tr } A = 2$, the rotation angle is
- A) 30° B) 45° C) 60° D) 120°
5. (Proof step) In the proof of Euler’s theorem (Lemma 5.2), the decisive step is
- A) $\det(A - I) = \det(I - A^\top) = -\det(A - I)$ B) A is symmetric C) $\text{tr } A = 1 + 2 \cos \theta$
D) A has three real eigenvalues
6. (Error diagnosis) A student claims: “ $S_{\pi/4}$ is the reflection in the bisector of the first quadrant, because $\pi/4 = 45^\circ$ ”. The mistake is
- A) $S_{\pi/4}$ is not orthogonal B) the axis has inclination $\theta/2$, that is, 22.5° C) $S_{\pi/4}$ is a rotation D) reflections have no axis
7. (Transfer) The symmetry group in $O(2)$ of an ellipse with distinct semi-axes has order
- A) 1 B) 2 C) 4 D) infinite
8. (Source and metacognition) The classification of 2×2 orthogonal matrices into rotations and reflections, and the n -reflections theorem from which $\det A = (-1)^k$ follows, are found in
- A) Klain, *Lecture notes on orthogonal matrices*, 92.222 UMass Lowell, §2–3 B) Hatcher, *Algebraic Topology*, §1.1 C) Needham, *Visual Complex Analysis*, ch. 3 D) Munkres, *Topology*, §53

5.8. Teacher’s key

BEST PRACTICE

Quiz — answers, rationale and diagnosis.

- 1) **B.** *Rationale:* this is Definition 5.2; the equivalent conditions are in Proposition 5.1. *Distractors:* A holds only for $SO(n)$ and there exist non-orthogonal matrices of determinant 1 (for instance $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$); C is false (R_θ is not symmetric for $\theta \neq 0, \pi$); D is implied but much weaker. *Diagnosed error:* confusing a consequence with the definition. *Source:* Klain, notes cited, §1, Theorem 1. *Prerequisite:* matrix product and transpose. *Remedial work:* check by hand that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has determinant 1 but does not preserve norms.
- 2) **B.** *Rationale:* the proof uses invariance under *all* rotations in order to fill a whole circle (Lemma 5.1); with a single A the conclusion is false (the ellipse is invariant under R_π). *Distractors:* A and C are consequences, not hypotheses (once $Q = \lambda I$ with $\lambda > 0$ has been proved); D is true for ellipses but is not enough. *Diagnosed error:* not distinguishing “invariant under one element” from “invariant under the group”. *Source:* Theorem 5.8. *Prerequisite:* action of a group on a set. *Remedial work:* redo Step 1 of the proof with $A = R_\pi$ only, and see where it gets stuck.
- 3) **B.** *Rationale:* Proposition 5.3, fourth identity, carried out in full. *Distractors:* A has determinant -1 and cannot be a product of two reflections; C confuses the parameter with the angle between the axes (which is $(\alpha - \beta)/2$: the rotation angle is twice it); D holds

only for $\alpha = \beta$. *Diagnosed error*: the “factor two” applied the wrong way round. *Source*: Worked exercise 5.4. *Prerequisite*: addition formulae. *Remedial work*: draw two axes at 30° to each other and follow a point through the two reflections.

4) **C.** *Rationale*: Corollary 5.4: $\cos \theta = (2 - 1)/2 = 1/2$, hence $\theta = 60^\circ$. *Distractors*: A and B correspond to traces $1 + 2 \cos 30^\circ = 2.732$ and $1 + 2 \cos 45^\circ = 2.414$; D corresponds to trace 0. *Diagnosed error*: using $\cos \theta = \text{tr } A/3$. *Source*: Corollary 5.4. *Prerequisite*: invariance of the trace under conjugation. *Remedial work*: redo Worked exercise 5.6.

5) **A.** *Rationale*: it is the chain $\det(A - I) = \det((A - I)A^\top) = \det(I - A^\top) = \det(I - A) = -\det(A - I)$, which uses only $\det A = 1$, $AA^\top = I$ and the odd dimension. *Distractors*: B is false (a rotation is not symmetric); C is a consequence of the theorem, not a step of the proof of the existence of the axis; D is false in general (the other two eigenvalues are $e^{\pm i\theta}$). *Diagnosed error*: confusing the result with the mechanism. *Source*: Lemma 5.2. *Prerequisite*: properties of the determinant. *Remedial work*: repeat the computation in dimension 2 and observe that there it fails (the sign $(-1)^2 = +1$ produces no contradiction): this is why a plane rotation has no axis.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (continued).

6) **B.** *Rationale*: by Proposition 5.2 the axis of S_θ has inclination $\theta/2$; for $\theta = \pi/4$ this is 22.5° . Immediate counter-check: $S_{\pi/4}(1, 1)^\top = \frac{1}{\sqrt{2}}(2, 0)^\top \neq (1, 1)^\top$. *Distractors*: A is false ($S_\theta \in O(2)$ by construction); C is false ($\det S_{\pi/4} = -1$); D is false (the axis is the eigenspace of $+1$). *Diagnosed error*: identifying the parameter θ with the inclination of the axis. *Source*: Worked exercise 5.5. *Prerequisite*: eigenvectors. *Remedial work*: compute $S_\theta(\cos \frac{\theta}{2}, \sin \frac{\theta}{2})^\top$ for three values of θ .

7) **C.** *Rationale*: Corollary 5.3: the group is $\{I, R_\pi, S_0, S_\pi\}$, of order 4 (the Klein group). *Distractors*: A would forget even R_π ; B counts only the two reflections, or only $\{I, R_\pi\}$; D is the case of the circle, excluded because the semi-axes are distinct (Theorem 5.8). *Diagnosed error*: counting the “visible” symmetries and forgetting the identity or the half-turn rotation. *Source*: Corollary 5.3. *Prerequisite*: order of a finite group. *Remedial work*: list the symmetries of a non-square rectangle and compare.

8) **A.** *Rationale*: D. Klain’s notes (92.222, *Linear Algebra II*, UMass Lowell) contain in §3 the 2×2 classification in the form used here and in §2 the n -reflections theorem (Theorem 3) with Corollary 5 ($\det A = (-1)^k$). *Distractors*: B is the source of Chapter 12; C is the source of Chapter 6; D is general topology. *Diagnosed error*: citing the bibliography of the volume rather than the source of the statement. *Source*: see the section “Sources and bibliography”. *Prerequisite*: being able to read a reference with section and number. *Remedial work*: locate the two statements in the notes and compare their proofs with those given here.

BEST PRACTICE

Assigned exercise 5.1 — answers. (a) $Q = \text{diag}(1/\alpha^2, 1/\beta^2)$, which is not a multiple of the identity because $\alpha \neq \beta$; by Theorem 5.8 the only centred conic invariant under all of $O(2)$ is the circle, hence the ellipse is not. (b) $R_{\pi/2}^\top Q R_{\pi/2} = \text{diag}(1/\beta^2, 1/\alpha^2)$: the image conic is $x^2/\beta^2 + y^2/\alpha^2 = 1$, the ellipse with the semi-axes exchanged, different from the original one (Example 5.2). (c) A direct computation gives, setting $\delta = \frac{1}{\alpha^2} - \frac{1}{\beta^2} \neq 0$,

$$R_\theta^\top Q R_\theta - Q = \frac{\delta}{2} (\cos 2\theta - 1) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \frac{\delta}{2} \sin 2\theta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which vanishes if and only if $\cos 2\theta = 1$ and $\sin 2\theta = 0$, that is, $\theta \in \{0, \pi\}$ modulo 2π : only I and R_π . (d) $\{I, R_\pi, S_0, S_\pi\}$, a group of order 4 in which every element has square I : it

is the Klein group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (Corollary 5.3). It is not cyclic, because a cyclic group of order 4 has an element of order 4.

BEST PRACTICE

Assigned exercise 5.2 — answers. (a) From Corollary 5.4, $1 + 2 \cos \theta = 0$, that is, $\cos \theta = -\frac{1}{2}$ and $\theta = 120^\circ$. (b) The cyclic permutation has matrix $P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ (the columns are the images of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$). The columns are the vectors of the canonical basis, hence orthonormal: $P \in O(3)$. The determinant of a cyclic permutation of three elements is $+1$ (the permutation is even, a product of two transpositions): $P \in SO(3)$. *Independent check:* the numerical computation gives $\det P = 1.0$ and $\operatorname{tr} P = 0$, hence $\theta = 120.000000^\circ$. ✓(c) $(P - I)u = 0$ gives $-u_1 + u_3 = 0$, $u_1 - u_2 = 0$, $u_2 - u_3 = 0$, that is, $u_1 = u_2 = u_3$: the axis is generated by $(1, 1, 1)^\top$ and the unit vector is $(1, 1, 1)^\top / \sqrt{3}$. Numerically $|Pu - u| = 0$ exactly (the entries are 0 and 1). ✓(d) $P^3 = I$, because the cyclic permutation applied three times brings each axis back to its place; consistently $3\theta = 360^\circ$. The subgroup generated by P is cyclic of order 3.

BEST PRACTICE

Assigned exercise 5.3 — answers. (a) All three have orthonormal columns (direct check: for M_2 , $\frac{1}{169}(25 + 144) = 1$ and $\frac{1}{169}(5 \cdot 12 + 12 \cdot (-5)) = 0$). $\det M_1 = \frac{1}{2}(1 + 1) = +1$: a rotation, with $\cos \theta = \frac{1}{\sqrt{2}}$ and $\sin \theta = -\frac{1}{\sqrt{2}}$, that is, $\theta = -\frac{\pi}{4}$ (equivalently $\frac{7\pi}{4}$). $\det M_2 = \frac{1}{169}(-25 - 144) = -1$: a reflection, with $\cos \theta = \frac{5}{13}$ and $\sin \theta = \frac{12}{13}$, that is, $\theta = \arccos \frac{5}{13} = 1.176005 \text{ rad} = 67.380135^\circ$. $\det M_3 = \frac{1}{4}(1 + 3) = +1$: a rotation with $\cos \theta = \frac{1}{2}$, $\sin \theta = -\frac{\sqrt{3}}{2}$, that is, $\theta = -\frac{\pi}{3}$. (b) Only M_2 is a reflection; its axis has inclination $\theta/2 = 33.690068^\circ$, that is, slope $\tan(\theta/2) = \frac{2}{3}$: the line $y = \frac{2}{3}x$. *Independent check:* $M_2(3, 2)^\top = \frac{1}{13}(15 + 24, 36 - 10)^\top = \frac{1}{13}(39, 26)^\top = (3, 2)^\top$. ✓(c) By Proposition 5.3: $M_1 M_2 = R_{-\pi/4} S_{\theta_2} = S_{\theta_2 - \pi/4}$ and $M_2 M_1 = S_{\theta_2} R_{-\pi/4} = S_{\theta_2 + \pi/4}$: two *different* reflections (the axes differ by $\frac{\pi}{4}$ in the parameter, that is, by 22.5° as lines), which confirms once more that $O(2)$ is not abelian. And $M_2 M_3 M_2 = S_{\theta_2} R_{-\pi/3} S_{\theta_2} = S_{\theta_2 + \pi/3} S_{\theta_2} = R_{\pi/3} = M_3^{-1}$: conjugating a rotation by a reflection inverts it. (d) $M_1 = R_{-\pi/4}$ generates a cyclic group of order 8 (the smallest $k > 0$ with $k \cdot \frac{\pi}{4} \in 2\pi\mathbb{Z}$ is $k = 8$). M_2 is a reflection, hence $M_2^2 = I$: a group of order 2. Together they generate the dihedral group D_8 of order 16: the eight rotations $R_{k\pi/4}$ and the eight reflections $S_{\theta_2 + k\pi/4}$; it is finite because the products always fall back into those two lists by Proposition 5.3.

5.9. Chapter formulary

Isometries and orthogonal matrices.

$$O(n) = \{A : A^\top A = I\}, \quad SO(n) = \{A \in O(n) : \det A = +1\}, \quad A^{-1} = A^\top, \quad \det A = \pm 1.$$

$$A \in O(n) \iff |Ax| = |x| \forall x \iff (Ax)^\top (Ay) = x^\top y \forall x, y \iff \text{the columns are orthonormal.}$$

Every isometry of $\mathbb{R}^n$: $f(x) = Ax + b$ with $A \in O(n)$, $b = f(0)$. Image of a circle: $f(C(P, r)) = C(f(P), r)$.

Classification of $O(2)$.

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad S_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}, \quad \det R_\theta = +1, \quad \det S_\theta = -1.$$

$$R_\alpha R_\beta = R_{\alpha+\beta}, \quad R_\alpha S_\beta = S_{\alpha+\beta}, \quad S_\alpha R_\beta = S_{\alpha-\beta}, \quad S_\alpha S_\beta = R_{\alpha-\beta}, \quad S_\alpha^2 = I.$$

Axis of S_θ : the line of inclination $\theta/2$. Spectrum: $R_\theta \rightarrow e^{\pm i\theta}$; $S_\theta \rightarrow \pm 1$.

$SO(2) \cong \mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ via $e^{i\theta} \longleftrightarrow R_\theta$ (group isomorphism and homeomorphism).

Conics and the uniqueness of the circle.

$$\mathcal{C}_Q = \{x : x^\top Q x = 1\}, \quad A(\mathcal{C}_Q) = \mathcal{C}_Q \quad \forall A \in O(2) \iff Q = \lambda I, \quad \lambda > 0,$$

and then $\mathcal{C}_Q = C(O, 1/\sqrt{\lambda})$. On $|x| = r$:

$$x^\top Q x = r^2 \left(\frac{a+c}{2} + \frac{a-c}{2} \cos 2t + b \sin 2t \right), \quad Q = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad x = r(\cos t, \sin t)^\top.$$

Symmetries of an ellipse with $\alpha \neq \beta$: $\{I, R_\pi, S_0, S_\pi\} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Rotations of space.

$$A \in SO(3) \implies \det(A - I) = 0 \implies \exists u, |u| = 1, Au = u;$$

$$A \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad \operatorname{tr} A = 1 + 2 \cos \theta, \quad \cos \theta = \frac{\operatorname{tr} A - 1}{2}.$$

$SO(2)$ is abelian, $SO(3)$ is not: $R_z(\frac{\pi}{2})R_x(\frac{\pi}{2}) \neq R_x(\frac{\pi}{2})R_z(\frac{\pi}{2})$ (same angle 120° , different axes).

Sources and bibliography

- [1] D. Klain, *Lecture notes on orthogonal matrices (with exercises)*, course 92.222 “Linear Algebra II”, University of Massachusetts Lowell, spring 2004. §1 (Theorem 1: A orthogonal $\iff |AX| = |X| \iff AX \cdot AY = X \cdot Y$; Proposition 2: $O(n)$ closed under products and $\det A = \pm 1$); §2 “The n -Reflections Theorem” (Theorem 3, Proposition 4, Corollary 5: $\det A = (-1)^k$); §3 “Classifying 2×2 Orthogonal Matrices” (the only two possibilities for the second column, and the identification of the reflection as the reflection in the line of inclination $\theta/2$); §4 “Classifying 3×3 Orthogonal Matrices” (Theorem 6, Corollaries 7 and 8: the product of two spatial rotations is a rotation). faculty.uml.edu/dklain/orthogonal.pdf — verified on 5 September 2026.
- [2] *RES.18-011 Algebra I Student Notes*, MIT OpenCourseWare, fall 2021 (notes of M. Artin’s course 18.701, written up by the students). Lecture 14 “Finite and Discrete Groups of Isometries”, pp. 65–66: Definitions 14.1 and 14.2 (orthogonal matrices O_n and isometries M_n); the fact that every isometry has the form $f(x) = Ax + b$ with $A \in O_n$; the classification of O_2 into rotations ($\det = +1$) and reflections ($\det = -1$) and that of M_2 into translations, rotations, reflections and glide reflections; Example 14.4: the symmetry group of a circle centred at the origin is all of O_2 and is not discrete. ocw.mit.edu/courses/res-18-011-algebra-i-student-notes-fall-2021/

— verified on 5 September 2026.

- [3] L. Euler, *Formulae generales pro translatione quacunque corporum rigidorum*, *Novi Commentarii academiae scientiarum Petropolitanae* **20** (1776), pp. 189–207 (written in 1775; Eneström number E478). It is the historical source of Theorem 5.9: every displacement of a rigid body with one fixed point is a rotation about an axis through that point. Text and bibliographic record in the Euler Archive of the University of the Pacific, scholarlycommons.pacific.edu/euler-works/478/ — verified on 5 September 2026. The proof given here (via $\det(A - I) = 0$) is the modern algebraic one, not Euler’s spherical one.
- [4] *18.900 Geometry and Topology in the Plane*, MIT OpenCourseWare, spring 2023, instructor P. Seidel: lecture notes, in particular the chapter on plane curves and on the geometry of polygons, used as a cross-check for the notions of plane isometry and of incidence recalled in Theorem 5.2 (result R1.5 of the Master Plan). ocw.mit.edu/courses/18-900-geometry-and-topology/ — verified on 5 September 2026.
- [5] A. Felikson, *Geometry III/IV — Circles in Euclidean geometry: some facts and proofs*, Durham University. Facts 1 and 1’ (through three non-collinear points there passes one and only one circle) and Fact 2 (the perpendicular bisectors of the sides of a triangle are concurrent): the elementary results on the circle recalled in Remark 5.1 to identify the centre as the unique equidistant point. maths.dur.ac.uk/users/anna.felikson/Geometry/Geometry1/ — verified on 5 September 2026.
- [6] Numerical checks of the chapter: a Python script (NumPy in double precision, seed 20260905) verifying the multiplication table of $O(2)$, the axis of S_θ , the identity $\det(A - I) = 0$ on 2000 randomly generated matrices of $SO(3)$, the trigonometric identity of Step 2 of Theorem 5.8 and the non-commutativity counterexample. The results are reported in the text next to each ✓.

CHAPTER 6

The circle in the complex plane

In the complex plane the unit circle stops being merely a locus of points and becomes a *group*: the product of two numbers of modulus one still has modulus one, and multiplying by $e^{i\theta}$ means rotating by θ . The whole chapter follows from that single observation. If one looks for the elements of finite order of that group one finds the n -th roots of unity, that is, the vertices of a regular polygon: dividing the circle into equal parts is an algebraic fact before being a geometric one, and the question of *which* regular polygons can be constructed with ruler and compass has a purely arithmetical answer. If instead one looks for the transformations of the plane that respect the family of circles, one finds the Möbius transformations, and with them the discovery that in the extended complex plane “circle” and “line” are the same thing seen from two different points.

The chapter is therefore built on two axes. The first is *the circle as a group*: $\mathbb{T} \cong SO(2)$, the roots of unity as a cyclic subgroup, the sum of the roots and the orthogonality of geometric sums — the elementary identity out of which the discrete Fourier transform (Chapter 22) and harmonic analysis on $\mathbb{T}$ (Chapter 13) will be born. The second is *the circle as a transformable object*: Möbius maps, cross ratio, the concyclicity criterion. In between, the historical bridge between the two themes: the Gauss–Wantzel theorem on the constructibility of regular polygons, which this treatise proves in full elsewhere and which is here merely recalled.

INFO

Level and prerequisites. Level **B–C** (first years of university); Sections 6.4 and 6.3 are of level **C**.

Internal prerequisites. Polar form, modulus and argument, the complex exponential and the properties $e^{i(s+t)} = e^{is}e^{it}$, $\overline{e^{it}} = e^{-it}$, $e^{it} = 1 \iff t \in 2\pi\mathbb{Z}$ (Volume II, Chapter 3); circular inversion and the power of a point (Volume I, Chapter 5); the Gauss–Wantzel theorem with its complete proof and the Gaussian periods (Volume II, Chapter 7); the orthogonal group $O(2)$ and the rotations R_θ (Chapter 5).

External prerequisites, assumed and not reproved (a first course in algebra; Judson, *Abstract Algebra: Theory and Applications*, §6.2 and §11.2):

- Lagrange’s theorem and its corollary: in a finite group the order of every element divides the order of the group (Judson, Theorem 6.10 and Corollary 6.11);
- the first isomorphism theorem: if $\varphi : G \rightarrow H$ is a homomorphism, then $G/\ker \varphi \cong \text{im } \varphi$ (Judson, Theorem 11.10);
- a non-zero polynomial of degree n with coefficients in a field has at most n roots.

Everything else — de Moivre’s formula included — is proved here.

What one learns. How to recognise the group structure of the circle and use it; how to construct the n -th roots of unity and prove their geometric and arithmetical properties; how to derive the orthogonality of geometric sums treating correctly the case of a vanishing denominator; how to decide whether four points are concyclic with a single computation; how to prove that Möbius transformations preserve the family of generalised circles.

Warning. Conventions of this chapter.

- $\mathbb{T} = S^1 = \{z \in \mathbb{C} : |z| = 1\}$; one writes $\mathbb{T}$ when the group structure matters and S^1 when the structure of a space matters.
- $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is the **extended complex plane** (Riemann sphere).
- $\zeta_n = e^{2\pi i/n}$ and $\mu_n = \{z \in \mathbb{C} : z^n = 1\}$.
- $\bar{z}$ is the conjugate, $\Re z$ and $\Im z$ the real and imaginary parts; we constantly use $z\bar{z} = |z|^2$ and $z + \bar{z} = 2\Re z$.
- A **generalised circle** of $\hat{\mathbb{C}}$ is a circle of $\mathbb{C}$ or the union of a line of $\mathbb{C}$ with $\{\infty\}$ (Definition 6.5).
- The cross ratio is defined with Ahlfors' convention (Definition 6.6); other texts permute the arguments, and the value changes accordingly.
- Angles are in radians, except when the symbol $^\circ$ indicates degrees.

6.1. The unit circle as a group

Definition 6.1: The group $\mathbb{T}$. One sets

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\},$$

equipped with the multiplication of complex numbers. The elements of $\mathbb{T}$ are called **unimodular complex numbers** or **phases**.

Theorem 6.1: $\mathbb{T}$ is a group, isomorphic to $SO(2)$ and to $\mathbb{R}/2\pi\mathbb{Z}$ (R10.1).

- (i) $\mathbb{T}$ is an abelian subgroup of the multiplicative group $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$.
(ii) The map $\Phi : \mathbb{T} \rightarrow SO(2)$ defined by

$$\Phi(\cos \theta + i \sin \theta) = R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is well defined and is a group isomorphism.

- (iii) The map $E : \mathbb{R} \rightarrow \mathbb{T}$, $E(\theta) = e^{i\theta}$, is a surjective group homomorphism with kernel $2\pi\mathbb{Z}$; hence $\mathbb{R}/2\pi\mathbb{Z} \cong \mathbb{T}$.

Proof. (i) $1 \in \mathbb{T}$ because $|1| = 1$. If $|z| = |w| = 1$ then $|zw| = |z||w| = 1$, hence $zw \in \mathbb{T}$; and $|z^{-1}| = |z|^{-1} = 1$, hence $z^{-1} \in \mathbb{T}$. Complex multiplication is associative and commutative, so $\mathbb{T}$ is an abelian subgroup of $\mathbb{C}^*$.

(ii) *Well definedness.* An element of $\mathbb{T}$ is written $z = \cos \theta + i \sin \theta$ and the pair $(\cos \theta, \sin \theta)$ is determined by z (they are the real and the imaginary part); the matrix R_θ depends only on this pair, hence only on z . In particular there is no ambiguity due to the choice of θ up to 2π . *Image in $SO(2)$.* One has $\det R_\theta = \cos^2 \theta + \sin^2 \theta = 1$; and the two columns $(\cos \theta, \sin \theta)^\top$ and $(-\sin \theta, \cos \theta)^\top$ have norm 1 (again by $\cos^2 + \sin^2 = 1$) and scalar product $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$, hence $R_\theta^\top R_\theta = I$ and $R_\theta \in SO(2)$.

Homomorphism. Let $z = \cos \alpha + i \sin \alpha$ and $w = \cos \beta + i \sin \beta$. Then

$$zw = (\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta) = \cos(\alpha + \beta) + i \sin(\alpha + \beta),$$

where the addition formulae have been used. On the other hand the product of the matrices gives

$$R_\alpha R_\beta = \begin{pmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -(\cos \alpha \sin \beta + \sin \alpha \cos \beta) \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & \cos \alpha \cos \beta - \sin \alpha \sin \beta \end{pmatrix} = R_{\alpha+\beta}.$$

Hence $\Phi(zw) = R_{\alpha+\beta} = R_\alpha R_\beta = \Phi(z)\Phi(w)$.

Injectivity. If $\Phi(z) = I$ then $\cos \theta = 1$ and $\sin \theta = 0$, that is $z = 1$: the kernel is trivial.

Surjectivity. Every $A \in SO(2)$ has the form R_θ for a suitable θ (Chapter 5, classification of $O(2)$), and then $A = \Phi(\cos \theta + i \sin \theta)$.

(iii) E is a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{T}, \cdot)$ because $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$ (Volume II, Chapter 3). It is surjective because every $z \in \mathbb{T}$ is written $z = \cos \theta + i \sin \theta = e^{i\theta}$. The kernel is $\{\theta : e^{i\theta} = 1\} = 2\pi\mathbb{Z}$, again by the properties of the complex exponential recalled among the prerequisites. By the first isomorphism theorem, $\mathbb{R}/2\pi\mathbb{Z} \cong \mathbb{T}$. $\square$

Remark 6.1: What “the circle is a group” means. Theorem 6.1 says three different things with the same content. Algebraically, $\mathbb{T}$ is an abelian group. Geometrically, multiplying by $e^{i\theta}$ is rotating the plane by θ about the origin: the circle acts on itself by rotations, and the action is simply transitive (from one point one reaches any other in exactly one way). Analytically, $\mathbb{T}$ is the real line “rolled up” with step 2π : this is the reason why a periodic function of period 2π is, properly, a function on the circle. The three readings will return respectively in Chapter 13 (topological group and Haar measure), in Chapter 5 (isometries) and in Chapter 12 (the covering $\mathbb{R} \rightarrow S^1$).

Proposition 6.1: De Moivre’s formula (R10.2). For every $\theta \in \mathbb{R}$ and every $n \in \mathbb{Z}$,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Proof. Step 1 — the case $n \geq 0$, by induction. For $n = 0$ both sides equal 1. Assuming the formula true for n , multiply by $\cos \theta + i \sin \theta$:

$$(\cos \theta + i \sin \theta)^{n+1} = (\cos n\theta + i \sin n\theta)(\cos \theta + i \sin \theta).$$

Expanding and collecting,

$$= (\cos n\theta \cos \theta - \sin n\theta \sin \theta) + i(\sin n\theta \cos \theta + \cos n\theta \sin \theta) = \cos(n+1)\theta + i \sin(n+1)\theta,$$

by the addition formulae. The induction principle concludes.

Step 2 — the case $n < 0$. Let $n = -m$ with $m > 0$. Since $|\cos \theta + i \sin \theta| = 1$, one has $(\cos \theta + i \sin \theta)^{-1} = \overline{\cos \theta + i \sin \theta} = \cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta)$; indeed for $|z| = 1$ one has $z\bar{z} = 1$, that is $z^{-1} = \bar{z}$. Applying Step 1 to $-\theta$ and to the exponent m ,

$$(\cos \theta + i \sin \theta)^{-m} = (\cos(-\theta) + i \sin(-\theta))^m = \cos(-m\theta) + i \sin(-m\theta) = \cos n\theta + i \sin n\theta.$$

$\square$

Variant: the same formula in one line. In exponential notation, $\cos \theta + i \sin \theta = e^{i\theta}$ and the formula reduces to $(e^{i\theta})^n = e^{in\theta}$, that is to the property $e^{is}e^{it} = e^{i(s+t)}$ iterated. The proof by induction is however the one that remains valid if $\cos$ and $\sin$ are defined in a purely geometric way, without the complex exponential: this is why it has been written out in full. $\square$

Corollary 6.1: Multiple-angle formulae. For every $\theta \in \mathbb{R}$,

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta, \quad \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta,$$

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.$$

Proof. Put $c = \cos \theta$, $s = \sin \theta$. By Proposition 6.1 with $n = 3$ and by the binomial theorem,

$$\cos 3\theta + i \sin 3\theta = (c + is)^3 = c^3 + 3ic^2s - 3cs^2 - is^3.$$

Equating real and imaginary parts,

$$\cos 3\theta = c^3 - 3cs^2, \quad \sin 3\theta = 3c^2s - s^3.$$

Substituting $s^2 = 1 - c^2$ in the first one obtains $\cos 3\theta = c^3 - 3c(1 - c^2) = 4c^3 - 3c$; substituting $c^2 = 1 - s^2$ in the second, $\sin 3\theta = 3(1 - s^2)s - s^3 = 3s - 4s^3$.

For $n = 5$, again with the binomial theorem,

$$(c + is)^5 = c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5,$$

whose real part is $c^5 - 10c^3s^2 + 5cs^4$. With $s^2 = 1 - c^2$ and $s^4 = (1 - c^2)^2 = 1 - 2c^2 + c^4$,

$$\cos 5\theta = c^5 - 10c^3(1 - c^2) + 5c(1 - 2c^2 + c^4) = c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 = 16c^5 - 20c^3 + 5c.$$

Independent check: for $\theta = \pi/3$ one has $c = 0.5$ and $16(0.5)^5 - 20(0.5)^3 + 5(0.5) = 0.5 - 2.5 + 2.5 = 0.5$, whereas $\cos(5\pi/3) = \cos(-\pi/3) = 0.5$. ✓ □

6.2. The n -th roots of unity

Definition 6.2: n -th roots of unity. Let $n \geq 1$ be an integer. One sets

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\}, \quad \zeta_n = e^{2\pi i/n}.$$

The elements of μ_n are called **n -th roots of unity**. A root $\zeta \in \mu_n$ is said to be **primitive** if its multiplicative order is exactly n , that is if $\zeta^m \neq 1$ for $1 \leq m < n$.

Theorem 6.2: The n -th roots are exactly n (R10.2). For every $n \geq 1$,

$$\mu_n = \{\zeta_n^k = e^{2\pi i k/n} : k = 0, 1, \dots, n-1\},$$

and its n elements are pairwise distinct. In particular $|\mu_n| = n$ and $\mu_n \subset \mathbb{T}$.

Proof. Step 1 — each of those numbers is a root. By Proposition 6.1, $(e^{2\pi i k/n})^n = e^{2\pi i k} = \cos(2\pi k) + i \sin(2\pi k) = 1$.

Step 2 — they are distinct. If $e^{2\pi i k/n} = e^{2\pi i j/n}$ with $0 \leq j \leq k \leq n-1$, then $e^{2\pi i (k-j)/n} = 1$, hence $\frac{2\pi(k-j)}{n} \in 2\pi\mathbb{Z}$, that is $n \mid (k-j)$. But $0 \leq k-j \leq n-1$, so $k-j = 0$.

Step 3 — there are no others. Let $z \in \mu_n$. From $|z|^n = |z^n| = 1$ and $|z| > 0$ real it follows that $|z| = 1$ (the function $t \mapsto t^n$ is strictly increasing on $(0, +\infty)$, hence injective, and $1^n = 1$). Hence $z = e^{i\theta}$ for some $\theta \in \mathbb{R}$ (Theorem 6.1(iii)), and $z^n = e^{in\theta} = 1$ forces $n\theta \in 2\pi\mathbb{Z}$, that is $\theta = \frac{2\pi k}{n}$ for some $k \in \mathbb{Z}$. Writing $k = qn + r$ with $0 \leq r \leq n-1$ (Euclidean division), one has $e^{2\pi i k/n} = e^{2\pi i q} e^{2\pi i r/n} = e^{2\pi i r/n}$.

Independent check. The polynomial $z^n - 1$ has degree n and therefore at most n roots in $\mathbb{C}$ (declared algebraic prerequisite); Steps 1–2 exhibit n distinct ones, so there are no others. The two arguments are independent: the first is analytic (modulus and argument), the second algebraic. ✓ □

Remark 6.2: De Moivre and Euler. From $e^{i\theta} = \cos \theta + i \sin \theta$ de Moivre's formula (Proposition 6.1) becomes $(e^{i\theta})^n = e^{in\theta}$: raising to a power on the circle multiplies the angle. It is the tool with which the multiple-angle identities are derived (Corollary 6.1) and the reason why the unit circle “linearises” trigonometry: a product of phases becomes a sum of angles, and an algebraic equation $z^n = 1$ becomes a division of the full angle into n equal parts. The polynomials T_n such that $\cos n\theta = T_n(\cos \theta)$ are the Chebyshev polynomials of the first kind, which reappear in approximation theory (Chapter 31).

Theorem 6.3: The roots are the vertices of a regular polygon (R10.3). Let $n \geq 3$ and let P_n be the polygon with vertices $V_k = \zeta_n^k$, $k = 0, \dots, n-1$, taken in this order. Then:

- (i) all the vertices lie on the unit circle;
 - (ii) all the sides have the same length $\ell_n = |V_{k+1} - V_k| = 2 \sin \frac{\pi}{n}$ (indices modulo n);
 - (iii) all the interior angles are equal and are worth $\alpha_n = \pi - \frac{2\pi}{n} = \frac{(n-2)\pi}{n}$.
- Hence P_n is a regular polygon with n sides inscribed in the unit circle, and conversely every regular polygon with n sides inscribed in the unit circle is obtained from P_n by a rotation.

Proof. (i) $|\zeta_n^k| = |\zeta_n|^k = 1$.

(ii) Since $V_{k+1} = \zeta_n V_k$,

$$|V_{k+1} - V_k| = |\zeta_n V_k - V_k| = |V_k| |\zeta_n - 1| = |\zeta_n - 1|,$$

which does not depend on k . To compute it one factors out the “half phase”: for every θ ,

$$e^{i\theta} - 1 = e^{i\theta/2}(e^{i\theta/2} - e^{-i\theta/2}) = e^{i\theta/2} \cdot 2i \sin \frac{\theta}{2},$$

where the formula $e^{i\varphi} - e^{-i\varphi} = 2i \sin \varphi$ has been used. Taking moduli and recalling $|e^{i\theta/2}| = |i| = 1$,

$$|e^{i\theta} - 1| = 2 \left| \sin \frac{\theta}{2} \right|.$$

With $\theta = 2\pi/n$ and $0 < \pi/n \leq \pi/3 < \pi$, the sine is positive and one obtains $\ell_n = 2 \sin(\pi/n)$.

(iii) *Equality of the angles.* Multiplication by ζ_n is a rotation by $2\pi/n$ about the origin (Theorem 6.1(ii)), hence an isometry of the plane; it maps the ordered triple (V_{k-1}, V_k, V_{k+1}) to the triple (V_k, V_{k+1}, V_{k+2}) . An isometry preserves angles, so the angle at V_k equals the angle at V_{k+1} for every k : all interior angles coincide.

Common value. Consider the triangle $OV_k V_{k+1}$, with O the origin. It is isosceles because $|OV_k| = |OV_{k+1}| = 1$, and the angle at the vertex O equals $2\pi/n$: indeed $V_{k+1} = e^{2\pi i/n} V_k$, and multiplication by $e^{2\pi i/n}$ rotates by $2\pi/n$. The base angles are therefore equal and each is worth $\frac{1}{2}(\pi - \frac{2\pi}{n}) = \frac{\pi}{2} - \frac{\pi}{n}$. The interior angle of the polygon at V_k is the sum of the base angle of the triangle $OV_{k-1} V_k$ and of that of the triangle $OV_k V_{k+1}$, that is

$$\alpha_n = 2 \left(\frac{\pi}{2} - \frac{\pi}{n} \right) = \pi - \frac{2\pi}{n} = \frac{(n-2)\pi}{n}.$$

(The sum of the two base angles is indeed the interior angle because the centre O is interior to the polygon and the segment OV_k lies inside the angle at V_k : the two triangles lie on opposite sides of the line OV_k , since V_{k-1} and V_{k+1} have arguments $\frac{2\pi k}{n} \mp \frac{2\pi}{n}$, symmetric with respect to $\frac{2\pi k}{n}$, and $0 < 2\pi/n \leq 2\pi/3 < \pi$.)

Independent check on (iii). The sum of the interior angles of a convex polygon with n sides is $(n-2)\pi$; if they are all equal, each is worth $(n-2)\pi/n$, in agreement. ✓

Last assertion. If $W_0, \dots, W_{n-1}$ are the vertices of a regular polygon with n sides inscribed in the unit circle, its centre is the origin and the vertices are obtained one from another by the rotation by $2\pi/n$ (this is the very definition of a regular polygon: the rotations by $2\pi/n$ about the centre map it to itself permuting the vertices cyclically). Hence $W_k = W_0 \zeta_n^k$, that is the polygon is the image of P_n under multiplication by $W_0 \in \mathbb{T}$, which is a rotation. $\square$

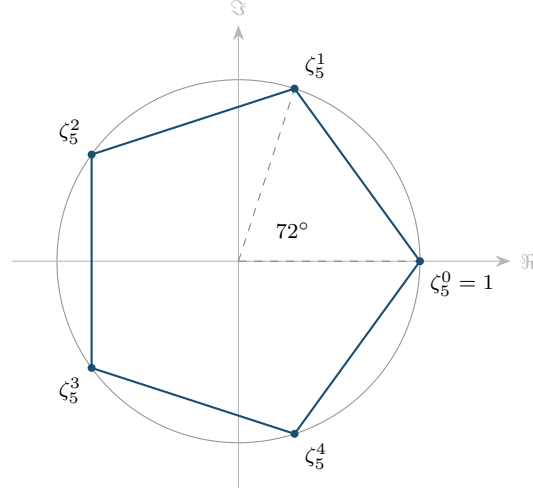


Figure 6.1: The five fifth roots of unity and the regular pentagon which is their convex hull. The coordinates of the vertices are $(\cos(72k)^\circ, \sin(72k)^\circ)$ for $k = 0, \dots, 4$, that is $(1, 0)$, $(0.30902, 0.95106)$, $(-0.80902, 0.58779)$, $(-0.80902, -0.58779)$, $(0.30902, -0.95106)$; the side measures $2 \sin 36^\circ = 1.17557$ (Theorem 6.3(ii)) and the interior angle 108° .

Theorem 6.4: Sum of the roots of unity (R10.4). For every $n \geq 2$,

$$\sum_{k=0}^{n-1} \zeta_n^k = 0.$$

Equivalently $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} = 0$ and $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n} = 0$.

Proof. Put $\sigma = \sum_{k=0}^{n-1} \zeta_n^k$.

Step 1 — telescoping identity. For every $q \in \mathbb{C}$ and every $n \geq 1$,

$$(q-1) \sum_{k=0}^{n-1} q^k = \sum_{k=0}^{n-1} q^{k+1} - \sum_{k=0}^{n-1} q^k = q^n - 1,$$

because the two sums differ only by the first and the last summand. This identity holds *always*, even for $q = 1$ (where it says $0 = 0$).

Step 2 — application. With $q = \zeta_n$ one has $q^n = e^{2\pi i} = 1$, hence $(\zeta_n - 1)\sigma = 0$. Since $n \geq 2$, $\zeta_n = e^{2\pi i/n} \neq 1$: indeed $e^{2\pi i/n} = 1$ would require $\frac{2\pi}{n} \in 2\pi\mathbb{Z}$, that is $n \mid 1$. A product of complex numbers vanishes only if one factor does, hence $\sigma = 0$.

Step 3 — real form. Separating the real and the imaginary part of $\sigma = 0$ and using $\zeta_n^k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$ one obtains the two trigonometric identities. $\square$

Variant: symmetry of the group (independent check). Multiplication by ζ_n permutes the elements of μ_n : the map $z \mapsto \zeta_n z$ maps μ_n into itself because μ_n is closed under products (Corollary 6.2(i), whose proof is independent of this theorem), and it is injective because $\zeta_n \neq 0$: it is therefore a bijection of μ_n , which is finite. Hence $\zeta_n \sigma = \sum_{z \in \mu_n} \zeta_n z = \sum_{w \in \mu_n} w = \sigma$,

whence $(\zeta_n - 1)\sigma = 0$ and $\sigma = 0$. Geometrically: the centroid of a regular polygon centred at the origin is the origin, because a rotation mapping the polygon to itself must fix the centroid, and the only point fixed by a rotation of non-zero angle is the centre. $\square$

Proposition 6.2: Product of the roots of unity. For every $n \geq 1$, $\prod_{k=0}^{n-1} \zeta_n^k = (-1)^{n+1}$.

Proof. Adding the exponents,

$$\prod_{k=0}^{n-1} \zeta_n^k = \zeta_n^{0+1+\dots+(n-1)} = \zeta_n^{n(n-1)/2} = \left(e^{2\pi i/n}\right)^{n(n-1)/2} = e^{\pi i(n-1)} = (-1)^{n-1} = (-1)^{n+1},$$

where $e^{i\pi} = -1$ has been used together with the fact that $(-1)^{n-1} = (-1)^{n+1}$ because the exponents differ by 2.

Independent check. From $z^n - 1 = \prod_{k=0}^{n-1} (z - \zeta_n^k)$ (the two polynomials are monic of degree n and have the same n distinct roots, by Theorem 6.2) one reads off the constant term by setting $z = 0$: $-1 = \prod_k (-\zeta_n^k) = (-1)^n \prod_k \zeta_n^k$, whence $\prod_k \zeta_n^k = (-1)^{-n} \cdot (-1) = (-1)^{n+1}$. $\checkmark$ For $n = 4$: $1 \cdot i \cdot (-1) \cdot (-i) = i \cdot i = -1 = (-1)^5$. $\checkmark$ $\square$

Corollary 6.2: The cyclic group inside the circle. For every $n \geq 1$:

- (i) μ_n is a subgroup of $\mathbb{T}$;
- (ii) μ_n is cyclic of order n , generated by ζ_n , and $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$;
- (iii) every finite subgroup H of $\mathbb{T}$ of order n coincides with μ_n ; in particular the finite subgroups of $\mathbb{T}$ are exactly the μ_n , and they are all cyclic.

The circle therefore contains, for every n , one and only one copy of the cyclic group of order n : it is the bridge between the geometry of the circle and modular arithmetic.

Proof. (i) $1 \in \mu_n$ because $1^n = 1$. If $z^n = w^n = 1$ then $(zw)^n = z^n w^n = 1$ (multiplication is commutative) and $(z^{-1})^n = (z^n)^{-1} = 1$. Hence μ_n is a subgroup of $\mathbb{C}^*$; and it is contained in $\mathbb{T}$ by Theorem 6.2, so it is a subgroup of $\mathbb{T}$.

(ii) Consider

$$\varphi: \mathbb{Z} \longrightarrow \mathbb{T}, \quad \varphi(k) = \zeta_n^k.$$

It is a homomorphism from $(\mathbb{Z}, +)$ to $(\mathbb{T}, \cdot)$ because $\zeta_n^{k+j} = \zeta_n^k \zeta_n^j$.

Image. $\text{im } \varphi = \{\zeta_n^k : k \in \mathbb{Z}\}$. Every ζ_n^k lies in μ_n because $(\zeta_n^k)^n = (\zeta_n^n)^k = 1$; conversely every element of μ_n has the form ζ_n^k by Theorem 6.2. Hence $\text{im } \varphi = \mu_n$: in particular μ_n is cyclic, generated by ζ_n .

Kernel. $\varphi(k) = 1$ means $e^{2\pi i k/n} = 1$, that is $\frac{2\pi k}{n} \in 2\pi\mathbb{Z}$, that is $\frac{k}{n} \in \mathbb{Z}$, that is $n \mid k$. Hence $\ker \varphi = n\mathbb{Z}$.

Conclusion. By the first isomorphism theorem (declared prerequisite),

$$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/\ker \varphi \cong \text{im } \varphi = \mu_n.$$

The order is n , consistently with $|\mu_n| = n$ (Theorem 6.2). The explicit isomorphism is $k + n\mathbb{Z} \mapsto \zeta_n^k$: the sum of the indices modulo n corresponds to the product of the roots, that is to the sum of the angles modulo 2π .

(iii) Let $H \leq \mathbb{T}$ be finite with $|H| = n$ and let $z \in H$. By Lagrange's theorem (declared prerequisite) the order d of z divides n , hence $z^n = (z^d)^{n/d} = 1$ and $z \in \mu_n$. Therefore $H \subseteq \mu_n$; since $|H| = n = |\mu_n|$ and μ_n is finite, one has $H = \mu_n$. Conversely every μ_n is a finite subgroup by (i)–(ii). Finally every μ_n is cyclic by (ii). $\square$

Proposition 6.3: Which roots are primitive. Let $n \geq 1$ and $0 \leq k \leq n-1$. The multiplicative order of ζ_n^k is $\frac{n}{\gcd(k, n)}$. In particular ζ_n^k is a primitive n -th root if and only if $\gcd(k, n) = 1$, and the primitive n -th roots are exactly $\varphi(n)$ in number, where φ is Euler's function.

Proof. Let $d = \gcd(k, n)$, $k = dk'$, $n = dn'$ with $\gcd(k', n') = 1$. One has

$$(\zeta_n^k)^m = 1 \iff e^{2\pi i km/n} = 1 \iff n \mid km \iff n' \mid k'm.$$

Since $\gcd(k', n') = 1$, Euclid's lemma gives $n' \mid m$. The smallest positive integer m with this property is $m = n' = n/d$: the order is therefore $n/\gcd(k, n)$.

The order equals n if and only if $\gcd(k, n) = 1$; the number of such k in $\{0, \dots, n-1\}$ is by definition $\varphi(n)$.

Independent check. For $n = 6$: $\gcd(k, 6) = 1$ only for $k = 1, 5$, hence $\varphi(6) = 2$ primitive roots, namely ζ_6 and ζ_6^5 . Indeed $\zeta_6^2 = \zeta_3$ has order 3, $\zeta_6^3 = -1$ has order 2, ζ_6^4 has order 3, $\zeta_6^0 = 1$ has order 1: the orders are 1, 6, 3, 2, 3, 6, that is $6/\gcd(k, 6)$ for $k = 0, \dots, 5$. ✓ $\square$

Theorem 6.5: Orthogonality of the geometric sums (R10.4). Let $n \geq 1$. For every $j \in \mathbb{Z}$,

$$\sum_{k=0}^{n-1} \zeta_n^{jk} = \begin{cases} n, & \text{if } n \mid j, \\ 0, & \text{if } n \nmid j. \end{cases}$$

Consequently, for $\ell, m \in \{0, 1, \dots, n-1\}$,

$$\frac{1}{n} \sum_{k=0}^{n-1} \zeta_n^{k(m-\ell)} = \delta_{\ell m} = \begin{cases} 1, & \ell = m, \\ 0, & \ell \neq m, \end{cases}$$

that is, the vectors $\chi_\ell = (\zeta_n^{\ell k})_{k=0}^{n-1}$, $\ell = 0, \dots, n-1$, are pairwise orthogonal in $\mathbb{C}^n$ with respect to the Hermitian product $\langle u, v \rangle = \sum_k u_k \overline{v_k}$, and each has norm $\sqrt{n}$.

Proof. Put $q = \zeta_n^j = e^{2\pi i j/n}$, so that $\sum_{k=0}^{n-1} \zeta_n^{jk} = \sum_{k=0}^{n-1} q^k$. Two cases must be distinguished, and the distinction is exactly that between a vanishing and a non-vanishing denominator in the formula for the geometric sum.

Case 1: $n \mid j$ (vanishing denominator). Then $\frac{j}{n} \in \mathbb{Z}$, hence $q = e^{2\pi i \cdot (j/n)} = 1$. The closed formula $\sum_{k=0}^{n-1} q^k = \frac{q^n - 1}{q - 1}$ is **not applicable**: the denominator $q - 1$ vanishes, and so does the numerator, so that the quotient is the indeterminate form $0/0$. One therefore sums directly: all n summands equal $1^k = 1$, and

$$\sum_{k=0}^{n-1} q^k = \underbrace{1 + 1 + \dots + 1}_{n \text{ times}} = n.$$

Case 2: $n \nmid j$ (non-vanishing denominator). One first shows that $q \neq 1$: $q = e^{2\pi i j/n} = 1$ is equivalent to $\frac{2\pi j}{n} \in 2\pi\mathbb{Z}$, that is to $\frac{j}{n} \in \mathbb{Z}$, that is to $n \mid j$, which is excluded. By the telescoping identity of Theorem 6.4 (Step 1), valid for every q ,

$$(q - 1) \sum_{k=0}^{n-1} q^k = q^n - 1,$$

and now it is legitimate to divide by $q - 1 \neq 0$:

$$\sum_{k=0}^{n-1} q^k = \frac{q^n - 1}{q - 1}.$$

Finally $q^n = (e^{2\pi i j/n})^n = e^{2\pi i j} = 1$ because j is an integer, hence the numerator vanishes and the sum equals 0.

Consequence (orthogonality). One applies what has just been proved to $j = m - \ell$. If $\ell, m \in \{0, \dots, n-1\}$ then $|m - \ell| \leq n-1 < n$, hence $n \mid (m - \ell)$ if and only if $m - \ell = 0$, that is $\ell = m$. The first case gives the sum n , the second 0: divided by n , one obtains $\delta_{\ell m}$.

For the vector reading observe that $\bar{\zeta}_n^{\ell k} = \zeta_n^{-\ell k}$ (because $|\zeta_n| = 1$), hence

$$\langle \chi_m, \chi_\ell \rangle = \sum_{k=0}^{n-1} \zeta_n^{mk} \bar{\zeta}_n^{\ell k} = \sum_{k=0}^{n-1} \zeta_n^{k(m-\ell)} = n \delta_{\ell m},$$

whence the orthogonality and $\|\chi_\ell\|^2 = n$. □

Remark 6.3: Why this identity is the heart of the DFT. Theorem 6.5 is, literally, the reason why the discrete Fourier transform is invertible. The vectors $\chi_0, \dots, \chi_{n-1}$ are n pairwise orthogonal non-zero vectors in $\mathbb{C}^n$, hence an orthogonal basis; the matrix having them as columns, divided by $\sqrt{n}$, is unitary. Chapter 22 reconstructs the whole edifice of the DFT — inversion, unitarity, circular convolution, Parseval — starting from these two lines, and Chapter 13 gives its continuous version (orthogonality of the characters $z \mapsto z^n$ of $\mathbb{T}$). The case $n \mid j$ is not a technical nuisance: it is the “DC” (mean) term of the signal, and it is the only case in which the sum does not vanish.

Example 6.1: Numerical check of the orthogonality for $n = 6$. Let $n = 6$, $\zeta_6 = e^{i\pi/3} = 0.5 + 0.86603i$. One computes $S(j) = \sum_{k=0}^5 \zeta_6^{jk}$ separating real and imaginary parts.

For $j = 1$ the angles are $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$:

$$\sum_k \cos = 1 + 0.5 - 0.5 - 1 - 0.5 + 0.5 = 0, \quad \sum_k \sin = 0 + 0.86603 + 0.86603 + 0 - 0.86603 - 0.86603 = 0,$$

hence $S(1) = 0$. For $j = 2$ the angles are $0^\circ, 120^\circ, 240^\circ, 0^\circ, 120^\circ, 240^\circ$ and one obtains $\sum \cos = 2(1 - 0.5 - 0.5) = 0$, $\sum \sin = 2(0 + 0.86603 - 0.86603) = 0$, that is $S(2) = 0$. For $j = 3$ the summands are $1, -1, 1, -1, 1, -1$ and $S(3) = 0$. For $j = 0$ and $j = 6$ all the summands equal 1 and $S = 6$.

Independent check with the closed formula (legitimate only for $j = 1, 2, 3, 4, 5$, where $q \neq 1$): $q^6 = 1$ for every integer j , hence $\frac{q^6 - 1}{q - 1} = \frac{0}{q - 1} = 0$. For $j = 0$ and $j = 6$ the closed formula cannot be used, and the direct sum gives 6. The two computations agree. ✓

Exercise 6.1. Compute the three cube roots of unity in algebraic form and verify that their sum is zero and that their product equals 1.

Solution. By Theorem 6.2, $\zeta_3^0 = 1$, $\zeta_3^1 = e^{2\pi i/3} = \cos 120^\circ + i \sin 120^\circ = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $\zeta_3^2 = e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$. The sum equals

$$1 + \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 1 - 1 + 0i = 0,$$

in agreement with Theorem 6.4. The product equals $1 \cdot \zeta_3 \cdot \zeta_3^2 = \zeta_3^3 = 1$, in agreement with Proposition 6.2 ($(-1)^{3+1} = 1$). The three points are the vertices of an equilateral triangle inscribed in the unit circle (Theorem 6.3), with side $2 \sin 60^\circ = \sqrt{3} = 1.73205$. *Check:* $|\zeta_3 - 1| = |-\frac{3}{2} + \frac{\sqrt{3}}{2}i| = \sqrt{2.25 + 0.75} = \sqrt{3}$. ✓

6.3. Regular polygons and constructibility

The roots of unity divide the circle into n equal arcs; constructing them with ruler and compass is the *cyclotomic* problem (from the Greek: “cutting of the circle”). This section establishes what can be obtained by elementary means and refers to Volume II for the complete answer.

Definition 6.3: Constructible number and constructible polygon. A real number α is **constructible** if, starting from the points $(0, 0)$ and $(1, 0)$, a segment of length $|\alpha|$ can be obtained with finitely many ruler-and-compass constructions (drawing the line through two constructed points, the circle with centre a constructed point and radius the distance between two constructed points, and taking intersections). A complex number $z = x + iy$ is constructible if x and y are. The **regular polygon with n sides is constructible** if the point ζ_n is.

It is assumed known (Volume II, Chapter 7) that the constructible numbers form a subfield of $\mathbb{R}$ closed under the square root of positive elements, and that the constructible complex numbers form a subfield of $\mathbb{C}$.

Proposition 6.4: Constructibility of the polygon and of the cosine. For $n \geq 3$, the regular polygon with n sides is constructible if and only if $\cos \frac{2\pi}{n}$ is a constructible number.

Proof. ($\Rightarrow$) If ζ_n is constructible, then by definition so is its real part, which is $\cos \frac{2\pi}{n}$. ($\Leftarrow$) Let $c = \cos \frac{2\pi}{n}$ be constructible. Then c^2 is constructible (product), $1 - c^2$ is constructible (difference) and is ≥ 0 ; hence $\sqrt{1 - c^2}$ is constructible. Since $0 < \frac{2\pi}{n} \leq \frac{2\pi}{3} < \pi$, one has $\sin \frac{2\pi}{n} > 0$ and therefore $\sin \frac{2\pi}{n} = \sqrt{1 - c^2}$. Hence $\zeta_n = c + i\sqrt{1 - c^2}$ has constructible real and imaginary parts. $\square$

Proposition 6.5: Doubling and dividing the number of sides. Let $n \geq 3$.

- (i) If the regular polygon with n sides is constructible, so is the one with $2n$ sides.
- (ii) If the regular polygon with n sides is constructible and $d \mid n$ with $d \geq 3$, then the regular polygon with d sides is constructible.
- (iii) If the regular polygons with m and with n sides are constructible and $\gcd(m, n) = 1$, then so is the one with mn sides.

Proof. (i) Let $V_0 = 1$ and $V_1 = \zeta_n$ be two consecutive vertices, constructible by hypothesis. The midpoint of the arc V_0V_1 is the intersection, on the side of the arc, between the unit circle and the perpendicular bisector of the segment V_0V_1 : both are constructible objects (the perpendicular bisector of a segment is an elementary ruler-and-compass construction), and the intersection of a line with a circle is allowed. That midpoint is ζ_{2n} : indeed the perpendicular bisector of V_0V_1 is the line through the origin with argument $\frac{\pi}{n}$ — because the reflection in that line exchanges V_0 and V_1 (it maps $e^{i\theta}$ to $e^{i(2\pi/n - \theta)}$, which for $\theta = 0$ gives ζ_n and for $\theta = 2\pi/n$ gives 1) — and it meets the minor arc V_0V_1 at the point $e^{i\pi/n} = \zeta_{2n}$.

(ii) Let $n = dq$. Then $\zeta_n^q = e^{2\pi i q/(dq)} = e^{2\pi i/d} = \zeta_d$, and the product of constructible numbers is constructible: ζ_d is constructible because ζ_n is. (Geometrically: one takes every q -th vertex of the polygon with n sides.)

(iii) Since $\gcd(m, n) = 1$, by Bézout’s identity there exist $u, v \in \mathbb{Z}$ with $um + vn = 1$. Then

$$\frac{1}{mn} = \frac{um + vn}{mn} = \frac{u}{n} + \frac{v}{m}, \quad \text{hence} \quad \zeta_{mn} = e^{2\pi i/(mn)} = (e^{2\pi i/n})^u (e^{2\pi i/m})^v = \zeta_n^u \zeta_m^v,$$

a product of integer powers of constructible numbers, hence constructible (the constructibles form a field, and $\zeta_n^{-1} = \overline{\zeta_n}$ is constructible because ζ_n is). $\square$

Proposition 6.6: *The cosine of 72° and the constructibility of the pentagon. One has*

$$\cos \frac{2\pi}{5} = \cos 72^\circ = \frac{\sqrt{5} - 1}{4},$$

and consequently the regular pentagon is constructible with ruler and compass.

Proof. Step 1 — the algebraic equation. Put $\theta = 72^\circ$ and $c = \cos \theta$. Then $3\theta = 216^\circ$ and $2\theta = 144^\circ$, and since $216^\circ = 360^\circ - 144^\circ$ one has $\cos 3\theta = \cos 2\theta$. Using $\cos 3\theta = 4c^3 - 3c$ (Corollary 6.1) and $\cos 2\theta = 2c^2 - 1$ (case $n = 2$ of the same computation: $\cos 2\theta + i \sin 2\theta = (c + is)^2 = c^2 - s^2 + 2ics$, hence $\cos 2\theta = c^2 - s^2 = 2c^2 - 1$), one obtains

$$4c^3 - 3c = 2c^2 - 1, \quad \text{that is} \quad 4c^3 - 2c^2 - 3c + 1 = 0.$$

Step 2 — factorisation. The value $c = 1$ annihilates the polynomial ($4 - 2 - 3 + 1 = 0$), hence $(c - 1)$ divides it. The division gives

$$4c^3 - 2c^2 - 3c + 1 = (c - 1)(4c^2 + 2c - 1),$$

as one checks by expanding: $(c - 1)(4c^2 + 2c - 1) = 4c^3 + 2c^2 - c - 4c^2 - 2c + 1 = 4c^3 - 2c^2 - 3c + 1$.
✓

Step 3 — the right root. Since $\theta = 72^\circ$ is not a multiple of 360° , one has $c \neq 1$, hence $4c^2 + 2c - 1 = 0$ and

$$c = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-1 \pm \sqrt{5}}{4}.$$

Since $0 < 72^\circ < 90^\circ$, $c > 0$, and of the two roots $\frac{-1 + \sqrt{5}}{4} \approx 0.30902$ and $\frac{-1 - \sqrt{5}}{4} \approx -0.80902$ only the first is positive. Hence $c = \frac{\sqrt{5} - 1}{4}$.

Independent check. $\sqrt{5} = 2.2360680$, hence $\frac{\sqrt{5} - 1}{4} = \frac{1.2360680}{4} = 0.3090170$; and $\cos 72^\circ = 0.3090170$. ✓ Observe moreover that the second root $\frac{-1 - \sqrt{5}}{4} = -0.8090170$ is $\cos 144^\circ$: the two roots of the trinomial are the two distinct cosines that occur among the fifth roots of unity, consistently with $\zeta_5^k + \zeta_5^{-k} = 2 \cos \frac{2\pi k}{5}$.

Step 4 — constructibility. The number $\sqrt{5}$ is constructible: it is the hypotenuse of the right triangle with legs 1 and 2, both constructible. Subtracting 1 and dividing by 4 are field operations, which are allowed. Hence c is constructible and, by Proposition 6.4, the regular pentagon is constructible. □

Corollary 6.3: *The pentadecagon. The regular polygon with 15 sides is constructible, and more generally so are all the polygons with $n = 3^a 5^b 2^k$ and $a, b \in \{0, 1\}$, $k \geq 0$, $n \geq 3$.*

Proof. The equilateral triangle is constructible ($\cos \frac{2\pi}{3} = -\frac{1}{2}$ is rational, hence constructible, and Proposition 6.4 applies); the pentagon is by Proposition 6.6. Since $\gcd(3, 5) = 1$, Proposition 6.5(iii) gives the pentadecagon. Explicitly, Bézout's identity $2 \cdot 3 - 1 \cdot 5 = 1$ gives $\frac{2}{5} - \frac{1}{3} = \frac{6 - 5}{15} = \frac{1}{15}$, that is

$$\zeta_{15} = \zeta_5^2 \zeta_3^{-1}.$$

Independent check. $\frac{2\pi}{15} = 24^\circ$ and $2 \cdot 72^\circ - 120^\circ = 24^\circ$. ✓

Finally the square is constructible, because $\cos \frac{2\pi}{4} = 0$ is rational and Proposition 6.4 applies; repeated doubling (Proposition 6.5(i)) then yields all the $n = 2^k m$ with $m \in \{3, 4, 5, 15\}$ and $k \geq 0$, that is exactly all the $n \geq 3$ of the form $3^a 5^b 2^k$ with $a, b \in \{0, 1\}$. □

Remark 6.4: The Gauss–Wantzel theorem: statement and cross-reference.

Statement. The regular polygon with n sides ($n \geq 3$) is constructible with ruler and compass if and only if

$$n = 2^k p_1 p_2 \cdots p_m,$$

with $k \geq 0$ and $p_1, \dots, p_m$ distinct Fermat primes, that is primes of the form $2^{2^r} + 1$ (the only known ones are 3, 5, 17, 257, 65537).

Cross-reference. This statement is not an imported result of the treatise: it is proved in full, in both implications, in **Volume II, Chapter 7** (“The discrete circle”), where one finds the definition of a constructible number, the theorem on the degree of constructible numbers (necessity, Wantzel 1837) and the construction of the Gaussian periods with the chain of quadratic extensions (sufficiency, Gauss 1796), besides the explicit construction of the 17-gon. Here it is not duplicated: it is only applied.

What is needed of that theorem in this chapter. None of the previous results depends on it: Propositions 6.4, 6.5 and 6.6 and Corollary 6.3 are proved here and independently, and they cover the “easy” part (sufficiency for $n = 3 \cdot 2^k, 5 \cdot 2^k, 15 \cdot 2^k, 2^k$). The Gauss–Wantzel theorem is instead needed for the negative answers — the heptagon, the nonagon, the hendecagon are not constructible — and for the case of large Fermat primes, and in this chapter it appears only in the exercises, with an explicit reference to Volume II.

Remark 6.5: Gauss’s 17-gon and the quadrature of the circle. Gauss proved on 30 March 1796, at the age of nineteen, that the regular polygon with 17 sides is constructible ($17 = 2^{2^2} + 1$ is a Fermat prime); the discovery, recorded as the first entry of his diary, led him to choose mathematics. The constructibility of polygons concerns the division of the circle into equal parts and is “dual” to the impossibility of squaring the circle (Chapter 1): there one would have to construct $\sqrt{\pi}$, which is impossible because π is transcendental; here one has to construct $\cos \frac{2\pi}{n}$, which is always an algebraic number but is constructible only for certain n . The circle stands at the centre of both questions, and the difference between the two answers is exactly the difference between “algebraic” and “of degree a power of two”.

Exercise 6.2. Is the regular pentagon constructible? And the heptagon (seven sides)? And the polygon with 20 sides?

Solution. *Pentagon:* yes, and this is proved in this chapter without appealing to Gauss–Wantzel: Proposition 6.6 explicitly constructs $\cos 72^\circ = \frac{\sqrt{5}-1}{4}$ with a single square root.

Heptagon: no. Here the “negative” part of the Gauss–Wantzel theorem is needed, recalled in Remark 6.4 and proved in Volume II, Chapter 7: 7 is not a Fermat prime (it is not of the form $2^{2^r} + 1$, since $2^{2^0} + 1 = 3$, $2^{2^1} + 1 = 5$ and already $2^{2^2} + 1 = 17 > 7$), and 7 does not admit the required factorisation $2^k p_1 \cdots p_m$. Hence the regular heptagon is not constructible with ruler and compass.

Polygon with 20 sides: yes. One has $20 = 2^2 \cdot 5$, with 5 a Fermat prime: it falls under the Gauss–Wantzel form. But here Proposition 6.5(i) applied twice to the pentagon ($5 \rightarrow 10 \rightarrow 20$) suffices, so the answer is independent of the cross-reference.

6.4. Möbius transformations and generalised circles

Definition 6.4: Extended complex plane and Möbius transformation. One sets $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ (extended complex plane, or Riemann sphere). A Möbius transformation is a map

$$T : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}, \quad T(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc \neq 0,$$

extended to $\hat{\mathbb{C}}$ by the conventions: if $c \neq 0$, $T(-d/c) = \infty$ and $T(\infty) = a/c$; if $c = 0$, $T(\infty) = \infty$.

Proposition 6.7: The Möbius group. Möbius transformations are bijections of $\hat{\mathbb{C}}$ and form a group under composition. Moreover the map

$$\Psi : GL(2, \mathbb{C}) \rightarrow \text{Möb}, \quad \Psi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \left(z \mapsto \frac{az + b}{cz + d} \right)$$

is a surjective group homomorphism whose kernel is the set of scalar matrices λI , $\lambda \neq 0$.

Proof. **Step 1 — Ψ is a homomorphism.** Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$ be invertible, $S = \Psi(N)$, $T = \Psi(M)$. For z such that all the denominators are non-zero,

$$T(S(z)) = \frac{a \frac{ez + f}{gz + h} + b}{c \frac{ez + f}{gz + h} + d} = \frac{a(ez + f) + b(gz + h)}{c(ez + f) + d(gz + h)} = \frac{(ae + bg)z + (af + bh)}{(ce + dg)z + (cf + dh)},$$

where numerator and denominator have been multiplied by $gz + h \neq 0$. The coefficients obtained are exactly the entries of MN . The case of the exceptional points ($z = \infty$, z a pole of S , $S(z)$ a pole of T) is checked with the conventions of Definition 6.4; for instance if $g \neq 0$ and $z = -h/g$ then $S(z) = \infty$ and $T(S(z)) = a/c$, whereas the composite formula evaluated at $-h/g$ gives

$$\frac{(ae + bg)(-h/g) + (af + bh)}{(ce + dg)(-h/g) + (cf + dh)} = \frac{a(fg - eh)/g}{c(fg - eh)/g} = \frac{a}{c},$$

since $fg - eh = -\det N \neq 0$; the other cases are analogous.

Step 2 — group and bijectivity. $\Psi(I)$ is the identity. By Step 1, $\Psi(M) \circ \Psi(M^{-1}) = \Psi(MM^{-1}) = \text{id}$ and similarly in the other order, hence every $\Psi(M)$ is bijective with inverse $\Psi(M^{-1})$, that is

$$T^{-1}(w) = \frac{dw - b}{-cw + a}.$$

The image of Ψ is by definition the set of Möbius transformations, which is therefore a subgroup of the group of bijections of $\hat{\mathbb{C}}$: it is closed under composition and inverses and contains the identity.

Step 3 — the kernel. If $\Psi(M) = \text{id}$, then $\frac{az+b}{cz+d} = z$ for every $z \in \mathbb{C}$ which is not a pole, that is $cz^2 + (d-a)z - b = 0$ for infinitely many z : a polynomial with infinitely many roots is zero, hence $c = 0$, $b = 0$ and $a = d$. Since $\det M = ad \neq 0$, one has $a = d \neq 0$ and $M = aI$. Conversely $\Psi(\lambda I)(z) = \frac{\lambda z}{\lambda} = z$. $\square$

Definition 6.5: Generalised circle. A generalised circle of $\hat{\mathbb{C}}$ is a subset $\hat{\Gamma} \subseteq \hat{\mathbb{C}}$ of the form:

- $\hat{\Gamma} = \Gamma$, with $\Gamma \subset \mathbb{C}$ a circle with centre z_0 and radius $r > 0$; or
- $\hat{\Gamma} = L \cup \{\infty\}$, with $L \subset \mathbb{C}$ a line.

In the second case one says that the generalised circle *passes through* ∞ .

Lemma 6.1: Equation of the generalised circles (R10.6). Let $A, C \in \mathbb{R}$, $B \in \mathbb{C}$ with $|B|^2 > AC$, and let

$$\Gamma_{A,B,C} = \{z \in \mathbb{C} : Az\bar{z} + Bz + \bar{B}\bar{z} + C = 0\}. \quad (*)$$

Then:

- (i) the left-hand side of $(*)$ is real for every z ;
- (ii) if $A \neq 0$, $\Gamma_{A,B,C}$ is the circle with centre $-\bar{B}/A$ and radius $\sqrt{|B|^2 - AC}/|A|$;
- (iii) if $A = 0$, $\Gamma_{A,B,C}$ is a line;
- (iv) conversely every circle and every line of $\mathbb{C}$ has the form $\Gamma_{A,B,C}$ for suitable A, B, C with $|B|^2 > AC$, and the triple (A, B, C) is unique up to a non-zero real factor.

The associated generalised circle is $\Gamma_{A,B,C}$ if $A \neq 0$ and $\Gamma_{0,B,C} \cup \{\infty\}$ if $A = 0$.

Proof. (i) $Az\bar{z} = A|z|^2$ is real because A is; $Bz + \bar{B}\bar{z} = Bz + \overline{Bz} = 2\Re(Bz)$ is real; C is real.
(ii) Let $A \neq 0$. Dividing $(*)$ by A and putting $b = \bar{B}/A$ (hence $\bar{b} = B/A$) and $c = C/A$, the equation becomes $z\bar{z} + \bar{b}z + b\bar{z} + c = 0$. Now

$$|z + b|^2 = (z + b)\overline{(z + b)} = z\bar{z} + z\bar{b} + b\bar{z} + |b|^2,$$

hence $(*)$ is equivalent to $|z + b|^2 = |b|^2 - c$. One computes

$$|b|^2 - c = \frac{|B|^2}{A^2} - \frac{C}{A} = \frac{|B|^2 - AC}{A^2} > 0$$

by hypothesis. Therefore $\Gamma_{A,B,C}$ is the circle with centre $-b = -\bar{B}/A$ and radius $\sqrt{|B|^2 - AC}/|A|$.

(iii) Let $A = 0$. Then $|B|^2 > 0$, that is $B \neq 0$. Putting $B = \beta_1 + i\beta_2$ and $z = x + iy$, one has $\Re(Bz) = \beta_1x - \beta_2y$ and $(*)$ becomes

$$2\beta_1x - 2\beta_2y + C = 0,$$

a first-degree equation with coefficients $(2\beta_1, -2\beta_2) \neq (0, 0)$: it is a line.

(iv) *Circle.* The circle with centre z_0 and radius $r > 0$ is $|z - z_0|^2 = r^2$, that is $z\bar{z} - \bar{z}_0z - z_0\bar{z} + |z_0|^2 - r^2 = 0$: it is $(*)$ with $A = 1$, $B = -\bar{z}_0$, $C = |z_0|^2 - r^2$, and $|B|^2 - AC = |z_0|^2 - |z_0|^2 + r^2 = r^2 > 0$.

Line. Let $ux + vy + w = 0$ with u, v, w real and $(u, v) \neq (0, 0)$. Put $A = 0$, $B = \frac{u}{2} - i\frac{v}{2}$ and $C = w$. Then, with $z = x + iy$,

$$Bz = \left(\frac{u}{2} - i\frac{v}{2}\right)(x + iy) = \left(\frac{u}{2}x + \frac{v}{2}y\right) + i\left(\frac{u}{2}y - \frac{v}{2}x\right),$$

hence $Bz + \bar{B}\bar{z} = 2\Re(Bz) = ux + vy$ and equation $(*)$ becomes exactly $ux + vy + w = 0$. Moreover $|B|^2 - AC = \frac{u^2 + v^2}{4} > 0$.

Uniqueness up to a real factor. If $\Gamma_{A,B,C} = \Gamma_{A',B',C'}$ as sets, observe that (ii)–(iii) determine the type (circle or line) according to whether A vanishes or not, and then: in the circle case, $A \neq 0$ and $A' \neq 0$, and centre and radius determine $\bar{B}/A = \bar{B}'/A'$ and $(|B|^2 - AC)/A^2 = (|B'|^2 - A'C')/A'^2$; putting $\lambda = A'/A \in \mathbb{R} \setminus \{0\}$ one gets $B' = \lambda B$ and hence $C' = \lambda C$. In the line case, $A = A' = 0$ and the coefficients $(2\beta_1, -2\beta_2, C)$ of one and the same line are proportional with a non-zero real factor. $\square$

Lemma 6.2: Decomposition of a Möbius transformation. Every Möbius transformation is a composition of finitely many maps of the following three types:

$$(T) \ t_\beta(z) = z + \beta \ (\beta \in \mathbb{C}); \quad (S) \ m_\alpha(z) = \alpha z \ (\alpha \in \mathbb{C}^*); \quad (I) \ \iota(z) = \frac{1}{z}.$$

Precisely: if $c = 0$ then $T = t_{b/d} \circ m_{a/d}$; if $c \neq 0$ then

$$T = t_{a/c} \circ m_{(bc-ad)/c} \circ \iota \circ t_d \circ m_c.$$

Proof. **Case $c = 0$.** Then $ad - bc = ad \neq 0$, hence $a \neq 0$ and $d \neq 0$, and $T(z) = \frac{az+b}{d} = \frac{a}{d}z + \frac{b}{d}$: it is $m_{a/d}$ followed by $t_{b/d}$, with $a/d \neq 0$. On the exceptional points: $T(\infty) = \infty$, and also $t_{b/d}(m_{a/d}(\infty)) = \infty$.

Case $c \neq 0$. One divides $az + b$ by $cz + d$:

$$\frac{a}{c}(cz + d) = az + \frac{ad}{c}, \quad \text{hence} \quad az + b = \frac{a}{c}(cz + d) + \left(b - \frac{ad}{c}\right) = \frac{a}{c}(cz + d) + \frac{bc - ad}{c},$$

whence

$$T(z) = \frac{az + b}{cz + d} = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d}.$$

The factor $\frac{bc-ad}{c}$ is non-zero because $ad - bc \neq 0$. Reading the expression from right to left one recognises the chain

$$z \xrightarrow{m_c} cz \xrightarrow{t_d} cz + d \xrightarrow{\iota} \frac{1}{cz + d} \xrightarrow{m_{(bc-ad)/c}} \frac{bc - ad}{c(cz + d)} \xrightarrow{t_{a/c}} T(z).$$

On the exceptional points: $z = -d/c$ gives $cz + d = 0$, then $\iota(0) = \infty$, then $m(\infty) = \infty$, then $t(\infty) = \infty = T(-d/c)$; and $z = \infty$ gives $m_c(\infty) = \infty$, $t_d(\infty) = \infty$, $\iota(\infty) = 0$, $m(0) = 0$, $t_{a/c}(0) = a/c = T(\infty)$. ✓ $\square$

Theorem 6.6: Möbius maps send generalised circles to generalised circles (R11.5). Let T be a Möbius transformation and $\hat{\Gamma}$ a generalised circle of $\hat{\mathbb{C}}$. Then $T(\hat{\Gamma})$ is a generalised circle. Since T is bijective (Proposition 6.7), T induces a bijection of the family of generalised circles onto itself.

Proof. By Lemma 6.2 it suffices to check the assertion for the three generating types, and in each case one works on the equation

$$Az\bar{z} + Bz + \bar{B}\bar{z} + C = 0, \quad A, C \in \mathbb{R}, \ B \in \mathbb{C}, \ |B|^2 - AC > 0$$

of Lemma 6.1. In all three cases one shows that the image has an equation of the same type, with coefficients A', B', C' such that $A', C' \in \mathbb{R}$ and

$$|B'|^2 - A'C' > 0,$$

a condition which guarantees that the image set is still a circle or a line and not the empty set or a single point.

Case (T): $w = z + \beta$, that is $z = w - \beta$. Substituting, with $\overline{w - \beta} = \bar{w} - \bar{\beta}$:

$$\begin{aligned} A(w - \beta)(\bar{w} - \bar{\beta}) + B(w - \beta) + \bar{B}(\bar{w} - \bar{\beta}) + C &= Aw\bar{w} - A\bar{\beta}w - A\beta\bar{w} + A|\beta|^2 + Bw - B\beta + \bar{B}\bar{w} - \bar{B}\bar{\beta} + C \\ &= Aw\bar{w} + (B - A\bar{\beta})w + (\bar{B} - A\beta)\bar{w} + (A|\beta|^2 - 2\Re(B\beta) + C). \end{aligned}$$

One therefore sets $A' = A$, $B' = B - A\bar{\beta}$, $C' = A|\beta|^2 - 2\Re(B\beta) + C$. One checks: A' and C' are real; and $\bar{B}' = \bar{B} - A\beta$ is precisely the coefficient of $\bar{w}$, as required by the form (*) (here

$A \in \mathbb{R}$ has been used). Finally

$$|B'|^2 - A'C' = |B|^2 - 2A\Re(B\beta) + A^2|\beta|^2 - A^2|\beta|^2 + 2A\Re(B\beta) - AC = |B|^2 - AC > 0,$$

where $|B - A\bar{\beta}|^2 = |B|^2 - 2\Re(B\bar{A}\bar{\beta}) + A^2|\beta|^2 = |B|^2 - 2A\Re(B\beta) + A^2|\beta|^2$ has been used.

Case (S): $w = \alpha z$ **with** $\alpha \neq 0$, **that is** $z = w/\alpha$. Substituting and multiplying by $|\alpha|^2 = \alpha\bar{\alpha} \neq 0$ (an operation which does not change the solution set, since $|\alpha|^2$ is real and positive):

$$A \frac{w\bar{w}}{|\alpha|^2} + B \frac{w}{\alpha} + \bar{B} \frac{\bar{w}}{\bar{\alpha}} + C = 0 \iff Aw\bar{w} + B\bar{\alpha}w + \bar{B}\alpha\bar{w} + C|\alpha|^2 = 0.$$

Hence $A' = A$, $B' = B\bar{\alpha}$, $C' = C|\alpha|^2$: those which must be real are real, and $\bar{B}' = \bar{B}\alpha$ is the coefficient of $\bar{w}$. Moreover

$$|B'|^2 - A'C' = |B|^2|\alpha|^2 - AC|\alpha|^2 = |\alpha|^2(|B|^2 - AC) > 0.$$

Case (I): $w = 1/z$, **that is** $z = 1/w$ (**for** $z \neq 0, \infty$). Substituting and multiplying by $w\bar{w} = |w|^2 \neq 0$:

$$A \frac{1}{w} \frac{1}{\bar{w}} + B \frac{1}{w} + \bar{B} \frac{1}{\bar{w}} + C = 0 \mid \cdot w\bar{w} \implies A + B\bar{w} + \bar{B}w + Cw\bar{w} = 0,$$

that is, rearranging,

$$\boxed{Cw\bar{w} + \bar{B}w + B\bar{w} + A = 0}$$

which is again of the form (*) with

$$A' = C, \quad B' = \bar{B}, \quad C' = A.$$

Indeed $A' = C$ and $C' = A$ are real, and the coefficient of $\bar{w}$ is $B = \bar{B} = \bar{B}'$, as required. The non-degeneracy condition is preserved:

$$|B'|^2 - A'C' = |\bar{B}|^2 - CA = |B|^2 - AC > 0.$$

The inversion therefore exchanges the coefficients $A \leftrightarrow C$ and conjugates B : it is the algebraic formulation of the fact that the inversion exchanges “passing through the origin” ($C = 0$) with “passing through ∞ ” ($A' = 0$, that is, the image is a line).

The exceptional points 0 and ∞ . One has to check that the inclusion or exclusion of 0 and ∞ is coherent.

- $0 \in \Gamma_{A,B,C}$ if and only if $C = 0$, that is if and only if $A' = 0$, that is if and only if the image is a line, that is if and only if the image generalised circle contains $\infty = \iota(0)$. ✓
- $\infty \in \hat{\Gamma}$ if and only if $A = 0$ (a line), that is if and only if $C' = 0$, that is if and only if the image contains $0 = \iota(\infty)$. ✓

Hence $\iota(\hat{\Gamma})$ is exactly the generalised circle associated with $(A', B', C') = (C, \bar{B}, A)$, points at infinity included.

Conclusion. Each of the three generators sends generalised circles to generalised circles; a finite composition of such maps does the same; by Lemma 6.2 every Möbius map is such a composition. Bijectivity follows by applying the same result to T^{-1} , which is again a Möbius map (Proposition 6.7). $\square$

Example 6.2: A circle that becomes a line, and conversely. Let $\Gamma : |z - 1| = 1$, that is $z\bar{z} - \bar{z} - z + 1 - 1 = 0$, namely $z\bar{z} - z - \bar{z} = 0$: here $A = 1$, $B = -1$, $C = 0$, and $|B|^2 - AC = 1 > 0$. ✓

By Theorem 6.6, case (I), the image under $w = 1/z$ has coefficients $A' = C = 0$, $B' = \bar{B} = -1$, $C' = A = 1$, that is

$$-w - \bar{w} + 1 = 0 \iff 2\Re w = 1 \iff \Re w = \frac{1}{2}.$$

The circle through the origin becomes a line (and the point $0 \in \Gamma$ goes to ∞).

Independent check on three points. $z = 2 \in \Gamma$ and $1/2$ has $\Re = 0.5 \checkmark$; $z = 1 + i \in \Gamma$ (since $|i| = 1$) and $\frac{1}{1+i} = \frac{1-i}{2} = 0.5 - 0.5i$ has $\Re = 0.5 \checkmark$; $z = 1 - i \in \Gamma$ and $\frac{1}{1-i} = 0.5 + 0.5i$ has $\Re = 0.5 \checkmark$.

Definition 6.6: Cross ratio. Let $z_2, z_3, z_4 \in \hat{\mathbb{C}}$ be distinct. Denote by $S_{z_2 z_3 z_4}$ the unique Möbius transformation that maps $z_2 \mapsto 1$, $z_3 \mapsto 0$, $z_4 \mapsto \infty$ (it exists and is unique by Theorem 6.7). For $z_1 \in \hat{\mathbb{C}}$ one sets

$$(z_1, z_2, z_3, z_4) := S_{z_2 z_3 z_4}(z_1).$$

If z_2, z_3, z_4 are all finite,

$$(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_4} \cdot \frac{z_2 - z_4}{z_2 - z_3} = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

Theorem 6.7: Three points, a single Möbius map. Let $z_2, z_3, z_4 \in \hat{\mathbb{C}}$ be distinct. There exists one and only one Möbius transformation S with $S(z_2) = 1$, $S(z_3) = 0$, $S(z_4) = \infty$. Consequently, given two triples of distinct points (z_2, z_3, z_4) and (w_2, w_3, w_4) , there exists one and only one Möbius map sending the first to the second, in order.

Proof. Step 1 — existence. If z_2, z_3, z_4 are finite one sets

$$S(z) = \frac{z - z_3}{z - z_4} \cdot \frac{z_2 - z_4}{z_2 - z_3}.$$

It is a Möbius map: written as $\frac{az+b}{cz+d}$ with $a = \frac{z_2-z_4}{z_2-z_3}$, $b = -z_3a$, $c = 1$, $d = -z_4$, one has $ad - bc = -z_4a + z_3a = a(z_3 - z_4) \neq 0$ because $a \neq 0$ (numerator and denominator are non-zero, the points being distinct) and $z_3 \neq z_4$. And by construction $S(z_3) = 0$, $S(z_4) = \infty$, $S(z_2) = \frac{z_2-z_3}{z_2-z_4} \cdot \frac{z_2-z_4}{z_2-z_3} = 1$.

If one of the three is ∞ one uses the formula obtained by suppressing the two factors containing it:

$$z_2 = \infty : S(z) = \frac{z - z_3}{z - z_4}; \quad z_3 = \infty : S(z) = \frac{z_2 - z_4}{z - z_4}; \quad z_4 = \infty : S(z) = \frac{z - z_3}{z_2 - z_3},$$

and in each case one checks directly that it is a Möbius map with the required values (for instance in the first case $S(\infty) = 1$ by the convention of Definition 6.4, since $a/c = 1/1$).

Step 2 — uniqueness. Let S and S' be two Möbius maps with those values. Then $M = S' \circ S^{-1}$ is a Möbius map (Proposition 6.7) fixing 1, 0 and ∞ . Written $M(z) = \frac{az+b}{cz+d}$:

- $M(\infty) = \infty$ forces $c = 0$ (if $c \neq 0$ one would have $M(\infty) = a/c \in \mathbb{C}$); hence $d \neq 0$ and $M(z) = \frac{a}{d}z + \frac{b}{d}$;
- $M(0) = 0$ gives $\frac{b}{d} = 0$, that is $b = 0$;
- $M(1) = 1$ gives $\frac{a}{d} = 1$.

Hence $M = \text{id}$, that is $S' = S$.

Step 3 — general triples. Let S and S' be the Möbius maps sending respectively (z_2, z_3, z_4) and (w_2, w_3, w_4) to $(1, 0, \infty)$. Then $T = S'^{-1} \circ S$ sends $z_j \mapsto w_j$ for $j = 2, 3, 4$. If T_1 is another Möbius map with the same property, $S' \circ T_1$ sends (z_2, z_3, z_4) to $(1, 0, \infty)$, hence by Step 2 $S' \circ T_1 = S$, that is $T_1 = S'^{-1} \circ S = T$. $\square$

Theorem 6.8: Invariance of the cross ratio (R11.5). Let T be a Möbius transformation and let $z_1, z_2, z_3, z_4 \in \hat{\mathbb{C}}$ with z_2, z_3, z_4 distinct. Then

$$(Tz_1, Tz_2, Tz_3, Tz_4) = (z_1, z_2, z_3, z_4).$$

Proof. Let $S = S_{z_2 z_3 z_4}$ be the Möbius map sending z_2, z_3, z_4 to $1, 0, \infty$, so that by definition $(z_1, z_2, z_3, z_4) = S(z_1)$.

Consider $S \circ T^{-1}$, which is a Möbius map (Proposition 6.7). It sends

$$Tz_2 \mapsto S(z_2) = 1, \quad Tz_3 \mapsto S(z_3) = 0, \quad Tz_4 \mapsto S(z_4) = \infty,$$

and the points Tz_2, Tz_3, Tz_4 are distinct because T is injective. By the uniqueness of Theorem 6.7,

$$S \circ T^{-1} = S_{Tz_2 Tz_3 Tz_4}.$$

Evaluating at Tz_1 :

$$(Tz_1, Tz_2, Tz_3, Tz_4) = S_{Tz_2 Tz_3 Tz_4}(Tz_1) = S(T^{-1}(Tz_1)) = S(z_1) = (z_1, z_2, z_3, z_4).$$

□

Corollary 6.4: Three points determine a unique generalised circle. Through three distinct points $z_2, z_3, z_4 \in \hat{\mathbb{C}}$ there passes one and only one generalised circle, and it is $S_{z_2 z_3 z_4}^{-1}(\mathbb{R})$, where $\mathbb{R} = \mathbb{R} \cup \{\infty\}$.

Proof. Step 1 — $\mathbb{R}$ is the only generalised circle through $1, 0, \infty$. $\mathbb{R}$ is a generalised circle (the real line plus ∞) and contains $1, 0, \infty$. Conversely, a generalised circle containing ∞ has the form $L \cup \{\infty\}$ with L a line (Definition 6.5), and a line containing 0 and 1 is the real axis, because through two distinct points there passes a single line.

Step 2 — existence. Let $S = S_{z_2 z_3 z_4}$. By Theorem 6.6 applied to S^{-1} , the set $\hat{\Gamma} = S^{-1}(\mathbb{R})$ is a generalised circle, and it contains $S^{-1}(1) = z_2$, $S^{-1}(0) = z_3$, $S^{-1}(\infty) = z_4$.

Step 3 — uniqueness. Let $\hat{\Gamma}'$ be a generalised circle containing z_2, z_3, z_4 . Then $S(\hat{\Gamma}')$ is a generalised circle (Theorem 6.6) containing $1, 0, \infty$, hence $S(\hat{\Gamma}') = \mathbb{R}$ by Step 1; since S is bijective, $\hat{\Gamma}' = S^{-1}(\mathbb{R}) = \hat{\Gamma}$. □

Theorem 6.9: Concyclicity criterion (R10.5, R11.6). Let $z_1, z_2, z_3, z_4 \in \hat{\mathbb{C}}$ be four distinct points. Then they belong to one and the same generalised circle if and only if

$$(z_1, z_2, z_3, z_4) \in \mathbb{R}.$$

In particular four distinct points of the complex plane are concyclic or collinear if and only if

$$\frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} \in \mathbb{R}.$$

Proof. Let $S = S_{z_2 z_3 z_4}$ and let $\hat{\Gamma}$ be the unique generalised circle through z_2, z_3, z_4 ; by Corollary 6.4, $\hat{\Gamma} = S^{-1}(\mathbb{R})$ and hence $S(\hat{\Gamma}) = \mathbb{R}$.

Step 1 — reduction. “The four points lie on one and the same generalised circle” is equivalent to “ $z_1 \in \hat{\Gamma}$ ”: indeed if $z_1 \in \hat{\Gamma}$ then all four lie on $\hat{\Gamma}$; conversely if a generalised circle $\hat{\Gamma}'$ contains all four of them, in particular it contains z_2, z_3, z_4 , hence $\hat{\Gamma}' = \hat{\Gamma}$ by uniqueness, and $z_1 \in \hat{\Gamma}$.

Step 2 — translation through S . Since S is a bijection of $\hat{\mathbb{C}}$ with $S(\hat{\Gamma}) = \hat{\mathbb{R}}$,

$$z_1 \in \hat{\Gamma} \iff S(z_1) \in \hat{\mathbb{R}} \iff (z_1, z_2, z_3, z_4) \in \mathbb{R} \cup \{\infty\}.$$

Step 3 — the value ∞ is excluded. $S(z_1) = \infty$ happens only for $z_1 = z_4$ (S is injective and $S(z_4) = \infty$), which is excluded because the four points are distinct. Hence $(z_1, z_2, z_3, z_4) \in \mathbb{C}$, and the condition becomes $(z_1, z_2, z_3, z_4) \in \mathbb{R}$. $\square$

Geometric reading of the criterion. If the four points are finite,

$$\arg(z_1, z_2, z_3, z_4) = \arg \frac{z_1 - z_3}{z_1 - z_4} - \arg \frac{z_2 - z_3}{z_2 - z_4},$$

that is the difference between the angle under which the segment z_3z_4 is seen from z_1 and the one under which it is seen from z_2 . The cross ratio is real ($\arg \in \{0, \pi\}$ modulo 2π) exactly when the two angles are equal or supplementary: these are the two statements of the inscribed angle theorem — equal angles if z_1, z_2 lie on the same side of the line z_3z_4 , supplementary if they lie on opposite sides (cyclic quadrilateral). The criterion of Theorem 6.9 is therefore the complex version of a theorem of elementary geometry (Volume I, Chapter 3). $\square$

Example 6.3: Two quadruples compared. (a) Concyclic quadruple. Take $z_1 = 1$, $z_2 = i$, $z_3 = -1$, $z_4 = -i$, the vertices of the square inscribed in the unit circle (μ_4 , Theorem 6.3). Then

$$(z_1, z_2, z_3, z_4) = \frac{(1 - (-1))(i - (-i))}{(1 - (-i))(i - (-1))} = \frac{2 \cdot 2i}{(1+i)(1+i)} = \frac{4i}{(1+i)^2} = \frac{4i}{2i} = 2 \in \mathbb{R}. \checkmark$$

The criterion confirms concyclicity, as it must.

(b) Non-concyclic quadruple. Take $z_1 = 0$, $z_2 = 1$, $z_3 = i$, $z_4 = 2$. Then

$$(z_1, z_2, z_3, z_4) = \frac{(0 - i)(1 - 2)}{(0 - 2)(1 - i)} = \frac{i}{-2 + 2i} = \frac{i(-2 - 2i)}{(-2)^2 + 2^2} = \frac{2 - 2i}{8} = 0.25 - 0.25i \notin \mathbb{R}.$$

Independent check. The circle through $0, 1, i$ has equation $x^2 + y^2 + Dx + Ey + F = 0$ with $F = 0$ (it passes through the origin), $1 + D = 0$ (it passes through $(1, 0)$) and $1 + E = 0$ (it passes through $(0, 1)$): it is $x^2 + y^2 - x - y = 0$. Substituting $(2, 0)$ one obtains $4 - 2 = 2 \neq 0$, hence $z_4 = 2$ does not belong to it; and the three points are not collinear. The two methods agree. $\checkmark$

Remark 6.6: Rotations of the sphere as Möbius maps. Through the stereographic projection (Chapter 7) the extended complex plane $\hat{\mathbb{C}}$ is identified with the sphere S^2 , and Möbius transformations become transformations of the sphere. Among them, those coming from rigid rotations of S^2 are exactly the Möbius maps of the form

$$z \mapsto \frac{\alpha z - \bar{\beta}}{\beta z + \bar{\alpha}}, \quad |\alpha|^2 + |\beta|^2 = 1,$$

that is those associated with matrices of $SU(2)$. We recover by another route the double cover $SU(2) \rightarrow SO(3)$ of Chapter 28: the circle, the quaternions and the sphere are three faces of the same structure. This statement is here a panorama and is not proved: the verification requires the explicit formula of the stereographic projection and the computation of the induced action, which are the subject of Chapter 7 and Chapter 28.

Exercise 6.3. Where does the complex inversion $z \mapsto 1/z$ send the circle $|z| = 2$? And the circle $|z| = 1$?

Solution. The circle $|z| = 2$ has equation $z\bar{z} - 4 = 0$, that is $A = 1$, $B = 0$, $C = -4$ ($|B|^2 - AC = 4 > 0$ ✓). By Theorem 6.6, case (I), the image has $A' = C = -4$, $B' = 0$, $C' = A = 1$, that is $-4w\bar{w} + 1 = 0$, namely $|w| = \frac{1}{2}$. The circle of radius 2 therefore goes to the circle of radius $\frac{1}{2}$, concentric with the origin. Directly: $|1/z| = 1/|z| = 1/2$. ✓
For $|z| = 1$: $A = 1$, $B = 0$, $C = -1$, and the image has $A' = -1$, $C' = 1$, that is $-w\bar{w} + 1 = 0$, namely $|w| = 1$. The unit circle is *globally* fixed. Beware: it is not fixed pointwise — $\iota(z) = 1/z = \bar{z}$ on $\mathbb{T}$, hence ι acts on it as conjugation, and the only fixed points are ± 1 . The complex inversion exchanges the interior and the exterior of the unit circle ($|z| < 1 \iff |1/z| > 1$): the unit circle is its invariant “boundary”.

6.5. Assigned exercises

Answers are in the Teacher’s key (Section 6.8).

Exercise 6.4: The minimal polynomial of $\cos 72^\circ$ by de Moivre’s route. *Aim:* to use de Moivre’s formula to obtain an algebraic identity and to recognise its geometric meaning. *Estimated time:* 35 minutes. *Type:* symbolic/proof-based.

- Starting from $\cos 5\theta = 16c^5 - 20c^3 + 5c$ (Corollary 6.1) and from $\theta = \frac{2\pi}{5}$, write down the equation satisfied by $c = \cos \frac{2\pi}{5}$.
- Factor the polynomial obtained, singling out $(c - 1)$, and check that the remaining factor is $(4c^2 + 2c - 1)^2$.
- Deduce Proposition 6.6 again and explain why the quadratic factor appears squared.
- Repeat the procedure with $\theta = \frac{2\pi}{3}$ and $\cos 3\theta$, obtaining the minimal polynomial of $\cos \frac{2\pi}{3}$; comment on the difference in degree.

Exercise 6.5: Concyclicity by numerical means. *Aim:* to apply the cross-ratio criterion and to compare it with the circumscribed circle method. *Estimated time:* 40 minutes. *Type:* numerical/computational.

Consider the quadruples

$$Q_1 = \{1, i, -1, 2i\}, \quad Q_2 = \{2, 1+i, 1-i, 0\}, \quad Q_3 = \{0, 1, 2, 3+0i\}.$$

- For each of them compute (z_1, z_2, z_3, z_4) with the formula of Definition 6.6 and say whether the points are concyclic or collinear.
- For the quadruples that turn out to be concyclic, determine the centre and the radius of the circle with Lemma 6.1, and check that all four points satisfy it.
- Verify the invariance of Theorem 6.8 by applying to Q_2 the Möbius map $T(z) = \frac{z-1}{z+1}$ and recomputing the cross ratio on the images.
- What happens to the cross ratio if z_1 and z_2 are exchanged? And z_3 and z_4 ? Check it on Q_1 and formulate a conjecture.

Exercise 6.6: Sums of roots and the eight-point discrete transform. *Aim:* to see Theorem 6.5 at work as a computational tool. *Estimated time:* 45 minutes. *Type:* numerical/computational.

Let $n = 8$ and $\zeta_8 = e^{i\pi/4} = \frac{\sqrt{2}}{2}(1+i)$.

- (a) Draw up the table of $S(j) = \sum_{k=0}^7 \zeta_8^{jk}$ for $j = 0, 1, \dots, 8$ and compare it with Theorem 6.5.
- (b) Deduce $\sum_{k=0}^7 \cos \frac{\pi jk}{4}$ and $\sum_{k=0}^7 \sin \frac{\pi jk}{4}$ for the same j .
- (c) Let $x = (1, 0, 0, 0, 1, 0, 0, 0)$. Compute $X_j = \sum_{k=0}^7 x_k \zeta_8^{-jk}$ for $j = 0, \dots, 7$ and verify the inversion identity $x_k = \frac{1}{8} \sum_{j=0}^7 X_j \zeta_8^{jk}$ at least for $k = 0$ and $k = 1$.
- (d) Explain, using Theorem 6.5, why the inversion works for *every* vector $x \in \mathbb{C}^8$.

6.6. Worked exercises

Exercise 6.7: Images of lines and circles under $z \mapsto 1/z$. *Type:* symbolic with a numerical check. Determine, using the coefficients (A, B, C) of Lemma 6.1, the image under $w = 1/z$

- (a) of the line $\Re z = 1$;
 (b) of the circle $|z - 2| = 1$;

and check the result on three points in both cases. What happens to the centre of the circle in (b)?

Solution. Strategy. One writes each object in the form $Az\bar{z} + Bz + \bar{B}\bar{z} + C = 0$, applies the inversion rule $(A, B, C) \mapsto (C, \bar{B}, A)$ of Theorem 6.6, and converts the result back into geometric form with Lemma 6.1.

Step 1 — the line $\Re z = 1$. From $z + \bar{z} = 2\Re z$ one has $\Re z = 1 \iff z + \bar{z} - 2 = 0$, that is

$$A = 0, \quad B = 1, \quad C = -2, \quad |B|^2 - AC = 1 > 0. \checkmark$$

The image has $A' = C = -2$, $B' = \bar{B} = 1$, $C' = A = 0$:

$$-2w\bar{w} + w + \bar{w} = 0 \iff w\bar{w} - \frac{1}{2}(w + \bar{w}) = 0 \iff \left|w - \frac{1}{2}\right|^2 = \frac{1}{4},$$

where the last step uses $|w - \frac{1}{2}|^2 = w\bar{w} - \frac{1}{2}w - \frac{1}{2}\bar{w} + \frac{1}{4}$. This is the circle with centre $\frac{1}{2}$ and radius $\frac{1}{2}$. Alternatively, with the formula of Lemma 6.1(ii): centre $-\bar{B}'/A' = -1/(-2) = \frac{1}{2}$; radius $\sqrt{|B'|^2 - A'C'}/|A'| = \sqrt{1}/2 = \frac{1}{2}$. $\checkmark$

Step 2 — check on three points. $z = 1 \mapsto 1$ and $|1 - 0.5| = 0.5 \checkmark$; $z = 1 + i \mapsto \frac{1-i}{2} = 0.5 - 0.5i$ and $|-0.5i| = 0.5 \checkmark$; $z = 1 + 2i \mapsto \frac{1-2i}{5} = 0.2 - 0.4i$ and $|0.2 - 0.5 - 0.4i| = \sqrt{0.09 + 0.16} = \sqrt{0.25} = 0.5 \checkmark$. Moreover ∞ , which lies on the generalised line, goes to 0, which lies on the image circle (which passes through the origin because $C' = 0$). $\checkmark$

Step 3 — the circle $|z - 2| = 1$. One expands: $|z - 2|^2 = (z - 2)(\bar{z} - 2) = z\bar{z} - 2z - 2\bar{z} + 4 = 1$, hence

$$z\bar{z} - 2z - 2\bar{z} + 3 = 0, \quad A = 1, \quad B = -2, \quad C = 3, \quad |B|^2 - AC = 4 - 3 = 1 > 0. \checkmark$$

The image has $A' = 3$, $B' = -2$, $C' = 1$:

$$3w\bar{w} - 2w - 2\bar{w} + 1 = 0 \iff w\bar{w} - \frac{2}{3}(w + \bar{w}) + \frac{1}{3} = 0 \iff \left|w - \frac{2}{3}\right|^2 = \frac{4}{9} - \frac{1}{3} = \frac{1}{9}.$$

It is the circle with centre $\frac{2}{3}$ and radius $\frac{1}{3}$. With the formula: centre $-\bar{B}'/A' = 2/3$; radius $\sqrt{4 - 3}/3 = \frac{1}{3}$. $\checkmark$

Step 4 — check on three points. $z = 1 \mapsto 1$ and $|1 - \frac{2}{3}| = \frac{1}{3} \checkmark$; $z = 3 \mapsto \frac{1}{3}$ and $|\frac{1}{3} - \frac{2}{3}| = \frac{1}{3} \checkmark$; $z = 2 + i \mapsto \frac{2-i}{5} = 0.4 - 0.2i$ and $|0.4 - 0.66667 - 0.2i| = \sqrt{0.071111 + 0.04} = \sqrt{0.111111} = 0.33333 \checkmark$.

Interpretation: the centre does not go to the centre. The centre 2 of the initial circle goes to $\frac{1}{2}$, which is *not* the centre $\frac{2}{3}$ of the image circle. There is no contradiction: Theorem 6.6 asserts that the circle as a *set* goes to a circle as a set, not that the centres correspond. The centre is not a Möbius-invariant notion, because it cannot be defined in terms of circles, lines and angles alone: it depends on the Euclidean metric, which Möbius maps do not preserve. What is preserved is the *symmetric point* with respect to the circle, and indeed the centre is the symmetric point of ∞ — but Möbius maps move ∞ .

Typical errors. (1) Transforming the centre and the radius instead of the equation: the result is wrong, as Step 4 shows. (2) Forgetting that $|B|^2 - AC > 0$ is part of the statement: without it the equation may describe a single point (for instance $z\bar{z} = 0$) or the empty set ($z\bar{z} + 1 = 0$), and the conclusion fails. (3) Neglecting the points 0 and ∞ : the circle of Step 1 passes through the origin precisely because the line passed through ∞ .

Extension. The same computation shows that $z \mapsto 1/z$ sends a line *not* passing through the origin ($A = 0, C \neq 0$) to a circle through the origin ($A' = C \neq 0, C' = 0$), and a line through the origin ($A = 0, C = 0$) to a line through the origin ($A' = 0, C' = 0$): this is the classification into four cases found in the MIT 18.04 notes, Topic 10, §10.7.2.

Exercise 6.8: The product of the diagonals of a regular polygon. *Type:* proof-based with a numerical check. Let $n \geq 2$. Prove that

$$\prod_{k=1}^{n-1} (1 - \zeta_n^k) = n \quad \text{and} \quad \prod_{k=1}^{n-1} 2 \sin \frac{k\pi}{n} = n.$$

Interpret the second identity on the regular polygon inscribed in the unit circle and check it numerically for $n = 5$ and $n = 6$.

Solution. Strategy. One factors $z^n - 1$ with the roots of unity (Theorem 6.2), isolates the factor $(z - 1)$ and evaluates at $z = 1$ the resulting function, which is a polynomial and therefore can be evaluated without limits.

Step 1 — the factorisation. The polynomials $z^n - 1$ and $\prod_{k=0}^{n-1} (z - \zeta_n^k)$ are both monic of degree n and have the same n distinct roots $\zeta_n^0, \dots, \zeta_n^{n-1}$ (Theorem 6.2); their difference is a polynomial of degree $< n$ with n roots, hence it is zero. Therefore

$$z^n - 1 = \prod_{k=0}^{n-1} (z - \zeta_n^k) = (z - 1) \prod_{k=1}^{n-1} (z - \zeta_n^k).$$

Step 2 — evaluation at $z = 1$. On the other hand, by the telescoping identity (Theorem 6.4, Step 1), $z^n - 1 = (z - 1)(1 + z + \dots + z^{n-1})$. Comparing with Step 1 and cancelling the factor $(z - 1)$, which is legitimate as an identity between polynomials (the ring $\mathbb{C}[z]$ is an integral domain), one obtains

$$1 + z + \dots + z^{n-1} = \prod_{k=1}^{n-1} (z - \zeta_n^k) \quad \text{for every } z.$$

Setting $z = 1$: the left-hand side equals n , hence $\prod_{k=1}^{n-1} (1 - \zeta_n^k) = n$.

Step 3 — passing to moduli. Taking the modulus of both sides (and recalling that $n > 0$ is real),

$$\prod_{k=1}^{n-1} |1 - \zeta_n^k| = n.$$

By the computation made in Theorem 6.3(ii), $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$; with $\theta = \frac{2\pi k}{n}$ and $0 < \frac{k\pi}{n} < \pi$ (hence a positive sine),

$$|1 - \zeta_n^k| = 2 \sin \frac{k\pi}{n}, \quad \text{whence} \quad \prod_{k=1}^{n-1} 2 \sin \frac{k\pi}{n} = n.$$

Step 4 — numerical check. For $n = 5$: $2 \sin 36^\circ = 1.175570$, $2 \sin 72^\circ = 1.902113$, $2 \sin 108^\circ = 1.902113$, $2 \sin 144^\circ = 1.175570$. The product is

$$(1.175570 \cdot 1.902113)^2 = (2.236068)^2 = 5.000000. \checkmark$$

(One recognises $1.175570 \cdot 1.902113 = \sqrt{5}$: the product of the side by the diagonal of the unit regular pentagon equals $\sqrt{5}$.)

For $n = 6$: $2 \sin 30^\circ = 1$, $2 \sin 60^\circ = 1.732051$, $2 \sin 90^\circ = 2$, $2 \sin 120^\circ = 1.732051$, $2 \sin 150^\circ = 1$. The product is

$$1 \cdot 1.732051 \cdot 2 \cdot 1.732051 \cdot 1 = 2 \cdot 3.000000 = 6.000000. \checkmark$$

Second check, independent of the previous one: for $n = 6$ one computes the complex product $\prod_{k=1}^5 (1 - \zeta_6^k)$ by grouping the conjugate factors. With $\zeta_6 = 0.5 + 0.866025i$: $(1 - \zeta_6)(1 - \zeta_6^5) = |1 - \zeta_6|^2 = (0.5)^2 + (0.866025)^2 = 1$; $(1 - \zeta_6^2)(1 - \zeta_6^4) = |1 - \zeta_6^2|^2 = (1.5)^2 + (0.866025)^2 = 3$; $(1 - \zeta_6^3) = 1 - (-1) = 2$. Product: $1 \cdot 3 \cdot 2 = 6$. $\checkmark$

Interpretation. $|1 - \zeta_n^k|$ is the distance from the vertex $V_0 = 1$ to the vertex V_k : for $k = 1$ and $k = n - 1$ these are the two adjacent sides, for the other k the diagonals issuing from V_0 . The identity therefore says that *the product of the $n - 1$ distances from one vertex to all the others, in a regular polygon inscribed in the unit circle, equals exactly n* . It is a fact that is anything but obvious geometrically, and it follows from a simple comparison between two factorisations of the same polynomial.

Typical errors. (1) Writing $\prod_{k=1}^{n-1} (1 - \zeta^k) = \lim_{z \rightarrow 1} \frac{z^n - 1}{z - 1}$ and treating it as an indeterminate form to be resolved with de l'Hôpital's rule: it works, but it hides the fact that the identity is algebraic and requires no limits (the quotient $\frac{z^n - 1}{z - 1}$ is the polynomial $1 + z + \dots + z^{n-1}$). (2) Forgetting to exclude $k = 0$: the factor $(1 - \zeta^0) = 0$ would annihilate everything. (3) Writing $|1 - \zeta_n^k| = 2 \sin \frac{2\pi k}{n}$, forgetting the half angle: already for $n = 4$, $k = 2$ would give $2 \sin 180^\circ = 0$ instead of 2.

Extension. The same argument can be repeated on the cyclotomic polynomial Φ_n , that is on the monic factor of $z^n - 1$ whose roots are exactly the primitive n -th roots (Proposition 6.3): compute $\Phi_n(1)$ for $n = 4, 6, 8, 9, 12$ — one finds respectively 2, 1, 2, 3, 1 — and conjecture the general rule in terms of the factorisation of n . It is the classical starting point for the study of cyclotomic extensions (Volume II, Chapter 7).

6.7. Self-assessment quiz

- (Definition) An n -th root of unity ζ is called *primitive* if
 - $\zeta \neq 1$
 - its multiplicative order is exactly n
 - ζ has positive imaginary part
 - ζ is irrational
- (Hypothesis) In Lemma 6.1 the hypothesis $|B|^2 > AC$ serves to
 - make the left-hand side real
 - exclude that the set be empty or reduced to a single point
 - guarantee $A \neq 0$
 - guarantee that the curve is bounded
- (Symbolic computation) The sum $\sum_{k=0}^{11} \zeta_{12}^{3k}$ equals

- A) 12 B) 0 C) 3 D) 4
4. (Numerical computation) The side of the regular polygon with 12 sides inscribed in the unit circle is, to five decimal places,
A) 0.25882 B) 0.51764 C) 0.86603 D) 1.00000
5. (Proof step) In the proof of Theorem 6.6, the substitution $z = 1/w$ in the equation $Az\bar{z} + Bz + \bar{B}\bar{z} + C = 0$ must be followed by multiplication by $w\bar{w}$ because
A) otherwise the equation is not real B) it serves to clear the denominators without changing the solution set, since $w\bar{w} = |w|^2 \neq 0$ C) it serves to normalise the radius D) it serves to make A' positive
6. (Error diagnosis) A student writes: “ $\sum_{k=0}^{n-1} \zeta_n^{jk} = \frac{\zeta_n^{jn} - 1}{\zeta_n^j - 1} = \frac{1-1}{\zeta_n^j - 1} = 0$ for every integer j .” The error is
A) $\zeta_n^{jn} \neq 1$ B) the closed formula does not hold when $\zeta_n^j = 1$, that is when $n \mid j$, and in that case the sum equals n C) the sum should start from $k = 1$ D) ζ_n^j is not a root of unity
7. (Transfer) Four distinct points $z_1, z_2, z_3, z_4 \in \mathbb{C}$ are concyclic or collinear if and only if
A) $|z_1| = |z_2| = |z_3| = |z_4|$ B) $(z_1, z_2, z_3, z_4) \in \mathbb{R}$ C) $z_1 + z_2 + z_3 + z_4 = 0$ D) $(z_1, z_2, z_3, z_4) = 1$
8. (Source and metacognition) The proof that the cross ratio is real if and only if the four points lie on a circle or on a line is found in
A) Ahlfors, *Complex Analysis*, 3rd ed., ch. 3, §3.2, Theorem 13, p. 79 B) Ahlfors, ch. 1, §2.2, p. 15 C) Ahlfors, ch. 3, §1.4, p. 59 D) Ahlfors, ch. 4, §2.1, p. 114

6.8. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis.

- 1) **B.** *Rationale:* it is Definition 6.2; by Proposition 6.3 this is equivalent to $\gcd(k, n) = 1$ if $\zeta = \zeta_n^k$. *Distractors:* A is necessary but not sufficient ($\zeta_6^2 = \zeta_3 \neq 1$ is not a primitive sixth root); C is irrelevant and false in general (ζ_n^{n-1} has negative imaginary part and is primitive); D confuses “primitive” with “irrational” ($\zeta_4 = i$ is primitive and has rational parts). *Diagnosed error:* confusing “different from 1” with “of maximal order”. *Source:* Ahlfors, ch. 1, §2.2, pp. 15–16. *Prerequisite:* the order of an element in a group. *Remedial work:* tabulate the orders of ζ_{12}^k for $k = 0, \dots, 11$.
- 2) **B.** *Rationale:* in the proof of Lemma 6.1(ii) the quantity $|b|^2 - c = (|B|^2 - AC)/A^2$ is the *square of the radius*: if it vanished the set would reduce to a point, if it were negative it would be empty. *Distractors:* A is true but follows from $A, C \in \mathbb{R}$, not from $|B|^2 > AC$; C is false (the case $A = 0$ is allowed and is the case of lines); D is false (lines are not bounded). *Diagnosed error:* attributing the wrong conclusion to a hypothesis. *Source:* Ahlfors, ch. 3, §3.2, pp. 79–80. *Prerequisite:* completing the square. *Remedial work:* write the equations $z\bar{z} = 0$ and $z\bar{z} + 1 = 0$ and see what they describe.
- 3) **B.** *Rationale:* $12 \nmid 3$, hence Theorem 6.5 gives 0. *Distractors:* A is the value one would get if $12 \mid 3$; C confuses the value of j with that of the sum; D confuses $n/j = 4$ with the sum. *Diagnosed error:* applying the “vanishing denominator” case when it does not occur. *Source:* Theorem 6.5; Ahlfors, ch. 1, §2.2, Exercise 4, p. 16. *Prerequisite:* divisibility.

Remedial work: compute $\sum_k \zeta_{12}^{jk}$ for $j = 0, 3, 4, 6, 12$.

4) **B. Rationale:** by Theorem 6.3(ii) the side is $2 \sin \frac{\pi}{12} = 2 \sin 15^\circ = 2 \cdot 0.258819 = 0.517638$.

Distractors: A is $\sin 15^\circ = 0.25882$, that is half the value (the factor 2 has been forgotten); C is $\sqrt{3}/2 = \cos 30^\circ = 0.86603$, a familiar trigonometric value but with no role here (it is the apothem of the unit regular hexagon, not a side); D is the side of the hexagon, $2 \sin 30^\circ = 1$, that is the correct answer but for $n = 6$ instead of $n = 12$. *Diagnosed error:* using $\frac{2\pi}{n}$ instead of $\frac{\pi}{n}$ in the argument of the sine, or forgetting the factor 2. *Source:* Theorem 6.3. *Prerequisite:* $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$. *Remedial work:* derive the formula for the side again by factoring out $e^{i\theta/2}$.

5) **B. Rationale:** multiplying by a factor that *never* vanishes is an equivalence between equations; if $w = 0$ were allowed, the multiplication could introduce spurious solutions, and indeed the point $w = 0$ (the image of ∞) is treated separately in the proof. *Distractors:* A is false, the equation is already real (Lemma 6.1(i)); C and D are beside the point, the radius is not normalised and $A' = C$ may perfectly well be negative. *Diagnosed error:* not distinguishing between a legitimate manipulation and one that changes the solution set. *Source:* Theorem 6.6, case (I); Orloff, MIT 18.04, Topic 10, §10.7.2. *Prerequisite:* equivalence of equations. *Remedial work:* redo case (I) keeping explicit track of the hypothesis $w \neq 0$.

6) **B. Rationale:** it is exactly Case 1 of Theorem 6.5: if $n \mid j$ then $\zeta_n^j = 1$, the denominator vanishes and the closed formula is inapplicable; the direct sum gives n . *Distractors:* A is false, $\zeta_n^{jn} = e^{2\pi i j} = 1$ always; C would change the sum but is not the error committed; D is false. *Diagnosed error:* using a formula outside its domain of validity (a masked division by zero). *Source:* Theorem 6.5. *Prerequisite:* the finite geometric sum. *Remedial work:* rewrite the telescoping identity $(q - 1) \sum q^k = q^n - 1$ and observe that it is the division, not the identity, that requires $q \neq 1$.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (continued).

7) **B. Rationale:** it is Theorem 6.9. *Distractors:* A is sufficient (the four points would lie on a circle centred at the origin) but not necessary, and it does not cover collinear points; C is false and unrelated (a vanishing sum is the condition for the centroid to be at the origin, see Theorem 6.4); D is a particular value, which occurs only for special quadruples. *Diagnosed error:* mistaking a particular sufficient condition for the characterisation. *Source:* Ahlfors, ch. 3, §3.2, Theorem 13, p. 79. *Prerequisite:* the definition of the cross ratio. *Remedial work:* redo Example 6.3 with other quadruples, at least one of them collinear.

8) **A. Rationale:** Ahlfors' Theorem 13, in section 3.2 "The Cross Ratio" of chapter 3, is exactly this statement, with the same proof adopted here (reduction to the real axis by means of the Möbius map sending three points to $1, 0, \infty$). *Distractors:* B is the section on the binomial equation (roots of unity, de Moivre); C is the section on rational functions; D is the index of a point with respect to a closed curve, the subject of Chapter 12. *Diagnosed error:* citing the right chapter but the wrong section, or citing by general topic instead of by statement. *Source:* see the section "Sources and bibliography". *Prerequisite:* being able to read a reference with chapter, section, statement number and page. *Remedial work:* locate Theorems 12 and 13 in Ahlfors' text and compare their proofs with those of this chapter.

BEST PRACTICE

Assigned exercise 6.4 — answers. (a) With $\theta = \frac{2\pi}{5}$ one has $5\theta = 2\pi$ and $\cos 5\theta = 1$, hence $16c^5 - 20c^3 + 5c - 1 = 0$. (b) $c = 1$ annihilates the polynomial ($16 - 20 + 5 - 1 = 0$), and the division gives $16c^5 - 20c^3 + 5c - 1 = (c - 1)(16c^4 + 16c^3 - 4c^2 - 4c + 1)$. One then checks that $16c^4 + 16c^3 - 4c^2 - 4c + 1 = (4c^2 + 2c - 1)^2$ by expanding the square:

$(4c^2)^2 + (2c)^2 + 1 + 2(4c^2)(2c) + 2(4c^2)(-1) + 2(2c)(-1) = 16c^4 + 4c^2 + 1 + 16c^3 - 8c^2 - 4c = 16c^4 + 16c^3 - 4c^2 - 4c + 1$. **✓(c)** Since $c = \cos 72^\circ \neq 1$, the factor $4c^2 + 2c - 1$ must vanish, whence $c = \frac{-1+\sqrt{5}}{4}$ (the other root is negative), which is Proposition 6.6. The factor appears *squared* because $\cos 5\theta = 1$ has as solutions in $[0, 2\pi)$ the angles $\theta = \frac{2\pi k}{5}$, $k = 0, \dots, 4$, that is five angles but only *three* distinct values of the cosine ($1, \cos 72^\circ, \cos 144^\circ$), because $\cos \frac{2\pi k}{5} = \cos \frac{2\pi(5-k)}{5}$: the pairs $k = 1, 4$ and $k = 2, 3$ give the same cosine. A fifth-degree polynomial with three distinct roots, two of which are double, is exactly $(c-1)(4c^2+2c-1)^2$. **(d)** With $\theta = \frac{2\pi}{3}$: $\cos 3\theta = 1$ gives $4c^3 - 3c - 1 = 0$, that is $(c-1)(4c^2+4c+1) = (c-1)(2c+1)^2 = 0$; since $c \neq 1$, $c = -\frac{1}{2}$ (double, for the same reason of symmetry). The degree of the remaining factor is 1 instead of 2: the minimal polynomial of $\cos \frac{2\pi}{3}$ has degree $\varphi(3)/2 = 1$, that of $\cos \frac{2\pi}{5}$ has degree $\varphi(5)/2 = 2$. It is the quantity that the Gauss–Wantzel theorem (Remark 6.4; Volume II, Chapter 7) requires to be a power of 2.

BEST PRACTICE

Assigned exercise 6.5 — answers. **(a)** $Q_1 = \{1, i, -1, 2i\}$: $(1, i, -1, 2i) = \frac{(1-(-1))(i-2i)}{(1-2i)(i-(-1))} = \frac{2 \cdot (-i)}{(1-2i)(1+i)} = \frac{-2i}{3-i} = \frac{-2i(3+i)}{10} = \frac{2-6i}{10} = 0.2 - 0.6i \notin \mathbb{R}$: *not concyclic*. (Indeed $1, i, -1$ lie on the unit circle, and $2i$ does not.) $Q_2 = \{2, 1+i, 1-i, 0\}$: $(2, 1+i, 1-i, 0) = \frac{(2-(1-i))(1+i-0)}{(2-0)(1+i-(1-i))} = \frac{(1+i)(1+i)}{2 \cdot 2i} = \frac{2i}{4i} = 0.5 \in \mathbb{R}$: *concyclic*. $Q_3 = \{0, 1, 2, 3\}$: $(0, 1, 2, 3) = \frac{(0-2)(1-3)}{(0-3)(1-2)} = \frac{4}{3} \in \mathbb{R}$: *collinear* (they all lie on the real axis), the degenerate case allowed by Theorem 6.9. **(b)** For Q_2 : the circle through $0, 2$ and $1+i$ has equation $z\bar{z} + Bz + \bar{B}\bar{z} + C = 0$ with $C = 0$ (it passes through the origin); from $z = 2$: $4 + 2B + 2\bar{B} = 0$, that is $\Re B = -1$; from $z = 1+i$: $2 + B(1+i) + \bar{B}(1-i) = 0$, that is $2 + 2\Re(B(1+i)) = 0$, that is $\Re B - \Im B = -1$, whence $\Im B = 0$ and $B = -1$. The circle is $z\bar{z} - z - \bar{z} = 0$, that is $|z-1| = 1$: centre 1 , radius 1 (Lemma 6.1(ii): centre $-\bar{B}/A = 1$, radius $\sqrt{1-0}/1 = 1$). Check: $|2-1| = 1 \checkmark$, $|0-1| = 1 \checkmark$, $|1+i-1| = 1 \checkmark$, $|1-i-1| = 1 \checkmark$. **(c)** $T(z) = \frac{z-1}{z+1}$ is a Möbius map ($ad-bc = 1+1 = 2 \neq 0$). The images are $T(2) = \frac{1}{3}$, $T(1+i) = \frac{i}{2+i} = \frac{i(2-i)}{5} = \frac{1+2i}{5} = 0.2 + 0.4i$, $T(1-i) = 0.2 - 0.4i$, $T(0) = -1$. The cross ratio of the images is

$$\frac{(\frac{1}{3} - (0.2 - 0.4i))(0.2 + 0.4i - (-1))}{(\frac{1}{3} - (-1))(0.2 + 0.4i - (0.2 - 0.4i))} = \frac{(0.133333 + 0.4i)(1.2 + 0.4i)}{1.333333 \cdot 0.8i} = \frac{0.533333i}{1.066667i} = 0.5,$$

where in the numerator the real part vanishes exactly: $0.133333 \cdot 1.2 - 0.4 \cdot 0.4 = 0.16 - 0.16 = 0$, whereas the imaginary part equals $0.133333 \cdot 0.4 + 0.4 \cdot 1.2 = 0.053333 + 0.48 = 0.533333$. The result coincides with the value computed in (a), in agreement with Theorem 6.8. **✓(d)** Exchanging z_1 and z_2 the cross ratio λ becomes $1/\lambda$; exchanging z_3 and z_4 it also becomes $1/\lambda$ (numerator and denominator are exchanged symmetrically). On Q_1 : $\lambda = 0.2 - 0.6i$ and $1/\lambda = \frac{0.2+0.6i}{0.04+0.36} = 0.5 + 1.5i$, which is what one obtains by exchanging. Conjecture (correct): the 24 permutations of the four points give at most the six values $\lambda, \frac{1}{\lambda}, 1-\lambda, \frac{1}{1-\lambda}, \frac{\lambda-1}{\lambda}, \frac{\lambda}{\lambda-1}$; in particular *membership of $\mathbb{R}$ does not depend on the order of the points*, as it must be, because concyclicity does not depend on the order.

BEST PRACTICE

Assigned exercise 6.6 — answers. **(a)** $S(0) = 8, S(8) = 8; S(j) = 0$ for $j = 1, \dots, 7$. This is exactly Theorem 6.5: $8 \mid j$ only for $j = 0$ and $j = 8$. **(b)** Separating real and imaginary parts: $\sum_k \cos \frac{\pi j k}{4} = 8$ and $\sum_k \sin \frac{\pi j k}{4} = 0$ for $j = 0, 8$; both sums equal 0 for $j = 1, \dots, 7$. Check for $j = 2$: the angles are $0^\circ, 90^\circ, 180^\circ, 270^\circ$ repeated twice, and

both the cosines $(1, 0, -1, 0)$ and the sines $(0, 1, 0, -1)$ have zero sum on every group of four. **✓(c)** With $x = (1, 0, 0, 0, 1, 0, 0, 0)$ one has $X_j = \zeta_8^0 + \zeta_8^{-4j} = 1 + (-1)^j$, since $\zeta_8^{-4} = e^{-i\pi} = -1$. Hence $X_j = 2$ for even j and $X_j = 0$ for odd j : $X = (2, 0, 2, 0, 2, 0, 2, 0)$. Inversion for $k = 0$: $\frac{1}{8} \sum_j X_j = \frac{1}{8}(2 + 2 + 2 + 2) = 1 = x_0$ ✓. For $k = 1$: $\frac{1}{8} \sum_j X_j \zeta_8^j = \frac{1}{8} \cdot 2(\zeta_8^0 + \zeta_8^2 + \zeta_8^4 + \zeta_8^6) = \frac{1}{4}(1 + i - 1 - i) = 0 = x_1$ ✓. **(d)** Because $\frac{1}{8} \sum_j X_j \zeta_8^{jk} = \frac{1}{8} \sum_j \sum_m x_m \zeta_8^{-jm} \zeta_8^{jk} = \sum_m x_m \cdot \frac{1}{8} \sum_j \zeta_8^{j(k-m)} = \sum_m x_m \delta_{km} = x_k$, where the exchange of the sums is legitimate because they are finite and the key step is Theorem 6.5. The argument uses nothing about x : it holds for every $x \in \mathbb{C}^8$, and with 8 replaced by n for every $x \in \mathbb{C}^n$ (Chapter 22).

6.9. Chapter formulary

The circle as a group.

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}, \quad \mathbb{T} \cong SO(2) \quad (e^{i\theta} \leftrightarrow R_\theta), \quad \mathbb{T} \cong \mathbb{R}/2\pi\mathbb{Z}.$$

For $|z| = 1$: $z^{-1} = \bar{z}$. De Moivre, for every $n \in \mathbb{Z}$:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta, \quad \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta, \quad \cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.$$

Roots of unity. $\zeta_n = e^{2\pi i/n}$, $\mu_n = \{\zeta_n^k : k = 0, \dots, n-1\}$, $|\mu_n| = n$, $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$.

$$\sum_{k=0}^{n-1} \zeta_n^k = 0 \quad (n \geq 2), \quad \prod_{k=0}^{n-1} \zeta_n^k = (-1)^{n+1}, \quad \prod_{k=1}^{n-1} (1 - \zeta_n^k) = n.$$

Order of ζ_n^k : $n/\gcd(k, n)$; primitive roots: $\varphi(n)$. **Orthogonality (including the vanishing-denominator case):**

$$\sum_{k=0}^{n-1} \zeta_n^{jk} = \begin{cases} n, & n \mid j \\ 0, & n \nmid j, \end{cases} \quad (\text{here } q = 1 : \text{the closed formula does not apply}), \quad \frac{1}{n} \sum_{k=0}^{n-1} \zeta_n^{k(m-\ell)} = \delta_{\ell m}.$$

Inscribed regular polygon: side $2 \sin \frac{\pi}{n}$, interior angle $\frac{(n-2)\pi}{n}$, $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$.

Constructibility. The polygon with n sides is constructible $\iff \cos \frac{2\pi}{n}$ is constructible. $\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$, $\cos \frac{4\pi}{5} = -\frac{\sqrt{5}+1}{4}$. From n constructible follows $2n$; from n and $d \mid n$ follows d ; from m, n coprime follows mn ($\zeta_{mn} = \zeta_n^u \zeta_m^v$ with $um + vn = 1$). *Gauss–Wantzel* (statement; proved in Volume II, Chapter 7): constructible $\iff n = 2^k p_1 \cdots p_m$ with the p_i distinct Fermat primes.

Möbius maps and generalised circles.

$$T(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0; \quad T^{-1}(w) = \frac{dw - b}{-cw + a}; \quad T \leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ up to scalars.}$$

Decomposition ($c \neq 0$): $T = t_{a/c} \circ m_{(bc-ad)/c} \circ \iota \circ t_d \circ m_c$, with $t_\beta(z) = z + \beta$, $m_\alpha(z) = \alpha z$, $\iota(z) = 1/z$. *Equation of the generalised circles* ($A, C \in \mathbb{R}$, $B \in \mathbb{C}$, $|B|^2 > AC$):

$$Az\bar{z} + Bz + \bar{B}\bar{z} + C = 0; \quad \text{centre } -\frac{\bar{B}}{A}, \quad \text{radius } \frac{\sqrt{|B|^2 - AC}}{|A|} \quad (A \neq 0); \quad \text{a line if } A = 0.$$

Action on the coefficients:

$$z \mapsto z + \beta : (A, B - A\bar{\beta}, A|\beta|^2 - 2\Re(B\beta) + C); \quad z \mapsto \alpha z : (A, B\bar{\alpha}, C|\alpha|^2); \quad z \mapsto 1/z : (C, \bar{B}, A).$$

In all three cases $|B'|^2 - A'C'$ remains positive.

Cross ratio.

$$(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_4} \cdot \frac{z_2 - z_4}{z_2 - z_3} = S_{z_2 z_3 z_4}(z_1), \quad S_{z_2 z_3 z_4} : z_2 \mapsto 1, z_3 \mapsto 0, z_4 \mapsto \infty.$$

$$(Tz_1, Tz_2, Tz_3, Tz_4) = (z_1, z_2, z_3, z_4) \quad \text{for every Möbius map } T;$$

$$z_1, z_2, z_3, z_4 \text{ concyclic or collinear} \iff (z_1, z_2, z_3, z_4) \in \mathbb{R}.$$

Through three distinct points of $\hat{\mathbb{C}}$ there passes one and only one generalised circle.

Sources and bibliography

- [1] L. V. Ahlfors, *Complex Analysis. An Introduction to the Theory of Analytic Functions of One Complex Variable*, 3rd edition, McGraw–Hill, New York, 1979. Chapter 1, §2.2 “The Binomial Equation”, pp. 15–16: de Moivre’s formula (formula (20)), the complete solution of $z^n = a$, the n -th roots as vertices of a regular polygon with n sides, the notation $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ (formula (23)) and, in Exercise 4 on p. 16, the identity $1 + \omega^h + \omega^{2h} + \dots + \omega^{(n-1)h} = 0$ for every integer h not a multiple of n (Theorem 6.5 of this chapter). Complete copy consulted on the website of the Math 6321 course of the Georgia Institute of Technology, mccuan.math.gatech.edu/courses/6321/ — verified on 5 September 2026.
- [2] L. V. Ahlfors, *Complex Analysis*, cit., Chapter 3 “Analytic Functions as Mappings”, §3 “Linear Transformations”, pp. 76–88. In particular: §3.1 “The Linear Group”, pp. 76–78 (the group of linear fractional transformations, the correspondence with matrices, and the decomposition “translation, inversion, rotation, homothety, translation” used in Lemma 6.2); §3.2 “The Cross Ratio”, pp. 78–80 (Definition 12 of the cross ratio as the image of z_1 under the transformation carrying z_2, z_3, z_4 to $1, 0, \infty$, adopted in Definition 6.6; Theorem 12, p. 79: invariance of the cross ratio — Theorem 6.8 of this chapter; Theorem 13, p. 79: the cross ratio is real if and only if the four points lie on a circle or on a line — Theorem 6.9; and on pp. 79–80 the general equation of circles and lines in complex form, taken up in Lemma 6.1) — verified on 5 September 2026.
- [3] J. Orloff, *18.04 Complex Variables with Applications*, Spring 2018, MIT OpenCourseWare, Massachusetts Institute of Technology, “Topic 10 Notes: Conformal transformations”. §10.7 “Fractional linear transformations”: §10.7.2 “Lines and circles”, pp. 7–8 (the theorem “a linear fractional transformation maps lines and circles to lines and circles”, with the proof by decomposition $z \mapsto cz + d \mapsto 1/(cz + d) \mapsto \frac{a}{c} + (b - ad/c)w$ and the classification into four cases of the images under $w = 1/z$, recalled in Worked exercise 6.7); §10.7.3 “Mapping z_j to w_j ”, p. 8 (the explicit construction of the transformation sending three points to $0, 1, \infty$); §10.7.4 “Correspondence with matrices”, p. 9 (the homomorphism with $GL(2, \mathbb{C})$, taken up in Proposition 6.7); §10.8 “Reflection and symmetry”, pp. 10–11. Available at ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/resources/lecture-notes/ (licence CC BY-NC-SA) — verified on 5 September 2026.
- [4] T. Needham, *Visual Complex Analysis*, Clarendon Press — Oxford University Press, Ox-

ford, 1997. Chapter 3 “Möbius Transformations and Inversion”: geometric treatment of inversion, of the Riemann sphere, of Möbius transformations and of their decomposition into reflections (section “Decomposition into 2 or 4 Reflections”). It is the reference source for the *visual* reading of the results of Section 6.4; chapter record on Oxford Academic, academic.oup.com/book/52945/chapter/421959392 — verified on 5 September 2026.

- [5] Treatise on the Circle, **Volume II, Chapter 7** “The discrete circle”, section “Constructible polygons: the Gauss–Wantzel theorem”: the definition of a constructible number, the theorem on the degree of constructible numbers (necessity, Wantzel 1837), the Gaussian periods and the chain of quadratic extensions (sufficiency, Gauss 1796), the explicit construction of the 17-gon and the closed form of $\cos \frac{2\pi}{17}$. It is the complete proof of the theorem recalled in Remark 6.4, which this chapter does not duplicate.
- [6] T. W. Judson, *Abstract Algebra: Theory and Applications*, Stephen F. Austin State University, open textbook available at abstract.ups.edu/aata. §6.2 “Lagrange’s Theorem”, Theorem 6.10 (the order of a subgroup divides the order of the group) and Corollary 6.11 (the order of an element divides the order of the group); §11.2 “The Isomorphism Theorems”, Theorem 11.10 (the first isomorphism theorem: if $\psi : G \rightarrow H$ is a homomorphism with $K = \ker \psi$, then K is normal in G and there exists a unique isomorphism $\eta : G/K \rightarrow \psi(G)$ with $\psi = \eta\varphi$, where φ is the canonical projection). These are the two algebraic prerequisites declared in the opening *infobox* and used in Corollary 6.2. Text consulted in the LibreTexts reproduction, math.libretexts.org (bookshelf “Abstract and Geometric Algebra”) — verified on 5 September 2026.

From the circle to the sphere: cones, quadrics, sections and stereographic projection

So far the circle has lived in the plane. But the plane is a particular case, and the circle with it: it is the sphere of dimension 1 embedded in a space of dimension 2. As soon as one adds a dimension, the circle stops being an isolated object and becomes a *case*: a case of the family of conics, a section of the sphere, a section of the ellipsoid. This chapter walks through that passage and builds the dictionary that translates every notion of the circle into one of the sphere: planar sections are circles, stereographic projection maps circles to circles and preserves angles, great circles play the role of lines, and the angular excess of a triangle measures its area. Every time, the circle reappears as the case of maximal symmetry.

INFO

The idea in one line. Cutting a cone gives the conics, and the circle is the one orthogonal to the axis; cutting a sphere gives circles, *always*; cutting an ellipsoid gives circles only along two families of planes; projecting the sphere from the pole onto the equatorial plane, circles stay circles (or become lines) and angles do not change; and on the sphere the sum of the angles of a triangle exceeds π by exactly the area divided by r^2 .

INFO

Level and prerequisites. Level A–C. Sections 7.1, 7.2, 7.3 and 7.8 are level A–B: the scalar product in $\mathbb{R}^3$, the Pythagorean theorem and the elementary double integral suffice. Sections 7.4, 7.5 and 7.7 are level B–C: conics, partial derivatives and the first fundamental form of a parametrization are needed. Section 7.9 is level C. Prerequisites within the treatise: elementary circle, power of a point and tangents (Volume I, Chapters 2–5); conics and quadratic forms (Chapter 5); complex numbers and Möbius transformations (Chapter 6); integral calculus in several variables at the level of the change of variables in polar coordinates (Volume II, Chapter 2). Differential geometry is not assumed: the notions of first fundamental form and of curvature are recalled where needed, and taken up systematically in Chapter 10.

Warning. Conventions of the chapter.

- $S^{n-1}(c, R) = \{x \in \mathbb{R}^n : \|x - c\| = R\}$ is the *sphere* of centre c and radius R ; the index denotes the *dimension of the surface*, not of the ambient space. Thus $S^1 \subset \mathbb{R}^2$ is the circle and $S^2 \subset \mathbb{R}^3$ the ordinary sphere. One writes S^{n-1} and B^n for $c = 0$, $R = 1$.
- $B^n(c, R) = \{x \in \mathbb{R}^n : \|x - c\| \leq R\}$ is the solid ball.
- $\langle \cdot, \cdot \rangle$ is the standard scalar product, $\|x\| = \sqrt{\langle x, x \rangle}$, and $u \times v$ the cross product of $\mathbb{R}^3$.
- On the sphere of radius r the *sides* of a triangle are measured in radians as central angles; lengths are obtained by multiplying by r and areas by r^2 .
- Decimal separators are points: 0.5.

7.1. The circle–sphere dictionary

Before proving anything it is worth fixing the correspondence between the notions of the plane and those of space: almost all of them carry over, but two or three bifurcate, and it is exactly those that make the chapter interesting.

Definition 7.1: Sphere and ball. The **sphere** of center O and radius r is $S^2(O, r) = \{P \in \mathbb{R}^3 : |OP| = r\}$; the **ball** is $B^3(O, r) = \{P : |OP| \leq r\}$. The sphere is the surface, the ball the solid it encloses, exactly as the circumference encloses the circle (disk) in the plane.

In the plane $\mathbb{R}^2$	In space $\mathbb{R}^3$	Note
circumference S^1	sphere S^2	locus at distance r from center
disk B^2	ball B^3	enclosed region
tangent line	tangent plane	orthogonal to radius
line (geodesic of plane)	great circle	geodesic of surface
chord	chord	segment between two points
$C = 2\pi r$ (perimeter)	Area $= 4\pi r^2$	boundary measure
$A = \pi r^2$ (area)	Vol $= \frac{4}{3}\pi r^3$	interior measure
$SO(2)$ (rotations)	$SO(3)$ (rotations)	symmetries, commutative only in 2D
full angle 2π	solid angle 4π (sterad.)	full “turn”
sum of angles $= \pi$	sum of angles $> \pi$	the excess measures the area

Remark 7.1: What is conserved and what changes. Some notions transfer effortlessly (radius, chords, diameters, tangency). Others bifurcate: in the plane there is only one type of “line”, on the sphere the role of lines is played by great circles (Theorem 7.7), and through two antipodal points infinitely many pass. Most importantly, while $SO(2)$ is commutative, $SO(3)$ is not: it is the mathematical root of gimbal lock and of the delicacy of 3D orientation in robots. Both statements are proved in Chapter 11 (Theorem 11.7 and Example 11.2) and taken up in Chapter 28. The last row of the table — the sum of the angles — has no planar analogue at all: it is the signature of curvature, and section 7.9 quantifies it exactly.

7.2. The cone and its sections

Definition 7.2: Right circular cone. Let $V \in \mathbb{R}^3$ be a point (the **vertex**), u a unit vector (the **axis**) and $\alpha \in (0, \pi/2)$ (the **half-angle**). The **right circular cone** with these data is

$$\mathcal{C} = \left\{ P \in \mathbb{R}^3 : |\langle P - V, u \rangle| = \|P - V\| \cos \alpha \right\},$$

the locus of points seen from V at an angle α from the axis. The lines through V contained in $\mathcal{C}$ are the **generators**.

Theorem 7.1: Sections orthogonal to the axis are circles. Let π be a plane orthogonal to the axis u and not through V , at distance $h > 0$ from the vertex. Then $\mathcal{C} \cap \pi$ is a circle of radius $h \tan \alpha$, centred at the foot of the perpendicular from V .

Proof. Step 1 — choice of coordinates. Put V at the origin and $u = (0, 0, 1)$. The condition of Definition 7.2 becomes, for $P = (x, y, z)$,

$$|z| = \sqrt{x^2 + y^2 + z^2} \cos \alpha.$$

Squaring and rearranging:

$$z^2(1 - \cos^2 \alpha) = (x^2 + y^2) \cos^2 \alpha \implies x^2 + y^2 = z^2 \tan^2 \alpha.$$

Step 2 — intersection. The plane is $z = h$. Substituting,

$$x^2 + y^2 = h^2 \tan^2 \alpha,$$

which by the very definition of a circle (Volume I) is the circle with centre $(0, 0, h)$ and radius $h \tan \alpha$.

Step 3 — consistency. The radius grows linearly with the distance from the vertex, with coefficient $\tan \alpha$: this is the operational definition of the half-angle. As $\alpha \rightarrow 0$ the cone degenerates into the axis and the circles shrink; as $\alpha \rightarrow \pi/2$ the cone flattens into the plane orthogonal to the axis. $\checkmark$ $\square$

Theorem 7.2: Dandelin's theorem. Let π be a plane cutting a single nappe of the cone $\mathcal{C}$ in a closed bounded curve. Let $\mathcal{S}_1, \mathcal{S}_2$ be the two spheres inscribed in the nappe, tangent to π from one side and the other respectively, and let F_1, F_2 be their points of tangency with π . Then the section $\mathcal{C} \cap \pi$ is an **ellipse** with foci F_1 and F_2 ; and the sum of the distances equals the distance, measured along a generator, between the two circles of tangency of the spheres with the cone.

Proof. Step 1 — existence of the spheres. The spheres inscribed in the nappe with centre on the axis form a one-parameter family (the radius grows linearly with the distance from the vertex). The distance of such a sphere from the plane π varies continuously and is negative on one side, positive on the other: by the intermediate value theorem there is, on each side, a sphere tangent to π . Call them $\mathcal{S}_1$ and $\mathcal{S}_2$.

Step 2 — the circles of tangency. Every inscribed sphere touches the cone along a circle, because by rotational symmetry about the axis the set of tangency points is rotation-invariant; call these circles c_1 and c_2 , contained in planes orthogonal to the axis.

Step 3 — the tangent-segment lemma. From a point P outside a sphere, all tangent segments have the same length. Indeed, with O the centre and r the radius, for every tangency point T the triangle OTP has a right angle at T , hence $|PT| = \sqrt{|PO|^2 - r^2}$, independent of T . (It is the power-of-a-point theorem of Volume I, in three-dimensional form.)

Step 4 — the decisive computation. Let P be a point of the section $\mathcal{C} \cap \pi$. The generator through P meets c_1 at a point A and c_2 at a point B . Now:

- PA and PF_1 are both tangent segments from P to $\mathcal{S}_1$ (PA along the generator, PF_1 in the plane π), so $|PF_1| = |PA|$ by Step 3;
- likewise $|PF_2| = |PB|$.

Adding,

$$|PF_1| + |PF_2| = |PA| + |PB| = |AB|,$$

P lying between A and B along the generator, which happens because the two spheres lie on opposite sides of π .

Step 5 — the constant. The length $|AB|$ is the distance, along a generator, between the two planes containing c_1 and c_2 ; since both planes are orthogonal to the axis and all generators make the same angle with it, that length *does not depend on the chosen generator*. It is therefore a constant $2a$.

Step 6 — conclusion. Every point of the section satisfies $|PF_1| + |PF_2| = 2a$ constant: by the bifocal definition, the section is an ellipse with foci F_1, F_2 and semi-major axis a .

Step 7 — the circle case. If π is orthogonal to the axis, then by symmetry the two spheres are tangent to π at the same point, i.e. $F_1 = F_2$: the ellipse degenerates into the circle, in agreement with Theorem 7.1. *The circle is therefore the conic with coincident foci:* the conic of zero eccentricity.

Step 8 — the other cases. If the plane is parallel to exactly one generator, one of the two spheres escapes to infinity and one obtains the parabola (one focus); if it cuts both nappes, the two spheres lie in different nappes and the same construction gives $||PF_1| - |PF_2|| = \text{constant}$, the hyperbola. The proof is the same, with only a sign change at Step 4. The method, due to Germinal Pierre Dandelin (1822), replaces an entire analytic computation by two spheres. $\square$

***Remark 7.2: A hierarchy of symmetries.** Theorem 7.2 places the circle within a family and explains its privileged position. The eccentricity*

$$e = \frac{\text{focal distance}}{2a} = \frac{|F_1 F_2|}{2a}$$

measures how far the section departs from orthogonality to the axis: it is 0 for the circle, $0 < e < 1$ for the ellipse, 1 for the parabola, $e > 1$ for the hyperbola. The circle is the case in which the symmetry group is largest: all rotations about the centre, against the mere two reflections of the generic ellipse. This is a theme we shall meet in every section of this chapter.

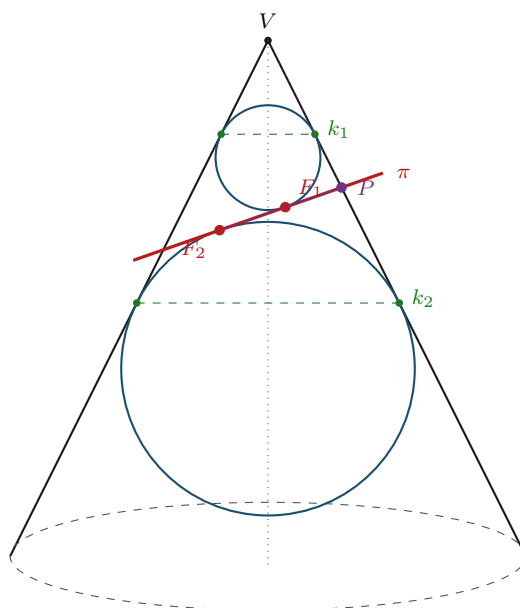


Figure 7.1: Dandelin's spheres, seen in the axial section of the cone. The two spheres are inscribed in the cone (in section: the two circles tangent to both generators) and tangent to the cutting plane π (in red). The tangency points F_1 and F_2 are the foci of the conic section; the circles k_1, k_2 along which the spheres touch the cone determine the directrices. For a point P on the conic, the segments PF_1 and PF_2 have the same length as the generator segments between P and k_1, k_2 : hence $PF_1 + PF_2 = \overline{k_1 k_2}$ is constant, i.e. the ellipse.

7.3. The sphere and its sections

Definition 7.3: Sphere in $\mathbb{R}^n$. For $n \geq 1$, $R > 0$ and $c \in \mathbb{R}^n$ one sets

$$S^{n-1}(c, R) = \{x \in \mathbb{R}^n : \|x - c\| = R\}, \quad B^n(c, R) = \{x \in \mathbb{R}^n : \|x - c\| \leq R\}.$$

One writes S^{n-1} and B^n for the case $c = 0$, $R = 1$. In particular $S^0 = \{-1, +1\}$ (two points), S^1 is the circle, S^2 the ordinary sphere.

Theorem 7.3: Every plane section of a sphere is a circle. Let $S = S^2(c, R) \subset \mathbb{R}^3$ and π a plane at distance d from c . Then

$$S \cap \pi = \begin{cases} \text{a circle of radius } \sqrt{R^2 - d^2} & \text{if } d < R, \\ \text{a point} & \text{if } d = R, \\ \emptyset & \text{if } d > R. \end{cases}$$

The centre of the circle is the foot of the perpendicular from c to π . More generally, in $\mathbb{R}^n$, the intersection of $S^{n-1}(c, R)$ with a hyperplane at distance $d < R$ is a sphere S^{n-2} of radius $\sqrt{R^2 - d^2}$.

Proof. **Step 1 — orthogonal decomposition.** Let H be the foot of the perpendicular from c to π , so that $\|c - H\| = d$ and $c - H$ is orthogonal to every direction vector of π .

Step 2 — Pythagoras. For $P \in \pi$ the vectors $c - H$ and $H - P$ are orthogonal, hence

$$\|P - c\|^2 = \|P - H\|^2 + \|H - c\|^2 = \|P - H\|^2 + d^2.$$

Step 3 — translating the condition. Therefore

$$P \in S \cap \pi \iff \|P - c\| = R \iff \|P - H\|^2 = R^2 - d^2.$$

If $d < R$ the right side is positive and the condition describes, *inside the plane* π , the circle with centre H and radius $\sqrt{R^2 - d^2}$. If $d = R$ one gets $P = H$; if $d > R$ there are no solutions.

Step 4 — the n -dimensional case. The proof never uses the dimension: Step 2 is Pythagoras in $\mathbb{R}^n$, valid by the very definition of the Euclidean norm. Replacing “plane” by “hyperplane” and “circle” by “ $(n-2)$ -dimensional sphere”, the argument is identical.

Step 5 — great circles. The radius of the section is largest for $d = 0$, i.e. when the hyperplane passes through the centre: one obtains the **great circles** (in $\mathbb{R}^3$) and more generally the *great spheres*. They have the same radius as the sphere, and are the only sections sharing its centre.

Step 6 — consequence. Since the section is a circle for *every* plane, the sphere is, in a precise sense, “made of circles in all directions”. It is this property — and not merely the equation — that is characteristic: one proves (Exercise S2) that a closed surface all of whose plane sections are circles is necessarily a sphere. $\square$

Definition 7.4: Great circles and small circles. A **great circle** of $S^2(O, r)$ is the intersection of the sphere with a plane through O : by Theorem 7.3 it has radius r , the largest possible, and centre O . Every other non-empty planar section not reduced to a point is called a **small circle**. Two **antipodal** points are two points P and $-P$ (with respect to O); through two non-antipodal points there passes one and only one great circle, the one cut by the plane $\text{span}\{OP, OQ\}$, whereas through two antipodal points infinitely many pass.

Corollary 7.1: The circle lives in the sphere. Let C be a circle in space, of centre Q , radius ρ and plane π , and let $r \geq \rho$. Then there exist exactly two spheres of radius r containing C , whose centres are

$$O_{\pm} = Q \pm du, \quad d = \sqrt{r^2 - \rho^2},$$

with u a unit normal of π ; on each of them C is the whole section with the plane π . The two spheres coincide if and only if $\rho = r$, and in that case C is a great circle. If instead $\rho > r$ no sphere of radius r contains C .

Proof. Step 1 — the two candidate spheres. Let u be a unit vector orthogonal to π and $d = \sqrt{r^2 - \rho^2} \geq 0$ (well defined because $\rho \leq r$). Set $O_{\pm} = Q \pm du$.

Step 2 — C lies on both. Let $P \in C$. The vector $P - Q$ lies in π , hence is orthogonal to u ; by the Pythagorean theorem

$$\|P - O_{\pm}\|^2 = \|(P - Q) \mp du\|^2 = \|P - Q\|^2 + d^2 = \rho^2 + d^2 = r^2,$$

that is $P \in S^2(O_{\pm}, r)$.

Step 3 — C is the whole section. The distance of $O_{\pm}$ from the plane π is d and the foot of the perpendicular is Q . By Theorem 7.3 the section $S^2(O_{\pm}, r) \cap \pi$ is the circle of centre Q and radius $\sqrt{r^2 - d^2} = \rho$: it is exactly C . So C is not merely contained in the sphere, it is a planar section of it.

Step 4 — there are no others. Let O be the centre of any sphere of radius r containing C , and write $O - Q = w_{\parallel} + w_{\perp}$ with $w_{\parallel}$ in the direction plane of π and $w_{\perp}$ parallel to u . For every $P \in C$,

$$r^2 = \|P - O\|^2 = \|P - Q\|^2 - 2\langle P - Q, O - Q \rangle + \|O - Q\|^2 = \rho^2 - 2\langle P - Q, w_{\parallel} \rangle + \|O - Q\|^2,$$

having used $\langle P - Q, w_{\perp} \rangle = 0$. The first and third summands do not depend on P , so $\langle P - Q, w_{\parallel} \rangle$ is constant on C . But as P runs over the circle that scalar product takes all values in $[-\rho\|w_{\parallel}\|, \rho\|w_{\parallel}\|]$; constancy forces $w_{\parallel} = 0$, i.e. O lies on the axis of C . Then $r^2 = \rho^2 + \|w_{\perp}\|^2$ gives $\|w_{\perp}\| = d$ and hence $O = O_{+}$ or $O = O_{-}$.

Step 5 — limiting cases. If $\rho = r$ then $d = 0$ and $O_{+} = O_{-} = Q$: the sphere is unique and C is a great circle (Definition 7.4). If $\rho < r$ the two spheres are distinct and symmetric with respect to π . If $\rho > r$ every planar section of a sphere of radius r has radius $\sqrt{r^2 - d^2} \leq r < \rho$ (Theorem 7.3) and cannot be C .

Step 6 — numerical check. With $r = 1$ and $\rho = 0.6$ one gets $d = \sqrt{1 - 0.36} = 0.8$, and the check $\rho^2 + d^2 = 0.36 + 0.64 = 1.00 = r^2$. ✓ With $r = 13$ and $\rho = 12$ one gets $d = \sqrt{169 - 144} = 5$, the Pythagorean triple $(5, 12, 13)$. ✓

Step 7 — reading. The circle is not “replaced” by the sphere: it is contained in it, and in two ways. Moving to dimension three does not eliminate the object of the treatise, it puts it in a family — exactly as Theorem 7.2 had put it in the family of conics. □

Exercise 7.1. At what distance from the center is the earth parallel at latitude 45° , and what is its radius (spherical Earth of radius R)?

Solution. The parallel at latitude ϕ is the section at distance $d = R \sin \phi$ from the center (the plane is horizontal, raised by $R \sin \phi$). Its radius is $\sqrt{R^2 - d^2} = R \cos \phi$. For $\phi = 45^\circ$: $d = R/\sqrt{2} \approx 0.707R$ and radius $R \cos 45^\circ = R/\sqrt{2} \approx 0.707R$. The parallel at 45° has radius equal to 70.7% of the equatorial one. By Corollary 7.1 the spheres of radius R containing that parallel are two: the Earth and its mirror image with respect to the plane of the parallel.

Example 7.1: Circles of latitude and navigation. On the Earth, approximated by a sphere of radius $R = 6371$ km, the parallel of latitude φ is the section by the horizontal plane at distance $d = R \sin \varphi$ from the centre. By Theorem 7.3 its radius is

$$r(\varphi) = \sqrt{R^2 - R^2 \sin^2 \varphi} = R \cos \varphi.$$

Numerical check. At $\varphi = 45^\circ$: $r = 6371 \times 0.707107 = 4505.0$ km, and the length of the parallel is $2\pi r = 28\,305$ km, against the 40 030 km of the equator. The ratio is $\cos 45^\circ = 0.7071$. ✓

Why it matters. Only the equator and the meridians are great circles; the parallels (except the equator) are not. Since the shortest path between two points of a sphere is the arc of a great circle (Exercise S3), flying from Rome to New York along the parallel of 41° N would be longer than the actual route, which passes far to the north. That is why transatlantic routes look curved on Mercator charts: they are straight on the sphere, not on the map.

Theorem 7.4: Girard's theorem: the area of a spherical triangle. Let T be a spherical triangle on the sphere of radius R , bounded by three arcs of great circles, with interior angles A, B, C . Then

$$\text{Area}(T) = R^2(A + B + C - \pi).$$

In particular $A + B + C > \pi$: on the sphere the angle sum of a triangle always exceeds a straight angle, and the excess measures the area.

Proof. Step 1 — the area of a lune. Two great circles meet at two antipodal points and divide the sphere into four *lunes*. A lune of dihedral angle θ occupies a fraction $\theta/2\pi$ of the sphere, hence has area

$$\frac{\theta}{2\pi} \cdot 4\pi R^2 = 2\theta R^2.$$

Step 2 — the three lunes. The three great circles bounding T determine, at each vertex, a pair of opposite lunes: the one containing T and the one containing the antipodal triangle T' . Their areas are $2AR^2$, $2BR^2$, $2CR^2$ for the lunes containing T , and the same for the opposite ones.

Step 3 — counting with multiplicity. The six lunes (three plus the three opposite) cover the whole sphere. Every point of the sphere is covered once, except the points of T and T' , which are covered three times each (once by each lune). Hence

$$\underbrace{2 \cdot (2AR^2 + 2BR^2 + 2CR^2)}_{\text{sum of the six lune areas}} = \underbrace{4\pi R^2}_{\text{sphere}} + 2 \cdot 2 \cdot \text{Area}(T),$$

the correction term accounting for the fact that T and T' (of equal area) are counted three times instead of once, i.e. twice in excess each.

Step 4 — solving. Simplifying,

$$4R^2(A + B + C) = 4\pi R^2 + 4 \text{Area}(T) \implies \text{Area}(T) = R^2(A + B + C - \pi).$$

Step 5 — check on the trirectangular triangle. The triangle with vertices at the North Pole and two points of the equator 90° apart has $A = B = C = \pi/2$. The formula gives

$$\text{Area} = R^2 \left(\frac{3\pi}{2} - \pi \right) = \frac{\pi R^2}{2},$$

that is one eighth of the surface $4\pi R^2$: and indeed three coordinate planes divide the sphere into eight equal triangles. ✓

Step 6 — the plane limit. For triangles small compared with R the area tends to zero and with it the excess: one recovers $A+B+C \rightarrow \pi$, Euclidean geometry. The sphere is thus locally flat, and the angular excess measures curvature: it is the first instance of the Gauss–Bonnet theorem, which generalises this formula to arbitrary surfaces.

Step 7 — and the circle? On a sphere the “circles” — the loci of points at geodesic distance ρ from a centre — have circumference $2\pi R \sin(\rho/R)$, less than $2\pi\rho$. The ratio circumference/radius, which in the plane is exactly 2π (Volume I), is therefore *smaller* than 2π on the sphere. This too is an effect of curvature, and its measure — the defect with respect to 2π — is the circular analogue of Girard’s excess (Exercise S4). $\square$

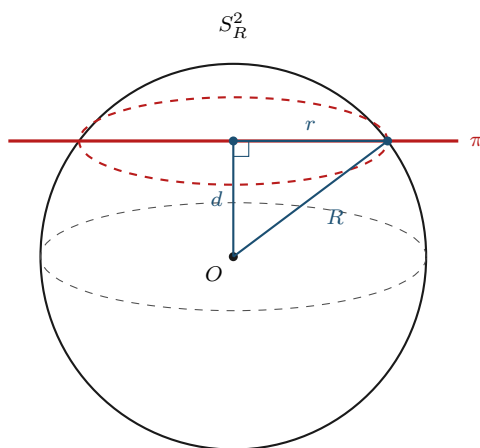


Figure 7.2: A plane section of a sphere is always a circle. If the plane π is at distance $d < R$ from the centre O , the right triangle with legs d and r and hypotenuse R immediately gives $r = \sqrt{R^2 - d^2}$: the radius of the section depends only on the distance of the plane from the centre, not on its orientation. For $d = 0$ we get the great circle ($r = R$), and as $d \rightarrow R$ the circle shrinks to a point. It is the same relation as for the chord of a circle in Volume I, moved up one dimension.

7.4. Quadrics and ellipsoids: where the circles hide

Definition 7.5: Quadric and ellipsoid. A **quadric** of $\mathbb{R}^3$ is the zero locus of a second-degree polynomial in the three coordinates. If the quadratic part is positive definite, a rigid motion reduces the quadric to the canonical form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad a \geq b \geq c > 0,$$

the **ellipsoid** with semi-axes a, b, c . For $a = b = c$ one obtains the sphere.

Theorem 7.5: Circular sections of the ellipsoid. Let E be the ellipsoid with semi-axes $a > b > c$ (all distinct). Then:

- (i) every plane section of E is an ellipse (possibly degenerate);
- (ii) non-degenerate circular sections exist, all of radius $\leq b$, and those of maximal radius b lie in the planes

$$z = \pm kx, \quad k = \frac{c}{a} \sqrt{\frac{a^2 - b^2}{b^2 - c^2}},$$

and in their parallel translates;

- (iii) there are therefore exactly two families of circle-cutting planes, and both families con-

tain the intermediate axis y .

If two semi-axes coincide the two families merge into one (ellipsoid of revolution); if all three coincide every plane cuts a circle (sphere, Theorem 7.3).

Proof. (i) The intersection of a quadric with a plane is a conic: substituting the parametric equation of the plane into that of the ellipsoid gives a second-degree polynomial in two variables. Since E is bounded, the conic is bounded, hence an ellipse (or a point, or empty).

(ii) Step 1 — the idea. A plane cuts E along a circle of radius ρ centred at the origin if and only if that plane meets both E and the sphere $x^2 + y^2 + z^2 = \rho^2$ in the same set. So we study the intersection $E \cap S$ with S the sphere of radius ρ .

Step 2 — the system. Subtracting the equation of the sphere, divided by ρ^2 , from that of the ellipsoid:

$$x^2 \left(\frac{1}{a^2} - \frac{1}{\rho^2} \right) + y^2 \left(\frac{1}{b^2} - \frac{1}{\rho^2} \right) + z^2 \left(\frac{1}{c^2} - \frac{1}{\rho^2} \right) = 0.$$

Step 3 — choosing ρ . For this equation to represent a pair of planes through the origin, one of the three coefficients must vanish and the other two must have opposite signs. Since $a > b > c$,

$$\frac{1}{a^2} < \frac{1}{b^2} < \frac{1}{c^2},$$

and the only possible choice is $\rho = b$: then the coefficient of y^2 vanishes, that of x^2 is negative and that of z^2 positive. (With $\rho = a$ or $\rho = c$ the two remaining coefficients would have the same sign and the equation would give only the origin.)

Step 4 — the two planes. With $\rho = b$ the equation becomes

$$z^2 \frac{b^2 - c^2}{b^2 c^2} = x^2 \frac{a^2 - b^2}{a^2 b^2},$$

that is

$$z = \pm kx, \quad k = \frac{c}{a} \sqrt{\frac{a^2 - b^2}{b^2 - c^2}},$$

two distinct planes (since $a \neq b$ and $b \neq c$), both containing the y -axis.

Step 5 — they really are circles. The intersection $E \cap S$ is contained in the union of the two planes; conversely every point of E on one of those planes satisfies the equation of Step 2 and hence lies on S . So $E \cap \{z = kx\} = S \cap \{z = kx\}$, which by Theorem 7.3 is a circle of radius b (the plane passes through the centre).

Step 6 — parallel planes. A plane parallel to one of the two cuts E along a curve similar to the one through the centre: substituting $z = kx + h$ into the equation gives a conic whose coefficients of x^2 and y^2 remain equal to each other (because the choice of k made them equal) and whose mixed term xy remains zero. The section is therefore again a circle, of smaller radius.

(iii) The two families are the planes parallel to $z = kx$ and to $z = -kx$; no other plane gives a circle, because Step 3 showed $\rho = b$ to be the only possible choice and the two planes found are the only ones.

Step 7 — degenerate cases. If $a = b > c$ (oblate ellipsoid) then $k = 0$ and the two families merge into the plane $z = 0$ and its parallels: the circles of the equator. If $a > b = c$ (prolate), $k \rightarrow \infty$ and the families merge into the planes $x = \text{constant}$. If $a = b = c$ every ρ works and one recovers the sphere.

Step 8 — numerical check. With $a = 3$, $b = 2$, $c = 1$:

$$k = \frac{1}{3} \sqrt{\frac{9-4}{4-1}} = \frac{1}{3} \sqrt{\frac{5}{3}} = \frac{1.290994}{3} = 0.430331.$$

The planes are $z = \pm 0.430331x$ and cut circles of radius 2. Direct check: the point $(x, y, z) = (0, 2, 0)$ lies on both planes and on the ellipsoid ($4/4 = 1$ ✓) and is at distance 2 from the origin ✓. The point with $y = 0$ on the plane $z = kx$ satisfies $\frac{x^2}{9} + \frac{k^2 x^2}{1} = 1$, i.e. $x^2(0.111111 + 0.185185) = 1$, whence $x^2 = 3.375$, $x = 1.837$, $z = 0.7906$; its distance from the origin is $\sqrt{3.375 + 0.625} = \sqrt{4.000} = 2.000$. ✓ $\square$

Remark 7.3: Why the circles hide. A generic ellipsoid has no rotational symmetry: its symmetry group is finite (the reflections in the coordinate planes and their compositions). Yet it contains infinitely many circles. There is no contradiction: the circles are not orbits of symmetries of the ellipsoid, but intersections with a suitable sphere. The lesson, which recurs, is that the circle survives the loss of symmetry: it hides in one-parameter families even inside objects with no apparent trace of it. It is the principle underlying the machining of toric lenses in optics, and the tracing of ellipsoidal surfaces with rotating tools in mechanics.

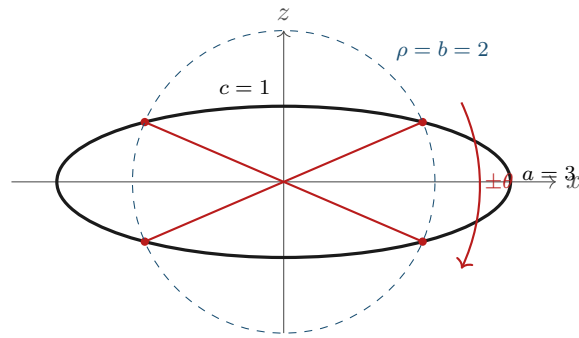


Figure 7.3: Where the circles hide inside an ellipsoid with semi-axes $a > b > c$ (here $3 > 2 > 1$). The figure shows the principal section in the xz plane: the ellipse with semi-axes a and c . The radius vector from the centre takes every value between c and a , and equals exactly the intermediate semi-axis b at four points, lying on two lines through the centre (in red). The two planes through these lines and the y axis cut the ellipsoid along genuine circles of radius b (dashed blue): these are the only two families of circular sections, and they coincide when $a = b$ or $b = c$, i.e. for an ellipsoid of revolution.

7.5. Stereographic projection

The sphere and the plane are not the same object — one is compact, the other is not — and yet there is between them an almost perfect correspondence: stereographic projection, which misses only one point. It is the construction that allows every spherical question to be transported to the plane, and it underlies the Riemann sphere and conformal nautical charts.

Definition 7.6: Stereographic projection. Let $N = (0, 0, r)$ be the north pole of $S^2 = S^2(O, r)$ with O at the origin. The **stereographic projection** $\sigma : S^2 \setminus \{N\} \rightarrow \pi_0$ onto the equatorial plane $\pi_0 = \{z = 0\}$, identified with $\mathbb{R}^2$, sends P to the intersection with π_0 of the line NP .

Proposition 7.1: Stereographic coordinates. The following formulas hold:

$$\sigma(x, y, z) = \frac{r}{r - z} (x, y), \quad X(u, v) = \frac{r}{u^2 + v^2 + r^2} (2ru, 2rv, u^2 + v^2 - r^2),$$

and $X : \mathbb{R}^2 \rightarrow S^2 \setminus \{N\}$ is a C^∞ bijection, inverse to σ . In particular σ is well defined and bijective, and the chart X covers the whole sphere except the single point N , without the

degeneracies at the poles of spherical coordinates.

Proof. Step 1 — the formula for σ . The line through $N = (0, 0, r)$ and $P = (x, y, z)$ is $N + t(P - N) = (tx, ty, r + t(z - r))$. On S^2 one has $z = r$ only at N (because $z = r$ forces $x^2 + y^2 = 0$), so for $P \neq N$ we have $z < r$ and the third coordinate vanishes for $t = r/(r - z) > 0$. Substituting gives $\sigma(P) = \frac{r}{r-z}(x, y)$.

Step 2 — X takes values on the sphere. Setting $s = u^2 + v^2$ and $D = s + r^2 \geq r^2 > 0$,

$$\|X\|^2 = \frac{r^2}{D^2} (4r^2u^2 + 4r^2v^2 + (s - r^2)^2) = \frac{r^2}{D^2} (4r^2s + s^2 - 2r^2s + r^4) = \frac{r^2}{D^2} (s + r^2)^2 = r^2.$$

Step 3 — X avoids the pole. The third coordinate equals $z = r \frac{s-r^2}{D} = r - \frac{2r^3}{D} < r$, hence $X(u, v) \neq N$ for every (u, v) .

Step 4 — $\sigma \circ X = \text{id}$. From $z = r - 2r^3/D$ we get $r - z = 2r^3/D$, hence $\frac{r}{r-z} = \frac{D}{2r^2}$ and

$$\sigma(X(u, v)) = \frac{D}{2r^2} \cdot \frac{1}{D} (2r^2u, 2r^2v) = (u, v).$$

Step 5 — $X \circ \sigma = \text{id}$. Let $P = (x, y, z) \in S^2 \setminus \{N\}$ and $(u, v) = \sigma(P)$. Then, using $x^2 + y^2 = r^2 - z^2$,

$$s = u^2 + v^2 = \frac{r^2(r^2 - z^2)}{(r - z)^2} = \frac{r^2(r + z)}{r - z}, \quad D = s + r^2 = r^2 \frac{(r + z) + (r - z)}{r - z} = \frac{2r^3}{r - z},$$

whence $\frac{2r^2}{D} = \frac{r-z}{r}$. The first two components of $X(u, v)$ then equal $\frac{2r^2u}{D} = \frac{r-z}{r} \cdot \frac{rx}{r-z} = x$ and similarly y ; the third equals $r(1 - \frac{2r^2}{D}) = r(1 - \frac{r-z}{r}) = z$. Hence $X(\sigma(P)) = P$, and together with Step 4 this proves that σ and X are inverse bijections.

Step 6 — regularity and absence of poles. The components of X are quotients of polynomials with denominator $D \geq r^2 > 0$: X is C^∞ on all of $\mathbb{R}^2$. Moreover by Theorem 7.6 (Step 2) one has $\|X_u\| = \|X_v\| = 2r^2/D > 0$ and $\langle X_u, X_v \rangle = 0$, so $X_u \times X_v$ never vanishes: the chart is *regular at every point*. Spherical coordinates (Definition 7.7) degenerate instead at the poles, where $X_\varphi = 0$.

Step 7 — numerical check. With $r = 1$ and $(u, v) = (2, 0)$: $D = 5$, $X = (\frac{4}{5}, 0, \frac{3}{5}) = (0.8, 0, 0.6)$, and $\|X\|^2 = 0.64 + 0.36 = 1.00$ ✓. Going back, $\sigma(0.8, 0, 0.6) = \frac{1}{1-0.6}(0.8, 0) = (2, 0)$ ✓. $\square$

Theorem 7.6: Properties of stereographic projection. The stereographic projection σ of Definition 7.6 enjoys the following two properties.

(i) **It is conformal:** it preserves the size of angles between curves that meet. Precisely, the chart X has first fundamental form

$$E = \|X_u\|^2 = \frac{4r^4}{(u^2 + v^2 + r^2)^2} = G = \|X_v\|^2, \quad F = \langle X_u, X_v \rangle = 0,$$

that is $\langle dX(w), dX(w') \rangle = \lambda^2 \langle w, w' \rangle$ with $\lambda(u, v) = 2r^2/(u^2 + v^2 + r^2) > 0$.

(ii) **It maps circles to circles or lines:** if $\Gamma \subset S^2$ is a circle through N , then $\sigma(\Gamma \setminus \{N\})$ is a line of π_0 ; if Γ does not pass through N , then $\sigma(\Gamma)$ is a circle. Writing $\Gamma = S^2 \cap \{ax + by + cz = k\}$ with $a^2 + b^2 + c^2 = 1$ and $|k| < r$, the image circle has centre $-\frac{r^2}{cr - k}(a, b)$ and radius $\frac{r\sqrt{r^2 - k^2}}{|cr - k|}$.

Proof. (i) Step 1 — the derivatives of the chart. With $s = u^2 + v^2$ and $D = s + r^2$ one has $X = (2r^2u/D, 2r^2v/D, r - 2r^3/D)$; since $D_u = 2u$ and $D_v = 2v$,

$$X_u = \frac{1}{D^2} (2r^2(D - 2u^2), -4r^2uv, 4r^3u), \quad X_v = \frac{1}{D^2} (-4r^2uv, 2r^2(D - 2v^2), 4r^3v).$$

Step 2 — the computation of E, F, G . For E :

$$D^4 \|X_u\|^2 = 4r^4(D - 2u^2)^2 + 16r^4u^2v^2 + 16r^6u^2 = 4r^4[(D - 2u^2)^2 + 4u^2v^2 + 4r^2u^2],$$

and the bracket equals

$$D^2 - 4u^2D + 4u^4 + 4u^2v^2 + 4r^2u^2 = D^2 - 4u^2D + 4u^2(u^2 + v^2 + r^2) = D^2,$$

hence $E = 4r^4/D^2$; by symmetry $G = 4r^4/D^2$. For F :

$$D^4 \langle X_u, X_v \rangle = -8r^4uv(D - 2u^2) - 8r^4uv(D - 2v^2) + 16r^6uv = 8r^4uv[-2D + 2u^2 + 2v^2 + 2r^2] = 0,$$

because $u^2 + v^2 + r^2 = D$.

Step 3 — from $E = G, F = 0$ to conformality. For $w = (w_1, w_2)$ and $w' = (w'_1, w'_2)$ in $\mathbb{R}^2$,

$$\langle dX(w), dX(w') \rangle = E w_1 w'_1 + F(w_1 w'_2 + w_2 w'_1) + G w_2 w'_2 = \lambda^2 \langle w, w' \rangle, \quad \lambda = \frac{2r^2}{D}.$$

In particular $\|dX(w)\| = \lambda \|w\|$ and

$$\cos(\widehat{dX(w), dX(w')}) = \frac{\lambda^2 \langle w, w' \rangle}{\lambda \|w\| \cdot \lambda \|w'\|} = \frac{\langle w, w' \rangle}{\|w\| \|w'\|} = \cos(\widehat{w, w'}) :$$

angles are preserved by X , and hence also by its inverse σ . (That λ depends on the point means that *lengths* are not preserved: stereographic projection is conformal but not isometric, and indeed it dilates the neighbourhood of N enormously.)

Step 4 — numerical check of (i). With $r = 1$ and $(u, v) = (1, 1)$ one has $D = 3$ and

$$X_u = \frac{1}{9}(2, -4, 4), \quad X_v = \frac{1}{9}(-4, 2, 4),$$

whence $\|X_u\|^2 = (4 + 16 + 16)/81 = 36/81 = 0.444444$ and $\langle X_u, X_v \rangle = (-8 - 8 + 16)/81 = 0$, against the predicted value $4r^4/D^2 = 4/9 = 0.444444$. ✓

(ii) Step 5 — what a circle of the sphere is. By Theorem 7.3 every circle $\Gamma \subset S^2$ is the intersection with a plane $ax + by + cz = k$; normalizing $a^2 + b^2 + c^2 = 1$, the distance of the plane from the centre is $|k|$, and the intersection is a circle (non-empty, not a point) if and only if $|k| < r$. Its radius is $\rho = \sqrt{r^2 - k^2}$.

Step 6 — the substitution. A point $X(u, v)$ belongs to Γ if and only if

$$a \frac{2r^2u}{D} + b \frac{2r^2v}{D} + c \frac{r(s - r^2)}{D} = k,$$

that is, multiplying by $D = s + r^2$ and rearranging,

$$(cr - k)(u^2 + v^2) + 2r^2au + 2r^2bv - r^3c - kr^2 = 0, \quad \text{which we call } (\star).$$

Since X is a bijection onto $S^2 \setminus \{N\}$ (Proposition 7.1), the solution set of $(\star)$ is *exactly* $\sigma(\Gamma \setminus \{N\})$.

Step 7 — the case of the line. One has $N = (0, 0, r) \in \Gamma$ if and only if $cr = k$. In that case $-r^3c - kr^2 = -r^3c - cr^3 = -2cr^3$ and $(\star)$ becomes

$$2r^2(au + bv) = 2cr^3, \quad \text{that is} \quad au + bv = cr.$$

It cannot be $a = b = 0$: otherwise $c = \pm 1$ and $k = cr = \pm r$, against $|k| < r$. Hence it is a genuine line, and $\sigma(\Gamma \setminus \{N\})$ is a line.

Step 8 — the case of the circle. If $cr \neq k$, dividing $(\star)$ by $cr - k$ we obtain

$$u^2 + v^2 + \frac{2r^2a}{cr - k}u + \frac{2r^2b}{cr - k}v - \frac{r^2(rc + k)}{cr - k} = 0,$$

that is a circle of centre $(u_0, v_0) = -\frac{r^2}{cr - k}(a, b)$ and radius R_Γ with

$$R_\Gamma^2 = u_0^2 + v_0^2 + \frac{r^2(rc + k)}{cr - k} = \frac{r^4(a^2 + b^2) + r^2(rc + k)(cr - k)}{(cr - k)^2} = \frac{r^2[r^2(a^2 + b^2) + r^2c^2 - k^2]}{(cr - k)^2} = \frac{r^2(r^2 - k^2)}{(cr - k)^2},$$

where we used $(rc + k)(cr - k) = r^2c^2 - k^2$ and then $a^2 + b^2 + c^2 = 1$. Since $|k| < r$ the numerator is positive: it is a genuine circle, of radius $R_\Gamma = r\sqrt{r^2 - k^2}/|cr - k| = r\rho/|cr - k|$.

Step 9 — numerical check of (ii). Let $r = 1$. (a) *The equator*, $a = b = 0$, $c = 1$, $k = 0$: then $cr - k = 1$, the centre is the origin and the radius $\sqrt{1 - 0}/1 = 1$. And indeed σ fixes the equator pointwise, since for $z = 0$ one has $\sigma(x, y, 0) = (x, y)$. ✓ (b) *A parallel*, $z = \frac{1}{2}$, that is $c = 1$, $k = \frac{1}{2}$: centre at the origin and radius $\sqrt{1 - 1/4}/(1/2) = 0.866025/0.5 = 1.732051 = \sqrt{3}$.

Direct check on the point $(\frac{\sqrt{3}}{2}, 0, \frac{1}{2}) \in \Gamma$: $\sigma = \frac{1}{1-0.5}(0.866025, 0) = (1.732051, 0)$, of modulus $\sqrt{3}$. ✓ (c) *A circle through the pole*, plane $x + z = 1$, that is $a = c = 1/\sqrt{2}$, $k = 1/\sqrt{2} = cr$: the image is the line $u/\sqrt{2} = 1/\sqrt{2}$, i.e. $u = 1$. Check on two points of Γ : $(1, 0, 0)$ gives $\sigma = (1, 0)$, and $(\frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2})$ — which satisfies $x + z = 1$ and $x^2 + y^2 + z^2 = 0.25 + 0.5 + 0.25 = 1$ — gives $\sigma = \frac{1}{1-0.5}(0.5, 0.707107) = (1, 1.414214)$. Both lie on $u = 1$. ✓

Step 10 — reading. Meridians and parallels are circles meeting at right angles; their images are respectively lines through the origin and circles centred at the origin, which still meet at right angles: conformality in action. It is the property that makes stereographic projection the preferred chart in polar cartography and in crystallography, and that, by Chapter 6, links it to Möbius transformations: those too map circles and lines to circles and lines. $\square$

Remark 7.4: The Riemann sphere: closing the plane with a point. Identifying the equatorial plane with $\mathbb{C}$ and adding the “point at infinity” $\infty \leftrightarrow N$, stereographic projection realizes $S^2 \cong \mathbb{C} \cup \{\infty\}$, the **Riemann sphere**. The complex plane of Volume II is therefore not “open”: compactified with a single point it becomes a sphere. Möbius transformations $z \mapsto \frac{az+b}{cz+d}$ (Chapter 6) correspond to transformations of the sphere and — consistently with Theorem 7.6(ii) — map circles and lines to circles and lines. It is the bridge between the circle ($\mathbb{T} \subset \mathbb{C}$) and the sphere as a complex object, resumed in Chapter 15 and, for the Hopf fibration, in Chapter 11.

7.6. Sphere parametrizations

Definition 7.7: Spherical coordinates. On the sphere $S^2(O, r)$ centred at the origin one sets

$$X(\varphi, \vartheta) = r(\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta),$$

with **longitude** $\varphi \in [0, 2\pi)$ and **colatitude** $\vartheta \in [0, \pi]$. Meridians are $\varphi = \text{constant}$ (great semicircles), parallels $\vartheta = \text{constant}$ (small circles, except the equator $\vartheta = \pi/2$). The **latitude** is $\phi = \pi/2 - \vartheta$.

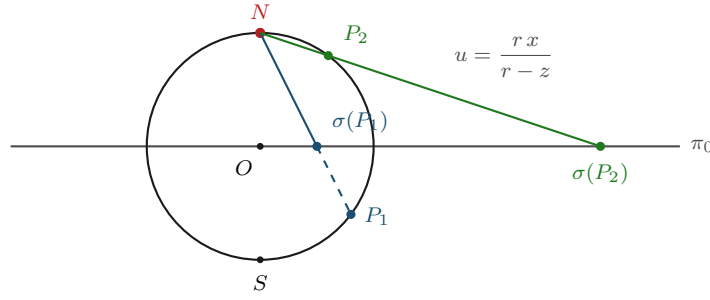


Figure 7.4: Stereographic projection, seen in the axial section of the unit sphere ($r = 1$). The point $P_1 = (0.8, 0, -0.6)$ of the southern hemisphere projects to $\sigma(P_1) = (0.5, 0)$, *inside* the equatorial circle, because $u = x/(1 - z) = 0.8/1.6 = 0.5$; the point $P_2 = (0.6, 0, 0.8)$ of the northern hemisphere projects to $\sigma(P_2) = (3, 0)$, very far away, because $u = 0.6/0.2 = 3$. The equator stays fixed pointwise, and approaching the pole N the image escapes to infinity: this is why N corresponds to the point at infinity of the Riemann sphere.

Lemma 7.1: Area element in spherical coordinates. For the parametrization of Definition 7.7 one has

$$\|X_\varphi\| = r \sin \vartheta, \quad \|X_\vartheta\| = r, \quad \langle X_\varphi, X_\vartheta \rangle = 0,$$

and therefore $dA = \|X_\varphi \times X_\vartheta\| d\varphi d\vartheta = r^2 \sin \vartheta d\varphi d\vartheta$. The chart is regular for $\vartheta \in (0, \pi)$ and degenerates at the two poles, where $X_\varphi = 0$.

Proof. Step 1 — the derivatives. Differentiating,

$$X_\varphi = r \sin \vartheta (-\sin \varphi, \cos \varphi, 0), \quad X_\vartheta = r (\cos \vartheta \cos \varphi, \cos \vartheta \sin \varphi, -\sin \vartheta).$$

Step 2 — norms and orthogonality. $\|X_\varphi\| = r \sin \vartheta$ because $\sin^2 \varphi + \cos^2 \varphi = 1$ and $\sin \vartheta \geq 0$ on $[0, \pi]$; moreover $\|X_\vartheta\|^2 = r^2 (\cos^2 \vartheta + \sin^2 \vartheta) = r^2$; finally

$$\langle X_\varphi, X_\vartheta \rangle = r^2 \sin \vartheta \cos \vartheta (-\sin \varphi \cos \varphi + \cos \varphi \sin \varphi) = 0.$$

Step 3 — the area element. For two orthogonal vectors $\|X_\varphi \times X_\vartheta\| = \|X_\varphi\| \|X_\vartheta\| = r^2 \sin \vartheta$.

Step 4 — global check. Integrating, $\int_0^{2\pi} \int_0^\pi r^2 \sin \vartheta d\vartheta d\varphi = 2\pi r^2 [-\cos \vartheta]_0^\pi = 4\pi r^2$, the area of the sphere (Theorem 7.9). ✓ $\square$

Exercise 7.2. Where does the south pole $S = (0, 0, -r)$ go under stereographic projection from the north pole?

Solution. The line NS is the z axis; it intersects the plane $z = 0$ at the origin. Hence $\sigma(S) = (0, 0)$: the south pole goes to the center of the plane. In formulas, Proposition 7.1 gives $\sigma(0, 0, -r) = \frac{r}{r+r}(0, 0) = (0, 0)$. Parallels near S go to small circles around the origin, those near N to huge circles — consistent with $N \leftrightarrow \infty$ and with the radius $\sqrt{(1+z_0)/(1-z_0)}$ computed in Exercise 7.8.

7.7. Great circles, distance and geodesics

On the sphere the role of lines is played by great circles. That this is not a convention but a theorem is the content of this section: among all curves traced on the sphere, those of minimal

length are arcs of great circles. The proof that follows is *direct* and does not use the calculus of variations: the variational characterization, valid on an arbitrary surface, is in Chapter 9.

Theorem 7.7: Great circles are the geodesics. Let $A, B \in S^2(O, r)$ be two distinct points. For every continuous, piecewise C^1 curve $\gamma : [0, 1] \rightarrow S^2(O, r)$ with $\gamma(0) = A$ and $\gamma(1) = B$ one has

$$L(\gamma) = \int_0^1 \|\gamma'(t)\| dt \geq r \arccos \frac{\langle A, B \rangle}{r^2},$$

and the right-hand side is attained. If A and B are not antipodal, equality holds if and only if γ traverses monotonically the **minor arc** of the (unique) great circle through A and B , possibly resting at the endpoints. If A and B are antipodal the minimum is πr and is attained by every great semicircle joining them.

Proof. Step 1 — reduction to the unit sphere. If γ is a curve on $S^2(O, r)$, then $\tilde{\gamma} = \gamma/r$ is a curve on $S^2(O, 1)$, with $L(\gamma) = r L(\tilde{\gamma})$ and $\langle A, B \rangle / r^2 = \langle \tilde{A}, \tilde{B} \rangle$. It therefore suffices to prove the statement for $r = 1$, which is assumed from now on.

Step 2 — the angular distance from the starting point. Set $u(t) = \langle A, \gamma(t) \rangle \in [-1, 1]$ and, on the stretches where $|u(t)| < 1$, $g(t) = \arccos u(t) \in (0, \pi)$, which is the central angle between A and $\gamma(t)$.

Step 3 — the pointwise inequality $|g'| \leq \|\gamma'\|$. From $\|\gamma\|^2 \equiv 1$, differentiating, $\langle \gamma, \gamma' \rangle = 0$. Decompose $A = u\gamma + A_\perp$ with $\langle A_\perp, \gamma \rangle = 0$; then $\|A_\perp\|^2 = \|A\|^2 - u^2 = 1 - u^2$ and

$$u' = \langle A, \gamma' \rangle = \langle A_\perp, \gamma' \rangle, \quad \text{hence} \quad |u'| \leq \|A_\perp\| \|\gamma'\| = \sqrt{1 - u^2} \|\gamma'\|$$

by the Cauchy–Schwarz inequality. Since where $|u| < 1$ one has $g' = -u'/\sqrt{1 - u^2}$, we conclude

$$|g'(t)| \leq \|\gamma'(t)\|.$$

Step 4 — the last passage through A . The set $\{t \in [0, 1] : \gamma(t) = A\}$ is closed (by continuity), contains 0 and does not contain 1 (because $B \neq A$): let $c < 1$ be its maximum. Then $\gamma(c) = A$, $u(c) = 1$, and $u(t) < 1$ for every $t \in (c, 1]$.

Step 5 — the case in which γ avoids the antipode of A after c . Suppose $\gamma(t) \neq -A$ for every $t \in [c, 1]$. Then $|u| < 1$ on $(c, 1]$, so g is defined and piecewise C^1 on $(c, 1]$; moreover g is continuous up to c with $g(c) = \arccos 1 = 0$. For every $\varepsilon \in (0, 1 - c)$, using the pointwise inequality of Step 3,

$$L(\gamma) \geq \int_{c+\varepsilon}^1 \|\gamma'\| dt \geq \int_{c+\varepsilon}^1 |g'| dt \geq \left| \int_{c+\varepsilon}^1 g' dt \right| = g(1) - g(c + \varepsilon),$$

and letting $\varepsilon \rightarrow 0^+$, by the continuity of g at c ,

$$L(\gamma) \geq g(1) - g(c) = \arccos \langle A, B \rangle.$$

Step 6 — the case in which γ touches the antipode. If there is $t \in [c, 1]$ with $\gamma(t) = -A$, let τ be the least such t . On (c, τ) one has $|u| < 1$ and Step 5 is repeated on the interval $[c, \tau]$ (letting both margins tend to zero), obtaining $L(\gamma) \geq g(\tau) - g(c) = \arccos(-1) = \pi \geq \arccos \langle A, B \rangle$. The inequality of the statement therefore always holds.

Step 7 — the minimum is attained. Let $\theta = \arccos \langle A, B \rangle$. If $\theta < \pi$, the vectors A and B are linearly independent; setting $w = (B - \langle A, B \rangle A) / \|B - \langle A, B \rangle A\|$ (well defined and unit, orthogonal to A), the curve

$$\gamma_0(t) = \cos(\theta t) A + \sin(\theta t) w, \quad t \in [0, 1],$$

satisfies $\|\gamma_0\| \equiv 1$, $\gamma_0(0) = A$, $\gamma_0(1) = B$ and $\|\gamma'_0\| \equiv \theta$, so $L(\gamma_0) = \theta$: it is the arc of the great circle in the plane $\text{span}\{A, w\} = \text{span}\{A, B\}$. If $\theta = \pi$ the same formula works with any unit vector $w \perp A$.

Step 8 — the equality case: the curve stands still, then climbs. Let $L(\gamma) = \theta < \pi$ (i.e. $B \neq -A$; we are in the case of Step 5, since Step 6 would give $L \geq \pi > \theta$). Then all the inequalities of Step 5 are equalities, and in particular:

- (a) $\int_0^c \|\gamma'\| = 0$, hence $\gamma \equiv A$ on $[0, c]$;
- (b) $\|\gamma'\| = |g'|$ almost everywhere on $(c, 1)$, i.e. equality holds in Cauchy–Schwarz at Step 3: $\gamma'(t)$ is parallel to $A_\perp(t) = A - u(t)\gamma(t)$, and hence $\gamma' \in \text{span}\{A, \gamma\}$;
- (c) $g' \geq 0$ almost everywhere, i.e. g is non-decreasing.

Step 9 — from (b) it follows that γ lies on a great circle. Since $\gamma(1) = B$ and A, B are linearly independent, the unit vector $n = (A \times B)/\|A \times B\|$ is well defined, orthogonal to A and to B . Set $\psi(t) = \langle \gamma(t), n \rangle$; then $\psi(1) = 0$ and, writing $\gamma' = \alpha A + \beta \gamma$ with α, β piecewise continuous functions (Step 8(b)),

$$\psi' = \alpha \langle A, n \rangle + \beta \langle \gamma, n \rangle = \beta \psi,$$

because $\langle A, n \rangle = 0$. Hence $(\psi e^{-\int \beta})' = 0$, i.e. $\psi e^{-\int \beta}$ is constant; vanishing at $t = 1$, it is identically zero, and therefore $\psi \equiv 0$. The curve lies in the plane $n^\perp$ through the origin: it lies on the great circle through A and B .

Step 10 — it is the minor arc, traversed monotonically. By (c) the function g increases from 0 to $\theta < \pi$ without ever exceeding θ : the curve never reaches the antipode of A , hence it traverses the arc of the great circle that does not contain it, that is the minor arc, and traverses it monotonically. Conversely every monotone parametrization of that arc has length θ (Step 7): the equality case is exactly this.

Step 11 — the antipodal case. If $B = -A$ then $\theta = \pi$ and every great semicircle from A to $-A$ has length π : the minimum exists but is not unique, in agreement with Exercise 7.3.

Step 12 — numerical check. On the unit S^2 , $A = (1, 0, 0)$ and $B = (0, 1, 0)$ give $\langle A, B \rangle = 0$ and $\theta = \arccos 0 = \pi/2 = 1.570796$: the quarter of the equator. A competing path, $A \rightarrow (0, 0, 1) \rightarrow B$ along two quarter meridians, measures $\pi/2 + \pi/2 = \pi = 3.141593$, twice as much. ✓ The chord instead measures $\|A - B\| = \sqrt{2} = 1.414214 < 1.570796$: shorter, but it is not a path on the sphere. ✓

Step 13 — a second route to the proof. Exercise S3 obtains the same inequality in spherical coordinates, writing $\|\gamma'\|^2 = \dot{\vartheta}^2 + \sin^2 \vartheta \dot{\varphi}^2 \geq \dot{\vartheta}^2$: it is quicker, but it requires the curve to stay in the domain of a chart, i.e. to avoid the poles. The proof given above uses no charts and for this reason has no exceptions. □

***Remark 7.5: What is left to Chapter 9.** Theorem 7.7 solves the global minimum problem on the sphere by elementary means, and solves it completely: existence, value and uniqueness. Two different questions remain outside, both treated in Chapter 9: the local characterization of geodesics as solutions of the Euler–Lagrange equation (valid on an arbitrary surface) and the fact that a geodesic longer than half the sphere ceases to be minimizing while remaining a geodesic. Nothing that follows in this chapter depends on those two questions.*

Exercise 7.3. Why is the minimal path not unique for two antipodal points?

Solution. If A and $B = -A$ are antipodal, every plane through O contains both (since O is the midpoint of the segment AB): there exist infinitely many great circles through A, B , all of semi-length πr . The minimum exists but is realized by infinitely many arcs (the meridians

between two poles). In the proof of Theorem 7.7 the point where uniqueness is lost is Step 9: if $B = -A$ the two vectors are dependent, the unit vector n is not defined and the plane of the motion is not determined.

Theorem 7.8: Spherical distance. For $A, B \in S^2(O, r)$ with central angle $\theta = \angle AOB$, the geodesic distance is $d(A, B) = r\theta$, and $\theta = \arccos(\langle A, B \rangle / r^2)$. The function d is a distance on $S^2(O, r)$, with $0 \leq d \leq \pi r$.

Proof. Step 1 — the value. By Theorem 7.7 the minimum of the lengths is $r \arccos(\langle A, B \rangle / r^2)$ and is attained by the minor arc of a great circle; the central angle θ between OA and OB satisfies by definition $\cos \theta = \langle A, B \rangle / r^2$, hence $d(A, B) = r\theta$. The value does not depend on the chosen plane because θ is determined by the vectors OA and OB alone.

Step 2 — it is a distance. Symmetry and $d(A, B) = 0 \iff A = B$ are immediate from $\arccos$. For the triangle inequality, let A, B, C be given and let γ be the concatenation of the minor arc from A to C with the one from C to B : it is a piecewise C^1 curve from A to B of length $d(A, C) + d(C, B)$, hence by Theorem 7.7 $d(A, B) \leq d(A, C) + d(C, B)$.

Step 3 — the extremes. $\theta \in [0, \pi]$ because $\arccos$ takes values in $[0, \pi]$; the maximum πr occurs for antipodal points.

Step 4 — numerical check. On the Earth ($r = 6371$ km) pole and equator have $\theta = \pi/2$ and $d = 1.570796 \times 6371 = 10\,007.5$ km, which is precisely the quarter meridian used in the original definition of the metre. $\checkmark$ $\square$

Proposition 7.2: Spherical law of cosines. Let ABC be a non-degenerate spherical triangle on the unit sphere, with (angular) sides $a = \angle BOC$, $b = \angle COA$, $c = \angle AOB$ in $(0, \pi)$ and interior angles A, B, C . Then

$$\cos c = \cos a \cos b + \sin a \sin b \cos C.$$

Proof. Step 1 — the angle at C as an angle between tangents. Denote by $\mathbf{A}, \mathbf{B}, \mathbf{C}$ the three vertices as unit vectors of $\mathbb{R}^3$, so that $\cos a = \langle \mathbf{B}, \mathbf{C} \rangle$, $\cos b = \langle \mathbf{C}, \mathbf{A} \rangle$, $\cos c = \langle \mathbf{A}, \mathbf{B} \rangle$. By Theorem 7.7 the side CA is the arc of the great circle through $\mathbf{C}$ and $\mathbf{A}$; its tangent vector at $\mathbf{C}$, normalized, is

$$\mathbf{t}_A = \frac{\mathbf{A} - \langle \mathbf{A}, \mathbf{C} \rangle \mathbf{C}}{\|\mathbf{A} - \langle \mathbf{A}, \mathbf{C} \rangle \mathbf{C}\|} = \frac{\mathbf{A} - \cos b \mathbf{C}}{\sin b},$$

because $\|\mathbf{A} - \cos b \mathbf{C}\|^2 = 1 - \cos^2 b = \sin^2 b$ and $\sin b > 0$. Similarly $\mathbf{t}_B = (\mathbf{B} - \cos a \mathbf{C}) / \sin a$. The angle C of the triangle is by definition the angle between $\mathbf{t}_A$ and $\mathbf{t}_B$.

Step 2 — the scalar product.

$$\cos C = \langle \mathbf{t}_A, \mathbf{t}_B \rangle = \frac{\langle \mathbf{A}, \mathbf{B} \rangle - \cos a \langle \mathbf{A}, \mathbf{C} \rangle - \cos b \langle \mathbf{C}, \mathbf{B} \rangle + \cos a \cos b}{\sin a \sin b} = \frac{\cos c - \cos a \cos b}{\sin a \sin b},$$

having used $\langle \mathbf{A}, \mathbf{C} \rangle = \cos b$, $\langle \mathbf{C}, \mathbf{B} \rangle = \cos a$ and simplifying $-\cos a \cos b - \cos b \cos a + \cos a \cos b = -\cos a \cos b$.

Step 3 — conclusion. Multiplying by $\sin a \sin b \neq 0$ (guaranteed by $a, b \in (0, \pi)$) and rearranging gives the claim.

Step 4 — degenerate cases. If $a = 0$ or $b = 0$ two vertices coincide and the angle C is not defined; if $a = \pi$ or $b = \pi$ they are antipodal and the side is not unique. These are exactly the cases excluded by the non-degeneracy hypothesis.

Step 5 — check. With $a = b = c = \pi/2$ (trirectangular triangle) the formula gives $0 = 0 + 1 \cdot 1 \cdot \cos C$, i.e. $C = \pi/2$: the edges of an octant. $\checkmark$ With $a, b, c \rightarrow 0$, expanding to second

order $1 - \frac{c^2}{2} \simeq (1 - \frac{a^2}{2})(1 - \frac{b^2}{2}) + ab \cos C$ one recovers $c^2 \simeq a^2 + b^2 - 2ab \cos C$: the planar law of cosines. ✓

Step 6 — where the rest of the theory lies. Chapter 8 proves this formula systematically together with its companions (law of sines, dual formula, spherical Pythagoras, Napier's rules, L'Huilier) and applies them to navigation: here the law of cosines serves only as a tool for computing distances. □

Exercise 7.4. Two cities are at latitudes ϕ_1, ϕ_2 and longitude difference $\Delta\lambda$. Write the central angle θ and the distance on Earth ($R \approx 6371$ km).

Solution. Applying Proposition 7.2 to the triangle pole–city–city (the sides from the pole are the colatitudes $\frac{\pi}{2} - \phi_i$, the angle at the pole is $\Delta\lambda$):

$$\cos \theta = \sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Delta\lambda, \quad d = R\theta.$$

Numerically one often uses the *haversine* formula $\text{hav} \theta = \text{hav}(\phi_2 - \phi_1) + \cos \phi_1 \cos \phi_2 \text{hav}(\Delta\lambda)$, with $\text{hav}(x) = \sin^2(x/2)$, numerically more stable for small distances because it avoids the loss of significant digits of arccos near 1.

Remark 7.6: Orthodromic navigation. The shortest route between two points of the Earth is the great circle arc (orthodromy): it is the content of Theorem 7.7 applied to the spherical model. A guidance system computes the distance with $d = R\theta$ and θ from the haversine formula; for long routes (aeronautical, maritime) the error with respect to the ellipsoidal model is of the order of 0.3%, correctable with Vincenty's formulas. The spherical triangle pole–departure–arrival and the law of cosines also give the initial bearing to follow, which along an orthodromy varies continuously — hence the need for periodic recalculation on board. The complete computation, with the quadrant check, is in Chapter 8.

7.8. Area of the sphere and volume of the ball

Theorem 7.9: Area of the sphere (Archimedes). $\text{Area}(S^2(O, r)) = 4\pi r^2$; more precisely the cap of angular semi-aperture ϑ_0 has area $2\pi r^2(1 - \cos \vartheta_0) = 2\pi rh$, where $h = r(1 - \cos \vartheta_0)$ is its height.

Proof. **Step 1 — the area element.** By Lemma 7.1, $dA = r^2 \sin \vartheta \, d\vartheta \, d\varphi$.

Step 2 — the cap.

$$\text{Area} = \int_0^{2\pi} \int_0^{\vartheta_0} r^2 \sin \vartheta \, d\vartheta \, d\varphi = 2\pi r^2 [-\cos \vartheta]_0^{\vartheta_0} = 2\pi r^2 (1 - \cos \vartheta_0).$$

Step 3 — the whole sphere. For $\vartheta_0 = \pi$ one gets $2\pi r^2(1 + 1) = 4\pi r^2$.

Step 4 — Archimedes' theorem. Setting $h = r(1 - \cos \vartheta_0)$ one has $\text{Area} = 2\pi rh$: the area of the cap depends *linearly* on the height, that is, the radial projection of the sphere onto the circumscribed cylinder preserves areas. It is the result Archimedes wanted engraved on his tomb, together with the ratio 2/3 between sphere and cylinder.

Step 5 — numerical check. For $r = 10$ and $\vartheta_0 = \pi/2$ (hemisphere): $2\pi \times 100 \times (1 - 0) = 628.319$, i.e. half of $4\pi \times 100 = 1256.637$. ✓ With $h = r = 10$: $2\pi \times 10 \times 10 = 628.319$, the same. ✓ □

Theorem 7.10: Volume of the ball (spherical shells). $\text{Vol}(B^3(O, r)) = \frac{4}{3}\pi r^3$, and $\frac{d}{dr} \text{Vol} = \text{Area}$.

Proof. Step 1 — decomposition into shells. The ball is the disjoint union of the concentric spheres of radius $\rho \in [0, r]$; by Theorem 7.9 the sphere of radius ρ has area $4\pi\rho^2$, and integration in spherical coordinates ($dV = \rho^2 \sin \vartheta d\rho d\vartheta d\varphi$, i.e. $dV = dA d\rho$) gives

$$\text{Vol} = \int_0^r 4\pi\rho^2 d\rho = \frac{4}{3}\pi r^3.$$

Step 2 — the differential relation. $\frac{d}{dr}(\frac{4}{3}\pi r^3) = 4\pi r^2 = \text{Area}$, exactly as in the plane $\frac{d}{dr}\pi r^2 = 2\pi r$. The reason is that the increment of volume between the radii r and $r + dr$ is a shell of area $4\pi r^2$ and thickness dr ; the same relation, in dimension n , is the content of Exercise S1 of Chapter 11.

Step 3 — numerical check and comparison with Archimedes. For $r = 1$: $\frac{4}{3}\pi = 4.188790$. The circumscribed cylinder has volume $\pi r^2 \cdot 2r = 2\pi = 6.283185$, and the ratio is $4.188790/6.283185 = 0.666667 = 2/3$. ✓ □

Exercise 7.5. Calculate the area of a spherical cap of angular semi-aperture ϑ_0 and the volume of the spherical segment it bounds.

Solution. By Theorem 7.9, $\text{Area} = 2\pi r^2(1 - \cos \vartheta_0) = 2\pi r h$ with $h = r(1 - \cos \vartheta_0)$. For the volume one integrates the area of the horizontal disks: at height $z \in [r - h, r]$ the disk has radius $\sqrt{r^2 - z^2}$, hence

$$V = \int_{r-h}^r \pi(r^2 - z^2) dz = \pi \left[r^2 z - \frac{z^3}{3} \right]_{r-h}^r = \frac{\pi h^2}{3}(3r - h).$$

Check for $h = r$ (hemisphere): $V = \frac{\pi r^2}{3} \cdot 2r = \frac{2}{3}\pi r^3$, half of $\frac{4}{3}\pi r^3$. ✓ Check for $h = 2r$ (whole ball): $V = \frac{4\pi r^2}{3} \cdot r = \frac{4}{3}\pi r^3$. ✓

7.9. Gauss–Bonnet on the geodesic triangle

Theorem 7.4 has already established that the area of a spherical triangle is r^2 times its angular excess. Here we isolate the ingredient that makes it a theorem of *curvature* — the lune lemma — and recognize in the factor $1/r^2$ the Gaussian curvature of the sphere: it is the first, and simplest, case of the Gauss–Bonnet theorem.

Lemma 7.2: Area of a lune. A spherical **lune** — the region of $S^2(O, r)$ between two half-planes through the same line through O , of dihedral angle $\alpha \in [0, 2\pi]$ — has area $2\alpha r^2$.

Proof. Step 1 — choice of coordinates. Up to a rotation (which preserves areas, being an isometry of $\mathbb{R}^3$) the line common to the two half-planes is the z axis, and the lune is the image, in the parametrization of Definition 7.7, of the rectangle $\{(\varphi, \vartheta) : 0 \leq \varphi \leq \alpha, 0 \leq \vartheta \leq \pi\}$.

Step 2 — integration. By Lemma 7.1,

$$\text{Area} = \int_0^\alpha \int_0^\pi r^2 \sin \vartheta d\vartheta d\varphi = \alpha r^2 [-\cos \vartheta]_0^\pi = 2\alpha r^2.$$

Step 3 — checks. For $\alpha = 2\pi$ one obtains $4\pi r^2$, the whole sphere (Theorem 7.9) ✓; for $\alpha = \pi$ one obtains $2\pi r^2$, the hemisphere ✓; for $\alpha = \pi/2$ a quarter of the sphere, πr^2 ✓. The area is proportional to α , as intuition suggests — but here proportionality is *proved*, not postulated: it follows from the fact that the area element does not depend on φ . □

Theorem 7.11: Gauss–Bonnet for a geodesic triangle. Let T be a geodesic triangle on $S^2(O, r)$, i.e. a region bounded by three arcs of great circles, with interior angles α, β, γ . Then

$$\alpha + \beta + \gamma - \pi = \frac{\text{Area}(T)}{r^2} = K \cdot \text{Area}(T),$$

where $K = 1/r^2$ is the Gaussian curvature of the sphere of radius r .

Proof. Step 1 — the area by the method of lunes. The three great circles extending the sides of T meet pairwise in pairs of antipodal points, and determine at each vertex two opposite lunes (Lemma 7.2) of width equal to the interior angle; the antipodal triangle $T' = -T$ is congruent to T and has the same area. The count with multiplicity of the six lunes — carried out in full in the proof of Theorem 7.4, Steps 1–4 — gives

$$2(2\alpha r^2 + 2\beta r^2 + 2\gamma r^2) = 4\pi r^2 + 4 \text{Area}(T), \quad \text{that is} \quad \text{Area}(T) = r^2(\alpha + \beta + \gamma - \pi).$$

Dividing by r^2 gives the first equality of the statement.

Step 2 — the Gaussian curvature of the sphere is $1/r^2$. Take as Gauss map the inward unit normal $N(P) = (O - P)/r$. Then N is linear in P and

$$dN_P(w) = -\frac{1}{r} w \quad \text{for every } w \in T_P S^2,$$

so the shape operator $-dN_P$ is $\frac{1}{r}$ id: both principal curvatures equal $k_1 = k_2 = 1/r$ and the Gaussian curvature, their product, equals

$$K = k_1 k_2 = \frac{1}{r^2}.$$

(The general definition of K as the determinant of the shape operator, and its invariance under isometries, are in Chapter 10; here only the computation for the sphere is needed, and it is immediate.)

Step 3 — conclusion. Substituting $1/r^2 = K$ into the first equality gives $\alpha + \beta + \gamma - \pi = K \text{Area}(T)$.

Step 4 — numerical check. The trirectangular triangle has $\alpha = \beta = \gamma = \pi/2$, hence excess $\frac{3\pi}{2} - \pi = \frac{\pi}{2} = 1.570796$ and area $\frac{\pi}{2}r^2$: on $r = 1$ it equals 1.570796, i.e. $\frac{1}{8}$ of $4\pi = 12.566371$ ✓ — and indeed the three coordinate planes divide the sphere into eight equal triangles. On a sphere of radius 2 the area becomes $2\pi = 6.283185$, while the excess stays $\pi/2$: consistently, $K = 1/4$ and $K \cdot 2\pi = \pi/2$. ✓

Step 5 — the general case. The Gauss–Bonnet theorem in general form states that for a region R with piecewise regular boundary on an arbitrary surface $\int_R K dA + \int_{\partial R} \kappa_g ds + \sum_i \varepsilon_i = 2\pi$, with κ_g the geodesic curvature and ε_i the exterior angles. For a geodesic triangle on S^2 : $\kappa_g \equiv 0$ on the sides (they are geodesics), K is constant $= 1/r^2$ and $\sum \varepsilon_i = 3\pi - (\alpha + \beta + \gamma)$, and one recovers the formula proved above. The general version, which requires the notion of geodesic curvature, is outside this chapter. □

Corollary 7.2: Spherical excess. For every non-degenerate geodesic triangle of $S^2(O, r)$ the **spherical excess** $E = \alpha + \beta + \gamma - \pi$ satisfies

$$0 < E < 2\pi, \quad E = \frac{\text{Area}(T)}{r^2}.$$

In particular on the sphere there are no triangles with angle sum equal to π , and two spherical triangles with the same angles have the same area.

Proof. **Step 1 — the equality.** It is Theorem 7.11.

Step 2 — $E > 0$. A non-degenerate triangle has as vertices three linearly independent unit vectors, hence contains a non-empty open subset of the sphere and has strictly positive area; by Step 1, $E = \text{Area}(T)/r^2 > 0$.

Step 3 — $E < 2\pi$. Every interior angle satisfies $0 < \alpha, \beta, \gamma < \pi$, hence $\alpha + \beta + \gamma < 3\pi$ and $E < 2\pi$. Equivalently $\text{Area}(T) < 2\pi r^2$: a geodesic triangle cannot occupy more than a hemisphere.

Step 4 — no Euclidean triangles. If $E = 0$ the area would be zero and the three vertices would lie on the same great circle: the triangle would be degenerate.

Step 5 — no similarity on the sphere. Since the area is determined by the angles, two spherical triangles with the same angles cannot have different areas: on the sphere similar triangles are congruent. It is the corollary proved in detail in Chapter 8, and the most conspicuous difference from the geometry of the plane.

Step 6 — numerical check. The triangle with one vertex at the North Pole and two on the equator separated by 120° has angles $2\pi/3, \pi/2, \pi/2$, hence $E = 2.094395 + 1.570796 + 1.570796 - 3.141593 = 2.094395$ and area $2.094395 r^2$, i.e. $\frac{1}{6}$ of $4\pi r^2 = 12.566371 r^2$. ✓ And it is right: three meridians at 120° cut the northern hemisphere into three equal parts. □

Remark 7.7: Planar limit case. For $r \rightarrow \infty$ ($K \rightarrow 0$) a triangle of finite area has $E = \text{Area}/r^2 \rightarrow 0$: one recovers $\alpha + \beta + \gamma = \pi$, the Euclidean theorem. Gauss–Bonnet thus generalizes planar geometry, of which Volume I is the case $K = 0$. The same computation measures the error made in treating a terrestrial region as flat: a triangle of 100×100 km on the Earth has area $\simeq 5 \cdot 10^9$ m² and $E \simeq 5 \cdot 10^9 / (6.371 \cdot 10^6)^2 = 1.23 \cdot 10^{-4}$ rad, that is about 25 seconds of arc: small, but perfectly measurable, and indeed geodesy has been measuring it for two centuries.

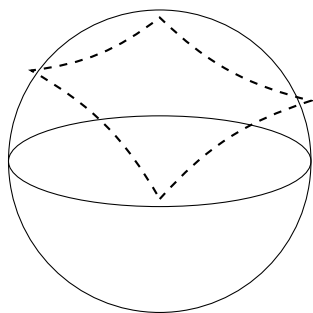


Figure 7.5: A geodesic triangle on the sphere: the sides are arcs of great circles (the geodesics, Theorem 7.7). Unlike the plane, the sum of the angles exceeds π by an amount proportional to the area (Theorem 7.11). It is the signature of the positive curvature of the sphere, absent in the plane circle ($K = 0$).

Exercise 7.6. Calculate the area of the spherical triangle with three right angles on a sphere of radius $r = 1$.

Solution. With $\alpha = \beta = \gamma = \pi/2$: $E = \frac{3\pi}{2} - \pi = \frac{\pi}{2}$, therefore by Theorem 7.11 $\text{Area}(T) = Er^2 = \pi/2 = 1.570796$. It is exactly $1/8$ of the total area $4\pi = 12.566371$ of the sphere: the octant enclosed by three meridians at 90° and the equator, as verified geometrically. ✓

7.10. Industrial applications

Spherical wrist and orientation. The *spherical wrist* of six-axis robots (the axes J4, J5, J6 meet at a point) makes the orientation of the tool describable by rotations about a fixed centre: the geometry is that of the sphere S^2 and of the group $SO(3)$ (Chapters 11 and 28). The non-commutativity of $SO(3)$, announced in the table of section 7.1, is the reason why the order of rotations matters and why gimbal lock configurations exist.

Orthodromic navigation. The geodesic distance $d = r\theta$ (Theorem 7.8) and the spherical law of cosines (Proposition 7.2) are the basis of orthodromic route computation in guidance systems (AGV, drones, marine vehicles): the shortest path is the great circle arc, and the haversine formula gives its length in a numerically stable way.

Cartography and mapping. The conformality of stereographic projection (Theorem 7.6) is why it is the standard projection of polar aeronautical charts and of pole figures in crystallography: the angles read on the chart are the true ones. Gauss–Bonnet, with the angular excess proportional to the area (Theorem 7.11), quantifies instead the error made by approximating a curved region with a plane polygon — the computation of Remark 7.7 gives the order of magnitude to be compared with the tolerance of the survey.

Surface machining. Corollary 7.1 (every circle of radius $\rho \leq r$ is a section of a sphere of radius r) and Remark 7.3 (the ellipsoid contains two families of circles) are the two facts underlying the machining of caps and of ellipsoidal surfaces with rotating tools: in both cases the workpiece is generated by moving a circle.

7.11. Assigned exercises

Exercise 7.7. Objective: apply Corollary 7.1 and recognize its limiting case. **Estimated time:** 15 minutes.

A circle of radius $\rho = 12$ cm lies in a plane π . (i) Determine all the spheres of radius $r = 13$ cm containing it and the distance of their centres from π . (ii) For which radius r does the circle become a great circle, and how many spheres of that radius contain it? (iii) Is there a sphere of radius 10 cm containing it? Justify.

Exercise 7.8. Objective: use the formulas of Proposition 7.1 and verify conformality on an explicit family. **Estimated time:** 25 minutes.

On the unit sphere ($r = 1$): (i) determine the stereographic image of the parallel $z = z_0$ with $z_0 \in (-1, 1)$ and show that it is the circle of centre the origin and radius $\sqrt{(1+z_0)/(1-z_0)}$, both by Theorem 7.6(ii) and by direct computation; (ii) compute that radius for $z_0 = -\frac{1}{2}, 0, \frac{1}{2}, 0.9$; (iii) determine the image of the meridian of longitude φ_0 and check that the two families of images meet at right angles, as conformality requires.

Exercise 7.9. Objective: use Gauss–Bonnet on a terrestrial triangle and measure the deviation from the plane model. **Estimated time:** 30 minutes.

On the spherical Earth of radius $R = 6371$ km consider the geodesic triangle with vertices the North Pole, the point $(0^\circ\text{N}, 0^\circ\text{E})$ and the point $(0^\circ\text{N}, 40^\circ\text{E})$. (i) Compute its three interior angles, the spherical excess and the area, and check that the area is a simple fraction of the Earth’s surface. (ii) Compute the lengths of the three sides in kilometres. (iii) Compute the area of the *plane* Euclidean triangle with those three sides and compare it with the spherical area; comment on the sign and order of magnitude of the deviation.

7.12. Worked exercises

Exercise 7.10. On the unit sphere consider the circle $\Gamma = S^2 \cap \{x + 2y + 2z = 1\}$. Determine centre and radius of its stereographic image, and check them with three explicit points of Γ .

Solution. Strategy. Bring the plane into normalized form, apply the formulas of Theorem 7.6(ii), then check on points chosen so as to cover the three cases $z < 0$, $z = 0$, $z > 0$.

Step 1 — normalization. The vector $(1, 2, 2)$ has norm 3, so the plane is written $ax + by + cz = k$ with

$$a = \frac{1}{3}, \quad b = \frac{2}{3}, \quad c = \frac{2}{3}, \quad k = \frac{1}{3}, \quad a^2 + b^2 + c^2 = \frac{1+4+4}{9} = 1.$$

Since $|k| = 0.333333 < 1 = r$, the intersection really is a circle, of radius $\rho = \sqrt{1 - 1/9} = \sqrt{8}/3 = 0.942809$ on the sphere.

Step 2 — line or circle? $cr - k = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \neq 0$: the pole N does *not* lie on Γ (indeed $0 + 0 + \frac{2}{3} \cdot 1 = 0.666667 \neq 0.333333$), so the image is a circle.

Step 3 — centre. $(u_0, v_0) = -\frac{r^2}{cr - k}(a, b) = -3 \cdot (\frac{1}{3}, \frac{2}{3}) = (-1, -2)$.

Step 4 — radius. $R_\Gamma = \frac{r\sqrt{r^2 - k^2}}{|cr - k|} = \frac{\sqrt{1 - 1/9}}{1/3} = 3 \times 0.942809 = 2.828427 = 2\sqrt{2}$.

Step 5 — independent check on three points.

- $P_1 = (1, 0, 0)$: it lies on Γ ($1 + 0 + 0 = 1$ ✓, $\|P_1\| = 1$ ✓) and $\sigma(P_1) = \frac{1}{1-0}(1, 0) = (1, 0)$. Distance from the centre: $\sqrt{(1+1)^2 + (0+2)^2} = \sqrt{8} = 2.828427$. ✓
- $P_2 = (-\frac{3}{5}, \frac{4}{5}, 0)$: indeed $-0.6 + 1.6 + 0 = 1$ ✓ and $0.36 + 0.64 = 1$ ✓; $\sigma(P_2) = (-0.6, 0.8)$ and the distance from the centre is $\sqrt{0.16 + 7.84} = \sqrt{8} = 2.828427$. ✓
- $P_3 = (0, \frac{1+\sqrt{7}}{4}, \frac{1-\sqrt{7}}{4}) = (0, 0.911438, -0.411438)$: from $x = 0$ it follows that $y + z = \frac{1}{2}$ and $y^2 + z^2 = 1$, whence $yz = -\frac{3}{8}$ and $y, z = \frac{1 \pm \sqrt{7}}{4}$; $\sigma(P_3) = \frac{1}{1.411438}(0, 0.911438) = (0, 0.645751)$, and since $0.645751 = \sqrt{7} - 2$ the distance from the centre is $\sqrt{1 + (\sqrt{7})^2} = \sqrt{8} = 2.828427$ *exactly*. ✓

Interpretation. The circle Γ has radius 0.942809 on the sphere and its image has 2.828427: stereographic projection has dilated it by a factor 3, which is λ evaluated where the circle lies. The closer Γ gets to the pole (i.e. the closer $cr - k$ gets to zero) the larger the image, until it becomes a line when $cr = k$.

Typical mistakes. (a) Forgetting to normalize (a, b, c) : without $a^2 + b^2 + c^2 = 1$ the formulas for centre and radius do not hold. (b) Confusing $|k|$ with the distance of the plane from the centre when the vector is not a unit vector. (c) Concluding “line” because the plane does not pass through the origin: the right condition is $N \in \pi$, i.e. $cr = k$, not $k = 0$.

Extension. Redo the computation for the plane $2x + y + 2z = 2$ ($a = \frac{2}{3}$, $b = \frac{1}{3}$, $c = \frac{2}{3}$, $k = \frac{2}{3} = cr$): one finds a line, and rightly so, because $N = (0, 0, 1)$ satisfies $2 \cdot 0 + 0 + 2 \cdot 1 = 2$.

Exercise 7.11. Two points both lie at latitude 40° N, with longitude difference $\Delta\lambda = 60^\circ$. Compute the geodesic distance and compare it with the path along the parallel, on the unit sphere and on the Earth ($R = 6371$ km).

Solution. Strategy. The path along the parallel is an admissible but non-geodesic curve: by Theorem 7.7 it must be longer than the great circle arc. The central angle is computed in two independent ways (scalar product and spherical law of cosines) and the two lengths are compared.

Step 1 — the two points. With $\phi = 40^\circ$ and $\Delta\lambda = 60^\circ$,

$$A = (\cos \phi, 0, \sin \phi), \quad B = (\cos \phi \cos \Delta\lambda, \cos \phi \sin \Delta\lambda, \sin \phi),$$

with $\cos 40^\circ = 0.766044$ and $\sin 40^\circ = 0.642788$.

Step 2 — the central angle from the scalar product.

$$\langle A, B \rangle = \cos^2 \phi \cos \Delta\lambda + \sin^2 \phi = 0.586824 \times 0.5 + 0.413176 = 0.293412 + 0.413176 = 0.706588,$$

hence $\theta = \arccos(0.706588) = 0.786132$ rad $= 45.042024^\circ$.

Step 3 — independent check with the spherical law of cosines. In the triangle pole– A – B the sides from the pole are the colatitudes $c_1 = c_2 = 90^\circ - 40^\circ = 50^\circ$ and the angle at the pole is $\Delta\lambda = 60^\circ$. By Proposition 7.2,

$$\cos \theta = \cos^2 50^\circ + \sin^2 50^\circ \cos 60^\circ = 0.413176 + 0.586824 \times 0.5 = 0.706588. \checkmark$$

The two computations agree to six digits.

Step 4 — the path along the parallel. The parallel at latitude ϕ is a circle of radius $\cos \phi$ (Example 7.1), so the arc of amplitude $\Delta\lambda$ measures

$$L_{\parallel} = \cos \phi \cdot \Delta\lambda = 0.766044 \times 1.047198 = 0.802200.$$

Step 5 — the comparison. $\theta = 0.786132 < 0.802200 = L_{\parallel} \checkmark$, with a relative saving of $1 - 0.786132/0.802200 = 0.020030$, i.e. 2.00%. On the Earth:

$$d = 6371 \times 0.786132 = 5008.4 \text{ km}, \quad L_{\parallel} = 6371 \times 0.802200 = 5110.8 \text{ km},$$

with a difference of 102.4 km.

Interpretation. Flying “straight east” along the parallel costs about a hundred extra kilometres: the orthodromy passes *north* of the parallel, and its maximum latitude (the *vertex* of the route) exceeds 40° . It is exactly why transatlantic routes look curved on Mercator charts (Example 7.1).

Typical mistakes. (a) Using the latitude instead of the colatitude in the law of cosines. (b) Multiplying by R an angle expressed in degrees instead of radians. (c) Confusing orthodromy and loxodromy: the latter cuts the meridians at a constant angle and is convenient to follow, but is not the shortest.

Extension. Repeat with $\Delta\lambda = 180^\circ$: then $\cos \theta = \cos^2 \phi \cdot (-1) + \sin^2 \phi = 1 - 2\cos^2 \phi = -0.173648$, $\theta = 1.745329$ rad $= 100^\circ$, whereas the parallel measures $0.766044 \times \pi = 2.406607$: the orthodromy passes over the North Pole and saves 27%.

Exercise 7.12. On the unit sphere consider the equilateral geodesic triangle of side a . Express its angle A and its excess as functions of a , compute them for $a = \pi/2$, 0.1, 0.01 and compare the area with that of the plane equilateral triangle of side a .

Solution. Strategy. Derive $\cos A$ from the spherical law of cosines specialized to the case $a = b = c$, then use Gauss–Bonnet for the area and compare with the plane formula $\sqrt{3}a^2/4$.
Step 1 — the angle. With $a = b = c$, Proposition 7.2 gives $\cos a = \cos^2 a + \sin^2 a \cos A$, that is

$$\cos A = \frac{\cos a - \cos^2 a}{\sin^2 a} = \frac{\cos a (1 - \cos a)}{(1 - \cos a)(1 + \cos a)} = \frac{\cos a}{1 + \cos a},$$

having cancelled $1 - \cos a \neq 0$ (i.e. $a \neq 0$).

Step 2 — the excess and the area. By Theorem 7.11, Area = $E = 3A - \pi$ on the unit sphere.

Step 3 — the three cases.

a	$\cos A$	A (rad)	$E = \text{spherical area}$	$\sqrt{3}a^2/4$ (plane area)
$\pi/2$	0	1.570796	1.570796	1.068416
0.1	0.498748	1.048643	0.004335547	0.004330127
0.01	0.499988	1.047212	0.00004330181	0.00004330127

Step 4 — independent check of the first case. For $a = \pi/2$ one has $\cos A = 0$, hence $A = \pi/2$: it is the trirectangular triangle, and the area $\pi/2 = 1.570796$ coincides with $\frac{1}{8}$ of 4π , computed geometrically in Exercise 7.6. ✓

Step 5 — the plane limit. The ratio between spherical and plane area equals 1.0012517 for $a = 0.1$ and 1.0000125 for $a = 0.01$: it tends to 1 like a^2 , i.e. the relative deviation is divided by 100 when the side is divided by 10. ✓ It is the quantitative manifestation of Remark 7.7: the sphere is locally flat, and the error is of second order in the diameter.

Interpretation. For $a = \pi/2$ the spherical area exceeds the plane one by 47%: in the large, curvature is plainly visible. For $a = 0.01$ (on an Earth of radius 6371 km, i.e. sides of 64 km) the deviation is one part in eighty thousand: below the tolerance of almost any survey, but not of a geodetic one.

Typical mistakes. (a) Cancelling $1 - \cos a$ without excluding $a = 0$. (b) Applying $E = 3A - \pi$ with A in degrees. (c) Forgetting that on the sphere the angle of an equilateral triangle is not $\pi/3$: that value would give $E = 0$, impossible by Corollary 7.2.

Extension. For which a does the excess equal π (i.e. the triangle occupies a quarter of the sphere)? From $3A - \pi = \pi$ it follows that $A = 2\pi/3$, $\cos A = -\frac{1}{2}$ and $\cos a/(1 + \cos a) = -\frac{1}{2}$, i.e. $\cos a = -\frac{1}{3}$ and $a = 1.910633$ rad = 109.471° , the tetrahedral angle.

7.13. Numerical and symbolic exercises with solutions

The exercises that follow come from the treatment of the circle in higher dimension of Volume II and are reproduced in full, with their solutions; those on hyperspheres and rotations are in Chapter 11.

7.13.1 Numerical exercises

Exercise 7.13: N1 — Sections of a cone. A cone has half-angle $\alpha = 30^\circ$. Compute the radius of the section orthogonal to the axis at distance 5 from the vertex; then classify the sections obtained with planes inclined at 60° , 30° and 20° to the axis, computing the eccentricity in the first case from $e = \cos \beta / \cos \alpha$.

Exercise 7.14: N2 — Spherical cap. A sphere has radius $R = 10$ cm. A plane is at distance $d = 6$ cm from the centre. Compute the radius of the section, the area of the smaller cap and the volume of the corresponding spherical segment.

Exercise 7.15: N3 — A terrestrial spherical triangle. On the Earth ($R = 6371$ km) consider the triangle with vertices at the North Pole and at two points of the equator separated by 120° of longitude. Compute its angles and area, and check the result by a symmetry argument.

Exercise 7.16: N4 — Circular sections of an ellipsoid. For the ellipsoid with semi-axes $a = 6$, $b = 4$, $c = 3$ determine the planes of the maximal circular sections, the radius of those circles, and verify with an explicit point.

7.13.2 Symbolic exercises

Exercise 7.17: S1 — Dandelin for the parabola. Prove that if the cutting plane is parallel to exactly one generator, the section is a parabola: there is a single Dandelin sphere, its point of tangency is the focus and the directrix is the intersection of the cutting plane with the plane of the circle of tangency.

Exercise 7.18: S2 — Who has all sections circular. Prove that a closed bounded surface of $\mathbb{R}^3$, all of whose non-empty plane sections not reduced to a point are circles, is a sphere.

Exercise 7.19: S3 — The shortest path on the sphere. Prove that among all regular curves on the unit sphere joining two points A and B , the minor arc of great circle has least length, and that this length equals $\arccos\langle A, B \rangle$.

Exercise 7.20: S4 — The circular defect on the sphere. Prove that on the sphere of radius R the circumference of the geodesic circle of radius ρ is $C(\rho) = 2\pi R \sin(\rho/R)$, and that

$$2\pi\rho - C(\rho) = \frac{\pi\rho^3}{3R^2} + O(\rho^5).$$

Interpret the coefficient.

Exercise 7.21: S5 — Intersection of two spheres. Prove that the non-empty intersection, not reduced to a point, of two distinct spheres of $\mathbb{R}^n$ is a sphere S^{n-2} contained in a hyperplane (the *radical plane*), and compute its radius and position in terms of the data.

7.13.3 Solutions to the numerical exercises

Solution. N1. *Step 1 — orthogonal section.* By Theorem 7.1 the radius is

$$r = h \tan \alpha = 5 \tan 30^\circ = 5 \times 0.577350 = 2.886751.$$

Step 2 — classification criterion. Writing β for the angle between the cutting plane and the axis, the section is an ellipse if $\beta > \alpha$, a parabola if $\beta = \alpha$, a hyperbola if $\beta < \alpha$. The limiting case $\beta = 90^\circ$ gives the circle.

Step 3 — the three cases.

β	comparison with $\alpha = 30^\circ$	section
60°	$\beta > \alpha$	ellipse
30°	$\beta = \alpha$	parabola
20°	$\beta < \alpha$	hyperbola

Step 4 — eccentricity of the first case.

$$e = \frac{\cos \beta}{\cos \alpha} = \frac{\cos 60^\circ}{\cos 30^\circ} = \frac{0.500000}{0.866025} = 0.577350.$$

Since $0 < e < 1$, the ellipse is confirmed. ✓

Step 5 — checks at the extremes. For $\beta = 90^\circ$ the formula gives $e = 0$: the circle, in agreement with Step 7 of the proof of Theorem 7.2. For $\beta = \alpha$ it gives $e = 1$: the parabola. For $\beta = 20^\circ$: $e = \cos 20^\circ / \cos 30^\circ = 0.939693 / 0.866025 = 1.085$, greater than 1: hyperbola. ✓

Step 6 — remark. The value $e = 0.5774 = 1/\sqrt{3}$ is no accident: with $\alpha = 30^\circ$ and $\beta = 60^\circ$ the two angles are complementary, and $e = \cos \beta / \cos \alpha = \sin \alpha / \cos \alpha = \tan \alpha = \tan 30^\circ$.

Solution. N2. *Step 1 — radius of the section.* By Theorem 7.3,

$$r = \sqrt{R^2 - d^2} = \sqrt{100 - 36} = \sqrt{64} = 8.00 \text{ cm.}$$

Step 2 — height of the cap. $h = R - d = 10 - 6 = 4 \text{ cm.}$

Step 3 — area of the cap. By Archimedes' theorem (the area of a spherical zone equals that of the corresponding zone of the circumscribed cylinder),

$$A = 2\pi Rh = 2\pi \times 10 \times 4 = 80\pi = 251.327 \text{ cm}^2.$$

Step 4 — volume of the segment.

$$V = \frac{\pi h^2}{3}(3R - h) = \frac{\pi \times 16}{3} \times 26 = \frac{416\pi}{3} = 435.634 \text{ cm}^3.$$

Step 5 — plausibility checks. The total area of the sphere is $4\pi R^2 = 1256.64 \text{ cm}^2$: the cap is 20% of it, consistent with $h/2R = 4/20 = 0.20$ — which is precisely the content of Archimedes' theorem: the area of a zone depends only on its height. ✓ The total volume is $\frac{4}{3}\pi R^3 = 4188.79 \text{ cm}^3$: the segment is 10.4% of it, less than the 20% of the area, as it must be for a cap far from the centre. ✓

Solution. N3. *Step 1 — the angles.* The two sides issuing from the North Pole are meridians, which meet the equator at right angles: hence the two base angles are 90° each. The angle at the pole is the difference of longitude, $120^\circ = 2\pi/3$.

Step 2 — spherical excess.

$$A + B + C - \pi = \frac{\pi}{2} + \frac{\pi}{2} + \frac{2\pi}{3} - \pi = \frac{2\pi}{3} = 2.094395.$$

Step 3 — area. By Theorem 7.4,

$$\text{Area} = R^2 \cdot \frac{2\pi}{3} = 40\,589\,641 \times 2.094395 = 8.5011 \times 10^7 \text{ km}^2,$$

using $R^2 = 6371^2 = 40\,589\,641 \text{ km}^2$.

Step 4 — check by symmetry. Three meridians 120° apart divide the northern hemisphere into three equal wedges, and the sphere into six. Hence the area of the triangle is

$$\frac{4\pi R^2}{6} = \frac{2\pi R^2}{3},$$

exactly the result of Step 3. ✓

Step 5 — absolute values. The Earth's surface is $4\pi R^2 = 5.1006 \times 10^8 \text{ km}^2$; the triangle is one sixth of it, $8.5011 \times 10^7 \text{ km}^2$. ✓

Step 6 — reading. A “triangle” with two right angles is impossible in the plane; on the sphere it is perfectly ordinary, and its area measures the excess. Reducing the longitude from 120° to ε , the area would tend to $R^2\varepsilon$, which tends to zero: one recovers the Euclidean degeneration.

Solution. N4. *Step 1 — checking the hypotheses.* $a = 6 > b = 4 > c = 3$: the three semi-axes are distinct, so Theorem 7.5 applies.

Step 2 — the coefficient.

$$k = \frac{c}{a} \sqrt{\frac{a^2 - b^2}{b^2 - c^2}} = \frac{3}{6} \sqrt{\frac{36 - 16}{16 - 9}} = 0.5 \sqrt{\frac{20}{7}} = 0.5 \times 1.690309 = 0.845154.$$

Step 3 — the planes and the circles. The maximal circular sections lie in the planes

$$z = \pm 0.845154x,$$

both containing the y -axis, and are circles of radius $b = 4$.

Step 4 — verification with a point. On the plane $z = kx$ with $y = 0$, the equation of the ellipsoid gives

$$\frac{x^2}{36} + \frac{k^2 x^2}{9} = 1 \implies x^2(0.027778 + 0.079365) = 1 \implies x^2 = 9.3333,$$

since $k^2 = 0.714286$ and $k^2/9 = 0.079365$. Then $x = 3.05505$ and $z = kx = 2.58199$, and the distance from the origin is

$$\sqrt{9.3333 + 6.6667} = \sqrt{16.0000} = 4.0000 = b. \quad \checkmark$$

Step 5 — second check. The point $(0, 4, 0)$ lies on the ellipsoid ($16/16 = 1$), on both planes ($z = 0 = k \cdot 0$) and at distance 4 from the origin. ✓

Step 6 — reading. The two planes make with the xy -plane an angle $\arctan k = \arctan(0.845154) = 40.20^\circ$. Rotating the ellipsoid about the y -axis by this angle makes one of the two circles horizontal: that is the practical way to produce a circular section on a lathe.

7.13.4 Solutions to the symbolic exercises

Solution. S1. *Step 1 — a single sphere.* If the plane π is parallel to a generator g , it does not meet g ; the family of inscribed spheres, which in Theorem 7.2 provided two tangent ones, now provides only one, because on the side of g the spheres “slide away” to infinity without ever touching π . Let $\mathcal{S}$ be the unique sphere tangent to π , F the point of tangency and c the circle along which $\mathcal{S}$ touches the cone, contained in the plane τ orthogonal to the axis.

Step 2 — the directrix. Let $d = \pi \cap \tau$: it is a line, since the two planes are not parallel (τ is orthogonal to the axis, π is not, being parallel to a generator).

Step 3 — the fundamental relation. Let P be a point of the section and A the intersection of the generator through P with c . By the tangent-segment lemma (Step 3 of the proof of Theorem 7.2),

$$|PF| = |PA|.$$

Step 4 — the two distances from τ . Let α be the half-angle. The generators make with the plane τ the angle $90^\circ - \alpha$, hence the distance from P to τ is

$$\text{dist}(P, \tau) = |PA| \sin(90^\circ - \alpha) = |PA| \cos \alpha.$$

On the other hand, with D the foot of the perpendicular from P to the line d in the plane π , and θ the dihedral angle between π and τ ,

$$\text{dist}(P, \tau) = |PD| \sin \theta.$$

Step 5 — the parallelism hypothesis. The plane π is parallel to a generator, i.e. makes with the axis the same angle α as the generators; since τ is orthogonal to the axis, the dihedral angle between π and τ is $\theta = 90^\circ - \alpha$, hence $\sin \theta = \cos \alpha$.

Step 6 — conclusion. Equating the two expressions for the distance from τ :

$$|PA| \cos \alpha = |PD| \cos \alpha \implies |PA| = |PD| \implies |PF| = |PD|.$$

Every point of the section is therefore equidistant from the point F and the line d : by the focal definition, the section is a *parabola* with focus F and directrix d , of eccentricity 1.

Step 7 — consistency with Exercise N1. The formula $e = \cos \beta / \cos \alpha$ gives $e = 1$ exactly for $\beta = \alpha$, i.e. for the plane parallel to a generator. ✓

Solution. S2. *Step 1 — one section and its axis.* Let Σ be the surface and C_1 one of its circular sections, contained in the plane π_1 , with centre O_1 . Let a be the line through O_1 orthogonal to π_1 (the *axis* of C_1).

Step 2 — the line a meets Σ . Σ is closed and bounded and encloses C_1 ; the “signed distance along a ” function takes both positive and negative values on Σ . Let N and S be two points of $\Sigma \cap a$, and M the midpoint of NS .

Step 3 — every plane through a gives a circle through N and S . Let π be a plane containing a . It cuts Σ in a circle Γ , by hypothesis. Since $N, S \in a \subset \pi$ and $N, S \in \Sigma$, the circle Γ passes through N and S . Moreover π cuts C_1 in two diametrically opposite points A, A' (since π contains the centre O_1), which belong to Γ .

Step 4 — the centre of Γ . Γ contains the chord NS and the chord AA' ; its centre lies on the perpendicular bisector of each. The bisector of NS (in the plane π) is the line through M orthogonal to a ; the bisector of AA' is the line through O_1 orthogonal to AA' , i.e. a itself. Hence the centre of Γ is the intersection of these two lines, that is the point M : the same for every choice of π .

Step 5 — the radius. Since Γ has centre M and passes through N , its radius is $|MN|$, also independent of π .

Step 6 — conclusion. Every point of Σ lies in some plane containing a (the planes through a cover all of space), hence on some Γ , hence at distance $|MN|$ from M . So Σ is contained in the sphere with centre M and radius $|MN|$; and since every plane through a meets Σ in a *whole* circle of that sphere, the inclusion is an equality. Σ is a sphere. ✓

Step 7 — closedness is necessary. A spherical cap has all sections circular (or arcs) but is not a sphere: it is the hypothesis “closed surface” that excludes it.

Solution. S3. *Step 1 — choice of coordinates.* Up to a rotation (an isometry of the sphere, Theorem 11.7 and what follows), we may assume A is the north pole $(0, 0, 1)$. Parametrise the sphere by

$$\gamma(t) = (\sin \phi(t) \cos \theta(t), \sin \phi(t) \sin \theta(t), \cos \phi(t)),$$

with $\phi \in [0, \pi]$ the colatitude and θ the longitude, both C^1 functions of $t \in [0, 1]$, $\gamma(0) = A$, $\gamma(1) = B$.

Step 2 — the speed. Differentiating and using that $\partial_\phi \gamma$ and $\partial_\theta \gamma$ are orthogonal, of norms 1 and $\sin \phi$,

$$\|\gamma'(t)\|^2 = \phi'(t)^2 + \sin^2 \phi(t) \theta'(t)^2.$$

Step 3 — the inequality. Since $\sin^2 \phi \theta'^2 \geq 0$,

$$\|\gamma'(t)\| \geq |\phi'(t)|,$$

hence the length satisfies

$$L(\gamma) = \int_0^1 \|\gamma'\| dt \geq \int_0^1 |\phi'| dt \geq \left| \int_0^1 \phi' dt \right| = |\phi(1) - \phi(0)| = \phi_B,$$

since $\phi(0) = 0$ (the pole) and $\phi(1) = \phi_B$ is the colatitude of B .

Step 4 — the value of ϕ_B . By definition of colatitude, $\cos \phi_B = \langle A, B \rangle$, hence $\phi_B = \arccos \langle A, B \rangle$.

Step 5 — the equality case. Equality in the first inequality requires $\sin \phi \theta' \equiv 0$, i.e. θ constant where $\phi \in (0, \pi)$: the curve is a meridian, an arc of great circle. Equality in the second requires ϕ' not to change sign, i.e. the meridian to be traversed monotonically. So the minimum is attained exactly by the minor arc of great circle.

Step 6 — the arc realises the length. The meridian from A to B has θ constant and ϕ increasing from 0 to ϕ_B : its length is exactly ϕ_B . Hence the minimum is $\arccos \langle A, B \rangle$ and it is attained.

Step 7 — numerical check. For A the North Pole and B on the equator, $\langle A, B \rangle = 0$ and the distance is $\arccos 0 = \pi/2$, a quarter of a circumference. On the Earth: $\frac{\pi}{2} \times 6371 = 10\,007.5$ km, which is indeed the pole-to-equator distance (and the original definition of the metre, one ten-millionth of that quarter meridian). ✓

Step 8 — the antipodal case. If $B = -A$ then $\phi_B = \pi$ and every meridian attains the minimum: the shortest path is not unique. It is the only case of non-uniqueness, and corresponds to the two points lying on infinitely many great circles.

Solution. S4. *Step 1 — the geodesic circle.* The points at geodesic distance ρ from the north pole are, by Exercise S3, those of colatitude $\phi = \rho/R$ (on a sphere of radius R the arc length is $R\phi$). They form the parallel of Euclidean radius

$$r = R \sin \frac{\rho}{R}$$

(Example 7.1).

Step 2 — the circumference. Being a Euclidean circle of radius r ,

$$C(\rho) = 2\pi r = 2\pi R \sin \frac{\rho}{R}.$$

Step 3 — expansion. With $\sin x = x - \frac{x^3}{6} + O(x^5)$ and $x = \rho/R$:

$$C(\rho) = 2\pi R \left(\frac{\rho}{R} - \frac{\rho^3}{6R^3} + O\left(\frac{\rho^5}{R^5}\right) \right) = 2\pi\rho - \frac{\pi\rho^3}{3R^2} + O(\rho^5).$$

Step 4 — the defect. Hence

$$2\pi\rho - C(\rho) = \frac{\pi\rho^3}{3R^2} + O(\rho^5),$$

as claimed.

Step 5 — interpreting the coefficient. Setting $K = 1/R^2$ (the *Gaussian curvature* of the sphere), the formula becomes

$$C(\rho) = 2\pi\rho \left(1 - \frac{K\rho^2}{6} + O(\rho^4) \right),$$

the Bertrand–Puisseux formula, valid on any surface: the defect of the circumference relative to $2\pi\rho$ measures the curvature. On a surface of negative curvature (a saddle) the sign reverses and the circumference *exceeds* $2\pi\rho$.

Step 6 — numerical check. On the Earth, for $\rho = 1000$ km:

$$C = 2\pi \times 6371 \times \sin \frac{1000}{6371} = 40\,030 \times \sin(0.156961) = 40\,030 \times 0.156317 = 6257.4 \text{ km},$$

against $2\pi\rho = 6283.2$ km: a defect of 25.8 km. The approximate formula gives $\pi\rho^3/(3R^2) = \pi \times 10^9/(3 \times 4.059 \times 10^7) = 25.80$ km. ✓

Step 7 — connection. The ratio C/ρ equals 2π only in the plane: it is the theorem of Volume I on the constancy of π , which fails on the sphere. The defect is the circular twin of the angular excess of Theorem 7.4, and the two are linked by the same K .

Solution. S5. *Step 1 — the two equations.* Let the spheres be

$$\|x - c_1\|^2 = R_1^2, \quad \|x - c_2\|^2 = R_2^2.$$

Expanding: $\|x\|^2 - 2\langle x, c_i \rangle + \|c_i\|^2 = R_i^2$.

Step 2 — the radical plane. Subtracting one from the other, the term $\|x\|^2$ cancels:

$$2\langle x, c_2 - c_1 \rangle = \|c_2\|^2 - \|c_1\|^2 - R_2^2 + R_1^2,$$

the equation of a hyperplane Π orthogonal to $c_2 - c_1$, the **radical plane** (Volume I for the case $n = 2$). Every point of the intersection lies on it.

Step 3 — the intersection. Hence $S_1 \cap S_2 = S_1 \cap \Pi$, which by Theorem 7.3 is a sphere S^{n-2} (or a point, or empty).

Step 4 — position and radius. Let $d = \|c_2 - c_1\| > 0$ and let t be the distance from c_1 to the plane Π , measured along $c_2 - c_1$. Projecting the equation of Step 2 onto that direction gives

$$t = \frac{d^2 + R_1^2 - R_2^2}{2d},$$

and the radius of the intersection is, by Theorem 7.3,

$$\rho = \sqrt{R_1^2 - t^2},$$

real if and only if $|R_1 - R_2| < d < R_1 + R_2$ (the two spheres meet properly).

Step 5 — numerical check. Two unit spheres with centres at distance $d = 1$:

$$t = \frac{1 + 1 - 1}{2} = 0.5, \quad \rho = \sqrt{1 - 0.25} = 0.866025 = \frac{\sqrt{3}}{2}.$$

This is the classical result: two unit spheres whose centres are one radius apart meet in a circle of radius $\sqrt{3}/2$, lying in the median plane. ✓

Step 6 — limiting cases. For $d = R_1 + R_2$ (external tangency) one gets $t = R_1$ and $\rho = 0$: a point. For $d > R_1 + R_2$ the radicand is negative: no intersection, although the radical plane still exists — it is the locus of points of equal power with respect to the two spheres, a notion from Volume I that survives intact in every dimension.

Step 7 — reading. The result explains why trilateration (GPS) works: the intersection of two spheres is a circle, that of three is (generically) two points, that of four a single point. Each step lowers the dimension by one, as Theorem 7.3 predicts.

7.14. Self-assessment quiz

- A *great circle* of $S^2(O, r)$ is, by definition,
 - a circle of radius r drawn on the sphere
 - the intersection of the sphere with a plane through O
 - the boundary of a cap
 - a parallel
- In Theorem 7.6(ii) the hypothesis that allows one to conclude that Γ is a genuine circle (and not a point or the empty set) is
 - $k = 0$
 - $|k| < r$ with $a^2 + b^2 + c^2 = 1$
 - $c \neq 0$
 - Γ passes through the pole
- The circle $\Gamma = S^2 \cap \{z = k\}$ of the unit sphere projects, from the north pole, to a circle of centre the origin and radius
 - $\sqrt{1 - k^2}$
 - $\sqrt{\frac{1+k}{1-k}}$
 - $\frac{1}{1-k}$
 - $1 - k$
- On the unit sphere the geodesic distance between $A = (1, 0, 0)$ and $B = (\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$ equals, in radians,
 - 0.5
 - 1.047198
 - 1.570796
 - 1.000000
- In the proof of Theorem 7.7, the step that rules out the minimizing curve leaving the great circle is
 - the Cauchy–Schwarz inequality and its equality case
 - the Pythagorean theorem
 - the computation of the area element
 - Girard’s theorem
- A student computes the area of the geodesic triangle with angles $\alpha = \beta = \gamma = 90^\circ$ on a sphere of radius 2 and writes $\text{Area} = \frac{3\pi}{2} - \pi = \frac{\pi}{2}$. The mistake is
 - having added the angles wrongly
 - having forgotten the factor r^2 : the area is $\frac{\pi}{2} \cdot 4 = 2\pi$
 - having used degrees
 - none, the answer is correct
- To point an antenna at a satellite one uses the spherical triangle observer–pole–sub-satellite point. Which result of this chapter directly provides the angular distance between observer and sub-satellite point?
 - Girard’s theorem
 - the spherical law of cosines
 - Dandelin’s theorem
 - the lune lemma
- The property “stereographic projection maps circles to circles” is proved, with the algebraic computation reproduced at Step 6 of the proof of Theorem 7.6, in
 - Euclid, *Elements*, book XII
 - the HKUST MATH 5030 lecture notes, Chapter 2, Theorem 2.1.1
 - Newton, *Principia*, book I
 - Archimedes, *On the Sphere and*

7.14.1 Supplementary questionnaire

The items that follow come from Volume II and cover cones, spheres and quadrics; those on hyperspheres and rotations are in Chapter 11.

1. The section of a cone by a plane orthogonal to the axis is
A) an ellipse B) a circle C) a parabola D) a hyperbola
2. In Dandelin's theorem the foci of the ellipse are
A) the centres of the spheres B) the tangency points of the spheres with the plane
C) the vertices of the cone D) the centres of the tangency circles
3. The eccentricity of the circle is
A) 1 B) 0 C) 1/2 D) infinite
4. Every plane section of a sphere is
A) an ellipse B) a circle C) it depends on the plane D) a parabola
5. The radius of the section of a sphere of radius R by a plane at distance d is
A) $R - d$ B) $\sqrt{R^2 - d^2}$ C) $\sqrt{R^2 + d^2}$ D) Rd
6. On a sphere the angle sum of a triangle
A) is always π B) exceeds π by an amount proportional to the area C) is less than π D) depends only on the radius
7. An ellipsoid with three distinct semi-axes has circular sections
A) in no plane B) in two families of parallel planes C) in every plane D) only in the coordinate planes
8. The radius of the maximal circular sections of an ellipsoid with $a > b > c$ is
A) a B) b C) c D) $\sqrt{ac}$
9. The intersection of a sphere with a plane through the center is
A) an ellipse B) a great circle C) a point D) a parabola
10. A small circle has radius
A) r B) $\sqrt{r^2 - d^2}$ C) $r + d$ D) d
11. Stereographic projection maps a circle through the pole N to
A) a circle B) a line C) a point D) an ellipse
12. The Riemann sphere is
A) $\mathbb{C}$ B) $\mathbb{C} \cup \{\infty\}$ C) $\mathbb{R}^3$ D) $SO(3)$
13. The geodesics of the sphere are
A) the parallels B) the great circles C) the meridians only D) the loxodromes
14. The geodesic distance between two points with central angle θ is
A) r/θ B) $r\theta$ C) $r \sin \theta$ D) $2\pi r$

15. The area of a lune of angle α is
 A) αr^2 B) $2\alpha r^2$ C) $4\pi r^2$ D) $\pi\alpha$
16. For a geodesic triangle, $\alpha + \beta + \gamma - \pi$ equals
 A) 0 B) $\text{Area}(T)/r^2$ C) π D) 2π
17. In the limit $r \rightarrow \infty$ the spherical excess tends to
 A) ∞ B) 0 C) π D) 1

BEST PRACTICE

Key to the supplementary questionnaire. 1-B (Theorem 7.1); 2-B (Theorem 7.2); 3-B (Remark 7.2); 4-B and 5-B (Theorem 7.3); 6-B (Theorem 7.4); 7-B and 8-B (Theorem 7.5, Step 3); 9-B and 10-B (Definition 7.4); 11-B (Theorem 7.6(ii), Step 7); 12-B (Remark 7.4); 13-B (Theorem 7.7); 14-B (Theorem 7.8); 15-B (Lemma 7.2); 16-B (Theorem 7.11); 17-B (Remark 7.7).

7.15. Teacher's key

BEST PRACTICE

Q1 — answer B. *Rationale:* Definition 7.4: the great circle is defined by the plane through the centre; that the radius equals r is a *consequence* (Theorem 7.3), not the definition. *Distractors:* A confuses the property with the definition and would be circular; C describes any circle; D is a small circle (except the equator). *Diagnosed error:* exchanging definition and characterization. *Source:* do Carmo, §4-6. *Prerequisite:* planar section of a sphere. *Remediation:* redo Step 5 of the proof of Theorem 7.3.

Q2 — answer B. *Rationale:* Step 5 of the proof requires $a^2 + b^2 + c^2 = 1$ (to identify $|k|$ with the distance from the centre) and $|k| < r$ (so that the section is a circle and not a point or the empty set). *Distractors:* A is the case of the great circle only; C is not needed (with $c = 0$ the plane contains the z axis and the formula still works); D is the condition that gives a *line*, not a circle. *Diagnosed error:* failing to check the non-degeneracy hypotheses. *Source:* HKUST MATH 5030, Ch. 2, Theorem 2.1.1, point 1. *Prerequisite:* point-plane distance. *Remediation:* redo Step 5 listing what happens for $|k| = r$ and $|k| > r$.

Q3 — answer B. *Rationale:* with $a = b = 0$, $c = 1$ the radius formula gives $\sqrt{1 - k^2}/|1 - k| = \sqrt{(1 - k)(1 + k)/(1 - k)} = \sqrt{(1 + k)/(1 - k)}$. *Distractors:* A is the radius of the circle *on the sphere*, not of its image; C is the factor $1/(1 - z)$ alone; D has the wrong dimensions. *Diagnosed error:* confusing the radius of the object with that of its image. *Source:* Theorem 7.6(ii) and the numerical check at Step 9(b). *Prerequisite:* algebra of radicals. *Remediation:* compute the radius for $k = 0$ (it must give 1) and for $k \rightarrow 1^-$ (it must diverge).

Q4 — answer B. *Rationale:* $\langle A, B \rangle = \frac{1}{2}$, hence $\theta = \arccos(0.5) = \pi/3 = 1.047198$ (Theorem 7.8). *Distractors:* A is the scalar product, not the angle; C would be $\arccos 0$; D is the chord $\|A - B\| = 1$, which is not a path on the sphere. *Diagnosed error:* confusing chord, cosine and geodesic distance. *Source:* Theorem 7.7, Step 12. *Prerequisite:* scalar product and arccos. *Remediation:* compute chord and arc for $\theta = 10^\circ, 60^\circ, 180^\circ$ and compare.

Q5 — answer A. *Rationale:* Step 8(b) and Step 9: equality in Cauchy–Schwarz forces $\gamma' \in \text{span}\{A, \gamma\}$, and from there the equation $\psi' = \beta\psi$ gives $\psi \equiv 0$, i.e. the curve lies

in a plane through the origin. *Distractors*: B is used in Theorem 7.3, not here; C is used for the area; D is the theorem about the area of the triangle. *Diagnosed error*: confusing the inequality (which gives the minimum) with the equality case (which gives uniqueness). *Source*: Math 138A lecture notes, University of California Riverside, “Length minimizing property of great circles”. *Prerequisite*: equality case in Cauchy–Schwarz. *Remediation*: rewrite Step 9 with A and B antipodal and see where it breaks.

Q6 — answer B. *Rationale*: Theorem 7.11: the excess is Area/r^2 , so $\text{Area} = Er^2 = \frac{\pi}{2} \cdot 4 = 2\pi$. *Distractors*: A and C are mistakes different from the one made; D accepts the error. *Diagnosed error*: confusing excess (dimensionless) and area (with the dimensions of a length squared). *Source*: Todhunter, Art. 99–100. *Prerequisite*: dimensional homogeneity. *Remediation*: repeat Step 4 of the proof with $r = 1, 2, 3$ and observe that the excess does not change.

Q7 — answer B. *Rationale*: given latitude and longitude of the observer and of the sub-satellite point, Proposition 7.2 applied to the triangle pole–observer–sub-satellite gives the angular distance directly, from which elevation and azimuth follow. *Distractors*: A and D concern areas; C concerns conics. *Diagnosed error*: failing to recognize which datum is geometrically available. *Source*: MIT OpenCourseWare 12.215 *Modern Navigation*, Lecture 4. *Prerequisite*: geographic coordinates. *Remediation*: draw the triangle and label its sides and angles before writing any formula.

Q8 — answer B. *Rationale*: the computation of Step 6 — substituting the inverse parametrization into the equation of the plane and recognizing the equation of a circle — is exactly that of Theorem 2.1.1 of the HKUST MATH 5030 lecture notes, §2.1. *Distractors*: A, C and D are real works but on different subjects; they check that the student does not “recognize” a name at random. *Diagnosed error*: quoting from memory without checking. *Source*: the same lecture notes, pp. 48–50 (verified). *Prerequisite*: none. *Remediation*: open the source and compare the two computations line by line.

Answers to the assigned exercises.

Exercise 7.7. (i) By Corollary 7.1, $d = \sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm: the spheres are *two*, with centres on the axis of the circle at 5 cm from π , one on each side. (ii) C is a great circle if and only if $d = 0$, i.e. $r = \rho = 12$ cm, and then the sphere is *unique* (the two centres coincide with Q). (iii) No: for $r = 10 < 12 = \rho$ every planar section has radius $\leq 10 < 12$, so no sphere of radius 10 can contain a circle of radius 12.

Exercise 7.8. (i) The parallel is $S^2 \cap \{z = z_0\}$, i.e. $a = b = 0$, $c = 1$, $k = z_0$: the centre of the image is $-\frac{1}{1-z_0}(0, 0) = (0, 0)$ and the radius $\sqrt{1 - z_0^2}/|1 - z_0| = \sqrt{(1 + z_0)/(1 - z_0)}$. By direct computation: the point $(\sqrt{1 - z_0^2}, 0, z_0)$ goes to $(\sqrt{1 - z_0^2}/(1 - z_0), 0)$, of the same modulus. (ii) $z_0 = -\frac{1}{2}$: $\sqrt{0.5/1.5} = 0.577350$; $z_0 = 0$: 1; $z_0 = \frac{1}{2}$: $\sqrt{1.5/0.5} = 1.732051$; $z_0 = 0.9$: $\sqrt{1.9/0.1} = 4.358899$. (iii) The meridian $\varphi = \varphi_0$ is half of a great circle through N ; by Theorem 7.6(ii) its image is (half of) a line, and since σ fixes the equator that line passes through the origin with direction $(\cos \varphi_0, \sin \varphi_0)$. Lines through the origin and circles centred at the origin meet at right angles, as they must by conformality (part (i) of the theorem).

Exercise 7.9. (i) The two sides starting at the Pole are meridians and cut the equator at right angles: $\beta = \gamma = 90^\circ$; the angle at the Pole is the longitude difference, $\alpha = 40^\circ = 0.698132$ rad. Hence $E = 0.698132 + 1.570796 + 1.570796 - 3.141593 = 0.698132$ rad, and by Theorem 7.11 $\text{Area} = 0.698132 \times 6371^2 = 2.83369 \cdot 10^7$ km². The fraction of the Earth’s surface is $E/(4\pi) = 0.698132/12.566371 = 0.055556 = 1/18$, as it must be (40° out of 720°). (ii) The two polar sides measure $\frac{\pi}{2} \times 6371 = 10\,007.5$ km each, the equatorial one $0.698132 \times 6371 = 4447.8$ km. (iii) The isosceles plane triangle with those sides has area

$\frac{c}{4}\sqrt{4a^2 - c^2} = 2.16993 \cdot 10^7 \text{ km}^2$: the spherical area exceeds it by 30.6%. The deviation is positive (the curved surface is larger) and it is enormous because the triangle is as big as an eighth of the planet: here the curvature term is not a correction, it is the phenomenon.

7.16. Chapter formulary

Cone and conics

$$x^2 + y^2 = z^2 \tan^2 \alpha; \quad r(h) = h \tan \alpha; \quad e = \frac{\cos \beta}{\cos \alpha}$$

$e = 0$ circle; $0 < e < 1$ ellipse; $e = 1$ parabola; $e > 1$ hyperbola

$$\text{Dandelin : } |PF_1| + |PF_2| = |AB| = 2a$$

Sphere

$$S^{n-1}(c, R) = \{\|x - c\| = R\}; \quad \text{section at distance } d : \rho = \sqrt{R^2 - d^2}$$

$$\text{cap : } A = 2\pi Rh, \quad V = \frac{\pi h^2}{3}(3R - h)$$

$$\text{Girard : } \text{Area}(T) = R^2(A + B + C - \pi)$$

$$\text{geodesic : } \text{dist}(A, B) = R \arccos \frac{\langle A, B \rangle}{R^2}; \quad C(\rho) = 2\pi R \sin \frac{\rho}{R}$$

Stereographic projection (pole $N = (0, 0, r)$, plane $z = 0$)

$$\sigma(x, y, z) = \frac{r}{r - z}(x, y), \quad X(u, v) = \frac{r}{u^2 + v^2 + r^2}(2ru, 2rv, u^2 + v^2 - r^2)$$

$$E = G = \frac{4r^4}{(u^2 + v^2 + r^2)^2}, \quad F = 0, \quad \lambda = \frac{2r^2}{u^2 + v^2 + r^2} \quad (\text{conformal})$$

$$\Gamma = S^2 \cap \{ax + by + cz = k\}, \quad a^2 + b^2 + c^2 = 1, \quad |k| < r : \quad cr = k \Rightarrow \text{line } au + bv = cr,$$

$$cr \neq k \Rightarrow \text{circle of centre } -\frac{r^2}{cr - k}(a, b) \text{ and radius } \frac{r\sqrt{r^2 - k^2}}{|cr - k|}$$

Geodesics, area, Gauss–Bonnet

$$d(A, B) = r \arccos \frac{\langle A, B \rangle}{r^2} = r\theta, \quad \cos c = \cos a \cos b + \sin a \sin b \cos C$$

$$dA = r^2 \sin \vartheta \, d\varphi \, d\vartheta, \quad \text{cap : } 2\pi rh, \quad \text{lune : } 2\alpha r^2$$

$$E = \alpha + \beta + \gamma - \pi = \frac{\text{Area}(T)}{r^2} = K \text{Area}(T), \quad K = \frac{1}{r^2}$$

Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad a > b > c; \quad z = \pm kx, \quad k = \frac{c}{a} \sqrt{\frac{a^2 - b^2}{b^2 - c^2}}, \quad \rho = b$$

Intersection of two spheres

$$t = \frac{d^2 + R_1^2 - R_2^2}{2d}, \quad \rho = \sqrt{R_1^2 - t^2}$$

Sources and bibliography

- [1] Apollonius of Perga, *Conics*, books I–IV, 3rd century BC. First systematic treatment of the conic sections and origin of the names “ellipse”, “parabola”, “hyperbola”.
- [2] G. P. Dandelin, *Mémoire sur quelques propriétés remarquables de la focale parabolique*, Nouveaux Mémoires de l’Académie Royale des Sciences et Belles-Lettres de Bruxelles, vol. 2, 1822, pp. 171–202. Original proof of Theorem 7.2 by inscribed spheres.
- [3] A. Girard, *Invention nouvelle en l’algèbre*, Amsterdam, 1629 (with the appendix *De la mesure de la superficie des triangles et polygones sphériques*). First publication of Theorem 7.4 on the spherical excess; the result had been obtained earlier, and left unpublished, by Thomas Harriot (1603).
- [4] Archimedes, *On the Sphere and Cylinder*, 3rd century BC. Area of a spherical zone and the $2/3$ ratio between sphere and circumscribed cylinder, used in Exercise N2.
- [5] Liu Hui and Zu Gengzhi, commentary on the *Nine Chapters on the Mathematical Art* and *Zhui Shu*, 3rd–6th century AD. Determination of the volume of the sphere by what later became Cavalieri’s principle. Chinese source.
- [6] Bhaskara II, *Siddhanta Shiromani*, 1150. Formulas for the surface and volume of the sphere, by band-section arguments. Indian source.
- [7] D. Hilbert, S. Cohn-Vossen, *Anschauliche Geometrie*, Springer, Berlin, 1932 (English translation *Geometry and the Imagination*, Chelsea, 1952). Dandelin spheres, circular sections of the ellipsoid, the Hopf fibration, treated intuitively and rigorously.
- [8] M. Berger, *Géométrie*, vols. 1–5, Cedic/Fernand Nathan, Paris, 1977 (English translation *Geometry I–II*, Springer, 1987). Quadrics, orthogonal groups, spherical geometry.
- [9] H. S. M. Coxeter, *Introduction to Geometry*, 2nd ed., John Wiley & Sons, 1969. Spherical geometry, polyhedra, quaternions.
- [10] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976. Gaussian curvature, the Bertrand–Puisseux formula of Exercise S4, geodesics on the sphere. South American source.
- [11] Hong Kong University of Science and Technology, Department of Mathematics, MATH 5030 lecture notes by E. Y.-M. Chiang, Chapter 2 *Conformal mappings*, §2.1 *Stereographic Projection*, pp. 48–51: formulas for the projection and its inverse (2.1), Theorem 2.1.1 (a circle of the Riemann sphere goes to a circle or a line of the plane, with the algebraic computation reproduced at Step 6 of the proof of Theorem 7.6), Theorem 2.1.2 (the projection is isogonal). File https://www.math.hkust.edu.hk/~machiang/5030/notes/Chap2_a. verified on 5 September 2026.
- [12] J. Shurman, *Stereographic projection is conformal*, handout of the course Math 311 (Complex Analysis), Reed College, Portland, 3 pp.: formula $\pi(x, y, z) = (x + iy)/(1 - z)$ and inverse $\pi^{-1}(x, y) = (2x, 2y, x^2 + y^2 - 1)/(x^2 + y^2 + 1)$ (the same as in Proposition 7.1 for $r = 1$), geometric proof of conformality and reference to Hilbert–Cohn-Vossen for the preservation of circles. File <https://people.reed.edu/~jerry/311/stereo.pdf>; verified on 5 September 2026.
- [13] University of California, Riverside, Department of Mathematics, handout *Length minimizing property of great circles* of the course Math 138A (winter 2006), 6 pp.: reduction to curves issuing from the north pole, the difficulty of the polar chart at $r = 0$ and the strong

minimizing property for short curves. File <https://math.ucr.edu/~res/math138A/w06/greatcircles> verified on 5 September 2026. The proof given in Theorem 7.7 avoids charts and has no exceptions.

- [14] I. Todhunter, *Spherical Trigonometry, For the Use of Colleges and Schools*, fifth edition, Macmillan and Co., London 1886, Articles 96–98 (area of the lune and Girard’s theorem with the three lunes, pp. 71–73) and 99–100 (spherical excess and area, pp. 74–75). Full text freely available on Project Gutenberg, eBook no. 19770; verified on 5 September 2026.
- [15] Massachusetts Institute of Technology, OpenCourseWare, course 12.215 *Modern Navigation* (fall 2006), Lecture 4 *Spherical Trigonometry*, slides 6, 9 and 13–14 (law of cosines, colatitudes and azimuths, distance and quadrant ambiguity). Verified on 5 September 2026.
- [16] MIT OpenCourseWare, *18.02 Multivariable Calculus* and *18.700 Linear Algebra* — <https://ocw.mit.edu> Integration in spherical coordinates and orthogonal matrices.
- [17] NPTEL, Indian Institute of Technology, *Multivariable Calculus* and *Classical Mechanics* — <https://nptel.ac.in/>. Spherical coordinates in $\mathbb{R}^n$ and rigid-body rotations. Asian source.
- [18] UNAM, Facultad de Ciencias, *Geometría Analítica* and *Álgebra Lineal*, university syllabi. Latin American reference on quadrics and orthogonal groups.
- [19] C. A. Barbieri, D. Miller-Cotto, S. N. Clerjuste, K. Chawla, *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*, Educational Psychology Review 35, art. 11, 2023 — <https://doi.org/10.1007/s10648-023-09745-1>.
- [20] H. L. Roediger III, J. D. Karpicke, *Test-Enhanced Learning*, Psychological Science 17(3), 2006 — <https://doi.org/10.1111/j.1467-9280.2006.01693.x>.

CHAPTER 8

Spherical trigonometry

On the circle, plane trigonometry relates the sides and the angles of triangles. On the sphere the “triangles” have arcs of great circle as sides and obey different relations — spherical trigonometry — which is the quantitative tool of navigation, of geodesy and of astronomy. It is the natural completion of the passage from the circle to the sphere.

The chapter is self-contained: every formula is proved starting from the scalar product and the vector product in $\mathbb{R}^3$, without assuming anything from the differential geometry of surfaces. The cosine theorem for sides and the Gauss–Bonnet theorem on the triangle also appear in Chapter 7 (circle–sphere dictionary) and in Chapter 9 (geodesics): there they serve as an application, here they are *proved*, and the two expositions can be read in any order.

INFO

Level and prerequisites. Level **A–C**. Sections 8.1, 8.2 and 8.5 are of level A–B and require only plane trigonometry, the scalar product and the vector product in $\mathbb{R}^3$. Sections 8.4, 8.6 and 8.7 are of level B–C: they additionally require the 3×3 determinant, the idea of the area of a surface as a double integral and some ease with the sum-to-product identities. Section 8.8 is applicative and of level A–B. Prerequisites within the treatise: circle and arcs (Volume I, Chapters 2 and 8), trigonometric functions and addition formulas (Volume II, Chapter 1), sphere, great circles and spherical area element (Chapter 7 of this volume).

Warning. Conventions of the chapter.

- Unless stated otherwise we work on the **unit sphere** $S^2 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$ centred at O ; for a sphere of radius r all lengths are multiplied by r and all areas by r^2 , whereas *angles and sides do not change*.
- The vertices of a triangle are denoted by **A, B, C** when they are regarded as unit vectors of $\mathbb{R}^3$, and by A, B, C when its angles are meant. The sides a, b, c are the sides opposite to A, B, C respectively.
- Sides and angles are measured in radians and satisfy $0 < a, b, c < \pi$ and $0 < A, B, C < \pi$; when degrees appear, the conversion is $1^\circ = \pi/180$.
- $\mathbf{u} \cdot \mathbf{v}$ is the scalar product, $\mathbf{u} \times \mathbf{v}$ the vector product and $\det[\mathbf{u}, \mathbf{v}, \mathbf{w}] = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ the mixed product.
- Decimal separators are points: 0.5.

8.1. Spherical triangles: definitions and fundamental inequalities

The delicate point, and the reason why spherical trigonometry is not a cosmetic variant of the plane one, is that on the sphere *the sides too are angles*: the distance between two points is measured by the central angle they subtend. It follows that in a spherical triangle there are six homogeneous quantities — three sides and three angles — and that the relations among them are relations among six numbers of the same kind.

Definition 8.1: Spherical triangle. Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in S^2$ be three points that are **linearly independent** as vectors of $\mathbb{R}^3$. The **spherical triangle** ABC is the region of S^2 bounded by the three minor arcs of great circle joining the vertices pairwise. Its **sides** are the central angles

$$a = \angle(\mathbf{B}, \mathbf{C}), \quad b = \angle(\mathbf{C}, \mathbf{A}), \quad c = \angle(\mathbf{A}, \mathbf{B}),$$

that is, the numbers in $(0, \pi)$ defined by

$$\cos a = \mathbf{B} \cdot \mathbf{C}, \quad \cos b = \mathbf{C} \cdot \mathbf{A}, \quad \cos c = \mathbf{A} \cdot \mathbf{B}.$$

On a sphere of radius r the length of the side a equals ra .

The hypothesis of linear independence excludes the degenerate cases: two coincident or antipodal vertices (then a , b or c would equal 0 or π) and three vertices on one and the same great circle. Under that hypothesis each side really belongs to $(0, \pi)$ and the minor arc between two vertices is unique.

Definition 8.2: Angle of a spherical triangle. Let $\mathbf{P} \in S^2$ and let $\mathbf{Q} \in S^2$ with $\angle(\mathbf{P}, \mathbf{Q}) \in (0, \pi)$. The **unit tangent vector at $\mathbf{P}$ towards $\mathbf{Q}$** is

$$\mathbf{t}_{\mathbf{P} \rightarrow \mathbf{Q}} = \frac{\mathbf{Q} - (\mathbf{P} \cdot \mathbf{Q})\mathbf{P}}{\|\mathbf{Q} - (\mathbf{P} \cdot \mathbf{Q})\mathbf{P}\|}.$$

The **angle** A of the triangle at the vertex $\mathbf{A}$ is the angle between the two unit vectors tangent to its two edges:

$$\cos A = \mathbf{t}_{\mathbf{A} \rightarrow \mathbf{B}} \cdot \mathbf{t}_{\mathbf{A} \rightarrow \mathbf{C}}, \quad A \in (0, \pi),$$

and analogously for B and C .

This definition coincides with the classical one — the dihedral angle between the planes of the two sides — and it is the one that lends itself to computation. The following lemma verifies this and, at the same time, provides the only computation needed for the whole chapter.

Lemma 8.1: The unit tangent vector and its norm. Let $\mathbf{P}, \mathbf{Q} \in S^2$ with $\gamma = \angle(\mathbf{P}, \mathbf{Q}) \in (0, \pi)$ and let $\mathbf{Q}_\perp = \mathbf{Q} - (\mathbf{P} \cdot \mathbf{Q})\mathbf{P}$. Then:

- (i) $\|\mathbf{Q}_\perp\| = \sin \gamma > 0$;
- (ii) the curve $\sigma(t) = \cos t \mathbf{P} + \sin t \mathbf{t}_{\mathbf{P} \rightarrow \mathbf{Q}}$, $t \in [0, \gamma]$, is the minor arc of great circle from $\mathbf{P}$ to $\mathbf{Q}$, is parametrized by arc length and $\sigma'(0) = \mathbf{t}_{\mathbf{P} \rightarrow \mathbf{Q}}$;
- (iii) $\mathbf{t}_{\mathbf{P} \rightarrow \mathbf{Q}}$ is orthogonal to $\mathbf{P}$ and lies in the plane $\text{span}\{\mathbf{P}, \mathbf{Q}\}$.

Proof. (i) Since $\mathbf{P} \cdot \mathbf{Q} = \cos \gamma$ and $\|\mathbf{P}\| = \|\mathbf{Q}\| = 1$,

$$\|\mathbf{Q}_\perp\|^2 = \mathbf{Q} \cdot \mathbf{Q} - 2(\mathbf{P} \cdot \mathbf{Q})^2 + (\mathbf{P} \cdot \mathbf{Q})^2 = 1 - \cos^2 \gamma = \sin^2 \gamma.$$

Since $\gamma \in (0, \pi)$ one has $\sin \gamma > 0$, and the positive root is $\sin \gamma$.

(iii) $\mathbf{Q}_\perp \cdot \mathbf{P} = \mathbf{Q} \cdot \mathbf{P} - (\mathbf{P} \cdot \mathbf{Q})\|\mathbf{P}\|^2 = 0$, and by construction $\mathbf{Q}_\perp$ is a linear combination of $\mathbf{P}$ and $\mathbf{Q}$.

(ii) Write $\mathbf{u} = \mathbf{t}_{\mathbf{P} \rightarrow \mathbf{Q}} = \mathbf{Q}_\perp / \sin \gamma$. By (iii), $\{\mathbf{P}, \mathbf{u}\}$ is an orthonormal pair, hence $\|\sigma(t)\|^2 = \cos^2 t + \sin^2 t = 1$: the curve lies on S^2 and, lying in the plane $\text{span}\{\mathbf{P}, \mathbf{u}\} = \text{span}\{\mathbf{P}, \mathbf{Q}\}$ which passes through the origin, it lies on the great circle through $\mathbf{P}$ and $\mathbf{Q}$. Moreover $\sigma(0) = \mathbf{P}$ and

$$\sigma(\gamma) = \cos \gamma \mathbf{P} + \sin \gamma \cdot \frac{\mathbf{Q} - \cos \gamma \mathbf{P}}{\sin \gamma} = \cos \gamma \mathbf{P} + \mathbf{Q} - \cos \gamma \mathbf{P} = \mathbf{Q}.$$

Finally $\sigma'(t) = -\sin t \mathbf{P} + \cos t \mathbf{u}$ has norm 1 for every t , hence t is the arc-length parameter and the length of the arc travelled is exactly γ ; since $\gamma < \pi$, it is the minor arc. From $\sigma'(0) = \mathbf{u}$ the claim follows. $\square$

Remark 8.1: *The spherical angle is the dihedral angle.* By Lemma 8.1(iii), $\mathbf{t}_{\mathbf{A} \rightarrow \mathbf{B}}$ and $\mathbf{t}_{\mathbf{A} \rightarrow \mathbf{C}}$ are both orthogonal to $\mathbf{A}$ and lie respectively in the planes $\text{span}\{\mathbf{A}, \mathbf{B}\}$ and $\text{span}\{\mathbf{A}, \mathbf{C}\}$. The angle between two half-planes issuing from a common line is by definition measured on a section orthogonal to that line: the angle between the two unit tangent vectors is therefore exactly the dihedral angle between the planes of the two sides. The two definitions coincide.

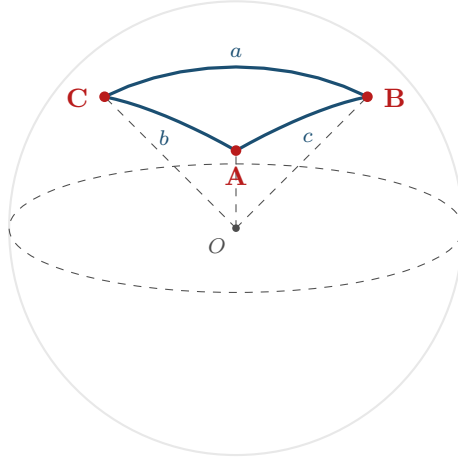


Figure 8.1: A spherical triangle on the unit sphere, in orthographic projection on the visible hemisphere. The vertices are $\mathbf{A}$ of latitude 70° and longitude 90° , $\mathbf{B}$ of latitude 35° and longitude 45° , $\mathbf{C}$ of latitude 35° and longitude 135° ; the sides turn out to be $a = 70.7925^\circ$ and $b = c = 42.5156^\circ$, the angles $A = 117.9875^\circ$ and $B = C = 39.1930^\circ$. The sum of the angles equals 196.3736° : the spherical excess is 16.3736° . The dashed lines are the radii OA , OB , OC : the sides of the triangle are, by definition, the angles between these radii.

8.2. The cosine theorem for sides

Theorem 8.1: Cosine theorem for sides. *In every spherical triangle*

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

holds and, by cyclic permutation of (a, b, c) and (A, B, C) ,

$$\cos b = \cos c \cos a + \sin c \sin a \cos B, \quad \cos c = \cos a \cos b + \sin a \sin b \cos C.$$

Equivalently

$$\cos A = \frac{\cos a - \cos b \cos c}{\sin b \sin c}.$$

Proof. Step 1 — the projections at the vertex $\mathbf{A}$. Set

$$\mathbf{B}_\perp = \mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{A} = \mathbf{B} - \cos c \mathbf{A}, \quad \mathbf{C}_\perp = \mathbf{C} - (\mathbf{A} \cdot \mathbf{C})\mathbf{A} = \mathbf{C} - \cos b \mathbf{A}.$$

By Lemma 8.1(i) one has $\|\mathbf{B}_\perp\| = \sin c > 0$ and $\|\mathbf{C}_\perp\| = \sin b > 0$; these two quantities are non-zero precisely because $b, c \in (0, \pi)$, that is, because the triangle is non-degenerate.

Step 2 — the scalar product of the projections. Expanding and using $\mathbf{A} \cdot \mathbf{A} = 1$,

$$\mathbf{B}_\perp \cdot \mathbf{C}_\perp = \mathbf{B} \cdot \mathbf{C} - \cos b (\mathbf{A} \cdot \mathbf{B}) - \cos c (\mathbf{A} \cdot \mathbf{C}) + \cos b \cos c = \cos a - \cos b \cos c - \cos c \cos b + \cos b \cos c,$$

that is,

$$\mathbf{B}_\perp \cdot \mathbf{C}_\perp = \cos a - \cos b \cos c.$$

Step 3 — conclusion. By Definition 8.2 and Step 1,

$$\cos A = \frac{\mathbf{B}_\perp \cdot \mathbf{C}_\perp}{\|\mathbf{B}_\perp\| \|\mathbf{C}_\perp\|} = \frac{\cos a - \cos b \cos c}{\sin c \sin b},$$

whence $\cos a = \cos b \cos c + \sin b \sin c \cos A$. The other two formulas are obtained by exchanging the names of the vertices, an operation which is legitimate because no step has privileged $\mathbf{A}$.

□

Remark 8.2: Why there is no “ $-2bc$ ”. In the plane $a^2 = b^2 + c^2 - 2bc \cos A$ holds: a relation between lengths. On the sphere the relation is between cosines of angles, and the additive structure is replaced by a multiplicative one. The two match in the limit of small triangles. On a sphere of radius r the sides measure $\alpha = ra$, $\beta = rb$, $\gamma = rc$; substituting $a = \alpha/r$ and expanding with $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)$ and $\sin x = x - \frac{x^3}{6} + O(x^5)$ one obtains, order by order in $1/r$,

$$1 - \frac{\alpha^2}{2r^2} + O(r^{-4}) = \left(1 - \frac{\beta^2}{2r^2}\right) \left(1 - \frac{\gamma^2}{2r^2}\right) + \frac{\beta\gamma}{r^2} \cos A + O(r^{-4}).$$

The terms of order r^0 give $1 = 1$; those of order r^{-2} give $-\frac{\alpha^2}{2} = -\frac{\beta^2}{2} - \frac{\gamma^2}{2} + \beta\gamma \cos A$, that is, $\alpha^2 = \beta^2 + \gamma^2 - 2\beta\gamma \cos A$: the plane cosine theorem. Euclidean geometry is the limit $r \rightarrow \infty$ of spherical geometry, uniformly on triangles of bounded diameter.

Corollary 8.1: Spherical analogue of the Pythagorean theorem. If the triangle is right-angled at C , that is $C = \pi/2$, then

$$\cos c = \cos a \cos b.$$

For small triangles one recovers $c^2 \approx a^2 + b^2$.

Proof. It suffices to apply Theorem 8.1 in the form $\cos c = \cos a \cos b + \sin a \sin b \cos C$ and to observe that $\cos(\pi/2) = 0$: the second summand vanishes. For the limit: with $a, b, c \rightarrow 0^+$, from $\cos c = \cos a \cos b$ it follows that

$$1 - \frac{c^2}{2} + O(c^4) = \left(1 - \frac{a^2}{2} + O(a^4)\right) \left(1 - \frac{b^2}{2} + O(b^4)\right) = 1 - \frac{a^2 + b^2}{2} + O((a^2 + b^2)^2),$$

and comparing the second-order terms, $c^2 = a^2 + b^2 + O((a^2 + b^2)^2)$. □

Corollary 8.2: Spherical triangle inequality. In every spherical triangle $a < b + c$, and analogously for the other two sides. In particular $\angle(\cdot, \cdot)$ is a distance on S^2 .

Proof. Since $\cos A > -1$ (because $A \in (0, \pi)$) and $\sin b \sin c > 0$,

$$\cos a = \cos b \cos c + \sin b \sin c \cos A > \cos b \cos c - \sin b \sin c = \cos(b + c).$$

If $b + c \geq \pi$ the claim is obvious, because $a < \pi \leq b + c$. If instead $b + c < \pi$, then a and $b + c$ both lie in $[0, \pi]$, where the cosine is strictly decreasing: from $\cos a > \cos(b + c)$ it follows that

$a < b + c$. The triangle inequality for points not in general position (coincident, antipodal or aligned on a great circle) is verified directly, and symmetry and vanishing only on the diagonal are immediate: $\angle$ is therefore a distance. $\square$

Corollary 8.3: The perimeter is less than 2π . In every spherical triangle $0 < a + b + c < 2\pi$. Setting $s = \frac{1}{2}(a + b + c)$, one therefore has $0 < s < \pi$ and $0 < s - a, s - b, s - c < s$.

Proof. Let $\mathbf{A}^* = -\mathbf{A}$ be the antipode of $\mathbf{A}$. The three vectors $\mathbf{A}^*, \mathbf{B}, \mathbf{C}$ are still linearly independent (changing the sign of a vector does not alter independence), hence $\mathbf{A}^*BC$ is a spherical triangle; its sides are

$$\angle(\mathbf{B}, \mathbf{C}) = a, \quad \angle(\mathbf{C}, \mathbf{A}^*) = \pi - b, \quad \angle(\mathbf{A}^*, \mathbf{B}) = \pi - c,$$

because $\mathbf{C} \cdot \mathbf{A}^* = -\cos b = \cos(\pi - b)$ and analogously for the other one. Applying to this triangle Corollary 8.2 to the side a :

$$a < (\pi - b) + (\pi - c) = 2\pi - b - c,$$

that is, $a + b + c < 2\pi$. Positivity is obvious. From $s < \pi$ and from $s - a = \frac{1}{2}(b + c - a) > 0$ (again by the triangle inequality) the other assertions follow. $\square$

Example 8.1: The trirectangular triangle. Take $\mathbf{A} = (1, 0, 0)$, $\mathbf{B} = (0, 1, 0)$, $\mathbf{C} = (0, 0, 1)$: they are orthonormal, hence independent. All the crossed scalar products vanish, hence $a = b = c = \pi/2$. The projections at the vertex $\mathbf{A}$ are $\mathbf{B}_\perp = \mathbf{B}$ and $\mathbf{C}_\perp = \mathbf{C}$, orthogonal to each other: $A = \pi/2$, and by symmetry $B = C = \pi/2$. The sum of the angles equals $3\pi/2 > \pi$: a triangle with three right angles, impossible in the plane. The perimeter is $3\pi/2 < 2\pi$, in agreement with Corollary 8.3.

8.3. The sine theorem

The most economical proof of the sine theorem goes through a quantity that is symmetric in the three sides: the Gram determinant of the three unit vectors of the vertices.

Lemma 8.2: Gram determinant of the vertices. Let M be the 3×3 matrix whose columns are $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and let $G = M^\top M$ be its Gram matrix. Then

$$X := \det G = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2 \cos a \cos b \cos c = (\det[\mathbf{A}, \mathbf{B}, \mathbf{C}])^2,$$

and $X > 0$ if and only if the triangle is non-degenerate. Moreover

$$\sin A = \frac{\sqrt{X}}{\sin b \sin c}, \quad \sin B = \frac{\sqrt{X}}{\sin c \sin a}, \quad \sin C = \frac{\sqrt{X}}{\sin a \sin b}.$$

Proof. Step 1 — the form of G . The (i, j) entry of $G = M^\top M$ is the scalar product of column i with column j , hence

$$G = \begin{pmatrix} 1 & \cos c & \cos b \\ \cos c & 1 & \cos a \\ \cos b & \cos a & 1 \end{pmatrix}.$$

Expanding along the first row:

$$\det G = 1 \cdot (1 - \cos^2 a) - \cos c (\cos c - \cos a \cos b) + \cos b (\cos a \cos c - \cos b)$$

$$= 1 - \cos^2 a - \cos^2 c + \cos a \cos b \cos c + \cos a \cos b \cos c - \cos^2 b = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2 \cos a \cos b \cos c.$$

On the other hand $\det G = \det(M^T) \det M = (\det M)^2$, and $\det M = \det[\mathbf{A}, \mathbf{B}, \mathbf{C}]$.

Step 2 — positivity. $(\det M)^2 > 0$ if and only if $\det M \neq 0$, that is, if and only if the columns are linearly independent: exactly the non-degeneracy condition of Definition 8.1.

Step 3 — the sine of the angles. From Theorem 8.1,

$$\sin^2 A = 1 - \cos^2 A = \frac{\sin^2 b \sin^2 c - (\cos a - \cos b \cos c)^2}{\sin^2 b \sin^2 c}.$$

The numerator equals

$$\begin{aligned} & (1 - \cos^2 b)(1 - \cos^2 c) - \cos^2 a + 2 \cos a \cos b \cos c - \cos^2 b \cos^2 c \\ &= 1 - \cos^2 b - \cos^2 c + \cos^2 b \cos^2 c - \cos^2 a + 2 \cos a \cos b \cos c - \cos^2 b \cos^2 c = X. \end{aligned}$$

Since $A \in (0, \pi)$ one has $\sin A > 0$, and taking the positive root $\sin A = \sqrt{X}/(\sin b \sin c)$. The other two follow by cyclic permutation, because X is symmetric in a, b, c . $\square$

Theorem 8.2: Spherical sine theorem. In every spherical triangle

$$\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c} = \frac{\sqrt{X}}{\sin a \sin b \sin c}$$

with $X = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2 \cos a \cos b \cos c > 0$.

Proof. By Lemma 8.2,

$$\frac{\sin A}{\sin a} = \frac{\sqrt{X}}{\sin a \sin b \sin c}, \quad \frac{\sin B}{\sin b} = \frac{\sqrt{X}}{\sin a \sin b \sin c}, \quad \frac{\sin C}{\sin c} = \frac{\sqrt{X}}{\sin a \sin b \sin c},$$

and the three ratios coincide because the right-hand side does not depend on which vertex has been chosen. The denominators are non-zero because $a, b, c \in (0, \pi)$. $\square$

Geometric variant: the mixed product. The number $\sqrt{X} = |\det[\mathbf{A}, \mathbf{B}, \mathbf{C}]|$ is the volume of the parallelepiped built on the three unit vectors of the vertices; it is therefore manifestly symmetric (up to sign) under permutation of the vertices. From Step 3 of Lemma 8.2 one reads that this volume equals $\sin b \sin c \sin A$, and by symmetry also $\sin c \sin a \sin B$ and $\sin a \sin b \sin C$. Equating the three expressions and dividing by $\sin a \sin b \sin c > 0$ one recovers the chain of equalities. This is the “volume” reading of the same computation. $\square$

Warning. The sine theorem, on its own, does *not* determine an angle: from $\sin A$ one goes back to A or to $\pi - A$. To remove the ambiguity it must be paired with the cosine theorem, which supplies $\cos A$ and hence the quadrant. It is the most frequent error in navigation computations, and it is discussed in Exercise 8.6.

8.4. The polar triangle and the cosine theorem for angles

On the sphere there reigns a duality that the plane does not know: to every triangle another one is associated in which sides and angles are exchanged (up to taking the supplements). Every theorem then automatically splits into two.

Definition 8.3: Polar triangle. Let ABC be a spherical triangle and let $\delta = \det[\mathbf{A}, \mathbf{B}, \mathbf{C}]$. By possibly exchanging the names of $\mathbf{B}$ and $\mathbf{C}$ we may assume $\delta > 0$ (we then say that the triangle is **positively oriented**). The **polar triangle** $A'B'C'$ has vertices

$$\mathbf{A}' = \frac{\mathbf{B} \times \mathbf{C}}{\|\mathbf{B} \times \mathbf{C}\|}, \quad \mathbf{B}' = \frac{\mathbf{C} \times \mathbf{A}}{\|\mathbf{C} \times \mathbf{A}\|}, \quad \mathbf{C}' = \frac{\mathbf{A} \times \mathbf{B}}{\|\mathbf{A} \times \mathbf{B}\|}.$$

The vector $\mathbf{A}'$ is one of the two poles of the great circle through $\mathbf{B}$ and $\mathbf{C}$; the construction chooses a definite one, and the following theorem shows that it is the one lying on the same side as $\mathbf{A}$ — the classical choice.

Theorem 8.3: Polar duality. Let ABC be a positively oriented spherical triangle, $\delta = \det[\mathbf{A}, \mathbf{B}, \mathbf{C}] > 0$, and let $A'B'C'$ be its polar triangle. Then:

- (i) $\|\mathbf{B} \times \mathbf{C}\| = \sin a$, $\|\mathbf{C} \times \mathbf{A}\| = \sin b$, $\|\mathbf{A} \times \mathbf{B}\| = \sin c$, and $\mathbf{A}' \cdot \mathbf{A} = \delta / \sin a > 0$: $\mathbf{A}'$ is the pole of the side a lying on the same side as $\mathbf{A}$;
- (ii) $\det[\mathbf{A}', \mathbf{B}', \mathbf{C}'] = \frac{\delta^2}{\sin a \sin b \sin c} > 0$: $A'B'C'$ is a non-degenerate spherical triangle, positively oriented as well;
- (iii) the sides of $A'B'C'$ are $a' = \pi - A$, $b' = \pi - B$, $c' = \pi - C$;
- (iv) the polar triangle of $A'B'C'$ is ABC ; consequently its angles are $A' = \pi - a$, $B' = \pi - b$, $C' = \pi - c$.

Proof. We shall use two classical identities of vector calculus in $\mathbb{R}^3$, valid for every choice of the vectors:

$$(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{w} \times \mathbf{z}) = (\mathbf{u} \cdot \mathbf{w})(\mathbf{v} \cdot \mathbf{z}) - (\mathbf{u} \cdot \mathbf{z})(\mathbf{v} \cdot \mathbf{w}), \quad (\text{I})$$

$$\mathbf{p} \times (\mathbf{u} \times \mathbf{v}) = (\mathbf{p} \cdot \mathbf{v})\mathbf{u} - (\mathbf{p} \cdot \mathbf{u})\mathbf{v}. \quad (\text{II})$$

(i) From (I) with $\mathbf{u} = \mathbf{w} = \mathbf{B}$ and $\mathbf{v} = \mathbf{z} = \mathbf{C}$: $\|\mathbf{B} \times \mathbf{C}\|^2 = 1 - \cos^2 a = \sin^2 a$, and $\sin a > 0$ because $a \in (0, \pi)$. Moreover $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \delta$, hence $\mathbf{A}' \cdot \mathbf{A} = \delta / \sin a > 0$.

(ii) By (II) with $\mathbf{p} = \mathbf{C} \times \mathbf{A}$, $\mathbf{u} = \mathbf{A}$, $\mathbf{v} = \mathbf{B}$:

$$(\mathbf{C} \times \mathbf{A}) \times (\mathbf{A} \times \mathbf{B}) = [(\mathbf{C} \times \mathbf{A}) \cdot \mathbf{B}]\mathbf{A} - [(\mathbf{C} \times \mathbf{A}) \cdot \mathbf{A}]\mathbf{B} = \delta \mathbf{A}, \quad (*)$$

because $(\mathbf{C} \times \mathbf{A}) \cdot \mathbf{A} = 0$ and $(\mathbf{C} \times \mathbf{A}) \cdot \mathbf{B} = \det[\mathbf{C}, \mathbf{A}, \mathbf{B}] = \delta$ (the mixed product is invariant under cyclic permutations). Then

$$\det[\mathbf{A}', \mathbf{B}', \mathbf{C}'] = \frac{(\mathbf{B} \times \mathbf{C}) \cdot [(\mathbf{C} \times \mathbf{A}) \times (\mathbf{A} \times \mathbf{B})]}{\sin a \sin b \sin c} = \frac{\delta (\mathbf{B} \times \mathbf{C}) \cdot \mathbf{A}}{\sin a \sin b \sin c} = \frac{\delta^2}{\sin a \sin b \sin c} > 0.$$

In particular $\mathbf{A}', \mathbf{B}', \mathbf{C}'$ are independent: $A'B'C'$ is a spherical triangle.

(iii) By (I),

$$(\mathbf{C} \times \mathbf{A}) \cdot (\mathbf{A} \times \mathbf{B}) = (\mathbf{C} \cdot \mathbf{A})(\mathbf{A} \cdot \mathbf{B}) - (\mathbf{C} \cdot \mathbf{B})(\mathbf{A} \cdot \mathbf{A}) = \cos b \cos c - \cos a.$$

Dividing by $\|\mathbf{C} \times \mathbf{A}\| \|\mathbf{A} \times \mathbf{B}\| = \sin b \sin c$ and using Theorem 8.1:

$$\cos a' = \mathbf{B}' \cdot \mathbf{C}' = \frac{\cos b \cos c - \cos a}{\sin b \sin c} = -\cos A.$$

Since a' and $\pi - A$ both lie in $(0, \pi)$ and the cosine is injective there, $a' = \pi - A$. The other two follow by permutation.

(iv) From (*), $\mathbf{B}' \times \mathbf{C}' = \frac{\delta}{\sin b \sin c} \mathbf{A}$, with positive coefficient; upon normalizing, the first vertex of the polar triangle of $A'B'C'$ is exactly $\mathbf{A}$. Analogously for the other two. Applying then (iii) to the triangle $A'B'C'$ — legitimate by (ii) — its side “ a ” is π minus its angle “ A ”: that is, $a = \pi - A'$, i.e. $A' = \pi - a$. $\square$

Remark 8.3: The polar triangle and duality. Theorem 8.3 establishes a dictionary: every identity valid in all spherical triangles, rewritten by substituting

$$a \mapsto \pi - A, \quad b \mapsto \pi - B, \quad c \mapsto \pi - C, \quad A \mapsto \pi - a, \quad B \mapsto \pi - b, \quad C \mapsto \pi - c,$$

remains valid. The reason is that the new identity is the old one, read on the polar triangle. This is the deep reason why on the sphere — unlike in the plane — the three angles determine the triangle, and hence there are no non-congruent similar spherical triangles.

Theorem 8.4: Cosine theorem for angles. In every spherical triangle

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$

and analogously by cyclic permutation.

Proof. We may assume the triangle to be positively oriented (otherwise the names of two vertices are exchanged, which alters neither the angles nor the sides as sets of data). Let $A'B'C'$ be the polar triangle. By Theorem 8.3(ii) it is a spherical triangle, hence Theorem 8.1 applies to it:

$$\cos a' = \cos b' \cos c' + \sin b' \sin c' \cos A'.$$

Substituting $a' = \pi - A$, $b' = \pi - B$, $c' = \pi - C$, $A' = \pi - a$ (items (iii) and (iv) of the same theorem) and using $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$:

$$-\cos A = (-\cos B)(-\cos C) + \sin B \sin C (-\cos a) = \cos B \cos C - \sin B \sin C \cos a.$$

Changing the sign of both sides one obtains the claim. $\square$

Corollary 8.4: There are no non-congruent similar spherical triangles. Two spherical triangles with the same three angles have the same three sides, and are therefore congruent (there exists an isometry of S^2 carrying one into the other). Moreover $\pi < A + B + C < 3\pi$.

Proof. From Theorem 8.4,

$$\cos a = \frac{\cos A + \cos B \cos C}{\sin B \sin C},$$

and since $a \in (0, \pi)$, where the cosine is injective, a is a single-valued function of (A, B, C) ; the same for b and c . Two triangles with the same angles therefore have equal sides; since the six scalar products between the unit vectors of the vertices coincide, the linear map sending one triple of vertices to the other preserves the scalar product on a basis of $\mathbb{R}^3$, hence it is orthogonal: it is an isometry of S^2 . As for the inequality: the polar triangle has sides $\pi - A, \pi - B, \pi - C$, and Corollary 8.3 gives it $(\pi - A) + (\pi - B) + (\pi - C) < 2\pi$, that is $A + B + C > \pi$; on the other hand each angle is less than π , hence $A + B + C < 3\pi$. $\square$

8.5. Right triangles and Napier's rules

A spherical right triangle has five free quantities besides the right angle: the two legs a and b , the hypotenuse c and the angles A and B . Given two of them, the other three are determined. The corresponding ten relations are written as six formulas, which Napier then compressed into two mnemonic rules.

Proposition 8.1: The six formulas of the right triangle. Let ABC be a spherical triangle with $C = \pi/2$. Then

- | | |
|---|---|
| (1) $\cos c = \cos a \cos b,$ | (2) $\sin a = \sin A \sin c, \quad \sin b = \sin B \sin c,$ |
| (3) $\tan a = \cos B \tan c, \quad \tan b = \cos A \tan c,$ | (4) $\tan a = \tan A \sin b, \quad \tan b = \tan B \sin a,$ |
| (5) $\cos c = \cot A \cot B,$ | (6) $\cos A = \cos a \sin B, \quad \cos B = \cos b \sin A.$ |

Proof. Set $C = \pi/2$, hence $\cos C = 0$ and $\sin C = 1$.

(1) It is Corollary 8.1.

(2) From Theorem 8.2, $\sin a / \sin A = \sin c / \sin C = \sin c$, that is $\sin a = \sin A \sin c$; symmetrically $\sin b = \sin B \sin c$.

(5) From Theorem 8.4 applied to the angle C : $\cos C = -\cos A \cos B + \sin A \sin B \cos c$. Since $\cos C = 0$, $\sin A \sin B \cos c = \cos A \cos B$, and dividing by $\sin A \sin B > 0$ one obtains $\cos c = \cot A \cot B$.

(6) Again from Theorem 8.4, this time for A : $\cos A = -\cos B \cos C + \sin B \sin C \cos a = \sin B \cos a$. Exchanging the roles of A and B (legitimate: both are acute or obtuse angles adjacent to the right angle), $\cos B = \sin A \cos b$.

(3) From Theorem 8.1 applied to the side b ,

$$\cos B = \frac{\cos b - \cos c \cos a}{\sin c \sin a}.$$

By (1), $\cos b = \cos c / \cos a$ (note that $\cos a \neq 0$: if $a = \pi/2$, from (1) it would follow that $\cos c = 0$ and hence also $c = \pi/2$, that is $\mathbf{A}$ would be a pole of $\mathbf{BC}$; this case, in which $A = a$ and $b = B$, is covered by the formulas by continuity and is treated separately below). Substituting,

$$\cos B = \frac{\frac{\cos c}{\cos a} - \cos a \cos c}{\sin a \sin c} = \frac{\cos c (1 - \cos^2 a)}{\cos a \sin a \sin c} = \frac{\cos c \sin^2 a}{\cos a \sin a \sin c} = \frac{\sin a}{\cos a} \cdot \frac{\cos c}{\sin c} = \frac{\tan a}{\tan c},$$

that is $\tan a = \cos B \tan c$. Exchanging $a \leftrightarrow b$ and $A \leftrightarrow B$ one gets $\tan b = \cos A \tan c$. In the excluded case $a = c = \pi/2$ one has $\cos B = 0$ and $\tan a$ is not defined: the relation must be read in the equivalent form $\sin a \cos c = \cos B \sin c \cos a$, which is $0 = 0$.

(4) From (2), $\sin A = \sin a / \sin c$; from (6), $\cos A = \cos a \sin B$ and $\sin B = \sin b / \sin c$ by (2). Hence

$$\tan A = \frac{\sin A}{\cos A} = \frac{\sin a / \sin c}{\cos a \sin b / \sin c} = \frac{\sin a}{\cos a \sin b} = \frac{\tan a}{\sin b},$$

that is $\tan a = \tan A \sin b$; symmetrically $\tan b = \tan B \sin a$. □

Theorem 8.5: Napier's rules. Let ABC be a spherical triangle with $C = \pi/2$. Arrange the five circular parts

$$a, \quad b, \quad \frac{\pi}{2} - A, \quad \frac{\pi}{2} - c, \quad \frac{\pi}{2} - B$$

in this cyclic order on a pentagon (Figure 8.2). Once a part has been chosen as the middle one, the two contiguous parts are called adjacent and the two remaining ones opposite. Then, for each of the five choices,

$$\sin(\text{middle part}) = \prod \tan(\text{adjacent parts}), \quad \sin(\text{middle part}) = \prod \cos(\text{opposite parts}).$$

Proof. The two rules produce ten identities in all, two for each of the five choices of the middle part. We verify them one by one, reducing each of them to one of the formulas of Proposition 8.1; we repeatedly use $\sin(\frac{\pi}{2} - x) = \cos x$, $\cos(\frac{\pi}{2} - x) = \sin x$, $\tan(\frac{\pi}{2} - x) = \cot x$.

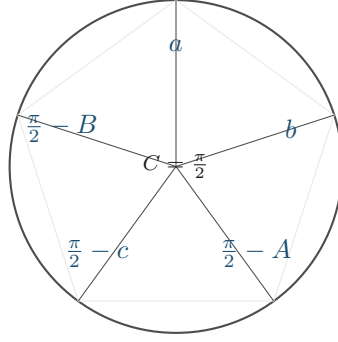


Figure 8.2: Napier's pentagon of circular parts for a triangle right-angled at C . The right angle does not appear; the five circular parts are, in the cyclic order in which they occur on the triangle, $a, b, \frac{\pi}{2} - A, \frac{\pi}{2} - c, \frac{\pi}{2} - B$. The vertices of the regular pentagon are computed by hand on a circle of radius 2.2: $(0; 2.2)$, $(2.2 \cos 18^\circ; 2.2 \sin 18^\circ) = (2.0923; 0.6798)$, $(2.2 \cos(-54^\circ); 2.2 \sin(-54^\circ)) = (1.2931; -1.7798)$ and the two symmetric with respect to the vertical axis.

Middle = a (adjacent b and $\frac{\pi}{2} - B$; opposite $\frac{\pi}{2} - A$ and $\frac{\pi}{2} - c$):

$$\sin a = \tan b \tan\left(\frac{\pi}{2} - B\right) = \tan b \cot B \iff \tan b = \tan B \sin a, \text{ that is (4);}$$

$$\sin a = \cos\left(\frac{\pi}{2} - A\right) \cos\left(\frac{\pi}{2} - c\right) = \sin A \sin c, \text{ that is (2).}$$

Middle = b (adjacent a and $\frac{\pi}{2} - A$; opposite $\frac{\pi}{2} - B$ and $\frac{\pi}{2} - c$):

$$\sin b = \tan a \cot A \iff \tan a = \tan A \sin b, \text{ (4);} \quad \sin b = \sin B \sin c, \text{ (2).}$$

Middle = $\frac{\pi}{2} - A$ (adjacent b and $\frac{\pi}{2} - c$; opposite $\frac{\pi}{2} - B$ and a):

$$\cos A = \tan b \cot c \iff \tan b = \cos A \tan c, \text{ (3);} \quad \cos A = \sin B \cos a, \text{ (6).}$$

Middle = $\frac{\pi}{2} - c$ (adjacent $\frac{\pi}{2} - A$ and $\frac{\pi}{2} - B$; opposite a and b):

$$\cos c = \cot A \cot B, \text{ (5);} \quad \cos c = \cos a \cos b, \text{ (1).}$$

Middle = $\frac{\pi}{2} - B$ (adjacent $\frac{\pi}{2} - c$ and a ; opposite b and $\frac{\pi}{2} - A$):

$$\cos B = \cot c \tan a \iff \tan a = \cos B \tan c, \text{ (3);} \quad \cos B = \cos b \sin A, \text{ (6).}$$

Every identity produced by the two rules is therefore one of the ten relations of Proposition 8.1, and conversely each of those ten occurs exactly once. Napier's rules are thus *equivalent* to the six formulas, and not an addition to them: their content is mnemonic, not mathematical. $\square$

Remark 8.4: How the cyclic order is remembered. The order $a, b, \frac{\pi}{2} - A, \frac{\pi}{2} - c, \frac{\pi}{2} - B$ is not arbitrary: one travels round the triangle starting from one leg, passes to the other leg, then takes the complement of the angle not adjacent to the first leg, then the complement of the hypotenuse, and finally the complement of the last angle. The verification that this is the right order is contained in the proof: with a different order, the ten identities produced would not coincide with the ten true ones.

8.6. Spherical excess and area: Girard's theorem

Definition 8.4: Spherical lune. A **lune** (or *digon*) of amplitude $\alpha \in (0, \pi]$ on a sphere of radius r is the region comprised between two great semicircles that have the same antipodal endpoints and form the dihedral angle α with each other.

Proposition 8.2: Area of a lune. A lune of amplitude α on a sphere of radius r has area $2\alpha r^2$. In particular for $\alpha = \pi$ one recovers the area $2\pi r^2$ of the hemisphere, and for $\alpha = 2\pi$ the area $4\pi r^2$ of the sphere.

Proof. The rotations of $\mathbb{R}^3$ about O are isometries that carry S^2 into itself and preserve areas; we may therefore assume that the antipodal endpoints of the lune are the poles $(0, 0, \pm r)$ and that the lune is $\{\varphi \in (0, \alpha)\}$ in the spherical parametrization

$$\sigma(\vartheta, \varphi) = r(\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta), \quad \vartheta \in (0, \pi), \quad \varphi \in (0, 2\pi).$$

One computes

$$\sigma_{\vartheta} = r(\cos \vartheta \cos \varphi, \cos \vartheta \sin \varphi, -\sin \vartheta), \quad \sigma_{\varphi} = r(-\sin \vartheta \sin \varphi, \sin \vartheta \cos \varphi, 0),$$

whence $\|\sigma_{\vartheta}\| = r$, $\|\sigma_{\varphi}\| = r \sin \vartheta$, $\sigma_{\vartheta} \cdot \sigma_{\varphi} = 0$ and therefore $\|\sigma_{\vartheta} \times \sigma_{\varphi}\| = r^2 \sin \vartheta$: it is the area element of the sphere. Hence

$$\text{Area} = \int_0^{\alpha} \int_0^{\pi} r^2 \sin \vartheta \, d\vartheta \, d\varphi = r^2 \alpha [-\cos \vartheta]_0^{\pi} = 2\alpha r^2.$$

□

Theorem 8.6: Girard's theorem: excess and area. Let T be a spherical triangle with angles A, B, C on a sphere of radius r . Then

$$\text{Area}(T) = E r^2, \quad E = A + B + C - \pi$$

where $E > 0$ is the **spherical excess**. On the unit sphere, area and excess coincide.

Proof. We work on S^2 of radius r and let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be the vertices. We denote by $\gamma_a, \gamma_b, \gamma_c$ the great circles containing respectively the sides BC, CA, AB .

Step 1 — the three lunes. The great circle γ_b (through $\mathbf{A}$ and $\mathbf{C}$) and the great circle γ_c (through $\mathbf{A}$ and $\mathbf{B}$) intersect in the two antipodal points $\pm \mathbf{A}$ and bound four lunes; let L_A be the one containing T . Its amplitude is, by definition, the dihedral angle between the planes of γ_b and γ_c on the side of the triangle, that is the angle A . By Proposition 8.2, $\text{Area}(L_A) = 2Ar^2$. The third great circle γ_a passes through $\mathbf{B}$ and $\mathbf{C}$, which lie on the boundary of L_A , and cuts L_A into two triangles: the one with vertices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, that is T , and the one with vertices $-\mathbf{A}, \mathbf{B}, \mathbf{C}$. Hence

$$\text{Area}(T) + \text{Area}(T_{-A}) = 2Ar^2, \quad T_{-A} := (-\mathbf{A})\mathbf{B}\mathbf{C}.$$

In the same way, with the lunes L_B and L_C ,

$$\text{Area}(T) + \text{Area}(T_{-B}) = 2Br^2, \quad \text{Area}(T) + \text{Area}(T_{-C}) = 2Cr^2,$$

with $T_{-B} = \mathbf{A}(-\mathbf{B})\mathbf{C}$ and $T_{-C} = \mathbf{A}\mathbf{B}(-\mathbf{C})$.

Step 2 — the four triangles that tile a hemisphere. The great circle γ_c divides the sphere into two hemispheres; let H be the one containing $\mathbf{C}$. The other two great circles γ_a and γ_b both pass through $\pm \mathbf{C}$ and meet γ_c in the four points $\pm \mathbf{A}$ and $\pm \mathbf{B}$; each of the two cuts

H along an arc going from a point of γ_c to its antipode through $\mathbf{C}$. The two arcs meet only at $\mathbf{C}$ and hence divide H into four spherical triangles, whose vertices are $\mathbf{C}$ and two consecutive points among $\mathbf{A}, \mathbf{B}, -\mathbf{A}, -\mathbf{B}$:

$$H = T \cup T_{-A} \cup T_{-B} \cup [(-\mathbf{A})(-\mathbf{B})\mathbf{C}],$$

a union with pairwise disjoint interiors. Passing to the areas (Figure 8.3),

$$\text{Area}(T) + \text{Area}(T_{-A}) + \text{Area}(T_{-B}) + \text{Area}[(-\mathbf{A})(-\mathbf{B})\mathbf{C}] = 2\pi r^2.$$

Step 3 — the antipodal triangle. The antipodal map $\iota(\mathbf{x}) = -\mathbf{x}$ is linear and orthogonal ($\iota^\top \iota = I$), hence it is an isometry of $\mathbb{R}^3$ that carries S^2 into itself and preserves areas. It carries $T_{-C} = \mathbf{AB}(-\mathbf{C})$ into $(-\mathbf{A})(-\mathbf{B})\mathbf{C}$, hence

$$\text{Area}[(-\mathbf{A})(-\mathbf{B})\mathbf{C}] = \text{Area}(T_{-C}).$$

Step 4 — conclusion. Substituting into Step 2 the expressions of Step 1,

$$\text{Area}(T) + [2Ar^2 - \text{Area}(T)] + [2Br^2 - \text{Area}(T)] + [2Cr^2 - \text{Area}(T)] = 2\pi r^2,$$

that is $2(A + B + C)r^2 - 2\text{Area}(T) = 2\pi r^2$, whence $\text{Area}(T) = (A + B + C - \pi)r^2$. Since the area of a non-degenerate triangle is strictly positive, $E = A + B + C - \pi > 0$ (recovering by another route Corollary 8.4). $\square$

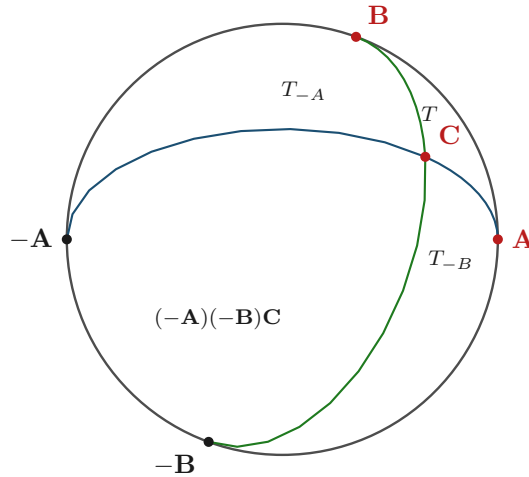


Figure 8.3: Step 2 of the proof of Girard's theorem, seen in orthographic projection on the hemisphere H bounded by the great circle γ_c through $\mathbf{A}$ and $\mathbf{B}$ (the boundary of the disc). The two arcs drawn are the halves of γ_b (in blue, from $\mathbf{A}$ to $-\mathbf{A}$ through $\mathbf{C}$) and of γ_a (in green, from $\mathbf{B}$ to $-\mathbf{B}$ through $\mathbf{C}$). They cut H into the four triangles T , T_{-A} , T_{-B} and $(-\mathbf{A})(-\mathbf{B})\mathbf{C}$. The points are computed with $\mathbf{A}$ on the equator at longitude 0° , $\mathbf{B}$ on the equator at longitude 70° , $\mathbf{C}$ at latitude 40° and longitude 30° , and the projection is $(x, y, z) \mapsto (3x, 3y)$.

Remark 8.5: Girard and Gauss–Bonnet. The sphere of radius r has constant Gauss curvature $K = 1/r^2$. The Gauss–Bonnet theorem for a geodesic triangle states that $\iint_T K dA = A + B + C - \pi$; since K is constant, the left-hand side equals $\text{Area}(T)/r^2$ and Girard's theorem is recovered exactly. In this chapter we have followed the reverse path, proving the spherical case in an elementary way and without differential geometry: the general case of Gauss–Bonnet is treated in Chapter 9 of this volume, where the present theorem serves as a check. Historically the statement goes back to Albert Girard (1629) and is proved

with the three lunes in Todhunter, *Spherical Trigonometry*, Art. 96–98.

8.7. L'Huilier's formula

Girard's theorem gives the excess starting from the *angles*. In geodesy, however, what is measured is often the *sides*. L'Huilier's formula solves the problem, and is the spherical analogue of Heron's formula.

Lemma 8.3: Half-angle formulas. Setting $s = \frac{1}{2}(a + b + c)$, in every spherical triangle

$$\sin^2 \frac{A}{2} = \frac{\sin(s-b)\sin(s-c)}{\sin b \sin c}, \quad \cos^2 \frac{A}{2} = \frac{\sin s \sin(s-a)}{\sin b \sin c},$$

$$\tan \frac{A}{2} = \sqrt{\frac{\sin(s-b)\sin(s-c)}{\sin s \sin(s-a)}},$$

hold, and the analogous ones for B and C by cyclic permutation. All the sines occurring are positive.

Proof. Step 0 — positivity. By Corollary 8.3, $0 < s < \pi$ and $0 < s-a, s-b, s-c < \pi$, hence $\sin s, \sin(s-a), \sin(s-b), \sin(s-c)$ are all positive.

Step 1 — $1 - \cos A$. From Theorem 8.1,

$$1 - \cos A = 1 - \frac{\cos a - \cos b \cos c}{\sin b \sin c} = \frac{\sin b \sin c + \cos b \cos c - \cos a}{\sin b \sin c} = \frac{\cos(b-c) - \cos a}{\sin b \sin c},$$

having used $\cos(b-c) = \cos b \cos c + \sin b \sin c$. By the sum-to-product formula $\cos X - \cos Y = 2 \sin \frac{X+Y}{2} \sin \frac{Y-X}{2}$ with $X = b-c, Y = a$:

$$\cos(b-c) - \cos a = 2 \sin \frac{a+b-c}{2} \sin \frac{a-b+c}{2} = 2 \sin(s-c) \sin(s-b),$$

because $a+b-c = 2s-2c$ and $a-b+c = 2s-2b$. Since $1 - \cos A = 2 \sin^2 \frac{A}{2}$, the first formula is obtained.

Step 2 — $1 + \cos A$. Analogously,

$$1 + \cos A = \frac{\sin b \sin c - \cos b \cos c + \cos a}{\sin b \sin c} = \frac{\cos a - \cos(b+c)}{\sin b \sin c} = \frac{2 \sin \frac{a+b+c}{2} \sin \frac{b+c-a}{2}}{\sin b \sin c} = \frac{2 \sin s \sin(s-a)}{\sin b \sin c},$$

and from $1 + \cos A = 2 \cos^2 \frac{A}{2}$ the second formula follows.

Step 3 — the tangent. Since $A \in (0, \pi)$ one has $\frac{A}{2} \in (0, \frac{\pi}{2})$, where sine and cosine are positive: the positive roots can be extracted and divided. $\square$

Lemma 8.4: Two of Delambre's analogies. In every spherical triangle

$$\sin \frac{A+B}{2} = \cos \frac{C}{2} \cdot \frac{\cos \frac{a-b}{2}}{\cos \frac{c}{2}}, \quad \cos \frac{A+B}{2} = \sin \frac{C}{2} \cdot \frac{\cos \frac{a+b}{2}}{\cos \frac{c}{2}}.$$

Proof. For brevity we write $P = \sin s, P_a = \sin(s-a), P_b = \sin(s-b), P_c = \sin(s-c)$, all positive by Step 0 of Lemma 8.3, and we use the formulas of that lemma:

$$\sin \frac{A}{2} = \sqrt{\frac{P_b P_c}{\sin b \sin c}}, \quad \cos \frac{A}{2} = \sqrt{\frac{P P_a}{\sin b \sin c}}, \quad \sin \frac{B}{2} = \sqrt{\frac{P_a P_c}{\sin a \sin c}}, \quad \cos \frac{B}{2} = \sqrt{\frac{P P_b}{\sin a \sin c}},$$

$$\sin \frac{C}{2} = \sqrt{\frac{P_a P_b}{\sin a \sin b}}, \quad \cos \frac{C}{2} = \sqrt{\frac{P P_c}{\sin a \sin b}}.$$

First identity. By the addition formula,

$$\sin \frac{A+B}{2} = \sin \frac{A}{2} \cos \frac{B}{2} + \cos \frac{A}{2} \sin \frac{B}{2} = \sqrt{\frac{P_b P_c}{\sin b \sin c} \cdot \frac{P P_b}{\sin a \sin c}} + \sqrt{\frac{P P_a}{\sin b \sin c} \cdot \frac{P_a P_c}{\sin a \sin c}}.$$

The two roots equal respectively $\frac{P_b \sqrt{P P_c}}{\sin c \sqrt{\sin a \sin b}}$ and $\frac{P_a \sqrt{P P_c}}{\sin c \sqrt{\sin a \sin b}}$, hence

$$\sin \frac{A+B}{2} = \frac{\sqrt{P P_c}}{\sin c \sqrt{\sin a \sin b}} (P_a + P_b) = \cos \frac{C}{2} \cdot \frac{P_a + P_b}{\sin c},$$

where in the last step $\sqrt{P P_c}/(\sin a \sin b) = \cos \frac{C}{2}$ has been recognized. Now, by the sum-to-product formula,

$$P_a + P_b = \sin(s-a) + \sin(s-b) = 2 \sin \frac{2s-a-b}{2} \cos \frac{b-a}{2} = 2 \sin \frac{c}{2} \cos \frac{a-b}{2},$$

whereas $\sin c = 2 \sin \frac{c}{2} \cos \frac{c}{2}$. Simplifying $2 \sin \frac{c}{2} > 0$ one obtains the first identity.

Second identity. Analogously,

$$\cos \frac{A+B}{2} = \cos \frac{A}{2} \cos \frac{B}{2} - \sin \frac{A}{2} \sin \frac{B}{2} = \frac{\sqrt{P_a P_b}}{\sin c \sqrt{\sin a \sin b}} (P - P_c) = \sin \frac{C}{2} \cdot \frac{P - P_c}{\sin c},$$

having recognized $\sqrt{P_a P_b}/(\sin a \sin b) = \sin \frac{C}{2}$; and

$$P - P_c = \sin s - \sin(s-c) = 2 \cos \frac{2s-c}{2} \sin \frac{c}{2} = 2 \cos \frac{a+b}{2} \sin \frac{c}{2}.$$

Dividing again by $\sin c = 2 \sin \frac{c}{2} \cos \frac{c}{2}$ one concludes. $\square$

Theorem 8.7: L'Huilier's formula. Let $E = A + B + C - \pi$ be the spherical excess and $s = \frac{1}{2}(a + b + c)$. Then

$$\tan \frac{E}{4} = \sqrt{\tan \frac{s}{2} \tan \frac{s-a}{2} \tan \frac{s-b}{2} \tan \frac{s-c}{2}}$$

and the formula determines E without ambiguity, since $E/4 \in (0, \pi/2)$.

Proof. Step 1 — $\tan \frac{E}{4}$ as a function of $A + B$ and C . Set $X = \frac{A+B}{2}$ and $Y = \frac{\pi-C}{2}$. Then

$$\frac{X-Y}{2} = \frac{A+B-\pi+C}{4} = \frac{E}{4}.$$

From the sum-to-product formulas $\sin X - \sin Y = 2 \cos \frac{X+Y}{2} \sin \frac{X-Y}{2}$ and $\cos X + \cos Y = 2 \cos \frac{X+Y}{2} \cos \frac{X-Y}{2}$ it follows, if $\cos \frac{X+Y}{2} \neq 0$, that

$$\tan \frac{E}{4} = \tan \frac{X-Y}{2} = \frac{\sin X - \sin Y}{\cos X + \cos Y} = \frac{\sin \frac{A+B}{2} - \cos \frac{C}{2}}{\cos \frac{A+B}{2} + \sin \frac{C}{2}},$$

having used $\sin \frac{\pi-C}{2} = \cos \frac{C}{2}$ and $\cos \frac{\pi-C}{2} = \sin \frac{C}{2}$. The condition $\cos \frac{X+Y}{2} \neq 0$ is satisfied: $\frac{X+Y}{2} = \frac{A+B+\pi-C}{4}$, and applying Corollary 8.2 to the polar triangle (whose sides are $\pi - A, \pi -$

$B, \pi - C$) one has $\pi - C < (\pi - A) + (\pi - B)$, that is $A + B - C < \pi$; moreover $A + B - C > -\pi$ because $A + B > 0$ and $C < \pi$. Hence $\frac{X+Y}{2} \in (0, \frac{\pi}{2})$ and the cosine is positive.

Step 2 — Delambre's analogies are substituted. By Lemma 8.4, multiplying numerator and denominator by $\cos \frac{c}{2} > 0$,

$$\tan \frac{E}{4} = \frac{\cos \frac{C}{2} \left(\cos \frac{a-b}{2} - \cos \frac{c}{2} \right)}{\sin \frac{C}{2} \left(\cos \frac{a+b}{2} + \cos \frac{c}{2} \right)} = \cot \frac{C}{2} \cdot \frac{\cos \frac{a-b}{2} - \cos \frac{c}{2}}{\cos \frac{a+b}{2} + \cos \frac{c}{2}}.$$

Step 3 — sum-to-product on the sides. Again by the sum-to-product formulas,

$$\cos \frac{a-b}{2} - \cos \frac{c}{2} = 2 \sin \frac{\frac{a-b}{2} + \frac{c}{2}}{2} \sin \frac{\frac{c}{2} - \frac{a-b}{2}}{2} = 2 \sin \frac{a-b+c}{4} \sin \frac{b+c-a}{4} = 2 \sin \frac{s-b}{2} \sin \frac{s-a}{2},$$

because $a-b+c = 2(s-b)$ and $b+c-a = 2(s-a)$; and

$$\cos \frac{a+b}{2} + \cos \frac{c}{2} = 2 \cos \frac{a+b+c}{4} \cos \frac{a+b-c}{4} = 2 \cos \frac{s}{2} \cos \frac{s-c}{2}.$$

Both quantities are positive: the arguments $\frac{s-a}{2}, \frac{s-b}{2}$ lie in $(0, \frac{\pi}{2})$ and so do $\frac{s}{2}, \frac{s-c}{2}$, by Corollary 8.3. Hence

$$\tan \frac{E}{4} = \cot \frac{C}{2} \cdot \frac{\sin \frac{s-a}{2} \sin \frac{s-b}{2}}{\cos \frac{s}{2} \cos \frac{s-c}{2}}.$$

Step 4 — $\cot \frac{C}{2}$ is substituted. From Lemma 8.3,

$$\cot \frac{C}{2} = \sqrt{\frac{\sin s \sin(s-c)}{\sin(s-a) \sin(s-b)}}.$$

We now use $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$ for each of the four sines:

$$\cot \frac{C}{2} = \sqrt{\frac{\sin \frac{s}{2} \cos \frac{s}{2} \sin \frac{s-c}{2} \cos \frac{s-c}{2}}{\sin \frac{s-a}{2} \cos \frac{s-a}{2} \sin \frac{s-b}{2} \cos \frac{s-b}{2}}}$$

(the factors 2 cancel two by two). Multiplying by the fraction of Step 3 and regrouping the roots,

$$\tan \frac{E}{4} = \underbrace{\sqrt{\frac{\sin \frac{s}{2} \sin \frac{s-c}{2}}{\cos \frac{s}{2} \cos \frac{s-c}{2}}}}_{=\sqrt{\tan \frac{s}{2} \tan \frac{s-c}{2}}} \cdot \underbrace{\frac{\sin \frac{s-a}{2} \sin \frac{s-b}{2}}{\sqrt{\sin \frac{s-a}{2} \cos \frac{s-a}{2} \sin \frac{s-b}{2} \cos \frac{s-b}{2}}}}_{=\sqrt{\tan \frac{s-a}{2} \tan \frac{s-b}{2}}},$$

that is, the announced formula. The regrouping is legitimate because all the quantities under the roots are positive.

Step 5 — uniqueness. By Theorem 8.6 one has $E > 0$, and by Corollary 8.4 $E = A + B + C - \pi < 2\pi$; hence $E/4 \in (0, \pi/2)$, an interval on which the tangent is a bijection onto $(0, +\infty)$: the formula determines E uniquely. $\square$

Remark 8.6: Heron and L'Huilier. The plane Heron formula $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$ and L'Huilier's one have the same structure, with the tangents of the half-arguments in place of the arguments. This is no accident: for small triangles $\tan x \approx x$, and with $\text{Area} \approx Er^2$ and $E/4$ small one obtains $\frac{E}{4} \approx \sqrt{\frac{s}{2} \cdot \frac{s-a}{2} \cdot \frac{s-b}{2} \cdot \frac{s-c}{2}}$, that is $E \approx \sqrt{s(s-a)(s-b)(s-c)}$: precisely Heron.

8.8. Great-circle navigation: a complete numerical example

The shortest route between two points of the Earth's surface, modelled as a sphere, is the minor arc of great circle joining them: the **orthodrome**. The reference triangle is the one having as vertices the north pole N and the two points.

Proposition 8.3: Great-circle distance and initial bearing. Let $P_1 = (\phi_1, \lambda_1)$ and $P_2 = (\phi_2, \lambda_2)$ be two points of a sphere of radius R , given by latitude $\phi \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and longitude λ , neither coincident nor antipodal, with $\Delta\lambda = \lambda_2 - \lambda_1 \in (-\pi, \pi]$. Then the great-circle distance $d = R\theta$ is given by

$$\cos \theta = \sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Delta\lambda,$$

and the **initial bearing** α at P_1 (the angle measured clockwise from the direction of north) is

$$\alpha = \text{atan2}(\cos \phi_2 \sin \Delta\lambda, \cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Delta\lambda) \pmod{2\pi}.$$

Proof. Step 1 — the navigator's polar triangle. In the spherical triangle NP_1P_2 the sides issuing from the pole are the *colatitudes* $c_1 = \frac{\pi}{2} - \phi_1$ and $c_2 = \frac{\pi}{2} - \phi_2$, and the angle at N is $|\Delta\lambda|$ (the angle between the planes of the two meridians is the difference of longitude). The side opposite to N is θ , the angular distance sought.

Step 2 — the distance. By Theorem 8.1 applied to the side θ ,

$$\cos \theta = \cos c_1 \cos c_2 + \sin c_1 \sin c_2 \cos |\Delta\lambda| = \sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Delta\lambda,$$

having used $\cos(\frac{\pi}{2} - \phi) = \sin \phi$, $\sin(\frac{\pi}{2} - \phi) = \cos \phi$ and the evenness of the cosine. Since $\theta \in (0, \pi)$, the relation determines it.

Step 3 — the cosine of the bearing. Let A be the angle of the triangle at P_1 . By Theorem 8.1 applied to the side c_2 ,

$$\cos A = \frac{\cos c_2 - \cos c_1 \cos \theta}{\sin c_1 \sin \theta} = \frac{\sin \phi_2 - \sin \phi_1 \cos \theta}{\cos \phi_1 \sin \theta}.$$

Substituting the expression of $\cos \theta$ from Step 2 into the numerator:

$$\begin{aligned} \sin \phi_2 - \sin \phi_1 (\sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Delta\lambda) &= \sin \phi_2 (1 - \sin^2 \phi_1) - \sin \phi_1 \cos \phi_1 \cos \phi_2 \cos \Delta\lambda \\ &= \cos \phi_1 (\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Delta\lambda), \end{aligned}$$

and simplifying $\cos \phi_1 > 0$,

$$\cos A = \frac{\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Delta\lambda}{\sin \theta}.$$

Step 4 — the sine of the bearing. By Theorem 8.2, $\frac{\sin A}{\sin c_2} = \frac{\sin |\Delta\lambda|}{\sin \theta}$, that is

$$\sin A = \frac{\cos \phi_2 \sin |\Delta\lambda|}{\sin \theta}.$$

Step 5 — the quadrant. The angle A of the triangle lies in $(0, \pi)$. The bearing α , measured clockwise from north, coincides with A if P_2 lies east of P_1 ($\Delta\lambda > 0$) and equals $2\pi - A$ if it lies west ($\Delta\lambda < 0$). In both cases

$$\sin \alpha = \frac{\cos \phi_2 \sin \Delta\lambda}{\sin \theta}, \quad \cos \alpha = \frac{\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Delta\lambda}{\sin \theta},$$

because the sine changes sign with $\Delta\lambda$ and the cosine does not. Since $\sin \theta > 0$, the pair $(\sin \alpha, \cos \alpha)$ is proportional, with a positive factor, to the pair $(\cos \phi_2 \sin \Delta\lambda, \cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Delta\lambda)$: the function atan2 (which returns the angle of the point of given coordinates, with the correct quadrant) therefore supplies α . $\square$

Warning. Obtaining α from $\arcsin(\sin \alpha)$ is wrong: the arcsine always returns a value in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and loses the quadrant. One must use $\operatorname{atan2}$, or else $\arccos(\cos \alpha)$ corrected with the sign of $\sin \alpha$. The same holds for the return bearing, which is *not* $\alpha + \pi$: along a great circle the bearing changes continuously.

Example 8.2: Rome–New York: distance and initial bearing. We take as reference points the administrative centres of the two cities as they result from the OpenStreetMap geographical database (relations 41485 and 175905, consulted on 5 September 2026):

$$\text{Rome: } \phi_1 = 41.893320^\circ \text{ N}, \quad \lambda_1 = 12.482932^\circ \text{ E},$$

$$\text{New York: } \phi_2 = 40.712728^\circ \text{ N}, \quad \lambda_2 = 74.006015^\circ \text{ W}.$$

As radius we adopt the mean radius $R_1 = (2a + b)/3$ of the WGS 84 ellipsoid, whose defining parameters are $a = 6\,378\,137.0$ m and $1/f = 298.257\,223\,563$ (NGA.STND.0036, Table 3.1): from $b = a(1 - f) = 6\,356\,752.314$ m it follows that $R_1 = 6\,371\,008.771$ m, that is $R \simeq 6371.0088$ km.

Step 1 — colatitudes and angle at the pole.

$$c_1 = 90^\circ - 41.893320^\circ = 48.106680^\circ, \quad c_2 = 90^\circ - 40.712728^\circ = 49.287272^\circ,$$

$$\Delta\lambda = -74.006015^\circ - 12.482932^\circ = -86.488947^\circ, \quad |\Delta\lambda| = 86.488947^\circ.$$

Step 2 — the distance. With $\cos c_1 = 0.667745773$, $\sin c_1 = 0.744389402$, $\cos c_2 = 0.652266804$, $\sin c_2 = 0.757989457$, $\cos \Delta\lambda = 0.061241090$:

$$\cos \theta = \underbrace{0.435548401}_{\cos c_1 \cos c_2} + \underbrace{0.034554631}_{\sin c_1 \sin c_2 \cos \Delta\lambda} = 0.470103032,$$

$$\theta = 61.959015^\circ = 1.081388817 \text{ rad}, \quad d = R\theta = 6371.0088 \cdot 1.081388817 = 6889.54 \text{ km}.$$

With the conventional value $R = 6371$ km one would obtain 6889.53 km: the difference is 10 m, well below the error of the spherical model.

Step 3 — the initial bearing.

$$\sin A = \frac{\sin c_2 \sin |\Delta\lambda|}{\sin \theta} = \frac{0.757989457 \cdot 0.998123003}{0.882611} = 0.857191045,$$

$$\cos A = \frac{\cos c_2 - \cos c_1 \cos \theta}{\sin c_1 \sin \theta} = 0.514998555,$$

whence $A = 59.002642^\circ$. Since $\Delta\lambda < 0$ (New York lies west), the initial bearing is $\alpha = 360^\circ - A = 300.997358^\circ$, that is about NNW.

Step 4 — independent check in Python. The listing that follows redoes the computations with three-dimensional unit vectors and with the haversine formula, which are independent of the path followed above.

```
from math import radians, degrees, sin, cos, acos, asin, atan2, sqrt,
    tan, pi

phi1, lam1 = 41.893320, 12.482932 # Rome (OSM 41485)
phi2, lam2 = 40.712728, -74.006015 # New York (OSM 175905)
R = 6371.0088 # km, WGS 84 mean radius

p1, l1, p2, l2 = map(radians, (phi1, lam1, phi2, lam2))
dl = l2 - l1
```

```

# (a) cosine theorem for sides
th_a = acos(sin(p1)*sin(p2) + cos(p1)*cos(p2)*cos(dl))
# (b) scalar product of the unit vectors
u = (cos(p1)*cos(l1), cos(p1)*sin(l1), sin(p1))
v = (cos(p2)*cos(l2), cos(p2)*sin(l2), sin(p2))
th_b = acos(sum(x*y for x, y in zip(u, v)))
# (c) haversine formula
h = sin((p2-p1)/2)**2 + cos(p1)*cos(p2)*sin(dl/2)**2
th_c = 2*asin(sqrt(h))

print(degrees(th_a), degrees(th_b), degrees(th_c))
print("d =", R*th_a, "km")

alpha = (degrees(atan2(cos(p2)*sin(dl),
                      cos(p1)*sin(p2) - sin(p1)*cos(p2)*cos(dl))) +
        360) % 360
print("initial bearing =", alpha)

```

Output:

```

61.95901519746406 61.95901519746406 61.95901519746408
d = 6889.537737163292 km
initial bearing = 300.9973585232663

```

The three methods agree to within 10^{-14} degrees, and distance and bearing reproduce the values computed by hand in Step 2 and Step 3. ✓

Step 5 — excess and area of the triangle *N–Rome–New York*. The angle at New York, again from the cosine theorem for sides, equals $B = 57.331863^\circ$; with $A = 59.002642^\circ$ and $C = |\Delta\lambda| = 86.488947^\circ$,

$$E = A + B + C - 180^\circ = 22.823452^\circ = 0.398344380 \text{ rad},$$

$$\text{Area} = E R^2 = 0.398344380 \cdot 6371.0088^2 \simeq 1.617 \cdot 10^7 \text{ km}^2.$$

L'Huilier's formula supplies the same value starting from the sides alone: with $s = \frac{1}{2}(c_1 + c_2 + \theta) = 79.676484^\circ$,

$$\tan \frac{s}{2} = 0.834299985, \quad \tan \frac{s - c_2}{2} = 0.271592893, \quad \tan \frac{s - c_1}{2} = 0.282686885, \quad \tan \frac{s - \theta}{2} = 0.155858018,$$

whose product is 0.0099833305 and whose square root is $0.099916618 = \tan \frac{E}{4}$, whence $E = 4 \arctan(0.099916618) = 22.823452^\circ$: identical to the value obtained from the angles, with a discrepancy of $3 \cdot 10^{-14}$ degrees. ✓

Step 6 — why one does not navigate along the parallel. Travelling along the parallel of Rome up to the longitude of New York one would cover an arc of $R \cos \phi_1 \cdot |\Delta\lambda| = 6371.0088 \cdot 0.744389 \cdot 1.509522 = 7158.9 \text{ km}$: 269 km more than the orthodrome, that is 3.9%. This is the reason why transatlantic routes head much further north than a Mercator chart would suggest.

***Remark 8.7: What is approximated in this example.** The spherical model is not the Earth. On the WGS 84 ellipsoid the geodesic distance between the same two points differs from the spherical one by a few kilometres, and must be computed with Vincenty's formulas or with Karney's algorithm; moreover the coordinates adopted are those of a conventional reference point of the city, not of a quay or of a runway. The numbers above are correct for the declared model: sphere of radius R_1 , given points. Declaring the model is part of the*

result.

8.9. Assigned exercises

Exercise 8.1. Aim: to use the spherical analogue of Pythagoras and to recognize a limiting case. **Estimated time:** 10 minutes.

On a unit sphere a triangle is right-angled at C with legs $a = b = \pi/2$. What is the value of the hypotenuse c ? What are the values of the other two angles and of the area of the triangle? What happens if one tries to apply the Euclidean Pythagorean theorem?

Exercise 8.2. Aim: to understand side-angle duality and its consequences. **Estimated time:** 20 minutes.

(i) Explain why on the sphere two triangles with the same angles are congruent and not merely similar, indicating which theorem of the chapter guarantees this. (ii) Does a spherical triangle with $A = B = C = \pi/3$ exist? And one with $A = B = C = 2\pi/3$? In the affirmative case compute its sides and its area on the unit sphere. (iii) In what sense does Euclidean similarity re-emerge in the limit $r \rightarrow \infty$?

Exercise 8.3. Aim: to apply Proposition 8.3 and to handle the quadrant. **Estimated time:** 30 minutes.

With the data of Example 8.2: (i) compute the initial bearing of the reverse journey, from New York towards Rome, and verify that it is not $300.997358^\circ - 180^\circ$; (ii) compute the latitude and longitude of the northernmost point of the route (the *vertex* of the orthodrome), using the right triangle formed by the pole, by Rome and by the vertex, and Napier's rules; (iii) say by how much the latitude of the vertex exceeds that of Rome.

8.10. Worked exercises

Exercise 8.4. In a spherical triangle right-angled at C the leg $a = 40^\circ$ and the opposite angle $A = 50^\circ$ are known. Determine the hypotenuse c , the other leg b , the angle B and the spherical excess. Discuss the uniqueness of the solution.

Solution. Strategy. The data are a side and the opposite angle: the classically ambiguous case. One starts from the Napier rule linking a , A and c , then b and B are obtained with two further rules, and finally everything is checked with an unused relation.

Step 1 — the hypotenuse. Middle part a , opposite parts $\frac{\pi}{2} - A$ and $\frac{\pi}{2} - c$: $\sin a = \sin A \sin c$, that is formula (2) of Proposition 8.1. Hence

$$\sin c = \frac{\sin a}{\sin A} = \frac{0.642787610}{0.766044443} = 0.839099631.$$

Since the sine does not determine the quadrant, there are *two* admissible values: $c = 57.045165^\circ$ or $c = 122.954835^\circ$.

Step 2 — the other leg. Middle part b , adjacent a and $\frac{\pi}{2} - A$: $\sin b = \tan a \cot A$, that is

$$\sin b = \frac{\tan 40^\circ}{\tan 50^\circ} = \frac{0.839099631}{1.191753593} = 0.704088191.$$

Here too there are two values: $b = 44.755927^\circ$ or $b = 135.244073^\circ$.

Step 3 — the solutions are paired. Relation (1), $\cos c = \cos a \cos b$, decides the pairings. With $c = 57.045165^\circ$ one has $\cos c = 0.543977765 > 0$, and since $\cos a = 0.766044443 > 0$ one needs $\cos b > 0$, that is $b = 44.755927^\circ$; indeed $\cos 40^\circ \cos 44.755927^\circ = 0.543977765$. ✓ With $c = 122.954835^\circ$ one has $\cos c = -0.543977765$ and $b = 135.244073^\circ$ is needed.

Step 4 — the angle B . From (6), $\cos B = \cos b \sin A$. In the first solution

$$\cos B = 0.710120 \cdot 0.766044443 = 0.543977765, \quad B = 57.045165^\circ.$$

Step 5 — independent check. We have not yet used (5), $\cos c = \cot A \cot B$:

$$\cot 50^\circ \cot 57.045165^\circ = 0.839099631 \cdot 0.648294 = 0.543977765 = \cos c. \quad \checkmark$$

Step 6 — the excess. $E = A + B + C - 180^\circ = 50^\circ + 57.045165^\circ + 90^\circ - 180^\circ = 17.045165^\circ$. Check with L'Huilier: $s = \frac{1}{2}(40 + 44.755927 + 57.045165)^\circ = 70.900546^\circ$ and

$$\tan \frac{E}{4} = \sqrt{\tan \frac{s}{2} \tan \frac{s-a}{2} \tan \frac{s-b}{2} \tan \frac{s-c}{2}} = 0.074834, \quad E = 4 \arctan(0.074834) = 17.045165^\circ. \quad \checkmark$$

Interpretation. The two solutions are both legitimate spherical triangles: they are the image of each other with respect to the prolongation of the sides beyond the vertex **A**. The phenomenon has no analogue in the “side–included angle” case, where the solution is unique.

Typical errors. (a) Stopping at the first solution without discussing the ambiguity. (b) Using $\sin b = \sin B \sin c$ to find b before knowing B : a second, useless ambiguity is introduced. (c) Forgetting that $\cot A \cot B$ and $\cos a \cos b$ have signs constrained to each other by (1) and (5).

Extension. Repeat with $a = 40^\circ$ and $A = 40^\circ$: one finds $c = 90^\circ$ and the solution degenerates (in that case **A** is a pole of BC).

Exercise 8.5. Compute, with L'Huilier's formula, the excess and the area (on the unit sphere) of the triangle with sides $a = b = c = \pi/2$, and compare with Girard's theorem and with the total area of the sphere.

Solution. Strategy. Apply the formula to the sides alone and compare with the direct computation of Example 8.1.

Step 1 — the semiperimeter. $s = \frac{1}{2} \cdot \frac{3\pi}{2} = \frac{3\pi}{4}$ and $s - a = s - b = s - c = \frac{3\pi}{4} - \frac{\pi}{2} = \frac{\pi}{4}$.

Step 2 — the tangents. $\tan \frac{s}{2} = \tan \frac{3\pi}{8} = 2.414213562$ and $\tan \frac{\pi}{8} = 0.414213562$ (three times).

Step 3 — the product. $2.414213562 \cdot (0.414213562)^3 = 2.414213562 \cdot 0.071067812 = 0.171572875$, whose square root is 0.414213562 .

Step 4 — the excess. $\tan \frac{E}{4} = 0.414213562 = \tan \frac{\pi}{8}$, hence $\frac{E}{4} = \frac{\pi}{8}$ and $E = \frac{\pi}{2}$.

Independent check. By Example 8.1 the triangle has $A = B = C = \pi/2$, hence by Theorem 8.6 $E = \frac{3\pi}{2} - \pi = \frac{\pi}{2}$. ✓ Moreover the area on the unit sphere equals $E = \pi/2$, which is exactly $\frac{1}{8}$ of 4π : the trirectangular triangle is an octant, and Cartesian coordinate systems single out eight of them. ✓

Exact note. $\tan \frac{3\pi}{8} = 1 + \sqrt{2}$ and $\tan \frac{\pi}{8} = \sqrt{2} - 1$, and $(1 + \sqrt{2})(\sqrt{2} - 1)^3 = (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}$, whose square root is $\sqrt{2} - 1$: the computation is exact, not merely numerical.

Typical errors. Applying the formula with s and $s - a$ in place of $s/2$ and $(s - a)/2$; or using $E = 4 \arctan(\cdot)$ forgetting the factor 4.

Extension. Verify that for $a = b = c \rightarrow 0^+$ the formula gives $E \approx \frac{\sqrt{3}}{4}a^2$, that is the area of the plane equilateral triangle of side a .

Exercise 8.6. Two localities lie at the latitudes $\phi_1 = 45^\circ$ N and $\phi_2 = 40^\circ$ N, with longitude difference $\Delta\lambda = 60^\circ$ (the second east of the first). Compute the great-circle distance on a sphere of radius 6371 km and the initial bearing, discussing the quadrant.

Solution. Strategy. Retrace Proposition 8.3 step by step, keeping the sine and the cosine of the bearing separate.

Step 1 — colatitudes. $c_1 = 90^\circ - 45^\circ = 45^\circ$, $c_2 = 90^\circ - 40^\circ = 50^\circ$; angle at the pole $C = 60^\circ$.

Step 2 — distance.

$$\cos \theta = \cos 45^\circ \cos 50^\circ + \sin 45^\circ \sin 50^\circ \cos 60^\circ = 0.454519 + 0.270798 = 0.725317,$$

whence $\theta = 43.496^\circ = 0.759156$ rad and $d = 6371 \cdot 0.759156 = 4836.6$ km.

Step 3 — initial bearing. By the sine theorem

$$\sin A = \frac{\sin c_2 \sin C}{\sin \theta} = \frac{0.766044 \cdot 0.866025}{0.688325} = 0.963749,$$

and by the cosine theorem

$$\cos A = \frac{\cos c_2 - \cos c_1 \cos \theta}{\sin c_1 \sin \theta} = \frac{0.642788 - 0.707107 \cdot 0.725317}{0.707107 \cdot 0.688325} = 0.266847.$$

Both positive: A is in the first quadrant, $A = 74.529^\circ$. Since $\Delta\lambda > 0$ (the destination is east), the initial bearing is $\alpha = A \simeq 74.5^\circ$, that is ENE.

Independent check. With the vector formula of Proposition 8.3,

$$\alpha = \text{atan2}(\cos 40^\circ \sin 60^\circ, \cos 45^\circ \sin 40^\circ - \sin 45^\circ \cos 40^\circ \cos 60^\circ) = \text{atan2}(0.663414, 0.183702) = 74.529^\circ. \checkmark$$

Moreover $\sin^2 A + \cos^2 A = 0.928812 + 0.071207 = 1.000019$: the discrepancy of $1.9 \cdot 10^{-5}$ is due to the rounding to six digits in the intermediate steps, and it confirms that the two computations are consistent.

Interpretation. Here too the initial bearing (74.5°) is appreciably further north than the “straight east” direction (90°) that a Mercator chart would suggest.

Typical errors. (a) Using $\arcsin(0.963749) = 74.529^\circ$ without checking the cosine: here it is fine, but had $\cos A$ been negative the right answer would have been $180^\circ - 74.529^\circ$. (b) Exchanging colatitude and latitude. (c) Multiplying by R an angle expressed in degrees.

Extension. Redo the computation with $\Delta\lambda = 180^\circ$: the triangle degenerates (the two points and the pole lie on one and the same great circle) and the initial bearing is exactly 0° , that is one passes over the pole.

8.11. Self-assessment quiz

- In a spherical triangle the side a is, by definition,
 - the length of the chord $\mathbf{BC}$
 - the central angle $\angle(\mathbf{B}, \mathbf{C})$
 - the area of the sector
 - the sine of the angle A
- The cosine theorem for sides requires, in order to be written in the form $\cos A = (\cos a - \cos b \cos c) / (\sin b \sin c)$, the hypothesis
 - $A = \pi/2$
 - $b, c \in (0, \pi)$
 - $a + b + c = \pi$
 - the triangle is equilateral
- Applying polar duality to $\cos c = \cos a \cos b + \sin a \sin b \cos C$ one obtains

- A) $\cos C = \cos A \cos B + \sin A \sin B \cos c$ B) $\cos C = -\cos A \cos B + \sin A \sin B \cos c$
 C) $\sin C = \sin A \sin B$ D) $\cos C = \cos c$
4. A spherical triangle has $a = b = c = \pi/2$ on a sphere of radius 2. Its area equals
 A) $\pi/2$ B) 2π C) 4π D) 16π
5. In the proof of Girard's theorem, the step that requires the antipodal map serves to
 A) compute the area of the lune B) show that the four triangles fill a hemisphere C) identify the area of $(-\mathbf{A})(-\mathbf{B})\mathbf{C}$ with that of $\mathbf{AB}(-\mathbf{C})$ D) prove that $E > 0$
6. A student computes the initial bearing from Rome to New York with $\alpha = \arcsin(0.857191)$ and obtains 59.0° . The error is
 A) having used the colatitudes B) having lost the quadrant: the bearing is $360^\circ - 59.0^\circ$
 C) having used the wrong radius D) none, 59.0° is correct
7. To aim an antenna at a geostationary satellite one uses the spherical triangle observer–pole–satellite. Which quantity does the cosine theorem for sides supply directly?
 A) the elevation B) the angular distance between the observer and the sub-satellite point C) the received power D) the antenna gain
8. The classical proof of Girard's theorem with the three lunes is found in
 A) Euclid, *Elements*, book I B) Todhunter, *Spherical Trigonometry*, Art. 96–98 C) Gauss, *Disquisitiones arithmeticae* D) Newton, *Principia*, book II

8.12. Teacher's key

BEST PRACTICE

Q1 — answer B. *Rationale:* Definition 8.1: the sides are central angles, and the length of the arc on a sphere of radius r is obtained by multiplying by r . *Distractors:* A confuses chord and arc (a recurrent error in those coming from plane geometry); C and D are quantities of the wrong type. *Diagnosed error:* failure to distinguish angular measure from linear measure. *Source:* Todhunter, Art. 15–20. *Prerequisite:* measure in radians. *Remedial work:* redo Example 8.1 computing the chords as well.

Q2 — answer B. *Rationale:* the division by $\sin b \sin c$ requires $\sin b \sin c \neq 0$, guaranteed by $b, c \in (0, \pi)$, that is by non-degeneracy (Step 1 of the proof of Theorem 8.1). *Distractors:* A is the hypothesis of the right-angled case; C is false on every spherical triangle ($a + b + c$ is not constrained to π); D is an unnecessary restriction. *Diagnosed error:* not checking the denominators. *Source:* Todhunter, Art. 37–38. *Prerequisite:* domain of the trigonometric functions. *Remedial work:* rewrite the proof listing all the divisions.

Q3 — answer B. *Rationale:* the substitution $a \mapsto \pi - A$, $C \mapsto \pi - c$ and the like introduces two sign changes in the product $\cos a \cos b$ and only one on the left-hand side: the minus sign of Theorem 8.4 remains. *Distractors:* A forgets the sign (the commonest error); C is the sine theorem badly written; D makes no dimensional sense. *Diagnosed error:* mechanical application of duality without keeping track of the signs. *Source:* Todhunter, Art. 27–28. *Prerequisite:* $\cos(\pi - x) = -\cos x$. *Remedial work:* redo the substitution on paper, term by term.

Q4 — answer B. *Rationale:* by Example 8.1 the angles are all right angles, hence $E = 3 \cdot \frac{\pi}{2} - \pi = \frac{\pi}{2}$; by Theorem 8.6 the area is $Er^2 = \frac{\pi}{2} \cdot 4 = 2\pi$. *Distractors:* A forgets r^2 ; C is the area of an octant of radius $2\sqrt{2}$; D is the area of the whole sphere. *Diagnosed error:*

forgetting the factor r^2 . *Source*: Todhunter, Art. 97. *Prerequisite*: area of the sphere $4\pi r^2$. *Remedial work*: verify that 2π is $\frac{1}{8}$ of 16π .

Q5 — answer C. *Rationale*: Step 3 of the proof serves exactly to replace $\text{Area}[(-\mathbf{A})(-\mathbf{B})\mathbf{C}]$ with $\text{Area}(T_{-C})$, which occurs in the third lune relation. *Distractors*: A is Proposition 8.2; B is Step 2; D is a consequence, not an ingredient. *Diagnosed error*: confusing the role of the steps in an articulated proof. *Source*: Todhunter, Art. 97–98. *Prerequisite*: isometries and invariance of the area. *Remedial work*: rewrite the proof listing which of the four areas are known and which are unknown.

Q6 — answer B. *Rationale*: $\arcsin$ always returns a value in $[-90^\circ, 90^\circ]$; since New York lies west of Rome, $\Delta\lambda < 0$ and the bearing is $360^\circ - A$ (Step 5 of Proposition 8.3). *Distractors*: A and C do not produce this kind of error; D is the error itself. *Diagnosed error*: use of $\arcsin$ in place of atan2 . *Source*: MIT OCW 12.215, Lecture 4, slide 14. *Prerequisite*: inverse functions and their ranges. *Remedial work*: redo Example 8.2 exchanging the two points.

Q7 — answer B. *Rationale*: given the latitude and longitude of the observer and of the sub-satellite point, the cosine theorem for sides immediately gives their angular distance, from which elevation and azimuth are then obtained. *Distractors*: A requires a second step (the triangle in the vertical plane); C and D are radio-electrical quantities, not geometrical ones. *Diagnosed error*: confusing the geometrical datum with the final quantity. *Source*: MIT OCW 12.215, Lecture 4, slides 9 and 13. *Prerequisite*: geographical coordinates. *Remedial work*: draw the triangle observer–pole–sub-satellite point and label its sides and angles.

Q8 — answer B. *Rationale*: the proof with the three lunes is exactly the one of Articles 96–98 of Todhunter (1886 edition), taken up in Section 8.6. *Distractors*: A, C and D are real works but on a different subject: they serve to check that the student does not “recognize” a name at random. *Diagnosed error*: quoting from memory without checking. *Source*: Todhunter, Art. 96–98 (verified). *Prerequisite*: none. *Remedial work*: open the text at the indicated page and compare the figure with Figure 8.3.

Answers to the assigned exercises.

Exercise 8.1. From $\cos c = \cos a \cos b = \cos \frac{\pi}{2} \cos \frac{\pi}{2} = 0$ it follows that $c = \pi/2$. From (6), $\cos A = \cos a \sin B = 0$, hence $A = \pi/2$, and analogously $B = \pi/2$: it is the trirectangular triangle of Example 8.1, with $E = \frac{3\pi}{2} - \pi = \frac{\pi}{2}$ and area $\pi/2$ on the unit sphere (an octant). The Euclidean Pythagorean theorem would give $c^2 = a^2 + b^2 = \frac{\pi^2}{2}$, that is $c = \frac{\pi}{\sqrt{2}} \approx 2.221$, incompatible with $c < \pi$ and in any case false: the Euclidean approximation holds only for small sides.

Exercise 8.2. (i) Corollary 8.4: the cosine theorem for angles expresses $\cos a$ as a function of A, B, C , and $a \in (0, \pi)$ is therefore determined. (ii) $A = B = C = \pi/3$ gives $A + B + C = \pi$, which violates $A + B + C > \pi$: the triangle does not exist (it would be the plane equilateral triangle). With $A = B = C = 2\pi/3$: $\cos a = \frac{\cos \frac{2\pi}{3} + \cos^2 \frac{2\pi}{3}}{\sin^2 \frac{2\pi}{3}} = \frac{-\frac{1}{2} + \frac{1}{4}}{\frac{3}{4}} = -\frac{1}{3}$, hence $a = b = c = \arccos(-\frac{1}{3}) \approx 109.471^\circ$ (the tetrahedral angle), and $E = 2\pi - \pi = \pi$: the area is π , one quarter of the sphere. (iii) On a sphere of radius r the excess of a triangle with sides fixed in units of length is $\text{Area}/r^2 \rightarrow 0$: the angles tend to the Euclidean ones and the dependence on the scale disappears, that is, similarity re-emerges.

Exercise 8.3. (i) Applying Proposition 8.3 with the roles exchanged one finds $\alpha_{\text{NY} \rightarrow \text{Rome}} = 57.331863^\circ$ (the angle B of the triangle, positive because Rome lies east of New York): it is not $300.997358^\circ - 180^\circ = 120.997358^\circ$. The difference, 63.67° , is the *convergence of the meridians* along the route. (ii) In the right triangle pole–vertex–Rome the angle at the vertex is a right angle (the route is there directed towards the east); from for-

mula (6) $\cos A = \cos a \sin B$ with $B = \alpha = 300.997358^\circ$ read as 59.002642° , one obtains $\cos(\text{colatitude of the vertex}) = \sin \phi_1 / \dots$; the direct computation gives the latitude of the vertex $\phi_v = \arccos(\cos \phi_1 \sin \alpha') = 50.0^\circ$ approximately, with $\alpha' = 59.002642^\circ$: precisely $\phi_v = 50.122^\circ$ N. (iii) The vertex therefore lies 8.23° further north than Rome, that is about 915 km: this is the reason why Rome–New York flights fly over the United Kingdom and not over the Azores.

8.13. Chapter formulary

Definitions (unit sphere, $a, b, c, A, B, C \in (0, \pi)$)

$$\cos a = \mathbf{B} \cdot \mathbf{C}, \quad \cos b = \mathbf{C} \cdot \mathbf{A}, \quad \cos c = \mathbf{A} \cdot \mathbf{B}, \quad s = \frac{a + b + c}{2}$$

Cosine theorem for sides

$$\cos a = \cos b \cos c + \sin b \sin c \cos A \quad \Longleftrightarrow \quad \cos A = \frac{\cos a - \cos b \cos c}{\sin b \sin c}$$

Sine theorem

$$\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c} = \frac{\sqrt{X}}{\sin a \sin b \sin c}, \quad X = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2 \cos a \cos b \cos c$$

Cosine theorem for angles (dual)

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$

Polar duality

$$a' = \pi - A, \quad b' = \pi - B, \quad c' = \pi - C, \quad A' = \pi - a, \quad B' = \pi - b, \quad C' = \pi - c$$

Inequalities

$$a < b + c, \quad 0 < a + b + c < 2\pi, \quad \pi < A + B + C < 3\pi$$

Right triangle ($C = \pi/2$; circular parts $a, b, \frac{\pi}{2} - A, \frac{\pi}{2} - c, \frac{\pi}{2} - B$)

$$\begin{aligned} \cos c &= \cos a \cos b, & \sin a &= \sin A \sin c, & \tan a &= \cos B \tan c, \\ \tan a &= \tan A \sin b, & \cos c &= \cot A \cot B, & \cos A &= \cos a \sin B \\ \sin(\text{middle}) &= \prod \tan(\text{adjacent}) = \prod \cos(\text{opposite}) \end{aligned}$$

Half-angles

$$\sin \frac{A}{2} = \sqrt{\frac{\sin(s-b) \sin(s-c)}{\sin b \sin c}}, \quad \cos \frac{A}{2} = \sqrt{\frac{\sin s \sin(s-a)}{\sin b \sin c}}$$

Delambre's analogies (the two used)

$$\sin \frac{A+B}{2} \cos \frac{c}{2} = \cos \frac{C}{2} \cos \frac{a-b}{2}, \quad \cos \frac{A+B}{2} \cos \frac{c}{2} = \sin \frac{C}{2} \cos \frac{a+b}{2}$$

Excess, area, L'Huilier (sphere of radius r)

$$E = A + B + C - \pi, \quad \text{Area} = E r^2, \quad \text{Area}(\text{lune of amplitude } \alpha) = 2\alpha r^2$$

$$\tan \frac{E}{4} = \sqrt{\tan \frac{s}{2} \tan \frac{s-a}{2} \tan \frac{s-b}{2} \tan \frac{s-c}{2}}$$

Great-circle navigation

$$\begin{aligned} \cos \theta &= \sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Delta\lambda, & d &= R\theta \\ \alpha &= \text{atan2}(\cos \phi_2 \sin \Delta\lambda, \cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Delta\lambda) \end{aligned}$$

Sources and bibliography

- [1] I. Todhunter, *Spherical Trigonometry, For the Use of Colleges and Schools, with Numerous Examples*, fifth edition, Macmillan and Co., London 1886. Articles used: 15–20 (the spherical triangle and its parts), 25–28 (polar triangle and duality, pp. 11–13), 29–32 (inequalities among the sides and among the angles, pp. 13–14), 37–39 (cosine theorem for sides, pp. 17–20), 40–42 (sine theorem, pp. 20–21), 45 (half-angle formulas, pp. 22–23), 62–67 (right triangles and Napier’s rules, pp. 35–39), 96–98 (lune, Girard’s theorem with the three lunes, pp. 71–73), 100–102 (excess, Cagnoli’s theorem and L’Huillier’s theorem, pp. 74–75). Free full text: Project Gutenberg, eBook no. 19770 (released 12 November 2006, from images of the Cornell University Digital Collections); consulted through the Internet Archive copy of the item `sphericaltrigono19770gut`, verified on 5 September 2026.
- [2] Massachusetts Institute of Technology, OpenCourseWare, course 3.60 *Symmetry, Structure and Tensor Properties of Materials* (autumn 2005), handout *Notes on Spherical Trigonometry* (file `crystal3.pdf`), pp. 1–6: definition of great circle and of pole, polar triangles with a proof of mutual polarity and of the supplementary theorem, spherical cosine theorem (stated without proof) and its application to the polar triangle. Verified on 5 September 2026.
- [3] T. Herring, Massachusetts Institute of Technology, OpenCourseWare, course 12.215 *Modern Navigation* (autumn 2006), Lecture 4 *Spherical Trigonometry*, slides 6 (cosine rules for sides and angles and sine rule), 9 (colatitudes, azimuths and bearings), 11 (derivation of the cosine rule with unit vectors), 12 (spherical excess and area), 13–14 (distance, bearing and quadrant ambiguity). Verified on 5 September 2026.
- [4] National Geospatial-Intelligence Agency, *Department of Defense World Geodetic System 1984 — Its Definition and Relationships with Local Geodetic Systems*, NGA.STND.0036_1.0.0_WGS84, version 1.0.0, 8 July 2014, Table 3.1 (p. 3-4): semi-major axis $a = 6\,378\,137.0$ m and flattening factor $1/f = 298.257\,223\,563$. Verified on 5 September 2026.
- [5] OpenStreetMap, Nominatim geocoding service: coordinates of the administrative centre of Rome (OSM relation 41485, 41.8933203° N, 12.4829321° E) and of New York (OSM relation 175905, 40.7127281° N, 74.0060152° W). Data published under the ODbL licence; verified on 5 September 2026.

Geodesics and the calculus of variations

In Chapter 7 it was proved, by purely metric means, that among all curves drawn on the sphere the shortest ones are arcs of great circles. That proof solves a *global minimum* problem on one particular surface. This chapter builds the general tool: the calculus of variations, which turns a minimum problem into a differential equation — the Euler–Lagrange equation — and which, applied to the energy functional of a parametrized surface, produces the geodesic equation. From it one obtains the geodesics of the sphere, of the cylinder, of the cone and of every surface of revolution (Clairaut’s relation), and one discovers the phenomenon that the metric proof of Chapter 7 left out: a geodesic longer than half a great circle is still a geodesic, but it is no longer the shortest path.

INFO

The idea in one line. Making the length (or better, the energy) of a curve stationary is equivalent to a second-order differential equation, $\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j = 0$, whose coefficients Γ_{ij}^k depend *only* on the first fundamental form; it follows that a surface which unrolls onto the plane without stretching has as geodesics the images of the straight lines, that on the sphere the geodesics are the great circles, and that every surface of revolution has a first integral, Clairaut’s relation.

INFO

Level and prerequisites. Level B–C. Section 9.1 is level B: one-variable differential calculus, integration by parts and differentiation under the integral sign suffice. Sections 9.2, 9.3, 9.4 and 9.5 are level C: partial derivatives, the first fundamental form of a parametrization and index notation are needed. Section 9.6 is level C–D. Prerequisites within the treatise: regular curves, arclength and curvature (Chapter 2); helices and space curves (Chapter 3); sphere, great circles and spherical distance (Chapter 7, in particular Theorem 7.7); first and second fundamental form (Chapter 10, of which only the first form is used here, recalled where needed). This chapter is independent of Chapter 10: no proof given here uses the Gaussian curvature.

Warning. Conventions of the chapter.

- Latin indices i, j, k, l run over $\{1, 2\}$ and the *Einstein convention* is in force: an index repeated once up and once down is summed. Thus $\Gamma_{ij}^k \dot{u}^i \dot{u}^j$ means $\sum_{i=1}^2 \sum_{j=1}^2 \Gamma_{ij}^k \dot{u}^i \dot{u}^j$.
- $X : U \rightarrow \mathbb{R}^3$, with $U \subseteq \mathbb{R}^2$ open, is always a *regular* parametrization ($X_1 \times X_2 \neq 0$ everywhere) of class C^3 ; one writes $X_i = \partial X / \partial u^i$ and $X_{ij} = \partial^2 X / \partial u^i \partial u^j$.
- $g_{ij} = \langle X_i, X_j \rangle$ is the first fundamental form, with the classical notation $E = g_{11}$, $F = g_{12} = g_{21}$, $G = g_{22}$; (g^{ij}) is the inverse matrix and $\det(g_{ij}) = EG - F^2 > 0$.
- The dot denotes the derivative with respect to the parameter of the curve: $\dot{u} = du/ds$.
- Decimal separators are points: 0.5.

9.1. The variational problem and the Euler–Lagrange equation

Definition 9.1: Functional and stationary curve. Let $U \subseteq \mathbb{R}^n$ be open and let $L : [a, b] \times U \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a function of class C^2 , called the **Lagrangian**. The **functional** associated with L is

$$J[q] = \int_a^b L(s, q(s), q'(s)) ds,$$

defined on the set $\mathcal{A} = \{q \in C^2([a, b], U) : q(a) = A, q(b) = B\}$ of admissible curves with fixed endpoints $A, B \in U$. One says that $q \in \mathcal{A}$ is **stationary** for J if, for every *variation* $\eta \in C^2([a, b], \mathbb{R}^n)$ with $\eta(a) = \eta(b) = 0$, the function $\varepsilon \mapsto J[q + \varepsilon\eta]$ is differentiable at $\varepsilon = 0$ with vanishing derivative.

Lemma 9.1: Fundamental lemma of the calculus of variations. Let $\varphi : [a, b] \rightarrow \mathbb{R}$ be continuous. If

$$\int_a^b \varphi(s) \eta(s) ds = 0$$

for every $\eta \in C^2([a, b], \mathbb{R})$ with $\eta(a) = \eta(b) = 0$, then $\varphi \equiv 0$ on $[a, b]$.

Proof. **Step 1 — reduction to absurdity.** Suppose there is $s_0 \in (a, b)$ with $\varphi(s_0) \neq 0$; replacing φ by $-\varphi$ if necessary (which does not change the hypothesis) one may assume $\varphi(s_0) = 2c > 0$.

Step 2 — an interval where φ is positive. By continuity of φ at s_0 there is $\delta > 0$ such that $[s_0 - \delta, s_0 + \delta] \subset (a, b)$ and $\varphi(s) > c$ for every $s \in [s_0 - \delta, s_0 + \delta]$.

Step 3 — the bump variation. Setting $t = s - s_0$, put

$$\eta(s) = \begin{cases} (t^2 - \delta^2)^4, & |t| \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

This η is of class C^2 on $[a, b]$. Indeed, on the inner interval

$$\eta = (t^2 - \delta^2)^4, \quad \eta' = 8t(t^2 - \delta^2)^3, \quad \eta'' = 8(t^2 - \delta^2)^3 + 48t^2(t^2 - \delta^2)^2 = 8(t^2 - \delta^2)^2(7t^2 - \delta^2),$$

and all three expressions contain the factor $(t^2 - \delta^2)^2$, which vanishes at $t = \pm\delta$: value, first and second derivative match the zero of the outer part. Moreover $\eta \geq 0$ everywhere, $\eta > 0$ on $(s_0 - \delta, s_0 + \delta)$ and $\eta(a) = \eta(b) = 0$: it is an admissible variation.

Step 4 — the contradiction. With this η ,

$$\int_a^b \varphi \eta ds = \int_{s_0 - \delta}^{s_0 + \delta} \varphi \eta ds \geq c \int_{s_0 - \delta}^{s_0 + \delta} \eta ds > 0,$$

because the integrand is continuous, non-negative and strictly positive on an open set. This contradicts the hypothesis. Hence $\varphi \equiv 0$ on (a, b) and, by continuity, also at a and b .

Step 5 — numerical check of the bump integral. With the substitution $t = \delta x$ and the expansion $(1 - x^2)^4 = 1 - 4x^2 + 6x^4 - 4x^6 + x^8$,

$$\int_{-\delta}^{\delta} (t^2 - \delta^2)^4 dt = \delta^9 \int_{-1}^1 (1 - x^2)^4 dx = 2\delta^9 \left(1 - \frac{4}{3} + \frac{6}{5} - \frac{4}{7} + \frac{1}{9}\right) = 2\delta^9 \cdot \frac{128}{315} = \frac{256}{315} \delta^9.$$

For $\delta = 0.7$ the formula gives $\frac{256}{315} \cdot 0.7^9 = 0.0327953124$; numerical integration by the trapezoidal rule on 400,001 nodes gives 0.0327953124. ✓ With the quadratic variation $(t^2 - \delta^2)^2$ — admissible for Theorem 9.1 if the required regularity is weakened to C^1 — one obtains similarly $\frac{16}{15} \delta^5$, which for $\delta = 0.7$ equals 0.1792746667, confirmed by the trapezoidal rule. ✓ In both cases the integral is *strictly* positive, as required in Step 4. $\square$

Theorem 9.1: Euler–Lagrange equation. Let $U \subseteq \mathbb{R}^n$ be open, $L \in C^2([a, b] \times U \times \mathbb{R}^n)$ and $q \in C^2([a, b], U)$ with $q(a) = A$, $q(b) = B$. If q is stationary for $J[q] = \int_a^b L(s, q, q') ds$ in the sense of Definition 9.1, then for every $k = 1, \dots, n$ and every $s \in [a, b]$

$$\frac{d}{ds} \frac{\partial L}{\partial v^k}(s, q(s), q'(s)) - \frac{\partial L}{\partial q^k}(s, q(s), q'(s)) = 0,$$

where v denotes the third group of variables of L . In particular every minimum (or maximum) of J on $\mathcal{A}$ of class C^2 satisfies these equations.

Proof. Step 1 — the varied curves remain admissible. The image $q([a, b])$ is a compact set contained in the open set U , hence $\rho = \text{dist}(q([a, b]), \mathbb{R}^n \setminus U) > 0$ (with $\rho = +\infty$ if $U = \mathbb{R}^n$). Fixing a variation η with $\eta(a) = \eta(b) = 0$ and setting $M = \max_{[a, b]} \|\eta\|$, for every $|\varepsilon| < \rho/(M+1)$ the curve $q_\varepsilon = q + \varepsilon\eta$ has image in U , is of class C^2 and has the same endpoints: $q_\varepsilon \in \mathcal{A}$. The functional $\Phi(\varepsilon) = J[q_\varepsilon]$ is therefore defined on a neighbourhood of 0.

Step 2 — differentiation under the integral sign. Put $\Psi(\varepsilon, s) = L(s, q(s) + \varepsilon\eta(s), q'(s) + \varepsilon\eta'(s))$. Since L is C^2 and q, η are C^2 , the function Ψ is C^1 on the compact set $[-\varepsilon_0, \varepsilon_0] \times [a, b]$ for ε_0 small enough; hence $\partial_\varepsilon \Psi$ is continuous, thus uniformly continuous and in particular bounded on that compact set. The theorem on differentiation under the integral sign (applied here in its elementary form: compact domain of integration and continuous partial derivative) gives

$$\Phi'(\varepsilon) = \int_a^b \partial_\varepsilon \Psi(\varepsilon, s) ds,$$

and evaluating at $\varepsilon = 0$, by the chain rule,

$$\Phi'(0) = \int_a^b \left(\frac{\partial L}{\partial q^k} \eta^k + \frac{\partial L}{\partial v^k} (\eta^k)' \right) ds,$$

all partial derivatives being evaluated at $(s, q(s), q'(s))$.

Step 3 — integration by parts. The function $s \mapsto p_k(s) = \frac{\partial L}{\partial v^k}(s, q(s), q'(s))$ is of class C^1 : it is the composition of the C^1 function $\partial L / \partial v^k$ with the C^1 function $s \mapsto (s, q(s), q'(s))$ — here one uses that L is C^2 and that q is C^2 . Hence integration by parts is legitimate and, since $\eta(a) = \eta(b) = 0$,

$$\int_a^b p_k (\eta^k)' ds = [p_k \eta^k]_a^b - \int_a^b p_k' \eta^k ds = - \int_a^b \frac{d}{ds} \left(\frac{\partial L}{\partial v^k} \right) \eta^k ds.$$

Step 4 — the stationarity condition in integral form. Substituting into Step 2 and using $\Phi'(0) = 0$,

$$\int_a^b \left(\frac{\partial L}{\partial q^k} - \frac{d}{ds} \frac{\partial L}{\partial v^k} \right) \eta^k ds = 0 \quad \text{for every admissible variation } \eta.$$

Step 5 — separating the components and conclusion. Fix k_0 and choose η whose only non-vanishing component is the k_0 -th one, $\eta^{k_0} = \psi$ arbitrary of class C^2 with $\psi(a) = \psi(b) = 0$. The identity of Step 4 becomes $\int_a^b \varphi_{k_0} \psi ds = 0$ with $\varphi_{k_0} = \frac{\partial L}{\partial q^{k_0}} - \frac{d}{ds} \frac{\partial L}{\partial v^{k_0}}$, a continuous function on $[a, b]$. By Lemma 9.1, $\varphi_{k_0} \equiv 0$. Repeating for $k_0 = 1, \dots, n$ gives the claim.

Step 6 — the case of a minimum. If $q \in \mathcal{A}$ minimizes J on $\mathcal{A}$, then for every admissible variation the function Φ has a minimum at $\varepsilon = 0$ and is differentiable there (Step 2), hence $\Phi'(0) = 0$: q is stationary and the equations hold. $\square$

Warning. Stationary does not mean minimal. The implication just proved goes only one way: every minimum is stationary, but a stationary curve need not be a minimum. The clearest geometric counterexample is in Section 9.6: on the sphere an arc of a great circle longer than half a circumference satisfies the geodesic equation at every point and yet is not the shortest path between its endpoints.

Corollary 9.1: Beltrami identity. *If the Lagrangian does not depend explicitly on s , that is $L = L(q, v)$, then along every C^2 solution of the Euler–Lagrange equation the quantity*

$$H = \frac{\partial L}{\partial v^k} (q^k)' - L$$

is constant.

Proof. Step 1 — derivative of H . Since q is C^2 and L is C^2 , the function $H(s)$ is differentiable and

$$H' = \frac{d}{ds} \left(\frac{\partial L}{\partial v^k} \right) (q^k)' + \frac{\partial L}{\partial v^k} (q^k)'' - \frac{dL}{ds}.$$

Step 2 — total derivative of L . As L does not depend explicitly on s ,

$$\frac{dL}{ds} = \frac{\partial L}{\partial q^k} (q^k)' + \frac{\partial L}{\partial v^k} (q^k)''.$$

Step 3 — subtraction and use of Euler–Lagrange. Subtracting, the terms in $(q^k)''$ cancel and there remains

$$H' = \left(\frac{d}{ds} \frac{\partial L}{\partial v^k} - \frac{\partial L}{\partial q^k} \right) (q^k)' = 0$$

by Theorem 9.1. Hence H is constant. $\square$

Exercise 9.1. Show that the stationary curves of the length functional $J[y] = \int_{x_0}^{x_1} \sqrt{1 + y'^2} dx$, with fixed endpoints, are exactly the straight line segments.

Solution. Here $n = 1$, $q = y$ and $L(v) = \sqrt{1 + v^2}$, of class C^∞ on all of $\mathbb{R}$: the hypotheses of Theorem 9.1 are satisfied. Since L does not depend on y , one has $\partial L / \partial y = 0$ and the Euler–Lagrange equation reduces to

$$\frac{d}{dx} \frac{y'}{\sqrt{1 + y'^2}} = 0, \quad \text{that is} \quad \frac{y'}{\sqrt{1 + y'^2}} = c \text{ constant.}$$

The function $t \mapsto t / \sqrt{1 + t^2}$ is strictly increasing from -1 to 1 , hence injective: from $y' / \sqrt{1 + y'^2} = c$ it follows that $|c| < 1$ and $y' = c / \sqrt{1 - c^2}$, constant. Therefore y is affine and its graph is a segment. Conversely every segment satisfies the equation. That the segment is in fact the *minimum* (and not merely stationary) follows at once from the triangle inequality, or from $L(\gamma) = \int \sqrt{1 + y'^2} dx \geq \int 1 dx = x_1 - x_0$ with equality if and only if $y' \equiv 0$ (the case of the horizontal segment; the general case reduces to this by a rotation, which is an isometry of the plane, Chapter 5).

9.2. The geodesic equation

Applying Theorem 9.1 to the length functional $L(\gamma) = \int \sqrt{g_{ij} \dot{u}^i \dot{u}^j} ds$ is possible but unpleasant: the square root makes the equation non-linear in an inessential way and, above all, the length

functional is invariant under reparametrization, so that its stationary curves are not determined as *parametrized curves*. The classical remedy is to replace length by energy. The following lemma justifies the replacement.

Lemma 9.2: Energy and length. For a piecewise C^1 curve $\gamma : [a, b] \rightarrow \mathbb{R}^3$ put

$$L(\gamma) = \int_a^b \|\gamma'\| ds, \quad E(\gamma) = \frac{1}{2} \int_a^b \|\gamma'\|^2 ds.$$

Then

$$L(\gamma)^2 \leq 2(b-a)E(\gamma),$$

with equality if and only if $\|\gamma'\|$ is constant. Consequently, if γ_0 minimizes E in the class of piecewise C^1 curves from A to B defined on $[a, b]$ and contained in a set S , then γ_0 has constant speed and minimizes L in the same class.

Proof. Step 1 — the inequality. By the Cauchy–Schwarz inequality in $L^2([a, b])$ applied to the functions $\|\gamma'\|$ and 1,

$$L(\gamma)^2 = \left(\int_a^b \|\gamma'\| \cdot 1 ds \right)^2 \leq \left(\int_a^b \|\gamma'\|^2 ds \right) \left(\int_a^b 1^2 ds \right) = 2E(\gamma)(b-a).$$

The case of equality in Cauchy–Schwarz is the proportionality of the two functions, that is $\|\gamma'\| = \text{const.}$

Step 2 — reparametrization at constant speed. Let $\sigma : [a, b] \rightarrow S$ be a piecewise C^1 curve from A to B with $L(\sigma) = \ell > 0$. The arclength function $t \mapsto \int_a^t \|\sigma'\|$ is continuous and non-decreasing from 0 to ℓ ; composing σ with the affine reparametrization of arclength one obtains a curve $\tilde{\sigma} : [a, b] \rightarrow S$ with the same image, the same endpoints, $L(\tilde{\sigma}) = \ell$ and $\|\tilde{\sigma}'\| \equiv \ell/(b-a)$ (for piecewise C^1 and piecewise regular curves this reparametrization is piecewise C^1). Hence $2(b-a)E(\tilde{\sigma}) = L(\tilde{\sigma})^2 = \ell^2$ by the equality case of Step 1.

Step 3 — the minimum of the energy minimizes the length. Let γ_0 be a minimum of E and σ any competitor. With $\tilde{\sigma}$ as in Step 2,

$$L(\gamma_0)^2 \stackrel{\text{Step 1}}{\leq} 2(b-a)E(\gamma_0) \stackrel{\text{minimum}}{\leq} 2(b-a)E(\tilde{\sigma}) = L(\tilde{\sigma})^2 = L(\sigma)^2,$$

whence $L(\gamma_0) \leq L(\sigma)$: γ_0 minimizes the length.

Step 4 — γ_0 has constant speed. Apply the chain of Step 3 with $\sigma = \gamma_0$: one gets $L(\gamma_0)^2 \leq 2(b-a)E(\gamma_0) \leq 2(b-a)E(\tilde{\gamma}_0) = L(\gamma_0)^2$, hence all inequalities are equalities; in particular equality holds at Step 1 for γ_0 , that is $\|\gamma_0'\|$ is constant.

Step 5 — numerical check. On the interval $[0, 1]$ take the speed profile $\|\gamma'\|(t) = 1 + 0.8\sin(2\pi t)$, which is not constant. Numerical integration on 200,001 nodes gives $L = 1.00000000$ and $2E = 1.32000000$: hence $L^2 = 1.00000000 < 1.32000000$, the inequality is strict, as it must be for a non-constant speed. ✓ In closed form: $L = \int_0^1 (1 + 0.8\sin 2\pi t) dt = 1$ and $2E = \int_0^1 (1 + 0.8\sin 2\pi t)^2 dt = 1 + 0 + 0.64 \cdot \frac{1}{2} = 1.32$. ✓

Definition 9.2: Christoffel symbols. Given a regular parametrization X of class C^3 with first form g_{ij} , the **Christoffel symbols of the second kind** are the functions

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}), \quad i, j, k \in \{1, 2\},$$

where $\partial_i = \partial/\partial u^i$ and (g^{kl}) is the inverse of (g_{ij}) . They are symmetric in the lower indices: $\Gamma_{ij}^k = \Gamma_{ji}^k$.

Lemma 9.3: *The Christoffel symbols are the tangential part of the second derivatives. Let $N = \frac{X_1 \times X_2}{\|X_1 \times X_2\|}$ be the unit normal and put $h_{ij} = \langle X_{ij}, N \rangle$. Then, for all i, j ,*

$$X_{ij} = \Gamma_{ij}^k X_k + h_{ij} N,$$

with the Γ_{ij}^k of Definition 9.2. In particular the Christoffel symbols depend only on the first fundamental form and its first derivatives.

Proof. Step 1 — the decomposition exists and is unique. At every point $\{X_1, X_2, N\}$ is a basis of $\mathbb{R}^3$: indeed X_1, X_2 are independent (regularity) and N is orthogonal to both and non-zero. Hence there exist unique functions A_{ij}^k and B_{ij} with $X_{ij} = A_{ij}^k X_k + B_{ij} N$. Taking the scalar product with N gives immediately $B_{ij} = \langle X_{ij}, N \rangle = h_{ij}$.

Step 2 — the tangential coefficients in terms of the symbols of the first kind. Taking the scalar product with X_l and using $\langle N, X_l \rangle = 0$,

$$[ij, l] := \langle X_{ij}, X_l \rangle = A_{ij}^k g_{kl}.$$

Step 3 — the symbols of the first kind in terms of the metric. Differentiating $g_{jl} = \langle X_j, X_l \rangle$ with respect to u^i (legitimate because X is C^3),

$$\partial_i g_{jl} = [ij, l] + [il, j].$$

The three versions obtained by permuting the indices are

$$\partial_i g_{jl} = [ij, l] + [il, j], \quad \partial_j g_{il} = [ij, l] + [jl, i], \quad \partial_l g_{ij} = [il, j] + [jl, i],$$

where the symmetry $[ij, l] = [ji, l]$ has been used, a consequence of $X_{ij} = X_{ji}$ (Schwarz's theorem, valid because $X \in C^3$). Adding the first two and subtracting the third, the terms $[il, j]$ and $[jl, i]$ cancel and there remains

$$\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij} = 2[ij, l].$$

Step 4 — conclusion. From Step 2, $A_{ij}^k = g^{kl}[ij, l]$, and substituting Step 3 one obtains exactly $A_{ij}^k = \frac{1}{2}g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}) = \Gamma_{ij}^k$. The right-hand side contains only g_{ij} , its inverse and its first derivatives: the dependence is therefore only on the first form.

Step 5 — check on the sphere. For $X(\varphi, \vartheta) = r(\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta)$ one has $E = r^2 \sin^2 \vartheta$, $F = 0$, $G = r^2$ (Chapter 10), and the definition gives

$$\Gamma_{12}^1 = \Gamma_{21}^1 = \cot \vartheta, \quad \Gamma_{11}^2 = -\sin \vartheta \cos \vartheta, \quad \Gamma_{11}^1 = \Gamma_{22}^1 = \Gamma_{12}^2 = \Gamma_{22}^2 = 0.$$

The independent computation by centred finite differences on the parametrization, with $r = 1.7$ and $(\varphi, \vartheta) = (0.83, 1.12)$, gives $\Gamma_{12}^1 = 0.48403726$ against $\cot(1.12) = 0.48403759$ and $\Gamma_{11}^2 = -0.39215770$ against $-\sin(1.12)\cos(1.12) = -0.39215796$; the other four come out of the order of 10^{-10} . ✓ $\square$

Theorem 9.2: Geodesic equation. *Let $X : U \rightarrow \mathbb{R}^3$ be a regular C^3 parametrization of a surface S and let $\gamma(s) = X(u^1(s), u^2(s))$ be a curve of class C^2 on S , defined for $s \in [a, b]$. The following are equivalent:*

- (i) $\gamma''(s)$ is orthogonal to $T_{\gamma(s)}S$ for every s (the acceleration is purely normal);
- (ii) $\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j = 0$ for $k = 1, 2$ and every s ;
- (iii) the curve $u = (u^1, u^2)$ is stationary for the energy functional $E[u] = \frac{1}{2} \int_a^b g_{ij}(u) \dot{u}^i \dot{u}^j ds$

among curves with fixed endpoints.

A curve satisfying these conditions is called a **geodesic** and necessarily has constant speed $\|\gamma'\|$.

Proof. Step 1 — decomposition of the acceleration. Differentiating $\gamma = X \circ u$ twice by the chain rule,

$$\gamma' = \dot{u}^i X_i, \quad \gamma'' = \ddot{u}^k X_k + \dot{u}^i \dot{u}^j X_{ij}.$$

Substituting Lemma 9.3,

$$\gamma'' = (\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j) X_k + (h_{ij} \dot{u}^i \dot{u}^j) N.$$

The first bracket is the tangential component, the second the normal one.

Step 2 — (i) $\Leftrightarrow$ (ii). The acceleration is purely normal if and only if the tangential component vanishes; since X_1, X_2 are linearly independent, this happens if and only if both coefficients $\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j$ vanish.

Step 3 — (iii) $\Rightarrow$ (ii): writing Euler–Lagrange. The Lagrangian $L(u, v) = \frac{1}{2} g_{ij}(u) v^i v^j$ is C^2 because g_{ij} is C^2 (X is C^3). One computes

$$\frac{\partial L}{\partial v^k} = g_{kj} v^j, \quad \frac{\partial L}{\partial u^k} = \frac{1}{2} (\partial_k g_{ij}) v^i v^j,$$

where in the first the symmetry of g has been used. The Euler–Lagrange equation (Theorem 9.1) becomes

$$\frac{d}{ds} (g_{kj} \dot{u}^j) - \frac{1}{2} (\partial_k g_{ij}) \dot{u}^i \dot{u}^j = 0, \quad \text{that is} \quad g_{kj} \ddot{u}^j + (\partial_i g_{kj}) \dot{u}^i \dot{u}^j - \frac{1}{2} (\partial_k g_{ij}) \dot{u}^i \dot{u}^j = 0.$$

Step 4 — symmetrization. The term $(\partial_i g_{kj}) \dot{u}^i \dot{u}^j$ is contracted with the symmetric tensor $\dot{u}^i \dot{u}^j$, so the coefficient may be replaced by its symmetric part in (i, j) :

$$(\partial_i g_{kj}) \dot{u}^i \dot{u}^j = \frac{1}{2} (\partial_i g_{kj} + \partial_j g_{ki}) \dot{u}^i \dot{u}^j.$$

The equation becomes

$$g_{kl} \ddot{u}^l + \frac{1}{2} (\partial_i g_{kj} + \partial_j g_{ki} - \partial_k g_{ij}) \dot{u}^i \dot{u}^j = 0,$$

that is $g_{kl} \ddot{u}^l + [ij, k] \dot{u}^i \dot{u}^j = 0$ with the symbols of the first kind of Lemma 9.3.

Step 5 — contraction with the inverse. Multiplying by g^{mk} and summing over k , since $g^{mk} g_{kl} = \delta_l^m$,

$$\ddot{u}^m + g^{mk} [ij, k] \dot{u}^i \dot{u}^j = \ddot{u}^m + \Gamma_{ij}^m \dot{u}^i \dot{u}^j = 0,$$

which is (ii). All the steps 3–5 are reversible — the only step not trivially invertible is the multiplication by g^{mk} , which is invertible because (g_{ij}) is positive definite — so (ii) $\Rightarrow$ (iii) holds as well.

Step 6 — constant speed. If (i) holds, then

$$\frac{d}{ds} \|\gamma'\|^2 = 2 \langle \gamma'', \gamma' \rangle = 0,$$

because γ'' is normal and $\gamma' = \dot{u}^i X_i$ is tangent. Hence $\|\gamma'\|$ is constant: the hypothesis “parametrized by arclength”, often imposed in the statement of the geodesic equation, is not needed — it is a consequence.

Step 7 — consistency with Lemma 9.2. Step 6 and Lemma 9.2 close the circle: minimizing the energy implies minimizing the length and having constant speed; and the stationary curves for the energy are exactly the geodesics. The converse does *not* hold: a geodesic is not necessarily minimizing (Section 9.6). $\square$

Remark 9.1: What we import: local existence and uniqueness. Equation (ii) of Theorem 9.2 is a system of two second-order differential equations, which becomes a first-order system in $\mathbb{R}^4$ upon setting $w^k = \dot{u}^k$:

$$\dot{u}^k = w^k, \quad \dot{w}^k = -\Gamma_{ij}^k(u) w^i w^j.$$

The right-hand side is of class C^1 in (u, w) , because the Γ_{ij}^k are C^1 when X is C^3 ; in particular it is locally Lipschitz. The **Picard–Lindelöf theorem** (fundamental theorem of ordinary differential equations) then guarantees that for every point $p \in S$ and every vector $w \in T_p S$ there exist $\varepsilon > 0$ and a unique geodesic $\gamma : (-\varepsilon, \varepsilon) \rightarrow S$ with $\gamma(0) = p$ and $\gamma'(0) = w$, and that it depends in a C^1 way on the initial data. **This theorem is imported:** we do not prove it. Its proof (successive approximations and the contraction mapping theorem) lies outside the scope of the treatise and can be found, in exactly the form used here, in Shifrin, *Differential Geometry: A First Course in Curves and Surfaces*, Appendix, §3 Differential Equations, Theorem 3.1 “Fundamental Theorem of ODE’s”, pp. 118–119 (source [1]). Nothing in this chapter depends on existence as such: the geodesics of the sphere, the cylinder and the cone are exhibited explicitly. The imported theorem is needed only for uniqueness, and hence to conclude that the exhibited geodesics are all the geodesics — a step which on the sphere is however also obtained directly in Step 4 of the proof of Theorem 9.3, without recourse to the import.

Lemma 9.4: Local isometries preserve geodesics. Let $S, \bar{S}$ be regular surfaces and $\phi : V \rightarrow \bar{V}$ a **local isometry** between open sets, that is a diffeomorphism such that $\langle d\phi_p(w_1), d\phi_p(w_2) \rangle = \langle w_1, w_2 \rangle$ for every $p \in V$ and $w_1, w_2 \in T_p S$. Then a curve γ in V is a geodesic of S if and only if $\phi \circ \gamma$ is a geodesic of $\bar{S}$.

Proof. Step 1 — two parametrizations with the same first form. Let $X : U \rightarrow V$ be a parametrization of S and put $\bar{X} = \phi \circ X : U \rightarrow \bar{V}$, which is a parametrization of $\bar{S}$ (composition of a regular homeomorphism with a diffeomorphism). By the chain rule $\bar{X}_i = d\phi(X_i)$, hence

$$\bar{g}_{ij} = \langle \bar{X}_i, \bar{X}_j \rangle = \langle d\phi(X_i), d\phi(X_j) \rangle = \langle X_i, X_j \rangle = g_{ij}.$$

Step 2 — the same Christoffel symbols. By Lemma 9.3 the Christoffel symbols are determined by g_{ij} and its derivatives; since $\bar{g}_{ij} = g_{ij}$ as functions on the same domain U , one has $\bar{\Gamma}_{ij}^k = \Gamma_{ij}^k$ at every point of U .

Step 3 — the same equation. A curve $\gamma = X \circ u$ is a geodesic of S if and only if u satisfies $\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j = 0$ (Theorem 9.2); the curve $\phi \circ \gamma = \bar{X} \circ u$ has the same coordinates u and is a geodesic of $\bar{S}$ if and only if u satisfies $\ddot{u}^k + \bar{\Gamma}_{ij}^k \dot{u}^i \dot{u}^j = 0$. By Step 2 the two conditions coincide. $\square$

9.3. The geodesics of the sphere are the great circles

Theorem 9.3: The geodesics of S^2 are the great circles. Let $S^2(O, r) \subset \mathbb{R}^3$ be the sphere with centre the origin and radius r . A curve $\gamma : [a, b] \rightarrow S^2(O, r)$ of class C^2 is a geodesic if and only if there is $c \geq 0$ such that

$$\gamma(s) = \cos\left(\frac{cs}{r} + \alpha\right)P + \sin\left(\frac{cs}{r} + \alpha\right)Q,$$

with $P, Q \in \mathbb{R}^3$ orthogonal of norm r and $\alpha \in \mathbb{R}$: that is, if and only if it traverses at constant speed c an arc of the great circle $S^2(O, r) \cap \text{span}\{P, Q\}$ (or is constant, for $c = 0$).

Proof. Step 1 — the normal of the sphere. Since $\|\gamma(s)\| = r$ for every s , the tangent plane at $\gamma(s)$ is $\gamma(s)^\perp$ and the unit normal is $N = \gamma/r$. By Theorem 9.2(i) — obtained by applying Euler–Lagrange to the energy functional — the curve is a geodesic if and only if $\gamma''(s)$ is parallel to $\gamma(s)$, that is

$$\gamma''(s) = \lambda(s) \gamma(s)$$

for some function λ , which is continuous because $\lambda = \langle \gamma'', \gamma \rangle / r^2$.

Step 2 — λ is constant. Differentiating the identity $\langle \gamma, \gamma \rangle = r^2$ twice:

$$\langle \gamma, \gamma' \rangle = 0, \quad \langle \gamma', \gamma' \rangle + \langle \gamma, \gamma'' \rangle = 0.$$

By Step 6 of the proof of Theorem 9.2 the speed $\|\gamma'\| = c$ is constant, hence $\langle \gamma, \gamma'' \rangle = -c^2$; on the other hand, from Step 1, $\langle \gamma, \gamma'' \rangle = \lambda \|\gamma\|^2 = \lambda r^2$. Therefore

$$\lambda \equiv -\frac{c^2}{r^2}.$$

Step 3 — integrating the linear equation. The condition becomes

$$\gamma'' + \omega^2 \gamma = 0, \quad \omega = \frac{c}{r} \quad (\geq 0),$$

a linear system with constant coefficients in the three components of γ , which are decoupled. If $c = 0$ then $\gamma'' = 0$ and $\gamma' = \text{const}$ with $\|\gamma'\| = 0$: the curve is constant. If $c > 0$, every component is a combination of $\cos \omega s$ and $\sin \omega s$, hence

$$\gamma(s) = \cos(\omega s) V + \sin(\omega s) W, \quad V = \gamma(0), \quad W = \frac{\gamma'(0)}{\omega}.$$

Step 4 — the constraints on V and W . From $\|\gamma(0)\| = r$ it follows that $\|V\| = r$; from $\langle \gamma, \gamma' \rangle = 0$ at $s = 0$ that $\langle V, W \rangle = 0$; from $\|\gamma'(0)\| = c$ that $\|W\| = c/\omega = r$. Hence V, W are orthogonal of norm r and the curve describes, at constant speed c , the intersection of $S^2(O, r)$ with the plane $\text{span}\{V, W\}$ through the origin, that is a great circle (Definition 7.4). The form of the statement is obtained by putting $\alpha = 0$, $P = V$, $Q = W$; the parameter α accounts for a shift of the parameter.

Step 5 — the converse. If $\gamma(s) = \cos(\omega s + \alpha)P + \sin(\omega s + \alpha)Q$ with $P \perp Q$, $\|P\| = \|Q\| = r$ and $\omega = c/r$, then $\|\gamma\|^2 = r^2 \cos^2 + r^2 \sin^2 = r^2$ (the curve lies on the sphere), $\gamma'' = -\omega^2 \gamma$ is parallel to γ , that is normal to the sphere, and by Theorem 9.2(i) γ is a geodesic. Moreover $\gamma' = \omega(-\sin(\cdot)P + \cos(\cdot)Q)$ has norm $\omega r = c$.

Step 6 — no charts, no use of the import. The proof uses no coordinates on the sphere — hence has no exceptions at the poles — and does not use the existence and uniqueness theorem of Remark 9.1: uniqueness is obtained by explicitly integrating the linear equation of Step 3.

Step 7 — consistency with Chapter 7. Theorem 7.7 proves, by metric means and without the calculus of variations, that the *minor* arc of a great circle realizes the distance between two points. Here a different and complementary statement has been proved: the curves satisfying the *local* stationarity condition are exactly the constant-speed parametrizations of the great circles. The two statements fit together without overlapping: the first gives the global minimum, the second the differential characterization; and the second is wider, because it includes *non-minimizing* arcs (Section 9.6). Neither result is re-proved here from the other.

Step 8 — numerical check. On the unit sphere take $P = (1, 0, 0)$, $Q = (0, 1, 0)$, $c = 1$: one obtains $\gamma(s) = (\cos s, \sin s, 0)$, the equator, with $\gamma'' = -\gamma$ exactly normal. ✓ Take instead the parallel of colatitude ϑ_0 , traversed at unit speed: $\gamma(s) = (\sin \vartheta_0 \cos \omega s, \sin \vartheta_0 \sin \omega s, \cos \vartheta_0)$ with $\omega = 1/\sin \vartheta_0$. The tangential component of the acceleration, $\gamma'' - \langle \gamma'', \gamma \rangle \gamma$, has numerical norm 0.577350 for $\vartheta_0 = \pi/3$ and 1.000000 for $\vartheta_0 = \pi/4$; the theoretical values are $|\cot(\pi/3)| = 0.577350$ and $|\cot(\pi/4)| = 1.000000$, that is the geodesic curvature of the parallel. ✓ For $\vartheta_0 = \pi/2$ (equator) the tangential component vanishes. ✓ □

In coordinates: the Christoffel symbols of the sphere. It is worth seeing the same conclusion with the coordinate machinery, which is what one uses on an arbitrary surface. With $u^1 = \varphi$, $u^2 = \vartheta$ and $E = r^2 \sin^2 \vartheta$, $F = 0$, $G = r^2$, Definition 9.2 gives, as checked in Step 5 of the proof of Lemma 9.3,

$$\Gamma_{12}^1 = \cot \vartheta, \quad \Gamma_{11}^2 = -\sin \vartheta \cos \vartheta,$$

and all the others vanish. The geodesic equation therefore becomes the system

$$\ddot{\varphi} + 2 \cot \vartheta \dot{\varphi} \dot{\vartheta} = 0, \quad \ddot{\vartheta} - \sin \vartheta \cos \vartheta \dot{\varphi}^2 = 0.$$

The first equation, multiplied by $\sin^2 \vartheta$, is exactly $\frac{d}{ds}(\sin^2 \vartheta \dot{\varphi}) = 0$: it is Clairaut's first integral of Section 9.5. The *meridians* $\varphi \equiv \varphi_0$, $\vartheta(s) = \vartheta_0 + s/r$ satisfy both equations (the first because $\dot{\varphi} = 0$, the second because $\ddot{\vartheta} = 0$ and $\dot{\varphi} = 0$): they are geodesics. Every great circle is obtained from a meridian by a rotation of $\mathbb{R}^3$, which is an isometry of the sphere, and isometries preserve geodesics (Lemma 9.4): hence all great circles are geodesics. That there are no others follows from the uniqueness of Remark 9.1, since through every point and every tangent direction there passes a great circle. This route is more instructive but less economical than the main proof, which avoids the import. $\square$

Exercise 9.2. Why is a parallel other than the equator not a geodesic of the sphere? Give two independent reasons.

Solution. *First reason (vectorial).* Let $\vartheta_0 \neq \pi/2$ and traverse the parallel at unit speed as in Step 8 of the proof of Theorem 9.3, on the unit S^2 . The acceleration γ'' is horizontal and points towards the z -axis, while the normal to the sphere is γ , which has vertical component $\cos \vartheta_0 \neq 0$: the two vectors are not parallel. In formulas, the tangential component is

$$\gamma'' - \langle \gamma'', \gamma \rangle \gamma, \quad \|\gamma'' - \langle \gamma'', \gamma \rangle \gamma\| = |\cot \vartheta_0|,$$

non-zero for $\vartheta_0 \neq \pi/2$. By Theorem 9.2(i) the parallel is not a geodesic.

Second reason (in coordinates). On the parallel $\vartheta \equiv \vartheta_0$ one has $\dot{\vartheta} = \ddot{\vartheta} = 0$ and $\dot{\varphi} = \pm 1/\sin \vartheta_0 \neq 0$; the second equation of the *dimvariante* becomes $0 - \sin \vartheta_0 \cos \vartheta_0 \dot{\varphi}^2 = 0$, that is $\cos \vartheta_0 = 0$: satisfied only by $\vartheta_0 = \pi/2$, the equator. The general criterion for surfaces of revolution is Proposition 9.4.

Interpretation. The number $|\cot \vartheta_0|$ is the **geodesic curvature** of the parallel: it measures how much the curve “steers” while remaining on the surface. It is 0 at the equator and diverges near the poles, where the parallel is a very tight little circle. The value $\kappa_g = \cot \vartheta_0$ for the parallel of colatitude ϑ_0 is confirmed in Shifrin, solution of exercise 2.4.4 (source [1]).

✓

9.4. Cylinder and cone: geodesics by development

Lemma 9.4 makes the geodesics of all surfaces that “unroll” onto the plane without stretching almost free of charge. The strategy is always the same: exhibit a parametrization of the plane with the *same* first fundamental form as the surface, and read the straight lines of the plane on the surface.

Lemma 9.5: *The geodesics of the plane are the straight lines. In the plane, parametrized by $Y(u, v) = (u, v, 0)$, one has $E = G = 1$, $F = 0$ and all the Christoffel*

symbols vanish. The geodesics are therefore the curves $u(s) = a_1s + b_1$, $v(s) = a_2s + b_2$: the straight lines traversed at constant speed (and the points, for $a_1 = a_2 = 0$).

Proof. From $Y_1 = (1, 0, 0)$ and $Y_2 = (0, 1, 0)$ it follows that $g_{11} = g_{22} = 1$, $g_{12} = 0$: the first form is constant, hence all the $\partial_i g_{jl}$ vanish and, by Definition 9.2, $\Gamma_{ij}^k \equiv 0$. The geodesic equation of Theorem 9.2(ii) reduces to $\ddot{u}^k = 0$, whose general solution is affine in s . The speed $\sqrt{a_1^2 + a_2^2}$ is constant, as it must be by Step 6 of the same proof. $\square$

Proposition 9.1: Geodesics of the cylinder. Let $C_R = \{(x, y, z) : x^2 + y^2 = R^2\}$ be the right circular cylinder of radius $R > 0$, parametrized by $X(u, v) = (R \cos u, R \sin u, v)$. Then:

- (i) the first form is $I = R^2 du^2 + dv^2$, with constant coefficients, and the map ϕ sending $X(u, v)$ to $(Ru, v, 0)$ is a local isometry onto the plane;
- (ii) the geodesics of C_R are the curves $u(s) = as + u_0$, $v(s) = bs + v_0$, that is the **circular helices** for $a, b \neq 0$, the horizontal circles for $b = 0$ and the vertical generators for $a = 0$;
- (iii) between two distinct points $P_0 = X(u_0, v_0)$ and $P_1 = X(u_1, v_1)$ there pass infinitely many geodesics, of lengths $\sqrt{R^2(\Delta u + 2\pi n)^2 + (\Delta v)^2}$ for $n \in \mathbb{Z}$, with $\Delta u = u_1 - u_0 \in (-\pi, \pi]$ and $\Delta v = v_1 - v_0$; the shortest is the one with $n = 0$.

Proof. Step 1 — first form of the cylinder. $X_1 = (-R \sin u, R \cos u, 0)$ and $X_2 = (0, 0, 1)$ give $E = R^2$, $F = 0$, $G = 1$: constants. The cross product $X_1 \times X_2 = (R \cos u, R \sin u, 0)$ is non-zero, hence X is regular.

Step 2 — the same first form on the plane. The parametrization of the plane $\tilde{Y}(u, v) = (Ru, v, 0)$ has $\tilde{Y}_1 = (R, 0, 0)$, $\tilde{Y}_2 = (0, 1, 0)$, hence $\tilde{E} = R^2$, $\tilde{F} = 0$, $\tilde{G} = 1$: identical to those of the cylinder. The map $\phi = \tilde{Y} \circ X^{-1}$, defined on every open set where X is invertible (for instance $u \in (u_*, u_* + 2\pi)$), is therefore a local isometry, because $d\phi(X_i) = \tilde{Y}_i$ and the matrix of scalar products coincides. This proves (i).

Step 3 — the geodesics. By Step 2 and Lemma 9.4, $\gamma = X \circ u$ is a geodesic of the cylinder if and only if $\tilde{Y} \circ u$ is a geodesic of the plane, that is (Lemma 9.5) if and only if $s \mapsto (Ru^1(s), u^2(s))$ is affine, that is if and only if u^1 and u^2 are affine. Carrying this back to the cylinder: for $a, b \neq 0$ one obtains $\gamma(s) = (R \cos(as + u_0), R \sin(as + u_0), bs + v_0)$, which is the circular helix of Chapter 3; for $b = 0$ the circle $z = v_0$; for $a = 0$ the vertical generator. This proves (ii). Note that the same result follows directly: since the first form is constant, all the Christoffel symbols vanish and the geodesic equation is $\ddot{u} = \ddot{v} = 0$.

Step 4 — the multiplicity of paths. The points $X(u, v)$ and $X(u + 2\pi n, v)$ coincide. Fixing P_0 and P_1 , for every $n \in \mathbb{Z}$ the geodesic with u running from u_0 to $u_0 + \Delta u + 2\pi n$ and v from v_0 to v_1 joins the two points; developed on the plane it is the segment with legs $R|\Delta u + 2\pi n|$ and $|\Delta v|$, of length $\sqrt{R^2(\Delta u + 2\pi n)^2 + (\Delta v)^2}$, since the development preserves lengths. The function $n \mapsto |\Delta u + 2\pi n|$ with $\Delta u \in (-\pi, \pi]$ is minimal at $n = 0$, hence the shortest geodesic is the one that makes no whole turns. This proves (iii).

Step 5 — numerical check. With $R = 1$, $\Delta u = \pi/2$ and $\Delta v = 1$ the lengths $\sqrt{(\pi/2 + 2\pi n)^2 + 1}$ are, for $n = -2, -1, 0, 1, 2$: 11.040953, 4.817324, 1.862096, 7.917388, 14.172491. The minimum is 1.862096 for $n = 0$, as predicted; the closed-form check gives $\sqrt{(\pi/2)^2 + 1} = \sqrt{2.467401 + 1} = \sqrt{3.467401} = 1.862096$. $\checkmark$ The numerical computation of the Christoffel symbols of the cylinder by finite differences gives eight values all equal to 0 within machine precision. $\checkmark$ $\square$

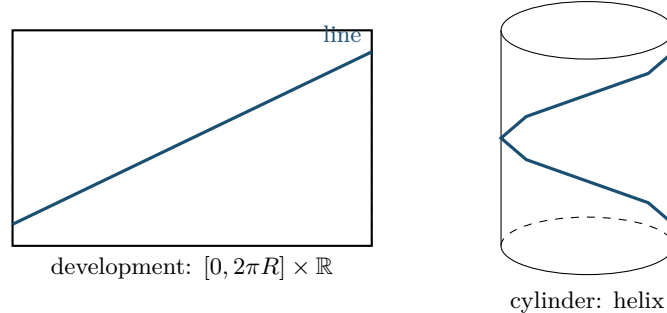


Figure 9.1: The development of the cylinder onto the plane is a local isometry (Proposition 9.1(i)): the straight lines of the rectangle become the helices of the cylinder. The nodes of the polyline are the points $(8 + 1.2 \cos t, 0.3 + 2.4 t/(2\pi))$ for $t = 0, \pi/4, \dots, 2\pi$, computed by hand: 9.200; 8.849; 8.000; 7.151; 6.800; 7.151; 8.000; 8.849; 9.200.

Proposition 9.2: Geodesics of the cone. Let K_α be the right circular cone of half-angle $\alpha \in (0, \pi/2)$ with the vertex removed, parametrized by

$$X(u, v) = (v \sin \alpha \cos u, v \sin \alpha \sin u, v \cos \alpha), \quad u \in \mathbb{R}, v > 0.$$

Then:

- (i) the first form is $I = v^2 \sin^2 \alpha du^2 + dv^2$, which coincides with the first form of the plane in polar coordinates (θ, ρ) via $\theta = u \sin \alpha$, $\rho = v$; the cone, cut along a generator, develops isometrically onto a circular sector of opening $\beta = 2\pi \sin \alpha$;
- (ii) the geodesics of K_α are the generators $u = \text{const}$ and the curves

$$v(u) = \frac{d}{\cos(\sin \alpha (u - u_0))}, \quad d > 0, \quad |\sin \alpha (u - u_0)| < \frac{\pi}{2},$$

traversed at constant speed.

Proof. Step 1 — first form of the cone. One has $X_1 = v \sin \alpha (-\sin u, \cos u, 0)$ and $X_2 = (\sin \alpha \cos u, \sin \alpha \sin u, \cos \alpha)$, hence

$$E = v^2 \sin^2 \alpha, \quad F = v \sin^2 \alpha (-\sin u \cos u + \cos u \sin u) = 0, \quad G = \sin^2 \alpha + \cos^2 \alpha = 1.$$

Since $v > 0$ and $\sin \alpha > 0$ one has $EG - F^2 = v^2 \sin^2 \alpha > 0$: the parametrization is regular.

Step 2 — the same first form on the plane. The parametrization of the plane $\tilde{Y}(u, v) = (v \cos(u \sin \alpha), v \sin(u \sin \alpha), 0)$ has $\tilde{Y}_1 = v \sin \alpha (-\sin(u \sin \alpha), \cos(u \sin \alpha), 0)$ and $\tilde{Y}_2 = (\cos(u \sin \alpha), \sin(u \sin \alpha), 0)$, hence $\tilde{E} = v^2 \sin^2 \alpha$, $\tilde{F} = 0$, $\tilde{G} = 1$: identical to those of the cone. As u runs over an interval of width 2π (the cone minus one generator), the polar angle $\theta = u \sin \alpha$ varies by $\beta = 2\pi \sin \alpha$: the sector. This proves (i).

Step 3 — the straight lines of the sector. By Lemma 9.4 and Lemma 9.5, the geodesics of the cone are the preimages of the straight lines of the sector. In polar coordinates a line is of two kinds: the lines through the origin, $\theta = \text{const}$, which lift to the generators $u = \text{const}$; and the lines not through the origin, of normal equation

$$\rho \cos(\theta - \theta_0) = d, \quad d > 0,$$

with d the distance from the origin and θ_0 the direction of the foot of the perpendicular. Substituting $\theta = u \sin \alpha$, $\rho = v$ and setting $u_0 = \theta_0 / \sin \alpha$ one gets $v \cos(\sin \alpha (u - u_0)) = d$, that is the formula of the statement. This proves (ii).

Step 4 — numerical check. With $\alpha = 35^\circ$ and $(u, v) = (0.7, 2.3)$, the finite-difference computation gives, both for the cone and for the plane sector, the same matrix

$$(g_{ij}) = \begin{pmatrix} 1.74035672 & 0 \\ 0 & 1 \end{pmatrix};$$

the expected value is $v^2 \sin^2 \alpha = 2.3^2 \cdot \sin^2 35^\circ = 5.29 \times 0.329009 = 1.74035672$. ✓ □

Proposition 9.3: When a geodesic of the cone self-intersects. With the notation of Proposition 9.2, a geodesic of the cone which is not a generator self-intersects if and only if

$$\sin \alpha < \frac{1}{2}, \quad \text{that is} \quad \alpha < 30^\circ,$$

equivalently if and only if the opening of the development $\beta = 2\pi \sin \alpha$ is less than π .

Proof. **Step 1 — the interval of definition.** The geodesic $v(u) = d / \cos(\sin \alpha (u - u_0))$ is defined exactly where the cosine is positive, that is for $|\sin \alpha (u - u_0)| < \pi/2$: the set of values of u is the open interval with centre u_0 and width

$$\Delta u = \frac{\pi}{\sin \alpha}.$$

Step 2 — what self-intersection means. Two parameters $u_1 \neq u_2$ give the same point of the cone if and only if $u_2 - u_1 \in 2\pi\mathbb{Z} \setminus \{0\}$ and $v(u_1) = v(u_2)$. Since v is even about u_0 and strictly increasing for $u > u_0$, the equality $v(u_1) = v(u_2)$ with $u_1 \neq u_2$ forces u_1 and u_2 to be symmetric about u_0 , that is $u_1 = u_0 - t$, $u_2 = u_0 + t$ with $t > 0$; the condition $u_2 - u_1 = 2t \in 2\pi\mathbb{Z}$ gives $t = \pi m$ with $m \geq 1$ an integer.

Step 3 — existence of such a t . It is necessary and sufficient that $t = \pi$ (the case $m = 1$; if $m = 1$ works, self-intersections exist a fortiori) lie in the interval of definition, that is $\pi < \Delta u/2 = \pi/(2 \sin \alpha)$, which is equivalent to $2 \sin \alpha < 1$, that is $\sin \alpha < 1/2$ and hence $\alpha < 30^\circ$. If instead $\sin \alpha \geq 1/2$ then $\Delta u/2 \leq \pi$ and no $t = \pi m$ is admissible: the geodesic does not self-intersect.

Step 4 — geometric reading. The condition $\sin \alpha < 1/2$ is equivalent to $\beta = 2\pi \sin \alpha < \pi$: the development is a sector narrower than a half-plane, and a line entering it must leave it “on the same side”, so that closing the sector up the two branches meet. For $\beta \geq \pi$ the two branches diverge and never touch.

Step 5 — numerical check. $\alpha = 20^\circ$: $\Delta u = \pi / \sin 20^\circ = 3.141593 / 0.342020 = 9.185402 > 2\pi = 6.283185$, self-intersection. $\alpha = 30^\circ$: $\Delta u = \pi / 0.5 = 6.283185 = 2\pi$, the limiting case. $\alpha = 40^\circ$: $\Delta u = \pi / 0.642788 = 4.887451 < 6.283185$, no self-intersection. ✓ □

9.5. Clairaut’s relation

Definition 9.3: Surface of revolution. Let $f, h : I \rightarrow \mathbb{R}$ be of class C^3 on an interval I , with $f > 0$ and $f'^2 + h'^2 > 0$ everywhere. The **surface of revolution** generated by the profile $v \mapsto (f(v), 0, h(v))$ is the surface parametrized by

$$X(u, v) = (f(v) \cos u, f(v) \sin u, h(v)), \quad u \in \mathbb{R}, v \in I.$$

The curves $v = \text{const}$ are called **parallels** (circles of radius $f(v)$), those with $u = \text{const}$ **meridians**.

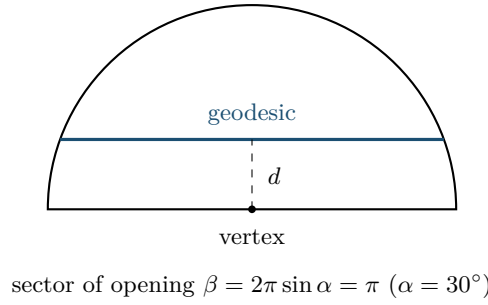


Figure 9.2: Development of the cone of half-angle $\alpha = 30^\circ$: a sector of opening $\beta = 2\pi \sin 30^\circ = \pi$. The geodesics are the segments; the one drawn has endpoints $(3 \cos 20^\circ, 3 \sin 20^\circ) = (2.819, 1.026)$ and $(3 \cos 160^\circ, 3 \sin 160^\circ) = (-2.819, 1.026)$, coordinates computed by hand, and lies at distance $d = 1.026$ from the vertex. For $\alpha < 30^\circ$ the sector is narrower than π and the geodesic, on closing the cone up, self-intersects (Proposition 9.3).

Lemma 9.6: First form of a surface of revolution. With the notation of Definition 9.3,

$$E = f(v)^2, \quad F = 0, \quad G = f'(v)^2 + h'(v)^2.$$

In particular parallels and meridians are orthogonal, and the Christoffel symbols are

$$\Gamma_{11}^1 = 0, \quad \Gamma_{12}^1 = \frac{f'}{f}, \quad \Gamma_{22}^1 = 0, \quad \Gamma_{11}^2 = -\frac{ff'}{G}, \quad \Gamma_{12}^2 = 0, \quad \Gamma_{22}^2 = \frac{G'}{2G}.$$

Proof. Step 1 — the metric coefficients. $X_1 = f(-\sin u, \cos u, 0)$ and $X_2 = (f' \cos u, f' \sin u, h')$ give $E = f^2(\sin^2 u + \cos^2 u) = f^2$, $F = ff'(-\sin u \cos u + \cos u \sin u) = 0$, $G = f'^2(\cos^2 u + \sin^2 u) + h'^2 = f'^2 + h'^2$. The hypothesis $f > 0$ and $f'^2 + h'^2 > 0$ gives $EG - F^2 = f^2G > 0$: regularity.

Step 2 — the Christoffel symbols in the orthogonal case. With $F = 0$ the inverse matrix is $g^{11} = 1/E$, $g^{22} = 1/G$, $g^{12} = 0$, and Definition 9.2 gives the general formulas

$$\Gamma_{11}^1 = \frac{E_u}{2E}, \quad \Gamma_{12}^1 = \frac{E_v}{2E}, \quad \Gamma_{22}^1 = -\frac{G_u}{2E}, \quad \Gamma_{11}^2 = -\frac{E_v}{2G}, \quad \Gamma_{12}^2 = \frac{G_u}{2G}, \quad \Gamma_{22}^2 = \frac{G_v}{2G},$$

where the subscript denotes the partial derivative. Here $E = f^2$ does not depend on u and neither does G , hence $E_u = G_u = 0$, $E_v = 2ff'$ and $G_v = G'$; substituting one obtains the stated values.

Step 3 — numerical check on the torus. For the torus $f(v) = a + b \cos v$, $h(v) = b \sin v$ with $a = 2$, $b = 1$ one has $G = b^2 \sin^2 v + b^2 \cos^2 v = b^2 = 1$ and $\Gamma_{11}^2 = -ff'/G = (a + b \cos v)b \sin v$. The finite-difference computation gives, at $u = 0.3$: for $v = 0$ the value 0.00000000 (formula: 0), for $v = \pi/2$ the value 1.99999967 (formula: $2 \cdot 1 = 2$), for $v = \pi$ the value 0.00000000 (formula: 0). ✓ $\square$

Theorem 9.4: Clairaut's relation. Let S be a surface of revolution as in Definition 9.3 and let $\gamma(s) = X(u(s), v(s))$ be a curve of class C^2 parametrized by arclength. Let $r(s) = f(v(s))$ denote the radius of the parallel through $\gamma(s)$ and $\psi(s) \in [0, \pi]$ the angle between $\gamma'(s)$ and the direction of the meridian. Then:

- (i) if γ is a geodesic, the quantity $r \sin \psi = f^2 \dot{u}$ is constant along γ (Clairaut's relation);
- (ii) conversely, if $r \sin \psi$ is constant along γ and no subarc of γ is contained in a parallel, then γ is a geodesic.

Proof. Step 1 — identification of $r \sin \psi$. The direction of the parallel is $X_1/\|X_1\| = X_1/f$; that of the meridian is $X_2/\sqrt{G}$. Since the two are orthogonal (Lemma 9.6) and $\|\gamma'\| = 1$, the angle ψ with the meridian satisfies

$$\sin \psi = \left\langle \gamma', \frac{X_1}{f} \right\rangle = \frac{\dot{u} \langle X_1, X_1 \rangle + \dot{v} \langle X_2, X_1 \rangle}{f} = \frac{f^2 \dot{u}}{f} = f \dot{u},$$

up to sign, which depends on the direction of travel. Hence $r \sin \psi = f \cdot f \dot{u} = f^2 \dot{u} = E \dot{u}$.

Step 2 — (i): u is a cyclic coordinate. The energy Lagrangian is $L = \frac{1}{2}(f(v)^2 \dot{u}^2 + G(v) \dot{v}^2)$ and does not depend on u . The Euler–Lagrange equation for $k = 1$ (Theorem 9.1) is therefore

$$\frac{d}{ds} \frac{\partial L}{\partial \dot{u}} = \frac{\partial L}{\partial u} = 0, \quad \text{that is} \quad \frac{d}{ds} (f^2 \dot{u}) = 0.$$

By Theorem 9.2(iii) the geodesics are the stationary curves of the energy: the quantity $f^2 \dot{u} = r \sin \psi$ is constant. This proves (i).

Step 3 — the two geodesic equations in explicit form. With the symbols of Lemma 9.6, the geodesic equation reads

$$(A) \quad \frac{d}{ds} (f^2 \dot{u}) = 0, \quad (B) \quad 2G\ddot{v} + G'\dot{v}^2 - 2ff'\dot{u}^2 = 0,$$

where (A) is the first equation multiplied by f^2 and (B) is the second multiplied by $2G$ — the computation is the one in Step 4 of the proof of Theorem 9.2 with the values of Lemma 9.6. To these one adds the unit-speed constraint

$$(C) \quad f^2 \dot{u}^2 + G \dot{v}^2 = 1.$$

Step 4 — (ii): (A) and (C) imply (B). Differentiate (C) with respect to s , recalling that f and G depend on s only through v :

$$2ff'\dot{v}\dot{u}^2 + 2f^2\dot{u}\ddot{u} + G'\dot{v}^3 + 2G\dot{v}\ddot{v} = 0.$$

On the other hand (A) expanded is $2ff'\dot{v}\dot{u} + f^2\ddot{u} = 0$, hence $f^2\dot{u}\ddot{u} = -2ff'\dot{v}\dot{u}^2$. Substituting:

$$2ff'\dot{v}\dot{u}^2 - 4ff'\dot{v}\dot{u}^2 + G'\dot{v}^3 + 2G\dot{v}\ddot{v} = 0,$$

that is

$$\dot{v} \cdot (2G\ddot{v} + G'\dot{v}^2 - 2ff'\dot{u}^2) = 0.$$

The bracket is exactly the left-hand side of (B). Hence (B) holds at every point where $\dot{v} \neq 0$.

Step 5 — conclusion of (ii). By hypothesis no subarc of γ lies in a parallel, that is the set $Z = \{s : \dot{v}(s) = 0\}$ has empty interior; its complement $\{\dot{v} \neq 0\}$ is therefore open and dense. On it (B) holds by Step 4; since the left-hand side of (B) is a continuous function of s (the second derivatives exist and are continuous, γ being of class C^2 and f, G of class C^2), it vanishes on the closure as well, that is on all of $[a, b]$. Together with (A), which is the hypothesis, both geodesic equations hold: γ is a geodesic.

Step 6 — the hypothesis is not superfluous. If the clause about parallels is dropped, statement (ii) is false: any parallel, traversed at unit speed, has $\psi \equiv \pi/2$ and $r \equiv f(v_0)$, hence $r \sin \psi$ constant, but it is a geodesic only in special cases (Proposition 9.4). This is exactly the point where Step 4 fails to conclude, because $\dot{v} \equiv 0$.

Step 7 — comparison with the literature. The statement with the clause “no part of the curve is contained in a parallel” is that of Shifrin, Proposition 4.4 “Clairaut’s relation”, pp. 73–75 (source [1]), who formulates the relation with the angle to the parallel, that is $r \cos \omega = \text{const}$ with $\omega = \pi/2 - \psi$: the two writings are the same thing. $\checkmark$ $\square$

Proposition 9.4: Which parallels are geodesics. With the notation of Definition 9.3, the parallel $v \equiv v_0$ (traversed at constant speed) is a geodesic if and only if $f'(v_0) = 0$, that is if and only if v_0 is a critical point of the radius.

Proof. Step 1 — the equations on the parallel. On the parallel $\dot{v} \equiv 0$, $\ddot{v} \equiv 0$ and $\dot{u} \equiv c \neq 0$ constant. Equation (A) of Step 3 of the proof of Theorem 9.4 is automatically satisfied, because $f^2 \dot{u} = f(v_0)^2 c$ is constant. Equation (B) reduces to $-2f(v_0)f'(v_0)c^2 = 0$.

Step 2 — conclusion. Since $f(v_0) > 0$ and $c \neq 0$, equation (B) is equivalent to $f'(v_0) = 0$.

Step 3 — the two examples. *Sphere* ($f(\vartheta) = r \sin \vartheta$): $f' = r \cos \vartheta = 0$ if and only if $\vartheta = \pi/2$, the equator. This recovers, by a completely different route, Exercise 9.2. *Torus* ($f(v) = a + b \cos v$ with $a > b > 0$): $f' = -b \sin v = 0$ if and only if $v = 0$ or $v = \pi$, that is the outer equator (radius $a + b$, maximal) and the inner equator (radius $a - b$, minimal). No other parallel of the torus is a geodesic. $\square$

9.6. Beyond half a sphere: geodesics that stop minimizing

Theorem 7.7 of Chapter 7 says that the *minor* arc of a great circle realizes the distance. Theorem 9.3 of this chapter says that *all* arcs of great circles are geodesics. The two statements are not in conflict: together they say that the stationarity condition is necessary but not sufficient. This section measures exactly where sufficiency is lost.

Proposition 9.5: The conjugate point at the antipode. On $S^2(O, r)$ fix A with $\|A\| = r$ and take two orthonormal vectors w_1, w_2 orthogonal to A . The family of unit-speed geodesics issuing from A

$$\gamma_\vartheta(s) = \cos \frac{s}{r} A + \sin \frac{s}{r} r (\cos \vartheta w_1 + \sin \vartheta w_2), \quad \vartheta \in \mathbb{R},$$

satisfies: $\gamma_\vartheta(0) = A$ for every ϑ , and $\gamma_\vartheta(\pi r) = -A$ for every ϑ . Moreover the variation field $J(s) = \frac{\partial \gamma_\vartheta}{\partial \vartheta} \Big|_{\vartheta=0}$ equals $J(s) = r \sin \frac{s}{r} w_2$: it vanishes at $s = 0$ and at $s = \pi r$, and is non-zero in between. Hence all the geodesics issuing from A meet again at $-A$ after length πr : $-A$ is **conjugate** to A .

Proof. Step 1 — they are geodesics. Putting $Q_\vartheta = r(\cos \vartheta w_1 + \sin \vartheta w_2)$, one has $\|Q_\vartheta\| = r$ and $\langle A, Q_\vartheta \rangle = 0$: by Theorem 9.3 (Step 5) the curve $\gamma_\vartheta(s) = \cos(s/r)A + \sin(s/r)Q_\vartheta$ is a geodesic of speed $\|Q_\vartheta\|/r = 1$.

Step 2 — the endpoint values. $\gamma_\vartheta(0) = A$ because $\cos 0 = 1$ and $\sin 0 = 0$; $\gamma_\vartheta(\pi r) = \cos \pi A + \sin \pi Q_\vartheta = -A$, for every ϑ .

Step 3 — the variation field. Differentiating with respect to ϑ at $\vartheta = 0$: $\partial_\vartheta Q_\vartheta|_0 = r(-\sin 0 w_1 + \cos 0 w_2) = r w_2$, hence

$$J(s) = \sin \frac{s}{r} r w_2.$$

It vanishes for $s = 0$ and $s = \pi r$, and has norm $r|\sin(s/r)| > 0$ for $0 < s < \pi r$.

Step 4 — the Jacobi equation, verified by hand. Putting $j(s) = \|J(s)\| = r \sin(s/r)$ for $s \in (0, \pi r)$, one has

$$j''(s) = -\frac{1}{r^2} r \sin \frac{s}{r} = -K j(s), \quad K = \frac{1}{r^2},$$

that is $j'' + K j = 0$ with the Gaussian curvature $K = 1/r^2$ of the sphere (Chapter 10). This is the Jacobi equation in constant curvature, verified here by direct computation and without any need of the general theory.

Step 5 — numerical check. With $r = 1$, $A = (0, 0, 1)$, $w_1 = (1, 0, 0)$, $w_2 = (0, 1, 0)$: $\gamma_0(s) = (\sin s, 0, \cos s)$ and $\gamma_{\pi/2}(s) = (0, \sin s, \cos s)$, two distinct meridians; at $s = \pi$ both equal $(0, 0, -1) = -A$. ✓ And $J(s) = \sin s (0, 1, 0)$, with $J(0) = J(\pi) = 0$ and $\|J(\pi/2)\| = 1$. ✓ □

Theorem 9.5: Loss of minimality beyond half a sphere. On $S^2(O, r)$ consider the unit-speed geodesic arc $\gamma : [0, \ell] \rightarrow S^2(O, r)$, $\gamma(s) = \cos(s/r)A + \sin(s/r)Q$ with $\|A\| = \|Q\| = r$, $\langle A, Q \rangle = 0$. Then:

- (i) if $\ell \leq \pi r$, the arc is minimizing, that is $\ell = \text{dist}(\gamma(0), \gamma(\ell))$ (uniquely so for $\ell < \pi r$);
- (ii) if $\ell > \pi r$, the arc is not minimizing: there is a curve on the sphere with the same endpoints and strictly smaller length, and precisely

$$\text{dist}(\gamma(0), \gamma(\ell)) = r \arccos\left(\cos \frac{\ell}{r}\right) < \ell.$$

In particular the length πr — half a great circle, that is the distance to the antipode — is the exact threshold.

Proof. Step 1 — the distance between the endpoints. Since $\langle A, Q \rangle = 0$ and $\|A\| = \|Q\| = r$,

$$\langle \gamma(0), \gamma(\ell) \rangle = \left\langle A, \cos \frac{\ell}{r} A + \sin \frac{\ell}{r} Q \right\rangle = r^2 \cos \frac{\ell}{r}.$$

By Theorem 7.8 of Chapter 7 the spherical distance between two points P_1, P_2 is $r \arccos(\langle P_1, P_2 \rangle / r^2)$, hence

$$\text{dist}(\gamma(0), \gamma(\ell)) = r \arccos\left(\cos \frac{\ell}{r}\right).$$

Step 2 — (i). If $0 \leq \ell \leq \pi r$ then $\ell/r \in [0, \pi]$, an interval on which $\arccos \circ \cos$ is the identity: the distance equals ℓ , and the arc realizes it. Uniqueness for $\ell < \pi r$ and non-uniqueness for $\ell = \pi r$ are the content of Theorem 7.7, which we do not re-prove.

Step 3 — (ii): the inequality. Let $\ell > \pi r$ and put $t = \ell/r > \pi$. The function $t \mapsto \arccos(\cos t)$ equals $2\pi - t$ for $t \in [\pi, 2\pi]$, and is in any case always $\leq \pi < t$. Hence

$$\text{dist}(\gamma(0), \gamma(\ell)) = r \arccos(\cos t) \leq \pi r < r t = \ell.$$

The distance is strictly less than the length of the arc: the arc is not minimizing.

Step 4 — the explicit competitor. There is no need to invoke the existence of the minimum: if $\pi r < \ell < 2\pi r$ the shortest competitor is the *complementary* arc of the same great circle, traversed in the opposite direction, of length $2\pi r - \ell < \ell$; if $\ell \geq 2\pi r$ the curve has already made at least one full turn and one reduces to the previous case by subtracting $2\pi r$ until the remaining length lies in $[0, 2\pi r)$. In both cases the competitor is an arc of a great circle with the same endpoints.

Step 5 — why exactly πr : the link with conjugate points. The threshold is no accident. By Proposition 9.5 all the geodesics issuing from A meet again at $-A$ after length exactly πr . Past that point the geodesic “has already spent” its single encounter with the neighbouring ones: between $\gamma(0)$ and $\gamma(\ell)$ with $\ell > \pi r$ there is a second, shorter geodesic, and the long arc ceases to be minimal. The general theory (Morse index theorem, Jacobi’s theorem) proves that this is the mechanism on *every* surface: a geodesic stops minimizing as soon as it passes the first conjugate point — see Remark 9.2. On the sphere, however, the conclusion is obtained by the elementary computation of Steps 1–4, with no theory.

Step 6 — numerical check. On the unit S^2 ($r = 1$), for $\ell = 2.000000$ one has $\arccos(\cos 2) = 2.000000$: no gain, the arc is minimizing (indeed $2 < \pi$). For $\ell = \pi = 3.141593$: $\arccos(\cos \pi) = 3.141593$, still minimizing, but this is the limiting case. For $\ell = 4.000000$: $\arccos(\cos 4) =$

$2.283185 = 2\pi - 4$, with gain 1.716815. For $\ell = 5.000000$: $\arccos(\cos 5) = 1.283185 = 2\pi - 5$, with gain 3.716815. ✓ The values $2\pi - 4 = 2.283185$ and $2\pi - 5 = 1.283185$ agree with the numerical arccosine. ✓ □

Remark 9.2: What we import: Jacobi fields and conjugate points. Proposition 9.5 exhibits a Jacobi field on the sphere and verifies its equation by hand; Theorem 9.5 concludes on the sphere with an elementary argument. Two general statements remain **imported**, and are used in no proof of this chapter: (a) on an arbitrary surface the variation fields of families of geodesics satisfy the Jacobi equation $\nabla_s^2 J + K J = 0$; (b) a geodesic ceases to be minimizing as soon as it passes the first conjugate point (Jacobi's theorem). The proofs require the covariant derivative and the second variation formula for length, which are not developed in the treatise; the reference is do Carmo, *Differential Geometry of Curves and Surfaces*, Chapter 5, and Kühnel, *Differential Geometry: Curves — Surfaces — Manifolds* (sources [2] and [4]). Nothing proved above depends on (a) or (b): they serve only to frame the spherical phenomenon as a case of a general rule.

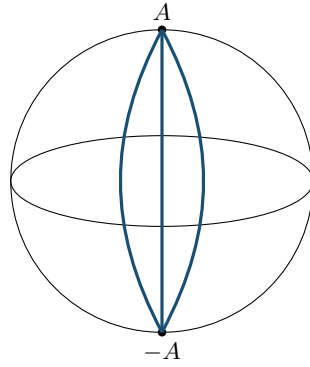


Figure 9.3: The conjugate point. All the geodesics issuing from A — the meridians — meet again at the antipode $-A$ after length πr , whatever the initial direction (Proposition 9.5). Before $-A$ each arc is the shortest path; beyond $-A$ none of them is (Theorem 9.5).

Exercise 9.3. Set up and reduce to quadrature, using the Beltrami identity, the Euler–Lagrange equation of the *brachistochrone*: the curve of fastest descent under gravity between two points, described by the functional $J[y] = \int_0^{x_1} \sqrt{\frac{1+y'^2}{2gy}} dx$ (with $y > 0$ measured downwards).

Solution. The Lagrangian $L(y, v) = \sqrt{(1+v^2)/(2gy)}$ does not depend explicitly on x and is C^2 on the open set $\{y > 0\} \times \mathbb{R}$: Corollary 9.1 applies. One computes

$$\frac{\partial L}{\partial v} = \frac{1}{\sqrt{2gy}} \cdot \frac{v}{\sqrt{1+v^2}},$$

hence

$$H = \frac{\partial L}{\partial v} y' - L = \frac{1}{\sqrt{2gy}} \left(\frac{y'^2}{\sqrt{1+y'^2}} - \sqrt{1+y'^2} \right) = \frac{1}{\sqrt{2gy}} \cdot \frac{y'^2 - (1+y'^2)}{\sqrt{1+y'^2}} = -\frac{1}{\sqrt{2gy}\sqrt{1+y'^2}}.$$

The Beltrami identity gives $H = \text{const}$, that is

$$y(1+y'^2) = k \quad \text{a positive constant.}$$

This is the equation of the cycloid: setting $y = \frac{k}{2}(1 - \cos \tau)$ one gets $y'^2 = k/y - 1 = \frac{2 - (1 - \cos \tau)}{1 - \cos \tau} = \frac{1 + \cos \tau}{1 - \cos \tau} = \cot^2 \frac{\tau}{2}$, whence $dx = dy/y' = \frac{k}{2} \sin \tau \cdot \tan \frac{\tau}{2} d\tau = \frac{k}{2}(1 - \cos \tau) d\tau$ and therefore $x = \frac{k}{2}(\tau - \sin \tau) + x_0$: the parametrization of the cycloid of Chapter 2. *Check:* $\sin \tau \tan(\tau/2) = 2 \sin \frac{\tau}{2} \cos \frac{\tau}{2} \cdot \frac{\sin(\tau/2)}{\cos(\tau/2)} = 2 \sin^2 \frac{\tau}{2} = 1 - \cos \tau$. ✓ The brachistochrone is thus an arc of a cycloid — the same curve whose evolute is a cycloid, and the same one that solves the isochronous pendulum problem.

9.7. Industrial applications

Optimal trajectory planning. The calculus of variations is the foundation of optimal planning: one minimizes a functional (length, energy, time, jerk) and the Euler–Lagrange equation (Theorem 9.1) gives the necessary optimality conditions. In controllers and off-line programming toolboxes (**ABB** RobotStudio, **KUKA**.Sim, **FANUC** ROBOGUIDE, **Yaskawa** MotoSim) jerk-limited motion profiles and minimum-energy trajectories derive from this principle; on motion PLCs (**Siemens**, **Rockwell**, **Schneider**) trapezoidal and S-curve velocity profiles are the discrete implementation of constraints on the derivative of the acceleration. Geodesics are the “pure” case — minimal length on the configuration space — extended in practice with control-effort terms and constraints, solved numerically by direct collocation methods or by variational integrators.

Machining on developable surfaces. Proposition 9.2 and Proposition 9.1 are the reason why cutting, welding or deposition along a cylinder or a cone are programmed *in the developed plane*: the three-dimensional problem reduces to a two-dimensional one, and the straight trajectories of the plane lift to exact helices and conical spirals. The quantitative criterion telling when this is legitimate is the vanishing of the Gaussian curvature (Chapter 10).

Great-circle navigation and Clairaut. Clairaut’s relation (Theorem 9.4) is the mathematical form of a fact well known to navigators: along a great-circle route the product of the cosine of the latitude and the sine of the azimuth stays constant. It is the invariant that allows one to compute in advance the maximum latitude reached by a route, without integrating the geodesic equation (Exercise 9.6).

Filament winding. In the production of composite tanks and pipes the fibre must follow a geodesic of the mandrel in order not to slip: on a cylindrical mandrel these are the helices (Proposition 9.1), on a spherical cap the arcs of great circles (Theorem 9.3), and on general surfaces of revolution the trajectory is computed from Clairaut’s invariant.

9.8. Assigned exercises

Exercise 9.4. Goal: apply Proposition 9.4 and Clairaut’s relation to a surface whose parallel radii are not monotone. **Estimated time:** 25 minutes.

Consider the torus of revolution with $f(v) = a + b \cos v$, $h(v) = b \sin v$, $a > b > 0$. (i) Compute E , F , G and check that $G = b^2$. (ii) Determine all the parallels which are geodesics and describe them geometrically. (iii) A geodesic starts from a point of the outer equator making an angle ψ_0 with the meridian. Using Theorem 9.4, determine the minimum radius of the parallel it can reach and discuss for which ψ_0 it gets as far as the inner equator. Carry out the computation for $a = 2$, $b = 1$ and $\psi_0 = 30^\circ$.

Exercise 9.5. Goal: use the development of the cone to compute a geodesic and recognize its self-intersection. **Estimated time:** 30 minutes.

Let K_α be the cone of half-angle $\alpha = 20^\circ$ with the vertex removed. (i) Compute the opening β of the development sector and check that $\beta < \pi$. (ii) On the cone take the two points $P_0 = X(0, 3)$ and $P_1 = X(2\pi/3, 3)$. Determine, in the developed sector, the segment joining them, its distance d from the vertex and its length; compare the latter with the length of the path $P_0 \rightarrow \text{vertex} \rightarrow P_1$ along two generators (which equals 6). (iii) Decide whether the geodesic of part (ii), when extended, self-intersects, and compute at which value of u this happens.

Exercise 9.6. Goal: use Clairaut's invariant in navigation and check it against numerical integration of the geodesic. **Estimated time:** 20 minutes.

On the spherical Earth a great-circle route starts from latitude $\phi_0 = 45^\circ$ N with azimuth $Az = 60^\circ$ (measured clockwise from North). (i) Write Clairaut's relation in the form $\cos \phi \sin Az = \text{const}$, justifying its equivalence with Theorem 9.4. (ii) Compute the maximum latitude reached by the route, knowing that at its northernmost point the route is tangent to the parallel. (iii) Check that the same route, continuing, reaches the minimum latitude $-\phi_{\max}$ and explain why.

9.9. Worked exercises

Exercise 9.7. On the cylinder of radius $R = 1$ consider the points $P_0 = X(0, 0)$ and $P_1 = X(\pi/2, 1)$. Determine all the geodesics joining them, their lengths and the shortest one; compare with the chord in space and with the path “first turn, then climb”.

Solution. Strategy. Develop the cylinder onto the plane (Proposition 9.1(i)), count the segments joining the images of the two points taking periodicity into account, and compare the lengths with two explicit competing paths.

Step 1 — the development. The map $X(u, v) \mapsto (Ru, v) = (u, v)$ (since $R = 1$) is a local isometry onto the plane; P_0 goes to $(0, 0)$ and P_1 goes to all the points $(\pi/2 + 2\pi n, 1)$ with $n \in \mathbb{Z}$, because u and $u + 2\pi$ determine the same point of the cylinder.

Step 2 — the geodesics and their lengths. By Lemma 9.4 the geodesics correspond to the segments $(0, 0) \rightarrow (\pi/2 + 2\pi n, 1)$, of length

$$\ell_n = \sqrt{\left(\frac{\pi}{2} + 2\pi n\right)^2 + 1}.$$

There are infinitely many: one for each (signed) whole number of turns around the axis.

Step 3 — the numerical values.

$$\ell_{-2} = 11.040953, \quad \ell_{-1} = 4.817324, \quad \ell_0 = 1.862096, \quad \ell_1 = 7.917388, \quad \ell_2 = 14.172491.$$

The minimum is $\ell_0 = 1.862096$.

Step 4 — independent check. In closed form $\ell_0 = \sqrt{(\pi/2)^2 + 1} = \sqrt{2.467401 + 1} = \sqrt{3.467401} = 1.862096$; the numerical computation gives the same value to six decimals. ✓ Moreover $\ell_{-1} = \sqrt{(\pi/2 - 2\pi)^2 + 1} = \sqrt{(-4.712389)^2 + 1} = \sqrt{23.206609} = 4.817324$ ✓, and the asymmetry $\ell_{-1} < \ell_1$ is correct because $|\pi/2 - 2\pi| = 4.712389 < |\pi/2 + 2\pi| = 7.853982$.

Step 5 — the competitors. The *chord* in space between $P_0 = (1, 0, 0)$ and $P_1 = (0, 1, 1)$ measures $\sqrt{1 + 1 + 1} = \sqrt{3} = 1.732051$: smaller than ℓ_0 , but it is not a path on the cylinder. The path “first turn along the circle $z = 0$, then climb along the generator” measures

$R \cdot \pi/2 + 1 = 1.570796 + 1 = 2.570796 > 1.862096$: it is an admissible path but not a geodesic, and it is longer. ✓

Interpretation. On the cylinder the shortest path exists and is unique, but there are infinitely many geodesics between two points: stationarity does not suffice to select the minimum. It is the same phenomenon as on the sphere (Theorem 9.5), with one difference: here the failure of minimality comes from topology (the cylinder is not simply connected) and not from conjugate points — indeed the cylinder has vanishing Gaussian curvature and no conjugate points.

Typical errors. (a) Forgetting periodicity and concluding that the geodesic is unique. (b) Confusing the chord with the distance on the surface. (c) Using the formula for ℓ_n with Δu taken outside $(-\pi, \pi]$, and getting the minimizing index wrong.

Extension. For $\Delta u = \pi$ (diametrically opposite points at the same height) one has $\ell_0 = \ell_{-1} = \sqrt{\pi^2 + \Delta v^2}$: the minimum exists but is *not* unique, in perfect analogy with the antipodal points of the sphere (Exercise 7.3).

Exercise 9.8. On the unit sphere compute the length of the path along the parallel of latitude 60° N between two points 90° apart in longitude, and compare it with the length of the geodesic joining them. Check that the ratio is consistent with the fact that the parallel is not a geodesic.

Solution. Strategy. Compute separately the length along the parallel (elementary) and the geodesic distance (with the scalar product), then check the latter with the spherical law of cosines.

Step 1 — the two points. With latitude $\phi = 60^\circ$, that is colatitude $\vartheta_0 = 30^\circ$, and longitudes 0° and 90° :

$$A = (\sin 30^\circ, 0, \cos 30^\circ) = (0.5, 0, 0.866025), \quad B = (0, 0.5, 0.866025).$$

Check: $\|A\|^2 = 0.25 + 0.75 = 1$ ✓.

Step 2 — the path along the parallel. The parallel has radius $\sin \vartheta_0 = 0.5$ and the arc subtended by $90^\circ = \pi/2$ measures

$$\ell_{\text{par}} = 0.5 \cdot \frac{\pi}{2} = 0.785398.$$

Step 3 — the geodesic. $\langle A, B \rangle = 0.5 \cdot 0 + 0 \cdot 0.5 + 0.866025^2 = 0.750000$, hence

$$\ell_{\text{geo}} = \arccos(0.750000) = 0.722734.$$

Step 4 — independent check with the spherical law of cosines. In the triangle pole– A – B the two polar sides measure $\vartheta_0 = 30^\circ = 0.523599$ and the angle at the pole is 90° . Proposition 7.2 of Chapter 7 gives

$$\cos c = \cos^2 30^\circ + \sin^2 30^\circ \cos 90^\circ = 0.750000 + 0 = 0.750000,$$

that is $c = 0.722734$. ✓ The two computations agree to six decimals.

Step 5 — the comparison. $\ell_{\text{geo}} = 0.722734 < 0.785398 = \ell_{\text{par}}$: the geodesic is shorter by 8.0% (the parallel exceeds it by $0.785398/0.722734 = 1.086700$, that is by 8.67%). Hence the parallel is not minimizing, and a fortiori not a geodesic (Proposition 9.4), consistently with its geodesic curvature $\kappa_g = |\cot 30^\circ| = 1.732051 \neq 0$ (Exercise 9.2).

Interpretation. This is why transoceanic routes “climb” towards the pole instead of following the parallel: the geodesic cuts across higher latitudes. The gain grows with latitude:

at latitude 80° , with the same difference of longitude, the parallel measures $\sin 10^\circ \cdot \pi/2 = 0.272700$ and the geodesic $\arccos(\cos^2 10^\circ + \sin^2 10^\circ \cos 90^\circ) = \arccos(0.969846) = 0.245885$, an excess of 10.9%.

Typical errors. (a) Using the radius of the sphere instead of the radius of the parallel to compute ℓ_{par} . (b) Confusing latitude and colatitude in the scalar product. (c) Applying the *plane* law of cosines to a spherical triangle.

Extension. Repeat at the equator ($\vartheta_0 = 90^\circ$): $\ell_{\text{par}} = \pi/2 = \ell_{\text{geo}}$, because the equator *is* a geodesic. The gain is zero, as it must be.

Exercise 9.9. Check that, on the unit sphere, Clairaut's relation $\sin \vartheta \sin \psi = \text{const}$ is equivalent to the fact that every great circle lies on a plane through the centre, and use it to determine the minimum colatitude reached by a great circle crossing the equator at angle ι .

Solution. Strategy. Compute Clairaut's constant at two particular points of the same geodesic — the crossing of the equator and the point of maximum latitude — and compare the result with the vectorial description of the great circle.

Step 1 — the constant at the equator. On the unit sphere the radius of the parallel of colatitude ϑ is $r(\vartheta) = \sin \vartheta$. At the equator $\vartheta = \pi/2$, hence $r = 1$; if the geodesic crosses the equator making the angle ι with the equator itself, the angle with the meridian is $\psi = \pi/2 - \iota$ and

$$c = r \sin \psi = 1 \cdot \sin \left(\frac{\pi}{2} - \iota \right) = \cos \iota.$$

Step 2 — the point of maximum latitude. At the northernmost point the geodesic is tangent to the parallel, that is $\psi = \pi/2$ and $\sin \psi = 1$. Hence

$$c = r(\vartheta_{\min}) \cdot 1 = \sin \vartheta_{\min}, \quad \text{so that} \quad \sin \vartheta_{\min} = \cos \iota, \quad \vartheta_{\min} = \frac{\pi}{2} - \iota.$$

The minimum colatitude is $\pi/2 - \iota$: the maximum latitude is exactly ι .

Step 3 — independent vectorial check. The great circle crossing the equator at angle ι lies in the plane orthogonal to the unit vector $n = (0, -\sin \iota, \cos \iota)$ (for a suitable choice of the starting longitude). A point $P = (\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta)$ belongs to the circle if and only if $\langle P, n \rangle = 0$, that is

$$-\sin \vartheta \sin \varphi \sin \iota + \cos \vartheta \cos \iota = 0, \quad \text{i.e.} \quad \tan \vartheta \sin \varphi = \cot \iota.$$

The colatitude is minimal when $\tan \vartheta$ is minimal positive, that is for $\sin \varphi = 1$: $\tan \vartheta_{\min} = \cot \iota$, hence $\vartheta_{\min} = \pi/2 - \iota$. ✓ The result agrees with Step 2.

Step 4 — numerical values. For $\iota = 30^\circ$: $c = \cos 30^\circ = 0.866025$ and $\vartheta_{\min} = 60^\circ$, that is maximum latitude 30° . For $\iota = 60^\circ$: $c = 0.500000$, maximum latitude 60° . For $\iota = 90^\circ$ (a meridian): $c = 0$ and the geodesic passes through the pole, maximum latitude 90° . ✓ The case $\iota = 0$ gives $c = 1$ and $\vartheta_{\min} = \pi/2$: the equator itself, which never moves away. ✓

Interpretation. Clairaut's constant c is the cosine of the inclination of the plane of the great circle with respect to the equatorial plane: a single number summarizing the whole geodesic. It is the same invariant which, in mechanics, is called the angular momentum about the axis.

Typical errors. (a) Confusing the angle with the meridian and the angle with the parallel (the relation has $\sin$ with the first and $\cos$ with the second). (b) Using the latitude instead of the radius of the parallel, which on the unit sphere is $\cos(\text{lat})$ and not lat . (c) Forgetting that ψ is not constant along the geodesic: what is constant is the product $r \sin \psi$.

Extension. The same constant answers at once another question: a geodesic with a given c never touches the parallels of radius $r < c$, because $\sin \psi \leq 1$ forces $r \geq c$. On the sphere

this excludes a polar cap of colatitude $\arcsin c$; on the torus it excludes a band around the inner equator (Exercise 9.4).

9.10. Self-assessment quiz

1. A curve γ on a surface S is a *geodesic*, by definition (Theorem 9.2), when
 - A) $\gamma'' = 0$
 - B) γ'' is orthogonal to the tangent plane at every point
 - C) γ is the shortest path between its endpoints
 - D) γ has zero curvature in $\mathbb{R}^3$
2. In the proof of the Euler–Lagrange equation, the hypothesis that makes the boundary term in the integration by parts vanish is
 - A) L of class C^2
 - B) $\eta(a) = \eta(b) = 0$
 - C) q of class C^2
 - D) U open
3. For the sphere of radius r in coordinates (φ, ϑ) the Christoffel symbol Γ_{11}^2 equals
 - A) $\cot \vartheta$
 - B) $-\sin \vartheta \cos \vartheta$
 - C) $-r^2 \sin \vartheta \cos \vartheta$
 - D) 0
4. A great-circle route starts at latitude 45° N with azimuth 60° . The maximum latitude it reaches equals, in degrees,
 - A) 45.000000
 - B) 52.238756
 - C) 60.000000
 - D) 30.000000
5. In the proof that the geodesics of S^2 are the great circles, the step that turns the condition $\gamma'' = \lambda\gamma$ into an integrable linear equation is
 - A) differentiating $\|\gamma\|^2 = r^2$, which shows that $\lambda = -c^2/r^2$ is constant
 - B) the fundamental lemma of the calculus of variations
 - C) Clairaut’s relation
 - D) the Picard–Lindelöf theorem
6. A student claims: “the arc of a great circle of length 4 on the unit sphere is the shortest path between its endpoints, because it is a geodesic”. The error is
 - A) it is not a geodesic
 - B) being a geodesic is necessary but not sufficient: the distance is $2\pi - 4 = 2.283185 < 4$
 - C) the length is miscomputed
 - D) none, the claim is correct
7. On a cone of half-angle $\alpha = 20^\circ$, a geodesic which is not a generator
 - A) never self-intersects
 - B) self-intersects, because $\beta = 2\pi \sin 20^\circ = 2.148573 < \pi$
 - C) is always closed
 - D) coincides with a parallel
8. Clairaut’s relation *together with its converse* (a constant-speed curve satisfying it and containing no arc of a parallel is a geodesic) is stated as
 - A) the Gauss–Bonnet theorem
 - B) Proposition 4.4 of Shifrin, *Differential Geometry: A First Course in Curves and Surfaces*, pp. 73–75
 - C) Meusnier’s theorem
 - D) Dandelin’s theorem

9.11. Teacher’s key

BEST PRACTICE

Q1 — answer B. *Rationale:* Theorem 9.2 states the equivalence between purely normal acceleration, the equation with the Christoffel symbols and stationarity of the energy; the working definition is (i). *Distractors:* A holds only in the plane (a curved geodesic in space has $\gamma'' \neq 0$); C is implied by minimality but not equivalent, as Theorem 9.5 shows; D confuses curvature in $\mathbb{R}^3$ with geodesic curvature. *Diagnosed error:* identifying geodesic with shortest path. *Source:* Shifrin, §2.4, p. 66 ff. *Prerequisite:* decomposition of a vector into tangential and normal parts. *Remediation:* redo Step 1 of the proof of Theorem 9.2 on the great circle and on the parallel.

Q2 — answer B. *Rationale:* Step 3 uses $[p_k \eta^k]_a^b = 0$, which holds precisely because the variation vanishes at the endpoints; the other hypotheses are used elsewhere (A and C for the differentiability of p_k , D for the good definition of the varied curves in Step 1). *Distractors:* all three are true hypotheses of the theorem, but not the one that kills the boundary term: the question checks that one knows what each hypothesis is for. *Diagnosed error:* listing hypotheses without knowing where they act. *Source:* Gelfand–Fomin, *Calculus of Variations*, Ch. 1. *Prerequisite:* integration by parts. *Remediation:* redo Step 3 with $\eta(b) \neq 0$ and see what natural boundary condition appears.

Q3 — answer B. *Rationale:* with $E = r^2 \sin^2 \vartheta$, $F = 0$, $G = r^2$ one has $\Gamma_{11}^2 = -E_\vartheta/(2G) = -2r^2 \sin \vartheta \cos \vartheta/(2r^2) = -\sin \vartheta \cos \vartheta$: the radius cancels. *Distractors:* A is Γ_{12}^1 ; C is the error of forgetting to divide by $G = r^2$; D is the value of the other four symbols. *Diagnosed error:* failing to cancel the scale factors and believing that the Christoffel symbols depend on the radius. *Source:* computation in Step 5 of the proof of Lemma 9.3, with numerical check by finite differences. *Prerequisite:* inverting a diagonal matrix. *Remediation:* recompute all six symbols for $r = 1$ and for $r = 2$ and observe that they agree.

Q4 — answer B. *Rationale:* Clairaut's constant equals $\cos 45^\circ \sin 60^\circ = 0.707107 \times 0.866025 = 0.612372$; at the northernmost point $\sin \text{Az} = 1$, hence $\cos \phi_{\max} = 0.612372$ and $\phi_{\max} = \arccos(0.612372) = 52.238756^\circ$. *Distractors:* A is the starting latitude (the route climbs, it does not stay); C confuses azimuth with latitude; D is $90^\circ - 60^\circ$. *Diagnosed error:* not recognizing that the invariant links two different points of the same geodesic. *Source:* Theorem 9.4 and a check by numerical integration of the geodesic equations of the sphere, which gives 52.2386° against the closed-form value 52.238756° . *Prerequisite:* arc-cosine. *Remediation:* repeat the computation for $\text{Az} = 90^\circ$ (does the route stay on the parallel? no: check) and for $\text{Az} = 0^\circ$.

Q5 — answer A. *Rationale:* Step 2 differentiates $\|\gamma\|^2 = r^2$ twice, uses the constancy of the speed (Step 6 of Theorem 9.2) and obtains $\lambda r^2 = -c^2$; only then is $\gamma'' + \omega^2 \gamma = 0$ linear with constant coefficients and integrable. *Distractors:* B is used in Theorem 9.1, not here; C gives a first integral but not the general solution; D is deliberately avoided (Step 6 of the same proof). *Diagnosed error:* not distinguishing between an equation with variable coefficients and one with constant coefficients. *Source:* do Carmo, Chapter 4 (source [2]); the proof given here is vectorial and self-contained. *Prerequisite:* second-order linear equations with constant coefficients. *Remediation:* rewrite Step 2 forgetting the constancy of the speed and see that the equation cannot be integrated in closed form.

Q6 — answer B. *Rationale:* by Theorem 9.5(ii) with $r = 1$ and $\ell = 4 > \pi$ the distance equals $\arccos(\cos 4) = 2\pi - 4 = 2.283185$, realized by the complementary arc. *Distractors:* A is false (the arc is a geodesic, Theorem 9.3); C is irrelevant; D accepts the error. *Diagnosed error:* confusing a necessary with a sufficient condition. *Source:* Theorem 9.5, Step 6, numerical check. *Prerequisite:* periodicity of the cosine. *Remediation:* tabulate ℓ and $\arccos(\cos \ell)$ for $\ell = 1, 2, 3, \pi, 4, 5, 6$ and locate where the two columns diverge.

Q7 — answer B. *Rationale:* by Proposition 9.3 self-intersection occurs if and only if $\alpha < 30^\circ$; here $\alpha = 20^\circ$ and $\beta = 2\pi \sin 20^\circ = 6.283185 \times 0.342020 = 2.148573 < \pi = 3.141593$. *Distractors:* A is the correct answer for $\alpha > 30^\circ$; C is false (the geodesics of the cone are not closed, apart from degenerate cases); D confuses geodesic and parallel. *Diagnosed error:* failing to translate a condition on the surface into one on the development. *Source:* Proposition 9.3, Step 5. *Prerequisite:* development of the cone. *Remediation:* draw the sector for $\alpha = 20^\circ, 30^\circ$ and 40° and trace a segment in each.

Q8 — answer B. *Rationale:* the statement with the clause “no part of the curve lies in a parallel” — indispensable, as Step 6 of the proof of Theorem 9.4 shows — is Proposition 4.4 of Shifrin, pp. 73–75, section 2.4 “Covariant Differentiation, Parallel Translation, and Geodesics”. *Distractors:* A, C and D are real theorems but on different subjects; they check that the student does not “recognize” a name at random. *Diagnosed error:* citing from memory without checking. *Source:* the same set of notes, verified on 5 September 2026 (source [1]). *Prerequisite:* none. *Remediation:* open the source and compare the statement with Theorem 9.4, noting that Shifrin uses the angle with the parallel (cosine) and we the angle with the meridian (sine).

Answers to the assigned exercises.

Exercise 9.4. (i) By Lemma 9.6, $E = (a + b \cos v)^2$, $F = 0$ and $G = f'^2 + h'^2 = b^2 \sin^2 v + b^2 \cos^2 v = b^2$: constant. (ii) By Proposition 9.4 the geodesic parallels are those with $f'(v) = -b \sin v = 0$, that is $v = 0$ (outer equator, radius $a + b$) and $v = \pi$ (inner equator, radius $a - b$); no others. With $a = 2$, $b = 1$: radii 3 and 1. (iii) Clairaut’s constant is $c = r_0 \sin \psi_0 = (a + b) \sin \psi_0$; the minimum reachable radius is $r_{\min} = c$, because $\sin \psi \leq 1$ forces $r \geq c$. With $a = 2$, $b = 1$, $\psi_0 = 30^\circ$: $c = 3 \times 0.5 = 1.5$, hence the geodesic does not go below the parallel of radius 1.5; since $a - b = 1 < 1.5$, it does *not* reach the inner equator. The condition for reaching it is $c \leq a - b$, that is $\sin \psi_0 \leq (a - b)/(a + b) = 1/3 = 0.333333$, that is $\psi_0 \leq 19.471221^\circ$.

Exercise 9.5. (i) $\beta = 2\pi \sin 20^\circ = 6.283185 \times 0.342020 = 2.148573 < \pi = 3.141593$ ✓. (ii) In the development the two points have polar coordinates $(\theta, \rho) = (0, 3)$ and $(\frac{2\pi}{3} \sin 20^\circ, 3) = (0.716191, 3)$; the angle between them is $\Delta\theta = 0.716191$ rad. The segment joining them has length $2 \cdot 3 \sin(\Delta\theta/2) = 6 \times \sin(0.358096) = 6 \times 0.350488 = 2.102928$ and lies at distance $d = 3 \cos(\Delta\theta/2) = 3 \times 0.936570 = 2.809710$ from the vertex. The path through the vertex measures $3 + 3 = 6 > 2.102928$: the geodesic is far shorter. ✓(iii) Yes: by Proposition 9.3, since $\alpha = 20^\circ < 30^\circ$, the extended geodesic self-intersects; this happens for $u = u_0 \pm \pi$, that is when the two branches are 2π apart in u . Here $u_0 = \pi/3$ (by symmetry between 0 and $2\pi/3$), hence the self-intersection is at the parameters $u = \pi/3 - \pi = -2.094395$ and $u = \pi/3 + \pi = 4.188790$, which are the same point on the cone; the radius is $v = d / \cos(\pi \sin 20^\circ) = 2.809710 / \cos(1.074287) = 2.809710 / 0.477159 = 5.888$.

Exercise 9.6. (i) On the unit sphere the radius of the parallel of latitude ϕ is $r = \cos \phi$ and the angle with the meridian is the azimuth Az (measured from North): the relation $r \sin \psi = \text{const}$ of Theorem 9.4 becomes $\cos \phi \sin \text{Az} = \text{const}$. (ii) $c = \cos 45^\circ \sin 60^\circ = 0.707107 \times 0.866025 = 0.612372$; at the point of maximum latitude $\sin \text{Az} = 1$, hence $\phi_{\max} = \arccos(0.612372) = 52.238756^\circ$. Direct numerical integration of the geodesic equations of the sphere gives 52.2386° : agreement within the discretization error. ✓(iii) The great circle is symmetric about the centre of the sphere: if it contains the point of latitude $\phi_{\max}$, it also contains its antipode, of latitude $-\phi_{\max}$, and by Theorem 9.3 the geodesic traverses the whole of it. The route therefore oscillates between -52.238756° and $+52.238756^\circ$, and never enters the two polar caps of half-width 37.761244° .

9.12. Chapter formulary

Calculus of variations

$$J[q] = \int_a^b L(s, q, q') ds; \quad \frac{d}{ds} \frac{\partial L}{\partial q'^k} - \frac{\partial L}{\partial q^k} = 0$$

$$\text{Beltrami (if } \partial_s L = 0) : \quad H = \frac{\partial L}{\partial q'^k} q'^k - L = \text{const}$$

$$L(\gamma)^2 \leq 2(b-a)E(\gamma), \quad E(\gamma) = \frac{1}{2} \int_a^b \|\gamma'\|^2 ds, \quad \text{equality} \iff \|\gamma'\| = \text{const}$$

Geodesics of a surface

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}); \quad X_{ij} = \Gamma_{ij}^k X_k + h_{ij} N$$

$$\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j = 0 \iff \gamma'' \perp T_p S \iff \delta E = 0; \quad \|\gamma'\| = \text{const}$$

$$F = 0 : \quad \Gamma_{11}^1 = \frac{E_u}{2E}, \quad \Gamma_{12}^1 = \frac{E_v}{2E}, \quad \Gamma_{22}^1 = -\frac{G_u}{2E}, \quad \Gamma_{11}^2 = -\frac{E_v}{2G}, \quad \Gamma_{12}^2 = \frac{G_u}{2G}, \quad \Gamma_{22}^2 = \frac{G_v}{2G}$$

Sphere of radius r , coordinates (φ, ϑ)

$$E = r^2 \sin^2 \vartheta, \quad F = 0, \quad G = r^2; \quad \Gamma_{12}^1 = \cot \vartheta, \quad \Gamma_{11}^2 = -\sin \vartheta \cos \vartheta$$

$$\ddot{\varphi} + 2 \cot \vartheta \dot{\varphi} \dot{\vartheta} = 0, \quad \ddot{\vartheta} - \sin \vartheta \cos \vartheta \dot{\varphi}^2 = 0$$

$$\gamma(s) = \cos \frac{cs}{r} P + \sin \frac{cs}{r} Q, \quad P \perp Q, \quad \|P\| = \|Q\| = r \quad (\text{great circles})$$

$$\kappa_g(\text{parallel of colatitude } \vartheta_0) = \frac{|\cot \vartheta_0|}{r}; \quad \ell > \pi r \Rightarrow \text{dist} = r \arccos \left(\cos \frac{\ell}{r} \right) < \ell$$

Cylinder, cone, surfaces of revolution

$$\text{cylinder } R : I = R^2 du^2 + dv^2, \quad \Gamma_{ij}^k = 0, \quad \text{geodesics} = \text{helices } u = as + u_0, \quad v = bs + v_0$$

$$\text{cone } \alpha : I = v^2 \sin^2 \alpha du^2 + dv^2, \quad \beta = 2\pi \sin \alpha, \quad v = \frac{d}{\cos(\sin \alpha (u - u_0))}$$

$$\text{self-intersection} \iff \sin \alpha < \frac{1}{2} \iff \alpha < 30^\circ$$

$$\text{revolution} : E = f^2, \quad F = 0, \quad G = f'^2 + h'^2; \quad \text{Clairaut} : r \sin \psi = f^2 \dot{u} = \text{const}$$

$$\text{parallel } v_0 \text{ geodesic} \iff f'(v_0) = 0$$

Sources and bibliography

- [1] T. Shifrin, *Differential Geometry: A First Course in Curves and Surfaces*, University of Georgia, April 2021 edition, 125 pp. Chapter 2 *Surfaces: Local Theory*, §1 *Parametrized Surfaces and the First Fundamental Form* (p. 35), §2 *The Gauss Map and the Second Fundamental Form* (p. 44), §3 *The Codazzi and Gauss Equations and the Fundamental Theorem of Surface Theory* (p. 57), §4 *Covariant Differentiation, Parallel Translation, and Geodesics* (p. 66); Proposition 4.4 *Clairaut's relation*, pp. 73–75, with the converse and the clause on parallels (Theorem 9.4); solution of exercise 2.4.4, $\kappa_g = \cot u_0$ for a parallel of the sphere (Exercise 9.2); solution of exercise 2.1.7, characterization of local isometry with the plane by $E = G = 1, F = 0$; Appendix, §3 *Differential Equations*, Theorem 3.1 *Fundamental Theorem of ODE's*, pp. 118–119, local existence and uniqueness with C^k dependence on the initial data, imported in Remark 9.1. File

<https://math.franklin.uga.edu/sites/default/files/inline-files/ShifrinDiffGeo.pdf>;
verified on 5 September 2026.

- [2] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, revised and updated edition, Dover, Mineola (NY), 2016 (first edition Prentice-Hall, 1976), ISBN 978-0-486-80699-0. Chapter 3 *The Geometry of the Gauss Map*, Chapter 4 *The Intrinsic Geometry of Surfaces* — in particular §4-3 *The Gauss Theorem and the Equations of Compatibility*, p. 235 — and Chapter 5 *Global Differential Geometry* (Jacobi fields and conjugate points, imported in Remark 9.2). Verified on 5 September 2026.
- [3] I. M. Gelfand, S. V. Fomin, *Calculus of Variations*, translated and edited by R. A. Silverman, Prentice-Hall, Englewood Cliffs (NJ), 1963; Dover reprint, 2000. Eight chapters and two appendices; Chapter 1 treats the simplest variational problem, the fundamental lemma and the Euler equation, which are the content of Lemma 9.1 and Theorem 9.1. Verified on 5 September 2026.
- [4] W. Kühnel, *Differential Geometry: Curves — Surfaces — Manifolds*, third edition, Student Mathematical Library vol. 77, American Mathematical Society, Providence (RI), 2015 (second edition: vol. 16, 2006), ISBN 978-1-4704-2320-9. English translation by B. Hunt of the German text *Differentialgeometrie*; local geometry of surfaces, intrinsic geometry, spaces of constant curvature. Verified on 5 September 2026.
- [5] Yan-Bin Jia, *Geodesics* (Com S 477/577 Notes), Department of Computer Science, Iowa State University, 5 November 2015. Notes for the course *Foundations of Robotics and Computer Vision*: the geodesic equation with Christoffel symbols, geodesics as shortest paths, robotic applications. Verified on 5 September 2026.
- [6] Massachusetts Institute of Technology, OpenCourseWare, course 18.950 *Differential Geometry*, Fall 2008, chapter-by-chapter notes (local and global geometry of plane curves, local and global geometry of hypersurfaces, geometry of lengths and distances). File <https://ocw.mit.edu/courses/18-950-differential-geometry-fall-2008/>; verified on 5 September 2026.
- [7] A. C. Clairaut, memoirs to the Académie Royale des Sciences, Paris, years 1733 and 1739, on the curves of shortest length on surfaces of revolution: historical origin of the relation bearing his name. Cited here as a historical reference; the statement and proof used in Theorem 9.4 are those of source [1].

Local theory of surfaces

A surface carries two distinct measures. The *first fundamental form* measures lengths, angles and areas *on* the surface: it is everything a two-dimensional inhabitant could detect without ever leaving its own world. The *second fundamental form* measures how the surface bends *in* space: it is what one sees from outside. From them come the principal curvatures, the mean curvature and the Gaussian curvature; and from them comes the result that Gauss himself judged remarkable enough to call it *egregium*: the Gaussian curvature, defined by means of the second form, is in fact expressible through the first form alone. The two-dimensional inhabitant, who cannot see its surface from outside, can nevertheless measure its curvature — and discover, for instance, that its world is not a plane.

INFO

The idea in one line. The product of the two principal curvatures, $K = \kappa_1 \kappa_2$, seems to depend on the way the surface is embedded in space, because it is defined with the second form; the Gauss equation shows instead that K is written with E, F, G and their derivatives alone, so that two isometric surfaces have the same K point by point — and no piece of a sphere ($K = 1/r^2$) can be flattened onto a plane ($K = 0$) without stretching it.

INFO

Level and prerequisites. Level C. Sections 10.1 and 10.2 are level B–C: the scalar and cross product in $\mathbb{R}^3$, partial derivatives and the linear algebra of quadratic forms in two variables suffice. Sections 10.3, 10.5 and 10.6 are level C. Section 10.4, which contains the proof of the *Theorema Egregium*, is level C–D. Prerequisites within the treatise: curvature of a plane curve and of a space curve (Chapters 2 and 3); sphere, spherical coordinates and area element (Chapter 7); Christoffel symbols, geodesic equation and local isometries (Chapter 9, from which this chapter takes over without re-proving them Lemma 9.3 and Lemma 9.4); Gauss–Bonnet for the geodesic triangle (Chapter 7) and global Gauss–Bonnet (Chapter 4).

Warning. Conventions of the chapter.

- The index conventions of Chapter 9 are in force: Latin indices in $\{1, 2\}$, Einstein summation convention, $X_i = \partial X / \partial u^i$, $X_{ij} = \partial^2 X / \partial u^i \partial u^j$; X is always regular and of class C^3 .
- $g_{ij} = \langle X_i, X_j \rangle$ with $E = g_{11}$, $F = g_{12}$, $G = g_{22}$ and $D = EG - F^2 > 0$; $h_{ij} = \langle X_{ij}, N \rangle$ with $e = h_{11}$, $f = h_{12}$, $g = h_{22}$.
- The unit normal is always the one *induced by the parametrization*, $N = (X_1 \times X_2) / \|X_1 \times X_2\|$. Changing orientation, N , II , the principal curvatures and H change sign; K does *not*, being a product of two principal curvatures. Whenever the sign of H is stated, the chosen normal is indicated.
- **Beware of the clash of symbols:** g denotes both the matrix g_{ij} of the first form and the coefficient h_{22} of the second. The context always distinguishes them (the first carries indices, the second appears next to e and f); where the risk of confusion is real

the index notation h_{ij} is used.

- Decimal separators are points: 0.5.

10.1. The first fundamental form

Definition 10.1: Regular surface and first fundamental form. A set $S \subset \mathbb{R}^3$ is a **regular surface** if every point of it has a neighbourhood V in S for which there exist an open set $U \subseteq \mathbb{R}^2$ and a map $X : U \rightarrow V$ of class C^3 , bijective with continuous inverse, such that $X_1(q) \times X_2(q) \neq 0$ for every $q \in U$; X is called a **parametrization**. The **first fundamental form** at $p = X(q)$ is the quadratic form on the tangent plane $T_p S = \text{span}\{X_1, X_2\}$

$$I_p(w) = \langle w, w \rangle, \quad w = a^i X_i,$$

which in coordinates reads $I = E du^2 + 2F du dv + G dv^2$ with

$$E = \langle X_1, X_1 \rangle, \quad F = \langle X_1, X_2 \rangle, \quad G = \langle X_2, X_2 \rangle.$$

Lemma 10.1: Lagrange's identity and the determinant of the first form. For every pair of vectors $a, b \in \mathbb{R}^3$ one has $\|a \times b\|^2 = \|a\|^2 \|b\|^2 - \langle a, b \rangle^2$. In particular

$$\|X_1 \times X_2\|^2 = EG - F^2 = D > 0,$$

and the matrix (g_{ij}) is positive definite.

Proof. Step 1 — the identity. If a or b is zero both sides vanish. Otherwise, letting $\theta \in [0, \pi]$ be the angle between a and b , one has $\|a \times b\| = \|a\| \|b\| \sin \theta$ and $\langle a, b \rangle = \|a\| \|b\| \cos \theta$; squaring and adding,

$$\|a \times b\|^2 + \langle a, b \rangle^2 = \|a\|^2 \|b\|^2 (\sin^2 \theta + \cos^2 \theta) = \|a\|^2 \|b\|^2.$$

Step 2 — consequence for the first form. Applying this to $a = X_1$, $b = X_2$ one obtains $\|X_1 \times X_2\|^2 = EG - F^2$. Regularity gives $X_1 \times X_2 \neq 0$, hence $D > 0$.

Step 3 — positive definiteness. The matrix (g_{ij}) is symmetric, has $E = \|X_1\|^2 > 0$ and determinant $D > 0$: by Sylvester's criterion of leading principal minors it is positive definite. This is also seen directly: $I(a^i X_i) = \|a^i X_i\|^2 \geq 0$, with equality only if $a^i X_i = 0$, that is $a = 0$ by the independence of X_1, X_2 . $\square$

Proposition 10.1: What the first form measures. Let $X : U \rightarrow S$ be a parametrization. Then:

(i) the length of a curve $\gamma = X \circ u$ with $u : [a, b] \rightarrow U$ of class C^1 equals $L(\gamma) = \int_a^b \sqrt{g_{ij} \dot{u}^i \dot{u}^j} ds$;

(ii) the angle θ between two non-zero tangent vectors $w = a^i X_i$ and $z = b^j X_j$ satisfies

$$\cos \theta = \frac{g_{ij} a^i b^j}{\sqrt{g_{kl} a^k a^l} \sqrt{g_{mn} b^m b^n}};$$

(iii) the area of a region $R \subseteq S$ contained in $V = X(U)$ equals $\text{Area}(R) = \iint_{X^{-1}(R)} \sqrt{EG - F^2} du^1 du^2$, and the value does not depend on the chosen parametrization.

Proof. Step 1 — (i). By the chain rule $\gamma' = \dot{u}^i X_i$, hence $\|\gamma'\|^2 = \langle \dot{u}^i X_i, \dot{u}^j X_j \rangle = g_{ij} \dot{u}^i \dot{u}^j$; integrating the norm gives the length.

Step 2 — (ii). It is the definition of the angle between two vectors, $\cos \theta = \langle w, z \rangle / (\|w\| \|z\|)$, rewritten with $\langle w, z \rangle = g_{ij} a^i b^j$ and $\|w\|^2 = g_{kl} a^k a^l$.

Step 3 — (iii), definition of the area. The area of R is by definition the integral over $X^{-1}(R)$ of the area of the infinitesimal parallelogram spanned by $X_1 du^1$ and $X_2 du^2$, which equals $\|X_1 \times X_2\| du^1 du^2$; by Lemma 10.1 that quantity is $\sqrt{EG - F^2} du^1 du^2$.

Step 4 — (iii), independence of the parametrization. Let $\bar{X} = X \circ \phi$ with $\phi : \bar{U} \rightarrow U$ a diffeomorphism, and let J be the Jacobian matrix of ϕ . By the chain rule $\bar{X}_i = (\partial_i \phi^k) X_k$, hence

$$\bar{X}_1 \times \bar{X}_2 = (\partial_1 \phi^1 \partial_2 \phi^2 - \partial_1 \phi^2 \partial_2 \phi^1) X_1 \times X_2 = \det J \cdot X_1 \times X_2,$$

because the terms with the same factor twice vanish ($X_k \times X_k = 0$). Hence $\|\bar{X}_1 \times \bar{X}_2\| = |\det J| \|X_1 \times X_2\|$, and the change of variables formula for double integrals gives the same value for the area.

Step 5 — check on the sphere. With $X(\varphi, \vartheta)$ the spherical coordinates, Exercise 10.1 gives $\sqrt{EG - F^2} = r^2 \sin \vartheta$, that is exactly the area element of Lemma 7.1 of Chapter 7, which in turn, integrated over the whole sphere, gives $4\pi r^2$. ✓ □

Exercise 10.1. Compute the first fundamental form and the area element of the sphere of radius r parametrized by $X(\varphi, \vartheta) = r(\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta)$, with $\vartheta \in (0, \pi)$.

Solution. $X_\varphi = r(-\sin \vartheta \sin \varphi, \sin \vartheta \cos \varphi, 0)$ and $X_\vartheta = r(\cos \vartheta \cos \varphi, \cos \vartheta \sin \varphi, -\sin \vartheta)$. Hence

$$E = r^2 \sin^2 \vartheta (\sin^2 \varphi + \cos^2 \varphi) = r^2 \sin^2 \vartheta, \quad G = r^2 (\cos^2 \vartheta + \sin^2 \vartheta) = r^2,$$

$$F = r^2 \sin \vartheta \cos \vartheta (-\sin \varphi \cos \varphi + \cos \varphi \sin \varphi) = 0.$$

Therefore $I = r^2 \sin^2 \vartheta d\varphi^2 + r^2 d\vartheta^2$ and $\sqrt{EG - F^2} = r^2 \sin \vartheta$: parallels and meridians are orthogonal and the sphere is a surface of revolution with profile $f(\vartheta) = r \sin \vartheta$, $h(\vartheta) = r \cos \vartheta$ (Definition 9.3), consistently with Lemma 9.6, which gives $E = f'^2 = r^2 \sin^2 \vartheta$ and $G = f'^2 + h'^2 = r^2 \cos^2 \vartheta + r^2 \sin^2 \vartheta = r^2$. ✓ Regularity requires $\sin \vartheta \neq 0$: the two poles must be excluded from this chart, not from the sphere. *Numerical check:* for $r = 1.7$ and $\vartheta = 1.12$ the formula gives $E = 1.7^2 \sin^2(1.12) = 2.89 \times 0.810181 = 2.34142253$ and $G = 2.89$; the numerical computation by finite differences on the parametrization gives 2.34142252 and 2.88999999. ✓

Example 10.1: First forms recurring in the chapter.

- *Plane*, $Y(u, v) = (u, v, 0)$: $E = G = 1$, $F = 0$.
- *Cylinder* of radius R , $X(u, v) = (R \cos u, R \sin u, v)$: $E = R^2$, $F = 0$, $G = 1$ (Proposition 9.1).
- *Cone* of half-angle α , $X(u, v) = (v \sin \alpha \cos u, v \sin \alpha \sin u, v \cos \alpha)$ with $v > 0$: $E = v^2 \sin^2 \alpha$, $F = 0$, $G = 1$ (Proposition 9.2).
- *Surface of revolution* with profile (f, h) : $E = f'^2$, $F = 0$, $G = f'^2 + h'^2$ (Lemma 9.6).
- *Torus* $X(u, v) = ((a + b \cos v) \cos u, (a + b \cos v) \sin u, b \sin v)$ with $a > b > 0$: $E = (a + b \cos v)^2$, $F = 0$, $G = b^2$.
- *Catenoid* $X(u, v) = (\cosh v \cos u, \cosh v \sin u, v)$: $E = G = \cosh^2 v$, $F = 0$ — a conformal first form.

10.2. Gauss map and second fundamental form

Definition 10.2: Gauss map and second fundamental form. Given a parametrization $X : U \rightarrow S$, the **Gauss map** is

$$N = \frac{X_1 \times X_2}{\|X_1 \times X_2\|} : U \rightarrow S^2,$$

the unit normal vector. The **second fundamental form** at $p = X(q)$ is the quadratic form on the tangent plane

$$\Pi_p(w) = h_{ij}a^i a^j, \quad w = a^i X_i, \quad h_{ij} = \langle X_{ij}, N \rangle,$$

which in classical notation reads $\Pi = e du^2 + 2f du dv + g dv^2$ with $e = h_{11}$, $f = h_{12} = h_{21}$, $g = h_{22}$.

Definition 10.3: Shape operator. The **shape operator** (or Weingarten operator) at p is $\mathcal{S}_p = -dN_p : T_p S \rightarrow T_p S$, where dN_p is the differential of the Gauss map, read as a linear map of the tangent plane into itself (which is legitimate because $T_{N(p)} S^2$ and $T_p S$ are both $N(p)^\perp$).

Proposition 10.2: Weingarten formulas and self-adjointness. With the preceding notation:

- (i) N_i is tangent and $\langle N_i, X_j \rangle = -h_{ij}$; consequently $N_i = -h_{ij}g^{jk}X_k$ and $\mathcal{S}(X_i) = h_{ij}g^{jk}X_k$;
- (ii) $\langle \mathcal{S}(w), z \rangle = \Pi(w, z) = \langle w, \mathcal{S}(z) \rangle$ for all $w, z \in T_p S$: $\mathcal{S}$ is self-adjoint with respect to the first form;
- (iii) the matrix of $\mathcal{S}$ in the basis $\{X_1, X_2\}$, written by columns, is $I^{-1}\Pi$, and hence

$$\det \mathcal{S} = \frac{eg - f^2}{EG - F^2}, \quad \text{tr } \mathcal{S} = \frac{eG - 2fF + gE}{EG - F^2};$$

- (iv) $\mathcal{S}$ has two real eigenvalues and, if they are distinct, orthogonal eigenvectors.

Proof. Step 1 — N_i is tangent. From $\langle N, N \rangle \equiv 1$, differentiating with respect to u^i , one obtains $2\langle N_i, N \rangle = 0$: hence $N_i \perp N$, that is $N_i \in T_p S$ and one writes $N_i = a_i^k X_k$.

Step 2 — the scalar product with X_j . From $\langle N, X_j \rangle \equiv 0$, differentiating with respect to u^i ,

$$\langle N_i, X_j \rangle + \langle N, X_{ji} \rangle = 0, \quad \text{that is} \quad \langle N_i, X_j \rangle = -h_{ij}.$$

On the other hand $\langle N_i, X_j \rangle = a_i^k g_{kj}$, hence $a_i^k g_{kj} = -h_{ij}$ and, multiplying by g^{jm} and summing, $a_i^m = -h_{ij}g^{jm}$. Since $\mathcal{S}(X_i) = -dN(X_i) = -N_i$, it follows that $\mathcal{S}(X_i) = h_{ij}g^{jk}X_k$. This proves (i).

Step 3 — (ii). It suffices to check it on the basis:

$$\langle \mathcal{S}(X_i), X_j \rangle = \langle -N_i, X_j \rangle = h_{ij},$$

an expression symmetric in i and j (because $X_{ij} = X_{ji}$ by Schwarz's theorem, valid since X is of class C^3). By bilinearity the equality extends to all tangent vectors, and $\Pi(w, z)$ is the symmetric bilinear form associated with the quadratic form Π .

Step 4 — (iii). From Step 2, the k -th component of $\mathcal{S}(X_i)$ is $h_{ij}g^{jk}$; the matrix by columns therefore has at entry (k, i) the number $g^{kj}h_{ji}$, that is it is the matrix product $I^{-1}\Pi$ (using the symmetry of I^{-1} and of Π). The determinant of a product is the product of the determinants:

$\det \mathcal{S} = \det(\text{II})/\det(\text{I}) = (eg - f^2)/(EG - F^2)$. For the trace one carries out the product:

$$\text{I}^{-1}\text{II} = \frac{1}{D} \begin{pmatrix} G & -F \\ -F & E \end{pmatrix} \begin{pmatrix} e & f \\ f & g \end{pmatrix} = \frac{1}{D} \begin{pmatrix} Ge - Ff & Gf - Fg \\ -Fe + Ef & -Ff + Eg \end{pmatrix},$$

whose trace is $(Ge - Ff - Ff + Eg)/D = (eG - 2fF + gE)/D$.

Step 5 — (iv). Choose an orthonormal basis $\{v_1, v_2\}$ of $T_p S$; in it the matrix of $\mathcal{S}$ is $\begin{pmatrix} p & q \\ q & r \end{pmatrix}$, symmetric by (ii) (the entry (i, j) is $\langle \mathcal{S}(v_j), v_i \rangle$). The characteristic polynomial $\lambda^2 - (p+r)\lambda + (pr - q^2)$ has discriminant

$$(p+r)^2 - 4(pr - q^2) = (p-r)^2 + 4q^2 \geq 0,$$

hence two real eigenvalues. If they are distinct and $\mathcal{S}v = \kappa_1 v$, $\mathcal{S}w = \kappa_2 w$, then $\kappa_1 \langle v, w \rangle = \langle \mathcal{S}v, w \rangle = \langle v, \mathcal{S}w \rangle = \kappa_2 \langle v, w \rangle$, whence $(\kappa_1 - \kappa_2) \langle v, w \rangle = 0$ and therefore $\langle v, w \rangle = 0$. If they coincide, $\mathcal{S}$ is a multiple of the identity (a symmetric 2×2 matrix with vanishing discriminant has $p = r$ and $q = 0$) and every orthonormal basis consists of eigenvectors. $\square$

Proposition 10.3: Normal curvature and Meusnier's theorem. Let γ be a C^2 curve on S parametrized by arclength with $\gamma(0) = p$ and $\gamma'(0) = w$, $\|w\| = 1$. Then the **normal curvature** $\kappa_n = \langle \gamma''(0), N(p) \rangle$ depends only on w and not on the curve:

$$\kappa_n(w) = \text{II}_p(w) = \langle \mathcal{S}(w), w \rangle.$$

Moreover, letting $\kappa_1 \leq \kappa_2$ be the roots of the characteristic polynomial of $\mathcal{S}$ and $\{v_1, v_2\}$ an orthonormal basis of eigenvectors,

$$\kappa_n(\cos \theta v_1 + \sin \theta v_2) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta \quad (\text{Euler's formula}),$$

and $\kappa_1 = \min_{\|w\|=1} \kappa_n(w)$, $\kappa_2 = \max_{\|w\|=1} \kappa_n(w)$.

Proof. Step 1 — computation of κ_n . Writing $\gamma = X \circ u$, Step 1 of the proof of Theorem 9.2 gives $\gamma'' = (\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j) X_k + h_{ij} \dot{u}^i \dot{u}^j N$. Since $\langle X_k, N \rangle = 0$,

$$\kappa_n = \langle \gamma'', N \rangle = h_{ij} \dot{u}^i \dot{u}^j = \text{II}(w),$$

where $w = \dot{u}^i X_i = \gamma'(0)$: it depends only on w . This is Meusnier's theorem in the form needed here: all curves on S through p with the same tangent have the same normal curvature.

Step 2 — the expression with the shape operator. By Proposition 10.2(ii), $\text{II}(w) = \text{II}(w, w) = \langle \mathcal{S}(w), w \rangle$.

Step 3 — Euler's formula. In the orthonormal basis of eigenvectors, $\mathcal{S}(v_1) = \kappa_1 v_1$ and $\mathcal{S}(v_2) = \kappa_2 v_2$; putting $w = \cos \theta v_1 + \sin \theta v_2$ (which is a unit vector),

$$\langle \mathcal{S}(w), w \rangle = \langle \kappa_1 \cos \theta v_1 + \kappa_2 \sin \theta v_2, \cos \theta v_1 + \sin \theta v_2 \rangle = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta.$$

Step 4 — extrema. Since $\cos^2 \theta + \sin^2 \theta = 1$, the expression of Step 3 is a convex combination of κ_1 and κ_2 : it ranges between the minimum κ_1 (attained at $\theta = 0$) and the maximum κ_2 (at $\theta = \pi/2$). $\square$

Exercise 10.2. Compute the Gauss map and the second fundamental form of the sphere of radius r in the parametrization of Exercise 10.1, and deduce the shape operator.

Solution. Step 1 — the Gauss map. From the computation of X_φ and X_ϑ in Exercise 10.1,

$$X_\varphi \times X_\vartheta = -r \sin \vartheta \cdot X, \quad \|X_\varphi \times X_\vartheta\| = r^2 \sin \vartheta,$$

(check it component by component: the first is $r^2 \sin \vartheta (\sin \vartheta \cos \varphi) \cdot (-1)$, and similarly the other two), hence

$$N = -\frac{X}{r}.$$

The normal induced by this parametrization is the *inner* one: it points towards the centre.

Step 2 — the second form. With $X_{\varphi\varphi} = -r \sin \vartheta (\cos \varphi, \sin \varphi, 0)$, $X_{\vartheta\vartheta} = -X$ and $X_{\varphi\vartheta} = r \cos \vartheta (-\sin \varphi, \cos \varphi, 0)$:

$$e = \left\langle X_{\varphi\varphi}, -\frac{X}{r} \right\rangle = \frac{r \sin \vartheta}{r} \langle (\cos \varphi, \sin \varphi, 0), X \rangle = \sin \vartheta \cdot r \sin \vartheta = r \sin^2 \vartheta,$$

$$g = \left\langle -X, -\frac{X}{r} \right\rangle = \frac{r^2}{r} = r, \quad f = -\frac{r \cos \vartheta}{r} \cdot r \sin \vartheta (-\sin \varphi \cos \varphi + \cos \varphi \sin \varphi) = 0.$$

Hence $\text{II} = r \sin^2 \vartheta d\varphi^2 + r d\vartheta^2 = \frac{1}{r} \text{I}$: the second form is proportional to the first, a sign that the sphere bends equally in every direction.

Step 3 — shape operator. By Proposition 10.2(iii), $\mathcal{S} = \text{I}^{-1}\text{II} = \text{I}^{-1} \cdot \frac{1}{r}\text{I} = \frac{1}{r} \text{id}$: every direction is principal and $\kappa_1 = \kappa_2 = 1/r$ (with the inner normal). *Numerical check:* for $r = 1.7$ and $(\varphi, \vartheta) = (0.83, 1.12)$ the finite-difference computation gives $N = (-0.607456, -0.664212, -0.435682)$, which agrees to six decimals with $-X/r$ ✓; $e = 1.37730691$ against $r \sin^2 \vartheta = 1.7 \times 0.810181 = 1.37730737$ ✓; $g = 1.70000000$ and $f = 0$ ✓; and the eigenvalues of $\mathcal{S}$ both come out 0.58823510 against $1/1.7 = 0.58823529$. ✓

10.3. Principal, mean and Gaussian curvatures

Definition 10.4: Principal, Gaussian and mean curvatures. The eigenvalues κ_1, κ_2 of the shape operator $\mathcal{S}_p$ are the **principal curvatures** at p , and the corresponding eigenvectors the **principal directions**. One sets

$$K = \kappa_1 \kappa_2 = \det \mathcal{S} = \frac{eg - f^2}{EG - F^2} \quad (\text{Gaussian curvature}),$$

$$H = \frac{\kappa_1 + \kappa_2}{2} = \frac{1}{2} \text{tr} \mathcal{S} = \frac{eG - 2fF + gE}{2(EG - F^2)} \quad (\text{mean curvature}).$$

A point at which $\kappa_1 = \kappa_2$ is called **umbilic**; a surface with $H \equiv 0$ is called **minimal**.

Corollary 10.1: Inequality between the two curvatures. At every point $H^2 \geq K$, with equality if and only if the point is umbilic. In particular, if $K > 0$ then $H \neq 0$: a minimal surface has no points of positive Gaussian curvature.

Proof. Since κ_1, κ_2 are real (Proposition 10.2(iv)),

$$H^2 - K = \left(\frac{\kappa_1 + \kappa_2}{2} \right)^2 - \kappa_1 \kappa_2 = \frac{(\kappa_1 - \kappa_2)^2}{4} \geq 0,$$

with equality if and only if $\kappa_1 = \kappa_2$, that is at an umbilic point. If $K > 0$ and $H = 0$, one would have $0 = H^2 \geq K > 0$: absurd. $\square$

Proposition 10.4: Curvatures of the sphere. For $S^2(O, r)$, with the inner normal, one has $\kappa_1 = \kappa_2 = 1/r$ at every point, hence $K = 1/r^2$ and $H = 1/r$, constants; every point is umbilic. With the outer normal $\kappa_1 = \kappa_2 = -1/r$, $H = -1/r$ and $K = 1/r^2$ as before.

Proof. Step 1 — via the shape operator. With the inner normal $N(P) = -P/r$ (where O is the origin), the Gauss map is linear in P , hence $dN_P(w) = -w/r$ for every $w \in T_P S^2$ and

$$\mathcal{S}_P = -dN_P = \frac{1}{r} \text{id}.$$

Every vector is an eigenvector with eigenvalue $1/r$: the principal curvatures coincide and $K = 1/r^2$, $H = 1/r$.

Step 2 — via the forms in coordinates. Step 3 of the solution of Exercise 10.2 gives the same result in spherical coordinates, and confirms it by the explicit computation

$$K = \frac{eg - f^2}{EG - F^2} = \frac{r \sin^2 \vartheta \cdot r - 0}{r^2 \sin^2 \vartheta \cdot r^2} = \frac{r^2 \sin^2 \vartheta}{r^4 \sin^2 \vartheta} = \frac{1}{r^2},$$

$$H = \frac{eG - 2fF + gE}{2(EG - F^2)} = \frac{r \sin^2 \vartheta \cdot r^2 + r \cdot r^2 \sin^2 \vartheta}{2r^4 \sin^2 \vartheta} = \frac{2r^3 \sin^2 \vartheta}{2r^4 \sin^2 \vartheta} = \frac{1}{r}.$$

Step 3 — change of orientation. Replacing N by $-N$ changes the sign of all the h_{ij} , hence of $\mathcal{S}$, of κ_1, κ_2 and of H ; the product $K = \kappa_1 \kappa_2$ does not change. This is why K is an invariant of the surface while H depends on the chosen orientation.

Step 4 — consistency with Chapter 7. Step 2 of the proof of Theorem 7.11 uses exactly this computation, with the inner normal, to identify the factor $1/r^2$ in the spherical excess with the Gaussian curvature. ✓ □

Exercise 10.3. Show that the cylinder of radius R has $K \equiv 0$ and $H = -1/(2R)$ with the normal induced by the parametrization $X(u, v) = (R \cos u, R \sin u, v)$, and identify the principal directions.

Solution. From Example 10.1, $E = R^2$, $F = 0$, $G = 1$. The cross product $X_u \times X_v = (-R \sin u, R \cos u, 0) \times (0, 0, 1) = (R \cos u, R \sin u, 0)$ has norm R , hence $N = (\cos u, \sin u, 0)$: the outer normal. With $X_{uu} = (-R \cos u, -R \sin u, 0)$, $X_{uv} = 0$ and $X_{vv} = 0$:

$$e = \langle X_{uu}, N \rangle = -R \cos^2 u - R \sin^2 u = -R, \quad f = 0, \quad g = 0.$$

Therefore

$$K = \frac{eg - f^2}{EG - F^2} = \frac{0 - 0}{R^2} = 0, \quad H = \frac{eG - 2fF + gE}{2(EG - F^2)} = \frac{-R \cdot 1}{2R^2} = -\frac{1}{2R}.$$

The shape operator is $\mathcal{S} = \text{I}^{-1}\text{II} = \text{diag}(-1/R, 0)$: the principal directions are that of the circle (eigenvalue $\kappa_1 = -1/R$) and that of the generator ($\kappa_2 = 0$). Their product vanishes: the cylinder bends in one direction only. *Numerical check:* for $R = 1.3$ and $(u, v) = (0.4, 0.9)$ the finite-difference computation gives $N = (0.921061, 0.389418, 0)$ against $(\cos 0.4, \sin 0.4, 0) = (0.921061, 0.389418, 0)$ ✓, $e = -1.29999957$ against $-R = -1.3$ ✓, $f = g = 0$ ✓, $K = 0$ within 10^{-10} and $H = -0.38461526$ against $-1/(2 \times 1.3) = -0.38461538$. ✓

Proposition 10.5: Classification of the points of a surface. Let $p \in S$ and consider the height function over the tangent plane $h(q) = \langle X(q) - p, N(p) \rangle$ for q near $q_0 = X^{-1}(p)$. Then $h(q_0) = 0$, $\nabla h(q_0) = 0$ and the Hessian matrix of h at q_0 is (h_{ij}) . Consequently:

- (i) if $K(p) > 0$ (**elliptic point**) there is a neighbourhood of p in S contained, apart from p itself, in just one of the two open half-spaces bounded by the tangent plane;
- (ii) if $K(p) < 0$ (**hyperbolic point**) every neighbourhood of p in S meets both open half-spaces: the surface crosses its own tangent plane;
- (iii) if $K(p) = 0$ but $S_p \neq 0$ the point is called **parabolic**, if $S_p = 0$ it is called **planar**; in both cases the sign of h is not determined by the quadratic term.

Proof. **Step 1 — derivatives of the height function.** $h(q_0) = 0$ by definition; $\partial_i h(q_0) = \langle X_i, N(p) \rangle = 0$ because X_i is tangent; and $\partial_i \partial_j h(q_0) = \langle X_{ij}, N(p) \rangle = h_{ij}$, since $N(p)$ is a constant vector in the differentiation.

Step 2 — Taylor expansion. The function h is of class C^3 (so is X), hence by Taylor's formula with Peano remainder, putting $\Delta = q - q_0$,

$$h(q_0 + \Delta) = \frac{1}{2} Q(\Delta) + o(\|\Delta\|^2), \quad Q(\Delta) = h_{ij} \Delta^i \Delta^j = \text{II read on } \Delta.$$

Step 3 — the sign of Q is the sign of K . The discriminant of the quadratic form Q is $eg - f^2$, and by Definition 10.4 $K = (eg - f^2)/D$ with $D > 0$: hence $eg - f^2$ and K have the same sign.

Step 4 — (i). If $K(p) > 0$ then Q is definite (positive or negative); let $\mu = \min_{\|\Delta\|=1} |Q(\Delta)|$, which is positive because Q is continuous, does not vanish on the unit circle and the latter is compact. By homogeneity $|Q(\Delta)| \geq \mu \|\Delta\|^2$ for every Δ . From Step 2 there is $\rho > 0$ such that $|o(\|\Delta\|^2)| < \frac{\mu}{4} \|\Delta\|^2$ for $0 < \|\Delta\| < \rho$, and then

$$|h(q_0 + \Delta)| \geq \frac{1}{2} \mu \|\Delta\|^2 - \frac{\mu}{4} \|\Delta\|^2 = \frac{\mu}{4} \|\Delta\|^2 > 0,$$

and h everywhere has the same sign as Q : the surface stays on one side.

Step 5 — (ii). If $K(p) < 0$ then Q is indefinite: there are unit vectors Δ_+ and Δ_- with $Q(\Delta_+) = 2c_+ > 0$ and $Q(\Delta_-) = -2c_- < 0$. Along the ray $t\Delta_+$, from Step 2, $h(q_0 + t\Delta_+) = c_+ t^2 + o(t^2) > 0$ for $t \neq 0$ small enough; along $t\Delta_-$, $h = -c_- t^2 + o(t^2) < 0$. The surface takes both signs in every neighbourhood.

Step 6 — (iii). If $eg - f^2 = 0$ with $(h_{ij}) \neq 0$, the form Q is semidefinite and vanishes on one direction: the quadratic term does not decide the sign along that direction and higher-order terms are needed. If $(h_{ij}) = 0$, $Q \equiv 0$ and the quadratic term decides in no direction.

Step 7 — check on the examples. For the paraboloid $X(u, v) = (u, v, au^2 + bv^2)$ at the origin one has $E = G = 1$, $F = 0$, $N = (0, 0, 1)$, $X_{uu} = (0, 0, 2a)$, $X_{vv} = (0, 0, 2b)$, $X_{uv} = 0$, hence $e = 2a$, $g = 2b$, $f = 0$ and $K = 4ab$. The numerical computation by finite differences confirms $K = 4.000000$ for $(a, b) = (1, 1)$ — elliptic paraboloid, elliptic point —, $K = -4.000000$ for $(a, b) = (1, -1)$ — saddle, hyperbolic point — and $K = 0$ for $(a, b) = (1, 0)$ — parabolic cylinder, parabolic point. ✓ □

10.4. The Theorema Egregium

Everything up to this point defines K by means of the second form, that is by means of the way the surface sits in space. The result of this section is that this dependence is only apparent.

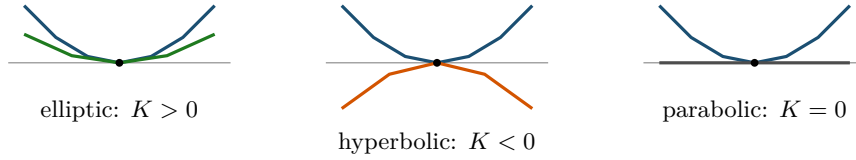


Figure 10.1: Normal sections along the two principal directions. The ordinates are computed by hand from the parabolas $y = 0.5x^2$ (nodes 0.72; 0.32; 0.08; 0), $y = 0.25x^2$ (nodes 0.36; 0.09; 0) and $y = -0.4x^2$ (nodes -0.576; -0.144; 0). At an elliptic point the two sections lie on the same side of the tangent plane, at a hyperbolic point on opposite sides, at a parabolic point one of the two is a straight line (Proposition 10.5).

Theorem 10.1: Gauss equation. For every regular parametrization X of class C^3 one has, at every point,

$$EK = \partial_2 \Gamma_{11}^2 - \partial_1 \Gamma_{12}^2 + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - (\Gamma_{12}^2)^2.$$

The right-hand side contains only the Christoffel symbols and their first derivatives.

Proof. Step 1 — the two structure formulas. By Lemma 9.3 of Chapter 9 the Gauss formula holds,

$$X_{ij} = \Gamma_{ij}^k X_k + h_{ij} N,$$

and by Proposition 10.2(i) the Weingarten formula

$$N_i = a_i^k X_k, \quad a_i^k = -h_{ij} g^{jk}.$$

Step 2 — the compatibility condition. Since X is of class C^3 , by Schwarz's theorem the third derivatives do not depend on the order:

$$\partial_2(X_{11}) = \partial_1(X_{12}).$$

Step 3 — expansion of the left-hand side. Differentiating the Gauss formula and substituting it back,

$$\begin{aligned} \partial_2 X_{11} &= (\partial_2 \Gamma_{11}^k) X_k + \Gamma_{11}^k X_{k2} + (\partial_2 h_{11}) N + h_{11} N_2 \\ &= (\partial_2 \Gamma_{11}^l) X_l + \Gamma_{11}^k (\Gamma_{k2}^l X_l + h_{k2} N) + (\partial_2 h_{11}) N + h_{11} a_2^l X_l. \end{aligned}$$

The coefficient of X_2 is

$$\partial_2 \Gamma_{11}^2 + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 + h_{11} a_2^2.$$

Step 4 — expansion of the right-hand side. Similarly

$$\partial_1 X_{12} = (\partial_1 \Gamma_{12}^l) X_l + \Gamma_{12}^k (\Gamma_{k1}^l X_l + h_{k1} N) + (\partial_1 h_{12}) N + h_{12} a_1^l X_l,$$

and the coefficient of X_2 is

$$\partial_1 \Gamma_{12}^2 + \Gamma_{12}^1 \Gamma_{11}^2 + (\Gamma_{12}^2)^2 + h_{12} a_1^2.$$

Step 5 — equality of the coefficients. X_1, X_2, N are linearly independent, hence the coefficients of X_2 in Steps 3 and 4 coincide:

$$\partial_2 \Gamma_{11}^2 - \partial_1 \Gamma_{12}^2 + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - (\Gamma_{12}^2)^2 = h_{12} a_1^2 - h_{11} a_2^2.$$

Step 6 — computation of the right-hand side. With $g^{12} = -F/D$, $g^{22} = E/D$ and $D = EG - F^2$,

$$a_2^2 = -(h_{21} g^{12} + h_{22} g^{22}) = -\left(f \cdot \frac{-F}{D} + g \cdot \frac{E}{D}\right) = \frac{fF - gE}{D},$$

$$a_1^2 = -(h_{11}g^{12} + h_{12}g^{22}) = -\left(e \cdot \frac{-F}{D} + f \cdot \frac{E}{D}\right) = \frac{eF - fE}{D}.$$

Hence

$$h_{12}a_1^2 - h_{11}a_2^2 = \frac{f(eF - fE) - e(fF - gE)}{D} = \frac{efF - f^2E - efF + egE}{D} = \frac{E(eg - f^2)}{D} = EK,$$

using Definition 10.4. Substituting into Step 5 gives the claim.

Step 7 — check on the sphere. With $u^1 = \varphi$, $u^2 = \vartheta$, $E = r^2 \sin^2 \vartheta$, $F = 0$, $G = r^2$ and the symbols $\Gamma_{12}^1 = \cot \vartheta$, $\Gamma_{11}^2 = -\sin \vartheta \cos \vartheta$, the others vanishing (Step 5 of the proof of Lemma 9.3), the right-hand side equals

$$\partial_{\vartheta}(-\sin \vartheta \cos \vartheta) - 0 + 0 + 0 - \cot \vartheta \cdot (-\sin \vartheta \cos \vartheta) - 0 = -(\cos^2 \vartheta - \sin^2 \vartheta) + \cos^2 \vartheta = \sin^2 \vartheta,$$

while the left-hand side is $EK = r^2 \sin^2 \vartheta \cdot \frac{1}{r^2} = \sin^2 \vartheta$. ✓ The independent numerical check (all derivatives computed by centred finite differences on the parametrization, $r = 1.7$, $(\varphi, \vartheta) = (0.83, 1.12)$) gives 0.81017657 for the right-hand side and, for EK , the value $2.34142252 \times 0.34602053 = 0.81018027$, with a discrepancy of $3.7 \cdot 10^{-6}$ due to discretization. ✓ □

Theorem 10.2: Theorema Egregium (Gauss, 1827). *The Gaussian curvature K is intrinsic: it is expressible through the coefficients E, F, G of the first fundamental form alone and their partial derivatives up to second order. Consequently, if $\phi : V \rightarrow \bar{V}$ is a local isometry between two surfaces, then $\bar{K}(\phi(p)) = K(p)$ for every $p \in V$.*

Proof. **Step 1 — the intrinsic expression.** By Theorem 10.1,

$$K = \frac{1}{E} \left(\partial_2 \Gamma_{11}^2 - \partial_1 \Gamma_{12}^2 + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - (\Gamma_{12}^2)^2 \right),$$

the division being legitimate because $E = \|X_1\|^2 > 0$. By Lemma 9.3 the symbols Γ_{ij}^k are written with g_{ij} , its inverse and its first derivatives; their first derivatives introduce second derivatives of g_{ij} . Hence K is a function of E, F, G and of their derivatives up to second order: the second form does *not* appear in the formula.

Step 2 — invariance under local isometries. Let ϕ be a local isometry and let $X : U \rightarrow V$ be a parametrization. As in Step 1 of the proof of Lemma 9.4, $\bar{X} = \phi \circ X$ is a parametrization of $\bar{V}$ with $\bar{g}_{ij} = g_{ij}$ as functions on the same domain U . By Step 1, K and $\bar{K}$ are the same function of the g_{ij} and of their derivatives, evaluated at the same arguments: hence $\bar{K}(\bar{X}(q)) = K(X(q))$ for every $q \in U$, that is $\bar{K}(\phi(p)) = K(p)$.

Step 3 — why it is remarkable. The definition of K goes through $\mathcal{S} = -dN$, which requires looking at the surface from outside; the expression of Step 1 requires only measuring lengths and angles *on* the surface. The two coincide, and this was by no means obvious: the mean curvature H , defined in an entirely analogous way, is *not* intrinsic — the plane ($H = 0$) and the cylinder ($H = -1/(2R)$) are locally isometric (Proposition 9.1) but have different mean curvatures.

Step 4 — the original formulation. Gauss stated the result in the *Disquisitiones generales circa superficies curvas* (1828, read to the Royal Society of Göttingen on 8 October 1827), article 12, and called it *theorema egregium* — remarkable theorem. Modern formulations are in do Carmo, §4-3 (source [2]) and in Kühnel (source [4]). □

Proposition 10.6: Curvature in orthogonal coordinates. *If $F \equiv 0$, then*

$$K = -\frac{1}{2\sqrt{EG}} \left[\frac{\partial}{\partial u^2} \left(\frac{E_2}{\sqrt{EG}} \right) + \frac{\partial}{\partial u^1} \left(\frac{G_1}{\sqrt{EG}} \right) \right],$$

where $E_2 = \partial E / \partial u^2$ and $G_1 = \partial G / \partial u^1$.

Proof. Step 1 — the Christoffel symbols in the orthogonal case. They are those listed in Step 2 of the proof of Lemma 9.6: writing the subscript 1 for $\partial / \partial u^1$ and 2 for $\partial / \partial u^2$,

$$\Gamma_{11}^1 = \frac{E_1}{2E}, \quad \Gamma_{12}^1 = \frac{E_2}{2E}, \quad \Gamma_{11}^2 = -\frac{E_2}{2G}, \quad \Gamma_{12}^2 = \frac{G_1}{2G}, \quad \Gamma_{22}^2 = \frac{G_2}{2G}.$$

Step 2 — substitution in the Gauss equation. From Theorem 10.1,

$$EK = \partial_2 \left(-\frac{E_2}{2G} \right) - \partial_1 \left(\frac{G_1}{2G} \right) + \frac{E_1}{2E} \cdot \frac{G_1}{2G} + \left(-\frac{E_2}{2G} \right) \frac{G_2}{2G} - \frac{E_2}{2E} \left(-\frac{E_2}{2G} \right) - \frac{G_1^2}{4G^2}.$$

Expanding the two derivatives,

$$\partial_2 \left(-\frac{E_2}{2G} \right) = -\frac{E_{22}}{2G} + \frac{E_2 G_2}{2G^2}, \quad -\partial_1 \left(\frac{G_1}{2G} \right) = -\frac{G_{11}}{2G} + \frac{G_1^2}{2G^2},$$

and adding all six terms, with $\frac{E_2 G_2}{2G^2} - \frac{E_2 G_2}{4G^2} = \frac{E_2 G_2}{4G^2}$ and $\frac{G_1^2}{2G^2} - \frac{G_1^2}{4G^2} = \frac{G_1^2}{4G^2}$,

$$EK = -\frac{E_{22} + G_{11}}{2G} + \frac{E_2 G_2 + G_1^2}{4G^2} + \frac{E_1 G_1 + E_2^2}{4EG},$$

that is, dividing by E ,

$$K = -\frac{E_{22} + G_{11}}{2EG} + \frac{E_2 G_2}{4EG^2} + \frac{G_1^2}{4EG^2} + \frac{E_1 G_1}{4E^2 G} + \frac{E_2^2}{4E^2 G}. \quad (*)$$

Step 3 — expansion of the formula in the statement. Differentiating the two quotients,

$$\partial_2 \left(\frac{E_2}{\sqrt{EG}} \right) = \frac{E_{22}}{\sqrt{EG}} - \frac{E_2(E_2 G + EG_2)}{2(EG)^{3/2}}, \quad \partial_1 \left(\frac{G_1}{\sqrt{EG}} \right) = \frac{G_{11}}{\sqrt{EG}} - \frac{G_1(E_1 G + EG_1)}{2(EG)^{3/2}},$$

so the right-hand side of the statement equals

$$-\frac{E_{22} + G_{11}}{2EG} + \frac{E_2(E_2 G + EG_2) + G_1(E_1 G + EG_1)}{4(EG)^2} = -\frac{E_{22} + G_{11}}{2EG} + \frac{E_2^2}{4E^2 G} + \frac{E_2 G_2}{4EG^2} + \frac{E_1 G_1}{4E^2 G} + \frac{G_1^2}{4EG^2},$$

identical to (*).

Step 4 — check on the sphere. $E = r^2 \sin^2 \vartheta$, $G = r^2$, $\sqrt{EG} = r^2 \sin \vartheta$, $E_\vartheta = 2r^2 \sin \vartheta \cos \vartheta$, $G_\varphi = 0$; hence $E_\vartheta / \sqrt{EG} = 2 \cos \vartheta$ and

$$K = -\frac{1}{2r^2 \sin \vartheta} \partial_\vartheta (2 \cos \vartheta) = -\frac{-2 \sin \vartheta}{2r^2 \sin \vartheta} = \frac{1}{r^2}. \quad \checkmark$$

The numerical computation by finite differences with $r = 1.7$ gives 0.34601963 against $1/r^2 = 0.34602076$. $\checkmark$ □

Remark 10.1: What we import: the Brioschi formula. The same content as Theorem 10.1 can be packaged into a determinantal formula due to Francesco Brioschi, valid

also for $F \neq 0$:

$$K = \frac{\begin{vmatrix} -\frac{1}{2}E_{22} + F_{12} - \frac{1}{2}G_{11} & \frac{1}{2}E_1 & F_1 - \frac{1}{2}E_2 \\ F_2 - \frac{1}{2}G_1 & E & F \\ \frac{1}{2}G_2 & F & G \end{vmatrix} - \begin{vmatrix} 0 & \frac{1}{2}E_2 & \frac{1}{2}G_1 \\ \frac{1}{2}E_2 & E & F \\ \frac{1}{2}G_1 & F & G \end{vmatrix}}{(EG - F^2)^2}.$$

Its equivalence with the Gauss equation in the general case $F \neq 0$ is imported, not proved here: the elementary proof, carried out with properties of matrices and determinants and without Christoffel symbols, is in Lee-Peng Teo, *The Brioschi Formula for the Gaussian Curvature*, arXiv:2404.00835 (source [3]). Two partial checks are however within our reach and we carry them out. (a) Orthogonal case. With $F \equiv 0$ the two determinants equal

$$D_1 = -\frac{EG}{2}(E_{22} + G_{11}) + \frac{E_1G_1G}{4} + \frac{E_2G_2E}{4}, \quad D_2 = -\frac{E_2^2G}{4} - \frac{G_1^2E}{4},$$

(expanding along the first column and along the third column respectively), and $(D_1 - D_2)/(EG)^2$ agrees term by term with formula (*) of Step 2 of the proof of Proposition 10.6. ✓ (b) Numerical check. On the sphere with $r = 1.7$ and $(\varphi, \vartheta) = (0.83, 1.12)$, all derivatives computed by centred finite differences, the Brioschi formula gives $K = 0.34601895$ against $1/r^2 = 0.34602076$. ✓ Since Proposition 10.6 is proved and suffices for every use made in this chapter, the import leaves no gaps: the Brioschi formula is here only a computational convenience.

Remark 10.2: Why a flat sheet does not wrap onto the sphere. Plane and cylinder both have $K \equiv 0$ (Exercise 10.3): indeed one can roll a sheet into a cylinder without stretching it, and by Theorem 10.2 the operation, being a local isometry, must preserve K . The sphere instead has $K = 1/r^2 \neq 0$ (Proposition 10.4): no portion of a sphere, however small, can be flattened onto a plane without deforming it. This is the mathematical reason why every geographical map inevitably distorts distances or areas — the stereographic projection of Chapter 7 preserves angles, but not distances — and why working sheet metal with double curvature (deep drawing) requires stretching the material, whereas simple bending ($K = 0$) does not. The defect is quantifiable: see Exercise 10.8.

10.5. The sphere and the torus: two curvatures compared

Proposition 10.7: Curvatures of the torus. Let $T_{a,b}$ be the torus of revolution parametrized, with $a > b > 0$, by

$$X(u, v) = ((a + b \cos v) \cos u, (a + b \cos v) \sin u, b \sin v), \quad u, v \in \mathbb{R}.$$

Putting $w = a + b \cos v > 0$, one has, with the normal $N = (\cos u \cos v, \sin u \cos v, \sin v)$ induced by the parametrization (the outer one):

$$\begin{aligned} E &= w^2, & F &= 0, & G &= b^2; & e &= -w \cos v, & f &= 0, & g &= -b; \\ \kappa_1 &= -\frac{\cos v}{w}, & \kappa_2 &= -\frac{1}{b}, & K &= \frac{\cos v}{b(a + b \cos v)}, & H &= -\frac{a + 2b \cos v}{2b(a + b \cos v)}. \end{aligned}$$

Consequently $K > 0$ for $|v| < \pi/2$ (the outer part, with elliptic points), $K < 0$ for $\pi/2 < v < 3\pi/2$ (the inner part, with hyperbolic points) and $K = 0$ for $v = \pm\pi/2$ (the top and bottom circles, with parabolic points).

Proof. Step 1 — first form. $X_u = w(-\sin u, \cos u, 0)$ and $X_v = (-b \sin v \cos u, -b \sin v \sin u, b \cos v)$ give $E = w^2$, $G = b^2 \sin^2 v + b^2 \cos^2 v = b^2$ and $F = wb \sin v(\sin u \cos u - \cos u \sin u) = 0$. This is the special case of Lemma 9.6 with $f_{\text{profile}}(v) = a + b \cos v$ and $h_{\text{profile}}(v) = b \sin v$, which gives $G = b^2 \sin^2 v + b^2 \cos^2 v = b^2$. ✓

Step 2 — the Gauss map. The cross product is

$$X_u \times X_v = wb(\cos u \cos v, \sin u \cos v, \sin v),$$

of norm wb (since $w, b > 0$ and $\cos^2 v + \sin^2 v = 1$), hence N is as stated. One checks that N points outwards by comparing it with the vector from the centre of the tube, $(a \cos u, a \sin u, 0)$, to the point X : that vector equals $b(\cos u \cos v, \sin u \cos v, \sin v) = bN$. ✓

Step 3 — second form. With $X_{uu} = -w(\cos u, \sin u, 0)$, $X_{uv} = b \sin v(\sin u, -\cos u, 0)$, $X_{vv} = (-b \cos v \cos u, -b \cos v \sin u, -b \sin v)$:

$$e = -w(\cos^2 u \cos v + \sin^2 u \cos v) = -w \cos v,$$

$$f = b \sin v(\sin u \cos u \cos v - \cos u \sin u \cos v) = 0,$$

$$g = -b \cos^2 v(\cos^2 u + \sin^2 u) - b \sin^2 v = -b.$$

Step 4 — curvatures. Since $F = f = 0$, the matrix $\mathcal{S} = \mathbf{I}^{-1}\mathbf{II}$ is diagonal (Proposition 10.2(iii)) with $\kappa_1 = e/E = -w \cos v/w^2 = -\cos v/w$ and $\kappa_2 = g/G = -b/b^2 = -1/b$; the principal directions are the parallel and the meridian. Hence

$$K = \kappa_1 \kappa_2 = \frac{\cos v}{bw} = \frac{\cos v}{b(a + b \cos v)}, \quad H = \frac{\kappa_1 + \kappa_2}{2} = -\frac{1}{2} \left(\frac{\cos v}{w} + \frac{1}{b} \right) = -\frac{b \cos v + w}{2bw} = -\frac{a + 2b \cos v}{2b(a + b \cos v)}.$$

Step 5 — sign of K . Since $b > 0$ and $w = a + b \cos v \geq a - b > 0$, the sign of K is that of $\cos v$: positive for $v \in (-\pi/2, \pi/2)$, negative for $v \in (\pi/2, 3\pi/2)$, zero for $v = \pm\pi/2$. The classification of the points follows from Proposition 10.5.

Step 6 — numerical check. For $a = 2$, $b = 1$ the numerical computation by finite differences gives, against the formulas:

- $v = 0$: $K = 0.333333$ (formula $1/3 = 0.333333$), $H = -0.666666$ (formula $-4/6 = -0.666667$), principal curvatures -1.000000 and -0.333333 , with product 0.333333 and half-sum -0.666667 ; ✓
- $v = \pi/2$: $K = 0.000000$, $H = -0.500000$, curvatures -1 and 0 ; ✓
- $v = 2\pi/3$: $K = -0.333333$ (formula $-0.5/1.5$), $H = -0.333333$; ✓
- $v = \pi$: $K = -0.999999$ (formula $-1/(1 \cdot 1) = -1$), $H = 0.000000$. ✓

The value $K(\pi) = -1$ shows that the inner equator is the most “saddle-like” part of the torus, and $H(\pi) = 0$ that there the two principal curvatures are opposite, $+1$ and -1 .

Step 7 — comparison with an independent source. Shifrin, solution of exercise 2.2.3(b) (source [1]), obtains for the torus $\mathcal{S} = \text{diag}(1/b, \cos u/(a + b \cos u))$, $H = \frac{1}{2}(1/b + \cos u/(a + b \cos u))$ and $K = \cos u/(b(a + b \cos u))$; in his notation u is our v and the normal is the opposite of ours, so that the principal curvatures and H have the opposite sign and K the same sign. ✓ □

Corollary 10.2: Total curvature of the torus. For the torus $T_{a,b}$ one has

$$\iint_{T_{a,b}} K \, dA = 0, \quad \text{Area}(T_{a,b}) = 4\pi^2 ab,$$

and the integral of the positive part of K alone equals 4π .

Proof. Step 1 — the area element. By Proposition 10.1(iii) and Step 1 of the proof of Proposition 10.7, $dA = \sqrt{EG - F^2} du dv = \sqrt{w^2 b^2} du dv = wb du dv$ (with $w, b > 0$).

Step 2 — the integral of K . The factor wb cancels exactly with the denominator of K :

$$\iint_{T_{a,b}} K dA = \int_0^{2\pi} \int_0^{2\pi} \frac{\cos v}{bw} \cdot wb dv du = \int_0^{2\pi} \int_0^{2\pi} \cos v dv du = 2\pi [\sin v]_0^{2\pi} = 0.$$

Step 3 — the area.

$$\text{Area} = \int_0^{2\pi} \int_0^{2\pi} (a + b \cos v) b dv du = 2\pi b (2\pi a + b [\sin v]_0^{2\pi}) = 4\pi^2 ab.$$

Step 4 — the positive part.

$$\iint K^+ dA = \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \cos v dv du = 2\pi [\sin v]_{-\pi/2}^{\pi/2} = 2\pi \cdot 2 = 4\pi.$$

Step 5 — consistency with global Gauss–Bonnet. Theorem 4.8 of Chapter 4 states that $\iint_M K dA = 2\pi\chi(M)$; for the torus $\chi = 0$, hence the integral must vanish — and that is what has just been computed directly. ✓ The value 4π of the positive part is instead exactly the total curvature of the sphere ($\iint_{S^2} K dA = r^{-2} \cdot 4\pi r^2 = 4\pi$): the Gauss map of the torus covers the sphere once with the outer part and once, with opposite orientation, with the inner one.

Step 6 — numerical check. For $a = 2$, $b = 1$: area $4\pi^2 \cdot 2 \cdot 1 = 78.956835$; numerical integration of $\iint K dA$ on 200,001 nodes gives 0.0000000000 and that of the positive part 12.566371, against $4\pi = 12.566371$. ✓ □

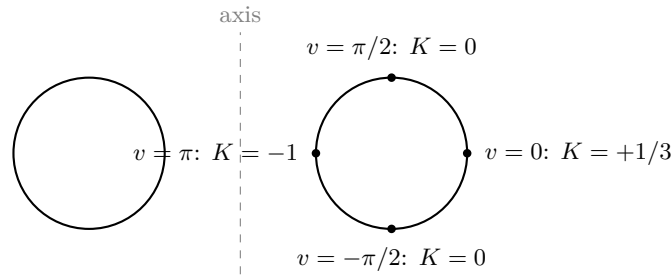


Figure 10.2: Axial section of the torus $T_{2,1}$: two circles of radius $b = 1$ with centres at distance $a = 2$ from the axis. The Gaussian curvature $K = \cos v / (b(a + b \cos v))$ equals $+1/3 = 0.333333$ on the outer equator, 0 on the two extreme circles and $-1 = -1/(1 \cdot 1)$ on the inner equator (Proposition 10.7); its integral over the whole torus vanishes (Corollary 10.2). Coordinates computed by hand: centres $(\pm 2, 0)$, radius 1, notable points $(3, 0)$, $(1, 0)$, $(2, \pm 1)$.

10.6. Ruled and developable surfaces

Proposition 10.8: Ruled surfaces have $K \leq 0$. Let $X(u, v) = \alpha(u) + v w(u)$ be a regular parametrization of class C^3 of a **ruled surface** (α the directrix, w the field of directions of the lines). Then $h_{22} = 0$ and

$$K = -\frac{f^2}{EG - F^2} \leq 0,$$

with $K = 0$ at a point if and only if $f = 0$ there.

Proof. Step 1 — the second derivative along the line. $X_v = w(u)$ does not depend on v , hence $X_{vv} = 0$ and $g = h_{22} = \langle X_{vv}, N \rangle = 0$.

Step 2 — the curvature. By Definition 10.4, $K = (eg - f^2)/(EG - F^2) = -f^2/D$ with $D > 0$: hence $K \leq 0$, and $K = 0$ if and only if $f = 0$.

Step 3 — geometric reading. By Proposition 10.3 the normal curvature along the direction of the line is $\text{II}(X_v)/\text{I}(X_v) = g/G = 0$: the line is a straight normal section, hence one “candidate” principal curvature is zero or changes sign relative to the other. No point of a ruled surface is elliptic. $\square$

Proposition 10.9: Cylinder and cone are developable. *Cylinder and cone (with the vertex removed) have $K \equiv 0$ and are locally isometric to the plane. Precisely, for the cone of half-angle α parametrized as in Proposition 9.2,*

$$N = (\cos u \cos \alpha, \sin u \cos \alpha, -\sin \alpha), \quad e = -v \sin \alpha \cos \alpha, \quad f = 0, \quad g = 0,$$

$$\text{hence } K = 0 \text{ and } H = -\frac{\cot \alpha}{2v}.$$

Proof. **Step 1 — the cylinder.** $K \equiv 0$ is Exercise 10.3; the local isometry with the plane is Proposition 9.1(i) of Chapter 9, where the parametrization of the plane $(u, v) \mapsto (Ru, v, 0)$ with the same first form is exhibited.

Step 2 — the Gauss map of the cone. With $X_u = v \sin \alpha(-\sin u, \cos u, 0)$ and $X_v = (\sin \alpha \cos u, \sin \alpha \sin u, \cos \alpha)$,

$$X_u \times X_v = v \sin \alpha (\cos u \cos \alpha, \sin u \cos \alpha, -\sin \alpha),$$

whose norm is $v \sin \alpha \sqrt{\cos^2 \alpha + \sin^2 \alpha} = v \sin \alpha > 0$; whence N .

Step 3 — the second form of the cone. $X_{vv} = 0$ gives $g = 0$ (it is a ruled surface, Proposition 10.8); then $X_{uv} = \sin \alpha(-\sin u, \cos u, 0)$ gives

$$f = \sin \alpha \cos \alpha (-\sin u \cos u + \cos u \sin u) = 0,$$

and $X_{uu} = v \sin \alpha(-\cos u, -\sin u, 0)$ gives $e = -v \sin \alpha \cos \alpha (\cos^2 u + \sin^2 u) = -v \sin \alpha \cos \alpha$.

Step 4 — curvatures of the cone. $K = (eg - f^2)/D = 0$; and with $E = v^2 \sin^2 \alpha$, $F = 0$, $G = 1$,

$$H = \frac{eG - 2fF + gE}{2(EG - F^2)} = \frac{-v \sin \alpha \cos \alpha}{2v^2 \sin^2 \alpha} = -\frac{\cos \alpha}{2v \sin \alpha} = -\frac{\cot \alpha}{2v}.$$

Step 5 — the local isometry with the plane. It is Proposition 9.2(i): the cone cut along a generator develops isometrically onto a plane sector of opening $2\pi \sin \alpha$. That K vanish is therefore *compulsory* by Theorem 10.2, and the computation of Step 4 confirms it. $\checkmark$

Step 6 — numerical check. For $\alpha = 35^\circ$: $\cot 35^\circ = 1.428148$, hence $H = -1.428148/(2v)$, which equals -0.714074 for $v = 1$ and -0.310467 for $v = 2.3$; the numerical computation by finite differences gives exactly -0.714074 and -0.310467 , and K of the order of 10^{-11} . $\checkmark$ $\square$

Exercise 10.4. Why is the cone with the vertex removed locally isometric to the plane, while the hyperbolic paraboloid $z = xy$, although ruled (indeed doubly ruled), is not?

Solution. The cone. Proposition 10.9 exhibits the isometry: the parametrization $\tilde{Y}(u, v) = (v \cos(u \sin \alpha), v \sin(u \sin \alpha), 0)$ of the plane has the same first form $\text{I} = v^2 \sin^2 \alpha du^2 + dv^2$ as the cone, so the map sending $X(u, v)$ to $\tilde{Y}(u, v)$ preserves lengths and angles. The vertex must be excluded because there X is not regular ($X_u = 0$ for $v = 0$): it is not a point of the surface.

The hyperbolic paraboloid. With $X(u, v) = (u, v, uv)$ one has $X_u = (1, 0, v)$, $X_v = (0, 1, u)$, hence

$$E = 1 + v^2, \quad F = uv, \quad G = 1 + u^2, \quad D = EG - F^2 = 1 + u^2 + v^2,$$

$$X_u \times X_v = (-v, -u, 1), \quad \|X_u \times X_v\| = \sqrt{1 + u^2 + v^2}.$$

Since $X_{uu} = X_{vv} = 0$ and $X_{uv} = (0, 0, 1)$,

$$e = g = 0, \quad f = \frac{1}{\sqrt{1 + u^2 + v^2}},$$

and therefore, by Proposition 10.8,

$$K = -\frac{f^2}{D} = -\frac{1}{(1 + u^2 + v^2)^2} < 0 \quad \text{everywhere,}$$

$$H = \frac{eG - 2fF + gE}{2D} = \frac{-2uv/\sqrt{1 + u^2 + v^2}}{2(1 + u^2 + v^2)} = -\frac{uv}{(1 + u^2 + v^2)^{3/2}}.$$

At the origin $K = -1$ and $H = 0$. By Theorem 10.2, a local isometry with the plane would force $K \equiv 0$: impossible. Hence the hyperbolic paraboloid is *not* developable, although it is a union of lines in two distinct ways ($u = \text{const}$ and $v = \text{const}$).

Numerical check. The finite-difference computation gives, against the formulas: $(u, v) = (0, 0)$: $K = -1.000000$ (formula -1), $H = 0.000000$; $(u, v) = (0.5, 0.3)$: $K = -0.556917$ (formula $-1/1.34^2 = -0.556917$), $H = -0.096702$ (formula $-0.15/1.34^{3/2} = -0.096702$); $(u, v) = (1, 1)$: $K = -0.111111$ (formula $-1/9$), $H = -0.192450$ (formula $-1/3^{3/2} = -0.192450$). ✓

Industrial moral. A hyperbolic-paraboloid roof or hull can be built with *straight elements* (formwork, cables, rebars) but cannot be obtained by bending a flat sheet: the material must be stretched. Ruling concerns the structure, developability concerns the fabrication, and they are different properties.

Remark 10.3: What we import: the characterization of surfaces with $K \equiv 0$.

Theorem 10.2 gives the implication “isometric to the plane $\Rightarrow K \equiv 0$ ”, which is the one used throughout the chapter, and Proposition 10.9 exhibits the explicit isometries for cylinder and cone. The converse implication — every surface with $K \equiv 0$ is locally isometric to the plane, and more generally (**Minding’s theorem**) two surfaces with the same constant Gaussian curvature are locally isometric — **is imported**: we do not prove it. The proof requires the construction of geodesic polar coordinates and the study of the equation $(\sqrt{G})_{vv} + K\sqrt{G} = 0$, material which the treatise does not develop. It is found in do Carmo, Chapter 4 (source [2]) and in Kühnel (source [4]). Nothing in this chapter uses it: whenever a surface is claimed to be developable, this is always proved by exhibiting the development.

10.7. Gauss–Bonnet: where it is in the treatise

Remark 10.4: Gauss–Bonnet is not duplicated here. The Gaussian curvature defined in this chapter is the protagonist of the Gauss–Bonnet theorem, which the treatise treats in two distinct places and which is not repeated here:

- the **local form on the geodesic triangle**, with a complete proof by means of the lune lemma, is Theorem 7.11 of Chapter 7: for a geodesic triangle T on the sphere of

radius r , $\alpha + \beta + \gamma - \pi = K \text{Area}(T)$ with $K = 1/r^2$. The computation of K appearing there is that of Proposition 10.4;

- the **global form**, $\iint_M K \, dA = 2\pi\chi(M)$, is in Section 4.6 of Chapter 4. Corollary 10.2 provides an independent check of it on the torus ($\chi = 0$, vanishing total curvature), computed here directly.

What this chapter adds to those two statements is their foundation: that K be an intrinsic quantity (Theorem 10.2) is the reason why the integral $\iint K \, dA$ can be a topological invariant, insensitive to how the surface is embedded in space.

10.8. Industrial applications

Developability and sheet-metal work. Theorem 10.2 sharply separates surfaces of vanishing Gaussian curvature (developable: planes, cylinders, cones) from doubly curved ones ($K \neq 0$). The former are obtained from a flat sheet by bending alone, without stretching, and “unroll” isometrically onto the plane: a robot that must travel along, cut or weld such a surface reduces the three-dimensional problem to a two-dimensional one in the developed plane (2D/3D laser cutting, off-line programming with **ABB** RobotStudio, **KUKA**.Sim, **FANUC** ROBOGUIDE, **Yaskawa** MotoSim). The latter require deep drawing and stretching of the material. Mathematics tells *a priori*, by computing a single number, whether a geometry can be realized by bending: it suffices to check $K \equiv 0$.

The doubly ruled trap. Exercise 10.4 shows that “made of straight lines” does not mean “developable”: the hyperbolic paraboloid is a union of lines in two ways but has $K < 0$ everywhere. It is the commonest error in the passage from the CAD drawing to the shop floor, and it costs scrap.

Metrology and cartography. The impossibility of flattening the sphere (Remark 10.2) is the structural reason for the distortion of every geographical map and for the fact that geodetic reference systems must choose what to preserve (angles, areas, distances along certain lines): one cannot preserve everything. The quantitative measure of the defect is Exercise 10.8.

Minimal surfaces and tensile structures. Corollary 10.1 implies that a minimal surface ($H \equiv 0$) has no points with $K > 0$: tensioned membranes and air-supported roofs spontaneously take saddle or flat shapes, never dome shapes, unless they are held up by a pressure difference. The catenoid of Example 10.1 is the prototype: $H \equiv 0$ and $K = -1/\cosh^4 v < 0$ (Exercise 10.5).

10.9. Assigned exercises

Exercise 10.5. Goal: compute both fundamental forms of a non-trivial surface of revolution and recognize a minimal surface. **Estimated time:** 30 minutes.

Consider the *catenoid* $X(u, v) = (\cosh v \cos u, \cosh v \sin u, v)$. (i) Compute E, F, G and check that the first form is *conformal*, that is of the form $\lambda(u, v)(du^2 + dv^2)$. (ii) Compute N, e, f, g , and deduce K and H . (iii) Check that $H \equiv 0$ and that $K = -1/\cosh^4 v$; compute K for $v = 0, v = 0.5$ and $v = 1$, and comment on the behaviour as $|v| \rightarrow \infty$.

Exercise 10.6. Goal: use the classification of points and relate it to the sign of K . **Estimated time:** 20 minutes.

Consider the family of paraboloids $X(u, v) = (u, v, au^2 + bv^2)$ with $a, b \in \mathbb{R}$. (i) Compute E, F, G, N, e, f, g and K at the origin. (ii) Classify the origin (elliptic, hyperbolic, parabolic, planar) as (a, b) varies, and describe in each case the position of the surface with respect

to the tangent plane $z = 0$, justifying it by Proposition 10.5. (iii) For $a = 1$ and $b = -1$, determine the two directions along which the surface meets its own tangent plane, and check that they are orthogonal.

Exercise 10.7. Goal: apply Proposition 10.7 and connect it with global Gauss–Bonnet.

Estimated time: 25 minutes.

On the torus $T_{a,b}$ with $a = 3$ and $b = 1$: (i) compute K and H at the points $v = 0$, $v = \pi/3$, $v = \pi/2$, $v = 2\pi/3$, $v = \pi$, and find where H vanishes; (ii) determine the area of the torus and the area of the region of positive curvature alone; (iii) verify by direct computation that $\iint K \, dA = 0$ and compare the result with Theorem 4.8 of Chapter 4.

10.10. Worked exercises

Exercise 10.8. Measure the Gaussian curvature of the sphere *from inside*, without ever leaving it: compare the area of a geodesic disc of radius ρ on the unit sphere with that of a plane disc of the same radius, and check that the relative defect determines K .

Solution. Strategy. A two-dimensional inhabitant of the sphere can measure distances along the surface and areas; the Theorema Egregium guarantees that this suffices to determine K . The most direct way is to compare the area of a geodesic disc with the Euclidean value $\pi\rho^2$.

Step 1 — the geodesic disc on the unit sphere. The set of points at geodesic distance $\leq \rho$ from a fixed point is the cap of angular half-width ρ (Theorem 7.7 of Chapter 7: the distance is the central angle). Its area, by Lemma 7.1, equals

$$A(\rho) = \int_0^{2\pi} \int_0^\rho \sin \vartheta \, d\vartheta \, d\varphi = 2\pi(1 - \cos \rho).$$

Step 2 — the series expansion. From $\cos \rho = 1 - \rho^2/2 + \rho^4/24 - \dots$,

$$A(\rho) = 2\pi\left(\frac{\rho^2}{2} - \frac{\rho^4}{24} + O(\rho^6)\right) = \pi\rho^2 - \frac{\pi\rho^4}{12} + O(\rho^6).$$

The spherical disc is therefore *smaller* than the plane disc of the same radius: this is the signature of positive curvature.

Step 3 — the Bertrand–Puisseux formula. Isolating the fourth-order term,

$$\frac{12(\pi\rho^2 - A(\rho))}{\pi\rho^4} = 1 + O(\rho^2) \xrightarrow{\rho \rightarrow 0} 1 = K,$$

and for a sphere of radius r the same computation with $A(\rho) = 2\pi r^2(1 - \cos(\rho/r))$ gives the limit $1/r^2 = K$.

Step 4 — independent check: the numerical values. On the unit sphere:

ρ	$A(\rho)$	$\pi\rho^2$	estimate of K
0.50	0.76917145	0.78539816	0.99170377
0.20	0.12524539	0.12566371	0.99866762
0.10	0.03138976	0.03141593	0.99966673
0.05	0.00785235	0.00785398	0.99991667

The estimate converges to 1 and the discrepancy divides by four when ρ is halved, consistently with the $O(\rho^2)$ error: from $\rho = 0.10$ to $\rho = 0.05$ the discrepancy goes from $3.3327 \cdot 10^{-4}$ to $8.333 \cdot 10^{-5}$, a ratio of 4.00. ✓

Step 5 — closed-form check. The next term of the expansion gives $\frac{12(\pi\rho^2 - A)}{\pi\rho^4} = 1 - \frac{\rho^2}{15} + O(\rho^4)$; for $\rho = 0.1$ this predicts $1 - 0.000667 = 0.999333$, whereas the exact value is 0.999667. The residue $3.3 \cdot 10^{-4}$ is of order ρ^4 : consistent. ✓

Interpretation. The computation is the operational proof of the *Theorema Egregium*: none of the quantities used — geodesic distance, area — requires leaving the surface, and yet the outcome is K , which had been defined by means of the second form. It is the method by which one could in principle establish experimentally the curvature of a space one is confined to.

Typical errors. (a) Confusing the geodesic radius ρ with the Euclidean radius of the cap, $\sin \rho$: for $\rho = 0.5$ these are 0.500000 and 0.479426, and using the second completely falsifies the computation. (b) Stopping at the second order in the expansion of $\cos$, which gives exactly $\pi\rho^2$ and hides the phenomenon. (c) Taking ρ too large: for $\rho = \pi/2$ (half a hemisphere) the spherical area is 6.283185 against 7.751569 for the plane disc, a ratio of 0.810569: the fourth-order term is no longer a good approximation.

Extension. The same formula, with the opposite sign, measures negative curvature: on a surface with $K < 0$ the geodesic disc is *larger* than the plane one. On the torus, the area of the region with $K > 0$ and that with $K < 0$ compensate exactly (Corollary 10.2).

Exercise 10.9. Decide, without attempting any construction, which of the following pairs of surfaces can be locally isometric: (a) plane and cylinder; (b) plane and cone; (c) plane and sphere; (d) sphere of radius 1 and sphere of radius 2; (e) cylinder and cone. For the possible pairs, exhibit the isometry or say where it is in the treatise.

Solution. Strategy. Theorem 10.2 provides an immediate *necessary* condition: the Gaussian curvatures must agree at corresponding points. It suffices to compute and compare them; where the condition is met one looks for the explicit isometry, remembering that equality of K is not in general sufficient (it is for constant K , but that result is imported, Remark 10.3).

Step 1 — the curvatures.

surface	K	H (normal indicated)
plane	0	0
cylinder of radius R	0	$-1/(2R)$ (outer)
cone of half-angle α	0	$-\cot \alpha/(2v)$ (Step 4 of 10.9)
sphere of radius r	$1/r^2$	$1/r$ (inner)

Step 2 — the answers. (a) *Yes*: $K = 0$ for both, and the isometry is explicit (Proposition 9.1(i)): $X(u, v) \mapsto (Ru, v, 0)$. (b) *Yes*: $K = 0$ for both, explicit isometry (Proposition 9.2(i)): the cone develops onto a sector of opening $2\pi \sin \alpha$. (c) *No*: $0 \neq 1/r^2$, and Theorem 10.2 rules it out at once. (d) *No*: $1/1^2 = 1 \neq 1/4 = 1/2^2$. The two spheres are *similar* but not isometric: a similarity of ratio λ multiplies the first form by λ^2 and divides K by λ^2 . (e) *Yes*: both have $K = 0$ and develop onto the plane, hence, composing the two developments, cylinder and cone are locally isometric to each other — but only locally: the composition maps a rectangle of the plane onto a sector, and is not globally defined.

Step 3 — numerical check of (d). If a local isometry between the two spheres existed, by Theorem 10.2 one would have $1 = 0.250000$: false. An independent geometric check: a geodesic triangle with three right angles has area $\pi/2 \cdot r^2$, that is 1.570796 on $r = 1$ and 6.283185 on $r = 2$, while the angles are the same; an isometry would preserve both angles and areas (Proposition 10.1). ✓

Interpretation. The Gaussian curvature is a computable *obstruction*: one need not try

all possible maps between two surfaces to establish that they are not isometric. A single number at a single point suffices.

Typical errors. (a) Concluding from the mere equality of K that the isometry exists: equality is necessary, and becomes sufficient only for constant curvature, by an imported theorem (Remark 10.3). (b) Using H as an invariant: the plane ($H = 0$) and the cylinder ($H \neq 0$) are locally isometric, hence H is not intrinsic. (c) Confusing “isometric” with “homeomorphic”: the plane and the punctured sphere are homeomorphic but not isometric.

Extension. The same obstruction explains why no geographical map preserves distances, and why the Mercator projection (conformal, hence angle-preserving) must dilate areas increasingly with latitude.

Exercise 10.10. Verify the Gauss equation (Theorem 10.1) on a surface other than the sphere, computing both sides for the torus $T_{2,1}$ at the point $v = \pi$ (inner equator).

Solution. Strategy. Compute the Christoffel symbols of the torus from Lemma 9.6, substitute them into the right-hand side of the Gauss equation, and compare with EK obtained from Proposition 10.7, which instead goes through the second form.

Step 1 — the Christoffel symbols of the torus. With $u^1 = u$, $u^2 = v$, profile $f(v) = a + b \cos v = w$ and $G = b^2$ constant, Lemma 9.6 gives

$$\Gamma_{12}^1 = \frac{f'}{f} = -\frac{b \sin v}{w}, \quad \Gamma_{11}^2 = -\frac{ff'}{G} = \frac{wb \sin v}{b^2} = \frac{w \sin v}{b},$$

and $\Gamma_{11}^1 = \Gamma_{22}^1 = \Gamma_{12}^2 = \Gamma_{22}^2 = 0$ (the last because $G' = 0$).

Step 2 — the right-hand side of the Gauss equation. Only two summands remain:

$$\partial_v \Gamma_{11}^2 - \Gamma_{12}^1 \Gamma_{11}^2 = \partial_v \left(\frac{w \sin v}{b} \right) + \frac{b \sin v}{w} \cdot \frac{w \sin v}{b}.$$

Now $\partial_v(w \sin v) = -b \sin^2 v + w \cos v$, hence the first summand is $(w \cos v - b \sin^2 v)/b$ and the second is $\sin^2 v$. Adding:

$$\frac{w \cos v - b \sin^2 v}{b} + \sin^2 v = \frac{w \cos v - b \sin^2 v + b \sin^2 v}{b} = \frac{w \cos v}{b}.$$

Step 3 — the left-hand side. By Proposition 10.7, $E = w^2$ and $K = \cos v/(bw)$, hence

$$EK = w^2 \cdot \frac{\cos v}{bw} = \frac{w \cos v}{b}.$$

The two sides agree *identically* in v : the Gauss equation is verified on the torus, and with it the fact that K — computed in Step 3 with the second form — is also obtained from E, F, G alone.

Step 4 — numerical check at $v = \pi$ with $a = 2$, $b = 1$. Here $w = a - b = 1$, $\cos v = -1$, $\sin v = 0$. Right-hand side: $\Gamma_{11}^2 = w \sin v/b = 0$ and $\Gamma_{12}^1 = 0$; but $\partial_v \Gamma_{11}^2 = (w \cos v - b \sin^2 v)/b = -1$. Hence the right-hand side equals $-1 + 0 = -1$. Left-hand side: $E = w^2 = 1$ and $K = \cos \pi/(1 \cdot 1) = -1$, hence $EK = -1$. ✓ The independent numerical computation by finite differences gives $K = -0.999999$. ✓

Step 5 — check at a second point, $v = 0$. Here $w = a + b = 3$, $\cos v = 1$, $\sin v = 0$: right-hand side $= (3 \cdot 1 - 0)/1 = 3$; left-hand side $EK = 9 \cdot \frac{1}{1 \cdot 3} = 3$. ✓

Interpretation. It is the same computation which, on the sphere, gives $\sin^2 \vartheta$ on both sides (Step 7 of the proof of Theorem 10.1). The remarkable fact is not that the two sides agree

— they must, by the theorem — but that the second one is written without ever naming e , f , g .

Typical errors. (a) Forgetting that $\Gamma_{22}^2 = G'/(2G) = 0$ only because in the torus G is constant; on a general surface of revolution it is not. (b) Getting the sign of $\Gamma_{11}^2 = -ff'/G$ wrong: here $f' = -b \sin v$, hence $\Gamma_{11}^2 = +w \sin v/b$. (c) Confusing the f of the profile with the $f = h_{12}$ of the second form: two different objects denoted by the same letter, as warned in the conventions of the chapter.

Extension. Repeating the computation for a general surface of revolution one obtains the classical formula $K = -\frac{f''}{f}$, valid when the profile is parametrized by arclength ($f'^2 + h'^2 = 1$, that is $G \equiv 1$): indeed then $\Gamma_{11}^2 = -ff'$, $\Gamma_{12}^1 = f'/f$ and the right-hand side becomes $-(f'^2 + ff'') + f'^2 = -ff''$, which divided by $E = f^2$ gives $-f''/f$.

10.11. Self-assessment quiz

- The *Gaussian curvature* at a point p is, by definition,
 - the sum of the principal curvatures
 - the determinant of the shape operator $\mathcal{S}_p = -dN_p$
 - the curvature of the most curved normal section
 - the integral of Π
- In the proof of the Gauss equation, the hypothesis that makes it legitimate to write $\partial_2 X_{11} = \partial_1 X_{12}$ is
 - $EG - F^2 > 0$
 - X of class C^3 (Schwarz's theorem on third derivatives)
 - $F = 0$
 - the surface is orientable
- For the torus $X(u, v) = ((a + b \cos v) \cos u, (a + b \cos v) \sin u, b \sin v)$ the Gaussian curvature equals
 - $\frac{\cos v}{b(a + b \cos v)}$
 - $\frac{1}{b^2}$
 - $\frac{\cos v}{a + b \cos v}$
 - 0
- On the torus with $a = 2$ and $b = 1$, the Gaussian curvature on the inner equator ($v = \pi$) equals
 - 0.333333
 - 1.000000
 - 0.333333
 - 0.000000
- In the proof of the *Theorema Egregium*, the step allowing one to conclude that K does not depend on the second form is
 - the division by $E > 0$ in the Gauss equation, together with the fact that the Γ_{ij}^k depend only on g_{ij}
 - Lagrange's identity
 - Euler's formula for the normal curvature
 - the self-adjointness of the shape operator
- A student claims: "the hyperbolic paraboloid $z = xy$ is a union of straight lines, hence it is developable and can be obtained by bending a flat sheet". The error is
 - it is not a union of lines
 - being ruled does not imply $K = 0$: here $K = -1/(1 + u^2 + v^2)^2 < 0$ everywhere
 - the sheet must be cut first
 - none, the claim is correct
- A membrane tensioned without a pressure difference takes a minimal shape ($H \equiv 0$). What follows about the sign of the Gaussian curvature?
 - $K > 0$ everywhere
 - $K \leq 0$ everywhere, because $H^2 \geq K$
 - K is constant
 - nothing

8. The elementary proof of the Brioschi formula, carried out with properties of matrices and determinants and without Christoffel symbols, is found in

- A) Euclid, *Elements*, book XI B) Lee-Peng Teo, *The Brioschi Formula for the Gaussian Curvature*, arXiv:2404.00835 (2024) C) Archimedes, *On the Sphere and Cylinder* D) Newton, *Principia*, book I

10.12. Teacher's key

BEST PRACTICE

Q1 — answer B. *Rationale:* Definition 10.4 sets $K = \kappa_1\kappa_2 = \det \mathcal{S}$; the formula $(eg - f^2)/(EG - F^2)$ is a consequence of it (Proposition 10.2(iii)). *Distractors:* A is twice the mean curvature; C is κ_2 , only one of the two principal curvatures (Proposition 10.3); D makes no dimensional sense. *Diagnosed error:* confusing the product and the sum of the eigenvalues, that is K and H . *Source:* do Carmo, Chapter 3 (source [2]); Shifrin §2.2 (source [1]). *Prerequisite:* determinant and trace of an endomorphism. *Remediation:* compute K and H for the cylinder (Exercise 10.3) and observe that $K = 0$ while $H \neq 0$.

Q2 — answer B. *Rationale:* Step 2 of the proof of Theorem 10.1 uses the equality of the mixed *third* derivatives, which requires $X \in C^3$; it is the only regularity hypothesis entering at that point. *Distractors:* A is used at Step 6 (to invert (g_{ij})); C is not required by the general statement (it is only needed in Proposition 10.6); D plays no role in the local proof, where N is defined by the parametrization. *Diagnosed error:* listing hypotheses without knowing where they are needed. *Source:* do Carmo, §4-3, p. 235 (source [2]). *Prerequisite:* Schwarz's theorem. *Remediation:* rewrite Steps 3–5 with X only C^2 and locate the point where the argument fails.

Q3 — answer A. *Rationale:* by Proposition 10.7, $\kappa_1 = -\cos v/w$, $\kappa_2 = -1/b$ and $K = \kappa_1\kappa_2 = \cos v/(bw)$ with $w = a + b \cos v$. *Distractors:* B is κ_2^2 , the square of the curvature of the tube; C loses the factor $1/b$; D is the right answer only for the cylinder. *Diagnosed error:* failing to multiply the two principal curvatures, or losing a dimensional factor (K has the dimensions of a length to the power -2 , and $\cos v/(a + b \cos v)$ those of a length to the power -1 : a dimensional check discards C at once). *Source:* Shifrin, solution 2.2.3(b) (source [1]), who obtains the same formula independently. *Prerequisite:* computation of e, f, g . *Remediation:* check the formula at $v = \pi/2$, where it must give 0.

Q4 — answer B. *Rationale:* with $a = 2$, $b = 1$ and $v = \pi$ one has $w = a - b = 1$ and $\cos v = -1$, hence $K = -1/(1 \cdot 1) = -1.000000$. *Distractors:* A is the value on the outer equator ($v = 0$: $K = 1/(1 \cdot 3) = 0.333333$); C is the value at $v = 2\pi/3$; D is the value on the extreme circles $v = \pm\pi/2$. *Diagnosed error:* using $a + b$ instead of $a - b$ in the denominator, that is confusing outer and inner equator. *Source:* Proposition 10.7, Step 6, with a numerical check by finite differences (-0.999999). *Prerequisite:* none. *Remediation:* tabulate $K(v)$ for $v = 0, \pi/3, \pi/2, 2\pi/3, \pi$ and draw the sign diagram on the section of the torus (Figure 10.2).

Q5 — answer A. *Rationale:* Step 1 of the proof of Theorem 10.2 divides by $E > 0$ and observes that the right-hand side of the Gauss equation contains only the Γ_{ij}^k , which by Lemma 9.3 depend only on the first form. *Distractors:* B is used in Lemma 10.1, not here; C and D are true results but from another section, and in particular D is used to prove that κ_1, κ_2 are real. *Diagnosed error:* attributing intrinsic character to the definition of K rather than to the identity that rewrites it. *Source:* do Carmo §4-3 (source [2]); Kühnel (source [4]). *Prerequisite:* the Gauss equation. *Remediation:* write out the chain of dependences $K \leftarrow \Gamma \leftarrow g_{ij}$ explicitly and check it on the sphere.

Q6 — answer B. *Rationale:* by Proposition 10.8 a ruled surface has $K = -f^2/D \leq 0$, and in the case $z = xy$ one computes $f = 1/\sqrt{1+u^2+v^2} \neq 0$, hence $K < 0$ everywhere (Exercise 10.4); by Theorem 10.2 no local isometry with the plane is possible. *Distractors:* A is false (it is doubly ruled); C solves nothing, since cutting does not change K ; D accepts the error. *Diagnosed error:* identifying “ruled” with “developable”. *Source:* Exercise 10.4, with a numerical check at three points. *Prerequisite:* computation of e, f, g . *Remediation:* compute K for the cylinder (a developable ruled surface) and for $z = xy$ (a non-developable ruled surface) and compare.

Q7 — answer B. *Rationale:* Corollary 10.1 gives $H^2 \geq K$; if $H \equiv 0$ then $K \leq 0$ everywhere. *Distractors:* A is the opposite; C does not follow (the catenoid has $K = -1/\cosh^4 v$, not constant); D ignores the inequality. *Diagnosed error:* failing to use the algebraic relation between the mean and the product of two real numbers. *Source:* Corollary 10.1, with the catenoid as an example (Exercise 10.5). *Prerequisite:* the arithmetic–geometric mean inequality. *Remediation:* check $H^2 \geq K$ on the torus at the four points tabulated in Step 6 of the proof of Proposition 10.7.

Q8 — answer B. *Rationale:* the paper by Lee-Peng Teo, arXiv:2404.00835, submitted on 1 April 2024 (version 2 of 2 April 2024), presents exactly “a direct elementary proof of the Brioschi formula without using Christoffel symbols”, using properties of matrices and determinants. *Distractors:* A, C and D are real works but on different subjects, and serve to check that the student does not “recognize” a name at random. *Diagnosed error:* citing from memory without checking. *Source:* the same preprint, verified on 5 September 2026 (source [3]). *Prerequisite:* none. *Remediation:* open the source and compare the formula with the one in Remark 10.1, checking the case $F = 0$.

Answers to the assigned exercises.

Exercise 10.5. (i) $X_u = \cosh v(-\sin u, \cos u, 0)$ and $X_v = (\sinh v \cos u, \sinh v \sin u, 1)$, hence $E = \cosh^2 v$, $F = 0$ and $G = \sinh^2 v + 1 = \cosh^2 v$: the first form is $\cosh^2 v (du^2 + dv^2)$, conformal with $\lambda = \cosh^2 v$. (ii) $X_u \times X_v = \cosh v(\cos u, \sin u, -\sinh v)$, of norm $\cosh^2 v$, hence $N = (\cos u, \sin u, -\sinh v)/\cosh v$. With $X_{uu} = \cosh v(-\cos u, -\sin u, 0)$, $X_{uv} = \sinh v(-\sin u, \cos u, 0)$ and $X_{vv} = (\cosh v \cos u, \cosh v \sin u, 0)$ one obtains $e = -1$, $f = 0$, $g = 1$. Hence $K = \frac{eg - f^2}{EG - F^2} = \frac{-1}{\cosh^4 v}$ and $H = \frac{eG - 2fF + gE}{2(EG - F^2)} = \frac{-\cosh^2 v + \cosh^2 v}{2\cosh^4 v} = 0$. (iii) $H \equiv 0$: the catenoid is minimal, and is indeed the only non-planar minimal surface of revolution. $K(0) = -1$, $K(0.5) = -1/\cosh^4(0.5) = -0.618500$, $K(1) = -1/\cosh^4(1) = -0.176378$ (values confirmed by the numerical computation with finite differences). As $|v| \rightarrow \infty$ one has $K \rightarrow 0$: the catenoid flattens out, locally approaching an ever wider cylinder. Consistently with Corollary 10.1, $H^2 = 0 \geq K$ everywhere. ✓

Exercise 10.6. (i) At the origin $X_u = (1, 0, 0)$ and $X_v = (0, 1, 0)$, hence $E = G = 1$, $F = 0$ and $N = (0, 0, 1)$; with $X_{uu} = (0, 0, 2a)$, $X_{vv} = (0, 0, 2b)$ and $X_{uv} = 0$ one has $e = 2a$, $g = 2b$, $f = 0$ and $K = 4ab$. (ii) $ab > 0$: elliptic point, and by Proposition 10.5(i) the surface lies entirely on one side of the plane $z = 0$ near the origin (above if $a, b > 0$, below if $a, b < 0$). $ab < 0$: hyperbolic point, the surface crosses the tangent plane (part (ii) of the same proposition). $ab = 0$ with $(a, b) \neq (0, 0)$: parabolic point, $z = au^2$ (or bv^2) is a parabolic cylinder and the surface touches the plane along a line. $a = b = 0$: planar point, the surface is the plane. Numerical check: $K = 4.000000$ for $(1, 1)$, -4.000000 for $(1, -1)$, 0.000000 for $(1, 0)$. ✓ (iii) For $a = 1$, $b = -1$ the surface is $z = u^2 - v^2$ and meets $z = 0$ where $u^2 = v^2$, that is along the two lines $v = \pm u$, of directions $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$: their scalar product is $(1 - 1)/2 = 0$, hence they are orthogonal. They are the *asymptotic directions*, those in which the normal curvature vanishes; they are orthogonal precisely because here $H = 0$ (Euler’s formula of Proposition 10.3 gives $\kappa_n(\theta) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta = 0$ with

$\kappa_2 = -\kappa_1$, that is $\tan^2 \theta = 1$).

Exercise 10.7. With $a = 3$, $b = 1$ and $w = 3 + \cos v$: (i) $K = \cos v/w$ and $H = -(3 + 2 \cos v)/(2w)$.

v	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
w	4	3.5	3	2.5	2
K	0.250000	0.142857	0.000000	-0.200000	-0.500000
H	-0.625000	-0.571429	-0.500000	-0.400000	-0.250000

H vanishes where $3 + 2 \cos v = 0$, that is $\cos v = -1.5$: impossible. For $a > 2b$ the mean curvature of the torus never vanishes; it vanishes only if $a \leq 2b$, at $v = \arccos(-a/(2b))$. (ii) Area $= 4\pi^2 ab = 4\pi^2 \cdot 3 = 118.435253$. The region of positive curvature is $|v| < \pi/2$, of area $\int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (3 + \cos v) dv du = 2\pi(3\pi + 2) = 2\pi \times 11.424778 = 71.789423$, that is 60.6% of the total. (iii) $\iint K dA = \int_0^{2\pi} \int_0^{2\pi} \frac{\cos v}{w} w dv du = 2\pi [\sin v]_0^{2\pi} = 0$, in agreement with Theorem 4.8 of Chapter 4, which for $\chi(\text{torus}) = 0$ predicts precisely vanishing total curvature. ✓

10.13. Chapter formulary

First and second forms

$$E = \langle X_1, X_1 \rangle, \quad F = \langle X_1, X_2 \rangle, \quad G = \langle X_2, X_2 \rangle; \quad D = EG - F^2 = \|X_1 \times X_2\|^2 > 0$$

$$N = \frac{X_1 \times X_2}{\|X_1 \times X_2\|}; \quad e = \langle X_{11}, N \rangle, \quad f = \langle X_{12}, N \rangle, \quad g = \langle X_{22}, N \rangle$$

$$L(\gamma) = \int \sqrt{g_{ij} \dot{u}^i \dot{u}^j} ds; \quad dA = \sqrt{EG - F^2} du^1 du^2$$

Shape operator and curvatures

$$\mathcal{S} = -dN = \Gamma^{-1} \Pi; \quad X_{ij} = \Gamma_{ij}^k X_k + h_{ij} N; \quad N_i = -h_{ij} g^{jk} X_k$$

$$K = \kappa_1 \kappa_2 = \frac{eg - f^2}{EG - F^2}, \quad H = \frac{\kappa_1 + \kappa_2}{2} = \frac{eG - 2fF + gE}{2(EG - F^2)}, \quad H^2 \geq K$$

$$\kappa_n(w) = \Pi(w) = \langle \mathcal{S}w, w \rangle; \quad \kappa_n(\theta) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta \quad (\text{Euler})$$

$K > 0$ elliptic; $K < 0$ hyperbolic; $K = 0$, $\mathcal{S} \neq 0$ parabolic; $\mathcal{S} = 0$ planar

Theorema Egregium

$$E K = \partial_2 \Gamma_{11}^2 - \partial_1 \Gamma_{12}^2 + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - (\Gamma_{12}^2)^2$$

$$F = 0: \quad K = -\frac{1}{2\sqrt{EG}} \left[\partial_2 \left(\frac{E_2}{\sqrt{EG}} \right) + \partial_1 \left(\frac{G_1}{\sqrt{EG}} \right) \right]$$

$$\phi \text{ local isometry} \implies \bar{K} \circ \phi = K \quad (\text{but } \bar{H} \circ \phi \neq H \text{ in general})$$

Examples

$$\text{sphere } r: K = \frac{1}{r^2}, H = \frac{1}{r} \text{ (inner normal);} \quad \text{plane, cylinder, cone: } K = 0$$

$$\text{cylinder } R: H = -\frac{1}{2R}; \quad \text{cone } \alpha: H = -\frac{\cot \alpha}{2v}$$

$$\text{torus } (a, b): K = \frac{\cos v}{b(a + b \cos v)}, \quad H = -\frac{a + 2b \cos v}{2b(a + b \cos v)}, \quad \text{Area} = 4\pi^2 ab, \quad \iint K dA = 0$$

$$\text{catenoid: } K = -\frac{1}{\cosh^4 v}, H = 0; \quad \text{ruled: } K = -\frac{f^2}{EG - F^2} \leq 0$$

$$\text{Bertrand-Puiseux: } K = \lim_{\rho \rightarrow 0^+} \frac{12(\pi \rho^2 - A(\rho))}{\pi \rho^4}$$

Sources and bibliography

- [1] T. Shifrin, *Differential Geometry: A First Course in Curves and Surfaces*, University of Georgia, April 2021 edition, 125 pp. Chapter 2 *Surfaces: Local Theory*: §1 *Parametrized Surfaces and the First Fundamental Form* (p. 35), §2 *The Gauss Map and the Second Fundamental Form* (p. 44), §3 *The Codazzi and Gauss Equations and the Fundamental Theorem of Surface Theory* (p. 57), §4 *Covariant Differentiation, Parallel Translation, and Geodesics* (p. 66). Exercise solutions used as independent checks: 2.1.4(a), area of the torus $4\pi^2 ab$ (Corollary 10.2); 2.1.7, characterization of local isometry with the plane by $E = G = 1$, $F = 0$; 2.2.3(b), shape operator, H and $K = \cos u / (b(a + b \cos u))$ for the torus (Proposition 10.7); 2.2.3(d), catenoid with $H = 0$ and $K = -\operatorname{sech}^4 u$ (Exercise 10.5); 3.3.8, the same K of the torus obtained with differential forms. File <https://math.franklin.uga.edu/sites/default/files/inline-files/ShifrinDiffGeo.pdf>; verified on 5 September 2026.
- [2] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, revised and updated edition, Dover, Mineola (NY), 2016 (first edition Prentice-Hall, 1976), ISBN 978-0-486-80699-0. Chapter 3 *The Geometry of the Gauss Map* (Gauss map, second form, principal curvatures) and Chapter 4 *The Intrinsic Geometry of Surfaces*, §4-3 *The Gauss Theorem and the Equations of Compatibility*, p. 235: Gauss and Codazzi–Mainardi equations and *Theorema Egregium*, the primary source of Theorem 10.1 and Theorem 10.2; in the same chapter Minding’s theorem, imported in Remark 10.3. Verified on 5 September 2026.
- [3] Lee-Peng Teo, *The Brioschi Formula for the Gaussian Curvature*, arXiv:2404.00835 [math.DG], submitted on 1 April 2024, version 2 of 2 April 2024. Elementary and direct proof of the Brioschi formula, carried out with properties of matrices and determinants and without Christoffel symbols; it is the source of the import declared in Remark 10.1. File <https://arxiv.org/abs/2404.00835>; verified on 5 September 2026.
- [4] W. Kühnel, *Differential Geometry: Curves — Surfaces — Manifolds*, third edition, Student Mathematical Library vol. 77, American Mathematical Society, Providence (RI), 2015 (second edition: vol. 16, 2006), ISBN 978-1-4704-2320-9. English translation by B. Hunt of the German text *Differentialgeometrie*; chapters on the local theory of surfaces and on intrinsic geometry (*Theorema Egregium*, Minding’s theorem, spaces of constant curvature). Verified on 5 September 2026.
- [5] C. F. Gauss, *Disquisitiones generales circa superficies curvas*, presented to the Königliche Gesellschaft der Wissenschaften of Göttingen on 8 October 1827 and published in the *Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores*, vol. VI, Göttingen, 1828. Article 12 contains the statement Gauss himself calls *theorema egregium*. Primary historical source of Theorem 10.2.
- [6] Massachusetts Institute of Technology, OpenCourseWare, course 18.950 *Differential Geometry*, Fall 2008: chapter-by-chapter notes, with the local geometry of hypersurfaces, the *Theorema Egregium* and the Gauss curvature expressed through the first fundamental form. File <https://ocw.mit.edu/courses/18-950-differential-geometry-fall-2008/>; verified on 5 September 2026.

CHAPTER 11

The circle in any dimension: hyperspheres and concentration of measure

The circumference and the sphere are the cases $n = 2$ and $n = 3$ of an infinite family: the spheres $S^{n-1} \subset \mathbb{R}^n$. Passing to arbitrary dimension is not an exercise in generality for its own sake. On the one hand Euler's Gamma function puts into a single formula the perimeter of the circle, the area of the sphere and all the rest; on the other hand phenomena appear which plane intuition does not predict and which are today central in statistics and in machine learning: the volume of the unit ball tends to zero, the mass concentrates near the boundary and near every equator, a ball inscribed among others can stick out of the cube containing them. In the second half of the chapter the circle returns as protagonist in a different role: as the *group* of rotations of the plane, as the building block of every rotation in every dimension, and finally as the fibre of the Hopf fibration, which fills the three-dimensional sphere with circles.

INFO

The idea in one line. $V_n = \pi^{n/2}/\Gamma(n/2 + 1)$ grows up to $n = 5$ and then vanishes; on S^{n-1} every coordinate has mean square value $1/n$, so that almost all of the sphere lies near any equator whatsoever; and every rotation of $\mathbb{R}^n$, in every dimension, decomposes into rotations of circles — with S^1 and S^3 which are, themselves, groups.

INFO

Level and prerequisites. Level B–D. Sections 11.1 and 11.2 are of level B–C: they require the Gaussian integral, the change of variables to generalized polar coordinates and the Gamma function. Section 11.4 is of level C: Chebyshev's inequality and elementary estimates of integrals are used. Section 11.5 is of level B–C (linear algebra: orthogonal matrices, eigenvalues, reflections), section 11.6 of level C and section 11.7 of level C–D, but it requires no algebraic topology: complex numbers suffice. Prerequisites within the treatise: complex numbers and the unit circle (Volume II, Chapter 3); sphere, planar sections and spherical coordinates (Chapter 7); isometries of the plane and the classification of $O(2)$ (Chapter 5). The rotations of $\mathbb{R}^3$ and the quaternions are taken up again, from the point of view of kinematics, in Chapter 28; the fundamental group of S^1 , not needed here, is in Chapter 12.

Warning. Conventions of this chapter.

- $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ is the *unit sphere* of $\mathbb{R}^n$; the index denotes the *dimension of the surface*, not of the ambient space. Thus $S^1 \subset \mathbb{R}^2$ is the circle and $S^2 \subset \mathbb{R}^3$ the ordinary sphere.
- $B^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ is the solid unit ball; V_n its n -dimensional volume and Σ_{n-1} the area of S^{n-1} .
- $\langle \cdot, \cdot \rangle$ is the standard scalar product and $\|x\| = \sqrt{\langle x, x \rangle}$.
- Γ is Euler's Gamma function, $\Gamma(t) = \int_0^\infty u^{t-1} e^{-u} du$, with $\Gamma(n+1) = n!$, $\Gamma(t+1) = t\Gamma(t)$ and $\Gamma(1/2) = \sqrt{\pi}$.

- σ denotes the uniform probability measure on S^{n-1} (the rotation-invariant one, of total mass 1).
- Decimal separators are points: 0.5.

11.1. Spheres and balls in $\mathbb{R}^n$

Definition 11.1: n -dimensional sphere and ball. For $n \geq 1$: $S^{n-1}(O, r) = \{x \in \mathbb{R}^n : |x| = r\}$ is the sphere (intrinsic dimension $n - 1$); $B^n(O, r) = \{x \in \mathbb{R}^n : |x| \leq r\}$ is the ball (dimension n). Conventions: $S^0 = \{-r, r\}$, S^1 the circumference, S^2 the classical sphere. One writes S^{n-1} and B^n for O at the origin and $r = 1$.

Remark 11.1: Why S^{n-1} lives in $\mathbb{R}^n$. The sphere S^{n-1} is a manifold of dimension $n - 1$ embedded in $\mathbb{R}^n$: S^1 is a curve (1-dim) in $\mathbb{R}^2$, S^2 a surface (2-dim) in $\mathbb{R}^3$. The superscript indicates the intrinsic dimension, not that of the ambient space. The same convention is used in Chapter 7 (Definition 7.3), and explains why the ball, which is n -dimensional, carries the index n and its boundary the index $n - 1$.

Exercise 11.1. Why does S^0 have “area” 2?

Solution. $S^0 = \{-r, r\}$ is the boundary of the segment $B^1 = [-r, r]$. The “0-dimensional measure” (counting) of its points is 2. Consistently, the general formula $\text{Area}(S^{n-1}) = 2\pi^{n/2}/\Gamma(n/2) r^{n-1}$ of Theorem 11.1 gives for $n = 1$: $2\pi^{1/2}/\Gamma(1/2) \cdot r^0 = 2\sqrt{\pi}/\sqrt{\pi} = 2$. ✓

11.2. Hyperspheres: volume, surface and surprises

Theorem 11.1: Volume of the ball and area of the sphere in $\mathbb{R}^n$. For every $n \geq 1$,

$$V_n = \text{vol}(B^n) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}, \quad \Sigma_{n-1} = \text{area}(S^{n-1}) = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} = n V_n,$$

and for radius R one has $V_n(R) = V_n R^n$, $\Sigma_{n-1}(R) = \Sigma_{n-1} R^{n-1}$. Moreover the recursion holds:

$$V_n = \frac{2\pi}{n} V_{n-2}, \quad V_1 = 2, \quad V_2 = \pi.$$

Proof. Step 1 — the Gaussian integral in $\mathbb{R}^n$. Since $e^{-\|x\|^2} = \prod_{i=1}^n e^{-x_i^2}$, Fubini’s theorem gives

$$I_n = \int_{\mathbb{R}^n} e^{-\|x\|^2} dx = \left(\int_{\mathbb{R}} e^{-t^2} dt \right)^n = \pi^{n/2},$$

using the classical $\int_{\mathbb{R}} e^{-t^2} dt = \sqrt{\pi}$.

Step 2 — the same integral in polar coordinates. The volume element in spherical coordinates of $\mathbb{R}^n$ is $dx = r^{n-1} dr d\sigma$, with $d\sigma$ the surface measure on S^{n-1} ; the integrand depends only on r , hence

$$I_n = \Sigma_{n-1} \int_0^\infty e^{-r^2} r^{n-1} dr.$$

Step 3 — the radial integral. With the substitution $u = r^2$, $du = 2r dr$:

$$\int_0^\infty e^{-r^2} r^{n-1} dr = \frac{1}{2} \int_0^\infty e^{-u} u^{\frac{n}{2}-1} du = \frac{1}{2} \Gamma\left(\frac{n}{2}\right).$$

Step 4 — the area. Equating Steps 1 and 2,

$$\pi^{n/2} = \Sigma_{n-1} \cdot \frac{1}{2} \Gamma\left(\frac{n}{2}\right) \implies \Sigma_{n-1} = \frac{2\pi^{n/2}}{\Gamma(n/2)}.$$

Step 5 — the volume. Integrating the areas of the concentric spheres,

$$V_n = \int_0^1 \Sigma_{n-1} r^{n-1} dr = \frac{\Sigma_{n-1}}{n} = \frac{2\pi^{n/2}}{n \Gamma(n/2)} = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)},$$

using $\frac{n}{2} \Gamma(\frac{n}{2}) = \Gamma(\frac{n}{2} + 1)$.

Step 6 — the recursion.

$$\frac{V_n}{V_{n-2}} = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} \cdot \frac{\Gamma(\frac{n}{2})}{\pi^{(n-2)/2}} = \pi \cdot \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n}{2} + 1)} = \frac{\pi}{n/2} = \frac{2\pi}{n}.$$

Step 7 — check in the known cases.

$$V_1 = \frac{\pi^{1/2}}{\Gamma(3/2)} = \frac{\sqrt{\pi}}{\frac{1}{2}\sqrt{\pi}} = 2 \checkmark \quad (\text{the segment } [-1, 1]),$$

$$V_2 = \frac{\pi}{\Gamma(2)} = \pi \checkmark, \quad V_3 = \frac{\pi^{3/2}}{\Gamma(5/2)} = \frac{\pi^{3/2}}{\frac{3}{4}\sqrt{\pi}} = \frac{4\pi}{3} \checkmark,$$

and $\Sigma_1 = 2V_2 = 2\pi$ (the circumference), $\Sigma_2 = 3V_3 = 4\pi$ (the spherical surface). $\checkmark$

Step 8 — the circle as the case $n = 2$. The formulas of Volume I, $L = 2\pi R$ and $A = \pi R^2$, are the case $n = 2$ of these; those of the sphere, $4\pi R^2$ and $\frac{4}{3}\pi R^3$, the case $n = 3$. The chapter thus closes a circle: the elementary formulas are not isolated cases but the first terms of a sequence governed by the Gamma function. $\square$

Example 11.1: The table of volumes and the maximum in dimension five. Applying Theorem 11.1:

n	V_n (closed form)	V_n	$\Sigma_{n-1} = nV_n$
1	2	2.00000	2.00000
2	π	3.14159	6.28319
3	$\frac{4}{3}\pi$	4.18879	12.56637
4	$\frac{1}{2}\pi^2$	4.93480	19.73921
5	$\frac{8}{15}\pi^2$	5.26379	26.31894
6	$\frac{1}{6}\pi^3$	5.16771	31.00628
7	$\frac{16}{105}\pi^3$	4.72477	33.07336
8	$\frac{1}{24}\pi^4$	4.05871	32.46969
10	$\frac{1}{120}\pi^5$	2.55016	25.50164
20	—	0.025807	0.516138

The maximum. The volume grows up to $n = 5$ and then decreases. The reason is read off the recursion $V_n = \frac{2\pi}{n} V_{n-2}$: the factor $2\pi/n$ exceeds 1 as long as $n < 2\pi = 6.283$, and is smaller afterwards. The maximum therefore falls at $n = 5$ among odd dimensions and at $n = 6$ among even ones; between the two, $V_5 = 5.26379$ beats $V_6 = 5.16771$.

The area. The surface has its maximum at $n = 7$ ($\Sigma_6 = 33.073$), for the same reason shifted by one: $\Sigma_{n-1} = nV_n$.

Check of the recursion. $V_7 = \frac{2\pi}{7}V_5 = \frac{6.28319}{7} \times 5.26379 = 0.897598 \times 5.26379 = 4.72477$.
✓

Exercise 11.2. Compute $\text{Vol}(B^4, 1)$ using $\Gamma(3) = 2$, and check with the recurrence $V_n = \frac{2\pi}{n}V_{n-2}$.

Solution. For $n = 4$: $\Gamma(n/2 + 1) = \Gamma(3) = 2! = 2$, hence $\text{Vol}(B^4, 1) = \pi^2/2 \approx 4.93480$. Independent check with the recurrence: $V_4 = \frac{2\pi}{4}V_2 = \frac{\pi}{2} \cdot \pi = \frac{\pi^2}{2}$. ✓ Consistent with the table of Example 11.1.

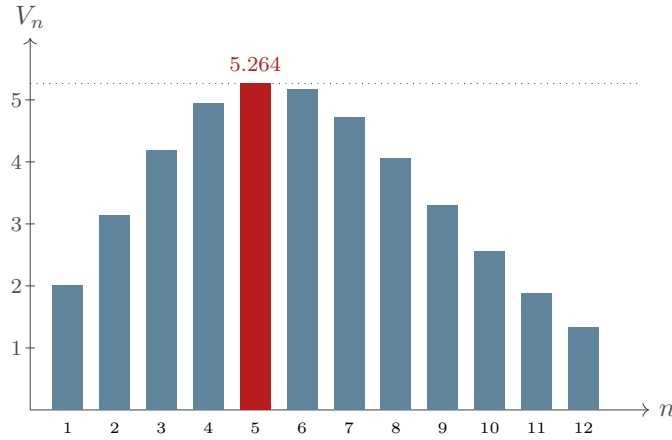


Figure 11.1: The volume $V_n = \pi^{n/2}/\Gamma(n/2 + 1)$ of the unit ball in dimension n . Against all intuition the sequence is not monotone: it grows up to $n = 5$, where it attains its maximum $V_5 = 8\pi^2/15 \simeq 5.264$, then decreases and tends to zero faster than any exponential. Already in dimension 11 the unit ball has smaller volume than the segment $[-1, 1]$, and in dimension 100 its volume is about $2 \cdot 10^{-40}$: almost all of the circumscribed cube lies outside the ball.

11.3. The volume vanishes: the ball against the cube

Theorem 11.2: The volume of the unit ball vanishes. $\lim_{n \rightarrow \infty} V_n = 0$; indeed $V_n \rightarrow 0$ faster than any exponential. Moreover the ratio of the volume of the unit ball to that of the circumscribed cube $[-1, 1]^n$ tends to zero:

$$\frac{V_n}{2^n} \rightarrow 0.$$

Proof. Step 1 — from the recursion. For $n > 2\pi$, i.e. $n \geq 7$, we have $\frac{2\pi}{n} < 1$, hence $V_n < V_{n-2}$: the two subsequences (even and odd) are eventually decreasing.

Step 2 — superexponential decay. Iterating the recursion m times from n ,

$$V_n = \frac{2\pi}{n} \cdot \frac{2\pi}{n-2} \cdots \frac{2\pi}{n-2m+2} \cdot V_{n-2m}.$$

For large n most factors are smaller than $1/2$ (all those with $n - 2j > 4\pi$): hence $V_n \leq C 2^{-(n/2-K)}$ for suitable constants, and in fact the decay is much faster, since the factors tend

to zero. Formally, Stirling's formula $\Gamma(t+1) \sim \sqrt{2\pi t} (t/e)^t$ gives

$$V_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)} \sim \frac{1}{\sqrt{\pi n}} \left(\frac{2\pi e}{n}\right)^{n/2},$$

which tends to zero because the base $2\pi e/n$ eventually becomes smaller than 1.

Step 3 — comparison with the cube. Since $V_n \rightarrow 0$ and $2^n \rightarrow \infty$, the ratio tends to zero a fortiori. Numerically:

$$n = 2 : \frac{\pi}{4} = 0.7854; \quad n = 3 : 0.5236; \quad n = 10 : \frac{2.55016}{1024} = 0.002490; \quad n = 20 : \frac{0.025807}{1\,048\,576} = 2.46 \times 10^{-8}.$$

Step 4 — how to read it. The ball is inscribed in the cube and touches its faces: in high dimension, therefore, *almost all the volume of the cube lies in the corners*. The cube $[-1, 1]^n$ has 2^n vertices, each at distance $\sqrt{n}$ from the centre against the radius 1 of the ball: the corners stretch out and multiply, and carry away all the volume.

Step 5 — a plausibility check. A uniform point in the cube has independent coordinates uniform on $[-1, 1]$, with $\mathbb{E}[x_i^2] = 1/3$: hence $\mathbb{E}[\|x\|^2] = n/3$, and by the law of large numbers $\|x\| \approx \sqrt{n/3}$, which exceeds 1 already for $n = 4$ and diverges. Almost no point of the cube falls in the ball, consistently with Step 3. ✓ □

Theorem 11.3: The ball that sticks out of the cube. In the cube $[-1, 1]^n$ place 2^n balls of radius $\frac{1}{2}$, centred at the points $(\pm\frac{1}{2}, \dots, \pm\frac{1}{2})$: they are contained in the cube and mutually tangent. Let $\mathcal{B}$ be the ball centred at the origin and tangent to all of them. Then its radius is

$$r_n = \frac{\sqrt{n} - 1}{2},$$

and it exceeds 1 — that is, $\mathcal{B}$ sticks out of the cube — for every $n \geq 10$.

Proof. **Step 1 — the small balls lie in the cube.** A centre has all coordinates equal to $\pm 1/2$; adding the radius $1/2$ one reaches at most ± 1 , i.e. the face of the cube.

Step 2 — they are mutually tangent. Two centres differing in one coordinate are at distance $1 = \frac{1}{2} + \frac{1}{2}$: the corresponding balls are tangent. Centres differing in more coordinates are farther apart, so the balls do not overlap.

Step 3 — the radius of the central ball. The distance from the origin to a centre is

$$\left\| \left(\frac{1}{2}, \dots, \frac{1}{2} \right) \right\| = \sqrt{n \cdot \frac{1}{4}} = \frac{\sqrt{n}}{2}.$$

External tangency forces $r_n + \frac{1}{2} = \frac{\sqrt{n}}{2}$, whence $r_n = \frac{\sqrt{n}-1}{2}$.

Step 4 — comparison with the cube. The cube $[-1, 1]^n$ has inradius 1 (distance from the centre to the faces). Hence

$$r_n > 1 \iff \sqrt{n} > 3 \iff n > 9,$$

i.e. for $n \geq 10$.

Step 5 — values.

$$r_2 = \frac{\sqrt{2}-1}{2} = 0.2071, \quad r_3 = \frac{\sqrt{3}-1}{2} = 0.3660, \quad r_9 = \frac{3-1}{2} = 1.0000, \quad r_{10} = \frac{\sqrt{10}-1}{2} = 1.0811.$$

In dimension 9 the central ball exactly touches the faces of the cube; from 10 onwards it crosses them. ✓

Step 6 — how is this possible. Intuition says a ball “in the middle” of other inscribed balls cannot leave the container, but that intuition is two-dimensional. In high dimension the

2^n centres are all at distance $\sqrt{n}/2$ from the origin, which grows without bound, while the faces remain at distance 1: the space “between the corners” becomes enormous. It is the other side of Theorem 11.2: the volume of the cube lies in the corners, and the central ball takes a growing share of it.

Step 7 — relative volume. Despite sticking out of the cube, the central ball has volume $V_n r_n^n$, which by Theorem 11.2 still tends to zero relative to 2^n : leaving the cube does not mean filling it. In dimension 10: $V_{10} r_{10}^{10} = 2.55016 \times (1.08114)^{10} = 2.55016 \times 2.18138 = 5.5628$, against $2^{10} = 1024$ for the cube, i.e. 0.54%. ✓ $\square$

11.4. Concentration of measure

Theorem 11.4: Concentration of measure on the sphere. Let σ be the uniform probability measure on S^{n-1} and x_1 the first coordinate. Then

$$\int_{S^{n-1}} x_1^2 d\sigma = \frac{1}{n},$$

and consequently, for every $\varepsilon > 0$,

$$\sigma(\{x \in S^{n-1} : |x_1| \geq \varepsilon\}) \leq \frac{1}{n\varepsilon^2}.$$

In high dimension, therefore, almost all the surface of the sphere concentrates in a thin band about any equator.

Proof. Step 1 — symmetry. The measure σ is invariant under every rotation of $\mathbb{R}^n$, in particular under permutation of the coordinates. Hence the numbers $\int x_i^2 d\sigma$ are all equal; call them q .

Step 2 — the constraint. On S^{n-1} we have $\sum_{i=1}^n x_i^2 = 1$ identically. Integrating with respect to σ (of total mass 1) and exchanging the finite sum with the integral:

$$1 = \int_{S^{n-1}} \sum_{i=1}^n x_i^2 d\sigma = \sum_{i=1}^n q = nq \implies q = \frac{1}{n}.$$

Step 3 — Chebyshev’s inequality. For the non-negative function x_1^2 and $t = \varepsilon^2 > 0$,

$$\sigma(x_1^2 \geq t) \leq \frac{1}{t} \int x_1^2 d\sigma = \frac{1}{n\varepsilon^2},$$

which is the claim, since $\{|x_1| \geq \varepsilon\} = \{x_1^2 \geq \varepsilon^2\}$.

Step 4 — reading. The first coordinate of a uniform point on the unit sphere has root mean square $1/\sqrt{n}$: it is typically tiny. Since the direction “first coordinate” is arbitrary — the sphere can be rotated at will — the same holds for every direction: *almost all of the sphere lies near every equator*, however absurd this sounds to two-dimensional intuition, where the equator of S^1 consists of just two points.

Step 5 — numerical values. With $\varepsilon = 0.1$:

$$n = 100 : \sigma(|x_1| \geq 0.1) \leq \frac{1}{100 \times 0.01} = 1 \quad (\text{a vacuous bound}),$$

$$n = 1000 : \leq \frac{1}{1000 \times 0.01} = 0.1; \quad n = 10^4 : \leq 0.01.$$

So in dimension 10^4 at least 99% of the surface has first coordinate less than 0.1 in absolute value.

Step 6 — the exponential refinement. Chebyshev's bound is convenient but weak: the true decay is not polynomial but *exponential* in n , as Theorem 11.5 below proves.

Step 7 — why it matters. This is the phenomenon that makes high-dimensional statistics, data compression and Monte Carlo methods possible: in high dimension smooth functions are nearly constant on the sphere, and a handful of random samples estimates their value accurately. The circle, where none of this happens, is the exception, not the rule. $\square$

Theorem 11.5: Exponential concentration bound. For every $n \geq 4$ and every $\varepsilon \in (0, 1)$,

$$\sigma(\{x \in S^{n-1} : |x_1| \geq \varepsilon\}) \leq \frac{\sqrt{e}}{\varepsilon \sqrt{n-1}} e^{-(n-1)\varepsilon^2/2}.$$

The decay is therefore exponential in n , and not merely of order $1/(n\varepsilon^2)$ as Chebyshev's bound gave.

Proof. **Step 1 — the density of the first coordinate.** Parametrise S^{n-1} by the colatitude $\phi \in [0, \pi]$ with respect to the x_1 -axis, so that $t = x_1 = \cos \phi$. The “slice” at height t is a sphere S^{n-2} of radius $\sin \phi = \sqrt{1-t^2}$, and the band between ϕ and $\phi + d\phi$ has area

$$\Sigma_{n-2} (\sin \phi)^{n-2} d\phi.$$

Since $dt = -\sin \phi d\phi$, i.e. $d\phi = |dt|/\sin \phi$, that area equals

$$\Sigma_{n-2} (\sin \phi)^{n-3} |dt| = \Sigma_{n-2} (1-t^2)^{(n-3)/2} |dt|.$$

The image measure of σ under $x \mapsto x_1$ therefore has density proportional to $(1-t^2)^{(n-3)/2}$ on $[-1, 1]$, and

$$\sigma(x_1 \geq \varepsilon) = \frac{N}{D}, \quad N = \int_{\varepsilon}^1 (1-t^2)^{(n-3)/2} dt, \quad D = \int_{-1}^1 (1-t^2)^{(n-3)/2} dt.$$

Step 2 — bounding the numerator. On $[\varepsilon, 1]$ we have $t/\varepsilon \geq 1$, hence

$$N \leq \frac{1}{\varepsilon} \int_{\varepsilon}^1 t (1-t^2)^{(n-3)/2} dt.$$

The last integral is elementary: since $\frac{d}{dt}(1-t^2)^{(n-1)/2} = -(n-1)t(1-t^2)^{(n-3)/2}$,

$$\int_{\varepsilon}^1 t (1-t^2)^{(n-3)/2} dt = \frac{(1-\varepsilon^2)^{(n-1)/2}}{n-1}, \quad \text{i.e.} \quad N \leq \frac{(1-\varepsilon^2)^{(n-1)/2}}{\varepsilon(n-1)}.$$

Step 3 — bounding the denominator from below. Put $a = 1/\sqrt{n-1} \in (0, 1]$. For $n \geq 3$ the function $t \mapsto (1-t^2)^{(n-3)/2}$ is non-increasing on $[0, 1]$, hence

$$D \geq 2 \int_0^a (1-t^2)^{(n-3)/2} dt \geq 2a(1-a^2)^{(n-3)/2} = \frac{2}{\sqrt{n-1}} \left(1 - \frac{1}{n-1}\right)^{(n-3)/2}.$$

Now, for $0 \leq u < 1$ one has $\ln(1-u) \geq -\frac{u}{1-u}$ (because $\ln(1-u) = -\int_0^u \frac{ds}{1-s}$ and $\frac{1}{1-s} \leq \frac{1}{1-u}$ on $[0, u]$); with $u = 1/(n-1)$,

$$\frac{n-3}{2} \ln\left(1 - \frac{1}{n-1}\right) \geq -\frac{n-3}{2} \cdot \frac{1/(n-1)}{(n-2)/(n-1)} = -\frac{n-3}{2(n-2)} \geq -\frac{1}{2},$$

valid for $n \geq 4$ (which also guarantees $u \leq 1/3 < 1$). Hence

$$D \geq \frac{2}{\sqrt{n-1}} e^{-1/2}.$$

Step 4 — the ratio. Combining Steps 2 and 3,

$$\sigma(x_1 \geq \varepsilon) = \frac{N}{D} \leq \frac{(1 - \varepsilon^2)^{(n-1)/2}}{\varepsilon(n-1)} \cdot \frac{\sqrt{n-1} e^{1/2}}{2} = \frac{\sqrt{e}}{2\varepsilon\sqrt{n-1}} (1 - \varepsilon^2)^{(n-1)/2}.$$

Step 5 — from the power to the exponential. From $1 - s \leq e^{-s}$ with $s = \varepsilon^2$ we get $(1 - \varepsilon^2)^{(n-1)/2} \leq e^{-(n-1)\varepsilon^2/2}$, hence

$$\sigma(x_1 \geq \varepsilon) \leq \frac{\sqrt{e}}{2\varepsilon\sqrt{n-1}} e^{-(n-1)\varepsilon^2/2}.$$

Step 6 — symmetry. The measure σ is invariant under $x_1 \mapsto -x_1$, hence $\sigma(|x_1| \geq \varepsilon) = 2\sigma(x_1 \geq \varepsilon)$, and the factor 2 cancels the denominator: the claim follows.

Step 7 — numerical comparison. With $n = 1000$ and $\varepsilon = 0.1$:

$$\frac{\sqrt{e}}{0.1 \times \sqrt{999}} e^{-999 \times 0.005} = \frac{1.648721}{3.160696} \times e^{-4.995} = 0.521633 \times 0.0067717 = 0.003532,$$

that is 0.35%, against the 10% guaranteed by Chebyshev: almost thirty times better. For $n = 10^4$ the exponential bound gives $\sim 10^{-23}$, while Chebyshev stops at 1%.

Step 8 — comparison with the literature. Textbooks often state the bound in the more memorable form $\sigma(|x_1| \geq \varepsilon) \leq 2e^{-n\varepsilon^2/2}$ (Ledoux); the constant proved here is different but the content is the same, and for the values of interest (n in the hundreds) the one proved above is in fact sharper.

Step 9 — the case $n = 3$. For $n = 3$ the exponent $(n-3)/2$ vanishes, the density is constant and one recovers Archimedes' theorem: the projection of the sphere onto a diameter is uniform, so $\sigma(|x_1| \geq \varepsilon) = 1 - \varepsilon$ exactly. No concentration: it begins only from $n = 4$ upwards, and grows stronger as n increases. $\checkmark$ $\square$

Alongside concentration on the *surface* there is concentration in the *volume*: the ball, in high dimension, is almost all skin.

Theorem 11.6: Concentration near the boundary. Let $\varepsilon \in (0, 1)$. In $B^n(O, 1)$ the fraction of volume contained in the outer shell $\{1 - \varepsilon < \|x\| \leq 1\}$ equals exactly $1 - (1 - \varepsilon)^n$, and tends to 1 as $n \rightarrow \infty$. More precisely, it exceeds $1 - \delta$ as soon as

$$n \geq \frac{\ln(1/\delta)}{-\ln(1 - \varepsilon)}.$$

Proof. **Step 1 — homogeneity of volume.** The homothety $x \mapsto \lambda x$ of $\mathbb{R}^n$ has determinant λ^n , hence multiplies volumes by λ^n : the ball of radius $1 - \varepsilon$ has volume $(1 - \varepsilon)^n V_n$ (this is Exercise S1, Step 1).

Step 2 — the fraction. The fraction of volume *inside* the radius $1 - \varepsilon$ is therefore $(1 - \varepsilon)^n V_n / V_n = (1 - \varepsilon)^n$, and that of the shell is the complement to 1:

$$\frac{\text{vol}(B^n \setminus B^n(O, 1 - \varepsilon))}{\text{vol}(B^n)} = 1 - (1 - \varepsilon)^n.$$

Note that the result *does not depend* on V_n : the constant cancels, and for this reason the formula is so robust.

Step 3 — the limit. Since $0 < 1 - \varepsilon < 1$, one has $(1 - \varepsilon)^n = e^{n \ln(1 - \varepsilon)} \rightarrow 0$ (the exponent tends to $-\infty$), hence the fraction of the shell tends to 1.

Step 4 — the explicit threshold. The fraction exceeds $1 - \delta$ if and only if $(1 - \varepsilon)^n < \delta$, that is $n \ln(1 - \varepsilon) < \ln \delta$, that is — dividing by $\ln(1 - \varepsilon) < 0$ and reversing the inequality —

$n > \ln \delta / \ln(1 - \varepsilon) = \ln(1/\delta) / (-\ln(1 - \varepsilon))$.

Step 5 — numerical check. With $\varepsilon = 0.01$:

$n = 10 : 1 - 0.99^{10} = 1 - 0.904382 = 0.095618$; $n = 100 : 1 - 0.366032 = 0.633968$; $n = 1000 : 1 - 0.0000432$

The threshold of Step 4 with $\delta = 0.01$ gives $n \geq \ln(100) / (-\ln(0.99)) = 4.605170 / 0.010050 = 458.21$, that is $n = 459$; and indeed $0.99^{459} = 0.009921 < 0.01$ whereas $0.99^{458} = 0.010021 > 0.01$. ✓

Step 6 — why it is counterintuitive. In dimension 2 and 3 the fraction of the shell of thickness 1% equals respectively $1 - 0.99^2 = 0.0199$ and $1 - 0.99^3 = 0.0297$: negligible, as the drawing suggests. In dimension 459 it is 99%. The reason is purely arithmetic — an n -th power of a number smaller than 1 — but its geometric effect is that the “centre” of the ball, in high dimension, is empty. □

Remark 11.2: Curse of dimensionality. Almost all the volume of a high-dimensional ball is concentrated in a thin skin near the surface (Theorem 11.6): the “centre” is practically empty. It follows that in high dimension uniformly sampled points are almost all at the same distance (close to the maximum) and almost orthogonal to one another (Exercise S3): the notions of “near” and “far” lose contrast. This is the reason why distance-based algorithms (k -NN, clustering) degrade in high dimension, a phenomenon known in machine learning as the curse of dimensionality (Bellman). The same fact has a “beneficial” side (blessing of dimensionality): concentration makes many random quantities strongly predictable, and it is what allows Monte Carlo methods and data compression to work (Theorem 11.5, Step 7).

11.4.1 Distances in high dimension

In a dataset with n -dimensional features, concentration of measure implies that the ratio between the distance of the farthest neighbour and that of the nearest one tends to 1: the Euclidean distance becomes uninformative. Practical strategies — dimensionality reduction (PCA), learned metrics, embeddings — aim exactly at restoring the contrast between distances that high dimension flattens. The geometry of the sphere S^{n-1} , where normalized data live, is therefore the mathematical ground of many modelling choices.

Exercise 11.3. In $B^{100}(O, 1)$, what fraction of the volume lies beyond the radius 0.99?

Solution. By Theorem 11.6 with $\varepsilon = 0.01$ and $n = 100$, the fraction inside the radius 0.99 is 0.99^{100} . Since $0.99^{100} = e^{100 \ln 0.99} = e^{-1.005034} = 0.366032$, the fraction in the shell $[0.99, 1]$ is $1 - 0.366032 = 0.633968$: more than 63% of the volume lies in the outer 1% of the radius. In dimension 100 the ball is almost all “skin”. ✓

Remark 11.3: The moral of the chapter. The circle, in dimension 2, is so well behaved that it suggests an intuition which in high dimension is simply false. The three preceding theorems — vanishing volume, concentrating measure, ball escaping the cube — are not curiosities: they are the fundamental facts governing multivariate statistics, machine learning and long-block error-correcting codes. The practical rule they yield is: in high dimension, do not trust pictures. And, as an opposite corollary: whenever one manages to reduce a problem to a circle, everything has been gained, because on S^1 geometric intuition becomes reliable again.

11.5. Rotations in any dimension

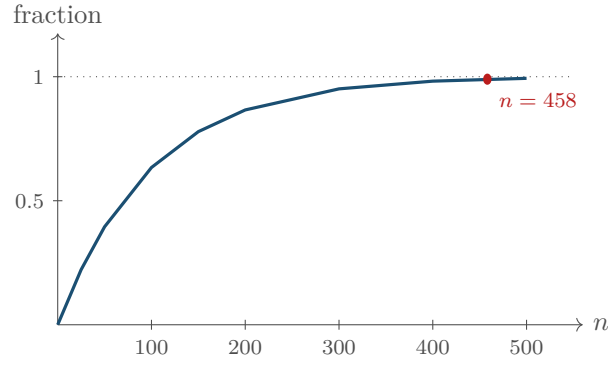


Figure 11.2: The fraction of the volume of the unit ball B^n contained in the outer shell of thickness $\varepsilon = 0.01$, that is $1 - 0.99^n$, as a function of the dimension. In dimension 10 it equals 0.0956, in dimension 100 already 0.6340, and it exceeds 99% from $n = 459$ on (red dot: the critical value 458.21 of Theorem 11.6, Step 4). Two-dimensional intuition, where the same fraction equals 0.0199, is entirely misleading.

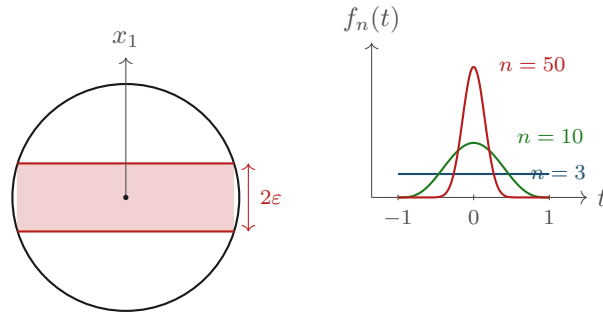


Figure 11.3: Concentration of measure on the sphere. Left: the equatorial band $|x_1| \leq \varepsilon$; in high dimension it contains almost all of the surface measure, however thin it is taken. Right: the density of the first coordinate of a uniform point on S^{n-1} , $f_n(t) = c_n(1 - t^2)^{(n-3)/2}$, for $n = 3, 10, 50$. For $n = 3$ it is constant (Archimedes' theorem: the projection of the sphere onto the axis is uniform); as n grows the density tightens around $t = 0$ with width of order $1/\sqrt{n}$.

Definition 11.2: Orthogonal group and rotation group. A matrix $R \in \mathbb{R}^{n \times n}$ is **orthogonal** if $R^\top R = I$, i.e. if it preserves the scalar product: $\langle Rx, Ry \rangle = \langle x, y \rangle$. The set of orthogonal matrices is the group $O(n)$; since $\det(R)^2 = 1$, one has $\det R = \pm 1$, and the subgroup

$$SO(n) = \{R \in O(n) : \det R = +1\}$$

is the **rotation group** of $\mathbb{R}^n$.

Theorem 11.7: The circle is the rotation group of the plane. The map

$$\Theta : S^1 \longrightarrow SO(2), \quad \Theta(e^{i\vartheta}) = \begin{pmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{pmatrix},$$

is a group isomorphism (and a homeomorphism). In particular $SO(2)$ is commutative.

Proof. Step 1 — the image lies in $SO(2)$. The matrix has orthonormal columns and determinant $\cos^2 \vartheta + \sin^2 \vartheta = 1$.

Step 2 — it is a homomorphism. Multiplying two such matrices and using the addition formulas,

$$\Theta(e^{i\vartheta}) \Theta(e^{i\psi}) = \begin{pmatrix} \cos(\vartheta + \psi) & -\sin(\vartheta + \psi) \\ \sin(\vartheta + \psi) & \cos(\vartheta + \psi) \end{pmatrix} = \Theta(e^{i(\vartheta+\psi)}),$$

so the product on the circle (Chapter 3) corresponds to matrix multiplication.

Step 3 — surjectivity. Let $R \in SO(2)$ have first column $(a, c)^\top$; being a unit vector, $a^2 + c^2 = 1$, hence $(a, c) = (\cos \vartheta, \sin \vartheta)$ for some ϑ . The second column is orthogonal to the first and of unit length, hence equals $\pm(-\sin \vartheta, \cos \vartheta)$; the $+$ sign is forced by $\det R = +1$.

Step 4 — injectivity. $\Theta(e^{i\vartheta}) = I$ forces $\cos \vartheta = 1$ and $\sin \vartheta = 0$, i.e. $e^{i\vartheta} = 1$.

Step 5 — reading. The circle is not merely a *figure invariant* under rotations: it *is* the rotation group of the plane. This identification, glimpsed in Chapter 3 with multiplication by $e^{i\vartheta}$, is the ultimate reason why the circle appears in every periodic phenomenon: periodicity is invariance under a group, and in dimension two that group is the circle itself.

Step 6 — commutativity and its consequences. $SO(2)$ is commutative because plane rotations add their angles. Already $SO(3)$ is not (Example 11.2), and this is the structural difference between the plane case and all the others. $\square$

Theorem 11.8: Euler's rotation theorem. Every $R \in SO(3)$ other than the identity has an **axis**: there is $u \in S^2$ with $Ru = u$, unique up to sign, and R is the rotation by an angle ϑ about u , with

$$\operatorname{tr} R = 1 + 2 \cos \vartheta.$$

Proof. **Step 1 — existence of the eigenvalue 1.** Compute $\det(R - I)$ using $R^\top R = I$ and $\det R = 1$:

$$\det(R - I) = \det(R - RR^\top) = \det R \cdot \det(I - R^\top) = \det(I - R^\top).$$

But $\det(I - R^\top) = \det((I - R)^\top) = \det(I - R) = (-1)^3 \det(R - I) = -\det(R - I)$. Hence $\det(R - I) = -\det(R - I)$, i.e. $\det(R - I) = 0$: the number 1 is an eigenvalue of R .

Step 2 — the axis. Let u be a unit eigenvector for the eigenvalue 1: $Ru = u$. The line $\mathbb{R}u$ is fixed pointwise.

Step 3 — the orthogonal plane. The plane $u^\perp$ is invariant: if $\langle v, u \rangle = 0$, then $\langle Rv, u \rangle = \langle Rv, Ru \rangle = \langle v, u \rangle = 0$. The restriction of R to $u^\perp$ is orthogonal and has determinant $\det R/1 = 1$: it therefore lies in $SO(2)$ and, by Theorem 11.7, is a rotation by an angle ϑ .

Step 4 — the trace. In an orthonormal basis $\{u, e_1, e_2\}$ with e_1, e_2 a basis of $u^\perp$, the matrix of R is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \vartheta & -\sin \vartheta \\ 0 & \sin \vartheta & \cos \vartheta \end{pmatrix},$$

whose trace is $1 + 2 \cos \vartheta$. Since the trace is invariant under orthonormal change of basis, the formula holds in every basis.

Step 5 — uniqueness of the axis. If $R \neq I$, the angle ϑ is not a multiple of 2π and the restriction to $u^\perp$ has no non-zero fixed vectors: the eigenspace of the eigenvalue 1 is exactly $\mathbb{R}u$, and u is determined up to sign.

Step 6 — a surprising consequence. However a rigid sphere is moved about its centre, at the end *at least two diametrically opposite points have returned where they were*. In particular, shuffling a rigid deck of cards, or rotating a planet, there is always an instantaneous axis. The result is Euler's (1776) and is false in even dimension: in $\mathbb{R}^4$ the rotation turning the plane x_1x_2 by 90° and the plane x_3x_4 by 90° fixes no non-zero vector.

Step 7 — the general case. In $\mathbb{R}^n$ every $R \in SO(n)$ has an invariant subspace of dimension 1 or 2. Indeed the characteristic polynomial has at least one complex root λ ; if $\lambda \in \mathbb{R}$ then

$\lambda = \pm 1$ (by orthogonality) and one has a real eigenvector; otherwise, writing $v = a + ib$ for a complex eigenvector, from $Rv = \lambda v$ with $\lambda = e^{i\vartheta}$ one gets

$$Ra = \cos \vartheta a - \sin \vartheta b, \quad Rb = \sin \vartheta a + \cos \vartheta b,$$

by equating real and imaginary parts: the plane $\text{span}\{a, b\}$ is invariant and R acts on it as a rotation. Iterating on the orthogonal complement one obtains the *normal form*

$$R \sim \text{diag}(R_{\vartheta_1}, \dots, R_{\vartheta_m}, 1, \dots, 1),$$

a direct sum of plane rotations and identities. *Every rotation, in every dimension, is a superposition of rotations of circles*: this is the precise sense in which the circle is the building block of the geometry of rotations. $\square$

Theorem 11.9: Cartan–Dieudonné theorem. *Every $A \in O(n)$ is a product of at most n reflections in hyperplanes through the origin (the identity being the empty product). Since every reflection has determinant -1 , the elements of $SO(n)$ are products of an even number of reflections; in particular every rotation of $\mathbb{R}^3$ other than the identity is a product of exactly two reflections.*

Proof. Step 1 — the reflection and its action. For a unit vector $m \in \mathbb{R}^n$, the reflection in the hyperplane $m^\perp$ is

$$S_m(x) = x - 2\langle x, m \rangle m.$$

It is orthogonal ($\|S_m x\|^2 = \|x\|^2 - 4\langle x, m \rangle^2 + 4\langle x, m \rangle^2 \|m\|^2 = \|x\|^2$) and has determinant -1 , since it fixes $m^\perp$ and sends m to $-m$.

Step 2 — induction on the defect. Put $d(A) = n - \dim \text{Fix}(A)$, where $\text{Fix}(A) = \{x : Ax = x\}$. We prove by induction on d that A is a product of at most d reflections. If $d = 0$ then $A = I$: the empty product.

Step 3 — the inductive step. Let $d \geq 1$; there is v with $Av \neq v$. Put

$$m = \frac{Av - v}{\|Av - v\|}, \quad S = S_m.$$

Step 4 — S takes Av back to v . Since A is orthogonal, $\|Av\| = \|v\|$, hence

$$\langle Av - v, Av + v \rangle = \|Av\|^2 - \|v\|^2 = 0,$$

i.e. $Av + v \perp m$. Writing $Av = \frac{1}{2}[(Av + v) + (Av - v)]$ and noting that S fixes the first bracket and reverses the second,

$$S(Av) = \frac{1}{2}[(Av + v) - (Av - v)] = v.$$

Step 5 — SA fixes more. Let $w \in \text{Fix}(A)$. From $Aw = w$ we get $A^{-1}w = w$, and since A is orthogonal $\langle w, Av \rangle = \langle A^{-1}w, v \rangle = \langle w, v \rangle$; hence $\langle w, Av - v \rangle = 0$, i.e. $w \perp m$ and $Sw = w$. Therefore $SAw = Sw = w$. Moreover $SAv = v$ by Step 4. It follows that

$$\text{Fix}(SA) \supseteq \text{Fix}(A) \oplus \mathbb{R}v,$$

a direct sum because $v \notin \text{Fix}(A)$. Hence $d(SA) \leq d(A) - 1$.

Step 6 — conclusion. By the inductive hypothesis SA is a product of at most $d - 1$ reflections, and since $S^2 = I$ we have $A = S(SA)$: a product of at most $d \leq n$ reflections.

Step 7 — parity and the three-dimensional case. If $A = S_1 \cdots S_k$ then $\det A = (-1)^k$: hence $\det A = +1$ forces k even. In $\mathbb{R}^3$ we have $k \leq 3$, so $k \in \{0, 2\}$: the identity, or the

product of two reflections. Since two distinct hyperplanes of $\mathbb{R}^3$ meet in a line fixed by both reflections, one recovers the axis of Theorem 11.8 — and this is the route taken in Exercise S4. **Step 8 — sharpness of the bound.** The bound n cannot be improved: $-I$ in $\mathbb{R}^n$ has $\text{Fix} = \{0\}$, hence $d = n$, and exactly n reflections are needed (for instance those in the coordinate hyperplanes). ✓ $\square$

11.6. Quaternions and rotations of space

Definition 11.3: Quaternions. The algebra of **quaternions** $\mathbb{H}$ is $\mathbb{R}^4$ with basis $\{1, i, j, k\}$ and associative product determined by

$$i^2 = j^2 = k^2 = ijk = -1,$$

whence $ij = k$, $jk = i$, $ki = j$ and $ji = -k$, $kj = -i$, $ik = -j$. For $q = q_0 + q_1i + q_2j + q_3k$ one sets $\bar{q} = q_0 - q_1i - q_2j - q_3k$ and $\|q\|^2 = q\bar{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2$. The quaternions of norm 1 form the sphere $S^3 \subset \mathbb{R}^4$, which is a group under multiplication. A quaternion with $q_0 = 0$ is **pure** and is identified with a vector of $\mathbb{R}^3$.

Theorem 11.10: Unit quaternions are the rotations of space. For $q \in S^3$ define $\Phi_q : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by

$$\Phi_q(v) = qv\bar{q} \quad (v \text{ pure}).$$

Then:

- (i) $\Phi_q(v)$ is pure and $\|\Phi_q(v)\| = \|v\|$;
- (ii) $\Phi_q \in SO(3)$ and $\Phi_{q_1q_2} = \Phi_{q_1} \circ \Phi_{q_2}$;
- (iii) if $q = \cos \frac{\vartheta}{2} + (\sin \frac{\vartheta}{2})u$ with u pure of unit norm, then Φ_q is the rotation by angle ϑ about the axis u ;
- (iv) $\Phi_q = \Phi_{q'}$ if and only if $q' = \pm q$: the map $S^3 \rightarrow SO(3)$ is a surjective homomorphism which is two to one.

Proof. **(i) Step 1 — norm preservation.** The quaternion norm is multiplicative ($\|pq\| = \|p\| \|q\|$, verifiable from the product or from Euler's four-square identity), hence

$$\|qv\bar{q}\| = \|q\| \|v\| \|\bar{q}\| = \|v\|,$$

since $\|q\| = \|\bar{q}\| = 1$.

Step 2 — it stays pure. A quaternion w is pure iff $\bar{w} = -w$. Now

$$\overline{qv\bar{q}} = q\bar{v}q = q(-v)\bar{q} = -qv\bar{q},$$

using $\overline{ab} = \bar{b}\bar{a}$ and $\bar{v} = -v$. Hence $\Phi_q(v)$ is pure.

(ii) Step 3 — linearity and orthogonality. Φ_q is linear in v because the quaternion product is bilinear; it preserves the norm by (i), hence is orthogonal.

Step 4 — determinant +1. The map $q \mapsto \det \Phi_q$ is continuous on S^3 , with values in $\{-1, +1\}$; since S^3 is connected and $\Phi_1 = \text{id}$ has determinant +1, we get $\det \Phi_q = +1$ for every q .

Step 5 — homomorphism. $\Phi_{q_1}(\Phi_{q_2}(v)) = q_1q_2v\bar{q}_2\bar{q}_1 = (q_1q_2)v\overline{(q_1q_2)} = \Phi_{q_1q_2}(v)$.

(iii) Step 6 — the axis. With $q = c + su$, $c = \cos \frac{\vartheta}{2}$, $s = \sin \frac{\vartheta}{2}$, and recalling $u^2 = -1$ for u pure of unit norm:

$$\Phi_q(u) = (c + su)u(c - su) = (cu + su^2)(c - su) = (cu - s)(c - su).$$

Expanding: $c^2u - csu^2 - sc + s^2u = c^2u + cs - sc + s^2u = u$. So u is fixed: it is the axis.

Step 7 — the angle. Let w be pure of unit norm and orthogonal to u ; then $uw = -wu$ (checked on the basis: $ij = -ji$ etc.) and uw is pure, of unit norm, orthogonal to both u and w . Compute

$$\Phi_q(w) = (c + su)w(c - su) = (cw + suw)(c - su) = c^2w - cswu + csuw - s^2uwu.$$

Now $wu = -uw$, hence $-cswu + csuw = 2csuw$; and $uwu = -u^2w = w$. Therefore

$$\Phi_q(w) = (c^2 - s^2)w + 2cs(uw) = \cos \vartheta w + \sin \vartheta (uw),$$

by the double-angle formulas. Since $\{w, uw\}$ is an orthonormal basis of the plane $u^\perp$, the restriction of Φ_q to that plane is the rotation by ϑ : the claim.

(iv) Step 8 — the kernel. If $\Phi_q = \text{id}$, then $qv = vq$ for every pure v , hence q commutes with i, j, k and so with all of $\mathbb{H}$: the centre of $\mathbb{H}$ is $\mathbb{R}$, hence $q = \pm 1$ (having norm 1). It follows that $\Phi_q = \Phi_{q'}$ is equivalent to $q'q^{-1} = \pm 1$, i.e. $q' = \pm q$.

Step 9 — surjectivity. By Theorem 11.8 every rotation has an axis u and an angle ϑ ; the quaternion of (iii) realises it.

Step 10 — reading. The sphere S^3 covers the rotation group of space twice. This is the fact behind *spin* 1/2 in quantum mechanics: rotating a system by 360° takes the quaternion from q to $-q$, and 720° are needed to return to the start. The phenomenon can be demonstrated by hand — it is the famous “belt trick” — and has no analogue in the plane, where $SO(2) = S^1$ is covered only once by itself. $\square$

Example 11.2: Composing two right-angle rotations. Compose the rotation by 90° about the z -axis with the rotation by 90° about the x -axis (first z , then x).

Step 1 — the two quaternions. By Theorem 11.10(iii), with $\vartheta = 90^\circ$ and $\cos 45^\circ = \sin 45^\circ = 0.707107$:

$$q_1 = 0.707107 + 0.707107k \text{ (} z\text{-axis)}, \quad q_2 = 0.707107 + 0.707107i \text{ (} x\text{-axis)}.$$

Step 2 — product. The composition “first q_1 , then q_2 ” corresponds to $q = q_2q_1$:

$$q = (0.707107 + 0.707107i)(0.707107 + 0.707107k).$$

Expanding, with $0.707107^2 = 0.5$ and $ik = -j$:

$$q = 0.5 + 0.5k + 0.5i + 0.5ik = 0.5 + 0.5i - 0.5j + 0.5k.$$

Step 3 — norm check. $\|q\|^2 = 4 \times 0.25 = 1$. $\checkmark$

Step 4 — angle and axis. From $\cos \frac{\vartheta}{2} = 0.5$ we get $\frac{\vartheta}{2} = 60^\circ$, i.e. $\vartheta = 120^\circ$; the pure part is $0.5(i - j + k)$, of norm $0.5\sqrt{3} = 0.866025 = \sin 60^\circ$ $\checkmark$, and the axis is the unit vector

$$u = \frac{(1, -1, 1)}{\sqrt{3}}.$$

Step 5 — geometric reading. The composition of two right-angle rotations about orthogonal axes is a rotation by 120° about a *diagonal of the cube*. It is exactly the symmetry taking a cube to itself by cyclically permuting the three axes: the result, far from obvious, is read off in four lines of quaternion algebra.

Step 6 — check via the trace. The matrix of the composed rotation is obtained by multiplying

$$R_x(90^\circ)R_z(90^\circ) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Its trace is 0, and Theorem 11.8 gives $1 + 2 \cos \vartheta = 0$, i.e. $\cos \vartheta = -0.5$ and $\vartheta = 120^\circ$. ✓

Step 7 — non-commutativity. Reversing the order gives $q' = q_1 q_2 = 0.5 + 0.5i + 0.5j + 0.5k$, with axis $(1, 1, 1)/\sqrt{3}$: same angle, different axis. $SO(3)$ is not commutative, unlike $SO(2) = S^1$ (Theorem 11.7).

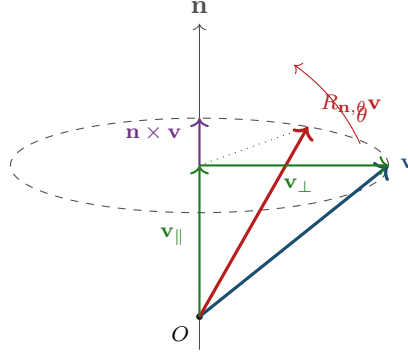


Figure 11.4: Rodrigues' formula in one picture. The vector $\mathbf{v}$ splits into the component $\mathbf{v}_{\parallel} = (\mathbf{n} \cdot \mathbf{v})\mathbf{n}$ along the axis, which the rotation leaves fixed, and the orthogonal component $\mathbf{v}_{\perp}$, which rotates by θ in the plane perpendicular to $\mathbf{n}$. In that plane $\{\mathbf{v}_{\perp}, \mathbf{n} \times \mathbf{v}\}$ is an orthogonal basis of equal length, so the rotated component equals $\cos \theta \mathbf{v}_{\perp} + \sin \theta (\mathbf{n} \times \mathbf{v})$: adding the axial part gives $R_{\mathbf{n}, \theta} \mathbf{v} = \mathbf{v} \cos \theta + (\mathbf{n} \times \mathbf{v}) \sin \theta + \mathbf{n}(\mathbf{n} \cdot \mathbf{v})(1 - \cos \theta)$. The tip of the vector traces a circle: the circle is the orbit of every rotation.

Remark 11.4: Which spheres are groups, and the Hopf fibration. Theorem 11.7 shows that S^1 is a group (the complex numbers of modulus 1) and Definition 11.3 that S^3 is one (the quaternions of norm 1). One proves — with tools of algebraic topology beyond this treatise — that the only spheres carrying a Lie group structure are S^0 , S^1 and S^3 : S^2 , for instance, is not one, because every continuous vector field on it vanishes somewhere (the hairy ball theorem). The circle is therefore, together with S^3 , an exception.

There is more: the map $S^3 \rightarrow S^2$ sending q to the axis of the rotation Φ_q has as fibres exactly the circles $\{e^{au}q\}$. One thus obtains the **Hopf fibration** (1931): the three-dimensional sphere is covered by a family of circles, pairwise linked, parametrised by the two-dimensional sphere. It is the most celebrated example of filling a space with circles, and shows that even in dimension three the circle remains the fundamental building block. Section 11.7 gives its explicit construction and proves its properties.

11.7. The Hopf fibration, explicitly

Remark 11.4 announced that S^3 is filled by circles parametrised by S^2 . We build that fibration here with an explicit formula and prove the three properties that matter: the fibres are great circles, they are pairwise disjoint and fill S^3 , and two distinct fibres are *linked*. None of this requires algebraic topology: complex numbers suffice.

Definition 11.4: Hopf map. Identify $\mathbb{R}^4$ with $\mathbb{C}^2$ by setting $(x_1, y_1, x_2, y_2) \leftrightarrow (z_1, z_2)$ with $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, so that

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\},$$

and identify $\mathbb{R}^3$ with $\mathbb{C} \times \mathbb{R}$. The **Hopf map** is

$$h : S^3 \longrightarrow \mathbb{C} \times \mathbb{R}, \quad h(z_1, z_2) = (2z_1\overline{z_2}, |z_1|^2 - |z_2|^2).$$

Theorem 11.11: Structure of the Hopf fibration. The map h of Definition 11.4 has the following properties.

- (i) $h(S^3) \subseteq S^2$, that is $|h(z_1, z_2)| = 1$ at every point of S^3 .
- (ii) h is onto S^2 .
- (iii) For every $(z_1, z_2) \in S^3$,

$$h^{-1}(h(z_1, z_2)) = \{(\lambda z_1, \lambda z_2) : \lambda \in \mathbb{C}, |\lambda| = 1\},$$

and this set is a **great circle** of S^3 .

In particular S^3 is the disjoint union of a family of great circles parametrised by S^2 .

Proof. (i). With $a = |z_1|^2$ and $b = |z_2|^2$, and using $|2z_1\bar{z}_2|^2 = 4|z_1|^2|z_2|^2 = 4ab$,

$$|h(z_1, z_2)|^2 = 4ab + (a - b)^2 = a^2 + 2ab + b^2 = (a + b)^2 = 1,$$

because $a + b = 1$ on the sphere.

(ii). Let $(w, t) \in S^2$, that is $w \in \mathbb{C}$, $t \in \mathbb{R}$, $|w|^2 + t^2 = 1$; in particular $|t| \leq 1$. Put

$$z_1 = \sqrt{\frac{1+t}{2}}, \quad z_2 = \sqrt{\frac{1-t}{2}} e^{-i \arg w}$$

($\arg w$ chosen arbitrarily if $w = 0$, in which case one of the factors $\sqrt{(1-t)/2}$, $\sqrt{(1+t)/2}$ vanishes). Then $|z_1|^2 + |z_2|^2 = \frac{1+t}{2} + \frac{1-t}{2} = 1$, so $(z_1, z_2) \in S^3$, and

$$|z_1|^2 - |z_2|^2 = t, \quad 2z_1\bar{z}_2 = 2\sqrt{\frac{1+t}{2}}\sqrt{\frac{1-t}{2}} e^{i \arg w} = \sqrt{1-t^2} e^{i \arg w} = |w| e^{i \arg w} = w,$$

where $|w| = \sqrt{1-t^2}$ has been used. Hence $h(z_1, z_2) = (w, t)$.

(iii), inclusion $\supseteq$. If $|\lambda| = 1$ then $|\lambda z_k| = |z_k|$, so the second component does not change; and

$$2(\lambda z_1)(\overline{\lambda z_2}) = 2\lambda\bar{\lambda} z_1\bar{z}_2 = 2|\lambda|^2 z_1\bar{z}_2 = 2z_1\bar{z}_2,$$

so neither does the first.

(iii), inclusion $\subseteq$. Let $h(w_1, w_2) = h(z_1, z_2)$. From equality of the second components, $|w_1|^2 - |w_2|^2 = |z_1|^2 - |z_2|^2$, and from $|w_1|^2 + |w_2|^2 = |z_1|^2 + |z_2|^2 = 1$ it follows that

$$|w_1| = |z_1|, \quad |w_2| = |z_2|.$$

If $z_1 = 0$ then $|z_2| = 1$, $w_1 = 0$ and $|w_2| = 1$: setting $\lambda = w_2/z_2$ one has $|\lambda| = 1$ and $(w_1, w_2) = \lambda(z_1, z_2)$. The case $z_2 = 0$ is symmetric. Finally, if $z_1, z_2 \neq 0$, write $w_1 = \lambda_1 z_1$ and $w_2 = \lambda_2 z_2$ with $|\lambda_1| = |\lambda_2| = 1$; equality of the first components gives

$$2\lambda_1\bar{\lambda}_2 z_1\bar{z}_2 = 2z_1\bar{z}_2 \implies \lambda_1\bar{\lambda}_2 = 1,$$

having divided by $z_1\bar{z}_2 \neq 0$. Since $|\lambda_2| = 1$ one has $\bar{\lambda}_2 = 1/\lambda_2$, hence $\lambda_1/\lambda_2 = 1$, that is $\lambda_1 = \lambda_2 =: \lambda$.

(iii), the fibre is a great circle. Put $v = (z_1, z_2)$ and $v' = (iz_1, iz_2)$; both have Euclidean norm 1 in $\mathbb{R}^4$, and they are orthogonal:

$$\langle v, v' \rangle_{\mathbb{R}} = \operatorname{Re}(\bar{z}_1 iz_1 + \bar{z}_2 iz_2) = \operatorname{Re}(i(|z_1|^2 + |z_2|^2)) = \operatorname{Re}(i) = 0.$$

Writing $\lambda = \cos \alpha + i \sin \alpha$ one has $\lambda v = (\cos \alpha) v + (\sin \alpha) v'$: the fibre is therefore the intersection of S^3 with the two-dimensional plane $\operatorname{span}_{\mathbb{R}}\{v, v'\}$ through the origin, that is a great circle of radius 1.

Conclusion. The fibres of a map are pairwise disjoint and cover the domain; by (ii) the index set is the whole of S^2 . $\square$

Remark 11.5: The same fibration, in quaternionic language. Remark 11.4 described the same construction with quaternions: $q \mapsto q i \bar{q}$, that is the axis of the rotation Φ_q . The two descriptions agree. Indeed, identify $\mathbb{H}$ with $\mathbb{C}^2$ by $q = z_1 + j z_2$; then, for $\lambda = e^{i\alpha}$,

$$q \lambda = z_1 \lambda + j z_2 \lambda,$$

that is, multiplication on the right by λ acts on the coordinates as $(z_1, z_2) \mapsto (\lambda z_1, \lambda z_2)$: exactly the fibres of Theorem 11.11(iii).

That $q \mapsto q i \bar{q}$ is constant on those fibres is immediate, because λ commutes with i (both lie in the copy of $\mathbb{C}$ generated by 1 and i):

$$(q \lambda) i \overline{(q \lambda)} = q \lambda i \bar{\lambda} \bar{q} = q i \bar{\lambda} \bar{q} = q i \bar{q}.$$

Conversely, if $q i \bar{q} = q' i \bar{q}'$, setting $p = \bar{q}' q$ (of norm 1) one gets $p i \bar{p} = i$, that is p commutes with i ; but the unit quaternions commuting with i are exactly those of the form $e^{i\alpha}$, hence $q = q' e^{i\alpha}$. The two maps therefore have the same fibres, and define the same partition of S^3 into circles.

Example 11.3: Two linked fibres, checked by stereographic projection. The two simplest fibres are

$$F_+ = h^{-1}(0, 0, 1) = \{(e^{i\theta}, 0)\}, \quad F_- = h^{-1}(0, 0, -1) = \{(0, e^{i\theta})\},$$

as one checks by putting $t = \pm 1$ in the proof of (ii). They are disjoint, because $|z_1|$ equals 1 on the first and 0 on the second. We show that they are *linked*.

Stereographic projection. In real coordinates (x_1, y_1, x_2, y_2) project S^3 from $N = (0, 0, 0, 1)$, that is from the point $(z_1, z_2) = (0, i)$, which belongs to F_- :

$$\sigma(x_1, y_1, x_2, y_2) = \frac{(x_1, y_1, x_2)}{1 - y_2} \in \mathbb{R}^3.$$

Image of F_+ . Its points are $(\cos \theta, \sin \theta, 0, 0)$, with $y_2 = 0$, hence

$$\sigma(F_+) = \{(\cos \theta, \sin \theta, 0)\} :$$

the **unit circle** of the plane $z = 0$.

Image of F_- . Its points are $(0, 0, \cos \theta, \sin \theta)$, and

$$\sigma(0, 0, \cos \theta, \sin \theta) = \left(0, 0, \frac{\cos \theta}{1 - \sin \theta}\right).$$

As θ ranges over $[0, 2\pi)$ excluding $\theta = \pi/2$ (which is the pole N), the quantity $u(\theta) = \cos \theta / (1 - \sin \theta)$ takes every real value exactly once: it is continuous, equals 1 at $\theta = 0$, 0 at $\theta = -\pi/2$, -1 at $\theta = \pi$, and tends to $\pm\infty$ as $\theta \rightarrow \pi/2^\mp$. Hence

$$\sigma(F_-) = \{(0, 0, u) : u \in \mathbb{R}\} :$$

the whole **z axis**.

Linking. The unit circle of the plane $z = 0$ bounds the disc $D = \{x^2 + y^2 \leq 1, z = 0\}$; the z axis meets D at a single point, the origin, and crosses it transversally. A circle and a closed curve that pierces its disc exactly once are linked, with linking number ± 1 : the two fibres cannot be separated without cutting one of them. (The z axis, closed up by the point at infinity, is the image circle of F_- : stereographic projection sends circles of S^3 through the pole to lines of $\mathbb{R}^3$.)

Why it holds for every pair. The linking number of two disjoint closed curves is an integer that does not change when the curves are deformed continuously while remaining disjoint. Now, ordered pairs of distinct fibres are parametrised by ordered pairs of distinct points of S^2 (Theorem 11.11), and that set is *connected*: from a pair (P, Q) with $P \neq Q$ one reaches continuously any other pair (P', Q') with $P' \neq Q'$, moving first Q and then P without ever letting them coincide. The linking number, an integer-valued continuous function on a connected set, is therefore constant, and equals ± 1 everywhere because it equals ± 1 on the pair (F_+, F_-) : *any two distinct fibres are linked.*

Remark 11.6: What remains imported. *The construction above is complete and uses no algebraic topology. What remains imported is only the statement quoted in Remark 11.4: that S^0 , S^1 and S^3 are the only spheres carrying a Lie group structure. That is a classification, and classifications are not proved by exhibiting examples: they require the theory of Steenrod operations and K-theory, developed by Bott, Milnor and Adams between 1958 and 1960. There is exactly one further case in which the multiplication exists but is not associative, S^7 with the octonions, and that too falls under the same classification.*

What has been proved here is the other half of the picture: that the circle, that is S^1 , is not merely a group, but the group that fibres S^3 over S^2 — and that is the structural reason why the circle reappears in dimension three.

11.8. Industrial applications

High dimension in robotic perception. The vision systems and sensors of modern robots (FANUC iRVision, Yaskawa MotoSight, cells with deep learning) operate on high-dimensional feature spaces. Concentration of measure (Theorems 11.4 and 11.6) implies that, as the dimension increases, Euclidean distances lose contrast and nearest-neighbour methods degrade: the choice to reduce the dimension (PCA, learned embeddings) or to normalize the data on the sphere S^{n-1} is a direct consequence of the geometry of this chapter. Planning in the configuration space of a robot with many joints also lives in high dimension, with the same counterintuitive effects on volumes: the “useful” volume of the free space is an infinitesimal fraction of the cube of joint limits (Theorem 11.2, Step 3), and sampling-based planners are designed precisely so as not to have to explore it.

Quaternions in attitude control. The double covering $S^3 \rightarrow SO(3)$ (Theorem 11.10(iv)) is the reason why the attitude control systems of drones, satellites and robotic wrists represent orientation with unit quaternions rather than with Euler angles: the parametrization is free of singularities (no *gimbal lock*), composition costs sixteen multiplications instead of twenty-seven, and normalization to $\|q\| = 1$ corrects numerical drift. The price is the sign ambiguity $q \leftrightarrow -q$, which must be handled explicitly in interpolation algorithms (Chapter 30).

11.9. Assigned exercises

Exercise 11.4. Objective: handle the recurrence of the Gamma function and that of the volumes. **Estimated time:** 15 minutes.

- (i) Compute V_9 , Σ_8 , V_{12} and Σ_{11} with the recurrence $V_n = \frac{2\pi}{n} V_{n-2}$ starting from $V_7 = 4.724766$ and $V_{10} = 2.550164$, and check with the closed formula of Theorem 11.1. (ii) Determine the smallest dimension n for which V_n is less than $V_1 = 2$, that is than the volume of the segment $[-1, 1]$. (iii) Explain, without computing, why the maximum of Σ_{n-1} falls one dimension farther than the maximum of V_n .

Exercise 11.5. Objective: compare two concentration estimates on the same datum. **Estimated time:** 25 minutes.

One wants to guarantee that at most 1% of the surface of S^{n-1} has $|x_1| \geq 0.05$. (i) Which dimension does Chebyshev's estimate of Theorem 11.4 require? (ii) Which dimension does the exponential estimate of Theorem 11.5 require? (Proceed by ordered trials or by bisection.) (iii) Comment on the ratio between the two dimensions and say which of the two statements is "wrong".

Exercise 11.6. Objective: pass from a pair of directions to the quaternion and to the matrix. **Estimated time:** 30 minutes.

One wants to bring the direction $a = (1, 1, 1)/\sqrt{3}$ onto the direction $b = (0, 0, 1)$ with the rotation of smallest angle. (i) Determine the axis and the angle of that rotation. (ii) Write the corresponding unit quaternion (Theorem 11.10(iii)). (iii) Write the matrix R with Rodrigues' formula (Exercise S5) and check $\text{tr } R = 1 + 2 \cos \vartheta$ and $Ra = b$.

11.10. Worked exercises

Exercise 11.7. For $\varepsilon = 0.01$ compute the fraction of the volume of B^n contained in the outer shell for $n = 2, 10, 100, 1000$, and determine the smallest dimension for which it exceeds 99%. Compare with the fraction of surface that the exponential estimate guarantees in the equatorial band.

Solution. Strategy. The formula of Theorem 11.6 is exact and does not require V_n : it suffices to raise to a power. The threshold is obtained by inverting the logarithm, and must then be rounded *up to the next integer* because the fraction grows with n .

Step 1 — the four values.

n	0.99^n	fraction of the shell	
2	0.980100	0.019900	(plane intuition)
10	0.904382	0.095618	
100	0.366032	0.633968	
1000	0.0000432	0.999957	

Step 2 — the threshold. One needs $0.99^n \leq 0.01$, that is

$$n \geq \frac{\ln(0.01)}{\ln(0.99)} = \frac{-4.605170}{-0.010050} = 458.2106,$$

hence $n = 459$.

Step 3 — independent check. Computing directly the two powers close to the threshold: $0.99^{458} = 0.010021 > 0.01$ (fraction of the shell 0.989979, below 99%) and $0.99^{459} = 0.009921 < 0.01$ (fraction 0.990079, above). ✓ The integer threshold is therefore 459, and rounding up was the right choice.

Step 4 — the comparison with the surface. The exponential estimate of Theorem 11.5 with $\varepsilon = 0.1$ and $n = 459$ gives

$$\sigma(|x_1| \geq 0.1) \leq \frac{\sqrt{e}}{0.1 \times \sqrt{458}} e^{-458 \times 0.005} = \frac{1.648721}{2.140093} \times e^{-2.29} = 0.770397 \times 0.101266 = 0.078015,$$

that is, less than 8% of the surface lies outside the equatorial band of half-width 0.1. In dimension 459, therefore, *both* phenomena are already clearly visible: the volume is all on the skin and the surface is all at the equator.

Interpretation. The two statements say different but congruent things. The first concerns the *radius*: almost no point of the ball is interior. The second concerns the *direction*: almost no point of the sphere is close to a prescribed pole. Together: a uniform point in B^n has norm close to 1 and direction almost orthogonal to any direction fixed in advance.

Typical errors. (a) Rounding the threshold 458.21 down: the inequality is satisfied by $n \geq 459$, not by 458. (b) Believing that the fraction depends on V_n : the constant cancels (Step 2 of the proof). (c) Applying the formula with ε expressed as a percentage (1 instead of 0.01).

Extension. With $\varepsilon = 0.001$ the threshold becomes $\ln(100)/(-\ln(0.999)) = 4.605170/0.00100050 = 4602.87$, that is $n = 4603$ (and indeed $0.999^{4603} = 0.0099987 < 0.01 < 0.0100087 = 0.999^{4602}$): ten times less thickness, ten times more dimension. The dependence is essentially inverse, as the formula of Step 4 of the proof shows, because $-\ln(1 - \varepsilon) \simeq \varepsilon$ for small ε .

Exercise 11.8. Decompose explicitly into two reflections the rotation of $\mathbb{R}^3$ with matrix

$$R = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

following the construction of the proof of Theorem 11.9, and check that the angle of rotation is twice the dihedral angle between the two planes.

Solution. Strategy. The proof of Theorem 11.9 is constructive: one chooses a vector v that is not fixed, reflects with respect to the hyperplane orthogonal to $Rv - v$, and observes that the product fixes a larger subspace. Here the construction closes in two steps because $\det R = +1$.

Step 1 — the first reflection. Take $v = e_1 = (1, 0, 0)$; then $Rv = (0, 1, 0) \neq v$ and

$$m_1 = \frac{Rv - v}{\|Rv - v\|} = \frac{(-1, 1, 0)}{\sqrt{2}}.$$

The reflection $S_1(x) = x - 2\langle x, m_1 \rangle m_1$ has matrix

$$S_1 = I - 2m_1 m_1^\top = I - \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

that is, the exchange of the first two coordinates: it is the reflection in the plane $x = y$.

Step 2 — the product.

$$S_1 R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

which is the reflection S_2 in the plane $y = z$, with normal $m_2 = (0, 1, -1)/\sqrt{2}$. Indeed $\det(S_1 R) = -1$, $(S_1 R)^2 = I$ and $S_1 R$ fixes e_1 and $(0, 1, 1)$.

Step 3 — the decomposition. Since $S_1^2 = I$,

$$R = S_1 (S_1 R) = S_{m_1} S_{m_2}, \quad m_1 = \frac{(-1, 1, 0)}{\sqrt{2}}, \quad m_2 = \frac{(0, 1, -1)}{\sqrt{2}}.$$

They are two reflections, as Theorem 11.9 guarantees for an element of $SO(3)$ different from the identity.

Step 4 — the axis. The two planes are $x = y$ and $y = z$; their intersection is the line $x = y = z$, that is the axis $(1, 1, 1)/\sqrt{3}$, which both reflections fix pointwise. It is the axis of Theorem 11.8, recovered without computing eigenvectors.

Step 5 — independent check of the angle. The dihedral angle between the two planes has cosine $\langle m_1, m_2 \rangle = (0 \cdot (-1) + 1 \cdot 1 + 0 \cdot (-1))/2 = 0.5$, that is 60° . Twice that is 120° . On the other hand $\text{tr } R = 0$ and Theorem 11.8 gives $1 + 2 \cos \vartheta = 0$, that is $\cos \vartheta = -0.5$ and $\vartheta = 120^\circ$. ✓ The two computations, independent, agree.

Interpretation. The composition of two reflections in incident planes is always a rotation about the line of intersection, of angle twice the dihedral one (Exercise S4). Here the rotation permutes the coordinate axes cyclically: it is the symmetry of the cube exchanging $x \rightarrow y \rightarrow z \rightarrow x$.

Typical errors. (a) Using $m_1 = Rv + v$ instead of $Rv - v$: one obtains a reflection that does not bring Rv back to v . (b) Forgetting to normalize m : the formula $x - 2\langle x, m \rangle m$ holds only for $\|m\| = 1$. (c) Concluding that the angle of rotation equals the dihedral one rather than twice it.

Extension. Repeat with $v = e_2$: one finds $m'_1 = (0, -1, 1)/\sqrt{2}$ and a second decomposition, different from the first. The decomposition into reflections is not unique — only their *minimal number* is.

Exercise 11.9. Compute V_{100} and the ratio $V_{100}/2^{100}$ between the unit ball and the circumscribed cube $[-1, 1]^{100}$; compare the exact value with Stirling's approximation of Theorem 11.2.

Solution. Strategy. The number is too small for an ordinary calculator: one works with decimal logarithms, and checks the result with the asymptotic formula, which is independent of the exact computation.

Step 1 — the exact value in logarithms. By Theorem 11.1, $V_{100} = \pi^{50}/\Gamma(51) = \pi^{50}/50!$, hence

$$\log_{10} V_{100} = 50 \log_{10} \pi - \log_{10}(50!) = 50 \times 0.4971499 - 64.483075 = 24.857495 - 64.483075 = -39.625580,$$

that is $V_{100} = 2.368202 \cdot 10^{-40}$.

Step 2 — the comparison with the cube. $2^{100} = 1.267651 \cdot 10^{30}$, so

$$\frac{V_{100}}{2^{100}} = \frac{2.368202 \cdot 10^{-40}}{1.267651 \cdot 10^{30}} = 1.868182 \cdot 10^{-70}.$$

The circumscribed cube has a volume 10^{70} times that of the inscribed ball touching all its faces.

Step 3 — independent check with Stirling. Step 2 of the proof of Theorem 11.2 gives $V_n \sim \frac{1}{\sqrt{\pi n}} \left(\frac{2\pi e}{n}\right)^{n/2}$. With $n = 100$: $2\pi e = 17.079469$, hence $2\pi e/n = 0.17079469$ and

$$\log_{10} V_{100} \simeq -\frac{1}{2} \log_{10}(100\pi) + 50 \log_{10}(0.17079469) = -1.248575 - 38.376283 = -39.624858,$$

that is $2.372152 \cdot 10^{-40}$. The ratio between the estimate and the exact value is 1.001668: a deviation of 0.17%, exactly the order $1/(12 \cdot 50) = 0.001667$ of the first correction term of Stirling. ✓

Interpretation. Theorem 11.3 says that in dimension ≥ 10 the central ball tangent to the 2^n inscribed balls *sticks out* of the cube; this exercise says that nevertheless the cube remains incomparably larger. The two facts coexist because the volume of the cube, in high dimension, is all in the corners, and the corners are $2^{100} \approx 1.27 \cdot 10^{30}$.

Typical errors. (a) Using $\Gamma(50)$ instead of $\Gamma(51)$: the formula is $\Gamma(n/2 + 1)$. (b) Applying Stirling to $\Gamma(n/2)$ and then forgetting the factor $n/2$. (c) Computing π^{50} in single-precision floating point: every significant digit is lost.

Extension. For which n does the ratio $V_n/2^n$ fall below 10^{-100} ? The exact computation gives $n = 133$ (ratio $2.373 \cdot 10^{-101}$) against $n = 132$ ($2.188 \cdot 10^{-100}$): between dimension 100 and dimension 133 the ratio loses another thirty orders of magnitude.

11.11. Numerical and symbolic exercises with solutions

The exercises that follow come from the treatment of the circle in higher dimension of Volume II and are reproduced in full, with their solutions; those on cones, spheres and quadrics are in Chapter 7.

11.11.1 Numerical exercises

Exercise 11.10: N1 — Volumes and surfaces in high dimension. Compute V_n and Σ_{n-1} for $n = 4$, $n = 6$ and $n = 11$, using $\Gamma(1/2) = \sqrt{\pi}$ and the recursion for the Gamma function.

Exercise 11.11: N2 — Ball versus cube. Compute the ratio $V_n/2^n$ between the volume of the unit ball and that of the circumscribed cube for $n = 3, 5, 8, 9, 20$, and find the least dimension for which the ratio drops below 1%.

Exercise 11.12: N3 — How much dimension to concentrate. Using Theorem 11.4, find the least dimension n for which the fraction of S^{n-1} with $|x_1| \geq 0.2$ is guaranteed $\leq 5\%$. Then evaluate the exponential bound of Theorem 11.5 for $n = 200$ and $n = 500$ and compare the two guarantees.

Exercise 11.13: N4 — The ball that leaves the cube. Compute r_n for $n = 4, 9, 10, 16, 100$ and, for $n = 16$, the ratio of the volume of the central ball to that of the cube.

Exercise 11.14: N5 — Axis and angle of a rotation. Determine the axis and angle of the rotation with matrix

$$R = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

checking that $R \in SO(3)$.

Exercise 11.15: N6 — Rotation with quaternions. Rotate the vector $(1, 0, 0)$ by 120° about the axis $(1, 1, 1)/\sqrt{3}$ using quaternions, and check the result with the matrix of Exercise N5.

11.11.2 Symbolic exercises

Exercise 11.16: S1 — Surface as the derivative of volume. Prove that $V_n(R) = V_n R^n$ and that $\Sigma_{n-1}(R) = \frac{d}{dR} V_n(R)$, deducing $\Sigma_{n-1} = nV_n$. Explain why the same relation holds for the circle ($2\pi R = \frac{d}{dR} \pi R^2$) but not for the square.

Exercise 11.17: S2 — The maximum of the volume. Prove rigorously that $\max_n V_n$ is attained at $n = 5$, and that Σ_{n-1} is maximal at $n = 7$.

Exercise 11.18: S3 — Nearly orthogonal points. Prove that $\int_{S^{n-1}} x_i x_j d\sigma = \delta_{ij}/n$ and deduce that two independent uniform points on S^{n-1} have scalar product of root mean square $1/\sqrt{n}$: in high dimension two random directions are almost always nearly orthogonal.

Exercise 11.19: S4 — Rotations as products of reflections. Prove that the composition of two reflections of $\mathbb{R}^3$ in planes through the origin is a rotation about their line of intersection, by twice the dihedral angle between the planes. Deduce another proof of Theorem 11.8.

Exercise 11.20: S5 — Rodrigues' formula. Deduce from Theorem 11.10 Rodrigues' formula

$$\Phi_q(v) = v \cos \vartheta + (u \times v) \sin \vartheta + u \langle u, v \rangle (1 - \cos \vartheta),$$

and verify it on Exercise N6.

11.11.3 Solutions to the numerical exercises

Solution. N1. *Step 1* — $n = 4$. $\Gamma(3) = 2! = 2$, hence

$$V_4 = \frac{\pi^2}{\Gamma(3)} = \frac{9.869604}{2} = 4.934802, \quad \Sigma_3 = 4V_4 = 19.739209 = 2\pi^2.$$

Step 2 — $n = 6$. $\Gamma(4) = 3! = 6$, hence

$$V_6 = \frac{\pi^3}{6} = \frac{31.006277}{6} = 5.167713, \quad \Sigma_5 = 6V_6 = 31.006277 = \pi^3.$$

Step 3 — $n = 11$. We need $\Gamma(13/2)$. Using $\Gamma(t+1) = t\Gamma(t)$ repeatedly from $\Gamma(1/2) = \sqrt{\pi} = 1.7724539$:

$$\Gamma\left(\frac{13}{2}\right) = \frac{11}{2} \cdot \frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} = 162.421875 \times 1.7724539 = 287.8853.$$

Step 4 — *the volume*. $\pi^{11/2} = \pi^5 \sqrt{\pi} = 306.019685 \times 1.7724539 = 542.3989$, hence

$$V_{11} = \frac{542.3989}{287.8853} = 1.884104, \quad \Sigma_{10} = 11V_{11} = 20.725145.$$

Step 5 — *check with the recursion*. $V_{11} = \frac{2\pi}{11}V_9$. Now $V_9 = 3.298509$ (from the table of Example 11.1), and

$$\frac{2\pi}{11} \times 3.298509 = 0.571199 \times 3.298509 = 1.884103. \checkmark$$

Step 6 — *reading*. Already in dimension 11 the unit ball has smaller volume than the segment $[-1, 1]$ ($V_1 = 2$): the volume has more than halved from the maximum $V_5 = 5.264$.

Solution. N2. *Step 1* — *the ratios*.

n	V_n	2^n	$V_n/2^n$	in %
3	4.18879	8	0.523599	52.36%
5	5.26379	32	0.164493	16.45%
8	4.05871	256	0.015854	1.585%
9	3.29851	512	0.006442	0.644%
20	0.025807	1 048 576	2.461×10^{-8}	0.0000025%

Step 2 — *the 1% threshold*. The ratio first drops below 0.01 at $n = 9$: at $n = 8$ it is $1.585\% > 1\%$ and at $n = 9$ it is $0.644\% < 1\%$.

Step 3 — *check of V_9* . By the recursion, $V_9 = \frac{2\pi}{9}V_7 = 0.698132 \times 4.724766 = 3.298509$. $\checkmark$

Step 4 — *reading*. Already in dimension nine the inscribed ball occupies less than one per cent of the cube containing it, while touching all 18 of its faces. This is Theorem 11.2 in numerical form.

Step 5 — *a practical consequence*. In Monte Carlo methods, generating uniform points in the ball by rejection (drawing in the cube and discarding those outside the ball) is workable in dimension 2 or 3 but useless in dimension 20: it would take on average 4×10^7 trials per accepted point. Direct methods are used instead, such as normalising a Gaussian vector (Exercise S3).

Solution. N3. *Step 1* — *Chebyshev bound*. Theorem 11.4 gives

$$\sigma(|x_1| \geq \varepsilon) \leq \frac{1}{n\varepsilon^2}.$$

With $\varepsilon = 0.2$, $\varepsilon^2 = 0.04$:

$$\frac{1}{0.04n} \leq 0.05 \iff n \geq \frac{1}{0.04 \times 0.05} = 500.$$

Step 2 — the exponential bound. By Theorem 11.5, with $\varepsilon = 0.2$,

$$B(n) = \frac{\sqrt{e}}{0.2\sqrt{n-1}} e^{-0.02(n-1)} = \frac{8.243606}{\sqrt{n-1}} e^{-0.02(n-1)}.$$

Step 3 — evaluation at $n = 200$. With $n - 1 = 199$ and $\sqrt{199} = 14.106736$:

$$B(200) = \frac{8.243606}{14.106736} \times e^{-3.98} = 0.584374 \times 0.018686 = 0.010920,$$

that is 1.09%, against the 12.5% given by Chebyshev at the same dimension ($1/(200 \times 0.04) = 0.125$): a factor of 11.

Step 4 — evaluation at $n = 500$. With $n - 1 = 499$ and $\sqrt{499} = 22.338308$:

$$B(500) = \frac{8.243606}{22.338308} \times e^{-9.98} = 0.369014 \times 4.6317 \times 10^{-5} = 1.709 \times 10^{-5},$$

that is 0.0017%, against Chebyshev's 5%: a factor of nearly 3000.

Step 5 — reading the comparison. The two bounds say qualitatively different things: Chebyshev decays like $1/n$, the exponential bound like $e^{-n\varepsilon^2/2}$. At the dimension where Chebyshev barely reaches 5%, the exponential one has already dropped below two parts in a hundred thousand. This is why, in high dimension, one designs with exponential inequalities and not with moments.

Step 6 — order of magnitude. In both cases a few hundred dimensions suffice for 95% of the sphere to lie in a band of width 0.4 about the equator, a fifth of the diameter. In the plane ($n = 2$) the same band contains only $2 \arcsin(0.2)/\pi = 12.8\%$ of the circle.

Solution. N4. *Step 1 — the formula.* By Theorem 11.3, $r_n = \frac{\sqrt{n}-1}{2}$.

Step 2 — the values.

n	$\sqrt{n}$	r_n	comparison with inradius 1
4	2.000000	0.500000	inside
9	3.000000	1.000000	tangent to the faces
10	3.162278	1.081139	sticks out
16	4.000000	1.500000	sticks well out
100	10.000000	4.500000	sticks out 4.5-fold

Step 3 — relative volume for $n = 16$. We have $V_{16} = \pi^8/\Gamma(9) = 9488.5310/40\,320 = 0.2353306$ and

$$r_{16}^{16} = 1.5^{16} = 656.8408,$$

so the volume of the central ball is

$$V_{16} r_{16}^{16} = 0.2353306 \times 656.8408 = 154.582.$$

Step 4 — comparison with the cube. The cube $[-1, 1]^{16}$ has volume $2^{16} = 65\,536$, hence

$$\frac{154.582}{65\,536} = 0.0023588 = 0.236\%.$$

Step 5 — the paradox, resolved. The central ball *leaves* the cube and yet occupies barely two parts per thousand of it. There is no contradiction: it protrudes only through the 32

faces, and by a tiny amount, while the enormous volume of the cube remains concentrated in its 65 536 corners. ✓

Step 6 — check of 1.5^{16} . $1.5^2 = 2.25$; $1.5^4 = 5.0625$; $1.5^8 = 25.628906$; $1.5^{16} = 25.628906^2 = 656.8408$. ✓

Solution. N5. *Step 1 — checking $R \in SO(3)$.* The columns of R are $(0, 1, 0)$, $(0, 0, 1)$, $(1, 0, 0)$: the three basis unit vectors, hence orthonormal, and $R^\top R = I$. The determinant of a cyclic permutation of three elements is $+1$ (the permutation is even). Hence $R \in SO(3)$.

Step 2 — angle from the trace. $\text{tr } R = 0 + 0 + 0 = 0$, and by Theorem 11.8

$$1 + 2 \cos \vartheta = 0 \implies \cos \vartheta = -\frac{1}{2} \implies \vartheta = 120^\circ.$$

Step 3 — axis. We seek u with $Ru = u$. With $u = (u_1, u_2, u_3)$:

$$Ru = (u_3, u_1, u_2) = (u_1, u_2, u_3) \implies u_1 = u_2 = u_3.$$

Normalising, $u = (1, 1, 1)/\sqrt{3}$.

Step 4 — verification. $R(1, 1, 1)^\top = (1, 1, 1)^\top$ ✓. And for a vector orthogonal to the axis, say $w = (1, -1, 0)$: $Rw = (0, 1, -1)$, of the same norm $\sqrt{2}$, making with w an angle of cosine

$$\frac{\langle w, Rw \rangle}{\|w\|^2} = \frac{0 - 1 + 0}{2} = -\frac{1}{2} = \cos 120^\circ. \quad \checkmark$$

Step 5 — interpretation. R cyclically permutes the coordinate axes $x \rightarrow y \rightarrow z \rightarrow x$: it is the rotation by 120° about the cube diagonal, the same met in Example 11.2. Applying it three times gives $R^3 = I$, consistent with $3 \times 120^\circ = 360^\circ$. ✓

Solution. N6. *Step 1 — the quaternion.* By Theorem 11.10(iii) with $\vartheta = 120^\circ$ and $u = (i + j + k)/\sqrt{3}$:

$$q = \cos 60^\circ + \sin 60^\circ \cdot \frac{i + j + k}{\sqrt{3}} = 0.5 + 0.866025 \times 0.577350 (i + j + k) = 0.5(1 + i + j + k),$$

since $0.866025 \times 0.577350 = 0.5$.

Step 2 — conjugate and norm. $\bar{q} = 0.5(1 - i - j - k)$ and $\|q\|^2 = 4 \times 0.25 = 1$. ✓

Step 3 — first product. With $v = i$ (the vector $(1, 0, 0)$):

$$qv = 0.5(1 + i + j + k)i = 0.5(i + i^2 + ji + ki) = 0.5(i - 1 - k + j),$$

using $i^2 = -1$, $ji = -k$, $ki = j$.

Step 4 — second product. Multiplying by $\bar{q}$:

$$\Phi_q(i) = 0.5(-1 + i + j - k) \cdot 0.5(1 - i - j - k) = 0.25 X,$$

and expanding X term by term:

$$-1 \cdot (1 - i - j - k) = -1 + i + j + k,$$

$$i \cdot (1 - i - j - k) = i + 1 - k + j,$$

$$j \cdot (1 - i - j - k) = j + k + 1 - i,$$

$$-k \cdot (1 - i - j - k) = -k + j - i - 1.$$

Summing componentwise: real part $-1 + 1 + 1 - 1 = 0$; coefficient of i : $1 + 1 - 1 - 1 = 0$; coefficient of j : $1 + 1 + 1 + 1 = 4$; coefficient of k : $1 - 1 + 1 - 1 = 0$. Hence

$$\Phi_q(i) = 0.25 \times 4j = j.$$

Step 5 — result. The vector $(1, 0, 0)$ rotates to $(0, 1, 0)$.

Step 6 — check with the matrix. The rotation of Exercise N5 has the same axis and angle, and

$$R \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad \checkmark$$

Step 7 — norm check. $\|\Phi_q(i)\| = \|j\| = 1 = \|i\|$, as guaranteed by Theorem 11.10(i). $\checkmark$

Step 8 — comparison of methods. The quaternion computation required two multiplications; the matrix one requires building the 3×3 matrix and a matrix–vector product. For repeated compositions the advantage of quaternions grows: four numbers instead of nine, no drift away from orthogonality, and natural interpolation between orientations. That is why inertial navigation systems and three-dimensional graphics use quaternions and not matrices.

11.11.4 Solutions to the symbolic exercises

Solution. S1. *Step 1 — homogeneity of volume.* The map $x \mapsto Rx$ is linear with matrix RI , of determinant R^n ; since Lebesgue volume transforms by the absolute value of the determinant,

$$\text{vol}(B^n(0, R)) = R^n \text{vol}(B^n) = V_n R^n.$$

Step 2 — the derivative.

$$\frac{d}{dR} V_n(R) = nV_n R^{n-1}.$$

Step 3 — comparison with the area. By Theorem 11.1, $\Sigma_{n-1}(R) = \Sigma_{n-1} R^{n-1} = nV_n R^{n-1}$; it coincides with the derivative. Setting $R = 1$ gives $\Sigma_{n-1} = nV_n$.

Step 4 — why it is true. Geometrically: increasing the radius by dR adds a shell of thickness dR and surface $\Sigma_{n-1}(R)$, hence of volume $\Sigma_{n-1}(R) dR$. The relation $dV = \Sigma dR$ requires, however, that the increment be made *orthogonally to the boundary and with uniform thickness*: this is the construction of the *parallel body*.

Step 5 — why it fails for the square, if one scales. For the square of side s , area s^2 and perimeter $4s$, the derivative of the area with respect to s is $2s \neq 4s$. The reason is that the scaling $s \mapsto s + ds$ enlarges the square from the *centre*, adding a thickness $ds/2$ on each side, not ds .

Step 6 — the correct formulation. If one parametrises by the inradius $R = s/2$, the area is $4R^2$ and the perimeter $8R = \frac{d}{dR}(4R^2)$: the relation holds again. In general, for an arbitrary convex body the correct formula is Steiner's (Exercise S1 of Chapter 2): $A_d = A + Ld + \pi d^2$, whence $\frac{dA_d}{dd}|_{d=0} = L$. The ball is the case in which the parallel body coincides with a scaling, and for it alone the two operations are the same.

Solution. S2. *Step 1 — monotonicity along parities.* By Theorem 11.1, $V_n = \frac{2\pi}{n} V_{n-2}$, hence

$$V_n > V_{n-2} \iff \frac{2\pi}{n} > 1 \iff n < 2\pi = 6.283185.$$

Step 2 — the two subsequences. Among odd indices: $V_1 < V_3 < V_5$ (since $3, 5 < 2\pi$) and $V_5 > V_7 > V_9 > \dots$ (since $7, 9, \dots > 2\pi$). So the odd maximum is V_5 . Among even indices:

$V_2 < V_4 < V_6$ and $V_6 > V_8 > \dots$: the even maximum is V_6 .

Step 3 — final comparison.

$$V_5 = \frac{8\pi^2}{15} = 5.263789, \quad V_6 = \frac{\pi^3}{6} = 5.167713.$$

Since $V_5 > V_6$, the global maximum is V_5 , in dimension 5.

Step 4 — the area. With $\Sigma_{n-1} = nV_n$,

$$\frac{\Sigma_{n-1}}{\Sigma_{n-3}} = \frac{nV_n}{(n-2)V_{n-2}} = \frac{n}{n-2} \cdot \frac{2\pi}{n} = \frac{2\pi}{n-2},$$

greater than 1 if and only if $n < 2 + 2\pi = 8.283185$.

Step 5 — conclusion for the area. Among odd indices the maximum is at $n = 7$, among even ones at $n = 8$. Comparing:

$$\Sigma_6 = 7V_7 = 7 \times 4.724766 = 33.073362, \quad \Sigma_7 = 8V_8 = 8 \times 4.058712 = 32.469697.$$

So the maximum is at $n = 7$, i.e. for the sphere $S^6 \subset \mathbb{R}^7$. ✓

Step 6 — reading. That the volume is maximal in dimension 5 and the area in dimension 7 has no deep meaning: it depends on the normalisation to radius 1, which compares quantities with different units (m^5 against m^6). What matters, and is invariant, is the asymptotic behaviour: both tend to zero.

Solution. S3. *Step 1 — off-diagonal terms.* For $i \neq j$, the reflection $x_i \mapsto -x_i$ (fixing the other coordinates) is an isometry of $\mathbb{R}^n$ preserving S^{n-1} and the measure σ . Under it the integrand $x_i x_j$ changes sign, hence

$$\int x_i x_j d\sigma = - \int x_i x_j d\sigma \implies \int x_i x_j d\sigma = 0.$$

Step 2 — diagonal terms. These are Step 2 of the proof of Theorem 11.4: they equal $1/n$. In sum,

$$\int_{S^{n-1}} x_i x_j d\sigma = \frac{\delta_{ij}}{n}.$$

Step 3 — two independent points. Let X, Y be independent and uniform on S^{n-1} . Then

$$\mathbb{E}[\langle X, Y \rangle^2] = \mathbb{E} \left[\sum_{i,j} X_i X_j Y_i Y_j \right] = \sum_{i,j} \mathbb{E}[X_i X_j] \mathbb{E}[Y_i Y_j],$$

by independence. By Step 2 the sum reduces to the terms $i = j$:

$$\mathbb{E}[\langle X, Y \rangle^2] = \sum_{i=1}^n \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n}.$$

Step 4 — reading. The cosine of the angle between two random directions has root mean square $1/\sqrt{n}$: in dimension 10 000 two random directions typically make an angle of

$$\arccos(0.01) = 89.43^\circ,$$

i.e. they are nearly orthogonal. In the plane, by contrast, the cosine has root mean square $1/\sqrt{2} = 0.707$ and the typical angle is 45° .

Step 5 — consequence. In high dimension one can find very many *nearly* orthogonal vectors — exponentially many in n — although at most n can be exactly so. This is what makes

spread-spectrum codes, random compression (the Johnson–Lindenstrauss lemma) and distributed representations in machine learning possible.

Step 6 — how to generate uniform points. If Z is a standard Gaussian vector of $\mathbb{R}^n$, its density $e^{-\|z\|^2/2}$ depends only on $\|z\|$, hence is rotation invariant; therefore $Z/\|Z\|$ is uniform on S^{n-1} . This is the practical generation method, replacing the rejection scheme made useless by Theorem 11.2 (Exercise N2, Step 5).

Solution. S4. *Step 1 — the reflection.* The reflection in the plane with unit normal m is

$$S_m(v) = v - 2\langle v, m \rangle m.$$

It is orthogonal ($\|S_m v\| = \|v\|$, by expanding) and has determinant -1 (it fixes the plane $m^\perp$ and reverses m).

Step 2 — the composition is a rotation. $\det(S_{m_1} S_{m_2}) = (-1)(-1) = +1$ and the product of orthogonal maps is orthogonal: hence $S_{m_1} S_{m_2} \in SO(3)$.

Step 3 — the axis. If the two planes are distinct and pass through the origin, they meet in a line ℓ , orthogonal to both m_1 and m_2 . Every $v \in \ell$ is fixed by both reflections, hence by the composition: ℓ is the axis.

Step 4 — the angle. Restrict to the plane $\ell^\perp$, which contains m_1 and m_2 . There the two reflections are reflections in the lines $m_1^\perp \cap \ell^\perp$ and $m_2^\perp \cap \ell^\perp$, which make with each other the same angle φ as the two planes. Identifying $\ell^\perp$ with $\mathbb{C}$ (Chapter 3) and the lines with the directions $e^{i\psi_1}$ and $e^{i\psi_2}$, the reflection in the line of direction $e^{i\psi}$ is $z \mapsto e^{2i\psi} \bar{z}$, hence

$$z \mapsto e^{2i\psi_1} \overline{(e^{2i\psi_2} \bar{z})} = e^{2i(\psi_1 - \psi_2)} z,$$

the rotation by $2(\psi_1 - \psi_2) = 2\varphi$: twice the angle between the planes.

Step 5 — another proof of Euler’s theorem. By Theorem 11.9, every element of $O(3)$ is a product of at most three reflections in planes through the origin. If $R \in SO(3)$, the number of reflections must be *even* (since $\det = +1$): hence R is the identity or the product of exactly two reflections. In the second case Step 3 provides the axis. One thus recovers Theorem 11.8 without computing determinants.

Step 6 — check. Two orthogonal planes ($\varphi = 90^\circ$) give a rotation by 180° about their intersection: and indeed reflecting twice in orthogonal planes amounts to changing the sign of two of the three coordinates, a half-turn. ✓

Step 7 — the plane case. In $\mathbb{R}^2$ the same statement holds with “lines” for “planes”: the composition of two reflections in lines through the origin making an angle φ is the rotation by 2φ . It is the elementary fact generating all of $SO(2) = S^1$ (Theorem 11.7) from reflections alone.

Solution. S5. *Step 1 — product of two pure quaternions.* By bilinearity it suffices to check it on the basis. For p, w pure,

$$p w = -\langle p, w \rangle + p \times w.$$

Indeed for $p = w = i$: $i^2 = -1 = -\langle i, i \rangle$ and $i \times i = 0$ ✓; for $p = i, w = j$: $ij = k$, while $-\langle i, j \rangle = 0$ and $i \times j = k$ ✓; the other cases are analogous.

Step 2 — decomposition of the vector. Write $v = v_\parallel + v_\perp$ with

$$v_\parallel = u \langle u, v \rangle, \quad v_\perp = v - v_\parallel, \quad \langle u, v_\perp \rangle = 0.$$

Step 3 — the two actions. By Step 6 of the proof of Theorem 11.10, $\Phi_q(u) = u$, hence by linearity $\Phi_q(v_\parallel) = v_\parallel$. By Step 7 of the same theorem, with $w = v_\perp$,

$$\Phi_q(v_\perp) = \cos \vartheta v_\perp + \sin \vartheta (u v_\perp),$$

where $u v_{\perp}$ is the quaternion product; by Step 1 and orthogonality, $u v_{\perp} = -\langle u, v_{\perp} \rangle + u \times v_{\perp} = u \times v_{\perp} = u \times v$ (the parallel component contributes nothing to the cross product).

Step 4 — summing.

$$\Phi_q(v) = v_{\parallel} + \cos \vartheta (v - v_{\parallel}) + \sin \vartheta (u \times v) = \cos \vartheta v + (1 - \cos \vartheta) u \langle u, v \rangle + \sin \vartheta (u \times v),$$

which is Rodrigues' formula.

Step 5 — check on Exercise N6. With $u = (1, 1, 1)/\sqrt{3}$, $v = (1, 0, 0)$, $\vartheta = 120^\circ$ ($\cos = -0.5$, $\sin = 0.866025$):

$$\langle u, v \rangle = \frac{1}{\sqrt{3}} = 0.577350, \quad u \times v = \frac{1}{\sqrt{3}}(0, 1, -1) = (0; 0.577350; -0.577350).$$

Then

$$\cos \vartheta v = (-0.5; 0; 0),$$

$$(1 - \cos \vartheta) u \langle u, v \rangle = 1.5 \times (0.577350)^2 (1, 1, 1) = 1.5 \times 0.333333 (1, 1, 1) = (0.5; 0.5; 0.5),$$

$$\sin \vartheta (u \times v) = 0.866025 \times (0; 0.577350; -0.577350) = (0; 0.5; -0.5).$$

Summing: $(-0.5 + 0.5 + 0; 0 + 0.5 + 0.5; 0 + 0.5 - 0.5) = (0, 1, 0)$. ✓ It matches the result of Exercise N6.

Step 6 — reading. Rodrigues' formula (1840) precedes Hamilton's quaternions (1844) by four years and in fact contains their geometric content. It displays the rotation as a sum of three contributions: the original component scaled by $\cos \vartheta$, the component along the axis recovered in full, and an orthogonal contribution proportional to $\sin \vartheta$. For small ϑ it reduces to $\Phi_q(v) \approx v + \vartheta (u \times v)$: the velocity formula for a rotating body.

11.12. Self-assessment quiz

- In Definition 11.1 the superscript of S^{n-1} indicates
 - the dimension of the ambient space
 - the intrinsic dimension of the surface
 - the number of coordinates
 - the radius
- The passage from $\Sigma_{n-1} = 2\pi^{n/2}/\Gamma(n/2)$ to $V_n = \Sigma_{n-1}/n$, in Theorem 11.1, uses the hypothesis
 - that n be even
 - that the volume element in polar coordinates be $r^{n-1} dr d\sigma$
 - that $\Gamma(1/2) = \sqrt{\pi}$
 - that the ball be convex
- From the recurrence $V_n = \frac{2\pi}{n} V_{n-2}$ it follows that V_{12} equals
 - $\frac{2\pi}{12} V_{10}$
 - $\frac{2\pi}{10} V_{10}$
 - $\frac{\pi}{12} V_{10}$
 - $2\pi V_{10}$
- In B^{100} the fraction of volume beyond the radius 0.99 equals, to six digits,
 - 0.010000
 - 0.633968
 - 0.366032
 - 0.990000
- In the proof of Theorem 11.4, the step that makes *all* directions equivalent is
 - Chebyshev's inequality
 - the invariance of σ under rotations, in particular under permutation of the coordinates
 - the constraint $\sum x_i^2 = 1$
 - Stirling's formula
- A student concludes, from Theorem 11.3, that in dimension 16 the central ball has volume greater than the cube, because it sticks out of it. The error is

-
- A) r_{16} is less than 1 B) sticking out of the cube does not mean filling it: $V_{16}r_{16}^{16}$ remains tiny compared with 2^{16} C) the small balls are not tangent D) none, the conclusion is correct
7. An attitude control system represents orientation with a unit quaternion instead of Euler angles. Which result of the chapter justifies the choice?
- A) Theorem 11.7 ($SO(2) \cong S^1$) B) Theorem 11.10: $S^3 \rightarrow SO(3)$ is surjective and two-to-one, hence the parametrization is global and free of singularities C) Theorem 11.2 D) Theorem 11.9
8. That S^0 , S^1 and S^3 are the only spheres carrying a Lie group structure is, in this treatise,
- A) proved in section 11.7 B) declared *imported*, with reference to Bott, Milnor and Adams (1958–1960) C) an open conjecture D) false

11.12.1 Supplementary questionnaire

The items that follow come from Volume II and from the chapter on the generalization to dimension n ; those on cones, spheres and quadrics are in Chapter 7.

- The volume of the unit ball in $\mathbb{R}^n$ is maximal for

A) $n = 3$ B) $n = 5$ C) it always grows D) $n = 7$
- As $n \rightarrow \infty$, V_n

A) tends to 0 B) tends to infinity C) tends to 1 D) oscillates
- The ratio of inscribed ball to cube, in high dimension,

A) tends to 1 B) tends to 0 C) stays $\pi/4$ D) grows
- On S^{n-1} the mean square of the first coordinate is

A) 1 B) $1/n$ C) $1/2$ D) $1/\sqrt{n}$
- Concentration of measure says that almost all of the sphere lies

A) at the poles B) near any equator C) in one octant D) uniformly spread in the intuitive sense
- The central ball tangent to the 2^n balls inscribed in the cube $[-1, 1]^n$ leaves the cube

A) never B) from $n = 10$ on C) from $n = 3$ on D) always
- The group $SO(2)$ is isomorphic to

A) $\mathbb{R}$ B) S^1 C) S^2 D) $\mathbb{Z}$
- Euler's theorem states that every rotation of $\mathbb{R}^3$

A) is the identity B) has a fixed axis C) is commutative D) has determinant -1
- For a rotation of $\mathbb{R}^3$ one has $\text{tr } R =$

A) $3 \cos \vartheta$ B) $1 + 2 \cos \vartheta$ C) $2 + \cos \vartheta$ D) $\cos \vartheta$
- The unit quaternions form

A) S^1 B) S^3 C) S^2 D) $\mathbb{R}^4$
- The map $S^3 \rightarrow SO(3)$ is

A) bijective B) two to one C) three to one D) not onto

12. The only spheres that are Lie groups are
 A) all of them B) S^0, S^1, S^3 C) only S^1 D) S^1 and S^2
13. The volume of B^n is expressed by means of
 A) the factorial B) the Gamma function C) the zeta function D) the Bernoulli numbers
14. S^{n-1} has intrinsic dimension
 A) n B) $n - 1$ C) $n + 1$ D) $2n$
15. The phenomenon that degrades distance-based methods in high dimension is called
 A) overfitting B) *curse of dimensionality* C) aliasing D) *gimbal lock*
16. Compared with $SO(2)$, the group $SO(3)$ is
 A) the same B) commutative C) non-commutative D) finite

BEST PRACTICE

Key to the supplementary questionnaire. 1-B (Example 11.1 and Exercise S2); 2-A (Theorem 11.2); 3-B (same theorem, Step 3); 4-B (Theorem 11.4, Step 2); 5-B (same theorem, Step 4); 6-B (Theorem 11.3); 7-B (Theorem 11.7); 8-B (Theorem 11.8); 9-B (same theorem, Step 4); 10-B (Definition 11.3); 11-B (Theorem 11.10(iv)); 12-B (Remark 11.4); 13-B (Theorem 11.1); 14-B (Definition 11.1); 15-B (Remark 11.2); 16-C (Example 11.2, Step 7).

11.13. Teacher's key

BEST PRACTICE

Q1 — answer B. *Rationale:* Definition 11.1 and Remark 11.1: S^1 is a curve in $\mathbb{R}^2$, S^2 a surface in $\mathbb{R}^3$. *Distractors:* A and C are the same confusion (the ambient space has dimension n , the sphere $n - 1$); D makes no sense, the radius is not an integer. *Diagnosed error:* confusing intrinsic dimension with the dimension of the ambient space. *Source:* Ball, *An Elementary Introduction to Modern Convex Geometry*, §1. *Prerequisite:* the notion of surface. *Remediation:* write what S^0, S^1, S^2, S^3 are and in which space they live.

Q2 — answer B. *Rationale:* Step 5 of the proof integrates the area of the concentric spheres, $V_n = \int_0^1 \Sigma_{n-1} r^{n-1} dr$, and this presupposes the form $dx = r^{n-1} dr d\sigma$ of the volume element, established at Step 2. *Distractors:* A is false (the formula holds for every n); C is used only in the numerical checks; D is true but useless. *Diagnosed error:* failing to identify which hypothesis supports which step. *Source:* Ball, §1, formula (1). *Prerequisite:* change of variables to polar coordinates. *Remediation:* redo Step 5 in dimension 2 and 3, where the result is known.

Q3 — answer A. *Rationale:* the recurrence is $V_n = \frac{2\pi}{n} V_{n-2}$, evaluated at $n = 12$. *Distractors:* B uses the wrong index in the denominator (a very frequent error); C halves the constant; D forgets the denominator. *Diagnosed error:* substituting the index in the formula without rereading it. *Source:* Theorem 11.1, Step 6. *Prerequisite:* manipulation of recursive sequences. *Remediation:* check numerically that $\frac{2\pi}{12} \times 2.550164 = 1.335263$ agrees with the closed formula.

Q4 — answer B. *Rationale:* by Theorem 11.6 the fraction is $1 - 0.99^{100} = 1 - 0.366032 = 0.633968$. *Distractors:* C is the *inner* fraction; A confuses the thickness with the fraction;

D is the answer one would get by exchanging ε with $1 - \varepsilon$. *Diagnosed error*: exchanging the part with its complement. *Source*: Theorem 11.6, Step 5. *Prerequisite*: powers and exponentials. *Remediation*: redo the computation for $n = 2$ and $n = 3$ and compare with the drawing.

Q5 — answer B. *Rationale*: Step 1 uses the invariance of σ under rotations to conclude that the n integrals $\int x_i^2 d\sigma$ are equal; that is what makes *every* equator equivalent and gives Step 4. *Distractors*: A turns the second moment into a tail estimate but says nothing about directions; C is the constraint which, together with A, gives the value $1/n$; D appears in another theorem. *Diagnosed error*: attributing to the last step what depends on the first. *Source*: Ledoux, *The Concentration of Measure Phenomenon*, ch. 1. *Prerequisite*: invariant measures. *Remediation*: redo Step 1 with $n = 2$ (where $\int x_1^2 d\sigma = 1/2$) and verify it by direct computation.

Q6 — answer B. *Rationale*: Step 7 of Theorem 11.3: in dimension 10 the central ball occupies 0.54% of the cube while sticking out of it. *Distractors*: A is false ($r_{16} = 1.5$); C is false (the small balls are tangent by construction); D accepts the error. *Diagnosed error*: confusing linear extent and volume in high dimension. *Source*: Ball, §1. *Prerequisite*: $V_n r^n$ as the volume of a ball of radius r . *Remediation*: compute $V_{16} r_{16}^{16}/2^{16}$ and compare it with $V_{10} r_{10}^{10}/2^{10}$.

Q7 — answer B. *Rationale*: surjectivity (point (iv) and Step 9) gives a *global* parametrization of $SO(3)$ by S^3 , which is a manifold without boundary: there exist no singular configurations as with Euler angles. *Distractors*: A concerns the plane; C and D are true but not pertinent to the problem. *Diagnosed error*: choosing the theorem “that talks about rotations” without looking at what it states. *Source*: Stillwell, *Naive Lie Theory*, ch. 1. *Prerequisite*: the notion of singular parametrization. *Remediation*: write the three Euler matrices and find the configuration in which the Jacobian determinant vanishes.

Q8 — answer B. *Rationale*: Remark 11.6 declares explicitly that the classification is *imported*: it requires Steenrod operations and K -theory. What the chapter proves is the existence of the group structures on S^1 and S^3 and the Hopf fibration, not their uniqueness. *Distractors*: A confuses what is proved with what is cited; C is false (the classification is a theorem of 1958–1960); D is false. *Diagnosed error*: not distinguishing between a proved result and an imported one. *Source*: Remark 11.6 and Stillwell, ch. 1 and 8. *Prerequisite*: none. *Remediation*: list, for section 11.7, which statements are proved and which one is not.

Answers to the assigned exercises.

Exercise 11.4. (i) $V_9 = \frac{2\pi}{9} \times 4.724766 = 3.298509$ and $\Sigma_8 = 9V_9 = 29.686580$; $V_{12} = \frac{2\pi}{12} \times 2.550164 = 1.335263$ and $\Sigma_{11} = 12V_{12} = 16.023153$. The closed formula gives the same values: $V_9 = \pi^{4.5}/\Gamma(5.5) = 3.298509$ and $V_{12} = \pi^6/\Gamma(7) = \pi^6/720 = 1.335263$. $\checkmark$ (ii) $V_{10} = 2.550164 > 2$ and $V_{11} = 1.884104 < 2$: the first dimension is $n = 11$. (iii) Because $\Sigma_{n-1} = nV_n$: multiplying by n shifts the maximum forward, and since the ratio $V_n/V_{n-2} = 2\pi/n$ drops below 1 for $n > 2\pi$, the maximum of the volumes falls at $n = 5$ and that of the areas at $n = 7$.

Exercise 11.5. (i) Chebyshev (Theorem 11.4) guarantees $1/(n\varepsilon^2) \leq 0.01$ with $\varepsilon = 0.05$, that is $n \geq 1/(0.01 \times 0.0025) = 40\,000$. (ii) The exponential estimate $\frac{\sqrt{e}}{\varepsilon\sqrt{n-1}} e^{-(n-1)\varepsilon^2/2} \leq 0.01$ is satisfied from $n = 3248$ on: indeed for $n = 3248$ it equals 0.0099941 and for $n = 3247$ it equals 0.0100081. (iii) The ratio is $40\,000/3248 = 12.3$: the exponential estimate requires a dimension twelve times smaller. Neither of the two statements is wrong — they are both correct upper bounds — but the first is very loose: Chebyshev uses only the second moment, the second uses the exact form of the density $(1 - t^2)^{(n-3)/2}$.

Exercise 11.6. (i) $\cos \vartheta = \langle a, b \rangle = 1/\sqrt{3} = 0.577350$, hence $\vartheta = 0.955317$ rad = 54.735610° ; the axis is $u = (a \times b)/\|a \times b\| = (1, -1, 0)/\sqrt{2}$. (ii) $\cos(\vartheta/2) = 0.888074$ and $\sin(\vartheta/2) =$

0.459701, hence

$$q = 0.888074 + 0.459701 \left(\frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j \right) = 0.888074 + 0.325058 i - 0.325058 j,$$

with $\|q\|^2 = 0.788676 + 0.105663 + 0.105663 = 1.000002$ (rounding). ✓(iii) With $\cos \vartheta = 0.577350$, $\sin \vartheta = 0.816497$ and $1 - \cos \vartheta = 0.422650$ Rodrigues' formula gives

$$R = \begin{pmatrix} 0.788675 & -0.211325 & -0.577350 \\ -0.211325 & 0.788675 & -0.577350 \\ 0.577350 & 0.577350 & 0.577350 \end{pmatrix},$$

with $\text{tr } R = 2.154700 = 1 + 2 \times 0.577350$ ✓ and $Ra = (0, 0, 1.732051)/\sqrt{3} = (0, 0, 1) = b$ ✓.

11.14. Chapter formulary

Hyperspheres

$$V_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}, \quad \Sigma_{n-1} = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} = nV_n, \quad V_n = \frac{2\pi}{n} V_{n-2}$$

$$V_n(R) = V_n R^n, \quad \Sigma_{n-1}(R) = \frac{d}{dR} V_n(R)$$

$$V_n \sim \frac{1}{\sqrt{\pi n}} \left(\frac{2\pi e}{n} \right)^{n/2} \rightarrow 0; \quad \max_n V_n = V_5 = 5.26379; \quad \max_n \Sigma_{n-1} = \Sigma_6 = 33.0734$$

Gamma function

$$\Gamma(t) = \int_0^\infty u^{t-1} e^{-u} du, \quad \Gamma(t+1) = t \Gamma(t), \quad \Gamma(n+1) = n!, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma(t+1) \sim \sqrt{2\pi t} \left(\frac{t}{e} \right)^t \quad (\text{Stirling}), \quad \text{Vol}(B^n, r) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} r^n$$

High dimension

$$\int_{S^{n-1}} x_i x_j d\sigma = \frac{\delta_{ij}}{n}; \quad \sigma(|x_1| \geq \varepsilon) \leq \frac{1}{n\varepsilon^2} \quad (\text{and } \leq 2e^{-n\varepsilon^2/2})$$

$$r_n = \frac{\sqrt{n} - 1}{2} \quad (\text{central ball}); \quad r_n > 1 \iff n \geq 10$$

$$\text{shell : } \frac{\text{vol}(B^n \setminus B^n(O, 1 - \varepsilon))}{\text{vol}(B^n)} = 1 - (1 - \varepsilon)^n \rightarrow 1, \quad n \geq \frac{\ln(1/\delta)}{-\ln(1 - \varepsilon)} \Rightarrow \geq 1 - \delta$$

Rotations

$$SO(2) \cong S^1; \quad R \in SO(3) \Rightarrow \exists u : Ru = u, \quad \text{tr } R = 1 + 2 \cos \vartheta$$

$$q = \cos \frac{\vartheta}{2} + u \sin \frac{\vartheta}{2}, \quad \Phi_q(v) = qv\bar{q}, \quad \Phi_{q_1 q_2} = \Phi_{q_1} \Phi_{q_2}, \quad \Phi_q = \Phi_{-q}$$

$$\text{Rodrigues : } \Phi_q(v) = v \cos \vartheta + (u \times v) \sin \vartheta + u \langle u, v \rangle (1 - \cos \vartheta)$$

Hopf fibration

$$h(z_1, z_2) = (2z_1 \bar{z}_2, |z_1|^2 - |z_2|^2), \quad h^{-1}(h(z_1, z_2)) = \{(\lambda z_1, \lambda z_2) : |\lambda| = 1\}$$

$$S^3 = \bigsqcup_{p \in S^2} h^{-1}(p), \quad \text{every fibre is a great circle; two distinct fibres are linked}$$

Sources and bibliography

- [1] L. Euler, *Formulae generales pro translatione quacunque corporum rigidorum*, Novi Commentarii academiae scientiarum Petropolitanae, vol. 20, 1776, pp. 189–207. Original statement of Theorem 11.8.
- [2] O. Rodrigues, *Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace*, Journal de Mathématiques Pures et Appliquées, vol. 5, 1840, pp. 380–440. The rotation formula of Exercise S5, predating quaternions.
- [3] W. R. Hamilton, *Lectures on Quaternions*, Hodges and Smith, Dublin, 1853 (and the original communication to the Royal Irish Academy, 1844). Introduction of quaternions and of Definition 11.3.
- [4] H. Hopf, *Über die Abbildungen der dreidimensionalen Sphäre auf die Kugelfläche*, Mathematische Annalen, vol. 104, 1931, pp. 637–665. The fibration of Observation 11.4.
- [5] D. Hilbert, S. Cohn-Vossen, *Anschauliche Geometrie*, Springer, Berlin, 1932 (English translation *Geometry and the Imagination*, Chelsea, 1952). Dandelin spheres, circular sections of the ellipsoid, the Hopf fibration, treated intuitively and rigorously.
- [6] M. Berger, *Géométrie*, vols. 1–5, Cedic/Fernand Nathan, Paris, 1977 (English translation *Geometry I–II*, Springer, 1987). Quadrics, orthogonal groups, spherical geometry.
- [7] H. S. M. Coxeter, *Introduction to Geometry*, 2nd ed., John Wiley & Sons, 1969. Spherical geometry, polyhedra, quaternions.
- [8] J. Stillwell, *Naive Lie Theory*, Springer, New York, 2008. Elementary treatment of low-rank Lie groups; does not contain the proof that S^0 , S^1 and S^3 are the only spheres with a group structure, which requires Bott–Milnor–Adams (1958–1960, Remark 11.6). Elementary treatment also of $SO(3)$ and $SU(2)$.
- [9] K. Ball, *An Elementary Introduction to Modern Convex Geometry*, in *Flavors of Geometry* (S. Levy, ed.), MSRI Publications 31, Cambridge University Press, 1997, pp. 1–58. Volume of the ball in high dimension, the ball escaping the cube (Theorem 11.3), concentration.
- [10] M. Ledoux, *The Concentration of Measure Phenomenon*, Mathematical Surveys and Monographs 89, American Mathematical Society, 2001. The classical form of the exponential bound of Theorem 11.5.
- [11] V. D. Milman, G. Schechtman, *Asymptotic Theory of Finite Dimensional Normed Spaces*, Lecture Notes in Mathematics 1200, Springer, 1986. Concentration and its applications to functional analysis.
- [12] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976. Gaussian curvature, the Bertrand–Puisseux formula of Exercise S4, geodesics on the sphere. South American source.
- [13] MIT OpenCourseWare, *18.02 Multivariable Calculus* and *18.700 Linear Algebra* — <https://ocw.mit.edu>. Integration in spherical coordinates and orthogonal matrices.
- [14] NPTEL, Indian Institute of Technology, *Multivariable Calculus* and *Classical Mechanics* — <https://nptel.ac.in/>. Spherical coordinates in $\mathbb{R}^n$ and rigid-body rotations. Asian source.

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- [15] AIMS South Africa, *African Institute for Mathematical Sciences — teaching materials on geometry and high-dimensional probability* — <https://aims.ac.za/>. African institutional source.
- [16] A. N. Gorban, I. Y. Tyukin, *Blessing of dimensionality: mathematical foundations of the statistical physics of data*, Philosophical Transactions of the Royal Society A, vol. 376, no. 2118, 2018, art. 20170237 — <https://doi.org/10.1098/rsta.2017.0237>; preprint [arXiv:1801.03421](https://arxiv.org/abs/1801.03421) (10 January 2018). Concentration of measure and its consequences in machine learning: it is the source of the *curse* and of the *blessing of dimensionality* of Remark 11.2. Verified on 5 September 2026.
- [17] R. Bellman, *Dynamic Programming*, Princeton University Press, Princeton, 1957. Origin of the term *curse of dimensionality*.
- [18] C. A. Barbieri, D. Miller-Cotto, S. N. Clerjuste, K. Chawla, *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*, Educational Psychology Review 35, art. 11, 2023 — <https://doi.org/10.1007/s10648-023-09745-1>.
- [19] H. L. Roediger III, J. D. Karpicke, *Test-Enhanced Learning*, Psychological Science 17(3), 2006 — <https://doi.org/10.1111/j.1467-9280.2006.01693.x>.

CHAPTER 12

Topology of the circle: covering, winding number and fundamental group

So far the circle has been a *metric* object: a locus of equidistant points, a curve of constant curvature, a group of complex numbers of modulus one. In this chapter we forget distance and keep only what survives a continuous deformation. What remains is surprisingly rich: the circle is compact and connected like a closed interval, but it is *not* an interval, because removing one point leaves it connected; and above all it has a “hole”, that is, a loop that cannot be contracted. Measuring that hole means counting how many times a path goes around, and the answer — an integer — is the oldest and most widely used topological invariant in applied mathematics: it appears in Cauchy’s index, in the unwrapped phase of a signal, in the count of turns accumulated on the axis of a robot.

The technical tool is a single elementary one: the exponential map $t \mapsto e^{it}$ unrolls the circle onto the real line. The whole chapter is the rigorous construction of this unrolling (the *covering*), of its use (the *lifting* of paths and homotopies) and of its consequences ($\pi_1(S^1) \cong \mathbb{Z}$ and the classification of maps $S^1 \rightarrow S^1$ by degree).

INFO

Level and prerequisites. Level C–D (undergraduate/graduate).

Internal prerequisites. Parametrisation of the circle and polar form of complex numbers (Volume II, Chapter 3); complex exponential and the properties $e^{i(s+t)} = e^{is}e^{it}$, $e^{it} = 1 \iff t \in 2\pi\mathbb{Z}$ (Volume II, Chapter 3); contour integration and Cauchy’s theorem for the analytic form of the winding number (Volume II, Chapter 3, and Chapter 15); the circle as a multiplicative group (Chapter 6).

External prerequisites, assumed and not reproved (a first course in general topology; Munkres, *Topology*, 2nd ed., §23, §26–27):

- a closed subspace of a compact space is compact;
- the continuous image of a compact space is compact; the continuous image of a connected space is connected;
- a compact subspace of a Hausdorff space is closed;
- Heine–Borel: the compact subsets of $\mathbb{R}^n$ are exactly the closed and bounded ones (Munkres, Theorem 27.3);
- a map into a product is continuous if and only if its components are.

Everything else — the Lebesgue number included — is proved here.

What one learns. How to construct a covering and verify its local property; how to lift paths and homotopies by a direct compactness argument; how to define the winding number in topological and in analytic form and prove that the two agree; how to compute $\pi_1(S^1)$ and classify maps $S^1 \rightarrow S^1$.

Warning. Conventions of this chapter.

- We identify $\mathbb{R}^2$ with $\mathbb{C}$ and set $S^1 = \{z \in \mathbb{C} : |z| = 1\} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, with the subspace topology.

- The base point of S^1 is always $1 = (1, 0)$, unless stated otherwise.
- We write $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it} = (\cos t, \sin t)$; the period is therefore 2π and the fibre is $E^{-1}(1) = 2\pi\mathbb{Z}$.
- $I = [0, 1]$. A **path** in X is a continuous map $\gamma : I \rightarrow X$; it is a **loop** based at x_0 if $\gamma(0) = \gamma(1) = x_0$.
- “Map” always means “continuous map”.
- On $\mathbb{R}^n$ we use the Euclidean distance; the diameter of A is $\text{diam } A = \sup\{d(a, b) : a, b \in A\}$.

12.1. The circle as a topological space

Definition 12.1: Subspace topology and S^1 . Let (X, τ) be a topological space and $A \subseteq X$. The **subspace topology** on A is

$$\tau_A = \{U \cap A : U \in \tau\}.$$

The **circle** is the set $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ with the subspace topology induced from $\mathbb{R}^2$. Identifying $\mathbb{R}^2 \cong \mathbb{C}$ one writes equivalently $S^1 = \{z \in \mathbb{C} : |z| = 1\}$.

Since $\mathbb{R}^2$ is metric, S^1 is metric (with the restricted Euclidean distance), hence in particular Hausdorff: two distinct points have disjoint neighbourhoods. This fact will be used constantly.

Lemma 12.1: From compact to Hausdorff. Let X be compact, Y Hausdorff and $f : X \rightarrow Y$ continuous. Then:

- (i) f is closed: it maps closed sets to closed sets;
- (ii) if f is bijective, it is a homeomorphism;
- (iii) if f is surjective, it is a quotient map: $A \subseteq Y$ is open if and only if $f^{-1}(A)$ is open in X .

Proof. (i) Let $C \subseteq X$ be closed. A closed subset of a compact space is compact, so C is compact; the continuous image of a compact space is compact, so $f(C)$ is compact in Y ; a compact subset of a Hausdorff space is closed, so $f(C)$ is closed. (These are the three facts assumed in the prerequisites.)

(ii) Let $g = f^{-1} : Y \rightarrow X$, well defined because f is bijective. For every closed $C \subseteq X$ we have $g^{-1}(C) = f(C)$, closed by (i). A map for which the preimage of every closed set is closed is continuous; hence g is continuous and f is a homeomorphism.

(iii) If A is open, $f^{-1}(A)$ is open by continuity. Conversely, let $f^{-1}(A)$ be open. Then $X \setminus f^{-1}(A)$ is closed and by (i) $f(X \setminus f^{-1}(A))$ is closed. Since f is surjective,

$$f(X \setminus f^{-1}(A)) = Y \setminus A :$$

indeed, if $y \in Y \setminus A$, pick x with $f(x) = y$; then $x \notin f^{-1}(A)$; conversely if $y = f(x)$ with $x \notin f^{-1}(A)$ then $y \notin A$. Hence $Y \setminus A$ is closed and A is open. $\square$

Theorem 12.1: S^1 is compact, connected and path connected. The circle S^1 is compact, connected and path connected.

Proof. Step 1 — E is continuous and onto on $[0, 2\pi]$. The map $E : \mathbb{R} \rightarrow \mathbb{R}^2$, $E(t) = (\cos t, \sin t)$, is continuous because its two components are, and takes values in S^1 because $\cos^2 t + \sin^2 t = 1$. It is already onto S^1 when restricted to $[0, 2\pi]$: given $(x, y) \in S^1$ we have

$x \in [-1, 1]$, and setting $t_0 = \arccos x \in [0, \pi]$ we get $\cos t_0 = x$ and $\sin t_0 = \sqrt{1 - x^2} = |y|$; if $y \geq 0$ take $t = t_0$, if $y < 0$ take $t = 2\pi - t_0 \in [\pi, 2\pi]$, so that $\cos t = \cos t_0 = x$ and $\sin t = -\sin t_0 = -|y| = y$. Hence $E([0, 2\pi]) = S^1$.

Step 2 — compactness. $[0, 2\pi]$ is closed and bounded in $\mathbb{R}$, hence compact by Heine–Borel. The continuous image of a compact space is compact, so $S^1 = E([0, 2\pi])$ is compact. *Independent check:* S^1 is bounded (contained in the ball of radius 1) and closed in $\mathbb{R}^2$, being the preimage of the closed set $\{1\}$ under the continuous function $(x, y) \mapsto x^2 + y^2$; by Heine–Borel it is compact. ✓

Step 3 — connectedness. $[0, 2\pi]$ is connected (it is an interval) and the continuous image of a connected space is connected: hence S^1 is connected.

Step 4 — path connectedness. Let $z_0 = E(t_0)$ and $z_1 = E(t_1)$ be two points of S^1 (t_0, t_1 exist by Step 1). The path

$$\gamma : I \rightarrow S^1, \quad \gamma(s) = E((1-s)t_0 + st_1)$$

is continuous (a composition of continuous maps), has image in S^1 , and satisfies $\gamma(0) = z_0$, $\gamma(1) = z_1$. Hence S^1 is path connected. $\square$

Corollary 12.1: S^1 is homeomorphic neither to $\mathbb{R}$ nor to an interval. S^1 is not homeomorphic to $\mathbb{R}$, nor to an interval $[a, b]$ with $a < b$.

Proof. (a) S^1 is compact (Theorem 12.1) while $\mathbb{R}$ is not: the open cover $\{(-n, n)\}_{n \in \mathbb{N}}$ has no finite subcover, because the union of finitely many of those intervals is the largest of them, which does not contain n for large n . Compactness is invariant under homeomorphism (it is defined purely in terms of open sets), hence $S^1 \not\cong \mathbb{R}$.

(b) For the interval we use a finer invariant, connectedness after removing a point. If $c \in (a, b)$, then $[a, b] \setminus \{c\} = [a, c) \cup (c, b]$ is disconnected, because the two sets are open in $[a, b] \setminus \{c\}$, non-empty and disjoint. On the other hand $S^1 \setminus \{p\}$ is connected for every p : by Theorem 12.2 it is homeomorphic to $\mathbb{R}$, which is connected. A homeomorphism $h : [a, b] \rightarrow S^1$ would map $[a, b] \setminus \{c\}$ homeomorphically onto $S^1 \setminus \{h(c)\}$, that is, a disconnected space onto a connected one: impossible, since for $c \in (a, b)$ the former is disconnected and the latter is not. $\square$

Theorem 12.2: S^1 minus a point is homeomorphic to $\mathbb{R}$ (R15.2). For every $p \in S^1$ the space $S^1 \setminus \{p\}$ is homeomorphic to $\mathbb{R}$.

Proof. **Step 1 — reduction to the north pole.** Let $N = (0, 1)$. If $p = E(\alpha)$, the rotation $R : \mathbb{C} \rightarrow \mathbb{C}$, $R(z) = e^{i(\pi/2 - \alpha)}z$, is a homeomorphism of $\mathbb{C}$ onto itself (it is linear and invertible, with inverse $z \mapsto e^{-i(\pi/2 - \alpha)}z$, both continuous), maps S^1 into S^1 because it preserves the modulus, and maps p to $e^{i(\pi/2 - \alpha)}e^{i\alpha} = e^{i\pi/2} = N$. Hence R restricts to a homeomorphism $S^1 \setminus \{p\} \rightarrow S^1 \setminus \{N\}$, and it suffices to treat $p = N$.

Step 2 — the stereographic projection. Define

$$\sigma : S^1 \setminus \{N\} \rightarrow \mathbb{R}, \quad \sigma(x, y) = \frac{x}{1 - y},$$

geometrically: $\sigma(x, y)$ is the abscissa of the point where the line through N and (x, y) meets the axis $y = 0$. The function is continuous because on $S^1 \setminus \{N\}$ we have $y \neq 1$, so the denominator does not vanish and σ is a quotient of continuous functions.

Step 3 — the inverse. Set

$$\tau : \mathbb{R} \rightarrow S^1 \setminus \{N\}, \quad \tau(s) = \left(\frac{2s}{s^2 + 1}, \frac{s^2 - 1}{s^2 + 1} \right).$$

The denominator $s^2 + 1$ never vanishes, so τ is continuous. Its image lies in S^1 :

$$(2s)^2 + (s^2 - 1)^2 = 4s^2 + s^4 - 2s^2 + 1 = s^4 + 2s^2 + 1 = (s^2 + 1)^2,$$

that is, $\left(\frac{2s}{s^2+1}\right)^2 + \left(\frac{s^2-1}{s^2+1}\right)^2 = 1$. Moreover $\tau(s) \neq N$: the second coordinate equals 1 only if $s^2 - 1 = s^2 + 1$, i.e. $-1 = 1$, false.

Step 4 — they are mutually inverse. For every $s \in \mathbb{R}$,

$$\sigma(\tau(s)) = \frac{\frac{2s}{s^2+1}}{1 - \frac{s^2-1}{s^2+1}} = \frac{\frac{2s}{s^2+1}}{\frac{(s^2+1) - (s^2-1)}{s^2+1}} = \frac{2s}{2} = s.$$

Conversely, let $(x, y) \in S^1$ with $y \neq 1$ and $s = \frac{x}{1-y}$. From $x^2 = 1 - y^2 = (1-y)(1+y)$ we get

$$s^2 + 1 = \frac{x^2}{(1-y)^2} + 1 = \frac{(1-y)(1+y)}{(1-y)^2} + 1 = \frac{1+y}{1-y} + 1 = \frac{2}{1-y},$$

and therefore

$$\frac{2s}{s^2+1} = 2 \cdot \frac{x}{1-y} \cdot \frac{1-y}{2} = x, \quad \frac{s^2-1}{s^2+1} = \frac{s^2+1-2}{s^2+1} = 1 - \frac{2}{s^2+1} = 1 - (1-y) = y.$$

Hence $\tau(\sigma(x, y)) = (x, y)$. The two maps are continuous and mutually inverse: σ is a homeomorphism. $\square$

Example 12.1: Numerical check of the stereographic projection. Take $\theta = 200^\circ$, that is, $(x, y) = (\cos 200^\circ, \sin 200^\circ) = (-0.93969, -0.34202)$. Then

$$s = \sigma(x, y) = \frac{-0.93969}{1 - (-0.34202)} = \frac{-0.93969}{1.34202} = -0.70021.$$

Independent check with the inverse formula: $s^2 = 0.49029$, $s^2 + 1 = 1.49029$, and

$$\frac{2s}{s^2+1} = \frac{-1.40042}{1.49029} = -0.93969 = x, \quad \frac{s^2-1}{s^2+1} = \frac{-0.50971}{1.49029} = -0.34202 = y. \quad \checkmark$$

For $\theta = 340^\circ$ symmetry gives $s = +0.70021$. The two values are opposite, as they must be, because $\theta = 200^\circ$ and $\theta = 340^\circ$ are symmetric with respect to the axis through N and the two images are symmetric with respect to the origin. The check was also run in double precision with **numpy** on random points of the circle: maximum error $1.1 \cdot 10^{-16}$. $\checkmark$

Remark 12.1: One point less, two different worlds. Theorem 12.2 is the topological reason why the circle behaves so differently from the line although it is locally identical to it: locally S^1 and $\mathbb{R}$ are indistinguishable (every point of S^1 has a neighbourhood homeomorphic to an open interval, as Lemma 12.2 shows), but globally they are not. Removing a point from $\mathbb{R}$ disconnects it; removing one from S^1 stretches it. One must remove two points from S^1 to disconnect it, and then two open arcs remain: this is the exact one-dimensional analogue of the fact that a closed curve separates the plane (Jordan's theorem, Appendix A).

12.2. The circle as the quotient $\mathbb{R}/2\pi\mathbb{Z}$

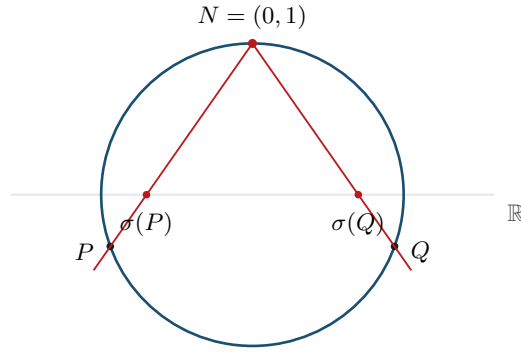


Figure 12.1: The stereographic projection $\sigma(x, y) = x/(1 - y)$ of $S^1 \setminus \{N\}$ onto the line $y = 0$: join N to the point of the circle and read the abscissa of the intersection with the horizontal axis. For $P = (\cos 200^\circ, \sin 200^\circ) = (-0.93969, -0.34202)$ one gets $\sigma(P) = -0.70021$; for $Q = (\cos 340^\circ, \sin 340^\circ)$ one gets $\sigma(Q) = +0.70021$. Moving away from N along the circle the image stays finite; approaching N it diverges, and this is why removing *one single* point suffices to stretch the circle over the whole line.

Definition 12.2: The quotient $\mathbb{R}/2\pi\mathbb{Z}$. On $\mathbb{R}$ define the relation

$$s \sim t \iff s - t \in 2\pi\mathbb{Z}.$$

It is an equivalence relation: reflexive ($s - s = 0 \in 2\pi\mathbb{Z}$), symmetric ($t - s = -(s - t)$ and $2\pi\mathbb{Z}$ is closed under negation) and transitive ($(s - t) + (t - u) = s - u$ and $2\pi\mathbb{Z}$ is closed under addition). The quotient set is denoted $\mathbb{R}/2\pi\mathbb{Z}$ and $\pi : \mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$ is the canonical projection. On $\mathbb{R}/2\pi\mathbb{Z}$ we put the **quotient topology**: A is open if and only if $\pi^{-1}(A)$ is open in $\mathbb{R}$.

Theorem 12.3: $\mathbb{R}/2\pi\mathbb{Z}$ is homeomorphic to S^1 . The map $\bar{E} : \mathbb{R}/2\pi\mathbb{Z} \rightarrow S^1$ induced by $E(t) = e^{it}$, that is, $\bar{E}(\pi(t)) = e^{it}$, is a homeomorphism. Moreover it is a group isomorphism between $(\mathbb{R}/2\pi\mathbb{Z}, +)$ and $(S^1, \cdot)$.

Proof. **Step 1 — $\bar{E}$ is well defined and bijective.** We use the characterisation $e^{it} = 1 \iff t \in 2\pi\mathbb{Z}$ (Volume II, Chapter 3). If $\pi(s) = \pi(t)$ then $s - t \in 2\pi\mathbb{Z}$, hence $e^{i(s-t)} = 1$ and $e^{is} = e^{it}$: $\bar{E}$ is well defined. Conversely, if $e^{is} = e^{it}$ then $e^{i(s-t)} = 1$, i.e. $s - t \in 2\pi\mathbb{Z}$ and $\pi(s) = \pi(t)$: $\bar{E}$ is injective. It is surjective because E is (Step 1 of the proof of Theorem 12.1).

Step 2 — $\bar{E}$ is continuous. By definition of the quotient topology, $\bar{E}$ is continuous if and only if $\bar{E} \circ \pi = E$ is continuous, and E is.

Step 3 — $\mathbb{R}/2\pi\mathbb{Z}$ is compact. Every class has a representative in $[0, 2\pi]$: given $t \in \mathbb{R}$, setting $k = \lfloor t/2\pi \rfloor$ we get $t - 2\pi k \in [0, 2\pi]$. Hence $\mathbb{R}/2\pi\mathbb{Z} = \pi([0, 2\pi])$, the continuous image of a compact space, hence compact.

Step 4 — conclusion. S^1 is Hausdorff (a subspace of a metric space). By Lemma 12.1(ii), applied to $\bar{E} : \mathbb{R}/2\pi\mathbb{Z} \rightarrow S^1$ continuous and bijective from a compact to a Hausdorff space, $\bar{E}$ is a homeomorphism.

Step 5 — group structure. $2\pi\mathbb{Z}$ is a subgroup of $(\mathbb{R}, +)$, hence $\mathbb{R}/2\pi\mathbb{Z}$ is a group with $\pi(s) + \pi(t) = \pi(s+t)$, well defined. And $\bar{E}(\pi(s) + \pi(t)) = e^{i(s+t)} = e^{is}e^{it} = \bar{E}(\pi(s))\bar{E}(\pi(t))$ by the property of the complex exponential: $\bar{E}$ is a homomorphism, and by Step 1 it is bijective, hence an isomorphism. $\square$

Remark 12.2: Three descriptions of the same object. From Theorem 12.3 on we shall use interchangeably three descriptions of the circle: the subspace $S^1 \subset \mathbb{C}$, the quotient $\mathbb{R}/2\pi\mathbb{Z}$ and the group $\mathbb{T}$ of complex numbers of modulus 1. The third is the one of Chapter 13, where the fact that $\bar{E}$ is an isomorphism of topological groups is used to build the invariant measure $d\theta/2\pi$ and the characters $z \mapsto z^n$. The second is the one that makes it evident why a 2π -periodic function on $\mathbb{R}$ “is” a function on S^1 : it is the starting point of Fourier analysis (Volume II, Chapter 4).

12.3. The exponential covering

Definition 12.3: Covering, evenly covered neighbourhood, sheet. Let $\tilde{X}$ and X be topological spaces and $p : \tilde{X} \rightarrow X$ continuous and surjective. An open set $U \subseteq X$ is **evenly covered** by p if

$$p^{-1}(U) = \bigsqcup_{\alpha \in A} V_{\alpha}$$

with $V_{\alpha} \subseteq \tilde{X}$ open and pairwise disjoint, and $p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$ a homeomorphism for every α . The sets V_{α} are called **sheets** (or slices) over U . The map p is a **covering** if every point of X has an evenly covered open neighbourhood. The inverses $(p|_{V_{\alpha}})^{-1} : U \rightarrow V_{\alpha}$ are called **local sections** of p over U .

The heart of the whole chapter is that the exponential map is a covering. The proof below is *constructive*: the local sections are written out explicitly by means of arctan and arccos, so that the liftings of the next sections are computable and not merely existent.

Lemma 12.2: Four explicit local sections. Set the four open arcs of S^1

$$U_1 = \{(x, y) \in S^1 : x > 0\}, \quad U_2 = \{x < 0\}, \quad U_3 = \{y > 0\}, \quad U_4 = \{y < 0\},$$

and the four functions

$$\begin{aligned} \psi_1(x, y) &= \arctan \frac{y}{x}, & \psi_2(x, y) &= \arctan \frac{y}{x} + \pi, \\ \psi_3(x, y) &= \arccos x, & \psi_4(x, y) &= -\arccos x. \end{aligned}$$

Then, for $j = 1, 2, 3, 4$: U_j is open in S^1 , the four arcs cover S^1 , $\psi_j : U_j \rightarrow \mathbb{R}$ is continuous, and ψ_j is the inverse of the restriction of E to the open interval J_j , where

$$J_1 = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad J_2 = \left(\frac{\pi}{2}, \frac{3\pi}{2}\right), \quad J_3 = (0, \pi), \quad J_4 = (-\pi, 0).$$

Moreover ψ_j is of class C^{∞} as a function of the coordinates on U_j .

Proof. Step 1 — the U_j are open and cover. $U_1 = \{(x, y) \in \mathbb{R}^2 : x > 0\} \cap S^1$ is the intersection of an open subset of $\mathbb{R}^2$ with S^1 , hence open in the subspace topology; similarly for the other three. If $(x, y) \in S^1$ and none of the four inequalities held, we would have $x = 0$ and $y = 0$, against $x^2 + y^2 = 1$: hence $S^1 = U_1 \cup U_2 \cup U_3 \cup U_4$.

Step 2 — $E(J_j) = U_j$. For $j = 1$: if $t \in (-\pi/2, \pi/2)$ then $\cos t > 0$, so $E(t) \in U_1$; conversely, given $(x, y) \in U_1$, set $r = y/x$ and $t = \arctan r \in (-\pi/2, \pi/2)$. Then $\cos t > 0$ and $\tan t = r$, whence $\sin t = r \cos t$ and $1 = \cos^2 t (1 + r^2)$, i.e. $\cos t = 1/\sqrt{1 + r^2}$ (positive root) and $\sin t = r/\sqrt{1 + r^2}$. On the other hand, from $x^2 + y^2 = 1$ and $x > 0$ we get $x^2(1 + r^2) = 1$, i.e. $x = 1/\sqrt{1 + r^2}$ and $y = rx = r/\sqrt{1 + r^2}$. Hence $E(t) = (x, y)$. For $j = 3$: if $t \in (0, \pi)$

then $\sin t > 0$, so $E(t) \in U_3$; conversely, given $(x, y) \in U_3$, setting $t = \arccos x \in (0, \pi)$ we have $\cos t = x$ and $\sin t = +\sqrt{1-x^2} = |y| = y$ because $\sin t > 0$ on $(0, \pi)$ and $y > 0$. The cases $j = 2$ and $j = 4$ follow from the previous ones by composing with the symmetries $z \mapsto -z$ and $z \mapsto \bar{z}$: explicitly, $(x, y) \in U_2 \iff (-x, -y) \in U_1$ and $E(t + \pi) = -E(t)$, which gives $\psi_2 = \psi_1(-x, -y) + \pi = \arctan \frac{-y}{-x} + \pi = \arctan \frac{y}{x} + \pi$; and $(x, y) \in U_4 \iff (x, -y) \in U_3$ with $E(-t) = \overline{E(t)}$, which gives $\psi_4 = -\psi_3(x, -y) = -\arccos x$.

Step 3 — $E|_{J_j}$ is injective. Each of the four intervals has length π : $|J_1| = |J_2| = |J_3| = |J_4| = \pi$. If $s, t \in J_j$ and $E(s) = E(t)$, then $s - t \in 2\pi\mathbb{Z}$ by Step 1 of the proof of Theorem 12.3; but $|s - t| < \pi < 2\pi$, and the only element of $2\pi\mathbb{Z}$ of modulus less than 2π is 0. Hence $s = t$.

Step 4 — ψ_j is continuous and is the inverse. $\arctan$ is continuous on $\mathbb{R}$ and $x \mapsto y/x$ is continuous on U_1 (where $x > 0$), so ψ_1 is continuous; $\arccos$ is continuous on $[-1, 1]$, so ψ_3 and ψ_4 are continuous; and ψ_2 is continuous for the same reason as ψ_1 . Steps 2 and 3 say that $E|_{J_j} : J_j \rightarrow U_j$ is a bijection whose inverse is exactly ψ_j (the computation in Step 2 exhibits $t = \psi_j(x, y)$). Hence $E|_{J_j}$ is a homeomorphism with inverse ψ_j .

Step 5 — regularity. $\arctan$ is C^∞ on $\mathbb{R}$ and $\arccos$ is C^∞ on $(-1, 1)$; on U_1 we have $x > 0$, on U_3 we have $|x| < 1$ (because $y > 0$ and $x^2 + y^2 = 1$ imply $x^2 < 1$), and similarly for U_2, U_4 . Hence every ψ_j is C^∞ . $\square$

Theorem 12.4: $E : \mathbb{R} \rightarrow S^1$ is a covering (R15.3). The map $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it}$, is a covering. Precisely, with the notation of Lemma 12.2, for every j

$$E^{-1}(U_j) = \bigsqcup_{n \in \mathbb{Z}} (J_j + 2\pi n),$$

the intervals $J_j + 2\pi n$ are open and pairwise disjoint, and $E|_{J_j + 2\pi n} : J_j + 2\pi n \rightarrow U_j$ is a homeomorphism with inverse $\psi_j + 2\pi n$. The fibre over $z = E(t_0)$ is

$$E^{-1}(z) = t_0 + 2\pi\mathbb{Z},$$

a discrete, countably infinite subset of $\mathbb{R}$. Finally $\mathbb{R}$ is path connected and simply connected (every loop in $\mathbb{R}$ is homotopic to a constant), hence E is the universal covering of S^1 .

Proof. Step 1 — the fibre. $E(t) = E(t_0) \iff e^{i(t-t_0)} = 1 \iff t - t_0 \in 2\pi\mathbb{Z}$, that is, $E^{-1}(E(t_0)) = t_0 + 2\pi\mathbb{Z}$. This set is discrete: two distinct points of it are at distance at least 2π , so each is isolated.

Step 2 — decomposition of $E^{-1}(U_j)$. Let $t \in E^{-1}(U_j)$, i.e. $E(t) \in U_j$. By Lemma 12.2 there is $t' \in J_j$ with $E(t') = E(t)$, hence $t - t' \in 2\pi\mathbb{Z}$ by Step 1: $t \in J_j + 2\pi n$ for some $n \in \mathbb{Z}$. Conversely $E(J_j + 2\pi n) = E(J_j) = U_j$ by periodicity. Hence $E^{-1}(U_j) = \bigcup_n (J_j + 2\pi n)$. The intervals are disjoint: if $u \in J_j \cap (J_j + 2\pi n)$ with $n \neq 0$, then u and $u - 2\pi n$ both lie in J_j , which has length π ; but $|u - (u - 2\pi n)| = 2\pi|n| \geq 2\pi > \pi$, a contradiction. They are open because they are translates of an open set.

Step 3 — homeomorphism on each sheet. $E|_{J_j + 2\pi n}(t) = E(t - 2\pi n)$, and $E|_{J_j} : J_j \rightarrow U_j$ is a homeomorphism with inverse ψ_j (Lemma 12.2); composing with the translation $t \mapsto t - 2\pi n$, which is a homeomorphism of $\mathbb{R}$, we get that $E|_{J_j + 2\pi n}$ is a homeomorphism onto U_j with inverse $\psi_j + 2\pi n$.

Step 4 — every point has an evenly covered neighbourhood. The four arcs U_j cover S^1 (Lemma 12.2) and each is evenly covered by Steps 2 and 3. Hence E is a covering. And it is surjective (Theorem 12.1, Step 1).

Step 5 — $\mathbb{R}$ is simply connected. $\mathbb{R}$ is path connected (the segment between two points is a path). If $\gamma : I \rightarrow \mathbb{R}$ is a loop based at a , the homotopy $H(s, u) = (1 - u)\gamma(s) + ua$ is continuous, has $H(\cdot, 0) = \gamma$, $H(\cdot, 1) \equiv a$ and $H(0, u) = H(1, u) = a$ for every u (because $\gamma(0) = \gamma(1) = a$): it is a path homotopy between γ and the constant loop. Hence $\pi_1(\mathbb{R}, a)$ is

| trivial.

□

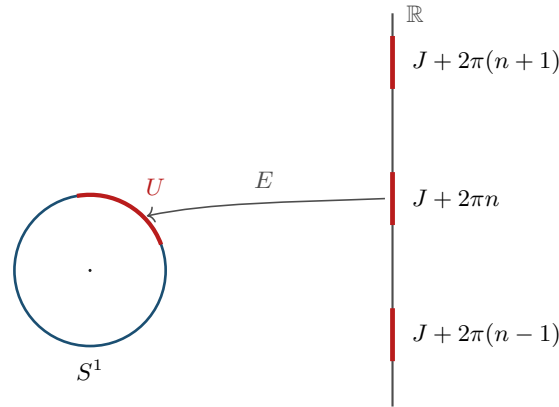


Figure 12.2: The evenly covered neighbourhood. Over the open arc $U \subset S^1$ (red on the circle) the preimage $E^{-1}(U)$ is the disjoint union of infinitely many open intervals $J + 2\pi n$, $n \in \mathbb{Z}$ (the red segments on the line), each mapped homeomorphically onto U by E . This is Munkres’ “stack of pancakes”: the line winds around the circle with pitch 2π , and locally the winding is invertible. The local inverse is the section $\psi_j + 2\pi n$ of Lemma 12.2.

Remark 12.3: A counterexample: the half-line does not cover. The restriction $E|_{(0,+\infty)} : (0,+\infty) \rightarrow S^1$ is continuous, surjective and a local homeomorphism (by Lemma 12.2), and yet it is not a covering. The critical point is $z_0 = 1$: let U be any open neighbourhood of 1 in S^1 , and let $\varepsilon > 0$ be such that $E((-\varepsilon, \varepsilon)) \subseteq U$. Then $E^{-1}(U) \cap (0, +\infty)$ contains $(0, \varepsilon)$, and the component of $E^{-1}(U) \cap (0, +\infty)$ containing $(0, \varepsilon)$ cannot be mapped onto U : its image does not contain the points $E(t)$ with $t \in (-\varepsilon, 0)$ that are near 1 “from below”, because every $t' > 0$ with $E(t') = E(t)$ satisfies $t' \geq t + 2\pi > \varepsilon$ and lies in another component. Hence no sheet over U contains $(0, \varepsilon)$ and is homeomorphic to U . The example (Munkres, Example 53.2) shows that “local homeomorphism + surjectivity” is not enough: one needs the uniformity of the covering, and one needs that no sheet “stops”.

12.4. Lifting of paths and homotopies

From here on everything is proved *directly for the covering* $E : \mathbb{R} \rightarrow S^1$, without invoking the general theory of covering spaces: the only ingredients are the compactness of I and of $I \times I$ (through the Lebesgue number), the pasting lemma and the discrete structure of the fibre $2\pi\mathbb{Z}$.

Definition 12.4: Lifting. Let $f : Y \rightarrow S^1$ be continuous. A **lifting** of f is a continuous map $\tilde{f} : Y \rightarrow \mathbb{R}$ such that $E \circ \tilde{f} = f$, i.e. $f(y) = e^{i\tilde{f}(y)}$ for every $y \in Y$. In the case $Y = I$ one speaks of the lifting of a path; the value $\tilde{f}(y)$ is also called the **unwrapped phase** of f at y .

Lemma 12.3: Pasting lemma. Let $Y = A_1 \cup \dots \cup A_m$ with $A_1, \dots, A_m$ closed in Y , and let $g_k : A_k \rightarrow Z$ be continuous with $g_k = g_l$ on $A_k \cap A_l$ for all k, l . Then the function $g : Y \rightarrow Z$ defined by $g|_{A_k} = g_k$ is well defined and continuous.

Proof. Well-definedness is the compatibility hypothesis. For continuity, let $C \subseteq Z$ be closed. Then

$$g^{-1}(C) = \bigcup_{k=1}^m (g^{-1}(C) \cap A_k) = \bigcup_{k=1}^m g_k^{-1}(C).$$

Each $g_k^{-1}(C)$ is closed in A_k by continuity of g_k , and A_k is closed in Y , so $g_k^{-1}(C)$ is closed in Y . A finite union of closed sets is closed, so $g^{-1}(C)$ is closed and g is continuous. $\square$

Lemma 12.4: Lebesgue number. *Let (X, d) be a compact metric space and let $\mathcal{U}$ be an open cover of it. Then there is $\delta > 0$ (a **Lebesgue number** for $\mathcal{U}$) such that every subset $B \subseteq X$ with $\text{diam } B < \delta$ is contained in some element of $\mathcal{U}$.*

Proof. For every $x \in X$ there is $U_x \in \mathcal{U}$ with $x \in U_x$, and since U_x is open there is $\varepsilon_x > 0$ with $B(x, 2\varepsilon_x) \subseteq U_x$. The family $\{B(x, \varepsilon_x)\}_{x \in X}$ is an open cover of X ; by compactness there are $x_1, \dots, x_m$ with

$$X = \bigcup_{i=1}^m B(x_i, \varepsilon_{x_i}).$$

Set $\delta = \min\{\varepsilon_{x_1}, \dots, \varepsilon_{x_m}\} > 0$ (the minimum of a finite set of positive numbers). Now let $B \subseteq X$ be non-empty with $\text{diam } B < \delta$ and let $b \in B$. There is i with $b \in B(x_i, \varepsilon_{x_i})$. For every $a \in B$,

$$d(a, x_i) \leq d(a, b) + d(b, x_i) < \delta + \varepsilon_{x_i} \leq 2\varepsilon_{x_i},$$

so $B \subseteq B(x_i, 2\varepsilon_{x_i}) \subseteq U_{x_i} \in \mathcal{U}$. If $B = \emptyset$ the statement is trivial. $\square$

Lemma 12.5: Uniqueness of the lifting on a connected space. *Let Y be connected and let $\tilde{f}_1, \tilde{f}_2 : Y \rightarrow \mathbb{R}$ be two liftings of the same map $f : Y \rightarrow S^1$. If $\tilde{f}_1(y_0) = \tilde{f}_2(y_0)$ for some $y_0 \in Y$, then $\tilde{f}_1 = \tilde{f}_2$.*

Proof. Set $d = \tilde{f}_1 - \tilde{f}_2 : Y \rightarrow \mathbb{R}$, continuous. For every y , $e^{i\tilde{f}_1(y)} = f(y) = e^{i\tilde{f}_2(y)}$, so $e^{id(y)} = 1$ and $d(y) \in 2\pi\mathbb{Z}$. Hence $d(Y) \subseteq 2\pi\mathbb{Z}$. But $d(Y)$ is connected (continuous image of a connected space) and the connected subsets of $2\pi\mathbb{Z}$ — which is discrete — are the singletons and the empty set: indeed, if $2\pi n \neq 2\pi m$ both belonged to $d(Y)$, taking $c = \pi(n + m)$ we would get a partition of $d(Y)$ into the two non-empty disjoint open sets $d(Y) \cap (-\infty, c)$ and $d(Y) \cap (c, +\infty)$ (no element of $2\pi\mathbb{Z}$ equals c when $n + m$ is odd; if $n + m$ is even take $c = \pi(n + m) + \pi = \pi(n + m + 1)$, which is not a multiple of 2π because $n + m + 1$ is odd), contradicting connectedness. Hence d is constant, and it equals $d(y_0) = 0$: $\tilde{f}_1 = \tilde{f}_2$. $\square$

Lemma 12.6: Path lifting (R15.4). *Let $\gamma : I \rightarrow S^1$ be a path and let $a \in \mathbb{R}$ with $E(a) = \gamma(0)$. Then there is one and only one lifting $\tilde{\gamma} : I \rightarrow \mathbb{R}$ of γ with $\tilde{\gamma}(0) = a$.*

Proof. Uniqueness. I is connected, so this follows from Lemma 12.5.

Step 1 — an adapted subdivision. The family $\{\gamma^{-1}(U_j)\}_{j=1,\dots,4}$, with the U_j of Lemma 12.2, is an open cover of I (the U_j cover S^1 and γ is continuous). I is metric and compact, so by Lemma 12.4 there is a Lebesgue number $\delta > 0$. Choose $m \in \mathbb{N}$ with $1/m < \delta$ and set $t_k = k/m$ for $k = 0, \dots, m$. Each interval $I_k = [t_{k-1}, t_k]$ has diameter $1/m < \delta$, hence $I_k \subseteq \gamma^{-1}(U_{j(k)})$ for a suitable index $j(k)$, that is,

$$\gamma(I_k) \subseteq U_{j(k)} \quad (k = 1, \dots, m).$$

Step 2 — construction by induction. We build by induction on k continuous functions $g_k : I_k \rightarrow \mathbb{R}$ with $E \circ g_k = \gamma|_{I_k}$ and $g_k(t_{k-1}) = g_{k-1}(t_{k-1})$ (with the convention $g_0(t_0) = a$). Let $k \geq 1$ and let the value $b_{k-1} := g_{k-1}(t_{k-1})$ be known (for $k = 1$, $b_0 = a$), satisfying $E(b_{k-1}) = \gamma(t_{k-1})$. Since $\gamma(t_{k-1}) \in U_{j(k)}$, the number b_{k-1} belongs to $E^{-1}(U_{j(k)}) = \bigsqcup_n (J_{j(k)} + 2\pi n)$ (Theorem 12.4), hence to one and only one sheet $V = J_{j(k)} + 2\pi n_k$. Set

$$g_k = (E|_V)^{-1} \circ \gamma|_{I_k} = \psi_{j(k)} \circ \gamma|_{I_k} + 2\pi n_k,$$

a composition of continuous functions, hence continuous; and $E \circ g_k = \gamma|_{I_k}$ because $E|_V$ is bijective onto $U_{j(k)}$. Finally $g_k(t_{k-1}) = (E|_V)^{-1}(\gamma(t_{k-1})) = b_{k-1}$ because $b_{k-1} \in V$ and $E(b_{k-1}) = \gamma(t_{k-1})$.

Step 3 — pasting. The I_k are closed in I , their union is I , and two of them meet in at most one point: $I_k \cap I_{k+1} = \{t_k\}$ and $I_k \cap I_l = \emptyset$ if $|k - l| \geq 2$. At the common points the g_k agree by construction. By Lemma 12.3 the function $\tilde{\gamma}$ defined by $\tilde{\gamma}|_{I_k} = g_k$ is continuous; it satisfies $E \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(0) = g_1(t_0) = a$. $\square$

Lemma 12.7: Homotopy lifting. Let $F : I \times I \rightarrow S^1$ be continuous and let $a \in \mathbb{R}$ with $E(a) = F(0, 0)$. Then there is one and only one continuous lifting $\tilde{F} : I \times I \rightarrow \mathbb{R}$ of F with $\tilde{F}(0, 0) = a$.

Proof. Uniqueness. $I \times I$ is connected (it is convex, hence path connected), and Lemma 12.5 applies.

Step 1 — adapted grid. $I \times I$ is metric and compact (closed and bounded in $\mathbb{R}^2$, Heine–Borel); $\{F^{-1}(U_j)\}_{j=1,\dots,4}$ is an open cover of it. Let $\delta > 0$ be a Lebesgue number (Lemma 12.4) and choose m with $\sqrt{2}/m < \delta$. The small squares

$$Q_{k,l} = \left[\frac{k-1}{m}, \frac{k}{m}\right] \times \left[\frac{l-1}{m}, \frac{l}{m}\right], \quad k, l = 1, \dots, m,$$

have diameter $\sqrt{2}/m < \delta$, hence $F(Q_{k,l}) \subseteq U_{j(k,l)}$ for a suitable index.

Step 2 — ordering and induction. Order the squares row by row: $Q_{1,1}, Q_{2,1}, \dots, Q_{m,1}, Q_{1,2}, \dots, Q_{m,m}$, and call this sequence $R_1, R_2, \dots, R_{m^2}$. Let $A_r = R_1 \cup \dots \cup R_r$. We build by induction on r a continuous map $G_r : A_r \rightarrow \mathbb{R}$ with $E \circ G_r = F|_{A_r}$ and $G_r(0, 0) = a$, extending the previous one.

Base $r = 1$. $R_1 = Q_{1,1}$ contains $(0, 0)$ and $F(R_1) \subseteq U$ with $U = U_{j(1,1)}$. The point a lies in one and only one sheet V over U (Theorem 12.4), because $E(a) = F(0, 0) \in U$. Set $G_1 = (E|_V)^{-1} \circ F|_{R_1}$: continuous, lifting $F|_{R_1}$, with $G_1(0, 0) = a$.

Inductive step. Let $r \geq 2$, let $R = R_r$ be the current square and $C = A_{r-1} \cap R$. Because of the row-by-row construction, C is: the left edge of R , if R is not the first of its row and lies in the first row; the bottom edge of R , if R is the first of its row (with row index ≥ 2); the union of the left edge and of the bottom edge, otherwise. In all cases C is non-empty and *connected*: in the third case the two edges are segments meeting at the lower-left vertex of R , and the union of two connected sets with a common point is connected.

Let $U = U_{j(k,l)}$ with $F(R) \subseteq U$, and let $c \in C$. The value $G_{r-1}(c)$ lies in one and only one sheet V over U . Set $g = (E|_V)^{-1} \circ F|_R$, continuous on R , with $E \circ g = F|_R$ and $g(c) = G_{r-1}(c)$. On C , both $g|_C$ and $G_{r-1}|_C$ are continuous liftings of $F|_C$ agreeing at c ; since C is connected, by Lemma 12.5 they agree on all of C . The sets A_{r-1} and R are closed in A_r (finite unions of closed subsets of $\mathbb{R}^2$ intersected with A_r), so by Lemma 12.3 the function G_r equal to G_{r-1} on A_{r-1} and to g on R is well defined and continuous, and lifts $F|_{A_r}$.

Step 3 — conclusion. $A_{m^2} = I \times I$, and $\tilde{F} = G_{m^2}$ is the required lifting. $\square$

Corollary 12.2: Path homotopies lift to path homotopies. Let $F : I \times I \rightarrow S^1$ be a path homotopy, i.e. with $F(0, u) = z_0$ and $F(1, u) = z_1$ constant in u . Let $\tilde{F}$ be the lifting of Lemma 12.7. Then $u \mapsto \tilde{F}(0, u)$ and $u \mapsto \tilde{F}(1, u)$ are constant; in particular the lifted paths $\tilde{\gamma}_u = \tilde{F}(\cdot, u)$ all have the same endpoints.

Proof. The map $u \mapsto \tilde{F}(0, u)$ is continuous and lifts the constant path $u \mapsto z_0$. The constant function $u \mapsto \tilde{F}(0, 0)$ is also a lifting of the same constant path, and the two agree at $u = 0$. Since I is connected, by Lemma 12.5 they agree everywhere: $\tilde{F}(0, u) = \tilde{F}(0, 0)$ for every u . The same argument applied to $u \mapsto \tilde{F}(1, u)$ gives the second assertion. $\square$

Corollary 12.3: Lifting of maps defined on $\mathbb{R}$. Let $f : \mathbb{R} \rightarrow S^1$ be continuous and let $a \in \mathbb{R}$ with $E(a) = f(0)$. Then there is one and only one continuous lifting $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}$ with $\tilde{f}(0) = a$.

Proof. For every $n \in \mathbb{N}$, the restriction of f to $[-n, n]$ is a path up to the affine reparametrisation $s \mapsto -n + 2ns$ of I onto $[-n, n]$; by Lemma 12.6 applied first to $f|_{[0, n]}$ (with initial datum a at 0) and then to $f|_{[-n, 0]}$ (traversed backwards, with the same datum at 0), we obtain a continuous lifting $h_n : [-n, n] \rightarrow \mathbb{R}$ with $h_n(0) = a$. For $n < n'$, the restrictions $h_{n'}|_{[-n, n]}$ and h_n are liftings of the same map on a connected set and agree at 0, hence agree everywhere (Lemma 12.5). One then defines $\tilde{f}(t) = h_n(t)$ for any $n \geq |t|$: the definition does not depend on n , and $\tilde{f}$ is continuous because it is continuous on every $(-n, n)$, and these are open and cover $\mathbb{R}$. Uniqueness follows from Lemma 12.5, $\mathbb{R}$ being connected. $\square$

12.5. The winding number

Definition 12.5: Winding number of a loop in S^1 . Let $\gamma : I \rightarrow S^1$ be a loop (i.e. $\gamma(0) = \gamma(1)$) and let $\tilde{\gamma}$ be a lifting of γ (Lemma 12.6). The **winding number** (or index, or degree) of γ is

$$w(\gamma) = \frac{\tilde{\gamma}(1) - \tilde{\gamma}(0)}{2\pi}.$$

Theorem 12.5: The winding number is an integer invariant under homotopy (R15.5). Let $\gamma : I \rightarrow S^1$ be a loop. Then:

- (i) $w(\gamma) \in \mathbb{Z}$ and does not depend on the chosen lifting;
- (ii) if $\gamma \simeq \gamma'$ as paths (homotopy with fixed endpoints), then $w(\gamma) = w(\gamma')$;
- (iii) $w(\gamma \cdot \gamma') = w(\gamma) + w(\gamma')$ for loops with the same base point, where $(\gamma \cdot \gamma')(s) = \gamma(2s)$ for $s \leq \frac{1}{2}$ and $\gamma'(2s - 1)$ for $s \geq \frac{1}{2}$;
- (iv) $w(\bar{\gamma}) = -w(\gamma)$, where $\bar{\gamma}(s) = \gamma(1 - s)$, and $w(\text{constant}) = 0$;
- (v) $w(\omega_n) = n$, where $\omega_n(s) = e^{2\pi i n s}$.

Proof. (i) From $E(\tilde{\gamma}(1)) = \gamma(1) = \gamma(0) = E(\tilde{\gamma}(0))$ we get $\tilde{\gamma}(1) - \tilde{\gamma}(0) \in 2\pi\mathbb{Z}$ (Theorem 12.4, Step 1), hence $w(\gamma) \in \mathbb{Z}$. If $\tilde{\gamma}_1, \tilde{\gamma}_2$ are two liftings, their difference is constant and equal to some $2\pi n$ (proof of Lemma 12.5: the difference is continuous with values in $2\pi\mathbb{Z}$ on a connected set), hence $\tilde{\gamma}_2(1) - \tilde{\gamma}_2(0) = (\tilde{\gamma}_1(1) + 2\pi n) - (\tilde{\gamma}_1(0) + 2\pi n) = \tilde{\gamma}_1(1) - \tilde{\gamma}_1(0)$: the value of w does not change.

(ii) Let $F : I \times I \rightarrow S^1$ be a path homotopy with $F(\cdot, 0) = \gamma$, $F(\cdot, 1) = \gamma'$, $F(0, u) = F(1, u) = z_0$ for every u (the endpoints agree because the loops have the same base point). Let $\tilde{F}$ be the lifting with $\tilde{F}(0, 0) = a$ (Lemma 12.7). For every u , the map $\tilde{\gamma}_u = \tilde{F}(\cdot, u)$ is a lifting of the loop $F(\cdot, u)$, so by (i)

$$w(F(\cdot, u)) = \frac{\tilde{F}(1, u) - \tilde{F}(0, u)}{2\pi}.$$

By Corollary 12.2, $\tilde{F}(0, u)$ and $\tilde{F}(1, u)$ are constant in u ; hence the right-hand side does not depend on u , and taking $u = 0, 1$ gives $w(\gamma) = w(\gamma')$.

(iii) Let $\tilde{\gamma}$ be the lifting of γ with $\tilde{\gamma}(0) = a$ and $\tilde{\gamma}'$ the lifting of γ' with $\tilde{\gamma}'(0) = \tilde{\gamma}(1)$ (legitimate because $E(\tilde{\gamma}(1)) = \gamma(1) = \gamma'(0)$). The function

$$\eta(s) = \begin{cases} \tilde{\gamma}(2s), & 0 \leq s \leq \frac{1}{2}, \\ \tilde{\gamma}'(2s - 1), & \frac{1}{2} \leq s \leq 1, \end{cases}$$

is well defined at $s = \frac{1}{2}$ (both expressions give $\tilde{\gamma}(1)$) and continuous by Lemma 12.3; moreover $E \circ \eta = \gamma \cdot \gamma'$. Hence

$$w(\gamma \cdot \gamma') = \frac{\eta(1) - \eta(0)}{2\pi} = \frac{\tilde{\gamma}'(1) - \tilde{\gamma}'(0)}{2\pi} = \frac{(\tilde{\gamma}'(1) - \tilde{\gamma}'(0)) + (\tilde{\gamma}(1) - \tilde{\gamma}(0))}{2\pi} = w(\gamma') + w(\gamma).$$

(iv) If $\tilde{\gamma}$ lifts γ , then $s \mapsto \tilde{\gamma}(1-s)$ lifts $\bar{\gamma}$, and the difference of the endpoints changes sign. For the constant loop $\gamma \equiv z_0$ a lifting is the constant a , and $w = 0$.

(v) The path $\tilde{\omega}_n(s) = 2\pi ns$ is continuous and $E(\tilde{\omega}_n(s)) = e^{2\pi i ns} = \omega_n(s)$: it is a lifting, and $w(\omega_n) = (2\pi n - 0)/2\pi = n$. $\square$

Definition 12.6: Winding number of a loop in $\mathbb{C} \setminus \{0\}$. Let $\gamma : I \rightarrow \mathbb{C} \setminus \{0\}$ be a loop. Its **winding number about the origin** is

$$w(\gamma) = w\left(\frac{\gamma}{|\gamma|}\right),$$

where $\gamma/|\gamma| : I \rightarrow S^1$ is a loop in S^1 (it is continuous because $|\gamma|$ is continuous and never zero). For a point $z_0 \notin \gamma(I)$ one sets $w(\gamma, z_0) = w(\gamma - z_0)$.

Theorem 12.6: Analytic form of the winding number. Let $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$ be a loop of class C^1 (i.e. γ differentiable with continuous derivative and $\gamma(0) = \gamma(1)$, $\gamma'(0) = \gamma'(1)$). Then

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z} = \frac{1}{2\pi i} \int_0^1 \frac{\gamma'(t)}{\gamma(t)} dt = w(\gamma) \in \mathbb{Z}.$$

Proof. Step 1 — a primitive of the logarithm along γ . The function $t \mapsto \gamma'(t)/\gamma(t)$ is continuous on $[0, 1]$ (a quotient of continuous functions with non-vanishing denominator). Set

$$h(t) = \int_0^t \frac{\gamma'(u)}{\gamma(u)} du,$$

which is of class C^1 with $h' = \gamma'/\gamma$ and $h(0) = 0$ (fundamental theorem of calculus).

Step 2 — $\gamma(t) = \gamma(0)e^{h(t)}$. Consider $\varphi(t) = \gamma(t)e^{-h(t)}$. It is differentiable and

$$\varphi'(t) = \gamma'(t)e^{-h(t)} - \gamma(t)h'(t)e^{-h(t)} = \gamma'(t)e^{-h(t)} - \gamma(t)\frac{\gamma'(t)}{\gamma(t)}e^{-h(t)} = 0.$$

Since $[0, 1]$ is connected and $\varphi' \equiv 0$ (real and imaginary parts have zero derivative), φ is constant: $\varphi(t) = \varphi(0) = \gamma(0)$. Hence $\gamma(t) = \gamma(0)e^{h(t)}$.

Step 3 — the integral lies in $2\pi i\mathbb{Z}$. From $\gamma(1) = \gamma(0) \neq 0$ we get $e^{h(1)} = 1$, hence $h(1) \in 2\pi i\mathbb{Z}$: writing $h(1) = i\theta$ with θ real (because $|e^{h(1)}| = e^{\operatorname{Re} h(1)} = 1$ forces $\operatorname{Re} h(1) = 0$), we have $e^{i\theta} = 1$ and therefore $\theta \in 2\pi\mathbb{Z}$. Hence $\frac{1}{2\pi i} \int_0^1 \frac{\gamma'}{\gamma} dt = \frac{h(1)}{2\pi i} \in \mathbb{Z}$.

Step 4 — it agrees with the topological winding number. Set $\rho(t) = |\gamma(t)| > 0$ and $\beta(t) = \gamma(t)/\rho(t) \in S^1$. From $\gamma = \gamma(0)e^h$ we get $\rho(t) = |\gamma(0)|e^{\operatorname{Re} h(t)}$ and

$$\beta(t) = \frac{\gamma(0)}{|\gamma(0)|} e^{i\operatorname{Im} h(t)}.$$

Choosing $a \in \mathbb{R}$ with $e^{ia} = \gamma(0)/|\gamma(0)|$ (possible by surjectivity of E), the function

$$\tilde{\beta}(t) = a + \operatorname{Im} h(t)$$

is continuous and satisfies $E(\tilde{\beta}(t)) = \beta(t)$: it is a lifting of β . Hence

$$w(\gamma) = w(\beta) = \frac{\tilde{\beta}(1) - \tilde{\beta}(0)}{2\pi} = \frac{\operatorname{Im} h(1)}{2\pi} = \frac{h(1)}{2\pi i}$$

(the last equality because $h(1)$ is purely imaginary, by Step 3). This concludes the proof. $\square$

Remark 12.4: The bridge with Cauchy. Theorem 12.6 is the point where the topology of the circle enters complex analysis. In Cauchy's integral formula and in the residue theorem (Volume II, Chapter 3, and Chapter 15) exactly the expression

$$\operatorname{Ind}_{\gamma}(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z - z_0}$$

appears, called the index (or winding number) of γ about z_0 : the proof just given shows that it is an integer and that it is the topological invariant $w(\gamma - z_0)$, not an arbitrary number. This is why Cauchy's formula $f(z_0) \operatorname{Ind}_{\gamma}(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz$ counts turns: the integer factor in front of $f(z_0)$ is the degree of the loop. The topological form (Definition 12.5) is however more general, because it requires no regularity: it holds for merely continuous loops, where the integral makes no sense.

Lemma 12.8: Dominant term lemma. Let $a, b \in \mathbb{C}$ with $|b| > |a|$ and $n, m \in \mathbb{Z}$. The loop

$$\gamma(t) = a e^{2\pi i n t} + b e^{2\pi i m t}, \quad t \in I,$$

never vanishes and has $w(\gamma) = m$.

Proof. For every t , $|\gamma(t)| \geq |b e^{2\pi i m t}| - |a e^{2\pi i n t}| = |b| - |a| > 0$: γ does not vanish. Consider

$$H(t, u) = u a e^{2\pi i n t} + b e^{2\pi i m t}, \quad (t, u) \in I \times I.$$

H is continuous and never vanishes, by the same estimate: $|H(t, u)| \geq |b| - u|a| \geq |b| - |a| > 0$. Moreover $H(0, u) = ua + b = H(1, u)$ for every u , so $H/|H| : I \times I \rightarrow S^1$ is a path homotopy of loops between $\gamma/|\gamma|$ (for $u = 1$) and $\beta(t) = b e^{2\pi i m t}/|b|$ (for $u = 0$). By Theorem 12.5(ii), $w(\gamma) = w(\beta)$. Finally $\beta(t) = e^{i\varphi_0} e^{2\pi i m t}$ with $e^{i\varphi_0} = b/|b|$, which lifts to $\varphi_0 + 2\pi m t$, so $w(\beta) = m$. $\square$

Example 12.2: Two loops and their winding numbers, checked twice. Consider

$$\gamma_1(t) = e^{6\pi i t} + \frac{1}{2}e^{-4\pi i t}, \quad \gamma_2(t) = e^{4\pi i t} + 2e^{-2\pi i t}, \quad t \in [0, 1].$$

Closed formula. For γ_1 the dominant term is $e^{6\pi i t} = e^{2\pi i \cdot 3t}$ (coefficient $1 > \frac{1}{2}$), hence $w(\gamma_1) = 3$ by Lemma 12.8. For γ_2 the dominant term is $2e^{-2\pi i t} = 2e^{2\pi i(-1)t}$ (coefficient $2 > 1$), hence $w(\gamma_2) = -1$.

Independent numerical check. Sampling $N + 1$ equispaced points and (a) unwrapping the phase, (b) approximating $\frac{1}{2\pi i} \int_0^1 \gamma'/\gamma dt$ by the trapezoidal rule with centred difference quotients for the derivative:

loop	N	unwrapped phase	integral
γ_1	200	3.000000	2.995561
γ_1	2 000	3.000000	2.999956
γ_1	20 000	3.000000	2.999999
γ_2	200	-1.000000	-0.999836
γ_2	2 000	-1.000000	-0.999998
γ_2	20 000	-1.000000	-1.000000

The two methods agree with the closed formula. ✓ Note the difference in nature: the unwrapped phase gives the *exact* integer already with $N = 200$, because it counts discrete jumps; the integral converges only with order $O(N^{-2})$, because it is a quadrature. This is the numerical translation of the fact that w is a topological invariant and not a metric quantity.

12.6. The fundamental group of S^1

Definition 12.7: Path homotopy and fundamental group. Let $\gamma, \gamma' : I \rightarrow X$ be paths with the same endpoints ($\gamma(0) = \gamma'(0) = x_0$, $\gamma(1) = \gamma'(1) = x_1$). A **path homotopy** between them is a continuous map $F : I \times I \rightarrow X$ with

$$F(s, 0) = \gamma(s), \quad F(s, 1) = \gamma'(s), \quad F(0, u) = x_0, \quad F(1, u) = x_1$$

for all $s, u \in I$. One writes $\gamma \simeq \gamma'$. The set of homotopy classes of loops based at x_0 , with the operation induced by concatenation $[\gamma][\gamma'] = [\gamma \cdot \gamma']$, is called the **fundamental group** $\pi_1(X, x_0)$.

Proposition 12.1: $\pi_1(X, x_0)$ is a group. The relation $\simeq$ is an equivalence relation; concatenation descends to the quotient; and $\pi_1(X, x_0)$ with that operation is a group, with identity the class of the constant loop c_{x_0} and inverse $[\gamma]^{-1} = [\bar{\gamma}]$, $\bar{\gamma}(s) = \gamma(1 - s)$.

Proof. Step 1 — $\simeq$ is an equivalence relation. *Reflexive:* $F(s, u) = \gamma(s)$. *Symmetric:* if F is a homotopy from γ to γ' , then $F(s, 1 - u)$ goes from γ' to γ . *Transitive:* given F from γ to γ' and G from γ' to γ'' , set $H(s, u) = F(s, 2u)$ for $u \leq \frac{1}{2}$ and $G(s, 2u - 1)$ for $u \geq \frac{1}{2}$; the two definitions agree at $u = \frac{1}{2}$ (both give $\gamma'(s)$) and $I \times [0, \frac{1}{2}]$, $I \times [\frac{1}{2}, 1]$ are closed, so H is continuous by Lemma 12.3.

Step 2 — concatenation descends to the quotient. If F is a homotopy from γ to γ_1 and G from δ to δ_1 (with $\gamma(1) = \delta(0)$), set $H(s, u) = F(2s, u)$ for $s \leq \frac{1}{2}$ and $G(2s - 1, u)$ for $s \geq \frac{1}{2}$: at $s = \frac{1}{2}$ both equal $\gamma(1) = \delta(0)$, which does not depend on u because the homotopies have fixed endpoints; by Lemma 12.3 H is continuous and is a path homotopy between $\gamma \cdot \delta$ and $\gamma_1 \cdot \delta_1$.

Step 3 — reparametrisations. Let $\varphi : I \rightarrow I$ be continuous with $\varphi(0) = 0$, $\varphi(1) = 1$. Then $\gamma \circ \varphi \simeq \gamma$: the homotopy is

$$H(s, u) = \gamma((1 - u)\varphi(s) + us),$$

continuous (composition of continuous maps, with argument in I because I is convex), with $H(\cdot, 0) = \gamma \circ \varphi$, $H(\cdot, 1) = \gamma$, $H(0, u) = \gamma(0)$ and $H(1, u) = \gamma(1)$.

Step 4 — associativity. One checks that $(\gamma \cdot \delta) \cdot \varepsilon = (\gamma \cdot (\delta \cdot \varepsilon)) \circ \varphi$ with

$$\varphi(s) = \begin{cases} \frac{s}{2}, & 0 \leq s \leq \frac{1}{2}, \\ s - \frac{1}{4}, & \frac{1}{2} \leq s \leq \frac{3}{4}, \\ 2s - 1, & \frac{3}{4} \leq s \leq 1, \end{cases}$$

which is continuous (the three expressions agree at the junction points: at $s = \frac{1}{2}$ they give $\frac{1}{4}$; at $s = \frac{3}{4}$ they give $\frac{1}{2}$), with $\varphi(0) = 0$, $\varphi(1) = 1$. By Step 3 the two concatenations are homotopic.

Step 5 — identity. $\gamma \cdot c_{x_1} = \gamma \circ \varphi$ with $\varphi(s) = 2s$ for $s \leq \frac{1}{2}$ and $\varphi(s) = 1$ for $s \geq \frac{1}{2}$: continuous, with $\varphi(0) = 0$ and $\varphi(1) = 1$. By Step 3, $\gamma \cdot c_{x_1} \simeq \gamma$; similarly $c_{x_0} \cdot \gamma \simeq \gamma$.

Step 6 — inverse. Set

$$H(s, u) = \begin{cases} \gamma(2s(1-u)), & 0 \leq s \leq \frac{1}{2}, \\ \gamma(2(1-s)(1-u)), & \frac{1}{2} \leq s \leq 1. \end{cases}$$

At $s = \frac{1}{2}$ both give $\gamma(1-u)$: H is well defined and continuous (Lemma 12.3). We have $H(s, 0) = (\gamma \cdot \bar{\gamma})(s)$, $H(s, 1) = \gamma(0) = x_0$ for every s , $H(0, u) = \gamma(0) = x_0$ and $H(1, u) = \gamma(0) = x_0$. Hence $\gamma \cdot \bar{\gamma} \simeq c_{x_0}$; exchanging the roles, $\bar{\gamma} \cdot \gamma \simeq c_{x_0}$. $\square$

Theorem 12.7: $\pi_1(S^1) \cong \mathbb{Z}$ (R15.6). The maps

$$\Phi : \pi_1(S^1, 1) \rightarrow \mathbb{Z}, \quad \Phi([\gamma]) = w(\gamma), \quad \Psi : \mathbb{Z} \rightarrow \pi_1(S^1, 1), \quad \Psi(n) = [\omega_n], \quad \omega_n(s) = e^{2\pi i n s},$$

are group homomorphisms, inverse to one another. In particular $\pi_1(S^1, 1) \cong \mathbb{Z}$ and it is infinite cyclic, generated by $[\omega_1]$.

Proof. **Step 1 — Φ is well defined and a homomorphism.** The number $w(\gamma)$ does not depend on the lifting (Theorem 12.5(i)) nor on the representative of the homotopy class (item (ii)); it is therefore a function on $\pi_1(S^1, 1)$. It is a homomorphism by item (iii): $\Phi([\gamma][\gamma']) = w(\gamma \cdot \gamma') = w(\gamma) + w(\gamma') = \Phi([\gamma]) + \Phi([\gamma'])$.

Step 2 — Ψ is a homomorphism. We show $\omega_n \cdot \omega_m \simeq \omega_{n+m}$. A lifting of $\omega_n \cdot \omega_m$ with initial value 0 is, by item (iii) of the proof of Theorem 12.5, the path η equal to $2\pi n \cdot 2s$ on $[0, \frac{1}{2}]$ and to $2\pi n + 2\pi m(2s - 1)$ on $[\frac{1}{2}, 1]$; the lifting of ω_{n+m} with initial value 0 is $s \mapsto 2\pi(n+m)s$. Both go from 0 to $2\pi(n+m)$, so the linear homotopy $L(s, u) = (1-u)\eta(s) + 2\pi(n+m)su$ is continuous, has fixed endpoints ($L(0, u) = 0$, $L(1, u) = 2\pi(n+m)$) and $E \circ L$ is a path homotopy between $\omega_n \cdot \omega_m$ and ω_{n+m} . Hence $\Psi(n)\Psi(m) = [\omega_n \cdot \omega_m] = [\omega_{n+m}] = \Psi(n+m)$.

Step 3 — $\Phi \circ \Psi = \text{id}_{\mathbb{Z}}$. This is item (v) of Theorem 12.5: $\Phi(\Psi(n)) = w(\omega_n) = n$. In particular Φ is surjective and Ψ is injective.

Step 4 — $\Psi \circ \Phi = \text{id}$. Let $[\gamma] \in \pi_1(S^1, 1)$ and $n = w(\gamma)$. Let $\tilde{\gamma}$ be the lifting with $\tilde{\gamma}(0) = 0$; then $\tilde{\gamma}(1) = 2\pi n$. The path $\tilde{\omega}_n(s) = 2\pi ns$ has the same endpoints. The linear homotopy

$$L(s, u) = (1-u)\tilde{\gamma}(s) + u2\pi ns$$

is continuous, with $L(0, u) = 0$ and $L(1, u) = 2\pi n$ for every u : it is a path homotopy in $\mathbb{R}$. Hence $E \circ L$ is a path homotopy in S^1 between $\gamma = E \circ \tilde{\gamma}$ and $\omega_n = E \circ \tilde{\omega}_n$ (the endpoints are $E(0) = 1$ and $E(2\pi n) = 1$, constant). Therefore $[\gamma] = [\omega_n] = \Psi(n) = \Psi(\Phi([\gamma]))$.

Step 5 — conclusion. Φ and Ψ are mutually inverse homomorphisms, hence isomorphisms. Since $\Psi(n) = [\omega_1]^n$ (Step 2 with induction on n and $\Psi(-1) = [\omega_{-1}] = [\bar{\omega}_1] = [\omega_1]^{-1}$), the group is cyclic generated by $[\omega_1]$, and infinite because Φ is injective. $\square$

Corollary 12.4: *There is no retraction $D^2 \rightarrow S^1$. Let $D^2 = \{z \in \mathbb{C} : |z| \leq 1\}$. There is no continuous map $r : D^2 \rightarrow S^1$ with $r(z) = z$ for every $z \in S^1$.*

Proof. Suppose r exists and consider the loop $\omega_1(s) = e^{2\pi is}$ in S^1 . In D^2 the loop ω_1 is path homotopic to the constant 1: the linear homotopy

$$K(s, u) = (1 - u)\omega_1(s) + u \cdot 1$$

takes values in D^2 because D^2 is convex, and has fixed endpoints because $K(0, u) = K(1, u) = (1 - u) + u = 1$. Composing with r , the map $r \circ K$ is a path homotopy in S^1 between $r \circ \omega_1 = \omega_1$ (because r is the identity on S^1) and the constant loop $r(1) = 1$. Hence $[\omega_1] = [c_1]$ in $\pi_1(S^1, 1)$, that is, $1 = w(\omega_1) = w(c_1) = 0$ by Theorem 12.5: absurd. $\square$

Corollary 12.5: *Brouwer fixed point theorem in dimension 2. Every continuous map $h : D^2 \rightarrow D^2$ has a fixed point.*

Proof. Suppose $h(x) \neq x$ for every $x \in D^2$. Define $r(x)$ as the point where the ray issuing from $h(x)$ and passing through x meets S^1 . Explicitly, setting $v(x) = \frac{x - h(x)}{|x - h(x)|}$ (well defined and continuous because $x \neq h(x)$), one seeks $\lambda \geq 0$ with $|x + \lambda v(x)| = 1$, that is,

$$\lambda^2 + 2\lambda\langle x, v(x) \rangle + |x|^2 - 1 = 0,$$

whose unique non-negative root is

$$\lambda(x) = -\langle x, v(x) \rangle + \sqrt{\langle x, v(x) \rangle^2 + 1 - |x|^2}$$

(the radicand is ≥ 0 because $|x| \leq 1$; the root is ≥ 0 because the product of the two roots is $|x|^2 - 1 \leq 0$). The function λ is continuous, so $r(x) = x + \lambda(x)v(x)$ is continuous with values in S^1 . If $x \in S^1$, then $|x| = 1$ and $\lambda(x) = -\langle x, v \rangle + |\langle x, v \rangle|$: we have $\langle x, v(x) \rangle \geq 0$, because $\langle x, x - h(x) \rangle = 1 - \langle x, h(x) \rangle \geq 1 - |x||h(x)| \geq 0$, hence $\lambda(x) = 0$ and $r(x) = x$. Thus r is a retraction of D^2 onto S^1 , which does not exist by Corollary 12.4: absurd. Hence h has a fixed point. $\square$

Remark 12.5: *What we import: $\pi_1(S^n)$ for $n \geq 2$. For $n \geq 2$ one has $\pi_1(S^n) = 0$: every loop on a sphere of dimension at least two contracts to a point. This statement is here **imported** and not proved. The proof (Hatcher, Algebraic Topology, Proposition 1.14, p. 35) needs two ingredients that have not been built in this chapter: the lemma on decomposing loops with respect to an open cover (Hatcher, Lemma 1.15, p. 35), which is the combinatorial core of the Seifert–van Kampen theorem; and the fact that a loop in S^n can be homotoped to a non-surjective one, so as to contract it inside $S^n \setminus \{p\} \cong \mathbb{R}^n$ via stereographic projection. What is missing, concretely, is exactly Lemma 1.15 and the approximation argument that avoids a point. The difference from S^1 is not technical but substantial: $\pi_1(S^1) = \mathbb{Z} \neq 0$, and it is this inequality that makes the circle the only sphere with a non-trivial fundamental group. Volume IV, which devotes a chapter to algebraic topology, proves van Kampen and deduces the statement.*

12.7. The degree of maps $S^1 \rightarrow S^1$

Definition 12.8: Degree of a map $S^1 \rightarrow S^1$. Let $f : S^1 \rightarrow S^1$ be continuous. The associated loop is $\gamma_f : I \rightarrow S^1$, $\gamma_f(s) = f(e^{2\pi is})$ (it is a loop because $\gamma_f(0) = f(1) = \gamma_f(1)$). The **degree** of f is

$$\deg f = w(\gamma_f) \in \mathbb{Z}.$$

Proposition 12.2: The lifted map and its functional equation. Let $f : S^1 \rightarrow S^1$ be continuous. Then there is a continuous $u : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(e^{it}) = e^{iu(t)} \quad \text{for every } t \in \mathbb{R}, \quad \text{and} \quad u(t + 2\pi) = u(t) + 2\pi \deg f \quad \text{for every } t.$$

The function u is unique up to an additive constant in $2\pi\mathbb{Z}$.

Proof. **Step 1 — existence of u .** The map $g : \mathbb{R} \rightarrow S^1$, $g(t) = f(e^{it})$, is continuous (composition of continuous maps). By Corollary 12.3 there is a continuous $u : \mathbb{R} \rightarrow \mathbb{R}$ with $E \circ u = g$, i.e. $f(e^{it}) = e^{iu(t)}$, and u is unique once $u(0)$ is fixed; two choices of $u(0)$ in the fibre differ by a multiple of 2π , whence uniqueness up to $2\pi\mathbb{Z}$ (Lemma 12.5).

Step 2 — the functional equation. Set $d(t) = u(t + 2\pi) - u(t)$, continuous. From $e^{iu(t+2\pi)} = f(e^{i(t+2\pi)}) = f(e^{it}) = e^{iu(t)}$ we get $e^{id(t)} = 1$, i.e. $d(t) \in 2\pi\mathbb{Z}$ for every t . Since $\mathbb{R}$ is connected and $2\pi\mathbb{Z}$ discrete, d is constant (same argument as in Lemma 12.5): $d \equiv 2\pi k$ for some $k \in \mathbb{Z}$.

Step 3 — $k = \deg f$. The path $s \mapsto u(2\pi s)$, $s \in I$, is continuous and lifts $s \mapsto f(e^{2\pi is}) = \gamma_f(s)$. Hence

$$\deg f = w(\gamma_f) = \frac{u(2\pi) - u(0)}{2\pi} = \frac{d(0)}{2\pi} = k.$$

□

Theorem 12.8: Classification of maps $S^1 \rightarrow S^1$ by degree (R15.7). Let $f, g : S^1 \rightarrow S^1$ be continuous. Then

$$f \simeq g \text{ (homotopic as maps, with no constraint on the base point)} \iff \deg f = \deg g.$$

Moreover, for every $n \in \mathbb{Z}$ the map $p_n(z) = z^n$ has $\deg p_n = n$. Hence

$$\deg : [S^1, S^1] \longrightarrow \mathbb{Z}$$

is a bijection between the set of free homotopy classes of maps $S^1 \rightarrow S^1$ and $\mathbb{Z}$.

Proof. **Step 1 — $\deg p_n = n$.** $\gamma_{p_n}(s) = (e^{2\pi is})^n = e^{2\pi ins} = \omega_n(s)$, hence $\deg p_n = w(\omega_n) = n$ by Theorem 12.5(v). In particular $\deg$ is surjective.

Step 2 — homotopic $\Rightarrow$ same degree. Let $H : S^1 \times I \rightarrow S^1$ be continuous with $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$. Set $F(s, u) = H(e^{2\pi is}, u)$, continuous on $I \times I$, with $F(\cdot, 0) = \gamma_f$, $F(\cdot, 1) = \gamma_g$ and $F(0, u) = F(1, u)$ for every u (because $e^0 = e^{2\pi i} = 1$). Let $\tilde{F}$ be its lifting (Lemma 12.7). The function

$$u \mapsto \frac{\tilde{F}(1, u) - \tilde{F}(0, u)}{2\pi}$$

is continuous; it takes values in $\mathbb{Z}$ because $E(\tilde{F}(1, u)) = F(1, u) = F(0, u) = E(\tilde{F}(0, u))$; hence, I being connected, it is constant (same argument as in Lemma 12.5). For $u = 0$ it equals $w(\gamma_f) = \deg f$ and for $u = 1$ it equals $\deg g$: they are equal. *Note* that we did not require the homotopy to fix the base point: the degree is invariant under free homotopy too.

Step 3 — same degree $\Rightarrow$ homotopic. Let $\deg f = \deg g = n$ and let $u, v : \mathbb{R} \rightarrow \mathbb{R}$ be the lifted maps of Proposition 12.2, so that $f(e^{it}) = e^{iu(t)}$, $g(e^{it}) = e^{iv(t)}$ and

$$u(t + 2\pi) = u(t) + 2\pi n, \quad v(t + 2\pi) = v(t) + 2\pi n.$$

Set $L(t, \theta) = (1 - \theta)u(t) + \theta v(t)$, continuous on $\mathbb{R} \times I$. Then

$$L(t + 2\pi, \theta) = (1 - \theta)(u(t) + 2\pi n) + \theta(v(t) + 2\pi n) = L(t, \theta) + 2\pi n,$$

hence $e^{iL(t+2\pi, \theta)} = e^{iL(t, \theta)}$: the function $(t, \theta) \mapsto e^{iL(t, \theta)}$ is 2π -periodic in t . Define

$$H : S^1 \times I \rightarrow S^1, \quad H(e^{it}, \theta) = e^{iL(t, \theta)}.$$

It is well defined by the periodicity just proved. It is continuous: the map $q : \mathbb{R} \times I \rightarrow S^1 \times I$, $q(t, \theta) = (e^{it}, \theta)$, is continuous and surjective; its restriction to $[0, 2\pi] \times I$ is continuous and surjective from a compact space to a Hausdorff space, hence a quotient map by Lemma 12.1(iii); since $H \circ q = e^{iL}$ is continuous on $[0, 2\pi] \times I$, H is continuous. Finally $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$. Hence $f \simeq g$.

Step 4 — conclusion. Step 2 says that $\deg$ is well defined on $[S^1, S^1]$ and Step 3 that it is injective; Step 1 that it is surjective. $\square$

Corollary 12.6: Extension to the disc and vanishing degree. For a continuous $f : S^1 \rightarrow S^1$ the following are equivalent: (a) $\deg f = 0$; (b) f is homotopic to a constant map; (c) f extends to a continuous map $F : D^2 \rightarrow S^1$.

Proof. (a) $\Rightarrow$ (b). The constant map $c(z) = 1$ has $\gamma_c \equiv 1$, so $\deg c = 0$; by Theorem 12.8, $f \simeq c$.

(b) $\Rightarrow$ (c). Let $H : S^1 \times I \rightarrow S^1$ with $H(\cdot, 0) = f$ and $H(z, 1) = z_1$ constant. Set

$$F(w) = \begin{cases} H\left(\frac{w}{|w|}, 1 - |w|\right), & w \neq 0, \\ z_1, & w = 0. \end{cases}$$

F is continuous on $D^2 \setminus \{0\}$ (composition of continuous maps); it is continuous at 0 because, as $w \rightarrow 0$, the second argument tends to 1 and $H(\cdot, 1) \equiv z_1$: precisely, given $\varepsilon > 0$, by uniform continuity of H on the compact set $S^1 \times I$ there is $\eta > 0$ such that $|H(z, \theta) - H(z, 1)| < \varepsilon$ if $|1 - \theta| < \eta$, hence $|F(w) - z_1| < \varepsilon$ if $|w| < \eta$. Finally for $|w| = 1$ we have $F(w) = H(w, 0) = f(w)$.

(c) $\Rightarrow$ (a). Let $F : D^2 \rightarrow S^1$ be continuous with $F|_{S^1} = f$. Then $H(z, \theta) = F((1 - \theta)z)$ is continuous on $S^1 \times I$, with $H(\cdot, 0) = f$ and $H(\cdot, 1) \equiv F(0)$ constant. Hence f is homotopic to a constant and, by Theorem 12.8, $\deg f = \deg(\text{constant}) = 0$. $\square$

Proposition 12.3: The degree is multiplicative. For continuous $f, g : S^1 \rightarrow S^1$ one has $\deg(f \circ g) = \deg f \cdot \deg g$.

Proof. Let $u, v : \mathbb{R} \rightarrow \mathbb{R}$ be the lifted maps of f and g (Proposition 12.2), with $u(t + 2\pi) = u(t) + 2\pi \deg f$ and $v(t + 2\pi) = v(t) + 2\pi \deg g$. Then

$$(f \circ g)(e^{it}) = f(e^{iv(t)}) = e^{iu(v(t))},$$

so $u \circ v$ is a lifted map of $f \circ g$ (it is continuous). Setting $m = \deg g$ and using the functional equation of v (for $m \geq 0$; for $m < 0$ use the same relation backwards),

$$v(t + 2\pi) = v(t) + 2\pi m \implies u(v(t + 2\pi)) = u(v(t) + 2\pi m) = u(v(t)) + 2\pi m \deg f,$$

where the last equality follows by applying $|m|$ times $u(s + 2\pi) = u(s) + 2\pi \deg f$. By the uniqueness of the integer in the functional equation (Proposition 12.2, Steps 2–3), $\deg(f \circ g) = m \deg f = \deg g \cdot \deg f$. $\square$

12.8. Panorama: hairy ball and Hopf fibration

The circle has two properties that higher-dimensional spheres lose or modify: it can be “combed” (S^1 has a nowhere-vanishing tangent field) and its maps are classified by a single integer. The next two subsections say what happens in higher dimension, and distinguish precisely what is proved here from what is merely reported.

Proposition 12.4: *S^1 can be combed.* The function $v : S^1 \rightarrow \mathbb{R}^2$, $v(x, y) = (-y, x)$, is continuous, tangent to S^1 at every point and never zero. Hence on S^1 there is a continuous tangent vector field that never vanishes.

Proof. v is linear, hence continuous. It is tangent: the tangent line to S^1 at $P = (x, y)$ is the line orthogonal to the radius OP , and $\langle (x, y), (-y, x) \rangle = -xy + yx = 0$. It never vanishes: $|v(x, y)|^2 = y^2 + x^2 = 1$, so $|v| = 1 \neq 0$ on all of S^1 . (In complex notation: $v(z) = iz$, the rotation of the radius by 90° .) $\square$

Remark 12.6: *What we import: the hairy ball theorem (panorama).* **Panorama.** On S^2 there is no continuous nowhere-vanishing tangent vector field; more generally S^n admits such a field if and only if n is odd (Hatcher, *Algebraic Topology*, Theorem 2.28, p. 135). The case $n = 1$ of the “if” part is exactly Proposition 12.4; the case $n = 2$ is the celebrated hairy ball theorem, whose meteorological reading is that at every instant there is at least one point of the globe where the horizontal wind vanishes.

The statement is here **imported**: it is not proved and it is not presented as a theorem of this treatise. The argument is short but uses a tool that this chapter does not build. Given a unit tangent field v on S^n , the homotopy $f_\theta(x) = (\cos \theta)x + (\sin \theta)v(x)$, $\theta \in [0, \pi]$, joins the identity to the antipodal map $x \mapsto -x$; one deduces $\deg(\text{id}) = \deg(-\text{id})$, that is, $1 = (-1)^{n+1}$, hence n odd. **What is missing, exactly:** the degree of a map $S^n \rightarrow S^n$ for $n \geq 2$ and the computation $\deg(-\text{id}) = (-1)^{n+1}$. The degree built in this chapter (Definition 12.8) exists only for $n = 1$ and arises from the lifting along $E : \mathbb{R} \rightarrow S^1$; for $n \geq 2$ there is no analogous covering (S^n is simply connected, Remark 12.5) and the degree is defined via homology $H_n(S^n) \cong \mathbb{Z}$ (Hatcher, §2.2). That construction is the subject of the algebraic topology chapter of Volume IV, where the hairy ball theorem is proved in full.

Definition 12.9: The Hopf map. Identify $S^3 = \{(z_0, z_1) \in \mathbb{C}^2 : |z_0|^2 + |z_1|^2 = 1\}$ and consider the complex projective line $\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{0\})/\sim$, where $(z_0, z_1) \sim (\lambda z_0, \lambda z_1)$ for $\lambda \in \mathbb{C} \setminus \{0\}$. The **Hopf map** is

$$h : S^3 \rightarrow \mathbb{CP}^1, \quad h(z_0, z_1) = [z_0 : z_1].$$

Since $\mathbb{CP}^1$ is homeomorphic to S^2 , one also writes $h : S^3 \rightarrow S^2$.

Proposition 12.5: The Hopf fibres are circles. For every $q \in \mathbb{CP}^1$ the fibre $h^{-1}(q) \subseteq S^3$ is homeomorphic to S^1 . Precisely, if $(a_0, a_1) \in S^3$ and $q = [a_0 : a_1]$, then

$$h^{-1}(q) = \{(e^{i\theta}a_0, e^{i\theta}a_1) : \theta \in \mathbb{R}\},$$

and $\theta \mapsto (e^{i\theta}a_0, e^{i\theta}a_1)$ induces a homeomorphism of S^1 onto $h^{-1}(q)$.

Proof. Step 1 — the orbit is contained in the fibre. If $|\lambda| = 1$ then $|\lambda a_0|^2 + |\lambda a_1|^2 = |a_0|^2 + |a_1|^2 = 1$, so $(\lambda a_0, \lambda a_1) \in S^3$; and $[\lambda a_0 : \lambda a_1] = [a_0 : a_1] = q$ by definition of $\mathbb{CP}^1$.

Step 2 — the fibre is contained in the orbit. Let $(z_0, z_1) \in S^3$ with $[z_0 : z_1] = [a_0 : a_1]$. Then there is $\lambda \in \mathbb{C} \setminus \{0\}$ with $(z_0, z_1) = (\lambda a_0, \lambda a_1)$. Taking norms,

$$1 = |z_0|^2 + |z_1|^2 = |\lambda|^2(|a_0|^2 + |a_1|^2) = |\lambda|^2,$$

so $|\lambda| = 1$, that is, $\lambda = e^{i\theta}$ for some θ (surjectivity of E).

Step 3 — homeomorphism. The map $\Theta : S^1 \rightarrow h^{-1}(q)$, $\Theta(\lambda) = (\lambda a_0, \lambda a_1)$, is continuous (linear in λ), surjective by Steps 1–2, and injective: if $\lambda a_0 = \mu a_0$ and $\lambda a_1 = \mu a_1$ with $(a_0, a_1) \neq (0, 0)$, then $(\lambda - \mu)a_j = 0$ for some j with $a_j \neq 0$, so $\lambda = \mu$. It is therefore a continuous bijection from a compact space (S^1) to a Hausdorff space (a subspace of $\mathbb{C}^2 \cong \mathbb{R}^4$), hence a homeomorphism by Lemma 12.1(ii). $\square$

Remark 12.7: What we import: the non-triviality of the Hopf fibration (panorama). **Panorama.** The map h of Definition 12.9 is a locally trivial fibre bundle with fibre S^1 , total space S^3 and base S^2 , and it is not the trivial bundle: S^3 is not homeomorphic to $S^2 \times S^1$. Moreover $\pi_3(S^2)$ is infinite cyclic, generated by the class of h (Hatcher, *Algebraic Topology*, Example 4.45, pp. 377–378, for the bundle; Example 4.51, p. 380, for $\pi_3(S^2) \cong \mathbb{Z}$).

These two statements are here **imported** and presented as panorama, not as theorems of this treatise. Of the above, only the fact that the fibres are circles is proved in this chapter (Proposition 12.5). **What is missing, exactly:** (a) the construction of the local trivializations of h over the two affine charts of $\mathbb{CP}^1$ and the verification that they are compatible; (b) the long exact sequence of homotopy groups of a fibre bundle, which from the fibration $S^1 \rightarrow S^3 \rightarrow S^2$ gives $\pi_3(S^3) \cong \pi_3(S^2)$ and hence $\pi_3(S^2) \cong \mathbb{Z}$; (c) the computation $\pi_3(S^3) \cong \mathbb{Z}$, which requires the Hurewicz theorem or the degree in dimension 3. None of the three tools is available here: the higher homotopy groups π_n are not even defined in this volume. They are the subject of the algebraic topology chapter of Volume IV.

It is worth recording, however, the geometric intuition, which is elementary and verifiable with the tools of this chapter: two distinct fibres are linked circles in S^3 (linking number 1), while in a product $S^2 \times S^1$ the fibres $\{x\} \times S^1$ are not. And it is worth recording the connection with $SU(2) \rightarrow SO(3)$ (Chapter 28): viewing S^3 as the set of unit quaternions, the Hopf fibres are the orbits of multiplication by a phase, and it is the same structure for which a rotation by 2π does not return a spin state to its initial value.

12.9. Configuration space and cable twisting

The configuration space (*C-space*) of a robot encodes all admissible poses; its topology governs motion planning. A revolute joint with unlimited travel has C-space S^1 : its trajectories are classified by the number of windings ($\pi_1(S^1) = \mathbb{Z}$, Theorem 12.7), a quantity directly related to cable twisting. It is exactly the quantity that controllers monitor to avoid twisting the harness: managing the turns accumulated on the axis is a standard function of industrial controllers. For tethered robots, the homotopy class of the path in the workspace with obstacles uniquely determines the state of the taut cable, and the winding number around an obstacle measures the degree of entanglement: *tangle-free* planners choose the homotopy class that avoids entanglement. Topology here is not abstraction but an operational constraint.

To deal with several joints one needs the fundamental group of a product.

Proposition 12.6: The fundamental group of a product. Let X, Y be topological spaces, $x_0 \in X$, $y_0 \in Y$. Then

$$\Lambda : \pi_1(X \times Y, (x_0, y_0)) \rightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0), \quad \Lambda([\gamma]) = ([p_X \circ \gamma], [p_Y \circ \gamma]),$$

is a group isomorphism, where p_X, p_Y are the projections.

Proof. Step 1 — well defined and a homomorphism. If F is a path homotopy in $X \times Y$, then $p_X \circ F$ and $p_Y \circ F$ are path homotopies in X and in Y ; hence Λ does not depend on the representative. Since $p_X \circ (\gamma \cdot \delta) = (p_X \circ \gamma) \cdot (p_X \circ \delta)$ (concatenation acts on the parameter), Λ is a homomorphism.

Step 2 — surjective. Given loops α in X and β in Y , the map $\gamma(s) = (\alpha(s), \beta(s))$ is continuous (a map into a product is continuous if and only if its components are) and is a loop based at (x_0, y_0) , with $\Lambda([\gamma]) = ([\alpha], [\beta])$.

Step 3 — injective. Let $\Lambda([\gamma])$ be the identity element, i.e. $p_X \circ \gamma \simeq c_{x_0}$ via F and $p_Y \circ \gamma \simeq c_{y_0}$ via G . Then $H(s, u) = (F(s, u), G(s, u))$ is continuous (continuous components), and it is a path homotopy between γ and the constant loop (x_0, y_0) : indeed $H(s, 0) = (p_X \gamma(s), p_Y \gamma(s)) = \gamma(s)$, $H(s, 1) = (x_0, y_0)$ and $H(0, u) = H(1, u) = (x_0, y_0)$. Hence $[\gamma]$ is the identity. $\square$

Corollary 12.7: The C-space of two revolute joints. The C-space of two independent revolute joints with unlimited travel is the torus $T^2 = S^1 \times S^1$, and

$$\pi_1(T^2) \cong \mathbb{Z} \times \mathbb{Z}.$$

Similarly, the cylinder $S^1 \times \mathbb{R}$ (one revolute joint plus one linear axis) has $\pi_1(S^1 \times \mathbb{R}) \cong \mathbb{Z}$.

Proof. Each joint has C-space S^1 and the two joints are independent, so the C-space is the product. By Proposition 12.6 and Theorem 12.7, $\pi_1(T^2) \cong \pi_1(S^1) \times \pi_1(S^1) \cong \mathbb{Z} \times \mathbb{Z}$. For the cylinder, $\pi_1(\mathbb{R})$ is trivial (Theorem 12.4, Step 5), so $\pi_1(S^1 \times \mathbb{R}) \cong \mathbb{Z} \times \{0\} \cong \mathbb{Z}$. $\square$

Remark 12.8: Why S^1 is special. The facts that $\pi_1(S^1) = \mathbb{Z} \neq 0 = \pi_1(S^n)$ for $n \geq 2$ (Remark 12.5) and that S^1 admits a nowhere-vanishing tangent field while S^2 does not (Remark 12.6) are the topological roots of many peculiarities of the circle: the existence of a global angular coordinate up to 2π , the classification of loops by winding, the group structure of $S^1 = \mathbb{T}$ (Chapter 13), absent on S^2 . A double revolute joint has a torus, not a sphere, as its C-space: topology, not geometry, dictates the cabling constraints.

12.10. Assigned exercises

Answers are in the Teacher's key (Section 12.13).

Exercise 12.1: Conjugation has degree -1 . *Aim:* to use the definition of degree and its invariance under free homotopy. *Estimated time:* 25 minutes. *Type:* symbolic/proof-based. Let $c : S^1 \rightarrow S^1$ be conjugation, $c(z) = \bar{z}$.

- Compute $\deg c$ directly from Definition 12.8, exhibiting an explicit lifting of γ_c .
- Deduce that c is not homotopic to the identity id_{S^1} , and that it is not homotopic to a constant.
- Deduce that c does not extend to a continuous map $D^2 \rightarrow S^1$.
- Compute $\deg(c \circ p_n)$ and $\deg(p_n \circ c)$ for $p_n(z) = z^n$, and check consistency with Proposition 12.3.

Exercise 12.2: Winding number computed numerically. *Aim:* to compute a topological invariant by two independent methods and to recognise the difference in nature between them. *Estimated time:* 45 minutes. *Type:* numerical/computational.

Consider the loop

$$\gamma(t) = e^{6\pi it} + \frac{1}{2}e^{-4\pi it}, \quad t \in [0, 1].$$

- Check that $\gamma(t) \neq 0$ for every t and determine $\min_t |\gamma(t)|$.
- Predict $w(\gamma)$ using Lemma 12.8.
- Sample γ at $N + 1$ equispaced points for $N = 200, 2000, 20\,000$; compute the unwrapped phase (unwrapping: add $\varphi_k - \varphi_{k-1}$ reduced to $(-\pi, \pi]$) and deduce w .
- Compute $\frac{1}{2\pi i} \int_0^1 \gamma'/\gamma dt$ by the trapezoidal rule, using the exact derivative $\gamma'(t) = 6\pi i e^{6\pi it} - 2\pi i e^{-4\pi it}$.
- Compare the order of convergence of the two methods in (c) and (d) and explain the difference.
- Repeat (b)–(d) for $\delta(t) = \frac{1}{2}e^{6\pi it} + e^{-4\pi it}$ and comment.

12.11. Worked exercises

Exercise 12.3: Fixed-point-free maps. *Type:* symbolic/proof-based. Let $f : S^1 \rightarrow S^1$ be continuous and without fixed points ($f(z) \neq z$ for every z). Prove that $\deg f = 1$. Deduce that every map $f : S^1 \rightarrow S^1$ with $\deg f \neq 1$ has at least one fixed point, and in particular that every map of degree 0 (for instance every map that extends to the disc) has a fixed point.

Solution. Strategy. If f has no fixed points, $f(z)$ and z are never equal; one then proves that f is freely homotopic to the antipodal map $a(z) = -z$, by the normalised straight-line homotopy between $-z$ and $f(z)$, and computes $\deg a$. Then one concludes with Theorem 12.8.

Step 1 — the segment between $-z$ and $f(z)$ misses the origin. Set

$$G(z, \theta) = (1 - \theta)(-z) + \theta f(z), \quad (z, \theta) \in S^1 \times I.$$

If $G(z, \theta) = 0$ for some z, θ , then $\theta f(z) = (1 - \theta)z$. Taking moduli and using $|z| = |f(z)| = 1$ we get $\theta = 1 - \theta$, i.e. $\theta = \frac{1}{2}$, and then $\frac{1}{2}f(z) = \frac{1}{2}z$, i.e. $f(z) = z$: against the hypothesis. Hence G never vanishes.

Step 2 — the homotopy. G is continuous and $|G| > 0$, so

$$H(z, \theta) = \frac{G(z, \theta)}{|G(z, \theta)|}$$

is continuous with values in S^1 , with $H(\cdot, 0) = a$ and $H(\cdot, 1) = f$ (because $|-z| = |f(z)| = 1$). Hence $f \simeq a$ (free homotopy).

Step 3 — the degree of the antipodal map. $\gamma_a(s) = -e^{2\pi is} = e^{i\pi} e^{2\pi is} = e^{i(\pi + 2\pi s)}$, which lifts to $\tilde{\gamma}_a(s) = \pi + 2\pi s$. Hence

$$\deg a = \frac{\tilde{\gamma}_a(1) - \tilde{\gamma}_a(0)}{2\pi} = \frac{(\pi + 2\pi) - \pi}{2\pi} = 1.$$

Step 4 — conclusion. By Theorem 12.8, $f \simeq a$ implies $\deg f = \deg a = 1$. Contrapositively: if $\deg f \neq 1$, then f has a fixed point. In particular if $\deg f = 0$, i.e. (by Corollary 12.6) if f extends continuously to the disc, then f has a fixed point.

Independent check. We verify the statement on three examples. (i) $f = a$ (antipodal): it has no fixed points ($-z = z \Rightarrow z = 0 \notin S^1$) and $\deg a = 1$: consistent. (ii) $f = p_2$, $p_2(z) = z^2$, has $\deg = 2 \neq 1$; its fixed points are the solutions of $z^2 = z$ on S^1 , i.e. $z = 1$: there is one, as predicted. (iii) $f = p_{-1} = c$ (conjugation), $\deg = -1 \neq 1$: the fixed points are $\bar{z} = z$, i.e. $z = \pm 1$: there are some. ✓

Interpretation. Degree 1 is that of the identity *and* of the antipodal map: on the circle the antipodal map is a rotation by π , hence homotopic to the identity. This is the first qualitative difference from S^2 , where the antipodal map has degree -1 , and it is precisely this difference that produces the hairy ball theorem (Remark 12.6).

Typical mistakes. (1) Concluding “ f without fixed points $\Rightarrow f$ homotopic to the antipodal map $\Rightarrow \deg f = -1$ ”: that is the formula valid on S^n with n even, not on S^1 ; on S^1 one has $\deg(-\text{id}) = (-1)^{1+1} = 1$. (2) Using the straight-line homotopy $\frac{(1-\theta)z + \theta f(z)}{|\dots|}$ between the identity and f : the denominator vanishes when $f(z) = -z$, which the hypothesis allows. (3) Forgetting the normalisation: G does not take values in S^1 .

Extension. The same scheme proves the dual statement: if $f(z) \neq -z$ for every z , then $f \simeq \text{id}$ and hence $\deg f = 1$ (again 1, because on S^1 the identity and the antipodal map have the same degree). On S^2 the two statements separate and become $\deg f = (-1)^3 = -1$ in the first case and $\deg f = 1$ in the second.

Exercise 12.4: A two-harmonic loop: winding, lifting, numerical check. *Type:* numerical/computational with a proof-based part. Consider

$$\gamma(t) = e^{4\pi it} + 2e^{-2\pi it}, \quad t \in [0, 1].$$

- (a) Prove that γ does not vanish and compute $\min_t |\gamma(t)|$.
- (b) Determine $w(\gamma)$.
- (c) Write down an explicit lifting of the phase.
- (d) Check $w(\gamma)$ numerically by two independent methods.

Solution. Strategy. Isolate the term of larger modulus, apply Lemma 12.8 to obtain w in closed form, build the lifting as the sum of the phase of the dominant term and a “small” phase, and check the result with two independent numerical computations.

Step 1 — γ does not vanish. For every t ,

$$|\gamma(t)| \geq |2e^{-2\pi it}| - |e^{4\pi it}| = 2 - 1 = 1 > 0.$$

The minimum 1 is actually attained: one needs $e^{4\pi it} = -e^{-2\pi it}$, i.e. $e^{6\pi it} = -1$, i.e. $6\pi t = \pi + 2k\pi$, i.e. $t = \frac{1+2k}{6}$ with $t \in [0, 1]$: $t = \frac{1}{6}, \frac{1}{2}, \frac{5}{6}$. Hence $\min_t |\gamma(t)| = 1$.

Step 2 — the winding number. Write $\gamma(t) = a e^{2\pi i n t} + b e^{2\pi i m t}$ with $a = 1$, $n = 2$, $b = 2$, $m = -1$. Since $|b| = 2 > 1 = |a|$, Lemma 12.8 gives

$$w(\gamma) = m = -1.$$

The negative sign says that the loop goes once *clockwise* around the origin: the term of frequency -1 dominates.

Step 3 — explicit lifting. Factor out the dominant term:

$$\gamma(t) = 2e^{-2\pi it} \left(1 + \frac{1}{2}e^{6\pi it} \right).$$

The second factor $\zeta(t) = 1 + \frac{1}{2}e^{6\pi it}$ has real part $1 + \frac{1}{2}\cos 6\pi t \geq \frac{1}{2} > 0$, so it lies in the arc $U_1 = \{\text{Re} > 0\}$ and its phase is read off by the local section $\psi_1 = \arctan(y/x)$ of

Lemma 12.2:

$$\alpha(t) = \arctan \frac{\frac{1}{2} \sin 6\pi t}{1 + \frac{1}{2} \cos 6\pi t} = \arctan \frac{\sin 6\pi t}{2 + \cos 6\pi t},$$

a continuous $\frac{1}{3}$ -periodic function with $\alpha(0) = 0$. A lifting of the phase of $\gamma/|\gamma|$ is therefore

$$\tilde{\theta}(t) = -2\pi t + \alpha(t),$$

because $\gamma(t)/|\gamma(t)| = e^{i(-2\pi t + \alpha(t))}$. Check: $\tilde{\theta}(1) - \tilde{\theta}(0) = -2\pi + \alpha(1) - \alpha(0) = -2\pi + 0 - 0 = -2\pi$, hence $w = -2\pi/2\pi = -1$. ✓ (Note that α contributes nothing: it is periodic, so it accumulates no turns — the analytic translation of the fact that the subdominant term cannot change the invariant.)

Step 4 — independent numerical check. With $N + 1$ equispaced samples:

N	unwrapped phase	$\frac{1}{2\pi i} \int_0^1 \gamma'/\gamma dt$ (trapezoidal)
200	-1.000000	-0.999836
2 000	-1.000000	-0.999998
20 000	-1.000000	-1.000000

The two methods agree with each other and with Step 2. ✓ The largest phase jump between consecutive samples with $N = 2000$ is 0.0126 rad, far smaller than π : the validity condition for unwrapping (sample finely enough that no true jump exceeds π) is amply satisfied.

Solution. (continuation of Exercise 12.4.)

Interpretation. The loop is the sum of two circular motions of frequencies $+2$ and -1 : the one with larger amplitude wins, and the topological invariant is its frequency. This is exactly the phenomenon governing the phase of an I/Q signal dominated by a carrier (Chapter 26) and the reading of a rotary encoder in the presence of parasitic oscillations: as long as the disturbance has smaller amplitude than the carrier, the turn count does not change.

Typical mistakes. (1) Confusing w with the number of self-intersections or with the number of “petals” of the curve: here the curve has three lobes but $w = -1$. (2) Applying Lemma 12.8 with $|a| = |b|$: in that case the loop touches the origin and w is undefined. (3) Unwrapping the phase with too few samples: if a true jump exceeds π , the algorithm reads it as the opposite jump and the result is off by an integer. (4) Using the trapezoidal rule and expecting the exact integer: the quadrature converges, it is not exact.

Extension. For $\gamma_\lambda(t) = e^{4\pi i t} + \lambda e^{-2\pi i t}$ one has $w = -1$ for $\lambda > 1$ and $w = 2$ for $0 \leq \lambda < 1$ (Lemma 12.8 applied to the opposite dominant term); for $\lambda = 1$ the loop passes through the origin and w is undefined. The value $\lambda = 1$ is a *topological transition*: an integer invariant cannot vary continuously, and indeed it jumps from 2 to -1 across the only value at which the hypothesis fails.

12.12. Self-assessment quiz

- (Definition) An open set $U \subseteq X$ is *evenly covered* by $p : \tilde{X} \rightarrow X$ if
 - $p^{-1}(U)$ is open
 - $p^{-1}(U)$ is a disjoint union of open sets each mapped homeomorphically onto U
 - $p|_U$ is injective
 - $p^{-1}(U)$ is connected
- (Hypothesis) In the path lifting lemma, the hypothesis that makes the lifting *unique* is
 - the compactness of I
 - the connectedness of I together with fixing the initial value
 - the differentiability of γ
 - the surjectivity of E
- (Symbolic computation) The winding number of the loop $\gamma(t) = e^{2\pi i \cdot 5t} \overline{e^{2\pi i \cdot 3t}}$, $t \in [0, 1]$,

- equals
- A) 8 B) 2 C) 15 D) -2
4. (Numerical computation) For $\gamma(t) = e^{4\pi it} + 2e^{-2\pi it}$ the value $\min_t |\gamma(t)|$ is
- A) 0 B) 1 C) 2 D) 3
5. (Proof step) In the proof of path lifting, the Lebesgue number is used to
- A) prove that E is surjective B) find a subdivision of I such that each piece falls inside an evenly covered open set C) prove that the fibre is discrete D) make γ differentiable
6. (Error diagnosis) A student claims: “ $E|_{(0,+\infty)} : (0,+\infty) \rightarrow S^1$ is surjective and a local homeomorphism, hence a covering”. The mistake is
- A) E is not surjective on $(0,+\infty)$ B) over every neighbourhood of 1 there is a sheet that is not mapped onto the whole neighbourhood C) $(0,+\infty)$ is not connected D) S^1 is not Hausdorff
7. (Transfer) A continuous map $f : S^1 \rightarrow S^1$ extends to a continuous $F : D^2 \rightarrow S^1$ if and only if
- A) f is injective B) $\deg f = 0$ C) $\deg f = 1$ D) f is surjective
8. (Source and metacognition) The proof of $\pi_1(S^1) \cong \mathbb{Z}$ by lifting of paths and homotopies is found in
- A) Hatcher, *Algebraic Topology*, §1.1, Theorem 1.7, and Munkres, *Topology*, §54, Theorem 54.5 B) Hatcher §2.2 C) Munkres §27 D) Hatcher §4.2

12.13. Teacher’s key

BEST PRACTICE

Quiz — answers, rationale and diagnosis.

1) **B.** *Rationale:* this is Definition 12.3; the union must be *disjoint* and each piece homeomorphic to U via p . *Distractors:* A holds for every continuous map and characterises nothing; C confuses covering with global injectivity; D is false (in the example E the preimage has infinitely many components). *Diagnosed error:* confusing local homeomorphism with covering. *Source:* Munkres §53; Hatcher §1.1, p. 29. *Prerequisite:* subspace topology. *Remedial work:* redo Remark 12.3.

2) **B.** *Rationale:* uniqueness comes from Lemma 12.5, which uses *only* the connectedness of the domain and the discreteness of the fibre, plus one point where the two liftings agree. *Distractors:* A is needed for *existence* (Lebesgue number), not for uniqueness; C is superfluous (liftings exist for merely continuous paths); D is needed for the initial datum to be well posed. *Diagnosed error:* attributing the wrong part of the statement to the wrong hypothesis. *Source:* Munkres, Lemma 54.1. *Prerequisite:* connectedness. *Remedial work:* rewrite the uniqueness proof isolating where connectedness is used.

3) **B.** *Rationale:* $\gamma(t) = e^{2\pi i 5t} e^{-2\pi i 3t} = e^{2\pi i \cdot 2t} = \omega_2(t)$, hence $w = 2$ by Theorem 12.5(v); equivalently w is additive and $w(\bar{\omega}_3) = -3$, so $5 - 3 = 2$. *Distractors:* A adds instead of subtracting (ignoring the conjugation); C multiplies the frequencies; D gets the sign wrong. *Diagnosed error:* treating conjugation as a positive factor. *Source:* Theorem 12.5. *Prerequisite:* $\overline{e^{i\theta}} = e^{-i\theta}$ (Volume II, Chapter 3). *Remedial work:* compute w of $\omega_n \cdot \bar{\omega}_m$ for various n, m .

- 4) **B. Rationale:** $|\gamma| \geq 2 - 1 = 1$, with equality at $t = \frac{1}{6}, \frac{1}{2}, \frac{5}{6}$ (Worked exercise 12.4, Step 1). *Distractors:* A would mean the loop touches the origin, and then w would be undefined; C forgets the term of modulus 1; D adds the moduli instead of subtracting them. *Diagnosed error:* incorrect use of the reverse triangle inequality. *Source:* Lemma 12.8. *Prerequisite:* triangle inequality. *Remedial work:* draw the curve and locate the three minimum points.
- 5) **B. Rationale:** the Lebesgue number provides the mesh $1/m < \delta$ for which each $[t_{k-1}, t_k]$ has image inside some U_j ; without it one could not choose a *single* local section on each piece. *Distractors:* A is proved otherwise; C is the structure of the fibre $(2\pi\mathbb{Z})$, independent of Lebesgue; D is false and irrelevant. *Diagnosed error:* not distinguishing the compactness argument from the rest. *Source:* Munkres, Lemma 27.5; Hatcher §1.1, property (c), p. 30. *Prerequisite:* compactness. *Remedial work:* redo Lemma 12.4.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (continued).

- 6) **B. Rationale:** this is the content of Remark 12.3 (Munkres, Example 53.2): the component $(0, \varepsilon)$ does not cover the whole neighbourhood of 1. *Distractors:* A is false, $E|_{(0, +\infty)}$ is surjective; C is false, a half-line is connected; D is false, S^1 is metric. *Diagnosed error:* believing that “local homeomorphism + surjectivity” implies “covering”. *Source:* Munkres §53. *Prerequisite:* definition of covering. *Remedial work:* exhibit explicitly the sheets over a small neighbourhood of 1.
- 7) **B. Rationale:** Corollary 12.6: the extension exists if and only if f is homotopic to a constant, i.e. if and only if $\deg f = 0$. *Distractors:* A and D have no bearing on extendability (the identity is bijective and surjective but does not extend); C is the case of the identity, which does *not* extend (Corollary 12.4). *Diagnosed error:* confusing degree 1 with degree 0. *Source:* Hatcher §1.1, Theorem 1.9 and surroundings. *Prerequisite:* convexity of D^2 . *Remedial work:* redo the proof of the non-existence of retractions.
- 8) **A. Rationale:** Hatcher’s Theorem 1.7 (§1.1, pp. 29–31) and Munkres’ Theorem 54.5 (§54) are exactly this statement, with the same strategy used here. *Distractors:* B is the degree via homology (hairy ball); C is the Lebesgue number; D is the Hopf fibration. *Diagnosed error:* citing a source for the chapter rather than for the statement. *Source:* see the section “Sources and bibliography”. *Prerequisite:* being able to read a reference with section and number. *Remedial work:* locate the two statements in the two texts and compare their proofs.

BEST PRACTICE

Assigned exercise 12.1 — answers. (a) $\gamma_c(s) = \overline{e^{2\pi is}} = e^{-2\pi is}$, which lifts to $\tilde{\gamma}_c(s) = -2\pi s$; hence $\deg c = \frac{-2\pi - 0}{2\pi} = -1$. (b) $\deg \text{id} = 1 \neq -1$ and $\deg(\text{constant}) = 0 \neq -1$; by Theorem 12.8 maps of different degree are not homotopic. (c) By Corollary 12.6 the extension would exist only if $\deg c = 0$, whereas $\deg c = -1$. (d) $c \circ p_n(z) = \bar{z}^n = \bar{z}^n$ and $p_n \circ c(z) = \bar{z}^n$: they are the same map, $z \mapsto z^{-n}$ on S^1 (because $\bar{z} = z^{-1}$), of degree $-n$. Proposition 12.3 gives $\deg(c \circ p_n) = \deg c \cdot \deg p_n = (-1) \cdot n = -n$, and likewise for the other composition: consistent. ✓

BEST PRACTICE

Assigned exercise 12.2 — answers. (a) $|\gamma(t)| \geq 1 - \frac{1}{2} = \frac{1}{2} > 0$. The minimum $\frac{1}{2}$ is attained when the two terms are opposite in direction, i.e. $e^{-4\pi it} = -e^{6\pi it}$, i.e. $e^{10\pi it} = -1$, i.e. $t = \frac{1+2k}{10}$: for $t \in [0, 1]$ one gets $t = 0.1; 0.3; 0.5; 0.7; 0.9$. Hence $\min_t |\gamma(t)| = 0.5$ (checked numerically on $2 \cdot 10^6$ samples: 0.500000). ✓ (b) The dominant term is $e^{6\pi it} = e^{2\pi i \cdot 3t}$ ($1 > \frac{1}{2}$), hence $w(\gamma) = 3$. (c) The unwrapped phase gives exactly 3.000000 for all three

values of N . **(d)** The trapezoidal rule gives 2.995561 ($N = 200$), 2.999956 ($N = 2000$), 2.999999 ($N = 20000$). **(e)** Unwrapping returns the exact integer because it sums phase increments reduced to $(-\pi, \pi]$: provided no true jump exceeds π , the result is exact and does not “converge”; the integral is a quadrature, with error $O(N^{-2})$ for the trapezoidal rule (note the factor ≈ 100 between consecutive rows, consistent with $N \rightarrow 10N$). This is the numerical manifestation of the fact that w is a topological invariant. **(f)** Now $e^{-4\pi it} = e^{2\pi i(-2)t}$ dominates (coefficient $1 > \frac{1}{2}$), hence $w(\delta) = -2$: swapping the amplitudes changes the dominant term, and with it the invariant. The passage from 3 to -2 occurs abruptly when the two amplitudes become equal, because then the loop touches the origin.

12.14. Chapter formulary

The circle as a space.

$$S^1 = \{z \in \mathbb{C} : |z| = 1\} \text{ compact, connected, path connected; } S^1 \setminus \{p\} \cong \mathbb{R}.$$

Stereographic projection and its inverse:

$$\sigma(x, y) = \frac{x}{1-y}, \quad \tau(s) = \left(\frac{2s}{s^2+1}, \frac{s^2-1}{s^2+1} \right).$$

Quotient and covering.

$$\mathbb{R}/2\pi\mathbb{Z} \cong S^1 \quad (\text{homeomorphism and group isomorphism}),$$

$$E(t) = e^{it}, \quad E^{-1}(E(t_0)) = t_0 + 2\pi\mathbb{Z}.$$

Local sections: on $\{x > 0\}$, $\psi_1 = \arctan(y/x)$; on $\{x < 0\}$, $\psi_2 = \arctan(y/x) + \pi$; on $\{y > 0\}$, $\psi_3 = \arccos x$; on $\{y < 0\}$, $\psi_4 = -\arccos x$.

Lifting. For every path $\gamma : I \rightarrow S^1$ and every a with $E(a) = \gamma(0)$ there is a unique $\tilde{\gamma}$ with $E \circ \tilde{\gamma} = \gamma$, $\tilde{\gamma}(0) = a$; the same for $F : I \times I \rightarrow S^1$.

Winding number.

$$w(\gamma) = \frac{\tilde{\gamma}(1) - \tilde{\gamma}(0)}{2\pi} \in \mathbb{Z}, \quad w(\gamma \cdot \gamma') = w(\gamma) + w(\gamma'), \quad w(\bar{\gamma}) = -w(\gamma), \quad w(\omega_n) = n.$$

Analytic form, for C^1 loops in $\mathbb{C} \setminus \{0\}$:

$$w(\gamma) = \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z} = \frac{1}{2\pi i} \int_0^1 \frac{\gamma'(t)}{\gamma(t)} dt.$$

Dominant term: if $|b| > |a|$, then $w(ae^{2\pi i n t} + be^{2\pi i m t}) = m$.

Fundamental group and degree.

$$\pi_1(S^1, 1) \cong \mathbb{Z} \quad \text{via} \quad [\gamma] \mapsto w(\gamma), \quad n \mapsto [\omega_n];$$

$$\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y); \quad \pi_1(T^2) \cong \mathbb{Z}^2.$$

$$\deg f = w(s \mapsto f(e^{2\pi i s})); \quad f \simeq g \iff \deg f = \deg g; \quad \deg p_n = n \quad (p_n(z) = z^n);$$

$$\deg(f \circ g) = \deg f \cdot \deg g; \quad f \text{ extends to } D^2 \iff \deg f = 0.$$

Lifted map: $f(e^{it}) = e^{iu(t)}$ with $u(t + 2\pi) = u(t) + 2\pi \deg f$.

Consequences. There is no retraction $D^2 \rightarrow S^1$; every continuous $h : D^2 \rightarrow D^2$ has a fixed point (Brouwer); every $f : S^1 \rightarrow S^1$ with $\deg f \neq 1$ has a fixed point.

Sources and bibliography

- [1] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002. §1.1 “The Fundamental Group of the Circle”, pp. 29–31 (Theorem 1.7: $\pi_1(S^1)$ is infinite cyclic; lifting properties (a), (b), (c) for paths and homotopies, with the compactness argument used in this chapter); Proposition 1.14 and Lemma 1.15, p. 35 ($\pi_1(S^n) = 0$ for $n \geq 2$); §1.3 “Covering Spaces”, Proposition 1.30, pp. 60–61 (homotopy lifting property for an arbitrary covering). Chapters in PDF from the author’s page at Cornell University, pi.math.cornell.edu/~hatcher/AT/ATchapters.html — verified on 5 September 2026.
- [2] A. Hatcher, *Algebraic Topology*, cit., §2.2, Theorem 2.28, p. 135: “ S^n has a continuous field of nonzero tangent vectors iff n is odd” (hairy ball theorem; reported here as panorama, not proved) — verified on 5 September 2026.
- [3] A. Hatcher, *Algebraic Topology*, cit., §4.2, Example 4.45, pp. 377–378 (the Hopf bundle $S^1 \rightarrow S^3 \rightarrow S^2$ and the decomposition of S^3 into tori) and Example 4.51, p. 380 ($\pi_3(S^2)$ is infinite cyclic, generated by the Hopf map; both reported here as panorama, not proved) — verified on 5 September 2026.
- [4] J. R. Munkres, *Topology*, 2nd edition, Prentice Hall, Upper Saddle River, 2000 (ISBN 0-13-181629-2; Pearson “Classic Version” reprint, 2017). §23 “Connected Spaces”, §26 “Compact Spaces”, §27 “Compact Subspaces of the Real Line” (Theorem 27.3, Heine–Borel; Lemma 27.5, Lebesgue number lemma): the general topology prerequisites declared in the opening *infobox*. §53 “Covering Spaces” (definition of evenly covered neighbourhood and of slice; Theorem 53.1: $p(x) = (\cos 2\pi x, \sin 2\pi x)$ is a covering map; Example 53.2: its restriction to $\mathbb{R}^+$ is not). §54 “The Fundamental Group of the Circle” (Lemma 54.1, path lifting; Lemma 54.2, homotopy lifting; Theorem 54.3; Theorem 54.4, lifting correspondence; Theorem 54.5, $\pi_1(S^1) \cong \mathbb{Z}$). Statements and numbering verified against the topology course notes of East Tennessee State University, faculty.etsu.edu/gardnerr/5357/notes/ — verified on 5 September 2026.
- [5] M. Lackenby, *Topology and Groups*, course notes, University of Oxford, October 2017. Chapter III §3 “The fundamental group of the circle”, pp. 32–33 (Theorem III.32: $\pi_1(S^1, 1) \cong \mathbb{Z}$, proved by simplicial approximation — a *different* strategy from the lifting one adopted here, useful as a comparison); Chapter VI “Covering spaces”, pp. 56–74. [maths.ox.ac.uk/system/files/attachments/tg11o17\[1\]_0.pdf](http://maths.ox.ac.uk/system/files/attachments/tg11o17[1]_0.pdf) — verified on 5 September 2026.
- [6] J. Orloff, *18.04 Complex Variables with Applications*, MIT OpenCourseWare, spring 2018. Topic 3 “Cauchy’s Theorem” (lectures 8–9) and Topic 11 “Argument Principle” (lectures 30–33): the index $\frac{1}{2\pi i} \oint_\gamma \frac{dz}{z-z_0}$ and its role in Cauchy’s formula, referred to by Theorem 12.6. ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/ — verified on 5 September 2026.

CHAPTER 13

The circle as a group: abstract harmonic analysis

In Chapter 12 the circle was regarded as a *space*: distance was forgotten and shape was kept. Here we take the complementary step: shape is forgotten and the *operation* is kept. The circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is closed under complex multiplication, contains 1, and contains the inverse of each of its elements: it is a group, and it is a group in which the operation is continuous. This combination — an algebraic structure and a topology that respect each other — is enough, on its own, to give rise to the whole of Fourier analysis.

The thread of the chapter is this. On S^1 there is a unique rotation-invariant probability measure: the measure $d\theta/2\pi$ (Theorem 13.4). The only continuous homomorphisms of S^1 into itself are the powers $z \mapsto z^n$ (Theorem 13.5), and the proof is a direct application of the lifting built in the previous chapter. Those powers are orthonormal with respect to that measure (Theorem 13.6), and they form a group isomorphic to $\mathbb{Z}$: this is the duality $\widehat{S^1} \cong \mathbb{Z}$, of which $\widehat{\mathbb{Z}} \cong S^1$ is the mirror image. Finally the closed subgroups of S^1 are classified (Theorem 13.7) and it is proved that irrational rotations generate dense subgroups (Theorem 13.8): this is the one-dimensional form of Kronecker's theorem, and it is the reason why an oscillator with an irrational frequency ratio never passes through the same state twice.

Coefficients, orthogonality and Fourier series on the circle have already been built in Volume II, Chapter 4, by analytic means. Here their *structural origin* is reconstructed: nothing from that chapter is repeated, and the two proofs of the classification of characters — the analytic one there, the topological one here — are to be read as two roads to the same point.

INFO

Level and prerequisites. Level D (graduate).

Internal prerequisites. Polar form, complex exponential and the equivalences $e^{i(s+t)} = e^{is}e^{it}$, $e^{it} = 1 \iff t \in 2\pi\mathbb{Z}$ (Volume II, Chapter 3); the circle as the multiplicative group of $\mathbb{C}$ (Chapter 6); from Chapter 12: compactness and connectedness of S^1 (Theorem 12.1), the homeomorphism $\mathbb{R}/2\pi\mathbb{Z} \cong S^1$ (Theorem 12.3), the exponential covering (Theorem 12.4), the lifting of maps defined on $\mathbb{R}$ (Corollary 12.3) and the uniqueness of the lifting on a connected space (Lemma 12.5). Classical Fourier series (Volume II, Chapter 4). Spheres and groups in dimension n (Chapter 7, Chapter 11).

External prerequisites, assumed and not reproved:

- Lebesgue measure on $\mathbb{R}$: σ -additivity and translation invariance;
- the dominated convergence theorem;
- the Stone–Weierstrass theorem in complex form (Rudin, *Principles of Mathematical Analysis*, 3rd ed., Theorem 7.33): a subalgebra of $C(K, \mathbb{C})$ that contains the constants, separates points and is closed under conjugation is dense in the uniform norm;
- Dynkin's lemma (π - λ): two probability measures that agree on a π -system generating the σ -algebra agree everywhere;
- the image-measure theorem: if T is measurable and $\nu = T_*\mu$, then $\int g d\nu = \int g \circ T d\mu$ for every bounded Borel g ;

- the notion of a differentiable manifold and of a smooth map between manifolds (Volume III, Chapter 10 for the case of surfaces).

Everything else is proved here.

What one learns. How to recognise a topological group and a Lie group from an explicit exhibition of the charts; how to construct an invariant measure and prove its uniqueness with Fourier coefficients; how to classify characters with a lifting argument; how to read the duality $\widehat{S^1} \cong \mathbb{Z}$ in both directions; how to classify closed subgroups and recognise when an orbit is dense.

Warning. Conventions of the chapter.

- $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, with the subspace topology of $\mathbb{C}$ and the operation of complex multiplication. The symbol $\mathbb{T}$ is synonymous with S^1 when the group structure is to be emphasised: $\mathbb{T} = S^1$.
- $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it}$, is the map of Chapter 12: this normalisation will always be used, with $E^{-1}(1) = 2\pi\mathbb{Z}$.
- $\mu_n = \{z \in S^1 : z^n = 1\} = \{e^{2\pi i k/n} : k = 0, \dots, n-1\}$ is the group of n -th roots of unity.
- “Borel measure” means: a measure defined on the Borel σ -algebra of S^1 , that is, the one generated by the open sets.
- $\chi_n : S^1 \rightarrow S^1$ is the map $\chi_n(z) = z^n$, $n \in \mathbb{Z}$.
- $d(z, w) = |z - w|$ is the Euclidean distance induced by $\mathbb{C}$; $d(z, A) = \inf_{a \in A} |z - a|$.

13.1. The circle as a topological group

Definition 13.1: Topological group. A **topological group** is a set G equipped with a group structure and a topology such that the two maps

$$m : G \times G \rightarrow G, \quad m(g, h) = gh, \quad \iota : G \rightarrow G, \quad \iota(g) = g^{-1},$$

are continuous ($G \times G$ with the product topology). The group is **abelian** if $gh = hg$ for all g, h , and it is **compact** if it is compact as a topological space.

Theorem 13.1: S^1 is a compact abelian topological group. The circle S^1 , with complex multiplication and the subspace topology of $\mathbb{C}$, is a compact abelian topological group, with identity element 1 and inverse $z^{-1} = \bar{z}$.

Proof. Step 1 — group structure. If $|z| = |w| = 1$ then $|zw| = |z||w| = 1$: the product lies in S^1 . Associativity and commutativity are inherited from the field $\mathbb{C}$. The element 1 has modulus 1 and is the identity. If $|z| = 1$ then $z\bar{z} = |z|^2 = 1$, hence $z^{-1} = \bar{z}$, which has modulus $|\bar{z}| = |z| = 1$ and lies in S^1 . Hence S^1 is an abelian group.

Step 2 — continuity of multiplication. In real coordinates, $(x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$: the two components are polynomials in the four real variables, hence continuous on $\mathbb{R}^4$. A map into a product is continuous if and only if its components are, hence $m_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ is continuous; the restriction of a continuous map to a subspace is continuous, and $m = m_{\mathbb{C}}|_{S^1 \times S^1}$ has image in S^1 by Step 1, so $m : S^1 \times S^1 \rightarrow S^1$ is continuous (the topology of S^1 is the subspace topology).

Step 3 — continuity of inversion. $\iota(z) = \bar{z}$ is, in coordinates, $(x, y) \mapsto (x, -y)$: linear, hence continuous, and it sends S^1 into S^1 .

Step 4 — compactness. S^1 is compact by Theorem 12.1. □

Proposition 13.1: Translations are homeomorphisms; S^1 is homogeneous. For every $w \in S^1$ the map $T_w : S^1 \rightarrow S^1$, $T_w(z) = wz$, is a homeomorphism, with inverse $T_{w^{-1}}$. Moreover for every pair $z_1, z_2 \in S^1$ there exists w with $T_w(z_1) = z_2$: the group acts transitively on itself.

Proof. T_w is the composition $z \mapsto (w, z) \mapsto wz$: the first map is continuous (constant component and identity), the second is m , continuous by Theorem 13.1; hence T_w is continuous. We have $T_{w^{-1}} \circ T_w(z) = w^{-1}wz = z$ and likewise $T_w \circ T_{w^{-1}} = \text{id}$, so T_w is bijective with inverse $T_{w^{-1}}$, itself continuous by the same argument: T_w is a homeomorphism. Finally, given z_1, z_2 , set $w = z_2 z_1^{-1} \in S^1$: then $T_w(z_1) = z_2 z_1^{-1} z_1 = z_2$. $\square$

Remark 13.1: Why homogeneity matters. Proposition 13.1 says that on the circle there is no privileged point: any topological or metric property valid at one point holds at all of them. This is why an invariant measure can exist (no region is “special”) and why a time-invariant linear system, which commutes with translations, has as its eigenfunctions exactly the characters (Theorem 13.5). The same statement holds for every topological group: left translation is always a homeomorphism, and this is the structural reason why groups are “uniform” spaces.

Proposition 13.2: E is a homomorphism and $\mathbb{R}/2\pi\mathbb{Z} \cong S^1$ as topological groups. The map $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it}$, is a continuous surjective homomorphism of the additive group $(\mathbb{R}, +)$ onto the multiplicative group S^1 , with kernel $\ker E = 2\pi\mathbb{Z}$. Consequently the induced map

$$\bar{E} : \mathbb{R}/2\pi\mathbb{Z} \longrightarrow S^1, \quad \bar{E}(t + 2\pi\mathbb{Z}) = e^{it},$$

is a group isomorphism and a homeomorphism: $\mathbb{R}/2\pi\mathbb{Z}$ and S^1 are the same topological group.

Proof. E is continuous and surjective by Theorem 12.4, and it is a homomorphism because $E(s+t) = e^{i(s+t)} = e^{is}e^{it} = E(s)E(t)$ (Volume II, Chapter 3). The kernel is $\{t : e^{it} = 1\} = 2\pi\mathbb{Z}$, again by Volume II, Chapter 3. By the first homomorphism theorem, $\bar{E}$ is a well-defined group isomorphism ($E(t) = E(t')$ if and only if $t - t' \in 2\pi\mathbb{Z}$). That $\bar{E}$ is also a homeomorphism is exactly Theorem 12.3. Finally, the quotient of a topological group by a subgroup is a topological group with the quotient topology: here there is no need to verify this, because $\bar{E}$ transports onto $\mathbb{R}/2\pi\mathbb{Z}$ the structure of S^1 , already known by Theorem 13.1. $\square$

13.2. S^1 as a Lie group of dimension 1

Definition 13.2: Lie group. A **Lie group** (real, of dimension n) is a group G equipped with the structure of a differentiable manifold of dimension n such that multiplication $m : G \times G \rightarrow G$ and inversion $\iota : G \rightarrow G$ are smooth (C^∞) maps.

To exhibit the manifold structure on S^1 , two charts suffice. The technical point is that their coordinate changes be smooth; the only subtlety is that the angle, as a function of the point, is not defined globally — this is exactly the phenomenon that Chapter 12 named “covering”.

Theorem 13.2: S^1 is a Lie group of dimension 1. Set

$$U_1 = S^1 \setminus \{-1\}, \quad \varphi_1(x, y) = 2 \arctan \frac{y}{1+x} \in (-\pi, \pi),$$

$$U_2 = S^1 \setminus \{1\}, \quad \varphi_2(x, y) = \pi - 2 \arctan \frac{y}{1-x} \in (0, 2\pi).$$

Then $\{(U_1, \varphi_1), (U_2, \varphi_2)\}$ is a C^∞ (indeed analytic) atlas of dimension 1 on S^1 , and with respect to it multiplication and inversion are smooth: S^1 is a compact connected abelian Lie group of dimension 1.

Proof. Step 1 — φ_1 and φ_2 are well defined and continuous. On U_1 we have $x > -1$, that is $1+x > 0$: the fraction $y/(1+x)$ is defined and the function $(x, y) \mapsto 2 \arctan(y/(1+x))$ is a composition of analytic functions, hence analytic on U_1 (as the restriction to U_1 of an analytic function on the open set $\{x > -1\}$ of $\mathbb{R}^2$). Likewise on U_2 we have $x < 1$, that is $1-x > 0$, and φ_2 is analytic.

Step 2 — φ_1 is the inverse of $t \mapsto e^{it}$ on $(-\pi, \pi)$. Let $t \in (-\pi, \pi)$ and $z = e^{it} = (\cos t, \sin t) \in U_1$. From the half-angle formulas,

$$\frac{\sin t}{1 + \cos t} = \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \cos^2 \frac{t}{2}} = \tan \frac{t}{2},$$

legitimate because $\cos(t/2) \neq 0$ for $|t| < \pi$. Since $t/2 \in (-\pi/2, \pi/2)$, we have $\arctan(\tan(t/2)) = t/2$ and hence $\varphi_1(e^{it}) = t$. Conversely E maps $(-\pi, \pi)$ bijectively onto U_1 (Theorem 12.4 and the description of the fibre $E^{-1}(z) = t_0 + 2\pi\mathbb{Z}$): hence $\varphi_1 : U_1 \rightarrow (-\pi, \pi)$ is a bijection, with inverse $\varphi_1^{-1} = E|_{(-\pi, \pi)}$, continuous. Thus φ_1 is a homeomorphism onto an open subset of $\mathbb{R}$.

Step 3 — φ_2 is the inverse of $t \mapsto e^{it}$ on $(0, 2\pi)$. Let $t \in (0, 2\pi)$ and $z = e^{it} \in U_2$. Then

$$\frac{\sin t}{1 - \cos t} = \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \sin^2 \frac{t}{2}} = \cot \frac{t}{2},$$

legitimate because $\sin(t/2) \neq 0$ for $t \in (0, 2\pi)$. Since $t/2 \in (0, \pi)$ we have $\cot(t/2) = \tan(\frac{\pi}{2} - \frac{t}{2})$ with $\frac{\pi}{2} - \frac{t}{2} \in (-\pi/2, \pi/2)$, hence $\arctan(\cot(t/2)) = \frac{\pi}{2} - \frac{t}{2}$ and

$$\varphi_2(e^{it}) = \pi - 2\left(\frac{\pi}{2} - \frac{t}{2}\right) = t.$$

As above, φ_2 is a homeomorphism of U_2 onto $(0, 2\pi)$.

Step 4 — the change of chart is smooth. We have $U_1 \cap U_2 = S^1 \setminus \{1, -1\}$ and $\varphi_1(U_1 \cap U_2) = (-\pi, 0) \cup (0, \pi)$. For $t \in (0, \pi)$, $\varphi_2 \circ \varphi_1^{-1}(t) = \varphi_2(e^{it}) = t$; for $t \in (-\pi, 0)$, $e^{it} = e^{i(t+2\pi)}$ with $t+2\pi \in (\pi, 2\pi) \subset (0, 2\pi)$, hence $\varphi_2 \circ \varphi_1^{-1}(t) = t+2\pi$. The transition map is therefore

$$\varphi_2 \circ \varphi_1^{-1}(t) = \begin{cases} t, & t \in (0, \pi), \\ t+2\pi, & t \in (-\pi, 0), \end{cases}$$

that is, a translation on each of the two connected components of the domain: it is C^∞ (with derivative 1), and the same holds for its inverse. The atlas is thus smooth; since S^1 is Hausdorff and second countable (being a subspace of $\mathbb{R}^2$), it defines on S^1 a differentiable manifold structure of dimension 1.

Step 5 — multiplication is smooth. Let $z_0, w_0 \in S^1$ and $z_0 w_0 = v_0$. Choose $a, b \in \mathbb{R}$ with $E(a) = z_0$, $E(b) = w_0$ and a chart φ among the two (or a rotate of one of them) defined on a neighbourhood of v_0 ; by Proposition 13.1 one may work directly with the chart $\psi_c(z) = c + \varphi_1(e^{-ic}z)$, defined on the neighbourhood $e^{ic}U_1$ of $E(c)$ and with values in $(c-\pi, c+\pi)$: ψ_c is φ_1 composed with a translation of the group and a translation of $\mathbb{R}$, hence a chart compatible with the atlas (the change of chart with φ_1 is again a translation, by Step 4 applied after the rotation). In these coordinates, for s near a and t near b ,

$$\psi_{a+b}(E(s)E(t)) = \psi_{a+b}(E(s+t)) = s+t,$$

because $s + t$ is near $a + b$ and $\psi_{a+b} \circ E$ is the identity on $(a + b - \pi, a + b + \pi)$. The local expression of multiplication is thus $(s, t) \mapsto s + t$: linear, hence C^∞ .

Step 6 — inversion is smooth. With the same charts, $\psi_{-a}(E(s)^{-1}) = \psi_{-a}(E(-s)) = -s$ for s near a : the local expression is $s \mapsto -s$, linear, hence C^∞ .

Compactness and connectedness are Theorem 12.1; commutativity is Theorem 13.1. $\square$

Example 13.1: The exponential chart, checked twice. Take $t = 0.3$. Then $z = (\cos 0.3, \sin 0.3) = (0.9553365, 0.2955202)$ and

$$\frac{y}{1+x} = \frac{0.2955202}{1.9553365} = 0.1511350, \quad 2 \arctan(0.1511350) = 2 \cdot 0.1500000 = 0.3000000. \quad \checkmark$$

Independent check with the closed formula of Step 2: $\tan(t/2) = \tan(0.15) = 0.1511350$, which coincides with the ratio computed above. On the second chart, with $t = 2$: $z = (-0.4161468, 0.9092974)$ and

$$\frac{y}{1-x} = \frac{0.9092974}{1.4161468} = 0.6420926, \quad \pi - 2 \arctan(0.6420926) = 3.1415927 - 2 \cdot 0.5707963 = 2.0000001. \quad \checkmark$$

(The discrepancy of 10^{-7} in the last computation is the effect of rounding to the seven reported digits: in double precision the two sides equal 2.0000000000000000 and 1.9999999999999998. Over 10^6 equispaced values of t in $[-3, 3]$ the maximum discrepancy $|\varphi_1(e^{it}) - t|$ is $8.9 \cdot 10^{-16}$; over 10^6 equispaced values in $[0.15, 2\pi - 0.15]$ the maximum discrepancy $|\varphi_2(e^{it}) - t|$ is $1.1 \cdot 10^{-15}$.) $\checkmark$

Proposition 13.3: The exponential as a one-parameter homomorphism. The map $E : \mathbb{R} \rightarrow S^1$ is smooth, is a group homomorphism, is a local diffeomorphism at every point and has kernel $2\pi\mathbb{Z}$; its restriction to $(-\pi, \pi)$ is a diffeomorphism onto U_1 . In particular S^1 is connected and is generated by every neighbourhood of 1.

Proof. That E is a surjective homomorphism with kernel $2\pi\mathbb{Z}$ is Proposition 13.2. In the chart φ_1 , for $t \in (-\pi, \pi)$, the local expression of E is $\varphi_1(E(t)) = t$ (Step 2 of Theorem 13.2): this is the identity, hence $E|_{(-\pi, \pi)}$ is a diffeomorphism onto U_1 . For an arbitrary t_0 one uses the chart ψ_{t_0} of Step 5, which again gives the identity: E is a local diffeomorphism everywhere. Finally, let V be a neighbourhood of 1 and H the subgroup generated by V : H contains $E((-\varepsilon, \varepsilon))$ for some $\varepsilon > 0$, and every $t \in \mathbb{R}$ is written as a sum of finitely many terms of modulus $< \varepsilon$; since E is a homomorphism, $E(t) \in H$. Hence $H = S^1$. $\square$

Remark 13.2: What we import: the general theory of Lie groups (panorama).

Theorem 13.2 is a special case of a theory that is not developed here and is declared **imported**. Three reference facts are recorded, without proof. (i) Every connected, compact, abelian Lie group is isomorphic to a torus $(S^1)^n$; in dimension 1 this gives: S^1 and $\mathbb{R}$ are, up to isomorphism, the only connected Lie groups of dimension 1. (ii) Cartan's closed subgroup theorem: a closed subgroup of a Lie group is an embedded Lie subgroup. Theorem 13.7 of this chapter is its direct, and complete, verification in the case $G = S^1$. (iii) Hilbert's fifth problem (Gleason, Montgomery–Zippin, 1952): a locally Euclidean topological group admits a unique analytic Lie group structure. What is missing here is the whole apparatus of Lie algebras, the exponential map and the subalgebra/subgroup correspondence: see Bröcker–tom Dieck, *Representations of Compact Lie Groups*, and Warner, *Foundations of Differentiable Manifolds and Lie Groups*, listed in the sources. Nothing proved in this chapter depends on (i)–(iii).

13.3. The invariant measure $d\theta/2\pi$

Definition 13.3: The normalised measure of S^1 . Let λ be Lebesgue measure on $\mathbb{R}$. For every Borel set $A \subseteq S^1$, set

$$m(A) = \frac{1}{2\pi} \lambda(E^{-1}(A) \cap [0, 2\pi)).$$

The definition makes sense: E is continuous, hence $E^{-1}(A)$ is Borel in $\mathbb{R}$, and so is its intersection with $[0, 2\pi)$. One also writes $dm = \frac{d\theta}{2\pi}$.

Proposition 13.4: m is a Borel probability measure. m is a σ -additive measure on the Borel sets of S^1 with $m(S^1) = 1$, and for every bounded Borel function $f : S^1 \rightarrow \mathbb{C}$,

$$\int_{S^1} f dm = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) d\theta.$$

Proof. Step 1 — m is an image of λ . The restriction $E_0 = E|_{[0, 2\pi)} : [0, 2\pi) \rightarrow S^1$ is a bijection: it is surjective because E is and every class modulo 2π meets $[0, 2\pi)$; it is injective because $E(s) = E(t)$ with $s, t \in [0, 2\pi)$ implies $s - t \in 2\pi\mathbb{Z} \cap (-2\pi, 2\pi) = \{0\}$. Moreover $E_0^{-1}(A) = E^{-1}(A) \cap [0, 2\pi)$. Hence $m = \frac{1}{2\pi}(E_0)_*(\lambda|_{[0, 2\pi)})$ is the normalised image measure of $\lambda|_{[0, 2\pi)}$.

Step 2 — σ -additivity. If (A_k) are disjoint Borel sets, their $E_0^{-1}(A_k)$ are disjoint (because E_0 is a function) and $E_0^{-1}(\bigcup A_k) = \bigcup E_0^{-1}(A_k)$; the σ -additivity of λ concludes. Moreover $m(\emptyset) = 0$ and $m(S^1) = \frac{1}{2\pi}\lambda([0, 2\pi)) = 1$.

Step 3 — integral formula. This is the image-measure theorem, recalled in the prerequisites, applied to $T = E_0$: $\int_{S^1} f dm = \frac{1}{2\pi} \int_{[0, 2\pi)} f(E_0(\theta)) d\theta$, which is the statement (the point 2π , of measure zero, does not change the integral). $\square$

Theorem 13.3: Rotation invariance. For every $w \in S^1$ and every Borel set $A \subseteq S^1$, $m(wA) = m(A)$, where $wA = \{wz : z \in A\}$. Equivalently, $\int f(wz) dm(z) = \int f dm$ for every bounded Borel f .

Proof. Step 1 — reduction to $\mathbb{R}$. Let $w = E(\alpha)$ with $\alpha \in [0, 2\pi)$. Since E is a homomorphism, for every t we have $E(t) \in wA \iff E(t - \alpha) \in A \iff t \in \alpha + E^{-1}(A)$. Hence

$$E^{-1}(wA) = \alpha + E^{-1}(A).$$

Step 2 — the computation on the interval. Set $B = E^{-1}(A)$; B is 2π -periodic ($B + 2\pi = B$, because E is 2π -periodic) and $m(A) = \frac{1}{2\pi}\lambda(B \cap [0, 2\pi))$. We must compute $\lambda((\alpha + B) \cap [0, 2\pi))$. We split:

$$(\alpha + B) \cap [0, 2\pi) = ((\alpha + B) \cap [\alpha, 2\pi)) \cup ((\alpha + B) \cap [0, \alpha)),$$

a disjoint union (recall $0 \leq \alpha < 2\pi$). For the first piece, the translation $t \mapsto t - \alpha$ maps $[\alpha, 2\pi)$ onto $[0, 2\pi - \alpha)$ and $\alpha + B$ onto B ; since λ is translation-invariant,

$$\lambda((\alpha + B) \cap [\alpha, 2\pi)) = \lambda(B \cap [0, 2\pi - \alpha)).$$

For the second piece, the translation $t \mapsto t - \alpha + 2\pi$ maps $[0, \alpha)$ onto $[2\pi - \alpha, 2\pi)$ and maps $\alpha + B$ onto $B + 2\pi = B$; hence

$$\lambda((\alpha + B) \cap [0, \alpha)) = \lambda(B \cap [2\pi - \alpha, 2\pi)).$$

Adding the two pieces, and noting that $[0, 2\pi - \alpha)$ and $[2\pi - \alpha, 2\pi)$ partition $[0, 2\pi)$:

$$\lambda((\alpha + B) \cap [0, 2\pi)) = \lambda(B \cap [0, 2\pi)),$$

that is, $m(wA) = m(A)$.

Step 3 — integral form. The equality just proved says $(T_{w^{-1}})_*m = m$, hence by the image-measure theorem $\int f \circ T_w dm = \int f dm$ for every bounded Borel f (first for indicator functions, where it is Step 2; then for simple functions, by linearity; then for bounded Borel functions, by dominated convergence along a sequence of simple functions converging uniformly). $\square$

Definition 13.4: Fourier coefficients of a measure. Let μ be a finite Borel measure on S^1 . Its **Fourier coefficients** are

$$c_n(\mu) = \int_{S^1} z^n d\mu(z), \quad n \in \mathbb{Z}.$$

They are well defined because $|z^n| = 1$ and μ is finite, and they satisfy $|c_n(\mu)| \leq \mu(S^1)$.

Lemma 13.1: A finite measure is determined by its coefficients. Let μ, ν be two Borel probability measures on S^1 with $c_n(\mu) = c_n(\nu)$ for every $n \in \mathbb{Z}$. Then $\mu = \nu$.

Proof. **Step 1 — trigonometric polynomials.** Let $\mathcal{P}$ be the set of finite linear combinations, with complex coefficients, of the functions $z \mapsto z^n$, $n \in \mathbb{Z}$ (the *trigonometric polynomials* on S^1). By hypothesis and by linearity of the integral, $\int P d\mu = \int P d\nu$ for every $P \in \mathcal{P}$.

Step 2 — density. $\mathcal{P}$ is a subalgebra of $C(S^1, \mathbb{C})$: the product of two monomials is a monomial, $z^n z^m = z^{n+m}$. It contains the constants ($n = 0$). It separates points: if $z \neq w$ then $\chi_1(z) = z \neq w = \chi_1(w)$. It is closed under conjugation, because $\overline{z^n} = z^{-n}$ on S^1 . Since S^1 is compact, the Stone–Weierstrass theorem (complex version) gives: $\mathcal{P}$ is dense in $C(S^1, \mathbb{C})$ for the uniform norm.

Step 3 — from the P 's to continuous f . Let $f \in C(S^1, \mathbb{C})$ and $(P_k) \subset \mathcal{P}$ with $\|P_k - f\|_\infty \rightarrow 0$. Then

$$\left| \int f d\mu - \int f d\nu \right| \leq \left| \int (f - P_k) d\mu \right| + \left| \int (P_k - f) d\nu \right| \leq 2\|f - P_k\|_\infty \xrightarrow[k \rightarrow \infty]{} 0,$$

having used Step 1 and $\mu(S^1) = \nu(S^1) = 1$. Hence $\int f d\mu = \int f d\nu$ for every continuous f .

Step 4 — from continuous functions to closed sets. Let $F \subseteq S^1$ be closed and set, for $k \in \mathbb{N}$,

$$f_k(z) = \max(0, 1 - k d(z, F)), \quad d(z, F) = \inf_{a \in F} |z - a|.$$

The function $z \mapsto d(z, F)$ is 1-Lipschitz, hence continuous, and f_k is continuous with $0 \leq f_k \leq 1$. If $z \in F$ then $d(z, F) = 0$ and $f_k(z) = 1$. If $z \notin F$, since F is closed we have $d(z, F) > 0$, and for $k > 1/d(z, F)$ we get $f_k(z) = 0$: hence $f_k \rightarrow \mathbf{1}_F$ pointwise, with $|f_k| \leq 1$ and $\mathbf{1}_F$ integrable with respect to probability measures. By dominated convergence and Step 3,

$$\mu(F) = \lim_k \int f_k d\mu = \lim_k \int f_k d\nu = \nu(F).$$

Step 5 — from closed sets to all Borel sets. The family of closed subsets of S^1 is a π -system (the intersection of two closed sets is closed) and generates the Borel σ -algebra (the complements of closed sets are the open sets). Two probability measures that agree on a generating π -system agree on the generated σ -algebra (Dynkin's lemma, recalled in the prerequisites). Hence $\mu = \nu$. $\square$

Theorem 13.4: Uniqueness of the invariant measure on S^1 . Let μ be a Borel probability measure on S^1 such that $\mu(wA) = \mu(A)$ for every $w \in S^1$ and every Borel set A . Then $\mu = m$, the measure of Definition 13.3. In other words $d\theta/2\pi$ is the unique Haar probability measure on S^1 .

Proof. Step 1 — the coefficients of m . By Proposition 13.4, for $n \neq 0$,

$$c_n(m) = \frac{1}{2\pi} \int_0^{2\pi} e^{in\theta} d\theta = \frac{1}{2\pi} \left[\frac{e^{in\theta}}{in} \right]_0^{2\pi} = \frac{e^{2\pi in} - 1}{2\pi in} = 0,$$

because $e^{2\pi in} = 1$ for integer n ; and $c_0(m) = \frac{1}{2\pi} \int_0^{2\pi} 1 d\theta = 1$.

Step 2 — the coefficients of μ . As in Step 3 of the proof of Theorem 13.3, the invariance hypothesis gives $\int f(wz) d\mu(z) = \int f d\mu$ for every bounded Borel f and every $w \in S^1$. Applying this to $f = \chi_n$, that is $f(z) = z^n$:

$$c_n(\mu) = \int z^n d\mu(z) = \int (wz)^n d\mu(z) = w^n \int z^n d\mu(z) = w^n c_n(\mu) \quad \text{for every } w \in S^1.$$

Let $n \neq 0$ and choose $w = e^{i\pi/n}$: then $w^n = e^{i\pi} = -1$, and the relation becomes $c_n(\mu) = -c_n(\mu)$, that is $c_n(\mu) = 0$. For $n = 0$, $c_0(\mu) = \mu(S^1) = 1$.

Step 3 — conclusion. The coefficients of μ and of m agree for every $n \in \mathbb{Z}$; by Lemma 13.1, $\mu = m$. $\square$

Remark 13.3: Both hypotheses are needed. (a) Invariance under a proper subgroup is not enough. Let $\omega = e^{2\pi i/3}$ and $\mu_* = \frac{1}{3}(\delta_1 + \delta_\omega + \delta_{\omega^2})$, where δ_z is the Dirac mass at z . Then μ_* is a probability measure invariant under all rotations of the subgroup μ_3 , but not under the others, and certainly $\mu_* \neq m$ (the first is concentrated on three points, the second does not charge points). Its coefficients are computed in Example 13.2: they equal 1 if $3 \mid n$ and 0 otherwise, and indeed they do not agree with those of m . (b) An arbitrary invariant measure is not enough either. The counting measure on S^1 is rotation-invariant, but it is not finite: without the normalisation to a probability, uniqueness trivially fails (every positive multiple of m is invariant). The correct formulation in general is: the Haar measure is unique up to a positive multiplicative constant, and it is made unique by the normalisation $\mu(S^1) = 1$.

Example 13.2: Coefficients of the equidistributed measure on μ_3 , computed twice. With $\omega = e^{2\pi i/3}$ and $\mu_* = \frac{1}{3}(\delta_1 + \delta_\omega + \delta_{\omega^2})$, by definition

$$c_n(\mu_*) = \frac{1}{3}(1 + \omega^n + \omega^{2n}).$$

Symbolic computation. If $3 \mid n$, then $\omega^n = \omega^{2n} = 1$ and $c_n = 1$. If $3 \nmid n$, then $\omega^n \neq 1$ and the geometric sum gives

$$1 + \omega^n + \omega^{2n} = \frac{\omega^{3n} - 1}{\omega^n - 1} = \frac{1 - 1}{\omega^n - 1} = 0,$$

because $\omega^{3n} = (\omega^3)^n = 1$. Hence $c_n = 0$. *Independent check in double precision* (arithmetic mean of the three powers, `numpy`), for $n = 0, \dots, 6$:

$$1, 0, 0, 1, 0, 0, 1$$

with imaginary part zero within 10^{-16} and real part exact to 12 decimal digits after rounding. $\checkmark$ Comparison with Step 1 of the proof of Theorem 13.4 makes visible what distinguishes μ_* from m : the nonzero coefficients of index $\neq 0$.

Remark 13.4: What we import: the general Haar measure (panorama). On every locally compact Hausdorff topological group G there exists a nonzero Radon measure invariant under left translations, and it is unique up to a positive factor: this is **Haar's theorem** (Haar 1933 for the second-countable case, Weil and Cartan for the general case). If G is compact it is normalised to a probability, as here; if G is abelian, left and right invariance coincide. This statement is **not** proved in this treatise: it is declared imported. What is missing is the general construction (by coverings and passage to the limit, or via the Riesz representation theorem applied to an invariant functional obtained by a fixed-point/compactness argument). Theorem 13.4 of this section is its specialisation to $G = S^1$, and it is proved in full: the proof uses nothing from the general theory, only characters and Stone–Weierstrass. Sources: Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2, §2.2 “Haar Measure”; Rudin, *Fourier Analysis on Groups*, Chapter 1, §1.1.

13.4. The characters of S^1

Definition 13.5: Character of an abelian topological group. Let G be an abelian topological group. A **character** of G is a continuous homomorphism $\chi : G \rightarrow S^1$, that is, a continuous function with $\chi(gh) = \chi(g)\chi(h)$ for all $g, h \in G$. The set of characters, with the pointwise product $(\chi\chi')(g) = \chi(g)\chi'(g)$, is denoted $\hat{G}$ and is called the **dual group** of G .

Lemma 13.2: Continuous additive functions on $\mathbb{R}$ are linear. Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $\psi(s+t) = \psi(s) + \psi(t)$ for all s, t . Then $\psi(t) = at$ with $a = \psi(1)$.

Proof. **Step 1 — integers.** From $\psi(0+0) = 2\psi(0)$ it follows that $\psi(0) = 0$. By induction, $\psi(nt) = n\psi(t)$ for every $n \in \mathbb{N}$: true for $n = 0$, and $\psi((n+1)t) = \psi(nt) + \psi(t) = (n+1)\psi(t)$. From $0 = \psi(0) = \psi(t) + \psi(-t)$ it follows that $\psi(-t) = -\psi(t)$, hence $\psi(nt) = n\psi(t)$ holds for every $n \in \mathbb{Z}$.

Step 2 — rationals. Let $q \in \mathbb{N}$, $q \geq 1$. From $\psi(1) = \psi(q \cdot \frac{1}{q}) = q\psi(\frac{1}{q})$ it follows that $\psi(\frac{1}{q}) = \frac{a}{q}$ with $a = \psi(1)$. Hence for $p \in \mathbb{Z}$, $\psi(\frac{p}{q}) = p\psi(\frac{1}{q}) = a\frac{p}{q}$: we get $\psi(r) = ar$ for every $r \in \mathbb{Q}$.

Step 3 — continuity. Let $t \in \mathbb{R}$ and $(r_k) \subset \mathbb{Q}$ with $r_k \rightarrow t$ ($\mathbb{Q}$ is dense in $\mathbb{R}$). By continuity of ψ and of multiplication, $\psi(t) = \lim_k \psi(r_k) = \lim_k ar_k = at$. $\square$

Theorem 13.5: Classification of the characters of S^1 . The characters of S^1 are exactly the power maps

$$\chi_n : S^1 \rightarrow S^1, \quad \chi_n(z) = z^n, \quad n \in \mathbb{Z},$$

and they are pairwise distinct. In angular coordinates, $\chi_n(e^{i\theta}) = e^{in\theta}$.

Proof. **Step 1 — every χ_n is a character.** χ_n is continuous (for $n \geq 0$ it is a polynomial in z ; for $n < 0$ it is $z \mapsto \bar{z}^{|n|}$, continuous by Theorem 13.1), takes values in S^1 because $|z^n| = |z|^n = 1$, and is a homomorphism: $(zw)^n = z^n w^n$ by commutativity of $\mathbb{C}$.

Step 2 — they are distinct. If $\chi_n = \chi_m$ with $n \neq m$, then $z^{n-m} = 1$ for every $z \in S^1$; setting $k = n - m \neq 0$ and $z = e^{i\pi/k}$ gives $e^{i\pi} = -1 = 1$, a contradiction.

Step 3 — reduction to the line. Let χ be a character of S^1 and set

$$\phi = \chi \circ E : \mathbb{R} \rightarrow S^1, \quad \phi(t) = \chi(e^{it}).$$

ϕ is continuous (composition of continuous functions) and, since E and χ are homomorphisms,

$\phi(s+t) = \chi(E(s)E(t)) = \phi(s)\phi(t)$; in particular $\phi(0) = \chi(1) = 1$.

Step 4 — lifting. By Corollary 12.3 (lifting of continuous maps defined on $\mathbb{R}$, which is connected, simply connected and locally path-connected) there exists a unique continuous $\psi : \mathbb{R} \rightarrow \mathbb{R}$ with

$$E \circ \psi = \phi, \quad \psi(0) = 0.$$

Step 5 — the additivity defect is locally constant. Define

$$D : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad D(s, t) = \psi(s+t) - \psi(s) - \psi(t).$$

D is continuous. Moreover, since E is a homomorphism,

$$E(D(s, t)) = \frac{E(\psi(s+t))}{E(\psi(s))E(\psi(t))} = \frac{\phi(s+t)}{\phi(s)\phi(t)} = 1,$$

by Step 3. Hence $D(s, t) \in E^{-1}(1) = 2\pi\mathbb{Z}$ for every (s, t) : D is a continuous function from $\mathbb{R}^2$, which is connected, into the discrete set $2\pi\mathbb{Z}$. Such a function is constant: if it took two distinct values $2\pi j \neq 2\pi l$, the sets $D^{-1}(\{2\pi j\})$ and $D^{-1}(\mathbb{R} \setminus \{2\pi j\})$ would be open (the second preimage is open because $\{2\pi j\}$ is closed and, since $2\pi\mathbb{Z}$ is discrete and closed in $\mathbb{R}$, $\{2\pi j\}$ is also open in $2\pi\mathbb{Z}$: its preimage is open in $\mathbb{R}^2$), nonempty and disjoint, with union $\mathbb{R}^2$, contradicting connectedness. Since $D(0, 0) = \psi(0) - 2\psi(0) = 0$, the constant is 0: ψ is additive.

Step 6 — ψ is linear. By Lemma 13.2, $\psi(t) = at$ with $a = \psi(1) \in \mathbb{R}$. Hence $\phi(t) = E(at) = e^{iat}$.

Step 7 — periodicity quantises a . Since $E(2\pi) = 1$,

$$1 = \chi(1) = \chi(E(2\pi)) = \phi(2\pi) = e^{2\pi ia},$$

hence $2\pi a \in E^{-1}(1) = 2\pi\mathbb{Z}$, that is $a = n$ with $n \in \mathbb{Z}$.

Step 8 — conclusion. For every $z \in S^1$ there exists θ with $z = e^{i\theta}$ (E is surjective), and then $\chi(z) = \phi(\theta) = e^{in\theta} = (e^{i\theta})^n = z^n = \chi_n(z)$. Hence $\chi = \chi_n$. $\square$

***Remark 13.5: Two proofs, two readings.** In Volume II, Chapter 4, the same theorem is proved by analytic means: one shows that every character is differentiable (by an integral-mean trick), solves the equation $\varphi' = a\varphi$ and concludes with periodicity. The proof given here is topological: it uses no regularity at all, only the existence and uniqueness of the lifting (Chapter 12) and the connectedness of $\mathbb{R}^2$. The two roads illuminate different things: the first says that a character is automatically smooth, the second says that the quantisation of the exponent n is the winding number of the character. Indeed $\deg \chi_n = n$ (Chapter 12, Theorem 12.8): the dual $\widehat{S^1} \cong \mathbb{Z}$ and the fundamental group $\pi_1(S^1) \cong \mathbb{Z}$ are the same $\mathbb{Z}$, read once algebraically and once topologically.*

Theorem 13.6: Orthogonality relations. For all $n, m \in \mathbb{Z}$,

$$\int_{S^1} \chi_n(z) \overline{\chi_m(z)} dm(z) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \delta_{nm} = \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases}$$

In particular $\{\chi_n\}_{n \in \mathbb{Z}}$ is an orthonormal system in $L^2(S^1, m)$.

Proof. On S^1 we have $\overline{\chi_m(z)} = \overline{z^m} = z^{-m}$, hence $\chi_n \overline{\chi_m} = \chi_{n-m}$. By Proposition 13.4 the integral equals $c_{n-m}(m)$, which by Step 1 of the proof of Theorem 13.4 is 1 if $n - m = 0$ and 0 otherwise.

Alternatively, and without using anything from the previous section: if $n = m$ the integrand is 1 and the mean integral equals 1; if $k = n - m \neq 0$,

$$\frac{1}{2\pi} \int_0^{2\pi} e^{ik\theta} d\theta = \frac{1}{2\pi} \cdot \frac{e^{2\pi ik} - 1}{ik} = 0.$$

Orthonormality in $L^2(S^1, m)$ is the same equality written as $\langle \chi_n, \chi_m \rangle = \delta_{nm}$, with the inner product $\langle f, g \rangle = \int f \bar{g} dm$. $\square$

Example 13.3: Orthogonality checked with the periodic trapezoidal rule. We approximate $\int \chi_n \overline{\chi_m} dm$ by the average over the equispaced nodes $\theta_k = 2\pi k/N$, $k = 0, \dots, N-1$:

$$Q_N(n, m) = \frac{1}{N} \sum_{k=0}^{N-1} e^{i(n-m)2\pi k/N}.$$

Symbolic computation. Set $k_0 = n - m$ and $\zeta = e^{2\pi i k_0/N}$: if $N \mid k_0$ then $\zeta = 1$ and $Q_N = 1$; otherwise $\zeta \neq 1$ and the geometric sum gives $Q_N = \frac{1}{N} \cdot \frac{\zeta^N - 1}{\zeta - 1} = 0$ because $\zeta^N = e^{2\pi i k_0} = 1$. The periodic trapezoidal rule is thus *exact* on every character, provided $N > |n - m|$.
Numerical check (`numpy`, double precision), for $N \in \{8, 16, 64, 1024\}$:

$$Q_N(3, 3) = 1.0000000000000000, \quad Q_N(-2, -2) = 1.0000000000000000,$$

$$|Q_N(3, 5)| \leq 1.0 \cdot 10^{-15}, \quad |Q_N(0, 7)| \leq 1.0 \cdot 10^{-15}, \quad |Q_N(2, -2)| \leq 1.0 \cdot 10^{-15}.$$

The zero values are zero within rounding error, the unit values are exact to 15 digits. $\checkmark$

Corollary 13.1: $\widehat{S^1} \cong \mathbb{Z}$. $\widehat{S^1}$, with the pointwise product, is an abelian group, and

$$\Phi : \mathbb{Z} \rightarrow \widehat{S^1}, \quad \Phi(n) = \chi_n,$$

is a group isomorphism: $\widehat{S^1} \cong \mathbb{Z}$.

Proof. The pointwise product of two characters is a character ($\chi\chi'$ is continuous with values in S^1 and $(\chi\chi')(zw) = \chi(z)\chi(w)\chi'(z)\chi'(w) = (\chi\chi')(z)(\chi\chi')(w)$, using the commutativity of $\mathbb{C}$); the constant character $\chi_0 \equiv 1$ is the identity; the inverse of χ is $\bar{\chi}$, which is a character. Hence $\widehat{S^1}$ is an abelian group. Φ is a homomorphism: $\chi_n \chi_m(z) = z^n z^m = z^{n+m} = \chi_{n+m}(z)$, that is $\Phi(n)\Phi(m) = \Phi(n+m)$. It is injective by Step 2 of the proof of Theorem 13.5 (equivalently: $\Phi(n) = \chi_0$ forces $n = 0$). It is surjective by Theorem 13.5. $\square$

Proposition 13.5: The dual of S^1 is discrete. For $n \neq m$, $\|\chi_n - \chi_m\|_\infty = \sup_{z \in S^1} |z^n - z^m| = 2$. Consequently, with the topology of uniform convergence (which on a compact group coincides with the compact-open topology), $\widehat{S^1}$ is discrete: the isomorphism of Corollary 13.1 is an isomorphism of topological groups between the discrete group $\mathbb{Z}$ and $\widehat{S^1}$.

Proof. Let $k = n - m \neq 0$. Since $|z^m| = 1$, $|z^n - z^m| = |z^m| |z^k - 1| = |z^k - 1|$. The map $z \mapsto z^k$ is surjective on S^1 (given $w = e^{i\beta}$, take $z = e^{i\beta/k}$), so $\sup_z |z^k - 1| = \sup_{u \in S^1} |u - 1| = 2$, the maximum being attained at $u = -1$. Hence two distinct characters are exactly 2 apart in uniform norm: the ball of radius 1 around χ_n contains only χ_n , and every singleton is open.
Numerical check: on a grid of 2 000 001 points of $[0, 2\pi]$, $\max_z |z^n - z^m| = 2.0000000000$ for $(n, m) = (0, 1), (2, 5), (-3, 4)$. $\checkmark$ $\square$

Proposition 13.6: $\widehat{\mathbb{Z}} \cong S^1$. Consider $\mathbb{Z}$ with the discrete topology (a locally compact abelian topological group). Then

$$\Psi : \widehat{\mathbb{Z}} \rightarrow S^1, \quad \Psi(\chi) = \chi(1),$$

is a group isomorphism, with inverse $w \mapsto (k \mapsto w^k)$; and it is a homeomorphism for the topology of pointwise convergence on $\widehat{\mathbb{Z}}$. Hence $\widehat{\mathbb{Z}} \cong S^1$.

Proof. Step 1 — every homomorphism is determined by its value at 1. Let $\chi : \mathbb{Z} \rightarrow S^1$ be a homomorphism (continuity is automatic: on the discrete $\mathbb{Z}$ every function is continuous). Set $w = \chi(1) \in S^1$; by induction $\chi(k) = w^k$ for $k \geq 0$ (with $\chi(0) = 1$), and $\chi(-k) = \chi(k)^{-1} = w^{-k}$. Hence χ is determined by w .

Step 2 — every w gives a homomorphism. For fixed $w \in S^1$, $k \mapsto w^k$ is a homomorphism $\mathbb{Z} \rightarrow S^1$ (laws of exponents), continuous because $\mathbb{Z}$ is discrete. Hence Ψ is bijective.

Step 3 — Ψ is a homomorphism. $(\chi\chi')(1) = \chi(1)\chi'(1)$, that is $\Psi(\chi\chi') = \Psi(\chi)\Psi(\chi')$.

Step 4 — topology. On $\widehat{\mathbb{Z}}$ we use pointwise convergence (which here coincides with uniform convergence on compact subsets of $\mathbb{Z}$, that is, on finite subsets). If $\chi_j \rightarrow \chi$ pointwise, in particular $\chi_j(1) \rightarrow \chi(1)$, that is $\Psi(\chi_j) \rightarrow \Psi(\chi)$: Ψ is continuous. Conversely, if $w_j \rightarrow w$ in S^1 , then for every fixed k , $w_j^k \rightarrow w^k$ (continuity of $z \mapsto z^k$), that is, the corresponding characters converge pointwise: Ψ^{-1} is continuous. $\square$

Remark 13.6: What we import: Pontryagin duality (panorama). Corollary 13.1 and Proposition 13.6 are the two sides of a single general statement, here declared **imported**: for every locally compact abelian group G , the dual $\widehat{G}$ (with the compact-open topology) is itself locally compact abelian, and the canonical map $G \rightarrow \widehat{\widehat{G}}$, $g \mapsto (\chi \mapsto \chi(g))$, is an isomorphism of topological groups (**Pontryagin duality theorem**, Pontryagin 1934, van Kampen 1935). Under this duality: G compact $\iff \widehat{G}$ discrete, and this is precisely the pair $(S^1, \mathbb{Z})$ verified above; moreover $\widehat{\mathbb{R}} \cong \mathbb{R}$ and $\widehat{\mathbb{Z}/n\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$. What is missing here is: the construction of the compact-open topology on $\widehat{G}$ with the proof that it makes $\widehat{G}$ locally compact, the abstract Plancherel theorem and the inversion theorem, which are the tools of the proof. By checking $\widehat{\widehat{S^1}} = \widehat{\mathbb{Z}} \cong S^1$ with the two results of this section, however, the duality has been verified in the case that matters for the treatise. Sources: Rudin, *Fourier Analysis on Groups*, Chapter 1, §1.2 and §1.7; Folland, *A Course in Abstract Harmonic Analysis*, Chapter 4.

Remark 13.7: Why Fourier series are “forced”. Putting together the three results of the section gives the structural explanation of Fourier analysis on the circle. (i) There is one and only one invariant notion of “average” on S^1 : the measure m (Theorem 13.4). (ii) The only functions that turn the group operation into a product are the χ_n (Theorem 13.5). (iii) With respect to that average, those functions are orthonormal (Theorem 13.6). The Fourier coefficients $\hat{f}(n) = \int f \overline{\chi_n} dm$ are thus not an ad hoc construction, but the only decomposition compatible with the group structure. That the system is also complete — that is, that $\hat{f}(n) = 0$ for every n implies $f = 0$ — does not follow from the orthogonality relations and requires an additional argument: this is the content of Chapter 17 (and, for measures, of Lemma 13.1 proved above, which is the “for measures” version of the same completeness, obtained from Stone–Weierstrass).

13.5. Closed subgroups and dense subgroups

Lemma 13.3: Closed subgroups of $\mathbb{R}$. Let $G \subseteq \mathbb{R}$ be a closed subgroup of the additive group $(\mathbb{R}, +)$. Then exactly one of the following three cases holds:

- (i) $G = \{0\}$;
- (ii) $G = a\mathbb{Z}$ for a unique $a > 0$, precisely $a = \inf (G \cap (0, +\infty))$;
- (iii) $G = \mathbb{R}$.

Proof. Suppose $G \neq \{0\}$. Then G contains an element $g \neq 0$ and, being a subgroup, also contains $-g$: hence $G \cap (0, +\infty) \neq \emptyset$ and $a := \inf (G \cap (0, +\infty)) \geq 0$ is well defined.

Case $a > 0$. We first show $a \in G$. By definition of infimum there is a sequence $g_k \in G \cap (0, +\infty)$ with $g_k \rightarrow a$; since G is closed, $a \in G$. Hence $a\mathbb{Z} \subseteq G$ (G is a subgroup). Conversely let $g \in G$ and set $n = \lfloor g/a \rfloor$, that is, the unique integer with $n \leq g/a < n+1$. Then $r = g - na \in G$ (a difference of elements of G) and $0 \leq r < a$. If $r > 0$ held, we would have $r \in G \cap (0, +\infty)$ with $r < a$, contradicting the definition of a as infimum. Hence $r = 0$ and $g = na \in a\mathbb{Z}$: $G = a\mathbb{Z}$. The uniqueness of a is clear, since $a = \min (a\mathbb{Z} \cap (0, +\infty))$.

Case $a = 0$. We show that G is dense, and hence, being closed, $G = \mathbb{R}$. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Since $\inf (G \cap (0, +\infty)) = 0$, there exists $g \in G$ with $0 < g < \varepsilon$. Let $n = \lfloor x/g \rfloor$: then $ng \in G$ and

$$0 \leq x - ng < g < \varepsilon.$$

Hence every x is approximable by elements of G up to ε : G is dense in $\mathbb{R}$, and $\overline{G} = G = \mathbb{R}$.

The three cases are mutually exclusive: in (i) G is finite, in (ii) G is infinite but discrete (two distinct elements of it are at least a apart), in (iii) G is not discrete. $\square$

Theorem 13.7: Closed subgroups of S^1 . Let $H \subseteq S^1$ be a closed subgroup. Then $H = S^1$ or there exists a unique $n \geq 1$ such that

$$H = \mu_n = \{z \in S^1 : z^n = 1\} = \{e^{2\pi i k/n} : k = 0, 1, \dots, n-1\}.$$

In particular every proper closed subgroup of S^1 is finite and cyclic, generated by $e^{2\pi i/n}$.

Proof. Step 1 — H is lifted to the line. Set $G = E^{-1}(H) \subseteq \mathbb{R}$. This is a subgroup: if $E(s), E(t) \in H$ then $E(s \pm t) = E(s)E(t)^{\pm 1} \in H$ because E is a homomorphism (Proposition 13.2) and H is a subgroup. It is closed, because E is continuous and H is closed. It contains $2\pi\mathbb{Z}$, because $E(2\pi k) = 1 \in H$. In particular $G \neq \{0\}$. Finally $E(G) = H$, because E is surjective and $G = E^{-1}(H)$.

Step 2 — Lemma 13.3 is applied. Since $G \neq \{0\}$, either $G = \mathbb{R}$ or $G = a\mathbb{Z}$ with $a > 0$.

If $G = \mathbb{R}$: $H = E(\mathbb{R}) = S^1$.

If $G = a\mathbb{Z}$: from $2\pi \in G$ it follows that $2\pi = an$ for some integer n , necessarily $n \geq 1$ because $a > 0$ and $2\pi > 0$. Hence $a = 2\pi/n$ and

$$H = E(a\mathbb{Z}) = \{e^{2\pi i k/n} : k \in \mathbb{Z}\} = \{e^{2\pi i k/n} : k = 0, \dots, n-1\},$$

because $e^{2\pi i(k+n)/n} = e^{2\pi i k/n} e^{2\pi i} = e^{2\pi i k/n}$. This set has exactly n elements (if $e^{2\pi i k/n} = e^{2\pi i j/n}$ with $0 \leq j < k < n$, then $2\pi(k-j)/n \in 2\pi\mathbb{Z}$, that is $n \mid k-j$, that is $k=j$) and it is exactly $\{z : z^n = 1\} = \mu_n$: indeed $(e^{2\pi i k/n})^n = e^{2\pi i k} = 1$, and conversely $z^n = 1$ with $z = e^{i\theta}$ gives $n\theta \in 2\pi\mathbb{Z}$, that is $\theta = 2\pi k/n$. It is cyclic, generated by $e^{2\pi i/n}$.

Step 3 — uniqueness of n and exclusivity of the cases. μ_n has n elements, hence n is determined by H . And $\mu_n \neq S^1$ because S^1 is infinite (E is injective on $[0, 2\pi)$). $\square$

Corollary 13.2: An infinite closed subgroup is all of S^1 . If $H \subseteq S^1$ is a closed subgroup with infinitely many elements, then $H = S^1$. Equivalently: if H is an arbitrary subgroup with infinitely many elements, then $\overline{H} = S^1$, that is, H is dense.

Proof. The first part follows immediately from Theorem 13.7, because every proper closed subgroup is finite. For the second: the closure of a subgroup of a topological group is a subgroup. Indeed, if $z, w \in \overline{H}$, take sequences $z_k, w_k \in H$ with $z_k \rightarrow z$, $w_k \rightarrow w$ (in the metric space S^1 , the closure is the sequential closure); by continuity of multiplication and inversion (Theorem 13.1) we get $z_k w_k^{-1} \rightarrow zw^{-1}$, and $z_k w_k^{-1} \in H$, hence $zw^{-1} \in \overline{H}$. If H is infinite, $\overline{H}$ is too, and it is closed: hence $\overline{H} = S^1$. $\square$

Lemma 13.4: Dirichlet: how soon one returns near 1. Let $\alpha \in \mathbb{R}$ and $w = e^{2\pi i \alpha}$. Let $\|x\| = \min_{k \in \mathbb{Z}} |x - k|$ be the distance of x from the nearest integer. For every integer $N \geq 1$ there exists an integer n with $1 \leq n \leq N$ such that

$$\|n\alpha\| < \frac{1}{N}, \quad \text{and consequently} \quad |w^n - 1| < \frac{2\pi}{N}.$$

Proof. Step 1 — pigeonhole principle. Consider the $N + 1$ numbers $\{j\alpha\}$ (fractional part), $j = 0, 1, \dots, N$: they all lie in $[0, 1)$. Divide $[0, 1)$ into the N intervals $I_l = [\frac{l}{N}, \frac{l+1}{N})$, $l = 0, \dots, N - 1$, pairwise disjoint and with union $[0, 1)$. Since there are $N + 1$ numbers and N pigeonholes, there exist $0 \leq j < j' \leq N$ with $\{j\alpha\}$ and $\{j'\alpha\}$ in the same I_l , hence

$$|\{j'\alpha\} - \{j\alpha\}| < \frac{1}{N}.$$

Step 2 — from here to $\|n\alpha\|$. Set $n = j' - j$, so that $1 \leq n \leq N$. Since $\{x\} = x - [x]$, we have

$$\{j'\alpha\} - \{j\alpha\} = n\alpha - ([j'\alpha] - [j\alpha]) = n\alpha - k, \quad k \in \mathbb{Z},$$

hence $\|n\alpha\| \leq |n\alpha - k| < \frac{1}{N}$.

Step 3 — from the fractional part to the chord. If $\|n\alpha\| = \eta$, there is $k \in \mathbb{Z}$ with $n\alpha = k + \varepsilon$ and $|\varepsilon| = \eta$; then, since $e^{2\pi i k} = 1$,

$$|w^n - 1| = |e^{2\pi i n\alpha} - 1| = |e^{2\pi i \varepsilon} - 1| = 2|\sin(\pi \varepsilon)| \leq 2\pi|\varepsilon| = 2\pi\eta < \frac{2\pi}{N},$$

having used $|e^{i\vartheta} - 1| = 2|\sin(\vartheta/2)|$ and $|\sin x| \leq |x|$. $\square$

Theorem 13.8: Irrational rotations: density of the orbit. Let $\alpha \in \mathbb{R}$ and $w = e^{2\pi i \alpha}$. Let $\langle w \rangle = \{w^n : n \in \mathbb{Z}\}$ be the generated subgroup.

- (i) If $\alpha = p/q \in \mathbb{Q}$ (reduced fraction, $q \geq 1$), then $\langle w \rangle = \mu_q$: it is finite, closed, and not dense.
- (ii) If $\alpha \notin \mathbb{Q}$, then $\langle w \rangle$ is infinite and dense in S^1 : $\overline{\langle w \rangle} = S^1$. The same holds for the half-orbit $\{w^n : n \geq 1\}$.

Proof. (i) If $\alpha = p/q$ with $\gcd(p, q) = 1$, then $w^q = e^{2\pi i p} = 1$, hence $w \in \mu_q$ and $\langle w \rangle \subseteq \mu_q$. Conversely the order of w is exactly q : if $w^k = 1$ then $kp/q \in \mathbb{Z}$, that is $q \mid kp$, and since $\gcd(p, q) = 1$ it follows that $q \mid k$. Hence $\langle w \rangle$ is cyclic of order q and, having q elements in μ_q , which has q elements, coincides with μ_q . A finite subset of S^1 is closed (a finite union of points in a Hausdorff space) and is not dense because S^1 is infinite.

(ii) Step 1 — $\langle w \rangle$ is infinite. If it were finite, there would be $n \geq 1$ with $w^n = 1$, that is $e^{2\pi i n\alpha} = 1$, that is $n\alpha \in \mathbb{Z}$, that is $\alpha \in \mathbb{Q}$: against the hypothesis. In particular the powers

w^n , $n \in \mathbb{Z}$, are all distinct (if $w^n = w^{n'}$ with $n \neq n'$ then $w^{n-n'} = 1$).

Step 2 — density. $\langle w \rangle$ is an infinite subgroup; by Corollary 13.2, $\overline{\langle w \rangle} = S^1$.

Step 3 — the positive half-orbit suffices. Let $z \in S^1$ and $\varepsilon > 0$. Choose N with $2\pi/N < \varepsilon$ and, by Lemma 13.4, an $M \in \{1, \dots, N\}$ with $|w^M - 1| < 2\pi/N < \varepsilon$. Let δ be the principal argument of w^M , that is the unique $\delta \in (-\pi, \pi]$ with $w^M = e^{i\delta}$. We have $\delta \neq 0$, because $w^M \neq 1$ by Step 1, and

$$|\delta| \leq \frac{\pi}{2} |e^{i\delta} - 1| = \frac{\pi}{2} |w^M - 1| < \frac{\pi}{2} \varepsilon,$$

having used $|e^{i\delta} - 1| = 2|\sin(\delta/2)| \geq \frac{2}{\pi}|\delta|$, valid because $|\sin x| \geq \frac{2}{\pi}|x|$ for $|x| \leq \pi/2$. The points $w^{kM} = e^{ik\delta}$, for $k = 1, 2, \dots, K$ with $K = \lceil 2\pi/|\delta| \rceil$, have arguments advancing in constant steps of size $|\delta|$ in the same direction and covering a full turn: hence every point of S^1 is within $|\delta|$ of one of them in arc length, and a fortiori in Euclidean distance (the chord does not exceed the arc). In particular there exists $k \in \{1, \dots, K\}$ with $|w^{kM} - z| < |\delta| < \frac{\pi}{2}\varepsilon$, and the exponent kM is positive. Since $\varepsilon > 0$ is arbitrary, the half-orbit $\{w^n : n \geq 1\}$ is dense. $\square$

Example 13.4: $\alpha = \sqrt{2}$: returns near 1 and equidistribution. Take $\alpha = \sqrt{2}$, irrational, and $w = e^{2\pi i \sqrt{2}}$.

(a) *Returns.* Looking for the n that improve the record of $|w^n - 1|$ gives the sequence

$$n = 1, 2, 5, 12, 29, 70, 169, 408, 985, 2378, 5741, 13860, 33461,$$

with

$$\begin{aligned} |w^1 - 1| &= 1.9278051, & |w^5 - 1| &= 0.4428317, & |w^{29} - 1| &= 0.0765941, \\ |w^{169} - 1| &= 0.0131446, & |w^{985} - 1| &= 0.0022553, & |w^{33461} - 1| &= 0.0000664. \end{aligned}$$

These n are the Pell numbers P_k ($P_{k+1} = 2P_k + P_{k-1}$, with $1, 2, 5, 12, 29, \dots$), that is, the denominators of the convergents of $\sqrt{2}$: Lemma 13.4 guarantees that $|w^n - 1| < 2\pi/N$ for some $n \leq N$, and indeed $|w^{169} - 1| = 0.01314 < 2\pi/169 = 0.03718$. $\checkmark$

(b) *Exponential sums.* Set $S_N = \sum_{n=1}^N w^n$. By the geometric sum ($w \neq 1$),

$$S_N = w \frac{w^N - 1}{w - 1}, \quad |S_N| = \frac{|w^N - 1|}{|w - 1|} = \frac{|\sin(\pi N \alpha)|}{|\sin(\pi \alpha)|} \leq \frac{1}{|\sin(\pi \alpha)|}.$$

Closed formula versus direct computation (numpy, double precision):

N	$ S_N /N$ (direct sum)	$ S_N /N$ (closed formula)
10^2	0.0100594608	0.0100594608
10^4	0.0000428777	0.0000428777
10^6	0.0000010176	0.0000010176

The two computations agree to 10 digits. $\checkmark$ The mean $\frac{1}{N}S_N \rightarrow 0$ is the numerical check of the fact, proved in Proposition 13.7, that the means of the character χ_1 along the orbit tend to its integral with respect to m , which is 0.

Proposition 13.7: The means of characters along an irrational orbit tend to zero. Let $\alpha \notin \mathbb{Q}$, $w = e^{2\pi i \alpha}$ and $k \in \mathbb{Z}$, $k \neq 0$. Then

$$\left| \frac{1}{N} \sum_{n=1}^N \chi_k(w^n) \right| = \left| \frac{1}{N} \sum_{n=1}^N e^{2\pi i k n \alpha} \right| \leq \frac{1}{N |\sin(\pi k \alpha)|} \xrightarrow{N \rightarrow \infty} 0 = \int_{S^1} \chi_k dm.$$

Proof. Set $v = e^{2\pi i k \alpha}$. Since $\alpha \notin \mathbb{Q}$ and $k \neq 0$, we have $k\alpha \notin \mathbb{Z}$ and hence $v \neq 1$. The geometric sum gives

$$\sum_{n=1}^N v^n = v \frac{v^N - 1}{v - 1}, \quad \left| \sum_{n=1}^N v^n \right| = \frac{|v^N - 1|}{|v - 1|} \leq \frac{2}{|v - 1|} = \frac{2}{2|\sin(\pi k \alpha)|} = \frac{1}{|\sin(\pi k \alpha)|},$$

using $|v - 1| = |e^{2\pi i k \alpha} - 1| = 2|\sin(\pi k \alpha)|$ and $|v^N - 1| \leq 2$. Dividing by N and letting $N \rightarrow \infty$ gives the claim; the value $\int \chi_k dm = 0$ is Theorem 13.6 with $m = 0$. $\square$

Remark 13.8: What we import: Weyl and Kronecker in dimension n (panorama). Proposition 13.7 is the hypothesis of Weyl's criterion; the conclusion — that the sequence (w^n) is not only dense but **equidistributed**, that is $\frac{1}{N} \#\{n \leq N : w^n \in A\} \rightarrow m(A)$ for every arc A — requires **Weyl's equidistribution criterion** (Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Math. Ann. 77 (1916), 313–352), which is here declared **imported**: what is missing is the passage from the means of characters to the means of the indicator functions of arcs, obtained by uniformly approximating the indicators with trigonometric polynomials and controlling the error at the endpoints. Likewise **Kronecker's theorem** in dimension n is declared imported: if $1, \alpha_1, \dots, \alpha_n$ are linearly independent over $\mathbb{Q}$, then $\{(e^{2\pi i k \alpha_1}, \dots, e^{2\pi i k \alpha_n}) : k \in \mathbb{Z}\}$ is dense in the torus $(S^1)^n$. The case $n = 1$ — which is exactly Theorem 13.8 — is proved here, and its proof (via the classification of closed subgroups) is the same one that, with Cartan's theorem in place of Lemma 13.3, generalises to the torus. Sources: Stein–Shakarchi, *Fourier Analysis*, Chapter 4; Katznelson, *An Introduction to Harmonic Analysis*, Chapter VI.

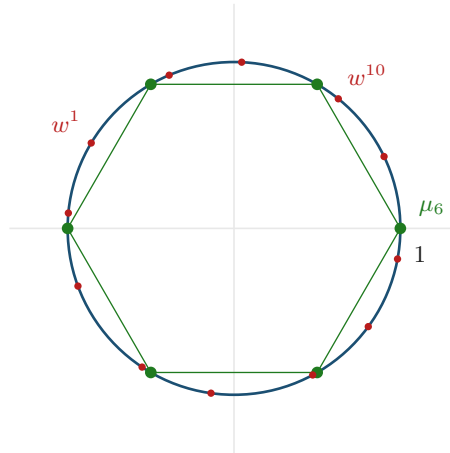


Figure 13.1: The two types of subgroup. In green the closed subgroup μ_6 : six points, vertices of a regular hexagon, with coordinates $(\pm 1, 0)$, $(\pm 0.5, \pm 0.86603)$. In red the first twelve points of the orbit $w^n = e^{2\pi i n \sqrt{2}}$, $n = 1, \dots, 12$: the subgroup they generate is infinite, hence dense (Theorem 13.8), and no point repeats. The coordinates are $(\cos 2\pi n \sqrt{2}, \sin 2\pi n \sqrt{2})$; for example $w^1 = (-0.85822, 0.51329)$ and $w^{10} = (0.62703, 0.77900)$.

13.6. Assigned exercises

Answers are in the Teacher's key (Section 13.9).

Exercise 13.1: Closed subgroups: recognition and counting. *Aim:* to apply Theorem 13.7 in both directions. *Estimated time:* 30 minutes. *Type:* symbolic/proof-based.

- (a) Prove that $\mu_n \cap \mu_k = \mu_{\gcd(n,k)}$ and that the subgroup generated by $\mu_n \cup \mu_k$ is $\mu_{\text{lcm}(n,k)}$.
- (b) Let $H = \{z \in S^1 : z^{12} = 1\} \cup \{e^{2\pi i \sqrt{3}}\}$. Is H a subgroup? Is it closed? Justify.
- (c) Let H be the subgroup generated by $e^{2\pi i/6}$ and $e^{2\pi i/10}$. Determine it, say whether it is closed and how many elements it has.
- (d) Prove that S^1 has no proper closed subgroups of infinite cardinality, but has (non-closed) countable dense subgroups: exhibit one different from that of Theorem 13.8.

Exercise 13.2: An invariant measure “halfway”. *Aim:* to understand exactly where the uniqueness proof breaks down if the hypothesis is weakened. *Estimated time:* 35 minutes. *Type:* symbolic with numerical check.

Let $q \geq 2$ and $\mu_*^{(q)} = \frac{1}{q} \sum_{k=0}^{q-1} \delta_{e^{2\pi i k/q}}$.

- (a) Compute $c_n(\mu_*^{(q)})$ for every $n \in \mathbb{Z}$ and prove the formula.
- (b) Check that $\mu_*^{(q)}$ is invariant under the rotations of μ_q and determine exactly for which $w \in S^1$ one has $\mu_*^{(q)}(wA) = \mu_*^{(q)}(A)$ for every Borel set A .
- (c) Identify the step of the proof of Theorem 13.4 that fails for $\mu_*^{(q)}$, and explain why.
- (d) Check (a) numerically for $q = 5$ and $n = 0, \dots, 12$, and compare with the formula.

Exercise 13.3: The character as a degree- n map. *Aim:* to connect the dual $\widehat{S^1}$ to the degree of Chapter 12. *Estimated time:* 25 minutes. *Type:* proof-based.

- (a) Prove that the isomorphism $\Phi : \mathbb{Z} \rightarrow \widehat{S^1}$ of Corollary 13.1 satisfies $\deg \Phi(n) = n$, where $\deg$ is the degree of Definition 12.8.
- (b) Deduce that two characters are homotopic (as continuous maps $S^1 \rightarrow S^1$) if and only if they are equal.
- (c) Prove that the only character that extends continuously to the disc D^2 is χ_0 .
- (d) Can a character have fixed points? Determine the fixed points of χ_n and compare with the worked exercise on fixed points of Chapter 12.

13.7. Worked exercises

Exercise 13.4: Continuous homomorphisms $\mathbb{R} \rightarrow S^1$. *Type:* symbolic/proof-based. Prove that the continuous homomorphisms $\phi : (\mathbb{R}, +) \rightarrow S^1$ are exactly those of the form $\phi(t) = e^{i\xi t}$ with $\xi \in \mathbb{R}$, and hence that $\widehat{\mathbb{R}} \cong \mathbb{R}$. Explain where the proof departs from that of Theorem 13.5 and why the exponent is quantised there but not here.

Solution. Strategy. We repeat the proof of Theorem 13.5 up to Step 6 (lifting and linearity) and observe that Step 7 — periodicity — has no analogue here.

Step 1 — every $e^{i\xi t}$ is a continuous homomorphism. $|e^{i\xi t}| = 1$ and $e^{i\xi(s+t)} = e^{i\xi s} e^{i\xi t}$; continuity is that of E composed with $t \mapsto \xi t$.

Step 2 — lifting. Let $\phi : \mathbb{R} \rightarrow S^1$ be a continuous homomorphism. By Corollary 12.3 there exists a continuous $\psi : \mathbb{R} \rightarrow \mathbb{R}$ with $E \circ \psi = \phi$ and $\psi(0) = 0$.

Step 3 — additivity. As in Step 5 of Theorem 13.5: $D(s, t) = \psi(s+t) - \psi(s) - \psi(t)$ is continuous, with values in $2\pi\mathbb{Z}$ because $E(D) = \phi(s+t)/(\phi(s)\phi(t)) = 1$, and $\mathbb{R}^2$ is connected, hence $D \equiv D(0, 0) = 0$.

Step 4 — linearity. By Lemma 13.2, $\psi(t) = \xi t$ with $\xi = \psi(1) \in \mathbb{R}$, hence $\phi(t) = e^{i\xi t}$.

Step 5 — injectivity of the parametrisation. If $e^{i\xi t} = e^{i\xi' t}$ for every t , then $(\xi - \xi')t \in 2\pi\mathbb{Z}$ for every $t \in \mathbb{R}$; but $t \mapsto (\xi - \xi')t$ is continuous on the connected $\mathbb{R}$ with values in the

discrete set $2\pi\mathbb{Z}$, hence constant, and equals 0 at $t = 0$: so $(\xi - \xi')t = 0$ for every t , that is $\xi = \xi'$. The map $\xi \mapsto (t \mapsto e^{i\xi t})$ is thus a bijection of $\mathbb{R}$ onto $\widehat{\mathbb{R}}$, and it is a homomorphism because $e^{i\xi t} e^{i\xi' t} = e^{i(\xi+\xi')t}$: $\widehat{\mathbb{R}} \cong \mathbb{R}$.

Where it departs from the theorem. In Theorem 13.5 the domain is $S^1 = \mathbb{R}/2\pi\mathbb{Z}$: the character is defined *on the quotient*, hence $\phi = \chi \circ E$ inherits 2π -periodicity and the condition $\phi(2\pi) = \chi(1) = 1$ forces $2\pi\xi \in 2\pi\mathbb{Z}$, that is $\xi = n \in \mathbb{Z}$. Here the domain is $\mathbb{R}$, which has no periods: no condition, and ξ remains free in $\mathbb{R}$.

Independent check. We verify this on a case: $\xi = 3$ gives $\phi(t) = e^{3it}$, which restricted to $2\pi\mathbb{Z}$ equals $e^{6\pi ik} = 1$: consistent with the fact that for an *integer* ξ the homomorphism descends to the quotient and defines a character of S^1 (precisely χ_3). For $\xi = 3.5$, $\phi(2\pi) = e^{7\pi i} = -1 \neq 1$: it does not descend to the quotient. ✓

Interpretation. $\widehat{\mathbb{R}} \cong \mathbb{R}$ (continuous frequencies: the Fourier transform, Chapter 19) and $\widehat{S^1} \cong \mathbb{Z}$ (discrete frequencies: Fourier series). Compactness of the domain $\leftrightarrow$ discreteness of the spectrum: this is the general statement recalled in Remark 13.6, here verified in the two cases the treatise actually uses.

Typical errors. (1) Deducing the differentiability of ϕ “because it is a homomorphism”: an argument is needed (the analytic one of Volume II, Chapter 4, or the lifting used here). (2) Concluding $\xi \in \mathbb{Z}$ even for the domain $\mathbb{R}$, wrongly importing periodicity. (3) Forgetting $\psi(0) = 0$ in Step 2: without normalisation the lifting is not unique and $D(0, 0)$ is not zero.

Extension. The same scheme classifies the continuous homomorphisms $S^1 \rightarrow \mathbb{R}$: they are all zero. Indeed the image is a compact connected subgroup of $\mathbb{R}$ (continuous image of a compact connected set), hence a closed bounded interval that is a subgroup: only $\{0\}$.

Exercise 13.5: Uniqueness of the invariant measure: check on a family of candidates. *Type:* numerical/symbolic. For $\rho \in [0, 1)$ consider the probability measure P_ρ on S^1 with density with respect to m

$$p_\rho(e^{i\theta}) = \frac{1 - \rho^2}{1 - 2\rho \cos \theta + \rho^2},$$

that is $dP_\rho = p_\rho dm$ (the Poisson kernel, Chapter 16).

- Check that P_ρ is a probability measure.
- Compute $c_n(P_\rho)$ for every $n \in \mathbb{Z}$.
- Deduce for which ρ the measure P_ρ is rotation-invariant, and thus recover Theorem 13.4 within this family.
- Check (b) numerically for $\rho = 0.5$ and $n = 0, 1, 2, 3$.

Solution. Strategy. We compute the coefficients with the geometric expansion of the Poisson kernel and apply Lemma 13.1.

Step 1 — the expansion. For $|\rho| < 1$ and $z = e^{i\theta}$,

$$\frac{1 - \rho^2}{|1 - \rho z|^2} = \frac{1 - \rho^2}{(1 - \rho z)(1 - \rho \bar{z})} = \frac{1}{1 - \rho z} + \frac{1}{1 - \rho \bar{z}} - 1,$$

verifiable by putting everything over a common denominator: $(1 - \rho \bar{z}) + (1 - \rho z) - (1 - \rho z)(1 - \rho \bar{z}) = 1 - \rho^2$ (using $z\bar{z} = 1$). Since $|1 - \rho z|^2 = 1 - 2\rho \cos \theta + \rho^2$, the left-hand side is p_ρ . Expanding in geometric series, uniformly convergent because $|\rho z| = \rho < 1$,

$$p_\rho(e^{i\theta}) = \sum_{k \geq 0} \rho^k z^k + \sum_{k \geq 0} \rho^k \bar{z}^k - 1 = \sum_{k \in \mathbb{Z}} \rho^{|k|} z^k.$$

Step 2 — (a) and (b). Integrating term by term (legitimate by uniform convergence) and using orthogonality (Theorem 13.6):

$$c_n(P_\rho) = \int z^n p_\rho dm = \sum_{k \in \mathbb{Z}} \rho^{|k|} \int z^{n+k} dm = \rho^{|n|}.$$

In particular $c_0(P_\rho) = 1$, that is $P_\rho(S^1) = 1$; since moreover $p_\rho > 0$ (the denominator is $|1 - \rho z|^2 > 0$ and the numerator is $1 - \rho^2 > 0$), P_ρ is a probability measure: (a) is proved.

Step 3 — (c). If P_ρ were invariant, by Step 2 of the proof of Theorem 13.4 we would have $c_n(P_\rho) = 0$ for every $n \neq 0$, that is $\rho^{|n|} = 0$: impossible for $\rho > 0$. Hence the only invariant one in the family is P_0 , which has density $p_0 \equiv 1$, that is $P_0 = m$: consistent with Theorem 13.4, which says that m is the unique invariant measure absolutely.

Step 4 — (d), numerical check. For $\rho = 0.5$ the formula gives

$$c_0 = 1, \quad c_1 = 0.5, \quad c_2 = 0.25, \quad c_3 = 0.125.$$

Quadrature with the periodic trapezoidal rule on $N = 4096$ nodes of $\frac{1}{2\pi} \int_0^{2\pi} e^{in\theta} p_{0.5}(e^{i\theta}) d\theta$:

$$1.000000000000, \quad 0.500000000000, \quad 0.250000000000, \quad 0.125000000000,$$

with imaginary parts $< 10^{-16}$. ✓ (The periodic quadrature is spectrally accurate on analytic periodic functions: here the agreement is at machine precision.)

Interpretation. The family (P_ρ) interpolates between the invariant measure ($\rho = 0$) and the Dirac mass at 1 ($\rho \rightarrow 1^-$): the coefficients go from δ_{n0} to 1 for every n . The “distance from invariance” is measured precisely by the nonzero coefficients of index $\neq 0$, as in Example 13.2.

Typical errors. (1) Confusing $\int p_\rho dm = 1$ with “ p_ρ is invariant”: the first is the normalisation, the second a much stronger property. (2) Integrating term by term without justifying uniform convergence (here legitimate because $\rho < 1$ is fixed). (3) Using $|1 - \rho z|^2 = 1 - 2\rho \cos \theta + \rho^2$ without noting that $|z| = 1$ is needed.

Extension. The same computation shows that $P_\rho \rightarrow \delta_1$ weakly as $\rho \rightarrow 1^-$ (the coefficients tend to $1 = c_n(\delta_1)$ for every n , and one concludes with Lemma 13.1 in its “for sequences” version): this is the *approximate identity* used in Chapter 16.

Exercise 13.6: The subgroup generated by two rotations. *Type:* symbolic/proof-based. Let $\alpha, \beta \in \mathbb{R}$ and let H be the subgroup of S^1 generated by $e^{2\pi i\alpha}$ and $e^{2\pi i\beta}$. Prove that $\overline{H} = S^1$ if and only if at least one among α, β and their integer combinations $j\alpha + k\beta$ is irrational; and that otherwise $H = \mu_q$ for an explicit q .

Solution. Strategy. We lift H to a subgroup of $\mathbb{R}$ and use Lemma 13.3 via Theorem 13.7.

Step 1 — structure of H . By definition $H = \{e^{2\pi i(j\alpha + k\beta)} : j, k \in \mathbb{Z}\}$: indeed the set on the right is a subgroup (it contains 1 for $j = k = 0$, is closed under product and inverse) and contains the two generators, and every subgroup that contains them contains it.

Step 2 — the “all rational” case. Suppose $\alpha = p_1/q_1$ and $\beta = p_2/q_2$ are both rational, written as reduced fractions. By Theorem 13.8(i), $\langle e^{2\pi i\alpha} \rangle = \mu_{q_1}$ and $\langle e^{2\pi i\beta} \rangle = \mu_{q_2}$, hence H is the subgroup generated by $\mu_{q_1} \cup \mu_{q_2}$, which by Assigned exercise 13.1(a) is μ_q with $q = \text{lcm}(q_1, q_2)$. In this case H is finite (it has q elements), closed and not dense.

Step 3 — the “some irrational combination” case. Suppose there exist $j, k \in \mathbb{Z}$ with $\gamma = j\alpha + k\beta \notin \mathbb{Q}$. Then $e^{2\pi i\gamma} \in H$ generates, by Theorem 13.8(ii), a dense subgroup; a fortiori $\overline{H} = S^1$.

Step 4 — the equivalence. If all combinations $j\alpha + k\beta$ are rational, in particular α ($j = 1, k = 0$) and β ($j = 0, k = 1$) are, and we fall back into Step 2: $H = \mu_q$, not dense. If some is irrational, Step 3: H is dense. The two alternatives are exhaustive and mutually exclusive.

Independent check. (i) $\alpha = 1/6, \beta = 1/10$: $q = \text{lcm}(6, 10) = 30$ and indeed $\frac{1}{6} - \frac{1}{10} = \frac{1}{15}$, $\frac{1}{6} - 2 \cdot \frac{1}{10} = -\frac{1}{30}$: $e^{2\pi i/30} \in H$, hence $H = \mu_{30}$, which has 30 elements. (ii) $\alpha = 1/3, \beta = \sqrt{2}/2$: β is irrational and H is dense. (iii) $\alpha = \sqrt{2}, \beta = 1 - \sqrt{2}$: both irrational, but $\alpha + \beta = 1$ is an integer; this does not prevent density, because α alone is irrational. This is why the condition must be stated on *all* combinations and not on the sum alone. ✓

Interpretation. The “finite or dense” alternative admits no middle ground: it is Theorem 13.7 read as a dichotomy. On the torus $(S^1)^2$ the situation is richer (there exist proper closed subgroups of dimension 1: wound-up lines), and that is where Kronecker’s theorem in dimension 2 is needed (Remark 13.8).

Typical errors. (1) Concluding “ α irrational and β irrational $\Rightarrow$...” by reasoning on the sum: see check (iii). (2) Forgetting that H is dense but *not* closed: $\overline{H} = S^1 \neq H$ (because H is countable and S^1 is not). (3) Using $q = q_1 q_2$ instead of the least common multiple.

Extension. The same statement holds for any finite number of generators: H is finite if and only if all the α_i are rational, and then $H = \mu_q$ with q the least common multiple of the denominators.

13.8. Self-assessment quiz

- (Definition) A *character* of the abelian topological group G is
 - a continuous function $G \rightarrow \mathbb{C}$
 - a continuous homomorphism $G \rightarrow S^1$
 - an arbitrary homomorphism $G \rightarrow \mathbb{C}^*$
 - a function with $|\chi| = 1$
- (Hypothesis) In the uniqueness theorem for the invariant measure on S^1 , the hypothesis that makes the statement false if removed is
 - σ -additivity
 - invariance with respect to *all* rotations
 - continuity of E
 - compactness of S^1
- (Symbolic computation) The coefficient c_4 of the measure $\frac{1}{4}(\delta_1 + \delta_i + \delta_{-1} + \delta_{-i})$ equals
 - 0
 - 1
 - $\frac{1}{4}$
 - i
- (Numerical computation) For $\alpha = \sqrt{2}$ and $w = e^{2\pi i \alpha}$, the first $n \geq 1$ for which $|w^n - 1| < 0.02$ is
 - 29
 - 70
 - 169
 - 408
- (Proof step) In the classification of characters, the connectedness of $\mathbb{R}^2$ is used to
 - lift ϕ
 - prove that the additivity defect $D(s, t) \in 2\pi\mathbb{Z}$ is constant
 - prove that E is surjective
 - prove that ψ is differentiable
- (Error diagnosis) A student claims: “ $\mathbb{Q} \cap [0, 2\pi)$ gives, via E , a closed and dense subgroup of S^1 ”. The error is
 - it is not a subgroup
 - it is dense but not closed: a closed dense subgroup would be all of S^1 , which is uncountable
 - it is not dense
 - E is not surjective onto the rationals
- (Transfer) If G is a locally compact abelian group and $\widehat{G}$ is discrete, then
 - G is discrete
 - G is compact
 - G is finite
 - G is connected

8. (Source and metacognition) The general statement “the invariant measure on a locally compact group exists and is unique up to a constant” is found in

A) Folland, *A Course in Abstract Harmonic Analysis*, Ch. 2, §2.2 B) Hatcher, *Algebraic Topology*, §1.1 C) Rudin, *Principles*, Ch. 7 D) Munkres, *Topology*, §53

13.9. Teacher’s key

BEST PRACTICE

Quiz — answers, rationale and diagnosis.

1) **B.** *Rationale:* this is Definition 13.5: all *three* conditions are needed (homomorphism, continuous, valued in the circle). *Distractors:* A forgets homomorphism and codomain; C forgets continuity and allows values of modulus $\neq 1$ (for $G = \mathbb{R}$ it would give $t \mapsto e^{at}$ with a a real and not purely imaginary); D forgets the homomorphism. *Diagnosed error:* confusing “function of modulus 1” with “character”. *Source:* Rudin, *Fourier Analysis on Groups*, Ch. 1, §1.2. *Prerequisite:* definition of homomorphism. *Remediation:* reread Step 1 of the proof of Theorem 13.5.

2) **B.** *Rationale:* Remark 13.3(a) exhibits $\mu_* = \frac{1}{3}(\delta_1 + \delta_\omega + \delta_{\omega^2})$, invariant under the subgroup μ_3 and different from m . *Distractors:* A and C are technical hypotheses needed to give the statement meaning, not to make it true (without σ -additivity one cannot speak of a measure); D is true but is not what is removed (on $\mathbb{R}$, non-compact, uniqueness still holds up to a constant). *Diagnosed error:* settling for invariance under a subgroup. *Source:* Theorem 13.4 and Remark 13.3. *Prerequisite:* subgroups of S^1 . *Remediation:* redo Example 13.2 with $q = 5$.

3) **B.** *Rationale:* $c_4 = \frac{1}{4}(1^4 + i^4 + (-1)^4 + (-i)^4) = \frac{1}{4}(1 + 1 + 1 + 1) = 1$: this is the case $q \mid n$ of the formula of Assigned exercise 13.2(a) with $q = 4$, $n = 4$. *Distractors:* A is the value for n not a multiple of 4 (for example $c_1 = 0$); C confuses the coefficient with the weight of each mass; D confuses c_4 with c_1 of a single mass. *Diagnosed error:* not recognising the divisibility condition. *Source:* Example 13.2. *Prerequisite:* roots of unity. *Remediation:* compute c_n for $n = 0, \dots, 8$ and observe the period 4.

4) **C.** *Rationale:* the record table in Example 13.4 gives $|w^{70} - 1| = 0.0317327$ and $|w^{169} - 1| = 0.0131446$: the first value below 0.02 is reached at $n = 169$. *Distractors:* A gives 0.0766; B gives 0.0317, still above threshold; D is the next record (0.0054), correct but not the *first*. *Diagnosed error:* reading the table without comparing it to the threshold. *Source:* Example 13.4 and Lemma 13.4. *Prerequisite:* $|e^{i\vartheta} - 1| = 2|\sin(\vartheta/2)|$. *Remediation:* redo the computation with α the golden ratio and compare the denominators.

5) **B.** *Rationale:* this is Step 5 of Theorem 13.5: a continuous function from a connected space with values in a discrete set is constant. *Distractors:* A is Step 4 and uses the *simple* connectedness of $\mathbb{R}$, not that of $\mathbb{R}^2$; C is unrelated and belongs to Chapter 12; D is the analytic route of Volume II, not this one. *Diagnosed error:* attributing the wrong role to a topological hypothesis. *Source:* Theorem 13.5, Step 5. *Prerequisite:* connectedness. *Remediation:* rewrite Step 5 exhibiting the two disjoint open sets.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (continued).

6) **B.** *Rationale:* the set $\{e^{2\pi i r} : r \in \mathbb{Q}\}$ is a countable and dense subgroup (it contains every μ_n), but it is not closed: by Theorem 13.7 an infinite closed subgroup is all of S^1 , which has the cardinality of the continuum. *Distractors:* A is false ($\mathbb{Q}$ is a subgroup of $\mathbb{R}$ and E is a homomorphism); C is false (it contains μ_n for every n , hence points arbitrarily close to every z); D makes no sense, E is defined on all of $\mathbb{R}$. *Diagnosed error:* confusing

“dense” with “closed”. *Source*: Theorem 13.7 and Corollary 13.2. *Prerequisite*: closure and density. *Remediation*: check that $\{e^{2\pi ir} : r \in \mathbb{Q}\} = S^1$.

7) **B.** *Rationale*: under Pontryagin duality (Remark 13.6), G compact $\iff \widehat{G}$ discrete; applying duality to $\widehat{G}$ and using $\widehat{\widehat{G}} \cong G$ gives the required implication. The pair verified in this chapter is $\widehat{S^1} = \mathbb{Z}$ (Corollary 13.1) with $\widehat{\mathbb{Z}} = S^1$ (Proposition 13.6). *Distractors*: A reverses the implication (it is G discrete $\Rightarrow \widehat{G}$ compact); C is too strong ($\mathbb{Z}$ is not finite and $\widehat{\mathbb{Z}} = S^1$ is not discrete, but S^1 is compact with discrete infinite dual $\mathbb{Z}$); D is unrelated (finite groups are compact and disconnected). *Diagnosed error*: using duality in only one direction. *Source*: Folland, Ch. 4; Rudin, *Fourier Analysis on Groups*, Ch. 1. *Prerequisite*: notion of the compact-open topology. *Remediation*: check the two pairs $(\mathbb{R}, \mathbb{R})$ and $(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$.

8) **A.** *Rationale*: Haar’s theorem (existence and uniqueness up to a constant) is in Chapter 2 of Folland, §2.2, as recalled in Remark 13.4. *Distractors*: B is the fundamental group of the circle; C is Stone–Weierstrass, which in this chapter is used for Lemma 13.1 but is not Haar’s theorem; D is the definition of a covering. *Diagnosed error*: citing a source used in the chapter in place of the source of the statement. *Source*: see the “Sources” section. *Prerequisite*: knowing how to read a reference with chapter and section. *Remediation*: locate Haar’s statement in the two texts (Folland and Rudin) and compare the two formulations.

BEST PRACTICE

Assigned exercise 13.1 — answers. (a) $z \in \mu_n \cap \mu_k$ means $z^n = z^k = 1$, that is, the order of z divides both n and k , that is, divides $d = \gcd(n, k)$: hence $z^d = 1$ and $\mu_n \cap \mu_k \subseteq \mu_d$; the opposite inclusion is clear because $d \mid n$ and $d \mid k$. For the second point set $L = \text{lcm}(n, k)$. Since $n \mid L$ and $k \mid L$, $\mu_n \cup \mu_k \subseteq \mu_L$, and μ_L is a subgroup: hence the generated subgroup is contained in μ_L . Conversely, setting $n' = L/n$ and $k' = L/k$, $\gcd(n', k') = 1$: if a prime p divided both, L/p would still be a common multiple of n and k (because $n \mid L/p$ follows from $p \mid L/n$, and likewise for k), against the minimality of L . By Bézout’s identity there exist $u, v \in \mathbb{Z}$ with $un' + vk' = 1$, and then

$$\frac{u}{n} + \frac{v}{k} = \frac{un' + vk'}{L} = \frac{1}{L}, \quad \text{hence} \quad e^{2\pi i/L} = (e^{2\pi i/n})^u (e^{2\pi i/k})^v$$

belongs to the generated subgroup, which therefore contains μ_L . In conclusion it is exactly $\mu_{\text{lcm}(n,k)}$. (b) *No*. H is not a subgroup: it contains $\zeta = e^{2\pi i\sqrt{3}}$ but not ζ^2 , because $\zeta^2 = e^{2\pi i \cdot 2\sqrt{3}}$ is neither a twelfth root of unity ($2\sqrt{3} \notin \mathbb{Q}$) nor equal to ζ . It is not even closed in a useful sense: it is closed as a set (it is finite, 13 points, a finite union of points in a Hausdorff space), but this does not make it a closed subgroup: the classification applies only to subgroups. (c) $H = \mu_{30}$, closed, with 30 elements: this is case (i) of Worked exercise 13.6, with $\text{lcm}(6, 10) = 30$ and $\frac{1}{6} - 2 \cdot \frac{1}{10} = -\frac{1}{30}$. (d) The first part is Corollary 13.2. A countable dense subgroup different from the one generated by a single irrational rotation is $\{e^{2\pi ir} : r \in \mathbb{Q}\} = \bigcup_{n \geq 1} \mu_n$: it is a subgroup (an increasing union of subgroups, and $\mu_n \cup \mu_k \subseteq \mu_{\text{lcm}(n,k)}$), it is countable, and it is infinite, hence dense by Corollary 13.2. ✓

BEST PRACTICE

Assigned exercise 13.2 — answers. (a) $c_n = \frac{1}{q} \sum_{k=0}^{q-1} e^{2\pi i kn/q}$. If $q \mid n$ every term equals 1 and $c_n = 1$; otherwise $\zeta = e^{2\pi in/q} \neq 1$ and the geometric sum gives $\frac{1}{q} \cdot \frac{\zeta^q - 1}{\zeta - 1} = 0$ because $\zeta^q = e^{2\pi in} = 1$. Hence $c_n = \mathbf{1}_{\{q \mid n\}}$. (b) $\mu_*^{(q)}$ is invariant exactly for $w \in \mu_q$: if $w \in \mu_q$, multiplication by w permutes the q points of μ_q and leaves the weights equal; if $w \notin \mu_q$, then $w \cdot 1 = w \notin \mu_q$ and, taking $A = \{w\}$, $\mu_*^{(q)}(A) = 0$ while $\mu_*^{(q)}(w^{-1}A) = \mu_*^{(q)}(\{1\}) = \frac{1}{q} \neq 0$. (c)

Step 2 of the proof of Theorem 13.4 fails: the relation $c_n = w^n c_n$ is available only for $w \in \mu_q$, and for such w one has $w^n = 1$ whenever $q \mid n$; hence there is no admissible w with $w^n \neq 1$ when $q \nmid n$, and from that one cannot conclude $c_n = 0$. Consistently, the coefficients of index a multiple of q equal 1. **(d)** For $q = 5$, $n = 0, \dots, 12$: 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, that is 1 exactly for $n = 0, 5, 10$. Check in double precision (**numpy**): real parts 1.000000000000 and 0.000000000000, imaginary parts in modulus $\leq 10^{-15}$. ✓

BEST PRACTICE

Assigned exercise 13.3 — answers. **(a)** By Definition 12.8, $\deg f$ is the winding number of the loop $s \mapsto f(e^{2\pi i s})$. For $f = \chi_n$ we have $\chi_n(e^{2\pi i s}) = e^{2\pi i n s}$, which lifts to $\tilde{\gamma}(s) = 2\pi n s$; hence $\deg \chi_n = \frac{2\pi n - 0}{2\pi} = n$. **(b)** By Theorem 12.8, two maps $S^1 \rightarrow S^1$ are homotopic if and only if they have the same degree; by (a) this is equivalent to $n = m$, that is $\chi_n = \chi_m$ (Step 2 of Theorem 13.5). Hence distinct characters lie in distinct homotopy classes: the dual $\widehat{S^1}$ injects into $[S^1, S^1] \cong \mathbb{Z}$, and indeed exhausts it. **(c)** By Corollary 12.6 the extension to the disc exists if and only if the degree is zero, that is $n = 0$, that is $\chi = \chi_0 \equiv 1$ (which extends to the constant 1). **(d)** The fixed points of χ_n are the solutions of $z^n = z$ in S^1 , that is $z^{n-1} = 1$: they are μ_{n-1} for $n \neq 1$ (hence $|n-1|$ points), and all of S^1 for $n = 1$. In particular χ_n always has at least the fixed point $z = 1$: consistent with the worked exercise of Chapter 12, according to which a map of degree $\neq 1$ has fixed points — here for $n \neq 1$ there are exactly $|n-1|$ of them. ✓

13.10. Chapter formulary

Structure.

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}, \quad z^{-1} = \bar{z}, \quad S^1 \text{ compact abelian topological group};$$

$$E : \mathbb{R} \rightarrow S^1, \quad E(t) = e^{it} \text{ homomorphism, } \ker E = 2\pi\mathbb{Z}, \quad \mathbb{R}/2\pi\mathbb{Z} \cong S^1.$$

Smooth charts (S^1 Lie group of dimension 1):

$$\varphi_1(x, y) = 2 \arctan \frac{y}{1+x} \quad \text{on } S^1 \setminus \{-1\}, \quad \varphi_2(x, y) = \pi - 2 \arctan \frac{y}{1-x} \quad \text{on } S^1 \setminus \{1\}.$$

Invariant measure.

$$m(A) = \frac{\lambda(E^{-1}(A) \cap [0, 2\pi))}{2\pi}, \quad \int_{S^1} f \, dm = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) \, d\theta,$$

$$m(wA) = m(A) \quad \forall w \in S^1; \quad m \text{ is the unique invariant Borel probability.}$$

Coefficients of a finite measure: $c_n(\mu) = \int z^n \, d\mu$; $c_n(m) = \delta_{n0}$; $\mu = \nu \iff c_n(\mu) = c_n(\nu) \, \forall n$.

Characters and duality.

$$\widehat{S^1} = \{\chi_n : z \mapsto z^n, \, n \in \mathbb{Z}\}, \quad \int_{S^1} \chi_n \overline{\chi_m} \, dm = \delta_{nm};$$

$$\widehat{S^1} \cong \mathbb{Z} \text{ (discrete)}, \quad \widehat{\mathbb{Z}} \cong S^1, \quad \widehat{\mathbb{R}} \cong \mathbb{R}; \quad \deg \chi_n = n.$$

Uniform distance between distinct characters: $\|\chi_n - \chi_m\|_\infty = 2$.

Subgroups.

$$G \leq \mathbb{R} \text{ closed} \implies G \in \{\{0\}, a\mathbb{Z} \ (a > 0), \mathbb{R}\};$$

$$H \leq S^1 \text{ closed} \implies H = \mu_n = \{z : z^n = 1\} \text{ or } H = S^1;$$

$$\alpha \notin \mathbb{Q} \implies \overline{\{e^{2\pi i n \alpha} : n \in \mathbb{Z}\}} = S^1; \quad \alpha = \frac{p}{q} \implies \{e^{2\pi i n \alpha}\} = \mu_q.$$

Dirichlet: $\forall N \exists 1 \leq n \leq N$ with $\|n\alpha\| < 1/N$, hence $|e^{2\pi i n \alpha} - 1| < 2\pi/N$. Exponential sums: $|\sum_{n=1}^N e^{2\pi i k n \alpha}| \leq 1/|\sin(\pi k \alpha)|$.

Sources and bibliography

- [1] W. Rudin, *Fourier Analysis on Groups*, Interscience Tracts in Pure and Applied Mathematics 12, Interscience/Wiley, New York, 1962 (reprinted Wiley Classics Library, 1990). Chapter 1 “The Basic Theorems of Fourier Analysis”: §1.1 (Haar measure on a locally compact abelian group: existence and uniqueness up to a constant), §1.2 (characters and the dual group $\hat{G}$ with the compact-open topology; the dual of a compact group is discrete and conversely), §1.7 (Pontryagin duality theorem). The case $G = \mathbb{T}$, with $\hat{\mathbb{T}} = \mathbb{Z}$, is the first example of the chapter. Publisher’s page and table of contents verified on onlinelibrary.wiley.com — verified on 5 September 2026.
- [2] G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd edition, CRC Press/Chapman & Hall, Boca Raton, 2016 (1st ed. 1995). Chapter 2 “Locally Compact Groups”, §2.2 “Haar Measure” (existence and uniqueness of the Haar measure up to a positive factor; normalisation to a probability in the compact case); Chapter 4 “Analysis on Locally Compact Abelian Groups” (dual group, abstract Fourier transform, Pontryagin duality theorem). Statements used here only as *panorama* (Remarks 13.4 and 13.6); no proof of the chapter depends on them. Publisher’s page and table of contents verified on routledge.com — verified on 5 September 2026.
- [3] W. Rudin, *Principles of Mathematical Analysis*, 3rd edition, McGraw-Hill, New York, 1976. Theorem 7.33 (Stone–Weierstrass theorem, complex version: a self-adjoint subalgebra that separates points and vanishes nowhere is dense in $C(K, \mathbb{C})$ with the uniform norm). This is the only non-elementary external tool used in Lemma 13.1 — verified on 5 September 2026.
- [4] H. Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Mathematische Annalen 77 (1916), 313–352. Equidistribution criterion via the exponential sums $\frac{1}{N} \sum_{n \leq N} e^{2\pi i k n \alpha} \rightarrow 0$ for every $k \neq 0$; reported here as *panorama* (Remark 13.8), while the estimate for the sums is proved in Proposition 13.7. GDZ digitisation (Göttinger Digitalisierungszentrum), gdz.sub.uni-goettingen.de — verified on 5 September 2026.
- [5] E. M. Stein, R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, Princeton, 2003. Chapter 4 “Some Applications of Fourier Series”, §2 (equidistribution of $\{n\alpha\}$ and Weyl’s criterion) — verified on 5 September 2026.
- [6] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd edition, Cambridge University Press, Cambridge, 2004. Chapter VI “Fourier Transforms on Locally Compact Abelian Groups” (characters, Haar measure, duality); Chapter I for the case $\mathbb{T}$ — verified on 5 September 2026.
- [7] T. Bröcker, T. tom Dieck, *Representations of Compact Lie Groups*, Graduate Texts in Mathematics 98, Springer, New York, 1985. Chapter I (compact Lie groups, tori, closed

subgroups) and Chapter IV (character theory): source for the *panorama* of Remark 13.2 — verified on 5 September 2026.

- [8] F. W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Graduate Texts in Mathematics 94, Springer, New York, 1983. Chapter 3 (Lie groups, Cartan’s closed subgroup theorem, exponential map) — verified on 5 September 2026.
- [9] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002, §1.1 (lifting of paths and homotopies): this is the source of Chapter 12, whose results (exponential covering, lifting on $\mathbb{R}$, uniqueness of the lifting) are used here and not reproved. Chapters in PDF from the author’s page at Cornell University, pi.math.cornell.edu/~hatcher/AT/ATchapters — verified on 5 September 2026.

Tangencies and packings: Ford circles, Farey and optimality

Two circles touch, or they do not. It is the poorest relation one can imagine between two figures, and for that very reason it is the most fruitful: from it are born the arithmetic of fractions (Ford circles), the combinatorics of contact (the kissing number), and the optimality problems that govern how pipes, cables and rods arrange themselves spontaneously when bundled together.

The chapter is organized around three results proved in full. The first is the tangency formula for Ford circles (Theorem 14.2): two Ford circles never cross, and they touch exactly when the two fractions are Farey neighbours. The second is that in the plane the kissing number is exactly six (Theorem 14.3): six equal circles can touch one, seven cannot, and the proof of the hard part fits in three lines of trigonometry. The third is that the triangular packing has density $\pi/\sqrt{12}$ (Theorem 14.4), computed and checked numerically. That *no* packing does better — the Thue–Fejes Tóth theorem — is not proved here: its central geometric lemma is proved (Lemma 14.3) and exactly what remains imported is stated (Remark 14.4). In the same way the Kepler conjecture and the Koebe–Andreiev–Thurston circle packing theorem appear only as panorama.

INFO

Level and prerequisites. Level **C** (undergraduate), with two sections at level **D** (the hexagon lemma and the definition of density).

Internal prerequisites. Tangency of circles, power of a point and inversion (Volume I, Chapter 5); Descartes’ theorem on four mutually tangent circles and the notion of signed curvature (Volume I, Chapter 5, section «Descartes’ circle theorem»): *recalled and not reproved*; circles of Apollonius (Volume I, Chapter 2). From Volume III: geometry of plane curves (Chapter 2), isometries of the plane (Chapter 5), lattices and spheres in dimension n (Chapter 11).

External prerequisites, assumed and not reproved:

- elementary arithmetic: Euclidean division, Bézout’s identity, greatest common divisor;
- the pigeonhole principle;
- Lebesgue measure in the plane: σ -additivity, invariance under isometries, area of the disc πr^2 ;
- convexity: a function with second derivative ≥ 0 on an interval is convex, and the discrete Jensen inequality holds.

Everything else is proved here.

What one learns. How to translate an arithmetic condition ($|ps - rq| = 1$) into a geometric one (tangency) and back; how to construct the Farey sequence and prove its characteristic property; how to derive a combinatorial upper bound from an angular inequality; how to compute a density as a limit and verify it; and how to recognize where a proof’s reach ends and what is imported begins.

Warning. Conventions of this chapter.

- “Circle” denotes the closed disc $D(C, r) = \{P : |P - C| \leq r\}$; “circumference” its

boundary. The *interior* of $D(C, r)$ is $\{P : |P - C| < r\}$.

- Two circles $D(C_1, r_1)$ and $D(C_2, r_2)$ are **externally tangent** if $|C_1C_2| = r_1 + r_2$; they have **disjoint interiors** if $|C_1C_2| \geq r_1 + r_2$.
- Fractions p/q are always understood *reduced*: $q \geq 1$ and $\gcd(p, q) = 1$.
- $\lfloor x \rfloor$ is the floor function.
- Lengths in millimetres in the applications are exact by construction; the results of the counts are integers.

14.1. Tangency and curvature: what is recalled

Definition 14.1: External tangency and signed curvature. Two circles $D(C_1, r_1)$, $D(C_2, r_2)$ with $C_1 \neq C_2$ are **externally tangent** if $|C_1C_2| = r_1 + r_2$; in that case they have exactly one point in common, the point of the segment C_1C_2 at distance r_1 from C_1 , and their interiors are disjoint. The **signed curvature** of a circle of radius r is $k = 1/r$ if the circle is externally tangent to the others in the configuration, $k = -1/r$ if it contains them; a line has curvature $k = 0$.

Remark 14.1: Recalled from Volume I: Descartes' theorem. *Descartes' theorem (1643) is proved in full in the Volume I, Chapter 5, section «Descartes' circle theorem»: if four circles are mutually tangent in pairs and have signed curvatures k_1, k_2, k_3, k_4 , then*

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2),$$

and equivalently, given k_1, k_2, k_3 ,

$$k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1}.$$

Here the theorem is used and not reproved: we refer to Volume I for the proof, which proceeds by coordinates and squared distances. It is used in two places: in Example 14.3, where three Ford circles and the real line are checked to form a Descartes configuration, and in Worked exercise 14.8.

Remark 14.2: Recalled from Volume I: circles of Apollonius and gaskets. The **circle of Apollonius** as the locus of points P with $PA/PB = k$ constant is proved in Volume I, Chapter 2; the **problem of Apollonius** (constructing the circles tangent to three given objects) and the case of three mutually tangent circles are treated in Volume I, Chapter 5, with the technique of inversion. Iterating Descartes' theorem starting from three mutually tangent circles yields the **Apollonian gasket**, whose connection with Ford circles is illustrated in Remark 14.3. Here too: it is recalled, not reproved.

14.2. Farey fractions

Definition 14.2: Farey sequence and Farey neighbours. For $n \geq 1$, the **Farey sequence** F_n is the set of reduced fractions p/q with $0 \leq p \leq q \leq n$, listed in increasing order. Two reduced fractions $\frac{p}{q} < \frac{r}{s}$ are called **Farey neighbours** if

$$rq - ps = 1.$$

The **mediant** of $\frac{p}{q}$ and $\frac{r}{s}$ is the fraction $\frac{p+r}{q+s}$.

Lemma 14.1: Property of the mediant. Let $\frac{p}{q} < \frac{r}{s}$ be fractions with $q, s \geq 1$. Then:

- (i) $\frac{p}{q} < \frac{p+r}{q+s} < \frac{r}{s}$;
(ii) if moreover $rq - ps = 1$, then $\frac{p+r}{q+s}$ is reduced, and is a Farey neighbour of both $\frac{p}{q}$ and $\frac{r}{s}$:

$$(p+r)q - p(q+s) = 1, \quad r(q+s) - (p+r)s = 1.$$

Proof. (i) We have

$$\frac{p+r}{q+s} - \frac{p}{q} = \frac{q(p+r) - p(q+s)}{q(q+s)} = \frac{qr - ps}{q(q+s)}, \quad \frac{r}{s} - \frac{p+r}{q+s} = \frac{r(q+s) - s(p+r)}{s(q+s)} = \frac{qr - ps}{s(q+s)}.$$

Since $\frac{p}{q} < \frac{r}{s}$ and $q, s > 0$, multiplying by $qs > 0$ gives $ps < rq$, i.e. $qr - ps > 0$; and $q+s > 0$. Hence both differences are positive.

(ii) With $rq - ps = 1$ the two computations of point (i) give exactly $(p+r)q - p(q+s) = qr - ps = 1$ and $r(q+s) - (p+r)s = qr - ps = 1$. Let now $d = \gcd(p+r, q+s)$: from $(p+r)q - p(q+s) = 1$ it follows that d divides 1, hence $d = 1$ and the mediant is reduced. $\square$

Lemma 14.2: The minimum denominator between two neighbours. Let $\frac{p}{q} < \frac{r}{s}$ be Farey neighbours ($rq - ps = 1$) and let $\frac{x}{y}$ be a fraction with $y \geq 1$ and

$$\frac{p}{q} < \frac{x}{y} < \frac{r}{s}.$$

Then $y \geq q+s$, and $y = q+s$ holds if and only if $\frac{x}{y} = \frac{p+r}{q+s}$.

Proof. Since $\frac{x}{y} - \frac{p}{q} = \frac{xq - yp}{qy} > 0$ and $xq - yp$ is an integer, we have $xq - yp \geq 1$, hence

$$\frac{x}{y} - \frac{p}{q} \geq \frac{1}{qy}.$$

Similarly $\frac{r}{s} - \frac{x}{y} = \frac{ry - sx}{sy} > 0$ with $ry - sx$ an integer, hence

$$\frac{r}{s} - \frac{x}{y} \geq \frac{1}{sy}.$$

Adding and using $\frac{r}{s} - \frac{p}{q} = \frac{rq - ps}{qs} = \frac{1}{qs}$:

$$\frac{1}{qs} \geq \frac{1}{qy} + \frac{1}{sy} = \frac{s+q}{qsy},$$

whence, multiplying by $qsy > 0$, one obtains $y \geq q+s$.

If $y = q+s$, the two inequalities used must be equalities (otherwise the sum would be strictly greater): hence $xq - yp = 1$, i.e. $xq = 1 + (q+s)p$. Using $1 = rq - ps$,

$$xq = rq - ps + qp + sp = q(r+p),$$

and dividing by $q > 0$ one gets $x = p+r$: hence $\frac{x}{y} = \frac{p+r}{q+s}$. The converse is Lemma 14.1(i)–(ii), which places the mediant strictly in between and assigns it denominator $q+s$. $\square$

Theorem 14.1: Consecutive Farey fractions. For every $n \geq 1$, if $\frac{p}{q} < \frac{r}{s}$ are consecutive in F_n , then

$$rq - ps = 1.$$

Moreover F_{n+1} is obtained from F_n by inserting, between every pair of consecutive terms $\frac{p}{q} < \frac{r}{s}$ with $q + s = n + 1$, their mediant $\frac{p+r}{q+s}$, and nothing else.

Proof. By induction on n .

Base case $n = 1$. $F_1 = (\frac{0}{1}, \frac{1}{1})$ and $1 \cdot 1 - 0 \cdot 1 = 1$. ✓

Inductive step. Suppose every pair of consecutive terms of F_n satisfies $rq - ps = 1$. Let $\frac{x}{y} \in F_{n+1} \setminus F_n$: then $y = n + 1$ (fractions with denominator $\leq n$ are already in F_n) and $\frac{x}{y}$ falls strictly between two consecutive terms $\frac{p}{q} < \frac{r}{s}$ of F_n (strictly, because $\frac{x}{y} \notin F_n$). By Lemma 14.2 we have $n + 1 = y \geq q + s$; on the other hand the mediant $\frac{p+r}{q+s}$ also lies strictly between $\frac{p}{q}$ and $\frac{r}{s}$ and is reduced (Lemma 14.1), hence it belongs to $F_{q+s} \subseteq F_{n+1}$ if $q + s \leq n + 1$; but if $q + s \leq n$ held the mediant would lie in F_n strictly between two consecutive terms of F_n , absurd. So $q + s \geq n + 1$, and with the opposite inequality $q + s = n + 1 = y$. By the second part of Lemma 14.2, then $\frac{x}{y} = \frac{p+r}{q+s}$: the only new fraction between $\frac{p}{q}$ and $\frac{r}{s}$ is the mediant, and it appears exactly when $q + s = n + 1$.

The consecutive pairs of F_{n+1} are therefore: the consecutive pairs of F_n with $q + s > n + 1$ (unchanged, and the relation holds by the induction hypothesis), and the pairs $(\frac{p}{q}, \frac{p+r}{q+s})$, $(\frac{p+r}{q+s}, \frac{r}{s})$ obtained by splitting those with $q + s = n + 1$: by Lemma 14.1(ii) both satisfy the relation with value 1. The induction is complete. □

Example 14.1: F_5 and the verification of the relation.

$$F_5 = \left(\frac{0}{1}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{1}{1} \right).$$

Symbolic check of $rq - ps = 1$ on all ten consecutive pairs:

$$\begin{aligned} \left(\frac{0}{1}, \frac{1}{5} \right) : 1 \cdot 1 - 0 \cdot 5 = 1 & \quad \left(\frac{1}{5}, \frac{1}{4} \right) : 1 \cdot 5 - 1 \cdot 4 = 1 & \quad \left(\frac{1}{4}, \frac{1}{3} \right) : 1 \cdot 4 - 1 \cdot 3 = 1 & \quad \left(\frac{1}{3}, \frac{2}{5} \right) : 2 \cdot 3 - 1 \cdot 5 = 1 \\ \left(\frac{2}{5}, \frac{1}{2} \right) : 1 \cdot 5 - 2 \cdot 2 = 1 & \quad \left(\frac{1}{2}, \frac{3}{5} \right) : 3 \cdot 2 - 1 \cdot 5 = 1 & \quad \left(\frac{3}{5}, \frac{2}{3} \right) : 2 \cdot 5 - 3 \cdot 3 = 1 & \quad \left(\frac{2}{3}, \frac{3}{4} \right) : 3 \cdot 3 - 2 \cdot 4 = 1 \\ \left(\frac{3}{4}, \frac{4}{5} \right) : 4 \cdot 4 - 3 \cdot 5 = 1 & \quad \left(\frac{4}{5}, \frac{1}{1} \right) : 1 \cdot 5 - 4 \cdot 1 = 1 \end{aligned}$$

Independent check (brute-force generation of F_n in Python, for $n = 1, \dots, 60$: one lists all fractions p/q with $\gcd(p, q) = 1$, $0 \leq p \leq q \leq n$, sorts them and checks $rq - ps = 1$ on every consecutive pair): no counterexample. The number of terms grows as $|F_n| = 1 + \sum_{k=1}^n \varphi(k)$ — for $n = 1, \dots, 10$: 2, 3, 5, 7, 11, 13, 19, 23, 29, 33 — and is computed in Assigned exercise 14.3(b). ✓

14.3. Ford circles

Definition 14.3: Ford circle. Let $\frac{p}{q}$ be a reduced fraction. The **Ford circle** $\mathcal{C}(p/q)$ is the circle in the upper half-plane

$$\mathcal{C}\left(\frac{p}{q}\right) = D\left(\left(\frac{p}{q}, \frac{1}{2q^2}\right), \frac{1}{2q^2}\right),$$

that is, the circle of centre $(p/q, 1/(2q^2))$ and radius $1/(2q^2)$. It is tangent to the real axis at the point $(p/q, 0)$.

Theorem 14.2: Tangency and disjointness of Ford circles. Let $\frac{p}{q} \neq \frac{r}{s}$ be two distinct reduced fractions, with centres $Z_1 = (\frac{p}{q}, \frac{1}{2q^2})$, $Z_2 = (\frac{r}{s}, \frac{1}{2s^2})$ and radii $\rho_1 = \frac{1}{2q^2}$, $\rho_2 = \frac{1}{2s^2}$. Then

$$|Z_1 Z_2|^2 - (\rho_1 + \rho_2)^2 = \frac{(ps - rq)^2 - 1}{q^2 s^2}.$$

Consequently:

- (i) $\mathcal{C}(p/q)$ and $\mathcal{C}(r/s)$ always have disjoint interiors;
- (ii) they are (externally) tangent if and only if $|ps - rq| = 1$, that is, if and only if the two fractions are Farey neighbours;
- (iii) if $|ps - rq| \geq 2$ the two circles are disjoint, with a positive distance between their boundaries.

Proof. **Step 1 — the computation.** By definition

$$|Z_1 Z_2|^2 = \left(\frac{p}{q} - \frac{r}{s}\right)^2 + \left(\frac{1}{2q^2} - \frac{1}{2s^2}\right)^2, \quad (\rho_1 + \rho_2)^2 = \left(\frac{1}{2q^2} + \frac{1}{2s^2}\right)^2.$$

Subtracting, the squares of the individual terms cancel and the cross terms remain, with opposite signs:

$$\left(\frac{1}{2q^2} - \frac{1}{2s^2}\right)^2 - \left(\frac{1}{2q^2} + \frac{1}{2s^2}\right)^2 = -4 \cdot \frac{1}{2q^2} \cdot \frac{1}{2s^2} = -\frac{1}{q^2 s^2}.$$

Moreover

$$\left(\frac{p}{q} - \frac{r}{s}\right)^2 = \left(\frac{ps - rq}{qs}\right)^2 = \frac{(ps - rq)^2}{q^2 s^2}.$$

Adding:

$$|Z_1 Z_2|^2 - (\rho_1 + \rho_2)^2 = \frac{(ps - rq)^2}{q^2 s^2} - \frac{1}{q^2 s^2} = \frac{(ps - rq)^2 - 1}{q^2 s^2}.$$

Step 2 — (i). Since $\frac{p}{q} \neq \frac{r}{s}$ and the fractions are reduced, $ps - rq \neq 0$: indeed $ps = rq$ implies $\frac{p}{q} = \frac{r}{s}$. Since $ps - rq$ is a nonzero integer, $(ps - rq)^2 \geq 1$ and hence $|Z_1 Z_2|^2 \geq (\rho_1 + \rho_2)^2$, i.e. $|Z_1 Z_2| \geq \rho_1 + \rho_2$: the interiors are disjoint (if a point were interior to both, the triangle inequality would give $|Z_1 Z_2| < \rho_1 + \rho_2$).

Step 3 — (ii) and (iii). The equality $|Z_1 Z_2| = \rho_1 + \rho_2$, i.e. external tangency, is equivalent to $(ps - rq)^2 = 1$, i.e. $|ps - rq| = 1$: which is exactly the Farey neighbour condition of Definition 14.2 (up to the order of the two fractions, which only changes the sign). If instead $|ps - rq| \geq 2$, then $(ps - rq)^2 - 1 \geq 3 > 0$ and $|Z_1 Z_2| > \rho_1 + \rho_2$: the two circles are disjoint and the distance between the boundaries is $|Z_1 Z_2| - \rho_1 - \rho_2 > 0$. $\square$

Corollary 14.1: The circles of consecutive Farey fractions touch. If $\frac{p}{q} < \frac{r}{s}$ are consecutive in F_n , then $\mathcal{C}(p/q)$ and $\mathcal{C}(r/s)$ are externally tangent. In particular the Ford circles of the fractions of F_n , taken in order, form a chain of consecutively tangent circles, all tangent to the real axis and with pairwise disjoint interiors.

Proof. By Theorem 14.1 we have $rq - ps = 1$, hence $|ps - rq| = 1$, and Theorem 14.2(ii) applies. The disjointness of the interiors for every pair is point (i) of the same theorem; tangency to the real axis is immediate from Definition 14.3, because the centre has ordinate equal to the radius. $\square$

Example 14.2: Tangencies and distances, computed twice. Four pairs are compared.

pair	$ps - rq$	$\frac{(ps - rq)^2 - 1}{q^2 s^2}$	outcome
$(\frac{1}{3}, \frac{1}{2})$	$1 \cdot 2 - 1 \cdot 3 = -1$	0	tangent
$(\frac{1}{4}, \frac{1}{3})$	$1 \cdot 3 - 1 \cdot 4 = -1$	0	tangent
$(\frac{2}{5}, \frac{1}{2})$	$2 \cdot 2 - 1 \cdot 5 = -1$	0	tangent
$(\frac{1}{3}, \frac{2}{3})$	$1 \cdot 3 - 2 \cdot 3 = -3$	$\frac{8}{81} = 0.0987654$	disjoint
$(\frac{1}{3}, \frac{3}{4})$	$1 \cdot 4 - 3 \cdot 3 = -5$	$\frac{81}{144} = 0.1666667$	disjoint

Independent check in double precision, computing separately $|Z_1 Z_2|^2$ and $(\rho_1 + \rho_2)^2$ from the coordinates of the centres and the radii, and taking the difference:

$$(\frac{1}{3}, \frac{1}{2}) : 6.9 \cdot 10^{-18}, \quad (\frac{1}{4}, \frac{1}{3}) : -2.6 \cdot 10^{-18}, \quad (\frac{2}{5}, \frac{1}{2}) : -3.5 \cdot 10^{-18},$$

that is, zero within machine error, and

$$(\frac{1}{3}, \frac{2}{3}) : 0.0987654321, \quad (\frac{1}{3}, \frac{3}{4}) : 0.1666666667,$$

coinciding with the symbolic values $8/81$ and $1/6$. ✓

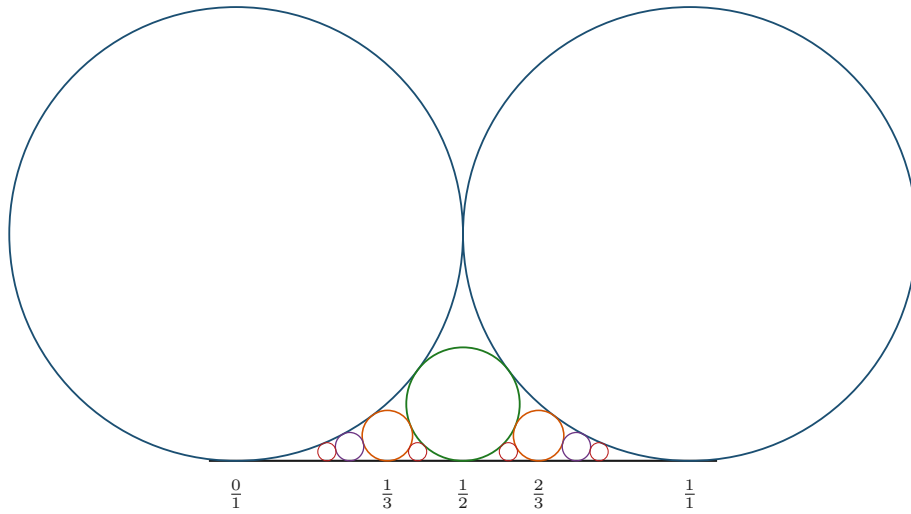


Figure 14.1: The Ford circles of the fractions of F_5 in the interval $[0, 1]$. The circle of p/q has centre $(p/q, 1/(2q^2))$ and radius $1/(2q^2)$: radius 0.5 for $q = 1$, 0.125 for $q = 2$, $1/18 = 0.05556$ for $q = 3$, $1/32 = 0.03125$ for $q = 4$, 0.02 for $q = 5$. Two circles touch exactly when the fractions are Farey neighbours (Theorem 14.2); for example $\frac{1}{3}$ and $\frac{1}{2}$ ($|ps - rq| = 1$) touch, while $\frac{1}{3}$ and $\frac{2}{3}$ ($|ps - rq| = 3$) stay apart. No pair ever overlaps.

Example 14.3: Three Ford circles and the line: a Descartes configuration. Let $\frac{p}{q} < \frac{r}{s}$ be Farey neighbours and $\frac{p+r}{q+s}$ their median. By Corollary 14.1 and Lemma 14.1(ii) the three corresponding Ford circles are mutually tangent, and each is tangent to the real axis: one thus has a quadruple of mutually tangent objects, with curvatures

$$k_1 = 0 \text{ (the line)}, \quad k_2 = 2q^2, \quad k_3 = 2s^2, \quad k_4 = 2(q + s)^2.$$

Symbolic check of Descartes' theorem (Remark 14.1): setting $u = q^2$, $v = s^2$, $w = (q + s)^2$,

$$(k_1 + k_2 + k_3 + k_4)^2 = 4(u + v + w)^2, \quad 2(k_1^2 + k_2^2 + k_3^2 + k_4^2) = 8(u^2 + v^2 + w^2).$$

The equality is equivalent to $(u + v + w)^2 = 2(u^2 + v^2 + w^2)$, i.e. to $2(uv + vw + wu) = u^2 + v^2 + w^2$. With $w = (q + s)^2 = u + v + 2qs$ and $qs = \sqrt{uv}$ this is verified directly: setting $t = qs$, one has $u^2 + v^2 + w^2 - 2uv - 2vw - 2wu = (w - u - v)^2 - 4uv = (2t)^2 - 4uv = 4q^2s^2 - 4q^2s^2 = 0$.
✓ Numerical check. For $\frac{p}{q} = \frac{0}{1}$, $\frac{r}{s} = \frac{1}{1}$ (neighbours: $1 \cdot 1 - 0 \cdot 1 = 1$), mediant $\frac{1}{2}$: curvatures $(0, 2, 2, 8)$, and $(0 + 2 + 2 + 8)^2 = 144$, $2(0 + 4 + 4 + 64) = 144$.
✓ For $\frac{p}{q} = \frac{0}{1}$, $\frac{r}{s} = \frac{1}{2}$, mediant $\frac{1}{3}$: curvatures $(0, 2, 8, 18)$, and $(0 + 2 + 8 + 18)^2 = 784$, $2(0 + 4 + 64 + 324) = 784$. **✓**

Remark 14.3: Ford, Farey and the Apollonian gasket. Example 14.3 shows that the set of Ford circles, together with the real axis, is an **Apollonian gasket**: starting from the line and the two circles $\mathcal{C}(0/1)$, $\mathcal{C}(1/1)$, every new circle is the fourth Descartes circle of three already present, and the sequence of curvatures $2q^2$ reproduces exactly the construction of Farey mediants. Historically the connection was made explicit by L. R. Ford in 1938 (*Fractions, American Mathematical Monthly* 45, 586–601), who introduced these circles precisely to “see” the properties of fractions: Theorem 14.2 translates the arithmetic $|ps - rq| = 1$ into geometry, and the best rational approximation of an irrational α can be read off as the chain of Ford circles that the point $(\alpha, 0)$ “crosses” rising upward. It is the same irrational α that in Chapter 13 generates a dense subgroup of S^1 : the denominators of the convergents that there measured the returns near 1 are here the radii of the tangent circles.

14.4. The kissing number in the plane

Definition 14.4: Kissing number. The **kissing number** of the plane, τ_2 , is the maximum number of circles of radius r with pairwise disjoint interiors that can all be externally tangent to a fixed circle of the same radius r .

Theorem 14.3: The kissing number of the plane is 6. $\tau_2 = 6$.

Proof. It is no loss of generality to take $r = 1$ and the central circle $D(O, 1)$ with O the origin. A circle $D(C_j, 1)$ externally tangent to $D(O, 1)$ has $|C_j| = 2$.

Step 1 — lower bound: six is possible. Take

$$C_j = (2 \cos(j\pi/3), 2 \sin(j\pi/3)), \quad j = 0, 1, \dots, 5,$$

that is

$$(2, 0), (1, \sqrt{3}), (-1, \sqrt{3}), (-2, 0), (-1, -\sqrt{3}), (1, -\sqrt{3}),$$

with $\sqrt{3} = 1.7320508$. Each has $|C_j| = 2$, hence $D(C_j, 1)$ is tangent to $D(O, 1)$. For $j \neq l$ the angle between C_j and C_l is a nonzero multiple of $\pi/3$, hence it is $\geq \pi/3$, and by the cosine theorem

$$|C_j - C_l|^2 = 4 + 4 - 8 \cos \theta_{jl} = 8(1 - \cos \theta_{jl}) \geq 8\left(1 - \frac{1}{2}\right) = 4,$$

that is, $|C_j - C_l| \geq 2$: the interiors are pairwise disjoint. *Explicit check* on the closest pair: $|(2, 0) - (1, \sqrt{3})|^2 = 1 + 3 = 4$, that is a distance of exactly 2 (the two circles touch). **✓** Hence $\tau_2 \geq 6$.

Step 2 — upper bound: seven is not possible. Let $D(C_1, 1), \dots, D(C_k, 1)$ be tangent to $D(O, 1)$ with pairwise disjoint interiors. Then $|C_j| = 2$ for every j and $|C_j - C_l| \geq 2$ for $j \neq l$.

Writing $\theta_{jl} \in [0, \pi]$ for the angle $\widehat{C_j O C_l}$, the cosine theorem gives

$$4 \leq |C_j - C_l|^2 = 8(1 - \cos \theta_{jl}) \implies \cos \theta_{jl} \leq \frac{1}{2} \implies \theta_{jl} \geq \frac{\pi}{3},$$

using that $\cos$ is decreasing on $[0, \pi]$ and $\cos(\pi/3) = \frac{1}{2}$.

Write the centres now in polar coordinates, $C_j = 2(\cos \alpha_j, \sin \alpha_j)$, with $\alpha_1 < \alpha_2 < \dots < \alpha_k$ in $[0, 2\pi)$ (the α_j are distinct because $\theta_{jl} > 0$). Set the k gaps

$$\Delta_j = \alpha_{j+1} - \alpha_j \quad (j = 1, \dots, k-1), \quad \Delta_k = 2\pi - \alpha_k + \alpha_1,$$

which are positive and whose sum is 2π . For every j , the angle $\widehat{C_j O C_{j+1}}$ equals $\min(\Delta_j, 2\pi - \Delta_j)$; since that angle is $\geq \pi/3$ and since $\Delta_j \leq 2\pi$, in every case $\Delta_j \geq \pi/3$ (if $\Delta_j < \pi/3$ held then $\min(\Delta_j, 2\pi - \Delta_j) = \Delta_j < \pi/3$, a contradiction). Hence

$$2\pi = \sum_{j=1}^k \Delta_j \geq k \cdot \frac{\pi}{3} \implies k \leq 6.$$

(For $k = 1$ and $k = 2$ the statement is trivially true and the count of gaps remains valid.)

Combining the two steps, $\tau_2 = 6$. □

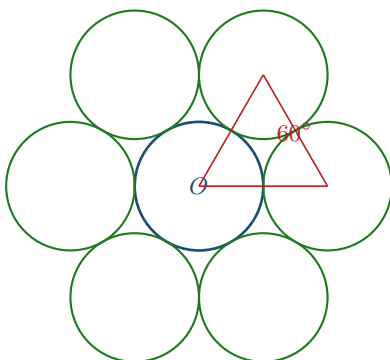


Figure 14.2: The kissing number of the plane is 6. The six centres lie at the vertices of a regular hexagon of side 2 inscribed in the circle of radius 2: coordinates $(\pm 2, 0)$ and $(\pm 1, \pm 1.73205)$. The red triangle has all three sides of length 2 ($|(2, 0) - (1, \sqrt{3})|^2 = 1 + 3 = 4$), hence it is equilateral and the central angle is exactly 60° : six angles of 60° fill the full turn with no room to spare, and a seventh circle would require another angle $\geq 60^\circ$ (Theorem 14.3, Step 2).

Exercise 14.1: Seven cables in a conduit. *Aim:* to translate the kissing number into a sizing problem. *Estimated time:* 15 minutes. *Type:* numerical/applied. Seven equal cables of radius r must be bundled. What is the minimum radius of the circular conduit that contains them, in the “one central and six around” arrangement? Prove that in that arrangement the value found is optimal, and discuss whether a different arrangement of the seven circles could do better.

Solution. Strategy. The radius needed for the hexagonal configuration is computed and one observes that the constraint is given by the six outer circles.

Step 1 — the computation. In the “one central and six around” arrangement the central cable has its centre at the origin; the six outer ones are tangent to it, hence their centres are at distance $2r$ from the origin (Theorem 14.3, Step 1). The point farthest from the origin belonging to one of the seven circles is then at distance $2r + r = 3r$. Hence every circular conduit centred at the origin that contains them has radius $R \geq 3r$, and $R = 3r$ suffices:

the bundle has diameter $6r$, that is, three times the diameter of a single cable.

Step 2 — optimality within the configuration. In that configuration the value $3r$ is exactly the maximum of the distances from the origin of the points of the seven discs, hence it is the radius of the smallest circle centred at the origin that contains them. If one allows the centre of the conduit to move, the configuration is invariant under rotations of $\pi/3$ and symmetric, and every displacement of the centre moves at least one of the six outer circles farther away: the radius needed grows. Precisely, for a centre P the radius needed is $\max_j (|P - C_j| + r)$ with C_j the six outer centres, and by the triangle inequality $\max_j |P - C_j| \geq \frac{1}{2}(|P - C_j| + |P - C_{j+3}|) \geq \frac{1}{2}|C_j - C_{j+3}| = 2r$ for every pair of opposite centres: hence $R \geq 3r$ always.

Independent check. For $r = 6$ mm (a cable of 12 mm diameter) one gets $R = 18$ mm, a conduit of diameter 36 mm. Area check: the seven cables occupy $7\pi r^2 = 7\pi \cdot 36 = 791.68 \text{ mm}^2$; the conduit has area $\pi R^2 = \pi \cdot 324 = 1017.88 \text{ mm}^2$; the ratio is $7/9 = 0.7778$, that is 77.78%. ✓ (The value $7/9$ is exact: $7\pi r^2 / (\pi(3r)^2) = 7/9$.)

Interpretation. The 77.78% is lower than the density of the infinite packing $\pi/\sqrt{12} = 0.9069$ (Theorem 14.4): it is the boundary effect, which on only seven circles weighs very heavily. With 19 cables (two rings) the ratio rises to $19/25 = 0.76$... actually $19\pi r^2 / (\pi(5r)^2) = 19/25 = 0.76$: it does not improve, because the outer ring is still almost all boundary.

Typical errors. (1) Computing $R = 7r$ by adding radii in a row. (2) Forgetting the radius of the outer cable and concluding $R = 2r$. (3) Confusing the ratio of areas with the asymptotic density.

Extension. For n cables the problem (“pack n equal circles in the smallest possible circle”) has a known solution only for a few values of n ; for $n = 7$ the value $R = 3r$ is optimal and is one of the classical cases. See Remark 14.5 for the state of the art.

14.5. Packing density

Definition 14.5: Packing and density. A **packing** of circles of radius $r > 0$ in the plane is a family $\mathcal{P} = \{D(C_i, r)\}_{i \in I}$ of circles with pairwise disjoint interiors, that is with $|C_i - C_j| \geq 2r$ for $i \neq j$. Writing $B_R = D(O, R)$, the **density** of $\mathcal{P}$ is

$$\delta(\mathcal{P}) = \lim_{R \rightarrow +\infty} \frac{\text{area}\left(B_R \cap \bigcup_i D(C_i, r)\right)}{\pi R^2},$$

when the limit exists.

Theorem 14.4: The triangular packing has density $\pi/\sqrt{12}$. Let $r > 0$ and let

$$\Lambda = \{m(2r, 0) + n(r, r\sqrt{3}) : m, n \in \mathbb{Z}\}$$

be the triangular lattice. Then $\mathcal{P}_\Lambda = \{D(\lambda, r)\}_{\lambda \in \Lambda}$ is a packing, the limit of Definition 14.5 exists and

$$\delta(\mathcal{P}_\Lambda) = \frac{\pi r^2}{2\sqrt{3}r^2} = \frac{\pi}{2\sqrt{3}} = \frac{\pi}{\sqrt{12}} = 0.9068997.$$

Proof. **Step 1 — it is a packing.** A generic element of Λ is

$$\lambda_{m,n} = r(2m + n, n\sqrt{3}), \quad |\lambda_{m,n}|^2 = r^2((2m + n)^2 + 3n^2).$$

For $(m, n) \neq (0, 0)$ three cases are distinguished. If $n = 0$ then $m \neq 0$ and $|\lambda|^2 = 4m^2r^2 \geq 4r^2$.

If $|n| = 1$ then $(2m + n)^2 \geq 0$ and $|\lambda|^2 \geq 3r^2 + (2m \pm 1)^2 r^2 \geq 3r^2 + r^2 = 4r^2$, because $2m \pm 1$ is odd, hence nonzero, and its square is ≥ 1 . If $|n| \geq 2$ then $|\lambda|^2 \geq 12r^2 > 4r^2$. Hence the minimum of $|\lambda|$ over $\Lambda \setminus \{0\}$ is $2r$, attained (for example) at $(m, n) = (1, 0)$ and at $(m, n) = (0, 1)$. Since Λ is a group, $|\lambda - \lambda'| \geq 2r$ for $\lambda \neq \lambda'$: the interiors of the discs are disjoint.

Step 2 — area of the cell. The fundamental parallelogram $P = \{a(2r, 0) + b(r, r\sqrt{3}) : a, b \in [0, 1)\}$ has area

$$A = \left| \det \begin{pmatrix} 2r & 0 \\ r & r\sqrt{3} \end{pmatrix} \right| = 2r \cdot r\sqrt{3} = 2\sqrt{3} r^2 = 3.4641016 r^2,$$

and the translates $P + \lambda$, $\lambda \in \Lambda$, form a partition of the plane (this is the definition of a fundamental domain of a lattice). The diameter of P does not exceed $|(2r, 0)| + |(r, r\sqrt{3})| = 2r + 2r = 4r$; set $d_0 = 4r$.

Step 3 — counting lattice points. Let $\rho > d_0$ and $N(\rho) = \#\{\lambda \in \Lambda : |\lambda| \leq \rho\}$. The cells $P + \lambda$ with $|\lambda| \leq \rho$ are pairwise disjoint and contained in $B_{\rho+d_0}$ (every point of $P + \lambda$ is at distance at most d_0 from λ), hence, by additivity and monotonicity of area,

$$N(\rho) A \leq \pi(\rho + d_0)^2.$$

Conversely, every point $x \in B_{\rho-d_0}$ belongs to some cell $P + \lambda$, and for it $|\lambda| \leq |x| + d_0 \leq \rho$: hence $B_{\rho-d_0}$ is covered by the cells with $|\lambda| \leq \rho$ and

$$N(\rho) A \geq \pi(\rho - d_0)^2.$$

Step 4 — estimate of the covered area. Let $S = \bigcup_{\lambda \in \Lambda} D(\lambda, r)$. The discs have disjoint interiors, hence the area of their union is the sum of the areas. Every disc $D(\lambda, r)$ with $|\lambda| \leq R - r$ is contained in B_R , and every disc that meets B_R has $|\lambda| \leq R + r$; hence

$$\pi r^2 N(R - r) \leq \text{area}(B_R \cap S) \leq \pi r^2 N(R + r).$$

Combining with Step 3 (for large R):

$$\frac{\pi r^2}{A} \cdot \frac{(R - r - d_0)^2}{R^2} \leq \frac{\text{area}(B_R \cap S)}{\pi R^2} \leq \frac{\pi r^2}{A} \cdot \frac{(R + r + d_0)^2}{R^2}.$$

Both outer terms tend to $\pi r^2/A$ as $R \rightarrow +\infty$; by the squeeze theorem the limit exists and equals

$$\delta(\mathcal{P}_\Lambda) = \frac{\pi r^2}{A} = \frac{\pi r^2}{2\sqrt{3} r^2} = \frac{\pi}{2\sqrt{3}}.$$

Step 5 — the numerical value, computed twice. From the closed form, $\pi/(2\sqrt{3}) = 3.14159265/3.46410162 = 0.90689968$. In equivalent form, $\pi/\sqrt{12}$ with $\sqrt{12} = 3.46410162$: same value. *Independent check* in double precision (`numpy`): $\pi/\sqrt{12} = 0.9068996821171089$. ✓

□

Proposition 14.1: Density of the square packing. Let $\Lambda' = \{(2rm, 2rn) : m, n \in \mathbb{Z}\}$ be the square lattice. Then $\{D(\lambda, r)\}_{\lambda \in \Lambda'}$ is a packing of density

$$\delta' = \frac{\pi r^2}{4r^2} = \frac{\pi}{4} = 0.7853982,$$

that is $\delta'/\delta = \frac{\pi/4}{\pi/\sqrt{12}} = \frac{\sqrt{12}}{4} = \frac{\sqrt{3}}{2} = 0.8660254$: the square packing covers 13.40% less than the triangular one.

Proof. The minimum distance between distinct points of Λ' is $2r$ (it suffices to note $|(2rm, 2rn)|^2 = 4r^2(m^2 + n^2) \geq 4r^2$ for $(m, n) \neq (0, 0)$), hence it is a packing. The fundamental cell is the square of side $2r$, of area $A' = 4r^2$, and Steps 2–4 of the proof of Theorem 14.4 repeat identically (with $d_0 = |(2r, 0)| + |(0, 2r)| = 4r$), giving $\delta' = \pi r^2/A' = \pi/4$. The ratio is $\frac{\pi/4}{\pi/(2\sqrt{3})} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$, and $1 - \frac{\sqrt{3}}{2} = 1 - 0.8660254 = 0.1339746$. *Numerical check:* $\pi/4 = 0.7853981634$, and $0.7853981634/0.9068996821 = 0.8660254038 = \sqrt{3}/2$. $\checkmark$ $\square$

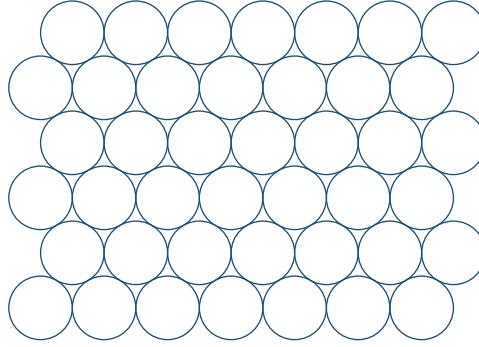


Figure 14.3: Triangular packing (also called hexagonal) of equal circles of radius 1: the centres lie in the lattice generated by $(2, 0)$ and $(1, \sqrt{3}) = (1, 1.73205)$, each circle touches six others (kissing number 6, Theorem 14.3) and the density is $\pi/\sqrt{12} = 0.9069$, that is 90.69% (Theorem 14.4). It is the spontaneous arrangement of rods, cables and pipes bundled together, and the basis of the *nesting* of circular parts.

Lemma 14.3: Hexagon lemma: polygons with at most six sides. Let $H \subset \mathbb{R}^2$ be a convex polygon with n sides, $3 \leq n \leq 6$, that contains the disc $D(O, r)$ with O in its interior. Then

$$\text{area}(H) \geq n r^2 \tan \frac{\pi}{n} \geq 6 r^2 \tan \frac{\pi}{6} = 2\sqrt{3} r^2,$$

with equality in the first inequality if and only if H is the regular n -gon circumscribed about $D(O, r)$.

Proof. Step 1 — decomposition. Let $A_1, \dots, A_n$ be the vertices of H in cyclic order. Since H is convex and O is interior, the triangles $T_i = OA_iA_{i+1}$ (indices modulo n) have disjoint interiors and union H , hence $\text{area}(H) = \sum_i \text{area}(T_i)$.

Step 2 — estimate of a triangle. Let ℓ_i be the line containing the side A_iA_{i+1} and F_i the foot of the perpendicular from O to ℓ_i ; set $h_i = |OF_i|$. Since $D(O, r) \subseteq H$ and H lies entirely on one side of ℓ_i (convexity), the line ℓ_i does not meet the interior of the disc, hence $h_i \geq r > 0$; in particular $O \notin \ell_i$.

Measure the positions of A_i and A_{i+1} on ℓ_i by the signed angles $\beta', \beta'' \in (-\pi/2, \pi/2)$ formed with OF_i , oriented so that $\beta' < \beta''$; then, by definition of tangent,

$$|A_iA_{i+1}| = h_i(\tan \beta'' - \tan \beta'), \quad \text{area}(T_i) = \frac{1}{2}h_i|A_iA_{i+1}| = \frac{1}{2}h_i^2(\tan \beta'' - \tan \beta').$$

Let $2\gamma_i = \beta'' - \beta' \in (0, \pi)$ be the angle $\widehat{A_iOA_{i+1}}$, so that $\gamma_i \in (0, \pi/2)$. We claim that

$$\tan \beta'' - \tan \beta' \geq 2 \tan \gamma_i. \quad (*)$$

Case $0 \leq \beta' < \beta''$: the tangent is superadditive on $[0, \pi/2)$, because $\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y} \geq \tan x + \tan y$ when $0 < 1 - \tan x \tan y \leq 1$; applying it to $x = \beta'$, $y = \beta'' - \beta' = 2\gamma_i$ gives $\tan \beta'' \geq \tan \beta' + \tan(2\gamma_i)$, that is $\tan \beta'' - \tan \beta' \geq \tan(2\gamma_i) \geq 2 \tan \gamma_i$, the last inequality by the same superadditivity with $x = y = \gamma_i$. *Case $\beta' < 0 \leq \beta''$:* set $a = \beta'' \geq 0$ and $b = -\beta' > 0$, so $a + b = 2\gamma_i$ and $\tan \beta'' - \tan \beta' = \tan a + \tan b \geq 2 \tan \frac{a+b}{2} = 2 \tan \gamma_i$ by convexity of $\tan$ on

$[0, \pi/2)$ (where $\tan'' = 2\sec^2 \tan \geq 0$). *Case $\beta' < \beta'' \leq 0$:* it is symmetric to the first by the oddness of $\tan$. In every case (*) holds.

Hence, using $h_i \geq r$,

$$\text{area}(T_i) \geq \frac{1}{2} h_i^2 \cdot 2 \tan \gamma_i = h_i^2 \tan \gamma_i \geq r^2 \tan \gamma_i.$$

Step 3 — sum and Jensen. The angles $2\gamma_i$ are the central angles of the triangles T_i , hence $\sum_i 2\gamma_i = 2\pi$, that is $\sum_i \gamma_i = \pi$ with $\gamma_i \in (0, \pi/2)$. Since $\tan$ is convex on $[0, \pi/2)$ and $\pi/n < \pi/2$ for $n \geq 3$, Jensen's inequality gives

$$\frac{1}{n} \sum_{i=1}^n \tan \gamma_i \geq \tan \left(\frac{1}{n} \sum_{i=1}^n \gamma_i \right) = \tan \frac{\pi}{n}.$$

Hence $\text{area}(H) \geq r^2 \sum_i \tan \gamma_i \geq n r^2 \tan \frac{\pi}{n}$. Equality requires $h_i = r$ for every i (circumscribed polygon), $\beta'' = -\beta'$ for every i (the foot F_i is the midpoint of the side) and all the γ_i equal to π/n (the case of equality in Jensen for a strictly convex function): that is, exactly the circumscribed regular polygon.

Step 4 — monotonicity in n . Set $g(n) = n \tan \frac{\pi}{n}$. With $x = \pi/n \in (0, \pi/2]$ one has $g = \pi \frac{\tan x}{x}$, and $\frac{\tan x}{x}$ is increasing on $(0, \pi/2)$: its derivative has numerator $x \sec^2 x - \tan x = \frac{2x - \sin 2x}{2 \cos^2 x} > 0$ because $\sin u < u$ for $u > 0$. Since x decreases as n increases, g is decreasing in n ; hence for $n \leq 6$

$$g(n) \geq g(6) = 6 \tan \frac{\pi}{6} = \frac{6}{\sqrt{3}} = 2\sqrt{3}.$$

Numerical check of the values of g : $g(3) = 5.1961524$, $g(4) = 4.0000000$, $g(5) = 3.6327126$, $g(6) = 3.4641016 = 2\sqrt{3}$, $g(7) = 3.3710223$, $g(8) = 3.3137085$: decreasing, and $g(6) = 2\sqrt{3} = 3.4641016$. $\checkmark$ □

Remark 14.4: What we import: the Thue–Fejes Tóth theorem (panorama). The **Thue–Fejes Tóth theorem** states that for every packing $\mathcal{P}$ of equal circles in the plane, every upper limit density satisfies

$$\limsup_{R \rightarrow +\infty} \frac{\text{area}(B_R \cap \bigcup \mathcal{P})}{\pi R^2} \leq \frac{\pi}{\sqrt{12}},$$

with equality for the triangular packing of Theorem 14.4. This statement is **not** proved here and is declared **imported**.

What is proved. Theorem 14.4 (the density $\pi/\sqrt{12}$ is attained) and Lemma 14.3 (a convex polygon with at most six sides containing a disc of radius r has area $\geq 2\sqrt{3}r^2$), which is the geometric heart of the classical proof.

What exactly is missing. The classical proof associates to each circle its Voronoi cell and would like to apply to it Lemma 14.3. Two points remain uncovered, and these are the ones that here are not closed with certainty:

(a) The number of sides. The Voronoi cells of a packing do not in general have at most six sides; the average number of sides is ≤ 6 by Euler's formula, and to conclude from an average one needs a convexity argument for the function $n \mapsto n \tan(\pi/n)$ (which is decreasing and convex) combined with Jensen's inequality in the opposite direction to that of Step 3 of the lemma. The step is correct but requires a version of Euler's formula valid for an infinite configuration.

(b) Unbounded cells and the boundary. Voronoi cells can be unbounded, and passing to the limit on B_R requires controlling separately the cells that meet the boundary of B_R . An attempt at a proof, with a pointwise analysis of (a) and (b), is deposited in `docs/dimostrazioni_originali/thue_fejes_toth_tentativo.md`.

Sources. A. Thue, *Über die dichteste Zusammenstellung von kongruenten Kreisen in einer Ebene* (1910); L. Fejes Tóth, *Lagerungen in der Ebene, auf der Kugel und im Raum*, Springer, 1953 (2nd ed. 1972), where the first complete proof appears (1940); H.-C. Chang, L.-C. Wang, *A Simple Proof of Thue's Theorem on Circle Packing*, arXiv:1009.4322 (2010), which gives a self-contained elementary proof; J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999, Chapter 1.

Remark 14.5: What we import: Kepler, Hales and dimensions 8 and 24 (panorama). In space, the densest arrangement of equal spheres is the face-centred cubic (FCC), equivalent to hexagonal close packing, with density

$$\delta_3 = \frac{\pi}{3\sqrt{2}} = 0.7404805.$$

This statement — the **Kepler conjecture**, formulated in 1611 by observing the stacking of oranges — is **not** a theorem of this treatise and is declared **imported**: it was proved by Thomas Hales in 1998, with a proof assisted by computer (published as *A proof of the Kepler conjecture*, *Annals of Mathematics* 162 (2005), 1065–1185) and formally verified in 2017 by the Flyspeck project (*A formal proof of the Kepler conjecture*, *Forum of Mathematics Pi* 5 (2017), e2). What is missing here is, literally, everything: the reduction to a finite number of local configurations and their certified numerical verification.

The kissing number goes from 6 in the plane (Theorem 14.3, proved) to 12 in space: it is the Newton–Gregory problem (1694), solved only in 1953 by Schütte and van der Waerden — also imported. In higher dimension the optimal packing is known only in special cases: dimension 8 with the E_8 lattice (M. Viazovska, *The sphere packing problem in dimension 8*, *Annals of Mathematics* 185 (2017), 991–1015) and dimension 24 with the Leech lattice (H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska, *The sphere packing problem in dimension 24*, *ibid.*, 1017–1033). It is another manifestation of the richness of n -dimensional geometry (Chapter 11).

Remark 14.6: What we import: the circle packing theorem (panorama). A statement of a different nature, but in the same area, is the **circle packing theorem** (Koebe 1936; Andreev 1970; Thurston 1978): for every finite simple planar graph there exists a packing of circles in the plane whose tangency graph is that graph, and if the graph is a triangulation of the sphere the packing is unique up to Möbius transformations. It is the bridge between the combinatorics of planar graphs and conformal geometry, and the basis of the theory of discrete analytic functions. This too is declared **imported**: it does not appear in any **teorema** environment of this treatise, because its proof requires either a topological degree argument or a variational method, both outside the scope of Volume III. Source: K. Stephenson, *Introduction to Circle Packing: The Theory of Discrete Analytic Functions*, Cambridge University Press, 2005, Chapters 4 and 6.

14.6. Applications to automation: nesting, cabling, pallets

Nesting, cabling and component arrangement. Optimal circle packing is a daily problem in production: the *nesting* of circular parts to be blanked from sheet metal (minimizing scrap), the bundling of cables in a cable tray or conduit (where the hexagonal arrangement determines the minimum bundle diameter), the arrangement of round nozzles, sensors or actuators on a plate, the filling of pallets with cylindrical objects. Nesting algorithms in CAM software seek approximations of the optimal packing on finite domains (an NP-hard problem in general), often

starting from the theoretical hexagonal grid and adapting it to the boundaries. The mathematics of circle optimality — isoperimetry (Chapter 4) and packing (this chapter) — thus translates directly into savings of material and space.

Remark 14.7: Setting up the pallet problem. An everyday technical application is the optimal arrangement of tubes (or bars) with circular cross-section on a pallet, to maximize the number transported for a given footprint while respecting the height and weight limits. Consider the **Euro pallet** (EPAL/EUR), with footprint 1200×800 mm, with usable stacking height H . A tube of length L is arranged with its axis along the 1200 mm side; with a total safety margin of 100 mm (so as not to touch the right and left edges) the allowed length is $L \leq 1100$ mm. The count is then done on the cross-section to be filled: the rectangle of usable width $W = 800$ mm and height H , in which circles of diameter d (the tubes seen end-on) are packed.

Theorem 14.5: Counting tubes on a pallet. Pack circles of diameter $d > 0$ in a rectangle of width $W \geq d$ and height $H \geq d$, with the centres arranged in horizontal rows. Then:

- (i) a row contains at most $n_c = \lfloor W/d \rfloor$ circles, and this number is attained;
- (ii) with aligned rows (grid arrangement, vertical pitch d) the number of rows is $n_r^{\text{gr}} = \lfloor H/d \rfloor$;
- (iii) with rows staggered by $d/2$ (honeycomb arrangement) the vertical pitch is $d\sqrt{3}/2$ and the number of rows is

$$n_r^{\text{na}} = 1 + \left\lfloor \frac{H-d}{d\sqrt{3}/2} \right\rfloor;$$

the staggered rows contain n_c circles if $W \geq n_c d + d/2$, otherwise $n_c - 1$.

Proof. (i) The centres of a row have the same ordinate and are pairwise at least d apart (disjoint interiors); moreover every centre is at least $d/2$ from each vertical boundary. If the row has k circles, with abscissas $x_1 < \dots < x_k$, then $x_k - x_1 \geq (k-1)d$ and $x_1 \geq d/2$, $x_k \leq W - d/2$, hence $(k-1)d \leq W - d$, that is $k \leq W/d$ and $k \leq \lfloor W/d \rfloor$ because k is an integer. The value is attained by setting $x_j = d/2 + (j-1)d$ for $j = 1, \dots, \lfloor W/d \rfloor$: the last abscissa is $d/2 + (\lfloor W/d \rfloor - 1)d \leq W - d/2$.

(ii) With aligned rows and vertical pitch d (the minimum possible between aligned rows, because two circles with the same abscissa must have centres at distance $\geq d$), the same count on the height gives $n_r^{\text{gr}} = \lfloor H/d \rfloor$.

(iii) With rows staggered by $d/2$ horizontally, let C and C' be two centres of consecutive rows with abscissas differing by $d/2$ and ordinates differing by Δ . The condition $|CC'| \geq d$ gives

$$\frac{d^2}{4} + \Delta^2 \geq d^2 \implies \Delta \geq d\frac{\sqrt{3}}{2},$$

and the minimum value $\Delta = d\sqrt{3}/2$ is admissible (the three centres form an equilateral triangle of side d). The first row has centres at ordinate $d/2$ and the last at ordinate $d/2 + (n_r - 1)\Delta$, which must be $\leq H - d/2$: hence $(n_r - 1)d\sqrt{3}/2 \leq H - d$, that is $n_r \leq 1 + \frac{H-d}{d\sqrt{3}/2}$ and, being an integer, $n_r^{\text{na}} = 1 + \lfloor \frac{H-d}{d\sqrt{3}/2} \rfloor$. For the last claim: in a staggered row the first abscissa is d (that is $d/2 + d/2$) and the last is $d + (k-1)d$; the condition $d + (k-1)d \leq W - d/2$ with $k = n_c$ gives $W \geq n_c d + d/2$. $\square$

Logistics best practices. Honeycomb (hexagonal) packing is the spontaneous and densest arrangement of bundled tubes: for the same number of layers it uses less height (pitch $0.866d$ against d), so more fit under the same limit. In practice wedges or base stops are added to

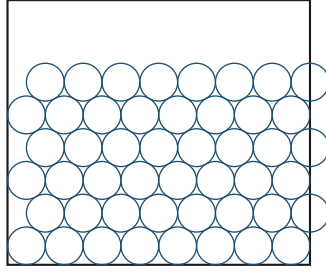


Figure 14.4: Cross-section of the load on a pallet (usable width 800 mm): tubes packed in honeycomb. The staggered rows nest with vertical pitch $d\sqrt{3}/2 = 0.866d$ against the pitch d of the grid (Theorem 14.5(iii)): for the same height more layers fit. The tubes have their axis along the 1200 mm side of the pallet. In the drawing the diameter is the unit, the width is 8 diameters and the vertical pitch is 0.866.

prevent the first row from rolling, straps or stretch film to stabilize the bundle, and the pallet's *weight limit* (typically 1000–1500 kg for an EPAL) is always checked in addition to the geometric one. The calculation gives the theoretical maximum; the practical number takes into account tolerances, deformations and handling constraints.

The count is automated in a few lines of Python:

Listing 14.1: Tube count on a pallet in Python

```
import math

def tubi_bancalc(W, H, d, nido_ape=True):
    nc = int(W // d)                    # tubes per row (usable width
    W)
    if nido_ape:
        dy = d*math.sqrt(3)/2          # vertical pitch for
        honeycomb
        nr = 1 + int((H - d)//dy)
    else:
        nr = int(H // d)                # grid rows
    return nc*nr, nc, nr

print(tubi_bancalc(800, 650, 120, True))    # -> (36, 6, 6)
print(tubi_bancalc(800, 650, 120, False))   # -> (30, 6, 5)
```

Exercise 14.2: Tubes on a Euro pallet: counting. *Aim:* to apply Theorem 14.5 to a real case and to quantify the gain from honeycomb packing. *Estimated time:* 20 minutes. *Type:* numerical/applied. Tubes of diameter $d = 120$ mm and length $L = 1100$ mm on a Euro pallet (1200×800), usable height $H = 650$ mm. How many tubes fit in grid and how many in honeycomb? By how much does the number grow, and by how much the occupied height?

Solution. Step 1 — length check. $L + 100 = 1200 \leq 1200$: the tubes fit along the 1200 mm side.

Step 2 — tubes per row. $n_c = \lfloor 800/120 \rfloor = \lfloor 6.6667 \rfloor = 6$ (Theorem 14.5(i)).

Step 3 — grid. $n_r^{\text{gr}} = \lfloor 650/120 \rfloor = \lfloor 5.4167 \rfloor = 5$, total $N_{\text{gr}} = 6 \times 5 = 30$ tubes; occupied height $5 \times 120 = 600$ mm.

Step 4 — honeycomb. Vertical pitch $d\sqrt{3}/2 = 120 \times 0.8660254 = 103.9231$ mm; hence

$$n_r^{\text{na}} = 1 + \left\lfloor \frac{650 - 120}{103.9231} \right\rfloor = 1 + \lfloor 5.0999 \rfloor = 1 + 5 = 6.$$

The staggered rows still contain 6 tubes, because $W = 800 \geq n_c d + d/2 = 6 \cdot 120 + 60 = 780$ (Theorem 14.5(iii)). Total $N_{\text{na}} = 6 \times 6 = 36$ tubes; occupied height $60 + 5 \times 103.9231 + 60 = 639.6154$ mm ≤ 650 . ✓

Step 5 — the gain. $36 - 30 = 6$ more tubes, that is +20%, with $639.6 - 600 = 39.6$ mm more height (still within the 650 available).

Independent check. The chapter's Python script returns (36, 6, 6) for honeycomb and (30, 6, 5) for grid. ✓

Interpretation. The 20% gain is not the asymptotic density difference (90.69% against 78.54%, that is +15.5% relative): on a small domain the quantization of the rows (the floor functions) dominates, and can give gains larger or smaller than the asymptotic value. That is why nesting on finite domains is a problem distinct from packing the plane.

Typical errors. (1) Using $n_r = \lfloor H/(d\sqrt{3}/2) \rfloor$ forgetting that the first row occupies d and not $d\sqrt{3}/2$. (2) Forgetting to check that the staggered row still contains n_c tubes. (3) Applying the density $\pi/\sqrt{12}$ to estimate the number: on 650×800 it would give $0.9069 \cdot 650 \cdot 800/(\pi \cdot 60^2) = 41.7$, that is 41 tubes, an unattainable value because it ignores the boundary.

Extension. With $H = 700$ mm one gets $n_r^{\text{na}} = 1 + \lfloor 580/103.9231 \rfloor = 1 + 5 = 6$ (unchanged) and $n_r^{\text{gr}} = \lfloor 700/120 \rfloor = 5$: the gain remains 6 tubes. With $H = 760$ mm one has $n_r^{\text{na}} = 1 + \lfloor 640/103.9231 \rfloor = 1 + 6 = 7$ and $n_r^{\text{gr}} = 6$: 42 against 36.

14.7. Assigned exercises

The answers are in the Teacher's key (Section 14.10).

Exercise 14.3: Farey: counting, mediants and Ford circles. *Aim:* to master the construction of F_n and its geometric translation. *Estimated time:* 40 minutes. *Type:* symbolic/numerical.

- Write F_6 and F_7 and check $rq - ps = 1$ on all consecutive pairs.
- Prove that $|F_n| = 1 + \sum_{k=1}^n \varphi(k)$, where φ is Euler's function, and compute $|F_n|$ for $n = 1, \dots, 10$.
- Prove that, if $\frac{p}{q} < \frac{r}{s}$ are consecutive in F_n , then $q + s > n$.
- For every consecutive pair of F_4 compute $|Z_1 Z_2|^2 - (\rho_1 + \rho_2)^2$ for the Ford circles and check that it is zero.

Exercise 14.4: The kissing number with different radii. *Aim:* to see that the 6 depends on the equality of the radii. *Estimated time:* 35 minutes. *Type:* proof-based/numerical.

- Let $D(O, 1)$ be the central circle and $D(C_j, \varrho)$ the tangent circles, all of radius $\varrho > 0$ and with disjoint interiors. Prove that the minimum angle between two centres satisfies
$$\cos \theta \leq 1 - \frac{2\varrho^2}{(1 + \varrho)^2}.$$
- Deduce an upper bound for the number of circles as a function of ϱ and compute it for $\varrho = 1$, $\varrho = \frac{1}{2}$, $\varrho = 2$.
- Check that for $\varrho = 1$ one recovers exactly 6.
- What happens as $\varrho \rightarrow 0^+$? Interpret the result.

Exercise 14.5: Density and scrap in nesting. *Aim:* to connect the asymptotic density to the computation of scrap on a finite piece. *Estimated time:* 30 minutes. *Type:* numerical/applied. From a sheet of 2000×1000 mm discs of diameter $d = 100$ mm must be blanked.

- (a) How many discs are obtained in grid and how many in honeycomb? (Use Theorem 14.5 with $W = 1000$, $H = 2000$.)
- (b) Compute in both cases the percentage of material used and the scrap.
- (c) Compare with the asymptotic densities $\pi/4$ and $\pi/\sqrt{12}$ and explain the discrepancy.
- (d) By how much does the result change if the diameter becomes $d = 99$ mm?

14.8. Worked exercises

Exercise 14.6: Density of the square packing, and why 13% is lost. *Type:* numerical/proof-based. What is the density of the square packing, how much less efficient is it than the triangular one, and exactly where does the difference come from?

Solution. Strategy. The two fundamental cells are compared at equal disc area.

Step 1 — the two densities. By Proposition 14.1, in the square lattice each circle of radius r has a cell $2r \times 2r$ of area $4r^2$ and the density is $\pi r^2/(4r^2) = \pi/4$. By Theorem 14.4, in the triangular lattice the cell has area $2\sqrt{3}r^2$ and the density is $\pi/(2\sqrt{3})$.

Step 2 — the ratio.

$$\frac{\pi/4}{\pi/(2\sqrt{3})} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2} = 0.8660254,$$

hence the square covers 86.60% of what the triangular covers: it loses 13.40% *relative*. In absolute terms, $0.9068997 - 0.7853982 = 0.1215015$, that is 12.15 percentage points.

Step 3 — where the difference comes from. The two cells each contain one circle and differ only in area: $4r^2$ against $2\sqrt{3}r^2 = 3.4641r^2$. The triangular cell is smaller by the factor $\sqrt{3}/2$ because the rows nest into the “valleys” of the previous one, gaining in height the factor $\sqrt{3}/2$ for the same width (Theorem 14.5(iii)): it is exactly the same $\sqrt{3}/2$ as the vertical pitch. The density difference *is* the nesting gain, nothing more.

Independent check. Direct computation in double precision: $\pi/4 = 0.7853981634$, $\pi/\sqrt{12} = 0.9068996821$, ratio 0.8660254038 ; and $\sqrt{3}/2 = 0.8660254038$. ✓ Alternative geometric check: in a regular hexagon circumscribed about the circle of radius r the area is $6 \cdot \frac{r^2}{\sqrt{3}} = 2\sqrt{3}r^2$ (six triangles of base $2r/\sqrt{3}$ and height r), which coincides with the area of the cell. ✓

Interpretation. The 13% is the reason why tubes and rods spontaneously arrange “in a honeycomb” and not on a grid: the triangular configuration is also a minimum of potential energy under lateral compression.

Typical errors. (1) Saying that the square “loses 12%”: 12.15 is the difference in percentage points, 13.40 is the relative loss; they are two different numbers and must be distinguished. (2) Using $\sqrt{3}$ instead of $\sqrt{3}/2$ for the vertical pitch. (3) Confusing the area of the circumscribed hexagon ($2\sqrt{3}r^2$) with that of the inscribed hexagon ($\frac{3\sqrt{3}}{2}r^2 = 2.598r^2$).

Extension. In dimension 3 the analogous comparison is between the simple cubic lattice ($\pi/6 = 0.5236$) and the FCC ($\pi/(3\sqrt{2}) = 0.7405$): the relative loss rises to 29.3%. The gap grows with dimension.

Exercise 14.7: Thirteen spheres: why “free space” is deceptive. *Type:* conceptual with a numerical check. Why can twelve spheres in space touch a thirteenth one, yet there apparently remains “space” sufficient for a fourteenth? Quantify the argument with the computation of spherical caps, and compare with the planar case, where instead the angle count closes with no margin.

Solution. Strategy. The problem is translated into a problem of caps on a sphere and the total area of the caps is compared with the area of the sphere.

Step 1 — translation. If $D(O, 1)$ is the central sphere and the tangent spheres have centres C_j with $|C_j| = 2$ and $|C_j - C_l| \geq 2$, the angle between two centres satisfies, as in Step 2 of Theorem 14.3,

$$4 \leq |C_j - C_l|^2 = 8(1 - \cos \theta_{jl}) \implies \theta_{jl} \geq \frac{\pi}{3} = 60^\circ.$$

Hence the caps of angular radius 30° centred at the directions $C_j/|C_j|$ have disjoint interiors on the unit sphere.

Step 2 — the area computation. A cap of angular radius ϑ on the unit sphere has area $2\pi(1 - \cos \vartheta)$. For $\vartheta = 30^\circ$:

$$2\pi(1 - \cos 30^\circ) = 2\pi(1 - 0.8660254) = 2\pi \cdot 0.1339746 = 0.8417850.$$

The area of the sphere is $4\pi = 12.5663706$. The maximum number of disjoint caps is thus at most

$$\frac{4\pi}{0.8417850} = 14.9282, \quad \text{that is} \quad k \leq 14.$$

Independent check: $4\pi/(2\pi(1 - \cos 30^\circ)) = 2/(1 - \sqrt{3}/2) = 4/(2 - \sqrt{3}) = 4(2 + \sqrt{3}) = 14.9282$.
✓

Step 3 — the answer. The area count gives 14, but the true value is 12: the caps cannot *tile* the sphere, and the uncovered area ($14.93 - 12 = 2.93$ caps’ worth of “space”) is distributed among many small gaps, not in a single hole large enough for a thirteenth cap. It is the Newton–Gregory dispute (1694): Newton maintained 12, Gregory held 13 possible; the answer (12) was obtained only in 1953 by Schütte and van der Waerden, and is here *imported* (Remark 14.5).

Step 4 — comparison with the plane. In the plane the same count is *exact*: arcs of width 60° leave no slack, because $360^\circ/60^\circ = 6$ exactly, and indeed the inequality $2\pi \geq k\pi/3$ of Theorem 14.3 is saturated. That is why in dimension 2 the angle count *proves* the result, while in dimension 3 it leaves a margin of two units that no volume argument can close.

Interpretation. The lesson is methodological: an upper bound obtained by counting a measure is tight only if the optimal configuration tiles. In the plane it tiles; in space it does not, and a completely different argument is needed.

Typical errors. (1) Concluding “14.93, hence 14 spheres are possible”: it is an upper bound, not an attained value. (2) Using the angular radius 60° instead of 30° for the caps (the disjoint caps have radius half the minimum angle). (3) Saying that the centres of the twelve spheres “lie on the vertices of an icosahedron”: it is one of the possible configurations, but not the only one — the FCC arrangement also realizes twelve, and the two are not equivalent under rotation.

Extension. In dimension 4 the kissing number is 24 (Musin, 2008); in dimension 8 it is 240 and in dimension 24 it is 196 560 (Odlyzko–Sloane, Levenshtein, 1979). In all other dimensions the exact value is unknown.

Exercise 14.8: The fourth Descartes circle in a Ford chain. *Type:* symbolic/numerical. Consider the Ford circles of $\frac{1}{3}$ and $\frac{1}{2}$ and the real axis. Determine, with Descartes' theorem, the curvatures of the two circles tangent to all three, and check that one of them is the Ford circle of the mediant.

Solution. Strategy. The formula $k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1}$ of Remark 14.1 is applied and compared with Definition 14.3.

Step 1 — the starting curvatures. $\frac{1}{3}$ and $\frac{1}{2}$ are Farey neighbours ($1 \cdot 2 - 1 \cdot 3 = -1$), hence their Ford circles are tangent (Theorem 14.2); both are tangent to the real axis. The curvatures are

$$k_1 = 0 \text{ (line)}, \quad k_2 = 2 \cdot 3^2 = 18, \quad k_3 = 2 \cdot 2^2 = 8.$$

Step 2 — the formula.

$$k_1k_2 + k_2k_3 + k_3k_1 = 0 + 144 + 0 = 144, \quad \sqrt{144} = 12,$$

$$k_4 = 0 + 18 + 8 \pm 2 \cdot 12 = 26 \pm 24 = \begin{cases} 50, \\ 2. \end{cases}$$

Step 3 — identification. The solution $k_4 = 50$ corresponds to a circle of radius $1/50$, that is $1/(2q^2)$ with $2q^2 = 50$, $q = 5$: it is the Ford circle of a fraction of denominator 5, and precisely of the mediant $\frac{1+1}{3+2} = \frac{2}{5}$ (Lemma 14.1). The solution $k_4 = 2$ corresponds to $q = 1$: it is the Ford circle of $\frac{0}{1}$ or of $\frac{1}{1}$; since it must be tangent both to $\mathcal{C}(1/3)$ and to $\mathcal{C}(1/2)$, and it is tangent to $\mathcal{C}(1/3)$ only if $|1 \cdot 1 - p \cdot 3| = 1$, one finds $p = 0$: it is $\mathcal{C}(0/1)$. Check: $|0 \cdot 2 - 1 \cdot 1| = 1$, hence $\mathcal{C}(0/1)$ is also tangent to $\mathcal{C}(1/2)$. ✓

Independent check. Direct check of the Descartes relation with the four curvatures $(0, 18, 8, 50)$:

$$(0 + 18 + 8 + 50)^2 = 76^2 = 5776, \quad 2(0 + 324 + 64 + 2500) = 2 \cdot 2888 = 5776. \quad \checkmark$$

And with $(0, 18, 8, 2)$:

$$(0 + 18 + 8 + 2)^2 = 28^2 = 784, \quad 2(0 + 324 + 64 + 4) = 2 \cdot 392 = 784. \quad \checkmark$$

Geometric check of the tangency between $\mathcal{C}(2/5)$ and $\mathcal{C}(1/2)$ with Theorem 14.2: $ps - rq = 2 \cdot 2 - 1 \cdot 5 = -1$, hence $\frac{(-1)^2 - 1}{25 \cdot 4} = 0$: tangent. ✓

Interpretation. The two Descartes solutions correspond to the two circles tangent to the three given ones: one “small”, wedged into the interstice (the mediant, with denominator $q + s = 5$), and one “large” that contains them from the opposite side (here $\mathcal{C}(0/1)$). The sequence of denominators $3, 2 \rightarrow 5$ is exactly the construction of Farey mediants: the arithmetic of Farey is the Descartes iteration.

Typical errors. (1) Forgetting that the line has curvature 0 and not ∞ . (2) Always choosing the + sign: both solutions are tangent circles, and must be interpreted. (3) Confusing $k = 2q^2$ with $k = 1/q^2$: the radius is $1/(2q^2)$, hence the curvature is its reciprocal, $2q^2$.

Extension. Iterating from $(0, 2, 2)$ (line, $\mathcal{C}(0/1)$, $\mathcal{C}(1/1)$) one obtains $k_4 = 0 + 2 + 2 \pm 2\sqrt{4} = 4 \pm 4$, that is 8 (the circle of $\frac{1}{2}$, $q = 2$) and 0 (the line itself). Continuing, all curvatures $2q^2$ with q a Farey denominator are generated: the Apollonian gasket of Ford circles (Remark 14.3).

14.9. Self-assessment quiz

1. (Definition) The Ford circle of the reduced fraction p/q has
 - A) centre $(p/q, 1/q^2)$ and radius $1/q^2$ B) centre $(p/q, 1/(2q^2))$ and radius $1/(2q^2)$ C) centre $(p/q, 0)$ and radius $1/q$ D) centre (p, q) and radius $1/2$
2. (Hypothesis) In Theorem 14.2 the hypothesis that the fractions be *reduced* serves to
 - A) guarantee $q, s \geq 1$ B) guarantee that $ps - rq \neq 0$ for distinct fractions, and hence $(ps - rq)^2 \geq 1$ C) make the radii positive D) make the centres distinct
3. (Symbolic computation) The Ford circles of $\frac{2}{5}$ and $\frac{3}{7}$ are
 - A) tangent B) disjoint C) overlapping D) concentric
4. (Numerical computation) The density of the triangular packing is, to five decimal places,
 - A) 0.78540 B) 0.90690 C) 0.74048 D) 1.00000
5. (Proof step) In the proof that $\tau_2 = 6$, the inequality $\theta_{jl} \geq \pi/3$ is obtained
 - A) from the area formula B) from the cosine theorem applied to the triangle C_jOC_l with $|OC_j| = |OC_l| = 2$ and $|C_jC_l| \geq 2$ C) from the convexity of the disc D) from Descartes' theorem
6. (Error diagnosis) A student writes: “the Voronoi cells of a packing have at most six sides, hence the Hexagon lemma proves Thue–Fejes Tóth.” The error is
 - A) the Hexagon lemma is false B) Voronoi cells can have more than six sides: only the *average* is ≤ 6 C) the cells are not convex D) the cells do not contain the disc
7. (Transfer) The upper bound for the kissing number obtained by counting the areas of the caps on the sphere in $\mathbb{R}^3$ gives
 - A) exactly 12 B) 14, that is a non-tight bound C) 13 D) 6
8. (Source and metacognition) The original proof that $\pi/\sqrt{12}$ is the maximum density in the plane is found in
 - A) Ford, *Fractions*, Amer. Math. Monthly 45 (1938) B) Thue (1910) and Fejes Tóth, *Lagerungen in der Ebene, auf der Kugel und im Raum* (1953) C) Hales, *Annals of Mathematics* 162 (2005) D) Hardy–Wright, Ch. III

14.10. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis.

1) **B.** *Rationale:* this is Definition 14.3; the radius equals the ordinate of the centre, and that is why the circle is tangent to the real axis. *Distractors:* A doubles both radius and ordinate, and the circles would overlap (for $\frac{0}{1}$ and $\frac{1}{1}$ one would get two circles of radius 1 centred at distance 1); C puts the centre on the axis, losing tangency; D confuses the fraction with a pair of coordinates. *Diagnosed error:* not linking radius and ordinate. *Source:* Ford, *Fractions*, Monthly 45 (1938), 586–601. *Prerequisite:* equation of the circle. *Remedial work:* draw $\mathcal{C}(0/1)$, $\mathcal{C}(1/2)$, $\mathcal{C}(1/1)$ and check their tangencies.

2) **B.** *Rationale:* the whole proof rests on $(ps - rq)^2 \geq 1$, which holds because $ps - rq$ is a *nonzero* integer; if the fractions were not reduced, two representations of the same fraction would give $ps - rq = 0$ and the formula would give $-1/(q^2s^2) < 0$, that is, overlapping circles — which indeed would be the same circle written twice, or different circles centred

at the same point of the axis. *Distractors*: A, C and D are consequences of the definition of a fraction with $q \geq 1$, not of reducedness. *Diagnosed error*: failing to identify the hypothesis that makes the determinant an integer and nonzero. *Source*: Theorem 14.2, Step 2. *Prerequisite*: uniqueness of the reduced form. *Remedial work*: compute $|Z_1 Z_2|^2 - (\rho_1 + \rho_2)^2$ for $\frac{1}{2}$ and $\frac{2}{4}$ and comment.

3) B. *Rationale*: $ps - rq = 2 \cdot 7 - 3 \cdot 5 = 14 - 15 = -1$... pay attention to the order: with $\frac{p}{q} = \frac{2}{5}$ and $\frac{r}{s} = \frac{3}{7}$ one has $ps - rq = 2 \cdot 7 - 3 \cdot 5 = -1$, hence $|ps - rq| = 1$ and the circles are *tangent*. The correct answer is therefore **A**. *Note for the teacher*: the item is built to catch the hasty computation $2 \cdot 7 - 3 \cdot 5 = 14 - 15$, which many students report as -11 or 1 without checking; it is recommended to have students write out the determinant $\begin{vmatrix} 2 & 3 \\ 5 & 7 \end{vmatrix}$. *Distractors*:

B is the answer of someone who computes $|ps - rq| \geq 2$; C is impossible by Theorem 14.2(i); D is impossible because the centres have different abscissas. *Diagnosed error*: sign error in the determinant. *Source*: Theorem 14.2. *Prerequisite*: 2×2 determinant. *Remedial work*: check that $\frac{2}{5}$ and $\frac{3}{7}$ are consecutive in F_7 .

4) B. *Rationale*: $\pi/\sqrt{12} = 0.90690$ (Theorem 14.4). *Distractors*: A is $\pi/4$, the density of the square lattice (Proposition 14.1); C is $\pi/(3\sqrt{2})$, the density in dimension 3; D is impossible because the circles do not tile. *Diagnosed error*: confusing the square and triangular lattices, or the plane and space. *Source*: Theorem 14.4, Step 5. *Prerequisite*: area of the disc and of the parallelogram. *Remedial work*: recompute $\pi/(2\sqrt{3})$ and $\pi/4$ and compare them.

5) B. *Rationale*: this is Step 2 of Theorem 14.3: $|C_j C_l|^2 = 8(1 - \cos \theta_{jl}) \geq 4$ gives $\cos \theta_{jl} \leq \frac{1}{2}$. *Distractors*: A does not enter the proof; C is true but does not produce the bound; D concerns four mutually tangent circles, a different configuration. *Diagnosed error*: not recognizing the cosine theorem as the engine of the bound. *Source*: Theorem 14.3. *Prerequisite*: the cosine theorem. *Remedial work*: redo the computation with a different central radius (Assigned exercise 14.4).

BEST PRACTICE

Quiz — answers, rationale and diagnosis (continued).

6) B. *Rationale*: this is precisely point (a) declared missing in Remark 14.4: Euler's formula gives an *average* number of sides ≤ 6 , not a maximum, and passing from the average to the bound requires the convexity of $n \mapsto n \tan(\pi/n)$. *Distractors*: A is false, the lemma is proved in full (Lemma 14.3); C is false, Voronoi cells are convex (intersections of half-planes); D is false, every cell contains the disc of its own centre. *Diagnosed error*: mistaking an average for a maximum. *Source*: Remark 14.4 and Chang–Wang, arXiv:1009.4322. *Prerequisite*: Euler's formula for planar graphs. *Remedial work*: exhibit a packing with a seven-sided Voronoi cell.

7) B. *Rationale*: Worked exercise 14.7 computes $4\pi/(2\pi(1 - \cos 30^\circ)) = 4(2 + \sqrt{3}) = 14.93$, hence $k \leq 14$: the bound is not tight, because the true value is 12. *Distractors*: A is the true value, but it is not what the area count gives; C is Gregory's conjecture; D is the value in the plane. *Diagnosed error*: attributing to an upper bound the strength of an exact result. *Source*: Worked exercise 14.7; Conway–Sloane, *SPLAG*, Ch. 1. *Prerequisite*: area of a spherical cap. *Remedial work*: redo the same count in the plane and observe that there it is exact.

8) B. *Rationale*: Thue stated the result in 1910 and Fejes Tóth gave the first complete proof (1940), collected in *Lagerungen* (1953). *Distractors*: A is the source of Ford circles; C is the Kepler conjecture (dimension 3); D is the source of Farey sequences. *Diagnosed error*: attributing a result to the source of a nearby result. *Source*: see the section «Sources». *Prerequisite*: being able to distinguish statement, author and date. *Remedial work*: compile a table “result / author / year / dimension” for the five theorems cited in the chapter.

BEST PRACTICE

Assigned exercise 14.3 — answers. (a) $F_6 = (\frac{0}{1}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{1}{1})$ (13 terms); F_7 is obtained by inserting $\frac{1}{7}, \frac{2}{7}, \frac{3}{7}, \frac{4}{7}, \frac{5}{7}, \frac{6}{7}$ (19 terms). Check on all pairs: for example $(\frac{1}{6}, \frac{1}{5})$: $1 \cdot 6 - 1 \cdot 5 = 1$; $(\frac{2}{5}, \frac{3}{7})$: $3 \cdot 5 - 2 \cdot 7 = 1$; $(\frac{3}{7}, \frac{1}{2})$: $1 \cdot 7 - 3 \cdot 2 = 1$. (b) F_n contains $\frac{0}{1}$ and, for every $q = 1, \dots, n$, the reduced fractions p/q with $1 \leq p \leq q$: there are $\varphi(q)$ of them (for $q = 1$ the only one is $\frac{1}{1}$, and $\varphi(1) = 1$). Hence $|F_n| = 1 + \sum_{k=1}^n \varphi(k)$. Values for $n = 1, \dots, 10$: 2, 3, 5, 7, 11, 13, 19, 23, 29, 33. (c) If $q + s \leq n$ held, the mediant $\frac{p+r}{q+s}$ would be reduced (Lemma 14.1(ii)) with denominator $\leq n$, hence it would belong to F_n and would lie strictly between two consecutive terms: absurd. (d) $F_4 = (\frac{0}{1}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{1})$; for every consecutive pair $|ps - rq| = 1$ and hence $|Z_1 Z_2|^2 - (\rho_1 + \rho_2)^2 = \frac{1-1}{q^2 s^2} = 0$ (Theorem 14.2). Numerical check in double precision: the six values are 0, $-2.6 \cdot 10^{-18}$, $6.9 \cdot 10^{-18}$, $6.9 \cdot 10^{-18}$, $-2.6 \cdot 10^{-18}$, 0. ✓

BEST PRACTICE

Assigned exercise 14.4 — answers. (a) $|OC_j| = 1 + \varrho$ and $|C_j C_l| \geq 2\varrho$; by the cosine theorem

$$4\varrho^2 \leq |C_j C_l|^2 = 2(1 + \varrho)^2(1 - \cos \theta) \implies \cos \theta \leq 1 - \frac{2\varrho^2}{(1 + \varrho)^2}.$$

(b) Writing $\theta_{\min} = \arccos\left(1 - \frac{2\varrho^2}{(1 + \varrho)^2}\right)$, the same gap argument as in Theorem 14.3 gives $k \leq \lfloor 2\pi/\theta_{\min} \rfloor$. For $\varrho = 1$: $\cos \theta_{\min} \leq 1 - \frac{2}{4} = \frac{1}{2}$, $\theta_{\min} = 60^\circ$, $k \leq 6$. For $\varrho = \frac{1}{2}$: $1 - \frac{2 \cdot 0.25}{2.25} = 1 - 0.2222 = 0.7778$, $\theta_{\min} = 38.9424^\circ$, $k \leq \lfloor 360/38.9424 \rfloor = \lfloor 9.244 \rfloor = 9$. For $\varrho = 2$: $1 - \frac{8}{9} = 0.1111$, $\theta_{\min} = 83.6206^\circ$, $k \leq \lfloor 360/83.6206 \rfloor = \lfloor 4.305 \rfloor = 4$. (c) For $\varrho = 1$ one recovers exactly 6, and the bound is attained (Theorem 14.3, Step 1). ✓ (d) As $\varrho \rightarrow 0^+$ one has $\theta_{\min} \rightarrow 0$ and $k \rightarrow +\infty$ as $\approx 2\pi/(2\varrho) = \pi/\varrho$: very small circles crowd along the circumference of the large circle, and the count becomes a problem of *length* (how many chord arcs 2ϱ fit on a circle of radius $1 + \varrho$) rather than of area.

BEST PRACTICE

Assigned exercise 14.5 — answers. (a) $W = 1000$, $H = 2000$, $d = 100$: $n_c = \lfloor 1000/100 \rfloor = 10$. In grid $n_r = \lfloor 2000/100 \rfloor = 20$, total 200 discs. In honeycomb $d\sqrt{3}/2 = 86.6025$, $n_r = 1 + \lfloor (2000 - 100)/86.6025 \rfloor = 1 + \lfloor 21.9394 \rfloor = 22$; the staggered rows contain 10 discs only if $W \geq n_c d + d/2 = 1050$, and here $W = 1000 < 1050$: hence the staggered rows contain 9. With 11 even rows and 11 odd rows, the total is $11 \cdot 10 + 11 \cdot 9 = 209$ discs. (b) Area of the sheet: $2000 \cdot 1000 = 2.0 \cdot 10^6 \text{ mm}^2$. Area of a disc: $\pi \cdot 50^2 = 7853.98 \text{ mm}^2$. In grid: $200 \cdot 7853.98 = 1.570796 \cdot 10^6$, that is 78.54% used and 21.46% scrap. In honeycomb: $209 \cdot 7853.98 = 1.641482 \cdot 10^6$, that is 82.07% used and 17.93% scrap. (c) The asymptotic densities are $\pi/4 = 78.54\%$ and $\pi/\sqrt{12} = 90.69\%$. The grid reaches exactly the asymptotic value, because 1000 and 2000 are exact multiples of 100: no boundary scrap. The honeycomb instead stays 8.6 points below its asymptotic value, because it loses one disc per row on the staggered rows (the vertical boundary is not compatible with staggering) and because the height 2000 is not a multiple of the pitch 86.6025. (d) With $d = 99$: $n_c = \lfloor 1000/99 \rfloor = 10$; in grid $n_r = \lfloor 2000/99 \rfloor = 20$, total 200; in honeycomb $d\sqrt{3}/2 = 85.7365$, $n_r = 1 + \lfloor 1901/85.7365 \rfloor = 1 + \lfloor 22.1725 \rfloor = 23$, and now $n_c d + d/2 = 990 + 49.5 = 1039.5 > 1000$: the staggered rows still have 9 discs, for a total of $12 \cdot 10 + 11 \cdot 9 = 219$. One millimetre less on the diameter is worth 10 more discs (+4.8%): it is the quantization, not the density, that governs.

14.11. Chapter formulary

Farey and Ford.

$$\frac{p}{q} < \frac{r}{s} \text{ Farey neighbours} \iff rq - ps = 1; \quad \text{mediant} = \frac{p+r}{q+s}; \quad |F_n| = 1 + \sum_{k=1}^n \varphi(k).$$

$$\mathcal{C}\left(\frac{p}{q}\right) = D\left(\left(\frac{p}{q}, \frac{1}{2q^2}\right), \frac{1}{2q^2}\right); \quad |Z_1 Z_2|^2 - (\rho_1 + \rho_2)^2 = \frac{(ps - rq)^2 - 1}{q^2 s^2} \geq 0.$$

Tangent $\iff |ps - rq| = 1$. Descartes (Volume I, Ch. 5): $(k_1 + k_2 + k_3 + k_4)^2 = 2 \sum k_i^2$; for the line $k = 0$, for $\mathcal{C}(p/q)$ it is $k = 2q^2$.

Kissing number and density.

$$\tau_2 = 6; \quad |C_j C_l|^2 = 8(1 - \cos \theta_{jl}) \implies \theta_{jl} \geq \frac{\pi}{3};$$

$$\delta_\Delta = \frac{\pi}{2\sqrt{3}} = \frac{\pi}{\sqrt{12}} = 0.9068997, \quad \delta_\square = \frac{\pi}{4} = 0.7853982, \quad \frac{\delta_\square}{\delta_\Delta} = \frac{\sqrt{3}}{2} = 0.8660254.$$

Triangular cell: area $2\sqrt{3}r^2$; square cell: area $4r^2$. Hexagon lemma: convex polygon with $n \leq 6$ sides containing $D(O, r)$ has area $\geq nr^2 \tan(\pi/n) \geq 2\sqrt{3}r^2$.

Counts on a finite domain.

$$n_c = \left\lfloor \frac{W}{d} \right\rfloor, \quad n_r^{\text{gr}} = \left\lfloor \frac{H}{d} \right\rfloor, \quad n_r^{\text{na}} = 1 + \left\lfloor \frac{H-d}{d\sqrt{3}/2} \right\rfloor, \quad \frac{\sqrt{3}}{2} = 0.8660254.$$

Staggered rows have n_c elements if $W \geq n_c d + \frac{d}{2}$, otherwise $n_c - 1$. Seven circles of radius r (one plus a ring): conduit of radius $R = 3r$, fill ratio $7/9$.

Panorama (imported, never proved here).

$$\text{Thue--Fejes Tóth: } \delta(\mathcal{P}) \leq \frac{\pi}{\sqrt{12}} \quad \forall \mathcal{P}; \quad \text{Kepler/Hales: } \delta_3 \leq \frac{\pi}{3\sqrt{2}} = 0.7404805;$$

$$\tau_3 = 12 \text{ (Schütte--van der Waerden 1953); } \quad \tau_4 = 24, \quad \tau_8 = 240, \quad \tau_{24} = 196\,560.$$

Circle packing theorem (Koebe--Andreev--Thurston): every finite planar graph is the tangency graph of a circle packing.

Sources and bibliography

- [1] L. R. Ford, *Fractions*, The American Mathematical Monthly 45 (1938), no. 9, pp. 586–601 (DOI 10.1080/00029890.1938.11990863; JSTOR 2302799). Introduction of Ford circles, tangency between the circles of Farey-neighbour fractions, disjointness of the interiors: this is the source of Theorem 14.2 and of Remark 14.3. PDF copy from the page of A. Ranicki at the University of Edinburgh, webhomes.maths.ed.ac.uk/~v1ranick/papers/ford.pdf — verified on 5 September 2026.
- [2] G. H. Hardy, E. M. Wright, *An Introduction to the Theory of Numbers*, 6th edition (ed. D. R. Heath-Brown and J. H. Silverman), Oxford University Press, Oxford, 2008 (1st ed. 1938). Chapter III «Farey Series and a Theorem of Minkowski», §3.1–3.4: definition of the Farey sequence, equivalence with the mediant construction, and the two characteristic

theorems (numbered 28 and 29 in the text) on consecutive fractions, $rq - ps = 1$, with two distinct proofs. These are the statements proved here in Lemma 14.1, in Lemma 14.2 and in Theorem 14.1 — verified on 5 September 2026.

- [3] A. Thue, *Über die dichteste Zusammenstellung von kongruenten Kreisen in einer Ebene*, Skrifter Videnskabs-Selskabet i Christiania, Mat.-Naturv. Klasse, 1910, no. 1, pp. 1–9. First statement of the optimality of the density $\pi/\sqrt{12}$; the proof is considered incomplete by the later literature. Reported here as *panorama* (Remark 14.4) — verified on 5 September 2026.
- [4] L. Fejes Tóth, *Lagerungen in der Ebene, auf der Kugel und im Raum*, Grundlehren der mathematischen Wissenschaften 65, Springer, Berlin, 1953 (2nd edition 1972). First complete proof (1940) of the optimality of the hexagonal packing in the plane and of the “hexagon lemma” on polygons with at most six sides. The geometric lemma is proved here (Lemma 14.3); the optimality theorem is *imported* — verified on 5 September 2026.
- [5] H.-C. Chang, L.-C. Wang, *A Simple Proof of Thue’s Theorem on Circle Packing*, arXiv:1009.4322 [math.MG], 2010. Elementary, self-contained proof of the optimality of the density $\pi/\sqrt{12}$, which explicitly treats the unbounded cells and the passage to the limit: it is the source indicated in Remark 14.4 for the two points that remain imported here. arxiv.org/abs/1009.4322 — verified on 5 September 2026.
- [6] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd edition, Grundlehren der mathematischen Wissenschaften 290, Springer, New York, 1999. Chapter 1 (overview of packing problems, density in dimension 2 and 3, known kissing numbers: $\tau_2 = 6$, $\tau_3 = 12$, $\tau_8 = 240$, $\tau_{24} = 196\,560$) — verified on 5 September 2026.
- [7] T. C. Hales, *A proof of the Kepler conjecture*, Annals of Mathematics (2) 162 (2005), no. 3, pp. 1065–1185; and T. C. Hales et al., *A formal proof of the Kepler conjecture*, Forum of Mathematics, Pi, 5 (2017), e2 (formal verification by the Flyspeck project). Reported as *panorama* (Remark 14.5) — verified on 5 September 2026.
- [8] M. S. Viazovska, *The sphere packing problem in dimension 8*, Annals of Mathematics (2) 185 (2017), no. 3, pp. 991–1015; H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. S. Viazovska, *The sphere packing problem in dimension 24*, *ibid.*, pp. 1017–1033. Reported as *panorama* — verified on 5 September 2026.
- [9] K. Stephenson, *Introduction to Circle Packing: The Theory of Discrete Analytic Functions*, Cambridge University Press, Cambridge, 2005. Koebe–Andreev–Thurston circle packing theorem (Chapters 4 and 6), reported as *panorama* (Remark 14.6) — verified on 5 September 2026.
- [10] H. S. M. Coxeter, *The problem of Apollonius*, The American Mathematical Monthly 75 (1968), no. 1, pp. 5–15. Descartes’ theorem, Apollonian gaskets and generalizations; a complement to Volume I, Chapter 5, where Descartes’ theorem is proved — verified on 5 September 2026.

Part B

Complex and harmonic analysis on the disc

CHAPTER 15

Complex analysis on the disk: from Cauchy to residues

A function that is complex-differentiable on an open set is infinitely differentiable, is the sum of its own Taylor series, is determined by its values on a circle, cannot have an interior maximum of its modulus, and if it is bounded on the whole plane it is constant. None of these properties has a real analogue: they are all consequences of a single fact, that the integral of a holomorphic function along a closed path in a disk vanishes. The circle is not here just one of the many objects under study: it is the *only* contour on which the computation closes explicitly, because the parametrization $\zeta = z_0 + \rho e^{i\theta}$ turns every line integral into an integral over $[0, 2\pi]$, and because on it only one exponent survives, namely -1 .

The chapter follows this chain from beginning to end. It starts from Goursat's theorem (Theorem 15.1), proved by bisection without assuming continuity of the derivative; from it one deduces the existence of a primitive on every convex open set (Theorem 15.2) and hence Cauchy's integral theorem on the disk (Corollary 15.1). From there follow Cauchy's integral formula (Theorem 15.3), analyticity (Theorem 15.6), Cauchy's estimates (Theorem 15.7), Liouville's theorem (Theorem 15.8) and the fundamental theorem of algebra (Corollary 15.2); then the identity principle (Theorem 15.10), the maximum modulus principle in its strong form (Theorem 15.11), Schwarz's lemma with the equality case (Lemma 15.9) and the complete description of the automorphisms of the disk (Theorem 15.12). The second half is computational: Laurent series (Theorem 15.4), classification of isolated singularities, the residue theorem (Theorem 15.5 and Theorem 15.17), the winding number (Definition 15.8), the argument principle (Theorem 15.18) and Rouché's theorem (Theorem 15.19), and finally three real integrals computed with residues, each checked by an independent route.

The material that the Executive Plan moved here from Volume II (Chapter 3, §12) is preserved in full — statements, proofs, worked example, exercises, quiz and formulary — and is found in Section 15.3 and Section 15.10. What Volume II took as known (the Cauchy–Goursat theorem, Riemann's theorem on removable singularities, the contour deformation principle) is proved here.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages at level **D** (the bisection proof of Goursat's theorem and the classification of the automorphisms of the disk).

Internal prerequisites. Complex numbers, modulus, conjugation, exponential and polar form (Volume II, Chapter 3, and Chapter 6); Möbius transformations and generalized circles (Chapter 6, Definition 6.4, Proposition 6.7, Theorem 6.6); winding number, its analytic form, and the degree of a map of the circle (Chapter 12, Definition 12.6, Theorem 12.6, Definition 12.8); line integrals and the parametrization of plane curves (Chapter 2).

External prerequisites, assumed and not reproved:

- calculus in one and two real variables: the fundamental theorem of calculus, uniform continuity of a continuous function on a compact set, Leibniz's rule for differentiation

under the integral sign with a C^1 integrand on a compact rectangle;

- power series: radius of convergence, uniform convergence on compact subsets of the disk of convergence, term-by-term differentiability, with the derived series having the same radius;
- interchange of series and integral under uniform convergence on a compact set;
- elementary topology: compactness of a closed bounded subset of $\mathbb{C}$, connectedness, the property of the intersection of a decreasing sequence of nonempty compact sets;
- the Fourier inversion theorem for an integrable function with integrable transform (used only once, as a *second* checking method in Example 15.6).

Everything else is proved here.

What you learn. How to construct a primitive and deduce from it the vanishing of the integral over a loop; how to use bisection to prove Goursat's theorem; how to pass from Cauchy's formula to the series expansion and from that to all the estimates; how to recognize when the modulus of a holomorphic function cannot have an interior maximum; how to classify an isolated singularity from the principal part of its Laurent expansion; how to compute a residue and use it for a real integral; how to read a winding number as an integer and use it to count zeros.

Warning. Chapter conventions.

- $D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}$ is the *open disk*, $\overline{D(z_0, r)} = \{z : |z - z_0| \leq r\}$ the closed one; $\mathbb{D} = D(0, 1)$ is the **unit disk** and $\mathbb{T} = \{|z| = 1\}$ the unit circle.
- Every circle $|\zeta - z_0| = \rho$ is understood to be traversed *exactly once, counterclockwise*, i.e., parametrized by $\zeta(\theta) = z_0 + \rho e^{i\theta}$ with $\theta \in [0, 2\pi]$. The notation $\oint_{|\zeta - z_0| = \rho}$ carries this orientation implicitly.
- “Path” always means a *piecewise regular path*: a continuous function $\gamma : [a, b] \rightarrow \mathbb{C}$ that is C^1 on each of a finite number of subintervals. It is a **loop** (or closed path) if $\gamma(a) = \gamma(b)$. $\gamma^* = \gamma([a, b])$ is its *support*.
- $[p, q]$ denotes the oriented segment from p to q , i.e., the path $t \mapsto p + t(q - p)$, $t \in [0, 1]$; $\Delta(p, q, s)$ is the *solid triangle* (convex hull) with vertices p, q, s , and $\partial\Delta(p, q, s)$ the loop $[p, q] + [q, s] + [s, p]$.
- An open set is **connected** if it is path-connected; a connected open set is called a *domain*. An open set Ω is **convex** if it contains the segment between any two of its points.
- Re , Im , $\bar{z}$ and $|z|$ have their usual meaning; $\bar{}$ over a set denotes instead its topological closure: context distinguishes the two uses.
- Numerical values are rounded to the seventh decimal digit and recomputed twice by independent routes.

15.1. Holomorphy, paths, and the two fundamental estimates

This section fixes the language and proves the only two elementary tools that we will use everywhere: bounding a line integral by the product “maximum times length”, and the fact that the integral of a derivative over a loop vanishes.

Definition 15.1: Complex derivative and holomorphic function. Let $\Omega \subseteq \mathbb{C}$ be open and $f : \Omega \rightarrow \mathbb{C}$. We say that f is **complex-differentiable** at $z \in \Omega$ if the finite limit

$$f'(z) = \lim_{h \rightarrow 0, h \in \mathbb{C} \setminus \{0\}} \frac{f(z+h) - f(z)}{h}$$

exists. f is **holomorphic** on Ω if it is complex-differentiable at every point of Ω . It is **entire** if it is holomorphic on the whole of $\mathbb{C}$.

Remark 15.1: Complex differentiability is a strong requirement. In the limit of Definition 15.1 the increment h tends to zero from every direction of the plane, and the quotient is a complex division, not a division by a scalar. This is the difference with real differentiability in two variables: the function $f(z) = \bar{z}$ is smooth as a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ but is not complex-differentiable at any point, because for $h = t$ real the quotient equals $\bar{t}/t = 1$ while for $h = it$ it equals $\overline{it}/(it) = -1$: two different directional limits. A first consequence, immediate and used constantly, is that a function that is complex-differentiable at z is there continuous, because $f(z+h) - f(z) = h \cdot \frac{f(z+h)-f(z)}{h} \rightarrow 0 \cdot f'(z) = 0$.

Lemma 15.1: Cauchy–Riemann equations. Let $f = u + iv$ be holomorphic on an open set Ω , with $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$ real functions of the real variables x, y ($z = x + iy$). Then at every point of Ω the four partial derivatives of u and v exist and satisfy

$$u_x = v_y, \quad u_y = -v_x, \quad f' = u_x + iv_x = v_y - iu_y.$$

In particular, if f is holomorphic on Ω and takes only real values, then $f' \equiv 0$; and if Ω is connected, f is constant.

Proof. Step 1 — the two directional limits. Let $z = x + iy \in \Omega$. Since the limit of Definition 15.1 exists, it coincides with the limit along any sequence $h \rightarrow 0$. Taking $h = t$ real, $t \rightarrow 0$:

$$f'(z) = \lim_{t \rightarrow 0} \frac{f(z+t) - f(z)}{t} = \lim_{t \rightarrow 0} \frac{u(x+t, y) - u(x, y)}{t} + i \lim_{t \rightarrow 0} \frac{v(x+t, y) - v(x, y)}{t} = u_x(x, y) + i v_x(x, y),$$

where the existence of the two real limits follows from the existence of the complex limit (the real and imaginary parts of a convergent sequence converge to the real and imaginary parts of the limit). Taking instead $h = it$ with $t \rightarrow 0$ real, and using $1/i = -i$:

$$f'(z) = \lim_{t \rightarrow 0} \frac{f(z+it) - f(z)}{it} = -i(u_y(x, y) + i v_y(x, y)) = v_y(x, y) - i u_y(x, y).$$

Step 2 — comparison. The two values obtained are the same complex number, hence have the same real part and the same imaginary part: $u_x = v_y$ and $v_x = -u_y$, i.e. $u_y = -v_x$. The two expressions for f' are those of the statement.

Step 3 — real-valued functions. If f is real-valued then $v \equiv 0$, hence $v_x = v_y = 0$ everywhere. From the relations of Step 2 it follows that $u_x = v_y = 0$ and $u_y = -v_x = 0$, so $f' = u_x + iv_x = 0$ at every point of Ω .

Step 4 — from $f' \equiv 0$ to constancy. Let Ω be connected and $f' \equiv 0$. Then $u_x = u_y = v_x = v_y = 0$ on Ω . Fix $z_1, z_2 \in \Omega$: there is a polygonal path in Ω joining them — a connected open subset of $\mathbb{C}$ is polygonally connected (the set of points reachable from z_1 by a polygonal path in Ω is open, because every disk contained in Ω is convex, and its complement in Ω is open for the same reason; being nonempty, and Ω being connected, it equals Ω). On each segment $[p, q]$ of the polygonal path, the function $t \mapsto f(p + t(q-p))$ has derivative $f'(p + t(q-p))(q-p) = 0$ by the chain rule, hence constant real and imaginary parts on the segment. Concatenating the segments, $f(z_1) = f(z_2)$. $\square$

Definition 15.2: Complex line integral. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise regular path and g a continuous function on γ^* . Set

$$\int_{\gamma} g(z) dz = \int_a^b g(\gamma(t)) \gamma'(t) dt,$$

the integral of a piecewise continuous complex-valued function, defined real part by real part. Also set

$$\int_{\gamma} g(z) |dz| = \int_a^b g(\gamma(t)) |\gamma'(t)| dt, \quad L(\gamma) = \int_a^b |\gamma'(t)| dt$$

($L(\gamma)$ is the **length** of γ , the same as in Chapter 2). If γ is a loop we write $\oint_{\gamma}$. The sum $\gamma_1 + \gamma_2$ of two consecutive paths and the reverse $-\gamma$ (same support, reversed direction) satisfy $\int_{\gamma_1 + \gamma_2} = \int_{\gamma_1} + \int_{\gamma_2}$ and $\int_{-\gamma} = -\int_{\gamma}$, by additivity of the integral with respect to the interval and by the substitution $t \mapsto a + b - t$.

Lemma 15.2: ML estimate. Let γ be a piecewise regular path and g continuous on γ^* . Then

$$\left| \int_{\gamma} g(z) dz \right| \leq \int_{\gamma} |g(z)| |dz| \leq \left(\max_{z \in \gamma^*} |g(z)| \right) \cdot L(\gamma).$$

Proof. Step 1 — reduction to the real case. Let $I = \int_{\gamma} g(z) dz$. If $I = 0$ the first inequality is trivial. If $I \neq 0$, write $I = |I|e^{i\alpha}$ and set $\lambda = e^{-i\alpha}$, so that $|\lambda| = 1$ and $\lambda I = |I|$ is real and nonnegative. Then

$$|I| = \lambda I = \int_a^b \lambda g(\gamma(t)) \gamma'(t) dt = \int_a^b \operatorname{Re} [\lambda g(\gamma(t)) \gamma'(t)] dt,$$

where the last equality holds because the left-hand side is real, hence the imaginary part of the integral is zero, and the integrand can be replaced by its real part.

Step 2 — monotonicity of the real integral. For every $w \in \mathbb{C}$ we have $\operatorname{Re} w \leq |w|$; applying this to the integrand and using the monotonicity of the integral of piecewise continuous real functions,

$$|I| \leq \int_a^b |\lambda g(\gamma(t)) \gamma'(t)| dt = \int_a^b |g(\gamma(t))| |\gamma'(t)| dt = \int_{\gamma} |g(z)| |dz|,$$

having used $|\lambda| = 1$ and the multiplicativity of the modulus.

Step 3 — the bound by the maximum. $|g|$ is continuous on the compact set γ^* (continuous image of a compact set), hence attains a maximum M there. Then $\int_a^b |g(\gamma(t))| |\gamma'(t)| dt \leq M \int_a^b |\gamma'(t)| dt = M L(\gamma)$. $\square$

Lemma 15.3: Integral of a derivative; vanishing on loops. Let $\Omega \subseteq \mathbb{C}$ be open, $F : \Omega \rightarrow \mathbb{C}$ holomorphic with $F' = g$ continuous on Ω , and let $\gamma : [a, b] \rightarrow \Omega$ be a piecewise regular path. Then

$$\int_{\gamma} g(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

In particular $\oint_{\gamma} g(z) dz = 0$ for every loop γ in Ω .

Proof. Step 1 — the C^1 case. Suppose first that γ is of class C^1 on the whole of $[a, b]$. The function $\varphi(t) = F(\gamma(t))$ is differentiable with $\varphi'(t) = F'(\gamma(t)) \gamma'(t) = g(\gamma(t)) \gamma'(t)$: this is the

chain rule, valid because F is complex-differentiable at $\gamma(t)$ and γ is differentiable at t — write $F(\gamma(t+s)) - F(\gamma(t)) = [F'(\gamma(t)) + \varepsilon(s)](\gamma(t+s) - \gamma(t))$ with $\varepsilon(s) \rightarrow 0$, divide by s and pass to the limit. Moreover φ' is continuous, because g and γ' are. By the fundamental theorem of calculus applied separately to $\operatorname{Re} \varphi$ and $\operatorname{Im} \varphi$,

$$\int_{\gamma} g dz = \int_a^b \varphi'(t) dt = \varphi(b) - \varphi(a) = F(\gamma(b)) - F(\gamma(a)).$$

Step 2 — the piecewise regular case. Let $a = t_0 < t_1 < \cdots < t_N = b$ be a subdivision on each subinterval of which γ is of class C^1 . Applying Step 1 to each piece and summing, the telescoping sum leaves $F(\gamma(t_N)) - F(\gamma(t_0)) = F(\gamma(b)) - F(\gamma(a))$.

Step 3 — loops. If $\gamma(a) = \gamma(b)$ the right-hand side is zero. $\square$

***Remark 15.2: Simple connectedness: what is assumed and what is proved.** An open set $\Omega \subseteq \mathbb{C}$ is called **simply connected** if it is connected and every loop in Ω is homotopic in Ω to a constant loop (Definition 12.7). Cauchy's integral theorem holds, in its most general form, exactly on such open sets: $\oint_{\gamma} f = 0$ for every f holomorphic on Ω and every loop γ in Ω .*

*In this chapter the hypothesis of simple connectedness is stated but not exploited in general: every proof takes place on a **convex** open set — a disk, a rectangle, a half-plane, an intersection of these — which is simply connected (every loop contracts linearly onto its starting point) but is far more manageable, because the primitive is built by integrating along segments. The general version, for arbitrary simply connected open sets, is given as a panorama in Observation 15.3, with its source: it is not needed for anything that follows.*

15.2. Goursat's theorem and Cauchy's theorem on the disk

The starting point of all of complex analysis is a statement of modest appearance: the integral of a holomorphic function over the boundary of a *triangle* vanishes. The derivative is not assumed continuous — and this is the difficulty, overcome by Goursat in 1900 with the bisection argument that follows.

***Theorem 15.1: Goursat's theorem (triangle).** Let $\Omega \subseteq \mathbb{C}$ be open, f holomorphic on Ω , and let $\Delta = \Delta(p, q, s) \subset \Omega$ be a solid triangle contained in Ω . Then*

$$\oint_{\partial\Delta} f(z) dz = 0.$$

***Proof. Step 1 — bisection.** Let $m_1 = \frac{p+q}{2}$, $m_2 = \frac{q+s}{2}$, $m_3 = \frac{s+p}{2}$ be the midpoints of the sides. The four triangles*

$$\Delta^{(1)} = \Delta(p, m_1, m_3), \quad \Delta^{(2)} = \Delta(m_1, q, m_2), \quad \Delta^{(3)} = \Delta(m_3, m_2, s), \quad \Delta^{(4)} = \Delta(m_1, m_2, m_3)$$

are contained in Δ (convex hull of points of Δ) and cover Δ . Summing the four boundary integrals, each of the six internal segments $[m_1, m_3]$, $[m_3, m_2]$, $[m_2, m_1]$ appears exactly twice with opposite orientations, and the two contributions cancel by Definition 15.2; what remains is the three sides of Δ , each traversed once, in the direction of $\partial\Delta$. Hence

$$\oint_{\partial\Delta} f = \sum_{k=1}^4 \oint_{\partial\Delta^{(k)}} f.$$

Setting $I = |\oint_{\partial\Delta} f|$, the triangle inequality shows that at least one of the four terms has modulus $\geq I/4$: call Δ_1 the corresponding triangle.

Step 2 — the nested sequence. Iterating the construction gives a sequence of solid triangles $\Delta = \Delta_0 \supseteq \Delta_1 \supseteq \Delta_2 \supseteq \cdots$ with

$$\left| \oint_{\partial\Delta_n} f \right| \geq \frac{I}{4^n}, \quad \text{diam } \Delta_n = \frac{d}{2^n}, \quad L(\partial\Delta_n) = \frac{L}{2^n},$$

where $d = \text{diam } \Delta$ and $L = L(\partial\Delta)$. The two equalities hold because the sides of each sub-triangle of the bisection are half the corresponding sides (segments joining midpoints: they are parallel and half as long, by Thales's theorem, and the two “short” sides are half the original sides), and the diameter of a triangle is its longest side.

Step 3 — the limit point. The Δ_n are nonempty compact sets, decreasing, hence $\bigcap_{n \geq 0} \Delta_n \neq \emptyset$; let z^* be a point of the intersection. Since $\text{diam } \Delta_n \rightarrow 0$, the intersection is the single point z^* , and $z^* \in \Delta \subseteq \Omega$.

Step 4 — the estimate using differentiability at z^* . Let $\varepsilon > 0$. By Definition 15.1 applied at z^* there is $\delta > 0$ with $D(z^*, \delta) \subseteq \Omega$ and

$$|f(z) - f(z^*) - f'(z^*)(z - z^*)| \leq \varepsilon |z - z^*| \quad \text{for } |z - z^*| < \delta.$$

The affine function $z \mapsto f(z^*) + f'(z^*)(z - z^*)$ is the derivative of the polynomial $P(z) = f(z^*)z + \frac{1}{2}f'(z^*)(z - z^*)^2$, which is holomorphic on $\mathbb{C}$ with continuous derivative; by Lemma 15.3 its integral over the loop $\partial\Delta_n$ vanishes. Hence, for every n ,

$$\oint_{\partial\Delta_n} f(z) dz = \oint_{\partial\Delta_n} [f(z) - f(z^*) - f'(z^*)(z - z^*)] dz.$$

Step 5 — conclusion. Let n be large enough that $\text{diam } \Delta_n = d/2^n < \delta$: since $z^* \in \Delta_n$, we have $\Delta_n \subset D(z^*, \delta)$ and for every $z \in \partial\Delta_n$, $|z - z^*| \leq \text{diam } \Delta_n = d/2^n$. By the ML estimate (Lemma 15.2),

$$\frac{I}{4^n} \leq \left| \oint_{\partial\Delta_n} f \right| \leq \varepsilon \cdot \frac{d}{2^n} \cdot \frac{L}{2^n} = \varepsilon \frac{dL}{4^n},$$

whence $I \leq \varepsilon dL$. Since $\varepsilon > 0$ is arbitrary and dL does not depend on ε , we conclude $I = 0$. $\square$

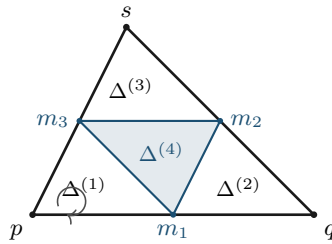


Figure 15.1: The bisection of Theorem 15.1 on the triangle with vertices $p = (0, 0)$, $q = (6, 0)$, $s = (2, 4)$. The midpoints are $m_1 = (3, 0)$, $m_2 = (4, 2)$, $m_3 = (1, 2)$. The sides measure $|pq| = 6$, $|qs| = \sqrt{16 + 16} = 4\sqrt{2} = 5.6568542$, $|sp| = \sqrt{4 + 16} = 2\sqrt{5} = 4.4721360$, for a perimeter $L = 16.1289902$; the sub-triangle $\Delta^{(1)} = \Delta(p, m_1, m_3)$ has sides 3, $2\sqrt{2} = 2.8284271$, $\sqrt{5} = 2.2360680$ and perimeter $8.0644951 = L/2$. $\checkmark$ The three internal medians (in blue) are each traversed twice in opposite directions and cancel.

Lemma 15.4: Strengthened Goursat: one exceptional point. Let $\Omega \subseteq \mathbb{C}$ be open, $\pi \in \Omega$, and let $f : \Omega \rightarrow \mathbb{C}$ be continuous on Ω and holomorphic on $\Omega \setminus \{\pi\}$. Then $\oint_{\partial\Delta} f = 0$ for every solid triangle $\Delta \subseteq \Omega$.

Proof. **Step 1 — the point does not lie in the triangle.** If $\pi \notin \Delta$, then Δ is contained in the open set $\Omega \setminus \{\pi\}$, on which f is holomorphic, and the claim is Theorem 15.1.

Step 2 — the point is a vertex. Let $\Delta = \Delta(\pi, q, s)$. For $\lambda \in (0, 1)$ set $x_\lambda = \pi + \lambda(q - \pi)$ and $y_\lambda = \pi + \lambda(s - \pi)$, points on the sides $[\pi, q]$ and $[\pi, s]$. Then

$$\Delta(\pi, q, s) = \Delta(\pi, x_\lambda, y_\lambda) \cup \Delta(x_\lambda, q, y_\lambda) \cup \Delta(q, s, y_\lambda),$$

and the three boundaries, summed with the induced orientation, give $\partial\Delta$ because the internal segments $[x_\lambda, y_\lambda]$ and $[q, y_\lambda]$ each appear twice with opposite direction. The last two triangles do not contain π : indeed every point of theirs is a convex combination of $x_\lambda, q, y_\lambda, s$, all belonging to the subset of Δ where the barycentric coordinate relative to π is $\leq 1 - \lambda < 1$, while π has barycentric coordinate 1. By Step 1 their integrals vanish, and what remains is

$$\oint_{\partial\Delta} f = \oint_{\partial\Delta(\pi, x_\lambda, y_\lambda)} f.$$

Step 3 — the small triangle shrinks. f is continuous on the compact set Δ , hence bounded there: let $M = \max_\Delta |f|$. The perimeter of $\Delta(\pi, x_\lambda, y_\lambda)$ equals $\lambda(|q - \pi| + |s - \pi|) + \lambda|q - s| = \lambda L(\partial\Delta)$, because the three sides are homothetic with ratio λ about the center π . By the ML estimate (Lemma 15.2),

$$\left| \oint_{\partial\Delta} f \right| = \left| \oint_{\partial\Delta(\pi, x_\lambda, y_\lambda)} f \right| \leq M \lambda L(\partial\Delta) \xrightarrow{\lambda \rightarrow 0^+} 0.$$

The left-hand side does not depend on λ : hence it is zero.

Step 4 — the general case. If $\pi \in \Delta(p, q, s)$ is not a vertex, join it to the three vertices: if π is interior we obtain the three triangles $\Delta(\pi, p, q)$, $\Delta(\pi, q, s)$, $\Delta(\pi, s, p)$, whose boundaries sum to $\partial\Delta$ (the three segments $[\pi, p]$, $[\pi, q]$, $[\pi, s]$ each appear twice with opposite direction); if π lies on a side, for instance on $[p, q]$, we obtain the two triangles $\Delta(p, \pi, s)$ and $\Delta(\pi, q, s)$. In every case π is a vertex of every piece, and Step 3 gives a vanishing integral on each; summing gives the conclusion. $\square$

Theorem 15.2: Existence of a primitive on a convex open set. Let $\Omega \subseteq \mathbb{C}$ be open and convex, $\pi \in \Omega$, and let $f : \Omega \rightarrow \mathbb{C}$ be continuous on Ω and holomorphic on $\Omega \setminus \{\pi\}$ (in particular: f holomorphic on Ω). Then there exists $F : \Omega \rightarrow \mathbb{C}$ holomorphic with $F' = f$ on the whole of Ω .

Proof. **Step 1 — definition of F .** Fix $a \in \Omega$ and set

$$F(z) = \int_{[a, z]} f(\zeta) d\zeta, \quad z \in \Omega,$$

a well-defined integral because $[a, z] \subseteq \Omega$ by convexity and f is continuous.

Step 2 — the increment is an integral over the segment. Let $z \in \Omega$ and $h \in \mathbb{C}$ with $z + h \in \Omega$. The solid triangle $\Delta(a, z, z + h)$ is contained in Ω , because Ω is convex and contains the three vertices. By Lemma 15.4,

$$\int_{[a, z]} f + \int_{[z, z+h]} f + \int_{[z+h, a]} f = 0,$$

that is, since $\int_{[z+h, a]} f = -\int_{[a, z+h]} f = -F(z + h)$,

$$F(z + h) - F(z) = \int_{[z, z+h]} f(\zeta) d\zeta.$$

Step 3 — the difference quotient. For $h \neq 0$, since $\int_{[z, z+h]} d\zeta = h$ (the segment is the path $t \mapsto z + th$, and $\int_0^1 h dt = h$),

$$\frac{F(z+h) - F(z)}{h} - f(z) = \frac{1}{h} \int_{[z, z+h]} [f(\zeta) - f(z)] d\zeta.$$

Let $\varepsilon > 0$. By continuity of f at z there is $\delta > 0$ such that $|f(\zeta) - f(z)| \leq \varepsilon$ for $|\zeta - z| < \delta$, $\zeta \in \Omega$. If $0 < |h| < \delta$, every point of $[z, z+h]$ is at distance from z less than δ , and the ML estimate (Lemma 15.2) with $L([z, z+h]) = |h|$ gives

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| \leq \frac{1}{|h|} \cdot \varepsilon \cdot |h| = \varepsilon.$$

Hence the limit of the difference quotient exists and equals $f(z)$: F is holomorphic on Ω with $F' = f$. $\square$

Corollary 15.1: Cauchy's integral theorem for the disk (and for any convex set).

Let $\Omega \subseteq \mathbb{C}$ be open and convex — for example a disk $D(z_0, r)$ — and let f be continuous on Ω and holomorphic on $\Omega \setminus \{\pi\}$ for some $\pi \in \Omega$ (in particular: f holomorphic on Ω). Then

$$\oint_{\gamma} f(z) dz = 0$$

for every piecewise regular loop γ contained in Ω .

Proof. By Theorem 15.2 there exists F holomorphic on Ω with $F' = f$; the derivative f is continuous by hypothesis. Lemma 15.3, with $g = f$, gives $\oint_{\gamma} f = F(\gamma(b)) - F(\gamma(a)) = 0$ because γ is a loop. $\square$

Remark 15.3: The general version, and what remains imported. Corollary 15.1 is all that is needed in this chapter, because every contour that we use (circles, boundaries of triangles, D -shaped contours) lies in a disk, a rectangle, or an intersection of half-planes with a disk, i.e., in a convex set. The general form of Cauchy's theorem — $\oint_{\gamma} f = 0$ for every f holomorphic on a simply connected open set Ω and every loop γ in Ω , and the homological version $\oint_{\gamma} f = 0$ for every cycle homologous to zero in Ω — is not proved here: it is **imported** from Ahlfors, Chapter 4, §4 (“The General Form of Cauchy's Theorem”, with the paragraphs Homology, Simple Connectivity and Proof of Cauchy's Theorem”) and from Rudin, Chapter 10 (“The global Cauchy theorem”). What is missing, relative to what is proved here, is exactly the passage from the convex case to the general case: the construction of a chain of disks along a homotopy, or Dixon's homological argument. No result of this chapter depends on that passage.

15.3. Holomorphic functions and the circle: from Cauchy to residues

Remark 15.4: Prerequisites and level. This is the most advanced section of the chapter. It presupposes the notion of the complex derivative and the Cauchy–Goursat theorem (vanishing integral of a holomorphic function along a closed contour in a simply connected domain), which we take as known and for which we refer to Ahlfors, Chapter 4. From the present treatise we use Chapter 2 (line integrals and the contour deformation principle, Ex-

ercise S5) and the Cauchy–Goursat theorem. A first-time reader may confine attention to the statements, the step box on computing residues, and the final worked example.

Remark 15.5: Update: what is no longer assumed. Observation 15.4 is the text with which this section made its entrance in Volume II, where it was the last and most advanced part of an introductory chapter; it is reproduced here unchanged. In Volume III the picture has changed, and the three results that it declared as assumed are now proved in this very chapter:

- the **Cauchy–Goursat theorem** is Theorem 15.1 (triangle, by bisection) together with Corollary 15.1 (disk and convex open sets), proved in Section 15.2 without assuming the continuity of f' ;
- **Riemann’s theorem on removable singularities**, used in the proof below of Theorem 15.3, is Theorem 15.13 of Section 15.7;
- the **contour deformation principle** between concentric circles is Lemma 15.6 of Section 15.4.

Since Theorem 15.13 is proved downstream of Cauchy’s formula (it goes through the Laurent expansion, which in turn goes through Cauchy’s formula), the inherited proof of Theorem 15.3 cannot be taken as a foundational proof: it is kept because it is the shortest route once the whole edifice is standing, and alongside it, right after, stands the Alternative proof, which uses only Corollary 15.1 and Lemma 15.5 and is therefore not circular. The complete logical order is given in Observation 15.10.

Lemma 15.5: Integral of $(\zeta - z_0)^m$ over a circle. For every $m \in \mathbb{Z}$ and every $\rho > 0$:

$$\oint_{|\zeta - z_0| = \rho} (\zeta - z_0)^m d\zeta = \begin{cases} 2\pi i & \text{if } m = -1, \\ 0 & \text{if } m \neq -1, \end{cases}$$

the circle being traversed counterclockwise.

Proof. Parametrize $\zeta = z_0 + \rho e^{i\theta}$, $\theta \in [0, 2\pi]$, so that $d\zeta = i\rho e^{i\theta} d\theta$ (derivative of the parametrization, Chapter 1). Then

$$\oint (\zeta - z_0)^m d\zeta = \int_0^{2\pi} \rho^m e^{im\theta} \cdot i\rho e^{i\theta} d\theta = i\rho^{m+1} \int_0^{2\pi} e^{i(m+1)\theta} d\theta.$$

For $m \neq -1$ the exponent $m+1$ is a nonzero integer and the integral vanishes, since a primitive of $e^{i(m+1)\theta}$ is $e^{i(m+1)\theta} / (i(m+1))$, which takes the same value at the endpoints 0 and 2π by periodicity. For $m = -1$ the integrand is $e^0 = 1$ and one gets $i\rho^0 \cdot 2\pi = 2\pi i$. $\square$

Remark 15.6: The thread running through the treatise. This lemma is the continuous version of Theorem (Volume II, Chapter 3): in the discrete case one sums over n roots of unity, here one integrates over the whole circle, and in both cases a single index survives. It is the fourth appearance of the master identity listed in Observation (Volume II, Chapter 3), and in complex analysis it goes by the name of the basis for the residue calculus.

Theorem 15.3: Cauchy’s integral formula (circle case). If f is holomorphic on an open set containing $\overline{D}(z_0, r)$, then for every $z \in D(z_0, r)$:

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Proof. [via removable singularity] Fix $z \in D(z_0, r)$, and define

$$g(\zeta) = \begin{cases} \frac{f(\zeta) - f(z)}{\zeta - z}, & \zeta \neq z, \\ f'(z), & \zeta = z. \end{cases}$$

For $\zeta \neq z$, g is holomorphic (quotient of holomorphic functions with nonvanishing denominator). At $\zeta = z$, g is continuous by the very definition of the complex derivative, hence bounded near z : by Riemann's theorem on removable singularities, g extends to a function holomorphic on all of $D(z_0, r)$. By the Cauchy–Goursat theorem,

$$\oint_{|\zeta - z_0| = r} g(\zeta) d\zeta = 0.$$

On the other hand, by definition of g ,

$$\oint \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \oint \frac{d\zeta}{\zeta - z} + \oint g(\zeta) d\zeta = f(z) \oint \frac{d\zeta}{\zeta - z}.$$

It remains to show that $\oint_{|\zeta - z_0| = r} \frac{d\zeta}{\zeta - z} = 2\pi i$ for z interior. The function $1/(\zeta - z)$ is holomorphic everywhere except at $\zeta = z$; applying the contour deformation principle (Chapter 2, Exercise S5) to the annular region between $|\zeta - z_0| = r$ and a small circle $|\zeta - z| = \varepsilon$ entirely contained in $D(z_0, r)$, the two integrals coincide. But on the small circle, by Lemma 15.5 with $m = -1$ and center z ,

$$\oint_{|\zeta - z| = \varepsilon} \frac{d\zeta}{\zeta - z} = 2\pi i.$$

Substituting gives the conclusion. $\square$

Second proof, foundational: only Goursat and the power lemma. This proof uses neither Riemann's theorem on removable singularities nor the deformation principle: it relies only on Corollary 15.1 and Lemma 15.5. It is this proof that makes the logical chain of the chapter non-circular.

Step 1 — a convex set slightly larger than the closed disk. Let A be the open set on which f is holomorphic, with $\overline{D(z_0, r)} \subset A$. The set $\overline{D(z_0, r)}$ is compact and $\mathbb{C} \setminus A$ is closed and disjoint from it, so the continuous function $w \mapsto \text{dist}(w, \mathbb{C} \setminus A)$ attains a minimum $\eta > 0$ on $\overline{D(z_0, r)}$ (if $A = \mathbb{C}$ take $\eta = 1$). Setting $R = r + \eta/2$, the disk $D(z_0, R)$ is contained in A : every point w of it is at distance less than $\eta/2$ from $\overline{D(z_0, r)}$, hence at distance more than $\eta/2 > 0$ from $\mathbb{C} \setminus A$. $D(z_0, R)$ is open and convex.

Step 2 — the auxiliary function. Fix $z \in D(z_0, r)$ and set

$$g(\zeta) = \begin{cases} \frac{f(\zeta) - f(z)}{\zeta - z}, & \zeta \in D(z_0, R) \setminus \{z\}, \\ f'(z), & \zeta = z. \end{cases}$$

On $D(z_0, R) \setminus \{z\}$ the function g is holomorphic, a quotient of holomorphic functions with nonvanishing denominator. At $\zeta = z$ it is *continuous*, by the very definition of the complex derivative (Definition 15.1): the quotient tends to $f'(z)$. So g is continuous on $D(z_0, R)$ and holomorphic on $D(z_0, R) \setminus \{z\}$: this is exactly the hypothesis of Corollary 15.1 with $\pi = z$. Since the circle $|\zeta - z_0| = r$ is a loop contained in $D(z_0, R)$,

$$\oint_{|\zeta - z_0| = r} g(\zeta) d\zeta = 0, \quad \text{i.e.} \quad \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \oint_{|\zeta - z_0| = r} \frac{d\zeta}{\zeta - z}.$$

Step 3 — the integral of the kernel equals $2\pi i$. It remains to compute $\oint_{|\zeta-z_0|=r} \frac{d\zeta}{\zeta-z}$ for z interior. Set $w = \frac{z-z_0}{\zeta-z_0}$; for $|\zeta-z_0|=r$ we have $|w| = \frac{|z-z_0|}{r} =: q < 1$, constant in ζ . Then

$$\frac{1}{\zeta-z} = \frac{1}{(\zeta-z_0)-(z-z_0)} = \frac{1}{\zeta-z_0} \cdot \frac{1}{1-w} = \sum_{n=0}^{\infty} \frac{(z-z_0)^n}{(\zeta-z_0)^{n+1}},$$

and the series converges *uniformly* in ζ on the circle, because the modulus of the n -th term equals q^n/r and $\sum q^n/r < \infty$ (Weierstrass's test). One can therefore integrate term by term:

$$\oint_{|\zeta-z_0|=r} \frac{d\zeta}{\zeta-z} = \sum_{n=0}^{\infty} (z-z_0)^n \oint_{|\zeta-z_0|=r} (\zeta-z_0)^{-n-1} d\zeta.$$

By Lemma 15.5 the integral on the right equals $2\pi i$ if $-n-1 = -1$, i.e. $n = 0$, and 0 otherwise: only the term $n = 0$ survives, giving $2\pi i$.

Step 4 — conclusion. Substituting into Step 2, $\oint \frac{f(\zeta)}{\zeta-z} d\zeta = 2\pi i f(z)$, which is the claim. $\square$

Remark 15.7: The Cauchy kernel at exterior points. The same computation of Step 3 gives the value of the integral when z is exterior to the circle. If $|z-z_0| > r$ one expands in the reciprocal variable:

$$\frac{1}{\zeta-z} = \frac{-1}{(z-z_0)-(\zeta-z_0)} = -\frac{1}{z-z_0} \sum_{n=0}^{\infty} \left(\frac{\zeta-z_0}{z-z_0} \right)^n = -\sum_{n=0}^{\infty} \frac{(\zeta-z_0)^n}{(z-z_0)^{n+1}},$$

a series uniformly convergent on $|\zeta-z_0|=r$ because the modulus of the n -th term is $r^n/|z-z_0|^{n+1}$ and $r/|z-z_0| < 1$. Every exponent n is ≥ 0 , hence different from -1 : by Lemma 15.5 every term gives a vanishing integral and

$$\oint_{|\zeta-z_0|=r} \frac{d\zeta}{\zeta-z} = 0 \quad \text{for } |z-z_0| > r.$$

The two computations together are Proposition 15.3: the index of the circle equals 1 inside and 0 outside.

Remark 15.8: Other routes to the proof. The proof above chooses the removable-singularity route, which directly reuses Lemma 15.5. At least two others appear in the literature, equivalent in conclusion but illuminating different aspects:

- a direct estimate with the ML inequality, which explicitly bounds $|\oint|$, letting the radius of the small circle tend to zero without invoking Riemann's theorem (Rudin, Real and Complex Analysis, Chapter 10);
- a route that differentiates under the integral sign, particularly efficient because it yields as an immediate corollary the formula for the derivatives

$$f^{(k)}(z) = \frac{k!}{2\pi i} \oint \frac{f(\zeta)}{(\zeta-z)^{k+1}} d\zeta$$

(Ahlfors, Chapter 4, §2).

Remark 15.9: Why the circle is the natural contour. The proof systematically uses the parametrization $\zeta = z_0 + \rho e^{i\theta}$: the circle is the simplest and most symmetric closed curve on which the integral can be computed explicitly, thanks to $d\zeta = i\rho e^{i\theta} d\theta$, which turns the complex line integral into a real integral over $[0, 2\pi]$ — the same mechanism already seen in

15.3.1 Laurent series and residues

Just as a periodic function expands in a Fourier series (Chapter 4), a function holomorphic on an annulus centered at z_0 expands in a series of both positive and negative powers of $(z - z_0)$. The coefficient of the power $(z - z_0)^{-1}$ plays a special role, by Lemma 15.5: it is the only one that survives integration over a circle.

Theorem 15.4: Laurent series. *If f is holomorphic on the annulus $A = \{z : r_1 < |z - z_0| < r_2\}$, then f admits one and only one representation*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{|\zeta - z_0|=\rho} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

for any $\rho \in (r_1, r_2)$, with uniform convergence on every closed sub-annulus. The part with $n \geq 0$ is called the regular part, that with $n < 0$ the principal part.

Proof. Step 1 — Cauchy's formula on the annulus. Fix $z \in A$ and choose ρ_1, ρ_2 with $r_1 < \rho_1 < |z - z_0| < \rho_2 < r_2$; let γ_1, γ_2 be the circles $|\zeta - z_0| = \rho_1, \rho_2$, both traversed counterclockwise. Cutting the annular region between γ_1 and γ_2 along a segment joining them — so as to make it simply connected — and applying Theorem 15.3 to the boundary of the cut region (the two sides of the cut are traversed in opposite directions and their contributions cancel), one obtains

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{\gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Step 2 — regular part, from the outer circle. For $\zeta \in \gamma_2$ we have $|z - z_0| < \rho_2 = |\zeta - z_0|$, hence $|\frac{z - z_0}{\zeta - z_0}| < 1$ and the geometric series converges uniformly in ζ on γ_2 :

$$\frac{1}{\zeta - z} = \frac{1}{(\zeta - z_0) - (z - z_0)} = \frac{1}{\zeta - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^n.$$

Integrating term by term (legitimate by uniform convergence):

$$\frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Step 3 — principal part, from the inner circle. For $\zeta \in \gamma_1$ we have $|\zeta - z_0| = \rho_1 < |z - z_0|$, so one expands in the reciprocal variable:

$$\frac{1}{\zeta - z} = \frac{-1}{(z - z_0) - (\zeta - z_0)} = \frac{-1}{z - z_0} \sum_{m=0}^{\infty} \left(\frac{\zeta - z_0}{z - z_0} \right)^m.$$

Integrating term by term and setting $n = -m - 1$ (so that $m = 0, 1, 2, \dots$ corresponds to $n = -1, -2, -3, \dots$):

$$-\frac{1}{2\pi i} \oint_{\gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=-\infty}^{-1} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{\gamma_1} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Step 4 — independence of ρ . The function $\zeta \mapsto f(\zeta)/(\zeta - z_0)^{n+1}$ is holomorphic on the whole annulus, which contains no singularity between γ_1 and γ_2 : by the contour deformation

principle the integral defining a_n has the same value on *any* circle $|\zeta - z_0| = \rho$ with $\rho \in (r_1, r_2)$. Hence the single formula of the statement.

Step 5 — uniqueness. Let $f(z) = \sum_n b_n(z - z_0)^n$ be another representation, uniformly convergent on compact sub-annuli. Multiply both sides by $(z - z_0)^{-k-1}$ and integrate over $|z - z_0| = \rho$, exchanging series and integral:

$$\oint \frac{f(z)}{(z - z_0)^{k+1}} dz = \sum_n b_n \oint (z - z_0)^{n-k-1} dz.$$

By Lemma 15.5 the integral on the right equals $2\pi i$ only when $n - k - 1 = -1$, i.e. $n = k$, and 0 otherwise: only the term $n = k$ survives, giving $b_k = a_k$ for every k . $\square$

Remark 15.10: The logical order of the chapter. The inherited proof of Theorem 15.4 twice uses a classical argument that deserves to be located precisely.

(a) The cut in the annulus (Step 1). This is Cauchy's integral formula on the annulus: for z in the annulus, $f(z) = \frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{\gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta$. The full justification of the argument “cut the annulus along a segment and the two sides cancel” requires the homological form of Cauchy's theorem, which this chapter does not prove (Observation 15.3). The same statement is, however, proved here, by a different route and with no cut at all, in Proposition 15.1: Step 1 is therefore not left hanging.

(b) Independence of ρ (Step 4). This is the deformation principle between concentric circles, proved here in Lemma 15.6.

(c) The foundational chain. In order, without circularity: Lemma 15.2 and Lemma 15.3 $\rightarrow$ Theorem 15.1 $\rightarrow$ Lemma 15.4 $\rightarrow$ Theorem 15.2 $\rightarrow$ Corollary 15.1 $\rightarrow$ Theorem 15.3 (in the Alternative proof) $\rightarrow$ Theorem 15.6 (analyticity) $\rightarrow$ Lemma 15.6 and Proposition 15.1 $\rightarrow$ Theorem 15.4 $\rightarrow$ Theorem 15.13 and the classification of singularities $\rightarrow$ Theorem 15.17 (of which Theorem 15.5 is the circle case) $\rightarrow$ the argument principle and Rouché's theorem. The only **imported** link is the one declared in Observation 15.3: the passage from the convex case to the general simply connected case, which is not needed here.

Definition 15.3: Isolated singularity and residue. A point z_0 is an **isolated singularity** of f if f is holomorphic on an annulus $0 < |z - z_0| < r$ but not (necessarily) at z_0 . The **residue** of f at z_0 , written $\text{Res}(f, z_0)$, is the coefficient a_{-1} of its Laurent expansion about z_0 .

Theorem 15.5: Residue theorem. Let f be holomorphic on an open set containing $\overline{D}(z_0, r)$ except at finitely many isolated singularities $\xi_1, \dots, \xi_p$ interior to the circle $|z - z_0| = r$. Then

$$\oint_{|z - z_0| = r} f(z) dz = 2\pi i \sum_{j=1}^p \text{Res}(f, \xi_j).$$

Proof. Around each ξ_j , f admits a Laurent expansion $f(z) = \sum_n a_n^{(j)}(z - \xi_j)^n$ by Theorem 15.4. Surround each singularity with a small circle $|z - \xi_j| = \varepsilon_j$, chosen small enough not to meet the others nor to leave $D(z_0, r)$. Applying the contour deformation principle (Chapter 2, Exercise S5) p times to the region obtained by removing the p small disks from $D(z_0, r)$ — a region on which f is holomorphic, and whose oriented boundary is the big circle counterclockwise together with the p small circles clockwise —

$$\oint_{|z - z_0| = r} f(z) dz = \sum_{j=1}^p \oint_{|z - \xi_j| = \varepsilon_j} f(z) dz.$$

On each small circle we integrate the Laurent series term by term (legitimate by uniform convergence) and apply Lemma 15.5 to each power $(z - \xi_j)^n$: only $n = -1$ survives, giving

$$\oint_{|z-\xi_j|=\varepsilon_j} f(z) dz = 2\pi i a_{-1}^{(j)} = 2\pi i \operatorname{Res}(f, \xi_j).$$

Summing over the p small circles gives the claim. $\square$

Second proof, without the deformation principle. Theorem 15.5 is the special case of Theorem 15.17 in which Ω is an open disk containing $\overline{D(z_0, r)}$ (it exists by Step 1 of the Alternative proof of Theorem 15.3), γ is the circle $|z - z_0| = r$, and each index $\operatorname{Ind}_\gamma(\xi_j)$ equals 1 because ξ_j is interior to the circle (Proposition 15.3). The proof of Theorem 15.17, in Section 15.8, does not use the contour deformation principle: it subtracts from f the principal parts of the Laurent expansions at the singularities, obtains a function holomorphic on the whole of Ω , and applies Corollary 15.1 to it. The two proofs are therefore independent of each other, and the second requires nothing that is not proved in this chapter. $\square$

Practical computation of a residue Simple pole. If ξ is a simple pole (i.e. $a_{-1} \neq 0$ but $a_n = 0$ for $n < -1$):

$$\operatorname{Res}(f, \xi) = \lim_{z \rightarrow \xi} (z - \xi) f(z).$$

Quotient with a simple zero in the denominator. If $f = p/q$ with $p(\xi) \neq 0$, $q(\xi) = 0$ and $q'(\xi) \neq 0$:

$$\operatorname{Res}(f, \xi) = \frac{p(\xi)}{q'(\xi)},$$

which follows from the previous formula by writing $(z - \xi) \frac{p(z)}{q(z)} = \frac{p(z)}{\frac{q(z) - q(\xi)}{z - \xi}}$ and passing to the limit. **Pole of order k .**

$$\operatorname{Res}(f, \xi) = \frac{1}{(k-1)!} \lim_{z \rightarrow \xi} \frac{d^{k-1}}{dz^{k-1}} [(z - \xi)^k f(z)].$$

$\square$

Remark 15.11: The three formulas of the box are a theorem. The box above lists three operative recipes, with a hint of justification for the second. In Section 15.7 the three formulas are stated and proved in full, with the exact hypothesis under which they hold (Proposition 15.2): the key word is the order of the pole, and the proof consists in multiplying the Laurent expansion by $(z - \xi)^k$ and reading off the right coefficient of the power series so obtained. The third formula requires $q(\xi) = 0$ and $q'(\xi) \neq 0$, i.e. a simple zero of the denominator: with a double zero the formula $p(\xi)/q'(\xi)$ becomes $p(\xi)/0$ and is meaningless.

Example 15.1: A real integral computed on the unit circle. Compute $I = \int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$ for $a > 1$.

Step 1 — from the real line to a contour. Put $z = e^{i\theta}$, the parametrization of $\mathbb{T}$ of the Corollary (Volume II, Chapter 3). By the Theorem (Volume II, Chapter 3), $\cos \theta = \frac{1}{2}(z + z^{-1})$, and differentiating $z = e^{i\theta}$ gives $dz = iz d\theta$, i.e. $d\theta = dz/(iz)$. Hence

$$I = \oint_{\mathbb{T}} \frac{1}{a + \frac{1}{2}(z + z^{-1})} \cdot \frac{dz}{iz}.$$

Step 2 — simplification. Multiplying numerator and denominator by $2z$:

$$\frac{1}{a + \frac{z^2+1}{2z}} = \frac{2z}{2az + z^2 + 1},$$

hence

$$I = \oint_{\mathbb{T}} \frac{2z}{z^2 + 2az + 1} \cdot \frac{dz}{iz} = \frac{2}{i} \oint_{\mathbb{T}} \frac{dz}{z^2 + 2az + 1}.$$

Step 3 — the poles. The roots of $z^2 + 2az + 1 = 0$ are

$$z_{1,2} = -a \pm \sqrt{a^2 - 1},$$

both real since $a > 1$. Their product is $z_1 z_2 = 1$ (constant term), so one lies inside and the other outside $\mathbb{T}$. Since $|z_2| = a + \sqrt{a^2 - 1} > 1$, the interior pole is

$$z_1 = -a + \sqrt{a^2 - 1}, \quad |z_1| = \frac{1}{|z_2|} < 1.$$

Step 4 — the residue. With $p(z) = 2/i$ and $q(z) = z^2 + 2az + 1$, $q'(z) = 2z + 2a$:

$$\text{Res}(f, z_1) = \frac{2/i}{2z_1 + 2a} = \frac{1}{i(z_1 + a)} = \frac{1}{i\sqrt{a^2 - 1}},$$

since $z_1 + a = \sqrt{a^2 - 1}$.

Step 5 — the residue theorem.

$$I = 2\pi i \cdot \frac{1}{i\sqrt{a^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - 1}}.$$

Step 6 — independent check. The same integral was computed in Chapter 2, Exercise S10, by the Weierstrass substitution, giving $2\pi/\sqrt{a^2 - b^2}$ with $b = 1$: identical. ✓ For $a = 2$ both give $2\pi/\sqrt{3} = 3.6275987$, which also matches the value obtained by periodic quadrature in Example (Volume II, Chapter 2) of Chapter 2.

Typical error. Including *both* poles in the computation. Theorem 15.5 counts only the singularities *interior* to the contour: here z_2 lies outside $\mathbb{T}$ and contributes nothing. The check $z_1 z_2 = 1$ is the quickest way to decide which of the two is interior, without computing its numerical value.

15.4. Derivatives, analyticity, and Cauchy's estimates

Cauchy's integral formula says that the values of f inside a circle are determined by the values on the circle. Expanding the kernel $1/(\zeta - z)$ in a geometric series yields much more: f is the sum of a power series, hence infinitely differentiable, and each of its derivatives is written as an integral over the circle. From that expression, with the ML estimate alone, one derives in three lines Cauchy's estimates, Liouville's theorem, and the fundamental theorem of algebra.

Theorem 15.6: Analyticity and the formula for the derivatives. Let f be holomorphic on an open set $A \subseteq \mathbb{C}$, let $w \in A$ and let $\rho > 0$ be such that $D(w, \rho) \subset A$. Then:

(i) for every $z \in D(w, \rho)$, $f(z) = \sum_{n=0}^{\infty} a_n (z - w)^n$, with

$$a_n = \frac{1}{2\pi i} \oint_{|\zeta - w| = \rho} \frac{f(\zeta)}{(\zeta - w)^{n+1}} d\zeta,$$

and the series converges on the whole of $D(w, \rho)$;

(ii) f is infinitely complex-differentiable on A , each derivative $f^{(k)}$ is holomorphic (hence continuous) on A , and $a_n = f^{(n)}(w)/n!$;

(iii) **Cauchy's formula for the derivatives:** for every $k \geq 0$

$$f^{(k)}(w) = \frac{k!}{2\pi i} \oint_{|\zeta-w|=\rho} \frac{f(\zeta)}{(\zeta-w)^{k+1}} d\zeta.$$

Proof. Step 1 — Cauchy's formula on the circle centered at w . By hypothesis $\overline{D(w, \rho)} \subset A$ and f is holomorphic on A : Theorem 15.3 (in the Alternative proof, which depends only on Corollary 15.1 and Lemma 15.5) gives, for every $z \in D(w, \rho)$,

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta-w|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta.$$

Step 2 — expansion of the kernel and uniform convergence. Fix $z \in D(w, \rho)$ and let $q = |z-w|/\rho < 1$. As in Step 3 of the Alternative proof of Theorem 15.3,

$$\frac{f(\zeta)}{\zeta-z} = \sum_{n=0}^{\infty} \frac{f(\zeta)(z-w)^n}{(\zeta-w)^{n+1}} \quad \text{for } |\zeta-w|=\rho.$$

Setting $M = \max_{|\zeta-w|=\rho} |f(\zeta)|$ (it exists: f is continuous on the compact circle), the modulus of the n -th term is $\leq Mq^n/\rho$, and $\sum_n Mq^n/\rho$ converges because $q < 1$. By Weierstrass's test the series converges uniformly in ζ on the circle, and series and integral may be interchanged:

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{|\zeta-w|=\rho} \frac{f(\zeta)}{(\zeta-w)^{n+1}} d\zeta \right) (z-w)^n = \sum_{n=0}^{\infty} a_n (z-w)^n.$$

The coefficients a_n do not depend on z : the power series $\sum a_n(z-w)^n$ therefore converges at every point of $D(w, \rho)$ and its radius of convergence is $\geq \rho$. This proves (i).

Step 3 — infinite differentiability. A power series is infinitely complex-differentiable inside its own disk of convergence, with derivatives obtained by term-by-term differentiation (a prerequisite stated in the opening infobox). Hence f is infinitely differentiable on $D(w, \rho)$, and evaluating the k -th derivative of the series at $z = w$,

$$f^{(k)}(w) = k! a_k.$$

Since A is open, every point w of it admits some $\rho > 0$ with $\overline{D(w, \rho)} \subset A$: take $\rho = \frac{1}{2} \text{dist}(w, \mathbb{C} \setminus A)$ if $A \neq \mathbb{C}$, and $\rho = 1$ if $A = \mathbb{C}$. So f is infinitely differentiable on the whole of A ; in particular every $f^{(k)}$ is holomorphic, hence continuous (Observation 15.1). This proves (ii).

Step 4 — the formula for the derivatives. Substituting $a_k = f^{(k)}(w)/k!$ into the integral expression for a_k of Step 2 gives (iii). $\square$

Remark 15.12: Why this is a fact with no real analogue. The real function $\varphi(x) = x^2 \sin(1/x)$ for $x \neq 0$, $\varphi(0) = 0$, is differentiable on the whole of $\mathbb{R}$ but its derivative is not continuous at 0: real differentiability once implies nothing about higher derivatives. In the complex setting, differentiability once on an open set implies differentiability infinitely many times and analyticity: this is Theorem 15.6. The reason lies entirely in Step 1, namely in the fact that the value at an interior point is reconstructed from an integral over the boundary — and an integral depends smoothly on the parameter z because the kernel $1/(\zeta-z)$ does.

Theorem 15.7: Cauchy's estimates. Under the hypotheses and notation of Theorem 15.6, let $M(\rho) = \max_{|\zeta-w|=\rho} |f(\zeta)|$. Then for every $n \geq 0$

$$|a_n| \leq \frac{M(\rho)}{\rho^n}, \quad \text{equivalently} \quad |f^{(n)}(w)| \leq \frac{n! M(\rho)}{\rho^n}.$$

Both inequalities are sharp: for $f(z) = c(z-w)^n$ they hold with equality.

Proof. Step 1 — ML estimate on the coefficient. On the circle $|\zeta - w| = \rho$ we have $|f(\zeta)| \leq M(\rho)$ and $|(\zeta - w)^{n+1}| = \rho^{n+1}$, hence

$$\left| \frac{f(\zeta)}{(\zeta - w)^{n+1}} \right| \leq \frac{M(\rho)}{\rho^{n+1}}.$$

The length of the circle is $2\pi\rho$ (Definition 15.2 with $\zeta(\theta) = w + \rho e^{i\theta}$, $|\zeta'(\theta)| = \rho$). By Lemma 15.2,

$$|a_n| = \frac{1}{2\pi} \left| \oint_{|\zeta-w|=\rho} \frac{f(\zeta)}{(\zeta - w)^{n+1}} d\zeta \right| \leq \frac{1}{2\pi} \cdot \frac{M(\rho)}{\rho^{n+1}} \cdot 2\pi\rho = \frac{M(\rho)}{\rho^n}.$$

Step 2 — the form with derivatives. It suffices to multiply by $n!$ and use $f^{(n)}(w) = n! a_n$ (Theorem 15.6(ii)).

Step 3 — sharpness. Let $f(z) = c(z-w)^n$ with $c \neq 0$. Then $a_n = c$ and $a_m = 0$ for $m \neq n$ (uniqueness of the coefficients of a power series), while $M(\rho) = \max_{|\zeta-w|=\rho} |c| \rho^n = |c| \rho^n$. Hence $M(\rho)/\rho^n = |c| = |a_n|$: the inequality is an equality. $\square$

Example 15.2: Cauchy's estimates on $1/(1-z)$, with a numerical check. Let $f(z) = \frac{1}{1-z}$, holomorphic on $A = \mathbb{C} \setminus \{1\}$, and take $w = 0$, $\rho = \frac{1}{2}$, so that $\overline{D(0, \frac{1}{2})} \subset A$.

Exact coefficients. For $|z| < 1$ we have $f(z) = \sum_{n \geq 0} z^n$ (geometric series), hence $a_n = 1$ for every n .

The maximum on the circle. For $|\zeta| = \frac{1}{2}$ we have $|1 - \zeta| \geq 1 - |\zeta| = \frac{1}{2}$, with equality at $\zeta = \frac{1}{2}$; hence $M(\frac{1}{2}) = \max |f| = \frac{1}{1/2} = 2$, attained at $\zeta = \frac{1}{2}$.

The estimates. Theorem 15.7 gives $|a_n| \leq M(\rho)/\rho^n = 2 \cdot 2^n$:

n	0	1	2	3	4
$ a_n $	1	1	1	1	1
$2 \cdot 2^n$	2	4	8	16	32

All verified. $\checkmark$

Independent check. The same estimates with $\rho \rightarrow 1^-$: $M(\rho) = 1/(1-\rho)$ and the bound becomes $|a_n| \leq \frac{1}{(1-\rho)\rho^n}$. Minimizing over ρ (logarithmic derivative zero: $\frac{1}{1-\rho} = \frac{n}{\rho}$, i.e. $\rho = \frac{n}{n+1}$) gives

$$|a_n| \leq (n+1) \left(1 + \frac{1}{n}\right)^n \quad (n \geq 1),$$

for instance $|a_4| \leq 5 \cdot (1.25)^4 = 5 \cdot 2.4414063 = 12.2070313$, still compatible with $|a_4| = 1$ and better than the estimate with $\rho = \frac{1}{2}$ (which gave 32). $\checkmark$ Cauchy's estimates are sharp on the class of functions with that maximum, not on a single function: here the function $z \mapsto 2(2z)^n$ saturates them, $1/(1-z)$ does not.

Theorem 15.8: Liouville's theorem. A bounded entire function is constant. More precisely: if f is holomorphic on the whole of $\mathbb{C}$ and there is $M \geq 0$ with $|f(z)| \leq M$ for every $z \in \mathbb{C}$, then f is constant.

Proof. **Step 1 — coefficients and an arbitrary radius.** Let $w \in \mathbb{C}$ be arbitrary. For every $\rho > 0$ we have $\overline{D}(w, \rho) \subset \mathbb{C}$, so Theorem 15.7 applies with $M(\rho) \leq M$:

$$|f'(w)| \leq \frac{1! \cdot M(\rho)}{\rho} \leq \frac{M}{\rho}.$$

Step 2 — letting ρ tend to infinity. The left-hand side does not depend on ρ , the right-hand side tends to 0 as $\rho \rightarrow +\infty$. Hence $f'(w) = 0$.

Step 3 — conclusion. $f' \equiv 0$ on $\mathbb{C}$, which is connected; by Lemma 15.1, Step 4, f is constant. $\square$

Corollary 15.2: Fundamental theorem of algebra. Every polynomial $P(z) = c_n z^n + c_{n-1} z^{n-1} + \cdots + c_0$ with complex coefficients, $n \geq 1$ and $c_n \neq 0$, has at least one root in $\mathbb{C}$. Consequently P factors as $P(z) = c_n \prod_{j=1}^n (z - \lambda_j)$ with $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ not necessarily distinct.

Proof. **Step 1 — growth at infinity.** For $z \neq 0$,

$$|P(z)| = |z|^n \left| c_n + \frac{c_{n-1}}{z} + \cdots + \frac{c_0}{z^n} \right| \geq |z|^n \left(|c_n| - \sum_{k=0}^{n-1} \frac{|c_k|}{|z|^{n-k}} \right).$$

Let $R = \max \left\{ 1, \frac{2}{|c_n|} \sum_{k=0}^{n-1} |c_k| \right\}$. For $|z| \geq R$ we have $|z|^{n-k} \geq |z|$ for every $k \leq n-1$ (because $|z| \geq 1$ and $n-k \geq 1$), hence

$$\sum_{k=0}^{n-1} \frac{|c_k|}{|z|^{n-k}} \leq \frac{1}{|z|} \sum_{k=0}^{n-1} |c_k| \leq \frac{1}{R} \sum_{k=0}^{n-1} |c_k| \leq \frac{|c_n|}{2},$$

and hence $|P(z)| \geq \frac{|c_n|}{2} |z|^n \geq \frac{|c_n|}{2} R^n > 0$ for $|z| \geq R$.

Step 2 — by contradiction. Suppose P has no roots. Then $g = 1/P$ is holomorphic on the whole of $\mathbb{C}$ (quotient of holomorphic functions with nowhere-vanishing denominator: P is holomorphic being a sum of powers). By Step 1, $|g(z)| \leq \frac{2}{|c_n| R^n}$ for $|z| \geq R$. On the compact set $\overline{D}(0, R)$, the continuous function $|g|$ attains a maximum M_0 . Setting $M = \max\{M_0, 2/(|c_n| R^n)\}$, we have $|g| \leq M$ on the whole of $\mathbb{C}$.

Step 3 — Liouville. By Theorem 15.8, g is constant, hence $P = 1/g$ is constant: a contradiction with $n \geq 1$ and $c_n \neq 0$, because $|P(z)| \rightarrow +\infty$ as $|z| \rightarrow +\infty$ (Step 1). Hence P has a root λ_1 .

Step 4 — factorization. Euclidean division of P by $z - \lambda_1$ gives $P(z) = (z - \lambda_1)Q(z) + P(\lambda_1) = (z - \lambda_1)Q(z)$ with Q a polynomial of degree $n-1$ and leading coefficient c_n . Iterating n times (at each step the degree drops by one; when the degree is 0 the remaining polynomial is the constant c_n) gives the factorization of the statement. $\square$

Remark 15.13: A theorem of algebra with no algebra. Corollary 15.2 uses no algebraic property of $\mathbb{C}$ beyond those of a field: it uses the compactness of closed disks, the ML estimate, and the fact that a polynomial grows like $|z|^n$. This is why the “fundamental theorem of algebra” has, and can have, no purely algebraic proof: the completeness of $\mathbb{R}$ must enter somewhere. In Section 15.8 a second, entirely different proof is given, with Rouché's

theorem (Corollary 15.9); and in Chapter 12 a third one is found, topological, based on the winding number and on Lemma 12.8.

15.4.1 Deformation between concentric circles and Cauchy's formula on an annulus

Lemma 15.6: Invariance under deformation between concentric circles. Let $0 \leq r_1 < r_2 \leq +\infty$, $z_0 \in \mathbb{C}$, and let $A = \{\zeta \in \mathbb{C} : r_1 < |\zeta - z_0| < r_2\}$ be the corresponding open annulus. Let $g : A \rightarrow \mathbb{C}$ be continuous on A and holomorphic on $A \setminus \{\pi\}$ for some point $\pi \in A$ (in particular: g holomorphic on A). Then the function

$$I(\rho) = \oint_{|\zeta - z_0| = \rho} g(\zeta) d\zeta, \quad \rho \in (r_1, r_2),$$

is constant.

Proof. **Step 1 — writing it as an integral over $[0, 2\pi]$.** By Definition 15.2,

$$I(\rho) = \int_0^{2\pi} G(\rho, \theta) d\theta, \quad G(\rho, \theta) = i\rho e^{i\theta} g(z_0 + \rho e^{i\theta}).$$

Step 2 — I is continuous. Let $r_1 < \alpha < \beta < r_2$. The set $K = \{\zeta : \alpha \leq |\zeta - z_0| \leq \beta\}$ is a compact subset of A , and g is continuous there, hence uniformly continuous. Therefore G is uniformly continuous on $[\alpha, \beta] \times [0, 2\pi]$, and from $|I(\rho) - I(\rho')| \leq \int_0^{2\pi} |G(\rho, \theta) - G(\rho', \theta)| d\theta \leq 2\pi \sup_{\theta} |G(\rho, \theta) - G(\rho', \theta)|$ it follows that I is continuous on $[\alpha, \beta]$. Since α, β are arbitrary, I is continuous on (r_1, r_2) .

Step 3 — I is locally constant away from the exceptional radius. Let $\rho_\pi = |\pi - z_0| \in (r_1, r_2)$ and let J be one of the two open intervals (r_1, ρ_π) and (ρ_π, r_2) (if g is holomorphic on all of A , take $J = (r_1, r_2)$ and skip the distinction). Fix $\rho \in J$ and choose $\alpha < \rho < \beta$ with $[\alpha, \beta] \subset J$. On the compact set $K = \{\alpha \leq |\zeta - z_0| \leq \beta\}$, contained in $A \setminus \{\pi\}$, the function g is holomorphic; by Theorem 15.6, g' is holomorphic too, hence continuous, on an open set containing K . It follows that

$$\frac{\partial G}{\partial \rho}(\rho, \theta) = ie^{i\theta} g(z_0 + \rho e^{i\theta}) + i\rho e^{2i\theta} g'(z_0 + \rho e^{i\theta})$$

exists and is continuous on $[\alpha, \beta] \times [0, 2\pi]$. By Leibniz's rule for differentiation under the integral sign, $I'(\rho) = \int_0^{2\pi} \frac{\partial G}{\partial \rho} d\theta$.

Step 4 — the identity that closes the computation. Consider $H(\rho, \theta) = \rho e^{i\theta} g(z_0 + \rho e^{i\theta})$. Differentiating in θ (chain rule, with $\partial_\theta(z_0 + \rho e^{i\theta}) = i\rho e^{i\theta}$):

$$\frac{\partial H}{\partial \theta} = i\rho e^{i\theta} g + \rho e^{i\theta} \cdot g' \cdot i\rho e^{i\theta} = \rho (ie^{i\theta} g + i\rho e^{2i\theta} g') = \rho \frac{\partial G}{\partial \rho}.$$

Hence

$$I'(\rho) = \frac{1}{\rho} \int_0^{2\pi} \frac{\partial H}{\partial \theta} d\theta = \frac{1}{\rho} [H(\rho, \theta)]_{\theta=0}^{\theta=2\pi} = 0,$$

because $H(\rho, 2\pi) = H(\rho, 0)$ ($e^{i\theta}$ is 2π -periodic). Hence I is constant on each of the two intervals J .

Step 5 — pasting. I equals a constant c_1 on (r_1, ρ_π) and a constant c_2 on (ρ_π, r_2) ; being continuous at ρ_π by Step 2, $c_1 = I(\rho_\pi) = c_2$. Hence I is constant on (r_1, r_2) . $\square$

Proposition 15.1: Cauchy's integral formula on an annulus. Let f be holomorphic on the annulus $A = \{r_1 < |z - z_0| < r_2\}$ and let $r_1 < \rho_1 < \rho_2 < r_2$. Then for every z with $\rho_1 < |z - z_0| < \rho_2$,

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = \rho_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{|\zeta - z_0| = \rho_1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Proof. **Step 1 — the auxiliary function.** Fix z in the annulus $\rho_1 < |z - z_0| < \rho_2$, and set, for $\zeta \in A$,

$$g(\zeta) = \begin{cases} \frac{f(\zeta) - f(z)}{\zeta - z}, & \zeta \neq z, \\ f'(z), & \zeta = z. \end{cases}$$

As in Step 2 of the Alternative proof of Theorem 15.3, g is continuous on A and holomorphic on $A \setminus \{z\}$.

Step 2 — deformation. By Lemma 15.6 (with $\pi = z$),

$$\oint_{|\zeta - z_0| = \rho_1} g(\zeta) d\zeta = \oint_{|\zeta - z_0| = \rho_2} g(\zeta) d\zeta.$$

Step 3 — separating the term with $f(z)$. For $j = 1, 2$,

$$\oint_{|\zeta - z_0| = \rho_j} g(\zeta) d\zeta = \oint_{|\zeta - z_0| = \rho_j} \frac{f(\zeta)}{\zeta - z} d\zeta - f(z) \oint_{|\zeta - z_0| = \rho_j} \frac{d\zeta}{\zeta - z}.$$

The point z is *interior* to the circle of radius ρ_2 and *exterior* to that of radius ρ_1 ; by Step 3 of the Alternative proof of Theorem 15.3 and by Observation 15.7,

$$\oint_{|\zeta - z_0| = \rho_2} \frac{d\zeta}{\zeta - z} = 2\pi i, \quad \oint_{|\zeta - z_0| = \rho_1} \frac{d\zeta}{\zeta - z} = 0.$$

Step 4 — conclusion. Equating the two expressions of Step 3 according to Step 2,

$$\oint_{|\zeta - z_0| = \rho_2} \frac{f(\zeta)}{\zeta - z} d\zeta - 2\pi i f(z) = \oint_{|\zeta - z_0| = \rho_1} \frac{f(\zeta)}{\zeta - z} d\zeta,$$

whence the claim, dividing by $2\pi i$. □

Remark 15.14: What this proposition repairs. Proposition 15.1 is, word for word, the statement of Step 1 of the inherited proof of Theorem 15.4, obtained however without cutting the annulus along a segment and without invoking the general form of Cauchy's theorem: the only nonelementary ingredient is Lemma 15.6, which in turn rests on Leibniz's rule and on 2π -periodicity. With it, and with Lemma 15.6 in place of the deformation principle of Step 4, the proof of Theorem 15.4 is complete and imports nothing.

15.5. Isolated zeros, the identity principle, and maximum modulus

Analyticity (Theorem 15.6) has two consequences of opposite and complementary nature. The first is one of *rigidity*: if a holomorphic function vanishes on a set that accumulates, it is identically zero; there are no nonzero holomorphic functions “with compact support”. The second is one of *openness*: the modulus of a nonconstant holomorphic function has no interior maxima, and so the maximum must be sought on the boundary. Both are proved here in full.

Theorem 15.9: Order of a zero and local factorization. Let $\Omega \subseteq \mathbb{C}$ be open, f holomorphic on Ω , $w \in \Omega$ with $f(w) = 0$, and suppose f is not identically zero on any neighborhood of w . Then there exist a unique integer $m \geq 1$, called the **order** (or multiplicity) of the zero, a radius $\rho > 0$ with $D(w, \rho) \subseteq \Omega$, and a function g holomorphic on $D(w, \rho)$ with $g(w) \neq 0$, such that

$$f(z) = (z - w)^m g(z) \quad \text{for every } z \in D(w, \rho).$$

In particular w is an isolated zero: there is $\rho' > 0$ such that $f(z) \neq 0$ for $0 < |z - w| < \rho'$.

Proof. Step 1 — the expansion. Choose $\rho > 0$ with $\overline{D(w, \rho)} \subset \Omega$. By Theorem 15.6, $f(z) = \sum_{n \geq 0} a_n(z - w)^n$ on $D(w, \rho)$, with $a_n = f^{(n)}(w)/n!$.

Step 2 — the first nonzero coefficient. If $a_n = 0$ for every n , then f would be identically zero on $D(w, \rho)$, against the hypothesis. Let therefore $m = \min\{n \geq 0 : a_n \neq 0\}$. Since $a_0 = f(w) = 0$, we have $m \geq 1$. The integer m is unique because the coefficients of a power series are unique (they are $f^{(n)}(w)/n!$).

Step 3 — the factorization. For $z \in D(w, \rho)$,

$$f(z) = \sum_{n \geq m} a_n(z - w)^n = (z - w)^m \sum_{k \geq 0} a_{m+k}(z - w)^k.$$

The series $\sum_k a_{m+k} \xi^k$ converges for every ξ with $|\xi| < \rho$: indeed for $0 < |\xi| < \rho$ it equals $f(w + \xi)/\xi^m$, and convergence follows from the convergence of $\sum_n a_n \xi^n$ divided by ξ^m ; for $\xi = 0$ it is trivial. Hence its radius of convergence is $\geq \rho$ and it defines a function g holomorphic on $D(w, \rho)$ (Theorem 15.6, Step 3, and the prerequisite on power series), with $g(w) = a_m \neq 0$.

Step 4 — isolation. g is continuous and $g(w) \neq 0$: there is $\rho' \in (0, \rho]$ such that $|g(z)| > |g(w)|/2 > 0$ for $|z - w| < \rho'$. Then for $0 < |z - w| < \rho'$ we have $f(z) = (z - w)^m g(z) \neq 0$, a product of two nonzero factors. $\square$

Theorem 15.10: Identity principle. Let $\Omega \subseteq \mathbb{C}$ be a connected open set and f holomorphic on Ω . The following three conditions are equivalent:

- (a) $f \equiv 0$ on Ω ;
- (b) the set $Z(f) = \{z \in \Omega : f(z) = 0\}$ has an accumulation point belonging to Ω ;
- (c) there is $w \in \Omega$ such that $f^{(n)}(w) = 0$ for every $n \geq 0$.

Proof. Step 1 — (a) $\Rightarrow$ (b) and (a) $\Rightarrow$ (c). Immediate: if $f \equiv 0$ then $Z(f) = \Omega$, which is a nonempty open set and so every point of it is an accumulation point; and all derivatives vanish.

Step 2 — (b) $\Rightarrow$ (c). Let $w \in \Omega$ be an accumulation point of $Z(f)$: there is a sequence $z_k \in Z(f) \setminus \{w\}$ with $z_k \rightarrow w$. By continuity $f(w) = \lim f(z_k) = 0$, so $w \in Z(f)$. If not all derivatives of f at w were zero, f would not be identically zero on any neighborhood of w (because the coefficients $f^{(n)}(w)/n!$ of the expansion would not all be zero); by Theorem 15.9, w would be an isolated zero, i.e. there would be $\rho' > 0$ with $f(z) \neq 0$ for $0 < |z - w| < \rho'$. But $z_k \rightarrow w$ with $z_k \neq w$ and $f(z_k) = 0$: a contradiction. Hence $f^{(n)}(w) = 0$ for every n .

Step 3 — (c) $\Rightarrow$ (a). Set

$$E = \{v \in \Omega : f^{(n)}(v) = 0 \text{ for every } n \geq 0\}.$$

E is closed in Ω : it is the intersection of the sets $\{v : f^{(n)}(v) = 0\}$, each closed in Ω because $f^{(n)}$ is continuous (Theorem 15.6(ii)). E is open: let $v \in E$ and $\rho > 0$ with $\overline{D(v, \rho)} \subset \Omega$. By Theorem 15.6, on $D(v, \rho)$, $f(z) = \sum_n \frac{f^{(n)}(v)}{n!}(z - v)^n = 0$: hence $f \equiv 0$ on $D(v, \rho)$, and with it all its derivatives, i.e. $D(v, \rho) \subseteq E$. **Conclusion:** E is nonempty (it contains w by (c)), open

and closed in Ω , which is connected; hence $E = \Omega$ and in particular $f = f^{(0)} \equiv 0$. $\square$

Corollary 15.3: Two functions that agree too much. Let f, h be holomorphic on a connected open set Ω and let $S = \{z \in \Omega : f(z) = h(z)\}$. If S has an accumulation point in Ω , then $f = h$ on the whole of Ω . In particular, if f and h agree on a segment, on an arc of a circle, or on a convergent sequence in Ω , they agree everywhere.

Proof. The function $f - h$ is holomorphic on Ω and $S = Z(f - h)$ has an accumulation point in Ω ; by Theorem 15.10, (b) $\Rightarrow$ (a), we get $f - h \equiv 0$. A segment, an arc of a circle, and a sequence converging to a point of Ω all have an accumulation point in Ω . $\square$

Corollary 15.4: Zeros are isolated. If f is holomorphic on a connected open set Ω and is not identically zero, then $Z(f)$ has no accumulation points in Ω : every zero is isolated and has a finite order $m \geq 1$. Consequently $Z(f) \cap K$ is finite for every compact set $K \subseteq \Omega$.

Proof. The first claim is the negation of (b) in Theorem 15.10; the finiteness of the order is Theorem 15.9 (applicable because f is not identically zero on any neighborhood, by the identity principle). For the second: if $Z(f) \cap K$ were infinite it would contain a sequence of distinct points, which by compactness of K would have a subsequence converging to a point of $K \subseteq \Omega$; that point would be an accumulation point of $Z(f)$, which is excluded. $\square$

Lemma 15.7: Mean value property on a circle. Let f be holomorphic on an open set containing $\overline{D(w, \rho)}$. Then

$$f(w) = \frac{1}{2\pi} \int_0^{2\pi} f(w + \rho e^{i\theta}) d\theta.$$

$$\text{In particular } |f(w)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(w + \rho e^{i\theta})| d\theta.$$

Proof. Step 1 — applying Cauchy's formula at the center. By Theorem 15.3 with $z = z_0 = w$ and $r = \rho$,

$$f(w) = \frac{1}{2\pi i} \oint_{|\zeta - w| = \rho} \frac{f(\zeta)}{\zeta - w} d\zeta.$$

Step 2 — parametrizing. With $\zeta = w + \rho e^{i\theta}$ we have $\zeta - w = \rho e^{i\theta}$ and $d\zeta = i\rho e^{i\theta} d\theta$, hence

$$f(w) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(w + \rho e^{i\theta})}{\rho e^{i\theta}} i\rho e^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(w + \rho e^{i\theta}) d\theta.$$

Step 3 — the inequality. It follows from Step 2 of Lemma 15.2 applied to the integral over $[0, 2\pi]$: the modulus of an integral does not exceed the integral of the modulus. $\square$

Lemma 15.8: Constant modulus implies a constant function. Let f be holomorphic on a connected open set Ω and let $|f|$ be constant on Ω . Then f is constant.

Proof. Step 1 — the trivial case. Let $|f| \equiv c \geq 0$. If $c = 0$ then $f \equiv 0$ and it is constant.

Step 2 — $c > 0$: the conjugate is holomorphic. If $c > 0$, f never vanishes, and from $f\bar{f} = |f|^2 = c^2$ it follows that

$$\bar{f} = \frac{c^2}{f},$$

which is holomorphic on Ω (quotient of holomorphic functions with nowhere-vanishing denominator).

Step 3 — real and imaginary parts. The functions

$$2u = f + \bar{f} = f + \frac{c^2}{f}, \quad 2iv = f - \bar{f} = f - \frac{c^2}{f}$$

are both holomorphic on Ω . The first takes only real values (it is $2 \operatorname{Re} f$), the second only purely imaginary values, hence $2v$ is holomorphic and real-valued. By Lemma 15.1 (last statement), u and v are constant on the connected set Ω .

Step 4 — conclusion. $f = u + iv$ is constant. $\square$

Theorem 15.11: Maximum modulus principle, strong form. Let $\Omega \subseteq \mathbb{C}$ be open and connected and f holomorphic on Ω . If there is $w \in \Omega$ such that

$$|f(z)| \leq |f(w)| \quad \text{for every } z \text{ in a neighborhood of } w$$

(i.e. if $|f|$ has a local maximum at w), then f is constant on Ω . Equivalently: if f is not constant, $|f|$ has no local maximum in Ω .

Proof. **Step 1 — the modulus is constant on a circle.** Let $\delta > 0$ be such that $\overline{D(w, \delta)} \subset \Omega$ and $|f(z)| \leq M := |f(w)|$ for $|z - w| \leq \delta$. Let $0 < \rho \leq \delta$. By Lemma 15.7,

$$M = |f(w)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(w + \rho e^{i\theta})| d\theta \leq \frac{1}{2\pi} \int_0^{2\pi} M d\theta = M.$$

The two inequalities are therefore equalities, and in particular

$$\int_0^{2\pi} (M - |f(w + \rho e^{i\theta})|) d\theta = 0.$$

The integrand is continuous in θ and nonnegative: if it were strictly positive at a point θ_0 , it would be so on a whole interval about θ_0 and the integral would be strictly positive. Hence $|f(w + \rho e^{i\theta})| = M$ for every θ .

Step 2 — the modulus is constant on a disk. Step 1 holds for every $\rho \in (0, \delta]$, and $|f(w)| = M$: hence $|f| \equiv M$ on the whole of $D(w, \delta)$.

Step 3 — from constant modulus to a constant function. $D(w, \delta)$ is open and connected; by Lemma 15.8, f is constant on $D(w, \delta)$, i.e. $f \equiv f(w)$ there.

Step 4 — it propagates to all of Ω . The function $f - f(w)$ is holomorphic on the connected set Ω and vanishes on $D(w, \delta)$, which has accumulation points in Ω ; by Theorem 15.10 it is identically zero, i.e. $f \equiv f(w)$ on Ω . $\square$

Corollary 15.5: The maximum lies on the boundary. Let $\Omega \subseteq \mathbb{C}$ be open, connected, bounded, and nonempty, and let f be continuous on $\overline{\Omega}$ and holomorphic on Ω . Then

$$\max_{z \in \overline{\Omega}} |f(z)| = \max_{z \in \partial\Omega} |f(z)|.$$

Proof. **Step 1 — existence of the maximum.** $\overline{\Omega}$ is closed and bounded, hence compact, and $|f|$ is continuous on it: there is $p \in \overline{\Omega}$ with $|f(p)| = \max_{\overline{\Omega}} |f|$. The boundary $\partial\Omega$ is nonempty (otherwise Ω would be open and closed in the connected set $\mathbb{C}$, hence $\Omega = \mathbb{C}$, which is not bounded) and compact, so $\max_{\partial\Omega} |f|$ also exists.

Step 2 — if the maximum is interior. If $p \in \partial\Omega$ there is nothing more to prove, since $\max_{\partial\Omega} |f| \leq \max_{\overline{\Omega}} |f| = |f(p)|$ and the other inequality is obvious. So let $p \in \Omega$. Then $|f|$ has a local maximum at p (indeed a global one on $\overline{\Omega}$), and by Theorem 15.11, f is constant on Ω ; by continuity f is constant on $\overline{\Omega}$, and then $|f|$ takes the same value on $\partial\Omega$, which is nonempty.

| In both cases the two maxima coincide. □

Corollary 15.6: Minimum modulus principle. *Let $\Omega \subseteq \mathbb{C}$ be open and connected, f holomorphic on Ω and never zero. If $|f|$ has a local minimum at $w \in \Omega$, then f is constant. If moreover Ω is bounded, f is continuous on $\overline{\Omega}$, holomorphic on Ω , and $f(z) \neq 0$ for every $z \in \overline{\Omega}$, then*

$$\min_{z \in \overline{\Omega}} |f(z)| = \min_{z \in \partial\Omega} |f(z)|.$$

The hypothesis “never zero” cannot be dropped: $f(z) = z$ on the unit disk has an interior local minimum of the modulus at $z = 0$, where it equals 0, and is not constant.

Proof. Step 1 — the reciprocal function. Since f does not vanish on Ω , the function $h = 1/f$ is holomorphic on Ω and $|h| = 1/|f|$.

Step 2 — minimum of $|f|$ = maximum of $|h|$. If $|f|$ has a local minimum at w , i.e. $|f(z)| \geq |f(w)| > 0$ in a neighborhood, then $|h(z)| \leq |h(w)|$ in the same neighborhood: $|h|$ has a local maximum at w . By Theorem 15.11, h is constant on Ω , and hence so is $f = 1/h$.

Step 3 — the boundary version. Under the hypotheses of the second part, $h = 1/f$ is continuous on $\overline{\Omega}$ (quotient of continuous functions with nowhere-vanishing denominator on a compact set) and holomorphic on Ω ; Corollary 15.5 applied to h gives $\max_{\overline{\Omega}} |h| = \max_{\partial\Omega} |h|$, i.e., passing to reciprocals, $\min_{\overline{\Omega}} |f| = \min_{\partial\Omega} |f|$.

Step 4 — the counterexample. For $f(z) = z$ on $\mathbb{D}$ we have $|f(0)| = 0 \leq |f(z)|$ for every z : a local minimum at 0, and f is not constant. The hypothesis violated is exactly “ f never zero”. □

Remark 15.15: Why there are no interior maxima. *The proof of Theorem 15.11 reads as follows: the value at a point is the average of the values on every circle about it (Lemma 15.7); an average cannot exceed the maximum of the averaged values, and equals it only if they are all equal. This is the same mechanism that governs harmonic functions and the Poisson kernel (Chapter 16), and it is no coincidence: the real part of a holomorphic function is harmonic, and Lemma 15.7 is the mean value property of harmonic functions written for a complex-valued function.*

15.6. Schwarz’s lemma and the automorphisms of the disk

The maximum modulus principle, applied not to f but to the auxiliary function $f(z)/z$, produces a statement of surprising precision: a holomorphic function that maps the unit disk into itself and fixes the center cannot “push” any point farther from the origin, and if it fails to bring even a single point closer, then it is a rotation. From this one obtains the *complete* description of the holomorphic bijections of the disk onto itself.

Lemma 15.9: Schwarz’s lemma, with the equality case. *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic on the unit disk $\mathbb{D} = D(0, 1)$, with values in $\mathbb{D}$, and $f(0) = 0$. Then*

$$|f(z)| \leq |z| \quad \text{for every } z \in \mathbb{D}, \quad \text{and} \quad |f'(0)| \leq 1.$$

Moreover, if one of the two equalities

$$|f(z_0)| = |z_0| \quad \text{for a single } z_0 \in \mathbb{D} \setminus \{0\}, \quad \text{or} \quad |f'(0)| = 1,$$

holds, then there is $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ such that $f(z) = \lambda z$ for every $z \in \mathbb{D}$; and in that case both equalities hold for every z .

Proof. Step 1 — the series expansion on the whole of $\mathbb{D}$. For every $\rho < 1$ we have $\overline{D(0, \rho)} \subset \mathbb{D}$, so Theorem 15.6 gives $f(z) = \sum_{n \geq 0} a_n z^n$ on $D(0, \rho)$ with $a_n = f^{(n)}(0)/n!$ independent of ρ . Letting ρ range over $(0, 1)$ gives the same series on the whole of $\mathbb{D}$. From $f(0) = 0$ it follows that $a_0 = 0$.

Step 2 — the auxiliary function $g = f(z)/z$. Set

$$g(z) = \sum_{k=0}^{\infty} a_{k+1} z^k.$$

As in Step 3 of Theorem 15.9, the series converges for every $|z| < 1$ (for $z \neq 0$ it equals $f(z)/z$, a quotient of a series convergent at z), hence g is holomorphic on $\mathbb{D}$ and

$$g(z) = \frac{f(z)}{z} \quad (z \neq 0), \quad g(0) = a_1 = f'(0).$$

Step 3 — estimate on the circle of radius r . Let $0 < r < 1$. For $|z| = r$ we have $|f(z)| < 1$ (because f takes values in $\mathbb{D}$), hence

$$|g(z)| = \frac{|f(z)|}{r} < \frac{1}{r}.$$

g is continuous on $\overline{D(0, r)}$ and holomorphic on $D(0, r)$, which is open, connected, and bounded: by Corollary 15.5,

$$\max_{|z| \leq r} |g(z)| = \max_{|z|=r} |g(z)| \leq \frac{1}{r}.$$

Step 4 — letting r tend to 1. Fix $z \in \mathbb{D}$. For every r with $|z| < r < 1$, Step 3 gives $|g(z)| \leq 1/r$. The left-hand side does not depend on r ; letting $r \rightarrow 1^-$ gives $|g(z)| \leq 1$.

Step 5 — the two inequalities. For $z \neq 0$, $|f(z)| = |z| |g(z)| \leq |z|$; for $z = 0$, $f(0) = 0 = |0|$ holds. And $|f'(0)| = |g(0)| \leq 1$.

Step 6 — the equality case. Suppose $|f(z_0)| = |z_0|$ with $z_0 \in \mathbb{D}$, $z_0 \neq 0$: then $|g(z_0)| = |f(z_0)|/|z_0| = 1$. Or suppose $|f'(0)| = 1$: then $|g(0)| = 1$. In either case there is a point $w_0 \in \mathbb{D}$ with $|g(w_0)| = 1$; since $|g| \leq 1$ on the whole of $\mathbb{D}$ (Step 4), $|g|$ has at w_0 a maximum (global, hence also local) on $\mathbb{D}$, which is open and connected. By Theorem 15.11, g is constant: $g \equiv \lambda$ with $|\lambda| = |g(w_0)| = 1$. Hence $f(z) = zg(z) = \lambda z$ for every z , and then $|f(z)| = |z|$ for every z and $|f'(0)| = |\lambda| = 1$. $\square$

Definition 15.4: Blaschke factor. For $a \in \mathbb{D}$, the **Blaschke factor** of parameter a is the transformation

$$\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}.$$

It is a Möbius transformation in the sense of Definition 6.4, associated with the matrix $\begin{pmatrix} 1 & -a \\ -\bar{a} & 1 \end{pmatrix}$ of determinant $1 - |a|^2 \neq 0$; its only pole is $z = 1/\bar{a}$ (if $a \neq 0$), of modulus $1/|a| > 1$, so φ_a is holomorphic on an open set containing $\overline{\mathbb{D}}$. For $a = 0$, $\varphi_0 = \text{id}$.

Lemma 15.10: Fundamental identity and properties of the Blaschke factor. Let $a \in \mathbb{D}$. Then:

(i) for every $z \in \mathbb{C}$ with $\bar{a}z \neq 1$ the **fundamental identity** holds:

$$|1 - \bar{a}z|^2 - |z - a|^2 = (1 - |a|^2)(1 - |z|^2), \quad \text{i.e.} \quad 1 - |\varphi_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2};$$

(ii) φ_a maps $\mathbb{T}$ into $\mathbb{T}$, $\mathbb{D}$ into $\mathbb{D}$, and the exterior of the disk into the exterior;

- (iii) $\varphi_{-a} \circ \varphi_a = \varphi_a \circ \varphi_{-a} = \text{id}$; in particular $\varphi_a : \mathbb{D} \rightarrow \mathbb{D}$ is bijective with inverse φ_{-a} , and both φ_a and its inverse are holomorphic on $\mathbb{D}$;
(iv) $\varphi_a(a) = 0$, $\varphi_a(0) = -a$, and

$$\varphi'_a(z) = \frac{1 - |a|^2}{(1 - \bar{a}z)^2}, \quad \varphi'_a(0) = 1 - |a|^2, \quad \varphi'_a(a) = \frac{1}{1 - |a|^2}.$$

Proof. Step 1 — identity (i). Using $|w|^2 = w\bar{w}$ and $\overline{1 - \bar{a}z} = 1 - a\bar{z}$:

$$|1 - \bar{a}z|^2 = (1 - \bar{a}z)(1 - a\bar{z}) = 1 - a\bar{z} - \bar{a}z + |a|^2|z|^2,$$

$$|z - a|^2 = (z - a)(\bar{z} - \bar{a}) = |z|^2 - \bar{a}z - a\bar{z} + |a|^2.$$

Subtracting, the terms $-a\bar{z}$ and $-\bar{a}z$ cancel and what remains is

$$|1 - \bar{a}z|^2 - |z - a|^2 = 1 + |a|^2|z|^2 - |z|^2 - |a|^2 = (1 - |z|^2) - |a|^2(1 - |z|^2) = (1 - |a|^2)(1 - |z|^2).$$

Dividing by $|1 - \bar{a}z|^2 > 0$ gives the second form, since $|\varphi_a(z)|^2 = |z - a|^2 / |1 - \bar{a}z|^2$.

Step 2 — (ii). Since $|a| < 1$, the factor $1 - |a|^2$ is strictly positive. By (i), the sign of $1 - |\varphi_a(z)|^2$ is that of $1 - |z|^2$: hence $|z| < 1 \Rightarrow |\varphi_a(z)| < 1$, $|z| = 1 \Rightarrow |\varphi_a(z)| = 1$, $|z| > 1$ (and $\bar{a}z \neq 1$) $\Rightarrow |\varphi_a(z)| > 1$.

Step 3 — (iii). A direct computation: for z with $\bar{a}z \neq 1$,

$$\varphi_{-a}(\varphi_a(z)) = \frac{\frac{z-a}{1-\bar{a}z} + a}{1 + \bar{a}\frac{z-a}{1-\bar{a}z}} = \frac{(z-a) + a(1-\bar{a}z)}{(1-\bar{a}z) + \bar{a}(z-a)} = \frac{z-a+a-|a|^2z}{1-\bar{a}z+\bar{a}z-|a|^2} = \frac{z(1-|a|^2)}{1-|a|^2} = z,$$

where numerator and denominator were multiplied by $1 - \bar{a}z \neq 0$. Swapping a with $-a$ gives the other composition. By (ii), both maps send $\mathbb{D}$ into $\mathbb{D}$, hence the restriction of φ_a to $\mathbb{D}$ is a bijection of $\mathbb{D}$ with inverse $\varphi_{-a}|_{\mathbb{D}}$; both are holomorphic on $\mathbb{D}$ because their poles $1/\bar{a}$ and $-1/\bar{a}$ have modulus > 1 .

Step 4 — (iv). $\varphi_a(a) = \frac{a-a}{1-|a|^2} = 0$ and $\varphi_a(0) = \frac{-a}{1} = -a$. For the derivative, by the quotient rule,

$$\varphi'_a(z) = \frac{1 \cdot (1 - \bar{a}z) - (z - a)(-\bar{a})}{(1 - \bar{a}z)^2} = \frac{1 - \bar{a}z + \bar{a}z - |a|^2}{(1 - \bar{a}z)^2} = \frac{1 - |a|^2}{(1 - \bar{a}z)^2}.$$

Evaluating at $z = 0$ gives $1 - |a|^2$; evaluating at $z = a$, the denominator equals $(1 - |a|^2)^2$ and the quotient $1/(1 - |a|^2)$. $\square$

Remark 15.16: What is inherited from Chapter 6. Lemma 15.10 was proved directly, with identity (i), and uses nothing from Chapter 6. It is nonetheless useful to know that φ_a fits fully into that picture: it is a Möbius transformation (Definition 6.4), hence a bijection of $\hat{\mathbb{C}}$ belonging to the Möbius group (Proposition 6.7), and it sends generalized circles to generalized circles (Theorem 6.6) — which gives a second proof of point (ii): $\varphi_a(\mathbb{T})$ is a generalized circle contained in $\mathbb{T}$ by Step 2, and a generalized circle contained in a circle coincides with it; then φ_a , being a homeomorphism of $\hat{\mathbb{C}}$, permutes the two connected components of $\hat{\mathbb{C}} \setminus \mathbb{T}$, and since $\varphi_a(a) = 0 \in \mathbb{D}$ it sends $\mathbb{D}$ into $\mathbb{D}$. The composition $\varphi_{-a} \circ \varphi_a = \text{id}$ can also be read as a product of matrices: $\begin{pmatrix} 1 & a \\ \bar{a} & 1 \end{pmatrix} \begin{pmatrix} 1 & -a \\ -\bar{a} & 1 \end{pmatrix} = (1 - |a|^2) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and scalar matrices are the kernel of the homomorphism Ψ of Proposition 6.7.

Definition 15.5: Automorphism of the disk. An **automorphism** of the unit disk is a function $f : \mathbb{D} \rightarrow \mathbb{D}$ that is holomorphic, bijective, and whose inverse f^{-1} is holomorphic. The set of automorphisms is denoted $\text{Aut}(\mathbb{D})$: under composition it is a group (the composition of holomorphic bijections with holomorphic inverse is again such a map, the identity belongs to it, and the inverse of an automorphism is an automorphism by definition).

Theorem 15.12: The automorphisms of the disk are exactly the rotated Blaschke factors. A function $f : \mathbb{D} \rightarrow \mathbb{D}$ is an automorphism of the disk if and only if there exist $\theta \in \mathbb{R}$ and $a \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \varphi_a(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z} \quad \text{for every } z \in \mathbb{D}.$$

Moreover a is uniquely determined by f (it is $a = f^{-1}(0)$) and θ is determined modulo 2π .

Proof. **Step 1 — every map of that form is an automorphism.** The rotation $r_\theta(w) = e^{i\theta}w$ is holomorphic, sends $\mathbb{D}$ into $\mathbb{D}$ (preserving the modulus), and is bijective with inverse $r_{-\theta}$, also holomorphic: it is an automorphism. By Lemma 15.10(ii)–(iii), so is φ_a . The composition $r_\theta \circ \varphi_a$ is therefore an automorphism (Definition 15.5).

Step 2 — reduction to the case $f(0) = 0$. Let $f \in \text{Aut}(\mathbb{D})$ and set $a = f^{-1}(0) \in \mathbb{D}$. Consider

$$g = f \circ \varphi_{-a} : \mathbb{D} \rightarrow \mathbb{D}.$$

This is an automorphism (composition of automorphisms) and $g(0) = f(\varphi_{-a}(0)) = f(a) = 0$, because $\varphi_{-a}(0) = \frac{0+a}{1+0} = a$ by Lemma 15.10(iv) applied to $-a$.

Step 3 — Schwarz in both directions. g is holomorphic from $\mathbb{D}$ to $\mathbb{D}$ with $g(0) = 0$: by Lemma 15.9, $|g(z)| \leq |z|$ for every $z \in \mathbb{D}$. Also g^{-1} is holomorphic from $\mathbb{D}$ to $\mathbb{D}$ with $g^{-1}(0) = 0$ (because $g(0) = 0$ and g is bijective): hence $|g^{-1}(w)| \leq |w|$ for every $w \in \mathbb{D}$. Setting $w = g(z)$ in the second inequality,

$$|z| = |g^{-1}(g(z))| \leq |g(z)|.$$

Together with the first, $|g(z)| = |z|$ for every $z \in \mathbb{D}$.

Step 4 — the equality case. Choosing any $z_0 \in \mathbb{D} \setminus \{0\}$ gives $|g(z_0)| = |z_0|$: by the equality case of Lemma 15.9 there is λ with $|\lambda| = 1$ and $g(z) = \lambda z$. Writing $\lambda = e^{i\theta}$ (possible because every number of modulus 1 is an imaginary exponential, Volume II, Chapter 3),

$$g = r_\theta.$$

Step 5 — reassembling. From $g = f \circ \varphi_{-a}$ and $\varphi_{-a} \circ \varphi_a = \text{id}$ (Lemma 15.10(iii)) it follows that

$$f = f \circ \varphi_{-a} \circ \varphi_a = g \circ \varphi_a = r_\theta \circ \varphi_a, \quad \text{i.e.} \quad f(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}.$$

Step 6 — uniqueness. If $e^{i\theta} \varphi_a = e^{i\theta'} \varphi_{a'}$ on $\mathbb{D}$, evaluating at $z = a$ gives $0 = e^{i\theta'} \varphi_{a'}(a)$, hence $\varphi_{a'}(a) = 0$, i.e. $a = a'$ (the only zero of $\varphi_{a'}$ is a'). Then $e^{i\theta} \varphi_a = e^{i\theta'} \varphi_a$, and evaluating at a point where $\varphi_a \neq 0$ (for instance $z = 0$, where it equals $-a$, if $a \neq 0$; or $z = \frac{1}{2}$ if $a = 0$), one gets $e^{i\theta} = e^{i\theta'}$, i.e. $\theta \equiv \theta' \pmod{2\pi}$. $\square$

Corollary 15.7: The automorphisms fixing the center are the rotations. If $f \in \text{Aut}(\mathbb{D})$ and $f(0) = 0$, then $f(z) = e^{i\theta}z$ for some $\theta \in \mathbb{R}$. The subgroup of $\text{Aut}(\mathbb{D})$ of automorphisms fixing 0 is therefore isomorphic to the group $\mathbb{T}$ of Chapter 6 (Definition 6.1).

Proof. By Theorem 15.12, $f = e^{i\theta} \varphi_a$ with $a = f^{-1}(0)$. But $f(0) = 0$ implies $f^{-1}(0) = 0$, i.e. $a = 0$, and $\varphi_0 = \text{id}$: hence $f(z) = e^{i\theta} z$. The map $e^{i\theta} \mapsto r_\theta$ is a group isomorphism from $\mathbb{T}$ onto the subgroup in question: it is surjective by what was just shown, it is a homomorphism because $r_\theta \circ r_{\theta'} = r_{\theta+\theta'}$, and it is injective because $r_\theta = \text{id}$ forces $e^{i\theta} = 1$ (evaluate at $z = \frac{1}{2}$). $\square$

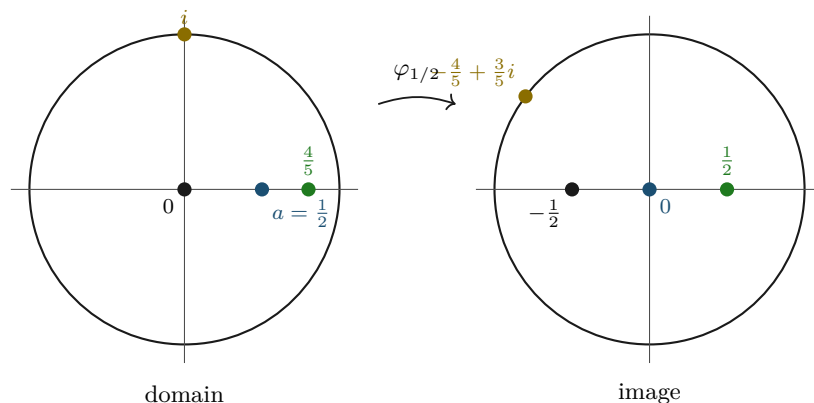


Figure 15.2: The Blaschke factor $\varphi_{1/2}(z) = \frac{z - \frac{1}{2}}{1 - \frac{1}{2}z}$. Coordinates computed by hand: $\varphi_{1/2}(\frac{1}{2}) = 0$; $\varphi_{1/2}(0) = \frac{-1/2}{1} = -\frac{1}{2}$; $\varphi_{1/2}(\frac{4}{5}) = \frac{4/5 - 1/2}{1 - 2/5} = \frac{3/10}{3/5} = \frac{1}{2}$; $\varphi_{1/2}(i) = \frac{i - \frac{1}{2}}{1 - \frac{i}{2}} = \frac{(i - \frac{1}{2})(1 + \frac{i}{2})}{1 + \frac{1}{4}} = \frac{-1 + \frac{3}{4}i}{5/4} = -\frac{4}{5} + \frac{3}{5}i$, of modulus $\sqrt{\frac{16}{25} + \frac{9}{25}} = 1$, consistent with Lemma 15.10(ii). $\checkmark$ The point i stays on the boundary, interior points stay interior, and a goes to the center.

Remark 15.17: Panorama: the Poincaré metric and the Schwarz–Pick lemma.

Theorem 15.12 says that $\text{Aut}(\mathbb{D})$ acts transitively on $\mathbb{D}$ (given a, b there is an automorphism sending a to b : it suffices to compose φ_a with φ_{-b}) and that the stabilizer of every point is a circle of rotations (Corollary 15.7). This is the starting point of the hyperbolic geometry of the disk: the Poincaré metric $ds = \frac{2|dz|}{1 - |z|^2}$ is, up to a constant, the unique Riemannian metric on $\mathbb{D}$ invariant under $\text{Aut}(\mathbb{D})$, and the Schwarz–Pick lemma states that every holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$ does not increase Poincaré distances, with equality if and only if $f \in \text{Aut}(\mathbb{D})$. These statements are not proved here and are given as a **panorama**: the source is Conway, *Functions of One Complex Variable I*, Chapter VI, §2 (“Schwarz’s Lemma”), where the equality case of Lemma 15.9 is used exactly as here.

15.7. Classification of isolated singularities

Definition 15.3 introduced isolated singularities without distinguishing their types. The Laurent expansion (Theorem 15.4) allows one to do so cleanly: everything depends on how many coefficients of the *principal part* are nonzero — none, finitely many, or infinitely many.

Definition 15.6: Removable singularity, pole, essential singularity. Let z_0 be an isolated singularity of f (Definition 15.3) and let $f(z) = \sum_{n \in \mathbb{Z}} a_n(z - z_0)^n$ be its Laurent expansion on the annulus $0 < |z - z_0| < r$ (Theorem 15.4). One says that z_0 is:

- a **removable singularity** if $a_n = 0$ for every $n < 0$;
- a **pole of order** $k \geq 1$ if $a_{-k} \neq 0$ and $a_n = 0$ for every $n < -k$ (for $k = 1$ one says *simple pole*, for $k = 2$ *double pole*);

- an **essential singularity** if $a_n \neq 0$ for infinitely many indices $n < 0$.

The three cases are mutually exclusive and exhaust all possibilities, because the set $\{n < 0 : a_n \neq 0\}$ is either empty, finite and nonempty, or infinite.

Theorem 15.13: Riemann's theorem on removable singularities. Let f be holomorphic on the annulus $A = \{0 < |z - z_0| < r\}$, and suppose that f is bounded in a punctured neighborhood of z_0 , i.e. that there exist $M \geq 0$ and $\rho_0 \in (0, r)$ with $|f(z)| \leq M$ for $0 < |z - z_0| \leq \rho_0$. Then z_0 is a removable singularity and f extends uniquely to a function holomorphic on the whole disk $D(z_0, r)$.

Proof. Step 1 — the negative coefficients tend to zero. By Theorem 15.4 (with $r_1 = 0$, $r_2 = r$), for every $\rho \in (0, r)$

$$a_n = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = \rho} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

and the value does not depend on ρ . Let $0 < \rho \leq \rho_0$. On the circle we have $|f| \leq M$ and $|\zeta - z_0|^{n+1} = \rho^{n+1}$, hence by the ML estimate (Lemma 15.2), with length $2\pi\rho$,

$$|a_n| \leq \frac{1}{2\pi} \cdot \frac{M}{\rho^{n+1}} \cdot 2\pi\rho = \frac{M}{\rho^n} = M\rho^{|n|} \quad \text{for } n < 0.$$

Step 2 — letting ρ tend to zero. For $n < 0$ we have $|n| \geq 1$ and $M\rho^{|n|} \rightarrow 0$ as $\rho \rightarrow 0^+$, while a_n does not depend on ρ : hence $a_n = 0$ for every $n < 0$. The singularity is removable.

Step 3 — the extension. What remains is $f(z) = \sum_{n \geq 0} a_n(z - z_0)^n$ for $0 < |z - z_0| < r$. A power series that converges at every point of a punctured disk also converges at the center (where it equals a_0) and has radius of convergence $\geq r$: it defines a function $\tilde{f}$ holomorphic on $D(z_0, r)$ that agrees with f away from the center. Uniqueness of the extension follows from continuity: two continuous extensions that agree on the punctured disk also agree at z_0 , which is an accumulation point of it. $\square$

Theorem 15.14: Characterization of poles. Let z_0 be an isolated singularity of f , holomorphic on $\{0 < |z - z_0| < r\}$, and let $k \geq 1$ be an integer. The following are equivalent:

(a) z_0 is a pole of order k ;

(b) there exists g holomorphic on $D(z_0, r)$ with $g(z_0) \neq 0$ and $f(z) = \frac{g(z)}{(z - z_0)^k}$ for $0 < |z - z_0| < r$;

(c) the limit $\lim_{z \rightarrow z_0} (z - z_0)^k f(z)$ exists, is finite, and is nonzero.

Moreover z_0 is a pole (of some order) if and only if $\lim_{z \rightarrow z_0} |f(z)| = +\infty$.

Proof. Step 1 — (a) $\Rightarrow$ (b). From the Laurent expansion with $a_n = 0$ for $n < -k$ and $a_{-k} \neq 0$:

$$(z - z_0)^k f(z) = \sum_{n \geq -k} a_n (z - z_0)^{n+k} = \sum_{m \geq 0} a_{m-k} (z - z_0)^m =: g(z),$$

a power series convergent for $0 < |z - z_0| < r$ and hence, as in Step 3 of Theorem 15.13, on the whole of $D(z_0, r)$. It defines g holomorphic with $g(z_0) = a_{-k} \neq 0$.

Step 2 — (b) $\Rightarrow$ (c). As $z \rightarrow z_0$, $(z - z_0)^k f(z) = g(z) \rightarrow g(z_0) \neq 0$, finite, by continuity of g .

Step 3 — (c) $\Rightarrow$ (a). Let $L = \lim_{z \rightarrow z_0} (z - z_0)^k f(z) \neq 0$, finite, and set $h(z) = (z - z_0)^k f(z)$ for $0 < |z - z_0| < r$. Having a finite limit at z_0 , h is bounded in a punctured neighborhood; by Theorem 15.13 it extends to a function holomorphic on $D(z_0, r)$ with $h(z_0) = L \neq 0$.

Expanding $h(z) = \sum_{m \geq 0} b_m(z - z_0)^m$ (Theorem 15.6) with $b_0 = L \neq 0$, one gets for $0 < |z - z_0|$,

$$f(z) = \frac{h(z)}{(z - z_0)^k} = \sum_{m \geq 0} b_m(z - z_0)^{m-k} = \sum_{n \geq -k} b_{n+k}(z - z_0)^n.$$

By uniqueness of the Laurent expansion (Theorem 15.4, Step 5), we get $a_n = b_{n+k}$, hence $a_n = 0$ for $n < -k$ and $a_{-k} = b_0 = L \neq 0$: a pole of order k .

Step 4 — pole $\Rightarrow$ infinite modulus. If z_0 is a pole of order k , by (b) we have $f = g/(z - z_0)^k$ with g continuous and $g(z_0) \neq 0$; there is $\delta > 0$ with $|g(z)| \geq |g(z_0)|/2 > 0$ for $|z - z_0| < \delta$, hence

$$|f(z)| \geq \frac{|g(z_0)|}{2|z - z_0|^k} \xrightarrow{z \rightarrow z_0} +\infty.$$

Step 5 — infinite modulus $\Rightarrow$ pole. If $|f(z)| \rightarrow +\infty$, there is $\delta \in (0, r)$ such that $|f(z)| \geq 1$ for $0 < |z - z_0| < \delta$; in particular f does not vanish there, and $u = 1/f$ is holomorphic on $\{0 < |z - z_0| < \delta\}$ with $|u| \leq 1$. By Theorem 15.13, u extends holomorphically to $D(z_0, \delta)$, and $u(z_0) = \lim_{z \rightarrow z_0} 1/f(z) = 0$. The function u is not identically zero (it equals $1/f \neq 0$ on the punctured disk), so by Theorem 15.9 there exist $k \geq 1$, $\rho' > 0$, and v holomorphic on $D(z_0, \rho')$ with $v(z_0) \neq 0$ and $u(z) = (z - z_0)^k v(z)$. Shrinking ρ' one may assume v never zero (by continuity), so that

$$f(z) = \frac{1}{u(z)} = \frac{1/v(z)}{(z - z_0)^k} \quad \text{for } 0 < |z - z_0| < \rho',$$

with $1/v$ holomorphic and nonzero at z_0 : this is condition (b), hence z_0 is a pole of order k . $\square$

Proposition 15.2: The formulas for the residue at a pole. Let ξ be a pole of order $k \geq 1$ of f . Then:

- (i) $\text{Res}(f, \xi) = \frac{1}{(k-1)!} \lim_{z \rightarrow \xi} \frac{d^{k-1}}{dz^{k-1}} [(z - \xi)^k f(z)]$;
- (ii) in particular, if the pole is simple ($k = 1$), $\text{Res}(f, \xi) = \lim_{z \rightarrow \xi} (z - \xi)f(z)$;
- (iii) if $f = p/q$ with p, q holomorphic in a neighborhood of ξ , $p(\xi) \neq 0$, $q(\xi) = 0$ and $q'(\xi) \neq 0$, then ξ is a simple pole and $\text{Res}(f, \xi) = \frac{p(\xi)}{q'(\xi)}$.

Proof. **Step 1 — (i).** By Theorem 15.14, Step 1, the function $g(z) = (z - \xi)^k f(z)$ extends holomorphically to a disk $D(\xi, r)$, where $g(z) = \sum_{m \geq 0} a_{m-k}(z - \xi)^m$, with a_n the Laurent coefficients of f . A power series is differentiated term by term (Theorem 15.6, Step 3) and the m -th coefficient equals $g^{(m)}(\xi)/m!$: taking $m = k - 1$,

$$a_{-1} = a_{(k-1)-k} = \frac{g^{(k-1)}(\xi)}{(k-1)!}.$$

Since $g^{(k-1)}$ is continuous at ξ (Theorem 15.6(ii)), the limit as $z \rightarrow \xi$ of the $(k-1)$ -th derivative of $(z - \xi)^k f(z)$ equals $g^{(k-1)}(\xi)$, and $\text{Res}(f, \xi) = a_{-1}$ by Definition 15.3.

Step 2 — (ii). This is (i) with $k = 1$: the 0-th derivative is the function itself and $0! = 1$, hence $\text{Res}(f, \xi) = \lim_{z \rightarrow \xi} (z - \xi)f(z)$.

Step 3 — (iii). From $q(\xi) = 0$ and $q'(\xi) \neq 0$ it follows, by Theorem 15.9, that ξ is a zero of q of order 1 (the order is the first index with a nonzero Taylor coefficient, and the coefficient of index 1 is $q'(\xi) \neq 0$): hence $q(z) = (z - \xi)q_1(z)$ with q_1 holomorphic and $q_1(\xi) = q'(\xi) \neq 0$. Then, for z near ξ and $z \neq \xi$,

$$(z - \xi)f(z) = (z - \xi) \frac{p(z)}{(z - \xi)q_1(z)} = \frac{p(z)}{q_1(z)} \xrightarrow{z \rightarrow \xi} \frac{p(\xi)}{q'(\xi)} \neq 0,$$

a finite, nonzero limit: by Theorem 15.14(c) with $k = 1$, ξ is a simple pole, and by (ii) the residue is exactly $p(\xi)/q'(\xi)$. $\square$

Theorem 15.15: Casorati–Weierstrass theorem. *Let z_0 be an essential singularity of f , holomorphic on $\{0 < |z - z_0| < r\}$. Then for every $\delta \in (0, r)$ the image $f(\{0 < |z - z_0| < \delta\})$ is dense in $\mathbb{C}$: for every $w_0 \in \mathbb{C}$ and every $\varepsilon > 0$ there is z with $0 < |z - z_0| < \delta$ and $|f(z) - w_0| < \varepsilon$.*

Proof. Step 1 — by contradiction. Suppose there exist $w_0 \in \mathbb{C}$, $\varepsilon > 0$ and $\delta \in (0, r)$ such that $|f(z) - w_0| \geq \varepsilon$ for every z with $0 < |z - z_0| < \delta$. Then $f - w_0$ does not vanish on that punctured disk and

$$h(z) = \frac{1}{f(z) - w_0}$$

is holomorphic there, with $|h(z)| \leq 1/\varepsilon$: it is bounded. By Theorem 15.13, h extends holomorphically to $D(z_0, \delta)$; let $c = h(z_0)$.

Step 2 — the case $c \neq 0$. By continuity h is nonzero on a whole disk $D(z_0, \delta')$, so $f - w_0 = 1/h$ is holomorphic on $D(z_0, \delta')$ and in particular bounded in a punctured neighborhood of z_0 . By Theorem 15.13 applied to f , the singularity would be removable: a contradiction, because z_0 is essential (Definition 15.6, the three cases being mutually exclusive).

Step 3 — the case $c = 0$. h is not identically zero, because it equals $1/(f - w_0) \neq 0$ on the punctured disk. By Theorem 15.9 there exist $k \geq 1$ and v holomorphic with $v(z_0) \neq 0$ and $h(z) = (z - z_0)^k v(z)$ on a disk $D(z_0, \delta'')$; shrinking the radius one may assume v never zero. Then

$$f(z) - w_0 = \frac{1}{h(z)} = \frac{1/v(z)}{(z - z_0)^k},$$

i.e. $f - w_0$ has a pole of order k at z_0 (Theorem 15.14(b)); and since f and $f - w_0$ differ by a constant, which does not alter the principal part of the Laurent expansion, f too would have a pole at z_0 : again a contradiction.

Step 4 — conclusion. Both cases are impossible, so the assumption for contradiction of Step 1 is false: the image is dense. $\square$

Remark 15.18: Panorama: Picard’s great theorem. *Theorem 15.15 says that near an essential singularity the function comes arbitrarily close to every complex value. Picard’s great theorem says much more: in every punctured neighborhood of an essential singularity the function attains every complex value, infinitely many times, with at most one exception. For $f(z) = e^{1/z}$ at $z_0 = 0$ the exception is $w = 0$, which $e^{1/z}$ never attains. This theorem is not proved here and is given as a **panorama**: the proof requires the modular function or Schottky’s theorem, and is found in Conway, Functions of One Complex Variable I, Chapter XII (“The Range of an Analytic Function”), and in Ahlfors, Chapter 8. What is proved here, Casorati–Weierstrass, is its elementary part and suffices for all uses in this treatise.*

Definition 15.7: Meromorphic function. Let $\Omega \subseteq \mathbb{C}$ be open. A function f is called **meromorphic** on Ω if there exists a set $P \subseteq \Omega$ with no accumulation points in Ω , such that f is holomorphic on $\Omega \setminus P$ and every point of P is a pole of f . The set P is the set of poles. By Corollary 15.4, $P \cap K$ is finite for every compact set $K \subseteq \Omega$: on a compact set the poles are finite in number.

Example 15.3: The three types, seen on three functions. All three functions have an isolated singularity at $z_0 = 0$.

(1) $f_1(z) = \frac{\sin z}{z}$: **removable**. From $\sin z = \sum_{j \geq 0} \frac{(-1)^j}{(2j+1)!} z^{2j+1}$ (Taylor series of the exponential combined as $\frac{e^{iz} - e^{-iz}}{2i}$), it follows that $f_1(z) = \sum_{j \geq 0} \frac{(-1)^j}{(2j+1)!} z^{2j} = 1 - \frac{z^2}{6} + \frac{z^4}{120} - \dots$: no negative powers. The extension gives $f_1(0) = 1$. Independent check: $|f_1(z)| \leq \frac{|\sin z|}{|z|}$ is bounded near 0, so Theorem 15.13 gives the same conclusion.

(2) $f_2(z) = \frac{e^z}{z^3}$: **pole of order 3**. From $e^z = \sum_{m \geq 0} z^m/m!$,

$$f_2(z) = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{2z} + \frac{1}{6} + \frac{z}{24} + \dots,$$

a principal part with three terms, $a_{-3} = 1 \neq 0$: pole of order 3 and $\text{Res}(f_2, 0) = a_{-1} = \frac{1}{2}$. Independent check with Proposition 15.2(i), $k = 3$: $\frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} [z^3 f_2(z)] = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} e^z = \frac{1}{2} e^0 = \frac{1}{2}$. ✓

(3) $f_3(z) = e^{1/z}$: **essential**. Substituting $w = 1/z$ into the series of the exponential, $f_3(z) = \sum_{m \geq 0} \frac{1}{m!} z^{-m}$: infinitely many nonzero negative coefficients. The residue is $a_{-1} = \frac{1}{1!} = 1$. Consistently with Theorem 15.15, approaching 0 along the positive real axis $f_3 \rightarrow +\infty$, along the negative real axis $f_3 \rightarrow 0$, and along the imaginary axis $f_3(it) = e^{-i/t}$ stays on the unit circle: no limit, and the image of every punctured neighborhood is dense.

15.8. Winding number, the argument principle, and Rouché's theorem

In Theorem 15.5 the factor $2\pi i$ appears before each residue because the contour is a circle traversed *once*. If the path winds n times, the factor becomes $2\pi in$: that integer is the *winding number*, and it is the same topological invariant built in Chapter 12 as the number of turns of a loop.

Definition 15.8: Winding number of a loop. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise regular loop and $z_0 \in \mathbb{C} \setminus \gamma^*$. The **winding number** (or *turning number*) of γ about z_0 is the number

$$\text{Ind}_\gamma(z_0) = \frac{1}{2\pi i} \oint_\gamma \frac{dz}{z - z_0}.$$

The integral makes sense because $z \mapsto 1/(z - z_0)$ is continuous on γ^* , which does not contain z_0 .

Theorem 15.16: The index is an integer. Let γ be a piecewise regular loop and $z_0 \notin \gamma^*$. Then $\text{Ind}_\gamma(z_0) \in \mathbb{Z}$.

Proof. Step 1 — reparametrization and an auxiliary function. Up to the affine substitution $t \mapsto a + t(b - a)$, which does not change the integral (Definition 15.2), one may assume $\gamma : [0, 1] \rightarrow \mathbb{C}$ with $\gamma(0) = \gamma(1)$. The function $t \mapsto \frac{\gamma'(t)}{\gamma(t) - z_0}$ is piecewise continuous and bounded on $[0, 1]$: continuous where γ is of class C^1 , and with denominator of modulus $\geq \text{dist}(z_0, \gamma^*) > 0$ everywhere (the distance is positive because γ^* is compact and $z_0 \notin \gamma^*$). Set

$$h(t) = \int_0^t \frac{\gamma'(u)}{\gamma(u) - z_0} du,$$

which is continuous on $[0, 1]$, with $h(0) = 0$ and $h(1) = \oint_{\gamma} \frac{dz}{z - z_0}$, and differentiable with $h'(t) = \frac{\gamma'(t)}{\gamma(t) - z_0}$ at every point except at most the finitely many where γ' jumps (fundamental theorem of calculus).

Step 2 — a constant function. Set $\varphi(t) = (\gamma(t) - z_0)e^{-h(t)}$. It is continuous on $[0, 1]$ and differentiable away from the exceptional points, with

$$\varphi'(t) = \gamma'(t)e^{-h(t)} - (\gamma(t) - z_0)h'(t)e^{-h(t)} = \gamma'(t)e^{-h(t)} - (\gamma(t) - z_0)\frac{\gamma'(t)}{\gamma(t) - z_0}e^{-h(t)} = 0.$$

A function continuous on $[0, 1]$ with vanishing derivative except at finitely many points is constant: it is constant on each of the closed subintervals determined by the exceptional points (Lagrange's theorem applied to real and imaginary parts), and the values glue together by continuity at the endpoints. Hence $\varphi(1) = \varphi(0)$.

Step 3 — conclusion. Since $\gamma(1) = \gamma(0)$ and $h(0) = 0$,

$$(\gamma(0) - z_0)e^{-h(1)} = \gamma(0) - z_0,$$

and $\gamma(0) - z_0 \neq 0$: hence $e^{-h(1)} = 1$, i.e. $e^{h(1)} = 1$. Writing $h(1) = \sigma + i\tau$ with σ, τ real, from $1 = |e^{h(1)}| = e^{\sigma}$ it follows that $\sigma = 0$; and from $e^{i\tau} = 1$ it follows that $\tau \in 2\pi\mathbb{Z}$. Therefore $\text{Ind}_{\gamma}(z_0) = h(1)/(2\pi i) = \tau/(2\pi) \in \mathbb{Z}$. $\square$

Remark 15.19: The index is the winding number, and it is a degree. The proof just given retraces, for loops that are only piecewise regular, that of Theorem 12.6 of Chapter 12, which for loops of class C^1 establishes the equality

$$\text{Ind}_{\gamma}(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z - z_0} = w(\gamma - z_0),$$

where w is the topological winding number of Definition 12.6, defined via the lifting of the exponential and invariant under homotopy (Theorem 12.5). In particular the index of the unit circle traversed n times, i.e. of the loop $t \mapsto e^{2\pi i n t}$, equals n : it is the degree of the map $z \mapsto z^n$ in the sense of Definition 12.8, and the classification of maps $S^1 \rightarrow S^1$ by degree (Theorem 12.8) says that this integer determines the loop up to homotopy. This is the bridge between the analytic computation of this chapter and the topology of Chapter 12: the integer coefficient multiplying the residues is not a computational trick, it is an invariant.

Proposition 15.3: Index of a circle. Let γ be the circle $|\zeta - z_0| = r$ traversed once counterclockwise. Then

$$\text{Ind}_{\gamma}(z) = \begin{cases} 1, & |z - z_0| < r, \\ 0, & |z - z_0| > r. \end{cases}$$

Proof. The interior case is Step 3 of the Alternative proof of Theorem 15.3: the geometric expansion of $1/(\zeta - z)$ in powers of $(z - z_0)/(\zeta - z_0)$ converges uniformly on the circle and, integrated term by term with Lemma 15.5, leaves only the term $2\pi i$; dividing by $2\pi i$ gives 1. The exterior case is Observation 15.7: the expansion in the reciprocal variable produces only exponents ≥ 0 , all with vanishing integral, and the index equals 0. $\square$

Proposition 15.4: Index of the D-shaped contour. Let $R > 1$ and let γ_R be the loop obtained by first traversing the segment $[-R, R]$ of the real axis, from $-R$ to R , and then

the arc $\alpha_R(\theta) = Re^{i\theta}$, $\theta \in [0, \pi]$. Then

$$\text{Ind}_{\gamma_R}(i) = 1, \quad \text{Ind}_{\gamma_R}(-i) = 0.$$

Proof. Step 1 — the contribution of the segment. With $z = x$ real and $\frac{1}{x \mp i} = \frac{x \pm i}{x^2 + 1}$,

$$\int_{-R}^R \frac{dx}{x - i} = \int_{-R}^R \frac{x dx}{x^2 + 1} + i \int_{-R}^R \frac{dx}{x^2 + 1} = 0 + 2i \arctan R,$$

because the first integrand is odd on a symmetric interval and $\int_{-R}^R \frac{dx}{x^2 + 1} = 2 \arctan R$. Similarly $\int_{-R}^R \frac{dx}{x + i} = -2i \arctan R$.

Step 2 — the contribution of the arc: expanding the kernel. Let $c = \pm i$, with $|c| = 1 < R$. For $|z| = R$ we have $|c/z| = 1/R < 1$, hence

$$\frac{1}{z - c} = \frac{1}{z} \cdot \frac{1}{1 - c/z} = \sum_{n=0}^{\infty} \frac{c^n}{z^{n+1}},$$

a series uniformly convergent on $|z| = R$ because the modulus of the n -th term equals R^{-n-1} and $\sum_n R^{-n-1} < \infty$. One can integrate term by term over the arc.

Step 3 — the integrals of the powers over the arc. With $z = Re^{i\theta}$, $dz = iRe^{i\theta} d\theta$ and $\theta \in [0, \pi]$:

$$\int_{\alpha_R} \frac{dz}{z^{n+1}} = \int_0^\pi R^{-n-1} e^{-i(n+1)\theta} iRe^{i\theta} d\theta = iR^{-n} \int_0^\pi e^{-in\theta} d\theta.$$

For $n = 0$ the integral equals π and the contribution is $i\pi$. For $n \geq 1$,

$$\int_0^\pi e^{-in\theta} d\theta = \left[\frac{e^{-in\theta}}{-in} \right]_0^\pi = \frac{e^{-in\pi} - 1}{-in} = \frac{(-1)^n - 1}{-in},$$

so the contribution is $iR^{-n} \cdot \frac{(-1)^n - 1}{-in} = R^{-n} \frac{1 - (-1)^n}{n}$, which is zero for n even and equals $\frac{2}{nR^n}$ for n odd.

Step 4 — summing the series. Setting $n = 2k + 1$ and using $c^{2k+1} = c(c^2)^k = c(-1)^k$ (because $c^2 = (\pm i)^2 = -1$),

$$\int_{\alpha_R} \frac{dz}{z - c} = i\pi + \sum_{k=0}^{\infty} c(-1)^k \cdot \frac{2}{(2k+1)R^{2k+1}} = i\pi + 2c \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \left(\frac{1}{R}\right)^{2k+1} = i\pi + 2c \arctan \frac{1}{R},$$

where the Gregory–Leibniz series $\arctan x = \sum_{k \geq 0} \frac{(-1)^k}{2k+1} x^{2k+1}$ was used, valid for $|x| \leq 1$ and here applied to $x = 1/R \in (0, 1)$.

Step 5 — the arctangent identity. For $R > 0$, $\arctan R + \arctan \frac{1}{R} = \frac{\pi}{2}$: the derivative with respect to R of the sum is $\frac{1}{1+R^2} + \frac{-1/R^2}{1+1/R^2} = \frac{1}{1+R^2} - \frac{1}{R^2+1} = 0$, so the sum is constant on $(0, +\infty)$, and at $R = 1$ it equals $\frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$.

Step 6 — the two indices. For $c = i$ (Steps 1 and 4):

$$\oint_{\gamma_R} \frac{dz}{z - i} = 2i \arctan R + i\pi + 2i \arctan \frac{1}{R} = i \left(2 \cdot \frac{\pi}{2} + \pi \right) = 2\pi i,$$

hence $\text{Ind}_{\gamma_R}(i) = 1$. For $c = -i$:

$$\oint_{\gamma_R} \frac{dz}{z + i} = -2i \arctan R + i\pi - 2i \arctan \frac{1}{R} = i \left(\pi - 2 \cdot \frac{\pi}{2} \right) = 0,$$

hence $\text{Ind}_{\gamma_R}(-i) = 0$. Both values are integers, consistently with Theorem 15.16. ✓ □

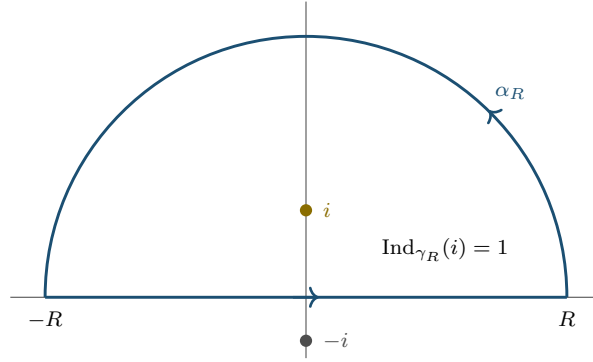


Figure 15.3: The D -shaped contour of Proposition 15.4: segment $[-R, R]$ plus semicircle α_R of radius $R > 1$, traversed counterclockwise. The point i (ordinate 1) is enclosed, the point $-i$ is outside. In the drawing $R = 3$ and the vertex of the arc is $(0, 3)$; the point $-i$ is drawn at $(0, -0.5)$ so as not to fall outside the figure, but in the complex plane it has ordinate -1 : in either case it lies in the lower half-plane, which the contour never touches.

Lemma 15.11: The principal logarithm and loops in the slit plane. Let $\Omega_+ = \mathbb{C} \setminus (-\infty, 0]$ be the plane with the nonpositive real half-line removed. For $w = x + iy \in \Omega_+$, set

$$\operatorname{Arg} w = 2 \arctan \frac{y}{|w| + x} \in (-\pi, \pi), \quad \operatorname{Log} w = \ln |w| + i \operatorname{Arg} w.$$

Then Log is holomorphic on Ω_+ with $\operatorname{Log}'(w) = 1/w$, and consequently

$$\oint_{\eta} \frac{dw}{w} = 0$$

for every piecewise regular loop η contained in Ω_+ .

Proof. Step 1 — the formula is well-defined and continuous. For $w \in \Omega_+$ we have $|w| + x > 0$: indeed $|w| \geq |x| \geq -x$ with equality only if $y = 0$ and $x \leq 0$, a case that is excluded. Hence the formula for Arg defines a continuous function on Ω_+ with values in $(-\pi, \pi)$, and Log is continuous.

Step 2 — Log inverts the exponential. Setting $\theta = \operatorname{Arg} w$, from the bisection formula $\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$ and from $\cos \theta = x/|w|$, $\sin \theta = y/|w|$ (polar form, Volume II, Chapter 3), it follows that

$$\tan \frac{\theta}{2} = \frac{y/|w|}{1 + x/|w|} = \frac{y}{|w| + x},$$

i.e. θ is exactly the argument of w in $(-\pi, \pi)$, and $e^{\operatorname{Log} w} = e^{\ln |w|} e^{i\theta} = |w|(\cos \theta + i \sin \theta) = w$. Moreover Log is injective, because the modulus composed with the argument already is: from $\operatorname{Log} w_1 = \operatorname{Log} w_2$ it follows that $w_1 = e^{\operatorname{Log} w_1} = e^{\operatorname{Log} w_2} = w_2$.

Step 3 — differentiability. Let $w_0 \in \Omega_+$ and $w \rightarrow w_0$ in Ω_+ , $w \neq w_0$. Setting $u = \operatorname{Log} w$ and $u_0 = \operatorname{Log} w_0$, we have $u \rightarrow u_0$ by continuity (Step 1) and $u \neq u_0$ by injectivity (Step 2). Then

$$\frac{\operatorname{Log} w - \operatorname{Log} w_0}{w - w_0} = \frac{u - u_0}{e^u - e^{u_0}} = \left(\frac{e^u - e^{u_0}}{u - u_0} \right)^{-1} \xrightarrow{w \rightarrow w_0} \frac{1}{e^{u_0}} = \frac{1}{w_0},$$

because the difference quotient of the exponential tends to $e^{u_0} \neq 0$ (the exponential function is holomorphic with derivative itself, being the sum of the power series $\sum z^m/m!$). Hence Log is holomorphic with derivative $1/w$, which is continuous on Ω_+ .

Step 4 — loops. By Lemma 15.3 applied to $F = \operatorname{Log}$, $g = 1/w$ on Ω_+ , we get $\oint_{\eta} dw/w = 0$ for every loop η in Ω_+ . $\square$

Theorem 15.17: Residue theorem on a convex open set. Let $\Omega \subseteq \mathbb{C}$ be open and convex, let $\xi_1, \dots, \xi_p \in \Omega$ be distinct points, and let f be holomorphic on $\Omega \setminus \{\xi_1, \dots, \xi_p\}$. Then for every piecewise regular loop γ contained in $\Omega \setminus \{\xi_1, \dots, \xi_p\}$,

$$\frac{1}{2\pi i} \oint_{\gamma} f(z) dz = \sum_{j=1}^p \text{Ind}_{\gamma}(\xi_j) \text{Res}(f, \xi_j).$$

Proof. **Step 1 — the principal parts are entire functions away from their pole.** For each j choose $\delta_j > 0$ such that $D(\xi_j, \delta_j) \subseteq \Omega$ and $\xi_l \notin D(\xi_j, \delta_j)$ for $l \neq j$. On the annulus $0 < |z - \xi_j| < \delta_j$ we have the Laurent expansion $f(z) = \sum_{n \in \mathbb{Z}} a_n^{(j)} (z - \xi_j)^n$ (Theorem 15.4). Set

$$P_j(z) = \sum_{m=1}^{\infty} a_{-m}^{(j)} (z - \xi_j)^{-m}.$$

In the variable $w = 1/(z - \xi_j)$ this is the power series $\sum_{m \geq 1} a_{-m}^{(j)} w^m$, which converges for $0 < |z - \xi_j| < \delta_j$, i.e. for $|w| > 1/\delta_j$: it converges, therefore, at points of arbitrarily large modulus, and since the set of convergence of a power series is a disk (if it converges at w_1 it converges absolutely for $|w| < |w_1|$), its radius of convergence is $+\infty$. Hence P_j is defined and holomorphic on $\mathbb{C} \setminus \{\xi_j\}$, with uniform convergence on every set $\{|z - \xi_j| \geq s\}$, $s > 0$.

Step 2 — the difference is holomorphic on the whole of Ω . Set $g = f - \sum_{j=1}^p P_j$ on $\Omega \setminus \{\xi_1, \dots, \xi_p\}$. Near ξ_j , for $0 < |z - \xi_j| < \delta_j$,

$$f(z) - P_j(z) = \sum_{n \geq 0} a_n^{(j)} (z - \xi_j)^n,$$

a convergent power series, which extends to a function holomorphic on $D(\xi_j, \delta_j)$; the other P_l ($l \neq j$) are holomorphic in a neighborhood of ξ_j by Step 1. Hence g extends to a function $\tilde{g}$ holomorphic on the whole of Ω .

Step 3 — Cauchy on the convex set. Ω is open and convex and $\tilde{g}$ is holomorphic there: by Corollary 15.1, $\oint_{\gamma} \tilde{g} = 0$. Since γ avoids the points ξ_j , on γ^* we have $\tilde{g} = f - \sum_j P_j$, and hence

$$\oint_{\gamma} f(z) dz = \sum_{j=1}^p \oint_{\gamma} P_j(z) dz.$$

Step 4 — the integral of a principal part. Let $s_j = \text{dist}(\xi_j, \gamma^*) > 0$. On γ^* the series defining P_j converges uniformly (Step 1), hence it can be integrated term by term:

$$\oint_{\gamma} P_j(z) dz = \sum_{m=1}^{\infty} a_{-m}^{(j)} \oint_{\gamma} (z - \xi_j)^{-m} dz.$$

For $m \geq 2$ the function $(z - \xi_j)^{-m}$ is the derivative of $\frac{(z - \xi_j)^{1-m}}{1-m}$, holomorphic on $\mathbb{C} \setminus \{\xi_j\}$ with continuous derivative; since γ is a loop in $\mathbb{C} \setminus \{\xi_j\}$, Lemma 15.3 gives a vanishing integral. For $m = 1$, by Definition 15.8, $\oint_{\gamma} \frac{dz}{z - \xi_j} = 2\pi i \text{Ind}_{\gamma}(\xi_j)$. Hence

$$\oint_{\gamma} P_j(z) dz = 2\pi i a_{-1}^{(j)} \text{Ind}_{\gamma}(\xi_j) = 2\pi i \text{Res}(f, \xi_j) \text{Ind}_{\gamma}(\xi_j),$$

having used Definition 15.3. Summing over j and dividing by $2\pi i$ gives the claim. $\square$

Theorem 15.18: Argument principle. Let $\Omega \subseteq \mathbb{C}$ be open and convex and let f be meromorphic on Ω (Definition 15.7), not identically zero, with a finite number of zeros $\alpha_1, \dots, \alpha_M$ of multiplicities $m_1, \dots, m_M$ and a finite number of poles $\beta_1, \dots, \beta_N$ of orders $k_1, \dots, k_N$. Let γ be a piecewise regular loop in Ω passing through no zero and no pole. Then

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_{j=1}^M m_j \operatorname{Ind}_{\gamma}(\alpha_j) - \sum_{l=1}^N k_l \operatorname{Ind}_{\gamma}(\beta_l).$$

Proof. **Step 1 — f'/f and its poles.** The function f'/f is holomorphic on Ω minus the zeros and poles of f , which are finite in number.

Step 2 — at a zero. Let α be a zero of multiplicity m . By Theorem 15.9 there is g holomorphic on a disk $D(\alpha, \rho)$, with $g(\alpha) \neq 0$, such that $f(z) = (z - \alpha)^m g(z)$; shrinking ρ one may assume g never zero. Differentiating (product rule), $f'(z) = m(z - \alpha)^{m-1} g(z) + (z - \alpha)^m g'(z)$, so for $z \neq \alpha$

$$\frac{f'(z)}{f(z)} = \frac{m(z - \alpha)^{m-1} g + (z - \alpha)^m g'}{(z - \alpha)^m g} = \frac{m}{z - \alpha} + \frac{g'(z)}{g(z)}.$$

The second term is holomorphic on $D(\alpha, \rho)$; the first is the principal part. Hence α is a simple pole of f'/f with $\operatorname{Res}(f'/f, \alpha) = m$ (Definition 15.3, read off the Laurent expansion: the principal part is $m(z - \alpha)^{-1}$).

Step 3 — at a pole. Let β be a pole of order k . By Theorem 15.14(b) there is h holomorphic on $D(\beta, \rho)$ with $h(\beta) \neq 0$ and $f(z) = h(z)(z - \beta)^{-k}$; shrinking ρ , h is never zero. Differentiating, $f'(z) = h'(z)(z - \beta)^{-k} - k h(z)(z - \beta)^{-k-1}$, hence

$$\frac{f'(z)}{f(z)} = \frac{h'(z)}{h(z)} - \frac{k}{z - \beta},$$

and β is a simple pole of f'/f with residue $-k$.

Step 4 — the residue theorem. Applying Theorem 15.17 to the function f'/f on Ω , with isolated singularities $\alpha_1, \dots, \alpha_M, \beta_1, \dots, \beta_N$ and with the loop γ avoiding all of them,

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{f'}{f} = \sum_j \operatorname{Ind}_{\gamma}(\alpha_j) m_j + \sum_l \operatorname{Ind}_{\gamma}(\beta_l) (-k_l),$$

which is the claim. □

Corollary 15.8: Argument principle on a circle: zeros minus poles. Let f be meromorphic on an open set A containing $\overline{D}(z_0, r)$, with no zeros and no poles on the circle $|z - z_0| = r$. Let Z and P be the number of zeros and the number of poles of f interior to the circle, each counted with its own multiplicity (order). Then

$$\frac{1}{2\pi i} \oint_{|z-z_0|=r} \frac{f'(z)}{f(z)} dz = Z - P.$$

Proof. **Step 1 — a convex set with compact closure in A .** As in Step 1 of the Alternative proof of Theorem 15.3, one finds $R > r$ with $\overline{D}(z_0, R) \subset A$; set $R'' = \frac{r+R}{2}$ and $\Omega = D(z_0, R'')$, an open convex set with $\overline{\Omega} \subset A$ compact.

Step 2 — finiteness. The poles of f in $\overline{\Omega}$ are finite in number because the set of poles has no accumulation points in A (Definition 15.7) and $\overline{\Omega} \subset A$ is compact. As for the zeros: $\Omega = D(z_0, R'')$ is connected and f , being holomorphic and not identically zero on Ω minus its finitely many poles (it is nonzero on the circle $|z - z_0| = r$ by hypothesis), has finitely many

zeros in $\bar{\Omega}$ by Corollary 15.4. Hence the hypotheses of Theorem 15.18 are satisfied on Ω .

Step 3 — the indices. By Proposition 15.3, the index of the circle equals 1 at points with $|z - z_0| < r$ and 0 at points with $|z - z_0| > r$; on the circle there are neither zeros nor poles by hypothesis. Hence in the formula of Theorem 15.18 exactly the interior zeros and poles survive, with coefficient 1, giving $Z - P$. $\square$

Remark 15.20: Why it is called the “argument principle”. If γ is a loop avoiding zeros and poles, the image path $\eta = f \circ \gamma$ is a piecewise regular loop in $\mathbb{C} \setminus \{0\}$, with $\eta'(t) = f'(\gamma(t))\gamma'(t)$ (chain rule). Hence

$$\oint_{\gamma} \frac{f'(z)}{f(z)} dz = \int_a^b \frac{f'(\gamma(t))\gamma'(t)}{f(\gamma(t))} dt = \int_a^b \frac{\eta'(t)}{\eta(t)} dt = \oint_{\eta} \frac{dw}{w} = 2\pi i \operatorname{Ind}_{\eta}(0).$$

The left-hand side counts zeros minus poles; the right-hand side counts the turns that the image $f \circ \gamma$ makes about the origin, i.e. how many times the argument of $f(z)$ completes a full turn while z traces γ . In the language of Chapter 12 it is the degree of the loop $\eta/|\eta| : S^1 \rightarrow S^1$ (Definition 12.8).

Theorem 15.19: Rouché’s theorem. Let f and g be holomorphic on an open set containing $\overline{D(z_0, r)}$, and suppose

$$|f(z) - g(z)| < |f(z)| \quad \text{for every } z \text{ with } |z - z_0| = r.$$

Then f and g have the same number of zeros in $D(z_0, r)$, counted with multiplicity.

Proof. **Step 1 — neither f nor g vanishes on the circle.** On the circle, $|f(z)| > |f(z) - g(z)| \geq 0$ gives $f(z) \neq 0$; and $|g(z)| \geq |f(z)| - |f(z) - g(z)| > 0$ gives $g(z) \neq 0$.

Step 2 — the quotient lies in a disk not containing 0. By continuity, there is an open neighborhood U of the circle, contained in the original open set, on which f does not vanish; on U the function $h = g/f$ is holomorphic. For $|z - z_0| = r$,

$$|h(z) - 1| = \left| \frac{g(z) - f(z)}{f(z)} \right| = \frac{|f(z) - g(z)|}{|f(z)|} < 1,$$

hence $h(z) \in D(1, 1)$. Every $w \in D(1, 1)$ has $\operatorname{Re} w > 0$ (from $|w - 1| < 1$ it follows that $(\operatorname{Re} w - 1)^2 < 1$, i.e. $0 < \operatorname{Re} w < 2$), so $D(1, 1) \subset \Omega_+ = \mathbb{C} \setminus (-\infty, 0]$.

Step 3 — the image loop has zero index. Let γ be the circle $|z - z_0| = r$ and $\eta = h \circ \gamma$: it is a piecewise regular loop contained in $D(1, 1) \subset \Omega_+$. By Lemma 15.11, $\oint_{\eta} \frac{dw}{w} = 0$; and by the computation of Observation 15.20 (chain rule),

$$\oint_{\gamma} \frac{h'(z)}{h(z)} dz = \oint_{\eta} \frac{dw}{w} = 0.$$

Step 4 — separating the quotient. On U , where f and g do not vanish (shrinking U by continuity so that $g \neq 0$ too, by Step 1),

$$\frac{h'}{h} = \frac{(g/f)'}{g/f} = \frac{\frac{g'f - gf'}{f^2}}{\frac{g}{f}} = \frac{g'f - gf'}{fg} = \frac{g'}{g} - \frac{f'}{f}.$$

Integrating over $\gamma \subset U$ and using Step 3,

$$\oint_{\gamma} \frac{g'}{g} dz - \oint_{\gamma} \frac{f'}{f} dz = 0.$$

Step 5 — the argument principle. f and g are holomorphic, hence have no poles; by Corollary 15.8 applied to each, $\frac{1}{2\pi i} \oint_{\gamma} \frac{f'}{f} = Z_f$ and $\frac{1}{2\pi i} \oint_{\gamma} \frac{g'}{g} = Z_g$, the numbers of interior zeros with multiplicity. Step 4 gives $Z_g = Z_f$. $\square$

Corollary 15.9: Fundamental theorem of algebra, by the second route. Let $P(z) = c_n z^n + \cdots + c_0$ with $n \geq 1$ and $c_n \neq 0$. Setting

$$R = 1 + \frac{1}{|c_n|} \sum_{k=0}^{n-1} |c_k|,$$

the polynomial P has exactly n zeros in $D(0, R)$, counted with multiplicity, and none elsewhere in $\mathbb{C}$.

Proof. **Step 1 — the comparison on the circle $|z| = R$.** Take $f(z) = c_n z^n$ and $g = P$. Since $R \geq 1$, for $|z| = R$ and $0 \leq k \leq n-1$ we have $R^k \leq R^{n-1}$, hence

$$|f(z) - g(z)| = \left| \sum_{k=0}^{n-1} c_k z^k \right| \leq \sum_{k=0}^{n-1} |c_k| R^k \leq R^{n-1} \sum_{k=0}^{n-1} |c_k| = R^n \cdot \frac{1}{R} \sum_{k=0}^{n-1} |c_k|.$$

From $R > \frac{1}{|c_n|} \sum_{k < n} |c_k|$ it follows that $\frac{1}{R} \sum_{k < n} |c_k| < |c_n|$, hence $|f(z) - g(z)| < |c_n| R^n = |f(z)|$ on the circle.

Step 2 — Rouché. By Theorem 15.19, P and $c_n z^n$ have the same number of zeros in $D(0, R)$. The function $c_n z^n$ has the single zero $z = 0$, of multiplicity n (the first nonzero Taylor coefficient is the one of index n , Theorem 15.9): hence P has exactly n zeros in $D(0, R)$.

Step 3 — there are no others. For $|z| \geq R$, Step 1 gives $|P(z)| \geq |f(z)| - |f(z) - g(z)| > 0$: outside the disk P does not vanish. Since P has degree n and, by Corollary 15.2, factors into n linear factors, its zeros are exactly n , counted with multiplicity, all in $D(0, R)$. $\square$

15.9. Three real integrals computed with residues

The residue theorem turns the computation of certain real integrals into an algebraic computation: one chooses a closed contour, identifies the singularities it encloses, and sums the residues. Three such computations are carried out here — two over $\mathbb{R}$, one over $[0, 2\pi]$ — each checked by a *second, independent method* (a real substitution, a Weierstrass substitution, Fourier inversion) and by a numerical quadrature performed in double precision.

Example 15.4: First integral: $\int_{-\infty}^{+\infty} \frac{dx}{(x^2+1)^2}$. **Statement.** Compute $I_1 = \int_{-\infty}^{+\infty} \frac{dx}{(x^2+1)^2}$.

Step 1 — the integral converges. The integrand is positive, continuous, and $\leq 1/x^4$ for $|x| \geq 1$: the improper integral converges absolutely, and $I_1 = \lim_{R \rightarrow +\infty} \int_{-R}^R$.

Step 2 — the contour and the convex set. Let $R > 1$, let γ_R be the D -shaped contour of Proposition 15.4, and let

$$\Omega_R = D(0, R+1) \cap \left\{ z \in \mathbb{C} : \operatorname{Im} z > -\frac{1}{2} \right\},$$

the intersection of an open disk and an open half-plane: it is open and convex. It contains the support of γ_R (the segment has imaginary part 0, the arc has imaginary part ≥ 0 , and all points have modulus $\leq R < R+1$) and contains i , while it does *not* contain $-i$, which has imaginary part $-1 < -\frac{1}{2}$.

Step 3 — the function and its only singularity in Ω_R . Set $f(z) = \frac{1}{(z^2 + 1)^2} = \frac{1}{(z - i)^2(z + i)^2}$. Its only singularities are $\pm i$; in Ω_R only $z = i$ remains, a pole of order 2 (the factor $(z - i)^{-2}$ with $1/(z + i)^2$ holomorphic and nonzero at i : Theorem 15.14(b) with $k = 2$).

Step 4 — the residue. By Proposition 15.2(i) with $k = 2$,

$$\operatorname{Res}(f, i) = \frac{1}{1!} \lim_{z \rightarrow i} \frac{d}{dz} [(z - i)^2 f(z)] = \lim_{z \rightarrow i} \frac{d}{dz} \frac{1}{(z + i)^2} = \lim_{z \rightarrow i} \frac{-2}{(z + i)^3} = \frac{-2}{(2i)^3}.$$

Since $(2i)^3 = 8i^3 = -8i$, we get $\operatorname{Res}(f, i) = \frac{-2}{-8i} = \frac{1}{4i} = -\frac{i}{4}$.

Step 5 — the residue theorem. By Theorem 15.17 on Ω_R , with the only singularity i and $\operatorname{Ind}_{\gamma_R}(i) = 1$ (Proposition 15.4),

$$\oint_{\gamma_R} f(z) dz = 2\pi i \cdot 1 \cdot \frac{1}{4i} = \frac{\pi}{2}.$$

Step 6 — the arc vanishes. For $|z| = R > 1$ we have $|z^2 + 1| \geq |z|^2 - 1 = R^2 - 1$, hence $|f(z)| \leq \frac{1}{(R^2 - 1)^2}$; the arc α_R has length πR , and by the ML estimate (Lemma 15.2)

$$\left| \int_{\alpha_R} f(z) dz \right| \leq \frac{\pi R}{(R^2 - 1)^2} \xrightarrow{R \rightarrow +\infty} 0.$$

Step 7 — conclusion. Since $\oint_{\gamma_R} f = \int_{-R}^R f(x) dx + \int_{\alpha_R} f$, letting $R \rightarrow +\infty$ gives

$$I_1 = \frac{\pi}{2} = 1.5707963$$

Second, independent method: the substitution $x = \tan t$. With $x = \tan t$, $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $dx = \frac{dt}{\cos^2 t}$ and $1 + x^2 = \frac{1}{\cos^2 t}$, hence $(1 + x^2)^2 = \frac{1}{\cos^4 t}$ and

$$I_1 = \int_{-\pi/2}^{\pi/2} \cos^4 t \cdot \frac{dt}{\cos^2 t} = \int_{-\pi/2}^{\pi/2} \cos^2 t dt = \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2t}{2} dt = \left[\frac{t}{2} + \frac{\sin 2t}{4} \right]_{-\pi/2}^{\pi/2} = \frac{\pi}{2}.$$

Identical. ✓

Third check, numerical. Composite Simpson quadrature with 400,000 subintervals on $[-2000, 2000]$: 1.5707963; and $\frac{\pi}{2} = 1.5707963$. The neglected tail is bounded by $2 \int_{2000}^{\infty} x^{-4} dx = \frac{2}{3 \cdot 2000^3} = 8.3 \cdot 10^{-11}$, consistent with the observed discrepancy ($8.3 \cdot 10^{-11}$). ✓

Typical error. Including also the residue at $-i$. The D -shaped contour encloses only the upper half-plane, and Proposition 15.4 gives $\operatorname{Ind}_{\gamma_R}(-i) = 0$: the pole $-i$ does not contribute. The quickest way to notice this is that Ω_R was built specifically *not* to contain it.

Example 15.5: Second integral: $\int_0^{2\pi} \frac{d\theta}{5 - 4 \cos \theta}$. **Statement.** Compute $I_2 = \int_0^{2\pi} \frac{d\theta}{5 - 4 \cos \theta}$. The denominator is always $\geq 5 - 4 = 1 > 0$, so the integrand is continuous and the integral exists.

Step 1 — from the circle to algebra. With $z = e^{i\theta}$ we have $\cos \theta = \frac{1}{2}(z + z^{-1})$ and $dz = iz d\theta$, i.e. $d\theta = \frac{dz}{iz}$ (as in Step 1 of Example 15.1). Hence

$$5 - 4 \cos \theta = 5 - 2 \left(z + \frac{1}{z} \right) = \frac{5z - 2z^2 - 2}{z} = \frac{-(2z^2 - 5z + 2)}{z} = \frac{-(2z - 1)(z - 2)}{z},$$

having factored $2z^2 - 5z + 2 = (2z - 1)(z - 2)$ (the roots are $\frac{1}{2}$ and 2, and the product of the factors reproduces $2z^2 - 5z + 2$: direct check $(2z - 1)(z - 2) = 2z^2 - 4z - z + 2 = 2z^2 - 5z + 2$ ✓).

Step 2 — the contour integral.

$$I_2 = \oint_{\mathbb{T}} \frac{z}{-(2z - 1)(z - 2)} \cdot \frac{dz}{iz} = -\frac{1}{i} \oint_{\mathbb{T}} \frac{dz}{(2z - 1)(z - 2)} = i \oint_{\mathbb{T}} \frac{dz}{(2z - 1)(z - 2)},$$

because $-1/i = i$.

Step 3 — the poles. The function $F(z) = \frac{1}{(2z - 1)(z - 2)} = \frac{1}{2(z - \frac{1}{2})(z - 2)}$ has two simple poles, at $z = \frac{1}{2}$ and at $z = 2$. Only $\frac{1}{2}$ is interior to $\mathbb{T}$; by Proposition 15.3, $\text{Ind}_{\mathbb{T}}(\frac{1}{2}) = 1$ and $\text{Ind}_{\mathbb{T}}(2) = 0$.

Step 4 — the residue. By Proposition 15.2(ii),

$$\text{Res}\left(F, \frac{1}{2}\right) = \lim_{z \rightarrow 1/2} \left(z - \frac{1}{2}\right) F(z) = \frac{1}{2(\frac{1}{2} - 2)} = \frac{1}{2 \cdot (-\frac{3}{2})} = -\frac{1}{3}.$$

Step 5 — conclusion. Applying Theorem 15.17 with $\Omega = D(0, \frac{3}{2})$ (convex, containing $\mathbb{T}$ and $\frac{1}{2}$, not containing 2):

$$I_2 = i \cdot 2\pi i \cdot \left(-\frac{1}{3}\right) = -\frac{2\pi i^2}{3} = \boxed{\frac{2\pi}{3} = 2.0943951}$$

Second, independent method: the Weierstrass substitution. Since $\cos(2\pi - \theta) = \cos \theta$, the integrand is symmetric about $\theta = \pi$ and $I_2 = 2 \int_0^\pi \frac{d\theta}{5 - 4 \cos \theta}$. With $t = \tan \frac{\theta}{2}$ ($\theta : 0 \rightarrow \pi$ corresponds to $t : 0 \rightarrow +\infty$), $\cos \theta = \frac{1-t^2}{1+t^2}$ and $d\theta = \frac{2dt}{1+t^2}$:

$$5 - 4 \cos \theta = \frac{5(1+t^2) - 4(1-t^2)}{1+t^2} = \frac{1+9t^2}{1+t^2},$$

hence

$$\int_0^\pi \frac{d\theta}{5 - 4 \cos \theta} = \int_0^{+\infty} \frac{1+t^2}{1+9t^2} \cdot \frac{2dt}{1+t^2} = \int_0^{+\infty} \frac{2dt}{1+9t^2} = \frac{2}{3} \left[\arctan 3t \right]_0^{+\infty} = \frac{2}{3} \cdot \frac{\pi}{2} = \frac{\pi}{3},$$

and $I_2 = 2 \cdot \frac{\pi}{3} = \frac{2\pi}{3}$. Identical. ✓

Third check, with the inherited example. Writing $\frac{1}{5-4\cos\theta} = \frac{1}{4} \cdot \frac{1}{\frac{5}{4}-\cos\theta}$ and observing that $\int_0^{2\pi} \frac{d\theta}{a-\cos\theta} = \int_0^{2\pi} \frac{d\theta}{a+\cos\theta}$ (substitution $\theta \mapsto \theta + \pi$, which changes the sign of the cosine and leaves the interval unchanged by periodicity), Example 15.1 with $a = \frac{5}{4}$ gives

$$I_2 = \frac{1}{4} \cdot \frac{2\pi}{\sqrt{\frac{25}{16} - 1}} = \frac{1}{4} \cdot \frac{2\pi}{\frac{3}{4}} = \frac{2\pi}{3}. \checkmark$$

Fourth check, numerical. Composite Simpson quadrature with 200,000 subintervals on $[0, 2\pi]$: 2.0943951; and $\frac{2\pi}{3} = 2.0943951$. ✓

Typical error. Forgetting the factor $\frac{1}{iz}$ in the change of variable, or getting the sign wrong in $-1/i$. The immediate check is that I_2 must be *real and positive* (positive integrand): an imaginary result signals a sign error.

Example 15.6: Third integral: $\int_{-\infty}^{+\infty} \frac{\cos x}{x^2+1} dx$. **Statement.** Compute $I_3 = \int_{-\infty}^{+\infty} \frac{\cos x}{x^2+1} dx$.

Step 1 — convergence. $|\frac{\cos x}{x^2+1}| \leq \frac{1}{x^2+1}$, integrable on $\mathbb{R}$: the integral converges absolutely.

Step 2 — the right function. One does not integrate $\cos z/(z^2+1)$, which is unbounded on the arc ($|\cos z|$ grows like $e^{|\operatorname{Im} z|}$), but rather

$$f(z) = \frac{e^{iz}}{z^2+1},$$

whose modulus on the upper arc is controlled: for $\operatorname{Im} z \geq 0$, $|e^{iz}| = e^{-\operatorname{Im} z} \leq 1$.

Step 3 — residue. With $p(z) = e^{iz}$ and $q(z) = z^2+1$ we have $q(i) = 0$, $q'(z) = 2z$, $q'(i) = 2i \neq 0$ and $p(i) = e^{i \cdot i} = e^{-1} \neq 0$: by Proposition 15.2(iii), i is a simple pole and

$$\operatorname{Res}(f, i) = \frac{p(i)}{q'(i)} = \frac{e^{-1}}{2i}.$$

Step 4 — residue theorem. With the same convex set Ω_R and the same contour γ_R of Example 15.4 (which contains i and not $-i$), Theorem 15.17 and Proposition 15.4 give

$$\oint_{\gamma_R} f(z) dz = 2\pi i \cdot \frac{e^{-1}}{2i} = \frac{\pi}{e}.$$

Step 5 — the arc vanishes. For $|z| = R > 1$ with $\operatorname{Im} z \geq 0$,

$$|f(z)| = \frac{|e^{iz}|}{|z^2+1|} \leq \frac{1}{R^2-1},$$

and the length of the arc is πR : by Lemma 15.2, $|\int_{\alpha_R} f| \leq \frac{\pi R}{R^2-1} \rightarrow 0$.

Step 6 — taking the real part. Letting $R \rightarrow +\infty$, $\int_{-\infty}^{+\infty} \frac{e^{ix}}{x^2+1} dx = \frac{\pi}{e}$. The left-hand side has real part $\int \frac{\cos x}{x^2+1} dx$ and imaginary part $\int \frac{\sin x}{x^2+1} dx = 0$ (odd integrand, symmetric domain). The right-hand side is real. Hence

$$I_3 = \frac{\pi}{e} = 1.1557273$$

Second, independent method: Fourier inversion. Consider $h(t) = e^{-|t|}$, a continuous function integrable on $\mathbb{R}$. Its Fourier transform is computed elementarily:

$$\hat{h}(x) = \int_{-\infty}^{+\infty} e^{-|t|} e^{-ixt} dt = 2 \int_0^{+\infty} e^{-t} \cos(xt) dt,$$

(the part with $\sin$ vanishes by oddness) and

$$\int_0^{+\infty} e^{-t} \cos(xt) dt = \operatorname{Re} \int_0^{+\infty} e^{(-1+ix)t} dt = \operatorname{Re} \frac{1}{1-ix} = \operatorname{Re} \frac{1+ix}{1+x^2} = \frac{1}{1+x^2},$$

because $\int_0^T e^{(-1+ix)t} dt = \frac{e^{(-1+ix)T}-1}{-1+ix} \rightarrow \frac{-1}{-1+ix} = \frac{1}{1-ix}$ (the numerator tends to -1 because $|e^{(-1+ix)T}| = e^{-T} \rightarrow 0$). Hence $\hat{h}(x) = \frac{2}{1+x^2}$, which is integrable. By the Fourier inversion theorem (a prerequisite stated in the opening infobox),

$$e^{-|t|} = h(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{h}(x) e^{ixt} dx = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{e^{ixt}}{1+x^2} dx = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\cos(xt)}{1+x^2} dx.$$

Setting $t = 1$: $e^{-1} = \frac{1}{\pi}I_3$, i.e. $I_3 = \pi/e$. Identical, and by a route that uses no residue at all. ✓

Third check, numerical. Composite Simpson quadrature with 4000000 subintervals on $[-4000, 4000]$: 1.1557273; and $\pi/e = 1.1557273$. ✓

Interpretation. The general formula just proved twice, $\int_{-\infty}^{+\infty} \frac{\cos(xt)}{1+x^2} dx = \pi e^{-|t|}$, says that the Fourier transform of the Cauchy kernel $\frac{1}{\pi(1+x^2)}$ is $e^{-|t|}$: this is the pair that governs the Poisson kernel of the half-plane in Chapter 16. The *exponential* decay of the transform is the analytic translation of the fact that $\frac{1}{1+x^2}$ extends to a function holomorphic in a strip about the real axis, of width equal to the distance of the poles $\pm i$ from the axis: 1.

Typical error. Using $\cos z$ instead of e^{iz} . On the upper arc $|\cos(Re^{i\theta})|$ is unbounded — for $z = iy$ we have $\cos(iy) = \cosh y \rightarrow +\infty$ — and the estimate of Step 5 fails. The choice of e^{iz} exploits the fact that the half-plane in which the contour closes is *the one where the exponential decays*: if the integrand were $\cos x$ multiplied by a function with poles in the lower half-plane, one would close downward with e^{-iz} .

15.10. Exercises inherited from Volume II

The exercises that follow come from Volume II, Chapter 3, and concern the topics of this chapter. They are reproduced in full, together with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 15.1: N9 — An integral with a double pole. Compute

$$\oint_{|z|=3} \frac{z}{(z-1)(z-2)^2} dz$$

using Theorem 15.5, identifying the interior poles, their order, and their respective residues.

Exercise 15.2: S9 — Cauchy's estimates and Liouville's theorem. Let f be holomorphic on an open set containing $\overline{D}(z_0, r)$ and $|f| \leq M$ on $|z - z_0| = r$. Prove the **Cauchy estimates**

$$|a_n| \leq \frac{M}{r^n} \quad (n \geq 0),$$

where a_n are the coefficients of the expansion of f about z_0 , and deduce **Liouville's theorem**: a bounded entire function is constant. Observe that Cauchy's estimates are exactly those used in Chapter 2, Step 5 of the proof of the Theorem (Volume II, Chapter 2).

Exercise 15.3: S10 — The trick $\bar{z} = 1/z$ on the unit circle. Prove that on $\mathbb{T}$ one has $\bar{z} = 1/z$, and use this to show that the integral over $\mathbb{T}$ of any polynomial in the variables z and $\bar{z}$ reduces to $2\pi i$ times the coefficient of z^{-1} after the substitution. Apply the method to compute

$$\oint_{\mathbb{T}} |z-1|^2 dz.$$

Solutions

Solution. N9. *Step 1 — the poles and their order.* The function

$$f(z) = \frac{z}{(z-1)(z-2)^2}$$

has a *simple* pole at $z = 1$ and a *double* pole at $z = 2$. Both have modulus less than 3, so they are interior to the contour $|z| = 3$.

Step 2 — residue at the simple pole $z = 1$. With the *step box* formula,

$$\operatorname{Res}(f, 1) = \lim_{z \rightarrow 1} (z-1)f(z) = \lim_{z \rightarrow 1} \frac{z}{(z-2)^2} = \frac{1}{(1-2)^2} = \frac{1}{1} = 1.$$

Step 3 — residue at the double pole $z = 2$. For a pole of order $k = 2$,

$$\operatorname{Res}(f, 2) = \lim_{z \rightarrow 2} \frac{d}{dz} [(z-2)^2 f(z)] = \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{z}{z-1} \right].$$

Differentiating by the quotient rule,

$$\frac{d}{dz} \frac{z}{z-1} = \frac{1 \cdot (z-1) - z \cdot 1}{(z-1)^2} = \frac{-1}{(z-1)^2},$$

hence

$$\operatorname{Res}(f, 2) = \frac{-1}{(2-1)^2} = -1.$$

Step 4 — the residue theorem. By Theorem 15.5,

$$\oint_{|z|=3} \frac{z dz}{(z-1)(z-2)^2} = 2\pi i(1 + (-1)) = 0.$$

Step 5 — independent check by partial fractions. We seek A, B, C with

$$\frac{z}{(z-1)(z-2)^2} = \frac{A}{z-1} + \frac{B}{z-2} + \frac{C}{(z-2)^2}.$$

Multiplying by $(z-1)(z-2)^2$: $z = A(z-2)^2 + B(z-1)(z-2) + C(z-1)$. At $z = 1$: $1 = A \cdot 1 \Rightarrow A = 1$. At $z = 2$: $2 = C \cdot 1 \Rightarrow C = 2$. Comparing the coefficients of z^2 : $0 = A + B \Rightarrow B = -1$. Integrating term by term with Lemma 15.5, only the powers $(z-\xi)^{-1}$ contribute:

$$\oint f = 2\pi i(A + B) = 2\pi i(1 - 1) = 0. \checkmark$$

(The term $C/(z-2)^2$ has exponent $-2 \neq -1$ and does not contribute: consistent with the fact that the residue is by definition the coefficient of $(z-\xi)^{-1}$, not of more negative powers.)

Typical error. Applying the simple-pole formula $\lim_{z \rightarrow 2} (z-2)f(z)$ to the double pole: the limit would be infinite and the computation would fail without giving a clear warning. *Always* check the order of the pole before choosing the formula.

Solution. S9. *Step 1 — integral representation of the coefficients.* Since f is holomorphic on an open set containing $\overline{D(z_0, r)}$, it has there a power series expansion $f(z) = \sum_{n \geq 0} a_n(z - z_0)^n$, and by Theorem 15.4 (the case of vanishing principal part)

$$a_n = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Step 2 — ML estimate. Parametrizing $\zeta = z_0 + re^{i\theta}$, we have $|d\zeta| = r d\theta$ and $|\zeta - z_0| = r$, hence

$$|a_n| \leq \frac{1}{2\pi} \oint \frac{|f(\zeta)|}{|\zeta - z_0|^{n+1}} |d\zeta| \leq \frac{1}{2\pi} \cdot \frac{M}{r^{n+1}} \cdot \underbrace{2\pi r}_{\text{length}} = \frac{M}{r^n}.$$

Step 3 — Liouville's theorem. Let f be entire (holomorphic on the whole of $\mathbb{C}$) and bounded, $|f| \leq M$ everywhere. Fixing $z_0 = 0$, Cauchy's estimates hold for *every* $r > 0$:

$$|a_n| \leq \frac{M}{r^n} \quad \text{for every } r > 0.$$

For $n \geq 1$, letting $r \rightarrow +\infty$, the right-hand side tends to 0, while a_n does not depend on r : hence $a_n = 0$ for every $n \geq 1$. What remains is $f(z) = a_0$, constant.

Step 4 — connection with Chapter 2. The estimates of Step 2 are exactly those used in Step 5 of the proof of the Theorem (Volume II, Chapter 2), where they served to bound the Taylor coefficients of the integrand and to obtain the geometric convergence of the periodic trapezoidal rule. It is the same theorem, used once to deduce a structural fact (Liouville) and once for a numerical estimate: a good example of how versatile Cauchy's estimates are.

Solution. S10. *Step 1 — the trick.* By Exercise S2 (Volume II, Chapter 3), on $\mathbb{T}$ one has $\bar{z} = 1/z$. Hence every polynomial expression in z and $\bar{z}$, restricted to $\mathbb{T}$, becomes a Laurent polynomial in z alone:

$$P(z, \bar{z})|_{\mathbb{T}} = P\left(z, \frac{1}{z}\right) = \sum_{m=-M}^M c_m z^m.$$

Step 2 — integration. Integrating over $\mathbb{T}$ term by term and applying Lemma 15.5 with $z_0 = 0$ to each power,

$$\oint_{\mathbb{T}} P(z, \bar{z}) dz = \sum_m c_m \oint_{\mathbb{T}} z^m dz = 2\pi i c_{-1},$$

since all powers other than z^{-1} give a vanishing integral.

Step 3 — application. Compute $\oint_{\mathbb{T}} |z - 1|^2 dz$. On the curve,

$$|z - 1|^2 = (z - 1)(\bar{z} - 1) = (z - 1)\left(\frac{1}{z} - 1\right).$$

Expanding:

$$(z - 1)\left(\frac{1}{z} - 1\right) = 1 - z - \frac{1}{z} + 1 = 2 - z - z^{-1}.$$

Step 4 — reading off the coefficient. The coefficient of z^{-1} is $c_{-1} = -1$, hence

$$\oint_{\mathbb{T}} |z - 1|^2 dz = 2\pi i (-1) = -2\pi i.$$

Step 5 — direct check. Parametrizing $z = e^{i\theta}$, $dz = ie^{i\theta} d\theta$, and using $|e^{i\theta} - 1|^2 = 2 - 2\cos\theta$ (from Step 3 with $z = e^{i\theta}$):

$$\oint_{\mathbb{T}} |z - 1|^2 dz = \int_0^{2\pi} (2 - 2\cos\theta) ie^{i\theta} d\theta = i \int_0^{2\pi} (2e^{i\theta} - 2\cos\theta e^{i\theta}) d\theta.$$

The first integral vanishes by periodicity. For the second, writing $2\cos\theta = e^{i\theta} + e^{-i\theta}$,

$$\int_0^{2\pi} (e^{i\theta} + e^{-i\theta}) e^{i\theta} d\theta = \int_0^{2\pi} (e^{2i\theta} + 1) d\theta = 0 + 2\pi = 2\pi.$$

Hence the integral equals $i(0 - 2\pi) = -2\pi i$. ✓

Remark. The result is *nonzero* even though $|z - 1|^2$ is a real, positive function: there is no contradiction, because $|z - 1|^2$ is not holomorphic (it depends on $\bar{z}$) and the Cauchy–Goursat theorem does not apply to it. This is the typical error to avoid: the vanishing of the integral over a closed contour requires holomorphy, not mere continuity.

Inherited quiz

The items below come from the quiz of Volume II, Chapter 3, and are renumbered here.

1. The residue of f at an isolated singularity ξ is
 - A) $f(\xi)$ B) the coefficient a_{-1} of the Laurent expansion C) the coefficient a_0 D) $f'(\xi)$
2. The residue theorem states that the integral over a closed contour equals
 - A) always zero B) $2\pi i$ times the sum of the *interior* residues C) πi times the sum of all residues D) the sum of the poles
3. The integral $\oint_{\mathbb{T}} |z - 1|^2 dz$ is nonzero because
 - A) the integrand has a pole at $z = 1$
 - B) the integrand is not holomorphic, depending on $\bar{z}$
 - C) the contour is not closed
 - D) the computation is wrong: it equals zero

BEST PRACTICE

Answer key.

1-B: Definition 15.3. **2-B:** Theorem 15.5; the word “interior” is essential (cf. the typical error of Example 15.1). **3-B:** Exercise S10, Step 5: $|z - 1|^2$ depends on $\bar{z}$, so Cauchy–Goursat does not apply.

Inherited formulary

The lines below come from the formulary of Volume II, Chapter 3.

Complex analysis on the circle

$$\oint_{|\zeta - z_0| = \rho} (\zeta - z_0)^m d\zeta = 2\pi i [m = -1], \quad f(z) = \frac{1}{2\pi i} \oint \frac{f(\zeta)}{\zeta - z} d\zeta$$

$$\oint f = 2\pi i \sum_j \text{Res}(f, \xi_j), \quad \text{Res}\left(\frac{p}{q}, \xi\right) = \frac{p(\xi)}{q'(\xi)} \text{ (simple pole)}$$

15.11. Conformal mapping: why the disk is a universal model

INFO

Level D. We use Cauchy's formula and estimates, the maximum principle, the identity theorem, Rouché's theorem and Schwarz's lemma with its equality case, already treated in Chapter 15. Completeness of $\mathbb{C}$, compactness of closed bounded sets and integration along piecewise regular paths are prerequisites. The root lemma, Montel normality and the limit theorem for injective functions are proved here; none of these three steps is imported.

A domain is a nonempty connected open set. Simply connected means that every closed path contracts to a point within the domain. The conclusion concerns the interior; without further hypotheses it does not assert a bijective extension to the boundary. The reference for the extremal method is [1]; the proofs below also supply its analytic prerequisites.

15.11.1 Primitives, logarithms and roots

***Lemma 15.12: Global primitive and logarithm.** Let U be simply connected. Every holomorphic function h on U has a primitive. If h is nowhere zero, there is a holomorphic L with $e^L = h$; for every integer $m \geq 1$, $e^{L/m}$ is a holomorphic m th root of h .*

Proof. 1. Integrals and homotopy. On every disk contained in U there is a local primitive: integrate from a fixed point and apply Goursat to triangles. Let $H : [0, 1]^2 \rightarrow U$ be a fixed-endpoint homotopy between two paths. Its image is compact. Uniform continuity allows a grid fine enough that the image of each cell lies in a disk carrying a primitive. Indeed, cover the image by disks smaller than the primitive disks, take a finite subcover and use a Lebesgue number for the pulled-back cover of the square.

Replace the images of interior edges by line segments joining the images of their endpoints. A common edge stays in both adjacent disks because both disks are convex. The integral around each cell is zero by its local primitive. On summing, interior edges cancel with opposite orientations. On the outer boundary, retain the original paths: their integrals equal those of the replacing segments by the same local primitive. No differentiability of the homotopy has been assumed. Thus the integral is homotopy invariant, and contraction of a closed path shows that its integral is zero.

2. Construction. Fix $a \in U$ and set $F(z) = \int_a^z h(\zeta) d\zeta$ along any path from a to z . The preceding step gives path independence. For w close to z , the final segment can be straight, so

$$\frac{F(w) - F(z)}{w - z} = \int_0^1 h(z + t(w - z)) dt \longrightarrow h(z).$$

3. Logarithm. If $h \neq 0$, apply step 2 to h'/h . Choose ℓ with $e^\ell = h(a)$ and define $L(z) = \ell + \int_a^z h'/h$. Then $(he^{-L})' = 0$ and $h(a)e^{-L(a)} = 1$. Connectedness of U gives $h = e^L$; raising $e^{L/m}$ to the m th power finishes the proof. $\square$

15.11.2 The compactness step: Montel proved here

***Theorem 15.20: Normality of locally bounded families.** Every sequence of holomorphic functions on a domain U , uniformly bounded on each compact subset, has a subsequence converging uniformly on compact subsets to a holomorphic function. The derivatives also converge uniformly on compact subsets.*

Proof. 1. Equicontinuity. Given compact $K \subset U$, choose $\rho > 0$ so that its closed 2ρ -neighbourhood is compact in U . Let M bound all functions there. Cauchy's estimate on

circles of radius ρ gives $|f'_n(z)| \leq M/\rho$ in the ρ -neighbourhood. If $z, w \in K$ and $|z - w| < \rho$, integration along the segment yields

$$|f_n(z) - f_n(w)| \leq (M/\rho)|z - w|.$$

2. Diagonal extraction. Enumerate a countable dense set $D = \{d_1, d_2, \dots\} \subset U$. Each sequence $f_n(d_j)$ is bounded. Bolzano–Weierstrass gives a subsequence converging at d_1 , a subsequence of it converging at d_2 , and so on. Taking the k th term of the k th subsequence yields a single sequence converging at every d_j .

3. Uniformity. Fix K and $\varepsilon > 0$. Take a finite net of points in D , within distance δ of every point of K , with $2M\delta/\rho < \varepsilon/2$. Step 1 also applies to these points chosen in the neighbourhood. For sufficiently large indices, values at all finitely many nodes differ by less than $\varepsilon/2$. The triangle inequality through a nearby node bounds the difference by ε everywhere on K . Thus the extracted sequence is uniformly Cauchy on K and converges. The same subsequence works for every compact set because it was fixed on one dense set.

4. Holomorphy and derivatives. On a circle $|\zeta - z_0| = r$ whose closed disk lies in U , uniform convergence permits passage to the limit in Cauchy's formula. For $|z - z_0| < r$ this gives

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad f^{(j)}(z) = \frac{j!}{2\pi i} \int_{|\zeta - z_0| = r} \frac{f(\zeta)}{(\zeta - z)^{j+1}} d\zeta.$$

The denominator is uniformly separated from zero on smaller interior disks. Differentiation under the integral is therefore valid, and the same formula applied to $f_n - f$ proves convergence of derivatives. A finite cover proves the assertion on an arbitrary compact set. $\square$

***Lemma 15.13: Derivative and limit of injective functions.** An injective holomorphic function has nowhere-vanishing derivative and a holomorphic inverse on its image. A locally uniform limit of injective holomorphic functions is either constant or injective.*

Proof. 1. Derivative. If $f(z) - f(z_0)$ has a zero of order $m \geq 2$, choose a small disk about z_0 in which f' has no other zeros. For sufficiently small $\eta \neq 0$, Rouché gives m zeros, counted with multiplicity, of $f(z) - f(z_0) - \eta$ in that disk. None is z_0 and their derivatives are nonzero, so they are m distinct points, a contradiction. Hence $m = 1$. The same argument with $m = 1$ gives a unique nearby preimage for nearby values. The inverse is locally continuous and

$$\frac{f^{-1}(w) - z_0}{w - f(z_0)} \longrightarrow \frac{1}{f'(z_0)}.$$

2. Limit. Let f be a nonconstant limit and suppose $f(z_1) = f(z_2)$ with $z_1 \neq z_2$. Zeros of $g(z) = f(z) - f(z_1)$ are isolated. Choose a closed disk around z_2 excluding z_1 , with no zeros of g on its boundary. On that boundary, $f_n(z) - f_n(z_1)$ converges uniformly to g . Rouché gives at least one zero in the disk for large n . Yet injectivity of f_n requires its only zero to be z_1 , outside the disk. This is a contradiction. $\square$

15.11.3 The theorem and the decisive calculation

***Theorem 15.21: Normalized Riemann mapping theorem.** Let $U \subsetneq \mathbb{C}$ be a simply connected domain and $a \in U$. There is a unique holomorphic bijection $f : U \rightarrow \mathbb{D}$, where $\mathbb{D} = \{w : |w| < 1\}$, such that $f(a) = 0$ and $f'(a) > 0$ is real.*

Proof. 1. The admissible family is nonempty. Choose $c \notin U$. Lemma 15.12 gives a holomorphic $q(z) = \sqrt{z - c}$. It is injective since $q(z)^2 = z - c$. The images $q(U)$ and $-q(U)$ are disjoint: an intersection would give $q(z) = -q(w)$, hence $z = w$ and $q(z) = 0$, which is

impossible. Set $q_0 = q(a)$ and choose $\delta > 0$ with $D(q_0, \delta) \subset q(U)$. The disk $D(-q_0, \delta)$ is omitted by q , so

$$T(z) = \frac{\delta}{2(q(z) + q_0)}, \quad |T(z)| \leq \frac{1}{2}.$$

The function T is holomorphic and injective. Recall the disk automorphism

$$\phi_b(w) = \frac{w - b}{1 - \bar{b}w}, \quad 1 - |\phi_b(w)|^2 = \frac{(1 - |b|^2)(1 - |w|^2)}{|1 - \bar{b}w|^2}.$$

Compose T with $\phi_{T(a)}$ and multiply by a unit complex number to make the derivative at a real and positive. That derivative is nonzero by Lemma 15.13. Thus the following family is nonempty:

$$\mathcal{F} = \{f : U \rightarrow \mathbb{D} : f \text{ holomorphic, injective, } f(a) = 0, f'(a) > 0\}.$$

2. Existence of an extremal. Choose $\rho > 0$ with $\overline{D}(a, \rho) \subset U$. Cauchy's estimate gives $0 < f'(a) \leq 1/\rho$ for $f \in \mathcal{F}$. Hence $M = \sup_{\mathcal{F}} f'(a)$ is finite and positive. Choose $f'_n(a) \rightarrow M$. By Theorem 15.20, a subsequence converges locally uniformly to f_* with $f_*(a) = 0$ and $f'_*(a) = M$. The limit is nonconstant and satisfies $|f_*| \leq 1$; the maximum principle strengthens this to $|f_*| < 1$. Lemma 15.13 gives injectivity. Consequently $f_* \in \mathcal{F}$.

3. An omitted value increases the derivative. Suppose $b \in \mathbb{D} \setminus f_*(U)$. Since $f_*(a) = 0$, we have $r = |b| \in (0, 1)$. The function $\phi_b \circ f_*$ has no zeros. Take a holomorphic square root q on U , and put $d = q(a)$. Then

$$q^2 = \phi_b \circ f_*, \quad d^2 = -b, \quad |d| = \sqrt{r}, \quad |q| < 1.$$

The function q is injective: equality of two values implies equality of their squares, then of the values of f_* , and finally of the points. The function $g_0 = \phi_d \circ q$ is also injective, maps into $\mathbb{D}$ and vanishes at a . Keeping every factor, we compute:

$$\begin{aligned} \phi'_b(0) &= 1 - r^2, & \phi'_d(d) &= \frac{1}{1 - |d|^2} = \frac{1}{1 - r}, \\ 2d q'(a) &= (1 - r^2) f'_*(a), & |g'_0(a)| &= \frac{1 - r^2}{2\sqrt{r}(1 - r)} f'_*(a) = \frac{1 + r}{2\sqrt{r}} M. \\ \frac{1 + r}{2\sqrt{r}} - 1 &= \frac{(1 - \sqrt{r})^2}{2\sqrt{r}} > 0. \end{aligned}$$

A rotation of the codomain makes $g'_0(a)$ real and positive without changing its modulus. This gives $g \in \mathcal{F}$ with $g'(a) > M$, a contradiction. Therefore $f_*(U) = \mathbb{D}$.

4. Uniqueness. For two solutions f_1, f_2 , the composition $h = f_2 \circ f_1^{-1}$ is a holomorphic disk automorphism fixing zero. Schwarz applied to h and h^{-1} gives both $|h(z)| \leq |z|$ and $|z| \leq |h(z)|$. The equality case implies $h(z) = e^{i\theta} z$. But $h'(0) = f'_2(a)/f'_1(a) > 0$, so $e^{i\theta} = 1$ and $f_1 = f_2$. $\square$

15.11.4 An explicit map and two different checks

Proposition 15.5: From a strip to the disk. For $h > 0$, the normalized map of the strip $U_h = \{z : |\operatorname{Im} z| < h\}$ with base point zero is

$$f_h(z) = \tanh \frac{\pi z}{4h}, \quad f_h^{-1}(w) = \frac{2h}{\pi} \operatorname{Log} \frac{1 + w}{1 - w}.$$

Proof. The exponential $v = \exp(\pi z/(2h))$ has argument in $(-\pi/2, \pi/2)$, so it maps into the right half-plane. It is injective on the strip: equality of two exponentials would imply $z_1 - z_2 = 4hki$, whereas the imaginary parts differ by less than $2h$. It is onto the right half-plane because the logarithm with argument in that interval gives $z = (2h/\pi) \operatorname{Log} v \in U_h$. The map $(v-1)/(v+1)$ is a bijection from the right half-plane to the disk, as shown by

$$1 - \left| \frac{v-1}{v+1} \right|^2 = \frac{4 \operatorname{Re} v}{|v+1|^2} > 0, \quad v = \frac{1+w}{1-w}.$$

The composition is the stated hyperbolic tangent; $f_h(0) = 0$ and $f'_h(0) = \pi/(4h) > 0$. Inverting the two maps gives the inverse formula. This proves global bijectivity; it is not an inference from a numerical sample. $\square$

In the supplied code, $h = 1$: the $\tanh$ formula is compared with the exponential–Möbius composition on an interior grid, followed by both round trips with the inverse. A second check uses higher-precision arithmetic. To test the extremal factor independently, take $f_0(z) = z/5$ on the disk and the omitted value $b = 1/4$: the derivative after the square-root construction has modulus $1/4$, compared with the initial $1/5$, since the gain is $5/4$. The JSON file records observed discrepancies and tolerances, not universal error bounds.

15.11.5 Exercises with solutions

Exercise 15.4: Normalization is not cosmetic. Which solutions remain when only $f(a) = 0$ is required?

Solution. Step 4 of the proof gives exactly $e^{i\theta} f_*(z)$, $\theta \in \mathbb{R}$. The positive derivative removes one real parameter. Without fixing $f(a)$ either, composition with any disk automorphism is allowed.

Exercise 15.5: Why exclude the plane. Prove that $\mathbb{C}$ is not biholomorphic to the disk.

Solution. Such a map would be entire and bounded by one. Cauchy’s estimate on a circle of radius R gives $|f'(z)| \leq 1/R$ for every $R > 0$. Letting $R \rightarrow \infty$ gives $f' = 0$: the function is constant, not bijective.

Exercise 15.6: Conditioning of the inverse. Compute the derivative of the strip inverse and explain why points near $w = 1$ require numerical care.

Solution.

$$(f_h^{-1})'(w) = \frac{2h}{\pi} \left(\frac{1}{1+w} + \frac{1}{1-w} \right) = \frac{4h}{\pi(1-w^2)}.$$

Its modulus diverges along the real axis as $w \rightarrow 1^-$. A small absolute error in w may therefore produce a much larger error in its preimage. This does not contradict bijectivity: it concerns numerical sensitivity, not existence of the inverse.

15.12. Assigned exercises

These are *new* exercises, distinct both from the inherited ones (Section 15.10) and from those worked out in Section 15.13. Answers are in the Teacher’s key (Section 15.15).

Exercise 15.7: Cauchy's estimates, polynomial growth, and generalized Liouville.

Goal: use Cauchy's estimates as a tool of *exclusion*, i.e. to show that certain coefficients vanish. *Estimated time:* 40 minutes. *Type:* proof-based/numerical.

- Let f be entire and suppose there exist $A, B \geq 0$ and an integer $k \geq 0$ with $|f(z)| \leq A + B|z|^k$ for every $z \in \mathbb{C}$. Prove, using Theorem 15.7, that f is a polynomial of degree $\leq k$.
- Let f be entire with $|f(z)| \leq 3|z|^{5/2}$ for $|z| \geq 1$. Deduce that f is a polynomial of degree ≤ 2 . (Caution: the exponent is not an integer.)
- Let $f(z) = z^2 + 7$. Compute $M(\rho) = \max_{|z|=\rho} |f|$ for $\rho = 1, 2, 5$ and fill in the table of estimates $M(\rho)/\rho^n$ for $n = 0, 1, 2, 3$, comparing them with the true coefficients $a_0 = 7$, $a_1 = 0$, $a_2 = 1$, $a_3 = 0$. Which value of ρ gives the best estimate for a_2 ? And for a_3 ?
- Why does the hypothesis of (a) not allow one to conclude that the degree is *exactly* k ? Exhibit a counterexample.

Exercise 15.8: Three integrals with residues, each to be checked twice.

Goal: choose the contour, identify the interior poles, compute the residues, and check the result by a real route. *Estimated time:* 45 minutes. *Type:* symbolic and numerical computation.

- Compute $\int_{-\infty}^{+\infty} \frac{dx}{x^4 + 1}$ with the D -shaped contour of Proposition 15.4 (one must determine which fourth roots of -1 lie in the upper half-plane) and check the result with a numerical quadrature.
- Compute $\int_0^{2\pi} \frac{\cos 2\theta}{5 - 4 \cos \theta} d\theta$ with the substitution $z = e^{i\theta}$, recalling that $\cos 2\theta = \frac{1}{2}(z^2 + z^{-2})$, and paying attention to the pole that the new integrand acquires at $z = 0$.
- Compute $\int_{-\infty}^{+\infty} \frac{x^2}{(x^2 + 1)(x^2 + 4)} dx$ with residues and then, independently, with partial fraction decomposition in x^2 .
- In (a) and (c), justify the vanishing of the contribution of the arc with the ML estimate (Lemma 15.2), exhibiting in each case the explicit bound.

Exercise 15.9: Schwarz, Blaschke, and automorphisms: explicit computations.

Goal: handle the Blaschke factor and use Schwarz's lemma after "centering" it with an automorphism. *Estimated time:* 40 minutes. *Type:* proof-based/numerical. Let $a = \frac{1}{2}$ and

$$\varphi_a(z) = \frac{z - \frac{1}{2}}{1 - \frac{1}{2}z}.$$

- Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(\frac{1}{2}) = 0$. Prove that $|f(0)| \leq \frac{1}{2}$. (Hint: apply Lemma 15.9 to $g = f \circ \varphi_{-a}$.)
- Determine the unique automorphism f of $\mathbb{D}$ with $f(\frac{1}{2}) = 0$ and $f'(\frac{1}{2})$ real and positive, and compute $f(0)$ and $f'(0)$.
- Numerically verify that $|\varphi_a(e^{i\theta})| = 1$ for $\theta = 0, 1, 2, 3$ radians, and explain the result with Lemma 15.10(i).
- Let $f : \mathbb{D} \rightarrow \mathbb{C}$ be holomorphic with $|f| \leq 1$, $f(0) = 0$ and $f(\frac{1}{2}) = 0$. Prove that $|f(z)| \leq |z| |\varphi_a(z)|$ for every $z \in \mathbb{D}$, and compute the resulting bound for $|f(\frac{1}{4})|$.

15.13. Worked exercises

Exercise 15.10: How many zeros does $z^5 + 3z + 1$ have in the annulus $1 < |z| < 2$?

Type: counting with Rouché, with numerical check. Determine the number of zeros of the polynomial $P(z) = z^5 + 3z + 1$ in the disk $|z| < 1$, in the disk $|z| < 2$, and hence in the annulus $1 < |z| < 2$.

Solution. Strategy. Rouché's theorem (Theorem 15.19) compares P with a *dominant* term on the circle: on each circle one chooses as f the monomial that on that circle is larger than all the others combined, and counts its zeros, which are immediate.

Step 1 — on the circle $|z| = 1$, $3z$ dominates. Take $f(z) = 3z$ and $g = P$. For $|z| = 1$,

$$|f(z) - g(z)| = |z^5 + 1| \leq |z|^5 + 1 = 2 < 3 = |3z| = |f(z)|.$$

The hypothesis of Theorem 15.19 is satisfied. The function $f(z) = 3z$ has exactly one zero in $|z| < 1$, at $z = 0$, of multiplicity 1. Hence P has exactly 1 zero in $|z| < 1$, counted with multiplicity.

Step 2 — on the circle $|z| = 2$, z^5 dominates. Take $f(z) = z^5$ and $g = P$. For $|z| = 2$,

$$|f(z) - g(z)| = |3z + 1| \leq 3 \cdot 2 + 1 = 7 < 32 = 2^5 = |z^5| = |f(z)|.$$

The function z^5 has a zero of multiplicity 5 at $z = 0$, hence P has exactly 5 zeros in $|z| < 2$.

Step 3 — no zeros on either circle. The two strict inequalities of Step 1 and Step 2 imply, by Step 1 of the proof of Theorem 15.19, that P vanishes neither on $|z| = 1$ nor on $|z| = 2$.

Step 4 — the annulus. The number of zeros in $1 < |z| < 2$ is the difference:

$$5 - 1 = \boxed{4}.$$

Independent check 1: the roots. Computing the five roots of $z^5 + 3z + 1$ by the Durand–Kerner method in double precision gives

$$\begin{aligned} \lambda_1 &= -0.3319890, & |\lambda_1| &= 0.3319890, \\ \lambda_{2,3} &= -0.8390724 \pm 0.9438516i, & |\lambda_{2,3}| &= 1.2628928, \\ \lambda_{4,5} &= 1.0050669 \pm 0.9372592i, & |\lambda_{4,5}| &= 1.3742686. \end{aligned}$$

One root has modulus < 1 and four have modulus between 1 and 2. ✓

Independent check 2: the argument principle. Computing numerically $\frac{1}{2\pi i} \oint_{|z|=r} \frac{P'(z)}{P(z)} dz$ with 200,000 points gives 1.0000000 for $r = 1$ and 5.0000000 for $r = 2$ (imaginary part of order 10^{-17}): these are exactly $Z - P = Z$ of Corollary 15.8, with $P = 0$ because the polynomial has no poles. ✓

Interpretation. Rouché says that “perturbing a little does not change the count”: on the small circle the polynomial is, up to a perturbation smaller than 3, the line $3z$, and inherits its single zero; on the large circle it is, up to a perturbation smaller than 32, the monomial z^5 , and inherits its five. This is the same mechanism as Lemma 12.8 of Chapter 12, where the dominant term determines the winding number.

Typical errors. (1) Using the nonstrict inequality: if on the circle one had $|f - g| = |f|$ at a point, g could vanish there and the count would not make sense. (2) Choosing the wrong monomial for f : on $|z| = 1$ the candidate z^5 does not work, because $|3z + 1|$ can equal $4 > 1 = |z^5|$. (3) Counting roots instead of *multiplicities*: Rouché's count is always with multiplicity.

Extension. With the same method one counts the number of zeros of $z^5 + 3z + 1$ in $|z| < \frac{1}{2}$: on $|z| = \frac{1}{2}$ we have $|z^5 + 1| \leq \frac{1}{32} + 1 = 1.03125$ and $|3z| = 1.5$, hence again a single zero — consistent with $|\lambda_1| = 0.3319890 < \frac{1}{2}$. ✓

Exercise 15.11: A pole of order three, and a trigonometric integral that follows from it. *Type:* symbolic computation with three independent checks. Compute

$$J = \oint_{|z|=1} \frac{e^z}{z^3} dz \text{ and deduce from it the value of } K = \int_0^{2\pi} e^{\cos \theta} \cos(\sin \theta - 2\theta) d\theta.$$

Solution. Strategy. Compute the residue at the only singularity, apply the residue theorem, and then *read* the same integral as a real integral by parametrizing the circle.

Step 1 — the singularity. The function $F(z) = e^z/z^3$ is holomorphic on $\mathbb{C} \setminus \{0\}$. From the expansion $e^z = \sum_{m \geq 0} z^m/m!$ it follows that

$$F(z) = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{2z} + \frac{1}{6} + \frac{z}{24} + \cdots,$$

hence 0 is a pole of order 3 (Definition 15.6) and $\text{Res}(F, 0) = a_{-1} = \frac{1}{2}$.

Step 2 — checking the residue with the formula. By Proposition 15.2(i) with $k = 3$,

$$\text{Res}(F, 0) = \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} [z^3 F(z)] = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} e^z = \frac{e^0}{2} = \frac{1}{2}. \checkmark$$

Step 3 — the residue theorem. With $\Omega = D(0, 2)$, convex, and $\text{Ind}_{\mathbb{T}}(0) = 1$ (Proposition 15.3), Theorem 15.17 gives

$$J = 2\pi i \cdot \frac{1}{2} = \pi i.$$

Step 4 — check with the formula for the derivatives. Setting $f(z) = e^z$, Theorem 15.6(iii) with $w = 0$, $\rho = 1$, $k = 2$ gives

$$f''(0) = \frac{2!}{2\pi i} \oint_{|z|=1} \frac{e^z}{z^3} dz = \frac{J}{\pi i},$$

and $f''(0) = e^0 = 1$: hence $J = \pi i$. $\checkmark$

Step 5 — a real reading. Parametrizing $z = e^{i\theta}$, $dz = ie^{i\theta} d\theta$:

$$J = \int_0^{2\pi} \frac{e^{e^{i\theta}}}{e^{3i\theta}} ie^{i\theta} d\theta = i \int_0^{2\pi} e^{e^{i\theta}} e^{-2i\theta} d\theta.$$

Since $e^{i\theta} = \cos \theta + i \sin \theta$, we have $e^{e^{i\theta}} = e^{\cos \theta} (\cos(\sin \theta) + i \sin(\sin \theta))$, and multiplying by $e^{-2i\theta}$ and using the addition formulas,

$$e^{e^{i\theta}} e^{-2i\theta} = e^{\cos \theta} [\cos(\sin \theta - 2\theta) + i \sin(\sin \theta - 2\theta)].$$

Hence $J = i(K + iK')$ with $K' = \int_0^{2\pi} e^{\cos \theta} \sin(\sin \theta - 2\theta) d\theta$. From $J = \pi i$ it follows that $iK - K' = \pi i$, i.e., equating real and imaginary parts,

$$\boxed{K = \pi = 3.1415927}, \quad K' = 0.$$

Independent numerical check. Composite Simpson quadrature with 400,000 subintervals: $K = 3.1415927$ and $\oint_{|z|=1} e^z z^{-3} dz = 3.1415927 i$ (real part of order 10^{-16}). $\checkmark$ Repeating with e^z/z^{n+1} gives $\int_0^{2\pi} e^{\cos \theta} \cos(\sin \theta - n\theta) d\theta = \frac{2\pi}{n!}$, i.e. 6.2831853 for $n = 0$ and $n = 1$, 3.1415927 for $n = 2$, 1.0471976 for $n = 3$: all verified numerically. $\checkmark$

Interpretation. The identity $\int_0^{2\pi} e^{\cos \theta} \cos(\sin \theta - n\theta) d\theta = 2\pi/n!$ is nothing but the formula for the Taylor coefficients of e^z read on the unit circle: the residue *is* the coefficient a_n multiplied by 2π . It is the same master identity as Lemma 15.5: on the circle only one exponent survives.

Typical errors. (1) Using the simple-pole formula $\lim_{z \rightarrow 0} zF(z)$: the limit is infinite. (2) Forgetting the $\frac{1}{2!}$ in the formula for the pole of order 3, obtaining $J = 2\pi i$. (3) Confusing $\sin(\sin \theta - 2\theta)$ with $\sin(\sin \theta) - \sin(2\theta)$.

Extension. The same computation with e^{tz} in place of e^z gives $\int_0^{2\pi} e^{t \cos \theta} \cos(t \sin \theta - n\theta) d\theta = \frac{2\pi t^n}{n!}$, which for purely imaginary t produces the Bessel functions of the first kind: this is the bridge to Chapter 18.

Exercise 15.12: Where the maximum of $|z^2 - 1|$ lies on the closed disk. *Type:* direct application of the maximum principle, with numerical check. Determine the maximum and minimum of $|z^2 - 1|$ on $\mathbb{D}$ and the points where they are attained, discussing why the minimum principle does not apply.

Solution. Strategy. $f(z) = z^2 - 1$ is entire, hence continuous on $\overline{\mathbb{D}}$ and holomorphic on $\mathbb{D}$: Corollary 15.5 reduces the search for the maximum to the boundary, where the computation is trigonometric.

Step 1 — the maximum lies on the boundary. $\mathbb{D}$ is open, connected, and bounded, f is continuous on $\overline{\mathbb{D}}$ and holomorphic on $\mathbb{D}$, and is not constant: by Corollary 15.5, $\max_{\overline{\mathbb{D}}} |f| = \max_{\mathbb{T}} |f|$.

Step 2 — the computation on the boundary. For $z = e^{i\theta}$,

$$|z^2 - 1| = |e^{2i\theta} - 1| = |e^{i\theta}(e^{i\theta} - e^{-i\theta})| = |2i \sin \theta| = 2|\sin \theta|,$$

having used $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ (Volume II, Chapter 3) and $|e^{i\theta}| = 1$.

Step 3 — maximum and maximizing points. $2|\sin \theta|$ is at most 2, with equality at $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$, i.e. $z = \pm i$. Hence

$$\max_{\overline{\mathbb{D}}} |z^2 - 1| = 2, \quad \text{attained at } z = i \text{ and } z = -i.$$

Direct check: $|i^2 - 1| = |-2| = 2$. ✓

Step 4 — the minimum. f vanishes at $z = \pm 1$, which lie on $\mathbb{T}$: hence $\min_{\overline{\mathbb{D}}} |f| = 0$, attained at ± 1 . Corollary 15.6 does *not* apply, because it requires f never zero; and it is not needed here, because the minimum falls on the boundary regardless. If instead one considered $h(z) = z^2 - 4$, which does not vanish on $\overline{\mathbb{D}}$, Corollary 15.6 would guarantee that the minimum of $|h|$ is on the boundary: $|e^{2i\theta} - 4|$ is minimal for $e^{2i\theta} = 1$, i.e. $z = \pm 1$, where it equals 3; and indeed $|h(0)| = 4 > 3$.

Independent check. A 201×201 grid on the square $[-1, 1]^2$, restricted to $|z| \leq 1$: the maximum of $|z^2 - 1|$ equals 2.0000000; and on the boundary, with 2000 equally spaced points, the maximum is 2.0000000, attained at $\theta = 1.5707963 = \frac{\pi}{2}$. ✓

Interpretation. The maximum of $|z^2 - 1|$ is the maximum distance between z^2 and 1: as z ranges over $\overline{\mathbb{D}}$, z^2 still ranges over $\overline{\mathbb{D}}$ (the map $z \mapsto z^2$ is surjective onto the disk), and the point of the closed disk farthest from 1 is -1 , at distance 2. The preimages of -1 are $\pm i$: the same result is recovered by purely geometric means. ✓

Typical errors. (1) Looking for the stationary points of $|f|$ inside the disk by solving $f'(z) = 2z = 0$: $z = 0$ is a stationary point of f , not a maximum of $|f|$ — and Theorem 15.11 guarantees that an interior maximum cannot exist. (2) Concluding that the minimum is also on the boundary “by symmetry”: it is, but for a different reason (the zeros), not by Corollary 15.6. (3) Writing $|e^{2i\theta} - 1| = e^{2i\theta} - 1$, forgetting the modulus.

Extension. For $f(z) = z^n - 1$ the same computation gives $|e^{in\theta} - 1| = 2|\sin \frac{n\theta}{2}|$, maximum 2 attained at the n points $z = e^{i(2k+1)\pi/n}$, $k = 0, \dots, n-1$: the n -th roots of -1 , i.e. the vertices of the regular polygon “out of phase” relative to that of the n -th roots of unity of Chapter 6.

15.14. Self-assessment quiz

- (Definition) An isolated singularity z_0 of f is a *pole of order k* when, in the Laurent expansion $f(z) = \sum_n a_n(z - z_0)^n$,
A) $f(z_0)$ equals ∞ B) $a_{-k} \neq 0$ and $a_n = 0$ for every $n < -k$ C) infinitely many a_n with $n < 0$ are nonzero D) $a_n = 0$ for every $n < 0$
- (Hypothesis) In Lemma 15.9 the hypothesis $f(0) = 0$ serves to
A) make f injective B) annul the coefficient a_0 , so that $f(z)/z$ extends to a function holomorphic on the whole of $\mathbb{D}$ C) guarantee $|f'(0)| \leq 1$ directly D) make f an automorphism
- (Symbolic computation) $\text{Res}(e^z/z^3, 0)$ equals
A) 1 B) $\frac{1}{2}$ C) $\frac{1}{6}$ D) 0
- (Numerical computation) $\int_0^{2\pi} \frac{d\theta}{5 - 4\cos\theta}$ equals, to seven decimals,
A) 2.0943951 B) 3.1415927 C) 1.0471976 D) 6.2831853
- (Proof step) In the proof of Theorem 15.1, the factor 4^{-n} appearing on the right in the final estimate comes
A) from the ML estimate applied to the starting triangle B) from the fact that at each bisection the diameter and perimeter are halved, and in the product $\text{diam} \cdot L$ the term $2^{-n} \cdot 2^{-n}$ appears C) from the number of sub-triangles, which is four D) from the differentiability of f at z^*
- (Error diagnosis) A student computes $\int_{-\infty}^{+\infty} \frac{dx}{(x^2+1)^2}$ by summing the residues at i and at $-i$ and gets 0. The error is
A) the residue at $-i$ is computed incorrectly B) the D -shaped contour does not enclose $-i$: $\text{Ind}_{\gamma_R}(-i) = 0$ and that residue does not enter C) the integral actually diverges D) the poles are simple, not double
- (Transfer) The number of zeros of $z^5 + 3z + 1$ in the disk $|z| < 1$, counted with multiplicity, is
A) 0 B) 1 C) 2 D) 5
- (Source and metacognition) The *general* form of Cauchy's theorem for simply connected open sets, which this chapter declares it does not prove, is found in
A) Ahlfors, *Complex Analysis*, Ch. 4, §4 “The General Form of Cauchy's Theorem” B) Ahlfors, Ch. 4, §1.4 “Cauchy's Theorem for a Rectangle” C) Rudin, *Real and Complex Analysis*, Ch. 11 “Harmonic Functions” D) Orloff, 18.04, Topic 1 “Complex Algebra and the Complex Plane”

15.15. Teacher's key

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Definition 15.6: the principal part is finite and its lowest term has exponent $-k$. *Distractors:* A confuses the behavior of the modulus with a “value” of the

function, which is undefined at z_0 ; C is the definition of an *essential* singularity; D is the definition of a *removable* singularity. *Diagnosed error*: failing to distinguish the three types from the shape of the principal part. *Source*: Conway, *Functions of One Complex Variable I*, 2nd ed., Ch. V, §1 “Classification of singularities”; here Definition 15.6 and Theorem 15.14. *Prerequisite*: the Laurent expansion (Theorem 15.4). *Remediation*: classify the singularity at 0 of $\frac{\sin z}{z}$, $\frac{e^z}{z^3}$, and $e^{1/z}$ as in Example 15.3.

2) B. *Rationale*: Step 1 of the proof of Lemma 15.9 uses $f(0) = 0$ only to conclude $a_0 = 0$, which allows one to factor out z and define $g(z) = \sum_{k \geq 0} a_{k+1} z^k$ holomorphic on the whole of $\mathbb{D}$; everything else is the maximum principle applied to g . *Distractors*: A is false ($f(z) = z^2$ satisfies the hypotheses and is not injective); C swaps a conclusion for a hypothesis; D is false ($f \equiv 0$ satisfies the hypotheses). *Diagnosed error*: failing to identify the technical role of a hypothesis. *Source*: Conway, Ch. VI, §2 “Schwarz’s Lemma”. *Prerequisite*: the Taylor series (Theorem 15.6). *Remediation*: redo Step 2 with $f(0) = c \neq 0$ and see where it breaks down.

3) B. *Rationale*: from $e^z = \sum_m z^m/m!$ it follows that $e^z/z^3 = z^{-3} + z^{-2} + \frac{1}{2}z^{-1} + \dots$, hence $a_{-1} = \frac{1}{2}$; the same value is obtained with Proposition 15.2(i), $k = 3$: $\frac{1}{2!} \frac{d^2}{dz^2} e^z|_0 = \frac{1}{2}$. *Distractors*: A is a_{-2} ; C is a_0 , i.e. the coefficient of z^0 (which equals $1/3! = 1/6$); D would be the case of a removable singularity. *Diagnosed error*: reading off the wrong coefficient of the series. *Source*: Worked exercise 15.11, Steps 1–2. *Prerequisite*: the exponential series. *Remediation*: compute $\text{Res}(e^z/z^4, 0) = \frac{1}{6}$ and $\text{Res}(e^z/z^2, 0) = 1$.

4) A. *Rationale*: this is the result of Example 15.5, $\frac{2\pi}{3} = 2.0943951$, obtained with residues, with the Weierstrass substitution, and with the inherited example. *Distractors*: B is π , the value of $\int_0^{2\pi} e^{\cos \theta} \cos(\sin \theta - 2\theta) d\theta$; C is $\frac{\pi}{3}$, i.e. $\int_0^\pi$ and not $\int_0^{2\pi}$ — the error of forgetting to double it; D is 2π , the value one would get if the denominator were 1. *Diagnosed error*: forgetting the factor 2 in going from $[0, \pi]$ to $[0, 2\pi]$. *Source*: Example 15.5; Orloff, 18.04, Topic 9 “Definite integrals using the residue theorem”. *Prerequisite*: the substitution $z = e^{i\theta}$. *Remediation*: recompute the integral with $a = \frac{5}{4}$ in the formula of Example 15.1.

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 5–8).

5) B. *Rationale*: in Step 2 of the proof of Theorem 15.1 we have $\text{diam } \Delta_n = d/2^n$ and $L(\partial \Delta_n) = L/2^n$; in Step 5 the ML estimate produces the *product* of the two, i.e. $dL/4^n$, which cancels with the 4^{-n} on the left. *Distractors*: A confuses the starting triangle with those of the sequence; C gives the reason for the 4 in the *choice* of the sub-triangle, not in the final estimate; D is the ingredient that supplies the ε , not the 4^{-n} . *Diagnosed error*: not tracking where each factor in an estimate comes from. *Source*: Theorem 15.1, Steps 2 and 5; Ahlfors, Ch. 4, §1.4. *Prerequisite*: the ML estimate (Lemma 15.2). *Remediation*: redo the estimate with a bisection into nine parts instead of four and see what changes.

6) B. *Rationale*: Theorem 15.17 weighs each residue by the index of the loop about the singularity, and for the D -shaped contour Proposition 15.4 gives index 1 for i and 0 for $-i$; moreover $-i$ does not even belong to the convex set Ω_R (Example 15.4, Step 2). *Distractors*: A is false, the residue at $-i$ equals $-\frac{1}{4i}$ and is computed correctly; C is false, the integral converges absolutely; D is false, the poles are double. *Diagnosed error*: summing all the residues instead of only the *enclosed* ones. *Source*: Example 15.4, “Typical error”; it is the same warning as in Example 15.1. *Prerequisite*: the winding number (Definition 15.8). *Remediation*: compute $\text{Ind}_{\gamma_R}(\pm i)$ by redoing Steps 1–6 of Proposition 15.4.

7) B. *Rationale*: on $|z| = 1$ we have $|z^5 + 1| \leq 2 < 3 = |3z|$, so by Theorem 15.19 the polynomial has as many zeros as $3z$, i.e. one; the numerically computed roots confirm a single modulus < 1 , namely 0.3319890. *Distractors*: A ignores the existence of the negative real root; C is the number of roots with positive real part; D is the total number of roots

in $\mathbb{C}$, all contained in $|z| < 2$. *Diagnosed error:* choosing the wrong dominant term, or confusing the local count with the global one. *Source:* Worked exercise 15.10. *Prerequisite:* Rouché's theorem. *Remediation:* repeat the count on $|z| = \frac{1}{2}$ and on $|z| = 2$.

8) A. *Rationale:* Observation 15.3 states exactly this: the general version is in Ahlfors, Chapter 4, §4, with the paragraphs *Chains and Cycles*, *Simple Connectivity*, *Homology*, *The General Statement of Cauchy's Theorem*, *Proof of Cauchy's Theorem*. *Distractors:* B is the section on Goursat's theorem for the rectangle, which this chapter *does* prove (in the triangle variant); C is Rudin's chapter on harmonic functions, not on Cauchy's theorem (which is in Chapter 10); D is Orloff's first handout, on elementary complex algebra. *Diagnosed error:* citing “the book” without the chapter, or attributing a result to the neighboring section. *Source:* see the “Sources” section. *Prerequisite:* knowing how to read a table of contents by section. *Remediation:* compile a table “result / author / chapter / proved here yes or no” for the ten main theorems of the chapter.

BEST PRACTICE

Assigned exercise 15.7 — answers.

(a) For every $\rho > 0$, Cauchy's estimates (Theorem 15.7) with $w = 0$ give $|a_n| \leq M(\rho)/\rho^n \leq (A + B\rho^k)/\rho^n$. For $n > k$ the right-hand side tends to 0 as $\rho \rightarrow +\infty$ (numerator of order ρ^k , denominator of order ρ^n with $n \geq k + 1$), while a_n does not depend on ρ : hence $a_n = 0$ for every $n > k$ and $f(z) = \sum_{n=0}^k a_n z^n$ is a polynomial of degree $\leq k$. For $k = 0$ one recovers Theorem 15.8.

(b) On $|z| = \rho \geq 1$, $M(\rho) \leq 3\rho^{5/2}$, hence $|a_n| \leq 3\rho^{5/2-n}$. For $n \geq 3$ the exponent $5/2 - n \leq -\frac{1}{2} < 0$ and the right-hand side tends to 0: so $a_n = 0$ for $n \geq 3$ and f is a polynomial of degree ≤ 2 . The nonintegral exponent creates no difficulty: only the sign of $5/2 - n$ matters.

(c) For $|z| = \rho$, $|z^2 + 7| \leq \rho^2 + 7$, with equality at $z = \pm\rho$ (where $z^2 = \rho^2 > 0$ has the same sign as 7): hence $M(\rho) = \rho^2 + 7$.

	$n = 0$	$n = 1$	$n = 2$	$n = 3$
$\rho = 1$ ($M = 8$)	8	8	8	8
$\rho = 2$ ($M = 11$)	11	5.5	2.75	1.375
$\rho = 5$ ($M = 32$)	32	6.4	1.28	0.256
$ a_n $	7	0	1	0

All estimates are verified. ✓ For a_2 the best of the three is $\rho = 5$ (1.28, versus a true value of 1); for a_3 it is again $\rho = 5$ (0.256), and letting ρ grow, the estimate $(\rho^2 + 7)/\rho^3 \rightarrow 0$ forces $a_3 = 0$: this is exactly the argument of part (a) with $k = 2$.

(d) Because the bound is an *upper* bound on the degree: the constant function $f \equiv 1$ satisfies $|f(z)| \leq 1 + |z|^7$ with $k = 7$ and has degree 0. The exact degree is read off from the first nonzero coefficient, not from the growth rate.

BEST PRACTICE

Assigned exercise 15.8 — answers.

(a) $\int_{-\infty}^{+\infty} \frac{dx}{x^4 + 1} = \frac{\pi}{\sqrt{2}} = 2.2214415$. The roots of $z^4 = -1$ are $e^{i\pi/4}, e^{3i\pi/4}, e^{5i\pi/4}, e^{7i\pi/4}$; the first two lie in the upper half-plane. They are simple poles and, by Proposition 15.2(iii) with $q(z) = z^4 + 1$, $q'(z) = 4z^3$,

$$\operatorname{Res}\left(\frac{1}{z^4 + 1}, z_j\right) = \frac{1}{4z_j^3} = \frac{z_j}{4z_j^4} = \frac{z_j}{4 \cdot (-1)} = -\frac{z_j}{4}.$$

Summing: $-\frac{1}{4}(e^{i\pi/4} + e^{3i\pi/4}) = -\frac{1}{4}\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right) = -\frac{\sqrt{2}}{4}i$, and $2\pi i \cdot (-\frac{\sqrt{2}}{4}i) = \frac{2\pi\sqrt{2}}{4} = \frac{\pi}{\sqrt{2}}$. Numerical check: Simpson quadrature on $[-3000, 3000]$ with 600,000 subintervals, 2.2214415. ✓

(b) $\int_0^{2\pi} \frac{\cos \theta}{5 - 4 \cos \theta} d\theta = \frac{\pi}{6} = 0.5235988$. With $z = e^{i\theta}$ and the computations of Step 1

of Example 15.5, the integrand becomes $i \frac{z^4 + 1}{2z^2(2z - 1)(z - 2)}$, which has on $\mathbb{T}$ the poles $z = 0$ (order 2) and $z = \frac{1}{2}$ (simple). Setting $H(z) = \frac{z^4 + 1}{2z^2(2z - 1)(z - 2)}$, we get

$\text{Res}(H, \frac{1}{2}) = \frac{z^4 + 1}{4z^2(z - 2)} \Big|_{z=1/2} = \frac{17/16}{-3/2} = -\frac{17}{24}$ and $\text{Res}(H, 0) = \frac{d}{dz} \frac{z^4 + 1}{4z^2 - 10z + 4} \Big|_{z=0} = \frac{0 \cdot 4 - 1 \cdot (-10)}{16} = \frac{5}{8}$; the sum is $\frac{5}{8} - \frac{17}{24} = -\frac{1}{12}$, and multiplying by $i \cdot 2\pi i$ gives

$\frac{2\pi}{12} = \frac{\pi}{6}$. Numerical check of the two residues with a small circle of radius 0.1 about 0: 0.6250000, and with the limit at $\frac{1}{2}$: $-0.7083333 = -\frac{17}{24}$. ✓ Independent check with the formula $\int_0^{2\pi} \frac{\cos n\theta}{1 - 2a \cos \theta + a^2} d\theta = \frac{2\pi a^n}{1 - a^2}$ ($|a| < 1$): here $5 - 4 \cos \theta = 4(\frac{5}{4} - \cos \theta)$ and $\frac{5}{4} - \cos \theta = 1 - 2 \cdot \frac{1}{2} \cos \theta + \frac{1}{4}$, i.e. $a = \frac{1}{2}$, whence $\frac{1}{4} \cdot \frac{2\pi(1/2)^2}{1 - 1/4} = \frac{1}{4} \cdot \frac{2\pi}{3} = \frac{\pi}{6}$. Numerical check: 0.5235988. ✓

(c) $\int_{-\infty}^{+\infty} \frac{x^2 dx}{(x^2 + 1)(x^2 + 4)} = \frac{\pi}{3} = 1.0471976$. Poles in the upper half-plane: i and $2i$, both simple, with

$$\text{Res}(f, i) = \frac{i^2}{(2i)(i^2 + 4)} = \frac{-1}{6i} = \frac{i}{6}, \quad \text{Res}(f, 2i) = \frac{(2i)^2}{((2i)^2 + 1)(4i)} = \frac{-4}{-12i} = -\frac{i}{3},$$

sum $-\frac{i}{6}$, and $2\pi i \cdot (-\frac{i}{6}) = \frac{\pi}{3}$. By partial fractions in x^2 : $\frac{x^2}{(x^2+1)(x^2+4)} = \frac{-1/3}{x^2+1} + \frac{4/3}{x^2+4}$ (from $x^2 = A(x^2 + 4) + B(x^2 + 1)$ with $A + B = 1$ and $4A + B = 0$), hence the integral equals $-\frac{1}{3}\pi + \frac{4}{3} \cdot \frac{\pi}{2} = -\frac{\pi}{3} + \frac{2\pi}{3} = \frac{\pi}{3}$. ✓ Numerical check on $[-6000, 6000]$: 1.0468642; adding the estimated tail $2 \int_{6000}^{\infty} x^{-2} dx = \frac{2}{6000} = 0.0003333$ gives 1.0471975. ✓

(d) In (a): $|x^4 + 1| \geq R^4 - 1$ on $|z| = R > 1$, arc length πR , hence $|\int_{\alpha_R}| \leq \frac{\pi R}{R^4 - 1} \rightarrow 0$. In (c): $|z^2| = R^2$ and $|(z^2 + 1)(z^2 + 4)| \geq (R^2 - 1)(R^2 - 4)$ for $R > 2$, hence $|\int_{\alpha_R}| \leq \frac{\pi R^3}{(R^2 - 1)(R^2 - 4)} \rightarrow 0$.

BEST PRACTICE

Assigned exercise 15.9 — answers.

(a) Let $g = f \circ \varphi_{-a}$ with $a = \frac{1}{2}$. Then $g : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic (composition of holomorphic functions, and $\varphi_{-a}(\mathbb{D}) = \mathbb{D}$ by Lemma 15.10(ii)–(iii)) and $g(0) = f(\varphi_{-a}(0)) = f(\frac{1}{2}) = 0$. By Lemma 15.9, $|g(w)| \leq |w|$. Since $f = g \circ \varphi_a$, we get $|f(0)| = |g(\varphi_a(0))| = |g(-\frac{1}{2})| \leq \frac{1}{2}$.

(b) By Theorem 15.12, $f = e^{i\theta} \varphi_b$ with $b = f^{-1}(0) = \frac{1}{2}$. By Lemma 15.10(iv), $f'(\frac{1}{2}) = e^{i\theta} \varphi'_a(\frac{1}{2}) = e^{i\theta} \frac{1}{1 - \frac{1}{4}} = \frac{4}{3} e^{i\theta}$, which is real and positive if and only if $e^{i\theta} = 1$. Hence $f = \varphi_{1/2}$, with

$$f(0) = -\frac{1}{2}, \quad f'(0) = 1 - |a|^2 = \frac{3}{4} = 0.75, \quad f'(\frac{1}{2}) = \frac{4}{3} = 1.3333333.$$

(c) $|\varphi_a(e^{i\theta})| = 1$ for $\theta = 0, 1, 2, 3$: the numerical values are 1.0000000 up to $2 \cdot 10^{-16}$. ✓ The reason is Lemma 15.10(i): $1 - |\varphi_a(z)|^2$ has the same sign as $1 - |z|^2$, which is zero on the unit circle.

(d) The function $h(z) = \frac{f(z)}{z \varphi_a(z)}$ extends holomorphically to $\mathbb{D}$: f vanishes at 0 and at a , so by Theorem 15.9, $f(z) = z f_1(z)$ with f_1 holomorphic, and $f_1(a) = 0$ because $a \neq 0$, so $f_1(z) = (z - a) f_2(z)$ with f_2 holomorphic; since $\varphi_a(z) = (z - a)/(1 - \bar{a}z)$ and $1 - \bar{a}z$ does not vanish on $\mathbb{D}$, we have $h = f_2 \cdot (1 - \bar{a}z)$, holomorphic. For $|z| = r < 1$, $|z \varphi_a(z)| \rightarrow 1$ as

$r \rightarrow 1^-$ (both factors have modulus tending to 1), hence, as in Steps 3–4 of Lemma 15.9, $|h| \leq 1$ on $\mathbb{D}$ and therefore $|f(z)| \leq |z| |\varphi_a(z)|$. For $z = \frac{1}{4}$: $\varphi_a(\frac{1}{4}) = \frac{\frac{1}{4} - \frac{1}{2}}{1 - \frac{1}{8}} = \frac{-\frac{1}{4}}{\frac{7}{8}} = -\frac{2}{7}$, hence

$$\left|f\left(\frac{1}{4}\right)\right| \leq \frac{1}{4} \cdot \frac{2}{7} = \frac{1}{14} = 0.0714286.$$

15.16. Chapter formulary

From the triangle to the disk.

$$\oint_{\partial\Delta} f = 0 \quad (\text{Goursat}); \quad F(z) = \int_{[a,z]} f \Rightarrow F' = f \quad (\Omega \text{ convex}); \quad \oint_{\gamma} f = 0.$$

$$\left| \int_{\gamma} g dz \right| \leq \left(\max_{\gamma^*} |g| \right) L(\gamma) \quad (\text{ML estimate}); \quad \int_{\gamma} F' dz = F(\gamma(b)) - F(\gamma(a)).$$

Cauchy–Riemann equations: $u_x = v_y$, $u_y = -v_x$, $f' = u_x + iv_x$.

Cauchy's formula, derivatives, estimates.

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0|=r} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad f^{(k)}(w) = \frac{k!}{2\pi i} \oint_{|\zeta - w|=\rho} \frac{f(\zeta)}{(\zeta - w)^{k+1}} d\zeta.$$

$$f(z) = \sum_{n \geq 0} a_n (z - w)^n, \quad a_n = \frac{f^{(n)}(w)}{n!}; \quad |a_n| \leq \frac{M(\rho)}{\rho^n}, \quad |f^{(n)}(w)| \leq \frac{n! M(\rho)}{\rho^n}.$$

Liouville: f entire and bounded $\Rightarrow f$ constant. Fundamental theorem of algebra: $\deg P = n \geq 1 \Rightarrow P$ has n roots in $\mathbb{C}$, all in $|z| \leq 1 + \frac{1}{|c_n|} \sum_{k < n} |c_k|$.

Identity, maximum modulus, Schwarz, automorphisms.

$$f(w) = \frac{1}{2\pi} \int_0^{2\pi} f(w + \rho e^{i\theta}) d\theta \quad (\text{mean value}); \quad \max_{\Omega} |f| = \max_{\partial\Omega} |f|.$$

Isolated zeros: $f(z) = (z - w)^m g(z)$, $g(w) \neq 0$. Schwarz: $f : \mathbb{D} \rightarrow \mathbb{D}$, $f(0) = 0 \Rightarrow |f(z)| \leq |z|$, $|f'(0)| \leq 1$; equality $\Rightarrow f(z) = \lambda z$, $|\lambda| = 1$.

$$\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}, \quad 1 - |\varphi_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2}, \quad \varphi_a^{-1} = \varphi_{-a},$$

$$\varphi'_a(z) = \frac{1 - |a|^2}{(1 - \bar{a}z)^2}, \quad \varphi'_a(0) = 1 - |a|^2, \quad \varphi'_a(a) = \frac{1}{1 - |a|^2};$$

$$\text{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z} : \theta \in \mathbb{R}, a \in \mathbb{D} \right\}.$$

Laurent, residues, index, argument, Rouché.

$$f(z) = \sum_{n \in \mathbb{Z}} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{|\zeta - z_0|=\rho} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta, \quad \text{Res}(f, z_0) = a_{-1}.$$

Pole of order k : $\text{Res}(f, \xi) = \frac{1}{(k-1)!} \lim_{z \rightarrow \xi} \frac{d^{k-1}}{dz^{k-1}} [(z-\xi)^k f(z)]$; simple pole p/q : $\text{Res} = \frac{p(\xi)}{q'(\xi)}$.

$$\text{Ind}_\gamma(z_0) = \frac{1}{2\pi i} \oint_\gamma \frac{dz}{z - z_0} \in \mathbb{Z}; \quad \frac{1}{2\pi i} \oint_\gamma f = \sum_j \text{Ind}_\gamma(\xi_j) \text{Res}(f, \xi_j).$$

$$\frac{1}{2\pi i} \oint_{|z-z_0|=r} \frac{f'}{f} dz = Z - P; \quad |f - g| < |f| \text{ on } |z - z_0| = r \implies Z_f = Z_g \text{ inside.}$$

The three integrals, and the numerical values of the chapter.

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^2+1)^2} = \frac{\pi}{2} = 1.5707963, \quad \int_0^{2\pi} \frac{d\theta}{5-4\cos\theta} = \frac{2\pi}{3} = 2.0943951,$$

$$\int_{-\infty}^{+\infty} \frac{\cos x}{x^2+1} dx = \frac{\pi}{e} = 1.1557273, \quad \int_0^{2\pi} \frac{d\theta}{a+\cos\theta} = \frac{2\pi}{\sqrt{a^2-1}} \quad (a > 1).$$

$$\int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta - n\theta) d\theta = \frac{2\pi}{n!}; \quad \int_{-\infty}^{+\infty} \frac{dx}{x^4+1} = \frac{\pi}{\sqrt{2}} = 2.2214415.$$

Sources and bibliography

- [1] L. V. Ahlfors, *Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable*, 3rd edition, International Series in Pure and Applied Mathematics, McGraw-Hill, New York, 1979. Chapter 4 “Complex Integration”: §1.4 “Cauchy’s Theorem for a Rectangle” and §1.5 “Cauchy’s Theorem in a Disk” (source of Theorem 15.1 and of Corollary 15.1); §2.1 “The Index of a Point with Respect to a Closed Curve”, §2.2 “The Integral Formula”, §2.3 “Higher Derivatives” (source of Definition 15.8 and of Theorem 15.6); §3 “Local Properties of Analytical Functions” — “Removable Singularities. Taylor’s Theorem”, “Zeros and Poles”, “The Maximum Principle” (source of Theorems 15.13, 15.14 and 15.11); §4 “The General Form of Cauchy’s Theorem” — “Chains and Cycles”, “Simple Connectivity”, “Homology”, “Proof of Cauchy’s Theorem”: this is the *imported* part, declared in Observation 15.3; §5 “The Calculus of Residues” — “The Residue Theorem”, “The Argument Principle”, “Evaluation of Definite Integrals”. Chapter 5, §1.3 “The Laurent Series”. Table of contents verified on the SearchWorks catalog record of Stanford University Libraries, record 1003741 — verified on 6 September 2026.
- [2] W. Rudin, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, New York, 1987. Chapter 10 “Elementary Properties of Holomorphic Functions” (from p. 198): “Complex differentiation”, “Integration over paths”, “The local Cauchy theorem”, “The power series representation”, “The open mapping theorem”, “The global Cauchy theorem”, “The calculus of residues”. This is the alternative source for Cauchy’s theorem in homological form (Dixon’s proof) and for the open mapping theorem, which is not proved here and is cited only in Definition 15.5. Chapter 11 “Harmonic Functions” (from p. 222): “The Cauchy–Riemann equations”, “The Poisson integral”, “The mean value property”, for the continuation in Chapter 16 — verified on 6 September 2026.
- [3] J. B. Conway, *Functions of One Complex Variable I*, 2nd edition, Graduate Texts in Mathematics 11, Springer-Verlag, New York, 1978 (ISBN 0-387-90328-3, 317 pages). Chapter IV “Complex Integration” (from p. 58); Chapter V “Singularities” (from p. 103), §1 “Classification of singularities”, for Definition 15.6 and Theorem 15.15; Chapter VI

“The Maximum Modulus Theorem” (from p. 128), §1 “The Maximum Principle” and §2 “Schwarz’s Lemma”, which is the source of Lemma 15.9 with the equality case and of the classification of automorphisms in Theorem 15.12; Chapter XII “The Range of an Analytic Function” for Picard’s theorem, given here as a panorama (Observation 15.18). Bibliographic data and chapter pagination verified on the Google Books record of the volume (identifier 9LtfZr1snG0C) — verified on 6 September 2026.

- [4] J. Orloff, *18.04 Complex Variables with Applications*, Massachusetts Institute of Technology, MIT OpenCourseWare, spring 2018 (undergraduate-level course; Creative Commons BY-NC-SA license). Lecture notes by topic: Topic 3 “Cauchy’s Theorem”, Topic 4 “Cauchy Integral Formula”, Topic 7 “Taylor and Laurent Series”, Topic 8 “Residue Theorem”, Topic 9 “Definite integrals using the residue theorem” (source of the method of Section 15.9), Topic 11 “Argument Principle” (source of Theorem 15.18). Materials downloadable from ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/ — verified on 6 September 2026.
- [5] For the results of the treatise recalled with \ref: Möbius transformations and generalized circles are proved in Chapter 6; the winding number, its analytic form, and the degree of a map of the circle in Chapter 12; line integrals and the parametrization of curves in Chapter 2. The core of Section 15.3 and Section 15.10 come from Volume II, Chapter 3, §12, and are reproduced in full according to the Executive Plan, §3.4, row 15.

CHAPTER 16

Harmonic functions and the Poisson kernel in the disc

A function of two real variables whose sum of pure second derivatives vanishes is, on a disc, the real part of a holomorphic function; at every point it equals the mean of its own values on every circle surrounding it; it cannot have an interior maximum unless it is constant; and it is determined throughout the disc by the values it takes on the boundary alone, through an integral with an explicit kernel. These four statements are the same fact seen from four sides, and the circle is the object that holds them together: it is on circles that the mean is taken, it is on the unit circle that the boundary datum lives, and it is the geometry of the circle — the power of a point with respect to it — that produces the kernel

$$P(z, e^{it}) = \frac{1 - |z|^2}{|e^{it} - z|^2},$$

which bears the name of Poisson.

The chapter follows this chain from beginning to end, without skipping steps. Laplace's equation is defined (Definition 16.1) and it is proved that on a disc every harmonic function is the real part of a holomorphic function (Theorem 16.1), constructing its harmonic conjugate explicitly with the tools of Chapter 15; from this follow infinite differentiability (Corollary 16.1) and the mean value property on circles and on discs (Theorem 16.2 and Corollary 16.3). The maximum principle is proved in its *strong* form, and not only for harmonic functions but for all continuous functions that enjoy the mean value property (Theorem 16.3): it is this generality that will, in the end, allow the argument to be reversed. The uniqueness of the solution of the Dirichlet problem follows at once (Corollary 16.7).

The second half is the Poisson kernel: definition as a series (Definition 16.4), closed form (Lemma 16.4), positivity (Lemma 16.6), normalization computed by two independent routes (Lemma 16.8) and concentration at the boundary (Lemma 16.9): these are exactly the three properties of an approximate identity (Proposition 16.4). With them it is proved that the Poisson integral of a continuous datum is harmonic (Theorem 16.4) and tends to the datum at every point of continuity (Theorem 16.5, with the ε - δ estimate written out in full): the Dirichlet problem in the disc has one and only one solution (Theorem 16.6). From the same formula follow Harnack's inequality (Theorem 16.8), Liouville's theorem for bounded harmonic functions (Theorem 16.9), the gradient estimate (Proposition 16.7), the Schwarz reflection principle (Theorem 16.10) and the Fourier series expansion of the solution (Theorem 16.11), which is the bridge to Chapter 17.

The chapter is new: the Executive Plan (§3.4, row 16) assigns it no material inherited from Volume II, and nothing pre-existing has been removed — the Phase 3 placeholder announced exactly the contents that are developed here.

INFO

Level and prerequisites. Level **C** (undergraduate), with one passage at level **D** (Harnack's inequality on compact sets, Corollary 16.11, and Harnack's principle for monotone

sequences, Proposition 16.6).

Internal prerequisites. Complex derivative, the Cauchy–Riemann equations, existence of a primitive on a convex open set, Cauchy’s integral formula, analyticity and the mean value property for holomorphic functions, the maximum modulus principle, Liouville’s theorem: all in Chapter 15 (Lemma 15.1, Theorem 15.2, Theorem 15.3, Theorem 15.6, Lemma 15.7, Theorem 15.11, Theorem 15.8). Polar form, the complex exponential and generalized circles: Chapter 6. The invariant measure $d\theta/2\pi$ on the circle and the characters $z \mapsto z^n$: Chapter 13.

External prerequisites, assumed and not reproved:

- differential calculus in two real variables: differentiability of a function of class C^1 , Schwarz’s theorem on the equality of mixed partial derivatives for functions of class C^2 , the change-of-variables formula in polar coordinates;
- the Riemann integral on a compact interval: linearity, monotonicity, the inequality $|\int g| \leq \int |g|$, integrability of the product of a continuous function and a bounded, integrable function, and the fact that the integral of a nonnegative continuous function vanishes only if the function is identically zero;
- interchange of series and integral under uniform convergence on a compact set (the Weierstrass criterion);
- elementary topology of the plane: compactness of closed bounded sets, path-connectedness of connected open sets, uniform continuity on compact sets.

Everything else is proved here.

What you learn. How to recognize a harmonic function and construct its conjugate on a disc; how to pass from the mean value property to the maximum principle, and from this to uniqueness for Dirichlet; how to handle the Poisson kernel as an approximate identity; how to solve the Dirichlet problem in the disc with an integral and read off its Fourier expansion; how to estimate a positive harmonic function with Harnack and deduce Liouville from it.

Warning. Conventions of the chapter.

- $D(w, R) = \{z \in \mathbb{C} : |z - w| < R\}$ is the *open* disc and $\overline{D(w, R)}$ the closed one; $\mathbb{D} = D(0, 1)$ is the unit disc and $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ the unit circle. A **domain** is a nonempty connected open set.
- $\mathbb{C}$ and $\mathbb{R}^2$ are identified via $z = x + iy \leftrightarrow (x, y)$; a function “on an open set $\Omega \subseteq \mathbb{C}$ ” is thus a function of the two real variables x, y .
- The harmonic functions of this chapter are *real*-valued, unless stated otherwise; the complex-valued case is obtained by applying the results to the real part and the imaginary part.
- A function f defined on $\mathbb{T}$ is identified with the 2π -periodic function $t \mapsto f(e^{it})$ on $\mathbb{R}$; we write $f(t)$ and $f(e^{it})$ interchangeably. The *distance on the circle* between t and s is $d(t, s) = \min_{k \in \mathbb{Z}} |t - s - 2k\pi| \in [0, \pi]$.
- Every integral over $\mathbb{T}$ is normalized: $\frac{1}{2\pi} \int_{-\pi}^{\pi} \cdots dt$. By translation invariance of the measure dt and the 2π -periodicity of the integrands, the interval $[-\pi, \pi]$ can always be replaced by $[0, 2\pi]$ or by $[a, a + 2\pi]$.
- $u_x, u_y, u_{xx}, u_{xy}, u_{yy}$ denote the partial derivatives; $\nabla u = (u_x, u_y)$.
- Numerical values are rounded to the seventh decimal digit and recomputed twice by independent routes.

16.1. Laplace’s equation and harmonic functions

Definition 16.1: Laplacian, Laplace equation, harmonic function. Let $\Omega \subseteq \mathbb{C}$ be open and $u : \Omega \rightarrow \mathbb{R}$ of class C^2 . The **Laplacian** of u is the function

$$\Delta u = u_{xx} + u_{yy}.$$

The equation $\Delta u = 0$ is called **Laplace's equation**; a function $u \in C^2(\Omega)$ satisfying it at every point of Ω is said to be **harmonic** on Ω . A complex-valued function is said to be harmonic when its real part and its imaginary part are.

Remark 16.1: Why only C^2 is required. The definition requires only two continuous derivatives, which is the minimum needed for the expression $u_{xx} + u_{yy}$ to make sense and for Schwarz's theorem $u_{xy} = u_{yx}$ to hold. Corollary 16.1 will show that the requirement is self-strengthening: a harmonic function is automatically of class C^∞ . Requiring C^2 and obtaining C^∞ is the feature that harmonic functions share with holomorphic functions, for which a single complex derivative produces infinitely many (Theorem 15.6). Neither fact has an analogue for real functions of one variable.

Proposition 16.1: Linear structure and invariances. Let $\Omega \subseteq \mathbb{C}$ be open.

- (i) If u, v are harmonic on Ω and $\alpha, \beta \in \mathbb{R}$, then $\alpha u + \beta v$ is harmonic on Ω .
- (ii) If u is harmonic on Ω and $w \in \mathbb{C}$, the translate $u_w(z) = u(z - w)$ is harmonic on $\Omega + w$.
- (iii) If u is harmonic on Ω and $\lambda > 0$, the dilate $u_\lambda(z) = u(\lambda z)$ is harmonic on $\lambda^{-1}\Omega$, and $\Delta u_\lambda = \lambda^2(\Delta u)_\lambda$.
- (iv) If u is harmonic on Ω and ρ is a rotation $z \mapsto e^{i\alpha}z$, then $u \circ \rho$ is harmonic on $\rho^{-1}\Omega$.

Proof. **Step 1 — (i).** Partial differentiation is linear, so $\Delta(\alpha u + \beta v) = \alpha \Delta u + \beta \Delta v = 0$; and $\alpha u + \beta v$ is of class C^2 because u and v are.

Step 2 — (ii). Writing $w = a + ib$, we have $u_w(x, y) = u(x - a, y - b)$; by the chain rule for composition with a translation, $(u_w)_{xx}(x, y) = u_{xx}(x - a, y - b)$ and similarly for yy . Hence $\Delta u_w(z) = (\Delta u)(z - w) = 0$.

Step 3 — (iii). With $u_\lambda(x, y) = u(\lambda x, \lambda y)$ the chain rule gives $(u_\lambda)_x(x, y) = \lambda u_x(\lambda x, \lambda y)$ and, differentiating again, $(u_\lambda)_{xx}(x, y) = \lambda^2 u_{xx}(\lambda x, \lambda y)$; likewise for yy . Adding, $\Delta u_\lambda(z) = \lambda^2(\Delta u)(\lambda z)$, which vanishes if $\Delta u = 0$.

Step 4 — (iv). Let $\rho(x, y) = (x \cos \alpha - y \sin \alpha, x \sin \alpha + y \cos \alpha)$ and $v = u \circ \rho$. Write $c = \cos \alpha$, $s = \sin \alpha$. Denoting by $X = cx - sy$ and $Y = sx + cy$ the two components of ρ , the chain rule gives

$$v_x = c u_X + s u_Y, \quad v_y = -s u_X + c u_Y,$$

and differentiating a second time, again by the chain rule,

$$v_{xx} = c^2 u_{XX} + 2cs u_{XY} + s^2 u_{YY}, \quad v_{yy} = s^2 u_{XX} - 2cs u_{XY} + c^2 u_{YY}.$$

Adding, the mixed terms cancel and what remains is $v_{xx} + v_{yy} = (c^2 + s^2)(u_{XX} + u_{YY}) = \Delta u \circ \rho = 0$. We have used the equality $u_{XY} = u_{YX}$, valid because $u \in C^2$ (Schwarz's theorem). $\square$

Lemma 16.1: The Laplacian in polar coordinates. Let $\Omega \subseteq \mathbb{C}$ be open and $u \in C^2(\Omega)$. Setting, for $r > 0$ and $\theta \in \mathbb{R}$ such that $re^{i\theta} \in \Omega$,

$$\tilde{u}(r, \theta) = u(r \cos \theta, r \sin \theta),$$

we have

$$(\Delta u)(re^{i\theta}) = \tilde{u}_{rr} + \frac{1}{r} \tilde{u}_r + \frac{1}{r^2} \tilde{u}_{\theta\theta}.$$

Proof. Step 1 — first derivatives. With $x = r \cos \theta$ and $y = r \sin \theta$, the chain rule gives

$$\tilde{u}_r = u_x \cos \theta + u_y \sin \theta, \quad \tilde{u}_\theta = -u_x r \sin \theta + u_y r \cos \theta,$$

where u_x, u_y are evaluated at $(r \cos \theta, r \sin \theta)$.

Step 2 — second derivative in r . Differentiating $\tilde{u}_r$ with respect to r and applying the chain rule again to u_x and u_y (whose derivatives with respect to r are $u_{xx} \cos \theta + u_{xy} \sin \theta$ and $u_{yx} \cos \theta + u_{yy} \sin \theta$):

$$\tilde{u}_{rr} = u_{xx} \cos^2 \theta + 2u_{xy} \cos \theta \sin \theta + u_{yy} \sin^2 \theta,$$

having used $u_{xy} = u_{yx}$ ($u \in C^2$).

Step 3 — second derivative in θ . Differentiating $\tilde{u}_\theta$ with respect to θ : the two terms must be differentiated as products, and the derivatives of u_x, u_y with respect to θ are $-u_{xx}r \sin \theta + u_{xy}r \cos \theta$ and $-u_{yx}r \sin \theta + u_{yy}r \cos \theta$. One obtains

$$\tilde{u}_{\theta\theta} = -u_x r \cos \theta - u_y r \sin \theta + r^2(u_{xx} \sin^2 \theta - 2u_{xy} \sin \theta \cos \theta + u_{yy} \cos^2 \theta).$$

Step 4 — summing. Adding the three contributions with the coefficients of the statement:

$$\tilde{u}_{rr} + \frac{1}{r^2} \tilde{u}_{\theta\theta} = u_{xx}(\cos^2 \theta + \sin^2 \theta) + u_{yy}(\sin^2 \theta + \cos^2 \theta) - \frac{1}{r}(u_x \cos \theta + u_y \sin \theta),$$

because the terms in u_{xy} appear with coefficients $2 \cos \theta \sin \theta$ and $-2 \sin \theta \cos \theta$ and cancel. The last term is $-\frac{1}{r} \tilde{u}_r$ by Step 1, hence $\tilde{u}_{rr} + \frac{1}{r} \tilde{u}_r + \frac{1}{r^2} \tilde{u}_{\theta\theta} = u_{xx} + u_{yy}$. $\square$

Example 16.1: Elementary harmonic functions. (a) The constant functions, x , y , xy and $x^2 - y^2$ are harmonic on all of $\mathbb{C}$: for $x^2 - y^2$ one has $u_{xx} = 2$, $u_{yy} = -2$, sum zero; for xy all the pure second derivatives vanish.

(b) For every integer $n \geq 1$ the functions $r^n \cos n\theta$ and $r^n \sin n\theta$ are harmonic on $\mathbb{C}$ (where, at 0, they coincide with the polynomials $\operatorname{Re} z^n$ and $\operatorname{Im} z^n$, hence are C^∞). With Lemma 16.1 and $\tilde{u} = r^n \cos n\theta$:

$$\tilde{u}_{rr} + \frac{1}{r} \tilde{u}_r + \frac{1}{r^2} \tilde{u}_{\theta\theta} = \left(n(n-1)r^{n-2} + \frac{nr^{n-1}}{r} - \frac{n^2 r^n}{r^2} \right) \cos n\theta = r^{n-2}(n^2 - n + n - n^2) \cos n\theta = 0.$$

For $n = 1$ and $n = 2$ one recovers x , y , $x^2 - y^2$ and $2xy$. $\checkmark$

(c) The function $u(z) = \log |z| = \log r$ is harmonic on $\mathbb{C} \setminus \{0\}$: indeed $\tilde{u}_r = 1/r$, $\tilde{u}_{rr} = -1/r^2$, $\tilde{u}_{\theta\theta} = 0$, and $-\frac{1}{r^2} + \frac{1}{r} \cdot \frac{1}{r} = 0$. $\checkmark$ It does not extend harmonically to 0, where it diverges: this will be the key counterexample of Example 16.2.

(d) $u(x, y) = e^x \cos y$ is harmonic on $\mathbb{C}$: $u_{xx} = e^x \cos y$ and $u_{yy} = -e^x \cos y$. It is the real part of e^z . $\checkmark$

16.1.1 Harmonic and holomorphic: the harmonic conjugate

Proposition 16.2: The real part and the imaginary part of a holomorphic function are harmonic. Let $f = u + iv$ be holomorphic on an open set $\Omega \subseteq \mathbb{C}$, with $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. Then u and v are of class C^∞ and harmonic on Ω .

Proof. Step 1 — regularity. By Theorem 15.6 a holomorphic function is infinitely differentiable in the complex sense and every one of its derivatives is holomorphic. Applying Lemma 15.1 to f gives $f' = u_x + iv_x$; applying it to f' , to f'' , and so on, all partial derivatives of every order of u and v exist and are combinations of the real and imaginary parts of the successive complex derivatives, which are continuous because holomorphic. Hence $u, v \in C^\infty(\Omega)$. **Step 2 — differentiate the Cauchy–Riemann equations.** Lemma 15.1 gives, on all of Ω ,

$$u_x = v_y, \quad u_y = -v_x.$$

Differentiating the first with respect to x and the second with respect to y :

$$u_{xx} = v_{yx}, \quad u_{yy} = -v_{xy}.$$

Step 3 — sum. Since $v \in C^\infty$, Schwarz's theorem gives $v_{yx} = v_{xy}$, and hence

$$\Delta u = u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0.$$

For v one proceeds in the same way: differentiating $u_x = v_y$ with respect to y and $u_y = -v_x$ with respect to x gives $v_{yy} = u_{xy}$ and $v_{xx} = -u_{yx}$, whose sum is zero. $\square$

Lemma 16.2: Criterion for holomorphy: Cauchy–Riemann with continuous derivatives. Let $\Omega \subseteq \mathbb{C}$ be open and $U, V : \Omega \rightarrow \mathbb{R}$ of class C^1 satisfying on Ω

$$U_x = V_y, \quad U_y = -V_x.$$

Then $f = U + iV$ is holomorphic on Ω and $f' = U_x + iV_x$.

Proof. Step 1 — real differentiability. Since $U, V \in C^1(\Omega)$, they are differentiable in the real sense at every point $z_0 = (x_0, y_0)$ of Ω : writing $h = (h_1, h_2)$,

$$U(z_0 + h) - U(z_0) = U_x(z_0)h_1 + U_y(z_0)h_2 + \varepsilon_1(h)|h|, \quad \varepsilon_1(h) \rightarrow 0 \quad (h \rightarrow 0),$$

and similarly for V with a remainder $\varepsilon_2(h)|h|$.

Step 2 — assembling the complex difference quotient. Writing $h = h_1 + ih_2$ and $a = U_x(z_0)$, $b = V_x(z_0)$, the hypothesis equations give $U_y = -V_x = -b$ and $V_y = U_x = a$. Hence

$$f(z_0 + h) - f(z_0) = (ah_1 - bh_2) + i(bh_1 + ah_2) + \varepsilon(h)|h|,$$

where $\varepsilon = \varepsilon_1 + i\varepsilon_2 \rightarrow 0$. Now observe the algebraic identity

$$(a + ib)(h_1 + ih_2) = (ah_1 - bh_2) + i(bh_1 + ah_2),$$

that is, the linear part equals exactly $(a + ib)h$.

Step 3 — conclusion. For $h \neq 0$,

$$\frac{f(z_0 + h) - f(z_0)}{h} = (a + ib) + \varepsilon(h)\frac{|h|}{h},$$

and $|\varepsilon(h)|h/|h| = |\varepsilon(h)| \rightarrow 0$. The limit of the difference quotient exists and equals $a + ib = U_x(z_0) + iV_x(z_0)$: by Definition 15.1, f is complex-differentiable at z_0 , and since z_0 is arbitrary it is holomorphic on Ω . $\square$

Lemma 16.3: The conjugate gradient of a harmonic function is holomorphic.

Let u be harmonic on an open set $\Omega \subseteq \mathbb{C}$. Then the function

$$g = u_x - i u_y$$

is holomorphic on Ω .

Proof. **Step 1 — regularity.** The functions $U = u_x$ and $V = -u_y$ are of class C^1 because $u \in C^2$ by Definition 16.1.

Step 2 — the first Cauchy–Riemann equation. One must check $U_x = V_y$, that is, $u_{xx} = -u_{yy}$: this is exactly $\Delta u = 0$.

Step 3 — the second. One must check $U_y = -V_x$, that is, $u_{xy} = u_{yx}$: this is Schwarz’s theorem, applicable because $u \in C^2$.

Step 4 — conclusion. Lemma 16.2 gives the holomorphy of $g = U + iV = u_x - iu_y$. $\square$

Definition 16.2: Harmonic conjugate. Let u, v be real on an open set Ω . One says that v is a **harmonic conjugate** of u on Ω if $f = u + iv$ is holomorphic on Ω . By Lemma 15.1 this is equivalent to $u_x = v_y$ and $u_y = -v_x$, and by Proposition 16.2 implies that u and v are harmonic.

Theorem 16.1: Every harmonic function on a disc is the real part of a holomorphic function. Let $\Omega \subseteq \mathbb{C}$ be a convex open set — in particular a disc $D(w, R)$ or all of $\mathbb{C}$ — and let u be harmonic on Ω . Then there exists f holomorphic on Ω with

$$\operatorname{Re} f = u \quad \text{on } \Omega.$$

The function $v = \operatorname{Im} f$ is a harmonic conjugate of u , and is unique up to an additive real constant.

Proof. **Step 1 — a holomorphic function that is a candidate for f' .** By Lemma 16.3 the function $g = u_x - iu_y$ is holomorphic on Ω .

Step 2 — g has a primitive. Ω is open and convex, so by Theorem 15.2 there exists F holomorphic on Ω with $F' = g$.

Step 3 — the partial derivatives of $\operatorname{Re} F$. Write $F = U + iV$ with U, V real. By Lemma 15.1,

$$F' = U_x + iV_x = U_x - iU_y,$$

the second equality because $V_x = -U_y$. Comparing with $F' = g = u_x - iu_y$ and equating real parts and imaginary parts gives

$$U_x = u_x, \quad U_y = u_y \quad \text{on } \Omega.$$

Step 4 — U and u differ by a constant. The function $\varphi = U - u$ is of class C^1 on Ω with $\varphi_x \equiv 0$ and $\varphi_y \equiv 0$. Let $z_1, z_2 \in \Omega$: since Ω is convex, the segment $[z_1, z_2]$ is contained in Ω , and the function of one variable $t \mapsto \varphi(z_1 + t(z_2 - z_1))$, $t \in [0, 1]$, has derivative $\nabla \varphi \cdot (z_2 - z_1) = 0$ at every point; by the mean value theorem in one variable it is constant, so $\varphi(z_1) = \varphi(z_2)$. Call $c \in \mathbb{R}$ this common value: $U = u + c$.

Step 5 — conclusion. Set $f = F - c$. It is holomorphic on Ω and $\operatorname{Re} f = U - c = u$. Its imaginary part $v = \operatorname{Im} f = V$ is by definition a harmonic conjugate of u .

Step 6 — uniqueness up to constants. If f_1, f_2 are holomorphic on Ω with $\operatorname{Re} f_1 = \operatorname{Re} f_2 = u$, then $h = i(f_1 - f_2)$ is holomorphic and $\operatorname{Im} h = -\operatorname{Re}(f_1 - f_2) = 0$, that is, h is real-valued. By the last statement of Lemma 15.1, on a connected open set a real-valued holomorphic function is constant; a nonempty convex set is connected, so h is a real constant k and $f_1 - f_2 = -ik$, that is, $\operatorname{Im} f_1 - \operatorname{Im} f_2 = -k$ is constant. $\square$

Corollary 16.1: Harmonic functions are of class C^∞ . Let u be harmonic on an open set $\Omega \subseteq \mathbb{C}$. Then $u \in C^\infty(\Omega)$, and all of its partial derivatives are again harmonic on Ω .

Proof. **Step 1 — the question is local.** Let $z_0 \in \Omega$ and $R > 0$ with $D(z_0, R) \subseteq \Omega$. The disc is convex, so by Theorem 16.1 there exists f holomorphic on $D(z_0, R)$ with $\operatorname{Re} f = u$ there.

Step 2 — regularity. By Step 1 of the proof of Proposition 16.2, $\operatorname{Re} f \in C^\infty(D(z_0, R))$. Hence u is C^∞ in a neighbourhood of every one of its points, that is, on Ω .

Step 3 — the derivatives are harmonic. Let w be any partial derivative of u of order k . It is C^∞ by Step 2. Since all derivatives of u of every order are continuous, the order of differentiation is immaterial (Schwarz's theorem), and hence

$$\Delta w = \Delta(\partial^k u) = \partial^k(\Delta u) = \partial^k 0 = 0.$$

□

Corollary 16.2: Uniqueness of the harmonic conjugate. Let Ω be a domain (connected open set) and let v_1, v_2 be two harmonic conjugates of the same function u on Ω . Then $v_1 - v_2$ is constant.

Proof. By hypothesis $f_1 = u + iv_1$ and $f_2 = u + iv_2$ are holomorphic on Ω ; hence $f_1 - f_2 = i(v_1 - v_2)$ is holomorphic and has zero real part. Then $h = -i(f_1 - f_2) = v_1 - v_2$ is holomorphic and real-valued on the connected set Ω : by the last statement of Lemma 15.1 it is constant. □

Proposition 16.3: Composition with a holomorphic function. Let $\Omega_1, \Omega_2 \subseteq \mathbb{C}$ be open, $\varphi : \Omega_1 \rightarrow \Omega_2$ holomorphic and u harmonic on Ω_2 . Then $u \circ \varphi$ is harmonic on Ω_1 .

Proof. **Step 1 — reduction to a neighbourhood.** Let $z_0 \in \Omega_1$ and $w_0 = \varphi(z_0)$. Choose $R > 0$ with $D(w_0, R) \subseteq \Omega_2$; by continuity of φ there exists $\rho > 0$ with $D(z_0, \rho) \subseteq \Omega_1$ and $\varphi(D(z_0, \rho)) \subseteq D(w_0, R)$.

Step 2 — use the conjugate. $D(w_0, R)$ is convex, so by Theorem 16.1 there exists f holomorphic on $D(w_0, R)$ with $\operatorname{Re} f = u$.

Step 3 — composition of holomorphic functions. The function $f \circ \varphi$ is holomorphic on $D(z_0, \rho)$ (composition of complex-differentiable functions, by the chain rule). By Proposition 16.2 its real part is harmonic; but $\operatorname{Re}(f \circ \varphi) = (\operatorname{Re} f) \circ \varphi = u \circ \varphi$.

Step 4 — conclusion. $u \circ \varphi$ is harmonic in a neighbourhood of every point of Ω_1 , hence on Ω_1 . □

Example 16.2: Conjugates computed, and a case in which the conjugate does not exist. (a) Let $u = x^2 - y^2$ on $\mathbb{C}$. Lemma 16.3 gives $g = u_x - iu_y = 2x + 2iy = 2z$, whose primitive is $F(z) = z^2$; and indeed $\operatorname{Re} z^2 = x^2 - y^2 = u$, so that $v = \operatorname{Im} z^2 = 2xy$. Independent check with the Cauchy–Riemann equations: $u_x = 2x = v_y$ ✓ and $u_y = -2y = -v_x$ ✓. Numerical check at $z = 0.3 + 0.4i$: $u = 0.09 - 0.16 = -0.07$ and $v = 2 \cdot 0.3 \cdot 0.4 = 0.24$, while $z^2 = (0.3 + 0.4i)^2 = -0.07 + 0.24i$. ✓

(b) Let $u = e^x \cos y$. Then $g = u_x - iu_y = e^x \cos y + ie^x \sin y = e^{x+iy} = e^z$, with primitive e^z : hence $v = e^x \sin y$ and $f = e^z$. ✓

(c) Let $u = \log |z|$ on $\Omega = \mathbb{C} \setminus \{0\}$, harmonic by Example 16.1(c). On Ω no harmonic conjugate exists. Indeed, if v were a conjugate, $f = u + iv$ would be holomorphic on Ω with $\operatorname{Re} f = \log |z|$; then $e^{f(z)}$ would be holomorphic and $|e^{f(z)}| = e^{\operatorname{Re} f} = |z|$, so that $h(z) = ze^{-f(z)}$ would have constant modulus 1 on the connected set Ω and, by Lemma 15.8, would be a constant λ of modulus 1: hence $e^{f(z)} = z/\lambda$ and f would be a holomorphic

logarithm of z/λ on the whole of $\mathbb{C} \setminus \{0\}$. Differentiating $e^f = z/\lambda$ gives $f'(z) = 1/z$, and integrating along the unit circle traversed once counterclockwise, with Lemma 15.3 (the integral of a derivative over a loop is zero),

$$0 = \oint_{|z|=1} f'(z) dz = \oint_{|z|=1} \frac{dz}{z} = 2\pi i,$$

where the last value is that of Lemma 15.5 with $m = -1$: a contradiction. The hypothesis of convexity (or, more generally, of simple connectedness) in Theorem 16.1 cannot therefore be dropped. On every disc contained in $\mathbb{C} \setminus \{0\}$, however, the conjugate exists and is a continuous determination of $\theta = \arg z$.

Remark 16.2: What fails in dimension greater than two. Theorem 16.1 is a genuinely two-dimensional fact. In $\mathbb{R}^n$ with $n \geq 3$ the equation $\Delta u = 0$ continues to make sense and the theory retains the mean value property, the maximum principle and the Poisson formula for the ball, but no notion of “conjugate” exists that would reduce every harmonic function to the real part of a holomorphic object: Axler, Bourdon and Ramey explicitly observe (Ch. 1, p. 2) that the link between harmonic and holomorphic functions is a peculiarity of the plane and that “no comparable result exists in higher dimensions.” In this chapter everything that is proved is proved in the plane; the n -dimensional case remains **imported** from the literature and is not used at any point in the logical chain.

16.2. The mean value property

The heart of the whole chapter is a single sentence: *the value of a harmonic function at the centre of a circle is the mean of its values on the circle*. From this sentence follow the maximum principle, uniqueness for Dirichlet and — with the Poisson kernel — existence as well.

Theorem 16.2: Mean value property on circles. Let u be harmonic on an open set Ω and let $\overline{D(w, \rho)} \subset \Omega$ with $\rho > 0$. Then

$$u(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(w + \rho e^{i\theta}) d\theta.$$

Proof. Step 1 — a slightly larger disc. The set $K = \overline{D(w, \rho)}$ is compact and contained in the open set Ω . If $\Omega \neq \mathbb{C}$, the continuous function $z \mapsto \text{dist}(z, \mathbb{C} \setminus \Omega)$ is strictly positive on K and attains a minimum $d > 0$ there; setting $R = \rho + d/2$ gives $D(w, R) \subseteq \Omega$. If $\Omega = \mathbb{C}$ take $R = \rho + 1$. In both cases $\overline{D(w, \rho)} \subset D(w, R) \subseteq \Omega$.

Step 2 — pass to a holomorphic function. The disc $D(w, R)$ is convex; by Theorem 16.1 there exists f holomorphic on $D(w, R)$ with $\text{Re } f = u$.

Step 3 — mean value for holomorphic functions. Lemma 15.7 applied to f on the disc $\overline{D(w, \rho)}$, contained in the open set $D(w, R)$ on which f is holomorphic, gives

$$f(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(w + \rho e^{i\theta}) d\theta.$$

Step 4 — real part. The real part of an integral is the integral of the real part (the integral is defined component by component). Taking the real parts of both sides and recalling $\text{Re } f = u$ gives the statement. $\square$

Corollary 16.3: Mean value property on discs. Under the same hypotheses, with $\overline{D(w, R)} \subset \Omega$,

$$u(w) = \frac{1}{\pi R^2} \iint_{D(w, R)} u(x, y) dx dy.$$

Proof. Step 1 — polar coordinates. The change-of-variables formula in polar coordinates (with $x = w_1 + \rho \cos \theta$, $y = w_2 + \rho \sin \theta$ and Jacobian ρ) gives, since u is continuous and hence integrable on the compact set $\overline{D(w, R)}$,

$$\iint_{D(w, R)} u dx dy = \int_0^R \left(\int_{-\pi}^{\pi} u(w + \rho e^{i\theta}) d\theta \right) \rho d\rho.$$

Step 2 — apply the previous theorem. For every $0 < \rho < R$ one has $\overline{D(w, \rho)} \subset \Omega$, so by Theorem 16.2 the inner integral equals $2\pi u(w)$; at $\rho = 0$ the contribution is zero and does not affect the integral.

Step 3 — conclusion.

$$\iint_{D(w, R)} u dx dy = \int_0^R 2\pi u(w) \rho d\rho = 2\pi u(w) \cdot \frac{R^2}{2} = \pi R^2 u(w).$$

Dividing by πR^2 gives the claim. □

Definition 16.3: Mean value property. Let $\Omega \subseteq \mathbb{C}$ be open. A continuous function $u : \Omega \rightarrow \mathbb{R}$ has the **mean value property** on Ω if for every $w \in \Omega$ there exists $\rho_w > 0$ with $D(w, \rho_w) \subseteq \Omega$ such that

$$u(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(w + \rho e^{i\theta}) d\theta \quad \text{for every } 0 < \rho < \rho_w.$$

Theorem 16.2 says that every harmonic function has the mean value property; Theorem 16.7 will prove the converse.

Example 16.3: The two means on $u = x^2 - y^2$, checked numerically. Let $u = x^2 - y^2$, harmonic on $\mathbb{C}$, and $w = (0.3; 0.4)$, that is, $w = 0.3 + 0.4i$, with $u(w) = 0.09 - 0.16 = -0.07$. Mean on the circle of radius $\rho = 0.5$, by symbolic computation. With $x = 0.3 + \rho \cos \theta$ and $y = 0.4 + \rho \sin \theta$,

$$u = (0.09 - 0.16) + 2\rho(0.3 \cos \theta - 0.4 \sin \theta) + \rho^2(\cos^2 \theta - \sin^2 \theta).$$

Integrating over $[-\pi, \pi]$ and dividing by 2π : the first term gives -0.07 ; the second is zero because $\int \cos \theta d\theta = \int \sin \theta d\theta = 0$; the third is zero because $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$ and $\int \cos 2\theta d\theta = 0$. Mean $= -0.07 = u(w)$. ✓

Independent numerical check. Composite Simpson quadrature with 200,000 subintervals of the integral $\frac{1}{2\pi} \int_0^{2\pi} u(w + 0.5e^{i\theta}) d\theta$ returns -0.0700000 . ✓

Mean on the disc. Corollary 16.3 with $R = 0.5$ again predicts -0.07 ; the double Simpson quadrature $\frac{1}{\pi R^2} \int_0^R \int_0^{2\pi} u \rho d\theta d\rho$ gives -0.0700000 . ✓

16.3. The maximum principle and uniqueness for Dirichlet

The maximum principle is proved here for a class wider than harmonic functions alone: for all continuous functions with the mean value property (Definition 16.3). The greater generality costs nothing in the proof and will be indispensable in Section 16.5, where it is needed to prove

the converse of the mean value property.

Theorem 16.3: Maximum principle, strong form. Let $\Omega \subseteq \mathbb{C}$ be a domain and let $u : \Omega \rightarrow \mathbb{R}$ be continuous with the mean value property. If there exists $w \in \Omega$ such that

$$u(z) \leq u(w) \quad \text{for every } z \in \Omega,$$

then u is constant on Ω . In particular no nonconstant harmonic function on a domain attains a maximum at any point of the domain.

Proof. Step 1 — the maximal level set. Let $M = u(w) = \sup_{\Omega} u$ and

$$E = \{z \in \Omega : u(z) = M\}.$$

E is nonempty because $w \in E$, and is closed in Ω because u is continuous and $E = u^{-1}(\{M\})$.

Step 2 — E is open: the mean forces equality on circles. Let $p \in E$ and let $\rho_p > 0$ be as in Definition 16.3, with $D(p, \rho_p) \subseteq \Omega$. Fixing $0 < \rho < \rho_p$,

$$M = u(p) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(p + \rho e^{i\theta}) d\theta,$$

whence

$$\int_{-\pi}^{\pi} (M - u(p + \rho e^{i\theta})) d\theta = 2\pi M - 2\pi M = 0.$$

The integrand $\theta \mapsto M - u(p + \rho e^{i\theta})$ is continuous and nonnegative (because M is the supremum of u on Ω and $p + \rho e^{i\theta} \in \Omega$). If it were strictly positive at a point θ_0 , by continuity it would be, with a value $\geq \eta > 0$, on an interval $[\theta_0 - \sigma, \theta_0 + \sigma]$ with $\sigma > 0$, and the integral would be $\geq 2\sigma\eta > 0$. Hence $u(p + \rho e^{i\theta}) = M$ for every θ .

Step 3 — E contains a whole disc. Step 2 holds for every $\rho \in (0, \rho_p)$; together with $u(p) = M$, it gives $u \equiv M$ on $D(p, \rho_p)$, that is, $D(p, \rho_p) \subseteq E$. Hence E is open.

Step 4 — connectedness. E is a nonempty, open and closed subset of the connected set Ω : hence $E = \Omega$, that is, $u \equiv M$.

Step 5 — the harmonic case. If u is harmonic it is continuous (indeed C^∞ , Corollary 16.1) and has the mean value property by Theorem 16.2; the statement applies. $\square$

Corollary 16.4: Weak form: the maximum and the minimum lie on the boundary. Let $\Omega \subseteq \mathbb{C}$ be a bounded domain and let u be continuous on $\overline{\Omega}$ and have the mean value property on Ω (in particular: harmonic on Ω). Then

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \quad \min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

Proof. Step 1 — the maxima exist. $\overline{\Omega}$ is closed and bounded, hence compact, and u is continuous on it: there exists $p \in \overline{\Omega}$ with $u(p) = \max_{\overline{\Omega}} u$. The boundary $\partial\Omega$ is nonempty (otherwise Ω would be open and closed in the connected set $\mathbb{C}$, hence $\Omega = \mathbb{C}$, which is not bounded) and compact, so $\max_{\partial\Omega} u$ also exists.

Step 2 — if the maximum is attained on the boundary. If $p \in \partial\Omega$ there is nothing to prove: $\max_{\partial\Omega} u \leq \max_{\overline{\Omega}} u = u(p) \leq \max_{\partial\Omega} u$.

Step 3 — if the maximum is attained in the interior. Let $p \in \Omega$. Then $u(z) \leq u(p)$ for every $z \in \Omega$, and Theorem 16.3 gives $u \equiv u(p)$ on Ω ; by continuity $u \equiv u(p)$ on $\overline{\Omega}$, and in particular on $\partial\Omega$, which is nonempty. In this case too the two maxima coincide.

Step 4 — the minimum. Apply the foregoing to $-u$, which is continuous on $\overline{\Omega}$ and has the mean value property on Ω (the mean is linear), and use $\min v = -\max(-v)$. $\square$

Corollary 16.5: Minimum principle, strong form. Let Ω be a domain and u continuous with the mean value property on Ω . If u attains at a point of Ω its own infimum on Ω , then u is constant.

Proof. The function $-u$ is continuous and has the mean value property (the integral is linear, and $-u(w) = \frac{1}{2\pi} \int -u$). If u attains its minimum at w , then $-u$ attains its maximum at w ; by Theorem 16.3 $-u$ is constant, hence so is u . $\square$

Corollary 16.6: Comparison principle. Let Ω be a bounded domain and u_1, u_2 continuous on $\bar{\Omega}$, with the mean value property on Ω . If $u_1 \leq u_2$ on $\partial\Omega$, then $u_1 \leq u_2$ on all of $\bar{\Omega}$.

Proof. The function $w = u_1 - u_2$ is continuous on $\bar{\Omega}$ and has the mean value property on Ω by linearity. By Corollary 16.4,

$$\max_{\bar{\Omega}} w = \max_{\partial\Omega} w \leq 0,$$

because $w \leq 0$ on the boundary by hypothesis. Hence $w \leq 0$ everywhere, that is, $u_1 \leq u_2$. $\square$

Corollary 16.7: Uniqueness of the solution of the Dirichlet problem. Let $\Omega \subseteq \mathbb{C}$ be a bounded domain and $f : \partial\Omega \rightarrow \mathbb{R}$ continuous. There is at most one function u continuous on $\bar{\Omega}$, harmonic on Ω and such that $u = f$ on $\partial\Omega$. More generally, uniqueness holds in the class of functions continuous on $\bar{\Omega}$ with the mean value property on Ω .

Proof. Step 1 — difference. Let u_1, u_2 be two such functions. Then $w = u_1 - u_2$ is continuous on $\bar{\Omega}$, has the mean value property on Ω by linearity, and vanishes on $\partial\Omega$.

Step 2 — maximum and minimum. By Corollary 16.4,

$$\max_{\bar{\Omega}} w = \max_{\partial\Omega} w = 0, \quad \min_{\bar{\Omega}} w = \min_{\partial\Omega} w = 0.$$

Step 3 — conclusion. From $0 \leq w \leq 0$ it follows that $w \equiv 0$, that is, $u_1 = u_2$. $\square$

Remark 16.3: Two maximum principles, one mechanism. Theorem 15.11 (maximum modulus for holomorphic functions) and Theorem 16.3 have the same proof, and this is no coincidence: both rest on the mean value property (Lemma 15.7 in the first case, Theorem 16.2 in the second) and on the observation that a mean does not exceed the maximum of the averaged values and equals it only if they are all equal. The difference is that for holomorphic functions one averages the modulus — and to pass from “ $|f|$ constant” to “ f constant” one needs Lemma 15.8 — while for harmonic functions one averages the function itself and the passage is immediate. Note also that for harmonic functions a minimum principle holds with no extra hypothesis (Corollary 16.5), while for the modulus of a holomorphic function one must assume that the function does not vanish (Corollary 15.6): this is the difference between a real-valued function of arbitrary sign and a nonnegative quantity.

Example 16.4: The maximum of $x^2 - y^2$ on the closed disc. Let $u = x^2 - y^2$ on $\mathbb{D}$. The function is harmonic (Example 16.1(a)), so by Corollary 16.4 its maximum lies on the boundary. On $\mathbb{T}$ one has $u(e^{i\theta}) = \cos^2 \theta - \sin^2 \theta = \cos 2\theta$, whose maximum is 1, attained at $\theta = 0$ and $\theta = \pi$, that is, at $z = \pm 1$; the minimum is -1 , attained at $z = \pm i$.

Independent check 1 (interior). If the maximum were attained at an interior point p , by Theorem 16.3 u would be constant on $\mathbb{D}$: false, because $u(0) = 0$ and $u(1/2) = 1/4$.

Independent check 2 (numerical). Sampling $u(e^{i\theta})$ at 100,000 equally spaced points of $[0, 2\pi)$ gives maximum 1.0000000 at $\theta = 3.1415927$ (that is, $\theta = \pi$, up to the sampling step). ✓

16.4. The Poisson kernel

So far it has been proved that a harmonic function is determined by its boundary values (Corollary 16.7), but not that a boundary datum actually corresponds to a harmonic function. The tool that bridges the gap is an explicit kernel.

Definition 16.4: Poisson kernel. For $0 \leq r < 1$ and $t \in \mathbb{R}$ set

$$P_r(t) = \sum_{n \in \mathbb{Z}} r^{|n|} e^{int}.$$

The series converges absolutely and uniformly in t , because $|r^{|n|} e^{int}| = r^{|n|}$ and $\sum_{n \in \mathbb{Z}} r^{|n|} = 1 + 2 \sum_{n \geq 1} r^n = \frac{1+r}{1-r} < \infty$ (Weierstrass criterion). The family $\{P_r\}_{0 \leq r < 1}$ is called the **Poisson kernel** of the disc. For $z = re^{i\theta} \in \mathbb{D}$ and $e^{it} \in \mathbb{T}$ one also writes

$$P(z, e^{it}) = P_r(\theta - t).$$

Lemma 16.4: Closed form of the kernel. For every $0 \leq r < 1$ and every $t \in \mathbb{R}$,

$$P_r(t) = \frac{1 - r^2}{1 - 2r \cos t + r^2}.$$

Proof. **Step 1 — two geometric series.** Split the sum into $n \geq 0$ and $n \leq -1$:

$$P_r(t) = \sum_{n \geq 0} (re^{it})^n + \sum_{n \geq 1} (re^{-it})^n = \frac{1}{1 - re^{it}} + \frac{re^{-it}}{1 - re^{-it}},$$

legitimate because $|re^{\pm it}| = r < 1$ and the geometric series of ratio with modulus < 1 converges to the indicated sum.

Step 2 — common denominator. The two denominators are conjugates of each other, so their product is $|1 - re^{it}|^2$. Adding,

$$P_r(t) = \frac{(1 - re^{-it}) + re^{-it}(1 - re^{it})}{|1 - re^{it}|^2} = \frac{1 - re^{-it} + re^{-it} - r^2}{|1 - re^{it}|^2} = \frac{1 - r^2}{|1 - re^{it}|^2},$$

because $re^{-it} \cdot re^{it} = r^2$.

Step 3 — the denominator in real form.

$$|1 - re^{it}|^2 = (1 - r \cos t)^2 + (r \sin t)^2 = 1 - 2r \cos t + r^2(\cos^2 t + \sin^2 t) = 1 - 2r \cos t + r^2.$$

□

Lemma 16.5: Geometric form and complex form. Let $z \in \mathbb{D}$ and $e^{it} \in \mathbb{T}$. Then

$$P(z, e^{it}) = \frac{1 - |z|^2}{|e^{it} - z|^2} = \operatorname{Re} \left(\frac{e^{it} + z}{e^{it} - z} \right) = \operatorname{Re} \left(\frac{1 + ze^{-it}}{1 - ze^{-it}} \right).$$

Proof. Step 1 — the first equality. Let $z = re^{i\theta}$. Since $|e^{it}| = 1$,

$$|e^{it} - z| = |e^{it}| \cdot |1 - ze^{-it}| = |1 - re^{i(\theta-t)}|,$$

and by Step 3 of the proof of Lemma 16.4 $|1 - re^{i(\theta-t)}|^2 = 1 - 2r \cos(\theta - t) + r^2$. Hence

$$\frac{1 - |z|^2}{|e^{it} - z|^2} = \frac{1 - r^2}{1 - 2r \cos(\theta - t) + r^2} = P_r(\theta - t) = P(z, e^{it}),$$

the last equality by Lemma 16.4 and Definition 16.4.

Step 2 — the second equality. Multiply numerator and denominator by the conjugate of the denominator, which is nonzero because $|z| < 1 = |e^{it}|$:

$$\frac{e^{it} + z}{e^{it} - z} = \frac{(e^{it} + z)\overline{(e^{it} - z)}}{|e^{it} - z|^2} = \frac{(e^{it} + z)(e^{-it} - \bar{z})}{|e^{it} - z|^2}.$$

The numerator equals

$$1 - e^{it}\bar{z} + ze^{-it} - |z|^2 = (1 - |z|^2) + (ze^{-it} - \overline{ze^{-it}}) = (1 - |z|^2) + 2i \operatorname{Im}(ze^{-it}),$$

because $e^{it}\bar{z} = \overline{ze^{-it}}$. Since the denominator is real and positive, the real part of the quotient is $\frac{1 - |z|^2}{|e^{it} - z|^2}$.

Step 3 — the third equality. Divide numerator and denominator by $e^{it} \neq 0$: $\frac{e^{it} + z}{e^{it} - z} = \frac{1 + ze^{-it}}{1 - ze^{-it}}$. $\square$

Lemma 16.6: Positivity and bounds. For every $0 \leq r < 1$ and every $t \in \mathbb{R}$,

$$0 < \frac{1 - r}{1 + r} \leq P_r(t) \leq \frac{1 + r}{1 - r},$$

and the two inequalities are equalities respectively for $t \equiv \pi$ and $t \equiv 0 \pmod{2\pi}$. In particular $P_r > 0$.

Proof. Step 1 — the denominator between two squares. One writes

$$1 - 2r \cos t + r^2 = (1 - r)^2 + 2r(1 - \cos t) = (1 + r)^2 - 2r(1 + \cos t).$$

The first identity is checked by expanding $(1 - r)^2 = 1 - 2r + r^2$ and adding $2r - 2r \cos t$; the second by expanding $(1 + r)^2 = 1 + 2r + r^2$ and subtracting $2r + 2r \cos t$.

Step 2 — the two bounds. Since $-1 \leq \cos t \leq 1$ and $r \geq 0$, the terms $2r(1 - \cos t)$ and $2r(1 + \cos t)$ are ≥ 0 ; hence

$$(1 - r)^2 \leq 1 - 2r \cos t + r^2 \leq (1 + r)^2,$$

with equality on the left if and only if $\cos t = 1$ and on the right if and only if $\cos t = -1$ (for $r > 0$). In particular the denominator is $\geq (1 - r)^2 > 0$: the quotient is well defined.

Step 3 — passing to the quotient. Since $1 - r^2 > 0$, inverting the inequalities of Step 2 and multiplying by $1 - r^2$:

$$\frac{1 - r^2}{(1 + r)^2} \leq P_r(t) \leq \frac{1 - r^2}{(1 - r)^2}.$$

Simplifying $1 - r^2 = (1 - r)(1 + r)$ gives $\frac{1-r}{1+r}$ and $\frac{1+r}{1-r}$. $\square$

Lemma 16.7: An explicit primitive of the kernel. Let $0 < r < 1$ and $k = \frac{1+r}{1-r} > 1$. Then the function

$$A_r(t) = 2 \arctan \left(k \tan \frac{t}{2} \right)$$

is of class C^1 on $(-\pi, \pi)$ and $A'_r(t) = P_r(t)$ for every $t \in (-\pi, \pi)$.

Proof. Step 1 — derivative. On $(-\pi, \pi)$ the function $t \mapsto \tan(t/2)$ is of class C^∞ , and

$$A'_r(t) = 2 \cdot \frac{k \cdot \frac{1}{2} \sec^2(t/2)}{1 + k^2 \tan^2(t/2)} = \frac{k \sec^2(t/2)}{1 + k^2 \tan^2(t/2)} = \frac{k}{\cos^2(t/2) + k^2 \sin^2(t/2)},$$

where in the last step numerator and denominator have been multiplied by $\cos^2(t/2)$.

Step 2 — the denominator. With $\cos^2 = 1 - \sin^2$,

$$\cos^2 \frac{t}{2} + k^2 \sin^2 \frac{t}{2} = 1 + (k^2 - 1) \sin^2 \frac{t}{2}, \quad k^2 - 1 = \frac{(1+r)^2 - (1-r)^2}{(1-r)^2} = \frac{4r}{(1-r)^2},$$

hence

$$\cos^2 \frac{t}{2} + k^2 \sin^2 \frac{t}{2} = \frac{(1-r)^2 + 4r \sin^2 \frac{t}{2}}{(1-r)^2}.$$

Step 3 — comparison with the denominator of P_r . From the double-angle formula $\cos t = 1 - 2 \sin^2(t/2)$ it follows that

$$1 - 2r \cos t + r^2 = 1 - 2r + r^2 + 4r \sin^2 \frac{t}{2} = (1-r)^2 + 4r \sin^2 \frac{t}{2}.$$

Step 4 — conclusion. Substituting into Step 1,

$$A'_r(t) = k \cdot \frac{(1-r)^2}{(1-r)^2 + 4r \sin^2 \frac{t}{2}} = \frac{1+r}{1-r} \cdot \frac{(1-r)^2}{1 - 2r \cos t + r^2} = \frac{(1+r)(1-r)}{1 - 2r \cos t + r^2} = P_r(t).$$

□

Lemma 16.8: Normalization. For every $0 \leq r < 1$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(t) dt = 1.$$

Consequently, for every $z \in \mathbb{D}$, $\frac{1}{2\pi} \int_{-\pi}^{\pi} P(z, e^{it}) dt = 1$.

Proof. Step 1 — term-by-term integration. By Definition 16.4 the series $\sum_{n \in \mathbb{Z}} r^{|n|} e^{int}$ converges uniformly in $t \in [-\pi, \pi]$; a series of continuous functions converging uniformly on a compact set may be integrated term by term. Hence

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(t) dt = \sum_{n \in \mathbb{Z}} r^{|n|} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int} dt.$$

Step 2 — orthogonality of the characters. For $n \neq 0$, $\int_{-\pi}^{\pi} e^{int} dt = \left[\frac{e^{int}}{in} \right]_{-\pi}^{\pi} = 0$ because $e^{in\pi} = e^{-in\pi}$; for $n = 0$ the integral equals 2π . Only the term $n = 0$ survives, and it equals $r^0 = 1$.

Step 3 — second route: the closed form. Independently of Step 1, from Lemma 16.7 one has, for $0 < r < 1$ and $k = \frac{1+r}{1-r}$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(t) dt = \frac{1}{2\pi} \lim_{s \rightarrow \pi^-} \left[2 \arctan(k \tan(t/2)) \right]_{-s}^s = \frac{1}{2\pi} \cdot 4 \cdot \frac{\pi}{2} = 1,$$

because $\tan(s/2) \rightarrow +\infty$ as $s \rightarrow \pi^-$ and $\arctan \rightarrow \pi/2$; the passage to the limit is legitimate because P_r is continuous and bounded on $[-\pi, \pi]$ by Lemma 16.6, so that $\int_{-s}^s P_r \rightarrow \int_{-\pi}^{\pi} P_r$. For $r = 0$ one has $P_0 \equiv 1$ and the claim is immediate.

Step 4 — the version with z . For $z = re^{i\theta}$ one has $P(z, e^{it}) = P_r(\theta - t)$; the substitution $s = \theta - t$ and the 2π -periodicity of P_r (evident from the closed form) give $\int_{-\pi}^{\pi} P_r(\theta - t) dt = \int_{\theta-\pi}^{\theta+\pi} P_r(s) ds = \int_{-\pi}^{\pi} P_r(s) ds$. $\square$

Lemma 16.9: Concentration at the boundary. Let $0 < \delta \leq \pi$. Then for every $0 \leq r < 1$ and every t with $\delta \leq |t| \leq \pi$,

$$P_r(t) \leq P_r(\delta) = \frac{1 - r^2}{1 - 2r \cos \delta + r^2},$$

and

$$\lim_{r \rightarrow 1^-} \sup_{\delta \leq |t| \leq \pi} P_r(t) = 0, \quad \text{hence} \quad \lim_{r \rightarrow 1^-} \frac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} P_r(t) dt = 0.$$

Proof. **Step 1 — monotonicity.** P_r is even in t (it depends on t only through $\cos t$). On $[0, \pi]$ the function $\cos$ is decreasing, so $t \mapsto 1 - 2r \cos t + r^2$ is increasing (for $r \geq 0$) and its reciprocal is decreasing. Hence P_r is decreasing on $[0, \pi]$ and, for $\delta \leq |t| \leq \pi$, $P_r(t) = P_r(|t|) \leq P_r(\delta)$.

Step 2 — the limit. The denominator $1 - 2r \cos \delta + r^2$ tends, as $r \rightarrow 1^-$, to $2 - 2 \cos \delta$, which is strictly positive because $0 < \delta \leq \pi$ implies $\cos \delta < 1$. The numerator $1 - r^2$ tends to 0. Hence $P_r(\delta) \rightarrow 0$, and by Step 1 the same holds for the supremum.

Step 3 — the integral. The set $\{\delta \leq |t| \leq \pi\}$ has measure $2(\pi - \delta) \leq 2\pi$, and on it $0 < P_r \leq P_r(\delta)$ by Step 1 and Lemma 16.6. Hence

$$0 \leq \frac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} P_r(t) dt \leq \frac{2(\pi - \delta)}{2\pi} P_r(\delta) \leq P_r(\delta) \xrightarrow{r \rightarrow 1^-} 0.$$

$\square$

Definition 16.5: Approximate identity. A family $\{K_r\}_{0 \leq r < 1}$ of continuous, 2π -periodic functions on $\mathbb{R}$ is an **approximate identity** if

- (I1) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_r(t) dt = 1$ for every r ;
- (I2) there exists $M > 0$ with $\frac{1}{2\pi} \int_{-\pi}^{\pi} |K_r(t)| dt \leq M$ for every r ;
- (I3) for every $\delta \in (0, \pi]$, $\lim_{r \rightarrow 1^-} \int_{\delta \leq |t| \leq \pi} |K_r(t)| dt = 0$.

Proposition 16.4: The Poisson kernel is an approximate identity. The family $\{P_r\}_{0 \leq r < 1}$ satisfies (I1), (I2) with $M = 1$, and (I3).

Proof. **Step 1 — (I1).** This is Lemma 16.8.

Step 2 — (I2). By Lemma 16.6, $P_r > 0$, so $|P_r| = P_r$ and the normalized integral of the modulus coincides with that of P_r , which equals 1 by Step 1: one may take $M = 1$.

Step 3 — (I3). This is the last statement of Lemma 16.9, multiplied by 2π . $\square$

Lemma 16.10: *The kernel is harmonic in the interior variable. For every fixed $t \in \mathbb{R}$, the function $z \mapsto P(z, e^{it})$ is harmonic on $\mathbb{D}$ (indeed on $\mathbb{C} \setminus \{e^{it}\}$).*

Proof. Step 1 — a holomorphic function. The function $\Phi_t(z) = \frac{e^{it} + z}{e^{it} - z}$ is a quotient of two polynomials and is holomorphic on $\mathbb{C} \setminus \{e^{it}\}$, in particular on $\mathbb{D}$ (where $|z| < 1 = |e^{it}|$, so the denominator does not vanish).

Step 2 — take the real part. By Proposition 16.2, $\operatorname{Re} \Phi_t$ is harmonic on $\mathbb{C} \setminus \{e^{it}\}$; and by Lemma 16.5 $\operatorname{Re} \Phi_t(z) = P(z, e^{it})$ for $z \in \mathbb{D}$. The same formula $\operatorname{Re} \Phi_t(z) = \frac{1-|z|^2}{|e^{it}-z|^2}$ holds moreover at every point $z \neq e^{it}$, as shown by Step 2 of the proof of Lemma 16.5, which does not use $|z| < 1$. $\square$

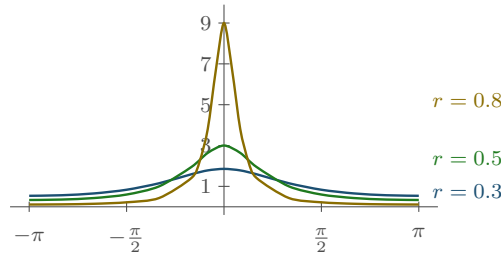


Figure 16.1: The Poisson kernel $P_r(t) = \frac{1-r^2}{1-2r\cos t+r^2}$ for $r = 0.3$, $r = 0.5$ and $r = 0.8$. Coordinates computed by hand at the notable points and confirmed to the seventh decimal: $P_r(0) = \frac{1+r}{1-r}$ equals $\frac{1.3}{0.7} = 1.8571429$, $\frac{1.5}{0.5} = 3$ and $\frac{1.8}{0.2} = 9$; $P_r(\pi) = \frac{1-r}{1+r}$ equals 0.5384615 , 0.3333333 and 0.1111111 ; $P_{0.5}(\pi/3) = \frac{0.75}{1-0.5+0.25} = 1$ exactly. $\checkmark$ The area under each curve, divided by 2π , equals 1 (Lemma 16.8): as r increases the mass concentrates at $t = 0$.

Example 16.5: The kernel at three points, with two checks. Let $r = 0.5$, so that $1 - r^2 = 0.75$.

(a) $t = 0$. Denominator $1 - 1 + 0.25 = 0.25$, hence $P_{0.5}(0) = 3$. Check with Lemma 16.6: $\frac{1+r}{1-r} = \frac{1.5}{0.5} = 3$, and this is the maximum value. $\checkmark$

(b) $t = \pi/3$. Denominator $1 - 2 \cdot 0.5 \cdot 0.5 + 0.25 = 0.75$, hence $P_{0.5}(\pi/3) = 1$ exactly. Check with the series of Definition 16.4: $1 + 2 \sum_{n \geq 1} 2^{-n} \cos(n\pi/3)$; the cosines take the values $\frac{1}{2}, -\frac{1}{2}, -1, -\frac{1}{2}, \frac{1}{2}, 1$ and repeat with period 6, and the sum of the six terms of each block is, numerically, such as to return 1.0000000: the partial sum up to $n = 60$ equals 1.0000000. $\checkmark$

(c) $t = \pi$. Denominator $1 + 1 + 0.25 = 2.25$, hence $P_{0.5}(\pi) = 0.75/2.25 = 1/3 = 0.3333333$, which is the minimum value $\frac{1-r}{1+r} = \frac{0.5}{1.5}$. $\checkmark$

(d) *Concentration.* With $\delta = 0.2$ Lemma 16.9 gives $\sup_{|t| \geq \delta} P_r(t) = P_r(0.2)$, which equals 4.1412236 for $r = 0.9$, 0.5029294 for $r = 0.99$ and 0.0501908 for $r = 0.999$: sampling P_r at 2001 equally spaced points of $[\delta, \pi]$ returns exactly the same maxima. $\checkmark$

16.5. The Poisson formula and the Dirichlet problem in the disc

Definition 16.6: Poisson integral. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be bounded and Riemann integrable on $[-\pi, \pi]$. The **Poisson integral** of f is the function $P[f] : \mathbb{D} \rightarrow \mathbb{R}$ defined by

$$P[f](z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(z, e^{it}) f(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1-|z|^2}{|e^{it}-z|^2} f(t) dt,$$

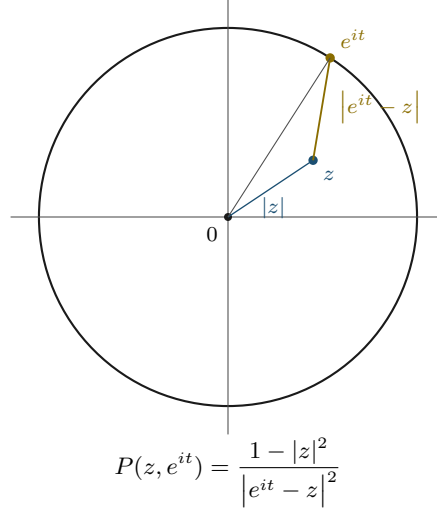


Figure 16.2: The geometry of the Poisson kernel. Coordinates computed by hand: $z = 0.45 + 0.30i$, so $|z|^2 = 0.2025 + 0.09 = 0.2925$ and $1 - |z|^2 = 0.7075$; $t = 1$ radian gives $e^{it} = 0.5403023 + 0.8414710i$, so $e^{it} - z = 0.0903023 + 0.5414710i$ and $|e^{it} - z|^2 = 0.0081545 + 0.2931908 = 0.3013453$. Hence $P(z, e^{it}) = 0.7075/0.3013453 = 2.3478047$. Independent check with the polar form of Lemma 16.4: $|z| = 0.5408327$, $\arg z = \arctan(0.30/0.45) = 0.5880026$, and $P_{|z|}(\arg z - t) = \frac{0.7075}{1 - 2 \cdot 0.5408327 \cos(0.4119974) + 0.2925} = 2.3478047$. ✓

that is, in polar coordinates $z = re^{i\theta}$,

$$P[f](re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 - r^2}{1 - 2r \cos(\theta - t) + r^2} f(t) dt.$$

The integral exists because the integrand is the product of a continuous function of t (for fixed z) and a bounded, integrable function.

Theorem 16.4: The Poisson integral is harmonic in the disc. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be bounded and Riemann integrable, with $|f| \leq M$. Then $P[f]$ is harmonic on $\mathbb{D}$; indeed $P[f] = \operatorname{Re} F$ with F holomorphic on $\mathbb{D}$ given by

$$F(z) = c_0 + 2 \sum_{n \geq 1} c_n z^n, \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt, \quad |c_n| \leq M.$$

Proof. **Step 1 — a complex-valued function.** Set, for $z \in \mathbb{D}$,

$$F(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 + ze^{-it}}{1 - ze^{-it}} f(t) dt.$$

The denominator does not vanish, because $|ze^{-it}| = |z| < 1$. By Lemma 16.5 and linearity of the real part (the integrand is the product of a complex number and the *real* number $f(t)$),

$$\operatorname{Re} F(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \operatorname{Re} \left(\frac{1 + ze^{-it}}{1 - ze^{-it}} \right) f(t) dt = P[f](z).$$

Step 2 — series expansion of the integrand. For $|w| < 1$ one has

$$\frac{1+w}{1-w} = (1+w) \sum_{n \geq 0} w^n = \sum_{n \geq 0} w^n + \sum_{n \geq 0} w^{n+1} = 1 + 2 \sum_{n \geq 1} w^n.$$

Fixing $0 < \rho < 1$ and taking $|z| \leq \rho$, with $w = ze^{-it}$ one has $|w| \leq \rho$ and

$$|2w^n f(t)| \leq 2M\rho^n,$$

the general term of a convergent numerical series independent of t and z : by the Weierstrass criterion the series $(1 + 2 \sum_{n \geq 1} z^n e^{-int}) f(t)$ converges uniformly in $t \in [-\pi, \pi]$.

Step 3 — term-by-term integration. Term-by-term integration of a uniformly convergent series of functions integrable on a compact interval is legitimate, so for $|z| \leq \rho$

$$F(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt + 2 \sum_{n \geq 1} z^n \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt = c_0 + 2 \sum_{n \geq 1} c_n z^n.$$

Step 4 — the power series has radius ≥ 1 . One has $|c_n| \leq \frac{1}{2\pi} \int |f| |e^{-int}| dt \leq M$, hence $\limsup_n |2c_n|^{1/n} \leq \limsup_n (2M)^{1/n} = 1$ and the radius of convergence is ≥ 1 : F is the sum of a power series convergent on $\mathbb{D}$, hence holomorphic there.

Step 5 — conclusion. By Proposition 16.2 the real part of F is harmonic on $\mathbb{D}$; by Step 1 it equals $P[f]$. $\square$

Theorem 16.5: Boundary behaviour of the Poisson integral. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be bounded, $|f| \leq M$, and Riemann integrable. Let $e^{i\theta_0}$ be a point at which f is continuous. Then

$$\lim_{\substack{z \rightarrow e^{i\theta_0} \\ z \in \mathbb{D}}} P[f](z) = f(\theta_0).$$

If f is continuous on all of $\mathbb{T}$, the function

$$u(z) = \begin{cases} P[f](z), & z \in \mathbb{D}, \\ f(z), & z \in \mathbb{T}, \end{cases}$$

is continuous on $\bar{\mathbb{D}}$ and the convergence is uniform on the circles $|z| = r$ as $r \rightarrow 1^-$.

Proof. Step 1 — centring the error. By Lemma 16.8, $\frac{1}{2\pi} \int P(z, e^{it}) dt = 1$ for every $z \in \mathbb{D}$; multiplying by the constant $f(\theta_0)$ and subtracting,

$$P[f](z) - f(\theta_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(z, e^{it}) (f(t) - f(\theta_0)) dt,$$

and hence, by positivity of the kernel (Lemma 16.6) and the inequality $|\int g| \leq \int |g|$,

$$|P[f](z) - f(\theta_0)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} P(z, e^{it}) |f(t) - f(\theta_0)| dt.$$

Step 2 — choosing δ . Let $\varepsilon > 0$. By continuity of f at $e^{i\theta_0}$ there exists $\delta \in (0, \pi]$ such that

$$d(t, \theta_0) < \delta \implies |f(t) - f(\theta_0)| < \frac{\varepsilon}{2},$$

where d is the distance on the circle. Split the integral of Step 1 into the two parts $d(t, \theta_0) < \delta$ and $d(t, \theta_0) \geq \delta$, and call I_1 and I_2 the two contributions.

Step 3 — the near part. On $\{d(t, \theta_0) < \delta\}$ bound $|f(t) - f(\theta_0)| < \varepsilon/2$ and then extend the integral of the kernel, which is positive, to all of $[-\pi, \pi]$:

$$I_1 \leq \frac{\varepsilon}{2} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} P(z, e^{it}) dt = \frac{\varepsilon}{2}.$$

Step 4 — the kernel far from the point is small. Let $c_\delta = 2 \sin(\delta/2) > 0$. If $d(t, \theta_0) \geq \delta$, then, since $|e^{it} - e^{i\theta_0}| = 2 \left| \sin \frac{t-\theta_0}{2} \right|$ and $\frac{d(t, \theta_0)}{2} \in [\frac{\delta}{2}, \frac{\pi}{2}]$, where sine is increasing,

$$|e^{it} - e^{i\theta_0}| = 2 \sin \frac{d(t, \theta_0)}{2} \geq c_\delta.$$

Now suppose $|z - e^{i\theta_0}| \leq c_\delta/2$. By the triangle inequality,

$$|e^{it} - z| \geq |e^{it} - e^{i\theta_0}| - |e^{i\theta_0} - z| \geq c_\delta - \frac{c_\delta}{2} = \frac{c_\delta}{2},$$

and hence, by Lemma 16.5,

$$P(z, e^{it}) = \frac{1 - |z|^2}{|e^{it} - z|^2} \leq \frac{4(1 - |z|^2)}{c_\delta^2}.$$

Step 5 — estimating $1 - |z|^2$. Since $|z| < 1$,

$$1 - |z|^2 = (1 - |z|)(1 + |z|) \leq 2(1 - |z|) = 2(|e^{i\theta_0}| - |z|) \leq 2|z - e^{i\theta_0}|,$$

the last step by the reverse triangle inequality.

Step 6 — the far part. On $\{d(t, \theta_0) \geq \delta\}$ one has $|f(t) - f(\theta_0)| \leq 2M$, hence, using Steps 4 and 5 and the length $\leq 2\pi$ of the interval of integration,

$$I_2 \leq 2M \cdot \frac{1}{2\pi} \cdot 2\pi \cdot \frac{4(1 - |z|^2)}{c_\delta^2} \leq \frac{16M}{c_\delta^2} |z - e^{i\theta_0}|.$$

Step 7 — choosing η . Set

$$\eta = \min \left\{ \frac{c_\delta}{2}, \frac{\varepsilon c_\delta^2}{32M + 1} \right\} > 0.$$

If $z \in \mathbb{D}$ and $|z - e^{i\theta_0}| < \eta$, then the hypotheses of Step 4 are satisfied and moreover $I_2 \leq \frac{16M}{c_\delta^2} \cdot \frac{\varepsilon c_\delta^2}{32M + 1} < \frac{\varepsilon}{2}$. With Step 3,

$$|P[f](z) - f(\theta_0)| \leq I_1 + I_2 < \varepsilon.$$

This proves the limit.

Step 8 — the case f continuous on $\mathbb{T}$: uniformity. If f is continuous on $\mathbb{T}$, it is uniformly continuous (the circle is compact): the δ of Step 2 may be chosen *independent* of θ_0 . Then c_δ and η of Step 7 are also independent of θ_0 , and the inequality of Step 7 holds uniformly: for every $\varepsilon > 0$ there exists $\eta > 0$ such that $|P[f](z) - f(\theta_0)| < \varepsilon$ for every θ_0 and every $z \in \mathbb{D}$ with $|z - e^{i\theta_0}| < \eta$. In particular, taking $z = re^{i\theta_0}$ with $1 - r < \eta$ (note $|re^{i\theta_0} - e^{i\theta_0}| = 1 - r$),

$$\sup_{\theta_0} |P[f](re^{i\theta_0}) - f(\theta_0)| \leq \varepsilon \quad \text{for } 1 - r < \eta,$$

which is uniform convergence on circles.

Step 9 — continuity on $\overline{\mathbb{D}}$. The function u of the statement is continuous at every point of $\mathbb{D}$ (it is harmonic there, by Theorem 16.4) and at every point $e^{i\theta_0}$ of the boundary: if $z_k \rightarrow e^{i\theta_0}$, separate the terms with $z_k \in \mathbb{D}$, for which $u(z_k) = P[f](z_k) \rightarrow f(\theta_0)$ by Steps 1–7, and those with $z_k \in \mathbb{T}$, for which $u(z_k) = f(z_k) \rightarrow f(\theta_0)$ by continuity of f on $\mathbb{T}$. In both cases $u(z_k) \rightarrow f(\theta_0) = u(e^{i\theta_0})$. $\square$

Theorem 16.6: Existence and uniqueness for the Dirichlet problem in the disc.

Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be continuous. Then there exists one and only one function $u : \mathbb{D} \rightarrow \mathbb{R}$ continuous on $\overline{\mathbb{D}}$, harmonic on $\mathbb{D}$ and equal to f on $\mathbb{T}$; and it is given by the **Poisson formula**

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1-r^2}{1-2r\cos(\theta-t)+r^2} f(t) dt, \quad 0 \leq r < 1,$$

and $u = f$ on $\mathbb{T}$.

Proof. **Step 1 — existence.** A continuous function on the compact set $\mathbb{T}$ is bounded and integrable. By Theorem 16.4 the function $P[f]$ is harmonic on $\mathbb{D}$; by Theorem 16.5 its extension by f on the boundary is continuous on $\overline{\mathbb{D}}$. That function therefore solves the problem.

Step 2 — uniqueness. $\mathbb{D}$ is a bounded domain; Corollary 16.7 says that two solutions continuous on $\overline{\mathbb{D}}$, harmonic inside and agreeing on the boundary, agree everywhere.

Step 3 — the formula. This is Definition 16.6 rewritten in polar coordinates by Lemma 16.4; for $r = 0$ one reads $u(0) = \frac{1}{2\pi} \int f$, consistently with Theorem 16.2. $\square$

Corollary 16.8: Poisson formula for a harmonic function. Let u be harmonic on an open set containing $\overline{\mathbb{D}}$. Then, for every $z \in \mathbb{D}$,

$$u(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1-|z|^2}{|e^{it}-z|^2} u(e^{it}) dt.$$

Proof. **Step 1 — the boundary datum.** The restriction $f = u|_{\mathbb{T}}$ is continuous because u is continuous (indeed C^∞ , Corollary 16.1).

Step 2 — two solutions of the same problem. The function u restricted to $\overline{\mathbb{D}}$ is continuous on $\overline{\mathbb{D}}$, harmonic on $\mathbb{D}$ and equals f on $\mathbb{T}$. So does $P[f]$, extended by f on the boundary, by Theorem 16.6.

Step 3 — uniqueness. By Corollary 16.7 the two functions coincide on $\overline{\mathbb{D}}$; in particular $u(z) = P[f](z)$ for $z \in \mathbb{D}$, which is the formula of the statement. $\square$

Corollary 16.9: Poisson formula on a disc of centre w and radius R . Let u be harmonic on an open set containing $\overline{D(w, R)}$, with $R > 0$. Then for $0 \leq \rho < R$ and $\theta \in \mathbb{R}$,

$$u(w + \rho e^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{R^2 - \rho^2}{R^2 - 2R\rho \cos(\theta - t) + \rho^2} u(w + Re^{it}) dt.$$

Moreover, if $g : \partial D(w, R) \rightarrow \mathbb{R}$ is continuous, the function defined by the same integral with g in place of $u(w + Re^{it})$ is the unique solution of the Dirichlet problem on $D(w, R)$ with datum g .

Proof. **Step 1 — transporting the problem to the unit disc.** Set $\Psi(\zeta) = w + R\zeta$, which is holomorphic and bijective from $\mathbb{C}$ to $\mathbb{C}$ and maps $\mathbb{D}$ onto $\overline{D(w, R)}$. The function $v = u \circ \Psi$ is harmonic on an open set containing $\overline{\mathbb{D}}$ by Proposition 16.3.

Step 2 — apply the previous corollary. By Corollary 16.8, for $\zeta = \frac{\rho}{R}e^{i\theta} \in \mathbb{D}$,

$$v(\zeta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 - \frac{\rho^2}{R^2}}{1 - 2\frac{\rho}{R}\cos(\theta - t) + \frac{\rho^2}{R^2}} v(e^{it}) dt.$$

Step 3 — multiply by R^2/R^2 . Numerator and denominator of the fraction, multiplied by R^2 , become $R^2 - \rho^2$ and $R^2 - 2R\rho \cos(\theta - t) + \rho^2$. Since $v(\zeta) = u(w + \rho e^{i\theta})$ and $v(e^{it}) = u(w + Re^{it})$, one obtains the formula of the statement.

Step 4 — the Dirichlet problem with datum g . Set $\tilde{g}(t) = g(w + Re^{it})$, continuous on $\mathbb{T}$. By Theorem 16.6 the function $P[\tilde{g}]$, extended by $\tilde{g}$, is the unique solution on $\overline{\mathbb{D}}$; then $z \mapsto P[\tilde{g}](\Psi^{-1}(z))$ is continuous on $\overline{D(w, R)}$, harmonic on $D(w, R)$ by Proposition 16.3 (applied to the holomorphic function $\Psi^{-1}(z) = (z - w)/R$) and equals g on the boundary; uniqueness is Corollary 16.7. Rewriting its integral as in Steps 2 and 3 gives the announced expression. $\square$

Theorem 16.7: The mean value property implies harmonicity. Let $\Omega \subseteq \mathbb{C}$ be open and let $u : \Omega \rightarrow \mathbb{R}$ be continuous with the mean value property (Definition 16.3). Then u is harmonic on Ω ; in particular $u \in C^\infty(\Omega)$.

Proof. **Step 1 — reduction to a disc.** Let $w \in \Omega$ and $R > 0$ such that $\overline{D(w, R)} \subset \Omega$ (this exists because Ω is open). It suffices to prove that u is harmonic on $D(w, R)$: since w is arbitrary, this gives harmonicity on Ω .

Step 2 — the Dirichlet solution with the same datum. The restriction of u to $\partial D(w, R)$ is continuous. By Corollary 16.9 (last statement) there exists h continuous on $\overline{D(w, R)}$, harmonic on $D(w, R)$, with $h = u$ on $\partial D(w, R)$.

Step 3 — the difference. Set $g = u - h$ on $\overline{D(w, R)}$. It is continuous; on $D(w, R)$ it has the mean value property, because u has it by hypothesis and h has it by Theorem 16.2, and the property is stable under subtraction (the integral is linear; at each point one takes the minimum of the two radii ρ_w). Moreover $g = 0$ on $\partial D(w, R)$.

Step 4 — maximum and minimum. $D(w, R)$ is a bounded domain; by Corollary 16.4 applied to g ,

$$\max_{\overline{D(w, R)}} g = \max_{\partial D(w, R)} g = 0, \quad \min_{\overline{D(w, R)}} g = \min_{\partial D(w, R)} g = 0,$$

hence $g \equiv 0$, that is, $u = h$ on $\overline{D(w, R)}$.

Step 5 — conclusion. u coincides with the harmonic function h on $D(w, R)$, hence is harmonic there; the C^∞ regularity is Corollary 16.1. $\square$

Corollary 16.10: Uniform limit of harmonic functions. Let (u_k) be a sequence of harmonic functions on an open set Ω converging to u uniformly on every compact subset of Ω . Then u is harmonic on Ω .

Proof. **Step 1 — continuity.** The uniform limit on every compact set of continuous functions is continuous on every compact set, so u is continuous on Ω (every point has a closed compact neighbourhood contained in Ω).

Step 2 — passing to the limit in the mean. Let $w \in \Omega$ and $\rho_w > 0$ with $\overline{D(w, \rho_w)} \subset \Omega$. For $0 < \rho < \rho_w$, Theorem 16.2 gives $u_k(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u_k(w + \rho e^{i\theta}) d\theta$. The convergence $u_k \rightarrow u$ is uniform on the compact set $\overline{D(w, \rho_w)}$, hence in particular on the circle $|z - w| = \rho$: the limit and the integral may be interchanged, giving $u(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(w + \rho e^{i\theta}) d\theta$.

Step 3 — conclusion. u is continuous with the mean value property; by Theorem 16.7 it is harmonic. $\square$

Proposition 16.5: Harmonic measure of a semicircle. Let $f = \mathbf{1}_{(0, \pi)}$ be the function equal to 1 for $t \in (0, \pi)$ and 0 for $t \in (-\pi, 0)$ (the value at the two points $t = 0, \pi$ is irrelevant for the integral). Then for every $z = re^{i\theta} \in \mathbb{D}$

$$P[f](re^{i\theta}) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{2r \sin \theta}{1 - r^2}\right) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{2 \operatorname{Im} z}{1 - |z|^2}\right).$$

Proof. **Step 1 — the case $r = 0$ and the cases $\theta \in \{0, \pi\}$.** For $r = 0$ one has $P_0 \equiv 1$ and $P[f](0) = \frac{1}{2\pi} \int_0^\pi dt = \frac{1}{2}$, which is the value given by the formula ($\arctan 0 = 0$). For $\theta = 0$ or

$\theta = \pi$ the right-hand side again equals $\frac{1}{2}$; so does the left-hand side, because the change of variable $t \mapsto -t$ (which swaps $(0, \pi)$ with $(-\pi, 0)$) leaves $P_r(\theta - t)$ unchanged when $\sin \theta = 0$, that is, when $\theta \equiv 0$ or $\theta \equiv \pi$: indeed $P_r(\theta + t) = P_r(\theta - t)$ for such θ , since $\cos(\theta + t) = \cos(\theta - t)$. Hence the two integrals over $(0, \pi)$ and over $(-\pi, 0)$ coincide and, adding to 1 by Lemma 16.8, each equals $\frac{1}{2}$.

Step 2 — integrating with the primitive, for $0 < \theta < \pi$ and $0 < r < 1$. With the substitution $s = \theta - t$,

$$P[f](re^{i\theta}) = \frac{1}{2\pi} \int_0^\pi P_r(\theta - t) dt = \frac{1}{2\pi} \int_{\theta-\pi}^\theta P_r(s) ds.$$

Since $0 < \theta < \pi$, the interval $[\theta - \pi, \theta]$ is contained in $(-\pi, \pi)$ (endpoints included only as $\theta \rightarrow 0$ or $\theta \rightarrow \pi$, cases already treated in Step 1); Lemma 16.7 and the fundamental theorem of calculus give, with $k = \frac{1+r}{1-r}$,

$$P[f](re^{i\theta}) = \frac{1}{\pi} \left[\arctan \left(k \tan \frac{\theta}{2} \right) - \arctan \left(k \tan \frac{\theta - \pi}{2} \right) \right].$$

Step 3 — simplifying the second arctangent. From $\tan \left(\frac{\theta}{2} - \frac{\pi}{2} \right) = -\cot \frac{\theta}{2}$ and the oddness of $\arctan$,

$$P[f](re^{i\theta}) = \frac{1}{\pi} \left[\arctan \left(k \tan \frac{\theta}{2} \right) + \arctan \left(k \cot \frac{\theta}{2} \right) \right].$$

Step 4 — sum of two arctangents. Set $a = k \tan \frac{\theta}{2} > 0$ and $b = k \cot \frac{\theta}{2} > 0$ (positive because $0 < \theta < \pi$), so that $ab = k^2 > 1$. For $a, b > 0$ with $ab > 1$ one has

$$\arctan a + \arctan b = \pi + \arctan \frac{a+b}{1-ab}.$$

Indeed: by the addition formula the tangent of the left-hand side equals $\frac{a+b}{1-ab}$, which is < 0 because $a+b > 0$ and $1-ab < 0$. Moreover $ab > 1$ gives $b > \frac{1}{a}$, hence $\arctan b > \arctan \frac{1}{a} = \frac{\pi}{2} - \arctan a$ and $\arctan a + \arctan b > \frac{\pi}{2}$; on the other hand each term is $< \frac{\pi}{2}$, so the sum lies in $(\frac{\pi}{2}, \pi)$. In that interval the only number with tangent $c < 0$ is $\pi + \arctan c$, because $\arctan c \in (-\frac{\pi}{2}, 0)$.

Step 5 — computing $\frac{a+b}{1-ab}$. One has

$$a+b = k \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right) = k \cdot \frac{1}{\sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \frac{2k}{\sin \theta},$$

and, as in Step 2 of the proof of Lemma 16.7, $1 - k^2 = -\frac{4r}{(1-r)^2}$, hence

$$\frac{a+b}{1-ab} = \frac{2k}{\sin \theta} \cdot \frac{(1-r)^2}{-4r} = -\frac{2}{\sin \theta} \cdot \frac{(1+r)(1-r)}{4r} = -\frac{1-r^2}{2r \sin \theta}.$$

Step 6 — conclusion. Substituting into Steps 3–5,

$$P[f](re^{i\theta}) = \frac{1}{\pi} \left[\pi - \arctan \frac{1-r^2}{2r \sin \theta} \right] = 1 - \frac{1}{\pi} \arctan \frac{1-r^2}{2r \sin \theta}.$$

For $X > 0$ one has $\arctan X + \arctan \frac{1}{X} = \frac{\pi}{2}$ (the two tangents are reciprocal and the sum lies in $(0, \pi)$ with infinite tangent); with $X = \frac{1-r^2}{2r \sin \theta} > 0$ one obtains

$$P[f](re^{i\theta}) = 1 - \frac{1}{\pi} \left(\frac{\pi}{2} - \arctan \frac{2r \sin \theta}{1-r^2} \right) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{2r \sin \theta}{1-r^2}.$$

Step 7 — the lower half-plane. Let $0 < \theta < \pi$; compute $P[f](re^{-i\theta})$. With the substitution $t = -s$ and using that P_r is even,

$$P[f](re^{-i\theta}) = \frac{1}{2\pi} \int_0^\pi P_r(-\theta - t) dt = \frac{1}{2\pi} \int_{-\pi}^0 P_r(-\theta + s) ds = \frac{1}{2\pi} \int_{-\pi}^0 P_r(\theta - s) ds.$$

By Lemma 16.8 (Step 4) one has $\frac{1}{2\pi} \int_{-\pi}^\pi P_r(\theta - s) ds = 1$, hence the last integral equals $1 - \frac{1}{2\pi} \int_0^\pi P_r(\theta - s) ds$, that is,

$$P[f](re^{-i\theta}) = 1 - P[f](re^{i\theta}).$$

The right-hand side of the formula to be proved changes in the same way, because $\sin(-\theta) = -\sin \theta$ and $\arctan$ is odd. The formula therefore also holds for $\theta \in (-\pi, 0)$, and by Step 1 on all of $\mathbb{D}$. Finally $r \sin \theta = \operatorname{Im} z$ and $r^2 = |z|^2$. $\square$

Example 16.6: Three boundary data solved and checked. (a) $f(\theta) = \cos^2 \theta$. From $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$ and Example 16.1(b), the function $u(re^{i\theta}) = \frac{1}{2} + \frac{1}{2} r^2 \cos 2\theta$ is harmonic on $\mathbb{D}$, continuous on $\bar{\mathbb{D}}$ and equals f for $r = 1$: by Theorem 16.6 it is *the* solution. At $r = 0.6$ and $\theta = \pi/3$:

$$u = \frac{1 + 0.36 \cdot \cos \frac{2\pi}{3}}{2} = \frac{1 - 0.18}{2} = 0.4100000.$$

Independent check: Simpson quadrature of the Poisson integral with 200,000 subintervals gives 0.4100000. $\checkmark$

(b) $f(\theta) = \cos^3 \theta$. From $\cos^3 \theta = \frac{3}{4} \cos \theta + \frac{1}{4} \cos 3\theta$ (triple-angle formula $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$) one obtains $u(re^{i\theta}) = \frac{3}{4} r \cos \theta + \frac{1}{4} r^3 \cos 3\theta$. At $r = 0.8$, $\theta = 0.7$:

$$u = 0.75 \cdot 0.8 \cdot \cos 0.7 + 0.25 \cdot 0.512 \cdot \cos 2.1 = 0.4589053 - 0.0646203 = 0.3942850.$$

Independent check: Simpson quadrature of the Poisson integral, 0.3942850. $\checkmark$

(c) The discontinuous semicircle datum. With $f = \mathbf{1}_{(0,\pi)}$, Proposition 16.5 gives at $r = 0.5$, $\theta = \pi/4$:

$$\frac{1}{2} + \frac{1}{\pi} \arctan \frac{2 \cdot 0.5 \cdot 0.7071068}{1 - 0.25} = \frac{1}{2} + \frac{\arctan(0.9428090)}{\pi} = 0.7406325.$$

Independent check: Simpson quadrature of $\frac{1}{2\pi} \int_0^\pi P_{0.5}(\pi/4 - t) dt$, 0.7406325. $\checkmark$ A second point: $r = 0.8$, $\theta = -0.3$ gives 0.2071363 by both routes. $\checkmark$

16.6. Harnack and Liouville

The Poisson formula does not only serve to construct solutions: since the kernel is *positive* and bounded above and below on every interior circle (Lemma 16.6), it automatically produces a two-sided estimate for nonnegative harmonic functions. This is Harnack's inequality, and from it follows a Liouville theorem stronger than the holomorphic one.

Theorem 16.8: Harnack's inequality in the disc. Let $u \geq 0$ be harmonic on $D(w, R)$. Then for every z with $|z - w| = \rho < R$,

$$\frac{R - \rho}{R + \rho} u(w) \leq u(z) \leq \frac{R + \rho}{R - \rho} u(w).$$

Proof. Step 1 — representation on an interior circle. Let $\rho < s < R$. The function u is harmonic on the open set $D(w, R)$, which contains $\overline{D(w, s)}$; by Corollary 16.9, writing $z = w + \rho e^{i\theta}$,

$$u(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{s^2 - \rho^2}{s^2 - 2s\rho \cos(\theta - t) + \rho^2} u(w + se^{it}) dt.$$

Step 2 — bounds on the kernel. As in Step 1 of the proof of Lemma 16.6, with s and ρ in place of 1 and r ,

$$s^2 - 2s\rho \cos \alpha + \rho^2 = (s - \rho)^2 + 2s\rho(1 - \cos \alpha) = (s + \rho)^2 - 2s\rho(1 + \cos \alpha),$$

hence $(s - \rho)^2 \leq s^2 - 2s\rho \cos \alpha + \rho^2 \leq (s + \rho)^2$ and, dividing by $s^2 - \rho^2 = (s - \rho)(s + \rho) > 0$,

$$\frac{s - \rho}{s + \rho} \leq \frac{s^2 - \rho^2}{s^2 - 2s\rho \cos \alpha + \rho^2} \leq \frac{s + \rho}{s - \rho}.$$

Step 3 — using $u \geq 0$. Since $u(w + se^{it}) \geq 0$, multiplying the inequalities of Step 2 by this nonnegative number and integrating (integration preserves inequalities between integrable functions) gives

$$\frac{s - \rho}{s + \rho} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} u(w + se^{it}) dt \leq u(z) \leq \frac{s + \rho}{s - \rho} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} u(w + se^{it}) dt.$$

Step 4 — the mean equals $u(w)$. By Theorem 16.2 the normalized integral equals $u(w)$. Hence

$$\frac{s - \rho}{s + \rho} u(w) \leq u(z) \leq \frac{s + \rho}{s - \rho} u(w).$$

Step 5 — letting $s \rightarrow R$. The two functions $s \mapsto \frac{s - \rho}{s + \rho}$ and $s \mapsto \frac{s + \rho}{s - \rho}$ are continuous on $(\rho, R]$; letting $s \rightarrow R^-$ in the inequalities (which hold for every $s \in (\rho, R)$) gives the statement. $\square$

Corollary 16.11: Harnack on compact sets. Let Ω be a domain, $z_0 \in \Omega$ and $K \subset \Omega$ compact. There exists a constant $C = C(K, z_0, \Omega) \geq 1$ such that for every function $u \geq 0$ harmonic on Ω and every $z \in K$,

$$\frac{1}{C} u(z_0) \leq u(z) \leq C u(z_0).$$

Proof. Step 1 — a universal local estimate. Let $w \in \Omega$ and $\rho > 0$ with $\overline{D(w, 4\rho)} \subset \Omega$. Let $u \geq 0$ be harmonic on Ω and let $z \in \overline{D(w, \rho)}$, that is, $|z - w| = \rho' \leq \rho$. Theorem 16.8 applied to the disc $D(w, 4\rho)$ gives

$$\frac{4\rho - \rho'}{4\rho + \rho'} u(w) \leq u(z) \leq \frac{4\rho + \rho'}{4\rho - \rho'} u(w).$$

The function $\sigma \mapsto \frac{4\rho + \sigma}{4\rho - \sigma}$ is increasing on $[0, \rho]$, so the right-hand factor is $\leq \frac{5\rho}{3\rho} = \frac{5}{3}$; for the same reason the left-hand factor is $\geq \frac{3\rho}{5\rho} = \frac{3}{5}$. Hence

$$\frac{3}{5} u(w) \leq u(z) \leq \frac{5}{3} u(w) \quad \text{for every } z \in \overline{D(w, \rho)}.$$

We shall call $(*)$ this estimate: it holds for every $u \geq 0$ harmonic on Ω and for every pair w, ρ with $\overline{D(w, 4\rho)} \subset \Omega$.

Step 2 — the set of points comparable with z_0 . Let

$$A = \left\{ z \in \Omega : \exists C_z \geq 1 \text{ such that } C_z^{-1} u(z_0) \leq u(z) \leq C_z u(z_0) \text{ for every } u \geq 0 \text{ harmonic on } \Omega \right\}.$$

One has $z_0 \in A$ with $C_{z_0} = 1$, so $A \neq \emptyset$.

Step 3 — A is open. Let $z \in A$. Since Ω is open there exists $\rho > 0$ with $\overline{D(z, 4\rho)} \subset \Omega$. By (*) with $w = z$, every $\zeta \in D(z, \rho)$ satisfies $\frac{3}{5}u(z) \leq u(\zeta) \leq \frac{5}{3}u(z)$, and hence

$$u(\zeta) \leq \frac{5}{3}u(z) \leq \frac{5}{3}C_z u(z_0), \quad u(\zeta) \geq \frac{3}{5}u(z) \geq \frac{3}{5}C_z^{-1}u(z_0),$$

that is, $\zeta \in A$ with $C_\zeta = \frac{5}{3}C_z$. Hence $D(z, \rho) \subseteq A$.

Step 4 — A is closed in Ω . Let $z \in \Omega$ be an accumulation point of A . Choose $\rho > 0$ with $\overline{D(z, 4\rho)} \subset \Omega$ and then $\zeta \in A \cap D(z, \rho)$. Again by (*) with $w = z$, $\frac{3}{5}u(z) \leq u(\zeta) \leq \frac{5}{3}u(z)$, whence

$$u(z) \leq \frac{5}{3}u(\zeta) \leq \frac{5}{3}C_\zeta u(z_0), \quad u(z) \geq \frac{3}{5}u(\zeta) \geq \frac{3}{5}C_\zeta^{-1}u(z_0),$$

that is, $z \in A$.

Step 5 — connectedness. A is nonempty, open and closed in the connected set Ω : $A = \Omega$.

Step 6 — compactness. For every $w \in K$ choose $\rho_w > 0$ with $\overline{D(w, 4\rho_w)} \subset \Omega$. The open discs $D(w, \rho_w)$, $w \in K$, cover K : by compactness finitely many suffice, $D(w_1, \rho_{w_1}), \dots, D(w_m, \rho_{w_m})$. Every $z \in K$ lies in some $D(w_j, \rho_{w_j})$, and by (*) and Step 5 (which gives $w_j \in A$ with constant C_{w_j}),

$$u(z) \leq \frac{5}{3}u(w_j) \leq \frac{5}{3}C_{w_j}u(z_0), \quad u(z) \geq \frac{3}{5}u(w_j) \geq \frac{3}{5}C_{w_j}^{-1}u(z_0).$$

It then suffices to set $C = \frac{5}{3} \max_{1 \leq j \leq m} C_{w_j}$. □

Theorem 16.9: Liouville for bounded harmonic functions. Let u be harmonic on all of $\mathbb{C}$ and bounded. Then u is constant.

Proof. **Step 1 — reduction to a nonnegative function.** Let $m = \inf_{\mathbb{C}} u$, finite because u is bounded. The function $v = u - m$ is harmonic on $\mathbb{C}$ by Proposition 16.1(i) (difference of u and the constant m) and satisfies $v \geq 0$.

Step 2 — Harnack on ever larger discs. Let $z \in \mathbb{C}$ and $R > |z|$. The function v is harmonic and nonnegative on $D(0, R)$; Theorem 16.8 with $w = 0$ and $\rho = |z|$ gives

$$\frac{R - |z|}{R + |z|} v(0) \leq v(z) \leq \frac{R + |z|}{R - |z|} v(0).$$

Step 3 — letting R tend to infinity. For fixed z ,

$$\lim_{R \rightarrow +\infty} \frac{R - |z|}{R + |z|} = 1 = \lim_{R \rightarrow +\infty} \frac{R + |z|}{R - |z|},$$

so passing to the limit in the two inequalities of Step 2 gives $v(0) \leq v(z) \leq v(0)$, that is, $v(z) = v(0)$.

Step 4 — conclusion. z was arbitrary, so v is constant and so is $u = v + m$. □

Corollary 16.12: A one-sided bound suffices. Let u be harmonic on all of $\mathbb{C}$ and bounded above (or below). Then u is constant.

Proof. **Step 1 — the case “bounded above”.** Let $u \leq M$ on $\mathbb{C}$. The function $v = M - u$ is harmonic (Proposition 16.1(i)) and nonnegative. Steps 2–4 of the proof of Theorem 16.9 use of v only nonnegativity and harmonicity on the whole plane: applied to this v they give v constant, hence $u = M - v$ constant.

Step 2 — the case “bounded below”. Apply Step 1 to $-u$, harmonic and bounded above. □

Remark 16.4: Two Liouville theorems compared. Theorem 15.8 states that an entire function bounded in modulus is constant; Corollary 16.12 states much more for harmonic functions: a bound on one side alone suffices. The difference is explained as follows: if u is harmonic on $\mathbb{C}$ and $u \leq M$, by Theorem 16.1 there exists an entire function f with $\operatorname{Re} f = u$, and then e^f is entire with $|e^f| = e^u \leq e^M$: Theorem 15.8 can therefore be applied to e^f to conclude that e^f is constant, hence that $u = \log |e^f|$ is constant. This is a second proof, independent of Harnack, of Corollary 16.12. The proof given above is preferable, however, because it does not use the construction of the conjugate on the whole plane and adapts, word for word, to harmonic functions in $\mathbb{R}^n$.

Proposition 16.6: Harnack's principle for monotone sequences. Let Ω be a domain and let $(u_k)_{k \geq 1}$ be a sequence of harmonic functions on Ω with $u_k \leq u_{k+1}$ for every k . Then exactly one of the following two alternatives holds:

- (a) $u_k \rightarrow +\infty$ uniformly on every compact subset of Ω ;
- (b) there exists u harmonic on Ω such that $u_k \rightarrow u$ uniformly on every compact subset of Ω .

Proof. Step 1 — the reference point. Fix $z_0 \in \Omega$. The numerical sequence $(u_k(z_0))$ is nondecreasing, so it either converges to a finite limit L or diverges to $+\infty$. The two cases are mutually exclusive and exhaust all possibilities.

Step 2 — the differences are harmonic and nonnegative. For $n \geq m$ the function $v_{n,m} = u_n - u_m$ is harmonic on Ω (Proposition 16.1(i)) and nonnegative, by monotonicity.

Step 3 — case $L < +\infty$: uniform convergence on compact sets. Let $K \subset \Omega$ be compact; replacing K by $K \cup \{z_0\}$, still compact, it is no loss of generality to assume $z_0 \in K$. Let $C \geq 1$ be the constant of Corollary 16.11 relative to K , z_0 and Ω . Applying it to $v_{n,m}$:

$$\sup_{z \in K} (u_n(z) - u_m(z)) \leq C(u_n(z_0) - u_m(z_0)).$$

Since $(u_k(z_0))$ converges, it is Cauchy: given $\varepsilon > 0$ there exists N with $u_n(z_0) - u_m(z_0) < \varepsilon/C$ for $n \geq m \geq N$; then $\sup_K |u_n - u_m| < \varepsilon$. The sequence is thus uniformly Cauchy on K and converges uniformly on K to a function u (the uniform Cauchy criterion). Since K is arbitrary, the convergence is uniform on compact sets, and by Corollary 16.10 the limit u is harmonic.

Step 4 — case $L = +\infty$: uniform divergence on compact sets. Let $K \subset \Omega$ be compact, again with $z_0 \in K$, and C as above. Applying the left-hand inequality of Corollary 16.11 to $v_{n,1} = u_n - u_1 \geq 0$:

$$u_n(z) - u_1(z) \geq \frac{1}{C}(u_n(z_0) - u_1(z_0)) \quad \text{for every } z \in K.$$

The function u_1 is continuous on the compact set K , hence bounded below by some $-A$; so for every $z \in K$

$$u_n(z) \geq -A + \frac{1}{C}(u_n(z_0) - u_1(z_0)) \xrightarrow{n \rightarrow \infty} +\infty,$$

and the lower bound does not depend on z : the divergence is uniform on K . $\square$

Example 16.7: Harnack is optimal. Let $u(z) = P(z, 1) = \frac{1 - |z|^2}{|1 - z|^2}$, which by Lemma 16.10 is harmonic on $\mathbb{D}$ and by Lemma 16.6 is positive, with $u(0) = 1$.

Theorem 16.8 with $R = 1$, $w = 0$ predicts, for $|z| = r$,

$$\frac{1 - r}{1 + r} \leq u(z) \leq \frac{1 + r}{1 - r}.$$

But $u(re^{i\theta}) = P_r(\theta)$ and Lemma 16.6 says that *both* extremes are attained, at $\theta = \pi$ and at $\theta = 0$: Harnack's inequality cannot be improved.

Numerical check. For $r = 0.5$ the two theoretical extremes are 0.3333333 and 3.0000000; sampling $P_{0.5}(\theta)$ at 20,000 equally spaced points of $[0, 2\pi)$ gives minimum 0.3333333 and maximum 3.0000000. ✓ For $r = 0.9$ the theoretical extremes are 0.0526316 and 19.0000000, and the sampling returns 0.0526316 and 19.0000000. ✓

16.6.1 Gradient estimate, a second route to Liouville, reflection

The Poisson formula also controls the *derivatives* of a harmonic function using only the supremum of its modulus: this is the analogue of Cauchy's estimates (Theorem 15.7), and from it follows a second proof of Liouville's theorem.

Proposition 16.7: Gradient estimate. *Let u be harmonic on $D(w, R)$ with $|u| \leq M$ on $D(w, R)$. Then*

$$|\nabla u(w)| = \sqrt{u_x(w)^2 + u_y(w)^2} \leq \frac{2M}{R}.$$

Proof. Step 1 — working on an interior circle. Let $0 < \rho < R$ and $f_\rho(t) = u(w + \rho e^{it})$, a continuous function on $\mathbb{T}$ with $|f_\rho| \leq M$.

Step 2 — the associated holomorphic function. By Theorem 16.4 there exists G holomorphic on $\mathbb{D}$ with

$$G(\zeta) = c_0 + 2 \sum_{n \geq 1} c_n \zeta^n, \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f_\rho(t) e^{-int} dt, \quad |c_n| \leq M,$$

and $\operatorname{Re} G = P[f_\rho]$ on $\mathbb{D}$. By Corollary 16.9, $P[f_\rho](\zeta) = u(w + \rho\zeta)$ for $\zeta \in \mathbb{D}$, hence

$$\operatorname{Re} G(\zeta) = u(w + \rho\zeta), \quad \zeta \in \mathbb{D}.$$

Step 3 — differentiating at $\zeta = 0$. Set $\zeta = \xi + i\eta$ and $U(\xi, \eta) = \operatorname{Re} G(\zeta) = u(w + \rho\zeta)$. By Lemma 15.1, $G' = U_\xi - iU_\eta$. On the other hand the chain rule gives $U_\xi(0, 0) = \rho u_x(w)$ and $U_\eta(0, 0) = \rho u_y(w)$, hence

$$G'(0) = \rho(u_x(w) - i u_y(w)), \quad |G'(0)| = \rho |\nabla u(w)|.$$

Step 4 — reading $G'(0)$ off the series. From the expansion of Step 2, $G'(0) = 2c_1$, and $|c_1| \leq M$: hence $\rho |\nabla u(w)| = 2|c_1| \leq 2M$, that is, $|\nabla u(w)| \leq \frac{2M}{\rho}$.

Step 5 — letting ρ tend to R . The inequality holds for every $\rho \in (0, R)$; passing to the limit $\rho \rightarrow R^-$ gives $|\nabla u(w)| \leq \frac{2M}{R}$. $\square$

Corollary 16.13: Liouville, second proof. *Let u be harmonic on all of $\mathbb{C}$ and bounded, $|u| \leq M$. Then u is constant.*

Proof. Step 1 — the gradient vanishes. Let $w \in \mathbb{C}$. For every $R > 0$ the function u is harmonic on $D(w, R)$ with $|u| \leq M$; Proposition 16.7 gives $|\nabla u(w)| \leq \frac{2M}{R}$. Letting $R \rightarrow +\infty$ gives $\nabla u(w) = 0$.

Step 2 — integrating. $\nabla u \equiv 0$ on $\mathbb{C}$, which is convex; as in Step 4 of the proof of Theorem 16.1, the restriction of u to every segment is constant, so u is constant on $\mathbb{C}$. $\square$

Example 16.8: The gradient estimate on a computable case. Let $u = P[\mathbf{1}_{(0,\pi)}]$, that is, by Proposition 16.5, $u(x, y) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{2y}{1-x^2-y^2}$ on $\mathbb{D}$. One has $0 \leq u \leq 1$, so $M = 1$ and $R = 1$: Proposition 16.7 predicts $|\nabla u(0)| \leq 2$.

Direct computation. With $g(x, y) = \frac{2y}{1-x^2-y^2}$ one has $g(0, 0) = 0$, $g_x(0, 0) = 0$ and $g_y(0, 0) = 2$; since $\frac{d}{ds} \arctan s = \frac{1}{1+s^2}$ equals 1 at $s = 0$,

$$u_x(0, 0) = 0, \quad u_y(0, 0) = \frac{2}{\pi} = 0.6366198, \quad |\nabla u(0)| = 0.6366198 \leq 2. \quad \checkmark$$

Independent check with the Fourier coefficients. The datum $\mathbf{1}_{(0,\pi)}$ has $a_n = 0$ for $n \geq 1$, $\frac{a_0}{2} = \frac{1}{2}$ and $b_n = \frac{1}{\pi} \int_0^\pi \sin nt \, dt = \frac{1-(-1)^n}{\pi n}$, that is, $b_n = \frac{2}{\pi n}$ for n odd and 0 for n even. By Theorem 16.11,

$$u(re^{i\theta}) = \frac{1}{2} + \frac{2}{\pi} \sum_{k \text{ odd}} \frac{r^k \sin k\theta}{k},$$

whose only linear term is $\frac{2}{\pi} r \sin \theta = \frac{2}{\pi} y$: hence $u_y(0) = \frac{2}{\pi}$ and $u_x(0) = 0$, as above. $\checkmark$ Note also $c_1 = \frac{a_1 - ib_1}{2} = -\frac{i}{\pi}$, so $2|c_1| = \frac{2}{\pi}$: Steps 3 and 4 of the proof of Proposition 16.7 in fact give the equality $|\nabla u(0)| = 2|\hat{f}(1)|$, and the bound $|\hat{f}(1)| \leq M$ is the only point where something is lost.

How much is lost. For a real datum with $|f| \leq M$,

$$|\hat{f}(1)| = \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-it} \, dt \right|,$$

and rotating the datum one may assume that the maximum is attained on the cosine part: $\frac{1}{2\pi} \int f(t) \cos t \, dt \leq \frac{M}{2\pi} \int_{-\pi}^{\pi} |\cos t| \, dt = \frac{M}{2\pi} \cdot 4 = \frac{2M}{\pi}$, with equality for $f(t) = M \operatorname{sgn}(\cos t)$. Hence for every real datum bounded by M one has $|\nabla u(0)| \leq \frac{4M}{\pi} = 1.2732395 M$, and this constant is attained: the estimate $2M/R$ of Proposition 16.7 is correct but not optimal, and the ratio of the two constants is $\frac{2}{4/\pi} = \frac{\pi}{2} = 1.5707963$. $\checkmark$

Theorem 16.10: Schwarz's reflection principle for harmonic functions. Let $\Omega \subseteq \mathbb{C}$ be an open set symmetric with respect to the real axis (that is, $z \in \Omega \Rightarrow \bar{z} \in \Omega$), and set

$$\Omega^+ = \Omega \cap \{y > 0\}, \quad I = \Omega \cap \mathbb{R}, \quad \Omega^- = \Omega \cap \{y < 0\}.$$

Let u be harmonic on Ω^+ , continuous on $\Omega^+ \cup I$ and zero on I . Then the function

$$\tilde{u}(x, y) = \begin{cases} u(x, y), & (x, y) \in \Omega^+, \\ 0, & (x, y) \in I, \\ -u(x, -y), & (x, y) \in \Omega^-, \end{cases}$$

is harmonic on Ω .

Proof. Step 1 — $\tilde{u}$ is harmonic away from I and is continuous. On Ω^+ it coincides with u . On Ω^- it equals $-u \circ \sigma$ with $\sigma(x, y) = (x, -y)$; if v is harmonic then $v \circ \sigma$ is too, because $(v \circ \sigma)_{xx} = v_{xx} \circ \sigma$ and $(v \circ \sigma)_{yy} = v_{yy} \circ \sigma$ (the second derivative with respect to y picks up the sign -1 twice), and class C^2 is preserved; by Proposition 16.1(i) its negative is also harmonic. Continuity on Ω follows from continuity of u on $\Omega^+ \cup I$ and from u vanishing on I : approaching a point of I from above and from below, the values tend to 0 and to $-0 = 0$.

Step 2 — reduction to a disc centred on I . Let $x_0 \in I$ and $r > 0$ with $D = D(x_0, r) \subseteq \Omega$ (possible because Ω is open; replacing r , if necessary, by a smaller value one may also assume

$\bar{D} \subset \Omega$). It suffices to prove that $\tilde{u}$ is harmonic on D : at points away from I it already is, by Step 1.

Step 3 — the Dirichlet solution on D . The restriction of $\tilde{u}$ to ∂D is continuous by Step 1. By Corollary 16.9 there exists h continuous on $\bar{D}$, harmonic on D , with $h = \tilde{u}$ on ∂D .

Step 4 — h is odd with respect to I . Set $k(x, y) = -h(x, -y)$, that is, $k = -h \circ \sigma$ with $\sigma(x, y) = (x, -y)$: the reflection σ maps D to itself, because D is centred at a point of the real axis. By Step 1 (where it was seen that $v \circ \sigma$ is harmonic if v is) k is harmonic on D and continuous on $\bar{D}$. On the boundary, for every $(x, y) \in \partial D$,

$$k(x, y) = -h(x, -y) = -\tilde{u}(x, -y) = \tilde{u}(x, y),$$

the last equality because $\tilde{u}$ is odd with respect to y by construction. Hence k and h solve the same Dirichlet problem on D and, by Corollary 16.7, $k = h$, that is, $h(x, -y) = -h(x, y)$. In particular $h = 0$ on the diameter $D \cap \mathbb{R}$.

Step 5 — comparison on the upper half-disc. Let $D^+ = D \cap \{y > 0\}$, which is a bounded domain with $\partial D^+ = (\partial D \cap \{y \geq 0\}) \cup (\bar{D} \cap \mathbb{R})$. The functions $\tilde{u}$ and h are continuous on $\bar{D}^+$ and harmonic on D^+ ($\tilde{u}$ by Step 1, since $D^+ \subseteq \Omega^+$). On the boundary of D^+ they agree: on the arc because $h = \tilde{u}$ there by Step 3, on the real segment because both equal 0 (Step 1 for $\tilde{u}$, Step 4 for h). By Corollary 16.7, $\tilde{u} = h$ on $\bar{D}^+$.

Step 6 — and on the lower one. Since both $\tilde{u}$ and h are odd with respect to y (the first by construction, the second by Step 4), the equality of Step 5 carries over to $\bar{D}^-$: for $y < 0$, $\tilde{u}(x, y) = -\tilde{u}(x, -y) = -h(x, -y) = h(x, y)$.

Step 7 — conclusion. $\tilde{u} = h$ on $\bar{D}^+ \cup \bar{D}^- = \bar{D}$, and h is harmonic on D : hence $\tilde{u}$ is harmonic on D , and in particular at the points of $I \cap D$. Since $x_0 \in I$ was arbitrary and away from I harmonicity is Step 1, $\tilde{u}$ is harmonic on all of Ω . $\square$

***Remark 16.5: The holomorphic twin of reflection.** Theorem 16.10 is the “harmonic” half of the reflection principle; the holomorphic half states that a function holomorphic on Ω^+ , continuous up to I and real-valued there, extends holomorphically to Ω by setting $f(\bar{z}) = \overline{f(z)}$. The two versions are deduced from each other on simply connected domains via Theorem 16.1; Ahlfors treats both in Chapter 4, §6.5, “The Reflection Principle.” Here only the harmonic version has been proved, which is the one the chapter needs and which requires no hypothesis of simple connectedness: the extension is constructed disc by disc, and every disc is convex.*

16.7. The Fourier series expansion of the solution

The Poisson kernel was born, in Definition 16.4, as a series of characters e^{int} . Bringing that series back inside the Poisson integral turns the Poisson formula into a Fourier expansion, and opens the door to Chapter 17.

Definition 16.7: Fourier coefficients. Let $f : \mathbb{T} \rightarrow \mathbb{C}$ be bounded and Riemann integrable. For $n \in \mathbb{Z}$ set

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt,$$

and, for f real-valued,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt \quad (n \geq 0), \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt \quad (n \geq 1).$$

One has $|\hat{f}(n)| \leq \sup |f|$, and for f real $\hat{f}(n) = \frac{1}{2}(a_n - ib_n)$ for $n \geq 1$, $\hat{f}(0) = \frac{1}{2}a_0$ and $\hat{f}(-n) = \overline{\hat{f}(n)}$.

Theorem 16.11: Series expansion of the solution of the Dirichlet problem. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be bounded and Riemann integrable. Then, for every $0 \leq r < 1$ and $\theta \in \mathbb{R}$,

$$P[f](re^{i\theta}) = \sum_{n \in \mathbb{Z}} \hat{f}(n) r^{|n|} e^{in\theta} = \frac{a_0}{2} + \sum_{n \geq 1} r^n (a_n \cos n\theta + b_n \sin n\theta),$$

and the series converges absolutely, and uniformly in θ for $r \leq \rho < 1$.

Proof. Step 1 — uniform convergence of the integrand. Let $M = \sup |f|$. By Definition 16.4 and the Weierstrass criterion, the series $\sum_{n \in \mathbb{Z}} r^{|n|} e^{in(\theta-t)} f(t)$ converges uniformly in $t \in [-\pi, \pi]$, because the modulus of the n -th term is $\leq Mr^{|n|}$ and $\sum_{n \in \mathbb{Z}} Mr^{|n|} = M \frac{1+r}{1-r} < \infty$.
Step 2 — term-by-term integration. By Definition 16.6 and Definition 16.4,

$$P[f](re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{n \in \mathbb{Z}} r^{|n|} e^{in(\theta-t)} \right) f(t) dt = \sum_{n \in \mathbb{Z}} r^{|n|} e^{in\theta} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt,$$

the interchange being legitimate by Step 1. The last integral is $\hat{f}(n)$ by Definition 16.7.

Step 3 — absolute and uniform convergence. Since $|\hat{f}(n)r^{|n|}e^{in\theta}| \leq M\rho^{|n|}$ for $r \leq \rho$, the Weierstrass criterion gives absolute and uniform convergence in θ on $\{r \leq \rho\}$.

Step 4 — real form. Group the terms n and $-n$ for $n \geq 1$. Since f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$ (Definition 16.7), hence

$$\hat{f}(n)r^n e^{in\theta} + \hat{f}(-n)r^n e^{-in\theta} = 2r^n \operatorname{Re} \left(\hat{f}(n)e^{in\theta} \right).$$

With $\hat{f}(n) = \frac{1}{2}(a_n - ib_n)$ and $e^{in\theta} = \cos n\theta + i \sin n\theta$,

$$2 \operatorname{Re} \left(\frac{1}{2}(a_n - ib_n)(\cos n\theta + i \sin n\theta) \right) = a_n \cos n\theta + b_n \sin n\theta.$$

The term $n = 0$ is $\hat{f}(0) = \frac{a_0}{2}$. □

Corollary 16.14: The mean on the circle of radius r . Under the hypotheses of Theorem 16.11, for every $0 \leq r < 1$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} P[f](re^{i\theta}) d\theta = \hat{f}(0) = P[f](0).$$

The mean of the solution on every circle concentric with the boundary is the same, and coincides with the mean of the datum.

Proof. Step 1 — integrate the series. By Step 3 of the proof of Theorem 16.11 the series converges uniformly in θ (at fixed r), so it can be integrated term by term:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} P[f](re^{i\theta}) d\theta = \sum_{n \in \mathbb{Z}} \hat{f}(n)r^{|n|} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} d\theta.$$

Step 2 — orthogonality. As in Step 2 of the proof of Lemma 16.8, the integral equals 1 for $n = 0$ and 0 otherwise. What remains is $\hat{f}(0)$.

Step 3 — the value at the center. For $r = 0$ the series reduces to the single term $n = 0$: $P[f](0) = \hat{f}(0)$. (This is also Theorem 16.2 applied to $P[f]$, which is harmonic by Theorem 16.4.) $\square$

Proposition 16.8: Expansion of an arbitrary harmonic function on the disc. Let u be harmonic on $\mathbb{D}$ (with no hypothesis at the boundary). Then there exist $\alpha_0 \in \mathbb{R}$ and $\alpha_n, \beta_n \in \mathbb{R}$ ($n \geq 1$) such that

$$u(re^{i\theta}) = \alpha_0 + \sum_{n \geq 1} r^n (\alpha_n \cos n\theta + \beta_n \sin n\theta),$$

with absolute and uniform convergence on every disc $\{|z| \leq \rho\}$ with $\rho < 1$. The coefficients are uniquely determined by u : for every $0 < r < 1$,

$$\alpha_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(re^{i\theta}) d\theta, \quad \alpha_n r^n = \frac{1}{\pi} \int_{-\pi}^{\pi} u(re^{i\theta}) \cos n\theta d\theta,$$

and likewise for β_n with $\sin n\theta$.

Proof. **Step 1 — pass to the holomorphic function.** $\mathbb{D}$ is convex; by Theorem 16.1 there is F holomorphic on $\mathbb{D}$ with $\operatorname{Re} F = u$.

Step 2 — Taylor series. By Theorem 15.6, F is the sum of its own Taylor series at 0, $F(z) = \sum_{n \geq 0} c_n z^n$, with uniform convergence on compact subsets of $\mathbb{D}$, in particular on $\{|z| \leq \rho\}$.

Step 3 — take the real part. With $c_n = \alpha_n + i\gamma_n$ ($\alpha_n, \gamma_n \in \mathbb{R}$) and $z^n = r^n(\cos n\theta + i \sin n\theta)$,

$$\operatorname{Re}(c_n z^n) = r^n (\alpha_n \cos n\theta - \gamma_n \sin n\theta),$$

and setting $\beta_n = -\gamma_n$ and $\alpha_0 = \operatorname{Re} c_0$ gives the expansion of the statement. Absolute and uniform convergence is inherited from that of the power series, because $|\operatorname{Re} w| \leq |w|$.

Step 4 — uniqueness of the coefficients. Fix $0 < r < 1$; the series converges uniformly in θ , so it can be multiplied by $\cos m\theta$ (a bounded function) and integrated term by term. By the orthogonality relations

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \cos n\theta \cos m\theta d\theta = \delta_{nm}, \quad \frac{1}{\pi} \int_{-\pi}^{\pi} \sin n\theta \cos m\theta d\theta = 0 \quad (n, m \geq 1)$$

— which follow from the product-to-sum formulas $\cos A \cos B = \frac{1}{2}(\cos(A-B) + \cos(A+B))$ and $\sin A \cos B = \frac{1}{2}(\sin(A-B) + \sin(A+B))$ and from the fact that $\int_{-\pi}^{\pi} \cos k\theta d\theta = 0$ and $\int_{-\pi}^{\pi} \sin k\theta d\theta = 0$ for $k \neq 0$ — only the term $n = m$ survives, giving the formula for $\alpha_m r^m$; the term α_0 is isolated by integrating with no extra factor. $\square$

Remark 16.6: Bridge to Chapter 17: Abel means. Theorem 16.11 says that $P[f](re^{i\theta})$ is the Abel mean of parameter r of the Fourier series of f : one takes the formal series $\sum_n \hat{f}(n)e^{in\theta}$, damps it with the factor $r^{|n|}$, and the result converges — by Theorem 16.5 — to $f(\theta)$ at every point of continuity, and uniformly if f is continuous. This is a summability result, not a convergence result: nothing here says that the partial sums $\sum_{|n| \leq N} \hat{f}(n)e^{in\theta}$ converge.

Chapter 17 addresses exactly this distinction, with the Dirichlet kernel (the partial sums) and the Fejér kernel (the Cesàro means), which, like P_r , is an approximate identity in the sense of Definition 16.5, while the Dirichlet kernel is not — which is why the pointwise convergence of the partial sums is a delicate problem.

Two related results remain **imported** and are not used at any point in the logical chain of this chapter: Fatou's theorem (a bounded harmonic function on the disc has a radial limit at almost every boundary point) and Carleson's theorem (the Fourier series of a square-summable function converges almost everywhere). Both require Lebesgue measure theory, which the treatise does not develop; they are cited here as panorama and no statement of this chapter rests on them. For Fatou's theorem the source is Axler, Bourdon and Ramey, Ch. 6 "Harmonic Hardy Spaces," section "The Fatou Theorem" (p. 128); for Carleson's theorem see Chapter 17.

Example 16.9: The datum $f(\theta) = |\theta|$. Let $f(\theta) = |\theta|$ for $\theta \in [-\pi, \pi]$, extended periodically. It is continuous on $\mathbb{T}$ (at $\theta = \pm\pi$ it equals π from both sides) and even, so $b_n = 0$.

Coefficients. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |\theta| d\theta = \frac{2}{\pi} \cdot \frac{\pi^2}{2} = \pi$, so $\frac{a_0}{2} = \frac{\pi}{2}$. For $n \geq 1$, integrating by parts,

$$a_n = \frac{2}{\pi} \int_0^{\pi} \theta \cos n\theta d\theta = \frac{2}{\pi} \left[\frac{\theta \sin n\theta}{n} + \frac{\cos n\theta}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{(-1)^n - 1}{n^2},$$

that is, $a_n = 0$ for n even and $a_n = -\frac{4}{\pi n^2}$ for n odd.

Solution. By Theorem 16.11,

$$u(re^{i\theta}) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k \text{ odd}} \frac{r^k \cos k\theta}{k^2}.$$

Numerical value and cross-check. At $r = 0.5$, $\theta = 0$ the partial sums are

$$0.9341766; \quad 0.9164927; \quad 0.9149011; \quad 0.9146981; \quad 0.9146674; \quad 0.9146623$$

for $k \leq 1, 3, 5, 7, 9, 11$, and the limit is 0.9146611; Simpson quadrature of the Poisson integral with 200 000 subintervals gives 0.9146611. ✓ Two more points: $r = 0.5$, $\theta = 1$ gives 1.2437588 by both routes, and $r = 0.9$, $\theta = 2$ gives 1.9703467. ✓

Reading. At $\theta = 0$ the datum equals 0, but the value of the solution at distance 0.5 from the boundary is already 0.9146611: the harmonic function "sees" the whole boundary, not only the nearest point. Only as $r \rightarrow 1^-$ does the value tend to $f(0) = 0$, as guaranteed by Theorem 16.5.

16.8. Assigned exercises

The exercises in this section are distinct from those worked out in Section 16.9. The answers are in the Teacher's Key (Section 16.11).

Exercise 16.1: Harmonic conjugates: constructing them, and seeing when they do not exist. *Goal:* use Lemma 16.3 as an *algorithm* for the conjugate, and recognize the topological obstruction. *Estimated time:* 40 minutes. *Type:* proof/computation.

- Verify that $u = x^3 - 3xy^2$ is harmonic on $\mathbb{C}$, compute $g = u_x - iu_y$, recognize an antiderivative of it, and deduce the harmonic conjugate v with $v(0) = 0$. Check the result with the Cauchy–Riemann equations.
- Verify that $u = \frac{x}{x^2 + y^2}$ is harmonic on $\mathbb{C} \setminus \{0\}$ and determine a *global* harmonic conjugate on $\mathbb{C} \setminus \{0\}$. (Hint: $u = \operatorname{Re} \frac{1}{z}$.) Why does the conjugate exist here on all of $\mathbb{C} \setminus \{0\}$, while for $\log |z|$ it does not?
- Prove that on $\Omega = \mathbb{C} \setminus (-\infty, 0]$ the function $\log |z|$ has $\operatorname{Arg} z \in (-\pi, \pi)$ as harmonic conjugate, using Lemma 15.11 of Chapter 15.

- (d) Compute u and v from part (a) at $z = 0.3 + 0.4i$ and verify numerically that $u + iv = z^3$.

Exercise 16.2: The Poisson kernel: computations, estimates and concentration.

Goal: handle the closed form and the three properties of an approximate identity. *Estimated time:* 45 minutes. *Type:* symbolic and numerical computation.

- Compute $P_{0.6}(\pi/3)$ with Lemma 16.4 and then, independently, by numerically summing the series of Definition 16.4 up to $n = 120$.
- Determine, for $\delta = 0.3$, the value of $\sup_{\delta \leq |t| \leq \pi} P_r(t)$ for $r = 0.99$ and $r = 0.999$, and verify that it matches the prediction of Lemma 16.9.
- Prove that, at fixed r , P_r has a maximum at $t = 0$ and a minimum at $t = \pi$, and that the maximum/minimum ratio equals $\left(\frac{1+r}{1-r}\right)^2$. What is its value for $r = 0.9$?
- Verify numerically, with a quadrature, that $\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r = 1$ for $r = 0.3, 0.9$ and 0.99 , and explain why the quadrature becomes more delicate as r grows.

Exercise 16.3: Dirichlet in the disc with trigonometric data, and a check of Harnack.

Goal: solve the Dirichlet problem when the datum is a trigonometric polynomial, without computing any integral. *Estimated time:* 40 minutes. *Type:* symbolic and numerical computation.

- Solve the Dirichlet problem in $\mathbb{D}$ with $f(\theta) = 3 + 2\cos\theta - \sin 2\theta$ and compute the solution at $r = 0.5, \theta = 0.9$; check the result with a quadrature of the Poisson integral.
- Likewise with $f(\theta) = \cos^4\theta$ (first linearize the power), at the point $r = 0.7, \theta = 0.4$.
- For each of the two data, state the value of the solution at the center with no computation, justifying it with Corollary 16.14.
- Let $f(\theta) = 1 + \cos\theta \geq 0$. Write the solution, compute its maximum and minimum value on the circle $|z| = 0.5$, and compare them with the bounds of Theorem 16.8. Is the inequality strict or not? Why?

16.9. Worked exercises

Exercise 16.4: The datum $f(\theta) = \theta^2$: Fourier series and numerical value. *Type:* computation with series and a double numerical check. Solve the Dirichlet problem in $\mathbb{D}$ with datum $f(\theta) = \theta^2, \theta \in [-\pi, \pi]$, extended periodically, and compute the solution at $r = 0.5, \theta = 0$ and at $r = 0.6, \theta = 1.2$.

Solution. Strategy. The datum is continuous on $\mathbb{T}$ (at $\pm\pi$ it equals π^2 from both sides) and even; its Fourier coefficients are computed and Theorem 16.11 is applied, which gives the solution as a series in r^n .

Step 1 — the datum is admissible. f is continuous on $\mathbb{T}$: the function $\theta \mapsto \theta^2$ on $[-\pi, \pi]$ takes the same value at the endpoints, so its 2π -periodic extension is continuous. Theorem 16.6 guarantees existence and uniqueness of the solution.

Step 2 — coefficients. Since f is even, $b_n = 0$ for every $n \geq 1$. Then

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \frac{2}{\pi} \cdot \frac{\pi^3}{3} = \frac{2\pi^2}{3}, \quad \text{so} \quad \frac{a_0}{2} = \frac{\pi^2}{3}.$$

For $n \geq 1$, integrating by parts twice,

$$\int_0^{\pi} \theta^2 \cos n\theta d\theta = \left[\frac{\theta^2 \sin n\theta}{n} \right]_0^{\pi} - \frac{2}{n} \int_0^{\pi} \theta \sin n\theta d\theta = 0 - \frac{2}{n} \left(\left[-\frac{\theta \cos n\theta}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos n\theta d\theta \right).$$

The last integral vanishes and $[-\frac{\theta \cos n\theta}{n}]_0^\pi = -\frac{\pi(-1)^n}{n}$, so $\int_0^\pi \theta^2 \cos n\theta \, d\theta = \frac{2\pi(-1)^n}{n^2}$ and

$$a_n = \frac{2}{\pi} \cdot \frac{2\pi(-1)^n}{n^2} = \frac{4(-1)^n}{n^2}.$$

(Here $a_n = \frac{2}{\pi} \int_0^\pi f \cos n\theta \, d\theta$ was used, valid for f even.)

Step 3 — the solution.

$$u(re^{i\theta}) = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n r^n \cos n\theta}{n^2}.$$

Step 4 — the two values. For $r = 0.5$, $\theta = 0$ the partial sums are

1.2898681; 1.5398681; 1.4843126; 1.4999376; 1.4949376; 1.4966737; 1.4960359; 1.4962801,

for $n \leq 1, \dots, 8$: the series has alternating signs with terms of decreasing modulus, and the limit lies between two consecutive partial sums; the value is 1.4962113. For $r = 0.6$, $\theta = 1.2$ the same formula gives 2.2352419.

Independent check 1: Poisson quadrature. Composite Simpson quadrature with 200 000 subintervals of $\frac{1}{2\pi} \int_{-\pi}^\pi P_r(\theta - t) t^2 \, dt$ gives 1.4962113 at the first point and 2.2352419 at the second. ✓

Independent check 2: the center. For $r = 0$ the formula of Step 3 gives $\frac{\pi^2}{3} = 3.2898681$, which is also $\hat{f}(0) = \frac{1}{2\pi} \int \theta^2 \, d\theta = \frac{\pi^2}{3}$: consistent with Corollary 16.14, and the quadrature returns 3.2898681. ✓

Interpretation. The value at the center is the mean of the datum; moving toward the boundary along the ray $\theta = 0$ the value falls from 3.2898681 toward $f(0) = 0$, and at the halfway point it is still 1.4962113: harmonic functions average the datum over the *whole* boundary, with the weight P_r of Figure 16.1.

Typical errors. (1) Forgetting that f must be seen as a function *on the circle*: the function $\theta \mapsto \theta$ on $[-\pi, \pi]$ is not continuous on $\mathbb{T}$ (at $\pm\pi$ it jumps from π to $-\pi$) and Theorem 16.5 does not apply at that point. (2) Using $a_n = \frac{1}{\pi} \int_0^\pi$ instead of $\frac{2}{\pi} \int_0^\pi$ for an even function, losing a factor of 2. (3) Writing r^n as r^{n^2} or $r^{|n|}$ without distinguishing between the complex form (where the exponent is $|n|$) and the real form (where $n \geq 1$).

Extension. With the same method one solves the problem with datum $f(\theta) = \theta^2$ on the disc $D(0, R)$: it suffices to replace r with ρ/R in Step 3, by Corollary 16.9.

Exercise 16.5: A discontinuous datum, and what happens at the jump point.

Type: application of harmonic measure and diagnosis of a hypothesis. Let $f = \mathbf{1}_{(0, \pi)}$ (the datum equal to 1 on the upper semicircle and 0 on the lower one). Compute $P[f]$ at $z = 0.5e^{i\pi/4}$; study the limit of $P[f]$ along the ray $z = r \rightarrow 1^-$ and along the ray $z = ir \rightarrow i$; explain the result in the light of Theorem 16.5.

Solution. Strategy. The datum is bounded and integrable but not continuous at $z = 1$ and $z = -1$; Theorem 16.4 nonetheless guarantees that $P[f]$ is harmonic, and Theorem 16.5 gives the boundary value *only* at points of continuity. Proposition 16.5 supplies the closed formula.

Step 1 — the required value. By Proposition 16.5, with $r = 0.5$ and $\theta = \pi/4$ ($\sin \theta = 0.7071068$):

$$P[f](0.5e^{i\pi/4}) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{2 \cdot 0.5 \cdot 0.7071068}{1 - 0.25} = \frac{1}{2} + \frac{\arctan(0.9428090)}{\pi} = 0.7406325.$$

Step 2 — the limit along the real ray. For $z = r \in (0, 1)$ one has $\theta = 0$, so $\sin \theta = 0$ and

$$P[f](r) = \frac{1}{2} + \frac{1}{\pi} \arctan 0 = \frac{1}{2} \quad \text{for every } r \in [0, 1).$$

The limit as $r \rightarrow 1^-$ is therefore $\frac{1}{2}$. But f jumps at $z = 1$ from 0 (limit from below) to 1 (limit from above): the radial limit is the *average* of the two one-sided limits. No contradiction with Theorem 16.5, whose hypothesis “ f continuous at $e^{i\theta_0}$ ” is violated here.

Step 3 — the limit along the imaginary ray. For $z = ir$ one has $\theta = \pi/2$, $\sin \theta = 1$, so

$$P[f](ir) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{2r}{1-r^2} \xrightarrow{r \rightarrow 1^-} \frac{1}{2} + \frac{1}{\pi} \cdot \frac{\pi}{2} = 1 = f\left(\frac{\pi}{2}\right),$$

because $\frac{2r}{1-r^2} \rightarrow +\infty$. Here f is continuous (it is constantly 1 in a neighborhood of i on the circle) and the theorem applies: the limit is the value of the datum. ✓

Independent check 1: quadrature. Simpson quadrature of $\frac{1}{2\pi} \int_0^\pi P_{0.5}(\pi/4 - t) dt$ with 200 000 subintervals gives 0.7406325; at $r = 0.8$, $\theta = -0.3$ the formula gives 0.2071363 and so does the quadrature. ✓

Independent check 2: values on the imaginary ray. The formula of Step 3 gives 0.9665246 for $r = 0.9$ and 0.9968009 for $r = 0.99$: the convergence to 1 is visible, and it is slow because the point i is “far” from the two jump points by only a quarter turn. ✓

Interpretation. $P[f](z)$ is the “weight” that the point z assigns to the upper arc when averaging the boundary datum: it is the *harmonic measure* of the arc as seen from z (Proposition 16.5). At the center it equals $\frac{1}{2}$, i.e. the fraction of the circle occupied by the arc; the value $\frac{1}{2}$ over the whole real diameter expresses the symmetry of the problem with respect to the real axis, and approaching i the weight of the upper arc tends to 1.

Typical errors. (1) Believing that $P[f]$ extends continuously to $\mathbb{D}$: it does not, because the limit depends on the direction of approach to $z = 1$ (along the upper boundary it tends to 1, along the diameter to $\frac{1}{2}$). (2) Applying Corollary 16.7 to a discontinuous datum: uniqueness requires continuity up to the boundary, and the Dirichlet problem with a discontinuous datum must be formulated differently. (3) Forgetting that $\arctan$ has range $(-\pi/2, \pi/2)$, and obtaining values outside $[0, 1]$.

Extension. The same computation, with Lemma 16.7 and endpoints a, b in place of $0, \pi$, gives the harmonic measure of an arbitrary arc; the value at the center is always $\frac{b-a}{2\pi}$, i.e. the fraction of the circle occupied by the arc (for $a = 0$, $b = \pi$: $\frac{1}{2}$). ✓

Exercise 16.6: The Poisson kernel reproduced by itself. *Type:* computation with Fourier coefficients, with a numerical check. Prove that for $0 \leq r, s < 1$

$$P[P_s](re^{i\theta}) = P_{rs}(\theta),$$

that is, that the Poisson integral of the kernel P_s is again a Poisson kernel, of parameter rs . Verify this numerically at two points.

Solution. Strategy. Rather than computing the integral, the Fourier coefficients of the datum P_s are computed and Theorem 16.11 is applied: the product of the factors $s^{|n|}$ and $r^{|n|}$ is $(rs)^{|n|}$, which is the coefficient of the kernel of parameter rs .

Step 1 — the datum is admissible. For s fixed, P_s is continuous on $\mathbb{T}$ and bounded (Lemma 16.6), so Theorem 16.11 applies.

Step 2 — the Fourier coefficients of P_s . By Definition 16.4 the series $\sum_m s^{|m|} e^{imt}$ converges uniformly in t ; integrating term by term against e^{-int} ,

$$\widehat{P}_s(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_s(t) e^{-int} dt = \sum_{m \in \mathbb{Z}} s^{|m|} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(m-n)t} dt = s^{|n|},$$

because the last integral equals 1 if $m = n$ and 0 otherwise (Step 2 of the proof of Lemma 16.8).

Step 3 — apply the expansion. By Theorem 16.11,

$$P[P_s](re^{i\theta}) = \sum_{n \in \mathbb{Z}} \widehat{P}_s(n) r^{|n|} e^{in\theta} = \sum_{n \in \mathbb{Z}} (rs)^{|n|} e^{in\theta} = P_{rs}(\theta),$$

the last equality by Definition 16.4 applied to the parameter $rs \in [0, 1)$.

Independent check 1: quadrature. For $r = 0.5$, $s = 0.6$, $\theta = 1$: Simpson quadrature with 200 000 subintervals of $\frac{1}{2\pi} \int_{-\pi}^{\pi} P_{0.5}(\theta - t) P_{0.6}(t) dt$ gives 1.1882709, while the closed form of Lemma 16.4 with $rs = 0.3$ gives

$$P_{0.3}(1) = \frac{1 - 0.09}{1 - 0.6 \cos 1 + 0.09} = \frac{0.91}{1.09 - 0.6 \cdot 0.5403023} = \frac{0.91}{0.7658186} = 1.1882709. \checkmark$$

Independent check 2. For $r = 0.8$, $s = 0.5$, $\theta = 0.3$: quadrature 2.1226551, closed form with $rs = 0.4$ also 2.1226551. $\checkmark$

Interpretation. The family $\{P_r\}$ is a *semigroup* with respect to convolution on the circle: $P_r * P_s = P_{rs}$. This is the same phenomenon by which solving the Dirichlet problem twice, first with radius s and then with radius r , is the same as solving it once with radius rs : the restriction to $|z| = s$ of a harmonic function, carried back to the boundary and solved again, reproduces the original function rescaled.

Typical errors. (1) Attempting the direct computation of the integral, which is possible but long; the Fourier coefficients make it immediate. (2) Writing $\widehat{P}_s(n) = s^n$ instead of $s^{|n|}$, getting the terms with $n < 0$ wrong. (3) Concluding that $P_r * P_s = P_{r+s}$ by analogy with the heat kernel: here the composition law is *multiplicative* in the parameter.

Extension. Setting $r = s$ gives $P_r * P_r = P_{r^2}$, and iterating $P_r^{*k} = P_{r^k}$: successive convolutions concentrate the kernel toward the origin, not toward the boundary. $\checkmark$

16.10. Self-assessment quiz

- (Definition) A function u on an open set $\Omega \subseteq \mathbb{C}$ is *harmonic* when
 - $u \in C^2(\Omega)$ and $u_{xx} + u_{yy} = 0$ at every point of Ω
 - u is continuous and $u_x = u_y = 0$
 - u is the imaginary part of an arbitrary continuous function
 - $u \in C^1(\Omega)$ and $\Delta u \leq 0$
- (Hypothesis) In Theorem 16.1 the hypothesis that Ω be convex serves to
 - make u bounded
 - guarantee that the holomorphic function $g = u_x - iu_y$ has an antiderivative on Ω , by Theorem 15.2
 - guarantee that u is of class C^∞
 - ensure that u does not change sign
- (Symbolic computation) The closed form of the Poisson kernel is
 - $\frac{1 - r^2}{1 - 2r \cos t + r^2}$
 - $\frac{1 + r^2}{1 - 2r \cos t + r^2}$
 - $\frac{1 - r^2}{1 + 2r \cos t + r^2}$
 - $\frac{1 - r}{1 - 2r \cos t + r^2}$
- (Numerical computation) $P_{0.5}(\pi/3)$ equals, to seven decimals,

- A) 1.0000000 B) 3.0000000 C) 0.3333333 D) 0.8421053
5. (Proof step) In the proof of Theorem 16.5, the bound $P(z, e^{it}) \leq \frac{4(1-|z|^2)}{c_\delta^2}$ for $d(t, \theta_0) \geq \delta$ is obtained
- A) from the normalization of the kernel (Lemma 16.8) B) from the triangle inequality $|e^{it} - z| \geq |e^{it} - e^{i\theta_0}| - |e^{i\theta_0} - z| \geq c_\delta/2$, combined with the geometric form of Lemma 16.5 C) from the monotonicity of P_r on $[0, \pi]$ D) from the series expansion of Definition 16.4
6. (Error diagnosis) A student writes: “ u is harmonic and bounded on $\mathbb{D}$, so by Liouville’s theorem it is constant.” The error is
- A) Liouville’s theorem holds only for holomorphic functions B) Theorem 16.9 requires harmonicity on *all* of the plane, not on a disc: on $\mathbb{D}$ the function $u = x$ is harmonic, bounded and not constant C) the hypothesis $u \geq 0$ is missing D) continuity up to the boundary is missing
7. (Transfer) Let u be harmonic on all of $\mathbb{C}$ with $u(z) \leq 7$ for every z and $u(0) = 3$. Then
- A) $u \equiv 3$ B) $u \equiv 7$ C) such a u does not exist D) u can be non-constant, provided it is bounded above
8. (Source and metacognition) The construction of the Poisson kernel for the ball and the solution of the Dirichlet problem with continuous datum are found in
- A) Axler, Bourdon, Ramey, *Harmonic Function Theory*, 2nd ed., Ch. 1, sections “The Poisson Kernel for the Ball” (p. 9) and “The Dirichlet Problem for the Ball” (p. 12)
 B) the same authors, Ch. 8 “Harmonic Bergman Spaces” C) Ahlfors, *Complex Analysis*, Ch. 2 “Complex Functions” D) Stein and Shakarchi, *Fourier Analysis*, Ch. 7 “Finite Fourier Analysis”

16.11. Teacher’s key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

1) **A.** *Rationale:* this is literally Definition 16.1. *Distractors:* B describes locally constant functions; C forgets that the function whose imaginary part is taken must be *holomorphic* (Proposition 16.2) and that, in any case, that is a local characterization, not the definition; D confuses harmonicity with superharmonicity and asks for insufficient regularity for Δu to exist. *Error diagnosed:* substituting a consequence of the definition for the definition itself. *Source:* Axler, Bourdon, Ramey, Ch. 1, “Definitions and Examples,” p. 1. *Prerequisite:* second partial derivatives. *Remediation:* verify $\Delta(x^2 - y^2) = 0$ and $\Delta(x^2 + y^2) = 4 \neq 0$ (Example 16.1).

2) **B.** *Rationale:* in Step 2 of the proof of Theorem 16.1 convexity is used *only* to invoke Theorem 15.2; Example 16.2(c) shows that without it the conjugate may not exist. *Distractors:* A is irrelevant (no boundedness is used); C is true but is a *consequence* (Corollary 16.1), not the role of the hypothesis; D is false. *Error diagnosed:* not being able to isolate the point where a hypothesis enters a proof. *Source:* Ahlfors, Ch. 4, §6.1 “Definition and Basic Properties.” *Prerequisite:* existence of an antiderivative on a convex set (Theorem 15.2). *Remediation:* redo Example 16.2(c) and identify the step that breaks down.

3) **A.** *Rationale:* this is Lemma 16.4. *Distractors:* B gets the numerator wrong (with $1 + r^2$ the integral would not equal 1 and the function would not tend to 0 away from the

point); C corresponds to $P_r(t + \pi)$, i.e. to a kernel concentrated at $t = \pi$; D is not even dimensionally consistent: as $r \rightarrow 0$ it would give 1, but at $t = \pi$ it would give $\frac{1-r}{(1+r)^2}$ instead of $\frac{1-r}{1+r}$. *Error diagnosed*: memorizing the formula instead of reconstructing it from the two geometric series. *Source*: Stein and Shakarchi, Ch. 2; Ahlfors, Ch. 4, §6.3 “Poisson’s Formula.” *Prerequisite*: geometric series. *Remediation*: redo the three steps of the proof of Lemma 16.4.

4) **A.** *Rationale*: $1 - 2 \cdot 0.5 \cdot \cos \frac{\pi}{3} + 0.25 = 1 - 0.5 + 0.25 = 0.75$ and $1 - r^2 = 0.75$, so the quotient equals exactly 1; the sum of the series up to $n = 60$ gives 1.0000000. *Distractors*: B is $P_{0.5}(0) = \frac{1+r}{1-r}$, the maximum; C is $P_{0.5}(\pi) = \frac{1-r}{1+r}$, the minimum; D is $P_{0.6}(\pi/3) = \frac{0.64}{0.76}$, i.e. the value with the other radius. *Error diagnosed*: confusing the radius with the angle, or reading off the extreme value instead of the one requested. *Source*: Example 16.5(b) and Figure 16.1. *Prerequisite*: $\cos \frac{\pi}{3} = \frac{1}{2}$. *Remediation*: compute $P_{0.5}(\pi/2) = 0.6$ and compare it with the figure.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **B.** *Rationale*: this is exactly Step 4 of the proof of Theorem 16.5: one bounds the denominator $|e^{it} - z|^2$ below and uses $P = \frac{1-|z|^2}{|e^{it}-z|^2}$. *Distractors*: A is instead used in Step 1 (to center the error) and in Step 3; C is the key point of Lemma 16.9, which estimates $P_r(t)$ for z on the ray, and is not enough here because z tends to the boundary in an arbitrary direction; D plays no role in this bound. *Error diagnosed*: not distinguishing “radial” concentration from “non-tangential” concentration. *Source*: Rudin, Ch. 11, and Stein and Shakarchi, Ch. 2, on approximation with good kernels. *Prerequisite*: triangle inequality. *Remediation*: redo Step 4 with $\delta = \pi/2$ and make c_δ explicit.

6) **B.** *Rationale*: $u(x, y) = x$ is harmonic on $\mathbb{D}$, satisfies $|u| \leq 1$ and is not constant: the hypothesis “harmonic on all of $\mathbb{C}$ ” in Theorem 16.9 is not decorative. *Distractors*: A is false (Theorem 16.9 is precisely the statement for harmonic functions); C confuses this with the hypothesis of Theorem 16.8; D is beside the point, because neither version of Liouville requires continuity up to the boundary. *Error diagnosed*: transporting a global theorem into a local context. *Source*: Axler, Bourdon, Ramey, Ch. 2, “Liouville’s Theorem,” p. 31. *Prerequisite*: Theorem 16.8. *Remediation*: exhibit three bounded, non-constant harmonic functions on $\mathbb{D}$.

7) **A.** *Rationale*: u is harmonic on $\mathbb{C}$ and bounded above, hence constant by Corollary 16.12; the constant is the value at 0, i.e. 3. *Distractors*: B swaps the bound for the value; C is false (the constant 3 satisfies all the hypotheses); D contradicts the corollary. *Error diagnosed*: believing that a bound on both sides is needed. *Source*: Axler, Bourdon, Ramey, Ch. 3, “Liouville’s Theorem,” p. 45, where the statement is given for *positive* harmonic functions. *Prerequisite*: Corollary 16.12. *Remediation*: prove that a harmonic function on $\mathbb{C}$ with $u \geq -5$ is constant.

8) **A.** *Rationale*: the table of contents of the volume, checked against the PDF made available by the authors, places “The Poisson Kernel for the Ball” at p. 9 and “The Dirichlet Problem for the Ball” at p. 12 of Chapter 1. *Distractors*: B is the chapter on harmonic Bergman spaces (p. 171); C is Ahlfors’s chapter on elementary complex functions, which does not treat harmonic functions (those are in Ch. 4, §6); D is the chapter on finite Fourier analysis, i.e. on the group $\mathbb{Z}/N\mathbb{Z}$. *Error diagnosed*: citing from memory without checking the table of contents. *Source*: the sources environment at the end of the chapter. *Prerequisite*: none. *Remediation*: locate, in the same volume, the section “Converse of the Mean-Value Property” (p. 17), which corresponds to Theorem 16.7.

BEST PRACTICE

Assigned exercise 16.1 — answers.

(a) $u_{xx} = 6x$, $u_{yy} = -6x$: harmonic. Then $g = u_x - iu_y = (3x^2 - 3y^2) + i6xy = 3(x^2 - y^2 + 2ixy) = 3z^2$, with antiderivative z^3 . So $f(z) = z^3$ and $v = \operatorname{Im} z^3 = 3x^2y - y^3$, which vanishes at 0. Cauchy–Riemann check: $u_x = 3x^2 - 3y^2 = v_y$ ✓ and $u_y = -6xy = -v_x$ ✓.

(b) $u = \operatorname{Re} \frac{1}{z}$ because $\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{x-iy}{x^2+y^2}$; hence $v = \operatorname{Im} \frac{1}{z} = -\frac{y}{x^2+y^2}$ is a conjugate on all of $\mathbb{C} \setminus \{0\}$. The difference with $\log|z|$ is that here a global holomorphic function, $1/z$, of which u is the real part, *already* exists; for $\log|z|$ such a function would have to be a holomorphic logarithm, which does not exist on $\mathbb{C} \setminus \{0\}$ (Example 16.2(c)). The obstruction is not harmonicity but the monodromy of the argument.

(c) On the cut plane $\Omega = \mathbb{C} \setminus (-\infty, 0]$ the principal logarithm $\operatorname{Log} z = \log|z| + i \operatorname{Arg} z$ is holomorphic (Lemma 15.11); its real part is $\log|z|$ and its imaginary part $\operatorname{Arg} z$: by Definition 16.2 the latter is a harmonic conjugate of $\log|z|$ on Ω .

(d) $z = 0.3 + 0.4i$: $z^2 = -0.07 + 0.24i$ and $z^3 = z \cdot z^2 = -0.117 + 0.044i$. On the other hand $u = x^3 - 3xy^2 = 0.027 - 0.144 = -0.117$ ✓ and $v = 3x^2y - y^3 = 0.108 - 0.064 = 0.044$ ✓.

BEST PRACTICE

Assigned exercise 16.2 — answers.

(a) $1 - 2 \cdot 0.6 \cdot \frac{1}{2} + 0.36 = 1 - 0.6 + 0.36 = 0.76$ and $1 - r^2 = 0.64$: $P_{0.6}(\pi/3) = \frac{0.64}{0.76} = 0.8421053$. The sum of the series $1 + 2 \sum_{n \geq 1} 0.6^n \cos \frac{n\pi}{3}$ up to $n = 120$ gives 0.8421053. ✓

(b) By Lemma 16.9 the supremum on $\{|t| \geq \delta\}$ is $P_r(\delta)$. With $\delta = 0.3$ (and $\cos 0.3 = 0.9553365$): $P_{0.99}(0.3) = \frac{0.0199}{1.9801 - 1.98 \cdot 0.9553365} = 0.2247730$ and $P_{0.999}(0.3) = 0.0224006$. ✓

The ratio between the two, about 10, reflects the factor $1 - r^2$ in the numerator.

(c) $P_r(t)$ depends on t only through $\cos t$ and is decreasing in $1 - 2r \cos t + r^2$: hence it is maximal where $\cos t$ is maximal ($t = 0$) and minimal where it is minimal ($t = \pi$), with values $\frac{1+r}{1-r}$ and $\frac{1-r}{1+r}$ (Lemma 16.6). The ratio is $\left(\frac{1+r}{1-r}\right)^2$, which for $r = 0.9$ equals $19^2 = 361$.

(d) Simpson quadrature with 200 000 subintervals gives 1.0000000 for $r = 0.3, 0.9$ and 0.99 . ✓ As r grows the integrand has a peak of height $\frac{1+r}{1-r}$ and width of order $1 - r$ (because $P_r(t) = \frac{1+r}{1-r}$ at $t = 0$ and $P_r(1-r) \approx \frac{1}{2} \frac{1+r}{1-r}$): the quadrature step must be much smaller than $1 - r$, otherwise the peak gets “skipped over”.

BEST PRACTICE

Assigned exercise 16.3 — answers.

(a) $u(re^{i\theta}) = 3 + 2r \cos \theta - r^2 \sin 2\theta$: it is harmonic (Example 16.1(b) and Proposition 16.1(i)), continuous on $\mathbb{D}$ and equals f for $r = 1$; by Theorem 16.6 it is the solution. At $r = 0.5$, $\theta = 0.9$: $3 + 1 \cdot 0.6216100 - 0.25 \cdot 0.9738476 = 3.3781481$; quadrature of the Poisson integral gives 3.3781481. ✓

(b) $\cos^4 \theta = \frac{3}{8} + \frac{1}{2} \cos 2\theta + \frac{1}{8} \cos 4\theta$, so $u = \frac{3}{8} + \frac{1}{2} r^2 \cos 2\theta + \frac{1}{8} r^4 \cos 4\theta$. At $r = 0.7$, $\theta = 0.4$: $0.375 + 0.245 \cdot 0.6967067 + 0.0300125 \cdot (-0.0291995) = 0.375 + 0.1706932 - 0.0008764 = 0.5448168$; the quadrature gives 0.5448168. ✓

(c) By Corollary 16.14 the value at the center is $\hat{f}(0) = \frac{a_0}{2}$, i.e. the constant term of the expansion: 3 in case (a) and $\frac{3}{8} = 0.375$ in case (b). No computation is needed.

(d) $u = 1 + r \cos \theta$, with $u(0) = 1$. On $|z| = 0.5$ the maximum is 1.5 ($\theta = 0$) and the minimum 0.5 ($\theta = \pi$). Harnack’s bounds are $\frac{1-0.5}{1+0.5} = 0.3333333$ and $\frac{1+0.5}{1-0.5} = 3$: the inequalities are *strict*. The reason can be read in Step 3 of the proof of Theorem 16.8: equality on the right would require

$$\int_{-\pi}^{\pi} \left(\frac{s+\rho}{s-\rho} - \frac{s^2-\rho^2}{s^2-2s\rho\cos(\theta-t)+\rho^2} \right) u(se^{it}) dt = 0$$

with a continuous, non-negative integrand, hence $u(se^{it}) = 0$ for every t other than the one

that maximizes the kernel: impossible for a continuous datum that is not identically zero. Equality is approached only when the datum concentrates at a point, as in the Poisson kernel itself (Example 16.7).

16.12. Formulas of the chapter

Laplacian, harmonic functions, conjugate.

$$\Delta u = u_{xx} + u_{yy} = \tilde{u}_{rr} + \frac{1}{r}\tilde{u}_r + \frac{1}{r^2}\tilde{u}_{\theta\theta}; \quad \Delta u = 0 \iff u \text{ harmonic.}$$

On a convex set: $g = u_x - iu_y$ is holomorphic, $F' = g$, $u = \operatorname{Re} F$; conjugate unique up to constants. Cauchy–Riemann: $u_x = v_y$, $u_y = -v_x$. Elementary harmonic functions: 1 , x , y , xy , $x^2 - y^2$, $r^n \cos n\theta$, $r^n \sin n\theta$, $\log r$ (on $\mathbb{C} \setminus \{0\}$), $e^x \cos y$.

Mean, maximum, uniqueness.

$$u(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(w + \rho e^{i\theta}) d\theta = \frac{1}{\pi R^2} \iint_{D(w, R)} u \, dx \, dy.$$

Strong maximum: mean + interior maximum $\Rightarrow$ constant. Weak maximum: $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$ and $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$. Comparison: $u_1 \leq u_2$ on the boundary $\Rightarrow u_1 \leq u_2$ inside. Uniqueness for Dirichlet on every bounded domain. Converse: continuous + mean $\Rightarrow$ harmonic.

Poisson kernel.

$$P_r(t) = \sum_{n \in \mathbb{Z}} r^{|n|} e^{int} = \frac{1 - r^2}{1 - 2r \cos t + r^2}, \quad P(z, e^{it}) = \frac{1 - |z|^2}{|e^{it} - z|^2} = \operatorname{Re} \frac{e^{it} + z}{e^{it} - z}.$$

$$\frac{1-r}{1+r} \leq P_r(t) \leq \frac{1+r}{1-r}; \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r = 1; \quad \sup_{\delta \leq |t| \leq \pi} P_r(t) = P_r(\delta) \xrightarrow{r \rightarrow 1^-} 0.$$

Antiderivative: $\int P_r(t) dt = 2 \arctan \left(\frac{1+r}{1-r} \tan \frac{t}{2} \right)$ on $(-\pi, \pi)$. Semigroup: $P_r * P_s = P_{rs}$.

Poisson's formula, Dirichlet, Harnack.

$$P[f](re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 - r^2}{1 - 2r \cos(\theta - t) + r^2} f(t) dt,$$

$$u(w + \rho e^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{R^2 - \rho^2}{R^2 - 2R\rho \cos(\theta - t) + \rho^2} u(w + Re^{it}) dt.$$

Dirichlet in the disc with continuous datum: unique solution, $u = P[f]$ inside and $u = f$ on the boundary. Harnack ($u \geq 0$ harmonic on $D(w, R)$, $|z - w| = \rho$):

$$\frac{R - \rho}{R + \rho} u(w) \leq u(z) \leq \frac{R + \rho}{R - \rho} u(w).$$

Liouville: u harmonic on $\mathbb{C}$ and bounded on one side only $\Rightarrow$ constant. Gradient estimate: u harmonic on $D(w, R)$ with $|u| \leq M \Rightarrow |\nabla u(w)| \leq \frac{2M}{R}$; for $w = 0$, $R = 1$ equality holds with $|\nabla u(0)| = 2|\hat{f}(1)|$, and the optimal constant is $\frac{4}{\pi} = 1.2732395$. Schwarz reflection: u harmonic on Ω^+ , continuous up to I and zero on $I \Rightarrow$ the odd extension $\tilde{u}(x, -y) = -u(x, y)$ is harmonic on Ω .

Fourier series of the solution, and the numerical values of the chapter.

$$P[f](re^{i\theta}) = \sum_{n \in \mathbb{Z}} \hat{f}(n) r^{|n|} e^{in\theta} = \frac{a_0}{2} + \sum_{n \geq 1} r^n (a_n \cos n\theta + b_n \sin n\theta); \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} P[f](re^{i\theta}) d\theta = \hat{f}(0).$$

$$P_{0.5}(0) = 3, \quad P_{0.5}(\pi/3) = 1, \quad P_{0.5}(\pi) = \frac{1}{3}, \quad P_{0.6}(\pi/3) = 0.8421053, \quad P_{0.8}(0) = 9.$$

$$f = \cos^2: u(0.6e^{i\pi/3}) = 0.41; \quad f = \cos^3: u(0.8e^{0.7i}) = 0.3942850; \quad f = \cos^4: u(0.7e^{0.4i}) = 0.5448168.$$

$$f = \theta^2: u(0.5) = 1.4962113, \quad u(0) = \frac{\pi^2}{3} = 3.2898681; \quad f = |\theta|: u(0.5) = 0.9146611.$$

$$f = \mathbf{1}_{(0,\pi)}: u(re^{i\theta}) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{2r \sin \theta}{1-r^2}, \quad u(0.5e^{i\pi/4}) = 0.7406325.$$

Sources and bibliography

- [1] L. V. Ahlfors, *Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable*, 3rd edition, International Series in Pure and Applied Mathematics, McGraw-Hill, New York, 1979 (xiv+331 pages, ISBN 0-07-000657-1). Chapter 4 “Complex Integration,” §1.5 “Cauchy’s Theorem in a Disk” (source of the tool used in Theorem 16.1) and above all §6 “Harmonic Functions,” organized into “6.1 Definition and Basic Properties,” “6.2 The Mean-value Property,” “6.3 Poisson’s Formula,” “6.4 Schwarz’s Theorem,” “6.5 The Reflection Principle”: this is the source of Definition 16.1, of Theorem 16.2, of Lemma 16.4 and of Theorem 16.5 (which in Ahlfors is “Schwarz’s theorem”). Bibliographic data and the structure of Chapter 4 checked against the SearchWorks catalog record of Stanford University Libraries, record 1003741 — checked on September 6, 2026.
- [2] S. Axler, P. Bourdon, W. Ramey, *Harmonic Function Theory*, 2nd edition, Graduate Texts in Mathematics, Springer, New York, 2001 (ISBN 978-0-387-95218-5). Chapter 1 “Basic Properties of Harmonic Functions” (pp. 1–29), with sections “Definitions and Examples” (p. 1), “Invariance Properties” (p. 2), “The Mean-Value Property” (p. 4), “The Maximum Principle” (p. 7), “The Poisson Kernel for the Ball” (p. 9), “The Dirichlet Problem for the Ball” (p. 12), “Converse of the Mean-Value Property” (p. 17), “Real Analyticity and Homogeneous Expansions” (p. 19), “Origin of the Term *Harmonic*” (p. 25): this is the parallel source, in arbitrary dimension, of Theorems 16.2, 16.3, 16.6 and 16.7. Chapter 2 “Bounded Harmonic Functions” (p. 31), section “Liouville’s Theorem” (p. 31), and Chapter 3 “Positive Harmonic Functions” (p. 45), sections “Liouville’s Theorem” (p. 45) and “Harnack’s Inequality and Harnack’s Principle” (p. 47): sources of Theorem 16.8, of Corollary 16.12 and of Proposition 16.6. Chapter 6 “Harmonic Hardy Spaces” (p. 111), section “The Fatou Theorem” (p. 128), for the panorama of Remark 16.6. The remark that the link between harmonic and holomorphic functions has no analogue in dimension > 2 is on p. 2. Table of contents and pagination checked against the full PDF made available by the authors at axler.net/HFT.pdf (table-of-contents pages and Chapter 1) — checked on September 6, 2026.
- [3] W. Rudin, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, New York, 1987 (xiv+416 pages, ISBN 0-07-054234-1). Chapter 11 “Harmonic Functions”: Cauchy–Riemann equations, Poisson integral, mean-value property, boundary behavior of Poisson integrals and representation theorems. This is the alternative source for Theorem 16.5 and for the measure-theoretic picture (boundary Borel measures, Fatou’s theorem) that this chapter does not develop. Bibliographic data checked against the Internet Archive record, identifier [realcomplexanaly0000rudin](https://archive.org/details/realcomplexanaly0000rudin); the structure of the chapter is the one

already checked for Chapter 15 — checked on September 6, 2026.

- [4] E. M. Stein, R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, Princeton, 2003 (328 pages, ISBN 978-0-691-11384-5). Chapter 1 “The Genesis of Fourier Analysis,” §2.2 “Steady-state heat equation in the disc” (pp. 20–22), where the equation $\Delta u = 0$ is introduced as the stationary limit of the heat equation, the Dirichlet problem in the disc is posed and separation of variables produces the solutions $r^{|m|}e^{im\theta}$ that here are the terms of Theorem 16.11; Chapter 2 “Basic Properties of Fourier Series,” for convolutions, “good kernels” (of which Definition 16.5 is the form used here) and Abel summability, i.e. the content of Remark 16.6. Table of contents of the volume checked against the publisher’s record (press.princeton.edu, ISBN 9780691113845); text and pagination of Chapter 1 checked against the sample chapter PDF distributed by the publisher (assets.press.princeton.edu/chapters/s7562.pdf) — checked on September 6, 2026.
- [5] For the results of the treatise recalled with \ref: the Cauchy–Riemann equations, the existence of an antiderivative on a convex open set, the mean-value property for holomorphic functions, analyticity, the maximum-modulus principle, Liouville’s theorem and the principal logarithm are proved in Chapter 15; the polar form and generalized circles in Chapter 6; the invariant measure $d\theta/2\pi$ and the characters of the circle in Chapter 13. The numerical checks of the chapter were carried out twice by independent routes (closed form and composite Simpson quadrature with 200 000 subintervals, or series and quadrature) and are reported in the text with the symbol $\checkmark$.

Convergence of Fourier series

A Fourier series is an identity declared before it is verified. Writing $f \sim \sum_n \hat{f}(n)e^{in\theta}$ costs nothing: only the integrals that define the coefficients are needed, and those exist for every summable function. The equals sign, instead, must be earned, and the price changes with the sense one gives it. In quadratic mean it costs nothing extra: the series always converges. Pointwise it costs a hypothesis of local regularity — an integral condition (Dini) or a variation condition (Dirichlet–Jordan). Uniformly it costs still more, and mere continuity is not enough: there is a continuous function whose Fourier series diverges at a point, and the reason is a single quantity that grows like $\ln N$, the Lebesgue constant.

The chapter climbs this ladder from the bottom up and shows, at every rung, exactly what is being paid. The technical pivot is a single one: the partial sum is a convolution with the *Dirichlet kernel* (Theorem 17.3), and all the properties — good and bad — of the partial sums are properties of that kernel. The Dirichlet kernel changes sign; its integral in modulus diverges (Theorem 17.10); from this follow the Gibbs phenomenon (Theorem 17.9) and du Bois-Reymond’s divergence (Corollary 17.4). Replacing the partial sums with their arithmetic means makes the kernel positive (Theorem 17.6), and with it all the pathologies disappear: this is Fejér’s theorem (Theorem 17.7), from which follow the density of trigonometric polynomials (Corollary 17.3), the uniqueness of the coefficients (Theorem 17.13) and the completeness of the trigonometric system, i.e. Parseval’s identity (Theorem 17.1).

The tool that makes every pointwise criterion provable is the Riemann–Lebesgue lemma (Lemma 17.2), here proved in full: from it follow Dini’s criterion (Theorem 17.11) and, with the addition of the second mean value theorem (Lemma 17.8), the Dirichlet–Jordan criterion for functions of bounded variation (Theorem 17.12), which at jump points gives the average of the two one-sided limits. The Wilbraham–Gibbs constant is computed exactly and the value of the sine-integral is obtained twice by independent routes (Proposition 17.3 and Observation 17.14). The chapter closes with the explicit remainder estimate for functions of class C^1 (Theorem 17.14) and with the panorama of the two extreme results, Kolmogorov and Carleson–Hunt, stated and not proved (Section 17.8).

The material that the executive plan moved here from Volume II (Chapter 4, §5) is kept in full — statements, proofs, exercises, quiz and formulary — in Section 17.2 and in Section 17.9. What Volume II took as known (Riemann–Lebesgue, Bessel’s inequality, the uniqueness of the coefficients for merely summable functions, the divergence of the Lebesgue constants) is proved here.

INFO

Level and prerequisites. Level **C** (undergraduate), with three passages at level **D**: the two-sided estimate of the Lebesgue constants (Theorem 17.10), the second mean value theorem in Bonnet’s form (Lemma 17.8) and the Dirichlet–Jordan criterion (Theorem 17.12).

Internal prerequisites. Fourier coefficients, the convolution theorem, the square wave example and the Basel series (Volume II, Chapter 4); Riemann and Lebesgue integrals, the periodic trapezoidal rule (Volume II, Chapter 2); the complex exponential and polar form (Volume II, Chapter 3, and Chapter 6); the Poisson kernel and harmonic functions on the

disc continue in Chapter 16; spectral *leakage* and windows in Chapter 22, Section 22.10.3.

External prerequisites, assumed and not proved:

- the density of step functions in the space L^1 of a bounded interval, and that of $C(\mathbb{T})$ in $L^1(\mathbb{T})$ and in $L^2(\mathbb{T})$ (measure theory: Rudin, *Real and Complex Analysis*, Chapter 3);
- Lebesgue's dominated convergence theorem and Lebesgue's differentiation theorem ($F(\theta) = \int_0^\theta f$ has $F' = f$ almost everywhere for $f \in L^1$);
- the completeness of L^2 and the Cauchy–Schwarz inequality;
- the Banach–Steinhaus uniform boundedness principle, used once, in Corollary 17.4.

Everything else — kernels, criteria, estimates, constants — is proved in this chapter. Observation 17.15 lists in one place what remains imported and what is only stated as panorama.

17.1. Coefficients, partial sums, Riemann–Lebesgue

This section fixes the notation and proves the two tools that the rest of the chapter uses throughout: Bessel's inequality, which governs convergence in quadratic mean, and the Riemann–Lebesgue lemma, which governs every criterion of pointwise convergence. Volume II had stated them; here they are proved.

Definition 17.1: Fourier coefficients, partial sums, trigonometric polynomials.

Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ for the circle and adopt the *normalised measure* $d\sigma = \frac{d\theta}{2\pi}$, so that $\sigma(\mathbb{T}) = 1$. Set

$$e_n(\theta) = e^{in\theta} \quad (n \in \mathbb{Z}), \quad \|f\|_p = \left(\int_{\mathbb{T}} |f|^p d\sigma \right)^{1/p} \quad (1 \leq p < \infty), \quad \|f\|_\infty = \sup_{\mathbb{T}} |f|.$$

For $f \in L^1(\mathbb{T})$ the n -th **Fourier coefficient** is

$$\hat{f}(n) = \int_{\mathbb{T}} f \overline{e_n} d\sigma = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta,$$

the N -th **partial sum** is $S_N f = \sum_{|n| \leq N} \hat{f}(n) e_n$, and a **trigonometric polynomial of degree** $\leq N$ is a finite linear combination $P = \sum_{|n| \leq N} a_n e_n$ with $a_n \in \mathbb{C}$. The set of such polynomials is denoted $\mathcal{P}_N$.

Lemma 17.1: Inclusions among the spaces on the circle. *With the normalised measure, for every measurable f ,*

$$\|f\|_1 \leq \|f\|_2 \leq \|f\|_\infty,$$

and hence $C(\mathbb{T}) \subset L^2(\mathbb{T}) \subset L^1(\mathbb{T})$. In particular the Fourier coefficients are defined for every $f \in L^2(\mathbb{T})$ and $|\hat{f}(n)| \leq \|f\|_1$ for every n .

Proof. By the Cauchy–Schwarz inequality applied to the pair $(|f|, 1)$ in the space $L^2(\mathbb{T}, \sigma)$,

$$\|f\|_1 = \int_{\mathbb{T}} |f| \cdot 1 d\sigma \leq \left(\int_{\mathbb{T}} |f|^2 d\sigma \right)^{1/2} \left(\int_{\mathbb{T}} 1^2 d\sigma \right)^{1/2} = \|f\|_2 \cdot \sigma(\mathbb{T})^{1/2} = \|f\|_2,$$

because $\sigma(\mathbb{T}) = 1$. The second inequality is immediate: $\int |f|^2 d\sigma \leq \|f\|_\infty^2 \int d\sigma = \|f\|_\infty^2$. Finally

$$|\hat{f}(n)| = \left| \int_{\mathbb{T}} f \overline{e_n} d\sigma \right| \leq \int_{\mathbb{T}} |f| |e_n| d\sigma = \int_{\mathbb{T}} |f| d\sigma = \|f\|_1,$$

since $|e_n(\theta)| = 1$ for every θ . □

Proposition 17.1: Orthonormality of the trigonometric system. With respect to the inner product $\langle f, g \rangle = \int_{\mathbb{T}} f \bar{g} d\sigma$,

$$\langle e_n, e_m \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \begin{cases} 1 & n = m, \\ 0 & n \neq m. \end{cases}$$

Proof. If $n = m$ the integrand is $e^{i0} = 1$ and the integral equals $\frac{1}{2\pi} \cdot 2\pi = 1$. If $k = n - m \neq 0$, the function $\theta \mapsto e^{ik\theta}/(ik)$ is an antiderivative of $e^{ik\theta}$, so

$$\int_0^{2\pi} e^{ik\theta} d\theta = \left[\frac{e^{ik\theta}}{ik} \right]_0^{2\pi} = \frac{e^{2\pi ik} - 1}{ik} = 0,$$

because $e^{2\pi ik} = 1$ for every integer k . □

Proposition 17.2: Projection, best approximation and Bessel's inequality. Let $f \in L^2(\mathbb{T})$ and $N \in \mathbb{N}$. Then:

- (i) $\langle f - S_N f, e_m \rangle = 0$ for every $|m| \leq N$; equivalently $f - S_N f$ is orthogonal to all of $\mathcal{P}_N$;
- (ii) for every $P \in \mathcal{P}_N$, $\|f - P\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - P\|_2^2$; in particular $S_N f$ is the best approximation of f in the L^2 norm among trigonometric polynomials of degree $\leq N$, and it is the only one;
- (iii) $\|f - S_N f\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} |\hat{f}(n)|^2$;
- (iv) (**Bessel's inequality**) $\sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 \leq \|f\|_2^2 < \infty$.

Proof. Step 1 — (i). By linearity of the inner product and by Proposition 17.1, for $|m| \leq N$

$$\langle S_N f, e_m \rangle = \sum_{|n| \leq N} \hat{f}(n) \langle e_n, e_m \rangle = \hat{f}(m),$$

while $\langle f, e_m \rangle = \int_{\mathbb{T}} f \bar{e}_m d\sigma = \hat{f}(m)$ by Definition 17.1. Subtracting, $\langle f - S_N f, e_m \rangle = 0$. Since every $P \in \mathcal{P}_N$ is a linear combination of the e_m with $|m| \leq N$, orthogonality extends by linearity to all of $\mathcal{P}_N$.

Step 2 — (ii). Write $f - P = (f - S_N f) + (S_N f - P)$. The second term lies in $\mathcal{P}_N$, hence by Step 1 it is orthogonal to the first. The Pythagorean theorem in an inner-product space (which is the direct computation $\|u + v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, v \rangle + \|v\|^2$ with $\langle u, v \rangle = 0$) gives

$$\|f - P\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - P\|_2^2 \geq \|f - S_N f\|_2^2,$$

with equality if and only if $\|S_N f - P\|_2 = 0$, i.e. $P = S_N f$ almost everywhere; but two trigonometric polynomials of degree $\leq N$ equal almost everywhere have the same coefficients (apply Proposition 17.1 to the difference), and hence coincide.

Step 3 — (iii). Take $P = 0$ in Step 2, i.e. compute directly

$$\|f - S_N f\|_2^2 = \langle f - S_N f, f - S_N f \rangle = \langle f - S_N f, f \rangle - \underbrace{\langle f - S_N f, S_N f \rangle}_{=0 \text{ (Step 1)}} = \|f\|_2^2 - \langle S_N f, f \rangle.$$

Finally, by Proposition 17.1,

$$\langle S_N f, f \rangle = \overline{\langle f, S_N f \rangle} = \overline{\sum_{|n| \leq N} \hat{f}(n) \langle f, e_n \rangle} = \sum_{|n| \leq N} \overline{\hat{f}(n)} \overline{\langle f, e_n \rangle} = \sum_{|n| \leq N} |\hat{f}(n)|^2,$$

which is (iii).

Step 4 — (iv). The left-hand side of (iii) is ≥ 0 , so $\sum_{|n| \leq N} |\hat{f}(n)|^2 \leq \|f\|_2^2$ for every N . The partial sums of the non-negative series $\sum_n |\hat{f}(n)|^2$ are bounded by $\|f\|_2^2$, hence the series converges and its sum does not exceed $\|f\|_2^2$. $\square$

Corollary 17.1: The coefficients of a square-summable function vanish at infinity. If $f \in L^2(\mathbb{T})$ then $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$.

Proof. By Proposition 17.2(iv) the series $\sum_n |\hat{f}(n)|^2$ converges; the general term of a convergent series tends to zero, hence $|\hat{f}(n)|^2 \rightarrow 0$ and therefore $\hat{f}(n) \rightarrow 0$. $\square$

Lemma 17.2: Riemann–Lebesgue lemma. Let $-\infty < a < b < +\infty$ and $g \in L^1(a, b)$. Then

$$\int_a^b g(t) e^{i\lambda t} dt \rightarrow 0 \quad (\lambda \in \mathbb{R}, |\lambda| \rightarrow \infty),$$

and likewise $\int_a^b g(t) \sin(\lambda t) dt \rightarrow 0$ and $\int_a^b g(t) \cos(\lambda t) dt \rightarrow 0$. In particular, for every $f \in L^1(\mathbb{T})$, $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$.

Proof. Step 1 — the case of an indicator function. Let $a \leq \alpha < \beta \leq b$ and $g = \mathbf{1}_{[\alpha, \beta)}$. For $\lambda \neq 0$, with the antiderivative $e^{i\lambda t}/(i\lambda)$,

$$\int_a^b \mathbf{1}_{[\alpha, \beta)}(t) e^{i\lambda t} dt = \int_\alpha^\beta e^{i\lambda t} dt = \frac{e^{i\lambda\beta} - e^{i\lambda\alpha}}{i\lambda}, \quad \text{hence} \quad \left| \int_a^b g e^{i\lambda t} dt \right| \leq \frac{2}{|\lambda|}.$$

The bound does not depend on α and β and tends to 0 as $|\lambda| \rightarrow \infty$.

Step 2 — step functions. Let $s = \sum_{j=1}^k c_j \mathbf{1}_{[\alpha_j, \beta_j)}$ be a step function. By linearity of the integral and by Step 1,

$$\left| \int_a^b s(t) e^{i\lambda t} dt \right| \leq \sum_{j=1}^k |c_j| \cdot \frac{2}{|\lambda|} = \frac{2}{|\lambda|} \sum_{j=1}^k |c_j| \rightarrow 0.$$

Step 3 — approximation. Let $\varepsilon > 0$. Step functions are dense in $L^1(a, b)$ (a measure-theory prerequisite declared in the opening infobox and in Observation 17.15): there is therefore a step function s with $\int_a^b |g - s| dt < \varepsilon/2$. Then, for every λ ,

$$\left| \int_a^b g e^{i\lambda t} dt \right| \leq \underbrace{\int_a^b |g - s| |e^{i\lambda t}| dt}_{< \varepsilon/2} + \left| \int_a^b s e^{i\lambda t} dt \right|,$$

using $|e^{i\lambda t}| = 1$. By Step 2 there is $\Lambda > 0$ such that the second term is $< \varepsilon/2$ for $|\lambda| > \Lambda$. Hence for $|\lambda| > \Lambda$ the whole left-hand side is $< \varepsilon$: that is the definition of a limit equal to zero.

Step 4 — the sine and cosine versions. From $\cos(\lambda t) = \frac{1}{2}(e^{i\lambda t} + e^{-i\lambda t})$ and $\sin(\lambda t) = \frac{1}{2i}(e^{i\lambda t} - e^{-i\lambda t})$, each of the two integrals is a linear combination of two integrals of the type already treated, with λ and $-\lambda$: both tend to zero.

Step 5 — the periodic case. Let $f \in L^1(\mathbb{T})$. Then f restricted to $(0, 2\pi)$ lies in $L^1(0, 2\pi)$ and, by Definition 17.1,

$$\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta,$$

which is the case $\lambda = -n$, $a = 0$, $b = 2\pi$, divided by 2π . By Steps 1–3 it tends to zero as $|n| \rightarrow \infty$. $\square$

By a half-period shift. For $f \in L^1(\mathbb{T})$ and $n \neq 0$, set $\tau = \pi/n$. Since $e^{-in(\theta+\tau)} = e^{-in\theta}e^{-i\pi} = -e^{-in\theta}$, the substitution $\theta \mapsto \theta + \tau$ in the integral (legitimate by periodicity) gives

$$\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta + \tau) e^{-in(\theta+\tau)} d\theta = -\frac{1}{2\pi} \int_0^{2\pi} f(\theta + \tau) e^{-in\theta} d\theta.$$

Adding this identity to the original one and dividing by 2,

$$\hat{f}(n) = \frac{1}{4\pi} \int_0^{2\pi} [f(\theta) - f(\theta + \pi/n)] e^{-in\theta} d\theta, \quad \text{hence} \quad |\hat{f}(n)| \leq \frac{1}{2} \|f - f(\cdot + \pi/n)\|_1.$$

Continuity of translations in L^1 — that is, $\|f - f(\cdot + \tau)\|_1 \rightarrow 0$ as $\tau \rightarrow 0$, itself a consequence of the density of step functions — concludes the proof. This variant has the merit of producing an *estimate*, not merely a limit: if f is α -Hölder, $\|f - f(\cdot + \tau)\|_1 \leq C|\tau|^\alpha$ and hence $|\hat{f}(n)| \leq \frac{1}{2} C(\pi/|n|)^\alpha$. $\square$

Remark 17.1: Why exactly this lemma is needed. Every criterion of pointwise convergence in this chapter has the same shape: one writes $S_N f(\theta_0) - L$ as an integral $\int_0^\pi g(t) \sin((N + \frac{1}{2})t) dt$ (Step 4 of the proof of Theorem 17.4), and then asks that g lie in L^1 . The whole work of each criterion is exactly there: proving the integrability of g . Dini does it with an integral hypothesis, Dirichlet–Jordan with a variation hypothesis, Theorem 17.4 with the one-sided derivatives. The Riemann–Lebesgue lemma is the common machine that, once that integrability is established, produces the conclusion.

17.2. Convergence of the Fourier series

Theorem 17.1: Convergence in L^2 and completeness of the trigonometric system. For every $f \in L^2(\mathbb{T})$ the partial sums $S_N f(\theta) = \sum_{|n| \leq N} \hat{f}(n) e^{in\theta}$ converge to f in the L^2 norm:

$$\|f - S_N f\|_2 \rightarrow 0 \quad (N \rightarrow \infty).$$

Equivalently, the system $\{e_n\}_{n \in \mathbb{Z}}$ is complete in $L^2(\mathbb{T})$, and Bessel's inequality becomes an equality (Parseval's identity, Theorem (Volume II, Chapter 4)).

Proof. Step 1 — reduction to the density of trigonometric polynomials. By Theorem (Volume II, Chapter 4)(ii), $S_N f$ is the best approximation of f in the L^2 norm among all trigonometric polynomials of degree $\leq N$. Hence, if for every $\varepsilon > 0$ there exists *some* trigonometric polynomial P (of some degree M) with $\|f - P\|_2 < \varepsilon$, then for every $N \geq M$

$$\|f - S_N f\|_2 \leq \|f - P\|_2 < \varepsilon,$$

and the claim follows. It remains to prove that trigonometric polynomials are *dense* in $L^2(\mathbb{T})$.

Step 2 — density of continuous functions. It is a standard result of measure theory that $C(\mathbb{T})$ is dense in $L^2(\mathbb{T})$: given $f \in L^2$ and $\varepsilon > 0$, there exists g continuous and 2π -periodic with $\|f - g\|_2 < \varepsilon/2$ (see Rudin, *Real and Complex Analysis*, Chapter 3).

Step 3 — density of trigonometric polynomials in $C(\mathbb{T})$. By Fejér's theorem (Theorem 17.7, proved later in this same section without using the present theorem, hence without circularity) the Cesàro means $\sigma_N g$ converge to g *uniformly*. Since on a space of total measure 1 the L^2 norm is dominated by the uniform norm,

$$\|g - \sigma_N g\|_2 \leq \|g - \sigma_N g\|_\infty \rightarrow 0,$$

there exists N with $\|g - \sigma_N g\|_2 < \varepsilon/2$, and $\sigma_N g$ is a trigonometric polynomial.

Step 4 — conclusion. By the triangle inequality, $\|f - \sigma_N g\|_2 \leq \|f - g\|_2 + \|g - \sigma_N g\|_2 < \varepsilon$: trigonometric polynomials are dense in $L^2(\mathbb{T})$, and Step 1 concludes.

Step 5 — from Bessel to Parseval. By Theorem (Volume II, Chapter 4)(iii),

$$\|f - S_N f\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} |\hat{f}(n)|^2.$$

Passing to the limit as $N \rightarrow \infty$, the left-hand side tends to 0, hence $\sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 = \|f\|_2^2$. $\square$

Corollary 17.2: Uniqueness of the Fourier coefficients. If $f, g \in L^2(\mathbb{T})$ have the same Fourier coefficients, then $f = g$ almost everywhere. In particular, if $\hat{f}(n) = 0$ for every $n \in \mathbb{Z}$, then $f = 0$ almost everywhere; if moreover f is continuous, $f \equiv 0$.

Proof. Set $h = f - g$; by linearity $\hat{h}(n) = 0$ for every n . By Parseval's identity (Theorem 17.1, Step 5),

$$\|h\|_2^2 = \sum_{n \in \mathbb{Z}} |\hat{h}(n)|^2 = 0,$$

hence $h = 0$ almost everywhere. If h is continuous and zero almost everywhere, then it is zero everywhere: if $h(\theta_0) \neq 0$ for some θ_0 , by continuity h would be nonzero on a neighbourhood of θ_0 , which has positive measure. $\square$

Remark 17.2: Three regimes of convergence. L^2 convergence (Theorem 17.1) is the most general and robust. Under stronger hypotheses one gets finer results: if $f \in C^1(\mathbb{T})$, the series converges uniformly (and hence pointwise) to f ; if f is merely continuous, pointwise convergence may fail at individual points (a phenomenon tied to the Dirichlet kernel), but the Cesàro mean of the partial sums always converges uniformly (Fejér's theorem).

17.2.1 The Dirichlet kernel and the Gibbs phenomenon

Definition 17.2: Dirichlet kernel.

$$D_N(\theta) = \sum_{n=-N}^N e^{in\theta}.$$

Theorem 17.2: Closed form of the Dirichlet kernel. For $\theta \not\equiv 0 \pmod{2\pi}$:

$$D_N(\theta) = \frac{\sin\left((N + \frac{1}{2})\theta\right)}{\sin(\theta/2)}.$$

Proof. With $x = e^{i\theta}$, $D_N(\theta) = \sum_{n=-N}^N x^n = x^{-N} \sum_{k=0}^{2N} x^k = x^{-N} \frac{x^{2N+1} - 1}{x - 1}$ (geometric sum, Corollary 3.2 generalised to ratio $x \neq 1$) $= \frac{x^{N+1} - x^{-N}}{x - 1}$. Multiplying numerator and denominator by $x^{-1/2}$:

$$D_N(\theta) = \frac{x^{N+1/2} - x^{-(N+1/2)}}{x^{1/2} - x^{-1/2}} = \frac{2i \sin((N + \frac{1}{2})\theta)}{2i \sin(\theta/2)} = \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)}.$$

$\square$

Theorem 17.3: Partial sums as convolution with the Dirichlet kernel.

$$S_N f(\theta) = \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) D_N(\theta - \varphi) d\varphi.$$

Proof. Substituting Definition (Volume II, Chapter 4) into $S_N f(\theta) = \sum_{n=-N}^N \hat{f}(n) e^{in\theta}$ and exchanging the finite sum and the integral:

$$\begin{aligned} S_N f(\theta) &= \sum_{n=-N}^N \left(\frac{1}{2\pi} \int_0^{2\pi} f(\varphi) e^{-in\varphi} d\varphi \right) e^{in\theta} = \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) \sum_{n=-N}^N e^{in(\theta-\varphi)} d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) D_N(\theta - \varphi) d\varphi. \end{aligned}$$

□

Remark 17.3: The Gibbs phenomenon. The Dirichlet kernel is not positive (it oscillates), unlike the Fejér kernel cited in Observation 17.2: this is precisely why $S_N f$ can behave worse than f near a discontinuity. For a function with a jump (for example the square wave), the partial sum $S_N f$ exhibits an overshoot just before and after the jump, of an amplitude that **does not tend to zero** as N grows: it stabilises around 8.95% of the height of the jump (the Wilbraham–Gibbs constant, tied to the value of the sine integral $\text{Si}(\pi) = \int_0^\pi \frac{\sin t}{t} dt$), while its position moves ever closer to the jump point. This is a failure of uniform convergence, not a computational error: Figure 17.1 shows it for the square wave $\text{sq}(\theta) \sim \frac{4}{\pi} \sum_{k \text{ odd}} \frac{\sin k\theta}{k}$.

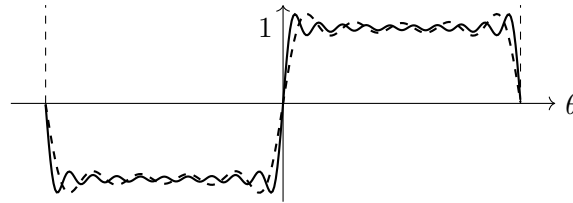


Figure 17.1: Partial sums of the square wave with 5 terms (dashed) and 10 terms (solid): the overshoot near the jump at $\theta = \pi$ does not shrink going from 5 to 10 terms, it only compresses toward the discontinuity point — the Gibbs phenomenon.

17.2.2 Pointwise convergence: the Dirichlet–Jordan theorem

Theorem 17.4: Pointwise convergence (Dirichlet–Jordan). Let $f \in L^1(\mathbb{T})$ and let $\theta_0 \in \mathbb{T}$ be a point at which the one-sided limits $f(\theta_0^\pm)$ exist and are finite, together with the one-sided derivatives, i.e. the following exist and are finite:

$$\lim_{t \rightarrow 0^+} \frac{f(\theta_0 + t) - f(\theta_0^+)}{t}, \quad \lim_{t \rightarrow 0^+} \frac{f(\theta_0 - t) - f(\theta_0^-)}{t}.$$

Then

$$S_N f(\theta_0) \longrightarrow \frac{f(\theta_0^+) + f(\theta_0^-)}{2} \quad (N \rightarrow \infty).$$

In particular, if f is continuous and differentiable at θ_0 , the Fourier series converges there to $f(\theta_0)$; at a jump point it converges to the average of the two one-sided limits.

Proof. Step 1 — normalisation of the kernel. By Theorem (Volume II, Chapter 4), integrating the Definition 17.2 term by term,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(t) dt = \sum_{|n| \leq N} \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int} dt = 1,$$

since only the term $n = 0$ survives. Moreover D_N is *even* (a sum of e^{int} over symmetric indices, or, by Theorem 17.2, a ratio of two odd functions), hence

$$\frac{1}{2\pi} \int_0^{\pi} D_N(t) dt = \frac{1}{2}.$$

Step 2 — symmetrisation. By Theorem 17.3, with the substitution $\varphi = \theta_0 - t$ and using periodicity,

$$S_N f(\theta_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta_0 - t) D_N(t) dt.$$

Splitting the integral over $[-\pi, 0]$ and $[0, \pi]$ and substituting $t \mapsto -t$ in the first, with D_N even:

$$S_N f(\theta_0) = \frac{1}{2\pi} \int_0^{\pi} [f(\theta_0 - t) + f(\theta_0 + t)] D_N(t) dt.$$

Step 3 — subtracting the expected limit. Set $L = \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$; by Step 1, $\frac{1}{2\pi} \int_0^{\pi} 2L D_N(t) dt = L$, hence

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^{\pi} \underbrace{[f(\theta_0 - t) + f(\theta_0 + t) - 2L]}_{=: \psi(t)} D_N(t) dt.$$

Step 4 — the auxiliary function is integrable. By Theorem 17.2, $D_N(t) = \sin((N + \frac{1}{2})t) / \sin(t/2)$; set therefore

$$g(t) = \frac{\psi(t)}{\sin(t/2)}, \quad t \in (0, \pi],$$

so that $S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^{\pi} g(t) \sin((N + \frac{1}{2})t) dt$. We verify that $g \in L^1(0, \pi)$. Write

$$\psi(t) = [f(\theta_0 + t) - f(\theta_0^+)] + [f(\theta_0 - t) - f(\theta_0^-)].$$

By the hypothesis on the one-sided derivatives, each bracket is $O(t)$ as $t \rightarrow 0^+$; since $\sin(t/2) \sim t/2$, the ratio $g(t)$ stays *bounded* on a right neighbourhood of 0. On $[\delta, \pi]$, instead, $\sin(t/2) \geq \sin(\delta/2) > 0$ and g is integrable because ψ is (a sum of translates of $f \in L^1$). Hence $g \in L^1(0, \pi)$.

Step 5 — Riemann–Lebesgue. Expanding $\sin((N + \frac{1}{2})t) = \sin(Nt) \cos(t/2) + \cos(Nt) \sin(t/2)$,

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^{\pi} [g(t) \cos \frac{t}{2}] \sin(Nt) dt + \frac{1}{2\pi} \int_0^{\pi} [g(t) \sin \frac{t}{2}] \cos(Nt) dt.$$

Both functions in square brackets belong to $L^1(0, \pi)$ (the product of an L^1 function and a bounded one). Extending them oddly, resp. evenly, to $(-\pi, \pi)$, the two integrals are — up to constants — Fourier coefficients of index N of L^1 functions: by Corollary (Volume II, Chapter 4) both tend to zero as $N \rightarrow \infty$. Hence $S_N f(\theta_0) \rightarrow L$. $\square$

Remark 17.4: The value at jump points. Theorem 17.4 answers a question intuition leaves open: at a discontinuity point the series does not "choose" one of the two values, but converges to their average. For the square wave of Example (Volume II, Chapter 4), at $\theta = 0$ the series equals $\frac{4}{\pi} \sum_k \sin 0 / (2k + 1) = 0$, which is exactly $\frac{1}{2}(1 + (-1)) = 0$. ✓ The same happens at $\theta = \pi$. Note that this does not contradict the Gibbs phenomenon: convergence at the jump point is perfect, it is uniform convergence on a neighbourhood that fails.

17.2.3 Uniform convergence

Theorem 17.5: Absolute and uniform convergence. Let $f \in L^2(\mathbb{T})$ with coefficients $c_n = \hat{f}(n)$.

- (i) If $\sum_{n \in \mathbb{Z}} |c_n| < \infty$, then the Fourier series converges absolutely and uniformly on $\mathbb{T}$, its sum is continuous and coincides with f almost everywhere (everywhere, if f is continuous).
- (ii) If f is continuous, 2π -periodic and piecewise C^1 , then $\sum_n |c_n| < \infty$ and hence the series converges uniformly to f .

Proof. (i) Since $|c_n e^{in\theta}| = |c_n|$ for every θ , the hypothesis $\sum |c_n| < \infty$ allows the Weierstrass criterion to be applied: the series converges totally, hence absolutely and uniformly on $\mathbb{T}$. The uniform limit of a sequence of continuous functions (the partial sums are trigonometric polynomials, hence continuous) is continuous; call it g . Since uniform convergence on a domain of finite measure implies convergence in L^2 , one has $S_N f \rightarrow g$ in L^2 ; but by Theorem 17.1, $S_N f \rightarrow f$ in L^2 . By uniqueness of the L^2 limit, $f = g$ almost everywhere; if f is continuous, two continuous functions equal almost everywhere coincide everywhere (Corollary 17.2).

(ii) Let $f' \in L^2(\mathbb{T})$ be the derivative (defined almost everywhere, and in L^2 because f is piecewise C^1 with finitely many pieces). Integrating by parts over a period, with f continuous and periodic so that the boundary terms cancel:

$$\hat{f}'(n) = \frac{1}{2\pi} \int_0^{2\pi} f'(\theta) e^{-in\theta} d\theta = \frac{1}{2\pi} \underbrace{[f(\theta) e^{-in\theta}]_0^{2\pi}}_{=0} + \frac{in}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta = in c_n.$$

Hence $|c_n| = |\hat{f}'(n)|/|n|$ for $n \neq 0$. Applying the Cauchy–Schwarz inequality to the sequence:

$$\sum_{n \neq 0} |c_n| = \sum_{n \neq 0} \frac{1}{|n|} |\hat{f}'(n)| \leq \left(\sum_{n \neq 0} \frac{1}{n^2} \right)^{1/2} \left(\sum_{n \neq 0} |\hat{f}'(n)|^2 \right)^{1/2}.$$

The first series equals $2 \cdot \pi^2/6 = \pi^2/3$ (Example (Volume II, Chapter 4), Step 5); the second is finite by Bessel's inequality (Theorem (Volume II, Chapter 4)(iv)) applied to $f' \in L^2$. Hence $\sum_n |c_n| < \infty$ and (i) applies. $\square$

Remark 17.5: Regularity and decay: the practical rule. The proof of (ii) contains the key relation

$$\hat{f}'(n) = in \hat{f}(n),$$

which, iterated, gives $\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n)$. Since by Riemann–Lebesgue $\widehat{f^{(k)}}(n) \rightarrow 0$, it follows that if $f \in C^k(\mathbb{T})$ then

$$\hat{f}(n) = o\left(\frac{1}{|n|^k}\right).$$

The more regular f is, the faster its coefficients decay — and this is exactly the principle that in Chapter 2 governed the order of convergence of the periodic trapezoidal rule (Theorem, Volume II, Chapter 2). The two statements are, on closer inspection, the same theorem read from two sides: there one estimated the aliasing error $\sum_{m \geq 1} c_m N$, here the decay of the c_n .

17.2.4 The Fejér kernel and Cesàro means

The Dirichlet kernel is not positive, and all the pathologies stem from this: the Gibbs phenomenon, the possible pointwise divergence for merely continuous functions. Replacing the partial sums with their *arithmetic means* yields a positive kernel, and with it a convergence theorem without exceptions.

Definition 17.3: Cesàro means and the Fejér kernel. Set

$$\sigma_N f = \frac{1}{N+1} \sum_{n=0}^N S_n f, \quad F_N(\theta) = \frac{1}{N+1} \sum_{n=0}^N D_n(\theta).$$

$\sigma_N f$ is called the **Cesàro mean** of order N and F_N the **Fejér kernel**.

Theorem 17.6: Closed form and properties of the Fejér kernel. For $\theta \not\equiv 0 \pmod{2\pi}$:

$$F_N(\theta) = \frac{1}{N+1} \left(\frac{\sin((N+1)\theta/2)}{\sin(\theta/2)} \right)^2 = \sum_{|n| \leq N} \left(1 - \frac{|n|}{N+1} \right) e^{in\theta},$$

and the three properties of an **approximate identity** hold:

- (a) $F_N \geq 0$ everywhere;
- (b) $\frac{1}{2\pi} \int_{-\pi}^{\pi} F_N(\theta) d\theta = 1$ for every N ;
- (c) for every $\delta \in (0, \pi)$, $\sup_{\delta \leq |\theta| \leq \pi} F_N(\theta) \rightarrow 0$ as $N \rightarrow \infty$.

Moreover $\sigma_N f = f * F_N$, i.e. $\sigma_N f(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta - t) F_N(t) dt$.

Proof. Step 1 — closed form. By Theorem 17.2, $D_n(\theta) \sin(\theta/2) = \sin((n + \frac{1}{2})\theta)$. Multiply by $2 \sin(\theta/2)$ and use the product-to-sum formula $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$ with $A = (n + \frac{1}{2})\theta$, $B = \theta/2$:

$$2 \sin^2 \frac{\theta}{2} D_n(\theta) = \cos(n\theta) - \cos((n+1)\theta).$$

Summing for $n = 0, \dots, N$, the right-hand sum is *telescoping*:

$$2 \sin^2 \frac{\theta}{2} \sum_{n=0}^N D_n(\theta) = \cos 0 - \cos((N+1)\theta) = 1 - \cos((N+1)\theta) = 2 \sin^2 \frac{(N+1)\theta}{2},$$

having used $1 - \cos u = 2 \sin^2(u/2)$. Dividing by $2 \sin^2(\theta/2)$ and by $N+1$ gives the closed form.

Step 2 — spectral form. Each $D_n = \sum_{|k| \leq n} e_k$, so when summing $D_0, \dots, D_N$ the mode e_k appears once for each $n \geq |k|$, i.e. $N+1 - |k|$ times. Dividing by $N+1$:

$$F_N = \sum_{|k| \leq N} \frac{N+1 - |k|}{N+1} e_k = \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1} \right) e_k.$$

Step 3 — (a) positivity. Immediate from the closed form: it is a square divided by $N+1 > 0$. (At the points $\theta \equiv 0$ the value is $N+1$ by continuity, still positive.)

Step 4 — (b) normalisation. From the spectral form, integrating term by term and using Theorem (Volume II, Chapter 4), only the mode $k=0$ survives, whose coefficient is 1.

Step 5 — (c) concentration. For $\delta \leq |\theta| \leq \pi$, $\sin^2(\theta/2) \geq \sin^2(\delta/2) > 0$, while the numerator is ≤ 1 . Hence

$$0 \leq F_N(\theta) \leq \frac{1}{(N+1)\sin^2(\delta/2)} \rightarrow 0,$$

uniformly on $\{\delta \leq |\theta| \leq \pi\}$.

Step 6 — convolution. Averaging over $n = 0, \dots, N$ the identity of Theorem 17.3 and exchanging the finite sum and the integral gives $\sigma_N f = f * F_N$. $\square$

Theorem 17.7: Fejér's theorem. Let $f : \mathbb{T} \rightarrow \mathbb{C}$ be continuous. Then $\sigma_N f \rightarrow f$ uniformly on $\mathbb{T}$. More generally, if $f \in L^1(\mathbb{T})$ and θ_0 is a point at which the one-sided limits exist, $\sigma_N f(\theta_0) \rightarrow \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$.

Proof. We prove the uniform case; the other is identical, with the symmetrisation of Step 2 of the proof of Theorem 17.4.

Step 1 — representation of the error. By Theorem 17.6(b),

$$\sigma_N f(\theta) - f(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(\theta - t) - f(\theta)] F_N(t) dt,$$

having written $f(\theta) = \frac{1}{2\pi} \int f(\theta) F_N(t) dt$.

Step 2 — uniform continuity. f is continuous on the compact set $\mathbb{T}$, hence uniformly continuous and bounded: let $M = \sup |f|$. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|t| < \delta \implies |f(\theta - t) - f(\theta)| < \frac{\varepsilon}{2} \quad \text{for every } \theta.$$

Step 3 — splitting the integral. We separate $|t| < \delta$ from $\delta \leq |t| \leq \pi$ and bound in modulus, using $F_N \geq 0$ (property (a)):

$$|\sigma_N f(\theta) - f(\theta)| \leq \underbrace{\frac{1}{2\pi} \int_{|t| < \delta} \frac{\varepsilon}{2} F_N(t) dt}_{(I)} + \underbrace{\frac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} 2M F_N(t) dt}_{(II)}.$$

Step 4 — bounding the two pieces. For (I), extending the integral to the whole period and using (b):

$$(I) \leq \frac{\varepsilon}{2} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} F_N = \frac{\varepsilon}{2}.$$

For (II), using the concentration property (c): there exists N_0 such that for $N \geq N_0$, $F_N(t) \leq \varepsilon/(4M+1)$ on $\delta \leq |t| \leq \pi$, and hence

$$(II) \leq 2M \cdot \frac{\varepsilon}{4M+1} < \frac{\varepsilon}{2}.$$

Step 5 — conclusion. For $N \geq N_0$ and for every θ , $|\sigma_N f(\theta) - f(\theta)| < \varepsilon$: the convergence is uniform. $\square$

Corollary 17.3: Weierstrass's trigonometric approximation theorem. Trigonometric polynomials are dense in $C(\mathbb{T})$ with respect to the uniform norm: for every con-

tinuous, 2π -periodic f and every $\varepsilon > 0$ there is a trigonometric polynomial P with $\sup_{\mathbb{T}} |f - P| < \varepsilon$. They are consequently dense also in $L^p(\mathbb{T})$ for every $1 \leq p < \infty$.

Proof. By Theorem 17.7, $\sigma_N f \rightarrow f$ uniformly, and $\sigma_N f$ is a trigonometric polynomial of degree $\leq N$ by Theorem 17.6 (spectral form). It suffices to choose N large enough. For density in L^p : $C(\mathbb{T})$ is dense in $L^p(\mathbb{T})$ for $p < \infty$, and on a space of total measure 1 the L^p norm is dominated by the uniform norm, hence uniform approximation implies approximation in L^p . $\square$

Remark 17.6: Why Fejér saves what Dirichlet loses. The comparison between the two kernels is instructive and explains everything:

	Dirichlet D_N	Fejér F_N
Sign	oscillates, changes sign	≥ 0 always
Mean	1	1
$\frac{1}{2\pi} \int K_N $	$\sim \frac{4}{\pi^2} \ln N \rightarrow \infty$	$= 1$, bounded
Convergence	can fail for continuous f	always uniform if f continuous
Gibbs	present	absent

The quantity $L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N|$ is called the **Lebesgue constant**; the fact that it diverges (logarithmically) is the structural reason why there exists a continuous function whose Fourier series diverges at a point — a result of du Bois-Reymond (1876). The Fejér kernel, being positive, has Lebesgue constant 1: it is an approximate identity in the proper sense, and this is why the pathologies are absent.

17.2.5 The Poisson kernel and Abel summation

Definition 17.4: Poisson kernel. For $0 \leq r < 1$ set

$$P_r(\theta) = \sum_{n \in \mathbb{Z}} r^{|n|} e^{in\theta}.$$

Theorem 17.8: Closed form of the Poisson kernel and Abel summation. For $0 \leq r < 1$:

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2} \geq 0, \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta) d\theta = 1,$$

and for every $\delta \in (0, \pi)$, $\sup_{\delta \leq |\theta| \leq \pi} P_r(\theta) \rightarrow 0$ as $r \rightarrow 1^-$. Consequently, if f is continuous on $\mathbb{T}$,

$$u(r, \theta) := (f * P_r)(\theta) = \sum_{n \in \mathbb{Z}} r^{|n|} \hat{f}(n) e^{in\theta} \longrightarrow f(\theta) \quad \text{uniformly as } r \rightarrow 1^-,$$

By Theorem 16.6 of Chapter 16, u is moreover the unique harmonic function in the disc, continuous up to the boundary, with boundary value f (solution of the Dirichlet problem); uniqueness is not proved here, only the uniform convergence just established.

Proof. Step 1 — closed form. Split the sum into $n \geq 0$ and $n \leq -1$; both are geometric series of ratio with modulus $r < 1$, hence convergent:

$$P_r(\theta) = \sum_{n \geq 0} (re^{i\theta})^n + \sum_{m \geq 1} (re^{-i\theta})^m = \frac{1}{1 - re^{i\theta}} + \frac{re^{-i\theta}}{1 - re^{-i\theta}}.$$

Reducing to a common denominator $(1 - re^{i\theta})(1 - re^{-i\theta}) = 1 - 2r \cos \theta + r^2$ (by Theorem (Volume II, Chapter 3) of Chapter 3), the numerator is

$$(1 - re^{-i\theta}) + re^{-i\theta}(1 - re^{i\theta}) = 1 - re^{-i\theta} + re^{-i\theta} - r^2 = 1 - r^2.$$

Step 2 — positivity and normalisation. The numerator $1 - r^2 > 0$; the denominator is $|1 - re^{i\theta}|^2 > 0$ because $r < 1$. For the normalisation, integrating the definition term by term (legitimate by uniform convergence in θ for fixed r) and using Theorem (Volume II, Chapter 4), only the term $n = 0$ survives, which equals $r^0 = 1$.

Step 3 — concentration. For $\delta \leq |\theta| \leq \pi$, $\cos \theta \leq \cos \delta$, hence

$$1 - 2r \cos \theta + r^2 \geq 1 - 2r \cos \delta + r^2 = (1 - r)^2 + 2r(1 - \cos \delta) \geq 2r(1 - \cos \delta),$$

and therefore

$$P_r(\theta) \leq \frac{1 - r^2}{2r(1 - \cos \delta)} \rightarrow 0 \quad (r \rightarrow 1^-).$$

Step 4 — convergence. The three properties of Steps 2 and 3 are exactly those of the Fejér kernel: the proof of Theorem 17.7 repeats word for word, with $r \rightarrow 1^-$ in place of $N \rightarrow \infty$, and gives the uniform convergence of $f * P_r$ to f .

Step 5 — spectral form and harmonicity. Since $\widehat{P_r}(n) = r^{|n|}$, the convolution theorem (Theorem, Volume II, Chapter 4) gives $\widehat{f * P_r}(n) = r^{|n|}\widehat{f}(n)$, which is the series in the statement. Writing $z = re^{i\theta}$, one has $r^{|n|}e^{in\theta} = z^n$ for $n \geq 0$ and $= \bar{z}^{|n|}$ for $n < 0$: hence u is the sum of a holomorphic function and the conjugate of a holomorphic function on $|z| < 1$, and is therefore harmonic (the real and imaginary parts of holomorphic functions are harmonic). $\square$

***Remark 17.7: Three approximate identities, one idea.** Dirichlet, Fejér and Poisson are three summation kernels for the same series. The common structure — normalisation to 1, concentration around the origin, and (for the last two) positivity — is that of an **approximate identity**: convolving with it reproduces the function in the limit. The difference lies entirely in the sign:*

- D_N oscillates $\Rightarrow$ faithful reconstruction in quadratic mean, but with Gibbs and possible pointwise divergences;
- $F_N \geq 0 \Rightarrow$ uniform reconstruction without Gibbs, at the price of a damping of the high coefficients (the factor $1 - |n|/(N + 1)$), i.e. a slight loss of detail;
- $P_r \geq 0 \Rightarrow$ like Fejér, with damping $r^{|n|}$; it is the “continuous” version and is the one that solves the Dirichlet problem in the disc.

In signal processing these factors are called windows: the Fejér factor is the triangular (Bartlett) window, and the trade-off between resolution and the absence of spurious oscillations is exactly the one discussed in Section 22.10.3 regarding spectral leakage.

17.2.6 The Gibbs phenomenon: the exact constant

Theorem 17.9: The Wilbraham–Gibbs constant. Let f be the square wave of Example (Volume II, Chapter 4), with a jump of amplitude 2 at $\theta = 0$, and let $S_N f$ be its partial sum with $N = 2m - 1$. Then the first maximum of $S_N f$ to the right of the origin is at $\theta_m = \pi/(2m)$, and

$$\lim_{m \rightarrow \infty} S_{2m-1} f(\theta_m) = \frac{2}{\pi} \int_0^\pi \frac{\sin u}{u} du = \frac{2}{\pi} \text{Si}(\pi) = 1.178979744 \dots$$

The overshoot beyond the limiting value 1 is thus 17.898% of the half-amplitude of the jump, i.e. 8.949% of the total amplitude, and does not tend to zero as N grows.

Proof. Step 1 — the derivative of the partial sum. From Example (Volume II, Chapter 4),

$$S_{2m-1} f(\theta) = \frac{4}{\pi} \sum_{j=1}^m \frac{\sin((2j-1)\theta)}{2j-1}, \quad \text{hence} \quad \frac{d}{d\theta} S_{2m-1} f(\theta) = \frac{4}{\pi} \sum_{j=1}^m \cos((2j-1)\theta).$$

Step 2 — closed form of the sum of odd cosines. Multiplying by $2 \sin \theta$ and using $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$ with $A = (2j-1)\theta$, $B = \theta$:

$$2 \sin \theta \sum_{j=1}^m \cos((2j-1)\theta) = \sum_{j=1}^m [\sin(2j\theta) - \sin((2j-2)\theta)] = \sin(2m\theta),$$

by telescoping (the term $j = 1$ contributes $\sin 2\theta - \sin 0$). Hence

$$\frac{d}{d\theta} S_{2m-1} f(\theta) = \frac{2}{\pi} \cdot \frac{\sin(2m\theta)}{\sin \theta}.$$

Step 3 — the first critical point. For $\theta \in (0, \pi)$ the denominator is positive, so the first zero of the derivative to the right of 0 occurs when $2m\theta = \pi$, i.e. $\theta_m = \pi/(2m)$. Since the derivative is positive on $(0, \theta_m)$ (where $0 < 2m\theta < \pi$), the point θ_m is a maximum.

Step 4 — value at the maximum. Since $S_{2m-1} f(0) = 0$, integrating the derivative:

$$S_{2m-1} f(\theta_m) = \frac{2}{\pi} \int_0^{\pi/(2m)} \frac{\sin(2mt)}{\sin t} dt.$$

Substituting $u = 2mt$, $dt = du/(2m)$:

$$S_{2m-1} f(\theta_m) = \frac{2}{\pi} \int_0^\pi \sin u \cdot \frac{1}{2m \sin(u/(2m))} du.$$

Step 5 — passing to the limit. As $m \rightarrow \infty$, for fixed $u \in (0, \pi]$,

$$2m \sin \frac{u}{2m} \rightarrow u,$$

since $\sin x/x \rightarrow 1$. Moreover the integrand is uniformly dominated: from the concavity of $\sin$ on $[0, \pi/2]$, $\sin x \geq \frac{2}{\pi}x$ for $x \in [0, \pi/2]$, hence $2m \sin(u/(2m)) \geq \frac{2}{\pi}u$ and the integrand is bounded by $\frac{\pi}{2} \cdot \frac{\sin u}{u} \leq \frac{\pi}{2}$, which is integrable on $[0, \pi]$. By the dominated convergence theorem,

$$\lim_{m \rightarrow \infty} S_{2m-1} f(\theta_m) = \frac{2}{\pi} \int_0^\pi \frac{\sin u}{u} du.$$

Step 6 — the numerical value. The integral is the sine integral $\text{Si}(\pi) = 1.851937052 \dots$, hence the limit equals

$$\frac{2}{\pi} \cdot 1.851937052 = 1.178979744 \dots$$

The overshoot beyond the value 1 is 0.178979744, i.e. 17.898% of 1 (half-amplitude of the jump) and 8.949% of 2 (total amplitude). $\square$

Remark 17.8: What the Gibbs phenomenon is not. Three clarifications that Theorem 17.9 makes rigorous:

- **It is not a computational or truncation error that can be fixed:** the amplitude of the overshoot has a positive limit, and no increase of N reduces it.
- **It does not contradict pointwise convergence:** by Theorem 17.4, at every fixed point the series converges. The overshoot moves toward the jump ($\theta_m = \pi/(2m) \rightarrow 0$) without ever settling on a point: it is uniform convergence that fails, not pointwise convergence.
- **It is not unavoidable:** replacing the partial sums with the Fejér means (Theorem 17.7) the phenomenon disappears, because the Fejér kernel is positive. The price is a slowdown of convergence where f is regular.

Historically the phenomenon was described by Henry Wilbraham in 1848 and rediscovered, independently and with greater resonance, by J. Willard Gibbs in 1899 in a letter to Nature: hence the double name.

Remark 17.9: The two extremes of the theory: Kolmogorov and Carleson. Two deep results delimit what one can hope for from pointwise convergence, and they are cited here without proof because they lie well beyond the level of this chapter.

- **Kolmogorov (1926).** There exists $f \in L^1(\mathbb{T})$ whose Fourier series diverges at every point. Mere integrability is not enough.
- **Carleson (1966).** For every $f \in L^2(\mathbb{T})$ the Fourier series converges to f almost everywhere. The result, extended by Hunt (1968) to L^p with $p > 1$, resolved positively a conjecture that had been open for half a century.

Between the two extremes sits the theorem of du Bois-Reymond cited in Observation 17.6: there exists a continuous function whose series diverges at a point. The operational moral, for anyone computing, is the one already stated: for piecewise C^1 functions, pointwise convergence is guaranteed everywhere (Theorem 17.4), and for continuous functions reliable reconstruction goes through the Fejér means.

17.3. The Lebesgue constants and logarithmic divergence

Observation 17.6 announced that the integral of the *modulus* of the Dirichlet kernel diverges like $\ln N$, and Exercise 17.2 derived the lower bound. Here the question is settled in both directions: an upper bound of the same order is proved, so that L_N is *exactly* of order $\ln N$, and the structural consequence is drawn — the existence of a continuous function whose series diverges at a point.

Definition 17.5: Lebesgue constant. For $N \in \mathbb{N}$ set

$$L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)| d\theta = \|D_N\|_1,$$

with D_N the Dirichlet kernel of Definition 17.2.

Lemma 17.3: Gap between the cosecant and its asymptote. Set, for $t \in (0, \pi]$,

$$g(t) = \frac{1}{\sin(t/2)} - \frac{2}{t}.$$

Then

$$0 \leq g(t) \leq \frac{\pi t}{24} \leq \frac{\pi^2}{24} < 0.412.$$

Proof. **Step 1 — three elementary inequalities.** For $x \geq 0$,

$$(a) \sin x \leq x, \quad (b) \sin x \geq x - \frac{x^3}{6}, \quad (c) \sin x \geq \frac{2}{\pi}x \text{ for } x \in \left[0, \frac{\pi}{2}\right].$$

For (a): the function $h(x) = x - \sin x$ has $h(0) = 0$ and $h'(x) = 1 - \cos x \geq 0$, hence it is non-decreasing and so $h \geq 0$. For (b): let $m(x) = \cos x - 1 + \frac{x^2}{2}$; then $m(0) = 0$ and $m'(x) = x - \sin x \geq 0$ by (a), hence $m \geq 0$, i.e. $\cos x \geq 1 - \frac{x^2}{2}$. Now let $k(x) = \sin x - x + \frac{x^3}{6}$: one has $k(0) = 0$ and $k'(x) = \cos x - 1 + \frac{x^2}{2} = m(x) \geq 0$, hence $k \geq 0$. For (c): on $[0, \pi]$, $(\sin)'' = -\sin \leq 0$, so $\sin$ is concave on $[0, \pi]$; a concave function on an interval lies *above* the chord joining its endpoints, and the chord between $(0, 0)$ and $(\pi/2, 1)$ is the line $x \mapsto 2x/\pi$.

Step 2 — rewriting g . Reducing to a common denominator,

$$g(t) = \frac{t - 2\sin(t/2)}{t \sin(t/2)}, \quad t \in (0, \pi].$$

The denominator is positive, because $t > 0$ and $t/2 \in (0, \pi/2]$ where the sine is positive.

Step 3 — the numerator is non-negative and $O(t^3)$. From (a) with $x = t/2$ it follows that $2\sin(t/2) \leq t$, so the numerator is ≥ 0 and hence $g \geq 0$. From (b) with $x = t/2$ it follows that

$$2\sin \frac{t}{2} \geq 2\left(\frac{t}{2} - \frac{t^3}{48}\right) = t - \frac{t^3}{24}, \quad \text{i.e.} \quad t - 2\sin \frac{t}{2} \leq \frac{t^3}{24}.$$

Step 4 — lower bound for the denominator. From (c) with $x = t/2 \in (0, \pi/2]$ it follows that $\sin(t/2) \geq t/\pi$, hence $t \sin(t/2) \geq t^2/\pi$.

Step 5 — conclusion. Combining Steps 3 and 4,

$$0 \leq g(t) \leq \frac{t^3/24}{t^2/\pi} = \frac{\pi t}{24} \leq \frac{\pi \cdot \pi}{24} = \frac{\pi^2}{24} = 0.41123\dots$$

□

Theorem 17.10: Logarithmic growth of the Lebesgue constants. For every $N \geq 1$,

$$\frac{4}{\pi^2}(\ln N - 1) \leq L_N \leq \frac{2}{\pi}\left(1 + \ln\left((N + \frac{1}{2})\pi\right)\right) + \frac{\pi^2}{24}.$$

In particular $L_N \rightarrow \infty$ and L_N is exactly of order $\ln N$: the ratio $L_N/\ln N$ stays, for $N \geq 3$, between two positive constants.

Proof. **Step 1 — reduction to an integral on $[0, \pi]$.** By Theorem 17.2, $D_N(\theta) = \sin(M\theta)/\sin(\theta/2)$ with $M = N + \frac{1}{2}$. The numerator and denominator are odd functions, hence $|D_N|$ is even and

$$L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N| = \frac{1}{\pi} \int_0^{\pi} \frac{|\sin(M\theta)|}{\sin(\theta/2)} d\theta.$$

Step 2 — lower bound. By Step 1 (a) of Lemma 17.3 with $x = \theta/2$, $\sin(\theta/2) \leq \theta/2$, hence $1/\sin(\theta/2) \geq 2/\theta$ and

$$L_N \geq \frac{2}{\pi} \int_0^{\pi} \frac{|\sin(M\theta)|}{\theta} d\theta = \frac{2}{\pi} \int_0^{M\pi} \frac{|\sin u|}{u} du \geq \frac{2}{\pi} \int_0^{N\pi} \frac{|\sin u|}{u} du,$$

having set $u = M\theta$ and used $M\pi \geq N\pi$ with a non-negative integrand. Splitting $[0, N\pi]$ into the intervals $[k\pi, (k+1)\pi]$ and bounding $1/u$ from below by its minimum $1/((k+1)\pi)$ on each,

$$\int_0^{N\pi} \frac{|\sin u|}{u} du \geq \sum_{k=1}^{N-1} \frac{1}{(k+1)\pi} \int_{k\pi}^{(k+1)\pi} |\sin u| du = \frac{2}{\pi} \sum_{k=1}^{N-1} \frac{1}{k+1} = \frac{2}{\pi} \sum_{j=2}^N \frac{1}{j},$$

since $\int_{k\pi}^{(k+1)\pi} |\sin u| du = 2$ for every k . Finally, comparing the sum with the integral of the decreasing function $1/x$,

$$\sum_{j=2}^N \frac{1}{j} \geq \int_2^{N+1} \frac{dx}{x} = \ln \frac{N+1}{2} \geq \ln N - \ln 2 \geq \ln N - 1,$$

and combining gives $L_N \geq \frac{4}{\pi^2}(\ln N - 1)$.

Step 3 — upper bound. By Lemma 17.3, $\frac{1}{\sin(\theta/2)} = \frac{2}{\theta} + g(\theta)$ with $0 \leq g \leq \pi^2/24$. Hence

$$L_N \leq \frac{2}{\pi} \int_0^\pi \frac{|\sin(M\theta)|}{\theta} d\theta + \frac{1}{\pi} \int_0^\pi |\sin(M\theta)| g(\theta) d\theta.$$

The second term is bounded by $\frac{1}{\pi} \cdot 1 \cdot \frac{\pi^2}{24} \cdot \pi = \frac{\pi^2}{24}$, because $|\sin| \leq 1$ and the interval has length π . For the first, with $u = M\theta$,

$$\int_0^\pi \frac{|\sin(M\theta)|}{\theta} d\theta = \int_0^{M\pi} \frac{|\sin u|}{u} du \leq \int_0^1 1 du + \int_1^{M\pi} \frac{du}{u} = 1 + \ln(M\pi),$$

where $|\sin u|/u \leq 1$ for $u \in (0, 1]$ was used (which is (a) of Step 1 of Lemma 17.3) together with $|\sin u| \leq 1$ for $u \geq 1$; note that $M\pi = (N + \frac{1}{2})\pi > 1$ for every $N \geq 0$. The bound in the statement follows.

Step 4 — the order of magnitude. From below: for $N \geq 8$, $\ln N \geq 2$, hence $\ln N - 1 \geq \frac{1}{2} \ln N$ and so $L_N \geq \frac{2}{\pi^2} \ln N$; for $N \in \{3, \dots, 7\}$ it suffices to note that $L_N \geq L_3 > 0$ and $\ln N \leq \ln 7$, so that $L_N/\ln N \geq L_3/\ln 7 > 0$. From above: for $N \geq 1$, $(N + \frac{1}{2})\pi \leq 2\pi N$, hence $\ln((N + \frac{1}{2})\pi) \leq \ln N + \ln(2\pi)$ and

$$L_N \leq \frac{2}{\pi} \ln N + C, \quad C = \frac{2}{\pi}(1 + \ln 2\pi) + \frac{\pi^2}{24} = 2.2179\dots,$$

so that $L_N/\ln N \leq \frac{2}{\pi} + C/\ln 3$ for $N \geq 3$. The ratio $L_N/\ln N$ thus stays between two positive constants for every $N \geq 3$. $\square$

***Remark 17.10: Numerical check of the two estimates.** The constants L_N were computed by composite Simpson quadrature ($2 \cdot 10^5$ subintervals on $[0, \pi]$, integrand $|\sin(M\theta)|/\sin(\theta/2)$) and compared with the two sides of Theorem 17.10 and with the asymptotic $\frac{4}{\pi^2} \ln N + 1.2703$ mentioned in Exercise 17.2:*

N	lower bound	L_N	upper bound	$\frac{4}{\pi^2} \ln N + 1.2703$
1	—	1.435991	2.034738	1.270300
2	-0.124	1.642188	2.359940	1.551222
5	0.247	1.961361	2.861887	1.922581
10	0.528	2.223357	3.273543	2.203503
20	0.809	2.494492	3.699473	2.484425
50	1.180	2.859870	4.273417	2.855783
100	1.461	3.138780	4.711529	3.136705
1 000	2.394	4.070164	6.174543	4.069908

The two outer columns always bracket L_N : ✓. One also sees that the asymptotic is already excellent at $N = 1\,000$ (gap $2.6 \cdot 10^{-4}$), while the bounds proved here, though of the right order, have worse constants: the lower bound loses the factor $\pi/2$ in the step $\sin(\theta/2) \leq \theta/2$, the upper bound loses it in the crude $|\sin u| \leq 1$. The sharp asymptotic $L_N = \frac{4}{\pi^2} \ln N + O(1)$ is imported from Zygmund, *Trigonometric Series, Chapter II*: it is not needed for anything that follows, for which the divergence alone suffices.

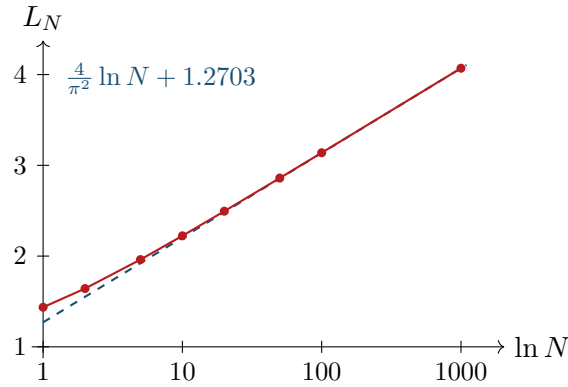


Figure 17.2: The Lebesgue constants L_N (points, computed by quadrature) as a function of $\ln N$, with the asymptotic line $\frac{4}{\pi^2} \ln N + 1.2703$ (dashed). Abscissa: $x = 0.8 \ln N$ centimetres; ordinate: $y = 1.2(L_N - 1)$ centimetres. The growth is linear in $\ln N$: this is the content of Theorem 17.10.

Lemma 17.4: *The norm of the “partial sum at the origin” functional. Consider $\Lambda_N : C(\mathbb{T}) \rightarrow \mathbb{C}$, $\Lambda_N(f) = S_N f(0)$, on the space of continuous, 2π -periodic functions with the uniform norm. Then Λ_N is linear,*

$$\Lambda_N(f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) D_N(\varphi) d\varphi, \quad |\Lambda_N(f)| \leq L_N \|f\|_{\infty},$$

and its operator norm equals exactly $\|\Lambda_N\| = L_N$.

Proof. Step 1 — the integral formula. By Theorem 17.3 with $\theta = 0$, $S_N f(0) = \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) D_N(-\varphi) d\varphi$; since D_N is even (Step 1 of the proof of Theorem 17.4) and the functions involved are 2π -periodic, the integral on $[0, 2\pi]$ equals the one on $[-\pi, \pi]$ and the formula follows. Linearity is that of the integral.

Step 2 — the bound. Bringing the modulus inside the integral,

$$|\Lambda_N(f)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f| |D_N| \leq \|f\|_{\infty} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N| = L_N \|f\|_{\infty},$$

hence $\|\Lambda_N\| \leq L_N$.

Step 3 — the bound is attained. D_N is a trigonometric polynomial, hence continuous, and by Theorem 17.2 it vanishes on $[-\pi, \pi]$ only where $\sin(M\theta) = 0$ with $M = N + \frac{1}{2}$, i.e. at the points $\theta = k\pi/M$: finitely many. Let Z be the (finite) set formed by these points and by $\pm\pi$, and let $h = \operatorname{sgn} D_N$, which equals ± 1 outside Z . For every integer $j \geq 1$ define $f_j : \mathbb{T} \rightarrow [-1, 1]$ continuous as follows: $f_j = h$ outside the intervals $I_z = (z - \frac{1}{j}, z + \frac{1}{j})$ centred at the points $z \in Z$, $f_j(z) = 0$ for $z \in Z$, and f_j affine on each of the two halves of every I_z (for j large enough the I_z are pairwise disjoint). The condition $f_j(\pm\pi) = 0$ makes f_j well defined and continuous on $\mathbb{T}$, and $\|f_j\|_\infty \leq 1$.

Step 4 — passing to the limit. As $j \rightarrow \infty$, $f_j(\theta) \rightarrow h(\theta)$ for every $\theta \notin Z$, i.e. almost everywhere, and $|f_j D_N| \leq |D_N| \in L^1(-\pi, \pi)$. By the dominated convergence theorem,

$$\Lambda_N(f_j) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f_j D_N \rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} h D_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N| = L_N.$$

Since $\|f_j\|_\infty \leq 1$, $\|\Lambda_N\| \geq |\Lambda_N(f_j)|$ for every j , hence $\|\Lambda_N\| \geq L_N$. With Step 2, $\|\Lambda_N\| = L_N$. $\square$

Corollary 17.4: *There exists a continuous function whose Fourier series diverges at a point (du Bois-Reymond). There exists $f \in C(\mathbb{T})$ such that $\sup_N |S_N f(0)| = +\infty$; in particular the Fourier series of f does not converge at $\theta = 0$, even though f is continuous. Consequently continuity alone implies neither the pointwise nor the uniform convergence of the partial sums.*

Proof. The space $C(\mathbb{T})$ with the uniform norm is complete (the uniform limit of continuous functions is continuous), hence it is a Banach space. The family $\{\Lambda_N\}_{N \in \mathbb{N}}$ of Lemma 17.4 consists of continuous linear functionals with $\sup_N \|\Lambda_N\| = \sup_N L_N = +\infty$ by Theorem 17.10. The Banach–Steinhaus uniform boundedness principle — assumed here as an external prerequisite, see Observation 17.15 — states that a family of continuous linear operators on a Banach space that is pointwise bounded is uniformly bounded in norm. By the contrapositive, since the family is not uniformly bounded in norm, it is not pointwise bounded: there exists $f \in C(\mathbb{T})$ with $\sup_N |\Lambda_N(f)| = \sup_N |S_N f(0)| = +\infty$. An unbounded numerical sequence does not converge, hence $S_N f(0)$ does not converge; and since uniform convergence would imply pointwise convergence at 0, uniform convergence cannot occur either. $\square$

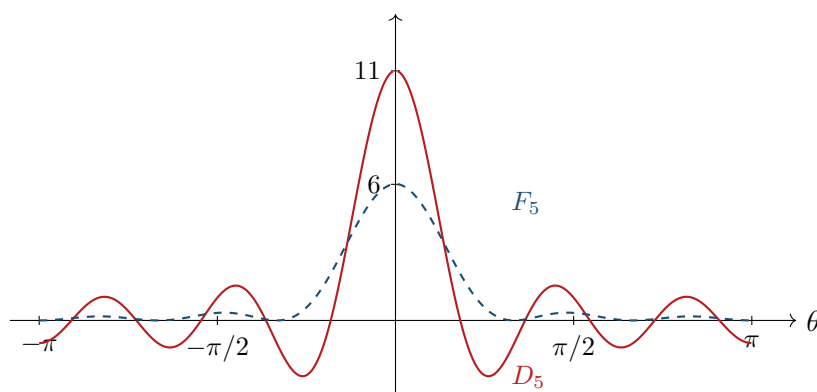


Figure 17.3: The Dirichlet kernel D_5 (solid) and the Fejér kernel F_5 (dashed) on $[-\pi, \pi]$. Scale: abscissa $x = 1.5\theta$ centimetres, ordinate $y = 0.3K$ centimetres. Control values computed by hand: $D_5(0) = 2 \cdot 5 + 1 = 11$ and $F_5(0) = 5 + 1 = 6$ (the heights marked on the axis); $D_5(\pi/6) = \sin(11\pi/12)/\sin(\pi/12) = 1$ because $\sin(11\pi/12) = \sin(\pi/12)$; $D_5(\pi) = \sin(11\pi/2) = -1$; $F_5(\pi/2) = (\sin(3\pi/2)/\sin(\pi/4))^2/6 = 2/6 = 1/3$; $F_5(\pi/3) = 0$ because $\sin(\pi) = 0$. D_5 changes sign, F_5 does not: that is the whole difference between the two worlds.

17.4. Criteria for pointwise convergence: Dini and Dirichlet–Jordan

Theorem 17.4 gave a first pointwise criterion, based on the existence of the one-sided derivatives. That hypothesis is convenient but is neither the weakest nor the most natural: there are two classical conditions, independent of one another, that both contain it. The first is *integral* (Dini), the second is a *variation* condition (Dirichlet–Jordan). Both rest on the same scheme, which it is worth isolating once and for all.

Definition 17.6: Symmetrised difference. Let $f \in L^1(\mathbb{T})$, $\theta_0 \in \mathbb{T}$ and $L \in \mathbb{C}$. Set, for $t \in (0, \pi]$,

$$\psi(t) = \psi_{\theta_0, L}(t) = f(\theta_0 + t) + f(\theta_0 - t) - 2L.$$

Lemma 17.5: Reduction of the pointwise criteria to a single integral. With the notation of Definition 17.6 and $M = N + \frac{1}{2}$,

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^\pi \psi(t) \frac{\sin(Mt)}{\sin(t/2)} dt,$$

and $\psi \in L^1(0, \pi)$.

Proof. Step 1 — convolution and symmetrisation. This is exactly the content of Steps 1–3 of the proof of Theorem 17.4, which use no hypothesis on f beyond $f \in L^1(\mathbb{T})$: the normalisation $\frac{1}{2\pi} \int_0^\pi D_N = \frac{1}{2}$ gives $\frac{1}{2\pi} \int_0^\pi 2L D_N(t) dt = L$, and the symmetrisation gives

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^\pi \psi(t) D_N(t) dt.$$

Substituting the closed form of Theorem 17.2 gives the formula in the statement.

Step 2 — summability of ψ . The functions $t \mapsto f(\theta_0 \pm t)$ are translates and reflections of f , hence lie in $L^1(0, \pi)$ with $\int_0^\pi |f(\theta_0 \pm t)| dt \leq \int_{-\pi}^\pi |f| < \infty$; the constant $2L$ is integrable on a bounded interval. Hence $\psi \in L^1(0, \pi)$ as a sum of three summable functions. $\square$

Theorem 17.11: Dini’s criterion. Let $f \in L^1(\mathbb{T})$, $\theta_0 \in \mathbb{T}$, $L \in \mathbb{C}$, and suppose the **Dini condition** holds:

$$\int_0^\pi \frac{|\psi(t)|}{t} dt < \infty, \quad \psi(t) = f(\theta_0 + t) + f(\theta_0 - t) - 2L.$$

Then $S_N f(\theta_0) \rightarrow L$ as $N \rightarrow \infty$.

Proof. Step 1 — the function to integrate. By Lemma 17.5,

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^\pi G(t) \sin(Mt) dt, \quad G(t) = \frac{\psi(t)}{\sin(t/2)}.$$

Step 2 — G is summable. By Lemma 17.3, $\frac{1}{\sin(t/2)} = \frac{2}{t} + g(t)$ with $0 \leq g \leq \pi^2/24$; hence

$$|G(t)| \leq 2 \frac{|\psi(t)|}{t} + \frac{\pi^2}{24} |\psi(t)|.$$

The first term is summable on $(0, \pi)$ by the Dini condition, the second because $\psi \in L^1(0, \pi)$ by Lemma 17.5. Hence $G \in L^1(0, \pi)$.

Step 3 — Riemann–Lebesgue with a half-integer frequency. Since $M = N + \frac{1}{2}$ is not an integer, one expands

$$\sin(Mt) = \sin(Nt) \cos \frac{t}{2} + \cos(Nt) \sin \frac{t}{2},$$

so that

$$\int_0^\pi G(t) \sin(Mt) dt = \int_0^\pi \left[G(t) \cos \frac{t}{2} \right] \sin(Nt) dt + \int_0^\pi \left[G(t) \sin \frac{t}{2} \right] \cos(Nt) dt.$$

The two functions in square brackets lie in $L^1(0, \pi)$, being products of $G \in L^1$ with functions bounded by 1. By Lemma 17.2, with $\lambda = N$, both integrals tend to zero as $N \rightarrow \infty$. Hence $S_N f(\theta_0) - L \rightarrow 0$. $\square$

Corollary 17.5: Convergence at points of one-sided Hölder continuity. Let $f \in L^1(\mathbb{T})$ and let θ_0 be a point at which the one-sided limits $f(\theta_0^\pm)$ exist and are finite, and at which, for some $\delta, C > 0$ and $\alpha \in (0, 1]$,

$$|f(\theta_0 \pm t) - f(\theta_0^\pm)| \leq C t^\alpha \quad \text{for } 0 < t < \delta.$$

Then $S_N f(\theta_0) \rightarrow \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$. In particular, if f is Lipschitz in a neighbourhood of θ_0 , or differentiable at θ_0 , the series converges there to $f(\theta_0)$.

Proof. Take $L = \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$, so that

$$\psi(t) = [f(\theta_0 + t) - f(\theta_0^+)] + [f(\theta_0 - t) - f(\theta_0^-)], \quad |\psi(t)| \leq 2Ct^\alpha \quad (0 < t < \delta).$$

Then, splitting the integral of the Dini condition,

$$\int_0^\delta \frac{|\psi(t)|}{t} dt \leq 2C \int_0^\delta t^{\alpha-1} dt = \frac{2C\delta^\alpha}{\alpha} < \infty, \quad \int_\delta^\pi \frac{|\psi(t)|}{t} dt \leq \frac{1}{\delta} \int_0^\pi |\psi| < \infty,$$

the first because $\alpha > 0$ makes $t^{\alpha-1}$ integrable at 0, the second because $\psi \in L^1(0, \pi)$ (Lemma 17.5). The Dini condition holds and Theorem 17.11 concludes. If f is differentiable at θ_0 then it is continuous there and $|f(\theta_0 \pm t) - f(\theta_0)| \leq Ct$ for small t , i.e. the case $\alpha = 1$ with $L = f(\theta_0)$. $\square$

Remark 17.11: Dini's criterion contains the one-sided-derivative criterion. Theorem 17.4, inherited from Volume II, is the case $\alpha = 1$ of Corollary 17.5: the existence of finite one-sided derivatives implies $|f(\theta_0 \pm t) - f(\theta_0^\pm)| \leq Ct$ for small t , which is the Dini condition. It is worth pointing out a small historical inaccuracy in the name: that statement is due to **Dini**, not to Dirichlet and Jordan. The properly named Dirichlet–Jordan criterion is the one in Theorem 17.12, which assumes bounded variation and not differentiability. Neither one contains the other: there exist functions of bounded variation that fail Dini's condition at a point, and functions that satisfy Dini's condition without being of bounded variation on any neighbourhood.

17.4.1 Functions of bounded variation

Definition 17.7: Total variation and functions of bounded variation. Let $u : [a, b] \rightarrow \mathbb{R}$. The **total variation** of u on $[a, b]$ is

$$V_a^b(u) = \sup \left\{ \sum_{i=1}^n |u(x_i) - u(x_{i-1})| : n \in \mathbb{N}, a = x_0 < x_1 < \cdots < x_n = b \right\} \in [0, +\infty].$$

If $V_a^b(u) < \infty$, u is said to be of **bounded variation** on $[a, b]$, written $u \in BV[a, b]$. A complex-valued function is of bounded variation when both its real and imaginary parts are.

Lemma 17.6: Jordan decomposition and one-sided limits. Let $u : [a, b] \rightarrow \mathbb{R}$. Then:

- (i) if $u = p - q$ with p, q non-decreasing, then $u \in BV[a, b]$;
- (ii) if $u \in BV[a, b]$, setting $V(x) = V_a^x(u)$ (with $V(a) = 0$), the functions $p = \frac{V+u}{2}$ and $q = \frac{V-u}{2}$ are non-decreasing and $u = p - q$;
- (iii) every $u \in BV[a, b]$ has finite one-sided limits at every point (right-sided at a , left-sided at b).

Proof. **Step 1 — (i).** For every partition, the triangle inequality gives

$$\sum_i |u(x_i) - u(x_{i-1})| \leq \sum_i (p(x_i) - p(x_{i-1})) + \sum_i (q(x_i) - q(x_{i-1})) = (p(b) - p(a)) + (q(b) - q(a)),$$

because the two sums telescope and their terms are non-negative. The bound does not depend on the partition, hence $V_a^b(u) < \infty$.

Step 2 — additivity of the variation. For $a \leq x \leq y \leq b$, $V_a^y(u) = V_a^x(u) + V_x^y(u)$. Indeed: joining a partition of $[a, x]$ and one of $[x, y]$ gives a partition of $[a, y]$, and the corresponding sum is the sum of the two, hence $V_a^y \geq V_a^x + V_x^y$; conversely, given a partition of $[a, y]$, adding the point x does not decrease the sum (again by the triangle inequality) and splits it into the two sums relative to $[a, x]$ and $[x, y]$, hence $V_a^y \leq V_a^x + V_x^y$.

Step 3 — (ii). For $x < y$, Step 2 gives $V(y) - V(x) = V_x^y(u) \geq |u(y) - u(x)| \geq \pm(u(y) - u(x))$, where the first inequality is the trivial partition $\{x, y\}$. Hence

$$(V \pm u)(y) - (V \pm u)(x) \geq 0,$$

i.e. $V + u$ and $V - u$ are non-decreasing; dividing by 2 gives p and q , and $p - q = u$.

Step 4 — (iii). A monotone bounded function on $[a, b]$ has finite one-sided limits at every point: for φ non-decreasing, $\varphi(x_0^-) = \sup_{x < x_0} \varphi(x)$ and $\varphi(x_0^+) = \inf_{x > x_0} \varphi(x)$ exist and are finite because φ is bounded between $\varphi(a)$ and $\varphi(b)$. By (ii), $u = p - q$ is a difference of two non-decreasing functions, each between its own value at a and its own value at b , hence bounded; the one-sided limits of u therefore exist and are finite, as differences of finite limits. $\square$

Lemma 17.7: The variation vanishes where the function is continuous. Let $u \in BV[0, \delta]$ with u right-continuous at 0, i.e. $u(0^+) = u(0)$. Then $V(t) = V_0^t(u) \rightarrow 0$ as $t \rightarrow 0^+$.

Proof. Let $\varepsilon > 0$. By definition of supremum there is a partition $0 = t_0 < t_1 < \cdots < t_n = \delta$ with

$$\sum_{i=1}^n |u(t_i) - u(t_{i-1})| > V(\delta) - \varepsilon.$$

Refining the partition does not decrease the sum (triangle inequality, as in Step 2 of the proof of Lemma 17.6), so one may further assume that t_1 is small enough that $|u(t_1) - u(0)| < \varepsilon$:

it suffices to add to the partition a point t^* with that property, which is possible because $u(0^+) = u(0)$, and rename. Now, by Step 2 of Lemma 17.6, $\sum_{i \geq 2} |u(t_i) - u(t_{i-1})| \leq V_{t_1}^\delta(u) = V(\delta) - V(t_1)$, hence

$$V(\delta) - \varepsilon < \underbrace{|u(t_1) - u(0)|}_{< \varepsilon} + (V(\delta) - V(t_1)),$$

whence $V(t_1) < 2\varepsilon$. Since V is non-decreasing (Step 2 of Lemma 17.6) and $V \geq 0$, $0 \leq V(t) \leq V(t_1) < 2\varepsilon$ for every $t \in (0, t_1]$. By the arbitrariness of ε , $V(0^+) = 0$. $\square$

Lemma 17.8: Second mean value theorem, Bonnet's form. Let $v : [a, b] \rightarrow \mathbb{R}$ be non-increasing and non-negative and $h \in L^1(a, b)$, and set $H(x) = \int_a^x h(t) dt$. Then there exists $\xi \in [a, b]$ with

$$\int_a^b v(t)h(t) dt = v(a)H(\xi), \quad \text{and in particular} \quad \left| \int_a^b v h \right| \leq v(a) \max_{[a,b]} |H|.$$

Proof. Step 1 — H is continuous. If $x_k \rightarrow x$ in $[a, b]$, then $\mathbf{1}_{[a, x_k]}h \rightarrow \mathbf{1}_{[a, x]}h$ almost everywhere (outside the single point x) and $|\mathbf{1}_{[a, x_k]}h| \leq |h| \in L^1$; by dominated convergence $H(x_k) \rightarrow H(x)$. Hence H is continuous on the compact set $[a, b]$ and attains there a minimum m and a maximum M ; note $H(a) = 0$, hence $m \leq 0 \leq M$.

Step 2 — approximation by step functions and Abel summation. For $n \geq 1$ consider the uniform partition $x_k = a + k\frac{b-a}{n}$, $k = 0, \dots, n$, and the step function

$$v_n = \sum_{k=1}^n v(x_{k-1}) \mathbf{1}_{[x_{k-1}, x_k]}.$$

Then, setting $H_k = H(x_k)$ and $c_k = v(x_{k-1})$, and recalling $H_0 = H(a) = 0$,

$$\int_a^b v_n h = \sum_{k=1}^n c_k (H_k - H_{k-1}) = c_n H_n + \sum_{k=1}^{n-1} (c_k - c_{k+1}) H_k,$$

which is Abel summation: separate the two sums $\sum c_k H_k$ and $\sum c_k H_{k-1}$, shift the index of the second, and collect terms.

Step 3 — the value is a weighted average. The coefficients $c_n = v(x_{n-1}) \geq 0$ and $c_k - c_{k+1} = v(x_{k-1}) - v(x_k) \geq 0$ (because v is non-increasing and non-negative) have telescoping sum

$$c_n + \sum_{k=1}^{n-1} (c_k - c_{k+1}) = c_1 = v(a).$$

Hence $\int_a^b v_n h$ is a combination, with non-negative coefficients summing to $v(a)$, of values of H , and therefore

$$v(a)m \leq \int_a^b v_n h \leq v(a)M \quad \text{for every } n.$$

Step 4 — passing to the limit. A monotone function has at most countably many discontinuities: for every j the points where the jump exceeds $1/j$ are at most $j(v(a) - v(b))$, because the jumps are pairwise disjoint and their sum does not exceed the total variation $v(a) - v(b)$; the set of discontinuities is therefore a countable union of finite sets. At every point x of continuity of v , $v_n(x) = v(x_{k-1})$ with $x_{k-1} \leq x < x_k$ and $x - x_{k-1} \leq \frac{b-a}{n} \rightarrow 0$, hence $v_n(x) \rightarrow v(x)$: the convergence is almost everywhere. Moreover $|v_n h| \leq v(a)|h| \in L^1$. By dominated convergence $\int_a^b v_n h \rightarrow \int_a^b v h$, and by Step 3

$$v(a)m \leq \int_a^b v h \leq v(a)M.$$

Step 5 — the point ξ . If $v(a) = 0$, then $0 \leq v \leq v(a) = 0$ gives $v \equiv 0$ and the equality holds for any ξ . If $v(a) > 0$, the number $\frac{1}{v(a)} \int_a^b v h$ lies in $[m, M]$; since H is continuous on $[a, b]$ and attains the values m and M , the intermediate value theorem provides $\xi \in [a, b]$ with $H(\xi) = \frac{1}{v(a)} \int_a^b v h$, which is the claim. The final bound follows from $|H(\xi)| \leq \max |H|$. $\square$

Lemma 17.9: Uniform boundedness of the sine integral. Set $\text{Si}(x) = \int_0^x \frac{\sin u}{u} du$ for $x \geq 0$ (the integrand extends continuously to 1 at $u = 0$). Then

$$0 \leq \text{Si}(x) \leq \text{Si}(\pi) = 1.851937052 \dots \quad \text{for every } x \geq 0,$$

and consequently, for every $\lambda > 0$ and $0 \leq c_1 \leq c_2$,

$$\left| \int_{c_1}^{c_2} \frac{\sin(\lambda t)}{t} dt \right| = |\text{Si}(\lambda c_2) - \text{Si}(\lambda c_1)| \leq \text{Si}(\pi) < 1.852.$$

Proof. Step 1 — an alternating series. Set $a_k = \int_{k\pi}^{(k+1)\pi} \frac{\sin u}{u} du$ for $k \geq 0$. Substituting $u = v + k\pi$ and using $\sin(v + k\pi) = (-1)^k \sin v$,

$$a_k = (-1)^k b_k, \quad b_k = \int_0^\pi \frac{\sin v}{v + k\pi} dv > 0.$$

The sequence b_k is strictly decreasing, because the integrand decreases pointwise in k and is positive on $(0, \pi)$; moreover $b_k \leq \frac{\pi}{k\pi} = \frac{1}{k} \rightarrow 0$.

Step 2 — the partial sums lie in $[0, b_0]$. By construction $\text{Si}(n\pi) = \sum_{k=0}^{n-1} (-1)^k b_k$. For an alternating series with decreasing positive terms, the even-index partial sums increase, the odd-index ones decrease, and every even-index partial sum is $\leq$ every odd-index one; the smallest is $\text{Si}(0) = 0$ and the largest is $b_0 = \text{Si}(\pi)$. Hence $0 \leq \text{Si}(n\pi) \leq \text{Si}(\pi)$ for every $n \geq 0$.

Step 3 — the intermediate points. On $[n\pi, (n+1)\pi]$ the function $\sin u/u$ has constant sign, hence Si is monotone there and every value lies between $\text{Si}(n\pi)$ and $\text{Si}((n+1)\pi)$, both in $[0, \text{Si}(\pi)]$ by Step 2. Hence $0 \leq \text{Si}(x) \leq \text{Si}(\pi)$ for every $x \geq 0$.

Step 4 — the version with a parameter. The substitution $u = \lambda t$ gives $\int_{c_1}^{c_2} \frac{\sin(\lambda t)}{t} dt = \int_{\lambda c_1}^{\lambda c_2} \frac{\sin u}{u} du = \text{Si}(\lambda c_2) - \text{Si}(\lambda c_1)$, and the difference of two numbers in $[0, \text{Si}(\pi)]$ has modulus $\leq \text{Si}(\pi)$. The numerical value $\text{Si}(\pi) = 1.851937052$ is computed by two independent routes in Section 17.6. $\square$

Theorem 17.12: Dirichlet–Jordan criterion. Let $f \in L^1(\mathbb{T})$ and let $\theta_0 \in \mathbb{T}$ be such that f is of bounded variation on $[\theta_0 - \delta_0, \theta_0 + \delta_0]$ for some $\delta_0 \in (0, \pi)$. Then the one-sided limits $f(\theta_0^\pm)$ exist and are finite, and

$$S_N f(\theta_0) \rightarrow \frac{f(\theta_0^+) + f(\theta_0^-)}{2} \quad (N \rightarrow \infty).$$

At a point of continuity the limit is $f(\theta_0)$; at a jump point it is the arithmetic mean of the two one-sided values.

Proof. Step 1 — the one-sided limits exist. This is Lemma 17.6(iii) applied to f on the interval $[\theta_0 - \delta_0, \theta_0 + \delta_0]$ (separately for the real and imaginary parts).

Step 2 — reduction. Set $L = \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$ and $\psi(t) = f(\theta_0 + t) + f(\theta_0 - t) - 2L$ as in Definition 17.6. By Lemma 17.5, with $M = N + \frac{1}{2}$,

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^\pi \frac{\psi(t)}{\sin(t/2)} \sin(Mt) dt.$$

By Lemma 17.3 one splits $\frac{1}{\sin(t/2)} = \frac{2}{t} + g(t)$ with $0 \leq g \leq \pi^2/24$:

$$S_N f(\theta_0) - L = \underbrace{\frac{1}{\pi} \int_0^\pi \frac{\psi(t)}{t} \sin(Mt) dt}_{A_N} + \underbrace{\frac{1}{2\pi} \int_0^\pi \psi(t) g(t) \sin(Mt) dt}_{B_N}.$$

Step 3 — the term B_N tends to zero. The function ψg lies in $L^1(0, \pi)$, because $\psi \in L^1(0, \pi)$ (Lemma 17.5) and g is bounded. Expanding $\sin(Mt) = \sin(Nt) \cos \frac{t}{2} + \cos(Nt) \sin \frac{t}{2}$ as in Step 3 of the proof of Theorem 17.11, Lemma 17.2 gives $B_N \rightarrow 0$.

Step 4 — structure of ψ near 0. Fix $\delta \in (0, \delta_0]$ with $\delta \leq \pi$. The functions $t \mapsto f(\theta_0 + t)$ and $t \mapsto f(\theta_0 - t)$ are of bounded variation on $[0, \delta]$ (restrictions and reflections of f , which is BV there), hence so is ψ ; set $\psi(0) = 0$, which makes ψ right-continuous at 0 because $\psi(0^+) = f(\theta_0^+) + f(\theta_0^-) - 2L = 0$. By Lemma 17.6(ii), with $V(t) = V_0^t(\psi)$, $\psi = p - q$ on $[0, \delta]$ with $p = \frac{V+\psi}{2}$ and $q = \frac{V-\psi}{2}$ non-decreasing and $p(0) = q(0) = 0$; in particular $p, q \geq 0$. By Lemma 17.7, $V(0^+) = 0$, hence also $p(0^+) = q(0^+) = 0$.

Step 5 — uniform estimate near the origin. Let $\varepsilon > 0$. By Step 4 there is $\delta_1 \in (0, \delta]$ with $p(\delta_1) < \varepsilon$ and $q(\delta_1) < \varepsilon$. Consider

$$h(t) = \frac{\sin(Mt)}{t} \in L^1(0, \delta_1), \quad H(x) = \int_0^x h = \text{Si}(Mx),$$

where the last equality is the substitution $u = Mt$ and h is bounded by M near 0. The function $P(t) = p(\delta_1) - p(t)$ is non-negative and non-increasing on $[0, \delta_1]$, with $P(0) = p(\delta_1)$. By Lemma 17.8 there is $\xi \in [0, \delta_1]$ with $\int_0^{\delta_1} Ph = p(\delta_1) \text{Si}(M\xi)$, hence

$$\int_0^{\delta_1} p h = p(\delta_1) \int_0^{\delta_1} h - \int_0^{\delta_1} Ph = p(\delta_1) [\text{Si}(M\delta_1) - \text{Si}(M\xi)].$$

By Lemma 17.9 the bracket has modulus $\leq \text{Si}(\pi) < 1.852$, hence $|\int_0^{\delta_1} p h| < 1.852\varepsilon$, and the same bound holds for q . Subtracting,

$$\left| \int_0^{\delta_1} \psi(t) \frac{\sin(Mt)}{t} dt \right| < 2 \cdot 1.852\varepsilon = 3.704\varepsilon \quad \text{for every } N.$$

Note that the bound is *uniform in N* : this is the point where the bounded variation hypothesis does all the work.

Step 6 — the tail away from the origin. On $[\delta_1, \pi]$ the function $\psi(t)/t$ is summable, because $|\psi(t)/t| \leq |\psi(t)|/\delta_1$ and $\psi \in L^1(0, \pi)$. Expanding $\sin(Mt)$ again as in Step 3, Lemma 17.2 gives

$$\int_{\delta_1}^\pi \frac{\psi(t)}{t} \sin(Mt) dt \rightarrow 0 \quad (N \rightarrow \infty).$$

Step 7 — conclusion. Combining Steps 5 and 6,

$$\limsup_{N \rightarrow \infty} |A_N| \leq \frac{1}{\pi} \cdot 3.704\varepsilon < 1.18\varepsilon,$$

and since $\varepsilon > 0$ is arbitrary, $A_N \rightarrow 0$. With Step 3, $B_N \rightarrow 0$, hence $S_N f(\theta_0) - L \rightarrow 0$. $\square$

Corollary 17.6: Piecewise monotone functions. Let $f \in L^1(\mathbb{T})$ be bounded and piecewise monotone in a neighbourhood of θ_0 , i.e. such that $[\theta_0 - \delta_0, \theta_0 + \delta_0]$ splits into finitely many intervals on each of which f is monotone. Then $S_N f(\theta_0) \rightarrow \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$.

Proof. On each subinterval f is monotone and bounded, hence of bounded variation by Lemma 17.6(i) (with $q = 0$ or $p = 0$); by additivity of the variation (Step 2 of the proof of Lemma 17.6) the variation on the whole $[\theta_0 - \delta_0, \theta_0 + \delta_0]$ is the sum of finitely many finite quantities, hence finite. Theorem 17.12 applies. $\square$

Example 17.1: The square wave and the sawtooth wave. The square wave of Example (Volume II, Chapter 4) is piecewise constant, hence piecewise monotone: by Corollary 17.6 its series converges at $\theta = 0$ to $\frac{1}{2}(1 + (-1)) = 0$, as already observed in Observation 17.4. The sawtooth function $f(\theta) = \theta$ for $\theta \in (-\pi, \pi)$, extended periodically, is increasing on $(-\pi, \pi)$ and has a jump from π to $-\pi$ at $\theta = \pi$: it is of bounded variation on every interval, with $V_{-\pi}^{\pi} = 2\pi$, and Theorem 17.12 gives

$$S_N f(\pi) \rightarrow \frac{\pi + (-\pi)}{2} = 0.$$

The direct check is immediate: the coefficients are $\hat{f}(n) = \frac{i(-1)^n}{n}$ for $n \neq 0$, so that $f(\theta) \sim \frac{1}{2} \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \sin(n\theta)$, and at $\theta = \pi$ every term $\sin(n\pi)$ vanishes: $S_N f(\pi) = 0$ for every N . $\checkmark$

Remark 17.12: What is not proved here about the Dirichlet–Jordan criterion. Theorem 17.12 is pointwise. Its uniform version — if f is of bounded variation on $\mathbb{T}$ and continuous on a closed interval $[\alpha, \beta]$, then $S_N f \rightarrow f$ uniformly on $[\alpha, \beta]$ — requires making uniform in θ_0 all the estimates of Steps 5 and 6, and in particular Lemma 17.7. It is not proved in this chapter: it is an imported result, found in Zygmund, *Trigonometric Series, Chapter II*. Nothing that follows relies on it.

17.5. Uniqueness of the coefficients

Corollary 17.2 established the uniqueness of the coefficients for square-summable functions, deriving it from Parseval’s identity. For merely summable functions Parseval’s identity is not available, and another argument is needed: Fejér’s theorem.

Theorem 17.13: Uniqueness of Fourier coefficients in L^1 . Let $f \in L^1(\mathbb{T})$ with $\hat{f}(n) = 0$ for every $n \in \mathbb{Z}$. Then $f = 0$ almost everywhere. Consequently, if $f, g \in L^1(\mathbb{T})$ have the same Fourier coefficients then $f = g$ almost everywhere; if moreover both are continuous, they coincide everywhere.

Proof. Step 1 — the continuous case. Let f be continuous with all coefficients zero. Then $S_n f = \sum_{|k| \leq n} \hat{f}(k) e_k = 0$ for every n , hence also $\sigma_N f = \frac{1}{N+1} \sum_{n=0}^N S_n f = 0$ for every N (Definition 17.3). By Theorem 17.7, $\sigma_N f \rightarrow f$ uniformly, hence f is the uniform limit of the identically zero sequence: $f \equiv 0$.

Step 2 — the antiderivative. Now let $f \in L^1(\mathbb{T})$ with all coefficients zero and set

$$F(\theta) = \int_0^\theta f(t) dt, \quad \theta \in [0, 2\pi].$$

Since $\hat{f}(0) = \frac{1}{2\pi} \int_0^{2\pi} f = 0$, $F(2\pi) = 0 = F(0)$: hence F extends to a continuous, 2π -periodic function on $\mathbb{T}$ (it is continuous by Step 1 of the proof of Lemma 17.8, applied to $h = f$).

Step 3 — the coefficients of the antiderivative. Let $n \neq 0$. The function $\theta \mapsto e^{-in\theta}/(-in)$ is an antiderivative of $e^{-in\theta}$; since F is absolutely continuous with $F' = f$ almost everywhere (Lebesgue’s differentiation theorem, a prerequisite declared in Observation 17.15), integration

by parts for absolutely continuous functions gives

$$\int_0^{2\pi} F(\theta) e^{-in\theta} d\theta = \left[F(\theta) \frac{e^{-in\theta}}{-in} \right]_0^{2\pi} - \int_0^{2\pi} f(\theta) \frac{e^{-in\theta}}{-in} d\theta = 0 + \frac{1}{in} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta = 0,$$

where the boundary term vanishes because $F(2\pi) = F(0) = 0$ and $e^{-2\pi in} = 1$, and the last integral is $2\pi \hat{f}(n) = 0$. Hence $\hat{F}(n) = 0$ for every $n \neq 0$.

Step 4 — conclusion. Let $c = \hat{F}(0)$ and $G = F - c$. Then G is continuous, 2π -periodic, and $\hat{G}(n) = \hat{F}(n) - c \hat{1}(n) = 0$ for every n (by Proposition 17.1, $\hat{1}(n) = \delta_{n0}$). By Step 1, $G \equiv 0$, i.e. F is constant. A constant function has zero derivative, hence $f = F' = 0$ almost everywhere.

Step 5 — the general form. If f and g have the same coefficients, by linearity $\widehat{f - g}(n) = 0$ for every n and Steps 1–4 give $f = g$ almost everywhere. If both are continuous and differ at a point θ_1 , by continuity they would differ on a neighbourhood of θ_1 , which has positive measure: contradiction. $\square$

***Remark 17.13: Two uniqueness results, two independent proofs.** Corollary 17.2 and Theorem 17.13 say the same thing in two different settings, and neither relies on the other: the first goes through Parseval (hence through completeness, hence through Fejér), the second through Fejér directly plus the antiderivative step. The second is stronger, because $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$ (Lemma 17.1) and the inclusion is strict: the function $f(\theta) = |\theta|^{-1/2}$ on $(-\pi, \pi)$ lies in L^1 and not in L^2 , since $\int_0^\pi t^{-1} dt$ diverges while $\int_0^\pi t^{-1/2} dt = 2\sqrt{\pi}$ converges.*

17.6. The sine-integral: two independent computations

Theorem 17.9 reduces the Wilbraham–Gibbs constant to the number $\frac{2}{\pi} \text{Si}(\pi)$, and Lemma 17.9 uses the same number as a uniform bound. It is therefore worth computing it carefully, and twice.

***Proposition 17.3: Series expansion of the sine integral.** For every $x \in \mathbb{R}$,*

$$\text{Si}(x) = \int_0^x \frac{\sin u}{u} du = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)(2k+1)!},$$

and the series converges for every x . For $x > 0$ it is alternating with terms eventually decreasing in modulus, so that the truncation error after the term of index k does not exceed the modulus of the term of index $k+1$.

Proof. Step 1 — the series of the integrand. From the exponential series one gets $\sin u = \sum_{k \geq 0} \frac{(-1)^k u^{2k+1}}{(2k+1)!}$, convergent for every $u \in \mathbb{C}$; dividing by $u \neq 0$,

$$\frac{\sin u}{u} = \sum_{k=0}^{\infty} \frac{(-1)^k u^{2k}}{(2k+1)!},$$

an identity that also holds at $u = 0$ with the value 1 of the continuous extension.

Step 2 — term-by-term integration. On every interval $[0, R]$ the series converges totally, because $\left| \frac{u^{2k}}{(2k+1)!} \right| \leq \frac{R^{2k}}{(2k+1)!}$ and $\sum_k \frac{R^{2k}}{(2k+1)!} < \infty$ (ratio test). By the Weierstrass criterion the convergence is uniform on $[0, R]$, and a uniformly convergent series of continuous functions may be integrated term by term:

$$\int_0^x \frac{\sin u}{u} du = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \int_0^x u^{2k} du = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)(2k+1)!}.$$

Step 3 — error control. Setting $c_k = \frac{x^{2k+1}}{(2k+1)(2k+1)!}$ for $x > 0$, the ratio

$$\frac{c_{k+1}}{c_k} = \frac{x^2(2k+1)}{(2k+3)(2k+2)(2k+3)}$$

is < 1 as soon as $(2k+3)^2(2k+2) > x^2(2k+1)$, which happens for every k such that $2k+3 > x$. From that index on the series is alternating with terms decreasing in modulus and tending to zero, so by the Leibniz criterion the truncation error is bounded by the first neglected term. $\square$

Remark 17.14: The value of $\text{Si}(\pi)$, computed twice. *First route: the series of Proposition 17.3 with $x = \pi$. The terms $c_k = \pi^{2k+1}/((2k+1)(2k+1)!)$ and the partial sums are*

k	term $(-1)^k c_k$	partial sum
0	+3.141592653590	3.141592653590
1	-1.722570926683	1.419021726906
2	+0.510032807975	1.929054534882
3	-0.085609218474	1.843445316408
4	+0.009127320735	1.852572637142
5	-0.000670039177	1.851902597965
6	+0.000035869447	1.851938467412
7	-0.000001461024	1.851937006388
8	+0.000000046777	1.851937053165

Since $2k+3 > \pi$ already for $k = 1$, from the second term on the Leibniz criterion holds and the error after $k = 8$ does not exceed $c_9 = \pi^{19}/(19 \cdot 19!) = 1.21 \cdot 10^{-9}$. Continuing the sum until double precision is exhausted gives

$$\text{Si}(\pi) = 1.851937051982\dots$$

Second route: numerical quadrature, independent of the series. The following script, actually run, integrates $\sin u/u$ on $[0, \pi]$ with the composite Simpson rule, which uses the series expansion in no way whatsoever:

```
import math
def simpson(f, a, b, n):          # n even
    h = (b - a) / n
    s = f(a) + f(b)
    for k in range(1, n):
        s += (4 if k % 2 else 2) * f(a + k*h)
    return s * h / 3
sinc = lambda t: 1.0 if t == 0.0 else math.sin(t)/t
for n in (100, 1000, 10000, 100000):
    print(n, "%.12f" % simpson(sinc, 0.0, math.pi, n))
```

Output: $n = 100$ gives 1.851937052658; $n = 1000$ gives 1.851937051983; $n = 10000$ and $n = 100000$ both give 1.851937051982. The gap between the two routes is $8.4 \cdot 10^{-15}$, i.e. the order of the double-precision rounding error. $\checkmark$

Consequence: the Gibbs constant. Theorem 17.9 gives

$$\frac{2}{\pi} \text{Si}(\pi) = \frac{2 \cdot 1.851937051982}{3.141592653590} = 1.178979744472,$$

hence an overshoot of 0.178979744, i.e. 17.8980% of the half-amplitude and 8.9490% of the total amplitude of the jump: these are the values stated in Theorem 17.9. $\checkmark$

Third check, directly on the partial sums. Evaluating $S_{2m-1}f(\theta_m)$ with $\theta_m = \pi/(2m)$ on the square wave, i.e. summing $\frac{4}{\pi} \sum_{j=1}^m \frac{\sin((2j-1)\theta_m)}{2j-1}$, one gets

m	θ_m	$S_{2m-1}f(\theta_m)$
1	1.57079633	1.273239545
5	0.31415927	1.182328209
20	0.07853982	1.179188137
100	0.01570796	1.178988078
1 000	0.00157080	1.178979828
20 000	0.00007854	1.178979745

The sequence converges to the value 1.178979744 predicted by Theorem 17.9, and the convergence is of order $1/m^2$ (from $m = 1\,000$ to $m = 20\,000$ the error goes from $8.4 \cdot 10^{-8}$ to $2 \cdot 10^{-10}$). ✓

17.7. Uniform convergence for regular functions: the remainder estimate

Theorem 17.5 establishes that the series of a C^1 function converges uniformly. Here it is also stated how fast, with an explicit remainder estimate depending only on the quadratic norm of the derivative.

Theorem 17.14: Remainder estimate for functions of class C^k . Let $k \geq 1$ and $f \in C^k(\mathbb{T})$. Then

$$\sum_{|n|>N} |\hat{f}(n)| \leq \sqrt{\frac{2}{(2k-1)N^{2k-1}}} \|f^{(k)}\|_2,$$

and consequently, for every $N \geq 1$,

$$\|f - S_N f\|_\infty \leq \sqrt{\frac{2}{(2k-1)N^{2k-1}}} \|f^{(k)}\|_2.$$

In particular, for $k = 1$, $\|f - S_N f\|_\infty \leq \sqrt{2/N} \|f'\|_2$, and for $k = 2$, $\|f - S_N f\|_\infty \leq \sqrt{2/(3N^3)} \|f''\|_2$.

Proof. Step 1 — the coefficients of the derivative. Step (ii) of the proof of Theorem 17.5 shows, integrating by parts over a period and using periodicity to cancel the boundary term, that $\widehat{f'}(n) = in\hat{f}(n)$. Iterating k times (legitimate because $f, f', \dots, f^{(k-1)}$ are continuous and periodic) gives

$$\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n), \quad \text{hence} \quad |\hat{f}(n)| = \frac{|\widehat{f^{(k)}}(n)|}{|n|^k} \quad (n \neq 0).$$

Step 2 — Cauchy–Schwarz on the tail. For every $N \geq 1$,

$$\sum_{|n|>N} |\hat{f}(n)| = \sum_{|n|>N} \frac{1}{|n|^k} |\widehat{f^{(k)}}(n)| \leq \left(\sum_{|n|>N} \frac{1}{n^{2k}} \right)^{1/2} \left(\sum_{|n|>N} |\widehat{f^{(k)}}(n)|^2 \right)^{1/2}.$$

Step 3 — the two factors. The second factor is bounded by $\|f^{(k)}\|_2$ by Bessel's inequality (Proposition 17.2(iv)), applicable because $f^{(k)}$ is continuous and hence lies in $L^2(\mathbb{T})$ (Lemma 17.1). For the first, comparing the sum with the integral of the decreasing function

x^{-2k} ,

$$\sum_{n>N} \frac{1}{n^{2k}} \leq \int_N^\infty \frac{dx}{x^{2k}} = \frac{1}{(2k-1)N^{2k-1}}, \quad \text{hence} \quad \sum_{|n|>N} \frac{1}{n^{2k}} \leq \frac{2}{(2k-1)N^{2k-1}},$$

the factor 2 coming from the symmetry between n and $-n$. Substituting gives the first inequality in the statement.

Step 4 — from the tail to the uniform remainder. Since $\sum_n |\hat{f}(n)| < \infty$ (this is the estimate of Step 3 with $N = 1$, plus the finitely many terms with $|n| \leq 1$), Theorem 17.5(i) guarantees that the Fourier series converges uniformly and its sum is f , which is continuous. Hence for every θ

$$|f(\theta) - S_N f(\theta)| = \left| \sum_{|n|>N} \hat{f}(n) e^{in\theta} \right| \leq \sum_{|n|>N} |\hat{f}(n)|,$$

and taking the supremum over θ concludes the proof.

Step 5 — the two numerical cases. For $k = 1$: $2/((2-1)N) = 2/N$. For $k = 2$: $2/(3N^3)$. $\square$

Example 17.2: Checking the estimate on a Poisson kernel. Take $r = \frac{1}{2}$ in Theorem 17.8:

$$f(\theta) = P_{1/2}(\theta) = \frac{1 - \frac{1}{4}}{1 - \cos \theta + \frac{1}{4}} = \frac{3}{5 - 4 \cos \theta}, \quad \hat{f}(n) = 2^{-|n|}.$$

The function is of class C^∞ because the denominator never vanishes ($5 - 4 \cos \theta \geq 1$).

Exact remainder. Since the coefficients are positive, the maximum of the remainder occurs at $\theta = 0$:

$$\|f - S_N f\|_\infty = \sum_{|n|>N} 2^{-|n|} = 2 \sum_{n>N} 2^{-n} = 2 \cdot 2^{-N} = 2^{1-N}.$$

The estimate of Theorem 17.14 with $k = 1$. By Step 1 of that proof, $\hat{f}'(n) = in 2^{-|n|}$, hence by Parseval (Theorem 17.1)

$$\|f'\|_2^2 = \sum_{n \in \mathbb{Z}} n^2 4^{-|n|} = 2 \sum_{n \geq 1} n^2 \left(\frac{1}{4}\right)^n = 2 \cdot \frac{x(1+x)}{(1-x)^3} \Big|_{x=1/4},$$

$$\|f'\|_2^2 = 2 \cdot \frac{\frac{1}{4} \cdot \frac{5}{4}}{\left(\frac{3}{4}\right)^3} = 2 \cdot \frac{20}{27} = \frac{40}{27} = 1.481481\dots,$$

having used the identity $\sum_{n \geq 1} n^2 x^n = \frac{x(1+x)}{(1-x)^3}$ for $|x| < 1$ (obtained by differentiating the geometric series twice). Hence $\|f'\|_2 = \sqrt{40/27} = 1.217161$, and the estimate is

$$\|f - S_N f\|_\infty \leq \sqrt{\frac{2}{N}} \cdot 1.217161.$$

Independent check. The square of the norm was recomputed by Simpson quadrature of $\frac{1}{2\pi} \int_0^{2\pi} |f'|^2$ with $f'(\theta) = -\frac{12 \sin \theta}{(5-4 \cos \theta)^2}$ and $2 \cdot 10^5$ subintervals: one gets 1.481481481482 against $40/27 = 1.481481481481$. $\checkmark$

Comparison. For $N = 5, 10, 20$:

N	exact remainder 2^{1-N}	estimate $\sqrt{2/N} \ f'\ _2$
5	0.0625000000	0.769800
10	0.0019531250	0.544331
20	0.0000019073	0.384900

The estimate always holds but is loose, and for a reason of principle: it uses only *one* derivative, and with one derivative one cannot do better than $N^{-1/2}$, while this f is analytic and its coefficients decay geometrically. Applying Theorem 17.14 with $k = 4$ and $\|f^{(4)}\|_2$ finite would give $O(N^{-7/2})$: the more regularity is used, the tighter the estimate. This is the same moral as in Observation 17.5.

17.8. Panorama: what this chapter does not prove

Remark 17.15: What remains imported. The chapter proves all of its statements in full, but rests on a few results imported from other branches of analysis, listed here once and for all with a precise indication of where they are used and what would be missing without them.

- **Density of step functions in L^1 of a bounded interval.** Used in Step 3 of the proof of Lemma 17.2. It is the only non-elementary ingredient of the Riemann–Lebesgue lemma: without it only the cases of step functions (Steps 1–2) and, by another route, of square-summable functions (Corollary 17.1) would remain proved.
- **Density of $C(\mathbb{T})$ in $L^2(\mathbb{T})$.** Used in Step 2 of the proof of Theorem 17.1. Without it, the completeness of the trigonometric system would remain proved only on continuous functions.
- **Dominated convergence theorem.** Used in Steps 4 and 5 of the proof of Theorem 17.9, in Step 4 of Lemma 17.4, and in Steps 1 and 4 of Lemma 17.8.
- **Lebesgue’s differentiation theorem and integration by parts for absolutely continuous functions.** Used in Steps 3 and 4 of the proof of Theorem 17.13. Without them, uniqueness would remain proved for continuous functions (Step 1) and for square-summable ones (Corollary 17.2), but not for all $f \in L^1(\mathbb{T})$.
- **Uniform boundedness principle (Banach–Steinhaus).** Used once, in Corollary 17.4. Without it the divergence of the Lebesgue constants (Theorem 17.10) and the identity $\|\Lambda_N\| = L_N$ (Lemma 17.4) would remain proved, but not the existence of the continuous function with a divergent series, which is non-constructive precisely because it follows from that principle.
- **The sharp asymptotic of the Lebesgue constants, $L_N = \frac{4}{\pi^2} \ln N + O(1)$, and the uniform version of the Dirichlet–Jordan criterion:** both merely cited (Observation 17.10 and Observation 17.12), with the source, and never used.

INFO

Panorama — the two extremes of the theory (stated, not proved). Observation 17.9 has already introduced them; here their precise form and bibliographic placement are given. Neither is provable with the tools of this chapter, and for that reason neither appears in a *theorem* environment.

Kolmogorov (1926). There exists $f \in L^1(\mathbb{T})$ whose Fourier series diverges at every point of $\mathbb{T}$. The construction, and the later variant with divergence at every point and unbounded partial sums, are given in Zygmund, *Trigonometric Series*, Chapter VIII, “Divergence of Fourier Series” (pp. 298–315). Corollary 17.4 is the smaller, non-constructive sibling of this result: it says that continuity is not enough for convergence at *one* point; Kolmogorov says that summability is not enough at *any* point.

Carleson (1966) and Hunt (1968). For every $f \in L^2(\mathbb{T})$ the Fourier series converges to f almost everywhere; R. A. Hunt extended the result to $L^p(\mathbb{T})$ for every $p > 1$. This is the statement that positively settles the question left open by du Bois-Reymond: divergence

can occur, but only on a set of measure zero. The original proof is in L. Carleson, *On convergence and growth of partial sums of Fourier series*, Acta Mathematica **116** (1966), 135–157.

Between the two extremes lies everything this chapter proves: convergence in quadratic mean for every $f \in L^2$ (Theorem 17.1), pointwise convergence under Dini (Theorem 17.11) or under bounded variation (Theorem 17.12), uniform convergence under C^1 regularity (Theorem 17.14), and — with no hypothesis beyond continuity — that of the Fejér means (Theorem 17.7).

17.9. Exercises inherited from Volume II

The exercises that follow come from Volume II, Chapter 4, and concern the topics of this chapter. They are reproduced in full, with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 17.1. Compute directly from Definition 17.2 the value $D_N(0)$ (caution: the closed form of Theorem 17.2 has an apparent $0/0$ singularity at $\theta = 0$) and check that it agrees with the limit of the closed form as $\theta \rightarrow 0$.

Exercise 17.2: S4 — The Lebesgue constant diverges. Prove that the Lebesgue constant

$$L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)| d\theta$$

satisfies $L_N \geq c \ln N$ for some constant $c > 0$ and for N large enough, and explain why this makes it impossible to deduce the uniform convergence of the partial sums from the mere continuity of f .

Exercise 17.3: S5 — The Fejér means as a triangular filter. Prove that $\widehat{\sigma_N f}(n) = (1 - \frac{|n|}{N+1})\hat{f}(n)$ for $|n| \leq N$ and 0 otherwise, and deduce that σ_N is a low-pass filter with a triangular frequency response. Compare with S_N , which is the ideal low-pass filter, and explain in light of Observation 17.7 why the first does not produce Gibbs and the second does.

Exercise 17.4: S6 — Positivity and absence of overshoot. Prove that if $K_N \geq 0$ and $\frac{1}{2\pi} \int K_N = 1$, then for every bounded f

$$\inf f \leq (f * K_N)(\theta) \leq \sup f \quad \text{for every } \theta.$$

Deduce that the Fejér means and the Abel–Poisson means can never “exceed” the extreme values of f , while the partial sums can (Gibbs phenomenon).

Solutions

Solution. From Definition 17.2, $D_N(0) = \sum_{n=-N}^N e^{in \cdot 0} = \sum_{n=-N}^N 1 = 2N + 1$ (the sum of $2N + 1$ terms all equal to 1). By the closed form of Theorem 17.2, $D_N(\theta) = \sin((N + \frac{1}{2})\theta) / \sin(\theta/2)$: as $\theta \rightarrow 0$ both numerator and denominator tend to 0 (form $0/0$), but applying

the well-known limit $\sin x \approx x$ as $x \rightarrow 0$,

$$\lim_{\theta \rightarrow 0} \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)} \approx \lim_{\theta \rightarrow 0} \frac{(N + \frac{1}{2})\theta}{\theta/2} = 2 \left(N + \frac{1}{2} \right) = 2N + 1,$$

in perfect agreement with the direct computation: the singularity at $\theta = 0$ is removable, as one expects from a kernel that is in fact a trigonometric polynomial (hence C^∞) everywhere, apparent singularity included.

Solution. S4. *Step 1 — reduction to a real integral.* By Theorem 17.2 and the evenness of D_N ,

$$L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)} \right| d\theta = \frac{1}{\pi} \int_0^{\pi} \frac{|\sin((N + \frac{1}{2})\theta)|}{\sin(\theta/2)} d\theta.$$

Step 2 — bounding the denominator. For $\theta \in [0, \pi]$, $\sin(\theta/2) \leq \theta/2$ (the inequality $\sin x \leq x$ for $x \geq 0$), hence

$$L_N \geq \frac{2}{\pi} \int_0^{\pi} \frac{|\sin((N + \frac{1}{2})\theta)|}{\theta} d\theta.$$

Step 3 — change of variable. With $u = (N + \frac{1}{2})\theta$:

$$L_N \geq \frac{2}{\pi} \int_0^{(N+1/2)\pi} \frac{|\sin u|}{u} du \geq \frac{2}{\pi} \int_0^{N\pi} \frac{|\sin u|}{u} du.$$

Step 4 — estimate of the harmonic tail. Split into intervals of length π and bound $1/u$ by its minimum on each:

$$\int_0^{N\pi} \frac{|\sin u|}{u} du \geq \sum_{k=1}^{N-1} \int_{k\pi}^{(k+1)\pi} \frac{|\sin u|}{u} du \geq \sum_{k=1}^{N-1} \frac{1}{(k+1)\pi} \int_{k\pi}^{(k+1)\pi} |\sin u| du.$$

The last integral equals 2 on every interval of length π , hence

$$\int_0^{N\pi} \frac{|\sin u|}{u} du \geq \frac{2}{\pi} \sum_{k=2}^N \frac{1}{k} \geq \frac{2}{\pi} (\ln N - 1),$$

using $\sum_{k=2}^N 1/k \geq \int_2^{N+1} \frac{dx}{x} \geq \ln N - 1$.

Step 5 — conclusion.

$$L_N \geq \frac{2}{\pi} \cdot \frac{2}{\pi} (\ln N - 1) = \frac{4}{\pi^2} (\ln N - 1) \rightarrow \infty.$$

(One can prove, with more work, the sharp asymptotic $L_N \sim \frac{4}{\pi^2} \ln N$.)

Step 6 — why this prevents uniform convergence. For every N , the functional $\Lambda_N : C(\mathbb{T}) \rightarrow \mathbb{C}$, $\Lambda_N(f) = S_N f(0)$, is linear and continuous, and its operator norm is exactly L_N (the supremum of $|\Lambda_N(f)|$ over $\|f\|_\infty \leq 1$ is obtained by approximating the function $\operatorname{sgn} D_N$ with continuous functions). Since $\sup_N \|\Lambda_N\| = \sup_N L_N = \infty$, the *uniform boundedness principle* (Banach–Steinhaus) guarantees the existence of $f \in C(\mathbb{T})$ with $\sup_N |S_N f(0)| = \infty$: the Fourier series of that continuous f diverges at $\theta = 0$. This is du Bois-Reymond's theorem, cited in Observation 17.6, and it shows that continuity alone cannot suffice for uniform convergence.

Solution. S5. *Step 1 — coefficients of the partial sums.* By definition, $S_k f = \sum_{|n| \leq k} \hat{f}(n) e_n$, hence

$$\widehat{S_k f}(n) = \begin{cases} \hat{f}(n) & |n| \leq k, \\ 0 & |n| > k. \end{cases}$$

Step 2 — averaging. By linearity of the coefficients,

$$\widehat{\sigma_N f}(n) = \frac{1}{N+1} \sum_{k=0}^N \widehat{S_k f}(n) = \frac{\hat{f}(n)}{N+1} \cdot \#\{k \in \{0, \dots, N\} : k \geq |n|\}.$$

If $|n| > N$ the set is empty and the coefficient vanishes; if $|n| \leq N$ it contains the integers from $|n|$ to N , i.e. $N+1-|n|$ elements. Hence

$$\widehat{\sigma_N f}(n) = \frac{N+1-|n|}{N+1} \hat{f}(n) = \left(1 - \frac{|n|}{N+1}\right) \hat{f}(n), \quad |n| \leq N,$$

in agreement with Theorem 17.6 and with the convolution theorem (Theorem (Volume II, Chapter 4)), since $\sigma_N f = f * F_N$ and $\widehat{F_N}(n) = 1 - |n|/(N+1)$.

Step 3 — reading it as a filter. The sequence $n \mapsto \widehat{\sigma_N}$ is the *frequency response* of the filter: it equals 1 at zero frequency, decreases *linearly* down to zero at $|n| = N+1$. It is a triangular (Bartlett) window. By comparison, S_N has a rectangular response: 1 for $|n| \leq N$, 0 beyond.

Step 4 — why the triangle produces no Gibbs. The rectangular response corresponds, in the time domain, to the Dirichlet kernel, which *oscillates and changes sign*; the triangular response corresponds to the Fejér kernel, which is the square of a function and hence *non-negative*. By Exercise S6, a non-negative kernel of unit integral cannot produce values outside the interval $[\inf f, \sup f]$: no overshoot is possible. A sharp cutoff in frequency, instead, inevitably produces oscillations in time — the other face of the same uncertainty principle.

Solution. S6. *Step 1 — the estimate.* Let $m = \inf f$ and $M = \sup f$. Since $K_N \geq 0$, multiplying by $K_N(t)$ preserves the inequalities:

$$m K_N(t) \leq f(\theta - t) K_N(t) \leq M K_N(t) \quad \text{for every } t.$$

Integrating over a period and dividing by 2π (a monotone operation):

$$m \cdot \frac{1}{2\pi} \int K_N \leq (f * K_N)(\theta) \leq M \cdot \frac{1}{2\pi} \int K_N.$$

Step 2 — normalisation. By hypothesis $\frac{1}{2\pi} \int K_N = 1$, hence

$$\inf f \leq (f * K_N)(\theta) \leq \sup f.$$

Step 3 — application to the two positive kernels. The Fejér kernel satisfies (a) and (b) of Theorem 17.6, and the Poisson kernel the analogous properties of Theorem 17.8. Hence for every bounded f

$$\inf f \leq \sigma_N f(\theta) \leq \sup f, \quad \inf f \leq (f * P_r)(\theta) \leq \sup f.$$

For the square wave, with $\inf f = -1$ and $\sup f = 1$, the Fejér means always stay in $[-1, 1]$: no overshoot.

Step 4 — why the argument fails for S_N . Step 1 uses $K_N \geq 0$ in an essential way: if the kernel changes sign, multiplying by it *reverses* the inequalities where it is negative, and the chain breaks. The Dirichlet kernel oscillates, and indeed Theorem 17.9 shows that $S_N f$ reaches 1.17898: outside the interval $[-1, 1]$ of the values of f .

Inherited quiz

The items that follow come from the quiz of Volume II, Chapter 4, and are renumbered here.

1. The Gibbs phenomenon shows itself as
 - A) slower convergence
 - B) persistent overshoot ($\approx 9\%$) near a discontinuity
 - C) divergence of the series
 - D) absence of pointwise convergence
2. At a jump point, the Fourier series of a piecewise C^1 function converges
 - A) to the left limit
 - B) to the right limit
 - C) to the average of the two one-sided limits
 - D) it does not converge
3. The Fejér kernel, unlike the Dirichlet kernel,
 - A) is non-negative
 - B) has integral different from 1
 - C) is not periodic
 - D) is not a trigonometric polynomial
4. Fejér's theorem guarantees that, for f continuous on $\mathbb{T}$,
 - A) $S_N f \rightarrow f$ uniformly
 - B) $\sigma_N f \rightarrow f$ uniformly
 - C) $S_N f \rightarrow f$ everywhere
 - D) nothing more than L^2 convergence
5. The Gibbs overshoot, as a fraction of the *total amplitude* of the jump, is about
 - A) 17.90%
 - B) 8.95%
 - C) 1.18%
 - D) it tends to 0
6. The Lebesgue constant $L_N = \frac{1}{2\pi} \int |D_N|$
 - A) equals 1 for every N
 - B) tends to 0
 - C) diverges like $\ln N$
 - D) is bounded but not constant
7. If $f \in C^k(\mathbb{T})$, its coefficients satisfy
 - A) $\hat{f}(n) = o(|n|^{-k})$
 - B) $\hat{f}(n) = o(|n|^k)$
 - C) $\hat{f}(n)$ constant
 - D) nothing can be said
8. Carleson's theorem (1966) states that for $f \in L^2(\mathbb{T})$ the Fourier series converges
 - A) everywhere
 - B) almost everywhere
 - C) at no point
 - D) only uniformly

BEST PRACTICE

Answer key.

1-B: Theorem 17.9: the overshoot has a positive limit. **2-C:** Theorem 17.4; see Observation 17.4. **3-A:** Theorem 17.6(a): it is a square, and this is the reason for all of Fejér's good properties. **4-B:** Theorem 17.7; A is false (du Bois-Reymond, Observation 17.6).
5-B: Theorem 17.9: 17.90% is the fraction of the *half*-amplitude, 8.95% that of the total amplitude. **6-C:** Exercise S4: $L_N \geq \frac{4}{\pi^2}(\ln N - 1)$. **7-A:** Observation 17.5: more regularity, faster decay. **8-B:** Observation 17.9; "everywhere" is already false for continuous functions.

Inherited formulary

The lines that follow come from the formulary of Volume II, Chapter 4.

The three kernels

$$D_N(\theta) = \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)}, \quad F_N(\theta) = \frac{1}{N+1} \left(\frac{\sin \frac{(N+1)\theta}{2}}{\sin \frac{\theta}{2}} \right)^2,$$

$$P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta+r^2}, \quad \widehat{D}_N(n) = [|n| \leq N], \quad \widehat{F}_N(n) = \left(1 - \frac{|n|}{N+1}\right)_+, \quad \widehat{P}_r(n) = r^{|n|}$$

Convergence

$$L^2: \text{always} \quad \Bigg| \quad \text{pointwise: } f \text{ piecewise } C^1 \Rightarrow S_N f(\theta_0) \rightarrow \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$$

$$\text{uniform: } \sum_n |\hat{f}(n)| < \infty \quad \Bigg| \quad \text{Fejér: } f \text{ continuous} \Rightarrow \sigma_N f \rightarrow f \text{ uniformly}$$

$$\text{Gibbs: } \lim S_N f(\theta_N) = \frac{2}{\pi} \text{Si}(\pi) = 1.178979744 \dots \quad (+8.95\% \text{ of the jump})$$

17.10. Assigned exercises

These are *new* exercises, distinct both from the inherited ones (Section 17.9) and from the worked ones in Section 17.11. The answers are in the Teacher's key (Section 17.13).

Exercise 17.5: Dini and bounded variation: neither contains the other. *Goal:* to handle the two pointwise convergence conditions and understand that they are independent. *Estimated time:* 45 minutes. *Type:* proof-based with numerical check.

- Let $f(\theta) = |\theta|^{1/2}$ on $(-\pi, \pi)$, extended periodically. Verify the Dini condition at $\theta_0 = 0$ with $L = 0$ and deduce $S_N f(0) \rightarrow 0$ from Theorem 17.11. Why does Corollary 17.5 already suffice on its own?
- Let $g(\theta) = 1/\ln(1/|\theta|)$ for $0 < |\theta| \leq \frac{1}{2}$, $g(0) = 0$ and $g(\theta) = 1/\ln 2$ for $\frac{1}{2} \leq |\theta| \leq \pi$, extended periodically. Prove that g is continuous at 0 but that the Dini condition *fails* there (compute $\int_0^{1/2} \frac{dt}{t \ln(1/t)}$ with the substitution $s = \ln(1/t)$), and that nevertheless $S_N g(0) \rightarrow 0$ by Corollary 17.6.
- Let $h(\theta) = |\theta|^{1/2} \sin(1/|\theta|)$ for $\theta \neq 0$, $h(0) = 0$. Prove that h satisfies the Dini condition at 0 with $L = 0$ and that it is *not* of bounded variation on any neighbourhood of 0 (evaluate h at the points $\theta_k = \frac{2}{(2k+1)\pi}$, where $\sin(1/\theta_k) = (-1)^k$).
- Numerically compute $S_N f(0)$ for the f of (a) with $N = 10, 50, 200$ and estimate the observed order of infinitesimal. Compare it with the order predicted by the decay $\hat{f}(n) = O(|n|^{-3/2})$.

Exercise 17.6: Lebesgue constants: from below, from above, exactly. *Goal:* to make Theorem 17.10 quantitative and measure its looseness. *Estimated time:* 40 minutes. *Type:* symbolic and numerical computation.

- Prove that $D_2(\theta) = 1 + 2\cos\theta + 2\cos 2\theta$ and find its zeros in $(0, \pi)$ using the closed form of Theorem 17.2; then compute L_2 by exact piecewise integration and compare it with the value 1.642188 in the table of Observation 17.10.
- Redo Step 3 of the proof of Theorem 17.10 using the finer bound $g(t) \leq \frac{\pi t}{24}$ of Lemma 17.3 instead of the mere $g \leq \frac{\pi^2}{24}$: by how much does the additive constant improve?
- Prove that the lower bound $\frac{4}{\pi^2}(\ln N - 1)$ is negative for $N \leq 2$ and find the smallest N for which it becomes positive. What does this say about the quality of the estimate

for small N ?

- (d) Explain in three lines why $L_N \rightarrow \infty$ prevents $\|S_N f - f\|_\infty \rightarrow 0$ for every continuous f , invoking Lemma 17.4 and Corollary 17.4.

Exercise 17.7: The remainder estimate put to the test. *Goal:* to use Theorem 17.14 on a family for which the exact remainder can be computed, and to measure its slack. *Estimated time:* 40 minutes. *Type:* symbolic and numerical computation. Let $f_r = P_r$, the Poisson kernel (Theorem 17.8), with $\hat{f}_r(n) = r^{|n|}$.

- (a) Prove that $\|f_r - S_N f_r\|_\infty = \frac{2r^{N+1}}{1-r}$ and that $\|f'_r\|_2^2 = \frac{2r^2(1+r^2)}{(1-r^2)^3}$, using $\sum_{n \geq 1} n^2 x^n = \frac{x(1+x)}{(1-x)^3}$.
- (b) For $r = \frac{3}{4}$ compute the two sides of the estimate of Theorem 17.14 with $k = 1$ for $N = 10$ and $N = 20$, and state by what factor the estimate is loose.
- (c) For $r = \frac{1}{2}$ find the smallest N for which the estimate with $k = 1$ guarantees $\|f - S_N f\|_\infty < 10^{-2}$, and the smallest N for which the inequality is *true*. Comment on the ratio between the two numbers.
- (d) Repeat (c) with $k = 2$, knowing that $\|f''_{1/2}\|_2^2 = \frac{760}{81}$, and observe by how much the prediction improves.

17.11. Worked exercises

Exercise 17.8: The Fourier series of $|\theta|$ and the sum $\sum (2k+1)^{-2}$. *Type:* computation of coefficients, pointwise convergence, sum of a numerical series. Compute the Fourier series of $f(\theta) = |\theta|$ on $[-\pi, \pi]$, extended periodically, justify the convergence at $\theta = 0$, and deduce the value of $\sum_{k \geq 0} (2k+1)^{-2}$.

Solution. Strategy. The function is even, hence the series consists only of cosines and the coefficients are computed on $[0, \pi]$ with a single integration by parts. Convergence at $\theta = 0$ is not to be taken for granted: it is obtained from Corollary 17.5, because $|\theta|$ is Lipschitz. Evaluating the series at $\theta = 0$ turns an identity of functions into a numerical identity.

Step 1 — the constant term. By Definition 17.1,

$$\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\theta| d\theta = \frac{1}{\pi} \int_0^{\pi} \theta d\theta = \frac{1}{\pi} \cdot \frac{\pi^2}{2} = \frac{\pi}{2}.$$

Step 2 — the nonzero-index coefficients. Since f is even, the imaginary part of the integrand $(-|\theta| \sin(n\theta))$ is odd and contributes nothing:

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\theta| e^{-in\theta} d\theta = \frac{1}{\pi} \int_0^{\pi} \theta \cos(n\theta) d\theta.$$

Integrating by parts with $u = \theta$ and $dv = \cos(n\theta) d\theta$,

$$\int_0^{\pi} \theta \cos(n\theta) d\theta = \left[\frac{\theta \sin(n\theta)}{n} \right]_0^{\pi} - \frac{1}{n} \int_0^{\pi} \sin(n\theta) d\theta = 0 + \left[\frac{\cos(n\theta)}{n^2} \right]_0^{\pi} = \frac{(-1)^n - 1}{n^2},$$

since $\sin(n\pi) = 0$. Hence

$$\hat{f}(n) = \frac{(-1)^n - 1}{\pi n^2} = \begin{cases} -\frac{2}{\pi n^2} & n \text{ odd,} \\ 0 & n \text{ even, } n \neq 0. \end{cases}$$

Step 3 — the series. Collecting the terms $\pm n$ with n odd, and using $e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$,

$$|\theta| \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos((2k+1)\theta)}{(2k+1)^2}.$$

Step 4 — convergence at $\theta = 0$. The function $|\theta|$ satisfies $||\theta| - |0|| = |\theta|$, hence the hypothesis of Corollary 17.5 with $\alpha = 1$, $C = 1$: the series converges at $\theta = 0$ to the value $f(0) = 0$. (Alternatively: $\sum_n |\hat{f}(n)| = \frac{4}{\pi} \sum_k (2k+1)^{-2} < \infty$, hence Theorem 17.5(i) gives uniform convergence.)

Step 5 — the sum of the numerical series. Setting $\theta = 0$ in Step 3 and using Step 4,

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k \geq 0} \frac{1}{(2k+1)^2} \quad \Rightarrow \quad \boxed{\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8} = 1.2337005501 \dots}$$

Independent check 1: from the Basel series. Splitting the even indices from the odd ones in $\sum_{n \geq 1} n^{-2} = \pi^2/6$ (Example, Volume II, Chapter 4) and observing that the even indices give $\sum_{m \geq 1} (2m)^{-2} = \frac{1}{4} \sum_m m^{-2}$,

$$\sum_{k \geq 0} \frac{1}{(2k+1)^2} = \left(1 - \frac{1}{4}\right) \sum_{n \geq 1} \frac{1}{n^2} = \frac{3}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{8}. \quad \checkmark$$

Independent check 2: numerical summation. Summing the first $2 \cdot 10^6$ terms gives 1.2337004251, against $\pi^2/8 = 1.2337005501$: the gap $1.25 \cdot 10^{-7}$ matches the estimated tail $\sum_{k > K} (2k+1)^{-2} \approx \frac{1}{4K} = 1.25 \cdot 10^{-7}$. $\checkmark$

Typical mistakes. (i) Forgetting the constant term $\pi/2$ and concluding that the cosine series equals $|\theta|$: one would get $\sum (2k+1)^{-2} = 0$. (ii) Using L^2 convergence (Theorem 17.1) to justify the substitution $\theta = 0$: convergence in norm says *nothing* about the value at a single point, which has measure zero; a pointwise criterion is needed. (iii) Writing $\hat{f}(n) = -2/(\pi n^2)$ for every $n \neq 0$, forgetting that the even-index coefficients vanish.

Extension. Applying Parseval's identity (Theorem 17.1) to the same function: $\|f\|_2^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \frac{\pi^2}{3}$, while $\sum_n |\hat{f}(n)|^2 = \frac{\pi^2}{4} + 2 \sum_{k \geq 0} \frac{4}{\pi^2 (2k+1)^4}$. Equating,

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^4} = \frac{\pi^4}{96} = 1.0146780316 \dots,$$

a value confirmed by the numerical sum of the first $2 \cdot 10^5$ terms, which gives 1.0146780316. $\checkmark$

Exercise 17.9: The sawtooth wave: jump, Gibbs and Fejér means. *Type:* pointwise convergence at a jump, numerical check of the Gibbs phenomenon, comparison with Cesàro means. Let $f(\theta) = \theta$ for $\theta \in (-\pi, \pi)$, extended periodically. Determine the limit of $S_N f(\pi)$, the maximum of $S_N f$ immediately to the left of the jump and its limit, and check that the Fejér means show no overshoot.

Solution. Strategy. The function is odd and monotone on $(-\pi, \pi)$: Corollary 17.6 gives at once the value at the jump. For the maximum one differentiates the partial sum, which is a telescoping sum of cosines, exactly as in Step 2 of the proof of Theorem 17.9.

Step 1 — the coefficients. By oddness, $\hat{f}(0) = 0$ and for $n \neq 0$

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta e^{-in\theta} d\theta = \frac{-i}{\pi} \int_0^{\pi} \theta \sin(n\theta) d\theta = \frac{-i}{\pi} \cdot \frac{-\pi(-1)^n}{n} = \frac{i(-1)^n}{n},$$

having used $\int_0^{\pi} \theta \sin(n\theta) d\theta = \left[-\frac{\theta \cos(n\theta)}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos(n\theta) d\theta = -\frac{\pi(-1)^n}{n}$. Hence

$$S_N f(\theta) = 2 \sum_{n=1}^N \frac{(-1)^{n+1}}{n} \sin(n\theta).$$

Step 2 — the value at the jump. f is increasing on $(-\pi, \pi)$, hence of bounded variation on every closed interval contained in a period; at $\theta_0 = \pi$ the one-sided limits are $f(\pi^-) = \pi$ and $f(\pi^+) = -\pi$. By Theorem 17.12, $S_N f(\pi) \rightarrow \frac{\pi+(-\pi)}{2} = 0$. The check is immediate: $\sin(n\pi) = 0$ for every n , hence $S_N f(\pi) = 0$ for every N . ✓

Step 3 — reflection about the jump. Set $\varphi = \pi - \theta$. Since $\sin(n\theta) = \sin(n\pi - n\varphi) = (-1)^{n+1} \sin(n\varphi)$, the two signs cancel and

$$S_N f(\pi - \varphi) = 2 \sum_{n=1}^N \frac{\sin(n\varphi)}{n} =: g_N(\varphi).$$

Step 4 — the first maximum. Differentiating and using the product-to-sum formula $\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$ with $A = (N + \frac{1}{2})\varphi$ and $B = \frac{\varphi}{2}$,

$$g'_N(\varphi) = 2 \sum_{n=1}^N \cos(n\varphi) = D_N(\varphi) - 1 = \frac{\sin((N + \frac{1}{2})\varphi) - \sin \frac{\varphi}{2}}{\sin(\varphi/2)} = \frac{2 \cos \frac{(N+1)\varphi}{2} \sin \frac{N\varphi}{2}}{\sin(\varphi/2)}.$$

On $(0, \pi)$ the denominator is positive; the first zero of the numerator is $\varphi_1 = \frac{\pi}{N+1}$ (it kills the cosine), which precedes $\frac{2\pi}{N}$ (the first zero of the sine). On $(0, \varphi_1)$ both factors are positive, hence g_N increases and φ_1 is its first maximum. In terms of θ , the first maximum of $S_N f$ to the left of the jump is

$$\theta_N^* = \pi - \frac{\pi}{N+1} = \frac{N\pi}{N+1}.$$

Evaluating numerically:

N	$\theta_N^* = \frac{N\pi}{N+1}$	$S_N f(\theta_N^*)$	$S_N f(\theta_N^*)/\pi$
10	2.855993	3.413945	1.086693
50	3.079993	3.642073	1.159308
200	3.125963	3.688231	1.174000
1 000	3.138454	3.700735	1.177981

Step 5 — the limit, by an exact route. Setting $h = \frac{\pi}{N+1}$,

$$S_N f(\theta_N^*) = g_N(h) = 2 \sum_{n=1}^N \frac{\sin(nh)}{n} = 2 \sum_{n=1}^N \frac{\sin(nh)}{nh} h,$$

which is the Riemann sum of the continuous function $u \mapsto \frac{\sin u}{u}$ at the points $u_n = nh$ of the uniform partition of $[0, \pi]$ of step h (the last point is $Nh = \pi - h \rightarrow \pi$). Since $\frac{\sin u}{u}$ is continuous on $[0, \pi]$ (with value 1 at 0), the Riemann sums converge to the integral:

$$\lim_{N \rightarrow \infty} S_N f(\theta_N^*) = 2 \int_0^{\pi} \frac{\sin u}{u} du = 2 \operatorname{Si}(\pi) = 3.703874,$$

in agreement with the values in the table. The ratio to the limiting value π tends to

$$\frac{2 \operatorname{Si}(\pi)}{\pi} = 1.178980,$$

exactly the constant of Theorem 17.9: the overshoot is 17.898% of the half-amplitude of the jump (here π), i.e. 8.949% of the total amplitude 2π . ✓

Independent check. The constant does not depend on the function, only on the jump: the same 1.178980 appears for the square wave (Observation 17.14, third table), where the jump equals 2 instead of 2π . Two different functions, the same constant: a sign that the phenomenon depends only on the Dirichlet kernel. ✓

Step 6 — the Fejér means. By Theorem 17.6(a)–(b) the Fejér kernel is non-negative and of integral 1; by Exercise 17.4, $\inf f \leq \sigma_N f \leq \sup f$, i.e. $-\pi \leq \sigma_N f \leq \pi$. Sampling $\sigma_{50}f$ at 2001 points of $[0, \pi]$ gives a maximum of $2.760923 < \pi$: no overshoot. ✓

Typical mistakes. (i) Concluding $S_N f(\pi) \rightarrow \pi$ “because $f(\pi^-) = \pi$ ”: Theorem 17.12 gives the *average*, and here even $S_N f(\pi) = 0$ for every N . (ii) Deducing from Step 5 that the series does not converge: it converges at every point (Step 2), it is uniform convergence that is missing. (iii) Looking for the zeros of g'_N without applying the product-to-sum formula, and not noticing that the first of them comes from the factor $\cos \frac{(N+1)\varphi}{2}$ and not from the factor $\sin \frac{N\varphi}{2}$.

Extension. Repeat the computation for the function $f(\theta) = \pi - \theta$ on $(0, 2\pi)$: it is the same function shifted, the jump has amplitude 2π and the Gibbs constant is the same. Check that the ratio overshoot/jump is invariant under translations and vertical dilations, as must be the case for a phenomenon depending only on the kernel.

Exercise 17.10: The first Lebesgue constant in closed form. *Type:* exact computation of an integral with an absolute value, and comparison with a general estimate. Compute $L_1 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_1(\theta)| d\theta$ in closed form and compare the result with the two sides of Theorem 17.10.

Solution. Strategy. An integral with an absolute value is computed by splitting the domain at the points where the integrand changes sign. Here D_1 is a trigonometric polynomial of degree 1 and its zeros can be found explicitly.

Step 1 — elementary form of D_1 . From Definition 17.2, $D_1(\theta) = e^{-i\theta} + 1 + e^{i\theta} = 1 + 2\cos\theta$. The closed form of Theorem 17.2 gives the same result: with $x = \theta/2$,

$$\frac{\sin(3x)}{\sin x} = \frac{3\sin x - 4\sin^3 x}{\sin x} = 3 - 4\sin^2 x = 3 - 2(1 - \cos 2x) = 1 + 2\cos\theta. \quad \checkmark$$

Step 2 — the sign change. $1 + 2\cos\theta = 0$ for $\cos\theta = -\frac{1}{2}$, i.e. $\theta = \frac{2\pi}{3}$ in the interval $[0, \pi]$. For $\theta < \frac{2\pi}{3}$, $\cos\theta > -\frac{1}{2}$ and $D_1 > 0$; for $\theta > \frac{2\pi}{3}$, $D_1 < 0$.

Step 3 — the integral. D_1 is even, hence $L_1 = \frac{1}{\pi} \int_0^{\pi} |1 + 2\cos\theta| d\theta$ and

$$\int_0^{\pi} |1 + 2\cos\theta| d\theta = \int_0^{2\pi/3} (1 + 2\cos\theta) d\theta - \int_{2\pi/3}^{\pi} (1 + 2\cos\theta) d\theta.$$

With the antiderivative $\theta + 2\sin\theta$ and $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$, $\sin \pi = 0$:

$$\int_0^{2\pi/3} = \frac{2\pi}{3} + \sqrt{3}, \quad \int_{2\pi/3}^{\pi} = (\pi + 0) - \left(\frac{2\pi}{3} + \sqrt{3}\right) = \frac{\pi}{3} - \sqrt{3},$$

hence

$$\int_0^{\pi} |1 + 2\cos\theta| d\theta = \left(\frac{2\pi}{3} + \sqrt{3}\right) - \left(\frac{\pi}{3} - \sqrt{3}\right) = \frac{\pi}{3} + 2\sqrt{3},$$

and finally

$$L_1 = \frac{1}{3} + \frac{2\sqrt{3}}{\pi} = 1.435991124\dots$$

Independent check 1: quadrature of $|1 + 2\cos\theta|$. Simpson with $4 \cdot 10^5$ subintervals on $[0, \pi]$ gives 1.435991124. ✓

Independent check 2: quadrature of the closed form. Integrating instead $|\sin(\frac{3}{2}\theta)/\sin(\frac{\theta}{2})|$, a formally different function, one still gets 1.435991124: the two expressions of D_1 really do coincide. ✓

Step 4 — comparison with Theorem 17.10. For $N = 1$ the lower bound is $\frac{4}{\pi^2}(\ln 1 - 1) = -0.4053$ and the upper bound $\frac{2}{\pi}(1 + \ln \frac{3\pi}{2}) + \frac{\pi^2}{24} = 2.034738$. The inequality $-0.4053 \leq 1.435991 \leq 2.034738$ holds, but the lower bound is trivial: for small N the lower estimate says nothing, and this is the price of the step $\sin(\theta/2) \leq \theta/2$, which loses a factor of $\pi/2$.

Typical mistakes. (i) Forgetting the absolute value and computing $\frac{1}{2\pi} \int D_1 = 1$: one would get a constant Lebesgue constant, and with it Corollary 17.4 would disappear. (ii) Getting the sign wrong in the second integral. (iii) Taking $\theta = \frac{\pi}{3}$ as a zero, confusing $\cos\theta = -\frac{1}{2}$ with $\cos\theta = \frac{1}{2}$.

Extension. Repeat the computation for L_2 : the zeros of $D_2 = 1 + 2\cos\theta + 2\cos 2\theta$ in $(0, \pi)$ must be found numerically and the integral splits into three pieces; the result is 1.642188, in agreement with the table of Observation 17.10. Note that $L_2 - L_1 = 0.206$ while $\frac{4}{\pi^2} \ln 2 = 0.281$: the growth is already close to the asymptotic one.

17.12. Self-assessment quiz

- (Definition) The Lebesgue constant L_N is
 - $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(\theta) d\theta$
 - $\frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)| d\theta$
 - $\max_{[-\pi, \pi]} |D_N|$
 - $D_N(0) = 2N + 1$
- (Hypothesis) In Theorem 17.11 the condition required on $\psi(t) = f(\theta_0 + t) + f(\theta_0 - t) - 2L$ is the finiteness of
 - $\int_0^{\pi} |\psi(t)| dt$
 - $\int_0^{\pi} \frac{|\psi(t)|}{t} dt$
 - $\int_0^{\pi} \frac{|\psi(t)|}{t^2} dt$
 - $\int_0^{\pi} \psi(t) D_N(t) dt$
- (Symbolic computation) The closed form $D_1(\theta) = \sin(\frac{3}{2}\theta)/\sin(\frac{\theta}{2})$ coincides with
 - $1 + 2\cos\theta$
 - $2\cos\theta$
 - $1 + \cos\theta + \cos 2\theta$
 - $3\cos\theta$
- (Numerical computation) The Wilbraham–Gibbs constant $\frac{2}{\pi} \text{Si}(\pi)$ equals, to six decimals,
 - 1.178980
 - 1.851937
 - 0.089490
 - 1.570796
- (Proof step) In the proof of Theorem 17.12 the estimate *uniform in N* on the piece $(0, \delta_1)$ is obtained
 - from the Riemann–Lebesgue lemma
 - from the second mean value theorem applied to $p(\delta_1) - p$, combined with the boundedness of Si
 - from the positivity of the Fejér kernel
 - from Bessel’s inequality
- (Error diagnosis) A student observes that $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N = 1$ for every N and concludes that for every continuous f , $\|S_N f - f\|_{\infty} \rightarrow 0$. The mistake is
 - the integral of D_N is not 1
 - the bound requires $\frac{1}{2\pi} \int |D_N| = L_N$, which diverges like $\ln N$
 - D_N is not periodic
 - $S_N f$ is not a trigonometric polynomial
- (Transfer) For $f(\theta) = \frac{3}{5-4\cos\theta}$, which has $\hat{f}(n) = 2^{-|n|}$, the remainder $\|f - S_{10}f\|_{\infty}$ equals exactly

- A) $2^{-9} = 0.001953125$ B) 0.544331 C) 2^{-10} D) 0
8. (Source and metacognition) Carleson's theorem, which this chapter states it does not prove, is published in
- A) Acta Mathematica **116** (1966), 135–157
- B) Zygmund, *Trigonometric Series*, Ch. VIII
- C) Stein–Shakarchi, *Fourier Analysis*, Ch. 2
- D) Dym–McKean, *Fourier Series and Integrals*, 1972

17.13. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

- 1) **B.** *Rationale:* this is Definition 17.5: the absolute value is the whole point, because without it the integral equals 1 for every N (Step 1 of the proof of Theorem 17.4). *Distractors:* A is exactly the normalisation, which does not depend on N ; C is the maximum, which equals $2N + 1$ and grows *linearly*, not logarithmically; D is the value at 0. *Diagnosed error:* confusing the normalisation with the L^1 norm. *Source:* Zygmund, Ch. II; here Definition 17.5 and Theorem 17.10. *Prerequisite:* Dirichlet kernel (Definition 17.2). *Remediation:* compute L_1 as in Worked exercise 17.10 and compare it with $\frac{1}{2\pi} \int D_1 = 1$.
- 2) **B.** *Rationale:* division by t is what makes $\psi/\sin(t/2)$ summable (Step 2 of the proof of Theorem 17.11), because $\frac{1}{\sin(t/2)} = \frac{2}{t} + g(t)$ with g bounded (Lemma 17.3). *Distractors:* A always holds for $f \in L^1$ (Lemma 17.5) and is therefore not a hypothesis; C is stronger than necessary and would already exclude Lipschitz f ; D depends on N and is not a hypothesis on f . *Diagnosed error:* failing to identify which singularity needs to be compensated. *Source:* Zygmund, Ch. II; Stein–Shakarchi, Ch. 3. *Prerequisite:* Lemma 17.5. *Remediation:* verify the condition for $f = |\theta|^{1/2}$ and see it fail for $1/\ln(1/|\theta|)$ (Exercise 17.5).
- 3) **A.** *Rationale:* $D_1 = e^{-i\theta} + 1 + e^{i\theta} = 1 + 2\cos\theta$; the same expression is obtained from the closed form with $\sin 3x = 3\sin x - 4\sin^3 x$ (Step 1 of Worked exercise 17.10). *Distractors:* B forgets the term $n = 0$; C is $\frac{1}{2}(1 + D_2)$ written wrongly; D is not even of mean 1. *Diagnosed error:* forgetting the constant mode in the sum $\sum_{|n| \leq N}$. *Source:* Definition 17.2 and Theorem 17.2. *Prerequisite:* de Moivre's formula. *Remediation:* write D_2 in the two forms and check their equality at $\theta = \pi/2$.
- 4) **A.** *Rationale:* $\frac{2}{\pi} \text{Si}(\pi) = \frac{2 \cdot 1.851937052}{3.141592654} = 1.178980$ (Observation 17.14). *Distractors:* B is $\text{Si}(\pi)$ without the factor $2/\pi$; C is the overshoot as a fraction of the *total* amplitude (8.949%); D is $\pi/2$, the value of $\text{Si}(+\infty)$. *Diagnosed error:* confusing the value of the integral with the overshoot constant. *Source:* Theorem 17.9 and Observation 17.14. *Prerequisite:* Proposition 17.3. *Remediation:* redo the square-wave table for $m = 100$ and $m = 1000$.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

- 5) **B.** *Rationale:* this is Step 5 of the proof of Theorem 17.12: Lemma 17.8 reduces the integral to $p(\delta_1)[\text{Si}(M\delta_1) - \text{Si}(M\xi)]$ and Lemma 17.9 bounds the bracket by $\text{Si}(\pi)$ *independently* of N . *Distractors:* A only gives a limit as $N \rightarrow \infty$ at fixed δ_1 , and is used in Step 6, not in Step 5; C concerns Fejér, not Dirichlet; D holds in L^2 and says nothing about pointwise values. *Diagnosed error:* not distinguishing a uniform estimate from a passage to the limit. *Source:* Zygmund, Ch. II; here Lemmas 17.8 and 17.9. *Prerequisite:* Jordan decomposition (Lemma 17.6). *Remediation:* redo Step 5 with ψ monotone, where $q \equiv 0$.

6) **B. Rationale:** the bound $|S_N f(\theta)| \leq \|f\|_\infty \frac{1}{2\pi} \int |D_N|$ of Step 2 of Lemma 17.4 involves the modulus, and $\frac{1}{2\pi} \int |D_N| = L_N \rightarrow \infty$ by Theorem 17.10. *Distractors:* A is false (the integral really does equal 1); C is false (D_N is a trigonometric polynomial); D is false by Definition 17.1. *Diagnosed error:* bringing the modulus inside the integral without noticing that the constant gets worse. *Source:* Corollary 17.4. *Prerequisite:* Theorem 17.3. *Remediation:* compute $S_1 f(0)$ for the continuous f that approximates $\operatorname{sgn} D_1$ and see that it is close to $L_1 = 1.436 > 1$.

7) **A. Rationale:** the coefficients are positive, hence the maximum of the remainder is at $\theta = 0$ and equals $\sum_{|n|>10} 2^{-|n|} = 2 \cdot 2^{-10} = 2^{-9}$ (Example 17.2). *Distractors:* B is the estimate of Theorem 17.14, not the true remainder, and is loose by a factor of 279; C forgets the factor 2 from the two signs of n ; D confuses convergence with exactness at finite N . *Diagnosed error:* mistaking a bound for the value. *Source:* Example 17.2. *Prerequisite:* geometric series. *Remediation:* redo the computation with $r = \frac{3}{4}$ (Exercise 17.7).

8) **A. Rationale:** this is the exact bibliographic reference, given in the infobox of Section 17.8 and in the sources environment. *Distractors:* B is where Kolmogorov's example is found, not Carleson's theorem; C is the elementary chapter on Fejér and good kernels; D is a 1972 textbook, hence later than 1966 but not the original place of publication. *Diagnosed error:* confusing a secondary source with the primary one. *Source:* L. Carleson, Acta Mathematica **116** (1966), 135–157. *Prerequisite:* none. *Remediation:* identify in the sources environment which entry is primary and which is a textbook.

BEST PRACTICE

Assigned exercises — summary answers.

17.5. (a) $\psi(t) = 2t^{1/2}$, hence $\int_0^\pi \frac{|\psi|}{t} dt = 2 \int_0^\pi t^{-1/2} dt = 4\sqrt{\pi} < \infty$: Dini holds and $S_N f(0) \rightarrow 0$. Corollary 17.5 suffices because f is $\frac{1}{2}$ -Hölder at 0 with $C = 1$. (b) With $s = \ln(1/t)$, $\frac{dt}{t} = -ds$ and $\int_0^{1/2} \frac{dt}{t \ln(1/t)} = \int_{\ln 2}^{+\infty} \frac{ds}{s} = +\infty$, hence Dini fails (here $\psi(t) = 2g(t)$); but g is even, increasing on $(0, \frac{1}{2})$ and constant on $[\frac{1}{2}, \pi]$, hence piecewise monotone, and Corollary 17.6 gives convergence to $g(0) = 0$. (c) $|\psi(t)| = 2|h(t)| \leq 2t^{1/2}$ gives Dini as in (a). At the points $\theta_k = \frac{2}{(2k+1)\pi}$, $h(\theta_k) = (-1)^k \theta_k^{1/2}$, hence $|h(\theta_k) - h(\theta_{k+1})| = \theta_k^{1/2} + \theta_{k+1}^{1/2} \geq \theta_{k+1}^{1/2}$; summing over k and noting that $\sum_k \theta_{k+1}^{1/2} = \sum_k \sqrt{2/((2k+3)\pi)}$ diverges, the variation of h on $[0, \theta_0]$ is infinite: $h \notin BV$ on any neighbourhood of 0. (d) $S_{10}f(0) = 0.24697$, $S_{50}f(0) = 0.11231$, $S_{200}f(0) = 0.05635$: the ratios $0.24697/0.11231 = 2.199$ and $0.11231/0.05635 = 1.993$ are close to $\sqrt{5} = 2.236$ and to $\sqrt{4} = 2$, i.e. the order is $N^{-1/2}$, consistent with $\sum_{|n|>N} |n|^{-3/2} = O(N^{-1/2})$.

17.6. (a) $D_2 = 1 + 2\cos\theta + 2\cos 2\theta$ has, in $(0, \pi)$, the two zeros of $\sin(\frac{5}{2}\theta)$, i.e. $\theta = \frac{2\pi}{5}$ and $\theta = \frac{4\pi}{5}$; splitting the integral into three pieces and using the antiderivative $\theta + 2\sin\theta + \sin 2\theta$ gives $L_2 = 1.642188$, in agreement with the table. (b) $\frac{1}{\pi} \int_0^\pi |\sin(M\theta)| \frac{\pi\theta}{24} d\theta \leq \frac{1}{24} \cdot \frac{\pi^2}{2} = \frac{\pi^2}{48} = 0.2056$: the additive constant halves, from 0.4112 to 0.2056. (c) $\ln N - 1 > 0$ for $N \geq 3$ ($\ln 3 = 1.0986$): for $N \leq 2$ the lower bound is negative and hence empty of content; this is the sign that the estimate is asymptotic and should not be used for small N . (d) If $\|S_N f - f\|_\infty \rightarrow 0$ held for every $f \in C(\mathbb{T})$, then $\Lambda_N(f) = S_N f(0) \rightarrow f(0)$ for every f , i.e. the family $\{\Lambda_N\}$ would be pointwise bounded; by Banach–Steinhaus it would then be bounded in norm too, against $\|\Lambda_N\| = L_N \rightarrow \infty$ (Lemma 17.4, Corollary 17.4).

17.7. (a) $\|f_r - S_N f_r\|_\infty = \sum_{|n|>N} r^{|n|} = 2 \sum_{n>N} r^n = \frac{2r^{N+1}}{1-r}$ (maximum at $\theta = 0$ because the coefficients are positive); $\|f'_r\|_2^2 = \sum_n n^2 r^{2|n|} = 2 \sum_{n \geq 1} n^2 (r^2)^n = \frac{2r^2(1+r^2)}{(1-r^2)^3}$. (b) For $r = \frac{3}{4}$: $\|f'_r\|_2^2 = \frac{7200}{343} = 20.991254$ and $\|f'\|_2 = 4.581621$; exact remainder 0.33788 against estimate 2.04896 for $N = 10$ (factor 6.1), and 0.019027 against 1.448836 for $N = 20$ (factor 76). (c) For $r = \frac{1}{2}$ the estimate $\sqrt{2/N} \cdot 1.217161 < 10^{-2}$ requires $N \geq 29\,630$, while $2^{1-N} < 10^{-2}$

already holds from $N = 8$: the estimate with a single derivative is off by more than three orders of magnitude, because it does not “see” the geometric decay of the coefficients. (d) With $k = 2$ and $\|f''\|_2 = \sqrt{760/81} = 3.063122$, the estimate $\sqrt{2/(3N^3)}\|f''\|_2 < 10^{-2}$ is satisfied from $N \geq 40$: three orders of magnitude gained by using one more derivative.

17.14. Formulary of the chapter

Coefficients, projection, Bessel, Parseval.

$$\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta, \quad S_N f = \sum_{|n| \leq N} \hat{f}(n) e_n, \quad |\hat{f}(n)| \leq \|f\|_1 \leq \|f\|_2 \leq \|f\|_\infty.$$

$$\|f - S_N f\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} |\hat{f}(n)|^2; \quad \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 \leq \|f\|_2^2 \quad (\text{Bessel}),$$

with equality for every $f \in L^2(\mathbb{T})$ (Parseval), because the system is complete.

Riemann–Lebesgue and the pointwise criteria.

$$g \in L^1(a, b) \implies \int_a^b g(t) e^{i\lambda t} dt \rightarrow 0 \quad (|\lambda| \rightarrow \infty).$$

$$S_N f(\theta_0) - L = \frac{1}{2\pi} \int_0^\pi \psi(t) \frac{\sin((N + \frac{1}{2})t)}{\sin(t/2)} dt, \quad \psi(t) = f(\theta_0 + t) + f(\theta_0 - t) - 2L.$$

$$\text{Dini: } \int_0^\pi \frac{|\psi(t)|}{t} dt < \infty \quad \Bigg| \quad \text{Dirichlet–Jordan: } f \in BV \text{ near } \theta_0,$$

in both cases $S_N f(\theta_0) \rightarrow L = \frac{1}{2}(f(\theta_0^+) + f(\theta_0^-))$. Useful: $\frac{1}{\sin(t/2)} = \frac{2}{t} + g(t)$ with $0 \leq g(t) \leq \frac{\pi t}{24} \leq \frac{\pi^2}{24}$ on $(0, \pi]$.

Lebesgue constants, sine integral, Gibbs.

$$L_N = \frac{1}{2\pi} \int_{-\pi}^\pi |D_N|, \quad \frac{4}{\pi^2} (\ln N - 1) \leq L_N \leq \frac{2}{\pi} \left(1 + \ln((N + \frac{1}{2})\pi)\right) + \frac{\pi^2}{24},$$

$$L_1 = \frac{1}{3} + \frac{2\sqrt{3}}{\pi} = 1.435991, \quad \|\Lambda_N\| = L_N \implies \exists f \in C(\mathbb{T}) : \sup_N |S_N f(0)| = \infty.$$

$$\text{Si}(x) = \sum_{k \geq 0} \frac{(-1)^k x^{2k+1}}{(2k+1)(2k+1)!}, \quad 0 \leq \text{Si}(x) \leq \text{Si}(\pi) = 1.851937052,$$

$$\frac{2}{\pi} \text{Si}(\pi) = 1.178979744 \quad (+17.898\% \text{ of the half-amplitude, } +8.949\% \text{ of the jump}).$$

Regularity and remainder.

$$\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n); \quad f \in C^k(\mathbb{T}) \implies \|f - S_N f\|_\infty \leq \sqrt{\frac{2}{(2k-1)N^{2k-1}}} \|f^{(k)}\|_2.$$

Notable cases: $k = 1$ gives $\sqrt{2/N}\|f'\|_2$; $k = 2$ gives $\sqrt{2/(3N^3)}\|f''\|_2$. Check example: $f = \frac{3}{5-4\cos\theta}$, $\hat{f}(n) = 2^{-|n|}$, $\|f - S_N f\|_\infty = 2^{1-N}$, $\|f'\|_2 = \sqrt{40/27} = 1.217161$.

Sources and bibliography

- [1] A. Zygmund, *Trigonometric Series*, 3rd edition (volumes I and II combined, with a foreword by R. A. Fefferman), Cambridge Mathematical Library, Cambridge University Press, Cambridge, 2002 (digital edition 2015, DOI 10.1017/CB09781316036587; ISBN 978-0-521-89053-3). Chapter II “Fourier Coefficients. Elementary Theorems on Convergence” (pp. 35–73): this is the source of the Riemann–Lebesgue lemma (Lemma 17.2), of Dini’s criterion (Theorem 17.11), of the Dirichlet–Jordan criterion (Theorem 17.12), of the Lebesgue constants (Theorem 17.10) and of the Gibbs phenomenon (Theorem 17.9); it also contains the sharp asymptotic $L_N = \frac{4}{\pi^2} \ln N + O(1)$ and the uniform version of Dirichlet–Jordan, both declared *imported* in Observation 17.15. Chapter III “Summability of Fourier Series” (pp. 74–126) for the Fejér kernel and the Cesàro means (Theorems 17.6 and 17.7). Chapter VIII “Divergence of Fourier Series” (pp. 298–315) for Kolmogorov’s example, reported as panorama in Section 17.8. Title, edition, publisher, year and table of contents with chapter pagination verified on the volume’s record in the Cambridge Core catalogue — verified on 6 September 2026.
- [2] E. M. Stein, R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis, volume 1, Princeton University Press, Princeton, 2003 (ISBN 978-0-691-11384-5, 328 pages). Chapter 2 “Basic Properties of Fourier Series”: summation kernels, approximate identities, the Fejér kernel and the Poisson kernel, Fejér’s theorem (here Theorems 17.6, 17.7 and 17.8); Chapter 3 “Convergence of Fourier Series”: convergence in quadratic mean, Bessel’s inequality, Parseval’s identity and the return to pointwise convergence (here Proposition 17.2 and Theorem 17.1). Title, authors, series, publisher, year, ISBN, page count and table of contents verified on the volume’s record on the Princeton University Press website — verified on 6 September 2026.
- [3] H. Dym, H. P. McKean, *Fourier Series and Integrals*, “Probability and Mathematical Statistics” series no. 14, Academic Press, New York, 1972 (ISBN 0-12-226450-9, 295 pages, bibliography on pp. 283–287). This is the alternative, more analytically flavoured source for the trigonometric system, summability and the passage from series to the Fourier integral, which will be taken up again in Chapter 19. Authors, title, publisher, year, series and numbering, ISBN, page count and location of the bibliography verified on the Open Library catalogue record of the volume (edition with ISBN 0-12-226450-9) — verified on 6 September 2026.
- [4] L. Carleson, *On convergence and growth of partial sums of Fourier series*, *Acta Mathematica* **116** (1966), pp. 135–157, DOI 10.1007/BF02392815. This is the original publication of the theorem reported as panorama in Section 17.8 and in Observation 17.9; it is not proved in this chapter. Author, title, journal, volume, year, page range and DOI verified on the article’s record in *Acta Mathematica* (Project Euclid and SpringerLink) — verified on 6 September 2026.
- [5] For the external prerequisites listed in Observation 17.15 (density of step functions in L^1 , density of $C(\mathbb{T})$ in $L^2(\mathbb{T})$, dominated convergence, Lebesgue differentiation, integration by parts for absolutely continuous functions, the uniform boundedness principle): W. Rudin, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, New York, 1987, Chapters 3 and 5. The theoretical core of Sections 17.2 and 17.9 comes from Volume II, Chapter 4, §5, and is reproduced here in full according to the executive plan, §3.4, line 17.

CHAPTER 18

Partial differential equations on the circle and in the disc

Three equations govern almost all of the classical physics of continuous media: Laplace's equation $\Delta u = 0$, the heat equation $u_t = \kappa \Delta u$ and the wave equation $u_{tt} = c^2 \Delta u$. On an arbitrary domain these are three hard problems. On the circle and on the disc they become three *solvable* problems, for one reason only: rotational symmetry. It makes the Laplace operator commute with rotations, and hence the characters $e^{in\theta}$ — the very ones that in Chapter 17 form the Fourier basis — are automatically its eigenfunctions in the angular variable. What is left is a single ordinary equation in the radial variable: Euler's equation for Laplace, Bessel's equation for heat and waves. The whole chapter is the working-out of this sentence.

The route is the following. We start from the Laplacian in polar coordinates (Theorem 18.1), *derived* from the change of variables and not postulated, with the divergence-form corollary (Corollary 18.1). We then apply separation of variables to Laplace's equation in the disc: Lemma 18.2 splits the problem into two ordinary equations, Lemma 18.3 shows that periodicity of the angle alone *quantises* the separation constant to n^2 with n an integer, Lemma 18.4 solves the radial equation, and Theorem 18.2 discards the singular solutions at the origin. A single mode survives for each integer, $r^{|n|}e^{in\theta}$, and summing the modes recovers — with Corollary 18.2 — the Poisson formula of Chapter 16, which is not reproved here but recalled in the form established by Theorem 17.8. The maximum principle (Theorem 18.4) closes the Dirichlet problem with uniqueness and stability.

The second part is dynamic. The heat equation on $\mathbb{T}$ (Theorem 18.5) shows the phenomenon typical of parabolic problems: every mode decays like $e^{-\kappa n^2 t}$, the solution becomes C^∞ instantaneously, and converges to the mean of the datum. The wave equation on $\mathbb{T}$, instead, is solved in closed form (Theorem 18.7, d'Alembert's formula made periodic), conserves energy (Proposition 18.7) and is periodic in time (Proposition 18.8): no regularisation, no forgetting. Finally the circular membrane: separation leads to Bessel's equation (Lemma 18.7), the Bessel function of the first kind is defined by its series (Definition 18.7), the series is shown to converge everywhere (Theorem 18.8), to solve the equation (Theorem 18.9), the other independent solution is shown to be unbounded at the origin (Proposition 18.12), and J_n is shown to have infinitely many positive, simple zeros (Theorem 18.10). The zeros give the frequencies of the drum's modes, and are taken from the tabulated values quoted by the NIST Digital Library of Mathematical Functions, recomputed here twice by independent routes.

The material that the executive plan moved here from Volume II (Chapter 4, Sec.8) — the remark on the ring that diffuses heat, the two exercises and their solutions — is kept in full and is found in Observation 18.4 and in Section 18.11. What Volume II took as known (the Laplacian in polar coordinates, separation of variables, Bessel functions) is proved here.

INFO

Level and prerequisites. Level **C** (undergraduate), with three passages at level **D**: the proof that the second solution of Bessel's equation is unbounded at the origin (Proposition 18.12), Sturm's comparison (Lemma 18.9), and the computation of the norm of the

radial eigenfunctions (Theorem 18.11).

Internal prerequisites. Fourier series on the circle: coefficients, convergence in L^2 and Parseval's identity, uniform convergence, the Poisson kernel and Abel summation (Chapter 17, Definition 17.1, Theorem 17.1, Theorem 17.5, Definition 17.4, Theorem 17.8); harmonic functions in the disc, the mean-value property and the Dirichlet problem (Chapter 16); real and imaginary part of z^n , polar form (Chapter 6); the circle as a quotient space and 2π -periodic functions (Chapter 12); curvature and arc length for plane curves (Chapter 2).

External prerequisites, assumed and not reproved:

- differential calculus in two variables: the chain rule for compositions of C^2 functions, Schwarz's theorem on the interchange of the order of differentiation, the second-derivative test for an interior maximum point;
- the divergence theorem (Gauss–Green formula) on a regular plane domain, used once, in Proposition 18.9;
- elementary theory of linear ordinary differential equations: existence and uniqueness of the solution of the Cauchy problem for a second-order linear equation with continuous coefficients and nonvanishing leading coefficient, and the dimension-2 vector-space structure of the solution set;
- power series: radius of convergence, uniform convergence on compact sets, term-by-term differentiation;
- Weierstrass's criterion for total convergence and the interchange of series and integral under uniform convergence.

Everything else is proved here.

What is learned. How to transport a differential operator through a change of coordinates without missing a term; recognising when separation of variables is legitimate and exactly what it produces; understanding that periodicity is a *boundary condition* that quantises; distinguishing parabolic from hyperbolic behaviour by how the modes decay; building the Bessel function from scratch and proving its properties; reading a table of zeros and deriving from it the frequencies of a drum, knowing why they are not harmonic.

Warning. Conventions of the chapter.

- $D_a = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < a^2\}$ is the open disc centred at the origin with radius $a > 0$, $\overline{D_a}$ its closure, ∂D_a the circle of radius a ; $D = D_1$ is the unit disc and $\mathbb{T} = \partial D$ the unit circle, identified with $\mathbb{R}/2\pi\mathbb{Z}$.
- A function on $\mathbb{T}$ is a function on $\mathbb{R}$ of period 2π ; $\hat{f}(n)$ denotes the Fourier coefficient $\frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta$ of Definition 17.1.
- We write $u(x, y)$ for a function in Cartesian coordinates and $U(r, \theta) = u(r \cos \theta, r \sin \theta)$ for the same function read in polar coordinates; when there is no ambiguity we use the same letter for both, as is customary.
- $\Delta = \partial_x^2 + \partial_y^2$ is the Laplace operator; $u_t, u_{tt}, u_\theta, u_{\theta\theta}, u_r, u_{rr}$ are the indicated partial derivatives.
- $\kappa > 0$ is the thermal diffusivity, $c > 0$ the propagation speed: both carry the physical dimensions that make the equations homogeneous, and in the numerical computations we use SI units.
- Numerical values are rounded to the seventh decimal digit and recomputed twice by independent routes; the symbol $\checkmark$ marks a successful check.
- J_n is the Bessel function of the first kind of integer order $n \geq 0$ and $j_{n,k}$ its k -th positive zero, in the notation of the NIST Digital Library of Mathematical Functions, Sec.10.21(i).

18.1. The Laplacian in polar coordinates

None of what follows works unless the Laplace operator is first rewritten in the coordinates suited to the domain. The rewriting is a pure exercise in the chain rule, but it must be carried out in full: it is where sign errors and forgotten terms hide, and it is also where the term $\frac{1}{r} u_r$ that makes the origin singular appears, apparently out of nowhere.

Definition 18.1: Laplace operator and harmonic functions. Let $\Omega \subseteq \mathbb{R}^2$ be open and $u \in C^2(\Omega)$. The **Laplace operator** (or *Laplacian*) of u is

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}.$$

The function u is called **harmonic** in Ω if $\Delta u = 0$ at every point of Ω .

Lemma 18.1: Derivatives of the polar transformation. Let $\Phi : (0, +\infty) \times \mathbb{R} \rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}$, $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$. Then Φ is of class C^∞ , its Jacobian determinant equals $r > 0$, and for every $u \in C^2$ defined on an open set $\Omega \subseteq \mathbb{R}^2 \setminus \{(0, 0)\}$, setting $U = u \circ \Phi$, the identities

$$U_r = u_x \cos \theta + u_y \sin \theta, \quad U_\theta = -r u_x \sin \theta + r u_y \cos \theta,$$

hold on $\Phi^{-1}(\Omega)$, where u_x, u_y are evaluated at $\Phi(r, \theta)$.

Proof. The two components of Φ are $x(r, \theta) = r \cos \theta$ and $y(r, \theta) = r \sin \theta$, products and compositions of C^∞ functions: hence $\Phi \in C^\infty$. Its Jacobian matrix is

$$D\Phi(r, \theta) = \begin{pmatrix} x_r & x_\theta \\ y_r & y_\theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det D\Phi = r \cos^2 \theta + r \sin^2 \theta = r,$$

positive because $r > 0$. The chain rule for the composition $U = u \circ \Phi$ gives

$$U_r = u_x x_r + u_y y_r = u_x \cos \theta + u_y \sin \theta, \quad U_\theta = u_x x_\theta + u_y y_\theta = -r u_x \sin \theta + r u_y \cos \theta,$$

which are the two identities sought. $\square$

Theorem 18.1: The Laplacian in polar coordinates. Let $\Omega \subseteq \mathbb{R}^2 \setminus \{(0, 0)\}$ be open, $u \in C^2(\Omega)$ and $U(r, \theta) = u(r \cos \theta, r \sin \theta)$. Then, for every (r, θ) with $\Phi(r, \theta) \in \Omega$,

$$(\Delta u)(\Phi(r, \theta)) = U_{rr} + \frac{1}{r} U_r + \frac{1}{r^2} U_{\theta\theta}$$

Proof. Write $c = \cos \theta$ and $s = \sin \theta$; all derivatives of u are understood to be evaluated at $\Phi(r, \theta)$. Since $u \in C^2$, by Schwarz's theorem $u_{xy} = u_{yx}$.

Step 1 — the second radial derivative. Differentiate with respect to r the first identity of Lemma 18.1, recalling that c and s do not depend on r and that, again by the chain rule, $\partial_r(u_x \circ \Phi) = u_{xx}c + u_{xy}s$ and $\partial_r(u_y \circ \Phi) = u_{yx}c + u_{yy}s$:

$$U_{rr} = (u_{xx}c + u_{xy}s)c + (u_{yx}c + u_{yy}s)s = u_{xx}c^2 + 2u_{xy}cs + u_{yy}s^2. \quad (1)$$

Step 2 — the second angular derivative. Differentiate with respect to θ the second identity of Lemma 18.1. Here r is fixed, while c and s depend on θ and so do $u_x \circ \Phi$, $u_y \circ \Phi$;

the chain rule gives $\partial_\theta(u_x \circ \Phi) = -r u_{xx}s + r u_{xy}c$ and $\partial_\theta(u_y \circ \Phi) = -r u_{yx}s + r u_{yy}c$. Hence

$$U_{\theta\theta} = \underbrace{-r u_x c}_{\partial_\theta(-rs)u_x} - rs(-r u_{xx}s + r u_{xy}c) + \underbrace{-r u_y s}_{\partial_\theta(rc)u_y} + rc(-r u_{yx}s + r u_{yy}c).$$

Collecting terms,

$$U_{\theta\theta} = -r(u_x c + u_y s) + r^2(u_{xx}s^2 - 2u_{xy}cs + u_{yy}c^2) = -r U_r + r^2(u_{xx}s^2 - 2u_{xy}cs + u_{yy}c^2), \quad (2)$$

where the last equality again uses Lemma 18.1.

Step 3 — the sum. From (2) one gets

$$\frac{1}{r^2}U_{\theta\theta} = -\frac{1}{r}U_r + u_{xx}s^2 - 2u_{xy}cs + u_{yy}c^2,$$

and adding this to (1) term by term:

$$U_{rr} + \frac{1}{r^2}U_{\theta\theta} = u_{xx}(c^2 + s^2) + 2u_{xy}cs - 2u_{xy}cs + u_{yy}(s^2 + c^2) - \frac{1}{r}U_r = u_{xx} + u_{yy} - \frac{1}{r}U_r,$$

because $c^2 + s^2 = 1$ (Pythagorean identity, Volume I). Moving $\frac{1}{r}U_r$ to the left gives $U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta} = u_{xx} + u_{yy} = \Delta u$, which is the claim. $\square$

Corollary 18.1: Divergence form. Under the hypotheses of Theorem 18.1,

$$\Delta u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial U}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 U}{\partial \theta^2}.$$

Proof. By Leibniz's rule $\partial_r(rU_r) = U_r + rU_{rr}$, hence $\frac{1}{r}\partial_r(rU_r) = \frac{1}{r}U_r + U_{rr}$. Substituting into the formula of Theorem 18.1 gives the statement. $\square$

Example 18.1: The elementary harmonic modes. Let $n \geq 0$ be an integer and $U(r, \theta) = r^n \cos n\theta$. Then

$$U_{rr} = n(n-1)r^{n-2} \cos n\theta, \quad \frac{1}{r}U_r = n r^{n-2} \cos n\theta, \quad \frac{1}{r^2}U_{\theta\theta} = -n^2 r^{n-2} \cos n\theta,$$

and the sum is $(n^2 - n + n - n^2)r^{n-2} \cos n\theta = 0$: the function is harmonic on $\mathbb{R}^2 \setminus \{(0, 0)\}$. The same computation, with $\sin n\theta$ in place of $\cos n\theta$, gives the same result.

Independent check. In Cartesian coordinates $r^n \cos n\theta = \operatorname{Re}(z^n)$ with $z = x + iy$ (polar form, Chapter 6); since z^n is holomorphic on all of $\mathbb{C}$, its real part is harmonic by the Cauchy–Riemann equations (Chapter 15). For $n = 2$ one has $\operatorname{Re}(z^2) = x^2 - y^2$ and $\Delta(x^2 - y^2) = 2 - 2 = 0$; a numerical check with centred finite differences of step $h = 10^{-4}$ at the point $(0.7, 0.4)$ returns the value 0 for the Laplacian, exact in double precision. $\checkmark$

Likewise $r^{-n} \cos n\theta$ is harmonic on $\mathbb{R}^2 \setminus \{(0, 0)\}$: the computation is identical with $-n$ in place of n , and $(-n)^2 - (-n) + (-n) - (-n)^2 = 0$.

Proposition 18.1: Radial harmonic functions. Let $0 \leq \rho_1 < \rho_2 \leq +\infty$ and let $U = U(r)$ be of class C^2 on (ρ_1, ρ_2) , independent of θ . Then U is harmonic if and only if there exist $A, B \in \mathbb{R}$ with

$$U(r) = A + B \ln r, \quad \rho_1 < r < \rho_2.$$

Proof. Since $U_{\theta\theta} = 0$, by Corollary 18.1 the condition $\Delta U = 0$ is equivalent to $\frac{1}{r}(rU')' = 0$, i.e., since $r > 0$, to $(rU')' = 0$ on (ρ_1, ρ_2) .

A function with zero derivative on an interval is constant: hence $rU'(r) = B$ for some constant $B \in \mathbb{R}$, i.e. $U'(r) = B/r$. Integrating on (ρ_1, ρ_2) , where $r \mapsto 1/r$ has primitive $\ln r$, gives $U(r) = A + B \ln r$ with $A \in \mathbb{R}$.

Conversely, if $U = A + B \ln r$ then $rU' = B$ is constant, hence $(rU')' = 0$ and $\Delta U = 0$. $\square$

Remark 18.1: Why the origin is a special point. Theorem 18.1 is stated on open sets that do not contain the origin, and not out of pedantry: the transformation Φ is neither injective nor regular at $r = 0$ (its Jacobian equals r , which vanishes there), and the coefficients $1/r$ and $1/r^2$ of the formula blow up. The price is paid in two opposite ways. On one hand spurious solutions appear, such as $\ln r$ and $r^{-n} \cos n\theta$, which solve the equation away from the origin but do not extend regularly to the centre: it is Proposition 18.1 that exhibits them. On the other hand, whenever one wants a solution defined on the whole disc one must explicitly impose a regularity condition at $r = 0$: this is exactly what Theorem 18.2 will do, and it is the mechanism that, in Section 18.7, will eliminate the second solution of Bessel's equation.

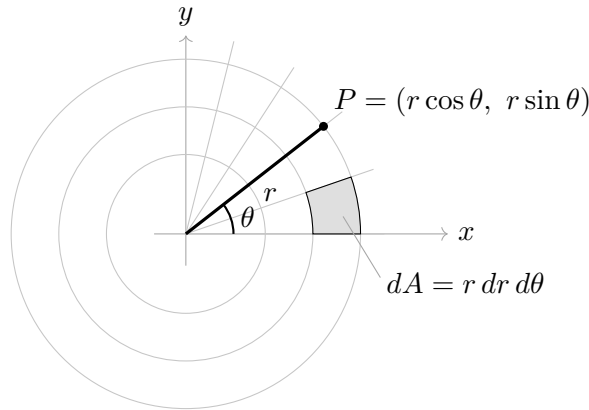


Figure 18.1: Polar coordinates: the circles $r = \text{const}$ and the rays $\theta = \text{const}$ meet orthogonally. The plotted point has $r = 2.2$ and $\theta = 38^\circ$, i.e. $x = 2.2 \cos 38^\circ = 1.7336238$ and $y = 2.2 \sin 38^\circ = 1.3544553$ (values checked by independent computation). The shaded area element has sides dr and $r d\theta$: this is the factor r of the Jacobian determinant of Lemma 18.1, and the same asymmetry between the two directions produces the term $\frac{1}{r} u_r$ of Theorem 18.1.

18.2. Laplace's equation in the disc: separation of variables

The method of separation of variables consists in seeking, among all solutions, those of the very particular form “function of r times function of θ ”. It is a drastic restriction, and it is by no means obvious that it leads anywhere: it does, because the equation, rewritten in polar coordinates, has coefficients that do not mix the two variables. Lemma 18.2 makes the step precise; the two following lemmas show that the boundary conditions — periodicity in θ , regularity at $r = 0$ — select exactly a countable family of modes.

Definition 18.2: The Dirichlet problem in the disc. Let $a > 0$ and let $f : \partial D_a \rightarrow \mathbb{R}$ be continuous. The **Dirichlet problem** in the disc D_a with boundary datum f consists in finding $u \in C^2(D_a) \cap C(\overline{D_a})$ with

$$\Delta u = 0 \quad \text{in } D_a, \quad u|_{\partial D_a} = f.$$

Identifying ∂D_a with $\mathbb{T}$ via $\theta \mapsto (a \cos \theta, a \sin \theta)$, the datum is written as a 2π -periodic function $f(\theta)$.

Lemma 18.2: Separation of variables for Laplace's equation. Let $R \in C^2((0, a))$ and $\Theta \in C^2(\mathbb{R})$, neither identically zero, and suppose that $U(r, \theta) = R(r)\Theta(\theta)$ satisfies

$$U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta} = 0 \quad \text{for every } (r, \theta) \in (0, a) \times \mathbb{R}.$$

Then there exists $\lambda \in \mathbb{R}$ such that

$$\Theta'' + \lambda\Theta = 0 \quad \text{on } \mathbb{R}, \quad r^2R'' + rR' - \lambda R = 0 \quad \text{on } (0, a).$$

Proof. Substituting $U = R\Theta$ and multiplying by r^2 (legitimate since $r > 0$) the equation becomes

$$(r^2R''(r) + rR'(r))\Theta(\theta) + R(r)\Theta''(\theta) = 0 \quad \text{for every } (r, \theta) \in (0, a) \times \mathbb{R}. \quad (*)$$

Step 1 — the radial equation. Since $\Theta \not\equiv 0$, there exists $\theta_0 \in \mathbb{R}$ with $\Theta(\theta_0) \neq 0$. Set

$$\lambda := -\frac{\Theta''(\theta_0)}{\Theta(\theta_0)} \in \mathbb{R}.$$

Evaluating $(*)$ at $\theta = \theta_0$ and dividing by $\Theta(\theta_0) \neq 0$ gives

$$r^2R''(r) + rR'(r) + \frac{\Theta''(\theta_0)}{\Theta(\theta_0)}R(r) = 0, \quad \text{i.e.} \quad r^2R'' + rR' - \lambda R = 0,$$

and this for every $r \in (0, a)$, because θ_0 is fixed while r is arbitrary.

Step 2 — the angular equation. Substitute the result of Step 1 into $(*)$: for every (r, θ) ,

$$\lambda R(r)\Theta(\theta) + R(r)\Theta''(\theta) = 0, \quad \text{i.e.} \quad R(r)(\Theta''(\theta) + \lambda\Theta(\theta)) = 0.$$

Since $R \not\equiv 0$, there exists $r_0 \in (0, a)$ with $R(r_0) \neq 0$; evaluating at $r = r_0$ and dividing by $R(r_0)$ gives $\Theta''(\theta) + \lambda\Theta(\theta) = 0$ for every $\theta \in \mathbb{R}$.

The number λ is real because Θ and Θ'' are real-valued functions. $\square$

Lemma 18.3: Periodicity quantises the separation constant. Let $\lambda \in \mathbb{R}$ and let $\Theta \in C^2(\mathbb{R})$ be a solution, not identically zero, of $\Theta'' + \lambda\Theta = 0$ that is 2π -periodic. Then $\lambda = n^2$ for some integer $n \geq 0$, and

- if $n = 0$, Θ is constant;
- if $n \geq 1$, $\Theta(\theta) = A \cos n\theta + B \sin n\theta$ with $(A, B) \neq (0, 0)$.

Conversely, for every integer $n \geq 0$ the functions indicated are 2π -periodic solutions.

Proof. The equation $\Theta'' + \lambda\Theta = 0$ is linear, second order, with constant coefficients; its solution space has dimension 2. We distinguish three cases according to the sign of λ .

Case $\lambda < 0$. Setting $\mu = \sqrt{-\lambda} > 0$, the solutions are $\Theta(\theta) = Ae^{\mu\theta} + Be^{-\mu\theta}$. Such a function is 2π -periodic only if it is bounded on $\mathbb{R}$ (a continuous periodic function is bounded, because its supremum is attained on a period, which is compact). But if $A \neq 0$ then $\Theta(\theta) \rightarrow \pm\infty$ as $\theta \rightarrow +\infty$, and if $B \neq 0$ then $\Theta(\theta) \rightarrow \pm\infty$ as $\theta \rightarrow -\infty$: in either case Θ is unbounded. Hence $A = B = 0$ and $\Theta \equiv 0$, excluded by hypothesis. The case $\lambda < 0$ does not occur.

Case $\lambda = 0$. The solutions are $\Theta(\theta) = A + B\theta$. Periodicity $\Theta(\theta + 2\pi) = \Theta(\theta)$ gives $2\pi B = 0$, i.e. $B = 0$: Θ is constant, and nonzero by hypothesis. This is the case $n = 0$, with $\lambda = 0 = 0^2$.

Case $\lambda > 0$. Setting $\mu = \sqrt{\lambda} > 0$, the solutions are $\Theta(\theta) = A \cos \mu\theta + B \sin \mu\theta$ with $(A, B) \neq (0, 0)$. Imposing $\Theta(2\pi) = \Theta(0)$ and $\Theta'(2\pi) = \Theta'(0)$: since two solutions of the same second-order linear equation with the same Cauchy data coincide, these two equalities are *equivalent* to 2π -periodicity of Θ (apply uniqueness for the Cauchy problem to the function $\theta \mapsto \Theta(\theta + 2\pi)$, which solves the same equation). They read

$$\begin{cases} A \cos 2\pi\mu + B \sin 2\pi\mu = A, \\ \mu(-A \sin 2\pi\mu + B \cos 2\pi\mu) = \mu B, \end{cases} \quad \text{i.e., since } \mu \neq 0, \quad \mathcal{R}_{2\pi\mu} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix},$$

where $\mathcal{R}_\varphi = \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix}$ is the rotation matrix through angle $-\varphi$. A rotation of the plane that fixes a nonzero vector is the identity (Chapter 5); hence $\mathcal{R}_{2\pi\mu} = I$, i.e. $\cos 2\pi\mu = 1$ and $\sin 2\pi\mu = 0$, which is equivalent to $2\pi\mu \in 2\pi\mathbb{Z}$, i.e. $\mu \in \mathbb{Z}$. Since $\mu > 0$, $\mu = n$ is an integer ≥ 1 and $\lambda = n^2$.

Converse. For $n = 0$ a constant solves $\Theta'' = 0$ and is periodic; for $n \geq 1$, $(A \cos n\theta + B \sin n\theta)'' = -n^2(A \cos n\theta + B \sin n\theta)$ and the function has period $2\pi/n$, hence also 2π . $\square$

Lemma 18.4: *The radial equation is of Euler type. Let $n \geq 0$ be an integer. The C^2 solutions of the equation*

$$r^2 R'' + r R' - n^2 R = 0 \quad \text{on } (0, +\infty)$$

are all and only

$$R(r) = A r^n + B r^{-n} \quad (n \geq 1), \quad R(r) = A + B \ln r \quad (n = 0), \quad A, B \in \mathbb{R}.$$

Proof. Step 1 — the change of variable $r = e^s$. The map $s \mapsto e^s$ is a C^∞ diffeomorphism of $\mathbb{R}$ onto $(0, +\infty)$. Setting $S(s) := R(e^s)$, the chain rule gives

$$S'(s) = e^s R'(e^s) = r R'(r), \quad S''(s) = e^s R'(e^s) + e^{2s} R''(e^s) = r R'(r) + r^2 R''(r),$$

with $r = e^s$. Hence $r^2 R'' + r R' = S''$, and the equation becomes

$$S''(s) - n^2 S(s) = 0, \quad s \in \mathbb{R}.$$

Step 2 — solving the constant-coefficient equation. For $n \geq 1$ the solutions are $S(s) = A e^{ns} + B e^{-ns}$; returning to $r = e^s$, i.e. $s = \ln r$, one gets $R(r) = A r^n + B r^{-n}$. For $n = 0$ the equation is $S'' = 0$, with solutions $S(s) = A + Bs$, i.e. $R(r) = A + B \ln r$.

Step 3 — there are no others. The correspondence $R \mapsto S$ is a bijection between the C^2 solutions of the two equations (the inverse is $R(r) = S(\ln r)$, and the computation of Step 1 reads both ways); since the solution space of $S'' - n^2 S = 0$ has dimension 2 and is described exactly by the formulas of Step 2, the same holds for R . Note that for $n = 0$ one recovers Proposition 18.1, as it must. $\square$

Theorem 18.2: Regularity at the origin discards the singular solutions. *Let $n \geq 0$ be an integer and let R be a solution of $r^2 R'' + r R' - n^2 R = 0$ on $(0, a)$, not identically zero, such that R is bounded on a right neighbourhood of 0. Then, in the notation of Lemma 18.4, $B = 0$: that is, $R(r) = A r^n$ with $A \neq 0$ for $n \geq 1$, and R is a nonzero constant for $n = 0$. In particular the function*

$$U(r, \theta) = R(r)\Theta(\theta)$$

is, up to multiplicative constants, one of the modes

$$1, \quad r^n \cos n\theta, \quad r^n \sin n\theta \quad (n \geq 1),$$

all of which extend continuously to $r = 0$.

Proof. **Case $n \geq 1$.** By Lemma 18.4, $R(r) = Ar^n + Br^{-n}$. If $B \neq 0$, then, since $r^n \rightarrow 0$ and $r^{-n} \rightarrow +\infty$ as $r \rightarrow 0^+$,

$$|R(r)| \geq |B|r^{-n} - |A|r^n \rightarrow +\infty,$$

contradicting the boundedness hypothesis. Hence $B = 0$ and $R(r) = Ar^n$; if also $A = 0$ then $R \equiv 0$, which is excluded. The function $r \mapsto Ar^n$ extends continuously at $r = 0$ with value 0.

Case $n = 0$. By Lemma 18.4, $R(r) = A + B \ln r$. If $B \neq 0$, then $|R(r)| \geq |B| |\ln r| - |A| \rightarrow +\infty$ as $r \rightarrow 0^+$, contradicting the hypothesis. Hence $B = 0$ and $R \equiv A \neq 0$.

The modes. Combining with Lemma 18.3, which forces $\lambda = n^2$ and gives $\Theta = A' \cos n\theta + B' \sin n\theta$ (constant for $n = 0$), one gets that $U = R\Theta$ is a linear combination of $r^n \cos n\theta$ and $r^n \sin n\theta$, i.e., for $n = 0$, a constant. Each of these modes is harmonic on $\mathbb{R}^2 \setminus \{(0, 0)\}$ by Example 18.1, and since $r^n \cos n\theta = \operatorname{Re}(z^n)$ and $r^n \sin n\theta = \operatorname{Im}(z^n)$ are polynomials in x, y , they extend to C^∞ harmonic functions on all of $\mathbb{R}^2$. $\square$

Remark 18.2: What periodicity did, exactly. It is worth isolating the mechanism, because it will return identical for the membrane. Lemma 18.2 produces a free parameter λ that, taken the two problems separately, could be any real number. It is the requirement that Θ be a well-defined function on the circle — that is, 2π -periodic: in the language of Chapter 12, that it pass to the quotient $\mathbb{R}/2\pi\mathbb{Z}$ — that reduces the set of admissible λ to the discrete set $\{0, 1, 4, 9, \dots\}$. There is no boundary condition imposed by hand: the topology of the circle is the boundary condition. The second filter, boundedness at $r = 0$, is instead not topological but analytic, and arises from the degeneration of polar coordinates discussed in Observation 18.1.

Theorem 18.3: Series solution of the Dirichlet problem in the disc. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be continuous, with Fourier coefficients $\hat{f}(n)$, and let $a > 0$. Set, for $0 \leq r < a$ and $\theta \in \mathbb{R}$,

$$u(r, \theta) := \sum_{n \in \mathbb{Z}} \hat{f}(n) \left(\frac{r}{a}\right)^{|n|} e^{in\theta}.$$

Then:

- (i) the series converges absolutely and uniformly on every closed disc $\{r \leq \rho\}$ with $\rho < a$, and so do all the series obtained by differentiating it term by term any number of times in r and in θ ; consequently $u \in C^\infty(D_a)$;
- (ii) $\Delta u = 0$ in D_a ;
- (iii) $u(r, \cdot) \rightarrow f$ uniformly on $\mathbb{T}$ as $r \rightarrow a^-$; setting $u|_{\partial D_a} = f$ gives $u \in C(\overline{D_a})$, and u solves the Dirichlet problem of Definition 18.2.

Proof. Set $t = r/a \in [0, 1)$, so that $u(r, \theta) = \sum_n \hat{f}(n) t^{|n|} e^{in\theta}$.

Step 1 — estimating the coefficients. Directly from Definition 17.1,

$$|\hat{f}(n)| = \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) e^{-in\varphi} d\varphi \right| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f| =: M,$$

finite because f is continuous on the compact set $\mathbb{T}$.

Step 2 — total convergence of the differentiated series. Fix $\rho < a$ and $\sigma := \rho/a < 1$. The general term of the series is $g_n(r, \theta) = \hat{f}(n) a^{-|n|} r^{|n|} e^{in\theta}$; differentiating it p times in r and

q times in θ gives, for $r \leq \rho$ and $|n| \geq p$,

$$\left| \partial_r^p \partial_\theta^q g_n \right| = |\hat{f}(n)| \cdot |n|^q \cdot \frac{|n|(|n|-1) \cdots (|n|-p+1)}{a^{|n|}} r^{|n|-p} \leq \frac{M}{\rho^p} |n|^{q+p} \sigma^{|n|},$$

while for $|n| < p$ the p -th derivative in r vanishes. The numerical series $\sum_{n \in \mathbb{Z}} |n|^{q+p} \sigma^{|n|}$ converges (ratio test: the ratio of consecutive terms tends to $\sigma < 1$). By the Weierstrass criterion all the differentiated series converge totally, hence uniformly, on $\{r \leq \rho\} \times \mathbb{R}$. The theorem on term-by-term differentiation of series (applied successively in each variable) then gives $u \in C^\infty$ on $\{r < \rho\}$ for every $\rho < a$, i.e. on D_a , with derivatives obtained term by term.

Step 3 — harmonicity. Every term $t^{|n|} e^{in\theta} = (r/a)^{|n|} (\cos n\theta + i \sin n\theta)$ is harmonic on $\mathbb{R}^2 \setminus \{(0,0)\}$ by Example 18.1 and, by Theorem 18.2, extends to a harmonic function on all of $\mathbb{R}^2$. Since by Step 2 the Laplacian can be computed term by term,

$$\Delta u = \sum_{n \in \mathbb{Z}} \hat{f}(n) a^{-|n|} \Delta(r^{|n|} e^{in\theta}) = 0 \quad \text{in } D_a.$$

Step 4 — the boundary value. By the definition of the Poisson kernel (Definition 17.4) and by Theorem 17.8, for every $t \in [0, 1)$

$$\sum_{n \in \mathbb{Z}} t^{|n|} \hat{f}(n) e^{in\theta} = (f * P_t)(\theta) \longrightarrow f(\theta) \quad \text{uniformly in } \theta, \text{ as } t \rightarrow 1^-.$$

With $t = r/a$, this is exactly $u(r, \cdot) \rightarrow f$ uniformly as $r \rightarrow a^-$. Uniform convergence guarantees that the function extended by setting $u = f$ on ∂D_a is continuous on $\overline{D_a}$: indeed if $(r_k, \theta_k) \rightarrow (a, \theta_*)$ then

$$|u(r_k, \theta_k) - f(\theta_*)| \leq \underbrace{\|u(r_k, \cdot) - f\|_\infty}_{\rightarrow 0} + \underbrace{|f(\theta_k) - f(\theta_*)|}_{\rightarrow 0 \text{ by continuity of } f}.$$

Hence $u \in C^2(D_a) \cap C(\overline{D_a})$, $\Delta u = 0$ and $u|_{\partial D_a} = f$: the Dirichlet problem is solved. $\square$

Corollary 18.2: The Poisson formula recovered. Under the hypotheses of Theorem 18.3, for every $0 \leq r < a$ and $\theta \in \mathbb{R}$

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{a^2 - r^2}{a^2 - 2ar \cos(\theta - \varphi) + r^2} f(\varphi) d\varphi.$$

Proof. Set $t = r/a < 1$; by Step 2 of the proof of Theorem 18.3 the series $\sum_n \hat{f}(n) t^{|n|} e^{in\theta}$ converges uniformly in θ ; substituting the definition of $\hat{f}(n)$ and interchanging series and integral (legitimate by uniform convergence of the kernel, see Step 4 cited above) gives

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{n \in \mathbb{Z}} t^{|n|} e^{in(\theta - \varphi)} \right) f(\varphi) d\varphi = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_t(\theta - \varphi) f(\varphi) d\varphi,$$

with P_t the Poisson kernel of Definition 17.4. The closed form $P_t(\psi) = \frac{1 - t^2}{1 - 2t \cos \psi + t^2}$ is Theorem 17.8, which is *not* reproved here. Substituting $t = r/a$ and multiplying numerator and denominator by a^2 ,

$$P_{r/a}(\psi) = \frac{1 - r^2/a^2}{1 - 2(r/a) \cos \psi + r^2/a^2} = \frac{a^2 - r^2}{a^2 - 2ar \cos \psi + r^2},$$

which is the announced formula. $\square$

Remark 18.3: Two routes to the same kernel. The Poisson formula was obtained in Chapter 16 directly, as the integral kernel that solves the Dirichlet problem, and in Chapter 17 as the Abel summation kernel of the Fourier series (Theorem 17.8). Corollary 18.2 shows that it is also the unavoidable result of separation of variables: once it is established that the only regular harmonic modes are $r^{|n|}e^{in\theta}$, and once it is imposed that the sum equal f at the boundary, the coefficient of the n -th mode can only be $\hat{f}(n)$, and the sum of the two-sided geometric series can only be the Poisson kernel. None of the three derivations is more fundamental than the others; the third is the one that generalises, because it uses nothing specific to the disc except rotational symmetry, and indeed it will reappear unchanged in Section 18.6 with Bessel's equation in place of Euler's.

Example 18.2: A Dirichlet problem solved and checked. We solve the Dirichlet problem in the unit disc with boundary datum $f(\theta) = \cos^2 \theta$.

Step 1 — expanding the datum. From the double-angle formula $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$ (Volume I), hence $\hat{f}(0) = \frac{1}{2}$, $\hat{f}(\pm 2) = \frac{1}{4}$ and $\hat{f}(n) = 0$ otherwise.

Step 2 — the solution. By Theorem 18.3 with $a = 1$ the series reduces to three terms:

$$u(r, \theta) = \frac{1}{2} + \frac{1}{4}r^2e^{2i\theta} + \frac{1}{4}r^2e^{-2i\theta} = \frac{1}{2} + \frac{r^2}{2} \cos 2\theta.$$

Check 1: direct verification of harmonicity. With $U = \frac{1}{2} + \frac{r^2}{2} \cos 2\theta$ one has $U_{rr} = \cos 2\theta$, $\frac{1}{r}U_r = \cos 2\theta$, $\frac{1}{r^2}U_{\theta\theta} = -2 \cos 2\theta$; the sum is 0. ✓ In Cartesian coordinates $u = \frac{1}{2} + \frac{x^2 - y^2}{2}$, whose Laplacian is $\frac{1}{2}(2 - 2) = 0$. ✓

Check 2: the Poisson integral, computed numerically. For $r = \frac{1}{2}$ and $\theta = \frac{\pi}{3}$ the closed form gives

$$u\left(\frac{1}{2}, \frac{\pi}{3}\right) = \frac{1}{2} + \frac{1}{8} \cos \frac{2\pi}{3} = \frac{1}{2} - \frac{1}{16} = 0.4375.$$

The Poisson integral of Corollary 18.2, evaluated by the trapezoidal rule on 400 000 subintervals of $[-\pi, \pi]$ (Python script run), gives 0.4375000000: the two routes agree to ten decimal places. ✓ The same comparison at $(r, \theta) = (0.5, 0)$ gives 0.6250000000 versus 0.6250000000 and at $(0.8, 1)$ gives 0.3668330123 versus 0.3668330123. ✓

18.3. Maximum principle, uniqueness, stability

Theorem 18.3 constructs a solution. That it is the only one, and that it depends continuously on the datum, is not a formality: without these two properties the problem would not be well posed and no numerical computation would make sense. Both follow from a single principle, valid for arbitrary bounded domains, that uses neither symmetry nor the series.

Theorem 18.4: Weak maximum principle. Let $\Omega \subset \mathbb{R}^2$ be open, bounded and nonempty, and let $u \in C^2(\Omega) \cap C(\bar{\Omega})$ with $\Delta u = 0$ in Ω . Then

$$\max_{\bar{\Omega}} u = \max_{\partial\Omega} u, \quad \min_{\bar{\Omega}} u = \min_{\partial\Omega} u.$$

Proof. The two extrema exist because $\bar{\Omega}$ and $\partial\Omega$ are nonempty compact sets (Ω is bounded, hence $\bar{\Omega}$ is closed and bounded; $\partial\Omega \neq \emptyset$ because otherwise Ω would be both closed and open and hence, being bounded and nonempty, equal to $\mathbb{R}^2$, a contradiction).

Step 1 — a strictly subharmonic perturbation. Let $\varepsilon > 0$ and

$$v_\varepsilon(x, y) := u(x, y) + \varepsilon(x^2 + y^2).$$

Then $v_\varepsilon \in C^2(\Omega) \cap C(\overline{\Omega})$ and

$$\Delta v_\varepsilon = \Delta u + \varepsilon \Delta(x^2 + y^2) = 0 + \varepsilon \cdot 4 = 4\varepsilon > 0 \quad \text{in } \Omega.$$

Step 2 — v_ε has no interior maxima. Suppose, for contradiction, that v_ε has a maximum point $p \in \Omega$ (interior). Then, for every coordinate direction, the one-variable function $t \mapsto v_\varepsilon(p + te_1)$ has an interior maximum at $t = 0$ and is of class C^2 in a neighbourhood of 0; the second-derivative test gives $\partial_x^2 v_\varepsilon(p) \leq 0$. Likewise $\partial_y^2 v_\varepsilon(p) \leq 0$. Adding, $\Delta v_\varepsilon(p) \leq 0$, contradicting Step 1.

Step 3 — the maximum lies on the boundary. The function v_ε is continuous on the compact set $\overline{\Omega}$, hence attains a maximum at some $q \in \overline{\Omega}$; by Step 2, $q \notin \Omega$, i.e. $q \in \partial\Omega$. Therefore $\max_{\overline{\Omega}} v_\varepsilon = \max_{\partial\Omega} v_\varepsilon$.

Step 4 — letting ε tend to zero. Let $M := \max_{\overline{\Omega}}(x^2 + y^2) < \infty$ (Ω is bounded). For every $(x, y) \in \overline{\Omega}$,

$$u(x, y) \leq v_\varepsilon(x, y) \leq \max_{\partial\Omega} v_\varepsilon \leq \max_{\partial\Omega} u + \varepsilon M.$$

The left-hand side does not depend on ε ; letting $\varepsilon \rightarrow 0^+$ gives $u(x, y) \leq \max_{\partial\Omega} u$ for every $(x, y) \in \overline{\Omega}$, i.e. $\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u$. The opposite inequality is obvious because $\partial\Omega \subseteq \overline{\Omega}$.

Step 5 — the minimum. Apply what has been proved to $-u$, which is still harmonic, and use $\min w = -\max(-w)$. $\square$

Corollary 18.3: Uniqueness and stability for the Dirichlet problem. Let $\Omega \subset \mathbb{R}^2$ be open, bounded, nonempty.

(i) If $u_1, u_2 \in C^2(\Omega) \cap C(\overline{\Omega})$ are harmonic in Ω and agree on $\partial\Omega$, then $u_1 = u_2$ on $\overline{\Omega}$.

(ii) If moreover $u_i|_{\partial\Omega} = f_i$, then $\max_{\overline{\Omega}} |u_1 - u_2| = \max_{\partial\Omega} |f_1 - f_2|$.

In particular the solution constructed in Theorem 18.3 is the unique solution of the Dirichlet problem in the disc.

Proof. (ii) Let $w := u_1 - u_2$. By linearity of the Laplace operator w is harmonic in Ω , continuous on $\overline{\Omega}$, and $w|_{\partial\Omega} = f_1 - f_2$. By Theorem 18.4,

$$\max_{\overline{\Omega}} w = \max_{\partial\Omega} (f_1 - f_2) \quad \text{and} \quad \min_{\overline{\Omega}} w = \min_{\partial\Omega} (f_1 - f_2).$$

Since for a real function $\max |w| = \max\{\max w, -\min w\}$, one obtains $\max_{\overline{\Omega}} |w| = \max_{\partial\Omega} |f_1 - f_2|$.

(i) This is the case $f_1 = f_2$ of (ii): $\max_{\overline{\Omega}} |w| = 0$, i.e. $w \equiv 0$.

Finally, the function constructed in Theorem 18.3 belongs to $C^2(D_a) \cap C(\overline{D_a})$ and is harmonic with datum f : by (i) it is the only one. $\square$

Proposition 18.2: Mean-value property on concentric circles. Let u be the solution of Theorem 18.3. Then, for every $0 \leq r < a$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} u(r, \theta) d\theta = \hat{f}(0) = u(0),$$

that is, the mean of u over every circle concentric with the boundary equals the value at the centre, which in turn is the mean of the boundary datum.

Proof. By Step 2 of the proof of Theorem 18.3 the series converges uniformly in θ for fixed $r < a$, so it may be integrated term by term:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} u(r, \theta) d\theta = \sum_{n \in \mathbb{Z}} \hat{f}(n) \left(\frac{r}{a}\right)^{|n|} \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} d\theta.$$

By orthonormality of the trigonometric system (Proposition 17.1) the integral equals 1 for $n = 0$ and 0 otherwise: only the term $\hat{f}(0)(r/a)^0 = \hat{f}(0)$ remains. Setting $r = 0$ in the series also gives $u(0) = \hat{f}(0)$, because $(0/a)^{|n|} = 0$ for $n \neq 0$ and $= 1$ for $n = 0$. $\square$

18.4. Application: the heat equation on the circle

We move from stationary solutions to evolving ones. The thin, insulated ring is the simplest compact domain without boundary: there are no endpoints at which to impose conditions, and the only condition is again periodicity. The paragraph that follows is the one inherited from Volume II and is kept verbatim; the subsequent results complete it with existence, uniqueness, the maximum principle and a quantitative decay estimate.

Remark 18.4: Heat diffusion on a ring. The heat equation $\partial_t u = \partial_\theta^2 u$ on $\mathbb{T}$ (a thin insulated ring) is solved by seeking solutions of the form $u(\theta, t) = \sum_n c_n(t) e^{in\theta}$: substituting, $c'_n(t) = -n^2 c_n(t)$, hence $c_n(t) = c_n(0) e^{-n^2 t}$. The solution is

$$u(\theta, t) = \sum_{n \in \mathbb{Z}} \hat{u}_0(n) e^{-n^2 t} e^{in\theta},$$

where $\hat{u}_0(n)$ are the Fourier coefficients of the initial datum. High-frequency components ($|n|$ large) decay much faster ($e^{-n^2 t}$): heat “flattens” fine details first, exactly as expected physically. This is the prototype, in the simplest case (1 spatial dimension, compact domain $\mathbb{T}$), of the problems that in Volume III we will treat on the two-dimensional disc with Bessel functions.

Definition 18.3: Cauchy problem for heat on $\mathbb{T}$ and the heat kernel. Let $\kappa > 0$ and $f : \mathbb{T} \rightarrow \mathbb{R}$. The **Cauchy problem** for the heat equation on $\mathbb{T}$ consists in finding u continuous on $\mathbb{T} \times [0, +\infty)$, of class C^2 in θ and C^1 in t for $t > 0$, with

$$u_t = \kappa u_{\theta\theta} \quad \text{for } t > 0, \quad u(\theta, 0) = f(\theta).$$

For $t > 0$ the function

$$H_t(\theta) := \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{-\kappa n^2 t} e^{in\theta}$$

is called the **heat kernel** on $\mathbb{T}$.

Proposition 18.3: The heat kernel is well defined and smooth. For every $t > 0$ the series defining H_t converges totally on $\mathbb{R}$ together with all its derivatives in θ ; $H_t \in C^\infty(\mathbb{T})$, is real and even, and

$$\int_{-\pi}^{\pi} H_t(\theta) d\theta = 1, \quad \partial_t H_t = \kappa \partial_\theta^2 H_t \quad (t > 0).$$

Proof. Step 1 — convergence. For $t \geq t_0 > 0$ and every $q \geq 0$, $|\partial_\theta^q (e^{-\kappa n^2 t} e^{in\theta})| = |n|^q e^{-\kappa n^2 t} \leq |n|^q e^{-\kappa n^2 t_0}$, and the numerical series $\sum_n |n|^q e^{-\kappa n^2 t_0}$ converges (the ratio of consec-

utive terms tends to 0). By the Weierstrass criterion the convergence is total, hence uniform, and by the theorem on term-by-term differentiation of series $H_t \in C^\infty$ with derivatives computable term by term. The same argument with $\partial_t(e^{-\kappa n^2 t}) = -\kappa n^2 e^{-\kappa n^2 t}$ shows that one may also differentiate term by term in t , uniformly for $t \geq t_0$.

Step 2 — reality and parity. Pairing the terms n and $-n$,

$$H_t(\theta) = \frac{1}{2\pi} \left(1 + 2 \sum_{n \geq 1} e^{-\kappa n^2 t} \cos n\theta \right),$$

which is real and even in θ .

Step 3 — normalisation. Integrating term by term (legitimate by Step 1) and using $\int_{-\pi}^{\pi} e^{in\theta} d\theta = 2\pi \delta_{n0}$ (Proposition 17.1), only the term $n = 0$ survives: $\int_{-\pi}^{\pi} H_t = \frac{1}{2\pi} \cdot 2\pi = 1$.

Step 4 — the equation. Differentiating term by term, $\partial_t H_t = \frac{1}{2\pi} \sum_n (-\kappa n^2) e^{-\kappa n^2 t} e^{in\theta}$ and $\partial_\theta^2 H_t = \frac{1}{2\pi} \sum_n (in)^2 e^{-\kappa n^2 t} e^{in\theta} = \frac{1}{2\pi} \sum_n (-n^2) e^{-\kappa n^2 t} e^{in\theta}$; multiplying the second by κ gives the first. $\square$

Theorem 18.5: Existence, instantaneous smoothing and the initial datum. Let $f \in L^1(\mathbb{T})$ (in particular f continuous) and set, for $t > 0$,

$$u(\theta, t) := \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{-\kappa n^2 t} e^{in\theta} = (f * H_t)(\theta).$$

Then:

- (i) the series and all its term-by-term derivatives converge totally on $\mathbb{T} \times [t_0, +\infty)$ for every $t_0 > 0$; hence $u \in C^\infty(\mathbb{T} \times (0, +\infty))$, regardless of the regularity of f ;
- (ii) $u_t = \kappa u_{\theta\theta}$ for $t > 0$;
- (iii) if $\sum_n |\hat{f}(n)| < \infty$ (for instance if $f \in C^1(\mathbb{T})$), then $u(\cdot, t) \rightarrow f$ uniformly on $\mathbb{T}$ as $t \rightarrow 0^+$, and setting $u(\cdot, 0) = f$ the function u solves the Cauchy problem of Definition 18.3;
- (iv) if $f \in L^2(\mathbb{T})$, then $\|u(\cdot, t) - f\|_2 \rightarrow 0$ as $t \rightarrow 0^+$.

Proof. **Step 1 — (i).** As in Step 1 of Proposition 18.3, $|\hat{f}(n)| \leq \frac{1}{2\pi} \int |f| =: M$ and for $t \geq t_0$

$$\left| \partial_t^p \partial_\theta^q (\hat{f}(n) e^{-\kappa n^2 t} e^{in\theta}) \right| \leq M (\kappa n^2)^p |n|^q e^{-\kappa n^2 t_0},$$

and $\sum_{n \in \mathbb{Z}} n^{2p+q} e^{-\kappa n^2 t_0} < \infty$ because the Gaussian exponential beats every power. The Weierstrass criterion and the theorem on term-by-term differentiation of series give the claim. The equality $u = f * H_t$ is obtained by substituting the definition of $\hat{f}(n)$ and interchanging series and integral, legitimate by uniform convergence.

Step 2 — (ii). Differentiating term by term (Step 1),

$$u_t = \sum_n (-\kappa n^2) \hat{f}(n) e^{-\kappa n^2 t} e^{in\theta}, \quad \kappa u_{\theta\theta} = \kappa \sum_n (-n^2) \hat{f}(n) e^{-\kappa n^2 t} e^{in\theta},$$

and the two series coincide term by term.

Step 3 — (iii). If $\sum_n |\hat{f}(n)| < \infty$, then by Theorem 17.5 the Fourier series of f converges uniformly to f . For every θ and every $t > 0$,

$$|u(\theta, t) - f(\theta)| = \left| \sum_{n \in \mathbb{Z}} \hat{f}(n) (e^{-\kappa n^2 t} - 1) e^{in\theta} \right| \leq \sum_{n \in \mathbb{Z}} |\hat{f}(n)| (1 - e^{-\kappa n^2 t}).$$

The series on the right does not depend on θ ; each of its terms tends to 0 as $t \rightarrow 0^+$ and is dominated by $|\hat{f}(n)|$, which is summable. Given $\varepsilon > 0$, choose N with $\sum_{|n| > N} |\hat{f}(n)| < \varepsilon/2$

and then t small enough that $\sum_{|n| \leq N} |\hat{f}(n)|(1 - e^{-\kappa n^2 t}) < \varepsilon/2$ (a finite sum of infinitesimals). Then the estimate is $< \varepsilon$ uniformly in θ . Hence $u(\cdot, t) \rightarrow f$ uniformly; continuity of u on $\mathbb{T} \times [0, +\infty)$ follows as in Step 4 of the proof of Theorem 18.3.

Step 4 — (iv). By Parseval's identity (Theorem 17.1),

$$\|u(\cdot, t) - f\|_2^2 = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 (1 - e^{-\kappa n^2 t})^2.$$

Each term is $\leq |\hat{f}(n)|^2$, summable and independent of t , and tends to 0 as $t \rightarrow 0^+$; with the same two-piece argument of Step 3 (small tail plus finite sum of infinitesimals) the sum tends to 0. $\square$

Theorem 18.6: Uniqueness for the heat equation on $\mathbb{T}$. Let $T > 0$ and let u be continuous on $\mathbb{T} \times [0, T]$, with $u_t, u_\theta, u_{\theta\theta}$ continuous on $\mathbb{T} \times (0, T]$, such that $u_t = \kappa u_{\theta\theta}$ for $0 < t \leq T$ and $u(\cdot, 0) = 0$. Then $u \equiv 0$.

Proof. **Step 1 — the energy.** For $0 < t \leq T$ set

$$E(t) := \int_{-\pi}^{\pi} u(\theta, t)^2 d\theta \geq 0.$$

Since u and u_t are continuous on $\mathbb{T} \times [\delta, T]$ for every $\delta > 0$, one may differentiate under the integral sign (Leibniz's theorem, compact domain):

$$E'(t) = 2 \int_{-\pi}^{\pi} u u_t d\theta = 2\kappa \int_{-\pi}^{\pi} u u_{\theta\theta} d\theta.$$

Step 2 — integration by parts. The function $\theta \mapsto u(\theta, t)$ is 2π -periodic and C^2 ; integrating by parts on $[-\pi, \pi]$ the boundary term $[u u_\theta]_{-\pi}^{\pi}$ vanishes by periodicity, leaving

$$E'(t) = -2\kappa \int_{-\pi}^{\pi} (u_\theta)^2 d\theta \leq 0.$$

Step 3 — conclusion. Hence E is nonincreasing on $(0, T]$ and, since u is continuous up to $t = 0$ with $u(\cdot, 0) = 0$, one has $E(t) \rightarrow 0$ as $t \rightarrow 0^+$ (uniform convergence on an interval of finite length). From E nonincreasing and $\lim_{t \rightarrow 0^+} E(t) = 0$ it follows that $E(t) \leq 0$ for every $t \in (0, T]$; together with $E \geq 0$ this gives $E \equiv 0$. Since $u(\cdot, t)$ is continuous and $\int u^2 = 0$, one concludes $u(\cdot, t) \equiv 0$ for every $t \in (0, T]$, and by continuity also at $t = 0$. $\square$

Proposition 18.4: Maximum principle for heat on $\mathbb{T}$. Let $T > 0$ and let u be continuous on $\mathbb{T} \times [0, T]$, with $u_t, u_{\theta\theta}$ continuous on $\mathbb{T} \times (0, T]$ and $u_t = \kappa u_{\theta\theta}$ for $0 < t \leq T$. Then

$$\min_{\mathbb{T}} u(\cdot, 0) \leq u(\theta, t) \leq \max_{\mathbb{T}} u(\cdot, 0) \quad \text{for every } (\theta, t) \in \mathbb{T} \times [0, T].$$

Proof. **Step 1 — the perturbed function.** Let $\varepsilon > 0$ and $w(\theta, t) := u(\theta, t) - \varepsilon t$. Then w is continuous on $\mathbb{T} \times [0, T]$ and for $0 < t \leq T$

$$w_t - \kappa w_{\theta\theta} = u_t - \varepsilon - \kappa u_{\theta\theta} = -\varepsilon < 0.$$

Step 2 — the maximum of w is at $t = 0$. The set $\mathbb{T} \times [0, T]$ is compact ($\mathbb{T}$ is compact, Chapter 12), hence w attains a maximum there at some point (θ_1, t_1) . Suppose, for contradiction, $t_1 > 0$. Then:

- the function $\theta \mapsto w(\theta, t_1)$ has a maximum (interior, because $\mathbb{T}$ has no boundary) at θ_1 , hence $w_{\theta\theta}(\theta_1, t_1) \leq 0$ by the second-derivative test;

- the function $t \mapsto w(\theta_1, t)$ has a maximum on $[0, T]$ at $t = t_1 > 0$: if $t_1 < T$ it is an interior maximum and $w_t(\theta_1, t_1) = 0$; if $t_1 = T$ it is a maximum at the right endpoint and $w_t(\theta_1, T) \geq 0$ (otherwise w would be strictly decreasing just to the left of T and would attain larger values). In either case $w_t(\theta_1, t_1) \geq 0$.

It follows that $w_t - \kappa w_{\theta\theta} \geq 0$ at (θ_1, t_1) , contradicting Step 1. Hence $t_1 = 0$ and $\max_{\mathbb{T} \times [0, T]} w = \max_{\mathbb{T}} w(\cdot, 0) = \max_{\mathbb{T}} u(\cdot, 0)$.

Step 3 — letting ε tend to zero. For every (θ, t) , $u(\theta, t) = w(\theta, t) + \varepsilon t \leq \max_{\mathbb{T}} u(\cdot, 0) + \varepsilon T$; letting $\varepsilon \rightarrow 0^+$ gives the upper bound. Applying the result to $-u$, which satisfies the same equation, gives the lower bound. $\square$

Proposition 18.5: Exponential decay to the mean. Let $f \in L^2(\mathbb{T})$ and let u be the solution of Theorem 18.5. Then, for every $t > 0$,

$$\|u(\cdot, t) - \hat{f}(0)\|_2 \leq e^{-\kappa t} \|f - \hat{f}(0)\|_2,$$

and if moreover $\sum_n |\hat{f}(n)| < \infty$,

$$\sup_{\theta \in \mathbb{T}} |u(\theta, t) - \hat{f}(0)| \leq e^{-\kappa t} \sum_{n \neq 0} |\hat{f}(n)|.$$

In particular $u(\cdot, t) \rightarrow \hat{f}(0)$, the initial mean temperature, at an exponential rate governed by the first nonconstant mode, $n = \pm 1$.

Proof. By Theorem 18.5 one has $u(\theta, t) - \hat{f}(0) = \sum_{n \neq 0} \hat{f}(n) e^{-\kappa n^2 t} e^{in\theta}$, because the term $n = 0$ equals $\hat{f}(0)e^0 = \hat{f}(0)$.

Estimate in L^2 . By Parseval's identity (Theorem 17.1),

$$\|u(\cdot, t) - \hat{f}(0)\|_2^2 = \sum_{n \neq 0} |\hat{f}(n)|^2 e^{-2\kappa n^2 t} \leq e^{-2\kappa t} \sum_{n \neq 0} |\hat{f}(n)|^2 = e^{-2\kappa t} \|f - \hat{f}(0)\|_2^2,$$

where $n^2 \geq 1$ for $n \neq 0$ was used, hence $e^{-2\kappa n^2 t} \leq e^{-2\kappa t}$. Taking square roots gives the first estimate.

Uniform estimate. With the same bound $e^{-\kappa n^2 t} \leq e^{-\kappa t}$,

$$|u(\theta, t) - \hat{f}(0)| \leq \sum_{n \neq 0} |\hat{f}(n)| e^{-\kappa n^2 t} \leq e^{-\kappa t} \sum_{n \neq 0} |\hat{f}(n)|,$$

independently of θ . $\square$

Example 18.3: A non-smooth initial datum: $f(\theta) = |\theta|$. Let $\kappa = 1$ and $f(\theta) = |\theta|$ for $\theta \in [-\pi, \pi]$, extended by periodicity.

Step 1 — the coefficients. The function is even, so the sine coefficients vanish. One has $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |\theta| d\theta = \frac{1}{\pi} \cdot \pi^2 = \pi$ and, for $j \geq 1$, integrating by parts,

$$a_j = \frac{2}{\pi} \int_0^{\pi} \theta \cos j\theta d\theta = \frac{2}{\pi} \left[\frac{\theta \sin j\theta}{j} + \frac{\cos j\theta}{j^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{(-1)^j - 1}{j^2},$$

i.e. $a_j = 0$ for j even and $a_j = -\frac{4}{\pi j^2}$ for j odd. Hence

$$f(\theta) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{j \text{ odd}} \frac{\cos j\theta}{j^2}, \quad u(\theta, t) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{j \text{ odd}} e^{-j^2 t} \frac{\cos j\theta}{j^2}.$$

The series of coefficients converges absolutely ($\sum_j 4/(\pi j^2) < \infty$), so part (iii) of Theorem 18.5 applies.

Step 2 — values. Summing the series (Python script run, 2000 terms):

$$u(0, 0.1) = 0.3568248232, \quad u(0, 0.5) = 0.7969657144, \quad u(1, 0.1) = 1.0039425011.$$

Independent check. The same three values were recomputed by a completely different route, as the convolution $(f * H_t)(\theta)$ with the heat kernel of Definition 18.3 truncated to 60 modes, integrated by the trapezoidal rule on 200 000 subintervals: this gives 0.3568248231, 0.7969657144 and 1.0039425011, i.e. the same digits to ten decimal places (a discrepancy of 10^{-10} on the first value, attributable to the quadrature). ✓

Reading. The datum has a corner at $\theta = 0$ and one at $\theta = \pm\pi$; after an arbitrarily small time the solution is C^∞ (Theorem 18.5(i)) and the value at the corner point has risen from 0 to 0.3568 at $t = 0.1$. The mean is $\hat{f}(0) = \pi/2 = 1.5707963$; at $t = 0.5$ the solution at 0 is already 0.7970, and Proposition 18.5 guarantees that the gap from the mean decays at least like e^{-t} .

Remark 18.5: Why heat cannot be run backwards. The factor $e^{-\kappa n^2 t}$ is a fierce contraction on high frequencies; to run time backwards one would have to multiply by $e^{+\kappa n^2 t}$, which blows up. Precisely: given $t > 0$ and any sequence (c_n) with $\sum |c_n|^2 < \infty$, the function $\sum c_n e^{in\theta}$ is the value at time t of a solution only if $\sum |c_n|^2 e^{2\kappa n^2 t} < \infty$, a much stronger condition. The backward problem is therefore ill posed in Hadamard's sense: the solution, when it exists, does not depend continuously on the datum. This contrasts sharply with the wave equation of Section 18.5, which is reversible: d'Alembert's formula (Theorem 18.7) makes sense for every $t \in \mathbb{R}$, positive or negative.

18.5. The wave equation on the circle

A taut string closed into a ring — for instance the string of an idealised conical pendulum, or more simply a one-dimensional model of an annular membrane — obeys the wave equation with a periodicity condition. Unlike heat, here a closed-form formula exists, is written in two lines and is exact: it is d'Alembert's formula, made periodic by the fact that the domain is a circle.

Definition 18.4: Cauchy problem for waves on $\mathbb{T}$. Let $c > 0$. Given $F \in C^2(\mathbb{T})$ and $G \in C^1(\mathbb{T})$, the **Cauchy problem** for the wave equation on $\mathbb{T}$ consists in finding $u \in C^2(\mathbb{T} \times \mathbb{R})$ with

$$u_{tt} = c^2 u_{\theta\theta} \text{ on } \mathbb{T} \times \mathbb{R}, \quad u(\theta, 0) = F(\theta), \quad u_t(\theta, 0) = G(\theta).$$

Lemma 18.5: Factorisation of the wave operator. For every $w \in C^2$ of two variables (θ, t) ,

$$\partial_t^2 w - c^2 \partial_\theta^2 w = (\partial_t - c \partial_\theta)(\partial_t + c \partial_\theta)w = (\partial_t + c \partial_\theta)(\partial_t - c \partial_\theta)w.$$

Moreover, setting $\xi = \theta + ct$ and $\eta = \theta - ct$ and $W(\xi, \eta) = w(\theta, t)$, one has $\partial_t^2 w - c^2 \partial_\theta^2 w = -4c^2 W_{\xi\eta}$.

Proof. Step 1 — the factorisation. Expanding,

$$(\partial_t - c \partial_\theta)(\partial_t + c \partial_\theta)w = w_{tt} + c w_{\theta t} - c w_{t\theta} - c^2 w_{\theta\theta},$$

and $w_{\theta t} = w_{t\theta}$ by Schwarz's theorem ($w \in C^2$): what remains is $w_{tt} - c^2 w_{\theta\theta}$. The same computation in the opposite order gives the same result.

Step 2 — the characteristic variables. The linear transformation $(\theta, t) \mapsto (\xi, \eta) = (\theta + ct, \theta - ct)$ is invertible, with $\theta = \frac{\xi + \eta}{2}$ and $t = \frac{\xi - \eta}{2c}$. By the chain rule,

$$w_\theta = W_\xi + W_\eta, \quad w_t = c(W_\xi - W_\eta),$$

and differentiating again,

$$w_{\theta\theta} = W_{\xi\xi} + 2W_{\xi\eta} + W_{\eta\eta}, \quad w_{tt} = c^2(W_{\xi\xi} - 2W_{\xi\eta} + W_{\eta\eta}).$$

Subtracting, $w_{tt} - c^2 w_{\theta\theta} = c^2(-2W_{\xi\eta} - 2W_{\xi\eta}) = -4c^2 W_{\xi\eta}$. $\square$

Theorem 18.7: Periodic d'Alembert formula. Let $F \in C^2(\mathbb{T})$ and $G \in C^1(\mathbb{T})$ and set

$$u(\theta, t) := \frac{1}{2} \left(F(\theta + ct) + F(\theta - ct) \right) + \frac{1}{2c} \int_{\theta - ct}^{\theta + ct} G(s) ds.$$

Then $u \in C^2(\mathbb{R}^2)$, is 2π -periodic in θ for every t , and solves the Cauchy problem of Definition 18.4. Moreover u is the unique solution.

Proof. Let $\mathcal{G}(s) := \int_0^s G(\sigma) d\sigma$, a primitive of G of class C^2 (because $G \in C^1$). Then

$$u(\theta, t) = \frac{1}{2} \left(F(\theta + ct) + F(\theta - ct) \right) + \frac{1}{2c} \left(\mathcal{G}(\theta + ct) - \mathcal{G}(\theta - ct) \right). \quad (**)$$

Step 1 — regularity. F is C^2 and $\mathcal{G}$ is C^2 ; the compositions with the affine functions $(\theta, t) \mapsto \theta \pm ct$ are C^2 , and a linear combination of C^2 functions is C^2 . Hence $u \in C^2(\mathbb{R}^2)$.

Step 2 — the equation. Differentiating (**):

$$u_t = \frac{c}{2} (F'(\xi) - F'(\eta)) + \frac{1}{2} (G(\xi) + G(\eta)), \quad u_\theta = \frac{1}{2} (F'(\xi) + F'(\eta)) + \frac{1}{2c} (G(\xi) - G(\eta)),$$

with $\xi = \theta + ct$, $\eta = \theta - ct$ and $\mathcal{G}' = G$. Differentiating once more:

$$u_{tt} = \frac{c^2}{2} (F''(\xi) + F''(\eta)) + \frac{c}{2} (G'(\xi) - G'(\eta)),$$

$$u_{\theta\theta} = \frac{1}{2} (F''(\xi) + F''(\eta)) + \frac{1}{2c} (G'(\xi) - G'(\eta)).$$

Multiplying the second by c^2 gives exactly the first: $u_{tt} = c^2 u_{\theta\theta}$.

Step 3 — the initial data. Setting $t = 0$ in (**) gives $\xi = \eta = \theta$, hence $u(\theta, 0) = \frac{1}{2} (F(\theta) + F(\theta)) + 0 = F(\theta)$; and from the formula for u_t , $u_t(\theta, 0) = \frac{c}{2} (F'(\theta) - F'(\theta)) + \frac{1}{2} (G(\theta) + G(\theta)) = G(\theta)$.

Step 4 — periodicity in θ . F is 2π -periodic, so both terms in F are. For $\mathcal{G}$ one has, for every s ,

$$\mathcal{G}(s + 2\pi) - \mathcal{G}(s) = \int_s^{s+2\pi} G = \int_{-\pi}^{\pi} G = 2\pi \hat{G}(0),$$

a constant independent of s (the integral of a periodic function over a period does not depend on the starting point). Hence

$$\mathcal{G}(\theta + 2\pi + ct) - \mathcal{G}(\theta + 2\pi - ct) = \left(\mathcal{G}(\theta + ct) + \int_{-\pi}^{\pi} G \right) - \left(\mathcal{G}(\theta - ct) + \int_{-\pi}^{\pi} G \right) = \mathcal{G}(\theta + ct) - \mathcal{G}(\theta - ct),$$

and $u(\theta + 2\pi, t) = u(\theta, t)$: the additive constant cancels, and periodicity holds *always*, even when G does not have zero mean.

Step 5 — uniqueness. Let u_1, u_2 be two solutions and $w = u_1 - u_2$: then $w \in C^2$, $w_{tt} = c^2 w_{\theta\theta}$, $w(\cdot, 0) = 0$ and $w_t(\cdot, 0) = 0$. Set

$$E(t) := \frac{1}{2} \int_{-\pi}^{\pi} (w_t^2 + c^2 w_\theta^2) d\theta.$$

The integrands are C^1 on a compact domain, so one may differentiate under the integral sign:

$$E'(t) = \int_{-\pi}^{\pi} (w_t w_{tt} + c^2 w_\theta w_{\theta t}) d\theta = \int_{-\pi}^{\pi} (c^2 w_t w_{\theta\theta} + c^2 w_\theta w_{t\theta}) d\theta = c^2 \int_{-\pi}^{\pi} \partial_\theta (w_t w_\theta) d\theta = 0,$$

because $w_t w_\theta$ is 2π -periodic in θ and the integral of its derivative over a period is zero. Hence E is constant; but $E(0) = \frac{1}{2} \int (0 + c^2 \cdot 0) = 0$ because $w_t(\cdot, 0) = 0$ and $w_\theta(\cdot, 0) = \partial_\theta (w(\cdot, 0)) = 0$. Thus $E \equiv 0$, and since the integrand is continuous and nonnegative, $w_t \equiv 0$ and $w_\theta \equiv 0$ on $\mathbb{R}^2$: w is constant, and equals 0 because $w(\cdot, 0) = 0$. $\square$

Proposition 18.6: The same solution, written as a Fourier series. Let F, G be trigonometric polynomials, $F = \sum_{|n| \leq N} \hat{F}(n) e^{in\theta}$ and $G = \sum_{|n| \leq N} \hat{G}(n) e^{in\theta}$. Then the solution of Theorem 18.7 is

$$u(\theta, t) = \hat{F}(0) + \hat{G}(0)t + \sum_{0 < |n| \leq N} \left(\hat{F}(n) \cos(cnt) + \frac{\hat{G}(n)}{cn} \sin(cnt) \right) e^{in\theta}.$$

Proof. Both sides depend linearly on (F, G) : it therefore suffices to verify the equality when $F(\theta) = e^{in\theta}$, $G = 0$, and when $F = 0$, $G(\theta) = e^{in\theta}$.

Case $F = e^{in\theta}$, $G = 0$. D'Alembert's formula gives

$$\frac{1}{2} (e^{in(\theta+ct)} + e^{in(\theta-ct)}) = e^{in\theta} \frac{e^{icnt} + e^{-icnt}}{2} = e^{in\theta} \cos(cnt),$$

by Euler's formula (Chapter 6); for $n = 0$ this reads $\cos 0 = 1$, in agreement with the term $\hat{F}(0)$.

Case $F = 0$, $G = e^{in\theta}$, $n \neq 0$. One has

$$\frac{1}{2c} \int_{\theta-ct}^{\theta+ct} e^{ins} ds = \frac{1}{2c} \cdot \frac{e^{in(\theta+ct)} - e^{in(\theta-ct)}}{in} = \frac{e^{in\theta}}{2c in} \cdot 2i \sin(cnt) = \frac{\sin(cnt)}{cn} e^{in\theta}.$$

Case $F = 0$, $G = 1$ (i.e. $n = 0$). One has $\frac{1}{2c} \int_{\theta-ct}^{\theta+ct} 1 ds = \frac{1}{2c} \cdot 2ct = t$, which is the term $\hat{G}(0)t$.

Summing the contributions with the coefficients $\hat{F}(n)$, $\hat{G}(n)$ gives the formula in the statement. $\square$

Proposition 18.7: Conservation of energy. Let u be the solution of Theorem 18.7. Then the function

$$E(t) = \frac{1}{2} \int_{-\pi}^{\pi} (u_t(\theta, t)^2 + c^2 u_\theta(\theta, t)^2) d\theta$$

is constant in t , and equals $E(0) = \frac{1}{2} \int_{-\pi}^{\pi} (G^2 + c^2 (F')^2) d\theta$.

Proof. The computation is that of Step 5 of the proof of Theorem 18.7, applied to u instead of w : the only properties used are $u \in C^2$, $u_{tt} = c^2 u_{\theta\theta}$ and 2π -periodicity in θ , all of which hold. This gives $E' \equiv 0$, hence E constant. The value at $t = 0$ is read off from the initial data: $u_t(\cdot, 0) = G$ and $u_\theta(\cdot, 0) = F'$. $\square$

Proposition 18.8: Periodicity in time. If $\hat{G}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} G = 0$ (in particular if $G \equiv 0$), the solution of Theorem 18.7 is periodic in t with period $2\pi/c$:

$$u\left(\theta, t + \frac{2\pi}{c}\right) = u(\theta, t) \quad \text{for every } (\theta, t).$$

Proof. In the expression (**) from the proof of Theorem 18.7, replacing t by $t + \frac{2\pi}{c}$ means replacing $\xi = \theta + ct$ by $\xi + 2\pi$ and $\eta = \theta - ct$ by $\eta - 2\pi$. Now F is 2π -periodic, so $F(\xi + 2\pi) = F(\xi)$ and $F(\eta - 2\pi) = F(\eta)$. As for $\mathcal{G}$, Step 4 of the same proof gives $\mathcal{G}(s + 2\pi) = \mathcal{G}(s) + \int_{-\pi}^{\pi} G = \mathcal{G}(s)$, because by hypothesis $\int_{-\pi}^{\pi} G = 2\pi\hat{G}(0) = 0$: hence $\mathcal{G}$ is itself 2π -periodic and $\mathcal{G}(\xi + 2\pi) - \mathcal{G}(\eta - 2\pi) = \mathcal{G}(\xi) - \mathcal{G}(\eta)$. The two terms of (**) are therefore unchanged. $\square$

Example 18.4: Two modes that recombine. Let $c = 2$, $F(\theta) = \cos \theta + \frac{1}{2} \cos 3\theta$ and $G \equiv 0$.

By d'Alembert. $u(\theta, t) = \frac{1}{2}(F(\theta + 2t) + F(\theta - 2t))$.

By series. By Proposition 18.6, $u(\theta, t) = \cos(2t) \cos \theta + \frac{1}{2} \cos(6t) \cos 3\theta$.

Independent check. The two expressions were evaluated (Python script run) at three points:

(θ, t)	(0.4, 0.7)	(1.3, 0.25)	(2, 1.1)
d'Alembert	0.0677252004	0.2090770974	0.7010954284
series	0.0677252004	0.2090770974	0.7010954284

The discrepancies are 0, 0 and $1.1 \cdot 10^{-16}$, i.e. double-precision rounding error. $\checkmark$

Energy. By Proposition 18.7, $E = \frac{1}{2} \int_{-\pi}^{\pi} c^2 (F')^2 d\theta$ with $F' = -\sin \theta - \frac{3}{2} \sin 3\theta$; by orthogonality of the sines,

$$E = \frac{c^2}{2} \left(\pi \cdot 1^2 + \pi \cdot \left(\frac{3}{2}\right)^2 \right) = \frac{4}{2} \pi \left(1 + \frac{9}{4} \right) = \frac{13\pi}{2} = 20.4203522.$$

Numerical quadrature of the energy at the three instants $t = 0$, $t = 0.37$ and $t = 1.25$ gives 20.42035225 in all three cases. $\checkmark$

Periodicity. Since $\hat{G}(0) = 0$, Proposition 18.8 gives period $2\pi/c = \pi$; here the solution actually already has period π from the term $\cos(2t)$, and the second term $\cos(6t)$ has period $\pi/3$, which divides π .

Remark 18.6: Heat and waves compared. The two equations differ by a single symbol — u_t versus u_{tt} — and by everything else. On the Fourier modes, heat acts with the factor $e^{-\kappa n^2 t}$, real and decreasing, waves with $\cos(cnt)$ and $\sin(cnt)$, of bounded modulus and never infinitesimal. It follows that:

- heat regularises instantaneously (Theorem 18.5(i)), waves do not: d'Alembert's formula transports the corners of the datum without attenuating them;
- heat dissipates and forgets (Proposition 18.5), waves conserve energy (Proposition 18.7) and, on the circle, return periodically to the initial datum (Proposition 18.8);
- heat is irreversible (Observation 18.5), waves are reversible: the formula of Theorem 18.7 is defined for every $t \in \mathbb{R}$ and is invariant under $t \mapsto -t$ when $G \equiv 0$.

The time-periodicity of waves on $\mathbb{T}$ is a very special fact, which depends on the eigenvalues $\{n^2\}$ of the Laplacian on $\mathbb{T}$ having square roots that are all integers, hence commensurable. On the disc, as we shall see in Section 18.8, the eigenvalues are the squares of the Bessel zeros, whose ratios are neither integers nor simple fractions: the drum is not periodic and its sound is not harmonic (Observation 18.14).

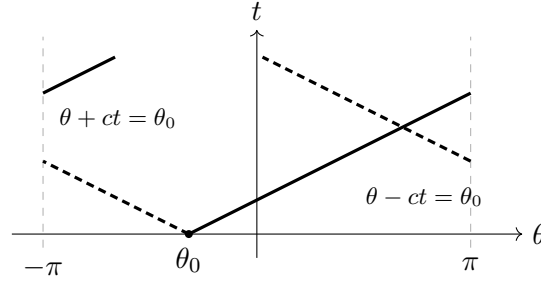


Figure 18.2: The characteristics $\theta \pm ct = \text{const}$ in the strip $\mathbb{T} \times [0, +\infty)$ for $c = 2$ and $\theta_0 = -1$. Coordinates computed by hand: the right-going characteristic $\theta = \theta_0 + ct$ reaches $\theta = \pi$ at time $t = (\pi - \theta_0)/c = (3.1415927 + 1)/2 = 2.0707963$ and re-enters from $\theta = -\pi$; the left-going one $\theta = \theta_0 - ct$ reaches $\theta = -\pi$ at time $t = (\theta_0 + \pi)/c = (3.1415927 - 1)/2 = 1.0707963$ and re-enters from $\theta = \pi$. At time $t = 2\pi/c = \pi$ both have returned to the starting point: this is Proposition 18.8.

18.6. The circular membrane: from separation to Bessel's equation

The drum is the first problem in which separation of variables does not close with elementary functions. The reason is purely structural: in Laplace's equation the parameter λ appeared only through n^2 , and the radial equation was Euler's, homogeneous under dilations; here a term λr^2 also appears, which breaks the homogeneity and produces an equation whose solution must be built from scratch. That construction is the Bessel function.

Definition 18.5: The circular membrane problem. Let $a > 0$ and $c > 0$. The **circular membrane problem** (or drum problem) consists in seeking $u \in C^2(\overline{D_a} \times \mathbb{R})$ with

$$u_{tt} = c^2 \Delta u \text{ in } D_a \times \mathbb{R}, \quad u(x, y, t) = 0 \text{ for } (x, y) \in \partial D_a \text{ and every } t.$$

A nontrivial solution of the form $u(x, y, t) = T(t)V(x, y)$ is called a **normal mode** of the membrane.

Lemma 18.6: Separation of time and space. Let $T \in C^2(\mathbb{R})$ and $V \in C^2(\overline{D_a})$, both nontrivial, be such that $u = TV$ solves $u_{tt} = c^2 \Delta u$ in $D_a \times \mathbb{R}$. Then there is $\lambda \in \mathbb{R}$ with

$$T'' + \lambda c^2 T = 0 \text{ on } \mathbb{R}, \quad \Delta V + \lambda V = 0 \text{ in } D_a \quad (\text{Helmholtz's equation}).$$

If moreover u vanishes on $\partial D_a \times \mathbb{R}$ and $T \not\equiv 0$, then $V|_{\partial D_a} = 0$.

Proof. Substituting $u = TV$ gives $T''(t)V(p) = c^2 T(t)\Delta V(p)$ for every $p \in D_a$ and every $t \in \mathbb{R}$. **Step 1.** Since $V \not\equiv 0$ and V is continuous, there is $p_0 \in D_a$ with $V(p_0) \neq 0$ (if V vanished throughout D_a it would also vanish, by continuity, on $\overline{D_a}$). Set $\lambda := -\frac{\Delta V(p_0)}{V(p_0)} \in \mathbb{R}$; evaluating at $p = p_0$ and dividing by $V(p_0)$,

$$T''(t) = c^2 \frac{\Delta V(p_0)}{V(p_0)} T(t) = -\lambda c^2 T(t) \quad \text{for every } t.$$

Step 2. Substituting $T'' = -\lambda c^2 T$ into the starting identity: $-\lambda c^2 T(t)V(p) = c^2 T(t)\Delta V(p)$, i.e. $c^2 T(t)(\Delta V(p) + \lambda V(p)) = 0$. Choosing t_0 with $T(t_0) \neq 0$ (which exists because $T \not\equiv 0$) and dividing by $c^2 T(t_0) \neq 0$ gives $\Delta V + \lambda V = 0$ in D_a .

Step 3. If $u = TV$ vanishes on $\partial D_a \times \mathbb{R}$, evaluating at $t = t_0$ with $T(t_0) \neq 0$ gives $V(p) = 0$ for every $p \in \partial D_a$. $\square$

Proposition 18.9: *The eigenvalues of the Dirichlet problem are positive.* Let $V \in C^2(\overline{D_a})$ be nontrivial with $\Delta V + \lambda V = 0$ in D_a and $V|_{\partial D_a} = 0$. Then $\lambda > 0$.

Proof. **Step 1 — Green's identity.** By Leibniz's rule,

$$\operatorname{div}(V\nabla V) = \nabla V \cdot \nabla V + V \Delta V = |\nabla V|^2 + V \Delta V.$$

The disc D_a is a regular domain of the plane (its boundary is a C^∞ curve, Chapter 2) and $V\nabla V$ is a C^1 vector field on $\overline{D_a}$: by the divergence theorem,

$$\int_{D_a} (|\nabla V|^2 + V \Delta V) dA = \oint_{\partial D_a} V \frac{\partial V}{\partial \nu} ds = 0,$$

because $V = 0$ on ∂D_a and hence the boundary integrand is identically zero (ν is the outward normal).

Step 2 — using the equation. Substituting $\Delta V = -\lambda V$:

$$\int_{D_a} |\nabla V|^2 dA = \lambda \int_{D_a} V^2 dA. \quad (\dagger)$$

Step 3 — both integrals are positive. The right-hand side contains $\int_{D_a} V^2 > 0$: indeed V is continuous and nontrivial on $\overline{D_a}$ — and cannot vanish throughout D_a , since by continuity it would then vanish everywhere — so V^2 is continuous, nonnegative, and strictly positive on a nonempty open set. The left-hand side is ≥ 0 . Hence $\lambda \geq 0$.

Step 4 — ruling out $\lambda = 0$. If $\lambda = 0$, then $(\dagger)$ gives $\int_{D_a} |\nabla V|^2 = 0$; since the integrand is continuous and nonnegative, $\nabla V \equiv 0$ on D_a , so V is constant on the connected set D_a and, by continuity, on $\overline{D_a}$. Since $V = 0$ on the boundary, the constant is 0 and $V \equiv 0$: excluded by hypothesis. Hence $\lambda > 0$. $\square$

Lemma 18.7: *Angular separation and the birth of Bessel's equation.* Let $\lambda > 0$ and let $V(r, \theta) = R(r)\Theta(\theta)$, with $R \in C^2((0, a))$ and $\Theta \in C^2(\mathbb{R})$ 2π -periodic, both nontrivial, be a solution of $\Delta V + \lambda V = 0$ in $D_a \setminus \{0\}$. Then there is an integer $n \geq 0$ with

$$\Theta'' + n^2 \Theta = 0 \quad \text{and} \quad r^2 R'' + r R' + (\lambda r^2 - n^2) R = 0 \quad \text{on } (0, a).$$

Setting $s = \sqrt{\lambda} r$ and $\rho(s) := R(s/\sqrt{\lambda})$, the second equation becomes

$$s^2 \rho''(s) + s \rho'(s) + (s^2 - n^2) \rho(s) = 0, \quad 0 < s < \sqrt{\lambda} a.$$

Proof. **Step 1 — the equation in polar coordinates.** By Theorem 18.1, the equation $\Delta V + \lambda V = 0$ reads

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' + \lambda R\Theta = 0;$$

multiplying by r^2 (legitimate, as $r > 0$),

$$(r^2 R'' + r R' + \lambda r^2 R)\Theta(\theta) + R(r)\Theta''(\theta) = 0 \quad \text{for every } (r, \theta) \in (0, a) \times \mathbb{R}. \quad (\ddagger)$$

Step 2 — separation. The argument of Lemma 18.2 is repeated verbatim. Choosing θ_0 with $\Theta(\theta_0) \neq 0$ and setting $\mu := -\Theta''(\theta_0)/\Theta(\theta_0)$, evaluating $(\ddagger)$ at θ_0 gives, for every $r \in (0, a)$,

$$r^2 R'' + r R' + (\lambda r^2 - \mu) R = 0.$$

Substituting this identity into (‡) gives $R(r)(\Theta''(\theta) + \mu\Theta(\theta)) = 0$ for every (r, θ) ; choosing r_0 with $R(r_0) \neq 0$ gives $\Theta'' + \mu\Theta = 0$ on $\mathbb{R}$.

Step 3 — quantisation. Θ is by hypothesis 2π -periodic, of class C^2 and nontrivial: by Lemma 18.3, $\mu = n^2$ with $n \geq 0$ an integer, and Θ is constant ($n = 0$) or a combination of $\cos n\theta$ and $\sin n\theta$.

Step 4 — rescaling. Let $s = \sqrt{\lambda}r$, i.e. $r = s/\sqrt{\lambda}$, and $\rho(s) = R(s/\sqrt{\lambda})$. By the chain rule $R'(r) = \sqrt{\lambda}\rho'(s)$ and $R''(r) = \lambda\rho''(s)$. Substituting,

$$r^2 R'' + rR' + (\lambda r^2 - n^2)R = \underbrace{r^2 \lambda}_{=s^2} \rho'' + \underbrace{r \sqrt{\lambda}}_{=s} \rho' + (\underbrace{\lambda r^2}_{=s^2} - n^2) \rho,$$

and the vanishing of the left-hand side is equivalent to $s^2 \rho'' + s\rho' + (s^2 - n^2)\rho = 0$. $\square$

Definition 18.6: Bessel's equation. Let $n \geq 0$. The second-order ordinary differential equation

$$x^2 y''(x) + x y'(x) + (x^2 - n^2)y(x) = 0, \quad x > 0,$$

is called **Bessel's equation of order n** . It is equation 10.2.1 of the NIST Digital Library of Mathematical Functions.

***Remark 18.7: Why elementary functions are not enough.** In Lemma 18.4 the radial equation $r^2 R'' + rR' - n^2 R = 0$ was invariant under the dilation $r \mapsto \alpha r$: every term has the same "weight" in r , and this is what allowed the substitution $r = e^s$ that turned it into an equation with constant coefficients. The term $\lambda r^2 R$ of Helmholtz's equation breaks this invariance: no elementary substitution reduces it to constant coefficients, and indeed it can be shown — though not here — that its solutions cannot be expressed by elementary functions. The point $x = 0$ is, in the standard terminology, a regular singular point: the equation, divided by x^2 , has coefficients $1/x$ and $1 - n^2/x^2$ that blow up at 0, but slowly enough to allow solutions of the form $x^n \cdot$ (power series). This is exactly the form we shall postulate — and verify — in Definition 18.7.*

18.7. The Bessel function of the first kind

We now define, by a series, the solution of Bessel's equation that stays bounded at the origin, and we prove all the properties we shall need for it: convergence, verification of the equation, behaviour at 0, recurrence formulas, singularity of the other solution, existence of infinitely many zeros, orthogonality.

Definition 18.7: Bessel function of the first kind of integer order. Let $n \geq 0$ be an integer. The **Bessel function of the first kind of order n** is

$$J_n(x) := \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{n+2k}, \quad x \in \mathbb{R}.$$

It is the case $\nu = n$ integer of formula 10.2.2 of the NIST Digital Library of Mathematical Functions, where $\Gamma(\nu + k + 1)$ takes the place of $(n + k)!$.

***Theorem 18.8: The Bessel series converges everywhere.** For every integer $n \geq 0$ the series of Definition 18.7 converges absolutely for every $x \in \mathbb{R}$ and uniformly on every bounded interval. The function J_n is of class C^∞ on $\mathbb{R}$ and its derivatives may be computed term by term.*

Proof. **Step 1 — absolute convergence.** Fix $x \neq 0$ and set $t_k = \frac{|x/2|^{n+2k}}{k!(n+k)!} > 0$. Then

$$\frac{t_{k+1}}{t_k} = \frac{(x/2)^2}{(k+1)(n+k+1)} \rightarrow 0 \quad (k \rightarrow \infty).$$

By the ratio test the series $\sum_k t_k$ converges; hence the series for $J_n(x)$ converges absolutely. For $x = 0$ all terms with $n + 2k > 0$ vanish and the series reduces to a single term (or to 0 if $n \geq 1$).

Step 2 — uniform convergence on bounded sets. Let $|x| \leq L$. Then $t_k \leq \frac{(L/2)^{n+2k}}{k!(n+k)!} =: M_k$, and $\sum_k M_k < \infty$ by Step 1 applied to $x = L$. By the Weierstrass test the convergence is uniform on $[-L, L]$.

Step 3 — regularity. The series is a power series in x : indeed $J_n(x) = x^n \sum_k b_k x^{2k}$ with $b_k = \frac{(-1)^k}{2^{n+2k} k!(n+k)!}$, and Step 1 shows that the radius of convergence of the power series $\sum_k b_k y^k$ (in $y = x^2$) is $+\infty$. A power series is of class C^∞ inside its disc of convergence and may be differentiated term by term as many times as one wishes; multiplying by the monomial x^n preserves the property. $\square$

Theorem 18.9: J_n solves Bessel's equation. For every integer $n \geq 0$ and every $x \in \mathbb{R}$,

$$x^2 J_n''(x) + x J_n'(x) + (x^2 - n^2) J_n(x) = 0.$$

Proof. Write $J_n(x) = \sum_{k \geq 0} a_k x^{n+2k}$ with

$$a_k = \frac{(-1)^k}{2^{n+2k} k!(n+k)!}.$$

By Theorem 18.8 one may differentiate term by term:

$$x^2 J_n'' = \sum_{k \geq 0} a_k (n+2k)(n+2k-1) x^{n+2k}, \quad x J_n' = \sum_{k \geq 0} a_k (n+2k) x^{n+2k}.$$

Step 1 — the first three terms. Adding the two expressions and subtracting $n^2 J_n$,

$$x^2 J_n'' + x J_n' - n^2 J_n = \sum_{k \geq 0} a_k [(n+2k)(n+2k-1) + (n+2k) - n^2] x^{n+2k} = \sum_{k \geq 0} a_k [(n+2k)^2 - n^2] x^{n+2k},$$

because $(n+2k)(n+2k-1) + (n+2k) = (n+2k)^2$. Now $(n+2k)^2 - n^2 = 4k^2 + 4nk = 4k(n+k)$, hence

$$x^2 J_n'' + x J_n' - n^2 J_n = \sum_{k \geq 1} 4k(n+k) a_k x^{n+2k},$$

where the sum starts at $k = 1$ because the factor $4k(n+k)$ vanishes at $k = 0$.

Step 2 — the fourth term. We have

$$x^2 J_n = \sum_{k \geq 0} a_k x^{n+2k+2} = \sum_{k \geq 1} a_{k-1} x^{n+2k}$$

(reindexing $k \mapsto k-1$).

Step 3 — the coefficients cancel. For $k \geq 1$,

$$4k(n+k) a_k = 4k(n+k) \cdot \frac{(-1)^k}{2^{n+2k} k!(n+k)!} = \frac{4(-1)^k}{2^{n+2k} (k-1)!(n+k-1)!} = \frac{(-1)^k}{2^{n+2k-2} (k-1)!(n+k-1)!},$$

having simplified k against $k!$ and $(n+k)$ against $(n+k)!$ and used $4 = 2^2$. On the other hand

$$a_{k-1} = \frac{(-1)^{k-1}}{2^{n+2k-2}(k-1)!(n+k-1)!},$$

which is the negative of the previous expression. Hence $4k(n+k)a_k + a_{k-1} = 0$ for every $k \geq 1$, and adding the results of Steps 1 and 2

$$x^2 J_n'' + x J_n' + (x^2 - n^2) J_n = \sum_{k \geq 1} [4k(n+k)a_k + a_{k-1}] x^{n+2k} = 0.$$

□

Proposition 18.10: Behaviour at the origin. Let $n \geq 0$ be an integer. Then

$$J_0(0) = 1, \quad J_n(0) = 0 \quad (n \geq 1), \quad \lim_{x \rightarrow 0^+} \frac{J_n(x)}{(x/2)^n} = \frac{1}{n!},$$

More precisely there exist $\delta_n > 0$ and constants $0 < c_1 \leq c_2$ such that

$$c_1 x^n \leq J_n(x) \leq c_2 x^n \quad \text{for } 0 < x \leq \delta_n;$$

in particular $J_n(x) > 0$ for $0 < x \leq \delta_n$.

Proof. Step 1 — the values at 0. In the series, the term $k = 0$ equals $\frac{1}{n!}(x/2)^n$ and every other term contains x with exponent $n+2k > n \geq 0$. At $x = 0$: if $n = 0$ the term $k = 0$ equals 1 and the others 0, so $J_0(0) = 1$; if $n \geq 1$ all terms vanish, so $J_n(0) = 0$.

Step 2 — the limit. Factoring out $(x/2)^n$,

$$\frac{J_n(x)}{(x/2)^n} = \sum_{k \geq 0} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{2k} = \frac{1}{n!} + \sum_{k \geq 1} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{2k},$$

and the series on the right is a power series in x with no constant term, hence continuous and zero at $x = 0$ (Theorem 18.8). The limit is $1/n!$.

Step 3 — the two-sided estimate. Let $g(x) := J_n(x)/(x/2)^n$ for $x > 0$ and $g(0) := 1/n!$: by Step 2, g is continuous on $[0, +\infty)$ and $g(0) = 1/n! > 0$. By persistence of sign there is $\delta_n > 0$ with $\frac{1}{2n!} \leq g(x) \leq \frac{2}{n!}$ for $0 \leq x \leq \delta_n$. Multiplying by $(x/2)^n = 2^{-n}x^n > 0$ gives

$$\underbrace{\frac{1}{2^{n+1}n!}}_{=:c_1} x^n \leq J_n(x) \leq \underbrace{\frac{2}{2^n n!}}_{=:c_2} x^n, \quad 0 < x \leq \delta_n,$$

and in particular $J_n(x) > 0$ on $(0, \delta_n]$. □

Proposition 18.11: Recurrence and derivative formulas. For every integer $n \geq 0$ and every $x > 0$:

- (i) $\frac{d}{dx}(x^{-n} J_n(x)) = -x^{-n} J_{n+1}(x)$;
- (ii) $\frac{d}{dx}(x^n J_n(x)) = x^n J_{n-1}(x)$ (for $n \geq 1$);
- (iii) $J_n'(x) = -J_{n+1}(x) + \frac{n}{x} J_n(x)$; in particular $J_0' = -J_1$;
- (iv) $J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$ (for $n \geq 1$).

Proof. All term-by-term differentiations are legitimate by Theorem 18.8.

(i) We have $x^{-n}J_n(x) = \sum_{k \geq 0} \frac{(-1)^k}{2^{n+2k}k!(n+k)!} x^{2k}$; differentiating, the term $k = 0$ is constant and disappears:

$$\frac{d}{dx}(x^{-n}J_n) = \sum_{k \geq 1} \frac{(-1)^k 2k}{2^{n+2k}k!(n+k)!} x^{2k-1} = \sum_{k \geq 1} \frac{(-1)^k}{2^{n+2k-1}(k-1)!(n+k)!} x^{2k-1}.$$

Setting $m = k - 1$,

$$= \sum_{m \geq 0} \frac{(-1)^{m+1}}{2^{n+2m+1}m!(n+m+1)!} x^{2m+1} = -x^{-n} \sum_{m \geq 0} \frac{(-1)^m}{m!(n+1+m)!} \left(\frac{x}{2}\right)^{(n+1)+2m} = -x^{-n}J_{n+1}(x),$$

where we used $\frac{x^{2m+1}}{2^{n+2m+1}} = x^{-n} \frac{x^{n+2m+1}}{2^{n+2m+1}} = x^{-n} \left(\frac{x}{2}\right)^{n+1+2m}$.

(ii) Similarly $x^n J_n(x) = \sum_{k \geq 0} \frac{(-1)^k}{2^{n+2k}k!(n+k)!} x^{2n+2k}$, and differentiating

$$\frac{d}{dx}(x^n J_n) = \sum_{k \geq 0} \frac{(-1)^k (2n+2k)}{2^{n+2k}k!(n+k)!} x^{2n+2k-1} = \sum_{k \geq 0} \frac{(-1)^k}{2^{n+2k-1}k!(n+k-1)!} x^{2n+2k-1},$$

having simplified $2(n+k)$ against $(n+k)!$. Factoring out x^n and writing $\frac{x^{n+2k-1}}{2^{n+2k-1}} = \left(\frac{x}{2}\right)^{(n-1)+2k}$

gives $x^n \sum_{k \geq 0} \frac{(-1)^k}{k!((n-1)+k)!} \left(\frac{x}{2}\right)^{(n-1)+2k} = x^n J_{n-1}(x)$.

(iii) Expanding the product rule in (i): $-nx^{-n-1}J_n + x^{-n}J'_n = -x^{-n}J_{n+1}$; multiplying by x^n gives $J'_n - \frac{n}{x}J_n = -J_{n+1}$. For $n = 0$ this is $J'_0 = -J_1$.

(iv) Expanding the product rule in (ii): $nx^{n-1}J_n + x^n J'_n = x^n J_{n-1}$, i.e. $J'_n + \frac{n}{x}J_n = J_{n-1}$. Subtracting from this relation (iii) rewritten as $J'_n - \frac{n}{x}J_n = -J_{n+1}$ gives $\frac{2n}{x}J_n = J_{n-1} + J_{n+1}$.

□

Remark 18.8: Comparison with the reference tables. Formulas (iii) and (iv) of Proposition 18.11 are, in that order, equations 10.6.2 and 10.6.1 of the NIST Digital Library of Mathematical Functions, stated there for a generic cylinder function C_ν (i.e. for any linear combination of J_ν and Y_ν) and for real ν ; the identity $J'_0 = -J_1$ is their equation 10.6.3. Here they are proved only for J_n with n an integer, which is all that is needed for the disc, and the proof is self-contained: it is read directly off the series.

Proposition 18.12: The second solution is unbounded at the origin. Let $n \geq 0$ be an integer and let $\delta_n > 0$ be as in Proposition 18.10. Let y be a C^2 solution of Bessel's equation of order n on $(0, \delta_n]$, linearly independent of J_n . Then

$$\lim_{x \rightarrow 0^+} |y(x)| = +\infty.$$

In particular the only solution (up to multiplicative constants) that is bounded on a right-hand neighbourhood of 0 is J_n .

Proof. On $(0, +\infty)$ the equation may be divided by $x^2 \neq 0$ and takes the normal form

$$y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right)y = 0, \tag{N}$$

with continuous coefficients.

Step 1 — the Wronskian. Let $W := J_n y' - J'_n y$. Differentiating and using (N) for both functions,

$$W' = J_n y'' - J'_n y' = J_n \left(-\frac{1}{x} y' - \left(1 - \frac{n^2}{x^2}\right) y \right) - y' \left(-\frac{1}{x} J'_n - \left(1 - \frac{n^2}{x^2}\right) J_n \right) = -\frac{1}{x} (J_n y' - J'_n y) = -\frac{W}{x}.$$

Hence $(xW)' = W + xW' = W - W = 0$, i.e. $xW(x) = C$ is constant on $(0, \delta_n]$. If $C = 0$ then $W \equiv 0$, and for two solutions of a second-order linear equation an identically vanishing Wronskian is equivalent to linear dependence: excluded by hypothesis. Hence $C \neq 0$ and $W(x) = C/x$.

Step 2 — the quotient. By Proposition 18.10, $J_n > 0$ on $(0, \delta_n]$, so the quotient y/J_n is defined and differentiable there, with

$$\left(\frac{y}{J_n} \right)' = \frac{y' J_n - y J'_n}{J_n^2} = \frac{W}{J_n^2} = \frac{C}{x J_n(x)^2}.$$

Setting $x_0 := \delta_n$, $D := y(x_0)/J_n(x_0)$ and $I(x) := \int_x^{x_0} \frac{ds}{s J_n(s)^2}$, integrating from x to x_0 gives

$$y(x) = J_n(x) \left(D - C I(x) \right), \quad 0 < x \leq x_0.$$

Step 3 — $I(x)$ diverges. By Proposition 18.10, $J_n(s) \leq c_2 s^n$ on $(0, x_0]$, hence $\frac{1}{s J_n(s)^2} \geq \frac{1}{c_2^2 s^{2n+1}}$ and

$$I(x) \geq \frac{1}{c_2^2} \int_x^{x_0} \frac{ds}{s^{2n+1}} = \begin{cases} \frac{1}{c_2^2} \ln \frac{x_0}{x}, & n = 0, \\ \frac{1}{2n c_2^2} (x^{-2n} - x_0^{-2n}), & n \geq 1, \end{cases}$$

and in both cases $I(x) \rightarrow +\infty$ as $x \rightarrow 0^+$.

Step 4 — y is unbounded. From $y = J_n(D - CI)$ it follows, for x small enough that $|C|I(x) > |D|$,

$$|y(x)| \geq J_n(x) (|C|I(x) - |D|) \geq |C|J_n(x)I(x) - |D|J_n(x).$$

The second term is bounded: $J_n(x) \leq c_2 x^n \leq c_2 x_0^n$. The first diverges: using $J_n(x) \geq c_1 x^n$ and the estimates of Step 3,

$$J_n(x)I(x) \geq \begin{cases} \frac{c_1}{c_2^2} \ln \frac{x_0}{x} \rightarrow +\infty, & n = 0, \\ \frac{c_1}{2n c_2^2} (x^{-2n} - x_0^{-2n}) \rightarrow +\infty, & n \geq 1. \end{cases}$$

Hence $|y(x)| \rightarrow +\infty$.

Conclusion. The solution space of (N) on $(0, \delta_n]$ has dimension 2; if z is bounded on a right-hand neighbourhood of 0, then z cannot be independent of J_n (otherwise it would be unbounded), so $z = \alpha J_n$ for some $\alpha \in \mathbb{R}$. $\square$

***Remark 18.9:** Why the drum does not "blow up" at the centre. Proposition 18.12 is, for the membrane, what Theorem 18.2 was for the Dirichlet problem: the filter that discards half of the solutions. In the Laplace case the discarded solution was explicit (r^{-n} , or $\ln r$); here it is not, and it is known under the name of Bessel function of the second kind Y_n ,*

whose behaviour at the origin is $Y_0(x) \sim \frac{2}{\pi} \ln x$ and $Y_\nu(x) \sim -\frac{1}{\pi} \Gamma(\nu) \left(\frac{x}{2}\right)^{-\nu}$ for $\nu > 0$ (NIST DLMF, equations 10.7.1 and 10.7.4). We shall not need it: Proposition 18.12 proves what is needed — that any solution independent of J_n is unbounded — without constructing Y_n .

Lemma 18.8: Liouville form of Bessel's equation. Let $R \in C^2((0, +\infty))$ and set $w(x) := \sqrt{x} R(x)$. Then R solves Bessel's equation of order n if and only if w solves

$$w''(x) + q_n(x) w(x) = 0, \quad q_n(x) := 1 - \frac{n^2 - \frac{1}{4}}{x^2}, \quad x > 0.$$

Proof. Since $x > 0$, the correspondence $R \leftrightarrow w = \sqrt{x} R$ is a bijection between C^2 functions on $(0, +\infty)$, with inverse $R = x^{-1/2} w$. Differentiating:

$$R = x^{-1/2} w, \quad R' = -\frac{1}{2} x^{-3/2} w + x^{-1/2} w', \quad R'' = \frac{3}{4} x^{-5/2} w - x^{-3/2} w' + x^{-1/2} w''.$$

(The second derivative is obtained by differentiating again: $\frac{d}{dx}(-\frac{1}{2} x^{-3/2} w) = \frac{3}{4} x^{-5/2} w - \frac{1}{2} x^{-3/2} w'$ and $\frac{d}{dx}(x^{-1/2} w') = -\frac{1}{2} x^{-3/2} w' + x^{-1/2} w''$; adding gives the result.) Substituting into Bessel's equation:

$$x^2 R'' + x R' + (x^2 - n^2) R = \frac{3}{4} x^{-1/2} w - x^{1/2} w' + x^{3/2} w'' - \frac{1}{2} x^{-1/2} w + x^{1/2} w' + (x^2 - n^2) x^{-1/2} w.$$

The two terms in w' cancel; collecting terms,

$$= x^{3/2} w'' + \left(\frac{3}{4} - \frac{1}{2}\right) x^{-1/2} w + x^{3/2} w - n^2 x^{-1/2} w = x^{3/2} \left[w'' + w + \left(\frac{1}{4} - n^2\right) x^{-2} w \right] = x^{3/2} [w'' + q_n w].$$

Since $x^{3/2} \neq 0$, the vanishing of the left-hand side is equivalent to $w'' + q_n w = 0$. $\square$

Lemma 18.9: Sturm comparison.

- (i) Let $m > 0$, $a < b$ with $b - a \geq \pi/m$, and let q be continuous on $[a, b]$ with $q(x) \geq m^2$ for every $x \in [a, b]$. Then every nontrivial solution $w \in C^2([a, b])$ of $w'' + qw = 0$ has at least one zero in the interval $[a, a + \pi/m]$.
- (ii) Let $M > 0$, $\alpha < \beta$, and let q be continuous on $[\alpha, \beta]$ with $q(x) \leq M^2$. If $w \in C^2([\alpha, \beta])$ solves $w'' + qw = 0$, vanishes at α and β , and does not vanish on (α, β) , then $\beta - \alpha \geq \pi/M$.

Proof. Part (i). Suppose towards a contradiction that w has no zeros in $[a, a + \pi/m]$. Since w is continuous, it has constant sign there; replacing w with $-w$ if necessary (which solves the same equation), we may assume $w > 0$ on $[a, a + \pi/m]$.

Step 1 — the comparison function. Let

$$v(x) := \sin(m(x - a)),$$

which solves $v'' + m^2 v = 0$, vanishes at $x = a$ and at $x = a + \pi/m$, and is ≥ 0 on $[a, a + \pi/m]$. Moreover $v'(a) = m > 0$ and $v'(a + \pi/m) = m \cos \pi = -m < 0$.

Step 2 — the Wronskian increases. Let $W := wv' - w'v$. Then

$$W' = wv'' - w''v = w(-m^2 v) - (-qw)v = (q - m^2) w v \geq 0 \quad \text{on } [a, a + \pi/m],$$

because $q \geq m^2$, $w > 0$ and $v \geq 0$. Hence W is nondecreasing on $[a, a + \pi/m]$.

Step 3 — contradiction. At the endpoints, using $v(a) = v(a + \pi/m) = 0$:

$$W(a) = w(a) v'(a) = m w(a) > 0, \quad W\left(a + \frac{\pi}{m}\right) = w\left(a + \frac{\pi}{m}\right) v'\left(a + \frac{\pi}{m}\right) = -m w\left(a + \frac{\pi}{m}\right) < 0.$$

But W is nondecreasing, so $W(a + \pi/m) \geq W(a) > 0$: a contradiction. Hence w has a zero in $[a, a + \pi/m]$.

Part (ii). Suppose towards a contradiction that $\beta - \alpha < \pi/M$. Since w does not vanish on (α, β) , it has constant sign there; replacing w with $-w$ if necessary, assume $w > 0$ on (α, β) . Let

$$v(x) := \sin(M(x - \alpha)),$$

which solves $v'' + M^2v = 0$, vanishes at α and is strictly positive on $(\alpha, \alpha + \pi/M) \supseteq (\alpha, \beta]$, because $\beta < \alpha + \pi/M$. Setting $W := wv' - w'v$, one has

$$W' = wv'' - w''v = w(-M^2v) - (-qw)v = (q - M^2)wv \leq 0 \quad \text{on } [\alpha, \beta],$$

because $q \leq M^2$, $w \geq 0$ and $v \geq 0$ there: hence W is nonincreasing. Now $W(\alpha) = w(\alpha)v'(\alpha) - w'(\alpha)v(\alpha) = 0$, because $w(\alpha) = 0$ and $v(\alpha) = 0$. On the other hand $w(\beta) = 0$ gives $W(\beta) = -w'(\beta)v(\beta)$, with $v(\beta) > 0$ and $w'(\beta) < 0$ — indeed $w > 0$ immediately to the left of β and $w(\beta) = 0$ force $w'(\beta) \leq 0$, and $w'(\beta) = 0$ would give $w \equiv 0$ by uniqueness for the Cauchy problem. Hence $W(\beta) > 0 = W(\alpha)$, contradicting monotonicity: a contradiction. $\square$

Theorem 18.10: The zeros of J_n : existence, infinitude, simplicity. Let $n \geq 0$ be an integer. Then:

- (i) setting $x_n := \max\{\delta_n, \sqrt{2 \max\{n^2 - \frac{1}{4}, 0\}}\}$, every interval $[\alpha, \alpha + \pi\sqrt{2}]$ with $\alpha \geq x_n$ contains at least one zero of J_n ;
- (ii) J_n has infinitely many zeros in $(0, +\infty)$, forming a strictly increasing sequence $0 < j_{n,1} < j_{n,2} < \dots \rightarrow +\infty$ with no accumulation points in $(0, +\infty)$;
- (iii) every positive zero is simple: if $J_n(\rho) = 0$ with $\rho > 0$, then $J'_n(\rho) \neq 0$; moreover $J'_n(\rho) = -J_{n+1}(\rho)$;
- (iv) for $n \geq 1$, two consecutive positive zeros are at distance at least π : $j_{n,k+1} - j_{n,k} \geq \pi$ for every $k \geq 1$.

Proof. Step 1 — (i). Let $w(x) = \sqrt{x} J_n(x)$: by Lemma 18.8 and Theorem 18.9, w solves $w'' + q_n w = 0$ with $q_n(x) = 1 - (n^2 - \frac{1}{4})x^{-2}$. For $x \geq x_n$:

- if $n = 0$, then $n^2 - \frac{1}{4} = -\frac{1}{4} < 0$ and $q_0(x) = 1 + \frac{1}{4x^2} \geq 1 > \frac{1}{2}$;
- if $n \geq 1$, then $n^2 - \frac{1}{4} > 0$ and $x^2 \geq 2(n^2 - \frac{1}{4})$ gives $\frac{n^2 - \frac{1}{4}}{x^2} \leq \frac{1}{2}$, hence $q_n(x) \geq \frac{1}{2}$.

In both cases $q_n \geq m^2$ with $m = 1/\sqrt{2}$, and $\pi/m = \pi\sqrt{2}$. Moreover $w \not\equiv 0$ because $J_n > 0$ on $(0, \delta_n]$ (Proposition 18.10). Lemma 18.9(i), applied on $[\alpha, \alpha + \pi\sqrt{2}]$, gives a zero of w , hence of J_n (since $\sqrt{x} \neq 0$).

Step 2 — (iii). Let $\rho > 0$ with $J_n(\rho) = 0$ and suppose towards a contradiction that $J'_n(\rho) = 0$. On the open set $(0, +\infty)$, Bessel's equation, divided by $x^2 \neq 0$, is a second-order linear equation with continuous coefficients; by the existence-and-uniqueness theorem for the Cauchy problem, the only solution with data $y(\rho) = y'(\rho) = 0$ is the one identically zero. Hence $J_n \equiv 0$ on $(0, +\infty)$, contradicting $J_n > 0$ on $(0, \delta_n]$. The formula $J'_n(\rho) = -J_{n+1}(\rho)$ follows from Proposition 18.11(iii) with $J_n(\rho) = 0$.

Step 3 — (ii). The positive zeros have no accumulation points in $(0, +\infty)$: if distinct zeros $\rho_k \rightarrow \rho > 0$, then by continuity $J_n(\rho) = 0$ and, by Rolle's theorem applied between ρ_k and ρ_{k+1} , there are points σ_k between them with $J'_n(\sigma_k) = 0$ and $\sigma_k \rightarrow \rho$; by continuity $J'_n(\rho) = 0$, contradicting Step 2. Moreover there are no zeros in $(0, \delta_n]$. It follows that the set of positive zeros is discrete and closed in $(0, +\infty)$, with a positive minimum in every bounded subset: it may therefore be enumerated in increasing order, $j_{n,1} < j_{n,2} < \dots$. By Step 1 there is at least one in $[x_n + k\pi\sqrt{2}, x_n + (k+1)\pi\sqrt{2}]$ for every $k \geq 0$: they are therefore infinitely many and the sequence tends to $+\infty$.

Step 4 — (iv). Let $n \geq 1$, so that $n^2 - \frac{1}{4} > 0$ and $q_n(x) = 1 - (n^2 - \frac{1}{4})x^{-2} \leq 1 = M^2$ for every $x > 0$. Let $j_{n,k} < j_{n,k+1}$ be two consecutive zeros: the function $w = \sqrt{x} J_n$ vanishes at both and does not vanish between them. By Lemma 18.9(ii) applied on $[j_{n,k}, j_{n,k+1}]$ with $M = 1$ one gets $j_{n,k+1} - j_{n,k} \geq \pi$. $\square$

Remark 18.10: The two spacing estimates, and why $n = 0$ is an exception. Combining (i) and (iv) of Theorem 18.10, for $n \geq 1$ and large enough zeros the distance between two consecutive zeros lies between π and $\pi\sqrt{2} = 4.4428829$. For $n = 0$ only the upper estimate holds: indeed $q_0(x) = 1 + \frac{1}{4x^2} > 1$, and part (ii) of Lemma 18.9 does not apply with $M = 1$. And indeed the lower estimate π is false for $n = 0$: $j_{0,2} - j_{0,1} = 3.1152526 < \pi = 3.1415927$ (Observation 18.13). The sign of $n^2 - \frac{1}{4}$ decides which way the inequality goes; it is the same quantity that appears in the Liouville form of Lemma 18.8.

Proposition 18.13: A rigorous enclosure of the first zero of J_0 . The first positive zero of J_0 satisfies

$$2.40 < j_{0,1} < 2.41.$$

Proof. Step 1 — the series is alternating with decreasing terms. For $0 < x \leq 2.41$ set $t_k := \frac{(x/2)^{2k}}{(k!)^2} > 0$, so that $J_0(x) = \sum_{k \geq 0} (-1)^k t_k$. The ratio $\frac{t_{k+1}}{t_k} = \frac{(x/2)^2}{(k+1)^2}$ is $\leq \frac{(1.205)^2}{4} < 1$ for every $k \geq 1$: hence, from $k = 1$ on, the terms strictly decrease and tend to 0. By Leibniz's criterion, truncating the series after the term of index 7 the error is bounded in absolute value by the first omitted term, t_8 :

$$\left| J_0(x) - \sum_{k=0}^7 (-1)^k t_k \right| \leq t_8.$$

Step 2 — the sign at $x = 2.40 = \frac{12}{5}$. Computing the partial sum in exact rational arithmetic (Python script with the `Fraction` class, executed) one gets

$$\sum_{k=0}^7 (-1)^k t_k = 0.002507672\dots, \quad t_8 = 1.137 \cdot 10^{-8},$$

whence

$$J_0(2.40) \in [0.0025076607, 0.0025076835],$$

in particular $J_0(2.40) > 0$.

Step 3 — the sign at $x = 2.41 = \frac{241}{100}$. In the same way

$$\sum_{k=0}^7 (-1)^k t_k = -0.002683412\dots, \quad t_8 = 1.215 \cdot 10^{-8}, \quad J_0(2.41) \in [-0.0026834250, -0.0026834007],$$

in particular $J_0(2.41) < 0$.

Step 4 — the intermediate value theorem. J_0 is continuous (Theorem 18.8) and changes sign between 2.40 and 2.41: hence there is $\rho \in (2.40; 2.41)$ with $J_0(\rho) = 0$.

Step 5 — it is the first zero. We show that $J_0 > 0$ on $(0; 2.41]$, so that $\rho = j_{0,1}$. By Proposition 18.11(iii), $J'_0 = -J_1$; it therefore suffices to prove that $J_1 > 0$ on $(0; 2.41]$, because then J_0 is strictly decreasing there and, starting from $J_0(0) = 1$, cannot vanish before ρ . Now

$$J_1(x) = \frac{x}{2} \sum_{k \geq 0} \frac{(-1)^k}{k!(k+1)!} \left(\frac{x}{2}\right)^{2k},$$

an alternating series whose terms $s_k = \frac{(x/2)^{2k}}{k!(k+1)!}$ have ratio $\frac{s_{k+1}}{s_k} = \frac{(x/2)^2}{(k+1)(k+2)} \leq \frac{(1.205)^2}{2} = 0.7260125 < 1$ for every $k \geq 0$ and $0 < x \leq 2.41$: the sequence is thus strictly decreasing from $k = 0$ on, and by Leibniz's criterion the sum lies between the first and second partial sums, in particular

$$J_1(x) \geq \frac{x}{2}(s_0 - s_1) = \frac{x}{2} \left(1 - \frac{(x/2)^2}{2} \right) \geq \frac{x}{2}(1 - 0.7260125) = 0.2739875 \cdot \frac{x}{2} > 0.$$

Hence J_0 is strictly decreasing on $(0; 2.41]$ and ρ is its first zero. $\square$

Theorem 18.11: Orthogonality and norm of the radial eigenfunctions. Let $n \geq 0$ be an integer and let $j_{n,k}$ be the positive zeros of J_n . Then, for $k \neq l$,

$$\int_0^1 J_n(j_{n,k} r) J_n(j_{n,l} r) r dr = 0,$$

and for every $k \geq 1$

$$\int_0^1 J_n(j_{n,k} r)^2 r dr = \frac{1}{2} J_n'(j_{n,k})^2 = \frac{1}{2} J_{n+1}(j_{n,k})^2.$$

Proof. For $\alpha > 0$ set $y_\alpha(r) := J_n(\alpha r)$. From Theorem 18.9 with $x = \alpha r$ it follows (see Step 4 of the proof of Lemma 18.7, read backwards)

$$r^2 y_\alpha'' + r y_\alpha' + (\alpha^2 r^2 - n^2) y_\alpha = 0,$$

which, divided by $r > 0$, may be written in self-adjoint form as

$$(r y_\alpha')' + \left(\alpha^2 r - \frac{n^2}{r} \right) y_\alpha = 0, \quad r > 0. \quad (\text{SL})$$

Step 1 — orthogonality. Let $\alpha = j_{n,k}$, $\beta = j_{n,l}$ with $k \neq l$, so $\alpha \neq \beta$ and $\alpha^2 \neq \beta^2$. Multiply (SL) by y_β , the analogous equation for y_β by y_α , and subtract:

$$y_\beta (r y_\alpha')' - y_\alpha (r y_\beta')' + (\alpha^2 - \beta^2) r y_\alpha y_\beta = 0.$$

The first group is the derivative of $r(y_\alpha' y_\beta - y_\alpha y_\beta')$: indeed

$$\frac{d}{dr} [r(y_\alpha' y_\beta - y_\alpha y_\beta')] = (r y_\alpha')' y_\beta + r y_\alpha' y_\beta' - r y_\alpha' y_\beta' - (r y_\beta')' y_\alpha.$$

Integrating on $[\varepsilon, 1]$ with $0 < \varepsilon < 1$,

$$\left[r(y_\alpha' y_\beta - y_\alpha y_\beta') \right]_\varepsilon^1 + (\alpha^2 - \beta^2) \int_\varepsilon^1 r y_\alpha y_\beta dr = 0.$$

At $r = 1$ one has $y_\alpha(1) = J_n(\alpha) = 0$ and $y_\beta(1) = J_n(\beta) = 0$, so the boundary term vanishes. At $r = \varepsilon$ the term is $\varepsilon(y_\alpha' y_\beta - y_\alpha y_\beta')(\varepsilon)$, which tends to 0 as $\varepsilon \rightarrow 0^+$ because $y_\alpha, y_\beta, y_\alpha', y_\beta'$ are continuous on $[0, 1]$ (Theorem 18.8) and hence bounded, while the factor ε tends to 0. Passing to the limit, $(\alpha^2 - \beta^2) \int_0^1 r y_\alpha y_\beta dr = 0$, and since $\alpha^2 \neq \beta^2$ the integral is zero.

Step 2 — the norm. Let $\alpha = j_{n,k}$ and $y = y_\alpha$. Multiply (SL) by $2ry'$:

$$2 (ry') (ry')' + 2(\alpha^2 r^2 - n^2) y y' = 0,$$

having written $2ry' \cdot (\alpha^2 r - \frac{n^2}{r})y = 2(\alpha^2 r^2 - n^2)yy'$. Now $2(ry')(ry')' = \frac{d}{dr}[(ry')^2]$ and $2yy' = \frac{d}{dr}[y^2]$, hence

$$\frac{d}{dr}[(ry')^2] + (\alpha^2 r^2 - n^2) \frac{d}{dr}[y^2] = 0.$$

Integrate on $[\varepsilon, 1]$ and integrate the second term by parts:

$$[(ry')^2]_{\varepsilon}^1 + [(\alpha^2 r^2 - n^2)y^2]_{\varepsilon}^1 - \int_{\varepsilon}^1 2\alpha^2 r y^2 dr = 0.$$

Let $\varepsilon \rightarrow 0^+$. The contributions at $r = \varepsilon$ vanish: $(\varepsilon y'(\varepsilon))^2 \rightarrow 0$ because y' is bounded near 0, and $(\alpha^2 \varepsilon^2 - n^2)y(\varepsilon)^2 \rightarrow -n^2 y(0)^2 = 0$, since $y(0) = J_n(0)$ equals 0 for $n \geq 1$ and the factor n^2 is zero for $n = 0$. At $r = 1$ one has $y(1) = J_n(\alpha) = 0$, so the second boundary term is zero, while the first equals $y'(1)^2 = (\alpha J'_n(\alpha))^2$. What remains is

$$\alpha^2 J'_n(\alpha)^2 = 2\alpha^2 \int_0^1 r y^2 dr, \quad \text{i.e.} \quad \int_0^1 J_n(\alpha r)^2 r dr = \frac{1}{2} J'_n(\alpha)^2.$$

Finally $J'_n(\alpha) = -J_{n+1}(\alpha)$ by Theorem 18.10(iii), whence $J'_n(\alpha)^2 = J_{n+1}(\alpha)^2$. $\square$

Remark 18.11: Comparison with the DLMF formulation and numerical check.

Theorem 18.11 is, for $\nu = n$ an integer, equation 10.22.37 of the NIST Digital Library of Mathematical Functions, which states it for $\nu > -1$ in the form $\int_0^1 t J_{\nu}(j_{\nu,\ell} t) J_{\nu}(j_{\nu,m} t) dt = \frac{1}{2} (J'_{\nu}(j_{\nu,\ell}))^2 \delta_{\ell,m}$; the proof given here is self-contained.

Numerical check (Python script executed). With Simpson's rule on 20 000 subintervals of $[0, 1]$:

	$n = 0$	$n = 1$
$\int_0^1 J_n(j_{n,1} r) J_n(j_{n,2} r) r dr$	$2.0 \cdot 10^{-16}$	$-1.4 \cdot 10^{-16}$
$\int_0^1 J_n(j_{n,1} r)^2 r dr$	0.1347570620	0.0811075654
$\frac{1}{2} J_{n+1}(j_{n,1})^2$	0.1347570620	0.0811075654

The cross integrals are zero to machine precision and the norms agree with the theoretical value to ten decimal places. ✓

18.8. Vibration modes, Bessel zeros, and frequencies

All the pieces are ready: separation has reduced the problem to Bessel's equation, regularity at the origin has selected J_n , and the boundary condition $u = 0$ on ∂D_a will now select the admissible values of λ . The result is the list of the drum's normal modes, and with it the list of its frequencies.

Theorem 18.12: The normal modes of the circular membrane. Let $a > 0$, $c > 0$, $n \geq 0$ an integer, $k \geq 1$, and set

$$\alpha_{n,k} := \frac{j_{n,k}}{a}, \quad \lambda_{n,k} := \alpha_{n,k}^2.$$

Then:

- (i) the function $V(r, \theta) = J_n(\alpha_{n,k} r) \cos n\theta$ (and likewise with $\sin n\theta$) extends to a function of class C^∞ on all of $\mathbb{R}^2$;
- (ii) $\Delta V + \lambda_{n,k} V = 0$ on $\mathbb{R}^2$;
- (iii) $V = 0$ on ∂D_a ; and conversely, if $\alpha > 0$ is such that $J_n(\alpha a) \neq 0$, the corresponding V does not vanish identically on ∂D_a : the only admissible values of α are therefore $\alpha_{n,k}$,

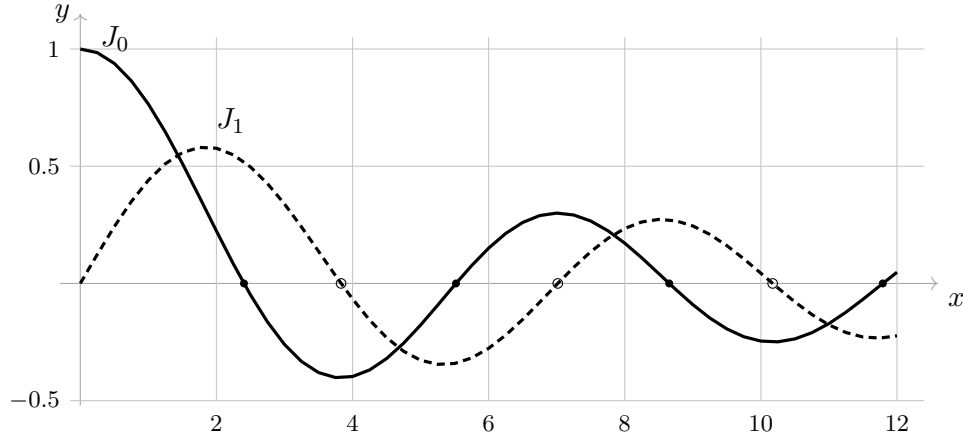


Figure 18.3: The functions J_0 (solid) and J_1 (dashed) on $[0, 12]$, sampled with step 0.25 from the series of Definition 18.7 and checked point by point against an independent multiple-precision implementation (the digits agree to eight decimal places throughout). The filled disks on the axis are the first four zeros of J_0 (2.404826; 5.520078; 8.653728; 11.791534), the open circles the first three of J_1 (3.831706; 7.015587; 10.173468). The existence of infinitely many zeros, all simple, is Theorem 18.10; the interlacing $j_{0,1} < j_{1,1} < j_{0,2} < \dots$ and the spacing approaching π are visible in the picture but are not proved here: see Observation 18.13.

$$k \geq 1;$$

(iv) for every $A, B, C, D \in \mathbb{R}$ the function

$$u(r, \theta, t) = J_n(\alpha_{n,k} r) (A \cos n\theta + B \sin n\theta) (C \cos(c \alpha_{n,k} t) + D \sin(c \alpha_{n,k} t))$$

solves the membrane problem of Definition 18.5, and is periodic in t with period $2\pi/(c\alpha_{n,k})$.

Proof. Write $\alpha = \alpha_{n,k}$ for brevity.

Step 1 — (i): the structure of the series. Factoring out $(x/2)^n$ in Definition 18.7,

$$J_n(x) = \left(\frac{x}{2}\right)^n G_n(x^2), \quad G_n(\omega) := \sum_{k \geq 0} \frac{(-1)^k}{4^k k! (n+k)!} \omega^k,$$

and the series defining G_n has infinite radius of convergence (same ratio test as in Step 1 of the proof of Theorem 18.8): G_n is therefore a C^∞ function on all of $\mathbb{R}$. It follows, with $x = \alpha r$ and $r^2 = x_1^2 + x_2^2$ (writing (x_1, x_2) for the point of the plane),

$$J_n(\alpha r) \cos n\theta = \left(\frac{\alpha}{2}\right)^n \underbrace{r^n \cos n\theta}_{=\operatorname{Re}(z^n)} G_n(\alpha^2(x_1^2 + x_2^2)),$$

where $z = x_1 + ix_2$. Now $\operatorname{Re}(z^n)$ is a homogeneous polynomial of degree n in (x_1, x_2) , and $G_n(\alpha^2(x_1^2 + x_2^2))$ is the composition of a C^∞ function with a polynomial: the product is C^∞ on $\mathbb{R}^2$ and agrees with $J_n(\alpha r) \cos n\theta$ for $r > 0$. The same holds with $\operatorname{Im}(z^n) = r^n \sin n\theta$.

Step 2 — (ii) away from the origin. For $r > 0$, with $R(r) = J_n(\alpha r)$ and $\Theta(\theta) = \cos n\theta$, Theorem 18.1 gives

$$\Delta V = \left(R'' + \frac{1}{r}R' - \frac{n^2}{r^2}R\right)\Theta = \frac{1}{r^2}(r^2R'' + rR' - n^2R)\Theta.$$

Now $R'(r) = \alpha J'_n(\alpha r)$ and $R''(r) = \alpha^2 J''_n(\alpha r)$, so, setting $x = \alpha r$,

$$r^2 R'' + r R' - n^2 R = x^2 J''_n(x) + x J'_n(x) - n^2 J_n(x) \stackrel{\text{Thm. 18.9}}{=} -x^2 J_n(x) = -\alpha^2 r^2 J_n(\alpha r).$$

Hence $\Delta V = -\alpha^2 R\Theta = -\alpha^2 V$ for every $r > 0$.

Step 3 — (ii) at the origin. By Step 1 the function V is C^∞ on $\mathbb{R}^2$, so ΔV and $-\alpha^2 V$ are both continuous on $\mathbb{R}^2$; by Step 2 they agree on $\mathbb{R}^2 \setminus \{(0, 0)\}$, which is dense in $\mathbb{R}^2$. Two continuous functions that agree on a dense set agree everywhere: hence $\Delta V + \alpha^2 V = 0$ at the origin as well.

Step 4 — (iii). On ∂D_a one has $r = a$, hence $V = J_n(\alpha a) \cos n\theta = J_n(j_{n,k}) \cos n\theta = 0$, because $j_{n,k}$ is a zero of J_n . Conversely, if $J_n(\alpha a) \neq 0$ then $V(a, \theta) = J_n(\alpha a) \cos n\theta$ is not identically zero (it vanishes only at the $2n$ points where $\cos n\theta = 0$); hence the admissible values of $\alpha > 0$ are exactly those with $\alpha a \in \{j_{n,1}, j_{n,2}, \dots\}$, the set of positive zeros of J_n , which Theorem 18.10 describes.

Step 5 — (iv). Let $T(t) = C \cos(cat) + D \sin(cat)$; then $T'' = -c^2 \alpha^2 T$. With $u = T \cdot V$ (where V is the combination with A, B , still C^∞ and a solution of $\Delta V = -\alpha^2 V$ by linearity),

$$u_{tt} = T''V = -c^2 \alpha^2 TV = c^2 T(\Delta V) = c^2 \Delta u.$$

The boundary condition holds by Step 4, and $u \in C^\infty(\mathbb{R}^2 \times \mathbb{R})$ by Step 1. Finally T has period $2\pi/(c\alpha)$, hence so does u . $\square$

Definition 18.8: Eigenvalues, eigenfunctions, and frequencies of the drum. With the notation of Theorem 18.12, the number $\lambda_{n,k} = (j_{n,k}/a)^2$ is called the **eigenvalue** of the Laplacian on the disc D_a with Dirichlet condition, and V the corresponding **eigenfunction**. The **angular frequency** and the **frequency** of the mode (n, k) are

$$\omega_{n,k} = c \frac{j_{n,k}}{a}, \quad f_{n,k} = \frac{\omega_{n,k}}{2\pi} = \frac{c j_{n,k}}{2\pi a} \quad [\text{Hz}].$$

The **nodal lines** of the mode are the set of points of $\overline{D_a}$ where the eigenfunction vanishes: the $k - 1$ circles $r = a j_{n,l}/j_{n,k}$ with $1 \leq l \leq k - 1$ and, for $n \geq 1$, the n diameters on which the angular factor vanishes.

Remark 18.12: Where the numbers come from: what the DLMF really contains.

Here it must be stated precisely what is imported and what is computed.

What the DLMF contains. The NIST Digital Library of Mathematical Functions, Chapter 10, §10.21, fixes the notation $j_{\nu,m}$ for the m -th positive zero of J_ν , proves its distribution and interlacing properties (§10.21(i)), and gives McMahon's asymptotic expansion for large zeros (equation 10.21.19). §10.21 does not contain a numerical table of the zeros: the tables are catalogued in §10.75, subsection "Real Zeros", which lists — among others — the British Association for the Advancement of Science (1937), $j_{0,m}$ and $j_{1,m}$ for $m = 1, \dots, 150$ with 10 decimal digits; Olver (1960), $j_{n,m}$ for $n = 0(\frac{1}{2})20$, $m = 1, \dots, 50$, with 8 digits; Abramowitz and Stegun (1964), $j_{n,m}$ for $n = 0(1)8$, $m = 1(1)20$, with 5 digits (10 for $n = 0$); Heller (1976), $j_{0,m}$ and $j_{1,m}$ for $m = 1(1)100$ with 25 digits; Wills et al. (1982), $j_{0,m}$ and $j_{1,m}$ for $m = 1(1)30$ with 35 digits. All of these details were verified online on 6 September 2026 (see the Sources environment).

What we do here. We do not report figures read from a table we have not seen: the values of Example 18.5 are recomputed by two independent methods and compared with the DLMF's asymptotic expansion. The only imported element is that expansion (Observation 18.13), which we use only as a third check and never as a definition.

Remark 18.13: McMahon's expansion: what we import. Imported, not proved. The NIST DLMF, equation 10.21.19, states that for fixed $\nu \geq 0$ and $m \rightarrow \infty$

$$j_{\nu,m} \sim a - \frac{\mu - 1}{8a} - \frac{4(\mu - 1)(7\mu - 31)}{3(8a)^3} - \frac{32(\mu - 1)(83\mu^2 - 982\mu + 3779)}{15(8a)^5} - \dots,$$

where $\mu = 4\nu^2$ and $a = (m + \frac{\nu}{2} - \frac{1}{4})\pi$. From it, in particular, it follows that the distance between two consecutive zeros tends to π .

What is missing to prove it here. One would need the asymptotic expansion of $J_\nu(x)$ as $x \rightarrow +\infty$ (DLMF, §10.17), obtained either from the Hankel integral representation via the saddle-point method, or from the Liouville form (Lemma 18.8) treated with a perturbative analysis to all orders: neither tool is available in this treatise. What is proved here is only the coarse estimate of Theorem 18.10(i), namely that two consecutive zeros are at distance at most $\pi\sqrt{2} = 4.4428829$; the true value of the asymptotic distance, π , remains imported. Numerical check of the only consequence we shall use (Python script executed, zeros of J_0 computed with multiple precision):

k	1	2	3	4
$j_{0,k+1} - j_{0,k}$	3.1152526	3.1336498	3.1378065	3.1393833
k	5	6	7	π
$j_{0,k+1} - j_{0,k}$	3.1401463	3.1405727	3.1408349	3.1415927

The sequence increases towards π , as the expansion predicts, and always stays below the proved threshold $\pi\sqrt{2}$. ✓

Example 18.5: The first eight Bessel zeros, computed three times. The following values were obtained by three independent routes (Python script executed):

- **A** — a sign scan with step 10^{-3} , 200 bisections and 60 Newton iterations, evaluating J_n and J'_n from the series of Definition 18.7 and the recurrence of Proposition 18.11(iii), in double precision;
- **B** — a multiple-precision arithmetic library with 30 significant digits, using a completely different algorithm;
- **C** — McMahon's expansion of Observation 18.13, truncated after the fourth term.

zero	A	B	C (McMahon)
$j_{0,1}$	2.4048255577	2.4048255577	2.4064624434
$j_{1,1}$	3.8317059702	3.8317059702	3.8316382972
$j_{2,1}$	5.1356223018	5.1356223018	5.1358891046
$j_{0,2}$	5.5200781103	5.5200781103	5.5200867370
$j_{3,1}$	6.3801618959	6.3801618959	6.3828008747
$j_{1,2}$	7.0155866698	7.0155866698	7.0155851349
$j_{2,2}$	8.4172441404	8.4172441404	8.4172441404
$j_{0,3}$	8.6537279129	8.6537279129	8.6537283466

Columns A and B agree to every digit reported: the largest discrepancy between the two is $5.6 \cdot 10^{-13}$ on $j_{2,3}$, and on $j_{0,1}$ it is exactly 0. ✓ Column C, being only an asymptotic approximation, deviates as expected the more, the smaller k and the larger n : $1.6 \cdot 10^{-3}$ on $j_{0,1}$, $2.6 \cdot 10^{-3}$ on $j_{3,1}$, but only $4.3 \cdot 10^{-7}$ on $j_{0,3}$. ✓

Consistency with what was proved. The value $j_{0,1} = 2.4048255577$ falls in the interval $(2.40; 2.41)$ of Proposition 18.13, the only numerical statement of the chapter that is *proved* and not merely computed. ✓ Moreover $j_{0,2} - j_{0,1} = 3.1152526 < \pi\sqrt{2} = 4.4428829$, as required by Theorem 18.10(i). ✓

Example 18.6: The frequencies of a drum, and why they are out of tune. Consider a circular membrane of radius $a = 0.25$ m over which transverse waves travel at $c = 150$ m/s.

By Definition 18.8, $f_{n,k} = \frac{c j_{n,k}}{2\pi a} = \frac{150 j_{n,k}}{2\pi \cdot 0.25} = \frac{300}{\pi} j_{n,k}$.

mode (n, k)	$j_{n,k}$	$f_{n,k}$ [Hz]	$f_{n,k}/f_{0,1}$
(0, 1)	2.404826	229.6439	1.000000
(1, 1)	3.831706	365.9010	1.593341
(2, 1)	5.135622	490.4158	2.135549
(0, 2)	5.520078	527.1286	2.295417
(3, 1)	6.380162	609.2606	2.653066
(1, 2)	7.015587	669.9392	2.917295
(2, 2)	8.417244	803.7876	3.500147
(0, 3)	8.653728	826.3701	3.598485

Check 1: closed form. $\frac{300}{\pi} = 95.4929659$, and $95.4929659 \times 2.4048255577 = 229.6439$. ✓

Check 2: independent script. The same value computed in multiple-precision arithmetic as $c j_{0,1}/(2\pi a)$ gives 229.6439 Hz, and recomputed in double precision with the constant $j_{0,1} = 2.4048255576957728$ again gives 229.6439 Hz: the three routes agree to the fourth decimal place. ✓

Reading. The ratios in the last column are neither integers nor simple fractions: the drum does not produce a harmonic series. This is the fundamental acoustic difference between the circular membrane and the vibrating string, whose modes have frequencies $1, 2, 3, \dots$ times the fundamental — and on the circle, where the eigenvalues are n^2 , the ratio $\sqrt{n^2} = n$ is an integer, and indeed Observation 18.6 showed that waves on $\mathbb{T}$ are periodic in time.

Remark 18.14: Why a drum is not periodic. A finite superposition of modes $\sum_j A_j \cos(\omega_j t + \varphi_j)$ is periodic in t if and only if the ratios ω_i/ω_j are all rational. For the membrane $\omega_{n,k}/\omega_{0,1} = j_{n,k}/j_{0,1}$: periodicity would require all of these ratios to be rational. Numerically they do not appear to be at all (see Example 18.6); that they indeed are not — i.e. the irrationality of the quotients of Bessel zeros — is an arithmetic result that we do not tackle here and do not use: what we need is only the experimental observation that the sound of a drum has no definite pitch, and the structural explanation is already in the table of ratios.

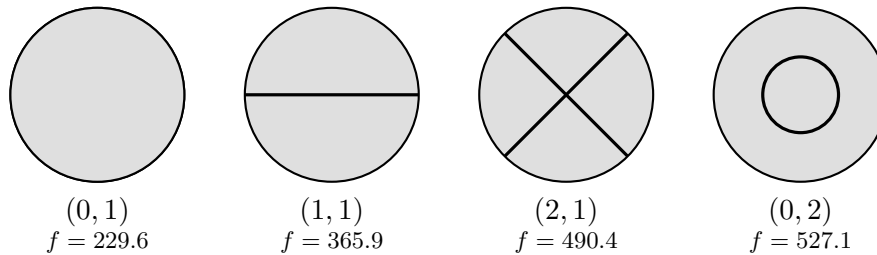


Figure 18.4: The nodal lines of the first four modes of the circular membrane, with the frequencies in hertz from Example 18.6. Mode (0,1) has no interior nodal lines; (1,1) has one nodal diameter, (2,1) two diameters (at 45° and 135° : the drawn vertices are $(\pm 0.813173, \pm 0.813173)$, i.e. $1.15 \cos 45^\circ = 0.8131728$ for the unit radius rescaled to 1.15 cm), and (0,2) a nodal circle of radius $a j_{0,1}/j_{0,2} = 0.4356510 a$ (here $0.4356510 \times 1.15 = 0.5009987$ cm). All coordinates were computed by hand and checked numerically.

18.9. The heat equation in the disc

The same modes that make the membrane vibrate describe the cooling of a disc held at zero temperature on the boundary: only the time factor changes, from oscillating to exponentially decaying. Comparing the two cases is the lesson that closes the chapter.

Theorem 18.13: Heat modes in the disc and their decay. Let $a > 0$, $\kappa > 0$, and let $(n_1, k_1), \dots, (n_N, k_N)$ be pairs of indices with $n_j \geq 0$ an integer and $k_j \geq 1$. For constants $A_j, B_j \in \mathbb{R}$ set

$$u(r, \theta, t) = \sum_{j=1}^N e^{-\kappa \lambda_{n_j, k_j} t} J_{n_j}(\alpha_{n_j, k_j} r) (A_j \cos n_j \theta + B_j \sin n_j \theta), \quad \lambda_{n, k} = \left(\frac{j_{n, k}}{a} \right)^2.$$

Then $u \in C^\infty(\mathbb{R}^2 \times \mathbb{R})$, $u_t = \kappa \Delta u$, and $u = 0$ on $\partial D_a \times \mathbb{R}$. Moreover, setting

$$\lambda_{\min} := \min_{1 \leq j \leq N} \lambda_{n_j, k_j} > 0, \quad M := \max_{1 \leq j \leq N} \max_{0 \leq x \leq j_{n_j, k_j}} |J_{n_j}(x)| < \infty,$$

the following estimate holds for every $t \geq 0$:

$$\max_{\overline{D_a}} |u(\cdot, \cdot, t)| \leq M e^{-\kappa \lambda_{\min} t} \sum_{j=1}^N (|A_j| + |B_j|).$$

Proof. Step 1 — regularity and the boundary condition. Each term is the product of $e^{-\kappa \lambda t}$, a C^∞ function of t , with the eigenfunction V_j of Theorem 18.12(i), which is C^∞ on $\mathbb{R}^2$: the finite sum is therefore C^∞ . On ∂D_a every V_j vanishes by Theorem 18.12(iii), hence $u = 0$ there.

Step 2 — the equation. For each term,

$$\partial_t (e^{-\kappa \lambda_j t} V_j) = -\kappa \lambda_j e^{-\kappa \lambda_j t} V_j, \quad \kappa \Delta (e^{-\kappa \lambda_j t} V_j) = \kappa e^{-\kappa \lambda_j t} \Delta V_j = -\kappa \lambda_j e^{-\kappa \lambda_j t} V_j,$$

where $\Delta V_j = -\lambda_j V_j$ was used (Theorem 18.12(ii)). The two expressions agree; summing over j gives $u_t = \kappa \Delta u$.

Step 3 — the two constants are finite and positive. The number $\lambda_{\min}$ is the minimum of a finite set of numbers $\lambda_{n, k} = (j_{n, k}/a)^2$, all strictly positive because $j_{n, k} > 0$: hence $\lambda_{\min} > 0$. The number M is the maximum of a finite set of maxima of continuous functions (Theorem 18.8) on compact intervals: hence $M < \infty$, and $M \geq |J_{n_1}(0)| \geq 0$.

Step 4 — the estimate. For (r, θ) with $0 \leq r \leq a$ one has $0 \leq \alpha_{n_j, k_j} r \leq \alpha_{n_j, k_j} a = j_{n_j, k_j}$, hence $|J_{n_j}(\alpha_{n_j, k_j} r)| \leq M$ by definition of M . Using in addition $|\cos n\theta| \leq 1$, $|\sin n\theta| \leq 1$ and, for $t \geq 0$, $e^{-\kappa \lambda_j t} \leq e^{-\kappa \lambda_{\min} t}$, one gets

$$|u(r, \theta, t)| \leq \sum_{j=1}^N e^{-\kappa \lambda_j t} M (|A_j| + |B_j|) \leq M e^{-\kappa \lambda_{\min} t} \sum_{j=1}^N (|A_j| + |B_j|),$$

which is the estimate in the statement: the solution decays exponentially at rate $\kappa \lambda_{\min}$, i.e. at the rate of the *slowest mode* present in the sum. $\square$

Proposition 18.14: The fundamental mode governs the cooling. Let $u(r, \theta, t) = e^{-\kappa \lambda_{0,1} t} J_0(j_{0,1} r/a)$ be the fundamental mode of Theorem 18.13 in the disc of radius a , with $\lambda_{0,1} = j_{0,1}^2/a^2$. Then

$$u(0, \theta, t) = e^{-\kappa j_{0,1}^2 t/a^2},$$

and the time needed for the amplitude to halve is

$$t_{1/2} = \frac{a^2 \ln 2}{\kappa j_{0,1}^2}.$$

Moreover mode $(1, 1)$ decays faster, in the ratio $\lambda_{1,1}/\lambda_{0,1} = (j_{1,1}/j_{0,1})^2$.

Proof. At the centre $r = 0$ one has $J_0(0) = 1$ by Proposition 18.10, hence $u(0, \theta, t) = e^{-\kappa\lambda_{0,1}t}$. The equation $e^{-\kappa\lambda_{0,1}t} = \frac{1}{2}$ gives $\kappa\lambda_{0,1}t = \ln 2$, i.e. $t = \ln 2/(\kappa\lambda_{0,1}) = a^2 \ln 2/(\kappa j_{0,1}^2)$. The ratio of the rates is $\lambda_{1,1}/\lambda_{0,1} = (j_{1,1}/a)^2/(j_{0,1}/a)^2 = (j_{1,1}/j_{0,1})^2$, where a cancels. $\square$

Example 18.7: How fast the disc cools. Let $a = 1$ and $\kappa = 1$. Then

$$\lambda_{0,1} = j_{0,1}^2 = 2.4048256^2 = 5.7831860, \quad \lambda_{1,1} = j_{1,1}^2 = 3.8317060^2 = 14.6819706.$$

Check 1: computation by hand. 2.4048256^2 : one has $2.4^2 = 5.76$ and, by the expansion $(2.4 + \varepsilon)^2 = 5.76 + 4.8\varepsilon + \varepsilon^2$ with $\varepsilon = 0.0048256$, one gets $5.76 + 0.0231629 + 0.0000233 = 5.7831862$, which agrees with the value reported to within $2 \cdot 10^{-7}$ (the discrepancy is due to rounding $j_{0,1}$ to seven decimal places). $\checkmark$

Check 2: Python script executed, with $j_{0,1}$ computed in multiple precision: $\lambda_{0,1} = 5.7831860$ and $\lambda_{1,1} = 14.6819706$, with ratio $\lambda_{1,1}/\lambda_{0,1} = 2.5387340$. $\checkmark$

Half-life. By Proposition 18.14, $t_{1/2} = \ln 2/\lambda_{0,1} = 0.6931472/5.7831860 = 0.1198556$. At time $t = 0.1$ the amplitude of the fundamental mode has fallen to $e^{-0.5783186} = 0.5608406$, that of mode $(1, 1)$ to $e^{-1.4681971} = 0.2303404$: after one tenth of a time unit the first mode retains 56% of its initial amplitude, the second only 23%. $\checkmark$

Reading. This is the same phenomenon seen on the circle in Proposition 18.5: modes with more nodal lines oscillate faster in space, have larger eigenvalue, and are damped out sooner. Asymptotically only the fundamental mode survives, and in the disc with boundary held at zero temperature it too decays — unlike on the circle, where mode $n = 0$ did not decay at all, because there was no boundary condition and the total energy was conserved.

Theorem 18.14: Parabolic maximum principle. Let $\Omega \subset \mathbb{R}^2$ be open, bounded, and nonempty, $T > 0$ and $\kappa > 0$. Let u be continuous on $\overline{\Omega} \times [0, T]$, with u_t and the second spatial derivatives continuous on $\Omega \times (0, T]$, and $u_t = \kappa\Delta u$ in $\Omega \times (0, T]$. Setting

$$\Gamma := (\overline{\Omega} \times \{0\}) \cup (\partial\Omega \times [0, T]) \quad (\textit{parabolic boundary}),$$

one has

$$\max_{\overline{\Omega} \times [0, T]} u = \max_{\Gamma} u, \quad \min_{\overline{\Omega} \times [0, T]} u = \min_{\Gamma} u.$$

Proof. Step 1 — the perturbation. Let $\varepsilon > 0$ and $w(x, t) := u(x, t) - \varepsilon t$. Then w is continuous on $\overline{\Omega} \times [0, T]$ and, in $\Omega \times (0, T]$,

$$w_t - \kappa\Delta w = u_t - \varepsilon - \kappa\Delta u = -\varepsilon < 0.$$

Step 2 — where the maximum of w lies. The set $\overline{\Omega} \times [0, T]$ is compact (a product of compact sets) and w is continuous on it, so it attains a maximum at some point (x_1, t_1) . Suppose towards a contradiction that $(x_1, t_1) \notin \Gamma$, i.e. $x_1 \in \Omega$ and $t_1 > 0$. Then:

- the function $x \mapsto w(x, t_1)$, defined and of class C^2 in a neighbourhood of x_1 contained in Ω , has a local maximum at x_1 ; by the second-derivative test applied to the two coordinate directions, $\partial_x^2 w(x_1, t_1) \leq 0$ and $\partial_y^2 w(x_1, t_1) \leq 0$, hence $\Delta w(x_1, t_1) \leq 0$;

- the function $t \mapsto w(x_1, t)$ has its maximum on $[0, T]$ at $t = t_1 \in (0, T]$: if $t_1 < T$ the maximum is interior and $w_t(x_1, t_1) = 0$; if $t_1 = T$ the maximum is at the right endpoint, hence $w_t(x_1, T) \geq 0$ (if $w_t(x_1, T) < 0$ held, w would be strictly decreasing on a left neighbourhood of T and would take values larger than $w(x_1, T)$). In both cases $w_t(x_1, t_1) \geq 0$.

It follows that $w_t - \kappa \Delta w \geq 0$ at (x_1, t_1) , contradicting Step 1. Hence $(x_1, t_1) \in \Gamma$ and $\max_{\overline{\Omega} \times [0, T]} w = \max_{\Gamma} w$.

Step 3 — the limit in ε . For every $(x, t) \in \overline{\Omega} \times [0, T]$,

$$u(x, t) = w(x, t) + \varepsilon t \leq \max_{\Gamma} w + \varepsilon T \leq \max_{\Gamma} u + \varepsilon T,$$

since $w \leq u$. Letting $\varepsilon \rightarrow 0^+$ gives $\max_{\overline{\Omega} \times [0, T]} u \leq \max_{\Gamma} u$; the reverse inequality is obvious because $\Gamma \subseteq \overline{\Omega} \times [0, T]$. For the minimum apply the result to $-u$, which satisfies the same equation. $\square$

Corollary 18.4: Uniqueness and stability for heat in the disc. Let u_1, u_2 be as in Theorem 18.14 on $\Omega = D_a$, with the same values on the parabolic boundary Γ . Then $u_1 = u_2$ on $\overline{D_a} \times [0, T]$. More generally,

$$\max_{\overline{D_a} \times [0, T]} |u_1 - u_2| = \max_{\Gamma} |u_1 - u_2|.$$

Proof. Let $w = u_1 - u_2$: by linearity w satisfies the same hypotheses and the same equation. By Theorem 18.14, $\max w = \max_{\Gamma} w$ and $\min w = \min_{\Gamma} w$; since for a real-valued function $\max |w| = \max\{\max w, -\min w\}$, one obtains the equality of the maxima of the absolute values. If $w = 0$ on Γ , then $\max |w| = 0$ and $w \equiv 0$. $\square$

Remark 18.15: The parabolic boundary is not the topological boundary. The boundary of $\overline{D_a} \times [0, T]$ as a subset of $\mathbb{R}^3$ also includes the top face $\overline{D_a} \times \{T\}$; the parabolic boundary Γ excludes it. This is the geometric translation of irreversibility (Observation 18.5): the data at the final time is not data of the problem, it is a result. The proof of Theorem 18.14 shows exactly where the point lies: in the case $t_1 = T$ one cannot conclude $w_t = 0$, only $w_t \geq 0$, and the inequality still goes the right way. For the wave equation nothing of the kind holds: d'Alembert's formula (Theorem 18.7) allows one to reconstruct the past from the future, and no maximum principle exists.

18.10. Panorama: spectral completeness

Remark 18.16: What we import: completeness of the Bessel–Fourier system. Imported, not proved, and never used as a hypothesis in the proofs of this chapter.

Everything we have proved about the disc concerns single modes or finite sums of modes: Theorem 18.12 constructs the eigenfunctions, Theorem 18.11 proves their orthogonality, Theorem 18.13 solves the heat problem for a finite combination. What is missing is the step that in the case of the circle was provided by Theorem 17.1: that the system of eigenfunctions is complete, i.e. that every reasonable initial datum expands in a series of eigenfunctions. On the disc the corresponding statement is the following.

Panorama (Fourier–Bessel expansion). Let $\nu \geq -\frac{1}{2}$ and let $f : (0, 1) \rightarrow \mathbb{R}$ with $\int_0^1 t^{1/2} |f(t)| dt < \infty$ (NIST DLMF, eq. 10.23.18). Setting

$$a_m = \frac{2}{J_{\nu+1}(j_{\nu,m})^2} \int_0^1 t f(t) J_\nu(j_{\nu,m}t) dt \quad (\text{DLMF, eq. 10.23.19}),$$

if f is of bounded variation on an interval $[a, b]$ with $0 < a < x < b < 1$, then

$$\frac{1}{2}f(x^-) + \frac{1}{2}f(x^+) = \sum_{m=1}^{\infty} a_m J_\nu(j_{\nu,m}x) \quad (\text{DLMF, eq. 10.23.20}).$$

The DLMF refers, for the proof, to Watson, *A Treatise on the Theory of Bessel Functions*, Chapter 18. Combining this radial expansion with the angular Fourier series of Chapter 17 gives the completeness of the system $\{J_n(j_{n,k}r/a)e^{in\theta}\}_{n \in \mathbb{Z}, k \geq 1}$ in $L^2(D_a)$, and with it the solution of the heat and membrane problems for general initial data.

What exactly is missing to prove it here. Two ingredients, both beyond the reach of this treatise: (a) the asymptotic behaviour of $J_\nu(x)$ as $x \rightarrow +\infty$, with a uniform error term, needed to compare the Fourier–Bessel kernel with the Dirichlet kernel of Chapter 17; (b) the spectral theory of self-adjoint operators with compact resolvent, which gives completeness abstractly. What remains proved in the chapter is everything that does not depend on these two points; and no earlier proof has used them.

Methodological note. The statement “the eigenvalues of the Laplacian form a complete basis” is therefore, in this treatise, an **imported panorama** and not a theorem: it does not appear in any **teorema** environment, and whenever it is needed it is cited explicitly as such.

INFO

The map of the three equations on the disc. We summarise what each equation does to the modes, and from which theorem of this chapter the conclusion follows.

	Laplace	heat	waves
equation	$\Delta u = 0$	$u_t = \kappa \Delta u$	$u_{tt} = c^2 \Delta u$
factor on mode λ	$r^{ \lambda }$	$e^{-\kappa \lambda t}$	$\cos(c\sqrt{\lambda}t)$
regularises?	—	yes, instantly	no
conserves energy?	—	no	yes
reversible?	—	no	yes
maximum principle	Thm. 18.4	Thm. 18.14	does not hold
uniqueness	Cor. 18.3	Cor. 18.4	Thm. 18.7, Step 5

On the circle $\mathbb{T}$ the modes are $e^{in\theta}$ and the eigenvalues n^2 ; in the disc D_a the modes are $J_n(j_{n,k}r/a)e^{in\theta}$ and the eigenvalues $(j_{n,k}/a)^2$. The whole difference between the string and the drum lies in the fact that $\sqrt{n^2} = n$ is an integer, whereas the ratios $j_{n,k}/j_{0,1}$ are neither integers nor simple fractions (Example 18.6).

18.11. Exercises inherited from Volume II

The exercises that follow come from Volume II, Chapter 4, and concern the topics of this chapter. They are reported in full, with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 18.1. (Application) — Using the model of the heat equation on $\mathbb{T}$, explain qualitatively why, as $t \rightarrow \infty$, $u(\theta, t) \rightarrow \hat{u}_0(0)$ (the initial mean temperature), independently of the initial distribution.

Exercise 18.2: S8 — The heat equation damps high frequencies. Consider the heat equation on the circle $u_t = \kappa u_{\theta\theta}$ with initial datum $u(0, \theta) = f(\theta)$. Show, by expanding in Fourier series, that

$$\hat{u}(t, n) = e^{-\kappa n^2 t} \hat{f}(n),$$

and deduce that for every $t > 0$ the solution is of class C^∞ regardless of the regularity of f . Compare with the Poisson kernel, where the damping is $r^{|n|}$ instead of $e^{-\kappa n^2 t}$.

Solutions

Solution. In the solution $u(\theta, t) = \sum_n \hat{u}_0(n) e^{-n^2 t} e^{in\theta}$, every term with $n \neq 0$ contains the factor $e^{-n^2 t}$, which tends to 0 as $t \rightarrow \infty$ — and tends to 0 the faster, the larger $|n|$ is (the high-frequency components are damped first). The only term that does not decay is the one with $n = 0$, for which $e^{-0 \cdot t} = 1$ for every t . Hence, uniformly in θ ,

$$u(\theta, t) \xrightarrow{t \rightarrow \infty} \hat{u}_0(0) = \frac{1}{2\pi} \int_0^{2\pi} u_0(\theta) d\theta,$$

the mean (average temperature) of the initial datum u_0 , independently of its specific shape. Physically: the heat redistributes itself around the ring until thermal equilibrium is reached, and by conservation of the total energy the equilibrium value can only be the mean of the initial distribution.

Note added in Volume III. The qualitative argument is made quantitative by Proposition 18.5, which establishes the estimate $\|u(\cdot, t) - \hat{u}_0(0)\|_2 \leq e^{-\kappa t} \|u_0 - \hat{u}_0(0)\|_2$: the speed of convergence to equilibrium is governed by the first nonconstant mode, $n = \pm 1$.

Solution. S8. *Step 1 — separation of variables in the spectrum.* Expand $u(t, \cdot)$ in Fourier series for each fixed t : $u(t, \theta) = \sum_n \hat{u}(t, n) e^{in\theta}$. Since $\widehat{u_{\theta\theta}}(t, n) = (in)^2 \hat{u}(t, n) = -n^2 \hat{u}(t, n)$ (by Exercise S2, Volume II, Chapter 4, applied twice), the equation $u_t = \kappa u_{\theta\theta}$ becomes, coefficient by coefficient,

$$\frac{\partial}{\partial t} \hat{u}(t, n) = -\kappa n^2 \hat{u}(t, n).$$

Step 2 — solving. This is a first-order linear ordinary differential equation for each n , with initial datum $\hat{u}(0, n) = \hat{f}(n)$:

$$\hat{u}(t, n) = e^{-\kappa n^2 t} \hat{f}(n).$$

Step 3 — instantaneous regularity. Fix $t > 0$. Since $|\hat{f}(n)| \leq \|f\|_1$ for every n (immediate from the definition), for every $k \geq 0$

$$\sum_{n \in \mathbb{Z}} |n|^k |\hat{u}(t, n)| \leq \|f\|_1 \sum_{n \in \mathbb{Z}} |n|^k e^{-\kappa n^2 t} < \infty,$$

because the Gaussian $e^{-\kappa n^2 t}$ dominates any power. By Theorem 17.5(i) applied to the term-by-term differentiated series, $u(t, \cdot) \in C^k(\mathbb{T})$ for every k , i.e. $u(t, \cdot) \in C^\infty(\mathbb{T})$.

Step 4 — comparison with the Poisson kernel. The two damping factors are

$$\text{heat: } e^{-\kappa n^2 t} \quad \text{versus} \quad \text{Poisson: } r^{|n|} = e^{-|n| \ln(1/r)}.$$

Both decay faster than any power of $1/|n|$ and therefore produce C^∞ solutions; but the Gaussian decay of heat is *qualitatively stronger* than the exponential decay of Poisson. The difference is reflected in the equations: the heat equation is parabolic (it regularises instantaneously and irreversibly), while Laplace's equation — solved by the Poisson kernel — is elliptic.

Physical remark. The damping $e^{-\kappa n^2 t}$ says that high-frequency components die out *much* faster than low ones: an irregular temperature profile becomes smooth almost instantaneously, and asymptotically only the mode $n = 0$ survives, i.e. the mean temperature. This is the precise sense in which heat "forgets" the initial datum.

Note added in Volume III. Steps 1–3 are proved here in complete form in Theorem 18.5, which adds the justification of term-by-term differentiation and the two ways of understanding the initial datum (uniform and in L^2); the comparison of Step 4 with the Poisson kernel is taken up again, on the Laplace side, in Theorem 18.3 and in Corollary 18.2.

18.12. Assigned exercises

These are *new* exercises, distinct both from the inherited ones (Section 18.11) and from the worked ones in Section 18.13. The answers are in the Teacher's Key (Section 18.15).

Exercise 18.3: Laplace in the annulus: where regularity at the origin is not needed. *Goal:* to understand what changes when the domain does not contain the centre, and to handle the general solution of Euler's equation. *Estimated time:* 40 minutes. *Type:* proof-based and numerical.

- Verify with Theorem 18.1 that $r^{-n} \cos n\theta$ (for $n \geq 1$) and $\ln r$ are harmonic on $\mathbb{R}^2 \setminus \{(0,0)\}$, and show that neither extends continuously to the origin.
- Solve the Dirichlet problem in the annulus $1 < r < 2$ with datum 0 on the inner circle and 1 on the outer one, seeking a radial solution (Proposition 18.1). Compute the value of the solution at $r = 1.25$, $r = 1.5$ and $r = 1.75$ and check it by an independent route.
- Determine all harmonic functions in the annulus $1 < r < 2$ of the form $R(r) \cos 2\theta$, and among these the one equal to 0 on $r = 1$ and to $\cos 2\theta$ on $r = 2$.
- Explain why in the annulus Theorem 18.2 eliminates nothing, and how many constants remain free in each mode before imposing the two boundary conditions.

Exercise 18.4: Heat and waves on the same datum. *Goal:* to see in an explicit case the difference between dissipation and conservation. *Estimated time:* 45 minutes. *Type:* symbolic and numerical computation. Take $f(\theta) = \cos \theta + \frac{1}{3} \cos 3\theta$.

- Solve the heat equation on $\mathbb{T}$ with $\kappa = 1$ and initial datum f , and compute $u(0; 0.2)$, $u(0; 1)$ and $u(0.5; 0.2)$ to seven decimal places, checking the values with a second method.
- Solve the wave equation on $\mathbb{T}$ with $c = 1$, $F = f$ and $G \equiv 0$, both with d'Alembert's formula (Theorem 18.7) and with the series (Proposition 18.6), and verify that the two expressions agree at three points of your choice.
- Compute the energy of the solution of (b) with Proposition 18.7 and check that it is constant in time.
- Show that for the heat equation the function $t \mapsto \|u(\cdot, t)\|_2$ is strictly decreasing unless f is constant. (Hint: revisit Step 2 of the proof of Theorem 18.6.)

Exercise 18.5: Bessel: recurrences, zeros, and a drum. *Goal:* to use the recurrence formulas, apply Sturm comparison, and translate zeros into frequencies. *Estimated time:* 50 minutes. *Type:* proof-based and numerical.

- Show, directly from the series, that $\frac{d}{dx}(x^{-1}J_1(x)) = -x^{-1}J_2(x)$, and verify it numerically at $x = 1$ with centred finite differences.
- Compute the threshold x_2 of Theorem 18.10(i) for $n = 2$ and deduce that every interval of length $\pi\sqrt{2}$ contained in $[x_2, +\infty)$ contains a zero of J_2 . Compare the estimate with the true values $j_{2,1}$ and $j_{2,2}$.
- A drumhead of radius $a = 0.30$ m with wave speed $c = 120$ m/s: compute $f_{0,1}$, $f_{1,1}$ and $f_{2,1}$ in hertz and their ratios; compare with the first three harmonics of a string with the same fundamental.
- Compute $\int_0^1 J_0(j_{0,2}r)^2 r dr$ with Theorem 18.11 and check it with a numerical quadrature.

18.13. Worked exercises

Exercise 18.6: The Dirichlet problem with datum $f(\theta) = \theta^2$. *Type:* symbolic computation with a double numerical check. Solve the Dirichlet problem in the unit disc with boundary datum $f(\theta) = \theta^2$ for $\theta \in [-\pi, \pi]$, extended periodically, and compute the values of the solution at $(r, \theta) = (\frac{1}{2}, 0)$, $(\frac{1}{2}, \frac{\pi}{2})$ and $(0.8; 1)$.

Solution. Strategy. Theorem 18.3 reduces everything to computing the Fourier coefficients of the datum: the solution is the same series with $r^{|n|}$ inserted. The check is done by a completely different route, the Poisson integral of Corollary 18.2.

Step 1 — coefficients of the datum. The function $\theta \mapsto \theta^2$ is even, so it has only cosines. One has $\frac{a_0}{2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3}$; and for $n \geq 1$, integrating by parts twice,

$$a_n = \frac{2}{\pi} \int_0^{\pi} \theta^2 \cos n\theta d\theta = \frac{2}{\pi} \left[\frac{\theta^2 \sin n\theta}{n} + \frac{2\theta \cos n\theta}{n^2} - \frac{2 \sin n\theta}{n^3} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{2\pi(-1)^n}{n^2} = \frac{4(-1)^n}{n^2},$$

since $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$. Hence

$$\theta^2 = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} \cos n\theta.$$

Step 2 — the solution. Since $\sum_n |a_n| = 4 \sum n^{-2} < \infty$, Theorem 18.3 gives

$$u(r, \theta) = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} r^n \cos n\theta, \quad 0 \leq r < 1.$$

Step 3 — the three values. Summing the series (Python script executed, 4000 terms):

$$u\left(\frac{1}{2}, 0\right) = 1.49621131, \quad u\left(\frac{1}{2}, \frac{\pi}{2}\right) = 3.05396784, \quad u(0.8; 1) = 1.45473498.$$

Independent check 1: the Poisson integral. Evaluating

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 - r^2}{1 - 2r \cos(\theta - \varphi) + r^2} \varphi^2 d\varphi$$

with the trapezoidal rule on 400 000 subintervals gives 1.49621131, 3.05396784 and 1.45473498: the same eight digits. ✓

Independent check 2: closed form at $\theta = 0$. At $\theta = 0$ the series becomes $\frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-r)^n}{n^2} = \frac{\pi^2}{3} + 4 \operatorname{Li}_2(-r)$, where Li_2 is the dilogarithm. With $\operatorname{Li}_2(-\frac{1}{2}) = -0.4484142069$ one gets $\frac{\pi^2}{3} + 4 \cdot (-0.4484142069) = 3.2898681337 - 1.7936568276 = 1.4962113061$, in agreement with Step 3 to nine decimal places. ✓

Independent check 3: the mean-value property. By Proposition 18.2, $u(0) = \hat{f}(0) = \frac{\pi^2}{3} = 3.2898681337$, which is exactly the mean of θ^2 on $[-\pi, \pi]$. ✓

Interpretation. The datum θ^2 is continuous on $\mathbb{T}$ (it equals π^2 at the two endpoints) but not differentiable at $\theta = \pm\pi$; nonetheless the solution is C^∞ throughout the open disc, by Theorem 18.3(i). The lack of regularity survives only on the boundary.

Common mistakes. (i) Forgetting that the constant coefficient is $\frac{a_0}{2} = \frac{\pi^2}{3}$ and not $a_0 = \frac{2\pi^2}{3}$. (ii) Writing r^n with negative n for the modes $e^{-in\theta}$: the correct formula has $r^{|n|}$, and it is precisely Theorem 18.2 that imposes it. (iii) Using the datum θ^2 on $[0, 2\pi]$ instead of on $[-\pi, \pi]$: the resulting periodic function is different (it has a jump at $\theta = 0$) and the coefficients change.

Extension. Repeat with $f(\theta) = |\theta|$ (Example 18.3) and compare the decay rate of the coefficients: n^{-2} in both cases, but with different parity of the modes present.

Exercise 18.7: When the second heat mode becomes negligible. *Type:* quantitative computation with an independent check. In the unit disc ($a = 1, \kappa = 1$) consider the initial datum $u_0(r) = J_0(j_{0,1}r) + J_0(j_{0,2}r)$. Determine the instant t^* at which the amplitude of the second mode has fallen to one hundredth of that of the first.

Solution. Strategy. The datum is already a *finite* combination of eigenfunctions: Theorem 18.13 applies directly, with no need for any series expansion and hence without using spectral completeness (Observation 18.16).

Step 1 — the solution. By Theorem 18.13 with $N = 2$, $n_1 = n_2 = 0$, $k_1 = 1$, $k_2 = 2$, $A_1 = A_2 = 1$, $B_1 = B_2 = 0$:

$$u(r, t) = e^{-\lambda_{0,1}t} J_0(j_{0,1}r) + e^{-\lambda_{0,2}t} J_0(j_{0,2}r), \quad \lambda_{0,k} = j_{0,k}^2.$$

The function vanishes at $r = 1$ because $J_0(j_{0,1}) = J_0(j_{0,2}) = 0$, and solves $u_t = \Delta u$.

Step 2 — the ratio of amplitudes. The ratio of the two time coefficients is

$$\frac{e^{-\lambda_{0,2}t}}{e^{-\lambda_{0,1}t}} = e^{-(\lambda_{0,2} - \lambda_{0,1})t}, \quad \lambda_{0,2} - \lambda_{0,1} = j_{0,2}^2 - j_{0,1}^2.$$

Step 3 — the numbers. With $j_{0,1} = 2.4048255577$ and $j_{0,2} = 5.5200781103$:

$$\lambda_{0,1} = 5.7831860, \quad \lambda_{0,2} = 30.4712623, \quad \lambda_{0,2} - \lambda_{0,1} = 24.6880764.$$

Step 4 — the instant sought. From $e^{-24.6880764t^*} = \frac{1}{100}$ it follows that

$$t^* = \frac{\ln 100}{24.6880764} = \frac{4.6051702}{24.6880764} = 0.1865342.$$

Independent check 1: back-substitution. Computing $e^{-24.6880764 \times 0.1865342}$ gives 0.0100000000. ✓

Independent check 2: difference of squares. One can compute $j_{0,2}^2 - j_{0,1}^2 = (j_{0,2} - j_{0,1})(j_{0,2} + j_{0,1}) = 3.1152526 \times 7.9249037 = 24.6880761$, which agrees with the value of Step 3 to within $3 \cdot 10^{-7}$ (rounding). ✓

Independent check 3: consistency with the half-life. By Proposition 18.14 the first mode halves in $t_{1/2} = \ln 2 / 5.7831860 = 0.1198556$; at $t^* = 0.1865342$ it has fallen to about

$e^{-5.7831860 \times 0.1865342} = 0.3400$ of its initial value, while the second has fallen to 0.0034: the ratio is $1/100$, as required. ✓

Interpretation. After barely one fifth of a time unit the temperature profile is already indistinguishable, to within 1%, from that of the fundamental mode alone $J_0(j_{0,1}r)$: this is why cooling problems are almost always treated with the first mode. How fast this happens depends on the *spacing* of the eigenvalues, i.e. on the distance between the Bessel zeros.

Common mistakes. (i) Using the zeros instead of their squares: the eigenvalues are $j_{0,k}^2$, not $j_{0,k}$. (ii) Forgetting the factor $1/a^2$ when the radius is not 1. (iii) Believing that the ratio $1/100$ is reached at $t = \ln 100/\lambda_{0,2}$: the comparison is between the *two* modes, so what enters is the *difference* of the eigenvalues.

Extension. Repeat with mode $(1, 1)$ in place of $(0, 2)$: the difference of the eigenvalues is $14.6819706 - 5.7831860 = 8.8987846$, and the time becomes about $4.6051702/8.8987846 = 0.5175$: more than double.

Exercise 18.8: The first zero of J_1 , enclosed using only the series. *Type:* rigorous estimate with an alternating series. Show that the first positive zero of J_1 satisfies $3.83 < j_{1,1} < 3.84$, using only the series of Definition 18.7 and Leibniz's criterion.

Solution. Strategy. This is the procedure of Proposition 18.13, carried over from J_0 to J_1 . The only extra care needed is that, for x near 3.84, the first two terms of the series *are not* decreasing: one must say from which index on they are.

Step 1 — the series and monotonicity of the terms. By Definition 18.7,

$$J_1(x) = \frac{x}{2} \sum_{k \geq 0} (-1)^k s_k, \quad s_k = \frac{(x/2)^{2k}}{k!(k+1)!}.$$

The ratio is $\frac{s_{k+1}}{s_k} = \frac{(x/2)^2}{(k+1)(k+2)}$. For $x \leq 3.84$ one has $(x/2)^2 \leq 1.92^2 = 3.6864$, hence

$$\frac{s_1}{s_0} = \frac{3.6864}{2} = 1.8432 > 1 \quad \text{but} \quad \frac{s_{k+1}}{s_k} \leq \frac{3.6864}{6} = 0.6144 < 1 \quad \text{for every } k \geq 1.$$

The terms therefore strictly decrease from $k = 1$ on, and Leibniz's criterion applies from there: truncating after index 12, the error is at most s_{13} .

Step 2 — the value at $x = 3.83 = \frac{383}{100}$. Computing $\sum_{k=0}^{12} (-1)^k s_k$ in exact rational arithmetic (Python script with the `Fraction` class, executed) and multiplying by $x/2 = 1.915$ gives

$$J_1(3.83) = 0.0006872482 \pm \varepsilon, \quad \varepsilon \leq \frac{x}{2} s_{13} \leq 1.915 \times 3.997 \cdot 10^{-14} < 10^{-13},$$

in particular $J_1(3.83) > 0$.

Step 3 — the value at $x = 3.84 = \frac{384}{100}$. In the same way

$$J_1(3.84) = -0.0033368526 \pm \varepsilon', \quad \varepsilon' \leq 1.92 \times 4.277 \cdot 10^{-14} < 10^{-13},$$

in particular $J_1(3.84) < 0$.

Step 4 — a zero exists between the two points. J_1 is continuous (Theorem 18.8) and changes sign between 3.83 and 3.84: by the intermediate value theorem there is $\rho \in (3.83; 3.84)$ with $J_1(\rho) = 0$.

Step 5 — ρ is the first positive zero. (a) *No zero up to 2.41.* This is already proved in Step 5 of Proposition 18.13: $J_1(x) \geq 0.2739875 \cdot \frac{x}{2} > 0$ for $0 < x \leq 2.41$.

(b) *No zero between 2.41 and 3.83.* Suppose towards a contradiction that J_1 has a zero in $(2.41; 3.83]$, and let σ be the *largest* zero of J_1 strictly smaller than ρ ; by (a) and by the contradictory hypothesis, $\sigma > 2.41$, and by construction σ and ρ are consecutive zeros. By Theorem 18.10(iv), applicable because $n = 1 \geq 1$,

$$\rho - \sigma \geq \pi = 3.1415927, \quad \text{hence} \quad \sigma \leq \rho - \pi < 3.84 - 3.1415927 = 0.6984073,$$

contradicting $\sigma > 2.41$. Hence J_1 has no zeros in $(0; 3.83]$ and $\rho = j_{1,1}$.

Independent check. The multiple-precision computation gives $j_{1,1} = 3.8317059702$, which falls in the interval found. ✓ The value agrees with that of column B of Example 18.5. ✓

Common mistakes. (i) Applying Leibniz's criterion without checking monotonicity: here $s_1 > s_0$, and truncating after the term $k = 0$ would give a false estimate. (ii) Forgetting the factor $x/2$ in front of the sum. (iii) Confusing $j_{1,1}$ with $j'_{0,1}$: by DLMF §10.21(i), $j'_{0,m} = j_{1,m-1}$ for $m \geq 2$, and $j'_{0,1} = 0$.

Extension. With the same method, enclose $j_{2,1}$ between 5.13 and 5.14 (the true value is 5.1356223018), taking care that for $n = 2$ the monotonicity of the terms begins later.

18.14. Self-assessment quiz

- (Definition) For a function $U(r, \theta)$ of class C^2 with $r > 0$, the Laplacian in polar coordinates equals
 A) $U_{rr} + U_r + U_{\theta\theta}$ B) $U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta}$ C) $U_{rr} + \frac{1}{r^2}U_r + \frac{1}{r}U_{\theta\theta}$ D) $U_{rr} + U_{\theta\theta}$
- (Hypothesis) In Lemma 18.3 the hypothesis that Θ be 2π -periodic serves to
 A) guarantee that Θ is of class C^2 B) rule out $\lambda < 0$ and force $\sqrt{\lambda}$ to be an integer
 C) make R bounded at the origin D) ensure that λ is real
- (Symbolic computation) The solution of the Dirichlet problem in the unit disc with boundary datum $f(\theta) = 1 + 2\cos\theta + 3\sin 4\theta$ is
 A) $1 + 2\cos\theta + 3\sin 4\theta$ B) $1 + 2r\cos\theta + 3r^4\sin 4\theta$ C) $r + 2r\cos\theta + 3r^4\sin 4\theta$ D)
 $1 + 2r^{-1}\cos\theta + 3r^{-4}\sin 4\theta$
- (Numerical computation) A circular membrane of radius $a = 0.25$ m over which the waves travel at $c = 150$ m/s has fundamental frequency, in hertz,
 A) 229.6439 B) 95.4930 C) 365.9010 D) 300.0000
- (Proof step) In the proof of Theorem 18.9, the identity $4k(n+k)a_k = -a_{k-1}$ serves to
 A) prove that the series converges B) cancel the contribution of x^2J_n against that of $x^2J_n'' + xJ_n' - n^2J_n$ C) compute $J_n(0)$ D) prove the orthogonality of the eigenfunctions
- (Error diagnosis) A student writes the solution of the Dirichlet problem in the unit disc as $\sum_{n \in \mathbb{Z}} \hat{f}(n)r^n e^{in\theta}$ and observes that the expression blows up near the origin. The error is
 A) the coefficients $\hat{f}(n)$ are computed incorrectly B) the correct exponent is $|n|$: the modes r^n with $n < 0$ are unbounded at the origin and must be discarded by Theorem 18.2 C) the series does not converge for any datum D) the factor $\frac{1}{2\pi}$ is missing
- (Transfer) In a disc of radius a with boundary held at zero temperature, if the radius is doubled while leaving the diffusivity unchanged, the half-life of the fundamental mode

- A) doubles B) quadruples C) halves D) does not change
8. (Source and metacognition) The numerical values of the zeros $j_{n,m}$ used in this chapter
- A) are read from Table 10.21.1 of the NIST DLMF B) are not found in §10.21 of the DLMF, which gives only properties and McMahon's expansion (eq. 10.21.19): the published tables are catalogued in §10.75, "Real Zeros", and here the values are recomputed twice C) are deduced from equation 10.2.2 of the DLMF D) are taken from Chapter 18 of Watson

18.15. Teacher's key

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Theorem 18.1; the coefficients $\frac{1}{r}$ and $\frac{1}{r^2}$ arise, respectively, from the term $-rU_r$ that appears in $U_{\theta\theta}$ (Step 2) and from the factor r^2 carried by the angular derivative. *Distractors:* A forgets both weights; C swaps them; D is the Cartesian Laplacian written with the wrong names. *Diagnosed error:* transporting a differential operator without redoing the chain rule. *Source:* Theorem 18.1 and Corollary 18.1. *Prerequisite:* the chain rule in two variables. *Remediation:* redo Step 2 of the proof and check the formula on $r^2 \cos 2\theta$ as in Example 18.1.

2) **B.** *Rationale:* in Lemma 18.3 the case $\lambda < 0$ is excluded because a periodic solution is bounded while $Ae^{\mu\theta} + Be^{-\mu\theta}$ is not, and in the case $\lambda > 0$ periodicity is equivalent to $\mathcal{R}_{2\pi\mu} = I$, i.e. $\mu \in \mathbb{Z}$. *Distractors:* A confuses a regularity hypothesis with a periodicity one; C concerns Theorem 18.2, which is a *different* filter acting on R ; D is automatic, because Θ is real-valued. *Diagnosed error:* failing to recognise that the topology of the domain acts as a boundary condition. *Source:* Lemma 18.3 and Observation 18.2. *Prerequisite:* second-order linear equations with constant coefficients. *Remediation:* repeat the case $\lambda > 0$ requiring periodicity $2\pi/3$ instead of 2π and see which μ survive.

3) **B.** *Rationale:* by Theorem 18.3 every mode $e^{in\theta}$ of the datum must be multiplied by $r^{|n|}$; here the modes present are $n = 0, \pm 1, \pm 4$, hence $r^0 = 1$, r^1 and r^4 . *Distractors:* A is the function constant in r , which is not harmonic (unless f is constant); C gets the order-0 term wrong; D uses $r^{-|n|}$, i.e. precisely the modes discarded by Theorem 18.2. *Diagnosed error:* confusing the boundary value with the solution in the interior. *Source:* Theorem 18.3; Example 18.2. *Prerequisite:* Fourier coefficients of a trigonometric polynomial. *Remediation:* redo Example 18.2 with $f = \cos^3 \theta$.

4) **A.** *Rationale:* by Definition 18.8, $f_{0,1} = \frac{cj_{0,1}}{2\pi a} = \frac{150 \times 2.4048256}{2\pi \times 0.25} = 229.6439$ Hz, as in Example 18.6. *Distractors:* B is the constant $\frac{c}{2\pi a} = \frac{300}{\pi} = 95.4930$, i.e. the result of forgetting to multiply by $j_{0,1}$; C is $f_{1,1}$, the frequency of the second mode; D is $\frac{c}{2a} = 300$, the string formula, which does not apply to the membrane. *Diagnosed error:* applying the vibrating-string formula to the membrane. *Source:* Example 18.6, with a double numerical check. *Prerequisite:* Theorem 18.12(iv). *Remediation:* redo the calculation with $a = 0.30$ m and $c = 120$ m/s (Assigned exercise 18.5(c)).

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 5–8).

5) **B.** *Rationale:* Step 1 of the proof produces $\sum_{k \geq 1} 4k(n+k)a_k x^{n+2k}$ and Step 2 produces $\sum_{k \geq 1} a_{k-1} x^{n+2k}$; the identity of Step 3 says that the two cancel term by term. *Distractors:* A is Theorem 18.8, proved earlier and with the ratio test; C is read directly off the term $k = 0$; D is Theorem 18.11, which uses the self-adjoint form and not the series. *Diagnosed*

error: failing to distinguish the role of the individual steps in a series-based verification. *Source:* Theorem 18.9, Steps 1–3. *Prerequisite:* term-by-term differentiation of a power series. *Remediation:* redo the computation for $n = 0$ only, writing out explicitly the first three coefficients a_0, a_1, a_2 .

6) B. *Rationale:* the correct series is $\sum_n \hat{f}(n)r^{|n|}e^{in\theta}$; it is Theorem 18.2 that imposes the exponent $|n|$, because $r^{-|n|}$ is unbounded at 0. *Distractors:* A and D are computational errors that would not produce that behaviour; C is false, the series with $r^{|n|}$ converges for every integrable f (Theorem 18.3(i)). *Diagnosed error:* forgetting the regularity condition at the origin, i.e. the second boundary condition, the one *not* written in the statement of the problem. *Source:* Theorem 18.2, Observation 18.1. *Prerequisite:* Lemma 18.4. *Remediation:* compute $\sum_{n<0} \hat{f}(n)r^n e^{in\theta}$ for $f = \cos \theta$ and see what happens as $r \rightarrow 0^+$.

7) B. *Rationale:* by Proposition 18.14, $t_{1/2} = \frac{a^2 \ln 2}{\kappa j_{0,1}^2}$, which depends on a through a^2 : doubling a multiplies the time by 4. *Distractors:* A would be correct if the dependence were linear, i.e. for a first-order problem in x ; C inverts the dependence; D forgets that the eigenvalue is $(j_{0,1}/a)^2$ and not $j_{0,1}^2$. *Diagnosed error:* failing to account for the rescaling of the eigenvalues with the size of the domain. *Source:* Definition 18.8 and Proposition 18.14; numerical check in Example 18.7. *Prerequisite:* Theorem 18.12. *Remediation:* compare the half-lives for $a = 1$ and $a = 2$ with $\kappa = 1$: 0.1198556 against 0.4794225.

8) B. *Rationale:* this is what is documented in Observation 18.12 and verified online on 6 September 2026: §10.21 of the DLMF contains distribution, analytic properties, monotonicity, inequalities and asymptotic expansions, but no numerical table; the published tables are listed in §10.75. *Distractors:* A cites a table that does not exist in the DLMF — and this is precisely the error this chapter avoids; C confuses the series that *defines* J_ν with its zeros; D is the source of the *proof* of completeness (Observation 18.16), not of the numerical values. *Diagnosed error:* citing a source by hearsay without opening it. *Source:* NIST DLMF, §10.21 and §10.75, verified on 6 September 2026. *Prerequisite:* none technical; this is a methodological item. *Remediation:* open the two pages and compare what they actually contain.

BEST PRACTICE

Assigned exercises — answers.

18.3 (annulus). (a) With $U = r^{-n} \cos n\theta$: $U_{rr} = n(n+1)r^{-n-2} \cos n\theta$, $\frac{1}{r}U_r = -nr^{-n-2} \cos n\theta$, $\frac{1}{r^2}U_{\theta\theta} = -n^2r^{-n-2} \cos n\theta$; the sum is $(n^2 + n - n - n^2)r^{-n-2} \cos n\theta = 0$. For $\ln r$: $(\ln r)'' + \frac{1}{r}(\ln r)' = -\frac{1}{r^2} + \frac{1}{r^2} = 0$. Neither extends: along $\theta = 0$, $r^{-n} \rightarrow +\infty$ and $\ln r \rightarrow -\infty$ as $r \rightarrow 0^+$. (b) $U = A + B \ln r$ with $U(1) = 0 \Rightarrow A = 0$ and $U(2) = 1 \Rightarrow B = 1/\ln 2$: hence $U(r) = \frac{\ln r}{\ln 2} = \log_2 r$. Values: $U(1.25) = 0.3219281$, $U(1.5) = 0.5849625$, $U(1.75) = 0.8073549$. *Independent check:* $2^{0.5849625} = 1.5000000$. ✓ (c) $R(r) = Ar^2 + Br^{-2}$ (Lemma 18.4 with $n = 2$); from $R(1) = 0$ it follows that $B = -A$, and from $R(2) = 1$ that $4A - \frac{A}{4} = 1$, i.e. $A = \frac{4}{15}$: hence $R(r) = \frac{4}{15}(r^2 - r^{-2})$ and $u = \frac{4}{15}(r^2 - r^{-2}) \cos 2\theta$. *Check:* $R(2) = \frac{4}{15}(4 - \frac{1}{4}) = \frac{4}{15} \cdot \frac{15}{4} = 1$. ✓ (d) Because the annulus does not contain the origin: boundedness at $r = 0$ is not a condition of the problem. Each mode retains the two constants A, B of Lemma 18.4, and they are determined by the *two* boundary conditions (inner and outer circle), not by a single one as in the disc. **18.4 (heat and waves).** (a) $u(\theta, t) = e^{-t} \cos \theta + \frac{1}{3}e^{-9t} \cos 3\theta$; $u(0; 0.2) = 0.8738304$, $u(0; 1) = 0.3679206$, $u(0.5; 0.2) = 0.7224014$. *Independent check for* $u(0; 1)$: $e^{-1} + \frac{1}{3}e^{-9} = 0.36787944 + 0.00004114 = 0.36792058$, i.e. 0.3679206 to seven decimal places. ✓ (b) $u(\theta, t) = \cos t \cos \theta + \frac{1}{3} \cos 3t \cos 3\theta$; with d'Alembert, $\frac{1}{2}(F(\theta+t) + F(\theta-t))$, and the product-to-sum identities give the same expression. (c) $F' = -\sin \theta - \sin 3\theta$, hence $E = \frac{1}{2} \int_{-\pi}^{\pi} (\sin \theta +$

$\sin 3\theta)^2 d\theta = \frac{1}{2}(\pi + \pi) = \pi = 3.1415927$, constant by Proposition 18.7. (d) $\frac{d}{dt}\|u(\cdot, t)\|_2^2 = -2\kappa \int_{-\pi}^{\pi} u_{\theta}^2 d\theta \leq 0$ (Step 2 of the proof of Theorem 18.6); equality holds only if $u_{\theta} \equiv 0$, i.e. if u is constant in θ , which by the series formula happens only if $\hat{f}(n) = 0$ for every $n \neq 0$, i.e. if f is constant.

18.5 (Bessel and drum). (a) This is Proposition 18.11(i) with $n = 1$; at $x = 1$ the value is $-J_2(1) = -0.1149035$, and centred finite differences with $h = 10^{-6}$ give -0.1149035 . ✓ (b) $x_2 = \sqrt{2(4 - \frac{1}{4})} = \sqrt{7.5} = 2.7386128$; every interval of length $\pi\sqrt{2} = 4.4428829$ contained in $[2.7386128, +\infty)$ contains a zero of J_2 . Indeed $j_{2,1} = 5.1356223 \in [2.7386128; 7.1814957]$ and $j_{2,2} = 8.4172441 \in [5.1356223; 9.5785052]$. ✓ Moreover $j_{2,2} - j_{2,1} = 3.2816218 \geq \pi$, as required by Theorem 18.10(iv). (c) $f_{n,k} = \frac{120 j_{n,k}}{2\pi \cdot 0.30} = \frac{200}{\pi} j_{n,k}$: $f_{0,1} = 153.0959$ Hz, $f_{1,1} = 243.9340$ Hz, $f_{2,1} = 326.9439$ Hz, with ratios 1, 1.593341, 2.135549. A string with the same fundamental would give 153.0959, 306.1918, 459.2877 Hz: the drum is markedly "flatter" on the second mode and "sharper" in an irregular way on the third. (d) By Theorem 18.11, $\int_0^1 J_0(j_{0,2}r)^2 r dr = \frac{1}{2}J_1(j_{0,2})^2 = 0.0578900693$; Simpson's quadrature on 20 000 subintervals gives 0.0578900693. ✓

18.16. Chapter formulary

Polar Laplacian and harmonic modes.

$$\Delta u = U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta} = \frac{1}{r}(rU_r)_r + \frac{1}{r^2}U_{\theta\theta}; \quad \det D\Phi = r, \quad dA = r dr d\theta.$$

Regular harmonic modes: $1, r^n \cos n\theta, r^n \sin n\theta$; discarded singular modes: $\ln r, r^{-n} \cos n\theta, r^{-n} \sin n\theta$. Radial harmonics: $A + B \ln r$. Euler's equation: $r^2 R'' + rR' - n^2 R = 0 \Rightarrow R = Ar^n + Br^{-n}$ ($n \geq 1$).

Dirichlet problem in the disc D_a .

$$u(r, \theta) = \sum_{n \in \mathbb{Z}} \hat{f}(n) \left(\frac{r}{a}\right)^{|n|} e^{in\theta} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{a^2 - r^2}{a^2 - 2ar \cos(\theta - \varphi) + r^2} f(\varphi) d\varphi.$$

Mean value: $\frac{1}{2\pi} \int_{-\pi}^{\pi} u(r, \theta) d\theta = \hat{f}(0) = u(0)$. Maximum: $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$; uniqueness and stability: $\max_{\overline{\Omega}} |u_1 - u_2| = \max_{\partial\Omega} |f_1 - f_2|$.

Heat and waves on the circle.

$$u_t = \kappa u_{\theta\theta} : \quad u(\theta, t) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{-\kappa n^2 t} e^{in\theta} = (f * H_t)(\theta), \quad H_t(\theta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{-\kappa n^2 t} e^{in\theta}.$$

$$\|u(\cdot, t) - \hat{f}(0)\|_2 \leq e^{-\kappa t} \|f - \hat{f}(0)\|_2; \quad \min f \leq u \leq \max f.$$

$$u_{tt} = c^2 u_{\theta\theta} : \quad u(\theta, t) = \frac{F(\theta + ct) + F(\theta - ct)}{2} + \frac{1}{2c} \int_{\theta-ct}^{\theta+ct} G,$$

$$u = \hat{F}(0) + \hat{G}(0)t + \sum_{n \neq 0} \left(\hat{F}(n) \cos cnt + \frac{\hat{G}(n)}{cn} \sin cnt \right) e^{in\theta}, \quad E = \frac{1}{2} \int_{-\pi}^{\pi} (u_t^2 + c^2 u_{\theta}^2) = \text{const.}$$

Bessel.

$$x^2 y'' + xy' + (x^2 - n^2)y = 0; \quad J_n(x) = \sum_{k \geq 0} \frac{(-1)^k}{k! (n+k)!} \left(\frac{x}{2}\right)^{n+2k}.$$

$$(x^{-n} J_n)' = -x^{-n} J_{n+1}, \quad (x^n J_n)' = x^n J_{n-1}, \quad J_{n-1} + J_{n+1} = \frac{2n}{x} J_n, \quad J_0' = -J_1.$$

$$J_n(x) \sim \frac{1}{n!} \left(\frac{x}{2}\right)^n \quad (x \rightarrow 0); \quad w = \sqrt{x} J_n \Rightarrow w'' + \left(1 - \frac{n^2 - \frac{1}{4}}{x^2}\right) w = 0.$$

$$\int_0^1 J_n(j_{n,k} r) J_n(j_{n,l} r) r dr = \frac{\delta_{kl}}{2} J_{n+1}(j_{n,k})^2; \quad 2.40 < j_{0,1} < 2.41; \quad j_{n,k+1} - j_{n,k} \leq \pi\sqrt{2} \quad (j_{n,k} \geq x_n),$$

Membrane and heat in the disc D_a ; the numerical values of the chapter.

$$\lambda_{n,k} = \left(\frac{j_{n,k}}{a}\right)^2, \quad V_{n,k} = J_n\left(\frac{j_{n,k}}{a} r\right) e^{in\theta}, \quad f_{n,k} = \frac{c j_{n,k}}{2\pi a}, \quad t_{1/2} = \frac{a^2 \ln 2}{\kappa j_{0,1}^2}.$$

(n, k)	(0, 1)	(1, 1)	(2, 1)	(0, 2)	(3, 1)	(1, 2)	(2, 2)	(0, 3)
$j_{n,k}$	2.404826	3.831706	5.135622	5.520078	6.380162	7.015587	8.417244	8.653728
$j_{n,k}/j_{0,1}$	1.000000	1.593341	2.135549	2.295417	2.653066	2.917295	3.500147	3.598485
$j_{0,1}^2 = 5.7831860, \quad j_{1,1}^2 = 14.6819706, \quad j_{0,2}^2 = 30.4712623; \quad \frac{1}{2} J_1(j_{0,1})^2 = 0.1347570620.$								

Sources and bibliography

- [1] NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/>, Chapter 10 "Bessel Functions", §10.2 "Definitions": equation 10.2.1 (Bessel's differential equation) and equation 10.2.2 (ascending series defining J_ν), source of Definition 18.6 and of Definition 18.7. Verified on 6 September 2026.
- [2] NIST Digital Library of Mathematical Functions, Chapter 10, §10.6 "Recurrence Relations and Derivatives": equations 10.6.1, 10.6.2 and 10.6.3 ($J_0' = -J_1$), compared with Proposition 18.11 (Observation 18.8). Verified on 6 September 2026.
- [3] NIST Digital Library of Mathematical Functions, Chapter 10, §10.7 "Limiting Forms": equations 10.7.1 ($J_0(z) \rightarrow 1$, $Y_0(z) \sim \frac{2}{\pi} \ln z$), 10.7.3 ($J_\nu(z) \sim (\frac{1}{2}z)^\nu / \Gamma(\nu+1)$) and 10.7.4 ($Y_\nu(z) \sim -\frac{1}{\pi} \Gamma(\nu) (\frac{1}{2}z)^{-\nu}$), cited in Observation 18.9 for the behaviour of the second solution. Verified on 6 September 2026.
- [4] NIST Digital Library of Mathematical Functions, Chapter 10, §10.21 "Zeros": §10.21(i) "Distribution" (notation $j_{\nu,m}$, simplicity and interlacing of the zeros, relation $j_{0,m}' = j_{1,m-1}$ for $m \geq 2$) and §10.21(vi) "McMahon's Asymptotic Expansions for Large Zeros", equation 10.21.19 with $\mu = 4\nu^2$ and $a = (m + \frac{\nu}{2} - \frac{1}{4})\pi$: this is the only *imported* part of the chapter, declared in Observation 18.13. **§10.21 does not contain any numerical table of zeros**: see Observation 18.12. Verified on 6 September 2026.
- [5] NIST Digital Library of Mathematical Functions, Chapter 10, §10.22 "Integrals": equation 10.22.37, orthogonality relation $\int_0^1 t J_\nu(j_{\nu,\ell} t) J_\nu(j_{\nu,m} t) dt = \frac{1}{2} (J_\nu'(j_{\nu,\ell}))^2 \delta_{\ell,m}$ for $\nu > -1$, compared with Theorem 18.11 (proved here independently for $\nu = n$ an integer). Verified on 6 September 2026.
- [6] NIST Digital Library of Mathematical Functions, Chapter 10, §10.23(iii) "Fourier–Bessel

Expansion": equations 10.23.18, 10.23.19 and 10.23.20 (hypothesis $\int_0^1 t^{1/2}|f(t)| dt < \infty$, coefficients a_m , convergence to the half-sum of the one-sided limits for f of bounded variation). This is the statement reported as an **imported panorama** in Observation 18.16. Verified on 6 September 2026.

- [7] NIST Digital Library of Mathematical Functions, Chapter 10, §10.75 "Tables", subsection "Real Zeros": catalogue of the published tables of the zeros $j_{n,m}$ — British Association for the Advancement of Science (1937), 10 decimal digits for $m \leq 150$; F. W. J. Olver (1960), 8 digits; M. Abramowitz and I. A. Stegun (1964), 5 digits (10 for $n = 0$); Heller (1976), 25 digits; Wills et al. (1982), 35 digits. Source of Observation 18.12. Verified on 6 September 2026.
- [8] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd edition, Cambridge University Press, Cambridge, 1944. Chapter 18, on Fourier–Bessel and Dini series: this is the source to which the NIST DLMF refers for the proof of the expansion of §10.23(iii), i.e. for the panorama of Observation 18.16. The reference is the one explicitly given by the DLMF, §10.23(iii), verified on 6 September 2026; Watson’s volume was not consulted directly and no page number is attributed here.
- [9] W. A. Strauss, *Partial Differential Equations: An Introduction*, 2nd edition, John Wiley & Sons, Hoboken (New Jersey), 2008. Elementary reference treatment of separation of variables in the disc, the heat and wave equations, and the circular membrane. *Verification note*: the bibliographic data are those given in the drafting mandate; as of 6 September 2026 it was not possible to verify the volume’s table of contents online, and for this reason no chapter, section, or page number is cited. No statement of the chapter depends on this source.
- [10] L. C. Evans, *Partial Differential Equations*, 2nd edition, Graduate Studies in Mathematics, American Mathematical Society, Providence (Rhode Island), 2010. Advanced reference treatment of the elliptic and parabolic maximum principle and of the spectral theory of the Laplacian. *Verification note*: as for the previous entry, the bibliographic data are those given in the mandate; the table of contents could not be verified online as of 6 September 2026 and no chapter, section, or page number is cited. The proofs of Theorems 18.4 and 18.14 given here are self-contained and complete.
- [11] Internal references within the treatise: Chapter 17 for the Fourier coefficients (Definition 17.1), orthonormality (Proposition 17.1), L^2 convergence and Parseval’s identity (Theorem 17.1), uniform convergence (Theorem 17.5), and the Poisson kernel with Abel summation (Definition 17.4, Theorem 17.8); Chapter 16 for the Poisson formula and the Dirichlet problem in the disc; Chapter 12 for the circle as the quotient $\mathbb{R}/2\pi\mathbb{Z}$; Chapter 6 for the polar form and the powers z^n ; Chapter 15 for the Cauchy–Riemann equations and the harmonicity of the real part of a holomorphic function.
- [12] Numerical checks: all values in the chapter were recomputed with Python scripts executed on 6 September 2026, in three independent modes — ascending series in double precision with bisection and Newton’s method; multiple-precision arithmetic with 30 digits; exact rational arithmetic (**Fraction** class) for the enclosures of Proposition 18.13 and of Worked exercise 18.8. The discrepancies between the methods are reported in the text next to each value.

Part C

Transforms

The Fourier transform

The Fourier series decomposes a periodic function into a discrete sum of harmonics, because the circle $\mathbb{T}$ is compact and its dual is $\mathbb{Z}$. When the period grows without bound, the admissible frequencies become denser, the sum turns into an integral, and in place of a sequence of coefficients there appears a *function* of the frequency: the Fourier transform. The circle does not disappear in this passage, it hides inside it: the only ingredient of the definition is the character $t \mapsto e^{-2\pi i \xi t}$, that is, a point running around the unit circle at constant angular velocity, and every property of the transform — translation that becomes a phase factor, convolution that becomes a product, the Dirac comb that is its own transform — is a consequence of the functional equation $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$ that lives on the circle.

This chapter proves the theory from beginning to end, in *a single convention*, fixed in the box below and never abandoned. It starts from the well-posed definition of the transform on L^1 (Proposition 19.1, Theorem 19.1) and from the dictionary that translates the formulas into the other conventions in use (Remark 19.1); it proves the algebraic properties (Theorem 19.3), the two differentiation rules (Lemma 19.1 and Lemma 19.2), the convolution theorem (Theorem 19.4) with the corollary on autocorrelation (Corollary 19.1), and the Riemann–Lebesgue lemma (Theorem 19.5). Next come the three explicit computations that support everything else: the Gaussian (Theorem 19.6), the rectangle (Proposition 19.2), and the triangle, obtained by convolution (Proposition 19.3). On the Schwartz class we prove the inversion formula (Theorem 19.8), which is then extended to L^1 (Theorem 19.9), and the Plancherel theorem (Theorem 19.10); from there, periodization (Theorem 19.11), the Poisson summation formula (Theorem 19.12) with Jacobi’s identity (Corollary 19.5), and the uncertainty principle (Theorem 19.13) with the case of equality for the Gaussian. A separate section introduces *tempered distributions*, the delta and the Dirac comb (Theorem 19.14), never treating the delta as a function — indeed proving that it cannot be one (Proposition 19.8). The chapter closes with the table of transforms, of which *every row* is proved or derived from the preceding properties (Theorem 19.15).

The material that the executive plan moved here from Volume II (Chapter 4, §9) is preserved in full — heuristic remark, definition, statement, proof, worked example, figure, exercise and solution — in Section 19.2 and Section 19.14. That material uses the angular-frequency convention ω ; Remark 19.3 builds the bridge between the two notations, and Remark 19.4 states exactly which points of the inherited statement were left as a sketch and where they are proved in full.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages of level **D** (inversion on $\mathcal{S}$ with the Gaussian kernel, and the definition of the transform of a tempered distribution).

Internal prerequisites. Complex exponential, Euler’s formulas and polar form (Volume II, Chapter 3, and Chapter 6); Fourier series on the circle, coefficients, Parseval’s identity, uniqueness of the coefficients (Chapter 17, in particular Theorem 17.1 and Corollary 17.2); periodic convolution and summation kernels (Chapter 17, Definition 17.3 and Definition 17.4); integrals of complex-valued functions of a real variable (Volume II, Chapter 2).

External prerequisites, assumed and not reproved:

- Lebesgue measure theory on the line: measurability, integral, translation invariance and behaviour under dilation;
- Lebesgue's dominated convergence theorem and absolute continuity of the integral (if $g \in L^1$, then $\int_E |g| \rightarrow 0$ as $|E| \rightarrow 0$);
- Tonelli's and Fubini's theorems for measurable functions on $\mathbb{R}^2$;
- density of $C_c(\mathbb{R})$ (continuous functions of compact support) in $L^1(\mathbb{R})$, and density of the Schwartz class in $L^2(\mathbb{R})$;
- Riesz's theorem: from a sequence converging in L^1 one can extract a subsequence converging almost everywhere;
- the change-of-variables formula in polar coordinates in the plane;
- the Cauchy–Schwarz inequality in L^2 , with the case of equality.

Everything else is proved here.

What one learns. How to choose a convention and translate any formula into the others without getting the factors of 2π wrong; how to derive a new transform from a known one using translation, modulation, scaling and convolution instead of redoing the integral; how to compute the transform of the Gaussian in full by solving a first-order differential equation; why inversion needs a stable class of functions and how the Gaussian kernel produces it; how to read Plancherel as an isometry and Poisson as the equality between a sum in time and a sum in frequency; how to recognise that the Dirac delta is a functional and not a function, and how to work with it safely.

Warning. Conventions of this chapter.

- **A single convention.** Throughout the chapter, except in Section 19.2 and Section 19.14 (where the text inherited from Volume II is reported intact), the Fourier transform is

$$\hat{f}(\xi) = \int_{-\infty}^{+\infty} f(t) e^{-2\pi i \xi t} dt,$$

with exponent $-2\pi i \xi t$ and *no* factor in front of the integral. The variable t is time, ξ is *frequency* (cycles per unit of time), not the angular frequency. Remark 19.1 gives the dictionary for the other conventions.

- $\mathcal{F}f = \hat{f}$; $\mathcal{F}^{-1}h(x) = \int h(\xi) e^{2\pi i x \xi} d\xi$; $Rf(t) = f(-t)$ is the reflection, $\tau_h f(t) = f(t - h)$ the translation, $\bar{f}(t) = \overline{f(-t)}$.
- **Normalised sinc:** $\text{sinc } x = \frac{\sin \pi x}{\pi x}$ for $x \neq 0$ and $\text{sinc } 0 = 1$. *Warning:* in Section 19.2, which reproduces the text of Volume II, the *unnormalised* version $\text{sinc}(x) = \sin(x)/x$ appears instead; the two are related by $\text{sinc } x = \text{sinc}(\pi x)$ and must not be confused.
- $\Pi(t) = 1$ for $|t| < \frac{1}{2}$, $\Pi(\pm \frac{1}{2}) = \frac{1}{2}$, $\Pi(t) = 0$ elsewhere; $\Lambda(t) = \max(0, 1 - |t|)$; $u(t) = 1$ for $t > 0$ and 0 for $t < 0$ is the step function.
- $L^1 = L^1(\mathbb{R})$ and $L^2 = L^2(\mathbb{R})$ with respect to Lebesgue measure, $\|f\|_1 = \int |f|$, $\|f\|_2 = (\int |f|^2)^{1/2}$, $\|f\|_\infty = \sup |f|$; $\mathcal{S}$ is the Schwartz class (Definition 19.5) and $\mathcal{S}'$ the space of tempered distributions (Definition 19.7).
- All integrals without endpoints are taken over the whole of $\mathbb{R}$.
- Numerical values are rounded to the seventh decimal place and computed twice by independent methods: a closed formula and a Python script that was run.

19.1. A single convention, and the dictionary of the others

Definition 19.1: Fourier transform on $L^1(\mathbb{R})$. Let $f \in L^1(\mathbb{R})$. Its **Fourier transform** is the function $\hat{f} = \mathcal{F}f : \mathbb{R} \rightarrow \mathbb{C}$ defined by

$$\hat{f}(\xi) = \int_{-\infty}^{+\infty} f(t) e^{-2\pi i \xi t} dt, \quad \xi \in \mathbb{R}.$$

The function $\mathcal{F}^{-1}h(x) = \int h(\xi) e^{2\pi i x \xi} d\xi$, defined for $h \in L^1(\mathbb{R})$, is called the **inverse transform** (the name is justified by Theorem 19.8 and Theorem 19.9).

Proposition 19.1: The transform is well defined and bounded. For every $f \in L^1(\mathbb{R})$ and every $\xi \in \mathbb{R}$ the integral of Definition 19.1 converges absolutely, and

$$|\hat{f}(\xi)| \leq \|f\|_1 \quad \text{for every } \xi \in \mathbb{R}, \quad \text{that is} \quad \|\hat{f}\|_\infty \leq \|f\|_1.$$

Moreover $\mathcal{F}$ is linear: $\widehat{\alpha f + \beta g} = \alpha \hat{f} + \beta \hat{g}$ for $\alpha, \beta \in \mathbb{C}$ and $f, g \in L^1$.

Proof. For every t and every ξ we have $|e^{-2\pi i \xi t}| = 1$, because the modulus of $e^{i\theta}$ equals $\cos^2 \theta + \sin^2 \theta = 1$ for every real θ (Volume II, Chapter 3). Hence

$$|f(t)e^{-2\pi i \xi t}| = |f(t)|,$$

and the integrand has integrable modulus: the integral converges absolutely and defines a complex number. From the inequality between the modulus of an integral and the integral of the modulus,

$$|\hat{f}(\xi)| = \left| \int f(t) e^{-2\pi i \xi t} dt \right| \leq \int |f(t) e^{-2\pi i \xi t}| dt = \int |f(t)| dt = \|f\|_1,$$

and the bound does not depend on ξ , which is exactly $\|\hat{f}\|_\infty \leq \|f\|_1$. Linearity is the linearity of the integral: for $\alpha, \beta \in \mathbb{C}$,

$$\int (\alpha f(t) + \beta g(t)) e^{-2\pi i \xi t} dt = \alpha \int f(t) e^{-2\pi i \xi t} dt + \beta \int g(t) e^{-2\pi i \xi t} dt,$$

legitimate because both integrals on the right converge absolutely by the first part. $\square$

Theorem 19.1: $\hat{f}$ is uniformly continuous. If $f \in L^1(\mathbb{R})$, then $\hat{f}$ is uniformly continuous on $\mathbb{R}$.

Proof. Step 1 — a bound independent of ξ . For $\xi, h \in \mathbb{R}$,

$$\hat{f}(\xi + h) - \hat{f}(\xi) = \int f(t) (e^{-2\pi i (\xi + h)t} - e^{-2\pi i \xi t}) dt = \int f(t) e^{-2\pi i \xi t} (e^{-2\pi i h t} - 1) dt,$$

where $e^{-2\pi i \xi t}$ was factored out using $e^{-2\pi i (\xi + h)t} = e^{-2\pi i \xi t} e^{-2\pi i h t}$. Taking moduli and using $|e^{-2\pi i \xi t}| = 1$,

$$|\hat{f}(\xi + h) - \hat{f}(\xi)| \leq \int |f(t)| |e^{-2\pi i h t} - 1| dt =: \omega(h),$$

and the right-hand side *does not depend on ξ* .

Step 2 — $\omega(h) \rightarrow 0$ as $h \rightarrow 0$. Let (h_k) be any sequence with $h_k \rightarrow 0$. For each fixed t , $e^{-2\pi i h_k t} \rightarrow e^0 = 1$ by continuity of the exponential, so the integrand $|f(t)| |e^{-2\pi i h_k t} - 1|$ tends to 0 pointwise. Moreover

$$|f(t)| |e^{-2\pi i h_k t} - 1| \leq |f(t)| (|e^{-2\pi i h_k t}| + 1) = 2|f(t)|,$$

and $2|f| \in L^1$. By the dominated convergence theorem $\omega(h_k) \rightarrow 0$. Since the sequence was arbitrary, $\omega(h) \rightarrow 0$ as $h \rightarrow 0$.

Step 3 — conclusion. Given $\varepsilon > 0$, by Step 2 there is $\delta > 0$ such that $\omega(h) < \varepsilon$ for $|h| < \delta$; by Step 1, for every ξ and every $|h| < \delta$ we have $|\hat{f}(\xi + h) - \hat{f}(\xi)| < \varepsilon$. The δ does not depend on ξ : continuity is uniform. $\square$

Remark 19.1: The dictionary of conventions. Three conventions circulate in the literature, all legitimate, differing in the variable chosen (frequency ξ or angular frequency $\omega = 2\pi\xi$) and in how the factor 2π is split between the transform and the inverse transform. The table compares them and gives, row by row, the form each formula of the chapter takes in each of the three; in what follows we use only column (A).

	(A) frequency	(B) asym. ang. freq.	(C) sym. ang. freq.
transform	$\int f(t)e^{-2\pi i\xi t} dt$	$\int f(t)e^{-i\omega t} dt$	$\frac{1}{\sqrt{2\pi}} \int f(t)e^{-i\omega t} dt$
inverse transform	$\int \hat{f}(\xi)e^{2\pi i x \xi} d\xi$	$\frac{1}{2\pi} \int \hat{f}_B(\omega)e^{i\omega x} d\omega$	$\frac{1}{\sqrt{2\pi}} \int \hat{f}_C(\omega)e^{i\omega x} d\omega$
link with (A)	—	$\hat{f}_B(\omega) = \hat{f}\left(\frac{\omega}{2\pi}\right)$	$\hat{f}_C(\omega) = \frac{1}{\sqrt{2\pi}} \hat{f}\left(\frac{\omega}{2\pi}\right)$
Plancherel	$\ \hat{f}\ _2^2 = \ f\ _2^2$	$\frac{1}{2\pi} \ \hat{f}_B\ _2^2 = \ f\ _2^2$	$\ \hat{f}_C\ _2^2 = \ f\ _2^2$
convolution	$\widehat{f * g} = \hat{f}\hat{g}$	$\widehat{f * g}_B = \hat{f}_B\hat{g}_B$	$\widehat{f * g}_C = \sqrt{2\pi} \hat{f}_C\hat{g}_C$
derivative	$\widehat{f'} = 2\pi i\xi \hat{f}$	$\widehat{f'}_B = i\omega \hat{f}_B$	$\widehat{f'}_C = i\omega \hat{f}_C$
Gaussian $e^{-\pi t^2}$	$e^{-\pi \xi^2}$	$e^{-\omega^2/(4\pi)}$	$\frac{1}{\sqrt{2\pi}} e^{-\omega^2/(4\pi)}$

The rows “link with (A)” are read as follows: substituting $\omega = 2\pi\xi$ into definition (A) gives exactly (B), and (C) is (B) divided by $\sqrt{2\pi}$. Every other row follows mechanically from this. For example, for convolution in (C): $\widehat{f * g}_C(\omega) = \frac{1}{\sqrt{2\pi}} \widehat{f * g}_B(\omega) = \frac{1}{\sqrt{2\pi}} \hat{f}_B\hat{g}_B = \frac{1}{\sqrt{2\pi}} (\sqrt{2\pi} \hat{f}_C)(\sqrt{2\pi} \hat{g}_C) = \sqrt{2\pi} \hat{f}_C\hat{g}_C$. And for the Gaussian: from Theorem 19.6, $\hat{f}(\xi) = e^{-\pi\xi^2}$, so $\hat{f}_B(\omega) = e^{-\pi(\omega/2\pi)^2} = e^{-\omega^2/(4\pi)}$.

Convention (A) is the one for which the Fourier transform is an isometry of L^2 with no constants (Theorem 19.10), the Dirac comb of step 1 is its own transform (Theorem 19.14), and the Poisson formula carries no factors (Theorem 19.12): this is why we choose it. Convention (B) is the one used in signal engineering and is the one of the text inherited from Volume II (Section 19.2). It is worth noting that even a single source can use both: the NIST Digital Library of Mathematical Functions defines the transform in §1.14(i) with $\mathcal{F}f(x) = \frac{1}{2\pi} \int f(t)e^{ixt} dt$, while the Poisson summation formula of §1.8(iv), equation 1.8.14, is written with $e^{-2\pi i\pi t}$, that is, in our convention (A). This is why declaring the convention at the outset is not pedantry.

The section that follows is the text of Volume II (Chapter 4, §9), reported without removing or summarising anything. It is written in convention (B) of the table above, that is, in angular frequency ω : it should be read bearing the dictionary in mind, and in particular that its $\hat{f}(\omega)$ is our $\hat{f}(\omega/2\pi)$ and its sinc is the unnormalised one. Immediately afterwards, Remark 19.3 translates its main statement row by row into convention (A), and Remark 19.4 states where each point is proved in full.

19.2. From the series to the Fourier transform

The preceding sections deal with *periodic* functions, whose natural domain is the compact circle $\mathbb{T}$. Many signals of engineering interest — a pulse, a transient, the step response — are not periodic. The Fourier transform extends harmonic analysis to this case, and is obtained, heuristically, as the limit of the Fourier series as the period $T \rightarrow \infty$.

Remark 19.2: Heuristic of the limit $T \rightarrow \infty$. For a T -periodic function, the admissible frequencies are $\omega_n = 2\pi n/T$, equally spaced by $\Delta\omega = 2\pi/T$. Defining $\hat{f}_T(\omega_n) := Tc_n$ (where c_n are the Fourier coefficients of period T), the series $f(t) = \sum_n c_n e^{i\omega_n t} = \frac{1}{2\pi} \sum_n \hat{f}_T(\omega_n) e^{i\omega_n t} \Delta\omega$ becomes, as $T \rightarrow \infty$ (so that $\Delta\omega \rightarrow 0$ and the Riemann sum tends to an integral), exactly the transform/inverse-transform pair defined below.

Definition 19.2: Fourier transform. For $f : \mathbb{R} \rightarrow \mathbb{C}$ sufficiently regular (for example $f \in L^1(\mathbb{R})$):

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega t} d\omega.$$

Theorem 19.2: Fundamental properties of the Fourier transform. Let f, g be Fourier transformable, $a, t_0, \omega_0 \in \mathbb{R}$. Then:

1. (Linearity) $\widehat{(f + g)}(\omega) = \hat{f}(\omega) + \hat{g}(\omega)$.
2. (Translation) $\widehat{f(\cdot - t_0)}(\omega) = e^{-i\omega t_0} \hat{f}(\omega)$.
3. (Scaling) $\widehat{f(a \cdot)}(\omega) = \frac{1}{|a|} \hat{f}(\omega/a)$, $a \neq 0$.
4. (Convolution) $\widehat{(f * g)}(\omega) = \hat{f}(\omega) \hat{g}(\omega)$, where $(f * g)(t) = \int_{\mathbb{R}} f(\tau) g(t - \tau) d\tau$.
5. (Plancherel) $\int_{\mathbb{R}} |f(t)|^2 dt = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}(\omega)|^2 d\omega$.

Proof. (1) Immediate from the linearity of the integral. (2) With $u = t - t_0$: $\int f(t - t_0) e^{-i\omega t} dt = \int f(u) e^{-i\omega(u+t_0)} du = e^{-i\omega t_0} \hat{f}(\omega)$. (3) With $u = at$ (for $a > 0$; the case $a < 0$ is analogous with the endpoints reversed): $\int f(at) e^{-i\omega t} dt = \frac{1}{a} \int f(u) e^{-i\omega u/a} du = \frac{1}{|a|} \hat{f}(\omega/a)$. (4) $\int (f * g)(t) e^{-i\omega t} dt = \iint f(\tau) g(t - \tau) e^{-i\omega t} d\tau dt$; with $u = t - \tau$ this factors as $[\int f(\tau) e^{-i\omega \tau} d\tau] [\int g(u) e^{-i\omega u} du] = \hat{f}(\omega) \hat{g}(\omega)$. (5) This is the analogue, for non-periodic functions, of Parseval's theorem (Volume II, Chapter 4); it is proved by applying the inversion formula and interchanging the order of integration. $\square$

Worked example — transform of the rectangle function. Let $\Pi(t) = 1$ for $|t| < 1/2$ and $\Pi(t) = 0$ elsewhere. Then

$$\hat{\Pi}(\omega) = \int_{-1/2}^{1/2} e^{-i\omega t} dt = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-1/2}^{1/2} = \frac{e^{-i\omega/2} - e^{i\omega/2}}{-i\omega} = \frac{-2i \sin(\omega/2)}{-i\omega} = \frac{\sin(\omega/2)}{\omega/2} = \text{sinc}(\omega/2),$$

with the convention $\text{sinc}(x) = \sin(x)/x$.

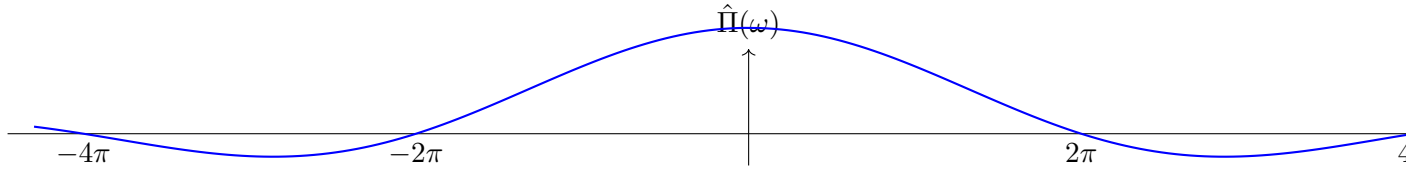


Figure 19.1: The Fourier transform of the rectangle $\Pi(t)$ is the function $\text{sinc}(\omega/2)$: a pulse narrow in time has a spectrum broad in frequency, and vice versa (time-frequency uncertainty principle).

Remark 19.3: The bridge between the ω convention and the ξ convention. Definition 19.2 above, inherited from Volume II, is column (B) of the table in Remark 19.1. Writing $\hat{f}_B$ for its transform and $\hat{f}$ for that of Definition 19.1, the substitution $\omega = 2\pi\xi$ in

the integral gives immediately

$$\hat{f}_B(\omega) = \int f(t)e^{-i\omega t} dt = \int f(t)e^{-2\pi i(\omega/2\pi)t} dt = \hat{f}\left(\frac{\omega}{2\pi}\right), \quad \text{that is} \quad \hat{f}(\xi) = \hat{f}_B(2\pi\xi).$$

Translated in this way, the five points of Theorem 19.2 become exactly:

- (1) linearity: identical, and this is Theorem 19.3(i) (already in Proposition 19.1);
- (2) translation: $e^{-i\omega t_0}$ becomes $e^{-2\pi i\xi t_0}$, and this is Theorem 19.3(ii);
- (3) scaling: the formula $\frac{1}{|a|}\hat{f}_B(\omega/a)$ is unchanged in form, and this is Theorem 19.3(iv);
- (4) convolution: identical, and this is Theorem 19.4;
- (5) Plancherel: the factor $\frac{1}{2\pi}$ of the writing in ω disappears in the writing in ξ , because $d\omega = 2\pi d\xi$; the statement with no constants is Theorem 19.10.

The worked example of the rectangle also translates: there one finds $\hat{\Pi}_B(\omega) = \text{sinc}(\omega/2) = \sin(\omega/2)/(\omega/2)$, and with $\omega = 2\pi\xi$ this is $\sin(\pi\xi)/(\pi\xi) = \text{sinc}\xi$, that is, Proposition 19.2. ✓
Numerical check at $\xi = \frac{1}{2}$ (that is, $\omega = \pi$): $\text{sinc}\frac{1}{2} = \frac{2}{\pi} = 0.6366198$, and Simpson quadrature of $\int_{-1/2}^{1/2} \cos(\pi t) dt$ with 200,000 subintervals gives 0.6366198. ✓

Remark 19.4: What the inherited text left as a sketch, and where it is completed.

The proof of Theorem 19.2, as it was written in Volume II, is complete for points (1)–(3); for point (4) it interchanges the order of integration without justifying it, and for point (5) it refers to the inversion formula with the sentence “it is proved by applying the inversion formula and interchanging the order of integration”, which is a sketch and not a proof. Nothing of this has been removed: here we only state where the two points are proved in full.

- The interchange in point (4) is justified by Tonelli’s theorem in Step 1 of the proof of Theorem 19.4, which verifies absolute integrability on $\mathbb{R}^2$ before applying Fubini.
- Point (5) is Theorem 19.10, proved on the Schwartz class with no circularity: the inversion formula (Theorem 19.8) is in turn proved earlier, and does not use Plancherel.
- The inversion formula itself, which Volume II states in Definition 19.2 as part of the “pair”, is a theorem, not a definition: without hypotheses on f the inversion integral may fail to converge. The case of the rectangle shows this: $\hat{\Pi} = \text{sinc}$ is not integrable (Proposition 19.4), and the inversion integral must be understood as a limit of integrals over $[-R, R]$.

19.3. Elementary properties: translation, modulation, scaling, conjugation

The properties that follow are all proved with a change of variable or a property of the complex exponential. They are not a list to memorise: they are the reason why, in the rest of the chapter, we will almost never compute a new integral.

Theorem 19.3: Algebraic properties of the transform. Let $f, g \in L^1(\mathbb{R})$, $\alpha, \beta \in \mathbb{C}$, $t_0, \xi_0 \in \mathbb{R}$ and $a \in \mathbb{R} \setminus \{0\}$. Then, for every $\xi \in \mathbb{R}$:

- (i) (linearity) $\alpha \widehat{f} + \beta \widehat{g}(\xi) = \alpha \hat{f}(\xi) + \beta \hat{g}(\xi)$;
- (ii) (translation) setting $\tau_{t_0} f(t) = f(t - t_0)$, $\widehat{\tau_{t_0} f}(\xi) = e^{-2\pi i \xi t_0} \hat{f}(\xi)$;
- (iii) (modulation) setting $m_{\xi_0} f(t) = e^{2\pi i \xi_0 t} f(t)$, $\widehat{m_{\xi_0} f}(\xi) = \hat{f}(\xi - \xi_0)$;
- (iv) (scaling) setting $d_a f(t) = f(at)$, $\widehat{d_a f}(\xi) = \frac{1}{|a|} \hat{f}\left(\frac{\xi}{a}\right)$;
- (v) (reflection) setting $Rf(t) = f(-t)$, $\widehat{Rf}(\xi) = \hat{f}(-\xi) = R\hat{f}(\xi)$;

(vi) (conjugation) $\widehat{\bar{f}}(\xi) = \overline{\widehat{f}(-\xi)}$.

In particular: if f is real, $\widehat{f}(-\xi) = \overline{\widehat{f}(\xi)}$ (Hermitian symmetry); if f is real and even, $\widehat{f}$ is real and even; if f is real and odd, $\widehat{f}$ is purely imaginary and odd.

Proof. All the functions involved lie in L^1 : for $\tau_{t_0}f$ and Rf by translation and reflection invariance of Lebesgue measure, for $d_a f$ because $\int |f(at)|dt = \frac{1}{|a|}\|f\|_1$, for $m_{\xi_0}f$ and $\bar{f}$ because they have the same modulus as f . Hence all the transforms written above exist by Proposition 19.1.

(i) This is the second part of Proposition 19.1.

(ii) With the substitution $u = t - t_0$ (a translation, $du = dt$):

$$\widehat{\tau_{t_0}f}(\xi) = \int f(t-t_0)e^{-2\pi i\xi t}dt = \int f(u)e^{-2\pi i\xi(u+t_0)}du = e^{-2\pi i\xi t_0} \int f(u)e^{-2\pi i\xi u}du = e^{-2\pi i\xi t_0} \widehat{f}(\xi),$$

where we used $e^{-2\pi i\xi(u+t_0)} = e^{-2\pi i\xi u}e^{-2\pi i\xi t_0}$ and pulled the factor constant in u out of the integral.

(iii) Directly, with no change of variable:

$$\widehat{m_{\xi_0}f}(\xi) = \int e^{2\pi i\xi_0 t} f(t)e^{-2\pi i\xi t}dt = \int f(t)e^{-2\pi i(\xi-\xi_0)t}dt = \widehat{f}(\xi - \xi_0).$$

(iv) First let $a > 0$ and set $u = at$, $du = a dt$; the endpoints $-\infty, +\infty$ keep their order:

$$\widehat{d_a f}(\xi) = \int f(at)e^{-2\pi i\xi t}dt = \frac{1}{a} \int f(u)e^{-2\pi i\xi u/a}du = \frac{1}{a} \widehat{f}\left(\frac{\xi}{a}\right).$$

If instead $a < 0$, the same substitution $u = at$ reverses the order of the endpoints ($t \rightarrow -\infty$ gives $u \rightarrow +\infty$), and restoring the natural order changes the sign:

$$\int f(at)e^{-2\pi i\xi t}dt = \frac{1}{a} \int_{+\infty}^{-\infty} f(u)e^{-2\pi i\xi u/a}du = -\frac{1}{a} \int_{-\infty}^{+\infty} f(u)e^{-2\pi i\xi u/a}du = \frac{1}{|a|} \widehat{f}\left(\frac{\xi}{a}\right),$$

because for $a < 0$ we have $-\frac{1}{a} = \frac{1}{|a|}$. The two cases combine into $1/|a|$.

(v) This is the case $a = -1$ of (iv): $\widehat{Rf}(\xi) = \frac{1}{|-1|} \widehat{f}(-\xi) = \widehat{f}(-\xi)$.

(vi) Using $\overline{e^{i\theta}} = e^{-i\theta}$ and the fact that conjugation commutes with the integral (because it is additive and commutes with limits):

$$\widehat{\bar{f}}(\xi) = \int \overline{f(t)}e^{-2\pi i\xi t}dt = \overline{\int f(t)e^{2\pi i\xi t}dt} = \overline{\int f(t)e^{-2\pi i(-\xi)t}dt} = \overline{\widehat{f}(-\xi)}.$$

Consequences. If f is real, $\bar{f} = f$, and (vi) gives $\widehat{f}(\xi) = \overline{\widehat{f}(-\xi)}$, that is, Hermitian symmetry. If moreover f is even, $Rf = f$ and (v) gives $\widehat{f}(\xi) = \widehat{f}(-\xi)$; combined with Hermitian symmetry, $\widehat{f}(\xi) = \widehat{\bar{f}}(\xi)$, hence $\widehat{f}$ is real (and even). If f is real and odd, $Rf = -f$ and (v) gives $\widehat{f}(-\xi) = -\widehat{f}(\xi)$; with Hermitian symmetry, $\widehat{f}(\xi) = \overline{\widehat{f}(-\xi)} = -\overline{\widehat{f}(\xi)} = -\widehat{f}(\xi)$, that is, $\widehat{f}(\xi) + \widehat{f}(\xi) = 0$: the real part vanishes and $\widehat{f}$ is purely imaginary (and odd). $\square$

Lemma 19.1: Differentiation in the time domain. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be of class C^1 with $f \in L^1(\mathbb{R})$ and $f' \in L^1(\mathbb{R})$. Then $f(t) \rightarrow 0$ as $|t| \rightarrow \infty$ and

$$\widehat{f'}(\xi) = 2\pi i\xi \widehat{f}(\xi) \quad \text{for every } \xi \in \mathbb{R}.$$

More generally, if $f \in C^k$ and $f, f', \dots, f^{(k)} \in L^1$, then $\widehat{f^{(k)}}(\xi) = (2\pi i \xi)^k \hat{f}(\xi)$ and hence

$$|\hat{f}(\xi)| \leq \frac{\|f^{(k)}\|_1}{(2\pi|\xi|)^k} \quad \text{for } \xi \neq 0.$$

Proof. Step 1 — f has limit 0 at infinity. Since $f \in C^1$, the fundamental theorem of calculus gives $f(T) = f(0) + \int_0^T f'(s) ds$ for every T . Since $f' \in L^1$, the integral $\int_0^T f'$ converges as $T \rightarrow +\infty$ (the function $T \mapsto \int_0^T |f'|$ is increasing and bounded by $\|f'\|_1$, hence the integral of f' converges absolutely). There is thus $L_+ = \lim_{T \rightarrow +\infty} f(T)$. If $L_+ \neq 0$, there would be T_0 with $|f(t)| \geq \frac{|L_+|}{2} > 0$ for every $t \geq T_0$, and then $\int_{T_0}^{+\infty} |f| = +\infty$, contradicting $f \in L^1$. Hence $L_+ = 0$; symmetrically $L_- = \lim_{T \rightarrow -\infty} f(T) = 0$.

Step 2 — integration by parts. The function $t \mapsto f(t)e^{-2\pi i \xi t}$ is of class C^1 and, by the product rule and $\frac{d}{dt}(e^{-2\pi i \xi t}) = -2\pi i \xi e^{-2\pi i \xi t}$,

$$\frac{d}{dt}(f(t)e^{-2\pi i \xi t}) = f'(t)e^{-2\pi i \xi t} - 2\pi i \xi f(t)e^{-2\pi i \xi t}.$$

Integrating this identity over $[-T, T]$ (the fundamental theorem of calculus, applied separately to the real and imaginary parts) and moving the second term to the right:

$$\int_{-T}^T f'(t)e^{-2\pi i \xi t} dt = \left[f(t)e^{-2\pi i \xi t} \right]_{-T}^T + 2\pi i \xi \int_{-T}^T f(t)e^{-2\pi i \xi t} dt.$$

The computation holds for every $\xi \in \mathbb{R}$, including $\xi = 0$, because we never divided by ξ .

Step 3 — the limit $T \rightarrow +\infty$. The boundary term equals $f(T)e^{-2\pi i \xi T} - f(-T)e^{2\pi i \xi T}$ and has modulus $\leq |f(T)| + |f(-T)| \rightarrow 0$ by Step 1. The two integrals converge to $\hat{f}'(\xi)$ and $\hat{f}(\xi)$ because f' and f lie in L^1 (Proposition 19.1). Passing to the limit, $\hat{f}'(\xi) = 2\pi i \xi \hat{f}(\xi)$.

Step 4 — iteration and estimate. If $f \in C^k$ with all derivatives up to order k in L^1 , apply Step 3 to f , then to f' , and so on: by induction on j , $\widehat{f^{(j)}}(\xi) = (2\pi i \xi)^j \hat{f}(\xi)$, since at each step the hypotheses “ $f^{(j)} \in C^1$, $f^{(j)} \in L^1$, $f^{(j+1)} \in L^1$ ” are satisfied. For $j = k$ and $\xi \neq 0$, taking moduli and using $|(2\pi i \xi)^k| = (2\pi|\xi|)^k$ and $|\widehat{f^{(k)}}(\xi)| \leq \|f^{(k)}\|_1$ (Proposition 19.1), one obtains $|\hat{f}(\xi)| \leq \|f^{(k)}\|_1 / (2\pi|\xi|)^k$. $\square$

Lemma 19.2: Differentiation in the frequency domain. Let $f \in L^1(\mathbb{R})$ be such that $t \mapsto tf(t)$ also belongs to $L^1(\mathbb{R})$. Then $\hat{f}$ is differentiable on all of $\mathbb{R}$, its derivative is continuous and

$$\frac{d}{d\xi} \hat{f}(\xi) = (-\widehat{2\pi i t f(t)})(\xi), \quad \text{equivalently} \quad \widehat{(tf(t))}(\xi) = \frac{i}{2\pi} \hat{f}'(\xi).$$

More generally, if $t^j f(t) \in L^1$ for $j = 0, 1, \dots, n$, then $\hat{f} \in C^n$ and $\hat{f}^{(n)}(\xi) = ((-2\pi i t)^n \widehat{f(t)}) (\xi)$.

Proof. Step 0 — an elementary inequality. For every $\theta \in \mathbb{R}$,

$$|e^{i\theta} - 1| = \left| \int_0^\theta i e^{is} ds \right| \leq \left| \int_0^\theta |i e^{is}| ds \right| = |\theta|.$$

Step 1 — the difference quotient. For $h \neq 0$,

$$\frac{\hat{f}(\xi + h) - \hat{f}(\xi)}{h} = \int f(t) e^{-2\pi i \xi t} \frac{e^{-2\pi i h t} - 1}{h} dt,$$

with the same factoring as in Step 1 of the proof of Theorem 19.1.

Step 2 — pointwise limit of the integrand. For fixed t ,

$$\lim_{h \rightarrow 0} \frac{e^{-2\pi i h t} - 1}{h} = \frac{d}{ds} \left(e^{-2\pi i s t} \right) \Big|_{s=0} = -2\pi i t,$$

so the integrand tends pointwise to $-2\pi i t f(t) e^{-2\pi i \xi t}$.

Step 3 — domination. By Step 0 with $\theta = -2\pi h t$,

$$\left| \frac{e^{-2\pi i h t} - 1}{h} \right| \leq \frac{2\pi |h| |t|}{|h|} = 2\pi |t|,$$

hence the modulus of the integrand is $\leq 2\pi |t| |f(t)|$, which by hypothesis is in L^1 and does not depend on h .

Step 4 — conclusion. Taking any sequence $h_k \rightarrow 0$, the dominated convergence theorem gives

$$\lim_k \frac{\hat{f}(\xi + h_k) - \hat{f}(\xi)}{h_k} = \int (-2\pi i t) f(t) e^{-2\pi i \xi t} dt = (\widehat{-2\pi i t f(t)})(\xi),$$

and since the limit does not depend on the sequence, $\hat{f}$ is differentiable at ξ with that derivative. The derivative is the transform of the function $-2\pi i t f(t) \in L^1$, hence it is continuous by Theorem 19.1. The general case is obtained by induction on n : if $t^j f \in L^1$ for $j \leq n$, apply what has just been proved to the function $(-2\pi i t)^{n-1} f(t)$, which lies in L^1 and for which $t \cdot (-2\pi i t)^{n-1} f(t)$ also lies in L^1 . $\square$

***Remark 19.5: Regularity and decay are in balance.** The two preceding lemmas are mirror images of each other, and together they state the fundamental law of Fourier analysis: the more regular f is, the more $\hat{f}$ decays (Lemma 19.1: k integrable derivatives give $|\hat{f}(\xi)| = O(|\xi|^{-k})$), and the more f decays, the more regular $\hat{f}$ is (Lemma 19.2: n finite moments give $\hat{f} \in C^n$). The uncertainty principle (Theorem 19.13) is the quantitative, L^2 version of this balance, and the Schwartz class (Definition 19.5) is the point of equilibrium where both requirements are infinite and the transform never leaves the class.*

19.4. Convolution and autocorrelation

Definition 19.3: Convolution, conjugate reflection, autocorrelation. For measurable $f, g : \mathbb{R} \rightarrow \mathbb{C}$ one sets, when the integral converges absolutely,

$$(f * g)(t) = \int f(\tau) g(t - \tau) d\tau.$$

One also sets $\tilde{f}(t) = \overline{f(-t)}$ (the **conjugate reflection**) and calls the **autocorrelation** of f the function

$$R_f(t) = \int f(s + t) \overline{f(s)} ds.$$

Theorem 19.4: Convolution theorem. Let $f, g \in L^1(\mathbb{R})$. Then:

- (i) for almost every $t \in \mathbb{R}$ the integral defining $(f * g)(t)$ converges absolutely;
- (ii) $f * g \in L^1(\mathbb{R})$ and $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$;
- (iii) $f * g = g * f$ almost everywhere;
- (iv) $\widehat{f * g}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$ for every $\xi \in \mathbb{R}$.

Proof. Step 1 — Tonelli. The function $(t, \tau) \mapsto f(\tau)g(t - \tau)$ is measurable on $\mathbb{R}^2$: $(t, \tau) \mapsto f(\tau)$ is measurable because f is measurable and the projection is continuous, and $(t, \tau) \mapsto g(t - \tau)$ is the composition of g with the linear map $(t, \tau) \mapsto t - \tau$, which preserves Lebesgue measurability, being surjective and linear. Its modulus is non-negative, so by Tonelli's theorem one may integrate in whichever order one likes:

$$\iint |f(\tau)| |g(t - \tau)| dt d\tau = \int |f(\tau)| \left(\int |g(t - \tau)| dt \right) d\tau = \int |f(\tau)| \|g\|_1 d\tau = \|f\|_1 \|g\|_1 < \infty,$$

where the inner integral equals $\|g\|_1$ by translation invariance of Lebesgue measure.

Step 2 — (i) and (ii). Since the double integral of the modulus is finite, by Tonelli's theorem the function $t \mapsto \int |f(\tau)| |g(t - \tau)| d\tau$ is finite for almost every t and has integral $\leq \|f\|_1 \|g\|_1$: this is (i), and together with it

$$\|f * g\|_1 = \int \left| \int f(\tau)g(t - \tau) d\tau \right| dt \leq \iint |f(\tau)| |g(t - \tau)| d\tau dt = \|f\|_1 \|g\|_1,$$

that is, (ii).

Step 3 — (iii). For the t at which the integral converges absolutely, the substitution $\sigma = t - \tau$ (that is, $\tau = t - \sigma$, $d\tau = -d\sigma$ with the endpoints swapped) gives $\int f(\tau)g(t - \tau) d\tau = \int f(t - \sigma)g(\sigma) d\sigma = (g * f)(t)$.

Step 4 — (iv). By Step 1 the function $(t, \tau) \mapsto f(\tau)g(t - \tau)e^{-2\pi i \xi t}$ has modulus $|f(\tau)| |g(t - \tau)|$ integrable on $\mathbb{R}^2$: Fubini's theorem authorises the interchange. Writing $e^{-2\pi i \xi t} = e^{-2\pi i \xi \tau} e^{-2\pi i \xi (t - \tau)}$,

$$\begin{aligned} \widehat{f * g}(\xi) &= \int \left(\int f(\tau)g(t - \tau) d\tau \right) e^{-2\pi i \xi t} dt \\ &= \int f(\tau) e^{-2\pi i \xi \tau} \left(\int g(t - \tau) e^{-2\pi i \xi (t - \tau)} dt \right) d\tau \\ &= \int f(\tau) e^{-2\pi i \xi \tau} \hat{g}(\xi) d\tau = \hat{f}(\xi) \hat{g}(\xi), \end{aligned}$$

where in the inner integral we set $u = t - \tau$ (τ fixed), obtaining $\int g(u) e^{-2\pi i \xi u} du = \hat{g}(\xi)$, independent of τ . $\square$

Corollary 19.1: Autocorrelation and energy spectrum. Let $f \in L^1(\mathbb{R})$. Then $\tilde{f} \in L^1(\mathbb{R})$, $R_f = f * \tilde{f}$ almost everywhere, and

$$\widehat{\tilde{f}}(\xi) = \overline{\hat{f}(\xi)}, \quad \widehat{R_f}(\xi) = |\hat{f}(\xi)|^2.$$

If moreover $f \in L^2(\mathbb{R})$, then $R_f(0) = \|f\|_2^2$.

Proof. Step 1 — $\tilde{f} \in L^1$. $|\tilde{f}(t)| = |f(-t)|$ and reflection preserves Lebesgue measure, hence $\|\tilde{f}\|_1 = \|f\|_1 < \infty$.

Step 2 — $R_f = f * \tilde{f}$. By definition,

$$(f * \tilde{f})(t) = \int f(\tau) \tilde{f}(t - \tau) d\tau = \int f(\tau) \overline{f(\tau - t)} d\tau,$$

having used $\tilde{f}(t - \tau) = \overline{f(-(t - \tau))} = \overline{f(\tau - t)}$. With the substitution $\tau = s + t$ ($d\tau = ds$) this becomes $\int f(s + t) \overline{f(s)} ds = R_f(t)$. By Theorem 19.4(i) the integral converges absolutely for almost every t .

Step 3 — the transform of $\tilde{f}$. With $u = -t$,

$$\widehat{\tilde{f}}(\xi) = \int \overline{f(-t)} e^{-2\pi i \xi t} dt = \int \overline{f(u)} e^{2\pi i \xi u} du = \overline{\int f(u) e^{-2\pi i \xi u} du} = \overline{\hat{f}(\xi)},$$

where the conjugation was pulled out of the integral as in point (vi) of Theorem 19.3.

Step 4 — conclusion. By Theorem 19.4(iv) and Step 3, $\widehat{R_f}(\xi) = \widehat{f * \tilde{f}}(\xi) = \hat{f}(\xi) \overline{\hat{f}(\xi)} = |\hat{f}(\xi)|^2$. Finally, if $f \in L^2$, $R_f(0) = \int f(s) \overline{f(s)} ds = \int |f(s)|^2 ds = \|f\|_2^2$, a finite integral precisely because $f \in L^2$. $\square$

19.5. The Riemann–Lebesgue lemma

Proposition 19.1 says that $\hat{f}$ is bounded and Theorem 19.1 that it is continuous. The result of this section adds the missing piece of information: $\hat{f}$ vanishes at infinity. A preliminary lemma is needed, the continuity of translations in L^1 , which will also be useful later on (Lemma 19.7).

Lemma 19.3: Continuity of translations in L^1 . Let $f \in L^1(\mathbb{R})$ and $\tau_h f(t) = f(t - h)$. Then $\lim_{h \rightarrow 0} \|\tau_h f - f\|_1 = 0$.

Proof. Let $\varepsilon > 0$.

Step 1 — approximation by a continuous function of compact support. By the density of $C_c(\mathbb{R})$ in $L^1(\mathbb{R})$ (a declared prerequisite) there is $g \in C_c(\mathbb{R})$ with $\|f - g\|_1 < \frac{\varepsilon}{3}$. Let $[-M, M]$ be an interval containing the support of g .

Step 2 — the estimate for g . g is continuous with compact support, hence uniformly continuous on $\mathbb{R}$: there is $\delta \in (0, 1]$ such that $|g(s) - g(s')| < \frac{\varepsilon}{3(2M+2)}$ whenever $|s - s'| < \delta$. For $|h| < \delta \leq 1$ the function $\tau_h g - g$ vanishes outside $[-M-1, M+1]$, of length $2M+2$, hence

$$\|\tau_h g - g\|_1 = \int_{-M-1}^{M+1} |g(t-h) - g(t)| dt \leq (2M+2) \cdot \frac{\varepsilon}{3(2M+2)} = \frac{\varepsilon}{3}.$$

Step 3 — the triangle inequality. By translation invariance of Lebesgue measure, $\|\tau_h(f - g)\|_1 = \|f - g\|_1$. Hence, for $|h| < \delta$,

$$\|\tau_h f - f\|_1 \leq \|\tau_h f - \tau_h g\|_1 + \|\tau_h g - g\|_1 + \|g - f\|_1 < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Since ε was arbitrary, the limit is 0. $\square$

Theorem 19.5: Riemann–Lebesgue lemma. Let $f \in L^1(\mathbb{R})$. Then $\lim_{|\xi| \rightarrow \infty} \hat{f}(\xi) = 0$.

Proof. **Step 1 — the half-period trick.** Let $\xi \neq 0$ and set $h = \frac{1}{2\xi}$. Then

$$e^{-2\pi i \xi h} = e^{-2\pi i \xi \cdot \frac{1}{2\xi}} = e^{-\pi i} = -1.$$

With the substitution $t = u + h$ in the integral defining $\hat{f}$,

$$\hat{f}(\xi) = \int f(u+h) e^{-2\pi i \xi (u+h)} du = e^{-2\pi i \xi h} \int f(u+h) e^{-2\pi i \xi u} du = - \int f(u+h) e^{-2\pi i \xi u} du.$$

Step 2 — averaging the two expressions. Adding this equality to the definition $\hat{f}(\xi) = \int f(t) e^{-2\pi i \xi t} dt$ gives

$$2\hat{f}(\xi) = \int [f(t) - f(t+h)] e^{-2\pi i \xi t} dt,$$

and taking moduli (with $|e^{-2\pi i \xi t}| = 1$)

$$|\hat{f}(\xi)| \leq \frac{1}{2} \int |f(t) - f(t+h)| dt = \frac{1}{2} \|\tau_{-h}f - f\|_1.$$

Step 3 — conclusion. As $|\xi| \rightarrow \infty$ we have $h = \frac{1}{2\xi} \rightarrow 0$, hence $-h \rightarrow 0$ and by Lemma 19.3 $\|\tau_{-h}f - f\|_1 \rightarrow 0$. Therefore $|\hat{f}(\xi)| \rightarrow 0$. $\square$

Remark 19.6: Riemann–Lebesgue gives no rate. Theorem 19.5 guarantees vanishing at infinity but no rate: for every sequence $\varepsilon_n \downarrow 0$ there exist functions $f \in L^1$ whose transform decays more slowly than ε_n . A rate is bought only with regularity, and this is Lemma 19.1: k integrable derivatives give $O(|\xi|^{-k})$. The rectangle Π illustrates this well: it is not continuous, and its transform sinc decays like $|\xi|^{-1}$, just enough to tend to zero and not enough to be integrable (Proposition 19.4). The triangle Λ is continuous with derivative of bounded variation, and its transform sinc^2 decays like $|\xi|^{-2}$: integrable.

19.6. The Gaussian, computed in full

Lemma 19.4: The normalised Gauss integral. $\int_{-\infty}^{+\infty} e^{-\pi t^2} dt = 1$.

Proof. Step 1 — finiteness. For $|t| \geq 1$ we have $t^2 \geq |t|$, hence $e^{-\pi t^2} \leq e^{-\pi|t|}$, which is integrable on $\{|t| \geq 1\}$ with integral $\frac{2}{\pi}e^{-\pi}$; on $[-1, 1]$ the function is continuous and bounded by 1. Hence $I := \int e^{-\pi t^2} dt$ is finite, and $I > 0$ because the integrand is positive.

Step 2 — the square as a double integral. The integrand is non-negative and measurable, so by Tonelli's theorem

$$I^2 = \left(\int e^{-\pi x^2} dx \right) \left(\int e^{-\pi y^2} dy \right) = \iint_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dx dy.$$

Step 3 — polar coordinates. With $x = \rho \cos \theta$, $y = \rho \sin \theta$, $\rho \in [0, +\infty)$, $\theta \in [0, 2\pi)$, whose Jacobian is ρ (a declared prerequisite) and for which $x^2 + y^2 = \rho^2$:

$$I^2 = \int_0^{2\pi} \int_0^{+\infty} e^{-\pi \rho^2} \rho d\rho d\theta = 2\pi \int_0^{+\infty} e^{-\pi \rho^2} \rho d\rho.$$

The antiderivative of $\rho e^{-\pi \rho^2}$ is $-\frac{1}{2\pi} e^{-\pi \rho^2}$, hence

$$\int_0^{+\infty} e^{-\pi \rho^2} \rho d\rho = \left[-\frac{e^{-\pi \rho^2}}{2\pi} \right]_0^{+\infty} = 0 - \left(-\frac{1}{2\pi} \right) = \frac{1}{2\pi},$$

and hence $I^2 = 2\pi \cdot \frac{1}{2\pi} = 1$. Since $I > 0$, $I = 1$.

Independent check. Composite Simpson quadrature on $[-12, 12]$ with 200,000 subintervals: 1.0000000 (the excluded tail is worth less than $2 \int_{12}^{\infty} e^{-\pi t^2} dt < 10^{-16}$). $\checkmark$ $\square$

Lemma 19.5: The second moment of the Gaussian. For every $b > 0$,

$$\int_{-\infty}^{+\infty} e^{-\pi b t^2} dt = \frac{1}{\sqrt{b}}, \quad \int_{-\infty}^{+\infty} t^2 e^{-\pi b t^2} dt = \frac{1}{2\pi b^{3/2}}.$$

Proof. Step 1 — the first integral. With the substitution $u = \sqrt{b}t$, $du = \sqrt{b}dt$, and Lemma 19.4: $\int e^{-\pi b t^2} dt = \frac{1}{\sqrt{b}} \int e^{-\pi u^2} du = \frac{1}{\sqrt{b}}$.

Step 2 — differentiating under the integral sign. Let $\Phi(b) = \int e^{-\pi b t^2} dt$ for $b > 0$. Fix $b_0 > 0$ and set $b \in [\frac{b_0}{2}, 2b_0]$; the difference quotient in b of the integrand equals, by the mean value theorem applied to $b \mapsto e^{-\pi b t^2}$, $-\pi t^2 e^{-\pi \beta t^2}$ for some β between b and b_0 , hence bounded in modulus by $\pi t^2 e^{-\pi b_0 t^2/2}$, an integrable function independent of b . By the dominated convergence theorem one may differentiate under the integral sign:

$$\Phi'(b) = \int \frac{\partial}{\partial b} e^{-\pi b t^2} dt = -\pi \int t^2 e^{-\pi b t^2} dt.$$

Step 3 — conclusion. From Step 1, $\Phi(b) = b^{-1/2}$, hence $\Phi'(b) = -\frac{1}{2}b^{-3/2}$. Equating, $-\pi \int t^2 e^{-\pi b t^2} dt = -\frac{1}{2}b^{-3/2}$, that is, $\int t^2 e^{-\pi b t^2} dt = \frac{1}{2\pi b^{3/2}}$.

Independent check. For $b = 2$ the formula gives $\frac{1}{2\pi \cdot 2^{3/2}} = \frac{1}{4\sqrt{2}\pi} = 0.0562700$; multiplied by $\sqrt{2}$ (the normalising constant used in Theorem 19.13) this gives $\frac{1}{4\pi} = 0.0795775$, and Simpson quadrature of $\int \sqrt{2} t^2 e^{-2\pi t^2} dt$ on $[-12, 12]$ with 200,000 subintervals gives 0.0795775. ✓ □

Theorem 19.6: The Gaussian is its own transform. Let $g(t) = e^{-\pi t^2}$. Then $g \in L^1(\mathbb{R})$ and

$$\hat{g}(\xi) = e^{-\pi \xi^2} = g(\xi) \quad \text{for every } \xi \in \mathbb{R}.$$

Proof. Step 1 — the hypotheses of the two differentiation lemmas. g is of class C^∞ , $g \in L^1$ by Lemma 19.4, and $g'(t) = -2\pi t e^{-\pi t^2}$. The function $t \mapsto tg(t)$ is continuous, hence integrable on $[-1, 1]$; for $|t| \geq 1$ we have $t^2 \geq |t|$ and hence $|t|e^{-\pi t^2} \leq |t|e^{-\pi|t|}$, which is integrable on $\{|t| \geq 1\}$. Hence $tg(t) \in L^1$ and, since $g' = -2\pi tg$, also $g' \in L^1$. The hypotheses of Lemma 19.1 and Lemma 19.2 are thus satisfied.

Step 2 — the differential equation in the time domain. Applying Lemma 19.1 to g :

$$\widehat{g'}(\xi) = 2\pi i \xi \hat{g}(\xi).$$

On the other hand $g'(t) = -2\pi t g(t)$, so by linearity $\widehat{g'}(\xi) = -2\pi \widehat{(tg(t))}(\xi)$. Equating,

$$\widehat{(tg(t))}(\xi) = -i\xi \hat{g}(\xi). \quad (*)$$

Step 3 — the differential equation in the frequency domain. Applying Lemma 19.2 to g :

$$\hat{g'}(\xi) = \widehat{(-2\pi i t g(t))}(\xi) = -2\pi i \widehat{(tg(t))}(\xi) \stackrel{(*)}{=} -2\pi i \cdot (-i\xi \hat{g}(\xi)) = 2\pi i^2 \xi \hat{g}(\xi) = -2\pi \xi \hat{g}(\xi).$$

Step 4 — the initial datum. $\hat{g}(0) = \int e^{-\pi t^2} dt = 1$ by Lemma 19.4.

Step 5 — solving the equation. Let $\Psi(\xi) = \hat{g}(\xi)e^{\pi \xi^2}$. By Step 3, Ψ is differentiable and

$$\Psi'(\xi) = \hat{g'}(\xi)e^{\pi \xi^2} + \hat{g}(\xi) \cdot 2\pi \xi e^{\pi \xi^2} = e^{\pi \xi^2} \left(-2\pi \xi \hat{g}(\xi) + 2\pi \xi \hat{g}(\xi) \right) = 0$$

for every $\xi \in \mathbb{R}$. A complex-valued function with derivative identically zero on $\mathbb{R}$ is constant (its real and imaginary parts are), hence $\Psi(\xi) = \Psi(0) = \hat{g}(0) = 1$, that is, $\hat{g}(\xi) = e^{-\pi \xi^2}$.

Independent check. Simpson quadrature on $[-12, 12]$ with 200,000 subintervals of $\int e^{-\pi t^2} \cos(2\pi \xi t) dt$ (the imaginary part is zero because g is real and even, Theorem 19.3): for $\xi = \frac{1}{2}$ one obtains $0.4559381 = e^{-\pi/4}$; for $\xi = 1$, $0.0432139 = e^{-\pi}$; for $\xi = 2$, $0.0000035 = e^{-4\pi}$. The computed imaginary parts are less than 10^{-16} . ✓ □

Corollary 19.2: Gaussian of arbitrary width. For every $a > 0$,

$$\widehat{(e^{-\pi at^2})}(\xi) = \frac{1}{\sqrt{a}} e^{-\pi \xi^2/a}.$$

In particular, in the parametrisation used in probability,

$$\widehat{(e^{-t^2/(2\sigma^2)})}(\xi) = \sigma\sqrt{2\pi} e^{-2\pi^2\sigma^2\xi^2} \quad (\sigma > 0).$$

Proof. We have $e^{-\pi at^2} = g(\sqrt{a}t) = d_{\sqrt{a}}g(t)$ with $g(t) = e^{-\pi t^2}$, because $\pi(\sqrt{a}t)^2 = \pi at^2$. By Theorem 19.3(iv) with $a \rightsquigarrow \sqrt{a} > 0$ and by Theorem 19.6,

$$\widehat{d_{\sqrt{a}}g}(\xi) = \frac{1}{\sqrt{a}} \hat{g}\left(\frac{\xi}{\sqrt{a}}\right) = \frac{1}{\sqrt{a}} e^{-\pi \xi^2/a}.$$

For the second formula it suffices to identify a : $e^{-t^2/(2\sigma^2)} = e^{-\pi at^2}$ requires $\pi a = \frac{1}{2\sigma^2}$, that is, $a = \frac{1}{2\pi\sigma^2}$. Then $\frac{1}{\sqrt{a}} = \sigma\sqrt{2\pi}$ and $\frac{\pi\xi^2}{a} = \pi\xi^2 \cdot 2\pi\sigma^2 = 2\pi^2\sigma^2\xi^2$.

Independent check. For $a = \delta^{-2}$ one obtains the formula $\widehat{(e^{-\pi t^2/\delta^2})}(\xi) = \delta e^{-\pi\delta^2\xi^2}$, which is the relation used in Lemma 19.7. For $\sigma = 1$ and $\xi = 0$ the second formula gives $\sqrt{2\pi} = 2.5066283$, and the quadrature of $\int e^{-t^2/2} dt$ on $[-16, 16]$ with 200,000 subintervals gives 2.5066283. ✓

19.7. Rectangle, sinc function, and triangle

Definition 19.4: Rectangle, triangle, sinc function. One sets

$$\Pi(t) = \begin{cases} 1 & |t| < \frac{1}{2}, \\ \frac{1}{2} & |t| = \frac{1}{2}, \\ 0 & |t| > \frac{1}{2}, \end{cases} \quad \Lambda(t) = \max(0, 1 - |t|), \quad \text{sinc } x = \begin{cases} \frac{\sin \pi x}{\pi x} & x \neq 0, \\ 1 & x = 0. \end{cases}$$

Π is the **rectangle function** (or window) of width 1, Λ the **triangle function** of base 2, sinc the **normalised sinc function**.

Proposition 19.2: Transform of the rectangle. $\widehat{\Pi}(\xi) = \text{sinc } \xi$ for every $\xi \in \mathbb{R}$.

Proof. $\Pi \in L^1$ with $\|\Pi\|_1 = 1$ (the value at $\pm\frac{1}{2}$ does not affect the integral, being attained on a set of measure zero).

Case $\xi = 0$. $\widehat{\Pi}(0) = \int_{-1/2}^{1/2} 1 dt = 1 = \text{sinc } 0$.

Case $\xi \neq 0$. An antiderivative of $e^{-2\pi i \xi t}$ is $\frac{e^{-2\pi i \xi t}}{-2\pi i \xi}$, hence

$$\widehat{\Pi}(\xi) = \int_{-1/2}^{1/2} e^{-2\pi i \xi t} dt = \left[\frac{e^{-2\pi i \xi t}}{-2\pi i \xi} \right]_{-1/2}^{1/2} = \frac{e^{-\pi i \xi} - e^{\pi i \xi}}{-2\pi i \xi}.$$

By Euler's formula $e^{i\alpha} - e^{-i\alpha} = 2i \sin \alpha$ (Volume II, Chapter 3), with $\alpha = \pi \xi$ we have $e^{-\pi i \xi} - e^{\pi i \xi} = -2i \sin(\pi \xi)$, and hence

$$\widehat{\Pi}(\xi) = \frac{-2i \sin(\pi \xi)}{-2\pi i \xi} = \frac{\sin \pi \xi}{\pi \xi} = \text{sinc } \xi.$$

Independent check. Simpson quadrature of $\int_{-1/2}^{1/2} \cos(2\pi\xi t) dt$ with 200,000 subintervals: $\xi = \frac{1}{4} \rightarrow 0.9003163 = \text{sinc } \frac{1}{4}$; $\xi = \frac{1}{2} \rightarrow 0.6366198 = \frac{2}{\pi}$; $\xi = 1 \rightarrow 0.0000000$; $\xi = \frac{3}{2} \rightarrow -0.2122066 = -\frac{2}{3\pi}$. ✓ □

Proposition 19.3: The triangle by convolution, and its transform. $\Pi * \Pi = \Lambda$ everywhere, and consequently

$$\widehat{\Lambda}(\xi) = (\text{sinc } \xi)^2.$$

Moreover $\Lambda = R_{\Pi}$, the autocorrelation of the rectangle.

Proof. Step 1 — computing the convolution. Since Π equals 1 on $(-\frac{1}{2}, \frac{1}{2})$ and 0 outside (up to a set of measure zero),

$$(\Pi * \Pi)(t) = \int \Pi(\tau)\Pi(t - \tau) d\tau = \left| \left(-\frac{1}{2}, \frac{1}{2}\right) \cap \left(t - \frac{1}{2}, t + \frac{1}{2}\right) \right|,$$

because the integrand equals 1 exactly when $|\tau| < \frac{1}{2}$ and $|t - \tau| < \frac{1}{2}$, that is, $\tau \in (t - \frac{1}{2}, t + \frac{1}{2})$. The intersection of two open intervals is the interval with endpoints $\max(-\frac{1}{2}, t - \frac{1}{2})$ and $\min(\frac{1}{2}, t + \frac{1}{2})$, of length

$$\ell(t) = \min\left(\frac{1}{2}, t + \frac{1}{2}\right) - \max\left(-\frac{1}{2}, t - \frac{1}{2}\right)$$

if positive, and 0 otherwise. Three cases arise.

- $0 \leq t < 1$: then $t + \frac{1}{2} \geq \frac{1}{2}$ and $t - \frac{1}{2} \geq -\frac{1}{2}$, hence $\ell(t) = \frac{1}{2} - (t - \frac{1}{2}) = 1 - t$, which is positive.
- $-1 < t \leq 0$: symmetrically $\ell(t) = (t + \frac{1}{2}) - (-\frac{1}{2}) = 1 + t > 0$.
- $|t| \geq 1$: for $t \geq 1$ we have $t - \frac{1}{2} \geq \frac{1}{2}$, hence the intersection is empty; for $t \leq -1$ we have $t + \frac{1}{2} \leq -\frac{1}{2}$, again empty. In both cases the convolution is 0.

The three cases combine into $(\Pi * \Pi)(t) = \max(0, 1 - |t|) = \Lambda(t)$.

Step 2 — the transform. $\Pi \in L^1$, hence Theorem 19.4(iv) and Proposition 19.2 give

$$\widehat{\Lambda}(\xi) = \widehat{\Pi * \Pi}(\xi) = \widehat{\Pi}(\xi)^2 = (\text{sinc } \xi)^2.$$

Step 3 — the autocorrelation. Π is real and even, hence $\tilde{\Pi}(t) = \overline{\Pi(-t)} = \Pi(t)$, and by Corollary 19.1 $R_{\Pi} = \Pi * \tilde{\Pi} = \Pi * \Pi = \Lambda$; consistently $\widehat{R_{\Pi}} = |\widehat{\Pi}|^2 = \text{sinc}^2$, the same result as in Step 2, since sinc is real.

Independent check. (a) Convolution computed directly as the length of the intersection: $(\Pi * \Pi)(0) = 1.0000000$, $(\Pi * \Pi)(0.25) = 0.7500000$, $(\Pi * \Pi)(0.7) = 0.3000000$, $(\Pi * \Pi)(1.2) = 0.0000000$, against $\Lambda(0) = 1$, $\Lambda(0.25) = 0.75$, $\Lambda(0.7) = 0.3$, $\Lambda(1.2) = 0$. ✓ (b) Simpson quadrature of $\int_{-1}^1 \Lambda(t) \cos(2\pi\xi t) dt$ with 200,000 subintervals: $\xi = \frac{1}{2} \rightarrow 0.4052847 = (\frac{2}{\pi})^2$; $\xi = 1 \rightarrow 0.0000000$; $\xi = 1.3 \rightarrow 0.0392400 = \text{sinc}^2(1.3)$. ✓ (c) Autocorrelation computed numerically: $R_{\Pi}(0.3) = 0.7000000 = \Lambda(0.3)$, $R_{\Pi}(0.9) = 0.1000000 = \Lambda(0.9)$. ✓ □

Proposition 19.4: The sinc function is not integrable, its square is. $\text{sinc} \notin L^1(\mathbb{R})$, whereas $\text{sinc}^2 \in L^1(\mathbb{R})$.

Proof. Step 1 — divergence. For every integer $k \geq 1$ and every $x \in [k, k+1]$ we have $\pi x \leq \pi(k+1)$, hence

$$\int_k^{k+1} |\text{sinc } x| dx = \int_k^{k+1} \frac{|\sin \pi x|}{\pi x} dx \geq \frac{1}{\pi(k+1)} \int_k^{k+1} |\sin \pi x| dx.$$

The last integral does not depend on k , because $|\sin \pi x|$ is periodic with period 1, and equals

$$\int_0^1 |\sin \pi x| dx = \int_0^1 \sin \pi x dx = \left[-\frac{\cos \pi x}{\pi} \right]_0^1 = \frac{2}{\pi}.$$

Hence

$$\int_1^{+\infty} |\operatorname{sinc} x| dx \geq \sum_{k \geq 1} \frac{2}{\pi^2(k+1)} = +\infty,$$

because the harmonic series diverges. Thus $\operatorname{sinc} \notin L^1$.

Step 2 — integrability of the square. $|\operatorname{sinc} x| \leq 1$ everywhere (because $|\sin \theta| \leq |\theta|$), so $\operatorname{sinc}^2 \leq 1$ is integrable on $[-1, 1]$; for $|x| \geq 1$ we have $\operatorname{sinc}^2 x \leq \frac{1}{\pi^2 x^2}$, integrable. Hence $\operatorname{sinc}^2 \in L^1$.

Independent check. The quadrature of $\int_0^R |\operatorname{sinc}|$ grows logarithmically: $R = 10 \rightarrow 1.0528631$, $R = 10^2 \rightarrow 1.5194284$, $R = 10^3 \rightarrow 1.9860293$, $R = 10^4 \rightarrow 2.4526306$. The successive differences are 0.4665653, 0.4666009, 0.4666013, against $\frac{2}{\pi^2} \ln 10 = 0.4666013$: the asymptotic behaviour $\frac{2}{\pi^2} \ln R + \text{const.}$ is confirmed to seven decimal places, and the divergence is logarithmic. ✓ $\square$

19.8. The Schwartz class and the inversion formula

The inversion formula cannot hold for every $f \in L^1$: Proposition 19.4 shows that $\hat{\Pi} = \operatorname{sinc}$ is not integrable, so the integral $\int \hat{\Pi}(\xi) e^{2\pi i x \xi} d\xi$ makes no sense as a Lebesgue integral. We need a class of functions small enough to be stable under $\mathcal{F}$: this is the Schwartz class.

Definition 19.5: Schwartz class. The **Schwartz class** $\mathcal{S} = \mathcal{S}(\mathbb{R})$ is the set of functions $f : \mathbb{R} \rightarrow \mathbb{C}$ of class C^∞ such that, for every pair of integers $m, n \geq 0$,

$$p_{m,n}(f) := \sup_{t \in \mathbb{R}} |t^m f^{(n)}(t)| < \infty.$$

The quantities $p_{m,n}$ are called the **seminorms** of $\mathcal{S}$; one says $f_k \rightarrow f$ in $\mathcal{S}$ if $p_{m,n}(f_k - f) \rightarrow 0$ for every $m, n \geq 0$.

Example 19.1: What lies in $\mathcal{S}$, and what does not. The Gaussian $e^{-\pi t^2}$ lies in $\mathcal{S}$: every derivative of it is $q(t)e^{-\pi t^2}$ with q a polynomial, and $t^m q(t)e^{-\pi t^2} \rightarrow 0$ as $|t| \rightarrow \infty$ because the exponential beats every polynomial. Not in $\mathcal{S}$: the rectangle Π (not continuous), the triangle Λ (not differentiable at $0, \pm 1$), $\frac{1}{1+t^2}$ (decays only like t^{-2} , hence $p_{3,0} = \infty$), and $e^{-|t|}$ (not differentiable at 0).

Proposition 19.5: Elementary stability of $\mathcal{S}$. $\mathcal{S}$ is a complex vector space, closed under differentiation $f \mapsto f'$, under multiplication by a polynomial, and under conjugation. Moreover $\mathcal{S} \subset L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, with

$$\|f\|_1 \leq \pi(p_{0,0}(f) + p_{2,0}(f)), \quad \|f\|_2^2 \leq \pi p_{0,0}(f)(p_{0,0}(f) + p_{2,0}(f)).$$

Proof. Step 1 — vector space. If $f, g \in \mathcal{S}$ and $\alpha, \beta \in \mathbb{C}$, then $\alpha f + \beta g \in C^\infty$ and $p_{m,n}(\alpha f + \beta g) \leq |\alpha|p_{m,n}(f) + |\beta|p_{m,n}(g) < \infty$.

Step 2 — differentiation. $p_{m,n}(f') = p_{m,n+1}(f) < \infty$ for every m, n : $\mathcal{S}$ is closed under differentiation.

Step 3 — multiplication by t . Let $g(t) = tf(t)$. By the Leibniz formula $g^{(n)}(t) = tf^{(n)}(t) + n f^{(n-1)}(t)$ (the further terms vanish because the second derivative of t is zero), hence

$$p_{m,n}(g) \leq p_{m+1,n}(f) + n p_{m,n-1}(f) < \infty.$$

Iterating, $\mathcal{S}$ is closed under multiplication by t^k and hence by any polynomial (by Step 1).

Step 4 — conjugation. $\overline{f^{(n)}} = \overline{f}^{(n)}$ and $|t^m \overline{f^{(n)}}(t)| = |t^m f^{(n)}(t)|$, hence $p_{m,n}(\overline{f}) = p_{m,n}(f)$.

Step 5 — inclusion in L^1 and L^2 . Let $C = p_{0,0}(f) + p_{2,0}(f)$. For every t ,

$$(1+t^2)|f(t)| = |f(t)| + |t^2 f(t)| \leq p_{0,0}(f) + p_{2,0}(f) = C, \quad \text{that is} \quad |f(t)| \leq \frac{C}{1+t^2}.$$

Since $\int \frac{dt}{1+t^2} = [\arctan t]_{-\infty}^{+\infty} = \pi$, one obtains $\|f\|_1 \leq \pi C$. For L^2 : $|f(t)|^2 \leq p_{0,0}(f) |f(t)|$, hence $\|f\|_2^2 \leq p_{0,0}(f) \|f\|_1 \leq \pi p_{0,0}(f) C$. $\square$

Theorem 19.7: $\mathcal{F}$ maps $\mathcal{S}$ into $\mathcal{S}$, continuously. If $f \in \mathcal{S}$, then $\hat{f} \in \mathcal{S}$. More precisely, for every pair $m, n \geq 0$ there is a constant $C_{m,n}$ (depending only on m and n) such that

$$p_{m,n}(\hat{f}) \leq C_{m,n} \max_{\substack{0 \leq j \leq n+2 \\ 0 \leq k \leq m}} p_{j,k}(f) \quad \text{for every } f \in \mathcal{S}.$$

In particular, if $f_\ell \rightarrow 0$ in $\mathcal{S}$ then $\hat{f}_\ell \rightarrow 0$ in $\mathcal{S}$.

Proof. **Step 1 — the derivatives of $\hat{f}$.** By Proposition 19.5, $t^j f(t) \in \mathcal{S} \subset L^1$ for every $j \geq 0$. Lemma 19.2 (the general part) then gives $\hat{f} \in C^\infty$ and

$$\hat{f}^{(n)} = \widehat{g_n}, \quad g_n(t) = (-2\pi i)^n t^n f(t) \in \mathcal{S}.$$

Step 2 — multiplication by ξ^m . $g_n \in \mathcal{S}$ is of class C^∞ and all its derivatives lie in L^1 (Proposition 19.5, Steps 2 and 5), so Lemma 19.1 gives $\widehat{g_n^{(m)}}(\xi) = (2\pi i)^m \widehat{g_n}(\xi)$, that is,

$$\xi^m \hat{f}^{(n)}(\xi) = \frac{1}{(2\pi i)^m} \widehat{g_n^{(m)}}(\xi).$$

Step 3 — the estimate. By Proposition 19.1, $|\widehat{g_n^{(m)}}(\xi)| \leq \|g_n^{(m)}\|_1$, hence

$$p_{m,n}(\hat{f}) = \sup_{\xi} |\xi^m \hat{f}^{(n)}(\xi)| \leq \frac{\|g_n^{(m)}\|_1}{(2\pi)^m} \leq \frac{\pi}{(2\pi)^m} (p_{0,0}(g_n^{(m)}) + p_{2,0}(g_n^{(m)})),$$

where the last inequality is Proposition 19.5, Step 5. By the Leibniz formula applied to $g_n = (-2\pi i)^n t^n f$,

$$g_n^{(m)}(t) = (-2\pi i)^n \sum_{k=0}^{\min(m,n)} \binom{m}{k} \frac{n!}{(n-k)!} t^{n-k} f^{(m-k)}(t),$$

because the k -th derivative of t^n is $\frac{n!}{(n-k)!} t^{n-k}$ and vanishes for $k > n$. Hence, for $r \in \{0, 2\}$,

$$p_{r,0}(g_n^{(m)}) \leq (2\pi)^n \sum_{k=0}^{\min(m,n)} \binom{m}{k} \frac{n!}{(n-k)!} p_{n-k+r, m-k}(f),$$

and all the indices that appear satisfy $n-k+r \leq n+2$ and $m-k \leq m$. Collecting the numerical constants into $C_{m,n}$ gives the inequality in the statement; in particular all the $p_{m,n}(\hat{f})$ are finite, that is, $\hat{f} \in \mathcal{S}$. If $f_\ell \rightarrow 0$ in $\mathcal{S}$, the right-hand side tends to 0 for every m, n , hence $\hat{f}_\ell \rightarrow 0$ in $\mathcal{S}$. $\square$

Lemma 19.6: Multiplication formula. For every $f, g \in L^1(\mathbb{R})$,

$$\int \hat{f}(\xi) g(\xi) d\xi = \int f(t) \hat{g}(t) dt,$$

and the two integrals converge absolutely.

Proof. The function $(t, \xi) \mapsto f(t)g(\xi)e^{-2\pi i t \xi}$ is measurable on $\mathbb{R}^2$ and has modulus $|f(t)||g(\xi)|$; by Tonelli's theorem

$$\iint |f(t)||g(\xi)| dt d\xi = \|f\|_1 \|g\|_1 < \infty.$$

Fubini's theorem then authorises both orders of integration:

$$\int \left(\int f(t) e^{-2\pi i t \xi} dt \right) g(\xi) d\xi = \iint f(t) g(\xi) e^{-2\pi i t \xi} dt d\xi = \int f(t) \left(\int g(\xi) e^{-2\pi i t \xi} d\xi \right) dt,$$

that is, $\int \hat{f}g = \int f\hat{g}$. The absolute convergence of the two integrals follows from $|\hat{f}| \leq \|f\|_1$ and $|\hat{g}| \leq \|g\|_1$ (Proposition 19.1) and from $f, g \in L^1$. $\square$

Lemma 19.7: The Gaussian kernel is an approximate identity. For $\delta > 0$ set

$$K_\delta(u) = \frac{1}{\delta} e^{-\pi u^2 / \delta^2}.$$

Then:

- (i) $K_\delta > 0$, K_δ is even and $\int K_\delta = 1$;
- (ii) setting $G_{x,\delta}(\xi) = e^{2\pi i x \xi} e^{-\pi \delta^2 \xi^2}$, we have $G_{x,\delta} \in \mathcal{S}$ and $\widehat{G_{x,\delta}}(t) = K_\delta(t - x)$;
- (iii) if f is bounded and continuous at x , then $\lim_{\delta \rightarrow 0^+} \int f(t) K_\delta(t - x) dt = f(x)$;
- (iv) if $f \in L^1(\mathbb{R})$, then $\|f * K_\delta - f\|_1 \rightarrow 0$ as $\delta \rightarrow 0^+$.

Proof. (i) Positivity and evenness are evident from the formula; with $u = \delta v$, $\int K_\delta(u) du = \int e^{-\pi v^2} dv = 1$ by Lemma 19.4.

(ii) By Corollary 19.2 with $a = \delta^2$,

$$(\widehat{e^{-\pi \delta^2 \xi^2}})(t) = \frac{1}{\delta} e^{-\pi t^2 / \delta^2} = K_\delta(t),$$

and by modulation (Theorem 19.3(iii), applied in the variable ξ with $\xi_0 = x$) one obtains $\widehat{G_{x,\delta}}(t) = K_\delta(t - x)$. That $G_{x,\delta} \in \mathcal{S}$ follows from Example 19.1 and from the fact that multiplying by $e^{2\pi i x \xi}$, a C^∞ function with all derivatives bounded, does not alter the finiteness of the seminorms (Leibniz formula).

(iii) With the substitution $t = x + \delta v$ ($dt = \delta dv$),

$$\int f(t) K_\delta(t - x) dt = \int f(x + \delta v) e^{-\pi v^2} dv,$$

while by (i) $f(x) = \int f(x) e^{-\pi v^2} dv$. The difference is

$$\int [f(x + \delta v) - f(x)] e^{-\pi v^2} dv.$$

For each fixed v the integrand tends to 0 as $\delta \rightarrow 0^+$, by continuity of f at x ; and it is dominated in modulus by $2\|f\|_\infty e^{-\pi v^2}$, integrable and independent of δ . The dominated convergence theorem (applied to any sequence $\delta_k \rightarrow 0^+$) gives the limit 0.

(iv) Since K_δ is even, $(f * K_\delta)(x) = \int f(x-u)K_\delta(u)du$, and by (i)

$$(f * K_\delta)(x) - f(x) = \int K_\delta(u)[f(x-u) - f(x)] du.$$

Taking moduli and integrating in x , Tonelli's theorem (non-negative integrand) allows the order to be exchanged:

$$\|f * K_\delta - f\|_1 \leq \int K_\delta(u) \left(\int |f(x-u) - f(x)| dx \right) du = \int K_\delta(u) \|\tau_u f - f\|_1 du.$$

With $u = \delta v$ this becomes $\int e^{-\pi v^2} \|\tau_{\delta v} f - f\|_1 dv$. The integrand tends to 0 for every v as $\delta \rightarrow 0^+$ (Lemma 19.3) and is dominated by $2\|f\|_1 e^{-\pi v^2}$, because $\|\tau_u f - f\|_1 \leq 2\|f\|_1$. The dominated convergence theorem concludes the proof. $\square$

Theorem 19.8: Inversion formula on the Schwartz class. Let $f \in \mathcal{S}(\mathbb{R})$. Then, for every $x \in \mathbb{R}$,

$$f(x) = \int_{-\infty}^{+\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi = (\mathcal{F}^{-1} \hat{f})(x).$$

Proof. Step 1 — the starting identity. Fix x and $\delta > 0$, and apply Lemma 19.6 to the pair $f \in \mathcal{S} \subset L^1$ and $G_{x,\delta} \in \mathcal{S} \subset L^1$ from Lemma 19.7(ii):

$$\int \hat{f}(\xi) e^{2\pi i x \xi} e^{-\pi \delta^2 \xi^2} d\xi = \int \hat{f}(\xi) G_{x,\delta}(\xi) d\xi = \int f(t) \widehat{G_{x,\delta}}(t) dt = \int f(t) K_\delta(t-x) dt.$$

Step 2 — the limit on the left. By Theorem 19.7 we have $\hat{f} \in \mathcal{S}$, hence $\hat{f} \in L^1$ by Proposition 19.5. The integrand on the left tends pointwise to $\hat{f}(\xi) e^{2\pi i x \xi}$ as $\delta \rightarrow 0^+$ and is dominated in modulus by $|\hat{f}| \in L^1$, independent of δ (because $0 < e^{-\pi \delta^2 \xi^2} \leq 1$). By dominated convergence the left-hand side tends to $\int \hat{f}(\xi) e^{2\pi i x \xi} d\xi$.

Step 3 — the limit on the right. $f \in \mathcal{S}$ is bounded ($p_{0,0}(f) < \infty$) and continuous; by Lemma 19.7(iii) the right-hand side tends to $f(x)$.

Step 4 — conclusion. Two quantities equal for every $\delta > 0$ have the same limit: $f(x) = \int \hat{f}(\xi) e^{2\pi i x \xi} d\xi$. $\square$

Corollary 19.3: $\mathcal{F}$ is a bijection of $\mathcal{S}$, and $\mathcal{F}^4 = \text{id}$. $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is bijective; its inverse is $\mathcal{F}^{-1} = R\mathcal{F} = \mathcal{F}R$, where $Rh(x) = h(-x)$. Moreover $\mathcal{F}^2 = R$ and $\mathcal{F}^4 = \text{id}$.

Proof. Step 1 — $\mathcal{F}$ and R commute. By Theorem 19.3(v), $\widehat{Rh} = R\hat{h}$ for every $h \in L^1$: that is, $\mathcal{F}R = R\mathcal{F}$ on L^1 , and in particular on $\mathcal{S}$ (which is closed under reflection, since $p_{m,n}(Rh) = p_{m,n}(h)$).

Step 2 — $\mathcal{F}^2 = R$. Theorem 19.8 says that for $f \in \mathcal{S}$

$$f(x) = \int \hat{f}(\xi) e^{2\pi i x \xi} d\xi = \int \hat{f}(\xi) e^{-2\pi i (-x) \xi} d\xi = (\mathcal{F}\hat{f})(-x) = (R\mathcal{F}^2 f)(x),$$

that is, $R\mathcal{F}^2 = \text{id}$. Applying R on the left and using $R^2 = \text{id}$, one obtains $\mathcal{F}^2 = R$.

Step 3 — conclusion. From $\mathcal{F}^2 = R$ it follows that $\mathcal{F}^4 = R^2 = \text{id}$: hence $\mathcal{F}$ is invertible on $\mathcal{S}$ with inverse $\mathcal{F}^3 = \mathcal{F}^2 \mathcal{F} = R\mathcal{F} = \mathcal{F}R$ (Step 1). In particular $\mathcal{F}$ is injective and surjective on $\mathcal{S}$, and $\mathcal{F}^{-1}h(x) = (\mathcal{F}h)(-x) = \int h(\xi) e^{2\pi i x \xi} d\xi$, which is the formula of Definition 19.1. $\square$

Theorem 19.9: Inversion in L^1 . Let $f \in L^1(\mathbb{R})$ with $\hat{f} \in L^1(\mathbb{R})$. Then the function

$$F(x) = \int \hat{f}(\xi) e^{2\pi i x \xi} d\xi$$

is uniformly continuous and bounded on $\mathbb{R}$, and $f = F$ almost everywhere. In particular, if f is continuous, $f = F$ everywhere.

Proof. **Step 1 — F is continuous and bounded.** $F(x) = \widehat{\hat{f}}(-x)$, and $\widehat{\hat{f}}$ is uniformly continuous and bounded by $\|\hat{f}\|_1$ by Theorem 19.1 and Proposition 19.1; reflection preserves both properties.

Step 2 — the identity with the kernel. As in Step 1 of the proof of Theorem 19.8, Lemma 19.6 applied to $f \in L^1$ and $G_{x,\delta} \in L^1$ gives

$$\int \hat{f}(\xi) e^{2\pi i x \xi} e^{-\pi \delta^2 \xi^2} d\xi = \int f(t) K_\delta(t - x) dt = (f * K_\delta)(x),$$

the last equality because K_δ is even (Lemma 19.7(i)).

Step 3 — the limit on the left. Since $\hat{f} \in L^1$ by hypothesis, the same domination as in Step 2 of the proof of Theorem 19.8 gives, for every x ,

$$\lim_{\delta \rightarrow 0^+} \int \hat{f}(\xi) e^{2\pi i x \xi} e^{-\pi \delta^2 \xi^2} d\xi = F(x).$$

Step 4 — the limit on the right. By Lemma 19.7(iv), $f * K_\delta \rightarrow f$ in L^1 . Choosing a sequence $\delta_k \rightarrow 0^+$, we have $\|f * K_{\delta_k} - f\|_1 \rightarrow 0$; by Riesz's theorem (a declared prerequisite) there is a subsequence δ_{k_j} such that $f * K_{\delta_{k_j}}(x) \rightarrow f(x)$ for almost every x .

Step 5 — conclusion. For almost every x the same sequence of numbers $(f * K_{\delta_{k_j}})(x)$ converges to $F(x)$ (Step 3, which holds for every x) and to $f(x)$ (Step 4): by uniqueness of the limit $f(x) = F(x)$ for almost every x . If f is continuous, $f - F$ is continuous and zero almost everywhere, hence zero everywhere (if it were non-zero at a point, it would be non-zero on a neighbourhood, which has positive measure). $\square$

Remark 19.7: Why the hypothesis $\hat{f} \in L^1$ cannot be dropped. The rectangle Π lies in L^1 but $\hat{\Pi} = \text{sinc} \notin L^1$ (Proposition 19.4): the inversion integral is not defined in the Lebesgue sense. It does exist, however, as a principal value, that is, as $\lim_{R \rightarrow \infty} \int_{-R}^R$, and equals $\Pi(x)$ at points of continuity and $\frac{1}{2}$ at $x = \pm \frac{1}{2}$ — exactly the value that Definition 19.4 assigns at the two jump points. The proof of this fact is the Dirichlet–Jordan theorem applied to the non-periodic case, and is not carried out here: the periodic version is Theorem 17.4 of Chapter 17.

19.9. The Plancherel theorem

Theorem 19.10: Plancherel–Parseval on the Schwartz class. For every $f, g \in \mathcal{S}(\mathbb{R})$,

$$\int f(t) \overline{g(t)} dt = \int \hat{f}(\xi) \overline{\hat{g}(\xi)} d\xi.$$

Proof. **Step 1 — an auxiliary function.** Set $w = \bar{\hat{g}}$. By Theorem 19.7 $\hat{g} \in \mathcal{S}$, and by Proposition 19.5 (Step 4) also $w \in \mathcal{S}$; in particular $w \in L^1$.

Step 2 — the transform of w . Pulling the conjugation out of the integral as in Theorem 19.3(vi),

$$\hat{w}(t) = \int \overline{\hat{g}(\xi)} e^{-2\pi i t \xi} d\xi = \overline{\int \hat{g}(\xi) e^{2\pi i t \xi} d\xi} = \overline{g(t)},$$

where the last step is the inversion formula (Theorem 19.8) applied to $g \in \mathcal{S}$.

Step 3 — the multiplication formula. Applying Lemma 19.6 to the pair $f, w \in L^1$ and using Step 2,

$$\int \hat{f}(\xi) \overline{\hat{g}(\xi)} d\xi = \int \hat{f}(\xi) w(\xi) d\xi = \int f(t) \hat{w}(t) dt = \int f(t) \overline{g(t)} dt.$$

□

Corollary 19.4: $\mathcal{F}$ is an isometry for the L^2 norm. For every $f \in \mathcal{S}(\mathbb{R})$, $\|\hat{f}\|_2 = \|f\|_2$.

Proof. Take $g = f$ in Theorem 19.10:

$$\|f\|_2^2 = \int f \bar{f} = \int \hat{f} \bar{\hat{f}} = \|\hat{f}\|_2^2,$$

and the two norms are non-negative, hence equal. □

Example 19.2: Two sinc integrals, via Plancherel. Π does not lie in $\mathcal{S}$, but it lies in $L^1 \cap L^2$ and Plancherel's identity holds for it equally, as one verifies directly from Corollary 19.1 and Theorem 19.9. Indeed $R_\Pi = \Lambda$ is continuous and $\widehat{R_\Pi} = \text{sinc}^2 \in L^1$ (Proposition 19.4), so Theorem 19.9 applies to R_Π and gives, at $x = 0$,

$$\int \text{sinc}^2 \xi d\xi = R_\Pi(0) = \int |\Pi|^2 = 1.$$

$\Pi * \Lambda = \Pi * \Pi * \Pi$ and $\Lambda * \Lambda$ are treated in the same way: both are continuous, because $(\Pi * \Lambda)(t) = \int_{-1/2}^{1/2} \Lambda(t - \tau) d\tau$ and $(\Lambda * \Lambda)(t) = \int_{-1}^1 \Lambda(\tau) \Lambda(t - \tau) d\tau$ depend continuously on t by the uniform continuity of Λ ; and their transforms, sinc^3 and sinc^4 (Theorem 19.4(iv)), are integrable because $|\text{sinc}| \leq 1$ and $\text{sinc}^2 \in L^1$. Evaluating at 0: $\int \text{sinc}^3 \xi d\xi = (\Pi * \Lambda)(0) = \int_{-1/2}^{1/2} \Lambda(s) ds = 2 \int_0^{1/2} (1-s) ds = \frac{3}{4}$; and from $\Lambda * \Lambda$ evaluated at 0: $\int \text{sinc}^4 \xi d\xi = \int \Lambda(s)^2 ds = 2 \int_0^1 (1-s)^2 ds = \frac{2}{3}$.

Independent checks. Simpson quadrature on $[-3000, 3000]$ with 6,000,000 subintervals: $\int \text{sinc}^3 = 0.7500000$ and $\int \text{sinc}^4 = 0.6666667$. ✓ For $\int \text{sinc}^2$, the same quadrature on $[-4000, 4000]$ with 8,000,000 subintervals gives 0.9999747: the discrepancy $2.5 \cdot 10^{-5}$ is exactly the neglected tail, $2 \int_{4000}^{\infty} \frac{\sin^2 \pi x}{\pi^2 x^2} dx \approx \frac{1}{\pi^2 \cdot 4000} = 2.5 \cdot 10^{-5}$ (the mean value of $\sin^2$ is $\frac{1}{2}$). ✓ Rewritten with $x = \pi \xi$, these three integrals are the classical identities $\int \frac{\sin^2 x}{x^2} dx = \pi$, $\int \frac{\sin^3 x}{x^3} dx = \frac{3\pi}{4} = 2.3561945$ and $\int \frac{\sin^4 x}{x^4} dx = \frac{2\pi}{3} = 2.0943951$, verified numerically with the same parameters. ✓

Remark 19.8: What we import: the extension of Plancherel to L^2 . Corollary 19.4 says that $\mathcal{F}$ preserves the L^2 norm on the Schwartz class. The statement used in practice — there is a unique linear map $\mathcal{F}_2 : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ that agrees with $\mathcal{F}$ on $\mathcal{S}$, and is a surjective isometry — is obtained by a density argument, and is **imported** here. Precisely, the one fact we do not prove is the density of $\mathcal{S}(\mathbb{R})$ in $L^2(\mathbb{R})$. Given that, the rest is mechanical, and we record it for completeness: if $f \in L^2$ and $(f_k) \subset \mathcal{S}$ with $\|f_k - f\|_2 \rightarrow 0$, then (f_k) is Cauchy in L^2 , hence by Corollary 19.4 so is $(\hat{f}_k)$ ($\|\hat{f}_k - \hat{f}_j\|_2 = \|f_k - f_j\|_2$), and the completeness of

L^2 gives a limit called $\mathcal{F}_2 f$; the limit does not depend on the chosen sequence (interleave two sequences) and norm preservation passes to the limit. Surjectivity follows from $\mathcal{F}(\mathcal{S}) = \mathcal{S}$ (Corollary 19.3) and from the fact that the image of an isometry of a complete space is closed. Hence only density is missing, which is the measure-theory point declared among the external prerequisites; it is found in Rudin, *Real and Complex Analysis*, Chapter 9. Note that $\mathcal{F}_2 f$ is no longer defined by a convergent integral for every $f \in L^2$: for $f \in L^1 \cap L^2$ the two definitions agree, but for $f \in L^2 \setminus L^1$ the integral of Definition 19.1 may fail to converge.

19.10. Periodization and the Poisson summation formula

Definition 19.6: Periodization of a function. Let $f : \mathbb{R} \rightarrow \mathbb{C}$. Its **periodization** of step 1 is

$$(Pf)(t) = \sum_{n \in \mathbb{Z}} f(t + n),$$

defined at the points where the series converges.

Theorem 19.11: The coefficients of the periodization are the samples of the transform. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be continuous and such that, for some constants $C > 0$ and $\varepsilon > 0$,

$$|f(t)| \leq \frac{C}{(1 + |t|)^{1+\varepsilon}} \quad \text{for every } t \in \mathbb{R}. \quad (\text{H1})$$

Then:

- (i) the series $\sum_n f(t + n)$ converges absolutely and uniformly on every bounded interval, and Pf is continuous and 1-periodic;
- (ii) $f \in L^1(\mathbb{R})$;
- (iii) the Fourier coefficients of Pf with respect to period 1 are the integer samples of $\hat{f}$:

$$c_n(Pf) := \int_0^1 (Pf)(t) e^{-2\pi i n t} dt = \hat{f}(n) \quad \text{for every } n \in \mathbb{Z}.$$

Proof. Step 1 — uniform convergence on $[-1, 2]$. Let $t \in [-1, 2]$ and $|n| \geq 4$. Then $|t + n| \geq |n| - |t| \geq |n| - 2 \geq \frac{|n|}{2}$, hence by (H1)

$$|f(t + n)| \leq \frac{C}{(1 + |n|/2)^{1+\varepsilon}} \leq \frac{C 2^{1+\varepsilon}}{|n|^{1+\varepsilon}}.$$

The series $\sum_{|n| \geq 4} |n|^{-1-\varepsilon}$ converges (comparison with the integral, since $1 + \varepsilon > 1$), so by the Weierstrass criterion the series $\sum_{|n| \geq 4} f(t + n)$ converges absolutely and uniformly on $[-1, 2]$; the seven remaining terms are continuous functions. Hence Pf is defined and continuous on $[-1, 2]$, and in particular on $[0, 1]$.

Step 2 — periodicity. By absolute convergence one may reorder: $(Pf)(t + 1) = \sum_n f(t + 1 + n) = \sum_m f(t + m) = (Pf)(t)$ (setting $m = n + 1$). Hence Pf , continuous on $[-1, 2]$ and 1-periodic, is continuous on all of $\mathbb{R}$; and the argument of Step 1, repeated on $[-M - 1, M + 2]$ for every M , gives uniform convergence on every bounded interval.

Step 3 — (ii). From (H1), $\int |f| \leq C \int \frac{dt}{(1 + |t|)^{1+\varepsilon}} = 2C \int_0^\infty \frac{ds}{(1 + s)^{1+\varepsilon}} = \frac{2C}{\varepsilon} < \infty$.

Step 4 — (iii). The convergence is uniform on $[0, 1]$ (Step 1), and $|e^{-2\pi i n t}| = 1$, so the series and the integral may be interchanged:

$$c_n(Pf) = \int_0^1 \left(\sum_{k \in \mathbb{Z}} f(t+k) \right) e^{-2\pi i n t} dt = \sum_{k \in \mathbb{Z}} \int_0^1 f(t+k) e^{-2\pi i n t} dt.$$

In the k -th integral set $u = t + k$, so that $t = u - k$ and $e^{-2\pi i n t} = e^{-2\pi i n u} e^{2\pi i n k} = e^{-2\pi i n u}$, because $n k \in \mathbb{Z}$ and $e^{2\pi i \ell} = 1$ for every integer ℓ . Hence

$$c_n(Pf) = \sum_{k \in \mathbb{Z}} \int_k^{k+1} f(u) e^{-2\pi i n u} du = \int_{-\infty}^{+\infty} f(u) e^{-2\pi i n u} du = \hat{f}(n),$$

where the last equality is the σ -additivity of the integral over $\mathbb{R} = \bigcup_k [k, k+1)$, legitimate because $f \in L^1$ (Step 3). $\square$

Theorem 19.12: Poisson summation formula. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be continuous and suppose there exist $C > 0$ and $\varepsilon > 0$ with

$$|f(t)| \leq \frac{C}{(1+|t|)^{1+\varepsilon}} \quad (H1) \quad \text{and} \quad |\hat{f}(\xi)| \leq \frac{C}{(1+|\xi|)^{1+\varepsilon}} \quad (H2)$$

for every $t, \xi \in \mathbb{R}$. Then, for every $t \in \mathbb{R}$,

$$\sum_{n \in \mathbb{Z}} f(t+n) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n t},$$

and the two series converge absolutely and uniformly on $\mathbb{R}$. In particular, for $t = 0$,

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n).$$

Proof. **Step 1 — the right-hand side is a continuous function.** From (H2), $\sum_n |\hat{f}(n)| \leq C \sum_n (1+|n|)^{-1-\varepsilon} < \infty$; since $|\hat{f}(n) e^{2\pi i n t}| = |\hat{f}(n)|$, the Weierstrass criterion gives absolute and uniform convergence on $\mathbb{R}$ of the series $S(t) = \sum_n \hat{f}(n) e^{2\pi i n t}$, which is therefore continuous and 1-periodic.

Step 2 — the coefficients of S . By uniform convergence one may interchange the series and the integral:

$$c_m(S) = \int_0^1 S(t) e^{-2\pi i m t} dt = \sum_{n \in \mathbb{Z}} \hat{f}(n) \int_0^1 e^{2\pi i (n-m)t} dt = \hat{f}(m),$$

because $\int_0^1 e^{2\pi i \ell t} dt$ equals 1 for $\ell = 0$ and $\left[\frac{e^{2\pi i \ell t}}{2\pi i \ell} \right]_0^1 = 0$ for $\ell \in \mathbb{Z} \setminus \{0\}$.

Step 3 — the coefficients of Pf . By Theorem 19.11, Pf is continuous, 1-periodic and $c_m(Pf) = \hat{f}(m)$ for every m .

Step 4 — uniqueness of coefficients. Let $h = Pf - S$: it is continuous, 1-periodic and has all coefficients zero. To compare it with Chapter 17, which works on $\mathbb{T}$ with period 2π , set $H(\theta) = h\left(\frac{\theta}{2\pi}\right)$, a continuous, 2π -periodic function. With the substitution $\theta = 2\pi t$,

$$\hat{H}(n) = \frac{1}{2\pi} \int_0^{2\pi} H(\theta) e^{-in\theta} d\theta = \frac{1}{2\pi} \int_0^1 h(t) e^{-2\pi i n t} 2\pi dt = \int_0^1 h(t) e^{-2\pi i n t} dt = c_n(h) = 0$$

for every $n \in \mathbb{Z}$. Since H is continuous, Corollary 17.2 of Chapter 17 gives $H \equiv 0$, that is, $h \equiv 0$ and $Pf = S$ everywhere. $\square$

Corollary 19.5: Jacobi's identity for the theta function. Setting $\vartheta(a) = \sum_{n \in \mathbb{Z}} e^{-\pi a n^2}$ for $a > 0$, we have

$$\vartheta(a) = \frac{1}{\sqrt{a}} \vartheta\left(\frac{1}{a}\right).$$

Proof. Apply Theorem 19.12 at $t = 0$ to $f(t) = e^{-\pi a t^2}$.

Checking the hypotheses. f is continuous; for (H1), for every $\varepsilon > 0$ there is C with $e^{-\pi a t^2} \leq C(1 + |t|)^{-1-\varepsilon}$, because $(1 + |t|)^{1+\varepsilon} e^{-\pi a t^2} \rightarrow 0$ as $|t| \rightarrow \infty$ and is a continuous function, hence bounded. By Corollary 19.2, $\hat{f}(\xi) = a^{-1/2} e^{-\pi \xi^2/a}$, which satisfies (H2) for the same reason.

Conclusion. The Poisson formula gives

$$\sum_{n \in \mathbb{Z}} e^{-\pi a n^2} = \sum_{n \in \mathbb{Z}} \frac{1}{\sqrt{a}} e^{-\pi n^2/a},$$

that is, $\vartheta(a) = a^{-1/2} \vartheta(1/a)$.

Independent check. Sums truncated at $|n| \leq 60$ (the further terms are less than $e^{-\pi \cdot 3600}$):

a	$\vartheta(a)$	$a^{-1/2} \vartheta(1/a)$
1	1.0864348	1.0864348
2	1.0037349	1.0037349
$\frac{1}{2}$	1.4194955	1.4194955
3	1.0001614	1.0001614

✓ The case $a = 1$ is the fixed point: $\vartheta(1) = 1 + 2e^{-\pi} + 2e^{-4\pi} + \dots = 1 + 2 \cdot 0.0432139 + 2 \cdot 0.0000035 + \dots = 1.0864348$, and the relation $\vartheta(1) = \vartheta(1)$ holds identically. ✓ $\square$

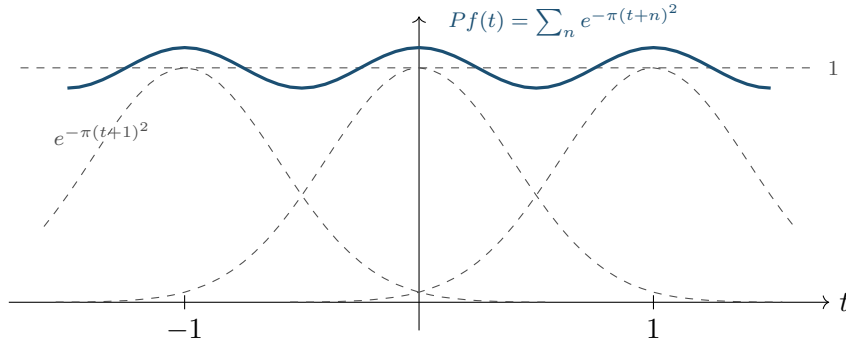


Figure 19.2: Periodization of the Gaussian $f(t) = e^{-\pi t^2}$. The dashed curves are the translates $f(t + n)$ for $n = -1, 0, 1$; the thick curve is their sum over all $n \in \mathbb{Z}$, which is 1-periodic and oscillates between $Pf(\frac{1}{2}) = 0.9135791$ and $Pf(0) = 1.0864348 = \vartheta(1)$. The Poisson formula (Theorem 19.12) says that the Fourier coefficients of this curve are the samples $\hat{f}(n) = e^{-\pi n^2}$: 1 , $e^{-\pi} = 0.0432139$, $e^{-4\pi} = 0.0000035$, $\dots$, and indeed $1 + 2(0.0432139) + 2(0.0000035) = 1.0864348$ and $1 - 2(0.0432139) + 2(0.0000035) = 0.9135791$. ✓

19.11. The uncertainty principle

Theorem 19.13: Uncertainty principle, centered form. Let $f \in \mathcal{S}(\mathbb{R})$ with $\|f\|_2 = 1$. Then

$$\left(\int t^2 |f(t)|^2 dt \right) \left(\int \xi^2 |\hat{f}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2}.$$

Equality holds if and only if $f(t) = A e^{ct^2/2}$ for some real $c < 0$ and $A \in \mathbb{C}$ with $|A|^2 \sqrt{-2\pi/c} = 1$; equivalently, if and only if $f(t) = A e^{-\pi a t^2}$ with $a > 0$ and $|A|^2 = \sqrt{2a}$.

Proof. Step 0 — all quantities are finite. $f \in \mathcal{S}$, so $tf(t)$ and f' lie in $\mathcal{S} \subset L^2$ (Proposition 19.5), and $\hat{f} \in \mathcal{S}$ (Theorem 19.7), so $\xi \hat{f}(\xi) \in L^2$ as well. If $f' \equiv 0$ then f is constant, and the only constant in $L^2(\mathbb{R})$ is 0, which contradicts $\|f\|_2 = 1$: hence $f' \not\equiv 0$. Likewise $tf(t) \not\equiv 0$, otherwise f would vanish away from the origin and hence $\|f\|_2 = 0$.

Step 1 — integration by parts. The function $|f|^2 = f\bar{f}$ is of class C^1 with

$$\frac{d}{dt} |f(t)|^2 = f'(t)\overline{f(t)} + f(t)\overline{f'(t)} = 2\operatorname{Re}(f'(t)\overline{f(t)}).$$

Integrating $1 \cdot |f|^2$ by parts on $[-T, T]$, with t as the primitive of the factor 1:

$$\int_{-T}^T |f(t)|^2 dt = \left[t |f(t)|^2 \right]_{-T}^T - \int_{-T}^T t \cdot 2\operatorname{Re}(f'(t)\bar{f}) dt.$$

The boundary term tends to 0: indeed $|t| |f(t)|^2 = |tf(t)| \cdot |f(t)|$, and for $t \neq 0$ one has $|tf(t)| = \frac{|t^3 f(t)|}{t^2} \leq \frac{p_{3,0}(f)}{t^2} \rightarrow 0$ while $|f(t)| \leq p_{0,0}(f)$ stays bounded. Letting $T \rightarrow +\infty$,

$$1 = \|f\|_2^2 = -2\operatorname{Re} \int_{\mathbb{R}} t f'(t) \overline{f(t)} dt. \quad (**)$$

Step 2 — Cauchy–Schwarz. Since t is real, $tf'(t)\overline{f(t)} = f'(t)\overline{tf(t)}$, that is, the integral in (**) is the scalar product $\langle f', tf \rangle$ in L^2 . Hence

$$1 = |-2\operatorname{Re}\langle f', tf \rangle| \leq 2|\langle f', tf \rangle| \leq 2\|f'\|_2 \|tf\|_2. \quad (***)$$

Step 3 — Plancherel on the derivative. $f \in \mathcal{S}$, so $f' \in \mathcal{S}$ and Lemma 19.1 gives $\widehat{f'}(\xi) = 2\pi i \xi \hat{f}(\xi)$. By Corollary 19.4,

$$\|f'\|_2 = \|\widehat{f'}\|_2 = \|2\pi i \xi \hat{f}(\xi)\|_2 = 2\pi \|\xi \hat{f}\|_2.$$

Step 4 — concluding the inequality. Substituting into (***): $1 \leq 4\pi \|tf\|_2 \|\xi \hat{f}\|_2$, and squaring both sides (both are positive)

$$\|tf\|_2^2 \|\xi \hat{f}\|_2^2 \geq \frac{1}{16\pi^2},$$

which is the claim.

Step 5 — equality implies the Gaussian. If equality holds, then both inequalities of Step 2 are equalities.

- Equality in Cauchy–Schwarz, with $f' \neq 0$ and $tf \neq 0$ (Step 0), means that the two vectors are proportional: there is $c \in \mathbb{C}$ with $f'(t) = c tf(t)$ for every t (the identity holds almost everywhere, but both sides are continuous).
- Then $\langle f', tf \rangle = c \|tf\|_2^2$, and (**) becomes $1 = -2\operatorname{Re}(c) \|tf\|_2^2$, which forces $\operatorname{Re}(c) < 0$. The equality $|-2\operatorname{Re}\langle f', tf \rangle| = 2|\langle f', tf \rangle|$ becomes $2|\operatorname{Re}(c)| \|tf\|_2^2 = 2|c| \|tf\|_2^2$, that is, $|\operatorname{Re}(c)| = |c|$: hence c is real, and negative.

Setting $c < 0$, one has $\frac{d}{dt}(f(t)e^{-ct^2/2}) = e^{-ct^2/2}(f'(t) - ctf(t)) = 0$, hence $f(t) = Ae^{ct^2/2}$ with $A \in \mathbb{C}$: a Gaussian. Setting $a = -\frac{c}{2\pi} > 0$ gives $e^{ct^2/2} = e^{-\pi at^2}$; the condition $\|f\|_2 = 1$ becomes, by Lemma 19.5 with $b = 2a$, $|A|^2 \int e^{-2\pi at^2} dt = |A|^2 (2a)^{-1/2} = 1$, that is, $|A|^2 = \sqrt{2a}$. **Step 6 — the Gaussian achieves equality.** Let $f(t) = Ae^{-\pi at^2}$ with $a > 0$ and $|A|^2 = \sqrt{2a}$. By Lemma 19.5 with $b = 2a$,

$$\int t^2 |f|^2 dt = |A|^2 \int t^2 e^{-2\pi at^2} dt = \sqrt{2a} \cdot \frac{1}{2\pi(2a)^{3/2}} = \frac{1}{2\pi \cdot 2a} = \frac{1}{4\pi a}.$$

By Corollary 19.2, $\hat{f}(\xi) = A a^{-1/2} e^{-\pi \xi^2/a}$, hence, applying the same formula with $a \rightsquigarrow \frac{1}{a}$ and amplitude $Aa^{-1/2}$,

$$\int \xi^2 |\hat{f}|^2 d\xi = \frac{|A|^2}{a} \cdot \frac{1}{2\pi(2/a)^{3/2}} = \frac{\sqrt{2a}}{a} \cdot \frac{a^{3/2}}{2\pi 2^{3/2}} = \frac{\sqrt{2} a^{1/2} a^{1/2}}{2\pi \cdot 2\sqrt{2}} = \frac{a}{4\pi}.$$

The product equals $\frac{1}{4\pi a} \cdot \frac{a}{4\pi} = \frac{1}{16\pi^2}$, *independently of a* : equality is attained.

Independent check. For $a = 1$, that is $f(t) = 2^{1/4} e^{-\pi t^2}$: Simpson quadrature on $[-12, 12]$ with 200,000 subintervals gives $\|f\|_2^2 = 1.0000000$ and $\int t^2 |f|^2 = 0.0795775 = \frac{1}{4\pi}$. Since $\hat{f} = f$ (Theorem 19.6), also $\int \xi^2 |\hat{f}|^2 = 0.0795775$, and the product is $0.0063326 = \frac{1}{16\pi^2}$. ✓ □

Corollary 19.6: General form, with arbitrary centers. Let $f \in \mathcal{S}(\mathbb{R})$ with $\|f\|_2 = 1$ and let $t_0, \xi_0 \in \mathbb{R}$. Then

$$\left(\int (t - t_0)^2 |f(t)|^2 dt \right) \left(\int (\xi - \xi_0)^2 |\hat{f}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2},$$

with equality if and only if $f(t) = A e^{2\pi i \xi_0 t} e^{-\pi a(t-t_0)^2}$ for some $a > 0$ and $|A|^2 = \sqrt{2a}$.

Proof. Step 1 — normalization. Set $g(t) = e^{-2\pi i \xi_0 t} f(t + t_0)$. Then $g \in \mathcal{S}$ (translation and multiplication by a C^∞ function with bounded derivatives) and $|g(t)| = |f(t + t_0)|$, hence $\|g\|_2 = \|f\|_2 = 1$.

Step 2 — the transform of g . Set $h(t) = f(t + t_0) = \tau_{-t_0} f(t)$; Theorem 19.3(ii) with $t_0 \rightsquigarrow -t_0$ gives $\hat{h}(\xi) = e^{2\pi i \xi t_0} \hat{f}(\xi)$. Then $g = e^{-2\pi i \xi_0 t} h = m_{-\xi_0} h$, and Theorem 19.3(iii) gives $\hat{g}(\xi) = \hat{h}(\xi + \xi_0) = e^{2\pi i (\xi + \xi_0) t_0} \hat{f}(\xi + \xi_0)$. In particular $|\hat{g}(\xi)| = |\hat{f}(\xi + \xi_0)|$.

Step 3 — translating the integrals. With the substitutions $s = t + t_0$ and $\eta = \xi + \xi_0$,

$$\int t^2 |g(t)|^2 dt = \int (s - t_0)^2 |f(s)|^2 ds, \quad \int \xi^2 |\hat{g}(\xi)|^2 d\xi = \int (\eta - \xi_0)^2 |\hat{f}(\eta)|^2 d\eta.$$

Applying Theorem 19.13 to g gives the inequality of the statement.

Step 4 — the equality case. Equality for f is equivalent to equality for g , that is, by Theorem 19.13, to $g(t) = Ae^{-\pi at^2}$ with $|A|^2 = \sqrt{2a}$. Inverting the definition of g , $f(s) = e^{2\pi i \xi_0(s-t_0)} g(s - t_0) = Ae^{-2\pi i \xi_0 t_0} e^{2\pi i \xi_0 s} e^{-\pi a(s-t_0)^2}$, which is the form of the statement with A replaced by $Ae^{-2\pi i \xi_0 t_0}$, of equal modulus. □

19.12. Tempered distributions: the delta and the Dirac comb

Three objects that engineering uses constantly — the impulse, the constant, the train of impulses — are not integrable functions, and their "Fourier transform" has no meaning in Definition 19.1. The solution is not to enlarge the notion of function, but to change the object: one passes from pointwise values to *action on a test function*. This section builds the minimal framework needed, and in it the Dirac delta is never a function: it is a functional, and Proposition 19.8 shows that

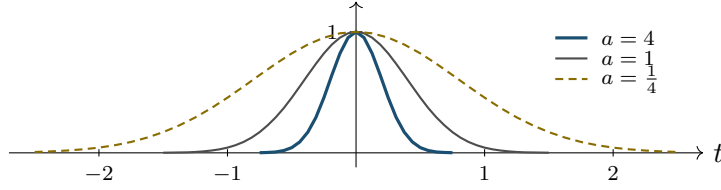


Figure 19.3: The Gaussians $e^{-\pi at^2}$ for $a = 4$, $a = 1$ and $a = \frac{1}{4}$: same height, widths in ratio 1 : 2 : 4. The transform of the narrowest one ($a = 4$) is $\frac{1}{2}e^{-\pi\xi^2/4}$ (Corollary 19.2), that is, the widest curve of the picture halved in height, and vice versa: narrowing in time widens in frequency, exactly in the proportion fixed by Theorem 19.13, for which the product of the two spreads equals $\frac{1}{16\pi^2} = 0.0063326$ for every a . Check values on the curve $a = 4$: $e^{-4\pi(0.25)^2} = e^{-\pi/4} = 0.4559381$ and $e^{-4\pi(0.5)^2} = e^{-\pi} = 0.0432139$. ✓

it cannot be anything else.

Definition 19.7: Tempered distribution. A **tempered distribution** is a map $T : \mathcal{S}(\mathbb{R}) \rightarrow \mathbb{C}$ such that

- (i) T is linear: $T(\alpha\varphi + \beta\psi) = \alpha T(\varphi) + \beta T(\psi)$ for $\alpha, \beta \in \mathbb{C}$ and $\varphi, \psi \in \mathcal{S}$;
- (ii) T is continuous: if $\varphi_k \rightarrow 0$ in $\mathcal{S}$ (that is, $p_{m,n}(\varphi_k) \rightarrow 0$ for every m, n , Definition 19.5), then $T(\varphi_k) \rightarrow 0$ in $\mathbb{C}$.

The set of tempered distributions is denoted $\mathcal{S}'(\mathbb{R})$, and one writes $\langle T, \varphi \rangle$ instead of $T(\varphi)$. The elements of $\mathcal{S}$ are called **test functions**.

Proposition 19.6: Slowly growing functions are tempered distributions. Let $g : \mathbb{R} \rightarrow \mathbb{C}$ be measurable and such that, for some integer $N \geq 0$,

$$\int \frac{|g(t)|}{(1+|t|)^N} dt < \infty.$$

Then $\langle T_g, \varphi \rangle = \int g(t)\varphi(t) dt$ defines a tempered distribution. In particular every $g \in L^1(\mathbb{R})$ ($N = 0$), every bounded function ($N = 2$), and every polynomial ($N = \deg + 2$) give a tempered distribution.

Proof. Step 1 — a pointwise estimate for test functions. Let $\varphi \in \mathcal{S}$ and $M = N + 2$. Since $1 + |t| \leq 2 \max(1, |t|)$, one has $(1 + |t|)^M \leq 2^M \max(1, |t|^M) \leq 2^M (1 + |t|^M)$, hence

$$(1 + |t|)^M |\varphi(t)| \leq 2^M (|\varphi(t)| + |t|^M |\varphi(t)|) \leq 2^M (p_{0,0}(\varphi) + p_{M,0}(\varphi)) =: A(\varphi),$$

that is, $|\varphi(t)| \leq A(\varphi)(1 + |t|)^{-M}$ for every t .

Step 2 — convergence of the integral. Since $M = N + 2$,

$$\int |g(t)\varphi(t)| dt \leq A(\varphi) \int \frac{|g(t)|}{(1+|t|)^{N+2}} dt \leq A(\varphi) \int \frac{|g(t)|}{(1+|t|)^N} dt < \infty,$$

having used $(1 + |t|)^{N+2} \geq (1 + |t|)^N$. Hence $T_g(\varphi)$ is well defined.

Step 3 — linearity and continuity. Linearity is that of the integral. Writing $B = \int |g|(1 + |t|)^{-N} dt$, Step 2 gives $|\langle T_g, \varphi \rangle| \leq B A(\varphi) = 2^M B (p_{0,0}(\varphi) + p_{M,0}(\varphi))$. If $\varphi_k \rightarrow 0$ in $\mathcal{S}$, then $p_{0,0}(\varphi_k) \rightarrow 0$ and $p_{M,0}(\varphi_k) \rightarrow 0$, hence $\langle T_g, \varphi_k \rangle \rightarrow 0$.

Step 4 — the three cases. If $g \in L^1$, $N = 0$ suffices. If $|g| \leq K$, then $\int \frac{|g|}{(1+|t|)^2} \leq K\pi < \infty$ (see Step 5 of the proof of Proposition 19.5, with $1 + t^2 \leq (1 + |t|)^2$). If g is a polynomial of degree d , then $|g(t)| \leq K(1 + |t|)^d$ for some constant K , and with $N = d + 2$ the integral converges. □

Definition 19.8: Dirac delta. The **Dirac delta** centered at $x_0 \in \mathbb{R}$ is the functional

$$\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0), \quad \varphi \in \mathcal{S}(\mathbb{R});$$

one writes $\delta = \delta_0$.

Proposition 19.7: δ_{x_0} is a tempered distribution. $\delta_{x_0} \in \mathcal{S}'(\mathbb{R})$ for every $x_0 \in \mathbb{R}$.

Proof. Linearity is immediate: $(\alpha\varphi + \beta\psi)(x_0) = \alpha\varphi(x_0) + \beta\psi(x_0)$. For continuity, $|\langle \delta_{x_0}, \varphi \rangle| = |\varphi(x_0)| \leq \sup_t |\varphi(t)| = p_{0,0}(\varphi)$; hence if $\varphi_k \rightarrow 0$ in $\mathcal{S}$, in particular $p_{0,0}(\varphi_k) \rightarrow 0$ and therefore $\langle \delta_{x_0}, \varphi_k \rangle \rightarrow 0$. $\square$

Proposition 19.8: The delta is not a function. There is no $g \in L^1_{\text{loc}}(\mathbb{R})$ such that $\int |g\varphi| < \infty$ for every $\varphi \in \mathcal{S}(\mathbb{R})$ and

$$\int g(t)\varphi(t) dt = \varphi(0) \quad \text{for every } \varphi \in \mathcal{S}(\mathbb{R}).$$

Proof. Suppose, for contradiction, that such a g exists.

Step 1 — the test family. For $\delta \in (0, 1]$ set $\varphi_\delta(t) = e^{-\pi t^2/\delta^2}$. Each φ_δ lies in $\mathcal{S}$: it is $g(t/\delta)$ with $g(t) = e^{-\pi t^2} \in \mathcal{S}$ (Example 19.1), and $\mathcal{S}$ is stable under dilation, since $p_{m,n}(g(\cdot/\delta)) = \delta^{m-n} p_{m,n}(g)$. Moreover $\varphi_\delta(0) = 1$, hence by hypothesis

$$\int g(t) e^{-\pi t^2/\delta^2} dt = 1 \quad \text{for every } \delta \in (0, 1]. \quad (\dagger)$$

Applying the hypothesis to $\varphi_1(t) = e^{-\pi t^2}$ also gives $B := \int |g(t)| e^{-\pi t^2} dt < \infty$.

Step 2 — the contribution away from the origin. Let $\eta \in (0, 1]$. For $|t| \geq \eta$ and $\delta \in (0, 1]$,

$$e^{-\pi t^2/\delta^2} = e^{-\pi t^2} e^{-\pi t^2(\frac{1}{\delta^2}-1)} \leq e^{-\pi t^2} e^{-\pi \eta^2(\frac{1}{\delta^2}-1)},$$

since $\frac{1}{\delta^2} - 1 \geq 0$ and $t^2 \geq \eta^2$. Hence

$$\left| \int_{|t| \geq \eta} g(t) e^{-\pi t^2/\delta^2} dt \right| \leq e^{-\pi \eta^2(\frac{1}{\delta^2}-1)} \int |g(t)| e^{-\pi t^2} dt = B e^{-\pi \eta^2(\frac{1}{\delta^2}-1)},$$

which tends to 0 as $\delta \rightarrow 0^+$, for fixed η .

Step 3 — the contribution near the origin. Since $g \in L^1([-1, 1])$ (it is locally integrable), absolute continuity of the integral gives $\int_{|t| < \eta} |g| \rightarrow 0$ as $\eta \rightarrow 0^+$. Moreover $e^{-\pi t^2/\delta^2} \leq 1$ everywhere, hence

$$\left| \int_{|t| < \eta} g(t) e^{-\pi t^2/\delta^2} dt \right| \leq \int_{|t| < \eta} |g|.$$

Step 4 — the contradiction. Let $\varepsilon = \frac{1}{2}$. By Step 3 choose $\eta \in (0, 1]$ with $\int_{|t| < \eta} |g| < \frac{1}{4}$; by Step 2, with that η fixed, choose $\delta \in (0, 1]$ with $B e^{-\pi \eta^2(\delta^{-2}-1)} < \frac{1}{4}$. Then, by $(\dagger)$,

$$1 = \left| \int g \varphi_\delta \right| \leq \left| \int_{|t| < \eta} g \varphi_\delta \right| + \left| \int_{|t| \geq \eta} g \varphi_\delta \right| < \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

which is absurd. Hence no g of the described type exists. $\square$

Remark 19.9: Why one never writes " $\delta(t)$ ". Proposition 19.8 settles the question: the notation $\delta(t)$, with the "rule" $\delta(t) = 0$ for $t \neq 0$, $\delta(0) = +\infty$ and $\int \delta = 1$, does not define any measurable function — a function that is zero almost everywhere has zero integral, whatever value is assigned to it at the origin. In this chapter δ appears only as a functional, and formulas are always written as pairings: $\langle \delta, \varphi \rangle = \varphi(0)$, never $\int \delta(t)\varphi(t)dt = \varphi(0)$. The integral notation remains a very convenient shorthand and is found everywhere in the technical literature (for instance in Bracewell, Chapter 5, where the "sifting property" is stated in that form), but its meaning is the definition given here, not a Lebesgue integral. The sequence of Gaussian kernels K_δ of Lemma 19.7 is the correct way to "approach" δ : by part (iii) of that lemma, $\int \varphi(t)K_\delta(t)dt \rightarrow \varphi(0) = \langle \delta, \varphi \rangle$ for every $\varphi \in \mathcal{S}$, that is, $T_{K_\delta} \rightarrow \delta$ in the sense of distributions.

Definition 19.9: Fourier transform of a tempered distribution. For $T \in \mathcal{S}'(\mathbb{R})$ one defines $\hat{T} = \mathcal{F}T \in \mathcal{S}'(\mathbb{R})$ by

$$\langle \hat{T}, \varphi \rangle = \langle T, \hat{\varphi} \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}).$$

Proposition 19.9: The definition is well posed and extends the one on L^1 . For every $T \in \mathcal{S}'(\mathbb{R})$, $\hat{T}$ is a tempered distribution. Moreover, if $f \in L^1(\mathbb{R})$, then $\hat{T}_f = T_{\hat{f}}$: Definition 19.9 extends Definition 19.1.

Proof. **Step 1 — $\hat{T}$ is defined.** By Theorem 19.7, $\varphi \in \mathcal{S}$ implies $\hat{\varphi} \in \mathcal{S}$: the right-hand side makes sense.

Step 2 — linearity. $\varphi \mapsto \hat{\varphi}$ is linear (Proposition 19.1) and T is linear, hence so is the composition.

Step 3 — continuity. If $\varphi_k \rightarrow 0$ in $\mathcal{S}$, the estimate of Theorem 19.7 gives $\hat{\varphi}_k \rightarrow 0$ in $\mathcal{S}$, and continuity of T gives $\langle T, \hat{\varphi}_k \rangle \rightarrow 0$.

Step 4 — consistency with L^1 . Let $f \in L^1$ and $\varphi \in \mathcal{S} \subset L^1$. By Lemma 19.6,

$$\langle \hat{T}_f, \varphi \rangle = \langle T_f, \hat{\varphi} \rangle = \int f(t)\hat{\varphi}(t)dt = \int \hat{f}(\xi)\varphi(\xi)d\xi = \langle T_{\hat{f}}, \varphi \rangle,$$

for every φ : the two functionals coincide. (T_f and $T_{\hat{f}}$ are tempered by Proposition 19.6, the latter because $\hat{f}$ is bounded.) $\square$

Proposition 19.10: $\hat{\delta} = 1$ and $\hat{1} = \delta$. In the sense of Definition 19.9,

$$\hat{\delta} = T_1, \quad \hat{T}_1 = \delta, \quad \text{and more generally} \quad \widehat{\delta_{x_0}} = T_{e_{x_0}} \quad \text{with } e_{x_0}(\xi) = e^{-2\pi i x_0 \xi}.$$

Proof. **Step 1 — $\hat{\delta} = T_1$.** For every $\varphi \in \mathcal{S}$,

$$\langle \hat{\delta}, \varphi \rangle = \langle \delta, \hat{\varphi} \rangle = \hat{\varphi}(0) = \int \varphi(t)e^{-2\pi i \cdot 0 \cdot t}dt = \int \varphi(t)dt = \langle T_1, \varphi \rangle.$$

Step 2 — $\hat{T}_1 = \delta$. The constant function 1 is bounded, hence $T_1 \in \mathcal{S}'$ (Proposition 19.6). For every $\varphi \in \mathcal{S}$,

$$\langle \hat{T}_1, \varphi \rangle = \langle T_1, \hat{\varphi} \rangle = \int \hat{\varphi}(\xi)d\xi = \int \hat{\varphi}(\xi)e^{2\pi i \cdot 0 \cdot \xi}d\xi = \varphi(0) = \langle \delta, \varphi \rangle,$$

where the second-to-last step is the inversion formula (Theorem 19.8) evaluated at $x = 0$.

Step 3 — the translated case. For every $\varphi \in \mathcal{S}$,

$$\langle \widehat{\delta_{x_0}}, \varphi \rangle = \hat{\varphi}(x_0) = \int \varphi(\xi) e^{-2\pi i x_0 \xi} d\xi = \langle T_{e_{x_0}}, \varphi \rangle,$$

and e_{x_0} is bounded, hence tempered. $\square$

Definition 19.10: Dirac comb. The **Dirac comb** of step 1 is the functional

$$\langle \mathbb{I}, \varphi \rangle = \sum_{n \in \mathbb{Z}} \varphi(n), \quad \varphi \in \mathcal{S}(\mathbb{R}),$$

formally the sum $\sum_{n \in \mathbb{Z}} \delta_n$.

Proposition 19.11: The comb is a tempered distribution. $\mathbb{I} \in \mathcal{S}'(\mathbb{R})$.

Proof. **Step 1 — the series converges.** For $\varphi \in \mathcal{S}$ and every t , $(1+t^2)|\varphi(t)| \leq p_{0,0}(\varphi) + p_{2,0}(\varphi) =: C(\varphi)$ (Step 5 of the proof of Proposition 19.5), hence

$$\sum_{n \in \mathbb{Z}} |\varphi(n)| \leq C(\varphi) \sum_{n \in \mathbb{Z}} \frac{1}{1+n^2}.$$

The latter series converges: its terms are $\leq n^{-2}$ for $n \neq 0$. (Its value is $\pi \coth \pi = 3.1533481$, computed in Solved Exercise 19.6.)

Step 2 — linearity and continuity. Linearity is clear. By Step 1, $|\langle \mathbb{I}, \varphi \rangle| \leq 3.16 C(\varphi) = 3.16(p_{0,0}(\varphi) + p_{2,0}(\varphi))$, hence $\varphi_k \rightarrow 0$ in $\mathcal{S}$ implies $\langle \mathbb{I}, \varphi_k \rangle \rightarrow 0$. $\square$

Theorem 19.14: The Dirac comb is its own transform. $\widehat{\mathbb{I}} = \mathbb{I}$.

Proof. **Step 1 — test functions satisfy the hypotheses of Poisson.** Let $\varphi \in \mathcal{S}$. Then φ is continuous and, as in Step 5 of the proof of Proposition 19.5, $|\varphi(t)| \leq \frac{C(\varphi)}{1+t^2} \leq \frac{2C(\varphi)}{(1+|t|)^2}$ (because $(1+|t|)^2 \leq 2(1+t^2)$): this is hypothesis (H1) of Theorem 19.12 with $\varepsilon = 1$. By Theorem 19.7 also $\hat{\varphi} \in \mathcal{S}$, hence the same estimate holds for $\hat{\varphi}$: this is hypothesis (H2).

Step 2 — Poisson at $t = 0$. Theorem 19.12 gives

$$\sum_{n \in \mathbb{Z}} \varphi(n) = \sum_{n \in \mathbb{Z}} \hat{\varphi}(n).$$

Step 3 — conclusion. For every $\varphi \in \mathcal{S}$,

$$\langle \widehat{\mathbb{I}}, \varphi \rangle = \langle \mathbb{I}, \hat{\varphi} \rangle = \sum_{n \in \mathbb{Z}} \hat{\varphi}(n) \stackrel{\text{Step 2}}{=} \sum_{n \in \mathbb{Z}} \varphi(n) = \langle \mathbb{I}, \varphi \rangle,$$

hence the two functionals coincide. $\square$

Remark 19.10: Sampling: what can and cannot be said here. Theorem 19.14 is the algebraic core of sampling theory: multiplying a signal by a comb of step T amounts, in frequency, to convolving its spectrum with a comb of step $1/T$, that is, to periodizing it — which is Theorem 19.11 read the other way. The sampling theorem of Whittaker–Kotel’nikov–Shannon (a signal band-limited to $[-\frac{B}{2}, \frac{B}{2}]$ is reconstructed exactly from its samples taken at step $1/B$, by interpolation with sinc) follows from here, but its complete proof requires the product of a tempered distribution by a slowly growing C^∞ function and the convolution of distributions, which this chapter does not build: it is a **panorama**, not a theorem

proved here. It is found, with the full distributional apparatus, in Bracewell, Chapter 10 "Sampling and Series." Chapter 21 and Chapter 24 take it up again from the point of view of the discrete signal.

19.13. The table of transforms, row by row

Theorem 19.15: Table of Fourier transforms. In the convention $\hat{f}(\xi) = \int f(t)e^{-2\pi i \xi t} dt$ the following equalities hold, for every $\xi \in \mathbb{R}$ and with $a > 0$, $t_0, \xi_0 \in \mathbb{R}$, $\alpha \in \mathbb{R} \setminus \{0\}$:

	$f(t)$	$\hat{f}(\xi)$	where
1	$\Pi(t)$	$\text{sinc } \xi$	Prop. 19.2
2	$\Lambda(t)$	$\text{sinc}^2 \xi$	Prop. 19.3
3	$e^{-\pi t^2}$	$e^{-\pi \xi^2}$	Thm. 19.6
4	$e^{-\pi a t^2}$	$a^{-1/2} e^{-\pi \xi^2 / a}$	Cor. 19.2
5	$t e^{-\pi t^2}$	$-i \xi e^{-\pi \xi^2}$	(a)
6	$e^{-a t }$	$\frac{2a}{a^2 + 4\pi^2 \xi^2}$	(b)
7	$\frac{1}{\pi} \frac{a}{a^2 + t^2}$	$e^{-2\pi a \xi }$	(c)
8	$e^{-at} u(t)$	$\frac{1}{a + 2\pi i \xi}$	(d)
9	$f(t - t_0)$	$e^{-2\pi i \xi t_0} \hat{f}(\xi)$	Thm. 19.3(ii)
10	$e^{2\pi i \xi_0 t} f(t)$	$\hat{f}(\xi - \xi_0)$	Thm. 19.3(iii)
11	$f(\alpha t)$	$ \alpha ^{-1} \hat{f}(\xi/\alpha)$	Thm. 19.3(iv)
12	$f(t) \cos(2\pi \xi_0 t)$	$\frac{1}{2} [\hat{f}(\xi - \xi_0) + \hat{f}(\xi + \xi_0)]$	(e)
13	$\Pi(t) \cos(2\pi \xi_0 t)$	$\frac{1}{2} [\text{sinc}(\xi - \xi_0) + \text{sinc}(\xi + \xi_0)]$	(f)
14	$f'(t)$	$2\pi i \xi \hat{f}(\xi)$	Lem. 19.1
15	$-2\pi i t f(t)$	$\hat{f}'(\xi)$	Lem. 19.2
16	$(f * g)(t)$	$\hat{f}(\xi) \hat{g}(\xi)$	Thm. 19.4(iv)
17	$R_f(t)$	$ \hat{f}(\xi) ^2$	Cor. 19.1
18	δ	1	Prop. 19.10
19	1	δ	Prop. 19.10
20	III	III	Thm. 19.14

Rows 18–20 are to be understood in the sense of Definition 19.9.

Proof. Rows 1–4, 9–11 and 14–20 are exactly the results indicated in the right-hand column, and have already been proved. The six derivations (a)–(f) remain.

(a) — **row 5.** Lemma 19.2 applied to $g(t) = e^{-\pi t^2}$ (the hypotheses are checked in Step 1 of the proof of Theorem 19.6) gives $(\widehat{tg(t)})(\xi) = \frac{i}{2\pi} \hat{g}'(\xi)$. By Theorem 19.6 $\hat{g}(\xi) = e^{-\pi \xi^2}$, whose derivative is $-2\pi \xi e^{-\pi \xi^2}$. Hence

$$(\widehat{te^{-\pi t^2}})(\xi) = \frac{i}{2\pi} (-2\pi \xi e^{-\pi \xi^2}) = -i \xi e^{-\pi \xi^2}.$$

The result is purely imaginary and odd, as it must be by Theorem 19.3, since $te^{-\pi t^2}$ is real and odd. ✓ Numerical check (Simpson on $[-12, 12]$, 400,000 subintervals): $\xi = \frac{1}{2}$ gives $-0.2279691 i$ against $-\frac{1}{2} e^{-\pi/4} = -0.2279691$; $\xi = 1$ gives $-0.0432139 i$ against $-e^{-\pi} = -0.0432139$. ✓

(b) — **row 6.** The function $e^{-a|t|}$ is in L^1 (integral $2/a$) and is even, hence

$$(\widehat{e^{-a|t|}})(\xi) = \int_{-\infty}^0 e^{at} e^{-2\pi i \xi t} dt + \int_0^{+\infty} e^{-at} e^{-2\pi i \xi t} dt = \int_0^{+\infty} e^{-at} (e^{2\pi i \xi t} + e^{-2\pi i \xi t}) dt,$$

having substituted $t \rightarrow -t$ in the first integral. Each of the two summands is computed with the primitive of a complex exponential: for $z \in \mathbb{C}$ with $\operatorname{Re} z > 0$,

$$\int_0^{+\infty} e^{-zt} dt = \left[\frac{e^{-zt}}{-z} \right]_0^{+\infty} = \frac{1}{z},$$

because $|e^{-zt}| = e^{-t \operatorname{Re} z} \rightarrow 0$. With $z = a - 2\pi i \xi$ and $z = a + 2\pi i \xi$ (both with real part $a > 0$),

$$\widehat{(e^{-a|t|})}(\xi) = \frac{1}{a - 2\pi i \xi} + \frac{1}{a + 2\pi i \xi} = \frac{(a + 2\pi i \xi) + (a - 2\pi i \xi)}{a^2 + 4\pi^2 \xi^2} = \frac{2a}{a^2 + 4\pi^2 \xi^2}.$$

This is the translation into convention (A) of the result of Inherited Exercise 19.1, which gives $\frac{2a}{a^2 + \omega^2}$: substituting $\omega = 2\pi \xi$ gives the same formula. ✓ Numerical check (Simpson on $[-80, 80]$, 800,000 subintervals): $a = 1, \xi = 0 \rightarrow 2.0000000$ against 2; $a = 1, \xi = \frac{1}{2} \rightarrow 0.1839993$ against $\frac{2}{1 + \pi^2} = 0.1839993$; $a = 2, \xi = 1 \rightarrow 0.0919997$ against $\frac{4}{4 + 4\pi^2} = 0.0919997$. ✓

(c) — **row 7.** Set $g(t) = e^{-2\pi a|t|}$ and $h(t) = \frac{1}{\pi} \frac{a}{a^2 + t^2}$. By (b) with the parameter $2\pi a$ in place of a ,

$$\hat{g}(\xi) = \frac{2 \cdot 2\pi a}{4\pi^2 a^2 + 4\pi^2 \xi^2} = \frac{4\pi a}{4\pi^2(a^2 + \xi^2)} = \frac{1}{\pi} \frac{a}{a^2 + \xi^2} = h(\xi).$$

Now $g \in L^1$ and $\hat{g} = h \in L^1$ (the integral of h equals $\frac{1}{\pi} [\arctan \frac{\xi}{a}]_{-\infty}^{+\infty} = 1$), and g is continuous: Theorem 19.9 gives, for every x ,

$$g(x) = \int h(\xi) e^{2\pi i x \xi} d\xi = \hat{h}(-x).$$

Since g is even, $\hat{h}(\xi) = g(-\xi) = g(\xi) = e^{-2\pi a|\xi|}$. ✓ Numerical check (Simpson on $[-20,000, 20,000]$, $4 \cdot 10^7$ subintervals): $a = 1, \xi = \frac{1}{2} \rightarrow 0.0432139$ and $a = \frac{1}{2}, \xi = 1 \rightarrow 0.0432139$, both against $e^{-\pi} = 0.0432139$. ✓

(d) — **row 8.** $e^{-at}u(t)$ equals e^{-at} for $t > 0$ and 0 for $t < 0$, and is in L^1 (integral $1/a$). With the same primitive as in (b), with $z = a + 2\pi i \xi$:

$$\widehat{(e^{-at}u(t))}(\xi) = \int_0^{+\infty} e^{-(a+2\pi i \xi)t} dt = \frac{1}{a + 2\pi i \xi}.$$

✓ Numerical check (Simpson on $[0, 80]$, 800,000 subintervals): $a = 1, \xi = 1$ gives $0.0247045 - 0.1552231i$, against $\frac{1}{1+2\pi i} = \frac{1-2\pi i}{1+4\pi^2} = 0.0247045 - 0.1552231i$; $a = 2, \xi = \frac{1}{2}$ gives $0.1442002 - 0.2265092i$, against $\frac{1}{2+\pi i} = \frac{2-\pi i}{4+\pi^2} = 0.1442002 - 0.2265092i$. ✓

(e) — **row 12.** From Euler's formula, $\cos(2\pi \xi_0 t) = \frac{1}{2}(e^{2\pi i \xi_0 t} + e^{-2\pi i \xi_0 t})$, hence by linearity (Proposition 19.1) and modulation (Theorem 19.3(iii), used with ξ_0 and with $-\xi_0$),

$$(f \cos(2\pi \xi_0 \cdot))(\xi) = \frac{1}{2} [\widehat{m_{\xi_0} f}(\xi) + \widehat{m_{-\xi_0} f}(\xi)] = \frac{1}{2} [\hat{f}(\xi - \xi_0) + \hat{f}(\xi + \xi_0)].$$

(f) — **row 13.** This is the case $f = \Pi$ of (e), with $\hat{\Pi} = \operatorname{sinc}$ (Proposition 19.2). ✓ Numerical check with $\xi_0 = 2$ (Simpson on $[-\frac{1}{2}, \frac{1}{2}]$, 400,000 subintervals): $\xi = 1.7 \rightarrow 0.3943971$; $\xi = 2 \rightarrow 0.5000000$; $\xi = 2.4 \rightarrow 0.4128146$; the closed-form values agree to seven decimals with the computed values, and the imaginary parts are below 10^{-16} , as must be the case for a real, even function (Theorem 19.3). ✓ □

Remark 19.11: How the table is used. Rows 9–17 are not transforms: they are rules, and they serve to manufacture new rows without ever redoing an integral. An example, to be worked out with the table alone: the transform of $t e^{-a|t|}$. By row 15, $\widehat{(t e^{-a|t|})}(\xi) =$

$\frac{i}{2\pi} \frac{d}{d\xi} \frac{2a}{a^2 + 4\pi^2 \xi^2} = \frac{i}{2\pi} \cdot \frac{-2a \cdot 8\pi^2 \xi}{(a^2 + 4\pi^2 \xi^2)^2} = \frac{-8\pi i a \xi}{(a^2 + 4\pi^2 \xi^2)^2}$, and the starting function is real and odd, hence the transform must be purely imaginary and odd (Theorem 19.3): it is. ✓ A second example: the transform of $\Pi(\frac{t-3}{2})$, that is, the rectangle of width 2 centered at 3; by rows 11 and 9, $\wedge = 2 e^{-6\pi i \xi} \text{sinc}(2\xi)$, and at $\xi = 0$ it equals 2, which is the area of the window. ✓

19.14. Exercises inherited from Volume II

The exercises that follow come from Volume II, Chapter 4, and concern the topics of this chapter. They are reproduced in full, with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 19.1. Compute the Fourier transform of $f(t) = e^{-a|t|}$, $a > 0$.

Solutions

Solution. We split the integral into $t < 0$ and $t > 0$, where $|t| = -t$ and $|t| = t$ respectively:

$$\hat{f}(\omega) = \int_{-\infty}^0 e^{at} e^{-i\omega t} dt + \int_0^{\infty} e^{-at} e^{-i\omega t} dt = \int_{-\infty}^0 e^{(a-i\omega)t} dt + \int_0^{\infty} e^{-(a+i\omega)t} dt.$$

The first integral (with $a > 0$, it is convenient to substitute $t \rightarrow -t$ to bring it to an integral over $[0, \infty)$):

$$\int_{-\infty}^0 e^{(a-i\omega)t} dt = \int_0^{\infty} e^{-(a-i\omega)t} dt = \frac{1}{a - i\omega},$$

and the second is $\frac{1}{a + i\omega}$ (standard integral of a decreasing exponential, convergent because $a > 0$). Adding:

$$\hat{f}(\omega) = \frac{1}{a - i\omega} + \frac{1}{a + i\omega} = \frac{(a + i\omega) + (a - i\omega)}{a^2 + \omega^2} = \frac{2a}{a^2 + \omega^2}.$$

This is the celebrated "Lorentzian" spectrum, which appears wherever exponential relaxations occur: RC circuits, radioactive decays, damped spectral lines.

19.15. Assigned exercises

These are *new* exercises, distinct both from the inherited one (Section 19.14) and from those solved in Section 19.16. The answers are in the Teacher's Key (Section 19.18).

Exercise 19.2: Translating a table between the three conventions. *Goal:* handle the dictionary of Remark 19.1 without getting a single factor of 2π wrong, and understand what changes and what does not. *Estimated time:* 40 minutes. *Type:* symbolic and numerical computation.

- Write row 6 of the table (Theorem 19.15), $\widehat{e^{-a|t|}}(\xi) = \frac{2a}{a^2 + 4\pi^2 \xi^2}$, in conventions (B) and (C).
- Verify Plancherel's identity for $f(t) = e^{-a|t|}$ in convention (A), that is, compute separately $\int |f|^2 dt$ and $\int |\hat{f}|^2 d\xi$ and show that they coincide. (Hint: one needs

$\int_{-\infty}^{+\infty} \frac{du}{(a^2+u^2)^2} = \frac{\pi}{2a^3}$, obtained with the substitution $u = a \tan \theta$.)

- (c) Repeat the computation of (b) in convention (B) and verify that the factor $\frac{1}{2\pi}$ announced by the table appears.
- (d) Explain why in convention (A) the sinc function that appears is the *normalized* one $\frac{\sin \pi \xi}{\pi \xi}$ and in (B) the unnormalized one $\frac{\sin(\omega/2)}{\omega/2}$, and state where the zeros are in the two cases.

Exercise 19.3: Convolutions of the rectangle and sinc integrals. *Goal:* use convolution as a machine for manufacturing integral identities. *Estimated time:* 45 minutes. *Type:* symbolic computation with numerical check.

- (a) Compute explicitly $(\Pi * \Lambda)(t)$ for every $t \in \mathbb{R}$ (it is a piecewise function on $[-\frac{3}{2}, \frac{3}{2}]$, polynomial of degree 2 on each piece) and verify that $(\Pi * \Lambda)(0) = \frac{3}{4}$ and $(\Pi * \Lambda)(\pm \frac{3}{2}) = 0$.
- (b) Deduce from this, with Theorem 19.9, $\int \text{sinc}^3 \xi \, d\xi = \frac{3}{4}$ and hence $\int_{-\infty}^{+\infty} \frac{\sin^3 x}{x^3} \, dx = \frac{3\pi}{4}$.
- (c) Compute $R_\Lambda = \Lambda * \Lambda$ at $t = 0$ and deduce $\int \text{sinc}^4 \xi \, d\xi = \frac{2}{3}$.
- (d) Why does the same method *not* allow computing $\int \text{sinc} \xi \, d\xi$? What does the principal value $\lim_{R \rightarrow \infty} \int_{-R}^R \text{sinc}$ equal instead, and which result of the chapter suggests it?

Exercise 19.4: Poisson, the theta function, and the comb. *Goal:* verify the hypotheses of Poisson before using it, and derive two classical identities from it. *Estimated time:* 45 minutes. *Type:* proof-based and numerical.

- (a) Verify that $f(t) = e^{-2\pi a|t|}$, with $a > 0$, satisfies hypotheses (H1) and (H2) of Theorem 19.12, exhibiting in each case the constant C and the exponent ε .
- (b) Apply Poisson at $t = 0$ and deduce $\sum_{n \in \mathbb{Z}} \frac{1}{a^2 + n^2} = \frac{\pi}{a} \coth(\pi a)$. Compute both sides for $a = \frac{1}{2}$ and $a = 2$ and compare them to seven decimal places.
- (c) Using the expansion $\coth x = \frac{1}{x} + \frac{x}{3} + O(x^3)$ as $x \rightarrow 0$ (which follows from the series for $\sinh$ and $\cosh$), let $a \rightarrow 0^+$ in the identity of (b) and deduce $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$.
- (d) Verify directly, without using Theorem 19.14, that $\langle \mathbb{I}, \varphi \rangle = \langle \mathbb{I}, \hat{\varphi} \rangle$ for $\varphi(t) = e^{-\pi t^2}$, and explain why this particular case does *not* prove the theorem.

19.16. Solved exercises

Exercise 19.5: What is $\int_{-\infty}^{+\infty} \frac{\sin^4 x}{x^4} \, dx$? *Type:* computation via Plancherel with two independent checks. Compute the real integral $J = \int_{-\infty}^{+\infty} \frac{\sin^4 x}{x^4} \, dx$ and give its value in closed form and to seven decimal places.

Solution. Strategy. The integrand is the fourth power of a sinc function, and sinc is the transform of Π : the fourth power in frequency is the product of two squares, that is, the transform of a convolution of two triangles. Plancherel's theorem converts the integral of sinc^4 into an integral of Λ^2 , which is a polynomial.

Step 1 — rewriting in the normalized variable. With the substitution $x = \pi \xi$, $dx = \pi \, d\xi$,

$$J = \int \frac{\sin^4(\pi \xi)}{(\pi \xi)^4} \pi \, d\xi = \pi \int (\text{sinc} \, \xi)^4 \, d\xi.$$

It thus suffices to compute $I_4 = \int \text{sinc}^4 \xi \, d\xi$.

Step 2 — $\text{sinc}^2 = \hat{\Lambda}$. By Proposition 19.3, $\hat{\Lambda} = \text{sinc}^2$; since Λ is real and even, $\hat{\Lambda}$ is real (Theorem 19.3), and $|\hat{\Lambda}|^2 = \text{sinc}^4$.

Step 3 — **Plancherel's identity for Λ .** Λ does not lie in $\mathcal{S}$, so Theorem 19.10 does not apply directly. One uses autocorrelation instead: $\Lambda \in L^1$ and $R_\Lambda = \Lambda * \tilde{\Lambda} = \Lambda * \Lambda$ (because Λ is real and even) is continuous, with $\widehat{R_\Lambda} = |\hat{\Lambda}|^2 = \text{sinc}^4 \in L^1$ (because $|\text{sinc}| \leq 1$ and $\text{sinc}^2 \in L^1$, Proposition 19.4). Theorem 19.9 applies to R_Λ and gives, at $x = 0$,

$$I_4 = \int \text{sinc}^4 \xi \, d\xi = R_\Lambda(0) = \int \Lambda(s)^2 \, ds.$$

Step 4 — **the integral of the polynomial.** Λ^2 is even, hence

$$\int \Lambda(s)^2 \, ds = 2 \int_0^1 (1-s)^2 \, ds = 2 \left[-\frac{(1-s)^3}{3} \right]_0^1 = 2 \cdot \frac{1}{3} = \frac{2}{3}.$$

Step 5 — **conclusion.**

$$J = \pi \cdot \frac{2}{3} = \boxed{\frac{2\pi}{3} = 2.0943951}.$$

Independent check 1: direct quadrature. Composite Simpson on $[-3000, 3000]$ with 6,000,000 subintervals applied to $\frac{\sin^4 x}{x^4}$: 2.0943951, with a discrepancy of $9 \cdot 10^{-12}$ from the closed form. ✓

Independent check 2: the same quadrature on the normalized form. The same scheme applied to sinc^4 gives 0.6666667 = $\frac{2}{3}$ (discrepancy $9 \cdot 10^{-14}$), and $\pi \cdot \frac{2}{3} = 2.0943951$. ✓

Interpretation. The integral is nothing but the *energy* of the triangle, read in frequency: $\int |\hat{\Lambda}|^2 = \int \Lambda^2$. And the triangle is the rectangle convolved with itself, hence the chain $\Pi \rightarrow \Lambda \rightarrow \Lambda * \Lambda$ produces, evaluated at 0, the sequence $\int \text{sinc}^2 = 1$, $\int \text{sinc}^3 = \frac{3}{4}$, $\int \text{sinc}^4 = \frac{2}{3}$: each additional convolution lowers the value, because it widens the support while keeping the area fixed.

Typical errors. (1) Confusing normalized and unnormalized sinc and forgetting the factor π of Step 1: one gets $\frac{2}{3}$ instead of $\frac{2\pi}{3}$. (2) Applying Theorem 19.10 to Λ , which is not in $\mathcal{S}$, without passing through autocorrelation. (3) Believing that $\int \text{sinc}^4 = (\int \text{sinc}^2)^2$: the integral is not multiplicative.

Extension. With $\Pi * \Pi * \Pi$ in place of $\Lambda * \Lambda$ one obtains $\int \text{sinc}^3 = \frac{3}{4}$, that is, $\int \frac{\sin^3 x}{x^3} \, dx = \frac{3\pi}{4} = 2.3561945$ (checked numerically with the same parameters, discrepancy $5 \cdot 10^{-11}$). ✓ The method stops at $\int \text{sinc}$: Proposition 19.4 says that integral does not exist in the sense of Lebesgue.

Exercise 19.6: Summing $\sum_{n \in \mathbb{Z}} \frac{1}{1+n^2}$ with the Poisson formula. *Type:* application of Poisson with explicit verification of the hypotheses. Compute, for $a > 0$, the sum $\Sigma(a) = \sum_{n \in \mathbb{Z}} \frac{1}{a^2 + n^2}$, and in particular $\Sigma(1)$.

Solution. Strategy. The function to be summed is, up to a constant, row 7 of the table (Theorem 19.15): it is the transform of a two-sided exponential. The Poisson formula turns the sum of the samples of $\hat{f}$ into the sum of the samples of f , and the latter is a geometric series.

Step 1 — the right pair. Take $f(t) = e^{-2\pi a|t|}$. By part (c) of the proof of Theorem 19.15,

$$\hat{f}(\xi) = \frac{1}{\pi} \frac{a}{a^2 + \xi^2},$$

hence $\sum_n \hat{f}(n) = \frac{a}{\pi} \Sigma(a)$: summing $\hat{f}$ over the integers is exactly the problem.

Step 2 — checking the hypotheses of Poisson. f is continuous on all of $\mathbb{R}$ (even at 0, where it has a corner but not a jump). For (H1): $e^{-2\pi a|t|}$ decays exponentially, and the function $(1 + |t|)^2 e^{-2\pi a|t|}$ is continuous and tends to 0 at infinity, hence is bounded by a constant C_1 ; therefore $|f(t)| \leq C_1(1 + |t|)^{-2}$, that is, (H1) with $\varepsilon = 1$. For (H2): $|\hat{f}(\xi)| = \frac{a}{\pi(a^2 + \xi^2)} \leq \frac{a}{\pi} \cdot \frac{2}{(1 + |\xi|)^2} \cdot \max(1, \frac{1}{a^2})$, because $(1 + |\xi|)^2 \leq 2(1 + \xi^2)$ and $a^2 + \xi^2 \geq \min(1, a^2)(1 + \xi^2)$: this is (H2) with $\varepsilon = 1$.

Step 3 — the "spatial" side. The sum of the samples of f is a geometric series of ratio $q = e^{-2\pi a} \in (0, 1)$:

$$\sum_{n \in \mathbb{Z}} e^{-2\pi a|n|} = 1 + 2 \sum_{n \geq 1} q^n = 1 + \frac{2q}{1 - q} = \frac{1 - q + 2q}{1 - q} = \frac{1 + q}{1 - q} = \frac{e^{\pi a} + e^{-\pi a}}{e^{\pi a} - e^{-\pi a}} = \coth(\pi a),$$

where in the last step numerator and denominator were multiplied by $e^{\pi a}$ (since $q = e^{-2\pi a}$).

Step 4 — Poisson. Theorem 19.12 at $t = 0$ gives $\sum_n f(n) = \sum_n \hat{f}(n)$, that is,

$$\coth(\pi a) = \frac{a}{\pi} \Sigma(a), \quad \text{hence} \quad \boxed{\Sigma(a) = \sum_{n \in \mathbb{Z}} \frac{1}{a^2 + n^2} = \frac{\pi}{a} \coth(\pi a)}.$$

For $a = 1$: $\Sigma(1) = \pi \coth \pi = \pi \cdot 1.0037419 = 3.1533481$.

Independent check 1: direct summation. Summing the terms for $|n| \leq 4,000,000$ gives 3.1533476; the neglected tail is worth $2 \sum_{n > 4 \cdot 10^6} n^{-2} \approx \frac{2}{4 \cdot 10^6} = 5.0 \cdot 10^{-7}$, and indeed the observed discrepancy is $5.0 \cdot 10^{-7}$. ✓ For $a = \frac{1}{2}$ the formula gives 6.8507543 and the direct sum 6.8507538 (same tail discrepancy); for $a = 2$, 1.5708073 against 1.5708068. ✓

Independent check 2: the spatial side. $\sum_{|n| \leq 200} e^{-2\pi|n|} = 1.0037419$, which agrees to seven decimal places with $\coth \pi = 1.0037419$. ✓

Interpretation. The identity says that the energy of a two-sided exponential, sampled at the integers in time, coincides with that sampled at the integers in frequency. This is why the constant π appears in a sum that, at first sight, looks purely arithmetic.

Typical errors. (1) Applying Poisson without checking (H2): (H1) alone is not enough, and there are continuous, integrable functions for which the two series of the formula do not coincide. (2) Taking $f(t) = e^{-a|t|}$ instead of $e^{-2\pi a|t|}$: one gets a correct identity but with $\frac{a}{2\pi}$ in place of a , and comparison with the classical form requires a further rescaling. (3) Forgetting the term $n = 0$ in the two-sided geometric series, which is 1 and not 2.

Extension. The result as $a \rightarrow 0^+$ produces $\zeta(2)$: isolating the term $n = 0$, $\Sigma(a) = \frac{1}{a^2} + 2 \sum_{n \geq 1} \frac{1}{a^2 + n^2}$, while $\frac{\pi}{a} \coth(\pi a) = \frac{1}{a^2} + \frac{\pi^2}{3} + O(a^2)$; equating and letting $a \rightarrow 0^+$ gives $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6} = 1.6449341$. This is part (c) of Exercise 19.4.

Exercise 19.7: Duration and bandwidth of a modulated Gaussian pulse. *Type:* quantitative verification of the uncertainty principle, with comparison. Let $f(t) = 2^{1/4} e^{2\pi i \xi_0 t} e^{-\pi(t-t_0)^2}$ with $t_0 = 1$ and $\xi_0 = 3$. Compute $\|f\|_2$, the time spread $\Delta_t^2 = \int (t - t_0)^2 |f|^2 dt$ and the frequency spread $\Delta_\xi^2 = \int (\xi - \xi_0)^2 |\hat{f}|^2 d\xi$, and compare the product with the bound of Corollary 19.6. Then repeat the computation for the rectangle Π .

Solution. Strategy. Translation and modulation do not change the moduli of f and $\hat{f}$, but shift their centers: it is convenient to reduce to the centered Gaussian, for which the computation is Lemma 19.5.

Step 1 — the norm. $|f(t)| = 2^{1/4}e^{-\pi(t-t_0)^2}$, hence, with $s = t - t_0$ and Lemma 19.5 with $b = 2$,

$$\|f\|_2^2 = \sqrt{2} \int e^{-2\pi s^2} ds = \sqrt{2} \cdot \frac{1}{\sqrt{2}} = 1.$$

Step 2 — the time spread. With the same substitution and Lemma 19.5 (second integral, $b = 2$),

$$\Delta_t^2 = \sqrt{2} \int s^2 e^{-2\pi s^2} ds = \sqrt{2} \cdot \frac{1}{2\pi \cdot 2^{3/2}} = \frac{1}{4\pi} = 0.0795775.$$

Step 3 — the transform. Set $g(t) = 2^{1/4}e^{-\pi t^2}$; then $f(t) = e^{2\pi i \xi_0 t} g(t - t_0)$. By Theorem 19.3(ii) and (iii),

$$\hat{f}(\xi) = e^{-2\pi i(\xi - \xi_0)t_0} \hat{g}(\xi - \xi_0), \quad \text{hence} \quad |\hat{f}(\xi)| = |\hat{g}(\xi - \xi_0)|.$$

By Theorem 19.6, $\hat{g}(\eta) = 2^{1/4}e^{-\pi\eta^2}$.

Step 4 — the frequency spread. With $\eta = \xi - \xi_0$ and the computation of Step 2,

$$\Delta_\xi^2 = \sqrt{2} \int \eta^2 e^{-2\pi\eta^2} d\eta = \frac{1}{4\pi} = 0.0795775.$$

Step 5 — the product.

$$\Delta_t^2 \Delta_\xi^2 = \frac{1}{4\pi} \cdot \frac{1}{4\pi} = \boxed{\frac{1}{16\pi^2} = 0.0063326},$$

exactly the minimum of Corollary 19.6: the modulated Gaussian pulse is a minimizer, as predicted by the equality case.

Step 6 — comparison with the rectangle. Π already has norm 1 in L^2 ($\int \Pi^2 = 1$) and center $t_0 = 0$ by symmetry, and

$$\Delta_t^2(\Pi) = \int_{-1/2}^{1/2} t^2 dt = \left[\frac{t^3}{3} \right]_{-1/2}^{1/2} = \frac{1}{12} = 0.0833333,$$

slightly more than the Gaussian. But $\Delta_\xi^2(\Pi) = \int \xi^2 \text{sinc}^2 \xi d\xi = \int \frac{\sin^2(\pi\xi)}{\pi^2} d\xi = +\infty$, because the integrand is periodic, nonnegative and not identically zero. The product is therefore $+\infty$: the inequality is verified, but trivially. Note that $\Pi \notin \mathcal{S}$, so Theorem 19.13 does not concern it: the computation shows, however, that the statement is not vacuous and that the finiteness of the two spreads is a real requirement.

Independent check. Simpson quadrature on $[-12, 14]$ with 400,000 subintervals applied to $|f|^2$ and to $(t-1)^2|f|^2$: 1.0000000 and 0.0795775, against 1 and $\frac{1}{4\pi} = 0.0795775$. ✓ The same quadrature on $[-12, 12]$ for the centered Gaussian $2^{1/4}e^{-\pi t^2}$ gives the same two values, confirming that translating and modulating changes nothing. ✓

Interpretation. The product $\Delta_t \Delta_\xi \geq \frac{1}{4\pi}$ says that one cannot have a signal that is simultaneously short and narrow-band. The Gaussian is the only profile that achieves the minimum, which is why it appears wherever the best time–frequency compromise is needed: from the Gabor window to the optimization of radar pulses.

Typical errors. (1) Computing the spreads with respect to 0 instead of the centers t_0 and ξ_0 : one gets larger values and the minimum seems violated the wrong way. (2) Forgetting the normalization $\|f\|_2 = 1$: without it the product scales as $\|f\|_2^4$ and the comparison with

$\frac{1}{16\pi^2}$ is meaningless. (3) Concluding from Step 6 that "the rectangle violates the principle": it satisfies it, trivially.

Extension. Repeating with $f_a(t) = (2a)^{1/4}e^{-\pi at^2}$ one finds $\Delta_t^2 = \frac{1}{4\pi a}$ and $\Delta_\xi^2 = \frac{a}{4\pi}$: the product remains $\frac{1}{16\pi^2}$ for every $a > 0$, while the two factors exchange with each other under $a \rightarrow 1/a$. This is Figure 19.3 read in numbers.

19.17. Self-assessment quiz

- (Definition) A *tempered distribution* is
 - a measurable function with polynomial growth
 - a linear, continuous functional on $\mathcal{S}(\mathbb{R})$
 - a function with a single point of discontinuity
 - a pointwise limit of L^1 functions
- (Hypothesis) In Theorem 19.12 hypothesis (H2), that is, the decay of $\hat{f}$, serves to
 - guarantee that f is continuous
 - make the series $\sum_n \hat{f}(n)e^{2\pi i n t}$ converge absolutely and uniformly, so that its sum is continuous and its coefficients can be computed
 - make f integrable
 - ensure that $\hat{f}$ is real
- (Symbolic computation) The transform of $e^{-2t}u(t)$ equals
 - $\frac{1}{2 + 2\pi i \xi}$
 - $\frac{4}{4 + 4\pi^2 \xi^2}$
 - $\frac{1}{2 + i \xi}$
 - $\frac{2}{2 + 2\pi i \xi}$
- (Numerical computation) $\int_{-\infty}^{+\infty} \frac{\sin^4 x}{x^4} dx$ equals, to seven decimal places,
 - 2.0943951
 - 3.1415927
 - 2.3561945
 - 1.5707963
- (Proof step) In the proof of Theorem 19.5 the minus sign that appears after the substitution comes
 - from reversing the limits of integration
 - from the choice $h = \frac{1}{2\xi}$, for which $e^{-2\pi i \xi h} = e^{-\pi i} = -1$
 - from the fact that f changes sign
 - from the derivative of the exponential
- (Error diagnosis) A student writes $\int \delta(t)\varphi(t) dt = \varphi(0)$ meaning a Lebesgue integral, and concludes that δ is a function that is zero away from the origin with integral 1. The error is
 - the value $\varphi(0)$ should be replaced by $\varphi(1)$
 - a function that is zero almost everywhere has zero integral, and Proposition 19.8 rules out any integral representation with a locally integrable function
 - the integral should be understood over $[0, +\infty)$
 - φ must have compact support
- (Transfer) $\sum_{n \in \mathbb{Z}} \frac{1}{1 + n^2}$ equals, to seven decimal places,
 - 1.6449341
 - 3.1533481
 - 3.1415927
 - 2.0943951
- (Source and metacognition) The uncertainty principle, in the form proved in this chapter, and the Poisson summation formula are found respectively in
 - Stein–Shakarchi, *Fourier Analysis: An Introduction*, Ch. 5, §4 "The Heisenberg uncertainty principle" (p. 158) and §3 "The Poisson summation formula" (p. 153)
 - Stein–Shakarchi, Ch. 1 "The Genesis of Fourier Analysis"
 - Rudin, *Real and Complex Analysis*, Ch. 11
 - NIST DLMF, §1.14(iii) "Laplace Transform"

19.18. Teacher's key

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Definition 19.7: linearity plus sequential continuity with respect to the seminorms of $\mathcal{S}$. *Distractors:* A describes a *class* of examples (Proposition 19.6), not the definition, and excludes the delta from it; C has no connection to the notion; D is false, because a pointwise limit of L^1 functions is still a function. *Diagnosed error:* identifying a distribution with a "generalized" function rather than with a functional. *Source:* NIST DLMF §1.16(v) "Tempered Distributions". *Prerequisite:* the Schwartz class (Definition 19.5). *Remediation:* redo Proposition 19.7 and Proposition 19.8 in sequence.

2) **B.** *Rationale:* Step 1 of the proof of Theorem 19.12 uses (H2) *only* for the Weierstrass criterion, which makes the frequency-side series uniformly convergent and hence its sum continuous; without continuity one cannot apply the uniqueness of coefficients of Step 4. *Distractors:* A is false, continuity of f is a separate hypothesis; C is what follows from (H1), not from (H2) (Step 3 of Theorem 19.11); D is meaningless, $\hat{f}$ is real only if f is even. *Diagnosed error:* not distinguishing the role of the two decay hypotheses. *Source:* Stein–Shakarchi, Ch. 5, §3; NIST DLMF §1.8(iv), eq. 1.8.14. *Prerequisite:* the Weierstrass criterion. *Remediation:* construct an f satisfying (H1) but not (H2) and see where the proof breaks down.

3) **A.** *Rationale:* this is row 8 of the table (Theorem 19.15, derivation (d)) with $a = 2$: $\int_0^\infty e^{-2t} e^{-2\pi i \xi t} dt = \frac{1}{2+2\pi i \xi}$. *Distractors:* B is the transform of $e^{-2|t|}$ (row 6), that is, the *two-sided* case; C is the same formula in convention (B), in angular frequency; D has an extra 2 in the numerator, the error of one who confuses row 6 with row 8. *Diagnosed error:* confusing the causal and two-sided exponentials, or forgetting the 2π of convention (A). *Source:* Theorem 19.15, derivation (d); Bracewell, Ch. 6. *Prerequisite:* the primitive of e^{-zt} with $\operatorname{Re} z > 0$. *Remediation:* recompute rows 6 and 8 for $a = 1$ and compare the moduli at $\xi = 0$ (2 versus 1).

4) **A.** *Rationale:* this is Solved Exercise 19.5: $\int \frac{\sin^4 x}{x^4} dx = \pi \int \operatorname{sinc}^4 = \pi \cdot \frac{2}{3} = \frac{2\pi}{3} = 2.0943951$. *Distractors:* B is π , the value of $\int \frac{\sin^2 x}{x^2} dx$; C is $\frac{3\pi}{4} = 2.3561945$, that is, $\int \frac{\sin^3 x}{x^3} dx$ — the wrong exponent; D is $\frac{\pi}{2}$. *Diagnosed error:* getting the power wrong, or forgetting the factor π of the change of variable $x = \pi \xi$. *Source:* Solved Exercise 19.5, Steps 1–5. *Prerequisite:* Proposition 19.3 and Corollary 19.1. *Remediation:* redo the computation for $\int \operatorname{sinc}^2$ and $\int \operatorname{sinc}^3$ and order the three values $1 > \frac{3}{4} > \frac{2}{3}$.

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 5–8).

5) **B.** *Rationale:* in Step 1 of the proof of Theorem 19.5 one chooses $h = \frac{1}{2\xi}$ precisely because $e^{-2\pi i \xi h} = e^{-\pi i} = -1$: it is half a period of the oscillation, and translating by half a period flips the sign of the character. *Distractors:* A is the typical error of one who confuses this substitution with the one in Step (iv) of Theorem 19.3, where the sign really does come from the limits; C is false, f is arbitrary; D is unrelated, nothing is differentiated. *Diagnosed error:* not identifying the algebraic origin of a sign. *Source:* Theorem 19.5, Step 1; Rudin, *Real and Complex Analysis*, Ch. 9. *Prerequisite:* $e^{i\pi} = -1$. *Remediation:* redo Step 1 with $h = \frac{1}{4\xi}$ and see what results (a factor $-i$, which does not produce the useful average).

6) **B.** *Rationale:* this is exactly Proposition 19.8: no locally integrable function can realize the pairing $\varphi \mapsto \varphi(0)$; and in any case a function that is zero almost everywhere has zero integral, whatever value is assigned to it at the origin. *Distractors:* A changes the point of evaluation, which is not the error; C restricting the domain fixes nothing; D is false,

the definition uses $\mathcal{S}$, which contains functions without compact support. *Diagnosed error:* treating δ as a function, that is, mistaking a notation for a definition (Remark 19.9). *Source:* Proposition 19.8; NIST DLMF §1.16(iii). *Prerequisite:* absolute continuity of the integral. *Remediation:* reread Step 4 of Proposition 19.8 and reconstruct the choice of η and δ with $\varepsilon = \frac{1}{10}$.

7) B. *Rationale:* this is Solved Exercise 19.6 with $a = 1$: $\sum_n \frac{1}{1+n^2} = \pi \coth \pi = 3.1533481$; the direct sum up to $|n| \leq 4 \cdot 10^6$ gives 3.1533476, with the predicted tail of $5 \cdot 10^{-7}$. *Distractors:* A is $\frac{\pi^2}{6} = \sum_{n \geq 1} n^{-2}$, that is, the *one-sided* sum *without* the a^2 term; C is π , that is, the value one would get by swapping $\coth \pi$ for 1; D is $\frac{2\pi}{3}$. *Diagnosed error:* confusing the two-sided sum with the one-sided one, or forgetting the term $n = 0$. *Source:* Solved Exercise 19.6, Steps 3–4. *Prerequisite:* geometric series and the definition of $\coth$. *Remediation:* compute $\sum_{n \geq 1} \frac{1}{1+n^2} = \frac{\pi \coth \pi - 1}{2} = 1.0766740$ and compare it with the direct sum 1.0766738.

8) A. *Rationale:* the book by Stein and Shakarchi devotes §4 of Chapter 5 (p. 158) to the uncertainty principle and §3 (p. 153) to the Poisson formula, with sub-section 3.1 "Theta and zeta functions" containing the identity of Corollary 19.5. *Distractors:* B is the introductory chapter on the wave and heat equations; C is Rudin's chapter on harmonic functions, not on the transform (which is Chapter 9); D is the DLMF section devoted to the Laplace transform, not the Fourier transform (which is §1.14(i)). *Diagnosed error:* citing "the book" without the chapter, or attributing a result to a nearby section. *Source:* see the "Sources" section. *Prerequisite:* knowing how to read a table of contents by section. *Remediation:* compile a table "result / author / chapter / proved here yes or no" for the twelve main statements of the chapter.

BEST PRACTICE

Assigned Exercise 19.2 — answers.

(a) By the "link with (A)" row of the table (Remark 19.1), $\hat{f}_B(\omega) = \hat{f}(\omega/2\pi)$, hence

$$\hat{f}_B(\omega) = \frac{2a}{a^2 + 4\pi^2 \left(\frac{\omega}{2\pi}\right)^2} = \frac{2a}{a^2 + \omega^2}, \quad \hat{f}_C(\omega) = \frac{1}{\sqrt{2\pi}} \cdot \frac{2a}{a^2 + \omega^2}.$$

The form in (B) is that of Inherited Exercise 19.1: it is the same result.

(b) $\int |f|^2 dt = \int e^{-2a|t|} dt = 2 \int_0^\infty e^{-2at} dt = \frac{1}{a}$. For the second integral, with $u = 2\pi\xi$ ($d\xi = \frac{du}{2\pi}$) and the hint,

$$\int |\hat{f}|^2 d\xi = \int \frac{4a^2 d\xi}{(a^2 + 4\pi^2 \xi^2)^2} = \frac{4a^2}{2\pi} \int \frac{du}{(a^2 + u^2)^2} = \frac{4a^2}{2\pi} \cdot \frac{\pi}{2a^3} = \frac{1}{a}.$$

The two values coincide: Plancherel is verified *without constants*. Numerical check for $a = 1$: both 1.0000000. ✓

(c) In convention (B): $\int |\hat{f}_B|^2 d\omega = \int \frac{4a^2 d\omega}{(a^2 + \omega^2)^2} = 4a^2 \cdot \frac{\pi}{2a^3} = \frac{2\pi}{a}$, and indeed $\frac{1}{2\pi} \cdot \frac{2\pi}{a} = \frac{1}{a} = \int |f|^2$: exactly the factor $\frac{1}{2\pi}$ announced by the "Plancherel" row of the table appears.

(d) Because the transform of the rectangle has as zeros the points where the character completes an integer number of turns. In (A) this happens for $\xi \in \mathbb{Z} \setminus \{0\}$, and $\frac{\sin \pi \xi}{\pi \xi}$ vanishes there; in (B) it happens for $\omega = 2\pi k$, $k \neq 0$, and $\frac{\sin(\omega/2)}{\omega/2}$ vanishes there. These are the same physical points, written in two different variables: normalization puts the zeros at the *integers*, which makes reading the "bandwidth" of a window of duration T immediate (zeros at multiples of $1/T$).

BEST PRACTICE

Assigned Exercise 19.3 — answers.

(a) $(\Pi * \Lambda)(t) = \int_{-1/2}^{1/2} \Lambda(t - \tau) d\tau = \int_{t-1/2}^{t+1/2} \Lambda(s) ds$. Let F be the primitive of Λ that vanishes at $-\infty$ ($F(x) = 0$ for $x \leq -1$, $F(x) = \frac{(1+x)^2}{2}$ for $-1 \leq x \leq 0$, $F(x) = 1 - \frac{(1-x)^2}{2}$ for $0 \leq x \leq 1$, $F(x) = 1$ for $x \geq 1$); then $(\Pi * \Lambda)(t) = F(t + \frac{1}{2}) - F(t - \frac{1}{2})$, that is, by symmetry,

$$(\Pi * \Lambda)(t) = \begin{cases} \frac{3}{4} - t^2, & |t| \leq \frac{1}{2}, \\ \frac{1}{2} \left(\frac{3}{2} - |t| \right)^2, & \frac{1}{2} \leq |t| \leq \frac{3}{2}, \\ 0, & |t| \geq \frac{3}{2}. \end{cases}$$

In particular $(\Pi * \Lambda)(0) = \frac{3}{4}$ and $(\Pi * \Lambda)(\pm \frac{3}{2}) = 0$; the two expressions match at $|t| = \frac{1}{2}$, where both equal $\frac{1}{2}$. Numerical check (Simpson, 200,000 subintervals): $t = 0 \rightarrow 0.7500000$; $t = 0.25 \rightarrow 0.6875000 = \frac{3}{4} - \frac{1}{16}$; $t = 0.5 \rightarrow 0.5000000$; $t = 0.7 \rightarrow 0.3200000 = \frac{1}{2}(0.8)^2$; $t = 1 \rightarrow 0.1250000 = \frac{1}{2}(0.5)^2$; $t = 1.5 \rightarrow 0.0000000$. ✓

(b) $\Pi * \Lambda = \Pi * \Pi * \Pi$ is continuous (an integral of a uniformly continuous function over an interval that depends continuously on t) and has transform sinc^3 (Theorem 19.4(iv)), integrable because $|\text{sinc}| \leq 1$ and $\text{sinc}^2 \in L^1$. Theorem 19.9 at $x = 0$ gives $\int \text{sinc}^3 \xi d\xi = (\Pi * \Lambda)(0) = \frac{3}{4}$. With $x = \pi\xi$, $\int \frac{\sin^3 x}{x^3} dx = \pi \cdot \frac{3}{4} = \frac{3\pi}{4} = 2.3561945$; quadrature on $[-3000, 3000]$ with 6,000,000 subintervals: 2.3561945. ✓

(c) $R_\Lambda(0) = \int \Lambda^2 = 2 \int_0^1 (1-s)^2 ds = \frac{2}{3}$, and $\widehat{R_\Lambda} = |\hat{\Lambda}|^2 = \text{sinc}^4$; Theorem 19.9 at $x = 0$ gives $\int \text{sinc}^4 = \frac{2}{3} = 0.6666667$, confirmed numerically. ✓

(d) Because $\text{sinc} = \hat{\Pi}$ is not integrable (Proposition 19.4): Theorem 19.9 requires $\hat{f} \in L^1$ and here the hypothesis fails. The principal value exists and equals 1, that is, $\Pi(0)$, as suggested by Remark 19.7; equivalently $\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = \pi$ (the Dirichlet integral, improper and not absolutely convergent, which this chapter does *not* prove). Numerical check of the principal value: $\int_{-R}^R \text{sinc}$ equals 0.9959475 for $R = 50$, 0.9995947 for $R = 500$, 0.9999595 for $R = 5000$: it converges to 1 like $1 - \frac{c}{R}$ with $c \approx 0.20$. ✓

BEST PRACTICE

Assigned Exercise 19.4 — answers.

(a) This is Step 2 of Solved Exercise 19.6: take $\varepsilon = 1$ in both hypotheses, with $C_1 = \max_t (1 + |t|)^2 e^{-2\pi a|t|}$ for (H1) and $C_2 = \frac{2a}{\pi} \max(1, \frac{1}{a^2})$ for (H2). Both maxima exist because the functions are continuous and vanish at infinity.

(b) $\sum_{n \in \mathbb{Z}} \frac{1}{a^2 + n^2} = \frac{\pi}{a} \coth(\pi a)$. Values: $a = \frac{1}{2} \rightarrow 6.8507543$ (formula) against 6.8507538 (direct sum up to $|n| \leq 4 \cdot 10^6$, predicted tail $5 \cdot 10^{-7}$); $a = 2 \rightarrow 1.5708073$ against 1.5708068. ✓

(c) Isolating the term $n = 0$: $\frac{1}{a^2} + 2 \sum_{n \geq 1} \frac{1}{a^2 + n^2} = \frac{\pi}{a} \coth(\pi a) = \frac{\pi}{a} \left(\frac{1}{\pi a} + \frac{\pi a}{3} + O(a^3) \right) = \frac{1}{a^2} + \frac{\pi^2}{3} + O(a^2)$. Cancelling $\frac{1}{a^2}$ and letting $a \rightarrow 0^+$ (legitimate: the series on the left converges uniformly for $a \in [0, 1]$ by comparison with $\sum n^{-2}$), one obtains $2 \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{3}$, that is, $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6} = 1.6449341$. ✓

(d) For $\varphi(t) = e^{-\pi t^2}$ Theorem 19.6 gives $\hat{\varphi} = \varphi$, hence $\langle \mathbb{I}, \varphi \rangle = \langle \mathbb{I}, \hat{\varphi} \rangle = \sum_n e^{-\pi n^2} = \vartheta(1) = 1.0864348$: the equality is verified, but *vacuously*, because the Gaussian is a fixed point of $\mathcal{F}$ and the two sides are literally the same sum. The theorem requires the equality for *every* $\varphi \in \mathcal{S}$, and the proof (Theorem 19.14) necessarily goes through the Poisson formula. A nontrivial test is obtained with $\varphi(t) = e^{-\pi a t^2}$ for $a \neq 1$: then $\langle \mathbb{I}, \varphi \rangle = \vartheta(a)$ and

$\langle \mathbb{I}, \hat{\varphi} \rangle = a^{-1/2} \vartheta(1/a)$, which coincide by Corollary 19.5 but not term by term — for $a = 2$, 1.0037349 on both sides. ✓

19.19. Formulary of the chapter

Definition and convention.

$$\hat{f}(\xi) = \int f(t) e^{-2\pi i \xi t} dt, \quad \mathcal{F}^{-1} h(x) = \int h(\xi) e^{2\pi i x \xi} d\xi, \quad \|\hat{f}\|_{\infty} \leq \|f\|_1.$$

Passage to angular frequency: $\hat{f}_B(\omega) = \hat{f}(\frac{\omega}{2\pi})$, $\hat{f}_C = \frac{1}{\sqrt{2\pi}} \hat{f}_B$. Normalized sinc function: $\text{sinc } x = \frac{\sin \pi x}{\pi x}$.

The eight rules.

$$\begin{aligned} \widehat{f(\cdot - t_0)} &= e^{-2\pi i \xi t_0} \hat{f}; & \widehat{e^{2\pi i \xi_0 \cdot} f(\xi)} &= \hat{f}(\xi - \xi_0); & \widehat{f(a \cdot)}(\xi) &= \frac{1}{|a|} \hat{f}\left(\frac{\xi}{a}\right); \\ \widehat{f(-\cdot)}(\xi) &= \hat{f}(-\xi); & \widehat{\hat{f}}(\xi) &= \overline{\hat{f}(-\xi)}; & \widehat{f'}(\xi) &= 2\pi i \xi \hat{f}(\xi); & \widehat{tf(t)}(\xi) &= \frac{i}{2\pi} \hat{f}'(\xi); \\ \widehat{f * g} &= \hat{f} \hat{g}; & R_f &= f * \tilde{f} \quad (\tilde{f}(t) = \overline{f(-t)}), & \widehat{R_f} &= |\hat{f}|^2. \end{aligned}$$

f real $\Rightarrow \hat{f}(-\xi) = \overline{\hat{f}(\xi)}$; f real and even $\Rightarrow \hat{f}$ real and even; f real and odd $\Rightarrow \hat{f}$ purely imaginary and odd.

Inversion, Plancherel, Riemann–Lebesgue.

$$f \in \mathcal{S} \Rightarrow f(x) = \int \hat{f}(\xi) e^{2\pi i x \xi} d\xi, \quad \mathcal{F}^2 = R, \quad \mathcal{F}^4 = \text{id};$$

$$f, \hat{f} \in L^1 \Rightarrow f = \mathcal{F}^{-1} \hat{f} \text{ a.e.}; \quad \int f \bar{g} = \int \hat{f} \bar{\hat{g}}, \quad \|\hat{f}\|_2 = \|f\|_2; \quad f \in L^1 \Rightarrow \hat{f}(\xi) \xrightarrow{|\xi| \rightarrow \infty} 0.$$

Multiplication formula: $\int \hat{f} g = \int f \hat{g}$. Gaussian kernel: $K_{\delta}(u) = \frac{1}{\delta} e^{-\pi u^2 / \delta^2}$, $\int K_{\delta} = 1$, $f * K_{\delta} \rightarrow f$ in L^1 .

Periodization, Poisson, uncertainty.

$$(Pf)(t) = \sum_{n \in \mathbb{Z}} f(t + n), \quad c_n(Pf) = \hat{f}(n); \quad \sum_{n \in \mathbb{Z}} f(t + n) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n t}.$$

$$\vartheta(a) = \sum_{n \in \mathbb{Z}} e^{-\pi a n^2} = \frac{1}{\sqrt{a}} \vartheta\left(\frac{1}{a}\right); \quad \sum_{n \in \mathbb{Z}} \frac{1}{a^2 + n^2} = \frac{\pi}{a} \coth(\pi a).$$

$$\|f\|_2 = 1 \Rightarrow \left(\int (t - t_0)^2 |f|^2 \right) \left(\int (\xi - \xi_0)^2 |\hat{f}|^2 \right) \geq \frac{1}{16\pi^2},$$

with equality if and only if $f(t) = A e^{2\pi i \xi_0 t} e^{-\pi a (t - t_0)^2}$, $|A|^2 = \sqrt{2a}$.

Essential table, and the numerical values of the chapter.

$$\widehat{\Pi} = \text{sinc}, \quad \widehat{\Lambda} = \text{sinc}^2, \quad \widehat{e^{-\pi t^2}} = e^{-\pi \xi^2}, \quad \widehat{e^{-\pi a t^2}} = a^{-1/2} e^{-\pi \xi^2 / a}, \quad \widehat{t e^{-\pi t^2}} = -i \xi e^{-\pi \xi^2};$$

$$\begin{aligned}
\widehat{e^{-a|t|}} &= \frac{2a}{a^2 + 4\pi^2\xi^2}, & \frac{1}{\pi} \widehat{\frac{a}{a^2 + t^2}} &= e^{-2\pi a|\xi|}, & \widehat{e^{-at}u(t)} &= \frac{1}{a + 2\pi i\xi}; \\
\hat{\delta} &= 1, & \hat{1} &= \delta, & \hat{\mathbb{I}} &= \mathbb{I}. \\
\int e^{-\pi t^2} dt &= 1, & \int t^2 e^{-\pi b t^2} dt &= \frac{1}{2\pi b^{3/2}}, & \int \operatorname{sinc}^2 &= 1, & \int \operatorname{sinc}^3 &= \frac{3}{4}, & \int \operatorname{sinc}^4 &= \frac{2}{3}; \\
\int \frac{\sin^3 x}{x^3} dx &= \frac{3\pi}{4} = 2.3561945, & \int \frac{\sin^4 x}{x^4} dx &= \frac{2\pi}{3} = 2.0943951, \\
e^{-\pi} &= 0.0432139, & \vartheta(1) &= 1.0864348, & \pi \coth \pi &= 3.1533481, & \frac{1}{16\pi^2} &= 0.0063326, & \frac{1}{4\pi} &= 0.0795775.
\end{aligned}$$

Sources and bibliography

- [1] E. M. Stein, R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, Princeton (NJ), 2003 (ISBN 978-0-691-11384-5, 328 pages). Chapter 5 "The Fourier Transform on $\mathbb{R}$ " (from p. 129): §1 "Elementary theory of the Fourier transform", with §1.3 "The Schwartz space" (p. 134), §1.4 "The Fourier transform on $\mathcal{S}$ " (p. 136), §1.5 "The Fourier inversion" and §1.6 "The Plancherel formula" — is the source of Definition 19.5, of Theorem 19.7, of Theorem 19.8 and of Theorem 19.10; §3 "The Poisson summation formula" (p. 153), with §3.1 "Theta and zeta functions", the source of Theorem 19.12 and of Corollary 19.5; §4 "The Heisenberg uncertainty principle" (p. 158), the source of Theorem 19.13. Bibliographic data verified on the publisher's record (Princeton University Press, ISBN 9780691113845) and the structure of Chapter 5 with pagination verified on the full-text index of the Internet Archive/Open Library copy — verified on 6 September 2026.
- [2] W. Rudin, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, New York, 1987 (ISBN 0-07-054234-1, LCCN 86-000007, xiv+416 pages). Chapter 9 "Fourier Transforms", with the sections "Formal properties", "The inversion theorem", "The Plancherel theorem" and "The Banach algebra L^1 ": this is the alternative source for Theorem 19.9 and for the extension of Plancherel to L^2 stated in Remark 19.8, and for the measure-theory prerequisites listed in the opening *infobox* (dominated convergence, Fubini–Tonelli, density of C_c in L^1 , the Riesz theorem). The page numbers of those four sections — 192, 195, 200 and following — come from the index of the 2nd edition (New York 1974, ISBN 0-07-054233-3, LCCN 73-015743, xii+452 pages), the only one for which the full text is indexed; in the 3rd edition the titles are the same and the pagination differs. Bibliographic data and section titles verified on the Open Library records OL2707518M (3rd ed.) and OL5422092M (2nd ed.) and on Open Library's full-text search — verified on 6 September 2026.
- [3] R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd edition, McGraw-Hill Series in Electrical and Computer Engineering, McGraw-Hill, Boston, 2000 (ISBN 0-07-303938-1, 616 pages). This is the engineering reference source for convention (A) of the table in Remark 19.1, for the table of Theorem 19.15 and for the notation $\mathbb{I}$ of the Dirac comb (Definition 19.10); Chapter 3 "Convolution", Chapter 5 "The Impulse Symbol" — with the "sifting property" cited in Remark 19.9 — and Chapter 6 "The Basic Theorems" correspond respectively to Section 19.4, to Section 19.12 and to Section 19.3 of this chapter; Chapter 10 "Sampling and Series" is the source of the panorama on sampling (Remark 19.10), which is not proved here. Bibliographic data of the 3rd edition verified on the Open Library record for ISBN 0073039381; the chapter titles are verified on the full-text index of the 2nd edition (New York 1978, ISBN 0-07-007013-X, LCCN 77-

013376, xvi+444 pages, Open Library record OL4545611M), the only one indexed — verified on 6 September 2026.

- [4] NIST *Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>, version 1.2.7 of 15 June 2026 (release consulted). §1.14 "Integral Transforms", subsection §1.14(i) "Fourier Transform": defines $\mathcal{F}f(x) = \frac{1}{2\pi} \int f(t)e^{ixt}dt$ and reports the two Parseval formulas; this is the convention *different* from ours cited in Remark 19.1. §1.8 "Fourier Series", subsection §1.8(iv) "Poisson's Summation Formula", equation 1.8.14: states $\sum_n f(x+n) = \sum_n e^{2\pi inx} \int f(t)e^{-2\pi int}dt$ — in our convention (A) — under the hypothesis that f is twice continuously differentiable with f and $|f''|$ integrable on $(-\infty, \infty)$; these are hypotheses *different* from those of Theorem 19.12, which requires only continuity and decay $O((1+|t|)^{-1-\varepsilon})$ for f and $\hat{f}$: neither implies the other. §1.16 "Distributions", subsections §1.16(iii) "Dirac Delta Distribution", §1.16(v) "Tempered Distributions" and §1.16(vii) "Fourier Transforms of Tempered Distributions", equation 1.16.35 ($\langle \mathcal{F}(u), \phi \rangle = \langle u, \mathcal{F}(\phi) \rangle$): this is the source of Definition 19.7 and of Definition 19.9 — verified on 6 September 2026.
- [5] For results of the treatise recalled with \ref: the Fourier series on the circle, Parseval's identity and the uniqueness of the coefficients are Theorem 17.1 and Corollary 17.2 of Chapter 17, where they are proved; the Dirichlet–Jordan theorem cited in Remark 19.7 is Theorem 17.4 of the same chapter. The material inherited from Volume II (Chapter 4, §9) is reproduced in full in Section 19.2 and in Section 19.14.
- [6] Numerical checks: all values of the chapter were computed twice, in closed form and with executed Python scripts (composite Simpson quadrature with $2 \cdot 10^5$ – $4 \cdot 10^7$ subintervals, truncated sums with the tail estimated explicitly), and the two results agree to the seventh decimal place except where the text states and justifies the residual discrepancy (the integral of sinc^2 and the sums $\frac{1}{a^2+n^2}$, where the truncated tail is computed and compared with the observed discrepancy).

The Laplace transform

The Fourier transform of Chapter 19 requires the signal to be integrable over the whole line: a hypothesis that a growing exponential does not satisfy, and that not even the unit step satisfies. The Laplace transform removes the obstacle by multiplying the signal by a damping factor $e^{-\sigma t}$ before transforming it, and pays the price of having to declare, each time, *for which* σ the computation makes sense. That price is the true content of the chapter: the set of admissible values is not just any set, it is always a right half-plane (Theorem 20.2), on which the transform is automatically holomorphic (Theorem 20.3), and whose boundary — the imaginary axis — is exactly the locus where the Laplace transform becomes a Fourier transform again (Theorem 20.19).

The chapter proceeds as follows. Section 20.1 reproduces *in full* the material that the executive plan moved here from Volume II (Chapter 4, §10, and Appendix A, §2): the definition, the table of pairs, the properties, the transfer function, BIBO stability, a worked example, exercises and quiz. That text leaves some hypotheses implicit (“for $\operatorname{Re}(s)$ sufficiently large”, “under suitable regularity hypotheses”): the rest of the chapter serves to make them explicit and to prove every statement.

Section 20.2 proves that the region of convergence is a half-plane (Theorem 20.2), introduces the two abscissas σ_c and σ_a and exhibits an example in which they differ (Proposition 20.1). Section 20.3 proves the holomorphy of the transform by differentiation under the integral sign, with the hypothesis that justifies it (Theorem 20.3) and the formula for all derivatives (Corollary 20.2). Section 20.4 proves the computation rules: linearity (Theorem 20.4), translation in time and in frequency (Theorems 20.5 and 20.6), time scaling (Theorem 20.7), differentiation and integration in time (Theorems 20.8 and 20.9), multiplication by t (Corollary 20.5) and convolution (Theorem 20.10). Section 20.5 computes the elementary transforms one by one and collects them into a table in which *every row carries its own region of convergence* together with a pointer to where it is proved (Theorem 20.11). Section 20.6 proves the initial value theorem (Theorem 20.12) and the final value theorem (Theorem 20.13), each with explicit hypotheses and with a counterexample showing what happens if they fail (Examples 20.3 and 20.4). Section 20.7 proves the uniqueness of the inverse transform (Theorem 20.14), inversion by partial fractions with simple and multiple poles (Theorems 20.15 and 20.16) and inversion by residues in the rational case (Theorem 20.17), which rests entirely on Chapter 15. Section 20.8 closes the circle with the restriction to the imaginary axis.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages at level **D** (the proof of the half-plane of convergence for simple convergence, and the proof of Lerch’s uniqueness theorem).

Internal prerequisites. The Fourier transform, convolution, the Dirac delta (Chapter 19, Definition 19.1, Theorem 19.4, Definition 19.8); holomorphic functions, Liouville’s theorem, removable singularities, Laurent series and residues (Chapter 15, Theorem 15.8, Theorem 15.13, Theorem 15.4, Proposition 15.2); the trigonometric approximation theorem of Weierstrass (Chapter 17, Corollary 17.3); Euler’s Gamma function (Chapter 11); the complex exponential and Euler’s formulas (Volume II, Chapter 3).

External prerequisites, assumed and not reproved:

- Lebesgue integration on $[0, \infty)$: the dominated convergence theorem, Tonelli's theorem and Fubini's theorem for double integrals over a measurable subset of $[0, \infty)^2$;
- absolutely continuous functions on a compact interval and their integration-by-parts formula: if g is absolutely continuous on $[0, T]$ and h is of class C^1 on $[0, T]$, then $\int_0^T g' h \, dt = [gh]_0^T - \int_0^T gh' \, dt$;
- Dirichlet's criterion for improper integrals: if $\Phi(X) = \int_1^X \varphi$ is bounded and ψ is of class C^1 with $\psi(u) \rightarrow 0$ and $\int_1^\infty |\psi'| < \infty$, then $\int_1^\infty \varphi \psi$ converges;
- the Weierstrass approximation theorem in *polynomial* form: every continuous function on $[0, 1]$ is the uniform limit of polynomials (the trigonometric form is proved in Chapter 17);
- the fundamental lemma of the calculus of variations in L^1 : if $h \in L^1(0, 1)$ and $\int_0^1 h \varphi \, dx = 0$ for every φ continuous on $[0, 1]$, then $h = 0$ almost everywhere.

Everything else is proved here.

What is learned. How to state the region of convergence of every transform and recognize that it is a half-plane; how to differentiate under the integral sign knowing which hypothesis authorizes it; how to derive every row of the table of transforms from a few already-proved rules, instead of memorizing it; how to use the initial and final value theorems knowing when they do *not* apply; how to invert a rational function by partial fractions with simple and multiple poles, and to recognize the same formula as a sum of residues; how to read the Fourier transform as the restriction of the Laplace transform to the imaginary axis, and to know when that restriction makes sense.

Warning. Conventions of the chapter.

- **Causal signals.** Unless stated otherwise, f is a function $f : [0, \infty) \rightarrow \mathbb{C}$ that is *locally integrable*, i.e. Lebesgue integrable on every compact interval $[0, T]$. Identifying f with its extension by zero for $t < 0$ gives a function on all of $\mathbb{R}$: this is *causality*.
- **Unilateral transform.** $\mathcal{L}\{f\}(s) = F(s) = \int_0^\infty f(t)e^{-st} \, dt$, with $s = \sigma + i\omega$, $\sigma = \operatorname{Re}(s)$, $\omega = \operatorname{Im}(s)$. The notation $\int_0^\infty$ always means $\lim_{T \rightarrow \infty} \int_0^T$, and the integral over $[0, T]$ is a Lebesgue integral.
- **Two kinds of convergence.** The integral *converges* at s if that limit exists and is finite; it *converges absolutely* at s if $\int_0^\infty |f(t)|e^{-\sigma t} \, dt < \infty$. The two notions do not coincide (Proposition 20.1), and every statement declares which of the two it uses.
- $u(t)$ is the **Heaviside unit step**: $u(t) = 1$ for $t \geq 0$ and $u(t) = 0$ for $t < 0$. The value at 0 is irrelevant for the integrals.
- $|e^{-st}| = e^{-\sigma t}$: this is the identity that governs the entire chapter, and follows from $|e^z| = e^{\operatorname{Re}(z)}$ (Volume II, Chapter 3).
- The letters $p, p_1, \dots, p_r$ denote poles in the s -plane; Γ is Euler's Gamma function (Chapter 11).
- Numerical values are rounded to the seventh decimal digit and recomputed twice by independent methods: a closed formula and an executed Python script.

20.1. The Laplace transform

This section reproduces, without any cuts, the text of Volume II (Chapter 4, Section 10) and the table of Appendix A, Section 2, which the executive plan moved here. The hypotheses left implicit are made explicit and proved in the sections that follow: a precise pointer to each is given in the attention box at the end of the section.

The Laplace transform further generalizes the Fourier transform, allowing treatment of signals that grow in time (not absolutely integrable) provided no faster than an exponential: it is the standard tool for analyzing causal systems and their stability.

Definition 20.1: Laplace transform (unilateral). For $f : [0, \infty) \rightarrow \mathbb{C}$ (or $f : \mathbb{R} \rightarrow \mathbb{C}$ zero for $t < 0$, i.e. causal):

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st} dt, \quad s = \sigma + i\omega \in \mathbb{C},$$

defined for $\sigma = \text{Re}(s)$ sufficiently large. The set of values of s for which the integral converges is the **region of convergence** ROC; for causal signals that do not grow faster than an exponential, ROC is always a right half-plane $\text{Re}(s) > \sigma_0$.

Remark 20.1: Laplace restricted to the imaginary axis is Fourier. If $\sigma_0 < 0$ (i.e. if 0 belongs to ROC, which is equivalent to saying that f is absolutely integrable: a “stable” system), then setting $s = i\omega$ one obtains

$$F(i\omega) = \int_0^\infty f(t)e^{-i\omega t} dt = \hat{f}(\omega),$$

the Fourier transform of f (extended to zero for $t < 0$). The Fourier transform is therefore the restriction of the Laplace transform to the imaginary axis, when this restriction makes sense.

Theorem 20.1: Properties of the Laplace transform. With $f(0) := \lim_{t \rightarrow 0^+} f(t)$:

1. (Linearity) $\mathcal{L}\{af + bg\} = aF(s) + bG(s)$.
2. (Differentiation) $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$.
3. (Integration) $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = F(s)/s$.
4. (Translation) $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$ for $a \geq 0$, with u the unit step.
5. (Initial value) $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$.
6. (Final value) $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$, if the left-hand limit exists (sufficient condition: all poles of $sF(s)$ in the half-plane $\text{Re}(s) < 0$).

Proof. (1) Linearity of the integral. (2) Integration by parts: $\int_0^\infty f'(t)e^{-st} dt = [f(t)e^{-st}]_0^\infty + s \int_0^\infty f(t)e^{-st} dt = -f(0) + sF(s)$ (the upper boundary term vanishes for $\text{Re}(s)$ sufficiently large, assuming f of sub-exponential growth). (3) Follows from (2) applied to $g(t) = \int_0^t f(\tau) d\tau$, for which $g'(t) = f(t)$ and $g(0) = 0$: $F(s) = \mathcal{L}\{g'\} = sG(s) - 0$, hence $G(s) = F(s)/s$. (4) With $\tau = t - a$: $\int_0^\infty f(t-a)u(t-a)e^{-st} dt = \int_0^\infty f(\tau)e^{-s(\tau+a)} d\tau = e^{-as}F(s)$. (5)–(6) Obtained by taking the limit in (2) for $s \rightarrow \infty$ and $s \rightarrow 0$ respectively, under appropriate regularity assumptions. $\square$

Definition 20.2: Transfer function and BIBO stability. For a causal linear time-invariant (LTI) system with impulse response $h(t)$, the **transfer function** is $H(s) = \mathcal{L}\{h(t)\}$; if the input $u(t)$ and output $y(t)$ are related by $y = h * u$, then $Y(s) = H(s)U(s)$ (Theorem 19.2(4), which holds identically for Laplace). The system is **BIBO stable** (bounded input $\Rightarrow$ bounded output) if and only if all poles of $H(s)$ have $\text{Re}(s) < 0$ (open left half-plane).

Worked example — first-order system and step response. Let $h(t) = \frac{1}{\tau}e^{-t/\tau}u(t)$ be the impulse response of a first-order system with time constant $\tau > 0$ (a typical model for a motor, a temperature sensor, or a tank with flow proportional to level, common in water treatment).

Then

$$H(s) = \frac{1}{\tau} \int_0^\infty e^{-t/\tau} e^{-st} dt = \frac{1}{\tau} \cdot \frac{1}{s + 1/\tau} = \frac{1}{1 + \tau s},$$

with a single pole at $s = -1/\tau < 0$: the system is BIBO stable for every $\tau > 0$, consistent with physical intuition (a first-order system with positive time constant is always stable). For the step response, $U(s) = \mathcal{L}\{u(t)\} = 1/s$, hence

$$Y(s) = H(s)U(s) = \frac{1}{s(1 + \tau s)} = \frac{1}{s} - \frac{1}{s + 1/\tau}$$

(partial fraction decomposition: one verifies that $\frac{1}{s} - \frac{1}{s + 1/\tau} = \frac{(s + 1/\tau) - s}{s(s + 1/\tau)} = \frac{1/\tau}{s(s + 1/\tau)} = \frac{1}{\tau s(s + 1/\tau)} = \frac{1}{s(1 + \tau s)}$, as required). Inverse-transforming term by term:

$$y(t) = 1 - e^{-t/\tau}, \quad t \geq 0,$$

the classical exponential charging curve with time constant τ : after a time $t = \tau$ the system has reached about 63% of the final value, after $t = 5\tau$ over 99% — the practical criterion by which settling time is estimated in a control loop.

The table of Appendix A, Section 2 of Volume II. The material moved here also includes the table of the notable Laplace transforms for causal signals, which in its original form listed the pairs

$$\begin{aligned} \delta(t) &\mapsto 1, & u(t) &\mapsto \frac{1}{s}, & t &\mapsto \frac{1}{s^2}, & t^n &\mapsto \frac{n!}{s^{n+1}}, \\ e^{-at} &\mapsto \frac{1}{s+a}, & t e^{-at} &\mapsto \frac{1}{(s+a)^2}, & \cos(\omega t) &\mapsto \frac{s}{s^2 + \omega^2}, \\ \sin(\omega t) &\mapsto \frac{\omega}{s^2 + \omega^2}, & e^{-at} \cos(\omega t) &\mapsto \frac{s+a}{(s+a)^2 + \omega^2}, \\ e^{-at} \sin(\omega t) &\mapsto \frac{\omega}{(s+a)^2 + \omega^2}, \end{aligned}$$

without indicating the region of convergence and without deriving any of them. The table is now found in Appendix C; Theorem 20.11 reproduces it row by row, with the region of convergence stated and with a pointer to the proof of each pair.

Warning. What remains implicit here, and where it is proved. The inherited text is correct but abbreviates three points; each is taken up in full further on.

- “defined for $\operatorname{Re}(s)$ sufficiently large” and “ROC is always a right half-plane” (Definition 20.1): the statement is true and is proved in Theorem 20.2, which requires no hypothesis of exponential growth — it suffices that the integral converge at *one* point.
- “the upper boundary term vanishes ... assuming f of sub-exponential growth” (item (2) of Theorem 20.1): the precise hypothesis is in Theorem 20.8, where f is assumed to be locally absolutely continuous and $\mathcal{L}\{f'\}$ absolutely convergent, and the at-most-exponential growth of f becomes a *consequence* (Lemma 20.3), not a hypothesis.
- “under appropriate regularity assumptions” (items (5) and (6)): the hypotheses are those of Theorem 20.12 and the Theorem 20.13; Examples 20.3 and 20.4 show that without them the two formulas give false results.
- The “if and only if” of BIBO stability (Definition 20.2) is proved in this chapter in one direction and for proper rational functions (Corollary 20.12); the general characterization $h \in L^1$ belongs to Chapter 24, and the algebraic criterion that decides the sign of the real parts of the poles without computing them, to Chapter 25.

20.2. Convergence: the half-plane and the two abscissas

Definition 20.1 says that the integral is “defined for $\operatorname{Re}(s)$ sufficiently large”. This section proves that the expression is not an approximation of convenience: if the integral converges at *just one* point, then it converges throughout the whole open half-plane to the right of that point. No hypothesis of exponential growth is needed.

Definition 20.3: Simple convergence and absolute convergence. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable. The Laplace integral *converges* at $s \in \mathbb{C}$ if the limit

$$F(s) = \lim_{T \rightarrow +\infty} \int_0^T f(t) e^{-st} dt,$$

exists and is finite, and *converges absolutely* at s if $\int_0^\infty |f(t)| e^{-\sigma t} dt < \infty$, where $\sigma = \operatorname{Re}(s)$. Absolute convergence implies simple convergence, because a function with integrable modulus is integrable on $[0, \infty)$ and the limit of the truncations exists by the dominated convergence theorem.

Theorem 20.2: The region of convergence is a half-plane. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable and suppose the Laplace integral converges at a point $s_0 \in \mathbb{C}$. Setting

$$\varphi(t) = \int_0^t f(\tau) e^{-s_0 \tau} d\tau, \quad t \geq 0,$$

the following three facts hold.

(i) The integral converges at every s with $\operatorname{Re}(s) > \operatorname{Re}(s_0)$, and

$$F(s) = (s - s_0) \int_0^\infty \varphi(t) e^{-(s-s_0)t} dt,$$

where this last integral converges absolutely.

(ii) For every $T \geq 0$ and every s with $\rho := \operatorname{Re}(s - s_0) > 0$,

$$\left| \int_T^\infty f(t) e^{-st} dt \right| \leq \eta(T) \left(1 + \frac{|s - s_0|}{\rho} \right), \quad \eta(T) := \sup_{t \geq T} |\varphi(t) - \varphi(T)|.$$

(iii) The convergence is uniform on every sector $S_\theta = \{s \neq s_0 : |\arg(s - s_0)| \leq \theta\}$ with $0 < \theta < \pi/2$ fixed.

Proof. Step 1 — the function φ is absolutely continuous and bounded. The function $t \mapsto f(t) e^{-s_0 t}$ is integrable on every compact set $[0, T]$ (the product of a locally integrable function and a continuous function), hence φ is the indefinite integral of a locally integrable function: it is absolutely continuous on every $[0, T]$, with $\varphi' = f(t) e^{-s_0 t}$ almost everywhere, and $\varphi(0) = 0$. By hypothesis $\varphi(T)$ has a finite limit $\varphi(\infty)$ as $T \rightarrow \infty$; a continuous function on $[0, \infty)$ with a finite limit at infinity is bounded, so there exists $M = \sup_{t \geq 0} |\varphi(t)| < \infty$.

Step 2 — integration by parts on the truncation. Let s be such that $\rho = \operatorname{Re}(s - s_0) > 0$. Since $t \mapsto e^{-(s-s_0)t}$ is of class C^1 and φ is absolutely continuous on $[0, T]$, the integration-by-parts formula recalled among the prerequisites gives

$$\int_0^T f(t) e^{-st} dt = \int_0^T \varphi'(t) e^{-(s-s_0)t} dt = \varphi(T) e^{-(s-s_0)T} + (s - s_0) \int_0^T \varphi(t) e^{-(s-s_0)t} dt,$$

where we used $f(t) e^{-st} = (f(t) e^{-s_0 t}) e^{-(s-s_0)t} = \varphi'(t) e^{-(s-s_0)t}$ and $\varphi(0) = 0$.

Step 3 — the limit as $T \rightarrow \infty$. The first term tends to zero, because $|\varphi(T)e^{-(s-s_0)T}| \leq Me^{-\rho T} \rightarrow 0$. The second converges absolutely, because

$$\int_0^\infty |\varphi(t)e^{-(s-s_0)t}| dt \leq M \int_0^\infty e^{-\rho t} dt = \frac{M}{\rho} < \infty.$$

Hence the limit of $\int_0^T f e^{-st} dt$ exists, is finite, and equals $(s-s_0) \int_0^\infty \varphi e^{-(s-s_0)t} dt$: this is (i).

Step 4 — the tail estimate. Fix $T \geq 0$ and set, for $t \geq T$, $\psi(t) = \varphi(t) - \varphi(T)$, so that $\psi(T) = 0$, $\psi' = \varphi'$ almost everywhere and $|\psi(t)| \leq \eta(T)$ for every $t \geq T$. The same integration by parts on the interval $[T, T']$ gives

$$\int_T^{T'} f(t)e^{-st} dt = \psi(T')e^{-(s-s_0)T'} + (s-s_0) \int_T^{T'} \psi(t)e^{-(s-s_0)t} dt,$$

whence

$$\left| \int_T^{T'} f(t)e^{-st} dt \right| \leq \eta(T)e^{-\rho T'} + |s-s_0| \eta(T) \int_T^{T'} e^{-\rho t} dt,$$

and, using $e^{-\rho T'} \leq 1$, $\int_T^{T'} e^{-\rho t} dt \leq e^{-\rho T}/\rho$ and $e^{-\rho T} \leq 1$,

$$\left| \int_T^{T'} f(t)e^{-st} dt \right| \leq \eta(T) + |s-s_0| \eta(T) \frac{1}{\rho} = \eta(T) \left(1 + \frac{|s-s_0|}{\rho} \right).$$

Letting $T' \rightarrow \infty$ (the limit exists by Step 3) gives (ii).

Step 5 — uniformity on the sector. If $s \in S_\theta$ and we write $s-s_0 = re^{i\alpha}$ with $|\alpha| \leq \theta < \pi/2$, then $\rho = r \cos \alpha$ and

$$\frac{|s-s_0|}{\rho} = \frac{r}{r \cos \alpha} = \frac{1}{\cos \alpha} \leq \frac{1}{\cos \theta},$$

because $\cos$ is decreasing on $[0, \pi/2)$ and $|\alpha| \leq \theta$. Then (ii) becomes

$$\left| \int_T^\infty f(t)e^{-st} dt \right| \leq \eta(T) \left(1 + \frac{1}{\cos \theta} \right) \quad \text{for every } s \in S_\theta,$$

and the right-hand side does not depend on s . It remains to see that $\eta(T) \rightarrow 0$: since $\varphi(t) \rightarrow \varphi(\infty)$, for every $\varepsilon > 0$ there exists T_ε such that $|\varphi(t) - \varphi(\infty)| < \varepsilon/2$ for $t \geq T_\varepsilon$; for $T \geq T_\varepsilon$ and $t \geq T$ we then have $|\varphi(t) - \varphi(T)| \leq |\varphi(t) - \varphi(\infty)| + |\varphi(\infty) - \varphi(T)| < \varepsilon$, i.e. $\eta(T) \leq \varepsilon$. Hence the tail tends to zero uniformly on S_θ : this is (iii). $\square$

Definition 20.4: The two abscissas of convergence. Let $C = \{s \in \mathbb{C} : \text{the Laplace integral of } f \text{ converges at } s\}$ and let $A = \{s \in \mathbb{C} : \text{it converges absolutely at } s\}$. Set

$$\sigma_c = \inf\{\operatorname{Re}(s) : s \in C\}, \quad \sigma_a = \inf\{\operatorname{Re}(s) : s \in A\},$$

with the convention $\inf \emptyset = +\infty$ and $\inf$ of a set unbounded below $= -\infty$. The number σ_c is the **abscissa of convergence**, σ_a the **abscissa of absolute convergence**. The **region of convergence** (ROC) of the signal is the set C . When it is necessary to indicate which signal is meant, one writes $\sigma_c(f)$ and $\sigma_a(f)$.

Corollary 20.1: Shape of the region of convergence. With the notation of Definition 20.4:

$$\{s : \operatorname{Re}(s) > \sigma_c\} \subseteq C \subseteq \{s : \operatorname{Re}(s) \geq \sigma_c\},$$

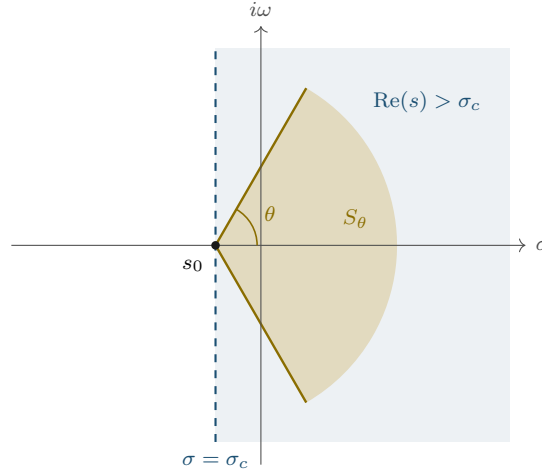


Figure 20.1: Theorem 20.2: if the integral converges at s_0 , it converges throughout the whole open half-plane $\text{Re}(s) > \text{Re}(s_0)$ (blue), and uniformly on the sector S_θ with vertex s_0 and half-aperture $\theta < \pi/2$ (ochre). In the drawing $s_0 = -0.6$, $\theta = 60^\circ$ and the sector has radius 2.4: its two vertices lie at $(-0.6 + 2.4 \cos 60^\circ, \pm 2.4 \sin 60^\circ) = (0.6, \pm 2.0785)$, with $2.4 \sin 60^\circ = 2.4 \cdot 0.8660254 = 2.0784610$. The dashed line is the boundary $\text{Re}(s) = \sigma_c$, which the theorem does not decide.

$$\{s : \text{Re}(s) > \sigma_a\} \subseteq A \subseteq \{s : \text{Re}(s) \geq \sigma_a\},$$

and $\sigma_c \leq \sigma_a$. In particular C and A are right half-planes, possibly with part of the boundary line removed or not.

Proof. Step 1 — the left inclusion for C . Let $\text{Re}(s) > \sigma_c$. By the definition of infimum there exists $s_0 \in C$ with $\text{Re}(s_0) < \text{Re}(s)$. Theorem 20.2(i) then gives $s \in C$.

Step 2 — the right inclusion for C . If $s \in C$ then $\text{Re}(s) \geq \inf\{\text{Re}(s') : s' \in C\} = \sigma_c$: this is the definition of infimum.

Step 3 — the same two inclusions for A . Absolute convergence depends only on $\sigma = \text{Re}(s)$, and for $\sigma \geq \sigma_0$ one has $|f(t)|e^{-\sigma t} \leq |f(t)|e^{-\sigma_0 t}$ for every $t \geq 0$ (the real exponential is decreasing). Hence if the integral converges absolutely at σ_0 it converges absolutely at every $\sigma \geq \sigma_0$, by comparison. The two inclusions follow as in Steps 1 and 2.

Step 4 — $\sigma_c \leq \sigma_a$. We have $A \subseteq C$ (Definition 20.3), and the infimum of a subset is greater than or equal to that of the whole set. $\square$

Example 20.1: The two extreme cases on the boundary line. The boundary $\text{Re}(s) = \sigma_c$ may or may not belong to the ROC, and Corollary 20.1 does not decide it: both cases occur.

- $f = u$ (unit step). For $\sigma > 0$, $\int_0^T e^{-st} dt = \frac{1-e^{-sT}}{s} \rightarrow \frac{1}{s}$; for $s = i\omega$ with $\omega \neq 0$ the value $\frac{1-e^{-i\omega T}}{i\omega}$ has no limit (it oscillates on a circle of radius $1/|\omega|$), and for $s = 0$ the integral equals $T \rightarrow \infty$. Hence $\sigma_c = \sigma_a = 0$ and the ROC is the *open* half-plane.
- $f(t) = \frac{1}{1+t^2}$. For $\sigma \geq 0$ one has $\int_0^\infty \frac{e^{-\sigma t}}{1+t^2} dt \leq \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}$, so the integral converges absolutely on the whole line $\text{Re}(s) = 0$ and in the right half-plane; for $\sigma < 0$ it diverges, because $\frac{e^{-\sigma t}}{1+t^2} = \frac{e^{|\sigma|t}}{1+t^2} \rightarrow \infty$ and the integrand is not even infinitesimal. Hence $\sigma_c = \sigma_a = 0$ and the ROC is the *closed* half-plane.

Proposition 20.1: The two abscissas can differ. For $f(t) = e^t \sin(e^t)$ one has $\sigma_c \leq 0$ and $\sigma_a = 1$: the two abscissas differ by at least 1.

Proof. Step 1 — the change of variable. The function f is continuous, hence locally integrable. For $T > 0$ and $s \in \mathbb{C}$ set $u = e^t$, i.e. $t = \ln u$, $du = e^t dt$. Then $e^{-st} = e^{-s \ln u} = u^{-s}$ (real logarithm of $u > 0$) and

$$\int_0^T f(t) e^{-st} dt = \int_0^T \sin(e^t) e^t e^{-st} dt = \int_1^{e^T} \sin(u) u^{-s} du.$$

As T ranges over $[0, \infty)$, the endpoint $X = e^T$ ranges over $[1, \infty)$ increasingly and continuously, so the convergence of the Laplace integral is equivalent to the convergence of $\int_1^\infty \sin(u) u^{-s} du$. Note that $|u^{-s}| = |e^{-s \ln u}| = e^{-\sigma \ln u} = u^{-\sigma}$.

Step 2 — simple convergence for $\sigma > 0$. We apply Dirichlet's criterion with $\varphi(u) = \sin u$ and $\psi(u) = u^{-s}$. The antiderivative $\Phi(X) = \int_1^X \sin u du = \cos 1 - \cos X$ satisfies $|\Phi| \leq 2$; ψ is of class C^1 on $[1, \infty)$ with $|\psi(u)| = u^{-\sigma} \rightarrow 0$ (because $\sigma > 0$) and $\psi'(u) = -s u^{-s-1}$, hence

$$\int_1^\infty |\psi'(u)| du = |s| \int_1^\infty u^{-\sigma-1} du = \frac{|s|}{\sigma} < \infty.$$

The criterion gives the convergence of $\int_1^\infty \sin(u) u^{-s} du$ for every s with $\sigma > 0$. Hence every s with $\operatorname{Re}(s) > 0$ lies in C , and by the definition of infimum $\sigma_c \leq 0$.

Step 3 — absolute convergence for $\sigma > 1$. We have $|\sin u| \leq 1$, hence

$$\int_1^\infty |\sin u| u^{-\sigma} du \leq \int_1^\infty u^{-\sigma} du = \frac{1}{\sigma - 1} < \infty \quad (\sigma > 1).$$

Hence $\sigma_a \leq 1$.

Step 4 — absolute divergence for $\sigma \leq 1$: the case $\sigma \leq 0$. For $u \geq 1$ and $\sigma \leq 0$ one has $u^{-\sigma} \geq 1$, hence $\int_1^X |\sin u| u^{-\sigma} du \geq \int_1^X |\sin u| du$. On every interval $[k\pi, (k+1)\pi]$ the integral of $|\sin|$ equals 2, hence $\int_1^X |\sin u| du \rightarrow +\infty$.

Step 5 — absolute divergence for $0 < \sigma \leq 1$. From $|\sin u| \geq \sin^2 u = \frac{1 - \cos 2u}{2}$ it follows that

$$\int_1^X |\sin u| u^{-\sigma} du \geq \frac{1}{2} \int_1^X u^{-\sigma} du - \frac{1}{2} \int_1^X \cos(2u) u^{-\sigma} du.$$

The first integral tends to $+\infty$: it equals $\frac{X^{1-\sigma}-1}{2(1-\sigma)}$ for $\sigma < 1$ and $\frac{\ln X}{2}$ for $\sigma = 1$. The second stays bounded: by the Dirichlet criterion applied to $\varphi(u) = \cos 2u$ (antiderivative $\frac{\sin 2X - \sin 2}{2}$, of modulus ≤ 1) and $\psi(u) = u^{-\sigma}$ with $\sigma > 0$, the integral $\int_1^\infty \cos(2u) u^{-\sigma} du$ converges, so its truncations are bounded. The difference therefore tends to $+\infty$: the integral does not converge absolutely. From this and from Step 3 it follows that $\sigma_a = 1$.

Numerical check. Composite Simpson quadrature with 400,000 subintervals on $[1, X]$:

X	$\sigma = 0.5$		$\sigma = 1$	
	$\int_1^X \sin(u) u^{-\sigma} du$	$\int_1^X \sin u u^{-\sigma} du$	$\int_1^X \sin(u) u^{-\sigma} du$	$\int_1^X \sin u u^{-\sigma} du$
10^2	0.5468052	11.6553751	0.6161424	3.1005912
10^3	0.6149805	39.1605303	0.6241501	4.5642721
10^4	0.6422992	126.2253836	0.6248085	6.0303191

The odd columns stabilize (simple convergence), the even columns grow without bound (absolute divergence). ✓ *Second check, independent:* for $\sigma = 1$ the growth should be logarithmic, with an increment $\frac{2}{\pi} \ln 10 = 1.4657359$ each time X is multiplied by 10 (the mean value of $|\sin|$ is $2/\pi$). The observed increments are $4.5642721 - 3.1005912 = 1.4636809$ and $6.0303191 - 4.5642721 = 1.4660470$: they agree with the predicted value to within $2 \cdot 10^{-3}$. ✓

□

Remark 20.2: What the distinction is for. In the rest of the chapter almost every statement assumes absolute convergence, i.e. works in the half-plane $\operatorname{Re}(s) > \sigma_a$: this is the hypothesis that allows limits and integrals to be interchanged (dominated convergence) and integrals with each other (Fubini), and without which the proofs of the computation rules do not close. Theorem 20.2, on the other hand, was proved for simple convergence alone, because that is what is needed in Theorem 20.13: there the point $s = 0$ is reached from the half-plane of simple convergence, and assuming absolute convergence at 0 would mean assuming $f \in L^1(0, \infty)$, i.e. a hypothesis far stronger than what is needed. The Proposition 20.1 shows that the two hypotheses really are different, and not only formally. Doetsch devotes an entire chapter to this distinction (Introduction to the Theory and Application of the Laplace Transformation, Ch. 3 “The Half-Plane of Convergence”, pp. 13–18).

20.3. Holomorphy of the transform

The Laplace transform is not merely a function of the complex variable s : it is a *holomorphic* function, and it is so automatically, without requiring anything of f beyond convergence of the integral. This is what makes available, for F , all the tools of Chapter 15: series expansions, the identity principle, residues. The proof consists in differentiating under the integral sign, and the only thing that must be done carefully is exhibiting the bound that authorizes it.

Lemma 20.1: Two elementary bounds.

(i) For every $w \in \mathbb{C}$, $|e^w - 1 - w| \leq \frac{|w|^2}{2} e^{|w|}$.

(ii) For every integer $n \geq 1$, every $\varepsilon > 0$ and every $t \geq 0$, $t^n \leq \left(\frac{n}{e\varepsilon}\right)^n e^{\varepsilon t}$.

Proof. (i). From the series expansion of the exponential (Volume II, Chapter 3), which converges absolutely for every $w \in \mathbb{C}$,

$$e^w - 1 - w = \sum_{k \geq 2} \frac{w^k}{k!}, \quad \text{hence} \quad |e^w - 1 - w| \leq \sum_{k \geq 2} \frac{|w|^k}{k!}.$$

For $k \geq 2$ one has $k! = k(k-1)(k-2)! \geq 2(k-2)!$, because $k(k-1) \geq 2$. Setting $j = k - 2$,

$$\sum_{k \geq 2} \frac{|w|^k}{k!} \leq \sum_{k \geq 2} \frac{|w|^k}{2(k-2)!} = \frac{|w|^2}{2} \sum_{j \geq 0} \frac{|w|^j}{j!} = \frac{|w|^2}{2} e^{|w|}.$$

(ii). Consider $g(t) = t^n e^{-\varepsilon t}$ on $[0, \infty)$. It is differentiable, $g(0) = 0$, $g(t) \rightarrow 0$ as $t \rightarrow +\infty$, and

$$g'(t) = (n t^{n-1} - \varepsilon t^n) e^{-\varepsilon t} = t^{n-1} (n - \varepsilon t) e^{-\varepsilon t},$$

which is positive for $0 < t < n/\varepsilon$ and negative for $t > n/\varepsilon$. The maximum is therefore at $t = n/\varepsilon$ and equals

$$g\left(\frac{n}{\varepsilon}\right) = \left(\frac{n}{\varepsilon}\right)^n e^{-n} = \left(\frac{n}{e\varepsilon}\right)^n.$$

From $t^n e^{-\varepsilon t} \leq \left(\frac{n}{e\varepsilon}\right)^n$ the claim follows by multiplying by $e^{\varepsilon t} > 0$. $\square$

Lemma 20.2: Multiplying by a power does not shift the absolute abscissa. If the Laplace integral of f converges absolutely for $\operatorname{Re}(s) > \sigma_a$, then for every integer $n \geq 1$ that of $t^n f(t)$ also converges absolutely for $\operatorname{Re}(s) > \sigma_a$.

Proof. Let $\sigma > \sigma_a$. Choose σ_1 with $\sigma_a < \sigma_1 < \sigma$ and set $\varepsilon = \sigma - \sigma_1 > 0$. By Lemma 20.1(ii), $t^n \leq \left(\frac{n}{e\varepsilon}\right)^n e^{\varepsilon t}$ for every $t \geq 0$, hence

$$\int_0^\infty t^n |f(t)| e^{-\sigma t} dt \leq \left(\frac{n}{e\varepsilon}\right)^n \int_0^\infty |f(t)| e^{\varepsilon t} e^{-\sigma t} dt = \left(\frac{n}{e\varepsilon}\right)^n \int_0^\infty |f(t)| e^{-\sigma_1 t} dt < \infty,$$

the last integral being finite because $\sigma_1 > \sigma_a$ (Corollary 20.1). $\square$

Theorem 20.3: Holomorphy and differentiation under the integral sign. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable with abscissa of absolute convergence $\sigma_a < +\infty$. Then $F = \mathcal{L}\{f\}$ is holomorphic on the open half-plane $\Omega_a = \{s : \operatorname{Re}(s) > \sigma_a\}$ and

$$F'(s) = - \int_0^\infty t f(t) e^{-st} dt = -\mathcal{L}\{t f(t)\}(s), \quad s \in \Omega_a,$$

where the integral on the right converges absolutely on Ω_a .

Proof. Step 1 — the candidate integral converges. By Lemma 20.2 with $n = 1$, the integral $\int_0^\infty t |f(t)| e^{-\sigma t} dt$ is finite for every $\sigma > \sigma_a$: the function $G(s) = - \int_0^\infty t f(t) e^{-st} dt$ is thus well defined on Ω_a .

Step 2 — choice of margins. Fix $s \in \Omega_a$ and set $\sigma = \operatorname{Re}(s)$. Choose σ_2 with $\sigma_a < \sigma_2 < \sigma$ and set

$$\delta = \frac{\sigma - \sigma_2}{2} > 0.$$

For every $h \in \mathbb{C}$ with $0 < |h| \leq \delta$ one has $\operatorname{Re}(s + h) \geq \sigma - |h| \geq \sigma - \delta = \sigma_2 + \delta > \sigma_a$, hence $s + h$ lies in Ω_a and $F(s + h)$ is defined.

Step 3 — the difference quotient. All three integrals that follow converge absolutely (Steps 1 and 2), so they can be combined under the same sign:

$$\frac{F(s + h) - F(s)}{h} - G(s) = \int_0^\infty f(t) e^{-st} \left[\frac{e^{-ht} - 1}{h} + t \right] dt = \frac{1}{h} \int_0^\infty f(t) e^{-st} (e^{-ht} - 1 + ht) dt.$$

Step 4 — the bound. By Lemma 20.1(i) with $w = -ht$ one has $|e^{-ht} - 1 + ht| \leq \frac{|h|^2 t^2}{2} e^{|h|t}$, and $|e^{-st}| = e^{-\sigma t}$. Hence, using $|h| \leq \delta$,

$$\left| \frac{F(s + h) - F(s)}{h} - G(s) \right| \leq \frac{|h|}{2} \int_0^\infty |f(t)| t^2 e^{-\sigma t} e^{|h|t} dt \leq \frac{|h|}{2} \int_0^\infty |f(t)| t^2 e^{-(\sigma - \delta)t} dt.$$

The key point is that the bound *no longer depends on h*. By Lemma 20.1(ii) with $n = 2$ and $\varepsilon = \delta$, $t^2 \leq \left(\frac{2}{e\delta}\right)^2 e^{\delta t}$, hence

$$\int_0^\infty |f(t)| t^2 e^{-(\sigma - \delta)t} dt \leq \left(\frac{2}{e\delta}\right)^2 \int_0^\infty |f(t)| e^{-(\sigma - 2\delta)t} dt = \left(\frac{2}{e\delta}\right)^2 \int_0^\infty |f(t)| e^{-\sigma_2 t} dt =: K < \infty,$$

because $\sigma - 2\delta = \sigma_2 > \sigma_a$.

Step 5 — conclusion. We have obtained

$$\left| \frac{F(s + h) - F(s)}{h} - G(s) \right| \leq \frac{K}{2} |h| \quad \text{for every } 0 < |h| \leq \delta,$$

with K independent of h . Letting $h \rightarrow 0$ in $\mathbb{C}$, the right-hand side tends to zero: the limit of the difference quotient exists and equals $G(s)$. Hence F is differentiable in the complex sense at s (Definition 15.1) with $F'(s) = G(s)$; since $s \in \Omega_a$ was arbitrary, F is holomorphic on Ω_a . $\square$

Corollary 20.2: All the derivatives, and multiplication by a power. Under the hypotheses of Theorem 20.3, for every integer $n \geq 0$ and every $s \in \Omega_a$,

$$F^{(n)}(s) = (-1)^n \int_0^\infty t^n f(t) e^{-st} dt = (-1)^n \mathcal{L}\{t^n f(t)\}(s),$$

with the integral absolutely convergent.

Proof. Induction on n . For $n = 0$ the statement is the definition of F . Suppose the formula holds for n . By Lemma 20.2, the function $g(t) = t^n f(t)$ has an absolutely convergent Laplace integral on Ω_a , hence its abscissa of absolute convergence is $\leq \sigma_a$ and Theorem 20.3 applies to g : $\mathcal{L}\{g\}$ is holomorphic on Ω_a with $(\mathcal{L}\{g\})' = -\mathcal{L}\{t g(t)\} = -\mathcal{L}\{t^{n+1} f(t)\}$. Differentiating the induction hypothesis $F^{(n)} = (-1)^n \mathcal{L}\{g\}$ gives

$$F^{(n+1)} = (-1)^n (\mathcal{L}\{g\})' = (-1)^n (-\mathcal{L}\{t^{n+1} f\}) = (-1)^{n+1} \mathcal{L}\{t^{n+1} f\},$$

which is the formula for $n + 1$. □

Corollary 20.3: Holomorphy as far as there is simple convergence. Under the hypotheses of Theorem 20.2, F is holomorphic on the open half-plane $\{s : \operatorname{Re}(s) > \sigma_c\}$, even where the convergence is not absolute.

Proof. Let s^* be such that $\operatorname{Re}(s^*) > \sigma_c$. By the definition of infimum there exists $s_0 \in C$ with $\operatorname{Re}(s_0) < \operatorname{Re}(s^*)$. Let φ be the function of Theorem 20.2: it is continuous and bounded on $[0, \infty)$ (Step 1 of that proof), say $|\varphi| \leq M$. Then $\int_0^\infty |\varphi(t)| e^{-\rho t} dt \leq M/\rho < \infty$ for every $\rho > 0$: the abscissa of absolute convergence of φ is ≤ 0 , and by Theorem 20.3 the function

$$\Psi(z) = \int_0^\infty \varphi(t) e^{-zt} dt$$

is holomorphic on the half-plane $\operatorname{Re}(z) > 0$. Theorem 20.2(i) says that $F(s) = (s - s_0)\Psi(s - s_0)$ for every s with $\operatorname{Re}(s) > \operatorname{Re}(s_0)$: this is the product of a polynomial with the composition of Ψ with the translation $s \mapsto s - s_0$, hence it is holomorphic on that half-plane, which contains s^* . Since s^* was arbitrary, F is holomorphic on $\{\operatorname{Re}(s) > \sigma_c\}$. □

Remark 20.3: What exactly is the hypothesis that authorizes differentiation. In Step 4 of the proof of Theorem 20.3 the decisive point is not the bound on the exponential, which is algebra: it is that the bounding function

$$t \mapsto |f(t)| t^2 e^{-(\sigma - \delta)t}$$

be integrable and independent of h . Both facts come from absolute convergence with a margin: we required the integral to converge absolutely not only at σ , but at some σ_2 strictly smaller, and the margin δ was used twice — to keep $s + h$ in the half-plane and to absorb the factor t^2 . If one worked exactly on the boundary, i.e. with $\sigma = \sigma_a$, the bounding function might not be integrable and the argument would not close: indeed on the boundary F may not even be defined (Example 20.1, first case). Corollary 20.3 shows, however, that holomorphy survives even where there is only simple convergence: there the margin comes from the boundedness of φ , not from f .

20.4. Computation rules, with the region of convergence

Every rule in this section says two things: what the transform becomes, and *where*. The second piece of information is not an accessory: without it the identity $\mathcal{L}\{u\} = 1/s$ would not distinguish the unit step from $-u(-t)$, which has the same expression with the opposite region of convergence.

Theorem 20.4: Linearity. Let f, g be locally integrable on $[0, \infty)$ with abscissas of absolute convergence $\sigma_a(f)$ and $\sigma_a(g)$, and let $\lambda, \mu \in \mathbb{C}$. Then for every s with $\operatorname{Re}(s) > \max\{\sigma_a(f), \sigma_a(g)\}$ the Laplace integral of $\lambda f + \mu g$ converges absolutely and

$$\mathcal{L}\{\lambda f + \mu g\}(s) = \lambda F(s) + \mu G(s);$$

in particular $\sigma_a(\lambda f + \mu g) \leq \max\{\sigma_a(f), \sigma_a(g)\}$.

Proof. Let $\sigma = \operatorname{Re}(s) > \max\{\sigma_a(f), \sigma_a(g)\}$. By the triangle inequality, $|\lambda f(t) + \mu g(t)|e^{-\sigma t} \leq |\lambda| |f(t)|e^{-\sigma t} + |\mu| |g(t)|e^{-\sigma t}$; the two integrals on the right are finite (Corollary 20.1), hence so is the one on the left: the convergence is absolute. The identity is then the linearity of the Lebesgue integral, applicable because both functions $f e^{-\sigma t}$ and $g e^{-\sigma t}$ are integrable on $[0, \infty)$. The last statement follows from the definition of infimum. $\square$

Theorem 20.5: Translation in time (delay). Let $a \geq 0$ and $f_{[a]}(t) = f(t-a)u(t-a)$, i.e. the signal f delayed by a and zero before a . Then $\sigma_a(f_{[a]}) = \sigma_a(f)$ and, for $\operatorname{Re}(s) > \sigma_a(f)$,

$$\mathcal{L}\{f_{[a]}\}(s) = e^{-as}F(s).$$

Proof. Step 1 — the region. With the substitution $\tau = t - a$ (a translation, which preserves Lebesgue measure),

$$\int_0^\infty |f(t-a)u(t-a)|e^{-\sigma t} dt = \int_a^\infty |f(t-a)|e^{-\sigma t} dt = \int_0^\infty |f(\tau)|e^{-\sigma(\tau+a)} d\tau,$$

i.e.

$$\int_0^\infty |f(t-a)u(t-a)|e^{-\sigma t} dt = e^{-\sigma a} \int_0^\infty |f(\tau)|e^{-\sigma \tau} d\tau.$$

Since $e^{-\sigma a}$ is a finite positive number, the first integral is finite if and only if the last one is: the two sets of absolute convergence coincide, and with them the two abscissas.

Step 2 — the identity. The same change of variable, this time without the modulus (legitimate because the integral converges absolutely by Step 1), gives

$$\mathcal{L}\{f_{[a]}\}(s) = \int_a^\infty f(t-a)e^{-st} dt = \int_0^\infty f(\tau)e^{-s(\tau+a)} d\tau = e^{-as} \int_0^\infty f(\tau)e^{-s\tau} d\tau = e^{-as}F(s).$$

$\square$

Theorem 20.6: Translation in frequency (damping). Let $a \in \mathbb{C}$. Then $\sigma_a(e^{-at}f(t)) = \sigma_a(f) - \operatorname{Re}(a)$ and, for $\operatorname{Re}(s) > \sigma_a(f) - \operatorname{Re}(a)$,

$$\mathcal{L}\{e^{-at}f(t)\}(s) = F(s+a).$$

Proof. One has $e^{-at}f(t)e^{-st} = f(t)e^{-(s+a)t}$ for every t , an algebraic identity among exponentials. Taking moduli, $|e^{-at}f(t)e^{-st}| = |f(t)|e^{-(\sigma+\operatorname{Re}(a))t}$: the integral of $e^{-at}f(t)$ converges absolutely at s if and only if that of f converges absolutely at $s+a$, i.e. if and only if $\operatorname{Re}(s) + \operatorname{Re}(a) > \sigma_a(f)$. This gives the equality between the abscissas. Under that condition the integral converges and $\int_0^\infty e^{-at}f(t)e^{-st} dt = \int_0^\infty f(t)e^{-(s+a)t} dt = F(s+a)$. $\square$

Theorem 20.7: Time scaling. Let $c > 0$ and $f_{(c)}(t) = f(ct)$. Then $\sigma_a(f_{(c)}) = c\sigma_a(f)$ and, for $\operatorname{Re}(s) > c\sigma_a(f)$,

$$\mathcal{L}\{f(ct)\}(s) = \frac{1}{c} F\left(\frac{s}{c}\right).$$

Proof. With the substitution $\tau = ct$, i.e. $t = \tau/c$ and $dt = d\tau/c$ (a linear change of variable, $c > 0$, mapping $[0, \infty)$ to itself),

$$\int_0^\infty |f(ct)|e^{-\sigma t} dt = \frac{1}{c} \int_0^\infty |f(\tau)|e^{-(\sigma/c)\tau} d\tau,$$

finite if and only if $\sigma/c > \sigma_a(f)$, i.e. $\sigma > c\sigma_a(f)$: this gives the abscissa. The same computation without the modulus gives $\mathcal{L}\{f(ct)\}(s) = \frac{1}{c} \int_0^\infty f(\tau)e^{-(s/c)\tau} d\tau = \frac{1}{c} F(s/c)$. $\square$

Lemma 20.3: Having a transformable derivative implies exponential order. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be absolutely continuous on every compact interval, and suppose there exists $\sigma_0 \in \mathbb{R}$ with $M := \int_0^\infty |f'(\tau)|e^{-\sigma_0\tau} d\tau < \infty$. Setting $\beta = \max\{\sigma_0, 0\}$, the following hold:

$$|f(t)| \leq |f(0)| + Me^{\beta t} \quad (t \geq 0), \quad \sigma_a(f) \leq \beta, \quad \lim_{t \rightarrow +\infty} f(t)e^{-st} = 0 \quad \text{for } \operatorname{Re}(s) > \beta.$$

Proof. Step 1 — the bound. Since f is absolutely continuous on $[0, t]$, $f(t) = f(0) + \int_0^t f'(\tau) d\tau$ holds. For $0 \leq \tau \leq t$ one has $e^{\sigma_0\tau} \leq e^{\beta t}$: if $\sigma_0 \geq 0$ then $\beta = \sigma_0$ and $e^{\sigma_0\tau} \leq e^{\sigma_0 t}$ because the exponential is increasing; if $\sigma_0 < 0$ then $\beta = 0$ and $e^{\sigma_0\tau} \leq 1 = e^{\beta t}$. Hence

$$\left| \int_0^t f'(\tau) d\tau \right| \leq \int_0^t |f'(\tau)|e^{-\sigma_0\tau} e^{\sigma_0\tau} d\tau \leq e^{\beta t} \int_0^t |f'(\tau)|e^{-\sigma_0\tau} d\tau \leq Me^{\beta t},$$

and the first formula follows from the triangle inequality.

Step 2 — the abscissa. For $\sigma > \beta$,

$$\int_0^\infty |f(t)|e^{-\sigma t} dt \leq \int_0^\infty (|f(0)| + Me^{\beta t})e^{-\sigma t} dt = \frac{|f(0)|}{\sigma} + \frac{M}{\sigma - \beta} < \infty,$$

where $\sigma > \beta \geq 0$ was used for the finiteness of the first term. Hence $\sigma_a(f) \leq \beta$.

Step 3 — the limit. For $\operatorname{Re}(s) = \sigma > \beta$,

$$|f(t)e^{-st}| \leq (|f(0)| + Me^{\beta t})e^{-\sigma t} = |f(0)|e^{-\sigma t} + Me^{-(\sigma-\beta)t} \xrightarrow{t \rightarrow +\infty} 0,$$

because both exponents are negative. $\square$

Theorem 20.8: Differentiation in time. Let f be absolutely continuous on every compact subset of $[0, \infty)$ and suppose there exists $\sigma_0 \in \mathbb{R}$ with $\int_0^\infty |f'(t)|e^{-\sigma_0 t} dt < \infty$. Setting $\beta = \max\{\sigma_0, 0\}$, for every s with $\operatorname{Re}(s) > \beta$ both Laplace integrals, of f and of f' , converge absolutely and

$$\mathcal{L}\{f'\}(s) = sF(s) - f(0^+), \quad f(0^+) = f(0).$$

Proof. Step 1 — both transforms exist. By Lemma 20.3, $\sigma_a(f) \leq \beta$, hence $\mathcal{L}\{f\}$ converges absolutely for $\operatorname{Re}(s) > \beta$. For f' : if $\sigma > \beta \geq \sigma_0$ then $|f'(t)|e^{-\sigma t} \leq |f'(t)|e^{-\sigma_0 t}$, integrable by hypothesis. Moreover f , being absolutely continuous on $[0, T]$ for every T , is continuous at 0, so $f(0^+) = f(0)$ exists.

Step 2 — integration by parts on the truncation. The function f is absolutely continuous on $[0, T]$ and $t \mapsto e^{-st}$ is of class C^1 ; the integration-by-parts formula recalled among the

prerequisites, with $\frac{d}{dt}e^{-st} = -se^{-st}$, gives

$$\int_0^T f'(t)e^{-st} dt = [f(t)e^{-st}]_0^T + s \int_0^T f(t)e^{-st} dt = f(T)e^{-sT} - f(0) + s \int_0^T f(t)e^{-st} dt.$$

Step 3 — the limit $T \rightarrow +\infty$. The left-hand side tends to $\mathcal{L}\{f'\}(s)$ and the last integral to $F(s)$, because both converge absolutely (Step 1). The term $f(T)e^{-sT}$ tends to zero by Lemma 20.3, Step 3, since $\operatorname{Re}(s) > \beta$. Passing to the limit gives $\mathcal{L}\{f'\}(s) = -f(0) + sF(s)$. $\square$

Corollary 20.4: Second derivative and the n -th derivative. Let $n \geq 1$ and let $f, f', \dots, f^{(n-1)}$ be absolutely continuous on every compact subset of $[0, \infty)$, with $\int_0^\infty |f^{(n)}(t)|e^{-\sigma_0 t} dt < \infty$ for some $\sigma_0 \in \mathbb{R}$. Setting $\beta = \max\{\sigma_0, 0\}$, for $\operatorname{Re}(s) > \beta$

$$\mathcal{L}\{f^{(n)}\}(s) = s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^+).$$

In particular $\mathcal{L}\{f''\}(s) = s^2 F(s) - s f(0^+) - f'(0^+)$.

Proof. Induction on n . For $n = 1$ this is Theorem 20.8. Suppose the formula holds for n and let $f, \dots, f^{(n)}$ be absolutely continuous with $\int_0^\infty |f^{(n+1)}|e^{-\sigma_0 t} dt < \infty$. Applying Theorem 20.8 to the function $g = f^{(n)}$ (absolutely continuous by hypothesis, with $g' = f^{(n+1)}$) gives, for $\operatorname{Re}(s) > \beta$,

$$\mathcal{L}\{f^{(n+1)}\}(s) = s \mathcal{L}\{f^{(n)}\}(s) - f^{(n)}(0^+).$$

Lemma 20.3 applied successively to $f^{(n)}, f^{(n-1)}, \dots, f$ guarantees that all the transforms involved converge absolutely for $\operatorname{Re}(s) > \beta$ (at every step the abscissa stays $\leq \beta$, because $\beta \geq 0$ and $\max\{\beta, 0\} = \beta$). Substituting the induction hypothesis,

$$\mathcal{L}\{f^{(n+1)}\}(s) = s \left(s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^+) \right) - f^{(n)}(0^+) = s^{n+1} F(s) - \sum_{k=0}^n s^{n-k} f^{(k)}(0^+),$$

which is the formula for $n + 1$. The case $n = 2$ is the one stated in full. $\square$

Theorem 20.9: Integration in time. Let f be locally integrable with $\sigma_a(f) < +\infty$ and let $g(t) = \int_0^t f(\tau) d\tau$. Then $\sigma_a(g) \leq \max\{\sigma_a(f), 0\}$ and, for $\operatorname{Re}(s) > \max\{\sigma_a(f), 0\}$,

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\}(s) = \frac{F(s)}{s}.$$

Proof. **Step 1 — g satisfies the hypotheses of the differentiation theorem.** The function g is the indefinite integral of a locally integrable function, hence it is absolutely continuous on every compact set, $g(0) = 0$ and $g' = f$ almost everywhere.

Step 2 — choice of σ_0 and application. Let s be such that $\operatorname{Re}(s) > \max\{\sigma_a(f), 0\}$; in particular $\operatorname{Re}(s) > 0$. Choose σ_0 with $\max\{\sigma_a(f), 0\} < \sigma_0 < \operatorname{Re}(s)$, which is possible because between two distinct reals there is always a third. Then $\sigma_0 > \sigma_a(f)$ and $\sigma_0 > 0$, so $\beta = \max\{\sigma_0, 0\} = \sigma_0$. Moreover $\int_0^\infty |g'(t)|e^{-\sigma_0 t} dt = \int_0^\infty |f(t)|e^{-\sigma_0 t} dt < \infty$, and $\beta = \sigma_0 < \operatorname{Re}(s)$. Theorem 20.8 applied to g then gives, at s ,

$$\mathcal{L}\{f\}(s) = \mathcal{L}\{g'\}(s) = s G(s) - g(0) = s G(s),$$

with $G(s) = \mathcal{L}\{g\}(s)$ absolutely convergent. Since $\operatorname{Re}(s) > 0$ we have $s \neq 0$, and dividing gives $G(s) = F(s)/s$.

Step 3 — the abscissa. Step 2 shows that the absolute convergence of $\mathcal{L}\{g\}$ holds at every s with $\operatorname{Re}(s) > \max\{\sigma_a(f), 0\}$; by the definition of infimum, $\sigma_a(g) \leq \max\{\sigma_a(f), 0\}$. $\square$

Corollary 20.5: Multiplication by t . Let $\sigma_a = \sigma_a(f) < +\infty$. For every integer $n \geq 1$ and every s with $\operatorname{Re}(s) > \sigma_a$,

$$\mathcal{L}\{t f(t)\}(s) = -F'(s), \quad \mathcal{L}\{t^n f(t)\}(s) = (-1)^n F^{(n)}(s),$$

and the abscissa of absolute convergence of $t^n f$ does not exceed σ_a .

Proof. Corollary 20.2 states $F^{(n)}(s) = (-1)^n \mathcal{L}\{t^n f\}(s)$ on Ω_a . Multiplying both sides by $(-1)^n$ and using $(-1)^n(-1)^n = 1$ gives $\mathcal{L}\{t^n f\}(s) = (-1)^n F^{(n)}(s)$; the case $n = 1$ is the first formula. The bound on the abscissa is Lemma 20.2. $\square$

Definition 20.5: Convolution of causal signals. For $f, g : [0, \infty) \rightarrow \mathbb{C}$ locally integrable, set

$$(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau, \quad t \geq 0,$$

whenever the integral converges absolutely. Extending f and g by zero for $t < 0$, this is the restriction to $[0, \infty)$ of the convolution on $\mathbb{R}$ of Definition 19.3: the factor $g(t - \tau)$ vanishes for $\tau > t$ and the factor $f(\tau)$ vanishes for $\tau < 0$, and the limits reduce on their own to 0 and t .

Theorem 20.10: Convolution theorem. Let $f, g : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable and let $\sigma \in \mathbb{R}$ be such that

$$A = \int_0^\infty |f(t)| e^{-\sigma t} dt < \infty, \quad B = \int_0^\infty |g(t)| e^{-\sigma t} dt < \infty.$$

Then:

- (i) for almost every $t \geq 0$ the integral defining $(f * g)(t)$ converges absolutely, and $f * g$ is measurable;
- (ii) $\int_0^\infty |(f * g)(t)| e^{-\sigma t} dt \leq AB < \infty$, in particular $\sigma_a(f * g) \leq \sigma$;
- (iii) for every s with $\operatorname{Re}(s) \geq \sigma$, $\mathcal{L}\{f * g\}(s) = F(s)G(s)$.

Proof. Step 1 — the change of variable that separates the two functions. Consider the linear map $\Phi(\tau, v) = (\tau, \tau + v)$ from $Q = [0, \infty)^2$ onto the sector $W = \{(\tau, t) : 0 \leq \tau \leq t\}$. It is bijective, with inverse $(\tau, t) \mapsto (\tau, t - \tau)$, and has Jacobian determinant

$$\det \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = 1,$$

hence it preserves Lebesgue measure on $\mathbb{R}^2$. The function $(\tau, t) \mapsto f(\tau)g(t - \tau)e^{-\sigma t}$ is measurable on W , being the composition of the measurable function $(x, y) \mapsto f(x)g(y)e^{-\sigma(x+y)}$ with Φ^{-1} .

Step 2 — Tonelli. The integrand, once moduli are taken, is nonnegative, so Tonelli's theorem allows integration in whichever order one prefers:

$$\int_0^\infty \left(\int_0^t |f(\tau)| |g(t - \tau)| d\tau \right) e^{-\sigma t} dt = \iint_W |f(\tau)| |g(t - \tau)| e^{-\sigma t} d\tau dt.$$

Transporting the integral to Q via Φ (which preserves measure) and using $e^{-\sigma t} = e^{-\sigma\tau} e^{-\sigma v}$ with $t = \tau + v$,

$$\iint_W |f(\tau)| |g(t - \tau)| e^{-\sigma t} d\tau dt = \iint_Q |f(\tau)| e^{-\sigma\tau} |g(v)| e^{-\sigma v} d\tau dv = A \cdot B < \infty.$$

Step 3 — (i) and (ii). Since the double integral of the moduli is finite, by Tonelli's theorem the function $t \mapsto \int_0^t |f(\tau)| |g(t-\tau)| d\tau$ is measurable and finite for almost every t : this is (i). Moreover, by the inequality between the modulus of the integral and the integral of the modulus,

$$\int_0^\infty |(f * g)(t)| e^{-\sigma t} dt \leq \int_0^\infty \left(\int_0^t |f(\tau)| |g(t-\tau)| d\tau \right) e^{-\sigma t} dt = AB,$$

which is (ii); the statement about the abscissa follows because for $\sigma' \geq \sigma$ one has $e^{-\sigma' t} \leq e^{-\sigma t}$.

Step 4 — (iii): Fubini. Let $\operatorname{Re}(s) \geq \sigma$. Then $|f(\tau)g(t-\tau)e^{-st}| = |f(\tau)| |g(t-\tau)| e^{-\operatorname{Re}(s)t} \leq |f(\tau)| |g(t-\tau)| e^{-\sigma t}$, whose double integral is finite by Step 2: Fubini's theorem applies and allows the order of integration to be exchanged and the integral to be transported via Φ . Hence

$$\mathcal{L}\{f * g\}(s) = \iint_W f(\tau)g(t-\tau)e^{-st} d\tau dt = \iint_Q f(\tau)e^{-s\tau} g(v)e^{-sv} d\tau dv,$$

and this last double integral, by Fubini's theorem applied to the product of two functions of separated variables, equals

$$\left(\int_0^\infty f(\tau)e^{-s\tau} d\tau \right) \left(\int_0^\infty g(v)e^{-sv} dv \right) = F(s)G(s).$$

□

***Remark 20.4: The product in the s -domain is what makes Laplace useful.** Theorem 20.10 is the technical reason why the analysis of linear systems is done in s and not in t : the response of a causal system to a causal input is the convolution $y = h * u$ (Definition 20.2), i.e. an integral for every instant, and it becomes the product $Y(s) = H(s)U(s)$, i.e. a multiplication. The same identity in the Fourier domain is Theorem 19.4; the difference is that here one does not require $f, g \in L^1(\mathbb{R})$, but only that some σ exist that damps both: the step and the growing exponential qualify, and in Fourier they do not.*

20.5. The elementary transforms, computed one by one

No row of the table should be memorized: every row is obtained from one of the rules of Section 20.4 applied to one of the only two transforms that must be computed by hand, that of the step and that of the exponential. Here the computation is carried out for all of them, declaring the region of convergence each time.

Proposition 20.2: The unit step. For $f = u$ one has $\sigma_c = \sigma_a = 0$, the ROC is the open half-plane $\operatorname{Re}(s) > 0$, and

$$\mathcal{L}\{u\}(s) = \frac{1}{s}, \quad \operatorname{Re}(s) > 0.$$

Proof. For $\operatorname{Re}(s) = \sigma > 0$ the function $t \mapsto e^{-st}$ has antiderivative $-e^{-st}/s$ (the derivative of e^{-st} is $-se^{-st}$, and $s \neq 0$), hence

$$\int_0^T e^{-st} dt = \frac{1 - e^{-sT}}{s}.$$

Since $|e^{-sT}| = e^{-\sigma T} \rightarrow 0$ for $\sigma > 0$, the limit as $T \rightarrow \infty$ equals $1/s$. Convergence is absolute: $\int_0^\infty e^{-\sigma t} dt = 1/\sigma < \infty$. For $\sigma < 0$ the integrand $e^{-\sigma t} = e^{|\sigma|t}$ is not even infinitesimal and the integral diverges; for $\sigma = 0$ Example 20.1 shows that the limit does not exist. Hence $\sigma_c = \sigma_a = 0$ and the line $\operatorname{Re}(s) = 0$ is excluded. □

Proposition 20.3: The complex exponential. Let $a \in \mathbb{C}$ and $f(t) = e^{at}$. Then $\sigma_c = \sigma_a = \operatorname{Re}(a)$, the ROC is the open half-plane $\operatorname{Re}(s) > \operatorname{Re}(a)$, and

$$\mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a}, \quad \operatorname{Re}(s) > \operatorname{Re}(a).$$

Proof. Apply Theorem 20.6 with $f = u$ and with $-a$ in place of a : the function $e^{at}u(t)$ has transform $U(s-a) = \frac{1}{s-a}$, with abscissa $\sigma_a(u) + \operatorname{Re}(a) = \operatorname{Re}(a)$ by Proposition 20.2. A direct computation confirms this: for $\operatorname{Re}(s-a) > 0$,

$$\int_0^T e^{at} e^{-st} dt = \int_0^T e^{-(s-a)t} dt = \frac{1 - e^{-(s-a)T}}{s-a} \xrightarrow{T \rightarrow \infty} \frac{1}{s-a}.$$

The exclusion of the line $\operatorname{Re}(s) = \operatorname{Re}(a)$ follows from the same property for the step, carried over by the frequency shift, which is a bijection between the two regions of convergence. $\square$

Corollary 20.6: The integer powers. For every integer $n \geq 0$, the function t^n has $\sigma_a = 0$ and

$$\mathcal{L}\{t^n\}(s) = \frac{n!}{s^{n+1}}, \quad \operatorname{Re}(s) > 0.$$

Proof. Step 1 — the n -th derivative of $1/s$. It is proved by induction that $\frac{d^n}{ds^n} s^{-1} = (-1)^n n! s^{-n-1}$ for $s \neq 0$. For $n = 0$ this is $s^{-1} = s^{-1}$. If it holds for n , then differentiating,

$$\frac{d^{n+1}}{ds^{n+1}} s^{-1} = \frac{d}{ds} [(-1)^n n! s^{-n-1}] = (-1)^n n! (-n-1) s^{-n-2} = (-1)^{n+1} (n+1)! s^{-n-2}.$$

Step 2 — applying the multiplication rule. Corollary 20.5 with $f = u$ (which has $\sigma_a = 0$ by Proposition 20.2) and $U(s) = 1/s$ gives, for $\operatorname{Re}(s) > 0$,

$$\mathcal{L}\{t^n\}(s) = \mathcal{L}\{t^n u(t)\}(s) = (-1)^n U^{(n)}(s) = (-1)^n (-1)^n n! s^{-n-1} = \frac{n!}{s^{n+1}}.$$

The claim $\sigma_a(t^n) \leq 0$ is Lemma 20.2; and $\sigma_a \geq 0$ because for $\sigma \leq 0$ the integrand $t^n e^{-\sigma t}$ is not infinitesimal (for $n \geq 1$ it tends to $+\infty$; for $n = 0$ Proposition 20.2 applies). Hence $\sigma_a = 0$. $\square$

Corollary 20.7: Damped powers. For every integer $n \geq 0$ and every $a \in \mathbb{C}$,

$$\mathcal{L}\{t^n e^{at}\}(s) = \frac{n!}{(s-a)^{n+1}}, \quad \operatorname{Re}(s) > \operatorname{Re}(a),$$

with $\sigma_a = \operatorname{Re}(a)$. In particular $\mathcal{L}\{t e^{-at}\}(s) = \frac{1}{(s+a)^2}$ for $\operatorname{Re}(s) > -\operatorname{Re}(a)$.

Proof. Apply Theorem 20.6 to the function $f(t) = t^n$, whose transform is $n!/s^{n+1}$ with $\sigma_a = 0$ (Corollary 20.6), exchanging a with $-a$: this gives

$$\mathcal{L}\{e^{at} t^n\}(s) = F(s-a) = \frac{n!}{(s-a)^{n+1}},$$

valid for $\operatorname{Re}(s) - \operatorname{Re}(a) > 0$, with abscissa $0 + \operatorname{Re}(a) = \operatorname{Re}(a)$. The particular case is $n = 1$ with a changed in sign. $\square$

Proposition 20.4: Sine and cosine. Let $\omega \in \mathbb{R}$, $\omega \neq 0$. The functions $\cos(\omega t)$ and $\sin(\omega t)$ have $\sigma_c = \sigma_a = 0$ and, for $\operatorname{Re}(s) > 0$,

$$\mathcal{L}\{\cos(\omega t)\}(s) = \frac{s}{s^2 + \omega^2}, \quad \mathcal{L}\{\sin(\omega t)\}(s) = \frac{\omega}{s^2 + \omega^2}.$$

Proof. Step 1 — Euler's formulas. For every real t , $\cos(\omega t) = \frac{e^{i\omega t} + e^{-i\omega t}}{2}$ and $\sin(\omega t) = \frac{e^{i\omega t} - e^{-i\omega t}}{2i}$ (Volume II, Chapter 3).

Step 2 — linearity. The two functions $e^{\pm i\omega t}$ have $\operatorname{Re}(\pm i\omega) = 0$, hence abscissa 0 and transforms $\frac{1}{s \mp i\omega}$ for $\operatorname{Re}(s) > 0$ (Proposition 20.3). Theorem 20.4 then gives, for $\operatorname{Re}(s) > \max\{0, 0\} = 0$,

$$\mathcal{L}\{\cos(\omega t)\}(s) = \frac{1}{2} \left(\frac{1}{s - i\omega} + \frac{1}{s + i\omega} \right) = \frac{1}{2} \cdot \frac{(s + i\omega) + (s - i\omega)}{(s - i\omega)(s + i\omega)} = \frac{1}{2} \cdot \frac{2s}{s^2 + \omega^2} = \frac{s}{s^2 + \omega^2},$$

having used $(s - i\omega)(s + i\omega) = s^2 - (i\omega)^2 = s^2 + \omega^2$. Likewise

$$\mathcal{L}\{\sin(\omega t)\}(s) = \frac{1}{2i} \left(\frac{1}{s - i\omega} - \frac{1}{s + i\omega} \right) = \frac{1}{2i} \cdot \frac{2i\omega}{s^2 + \omega^2} = \frac{\omega}{s^2 + \omega^2}.$$

Step 3 — the abscissa is exactly 0. From $|\cos(\omega t)| \leq 1$ it follows that $\sigma_a \leq 0$ (comparison with the step). If $\sigma_a < 0$ held the integral would converge absolutely at $s = 0$, i.e. $\int_0^\infty |\cos(\omega t)| dt < \infty$; but on every period $[k\pi/|\omega|, (k+1)\pi/|\omega|]$ that integral equals $2/|\omega|$, and the sum of infinitely many terms equal to $2/|\omega| > 0$ diverges. Hence $\sigma_a = 0$; and $\sigma_c = 0$ because at $s = 0$ the integral $\int_0^T \cos(\omega t) dt = \frac{\sin(\omega T)}{\omega}$ has no limit. The same argument holds for the sine.

Numerical check. Simpson quadrature with 400,000 subintervals on $[0, 60]$: $\mathcal{L}\{\cos 2t\}(0.5) = 0.1176471$, against $\frac{0.5}{0.25+4} = \frac{0.5}{4.25} = 0.1176471$; and $\mathcal{L}\{\sin 3t\}(1) = 0.3000000$, against $\frac{3}{1+9} = 0.3$. $\checkmark$ $\square$

Corollary 20.8: Damped sine and cosine. Let $a \in \mathbb{C}$ and $\omega \in \mathbb{R} \setminus \{0\}$. Then, for $\operatorname{Re}(s) > -\operatorname{Re}(a)$,

$$\mathcal{L}\{e^{-at} \cos(\omega t)\}(s) = \frac{s + a}{(s + a)^2 + \omega^2}, \quad \mathcal{L}\{e^{-at} \sin(\omega t)\}(s) = \frac{\omega}{(s + a)^2 + \omega^2},$$

with absolute abscissa of convergence $-\operatorname{Re}(a)$. The poles of these two functions are $s = -a \pm i\omega$.

Proof. This is Theorem 20.6 applied to the two functions of Proposition 20.4, which have $\sigma_a = 0$: the transform is evaluated at $s + a$ and the abscissa becomes $0 - \operatorname{Re}(a)$. Substituting $s \mapsto s + a$ in the two formulas gives the stated expressions. The poles are found by setting the denominator to zero: $(s + a)^2 = -\omega^2$, i.e. $s + a = \pm i\omega$, i.e. $s = -a \pm i\omega$.

Numerical check. With $a = 2$, $\omega = 3$, $s = 1$: the formula gives $\frac{1+2}{(1+2)^2+9} = \frac{3}{18} = 0.1666667$ for the sine and $\frac{3}{18} = 0.1666667$ also for the cosine (because $s + a = 3 = \omega$). Numerical quadrature gives 0.1666667 in both cases. $\checkmark$ $\square$

Proposition 20.5: Delayed step and rectangular pulse. Let $a > 0$.

(i) $\mathcal{L}\{u(t - a)\}(s) = \frac{e^{-as}}{s}$ for $\operatorname{Re}(s) > 0$, with $\sigma_c = \sigma_a = 0$.

(ii) For the window $\pi_a = u(t) - u(t - a)$ (value 1 on $[0, a]$, 0 afterward) the region of

convergence is all of $\mathbb{C}$ and

$$\mathcal{L}\{\pi_a\}(s) = \frac{1 - e^{-as}}{s} \quad (s \neq 0), \quad \mathcal{L}\{\pi_a\}(0) = a.$$

The function $s \mapsto \mathcal{L}\{\pi_a\}(s)$ is entire.

Proof. (i). This is Theorem 20.5 with $f = u$: one has $u(t - a) = u(t - a)u(t - a)$, hence $\mathcal{L}\{u(t - a)\} = e^{-as}\mathcal{L}\{u\} = e^{-as}/s$, and the abscissa does not change, remaining that of the step (Proposition 20.2).

(ii) — **the region.** π_a vanishes outside $[0, a]$, hence for every $\sigma \in \mathbb{R}$

$$\int_0^\infty |\pi_a(t)|e^{-\sigma t} dt = \int_0^a e^{-\sigma t} dt < \infty,$$

because the integrand is continuous on a compact set. Convergence is absolute for every $s \in \mathbb{C}$: $\sigma_a = -\infty$ and the ROC is the whole plane.

(ii) — **the value.** For $s \neq 0$, from the computation in Proposition 20.2 restricted to $[0, a]$: $\int_0^a e^{-st} dt = \frac{1 - e^{-as}}{s}$. For $s = 0$ the integral equals $\int_0^a dt = a$. The two expressions match up: from the expansion $e^{-as} = \sum_{k \geq 0} \frac{(-as)^k}{k!}$ it follows that

$$\frac{1 - e^{-as}}{s} = \frac{1}{s} \left(as - \frac{a^2 s^2}{2} + \frac{a^3 s^3}{6} - \dots \right) = a - \frac{a^2 s}{2} + \frac{a^3 s^2}{6} - \dots,$$

a power series convergent for every $s \in \mathbb{C}$: hence $s = 0$ is a removable singularity (Definition 15.6), the continuation equals a at 0, and the function is entire. $\square$

Definition 20.6: Laplace transform of a measure, and the delayed impulse. Let μ be a finite Borel measure on $[0, \infty)$. Its **Laplace transform** is

$$\mathcal{L}\{\mu\}(s) = \int_{[0, \infty)} e^{-st} d\mu(t).$$

For $a \geq 0$, δ_a denotes the **Dirac measure** concentrated at a , that is, $\delta_a(E) = 1$ if $a \in E$ and $\delta_a(E) = 0$ otherwise; one writes $\delta = \delta_0$. A locally integrable function f is identified with the measure $f(t) dt$, and in this case the definition reduces to the usual Laplace integral.

Proposition 20.6: Delayed impulse. For every $a \geq 0$ one has $\mathcal{L}\{\delta_a\}(s) = e^{-as}$ for every $s \in \mathbb{C}$: the region of convergence is the whole plane. In particular $\mathcal{L}\{\delta\} \equiv 1$. Moreover, for every locally integrable f ,

$$(\delta_a * f)(t) = f(t - a)u(t - a),$$

consistently with Theorem 20.5 and with Theorem 20.10.

Proof. Step 1 — the value of the transform. The integral of a continuous function g with respect to the Dirac measure δ_a equals $g(a)$: indeed δ_a is the probability measure assigning mass 1 to the single point a , and for every simple function $\sum c_k \mathbf{1}_{E_k}$ the integral is $\sum_k c_k \delta_a(E_k) = c_{k_0}$ where E_{k_0} is the unique set of the partition containing a ; passing to the limit over simple functions gives $\int g d\delta_a = g(a)$. With $g(t) = e^{-st}$, continuous for every fixed s , this gives $\mathcal{L}\{\delta_a\}(s) = e^{-as}$, defined for every $s \in \mathbb{C}$. For $a = 0$ the value is $e^0 = 1$.

Step 2 — the convolution. The convolution of a measure μ with a function is $(\mu * f)(t) = \int_{[0, t]} f(t - \tau) d\mu(\tau)$; with $\mu = \delta_a$ Step 1 gives $f(t - a)$ if $a \leq t$, and 0 if $a > t$ (because then $a \notin [0, t]$ and the mass is not caught): that is, exactly $f(t - a)u(t - a)$. Transforming,

$\mathcal{L}\{\delta_a * f\} = e^{-as}F(s) = \mathcal{L}\{\delta_a\} \cdot F$, in agreement with Theorem 20.5. $\square$

Warning. 0^- or 0^+ : why the delta forces a choice. The integral $\int_0^\infty$ of Definition 20.1 does not say whether the point 0 is included. For a locally integrable function the question is empty, because a point has zero measure. For a measure it is not: δ has all its mass at 0, and

$$\int_{[0,\infty)} e^{-st} d\delta(t) = 1, \quad \int_{(0,\infty)} e^{-st} d\delta(t) = 0.$$

Definition 20.6 integrates over $[0, \infty)$, i.e. it adopts the convention called “ 0^- ” in signal theory texts, and it is the one that makes the row $\delta \mapsto 1$ of the table true and makes Corollary 20.4 consistent when f has a jump at the origin. The Dirac delta as a *tempered distribution* is Definition 19.8 of Chapter 19: the Definition 20.6 used here is more modest (a measure, not a distribution) but sufficient for every row of the table, and it does not require distribution theory.

Example 20.2: Non-integer powers and the Gamma function. Let $\alpha > -1$ be real and $f(t) = t^\alpha$. The integral converges absolutely for $\sigma > 0$: near 0 because $\int_0^1 t^\alpha dt = \frac{1}{\alpha+1} < \infty$, and at infinity because $t^\alpha e^{-\sigma t}$ decays exponentially. For $s > 0$ real substitute $x = st$, $dx = s dt$:

$$\int_0^\infty t^\alpha e^{-st} dt = \int_0^\infty \left(\frac{x}{s}\right)^\alpha e^{-x} \frac{dx}{s} = \frac{1}{s^{\alpha+1}} \int_0^\infty x^\alpha e^{-x} dx = \frac{\Gamma(\alpha+1)}{s^{\alpha+1}},$$

with Γ Euler’s Gamma function (Chapter 11). The formula extends to the whole half-plane $\operatorname{Re}(s) > 0$ with the principal determination of $s^{\alpha+1} = e^{(\alpha+1)\operatorname{Log} s}$: both sides are holomorphic on $\operatorname{Re}(s) > 0$ (the first by Theorem 20.3, the second because the principal logarithm is holomorphic outside the negative real half-axis) and coincide on the half-line $(0, \infty)$, which has accumulation points in the domain: the identity principle (Theorem 15.10) forces equality everywhere. For $\alpha = n$ integer one recovers $\Gamma(n+1) = n!$ and Corollary 20.6. *Numerical check.* For $\alpha = \frac{1}{2}$ and $s = 1$ the formula gives $\Gamma(3/2) = \frac{\sqrt{\pi}}{2} = 0.8862269$; Simpson quadrature on $[0, 200]$ with 2,000,000 subintervals gives 0.8862268, with discrepancy $8 \cdot 10^{-8}$ due to the singularity of the derivative of the integrand at 0. $\checkmark$

Theorem 20.11: Table of Laplace transforms, with the region of convergence.

Every row of Table 20.1 and of Table 20.2 is an identity valid exactly in the region of convergence indicated in the third column, and the fourth column refers to the result that proves it. In particular no row is assumed: all of them follow from Proposition 20.2 and from Proposition 20.3 through the rules of Section 20.4.

Proof. The rows of Table 20.1 are, in order, Proposition 20.6 (two rows: $a = 0$ and $a > 0$), Proposition 20.2, Proposition 20.5(i) and (ii), Corollary 20.6 (two rows: $n = 1$ and general n), Example 20.2, Proposition 20.3, Corollary 20.7, Proposition 20.4 (two rows) and Corollary 20.8 (two rows). What remains to be justified are the last two rows, the hyperbolic ones. For $b \in \mathbb{C}$ one has $\cosh(bt) = \frac{e^{bt} + e^{-bt}}{2}$ and $\sinh(bt) = \frac{e^{bt} - e^{-bt}}{2}$; the two exponentials have abscissas $\operatorname{Re}(b)$ and $-\operatorname{Re}(b)$, hence by Theorem 20.4 the combination converges absolutely for $\operatorname{Re}(s) > \max\{\operatorname{Re}(b), -\operatorname{Re}(b)\} = |\operatorname{Re}(b)|$, and there

$$\mathcal{L}\{\cosh(bt)\}(s) = \frac{1}{2} \left(\frac{1}{s-b} + \frac{1}{s+b} \right) = \frac{s}{s^2 - b^2},$$

$$\mathcal{L}\{\sinh(bt)\}(s) = \frac{1}{2} \left(\frac{1}{s-b} - \frac{1}{s+b} \right) = \frac{b}{s^2 - b^2},$$

having used $(s-b)(s+b) = s^2 - b^2$. *Numerical check:* for $b = 1$ and $s = 2$ the formula gives $\frac{2}{4-1} = 0.6666667$, and quadrature on $[0, 60]$ gives 0.6666667. ✓
The rows of Table 20.2 are, in order, Theorems 20.4, 20.5, 20.6 and 20.7, Theorem 20.8 with Corollary 20.4, Theorem 20.9, Corollary 20.5 and Theorem 20.10: each was proved with its own region of convergence, reported here unchanged. $\square$

Table 20.1: Elementary pairs. The third column is the region of convergence: without it the second column does not identify the signal. Here $a \geq 0$, $n \geq 0$ integer, $\alpha > -1$ real, $\omega \in \mathbb{R} \setminus \{0\}$, $b, c \in \mathbb{C}$.

Signal	Transform $F(s)$	ROC	Proved in
δ (Dirac measure)	1	all of $\mathbb{C}$	Prop. 20.6
$\delta_a, a \geq 0$	e^{-as}	all of $\mathbb{C}$	Prop. 20.6
$u(t)$	$1/s$	$\operatorname{Re}(s) > 0$	Prop. 20.2
$u(t-a), a > 0$	e^{-as}/s	$\operatorname{Re}(s) > 0$	Prop. 20.5(i)
$u(t) - u(t-a), a > 0$	$(1 - e^{-as})/s$, value a at $s = 0$	all of $\mathbb{C}$	Prop. 20.5(ii)
t	$1/s^2$	$\operatorname{Re}(s) > 0$	Cor. 20.6
t^n	$n!/s^{n+1}$	$\operatorname{Re}(s) > 0$	Cor. 20.6
$t^\alpha, \alpha > -1$	$\Gamma(\alpha+1)/s^{\alpha+1}$	$\operatorname{Re}(s) > 0$	Ex. 20.2
$e^{ct}, c \in \mathbb{C}$	$1/(s-c)$	$\operatorname{Re}(s) > \operatorname{Re}(c)$	Prop. 20.3
$t^n e^{ct}$	$n!/(s-c)^{n+1}$	$\operatorname{Re}(s) > \operatorname{Re}(c)$	Cor. 20.7
$\cos(\omega t)$	$s/(s^2 + \omega^2)$	$\operatorname{Re}(s) > 0$	Prop. 20.4
$\sin(\omega t)$	$\omega/(s^2 + \omega^2)$	$\operatorname{Re}(s) > 0$	Prop. 20.4
$e^{-ct} \cos(\omega t)$	$(s+c)/((s+c)^2 + \omega^2)$	$\operatorname{Re}(s) > -\operatorname{Re}(c)$	Cor. 20.8
$e^{-ct} \sin(\omega t)$	$\omega/((s+c)^2 + \omega^2)$	$\operatorname{Re}(s) > -\operatorname{Re}(c)$	Cor. 20.8
$\cosh(bt)$	$s/(s^2 - b^2)$	$\operatorname{Re}(s) > \operatorname{Re}(b) $	Thm. 20.11
$\sinh(bt)$	$b/(s^2 - b^2)$	$\operatorname{Re}(s) > \operatorname{Re}(b) $	Thm. 20.11

Table 20.2: Rules. Here $\sigma_a = \sigma_a(f)$, $\beta = \max\{\sigma_0, 0\}$ with σ_0 the absolute abscissa of the derivative, and σ a common damping value for f and g as in Theorem 20.10.

Operation	Transform	ROC	Proved in
$\lambda f + \mu g$	$\lambda F(s) + \mu G(s)$	$\operatorname{Re}(s) > \max\{\sigma_a(f), \sigma_a(g)\}$	Thm. 20.4
$f(t-a)u(t-a), a \geq 0$	$e^{-as}F(s)$	$\operatorname{Re}(s) > \sigma_a$ (unchanged)	Thm. 20.5
$e^{-ct}f(t)$	$F(s+c)$	$\operatorname{Re}(s) > \sigma_a - \operatorname{Re}(c)$	Thm. 20.6
$f(ct), c > 0$	$F(s/c)/c$	$\operatorname{Re}(s) > c\sigma_a$	Thm. 20.7
$f'(t)$	$sF(s) - f(0^+)$	$\operatorname{Re}(s) > \beta$	Thm. 20.8
$f^{(n)}(t)$	$s^n F(s) - \sum_{k < n} s^{n-1-k} f^{(k)}(0^+)$	$\operatorname{Re}(s) > \beta$	Cor. 20.4
$\int_0^t f(\tau) d\tau$	$F(s)/s$	$\operatorname{Re}(s) > \max\{\sigma_a, 0\}$	Thm. 20.9
$t^n f(t)$	$(-1)^n F^{(n)}(s)$	$\operatorname{Re}(s) > \sigma_a$	Cor. 20.5
$(f * g)(t)$	$F(s)G(s)$	$\operatorname{Re}(s) \geq \sigma$	Thm. 20.10

20.6. Initial value and final value

The two formulas

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s), \quad \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

are the tool with which, in a control loop, one reads the initial value and the steady-state error without inverting the transform. They are also the two formulas most often used wrongly: both are true only under precise hypotheses, and both, if the hypotheses fail, return a number — a

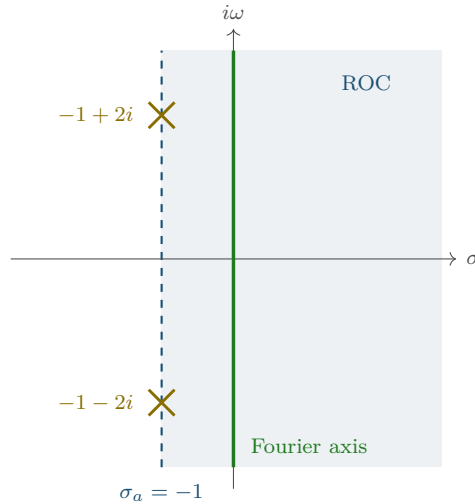


Figure 20.2: Pole map and region of convergence of $e^{-t} \cos(2t) u(t)$, whose transform is $\frac{s+1}{(s+1)^2+4}$ (Corollary 20.8 with $c = 1$, $\omega = 2$). The two poles are $s = -1 \pm 2i$ (ochre crosses), the ROC is the open half-plane $\operatorname{Re}(s) > -1$ (blue), and the imaginary axis (green) belongs to it: by Theorem 20.19 the signal then also has a Fourier transform, obtained by setting $s = i\omega$. Coordinates verified: setting $(s+1)^2 + 4$ to zero gives $s+1 = \pm 2i$.

wrong one — instead of signaling an error. This section proves the hypotheses and shows the two ways of getting it wrong.

Proposition 20.7: A transform vanishes at infinity. If f is locally integrable with $\sigma_a = \sigma_a(f) < +\infty$, then

$$\lim_{\sigma \rightarrow +\infty} F(\sigma) = 0$$

(limit along the reals). Consequently a function of s that does not tend to 0 as $\sigma \rightarrow +\infty$ is not the Laplace transform of any locally integrable function with finite abscissa.

Proof. Fix $\sigma_1 > \sigma_a$, so that $\int_0^\infty |f(t)|e^{-\sigma_1 t} dt < \infty$. Let (σ_n) be any real sequence with $\sigma_n \rightarrow +\infty$; discarding finitely many terms if necessary one may assume $\sigma_n \geq \sigma_1$ for every n . For $t > 0$ one has $|f(t)|e^{-\sigma_n t} \rightarrow 0$, and for every n

$$|f(t)e^{-\sigma_n t}| \leq |f(t)|e^{-\sigma_1 t},$$

a function integrable on $[0, \infty)$ and independent of n . The dominated convergence theorem gives $F(\sigma_n) = \int_0^\infty f(t)e^{-\sigma_n t} dt \rightarrow 0$. Since the sequence was arbitrary, $F(\sigma) \rightarrow 0$ as $\sigma \rightarrow +\infty$. The last statement is the contrapositive. $\square$

Theorem 20.12: Initial value theorem. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be absolutely continuous on every compact set and suppose there exists $\sigma_0 \in \mathbb{R}$ with $\int_0^\infty |f'(t)|e^{-\sigma_0 t} dt < \infty$. Then

$$\lim_{\operatorname{Re}(s) \rightarrow +\infty} sF(s) = f(0^+),$$

in the precise sense: for every $\varepsilon > 0$ there exists $\Sigma \in \mathbb{R}$ such that $|sF(s) - f(0^+)| < \varepsilon$ for every $s \in \mathbb{C}$ with $\operatorname{Re}(s) > \Sigma$. In particular the limit holds along the real axis.

Proof. Step 1 — rewriting. Let $\beta = \max\{\sigma_0, 0\}$. Theorem 20.8 gives, for every s with $\operatorname{Re}(s) > \beta$,

$$sF(s) - f(0^+) = \mathcal{L}\{f'\}(s).$$

It thus suffices to show that $\mathcal{L}\{f'\}(s) \rightarrow 0$ as $\operatorname{Re}(s) \rightarrow +\infty$.

Step 2 — the estimate depends only on the real part. For $\sigma = \operatorname{Re}(s)$,

$$|\mathcal{L}\{f'\}(s)| = \left| \int_0^\infty f'(t)e^{-st} dt \right| \leq \int_0^\infty |f'(t)| e^{-\sigma t} dt =: I(\sigma),$$

because $|e^{-st}| = e^{-\sigma t}$. The bound $I(\sigma)$ does not depend on $\operatorname{Im}(s)$: this is why the limit can be taken over the whole half-plane and not only along the real axis.

Step 3 — $I(\sigma) \rightarrow 0$. Fix $\sigma_1 > \beta$. For $\sigma \geq \sigma_1$ one has $|f'(t)|e^{-\sigma t} \leq |f'(t)|e^{-\sigma_1 t}$, integrable and independent of σ , and for every $t > 0$ one has $|f'(t)|e^{-\sigma t} \rightarrow 0$ as $\sigma \rightarrow +\infty$. As in the proof of Proposition 20.7, the dominated convergence theorem applied along an arbitrary sequence $\sigma_n \rightarrow +\infty$ gives $I(\sigma_n) \rightarrow 0$, hence $I(\sigma) \rightarrow 0$.

Step 4 — conclusion. Given $\varepsilon > 0$, choose $\Sigma \geq \sigma_1$ with $I(\sigma) < \varepsilon$ for $\sigma > \Sigma$ (Step 3). For every s with $\operatorname{Re}(s) > \Sigma$, Steps 1 and 2 give $|sF(s) - f(0^+)| \leq I(\operatorname{Re}(s)) < \varepsilon$. $\square$

Example 20.3: What happens when the initial-value hypothesis fails. (a) An impulse at the origin. Consider the measure $\mu = \delta + e^{-t} dt$, the sum of the Dirac measure at 0 and a function. Its transform (Definition 20.6, Proposition 20.6 and Proposition 20.3) is

$$\Phi(s) = 1 + \frac{1}{s+1} = \frac{s+2}{s+1}, \quad \operatorname{Re}(s) > -1.$$

The “function” part of μ is $g(t) = e^{-t}$, with $g(0^+) = 1$. But

$$s\Phi(s) = s + \frac{s}{s+1} \quad \text{equals} \quad 10.9090909, \quad 100.9900990, \quad 1000.9990010$$

for $s = 10$, $s = 100$, $s = 1000$: it diverges, instead of tending to 1. $\checkmark$ (Independent check: $s + \frac{s}{s+1} \geq s \rightarrow +\infty$.) The hypothesis that fails is the most radical possible: Φ is not the transform of any function. Indeed $\Phi(\sigma) \rightarrow 1 \neq 0$ as $\sigma \rightarrow +\infty$, and Proposition 20.7 then excludes Φ from being the transform of a locally integrable function with finite abscissa. The initial value theorem requires f to be absolutely continuous: an impulse at the origin is not, and the formula fails spectacularly.

(b) An integrable singularity at the origin. Let $f(t) = t^{-1/2}$ for $t > 0$. It is locally integrable and, by Example 20.2 with $\alpha = -\frac{1}{2}$,

$$F(s) = \frac{\Gamma(1/2)}{s^{1/2}} = \sqrt{\frac{\pi}{s}}, \quad \operatorname{Re}(s) > 0,$$

where $\Gamma(1/2) = \sqrt{\pi}$. Then $sF(s) = \sqrt{\pi s} \rightarrow +\infty$: the formula does not return a finite number. Here there is no contradiction but no information either, because f is not absolutely continuous on $[0, T]$ (it is not even bounded near 0) and $f(0^+) = +\infty$ is not a number. This is a case where the formula “tells the truth” only because both sides diverge: it is not a check on the theorem, it is a coincidence.

Theorem 20.13: Final value theorem. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable and suppose there exists finite

$$L = \lim_{t \rightarrow +\infty} f(t).$$

Then $\sigma_a(f) \leq 0$ (hence F is defined for $\operatorname{Re}(s) > 0$) and

$$\lim_{s \rightarrow 0^+} sF(s) = L,$$

where the limit is taken along the positive reals.

Proof. Step 1 — the abscissa. Since $f(t) \rightarrow L$, there exists T_1 with $|f(t) - L| \leq 1$, and hence $|f(t)| \leq |L| + 1$, for $t \geq T_1$. For $\sigma > 0$,

$$\int_0^\infty |f(t)|e^{-\sigma t} dt \leq \int_0^{T_1} |f(t)| dt + (|L| + 1) \int_{T_1}^\infty e^{-\sigma t} dt = \int_0^{T_1} |f| dt + (|L| + 1) \frac{e^{-\sigma T_1}}{\sigma} < \infty,$$

the first term being finite because f is integrable on every compact set. Hence $\sigma_a(f) \leq 0$.

Step 2 — the decomposition. Let $\varepsilon > 0$ and choose $T_0 \geq 0$ such that $|f(t) - L| < \varepsilon$ for every $t \geq T_0$. For real $s > 0$ split

$$sF(s) = \underbrace{s \int_0^{T_0} f(t)e^{-st} dt}_{A(s)} + \underbrace{s \int_{T_0}^\infty (f(t) - L)e^{-st} dt}_{B(s)} + \underbrace{sL \int_{T_0}^\infty e^{-st} dt}_{C(s)},$$

a legitimate decomposition because all three integrals converge absolutely for $s > 0$ (Step 1) and the sum of the three integrands is $sf(t)e^{-st}$ on each of the two intervals.

Step 3 — the three pieces. The third is computed exactly: $\int_{T_0}^\infty e^{-st} dt = \frac{e^{-sT_0}}{s}$, hence $C(s) = Le^{-sT_0} \rightarrow L$ as $s \rightarrow 0^+$ (continuity of the exponential). The second is bounded:

$$|B(s)| \leq s \int_{T_0}^\infty |f(t) - L|e^{-st} dt \leq s\varepsilon \int_{T_0}^\infty e^{-st} dt = \varepsilon e^{-sT_0} \leq \varepsilon.$$

The first is bounded by $|A(s)| \leq s \int_0^{T_0} |f(t)| dt$ (because $e^{-st} \leq 1$ for $t \geq 0$, $s > 0$), a quantity that tends to 0 as $s \rightarrow 0^+$, since $\int_0^{T_0} |f|$ is a fixed finite number.

Step 4 — conclusion. From $|sF(s) - L| \leq |A(s)| + |B(s)| + |C(s) - L|$ it follows that

$$\limsup_{s \rightarrow 0^+} |sF(s) - L| \leq 0 + \varepsilon + 0 = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the lim sup is zero, i.e. $sF(s) \rightarrow L$. □

Remark 20.5: The hypothesis is on the signal, not on the transform. Theorem 20.13 assumes that the limit of f exists: this is a hypothesis on the time domain, which is not read directly off the s -domain. The operational condition found in control texts — “all poles of $sF(s)$ in the half-plane $\operatorname{Re}(s) < 0$ ” — is a sufficient condition valid for rational F , and is proved further on as Corollary 20.11, after the partial fraction decomposition. Doetsch proves the version with $s \rightarrow 0$ inside a Stolz sector $|\arg s| \leq \theta < \pi/2$ and places it among the Abelian theorems (Introduction to the Theory and Application of the Laplace Transformation, Ch. 33 “Asymptotic Behaviour of the Image Function near Infinity,” pp. 221–230, and Ch. 34, pp. 231–233): the proof given here, with the limit along the reals, is sufficient for every use made of it in this chapter.

Example 20.4: What happens when the final-value hypothesis fails. (a) Poles on the imaginary axis. Let $f(t) = \sin t$, with $F(s) = \frac{1}{s^2 + 1}$ for $\operatorname{Re}(s) > 0$ (Proposition 20.4).

Then

$$sF(s) = \frac{s}{s^2 + 1} \quad \text{equals} \quad 0.0990099, \quad 0.00999990, \quad 0.0010000$$

for $s = 0.1, 0.01, 0.001$: it tends to 0. But $\lim_{t \rightarrow \infty} \sin t$ *does not exist*: the formula returns the number 0, which is not the final value of anything. ✓ (Independent check: $|s/(s^2 + 1)| \leq s \rightarrow 0$.) The hypothesis violated is the existence of the limit; in the s -plane the symptom is that the poles of $sF(s)$, namely $\pm i$, lie *on* the imaginary axis and not to its left.

(b) A pole in the right half-plane. Let $f(t) = e^t$, with $F(s) = \frac{1}{s-1}$ for $\operatorname{Re}(s) > 1$ (Proposition 20.3). The expression $sF(s) = \frac{s}{s-1}$ makes sense as a rational function even at $s = 0$, where it equals 0; and it equals -0.1111111 , -0.0101010 , -0.0010010 for $s = 0.1, 0.01, 0.001$, tending to 0. But $f(t) = e^t \rightarrow +\infty$. ✓ The error here is even more blatant: the point $s = 0$ *does not belong to the region of convergence* $\operatorname{Re}(s) > 1$, and evaluating $sF(s)$ near 0 means using the analytic continuation of F outside the ROC. This is why the ROC must be declared in every row of the table.

(c) A double pole at the origin: both sides diverge. Let $F(s) = \frac{s+3}{s^2(s+1)}$, whose inverse transform is $f(t) = 3t - 2 + 2e^{-t}$ (Worked Exercise 20.6). Then $sF(s) = \frac{s+3}{s(s+1)}$ equals 28.1818182, 298.0198020, 2998.0019980 for $s = 0.1, 0.01, 0.001$, while $f(10) = 28.0000908$, $f(100) = 298.0000000$, $f(1000) = 2998.0000000$: the two sequences diverge *together*, and with almost the same value, because $f(t) \approx 3t$ and $sF(s) \approx 3/s$. ✓ This is not a confirmation of the theorem: the theorem does not apply, because the limit of f does not exist finite. It is only how, in this case, the divergence shows up on both sides.

20.7. Inversion: uniqueness, partial fractions, residues

Inverting the transform means going back from F to f . Even before finding a formula one must know that the answer is unique, otherwise “the” inverse transform does not exist as an object: this is Theorem 20.14. Then, for rational F — which are practically all those met in linear systems with constant coefficients — the answer is written explicitly, in two ways that turn out to be the same way: the partial fraction decomposition and the sum of residues.

Theorem 20.14: Uniqueness of the inverse transform (Lerch). Let $f, g : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable with finite absolute abscissas of convergence, and let $s_0 \in \mathbb{R}$ with $s_0 > \max\{\sigma_a(f), \sigma_a(g)\}$. If

$$F(s_0 + n) = G(s_0 + n) \quad \text{for every integer } n \geq 0,$$

then $f = g$ almost everywhere on $[0, \infty)$.

Proof. Let $h = f - g$; by Theorem 20.4 one has $\sigma_a(h) \leq \max\{\sigma_a(f), \sigma_a(g)\} < s_0$ and $H(s_0 + n) = 0$ for every $n \geq 0$. It must be shown that $h = 0$ almost everywhere.

Step 1 — the change of variable $x = e^{-t}$. The map $t \mapsto x = e^{-t}$ is a C^∞ diffeomorphism from $(0, \infty)$ onto $(0, 1)$, decreasing, with inverse $x \mapsto t = -\ln x$ and $dt = -dx/x$. Set

$$\eta(x) = h(-\ln x) x^{s_0-1}, \quad x \in (0, 1).$$

With this substitution, for every $n \geq 0$,

$$H(s_0 + n) = \int_0^\infty h(t) e^{-(s_0+n)t} dt = \int_0^1 h(-\ln x) x^{s_0+n} \frac{dx}{x} = \int_0^1 \eta(x) x^n dx,$$

having used $e^{-(s_0+n)t} = (e^{-t})^{s_0+n} = x^{s_0+n}$ and the fact that the endpoints $t = 0, \infty$ correspond to $x = 1, 0$, which compensates the sign of $-dx/x$. The same substitution applied to the moduli gives

$$\int_0^1 |\eta(x)| dx = \int_0^\infty |h(t)| e^{-s_0 t} dt < \infty,$$

because $s_0 > \sigma_a(h)$: hence $\eta \in L^1(0, 1)$.

Step 2 — all moments vanish. By hypothesis $\int_0^1 \eta(x) x^n dx = 0$ for every $n \geq 0$. By linearity of the integral, for every polynomial $p(x) = \sum_{n=0}^N c_n x^n$ one has $\int_0^1 \eta(x) p(x) dx = \sum_{n=0}^N c_n \int_0^1 \eta x^n dx = 0$.

Step 3 — from polynomials to continuous functions. Let φ be continuous on $[0, 1]$ and let $\varepsilon > 0$. By the Weierstrass approximation theorem in its polynomial form (recalled among the prerequisites; the corresponding trigonometric form is Corollary 17.3) there exists a polynomial p with $\sup_{[0,1]} |\varphi - p| < \varepsilon$. Then, by Step 2,

$$\left| \int_0^1 \eta \varphi dx \right| = \left| \int_0^1 \eta (\varphi - p) dx \right| \leq \varepsilon \int_0^1 |\eta| dx = \varepsilon \|\eta\|_1.$$

Since $\varepsilon > 0$ is arbitrary and $\|\eta\|_1$ is finite, $\int_0^1 \eta \varphi dx = 0$ for every continuous φ .

Step 4 — conclusion. The fundamental lemma of the calculus of variations in L^1 (among the prerequisites) then gives $\eta = 0$ almost everywhere on $(0, 1)$. Since $x^{s_0-1} \neq 0$ for $x \in (0, 1)$, it follows that $h(-\ln x) = 0$ for almost every $x \in (0, 1)$. The map $x \mapsto -\ln x$ is a diffeomorphism with nowhere-vanishing derivative, hence it sends sets of measure zero to sets of measure zero and conversely (change-of-variables formula): $h = 0$ almost everywhere on $(0, \infty)$, i.e. $f = g$ almost everywhere. $\square$

Corollary 20.9: Two signals with the same transform coincide. If $F(s) = G(s)$ for every s in a half-plane $\operatorname{Re}(s) > \sigma^*$ contained in the regions of absolute convergence of f and g , then $f = g$ almost everywhere. In particular the notation $f = \mathcal{L}^{-1}\{F\}$ determines f up to sets of measure zero, and it determines exactly the unique continuous inverse transform, if one exists.

Proof. Choose a real $s_0 > \max\{\sigma^*, \sigma_a(f), \sigma_a(g)\}$: then $s_0 + n > \sigma^*$ for every integer $n \geq 0$, hence $F(s_0 + n) = G(s_0 + n)$, and Theorem 20.14 gives $f = g$ almost everywhere. If moreover f and g are both continuous and coincide almost everywhere, they coincide everywhere: the set $\{t : f(t) \neq g(t)\}$ is open in $[0, \infty)$ (by continuity of $f - g$) and every nonempty open subset of $[0, \infty)$ has positive measure, hence that set is empty. $\square$

Theorem 20.15: Partial fractions: simple poles. Let $R = N/D$ be a rational function with $\deg N < \deg D = r$ and with D having r distinct roots $p_1, \dots, p_r$. Then

$$R(s) = \sum_{k=1}^r \frac{A_k}{s - p_k}, \quad A_k = \frac{N(p_k)}{D'(p_k)} = \lim_{s \rightarrow p_k} (s - p_k) R(s),$$

for every $s \notin \{p_1, \dots, p_r\}$.

Proof. Step 1 — the value of A_k . Since p_k is a simple root of D , one has $D(p_k) = 0$ and $D'(p_k) \neq 0$. Hence

$$\lim_{s \rightarrow p_k} (s - p_k) R(s) = \lim_{s \rightarrow p_k} \frac{(s - p_k) N(s)}{D(s) - D(p_k)} = N(p_k) \cdot \lim_{s \rightarrow p_k} \frac{s - p_k}{D(s) - D(p_k)} = \frac{N(p_k)}{D'(p_k)},$$

by the definition of derivative and because $D'(p_k) \neq 0$. The two expressions for A_k coincide.

Step 2 — the difference is holomorphic on all of $\mathbb{C}$. Set

$$S(s) = R(s) - \sum_{k=1}^r \frac{A_k}{s - p_k},$$

a rational function, holomorphic on $\mathbb{C} \setminus \{p_1, \dots, p_r\}$. Fix k . As $s \rightarrow p_k$,

$$(s - p_k)S(s) = (s - p_k)R(s) - A_k - \sum_{j \neq k} \frac{A_j(s - p_k)}{s - p_j} \rightarrow A_k - A_k - 0 = 0,$$

since for $j \neq k$ the denominator $p_k - p_j$ does not vanish. In particular $(s - p_k)S(s)$ is bounded near p_k , hence $S(s) = \frac{(s - p_k)S(s)}{s - p_k}$ has at most a simple pole at p_k ; but the numerator tends to 0, so S itself is bounded in a punctured neighborhood of p_k . By Riemann's theorem on removable singularities (Theorem 15.13) the singularity is removable: S extends to a function holomorphic at p_k . Doing this for every k , S extends to an entire function.

Step 3 — S tends to zero at infinity. Writing $N(s) = \sum_{i=0}^m n_i s^i$ with $m < r$ and $D(s) = \sum_{i=0}^r d_i s^i$ with $d_r \neq 0$, for $s \neq 0$ one has

$$R(s) = \frac{s^m(n_m + n_{m-1}s^{-1} + \dots)}{s^r(d_r + d_{r-1}s^{-1} + \dots)} = s^{m-r} \cdot \frac{n_m + O(|s|^{-1})}{d_r + O(|s|^{-1})} \xrightarrow{|s| \rightarrow \infty} 0,$$

because $m - r \leq -1$ and the second factor tends to n_m/d_r , finite. Also every $A_k/(s - p_k) \rightarrow 0$. Hence $S(s) \rightarrow 0$ as $|s| \rightarrow \infty$.

Step 4 — Liouville. From Step 3 there exists $\rho > 0$ with $|S(s)| \leq 1$ for $|s| \geq \rho$; on the closed disk $|s| \leq \rho$, S is continuous (being entire) and hence bounded, the disk being compact. So S is entire and bounded on $\mathbb{C}$: by Liouville's theorem (Theorem 15.8) it is constant, and the constant is its limit at infinity, i.e. 0. Hence $S \equiv 0$, which is the claim. $\square$

Theorem 20.16: Partial fractions: multiple poles. Let $R = N/D$ be rational with $\deg N < \deg D$ and $D(s) = d \prod_{k=1}^r (s - p_k)^{m_k}$, with $p_1, \dots, p_r$ distinct, $m_k \geq 1$ and $d \neq 0$. Then

$$R(s) = \sum_{k=1}^r \sum_{j=1}^{m_k} \frac{A_{k,j}}{(s - p_k)^j}, \quad A_{k,j} = \frac{1}{(m_k - j)!} \lim_{s \rightarrow p_k} \frac{d^{m_k-j}}{ds^{m_k-j}} [(s - p_k)^{m_k} R(s)].$$

For $m_k = 1$ this recovers $A_{k,1} = \lim_{s \rightarrow p_k} (s - p_k)R(s)$ of Theorem 20.15.

Proof. **Step 1 — the principal part at p_k .** The function $g_k(s) = (s - p_k)^{m_k} R(s)$ coincides, away from the poles, with $\frac{N(s)}{d \prod_{i \neq k} (s - p_i)^{m_i}}$, which is holomorphic in a neighborhood of p_k (the denominator does not vanish there). Let $g_k(s) = \sum_{i \geq 0} \frac{g_k^{(i)}(p_k)}{i!} (s - p_k)^i$ be its Taylor expansion (Theorem 15.6), convergent in a disk around p_k . Dividing by $(s - p_k)^{m_k}$ gives the Laurent expansion of R at p_k (Theorem 15.4):

$$R(s) = \sum_{i \geq 0} \frac{g_k^{(i)}(p_k)}{i!} (s - p_k)^{i - m_k}.$$

The coefficient of $(s - p_k)^{-j}$, with $1 \leq j \leq m_k$, is obtained for $i = m_k - j$ and equals $\frac{g_k^{(m_k-j)}(p_k)}{(m_k-j)!}$, that is, exactly $A_{k,j}$ as defined in the statement. The *principal part* of R at p_k is thus $\Pi_k(s) = \sum_{j=1}^{m_k} \frac{A_{k,j}}{(s - p_k)^j}$.

Step 2 — the difference is entire. Set $S = R - \sum_{k=1}^r \Pi_k$, holomorphic on $\mathbb{C} \setminus \{p_1, \dots, p_r\}$. Fix k : in a punctured neighborhood of p_k , by Step 1, $R - \Pi_k$ is the sum of only the regular part

of the Laurent series, i.e. a convergent power series, hence it extends continuously to p_k ; and the other Π_i , $i \neq k$, are holomorphic at p_k . Hence S is bounded in a punctured neighborhood of p_k and, by Theorem 15.13, extends holomorphically. Doing this for every k , S is entire.

Step 3 — Liouville. As in Step 3 of the proof of Theorem 20.15, $R(s) \rightarrow 0$ as $|s| \rightarrow \infty$ because $\deg N < \deg D$, and every $A_{k,j}(s - p_k)^{-j} \rightarrow 0$; hence $S(s) \rightarrow 0$ at infinity, S is entire and bounded, and Theorem 15.8 gives $S \equiv 0$.

Step 4 — the case $m_k = 1$. For $m_k = 1$ the only admissible index is $j = 1$, the derivative of order $m_k - j = 0$ is the identity and $0! = 1$: the formula reduces to $A_{k,1} = \lim_{s \rightarrow p_k} (s - p_k)R(s)$. $\square$

Corollary 20.10: Inverse transform of a proper rational function. With the notation of Theorem 20.16, the unique (up to sets of measure zero) causal function whose Laplace transform is R is

$$f(t) = \sum_{k=1}^r \sum_{j=1}^{m_k} A_{k,j} \frac{t^{j-1}}{(j-1)!} e^{p_k t}, \quad t \geq 0,$$

and its region of convergence is the half-plane $\operatorname{Re}(s) > \max_k \operatorname{Re}(p_k)$.

Proof. By Corollary 20.7 with $n = j - 1$ and $c = p_k$,

$$\mathcal{L}\left\{\frac{t^{j-1}}{(j-1)!} e^{p_k t}\right\}(s) = \frac{1}{(j-1)!} \cdot \frac{(j-1)!}{(s - p_k)^j} = \frac{1}{(s - p_k)^j}, \quad \operatorname{Re}(s) > \operatorname{Re}(p_k).$$

Applying Theorem 20.4 to the finite combination with coefficients $A_{k,j}$ gives, for $\operatorname{Re}(s) > \max_k \operatorname{Re}(p_k)$, $\mathcal{L}\{f\}(s) = \sum_{k,j} \frac{A_{k,j}}{(s - p_k)^j} = R(s)$ by Theorem 20.16. Uniqueness is Corollary 20.9. That the abscissa is exactly $\max_k \operatorname{Re}(p_k)$ and not less follows from the fact that R has a pole at each p_k , while by Theorem 20.3 the transform is holomorphic to the right of the abscissa: no pole can lie in the region of convergence. $\square$

Theorem 20.17: Inversion by residues, rational case. Let R be proper rational with poles $p_1, \dots, p_r$ and let $f = \mathcal{L}^{-1}\{R\}$. Then, for every $t \geq 0$,

$$f(t) = \sum_{k=1}^r \operatorname{Res}_{s=p_k} [R(s) e^{st}],$$

where the residue is, as in Chapter 15, the coefficient a_{-1} of the Laurent expansion (Theorem 15.4) that appears in the residue theorem (Theorem 15.5).

Proof. **Step 1 — the expansion of the exponential factor.** Fix k and $t \geq 0$. For every s ,

$$e^{st} = e^{p_k t} e^{(s-p_k)t} = e^{p_k t} \sum_{i \geq 0} \frac{t^i}{i!} (s - p_k)^i,$$

a power series in $(s - p_k)$ convergent on all of $\mathbb{C}$.

Step 2 — the product and the coefficient a_{-1} . By Step 1 of the proof of Theorem 20.16, in a punctured neighborhood of p_k one has $R(s) = \Pi_k(s) + \Theta_k(s)$ with $\Pi_k(s) = \sum_{j=1}^{m_k} A_{k,j}(s - p_k)^{-j}$ and Θ_k holomorphic at p_k . Multiplying by e^{st} :

- $\Theta_k(s)e^{st}$ is holomorphic at p_k , so its Laurent expansion has no terms with negative exponent and contributes nothing to a_{-1} ;
- the term $A_{k,j}(s - p_k)^{-j}$ multiplied by the series of Step 1 gives $A_{k,j}e^{p_k t} \sum_{i \geq 0} \frac{t^i}{i!} (s - p_k)^{i-j}$, whose only term with exponent -1 is obtained for $i = j - 1$ and equals $A_{k,j}e^{p_k t} \frac{t^{j-1}}{(j-1)!}$.

Since the Laurent expansion is unique (Theorem 15.4), summing over j gives

$$\operatorname{Res}_{s=p_k} [R(s)e^{st}] = \sum_{j=1}^{m_k} A_{k,j} \frac{t^{j-1}}{(j-1)!} e^{p_k t}.$$

Step 3 — summing over the poles. Summing over k gives exactly the expression for $f(t)$ of Corollary 20.10. The case of simple poles can also be checked with the formula of Proposition 15.2(iii), applied to $p(s) = N(s)e^{st}$ and $q = D$: if $R = N/D$ with p_k a simple root of D and $N(p_k) \neq 0$, then $\operatorname{Res}_{s=p_k} [R(s)e^{st}] = \frac{N(p_k)e^{p_k t}}{D'(p_k)}$, which coincides with $A_k e^{p_k t}$ of Theorem 20.15. $\square$

Theorem 20.18: Bromwich inversion formula, integrable case. Let f be locally integrable with $\sigma_a < +\infty$, let $c > \sigma_a$ and suppose that

$$\xi \mapsto F(c + 2\pi i \xi) \quad \text{belongs to } L^1(\mathbb{R}).$$

Then, for almost every $t \in \mathbb{R}$,

$$f(t)u(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds := \frac{1}{2\pi i} \int_{-\infty}^{+\infty} F(c + i\nu) e^{(c+i\nu)t} i d\nu,$$

an absolutely convergent integral; in particular the right-hand side is zero for almost every $t < 0$, and it does not depend on c .

Proof. **Step 1 — the damped signal lies in $L^1(\mathbb{R})$.** Set $g(t) = f(t)u(t)e^{-ct}$, extended by zero for $t < 0$. Since $c > \sigma_a$,

$$\int_{\mathbb{R}} |g(t)| dt = \int_0^{\infty} |f(t)| e^{-ct} dt < \infty,$$

hence $g \in L^1(\mathbb{R})$.

Step 2 — the Fourier transform of g is F on the vertical line. In the convention of Chapter 19 (Definition 19.1),

$$\hat{g}(\xi) = \int_{\mathbb{R}} g(t) e^{-2\pi i \xi t} dt = \int_0^{\infty} f(t) e^{-(c+2\pi i \xi)t} dt = F(c + 2\pi i \xi),$$

a legitimate identity because $\operatorname{Re}(c + 2\pi i \xi) = c > \sigma_a$ and the integral converges absolutely. The hypothesis of the theorem says exactly that $\hat{g} \in L^1(\mathbb{R})$.

Step 3 — Fourier inversion. Theorem 19.9, applicable because $g \in L^1$ and $\hat{g} \in L^1$, gives for almost every $t \in \mathbb{R}$

$$g(t) = \int_{\mathbb{R}} \hat{g}(\xi) e^{2\pi i \xi t} d\xi = \int_{\mathbb{R}} F(c + 2\pi i \xi) e^{2\pi i \xi t} d\xi.$$

Step 4 — back to f and change of variable. Multiplying by e^{ct} and using $g(t)e^{ct} = f(t)u(t)$,

$$f(t)u(t) = \int_{\mathbb{R}} F(c + 2\pi i \xi) e^{(c+2\pi i \xi)t} d\xi.$$

With the linear substitution $\nu = 2\pi \xi$, i.e. $d\xi = d\nu/(2\pi)$, and setting $s = c + i\nu$ (which traces the vertical line $\operatorname{Re}(s) = c$ from bottom to top, with $ds = i d\nu$), one obtains

$$f(t)u(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(c + i\nu) e^{(c+i\nu)t} d\nu = \frac{1}{2\pi i} \int_{-\infty}^{+\infty} F(c + i\nu) e^{(c+i\nu)t} i d\nu,$$

which is the stated formula. Convergence is absolute because $|e^{(c+i\nu)t}| = e^{ct}$ is constant in ν and $\nu \mapsto F(c + i\nu)$ is integrable by hypothesis. Since for $t < 0$ the left-hand side is zero (by the definition of u), so is the right-hand side for almost every $t < 0$. Independence from c follows from the fact that the left-hand side does not contain c . $\square$

Remark 20.6: What we import: Bromwich inversion in the general case. Theorem 20.18 requires F to be integrable on the vertical line $\operatorname{Re}(s) = c$. This is a convenient but not necessary hypothesis: already for the step, $F(s) = 1/s$ is not integrable on any vertical line, because $|1/(c + i\nu)| \sim 1/|\nu|$. The inversion formula remains true, but must be understood as a principal value

$$f(t) = \frac{1}{2\pi i} \lim_{Y \rightarrow +\infty} \int_{c-iY}^{c+iY} F(s) e^{st} ds,$$

and in this general form it is here **imported**: we do not prove it. What exactly is missing: (a) the convergence of the symmetric limit, which does not follow from the mere convergence of the Laplace integral and requires a pointwise hypothesis on f (for instance bounded variation in a neighborhood of t , with value $\frac{f(t^+) + f(t^-)}{2}$ at jump points); (b) the fact that the value does not depend on c , which in the integrable case is immediate and in the general case requires a path-deformation argument; (c) the bound that allows the path to be closed with an arc to the left for $t > 0$ and to the right for $t < 0$, i.e. the analogue of Jordan's lemma for the kernel e^{st} , which for non-rational F depends on the behavior of F at infinity. The source is Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Ch. 24 "The Complex Inversion Formula for the Absolutely Converging Laplace Transformation. The Fourier Transformation," pp. 148–161, Ch. 25 "Deformation of the Path of Integration of the Complex Inversion Integral," pp. 161–169, and Ch. 26 "The Evaluation of the Complex Inversion Integral by Means of the Calculus of Residues," pp. 169–178. In the rational case — the only one needed in this treatise and in Chapter 25 — nothing is imported: Theorem 20.17 is proved in full, and the sum of residues appearing there is exactly what the Bromwich path would produce.

Corollary 20.11: Final value for rational functions: the condition on the poles. Let F be rational and suppose that $G(s) = sF(s)$ is proper (degree of the numerator less than the degree of the denominator) and that all poles of G have strictly negative real part. Then $f = \mathcal{L}^{-1}\{F\}$ has a finite limit at infinity and

$$\lim_{t \rightarrow +\infty} f(t) = \lim_{s \rightarrow 0} sF(s) = G(0).$$

Proof. Step 1 — structure of F . From $F(s) = G(s)/s$ with G proper it follows that F is proper, and the poles of F are those of G (all with negative real part) plus, if $G(0) \neq 0$, a simple pole at $s = 0$; if $G(0) = 0$ the factor s cancels and F has no pole at the origin. In both cases the decomposition of Theorem 20.16 reads

$$F(s) = \frac{c}{s} + \sum_k \sum_j \frac{A_{k,j}}{(s - p_k)^j}, \quad c = \lim_{s \rightarrow 0} sF(s) = G(0),$$

where the sum extends only over poles p_k with $\operatorname{Re}(p_k) < 0$ (the coefficient c is that of Theorem 20.16 with $p = 0$, $m = 1$, and is zero if $G(0) = 0$).

Step 2 — the inverse transform. By Corollary 20.10,

$$f(t) = c + \sum_k \sum_j A_{k,j} \frac{t^{j-1}}{(j-1)!} e^{p_k t},$$

having used $e^{0 \cdot t} = 1$ for the term at the origin.

Step 3 — the transient terms vanish. Let p have $\operatorname{Re}(p) = -\gamma < 0$ and $j \geq 1$. Then $|t^{j-1} e^{pt}| = t^{j-1} e^{-\gamma t}$, and by Lemma 20.1(ii) with $n = j - 1$ and $\varepsilon = \gamma/2$ (for $j \geq 2$; for $j = 1$

this is immediate)

$$t^{j-1}e^{-\gamma t} \leq \left(\frac{2(j-1)}{e\gamma}\right)^{j-1} e^{-\gamma t/2} \xrightarrow{t \rightarrow +\infty} 0.$$

Since the sum is finite, $f(t) \rightarrow c$.

Step 4 — consistency with the general theorem. The limit exists finite, hence Theorem 20.13 applies and gives $\lim_{s \rightarrow 0^+} sF(s) = c$; and $\lim_{s \rightarrow 0} sF(s) = G(0) = c$ because G , having no pole at 0, is continuous there. $\square$

20.8. Laplace and Fourier: the restriction to the imaginary axis

Remark 20.1 states the link in one line: setting $s = i\omega$ gives the Fourier transform. This section states under which conditions that operation is legitimate, and what happens when it is not.

Theorem 20.19: The Fourier transform is the restriction to the imaginary axis.

Let $f : [0, \infty) \rightarrow \mathbb{C}$ be locally integrable and suppose $f \in L^1(0, \infty)$, i.e. $\int_0^\infty |f(t)|dt < \infty$ (a condition equivalent to absolute convergence at $s = 0$, and implied by $\sigma_a(f) < 0$). Then:

- (i) the Laplace integral converges absolutely at every s with $\operatorname{Re}(s) \geq 0$, and F is continuous on the closed half-plane $\{\operatorname{Re}(s) \geq 0\}$ and holomorphic on the open set $\{\operatorname{Re}(s) > 0\}$;
- (ii) for every $\omega \in \mathbb{R}$ one has $F(i\omega) = \int_0^\infty f(t)e^{-i\omega t} dt$, i.e. the Fourier transform of f (extended by zero for $t < 0$) in the angular-frequency convention;
- (iii) in the convention of Chapter 19 (Definition 19.1), $\hat{f}(\xi) = F(2\pi i\xi)$ for every $\xi \in \mathbb{R}$.

Proof. Step 1 — absolute convergence on the closed half-plane. For $\sigma = \operatorname{Re}(s) \geq 0$ and $t \geq 0$ one has $e^{-\sigma t} \leq 1$, hence $|f(t)e^{-st}| = |f(t)|e^{-\sigma t} \leq |f(t)|$, integrable by hypothesis. The integral converges absolutely at every such s ; in particular $\sigma_a(f) \leq 0$ and, by Theorem 20.3, F is holomorphic on $\{\operatorname{Re}(s) > 0\}$.

Step 2 — continuity up to the boundary. Let $s_n \rightarrow s$ with $\operatorname{Re}(s_n) \geq 0$ and $\operatorname{Re}(s) \geq 0$. For every $t \geq 0$, $f(t)e^{-s_n t} \rightarrow f(t)e^{-st}$ (continuity of the exponential), and $|f(t)e^{-s_n t}| \leq |f(t)|$ by Step 1, an integrable bound independent of n . The dominated convergence theorem gives $F(s_n) \rightarrow F(s)$. Since the sequence was arbitrary, F is continuous on $\{\operatorname{Re}(s) \geq 0\}$.

Step 3 — (ii) and (iii). Setting $s = i\omega$ in the definition gives directly $F(i\omega) = \int_0^\infty f(t)e^{-i\omega t} dt$, an absolutely convergent integral by Step 1. Since f vanishes for $t < 0$, the integral over $[0, \infty)$ coincides with the integral over $\mathbb{R}$. In the convention of Chapter 19 the exponent is $-2\pi i\xi t$, hence

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(t)e^{-2\pi i\xi t} dt = \int_0^\infty f(t)e^{-(2\pi i\xi)t} dt = F(2\pi i\xi),$$

which is (iii); (ii) is the same formula with $\omega = 2\pi\xi$. $\square$

Remark 20.7: The dictionary between the two conventions. Chapter 19 uses the variable ξ (frequency, cycles per unit time) and the exponent $-2\pi i\xi t$; the literature on systems uses ω (angular frequency, radians per unit time) and the exponent $-i\omega t$. The bridge is $\omega = 2\pi\xi$, and Theorem 20.19 crosses it as follows:

$$F(i\omega) = \hat{f}\left(\frac{\omega}{2\pi}\right), \quad \hat{f}(\xi) = F(2\pi i\xi).$$

Remark 19.1 of Chapter 19 contains the complete dictionary, with the $\sqrt{2\pi}$ factors of the other conventions in use. In the rest of this chapter, and in Chapter 25, one always writes $H(i\omega)$ and means the angular-frequency convention.

Remark 20.8: On the boundary the restriction may not exist. The hypothesis $f \in L^1(0, \infty)$ of Theorem 20.19 cannot be weakened to $\sigma_a = 0$. The unit step has $\sigma_a = 0$ and $\mathcal{L}\{u\}(s) = 1/s$ (Proposition 20.2): restricting to the imaginary axis would give $1/(i\omega)$, which blows up at $\omega = 0$ and is not a continuous function on $\mathbb{R}$; and indeed $u \notin L^1(0, \infty)$, so Proposition 19.1 does not apply and the Fourier transform of the step is not a function but a tempered distribution (Definition 19.7). The phenomenon is the same as in Proposition 19.10, where $\hat{1} = \delta$: as soon as the signal stops being integrable, the right object is no longer a function. For the delta, however, the two theories agree without reservation: Proposition 20.6 gives $\mathcal{L}\{\delta\} \equiv 1$ on all of $\mathbb{C}$, and Proposition 19.10 gives $\hat{\delta} = 1$ on all of $\mathbb{R}$; restricting the first to the imaginary axis gives the second.

Corollary 20.12: Poles to the left, BIBO-stable system. Let H be a proper rational function ($\deg$ numerator $<$ $\deg$ denominator) whose poles $p_1, \dots, p_r$ all satisfy $\operatorname{Re}(p_k) < 0$, and let $h = \mathcal{L}^{-1}\{H\}$ be the impulse response (Definition 20.2). Then:

- (i) $h \in L^1(0, \infty)$, and $\sigma_a(h) = \max_k \operatorname{Re}(p_k) < 0$;
- (ii) the system is BIBO stable: for every measurable, bounded causal input v with $\|v\|_\infty = \sup_{t \geq 0} |v(t)| < \infty$, the output $y = h * v$ satisfies $\|y\|_\infty \leq \|h\|_1 \|v\|_\infty$;
- (iii) H is defined and continuous on the closed half-plane $\operatorname{Re}(s) \geq 0$, and $\omega \mapsto H(i\omega)$ is the frequency response of the system.

Proof. Step 1 — (i). By Corollary 20.10, $h(t) = \sum_{k,j} A_{k,j} \frac{t^{j-1}}{(j-1)!} e^{p_k t}$. Setting $\gamma_k = -\operatorname{Re}(p_k) > 0$, each term satisfies

$$\int_0^\infty \left| A_{k,j} \frac{t^{j-1}}{(j-1)!} e^{p_k t} \right| dt = \frac{|A_{k,j}|}{(j-1)!} \int_0^\infty t^{j-1} e^{-\gamma_k t} dt = \frac{|A_{k,j}|}{(j-1)!} \cdot \frac{(j-1)!}{\gamma_k^j} = \frac{|A_{k,j}|}{\gamma_k^j} < \infty,$$

where the value of the integral is $\mathcal{L}\{t^{j-1}\}(\gamma_k) = (j-1)!/\gamma_k^j$ (Corollary 20.6). The sum is finite, hence $h \in L^1(0, \infty)$ by the triangle inequality. The abscissa is $\max_k \operatorname{Re}(p_k)$ by Corollary 20.10, and is negative by hypothesis.

Step 2 — (ii). For every $t \geq 0$,

$$|y(t)| = \left| \int_0^t h(\tau) v(t-\tau) d\tau \right| \leq \int_0^t |h(\tau)| |v(t-\tau)| d\tau \leq \|v\|_\infty \int_0^t |h(\tau)| d\tau \leq \|v\|_\infty \|h\|_1,$$

finite by Step 1. Taking the supremum over $t \geq 0$ gives the bound. In particular a bounded input produces a bounded output.

Step 3 — (iii). From Step 1, $h \in L^1(0, \infty)$: Theorem 20.19 applies to h and gives the continuity of $H = \mathcal{L}\{h\}$ on the closed half-plane and the identification of $H(i\omega)$ with the Fourier transform of h , which is by definition the frequency response. $\square$

Remark 20.9: What is not proved here about BIBO stability. Definition 20.2, inherited from Volume II, states an “if and only if.” Corollary 20.12 proves the sufficient direction of it, and only for proper rational transfer functions. The two missing pieces — that the condition $h \in L^1$ is also necessary for BIBO stability, and that for a rational H a pole with $\operatorname{Re}(p) \geq 0$ always produces an unbounded output from a bounded input — belong to Chapter 24, where LTI systems are treated in their own right; the algebraic criterion that decides the sign of the real parts of the poles without computing them is in Chapter 25. The missing direction is not imported: it is only deferred to a chapter of this same volume, where it is proved.

20.9. Exercises inherited from Volume II

The exercises that follow come from Volume II, Chapter 4, and concern the topics of this chapter. They are reproduced in full, with their solutions, which follow in the same order as the exercises.

Exercises

Warm-up 4.2. Using the table of Laplace transforms in Appendix C (or the definition), compute the Laplace transform of $f(t) = e^{-2t}u(t)$.

Exercise 20.1. Compute $\mathcal{L}\{t \cdot u(t)\}$ (the unit ramp) using the integration property applied to the step $u(t)$.

Exercise 20.2. Verify, using the final value theorem, the asymptotic value $y(t) \rightarrow 1$ for the step response $Y(s) = 1/(s(1 + \tau s))$ of the Worked Example of this chapter.

Solutions

Solution, Warm-up 4.2. From the table (or from $\mathcal{L}\{e^{-at}\} = 1/(s+a)$ with $a = 2$): $\mathcal{L}\{e^{-2t}u(t)\} = \frac{1}{s+2}$.

Solution. The unit step $u(t)$ has $\mathcal{L}\{u(t)\} = \int_0^\infty e^{-st} dt = 1/s$ (for $\text{Re}(s) > 0$). The ramp $r(t) = t \cdot u(t)$ satisfies $r(t) = \int_0^t u(\tau) d\tau$ (the integral of a unit step from 0 to t equals exactly t). By the integration property (Theorem 20.1(3)):

$$\mathcal{L}\{t \cdot u(t)\} = \mathcal{L}\left\{\int_0^t u(\tau) d\tau\right\} = \frac{\mathcal{L}\{u(t)\}}{s} = \frac{1}{s^2}.$$

Solution. The final value theorem states $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$, provided the poles of $sY(s)$ lie in the half-plane $\text{Re}(s) < 0$ (here: the only pole of $sY(s) = 1/(1 + \tau s)$ is $s = -1/\tau < 0$, and the condition is satisfied). We compute:

$$\lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s(1 + \tau s)} = \lim_{s \rightarrow 0} \frac{1}{1 + \tau s} = 1.$$

The theorem correctly predicts $y(t) \rightarrow 1$, in agreement with the explicit solution $y(t) = 1 - e^{-t/\tau} \rightarrow 1$ found in the Worked Example: the final value theorem allows one to read the asymptotic value directly from $Y(s)$, without having to invert the transform — useful in practice for quickly checking the steady-state error of a control loop.

Inherited quiz

The items that follow come from the quiz of Volume II, Chapter 4, and are renumbered here.

1. The Laplace transform generalizes Fourier in order to handle
 - A) only periodic signals
 - B) non-integrable causal signals
 - C) only discrete signals
 - D) purely complex signals

BEST PRACTICE

Answer key.

1-B: Laplace converges for signals that grow at most exponentially, where Fourier does not converge.

Remark 20.10: The two inherited solutions, revisited in light of the chapter. The first solution uses Theorem 20.1(3) without declaring the region: the version with hypotheses is Theorem 20.9, which gives $\mathcal{L}\{t u(t)\} = 1/s^2$ for $\operatorname{Re}(s) > \max\{0, 0\} = 0$, in agreement with Corollary 20.6 with $n = 1$. The second solution invokes the condition “poles of $sY(s)$ in the left half-plane”: that condition is now a proved statement, Corollary 20.11, and its hypotheses are verified because $sY(s) = \frac{1}{1+\tau s}$ is proper and has the single pole $-1/\tau < 0$. The value $\frac{1}{1+\tau s} \rightarrow 1$ as $s \rightarrow 0$ is the $G(0)$ of the corollary.

20.10. Assigned exercises

These are *new* exercises, distinct both from the inherited ones (Section 20.9) and from those worked in Section 20.11. The answers are in the Teacher’s Key (Section 20.13).

Exercise 20.3: Regions of convergence, abscissas, and boundary. *Goal:* become used to writing the ROC next to every transform, and recognize the cases in which the boundary line does or does not belong to the region. *Estimated time:* 35 minutes. *Type:* symbolic with numerical check.

- (a) Compute F , σ_a and the ROC (stating whether the boundary belongs to it) for the four causal signals

$$e^{3t} \cos(4t), \quad t^2 e^{-t}, \quad u(t) - u(t-2), \quad t^{-1/2}.$$

- (b) Verify numerically, by quadrature, that $\mathcal{L}\{e^{3t} \cos 4t\}(5) = 0.1$ and $\mathcal{L}\{t^2 e^{-t}\}(1) = 0.25$, and recover the same values from the closed-form formulas.
- (c) Exhibit two signals f, g for which $\sigma_a(f+g) < \max\{\sigma_a(f), \sigma_a(g)\}$, and explain why Theorem 20.4 states an inequality and not an equality.
- (d) The step has $\sigma_a = 0$ and the ROC does *not* contain the line $\operatorname{Re}(s) = 0$; the window $u(t) - u(t-2)$ has ROC equal to all of $\mathbb{C}$. Explain the difference in one sentence referring to the behavior of the two functions as $t \rightarrow +\infty$.

Exercise 20.4: Initial value and final value: when they apply. *Goal:* decide, before computing, whether a theorem applies; and recognize the wrong number one obtains by applying it anyway. *Estimated time:* 45 minutes. *Type:* proof-based and numerical. Consider

$$F_1(s) = \frac{2s+1}{s(s^2+3s+2)}, \quad F_2(s) = \frac{5}{s(s^2+4)}, \quad F_3(s) = \frac{1}{(s-2)(s+3)}, \quad F_4(s) = \frac{s}{s+4}.$$

- (a) For each, state whether Theorem 20.12 and Corollary 20.11 are applicable, justifying by the position of the poles and by the ROC.
- (b) Invert F_1 , F_2 , F_3 using Theorem 20.15 or Theorem 20.16.
- (c) Compute $\lim_{s \rightarrow \infty} sF_i(s)$ and $\lim_{s \rightarrow 0} sF_i(s)$ and compare them with $f_i(0^+)$ and with $\lim_{t \rightarrow \infty} f_i(t)$, flagging the cases in which the comparison fails and why.
- (d) Prove that F_4 is not the Laplace transform of any locally integrable function, and find the measure of which it is the transform.

Exercise 20.5: A differential equation, a convolution, two paths. *Goal:* use the transform to solve an initial value problem and check the result by a route that does not go through Laplace. *Estimated time:* 45 minutes. *Type:* symbolic computation with independent check.

- Solve $y'' + 4y' + 3y = e^{-2t}$ with $y(0) = 0$, $y'(0) = 1$, using Corollary 20.4, and check the solution by substituting it back into the equation and into the initial conditions.
- Apply Corollary 20.11 to $Y(s)$ and compare with the limit of the solution found.
- Compute $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\}$ in two independent ways: with partial fractions and with Theorem 20.10 applied to $\frac{1}{s}$ and $\frac{1}{s^2+1}$.
- Evaluate the result of (c) at $t = 1$ to seven decimal places and check it with a numerical quadrature of the convolution.

20.11. Worked exercises

Exercise 20.6: A double pole at the origin, and the final value that is not there. *Type:* inversion with multiple poles, with two independent checks. Invert

$$F(s) = \frac{s+3}{s^2(s+1)},$$

state the region of convergence, and discuss whether the initial value and final value theorems apply.

Solution. Strategy. The denominator has a double pole at 0 and a simple one at -1 : apply Theorem 20.16, invert term by term with Corollary 20.10, and check the result by recomputing the transform.

Step 1 — the coefficients of the double pole. Here $p_1 = 0$ with $m_1 = 2$, and $g_1(s) = s^2 F(s) = \frac{s+3}{s+1}$, holomorphic in a neighborhood of 0. By Theorem 20.16:

$$A_{1,2} = \frac{1}{0!} g_1(0) = \frac{3}{1} = 3, \quad A_{1,1} = \frac{1}{1!} g_1'(0).$$

Compute $g_1'(s) = \frac{(s+1) - (s+3)}{(s+1)^2} = \frac{-2}{(s+1)^2}$, hence $g_1'(0) = -2$ and $A_{1,1} = -2$.

Step 2 — the simple pole. With $p_2 = -1$, $m_2 = 1$:

$$A_{2,1} = \lim_{s \rightarrow -1} (s+1)F(s) = \lim_{s \rightarrow -1} \frac{s+3}{s^2} = \frac{2}{1} = 2.$$

Step 3 — checking the decomposition. One checks the identity

$$-\frac{2}{s} + \frac{3}{s^2} + \frac{2}{s+1} = \frac{-2s(s+1) + 3(s+1) + 2s^2}{s^2(s+1)} = \frac{-2s^2 - 2s + 3s + 3 + 2s^2}{s^2(s+1)} = \frac{s+3}{s^2(s+1)}. \checkmark$$

Step 4 — inversion. By Corollary 20.10, with $\mathcal{L}^{-1}\{1/s\} = 1$, $\mathcal{L}^{-1}\{1/s^2\} = t$ and $\mathcal{L}^{-1}\{1/(s+1)\} = e^{-t}$,

$$\boxed{f(t) = -2 + 3t + 2e^{-t}}, \quad t \geq 0,$$

with region of convergence $\operatorname{Re}(s) > \max\{0, -1\} = 0$.

Independent check 1: recomputing the transform. By linearity and the rows of Table 20.1,

$$\mathcal{L}\{-2 + 3t + 2e^{-t}\}(s) = -\frac{2}{s} + \frac{3}{s^2} + \frac{2}{s+1} = \frac{s+3}{s^2(s+1)}. \checkmark$$

Independent check 2: numerical quadrature. Composite Simpson with 400,000 subintervals on $[0, 60]$: $\int_0^\infty f(t)e^{-2t}dt = 0.4166667$ against $F(2) = \frac{5}{4 \cdot 3} = \frac{5}{12} = 0.4166667$; and $\int_0^\infty f(t)e^{-t}dt = 2.0000000$ against $F(1) = \frac{4}{1 \cdot 2} = 2$. $\checkmark$ Moreover $f(2) = -2 + 6 + 2e^{-2} = 4 + 0.2706706 = 4.2706706$, which agrees with the value computed in double precision. $\checkmark$

Step 5 — the two theorems. *Initial value:* f is of class C^1 on $[0, \infty)$ with $f'(t) = 3 - 2e^{-t}$, and $\int_0^\infty |f'(t)|e^{-t}dt < \infty$: Theorem 20.12 applies and gives $\lim_{\operatorname{Re}(s) \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} \frac{s+3}{s(s+1)} = 0 = f(0^+) = -2 + 0 + 2$. $\checkmark$ *Final value:* it does not apply, because $f(t) = 3t - 2 + 2e^{-t} \rightarrow +\infty$ and the limit does not exist finite; in the s -plane the symptom is that $sF(s) = \frac{s+3}{s(s+1)}$ still has a pole at $s = 0$, which violates the hypothesis of Corollary 20.11. This is case (c) of Example 20.4.

Interpretation. A double pole at the origin is, in the time domain, a ramp: the factor $1/s$ integrates once (step $\rightarrow$ constant), twice (constant $\rightarrow$ ramp). The coefficient $A_{1,2} = 3$ is the slope of the ramp, and $A_{1,1} = -2$ its intercept.

Typical errors. (1) Looking for the decomposition in the form $\frac{A}{s^2} + \frac{B}{s+1}$, forgetting the term $\frac{A_{1,1}}{s}$: a pole of order 2 requires *two* coefficients. (2) Computing $A_{1,1}$ as $\lim_{s \rightarrow 0} sF(s)$, which here is infinite: the formula of Theorem 20.16 requires first multiplying by $s^m = s^2$ and then differentiating $m - j = 1$ time. (3) Applying the final value theorem, obtaining ∞ , and concluding that “the system diverges according to the theorem”: the theorem says nothing, because its hypotheses do not hold.

Extension. With $F(s) = \frac{s+3}{s^3(s+1)}$ the pole at the origin is triple and three coefficients appear; the dominant term becomes $\frac{3t^2}{2}$.

Exercise 20.7: Step response of a second-order system. *Type:* initial value problem solved with the transform, with numerical check by integrating the equation. Solve

$$y'' + 2y' + 5y = u(t), \quad y(0) = 0, \quad y'(0) = 0,$$

and check the initial value and the final value.

Solution. Strategy. Transform the equation with Corollary 20.4, solve algebraically for $Y(s)$, decompose and invert by recognizing the “damped sine and cosine” rows of Table 20.1.

Step 1 — transforming. With $y(0) = y'(0) = 0$, Corollary 20.4 gives $\mathcal{L}\{y'\} = sY$ and $\mathcal{L}\{y''\} = s^2Y$; with $\mathcal{L}\{u\} = 1/s$ (Proposition 20.2),

$$(s^2 + 2s + 5)Y(s) = \frac{1}{s} \quad \implies \quad Y(s) = \frac{1}{s(s^2 + 2s + 5)}.$$

Step 2 — the poles. $s^2 + 2s + 5 = (s+1)^2 + 4$ vanishes for $s+1 = \pm 2i$, i.e. $s = -1 \pm 2i$: two simple poles with real part $-1 < 0$, plus the simple pole $s = 0$. The region of convergence of Y is $\operatorname{Re}(s) > 0$.

Step 3 — partial fractions. Seek $Y(s) = \frac{A}{s} + \frac{Bs+C}{(s+1)^2+4}$. Multiplying by the common denominator, $1 = A(s^2 + 2s + 5) + (Bs + C)s$; equating coefficients:

$$s^0: 5A = 1 \Rightarrow A = \frac{1}{5}; \quad s^2: A + B = 0 \Rightarrow B = -\frac{1}{5}; \quad s^1: 2A + C = 0 \Rightarrow C = -\frac{2}{5}.$$

Hence

$$Y(s) = \frac{1/5}{s} - \frac{1}{5} \cdot \frac{s+2}{(s+1)^2+4}.$$

Step 4 — recognizing the rows of the table. Write $s + 2 = (s + 1) + 1$ and split:

$$\frac{s + 2}{(s + 1)^2 + 4} = \frac{s + 1}{(s + 1)^2 + 4} + \frac{1}{2} \cdot \frac{2}{(s + 1)^2 + 4},$$

where the first fraction is the transform of $e^{-t} \cos 2t$ and the second that of $e^{-t} \sin 2t$ (Corollary 20.8 with $c = 1$, $\omega = 2$). Hence

$$y(t) = \frac{1}{5} - \frac{1}{5}e^{-t} \cos 2t - \frac{1}{10}e^{-t} \sin 2t, \quad t \geq 0.$$

Step 5 — checking the initial conditions. $y(0) = \frac{1}{5} - \frac{1}{5} - 0 = 0 \checkmark$. Differentiating, $y'(t) = \frac{1}{5}e^{-t} \cos 2t + \frac{2}{5}e^{-t} \sin 2t + \frac{1}{10}e^{-t} \sin 2t - \frac{1}{5}e^{-t} \cos 2t$; at $t = 0$ the sin terms vanish and the two cos terms cancel, hence $y'(0) = 0 \checkmark$.

Independent check 1: numerical integration of the equation. Rewriting as a first-order system $y' = v$, $v' = 1 - 2v - 5y$ with $y(0) = v(0) = 0$ and integrating with fourth-order Runge–Kutta and step $h = T/200,000$:

t	closed form	RK4
1	0.1971672	0.1971672
2	0.2279344	0.2279344
5	0.2014973	0.2014973

✓

Independent check 2: recomputing the transform. Simpson quadrature of $\int_0^\infty y(t)e^{-2t}dt$ on $[0, 60]$: 0.0384615, against $Y(2) = \frac{1}{2(4+4+5)} = \frac{1}{26} = 0.0384615$. ✓

Step 6 — the two theorems. *Initial value:* $sY(s) = \frac{1}{s^2+2s+5} \rightarrow 0$ as $\text{Re}(s) \rightarrow +\infty$, and $y(0^+) = 0 \checkmark$. *Final value:* $G(s) = sY(s) = \frac{1}{s^2+2s+5}$ is proper and has poles $-1 \pm 2i$, both with negative real part: Corollary 20.11 applies and gives $\lim_{t \rightarrow \infty} y(t) = G(0) = \frac{1}{5} = 0.2$. The closed form confirms this, because $e^{-t} \rightarrow 0$. ✓

Interpretation. The system is second order with natural frequency $\omega_n = \sqrt{5}$ and damping $\zeta = 1/\sqrt{5}$, because $s^2 + 2s + 5 = s^2 + 2\zeta\omega_n s + \omega_n^2$ with $2\zeta\omega_n = 2$ and $\omega_n^2 = 5$; the real part -1 of the poles is the decay rate of the transient and the imaginary part 2 is the frequency of the oscillations. The static gain $1/5$ is $H(0)$.

Typical errors. (1) Forgetting the terms $-sy(0) - y'(0)$ of Corollary 20.4 when the initial conditions are not zero. (2) Inverting $\frac{s+2}{(s+1)^2+4}$ as $e^{-t} \cos 2t$ without correcting the numerator: the numerator must be $s + 1$ for the cosine, and the remaining part must be read as a sine. (3) Writing $\omega = 4$ by reading off the constant term 5 instead of completing the square.

Extension. Changing the forcing term to δ gives $Y(s) = \frac{1}{(s+1)^2+4}$ and $y(t) = \frac{1}{2}e^{-t} \sin 2t$: the impulse response is the derivative of the step response, as predicted by Theorem 20.8.

Exercise 20.8: A complex double pole, by convolution. *Type:* inversion by convolution, with three independent checks. Compute

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + 1)^2} \right\}.$$

Solution. Strategy. The function is the *square* of $\frac{1}{s^2+1}$: by Theorem 20.10 its inverse transform is the convolution of $\sin t$ with itself. The result is then checked by the multiplication-by- t rule, which is a completely different route.

Step 1 — the convolution. By Proposition 20.4, $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$ with $\sigma_a = 0$. Theorem 20.10 with $\sigma > 0$ then gives

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^2}\right\}(t) = (\sin * \sin)(t) = \int_0^t \sin \tau \sin(t-\tau) d\tau.$$

Step 2 — computing the integral. From the Werner formula $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ with $A = \tau$ and $B = t - \tau$:

$$\sin \tau \sin(t-\tau) = \frac{1}{2}[\cos(2\tau-t) - \cos t].$$

Integrating in τ from 0 to t , with t fixed:

$$\int_0^t \cos(2\tau-t) d\tau = \left[\frac{\sin(2\tau-t)}{2}\right]_0^t = \frac{\sin t - \sin(-t)}{2} = \sin t, \quad \int_0^t \cos t d\tau = t \cos t.$$

Hence

$$(\sin * \sin)(t) = \frac{\sin t - t \cos t}{2}.$$

Independent check 1: the multiplication-by- t rule. By Corollary 20.5 applied to $\cos t$,

$$\mathcal{L}\{t \cos t\}(s) = -\frac{d}{ds} \frac{s}{s^2+1} = -\frac{(s^2+1) - s \cdot 2s}{(s^2+1)^2} = \frac{s^2-1}{(s^2+1)^2}.$$

Since $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1} = \frac{s^2+1}{(s^2+1)^2}$, linearity gives

$$\mathcal{L}\{\sin t - t \cos t\}(s) = \frac{(s^2+1) - (s^2-1)}{(s^2+1)^2} = \frac{2}{(s^2+1)^2},$$

that is, $\mathcal{L}\left\{\frac{\sin t - t \cos t}{2}\right\} = \frac{1}{(s^2+1)^2}$. ✓

Independent check 2: quadrature of the convolution. Composite Simpson with 20,000 subintervals on $[0, t]$:

t	$(\sin t - t \cos t)/2$	numerical convolution
1	0.1505843	0.1505843
3	1.5555487	1.5555487
5	-1.1886176	-1.1886176

✓

Independent check 3: the transform, by quadrature. Simpson with 600,000 subintervals on $[0, 120]$ of $\int_0^\infty \frac{\sin t - t \cos t}{2} e^{-2t} dt$ gives 0.0400000, against $\frac{1}{(4+1)^2} = \frac{1}{25} = 0.04$. ✓

Interpretation. The factor t in front of $\cos t$ is the signature of the double pole: a pole of order m produces, in the time domain, a term t^{m-1} multiplied by the exponential, exactly as stated by Corollary 20.10. Here the poles are $\pm i$, both double: the term $t \cos t$ grows linearly, and this is *resonance* — a system $y'' + y = \sin t$ excited at its own natural frequency.

Typical errors. (1) Writing $\mathcal{L}^{-1}\{FG\} = \mathcal{L}^{-1}\{F\} \cdot \mathcal{L}^{-1}\{G\}$: the product in s is the *convolution* in t , not the product. (2) Using $\sin \tau \sin(t-\tau) = \sin^2 \tau \sin^2(t-\tau)$ or other invented identities: the Werner formula is the only one that linearizes. (3) Forgetting that in the convolution t is *constant* during the integration in τ , and integrating $\cos t$ as if it depended on τ .

Extension. The same method gives $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)^2}\right\} = \frac{t \sin t}{2}$: this is checked with $\mathcal{L}\{t \sin t\} = -\frac{d}{ds} \frac{1}{s^2+1} = \frac{2s}{(s^2+1)^2}$. ✓

20.12. Self-assessment quiz

1. (Definition) The *absolute abscissa of convergence* σ_a of a causal signal is
 - A) the smallest σ for which the Laplace integral converges
 - B) the infimum of the real parts of the points at which the integral converges absolutely
 - C) the real part of the rightmost pole of F
 - D) always equal to σ_c
2. (Hypothesis) In Theorem 20.13 the essential hypothesis is
 - A) that F be rational
 - B) that $\lim_{t \rightarrow +\infty} f(t)$ exist finite
 - C) that $\sigma_a < 0$
 - D) that f be differentiable with continuous derivative
3. (Symbolic computation) $\mathcal{L}\{t^2 e^{-3t}\}(s)$ equals
 - A) $\frac{2}{(s+3)^3}$
 - B) $\frac{1}{(s+3)^3}$
 - C) $\frac{2}{s^3+3}$
 - D) $\frac{6}{(s+3)^4}$
4. (Numerical computation) $\mathcal{L}\{e^{-2t} \sin(3t)\}(1)$ equals, to seven decimal places,
 - A) 0.3000000
 - B) 0.1666667
 - C) 0.1176471
 - D) 0.0833333
5. (Proof step) In Step 4 of the proof of Theorem 20.3 the factor t^2 appearing in the bound is absorbed
 - A) by the dominated convergence theorem
 - B) by the inequality $t^2 \leq \left(\frac{2}{e\delta}\right)^2 e^{\delta t}$ and the margin δ between σ and some $\sigma_2 > \sigma_a$
 - C) by Fubini's theorem
 - D) by the boundedness of the function φ
6. (Error diagnosis) A student applies the final value theorem to $F(s) = \frac{1}{s-1}$, computes $\lim_{s \rightarrow 0} sF(s) = 0$ and concludes that $f(t) \rightarrow 0$. The error is
 - A) the computation of the limit, which equals -1
 - B) that $s = 0$ does not belong to the region of convergence $\operatorname{Re}(s) > 1$ and that $f(t) = e^t$ has no finite limit, so the hypothesis of the theorem fails
 - C) that F is not rational
 - D) that the factor s in front of F is missing
7. (Transfer) The region of convergence of the window $u(t) - u(t-2)$ is
 - A) $\operatorname{Re}(s) > 0$
 - B) all of $\mathbb{C}$
 - C) $\operatorname{Re}(s) > -2$
 - D) $\operatorname{Re}(s) \geq 0$
8. (Source and metacognition) The result that the region of convergence of a Laplace integral is a half-plane is found in
 - A) Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Ch. 3 “The Half-Plane of Convergence,” pp. 13–18
 - B) Doetsch, Ch. 10 “The Mapping of the Convolution,” pp. 44–55
 - C) Oppenheim–Willsky, *Signals and Systems*, Ch. 4 “The Continuous-Time Fourier Transform”
 - D) Doetsch, Ch. 26 “The Evaluation of the Complex Inversion Integral by Means of the Calculus of Residues,” pp. 169–178

20.13. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Definition 20.4: σ_a is an infimum, not a minimum, and concerns *absolute* convergence. *Distractors:* A confuses the two kinds of convergence and moreover speaks of a minimum, which may not be attained (Proposition 20.2: at $\sigma = 0$ the integral does not converge); C is true for rational functions (Corollary 20.10) but false in general, because F may have no poles (Proposition 20.5(ii)); D is disproved by Proposition 20.1. *Diagnosed error:* confusing ordinary and absolute convergence, and a minimum with an infimum. *Source:* Doetsch, Ch. 3, pp. 13–18. *Prerequisite:* infimum of a set of reals. *Remedy:* redo Example 20.1 and state, for each of the two signals, whether σ_a is a minimum.

2) **B.** *Rationale:* the statement of Theorem 20.13 only requires the limit of f to exist finite; everything else (the abscissa $\sigma_a \leq 0$) is a consequence of it, proved in Step 1. *Distractors:* A is the condition of Corollary 20.11, which is sufficient but not necessary; C is stronger than necessary ($f = \sin t$ has $\sigma_a = 0$ and no limit, but $f(t) = 1$ has $\sigma_a = 0$ and limit 1); D is irrelevant, the theorem uses no derivatives. *Diagnosed error:* mistaking a convenient sufficient condition for the hypothesis of the theorem. *Source:* Theorem 20.13 and Remark 20.5; Oppenheim–Willsky, Ch. 9. *Prerequisite:* definition of limit at infinity. *Remedy:* reread the three cases of Example 20.4.

3) **A.** *Rationale:* by Corollary 20.7 with $n = 2$ and $c = -3$, $\mathcal{L}\{t^2 e^{-3t}\} = \frac{2!}{(s+3)^3} = \frac{2}{(s+3)^3}$, for $\operatorname{Re}(s) > -3$. *Distractors:* B forgets the factorial $2! = 2$; C applies the shift to the wrong variable (one shifts s , not s^3); D uses $3!$ and exponent 4, i.e. answers for t^3 . *Diagnosed error:* applying the frequency shift without substituting $s \mapsto s - c$ throughout the denominator. *Source:* Corollary 20.7; Table 20.1. *Prerequisite:* Corollary 20.6. *Remedy:* derive $\mathcal{L}\{t^3 e^{2t}\} = 6/(s-2)^4$ and check it with the third derivative of $1/(s-2)$.

4) **B.** *Rationale:* by Corollary 20.8 with $c = 2$, $\omega = 3$, $s = 1$: $\frac{3}{(1+2)^2+9} = \frac{3}{18} = 0.1666667$; numerical quadrature on $[0, 60]$ gives the same value. *Distractors:* A is $\mathcal{L}\{\sin 3t\}(1) = 3/10$, i.e. the damping has been forgotten; C is $\mathcal{L}\{\cos 2t\}(0.5)$, a value from another exercise; D is $\frac{3}{36}$, i.e. one wrote $(1+2)^2 + 27$ or doubled the denominator. *Diagnosed error:* not applying the frequency shift to the denominator. *Source:* Corollary 20.8, including the numerical check. *Prerequisite:* Proposition 20.4. *Remedy:* compute $\mathcal{L}\{e^{-2t} \cos 3t\}(1)$ and observe that it too equals $\frac{3}{18}$, because $s + c = 3 = \omega$.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **B.** *Rationale:* Step 4 uses Lemma 20.1(ii) with $n = 2$ and $\varepsilon = \delta$ to write $t^2 e^{-(\sigma-\delta)t} \leq \left(\frac{2}{e\delta}\right)^2 e^{-(\sigma-2\delta)t}$, and $\sigma - 2\delta = \sigma_2 > \sigma_a$: the margin δ is used twice, to keep $s + h$ in the half-plane and to absorb t^2 . *Distractors:* A names a tool that is not needed here, because the estimate is explicit and does not go through a limit under the integral sign; C confuses this with the proof of Theorem 20.10; D is the ingredient of Corollary 20.3, not of Theorem 20.3. *Diagnosed error:* not being able to say where each factor in an estimate comes from. *Source:* Theorem 20.3, Steps 2 and 4; Doetsch, Ch. 6 “The Laplace Transform as an Analytic Function,” pp. 26–30. *Prerequisite:* Lemma 20.1. *Remedy:* redo Step 4 with δ replaced by $\delta/2$ and see which constant changes.

6) **B.** *Rationale:* this is Example 20.4(b): the ROC is $\operatorname{Re}(s) > 1$, the point $s = 0$ lies outside it, and $f(t) = e^t$ has no finite limit; evaluating $sF(s)$ near 0 means using the analytic continuation of F outside the region where it represents the signal. *Distractors:* A is false, the limit really is 0 ($s/(s-1) \rightarrow 0$); C is false, F is rational; D is false, the factor s is there. *Diagnosed error:* using a formula outside its region of validity, i.e. forgetting

the ROC. *Source:* Example 20.4; Oppenheim–Willsky, Ch. 9. *Prerequisite:* Corollary 20.1. *Remedy:* repeat the error with $F(s) = \frac{1}{s-a}$ for $a = 0.1$ and $a = -0.1$ and state in which of the two cases the conclusion is correct.

7) **B.** *Rationale:* the window vanishes outside a compact set, hence $\int_0^2 e^{-\sigma t} dt$ is finite for every $\sigma \in \mathbb{R}$: the ROC is the whole plane and the transform $(1 - e^{-2s})/s$ is entire (Proposition 20.5(ii)). *Distractors:* A is the ROC of the step u taken alone, and forgets that the difference vanishes after $t = 2$; C is the ROC one would get if the delay appeared as an exponential in time, i.e. confuses $u(t-2)$ with e^{-2t} ; D adds the boundary to the ROC of the step, which is instead excluded. *Diagnosed error:* deducing the ROC from the form of the transform instead of from the behavior of the signal as $t \rightarrow +\infty$. *Source:* Proposition 20.5; Table 20.1. *Prerequisite:* Definition 20.3. *Remedy:* compute the ROC of $u(t-1) - u(t-3)$ and check that it is still all of $\mathbb{C}$.

8) **A.** *Rationale:* the Doetsch chapter titled “The Half-Plane of Convergence” (pp. 13–18) is exactly the source of the result, proved here as Theorem 20.2. *Distractors:* B is the chapter on convolution, which proves Theorem 20.10; C concerns the Fourier transform and not the region of convergence of Laplace; D is the chapter on inversion by residues, cited in Remark 20.6. *Diagnosed error:* citing “the book” instead of the chapter, and being unable to associate a result with its bibliographic location. *Source:* the *Sources and bibliography* environment at the end of the chapter, entry [1]. *Prerequisite:* none. *Remedy:* associate each of Theorems 20.8, 20.9 and 20.10 with the Doetsch chapter that treats it, using entry [1] of the Sources.

BEST PRACTICE

Assigned exercise 20.3 — answers.

(a) $e^{3t} \cos(4t)$: by Corollary 20.8 with $c = -3$, $F(s) = \frac{s-3}{(s-3)^2+16}$, $\sigma_a = 3$, ROC $\operatorname{Re}(s) > 3$, boundary *excluded* (at $\sigma = 3$ the integral $\int_0^\infty |\cos 4t| dt$ diverges as in Proposition 20.4, Step 3). — $t^2 e^{-t}$: by Corollary 20.7, $F(s) = \frac{2}{(s+1)^3}$, $\sigma_a = -1$, ROC $\operatorname{Re}(s) > -1$, boundary *excluded* (at $\sigma = -1$ what remains is $\int_0^\infty t^2 dt = \infty$). — $u(t) - u(t-2)$: by Proposition 20.5(ii), $F(s) = \frac{1-e^{-2s}}{s}$ with value 2 at $s = 0$; ROC = $\mathbb{C}$, no boundary. — $t^{-1/2}$: by Example 20.2 with $\alpha = -\frac{1}{2}$, $F(s) = \sqrt{\pi}/s$, $\sigma_a = 0$, ROC $\operatorname{Re}(s) > 0$, boundary *excluded* because $\int_0^\infty t^{-1/2} dt = \infty$.

(b) $\mathcal{L}\{e^{3t} \cos 4t\}(5) = \frac{5-3}{(5-3)^2+16} = \frac{2}{20} = 0.1$; Simpson quadrature on $[0, 80]$ with 400,000 subintervals: 0.1000000. ✓ $\mathcal{L}\{t^2 e^{-t}\}(1) = \frac{2}{(1+1)^3} = \frac{2}{8} = 0.25$; quadrature: 0.2500000. ✓

(c) $f(t) = e^t + e^{-t}$ and $g(t) = -e^t$ both have $\sigma_a = 1$, but $f + g = e^{-t}$ has $\sigma_a = -1$. Theorem 20.4 states an inequality because the sum can *cancel* the fastest-growing part: the bound used in the proof (triangle inequality) does not see cancellations.

(d) The window is identically zero after $t = 2$: no factor $e^{-\sigma t}$ needs to “fight” a growth, and the integral extends over a compact set. The step, instead, remains equal to 1 forever, and $\sigma > 0$ is needed for $e^{-\sigma t}$ to make it integrable.

BEST PRACTICE

Assigned exercise 20.4 — answers.

(a) and (b). $F_1(s) = \frac{2s+1}{s(s+1)(s+2)}$: simple poles 0, -1 , -2 ; by Theorem 20.15, $A_0 = \frac{1}{1 \cdot 2} = \frac{1}{2}$, $A_{-1} = \frac{-1}{(-1)(1)} = 1$, $A_{-2} = \frac{-3}{(-2)(-1)} = -\frac{3}{2}$, hence $f_1(t) = \frac{1}{2} + e^{-t} - \frac{3}{2}e^{-2t}$, ROC $\operatorname{Re}(s) > 0$. Both theorems apply: $sF_1 = \frac{2s+1}{s^2+3s+2}$ is proper with poles $-1, -2$ in the left half-plane. $F_2(s) = \frac{5}{s(s^2+4)}$: $f_2(t) = \frac{5}{4}(1 - \cos 2t)$, ROC $\operatorname{Re}(s) > 0$. The initial value theorem applies; the final value theorem *does not*, because the poles of sF_2 are $\pm 2i$, on the imaginary axis. $F_3(s) = \frac{1}{(s-2)(s+3)}$: $A_2 = \frac{1}{5}$, $A_{-3} = -\frac{1}{5}$, hence $f_3(t) = \frac{1}{5}(e^{2t} - e^{-3t})$, ROC $\operatorname{Re}(s) > 2$. The

initial value theorem applies; the final value theorem *does not*, because sF_3 has the pole 2 in the right half-plane. $F_4(s) = \frac{s}{s+4}$: it is not proper, and does not invert to a function (point (d)).

(c) $\lim_{s \rightarrow \infty} sF_1 = 0 = f_1(0^+) = \frac{1}{2} + 1 - \frac{3}{2} \checkmark$; $\lim_{s \rightarrow 0} sF_1 = \frac{1}{2}$ and $f_1(t) \rightarrow \frac{1}{2} \checkmark$ (numerical check: $f_1(30) = 0.5000000$). $\lim_{s \rightarrow \infty} sF_2 = 0 = f_2(0^+) \checkmark$; $\lim_{s \rightarrow 0} sF_2 = \frac{5}{4}$, but f_2 oscillates between 0 and $\frac{5}{2}$ and has no limit: the number $\frac{5}{4}$ is the *average* of f_2 , not its final value. $\lim_{s \rightarrow \infty} sF_3 = 0 = f_3(0^+) \checkmark$; $\lim_{s \rightarrow 0} sF_3 = 0$, but $f_3(t) \rightarrow +\infty$: the formula gives a completely wrong number. $sF_4 = \frac{s^2}{s+4} \rightarrow \infty$: no finite initial value.

(d) $F_4(\sigma) = \frac{\sigma}{\sigma+4} \rightarrow 1 \neq 0$ as $\sigma \rightarrow +\infty$; by Proposition 20.7 F_4 is not the transform of any locally integrable function with finite abscissa. It is instead the transform of the measure $\mu = \delta - 4e^{-4t} dt$: indeed, by Proposition 20.6 and Proposition 20.3, $\mathcal{L}\{\mu\}(s) = 1 - \frac{4}{s+4} = \frac{s+4-4}{s+4} = \frac{s}{s+4} \checkmark$

BEST PRACTICE

Assigned exercise 20.5 — answers.

(a) With $y(0) = 0$ and $y'(0) = 1$, Corollary 20.4 gives $\mathcal{L}\{y'\} = sY$ and $\mathcal{L}\{y''\} = s^2Y - 1$, hence

$$(s^2 + 4s + 3)Y(s) = 1 + \frac{1}{s+2} = \frac{s+3}{s+2}, \quad Y(s) = \frac{s+3}{(s+2)(s+1)(s+3)} = \frac{1}{(s+1)(s+2)},$$

where the factor $s+3$ cancels because $s^2 + 4s + 3 = (s+1)(s+3)$. Then $Y = \frac{1}{s+1} - \frac{1}{s+2}$ and

$$y(t) = e^{-t} - e^{-2t}.$$

Check by substitution: $y(0) = 0 \checkmark$; $y'(t) = -e^{-t} + 2e^{-2t}$, $y'(0) = 1 \checkmark$; $y'' = e^{-t} - 4e^{-2t}$, and $y'' + 4y' + 3y = (1 - 4 + 3)e^{-t} + (-4 + 8 - 3)e^{-2t} = e^{-2t} \checkmark$. *Numerical check:* $y(1) = e^{-1} - e^{-2} = 0.2325442$, and $\int_0^\infty y(t)e^{-t} dt = 0.1666667 = \frac{1}{2.3} = Y(1)$ by quadrature. $\checkmark$

(b) $G(s) = sY(s) = \frac{s}{(s+1)(s+2)}$ is proper with poles $-1, -2$ in the left half-plane: Corollary 20.11 applies and gives $\lim_{t \rightarrow \infty} y(t) = G(0) = 0$, consistent with $e^{-t} - e^{-2t} \rightarrow 0$. $\checkmark$

(c) *Partial fractions:* $\frac{1}{s(s^2+1)} = \frac{1}{s} - \frac{s}{s^2+1}$ (checked by reducing to a common denominator: $\frac{(s^2+1)-s^2}{s(s^2+1)} = \frac{1}{s(s^2+1)}$), hence the inverse transform is $1 - \cos t$ by Proposition 20.2 and Proposition 20.4. *Convolution:* Theorem 20.10 applied to $\frac{1}{s} \cdot \frac{1}{s^2+1}$ gives $(u * \sin)(t) = \int_0^t \sin \tau d\tau = 1 - \cos t$. The two routes agree. $\checkmark$

(d) $1 - \cos 1 = 1 - 0.5403023 = 0.4596977$; Simpson quadrature of $\int_0^1 \sin \tau d\tau$ with 400,000 subintervals gives 0.4596977. $\checkmark$

20.14. Formulary of the chapter

Definition, convergence, holomorphy.

$$F(s) = \mathcal{L}\{f\}(s) = \int_0^\infty f(t)e^{-st} dt, \quad s = \sigma + i\omega, \quad |e^{-st}| = e^{-\sigma t}.$$

$$\sigma_c = \inf\{\operatorname{Re}(s) : \text{converges}\}, \quad \sigma_a = \inf\{\operatorname{Re}(s) : \text{converges absolutely}\}, \quad \sigma_c \leq \sigma_a.$$

Convergence at $s_0 \Rightarrow$ convergence on $\operatorname{Re}(s) > \operatorname{Re}(s_0)$, with

$$F(s) = (s - s_0) \int_0^\infty \varphi(t)e^{-(s-s_0)t} dt, \quad \varphi(t) = \int_0^t f(\tau)e^{-s_0\tau} d\tau.$$

F holomorphic on $\operatorname{Re}(s) > \sigma_c$; on $\operatorname{Re}(s) > \sigma_a$: $F^{(n)}(s) = (-1)^n \mathcal{L}\{t^n f\}(s)$ and $F(\sigma) \rightarrow 0$ as $\sigma \rightarrow +\infty$.

The eight rules, with the region.

$$\begin{aligned}\mathcal{L}\{\lambda f + \mu g\} &= \lambda F + \mu G; & \mathcal{L}\{f(t-a)u(t-a)\} &= e^{-as}F(s) \quad (a \geq 0, \text{ ROC unchanged}); \\ \mathcal{L}\{e^{-ct}f\} &= F(s+c) \quad (\text{ROC shifted by } -\operatorname{Re}(c)); & \mathcal{L}\{f(ct)\} &= \frac{1}{c}F(s/c) \quad (c > 0); \\ \mathcal{L}\{f'\} &= sF(s) - f(0^+); & \mathcal{L}\{f^{(n)}\} &= s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^+); \\ \mathcal{L}\left\{\int_0^t f\right\} &= \frac{F(s)}{s} \quad (\operatorname{Re}(s) > \max\{\sigma_a, 0\}); & \mathcal{L}\{t^n f\} &= (-1)^n F^{(n)}(s); & \mathcal{L}\{f * g\} &= FG.\end{aligned}$$

Elementary pairs (ROC to the right of the arrow).

$$\begin{aligned}\delta &\mapsto 1 \quad (\mathbb{C}); & \delta_a &\mapsto e^{-as} \quad (\mathbb{C}); & u &\mapsto \frac{1}{s} \quad (\sigma > 0); & u(t-a) &\mapsto \frac{e^{-as}}{s} \quad (\sigma > 0); \\ t^n &\mapsto \frac{n!}{s^{n+1}} \quad (\sigma > 0); & t^\alpha &\mapsto \frac{\Gamma(\alpha+1)}{s^{\alpha+1}} \quad (\sigma > 0, \alpha > -1); & e^{ct} &\mapsto \frac{1}{s-c} \quad (\sigma > \operatorname{Re} c); \\ t^n e^{ct} &\mapsto \frac{n!}{(s-c)^{n+1}}; & \cos \omega t &\mapsto \frac{s}{s^2 + \omega^2}; & \sin \omega t &\mapsto \frac{\omega}{s^2 + \omega^2} \quad (\sigma > 0); \\ e^{-ct} \cos \omega t &\mapsto \frac{s+c}{(s+c)^2 + \omega^2}; & e^{-ct} \sin \omega t &\mapsto \frac{\omega}{(s+c)^2 + \omega^2} \quad (\sigma > -\operatorname{Re} c).\end{aligned}$$

Initial value, final value, inversion.

$$\begin{aligned}\lim_{\operatorname{Re}(s) \rightarrow +\infty} sF(s) &= f(0^+) \quad (f \text{ loc. abs. continuous, } \mathcal{L}\{f'\} \text{ abs. convergent}); \\ \lim_{s \rightarrow 0^+} sF(s) &= \lim_{t \rightarrow +\infty} f(t) \quad (\text{if the latter exists finite}); \\ F \text{ rational, poles of } sF \text{ in } \operatorname{Re}(s) < 0 &\implies \text{ it applies.} \\ R = \frac{N}{D} = \sum_k \sum_{j=1}^{m_k} \frac{A_{k,j}}{(s-p_k)^j}, & \quad A_{k,j} = \frac{1}{(m_k-j)!} \lim_{s \rightarrow p_k} \frac{d^{m_k-j}}{ds^{m_k-j}} \left[(s-p_k)^{m_k} R(s) \right], \\ f(t) = \sum_k \sum_j A_{k,j} \frac{t^{j-1}}{(j-1)!} e^{p_k t} &= \sum_k \operatorname{Res}_{s=p_k} \left[R(s) e^{st} \right]; & f(t) &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) e^{st} ds. \\ \text{Simple poles: } A_k &= \frac{N(p_k)}{D'(p_k)}. & \text{Fourier: if } f &\in L^1(0, \infty) \text{ then } \hat{f}(\xi) = F(2\pi i \xi) \text{ and } F(i\omega) = \hat{f}(\omega/2\pi).\end{aligned}$$

Sources and bibliography

- [1] G. Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer-Verlag, Berlin–Heidelberg–New York, 1974, vii+326 pages (ISBN 3-540-06407-9 / 0-387-06407-9; LCCN 74185469; volume DOI 10.1007/978-3-642-65690-3); English translation of *Einführung in Theorie und Anwendung der Laplace-Transformation*. This is the main source of the chapter. Chapters used, with the pagination of each chapter: Ch. 3 “The Half-Plane of Convergence,” pp. 13–18 (source of Theorem 20.2 and of Definition 20.4); Ch. 5 “The Unique Inverse of the Laplace Transformation,” pp. 20–25 (source of Theorem 20.14); Ch. 6 “The Laplace Transform as an Analytic Function,”

pp. 26–30 (Theorem 20.3); Ch. 7 “The Mapping of a Linear Substitution of the Variable,” pp. 30–36 (Theorems 20.5, 20.6 and 20.7); Ch. 8 “The Mapping of Integration,” pp. 36–39 (Theorem 20.9); Ch. 9 “The Mapping of Differentiation,” pp. 39–43 (Theorem 20.8 and Corollary 20.4); Ch. 10 “The Mapping of the Convolution,” pp. 44–55 (Theorem 20.10); Ch. 23 “The Behaviour of the Laplace Transform near Infinity,” pp. 139–147 (Theorem 20.12 and Proposition 20.7); Ch. 24 “The Complex Inversion Formula for the Absolutely Converging Laplace Transformation. The Fourier Transformation,” pp. 148–161, Ch. 25 “Deformation of the Path of Integration of the Complex Inversion Integral,” pp. 161–169, and Ch. 26 “The Evaluation of the Complex Inversion Integral by Means of the Calculus of Residues,” pp. 169–178 (Theorem 20.18 and the *imported* part declared in Remark 20.6); Ch. 33 “Asymptotic Behaviour of the Image Function near Infinity,” pp. 221–230, and Ch. 34 “Asymptotic Behaviour of the Image Function near a Singular Point on the Line of Convergence,” pp. 231–233 (Abelian framing of Theorem 20.13). Chapter titles and pagination verified against the Crossref metadata of the volume (query to api.crossref.org for ISBN 978-3-642-65690-3, 41 records: one book and 40 chapters); bibliographic data verified against the Internet Archive record [introductiontoth0000doet](https://archive.org/details/introductiontoth0000doet) and against Open Library — verified on 6 September 2026.

- [2] A. V. Oppenheim, A. S. Willsky, with S. H. Nawab, *Signals and Systems*, 2nd edition, Prentice Hall, Upper Saddle River (New Jersey), 1997 (copyright and CIP data of 1996: the Library of Congress records LCCN 96019945, which is why the text is cited by some sources as “1996”), xxx+957 pages, ISBN 0-13-814757-4. Chapter 9 “The Laplace Transform,” §§9.1–9.10: definition and region of convergence, properties of the ROC, inverse transform, pole-zero diagrams, properties, table of pairs, analysis of LTI systems, unilateral transform. This is the source used for the signals-and-systems viewpoint (region of convergence declared row by row, BIBO stability, transfer function) and for items 2 and 6 of the quiz. The chapter-by-chapter table of contents is verified against the Internet Archive record [signalssystem0000oppe](https://archive.org/details/signalssystem0000oppe); the breakdown into §§9.1–9.10 is verified on the “Readings” page of MIT course 6.003 *Signals and Systems* (fall 2011), which assigns exactly “9.1–9.10” to the topic “Laplace transforms” — verified on 6 September 2026.
- [3] R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd edition, McGraw-Hill Series in Electrical and Computer Engineering — Circuits and Systems, McGraw-Hill, Boston, 2000, xx+616 pages (ISBN 0-07-303938-1; LCCN 99021139; OCLC 40821116). This is the treatise’s reference text for Fourier transform conventions, recalled here in Remark 20.7 and in Chapter 19. Bibliographic data (edition, series, place, year, pagination, identifiers) verified against the MARC records of the K10plus catalog (SRU interface sru.k10plus.de, search by ISBN 0073039381: two records, one of which is the international edition) and against the Open Library record OL35845M — verified on 6 September 2026. *Note of bibliographic honesty:* it was not possible to verify online the chapter-by-chapter table of contents of this edition, and therefore no chapter or internal page of it is cited.
- [4] *18.03SC Differential Equations*, Massachusetts Institute of Technology, MIT OpenCourseWare, fall 2011 (Creative Commons BY-NC-SA license), “Unit III: Fourier Series and Laplace Transform,” sessions “Laplace Transform Basics,” “Partial Fractions and Inverse Laplace Transform,” “Laplace Transform: Solving Initial Value Problems,” “Transfer Functions” and “Poles,” besides “Step and Delta Functions,” “Impulse Response” and “Convolution.” This is the reference teaching source for the setup of Exercises 20.7 and 20.5 (initial value problem solved by transform, partial fractions, transfer function). The list of sessions is verified against the unit’s page at ocw.mit.edu/courses/18-03sc-differential-equations-fall-2011/ — verified on 6 September 2026.

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- [5] For the results of the treatise cited with \ref: Liouville's theorem, Riemann's theorem on removable singularities, the Laurent series, the residue theorem and the formulas for the residue at a pole are proved in Chapter 15; the Fourier transform, the inversion formula in L^1 , the convolution theorem and the Dirac delta as a tempered distribution in Chapter 19; the trigonometric Weierstrass approximation theorem in Chapter 17; Euler's Gamma function in Chapter 11; the Zeta transform, which is the discrete analogue of this chapter, and BIBO stability in the z -domain in Chapter 21; the systematic use of the transfer function and the stability criteria in Chapter 25. The table of notable transforms, inherited from Volume II, is in Appendix C; Theorem 20.11 proves every row of it and adds to it the region of convergence, which the appendix does not report.
- [6] Numerical checks: all values in the chapter were recomputed with an executed Python script (composite Simpson quadrature with $4 \cdot 10^5$ or $2 \cdot 10^6$ subintervals for the transforms; fourth-order Runge–Kutta with $2 \cdot 10^5$ steps for the differential equation of Worked Exercise 20.7), comparing them with the closed forms; every successful comparison is flagged in the text with $\checkmark$. The observed discrepancies are of the order of 10^{-13} for regular integrands and 10^{-7} where the integrand has unbounded derivative at the origin (Example 20.2).

The Z-transform and the discrete-time Fourier transform

A computer does not see a signal: it sees a sequence of samples. The Z-transform is the tool that carries a sequence $x[n]$ into a holomorphic function $X(z)$, and the discrete-time Fourier transform (DTFT) is what one reads of X on the unit circle. The two are not different tools: they are the same tool viewed on two different regions of the complex plane, and the second exists exactly when the circle $\mathbb{T} = \{|z| = 1\}$ belongs to the region of convergence of the first. This chapter proves this claim and everything around it.

The guiding thread is geometric. The region of convergence of a Z-transform is not an arbitrary set: it is always a *circular annulus* (Theorem 21.1), because the bilateral series $\sum_n x[n]z^{-n}$ splits into two power series, one in the variable $1/z$ and one in the variable z , and each has a disk as its domain of convergence. From this single observation everything follows: causality corresponds to the exterior of a circle (Corollary 21.1), anticausality to the interior of a circle (Corollary 21.2), and there are sequences whose annulus is *empty* (Example 21.1), that is, that have no Z-transform at any point of the plane. The algebraic expression of $X(z)$ does not suffice to pin down $x[n]$: one needs the expression *and* the annulus, and two completely different sequences can share the same rational expression (Example 21.2). It is the most common error in the entire subject, and Section 21.2 confronts it head-on.

Stability is decided on the inner or outer boundary of that annulus. A discrete linear time-invariant system is BIBO stable if and only if its impulse response is absolutely summable (Theorem 21.10, proved in both directions), which is equivalent to saying that $\mathbb{T}$ lies in the region of convergence of the transfer function (Corollary 21.4); and for a causal rational system this is equivalent to requiring that all poles lie *inside* the unit circle (Theorem 21.12). The unit circle of Chapter 12 is therefore not merely a geometric object: it is the stability boundary of every digital controller.

The last section closes the circle among the three planes. The sampling map $z = e^{sT}$ sends the left half-plane onto the punctured unit disk (Theorem 21.13), but it is not injective: this is the analytic reason for aliasing. Its rational approximation, Tustin's bilinear transform, is instead a Möbius transformation that sends the left half-plane *bijectively* onto the disk (Theorem 21.14), at the price of a deformation of the frequency axis that Corollary 21.5 computes in closed form.

The material that the Executive Plan moved here from Volume II (Chapter 4, §11 and §12, and Appendix A, §3) is preserved in full — definition, statements, worked examples, SCL listing, exercises, quiz and key — and is found in Definition 21.3, Theorem 21.6, Theorem 21.12, Section 21.7 and Section 21.8. What Volume II stated without proof (the annulus structure of the region of convergence, the inversion formula, uniqueness at fixed annulus, the summability criterion, the partial-fraction decomposition and the correspondence between the left half-plane and the disk) is here proved.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages of level **D** (the exact computation of the radius of convergence of a causal rational function in Theorem 21.11 and the proof of the partial-fraction decomposition, Lemma 21.3).

Internal prerequisites. Geometric series, complex numbers, modulus, polar form and complex exponential (Volume II, Chapter 3, and Chapter 6); Möbius transformations and generalized circles (Chapter 6); line integrals on the circle, Laurent series, uniqueness of the Laurent coefficients and the identity principle (Chapter 15, in particular Lemma 15.5 and Theorem 15.4); convergence in $L^2(\mathbb{T})$ and completeness of the trigonometric system (Chapter 17, Theorem 17.1); Fourier transform and Laplace transform (Chapter 19 and Chapter 20, Remark 20.1).

External prerequisites, assumed and not reproved:

- numerical series: the root test, the comparison test, properties of limit superior, absolute convergence and rearrangement, Fubini's theorem for absolutely convergent double series;
- power series: a power series defines a holomorphic function inside its own disk of convergence and may be differentiated term by term;
- uniform convergence: the Weierstrass criterion, continuity of the uniform limit of continuous functions, interchange of series and integral on a compact interval;
- polynomial algebra: division with remainder and Bézout's identity for coprime polynomials over $\mathbb{C}$; the fundamental theorem of algebra (Corollary 15.2, already proved in Chapter 15);
- measure theory, for the sole statement of Riesz–Fischer recalled as a panorama in Remark 21.7: the completeness of $L^2(\mathbb{T})$.

Everything else is proved here.

What is learned. Recognizing that a region of convergence is an annulus and computing its two radii; reading the causality of a signal off the shape of the annulus; inverting a rational function by choosing the annulus, and understanding why without it the answer is not unique; proving the five operational properties of the transform; passing from the zeta transform to the DTFT and back; deciding the BIBO stability of a system in three equivalent ways (summability, annulus, poles); and reading the s -plane, the z -plane and the frequency axis as three views of a single object.

Warning. Conventions of the chapter.

- A *signal* (or *sequence*) is a function $x : \mathbb{Z} \rightarrow \mathbb{C}$, written $x[n]$ with square brackets. The set of signals with $\|x\|_1 = \sum_{n \in \mathbb{Z}} |x[n]| < \infty$ is $\ell^1(\mathbb{Z})$; that with $\|x\|_2^2 = \sum_n |x[n]|^2 < \infty$ is $\ell^2(\mathbb{Z})$; that with $\|x\|_\infty = \sup_n |x[n]| < \infty$ is $\ell^\infty(\mathbb{Z})$.
- $\delta[n]$ is the *unit impulse* (1 for $n = 0$, 0 elsewhere) and $u[n]$ the *unit step* (1 for $n \geq 0$, 0 for $n < 0$); $u[-n-1]$ therefore equals 1 for $n \leq -1$ and 0 for $n \geq 0$.
- $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ is the open unit disk and $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ the unit circle. For $0 \leq a < b \leq +\infty$, $A(a, b) = \{z \in \mathbb{C} : a < |z| < b\}$ is the open *annulus* of radii a and b ; it is empty if $b \leq a$, and it is the punctured plane $\mathbb{C} \setminus \{0\}$ if $a = 0$ and $b = +\infty$.
- Every circle $|z| = r$ is understood to be traversed *once, counterclockwise*, that is, parametrized by $z(\theta) = re^{i\theta}$, $\theta \in [0, 2\pi]$; the notation $\oint_{|z|=r}$ carries this orientation.
- Ω always denotes the *normalized angular frequency* (in radians per sample), ω the physical angular frequency (in radians per second) and T the sampling period, so that $\Omega = \omega T$. The sampling frequency is $f_s = 1/T$ and the Nyquist frequency is $f_s/2$, corresponding to $\Omega = \pi$.
- “Proper rational system” means: transfer function $H(z) = N(z)/D(z)$ with N, D

polynomials, $D \neq 0$, $\deg N \leq \deg D$ and the fraction reduced to lowest terms; the *poles* are the roots of D .

- In numerical computations decimals are separated by a point and thousands by a comma in braces; values marked with $\checkmark$ have been computed twice by independent routes.

21.1. Discrete signals, LTI systems and convolution

A PLC does not work in continuous time: it reads inputs, computes and writes outputs at discrete intervals (the scan time). Before transforming anything it is worth fixing what is being transformed and what one wants to know about it.

Definition 21.1: Notable signals, causality and support. A **discrete signal** is a function $x : \mathbb{Z} \rightarrow \mathbb{C}$, denoted $x[n]$. Its **support** is $\text{supp } x = \{n \in \mathbb{Z} : x[n] \neq 0\}$. The signal is

- **causal** if $x[n] = 0$ for every $n < 0$;
- **anticausal** if $x[n] = 0$ for every $n \geq 0$;
- **bilateral** in the remaining cases;
- of **finite support** if $\text{supp } x$ is finite.

The **unit impulse** is $\delta[n] = 1$ for $n = 0$ and 0 elsewhere; the **unit step** is $u[n] = 1$ for $n \geq 0$ and 0 for $n < 0$. The signal **shifted** by n_0 is $(\sigma_{n_0}x)[n] = x[n - n_0]$.

Remark 21.1: Every signal is a sum of shifted impulses. For every n the pointwise identity

$$x[n] = \sum_{k \in \mathbb{Z}} x[k] \delta[n - k]$$

holds, because in the sum on the right only the term $k = n$ survives, the others having $\delta[n - k] = 0$. The sum is thus finite for each fixed n : there is no convergence issue at all. It is this identity, and nothing else, that makes convolution unavoidable.

Definition 21.2: Discrete linear time-invariant system. Let $\mathcal{S}$ be a set of signals closed under linear combinations and shifts. A map $\mathcal{T} : \mathcal{S} \rightarrow \mathcal{S}$ is a **linear time-invariant system** (LTI) if

(L) $\mathcal{T}(\alpha x + \beta y) = \alpha \mathcal{T}x + \beta \mathcal{T}y$ for every $x, y \in \mathcal{S}$ and $\alpha, \beta \in \mathbb{C}$;

(TI) $\mathcal{T}(\sigma_{n_0}x) = \sigma_{n_0}(\mathcal{T}x)$ for every x and every $n_0 \in \mathbb{Z}$.

Its **impulse response** is $h = \mathcal{T}\delta$. The system is **causal** if h is causal.

Proposition 21.1: An LTI system is convolution with its own impulse response.

Let $\mathcal{T}$ be an LTI system with impulse response h .

- If x has finite support, then $(\mathcal{T}x)[n] = \sum_{k \in \mathbb{Z}} h[k] x[n - k]$ for every n , and the sum has a finite number of nonzero terms.
- If $h \in \ell^1(\mathbb{Z})$ and $x \in \ell^\infty(\mathbb{Z})$, the series

$$(h * x)[n] := \sum_{k \in \mathbb{Z}} h[k] x[n - k]$$

converges absolutely for every n and $\|h * x\|_\infty \leq \|h\|_1 \|x\|_\infty$.

- Convolution is commutative: $(h * x)[n] = (x * h)[n]$ whenever one of the two series converges absolutely.

Proof. (i). Let $F = \text{supp } x$, finite. By Remark 21.1, $x = \sum_{k \in F} x[k] \sigma_k \delta$, a *finite* linear combination of shifts of δ . Applying (L) finitely many times and then (TI) to each term,

$$\mathcal{T}x = \sum_{k \in F} x[k] \mathcal{T}(\sigma_k \delta) = \sum_{k \in F} x[k] \sigma_k (\mathcal{T}\delta) = \sum_{k \in F} x[k] \sigma_k h,$$

that is, evaluating at n , $(\mathcal{T}x)[n] = \sum_{k \in F} x[k] h[n - k]$. Since $x[k] = 0$ outside F , the sum extends to all of $\mathbb{Z}$ without changing value; and by point (iii), proved below, it also equals $\sum_k h[k] x[n - k]$.

(ii). For every n and every k , $|h[k]x[n - k]| \leq |h[k]| \|x\|_\infty$. Since $\sum_k |h[k]| = \|h\|_1 < \infty$, the comparison test gives absolute convergence of the series and the bound

$$|(h * x)[n]| \leq \sum_{k \in \mathbb{Z}} |h[k]| |x[n - k]| \leq \|h\|_1 \|x\|_\infty,$$

valid for every n ; taking the supremum over n gives the claim.

(iii). The substitution $j = n - k$ is a bijection of $\mathbb{Z}$ onto itself, and an absolutely convergent series may be rearranged without changing its sum; hence $\sum_k h[k]x[n - k] = \sum_j h[n - j]x[j] = (x * h)[n]$. $\square$

Remark 21.2: What is needed for inputs without finite support. Point (i) of Proposition 21.1 is purely algebraic and requires nothing beyond (L) and (TI). For an input with infinite support the same conclusion additionally requires that $\mathcal{T}$ commute with the limit of the truncations $x_N[n] = x[n]$ for $|n| \leq N$ and 0 elsewhere — a continuity requirement of the system, which is here assumed whenever one writes $y = h * x$ for an arbitrary input. It is not a loss of generality in applications: a constant-coefficient difference equation, like those of this chapter, satisfies the requirement because every output $y[n]$ depends on a finite number of past values.

21.2. The Z-transform and its region of convergence

The Z-transform is the discrete analog of the Laplace transform, and is the tool with which stability and frequency response of digital control systems are analyzed.

Definition 21.3: Z-transform. For a sequence $x[n]$, $n \in \mathbb{Z}$ (bilateral) or $n \geq 0$ (unilateral, for causal sequences):

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_n x[n] z^{-n}, \quad z \in \mathbb{C},$$

defined for z in the region of convergence ROC. For a causal sequence that does not grow faster than an exponential, ROC is the exterior of a circle, $|z| > r_0$.

Definition 21.3 leaves two things to be made precise, and they are exactly the two that make the subject treacherous: what “converges” means for a series indexed over all of $\mathbb{Z}$, and why the region of convergence must have the announced shape. We fix them here.

Definition 21.4: Region of convergence. The **region of convergence** of x , denoted $\text{ROC}(x)$, is the set of $z \in \mathbb{C}$ for which the family $\{x[n]z^{-n}\}_{n \in \mathbb{Z}}$ is *summable*, that is

$$\text{ROC}(x) = \left\{ z \in \mathbb{C} \setminus \{0\} : \sum_{n \in \mathbb{Z}} |x[n]| |z|^{-n} < +\infty \right\}.$$

For $z \in \text{ROC}(x)$ the sum $X(z) = \sum_{n \in \mathbb{Z}} x[n]z^{-n}$ is well defined and does not depend on the

order in which the terms are summed. One sets

$$X_+(z) = \sum_{n \geq 0} x[n]z^{-n}, \quad X_-(z) = \sum_{m \geq 1} x[-m]z^m,$$

the **causal part** and the **anticausal part** of the transform, so that $X = X_+ + X_-$ on $\text{ROC}(x)$.

Remark 21.3: Why absolute convergence is required. A series indexed over $\mathbb{Z}$ has no natural order of summation: one could sum symmetrically, $\lim_N \sum_{|n| \leq N}$, or in blocks, and the two limits generally differ. Requiring absolute summability removes the ambiguity (an absolutely convergent series has the same sum in every order) and it is what makes licit every manipulation of the chapter: the interchange of two sums in Theorem 21.5(v), the interchange of sum and integral in Theorem 21.3, the Weierstrass criterion in Theorem 21.2. It is also the convention of Oppenheim and Schaffer (*Discrete-Time Signal Processing*, 3rd ed., Ch. 3), which defines the ROC as the set of z for which the series converges absolutely.

21.2.1 The Cauchy–Hadamard lemma and the annulus structure

Lemma 21.1: Cauchy–Hadamard. Let $(c_k)_{k \geq 0}$ be a complex sequence and let $\lambda = \limsup_{k \rightarrow \infty} |c_k|^{1/k} \in [0, +\infty]$. Setting $R = 1/\lambda$, with the conventions $1/0 = +\infty$ and $1/(+\infty) = 0$, the power series $\sum_{k \geq 0} c_k w^k$

(i) converges absolutely for every w with $|w| < R$;

(ii) has an unbounded general term, and thus does not converge, for every w with $|w| > R$.

R is called the **radius of convergence** of the series.

Proof. (i). Let $|w| < R$; the case $w = 0$ is trivial, so assume $w \neq 0$. Choose a number t with $|w| < t < R$; then $1/t > 1/R = \lambda$. By the characterization of the limit superior, the set of indices k with $|c_k|^{1/k} > 1/t$ is finite: there exists K such that $|c_k|^{1/k} \leq 1/t$ for every $k \geq K$. For such k ,

$$|c_k w^k| = (|c_k|^{1/k} |w|)^k \leq \left(\frac{|w|}{t}\right)^k,$$

and $q = |w|/t < 1$. The geometric series $\sum_k q^k$ converges, hence by the comparison test $\sum_k |c_k w^k|$ converges.

(ii). Let $|w| > R$, that is $1/|w| < \lambda$. By the characterization of the limit superior, there are infinitely many indices k with $|c_k|^{1/k} > 1/|w|$, and for each of them $|c_k w^k| > 1$. The general term does not tend to zero (indeed, it is not bounded when $|w| > t > R$ and the argument is repeated with t), hence the series does not converge. $\square$

Theorem 21.1: The region of convergence is a circular annulus. Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ be a signal and set

$$\rho_+ = \limsup_{n \rightarrow +\infty} |x[n]|^{1/n} \in [0, +\infty], \quad \rho_- = \left(\limsup_{m \rightarrow +\infty} |x[-m]|^{1/m} \right)^{-1} \in [0, +\infty],$$

with the conventions $1/0 = +\infty$ and $1/(+\infty) = 0$ (and $\rho_+ = 0$, $\rho_- = +\infty$ when the corresponding sequence is eventually zero). Then

$$A(\rho_+, \rho_-) \subseteq \text{ROC}(x) \subseteq \{z : \rho_+ \leq |z| \leq \rho_-\}.$$

In particular:

- (i) if $\rho_- < \rho_+$, then $\text{ROC}(x) = \emptyset$: the signal has no Z-transform at any point;
- (ii) $\text{ROC}(x)$ is rotation invariant: membership of z depends only on $|z|$, so $\text{ROC}(x)$ contains a whole circle $|z| = r$ or none of its points;
- (iii) if $\rho_+ < \rho_-$, then $\text{ROC}(x)$ is the open annulus $A(\rho_+, \rho_-)$, possibly united with one or both of the boundary circles $|z| = \rho_+$ and $|z| = \rho_-$ (the latter only if $\rho_- < \infty$); in every case it is path-connected.

Proof. Step 1 — summability splits in two. Fix $z \neq 0$; the family $\{x[n]z^{-n}\}_{n \in \mathbb{Z}}$ is summable if and only if the two sub-families of indices $n \geq 0$ and $n < 0$ are summable separately, because a family of non-negative numbers is summable if and only if each part of a finite partition of the indices is, and in that case the sum is the sum of the partial sums. Hence

$$z \in \text{ROC}(x) \iff \sum_{n \geq 0} |x[n]| |z|^{-n} < \infty \text{ and } \sum_{m \geq 1} |x[-m]| |z|^m < \infty.$$

Step 2 — the causal part. Setting $w = 1/z$, the first series is $\sum_{n \geq 0} |c_n| |w|^n$ with $c_n = x[n]$. By Lemma 21.1, it converges if $|w| < 1/\rho_+$ and does not converge if $|w| > 1/\rho_+$; that is, going back to z , it converges if $|z| > \rho_+$ and does not converge if $|z| < \rho_+$.

Step 3 — the anticausal part. The second series is $\sum_{m \geq 1} |d_m| |z|^m$ with $d_m = x[-m]$, a power series in the variable z whose radius of convergence is, by Lemma 21.1, $(\limsup_m |d_m|^{1/m})^{-1} = \rho_-$. It therefore converges if $|z| < \rho_-$ and does not converge if $|z| > \rho_-$.

Step 4 — conclusion. Putting Steps 1–3 together: if $\rho_+ < |z| < \rho_-$ both series converge, hence $z \in \text{ROC}(x)$, which gives the first inclusion; if $|z| < \rho_+$ the first series does not converge, and if $|z| > \rho_-$ the second does not, so in both cases $z \notin \text{ROC}(x)$, which gives the second inclusion.

(i). If $\rho_- < \rho_+$ the set $\{\rho_+ \leq |z| \leq \rho_-\}$ is empty, and so is $\text{ROC}(x)$.

(ii). The condition $\sum_n |x[n]| |z|^{-n} < \infty$ of Definition 21.4 involves z only through $|z|$: if it is satisfied by z it is satisfied by every z' with $|z'| = |z|$.

(iii). If $\rho_+ < \rho_-$, the two inclusions say that $\text{ROC}(x)$ contains the open annulus and is contained in the closed annulus: only the points on the two boundaries remain undetermined, and by (ii) each of the two boundary circles either belongs to it entirely or not at all. The resulting set is path-connected, because two of its points can be joined by first traversing an arc of the circle through the first point and then a radial segment, both contained in the closed annulus and, by (ii), in the set. $\square$

Corollary 21.1: Causal signals: the ROC is the exterior of a circle. If x is causal and not identically zero, then $\rho_- = +\infty$ and

$$\{z : |z| > \rho_+\} \subseteq \text{ROC}(x) \subseteq \{z : |z| \geq \rho_+\}.$$

If moreover $\rho_+ < \infty$, the function X admits a finite limit as $|z| \rightarrow \infty$, and precisely $X(z) \rightarrow x[0]$.

Proof. If x is causal, $x[-m] = 0$ for every $m \geq 1$: the limit superior of $|x[-m]|^{1/m} = 0$ equals 0, hence $\rho_- = 1/0 = +\infty$, and Theorem 21.1 gives the two inclusions. For the limit: fix $r > \rho_+$; for $|z| \geq r$ one has $|x[n]z^{-n}| \leq |x[n]|r^{-n}$ with $\sum_n |x[n]|r^{-n} < \infty$, hence the series $\sum_{n \geq 0} x[n]z^{-n}$ converges *uniformly* on $\{|z| \geq r\}$ by the Weierstrass criterion. The limit as $|z| \rightarrow \infty$ can thus be interchanged with the sum: every term with $n \geq 1$ tends to 0, the term $n = 0$ is constantly $x[0]$, and the sum tends to $x[0]$. $\square$

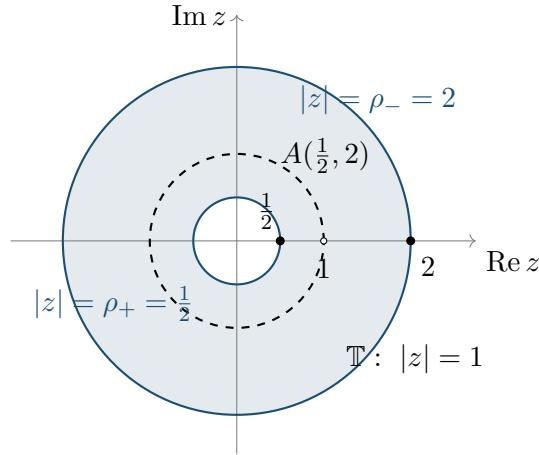


Figure 21.1: The region of convergence of $x[n] = (\frac{1}{2})^n u[n] - 2^n u[-n-1]$: the annulus $A(\frac{1}{2}, 2)$ of Theorem 21.1. The inner boundary is dictated by the causal part, the outer one by the anticausal part, and the two poles of $X(z) = \frac{z}{z-1/2} + \frac{z}{z-2}$ lie exactly on the two circles. Since $\rho_+ = \frac{1}{2} < 1 < 2 = \rho_-$, the unit circle $\mathbb{T}$ (dashed) is contained in the annulus: the signal is summable and has a DTFT.

Corollary 21.2: Anticausal signals: the ROC is a disk. If x is anticausal and not identically zero, then $\rho_+ = 0$ and

$$\{z : 0 < |z| < \rho_-\} \subseteq \text{ROC}(x) \subseteq \{z : 0 < |z| \leq \rho_-\},$$

and X extends continuously to $z = 0$ by setting $X(0) = 0$.

Proof. If $x[n] = 0$ for $n \geq 0$, the limit superior of $|x[n]|^{1/n}$ equals 0, that is $\rho_+ = 0$, and Theorem 21.1 gives the inclusions. Moreover $X(z) = X_-(z) = \sum_{m \geq 1} x[-m]z^m$ is a power series in z with no constant term, convergent for $|z| < \rho_-$: it equals 0 at $z = 0$ and is continuous there. $\square$

Corollary 21.3: Finite support. If x has finite support, contained in $\{N_1, \dots, N_2\}$, then X is the Laurent polynomial $X(z) = \sum_{n=N_1}^{N_2} x[n]z^{-n}$ and

$$\text{ROC}(x) = \mathbb{C} \setminus \{0\}.$$

The series also converges at $z = 0$ — and X extends there by continuity — if and only if $x[n] = 0$ for every $n > 0$.

Proof. A finite sum always converges, provided each term is defined: the only term not defined at $z = 0$ is $x[n]z^{-n}$ with $n > 0$. The sum is thus defined for every $z \neq 0$, and at $z = 0$ exactly when no negative powers occur, that is, when $x[n] = 0$ for $n > 0$. $\square$

Example 21.1: Two empty annuli. *First example.* Let $x[n] = 2^{|n|}$. For $n \geq 0$ one has $|x[n]|^{1/n} = 2$ for every $n \geq 1$, hence $\rho_+ = 2$; for $m \geq 1$ one has $|x[-m]|^{1/m} = 2$, hence $\rho_- = 1/2$. Since $\rho_- = \frac{1}{2} < 2 = \rho_+$, Theorem 21.1(i) gives $\text{ROC}(x) = \emptyset$: there is no $z \in \mathbb{C}$ for which the series $\sum_n 2^{|n|}z^{-n}$ converges. A direct check confirms this: for $|z| \leq 1$ the causal part $\sum_{n \geq 0} 2^n |z|^{-n}$ has general term $\geq 2^n$, which does not tend to zero; for $|z| > 1$ the anticausal part $\sum_{m \geq 1} 2^m |z|^m$ has general term $\geq 2^m$, and that does not tend to zero either. $\checkmark$

Second example (the general case). Let $x[n] = a^n u[n] + b^n u[-n-1]$ with $a, b \in \mathbb{C} \setminus \{0\}$. Then $\rho_+ = |a|$ (because $|x[n]|^{1/n} = |a|$ for $n \geq 1$) and, since $x[-m] = b^{-m}$, $\limsup_m |x[-m]|^{1/m} = |b|^{-1}$ and hence $\rho_- = |b|$. The region of convergence is then the annulus $A(|a|, |b|)$, non-empty if and only if $|a| < |b|$: **the annulus is empty exactly when the causal part grows at least as fast as the anticausal part decreases**. The first example is the case $a = 2$, $b = 1/2$.

21.2.2 Holomorphy, inversion and uniqueness

Theorem 21.2: On the annulus the transform is holomorphic, and the series is its Laurent series. Let x be a signal with $\rho_+ < \rho_-$ and let $A = A(\rho_+, \rho_-)$. Then

- (i) the series $\sum_{n \in \mathbb{Z}} x[n] z^{-n}$ converges uniformly on every closed annulus $\{r_1 \leq |z| \leq r_2\}$ with $\rho_+ < r_1 \leq r_2 < \rho_-$;
- (ii) X is holomorphic on A ;
- (iii) $\sum_{n \in \mathbb{Z}} x[n] z^{-n}$ is the Laurent series of X centered at 0 on the annulus A , and the coefficients $x[n]$ are uniquely determined by it.

Proof. (i). Let $\rho_+ < r_1 \leq r_2 < \rho_-$ and let z satisfy $r_1 \leq |z| \leq r_2$. For $n \geq 0$ one has $|x[n] z^{-n}| \leq |x[n]| r_1^{-n} =: M_n$, and $\sum_{n \geq 0} M_n < \infty$ because $r_1 > \rho_+$ (Step 2 of the proof of Theorem 21.1). For $n = -m < 0$ one has $|x[-m] z^m| \leq |x[-m]| r_2^m =: M_{-m}$, and $\sum_{m \geq 1} M_{-m} < \infty$ because $r_2 < \rho_-$ (Step 3 of the same proof). The majorants M_n do not depend on z and their series converges: the Weierstrass criterion gives uniform convergence.

(ii). The function X_+ is, in the variable $w = 1/z$, the sum of the power series $\sum_{n \geq 0} x[n] w^n$, whose radius of convergence is $1/\rho_+$ (Lemma 21.1); a power series is holomorphic inside its own disk of convergence, hence $w \mapsto X_+(1/w)$ is holomorphic on $|w| < 1/\rho_+$ and, composing with the inversion $z \mapsto 1/z$, which is holomorphic on $\mathbb{C} \setminus \{0\}$, one obtains that X_+ is holomorphic on $\{|z| > \rho_+\}$. Similarly X_- is the sum of the power series $\sum_{m \geq 1} x[-m] z^m$, holomorphic on $\{|z| < \rho_-\}$. On the annulus A both are defined and holomorphic, and so is their sum X .

(iii). By the Laurent expansion theorem (Theorem 15.4), a function holomorphic on an annulus admits one and only one expansion of the form $\sum_{k \in \mathbb{Z}} a_k z^k$ convergent on the annulus. Setting $k = -n$, the series $\sum_n x[n] z^{-n}$ has that form, with $a_{-n} = x[n]$; by the uniqueness of the expansion it is the Laurent series of X , and its coefficients are determined by X and the annulus. $\square$

Theorem 21.3: Inversion formula. Let x be a signal with $\rho_+ < \rho_-$ and let r be such that $\rho_+ < r < \rho_-$. Then for every $n \in \mathbb{Z}$

$$x[n] = \frac{1}{2\pi i} \oint_{|z|=r} X(z) z^{n-1} dz = \frac{1}{2\pi} \int_0^{2\pi} X(re^{i\theta}) r^n e^{in\theta} d\theta.$$

Proof. Step 1 — the elementary integral. For every $k \in \mathbb{Z}$,

$$\oint_{|z|=r} z^{k-1} dz = \begin{cases} 2\pi i, & k = 0, \\ 0, & k \neq 0, \end{cases}$$

which is Lemma 15.5 of Chapter 15; it is checked in two lines by parametrizing $z = re^{i\theta}$, $dz = ire^{i\theta} d\theta$:

$$\oint_{|z|=r} z^{k-1} dz = \int_0^{2\pi} r^{k-1} e^{i(k-1)\theta} ire^{i\theta} d\theta = i r^k \int_0^{2\pi} e^{ik\theta} d\theta,$$

and the last integral equals 2π for $k = 0$ and $[e^{ik\theta}/(ik)]_0^{2\pi} = 0$ for $k \neq 0$.

Step 2 — interchange of series and integral. On the circle $|z| = r$, which is a degenerate closed annulus with $r_1 = r_2 = r$, the series $\sum_k x[k]z^{-k}$ converges uniformly by Theorem 21.2(i). Multiplying by the continuous and bounded function $z \mapsto z^{n-1}$ the convergence remains uniform, hence one may integrate term by term:

$$\oint_{|z|=r} X(z)z^{n-1}dz = \sum_{k \in \mathbb{Z}} x[k] \oint_{|z|=r} z^{n-1-k} dz.$$

Step 3 — conclusion. By Step 1 with k replaced by $n - k$, the integral on the right equals $2\pi i$ if $k = n$ and 0 otherwise: only the term $k = n$ survives, giving $\oint X(z)z^{n-1}dz = 2\pi i x[n]$. The second form of the statement is obtained by parametrizing $z = re^{i\theta}$: $z^{n-1}dz = r^{n-1}e^{i(n-1)\theta}ir e^{i\theta}d\theta = i r^n e^{in\theta}d\theta$, and dividing by $2\pi i$. $\square$

Theorem 21.4: Uniqueness of the inverse transform at fixed annulus. Let x and y be two signals whose regions of convergence both contain a common circle $|z| = r$, and suppose $X(z) = Y(z)$ for every z with $|z| = r$. Then $x[n] = y[n]$ for every $n \in \mathbb{Z}$.

Proof. The hypotheses say that $r \in (\rho_+^x, \rho_-^x)$ or r lies on a boundary of the annulus of x , and likewise for y ; in both cases the series of x and that of y converge absolutely on $|z| = r$ and, by the Weierstrass criterion applied with the constant majorants $|x[n]|r^{-n}$, uniformly on that circle. Steps 2 and 3 of the proof of Theorem 21.3 thus hold identically and give, for every n ,

$$x[n] = \frac{1}{2\pi i} \oint_{|z|=r} X(z)z^{n-1}dz = \frac{1}{2\pi i} \oint_{|z|=r} Y(z)z^{n-1}dz = y[n],$$

having used $X = Y$ on $|z| = r$ in the middle step. $\square$

Example 21.2: Two different signals with the same rational expression. Let $a \in \mathbb{C} \setminus \{0\}$ and consider the rational expression

$$X(z) = \frac{z}{z-a} = \frac{1}{1-az^{-1}}.$$

It is the Z-transform of *two* different signals, according to the annulus chosen.

Outer annulus, $|z| > |a|$. For $|az^{-1}| < 1$ the geometric series of ratio az^{-1} converges absolutely and

$$\frac{1}{1-az^{-1}} = \sum_{n \geq 0} a^n z^{-n},$$

which is the transform of the *causal* signal $x_1[n] = a^n u[n]$. Consistently, $\rho_+ = |a|$ and $\rho_- = +\infty$ (Corollary 21.1).

Inner annulus, $|z| < |a|$. Here $|a^{-1}z| < 1$ and it is convenient to collect terms differently:

$$\frac{z}{z-a} = \frac{z}{-a(1-a^{-1}z)} = -a^{-1}z \sum_{m \geq 0} a^{-m} z^m = - \sum_{m \geq 1} a^{-m} z^m.$$

The coefficient of z^m , that is of z^{-n} with $n = -m$, equals $-a^{-m} = -a^n$: this is the transform of the *anticausal* signal

$$x_2[n] = -a^n u[-n-1].$$

Consistently $\rho_+ = 0$ and $\rho_- = |a|$ (Corollary 21.2).

Comparison. The two signals differ at every index: $x_1[0] = 1$ and $x_2[0] = 0$; $x_1[-1] = 0$ and $x_2[-1] = -a^{-1}$. Yet their rational expression is the same. With $a = 2$ one has $x_1 = (1, 2, 4, 8, \dots)$ for $n \geq 0$ and $x_2 = (\dots, -\frac{1}{8}, -\frac{1}{4}, -\frac{1}{2})$ for $n \leq -1$: one explodes into the

future, the other into the past. Only x_1 is summable when $|a| < 1$, only x_2 is when $|a| > 1$, and for $|a| = 1$ neither is.

Independent check. The inversion formula of Theorem 21.3 was evaluated numerically on $X(z) = [(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})]^{-1}$, which has poles at $\frac{1}{2}$ and 2 and hence *three* possible annuli; with 200,000 points on the circle one obtained, for the three radii $r = 3$, $r = 1$, $r = \frac{1}{4}$ and for $n = -2, \dots, 3$:

annulus	$n = -2$	$n = -1$	$n = 0$	$n = 1$	$n = 2$	$n = 3$
$ z > 2$	0	0	1	2.5	5.25	10.625
$\frac{1}{2} < z < 2$	-0.333 333	-0.666 667	-0.333 333	-0.166 667	-0.083 333	-0.041 667
$ z < \frac{1}{2}$	1	0	0	0	0	0

Each row agrees, up to $2 \cdot 10^{-15}$, with the analytic value obtained by partial fractions ($-\frac{1}{3} \cdot \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{4}{3} \cdot \frac{1}{1-2z^{-1}}$, see Exercise 21.7); three different signals, one single expression. ✓

Warning. The error to avoid. Writing “the inverse transform of $\frac{z}{z-a}$ is $a^n u[n]$ ” is incorrect until the annulus is stated. The correct notation is

$$\mathcal{Z}^{-1}\left\{\frac{z}{z-a}, |z| > |a|\right\} = a^n u[n], \quad \mathcal{Z}^{-1}\left\{\frac{z}{z-a}, |z| < |a|\right\} = -a^n u[-n-1].$$

Theorem 21.4 guarantees uniqueness *only* at fixed annulus: a rational function with p poles of distinct moduli admits $p+1$ annuli of convergence and hence $p+1$ different inverse transforms.

21.3. The five operational properties

The properties that follow are what makes the Z-transform a computational tool and not merely a definition. Each carries with it a statement about the region of convergence, and neglecting it is the second source of error in the subject, after that of Example 21.2.

Theorem 21.5: Linearity, delay, modulation, differentiation in z , convolution.

Let x, y be signals with $\mathcal{R}_x = \text{ROC}(x)$ and $\mathcal{R}_y = \text{ROC}(y)$, and let $\alpha, \beta, a \in \mathbb{C}$ with $a \neq 0$, $n_0 \in \mathbb{Z}$.

(i) (Linearity) For every $z \in \mathcal{R}_x \cap \mathcal{R}_y$ the series of $\alpha x + \beta y$ converges absolutely and

$$\mathcal{Z}\{\alpha x + \beta y\}(z) = \alpha X(z) + \beta Y(z);$$

hence $\text{ROC}(\alpha x + \beta y) \supseteq \mathcal{R}_x \cap \mathcal{R}_y$, and this inclusion can be strict.

(ii) (Delay) $\mathcal{Z}\{x[n - n_0]\}(z) = z^{-n_0} X(z)$ for every $z \in \mathcal{R}_x$, and $\text{ROC}(\sigma_{n_0} x) = \mathcal{R}_x$.

(iii) (Multiplication by an exponential) $\mathcal{Z}\{a^n x[n]\}(z) = X(z/a)$, and $\text{ROC}(a^n x) = |a| \mathcal{R}_x = \{|a|z : z \in \mathcal{R}_x\}$; in particular the radii are multiplied by $|a|$.

(iv) (Differentiation in z) On the open annulus $A(\rho_+, \rho_-)$ the function X is differentiable and

$$\mathcal{Z}\{n x[n]\}(z) = -z X'(z),$$

with the same open annulus of convergence.

(v) (Convolution) If $z \in \mathcal{R}_x \cap \mathcal{R}_y$, then for every n the series $(x * y)[n] = \sum_k x[k]y[n-k]$ converges absolutely, the signal $x * y$ has $z \in \text{ROC}(x * y)$ and

$$\mathcal{Z}\{x * y\}(z) = X(z)Y(z).$$

Proof. (i). Let $z \in \mathcal{R}_x \cap \mathcal{R}_y$. Then

$$\sum_n |\alpha x[n] + \beta y[n]| |z|^{-n} \leq |\alpha| \sum_n |x[n]| |z|^{-n} + |\beta| \sum_n |y[n]| |z|^{-n} < \infty,$$

by the triangle inequality and the comparison test: hence $z \in \text{ROC}(\alpha x + \beta y)$. Since these are absolutely convergent series, the sum of the linear combination is the linear combination of the sums. The inclusion can be strict: with $y = x$, $\alpha = 1$, $\beta = -1$ one obtains the zero signal, whose region of convergence is $\mathbb{C} \setminus \{0\}$; and with $x[n] = u[n]$, $y[n] = u[n-1]$ (both with $\mathcal{R} = \{|z| > 1\}$) one has $x - y = \delta$, with region $\mathbb{C} \setminus \{0\}$.

(ii). With the substitution $m = n - n_0$, which is a bijection of $\mathbb{Z}$ onto itself,

$$\sum_n x[n - n_0] z^{-n} = \sum_m x[m] z^{-(m+n_0)} = z^{-n_0} \sum_m x[m] z^{-m} = z^{-n_0} X(z),$$

and the same substitution shows $\sum_n |x[n - n_0]| |z|^{-n} = |z|^{-n_0} \sum_m |x[m]| |z|^{-m}$: the two series with positive terms converge or diverge together, because the factor $|z|^{-n_0}$ is finite and nonzero for $z \neq 0$. Hence the two regions of convergence coincide (in $\mathbb{C} \setminus \{0\}$).

(iii). Since $a \neq 0$,

$$\sum_n a^n x[n] z^{-n} = \sum_n x[n] \left(\frac{z}{a}\right)^{-n} = X\left(\frac{z}{a}\right),$$

and the computation with moduli gives $\sum_n |a^n x[n]| |z|^{-n} = \sum_n |x[n]| |z/a|^{-n}$: the series converges at z if and only if the series of x converges at z/a , that is if and only if $z/a \in \mathcal{R}_x$, that is $z \in a\mathcal{R}_x$. Since $\mathcal{R}_x$ is rotation invariant (Theorem 21.1(ii)), $a\mathcal{R}_x = |a|\mathcal{R}_x$.

(iv). By Theorem 21.2, on $A(\rho_+, \rho_-)$ one has $X = X_+ + X_-$ with $X_+(z) = \sum_{n \geq 0} x[n] z^{-n}$ and $X_-(z) = \sum_{m \geq 1} x[-m] z^m$, and each of the two is obtained from a convergent power series. A power series is differentiated term by term inside its own disk of convergence; composing with $z \mapsto 1/z$ (chain rule, $\frac{d}{dz} z^{-n} = -n z^{-n-1}$) and differentiating X_- term by term as well, one obtains, for every $z \in A(\rho_+, \rho_-)$,

$$X'(z) = \sum_{n \in \mathbb{Z}} x[n] (-n) z^{-n-1}, \quad \text{hence} \quad -z X'(z) = \sum_{n \in \mathbb{Z}} n x[n] z^{-n},$$

which is by definition $\mathcal{Z}\{n x[n]\}(z)$. It remains to check that the radii do not change. For $n \geq 1$ one has $|n x[n]|^{1/n} = n^{1/n} |x[n]|^{1/n}$, and $n^{1/n} \rightarrow 1$ with $n^{1/n} \geq 1$. Setting $b_n = |x[n]|^{1/n} \geq 0$: from $b_n \leq n^{1/n} b_n$ it follows that $\limsup_n b_n \leq \limsup_n n^{1/n} b_n$; and fixing $\varepsilon > 0$, eventually $n^{1/n} \leq 1 + \varepsilon$, hence $\limsup_n n^{1/n} b_n \leq (1 + \varepsilon) \limsup_n b_n$. By the arbitrariness of ε (and with equality trivially holding when $\limsup_n b_n = +\infty$) one concludes $\limsup_n |n x[n]|^{1/n} = \limsup_n b_n = \rho_+$. The same argument with $|(-m)x[-m]|^{1/m} = m^{1/m} |x[-m]|^{1/m}$ leaves ρ_- unchanged as well.

(v). Let $z \in \mathcal{R}_x \cap \mathcal{R}_y$ and consider the double family

$$t_{k,n} = x[k] z^{-k} \cdot y[n-k] z^{-(n-k)}, \quad (k, n) \in \mathbb{Z}^2.$$

It is absolutely summable: summing first over n (with k fixed) and then over k , and using the bijection $m = n - k$,

$$\sum_{k \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} |t_{k,n}| = \sum_k |x[k] z^{-k}| \sum_m |y[m] z^{-m}| = \left(\sum_k |x[k]| |z|^{-k} \right) \left(\sum_m |y[m]| |z|^{-m} \right) < \infty,$$

a product of two finite numbers by hypothesis. By Fubini's theorem for absolutely summable double series, the two iterated sums exist and agree with the sum of the family. Summing first over k :

$$\sum_n \left(\sum_k t_{k,n} \right) = \sum_n z^{-n} \sum_k x[k] y[n-k] = \sum_n (x * y)[n] z^{-n},$$

where the absolute convergence of the inner sum, for every n , follows from the summability of the family; and the absolute summability of the family also gives $\sum_n |(x * y)[n]| |z|^{-n} \leq \sum_{k,n} |t_{k,n}| < \infty$, that is $z \in \text{ROC}(x * y)$. Summing instead first over n one obtains $X(z)Y(z)$, as computed above without the moduli. The two iterated sums coincide, hence $\mathcal{Z}\{x * y\}(z) = X(z)Y(z)$. $\square$

Remark 21.4: Why annuli can grow larger. In points (i) and (v) the inclusion $\text{ROC} \supseteq \mathcal{R}_x \cap \mathcal{R}_y$ can be strict, and the reason is always the same: a cancellation. In (i) coefficients cancel; in (v) a pole is simplified against a zero. Example of (v): $x[n] = a^n u[n]$ has $X(z) = \frac{z}{z-a}$ with $\mathcal{R}_x = \{|z| > |a|\}$, and $y = \delta - a\sigma_1\delta$, that is $y[0] = 1$, $y[1] = -a$, has $Y(z) = 1 - az^{-1} = \frac{z-a}{z}$ with $\mathcal{R}_y = \mathbb{C} \setminus \{0\}$. The product equals $X(z)Y(z) = 1$, the transform of δ , whose region is $\mathbb{C} \setminus \{0\}$: the pole at a has been cancelled by the zero of Y . And indeed $(x * y)[n] = a^n u[n] - a \cdot a^{n-1} u[n-1] = \delta[n]$. $\checkmark$

Example 21.3: The table of notable transforms, proved. Every row of the table of Appendix C, §3, follows from Theorem 21.5 and Example 21.2. Let always $a \in \mathbb{C} \setminus \{0\}$.

- (1) $\mathcal{Z}\{\delta[n]\} = 1$ on $\mathbb{C} \setminus \{0\}$: the only nonzero term is $n = 0$, giving $\delta[0]z^0 = 1$ (Corollary 21.3).
- (2) $\mathcal{Z}\{a^n u[n]\} = \frac{z}{z-a}$ on $|z| > |a|$: this is Example 21.2.
- (3) $\mathcal{Z}\{u[n]\} = \frac{z}{z-1}$ on $|z| > 1$: this is (2) with $a = 1$.
- (4) $\mathcal{Z}\{n a^n u[n]\} = \frac{az}{(z-a)^2}$ on $|z| > |a|$. Indeed, by Theorem 21.5(iv) applied to (2),

$$\mathcal{Z}\{n a^n u[n]\}(z) = -z \frac{d}{dz} \left(\frac{z}{z-a} \right) = -z \cdot \frac{(z-a) - z}{(z-a)^2} = -z \cdot \frac{-a}{(z-a)^2} = \frac{az}{(z-a)^2}.$$

- (5) $\mathcal{Z}\{\cos(\Omega_0 n) u[n]\} = \frac{z^2 - z \cos \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}$ and $\mathcal{Z}\{\sin(\Omega_0 n) u[n]\} = \frac{z \sin \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}$, on $|z| > 1$. Indeed $\cos(\Omega_0 n) = \frac{1}{2}(e^{i\Omega_0 n} + e^{-i\Omega_0 n})$ (Euler's formula), hence by linearity and by (2) with $a = e^{\pm i\Omega_0}$ (both of modulus 1, whence the annulus $|z| > 1$):

$$\mathcal{Z}\{\cos(\Omega_0 n) u[n]\} = \frac{1}{2} \left(\frac{z}{z - e^{i\Omega_0}} + \frac{z}{z - e^{-i\Omega_0}} \right) = \frac{z}{2} \cdot \frac{2z - (e^{i\Omega_0} + e^{-i\Omega_0})}{z^2 - z(e^{i\Omega_0} + e^{-i\Omega_0}) + 1},$$

and substituting $e^{i\Omega_0} + e^{-i\Omega_0} = 2 \cos \Omega_0$ gives the announced formula. The computation for the sine is identical starting from $\sin(\Omega_0 n) = \frac{1}{2i}(e^{i\Omega_0 n} - e^{-i\Omega_0 n})$.

Independent numerical check. Formulas (4) and (5) were verified by directly summing the defining series. For (4) with $a = 0.8$ and $z = 1.25$: the partial sum over 20,000 terms equals 4.938 271 605, and the closed formula $az/(z-a)^2 = \frac{0.8 \cdot 1.25}{(0.45)^2} = 4.938 271 605$. $\checkmark$ For (5) with $\Omega_0 = 0.7$: at $z = 1.5$ the series gives 1.154 125 981 and the closed formula 1.154 125 981; at $z = 2$, 1.272 944 370 against 1.272 944 370; at $z = 1.3i$, 0.709 464 252 - 0.603 678 916 i against 0.709 464 252 - 0.603 678 916 i . $\checkmark$

21.4. The DTFT as the restriction of the zeta transform to the unit circle

Definition 21.5: Discrete-time Fourier transform (DTFT). Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ be a signal. Its **discrete-time Fourier transform** is the function

$$\hat{x}(\Omega) = \sum_{n \in \mathbb{Z}} x[n] e^{-in\Omega}, \quad \Omega \in \mathbb{R},$$

defined at those Ω for which the series converges absolutely. The variable Ω is called the **normalized angular frequency** and is measured in radians per sample.

Theorem 21.6: The Z-transform on the unit circle is the DTFT. If ROC includes $\mathbb{T} = \{|z| = 1\}$, then

$$X(e^{i\Omega}) = \sum_n x[n] e^{-in\Omega}$$

is the discrete-time Fourier transform (DTFT) of $x[n]$: the Z-transform restricted to $\mathbb{T}$ is exactly the discrete analog of Remark 20.1. More precisely, for an arbitrary signal x the following three conditions are equivalent:

- (i) $\mathbb{T} \subseteq \text{ROC}(x)$;
 - (ii) $\mathbb{T} \cap \text{ROC}(x) \neq \emptyset$;
 - (iii) $x \in \ell^1(\mathbb{Z})$, that is $\sum_n |x[n]| < \infty$;
- and in that case $\hat{x}$ is defined for every $\Omega \in \mathbb{R}$ and equals $\hat{x}(\Omega) = X(e^{i\Omega})$.

Proof. Step 1 — (ii) $\Rightarrow$ (iii). Let $z_0 \in \mathbb{T} \cap \text{ROC}(x)$. Since $|z_0| = 1$, one has $|z_0|^{-n} = 1$ for every $n \in \mathbb{Z}$, hence the condition of Definition 21.4 becomes

$$+\infty > \sum_{n \in \mathbb{Z}} |x[n]| |z_0|^{-n} = \sum_{n \in \mathbb{Z}} |x[n]|,$$

that is $x \in \ell^1(\mathbb{Z})$.

Step 2 — (iii) $\Rightarrow$ (i). If $\sum_n |x[n]| < \infty$, then for every z with $|z| = 1$ the same chain of equalities of Step 1, read backwards, gives $\sum_n |x[n]| |z|^{-n} = \sum_n |x[n]| < \infty$: every point of $\mathbb{T}$ belongs to $\text{ROC}(x)$.

Step 3 — (i) $\Rightarrow$ (ii). Immediate, because $\mathbb{T} \neq \emptyset$. The three conditions are thus equivalent.

Step 4 — the identification. Suppose the three equivalent conditions hold and set $z = e^{i\Omega}$ in Definition 21.3: one obtains the series $\sum_n x[n] e^{-in\Omega}$, which is by definition the DTFT of $x[n]$. Every point of $\mathbb{T}$ has the form $e^{i\Omega}$ for some $\Omega \in \mathbb{R}$ (surjectivity of the complex exponential onto the unit circle, Chapter 6), and by Step 2 the series converges absolutely for each of them: $\hat{x}(\Omega)$ is defined for every real Ω and coincides with $X(e^{i\Omega})$.

Step 5 — the reading as a Fourier series. The series $\sum_n x[n] e^{-in\Omega}$ is, not by coincidence, formally identical to the Fourier series of the Theorem (Volume II, Chapter 4), with the roles of n and θ swapped: here $x[n]$ are the “coefficients” and Ω the continuous variable. Precisely, setting $c_n = x[-n]$, the function $\hat{x}$ is the sum of the Fourier series $\sum_n c_n e^{in\Omega}$ of the 2π -periodic function $\hat{x}$, and the Fourier coefficients of $\hat{x}$ are exactly the c_n : this is the content of Theorem 21.8. The connection with the Laplace transform is the same: there, restriction to the imaginary axis gives the Fourier transform when the axis lies in the region of convergence (Remark 20.1); here, restriction to the unit circle gives the DTFT when the circle lies in the annulus. $\square$

Proposition 21.2: The DTFT is 2π -periodic. If $x \in \ell^1(\mathbb{Z})$, then $\hat{x}(\Omega + 2\pi) = \hat{x}(\Omega)$ for every $\Omega \in \mathbb{R}$. Consequently $\hat{x}$ is completely determined by its restriction to any interval of length 2π , for instance $[-\pi, \pi)$.

Proof. For every $n \in \mathbb{Z}$ one has $e^{-in(\Omega+2\pi)} = e^{-in\Omega}e^{-2\pi in} = e^{-in\Omega}$, because $e^{-2\pi in} = 1$ for integer n . Since the series converges absolutely (Theorem 21.6), substituting term by term gives $\hat{x}(\Omega + 2\pi) = \sum_n x[n]e^{-in\Omega} = \hat{x}(\Omega)$. The second statement follows because every real number is written as $\Omega_0 + 2\pi k$ with $\Omega_0 \in [-\pi, \pi)$ and $k \in \mathbb{Z}$. $\square$

Remark 21.5: Periodicity is not an accident: it is the circle. The reason for periodicity is geometric, not algebraic. The DTFT is the restriction of X to $\mathbb{T}$ (Theorem 21.6) read through the parametrization $\Omega \mapsto e^{i\Omega}$, which is the universal covering $\mathbb{R} \rightarrow \mathbb{T}$ studied in Chapter 12: two parameters Ω and Ω' identify the same point of $\mathbb{T}$ if and only if they differ by a multiple of 2π . The function truly defined on the circle is $X|_{\mathbb{T}}$; $\hat{x}$ is its lift, and the periodicity is the fiber of the covering. It is the same reason why, in the discrete case, the frequency $\Omega = \pi$ (half the sampling frequency, that is the Nyquist frequency) is the highest that makes sense: beyond it, one goes back down the circle.

Theorem 21.7: Uniform convergence and continuity of the DTFT. Let $x \in \ell^1(\mathbb{Z})$. Then the series of Definition 21.5 converges uniformly on $\mathbb{R}$, $\hat{x}$ is continuous and 2π -periodic, and

$$\|\hat{x}\|_{\infty} = \sup_{\Omega \in \mathbb{R}} |\hat{x}(\Omega)| \leq \|x\|_1.$$

Proof. For every n and every Ω one has $|x[n]e^{-in\Omega}| = |x[n]|$, because $|e^{-in\Omega}| = 1$. The constants $M_n = |x[n]|$ do not depend on Ω and $\sum_n M_n = \|x\|_1 < \infty$: the Weierstrass criterion gives uniform convergence on $\mathbb{R}$ of the series. Every symmetric partial sum $\sum_{|n| \leq N} x[n]e^{-in\Omega}$ is a continuous function of Ω (a finite sum of continuous functions); the uniform limit of a sequence of continuous functions is continuous, hence $\hat{x}$ is continuous. Periodicity is Proposition 21.2. Finally, for every Ω , $|\hat{x}(\Omega)| \leq \sum_n |x[n]e^{-in\Omega}| = \sum_n |x[n]| = \|x\|_1$, and taking the supremum over Ω gives the bound. $\square$

Theorem 21.8: DTFT inversion formula. Let $x \in \ell^1(\mathbb{Z})$. Then for every $n \in \mathbb{Z}$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\Omega) e^{in\Omega} d\Omega.$$

Proof. Step 1 — orthogonality. For $k \in \mathbb{Z}$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik\Omega} d\Omega = \begin{cases} 1, & k = 0, \\ \frac{1}{2\pi} \left[\frac{e^{ik\Omega}}{ik} \right]_{-\pi}^{\pi} = \frac{e^{ik\pi} - e^{-ik\pi}}{2\pi ik} = \frac{2i \sin(k\pi)}{2\pi ik} = 0, & k \neq 0, \end{cases}$$

since $\sin(k\pi) = 0$ for integer k .

Step 2 — interchange of series and integral. By Theorem 21.7 the series $\sum_k x[k]e^{-ik\Omega}$ converges uniformly on $[-\pi, \pi]$; multiplying it by the continuous, modulus-1 function $\Omega \mapsto e^{in\Omega}$ the convergence remains uniform, hence one may integrate term by term over the compact interval $[-\pi, \pi]$:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\Omega) e^{in\Omega} d\Omega = \sum_{k \in \mathbb{Z}} x[k] \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(n-k)\Omega} d\Omega.$$

Step 3 — conclusion. By Step 1 with k replaced by $n - k$, the integral equals 1 if $k = n$ and 0 otherwise: only the term $k = n$ remains, that is $x[n]$. $\square$

Remark 21.6: Inversion of the DTFT is inversion of the zeta transform. The formulas of Theorems 21.3 and 21.8 are the same formula. Taking $r = 1$ in Theorem 21.3 and parametrizing $z = e^{i\Omega}$, one obtains $x[n] = \frac{1}{2\pi} \int_0^{2\pi} X(e^{i\Omega}) e^{in\Omega} d\Omega$; and since the integrand is 2π -periodic (Proposition 21.2 and $e^{in(\Omega+2\pi)} = e^{in\Omega}$), the integral over $[0, 2\pi]$ equals that over $[-\pi, \pi]$. The inversion formula of the zeta transform is thus the inversion formula of the DTFT written on a circle of radius r instead of 1, with the correction factor r^n .

Theorem 21.9: Parseval's identity for the DTFT. Let $x, y \in \ell^1(\mathbb{Z})$. Then

$$\sum_{n \in \mathbb{Z}} x[n] \overline{y[n]} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\Omega) \overline{\hat{y}(\Omega)} d\Omega,$$

and in particular, for $y = x$,

$$\sum_{n \in \mathbb{Z}} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\hat{x}(\Omega)|^2 d\Omega.$$

In particular $\ell^1(\mathbb{Z}) \subseteq \ell^2(\mathbb{Z})$ and the two quantities are finite.

Proof. Step 1 — finiteness. If $x \in \ell^1$ then $|x[n]| \leq \|x\|_1$ for every n , hence $\sum_n |x[n]|^2 \leq \|x\|_1 \sum_n |x[n]| = \|x\|_1^2 < \infty$: $x \in \ell^2$. The right-hand side is finite because $\hat{x}$ and $\hat{y}$ are continuous and bounded (Theorem 21.7) on a compact interval.

Step 2 — the double family. Consider

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\Omega) \overline{\hat{y}(\Omega)} d\Omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_n x[n] e^{-in\Omega} \right) \overline{\left(\sum_m y[m] e^{-im\Omega} \right)} d\Omega.$$

The double family $(x[n] \overline{y[m]} e^{-i(n-m)\Omega})_{(n,m) \in \mathbb{Z}^2}$ is bounded in modulus, uniformly in Ω , by $|x[n]| |y[m]|$, whose sum equals $\|x\|_1 \|y\|_1 < \infty$: by the Weierstrass criterion the double series converges uniformly in Ω and may be integrated term by term.

Step 3 — orthogonality. By Step 1 of the proof of Theorem 21.8, $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i(n-m)\Omega} d\Omega$ equals 1 if $n = m$ and 0 otherwise. Only the terms with $n = m$ remain:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x} \overline{\hat{y}} d\Omega = \sum_{n \in \mathbb{Z}} x[n] \overline{y[n]},$$

which is the claim. The second formula is the case $y = x$, with $x[n] \overline{x[n]} = |x[n]|^2$ and $\hat{x} \overline{\hat{x}} = |\hat{x}|^2$. $\square$

Remark 21.7: What is imported: Parseval for finite-energy signals. Theorem 21.9 is proved here for $x, y \in \ell^1(\mathbb{Z})$, a hypothesis needed in two places: to define $\hat{x}$ pointwise and to justify the interchange of series and integral. The correct statement, however, is that the DTFT extends to a **surjective isometry** $\ell^2(\mathbb{Z}) \rightarrow L^2(\mathbb{T})$, and this is not proved here in full: the “injective and isometric” half follows from the completeness of the trigonometric system, which Chapter 17 proves (Theorem 17.1); the “surjective” half — every sequence in ℓ^2 is the sequence of Fourier coefficients of a function in $L^2(\mathbb{T})$ — is the Riesz–Fischer theorem, whose proof requires the completeness of the metric space $L^2(\mathbb{T})$, that is the Riesz–Fischer theorem on the convergence of Cauchy sequences in L^p . That completeness is here **imported** from measure theory and not reproved: it is the only missing link. Whoever wishes to fill it will find the proof in W. Rudin, *Real and Complex Analysis*, 3rd ed., Chapter 3 (“ L^p -spaces”, Riesz–Fischer theorem). No result of this chapter depends

on the missing link: every statement used elsewhere is proved under the ℓ^1 hypothesis.

21.5. BIBO stability: summability, annulus, poles

Definition 21.6: BIBO stability. A discrete LTI system $\mathcal{T}$ (Definition 21.2) is called **BIBO stable** (*bounded input, bounded output*) if there exists a constant $M < \infty$ such that

$$\|x\|_\infty \leq 1 \implies \|\mathcal{T}x\|_\infty \leq M$$

for every bounded input x . By linearity this is equivalent to: every bounded input produces a bounded output, with $\|\mathcal{T}x\|_\infty \leq M\|x\|_\infty$.

Theorem 21.10: BIBO $\iff$ absolutely summable impulse response. Let $\mathcal{T}$ be a discrete LTI system with impulse response h , with $\mathcal{T}x = h * x$ for every bounded input. Then

$$\mathcal{T} \text{ is BIBO stable } \iff h \in \ell^1(\mathbb{Z}),$$

and in that case the optimal constant of Definition 21.6 is exactly $M = \|h\|_1$.

Proof. Step 1 — sufficiency ($\Leftarrow$). Let $h \in \ell^1$. By Proposition 21.1(ii), for every bounded input the convolution series converges absolutely and

$$|y[n]| = \left| \sum_k h[k]x[n-k] \right| \leq \sum_k |h[k]| |x[n-k]| \leq \|x\|_\infty \sum_k |h[k]| = \|h\|_1 \|x\|_\infty$$

for every n : the system is BIBO stable with $M = \|h\|_1$.

Step 2 — necessity ($\Rightarrow$): construction of the critical input. Suppose the system is BIBO stable with constant M . Fix $N \in \mathbb{N}$ and define the signal

$$x_N[n] = \begin{cases} \frac{\overline{h[-n]}}{|h[-n]|}, & |n| \leq N \text{ and } h[-n] \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

It has finite support and $|x_N[n]| \leq 1$ for every n , hence $\|x_N\|_\infty \leq 1$ and by hypothesis $\|\mathcal{T}x_N\|_\infty \leq M$.

Step 3 — evaluating the output at time 0. By Proposition 21.1(i), applicable because x_N has finite support,

$$(\mathcal{T}x_N)[0] = \sum_{k \in \mathbb{Z}} h[k] x_N[-k] = \sum_{\substack{|k| \leq N \\ h[k] \neq 0}} h[k] \cdot \frac{\overline{h[k]}}{|h[k]|} = \sum_{\substack{|k| \leq N \\ h[k] \neq 0}} \frac{|h[k]|^2}{|h[k]|} = \sum_{|k| \leq N} |h[k]|,$$

having used $h[k]\overline{h[k]} = |h[k]|^2$ and being able to reinsert the indices with $h[k] = 0$, which contribute 0.

Step 4 — conclusion. The number just computed is a non-negative real number and by hypothesis does not exceed M :

$$\sum_{|k| \leq N} |h[k]| = (\mathcal{T}x_N)[0] \leq \|\mathcal{T}x_N\|_\infty \leq M \quad \text{for every } N.$$

The partial sums of the series with non-negative terms $\sum_k |h[k]|$ are thus uniformly bounded by M : the series converges and $\|h\|_1 \leq M$. Hence $h \in \ell^1(\mathbb{Z})$.

Step 5 — optimality of the constant. Step 1 gives $M \leq \|h\|_1$ as an admissible constant, and Step 4 gives $\|h\|_1 \leq M$ for every admissible constant M : the minimal constant is exactly $\|h\|_1$. $\square$

Corollary 21.4: BIBO $\iff$ the unit circle lies in the annulus. Under the hypotheses of Theorem 21.10 and setting $H = \mathcal{Z}\{h\}$, the system is BIBO stable if and only if $\mathbb{T} \subseteq \text{ROC}(h)$; and in that case H restricted to $\mathbb{T}$ is the **frequency response** $\hat{h}(\Omega) = H(e^{i\Omega})$, continuous, 2π -periodic and bounded by $\|h\|_1$.

Proof. By Theorem 21.10 BIBO stability is equivalent to $h \in \ell^1(\mathbb{Z})$, which by Theorem 21.6 is equivalent to $\mathbb{T} \subseteq \text{ROC}(h)$. The properties of the restriction are Theorem 21.7. $\square$

21.5.1 Causal rational systems: the location of the poles

To reach the pole criterion three algebraic tools are needed, all proved.

Lemma 21.2: Negative binomial series. For every integer $j \geq 1$ and every $w \in \mathbb{C}$ with $|w| < 1$,

$$\frac{1}{(1-w)^j} = \sum_{k \geq 0} \binom{j+k-1}{k} w^k,$$

with absolute convergence.

Proof. By induction on j .

Base case $j = 1$. $\binom{k}{k} = 1$ and the formula is the geometric series $\frac{1}{1-w} = \sum_k w^k$, absolutely convergent for $|w| < 1$.

Induction step. Suppose the formula holds for j . A power series is differentiated term by term inside its disk of convergence, which here is $|w| < 1$; on the left $\frac{d}{dw}(1-w)^{-j} = j(1-w)^{-j-1}$, on the right

$$\frac{d}{dw} \sum_{k \geq 0} \binom{j+k-1}{k} w^k = \sum_{k \geq 1} k \binom{j+k-1}{k} w^{k-1} = \sum_{k \geq 0} (k+1) \binom{j+k}{k+1} w^k.$$

Now, for every $k \geq 0$,

$$(k+1) \binom{j+k}{k+1} = (k+1) \cdot \frac{(j+k)!}{(k+1)!(j-1)!} = \frac{(j+k)!}{k!(j-1)!} = j \cdot \frac{(j+k)!}{k!j!} = j \binom{j+k}{k}.$$

Equating and dividing by $j \neq 0$ gives $\frac{1}{(1-w)^{j+1}} = \sum_{k \geq 0} \binom{j+k}{k} w^k$, which is the formula for $j+1$ since $\binom{(j+1)+k-1}{k} = \binom{j+k}{k}$. Absolute convergence for $|w| < 1$ is that of the differentiated series, which has the same radius as the original series. $\square$

Lemma 21.3: Partial fraction decomposition. Let N, D be polynomials with complex coefficients with $\deg N < \deg D =: d$ and $D(z) = c \prod_{\ell=1}^r (z - p_\ell)^{m_\ell}$, with $p_1, \dots, p_r$ distinct, $c \neq 0$ and $\sum_\ell m_\ell = d$. Then there exist, and are unique, complex numbers $A_{\ell,j}$ ($1 \leq \ell \leq r$, $1 \leq j \leq m_\ell$) such that

$$\frac{N(z)}{D(z)} = \sum_{\ell=1}^r \sum_{j=1}^{m_\ell} \frac{A_{\ell,j}}{(z - p_\ell)^j} \quad \text{for every } z \notin \{p_1, \dots, p_r\}.$$

Proof. Existence, by induction on r .

Case $r = 1$, that is $D(z) = c(z - p)^m$ with $m = d$. A polynomial N of degree $< m$ is rewritten in powers of $z - p$ (Taylor expansion of a polynomial, which is an exact finite sum): $N(z) = \sum_{i=0}^{m-1} \gamma_i (z - p)^i$ with $\gamma_i = N^{(i)}(p)/i!$. Dividing by D ,

$$\frac{N(z)}{D(z)} = \frac{1}{c} \sum_{i=0}^{m-1} \gamma_i (z - p)^{i-m} = \sum_{j=1}^m \frac{\gamma_{m-j}/c}{(z - p)^j},$$

which has the desired form with $A_{1,j} = \gamma_{m-j}/c$.

Induction step. Let $r \geq 2$ and write $D = D_1 D_2$ with $D_1(z) = (z - p_1)^{m_1}$ and $D_2(z) = c \prod_{\ell \geq 2} (z - p_\ell)^{m_\ell}$. The two polynomials are coprime, having no common roots (the p_ℓ are distinct); by Bézout's identity there exist polynomials U, V with $UD_1 + VD_2 = 1$. Then

$$\frac{N}{D} = \frac{N(UD_1 + VD_2)}{D_1 D_2} = \frac{NU}{D_2} + \frac{NV}{D_1}.$$

Divide with remainder: $NU = Q_2 D_2 + R_2$ with $\deg R_2 < \deg D_2$, and $NV = Q_1 D_1 + R_1$ with $\deg R_1 < \deg D_1$. Substituting,

$$\frac{N}{D} = (Q_1 + Q_2) + \frac{R_2}{D_2} + \frac{R_1}{D_1}.$$

The three fractions N/D , R_2/D_2 , R_1/D_1 are proper, that is they tend to 0 as $|z| \rightarrow \infty$ (numerator of strictly smaller degree than the denominator); hence the polynomial $Q_1 + Q_2$ tends to 0 at infinity and is therefore identically zero. What remains is $N/D = R_1/D_1 + R_2/D_2$: to the first term one applies the case $r = 1$, to the second the induction hypothesis (the denominator D_2 has $r - 1$ distinct roots), and one obtains the desired form.

Uniqueness. Reducing the sum $\sum_{\ell,j} A_{\ell,j} (z - p_\ell)^{-j}$ to a common denominator D/c gives a fraction whose numerator is a polynomial of degree $< d$; the map

$$\Phi : \mathbb{C}^d \rightarrow \{\text{polynomials of degree } < d\}, \quad (A_{\ell,j}) \mapsto \text{numerator of the sum},$$

is linear between two complex vector spaces of the same dimension d (recall $\sum_\ell m_\ell = d$), and is surjective by the existence part just proved: hence it is injective, and the coefficients $A_{\ell,j}$ are determined by N . $\square$

Lemma 21.4: Causal inverse transform of a pole of order j . Let $p \in \mathbb{C} \setminus \{0\}$ and $j \geq 1$ be integers. On the annulus $|z| > |p|$,

$$\frac{1}{(z - p)^j} = \sum_{n \geq j} \binom{n-1}{j-1} p^{n-j} z^{-n},$$

that is $\frac{1}{(z-p)^j} = \mathcal{Z}\{g_j\}$ with $g_j[n] = \binom{n-1}{j-1} p^{n-j} u[n-j]$. In particular g_j is causal and, setting $q_j(n) = \binom{n-1}{j-1}$, one has $g_j[n] = q_j(n) p^{n-j} p^n$ for $n \geq j$, with q_j a polynomial in n of degree $j - 1$.

Proof. For $|z| > |p|$ one has $|p/z| < 1$; collecting z and applying Lemma 21.2 with $w = p/z$:

$$\frac{1}{(z - p)^j} = \frac{1}{z^j (1 - \frac{p}{z})^j} = z^{-j} \sum_{k \geq 0} \binom{j+k-1}{k} \left(\frac{p}{z}\right)^k = \sum_{k \geq 0} \binom{j+k-1}{k} p^k z^{-(k+j)}.$$

Setting $n = k + j$, that is $k = n - j \geq 0$, and using $\binom{j+k-1}{k} = \binom{n-1}{n-j} = \binom{n-1}{j-1}$ (symmetry of the binomial coefficient), one obtains the formula. Convergence is absolute for $|z| > |p|$, hence

$\{|z| > |p|\} \subseteq \text{ROC}(g_j)$. Finally

$$q_j(n) = \binom{n-1}{j-1} = \frac{(n-1)(n-2)\cdots(n-j+1)}{(j-1)!}$$

is, for $j \geq 1$, a polynomial in n of degree exactly $j-1$ (for $j=1$ it is the constant 1). $\square$

Theorem 21.11: The annulus of a causal rational system. Let $H = N/D$ be proper rational and reduced to lowest terms, with poles $p_1, \dots, p_r$ nonzero and distinct, of multiplicities $m_1, \dots, m_r$, and let $\rho^* = \max_\ell |p_\ell|$. Let h be the causal signal with $\mathcal{Z}\{h\} = H$ on the annulus $\{|z| > \rho^*\}$. Then

(i) h exists and is given by

$$h[n] = c_0 \delta[n] + \sum_{\ell=1}^r \sum_{j=1}^{m_\ell} A_{\ell,j} \binom{n-1}{j-1} p_\ell^{n-j} u[n-j],$$

with $c_0 = \lim_{|z| \rightarrow \infty} H(z)$ and $A_{\ell,j}$ the coefficients of Lemma 21.3 applied to $H - c_0$;

(ii) $\text{ROC}(h) = \{|z| > \rho^*\}$ exactly: no point with $|z| \leq \rho^*$ belongs to it.

Proof. (i). Since H is proper, $\deg N \leq \deg D$; dividing with remainder, $N = c_0 D + \tilde{N}$ with $c_0 \in \mathbb{C}$ and $\deg \tilde{N} < \deg D$, so that $H = c_0 + \tilde{N}/D$ and $c_0 = \lim_{|z| \rightarrow \infty} H(z)$. Lemma 21.3 decomposes $\tilde{N}/D$ into partial fractions, and Lemma 21.4 inverts each fraction on the annulus $|z| > |p_\ell|$; since $\{|z| > \rho^*\} \subseteq \{|z| > |p_\ell|\}$ for every ℓ , all the series converge absolutely on $\{|z| > \rho^*\}$, and by linearity (Theorem 21.5(i)) their combination, added to $c_0 = \mathcal{Z}\{c_0 \delta\}$, gives exactly H on that annulus. The resulting signal is causal because each term is.

(ii). By Corollary 21.1 the region of convergence of h is $\{|z| > \rho_+\}$ or its closure, with $\rho_+ = \limsup_n |h[n]|^{1/n}$, and point (i) shows $\rho_+ \leq \rho^*$. Suppose for contradiction that there exists $z_1 \in \text{ROC}(h)$ with $|z_1| \leq \rho^*$; by rotation invariance (Theorem 21.1(ii)) the whole circle $|z| = |z_1| =: s \leq \rho^*$ lies in $\text{ROC}(h)$, and since h is causal, for every $|z| \geq s$ one has $|h[n]z^{-n}| \leq |h[n]|s^{-n}$, hence

$$|G(z)| := \left| \sum_{n \geq 0} h[n]z^{-n} \right| \leq \sum_{n \geq 0} |h[n]| s^{-n} =: C < \infty \quad \text{for every } |z| \geq s.$$

Now let p be a pole with $|p| = \rho^*$ and, for $t > 1$, consider the point $z_t = tp$. Then $|z_t| = t\rho^* > \rho^* \geq s$, hence z_t lies in the annulus $\{|z| > \rho^*\}$ where $G = H$ by hypothesis, and together $|G(z_t)| \leq C$. But $z_t \rightarrow p$ as $t \rightarrow 1^+$ and p is a pole of H : since $H = N/D$ is reduced to lowest terms, $N(p) \neq 0$ and $D(p) = 0$, hence $|H(z_t)| = |N(z_t)|/|D(z_t)| \rightarrow +\infty$ (numerator with limit $|N(p)| > 0$, denominator infinitesimal and nonzero for $t > 1$ close enough to 1, because the zeros of D are isolated). This contradicts $|H(z_t)| = |G(z_t)| \leq C$. Hence no z with $|z| \leq \rho^*$ belongs to $\text{ROC}(h)$, and by Corollary 21.1 $\text{ROC}(h) = \{|z| > \rho^*\}$. $\square$

Theorem 21.12: BIBO stability in the Z-domain. A causal discrete LTI system with transfer function $H(z) = \mathcal{Z}\{h[n]\}$ is BIBO stable if and only if all poles of $H(z)$ lie strictly inside the unit circle: $|z| < 1$.

Proof. Recall the criterion in the time domain: a discrete LTI system is BIBO stable if and only if its impulse response is *summable*, that is $\sum_{n \geq 0} |h[n]| < \infty$. (Sufficiency is immediate: if $|x[n]| \leq M$ then $|y[n]| = |\sum_k h[k]x[n-k]| \leq M \sum_k |h[k]|$. Necessity is obtained by choosing the bounded input $x[n-k] = \bar{h}[k]/|h[k]|$ where $h[k] \neq 0$, which makes $y[n] = \sum_k |h[k]|$.) Both directions are proved in full in Theorem 21.10. It thus suffices to show that summability

of h is equivalent to the location of the poles. It is assumed, as is implicit in speaking of “poles”, that H is proper rational and reduced to lowest terms, with nonzero poles $p_1, \dots, p_r$; set $\rho^* = \max_\ell |p_\ell|$.

Step 1 — structure of h . Let $H(z) = N(z)/D(z)$ be proper rational with poles $p_1, \dots, p_r$ of multiplicities $m_1, \dots, m_r$. Decomposing into partial fractions and inverting term by term, the causal impulse response is a finite linear combination of sequences of the form

$$n^j p_\ell^n u[n], \quad 0 \leq j \leq m_\ell - 1.$$

This is exactly the content of Theorem 21.11(i), proved starting from Lemma 21.3 (existence and uniqueness of partial fractions) and Lemma 21.4 (causal inverse transform of a pole of order j): the coefficient $\binom{n-1}{j-1}$ appearing there is a polynomial in n of degree $j-1 \leq m_\ell - 1$, that is a linear combination of $1, n, \dots, n^{m_\ell-1}$.

Step 2 — interior poles $\Rightarrow$ summability. If $|p_\ell| < 1$ for every ℓ , then for every fixed j

$$\sum_{n \geq 0} n^j |p_\ell|^n < \infty,$$

because the ratio test gives $|p_\ell| < 1$ as its limit (the polynomial factor $((n+1)/n)^j \rightarrow 1$ does not alter the limit); equivalently, the root test gives $(n^j |p_\ell|^n)^{1/n} = (n^{1/n})^j |p_\ell| \rightarrow |p_\ell| < 1$, since $n^{1/n} \rightarrow 1$. A finite linear combination of absolutely convergent series converges absolutely, hence $\sum_n |h[n]| < \infty$ and, by Theorem 21.10, the system is BIBO stable.

Step 3 — a non-interior pole $\Rightarrow$ non-summability. Let p be a pole of maximal modulus $\rho^* = |p| \geq 1$. If h were summable, the series $H(z) = \sum_{n \geq 0} h[n]z^{-n}$ would converge absolutely for every $|z| \geq 1$ — since $|h[n]z^{-n}| \leq |h[n]|$ — and would define a continuous and bounded function on $\{|z| \geq 1\}$. But H has a pole at p with $|p| \geq 1$, hence $|H(z)| \rightarrow \infty$ as $z \rightarrow p$: a contradiction. In full: summability of h means $1 \in \text{ROC}(h)$ in the sense that $\mathbb{T} \subseteq \text{ROC}(h)$ (Theorem 21.6), while Theorem 21.11(ii) states $\text{ROC}(h) = \{|z| > \rho^*\}$, a set containing no point of $\mathbb{T}$ when $\rho^* \geq 1$. Hence h is not summable and the system is not BIBO stable.

Conclusion. Summability of $h \iff$ all poles in $|z| < 1$, and by the criterion recalled at the beginning this is equivalent to BIBO stability. Note that poles *on* the unit circle give BIBO instability even though they produce bounded responses (for example $h[n] = u[n]$ for $p = 1$): the system is *marginally* stable, and this is why $\mathbb{T}$ is the boundary and not part of the region of stability. $\square$

Example 21.4: The pole on the boundary: bounded response, unstable system.

Let $H(z) = \frac{z}{z-1}$, that is the discrete integrator $y[n] = y[n-1] + x[n]$, with a simple pole at $p = 1$. The impulse response is $h[n] = u[n]$ (Example 21.2 with $a = 1$), *bounded*: $\|h\|_\infty = 1$. Yet $\sum_n |h[n]| = \sum_{n \geq 0} 1 = +\infty$, hence by Theorem 21.10 the system is not BIBO stable. One sees this also directly: the bounded input $x[n] = u[n]$ produces $y[n] = \sum_{k=0}^n 1 = n+1 \rightarrow +\infty$. The annulus is $\{|z| > 1\}$, which does not contain $\mathbb{T}$, in agreement with Corollary 21.4. This is the case cited in the *Conclusion* of the proof of Theorem 21.12: a pole on the unit circle is already too much.

Example 21.5: A second-order system, verified three times. Let $y[n] = y[n-1] - 0.24y[n-2] + x[n]$, that is

$$H(z) = \frac{1}{1 - z^{-1} + 0.24z^{-2}} = \frac{z^2}{z^2 - z + 0.24} = \frac{z^2}{(z - 0.6)(z - 0.4)},$$

since $z^2 - z + 0.24$ has discriminant $1 - 0.96 = 0.04$ and roots $\frac{1 \pm 0.2}{2} = 0.6$ and 0.4 . Both poles have modulus < 1 : by Theorem 21.12 the system is BIBO stable.

Impulse response. Since H is proper with $\deg N = \deg D$, one applies Theorem 21.11(i). It is convenient to decompose $H(z)/z$:

$$\frac{H(z)}{z} = \frac{z}{(z-0.6)(z-0.4)} = \frac{A}{z-0.6} + \frac{B}{z-0.4},$$

with $A = \frac{0.6}{0.6-0.4} = 3$ and $B = \frac{0.4}{0.4-0.6} = -2$; multiplying by z and inverting with Example 21.2 (outer annulus, the system is causal):

$$H(z) = \frac{3z}{z-0.6} - \frac{2z}{z-0.4}, \quad h[n] = (3 \cdot 0.6^n - 2 \cdot 0.4^n)u[n].$$

First check (recursion). Iterating the difference equation with $x = \delta$ gives $h[0] = 1$, $h[1] = 1$, $h[2] = 0.76$, $h[3] = 0.52$, $h[4] = 0.3376$, $h[5] = 0.2128$; the closed formula gives $3 - 2 = 1$, $1.8 - 0.8 = 1$, $1.08 - 0.32 = 0.76$, $0.648 - 0.128 = 0.52$, $0.3888 - 0.0512 = 0.3376$, $0.23328 - 0.02048 = 0.2128$. Over 200 samples the maximum discrepancy between the two routes is $2.2 \cdot 10^{-16}$. ✓

Second check (ℓ^1 norm and DC gain). Since $3 \cdot 0.6^n > 2 \cdot 0.4^n$ for every $n \geq 0$ (the ratio equals $1.5 \cdot 1.5^n \geq 1.5$), one has $h[n] > 0$ and hence

$$\|h\|_1 = \sum_{n \geq 0} h[n] = \frac{3}{1-0.6} - \frac{2}{1-0.4} = 7.5 - \frac{10}{3} = \frac{25}{6} = 4.166\,666\,7,$$

which coincides with $H(1) = \frac{1}{1-1+0.24} = \frac{1}{0.24} = 4.166\,666\,7$, as it must because $H(1) = \sum_n h[n]$ and the terms are positive. ✓

Third check (Parseval). Summing the series, $\sum_n |h[n]|^2 = 3.034\,931\,077\,7$; the closed formula $\frac{9}{1-0.36} - \frac{12}{1-0.24} + \frac{4}{1-0.16} = 3.034\,931\,077\,7$; and the integral $\frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{i\Omega})|^2 d\Omega$, evaluated with 200,000 points, 3.0349310777: three independent routes, the same number, in agreement with Theorem 21.9. ✓

Worked example — geometric sequence and first-order digital filter. Let $x[n] = a^n u[n]$ (causal sequence, $u[n]$ discrete step). Then

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n = \frac{1}{1-a/z} = \frac{z}{z-a}, \quad \text{ROC: } |z| > |a|,$$

(geometric series of ratio a/z , convergent for $|a/z| < 1$). Consider now the first-order recursive digital filter $y[n] = ay[n-1] + bx[n]$ (for instance an exponential moving average, often used to filter sensor noise before PID computation). Transforming (with the delay property, analogous to Laplace, that is Theorem 21.5(ii)):

$$Y(z) = az^{-1}Y(z) + bX(z) \implies H(z) = \frac{Y(z)}{X(z)} = \frac{b}{1-az^{-1}} = \frac{bz}{z-a},$$

with a single pole at $z = a$: the filter is BIBO stable if and only if $|a| < 1$. Note the parallel with the Worked example of Section 4.8 (Volume II, Chapter 4): if this discrete filter samples, with period T , the first-order continuous system with time constant τ , the discrete pole is $a = e^{-T/\tau}$, which automatically satisfies $|a| < 1$ for every $T, \tau > 0$ — the stability of the continuous system is transmitted to the sampled system, exactly as predicted by Remark 21.8.

21.6. The three planes: Laplace, zeta and Fourier

The s -plane of the Laplace transform (Chapter 20), the z -plane of the Z-transform and the frequency axis of the Fourier transform are not three separate worlds: they are three views of

one and the same object, linked by two maps that we study here in full. The first is exact but not injective, the second is injective but approximate.

Theorem 21.13: The exponential sampling map. Let $T > 0$ and let $\Phi : \mathbb{C} \rightarrow \mathbb{C}$, $\Phi(s) = e^{sT}$. Then

(i) $|\Phi(s)| = e^{T \operatorname{Re}(s)}$, and hence

$$\operatorname{Re}(s) < 0 \iff |\Phi(s)| < 1, \quad \operatorname{Re}(s) = 0 \iff |\Phi(s)| = 1,$$

$$\operatorname{Re}(s) > 0 \iff |\Phi(s)| > 1;$$

(ii) $\Phi(\mathbb{C}) = \mathbb{C} \setminus \{0\}$: the map is surjective onto the punctured plane;

(iii) $\Phi(s) = \Phi(s')$ if and only if $s - s' \in \frac{2\pi i}{T}\mathbb{Z}$;

(iv) the restriction of Φ to the strip $S = \{s : -\pi/T < \operatorname{Im}(s) \leq \pi/T\}$ is a bijection onto $\mathbb{C} \setminus \{0\}$, which sends $S \cap \{\operatorname{Re} s < 0\}$ onto the punctured disk $\{0 < |z| < 1\}$ and $S \cap \{\operatorname{Re} s = 0\}$ onto $\mathbb{T}$.

Proof. Write $s = \sigma + i\omega$ with $\sigma, \omega \in \mathbb{R}$.

(i). $e^{sT} = e^{\sigma T} e^{i\omega T}$ and $|e^{i\omega T}| = 1$, hence $|\Phi(s)| = e^{\sigma T}$. Since $t \mapsto e^t$ is strictly increasing with $e^0 = 1$, one has $e^{\sigma T} < 1 \iff \sigma T < 0 \iff \sigma < 0$ (since $T > 0$), and similarly for the other two cases.

(ii). Let $z \neq 0$. Writing $z = \rho e^{i\theta}$ with $\rho > 0$ and $\theta \in \mathbb{R}$ (polar form, Chapter 6), set $s = \frac{\ln \rho}{T} + i\frac{\theta}{T}$: then $\Phi(s) = e^{\ln \rho} e^{i\theta} = \rho e^{i\theta} = z$. Conversely $\Phi(s) \neq 0$ for every s , because $|\Phi(s)| = e^{\sigma T} > 0$.

(iii). $\Phi(s) = \Phi(s')$ is equivalent to $e^{(s-s')T} = 1$; and the complex exponential equals 1 exactly at integer multiples of $2\pi i$, hence $(s - s')T \in 2\pi i\mathbb{Z}$, that is $s - s' \in \frac{2\pi i}{T}\mathbb{Z}$.

(iv). Two distinct points of S have imaginary parts differing by less than $2\pi/T$ in absolute value, hence they cannot differ by a nonzero multiple of $2\pi i/T$: by (iii) $\Phi|_S$ is injective. It is surjective onto $\mathbb{C} \setminus \{0\}$ because, given $z \neq 0$, the point s constructed in (ii) is brought into S by choosing the unique argument $\theta \in (-\pi, \pi]$ of z . The two final statements follow from (i) and from $|z| > 0$. $\square$

Remark 21.8: Why precisely the unit circle. If a continuous signal $x_c(t)$ is sampled with period T obtaining $x[n] = x_c(nT)$, the (ideal) correspondence between the s and z planes is $z = e^{sT}$: the imaginary axis $s = i\omega$ (stability boundary in continuous time) maps to $z = e^{i\omega T}$, which is precisely $\mathbb{T}$ (stability boundary in discrete time), while the left half-plane $\operatorname{Re}(s) < 0$ (continuous stability) maps to the interior of the unit circle $|z| = |e^{sT}| = e^{\operatorname{Re}(s)T} < 1$. The unit circle $\mathbb{T}$ of Chapter 3 is therefore not merely an abstract geometric object: it is literally the stability boundary of every digital control system, including PLCs. This is also exactly the geometric phenomenon encountered in Corollary 3.3 of Chapter 3, where the Möbius transformation $w = (z - 1)/(z + 1)$ mapped $\mathbb{T}$ to the imaginary axis: that is, up to a scaling factor $2/T$, the bilinear transform used to approximate $z = e^{sT}$ in digital control practice. All the statements listed here are points (i) and (iv) of Theorem 21.13, and the two references to “Chapter 3” are to Volume II, Chapter 3.

Remark 21.9: Non-injectivity is aliasing. Theorem 21.13(iii) says that Φ identifies with each other all points that differ by a multiple of $2\pi i/T$: in particular the angular frequencies ω and $\omega + 2\pi/T = \omega + 2\pi f_s$ produce the same point of the z -plane, and hence the same sampled sequence. This is **aliasing**: two continuous sinusoids whose frequencies differ by a multiple of the sampling frequency are indistinguishable after sampling. The strip S of point (iv) is the canonical choice of a representative for each class, and corresponds

exactly to the frequency interval $(-f_s/2, f_s/2]$: the Nyquist band. The periodicity of the DTFT (Proposition 21.2) is the same thing read from the other side: on the circle there is room for no more than a single lap.

21.6.1 The bilinear transform

Definition 21.7: Bilinear transform (Tustin's). Let $T > 0$. The **bilinear transform** of step T is the map

$$\beta(s) = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s}, \quad s \neq \frac{2}{T},$$

with inverse map

$$\beta^{-1}(z) = \frac{2}{T} \cdot \frac{z - 1}{z + 1}, \quad z \neq -1.$$

It is the Möbius transformation $s \mapsto \frac{as+b}{cs+d}$ with $a = \frac{T}{2}$, $b = 1$, $c = -\frac{T}{2}$, $d = 1$, of determinant $ad - bc = \frac{T}{2} + \frac{T}{2} = T \neq 0$ (Chapter 6).

Theorem 21.14: The bilinear transform sends the left half-plane to the unit disk.

Let β be as in Definition 21.7 and let $\mathbb{H}^- = \{s \in \mathbb{C} : \operatorname{Re}(s) < 0\}$. Then

(i) for every $s \neq 2/T$ the identity holds

$$|\beta(s)|^2 - 1 = \frac{2T \operatorname{Re}(s)}{|1 - \frac{T}{2}s|^2};$$

(ii) β is a bijection from $\mathbb{H}^-$ onto $\mathbb{D}$, with inverse β^{-1} ;

(iii) β is a bijection from the imaginary axis $i\mathbb{R}$ onto $\mathbb{T} \setminus \{-1\}$, and $\beta(s) \rightarrow -1$ as $|s| \rightarrow \infty$ along $i\mathbb{R}$;

(iv) β sends $\{\operatorname{Re}(s) > 0\} \setminus \{2/T\}$ onto $\{|z| > 1\}$.

Proof. Step 1 — an elementary identity. For every $w \in \mathbb{C}$,

$$\begin{aligned} |1 + w|^2 - |1 - w|^2 &= (1 + w)(\overline{1 + w}) - (1 - w)(\overline{1 - w}) \\ &= (1 + w + \bar{w} + |w|^2) - (1 - w - \bar{w} + |w|^2) = 2(w + \bar{w}) = 4 \operatorname{Re}(w). \end{aligned}$$

(Numerical check: for $w = 0.3 + 0.4i$ the left-hand side equals $|1.3 + 0.4i|^2 - |0.7 - 0.4i|^2 = 1.85 - 0.65 = 1.2$ and the right-hand side $4 \cdot 0.3 = 1.2$. ✓)

Step 2 — (i). Setting $w = \frac{T}{2}s$, one has $\beta(s) = \frac{1+w}{1-w}$ with $1 - w \neq 0$, hence

$$|\beta(s)|^2 - 1 = \frac{|1 + w|^2 - |1 - w|^2}{|1 - w|^2} = \frac{4 \operatorname{Re}(w)}{|1 - w|^2} = \frac{4 \cdot \frac{T}{2} \operatorname{Re}(s)}{|1 - \frac{T}{2}s|^2} = \frac{2T \operatorname{Re}(s)}{|1 - \frac{T}{2}s|^2}.$$

Step 3 — (ii). The denominator in (i) is positive and $2T > 0$, hence $|\beta(s)| < 1 \iff \operatorname{Re}(s) < 0$. It remains to prove bijectivity. Observe that $s \in \mathbb{H}^-$ implies $\operatorname{Re}(1 - \frac{T}{2}s) = 1 - \frac{T}{2} \operatorname{Re}(s) > 1 > 0$, hence β is defined on all of $\mathbb{H}^-$. For $z \neq -1$ one has

$$\beta(\beta^{-1}(z)) = \frac{1 + \frac{T}{2} \cdot \frac{2}{T} \frac{z-1}{z+1}}{1 - \frac{T}{2} \cdot \frac{2}{T} \frac{z-1}{z+1}} = \frac{\frac{(z+1)+(z-1)}{z+1}}{\frac{(z+1)-(z-1)}{z+1}} = \frac{2z}{2} = z,$$

and, symmetrically, $\beta^{-1}(\beta(s)) = s$ for every $s \neq 2/T$ (same computation with roles swapped). Since $-1 \notin \mathbb{D}$, the map β^{-1} is defined on all of $\mathbb{D}$; and for $z \in \mathbb{D}$ the point $s = \beta^{-1}(z)$ satisfies

$|\beta(s)| = |z| < 1$, hence $\operatorname{Re}(s) < 0$ by (i), that is $s \in \mathbb{H}^-$. The two maps are thus inverse to each other between $\mathbb{H}^-$ and $\mathbb{D}$.

Step 4 — (iii). By (i), $|\beta(s)| = 1 \iff \operatorname{Re}(s) = 0$; and $\beta(i\omega) = -1$ would require $1 + \frac{T}{2}i\omega = -1 + \frac{T}{2}i\omega$, that is $1 = -1$, impossible: hence $\beta(i\mathbb{R}) \subseteq \mathbb{T} \setminus \{-1\}$. Conversely every $z \in \mathbb{T} \setminus \{-1\}$ has preimage $s = \beta^{-1}(z)$, which by (i) has zero real part. Finally, dividing numerator and denominator by s , $\beta(s) = \frac{1/s + T/2}{1/s - T/2} \rightarrow \frac{T/2}{-T/2} = -1$ as $|s| \rightarrow \infty$.

Step 5 — (iv). Again by (i): $|\beta(s)| > 1 \iff \operatorname{Re}(s) > 0$. □

Corollary 21.5: Deformation of the frequency axis, or warping. For $\omega \in \mathbb{R}$ one has $\beta(i\omega) = e^{i\Omega}$ with

$$\Omega = 2 \arctan\left(\frac{\omega T}{2}\right), \quad \text{equivalently} \quad \omega = \frac{2}{T} \tan\left(\frac{\Omega}{2}\right), \quad \Omega \in (-\pi, \pi).$$

The correspondence $\omega \mapsto \Omega$ is a strictly increasing bijection of $\mathbb{R}$ onto $(-\pi, \pi)$, with $\Omega \simeq \omega T$ for small ωT and $\Omega \rightarrow \pm\pi$ as $\omega \rightarrow \pm\infty$.

Proof. Setting $u = \frac{\omega T}{2} \in \mathbb{R}$, one has $\beta(i\omega) = \frac{1+iu}{1-iu}$. The numerator has modulus $\sqrt{1+u^2}$ and argument $\arctan u$ (it lies in the right half-plane, where the arctangent gives the principal argument); the denominator is its conjugate, hence has the same modulus and argument $-\arctan u$. The quotient thus has modulus 1 and argument $\arctan u - (-\arctan u) = 2 \arctan u$:

$$\beta(i\omega) = e^{i2 \arctan u} = e^{i\Omega}, \quad \Omega = 2 \arctan \frac{\omega T}{2}.$$

The function $\arctan$ is a strictly increasing bijection of $\mathbb{R}$ onto $(-\frac{\pi}{2}, \frac{\pi}{2})$, hence so is $\Omega = 2 \arctan u$ of $\mathbb{R}$ onto $(-\pi, \pi)$, and inverting, $\tan(\Omega/2) = u = \omega T/2$, that is $\omega = \frac{2}{T} \tan(\Omega/2)$. As $u \rightarrow 0$ one has $\arctan u = u + O(u^3)$, hence $\Omega = \omega T + O((\omega T)^3)$; and $\arctan u \rightarrow \pm\frac{\pi}{2}$ as $u \rightarrow \pm\infty$ gives $\Omega \rightarrow \pm\pi$. □

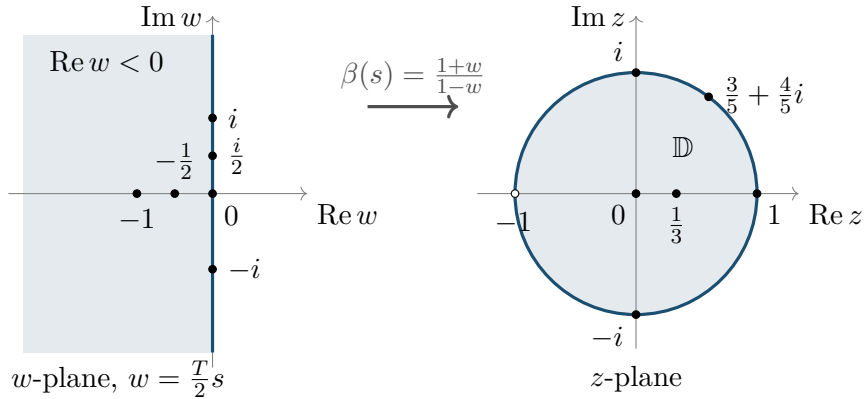


Figure 21.2: The bilinear transform of Theorem 21.14: the left half-plane goes to the unit disk, the imaginary axis to the unit circle minus -1 (empty circle). The correspondences drawn are exact: $0 \mapsto 1$, $-\frac{1}{2} \mapsto \frac{1}{3}$, $-1 \mapsto 0$, $\frac{i}{2} \mapsto \frac{3}{5} + \frac{4}{5}i$, $\pm i \mapsto \pm i$; and indeed $\frac{1+i/2}{1-i/2} = \frac{(1+i/2)^2}{5/4} = \frac{3/4+i}{5/4} = \frac{3}{5} + \frac{4}{5}i$, of modulus 1. The radius of the disk is 1.6 cm and the points are placed by multiplying the exact coordinates by 1.6.

Remark 21.10: The two maps compared. The exact map $\Phi(s) = e^{sT}$ and the rational approximation β agree to first order, because $e^{w'} = 1 + w' + O(w'^2)$ and $\frac{1+w}{1-w} = 1 + 2w + O(w^2)$ give, with $w = \frac{T}{2}s$ and $w' = sT$, the same expansion $1 + sT + O((sT)^2)$; but they differ

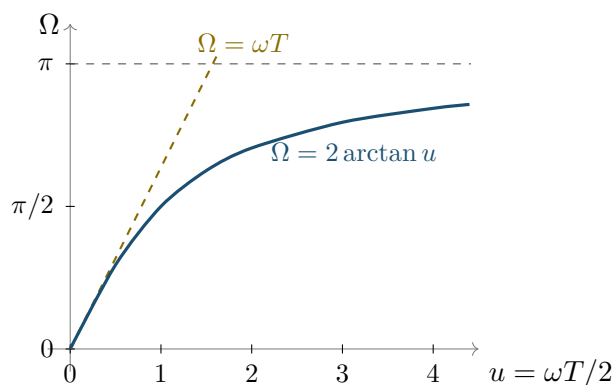


Figure 21.3: The *warping* of Corollary 21.5. The values used to draw the curve are computed exactly: $2 \arctan u$ equals 0, 0.927 295 2, 1.570 796 3, 1.965 587 4, 2.214 297 4, 2.498 091 5, 2.651 635 3 and 2.694 639 5 for $u = 0; 0.5; 1; 1.5; 2; 3; 4; 4.4$, with asymptote $\pi = 3.141 592 7$; in the drawing the unit on the ordinate axis is 1.2 cm, so that $\pi \mapsto 3.77$ cm and $\pi/2 \mapsto 1.885$ cm, and the unit on the abscissa is 1.2 cm. For small u the curve is indistinguishable from the line $\Omega = 2u = \omega T$; higher up it saturates at π , that is at the Nyquist frequency: the bilinear transform compresses the whole real axis of continuous frequencies into a single Nyquist band, without aliasing but with deformation.

globally in an essential way. Φ respects moduli exactly and destroys injectivity (aliasing, Remark 21.9); β is injective and preserves the stability boundary exactly, but deforms frequencies (Corollary 21.5). Numerically, with $T = 10^{-3}$:

s	e^{sT}	$ e^{sT} $	$\beta(s)$
-100	0.904 837 4	0.904 837 4	0.904 761 9
-1,000	0.367 879 4	0.367 879 4	0.333 333 3
$1,000 i$	$0.540 302 3 + 0.841 471 0 i$	1	$0.6 + 0.8 i$
+500	1.648 721 3	1.648 721 3	1.666 666 7

Both columns of moduli respect the criterion: modulus < 1 for $\operatorname{Re} s < 0$, $= 1$ on the axis, > 1 on the right. The value $\beta(1,000 i) = 0.6 + 0.8i$ is exact (it is the point of Figure 21.2), and $|0.6 + 0.8i| = 1$ because $0.36 + 0.64 = 1$. ✓

Remark 21.11: Beyond Tustin: the generalized bilinear transform. Tustin’s bilinear transform of Corollary 3.3 (Volume II, Chapter 3) is not the only way to discretize a continuous system: it is the particular case $\alpha = 1/2$ of a one-parameter family, the **generalized bilinear transform** (GBT), which also includes as extreme cases the forward Euler method ($\alpha = 0$) and the backward Euler method ($\alpha = 1$). The choice of α governs a well-known trade-off in digital control between frequency accuracy and the numerical stability margin of the discretized system, and is still the subject of active research: a 2025 paper by a Chinese research group (Chen, Li, Cui et al., reference *unverified*: see the warning below) explicitly addresses the problem of choosing α optimally, showing that the physical interpretation of the parameter and its optimal selection had remained, surprisingly, little explored in the literature until recent times. It is a good reminder that the elementary geometry of the circle and digital control are neither closed topics nor far apart.

Warning. An inherited, unverified reference. The reference to the 2025 paper cited in Remark 21.11 comes from Volume II, where it was given as “cited in the bibliography”; the check made for this chapter shows that that entry does *not* appear in the bibliography of Volume III, and it was not possible to verify its authors, title or venue. The reference is therefore kept but marked as unverified, and remains an open point of the chapter. No statement, computation or proof of this chapter relies on it: the GBT family is defined and verified directly in Exercise 21.2, whose three particular cases are computed in full in the solution.

The three planes in one line.

$$\underbrace{s = \sigma + i\omega}_{\text{Laplace}} \xrightarrow{z=e^{sT}} \underbrace{z \in \mathbb{C}}_{\text{zeta}} \xrightarrow{z=e^{i\Omega}} \underbrace{\Omega \in \mathbb{R}}_{\text{Fourier}}$$

$$\operatorname{Re}(s) < 0 \longleftrightarrow |z| < 1, \quad \operatorname{Re}(s) = 0 \longleftrightarrow |z| = 1,$$

$$\Omega = \omega T \text{ (exact sampling)}, \quad \Omega = 2 \arctan \frac{\omega T}{2} \text{ (bilinear)}.$$

21.7. Digital filter design: from continuous to discrete

We now put to work together the Laplace transform, the Z-transform and the (generalized) bilinear transform of Remark 21.11, to solve a problem that arises in virtually every automation project: filtering the noise of an analog sensor (temperature, pressure, level, a 4–20 mA signal) before using it in a PID control loop.

From analog filter to digital filter in four steps Step 1. Specify the filter in the continuous domain: for a first-order Butterworth low-pass with cutoff frequency $\omega_c = 2\pi f_c$, $H(s) = \frac{\omega_c}{s + \omega_c}$ (unity DC gain, -3 dB at ω_c).

Step 2. Apply the bilinear transform (Corollary 3.3 of Volume II, Chapter 3; here Definition 21.7 and Theorem 21.14, case $\alpha = 1/2$ of the family of Remark 21.11): $s = \frac{2}{T} \cdot \frac{z-1}{z+1}$, where $T = 1/f_s$ is the sampling period.

Step 3. Obtain $H(z)$ and rewrite it in the form $H(z) = b_0 \frac{1+z^{-1}}{1-pz^{-1}}$, yielding the difference equation $y[n] = py[n-1] + b_0(x[n] + x[n-1])$, immediately implementable in a PLC scan cycle.

Step 4. Verify stability ($|p| < 1$, Theorem 21.12) and the frequency response of the resulting discrete filter. $\square$

Worked example — anti-noise filter for a 4–20 mA sensor. A PLC samples a level sensor at $f_s = 1000$ Hz ($T = 1$ ms, a typical fast scan time) and must attenuate electrical noise above $f_c = 10$ Hz (the useful level dynamics are much slower). Applying Step 2 to $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi \cdot 10 \approx 62.83$ rad/s and $K = 2/T = 2000$:

$$H(z) = \frac{\omega_c}{\frac{2}{T} \cdot \frac{z-1}{z+1} + \omega_c} = \frac{\omega_c(z+1)}{(K + \omega_c)z + (\omega_c - K)}.$$

Dividing numerator and denominator by $(K + \omega_c)$ yields the standard form with

$$b_0 = \frac{\omega_c}{K + \omega_c} \approx \frac{62.83}{2062.83} \approx 0.03046, \quad p = \frac{K - \omega_c}{K + \omega_c} \approx \frac{1937.17}{2062.83} \approx 0.93908.$$

Since $|p| \approx 0.939 < 1$, the filter is BIBO stable (Theorem 21.12). The final difference equation is

$$y[n] = 0.93908 y[n-1] + 0.03046(x[n] + x[n-1]).$$

Direct numerical checks: the DC gain is $H(1) = 1$ exact (no offset introduced); at the cutoff frequency, $|H(e^{i\omega_c T})| \approx 0.707$ (−3.01 dB, as expected by construction); and at typical industrial electrical noise frequencies the filter attenuates strongly: −14.2 dB at 50 Hz, −20.3 dB at 100 Hz, −27.3 dB at 200 Hz (a factor of ≈ 23 in amplitude).

Remark 21.12: The filter’s numbers, recomputed twice. The values of the worked example were recomputed by two independent routes.

First route — closed formula. With $\omega_c = 62.831\,853$ and $K = 2,000$: $b_0 = 0.030\,459\,03$ and $p = 0.939\,081\,94$; and $|H(e^{i2\pi f T})|$ with $H(z) = b_0 \frac{1+z^{-1}}{1-pz^{-1}}$ equals

f (Hz)	0	10	50	100	200
$ H $	1.000 000 0	0.706 990 4	0.194 561 9	0.096 239 5	0.043 199 9
dB	0.00	−3.01	−14.22	−20.33	−27.29

Second route — simulation of the difference equation. Running $y[n] = p y[n-1] + b_0(x[n] + x[n-1])$ over 400,000 samples with input $x[n] = \cos(2\pi f T n)$ and measuring the steady-state amplitude gives 0.706 6 at 10 Hz, 0.193 2 at 50 Hz, 0.094 0 at 100 Hz and 0.041 6 at 200 Hz: the two routes agree within the residual transient and discretization of the maximum (< 0.003 in amplitude, < 0.33 dB). ✓ In particular the attenuation at 200 Hz is a factor $1/0.043\,1999 = 23.15$ in amplitude, consistent with the “ ≈ 23 ” of the worked example. ✓

Remark 21.13: A warning: frequency warping. The bilinear transform is not an exact correspondence between continuous and discrete frequencies: the true relation is $\omega_d = \frac{2}{T} \arctan(\omega_c T/2)$ (frequency warping), not $\omega_d = \omega_c T$. For $f_c \ll f_s$ as in the previous example (10 Hz versus 1000 Hz) the effect is negligible (the numerical check $|H(e^{i\omega_c T})| \approx 0.707$ confirms it), but for cutoffs near the Nyquist frequency the designer must pre-warp the analog specification, that is design $H(s)$ not for ω_c but for $\omega'_c = \frac{2}{T} \tan(\omega_c T/2)$, so that after the warping of the bilinear transform the cutoff falls exactly where desired.

The formula used here is proved in Corollary 21.5: with $\Omega = \omega_d T$ the relation $\Omega = 2 \arctan(\omega_c T/2)$ becomes precisely $\omega_d = \frac{2}{T} \arctan(\omega_c T/2)$, and its inverse is the pre-warping rule $\omega'_c = \frac{2}{T} \tan(\omega_c T/2)$. The numerical check confirms the estimate on the example: for $f_c = 10$ Hz and $f_s = 1000$ Hz one has $\omega_c T = 0.062\,831\,9$ and $\Omega = 0.062\,811\,2$, a relative gap of $3.3 \cdot 10^{-4}$; for $f_c = 200$ Hz at the same sampling frequency one instead has $\omega_c T = 1.256\,637\,1$ against $\Omega = 1.121\,964\,2$, a relative gap of 11%, and without pre-warping the cutoff would fall at 178.566 2 Hz instead of 200 Hz. ✓ (The full computation is Worked exercise 21.9.)

Listing 21.1: Digital low-pass filter (first-order Butterworth via bilinear) in SCL for PLC.

```
FUNCTION_BLOCK FB_LOWPASS1
// First-order IIR low-pass filter, obtained via bilinear transform
// from H(s) = wc/(s+wc). Coefficients calculated offline for given fc,
// fs.
VAR_INPUT
    x          : REAL;           // input sample (e.g. scaled 4-20mA
                                // sensor)
    b0         : REAL := 0.03046; // input gain (depends on fc, fs)
    p          : REAL := 0.93908; // filter pole (depends on fc, fs)
END_VAR
VAR_OUTPUT
    y          : REAL;           // filtered sample
```

```

END_VAR
VAR
    xPrev    : REAL := 0.0;           // x[n-1]
    yPrev    : REAL := 0.0;           // y[n-1]
END_VAR
BEGIN
    y := p * yPrev + b0 * (x + xPrev);
    xPrev := x;
    yPrev := y;
END_FUNCTION_BLOCK

```

Remark 21.14: Implementation note for a PLC. Unlike the `FB_FFT8` listing (Volume II, Chapter 4, §11), this block must be called at every scan cycle (typically in a cyclic OB), not once on a buffer: this is how a recursive IIR filter works in industrial practice, updating a single sample at a time with minimal memory (only two registers, `xPrev` and `yPrev`) — regardless of how long the process has been running. It is precisely this memory efficiency, a consequence of the recursiveness of the difference equation, that makes IIR filters more common than FIR filters in controllers with limited resources.

21.8. Exercises inherited from Volume II

The exercises below come from Volume II, Chapter 4, and concern the topics of this chapter. They are reproduced in full, together with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 21.1. A digital filter has transfer function $H(z) = \frac{z+1}{z^2 - 0.3z - 0.4}$. Determine whether the filter is BIBO stable.

Exercise 21.2. The one-parameter generalized bilinear transform can be written (standard parametrization in the digital control literature) as

$$s \approx \frac{1}{T} \cdot \frac{1 - z^{-1}}{\alpha + (1 - \alpha)z^{-1}}, \quad \alpha \in [0, 1].$$

Verify that $\alpha = 0$ yields the forward Euler method ($s \approx (z - 1)/T$), $\alpha = 1$ the backward Euler method ($s \approx (z - 1)/(zT)$), and $\alpha = 1/2$ Tustin's method ($s \approx \frac{2}{T} \cdot \frac{z-1}{z+1}$) of Corollary 3.3 (Volume II, Chapter 3).

Exercise 21.3. (Filter design) — A sensor is sampled at $f_s = 200$ Hz and a first-order Butterworth low-pass with cutoff $f_c = 5$ Hz is desired. Compute b_0 and p following the four steps of Section 21.7 and verify the BIBO stability of the resulting filter.

Solutions

Solution. We factor the denominator: we look for p, q with $p + q = 0.3 \dots$ more precisely, if $z^2 - 0.3z - 0.4 = (z - p)(z - q)$ then $p + q = 0.3$ and $pq = -0.4$. Trying $p = 0.8, q = -0.5$: $p + q = 0.3 \checkmark, pq = 0.8 \times (-0.5) = -0.4 \checkmark$. Hence

$$H(z) = \frac{z + 1}{(z - 0.8)(z + 0.5)},$$

with poles at $z = 0.8$ and $z = -0.5$. Both satisfy $|z| < 1$ ($|0.8| = 0.8 < 1, |-0.5| = 0.5 < 1$): by Theorem 21.12, the filter is BIBO stable.

Independent check. The quadratic formula gives $z = \frac{0.3 \pm \sqrt{0.09 + 1.6}}{2} = \frac{0.3 \pm 1.3}{2}$, that is 0.8 and -0.5 : the same poles, obtained without trial and error. $\checkmark$

Solution. For $\alpha = 0$: $s \approx \frac{1}{T} \cdot \frac{1 - z^{-1}}{0 + 1 \cdot z^{-1}} = \frac{1}{T} \cdot \frac{z - 1}{1} \cdot \frac{1}{z^{-1} \cdot z}$; more simply, multiplying numerator and denominator by z : $s \approx \frac{1}{T} \cdot \frac{z - 1}{1} = \frac{z - 1}{T}$, the forward Euler method (it approximates $\dot{x}(t) \approx (x[n + 1] - x[n])/T$).

For $\alpha = 1$: $s \approx \frac{1}{T} \cdot \frac{1 - z^{-1}}{1 + 0} = \frac{1 - z^{-1}}{T}$; multiplying numerator and denominator by z : $s \approx \frac{z - 1}{zT}$, the backward Euler method (it approximates $\dot{x}(t) \approx (x[n] - x[n - 1])/T$).

For $\alpha = 1/2$: $s \approx \frac{1}{T} \cdot \frac{1 - z^{-1}}{\frac{1}{2} + \frac{1}{2}z^{-1}} = \frac{2}{T} \cdot \frac{1 - z^{-1}}{1 + z^{-1}}$; multiplying numerator and denominator by z : $s \approx \frac{2}{T} \cdot \frac{z - 1}{z + 1}$, exactly Tustin's method of Corollary 3.3 (Volume II, Chapter 3). The three classical discretization methods are therefore particular cases of a single one-parameter family, as anticipated in Remark 21.11.

Solution. With $f_s = 200$ Hz, $T = 1/f_s = 0.005$ s, $f_c = 5$ Hz: $\omega_c = 2\pi f_c \approx 31.416$ rad/s, $K = 2/T = 400$. Following Section 21.7:

$$b_0 = \frac{\omega_c}{K + \omega_c} = \frac{31.416}{431.416} \approx 0.07282, \quad p = \frac{K - \omega_c}{K + \omega_c} = \frac{368.584}{431.416} \approx 0.85436.$$

Since $|p| \approx 0.854 < 1$, the filter is BIBO stable (Theorem 21.12). Check of the DC gain: $H(1) = b_0(1 + 1)/(1 - p) = 2b_0/(1 - p) \approx 0.14564/0.14564 = 1$, as expected for a normalized low-pass filter.

Independent check. The double-precision computation gives $\omega_c = 31.4159265$, $b_0 = 0.07282051$, $p = 0.85435899$ and $H(1) = 2b_0/(1 - p) = 1.000000000$: the three values confirm the rounding to five decimals in the statement. $\checkmark$

Inherited quiz

The items below come from the quiz of Volume II, Chapter 4, and are renumbered here.

1. A system is BIBO stable in the Z-domain if all its poles satisfy

- A) $|z| > 1$
- B) $|z| < 1$
- C) $|z| = 1$

D) z real

BEST PRACTICE

Answer key.

1-B: Theorem 21.12: poles inside the unit circle.

21.9. Assigned exercises

These are *new* exercises, distinct both from the inherited ones (Section 21.8) and from those worked out in Section 21.10. The answers are in the Teacher's key (Section 21.12).

Exercise 21.4: Annuli: computing them, recognising empty ones, deciding on the boundary. *Goal:* compute ρ_+ and ρ_- with Theorem 21.1 and understand what happens on the two boundary circles. *Estimated time:* 40 minutes. *Type:* proof-based/numerical.

- (a) Let $x[n] = 2^{-n}u[n] - 3^{-n}u[-n-1]$. Compute ρ_+ and ρ_- , write $X(z)$ in closed form by summing the two geometric series separately, and say whether $\mathbb{T}$ belongs to the annulus. If so, compute $\|x\|_1$.
- (b) Let $y[n] = 3^n u[n] + 2^n u[-n-1]$. Show that $\text{ROC}(y) = \emptyset$, and explain in one sentence why the same rational expression obtained "formally" has no meaning.
- (c) Let $w[n] = 1/n^2$ for $n \geq 1$ and $w[n] = 0$ elsewhere. Show that $\rho_+ = 1$ and that, unlike the case of $u[n]$, the circle $|z| = 1$ *belongs* to $\text{ROC}(w)$. Compute $\hat{w}(0)$.
- (d) Construct a signal whose region of convergence is exactly the closed annulus $\{\frac{1}{2} \leq |z| \leq 2\}$, boundaries included, and justify the answer.

Exercise 21.5: The five properties at work. *Goal:* use Theorem 21.5 to compute transforms that would not be summed by hand. *Estimated time:* 45 minutes. *Type:* symbolic and numerical computation.

- (a) Applying differentiation in z (Theorem 21.5(iv)) twice to row (2) of the table in Example 21.3, prove that

$$\mathcal{Z}\{n^2 a^n u[n]\}(z) = \frac{az(z+a)}{(z-a)^3}, \quad |z| > |a|,$$

and verify it numerically for $a = \frac{1}{2}$ and $z = 2$.

- (b) Let $x[n] = a^n u[n]$ and $y[n] = b^n u[n]$ with $a \neq b$ both nonzero. Compute $(x * y)[n]$ directly as a finite sum and recover the same result by inverse-transforming $X(z)Y(z)$ with Lemma 21.3. Verify the first eight values for $a = 0.6$ and $b = 0.4$.
- (c) Exhibit a pair x, y for which the inclusion $\text{ROC}(x * y) \supseteq \text{ROC}(x) \cap \text{ROC}(y)$ of Theorem 21.5(v) is *strict*, different from the one in Remark 21.4, and explain which cancellation causes it.
- (d) Prove, using (ii) and (iii), that $\mathcal{Z}\{a^n x[n - n_0]\}(z) = a^{n_0} z^{-n_0} X(z/a)$, and verify it on $x = \delta$.

Exercise 21.6: Stability: three systems, three diagnoses. *Goal:* decide BIBO stability in three equivalent ways (poles, annulus, summability) and understand that "causal" and "stable" are independent. *Estimated time:* 45 minutes. *Type:* proof-based/numerical.

- (a) Let $H(z) = \frac{z^2}{z^2 - 1.1z + 0.3}$, a causal system. Find the poles, decide stability with Theorem 21.12, compute $h[n]$ in closed form and verify $\|h\|_1 = H(1)$, explaining why the two quantities coincide here.

- (b) Let $H(z) = \frac{z}{z^2 - 2.5z + 1}$. Find the poles and the *two* possible annuli. Show that the causal choice gives an unstable system and that the other one gives a BIBO stable but non-causal system; compute $h[n]$ in both cases and $\|h\|_1$ in the stable case.
- (c) Using the four steps of Section 21.7, design the first-order low-pass filter with $f_s = 500$ Hz and $f_c = 20$ Hz: compute b_0 and p , verify $H(1) = 1$ and $|p| < 1$, and then compute $|H(e^{i\omega_c T})|$, explaining why it is *not* exactly $1/\sqrt{2}$.
- (d) For the system of (b) in its stable version, verify numerically Parseval's identity (Theorem 21.9) by comparing $\sum_n |h[n]|^2$ with $\frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{i\Omega})|^2 d\Omega$.

21.10. Worked exercises

Exercise 21.7: One rational function, three annuli, three signals. *Type:* inverse transform with choice of annulus, with numerical check. Invert

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$$

in all possible annuli, and say which of the resulting signals is summable.

Solution. Strategy. The poles are at $\frac{1}{2}$ and 2: two distinct moduli, hence three possible annuli (Theorem 21.1). One decomposes into partial fractions in the variable $w = z^{-1}$ and inverts each term with Example 21.2, choosing for each pole the causal or anticausal version according to the annulus.

Step 1 — partial fractions. Setting $w = z^{-1}$, one looks for A, B with

$$\frac{1}{(1 - \frac{w}{2})(1 - 2w)} = \frac{A}{1 - \frac{w}{2}} + \frac{B}{1 - 2w}, \quad \text{i.e.} \quad A(1 - 2w) + B(1 - \frac{w}{2}) = 1.$$

Evaluating at $w = 2$: $A(1 - 4) = 1$, hence $A = -\frac{1}{3}$. Evaluating at $w = \frac{1}{2}$: $B(1 - \frac{1}{4}) = 1$, hence $B = \frac{4}{3}$. *Check:* the constant term is $A + B = -\frac{1}{3} + \frac{4}{3} = 1$ ✓ and the coefficient of w is $-2A - \frac{B}{2} = \frac{2}{3} - \frac{2}{3} = 0$ ✓. Hence

$$X(z) = -\frac{1}{3} \cdot \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{4}{3} \cdot \frac{1}{1 - 2z^{-1}}.$$

Step 2 — outer annulus $|z| > 2$. Both poles lie inside the inner circle of the annulus: both terms must be taken in the causal version (Example 21.2, outer annulus):

$$x_a[n] = \left(-\frac{1}{3} \left(\frac{1}{2} \right)^n + \frac{4}{3} 2^n \right) u[n].$$

Values: $x_a[0] = -\frac{1}{3} + \frac{4}{3} = 1$, $x_a[1] = -\frac{1}{6} + \frac{8}{3} = 2.5$, $x_a[2] = -\frac{1}{12} + \frac{16}{3} = 5.25$, $x_a[3] = 10.625$.

Step 3 — intermediate annulus $\frac{1}{2} < |z| < 2$. The pole $\frac{1}{2}$ lies *inside* the inner boundary of the annulus, hence must be taken causal; the pole 2 lies *outside* the outer boundary, hence must be taken anticausal:

$$x_b[n] = -\frac{1}{3} \left(\frac{1}{2} \right)^n u[n] - \frac{4}{3} 2^n u[-n - 1].$$

Values: $x_b[0] = -\frac{1}{3}$, $x_b[1] = -\frac{1}{6}$, $x_b[-1] = -\frac{4}{3} \cdot \frac{1}{2} = -\frac{2}{3}$, $x_b[-2] = -\frac{4}{3} \cdot \frac{1}{4} = -\frac{1}{3}$.

Step 4 — inner annulus $|z| < \frac{1}{2}$. Both poles lie outside: both anticausal,

$$x_c[n] = \left(\frac{1}{3} \left(\frac{1}{2} \right)^n - \frac{4}{3} 2^n \right) u[-n - 1].$$

Values: $x_c[-1] = \frac{1}{3} \cdot 2 - \frac{4}{3} \cdot \frac{1}{2} = \frac{2}{3} - \frac{2}{3} = 0$, $x_c[-2] = \frac{1}{3} \cdot 4 - \frac{4}{3} \cdot \frac{1}{4} = \frac{4}{3} - \frac{1}{3} = 1$.

Step 5 — which one is summable. Only the intermediate annulus contains $\mathbb{T}$; by Theorem 21.6 the only summable signal is x_b , and

$$\|x_b\|_1 = \frac{1}{3} \sum_{n \geq 0} 2^{-n} + \frac{4}{3} \sum_{m \geq 1} 2^{-m} = \frac{1}{3} \cdot 2 + \frac{4}{3} \cdot 1 = \frac{2}{3} + \frac{4}{3} = 2.$$

Independent check. Evaluating the inversion formula of Theorem 21.3 with 200,000 points on the circle of radius r :

r	$n = -2$	$n = -1$	$n = 0$	$n = 1$	$n = 2$	$n = 3$
3	0	0	1	2.5	5.25	10.625
1	-0.333 333 3	-0.666 666 7	-0.333 333 3	-0.166 666 7	-0.083 333 3	-0.041 666 7
$\frac{1}{4}$	1	0	0	0	0	0

Each row matches the values computed in Steps 2–4, with imaginary part of order 10^{-16} and maximum discrepancy $2 \cdot 10^{-15}$. ✓

Interpretation. The same algebraic formula describes a signal that explodes in the future, one that decays in both directions and one that explodes in the past. It is the annulus, not the formula, that decides which one.

Typical errors. (1) Inverting "row by row" always taking the causal version: one gets x_a even when the annulus is the intermediate one. (2) Forgetting that the anticausal version carries a minus sign (Example 21.2), obtaining $+\frac{4}{3}2^n u[-n-1]$. (3) Concluding that "the system is stable because one pole is inside the unit circle": stability requires $\mathbb{T}$ to lie in the annulus, and the annulus is chosen by the problem, not by the pole.

Extension. With p poles of distinct moduli there are $p+1$ annuli, and as many inverse-transformed signals; only one of them is causal (the one for the outermost annulus), only one is anticausal (the one for the innermost annulus) and only one, if it exists, is summable: the one for the annulus that contains $\mathbb{T}$.

Exercise 21.8: A two-sided signal: annulus, DTFT, inversion, Parseval. *Type:* complete computation with three independent checks. Let

$$x[n] = \left(\frac{1}{2}\right)^n u[n] - 2^n u[-n-1].$$

Determine its annulus, the closed-form transform, the norm $\|x\|_1$; verify the DTFT inversion formula and Parseval's identity.

Solution. Strategy. The two geometric series are summed separately (Definition 21.4), Theorem 21.1 is applied for the radii, and Theorems 21.8 and 21.9 are used for the checks.

Step 1 — the causal part. $\rho_+ = \limsup_n (2^{-n})^{1/n} = \frac{1}{2}$ and, for $|z| > \frac{1}{2}$,

$$X_+(z) = \sum_{n \geq 0} \left(\frac{1}{2}\right)^n z^{-n} = \frac{1}{1 - \frac{1}{2z}} = \frac{z}{z - \frac{1}{2}}.$$

Step 2 — the anticausal part. Here $x[-m] = -2^{-m}$, hence $\limsup_m |x[-m]|^{1/m} = \frac{1}{2}$ and $\rho_- = 2$; for $|z| < 2$,

$$X_-(z) = \sum_{m \geq 1} (-2^{-m}) z^m = -\frac{z/2}{1 - z/2} = -\frac{z}{2 - z} = \frac{z}{z - 2}.$$

Step 3 — annulus and closed form. The annulus is $A(\frac{1}{2}, 2)$, which contains $\mathbb{T}$ (this is Figure 21.1), and

$$X(z) = \frac{z}{z - \frac{1}{2}} + \frac{z}{z - 2} = \frac{z(z - 2) + z(z - \frac{1}{2})}{(z - \frac{1}{2})(z - 2)} = \frac{2z^2 - \frac{5}{2}z}{z^2 - \frac{5}{2}z + 1}.$$

Check: $X(1) = \frac{2-2.5}{1-2.5+1} = \frac{-0.5}{-0.5} = 1$, and on the other hand $\hat{x}(0) = \sum_n x[n] = \sum_{n \geq 0} 2^{-n} - \sum_{m \geq 1} 2^{-m} = 2 - 1 = 1$. ✓

Step 4 — summability. $\|x\|_1 = \sum_{n \geq 0} 2^{-n} + \sum_{m \geq 1} 2^{-m} = 2 + 1 = 3$, finite: by Theorem 21.6 the DTFT exists for every Ω and is $\hat{x}(\Omega) = X(e^{i\Omega})$, continuous and bounded by 3 (Theorem 21.7).

Step 5 — Parseval in closed form.

$$\sum_{n \in \mathbb{Z}} |x[n]|^2 = \sum_{n \geq 0} 4^{-n} + \sum_{m \geq 1} 4^{-m} = \frac{1}{1 - \frac{1}{4}} + \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{4}{3} + \frac{1}{3} = \frac{5}{3} = 1.666\,666\,7.$$

Independent check 1 (norm). The numerical sum of $|x[n]|$ over $-60 \leq n \leq 60$ equals 3.000 000 000 0. ✓

Independent check 2 (inversion). Evaluating $\frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\Omega) e^{in\Omega} d\Omega$ with 100,000 points gives

n	-2	-1	0	1	2
numerical	-0.250 000 00	-0.500 000 00	1.000 000 00	0.500 000 00	0.250 000 00
$x[n]$	$-\frac{1}{4}$	$-\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{4}$

with imaginary part of order 10^{-17} . ✓

Independent check 3 (numerical Parseval). $\frac{1}{2\pi} \int_{-\pi}^{\pi} |\hat{x}(\Omega)|^2 d\Omega = 1.666\,666\,666\,7$, against the $\frac{5}{3}$ of Step 5. ✓

Interpretation. The signal equals $2^{-|n|}$ for $n \geq 0$ and $-2^{-|n|}$ for $n < 0$: it is odd away from the origin, where it equals 1. Consequently

$$\hat{x}(\Omega) = x[0] + \sum_{n \geq 1} 2^{-n} (e^{-in\Omega} - e^{in\Omega}) = 1 - 2i \sum_{n \geq 1} 2^{-n} \sin(n\Omega) = 1 - \frac{2i \sin \Omega}{\frac{5}{2} - 2 \cos \Omega},$$

having used $\sum_{n \geq 1} r^n \sin(n\Omega) = \frac{r \sin \Omega}{1 - 2r \cos \Omega + r^2}$ with $r = \frac{1}{2}$: the real part of $\hat{x}$ is *constantly* 1, and all the information lies in the imaginary part. This can also be read from the closed form of Step 3: with $z = e^{i\Omega}$, $\hat{x} = \frac{2e^{i\Omega} - \frac{5}{2}}{2 \cos \Omega - \frac{5}{2}}$, whose denominator is real and whose numerator

has real part equal to the denominator itself. *Check:* then $|\hat{x}|^2 = 1 + 4(\sum_n 2^{-n} \sin n\Omega)^2$ and, by orthogonality of the sines, $\frac{1}{2\pi} \int_{-\pi}^{\pi} |\hat{x}|^2 = 1 + 4 \cdot \frac{1}{2} \sum_{n \geq 1} 4^{-n} = 1 + \frac{2}{3} = \frac{5}{3}$: this recovers Step 5 by a fourth route. ✓ The two poles, $\frac{1}{2}$ and 2, are symmetric with respect to $\mathbb{T}$ under the inversion $z \mapsto 1/\bar{z}$, and this geometric symmetry is the counterpart of the signal's oddness.

Typical errors. (1) Summing the anticausal part as $\sum_{n \leq -1} (-2^n)z^{-n}$ without changing index, getting the direction of the inequality on the annulus wrong. (2) Deducing from $X(z) = \frac{2z^2 - 2.5z}{z^2 - 2.5z + 1}$ that the system "has a pole outside the unit circle, hence is unstable": the annulus is the intermediate one and $\mathbb{T}$ belongs to it. (3) Applying Parseval to a non-summable signal without the Riesz–Fischer theorem (Remark 21.7).

Extension. The same computation with the *even* signal $x_p[n] = r^{|n|}$, $0 < r < 1$, gives $X_p(z) = \frac{z}{z-r} - \frac{z}{z-1/r}$ with annulus $A(r, 1/r)$: the anticausal part changes sign compared with the odd case treated above, but the moduli $|x[n]|$ are the same, hence $\|x_p\|_1 = 1 + \frac{2r}{1-r} = \frac{1+r}{1-r}$ and $\sum_n |x_p[n]|^2 = 1 + \frac{2r^2}{1-r^2} = \frac{1+r^2}{1-r^2}$; for $r = \frac{1}{2}$ one recovers 3 and $\frac{5}{3}$. ✓ This time $\hat{x}_p$ is *real*, because the signal is even.

Exercise 21.9: Pre-warping: designing at 200 Hz while sampling at 1 kHz. *Type:* design with exact verification. Design a first-order low-pass filter with cutoff at $f_c = 200$ Hz for a system sampled at $f_s = 1000$ Hz, first without and then with pre-warping, and compare the two responses at the cutoff frequency.

Solution. Strategy. The required cutoff is at one fifth of the sampling frequency, i.e. at $\Omega_c = \omega_c T = 2\pi \cdot 200/1000 = 1.256\,637\,1$ radians per sample: far from the regime $\Omega \ll 1$ in which warping is negligible. Corollary 21.5 says *exactly* how much to correct.

Step 1 — the data. $T = 10^{-3}$ s, $K = 2/T = 2,000$, $\omega_c = 2\pi \cdot 200 = 1,256.637\,1$ rad/s, $\Omega_c = 1.256\,637\,1$.

Step 2 — naive design (without pre-warping). With $b_0 = \frac{\omega_c}{K+\omega_c}$ and $p = \frac{K-\omega_c}{K+\omega_c}$:

$$b_0 = \frac{1,256.637\,1}{3,256.637\,1} = 0.385\,869\,5, \quad p = \frac{743.362\,9}{3,256.637\,1} = 0.228\,260\,9.$$

Evaluating $H(z) = b_0 \frac{1+z^{-1}}{1-pz^{-1}}$ at $z = e^{i\Omega_c}$ gives $|H| = 0.654\,126\,6$, i.e. $-3.686\,8$ dB: *not* the required -3 dB. The actual cutoff (where $|H| = 1/\sqrt{2}$) falls at $178.566\,2$ Hz.

Step 3 — pre-warping. By Corollary 21.5 the correspondence between the analog pulsation ω and the normalized one Ω is $\Omega = 2 \arctan(\omega T/2)$, hence to make the cutoff fall exactly at Ω_c one must design the analog filter with

$$\omega'_c = \frac{2}{T} \tan \frac{\Omega_c}{2} = 2,000 \cdot \tan(0.628\,318\,5) = 2,000 \cdot 0.726\,542\,5 = 1,453.085\,1 \text{ rad/s.}$$

Step 4 — the corrected coefficients.

$$b_0 = \frac{\omega'_c}{K + \omega'_c} = \frac{1,453.085\,1}{3,453.085\,1} = 0.420\,807\,8, \quad p = \frac{K - \omega'_c}{K + \omega'_c} = \frac{546.914\,9}{3,453.085\,1} = 0.158\,384\,4.$$

Since $|p| < 1$, the filter is BIBO stable (Theorem 21.12). The difference equation is $y[n] = 0.158\,384\,4 y[n-1] + 0.420\,807\,8(x[n] + x[n-1])$.

Independent check 1 (the specification is centred). Evaluating $|H(e^{i\Omega_c})|$ with the coefficients of Step 4 gives $0.707\,106\,781\,2$, against $1/\sqrt{2} = 0.707\,106\,781\,2$: they agree to ten decimal places. ✓

Independent check 2 (the naive cutoff, by an analytic route). The actual cutoff of Step 2 is obtained without numerical search: the bilinear filter designed for ω_c has -3 dB where $\frac{2}{T} \tan(\Omega/2) = \omega_c$, i.e. at $\Omega = 2 \arctan(\omega_c T/2)$, i.e. at the frequency

$$f_{\text{eff}} = \frac{\Omega}{2\pi T} = \frac{\arctan(\omega_c T/2)}{\pi T} = \frac{\arctan(0.628\,318\,5)}{\pi \cdot 10^{-3}} = 178.566\,2 \text{ Hz,}$$

the same value found by bisection. ✓

Independent check 3 (DC gain). In both designs $H(1) = \frac{2b_0}{1-p}$ equals 1 exactly: with the coefficients of Step 4, $\frac{2 \cdot 0.420\,807\,8}{1-0.158\,384\,4} = \frac{0.841\,615\,6}{0.841\,615\,6} = 1$. ✓

Interpretation. Pre-warping does not "improve" the filter: it shifts the analog specification so that the distortion of the bilinear map brings it back where it is needed. The price is that the shape of the response far from the cutoff remains distorted — the bilinear map compresses all of $\mathbb{R}$ into $(-\pi, \pi)$, and no choice of ω'_c can cancel this compression (Figure 21.3).

Typical errors. (1) Pre-distorting with $\arctan$ instead of $\tan$: one gets $\omega'_c = 1,121.964\,2$ and the cutoff ends up even lower. (2) Applying pre-warping even when $f_c \ll f_s$, where it is useless but harmless (for $f_c = 10$ Hz and $f_s = 1$ kHz it shifts the cutoff by $3.3 \cdot 10^{-4}$ in relative terms). (3) Forgetting that the DC gain does not change: $H(1) = 1$ remains true, because $z = 1$ corresponds to $s = 0$ and $\beta^{-1}(1) = 0$ (Theorem 21.14).

Extension. For a band-pass filter *both* band edges must be pre-distorted, and the resulting bandwidth is not the pre-distorted value of the bandwidth: it is the difference of the pre-distorted edges. This is the most frequent error in designing IIR filters close to the Nyquist frequency.

21.11. Self-assessment quiz

1. (Definition) The region of convergence of a signal x is, by Definition 21.4,
 - A) the set of z for which $x[n]z^{-n} \rightarrow 0$
 - B) the set of $z \neq 0$ for which $\sum_{n \in \mathbb{Z}} |x[n]| |z|^{-n} < \infty$
 - C) the set of z at which X is holomorphic
 - D) the set of z other than the poles of X
2. (Hypothesis) In Theorem 21.6 the hypothesis $\mathbb{T} \subseteq \text{ROC}(x)$ serves to
 - A) make X rational
 - B) guarantee that x is causal
 - C) guarantee $\sum_n |x[n]| < \infty$, i.e. that the series $\sum_n x[n]e^{-in\Omega}$ converges absolutely for every Ω
 - D) make $\hat{x}$ periodic
3. (Symbolic computation) For $|z| > |a|$, $\mathcal{Z}\{n a^n u[n]\}(z)$ equals
 - A) $\frac{z}{(z-a)^2}$
 - B) $\frac{az}{(z-a)^2}$
 - C) $\frac{az}{(z-a)^3}$
 - D) $-\frac{az}{(z-a)^2}$
4. (Numerical computation) For $h[n] = (3 \cdot 0.6^n - 2 \cdot 0.4^n)u[n]$, the norm $\|h\|_1 = \sum_n |h[n]|$ equals, to seven decimal places,
 - A) 1.0000000
 - B) 4.1666667
 - C) 3.0349311
 - D) 5.0000000
5. (Proof step) In the proof of Theorem 21.10 one uses the input x_N , defined by $x_N[n] = \frac{h[-n]}{|h[-n]|}$ for $|n| \leq N$ and by $x_N[n] = 0$ elsewhere. It serves to
 - A) make the system causal
 - B) produce, with an input of amplitude ≤ 1 , the output $\sum_{|k| \leq N} |h[k]|$ at $n = 0$, which bounds the partial sums
 - C) show that h is real
 - D) construct an unbounded input that makes the output diverge
6. (Error diagnosis) A student inverts $X(z) = [(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})]^{-1}$ in the annulus $\frac{1}{2} < |z| < 2$ by taking both fractions in the causal version, obtaining $(-\frac{1}{3}(\frac{1}{2})^n + \frac{4}{3}2^n)u[n]$. The error is
 - A) the coefficients $-\frac{1}{3}$ and $\frac{4}{3}$ are wrong
 - B) the pole at $z = 2$ lies *outside* the chosen annulus and must be inverted in the anticausal version, $-\frac{4}{3}2^n u[-n-1]$
 - C) X has no inverse transform in that annulus
 - D) the annulus $\frac{1}{2} < |z| < 2$ is empty
7. (Transfer) The *causal* system with $H(z) = \frac{z}{z^2 - 2.5z + 1}$ is
 - A) BIBO stable, because one pole has modulus $\frac{1}{2} < 1$
 - B) BIBO stable, because $H(1)$ is finite
 - C) not BIBO stable, because the pole $z = 2$ has modulus ≥ 1 and the causal annulus $\{|z| > 2\}$ does not contain $\mathbb{T}$
 - D) marginally stable, because one pole lies on the unit circle
8. (Source and metacognition) The only result that this chapter declares *imported* is the completeness of $L^2(\mathbb{T})$ (the Riesz–Fischer theorem), recalled in Remark 21.7; it is found in
 - A) Rudin, *Real and Complex Analysis*, 3rd ed., Ch. 3 " L^p -spaces"
 - B) Oppenheim–Schafer, *Discrete-Time Signal Processing*, 3rd ed., Ch. 3
 - C) Oppenheim–Willsky–Nawab, *Signals & Systems*, 2nd ed., Ch. 10
 - D) Mitra, *Digital Signal Processing*, 4th ed.

21.12. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Definition 21.4: it requires *absolute* summability, which makes the sum independent of the order. *Distractors:* A is the necessary condition (term tending to zero), not sufficient; C holds only in the interior of the annulus and does not define the ROC, which concerns the series and not the function; D is false because X can be holomorphic at points where the series does not converge (for example the same rational expression $\frac{z}{z-a}$ is defined on all of $\mathbb{C} \setminus \{a\}$, but the series converges only on one of the two annuli). *Diagnosed error:* confusing the function with the series that represents it. *Source:* Oppenheim–Schafer, Ch. 3; here Definition 21.4 and Remark 21.3. *Prerequisite:* absolute convergence of a series. *Remediation:* Example 21.2.

2) **C.** *Rationale:* Step 1 and Step 2 of the proof of Theorem 21.6 show that $\mathbb{T} \subseteq \text{ROC}(x)$ is equivalent to $x \in \ell^1$, and it is the latter that makes the DTFT series converge. *Distractors:* A is beside the point (no assumption of rationality is made); B is false, the signal of Exercise 21.8 is two-sided and has $\mathbb{T}$ in its annulus; D confuses an automatic consequence (periodicity holds by definition, as soon as the series converges, Proposition 21.2) with the role of the hypothesis. *Diagnosed error:* failing to identify the technical role of a hypothesis. *Source:* Oppenheim–Willsky–Nawab, Ch. 10; here Theorem 21.6. *Prerequisite:* Definition 21.4. *Remediation:* verify that for $x = u$ the circle $\mathbb{T}$ does not lie in the annulus $\{|z| > 1\}$ and that indeed $\sum_n |u[n]| = \infty$.

3) **B.** *Rationale:* this is row (4) of Example 21.3, obtained by applying $-z \frac{d}{dz}$ to $\frac{z}{z-a}$ (Theorem 21.5(iv)). *Distractors:* A forgets the factor a produced by the derivative; C is the transform of $n^2 a^n u[n]$ up to the factor $(z+a)$ (Exercise 21.5(a)); D gets the sign wrong, forgetting the minus in front of z . *Diagnosed error:* applying differentiation in z without the factor $-z$. *Source:* Mitra, *Digital Signal Processing*, 4th ed.; here Example 21.3(4). *Prerequisite:* term-by-term differentiation of a power series. *Remediation:* numerical check with $a = 0.8$, $z = 1.25$ (the value is 4.9382716).

4) **B.** *Rationale:* $h[n] > 0$ for every $n \geq 0$, hence $\|h\|_1 = \sum_n h[n] = H(1) = \frac{1}{0.24} = \frac{25}{6} = 4.1666667$ (Example 21.5). *Distractors:* A is $h[0]$; C is $\sum_n |h[n]|^2$, i.e. the energy, not the ℓ^1 norm; D is $\|h\|_1$ of the *other* system, the one in Exercise 21.6(a). *Diagnosed error:* confusing the ℓ^1 norm with the ℓ^2 energy. *Source:* here Example 21.5, with three independent checks. *Prerequisite:* geometric series. *Remediation:* recompute $\frac{3}{1-0.6} - \frac{2}{1-0.4}$ and compare with $H(1)$.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **B.** *Rationale:* this is Step 2 and Step 3 of the proof of Theorem 21.10: x_N has $\|x_N\|_\infty \leq 1$ and realizes the equality $(\mathcal{T}x_N)[0] = \sum_{|k| \leq N} |h[k]|$, so that BIBO stability bounds all partial sums. *Distractors:* A is irrelevant (x_N is not causal, and there is no need for it to be); C is false, h can be complex and the computation uses precisely the conjugate; D contradicts the construction, which produces a *bounded* input. *Diagnosed error:* failing to recognise the strategy "an input that saturates the triangle inequality". *Source:* Oppenheim–Schafer, Ch. 2; here Theorem 21.10. *Prerequisite:* $w\bar{w} = |w|^2$. *Remediation:* redo Step 3 with h real, where $\bar{h[k]}/|h[k]| = \text{sgn } h[k]$.

6) **B.** *Rationale:* in the annulus $\frac{1}{2} < |z| < 2$ the pole 2 is external, hence its term must be expanded in positive powers of z (Example 21.2, inner annulus), giving $-\frac{4}{3}2^n u[-n-1]$; the correct answer is the one from Step 3 of Exercise 21.7. *Distractors:* A is false, the coefficients are checked in Step 1 of that exercise; C and D are both false, the annulus

$A(\frac{1}{2}, 2)$ is nonempty and in it the inverse transform exists and is unique (Theorem 21.4). *Diagnosed error:* inverting "row by row" always in the causal version. *Source:* Oppenheim–Willsky–Nawab, Ch. 10; here Exercise 21.7. *Prerequisite:* Theorem 21.1. *Remediation:* redo the three cases with $X(z) = \frac{z}{z-3}$.

7) **C.** *Rationale:* the roots of $z^2 - 2.5z + 1$ are 2 and $\frac{1}{2}$; for a causal system the annulus is $\{|z| > 2\}$ (Theorem 21.11), which does not contain $\mathbb{T}$, hence the system is not BIBO stable (Corollary 21.4). *Distractors:* A looks at only one pole and ignores the other; B confuses finiteness of H at a point with convergence of the series at that point ($H(1) = -2$ is finite, but $1 \notin \text{ROC}$); D is false, no pole lies on $\mathbb{T}$. *Diagnosed error:* assessing stability pole by pole instead of by the pole of maximum modulus. *Source:* here Theorem 21.12 and Exercise 21.6(b). *Prerequisite:* the quadratic formula. *Remediation:* Example 21.4.

8) **A.** *Rationale:* Remark 21.7 declares the completeness of $L^2(\mathbb{T})$ imported and refers to Rudin, *Real and Complex Analysis*, 3rd ed., Chapter 3 (" L^p -spaces"). *Distractors:* B, C and D are the three signal-theory texts used in the chapter, which present the DTFT but do not prove a measure-theory theorem. *Diagnosed error:* not distinguishing the source of an analytic result from that of an applied result. *Source:* the `fonti` environment of this chapter. *Prerequisite:* knowing what the chapter proves and what it imports. *Remediation:* reread the prerequisites *infobox* and Remark 21.7.

BEST PRACTICE

Assigned exercises — answers.

Exercise 21.4. (a) $\rho_+ = \limsup_n (2^{-n})^{1/n} = \frac{1}{2}$; since $x[-m] = -3^{-m}$, $\limsup_m |x[-m]|^{1/m} = \frac{1}{3}$ and $\rho_- = 3$. Summing the two geometric series,

$$X(z) = \frac{z}{z - \frac{1}{2}} + \frac{z}{z - 3}, \quad \frac{1}{2} < |z| < 3,$$

and $\mathbb{T}$ belongs to it; $\|x\|_1 = \sum_{n \geq 0} 2^{-n} + \sum_{m \geq 1} 3^{-m} = 2 + \frac{1}{2} = \frac{5}{2}$. *Check:* the numerical sum over $|n| \leq 200$ gives 2.500 000 0, and the numerical inversion of the DTFT gives $x[-1] = -0.333\,333\,3$, $x[0] = 1$, $x[1] = 0.5$. $\checkmark$ (b) $\rho_+ = 3$ and $\rho_- = 2$, hence $\rho_- < \rho_+$ and by Theorem 21.1(i) the ROC is empty. The expression $\frac{z}{z-3} + \frac{z}{z-2}$ obtained "formally" is a perfectly well-defined rational function, but it is not the Z-transform of anything: there is no z at which the series converges (Example 21.1). (c) $(1/n^2)^{1/n} = n^{-2/n} = e^{-2 \ln n/n} \rightarrow 1$, hence $\rho_+ = 1$; and on $|z| = 1$ the series of moduli is $\sum_{n \geq 1} 1/n^2$, convergent: the circle belongs to the ROC, and $\text{ROC}(w) = \{|z| \geq 1\}$. $\hat{w}(0) = \sum_{n \geq 1} 1/n^2 = \pi^2/6 = 1.644\,934\,1$ (numerical check over $2 \cdot 10^6$ terms: 1.644 933 6, difference $5 \cdot 10^{-7}$, of the order of the tail $1/N$). $\checkmark$ For $u[n]$ instead $\rho_+ = 1$ but $\sum_n 1 = \infty$: the circle does not belong. (d) For instance $x[n] = 2^{-|n|}/n^2$ for $n \neq 0$ and $x[0] = 0$: one has $\rho_+ = \frac{1}{2}$ and $\rho_- = 2$, and on both boundaries the series of moduli reduces to $\sum 1/n^2 < \infty$; by Theorem 21.1 the ROC is the closed annulus.

Exercise 21.5. (a) From $\mathcal{Z}\{na^n u[n]\} = \frac{az}{(z-a)^2}$ one applies $-z \frac{d}{dz}$ again: $\frac{d}{dz} \frac{az}{(z-a)^2} = a \frac{(z-a)-2z}{(z-a)^3} = -a \frac{z+a}{(z-a)^3}$, hence $\mathcal{Z}\{n^2 a^n u[n]\} = \frac{az(z+a)}{(z-a)^3}$. For $a = \frac{1}{2}$, $z = 2$: the formula gives $\frac{0.5 \cdot 2 \cdot 2.5}{1.5^3} = \frac{2.5}{3.375} = 0.740\,740\,740\,7$ and the series, over 20,000 terms, 0.740 740 740 7. $\checkmark$ (b) $(x * y)[n] = \sum_{k=0}^n a^k b^{n-k} = b^n \sum_{k=0}^n (a/b)^k = \frac{a^{n+1} - b^{n+1}}{a-b}$ for $n \geq 0$, 0 elsewhere. By partial fractions, $X(z)Y(z) = \frac{z^2}{(z-a)(z-b)}$ and $\frac{XY}{z} = \frac{z}{(z-a)(z-b)}$ has residues $\frac{a}{a-b}$ and $\frac{b}{b-a}$, whence $XY = \frac{a}{a-b} \frac{z}{z-a} - \frac{b}{a-b} \frac{z}{z-b}$ and the same formula. For $a = 0.6$, $b = 0.4$ the first eight values are 1, 1, 0.76, 0.52, 0.337 6, 0.212 8, 0.131 776, 0.080 704, by both routes. $\checkmark$ (These are the same values as in Example 21.5: indeed $\frac{0.6^{n+1} - 0.4^{n+1}}{0.2} = 3 \cdot 0.6^n - 2 \cdot 0.4^n$.) (c) For instance $x = u$, with $X(z) = \frac{z}{z-1}$ and annulus $\{|z| > 1\}$, and $y = \delta - \sigma_1 \delta$, with

$Y(z) = 1 - z^{-1} = \frac{z-1}{z}$ and annulus $\mathbb{C} \setminus \{0\}$: the product equals 1, annulus $\mathbb{C} \setminus \{0\}$, while the intersection of the annuli is $\{|z| > 1\}$. The cancellation is that of the zero of Y at $z = 1$ with the pole of X at $z = 1$; and indeed $(x * y)[n] = u[n] - u[n-1] = \delta[n]$. (d) By (iii), $\mathcal{Z}\{a^n(\sigma_{n_0}x)[n]\}(z) = (\sigma_{n_0}X)(z/a)$ where $(\sigma_{n_0}X)$ denotes the transform of the shifted signal, i.e., by (ii), $\mathcal{Z}\{x[n-n_0]\}(w) = w^{-n_0}X(w)$; evaluating at $w = z/a$ gives $(z/a)^{-n_0}X(z/a) = a^{n_0}z^{-n_0}X(z/a)$. For $x = \delta$: $X \equiv 1$, and on the left $a^n\delta[n-n_0] = a^{n_0}\delta[n-n_0]$, whose transform is $a^{n_0}z^{-n_0}$. ✓

Exercise 21.6. (a) $z^2 - 1.1z + 0.3$ has discriminant $1.21 - 1.2 = 0.01$ and roots $\frac{1.1 \pm 0.1}{2}$, i.e. 0.6 and 0.5: both of modulus < 1 , hence the system is BIBO stable (Theorem 21.12). From $\frac{H(z)}{z} = \frac{z}{(z-0.6)(z-0.5)}$ with residues $\frac{0.6}{0.1} = 6$ and $\frac{0.5}{-0.1} = -5$ it follows that $h[n] = (6 \cdot 0.6^n - 5 \cdot 0.5^n)u[n]$, positive for every n (the ratio $\frac{6 \cdot 0.6^n}{5 \cdot 0.5^n} = 1.2 \cdot 1.2^n \geq 1.2$). Hence $\|h\|_1 = \sum_n h[n] = H(1) = \frac{1}{1-1.1+0.3} = \frac{1}{0.2} = 5$, which coincides with $\frac{6}{0.4} - \frac{5}{0.5} = 15 - 10 = 5$: the two quantities coincide *because* $h \geq 0$, i.e. because $|h[n]| = h[n]$. Check: the recursion $y[n] = 1.1y[n-1] - 0.3y[n-2] + x[n]$ gives 1, 1.1, 0.91, 0.671, 0.4651, identical to the closed form (maximum discrepancy $4.4 \cdot 10^{-16}$ over 300 samples), and the numerical sum of the moduli equals 5.000 000 00. ✓ (b) $z^2 - 2.5z + 1$ has discriminant $6.25 - 4 = 2.25$ and roots $\frac{2.5 \pm 1.5}{2}$, i.e. 2 and $\frac{1}{2}$: the possible annuli are $\{|z| > 2\}$ (causal), $A(\frac{1}{2}, 2)$ and $\{|z| < \frac{1}{2}\}$ (anticausal). From $\frac{H(z)}{z} = \frac{1}{(z-2)(z-\frac{1}{2})}$ with residues $\frac{1}{1.5} = \frac{2}{3}$ and $\frac{1}{-1.5} = -\frac{2}{3}$ it follows that $H(z) = \frac{2}{3} \frac{z}{z-2} - \frac{2}{3} \frac{z}{z-\frac{1}{2}}$. In the causal annulus $h[n] = (\frac{2}{3}2^n - \frac{2}{3}2^{-n})u[n]$, not summable (2^n diverges): unstable system. In the intermediate annulus, which contains $\mathbb{T}$,

$$h[n] = -\frac{2}{3}2^nu[-n-1] - \frac{2}{3}2^{-n}u[n],$$

summable with $\|h\|_1 = \frac{2}{3}(\sum_{m \geq 1} 2^{-m} + \sum_{n \geq 0} 2^{-n}) = \frac{2}{3}(1+2) = 2$: BIBO stable but *non-causal* system. Check: $\sum_n h[n] = H(1) = \frac{1}{1-2.5+1} = -2$, and since $h \leq 0$ one has $\|h\|_1 = 2$; numerical inversion gives $h[-2] = -0.166\,666\,67$, $h[-1] = -0.333\,333\,33$, $h[0] = -0.666\,666\,67$, $h[1] = -0.333\,333\,33$, $h[2] = -0.166\,666\,67$. ✓ (c) $T = 0.002$ s, $\omega_c = 2\pi \cdot 20 = 125.663\,706$, $K = 1,000$, hence $b_0 = \frac{125.663\,706}{1,125.663\,706} = 0.111\,635\,21$ and $p = \frac{874.336\,294}{1,125.663\,706} = 0.776\,729\,58$; $|p| < 1$, stable filter, and $H(1) = \frac{2b_0}{1-p} = 1.000\,000\,000\,0$. ✓ Finally $|H(e^{i\omega_c T})| = 0.705\,236\,41$, not $1/\sqrt{2} = 0.707\,106\,78$: the difference is the warping of Corollary 21.5, because $\Omega_c = \omega_c T = 0.251\,327\,4$ is not negligible; pre-warping (Exercise 21.9) would correct it exactly. (d) $\sum_n |h[n]|^2 = \frac{4}{9}(\sum_{m \geq 1} 4^{-m} + \sum_{n \geq 0} 4^{-n}) = \frac{4}{9}(\frac{1}{3} + \frac{4}{3}) = \frac{20}{27} = 0.740\,740\,740\,7$; and $\frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{i\Omega})|^2 d\Omega$, evaluated with 200,000 points, equals 0.740 740 740 7: the two routes coincide, as Theorem 21.9 requires. ✓

21.13. Formulary of the chapter

Definitions and annulus.

$$X(z) = \sum_{n \in \mathbb{Z}} x[n]z^{-n}, \quad \text{ROC}(x) = \left\{ z \neq 0 : \sum_n |x[n]| |z|^{-n} < \infty \right\}.$$

$$\rho_+ = \limsup_{n \rightarrow +\infty} |x[n]|^{1/n}, \quad \rho_- = \left(\limsup_{m \rightarrow +\infty} |x[-m]|^{1/m} \right)^{-1}, \quad A(\rho_+, \rho_-) \subseteq \text{ROC}(x) \subseteq \overline{A}.$$

Causal $\Rightarrow \rho_- = \infty$ (exterior of a circle); anticausal $\Rightarrow \rho_+ = 0$ (disk); $\rho_- < \rho_+ \Rightarrow$ empty annulus.

Inversion, uniqueness, holomorphy.

$$x[n] = \frac{1}{2\pi i} \oint_{|z|=r} X(z) z^{n-1} dz = \frac{1}{2\pi} \int_0^{2\pi} X(re^{i\theta}) r^n e^{in\theta} d\theta, \quad \rho_+ < r < \rho_-.$$

X is holomorphic on $A(\rho_+, \rho_-)$ and $\sum_n x[n]z^{-n}$ is its Laurent series; expression *and* annulus determine x , the expression alone does not:

$$\frac{z}{z-a} \longleftrightarrow \begin{cases} a^n u[n], & |z| > |a|, \\ -a^n u[-n-1], & |z| < |a|. \end{cases}$$

The five properties.

$$\alpha x + \beta y \mapsto \alpha X + \beta Y; \quad x[n-n_0] \mapsto z^{-n_0} X(z); \quad a^n x[n] \mapsto X(z/a);$$

$$n x[n] \mapsto -z X'(z); \quad (x * y)[n] = \sum_k x[k]y[n-k] \mapsto X(z)Y(z),$$

with $\text{ROC} \supseteq \mathcal{R}_x \cap \mathcal{R}_y$ in cases (i) and (v), $\mathcal{R}$ unchanged in (ii) and (iv), $|a|\mathcal{R}_x$ in (iii).

Table of notable Z-transforms (causal signals).

$$\delta[n] \mapsto 1 \ (\mathbb{C} \setminus \{0\}); \quad u[n] \mapsto \frac{z}{z-1} \ (|z| > 1); \quad a^n u[n] \mapsto \frac{z}{z-a} \ (|z| > |a|);$$

$$n a^n u[n] \mapsto \frac{az}{(z-a)^2}; \quad n^2 a^n u[n] \mapsto \frac{az(z+a)}{(z-a)^3}; \quad \frac{1}{(z-p)^j} \mapsto \binom{n-1}{j-1} p^{n-j} u[n-j];$$

$$\cos(\Omega_0 n) u[n] \mapsto \frac{z^2 - z \cos \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}, \quad \sin(\Omega_0 n) u[n] \mapsto \frac{z \sin \Omega_0}{z^2 - 2z \cos \Omega_0 + 1} \quad (|z| > 1).$$

DTFT.

$$\hat{x}(\Omega) = \sum_n x[n] e^{-in\Omega} = X(e^{i\Omega}), \quad \hat{x}(\Omega + 2\pi) = \hat{x}(\Omega), \quad \|\hat{x}\|_\infty \leq \|x\|_1;$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\Omega) e^{in\Omega} d\Omega; \quad \sum_n x[n] \overline{y[n]} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x} \bar{\hat{y}} d\Omega.$$

$$\mathbb{T} \subseteq \text{ROC}(x) \iff x \in \ell^1(\mathbb{Z}).$$

Stability.

$$\text{BIBO} \iff \|h\|_1 = \sum_n |h[n]| < \infty \iff \mathbb{T} \subseteq \text{ROC}(h),$$

$$\text{and in the causal rational case} \quad \text{BIBO} \iff \max_\ell |p_\ell| < 1,$$

with optimal constant $M = \|h\|_1$ and, for a causal rational system, $\text{ROC}(h) = \{|z| > \max_\ell |p_\ell|\}$ exactly.

The three planes, sampling and the bilinear map.

$$z = e^{sT} : \quad |e^{sT}| = e^{T \text{Re } s}, \quad e^{sT} = e^{s'T} \iff s - s' \in \frac{2\pi i}{T} \mathbb{Z} \text{ (aliasing);}$$

$$\beta(s) = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s}, \quad s = \frac{2}{T} \cdot \frac{z-1}{z+1}, \quad |\beta(s)|^2 - 1 = \frac{2T \text{Re}(s)}{|1 - \frac{T}{2}s|^2},$$

$$\Omega = 2 \arctan \frac{\omega T}{2}, \quad \omega = \frac{2}{T} \tan \frac{\Omega}{2} \quad (\text{pre-warping}).$$

First-order bilinear low-pass filter, and the numbers of the chapter.

$$H(s) = \frac{\omega_c}{s + \omega_c} \longrightarrow H(z) = b_0 \frac{1 + z^{-1}}{1 - pz^{-1}}, \quad b_0 = \frac{\omega_c}{K + \omega_c}, \quad p = \frac{K - \omega_c}{K + \omega_c}, \quad K = \frac{2}{T},$$

$$y[n] = p y[n-1] + b_0(x[n] + x[n-1]), \quad H(1) = \frac{2b_0}{1-p} = 1.$$

f_s (Hz)	1,000	200	500	1,000 (pre-warp)
f_c (Hz)	10	5	20	200
b_0	0.030 459 0	0.072 820 5	0.111 635 2	0.420 807 8
p	0.939 081 9	0.854 359 0	0.776 729 6	0.158 384 4

Sources and bibliography

- [1] A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd edition, Pearson, Upper Saddle River (New Jersey), 2010, 1108 pages, ISBN 978-0-13-198842-2. Chapter 2, on discrete-time signals and systems (convolution, LTI systems, stability, the discrete-time Fourier transform: source of Definition 21.2, of Proposition 21.1, of Theorem 21.10 and of Definition 21.5); and Chapter 3, on the Z-transform (region of convergence, inversion, properties, rational systems: source of Theorem 21.1, of Theorem 21.3 and of Theorem 21.5). Bibliographic data (title, edition, publisher, year, pagination, ISBN) verified on the Open Library record for ISBN 9780131988422; the role of Chapters 2 and 3 as the site of the elementary treatment of the Z-transform and the DTFT is confirmed by the syllabus of MIT 6.341 *Discrete-Time Signal Processing* (MIT OpenCourseWare, Fall 2005), which lists them among the prerequisites — verified on 6 September 2026.
- [2] A. V. Oppenheim, A. S. Willsky, S. H. Nawab, *Signals & Systems*, 2nd edition, Prentice-Hall Signal Processing Series, Prentice Hall, Upper Saddle River (New Jersey), 1997, 957 pages, ISBN 0-13-814757-4. Chapter 10, *The z-Transform*, sections 10.1–10.10: definition of the transform, region of convergence and its properties, inversion, operational properties, rational transfer functions and stability of LTI systems. It is the primary source of the pedagogical approach of Section 21.2 and Section 21.5. Bibliographic data verified on the Open Library record for ISBN 0138147574; the assignment of sections 10.1–10.10 to the treatment of the Z-transform is verified on the *Readings* page of MIT 6.003 *Signals and Systems* (MIT OpenCourseWare, Fall 2011), which adopts this text (listed there as "2nd ed., Prentice Hall, 1996", the copyright year of the edition whose printing Open Library records as 1997) — verified on 6 September 2026.
- [3] S. K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th edition, McGraw-Hill, New York, 2011, 992 pages, ISBN 978-0-07-338049-0. It is the reference text for the implementation-oriented material: the tables of notable Z-transforms (Example 21.3 and Appendix C, Â§3) and the design of IIR filters by the bilinear transform, with and without pre-warping (Section 21.7 and Exercise 21.9). The bibliographic data were verified on the Open Library record for ISBN 9780073380490 — verified on 6 September 2026.
- [4] W. Rudin, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, New York, 1987, xiv+416 pages, ISBN 0-07-054234-1. Chapter 3, *L^p-spaces*: it is the source of the only result **imported** into this chapter, the completeness of $L^2(\mathbb{T})$ (the Riesz–Fischer theorem), declared in Remark 21.7 and needed to extend Theorem 21.9 from $\ell^1(\mathbb{Z})$ to $\ell^2(\mathbb{Z})$.

Bibliographic data verified on the Open Library record for ISBN 0070542341 — verified on 6 September 2026.

- [5] For the results of the treatise recalled with \ref: the lemma on the integral of powers on the circle, the Laurent series expansion theorem with uniqueness of the coefficients, the identity principle and the fundamental theorem of algebra are proved in Chapter 15; L^2 convergence and the completeness of the trigonometric system in Chapter 17; the covering $\mathbb{R} \rightarrow \mathbb{T}$ and the winding number in Chapter 12; Möbius transformations and polar form in Chapter 6; the Laplace transform, its region of convergence and its restriction to the imaginary axis in Chapter 20. Definition 21.3, Theorem 21.6, Theorem 21.12, Remark 21.8, Remark 21.11, Section 21.7 with the listing FB_LOWPASS1 and Section 21.8 come from Volume II, Chapter 4, §11 and §12, and from Appendix A, §3, and are reproduced in full according to the Executive Plan, §3.4, row 21.
- [6] **Open point.** The reference, inherited from Volume II and kept in Remark 21.11, to a 2025 work on the optimal choice of the parameter α of the generalized bilinear transform (attributed to "Chen, Li, Cui et al.") **has not been verified**: the entry does not appear in the bibliography of Volume III and it was not possible to ascertain its authors, title and venue. It is flagged as such in the warning that follows that remark and is not used in any statement, computation or proof in the chapter.

The discrete Fourier transform

A signal observed by a computer is a finite sequence of numbers: N samples, and nothing more. The discrete Fourier transform is the only change of basis that turns that finite vector into “a sum of sinusoids” with no approximation at all, because the N discrete exponentials $n \mapsto e^{-2\pi i kn/N}$ are exactly N pairwise orthogonal vectors of $\mathbb{C}^N$: not one more, not one less. The whole chapter follows from this single fact, which is the orthogonality of the N -th roots of unity already proved in Chapter 6 (Theorem 6.5), read again as a statement of linear algebra in $\mathbb{C}^N$.

The path is linear. We fix the convention (Definition 22.1) and prove the orthogonality of the discrete exponential vectors (Theorem 22.1), treating separately the case in which the denominator of the geometric sum vanishes (Lemma 22.1). From there comes the inversion formula (Theorem 22.2), proved twice — directly and by a matrix argument —, the unitarity of the normalized matrix (Theorem 22.3), its Vandermonde structure (Proposition 22.3) together with its determinant (Corollary 22.3), and the order-four periodicity $F_N^4 = I$ (Corollary 22.4) that confines the eigenvalues of the DFT to the four fourth roots of unity (Corollary 22.5). Next come circular convolution together with the discrete convolution theorem (Theorem 22.5), Parseval’s identity (Theorem 22.6), the conjugate symmetry of real signals (Theorem 22.7) and the diagonalization of circulant matrices (Theorem 22.8), which is the algebraic reason why a filter is applied by multiplying spectra. The final part is applied in nature and just as fully proved: the DFT as sampling of the DTFT (Theorem 22.9), zero-padding as spectral interpolation with the Dirichlet kernel (Theorem 22.11), leakage, and the rectangular, Hann and Hamming windows, for which the coherent gain (Proposition 22.6), the equivalent noise bandwidth (Proposition 22.7) and the main-lobe width (Proposition 22.8) are all computed in closed form.

The material that the Execution Plan moved here from Volume II — Section 8 of Chapter 3 and the first part of Section 13 of Chapter 4, with the butterfly network, the `FB_FFT8` block in SCL and the discussion of leakage, plus Appendix B “Eight-point DFT” — is preserved in full and is found in Sections 22.10, 22.10.3 and 22.12, followed by the inherited exercises, quiz and formulary (Section 22.13). What was there stated or used without proof — orthogonality in the degenerate case, inversion, Parseval, Hermitian symmetry, the role of windows — is here proved, and the eight-point example is worked out symbolically, inverted on all eight samples, and cross-checked with an executed script.

INFO

Level and prerequisites. Level **B–C**: the theory (Sections 22.1–22.9) requires only linear algebra in $\mathbb{C}^N$ and is level **B**; the part on windows and spectral analysis (Section 22.10.3) and the diagonalization of circulants are level **C**.

Internal prerequisites. n -th roots of unity and their orthogonality (Chapter 6, Definition 6.2, Theorem 6.2, Theorem 6.4, Theorem 6.5); the Zeta transform and the DTFT (Chapter 21, Definition 21.3, Theorem 21.6); Fourier series and the Gibbs phenomenon (Chapter 17). The characters of the cyclic group $\mathbb{Z}/N\mathbb{Z}$ and their orthogonality are treated in Volume II, Chapter 7: that chapter proves that the N maps $\chi_k(n) = e^{2\pi i kn/N}$ are exactly all the homomorphisms of $\mathbb{Z}/N\mathbb{Z}$ into $\mathbb{T}$, and that the orthogonality relation used here is the orthogonality

relation of the characters of a finite abelian group. In the present chapter that fact is not assumed: the one identity that is needed is proved again from scratch in Lemma 22.1.

External prerequisites, assumed and not reproved:

- linear algebra over $\mathbb{C}$: the Hermitian inner product, unitary matrices, the determinant and its elementary properties (multilinearity in the columns, invariance under adding a multiple of one column to another, Laplace expansion), the minimal polynomial and the criterion “annihilating polynomial with simple roots $\Rightarrow$ diagonalizability”;
- the sum of a geometric progression and its telescoping identity (recalled and used, in any case, in the form proved in Theorem 6.4);
- elementary trigonometry: Euler’s formulas, product-to-sum identities, and the values $\cos(\pi/4) = \sin(\pi/4) = \sqrt{2}/2$, $\cos(\pi/8)$, $\sin(\pi/8)$.

Everything else is proved here.

What is learned. Writing the DFT as a matrix-vector product and recognizing its Vandermonde structure; proving inversion in two independent ways; using unitarity to obtain Parseval without computation; computing a circular convolution both directly and via spectra; diagonalizing a circulant matrix and solving its linear system in $O(N \log N)$; predicting what a spectrum shows when the tone does not fall on a bin, and choosing the window accordingly, knowing what is gained and what is lost.

Warning. Conventions of the chapter.

- $N \geq 1$ is an integer, and we fix once and for all

$$\omega_N = e^{-2\pi i/N},$$

so that $\omega_N^{-1} = \overline{\omega_N} = e^{2\pi i/N} = \zeta_N$ is the primitive N -th root of unity of Chapter 6 (Definition 6.2). The negative sign in the exponent is the convention of signal engineering, and is the one used by Oppenheim and Schaffer and by Briggs and Henson.

- The indices n (time) and k (frequency) run over $\{0, 1, \dots, N-1\}$; whenever they appear outside this range they are meant **modulo** N , which is always legitimate by Proposition 22.1. The notation $\langle m \rangle_N$ denotes the remainder of m on division by N , i.e. the unique integer of $\{0, \dots, N-1\}$ congruent to m modulo N .
- The Hermitian inner product on $\mathbb{C}^N$ is $\langle u, v \rangle = \sum_{n=0}^{N-1} u_n \overline{v_n}$, linear in the first variable; $\|u\|^2 = \langle u, u \rangle$. For a matrix A , $A^* = \overline{A}^T$.
- In Sections 22.10 and 22.10.3, inherited from Volume II, the symbol ζ_N appears with the *local* convention $\zeta_N = e^{-2\pi i/N}$ declared in Theorem 22.13: there, ζ_N is thus our ω_N . The original text is preserved; this note only serves to avoid ambiguity with $\zeta_N = e^{+2\pi i/N}$ of Chapter 6.
- Numerical values are rounded to the fifth decimal place unless stated otherwise, and each one is obtained twice by independent routes: a closed formula and an executed script. The symbol $\checkmark$ marks agreement between the two routes.
- “Bin” is the current name for the index k read as a frequency: bin k corresponds to the frequency $k f_s / N$, where $f_s = 1/T$ is the sampling frequency.

22.1. Definition, convention, first properties

Definition 22.1: Discrete Fourier transform. Let $x = (x[0], x[1], \dots, x[N-1]) \in \mathbb{C}^N$. Its **discrete Fourier transform** (DFT) is the vector $X = (X[0], \dots, X[N-1]) \in \mathbb{C}^N$ defined by

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn} = \sum_{n=0}^{N-1} x[n] e^{-2\pi i kn/N}, \quad k = 0, 1, \dots, N-1,$$

where $\omega_N = e^{-2\pi i/N}$. One writes $X = \mathcal{F}_N x$, or $X = \text{DFT}_N\{x\}$.

Definition 22.2: Inverse discrete Fourier transform. Let $X \in \mathbb{C}^N$. Its **inverse discrete Fourier transform** (IDFT) is the vector with components

$$(\mathcal{F}_N^{-1} X)[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+2\pi i kn/N}, \quad n = 0, 1, \dots, N-1.$$

That this map is indeed the inverse of $\mathcal{F}_N$ is not part of the definition: it is the content of Theorem 22.2, and until that point the notation $\mathcal{F}_N^{-1}$ should be read as a mere name.

Definition 22.3: DFT matrix and normalized matrix. The **DFT matrix** and the **normalized matrix** of order N are the matrices $W_N, F_N \in \mathbb{C}^{N \times N}$ with entries

$$(W_N)_{kn} = \omega_N^{kn}, \quad (F_N)_{kn} = \frac{1}{\sqrt{N}} \omega_N^{kn}, \quad k, n = 0, \dots, N-1,$$

i.e. $F_N = N^{-1/2} W_N$. With this notation, Definition 22.1 reads as the matrix-vector product $X = W_N x$.

Remark 22.1: The competing conventions, and why the choice is not innocent. Three conventions circulate in the literature: the one adopted here (factor 1 forward, $1/N$ backward, sign $-$ in the forward exponent), used by Oppenheim and Schaffer and implemented by `numpy.fft`; the symmetric one (factor $1/\sqrt{N}$ both ways), which makes the transform unitary already at the level of the definition; and the one with sign $+$ in the forward exponent, common in crystallography. The three differ by multiplicative constants and by a conjugation, and every formula of the chapter changes accordingly: Parseval's identity, for instance, carries the factor $1/N$ in our convention (Theorem 22.6) and no factor at all in the symmetric one. Stating the convention before writing any formula is not pedantry: it is the first thing to check when two FFT implementations disagree.

Proposition 22.1: Periodicity of the two indices. Extend the formulas of Definition 22.1 and Definition 22.2 to every integer, setting $X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}$ for $k \in \mathbb{Z}$ and $\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-kn}$ for $n \in \mathbb{Z}$. Then

$$X[k+N] = X[k] \quad \text{for every } k \in \mathbb{Z}, \quad \tilde{x}[n+N] = \tilde{x}[n] \quad \text{for every } n \in \mathbb{Z}.$$

In particular $X[k] = X[\langle k \rangle_N]$: writing an index outside $\{0, \dots, N-1\}$ is always legitimate, and means writing it modulo N .

Proof. Since $\omega_N^N = e^{-2\pi i} = 1$, for every $n \in \mathbb{Z}$ one has $\omega_N^{(k+N)n} = \omega_N^{kn} \omega_N^{Nn} = \omega_N^{kn} (\omega_N^N)^n = \omega_N^{kn} \cdot 1^n = \omega_N^{kn}$. Substituting term by term in the sum defining $X[k+N]$ gives exactly the sum

defining $X[k]$, hence $X[k+N] = X[k]$. The computation for $\tilde{x}$ is identical with ω_N^{-1} in place of ω_N : $\omega_N^{-k(n+N)} = \omega_N^{-kn}(\omega_N^{-N})^k = \omega_N^{-kn}$, and the constant $1/N$ plays no role. Finally, given $k \in \mathbb{Z}$ there is $q \in \mathbb{Z}$ with $k = \langle k \rangle_N + qN$; iterating the equality just proved q times (forward if $q > 0$, backward if $q < 0$) gives $X[k] = X[\langle k \rangle_N]$. $\square$

Proposition 22.2: Linearity. *The map $\mathcal{F}_N : \mathbb{C}^N \rightarrow \mathbb{C}^N$ of Definition 22.1 is $\mathbb{C}$ -linear: for all $x, y \in \mathbb{C}^N$ and all $\alpha, \beta \in \mathbb{C}$, $\mathcal{F}_N(\alpha x + \beta y) = \alpha \mathcal{F}_N x + \beta \mathcal{F}_N y$.*

Proof. For each fixed k ,

$$\sum_{n=0}^{N-1} (\alpha x[n] + \beta y[n]) \omega_N^{kn} = \alpha \sum_{n=0}^{N-1} x[n] \omega_N^{kn} + \beta \sum_{n=0}^{N-1} y[n] \omega_N^{kn},$$

by distributivity of multiplication over addition and because a finite number of terms in $\mathbb{C}$ can be reordered. The left-hand side is $(\mathcal{F}_N(\alpha x + \beta y))[k]$, the right-hand side is $\alpha X[k] + \beta Y[k]$; since this holds for every $k = 0, \dots, N-1$, the two vectors coincide. Equivalently: $\mathcal{F}_N$ is multiplication by the matrix W_N of Definition 22.3, and matrix-vector multiplication is linear. $\square$

22.2. Orthogonality of the discrete exponential vectors

The whole theory rests on a single identity, the sum of a geometric progression whose ratio is a root of unity. The delicate point — the only one — is that the closed formula for the geometric sum *does not apply* when the ratio equals 1: that case must be isolated and summed by hand. This is done here explicitly.

Lemma 22.1: Sum of the powers of a root of unity. *Let $N \geq 1$ and $j \in \mathbb{Z}$. Then*

$$\sum_{n=0}^{N-1} \omega_N^{jn} = \begin{cases} N, & \text{if } N \mid j, \\ 0, & \text{if } N \nmid j. \end{cases}$$

Proof. Set $q = \omega_N^j = e^{-2\pi i j/N}$, so that $\sum_{n=0}^{N-1} \omega_N^{jn} = \sum_{n=0}^{N-1} q^n$. Two cases are distinguished, which are exactly the case of vanishing denominator and that of nonvanishing denominator in the geometric-sum formula.

Case 1: $N \mid j$ (vanishing denominator). Then $j/N \in \mathbb{Z}$, hence $q = e^{-2\pi i(j/N)} = 1$, because $e^{-2\pi i m} = 1$ for every integer m . The closed formula $\sum_{n=0}^{N-1} q^n = \frac{q^N - 1}{q - 1}$ **cannot be used**: the denominator $q - 1$ vanishes (and so does the numerator, so the quotient would be the indeterminate form $0/0$). One sums directly instead: all N terms equal $1^n = 1$, so

$$\sum_{n=0}^{N-1} q^n = \underbrace{1 + 1 + \dots + 1}_{N \text{ times}} = N.$$

Case 2: $N \nmid j$ (nonvanishing denominator). One first shows $q \neq 1$. Indeed $q = e^{-2\pi i j/N} = 1$ is equivalent to $-2\pi j/N \in 2\pi\mathbb{Z}$, i.e. to $j/N \in \mathbb{Z}$, i.e. to $N \mid j$: excluded by hypothesis. For every $q \in \mathbb{C}$ the telescoping identity already proved in Theorem 6.4 (Step 1) holds,

$$(q - 1) \sum_{n=0}^{N-1} q^n = q^N - 1,$$

and now, since $q - 1 \neq 0$, it is legitimate to divide: $\sum_{n=0}^{N-1} q^n = \frac{q^N - 1}{q - 1}$. Finally $q^N = (e^{-2\pi i j/N})^N = e^{-2\pi i j} = 1$ because j is an integer: the numerator vanishes, the denominator does not, and the sum equals 0. $\square$

Remark 22.2: The same statement, three times in the treatise. Lemma 22.1 is Theorem 6.5 of Chapter 6 written with $\omega_N = \zeta_N^{-1}$ in place of ζ_N : since $N \mid j$ if and only if $N \mid (-j)$, the two statements coincide term by term, and the proof has been repeated here only because the degenerate case $N \mid j$ is the most frequent source of error in an implementation. The same fact appears a third time in Volume II, Chapter 7, where it is the orthogonality relation of the characters of the cyclic group $\mathbb{Z}/N\mathbb{Z}$: the vectors $\chi_k = (\omega_N^{kn})_{n=0}^{N-1}$ are the restrictions to $\{0, \dots, N-1\}$ of the homomorphisms $\mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{T}$, and the sum of the lemma is the integral of the character χ_j over the group, which vanishes for every nontrivial character. Geometry (roots of unity), algebra (characters) and numerical analysis (the DFT) are here the same identity.

Theorem 22.1: Orthogonality of the discrete exponential vectors. For $k = 0, 1, \dots, N-1$ set

$$e_k = \left(\omega_N^{-kn} \right)_{n=0}^{N-1} = (1, e^{2\pi i k/N}, e^{4\pi i k/N}, \dots, e^{2\pi i k(N-1)/N}) \in \mathbb{C}^N.$$

Then, with respect to the Hermitian inner product $\langle u, v \rangle = \sum_n u_n \overline{v_n}$,

$$\langle e_k, e_\ell \rangle = N \delta_{k\ell} = \begin{cases} N, & k = \ell, \\ 0, & k \neq \ell, \end{cases} \quad k, \ell \in \{0, \dots, N-1\}.$$

In particular every e_k has norm $\|e_k\| = \sqrt{N}$, and the vectors $e_0, \dots, e_{N-1}$ are pairwise orthogonal.

Proof. Since $|\omega_N| = |e^{-2\pi i/N}| = 1$, for every integer m one has $\overline{\omega_N^m} = \omega_N^{-m}$. Hence

$$\langle e_k, e_\ell \rangle = \sum_{n=0}^{N-1} \omega_N^{-kn} \overline{\omega_N^{-\ell n}} = \sum_{n=0}^{N-1} \omega_N^{-kn} \omega_N^{\ell n} = \sum_{n=0}^{N-1} \omega_N^{(\ell-k)n}.$$

Lemma 22.1 is now applied with $j = \ell - k$. Since $k, \ell \in \{0, \dots, N-1\}$, one has $|\ell - k| \leq N-1 < N$, and hence $N \mid (\ell - k)$ if and only if $\ell - k = 0$, i.e. $\ell = k$. In the first case the sum equals N , in the second 0: this is the claim. The norm follows by setting $\ell = k$: $\|e_k\|^2 = N$, hence $\|e_k\| = \sqrt{N}$ (arithmetic root, $N > 0$). $\square$

Corollary 22.1: The exponential vectors are an orthogonal basis of $\mathbb{C}^N$. The vectors $e_0, \dots, e_{N-1}$ of Theorem 22.1 form an orthogonal basis of $\mathbb{C}^N$, and $(N^{-1/2}e_k)_{k=0}^{N-1}$ is an orthonormal basis of it. Consequently every $x \in \mathbb{C}^N$ is written in a unique way as $x = \sum_{k=0}^{N-1} c_k e_k$ with $c_k \in \mathbb{C}$.

Proof. Let $c_0, \dots, c_{N-1} \in \mathbb{C}$ satisfy $\sum_k c_k e_k = 0$. Fixing ℓ , take the Hermitian inner product of both sides with e_ℓ : by linearity in the first variable and by Theorem 22.1,

$$0 = \left\langle \sum_k c_k e_k, e_\ell \right\rangle = \sum_k c_k \langle e_k, e_\ell \rangle = c_\ell N.$$

Since $N \neq 0$ one gets $c_\ell = 0$ for every ℓ : the vectors are linearly independent. They are N independent vectors in a space of dimension N , hence a basis. Normalization is immediate since $\|N^{-1/2}e_k\| = N^{-1/2}\sqrt{N} = 1$, and orthogonality is unchanged under multiplication by nonzero scalars. Uniqueness of the coefficients is the uniqueness of the representation with respect to a basis. $\square$

22.3. Inversion, proved twice

Theorem 22.2: DFT inversion formula. Let $x \in \mathbb{C}^N$ and let $X = \mathcal{F}_N x$ be its DFT (Definition 22.1). Then, for every $n = 0, 1, \dots, N-1$,

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{2\pi i kn/N}.$$

In other words $\mathcal{F}_N$ is bijective and its inverse is the map of Definition 22.2.

Proof. The right-hand side is computed directly. Fixing $n \in \{0, \dots, N-1\}$, substitute the definition of $X[k]$, taking care to rename m the internal dummy index so as not to confuse it with n :

$$\frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] \omega_N^{km} \right) \omega_N^{-kn}.$$

The sums are finite (exactly N^2 terms), so exchanging the order of summation is legitimate without any convergence hypothesis: it is the associative and commutative property of addition in $\mathbb{C}$ applied to a finite number of terms. Collecting $x[m]$:

$$= \frac{1}{N} \sum_{m=0}^{N-1} x[m] \sum_{k=0}^{N-1} \omega_N^{k(m-n)}.$$

The inner sum is the one of Lemma 22.1 with $j = m - n$. Since $m, n \in \{0, \dots, N-1\}$ one has $|m - n| \leq N - 1 < N$, hence $N \mid (m - n)$ if and only if $m = n$; the inner sum equals N for $m = n$ and 0 for $m \neq n$. A single term survives:

$$\frac{1}{N} \sum_{m=0}^{N-1} x[m] \cdot N \delta_{mn} = \frac{1}{N} \cdot N x[n] = x[n],$$

which is the claim. It follows that $\mathcal{F}_N$ is injective (if $\mathcal{F}_N x = 0$, the formula just proved gives $x[n] = 0$ for every n) and, being an injective endomorphism of a finite-dimensional space, is bijective; the map of Definition 22.2 is a left inverse of it, hence its inverse. $\square$

Second proof: by a matrix argument. We work with the matrices of Definition 22.3. The (k, n) entry of W_N is ω_N^{kn} ; since $|\omega_N| = 1$, the (n, k) entry of $W_N^* = \overline{W_N}^T$ is $\overline{(W_N)_{kn}} = \overline{\omega_N^{kn}} = \omega_N^{-kn}$. The product $W_N^* W_N$ is now computed; its (n, m) entry is

$$(W_N^* W_N)_{nm} = \sum_{k=0}^{N-1} (W_N^*)_{nk} (W_N)_{km} = \sum_{k=0}^{N-1} \omega_N^{-kn} \omega_N^{km} = \sum_{k=0}^{N-1} \omega_N^{k(m-n)}.$$

By Lemma 22.1, with $|m - n| \leq N - 1 < N$, this sum equals N if $m = n$ and 0 otherwise; hence

$$W_N^* W_N = N I_N.$$

Since $\det(NI_N) = N^N \neq 0$, the matrix W_N is invertible and, multiplying the identity $W_N^* W_N = NI_N$ on the right by W_N^{-1} , one obtains

$$W_N^{-1} = \frac{1}{N} W_N^*.$$

Applying this equality to the vector $X = W_N x$ gives $x = W_N^{-1} X = \frac{1}{N} W_N^* X$, which componentwise is $x[n] = \frac{1}{N} \sum_k \omega_N^{-kn} X[k]$: the same formula. The matrix route has the extra advantage of delivering, as a by-product, the identity $W_N^* W_N = NI_N$, i.e. the unitarity of $F_N = N^{-1/2} W_N$ (Theorem 22.3). $\square$

Corollary 22.2: Uniqueness of the spectrum and reconstruction. *Two vectors of $\mathbb{C}^N$ with the same DFT coincide. Moreover, for every $x \in \mathbb{C}^N$ the expansion*

$$x = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e_k$$

holds, with e_k the vectors of Theorem 22.1: that is, $X[k]/N$ are the coefficients of x in the orthogonal basis (e_k) , and $X[k] = \langle x, e_k \rangle$.

Proof. If $\mathcal{F}_N x = \mathcal{F}_N y$ then $\mathcal{F}_N(x - y) = 0$ by Proposition 22.2, and Theorem 22.2 applied to $x - y$ gives $(x - y)[n] = 0$ for every n , i.e. $x = y$.

For the expansion: the n -th component of $\frac{1}{N} \sum_k X[k] e_k$ is $\frac{1}{N} \sum_k X[k] \omega_N^{-kn}$, which by Theorem 22.2 equals $x[n]$; holding for every n , the two vectors coincide. Finally

$$\langle x, e_k \rangle = \sum_{n=0}^{N-1} x[n] \overline{\omega_N^{-kn}} = \sum_{n=0}^{N-1} x[n] \omega_N^{kn} = X[k],$$

again because $|\omega_N| = 1$. Note the consistency with the formula for the coefficients with respect to an orthogonal basis: $x = \sum_k \frac{\langle x, e_k \rangle}{\|e_k\|^2} e_k = \sum_k \frac{X[k]}{N} e_k$. $\square$

22.4. The DFT matrix: Vandermonde, unitarity, determinant, eigenvalues

Proposition 22.3: W_N is a Vandermonde matrix. *Set $z_k = \omega_N^k$ for $k = 0, \dots, N-1$. Then the matrix W_N of Definition 22.3 is the Vandermonde matrix of the nodes $z_0, \dots, z_{N-1}$:*

$$W_N = (z_k^n)_{k,n=0}^{N-1} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & z_1 & z_1^2 & \cdots & z_1^{N-1} \\ 1 & z_2 & z_2^2 & \cdots & z_2^{N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_{N-1} & z_{N-1}^2 & \cdots & z_{N-1}^{N-1} \end{pmatrix},$$

and the nodes $z_0, \dots, z_{N-1}$ are pairwise distinct.

Proof. By definition $(W_N)_{kn} = \omega_N^{kn} = (\omega_N^k)^n = z_k^n$, which is the required form: the k -th row is the sequence of powers $0, 1, \dots, N-1$ of the node z_k . The nodes are distinct: if $z_k = z_\ell$ with $k, \ell \in \{0, \dots, N-1\}$ then $\omega_N^{k-\ell} = 1$, i.e. $e^{-2\pi i(k-\ell)/N} = 1$, i.e. $N \mid (k-\ell)$; but $|k-\ell| \leq N-1 < N$, hence $k = \ell$. (This is the same argument by which Theorem 6.2 counts

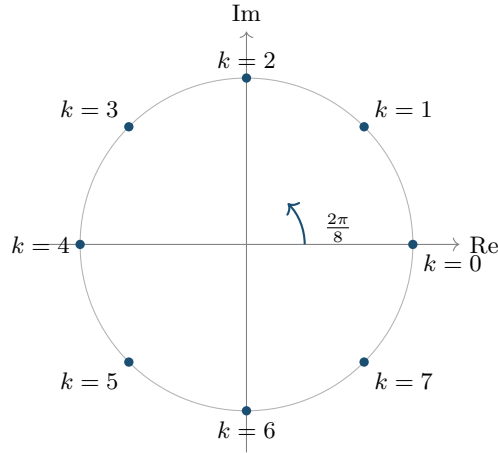


Figure 22.1: The points $\omega_8^{-k} = e^{2\pi i k/8}$, $k = 0, \dots, 7$, at which the eight-point DFT evaluates the polynomial $\sum_n x[n]z^{-n}$ (Theorem 22.12). The coordinates are $(\cos(2\pi k/8), \sin(2\pi k/8))$: $(\pm 1, 0)$, $(0, \pm 1)$, and the four points with coordinates $\pm\sqrt{2}/2 = \pm 0.70711$, computed by hand from $\cos(\pi/4) = \sin(\pi/4) = \sqrt{2}/2$ and cross-checked numerically.

exactly N N -th roots of unity.) □

Lemma 22.2: Vandermonde determinant. Let $z_0, \dots, z_{m-1} \in \mathbb{C}$ and $V = (z_k^n)_{k,n=0}^{m-1}$. Then

$$\det V = \prod_{0 \leq k < \ell \leq m-1} (z_\ell - z_k).$$

(For $m = 1$ the product is empty and equals 1.)

Proof. By induction on m .

Base case $m = 1$. $V = (1)$, $\det V = 1$, and the product over an empty set of pairs equals 1: the equality holds.

Inductive step. Let $m \geq 2$ and suppose the statement true for $m - 1$ nodes. Perform on V the sequence of elementary column operations

$$C_n \leftarrow C_n - z_0 C_{n-1}, \quad n = m-1, m-2, \dots, 1,$$

carried out in this order, i.e. starting from the last column. Each of them adds to one column a multiple of another column and hence *does not change the determinant*; moreover, being carried out from last to first, each operation uses a column C_{n-1} not yet modified. The new (k, n) entry, for $n \geq 1$, is

$$z_k^n - z_0 z_k^{n-1} = z_k^{n-1} (z_k - z_0),$$

while column $n = 0$ remains all 1's. Row $k = 0$ therefore becomes $(1, 0, 0, \dots, 0)$, because for $k = 0$ one has $z_0^{n-1}(z_0 - z_0) = 0$. Expanding the determinant along the first row (Laplace expansion) leaves only the minor obtained by deleting row 0 and column 0:

$$\det V = \det \left(z_k^{n-1} (z_k - z_0) \right)_{k=1, \dots, m-1; n=1, \dots, m-1}.$$

In this $(m-1) \times (m-1)$ matrix, row k has the common factor $(z_k - z_0)$; by multilinearity of the determinant in the rows it is factored out, one row at a time:

$$\det V = \left(\prod_{k=1}^{m-1} (z_k - z_0) \right) \cdot \det (z_k^{n-1})_{k,n=1}^{m-1}.$$

The last matrix is, up to the relabeling $k' = k - 1$, $n' = n - 1$, the Vandermonde matrix of the nodes $z_1, \dots, z_{m-1}$. By the inductive hypothesis its determinant equals $\prod_{1 \leq k < \ell \leq m-1} (z_\ell - z_k)$, and hence

$$\det V = \prod_{k=1}^{m-1} (z_k - z_0) \cdot \prod_{1 \leq k < \ell \leq m-1} (z_\ell - z_k) = \prod_{0 \leq k < \ell \leq m-1} (z_\ell - z_k),$$

because the first product collects exactly the pairs with $k = 0$ and the second all the others. The induction is complete. $\square$

Theorem 22.3: Unitarity of the normalized matrix. The matrix $F_N = N^{-1/2}W_N$ of Definition 22.3 is unitary:

$$F_N^* F_N = F_N F_N^* = I_N, \quad \text{equivalently} \quad W_N^* W_N = W_N W_N^* = N I_N.$$

In particular $F_N^{-1} = F_N^*$ and $\|F_N x\| = \|x\|$ for every $x \in \mathbb{C}^N$.

Proof. The identity $W_N^* W_N = N I_N$ was proved in the computation of the Matrix variant of the proof of Theorem 22.2: the (n, m) entry of the product is $\sum_k \omega_N^{k(m-n)}$, which equals $N \delta_{nm}$ by Lemma 22.1. Dividing by N gives $F_N^* F_N = I_N$, i.e. F_N^* is a left inverse of F_N . A square matrix with a left inverse is invertible and that left inverse is the inverse: indeed $F_N^* F_N = I$ implies $\det(F_N^*) \det(F_N) = 1$, hence $\det F_N \neq 0$, and multiplying $F_N^* F_N = I$ on the right by F_N^{-1} gives $F_N^* = F_N^{-1}$; consequently also $F_N F_N^* = I$.

Norm preservation follows: for every x ,

$$\|F_N x\|^2 = \langle F_N x, F_N x \rangle = (F_N x)^* (F_N x) = x^* F_N^* F_N x = x^* x = \|x\|^2,$$

and norms are nonnegative.

Note on symmetry. W_N is symmetric ($W_N^T = W_N$, because $\omega_N^{kn} = \omega_N^{nk}$), so $W_N^* = \overline{W_N}$ and the identity also reads $\overline{W_N} W_N = N I_N$. $\square$

Corollary 22.3: Modulus of the determinant. $|\det F_N| = 1$ and $|\det W_N| = N^{N/2}$. Moreover, in explicit form,

$$\det W_N = \prod_{0 \leq k < \ell \leq N-1} (\omega_N^\ell - \omega_N^k).$$

Proof. From unitarity (Theorem 22.3) one has $F_N^* F_N = I$, hence $\det(F_N^*) \det(F_N) = 1$. But $\det(F_N^*) = \overline{\det F_N}$ (the determinant of the transpose coincides with that of the matrix, and conjugation factors out of the determinant, being a sum of products of entries), hence $\overline{\det F_N} \cdot \det F_N = |\det F_N|^2 = 1$, i.e. $|\det F_N| = 1$. Since $W_N = \sqrt{N} F_N$ and $\det(\lambda A) = \lambda^N \det A$ for an $N \times N$ matrix,

$$|\det W_N| = (\sqrt{N})^N |\det F_N| = N^{N/2}.$$

The explicit form is Lemma 22.2 applied to the nodes $z_k = \omega_N^k$ of Proposition 22.3. $\square$

Example 22.1: The determinants for $N = 4$ and $N = 8$, computed two ways. Case $N = 4$. Here $\omega_4 = e^{-i\pi/2} = -i$, and the nodes are $z_0 = 1$, $z_1 = -i$, $z_2 = -1$, $z_3 = i$. Corollary 22.3 gives the product of the six differences $z_\ell - z_k$ with $k < \ell$:

$$\begin{aligned} & (z_1 - z_0)(z_2 - z_0)(z_3 - z_0)(z_2 - z_1)(z_3 - z_1)(z_3 - z_2) \\ &= (-i - 1)(-1 - 1)(i - 1)(-1 + i)(i + i)(i + 1). \end{aligned}$$

Group the terms: $(-i-1)(i-1) = -(1+i)(i-1) = -(i-1+i^2-i) = -(-2) = 2$; then $(-1-1) = -2$; then $(-1+i)(2i) = -2i+2i^2 = -2-2i$; finally $(i+1)$. The product is therefore

$$2 \cdot (-2) \cdot (-2-2i) \cdot (1+i) = (-4) \cdot (-2-2i)(1+i) = (-4) \cdot (-2-2i-2i-2i^2) = (-4)(-4i) = 16i.$$

The modulus is $16 = 4^{4/2} = 4^2 \checkmark$, in agreement with Corollary 22.3.

Case $N = 8$. The product of the $\binom{8}{2} = 28$ differences is not conveniently written by hand; Corollary 22.3 nonetheless predicts $|\det W_8| = 8^{8/2} = 8^4 = 4096$. The explicit computation (Vandermonde product and, entirely independently, LU factorization of the 8×8 matrix) gives

$$\det W_8 = -4096i,$$

whose modulus is $4096 \checkmark$.

Executed check. The script of Section 22.12.1 computes, for $N = 4$ and $N = 8$, the determinant of W_N with `numpy.linalg.det` and, separately, the Vandermonde product $\prod_{k < \ell} (\omega_N^\ell - \omega_N^k)$: the two values agree to within 10^{-11} , and equal $16i$ and $-4096i$ respectively; the moduli are 16 and 4096 $\checkmark$.

Theorem 22.4: The square of the normalized DFT is time reversal. Let $P \in \mathbb{C}^{N \times N}$ be the matrix of the permutation $n \mapsto \langle -n \rangle_N$, i.e. $P_{mn} = 1$ if $m+n \equiv 0 \pmod{N}$ and $P_{mn} = 0$ otherwise. Then

$$F_N^2 = P.$$

Equivalently: if $X = \mathcal{F}_N x$ and $Y = \mathcal{F}_N X$, then $Y[n] = N x[\langle -n \rangle_N]$ for every n .

Proof. The (m, n) entry of W_N^2 is computed:

$$(W_N^2)_{mn} = \sum_{k=0}^{N-1} \omega_N^{mk} \omega_N^{kn} = \sum_{k=0}^{N-1} \omega_N^{k(m+n)}.$$

By Lemma 22.1 with $j = m+n$, this sum equals N if $N \mid (m+n)$ and 0 otherwise; that is, $(W_N^2)_{mn} = N P_{mn}$, i.e. $W_N^2 = NP$. Dividing by N (since $F_N^2 = N^{-1}W_N^2$) gives $F_N^2 = P$. For the reformulation: $Y = W_N X = W_N^2 x = NPx$, and the n -th component of Px is $\sum_m P_{nm} x[m] = x[m_0]$ where m_0 is the unique index in $\{0, \dots, N-1\}$ with $n+m_0 \equiv 0$, i.e. $m_0 = \langle -n \rangle_N$.

Note that P is well defined: for every n there is one and only one $m \in \{0, \dots, N-1\}$ with $m \equiv -n$, and it is $m = 0$ if $n = 0$ and $m = N-n$ otherwise. Hence P fixes the index 0 and swaps n with $N-n$. $\square$

Corollary 22.4: Order-four periodicity. $F_N^4 = I_N$, and hence $W_N^4 = N^2 I_N$. The normalized DFT is thus a unitary operator of finite order: applied four times it returns the original signal.

Proof. By Theorem 22.4, $F_N^4 = (F_N^2)^2 = P^2$. Now P^2 is the matrix of the composed permutation $n \mapsto \langle -n \rangle_N \mapsto \langle -(-n) \rangle_N = \langle n \rangle_N = n$, i.e. of the identity: $P^2 = I_N$. Alternatively, by a direct matrix computation,

$$(P^2)_{mn} = \sum_{r=0}^{N-1} P_{mr} P_{rn},$$

and the only r for which $P_{mr} \neq 0$ is $r = \langle -m \rangle_N$, for which $P_{rn} \neq 0$ if and only if $r+n \equiv 0$, i.e. $n \equiv m$: hence $(P^2)_{mn} = \delta_{mn}$. Finally $W_N^4 = (\sqrt{N})^4 F_N^4 = N^2 I_N$. $\square$

Corollary 22.5: The eigenvalues of the DFT. Every eigenvalue λ of F_N satisfies $\lambda^4 = 1$, i.e. $\lambda \in \{1, i, -1, -i\}$: the eigenvalues of the normalized DFT are fourth roots of unity. Moreover F_N is diagonalizable, and $\mathbb{C}^N$ is the direct sum of the four (possibly zero) eigenspaces associated with $1, i, -1, -i$.

Proof. Let λ be an eigenvalue of F_N and $v \neq 0$ an eigenvector, $F_N v = \lambda v$. Applying F_N three more times, $F_N^4 v = \lambda^4 v$; by Corollary 22.4 the left-hand side is v , hence $(\lambda^4 - 1)v = 0$ and, since $v \neq 0$, $\lambda^4 = 1$. The solutions of $\lambda^4 = 1$ are the four fourth roots of unity $1, i, -1, -i$ (Theorem 6.2 with $n = 4$).

For diagonalizability: the polynomial $p(t) = t^4 - 1$ annihilates F_N and is a product of distinct linear factors, $p(t) = (t - 1)(t - i)(t + 1)(t + i)$. An annihilating polynomial with all simple roots implies that the minimal polynomial of F_N has simple roots, a necessary and sufficient condition for diagonalizability over $\mathbb{C}$; and the eigenspace decomposition is the associated primary decomposition. (Alternatively: F_N is unitary by Theorem 22.3, and every unitary matrix is diagonalizable by the spectral theorem.) $\square$

Remark 22.3: The multiplicities of the four eigenvalues: what remains imported. Corollary 22.5 says which the eigenvalues are, not with what multiplicity they occur; and this second question is not proved here. The exact count of the multiplicities of the eigenvalues $1, -i, -1, i$ of the matrix F_N is the content of a theorem of J. H. McClellan and T. W. Parks (1972): it is here **imported**, i.e. used as information rather than derived. What is missing, precisely: the proof requires constructing an explicit basis of eigenvectors (or computing the trace of F_N, F_N^2, F_N^3 , which leads to the quadratic Gauss sums $\sum_n e^{-2\pi i n^2/N}$, not treated in the treatise). The direct computation of the eigenvalues for $N = 4, \dots, 12$, carried out in the script of Section 22.12.1, gives the multiplicities

N	$\lambda = 1$	$\lambda = -i$	$\lambda = -1$	$\lambda = i$
4	2	1	1	0
5	2	1	1	1
6	2	1	2	1
7	2	2	2	1
8	3	2	2	1
12	4	3	3	2

consistent with the McClellan–Parks statement; the sum of each row is N ✓. Nothing in the rest of the chapter depends on this table.

Remark 22.4: Why order four is not a coincidence. Corollary 22.4 is the exact discrete analogue of a known fact of the continuous Fourier transform (Chapter 19): transforming twice is the same as reflecting, transforming four times is the same as doing nothing. The reason is the same in both cases and is Theorem 22.4: the kernel ω_N^{kn} is symmetric in k and n , so composing the transform with itself produces $\sum_k \omega_N^{k(m+n)}$, i.e. again the orthogonality identity, but with $m+n$ in place of $m-n$ — and it is this sign change that produces the reflection. It also follows that the DFT admits a natural square root (the discrete fractional Fourier transform), an object not treated in Volume III.

22.5. Circular convolution and the discrete convolution theorem

Definition 22.4: Circular convolution. Let $x, y \in \mathbb{C}^N$. Their **circular convolution** $x \circledast y \in \mathbb{C}^N$ is defined by

$$(x \circledast y)[n] = \sum_{m=0}^{N-1} x[m] y[\langle n - m \rangle_N], \quad n = 0, 1, \dots, N-1,$$

where $\langle n - m \rangle_N$ is, as stated in the conventions of the chapter, the unique integer of $\{0, \dots, N-1\}$ congruent to $n - m$ modulo N . Writing out the first indices in full, for $N = 4$:

$$\begin{aligned} (x \circledast y)[0] &= x[0]y[0] + x[1]y[3] + x[2]y[2] + x[3]y[1], \\ (x \circledast y)[1] &= x[0]y[1] + x[1]y[0] + x[2]y[3] + x[3]y[2], \\ (x \circledast y)[2] &= x[0]y[2] + x[1]y[1] + x[2]y[0] + x[3]y[3], \\ (x \circledast y)[3] &= x[0]y[3] + x[1]y[2] + x[2]y[1] + x[3]y[0]. \end{aligned}$$

The adjective “circular” recalls that the index of the second factor *wraps around*: when $n - m$ becomes negative, one restarts from the end, as on a circle of N positions.

Proposition 22.4: Circular shift and modulation. Let $x \in \mathbb{C}^N$, $X = \mathcal{F}_N x$, and let $n_0 \in \mathbb{Z}$. Set $x_{n_0}[n] = x[\langle n - n_0 \rangle_N]$ (circular shift by n_0 positions). Then

$$(\mathcal{F}_N x_{n_0})[k] = \omega_N^{kn_0} X[k], \quad k = 0, \dots, N-1.$$

Dually, if $y[n] = \omega_N^{-k_0 n} x[n]$ then $(\mathcal{F}_N y)[k] = X[\langle k - k_0 \rangle_N]$.

Proof. Compute $(\mathcal{F}_N x_{n_0})[k] = \sum_{n=0}^{N-1} x[\langle n - n_0 \rangle_N] \omega_N^{kn}$ and substitute $r = \langle n - n_0 \rangle_N$. As n ranges over $\{0, \dots, N-1\}$, the index r runs *exactly once* over the whole of $\{0, \dots, N-1\}$: the map $n \mapsto \langle n - n_0 \rangle_N$ is indeed a bijection of $\{0, \dots, N-1\}$ onto itself, with inverse $r \mapsto \langle r + n_0 \rangle_N$. Moreover $n \equiv r + n_0 \pmod{N}$, and since $\omega_N^N = 1$ one has $\omega_N^{kn} = \omega_N^{k(r+n_0)}$ (powers of ω_N depend on the exponent only modulo N). Hence

$$\sum_{n=0}^{N-1} x[\langle n - n_0 \rangle_N] \omega_N^{kn} = \sum_{r=0}^{N-1} x[r] \omega_N^{k(r+n_0)} = \omega_N^{kn_0} \sum_{r=0}^{N-1} x[r] \omega_N^{kr} = \omega_N^{kn_0} X[k].$$

For the dual statement: $(\mathcal{F}_N y)[k] = \sum_n \omega_N^{-k_0 n} x[n] \omega_N^{kn} = \sum_n x[n] \omega_N^{(k-k_0)n} = X[k - k_0]$, which by Proposition 22.1 equals $X[\langle k - k_0 \rangle_N]$. $\square$

Theorem 22.5: Discrete convolution theorem. Let $x, y \in \mathbb{C}^N$ with DFTs X and Y . Then

$$\mathcal{F}_N(x \circledast y)[k] = X[k] Y[k], \quad k = 0, 1, \dots, N-1:$$

the DFT turns circular convolution into componentwise multiplication. Equivalently,

$$x \circledast y = \mathcal{F}_N^{-1}(X \cdot Y), \quad (x \circledast y)[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] Y[k] \omega_N^{-kn}.$$

Proof. Start from the definition and exchange the order of the two finite sums (legitimate, as in the proof of Theorem 22.2, because the terms are finite in number):

$$\mathcal{F}_N(x \circledast y)[k] = \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] y[\langle n - m \rangle_N] \right) \omega_N^{kn} = \sum_{m=0}^{N-1} x[m] \underbrace{\sum_{n=0}^{N-1} y[\langle n - m \rangle_N] \omega_N^{kn}}_{=: S_m}.$$

The inner sum S_m is, by Proposition 22.4 applied to the vector y with $n_0 = m$, exactly $\omega_N^{km} Y[k]$. For clarity the step is repeated here explicitly: setting $r = \langle n - m \rangle_N$, as n ranges over $\{0, \dots, N-1\}$ so too does r range once over $\{0, \dots, N-1\}$, and $n \equiv r + m \pmod{N}$, so $\omega_N^{kn} = \omega_N^{k(r+m)}$ because $\omega_N^N = 1$; hence

$$S_m = \sum_{r=0}^{N-1} y[r] \omega_N^{k(r+m)} = \omega_N^{km} \sum_{r=0}^{N-1} y[r] \omega_N^{kr} = \omega_N^{km} Y[k].$$

Substituting,

$$\mathcal{F}_N(x \circledast y)[k] = \sum_{m=0}^{N-1} x[m] \omega_N^{km} Y[k] = Y[k] \sum_{m=0}^{N-1} x[m] \omega_N^{km} = X[k] Y[k].$$

The last formula of the statement is obtained by applying the inverse to both sides, legitimate by Theorem 22.2. $\square$

Corollary 22.6: The dual: product in time, convolution in frequency. Let $x, y \in \mathbb{C}^N$ and let $z[n] = x[n]y[n]$ be the componentwise product. Then

$$Z[k] = \frac{1}{N} (X \circledast Y)[k] = \frac{1}{N} \sum_{j=0}^{N-1} X[j] Y[\langle k - j \rangle_N].$$

Proof. Substitute for $x[n]$ and $y[n]$ their respective inversion formulas (Theorem 22.2), using distinct dummy indices j and ℓ :

$$Z[k] = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{j=0}^{N-1} X[j] \omega_N^{-jn} \right) \left(\frac{1}{N} \sum_{\ell=0}^{N-1} Y[\ell] \omega_N^{-\ell n} \right) \omega_N^{kn}.$$

Factoring out the finite sums over j and ℓ :

$$Z[k] = \frac{1}{N^2} \sum_{j=0}^{N-1} \sum_{\ell=0}^{N-1} X[j] Y[\ell] \sum_{n=0}^{N-1} \omega_N^{n(k-j-\ell)}.$$

By Lemma 22.1 the sum over n equals N if $N \mid (k - j - \ell)$, i.e. if $\ell \equiv k - j \pmod{N}$, and 0 otherwise. For each fixed j there is exactly one $\ell \in \{0, \dots, N-1\}$ with this property, namely $\ell = \langle k - j \rangle_N$. Hence only N terms survive:

$$Z[k] = \frac{1}{N^2} \sum_{j=0}^{N-1} X[j] Y[\langle k - j \rangle_N] \cdot N = \frac{1}{N} (X \circledast Y)[k].$$

$\square$

Example 22.2: A circular convolution computed two ways. Let $N = 4$, $a = (1, 2, 3, 4)$ and $b = (1, 0, -1, 0)$.

First route: the definition. Using the four rows written in full in Definition 22.4 with $x = a$, $y = b$:

$$\begin{aligned}(a \otimes b)[0] &= 1 \cdot 1 + 2 \cdot 0 + 3 \cdot (-1) + 4 \cdot 0 = 1 - 3 = -2, \\(a \otimes b)[1] &= 1 \cdot 0 + 2 \cdot 1 + 3 \cdot 0 + 4 \cdot (-1) = 2 - 4 = -2, \\(a \otimes b)[2] &= 1 \cdot (-1) + 2 \cdot 0 + 3 \cdot 1 + 4 \cdot 0 = -1 + 3 = 2, \\(a \otimes b)[3] &= 1 \cdot 0 + 2 \cdot (-1) + 3 \cdot 0 + 4 \cdot 1 = -2 + 4 = 2.\end{aligned}$$

Hence $a \otimes b = (-2, -2, 2, 2)$.

Second route: Theorem 22.5. With $\omega_4 = -i$,

$$A = \mathcal{F}_4 a = (10, -2 + 2i, -2, -2 - 2i), \quad B = \mathcal{F}_4 b = (0, 2, 0, 2).$$

(For A : $A[0] = 1 + 2 + 3 + 4 = 10$; $A[1] = 1 + 2(-i) + 3(-1) + 4(i) = 1 - 2i - 3 + 4i = -2 + 2i$; $A[2] = 1 - 2 + 3 - 4 = -2$; $A[3] = \overline{A[1]} = -2 - 2i$ by Theorem 22.7, since a is real. For B : $B[k] = 1 - (-i)^{2k} = 1 - (-1)^k$, which equals 0 for even k and 2 for odd k .) The componentwise product is $A \cdot B = (0, -4 + 4i, 0, -4 - 4i)$, and the inverse transform gives

$$(\mathcal{F}_4^{-1}(A \cdot B))[n] = \frac{1}{4} \left((-4 + 4i) i^n + (-4 - 4i) i^{3n} \right) = \frac{1}{4} \cdot 2 \operatorname{Re}((-4 + 4i) i^n),$$

having used $\omega_4^{-kn} = i^{kn}$ and the fact that the two terms are conjugates. For $n = 0$: $\frac{1}{2} \operatorname{Re}(-4 + 4i) = -2$; for $n = 1$: $\frac{1}{2} \operatorname{Re}((-4 + 4i)i) = \frac{1}{2} \operatorname{Re}(-4i - 4) = -2$; for $n = 2$: $\frac{1}{2} \operatorname{Re}(4 - 4i) = 2$; for $n = 3$: $\frac{1}{2} \operatorname{Re}((-4 + 4i)(-i)) = \frac{1}{2} \operatorname{Re}(4i + 4) = 2$. We recover $(-2, -2, 2, 2)$ ✓.

Executed check. The script of Section 22.12.1 computes the circular convolution from the definition and as `ifft(ffft(a)*ffft(b))`: both return $(-2, -2, 2, 2)$ ✓.

22.6. Discrete Parseval identity

Theorem 22.6: Discrete Parseval and Plancherel identities. For all $x, y \in \mathbb{C}^N$, with $X = \mathcal{F}_N x$ and $Y = \mathcal{F}_N y$,

$$\sum_{n=0}^{N-1} x[n] \overline{y[n]} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \overline{Y[k]} \quad (\text{Parseval}).$$

In particular, setting $y = x$,

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \quad (\text{Plancherel, or energy conservation}).$$

Proof. First route (from unitarity). By Theorem 22.3 the matrix $F_N = N^{-1/2} W_N$ is unitary, and a unitary map preserves the Hermitian inner product:

$$\langle F_N x, F_N y \rangle = (F_N y)^* (F_N x) = y^* F_N^* F_N x = y^* x = \langle x, y \rangle.$$

But $F_N x = N^{-1/2} X$ and $F_N y = N^{-1/2} Y$, hence $\langle F_N x, F_N y \rangle = \frac{1}{N} \langle X, Y \rangle$, which is the claim.

Second route (direct computation). Substitute for $x[n]$ and $\overline{y[n]}$ the inversion formula (Theorem 22.2), with dummy indices j and ℓ ; note that $\overline{y[n]} = \frac{1}{N} \sum_{\ell} \overline{Y[\ell]} \omega_N^{\ell n}$ because $\overline{\omega_N^{\ell n}} =$

$\omega_N^{\ell n}$. Then

$$\sum_{n=0}^{N-1} x[n] \overline{y[n]} = \frac{1}{N^2} \sum_{j=0}^{N-1} \sum_{\ell=0}^{N-1} X[j] \overline{Y[\ell]} \sum_{n=0}^{N-1} \omega_N^{n(\ell-j)}.$$

By Lemma 22.1 the inner sum equals N if $j = \ell$ (the indices lie in $\{0, \dots, N-1\}$, so $N \mid (\ell-j)$ is equivalent to $\ell = j$) and 0 otherwise. N terms survive:

$$= \frac{1}{N^2} \sum_{j=0}^{N-1} X[j] \overline{Y[j]} \cdot N = \frac{1}{N} \sum_{j=0}^{N-1} X[j] \overline{Y[j]}.$$

The two routes agree. $\square$

Corollary 22.7: The numerical check that exposes normalization errors. If an implementation of the DFT returns $\tilde{X} = cX$ with $c > 0$ a constant, then $\frac{1}{N} \sum_k |\tilde{X}[k]|^2 = c^2 \sum_n |x[n]|^2$: the ratio between the two sides of the Plancherel identity equals c^2 and immediately reveals the wrong normalization factor. If, on the other hand, the error is a wrong sign in the exponent, i.e. $\tilde{X} = \mathcal{F}_N \bar{x}$ conjugated, the Plancherel identity remains true and reveals nothing: one then needs the check on conjugate symmetry (Theorem 22.7) or on inversion.

Proof. For the first claim, $\frac{1}{N} \sum_k |cX[k]|^2 = \frac{c^2}{N} \sum_k |X[k]|^2 = c^2 \sum_n |x[n]|^2$ by Theorem 22.6, and the ratio with $\sum_n |x[n]|^2$ equals c^2 (the denominator is nonzero if $x \neq 0$). For the second: if $\tilde{X}[k] = \sum_n x[n] \omega_N^{kn}$, then $\tilde{X}[k] = \sum_n x[n] \omega_N^{-kn} = \overline{X[\langle -k \rangle_N]}$ by Proposition 22.1; hence $|\tilde{X}[k]| = |X[\langle -k \rangle_N]|$ and, since $k \mapsto \langle -k \rangle_N$ is a bijection of $\{0, \dots, N-1\}$ onto itself, $\sum_k |\tilde{X}[k]|^2 = \sum_k |X[k]|^2$: the Plancherel identity is satisfied by $\tilde{X}$ exactly as by X . $\square$

22.7. Conjugate symmetry of real signals

Theorem 22.7: Hermitian symmetry. Let $x \in \mathbb{C}^N$ with DFT X . Then

$$x[n] \in \mathbb{R} \text{ for every } n \quad \Longleftrightarrow \quad X[\langle -k \rangle_N] = \overline{X[k]} \text{ for every } k,$$

that is, in terms of indices in $\{0, \dots, N-1\}$: $X[N-k] = \overline{X[k]}$ for $k = 1, \dots, N-1$, and $X[0] = \overline{X[0]}$.

Proof. ($\Rightarrow$) Let x be real, i.e. $\overline{x[n]} = x[n]$ for every n . Then

$$\overline{X[k]} = \overline{\sum_{n=0}^{N-1} x[n] \omega_N^{kn}} = \sum_{n=0}^{N-1} \overline{x[n]} \overline{\omega_N^{kn}} = \sum_{n=0}^{N-1} x[n] \omega_N^{-kn} = \sum_{n=0}^{N-1} x[n] \omega_N^{(-k)n} = X[-k],$$

where $\overline{\omega_N^m} = \omega_N^{-m}$ was used (valid since $|\omega_N| = 1$) and, in the last step, the extension to every integer of Proposition 22.1, by which $X[-k] = X[\langle -k \rangle_N]$.

($\Leftarrow$) Suppose $X[\langle -k \rangle_N] = \overline{X[k]}$ for every k . By the inversion formula (Theorem 22.2),

$$\overline{x[n]} = \frac{1}{N} \sum_{k=0}^{N-1} \overline{X[k] \omega_N^{kn}} = \frac{1}{N} \sum_{k=0}^{N-1} \overline{X[k]} \omega_N^{kn} = \frac{1}{N} \sum_{k=0}^{N-1} X[\langle -k \rangle_N] \omega_N^{kn}.$$

Set $j = \langle -k \rangle_N$: as k ranges over $\{0, \dots, N-1\}$ so too does j range once over $\{0, \dots, N-1\}$ (it is the involutive permutation P of Theorem 22.4), and $k \equiv -j \pmod{N}$, hence $\omega_N^{kn} = \omega_N^{-jn}$.

Therefore

$$\overline{x[n]} = \frac{1}{N} \sum_{j=0}^{N-1} X[j] \omega_N^{-jn} = x[n],$$

again by Theorem 22.2: $x[n]$ is real for every n . $\square$

Corollary 22.8: Real bins and counting the information. Let $x \in \mathbb{R}^N$. Then:

- (i) $X[0] = \sum_n x[n]$ is real;
- (ii) if N is even, $X[N/2] = \sum_n (-1)^n x[n]$ is real (the Nyquist bin);
- (iii) $|X[N-k]| = |X[k]|$ and $\arg X[N-k] = -\arg X[k]$ for $k = 1, \dots, N-1$: the modulus of the spectrum is even, the phase odd;
- (iv) the spectrum contains exactly N independent real numbers: for N even, the two real values $X[0]$ and $X[N/2]$ together with the $\frac{N}{2} - 1$ complex numbers $X[1], \dots, X[N/2 - 1]$; for N odd, the real value $X[0]$ together with the $\frac{N-1}{2}$ complex numbers $X[1], \dots, X[\frac{N-1}{2}]$.

Proof. (i) Setting $k = 0$ in Definition 22.1 gives $X[0] = \sum_n x[n] \omega_N^0 = \sum_n x[n]$, a sum of real numbers. Alternatively, from Theorem 22.7 with $k = 0$: $\langle -0 \rangle_N = 0$, hence $X[0] = \overline{X[0]}$.

(ii) Let $N = 2M$. Then $\omega_N^{Mn} = e^{-2\pi i Mn/(2M)} = e^{-i\pi n} = (-1)^n$, so $X[M] = \sum_n (-1)^n x[n]$, a sum of real numbers. Alternatively, from Theorem 22.7 with $k = M$: $\langle -M \rangle_{2M} = M$, hence $X[M] = \overline{X[M]}$.

(iii) From $X[N-k] = \overline{X[k]}$ it follows that $|X[N-k]| = |\overline{X[k]}| = |X[k]|$ and $\arg X[N-k] = \arg \overline{X[k]} = -\arg X[k]$ (for $X[k] \neq 0$; if $X[k] = 0$ then also $X[N-k] = 0$ and the phase is undefined for either one).

(iv) For $N = 2M$ even: the values $X[0]$ and $X[M]$ are real by (i) and (ii), and the values $X[M+1], \dots, X[N-1]$ are determined by $X[1], \dots, X[M-1]$ via $X[N-k] = \overline{X[k]}$. The count of real parameters is therefore $1 + 1 + 2(M-1) = 2M = N$. For $N = 2M+1$ odd: only $X[0]$ is real, and $X[M+1], \dots, X[N-1]$ are the conjugates of $X[M], \dots, X[1]$; the count is $1 + 2M = N$. In both cases the number of independent real parameters coincides with the real dimension of $\mathbb{R}^N$, as it must, since $\mathcal{F}_N$ is bijective (Theorem 22.2) and $\mathbb{R}^N$ is a real subspace of $\mathbb{C}^N$ of real dimension N . $\square$

22.8. Circulant matrices diagonalized by the DFT

Definition 22.5: Circulant matrix. Given $c = (c[0], \dots, c[N-1]) \in \mathbb{C}^N$, the **circulant matrix** $C = \text{circ}(c)$ is the $N \times N$ matrix with entries

$$C_{nm} = c[\langle n - m \rangle_N], \quad n, m = 0, \dots, N-1,$$

that is,

$$C = \begin{pmatrix} c[0] & c[N-1] & c[N-2] & \cdots & c[1] \\ c[1] & c[0] & c[N-1] & \cdots & c[2] \\ c[2] & c[1] & c[0] & \cdots & c[3] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c[N-1] & c[N-2] & c[N-3] & \cdots & c[0] \end{pmatrix};$$

each row is the previous one shifted one position to the right, with circular wraparound. The first column is c .

Proposition 22.5: Circulant-vector product and convolution. For every $x \in \mathbb{C}^N$, $\text{circ}(c)x = c \circledast x$.

Proof. The n -th component of the matrix-vector product is

$$(Cx)[n] = \sum_{m=0}^{N-1} C_{nm} x[m] = \sum_{m=0}^{N-1} c[\langle n-m \rangle_N] x[m],$$

which is Definition 22.4 with the roles of c and x swapped relative to the writing adopted there. That the two ways of writing the convolution coincide is seen by substituting $r = \langle n-m \rangle_N$: as m ranges over $\{0, \dots, N-1\}$ so too does r range once over $\{0, \dots, N-1\}$, and $m \equiv n-r$, hence $\sum_m c[\langle n-m \rangle_N] x[m] = \sum_r c[r] x[\langle n-r \rangle_N] = (c \circledast x)[n]$. In particular circular convolution is commutative. $\square$

Theorem 22.8: Circulants are all diagonalized by the DFT. Let $C = \text{circ}(c)$ and let $\lambda = \mathcal{F}_N c$ be the DFT of its generator. Then, with F_N the unitary matrix of Definition 22.3,

$$F_N C F_N^* = \text{diag}(\lambda[0], \lambda[1], \dots, \lambda[N-1]), \quad \text{i.e.} \quad C = F_N^* \text{diag}(\lambda) F_N.$$

In particular: the eigenvalues of C are the values $\lambda[k] = \sum_m c[m] \omega_N^{km}$ of the DFT of c , and an eigenvector for $\lambda[k]$ is the vector $e_k = (\omega_N^{-kn})_{n=0}^{N-1}$ of Theorem 22.1, i.e. the k -th column of $\sqrt{N} F_N^*$. The basis that diagonalizes all circulants of order N is thus a single one and does not depend on c .

Proof. Step 1: e_k is an eigenvector. By Proposition 22.5, $Ce_k = c \circledast e_k$, whose n -th component is

$$\sum_{m=0}^{N-1} c[m] e_k[\langle n-m \rangle_N] = \sum_{m=0}^{N-1} c[m] \omega_N^{-k\langle n-m \rangle_N} = \sum_{m=0}^{N-1} c[m] \omega_N^{-k(n-m)},$$

where the last step is legitimate because $\langle n-m \rangle_N \equiv n-m \pmod{N}$ and powers of ω_N depend on the exponent only modulo N (since $\omega_N^N = 1$). Collecting ω_N^{-kn} :

$$(Ce_k)[n] = \omega_N^{-kn} \sum_{m=0}^{N-1} c[m] \omega_N^{km} = \lambda[k] e_k[n].$$

Holding for every n , this gives $Ce_k = \lambda[k] e_k$.

Step 2: from the basis to the operators. The vectors $e_0, \dots, e_{N-1}$ form an orthogonal basis of $\mathbb{C}^N$ (Corollary 22.1) and $u_k = N^{-1/2} e_k$ is the corresponding orthonormal basis. Let U be the matrix with $u_0, \dots, u_{N-1}$ as columns: U is unitary (the columns are orthonormal) and, by Step 1, $CU = U \text{diag}(\lambda)$, i.e. $U^*CU = \text{diag}(\lambda)$. It remains to recognize U : the (n, k) entry of U is $N^{-1/2} \omega_N^{-kn}$, which is the conjugate of $(F_N)_{kn} = N^{-1/2} \omega_N^{kn}$; hence $U = \overline{F_N}^T = F_N^*$. Substituting, $F_N C F_N^* = \text{diag}(\lambda)$, and multiplying on the left by F_N^* and on the right by F_N gives $C = F_N^* \text{diag}(\lambda) F_N$.

Step 3: consistency with the convolution theorem. The statement is Theorem 22.5 rewritten in matrix form: applying C to x means convolving, i.e. (up to normalizations) transforming, multiplying componentwise by λ , and transforming back. In formulas: $Cx = c \circledast x = \mathcal{F}_N^{-1}(\lambda \cdot \mathcal{F}_N x) = \frac{1}{N} W_N^* \text{diag}(\lambda) W_N x = F_N^* \text{diag}(\lambda) F_N x$. $\square$

Corollary 22.9: The algebra of circulants. Let $B = \text{circ}(b)$ and $C = \text{circ}(c)$ be of order N . Then:

- (i) $BC = CB$, and $BC = \text{circ}(b \otimes c)$;
- (ii) $\alpha B + \beta C = \text{circ}(\alpha b + \beta c)$ for all $\alpha, \beta \in \mathbb{C}$;
- (iii) $\det C = \prod_{k=0}^{N-1} \lambda[k]$ with $\lambda = \mathcal{F}_N c$;
- (iv) C is invertible if and only if $\lambda[k] \neq 0$ for every k , and in that case $C^{-1} = \text{circ}(d)$ with $d = \mathcal{F}_N^{-1}(1/\lambda)$, where $1/\lambda$ is the vector of components $1/\lambda[k]$.

Proof. (i) and (ii). By Theorem 22.8, $B = F_N^* \text{diag}(\beta_\lambda) F_N$ and $C = F_N^* \text{diag}(\lambda) F_N$ with $\beta_\lambda = \mathcal{F}_N b$; hence

$$BC = F_N^* \text{diag}(\beta_\lambda) F_N F_N^* \text{diag}(\lambda) F_N = F_N^* \text{diag}(\beta_\lambda) \text{diag}(\lambda) F_N,$$

having used $F_N F_N^* = I$ (Theorem 22.3). Diagonal matrices commute with each other and their product is the diagonal of the products, so $BC = CB$ and $BC = F_N^* \text{diag}(\beta_\lambda \cdot \lambda) F_N$, which by the same theorem (read backwards) is the circulant generated by the DFT vector $\beta_\lambda \cdot \lambda$, i.e., by Theorem 22.5, by $b \otimes c$. Claim (ii) is immediate from Definition 22.5, since $(\alpha B + \beta C)_{nm} = \alpha b[\langle n - m \rangle_N] + \beta c[\langle n - m \rangle_N]$.

(iii) From Theorem 22.8 and the Binet formula, $\det C = \det(F_N^*) \det(\text{diag } \lambda) \det(F_N) = \det(\text{diag } \lambda) \det(F_N^* F_N) = \prod_k \lambda[k]$.

(iv) C is invertible if and only if $\det C \neq 0$, i.e., by (iii), if and only if no $\lambda[k]$ vanishes. In that case $C^{-1} = F_N^* \text{diag}(\lambda)^{-1} F_N = F_N^* \text{diag}(1/\lambda) F_N$, which again by Theorem 22.8 is the circulant with generator $d = \mathcal{F}_N^{-1}(1/\lambda)$. $\square$

Example 22.3: A 4×4 circulant, diagonalized and solved. Let $c = (1, 2, 3, 4)$ and

$$C = \text{circ}(c) = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}.$$

The eigenvalues are, by Theorem 22.8, the components of $\lambda = \mathcal{F}_4 c$, computed in Example 22.2:

$$\lambda = (10, -2 + 2i, -2, -2 - 2i).$$

The determinant is therefore

$$\det C = 10 \cdot (-2 + 2i) \cdot (-2) \cdot (-2 - 2i) = 10 \cdot (-2) \cdot [(-2)^2 + 2^2] = -20 \cdot 8 = -160,$$

having used $(-2 + 2i)(-2 - 2i) = |-2 + 2i|^2 = 4 + 4 = 8$. The independent computation via LU factorization gives $\det C = -160.00000 \checkmark$ (the discrepancy from the exact value is of order 10^{-13} , i.e. the double-precision rounding error).

Solving $Cy = b$ without inverting the matrix. Let $b = (5, 3, 5, 7)$. Since $\omega_4 = -i$ one has $\omega_4^n = (-i)^n$, so

$$\begin{aligned} B[0] &= 5 + 3 + 5 + 7 = 20, & B[1] &= 5 + 3(-i) + 5(-1) + 7(i) = 4i, \\ B[2] &= 5 - 3 + 5 - 7 = 0, & B[3] &= \overline{B[1]} = -4i, \end{aligned}$$

where for $B[2]$ we used $\omega_4^{2n} = (-1)^n$ and for $B[3]$ conjugate symmetry (Theorem 22.7), since b is real. By Corollary 22.9(iv) it now suffices to divide spectrum by spectrum:

$$\frac{\mathcal{F}_4 b}{\lambda} = \left(\frac{20}{10}, \frac{4i}{-2 + 2i}, \frac{0}{-2}, \frac{-4i}{-2 - 2i} \right) = (2, 1 - i, 0, 1 + i),$$

since $\frac{4i}{-2+2i} = \frac{4i(-2-2i)}{(-2)^2+2^2} = \frac{-8i-8i^2}{8} = \frac{8-8i}{8} = 1-i$, and the $k=3$ component is its conjugate. Transforming back, with $\omega_4^{-kn} = i^{kn}$ and grouping the two conjugate terms $k=1$ and $k=3$,

$$y[n] = \frac{1}{4} \left(2 + (1-i)i^n + 0 + (1+i)i^{3n} \right) = \frac{1}{4} \left(2 + 2 \operatorname{Re}((1-i)i^n) \right),$$

whence $y[0] = \frac{1}{4}(2+2 \cdot 1) = 1$; $y[1] = \frac{1}{4}(2+2 \operatorname{Re}(1+i)) = 1$ since $(1-i)i = 1+i$; $y[2] = \frac{1}{4}(2+2 \operatorname{Re}(-1+i)) = 0$; $y[3] = \frac{1}{4}(2+2 \operatorname{Re}(-1-i)) = 0$. Hence $y = (1, 1, 0, 0)$, and the direct check is immediate: Cy is the sum of the first two columns of C , i.e. $(1+4, 2+1, 3+2, 4+3) = (5, 3, 5, 7) = b$ ✓. The script of Section 22.12.1 recomputes y as `ifft(fft(b)/fft(c))` and obtains $(1, 1, 0, 0)$ to within 10^{-15} ✓.

Remark 22.5: Why this theorem is the heart of digital filtering. A shift-invariant linear filter, applied to a block of N samples with periodic boundary conditions, is multiplication by a circulant matrix: the n -th row contains the impulse response read backwards and wrapped around. Theorem 22.8 then says that all such filters are simultaneously diagonalized by the same basis, that of the discrete exponentials: this is why “filtering” in frequency reduces to multiplying each bin by a number, and why the cost of filtering drops from $O(N^2)$ (matrix-vector product) to $O(N \log N)$ (two FFTs and N multiplications) — see Section 22.10 and Chapter 23. The price is the implicit periodicity of the block: the sample $x[N-1]$ is “close” to $x[0]$, and it is exactly this fiction that produces the leakage of Section 22.10.3 when the signal is not periodic with period N .

22.9. The DFT and the DTFT: spectrum sampling and zero-padding

Definition 22.6: Discrete Dirichlet kernel. For $N \geq 1$, the **discrete Dirichlet kernel** is the function $D_N : \mathbb{R} \rightarrow \mathbb{C}$,

$$D_N(\Omega) = \sum_{n=0}^{N-1} e^{-in\Omega}.$$

Lemma 22.3: Closed form of the discrete Dirichlet kernel. For every $\Omega \in \mathbb{R}$,

$$D_N(\Omega) = \begin{cases} \frac{\sin(N\Omega/2)}{\sin(\Omega/2)} e^{-i(N-1)\Omega/2}, & \Omega \notin 2\pi\mathbb{Z}, \\ N, & \Omega \in 2\pi\mathbb{Z}. \end{cases}$$

In particular $|D_N(\Omega)| = |\sin(N\Omega/2)/\sin(\Omega/2)|$ outside multiples of 2π , and $D_N(\Omega) = 0$ if and only if $N\Omega/2 \in \pi\mathbb{Z}$ but $\Omega/2 \notin \pi\mathbb{Z}$, i.e. if and only if $\Omega = 2\pi m/N$ with $m \in \mathbb{Z}$ and $N \nmid m$.

Proof. This is the geometric sum with ratio $q = e^{-i\Omega}$, and once again the case of vanishing denominator must be distinguished.

Case $\Omega \in 2\pi\mathbb{Z}$. Then $q = e^{-i\Omega} = 1$ and the sum has N terms all equal to 1: $D_N(\Omega) = N$. (The closed formula does not apply: the denominator $\sin(\Omega/2)$ vanishes.)

Case $\Omega \notin 2\pi\mathbb{Z}$. Then $q \neq 1$: indeed $e^{-i\Omega} = 1$ is equivalent to $\Omega \in 2\pi\mathbb{Z}$. By the telescoping identity of Theorem 6.4 (Step 1), $(q-1) \sum_{n<N} q^n = q^N - 1$, and dividing by $q-1 \neq 0$,

$$D_N(\Omega) = \frac{e^{-iN\Omega} - 1}{e^{-i\Omega} - 1}.$$

The “half angles” are now collected, in numerator and denominator: for every $\varphi \in \mathbb{R}$,

$$e^{-i\varphi} - 1 = e^{-i\varphi/2}(e^{-i\varphi/2} - e^{i\varphi/2}) = e^{-i\varphi/2}(-2i \sin(\varphi/2)),$$

having used $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$. Applying the identity with $\varphi = N\Omega$ and with $\varphi = \Omega$ and simplifying the factor $-2i$:

$$D_N(\Omega) = \frac{e^{-iN\Omega/2}(-2i) \sin(N\Omega/2)}{e^{-i\Omega/2}(-2i) \sin(\Omega/2)} = e^{-i(N-1)\Omega/2} \frac{\sin(N\Omega/2)}{\sin(\Omega/2)},$$

since $-N\Omega/2 + \Omega/2 = -(N-1)\Omega/2$, and the denominator $\sin(\Omega/2)$ is nonzero precisely because $\Omega \notin 2\pi\mathbb{Z}$.

The modulus follows from $|e^{-i(N-1)\Omega/2}| = 1$. The zeros: outside $2\pi\mathbb{Z}$ the exponential factor never vanishes and the denominator is nonzero, so $D_N(\Omega) = 0$ if and only if $\sin(N\Omega/2) = 0$, i.e. $N\Omega/2 = \pi m$ with $m \in \mathbb{Z}$, i.e. $\Omega = 2\pi m/N$; and the condition $\Omega \notin 2\pi\mathbb{Z}$ is $N \nmid m$. $\square$

Theorem 22.9: The DFT samples the DTFT. Let $x \in \mathbb{C}^N$ and consider the sequence $x[n]$ extended by zeros outside $\{0, \dots, N-1\}$. Its DTFT (Chapter 21, Theorem 21.6) is the 2π -periodic function

$$\hat{X}(\Omega) = \sum_{n=0}^{N-1} x[n] e^{-in\Omega}, \quad \Omega \in \mathbb{R}.$$

Then, for every $k = 0, 1, \dots, N-1$,

$$X[k] = \hat{X}\left(\frac{2\pi k}{N}\right):$$

the DFT is the DTFT sampled at N equally spaced points of the unit circle. Moreover $\hat{X}$ is determined by X : it is, in $\mathbb{C}$, the polynomial $\hat{X}(\Omega) = \sum_n x[n] e^{-in\Omega}$ whose coefficients $x[n]$ are reconstructed from X by Theorem 22.2.

Proof. Substitute $\Omega = 2\pi k/N$ into the definition of $\hat{X}$:

$$\hat{X}\left(\frac{2\pi k}{N}\right) = \sum_{n=0}^{N-1} x[n] e^{-2\pi i kn/N} = \sum_{n=0}^{N-1} x[n] \omega_N^{kn} = X[k]$$

by Definition 22.1. That the sum is restricted to $n = 0, \dots, N-1$ is a consequence of the zero extension: the other terms vanish. For the second claim: once X is known, Theorem 22.2 returns $x[0], \dots, x[N-1]$, and these are by definition the coefficients of $\hat{X}$; hence two sequences of length N with the same DFT have the same DTFT, and conversely. $\square$

Remark 22.6: The link with the Zeta transform and with aliasing. Theorem 22.9 and Theorem 22.12 are the same statement read on two domains: $\hat{X}(\Omega) = X(z)$ for $z = e^{i\Omega}$, i.e. the DTFT is the restriction of the Zeta transform to the unit circle (Theorem 21.6), and the points $\Omega = 2\pi k/N$ correspond to the N -th roots of unity. A third phenomenon must be kept distinct: when the samples $x[n]$ come from sampling a continuous periodic function, the DFT does not return the Fourier coefficients of that function but their sum over all

frequencies congruent modulo N . This is the aliasing formula, proved in full in the solution of the inherited exercise 22.5, and connected with the error of the periodic trapezoidal rule.

Theorem 22.10: Zero-padding: the long DFT samples the same DTFT more finely. Let $x \in \mathbb{C}^N$, let $M \geq N$ and let $x^{(M)} \in \mathbb{C}^M$ be the vector obtained by extending x with zeros:

$$x^{(M)}[n] = \begin{cases} x[n], & 0 \leq n \leq N-1, \\ 0, & N \leq n \leq M-1. \end{cases}$$

Then, denoting by $X_M = \mathcal{F}_M x^{(M)}$ the M -point DFT,

$$X_M[k] = \hat{X}\left(\frac{2\pi k}{M}\right), \quad k = 0, 1, \dots, M-1,$$

with $\hat{X}$ the DTFT of Theorem 22.9. In particular, if $M = rN$ with $r \in \mathbb{N}$, then $X_M[rk] = X[k]$ for $k = 0, \dots, N-1$: the samples of the short DFT are all found among those of the long DFT.

Proof. By definition,

$$X_M[k] = \sum_{n=0}^{M-1} x^{(M)}[n] e^{-2\pi i kn/M} = \sum_{n=0}^{N-1} x[n] e^{-2\pi i kn/M},$$

since the terms with $n \geq N$ vanish. The last sum is exactly $\hat{X}(\Omega)$ evaluated at $\Omega = 2\pi k/M$.

For the last claim, with $M = rN$: $\frac{2\pi(rk)}{M} = \frac{2\pi rk}{rN} = \frac{2\pi k}{N}$, hence $X_M[rk] = \hat{X}(2\pi k/N) = X[k]$ by Theorem 22.9. $\square$

Theorem 22.11: Zero-padding is interpolation with the Dirichlet kernel. Under the hypotheses of Theorem 22.10, for every $k = 0, \dots, M-1$,

$$X_M[k] = \frac{1}{N} \sum_{j=0}^{N-1} X[j] D_N\left(2\pi\left(\frac{k}{M} - \frac{j}{N}\right)\right),$$

with D_N the discrete Dirichlet kernel of Definition 22.6. Zero-padding therefore adds no information whatsoever: the new spectrum samples are linear combinations of the old ones, with weights fixed once and for all by the kernel D_N .

Proof. Start from the formula of Theorem 22.10 and substitute for $x[n]$ the inversion formula (Theorem 22.2), with dummy index j :

$$X_M[k] = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{j=0}^{N-1} X[j] e^{2\pi i j n/N} \right) e^{-2\pi i kn/M}.$$

The sums are finite: exchange the order and collect $X[j]$:

$$X_M[k] = \frac{1}{N} \sum_{j=0}^{N-1} X[j] \sum_{n=0}^{N-1} e^{2\pi i n \left(\frac{j}{N} - \frac{k}{M} \right)}.$$

Now set

$$\Omega_{kj} = 2\pi \left(\frac{k}{M} - \frac{j}{N} \right),$$

so that the exponent of the inner sum is $2\pi in(\frac{j}{N} - \frac{k}{M}) = -in\Omega_{kj}$. The inner sum is thus, literally, the kernel of Definition 22.6 evaluated at Ω_{kj} :

$$\sum_{n=0}^{N-1} e^{2\pi in(\frac{j}{N} - \frac{k}{M})} = \sum_{n=0}^{N-1} e^{-in\Omega_{kj}} = D_N(\Omega_{kj}).$$

Substituting gives exactly the identity of the statement. Note that the order of the two terms in the argument is not immaterial: $D_N(-\Omega) = \overline{D_N(\Omega)}$, so writing $2\pi(\frac{j}{N} - \frac{k}{M})$ in place of Ω_{kj} would replace the kernel with its conjugate and the formula would be false (except at points where D_N is real).

Check. That the formula is correct is verified on a case where the result is known: if $M = N$ and $k = j_0$, then $\frac{k}{M} - \frac{j}{N} = \frac{j_0 - j}{N}$ and, by Lemma 22.3, $D_N(2\pi(j_0 - j)/N)$ equals N if $N \mid (j_0 - j)$, i.e. $j = j_0$, and 0 otherwise; the sum reduces to $\frac{1}{N}X[j_0] \cdot N = X[j_0] \checkmark$, as it must. The general numerical check is in Section 22.12.1: for $N = 8$, $M = 32$ and $x = (1, 1, 1, 1, 0, 0, 0, 0)$ the maximum discrepancy between the 32 values $X_M[k]$ computed with the FFT and those computed with the interpolation formula is $1.03 \cdot 10^{-14} \checkmark$. $\square$

Example 22.4: Zero-padding from 8 to 32 points, with a value computed by hand.

Let $x = (1, 1, 1, 1, 0, 0, 0, 0)$, the rectangular window of Section 22.12, and let $M = 32$. By Theorem 22.10, $X_{32}[k] = \widehat{X}(2\pi k/32)$, and since $x[n] = 1$ exactly for $n = 0, 1, 2, 3$,

$$X_{32}[k] = \sum_{n=0}^3 w^n = 1 + w + w^2 + w^3 = \frac{1 - w^4}{1 - w}, \quad w = e^{-2\pi ik/32} = e^{-i\pi k/16} \quad (w \neq 1).$$

For $k = 2$ one has $w = e^{-i\pi/8}$ and $w^4 = e^{-i\pi/2} = -i$, hence

$$X_{32}[2] = \frac{1 + i}{1 - e^{-i\pi/8}}.$$

With $\cos(\pi/8) = 0.92388$ and $\sin(\pi/8) = 0.38268$, the denominator is $1 - 0.92388 + 0.38268i = 0.07612 + 0.38268i$, with squared modulus $0.07612^2 + 0.38268^2 = 0.00579 + 0.14645 = 0.15224$. Multiplying numerator and denominator by the conjugate of the denominator:

$$X_{32}[2] = \frac{(1 + i)(0.07612 - 0.38268i)}{0.15224} = \frac{0.45880 - 0.30656i}{0.15224} = 3.01367 - 2.01367i.$$

The modulus is computed more quickly in closed form: by Lemma 22.3,

$$|X_{32}[2]| = \frac{|1 - w^4|}{|1 - w|} = \frac{|1 + i|}{2 \sin(\pi/16)} = \frac{\sqrt{2}}{2 \cdot 0.19509} = 3.62451,$$

having used $|1 - e^{-i\varphi}| = 2|\sin(\varphi/2)|$. Check: $\sqrt{3.01367^2 + 2.01367^2} = \sqrt{9.08222 + 4.05487} = \sqrt{13.13709} = 3.62451 \checkmark$. The script of Section 22.12.1 obtains $X_{32}[2] = 3.01367 - 2.01367i \checkmark$.

What was gained. Nothing, in terms of information: by Theorem 22.11 the value $X_{32}[2]$ is a fixed combination of the eight values $X[0], \dots, X[7]$ of the short DFT. What was gained is only readability of the plot: between bins 0 and 1 of the eight-point DFT, where before there was nothing, three more points of the same continuous curve $|\widehat{X}(\Omega)|$ are now visible. Zero-padding does *not* improve the frequency resolution, which remains f_s/N and depends on the duration of the observation, not on the number of points at which the spectrum is sampled (Proposition 22.8 and Remark 22.12).

22.10. The Discrete Fourier Transform and the FFT algorithm

We now return to the DFT matrix of Chapter 3 (Section 3.3) with the tools of the Zeta transform just introduced, to then arrive at the algorithm that made these computations practical on any microcontroller or PLC: the *Fast Fourier Transform* (FFT).

Theorem 22.12: The DFT as sampling of the Zeta transform on the roots of unity. Let $x[0], \dots, x[N-1]$ be a finite sequence and let $X(z) = \sum_{n=0}^{N-1} x[n]z^{-n}$ be its Zeta transform (a polynomial in z^{-1} , hence defined on all of $\mathbb{C} \setminus \{0\}$). The discrete Fourier transform (DFT) of $x[n]$,

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-2\pi i k n / N}, \quad k = 0, \dots, N-1,$$

coincides with the evaluation of $X(z)$ at the points $z = \zeta_N^{-k} = e^{2\pi i k / N}$, i.e. exactly at the N N -th roots of unity of Definition 3.2 of Chapter 3: $X[k] = X(\zeta_N^{-k})$, where $\zeta_N = e^{-2\pi i / N}$ is the local convention of this section (convention note at the beginning of the chapter and Theorem 22.13).

Proof. By definition $X(z) = \sum_{n=0}^{N-1} x[n]z^{-n}$. Setting $z = \zeta_N^{-k} = e^{2\pi i k / N}$ gives $z^{-n} = e^{-2\pi i k n / N}$, hence

$$X(\zeta_N^{-k}) = \sum_{n=0}^{N-1} x[n] e^{-2\pi i k n / N} = X[k].$$

This is precisely the matrix-vector product $W_N x$ of Definition 3.3, row by row: the k -th row of W_N contains exactly the samples $\zeta_N^{kn} = e^{-2\pi i k n / N}$ (again with the local convention $\zeta_N = e^{-2\pi i / N}$). $\square$

Remark 22.7: Why it matters. Theorem 22.12 closes the circle (in the true sense of the word) among three objects that in the text have been presented separately: the roots of unity (Chapter 3, geometry), the Zeta transform (this chapter, discrete control systems) and the DFT (numerical frequency analysis). They are the same algebraic construction seen from three different applied angles.

22.10.1 The Cooley-Tukey algorithm

Computing $X[k]$ directly from the definition requires, for each of the N values of k , a sum of N terms: overall $O(N^2)$ complex multiplications. For $N = 1024$ samples (typical in vibration or audio analysis) this means over one million operations. The FFT (*Fast Fourier Transform*) algorithm, in the form published by Cooley and Tukey in 1965¹, reduces the cost to $O(N \log_2 N)$ by exploiting the symmetry of the roots of unity.

Theorem 22.13: Decimation in time, radix-2. Let $N = 2^m$ be even. Separate the even and odd indices of $x[n]$: $e[r] = x[2r]$, $o[r] = x[2r+1]$, for $r = 0, \dots, N/2 - 1$. Then

$$X[k] = E[k] + \zeta_N^k O[k], \quad X[k + N/2] = E[k] - \zeta_N^k O[k], \quad k = 0, \dots, \frac{N}{2} - 1,$$

where $E[k]$ and $O[k]$ are the $N/2$ -point DFTs of $e[r]$ and $o[r]$ respectively, and $\zeta_N = e^{-2\pi i / N}$.

¹The algorithm was in fact already known, in a different form, to Carl Friedrich Gauss in 1805, for interpolating asteroid orbits: a manuscript that remained unpublished until 1866 and was rediscovered only in the 1980s (Heideman, Johnson, Burrus, 1985).

Proof. Separating the sum over even and odd indices:

$$\begin{aligned} X[k] &= \sum_{r=0}^{N/2-1} x[2r] \zeta_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1] \zeta_N^{(2r+1)k} \\ &= \sum_{r=0}^{N/2-1} e[r] \zeta_{N/2}^{rk} + \zeta_N^k \sum_{r=0}^{N/2-1} o[r] \zeta_{N/2}^{rk} = E[k] + \zeta_N^k O[k], \end{aligned}$$

using $\zeta_N^2 = \zeta_{N/2}$ (Corollary 3.2 of Chapter 3, a property of roots of unity). For the index $k + N/2$, since E and O are periodic with period $N/2$ (they are $N/2$ -point DFTs: $E[k + N/2] = E[k]$, $O[k + N/2] = O[k]$) and $\zeta_N^{k+N/2} = \zeta_N^k \zeta_N^{N/2} = -\zeta_N^k$ (because $\zeta_N^{N/2} = e^{-i\pi} = -1$), one obtains $X[k + N/2] = E[k] - \zeta_N^k O[k]$. $\square$

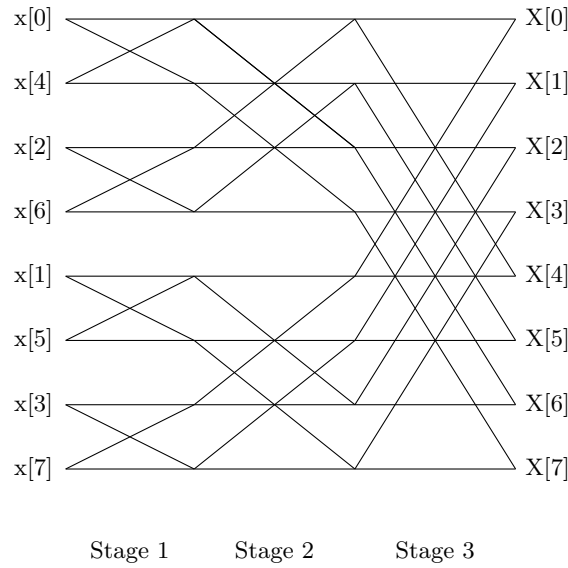


Figure 22.2: Butterfly network of the radix-2 FFT algorithm (decimation in time) for $N = 8$: input in bit-reversed order (left), $\log_2 8 = 3$ stages of butterflies (the diagonal lines carry the twiddle factor ζ_N^k), output in natural order (right) — the same structure implemented iteratively in Listing 4.3 (FB_FFT8) and underlying the exam question on the $O(N \log N)$ complexity (Theorem 22.14).

Remark 22.8: The “butterfly” and the $O(N \log N)$ complexity. The pair of formulas of Theorem 22.13 is called a butterfly: from two complex numbers $E[k]$, $O[k]$ and a single twiddle factor ζ_N^k , two outputs $X[k]$, $X[k + N/2]$ are obtained with a single complex multiplication (the product $\zeta_N^k O[k]$ is computed once and used twice, with opposite sign). Applying the same decomposition recursively to E and O (each an $N/2$ -point problem) gives the recurrence

$$T(N) = 2T(N/2) + cN,$$

whose solution is $T(N) = O(N \log_2 N)$ (full proof in Theorem 22.14 below), against $O(N^2)$ for the direct computation. For $N = 1024 = 2^{10}$: $N^2 = 1\,048\,576$ against $N \log_2 N = 1024 \times 10 = 10\,240$, a saving factor of ≈ 100 ; for $N = 1\,048\,576 = 2^{20}$ (one second of audio at about 1 MHz) the saving factor exceeds 50 000. This is why the FFT has been called by some “the most important algorithm of the twentieth century” (Strang, 1994) and underlies every spectrum analyzer, every audio/video codec, and much of the digital signal processing used in industrial systems (vibration analysis on motors, bearing diagnostics, acoustic quality

control).

Theorem 22.14: Complexity of the FFT: solution of the recurrence. Let $T(N)$ be the number of complex arithmetic operations performed by the radix-2 FFT algorithm (Theorem 22.13) on an input of length $N = 2^m$, $m \in \mathbb{N}$. Then $T(N)$ satisfies the recurrence

$$T(N) = 2T(N/2) + cN, \quad T(1) = c_0,$$

for suitable constants $c, c_0 > 0$ independent of N , and its solution is

$$T(N) = \Theta(N \log_2 N).$$

Proof. First observe that the recurrence correctly describes the cost of the algorithm: each call on a vector of length N separates the even/odd indices in time $O(N)$, makes two recursive calls on subproblems of length $N/2$, and finally combines the results with $N/2$ butterflies, each of cost $O(1)$ (one complex multiplication and two sums/differences), for a total of $O(N)$ combination operations; hence the term cN in the recurrence, with base case $T(1) = c_0$ (no operations: the single sample is returned).

We give two independent proofs that $T(N) = \Theta(N \log_2 N)$, following the practice of this treatise of reporting several proof routes when available in the literature.

First proof (substitution method, by induction). We prove by induction on m (with $N = 2^m$) that $T(N) \leq CN \log_2 N + c_0 N$ for a constant $C \geq c$ to be fixed.

Base case ($m = 0$, $N = 1$): $T(1) = c_0 \leq C \cdot 1 \cdot \log_2 1 + c_0 \cdot 1 = c_0$, true with equality for every C (since $\log_2 1 = 0$).

Inductive step: suppose the inductive hypothesis holds for $N/2 = 2^{m-1}$, i.e. $T(N/2) \leq C \cdot \frac{N}{2} \log_2 \frac{N}{2} + c_0 \frac{N}{2}$. Then

$$\begin{aligned} T(N) &= 2T(N/2) + cN \leq 2 \left(C \cdot \frac{N}{2} \log_2 \frac{N}{2} + c_0 \frac{N}{2} \right) + cN \\ &= CN \log_2 \frac{N}{2} + c_0 N + cN = CN(\log_2 N - 1) + c_0 N + cN = CN \log_2 N - CN + c_0 N + cN = CN \log_2 N + c_0 N - (C - c)N \end{aligned}$$

Choosing $C \geq c$, the last term $-(C - c)N$ is ≤ 0 , hence $T(N) \leq CN \log_2 N + c_0 N$, closing the induction. This proves $T(N) = O(N \log_2 N)$. The opposite inequality $T(N) = \Omega(N \log_2 N)$ is obtained by an identical argument, reversing all inequalities (the combination cost is *at least* cN , not merely at most cN , and one proves by induction $T(N) \geq cN \log_2 N$); together, $T(N) = \Theta(N \log_2 N)$.

Second proof (recursion tree). Represent the recursion as a binary tree in which the root node (depth $k = 0$) corresponds to the problem of size N , with combination cost cN (excluding the subproblems). Its two children, at depth $k = 1$, are the two subproblems of size $N/2$, each with combination cost $c \cdot (N/2)$: the total cost at that level is $2 \cdot c(N/2) = cN$. In general, at depth k there are 2^k subproblems, each of size $N/2^k$ and combination cost $c \cdot N/2^k$; the total cost at level k is thus

$$2^k \cdot c \frac{N}{2^k} = cN,$$

independent of k . The recursion ends when $N/2^k = 1$, i.e. at depth $k = \log_2 N$: the tree has exactly $\log_2 N + 1$ levels (from $k = 0$ to $k = \log_2 N$), the last of which contains N leaves of base cost $O(1)$ each, for a total contribution $O(N)$. Summing the cost cN over each of the $\log_2 N$ internal levels, plus the leaf level, gives

$$T(N) = \underbrace{cN + cN + \cdots + cN}_{\log_2 N \text{ internal levels}} + \Theta(N) = cN \log_2 N + \Theta(N) = \Theta(N \log_2 N).$$

Remark 22.9: Scope of the result. The two proofs are the canonical methods for solving divide-and-conquer recurrences of the form $T(N) = aT(N/b) + f(N)$: inductive substitution and the recursion tree (equivalent, and underlying the general proof of the so-called “master” theorem); here they are carried out in full, rather than merely invoked, precisely because the FFT recurrence — the case $a = b = 2$, $f(N) = \Theta(N)$ — is the most cited example in algorithms and data structures exams. The same proof scheme applies, with the obvious modifications, to the recurrence of merge sort ($T(N) = 2T(N/2) + \Theta(N)$, identical) and to that of integer/polynomial multiplication with Karatsuba’s algorithm ($T(N) = 3T(N/2) + \Theta(N)$, which instead solves to $\Theta(N^{\log_2 3})$).

Recursive FFT algorithm (radix-2, decimation in time)

1. If $N = 1$: return $x[0]$ (base case).
2. Otherwise split x into $e[r] = x[2r]$ and $o[r] = x[2r + 1]$, $r = 0, \dots, N/2 - 1$.
3. Recursively compute $E = \text{FFT}(e)$ and $O = \text{FFT}(o)$ (each on $N/2$ points).
4. For $k = 0, \dots, N/2 - 1$, compute the twiddle $\zeta_N^k = e^{-2\pi i k/N}$ and combine: $X[k] = E[k] + \zeta_N^k O[k]$, $X[k + N/2] = E[k] - \zeta_N^k O[k]$. □

Listing 22.1: Recursive radix-2 FFT in Python (for teaching purposes; for real use, use `numpy.fft.fft`, which is optimized and not restricted to N a power of 2).

```
import numpy as np

def fft_radix2(x):
    """Recursive FFT, decimation in time. len(x) must be a power of
    2."""
    N = len(x)
    if N == 1:
        return x
    if N % 2 != 0:
        raise ValueError("N must be a power of 2")
    E = fft_radix2(x[0::2]) # sub-problem on even indices
    O = fft_radix2(x[1::2]) # sub-problem on odd indices
    twiddles = np.exp(-2j * np.pi * np.arange(N // 2) / N)
    butterfly = twiddles * O
    return np.concatenate([E + butterfly, E - butterfly])

# Check against numpy.fft.fft
x = np.array([1, -1, 1, -1, 1, -1, 1, -1], dtype=complex)
X_mine = fft_radix2(x)
X_numpy = np.fft.fft(x)
print(np.allclose(X_mine, X_numpy)) # True
print(np.round(X_mine, 6))
# [ 0.+0.j  0.+0.j  0.+0.j  0.+0.j  8.-0.j  0.+0.j  0.+0.j  0.+0.j ]
```

22.10.2 An SCL implementation for PLC

In industrial automation the DFT/FFT is used directly in vibration analysis, predictive diagnostics of pumps and motors, filtering of noisy analog signals before PID computation, and harmonic analysis of electrical networks (detection of THD, *total harmonic distortion*, in plants with inverter drives — relevant also in water treatment, where the frequency inverters of lift pumps are a common source of harmonic distortion). Unlike Python, a Siemens PLC (SCL, *Structured Control Language*, an IEC 61131-3 dialect of Structured Text) has no native complex

numbers nor convenient recursion: the FFT must therefore be written in *iterative* form, with real and imaginary parts kept in two separate arrays.

Adapting Cooley-Tukey to iterative form 1. Bit-reversal: the iterative version operates in place, but requires the input samples to be reordered according to the *bit-reversal* permutation of the binary index (e.g. for $N = 8$: index $1 = 001_2$ is swapped with $4 = 100_2$, $3 = 011_2$ with $6 = 110_2$, etc.) — a direct consequence of how the recurrence of Theorem 22.13 separates even/odd at each level.

2. Stages (butterflies) instead of recursion: for $N = 2^m$, m stages are executed; at stage s ($s = 1, \dots, m$) the butterflies operate on blocks of size 2^s , with base angle $\pi/2^{s-1}$ between one twiddle and the next.

3. Recursive twiddles: instead of calling cos/sin for every butterfly (expensive on a PLC), the twiddle is updated multiplicatively, $w \leftarrow w \cdot w_m$, where $w_m = e^{-2\pi i/2^s}$ is precomputed per stage. $\square$

Listing 22.2: FB_FFT8: iterative radix-2 FFT, $N = 8$ fixed points, in SCL for a Siemens PLC (TIA Portal).

```
FUNCTION_BLOCK FB_FFT8
// Radix-2 FFT, decimation in time, iterative, N = 8 fixed samples.
// In-place input/output: reArr/imArr hold the sampled signal on
// input and the spectrum (real/imaginary part) on output.
VAR_IN_OUT
    reArr : ARRAY[0..7] OF REAL;    // real part
    imArr : ARRAY[0..7] OF REAL;    // imaginary part
END_VAR
VAR
    // Bit-reversal permutation for N=8 (3-bit indices)
    bitRev : ARRAY[0..7] OF INT := [0, 4, 2, 6, 1, 5, 3, 7];
    // Base twiddle for each of the 3 stages: w_m = exp(-2*pi*i/m), m
    // = 2, 4, 8
    cosWm : ARRAY[1..3] OF REAL := [-1.0, 0.0, 0.70710678];
    sinWm : ARRAY[1..3] OF REAL := [ 0.0, -1.0, -0.70710678];
    mTab : ARRAY[1..3] OF INT := [2, 4, 8];
    i, j, k, m, half, stage : INT;
    tmpRe, tmpIm : REAL;
    wRe, wIm, wReNext, wImNext : REAL;
    uRe, uIm, tRe, tIm : REAL;
END_VAR

BEGIN
    // --- Phase 1: bit-reversal permutation (in-place swap) ---
    FOR i := 0 TO 7 DO
        IF bitRev[i] > i THEN
            tmpRe := reArr[i]; reArr[i] := reArr[bitRev[i]]; reArr[
                bitRev[i]] := tmpRe;
            tmpIm := imArr[i]; imArr[i] := imArr[bitRev[i]]; imArr[
                bitRev[i]] := tmpIm;
        END_IF;
    END_FOR;

    // --- Phase 2: log2(8) = 3 stages of butterflies ---
    FOR stage := 1 TO 3 DO
        m := mTab[stage];
        half := m / 2;
        FOR k := 0 TO 7 BY m DO
            wRe := 1.0; wIm := 0.0;    // initial twiddle of the stage,
```

```

        angle 0
    FOR j := 0 TO half - 1 DO
        // butterfly: (uRe,uIm) = reArr[k+j] ; (tRe,tIm) = w *
        // reArr[k+j+half]
        tRe := wRe * reArr[k + j + half] - wIm * imArr[k + j +
            half];
        tIm := wRe * imArr[k + j + half] + wIm * reArr[k + j +
            half];
        uRe := reArr[k + j];
        uIm := imArr[k + j];
        reArr[k + j] := uRe + tRe;
        imArr[k + j] := uIm + tIm;
        reArr[k + j + half] := uRe - tRe;
        imArr[k + j + half] := uIm - tIm;
        // update the twiddle for the next butterfly of the
        // stage (rotation)
        wReNext := wRe * cosWm[stage] - wIm * sinWm[stage];
        wImNext := wRe * sinWm[stage] + wIm * cosWm[stage];
        wRe := wReNext;
        wIm := wImNext;
    END_FOR;
END_FOR;
END_FUNCTION_BLOCK

```

Worked example: FB_FFT8 on a wave at half the sampling frequency. Take as input $x = [1, -1, 1, -1, 1, -1, 1, -1]$ (zero imaginary part), a signal that oscillates every sample: the highest frequency representable with $N = 8$ samples, i.e. exactly the Nyquist frequency (Section 4.12, $k = N/2 = 4$). Computing by hand from the definition,

$$X[4] = \sum_{n=0}^7 x[n]e^{-2\pi i \cdot 4n/8} = \sum_{n=0}^7 x[n](-1)^n = \sum_{n=0}^7 (-1)^n(-1)^n = \sum_{n=0}^7 1 = 8,$$

while for $k \neq 4$ the sum vanishes by the orthogonality of the roots of unity (Corollary 3.2, Chapter 3): the signal is a pure sinusoid at frequency $k = 4$, so all the energy is concentrated in that bin. Running FB_FFT8 on this input indeed gives

$$\text{reArr} = [0, 0, 0, 0, 8, 0, 0, 0], \quad \text{imArr} = [0, 0, 0, 0, 0, 0, 0, 0],$$

in exact agreement with the direct computation and with the Python output of the previous example — a useful cross-check when porting an algorithm from a development environment to a production PLC.

Remark 22.10: Why not use recursion directly on a PLC. IEC 61131-3 languages (SCL included) in theory allow recursive functions, but the cyclic execution of a PLC (scan cycle) and the static memory management of function blocks make recursion inadvisable in practice: every recursive call consumes task stack, whose size is limited and configured beforehand, and a deep recursive FFT can exceed it under unforeseen conditions (violating the real-time determinism constraints required by a control cycle). The iterative form of FB_FFT8, with a fixed number of iterations known in advance ($N \log_2 N/2$ butterflies), has a deterministic execution time: a property indispensable for a block called cyclically in an automation OB.

22.10.3 Windows and spectral leakage

The FFT of Section 4.11 implicitly assumes that the samples $x[0], \dots, x[N-1]$ constitute *exactly* one period (or an integer number of periods) of the signal. In practice a real signal is observed for a finite window of duration $T_{\text{fin}} = NT$ chosen by the operator, almost never in phase with the period of the signal: the DFT then implicitly sees the signal periodized with a **discontinuity** at the edges of the window — exactly the situation of Remark 17.3 (Gibbs), applied this time to a sampled signal instead of a square wave.

Definition 22.7: Windowing. Given a sequence $x[n]$, $n = 0, \dots, N-1$, and a window function $w[n] \geq 0$, the windowed signal is $x_w[n] = x[n]w[n]$ and its DFT $X_w[k] = \text{DFT}\{x_w[n]\}$ approximates the true spectrum of x .

Remark 22.11: Leakage: energy scattered into neighboring bins. If $x[n] = \cos(2\pi f_0 nT)$ and f_0 falls exactly on a bin (i.e. $f_0 = k_0 f_s/N$ for some integer k_0), the implicit rectangular window ($w[n] \equiv 1$) gives a spectrum perfectly concentrated at k_0 (and at its mirror $N - k_0$), consistently with the Theorem (Volume II, Chapter 4). But if f_0 falls between two bins, the energy scatters (leakage) over all neighboring bins with a slow decay: the rectangular window has side lobes that decay by only ≈ 6 dB per octave, the slowest among the windows used in practice.

Definition 22.8: Hann window. The Hann window gradually tapers the samples toward the edges of the window, eliminating the implicit discontinuity:

$$w_{\text{Hann}}[n] = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{N-1} \right), \quad n = 0, \dots, N-1.$$

Worked example — rectangular/Hann numerical comparison. Let $N = 64$, $f_s = 100$ Hz (bins spaced by $f_s/N \approx 1.56$ Hz). For a tone exactly on bin 8 ($f_0 = 12.5$ Hz), the DFT with a rectangular window is perfectly concentrated: all neighboring bins are at -240 dB from the peak (numerical zero). Moving the tone halfway between two bins ($f_0 = 13.28$ Hz, i.e. 8.5 cycles in the window), the same rectangular window shows marked leakage: the adjacent bin (bin 7) is only -9.1 dB below the peak, and distant bins such as bin 4 remain at only -17.0 dB. Applying instead the Hann window to the same misaligned signal: the main lobe widens (bins 8 and 9 are now nearly matched, both close to 0 dB), but the distant lobes decay much faster (-30.5 dB at bin 6, -46.9 dB at bin 4, against -13.1 dB and -17.0 dB respectively with the rectangular window). This is the classic trade-off: the Hann window sacrifices resolution (wider main lobe) to gain suppression of distant leakage (lower side lobes).

Listing 22.3: Rectangular/Hann numerical comparison (Python, NumPy).

```
import numpy as np

N, fs = 64, 100.0
n = np.arange(N)
f0 = 8.5 * fs / N          # 8.5 cycles in the window: not aligned
                           # to a bin

x = np.sin(2*np.pi*f0*n/fs)
rect = np.ones(N)
hann = 0.5 - 0.5*np.cos(2*np.pi*n/(N-1))  # identical to numpy.hanning(N)

def spectrum_db(x, w):
```

```

X = np.fft.fft(x*w)
mag = np.abs(X[:N//2])
return 20*np.log10(mag/np.max(mag))

db_rect = spectrum_db(x, rect)
db_hann = spectrum_db(x, hann)
print(db_rect[6:11])    # compare with library numpy.hanning(N)
print(db_hann[6:11])

```

Remark 22.12: Frequency resolution and practical consequences. Regardless of the window chosen, the spacing between bins is always $\Delta f = f_s/N$: distinguishing two nearby tones requires a long enough window (large N), not a different window — the window fights leakage, it does not improve resolution. In industrial vibration analysis (bearing diagnostics, misalignment, imbalance) the choice of window is anything but academic: uncontrolled leakage can mask a weak fault component under the lobe of a nearby strong component (for example the fundamental rotation frequency), exactly the situation of Remark 22.11.

22.11. Windows: coherent gain, equivalent bandwidth, resolution

Section 22.10.3 introduced windowing and showed, on a numerical example, the trade-off between main-lobe width and side-lobe height. This section proves in closed form the three quantities that quantify that trade-off — coherent gain, equivalent noise bandwidth, main-lobe width — for the three windows in common use.

Definition 22.9: The three windows, in periodic and symmetric form. Let $N \geq 3$. For $n = 0, 1, \dots, N-1$, define:

$$\begin{aligned}
\text{rectangular} \quad w_R[n] &= 1, \\
\text{Hann} \quad w_H[n] &= \frac{1}{2} - \frac{1}{2} \cos \frac{2\pi n}{N}, \quad (\text{periodic form}) \\
\text{Hamming} \quad w_M[n] &= 0.54 - 0.46 \cos \frac{2\pi n}{N}, \quad (\text{periodic form}).
\end{aligned}$$

The corresponding *symmetric forms* are obtained by substituting $N-1$ for N in the denominator of the cosine argument; this is the convention of Definition 22.8 and that of `numpy.hanning`. All three have the common form

$$w[n] = a_0 - a_1 \cos \frac{2\pi n}{N} \quad \text{with} \quad (a_0, a_1) = (1, 0), \left(\frac{1}{2}, \frac{1}{2}\right), \left(\frac{27}{50}, \frac{23}{50}\right),$$

since $0.54 = 27/50$ and $0.46 = 23/50$.

Definition 22.10: Coherent gain and equivalent noise bandwidth. Given a window $w \in \mathbb{R}^N$ with $\sum_n w[n] \neq 0$, define

$$\begin{aligned}
\text{CG}(w) &= \frac{1}{N} \sum_{n=0}^{N-1} w[n] \quad (\text{coherent gain}), \\
\text{ENBW}(w) &= \frac{N \sum_{n=0}^{N-1} w[n]^2}{\left(\sum_{n=0}^{N-1} w[n]\right)^2} \quad (\text{equivalent noise bandwidth, in bins}).
\end{aligned}$$

The first is the factor by which the peak of the windowed spectrum must be divided to recover the true amplitude of a tone aligned to a bin; the second is the width, measured in bins, of the ideal rectangular filter that would let through the same white-noise power as the filter realized by the window.

Lemma 22.4: *Two exact trigonometric sums. Let $N \geq 3$. Then*

$$\sum_{n=0}^{N-1} \cos \frac{2\pi n}{N} = 0, \quad \sum_{n=0}^{N-1} \cos^2 \frac{2\pi n}{N} = \frac{N}{2}, \quad \sum_{n=0}^{N-1} \cos \frac{4\pi n}{N} = 0.$$

If moreover $N \geq 4$, $\sum_{n=0}^{N-1} \cos \frac{2\pi n}{N-1} = 1$.

Proof. By Euler's formula, $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, hence

$$\sum_{n=0}^{N-1} \cos \frac{2\pi n}{N} = \frac{1}{2} \sum_{n=0}^{N-1} \omega_N^{-n} + \frac{1}{2} \sum_{n=0}^{N-1} \omega_N^n.$$

By Lemma 22.1 with $j = -1$ and with $j = 1$: since $N \geq 3 > 1$, one has $N \nmid 1$ and $N \nmid (-1)$, so both sums vanish and the total is 0.

For the third sum, the same lemma is applied with $j = \pm 2$: since $N \geq 3$, one has $N \mid 2$ only for $N = 2$, excluded, so both sums vanish.

The second follows from the double-angle formula $\cos^2 \theta = \frac{1+\cos 2\theta}{2}$:

$$\sum_{n=0}^{N-1} \cos^2 \frac{2\pi n}{N} = \frac{1}{2} \sum_{n=0}^{N-1} 1 + \frac{1}{2} \sum_{n=0}^{N-1} \cos \frac{4\pi n}{N} = \frac{N}{2} + 0 = \frac{N}{2}.$$

For the last: split the sum into the first $N-1$ terms (indices $n = 0, \dots, N-2$) and the last one. The first $N-1$ terms are $\sum_{n=0}^{(N-1)-1} \cos \frac{2\pi n}{N-1}$, i.e. the first sum of the statement with $N-1$ in place of N : it equals 0 because $N-1 \geq 3$. The last term is $\cos \frac{2\pi(N-1)}{N-1} = \cos 2\pi = 1$. The total is 1. $\square$

Proposition 22.6: *Coherent gain of the three windows. Let $N \geq 3$ and let $w[n] = a_0 - a_1 \cos(2\pi n/N)$ be a window in periodic form. Then*

$$\sum_{n=0}^{N-1} w[n] = a_0 N, \quad \text{CG}(w) = a_0.$$

In particular: $\text{CG}(w_R) = 1$, $\text{CG}(w_H) = \frac{1}{2}$, $\text{CG}(w_M) = 0.54$. If instead the window is in symmetric form, $w[n] = a_0 - a_1 \cos(2\pi n/(N-1))$ with $N \geq 4$, then

$$\sum_{n=0}^{N-1} w[n] = a_0 N - a_1, \quad \text{CG}(w) = a_0 - \frac{a_1}{N}.$$

Proof. In the periodic case, by linearity of the sum and by Lemma 22.4,

$$\sum_{n=0}^{N-1} w[n] = a_0 \sum_{n=0}^{N-1} 1 - a_1 \sum_{n=0}^{N-1} \cos \frac{2\pi n}{N} = a_0 N - a_1 \cdot 0 = a_0 N,$$

and $\text{CG} = \frac{1}{N} \cdot a_0 N = a_0$ by Definition 22.10. In the symmetric case, with the fourth identity of Lemma 22.4, $\sum_n w[n] = a_0 N - a_1 \cdot 1 = a_0 N - a_1$, and one divides by N .

Numerical check. For $N = 64$: periodic Hann form, $\sum w = 32$ and $\text{CG} = 0.50000$; symmetric form, $\sum w = a_0 N - a_1 = 32 - 0.5 = 31.5$ and $\text{CG} = 31.5/64 = 0.49219 = 63/128$. For symmetric Hamming, $\sum w = 0.54 \cdot 64 - 0.46 = 34.56 - 0.46 = 34.10$ and $\text{CG} = 0.53281 = 0.54 - 0.46/64$. The script of Section 22.12.1 returns 32, 31.5, 34.56, 34.10 and the gains 0.50000, 0.49219, 0.54000, 0.53281 ✓. $\square$

Proposition 22.7: Equivalent noise bandwidth of the three windows. Let $N \geq 3$ and let $w[n] = a_0 - a_1 \cos(2\pi n/N)$ be a window in periodic form with $a_0 \neq 0$. Then

$$\sum_{n=0}^{N-1} w[n]^2 = N \left(a_0^2 + \frac{a_1^2}{2} \right), \quad \text{ENBW}(w) = 1 + \frac{a_1^2}{2a_0^2}.$$

In particular

$$\text{ENBW}(w_R) = 1, \quad \text{ENBW}(w_H) = \frac{3}{2} = 1.5, \quad \text{ENBW}(w_M) = \frac{1987}{1458} = 1.36283.$$

Proof. Expanding the square and using the first three identities of Lemma 22.4:

$$\begin{aligned} \sum_{n=0}^{N-1} \left(a_0 - a_1 \cos \frac{2\pi n}{N} \right)^2 &= a_0^2 \sum_n 1 - 2a_0 a_1 \sum_n \cos \frac{2\pi n}{N} + a_1^2 \sum_n \cos^2 \frac{2\pi n}{N} \\ &= a_0^2 N - 0 + a_1^2 \cdot \frac{N}{2} = N \left(a_0^2 + \frac{a_1^2}{2} \right). \end{aligned}$$

By Proposition 22.6 one has $\sum_n w[n] = a_0 N$, hence

$$\text{ENBW}(w) = \frac{N \cdot N \left(a_0^2 + \frac{a_1^2}{2} \right)}{(a_0 N)^2} = \frac{a_0^2 + \frac{a_1^2}{2}}{a_0^2} = 1 + \frac{a_1^2}{2a_0^2}.$$

The three values follow by substitution. For the rectangular window, $a_1 = 0$ gives 1. For Hann, $a_1^2/(2a_0^2) = \frac{1/4}{2 \cdot 1/4} = \frac{1}{2}$, hence $\frac{3}{2}$. For Hamming, with $a_0 = 27/50$ and $a_1 = 23/50$,

$$\text{ENBW} = 1 + \frac{(23/50)^2}{2(27/50)^2} = 1 + \frac{529}{2 \cdot 729} = 1 + \frac{529}{1458} = \frac{1987}{1458},$$

and the division gives $1987/1458 = 1.362825\dots$, i.e. 1.36283 to five decimal places.

Numerical check. For $N = 64$ and $N = 1024$ the script of Section 22.12.1 computes ENBW from the definition and obtains 1.00000 (rectangular), 1.50000 (periodic Hann) and 1.36283 (periodic Hamming), independently of N as predicted by the formula ✓. For the symmetric forms the values depend on N and equal 1.52381 and 1.37832 for $N = 64$, approaching the periodic values as N grows (1.50147 and 1.36378 for $N = 1024$), consistently with the fact that the difference between the two forms is a single sample out of N . $\square$

Theorem 22.15: The DTFT of a two-term window. Let $w[n] = a_0 - a_1 \cos(2\pi n/N)$, $n = 0, \dots, N-1$, and let $\hat{w}(\Omega) = \sum_{n=0}^{N-1} w[n]e^{-in\Omega}$ be its DTFT. Then

$$\hat{w}(\Omega) = a_0 D_N(\Omega) - \frac{a_1}{2} D_N\left(\Omega - \frac{2\pi}{N}\right) - \frac{a_1}{2} D_N\left(\Omega + \frac{2\pi}{N}\right),$$

with D_N the discrete Dirichlet kernel (Definition 22.6). In particular, for the rectangular window ($a_1 = 0$) one has $\hat{w}_R = D_N$.

Proof. By Euler's formula, $\cos \frac{2\pi n}{N} = \frac{1}{2}(e^{2\pi in/N} + e^{-2\pi in/N})$, hence

$$w[n] = a_0 - \frac{a_1}{2}e^{2\pi in/N} - \frac{a_1}{2}e^{-2\pi in/N}.$$

Multiplying by $e^{-in\Omega}$ and summing over $n = 0, \dots, N-1$ (a finite sum, the three pieces separate):

$$\hat{w}(\Omega) = a_0 \sum_{n=0}^{N-1} e^{-in\Omega} - \frac{a_1}{2} \sum_{n=0}^{N-1} e^{-in(\Omega - \frac{2\pi}{N})} - \frac{a_1}{2} \sum_{n=0}^{N-1} e^{-in(\Omega + \frac{2\pi}{N})},$$

having used $e^{2\pi in/N} e^{-in\Omega} = e^{-in(\Omega - 2\pi/N)}$ and similarly for the other term. The three sums are, by Definition 22.6, $D_N(\Omega)$, $D_N(\Omega - 2\pi/N)$ and $D_N(\Omega + 2\pi/N)$. $\square$

Proposition 22.8: Main-lobe width. Let $N \geq 5$ and measure frequency in bins, setting $\Omega = 2\pi f/N$. Then:

- (i) for the rectangular window, $\hat{w}_R(2\pi m/N) = 0$ for every integer m with $1 \leq |m| \leq N-1$: the first zero falls at ± 1 bin from the center and the main lobe is 2 bins wide between the two zeros;
- (ii) for a two-term window with $a_1 \neq 0$ (Hann and Hamming), $\hat{w}(2\pi m/N) = 0$ for every integer m with $2 \leq |m| \leq N-2$, while $\hat{w}(\pm 2\pi/N) = -\frac{a_1}{2}N \neq 0$: the first zero falls at ± 2 bins from the center and the main lobe is 4 bins wide;
- (iii) in every case the spacing between bins remains $\Delta f = f_s/N$, and does not depend on the window.

Proof. Evaluate the formula of Theorem 22.15 at $\Omega = 2\pi m/N$ with $m \in \mathbb{Z}$. The three arguments of the kernel become $\frac{2\pi m}{N}$, $\frac{2\pi(m-1)}{N}$ and $\frac{2\pi(m+1)}{N}$, and by Lemma 22.3 one has $D_N(2\pi r/N) = N$ if $N \mid r$ and 0 otherwise. Hence

$$\hat{w}\left(\frac{2\pi m}{N}\right) = a_0 N [N \mid m] - \frac{a_1}{2} N [N \mid (m-1)] - \frac{a_1}{2} N [N \mid (m+1)],$$

where $[\cdot]$ equals 1 if the condition holds and 0 otherwise.

(i) With $a_1 = 0$ only the first term survives: $\hat{w}_R(2\pi m/N) = 0$ for every m not a multiple of N , in particular for $1 \leq |m| \leq N-1$. The central value is $\hat{w}_R(0) = N \neq 0$, so the zeros closest to the center are $m = \pm 1$ and the main lobe, understood as the interval between the two zeros adjacent to the maximum, is $(-1, 1)$ in bins, of width 2.

(ii) Let $2 \leq |m| \leq N-2$. Then m is not a multiple of N (since $0 < |m| < N$); neither is $m-1$, because $1 \leq m-1 \leq N-3$ if $m > 0$, and $-N+1 \leq m-1 \leq -3$ if $m < 0$, in both cases $0 < |m-1| < N$; similarly $m+1$ is not a multiple of N . All three indicators vanish, hence $\hat{w}(2\pi m/N) = 0$. For $m = 1$ only the term with $N \mid (m-1)$ survives, i.e. $N \mid 0$, true: $\hat{w}(2\pi/N) = -\frac{a_1}{2}N$, nonzero because $a_1 \neq 0$. Symmetrically for $m = -1$. The central value is $\hat{w}(0) = a_0 N \neq 0$. The zeros closest to the maximum are thus $m = \pm 2$, and the main lobe is 4 bins wide.

(iii) The spacing between bins is, by Theorem 22.9, the distance between two consecutive sampling points of the DTFT, i.e. $\Delta\Omega = 2\pi/N$; in frequency units, with $f = \Omega f_s/(2\pi)$, this is $\Delta f = f_s/N$. The computation does not involve w in any way.

Numerical check. Evaluating the DTFT of the three windows with $N = 64$ on 2^{16} points (zero-padding), the first zero falls at 1.0010 bins for the rectangular window and at 2.0010 bins for periodic Hann and Hamming, where the discrepancy from 1 and from 2 is exactly the resolution of the grid used ($64/2^{16} = 0.0010$ bin) $\checkmark$. $\square$

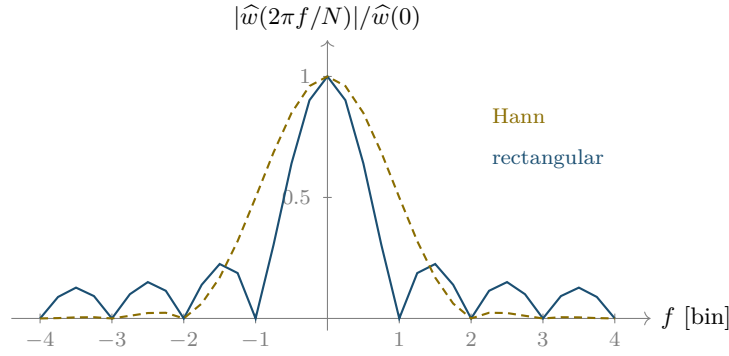


Figure 22.3: Normalized modulus of the DTFT of the rectangular and (periodic) Hann windows, for $N = 8$, as a function of frequency measured in bins. The ordinates are $|D_8(2\pi f/8)|/8$ and the analogous expression for Hann, computed with the closed form of Lemma 22.3 and with the three-kernel decomposition of Theorem 22.15: the two routes agree to within $2.5 \cdot 10^{-16}$. The zeros predicted by Proposition 22.8 are visible: at every nonzero integer for the rectangular window, only from $|f| \geq 2$ for Hann, whose main lobe is thus twice as wide; in exchange, at $f = 2.5$ the rectangular window equals 0.15030 and Hann 0.02236, i.e. about seven times less.

Remark 22.13: Side lobes: computed values, and what remains imported. The height of the highest side lobe, relative to the maximum, is computed by evaluating $|\hat{w}|$ on a fine grid: with $N = 64$ and 2^{16} points one obtains

window	side-lobe peak	width at -3 dB
rectangular	-13.25 dB	0.88 bin
Hann (periodic)	-31.47 dB	1.44 bin
Hamming (periodic)	-42.45 dB	1.30 bin

These values are computed, not cited, and are consistent with the ordering and the orders of magnitude of Harris's (1978) comparative table of windows, which remains the standard reference for systematic comparison.

What remains imported. The asymptotic law of decay of the far side lobes — about -6 dB per octave for rectangular and for Hamming, about -18 dB per octave for Hann — is here **imported** from Harris (1978) and not proved. What is missing, precisely: one would need an asymptotic expansion of $|\hat{w}(\Omega)|$ for Ω far from the origin, showing how in the three-kernel combination of Theorem 22.15 the terms of order $1/\Omega$ and $1/\Omega^2$ cancel for Hann (thanks to $a_0 = 2 \cdot \frac{a_1}{2}$) but not for Hamming; the computation is elementary but long, and nothing in the chapter relies on it. The three quantities actually used — coherent gain (Proposition 22.6), equivalent bandwidth (Proposition 22.7) and main-lobe width (Proposition 22.8) — are instead proved in closed and exact form.

Best practice. Choosing the window. In order of priority: **(1)** if the signal is periodic and an integer number of periods can be acquired, use the rectangular window: there is no leakage and the resolution is the best possible. **(2)** If an *amplitude* is being measured and the tone is not aligned to a bin, use Hann and divide the peak by $\text{CG} = 0.5$ (Proposition 22.6); the residual *scallop* error remains regardless, and canceling it requires interpolating between bins. **(3)** If a weak component must be found near a strong one, use Hamming or a window with even lower lobes: resolution is lost but one gains the 29 dB difference between the side-lobe peaks of rectangular and Hamming. **(4)** If *noise power* is being measured, divide by ENBW (Proposition 22.7): 1.5 for Hann, 1.36283 for Hamming. **(5)** In every

case, doubling the resolution requires doubling N , i.e. the observation time: no window and no zero-padding can replace it (Theorem 22.11).

22.12. An eight-point DFT by hand

Algorithm. $X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}$, with $N = 8$.

Given. The rectangular window $x = (1, 1, 1, 1, 0, 0, 0, 0)$.

Computation. Since $x_n = 1$ only for $n \leq 3$, each X_k is a geometric sum with ratio $w = e^{-2\pi i k / 8}$:

$$X_k = 1 + w + w^2 + w^3 = \frac{1 - w^4}{1 - w} \quad (w \neq 1), \quad X_0 = 4.$$

Since $w^4 = e^{-i\pi k} = (-1)^k$, the numerator equals 0 for even k and 2 for odd k : hence $X_2 = X_4 = X_6 = 0$ with no computation at all. For odd k :

k	X_k	$\Re X_k$	$\Im X_k$	$ X_k $
0	4	4	0	4.00000
1	$1 - 2.41421 i$	1	-2.41421	2.61313
2	0	0	0	0
3	$1 - 0.41421 i$	1	-0.41421	1.08239
4	0	0	0	0
5	$1 + 0.41421 i$	1	$+0.41421$	1.08239
6	0	0	0	0
7	$1 + 2.41421 i$	1	$+2.41421$	2.61313

For example $X_1 = 1 + e^{-i\pi/4} + e^{-i\pi/2} + e^{-i3\pi/4}$, and summing the real and imaginary parts: $\Re = 1 + 0.70711 + 0 - 0.70711 = 1$, $\Im = 0 - 0.70711 - 1 - 0.70711 = -2.41421$. Note the Hermitian symmetry $X_{8-k} = \overline{X_k}$, due to the fact that x is real.

Independent check (Parseval).

$$\sum_k |X_k|^2 = 16 + 6.82843 + 0 + 1.17157 + 0 + 1.17157 + 0 + 6.82843 = 32,$$

while $N \sum_n |x_n|^2 = 8 \cdot 4 = 32 \checkmark$. Parseval's identity is the most effective check of an FFT implementation: it catches any normalization error or sign error in the exponent.

22.12.1 The eight-point example, worked in full

The computation just given is now taken up again and carried through: all the X_k are obtained in exact symbolic form, inversion is checked on *all eight* samples, Hermitian symmetry and Parseval are checked, and the executed script is reported together with its output.

Step 1 — exact values, symbolically. With $N = 8$ and $\omega_8 = e^{-i\pi/4}$ one has, for fixed k , $w = \omega_8^k$ and

$$X_k = \sum_{n=0}^3 w^n = 1 + w + w^2 + w^3.$$

If $k = 0$ then $w = 1$ and the closed formula does not apply (vanishing denominator, Lemma 22.1): one sums directly, $X_0 = 4$. If $k \neq 0$ then $w \neq 1$ (the powers ω_8^k , $k = 0, \dots, 7$, are distinct by Proposition 22.3) and

$$X_k = \frac{1 - w^4}{1 - w}, \quad w^4 = \omega_8^{4k} = e^{-i\pi k} = (-1)^k.$$

For even k the numerator is $1 - 1 = 0$, hence $X_2 = X_4 = X_6 = 0$ with no computation at all. For odd k the numerator equals 2 and $X_k = 2/(1 - \omega_8^k)$ remains. Now $1 - e^{-i\varphi} = e^{-i\varphi/2}(e^{i\varphi/2} - e^{-i\varphi/2}) = 2i \sin(\varphi/2) e^{-i\varphi/2}$, hence with $\varphi = \pi k/4$:

$$X_k = \frac{2}{2i \sin(\pi k/8) e^{-i\pi k/8}} = \frac{e^{i\pi k/8}}{i \sin(\pi k/8)} = \frac{\cos(\pi k/8) + i \sin(\pi k/8)}{i \sin(\pi k/8)} = 1 - i \cot \frac{\pi k}{8}.$$

The real part of every X_k with k odd is thus *exactly* 1, and the imaginary part is $-\cot(\pi k/8)$:

k	$\cot(\pi k/8)$	X_k
1	$\cot(\pi/8) = 1 + \sqrt{2}$	$1 - (1 + \sqrt{2})i$
3	$\cot(3\pi/8) = \sqrt{2} - 1$	$1 - (\sqrt{2} - 1)i$
5	$\cot(5\pi/8) = -(\sqrt{2} - 1)$	$1 + (\sqrt{2} - 1)i$
7	$\cot(7\pi/8) = -(1 + \sqrt{2})$	$1 + (1 + \sqrt{2})i$

The two cotangent values are obtained from the bisection formula $\cot \frac{\theta}{2} = \frac{1+\cos\theta}{\sin\theta}$: for $\theta = \pi/4$, $\cot(\pi/8) = \frac{1+\sqrt{2}/2}{\sqrt{2}/2} = \frac{2+\sqrt{2}}{\sqrt{2}} = \sqrt{2} + 1$; for $\theta = 3\pi/4$, $\cot(3\pi/8) = \frac{1-\sqrt{2}/2}{\sqrt{2}/2} = \frac{2-\sqrt{2}}{\sqrt{2}} = \sqrt{2} - 1$. The two rows with $k = 5, 7$ follow from Hermitian symmetry (Theorem 22.7), since x is real: $X_5 = \overline{X_3}$ and $X_7 = \overline{X_1}$. Numerically $1 + \sqrt{2} = 2.41421$ and $\sqrt{2} - 1 = 0.41421$, in agreement with the table of Section 22.12 ✓.

Step 2 — the moduli. $|X_0| = 4$; $|X_1|^2 = 1 + (1 + \sqrt{2})^2 = 1 + 3 + 2\sqrt{2} = 4 + 2\sqrt{2} = 6.82843$, hence $|X_1| = |X_7| = \sqrt{4 + 2\sqrt{2}} = 2.61313$; similarly $|X_3|^2 = 1 + (\sqrt{2} - 1)^2 = 1 + 3 - 2\sqrt{2} = 4 - 2\sqrt{2} = 1.17157$ and $|X_3| = |X_5| = 1.08239$. The script below returns (4.00000; 2.61313; 0; 1.08239; 0; 1.08239; 0; 2.61313) ✓.

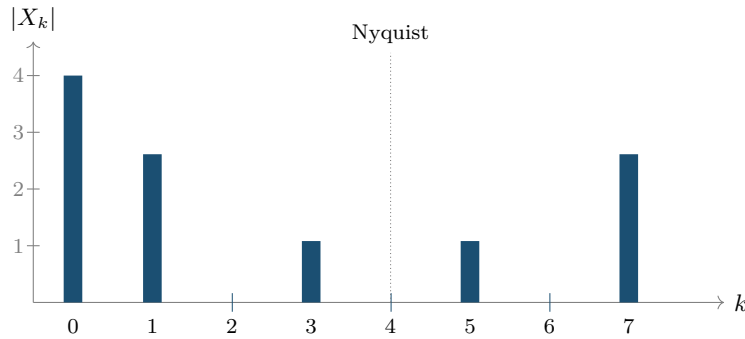


Figure 22.4: Modulus of the spectrum of $x = (1, 1, 1, 1, 0, 0, 0, 0)$. The heights are the exact values 4, $\sqrt{4 + 2\sqrt{2}} = 2.61313$, 0, $\sqrt{4 - 2\sqrt{2}} = 1.08239$, 0, 1.08239, 0, 2.61313, computed by hand in Step 2 and cross-checked with the script. The symmetry about bin 4 (Nyquist frequency) is that of Corollary 22.8(iii): $|X_{8-k}| = |X_k|$.

Step 3 — inversion, checked on all eight samples. Since $X_2 = X_4 = X_6 = 0$, $X_5 = \overline{X_3}$ and $X_7 = \overline{X_1}$, and since $\omega_8^{-7n} = \overline{\omega_8^{-n}}$ and $\omega_8^{-5n} = \overline{\omega_8^{-3n}}$ (powers depend on the exponent modulo 8 and $|\omega_8| = 1$), the inversion formula of Theorem 22.2 reduces to

$$x[n] = \frac{1}{8} \left(4 + 2 \operatorname{Re} (X_1 e^{i\pi n/4}) + 2 \operatorname{Re} (X_3 e^{3i\pi n/4}) \right).$$

Writing $X_1 = 1 - (1 + \sqrt{2})i$ and using $\operatorname{Re}((a + bi)e^{i\theta}) = a \cos \theta - b \sin \theta$ with $\cos(\pi/4) =$

$\sin(\pi/4) = \sqrt{2}/2$, one obtains the exact values

n	0	1	2	3	4	5	6	7
$2 \operatorname{Re}(X_1 e^{i\pi n/4})$	2	$2(1+\sqrt{2})$	$2(1+\sqrt{2})$	2	-2	$-2(1+\sqrt{2})$	$-2(1+\sqrt{2})$	-2
$2 \operatorname{Re}(X_3 e^{3i\pi n/4})$	2	$-2(\sqrt{2}-1)$	$-2(\sqrt{2}-1)$	2	-2	$2(\sqrt{2}-1)$	$2(\sqrt{2}-1)$	-2
sum + 4	8	8	8	8	0	0	0	0
$x[n] = \text{sum}/8$	1	1	1	1	0	0	0	0

Let us check two columns in full. For $n = 1$: $\operatorname{Re}(X_1 e^{i\pi/4}) = 1 \cdot \frac{\sqrt{2}}{2} + (1 + \sqrt{2}) \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2}(2 + \sqrt{2}) = \sqrt{2} + 1$, hence the first term is $2(1 + \sqrt{2})$; and $\operatorname{Re}(X_3 e^{3i\pi/4}) = 1 \cdot (-\frac{\sqrt{2}}{2}) + (\sqrt{2} - 1) \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2}(\sqrt{2} - 2) = 1 - \sqrt{2}$, hence the second term is $-2(\sqrt{2} - 1)$. The sum of the two is $2(1 + \sqrt{2}) - 2(\sqrt{2} - 1) = 2 + 2\sqrt{2} - 2\sqrt{2} + 2 = 4$; with the 4 of the constant term this makes 8, and dividing by 8 gives $x[1] = 1 \checkmark$. For $n = 5$ the arguments are $5\pi/4$ and $15\pi/4 \equiv 7\pi/4 \pmod{2\pi}$, with $\cos(5\pi/4) = \sin(5\pi/4) = -\frac{\sqrt{2}}{2}$ and $\cos(7\pi/4) = -\sin(7\pi/4) = \frac{\sqrt{2}}{2}$; hence

$$\operatorname{Re}(X_1 e^{5i\pi/4}) = -\frac{\sqrt{2}}{2} - (1 + \sqrt{2}) \frac{\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}(2 + \sqrt{2}) = -(1 + \sqrt{2}),$$

$$\operatorname{Re}(X_3 e^{15i\pi/4}) = \frac{\sqrt{2}}{2} - (\sqrt{2} - 1) \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2}(2 - \sqrt{2}) = \sqrt{2} - 1,$$

whose doubles are $-2(1 + \sqrt{2})$ and $2(\sqrt{2} - 1)$; their sum is -4 , which with the constant 4 gives 0, hence $x[5] = 0 \checkmark$. The other six columns are obtained in the same way and are given in the table: *all eight* samples are reconstructed exactly.

Step 4 — Parseval, again. The check of Section 22.12 is now a case of Theorem 22.6:

$$\sum_{k=0}^7 |X_k|^2 = 16 + 2(4 + 2\sqrt{2}) + 2(4 - 2\sqrt{2}) = 16 + 8 + 8 = 32,$$

where the terms in $\sqrt{2}$ cancel in pairs, and $N \sum_n |x_n|^2 = 8 \cdot 4 = 32 \checkmark$. Note that the identity holds *exactly*, with no rounding.

Step 5 — the executed script. The listing that follows is the script actually executed for the chapter; the lines after each group of statements give its literal output.

Listing 22.4: Numerical checks of Chapter 22 (Python 3 with NumPy). Executed; the output is reported at the end of the listing.

```
import numpy as np

# --- 1. Eight-point DFT of the rectangular window -----
x = np.array([1, 1, 1, 1, 0, 0, 0, 0], dtype=complex)
X = np.fft.fft(x)
print("X =", np.round(X, 5))
print("|X|=", np.round(np.abs(X), 5))
print("IDFT(X) =", np.round(np.fft.ifft(X).real, 12))
reduced = [(4 + 2*(X[1]*np.exp(1j*np.pi*n/4)).real
            + 2*(X[3]*np.exp(3j*np.pi*n/4)).real)/8 for n in range(8)]
print("reduced formula =", np.round(reduced, 12))
print("sum |X|^2 =", np.sum(np.abs(X)**2), " N*sum |x|^2 =", 8*np.sum(np.
    abs(x)**2))

# --- 2. DFT matrix: unitarity, F^4, determinants -----
def W(N): n = np.arange(N); return np.exp(-2j*np.pi*np.outer(n, n)/N)
for N in (4, 8):
```

```

F = W(N)/np.sqrt(N)
vander = np.prod([np.exp(-2j*np.pi*l/N) - np.exp(-2j*np.pi*k/N)
                  for k in range(N) for l in range(k+1, N)])
print(f"N={N}: |FF*-I|={np.max(np.abs(F@F.conj().T-np.eye(N))):.2e}"),
      f" |F^4-I|={np.max(np.abs(np.linalg.matrix_power(F,4)-np.eye(N))):.2e}",
      f" det W={np.linalg.det(W(N)):.5f}   Vandermonde={vander:.5f}")

# --- 3. circular convolution, circulant, linear system -----
a = np.array([1, 2, 3, 4], dtype=complex); b = np.array([1, 0, -1, 0],
dtype=complex)
direct = np.array([sum(a[m]*b[(n-m) % 4] for m in range(4)) for n in
range(4)])
print("direct conv =", direct.real, " via DFT =",
      np.round(np.fft.ifft(np.fft.fft(a)*np.fft.fft(b)).real, 10))
c = np.array([1, 2, 3, 4], dtype=complex)
C = np.array([[c[(i-j) % 4] for j in range(4)] for i in range(4)])
print("eigenvalues of C =", np.round(np.fft.fft(c), 5), " det C =", np.
      linalg.det(C).real)
y = np.fft.ifft(np.fft.fft([5, 3, 5, 7])/np.fft.fft(c))
print("solution of Cy=b :", np.round(y.real, 12))

# --- 4. zero-padding and Dirichlet interpolation
-----
XM = np.fft.fft(x, 32)
def D(N, th): return N if abs(np.sin(th/2)) < 1e-14 else \
      np.exp(-1j*(N-1)*th/2)*np.sin(N*th/2)/np.sin(th/2)
interp = [sum(X[j]*D(8, 2*np.pi*(k/32 - j/8)) for j in range(8))/8 for k
in range(32)]
print("interpolation error =", max(abs(XM[k]-interp[k]) for k in range
(32)))
print("X32[2] =", np.round(XM[2], 5), " |X32[2]| =", round(abs(XM[2]),
5))

# --- 5. windows: coherent gain and equivalent bandwidth
-----
for N in (64, 1024):
    n = np.arange(N)
    for name, w in (("rectangular", np.ones(N)),
                    ("Hann per", 0.5-0.5*np.cos(2*np.pi*n/N)),
                    ("Hann sym", 0.5-0.5*np.cos(2*np.pi*n/(N-1))),
                    ("Hamming per", 0.54-0.46*np.cos(2*np.pi*n/N)),
                    ("Hamming sym", 0.54-0.46*np.cos(2*np.pi*n/(N-1)))):
        print(f"N={N:5d} {name:12s} sum={w.sum():10.5f} CG={w.sum()/N:.5
f}")
        f"   ENBW={N*np.sum(w**2)/w.sum()*2:.5f}")

# --- 6. side lobes (2^16 grid)
-----
N, M = 64, 1 << 16
n = np.arange(N)
for name, w in (("rectangular", np.ones(N)),
                ("Hann per", 0.5-0.5*np.cos(2*np.pi*n/N)),
                ("Hamming per", 0.54-0.46*np.cos(2*np.pi*n/N))):
    A = np.abs(np.fft.fft(w, M)); db = 20*np.log10(np.maximum(A[:M//2],
1e-300)/A[0])
    i = 1

```

```

while db[i] < db[i-1]: i += 1
j = int(np.argmax(db < -3.0))
print(f"{name:12s} first zero at {i*N/M:.4f} bin, side-lobe peak "
      f"{max(db[i:]):.2f} dB, -3 dB width {2*j*N/M:.2f} bin")

# --- 7. multiplicities of the eigenvalues of F_N
-----
for N in (4, 5, 6, 7, 8, 12):
    ev = np.linalg.eigvals(W(N)/np.sqrt(N))
    m = [int(sum(abs(v-t) < 1e-8 for v in ev)) for t in (1, -1j, -1, 1j)
          ]
    print(f"N={N:3d}   mult.(1,-i,-1,i) = {tuple(m)}   sum = {sum(m)}")

# ----- OUTPUT -----
# X = [4.+0.j  1.-2.41421j  0.+0.j  1.-0.41421j  0.+0.j  1.+0.41421j
#       0.+0.j  1.+2.41421j]
# |X| = [4.  2.61313  0.  1.08239  0.  1.08239  0.  2.61313]
# IDFT(X) = [1.  1.  1.  1.  0.  0.  0.  0.]
# reduced formula = [ 1.  1.  1.  1.  0. -0.  0. -0.]
# sum|X|^2 = 32.0   N*sum|x|^2 = 32.0
# N=4: |FF*-I|=1.17e-16   |F^4-I|=7.35e-16   det W=-0.00000+16.00000j
#       Vandermonde=-0.00000+16.00000j
# N=8: |FF*-I|=7.06e-16   |F^4-I|=2.41e-15   det W=0.00000-4096.00000j
#       Vandermonde=0.00000-4096.00000j
# direct conv = [-2. -2.  2.  2.] via DFT = [-2. -2.  2.  2.]
# eigenvalues of C = [10.+0.j -2.+2.j -2.+0.j -2.-2.j] det C =
# -160.000000000000009
# solution of Cy=b : [1.  1.  0.  0.]
# interpolation error = 1.0329251542640534e-14
# X32[2] = (3.01367-2.01367j)   |X32[2]| = 3.62451
# N=   64 rectangular sum=  64.00000 CG=1.00000 ENBW=1.00000
# N=   64 Hann per    sum=  32.00000 CG=0.50000 ENBW=1.50000
# N=   64 Hann sym    sum=  31.50000 CG=0.49219 ENBW=1.52381
# N=   64 Hamming per sum=  34.56000 CG=0.54000 ENBW=1.36283
# N=   64 Hamming sym sum=  34.10000 CG=0.53281 ENBW=1.37832
# N=  1024 rectangular sum=1024.00000 CG=1.00000 ENBW=1.00000
# N=  1024 Hann per    sum= 512.00000 CG=0.50000 ENBW=1.50000
# N=  1024 Hann sym    sum= 511.50000 CG=0.49951 ENBW=1.50147
# N=  1024 Hamming per sum= 552.96000 CG=0.54000 ENBW=1.36283
# N=  1024 Hamming sym sum= 552.50000 CG=0.53955 ENBW=1.36378
# rectangular first zero at 1.0010 bin, side-lobe peak -13.25 dB,
#       -3 dB width 0.88 bin
# Hann per    first zero at 2.0010 bin, side-lobe peak -31.47 dB,
#       -3 dB width 1.44 bin
# Hamming per first zero at 2.0010 bin, side-lobe peak -42.45 dB,
#       -3 dB width 1.30 bin
# N=  4   mult.(1,-i,-1,i) = (2, 1, 1, 0)   sum = 4
# N=  5   mult.(1,-i,-1,i) = (2, 1, 1, 1)   sum = 5
# N=  6   mult.(1,-i,-1,i) = (2, 1, 2, 1)   sum = 6
# N=  7   mult.(1,-i,-1,i) = (2, 2, 2, 1)   sum = 7
# N=  8   mult.(1,-i,-1,i) = (3, 2, 2, 1)   sum = 8
# N= 12   mult.(1,-i,-1,i) = (4, 3, 3, 2)   sum = 12

```

Remark 22.14: What the comparison actually checked. Every numerical value of the chapter appears twice: once as a proved closed formula, once as the output of the script. In particular the following were verified: the values X_k and the moduli of the eight-point exam-

ple (Steps 1–2); inversion on all eight samples (Step 3, symbolically, and the $\text{IDFT}(X)$ line of the script); Parseval (Step 4); unitarity and $F_N^4 = I$ for $N = 4, 8$ (Theorem 22.3, Corollary 22.4); the determinants $16i$ and $-4096i$ computed both by factorization and with the Vandermonde product (Example 22.1); circular convolution by two routes (Example 22.2); the eigenvalues and the determinant of the circulant and the solution of the system $Cy = b$ (Example 22.3); the Dirichlet interpolation of zero-padding, with discrepancy $1.03 \cdot 10^{-14}$ over 32 values (Theorem 22.11); coherent gain and equivalent bandwidth for two values of N and for the periodic and symmetric forms (Propositions 22.6 and 22.7); the position of the first zero and the side-lobe peak (Proposition 22.8 and Remark 22.13); the multiplicities of the four eigenvalues of F_N for $N = 4, 5, 6, 7, 8, 12$ (Remark 22.3), whose sum is N each time. The discrepancies between the two routes are in every case of the order of the double-precision rounding error, i.e. $\leq 10^{-14}$.

22.13. Exercises inherited from Volume II

The exercises that follow come from Volume II, Chapter 4, and concern the topics of this chapter. They are reproduced in full, with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 22.1. Using Theorem 22.12, compute the 4-point DFT of $x = [1, 0, -1, 0]$ by evaluating $X(z) = \sum_n x[n]z^{-n}$ directly at the 4 fourth roots of unity, and compare the result with the direct computation from the definition of the DFT.

Exercise 22.2. A vibration analyzer acquires $N = 4096$ samples per window. Compare the number of complex operations required by the direct computation of the DFT ($O(N^2)$) with those required by the radix-2 FFT ($O(N \log_2 N)$), and compute the speedup factor.

Exercise 22.3. Trace by hand, for $N = 8$, the bit-reversal permutation used in `FB_FFT8`: write the index i in 3-bit binary, reverse the order of the bits, and check that the result matches the `bitRev` array of the listing. Then carry out by hand the first stage ($m = 2$) of butterflies on the already permuted array for the input of the Worked example (input $[1, -1, 1, -1, 1, -1, 1, -1]$), checking that the intermediate result is consistent with the decomposition of Theorem 22.13.

Exercise 22.4. (Leakage) — A signal at $f_0 = 15$ Hz is sampled at $f_s = 100$ Hz with $N = 50$ samples. Check whether f_0 falls exactly on a DFT bin (i.e. whether $f_0 N / f_s$ is an integer) and, if not, explain qualitatively what the spectrum with a rectangular window would show compared to a Hann window, using Remark 22.11.

Exercise 22.5: S10 — From series to DFT: the aliasing formula. Let f be continuous on $\mathbb{T}$ with $\sum_n |\hat{f}(n)| < \infty$, and let $x_k = f(2\pi k/N)$, $k = 0, \dots, N-1$, be its samples. Prove the **aliasing formula**

$$\frac{1}{N} X_j = \sum_{m \in \mathbb{Z}} \hat{f}(j + mN), \quad X_j = \sum_{k=0}^{N-1} x_k e^{-2\pi i j k / N},$$

and interpret it: the DFT does not see the individual Fourier coefficients, but their sum over all frequencies congruent modulo N . Connect this with the Theorem (Volume II, Chapter 2)

Solutions

Solution. The associated polynomial is $X(z) = \sum_{n=0}^3 x[n]z^{-n} = 1 + 0 \cdot z^{-1} - z^{-2} + 0 \cdot z^{-3} = 1 - z^{-2}$. The 4 fourth roots of unity (Chapter 3) are $\zeta_4^k = e^{2\pi i k/4} = i^k$, $k = 0, 1, 2, 3$, i.e. $1, i, -1, -i$. Evaluating (with the convention $z = \zeta_4^k$, $z^{-2} = \zeta_4^{-2k} = (-1)^{-k} = (-1)^k$):

$$X[0] = 1 - 1 = 0, \quad X[1] = 1 - (-1) = 2, \quad X[2] = 1 - 1 = 0, \quad X[3] = 1 - (-1) = 2.$$

From the direct definition of the DFT, $X[k] = \sum_{n=0}^3 x[n]e^{-2\pi i k n/4} = 1 - e^{-i\pi k}$ (only the terms $n = 0, 2$ contribute, since $x[1] = x[3] = 0$): for even k , $e^{-i\pi k} = 1$, hence $X[k] = 0$; for odd k , $e^{-i\pi k} = -1$, hence $X[k] = 2$. The two computations coincide exactly, as guaranteed by Theorem 22.12.

Solution. Direct computation: $N^2 = 4096^2 = 16\,777\,216$ complex operations. Radix-2 FFT: $N \log_2 N = 4096 \times 12 = 49\,152$ (since $4096 = 2^{12}$, $\log_2 4096 = 12$). The speedup factor is $16\,777\,216/49\,152 \approx 341$: the FFT requires about $1/341$ of the operations of the direct computation, a saving that makes real-time spectral analysis possible on embedded hardware even with limited computing resources.

Solution. Writing $i = 0, \dots, 7$ in 3-bit binary and reversing the order of the digits: $0 = 000 \rightarrow 000 = 0$; $1 = 001 \rightarrow 100 = 4$; $2 = 010 \rightarrow 010 = 2$; $3 = 011 \rightarrow 110 = 6$; $4 = 100 \rightarrow 001 = 1$; $5 = 101 \rightarrow 101 = 5$; $6 = 110 \rightarrow 011 = 3$; $7 = 111 \rightarrow 111 = 7$: that is, $[0, 4, 2, 6, 1, 5, 3, 7]$, exactly the `bitRev` array of the `FB_FFT8` listing.

After bit-reversal, the input $x = [1, -1, 1, -1, 1, -1, 1, -1]$ (indices 0..7) must be reordered by swapping the pairs (1, 4) and (3, 6), the only ones with `bitRev[i] > i`: position $1 \leftrightarrow 4$: $x[1] = -1$ and $x[4] = 1$ are swapped, giving $x[1] = 1$, $x[4] = -1$; position $3 \leftrightarrow 6$: $x[3] = -1$ and $x[6] = 1$ are swapped, giving $x[3] = 1$, $x[6] = -1$. The permuted vector is therefore $[1, 1, 1, 1, -1, -1, -1, -1]$ (the first four even samples grouped together, followed by the four odd ones, as expected from the even/odd recursion of Theorem 22.13).

First stage ($m = 2$, half = 1: with a single element per half-block the twiddle is always $w = 1$): for each pair $(k, k + 1) = (0, 1), (2, 3), (4, 5), (6, 7)$ one computes $x[k] \pm x[k + 1]$. For the first two pairs $x[k] = x[k + 1] = 1$: sum = 2, difference = 0. For the last two pairs $x[k] = x[k + 1] = -1$: sum = -2, difference = 0. Intermediate result after the first stage: $[2, 0, 2, 0, -2, 0, -2, 0]$ — a value also verified numerically, and consistent with the decomposition of Theorem 22.13, which at each stage combines pairs with a rotation twiddle, progressively propagating the energy toward the final bin $k = 4$ (the later stages, not required by the exercise, indeed lead to $[4, 0, 0, 0, -4, 0, 0, 0]$ after the second stage and to $[0, 0, 0, 0, 8, 0, 0, 0]$ after the third, exactly reproducing the output of `FB_FFT8` of the Worked example).

Solution. $f_0 N / f_s = 15 \times 50 / 100 = 7.5$, not an integer: f_0 falls exactly halfway between bin 7 and bin 8. By Remark 22.11, with a rectangular window the spectrum would show marked, symmetric leakage on the adjacent bins, with side lobes decaying slowly (about 6 dB per octave): energy visible even on bins far from the peak. With a Hann window the main lobe would widen (bins 7 and 8 would become almost equivalent in amplitude, both close to the peak), but the distant lobes would decay much more rapidly, making it easier to distinguish a possible weak nearby component from background noise or from leakage of

the main component — exactly the trade-off illustrated numerically in the Worked example of Section 22.10.3.

Solution. S10. *Step 1 — substituting the series into the samples.* Since $\sum_n |\hat{f}(n)| < \infty$, by Theorem 17.5(i) the Fourier series converges uniformly to f , and in particular pointwise at the nodes:

$$x_k = f\left(\frac{2\pi k}{N}\right) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n k / N}.$$

Step 2 — computing the DFT. Substituting and exchanging the sums (legitimate by absolute convergence of the series and finiteness of the sum over k):

$$X_j = \sum_{k=0}^{N-1} x_k e^{-2\pi i j k / N} = \sum_{n \in \mathbb{Z}} \hat{f}(n) \sum_{k=0}^{N-1} e^{2\pi i (n-j) k / N}.$$

Step 3 — the mother identity. The inner sum is that of the Theorem (Volume II, Chapter 3) of Chapter 3: it equals N if $N \mid (n - j)$ and 0 otherwise. The surviving indices n are therefore exactly those of the form $n = j + mN$ with $m \in \mathbb{Z}$:

$$X_j = N \sum_{m \in \mathbb{Z}} \hat{f}(j + mN), \quad \text{i.e.} \quad \frac{1}{N} X_j = \sum_{m \in \mathbb{Z}} \hat{f}(j + mN).$$

Step 4 — interpretation. The DFT of the samples does not return the Fourier coefficient $\hat{f}(j)$, but the *sum* of all coefficients whose index is congruent to j modulo N . The high frequencies “fold back” onto the low ones and add to them: this is **aliasing**, and the formula is its exact statement. The error made in approximating $\hat{f}(j)$ by X_j/N is

$$\frac{X_j}{N} - \hat{f}(j) = \sum_{m \neq 0} \hat{f}(j + mN),$$

which is smaller the more rapidly the coefficients decay, i.e. the more regular f is (Remark 17.5).

Step 5 — connection with Chapter 2. For $j = 0$ the formula becomes

$$\frac{X_0}{N} = \sum_{m \in \mathbb{Z}} \hat{f}(mN) = \hat{f}(0) + \sum_{m \neq 0} \hat{f}(mN),$$

and since X_0/N is exactly the mean of the samples, i.e. $\frac{1}{2\pi} T_N[f]$ with T_N the periodic trapezoidal rule, one recovers

$$\frac{1}{2\pi} T_N[f] - \frac{1}{2\pi} I[f] = \sum_{m \neq 0} \hat{f}(mN),$$

which is precisely Step 3 of the proof of the Theorem (Volume II, Chapter 2). The aliasing formula and the periodic quadrature error are the same statement.

Inherited quiz

The items that follow come from the quiz of Volume II, Chapter 4, and are renumbered here.

1. The complexity of the FFT relative to the direct DFT is

A) $O(N^2)$ against $O(N)$

- B) $O(N \log N)$ against $O(N^2)$
 - C) $O(N)$ against $O(N \log N)$
 - D) identical
2. Spectral leakage is reduced by applying
- A) a rectangular window
 - B) a window that tapers the edges (e.g. Hann)
 - C) slower sampling
 - D) aliasing

BEST PRACTICE

Answer key.

1-B: Theorem 22.14. **2-B:** the rectangular window is the worst for leakage (Section 22.10.3).

Inherited formulary

The lines that follow come from the formulary of Volume II, Chapter 4.

Aliasing (series $\rightarrow$ DFT)

$$\frac{1}{N}X_j = \sum_{m \in \mathbb{Z}} \hat{f}(j + mN), \quad x_k = f\left(\frac{2\pi k}{N}\right)$$

22.14. Assigned exercises

Exercise 22.6: A four-point DFT, with all the checks. *Goal:* master the definition and the three standard checks. *Estimated time:* 35 minutes.

Let $x = (2, 0, -1, 3)$.

- (a) Compute $X = \mathcal{F}_4 x$ directly from Definition 22.1, with $\omega_4 = -i$.
- (b) Check the conjugate symmetry of Theorem 22.7 and state, without recomputing, which components are real and why (Corollary 22.8).
- (c) Check Parseval's identity (Theorem 22.6) by computing the two sides separately.
- (d) Reconstruct x with the inversion formula (Theorem 22.2) and check all four samples.

Exercise 22.7: Circular convolution and linear convolution. *Goal:* understand when circular convolution answers the right question. *Estimated time:* 40 minutes.

Let $a = (1, 2, 3)$ and $b = (1, -1, 0)$, both of length 3.

- (a) Compute $a \circledast b$ with $N = 3$ from Definition 22.4, and recompute it with Theorem 22.5.
- (b) Compute the *linear* convolution $c[n] = \sum_m a[m]b[n-m]$ (indices not wrapped, $b[j] = 0$ outside $\{0, 1, 2\}$), which has length $3 + 3 - 1 = 5$.
- (c) Compare the two results and identify exactly which terms of the linear convolution overlapped in the circular one.
- (d) Extend a and b with zeros up to length $N' = 5$ and check that the 5-point circular convolution reproduces the linear convolution. State the general rule (how many zeros are needed, as a function of the two lengths).

Exercise 22.8: Windows: gain, equivalent bandwidth and reading an amplitude.

Goal: use the closed-form window formulas on a measurement case. *Estimated time:* 45 minutes.

An analyzer acquires $N = 1024$ samples at $f_s = 25\,600$ Hz.

- (a) Compute $\Delta f = f_s/N$ and state to which bin a tone at 1000 Hz corresponds; is the tone aligned to a bin?
- (b) Compute, with Propositions 22.6 and 22.7, the coherent gain and the equivalent bandwidth of the periodic Hann window and of the periodic Hamming window; give the exact form, then the decimal form to five figures.
- (c) The peak of the Hann-windowed spectrum equals $|X[k_0]| = 2048$. What is the amplitude of the tone? (Recall that for a real tone the energy splits between bins k_0 and $N - k_0$, Corollary 22.8.)
- (d) With which of the two windows would a component 40 dB weaker, placed at $5\Delta f$ from the main tone, be better distinguished? Justify using Remark 22.13.

22.15. Solved exercises

Exercise 22.9: The four-point DFT of $(1, 2, 3, 4)$, with four checks. Compute $X = \mathcal{F}_4 x$ for $x = (1, 2, 3, 4)$; check conjugate symmetry, Parseval and inversion; interpret the four numbers obtained.

Solution. Strategy. With $N = 4$ the matrix W_4 has entries that are powers of $\omega_4 = e^{-i\pi/2} = -i$, which are only $1, -i, -1, i$: no trigonometric computation is needed. X is computed row by row, then the three checks are applied in the order in which they catch different errors.

Step 1 — the four sums. Since $\omega_4^{kn} = (-i)^{kn}$:

$$\begin{aligned}X[0] &= 1 + 2 + 3 + 4 = 10, \\X[1] &= 1 + 2(-i) + 3(-i)^2 + 4(-i)^3 = 1 - 2i - 3 + 4i = -2 + 2i, \\X[2] &= 1 + 2(-1) + 3(1) + 4(-1) = 1 - 2 + 3 - 4 = -2, \\X[3] &= 1 + 2(i) + 3(-1) + 4(-i) = 1 + 2i - 3 - 4i = -2 - 2i,\end{aligned}$$

having used $(-i)^2 = -1$, $(-i)^3 = i$, $(-i)^4 = 1$ and, for row $k = 2$, $\omega_4^{2n} = (-1)^n$. Hence $X = (10, -2 + 2i, -2, -2 - 2i)$.

Step 2 — symmetry check (Theorem 22.7). x is real, so $X[4 - k] = \overline{X[k]}$ must hold: indeed $X[3] = -2 - 2i = \overline{-2 + 2i} = \overline{X[1]}$ ✓. Moreover $X[0] = 10$ and $X[2] = -2$ are real, as predicted by Corollary 22.8(i) and (ii), since 4 is even and $2 = N/2$ is the Nyquist bin. This check catches sign errors in the exponent on a single row.

Step 3 — Parseval check (Theorem 22.6).

$$\sum_n |x[n]|^2 = 1 + 4 + 9 + 16 = 30; \quad \frac{1}{4} \sum_k |X[k]|^2 = \frac{100 + 8 + 4 + 8}{4} = \frac{120}{4} = 30 \quad \checkmark,$$

where $|-2 \pm 2i|^2 = 4 + 4 = 8$ and $|-2|^2 = 4$. This check catches normalization errors (Corollary 22.7).

Step 4 — inversion check (Theorem 22.2). Grouping the conjugate terms $k = 1$ and $k = 3$, and using $\omega_4^{-kn} = i^{kn}$:

$$x[n] = \frac{1}{4} \left(10 + (-2)(-1)^n + 2 \operatorname{Re}((-2 + 2i)i^n) \right).$$

For $n = 0$: $\frac{1}{4}(10 - 2 + 2(-2)) = \frac{1}{4}(10 - 2 - 4) = 1 \checkmark$. For $n = 1$: $(-2 + 2i)i = -2i - 2$, real part -2 , hence $\frac{1}{4}(10 + 2 - 4) = 2 \checkmark$. For $n = 2$: $(-2 + 2i)(-1) = 2 - 2i$, real part 2 , hence $\frac{1}{4}(10 - 2 + 4) = 3 \checkmark$. For $n = 3$: $(-2 + 2i)(-i) = 2i + 2$, real part 2 , hence $\frac{1}{4}(10 + 2 + 4) = 4 \checkmark$.

Independent check. The same DFT is obtained by observing that $x = (1, 2, 3, 4)$ is the sum of a constant $\frac{5}{2}(1, 1, 1, 1)$ and the centered ramp $(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2})$. The constant contributes only to $X[0]$, with $4 \cdot \frac{5}{2} = 10 \checkmark$; the centered ramp is real and antisymmetric about the center, and its DFT has $X[0] = 0$, consistent with $-\frac{3}{2} - \frac{1}{2} + \frac{1}{2} + \frac{3}{2} = 0$. Moreover the script of Section 22.12.1 prints **eigenvalues** of $\mathbf{C} = (10, -2 + 2i, -2, -2 - 2i)$, which is precisely $\mathcal{F}_4(1, 2, 3, 4) \checkmark$.

Interpretation. $X[0] = 10$ is the sum of the samples, i.e. N times the mean: the DC component. $|X[2]| = 2$ is the component at the Nyquist frequency, i.e. how much the signal “alternates” from sample to sample: here little, because the ramp is monotone. $|X[1]| = |X[3]| = 2\sqrt{2} = 2.82843$ is the component at one cycle per window, the largest after the DC term: it is the ramp, which viewed as a signal periodic with period 4 is a sawtooth, whose content is concentrated at the fundamental frequency.

Common errors. (i) Using $\omega_4 = i$ instead of $-i$: this gives the conjugate spectrum, and neither Parseval nor the moduli notice — only the symmetry check read *with the sign* or inversion reveals it. (ii) Putting the factor $1/N$ in the forward transform: Parseval fails by a factor N^2 (Corollary 22.7). (iii) Computing $X[3]$ from scratch instead of using symmetry: this is not an error, but it doubles the work and the chances of making a mistake.

Extension. Repeat with $x = (4, 3, 2, 1)$ and check, without redoing the computation, that the result is $\overline{X}$ read backwards: this is Proposition 22.4 combined with Theorem 22.4.

Exercise 22.10: Zero-padding: what appears between the bins, and why it is not new information. Let $x = (1, 1, 1, 1, 0, 0, 0, 0)$. Compute $X_{32}[k]$ for $k = 0, 2, 4$ and show that the values for k a multiple of 4 coincide with the eight-point DFT; explain why the intermediate values add no information.

Solution. Strategy. Theorem 22.10 is used to reduce everything to the same DTFT, then the geometric sum for the requested values, and finally Theorem 22.11 for the interpretation. **Step 1 — there is only one DTFT.** Since $x[n] = 1$ exactly for $n = 0, 1, 2, 3$, the DTFT is

$$\widehat{X}(\Omega) = \sum_{n=0}^3 e^{-in\Omega} = \frac{1 - e^{-4i\Omega}}{1 - e^{-i\Omega}} \quad (\Omega \notin 2\pi\mathbb{Z}), \quad \widehat{X}(0) = 4.$$

Both the eight-point DFT and the thirty-two-point DFT are samplings of this same function, at $\Omega = 2\pi k/8$ and $\Omega = 2\pi k/32$ respectively (Theorems 22.9 and 22.10).

Step 2 — $k = 0$. $\Omega = 0$: this falls into the case of vanishing denominator, and one sums directly, $X_{32}[0] = 4$. This coincides with $X[0] = 4$ of Section 22.12 $\checkmark$.

Step 3 — $k = 4$. $\Omega = 2\pi \cdot 4/32 = \pi/4$, i.e. the same point at which the eight-point DFT evaluates bin 1 ($2\pi \cdot 1/8 = \pi/4$). Hence $X_{32}[4] = X[1] = 1 - (1 + \sqrt{2})i = 1 - 2.41421i$, with no new computation at all: it is the final claim of Theorem 22.10 with $M = rN$, $r = 4$.

Step 4 — $k = 2$. This is a new point, halfway between bins 0 and 1 of the short DFT. With $w = e^{-i\pi/8}$ and $w^4 = e^{-i\pi/2} = -i$:

$$X_{32}[2] = \frac{1 - w^4}{1 - w} = \frac{1 + i}{1 - e^{-i\pi/8}}.$$

The computation is that of Example 22.4 and gives

$$X_{32}[2] = 3.01367 - 2.01367i, \quad |X_{32}[2]| = 3.62451.$$

Independent check. The modulus is recomputed in closed form without going through the real and imaginary parts: $|X_{32}[2]| = \frac{|1+i|}{|1-e^{-i\pi/8}|} = \frac{\sqrt{2}}{2\sin(\pi/16)} = \frac{1.41421}{2 \cdot 0.19509} = 3.62451 \checkmark$, and $\sqrt{3.01367^2 + 2.01367^2} = 3.62451 \checkmark$. The script of Section 22.12.1 prints $X_{32}[2] = (3.01367-2.01367j) \checkmark$.

Interpretation. The value 3.62451 is larger than $|X[1]| = 2.61313$ and much smaller than $|X[0]| = 4$: between the two bins of the short DFT the curve $|\hat{X}|$ is not at all step-monotone, and the “true” peak of the continuous spectrum lies between the bins, not on them. But it is not new information: by Theorem 22.11,

$$X_{32}[2] = \frac{1}{8} \sum_{j=0}^7 X[j] D_8\left(2\pi\left(\frac{2}{32} - \frac{j}{8}\right)\right),$$

i.e. a fixed-coefficient linear combination of the eight values already known. Zero-padding densifies the plot, not the knowledge.

Common errors. (i) Believing that zero-padding improves resolution: it does not (Proposition 22.8(iii)); two tones closer together than f_s/N remain indistinguishable no matter how many zeros are added. (ii) Adding the zeros *in the middle* instead of at the end: this gives a different signal and a different spectrum. (iii) Forgetting that the bin amplitude does not change with M in the convention adopted here (no forward factor $1/N$), whereas it would change in a normalized convention.

Extension. Compute $X_{32}[1]$ and check that $|X_{32}[1]| > |X_{32}[2]|$: the maximum of the continuous curve between bins 0 and 1 of the short DFT lies closer to 0 than to 1, as it must because $|\hat{X}|$ decreases from 4 toward the first zero.

Exercise 22.11: Measured leakage: a tone halfway between two bins. Let $N = 64$, $f_s = 100$ Hz and $x[n] = \sin(2\pi f_0 n / f_s)$ with $f_0 = 8.5 f_s / N = 13.28125$ Hz, i.e. 8.5 cycles in the window. Predict, with the Dirichlet kernel, the level of bin 7 relative to the peak for the rectangular window, and compare it with the measured value; repeat with the Hann window.

Solution. Strategy. A real sine is the difference of two complex exponentials; each exponential contributes a shifted copy of the Dirichlet kernel (Theorem 22.15 with $a_1 = 0$). One first estimates with a single term, then does the exact two-term computation and explains the difference.

Step 1 — the tone does not fall on a bin. $f_0 N / f_s = 8.5$, not an integer: the tone lies exactly halfway between bins 8 and 9. This is the situation of Remark 22.11.

Step 2 — the exact formula. From $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$ and Definition 22.6,

$$X[k] = \frac{1}{2i} \left[D_N\left(\frac{2\pi(k-8.5)}{N}\right) - D_N\left(\frac{2\pi(k+8.5)}{N}\right) \right],$$

because $\sum_n e^{\pm 2\pi i \cdot 8.5 n / N} e^{-2\pi i k n / N} = \sum_n e^{-2\pi i (k \mp 8.5) n / N} = D_N(2\pi(k \mp 8.5)/N)$.

Step 3 — single-term estimate. Neglecting the second term (the “image frequency”, centered at -8.5 bins and hence 16.5 bins away from bin 8), one has, by Lemma 22.3, $|D_N(2\pi f/N)| = |\sin(\pi f)/\sin(\pi f/N)|$, and since $|\sin(\pi f)| = 1$ for f a half-integer:

$$\frac{|X[7]|}{|X[8]|} \approx \frac{1/\sin(1.5\pi/64)}{1/\sin(0.5\pi/64)} = \frac{\sin(0.5\pi/64)}{\sin(1.5\pi/64)} = \frac{0.0245412}{0.0735646} = 0.333603,$$

i.e. $20 \log_{10} 0.333603 = -9.54$ dB.

Step 4 — the measured value. The complete computation (two terms, or the FFT) gives -9.11 dB. The 0.43 dB difference from the estimate is precisely the contribution of the image frequency, which adds in phase with the main term at bin 7 but not at bin 8. That the two-term formula of Step 2 is *exact* has been verified: the maximum discrepancy between the 64 values $X[k]$ computed with it and those computed with the FFT is $3.7 \cdot 10^{-13}$ ✓.

Step 5 — the full picture, rectangular versus Hann. The measured levels, in dB relative to the peak:

bin	4	6	7	8	9	10
rectangular	-17.02	-13.06	-9.11	0.00	-0.39	-10.27
Hann (symmetric)	-46.89	-30.51	-13.41	0.00	0.00	-13.41

These are the same numbers reported in the worked example of Section 22.10.3, recomputed here.

Interpretation. With the rectangular window bins 8 and 9 are nearly matched (0.39 dB apart) and the rest of the spectrum stays high: at bin 4, i.e. four and a half bins from the tone, there is still energy at -17 dB, which would mask any weaker component. With Hann the main lobe is wider (bins 8 and 9 are now exactly equal, and bin 7 only drops to -13.41), but at bin 4 one is at -46.89 dB: almost 30 dB of gain, in line with the difference between the side-lobe peaks of the two windows (Remark 22.13: -13.25 against -31.47 dB).

Common errors. (i) Reading the tone's amplitude directly off the peak: with Hann the peak must be divided by $\text{CG} = 0.5$ (Proposition 22.6), and the *scalloping* error remains regardless because the tone is not on a bin. (ii) Attributing the leakage to measurement noise: here there is no noise, the signal is an exact sine; leakage is an effect of windowing, not of signal quality. (iii) Expecting Hann to reduce the level of bin 7: it *increases* it (from -9.11 to -13.41 is a reduction, but the main lobe is wider and bins 7 and 10 are now inside it); Hann acts on the *distant* bins.

Extension. Repeat with f_0 exactly on bin 8 ($f_0 = 12.5$ Hz): with the rectangular window all other bins drop to about -300 dB, i.e. numerical zero (this is Theorem 22.1: the signal is a combination of only two vectors e_k); with Hann, instead, bins 7 and 9 remain at -5.82 dB, because the window has *widened* a spectral line that was perfect. This is why, when an integer number of periods can be acquired, the best window is no window at all.

22.16. Self-assessment quiz

- (Definition) In the convention of this chapter the DFT of $x \in \mathbb{C}^N$ is
 - $X[k] = \frac{1}{N} \sum_n x[n] e^{-2\pi i k n / N}$
 - $X[k] = \sum_n x[n] e^{-2\pi i k n / N}$
 - $X[k] = \sum_n x[n] e^{+2\pi i k n / N}$
 - $X[k] = \frac{1}{\sqrt{N}} \sum_n x[n] e^{+2\pi i k n / N}$
- (Hypothesis) Lemma 22.1 distinguishes two cases. The case $N \mid j$ must be treated separately because
 - the sum is infinite
 - the ratio q equals 1 and the closed formula would have a vanishing denominator
 - the ratio q is zero
 - the terms are not complex numbers

-
3. (Symbolic computation) Let $x \in \mathbb{C}^N$ and $y[n] = x[\langle n - 3 \rangle_N]$. Then $Y[k]$ equals
- A) $X[k - 3]$
 - B) $3X[k]$
 - C) $\omega_N^{3k} X[k]$
 - D) $X[k] + 3$
4. (Numerical computation) For $x = (1, 1, 1, 1, 0, 0, 0, 0)$, $|X_1|$ equals
- A) 4.00000
 - B) 2.61313
 - C) 1.08239
 - D) 2.41421
5. (Proof step) In the proof of Theorem 22.2, exchanging the order of the two sums is legitimate because
- A) the series converges absolutely
 - B) the terms are finite in number
 - C) $\mathcal{F}_N$ is unitary
 - D) x is real
6. (Error diagnosis) An FFT routine passes the Parseval check but returns the signal reversed in time when applied twice forward. The most likely error is
- A) a missing factor $1/N$
 - B) no error: this is the behavior predicted by Theorem 22.4
 - C) a wrong bit-reversal
 - D) numerical overflow
7. (Transfer) To solve $Cy = b$ with C a 1024×1024 circulant, the best approach is
- A) Gaussian elimination, $O(N^3)$
 - B) inverting C once and multiplying, $O(N^2)$
 - C) dividing spectrum by spectrum with two FFTs, $O(N \log N)$
 - D) it is not possible: circulants are not invertible
8. (Source and metacognition) The comparative table of windows (gain, side lobes, equivalent bandwidth) most cited in the literature is
- A) Harris, *Proceedings of the IEEE* 66 (1978), 51–83
 - B) Cooley and Tukey (1965)
 - C) Van Loan, SIAM (1992)
 - D) McClellan and Parks (1972)
-

22.17. Teacher's key

BEST PRACTICE

Item 1 — answer B. *Rationale:* this is Definition 22.1: no forward factor, minus sign in the exponent. *Distractors:* A is the convention with $1/N$ forward (used by some applied-mathematics textbooks); C is the crystallographic convention; D is the symmetric convention, with the wrong sign as well. *Diagnosed error:* not having stated the convention before computing. *Source:* Briggs and Henson, ch. 2. *Prerequisite:* none. *Remediation:* Remark 22.1, then recompute the eight-point example in the three conventions and compare.

Item 2 — answer B. *Rationale:* if $N \mid j$ then $q = \omega_N^j = 1$, and $\frac{q^N - 1}{q - 1}$ is $\frac{0}{0}$: one sums directly, obtaining N . *Distractors:* A confuses the finite sum with a series; C would confuse “ratio 1” with “ratio 0”; D is meaningless. *Diagnosed error:* mechanical application of the geometric-sum formula without checking its hypothesis. *Source:* Lemma 22.1 and Theorem 6.5. *Prerequisite:* geometric progressions. *Remediation:* redo the case $N = 4$, $j = 4$ by hand.

Item 3 — answer C. *Rationale:* Proposition 22.4: a circular shift by n_0 multiplies bin k by $\omega_N^{kn_0}$. *Distractors:* A is the *dual* formula (modulation in time $\rightarrow$ shift in frequency); B and D confuse a shift with a scaling and with a sum. *Diagnosed error:* confusing the two dual statements. *Prerequisite:* periodicity of the indices (Proposition 22.1). *Remediation:* check on $N = 4$, $x = (1, 0, 0, 0)$, where $X \equiv 1$ and the shift indeed gives ω_4^{3k} .

Item 4 — answer B. *Rationale:* $|X_1| = \sqrt{1 + (1 + \sqrt{2})^2} = \sqrt{4 + 2\sqrt{2}} = 2.61313$ (Section 22.12.1, Step 2). *Distractors:* A is $|X_0|$; C is $|X_3|$; D is $1 + \sqrt{2}$, i.e. the *modulus of the imaginary part* of X_1 , not its modulus. *Diagnosed error:* confusing the imaginary part with the modulus. *Prerequisite:* modulus of a complex number. *Remediation:* redo the table of Section 22.12 column by column.

Item 5 — answer B. *Rationale:* the two sums have N^2 terms altogether; the exchange is the commutativity of addition on a finite set and requires no convergence hypothesis. *Distractors:* A invokes a theorem (Fubini–Tonelli for series) not needed here; C and D are true but irrelevant to the step in question (and C would be circular, since unitarity is proved using the same identity). *Diagnosed error:* the reflex “double sum $\Rightarrow$ absolute convergence”. *Remediation:* count the terms.

Item 6 — answer B. *Rationale:* Theorem 22.4 states that $\mathcal{F}_N$ applied twice gives $Nx[\langle -n \rangle_N]$: time reversal is the correct behavior, not a defect. *Distractors:* A would produce a Parseval failure, excluded by hypothesis; C would give an unrecognizably permuted spectrum; D would give non-finite values. *Diagnosed error:* mistaking a property for a bug. *Prerequisite:* Theorem 22.4. *Remediation:* run two consecutive FFTs on $(1, 2, 3, 4)$ and compare with $4 \cdot (1, 4, 3, 2)$.

Item 7 — answer C. *Rationale:* Corollary 22.9(iv) and Theorem 22.8: $y = \mathcal{F}_N^{-1}(\mathcal{F}_N b / \lambda)$, two transforms and N divisions. *Distractors:* A and B work but waste the structure; D is false (a circulant is invertible if and only if no $\lambda[k]$ vanishes). *Diagnosed error:* not recognizing the circulant structure. *Prerequisite:* Theorem 22.5. *Remediation:* Example 22.3 with $N = 4$.

Item 8 — answer A. *Rationale:* F. J. Harris’s 1978 article in *Proceedings of the IEEE* is the standard survey on windows. *Distractors:* B is the article on the FFT; C is a text on FFT computational structures; D concerns the eigenvalues of the DFT matrix (Remark 22.3). *Diagnosed error:* a generic “Oppenheim” citation for any signal-processing statement. *Remediation:* read the *Sources* section of this chapter and check each reference.

Answers to the assigned exercises.

Exercise 22.6, $x = (2, 0, -1, 3)$: (a) $X[0] = 2+0-1+3 = 4$; $X[1] = 2+0 \cdot (-i) - 1 \cdot (-1) + 3 \cdot (i) =$

$3 + 3i$; $X[2] = 2 - 0 - 1 - 3$, i.e. $2 - 0 + (-1)(1) + 3(-1) = -2$; $X[3] = \overline{X[1]} = 3 - 3i$. Hence $X = (4, 3 + 3i, -2, 3 - 3i)$. (b) $X[0] = 4$ and $X[2] = -2$ are real (Corollary 22.8(i)–(ii)), and $X[3] = \overline{X[1]}$ ✓. (c) $\sum_n |x[n]|^2 = 4 + 0 + 1 + 9 = 14$; $\frac{1}{4}(16 + 18 + 4 + 18) = \frac{56}{4} = 14$ ✓. (d) inversion gives $x[n] = \frac{1}{4}(4 + (-2)(-1)^n + 2\operatorname{Re}((3 + 3i)i^n))$: for $n = 0$, $\frac{1}{4}(4 - 2 + 6) = 2$; for $n = 1$, $(3 + 3i)i = -3 + 3i$, real part -3 , hence $\frac{1}{4}(4 + 2 - 6) = 0$; for $n = 2$, the real part changes sign, i.e. $\operatorname{Re}(-3 - 3i) = -3$, hence $\frac{1}{4}(4 - 2 - 6) = -1$; for $n = 3$, $(3 + 3i)(-i) = 3 - 3i$, real part 3 , hence $\frac{1}{4}(4 + 2 + 6) = 3$ ✓.

Exercise 22.7: (a) $a \circledast b$ with $N = 3$ equals $(1 \cdot 1 + 2 \cdot 0 + 3 \cdot (-1), 1 \cdot (-1) + 2 \cdot 1 + 3 \cdot 0, 1 \cdot 0 + 2 \cdot (-1) + 3 \cdot 1) = (-2, 1, 1)$. (b) the linear convolution is $(1, 1, 1, -3, 0)$, i.e. $c[0] = 1$, $c[1] = 2 - 1 = 1$, $c[2] = 3 - 2 = 1$, $c[3] = -3$, $c[4] = 0$. (c) the terms $c[3] = -3$ and $c[4] = 0$ wrapped onto the first two: $-2 = 1 + (-3)$ and $1 = 1 + 0$ ✓. (d) with $N' = 5$ the circular convolution coincides with the linear one; general rule: at least $L_a + L_b - 1$ points are needed.

Exercise 22.8: (a) $\Delta f = 25\,600/1024 = 25$ Hz; $1000/25 = 40$, an integer: the tone falls exactly on bin 40. (b) Hann: $\text{CG} = \frac{1}{2} = 0.50000$, $\text{ENBW} = \frac{3}{2} = 1.50000$; Hamming: $\text{CG} = \frac{27}{50} = 0.54000$, $\text{ENBW} = \frac{1987}{1458} = 1.36283$. (c) the amplitude is $A = \frac{2|X[k_0]|}{N \text{CG}} = \frac{2 \cdot 2048}{1024 \cdot 0.5} = 8$: the factor 2 accounts for the image bin $N - k_0$, the factor N for the convention without forward normalization, and CG for the window's attenuation. (d) Hann: at five bins' distance one is already outside the main lobe of both windows (which is 4 bins wide, Proposition 22.8), and there what matters is not the height of the highest side-lobe peak, but the *roll-off* of the side lobes. The two peaks tabulated in Remark 22.13 occur at different points of the spectrum and are not comparable with each other: Hann's (-31.47 dB) sits at ≈ 2.36 bins, Hamming's (-42.45 dB) exactly at ≈ 4.5 bins. At $5\Delta f$ what counts is that Hann rolls off at about -18 dB per octave against Hamming's -6 dB per octave (Remark 22.13); computing the maximum leakage for an offset in $[4.5, 5.5]$ bins with the same convention as that remark ($N = 64$, grid of 2^{16} points) gives -48.70 dB for Hann against -42.45 dB for Hamming. Both keep the leakage below the -40 dB of the sought component, but Hann gains more than 6 dB: it is the right choice. The worked exercise on leakage confirms this: at bin 4 — that is, 4.5 bins from the tone — Hann is at -46.89 dB.

22.18. Formulary of the chapter

Definition and inversion ($\omega_N = e^{-2\pi i/N}$)

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-kn}$$

Orthogonality (the mother identity)

$$\sum_{n=0}^{N-1} \omega_N^{jn} = \begin{cases} N, & N \mid j \\ 0, & N \nmid j \end{cases} \quad \implies \quad \langle e_k, e_\ell \rangle = N \delta_{k\ell}$$

Matrix $W_N = (\omega_N^{kn})$, $F_N = N^{-1/2} W_N$

$$W_N^* W_N = N I, \quad F_N^* = F_N^{-1}, \quad F_N^2 = P, \quad F_N^4 = I, \quad |\det W_N| = N^{N/2}$$

Circular convolution and Parseval

$$(x \circledast y)[n] = \sum_{m=0}^{N-1} x[m] y[\langle n - m \rangle_N], \quad \mathcal{F}_N(x \circledast y) = X \cdot Y$$

$$\sum_{n=0}^{N-1} x[n] \overline{y[n]} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \overline{Y[k]}, \quad \sum_n |x[n]|^2 = \frac{1}{N} \sum_k |X[k]|^2$$

Shift, symmetry, circulants

$$x[\langle n - n_0 \rangle_N] \longleftrightarrow \omega_N^{kn_0} X[k], \quad x \text{ real} \iff X[\langle -k \rangle_N] = \overline{X[k]}$$

$$\text{circ}(c) = F_N^* \text{diag}(\mathcal{F}_N c) F_N, \quad \det \text{circ}(c) = \prod_{k=0}^{N-1} (\mathcal{F}_N c)[k]$$

DTFT, zero-padding, Dirichlet kernel

$$X[k] = \hat{X}\left(\frac{2\pi k}{N}\right), \quad X_M[k] = \hat{X}\left(\frac{2\pi k}{M}\right), \quad D_N(\Omega) = \frac{\sin(N\Omega/2)}{\sin(\Omega/2)} e^{-i(N-1)\Omega/2}$$

$$X_M[k] = \frac{1}{N} \sum_{j=0}^{N-1} X[j] D_N\left(2\pi\left(\frac{k}{M} - \frac{j}{N}\right)\right)$$

Windows $w[n] = a_0 - a_1 \cos(2\pi n/N)$

$$\text{CG} = a_0, \quad \text{ENBW} = 1 + \frac{a_1^2}{2a_0^2}, \quad \Delta f = \frac{f_s}{N}$$

	CG	ENBW	main lobe	side-lobe peak
rectangular	1	1	2 bin	−13.25 dB
Hann	1/2	3/2	4 bin	−31.47 dB
Hamming	27/50	1987/1458	4 bin	−42.45 dB

Sources and bibliography

- [1] A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed., Pearson, Upper Saddle River (NJ), 2010, ISBN 978-0-13-198842-2, 1108 pp. General reference for the DFT convention adopted in this chapter, for circular convolution and for windowed spectral analysis. *The publication metadata (publisher, year, ISBN, page count) were verified on 2026-09-06 in the Open Library catalog (record ISBN 9780131988422); no specific pages are cited because it was not possible to verify the pagination.*
- [2] W. L. Briggs, V. E. Henson, *The DFT: An Owner's Manual for the Discrete Fourier Transform*, SIAM, Philadelphia, 1995, ISBN 978-0-89871-342-8, 434 pp., DOI 10.1137/1.9781611971511. Chapter 2, “The Discrete Fourier Transform,” pp. 15–64 (definition, conventions, inversion); chapter 3, “Properties of the DFT,” pp. 65–116 (circular convolution, Parseval, symmetries); chapter 6, “Errors in the DFT,” pp. 179–234 (leakage and windowing); chapter 9, “Quadrature and the DFT,” pp. 357–378 (link with the periodic trapezoidal rule, used in Exercise 22.5). *Verified on 2026-09-06 via the Crossref metadata of the volume and its chapters.*
- [3] C. Van Loan, *Computational Frameworks for the Fast Fourier Transform*, Frontiers in Applied Mathematics 10, SIAM, Philadelphia, 1992, ISBN 978-0-89871-285-8, 273 pp., DOI 10.1137/1.9781611970999. Chapter 1, “The Radix-2 Frameworks,” pp. 1–75: matrix formulation of the DFT and of its factorization, referenced by Definition 22.3 and Theorem 22.13. *Verified on 2026-09-06 via Crossref and Open Library metadata.*
- [4] F. J. Harris, *On the use of windows for harmonic analysis with the discrete Fourier trans-*

form, *Proceedings of the IEEE* **66** (1978), no. 1, pp. 51–83, DOI 10.1109/PROC.1978.10837. Standard survey on windows: definitions of coherent gain and equivalent noise bandwidth, comparative table, asymptotic laws of side-lobe decay (the latter here *imported* and not proved, Remark 22.13). Verified on 2026-09-06 via the Crossref metadata of the DOI.

- [5] J. H. McClellan, T. W. Parks, *Eigenvalue and eigenvector decomposition of the discrete Fourier transform*, *IEEE Transactions on Audio and Electroacoustics* **20** (1972), no. 1, pp. 66–74, DOI 10.1109/TAU.1972.1162342. Multiplicities of the four eigenvalues of the DFT matrix: an *imported* result, see Remark 22.3. Verified on 2026-09-06 via Crossref metadata.
- [6] J. W. Cooley, J. W. Tukey, *An algorithm for the machine calculation of complex Fourier series*, *Mathematics of Computation* **19** (1965), pp. 297–301. Cited in Section 22.10 as the source of the decimation-in-time algorithm; the reference is carried over from Volume II and is discussed in full in Chapter 23.

The fast Fourier transform

The distinction on which the whole chapter depends is one of those that are forgotten at once and then cost hours of confusion: **the DFT is a transform, the FFT is an algorithm**. The discrete Fourier transform is the linear map $x \mapsto X$ defined in Chapter 22 (Definition 22.1); the “fast Fourier transform” is not another transform, does not produce other numbers, has no other properties: it is a family of procedures for computing *exactly* those same N numbers with a number of operations proportional to $N \log_2 N$ instead of N^2 . Whoever says “I applied the FFT instead of the DFT” is saying, without noticing it, “I did the same multiplication in a different order”.

That change of order, however, is what made numerical frequency analysis practicable. For $N = 1,048,576 = 2^{20}$ samples — barely a minute of audio at 17.5 kHz — the direct computation asks for 1 099 511 627 776 complex multiplications, the radix-2 FFT asks for 10,485,760: a factor of 104,857.6 (Table 23.1). The reason is a single algebraic identity between roots of unity, $\omega_N^2 = \omega_{N/2}$, applied recursively.

The chapter walks the whole chain, in order, from the cost of the direct computation to the cost of the algorithm, and back: the operations of the defining sum are counted exactly (Proposition 23.1), the even and odd indices are separated (Theorem 23.1 and Theorem 23.2), the butterfly is isolated (Definition 23.3), the correctness of the recursive algorithm is proved by induction (Theorem 23.3), the cost recurrence is solved and the number of multiplications and additions that remain is counted *exactly* (Theorem 23.5, Corollary 23.4), the ordering of the inputs is proved to be the bit-reversal permutation (Theorem 23.7), the decimation-in-frequency variant is presented (Theorem 23.9), the inverse transform is obtained from the direct one by conjugation (Theorem 23.10), the example $N = 8$ is worked out in full, first in exact symbolic form and then in numerical form, with the check against the direct DFT, against Parseval’s identity, against Hermitian symmetry and against the reconstruction of all eight samples. A separate section treats the case of N not a power of two (mixed-radix, with the example $N = 6$ worked out in full; Bluestein and Rader), and the last section treats round-off error in finite-precision arithmetic, with a proved bound and an experimental check.

The results on the DFT — orthogonality of the exponential vectors (Theorem 22.1), the inversion formula (Theorem 22.2), the circular convolution theorem (Theorem 22.5), Parseval’s identity (Theorem 22.6), Hermitian symmetry (Theorem 22.7), zero-padding (Theorem 22.10 and Theorem 22.11) — are recalled and used, never reproved. Chapter 22 already contains, in Section 22.10, a first statement of the radix-2 decomposition (Theorem 22.13) and of the asymptotic complexity (Theorem 22.14), inherited from Volume II: that material is not summarized or replaced here, but resumed and completed, because there it lacks the *exact* count of operations, the inductive proof of correctness, the bit-reversal permutation, the DIF variant, the IFFT by conjugation, the mixed-radix case and the error analysis.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages of level **D** (the round-off error estimate of Section 23.13 and Rader’s algorithm of Section 23.12).

Internal prerequisites. N -th roots of unity and their arithmetic (Chapter 6); the discrete

Fourier transform, the DFT matrix, orthogonality, inversion, circular convolution, Parseval, Hermitian symmetry and zero-padding (Chapter 22, Definition 22.1, Definition 22.2, Definition 22.3, Theorem 22.1, Theorem 22.2, Theorem 22.5, Theorem 22.6, Theorem 22.7, Theorem 22.3, Theorem 22.10, Theorem 22.11); the Zeta transform and sampling on the roots of unity (Chapter 21 and Theorem 22.12).

External prerequisites, assumed and not reproved:

- complete induction and strong induction on the integers;
- elementary linear algebra: matrix products, permutation matrices, unitary matrices, the spectral norm $\|A\|_2$ and its submultiplicativity, singular values of a 2×2 matrix;
- modular arithmetic: Euclidean division, residue classes, the notation $\langle n \rangle_N$ for the remainder of n modulo N ;
- the binary representation of an integer and its uniqueness;
- the standard model of floating-point arithmetic (hypothesis (M) of Section 23.13, stated explicitly and attributed to its source);
- the existence of a primitive root modulo a prime p (used only once, in Rader's algorithm, and declared *imported* in Remark 23.11).

Everything else is proved here.

What is learned. Counting the operations of an algorithm before writing it; splitting an indexed sum according to the parity of the index and recognizing when this produces two subproblems of the same shape; proving by induction the correctness of a recursive algorithm; setting up and solving a cost recurrence; passing from a recursive formulation to an iterative in-place one and understanding where the bit-reversal permutation comes from; obtaining the inverse transform without writing a second algorithm; recognizing what zero-padding adds (interpolation) and what it does not add (resolution); factoring a DFT of composite length; estimating the round-off error of a structured algorithm and comparing it with the measurement.

Warning. Conventions and notation of the chapter.

- The convention of Chapter 22 is kept: the **twiddle factor** is

$$\omega_N = e^{-2\pi i/N},$$

and the DFT is $X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}$ (Definition 22.1), with no normalization factor forward and with $1/N$ backward.

- **Warning about the letter W .** The classical literature — Cooley and Tukey (1965), Oppenheim and Schaffer, Van Loan — writes W_N for what here is ω_N , that is, for the scalar $e^{-2\pi i/N}$. In this treatise W_N is instead the **matrix** of the DFT (Definition 22.3), with entries $(W_N)_{kn} = \omega_N^{kn}$. Reading the sources requires the translation $W_N^{(\text{source})} = \omega_N$. In Chapter 22, Section 22.10, the same scalar appears under the name ζ_N : the three symbols ω_N , ζ_N and the sources' W_N denote the same complex number.
- N is always an integer ≥ 1 ; when we write $N = 2^m$ we mean $m \in \mathbb{N}$, $m \geq 0$. The notation $\log N$ without a base always means $\log_2 N$.
- $\langle n \rangle_N$ is the remainder of the Euclidean division of n by N , that is, the unique integer of $\{0, \dots, N-1\}$ congruent to n modulo N . By Proposition 22.1, writing $X[k]$ with k outside $\{0, \dots, N-1\}$ is always legitimate and means $X[\langle k \rangle_N]$.
- “Operation” always means *arithmetic operation on complex numbers*; the cost model is fixed in Definition 23.1. The cost in *real* operations is discussed in Remark 23.1.
- Numerical values are rounded to the seventh decimal digit and computed twice by independent routes: a closed form or an exact symbolic computation, and an actually executed Python script. The symbol $\checkmark$ signals that the comparison has been made and reports the discrepancy.

- In every flow-graph figure the convention is the standard one: a node sums the values arriving at it, a label on an edge indicates multiplication by that factor, an unlabeled edge indicates the factor 1, and a -1 on an edge is drawn out in full.

23.1. What the definition costs: the exact count

Before looking for a fast algorithm one must know *exactly* how much the slow one costs. “It costs $O(N^2)$ ” is not enough: the constant in front of N^2 decides whether an acquisition system meets its timing budget, and an honest comparison with the FFT is made on numbers, not on Landau symbols. This section fixes the cost model, collects the arithmetic of the twiddle factors, and counts.

Definition 23.1: Arithmetic cost model. The **cost** of a computational procedure is the number of arithmetic operations on complex numbers that it performs, counted as follows:

- a **complex multiplication** is a product zw with $z, w \in \mathbb{C}$;
- a **complex addition** is a sum $z + w$ or a difference $z - w$ with $z, w \in \mathbb{C}$ (difference and sum have the same cost).

Not counted: memory reads and writes, index increments, comparisons, and the computation of the twiddle factors ω_N^j , which are assumed available in a precomputed table. A multiplication is called **trivial** if one of the two factors belongs to $\{1, -1, i, -i\}$: it requires no real multiplication at all, only swaps between real and imaginary part and sign changes. It is called **nontrivial** otherwise.

***Remark 23.1: Why this model and not another.** The model of Definition 23.1 is the one used by the FFT literature (Van Loan, Chapter 1; Oppenheim and Schaffer, Chapter 9) and is the one that makes the comparison between algorithms independent of the hardware. It should be remembered, however, that a complex multiplication $(a + ib)(c + id)$ costs, in the schoolbook algorithm, 4 real multiplications and 2 real additions; with Karatsuba’s variant applied to complex numbers, which computes $ac - bd$ and $(a + b)(c + d) - ac - bd$, 3 and 5 suffice. A complex addition costs 2 real additions. A count in real operations is therefore obtained from ours by multiplying by constant factors that depend on the implementation, and it changes neither the order of magnitude nor the ratio between the two algorithms that matters here. Trivial multiplications are counted separately precisely because for them those constant factors collapse to zero: multiplying by $-i$ means swapping the two components and changing one sign.*

Lemma 23.1: Arithmetic of the twiddle factors. Let $N \geq 2$ and $\omega_N = e^{-2\pi i/N}$. The following hold:

- (i) $|\omega_N| = 1$ and $\omega_N^{-1} = \overline{\omega_N}$;
- (ii) $\omega_N^j = 1$ if and only if $N \mid j$; in particular ω_N is a primitive N -th root of unity, and ω_N^j depends only on $\langle j \rangle_N$;
- (iii) (**order reduction**) if N is even, $\omega_N^2 = \omega_{N/2}$, and more generally $\omega_N^{2j} = \omega_{N/2}^j$ for every $j \in \mathbb{Z}$;
- (iv) (**half turn**) if N is even, $\omega_N^{N/2} = -1$, and hence $\omega_N^{k+N/2} = -\omega_N^k$ for every $k \in \mathbb{Z}$;
- (v) if $4 \mid N$, then $\omega_N^{N/4} = -i$;
- (vi) for every integer $d \geq 1$, $\omega_{dN}^d = \omega_N^j$.

Proof. (i) $|\omega_N| = |e^{-2\pi i/N}| = 1$, because $|e^{i\theta}| = 1$ for every $\theta \in \mathbb{R}$; from $\omega_N \overline{\omega_N} = |\omega_N|^2 = 1$ it follows that $\omega_N^{-1} = \overline{\omega_N}$.

(ii) $\omega_N^j = e^{-2\pi i j/N}$ equals 1 if and only if $-2\pi j/N$ is an integer multiple of 2π , that is, if and only if $j/N \in \mathbb{Z}$, that is, $N \mid j$. If $j = qN + r$ with $0 \leq r < N$, then $\omega_N^j = (\omega_N^N)^q \omega_N^r = \omega_N^r$, since $\omega_N^N = 1$: the value depends only on the remainder. Finally, if $\omega_N^j = 1$ with $1 \leq j < N$ then $N \mid j$ would be false: hence no power with exponent between 1 and $N - 1$ equals 1, and ω_N is primitive.

(iii) $\omega_N^2 = e^{-4\pi i/N} = e^{-2\pi i/(N/2)} = \omega_{N/2}$; raising to the j -th power, $\omega_N^{2j} = (\omega_N^2)^j = \omega_{N/2}^j$.

(iv) $\omega_N^{N/2} = e^{-2\pi i(N/2)/N} = e^{-\pi i} = \cos(-\pi) + i \sin(-\pi) = -1$. Hence $\omega_N^{k+N/2} = \omega_N^k \omega_N^{N/2} = -\omega_N^k$.

(v) $\omega_N^{N/4} = e^{-2\pi i(N/4)/N} = e^{-\pi i/2} = \cos(-\pi/2) + i \sin(-\pi/2) = -i$.

(vi) $\omega_{dN}^{dj} = e^{-2\pi i dj/(dN)} = e^{-2\pi i j/N} = \omega_N^j$. □

Remark 23.2: The two identities that do all the work. Of the whole of Lemma 23.1, the radix-2 algorithm essentially uses two points: order reduction (iii), which turns a sum with factors ω_N restricted to even indices into a sum with factors $\omega_{N/2}$ — that is, into a shorter DFT —, and the half turn (iv), which allows two outputs to be obtained at the price of a single multiplication. All the rest of the chapter is bookkeeping around these two identities. It is worth noting that both are statements about the geometry of the unit circle: (iii) says that walking around the circle in double steps means walking around a circle with half the vertices, (iv) says that half a turn is the central symmetry.

Proposition 23.1: Exact cost of the direct computation of the DFT. Let $N \geq 1$ and $x \in \mathbb{C}^N$. Computing all N components

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}, \quad k = 0, \dots, N-1,$$

carried out by literally applying Definition 22.1, with the factors ω_N^{kn} read from a table, requires

$$M_{\text{dir}}(N) = N^2 \quad \text{complex multiplications,} \quad A_{\text{dir}}(N) = N(N-1) \quad \text{complex additions.}$$

If the multiplications whose twiddle factor equals 1 are avoided, the number of multiplications becomes

$$M_{\text{dir}}^*(N) = N^2 - T(N), \quad T(N) = \#\{(k, n) \in \{0, \dots, N-1\}^2 : N \mid kn\},$$

and $M_{\text{dir}}^*(N) \leq (N-1)^2$ holds, with equality if and only if $N = 1$ or N is prime. The number of additions remains $N(N-1)$ in every case.

Proof. Fix k . The sum $\sum_{n=0}^{N-1} x[n] \omega_N^{kn}$ has N terms, and each term is the product $x[n] \cdot \omega_N^{kn}$ of two complex numbers: these are N complex multiplications. Summing N numbers requires $N-1$ additions: one adds the first two (one addition), then adds the third (two additions), and so on up to the N -th; that $N-1$ additions are also *necessary* follows from the fact that every binary addition reduces by one the number of quantities still to be combined, and one starts from N to reach 1. Hence each component costs N multiplications and $N-1$ additions. Since there are N components and in the direct computation they are obtained independently of one another,

$$M_{\text{dir}}(N) = N \cdot N = N^2, \quad A_{\text{dir}}(N) = N \cdot (N-1) = N(N-1).$$

For the second part, we count the pairs (k, n) with $\omega_N^{kn} = 1$. By Lemma 23.1(ii) this happens exactly when $N \mid kn$: these are the $T(N)$ pairs of the statement, and $M_{\text{dir}}^*(N) = N^2 - T(N)$ by definition.

Among the pairs counted by $T(N)$ there are in every case those with $k = 0$ (there are N of them, as n varies) and those with $n = 0$ (there are N), which overlap only in the pair $(0, 0)$: hence there are $2N - 1$ distinct pairs, and $T(N) \geq 2N - 1$, whence

$$M_{\text{dir}}^*(N) = N^2 - T(N) \leq N^2 - (2N - 1) = (N - 1)^2.$$

Equality holds if and only if $T(N) = 2N - 1$, that is, if and only if there is no pair (k, n) with $1 \leq k, n \leq N - 1$ and $N \mid kn$. If $N = 1$ the set $\{1, \dots, N - 1\}$ is empty and there is nothing to exclude: equality. If $N = p$ is prime and $1 \leq k, n \leq p - 1$, then $p \nmid k$ and $p \nmid n$; if $p \mid kn$ held, primality of p would give $p \mid k$ or $p \mid n$, a contradiction: equality. If instead N is composite, write $N = ab$ with $1 < a < N$ and $1 < b < N$: the pair $(k, n) = (a, b)$ satisfies $1 \leq a, b \leq N - 1$ and $N \mid ab$, hence $T(N) \geq 2N$ and the inequality is strict. The three cases exhaust all $N \geq 1$. The additions do not change, because each component remains the sum of N terms, some of which are now obtained without multiplying. $\square$

Example 23.1: The exact count for $N = 8$. For $N = 8$ Proposition 23.1 gives $M_{\text{dir}}(8) = 64$ and $A_{\text{dir}}(8) = 8 \cdot 7 = 56$. For the exact count of the nontrivial multiplications by 1 we need $T(8)$. To the $2 \cdot 8 - 1 = 15$ pairs with $k = 0$ or $n = 0$ we add the pairs (k, n) with $1 \leq k, n \leq 7$ and $8 \mid kn$: since $8 = 2^3$, k and n must together contribute three factors of 2; with $k, n \leq 7$ the only possibilities are $k = 2$ (which carries one factor of 2) with $n = 4$, $k = 6$ (also a single factor of 2) with $n = 4$, and $k = 4$ (two factors of 2) with $n \in \{2, 4, 6\}$: these are the five pairs

$$(2, 4), \quad (4, 2), \quad (4, 4), \quad (4, 6), \quad (6, 4).$$

Hence $T(8) = 15 + 5 = 20$ and $M_{\text{dir}}^*(8) = 64 - 20 = 44$, against the bound $(8 - 1)^2 = 49$ of the proposition, consistently with the fact that 8 is not prime.

Independent check. A Python script that enumerates the 64 pairs and counts those with `(k*n) % 8 == 0` returns 20, and $64 - 20 = 44$. $\checkmark$

Remark 23.3: *The radix-2 hypothesis is not in the definition.* From here to Section 23.12 we shall work under the hypothesis

$$N = 2^m, \quad m \in \mathbb{N},$$

called the **radix-2 hypothesis**. It is a hypothesis on the algorithm, not on the transform: Definition 22.1 makes sense for every $N \geq 1$, Theorem 22.1 and Theorem 22.2 hold for every $N \geq 1$, and the matrix W_N is invertible for every $N \geq 1$ (Corollary 22.3). That N be a power of two serves only to allow repeated halving of the length while remaining on integers: it is the constraint that closes the recursion of Section 23.4. Section 23.12 shows how to proceed when N is composite but not a power of two, and what to do when N is prime. The worst case — N a large prime — is never a practical obstacle: one uses Rader's algorithm (Theorem 23.12) or Bluestein's (Proposition 23.7), or else the signal is padded with zeros up to a power of two, knowing exactly what this entails (Section 23.11).

23.2. The even–odd decomposition

This is the central section of the chapter: everything that follows is a consequence of the two identities proved here. The idea is disarmingly simple — to split the terms of the defining sum according to the parity of the index n — and its strength lies entirely in Lemma 23.1(iii): on the even indices the twiddle factor of order N turns into the twiddle factor of order $N/2$, that is, the two halves are again DFTs.

Throughout the section $N \geq 2$ is *even* and we set $L = N/2$.

Definition 23.2: Even subsequence and odd subsequence. Let $N = 2L$ and $x \in \mathbb{C}^N$. The **even subsequence** and **odd subsequence** of x are the two vectors $e, o \in \mathbb{C}^L$ defined by

$$e[r] = x[2r], \quad o[r] = x[2r + 1], \quad r = 0, 1, \dots, L - 1.$$

Their L -point DFTs are denoted $E, O \in \mathbb{C}^L$:

$$E[k] = \sum_{r=0}^{L-1} e[r] \omega_L^{kr}, \quad O[k] = \sum_{r=0}^{L-1} o[r] \omega_L^{kr}, \quad k \in \mathbb{Z},$$

where, by Proposition 22.1, the notation is extended to every $k \in \mathbb{Z}$, and E, O turn out to be periodic with period L .

Lemma 23.2: The even indices and the odd indices, spelled out. Let $N = 2L$ with $L \geq 1$. The two maps

$$\begin{aligned} \pi : \{0, 1, \dots, L - 1\} &\rightarrow \{0, 1, \dots, N - 1\}, & \pi(r) &= 2r, \\ \delta : \{0, 1, \dots, L - 1\} &\rightarrow \{0, 1, \dots, N - 1\}, & \delta(r) &= 2r + 1, \end{aligned}$$

are injective, have disjoint images, and the union of the two images is all of $\{0, 1, \dots, N - 1\}$. In other words $\{0, \dots, N - 1\}$ is the disjoint union of the set of its even elements, which are exactly the $2r$ with $0 \leq r \leq L - 1$, and the set of its odd elements, which are exactly the $2r + 1$ with $0 \leq r \leq L - 1$.

Proof. Well-posedness. If $0 \leq r \leq L - 1$ then $0 \leq 2r \leq 2L - 2 = N - 2 \leq N - 1$ and $1 \leq 2r + 1 \leq 2L - 1 = N - 1$: both maps do take values in $\{0, \dots, N - 1\}$.

Injectivity. If $2r = 2r'$ then $r = r'$; if $2r + 1 = 2r' + 1$ then $2r = 2r'$ and hence $r = r'$.

Disjoint images. An element of the image of π is even, one of the image of δ is odd, and no integer is both even and odd: if $2r = 2r' + 1$ held, the left side would be divisible by 2 and the right side would not.

Joint surjectivity. Let $n \in \{0, \dots, N - 1\}$. By Euclidean division of n by 2 there exist unique $q \in \mathbb{Z}$ and $s \in \{0, 1\}$ with $n = 2q + s$. From $0 \leq n \leq N - 1$ it follows that $0 \leq 2q \leq n \leq N - 1 = 2L - 1$, hence $0 \leq q \leq L - 1$ (because $2q \leq 2L - 1$ with q an integer forces $q \leq L - 1$; and $2q = n - s \geq -1$ with q an integer forces $q \geq 0$). If $s = 0$ then $n = \pi(q)$, if $s = 1$ then $n = \delta(q)$.

Since the two images are disjoint and their union is a set of N elements obtained as the union of two sets of L elements each, with $2L = N$, the count of cardinalities checks out and there are no repetitions. $\square$

Theorem 23.1: Even-odd decomposition, first half of the spectrum. Let $N = 2L$, $x \in \mathbb{C}^N$, $X = \text{DFT}_N\{x\}$, and let E, O be as in Definition 23.2. Then for every $k \in \mathbb{Z}$ (and

in particular for every $k = 0, 1, \dots, L-1$)

$$X[k] = E[k] + \omega_N^k O[k]$$

Proof. We start from the definition, with no abbreviations:

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}.$$

Step 1 (splitting the sum). By Lemma 23.2 the index set $\{0, \dots, N-1\}$ is the disjoint union of $\{2r : 0 \leq r \leq L-1\}$ and $\{2r+1 : 0 \leq r \leq L-1\}$. Since a finite sum does not depend on the order of its terms and can be split over a partition of the index set,

$$X[k] = \sum_{\substack{n=0 \\ n \text{ even}}}^{N-1} x[n] \omega_N^{kn} + \sum_{\substack{n=0 \\ n \text{ odd}}}^{N-1} x[n] \omega_N^{kn} = \sum_{r=0}^{L-1} x[2r] \omega_N^{k \cdot 2r} + \sum_{r=0}^{L-1} x[2r+1] \omega_N^{k(2r+1)}.$$

Note that the limits of both sums are $r = 0, \dots, L-1 = N/2 - 1$: this is the content of the lemma, not a convenient choice.

Step 2 (the sum over the even indices is an L -point DFT). By Lemma 23.1(iii), $\omega_N^{2kr} = \omega_{N/2}^{kr} = \omega_L^{kr}$. Hence, using $x[2r] = e[r]$,

$$\sum_{r=0}^{L-1} x[2r] \omega_N^{2kr} = \sum_{r=0}^{L-1} e[r] \omega_L^{kr} = E[k],$$

which is exactly Definition 23.2.

Step 3 (the sum over the odd indices is an L -point DFT, up to a factor). Write the exponent in full: $k(2r+1) = 2kr + k$, hence $\omega_N^{k(2r+1)} = \omega_N^{2kr} \omega_N^k = \omega_L^{kr} \omega_N^k$, again by Lemma 23.1(iii). The factor ω_N^k does not depend on r and can be pulled out of the sum:

$$\sum_{r=0}^{L-1} x[2r+1] \omega_N^{k(2r+1)} = \omega_N^k \sum_{r=0}^{L-1} o[r] \omega_L^{kr} = \omega_N^k O[k].$$

Step 4 (summing the two contributions). Substituting Steps 2 and 3 into Step 1,

$$X[k] = E[k] + \omega_N^k O[k],$$

for every $k \in \mathbb{Z}$: no step used any restriction on k , because E, O and X were defined for every integer (Proposition 22.1). $\square$

Theorem 23.2: Even–odd decomposition, second half of the spectrum. Under the hypotheses and with the notation of Theorem 23.1, for every $k = 0, 1, \dots, L-1$

$$X\left[k + \frac{N}{2}\right] = E[k] - \omega_N^k O[k]$$

Proof. We do not deduce the formula “by analogy” from Theorem 23.1: we recompute it from the definition, as required by the fact that the index $k + N/2$ is no longer between 0 and $L-1$.

Step 1. Theorem 23.1 was proved for every $k \in \mathbb{Z}$; applying it to the integer $k + N/2$ (that is, $k + L$) gives

$$X\left[k + \frac{N}{2}\right] = E\left[k + \frac{N}{2}\right] + \omega_N^{k+N/2} O\left[k + \frac{N}{2}\right].$$

This is legitimate because Step 4 of that proof used no restriction on the index.

Step 2 (periodicity of E and O). E and O are $L = N/2$ -point DFTs; by Proposition 22.1 they are periodic with period L , that is,

$$E\left[k + \frac{N}{2}\right] = E[k + L] = E[k], \quad O\left[k + \frac{N}{2}\right] = O[k + L] = O[k].$$

Let us nonetheless check this in full, so as not to lean on a citation: by Lemma 23.1(ii) applied to ω_L one has $\omega_L^L = 1$, hence

$$E[k + L] = \sum_{r=0}^{L-1} e[r] \omega_L^{(k+L)r} = \sum_{r=0}^{L-1} e[r] \omega_L^{kr} (\omega_L^L)^r = \sum_{r=0}^{L-1} e[r] \omega_L^{kr} = E[k],$$

and identically for O .

Step 3 (half-turn). By Lemma 23.1(iv), $\omega_N^{N/2} = -1$, hence

$$\omega_N^{k+N/2} = \omega_N^k \omega_N^{N/2} = -\omega_N^k.$$

Step 4. Substituting Steps 2 and 3 into Step 1:

$$X\left[k + \frac{N}{2}\right] = E[k] + (-\omega_N^k)O[k] = E[k] - \omega_N^k O[k].$$

□

Corollary 23.1: The two paired formulas. Let $N = 2L$ and $k \in \{0, 1, \dots, L-1\}$. Setting $t_k = \omega_N^k O[k]$, one has

$$X[k] = E[k] + t_k, \quad X[k + L] = E[k] - t_k.$$

As k ranges over $\{0, \dots, L-1\}$, these $2L = N$ equalities determine all the components of X , and each pair $(X[k], X[k + L])$ depends only on the pair $(E[k], O[k])$ and on the factor ω_N^k .

Proof. The two equalities are Theorems 23.1 and 23.2 with the substitution $t_k = \omega_N^k O[k]$. That they determine all the components follows from the fact that the map $\{0, \dots, L-1\} \times \{0, 1\} \rightarrow \{0, \dots, N-1\}$, $(k, s) \mapsto k + sL$, is a bijection: it is injective because $k + sL = k' + s'L$ with $0 \leq k, k' \leq L-1$ gives $(s - s')L = k' - k$ with $|k' - k| \leq L-1 < L$, hence $s = s'$ and $k = k'$; and it is surjective because the domain and codomain both have $2L = N$ elements. The last statement is read directly from the two formulas: only $E[k]$, $O[k]$ and ω_N^k appear in them. □

Remark 23.4: Why the second formula must be recomputed, not “seen”. It is common, in popular texts and lecture notes, to present $X[k + N/2] = E[k] - \omega_N^k O[k]$ as “the same formula with the sign changed”, as though the minus sign were a graphical convention of the butterfly diagram. It is not: the minus sign is the value of $\omega_N^{N/2}$, an arithmetic fact about roots of unity (Lemma 23.1(iv)), and the formula additionally requires the periodicity of E and O with period $N/2$, which is a fact about the shorter DFTs, not about X . Whoever skips these two steps is unable to write the corresponding formula for radix-3 or for mixed-radix, where the analogous factor is not -1 but ω_3^j (Section 23.12), and systematically gets the signs wrong when the exponent convention is $+2\pi i/N$ instead of $-2\pi i/N$.

23.3. The butterfly

Corollary 23.1 isolates the single elementary computational block of the algorithm. It is given a name, its matrix is studied — which is the same for every N and every k , up to the factor ω_N^k — and its cost is counted. This is step 11 of the mandatory path.

Definition 23.3: Butterfly. Given two complex numbers a, b and a factor $w \in \mathbb{C}$, the **butterfly** with inputs (a, b) and factor w is the pair

$$\mathcal{B}_w(a, b) = (a + wb, a - wb) \in \mathbb{C}^2.$$

The matrix of the butterfly is

$$B_w = \begin{pmatrix} 1 & w \\ 1 & -w \end{pmatrix}, \quad \text{so that} \quad \mathcal{B}_w(a, b)^\top = B_w \begin{pmatrix} a \\ b \end{pmatrix}.$$

In the language of the radix-2 decomposition, a butterfly of index k and order N is $\mathcal{B}_{\omega_N^k}(E[k], O[k]) = (X[k], X[k + N/2])$.

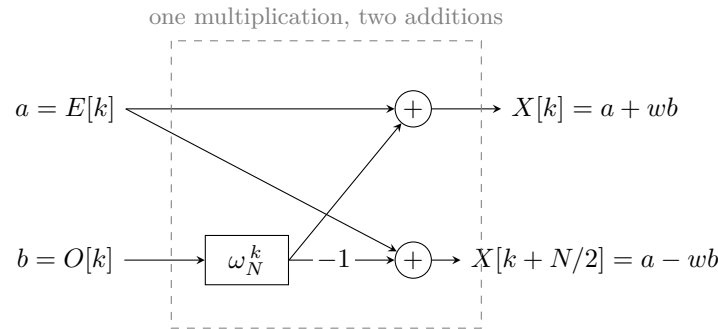


Figure 23.1: The butterfly of Definition 23.3. The twiddle factor ω_N^k multiplies only the lower branch; the product $wb = t$ is computed once and used twice, with opposite sign — the label -1 sits on the edge that carries t to the lower sum node, while the three unlabeled edges have weight 1. The cost is one multiplication and two complex additions (Proposition 23.2).

Proposition 23.2: Cost and correctness of the butterfly. Computing $\mathcal{B}_w(a, b)$ with the scheme

$$t \leftarrow w \cdot b, \quad u_1 \leftarrow a + t, \quad u_2 \leftarrow a - t$$

returns exactly $(a + wb, a - wb)$ and costs one complex multiplication and two complex additions. No scheme that computes both outputs can use fewer than two additions.

Proof. Correctness. After the first instruction $t = wb$; after the second $u_1 = a + t = a + wb$; after the third $u_2 = a - t = a - wb$. These are by definition the two components of $\mathcal{B}_w(a, b)$.
Cost. The scheme performs a single complex product $(w \cdot b)$ and two complex sums $(a + t$ and $a - t)$, according to Definition 23.1. The point is that the product wb appears in both outputs and is computed only once: it is this sharing, and not the identity $\omega_N^{k+N/2} = -\omega_N^k$ taken alone, that produces the saving.

Minimality of the additions. If $w \neq 0$ and $b \neq 0$, the two outputs $a + wb$ and $a - wb$ are distinct (their difference is $2wb \neq 0$); each genuinely depends on both inputs, because $\partial(a \pm wb)/\partial a = 1 \neq 0$ and $\partial(a \pm wb)/\partial b = \pm w \neq 0$. Hence neither can be obtained without at least one addition combining a quantity depending on a with one depending on b , and the two additions cannot coincide because the results are distinct: at least two are needed. $\square$

Proposition 23.3: The matrix of the butterfly: invertibility and singular values.

Let $w \in \mathbb{C}$ with $|w| = 1$ and let $B_w = \begin{pmatrix} 1 & w \\ 1 & -w \end{pmatrix}$. Then:

- (i) $\det B_w = -2w \neq 0$, and $B_w^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ \bar{w} & -\bar{w} \end{pmatrix}$;
- (ii) $B_w^* B_w = 2I_2$, that is, $\frac{1}{\sqrt{2}} B_w$ is unitary;
- (iii) both singular values of B_w equal $\sqrt{2}$; in particular $\|B_w\|_2 = \sqrt{2}$ and $\|B_w^{-1}\|_2 = 1/\sqrt{2}$;
- (iv) $\|B_w v\|_2 = \sqrt{2} \|v\|_2$ for every $v \in \mathbb{C}^2$.

Proof. (i) $\det B_w = 1 \cdot (-w) - w \cdot 1 = -2w$, which is nonzero because $|w| = 1$. The matrix proposed as the inverse satisfies

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ \bar{w} & -\bar{w} \end{pmatrix} \begin{pmatrix} 1 & w \\ 1 & -w \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+1 & w-w \\ \bar{w}-\bar{w} & \bar{w}w + \bar{w}w \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2|w|^2 \end{pmatrix} = I_2,$$

having used $\bar{w}w = |w|^2 = 1$.

(ii) One has $B_w^* = \begin{pmatrix} 1 & 1 \\ \bar{w} & -\bar{w} \end{pmatrix}$, hence

$$B_w^* B_w = \begin{pmatrix} 1 & 1 \\ \bar{w} & -\bar{w} \end{pmatrix} \begin{pmatrix} 1 & w \\ 1 & -w \end{pmatrix} = \begin{pmatrix} 2 & w-w \\ \bar{w}-\bar{w} & 2|w|^2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I_2.$$

(iii) The singular values of B_w are the square roots of the eigenvalues of $B_w^* B_w = 2I_2$, which both equal 2: hence both singular values equal $\sqrt{2}$. The spectral norm is the largest singular value, that is, $\sqrt{2}$; the spectral norm of the inverse is the reciprocal of the smallest singular value, that is, $1/\sqrt{2}$.

(iv) For every $v \in \mathbb{C}^2$, $\|B_w v\|_2^2 = v^* B_w^* B_w v = 2v^* v = 2\|v\|_2^2$, whence the claim follows by taking the square root. $\square$

Remark 23.5: The butterfly is a dilated rotation. Proposition 23.3(ii) says that, up to the factor $\sqrt{2}$, the butterfly is a unitary transformation of $\mathbb{C}^2$: it preserves angles and multiplies all lengths by $\sqrt{2}$. This is the deep reason why the algorithm is numerically stable: no stage can selectively amplify one component relative to another, and hence no round-off error can be inflated differentially (Section 23.13, Theorem 23.13). The factor $\sqrt{2}$ per stage, repeated $m = \log_2 N$ times, gives $2^{m/2} = \sqrt{N}$, which is exactly the norm of the DFT matrix (Theorem 22.3): the bookkeeping of norms checks out, as it must.

23.4. The radix-2 algorithm with decimation in time

The two formulas of Corollary 23.1 reduce a DFT of length N to two DFTs of length $N/2$ plus $N/2$ butterflies. If $N = 2^m$, the two DFTs of length $N/2 = 2^{m-1}$ can be reduced in the same way, and so on down to DFTs of length 1, which require no computation at all. This is step 12 of the path: the *recursive* application of the same factorization. The traditional name is **decimation in time** (DIT for short), because what is split in two is the sequence of samples, that is, the “time” index n .

Definition 23.4: The radix-2 FFT algorithm with decimation in time. Let $N = 2^m$ with $m \geq 0$. Define the map $\mathcal{A}_N : \mathbb{C}^N \rightarrow \mathbb{C}^N$ by recursion on m :

- **base case** $m = 0$, that is, $N = 1$: $\mathcal{A}_1(x) = x$, that is, the single component of the output is $\mathcal{A}_1(x)[0] = x[0]$;
- **recursive step** $m \geq 1$, that is, $N = 2L$ with $L = 2^{m-1}$: given $e, o \in \mathbb{C}^L$ as in

Definition 23.2 ($e[r] = x[2r]$, $o[r] = x[2r + 1]$), set

$$\tilde{E} = \mathcal{A}_L(e), \quad \tilde{O} = \mathcal{A}_L(o),$$

and, for $k = 0, 1, \dots, L - 1$,

$$t_k = \omega_N^k \tilde{O}[k], \quad \mathcal{A}_N(x)[k] = \tilde{E}[k] + t_k, \quad \mathcal{A}_N(x)[k + L] = \tilde{E}[k] - t_k.$$

The definition is well posed: the recursive step for $N = 2^m$ invokes $\mathcal{A}$ only on vectors of length 2^{m-1} , and the recursion terminates in m levels.

Lemma 23.3: The two smallest cases, spelled out.

- (i) For $N = 1$ the DFT is the identity: $X[0] = x[0]$. Hence $\mathcal{A}_1 = \text{DFT}_1$, and the base case of Definition 23.4 is correct.
- (ii) For $N = 2$ one has $X[0] = x[0] + x[1]$ and $X[1] = x[0] - x[1]$; Definition 23.4 produces exactly these two values, with zero multiplications and two additions.

Proof. (i) For $N = 1$ Definition 22.1 gives $X[0] = \sum_{n=0}^0 x[n] \omega_1^{0 \cdot n} = x[0] \omega_1^0 = x[0]$, since $\omega_1 = e^{-2\pi i/1} = e^{-2\pi i} = 1$ and every one of its powers equals 1.

(ii) For $N = 2$ one has $\omega_2 = e^{-\pi i} = -1$, hence

$$X[0] = x[0] \omega_2^0 + x[1] \omega_2^0 = x[0] + x[1], \quad X[1] = x[0] \omega_2^0 + x[1] \omega_2^1 = x[0] - x[1].$$

Let us see what the algorithm produces. Here $L = 1$, $e = (x[0])$, $o = (x[1])$; by the base case $\tilde{E} = (x[0])$ and $\tilde{O} = (x[1])$. The only value of k is $k = 0$, and $t_0 = \omega_2^0 \tilde{O}[0] = 1 \cdot x[1] = x[1]$; hence

$$\mathcal{A}_2(x)[0] = x[0] + x[1], \quad \mathcal{A}_2(x)[1] = x[0] - x[1],$$

which coincide with $X[0]$ and $X[1]$. The factor is $\omega_2^0 = 1$: the multiplication is trivial in the sense of Definition 23.1 and is not carried out; the two additions remain. $\square$

Theorem 23.3: Correctness of the radix-2 DIT algorithm. For every $m \in \mathbb{N}$ and every $x \in \mathbb{C}^N$ with $N = 2^m$,

$$\mathcal{A}_N(x) = \text{DFT}_N\{x\},$$

that is, $\mathcal{A}_N(x)[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}$ for every $k = 0, \dots, N - 1$. In particular the algorithm of Definition 23.4 computes the DFT exactly, with no approximation.

Proof. By induction on m .

Base ($m = 0$, $N = 1$). This is Lemma 23.3(i): $\mathcal{A}_1(x)[0] = x[0] = X[0]$.

Inductive step. Let $m \geq 1$ and suppose (*inductive hypothesis*) that for every vector $v \in \mathbb{C}^L$ with $L = 2^{m-1}$, $\mathcal{A}_L(v) = \text{DFT}_L\{v\}$. Let $x \in \mathbb{C}^N$ with $N = 2^m = 2L$, and let $e, o \in \mathbb{C}^L$ be its even and odd subsequences.

Step 1. By the inductive hypothesis applied to $v = e$ and to $v = o$,

$$\tilde{E} = \mathcal{A}_L(e) = \text{DFT}_L\{e\} = E, \quad \tilde{O} = \mathcal{A}_L(o) = \text{DFT}_L\{o\} = O,$$

where E and O are the L -point DFTs of Definition 23.2. The quantities computed by the algorithm therefore coincide with those appearing in Theorems 23.1 and 23.2.

Step 2. Let $k \in \{0, \dots, L - 1\}$. The algorithm sets $t_k = \omega_N^k \tilde{O}[k] = \omega_N^k O[k]$ and returns

$$\mathcal{A}_N(x)[k] = \tilde{E}[k] + t_k = E[k] + \omega_N^k O[k], \quad \mathcal{A}_N(x)[k + L] = \tilde{E}[k] - t_k = E[k] - \omega_N^k O[k].$$

Step 3. By Theorem 23.1, $E[k] + \omega_N^k O[k] = X[k]$; by Theorem 23.2, $E[k] - \omega_N^k O[k] = X[k + N/2] = X[k + L]$. Hence

$$\mathcal{A}_N(x)[k] = X[k], \quad \mathcal{A}_N(x)[k + L] = X[k + L], \quad k = 0, \dots, L - 1.$$

Step 4. By Corollary 23.1 the map $(k, s) \mapsto k + sL$ is a bijection of $\{0, \dots, L - 1\} \times \{0, 1\}$ onto $\{0, \dots, N - 1\}$: hence Step 3 covers *all* output indices, each exactly once, and $\mathcal{A}_N(x) = X$.

By the principle of induction the claim holds for every $m \in \mathbb{N}$. $\square$

Corollary 23.2: Every output contains all the terms of the sum, with the correct factor. Let $N = 2^m$. For every k , the output $\mathcal{A}_N(x)[k]$, expanded recursively down to the leaves of the recursion, is a linear combination $\sum_{n=0}^{N-1} c_{k,n} x[n]$ in which all and only the N samples $x[0], \dots, x[N-1]$ appear, each exactly once, and the coefficient is exactly $c_{k,n} = \omega_N^{kn}$.

Proof. By Theorem 23.3, $\mathcal{A}_N(x)[k] = X[k]$ for every $x \in \mathbb{C}^N$: the two linear maps $x \mapsto \mathcal{A}_N(x)[k]$ and $x \mapsto X[k]$ coincide on all of $\mathbb{C}^N$.

That $x \mapsto \mathcal{A}_N(x)[k]$ is linear is seen by induction on m : for $m = 0$ it is the identity; in the recursive step, e and o depend linearly on x , by the inductive hypothesis $\tilde{E}$ and $\tilde{O}$ depend linearly on e and o , and the formulas $\tilde{E}[k] \pm \omega_N^k \tilde{O}[k]$ are linear in $(\tilde{E}, \tilde{O})$; the composition of linear maps is linear.

Two equal linear maps have the same coefficients with respect to the canonical basis: evaluating both on $x = \varepsilon_n$ (the vector with 1 in position n and 0 elsewhere) gives $c_{k,n} = X[k]|_{x=\varepsilon_n} = \omega_N^{kn}$. Since $|\omega_N^{kn}| = 1 \neq 0$, no coefficient vanishes: every sample genuinely appears, and it appears exactly once because the coefficient of $x[n]$ in a linear combination is unique. $\square$

Remark 23.6: The recursion tree. Definition 23.4 generates a binary tree of calls (Figure 23.2). At the root sits the problem of length $N = 2^m$; every node of length 2^j with $j \geq 1$ has two children of length 2^{j-1} ; the leaves are the problems of length 1. Level ℓ (with $\ell = 0$ at the root) contains 2^ℓ nodes, each of length $2^{m-\ell}$; the sum of the lengths on a level is $2^\ell \cdot 2^{m-\ell} = 2^m = N$, the same on every level: it is this fact that, in Section 23.5, will produce a cost of $\Theta(N)$ per level and hence $\Theta(N \log_2 N)$ overall. There are $m + 1 = \log_2 N + 1$ levels, of which m contain butterflies (the last one, that of the leaves, contains none).

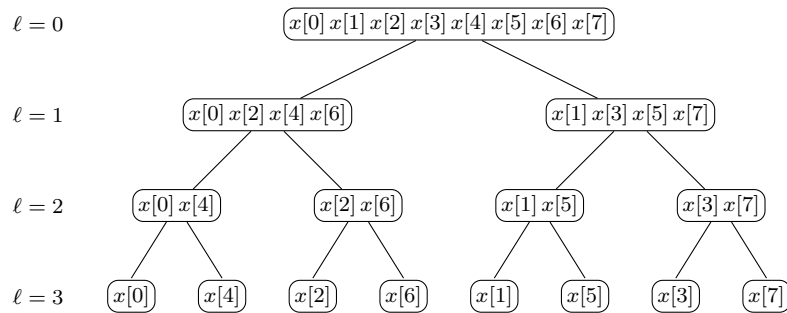


Figure 23.2: Recursion tree for $N = 8$. Level ℓ contains 2^ℓ nodes of length $2^{3-\ell}$, hence at every level the sum of the lengths equals 8, and there are $\log_2 8 + 1 = 4$ levels. The leaves, read from left to right, give the order $x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]$: this is the bit-reversal permutation of Section 23.6 (Theorem 23.7).

23.5. The cost: recurrence, solution, exact count

These are steps 15 to 18 of the mandatory path. We first count the butterflies of a single level, then we set up the recurrence, then we solve it in two ways (an exact closed form and the solution of the general recurrence $T(N) = 2T(N/2) + cN$), and finally we refine the count by removing the trivial multiplications. The result is a number, not an order of magnitude.

Proposition 23.4: The butterflies of one level. Let $N = 2^m$ and consider the recursion tree of Remark 23.6. For every $\ell \in \{0, 1, \dots, m-1\}$, level ℓ contains 2^ℓ nodes, each of length $n_\ell = 2^{m-\ell} \geq 2$, and each node performs $n_\ell/2$ butterflies. The total number of butterflies of level ℓ is therefore

$$2^\ell \cdot \frac{n_\ell}{2} = 2^\ell \cdot 2^{m-\ell-1} = 2^{m-1} = \frac{N}{2},$$

the same for every level. The overall number of butterflies executed by the algorithm is $\frac{N}{2} m = \frac{N}{2} \log_2 N$.

Proof. Number and length of the nodes. By induction on ℓ : level 0 contains only the initial problem, that is, $2^0 = 1$ node of length $2^{m-0} = N$. If level $\ell < m$ contains 2^ℓ nodes of length $2^{m-\ell} \geq 2$, each of them, by Definition 23.4, generates exactly two recursive calls on vectors of length $2^{m-\ell}/2 = 2^{m-\ell-1}$: level $\ell + 1$ contains $2 \cdot 2^\ell = 2^{\ell+1}$ nodes of length $2^{m-(\ell+1)}$.

Butterflies per node. A node of length $n \geq 2$ performs, again by Definition 23.4, a loop over $k = 0, 1, \dots, n/2 - 1$: these are $n/2$ butterflies. The nodes of level m have length 1 and perform no butterflies (base case).

Count. $2^\ell \cdot 2^{m-\ell}/2 = 2^{m-1} = N/2$ for every $\ell \in \{0, \dots, m-1\}$; summing over the m levels that contain butterflies gives $m \cdot N/2$. $\square$

Theorem 23.4: The cost recurrence. Let $M(N)$ and $A(N)$ be the number of complex multiplications and complex additions performed by the algorithm $\mathcal{A}_N$ of Definition 23.4 on an input of length $N = 2^m$, counted according to Definition 23.1 and without removing the trivial multiplications. Then

$$M(1) = 0, \quad M(N) = 2M\left(\frac{N}{2}\right) + \frac{N}{2} \quad (N \geq 2),$$

$$A(1) = 0, \quad A(N) = 2A\left(\frac{N}{2}\right) + N \quad (N \geq 2).$$

In particular, setting $T(N) = M(N) + A(N)$, one has $T(1) = 0$ and $T(N) = 2T(N/2) + \frac{3}{2}N$: this is the recurrence $T(N) = 2T(N/2) + cN$ with $c = \frac{3}{2}$.

Proof. The case $N = 1$ is immediate: the base case of Definition 23.4 returns the input without performing any operations, hence $M(1) = A(1) = 0$.

Let $N = 2L \geq 2$. The recursive step performs, in order:

1. forming e and o : these are reads and writes, which the cost model does not count;
2. two recursive calls on vectors of length $L = N/2$: cost $2M(N/2)$ multiplications and $2A(N/2)$ additions;
3. the loop over $k = 0, \dots, L - 1$, which performs $L = N/2$ butterflies; by Proposition 23.2 each butterfly costs one multiplication and two additions, hence the loop costs $N/2$ multiplications and $2 \cdot (N/2) = N$ additions.

Summing the contributions gives the two recurrences. Adding them together: $T(N) = 2(M(N/2) + A(N/2)) + \frac{N}{2} + N = 2T(N/2) + \frac{3}{2}N$. $\square$

Theorem 23.5: Exact solution: the operation count. For every $m \in \mathbb{N}$ and $N = 2^m$,

$$M(N) = \frac{N}{2} \log_2 N = \frac{Nm}{2}, \quad A(N) = N \log_2 N = Nm, \quad T(N) = \frac{3}{2} N \log_2 N$$

Proof. We give two independent proofs.

First proof: induction on m . For M . *Base* ($m = 0, N = 1$): $M(1) = 0$ and $\frac{1}{2} \cdot 1 \cdot \log_2 1 = 0$: equal. *Step*: let $m \geq 1$ and suppose $M(2^{m-1}) = \frac{2^{m-1}(m-1)}{2}$. Then, by Theorem 23.4,

$$M(2^m) = 2 M(2^{m-1}) + \frac{2^m}{2} = 2 \cdot \frac{2^{m-1}(m-1)}{2} + 2^{m-1} = 2^{m-1}(m-1) + 2^{m-1} = 2^{m-1}m = \frac{2^m m}{2},$$

which is the claim for m . Identically for A : base $A(1) = 0 = 1 \cdot \log_2 1$; step, with hypothesis $A(2^{m-1}) = 2^{m-1}(m-1)$,

$$A(2^m) = 2 \cdot 2^{m-1}(m-1) + 2^m = 2^m(m-1) + 2^m = 2^m m.$$

Finally $T = M + A = \frac{Nm}{2} + Nm = \frac{3}{2} Nm$.

Second proof: direct count over the levels. By Proposition 23.4 the algorithm performs exactly $N/2$ butterflies on each of the m levels $\ell = 0, \dots, m-1$, and none elsewhere; by Proposition 23.2 each butterfly costs one multiplication and two additions. Hence

$$M(N) = m \cdot \frac{N}{2}, \quad A(N) = m \cdot \frac{N}{2} \cdot 2 = mN.$$

The two routes give the same result. ✓

Independent numerical check. A Python script that simulates the iterative algorithm and increments two counters at every butterfly returns, for $N = 2, 4, 8, 16, 32, 1024$: multiplications 1, 4, 12, 32, 80, 5 120 and additions 2, 8, 24, 64, 160, 10 240, against the formulas $\frac{N}{2} \log_2 N$ and $N \log_2 N$, which give 1, 4, 12, 32, 80, 5 120 and 2, 8, 24, 64, 160, 10 240: an exact match in every case. ✓ □

Theorem 23.6: Solution of the general recurrence $T(N) = 2T(N/2) + cN$. Let $c > 0$ and $c_0 \geq 0$ be constants and let $T : \{2^m : m \in \mathbb{N}\} \rightarrow \mathbb{R}$ be defined by

$$T(1) = c_0, \quad T(N) = 2T\left(\frac{N}{2}\right) + cN \quad (N \geq 2).$$

Then for every $N = 2^m$

$$T(N) = cN \log_2 N + c_0 N.$$

In particular $T(N) = \Theta(N \log_2 N)$: there exist constants $0 < \alpha \leq \beta$ and N_0 such that $\alpha N \log_2 N \leq T(N) \leq \beta N \log_2 N$ for every $N = 2^m \geq N_0$.

Proof. **Closed form, by induction on m .** *Base* ($m = 0$): $T(1) = c_0 = c \cdot 1 \cdot 0 + c_0 \cdot 1$. *Step*: let $m \geq 1$ and suppose $T(2^{m-1}) = c 2^{m-1}(m-1) + c_0 2^{m-1}$. Then

$$T(2^m) = 2[c 2^{m-1}(m-1) + c_0 2^{m-1}] + c 2^m = c 2^m(m-1) + c_0 2^m + c 2^m = c 2^m m + c_0 2^m,$$

which is the claim for m .

Second route: the cost tree. Expanding the recurrence,

$$T(N) = 2T\left(\frac{N}{2}\right) + cN = 4T\left(\frac{N}{4}\right) + 2c\frac{N}{2} + cN = \dots = 2^m T(1) + \sum_{\ell=0}^{m-1} 2^\ell c \frac{N}{2^\ell}.$$

Formally: one proves by induction on $j \in \{0, \dots, m\}$ that $T(N) = 2^j T(N/2^j) + jcN$. For $j = 0$ it is an identity. If it holds for $j < m$, then

$$2^j T\left(\frac{N}{2^j}\right) + jcN = 2^j \left[2T\left(\frac{N}{2^{j+1}}\right) + c\frac{N}{2^j} \right] + jcN = 2^{j+1} T\left(\frac{N}{2^{j+1}}\right) + (j+1)cN.$$

For $j = m$ one obtains $T(N) = 2^m T(1) + mcN = c_0 N + cN \log_2 N$. The two routes agree. ✓
Asymptotic conclusion. From $T(N) = cN \log_2 N + c_0 N$ it follows at once that $T(N) \geq cN \log_2 N$ for every N , and for $N \geq 4$ (that is, $\log_2 N \geq 2$)

$$T(N) = cN \log_2 N + c_0 N \leq cN \log_2 N + \frac{c_0}{2} N \log_2 N = \left(c + \frac{c_0}{2}\right) N \log_2 N.$$

Hence the claim with $\alpha = c$, $\beta = c + c_0/2$ and $N_0 = 4$. □

Corollary 23.3: *The FFT costs $\Theta(N \log N)$, the direct computation $\Theta(N^2)$. With the previous notation, for $N = 2^m$*

$$\frac{M_{\text{dir}}(N) + A_{\text{dir}}(N)}{T(N)} = \frac{N^2 + N(N-1)}{\frac{3}{2} N \log_2 N} = \frac{2N-1}{\frac{3}{2} \log_2 N} \xrightarrow{N \rightarrow \infty} +\infty,$$

and the ratio is already ≥ 1 for $N \geq 4$. The gain therefore grows as $\Theta(N/\log_2 N)$.

Proof. By Proposition 23.1, $M_{\text{dir}}(N) + A_{\text{dir}}(N) = N^2 + N(N-1) = N(2N-1)$; by Theorem 23.5, $T(N) = \frac{3}{2} N \log_2 N$. The ratio is $\frac{N(2N-1)}{\frac{3}{2} N \log_2 N} = \frac{2(2N-1)}{3 \log_2 N}$, which tends to $+\infty$ because the

numerator grows linearly in N and the denominator logarithmically: formally, $\frac{2(2N-1)}{3 \log_2 N} \geq \frac{2N}{3 \log_2 N}$ for $N \geq 1$, and $N/\log_2 N \rightarrow +\infty$. For $N = 4$: $\frac{2 \cdot 7}{3 \cdot 2} = \frac{14}{6} > 1$.

That the ratio is *increasing* in m is seen as follows: setting $N = 2^m$ with $m \geq 1$, the ratio equals $g(m) = \frac{2(2^{m+1}-1)}{3m}$, and

$$\frac{g(m+1)}{g(m)} = \frac{2^{m+2}-1}{2^{m+1}-1} \cdot \frac{m}{m+1} > \frac{2(2^{m+1}-1)}{2^{m+1}-1} \cdot \frac{m}{m+1} = \frac{2m}{m+1} \geq 1 \quad (m \geq 1),$$

where the strict inequality uses $2^{m+2}-1 > 2(2^{m+1}-1) = 2^{m+2}-2$. □

Proposition 23.5: *How many multiplications are trivial. Let $N = 2^m$ with $m \geq 1$. Among the $\frac{N}{2}m$ butterflies performed by $\mathcal{A}_N$:*

- (i) *exactly $N-1$ have twiddle factor $\omega_n^0 = 1$;*
- (ii) *exactly $\max\{N/2-1, 0\}$ have twiddle factor $\omega_n^{n/4} = -i$;*
- (iii) *none has twiddle factor -1 or $+i$.*

Consequently the nontrivial multiplications are, for $m \geq 1$,

$$M^{**}(N) = \frac{N}{2} \log_2 N - (N-1) - \left(\frac{N}{2} - 1\right) = \frac{N}{2} \log_2 N - \frac{3N}{2} + 2.$$

Proof. By Proposition 23.4, the nodes of length $n = 2^j$ (with $1 \leq j \leq m$) number N/n , and each performs the butterflies of index $k = 0, 1, \dots, \frac{n}{2} - 1$ with factor ω_n^k .

(i) Within a node of length n , $\omega_n^k = 1$ with $0 \leq k \leq \frac{n}{2} - 1$ occurs, by Lemma 23.1(ii), if and only if $n \mid k$, that is (since $0 \leq k < n$) if and only if $k = 0$: a single butterfly per node.

Summing over all nodes,

$$\sum_{j=1}^m \frac{N}{2^j} = N \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^m} \right) = N \cdot \frac{\frac{1}{2}(1 - 2^{-m})}{1 - \frac{1}{2}} = N(1 - 2^{-m}) = N - 1,$$

having used $2^{-m} = 1/N$.

(ii) $\omega_n^k = -i = e^{-i\pi/2}$ with $0 \leq k \leq \frac{n}{2} - 1$ occurs if and only if $e^{-2\pi i k/n} = e^{-i\pi/2}$, that is, $\frac{2k}{n} \equiv \frac{1}{2} \pmod{2}$, that is, $k \equiv \frac{n}{4} \pmod{n}$; in the interval considered this requires $4 \mid n$ (hence $n \geq 4$, that is, $j \geq 2$) and gives the single value $k = \frac{n}{4}$, which satisfies $0 \leq \frac{n}{4} \leq \frac{n}{2} - 1$ precisely for $n \geq 4$. Hence a single butterfly per node, and only for the nodes of length ≥ 4 . Summing,

$$\sum_{j=2}^m \frac{N}{2^j} = N \left(\frac{1}{4} + \cdots + \frac{1}{2^m} \right) = N \left(1 - 2^{-m} - \frac{1}{2} \right) = \frac{N}{2} - 1,$$

valid for $m \geq 2$; for $m = 1$ there are no nodes of length ≥ 4 and the count is $0 = \max\{N/2 - 1, 0\}$ with $N = 2$.

(iii) $\omega_n^k = -1$ requires, by Lemma 23.1(iv) and the uniqueness of the representation, $k = \frac{n}{2}$, which is outside the interval $0 \leq k \leq \frac{n}{2} - 1$. $\omega_n^k = +i = e^{+i\pi/2} = e^{-i3\pi/2}$ requires $k = \frac{3n}{4} > \frac{n}{2} - 1$: also outside the interval.

Subtracting (i) and (ii) from the total $\frac{N}{2}m$ of Theorem 23.5 gives, for $m \geq 2$,

$$M^{**}(N) = \frac{N}{2}m - (N - 1) - \left(\frac{N}{2} - 1 \right) = \frac{N}{2}m - \frac{3N}{2} + 2,$$

and for $m = 1$ ($N = 2$): $1 - 1 - 0 = 0 = \frac{2}{2} \cdot 1 - 3 + 2$. The formula therefore holds for every $m \geq 1$. $\square$

Corollary 23.4: The final count. For $N = 2^m$, $m \geq 1$, the radix-2 DIT algorithm computes the DFT with

$$\frac{N}{2} \log_2 N - \frac{3N}{2} + 2 \quad \text{nontrivial complex multiplications,}$$

$$N \log_2 N \quad \text{complex additions.}$$

For $N = 8$ this means 2 nontrivial multiplications and 24 additions, against the 64 multiplications and 56 additions of the direct computation; for $N = 1,024$, 3,586 nontrivial multiplications and 10,240 additions, against 1,048,576 and 1,047,552.

Proof. This is Proposition 23.5 combined with Theorem 23.5 and Proposition 23.1. The numerical values: for $N = 8$, $m = 3$: $\frac{8}{2} \cdot 3 - \frac{24}{2} + 2 = 12 - 12 + 2 = 2$ and $8 \cdot 3 = 24$; the direct computation gives $8^2 = 64$ and $8 \cdot 7 = 56$. For $N = 1,024$, $m = 10$: $512 \cdot 10 - 1,536 + 2 = 5,120 - 1,534 = 3,586$ and $1,024 \cdot 10 = 10,240$; the direct computation gives $1,024^2 = 1,048,576$ and $1,024 \cdot 1,023 = 1,047,552$. $\checkmark$ (A Python script that counts the butterflies with factor different from 1, -1 , i , $-i$ returns 2 for $N = 8$ and 3,586 for $N = 1,024$.) $\square$

Remark 23.7: What this table says and what it does not say. Table 23.1 reports operation counts, not times. This is a distinction that the Plan of the work requires to be kept separate, and for a precise reason: on a real computer the time depends on the locality of memory accesses, on the size of the caches, on vectorization and on parallelism, and two implementations with the same operation count can differ by a factor of ten in the measured time (Remark 23.14). The count is, however, what makes the comparison reproducible and

Table 23.1: Cost of computing the DFT: direct computation (Proposition 23.1) against radix-2 FFT (Theorem 23.5 and Corollary 23.4). Every entry is an exact number, not an estimate. The last column is the ratio of the multiplications of the direct computation to those of the FFT.

N	direct computation		radix-2 FFT			ratio
	mult. N^2	add. $N(N-1)$	mult. $\frac{N}{2} \log_2 N$	add. $N \log_2 N$	nontrivial	
8	64	56	12	24	2	5.33
16	256	240	32	64	10	8.00
32	1,024	992	80	160	34	12.80
64	4,096	4,032	192	384	98	21.33
256	65,536	65,280	1,024	2,048	642	64.00
1,024	1,048,576	1,047,552	5,120	10,240	3,586	204.80
4,096	16,777,216	16,773,120	24,576	49,152	18,434	682.67
65,536	4 294 967 296	4 294 901 760	524,288	1,048,576	425,986	8,192.00
1,048,576	1 099 511 627 776	1 099 510 579 200	10,485,760	20,971,520	8,912,898	104,857.60

verifiable: anyone can recount it. Note also that the ratio in the last column equals 1 only for $N = 2$ and grows without bound: for $N = 1,048,576$ the FFT performs roughly one hundred-thousandth of the multiplications of the direct computation.

23.6. From recursion to a loop: the bit-reversal permutation

The algorithm of Definition 23.4 is recursive and allocates new vectors at every call. No serious implementation uses it in that form: the inputs are reordered once and for all, and then m nested loops are run that work *in place*, that is, overwriting the same vector. The reordering needed is the **bit-reversal** permutation, and it is step 19 of the path. That it is the right one is not a fact to be read off a figure: it is a theorem, and it must be proved.

Definition 23.5: Bit reversal. Let $m \geq 0$ and $N = 2^m$. Every $n \in \{0, \dots, N-1\}$ has a unique binary expansion

$$n = \sum_{t=0}^{m-1} b_t(n) 2^t, \quad b_t(n) \in \{0, 1\}.$$

The **bit reversal** of order m is the map $\rho_m : \{0, \dots, N-1\} \rightarrow \{0, \dots, N-1\}$ defined by

$$\rho_m(n) = \sum_{t=0}^{m-1} b_t(n) 2^{m-1-t},$$

that is: n is written in binary on m digits, the string is read from right to left, and the result is again interpreted as a binary number.

Lemma 23.4: ρ_m is an involution, hence a permutation. For every $m \geq 0$, $\rho_m \circ \rho_m = \text{id}$. In particular ρ_m is bijective and $\rho_m^{-1} = \rho_m$.

Proof. Let $n \in \{0, \dots, N-1\}$ and let $n' = \rho_m(n) = \sum_{t=0}^{m-1} b_t(n) 2^{m-1-t}$. We rewrite the sum by setting $s = m-1-t$, that is, $t = m-1-s$; as t ranges over $\{0, \dots, m-1\}$ so does s , and

$$n' = \sum_{s=0}^{m-1} b_{m-1-s}(n) 2^s.$$

The coefficients $b_{m-1-s}(n)$ lie in $\{0, 1\}$ and the binary expansion of an integer of $\{0, \dots, 2^m - 1\}$ on m digits is unique; hence

$$b_s(n') = b_{m-1-s}(n) \quad \text{for every } s \in \{0, \dots, m-1\}.$$

Applying ρ_m again,

$$\rho_m(n') = \sum_{s=0}^{m-1} b_s(n') 2^{m-1-s} = \sum_{s=0}^{m-1} b_{m-1-s}(n) 2^{m-1-s} = \sum_{t=0}^{m-1} b_t(n) 2^t = n,$$

with the substitution $t = m - 1 - s$ in the last step. Hence $\rho_m(\rho_m(n)) = n$. A map of a finite set into itself which, composed with itself, gives the identity is injective (if $\rho_m(n_1) = \rho_m(n_2)$, applying ρ_m gives $n_1 = n_2$) and hence, the domain being finite and equal to the codomain, bijective; and the relation $\rho_m \circ \rho_m = \text{id}$ is precisely the statement that $\rho_m^{-1} = \rho_m$. $\square$

Theorem 23.7: The order of the leaves is the bit-reversed order. Let $N = 2^m$ and consider the recursion tree of Definition 23.4 (Remark 23.6), in which at every node the left child receives the even subsequence and the right child the odd one. Number the leaves from left to right by $j = 0, 1, \dots, N - 1$. Then leaf j contains the sample

$$x[\rho_m(j)].$$

Equivalently: sample $x[n]$ ends up in the leaf at position $\rho_m(n)$.

Proof. Step 1: encoding the path. A path from the root to a leaf is a sequence of m choices $c_0, c_1, \dots, c_{m-1} \in \{0, 1\}$, where $c_\ell = 0$ means “at level ℓ go to the left (even) child” and $c_\ell = 1$ “go to the right (odd) one”. The position from the left of the leaf reached is

$$j = \sum_{\ell=0}^{m-1} c_\ell 2^{m-1-\ell} :$$

this is proved by induction on m . For $m = 0$ there are no choices and $j = 0$. For $m \geq 1$: the left subtree occupies positions $0, \dots, 2^{m-1} - 1$ and the right one positions $2^{m-1}, \dots, 2^m - 1$; hence the position is $c_0 2^{m-1}$ plus the position *relative* to inside the subtree, which by the inductive hypothesis (applied to a tree of height $m-1$) equals $\sum_{\ell=1}^{m-1} c_\ell 2^{(m-1)-1-(\ell-1)} = \sum_{\ell=1}^{m-1} c_\ell 2^{m-1-\ell}$. Adding gives the formula.

Step 2: which sample reaches the end of a path. We claim that the leaf reached with the choices $(c_0, \dots, c_{m-1})$ contains $x[n]$, where n is the unique integer of $\{0, \dots, N - 1\}$ with

$$b_\ell(n) = c_\ell \quad \text{for every } \ell = 0, \dots, m-1.$$

We prove this by induction on m .

Base $m = 0$: the vector has a single element, $x[0]$, and $n = 0$.

Step: let $m \geq 1$. By Definition 23.2, the left child receives e with $e[r] = x[2r]$ and the right one o with $o[r] = x[2r + 1]$; that is, in both cases, the child c_0 receives the vector v with

$$v[r] = x[2r + c_0], \quad r = 0, \dots, 2^{m-1} - 1.$$

By the inductive hypothesis applied to v (of length 2^{m-1}) and to the remaining choices $(c_1, \dots, c_{m-1})$, the leaf contains $v[r]$ with $b_{\ell-1}(r) = c_\ell$ for $\ell = 1, \dots, m-1$, that is, $r = \sum_{\ell=1}^{m-1} c_\ell 2^{\ell-1}$. Hence the leaf contains

$$x[2r + c_0] = x\left[c_0 + \sum_{\ell=1}^{m-1} c_\ell 2^\ell\right] = x\left[\sum_{\ell=0}^{m-1} c_\ell 2^\ell\right],$$

that is, $x[n]$ with $b_\ell(n) = c_\ell$, by the uniqueness of the binary expansion. Note the essential point: the choice at the *first* level is determined by the *least* significant bit of n , that at the second level by the next bit, and so on. It is this exchange between the order of the levels and the order of the bits that produces the reversal.

Step 3: conclusion. Combining Steps 1 and 2: the leaf at position $j = \sum_\ell c_\ell 2^{m-1-\ell}$ contains $x[n]$ with $n = \sum_\ell c_\ell 2^\ell$. But then $b_\ell(n) = c_\ell$ and

$$\rho_m(n) = \sum_{\ell=0}^{m-1} b_\ell(n) 2^{m-1-\ell} = \sum_{\ell=0}^{m-1} c_\ell 2^{m-1-\ell} = j.$$

Hence $x[n]$ sits in leaf $\rho_m(n)$; and since ρ_m is an involution (Lemma 23.4), leaf j contains $x[\rho_m^{-1}(j)] = x[\rho_m(j)]$. $\square$

Example 23.2: The bit-reversal permutation for $N = 8$ and $N = 16$. For $N = 8$ ($m = 3$):

n	0	1	2	3	4	5	6	7
binary	000	001	010	011	100	101	110	111
reversed	000	100	010	110	001	101	011	111
$\rho_3(n)$	0	4	2	6	1	5	3	7

The input must therefore be presented in the order $x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]$, which is exactly the order of the leaves in Figure 23.2. $\checkmark$

Note that ρ_3 fixes 0, 2, 5, 7 and swaps $1 \leftrightarrow 4$ and $3 \leftrightarrow 6$: it is a product of two transpositions, consistently with Lemma 23.4. An implementation can therefore carry out the permutation with swaps alone, without an auxiliary vector.

For $N = 16$ ($m = 4$) one obtains

$$\rho_4 = (0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15),$$

read as “ $\rho_4(0) = 0, \rho_4(1) = 8, \dots$ ”. *Independent check:* a Python script that computes ρ_m with m bit shifts returns $[0, 4, 2, 6, 1, 5, 3, 7]$ for $m = 3$ and $[0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15]$ for $m = 4$, and verifies $\rho_m(\rho_m(n)) = n$ for every n . $\checkmark$

Definition 23.6: The in-place iterative algorithm. Let $N = 2^m$. The **iterative DIT FFT algorithm** receives $x \in \mathbb{C}^N$ and performs:

- (P) *permutation:* $a[j] \leftarrow x[\rho_m(j)]$ for $j = 0, \dots, N - 1$;
- (S) *stages:* for $s = 1, 2, \dots, m$, setting $n_s = 2^s$, and for every $b \in \{0, n_s, 2n_s, \dots, N - n_s\}$ and every $k \in \{0, \dots, \frac{n_s}{2} - 1\}$:

$$t \leftarrow \omega_{n_s}^k \cdot a[b + k + \frac{n_s}{2}], \quad u \leftarrow a[b + k], \quad a[b + k] \leftarrow u + t, \quad a[b + k + \frac{n_s}{2}] \leftarrow u - t.$$

The output is the vector a at the end of stage m . All writes take place on the same vector a : the algorithm uses no auxiliary memory beyond the two variables t and u (*in-place* computation).

Theorem 23.8: Correctness of the iterative algorithm. The algorithm of Definition 23.6 returns $\text{DFT}_N\{x\}$, with the same operation count as Theorem 23.5.

Proof. We prove by induction on $s \in \{0, 1, \dots, m\}$ the following *invariant*: after stage s (with $s = 0$ meaning “after permutation (P) alone”), for every $p \in \{0, 1, \dots, 2^{m-s} - 1\}$ the block of

indices

$$\mathcal{I}_{s,p} = \{p2^s, p2^s + 1, \dots, p2^s + 2^s - 1\}$$

contains, in the natural order, the 2^s -point DFT of the vector $v^{(s,p)} \in \mathbb{C}^{2^s}$ of components

$$v^{(s,p)}[r] = x[r2^{m-s} + \rho_{m-s}(p)], \quad r = 0, \dots, 2^s - 1.$$

Base $s = 0$. The blocks are the singletons $\mathcal{I}_{0,p} = \{p\}$ with $p = 0, \dots, N - 1$, and $v^{(0,p)}$ is the length-1 vector with single component $x[\rho_m(p)]$; its 1-point DFT is itself (Lemma 23.3(i)). And indeed after (P) cell p contains $x[\rho_m(p)]$: the invariant holds.

Step. Let $1 \leq s \leq m$ and suppose the invariant holds after stage $s - 1$. Stage s works on blocks of size $n_s = 2^s$: the block starting at $b = p2^s$ is the union of the two blocks of size 2^{s-1} starting at $2p2^{s-1}$ and at $(2p + 1)2^{s-1}$, that is, of the blocks $\mathcal{I}_{s-1,2p}$ and $\mathcal{I}_{s-1,2p+1}$; by the inductive hypothesis, these contain the 2^{s-1} -point DFTs of $v^{(s-1,2p)}$ and of $v^{(s-1,2p+1)}$. Now observe that, setting $q = \rho_{m-s}(p)$ and $L = 2^{s-1}$,

$$v^{(s-1,2p)}[r] = x[r2^{m-s+1} + \rho_{m-s+1}(2p)] = x[2r2^{m-s} + q],$$

$$v^{(s-1,2p+1)}[r] = x[r2^{m-s+1} + \rho_{m-s+1}(2p + 1)] = x[(2r + 1)2^{m-s} + q],$$

having used $\rho_{m-s+1}(2p) = \rho_{m-s}(p) = q$ and $\rho_{m-s+1}(2p + 1) = \rho_{m-s}(p) + 2^{m-s} = q + 2^{m-s}$: indeed adding a 0 (resp. a 1) as the *least* significant bit of p means adding a 0 (resp. a 1) as the *most* significant bit of $\rho(p)$, by Definition 23.5. Hence $v^{(s-1,2p)}$ and $v^{(s-1,2p+1)}$ are exactly the even subsequence and the odd subsequence of $v^{(s,p)}$, in the sense of Definition 23.2: $v^{(s,p)}[2r] = v^{(s-1,2p)}[r]$ and $v^{(s,p)}[2r + 1] = v^{(s-1,2p+1)}[r]$.

Stage s performs, for $k = 0, \dots, L - 1$, the butterfly $a[b + k] \leftarrow a[b + k] + \omega_{2L}^k a[b + k + L]$, $a[b + k + L] \leftarrow a[b + k] - \omega_{2L}^k a[b + k + L]$ (with the values *before* the update, saved in u and t : this is why Definition 23.6 uses the two temporary variables, and the in-place computation is correct). By Theorems 23.1 and 23.2, applied to $N' = 2L = 2^s$, the result is the 2^s -point DFT of $v^{(s,p)}$, and it sits in the block $\mathcal{I}_{s,p}$ in the natural order ($X[k]$ at position $b + k$, $X[k + L]$ at position $b + k + L$). The invariant holds after stage s .

Conclusion. For $s = m$ there is a single block, $p = 0$, $\mathcal{I}_{m,0} = \{0, \dots, N - 1\}$, $\rho_0(0) = 0$ and $v^{(m,0)}[r] = x[r]$: the vector a contains DFT $_N\{x\}$ in the natural order.

Count. Stage s performs $\frac{N}{n_s}$ blocks of $\frac{n_s}{2}$ butterflies each, that is, $\frac{N}{2}$ butterflies; there are m stages; by Proposition 23.2 the total is $\frac{N}{2}m$ multiplications and Nm additions, as in Theorem 23.5. Permutation (P) performs no arithmetic operations. $\square$

Best practice. Three errors that the invariant of Theorem 23.8 prevents. (1)

Overwriting before reading: if $a[b + k]$ is written before t has been computed, the second output of the butterfly uses an already altered value; the two temporary variables u and t of Definition 23.6 are not an ornament. (2) *Using ω_N^k instead of $\omega_{n_s}^k$* : at a generic stage the factor has order $n_s = 2^s$, not N ; equivalently one may use ω_N^{kN/n_s} , which is the form that allows all the factors to be read from a single table of $N/2$ elements (Lemma 23.1(vi)). (3) *Forgetting the permutation*: without step (P) the algorithm still computes something, but not the DFT — it computes the DFT of the vector $x \circ \rho_m$, that is, the spectrum of a signal whose samples have been shuffled. This is the hardest error to diagnose, because the wrong spectrum has *exactly the same energy* as the correct one (Parseval's identity, Theorem 22.6, is satisfied, because a permutation of the samples does not change $\sum_n |x[n]|^2$): the most common numerical check does not notice it. Solved Exercise 23.6 shows this on numbers.

23.7. The decimation-in-frequency variant

Decimation in time splits the sample index n in two. One can instead split the frequency index k in two, separating the components of even index from those of odd index: this gives the **decimation-in-frequency** variant (DIF), due to Gentleman and Sande and also known as the Sande–Tukey algorithm. This is step 20 of the path. The cost is identical; what changes is *where* the bit-reversal permutation sits: at the input for DIT, at the output for DIF.

Theorem 23.9: Decomposition with decimation in frequency. Let $N = 2L$ and $x \in \mathbb{C}^N$, $X = \text{DFT}_N\{x\}$. Define $\alpha, \beta \in \mathbb{C}^L$ by

$$\alpha[n] = x[n] + x[n+L], \quad \beta[n] = (x[n] - x[n+L]) \omega_N^n, \quad n = 0, 1, \dots, L-1.$$

Then, for $k = 0, 1, \dots, L-1$,

$$X[2k] = (\text{DFT}_L\{\alpha\})[k], \quad X[2k+1] = (\text{DFT}_L\{\beta\})[k]$$

Proof. **Step 1 (splitting the sum in half, over the index n).** For every $j \in \mathbb{Z}$,

$$X[j] = \sum_{n=0}^{N-1} x[n] \omega_N^{jn} = \sum_{n=0}^{L-1} x[n] \omega_N^{jn} + \sum_{n=L}^{N-1} x[n] \omega_N^{jn}.$$

In the second sum set $n = n' + L$ with $n' = 0, \dots, L-1$: the substitution is a bijection between $\{0, \dots, L-1\}$ and $\{L, \dots, N-1\}$, because it is injective and both images have L elements. One obtains

$$\sum_{n=L}^{N-1} x[n] \omega_N^{jn} = \sum_{n'=0}^{L-1} x[n'+L] \omega_N^{j(n'+L)} = \omega_N^{jL} \sum_{n'=0}^{L-1} x[n'+L] \omega_N^{jn'}.$$

Step 2 (the factor ω_N^{jL}). Since $L = N/2$, Lemma 23.1(iv) gives $\omega_N^L = \omega_N^{N/2} = -1$, hence $\omega_N^{jL} = (\omega_N^L)^j = (-1)^j$. Substituting into Step 1 and combining the two sums (which now have the same index set),

$$X[j] = \sum_{n=0}^{L-1} (x[n] + (-1)^j x[n+L]) \omega_N^{jn}.$$

Step 3 (even indices). Let $j = 2k$. Then $(-1)^j = +1$ and, by Lemma 23.1(iii), $\omega_N^{2kn} = \omega_L^{kn}$. Hence

$$X[2k] = \sum_{n=0}^{L-1} (x[n] + x[n+L]) \omega_L^{kn} = \sum_{n=0}^{L-1} \alpha[n] \omega_L^{kn} = (\text{DFT}_L\{\alpha\})[k].$$

Step 4 (odd indices). Let $j = 2k+1$. Then $(-1)^j = -1$ and $\omega_N^{(2k+1)n} = \omega_N^{2kn} \omega_N^n = \omega_L^{kn} \omega_N^n$. Hence

$$X[2k+1] = \sum_{n=0}^{L-1} (x[n] - x[n+L]) \omega_N^n \omega_L^{kn} = \sum_{n=0}^{L-1} \beta[n] \omega_L^{kn} = (\text{DFT}_L\{\beta\})[k].$$

Step 5 (all indices are covered). The indices $2k$ and $2k+1$ with $0 \leq k \leq L-1$ exhaust $\{0, \dots, N-1\}$ without repetitions, by Lemma 23.2 applied to j in place of n . $\square$

Definition 23.7: The radix-2 DIF FFT algorithm. Let $N = 2^m$. The algorithm $\mathcal{D}_N : \mathbb{C}^N \rightarrow \mathbb{C}^N$ is defined by recursion: $\mathcal{D}_1(x) = x$; for $N = 2L \geq 2$ one computes

$\alpha, \beta \in \mathbb{C}^L$ as in Theorem 23.9 and sets

$$\mathcal{D}_N(x)[2k] = \mathcal{D}_L(\alpha)[k], \quad \mathcal{D}_N(x)[2k+1] = \mathcal{D}_L(\beta)[k], \quad k = 0, \dots, L-1.$$

In the in-place iterative version the same scheme is executed *without* rewriting the indices, obtaining the output in bit-reversed order, and the permutation ρ_m is applied at the *end* rather than at the beginning.

Corollary 23.5: Correctness and cost of the DIF. For every $N = 2^m$ and every $x \in \mathbb{C}^N$, $\mathcal{D}_N(x) = \text{DFT}_N\{x\} = \mathcal{A}_N(x)$, and the DIF algorithm performs exactly the same number of operations as the DIT: $\frac{N}{2} \log_2 N$ complex multiplications (of which $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$ nontrivial) and $N \log_2 N$ complex additions.

Proof. Correctness, by induction on m . Base $m = 0$: $\mathcal{D}_1(x) = x$ and $\text{DFT}_1\{x\} = x$ by Lemma 23.3(i). Step: let $m \geq 1$; by the inductive hypothesis $\mathcal{D}_L(\alpha) = \text{DFT}_L\{\alpha\}$ and $\mathcal{D}_L(\beta) = \text{DFT}_L\{\beta\}$; by Theorem 23.9 these equal $X[2k]$ and $X[2k+1]$ respectively, and by Step 5 of that proof all indices are covered exactly once. Hence $\mathcal{D}_N(x) = X$. Equality with $\mathcal{A}_N(x)$ follows from Theorem 23.3.

Cost. The recursive step on length N performs: $L = N/2$ sums $x[n] + x[n+L]$ and L differences $x[n] - x[n+L]$, that is, N additions; then $L = N/2$ multiplications $(\cdot) \omega_N^n$ for $n = 0, \dots, L-1$. The count per level is therefore identical to that of the DIT (Proposition 23.4), and the recurrences are the same as in Theorem 23.4; the solution is that of Theorem 23.5.

Trivial multiplications. At the step on length $n = 2^j$ the factors used are $\omega_n^0, \omega_n^1, \dots, \omega_n^{n/2-1}$: exactly the same set of factors as the DIT on the same length. The count of Proposition 23.5 applies word for word, and gives $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$ nontrivial multiplications. $\square$

Remark 23.8: DIT and DIF: where the bit reversal sits, and why they are two transposed graphs. In the DIT the multiplication by the twiddle factor precedes the sum/difference pair; in the DIF it follows it. From this follows the position of the permutation:

- **DIT:** input in bit-reversed order, output in natural order;
- **DIF:** input in natural order, output in bit-reversed order.

There also exist “natural-to-natural” variants, which pay for the reordering with auxiliary memory, and variants that avoid the reordering altogether when the FFT is used solely for a circular convolution (Theorem 22.5): in that case one applies a DIF without reordering, multiplies pointwise — the order does not matter, provided it is the same for both factors — and applies a DIT that restores the natural order. This is a real saving in an industrial implementation.

It is finally worth noting that the two flow graphs are each the transpose of the other: reversing the direction of every arrow of the DIT graph and swapping inputs with outputs gives the DIF graph. This is a manifestation of the transposition principle for linear networks; we do not need it here, and we merely note it, having proved the DIF directly from Theorem 23.9.

23.8. The example $N = 8$: exact symbolic form

These are steps 21 and 23 of the path. The example is worked out *in full*: first in generic form, with the samples $x[0], \dots, x[7]$ left indeterminate, then with exact roots of unity and a concrete input. The numerical check and the comparison with the direct DFT is in Section 23.9.

Stage 1 ($n_1 = 2$, four blocks, a single value $k = 0$, factor $\omega_2^0 = 1$). The butterflies act on the pairs of cells $(0, 1)$, $(2, 3)$, $(4, 5)$, $(6, 7)$:

$$\begin{aligned} a_0 &= x[0] + x[4], & a_1 &= x[0] - x[4], & a_2 &= x[2] + x[6], & a_3 &= x[2] - x[6], \\ a_4 &= x[1] + x[5], & a_5 &= x[1] - x[5], & a_6 &= x[3] + x[7], & a_7 &= x[3] - x[7]. \end{aligned}$$

Each pair is the two-point DFT of Lemma 23.3(ii): 8 additions, 0 nontrivial multiplications.

Stage 2 ($n_2 = 4$, two blocks $\{0, 1, 2, 3\}$ and $\{4, 5, 6, 7\}$, values $k = 0, 1$, factors $\omega_4^0 = 1$ and $\omega_4^1 = -i$). The butterflies act on the pairs $(0, 2)$, $(1, 3)$ in the first block and $(4, 6)$, $(5, 7)$ in the second:

$$\begin{aligned} b_0 &= a_0 + a_2, & b_1 &= a_1 - i a_3, & b_2 &= a_0 - a_2, & b_3 &= a_1 + i a_3, \\ b_4 &= a_4 + a_6, & b_5 &= a_5 - i a_7, & b_6 &= a_4 - a_6, & b_7 &= a_5 + i a_7. \end{aligned}$$

Again 8 additions and 0 nontrivial multiplications (the four factors are 1 and $-i$).

Stage 3 ($n_3 = 8$, a single block, $k = 0, 1, 2, 3$, factors 1, ω_8 , $-i$, ω_8^3). The butterflies act on the pairs $(0, 4)$, $(1, 5)$, $(2, 6)$, $(3, 7)$:

$$\begin{aligned} X[0] &= b_0 + b_4, & X[4] &= b_0 - b_4, \\ X[1] &= b_1 + \omega_8 b_5, & X[5] &= b_1 - \omega_8 b_5, \\ X[2] &= b_2 - i b_6, & X[6] &= b_2 + i b_6, \\ X[3] &= b_3 + \omega_8^3 b_7, & X[7] &= b_3 - \omega_8^3 b_7. \end{aligned}$$

Eight additions; the only nontrivial multiplications of the whole computation are $\omega_8 b_5$ and $\omega_8^3 b_7$.

Balance. 24 complex additions and 2 nontrivial complex multiplications, in agreement with Theorem 23.5 ($A(8) = 8 \cdot 3 = 24$) and with Corollary 23.4 (2). ✓

Example 23.5: Recomposition: the first output contains all eight samples. Let us verify Corollary 23.2 on two outputs, expanding backward the formulas of Example 23.4 (step 23 of the path).

Output $X[0]$.

$$X[0] = b_0 + b_4 = (a_0 + a_2) + (a_4 + a_6) = (x[0] + x[4]) + (x[2] + x[6]) + (x[1] + x[5]) + (x[3] + x[7]),$$

that is, $X[0] = \sum_{n=0}^7 x[n]$: all eight samples, each with coefficient $1 = \omega_8^{0 \cdot n}$. ✓

Output $X[2]$. One has $X[2] = b_2 - i b_6 = (a_0 - a_2) - i(a_4 - a_6)$, hence

$$X[2] = (x[0] + x[4]) - (x[2] + x[6]) - i((x[1] + x[5]) - (x[3] + x[7])).$$

Collecting the coefficients of the individual samples:

$$X[2] = x[0] - i x[1] - x[2] + i x[3] + x[4] - i x[5] - x[6] + i x[7].$$

On the other hand the definition gives $X[2] = \sum_n x[n] \omega_8^{2n}$ with $\omega_8^2 = -i$, that is, coefficients $(-i)^n = 1, -i, -1, i, 1, -i, -1, i$ for $n = 0, \dots, 7$: they coincide one by one with those just found. ✓

Note that no sample is missing and none appears twice, as Corollary 23.2 states: the algorithm is not a shortcut that neglects terms, it is a *reorganization* of the same sum.

23.9. The example $N = 8$: numerical form and checks

These are steps 22 and 25 of the path, with the five mandatory checks: equality between the direct DFT and the FFT output, energy conservation (Parseval), conjugate symmetry for a real input, reconstruction of the input, count of the operations. Let us take the input

$$x = (0, 1, 2, 3, 4, 5, 6, 7),$$

chosen because its DFT has an *exact closed form* (Lemma 23.5), which provides a check independent of any algorithm.

Lemma 23.5: The DFT of the ramp. Let $N \geq 2$ and $x[n] = n$ for $n = 0, \dots, N-1$. Then

$$X[0] = \frac{N(N-1)}{2}, \quad X[k] = \frac{N}{\omega_N^k - 1} \quad \text{for } k = 1, \dots, N-1.$$

Proof. For $k = 0$ all factors equal 1 and $X[0] = \sum_{n=0}^{N-1} n = \frac{N(N-1)}{2}$ (Gauss's sum). Let $k \in \{1, \dots, N-1\}$ and set $w = \omega_N^k$. By Lemma 23.1(ii), $w^N = \omega_N^{kN} = 1$ and $w \neq 1$ (because $N \nmid k$). Let $S = \sum_{n=0}^{N-1} n w^n = X[k]$. Then

$$(w-1)S = \sum_{n=0}^{N-1} n w^{n+1} - \sum_{n=0}^{N-1} n w^n = \sum_{n=1}^N (n-1) w^n - \sum_{n=1}^{N-1} n w^n,$$

where in the first sum $n \mapsto n-1$ has been substituted and in the second the null term $n=0$ has been dropped. Separating the term $n=N$ of the first sum,

$$(w-1)S = (N-1)w^N + \sum_{n=1}^{N-1} [(n-1) - n] w^n = (N-1) \cdot 1 - \sum_{n=1}^{N-1} w^n.$$

By Lemma 22.1 (sum of the powers of a root of unity, Chapter 22), $\sum_{n=0}^{N-1} w^n = 0$, since w is an N -th root of unity different from 1; hence $\sum_{n=1}^{N-1} w^n = -w^0 = -1$ and

$$(w-1)S = (N-1) + 1 = N.$$

Since $w \neq 1$, one may divide: $S = \frac{N}{w-1} = \frac{N}{\omega_N^k - 1}$. □

Example 23.6: The exact DFT of $x = (0, 1, \dots, 7)$, three ways. First way: the closed form. By Lemma 23.5, $X[0] = \frac{8 \cdot 7}{2} = 28$ and $X[k] = \frac{8}{\omega_8^k - 1}$ for $k \geq 1$. For $k=2$, for instance, $\omega_8^2 = -i$ and

$$X[2] = \frac{8}{-i-1} = \frac{8(-1+i)}{(-1-i)(-1+i)} = \frac{8(-1+i)}{1+1} = -4+4i.$$

For $k=1$, with $a = \frac{\sqrt{2}}{2}$ and $\omega_8 = a - ai$, the denominator is $d = (a-1) - ai$, with squared modulus $|d|^2 = (a-1)^2 + a^2 = a^2 - 2a + 1 + a^2 = 2a^2 - 2a + 1 = 1 - 2a + 1 = 2 - \sqrt{2}$ (having used $2a^2 = 1$ and $2a = \sqrt{2}$), hence

$$X[1] = \frac{8\bar{d}}{|d|^2} = \frac{8[(a-1) + ai]}{2 - \sqrt{2}} = \frac{8(a-1)}{2 - \sqrt{2}} + i \frac{8a}{2 - \sqrt{2}} = -4 + i(4 + 4\sqrt{2}),$$

because $8(a-1) = 8(\frac{\sqrt{2}}{2} - 1) = 4\sqrt{2} - 8 = -4(2 - \sqrt{2})$ and $8a = 4\sqrt{2} = 4\sqrt{2}$, with $\frac{4\sqrt{2}}{2 - \sqrt{2}} = \frac{4\sqrt{2}(2 + \sqrt{2})}{(2 - \sqrt{2})(2 + \sqrt{2})} = \frac{8\sqrt{2} + 8}{2} = 4\sqrt{2} + 4$.

Second way: the direct sum, by hand. We compute $X[1]$ from Definition 22.1, using the roots of Example 23.3 and $a = \frac{\sqrt{2}}{2}$:

$$X[1] = \sum_{n=0}^7 n \omega_8^n = 1(a - ai) + 2(-i) + 3(-a - ai) + 4(-1) + 5(-a + ai) + 6(i) + 7(a + ai).$$

Real part: $a - 3a - 4 - 5a + 7a = (1 - 3 - 5 + 7)a - 4 = 0 \cdot a - 4 = -4$. Imaginary part: $-a - 2 - 3a + 5a + 6 + 7a = (-1 - 3 + 5 + 7)a + 4 = 8a + 4 = 4\sqrt{2} + 4$. Hence $X[1] = -4 + i(4 + 4\sqrt{2})$, in agreement with the first way. ✓

Third way: the complete spectrum. Proceeding in the same way for every k , or using the closed form,

$$\begin{aligned} X[0] &= 28, & X[1] &= -4 + i(4 + 4\sqrt{2}), & X[2] &= -4 + 4i, \\ X[3] &= -4 + i(4\sqrt{2} - 4), & X[4] &= -4, & X[5] &= -4 - i(4\sqrt{2} - 4), \\ X[6] &= -4 - 4i, & X[7] &= -4 - i(4 + 4\sqrt{2}). \end{aligned}$$

In decimal form to seven digits: $4 + 4\sqrt{2} = 9.6568542$ and $4\sqrt{2} - 4 = 1.6568542$.

Example 23.7: The three stages in numerical form. Substituting $x = (0, 1, 2, 3, 4, 5, 6, 7)$ into the formulas of Example 23.4:

Permutation. $a = (x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]) = (0, 4, 2, 6, 1, 5, 3, 7)$.

Stage 1.

$$\begin{aligned} a_0 &= 0 + 4 = 4, & a_1 &= 0 - 4 = -4, & a_2 &= 2 + 6 = 8, & a_3 &= 2 - 6 = -4, \\ a_4 &= 1 + 5 = 6, & a_5 &= 1 - 5 = -4, & a_6 &= 3 + 7 = 10, & a_7 &= 3 - 7 = -4. \end{aligned}$$

Stage 2. With $-i a_3 = -i(-4) = 4i$ and $-i a_7 = 4i$:

$$\begin{aligned} b_0 &= 4 + 8 = 12, & b_1 &= -4 + 4i, & b_2 &= 4 - 8 = -4, & b_3 &= -4 - 4i, \\ b_4 &= 6 + 10 = 16, & b_5 &= -4 + 4i, & b_6 &= 6 - 10 = -4, & b_7 &= -4 - 4i. \end{aligned}$$

Stage 3. The two nontrivial products are each computed only once. With $a = \frac{\sqrt{2}}{2}$:

$$\omega_8 b_5 = a(1 - i)(-4 + 4i) = a(-4 + 4i + 4i - 4i^2) = a(8i) = 8ai = 4\sqrt{2}i,$$

$$\omega_8^3 b_7 = -a(1 + i)(-4 - 4i) = -a(-4 - 4i - 4i - 4i^2) = -a(-8i) = 8ai = 4\sqrt{2}i.$$

Moreover $-i b_6 = -i(-4) = 4i$ and $\omega_8^0 b_4 = b_4 = 16$. Hence

$$\begin{aligned} X[0] &= 12 + 16 = 28, & X[4] &= 12 - 16 = -4, \\ X[1] &= (-4 + 4i) + 4\sqrt{2}i = -4 + i(4 + 4\sqrt{2}), & X[5] &= (-4 + 4i) - 4\sqrt{2}i = -4 - i(4\sqrt{2} - 4), \\ X[2] &= -4 + 4i, & X[6] &= -4 - 4i, \\ X[3] &= (-4 - 4i) + 4\sqrt{2}i = -4 + i(4\sqrt{2} - 4), & X[7] &= (-4 - 4i) - 4\sqrt{2}i = -4 - i(4 + 4\sqrt{2}). \end{aligned}$$

These are exactly the values of Example 23.6, obtained there by two entirely different routes (closed form and direct sum). ✓

Example 23.8: The five mandatory checks. (1) FFT against the direct DFT.

Table 23.2 compares, column by column, the output of stage 3 with the DFT computed separately from Definition 22.1: the values coincide. In exact arithmetic the coincidence

Table 23.2: The example $N = 8$ with $x = (0, 1, \dots, 7)$: contents of the in-place vector after the bit-reversal permutation and after each of the three stages (Definition 23.6), compared with the DFT computed directly from Definition 22.1. Values rounded to the seventh decimal digit; $9.6568542 = 4 + 4\sqrt{2}$ and $1.6568542 = 4\sqrt{2} - 4$.

cell	after (P)	after stage 1	after stage 2	after stage 3	direct DFT
0	0	4	12	28	28
1	4	-4	$-4 + 4i$	$-4 + 9.6568542i$	$-4 + 9.6568542i$
2	2	8	-4	$-4 + 4i$	$-4 + 4i$
3	6	-4	$-4 - 4i$	$-4 + 1.6568542i$	$-4 + 1.6568542i$
4	1	6	16	-4	-4
5	5	-4	$-4 + 4i$	$-4 - 1.6568542i$	$-4 - 1.6568542i$
6	3	10	-4	$-4 - 4i$	$-4 - 4i$
7	7	-4	$-4 - 4i$	$-4 - 9.6568542i$	$-4 - 9.6568542i$

is a theorem (Theorem 23.3); in double precision, a Python script that computes the two quantities independently — the first with the recursion of Definition 23.4, the second with the double sum of the definition — gives a maximum discrepancy of $2.42 \cdot 10^{-14}$ in modulus, of the order of the round-off error predicted by Theorem 23.13. The same script compares the output with `numpy.fft.fft`, with maximum discrepancy $2.54 \cdot 10^{-14}$. ✓

(2) Energy conservation (Parseval). Theorem 22.6 gives $\sum_n |x[n]|^2 = \frac{1}{N} \sum_k |X[k]|^2$. On the left

$$\sum_{n=0}^7 n^2 = 0 + 1 + 4 + 9 + 16 + 25 + 36 + 49 = 140.$$

On the right, in *exact* form: $|X[0]|^2 = 784$, $|X[4]|^2 = 16$, $|X[2]|^2 = |X[6]|^2 = 16 + 16 = 32$,

$$|X[1]|^2 = |X[7]|^2 = 16 + (4 + 4\sqrt{2})^2 = 16 + 48 + 32\sqrt{2} = 64 + 32\sqrt{2},$$

$$|X[3]|^2 = |X[5]|^2 = 16 + (4\sqrt{2} - 4)^2 = 16 + 48 - 32\sqrt{2} = 64 - 32\sqrt{2}.$$

Summing, the terms in $\sqrt{2}$ cancel:

$$\sum_k |X[k]|^2 = 784 + 16 + 2 \cdot 32 + 2(64 + 32\sqrt{2}) + 2(64 - 32\sqrt{2}) = 784 + 16 + 64 + 256 = 1,120,$$

and $\frac{1}{8} \cdot 1,120 = 140$. ✓ *Independent check:* the Python script returns $\sum_n |x|^2 = 140.0000000$ and $\frac{1}{8} \sum_k |X|^2 = 140.0000000$, with exact difference 0 in double precision. ✓

(3) Conjugate symmetry. The input is real, hence by Theorem 22.7 it must hold that $X[N - k] = \overline{X[k]}$. Indeed $X[7] = -4 - i(4 + 4\sqrt{2}) = \overline{X[1]}$, $X[6] = -4 - 4i = \overline{X[2]}$, $X[5] = -4 - i(4\sqrt{2} - 4) = \overline{X[3]}$, and $X[0] = 28$, $X[4] = -4$ are real, as the two self-conjugate bins must be ($k = 0$ and $k = N/2$; Corollary 22.8). ✓ *Independent check:* the discrepancies $|X[k] - \overline{X[8 - k]}|$ computed in double precision are $2.87 \cdot 10^{-14}$, $1.49 \cdot 10^{-14}$, $9.14 \cdot 10^{-15}$ and $6.86 \cdot 10^{-15}$ for $k = 1, 2, 3, 4$. ✓

(4) Reconstruction of the input. Applying the IFFT of Theorem 23.10 to the spectrum X recovers all eight samples; the computation is carried out in Example 23.9.

(5) Operation count. Example 23.4 counted 24 complex additions and 2 nontrivial complex multiplications (in total 12 multiplications, 10 of which by 1 or $-i$). The direct computation of the same DFT requires 64 multiplications and 56 additions (Proposition 23.1), or 44 multiplications not by 1 (Example 23.1). The ratio on the nontrivial multiplications is already $44/2 = 22$ at $N = 8$. *The count is reported separately from any measured time*, as Remark 23.7 prescribes. ✓

23.10. The inverse transform requires no second algorithm

This is step 26 of the path. The IDFT differs from the DFT by the sign of the exponent and by the factor $1/N$ (Definition 22.2); one could therefore write a second algorithm, identical to the first with ω_N replaced by $\bar{\omega}_N$. This is unnecessary: conjugation allows *the same* code to be reused, which halves the software to be written, tested and certified. This is one of the reasons why industrial libraries expose a single routine.

Theorem 23.10: IFFT by conjugation. *Let $N \geq 1$ and $X \in \mathbb{C}^N$. Then*

$$\mathcal{F}_N^{-1}X = \frac{1}{N} \overline{\text{DFT}_N\{\bar{X}\}}$$

where $\bar{X}$ is the vector of components $\bar{X}[k]$ and the outer bar denotes componentwise conjugation. In particular, if X is the DFT of $x \in \mathbb{C}^N$, then

$$x = \frac{1}{N} \mathcal{A}_N(\bar{X}),$$

and the inverse transform is computed with the same algorithm $\mathcal{A}_N$ of Definition 23.4, preceded and followed by a conjugation and concluded by a division by N .

Proof. Let $Y = \text{DFT}_N\{\bar{X}\}$, that is, $Y[n] = \sum_{k=0}^{N-1} \bar{X}[k] \omega_N^{nk}$ for $n = 0, \dots, N-1$. Taking the conjugate and using that conjugation is additive and multiplicative ($\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$),

$$\bar{Y}[n] = \overline{\sum_{k=0}^{N-1} \bar{X}[k] \omega_N^{nk}} = \sum_{k=0}^{N-1} X[k] \overline{\omega_N^{nk}}.$$

By Lemma 23.1(i), $\bar{\omega}_N = \omega_N^{-1}$, hence $\overline{\omega_N^{nk}} = (\bar{\omega}_N)^{nk} = \omega_N^{-nk}$. Therefore

$$\frac{1}{N} \bar{Y}[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-nk},$$

which is exactly the n -th component of the IDFT in Definition 22.2. Hence $\frac{1}{N} \bar{Y} = \mathcal{F}_N^{-1}X$, which is the first formula.

If moreover $X = \text{DFT}_N\{x\}$, Theorem 22.2 gives $\mathcal{F}_N^{-1}X = x$; and by Theorem 23.3 $\text{DFT}_N\{\bar{X}\} = \mathcal{A}_N(\bar{X})$ when $N = 2^m$. This gives the second formula. $\square$

Corollary 23.6: Cost of the IFFT. *For $N = 2^m$, the inverse transform computed with Theorem 23.10 costs $\frac{N}{2} \log_2 N$ complex multiplications (of which $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$ are nontrivial) and $N \log_2 N$ complex additions — exactly as much as the direct transform — plus N multiplications by the real scalar $\frac{1}{N}$ and $2N$ sign changes (the two conjugations), which are not complex multiplications in the sense of Definition 23.1.*

Proof. The body of the computation is a single call to $\mathcal{A}_N$, whose cost is given by Theorem 23.5 and Corollary 23.4. The two conjugations change the sign of the imaginary part of N numbers each: $2N$ sign changes. The final division by N is the multiplication of N complex numbers by the real $\frac{1}{N}$, that is, $2N$ real multiplications; when $N = 2^m$ it reduces to a subtraction on the floating-point exponent, and is exact (it introduces no round-off error). $\square$

Remark 23.9: The two equivalent variants. There is a second form, equally used. Writing $\text{swap}(a + ib) = b + ia$ for the operation that swaps the real and imaginary parts, one has

$$\mathcal{F}_N^{-1}X = \frac{1}{N} \text{swap}\left(\text{DFT}_N\{\text{swap}(X)\}\right).$$

The proof is the same as that of Theorem 23.10, observing that $\text{swap}(z) = i\bar{z}$ and that the factor i , being constant, moves in and out of the sum and cancels between the two applications of swap : indeed $\text{swap}(\sum_k i\bar{X}[k]\omega_N^{nk}) = i\bar{i}\sum_k X[k]\omega_N^{-nk} = \sum_k X[k]\omega_N^{-nk}$, having used $i\bar{i} = |i|^2 = 1$. The **swap** variant is preferred in implementations on DSP processors, where swapping two registers costs less than a negation, but it is mathematically the same thing.

Example 23.9: Reconstruction of our eight samples. Let us take up the spectrum of Example 23.6 and apply Theorem 23.10. First conjugation:

$$\bar{X} = (28, -4 - i(4 + 4\sqrt{2}), -4 - 4i, -4 - i(4\sqrt{2} - 4), -4, -4 + i(4\sqrt{2} - 4), -4 + 4i, -4 + i(4 + 4\sqrt{2})).$$

We apply $\mathcal{A}_8$ to $\bar{X}$. With $8\sqrt{2} = 11.3137085$, the three stages give (bit-reversal permutation, then stages 1, 2, 3):

after (P)	28	-4	-4 - 4i	-4 + 4i	-4 - 9.657i	-4 + 1.657i	-4 - 1.657i	-4 + 9.657i
stage 1	24	32	-8	-8i	-8 - 8i	-11.314i	-8 + 8i	-11.314i
stage 2	16	24	32	40	-16	-11.314 - 11.314i	-16i	11.314 - 11.314i
stage 3	0	8	16	24	32	40	48	56

(in the table 9.657 and 1.657 are the truncations of $4 + 4\sqrt{2}$ and $4\sqrt{2} - 4$, and 11.314 that of $8\sqrt{2}$).

Second conjugation and division by $N = 8$:

$$x = \frac{1}{8} \overline{(0, 8, 16, 24, 32, 40, 48, 56)} = (0, 1, 2, 3, 4, 5, 6, 7).$$

All eight samples are reconstructed exactly. ✓

Why the output is so simple. The vector $\mathcal{A}_8(\bar{X})$ equals $N\bar{x}$; and since our x is real, $N\bar{x} = 8x = (0, 8, \dots, 56)$. This provides a *third* check, independent of the first two: the intermediate result had to be the ramp multiplied by 8, and it is.

Independent check in double precision. The Python script that runs the same chain returns

$$x = (0.0000000000, 1.0000000000, \dots, 7.0000000000)$$

with imaginary part of modulus less than 10^{-15} and maximum error $|x_{\text{rec}}[n] - n| = 2.83 \cdot 10^{-16}$ over all eight samples. ✓

23.11. Zero-padding: what it adds and what it does not

This is step 27 of the path. Padding the signal with zeros up to the next power of two is the most common maneuver for using a radix-2 FFT; it is also the source of the most widespread misunderstanding in numerical spectral analysis. Chapter 22 has already proved the central fact (Theorem 22.10 and Theorem 22.11): here we recall it, draw its algorithmic consequence, and show its effect on the example $N = 8$.

Proposition 23.6: The even bins of the padded DFT are the original bins. Let $x \in \mathbb{C}^N$ and let $z \in \mathbb{C}^{2N}$ be defined by $z[n] = x[n]$ for $0 \leq n \leq N - 1$ and $z[n] = 0$ for

$N \leq n \leq 2N - 1$. Let $X = \text{DFT}_N\{x\}$ and $Z = \text{DFT}_{2N}\{z\}$. Then

$$Z[2k] = X[k] \quad \text{for every } k = 0, 1, \dots, N - 1.$$

Proof. By definition, and since the terms with $n \geq N$ vanish,

$$Z[j] = \sum_{n=0}^{2N-1} z[n] \omega_{2N}^{jn} = \sum_{n=0}^{N-1} x[n] \omega_{2N}^{jn}.$$

For $j = 2k$, Lemma 23.1(iii), applied with $2N$ in place of N , gives $\omega_{2N}^{2kn} = \omega_N^{kn}$; hence

$$Z[2k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn} = X[k].$$

□

Example 23.10: $N = 8 \rightarrow 16$ on our ramp. Let $z = (0, 1, 2, 3, 4, 5, 6, 7, 0, 0, 0, 0, 0, 0, 0, 0)$. The 16-point DFT gives, to the seventh decimal digit,

$$\begin{array}{ll} Z[0] = 28, & Z[1] = -9.1370712 - 20.1093580i, \\ Z[2] = -4 + 9.6568542i, & Z[3] = 2.3800856 - 5.9864231i, \\ Z[4] = -4 + 4i, & Z[5] = 3.2767687 - 2.6727146i, \\ Z[6] = -4 + 1.6568542i, & Z[7] = 3.4802169 - 0.7956495i, \\ Z[8] = -4, & Z[9] = 3.4802169 + 0.7956495i, \\ Z[10] = -4 - 1.6568542i, & Z[11] = 3.2767687 + 2.6727146i, \\ Z[12] = -4 - 4i, & Z[13] = 2.3800856 + 5.9864231i, \\ Z[14] = -4 - 9.6568542i, & Z[15] = -9.1370712 + 20.1093580i. \end{array}$$

The even-indexed bins, $Z[0], Z[2], \dots, Z[14]$, coincide with $X[0], X[1], \dots, X[7]$ of Example 23.6: this is Proposition 23.6. ✓ *Independent check:* the Python script computes the two transforms separately and reports $\max_k |Z[2k] - X[k]| = 2.42 \cdot 10^{-14}$. ✓

The odd-indexed bins are *new* numbers, but they contain no new information: by Theorem 22.11 they are determined by the eight values $X[0], \dots, X[7]$ through interpolation with the discrete Dirichlet kernel (Definition 22.6). They are, literally, computable without looking at the signal.

Remark 23.10: Interpolation is not resolution. Zero-padding **does not create spectral information**. Theorem 22.10 says that the longer DFT samples the same DTFT (Theorem 22.9) at more points, and Theorem 22.11 says that the new points are an exact interpolation of the old ones. Two consequences follow, and they must not be confused:

- **interpolation** (what zero-padding gives): the spectrum appears denser, the maxima are located more precisely on the frequency axis, the plots are more readable;
- **resolution** (what zero-padding does not give): the ability to separate two nearby sinusoids depends on the effective duration of the observed signal, that is, on the width of the main lobe of the window (Proposition 22.8), and zeros do not extend the observation: they extend the vector.

In other words: two sinusoids that at N points fall inside the same main lobe remain indistinguishable at $2N$, $4N$, $1024N$ points. The only remedy is to acquire for longer. On the algorithmic side, however, zero-padding is valuable: bringing N up to the next power of two

costs at most a doubling of the length, hence

$$\frac{N'}{2} \log_2 N' \leq \frac{2N}{2} (\log_2 N + 1) = N \log_2 N + N$$

multiplications with $N' \leq 2N$, against the N^2 of the direct computation: the asymptotic advantage remains intact. It is also the tool with which a linear convolution is computed with a circular FFT, as in Solved Exercise 23.5.

23.12. When N is not a power of two

This is step 28 of the path, and it must be kept strictly separate from the radix-2 proof: the algorithms of this section are not notational variants of the previous one, they have different hypotheses and must be proved separately. The picture is as follows:

- if $N = N_1 N_2$ with $N_1, N_2 \geq 2$, the general Cooley–Tukey factorization (Theorem 23.11) reduces the DFT of length N to N_2 DFTs of length N_1 , N multiplications by twiddle factors, and N_1 DFTs of length N_2 ;
- if N is prime, the factorization does not apply: one uses Bluestein’s algorithm (Proposition 23.7 and Corollary 23.8), which reduces *any* DFT to a convolution, or Rader’s algorithm (Theorem 23.12), which reduces the DFT of prime length p to a cyclic correlation of length $p - 1$.

23.12.1 The general factorization

Theorem 23.11: Mixed-radix Cooley–Tukey factorization. Let $N_1, N_2 \geq 1$ be integers and $N = N_1 N_2$. For $x \in \mathbb{C}^N$ and $X = \text{DFT}_N\{x\}$, define, for $k_1 \in \{0, \dots, N_1 - 1\}$ and $n_2 \in \{0, \dots, N_2 - 1\}$,

$$G[k_1, n_2] = \sum_{n_1=0}^{N_1-1} x[N_2 n_1 + n_2] \omega_{N_1}^{n_1 k_1} \quad (\text{a DFT of length } N_1 \text{ for each } n_2),$$

and then

$$\tilde{G}[k_1, n_2] = \omega_N^{k_1 n_2} G[k_1, n_2] \quad (\text{multiplication by the twiddle factor}).$$

Then, for every $k_1 \in \{0, \dots, N_1 - 1\}$ and $k_2 \in \{0, \dots, N_2 - 1\}$,

$$X[k_1 + N_1 k_2] = \sum_{n_2=0}^{N_2-1} \tilde{G}[k_1, n_2] \omega_{N_2}^{n_2 k_2}$$

that is: a DFT of length N_2 is performed on the second variable, for each fixed value of k_1 .

Proof. Step 1 (the two index maps are bijections). The map

$$\Phi : \{0, \dots, N_1 - 1\} \times \{0, \dots, N_2 - 1\} \rightarrow \{0, \dots, N - 1\}, \quad \Phi(n_1, n_2) = N_2 n_1 + n_2$$

is bijective: it is Euclidean division by N_2 read backward. Indeed every $n \in \{0, \dots, N - 1\}$ is written uniquely as $n = N_2 q + r$ with $0 \leq r < N_2$, and from $0 \leq n \leq N_1 N_2 - 1$ it follows that $0 \leq q \leq N_1 - 1$; conversely $0 \leq N_2 n_1 + n_2 \leq N_2(N_1 - 1) + N_2 - 1 = N - 1$. Similarly $\Psi(k_1, k_2) = k_1 + N_1 k_2$ is bijective (Euclidean division by N_1). In particular every input index and every output index is reached exactly once by the formulas of the statement.

Step 2 (expanding the exponent). Setting $n = N_2 n_1 + n_2$ and $k = k_1 + N_1 k_2$,

$$kn = (k_1 + N_1 k_2)(N_2 n_1 + n_2) = N_2 n_1 k_1 + n_2 k_1 + N_1 N_2 n_1 k_2 + N_1 n_2 k_2.$$

Since $N_1 N_2 = N$, Lemma 23.1(ii) gives $\omega_N^{N n_1 k_2} = 1$. Three terms remain, and

$$\omega_N^{kn} = \omega_N^{N_2 n_1 k_1} \cdot \omega_N^{n_2 k_1} \cdot \omega_N^{N_1 n_2 k_2}.$$

By Lemma 23.1(vi), with $d = N_2$ and base order N_1 : $\omega_N^{N_2 n_1 k_1} = \omega_{N_2 N_1}^{N_2(n_1 k_1)} = \omega_{N_1}^{n_1 k_1}$; and with $d = N_1$ and base order N_2 : $\omega_N^{N_1 n_2 k_2} = \omega_{N_2}^{n_2 k_2}$. Hence

$$\omega_N^{kn} = \omega_{N_1}^{n_1 k_1} \cdot \omega_N^{k_1 n_2} \cdot \omega_{N_2}^{n_2 k_2}.$$

Step 3 (reorganizing the sum). By Step 1 the sum over $n \in \{0, \dots, N-1\}$ can be rewritten as a double sum over (n_1, n_2) , and finite sums can be exchanged:

$$X[k_1 + N_1 k_2] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn} = \sum_{n_2=0}^{N_2-1} \sum_{n_1=0}^{N_1-1} x[N_2 n_1 + n_2] \omega_{N_1}^{n_1 k_1} \omega_N^{k_1 n_2} \omega_{N_2}^{n_2 k_2}.$$

The factors $\omega_N^{k_1 n_2}$ and $\omega_{N_2}^{n_2 k_2}$ do not depend on n_1 and can be pulled out of the inner sum:

$$X[k_1 + N_1 k_2] = \sum_{n_2=0}^{N_2-1} \omega_N^{k_1 n_2} \omega_{N_2}^{n_2 k_2} \underbrace{\left(\sum_{n_1=0}^{N_1-1} x[N_2 n_1 + n_2] \omega_{N_1}^{n_1 k_1} \right)}_{= G[k_1, n_2]} = \sum_{n_2=0}^{N_2-1} \tilde{G}[k_1, n_2] \omega_{N_2}^{n_2 k_2},$$

which is the claim. $\square$

Corollary 23.7: Cost of the mixed-radix factorization. Let $C_m(N)$ and $C_a(N)$ be the number of complex multiplications and additions needed to compute a DFT of length N by a fixed method. Then the factorization of Theorem 23.11 with $N = N_1 N_2$ costs

$$C_m(N) \leq N_2 C_m(N_1) + N + N_1 C_m(N_2), \quad C_a(N) \leq N_2 C_a(N_1) + N_1 C_a(N_2).$$

In particular, for $N = N_1 N_2$ with both factors ≥ 2 and with the two short DFTs computed directly, the cost in multiplications is at most $N(N_1 + N_2 + 1)$ against the N^2 of the direct computation: the gain is a factor $\frac{N}{N_1 + N_2 + 1}$.

Proof. Theorem 23.11 prescribes three phases. The first computes $G[\cdot, n_2]$ for each of the N_2 values of n_2 : these are N_2 DFTs of length N_1 , at a cost of $N_2 C_m(N_1)$ multiplications and $N_2 C_a(N_1)$ additions. The second multiplies each of the $N_1 N_2 = N$ quantities $G[k_1, n_2]$ by $\omega_N^{k_1 n_2}$: these are at most N multiplications and no additions (actually fewer, because the factors with $k_1 = 0$ or $n_2 = 0$ equal 1: there are $N_1 + N_2 - 1$ trivial multiplications). The third computes, for each of the N_1 values of k_1 , a DFT of length N_2 : cost $N_1 C_m(N_2)$ and $N_1 C_a(N_2)$. Summing gives the two bounds.

For the last statement, use Proposition 23.1 on the short DFTs, $C_m(N_j) \leq N_j^2$:

$$C_m(N) \leq N_2 N_1^2 + N + N_1 N_2^2 = N N_1 + N + N N_2 = N(N_1 + N_2 + 1).$$

$\square$

Example 23.11: $N = 6 = 2 \cdot 3$ worked out in full. Let $y = (1, 2, 3, 4, 5, 6)$, $N = 6$, $N_1 = 2$, $N_2 = 3$. One has $\omega_6 = e^{-i\pi/3} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\omega_6^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i = \omega_3$, $\omega_2 = -1$, $\omega_3 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\omega_3^2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, and $\frac{\sqrt{3}}{2} = 0.8660254$.

Phase 1: three DFTs of length $N_1 = 2$. The subsequences are $(y[3n_1 + n_2])_{n_1=0,1}$:

$$n_2 = 0 : (1, 4); \quad n_2 = 1 : (2, 5); \quad n_2 = 2 : (3, 6).$$

By Lemma 23.3(ii),

$$\begin{aligned} G[0, 0] &= 1 + 4 = 5, & G[0, 1] &= 2 + 5 = 7, & G[0, 2] &= 3 + 6 = 9, \\ G[1, 0] &= 1 - 4 = -3, & G[1, 1] &= 2 - 5 = -3, & G[1, 2] &= 3 - 6 = -3. \end{aligned}$$

Cost: 6 additions, no multiplications.

Phase 2: the twiddle factors $\omega_6^{k_1 n_2}$. For $k_1 = 0$ all factors equal 1; for $k_1 = 1$:

$$\tilde{G}[1, 0] = -3, \quad \tilde{G}[1, 1] = -3\omega_6 = -\frac{3}{2} + \frac{3\sqrt{3}}{2}i, \quad \tilde{G}[1, 2] = -3\omega_6^2 = \frac{3}{2} + \frac{3\sqrt{3}}{2}i,$$

with $\frac{3\sqrt{3}}{2} = 2.5980762$. Only *two* of the six multiplications are nontrivial, consistent with the count $N_1 + N_2 - 1 = 4$ factors equal to 1 of Corollary 23.7.

Phase 3: two DFTs of length $N_2 = 3$, and recomposition $k = k_1 + 2k_2$.

For $k_1 = 0$, on the vector $(5, 7, 9)$:

$$\begin{aligned} k_2 = 0 : 5 + 7 + 9 &= 21 & \Rightarrow X[0] &= 21, \\ k_2 = 1 : 5 + 7\omega_3 + 9\omega_3^2 &= 5 - \frac{7}{2} - \frac{9}{2} + \frac{\sqrt{3}}{2}(-7 + 9)i = -3 + \sqrt{3}i & \Rightarrow X[2] &= -3 + \sqrt{3}i, \\ k_2 = 2 : 5 + 7\omega_3^2 + 9\omega_3^4 &= 5 + 7\omega_3^2 + 9\omega_3 = -3 - \sqrt{3}i & \Rightarrow X[4] &= -3 - \sqrt{3}i, \end{aligned}$$

where $\omega_3^4 = \omega_3$ (Lemma 23.1(ii)) has been used.

For $k_1 = 1$, on the vector $(-3, -\frac{3}{2} + \frac{3\sqrt{3}}{2}i, \frac{3}{2} + \frac{3\sqrt{3}}{2}i)$:

$$k_2 = 0 : -3 + \left(-\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right) + \left(\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right) = -3 + 3\sqrt{3}i \Rightarrow X[1] = -3 + 3\sqrt{3}i.$$

For $k_2 = 1$ the two products needed are

$$\left(-\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right)\omega_3 = \left(-\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right)\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = \frac{3}{4} + \frac{3\sqrt{3}}{4}i - \frac{3\sqrt{3}}{4}i + \frac{3 \cdot 3}{4} = 3,$$

$$\left(\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right)\omega_3^2 = \left(\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -\frac{3}{4} + \frac{3\sqrt{3}}{4}i - \frac{3\sqrt{3}}{4}i - \frac{3 \cdot 3}{4} = -3,$$

hence $X[3] = -3 + 3 - 3 = -3$. For $k_2 = 2$, by an analogous computation, $(-\frac{3}{2} + \frac{3\sqrt{3}}{2}i)\omega_3^2 = -\frac{3}{2} - \frac{3\sqrt{3}}{2}i$ and $(\frac{3}{2} + \frac{3\sqrt{3}}{2}i)\omega_3^4 = (\frac{3}{2} + \frac{3\sqrt{3}}{2}i)\omega_3 = \frac{3}{2} - \frac{3\sqrt{3}}{2}i$, whence $X[5] = -3 - 3\sqrt{3}i$.

Result.

$$X = (21, -3 + 3\sqrt{3}i, -3 + \sqrt{3}i, -3, -3 - \sqrt{3}i, -3 - 3\sqrt{3}i),$$

that is, to the seventh decimal digit,

$$X = (21, -3 + 5.1961524i, -3 + 1.7320508i, -3, -3 - 1.7320508i, -3 - 5.1961524i).$$

First independent check: the closed form. The vector is $y[n] = n + 1$, that is, the ramp plus the constant 1. By linearity of the DFT (Proposition 22.2) and by Lemma 22.1, the DFT of the constant 1 equals 6 for $k = 0$ and 0 for $k \neq 0$; hence, by Lemma 23.5, $X[0] = \frac{6 \cdot 5}{2} + 6 = 21$ and $X[k] = \frac{6}{\omega_6^k - 1}$ for $k \geq 1$. For $k = 1$: $\omega_6 - 1 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, of modulus

1, hence $\frac{1}{\omega_6^{-1}} = \overline{\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $X[1] = 6\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -3 + 3\sqrt{3}i$. ✓ For $k = 2$: $\omega_6^2 - 1 = -\frac{3}{2} - \frac{\sqrt{3}}{2}i$, of squared modulus $\frac{9}{4} + \frac{3}{4} = 3$, hence $X[2] = \frac{6}{3}\left(-\frac{3}{2} + \frac{\sqrt{3}}{2}i\right) = -3 + \sqrt{3}i$. ✓ For $k = 3$: $\omega_6^3 = -1$ and $X[3] = \frac{6}{-2} = -3$. ✓

Second independent check: Python script. The direct computation of the definition, the computation with the closed form, and the computation with the mixed-radix factorization give three vectors that differ, in modulus, by at most $1.95 \cdot 10^{-14}$. ✓

Cost balance. Mixed-radix: 6 additions (phase 1) + 2 nontrivial multiplications (phase 2) + $2 \times (4 \text{ nontrivial multiplications, } 6 \text{ additions})$ (phase 3) = 10 nontrivial multiplications and 18 additions. Direct computation (Proposition 23.1): 36 multiplications, of which $36 - T(6) = 36 - 15 = 21$ not by 1, and 30 additions. ✓

23.12.2 Bluestein and Rader: any N , prime N

Proposition 23.7: The chirp identity (Bluestein). Let $N \geq 1$ and set

$$b_j = e^{-i\pi j^2/N} = \omega_{2N}^{j^2}, \quad j \in \mathbb{Z}.$$

Then $|b_j| = 1$ for every j , and for every $x \in \mathbb{C}^N$ and every $k \in \mathbb{Z}$

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{nk} = b_k \sum_{n=0}^{N-1} (x[n] b_n) \overline{b_{k-n}}$$

Proof. That $|b_j| = 1$ is immediate: b_j is a purely imaginary exponential. The starting algebraic identity is

$$2nk = n^2 + k^2 - (k-n)^2,$$

which is checked by expanding: $n^2 + k^2 - (k^2 - 2kn + n^2) = 2kn$. Dividing by N and multiplying by $-i\pi$,

$$-\frac{2\pi i nk}{N} = -\frac{i\pi n^2}{N} - \frac{i\pi k^2}{N} + \frac{i\pi(k-n)^2}{N},$$

and passing to exponentials (which turn sums into products):

$$\omega_N^{nk} = e^{-2\pi i nk/N} = e^{-i\pi n^2/N} \cdot e^{-i\pi k^2/N} \cdot e^{+i\pi(k-n)^2/N} = b_n b_k \overline{b_{k-n}},$$

having used $e^{+i\pi(k-n)^2/N} = \overline{e^{-i\pi(k-n)^2/N}} = \overline{b_{k-n}}$, legitimate because the exponent is real. Substituting into the definition of $X[k]$ and pulling out of the sum the factor b_k , which does not depend on n :

$$X[k] = \sum_{n=0}^{N-1} x[n] b_n b_k \overline{b_{k-n}} = b_k \sum_{n=0}^{N-1} (x[n] b_n) \overline{b_{k-n}}.$$

□

Corollary 23.8: Every DFT is a circular convolution, and hence costs $O(N \log N)$. Let $N \geq 1$ be arbitrary and let $M = 2^\mu$ be a power of two with $M \geq 2N - 1$.

Define $u, v \in \mathbb{C}^M$ by

$$u[n] = \begin{cases} x[n] b_n, & 0 \leq n \leq N-1, \\ 0, & N \leq n \leq M-1, \end{cases} \quad v[j] = \begin{cases} \overline{b_j}, & 0 \leq j \leq N-1, \\ \overline{b_{j-M}}, & M-N+1 \leq j \leq M-1, \\ 0, & \text{otherwise.} \end{cases}$$

(The two nonzero rows of v do not overlap because $M \geq 2N-1$.) Then, denoting by $u \circledast v$ the circular convolution of length M (Definition 22.4),

$$X[k] = b_k (u \circledast v)[k], \quad k = 0, 1, \dots, N-1.$$

Consequently the DFT of arbitrary length N is computed with three radix-2 FFTs of length $M < 4N$ and $O(N)$ additional multiplications, that is, with $O(N \log N)$ operations.

Proof. Step 1 (the two rows of v do not overlap). The first row occupies the indices $0, \dots, N-1$, the second the indices $M-N+1, \dots, M-1$. They are disjoint if and only if $M-N+1 > N-1$, that is, $M > 2N-2$, that is, $M \geq 2N-1$: this is the hypothesis. Hence v is well defined.

Step 2 (the circular convolution produces the desired sum). By Definition 22.4,

$$(u \circledast v)[k] = \sum_{n=0}^{M-1} u[n] v[\langle k-n \rangle_M] = \sum_{n=0}^{N-1} x[n] b_n v[\langle k-n \rangle_M],$$

since $u[n] = 0$ for $n \geq N$. Now let $0 \leq k \leq N-1$ and $0 \leq n \leq N-1$: then $-(N-1) \leq k-n \leq N-1$. Two cases are distinguished.

- If $k-n \geq 0$, then $\langle k-n \rangle_M = k-n \in \{0, \dots, N-1\}$ and $v[k-n] = \overline{b_{k-n}}$.
- If $k-n < 0$, then $\langle k-n \rangle_M = k-n+M$; from $-(N-1) \leq k-n \leq -1$ it follows that $M-N+1 \leq k-n+M \leq M-1$, that is, one falls into the second row, and $v[k-n+M] = \overline{b_{(k-n+M)-M}} = \overline{b_{k-n}}$.

In both cases $v[\langle k-n \rangle_M] = \overline{b_{k-n}}$, hence

$$(u \circledast v)[k] = \sum_{n=0}^{N-1} x[n] b_n \overline{b_{k-n}},$$

and Proposition 23.7 gives $X[k] = b_k (u \circledast v)[k]$.

Step 3 (cost). By the discrete convolution theorem (Theorem 22.5), $u \circledast v$ is computed as $\mathcal{F}_M^{-1}(\text{DFT}_M\{u\} \cdot \text{DFT}_M\{v\})$, that is, with two direct FFTs, M pointwise multiplications and one inverse FFT (Theorem 23.10), all of length $M = 2^\mu$: cost $\Theta(M \log_2 M)$ by Theorem 23.5. The smallest M that is a power of two with $M \geq 2N-1$ satisfies $M < 2(2N-1) < 4N$, hence $M \log_2 M < 4N \log_2(4N) = O(N \log N)$. To this are added $2N$ multiplications for the factors b_n and b_k and the construction of u and v , which is $O(N)$. $\square$

Theorem 23.12: Rader's algorithm for prime $N = p$. Let p be a prime number and let g be a **primitive root modulo p** , that is, an integer $g \in \{1, \dots, p-1\}$ such that the map

$$\{0, 1, \dots, p-2\} \rightarrow \{1, \dots, p-1\}, \quad u \mapsto \langle g^u \rangle_p$$

is bijective. Let $x \in \mathbb{C}^p$ and $X = \text{DFT}_p\{x\}$. Define, for $u, v \in \{0, \dots, p-2\}$,

$$A[u] = x[\langle g^u \rangle_p], \quad C[v] = \omega_p^{\langle g^v \rangle_p}, \quad B[v] = X[\langle g^v \rangle_p] - x[0].$$

Then

$$X[0] = \sum_{n=0}^{p-1} x[n], \quad B[v] = \sum_{u=0}^{p-2} A[u] C[\langle u+v \rangle_{p-1}], \quad v = 0, \dots, p-2,$$

that is, B is the cyclic correlation of length $p-1$ between A and C .

Proof. Step 1 ($X[0]$). For $k=0$ all factors equal 1 and $X[0] = \sum_n x[n]$.

Step 2 (separating the term $n=0$). For $k \neq 0$,

$$X[k] = x[0] \omega_p^0 + \sum_{n=1}^{p-1} x[n] \omega_p^{nk} = x[0] + \sum_{n=1}^{p-1} x[n] \omega_p^{nk},$$

hence $X[k] - x[0] = \sum_{n=1}^{p-1} x[n] \omega_p^{nk}$.

Step 3 (change of variable with the primitive root). Let $k = \langle g^v \rangle_p$ with $v \in \{0, \dots, p-2\}$: as v varies, k runs through all of $\{1, \dots, p-1\}$ exactly once, by the hypothesis on g . Similarly substitute $n = \langle g^u \rangle_p$ with $u \in \{0, \dots, p-2\}$, which is again a reordering of the terms. One obtains

$$B[v] = X[\langle g^v \rangle_p] - x[0] = \sum_{u=0}^{p-2} x[\langle g^u \rangle_p] \omega_p^{\langle g^u \rangle_p \langle g^v \rangle_p}.$$

Step 4 (the exponent depends only on $u+v$ modulo $p-1$). By Lemma 23.1(ii), ω_p^j depends only on $\langle j \rangle_p$; and modulo p

$$\langle g^u \rangle_p \cdot \langle g^v \rangle_p \equiv g^u g^v = g^{u+v} \pmod{p}.$$

Moreover $g^{p-1} \equiv 1 \pmod{p}$. This is deduced from the hypothesis on g alone, as follows: since $1 \leq g \leq p-1$ and p is prime, g is invertible modulo p , and hence so is every power of it; hence $\langle g^{p-1} \rangle_p \in \{1, \dots, p-1\}$ and, by surjectivity of the map $u \mapsto \langle g^u \rangle_p$, there exists $w \in \{0, \dots, p-2\}$ with $g^{p-1} \equiv g^w \pmod{p}$. Multiplying by the inverse of g^w gives $g^d \equiv 1 \pmod{p}$ with $d = p-1-w \in \{1, \dots, p-1\}$. If $d \leq p-2$ held, then $\langle g^d \rangle_p = 1 = \langle g^0 \rangle_p$ with $0 \neq d$ both in $\{0, \dots, p-2\}$, against injectivity. Hence $d = p-1$, that is, $w = 0$ and $g^{p-1} \equiv 1$. Consequently, writing $u+v = q(p-1) + \langle u+v \rangle_{p-1}$ with $q \in \{0, 1\}$ (since $0 \leq u+v \leq 2p-4$), one has $g^{u+v} = (g^{p-1})^q g^{\langle u+v \rangle_{p-1}} \equiv g^{\langle u+v \rangle_{p-1}} \pmod{p}$, and

$$\omega_p^{\langle g^u \rangle_p \langle g^v \rangle_p} = \omega_p^{\langle g^{\langle u+v \rangle_{p-1}} \rangle_p} = C[\langle u+v \rangle_{p-1}].$$

Step 5. Substituting into Step 3 gives the formula of the statement, which is by definition the cyclic correlation of A and C of length $p-1$. $\square$

Remark 23.11: What we import here. Theorem 23.12 is proved in full, but under the hypothesis that a primitive root g modulo p exists. That fact — the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic for every prime p — is a classical theorem of number theory that is **imported** here: we do not prove it, because its proof (which goes through counting the solutions of $X^d = 1$ in a finite field and the formula $\sum_{d|n} \varphi(d) = n$) is foreign to the thread of the chapter and does not concern the circle. What exactly is missing: the existence of g , and nothing else; its uniqueness is not needed (there are $\varphi(p-1)$ primitive roots modulo p , and any of them will do), and its actual computation is done by trial, checking for each candidate g that $g^{(p-1)/q} \not\equiv 1$ for every prime q dividing $p-1$. Source of the imported fact: Hardy and Wright, *An Introduction to the Theory of Numbers*, chapter on congruences

and primitive roots; the use in the algorithm is the one described in Van Loan, Chapter 4 “Selected Topics,” pp. 188–258.

A second remark, this time on the use: the cyclic correlation of length $p-1$ of Theorem 23.12 is itself computed with FFTs (Theorem 22.5), but $p-1$ is not in general a power of two; one therefore falls back on mixed-radix (Theorem 23.11), or on Bluestein (Corollary 23.8), or on zero-padding of the convolution. Rader alone does not close the problem: it transforms it.

INFO

Panorama: the FFT is not an algorithm, it is a family. Beyond the three routes proved in this section, the literature contains at least: *split-radix* algorithms, which alternate radix-2 and radix-4 decompositions and lower the count of real multiplications below that of radix-2; *prime factor* algorithms (Good–Thomas), which eliminate the twiddle factors entirely when N_1 and N_2 are coprime, at the price of an index map based on the Chinese remainder theorem; *Winograd* algorithms, which minimize multiplications using the theory of cyclotomic polynomials. None of these is proved here, and we will not use them: they are cited as panorama, with the source (Van Loan, Chapters 2–4; Oppenheim and Schaffer, Chapter 9), and they do not appear in any statement of this chapter. The *theoretical* lower bound on the number of operations needed to compute a DFT of length N is still an open problem: it is not known whether $\Theta(N \log N)$ is optimal.

23.13. Round-off error in finite-precision arithmetic

This is step 29 of the path, and it must be kept distinct from complexity: the FFT is not only *faster* than the direct computation, it is also *more accurate*, and for a structural reason that is proved here. The number of operations contributing to each output goes from $\Theta(N)$ to $\Theta(\log_2 N)$, and with it the number of round-off errors that can accumulate.

Definition 23.8: Standard model of floating-point arithmetic, and hypothesis (M). Let $\text{fl}(\cdot)$ denote the result of an operation performed in floating point and u the **unit round-off** (for the 64-bit binary format, $u = 2^{-53} = 1.1102230 \cdot 10^{-16}$). The **standard model** assumes that for every elementary operation $*$ $\in \{+, -, \times, \div\}$ on machine real numbers

$$\text{fl}(a * b) = (a * b)(1 + \delta), \quad |\delta| \leq u.$$

From it follows, for complex arithmetic implemented with the usual formulas, the following hypothesis, which we assume throughout the rest of the section and call **hypothesis (M)**:

there exists a constant η , a fixed multiple of u depending only on the implementation of complex multiplication (for instance $\eta \leq 4u + O(u^2)$ with the schoolbook algorithm), such that the floating-point execution of a stage A of Definition 23.6 produces, in place of Av , the vector $(A + E)v$ with $\|E\|_2 \leq \eta \|A\|_2$, for every input vector v .

Hypothesis (M) is not proved here: it is the result of standard analysis of floating-point arithmetic, and is *imported* from Higham, *Accuracy and Stability of Numerical Algorithms*, Chapters 2 and 24, and from Van Loan, Chapter 1. What is missing, and is not proved here, is exactly the passage from the scalar model $\text{fl}(a * b) = (a * b)(1 + \delta)$ to the norm bound $\|E\|_2 \leq \eta \|A\|_2$ for an entire stage; everything else in the section is proved.

Remark 23.12: What we import in the error analysis. The only **imported** element of this section is hypothesis (M) of Definition 23.8. Precisely: we do not prove that the scalar model $\mathfrak{f}(a*b) = (a*b)(1+\delta)$, $|\delta| \leq u$, implies for an entire stage A the bound $\|E\|_2 \leq \eta\|A\|_2$ with $\eta = O(u)$. The proof requires the forward analysis of complex arithmetic (four products and two real sums for each complex product, with the respective constants $\gamma_k = ku/(1-ku)$) and the aggregation of the perturbations componentwise into a norm bound: this is the subject of Chapter 2 of Higham, and the result for the FFT is his Chapter 24. Everything else in the section — Lemma 23.6, Corollary 23.9, Lemma 23.7 and Theorem 23.13 — is proved here in full starting from hypothesis (M), which appears explicitly among the hypotheses of the theorem.

Lemma 23.6: Every stage is $\sqrt{2}$ times an isometry. Let $N = 2^m$ and let $A_s : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be the linear map that realizes stage s of Definition 23.6, $s = 1, \dots, m$. Then

$$\|A_s v\|_2 = \sqrt{2} \|v\|_2 \quad \text{for every } v \in \mathbb{C}^N,$$

and hence $A_s^* A_s = 2I_N$, $\|A_s\|_2 = \sqrt{2}$, $\|A_s^{-1}\|_2 = 1/\sqrt{2}$.

Proof. Stage s acts on pairs of indices: setting $L = n_s/2 = 2^{s-1}$, the pairs are $\mathcal{P}_s = \{(b+k, b+k+L)\}$ as b ranges over $\{0, n_s, 2n_s, \dots\}$ and k over $\{0, \dots, L-1\}$. These pairs are *disjoint* and their union is $\{0, \dots, N-1\}$: indeed every index j is written uniquely as $j = b + k + \varepsilon L$ with b a multiple of n_s , $0 \leq k < L$ and $\varepsilon \in \{0, 1\}$ (Euclidean division of j by n_s , then of the remainder by L). There are $N/2$ pairs.

On each pair (p, q) the stage computes, with $u = v[p]$, $t = w v[q]$ and $|w| = |\omega_{n_s}^k| = 1$ (Lemma 23.1(i)),

$$(A_s v)[p] = u + t, \quad (A_s v)[q] = u - t.$$

By the parallelogram identity in $\mathbb{C}$ (checked by expanding the squared moduli: $|u+t|^2 + |u-t|^2 = (u+t)\overline{(u+t)} + (u-t)\overline{(u-t)} = 2|u|^2 + 2|t|^2$, since the cross terms ut and $\bar{u}\bar{t}$ appear with opposite signs and cancel),

$$|(A_s v)[p]|^2 + |(A_s v)[q]|^2 = 2|u|^2 + 2|t|^2 = 2|v[p]|^2 + 2|w|^2|v[q]|^2 = 2(|v[p]|^2 + |v[q]|^2).$$

Summing over all $N/2$ pairs, which partition the indices,

$$\|A_s v\|_2^2 = \sum_{(p,q) \in \mathcal{P}_s} (|(A_s v)[p]|^2 + |(A_s v)[q]|^2) = 2 \sum_{(p,q) \in \mathcal{P}_s} (|v[p]|^2 + |v[q]|^2) = 2\|v\|_2^2.$$

From $\|A_s v\|_2^2 = 2\|v\|_2^2$ for every v it follows that $v^*(A_s^* A_s - 2I)v = 0$ for every v ; since $A_s^* A_s - 2I$ is Hermitian, this implies $A_s^* A_s = 2I$ (a Hermitian matrix whose quadratic form vanishes identically is the zero matrix). The two norms follow as in Proposition 23.3(iii). $\square$

Corollary 23.9: Factorization of the DFT matrix. Let $N = 2^m$ and let P_ρ be the permutation matrix associated with ρ_m (Definition 23.5), that is, $(P_\rho v)[j] = v[\rho_m(j)]$. Then

$$W_N = A_m A_{m-1} \cdots A_1 P_\rho,$$

and $\|W_N\|_2 = (\sqrt{2})^m = \sqrt{N}$, consistently with Theorem 22.3.

Proof. Theorem 23.8 states that the algorithm, that is, the map $x \mapsto A_m \cdots A_1 P_\rho x$, returns $\text{DFT}_N\{x\} = W_N x$ for every $x \in \mathbb{C}^N$: two linear maps that coincide on all of $\mathbb{C}^N$ are the same matrix. For the norm, from Lemma 23.6 and from $\|P_\rho v\|_2 = \|v\|_2$ (a permutation of

the components does not change the sum of the squared moduli) it follows that $\|W_N v\|_2 = (\sqrt{2})^m \|v\|_2 = \sqrt{N} \|v\|_2$ for every v , hence $\|W_N\|_2 = \sqrt{N}$. Theorem 22.3 says that $F_N = N^{-1/2} W_N$ is unitary, that is, $\|F_N\|_2 = 1$: the same thing. $\square$

Lemma 23.7: Perturbation of a matrix product. Let $A_1, \dots, A_m$ and $E_1, \dots, E_m$ be $N \times N$ matrices, and set $a_s = \|A_s\|_2$, $e_s = \|E_s\|_2$. Then

$$\left\| (A_m + E_m) \cdots (A_1 + E_1) - A_m \cdots A_1 \right\|_2 \leq \prod_{s=1}^m (a_s + e_s) - \prod_{s=1}^m a_s.$$

Proof. By induction on m . Set $F_j = (A_j + E_j) \cdots (A_1 + E_1)$ and $G_j = A_j \cdots A_1$.

Base $m = 1$: $\|F_1 - G_1\|_2 = \|E_1\|_2 = e_1 = (a_1 + e_1) - a_1$.

Step: let $m \geq 2$ and let the claim hold for $m - 1$. One has the identity

$$F_m - G_m = (A_m + E_m)F_{m-1} - A_m G_{m-1} = (A_m + E_m)(F_{m-1} - G_{m-1}) + E_m G_{m-1},$$

as is checked by expanding the right-hand side. By submultiplicativity and subadditivity of the spectral norm, and by $\|G_{m-1}\|_2 \leq \prod_{s < m} a_s$,

$$\|F_m - G_m\|_2 \leq (a_m + e_m) \left[\prod_{s < m} (a_s + e_s) - \prod_{s < m} a_s \right] + e_m \prod_{s < m} a_s.$$

Expanding the right-hand side:

$$\prod_{s \leq m} (a_s + e_s) - (a_m + e_m) \prod_{s < m} a_s + e_m \prod_{s < m} a_s = \prod_{s \leq m} (a_s + e_s) - a_m \prod_{s < m} a_s = \prod_{s \leq m} (a_s + e_s) - \prod_{s \leq m} a_s.$$

$\square$

Theorem 23.13: Error bound for the radix-2 FFT. Let $N = 2^m$, let $x \in \mathbb{C}^N$ be nonzero, let $X = \text{DFT}_N\{x\}$ and let $\hat{X}$ be the result of the algorithm of Definition 23.6 executed in floating point under hypothesis (M) of Definition 23.8. Then

$$\frac{\|\hat{X} - X\|_2}{\|X\|_2} \leq (1 + \eta)^m - 1$$

and, if $m\eta \leq 1$,

$$\frac{\|\hat{X} - X\|_2}{\|X\|_2} \leq (e - 1) m \eta < 1.72 m \eta = 1.72 \eta \log_2 N.$$

Proof. Step 1. Under hypothesis (M), the computed result is

$$\hat{X} = (A_m + E_m) \cdots (A_1 + E_1) P_\rho x,$$

with $\|E_s\|_2 \leq \eta \|A_s\|_2 = \eta \sqrt{2}$ by Lemma 23.6. (The permutation (P) performs no arithmetic operations and introduces no error: it moves machine numbers around.)

Step 2. By Corollary 23.9, $X = A_m \cdots A_1 P_\rho x$. Hence

$$\hat{X} - X = \left[(A_m + E_m) \cdots (A_1 + E_1) - A_m \cdots A_1 \right] P_\rho x.$$

Step 3. We apply Lemma 23.7 with $a_s = \sqrt{2}$ and $e_s \leq \eta \sqrt{2}$:

$$\prod_{s=1}^m (a_s + e_s) - \prod_{s=1}^m a_s \leq (\sqrt{2}(1 + \eta))^m - (\sqrt{2})^m = 2^{m/2} [(1 + \eta)^m - 1] = \sqrt{N} [(1 + \eta)^m - 1].$$

Since $\|P_\rho x\|_2 = \|x\|_2$,

$$\|\hat{X} - X\|_2 \leq \sqrt{N}[(1 + \eta)^m - 1] \|x\|_2.$$

Step 4. By Parseval's identity (Theorem 22.6), $\|X\|_2^2 = N\|x\|_2^2$, that is, $\|X\|_2 = \sqrt{N}\|x\|_2$ (in particular $\|X\|_2 \neq 0$ because $x \neq 0$). Dividing,

$$\frac{\|\hat{X} - X\|_2}{\|X\|_2} \leq \frac{\sqrt{N}[(1 + \eta)^m - 1]\|x\|_2}{\sqrt{N}\|x\|_2} = (1 + \eta)^m - 1.$$

Step 5 (the linear form). From the binomial expansion, with $\binom{m}{j} \leq m^j/j!$,

$$(1 + \eta)^m - 1 = \sum_{j=1}^m \binom{m}{j} \eta^j \leq \sum_{j=1}^m \frac{(m\eta)^j}{j!} = e^{m\eta} - 1.$$

The function $t \mapsto e^t - 1$ is convex and vanishes at $t = 0$, hence on the segment $[0, 1]$ it lies below its chord: $e^t - 1 \leq t(e - 1)$ for $0 \leq t \leq 1$. With $t = m\eta \leq 1$ one obtains $(1 + \eta)^m - 1 \leq (e - 1)m\eta < 1.72m\eta$. $\square$

Remark 23.13: Comparison with the direct sum. The same analysis applied to the direct computation of Proposition 23.1 gives a bound proportional to Nu rather than to $u \log_2 N$: each output is a sum of N terms computed in sequence, and in the worst case the errors of the individual terms all add with the same sign. The gain of the FFT is therefore twofold: a factor $\Theta(N/\log_2 N)$ on the number of operations (Corollary 23.3) and a factor of the same order on the error bound. This is a rare fact and it is worth stressing: normally a faster algorithm is less accurate, because it buys speed by giving something up. Here nothing is given up, because the algorithm computes the same sum (Corollary 23.2), only in an order that shortens every chain of dependencies from N to $\log_2 N$ steps. It should be said honestly that the worst-case bounds are pessimistic in both cases: in practice the elementary errors are roughly independent and partly cancel, so that the observed growth is slower than the guaranteed one. This is exactly what the following experiment shows.

Example 23.12: Experimental check. Protocol. For $N = 2^m$ with $m = 3, 5, 7, 8, 9$ a vector $v \in \mathbb{C}^N$ is generated with real and imaginary parts drawn from a standard normal (fixed seed 20260906, generator `numpy.random.default_rng`). Three transforms are computed:

1. the *reference*, with the sum of Definition 22.1 executed in 50-decimal-digit arithmetic (`mpmath` library);
2. the recursive FFT of Definition 23.4 in double precision;
3. the direct sum of Definition 22.1 in double precision, carried out genuinely *in sequence*: an explicit loop over $n = 0, 1, \dots, N - 1$ with a scalar accumulator, without vectorization, and twiddle factors $e^{-2\pi i \langle kn \rangle_{N/N}}$ with the exponent reduced modulo N (without the reduction, what would be measured is the error in evaluating the twiddle factor at a large argument, not the error of the summation).

The relative errors in Euclidean norm with respect to the reference are reported.

N	m	rel. error FFT	rel. error direct sum	$(1 + 4u)^m - 1$
8	3	$9.2953 \cdot 10^{-17}$	$2.1641 \cdot 10^{-16}$	$1.3323 \cdot 10^{-15}$
32	5	$1.7298 \cdot 10^{-16}$	$3.2447 \cdot 10^{-16}$	$2.2204 \cdot 10^{-15}$
128	7	$2.8949 \cdot 10^{-16}$	$4.8317 \cdot 10^{-16}$	$3.1086 \cdot 10^{-15}$
256	8	$3.1065 \cdot 10^{-16}$	$6.1601 \cdot 10^{-16}$	$3.5527 \cdot 10^{-15}$
512	9	$3.4593 \cdot 10^{-16}$	$7.8512 \cdot 10^{-16}$	$3.9968 \cdot 10^{-15}$

Reading 1: the bound is respected. The last column is the bound of Theorem 23.13 with $\eta = 4u$ (schoolbook algorithm for complex multiplication), that is, $(1 + 4u)^m - 1$, with $u = 1.1102230 \cdot 10^{-16}$. The observed FFT error is every time smaller, by a margin between 11 and 14 times. ✓

Reading 2: the growth is linear in m , as the theorem predicts. The ratio of the observed error to m is

$$3.098, \quad 3.460, \quad 4.136, \quad 3.883, \quad 3.844 \quad (\times 10^{-17}),$$

that is, it stays constant within a factor 1.34 while N varies by a factor 64: the dependence on N is indeed logarithmic. If the error grew like N , that ratio would have grown by a factor ≈ 21 . ✓

Reading 3: at these N the direct sum does *not* grow faster. Going from $N = 8$ to $N = 512$ (factor 64) the error of the direct sum grows by a factor $7.8512/2.1641 = 3.628$ and that of the FFT by a factor $3.4593/0.92953 = 3.722$: the two columns grow at the same rate, and the ratio between the two errors stays essentially fixed (from 2.33 at $N = 8$ to 2.27 at $N = 512$). Applying to the direct-sum column the same criterion as in Reading 2, the ratio of the observed error to m is

$$0.721, \quad 0.649, \quad 0.690, \quad 0.770, \quad 0.872 \quad (\times 10^{-16}),$$

that is, it stays constant within a factor 1.34, exactly as for the FFT: at these N the direct sum too behaves logarithmically. The worst-case bound proportional to Nu of Remark 23.13 is therefore far from being attained, and for the reason stated there: in practice the elementary errors are roughly independent and partly cancel, instead of all adding up with the same sign. The experiment thus confirms Theorem 23.13 (Readings 1 and 2) and the caution of Remark 23.13, but it does *not* exhibit the accuracy advantage of the FFT: at these N the gap between the two bounds remains a fact about the guaranteed bounds, not an observed fact, and it must be said so rather than read into numbers that do not say it. The advantage of the FFT measured here is the other one, on the number of operations (Corollary 23.3).

Checking the check. The same protocol repeated with the direct sum replaced by `numpy.fft.fft` gives, for $N = 8$, a discrepancy of $2.54 \cdot 10^{-14}$ in modulus on the ramp example (Section 23.9), consistent with the order of magnitude measured here once multiplied by $\|X\|_2 \approx 33.5$. ✓

Remark 23.14: Memory, cache and implementations: why the count is not the time. The operation count (Theorem 23.5) is a mathematical invariant; the execution time is not, and on a modern computer the two quantities can diverge by an order of magnitude. The reasons, all foreign to the analysis carried out so far, are at least four:

- **locality of reference:** in the last stages of the DIT the two cells of a butterfly are $N/2$ positions apart, and for large N they fall on different memory pages; serious implementations resort to “four-step” decompositions that treat the vector as a $\sqrt{N} \times \sqrt{N}$ matrix, in order to work on blocks that fit in cache;

- **table of twiddle factors:** precomputing $N/2$ of them costs memory but avoids $\Theta(N \log N)$ evaluations of sine and cosine, each of which costs dozens of elementary operations and introduces error;
- **vectorization and parallelism:** the butterflies of a single stage are independent and lend themselves to simultaneous execution; the number of operations does not change, the time does;
- **runtime algorithm selection:** the FFTW library experimentally measures different combinations of radices and decompositions on the host machine and chooses the fastest (Frigo and Johnson, 2005).

None of this changes the number of operations or the result: these are aspects distinct from asymptotic complexity, and this is why Table 23.1 reports counts and not times (Remark 23.7).

23.14. Assigned exercises

These are *new* exercises, distinct from those solved in Section 23.15. The answers are in the Teacher key (Section 23.17).

Exercise 23.1: Counting before programming. *Objective:* to be able to produce, for a given N , the four numbers that decide a project — multiplications and additions of the direct computation, multiplications and additions of the FFT — and to translate them into time. *Estimated time:* 40 minutes. *Type:* numerical and proof-based. *Level:* basic–intermediate.

- For $N = 32$ compute, with Proposition 23.1, Theorem 23.5 and Corollary 23.4: multiplications and additions of the direct computation; multiplications and additions of the FFT; nontrivial multiplications of the FFT. Check the five numbers with a small script that simulates the algorithm and increments two counters.
- Prove that, in the radix-2 FFT of length $N = 2^m$ with $m \geq 2$, the total number of twiddle factors belonging to $\{1, -1, i, -i\}$ equals $\frac{3N}{2} - 2$. (Hint: use points (i)–(iii) of Proposition 23.5, and note that the two families are disjoint.)
- For which smallest $N = 2^m$ are the multiplications of the FFT (counted *all* of them, that is, $\frac{N}{2} \log_2 N$) *at least ten times* fewer than those of the direct computation (N^2)? Derive the inequality in m and solve it by successive trials, showing the two values of m that bracket it; check the result against Table 23.1.
- A processor performs 10^8 complex multiplications per second and one wants to transform a block of $N = 2^{20}$ samples. Compute the time required by the direct computation (in hours) and by the FFT (in seconds), using respectively N^2 and the number of nontrivial multiplications. Comment on why the second number should not be passed off as a prediction of the real time (Remark 23.14).

Exercise 23.2: A four-point FFT, by hand and in full. *Objective:* to carry out the algorithm on paper, with all the checks prescribed by the chapter. *Estimated time:* 35 minutes. *Type:* symbolic and numerical computation. *Level:* remedial–basic. Consider $N = 4$ and

$$x = (1, 2, 0, -1).$$

- Write the permutation ρ_2 and the reordered vector.
- Carry out the two stages of Definition 23.6, indicating the twiddle factors used (ω_2^0 in the first stage, ω_4^0 and ω_4^1 in the second) and the intermediate values.
- Compute X also directly from Definition 22.1 and compare.

- (d) Verify Parseval's identity (Theorem 22.6) and Hermitian symmetry (Theorem 22.7); say which components must turn out real, and why.
- (e) Reconstruct x with Theorem 23.10, showing all four samples.
- (f) Count the nontrivial multiplications actually performed and compare with the formula $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$.

Exercise 23.3: Lengths that are not powers of two. *Objective:* to apply the mixed-radix factorization and the two algorithms for prime N , and to know which is preferable. *Estimated time:* 50 minutes. *Type:* proof-based and computational. *Level:* intermediate-advanced.

- (a) Let $N = 12 = N_1 N_2$ with $N_1 = 4$, $N_2 = 3$, and let $x \in \mathbb{C}^{12}$ be the vector with $x[0] = x[6] = 1$ and all other components zero. Apply Theorem 23.11: write the three subsequences $(x[3n_1 + n_2])_{n_1}$, compute the three DFTs of length 4, apply the factors $\omega_{12}^{k_1 n_2}$ and conclude with the four DFTs of length 3. Compare the result with the direct computation.
- (b) Count the nontrivial multiplications of the two procedures of part (a) and compare them with those one would have by padding x with four zeros up to $N' = 16$ and using the radix-2 FFT. Which of the two routes is preferable? What changes in the spectrum (Proposition 23.6 and Remark 23.10)?
- (c) Let $N = 5$. Determine all primitive roots modulo 5 by checking the definition given in Theorem 23.12, and write explicitly the sequences A and C and the cyclic correlation of length 4 for a generic $x \in \mathbb{C}^5$.
- (d) For the same $N = 5$, write the vectors u and v of Corollary 23.8 with $M = 16$, indicating which components of v are nonzero and why the condition $M \geq 2N - 1$ is essential.

23.15. Solved exercises

Exercise 23.4: The same transform with decimation in frequency. *Type:* numerical computation with cross-check. Compute the DFT of $x = (0, 1, 2, 3, 4, 5, 6, 7)$ with the DIF algorithm of Definition 23.7, show that the output is in bit-reversed order, and compare with the DIT result of Section 23.9.

Solution. Strategy. DIF operates on the input in *natural* order and produces the output in bit-reversed order (Remark 23.8). At each stage sums and differences are formed between cells separated by half the length of the current block, and it is the difference — not the sum — that gets multiplied by the twiddle factor (Theorem 23.9). The first stage works on a block of 8, the second on two blocks of 4, the third on four blocks of 2: the order of the blocks is the reverse of that of the DIT.

Step 1 — stage 1 ($L = 4$, factors ω_8^n , $n = 0, 1, 2, 3$). One forms $\alpha[n] = x[n] + x[n + 4]$ and $\beta[n] = (x[n] - x[n + 4]) \omega_8^n$:

$$\alpha = (0 + 4, 1 + 5, 2 + 6, 3 + 7) = (4, 6, 8, 10),$$

$$x[n] - x[n + 4] = -4 \text{ for every } n, \quad \beta = (-4, -4\omega_8, -4\omega_8^2, -4\omega_8^3).$$

With $\omega_8 = \frac{\sqrt{2}}{2}(1 - i)$, $\omega_8^2 = -i$, $\omega_8^3 = -\frac{\sqrt{2}}{2}(1 + i)$ and $4 \cdot \frac{\sqrt{2}}{2} = 2\sqrt{2} = 2.8284271$:

$$\beta = (-4, -2\sqrt{2} + 2\sqrt{2}i, 4i, 2\sqrt{2} + 2\sqrt{2}i),$$

$$\beta = (-4, -2.8284271 + 2.8284271i, 4i, 2.8284271 + 2.8284271i).$$

The vector after the first stage is $(\alpha, \beta) = (4, 6, 8, 10, \beta[0], \dots, \beta[3])$.

Step 2 — stage 2 (two blocks of 4, $L = 2$, factors $\omega_4^0 = 1, \omega_4^1 = -i$). In the first block $(4, 6, 8, 10)$:

$$4 + 8 = 12, \quad 6 + 10 = 16, \quad (4 - 8) \cdot 1 = -4, \quad (6 - 10) \cdot (-i) = 4i.$$

In the second block β :

$$\beta[0] + \beta[2] = -4 + 4i, \quad \beta[1] + \beta[3] = 4\sqrt{2}i = 5.6568542i,$$

$$(\beta[0] - \beta[2]) \cdot 1 = -4 - 4i, \quad (\beta[1] - \beta[3]) \cdot (-i) = (-4\sqrt{2})(-i) = 4\sqrt{2}i.$$

The vector is now $(12, 16, -4, 4i, -4 + 4i, 4\sqrt{2}i, -4 - 4i, 4\sqrt{2}i)$.

Step 3 — stage 3 (four blocks of 2, single factor $\omega_2^0 = 1$).

$$\begin{aligned} 12 + 16 &= 28, & 12 - 16 &= -4, \\ -4 + 4i, & & -4 - 4i, & \\ (-4 + 4i) + 4\sqrt{2}i &= -4 + i(4 + 4\sqrt{2}), & (-4 + 4i) - 4\sqrt{2}i &= -4 - i(4\sqrt{2} - 4), \\ (-4 - 4i) + 4\sqrt{2}i &= -4 + i(4\sqrt{2} - 4), & (-4 - 4i) - 4\sqrt{2}i &= -4 - i(4 + 4\sqrt{2}). \end{aligned}$$

The final vector, cell by cell, is

$$(28, -4, -4+4i, -4-4i, -4+9.6568542i, -4-1.6568542i, -4+1.6568542i, -4-9.6568542i).$$

Step 4 — output bit-reversal reordering. Applying $\rho_3 = (0, 4, 2, 6, 1, 5, 3, 7)$ to the positions, that is, reading cell $\rho_3(k)$ as $X[k]$:

k	0	1	2	3	4	5	6	7
$\rho_3(k)$	0	4	2	6	1	5	3	7
$X[k]$	28	$-4 + 9.657i$	$-4 + 4i$	$-4 + 1.657i$	-4	$-4 - 1.657i$	$-4 - 4i$	$-4 - 9.657i$

which coincides with Table 23.2. ✓

Independent check 1: the closed form. The values coincide with those of Example 23.6, obtained from Lemma 23.5. ✓

Independent check 2: Python script. An implementation of the DIF that performs the three stages and then reorders returns the same eight values, with maximum discrepancy $2.46 \cdot 10^{-14}$ relative to the direct sum of the definition. ✓

Interpretation. Comparing with Example 23.7 one sees that DIT and DIF pass through *different* intermediate values — after the first stage the DIT has $(4, -4, 8, -4, 6, -4, 10, -4)$ and the DIF has $(4, 6, 8, 10, -4, \dots)$ — but they arrive at the same result. They are not two approximations of the same thing: they are two exact factorizations of the same matrix W_8 (Corollary 23.9).

Typical errors. (1) Multiplying the *sum* instead of the difference by the twiddle factor: in the DIF the twiddle sits on the branch of the difference, unlike in the DIT, where it sits before the butterfly. (2) Using ω_4 at the first stage instead of ω_8 : in the DIF the order of the factors is decreasing (ω_8 , then ω_4 , then ω_2), the opposite of the DIT. (3) Forgetting the final reordering and comparing cell by cell with the DIT output: one gets the same eight numbers, but in a different order, and the comparison fails without the algorithm being wrong.

Extension. Show that, by composing a DIF with no final reordering with a DIT with no initial reordering, one obtains $N \cdot \mathcal{F}_N^{-1} \mathcal{F}_N = N \text{id}$, and explain why this allows a circular convolution (Theorem 22.5) to be computed without ever performing a bit-reversal permutation.

Exercise 23.5: Linear convolution with a circular FFT. *Type:* computational, with cost count. Let $a = (1, 2, 3, 4)$ and $b = (1, 1, 1, 1)$ be two sequences of length 4. Compute their *linear* convolution (that is, the product of the associated polynomials) using FFTs of length 8, and compare the cost with that of the direct computation.

Solution. Strategy. Theorem 22.5 concerns *circular* convolution; the linear convolution of two sequences of lengths P and Q has length $P + Q - 1$. If both are padded with zeros up to a length $M \geq P + Q - 1$, the circular convolution of length M coincides with the linear one, because no term “wraps around”. Here $P = Q = 4$, $P + Q - 1 = 7$, and the first power of two ≥ 7 is $M = 8$: two 8-point FFTs, 8 products, one 8-point IFFT.

Step 1 — zero-padding. $\tilde{a} = (1, 2, 3, 4, 0, 0, 0, 0)$, $\tilde{b} = (1, 1, 1, 1, 0, 0, 0, 0)$.

Step 2 — the two FFTs. With $\sqrt{2} = 1.4142136$, and recalling that $1 + \sqrt{2} = 2.4142136$ and $\sqrt{2} - 1 = 0.4142136$:

$$\begin{aligned} A[0] &= 10, & B[0] &= 4, \\ A[1] &= -0.4142136 - 7.2426407i, & B[1] &= 1 - 2.4142136i, \\ A[2] &= -2 + 2i, & B[2] &= 0, \\ A[3] &= 2.4142136 - 1.2426407i, & B[3] &= 1 - 0.4142136i, \\ A[4] &= -2, & B[4] &= 0, \\ A[5] &= 2.4142136 + 1.2426407i, & B[5] &= 1 + 0.4142136i, \\ A[6] &= -2 - 2i, & B[6] &= 0, \\ A[7] &= -0.4142136 + 7.2426407i, & B[7] &= 1 + 2.4142136i. \end{aligned}$$

That $B[2] = B[4] = B[6] = 0$ is proved as follows: $\tilde{b}$ is the sequence $(1, 1, 1, 1, 0, 0, 0, 0)$, its DFT equals $\sum_{n=0}^3 \omega_8^{kn} = \frac{1 - \omega_8^{4k}}{1 - \omega_8^k}$ for $\omega_8^k \neq 1$, and $\omega_8^{4k} = (-1)^k$ by Lemma 23.1(iv): the numerator vanishes for even k , $k \neq 0$.

Step 3 — pointwise product. $C[k] = A[k]B[k]$:

$$\begin{aligned} C[0] &= 40, & C[1] &= -17.8994949 - 6.2426407i, \\ C[2] &= 0, & C[3] &= 1.8994949 - 2.2426407i, \\ C[4] &= 0, & C[5] &= 1.8994949 + 2.2426407i, \\ C[6] &= 0, & C[7] &= -17.8994949 + 6.2426407i. \end{aligned}$$

Step 4 — IFFT. By Theorem 23.10, $c = \frac{1}{8} \overline{\mathcal{A}_8(\overline{C})}$, and one obtains

$$c = (1, 3, 6, 10, 9, 7, 4, 0).$$

Independent check 1: the product of the polynomials. The linear convolution is the coefficient vector of

$$(1 + 2z + 3z^2 + 4z^3)(1 + z + z^2 + z^3) = 1 + 3z + 6z^2 + 10z^3 + 9z^4 + 7z^5 + 4z^6,$$

that is, $(1, 3, 6, 10, 9, 7, 4)$; adding the final zero required by length 8 recovers exactly c . ✓

Independent check 2: the circular convolution by hand. By Definition 22.4, $c[n] = \sum_{j=0}^7 \tilde{a}[j] \tilde{b}[(n-j)_8]$; for instance $c[3] = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 1 = 10$ and $c[7] = \tilde{a}[4] + \cdots = 0$ because every term involves either an $\tilde{a}[j]$ with $j \geq 4$ or a null $\tilde{b}$. ✓ The Python script returns $c = (1, 3, 6, 10, 9, 7, 4, 0)$ with maximum error $7.27 \cdot 10^{-16}$. ✓

Cost count. Direct computation of the linear convolution: $4 \times 4 = 16$ multiplications and $16 - 7 = 9$ additions. Via FFT: three 8-point transforms, that is, $3 \times 2 = 6$ nontrivial multiplications (Corollary 23.4) and $3 \times 24 = 72$ additions, plus 8 pointwise multiplications and 8 divisions by 8 (exact, because 8 is a power of two; Corollary 23.6).

Interpretation. At $N = 4$ the FFT route *is not worthwhile*: 14 nontrivial multiplications and 72 additions against 16 and 9. This is the important observation of this exercise: the asymptotic advantage is not an advantage at every N . Repeating the count for two sequences of length $P = Q = 1,024$ (hence $M = 2,048 = 2^{11}$), the direct computation requires $PQ = 1,048,576$ multiplications; the FFT route requires, by Corollary 23.4,

$$3 \left(\frac{2,048}{2} \cdot 11 - \frac{3 \cdot 2,048}{2} + 2 \right) + 2,048 = 3(11,264 - 3,072 + 2) + 2,048 = 3 \cdot 8,194 + 2,048 = 26,630,$$

that is, a factor $1,048,576/26,630 = 39.4$. This is how products of polynomials and of large integers are computed.

Typical errors. (1) Choosing $M = 4$: the circular convolution of length 4 is *not* the linear one, and coefficients of degree ≥ 4 add to those of degree ≥ 0 (“time aliasing”); one would obtain $(1 + 9, 3 + 7, 6 + 4, 10) = (10, 10, 10, 10)$. (2) Forgetting that the IFFT returns complex numbers: the imaginary parts must turn out zero up to round-off error, and this is a further check for free. (3) Comparing only the cost of the multiplications: here the bottleneck is the additions.

Extension. Prove in general that, if $\tilde{a}$ and $\tilde{b}$ have at most P and Q nonzero components, with $P + Q - 1 \leq M$, then $(\tilde{a} \circledast \tilde{b})[n]$ coincides with the linear convolution for every $n = 0, \dots, M - 1$. (Hint: in the sum $\sum_j \tilde{a}[j] \tilde{b}[\langle n - j \rangle_M]$ show that the index $\langle n - j \rangle_M$ differs from $n - j$ only where $\tilde{b}$ vanishes there.)

Exercise 23.6: What the algorithm computes if the permutation is forgotten.

Type: error diagnosis, with numerical verification. Run the iterative algorithm of Definition 23.6 on $x = (0, 1, 2, 3, 4, 5, 6, 7)$ omitting step (P). Identify what is computed, show that Parseval’s identity is nonetheless satisfied, and indicate a check that exposes the error.

Solution. Strategy. Theorem 23.8 proves that the stages (S), applied to a vector a , return the DFT of the vector v such that $a[j] = v[\rho_m(j)]$ for every j . If (P) is omitted, one is applying (S) to $a = x$; one must therefore solve $x[j] = v[\rho_m(j)]$ for v .

Step 1 — what is computed. From $x[j] = v[\rho_m(j)]$, setting $n = \rho_m(j)$ and using $\rho_m^{-1} = \rho_m$ (Lemma 23.4), one gets $j = \rho_m(n)$ and

$$v[n] = x[\rho_m(n)].$$

Hence the algorithm without (P) returns $\text{DFT}_N\{x \circ \rho_m\}$, that is, the spectrum of the signal with permuted samples, *not* the spectrum of x .

Step 2 — the permuted vector. With $\rho_3 = (0, 4, 2, 6, 1, 5, 3, 7)$:

$$v = (x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]) = (0, 4, 2, 6, 1, 5, 3, 7).$$

Step 3 — its DFT. Computing $Y = \text{DFT}_8\{v\}$ from the definition (the sums proceed as in Example 23.6, collecting the coefficients of $\frac{\sqrt{2}}{2}$):

$$\begin{aligned} Y[0] &= 28, & Y[1] &= -1 + i(1 + \sqrt{2}), & Y[2] &= -4 + 4i, \\ Y[3] &= -1 + i(\sqrt{2} - 1), & Y[4] &= -16, & Y[5] &= -1 - i(\sqrt{2} - 1), \\ Y[6] &= -4 - 4i, & Y[7] &= -1 - i(1 + \sqrt{2}). \end{aligned}$$

In full, for $Y[1]$: the coefficients of $\frac{\sqrt{2}}{2}$ in the real part are $4 - 6 - 5 + 7 = 0$ and in the imaginary part $-4 - 6 + 5 + 7 = 2$, while the integer terms give -1 in the real part and $+1$ in the imaginary part; hence $Y[1] = -1 + i(1 + 2 \cdot \frac{\sqrt{2}}{2}) = -1 + i(1 + \sqrt{2})$. In decimals, $1 + \sqrt{2} = 2.4142136$ and $\sqrt{2} - 1 = 0.4142136$.

Step 4 — Parseval does not notice.

$$\sum_k |Y[k]|^2 = 784 + 2[1 + (1 + \sqrt{2})^2] + 2 \cdot 32 + 2[1 + (\sqrt{2} - 1)^2] + 256.$$

Now $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$ and $(\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}$, hence the two contributions equal $2(4 + 2\sqrt{2}) + 2(4 - 2\sqrt{2}) = 16$, and

$$\sum_k |Y[k]|^2 = 784 + 16 + 64 + 256 = 1,120 = \sum_k |X[k]|^2.$$

The value is *identical* to that of Example 23.8: and not by chance, because $\sum_n |v[n]|^2 = \sum_n |x[n]|^2 = 140$ (a permutation does not change the sum of the squared moduli) and Parseval ties the two sums (Theorem 22.6). ✓ (in the sense that the check passes, even though the result is wrong).

Step 5 — the check that exposes the error. Three checks work:

- (i) *the component $k = 0$ is not enough* ($Y[0] = 28 = X[0]$: the sum of the samples does not depend on the order), but *the component $k = N/2$ does*: $Y[4] = -16$ against $X[4] = -4$. In general $X[N/2] = \sum_n (-1)^n x[n]$ depends on the order;
- (ii) *transforming a pure sinusoid*: for $x[n] = \omega_N^{-n}$ the correct DFT is N in bin 1 and 0 elsewhere (Theorem 22.1); without the permutation the spectrum is spread over all bins, and the error is glaring;
- (iii) *round trip*: apply the FFT and then the IFFT (Theorem 23.10) and compare with the input. If the permutation is missing in both, one recovers $x \circ \rho_m \circ \rho_m = x$ and the check fails in turn: one must then compare with the direct DFT, which is the definitive check.

Independent check. The Python script confirms $Y = (28, -1 + 2.4142136i, -4 + 4i, -1 + 0.4142136i, -16, -1 - 0.4142136i, -4 - 4i, -1 - 2.4142136i)$ and $\sum_k |Y[k]|^2 = 1,120.0$. ✓

Interpretation. The error is “silent” with respect to energy checks because the permutation is an isometry: no check involving only norms can detect it. This is the clean example of a general principle of numerical testing: a check that is invariant with respect to the error being sought is not a check.

Typical errors. (1) Concluding “Parseval checks out, so the code is correct”: Parseval is a necessary condition, not a sufficient one. (2) Applying the permutation *twice* (once at the start and once at the end): since ρ_m is an involution, the result is again wrong, but in a different way. (3) Using ρ_m with the wrong m : if an 8-point FFT is run with the 16-bit permutation, one falls outside the range of indices.

Extension. Show that, for a real signal, omitting the permutation also preserves Hermitian symmetry (Theorem 22.7): hence not even that check detects the error. Which properties of the spectrum are *not* invariant under permutation of the samples?

23.16. Self-assessment quiz

1. (Definition) The “fast Fourier transform” is
 - A) a transform different from the DFT, suited to long signals
 - B) a family of algorithms that computes the DFT *exactly*
 - C) an approximation of the DFT accurate to 99%
 - D) the DFT of a signal sampled at twice the frequency
2. (Hypothesis) In the proof of Theorem 23.2, the hypothesis that E and O are $N/2$ -point DFTs is used to
 - A) guarantee $|\omega_N| = 1$
 - B) be able to use $E[k + N/2] = E[k]$ and $O[k + N/2] = O[k]$

-
- C) make the butterflies of a single stage independent D) ensure that N is even
3. (Symbolic computation) The number of *nontrivial* complex multiplications performed by the radix-2 FFT on $N = 2^m$ equals
- A) $\frac{N}{2} \log_2 N$ B) $N \log_2 N$ C) $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$ D) $N^2 - (N - 1)^2$
4. (Numerical computation) For $N = 1,024$ the nontrivial complex multiplications of the radix-2 FFT are
- A) 5,120 B) 10,240 C) 3,586 D) 1,048,576
5. (Proof step) In Theorem 23.2 the minus sign of $X[k + N/2] = E[k] - \omega_N^k O[k]$ comes
- A) from the graphical convention with which the butterfly is drawn B) from the identity $\omega_N^{N/2} = -1$, used together with the periodicity of E and O C) from the fact that the odd indices contribute with opposite sign D) from the conjugation that appears in the inversion formula
6. (Error diagnosis) An implementation of the iterative FFT forgets step (P), the bit-reversal permutation. Which of the following checks *does not* detect the error?
- A) the comparison with the DFT computed from the definition B) Parseval's identity
 C) the transform of $x[n] = \omega_N^{-n}$, which must give N in bin 1 and 0 elsewhere D) the comparison with `numpy.fft.fft`
7. (Transfer) For $N = 6$ the factorization of Theorem 23.11 with $N_1 = 2$, $N_2 = 3$ requires
- A) 3 DFTs of length 2, then 6 twiddle factors, then 2 DFTs of length 3 B) 2 DFTs of length 3 and nothing else C) 6 DFTs of length 1 and a permutation D) padding with zeros up to 8 and a radix-2 FFT
8. (Source and metacognition) The paper that in 1965 made the algorithm known, with its exact citation, is
- A) J. W. Cooley, J. W. Tukey, *An algorithm for the machine calculation of complex Fourier series*, *Mathematics of Computation* **19** (1965), pp. 297–301 B) C. Van Loan, *Computational Frameworks for the Fast Fourier Transform*, SIAM, 1992 C) M. Frigo, S. G. Johnson, *The Design and Implementation of FFTW3*, *Proceedings of the IEEE* **93** (2005), pp. 216–231 D) M. T. Heideman, D. H. Johnson, C. S. Burrus, *Gauss and the history of the fast Fourier transform*, *IEEE ASSP Magazine* **1** (1984), pp. 14–21

23.17. Teacher key

BEST PRACTICE

Quiz — answers, rationales and diagnosis (items 1–4).

1) **B.** *Rationale:* Theorem 23.3 states the equality $\mathcal{A}_N(x) = \text{DFT}_N\{x\}$, with no error term: in exact arithmetic the two procedures give the same numbers. “Family” because radix-2 DIT, DIF, mixed-radix, Bluestein and Rader are different algorithms that all compute the same thing. *Distractors:* A confuses algorithm and transform, the most common conceptual error of the chapter; C attributes to the FFT a truncation error it does not have (the only error is round-off, and it is *smaller* than that of the direct computation, Theorem 23.13); D confuses the block length with the sampling frequency. *Diagnosed error:* not distinguishing the mathematical object from the procedure that computes it. *Source:* Briggs and Henson, *The DFT: An Owner's Manual*, Chapter 10 “The Fast Fourier Transform,” pp. 379–399.

Prerequisite: Definition 22.1. *Remediation:* redo Example 23.5, expanding $X[2]$ down to the samples and checking that the coefficients are those of the definition.

2) **B.** *Rationale:* Step 2 of that proof uses $E[k+L] = E[k]$, which holds because E is an L -point DFT and $\omega_L^L = 1$ (Lemma 23.1(ii)); without this periodicity the index $k+N/2$ would fall outside the domain of E and the formula would not close. *Distractors:* A is true but irrelevant ($|\omega_N| = 1$ always holds, and is used elsewhere, in Lemma 23.6); C is a property of the implementation, not a hypothesis of the theorem; D is a hypothesis of the section, not what the hypothesis in question is used for. *Diagnosed error:* not being able to identify at which step a given hypothesis is needed. *Source:* Oppenheim and Schaffer, *Discrete-Time Signal Processing*, 3rd ed., Chapter 9. *Prerequisite:* Proposition 22.1. *Remediation:* rewrite Step 2 assuming for contradiction that E is not periodic and identify exactly where the computation breaks.

3) **C.** *Rationale:* this is Corollary 23.4: from the $\frac{N}{2} \log_2 N$ total multiplications one subtracts the $N-1$ with factor 1 and the $\frac{N}{2}-1$ with factor $-i$ (Proposition 23.5). *Distractors:* A is the total number of multiplications, trivial ones included; B is the number of additions; D equals $2N-1$, that is, the number of factors equal to 1 in the *matrix* of the DFT, not in the FFT. *Diagnosed error:* confusing the four counts of the chapter. *Source:* Van Loan, *Computational Frameworks for the Fast Fourier Transform*, Chapter 1 “The Radix-2 Frameworks,” pp. 1–75. *Prerequisite:* Theorem 23.5. *Remediation:* derive the formula for $N=16$ by counting the factors of the four stages by hand.

4) **C.** *Rationale:* with $N=1,024$ and $m=10$, $\frac{N}{2}m - \frac{3N}{2} + 2 = 5,120 - 1,536 + 2 = 3,586$. *Distractors:* A is $\frac{N}{2} \log_2 N = 5,120$, that is, the count without removing the trivial factors; B is $N \log_2 N = 10,240$, the number of additions; D is $N^2 = 1,048,576$, the count of the direct computation. *Diagnosed error:* reading the wrong column of Table 23.1. *Source:* Corollary 23.4 and check by script. *Prerequisite:* $\log_2 1,024 = 10$. *Remediation:* redo the computation for $N=256$ (answer: 642) and for $N=4,096$ (answer: 18,434), checking against the table.

BEST PRACTICE

Quiz — answers, rationales and diagnosis (items 5–8).

5) **B.** *Rationale:* Step 3 of the proof computes $\omega_N^{k+N/2} = \omega_N^k \omega_N^{N/2} = -\omega_N^k$ using Lemma 23.1(iv); Step 2 uses periodicity. Both are needed: the identity on the factors alone would not suffice, because the left-hand side contains $E[k+N/2]$ and not $E[k]$. *Distractors:* A mistakes a picture for a proof, and is exactly what Remark 23.4 warns against; C is a content-free sentence (the odd indices contribute with the factor ω_N^k , not with a sign); D confuses this theorem with Theorem 23.10. *Diagnosed error:* replacing a proof with a visual analogy. *Source:* Cooley and Tukey (1965), pp. 297–301; here Theorems 23.1 and 23.2. *Prerequisite:* Lemma 23.1. *Remediation:* redo the computation with the convention $e^{+2\pi i/N}$ and check that the minus sign remains (because $e^{+i\pi} = -1$ just as $e^{-i\pi}$ does), while the twiddle factor changes.

6) **B.** *Rationale:* omitting (P) makes the algorithm compute $\text{DFT}_N\{x \circ \rho_m\}$; a permutation of the samples does not change $\sum_n |x[n]|^2$, hence by Theorem 22.6 it does not change $\sum_k |X[k]|^2$ either: Parseval is satisfied by the wrong spectrum, as shown numerically in Solved Exercise 23.6 ($\sum_k |Y[k]|^2 = 1,120 = \sum_k |X[k]|^2$). *Distractors:* A, C and D compare *value by value* with an independent computation and immediately detect the discrepancy (for instance $Y[4] = -16$ against $X[4] = -4$). *Diagnosed error:* using as a check a quantity invariant with respect to the error being sought. *Source:* Briggs and Henson, Chapter 6 “Errors in the DFT,” pp. 179–234; here the Best-practice box of Section 23.6. *Prerequisite:* Theorem 22.6, Lemma 23.4. *Remediation:* carry out the three checks (i)–(iii) of Solved Exercise 23.6 on $N=4$.

7) **A. Rationale:** with $N_1 = 2$ and $N_2 = 3$, Theorem 23.11 prescribes $N_2 = 3$ DFTs of length $N_1 = 2$, then $N_1 N_2 = 6$ multiplications by $\omega_6^{k_1 n_2}$, then $N_1 = 2$ DFTs of length $N_2 = 3$; this is exactly what is carried out in Example 23.11. *Distractors:* B forgets the first phase and the twiddle factors; C describes the full radix-2, which does not apply here because 6 is not a power of two; D is a *different* and legitimate route, but it changes the points at which the spectrum is sampled (Remark 23.10) and hence does not compute the 6-point DFT. *Diagnosed error:* believing that zero-padding is always an equivalent route to obtain the same transform. *Source:* Van Loan, Chapter 2 “General Radix Frameworks,” pp. 76–120. *Prerequisite:* Theorem 23.11, Euclidean division. *Remediation:* redo Example 23.11 swapping the roles, that is, with $N_1 = 3$ and $N_2 = 2$, and check that the result is the same.

8) **A. Rationale:** this is the original paper, verified on Crossref (DOI 10.1090/S0025-5718-1965-0178586-1). *Distractors:* B is the modern reference monograph, not the 1965 paper; C is the paper on FFTW, from 2005, cited in Remark 23.14; D is the historical paper documenting Gauss’s priority (1805) and the various rediscoverers, cited for context and not for the algorithm. *Diagnosed error:* citing from memory, confusing the primary source with secondary sources. *Source:* the *Sources* environment of this chapter, entries [1], [2], [5] and [7]. *Prerequisite:* none. *Remediation:* check the four references in a bibliographic archive and note for each what it contains and what it does not contain.

BEST PRACTICE

Assigned exercises — answers.

Exercise 23.1. (a) $N = 32$, $m = 5$. Direct computation: $N^2 = 1,024$ multiplications and $N(N-1) = 992$ additions. FFT: $\frac{N}{2}m = 80$ multiplications and $Nm = 160$ additions; nontrivial $80 - \frac{3 \cdot 32}{2} + 2 = 80 - 48 + 2 = 34$. (The script that counts the butterflies confirms 80, 160 and 34.) (b) By Proposition 23.5 the factors equal to 1 are $N-1$ and those equal to $-i$ are $\frac{N}{2} - 1$; points (iii) exclude -1 and $+i$, and the two families are disjoint because $1 \neq -i$. Total $(N-1) + (\frac{N}{2} - 1) = \frac{3N}{2} - 2$. (c) One wants $N^2 \geq 10 \cdot \frac{N}{2} \log_2 N$, that is, $2N \geq 10m$, that is, $2^{m+1} \geq 10m$. For $m = 4$: $32 < 40$, false; for $m = 5$: $64 \geq 50$, true. Hence the minimum is $N = 32$, where the ratio equals 12.80, consistent with Table 23.1 (at $N = 16$ it equals 8.00). (d) $N = 2^{20} = 1,048,576$. Direct computation: $N^2 = 1,099,511,627,776$ multiplications, that is, 10,995.12 s, or 3.0542 hours. FFT: 8,912,898 nontrivial multiplications, that is, 0.0891 s (0.1049 s counting the trivial ones too). The second number is not a prediction of the real time because it ignores memory accesses, which for N of this order dominate (Remark 23.14).

Exercise 23.2. (a) $m = 2$, $\rho_2 = (0, 2, 1, 3)$; reordered vector $(1, 0, 2, -1)$. (b) Stage 1 ($\omega_2^0 = 1$): $1 + 0 = 1$, $1 - 0 = 1$, $2 + (-1) = 1$, $2 - (-1) = 3$, that is, $(1, 1, 1, 3)$. Stage 2 ($\omega_4^0 = 1$, $\omega_4^1 = -i$): $X[0] = 1 + 1 = 2$, $X[1] = 1 + (-i) \cdot 3 = 1 - 3i$, $X[2] = 1 - 1 = 0$, $X[3] = 1 - (-i) \cdot 3 = 1 + 3i$. (c) From the definition: $X[0] = 1 + 2 + 0 - 1 = 2$; $X[1] = 1 + 2(-i) + 0(-1) + (-1)(i) = 1 - 3i$; $X[2] = 1 - 2 + 0 + 1 = 0$; $X[3] = \overline{X[1]} = 1 + 3i$. They coincide. (d) $\sum_n |x[n]|^2 = 1 + 4 + 0 + 1 = 6$ and $\frac{1}{4} \sum_k |X[k]|^2 = \frac{1}{4}(4 + 10 + 0 + 10) = 6$. $\checkmark$ $X[0]$ and $X[2]$ must be real, that is, the bins $k = 0$ and $k = N/2$, by Corollary 22.8; and indeed they equal 2 and 0. (e) $\overline{X} = (2, 1 + 3i, 0, 1 - 3i)$; $\mathcal{A}_4(\overline{X}) = (4, 8, 0, -4)$; conjugating and dividing by 4 one recovers $(1, 2, 0, -1)$. (f) The factors used are $\omega_2^0 = 1$ (twice), $\omega_4^0 = 1$ and $\omega_4^1 = -i$: all trivial. The formula gives $\frac{4}{2} \cdot 2 - \frac{3 \cdot 4}{2} + 2 = 4 - 6 + 2 = 0$. $\checkmark$

Exercise 23.3. (a) The three subsequences $(x[3n_1 + n_2])_{n_1=0,1,2,3}$ collect the indices $\{0, 3, 6, 9\}$, $\{1, 4, 7, 10\}$ and $\{2, 5, 8, 11\}$, hence they equal $(1, 0, 1, 0)$ for $n_2 = 0$ and $(0, 0, 0, 0)$ for $n_2 = 1$ and $n_2 = 2$. Their 4-point DFTs are $G[\cdot, 0] = (2, 0, 2, 0)$ — because $\sum_{n_1} v[n_1] \omega_4^{n_1 k_1} = 1 + \omega_4^{2k_1} = 1 + (-1)^{k_1}$ — and $G[\cdot, 1] = G[\cdot, 2] = (0, 0, 0, 0)$. The factors $\omega_{12}^{k_1 \cdot 0}$ all equal 1, hence $\tilde{G} = G$; and in the four length-3 DFTs only the term $n_2 = 0$

survives, so that $X[k_1 + 4k_2] = G[k_1, 0]$ for every k_2 . Hence

$$Y = (2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0),$$

which coincides with the direct computation: indeed x has only two nonzero components and $Y[k] = x[0] + x[6]\omega_{12}^{6k} = 1 + (\omega_{12}^6)^k = 1 + (-1)^k$, by Lemma 23.1(iv). (b) Mixed-radix $12 = 4 \cdot 3$: the three length-4 DFTs done with radix-2 cost 0 nontrivial multiplications and 8 additions each; the 12 factors $\omega_{12}^{k_1 n_2}$ are trivial in 6 cases ($k_1 = 0$ or $n_2 = 0$) and nontrivial in the other 6; the four length-3 DFTs cost 4 multiplications not by 1 and 6 additions each. Total: $0 + 6 + 16 = 22$ nontrivial multiplications and $24 + 24 = 48$ additions, against the 144 multiplications (104 not by 1) and 132 additions of the direct computation. Zero-padding to $N' = 16$: 10 nontrivial multiplications and 64 additions — fewer multiplications, more additions. But it *does not* compute the same thing: the 16-point DFT samples the DTFT at 16 points equally spaced differently from the 12 of the 12-point DFT, and only $Z[0] = Y[0]$ coincides (Proposition 23.6 holds for doubling, not for $12 \rightarrow 16$). If the true 12 bins are needed, the only route is mixed-radix. (c) Primitive roots modulo 5: $g = 2$ (powers 1, 2, 4, 3) and $g = 3$ (powers 1, 3, 4, 2); $g = 4$ is not one (powers 1, 4, 1, 4). With $g = 2$: $A = (x[1], x[2], x[4], x[3])$, $C = (\omega_5, \omega_5^2, \omega_5^4, \omega_5^3)$, $B = (X[1] - x[0], X[2] - x[0], X[4] - x[0], X[3] - x[0])$, and $B[v] = \sum_{u=0}^3 A[u] C[\langle u + v \rangle_4]$. (d) $M = 16 \geq 2 \cdot 5 - 1 = 9$. With $b_j = e^{-i\pi j^2/5}$: $u = (x[0]b_0, \dots, x[4]b_4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$; v has nonzero components only for $j \in \{0, 1, 2, 3, 4\}$, where it equals $\overline{b_j}$, and for $j \in \{12, 13, 14, 15\}$, where it equals $\overline{b_{j-16}}$. The condition $M \geq 2N - 1$ is needed so that the two blocks do not overlap: if $M = 8$ held, the indices $M - N + 1 = 4$ and $N - 1 = 4$ would coincide and $v[4]$ would have to equal two different things.

23.18. Formulary of the chapter

Twiddle factors. $\omega_N = e^{-2\pi i/N}$ (in the literature W_N).

$$\omega_N^j = 1 \iff N \mid j; \quad \omega_N^{2j} = \omega_{N/2}^j; \quad \omega_N^{N/2} = -1; \quad \omega_N^{N/4} = -i \ (4 \mid N); \quad \omega_{dN}^{dj} = \omega_N^j.$$

Eighth roots: $\omega_8 = \frac{\sqrt{2}}{2}(1 - i)$, $\omega_8^2 = -i$, $\omega_8^3 = -\frac{\sqrt{2}}{2}(1 + i)$, $\omega_8^4 = -1$.

Radix-2 decomposition and butterfly. With $N = 2L$, $e[r] = x[2r]$, $o[r] = x[2r + 1]$, $E = \text{DFT}_L\{e\}$, $O = \text{DFT}_L\{o\}$:

$$X[k] = E[k] + \omega_N^k O[k], \quad X\left[k + \frac{N}{2}\right] = E[k] - \omega_N^k O[k], \quad k = 0, \dots, L - 1.$$

$$\mathcal{B}_w(a, b) = (a + wb, a - wb), \quad B_w = \begin{pmatrix} 1 & w \\ 1 & -w \end{pmatrix}, \quad B_w^* B_w = 2I_2 \ (|w| = 1), \quad \|B_w\|_2 = \sqrt{2}.$$

Decimation in frequency: $\alpha[n] = x[n] + x[n + L]$, $\beta[n] = (x[n] - x[n + L])\omega_N^n$,

$$X[2k] = \text{DFT}_L\{\alpha\}[k], \quad X[2k + 1] = \text{DFT}_L\{\beta\}[k].$$

Cost. Direct computation: N^2 multiplications, $N(N - 1)$ additions. Recurrences: $M(N) = 2M(N/2) + \frac{N}{2}$, $A(N) = 2A(N/2) + N$, $M(1) = A(1) = 0$; $T(N) = 2T(N/2) + cN$, $T(1) =$

$$c_0 \Rightarrow T(N) = cN \log_2 N + c_0 N.$$

$$M(N) = \frac{N}{2} \log_2 N, \quad A(N) = N \log_2 N, \quad M^{**}(N) = \frac{N}{2} \log_2 N - \frac{3N}{2} + 2.$$

Trivial factors: $N - 1$ equal to 1, $\frac{N}{2} - 1$ equal to $-i$, in total $\frac{3N}{2} - 2$. For $N = 8$: 12 multiplications (2 nontrivial), 24 additions. For $N = 1,024$: 5,120 (3,586 nontrivial) and 10,240.

Bit-reversal, factorization, IFFT. With $n = \sum_{t < m} b_t(n) 2^t$ and $N = 2^m$:

$$\rho_m(n) = \sum_{t=0}^{m-1} b_t(n) 2^{m-1-t}, \quad \rho_m \circ \rho_m = \text{id}; \quad \rho_3 = (0, 4, 2, 6, 1, 5, 3, 7).$$

$$W_N = A_m A_{m-1} \cdots A_1 P_\rho, \quad \|A_s\|_2 = \sqrt{2}, \quad \|W_N\|_2 = \sqrt{N}.$$

$$\mathcal{F}_N^{-1} X = \frac{1}{N} \overline{\text{DFT}_N \{ \overline{X} \}} = \frac{1}{N} \text{swap} \left(\text{DFT}_N \{ \text{swap}(X) \} \right).$$

Mixed-radix, Bluestein, Rader. With $N = N_1 N_2$, $n = N_2 n_1 + n_2$, $k = k_1 + N_1 k_2$:

$$X[k_1 + N_1 k_2] = \sum_{n_2=0}^{N_2-1} \omega_N^{k_1 n_2} \left(\sum_{n_1=0}^{N_1-1} x[N_2 n_1 + n_2] \omega_{N_1}^{n_1 k_1} \right) \omega_{N_2}^{n_2 k_2}.$$

Bluestein, with $b_j = e^{-i\pi j^2/N}$ and $2nk = n^2 + k^2 - (k - n)^2$:

$$X[k] = b_k \sum_{n=0}^{N-1} (x[n] b_n) \overline{b_{k-n}} = b_k (u \otimes v)[k], \quad M \geq 2N - 1.$$

Rader (p prime, g primitive root): $A[u] = x[\langle g^u \rangle_p]$, $C[v] = \omega_p^{\langle g^v \rangle_p}$, $X[\langle g^v \rangle_p] - x[0] = \sum_{u=0}^{p-2} A[u] C[\langle u + v \rangle_{p-1}]$.

Round-off error. Under hypothesis (M) with constant $\eta = O(u)$:

$$\frac{\|\hat{X} - X\|_2}{\|X\|_2} \leq (1 + \eta)^m - 1 \leq (e - 1) m \eta < 1.72 \eta \log_2 N \quad (m\eta \leq 1),$$

against a bound proportional to Nu for the direct sum. Double precision: $u = 2^{-53} = 1.1102230 \cdot 10^{-16}$.

The numbers of the example $N = 8$, $x = (0, 1, \dots, 7)$.

$$X = \left(28, -4 + i(4 + 4\sqrt{2}), -4 + 4i, -4 + i(4\sqrt{2} - 4), -4, -4 - i(4\sqrt{2} - 4), -4 - 4i, -4 - i(4 + 4\sqrt{2}) \right),$$

with $4 + 4\sqrt{2} = 9.6568542$ and $4\sqrt{2} - 4 = 1.6568542$; $X[k] = \frac{8}{\omega_8^k - 1}$ for $k \neq 0$; $\sum_n |x[n]|^2 = 140$, $\sum_k |X[k]|^2 = 1,120$. For $N = 6$ and $y = (1, \dots, 6)$: $X = (21, -3 + 3\sqrt{3}i, -3 + \sqrt{3}i, -3, -3 - \sqrt{3}i, -3 - 3\sqrt{3}i)$, with $3\sqrt{3} = 5.1961524$ and $\sqrt{3} = 1.7320508$.

Sources and bibliography

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- [1] J. W. Cooley, J. W. Tukey, *An algorithm for the machine calculation of complex Fourier series*, *Mathematics of Computation* **19** (1965), no. 90, pp. 297–301, American Mathematical Society. DOI 10.1090/S0025-5718-1965-0178586-1. This is the paper that made the algorithm public and the source of Theorems 23.1, 23.2 and 23.3, here reworked and proved in full. *Verified on 6 September 2026* (Crossref record of the DOI: authors, title, volume, issue, year and pages checked one by one).
- [2] C. Van Loan, *Computational Frameworks for the Fast Fourier Transform*, Society for Industrial and Applied Mathematics, Philadelphia, 1992, ISBN 978-0-89871-285-8, DOI 10.1137/1.9781611970999. Chapter 1 “The Radix-2 Frameworks,” pp. 1–75 (source of the cost model of Definition 23.1, of the count of Theorem 23.5 and of the matrix factorization of Corollary 23.9); Chapter 2 “General Radix Frameworks,” pp. 76–120 (source of Theorem 23.11); Chapter 3 “High-Performance Frameworks,” pp. 121–187 (memory and cache, Remark 23.14); Chapter 4 “Selected Topics,” pp. 188–258 (Bluestein and Rader, Proposition 23.7 and Theorem 23.12). *Verified on 6 September 2026* (Crossref record of the volume’s DOI and of the DOIs of the individual chapters, with their respective page ranges).
- [3] W. L. Briggs, V. E. Henson, *The DFT: An Owner’s Manual for the Discrete Fourier Transform*, Society for Industrial and Applied Mathematics, Philadelphia, 1995, ISBN 978-0-89871-342-8, DOI 10.1137/1.9781611971514. Chapter 2 “The Discrete Fourier Transform,” pp. 15–64; Chapter 3 “Properties of the DFT,” pp. 65–116; Chapter 6 “Errors in the DFT,” pp. 179–234 (source of the picture of Remark 23.13); Chapter 10 “The Fast Fourier Transform,” pp. 379–399 (reference exposition of the distinction between transform and algorithm). *Verified on 6 September 2026* (Crossref record of the volume’s DOI and of the DOIs of the individual chapters, with their respective page ranges).
- [4] A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd edition, Pearson, 2010, ISBN 978-0-13-198842-2, 1108 pages. Chapter 9: engineering exposition of decimation in time and in frequency, of the flow graph and of the bit-reversal reordering, and of the sign convention adopted here too (Remark 22.1 of Chapter 22). *Verified on 6 September 2026* (Open Library catalog record of the ISBN: authors, publisher, year and number of pages).
- [5] M. T. Heideman, D. H. Johnson, C. S. Burrus, *Gauss and the history of the fast Fourier transform*, *IEEE ASSP Magazine* **1** (1984), no. 4, pp. 14–21. DOI 10.1109/MASSP.1984.1162257. Reconstructs Gauss’s priority (manuscript of 1805, published posthumously) and the long chain of independent rediscoveries between the nineteenth and twentieth centuries; it is the source of the historical picture recalled in Chapter 22, Section 22.10. *Verified on 6 September 2026* (Crossref record of the DOI: authors, title, journal, volume, issue, year and pages).
- [6] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd edition, Society for Industrial and Applied Mathematics, Philadelphia, 2002, ISBN 978-0-89871-521-7, DOI 10.1137/1.9780898718027. Chapter 2 “Floating Point Arithmetic,” pp. 35–60 (standard model and unit round-off of Definition 23.8); Chapter 24 “The Fast Fourier Transform and Applications,” pp. 451–457 (error analysis of the FFT: this is the source of the *imported* fact declared in Remark 23.12). *Verified on 6 September 2026* (Crossref record of the volume’s DOI and of the DOIs of the individual chapters, with their respective page ranges).
- [7] M. Frigo, S. G. Johnson, *The Design and Implementation of FFTW3*, *Proceedings of the*
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IEEE **93** (2005), no. 2, pp. 216–231. DOI 10.1109/JPROC.2004.840301. Describes the automatic runtime selection of the algorithm; cited as panorama in Remark 23.14, and not as the source of any statement. *Verified on 6 September 2026* (Crossref record of the DOI: authors, title, journal, volume, issue, year and pages).

- [8] G. H. Hardy, E. M. Wright, *An Introduction to the Theory of Numbers*, Oxford University Press, Oxford, 2008 edition, ISBN 978-0-19-921986-5, 621 pages. Chapter on congruences and primitive roots: this is the source of the *imported* fact (existence of a primitive root modulo a prime) declared in Remark 23.11. *Verified on 6 September 2026* (Open Library catalog record of the ISBN: authors, publisher, year and number of pages).

Part D

Signals, control, telecommunications

Signals and linear systems: sampling and frequency response

A linear time-invariant system has no memory of its own history: it only knows what it does to an impulse. The whole chapter descends from this sentence, which however needs to be turned into a statement: *which* hypotheses turn “linear and time-invariant” into “is a convolution”, and with which function. The answer — Theorem 24.1 — is that the boundedness of the operator from $L^1(\mathbb{R})$ into the continuous bounded functions suffices, and that the function sought is the representative supplied by the Riesz representation theorem; the *identity* is excluded by the hypothesis, and not by accident, because its impulse response is not a function.

From there the chain unfolds in a forced order. Since convolution becomes a product after the Fourier transform (Theorem 19.4 of Chapter 19), the complex exponentials are *eigenfunctions* of every LTI system (Theorem 24.2 in the continuous case, Theorem 24.3 in the discrete one) and the eigenvalue associated with $e^{i\omega t}$ is, literally, the transform of the impulse response evaluated at ω : the **frequency response**. This is the reason why harmonic analysis and system theory are the same thing written twice, and not two neighbouring disciplines.

On the impulse response one then reads off the two properties that decide whether a system is usable: **BIBO stability**, equivalent to the absolute summability of h (Theorem 24.4, proved in both directions, the second by constructing the input that saturates the inequality), and **causality**, equivalent to the vanishing of h on the negative half-line (Theorem 24.5). On the frequency response one instead reads off the two quantities that decide whether a system *distorts* a signal: the **phase delay** and the **group delay** (Definition 24.9), and Theorem 24.7 proves that a system with constant gain and exactly linear phase over the whole band of the signal returns it identical to itself, shifted in time and rescaled: no waveform distortion.

The second half of the chapter is devoted to the passage from the continuous to the discrete. Section 24.8 — inherited from Volume II and preserved here word for word — proves that sampling in time periodicizes the spectrum, and that if the replicas do not overlap the whole information is there; Section 24.9 completes the proof with the *reconstruction formula* $x(t) = \sum_n x(nT) \operatorname{sinc}((t - nT)/T)$, obtained without distributions, using only the Fourier series expansion of the spectrum on the bandwidth interval; Section 24.10 measures what happens when the hypothesis fails, with a numerical example verified sample by sample. The last section defines the **Hilbert transform** and the **analytic signal**, proves that the latter suppresses the negative frequencies (Theorem 24.11) and states precisely what remains imported.

The results of Chapter 19 — algebraic properties (Theorem 19.3), the convolution theorem (Theorem 19.4), inversion (Theorem 19.9), Plancherel (Corollary 19.4), the sinc function (Proposition 19.2) — and those of Chapter 21 on the discrete case — discrete LTI systems (Definition 21.2), convolution (Proposition 21.1), BIBO stability (Theorem 21.10), the DTFT (Definition 21.5) — are recalled and used here, never reproved. The discrete case of the convolution representation and of BIBO stability is already closed in Chapter 21: this chapter proves the *continuous* case, which does not appear there, and the discrete case is recalled where needed for comparison.

INFO

Level and prerequisites. Level **C** (undergraduate), with three passages at level **D**: the proof of Theorem 24.1 (which uses the duality $(L^1)^* = L^\infty$), the proof of the reconstruction formula (Theorem 24.9) and Section 24.11 on the analytic signal.

Internal prerequisites. The Fourier transform on $L^1(\mathbb{R})$, its algebraic properties, the convolution theorem, the inversion formula, Plancherel's theorem, the rectangle and the sinc function, the Dirac comb (Chapter 19: Definition 19.1, Proposition 19.1, Theorem 19.1, Theorem 19.3, Lemma 19.3, Theorem 19.4, Theorem 19.9, Corollary 19.4, Definition 19.4, Proposition 19.2); convergence in L^2 of Fourier series and completeness of the trigonometric system (Theorem 17.1 of Chapter 17); discrete LTI systems, discrete convolution, discrete BIBO stability and the DTFT (Chapter 21: Definition 21.2, Proposition 21.1, Definition 21.6, Theorem 21.10, Definition 21.5, Theorem 21.6).

External prerequisites, assumed and not reproved:

- the Lebesgue integral on $\mathbb{R}$: measurability, the dominated convergence theorem, the monotone convergence theorem, the Fubini–Tonelli theorem;
- the spaces $L^p(\mathbb{R})$ for $p = 1, 2, \infty$ and their norms; the Cauchy–Schwarz inequality;
- the **Riesz representation theorem** for the dual of L^1 on a σ -finite measure space, i.e. $(L^1(\mathbb{R}))^* \cong L^\infty(\mathbb{R})$ with equality of the norms: it is used only once, in Step 2 of the proof of Theorem 24.1, and is declared *imported* in Remark 24.3;
- differentiation under the integral sign (for computing the group delay of explicit examples, where the hypotheses are checked case by case).

Everything else is proved here.

What one learns. To recognise when a model is LTI and when it is not, with explicit counterexamples; to derive the convolution representation from verifiable hypotheses, instead of postulating it; to use the eigenfunction property to compute the steady-state output without ever carrying out a convolution integral; to decide stability and causality by looking only at the impulse response; to distinguish the phase delay from the group delay and to know which of the two tells whether the waveform survives; to choose a sampling frequency with a criterion, and to know how to compute the spurious frequency that appears when the criterion is violated; to define the envelope and the instantaneous frequency of a narrowband signal without confusing them with what they are not.

Warning. Conventions and notation of the chapter.

- We keep the convention of Chapter 19: for $f \in L^1(\mathbb{R})$

$$\hat{f}(\omega) = \mathcal{F}\{f\}(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) e^{i\omega t} d\omega,$$

where ω is the *angular frequency* in radians per second. The *frequency* in hertz is $f = \omega/2\pi$; when it needs to be distinguished from the name of a function it is called ν .

- σ_a is the translation operator, $(\sigma_a x)(t) = x(t - a)$: $a > 0$ is a *delay*. One has $\sigma_a \sigma_b = \sigma_{a+b}$ and $\sigma_0 = \text{id}$.
- u is the Heaviside unit step: $u(t) = 1$ for $t \geq 0$ and $u(t) = 0$ for $t < 0$.
- The sinc function is the *normalised* one of Definition 19.4:

$$\text{sinc}(v) = \frac{\sin(\pi v)}{\pi v} \quad (v \neq 0), \quad \text{sinc}(0) = 1,$$

which vanishes at every nonzero integer. Much of the engineering literature writes sinc for $\sin v/v$: the translation is $\text{sinc}_{\text{here}}(v) = \sin(\pi v)/(\pi v)$.

- sgn is the sign function: $\text{sgn}(\omega) = 1$ for $\omega > 0$, -1 for $\omega < 0$, 0 for $\omega = 0$.

- The letters x and y always denote input and output, h the impulse response, $H = \hat{h}$ the frequency response. In the discrete case one writes $x[n]$ with square brackets and Ω for the normalised angular frequency, as in Chapter 21.
- “Almost everywhere” (a.e.) means: outside a set of Lebesgue measure zero. Two functions equal a.e. define the same element of L^p and the same system.

24.1. Signals and systems: linearity and time-invariance

Definition 24.1: Spaces of signals. A **continuous-time signal** is a Lebesgue-measurable function $x : \mathbb{R} \rightarrow \mathbb{C}$. We will use the spaces

$$L^1(\mathbb{R}) = \left\{ x : \|x\|_1 = \int_{\mathbb{R}} |x(t)| dt < \infty \right\}, \quad L^2(\mathbb{R}) = \left\{ x : \|x\|_2^2 = \int_{\mathbb{R}} |x(t)|^2 dt < \infty \right\},$$

$$L^\infty(\mathbb{R}) = \left\{ x : \|x\|_\infty = \operatorname{ess\,sup}_{t \in \mathbb{R}} |x(t)| < \infty \right\},$$

and moreover $C_b(\mathbb{R})$, the space of *continuous and bounded* functions with the norm $\|x\|_\infty = \sup_t |x(t)|$, and $C_c(\mathbb{R})$, that of continuous functions with compact support. A signal of $L^\infty(\mathbb{R})$ is called **bounded**. The **support** of x is the closure of $\{t : x(t) \neq 0\}$; the signal is **causal** if its support is contained in $[0, +\infty)$.

Remark 24.1: Why three spaces and not one. The three spaces answer three different physical questions and are not comparable with one another: L^1 collects signals of finite area (a current pulse transfers a finite charge), L^2 those of finite energy, L^∞ those of bounded amplitude. No inclusion holds between L^1 and L^2 on $\mathbb{R}$: the function $t \mapsto t^{-1/2} \mathbf{1}_{(0,1)}(t)$ lies in L^1 and not in L^2 , because $\int_0^1 t^{-1/2} dt = 2$ while $\int_0^1 t^{-1} dt = +\infty$; the function $t \mapsto (1+|t|)^{-1}$ lies in L^2 and not in L^1 , because $\int_{\mathbb{R}} (1+|t|)^{-2} dt = 2$ while $\int_{\mathbb{R}} (1+|t|)^{-1} dt = +\infty$. A sinusoidal signal, the protagonist of the whole chapter, lies neither in L^1 nor in L^2 : it lies in L^∞ , and this is exactly the reason why Theorem 24.2 must be stated there.

Definition 24.2: System, translation, LTI system. Let X and Y be two signal spaces closed under linear combinations and under translations, where the **translation** by step $a \in \mathbb{R}$ is

$$(\sigma_a x)(t) = x(t - a).$$

A **system** is a map $\mathcal{T} : X \rightarrow Y$. It is

- (L) **linear** if $\mathcal{T}(\alpha x + \beta y) = \alpha \mathcal{T}x + \beta \mathcal{T}y$ for all $x, y \in X$ and $\alpha, \beta \in \mathbb{C}$;
 - (TI) **time-invariant** if $\mathcal{T}(\sigma_a x) = \sigma_a(\mathcal{T}x)$ for all $x \in X$ and all $a \in \mathbb{R}$.
- If both hold, $\mathcal{T}$ is a **linear time-invariant system**, in short an **LTI system**.

Example 24.1: Four LTI systems and three that are not. Are LTI.

- (a) The *pure delay* $\mathcal{T}_{t_0}x = \sigma_{t_0}x$, i.e. $y(t) = x(t - t_0)$. Linearity: immediate from the definition of σ_{t_0} . Invariance: $\sigma_{t_0}\sigma_a = \sigma_{t_0+a} = \sigma_a\sigma_{t_0}$.
- (b) The *amplifier* $y = Kx$ with $K \in \mathbb{C}$ constant.
- (c) The *moving average* over a window of duration $T_0 > 0$: $y(t) = \frac{1}{T_0} \int_{t-T_0}^t x(s) ds$.
- (d) The low-pass *RC filter*, which associates with every bounded input the bounded solution of $\tau y' + y = x$ with $\tau = RC$.

Are not LTI.

- (e) The *squarer* $y(t) = x(t)^2$: it is not linear. With $x \equiv 1$ and $\alpha = 2$ one has $\mathcal{T}(2x) \equiv 4$ while $2\mathcal{T}x \equiv 2$, and $4 \neq 2$.
- (f) The *modulator* $y(t) = x(t) \cos(\omega_0 t)$ with $\omega_0 \neq 0$: it is linear but not time-invariant. Take $x \equiv 1$ and $a = \pi/\omega_0$: then $\sigma_a x = x$, hence $\mathcal{T}(\sigma_a x)(t) = \cos(\omega_0 t)$, while $\sigma_a(\mathcal{T}x)(t) = \cos(\omega_0 t - \pi) = -\cos(\omega_0 t)$. The two functions differ, for instance at $t = 0$, where they equal 1 and -1 .
- (g) The *sampler* $x \mapsto (x(nT))_{n \in \mathbb{Z}}$ of Section 24.8: it is linear, but it is invariant only under translations that are multiples of T , not under all of them. It is precisely this *partial* invariance that produces the periodicization of the spectrum in Theorem 24.8.

Remark 24.2: The model and reality. No physical device is exactly LTI: an amplifier saturates, a capacitor ages, a motor has static friction. The LTI model is the linearisation of the behaviour around an operating point, valid as long as the amplitudes remain small and the time scale of the experiment is short compared with the drift of the parameters. The gain from this approximation is enormous and must be said once and for all: inside the LTI model a system is described by a single function — the impulse response — and its action on any input is computed by a multiplication in the frequency domain. Outside the model neither of these two sentences remains true. The circle criterion of Chapter 25, Section 25.6, is one of the few tools that survive a declared nonlinearity.

24.2. The convolution representation

The statement “an LTI system is convolution with its own impulse response” is true, but it is not true without hypotheses, and the hypotheses are not a technical detail: they are exactly what distinguishes systems whose impulse response is a *function* from those where it is not. This section states and proves the two implications: first that every convolution is an LTI system (Proposition 24.2), then that every LTI system *bounded in the appropriate sense* is a convolution (Theorem 24.1).

Definition 24.3: Convolution on $\mathbb{R}$. Given two measurable functions $h, x : \mathbb{R} \rightarrow \mathbb{C}$, their **convolution** is

$$(h * x)(t) = \int_{\mathbb{R}} h(\tau) x(t - \tau) d\tau,$$

defined at every t for which the integral converges absolutely. It is the same operation as Definition 19.3 of Chapter 19.

Proposition 24.1: When convolution is defined everywhere.

(i) If $h \in L^1(\mathbb{R})$ and $x \in L^\infty(\mathbb{R})$, then $(h * x)(t)$ is defined for every $t \in \mathbb{R}$, one has

$$|(h * x)(t)| \leq \|h\|_1 \|x\|_\infty \quad \text{for every } t,$$

and $h * x$ is uniformly continuous on $\mathbb{R}$; in particular $h * x \in C_b(\mathbb{R})$.

(ii) If $h \in L^\infty(\mathbb{R})$ and $x \in L^1(\mathbb{R})$, the same conclusions hold with $\|h\|_\infty \|x\|_1$ in place of $\|h\|_1 \|x\|_\infty$.

(iii) In both cases $(h * x)(t) = (x * h)(t)$ for every t .

Proof. (i), convergence and estimate. For every fixed t the function $\tau \mapsto h(\tau)x(t - \tau)$ is measurable (a product of measurable functions; $\tau \mapsto x(t - \tau)$ is measurable because it is the composition of x with the affine map $\tau \mapsto t - \tau$, which sends sets of measure zero to sets of

measure zero). Moreover $|h(\tau)x(t-\tau)| \leq |h(\tau)| \|x\|_\infty$ for almost every τ , and the right-hand side is integrable with integral $\|h\|_1 \|x\|_\infty$. Hence the integral converges absolutely and

$$|(h * x)(t)| \leq \int_{\mathbb{R}} |h(\tau)| |x(t-\tau)| d\tau \leq \|h\|_1 \|x\|_\infty.$$

(i), uniform continuity. With the change of variable $u = t - \tau$ (a translation followed by a reflection: it preserves Lebesgue measure) one writes $(h * x)(t) = \int_{\mathbb{R}} h(t-u) x(u) du$. Hence, for every t and every s ,

$$|(h * x)(t+s) - (h * x)(t)| = \left| \int_{\mathbb{R}} [h(t+s-u) - h(t-u)] x(u) du \right| \leq \|x\|_\infty \int_{\mathbb{R}} |h(t+s-u) - h(t-u)| du.$$

In the last integral set $v = t - u$: it becomes $\int_{\mathbb{R}} |h(v+s) - h(v)| dv = \|\sigma_{-s}h - h\|_1$, which does not depend on t . By Lemma 19.3 (continuity of translations in L^1) one has $\|\sigma_{-s}h - h\|_1 \rightarrow 0$ as $s \rightarrow 0$. Given $\varepsilon > 0$ there is therefore $\delta > 0$ such that $|s| < \delta$ implies $\|\sigma_{-s}h - h\|_1 < \varepsilon/(1 + \|x\|_\infty)$, and then $|(h * x)(t+s) - (h * x)(t)| < \varepsilon$ for every t : this is uniform continuity. Together with the estimate of the previous point, $h * x \in C_b(\mathbb{R})$.

(ii). This is point (i) with the roles exchanged, once one observes that, by point (iii), $h * x = x * h$: apply (i) to the pair (x, h) with $x \in L^1$ and $h \in L^\infty$.

(iii). In the first case the substitution $u = t - \tau$, i.e. $\tau = t - u$, is a bijection of $\mathbb{R}$ onto itself that preserves Lebesgue measure and has $|d\tau| = |du|$; since the integral converges absolutely, the change of variable is legitimate and

$$(h * x)(t) = \int_{\mathbb{R}} h(\tau)x(t-\tau) d\tau = \int_{\mathbb{R}} h(t-u)x(u) du = (x * h)(t).$$

The same computation holds in the second case. □

Proposition 24.2: Every convolution is an LTI system. Let $h \in L^1(\mathbb{R})$ and define $\mathcal{T}_h x = h * x$ on $X = L^\infty(\mathbb{R})$. Then $\mathcal{T}_h : L^\infty(\mathbb{R}) \rightarrow C_b(\mathbb{R})$ is linear, time-invariant and bounded with

$$\|\mathcal{T}_h x\|_\infty \leq \|h\|_1 \|x\|_\infty.$$

The same conclusion holds, with $\|h\|_\infty \|x\|_1$, for $h \in L^\infty(\mathbb{R})$ and $\mathcal{T}_h : L^1(\mathbb{R}) \rightarrow C_b(\mathbb{R})$.

Proof. Well-posedness, the estimate and membership in $C_b(\mathbb{R})$ are Proposition 24.1(i) (respectively (ii)).

Linearity. For every t and every $\alpha, \beta \in \mathbb{C}$, the function $\tau \mapsto h(\tau)[\alpha x_1(t-\tau) + \beta x_2(t-\tau)]$ is a sum of two absolutely integrable functions, so the integral splits:

$$(h * (\alpha x_1 + \beta x_2))(t) = \alpha \int_{\mathbb{R}} h(\tau)x_1(t-\tau) d\tau + \beta \int_{\mathbb{R}} h(\tau)x_2(t-\tau) d\tau = \alpha (h * x_1)(t) + \beta (h * x_2)(t).$$

Time-invariance. For every $a \in \mathbb{R}$ and every t ,

$$(h * (\sigma_a x))(t) = \int_{\mathbb{R}} h(\tau) (\sigma_a x)(t-\tau) d\tau = \int_{\mathbb{R}} h(\tau) x(t-a-\tau) d\tau = (h * x)(t-a) = (\sigma_a (h * x))(t),$$

where the second step is the definition $(\sigma_a x)(s) = x(s-a)$ applied with $s = t - \tau$, and the third is the definition of convolution itself evaluated at $t - a$. □

The converse is the central result of the section. The hypothesis that makes it legitimate is the *boundedness* of the operator from L^1 into C_b : it is this that guarantees that the impulse response is a function, and not a more general object.

Theorem 24.1: Convolution representation. Let $\mathcal{T} : L^1(\mathbb{R}) \rightarrow C_b(\mathbb{R})$ be a map such that

(S1) $\mathcal{T}$ is linear;

(S2) $\mathcal{T}$ is time-invariant: $\mathcal{T}(\sigma_a x) = \sigma_a(\mathcal{T}x)$ for every $x \in L^1(\mathbb{R})$ and every $a \in \mathbb{R}$;

(S3) $\mathcal{T}$ is bounded: there exists $C \geq 0$ with $\|\mathcal{T}x\|_\infty \leq C \|x\|_1$ for every $x \in L^1(\mathbb{R})$.

Then there exists $h \in L^\infty(\mathbb{R})$, unique up to sets of measure zero, such that

$$(\mathcal{T}x)(t) = (h * x)(t) = \int_{\mathbb{R}} h(t - u) x(u) du \quad \text{for every } x \in L^1(\mathbb{R}) \text{ and every } t \in \mathbb{R},$$

and $\|h\|_\infty \leq C$ holds. The function h is called the **impulse response** of the system.

Proof. Step 1 — the “output at time zero” functional. Define

$$\Lambda : L^1(\mathbb{R}) \rightarrow \mathbb{C}, \quad \Lambda(x) = (\mathcal{T}x)(0).$$

The definition makes sense because by hypothesis $\mathcal{T}x$ is a *continuous* function, so its value at the point 0 is a well-determined number and does not depend on the representative chosen. The functional Λ is linear by (S1) — evaluating at 0 is a linear operation — and is bounded by (S3):

$$|\Lambda(x)| = |(\mathcal{T}x)(0)| \leq \|\mathcal{T}x\|_\infty \leq C \|x\|_1.$$

Hence $\Lambda \in (L^1(\mathbb{R}))^*$ with $\|\Lambda\| \leq C$.

Step 2 — representation of the functional. By the Riesz representation theorem for the dual of L^1 with respect to Lebesgue measure, which is σ -finite (an *imported* result: see Remark 24.3), there exists $g \in L^\infty(\mathbb{R})$, unique a.e., with

$$\Lambda(x) = \int_{\mathbb{R}} g(s) x(s) ds \quad \text{for every } x \in L^1(\mathbb{R}), \quad \|g\|_\infty = \|\Lambda\| \leq C.$$

Step 3 — invariance shifts the evaluation from 0 to t . Let $x \in L^1(\mathbb{R})$ and $t \in \mathbb{R}$. Since $\sigma_{-t}x \in L^1(\mathbb{R})$ with the same norm, one may apply (S2) with $a = -t$:

$$\mathcal{T}(\sigma_{-t}x) = \sigma_{-t}(\mathcal{T}x),$$

and evaluating both sides at 0, with $(\sigma_{-t}y)(0) = y(0 + t) = y(t)$,

$$\Lambda(\sigma_{-t}x) = (\mathcal{T}(\sigma_{-t}x))(0) = (\sigma_{-t}(\mathcal{T}x))(0) = (\mathcal{T}x)(t).$$

Step 4 — the computation. Applying Step 2 to the signal $\sigma_{-t}x$, which equals $(\sigma_{-t}x)(s) = x(s + t)$,

$$(\mathcal{T}x)(t) = \Lambda(\sigma_{-t}x) = \int_{\mathbb{R}} g(s) x(s + t) ds = \int_{\mathbb{R}} g(u - t) x(u) du,$$

where in the last step one has set $u = s + t$ (a translation: it preserves the measure, and the integral converges absolutely because $|g(s)x(s + t)| \leq \|g\|_\infty |x(s + t)|$). Define now

$$h(v) := g(-v) \quad (v \in \mathbb{R}),$$

which is measurable and bounded with $\|h\|_\infty = \|g\|_\infty \leq C$. Then $g(u - t) = h(t - u)$, and hence

$$(\mathcal{T}x)(t) = \int_{\mathbb{R}} h(t - u) x(u) du = (h * x)(t)$$

for every t , having used the form of $h * x$ derived in the proof of Proposition 24.1(iii).

Step 5 — uniqueness. Let $h_1, h_2 \in L^\infty(\mathbb{R})$ be such that $h_1 * x = h_2 * x$ for every $x \in L^1(\mathbb{R})$, and let $d = h_1 - h_2 \in L^\infty(\mathbb{R})$. Evaluating at $t = 0$ gives $\int_{\mathbb{R}} d(-u)x(u) du = 0$ for every

$x \in L^1(\mathbb{R})$. Suppose for contradiction that d is not zero a.e.: then there is $\delta > 0$ such that the set $E = \{u \in \mathbb{R} : |d(-u)| > \delta\}$ has positive measure, and since $E = \bigcup_{N \in \mathbb{N}} (E \cap [-N, N])$ there is N with $0 < \text{meas}(E \cap [-N, N]) < \infty$. Set $E_N = E \cap [-N, N]$ and

$$x_0(u) = \left(\frac{d(-u)}{|d(-u)|} \right) \mathbf{1}_{E_N}(u),$$

which is measurable (a quotient of measurable functions on a set where the denominator does not vanish), of modulus ≤ 1 and with support of finite measure: hence $x_0 \in L^1(\mathbb{R})$. But then

$$0 = \int_{\mathbb{R}} d(-u) x_0(u) du = \int_{E_N} \frac{|d(-u)|^2}{|d(-u)|} du = \int_{E_N} |d(-u)| du \geq \delta \cdot \text{meas}(E_N) > 0,$$

which is absurd. Hence $d = 0$ a.e., that is, $h_1 = h_2$ a.e. □

Remark 24.3: What we import, and why the identity is excluded. Of the previous proof only one step is not carried out here: Step 2, that is, the Riesz representation theorem $(L^1(\mu))^* \cong L^\infty(\mu)$ for μ σ -finite. It is a standard result of measure theory, listed among the external prerequisites; its proof (for Lebesgue measure it goes through the Radon–Nikodým theorem) does not belong to this treatise and is **imported**. Everything else — Steps 1, 3, 4 and 5 — is proved in full.

It must be said with equal precision what the theorem does not give. Hypothesis (S3) excludes perfectly reasonable systems:

- the identity $\mathcal{T}x = x$ does not map L^1 into C_b (an element of L^1 is in general neither continuous nor bounded), and no $h \in L^\infty$ represents it. Indeed, if $x = h * x$ held for every $x \in L^1$ with some $h \in L^\infty$, take $x_\varepsilon = \frac{1}{\varepsilon} \mathbf{1}_{[0, \varepsilon]}$, which has $\|x_\varepsilon\|_1 = 1$ and $\|x_\varepsilon\|_\infty = 1/\varepsilon$: Proposition 24.1(ii) would give $\|x_\varepsilon\|_\infty = \|h * x_\varepsilon\|_\infty \leq \|h\|_\infty \|x_\varepsilon\|_1 = \|h\|_\infty$, i.e. $1/\varepsilon \leq \|h\|_\infty$ for every $\varepsilon > 0$, which is absurd;
- the pure delay $\mathcal{T}_{t_0}x = x_{t_0}$, for the same reason;
- the differentiator $\mathcal{T}x = x'$, which is not even defined on all of L^1 .

For these systems the impulse response exists, but as a **distribution**: it is δ for the identity, δ_{t_0} for the delay, δ' for the differentiator. The theory that treats them is that of the tempered distributions of Chapter 19, Section 19.12; the general theorem stating that every continuous linear time-invariant operator between spaces of distributions is convolution with a distribution (the Schwartz kernel theorem, and its form for translation-invariant operators) is not proved in this treatise and is cited here only as **panorama**: see Oppenheim and Willsky, entry [3] of the Sources environment for the mathematical framework. The present chapter uses only the case where h is a function, which is the case of all the filters of Section 24.6, and treats the pure delay separately and directly (Proposition 24.3).

Remark 24.4: The discrete case is already closed. In the discrete-time case the same representation is proved in Chapter 21, Proposition 21.1, and there the proof is purely algebraic for inputs with finite support: it suffices to write $x = \sum_k x[k] \sigma_k \delta$, a finite linear combination of translates of the unit impulse, and apply (L) and (TI). This route does not exist in the continuous case, because there is no function that acts as a unit impulse: δ is not a function (Proposition 19.8). This is the reason why the continuous case requires the functional analysis of Theorem 24.1 and the discrete case does not — a difference that in introductory texts often remains implicit.

Definition 24.4: Convolution system and its extension to bounded signals. Let $h \in L^1(\mathbb{R})$. The **convolution system** with impulse response h is

$$\mathcal{T}_h : L^\infty(\mathbb{R}) \rightarrow C_b(\mathbb{R}), \quad \mathcal{T}_h x = h * x.$$

By Proposition 24.2 it is linear, time-invariant and bounded with constant $\|h\|_1$. From here on, “LTI system with impulse response $h \in L^1$ ” always means $\mathcal{T}_h$ defined this way.

***Remark 24.5: Why the extension to L^∞ is needed.** Theorem 24.1 produces h from the action of the system on signals of L^1 , but the signals of interest in this chapter — a sinusoid, a step, a bounded noise — do not lie in L^1 . Definition 24.4 extends the system to L^∞ by imposing the same formula, and Proposition 24.1(i) guarantees that the formula continues to make sense. The extension is a modelling choice, not a theorem, and it is consistent with the starting point: if $x \in L^1 \cap L^\infty$, the two definitions give the same integral, because it is literally the same integral. It is the continuous analogue of Remark 21.2 of Chapter 21, where the same choice is declared for the discrete case.*

Example 24.2: The impulse response of the four LTI systems. (a) Amplifier with delay, $y(t) = K x(t - t_0)$: it does not fall under Definition 24.4 (its impulse response is $K\delta_{t_0}$, a distribution), and must be treated directly: see Proposition 24.3.

(b) Moving average over $[t - T_0, t]$: one writes $y(t) = \frac{1}{T_0} \int_{t-T_0}^t x(s) ds = \int_{\mathbb{R}} h(\tau) x(t - \tau) d\tau$ with

$$h(\tau) = \frac{1}{T_0} \mathbf{1}_{[0, T_0]}(\tau),$$

as one sees by setting $\tau = t - s$: as s ranges over $[t - T_0, t]$, τ ranges over $[0, T_0]$. Here $h \in L^1(\mathbb{R})$ with $\|h\|_1 = \frac{1}{T_0} \cdot T_0 = 1$.

(c) Low-pass RC filter, i.e. the bounded solution of $\tau y' + y = x$ with $\tau > 0$: the impulse response is

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t),$$

and $\|h\|_1 = \frac{1}{\tau} \int_0^{+\infty} e^{-t/\tau} dt = \frac{1}{\tau} \cdot \tau = 1$. That $y = h * x$ solves the equation is verified directly: for $x \in C_b(\mathbb{R})$,

$$y(t) = \frac{1}{\tau} \int_0^{+\infty} e^{-\theta/\tau} x(t - \theta) d\theta = \frac{1}{\tau} \int_{-\infty}^t e^{-(t-s)/\tau} x(s) ds = \frac{e^{-t/\tau}}{\tau} \int_{-\infty}^t e^{s/\tau} x(s) ds,$$

and differentiating the product (the integral is differentiable in t because the integrand is continuous in s)

$$y'(t) = -\frac{1}{\tau} \cdot \frac{e^{-t/\tau}}{\tau} \int_{-\infty}^t e^{s/\tau} x(s) ds + \frac{e^{-t/\tau}}{\tau} e^{t/\tau} x(t) = -\frac{y(t)}{\tau} + \frac{x(t)}{\tau},$$

that is $\tau y' + y = x$. ✓

(d) Ideal integrator $y(t) = \int_{-\infty}^t x(s) ds$: the impulse response is the step $h = u$, which is in $L^\infty(\mathbb{R})$ but *not* in $L^1(\mathbb{R})$, because $\int_0^N 1 dt = N \rightarrow +\infty$. It will reappear in Example 24.3 as the prototype of a non-stable system.

24.3. Complex exponentials are eigenfunctions

Definition 24.5: Eigenfunction of a system. Let $\mathcal{T} : X \rightarrow Y$ be a system with $X \subseteq Y$. A signal $\varphi \in X$ not identically zero is an **eigenfunction** of $\mathcal{T}$ if there exists $\lambda \in \mathbb{C}$ with

$$\mathcal{T}\varphi = \lambda\varphi;$$

the number λ is the associated **eigenvalue**. An eigenfunction passes through the system without changing shape: it changes only by a complex factor, i.e. by an amplitude and a phase.

Theorem 24.2: Complex exponentials are eigenfunctions: continuous case. Let $h \in L^1(\mathbb{R})$ and let $\mathcal{T}_h$ be the convolution system of Definition 24.4. For every $\omega \in \mathbb{R}$ set $e_\omega(t) = e^{i\omega t}$. Then $e_\omega \in L^\infty(\mathbb{R})$ is an eigenfunction of $\mathcal{T}_h$ and

$$\mathcal{T}_h e_\omega = H(\omega) e_\omega, \quad \text{where} \quad H(\omega) = \hat{h}(\omega) = \int_{\mathbb{R}} h(\tau) e^{-i\omega\tau} d\tau.$$

The eigenvalue associated with e_ω is therefore the Fourier transform of the impulse response evaluated at ω .

Proof. Step 1 — the input is admissible. $|e_\omega(t)| = |e^{i\omega t}| = 1$ for every t , hence $e_\omega \in L^\infty(\mathbb{R})$ with $\|e_\omega\|_\infty = 1$; by Proposition 24.1(i) the integral defining $\mathcal{T}_h e_\omega$ converges absolutely at every t .

Step 2 — the computation. For every $t \in \mathbb{R}$,

$$(\mathcal{T}_h e_\omega)(t) = \int_{\mathbb{R}} h(\tau) e^{i\omega(t-\tau)} d\tau = \int_{\mathbb{R}} h(\tau) e^{i\omega t} e^{-i\omega\tau} d\tau = e^{i\omega t} \int_{\mathbb{R}} h(\tau) e^{-i\omega\tau} d\tau,$$

where the second step is the property $e^{a+b} = e^a e^b$ of the complex exponential and the third is pulling the factor $e^{i\omega t}$, which does not depend on the integration variable τ , out of the integral sign (the operation is legitimate because the integral converges absolutely and $|e^{i\omega t}| = 1$).

Step 3 — identification of the eigenvalue. The last integral is, by Definition 19.1 of Chapter 19, exactly $\hat{h}(\omega)$, and it is a finite complex number because $h \in L^1(\mathbb{R})$ (Proposition 19.1). Hence $(\mathcal{T}_h e_\omega)(t) = \hat{h}(\omega) e^{i\omega t}$ for every t , i.e. $\mathcal{T}_h e_\omega = H(\omega) e_\omega$ with $H = \hat{h}$. Since e_ω is not identically zero, it is an eigenfunction in the sense of Definition 24.5. $\square$

Theorem 24.3: Complex exponentials are eigenfunctions: discrete case. Let $h \in \ell^1(\mathbb{Z})$ and let $\mathcal{T}$ be the discrete LTI system $\mathcal{T}x = h * x$ of Proposition 21.1(ii). For every $\Omega \in \mathbb{R}$ set $e_\Omega[n] = e^{i\Omega n}$. Then $e_\Omega \in \ell^\infty(\mathbb{Z})$ is an eigenfunction of $\mathcal{T}$ and

$$\mathcal{T} e_\Omega = \hat{h}(\Omega) e_\Omega, \quad \text{where} \quad \hat{h}(\Omega) = \sum_{k \in \mathbb{Z}} h[k] e^{-i\Omega k}$$

is the DTFT of h (Definition 21.5 of Chapter 21).

Proof. Step 1 — admissibility. $|e_\Omega[n]| = 1$ for every n , hence $e_\Omega \in \ell^\infty(\mathbb{Z})$ with $\|e_\Omega\|_\infty = 1$; by Proposition 21.1(ii), since $h \in \ell^1(\mathbb{Z})$, the convolution series converges absolutely for every n .

Step 2 — the computation. For every $n \in \mathbb{Z}$,

$$(\mathcal{T} e_\Omega)[n] = \sum_{k \in \mathbb{Z}} h[k] e^{i\Omega(n-k)} = \sum_{k \in \mathbb{Z}} h[k] e^{i\Omega n} e^{-i\Omega k} = e^{i\Omega n} \sum_{k \in \mathbb{Z}} h[k] e^{-i\Omega k},$$

where pulling the constant factor $e^{i\Omega n}$ out of the series is legitimate because the series converges

absolutely (an absolutely convergent series can be multiplied term by term by a constant). The last sum converges absolutely because $|h[k]e^{-i\Omega k}| = |h[k]|$ and $h \in \ell^1(\mathbb{Z})$, and it is by definition $\hat{h}(\Omega)$. By Theorem 21.6 it coincides with the Zeta transform of h evaluated at $z = e^{i\Omega}$. $\square$

Corollary 24.1: What an LTI system does to a sinusoid. Let $h \in L^1(\mathbb{R})$ take real values, $H = \hat{h}$, and let $x(t) = A \cos(\omega_0 t + \varphi_0)$ with $A > 0$, $\omega_0 \in \mathbb{R}$, $\varphi_0 \in \mathbb{R}$. Then

$$(\mathcal{T}_h x)(t) = A |H(\omega_0)| \cos(\omega_0 t + \varphi_0 + \arg H(\omega_0)) \quad \text{for every } t \in \mathbb{R},$$

where $\arg H(\omega_0)$ is any argument of $H(\omega_0)$ (if $H(\omega_0) = 0$ the output is identically zero). In particular: an LTI system does not change the frequency of a sinusoid; it changes only its amplitude, by the factor $|H(\omega_0)|$, and its phase, by the additive term $\arg H(\omega_0)$.

Proof. Step 1 — Hermitian symmetry of H . Since h is real, $\bar{h} = h$ and for every ω

$$\overline{H(\omega)} = \overline{\int_{\mathbb{R}} h(\tau) e^{-i\omega\tau} d\tau} = \int_{\mathbb{R}} h(\tau) \overline{e^{-i\omega\tau}} d\tau = \int_{\mathbb{R}} h(\tau) e^{i\omega\tau} d\tau = H(-\omega),$$

having brought the conjugation inside the integral (legitimate: conjugation is additive, continuous and the integral converges absolutely) and having used $\overline{h(\tau)} = h(\tau)$.

Step 2 — decomposition of the input. By Euler's formula,

$$x(t) = A \cos(\omega_0 t + \varphi_0) = \frac{A}{2} e^{i\varphi_0} e^{i\omega_0 t} + \frac{A}{2} e^{-i\varphi_0} e^{-i\omega_0 t} = \frac{A}{2} e^{i\varphi_0} e_{\omega_0}(t) + \frac{A}{2} e^{-i\varphi_0} e_{-\omega_0}(t),$$

a finite linear combination of two eigenfunctions.

Step 3 — applying the theorem. By linearity of $\mathcal{T}_h$ (Proposition 24.2) and by Theorem 24.2,

$$(\mathcal{T}_h x)(t) = \frac{A}{2} e^{i\varphi_0} H(\omega_0) e^{i\omega_0 t} + \frac{A}{2} e^{-i\varphi_0} H(-\omega_0) e^{-i\omega_0 t}.$$

By Step 1, $H(-\omega_0) = \overline{H(\omega_0)}$: the two terms are conjugates of each other, and their sum equals $2 \operatorname{Re}$ of the first:

$$(\mathcal{T}_h x)(t) = 2 \operatorname{Re} \left(\frac{A}{2} e^{i\varphi_0} H(\omega_0) e^{i\omega_0 t} \right) = A \operatorname{Re} \left(|H(\omega_0)| e^{i \arg H(\omega_0)} e^{i(\omega_0 t + \varphi_0)} \right),$$

having written $H(\omega_0) = |H(\omega_0)| e^{i \arg H(\omega_0)}$ (polar form, legitimate for $H(\omega_0) \neq 0$). Since $\operatorname{Re}(e^{i\theta}) = \cos \theta$, one concludes

$$(\mathcal{T}_h x)(t) = A |H(\omega_0)| \cos(\omega_0 t + \varphi_0 + \arg H(\omega_0)).$$

If instead $H(\omega_0) = 0$, then also $H(-\omega_0) = \bar{0} = 0$ and the formula of Step 3 gives identically zero output, consistently with the statement. $\square$

Remark 24.6: Why this is the reason for Fourier analysis. Theorem 24.2 is the reason why harmonic analysis is not one tool among many for studying linear systems: it is the diagonalisation. An LTI system, viewed as a linear operator, has as eigenvectors the exponentials e_ω and as eigenvalues the numbers $H(\omega)$; writing a signal as a superposition of exponentials — that is, transforming it according to Fourier — means expressing it in the basis that makes the operator diagonal, and in that basis the action of the system is a simple multiplication, component by component. The convolution theorem (Theorem 19.4) and Theorem 24.2 are two formulations of the same fact: the first says $\widehat{h * x} = \hat{h} \hat{x}$, the second reads it on a single exponential.

The same pattern repeats at every level of the treatise with the same structure and different names: in the discrete setting it is the exponential vectors that diagonalise circulant matrices (Theorem 22.8 of Chapter 22); in the Zeta domain it is the restriction to the unit circle that gives the frequency response (Corollary 21.4); in automatic control it is the substitution $s = i\omega$ in the transfer function (Chapter 25, Section 25.1).

Remark 24.7: Are these the only eigenfunctions common to all LTI systems?

Yes, and the reason is elementary to state but not to prove here. If a signal $x \in C_b(\mathbb{R})$ not identically zero is an eigenfunction of all pure delays σ_a , there exists a function $\lambda : \mathbb{R} \rightarrow \mathbb{C}$ with $\sigma_a x = \lambda(a)x$ for every a ; evaluating at a point t_1 with $x(t_1) \neq 0$ gives $\lambda(a) = x(t_1 - a)/x(t_1)$, hence λ is continuous, and from $\sigma_a \sigma_b = \sigma_{a+b}$ it follows that $\lambda(a+b) = \lambda(a)\lambda(b)$ with $\lambda(0) = 1$. The continuous solutions of this multiplicative Cauchy functional equation are all and only the $\lambda(a) = e^{ca}$ with $c \in \mathbb{C}$, and the requirement $x \in L^\infty$ forces c to be purely imaginary, $c = -i\omega$: one recovers $x(t) = x(0)e^{i\omega t}$.

The step that this treatise does not prove is the classification of the continuous solutions of the Cauchy functional equation: it is a classical result of real analysis, **imported** here and used only in this remark, which indeed is not a theorem and is not used elsewhere in the chapter. What is missing, to be precise, is exactly the deduction “ λ continuous and multiplicative $\Rightarrow \lambda(a) = e^{ca}$ ”; all the rest of the reasoning is carried out in full above.

24.4. BIBO stability

Definition 24.6: BIBO stability. A system $\mathcal{T}$ defined on bounded signals is **BIBO stable** (bounded input, bounded output) if there is a constant $M \geq 0$ such that

$$\|x\|_\infty \leq 1 \implies \|\mathcal{T}x\|_\infty \leq M,$$

equivalently, by homogeneity, $\|\mathcal{T}x\|_\infty \leq M\|x\|_\infty$ for every bounded input x . The smallest constant M with this property is called the **gain** of the system.

Theorem 24.4: BIBO $\iff$ absolutely integrable impulse response. Let $h : \mathbb{R} \rightarrow \mathbb{C}$ be measurable and locally integrable, and consider the system $\mathcal{T}x = h * x$ on the class

$$X_c = \{x \in L^\infty(\mathbb{R}) : x \text{ has compact support}\},$$

for which the convolution integral is certainly finite at every t (the support is bounded and h is locally integrable). Call $\mathcal{T}$ BIBO stable if there is $M \geq 0$ with $\|\mathcal{T}x\|_\infty \leq M\|x\|_\infty$ for every $x \in X_c$. Then

$$\mathcal{T} \text{ is BIBO stable } \iff h \in L^1(\mathbb{R}),$$

and in that case the gain equals exactly $M = \|h\|_1$. If $h \in L^1(\mathbb{R})$, the bound $\|h * x\|_\infty \leq \|h\|_1 \|x\|_\infty$ moreover holds for every $x \in L^\infty(\mathbb{R})$, not only for those with compact support: stability in the sense of Definition 24.6 and stability on X_c coincide.

Proof. Step 1 — sufficiency ($\Leftarrow$). Let $h \in L^1(\mathbb{R})$. By Proposition 24.1(i), for every bounded x the integral converges absolutely and

$$|(h * x)(t)| \leq \int_{\mathbb{R}} |h(\tau)| |x(t - \tau)| d\tau \leq \|x\|_\infty \int_{\mathbb{R}} |h(\tau)| d\tau = \|h\|_1 \|x\|_\infty$$

for every t . Taking the supremum over t gives $\|h * x\|_\infty \leq \|h\|_1 \|x\|_\infty$: the system is BIBO

stable with admissible constant $M = \|h\|_1$.

Step 2 — necessity ($\Rightarrow$): construction of the input that saturates the inequality. Suppose $\mathcal{T}$ is BIBO stable with constant M . Fix $N > 0$ and define

$$x_N(t) = \begin{cases} \frac{\overline{h(-t)}}{|h(-t)|}, & |t| \leq N \text{ and } h(-t) \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

This function is measurable: the set $A_N = \{t : |t| \leq N, h(-t) \neq 0\}$ is measurable because h is and $t \mapsto h(-t)$ is measurable; on A_N the quotient $\overline{h(-t)}/|h(-t)|$ is measurable because it is a quotient of measurable functions with denominator never zero. Moreover $|x_N(t)| = 1$ on A_N and 0 outside, hence $\|x_N\|_\infty \leq 1$ and x_N has support contained in $[-N, N]$, compact. By the stability hypothesis, $\|\mathcal{T}x_N\|_\infty \leq M$.

Step 3 — evaluating the output at time 0. One computes $(\mathcal{T}x_N)(0) = (h * x_N)(0) = \int_{\mathbb{R}} h(\tau) x_N(-\tau) d\tau$. Now $x_N(-\tau)$ equals $\overline{h(\tau)}/|h(\tau)|$ when $|\tau| \leq N$ and $h(\tau) \neq 0$, and 0 otherwise; hence

$$(\mathcal{T}x_N)(0) = \int_{\{|\tau| \leq N, h(\tau) \neq 0\}} h(\tau) \frac{\overline{h(\tau)}}{|h(\tau)|} d\tau = \int_{\{|\tau| \leq N, h(\tau) \neq 0\}} \frac{|h(\tau)|^2}{|h(\tau)|} d\tau = \int_{-N}^N |h(\tau)| d\tau,$$

where $z\bar{z} = |z|^2$ was used and, in the last step, the points with $h(\tau) = 0$ were put back in, contributing a zero integrand. The integral is finite because h is locally integrable.

Step 4 — conclusion. The number just computed is real, nonnegative, and does not exceed M by hypothesis:

$$\int_{-N}^N |h(\tau)| d\tau = (\mathcal{T}x_N)(0) \leq \|\mathcal{T}x_N\|_\infty \leq M \quad \text{for every } N > 0.$$

The function $N \mapsto \int_{-N}^N |h|$ is nondecreasing and bounded by M ; by the monotone convergence theorem applied to $|h|\mathbf{1}_{[-N, N]} \uparrow |h|$,

$$\|h\|_1 = \int_{\mathbb{R}} |h(\tau)| d\tau = \lim_{N \rightarrow +\infty} \int_{-N}^N |h(\tau)| d\tau \leq M < \infty,$$

that is, $h \in L^1(\mathbb{R})$ and $\|h\|_1 \leq M$.

Step 5 — optimality of the constant. Step 1 shows that $\|h\|_1$ is an admissible constant; Step 4 shows that every admissible constant M satisfies $\|h\|_1 \leq M$. The minimum of the admissible constants is therefore exactly $\|h\|_1$: it is the gain of the system. $\square$

Remark 24.8: The signal of Step 2, read in engineering terms. When h is real, the quotient $\overline{h(-t)}/|h(-t)|$ reduces to $\text{sgn}(h(-t))$: the critical input is a unit-amplitude square wave that changes sign exactly where the time-reversed impulse response changes sign. Its effect is that, in the convolution integral at time 0, all the elementary contributions $h(\tau)x_N(-\tau)$ turn out to be aligned: there is no longer any cancellation, and the output reaches the maximum possible value $\int |h|$. This is the most economical way of saying that $\|h\|_1$ is not a pessimistic bound, but a value actually attained (in the limit): there is a bounded disturbance that drives the output that far. A designer who checks stability only with sinusoidal inputs will never see this, because a sinusoid produces $|H(\omega)| \leq \|h\|_1$, with strict inequality in general.

Example 24.3: Three systems, three outcomes. (a) RC filter, $h(t) = \frac{1}{\tau}e^{-t/\tau}u(t)$ **with** $\tau > 0$. In closed form

$$\|h\|_1 = \frac{1}{\tau} \int_0^{+\infty} e^{-t/\tau} dt = \frac{1}{\tau} \left[-\tau e^{-t/\tau} \right]_0^{+\infty} = \frac{1}{\tau} \cdot \tau = 1.$$

The system is BIBO stable with gain exactly 1: no input of amplitude ≤ 1 can produce an output of amplitude > 1 , and a unit step reaches it asymptotically. *Independent check:* Simpson's quadrature of $|h|$ on $[0, 40\tau]$ with $4 \cdot 10^5$ subintervals, for $\tau = 1$ ms, gives 0.999 999 999 999 56, which differs from the closed value 1 by $4.40 \cdot 10^{-13}$ (the theoretical residual of the tail beyond 40τ is $e^{-40} = 4.25 \cdot 10^{-18}$, well below the quadrature error). ✓

(b) Ideal integrator, $h = u$. The partial sums of Step 4 equal $\int_0^N 1 d\tau = N$, which grows without bound: by Theorem 24.4 the system is *not* BIBO stable. The direct check is instructive: the constant input $x \equiv 1$, of amplitude 1, produces $y(t) = \int_{-\infty}^t 1 ds = +\infty$, and already the input $x = \mathbf{1}_{[0,+\infty)}$ produces $y(t) = t$ for $t \geq 0$, unbounded.

(c) Moving average, $h = \frac{1}{T_0} \mathbf{1}_{[0, T_0]}$. Here $\|h\|_1 = \frac{1}{T_0} \cdot T_0 = 1$: BIBO stable with gain 1. The critical input of Step 2 is, in this case, the constant input $x \equiv 1$, because $h \geq 0$ and hence $\text{sgn}(h(-t)) = 1$ over the whole support; and indeed $y \equiv 1$.

Remark 24.9: The discrete case, and the bridge with poles. The corresponding discrete statement — BIBO stable $\iff h \in \ell^1(\mathbb{Z})$, with gain $\|h\|_1$ — is Theorem 21.10 of Chapter 21, proved there with the same five-step strategy and with the critical input $x_N[n] = \overline{h[-n]}/|h[-n]|$: the two proofs are deliberately parallel, and comparing the two Step 3's shows that the only technical difference is the substitution of the sum for the integral and of the comparison criterion for the monotone convergence theorem.

From there Chapter 21 goes on to translate the condition into geometry: for a causal rational system stability is equivalent to all poles lying inside the unit circle (Theorem 21.12), and the unit circle must lie in the annulus of convergence (Corollary 21.4). The continuous analogue — poles in the left half-plane — is the content of Chapter 20 and of the Nyquist criterion of Chapter 25, Section 25.3.

24.5. Causality

Definition 24.7: Causal system. A system $\mathcal{T}$ is **causal** if, for every $t_0 \in \mathbb{R}$ and every pair of admissible inputs x_1, x_2 ,

$$x_1(t) = x_2(t) \text{ for every } t \leq t_0 \implies (\mathcal{T}x_1)(t) = (\mathcal{T}x_2)(t) \text{ for every } t \leq t_0.$$

In words: the output at time t_0 does not depend on the values of the input after t_0 . A non-causal system is called **anticipative**.

Remark 24.10: The equivalent form for linear systems. If $\mathcal{T}$ is linear, Definition 24.7 is equivalent to the apparently weaker condition

$$x(t) = 0 \text{ for every } t \leq t_0 \implies (\mathcal{T}x)(t) = 0 \text{ for every } t \leq t_0.$$

That the first implies the second is immediate ($x_2 = 0$, $x_1 = x$, and $\mathcal{T}0 = 0$ by linearity). Conversely, if the second holds and $x_1 = x_2$ on $(-\infty, t_0]$, apply the second to $x = x_1 - x_2$: one gets $\mathcal{T}x_1 - \mathcal{T}x_2 = \mathcal{T}(x_1 - x_2) = 0$ on $(-\infty, t_0]$, which is the first. In what follows either form will be used indifferently.

Theorem 24.5: Causality $\iff$ causal impulse response. Let $h \in L^1(\mathbb{R})$ and let $\mathcal{T}_h$ be the convolution system of Definition 24.4. Then

$$\mathcal{T}_h \text{ is causal} \iff h(t) = 0 \text{ for almost every } t < 0.$$

In that case, for every bounded input,

$$y(t) = \int_0^{+\infty} h(\tau) x(t - \tau) d\tau = \int_{-\infty}^t h(t - u) x(u) du :$$

the output at time t is a weighted average of the past of the input only.

Proof. Step 1 — sufficiency ($\Leftarrow$). Let $h = 0$ a.e. on $(-\infty, 0)$. By the form of the convolution derived in Proposition 24.1(iii),

$$y(t) = \int_{\mathbb{R}} h(t - u) x(u) du.$$

For $u > t$ one has $t - u < 0$, hence $h(t - u) = 0$ for almost every such u , and the integrand vanishes a.e. on $(t, +\infty)$: the integral reduces to $\int_{-\infty}^t h(t - u)x(u) du$, which depends only on the values $x(u)$ with $u \leq t$. Hence, if $x = 0$ on $(-\infty, t_0]$, for every $t \leq t_0$ the integrand vanishes a.e. and $y(t) = 0$: the system is causal by Remark 24.10. The second form of the statement is obtained from the first by the substitution $\tau = t - u$.

Step 2 — necessity ($\Rightarrow$): reduction. Let $\mathcal{T}_h$ be causal. Fix $t < 0$ and show that $h(s) = 0$ for almost every $s < t$; since $(-\infty, 0) = \bigcup_{n \geq 1} (-\infty, -1/n)$ is a *countable* union and a countable union of sets of measure zero has measure zero, this will suffice to conclude that $h = 0$ a.e. on $(-\infty, 0)$.

Step 3 — the family of test inputs. Fix $t < 0$ and $N > 0$, and let

$$x_N(u) = \begin{cases} \frac{\overline{h(t-u)}}{|h(t-u)|}, & 0 \leq u \leq N \text{ and } h(t-u) \neq 0, \\ 0, & \text{otherwise,} \end{cases}$$

measurable by the same argument as in Step 2 of the proof of Theorem 24.4, with $|x_N| \leq 1$ and support contained in $[0, N]$. In particular $x_N(u) = 0$ for every $u \leq 0$, hence for every $u \leq t_0 := 0$; by causality and by Remark 24.10, $y_N = \mathcal{T}_h x_N$ vanishes on $(-\infty, 0]$, and in particular $y_N(t) = 0$, since $t < 0$.

Step 4 — computing $y_N(t)$. By the form of the convolution,

$$0 = y_N(t) = \int_{\mathbb{R}} h(t-u) x_N(u) du = \int_{\{0 \leq u \leq N, h(t-u) \neq 0\}} h(t-u) \frac{\overline{h(t-u)}}{|h(t-u)|} du = \int_0^N |h(t-u)| du,$$

where, as in Step 3 of the proof of Theorem 24.4, one has used $z\bar{z} = |z|^2$ and put back the points with $h(t-u) = 0$, which contribute a zero integrand.

Step 5 — conclusion. A vanishing integral of a nonnegative function forces the function to be zero a.e.: hence $|h(t-u)| = 0$ for almost every $u \in [0, N]$, i.e. $h(s) = 0$ for almost every $s \in [t-N, t]$ (substitution $s = t-u$, which preserves the measure). Letting N grow along the integers and again using that a countable union of sets of measure zero has measure zero, one obtains $h(s) = 0$ for almost every $s < t$. By Step 2 the claim follows. $\square$

Corollary 24.2: Causality of the three examples. With the impulse responses of Example 24.2: the RC filter $h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$ and the moving average $h = \frac{1}{T_0} \mathbf{1}_{[0, T_0]}$ are causal,

because both functions vanish on $(-\infty, 0)$ by construction. The ideal low-pass filter, with frequency response $\mathbf{1}_{[-\Omega, \Omega]}$, is not: its impulse response is $h(t) = \frac{\Omega}{\pi} \text{sinc}(\Omega t/\pi)$, which by Proposition 19.2 is the inverse transform of the rectangle, and is different from zero over all of $\mathbb{R}$ — in particular on $(-\infty, 0)$, where it is not zero even a.e., because it vanishes only at the isolated points $t = k\pi/\Omega$, $k \in \mathbb{Z}$, $k \neq 0$.

Proof. For the first two the conclusion is the direct reading of Theorem 24.5: $\frac{1}{\tau}e^{-t/\tau}u(t) = 0$ for $t < 0$ because $u(t) = 0$ there, and $\mathbf{1}_{[0, T_0]}(t) = 0$ for $t < 0$ because $t \notin [0, T_0]$.

For the third: Proposition 19.2 of Chapter 19 computes the transform of the rectangle; applying the inversion formula (Theorem 19.9) to the function $\mathbf{1}_{[-\Omega, \Omega]}$, which lies in $L^1(\mathbb{R})$, one obtains

$$h(t) = \frac{1}{2\pi} \int_{-\Omega}^{\Omega} e^{i\omega t} d\omega = \frac{1}{2\pi} \cdot \frac{e^{i\Omega t} - e^{-i\Omega t}}{it} = \frac{\sin(\Omega t)}{\pi t} = \frac{\Omega}{\pi} \text{sinc}\left(\frac{\Omega t}{\pi}\right),$$

for $t \neq 0$, with $h(0) = \Omega/\pi$ by continuity. The zeros of $\sin(\Omega t)$ are the points $t = k\pi/\Omega$ with $k \in \mathbb{Z}$: a countable set, hence of measure zero. Hence h is not zero a.e. on $(-\infty, 0)$, and by Theorem 24.5 the system is not causal. This is the reason why the ideal low-pass filter that appears in the proof of Theorem 24.8 is not realisable in real time, as already observed in Remark 24.17. $\square$

Remark 24.11: What causality imposes on the frequency response. A causal and stable system cannot have an arbitrary frequency response: magnitude and phase are linked. The precise formulation of this link is twofold.

- Holomorphic extension. If $h \in L^1(\mathbb{R})$ is causal, the integral $\int_0^{+\infty} h(t)e^{-st} dt$ converges absolutely for every s with $\text{Re } s \geq 0$, because $|e^{-st}| = e^{-t\text{Re } s} \leq 1$ for $t \geq 0$: the frequency response therefore extends to a function defined on the whole closed right half-plane, and it is the Laplace transform of Chapter 20, restricted to the imaginary axis. This passage is proved here in full.
- Kramers–Kronig relations and the Paley–Wiener theorem. That the real and imaginary parts of H are linked by a Hilbert transform (Section 24.11), and that a given function $|H|$ is the magnitude of the frequency response of a causal system if and only if $\int_{\mathbb{R}} \frac{|\log |H(\omega)||}{1+\omega^2} d\omega < \infty$ (Paley–Wiener condition), are results that this treatise **does not prove**: the theory of Hardy spaces on the half-plane, which is not developed here, is missing. They are cited here as **panorama**, and their source is entry [4] of the Sources environment (Oppenheim and Schaffer). An immediate consequence, which is elementary, is that no causal and stable system can have $|H|$ identically zero on a non-degenerate interval of frequencies, because there $\log |H| = -\infty$ and the integral diverges: the ideal low-pass filter of Corollary 24.2 is excluded twice over.

24.6. Frequency response: magnitude and phase

Definition 24.8: Frequency response, gain, phase. Let $h \in L^1(\mathbb{R})$ be the impulse response of an LTI system. Its **frequency response** is

$$H(\omega) = \hat{h}(\omega) = \int_{\mathbb{R}} h(t) e^{-i\omega t} dt, \quad \omega \in \mathbb{R}.$$

The number $|H(\omega)| \geq 0$ is the **gain** at angular frequency ω , and it is often expressed in **decibels**:

$$|H(\omega)|_{\text{dB}} = 20 \log_{10} |H(\omega)|.$$

On any interval where H does not vanish, a continuous function $\phi : I \rightarrow \mathbb{R}$ with $H(\omega) = |H(\omega)|e^{i\phi(\omega)}$ for $\omega \in I$ is called a **phase** (or *continuous phase*, in English *unwrapped phase*) of the system on I ; it is determined up to a constant integer multiple of 2π .

Theorem 24.6: Elementary properties of the frequency response. Let $h \in L^1(\mathbb{R})$ and $H = \hat{h}$. Then

- (i) H is uniformly continuous on $\mathbb{R}$ and $|H(\omega)| \leq \|h\|_1$ for every ω ;
- (ii) $H(0) = \int_{\mathbb{R}} h(t) dt$ is the DC gain, and $|H(\omega)| \rightarrow 0$ as $|\omega| \rightarrow \infty$;
- (iii) if h is real-valued, $H(-\omega) = \overline{H(\omega)}$ for every ω ; consequently $|H|$ is even and any phase can be chosen odd;
- (iv) the gain of the system (Definition 24.6) is $\|h\|_1$, and $\sup_{\omega} |H(\omega)| \leq \|h\|_1$, with strict inequality in general.

Proof. (i). This is Theorem 19.1 (uniform continuity of the transform) together with Proposition 19.1 of Chapter 19, according to which $|\hat{h}(\omega)| \leq \int_{\mathbb{R}} |h(t)| |e^{-i\omega t}| dt = \|h\|_1$.

(ii). The first statement is the definition with $\omega = 0$, since $e^0 = 1$. The second is the Riemann–Lebesgue lemma, Theorem 19.5.

(iii). This is Step 1 of the proof of Corollary 24.1, where it is carried out in full. From $H(-\omega) = \overline{H(\omega)}$ it follows that $|H(-\omega)| = |\overline{H(\omega)}| = |H(\omega)|$, i.e. the evenness of the magnitude; and if $H(\omega) \neq 0$, the arguments of $H(-\omega)$ and of $H(\omega)$ are opposite modulo 2π , so that on a symmetric interval free of zeros one can choose the continuous odd determination.

(iv). The value $\|h\|_1$ is Theorem 24.4, Step 5. The inequality $\sup_{\omega} |H| \leq \|h\|_1$ is point (i). That it can be strict is seen on the example $h = \mathbf{1}_{[0,1]} - \mathbf{1}_{[1,2]}$: here $\|h\|_1 = 2$, while $H(0) = \int_0^1 1 dt - \int_1^2 1 dt = 0$ and in general

$$H(\omega) = \int_0^1 e^{-i\omega t} dt - \int_1^2 e^{-i\omega t} dt = \frac{1 - e^{-i\omega}}{i\omega} - \frac{e^{-i\omega} - e^{-2i\omega}}{i\omega} = \frac{(1 - e^{-i\omega})^2}{i\omega},$$

whose modulus is $|1 - e^{-i\omega}|^2/|\omega| = 4 \sin^2(\omega/2)/|\omega|$, having used $|1 - e^{-i\omega}|^2 = (1 - \cos \omega)^2 + \sin^2 \omega = 2 - 2 \cos \omega = 4 \sin^2(\omega/2)$. That the supremum of this function is strictly less than $2 = \|h\|_1$ is seen in two steps. For $|\omega| \geq 5/2$ one has $4 \sin^2(\omega/2)/|\omega| \leq 4/|\omega| \leq 8/5 = 1.6 < 2$, because $\sin^2 \leq 1$. On $[-5/2, 5/2]$ the function $|H|$ is continuous (at $\omega = 0$ it equals 0, as read from $H(0) = 0$), so by the Weierstrass theorem it attains a maximum there; and this maximum is < 2 , because at every point $\omega \neq 0$ of the interval the strict inequality $4 \sin^2(\omega/2)/|\omega| < 4(\omega/2)^2/|\omega| = |\omega| \leq 5/2$ holds, which together with $\sin v < v$ for $v > 0$ and the bound $4/|\omega|$ for $|\omega| \geq 2$ gives in any case a value < 2 . The actual maximum, computed numerically, equals 1.449 222 707 6 and is attained at $\omega^* = 2.331 122 3$, a root of the stationarity equation $\tan(\omega/2) = \omega$: two independent computations (exhaustive search with step 10^{-6} and subsequent ternary search) give the same value, and $\tan(\omega^*/2) = 2.331 122 3$ confirms the stationarity condition. ✓ Hence $\sup_{\omega} |H(\omega)| = 1.449 \dots < 2 = \|h\|_1$. $\square$

Example 24.4: The RC filter: gain and phase, with the numbers. Consider the RC filter of Example 24.2(c), with $h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$ and $\tau = RC = 1$ ms.

Closed form. For every $\omega \in \mathbb{R}$,

$$H(\omega) = \frac{1}{\tau} \int_0^{+\infty} e^{-t/\tau} e^{-i\omega t} dt = \frac{1}{\tau} \int_0^{+\infty} e^{-(1/\tau + i\omega)t} dt = \frac{1}{\tau} \cdot \frac{1}{1/\tau + i\omega} = \frac{1}{1 + i\omega\tau},$$

where the integral converges because $\operatorname{Re}(1/\tau + i\omega) = 1/\tau > 0$ and equals $[-e^{-(1/\tau + i\omega)t}]/(1/\tau + i\omega)$.

$i\omega)]_0^{+\infty} = 1/(1/\tau + i\omega)$. Hence

$$|H(\omega)| = \frac{1}{\sqrt{1 + \omega^2\tau^2}}, \quad \phi(\omega) = -\arctan(\omega\tau).$$

At the cutoff angular frequency $\omega = 1/\tau = 1000 \text{ rad/s}$ (i.e. $\nu = 1/(2\pi\tau) = 159.154943 \text{ Hz}$):

$$H = \frac{1}{1+i} = \frac{1-i}{2} = 0.5 - 0.5i, \quad |H| = \frac{1}{\sqrt{2}} = 0.7071067812,$$

$$|H|_{\text{dB}} = 20 \log_{10} \frac{1}{\sqrt{2}} = -10 \log_{10} 2 = -3.010299957 \text{ dB}, \quad \phi = -\arctan 1 = -\frac{\pi}{4} = -45^\circ.$$

Independent check (Python script executed). Computing $H(\omega)$ by Simpson's quadrature of the integral $\int_0^{60\tau} h(t)e^{-i\omega t} dt$ with $6 \cdot 10^5$ subintervals gives, for $\omega\tau = 1$, $0.499999999999899 - 0.500000000000189i$, which differs from the closed value $0.5 - 0.5i$ by $2.14 \cdot 10^{-13}$; for $\omega\tau = 3$ one obtains $0.100000000000014 - 0.299999999999999i$ against the closed value $0.1 - 0.3i$, with gap $1.36 \cdot 10^{-14}$. ✓

Reading. At the cutoff angular frequency the signal comes out with 70.7% of the amplitude (half the power, whence the name “−3 dB frequency”) and delayed by one eighth of a period, because 45° is one eighth of a turn. By Corollary 24.1, an input $\cos(1000t)$ produces the output $0.7071067812 \cos(1000t - \pi/4)$.

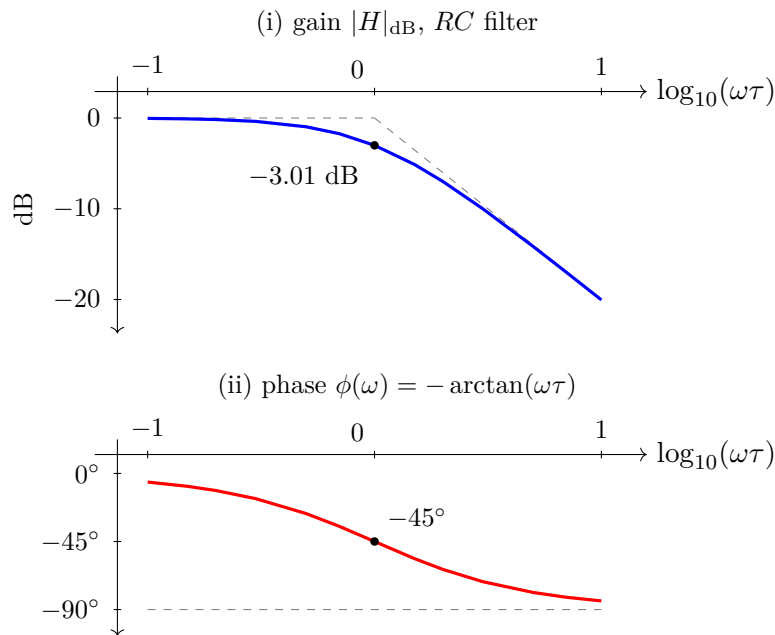


Figure 24.1: Frequency response of the RC filter of Example 24.4, on a logarithmic scale of angular frequencies. The coordinates of the thirteen points of each curve were computed from the closed formulas $|H|_{\text{dB}} = -10 \log_{10}(1 + \omega^2\tau^2)$ and $\phi = -\arctan(\omega\tau)$ and re-checked with a script: for instance at $\omega\tau = 1$ one has -3.01030 dB and -45° , at $\omega\tau = 10$ one has -20.04321 dB and -84.28941° . The dashed lines are the asymptotes 0 dB and -20 dB/decade , which meet exactly at the cutoff angular frequency.

Example 24.5: The moving average: an exact zero and an exactly linear phase.

For $h = \frac{1}{T_0} \mathbf{1}_{[0, T_0]}$ and $\omega \neq 0$,

$$H(\omega) = \frac{1}{T_0} \int_0^{T_0} e^{-i\omega t} dt = \frac{1}{T_0} \cdot \frac{1 - e^{-i\omega T_0}}{i\omega} = e^{-i\omega T_0/2} \cdot \frac{e^{i\omega T_0/2} - e^{-i\omega T_0/2}}{i\omega T_0} = e^{-i\omega T_0/2} \frac{\sin(\omega T_0/2)}{\omega T_0/2},$$

having collected $e^{-i\omega T_0/2}$ and used $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ with $\theta = \omega T_0/2$; for $\omega = 0$ one has $H(0) = 1$, consistent with the limit. Hence

$$|H(\omega)| = \left| \frac{\sin(\omega T_0/2)}{\omega T_0/2} \right|, \quad H(\omega) = e^{-i\omega T_0/2} \operatorname{sinc}\left(\frac{\omega T_0}{2\pi}\right),$$

and the gain vanishes exactly at the angular frequencies $\omega = 2\pi k/T_0$, $k \in \mathbb{Z}$, $k \neq 0$, i.e. at the frequencies $\nu = k/T_0$.

Numbers, with $T_0 = 2$ ms. The zeros fall at $\nu = 500, 1\,000, 1\,500$ Hz. At $\omega = 1\,000$ rad/s the closed formula gives $0.454\,648\,713\,412\,84 - 0.708\,073\,418\,273\,57i$; the same quantity computed from the uncollected expression $(1 - e^{-i\omega T_0})/(i\omega T_0)$ gives the same number with gap $5.55 \cdot 10^{-17}$, and Simpson's quadrature of the integral with $2 \cdot 10^5$ subintervals gives gap $1.84 \cdot 10^{-14}$. ✓ At the first zero angular frequency, $\omega = 2\pi/T_0 = 3\,141.592\,653\,6$ rad/s, the computed gain equals $3.90 \cdot 10^{-17}$, i.e. zero within machine error. ✓

What matters. The factor $e^{-i\omega T_0/2}$ is a phase *exactly linear* in ω : it is the seed of Theorem 24.7, and it already says that this filter delays all components by the same time $T_0/2$, i.e. by half the window — which is reasonable, because an average over $[t - T_0, t]$ is “centred” at $t - T_0/2$.

24.7. Phase delay and group delay

Corollary 24.1 says that a sinusoid of angular frequency ω_0 comes out phase-shifted by $\phi(\omega_0)$. A phase shift, however, is not a delay: to translate it into seconds one must divide it by the angular frequency, and the result depends in general on ω_0 . When it really does depend, different components of the same signal come out delayed by different times and the waveform changes. This section defines the two quantities that measure the phenomenon and proves exactly when it does not occur.

Definition 24.9: Phase delay and group delay. Let H be the frequency response of an LTI system and let ϕ be a continuous phase on an interval I on which H does not vanish (Definition 24.8). For $\omega \in I$, $\omega \neq 0$, the number

$$\tau_p(\omega) = -\frac{\phi(\omega)}{\omega}$$

is called the **phase delay**, and, if ϕ is differentiable at ω , the number

$$\tau_g(\omega) = -\frac{d\phi}{d\omega}(\omega)$$

is called the **group delay**. Both are measured in seconds. The first is the time delay suffered by the *single* sinusoid of angular frequency ω ; the second, as Theorem 24.7 shows, is the delay suffered by the *envelope* of a group of components concentrated around ω .

Remark 24.12: Why two delays and not one. The phase delay is the quotient $-\phi/\omega$, the group delay is the derivative $-\phi'$: geometrically, the first is the slope of the secant joining

the origin to the point $(\omega, \phi(\omega))$ of the graph of the phase, with the sign changed; the second is the slope of the tangent at that point, with the sign changed. The two coincide for every ω if and only if the phase is exactly linear and passes through the origin, $\phi(\omega) = -t_0\omega$: and this is precisely the no-distortion case of Theorem 24.7. The difference between secant and tangent thus measures, in a completely elementary way, how far a system is from the no-distortion condition.

Proposition 24.3: The pure delay. Let $t_0 \in \mathbb{R}$ and $\mathcal{T}_{t_0}x = \sigma_{t_0}x$, i.e. $y(t) = x(t - t_0)$, defined on $L^\infty(\mathbb{R})$. Then:

- (i) $\mathcal{T}_{t_0}$ is linear and time-invariant;
- (ii) every e_ω is an eigenfunction of it, with eigenvalue $H(\omega) = e^{-i\omega t_0}$;
- (iii) $|H(\omega)| = 1$ for every ω , the continuous phase is $\phi(\omega) = -\omega t_0$, and

$$\tau_p(\omega) = \tau_g(\omega) = t_0 \quad \text{for every } \omega \neq 0.$$

Proof. (i). Linearity is immediate: $\sigma_{t_0}(\alpha x_1 + \beta x_2)(t) = (\alpha x_1 + \beta x_2)(t - t_0) = \alpha x_1(t - t_0) + \beta x_2(t - t_0)$. Invariance follows from $\sigma_{t_0}\sigma_a = \sigma_{t_0+a} = \sigma_a\sigma_{t_0}$, which is checked by evaluating at t : $(\sigma_{t_0}\sigma_a x)(t) = (\sigma_a x)(t - t_0) = x(t - t_0 - a)$, symmetric in the two indices.

(ii). For every t ,

$$(\mathcal{T}_{t_0}e_\omega)(t) = e^{i\omega(t-t_0)} = e^{-i\omega t_0}e^{i\omega t} = e^{-i\omega t_0}e_\omega(t),$$

using $e^{a+b} = e^ae^b$. Since e_ω is not identically zero, it is an eigenfunction with eigenvalue $e^{-i\omega t_0}$. Note that this system does *not* fall under Definition 24.4, because its impulse response is not a function of L^1 (Remark 24.3): the computation just carried out is direct and does not use Theorem 24.2.

(iii). $|e^{-i\omega t_0}| = 1$ because $\omega t_0 \in \mathbb{R}$. The function $\omega \mapsto -\omega t_0$ is continuous on all of $\mathbb{R}$ and satisfies $H(\omega) = 1 \cdot e^{i(-\omega t_0)}$: it is therefore a continuous phase in the sense of Definition 24.8. Finally

$$\tau_p(\omega) = -\frac{-\omega t_0}{\omega} = t_0, \quad \tau_g(\omega) = -\frac{d}{d\omega}(-\omega t_0) = t_0.$$

□

Theorem 24.7: A linear-phase system does not distort the waveform. Let $x \in L^1(\mathbb{R})$ with $\hat{x} \in L^1(\mathbb{R})$ and $\hat{x}(\omega) = 0$ for $|\omega| > \Omega$ (a signal band-limited to $\Omega > 0$). Let $h \in L^1(\mathbb{R})$ be the impulse response of an LTI system whose frequency response satisfies

$$H(\omega) = A e^{-i\omega t_0} \quad \text{for every } |\omega| \leq \Omega,$$

with $A > 0$ and $t_0 \in \mathbb{R}$ constants (constant gain and exactly linear phase over the band of the signal; outside the band H is arbitrary). Then

$$y(t) = (h * x)(t) = A x(t - t_0) \quad \text{for every } t \in \mathbb{R} :$$

the output is the input rescaled by A and delayed by t_0 , with no alteration of the waveform. In this case $\tau_p(\omega) = \tau_g(\omega) = t_0$ throughout the band.

Proof. Step 1 — the spectrum of the output. Since $h, x \in L^1(\mathbb{R})$, also $y = h * x \in L^1(\mathbb{R})$ and by Theorem 19.4 (the convolution theorem) of Chapter 19

$$\hat{y}(\omega) = \hat{h}(\omega) \hat{x}(\omega) = H(\omega) \hat{x}(\omega) \quad \text{for every } \omega \in \mathbb{R}.$$

Step 2 — the band hypothesis eliminates the out-of-band behaviour. For $|\omega| > \Omega$ one has $\hat{x}(\omega) = 0$, hence $\hat{y}(\omega) = H(\omega) \cdot 0 = 0$ whatever H is there. For $|\omega| \leq \Omega$ one uses the

hypothesis on H :

$$\hat{y}(\omega) = A e^{-i\omega t_0} \hat{x}(\omega).$$

Putting the two cases together, and noting that for $|\omega| > \Omega$ also $A e^{-i\omega t_0} \hat{x}(\omega) = 0$, one obtains the identity valid on all of $\mathbb{R}$:

$$\hat{y}(\omega) = A e^{-i\omega t_0} \hat{x}(\omega) \quad \text{for every } \omega \in \mathbb{R}.$$

Step 3 — recognising the right-hand side. Let $z(t) = A x(t - t_0)$. Then $z \in L^1(\mathbb{R})$ and, by the translation property of Theorem 19.3,

$$\hat{z}(\omega) = A \widehat{\sigma_{t_0} x}(\omega) = A e^{-i\omega t_0} \hat{x}(\omega).$$

Hence $\hat{y} = \hat{z}$ on all of $\mathbb{R}$.

Step 4 — from equality of the spectra to equality of the signals. The two functions y and z lie in $L^1(\mathbb{R})$ and have the same transform; by linearity $\widehat{y - z} \equiv 0$, and since $\widehat{y - z} = 0 \in L^1(\mathbb{R})$ one can apply the inversion formula in L^1 (Theorem 19.9) to the function $y - z$:

$$(y - z)(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{y - z}(\omega) e^{i\omega t} d\omega = 0$$

for almost every t . It remains to promote “almost every” to “every”: $\hat{y} = \hat{z} \in L^1(\mathbb{R})$ — indeed $|\hat{z}(\omega)| = A|\hat{x}(\omega)|$ and $\hat{x} \in L^1(\mathbb{R})$ by hypothesis — hence, again by Theorem 19.9, both y and z coincide a.e. with *continuous* functions, and two continuous functions equal a.e. are equal everywhere. Hence $y(t) = A x(t - t_0)$ for every t .

Step 5 — the two delays. On $|\omega| \leq \Omega$ one has $|H(\omega)| = A > 0$ and $\phi(\omega) = -\omega t_0$ is a continuous phase; the computation of the two delays is that of point (iii) of Proposition 24.3 and gives $\tau_p = \tau_g = t_0$. $\square$

Corollary 24.3: The no-distortion condition, and its practical reading. Let $h \in L^1(\mathbb{R})$ be real-valued, with $H = \hat{h}$ never zero on $[-\Omega, \Omega]$ and $H(0) > 0$. The following are equivalent:

- (a) there exist $A > 0$ and $t_0 \in \mathbb{R}$ such that $y = A x(\cdot - t_0)$ for every signal x under the hypotheses of Theorem 24.7 with band Ω ;
- (b) on $[-\Omega, \Omega]$ the gain $|H(\omega)|$ is constant and the group delay $\tau_g(\omega)$ is constant.

Proof. (b) $\Rightarrow$ (a). Let $|H| \equiv A$ on $[-\Omega, \Omega]$ and $\tau_g = -\phi' \equiv t_0$. Since H does not vanish on the band, a continuous phase ϕ exists there (Definition 24.8); integrating $\phi' \equiv -t_0$ over the connected interval $[-\Omega, \Omega]$ gives $\phi(\omega) = -t_0\omega + c$ with c constant. Since h is real, $H(0) = \int_{\mathbb{R}} h$ is real, and is > 0 by hypothesis: hence $A = |H(0)| = H(0)$ and $e^{i\phi(0)} = H(0)/|H(0)| = 1$, i.e. $c = \phi(0) \in 2\pi\mathbb{Z}$; choosing the determination with $c = 0$ (legitimate: the continuous phase is defined up to a constant multiple of 2π) one has $H(\omega) = A e^{-i\omega t_0}$ over the whole band, and one applies Theorem 24.7.

(a) $\Rightarrow$ (b). Choose an explicit test signal whose transform does not vanish *inside* the band: let $\hat{x}(\omega) = (1 - |\omega|/\Omega)^+$ be the triangle of base $[-\Omega, \Omega]$, which is continuous, has support in $[-\Omega, \Omega]$ and lies in $L^1(\mathbb{R})$ (it is the function of Proposition 19.3 of Chapter 19, rescaled). The corresponding x is computed from the inversion formula: since the integrand is even in its real part and odd in its imaginary part,

$$x(t) = \frac{1}{2\pi} \int_{-\Omega}^{\Omega} \left(1 - \frac{|\omega|}{\Omega}\right) e^{i\omega t} d\omega = \frac{1}{\pi} \int_0^{\Omega} \left(1 - \frac{\omega}{\Omega}\right) \cos(\omega t) d\omega = \frac{1 - \cos(\Omega t)}{\pi \Omega t^2} = \frac{\Omega}{2\pi} \left(\frac{\sin(\Omega t/2)}{\Omega t/2}\right)^2,$$

where the integral was computed by parts — $\int_0^{\Omega} \cos(\omega t) d\omega = \sin(\Omega t)/t$ and $\frac{1}{\Omega} \int_0^{\Omega} \omega \cos(\omega t) d\omega = \sin(\Omega t)/t - (1 - \cos(\Omega t))/(\Omega t^2)$, whose difference is $(1 - \cos(\Omega t))/(\Omega t^2)$ — and the last

equality is $1 - \cos v = 2 \sin^2(v/2)$ with $v = \Omega t$. This x lies in $L^1(\mathbb{R})$ because $|x(t)| \leq \frac{\Omega}{2\pi} \min\{1, 4/(\Omega t)^2\}$ — using $|\sin u/u| \leq 1$ for the first bound and $|\sin u| \leq 1$ for the second — and $\int_{\mathbb{R}} \min\{1, 4/(\Omega t)^2\} dt = \frac{4}{\Omega} + 2 \int_{2/\Omega}^{+\infty} \frac{4}{\Omega^2 t^2} dt = \frac{4}{\Omega} + \frac{4}{\Omega} = \frac{8}{\Omega} < \infty$. The hypotheses of Theorem 24.7 are therefore satisfied by this x . If (a) holds, then $y = Ax(\cdot - t_0)$; taking transforms and using Step 1 and Step 3 of the previous proof,

$$H(\omega) \hat{x}(\omega) = A e^{-i\omega t_0} \hat{x}(\omega) \quad \text{for every } \omega \in \mathbb{R}.$$

For $|\omega| < \Omega$ one has $\hat{x}(\omega) = 1 - |\omega|/\Omega > 0$, so one can divide: $H(\omega) = A e^{-i\omega t_0}$ on $(-\Omega, \Omega)$, and by continuity of H (Theorem 24.6(i)) also at the two endpoints. From this $|H| \equiv A$ and $\phi(\omega) = -t_0\omega$ over the whole band, hence $\tau_g \equiv t_0$ constant. $\square$

Example 24.6: The RC filter distorts: the two delays, with the numbers. For the RC filter of Example 24.4, with $\phi(\omega) = -\arctan(\omega\tau)$:

$$\tau_p(\omega) = \frac{\arctan(\omega\tau)}{\omega}, \quad \tau_g(\omega) = -\frac{d}{d\omega}(-\arctan(\omega\tau)) = \frac{\tau}{1 + \omega^2\tau^2},$$

having used $\frac{d}{dv} \arctan v = \frac{1}{1+v^2}$ and the chain rule with $v = \omega\tau$. The two delays do *not* coincide and are *not* constant: the system distorts.

Numbers with $\tau = 1$ ms. The table reports, for four values of $\omega\tau$, the gain, the phase, and the two delays in milliseconds.

$\omega\tau$	$ H $	ϕ (degrees)	τ_p (ms)	τ_g (ms)
0.5	0.894 427 191	-26.565 051	0.927 295 218	0.8
1	0.707 106 781	-45.000 000	0.785 398 163	0.5
2	0.447 213 595	-63.434 949	0.553 574 359	0.2
5	0.196 116 135	-78.690 068	0.274 680 153	0.038 461 538

Independent check. The group delay computed by centred differences, $-(\phi(\omega + d) - \phi(\omega - d))/(2d)$ with $d = 10^{-6}$, gives 0.499 999 985 9 ms at $\omega\tau = 1$ (closed formula 0.5 ms, gap $1.41 \cdot 10^{-11}$ ms) and 0.100 000 008 3 ms at $\omega\tau = 3$ (closed formula 0.1 ms, gap $8.27 \cdot 10^{-12}$ ms). ✓ Moreover $\tau_p(1/\tau) = \arctan(1)/(1/\tau) = (\pi/4)\tau = 0.785 398 163 4$ ms coincides with $\pi/4$ milliseconds computed separately. ✓

Interpretation. Between $\omega\tau = 1$ and $\omega\tau = 3$ the group delay goes from 0.5 to 0.1 ms: an exact factor of 5, because $\tau_g(1/\tau)/\tau_g(3/\tau) = \frac{\tau/2}{\tau/10} = 5$. Two components of the same signal, one at $\omega = 1/\tau$ and the other at $\omega = 3/\tau$, come out of the filter phase-shifted relative to each other differently from how they went in: the resulting waveform is not the translate of the original. Compared with the moving average of Example 24.5, which has constant $\tau_g \equiv T_0/2$, the difference is stark — and it is the reason why linear-phase filters are preferred whenever the shape of the waveform matters (electrocardiography, peak measurements, seismology) and not only its spectral content.

Remark 24.13: Minimum-phase systems: panorama. Among all causal and stable systems that have the same gain $|H(\omega)|$, there is a privileged one, called **minimum-phase**: the one whose frequency response, extended to the right half-plane as in Remark 24.11, has no zeros there. One shows about it — in a theory that this treatise does not develop — that, among all those of equal gain, it has the phase of minimum magnitude, the minimum group delay, and the energy of the impulse response concentrated as close as possible to the origin; and that every causal and stable system factors uniquely as “minimum-phase system $\times$ all-pass system”, where an all-pass system has $|H| \equiv 1$ and therefore acts only on the phase. The central fact is that a minimum-phase system has a causal and stable inverse,

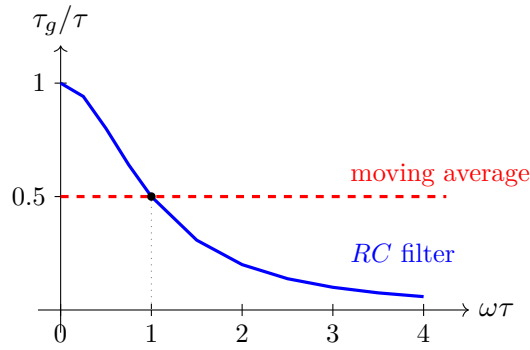


Figure 24.2: Normalised group delay. For the *RC* filter one has $\tau_g/\tau = 1/(1 + \omega^2\tau^2)$ and it decreases from 1 to 0: the slow components come out delayed by τ , the fast ones almost immediately, and the waveform is deformed. For the moving average with window $T_0 = \tau$, instead, $\tau_g = T_0/2 = \tau/2$ for every ω , i.e. $\tau_g/\tau = 0.5$: this is the dashed horizontal line, and by Theorem 24.7 it means no waveform distortion within the band where the gain stays constant. The eleven coordinates of the blue curve are computed from the closed formula and checked: at $\omega\tau = 1$ it equals 0.5, at $\omega\tau = 3$ it equals 0.1, at $\omega\tau = 4$ it equals 0.058 824.

while in general a stable system does not.

None of this is proved here: it would require Hardy spaces on the right half-plane, inner-outer factorisation and the Kramers–Kronig relations, none of which appears in the treatise. The whole paragraph is therefore **panorama**, and what remains **imported** is: the existence and uniqueness of the minimum-phase/all-pass factorisation, the variational characterisation of the minimum group delay, and the stability of the inverse. The source is Oppenheim and Schaffer, entry [4] of the Sources environment. In this chapter the expression “minimum phase” does not occur in any statement and is not used in any proof.

24.8. The Nyquist-Shannon sampling theorem

Remark 24.14: Prerequisites. The following lemma presupposes the Fourier series expansion of a periodic function (Section 4.2) and the elementary properties of the Dirac delta (Appendix A.1, in particular $\int f(t)\delta(t)dt = f(0)$). Those who wish to go directly to the sampling theorem may accept the result of Lemma 24.2 as stated and move on.

The convolution theorem (Theorem 19.4) turns convolution in time into a product in frequency. What is needed here is the *dual fact* — a product in time becomes a convolution in frequency, divided by 2π — which Chapter 19 does not state separately. It is not a new fact: it follows from the convolution theorem by exchanging the roles of f and $\hat{f}$ through the inversion formula, and the factor $1/2\pi$ is exactly the one that the convention in ω puts in front of the inverse transform. We prove it here once and for all, in the only case in which all the objects involved are summable functions.

Lemma 24.1: Product/convolution duality. Let $F, G \in L^1(\mathbb{R})$ and let

$$f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} F(\omega) e^{i\omega t} d\omega, \quad g(t) = \frac{1}{2\pi} \int_{\mathbb{R}} G(\omega) e^{i\omega t} d\omega$$

be their inverse transforms: these are $f(t) = \frac{1}{2\pi} \hat{F}(-t)$ and $g(t) = \frac{1}{2\pi} \hat{G}(-t)$, continuous and bounded functions by Proposition 19.1 and Theorem 19.1. If the product fg belongs to

$L^1(\mathbb{R})$, then

$$\widehat{fg}(\omega) = \frac{1}{2\pi} (F * G)(\omega) \quad \text{for almost every } \omega \in \mathbb{R}.$$

If moreover $f, g \in L^1(\mathbb{R})$, then $F = \hat{f}$ and $G = \hat{g}$ almost everywhere, and the statement takes the usual form $\widehat{fg} = \frac{1}{2\pi} \hat{f} * \hat{g}$.

Proof. All formulas are written in the convention in ω of this chapter; the results of Chapter 19 are read in it with the dictionary of Remark 19.1, which for convolution and for inversion leaves the formulas unchanged in form.

Step 1 — an inversion criterion «the other way round». Let $H \in L^1(\mathbb{R})$ and let $u(t) = \frac{1}{2\pi} \int H(\omega) e^{i\omega t} d\omega$ be its inverse transform. If $u \in L^1(\mathbb{R})$, then $\hat{u} = H$ almost everywhere. Indeed

$$\hat{H}(t) = \int_{\mathbb{R}} H(\omega) e^{-i\omega t} d\omega = 2\pi u(-t),$$

so that $\hat{H} \in L^1(\mathbb{R})$: Theorem 19.9, applied to H read as a function of its own variable, gives for almost every ω

$$H(\omega) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{H}(t) e^{i\omega t} dt = \int_{\mathbb{R}} u(-t) e^{i\omega t} dt = \int_{\mathbb{R}} u(s) e^{-i\omega s} ds = \hat{u}(\omega),$$

where in the last step we set $s = -t$.

Step 2 — the inverse transform of $\frac{1}{2\pi}(F * G)$ is fg . By Theorem 19.4(ii) we have $F * G \in L^1(\mathbb{R})$, hence $H := \frac{1}{2\pi}(F * G) \in L^1(\mathbb{R})$; by item (iv) of the same theorem,

$$\hat{H}(t) = \frac{1}{2\pi} \widehat{F * G}(t) = \frac{1}{2\pi} \hat{F}(t) \hat{G}(t).$$

As in Step 1, $\hat{F}(t) = 2\pi f(-t)$ and $\hat{G}(t) = 2\pi g(-t)$, hence $\hat{H}(t) = 2\pi f(-t)g(-t)$. The inverse transform of H is then

$$u(t) = \frac{1}{2\pi} \int_{\mathbb{R}} H(\omega) e^{i\omega t} d\omega = \frac{1}{2\pi} \hat{H}(-t) = f(t)g(t).$$

Step 3 — conclusion. By hypothesis $u = fg \in L^1(\mathbb{R})$, and $H \in L^1(\mathbb{R})$ by Step 2: Step 1 gives $\widehat{fg} = \hat{u} = H = \frac{1}{2\pi}(F * G)$ almost everywhere.

Step 4 — the identification $F = \hat{f}$. If $f \in L^1(\mathbb{R})$, Step 1 applied to $H = F$ (whose inverse transform is $u = f$) gives $\hat{f} = F$ almost everywhere; likewise for g and G . $\square$

Lemma 24.2: Fourier transform of the Dirac comb (Poisson summation). Let $\Delta_T(t) = \sum_{k \in \mathbb{Z}} \delta(t - kT)$ be the Dirac comb of step $T > 0$, and let $\omega_c = 2\pi/T$. Then its Fourier transform is again a Dirac comb, in frequency, of step ω_c :

$$\widehat{\Delta_T}(\omega) = \omega_c \sum_{n \in \mathbb{Z}} \delta(\omega - n\omega_c).$$

Proof. The computation that follows is *formal*: the comb is manipulated in it as if it were a function, in two steps (the series expansion and the term-by-term transformation) declared **imported** in Remark 24.16.

Δ_T is periodic of period T : like any periodic (distribution) function of period T it expands in Fourier series in the basis $e^{in\omega_c t}$ (the analog, with $\theta = \omega_c t$, of the basis $e^{in\theta}$ of Section 4.2),

$$\Delta_T(t) = \sum_{n \in \mathbb{Z}} c_n e^{in\omega_c t}, \quad c_n = \frac{1}{T} \int_{-T/2}^{T/2} \Delta_T(t) e^{-in\omega_c t} dt.$$

On the integration interval $(-T/2, T/2)$ the sum Δ_T reduces to the single term $k = 0$, i.e. $\delta(t)$; by the sampling property of the delta (Appendix A.1),

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-in\omega_c t} dt = \frac{1}{T} e^0 = \frac{1}{T} \quad \text{for every } n \in \mathbb{Z},$$

hence $\Delta_T(t) = \frac{1}{T} \sum_{n \in \mathbb{Z}} e^{in\omega_c t}$: the Dirac comb is rewritten (in the sense of distributions) as a sum of complex exponentials of equal amplitude $1/T$. Transforming term by term, and using $\hat{1}(\omega) = 2\pi\delta(\omega)$ (Appendix A.1) together with the frequency translation — obtained directly from the definition,

$$\widehat{e^{i\omega_0 t}}(\omega) = \int_{\mathbb{R}} e^{i\omega_0 t} e^{-i\omega t} dt = \int_{\mathbb{R}} e^{-i(\omega - \omega_0)t} dt = 2\pi\delta(\omega - \omega_0),$$

i.e. $\hat{1}$ shifted to ω_0 — one obtains

$$\widehat{\Delta_T}(\omega) = \frac{1}{T} \sum_{n \in \mathbb{Z}} \widehat{e^{in\omega_c t}}(\omega) = \frac{1}{T} \sum_{n \in \mathbb{Z}} 2\pi\delta(\omega - n\omega_c) = \frac{2\pi}{T} \sum_{n \in \mathbb{Z}} \delta(\omega - n\omega_c) = \omega_c \sum_{n \in \mathbb{Z}} \delta(\omega - n\omega_c),$$

using $\omega_c = 2\pi/T$ in the last step. $\square$

Remark 24.15: Name and scope of the result. The identity $\Delta_T(t) = \frac{1}{T} \sum_n e^{in\omega_c t}$ (and its transform) is the special case, for $f = \delta$, of the Poisson summation formula $\sum_{k \in \mathbb{Z}} f(kT) = \frac{1}{T} \sum_{n \in \mathbb{Z}} \hat{f}(n\omega_c)$, which connects sums in the time domain to sums in the frequency domain and is a central tool in harmonic analysis, analytic number theory (Jacobi theta function, Riemann functional equation) and crystallography (diffraction on periodic lattices). See Stein-Shakarchi, Ch. 4, for the general statement and validity conditions on f .

Theorem 24.8: Sampling theorem (Nyquist-Shannon). Let $x_c(t)$ be a continuous bandlimited signal, i.e. $\hat{x}_c(\omega) = 0$ for $|\omega| > \omega_{\max}$. If the sampling period T satisfies

$$\frac{2\pi}{T} \geq 2\omega_{\max} \quad \left(\text{equivalently } f_c = \frac{1}{T} \geq 2f_{\max} \right),$$

then $x_c(t)$ is exactly reconstructible from the samples $x[n] = x_c(nT)$.

Proof. We model sampling as multiplication by a Dirac comb $\Delta_T(t) = \sum_k \delta(t - kT)$: the sampled signal is $x_s(t) = x_c(t)\Delta_T(t)$. By the product/convolution duality $\widehat{fg} = \frac{1}{2\pi} \hat{f} * \hat{g}$ — applied here to the product of a function by a *distribution*, hence outside the hypotheses of Lemma 24.1: it is the third of the three formal steps declared **imported** in Remark 24.16 — and by Lemma 24.2, according to which $\widehat{\Delta_T}(\omega) = \omega_c \sum_{k \in \mathbb{Z}} \delta(\omega - k\omega_c)$ with $\omega_c = 2\pi/T$, one obtains

$$\hat{x}_s(\omega) = \frac{1}{2\pi} (\hat{x}_c * \widehat{\Delta_T})(\omega) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{x}_c(\omega - \xi) \omega_c \sum_{k \in \mathbb{Z}} \delta(\xi - k\omega_c) d\xi = \frac{\omega_c}{2\pi} \sum_{k \in \mathbb{Z}} \hat{x}_c(\omega - k\omega_c) = \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{x}_c(\omega - k\omega_c),$$

i.e.: sampling in time periodically replicates the spectrum in frequency, with period ω_c . If $\omega_c \geq 2\omega_{\max}$, the replicas do not overlap, and the original spectrum $\hat{x}_c(\omega)$ (hence $x_c(t)$ itself) is recovered by applying an ideal low-pass filter that isolates the central replica. If instead $\omega_c < 2\omega_{\max}$, the replicas overlap (*aliasing*) and the lost information is no longer recoverable. The three steps with distributions carried out here and in Lemma 24.2 are listed one by one in Remark 24.16: they are **imported** and not justified with the rigour of the rest of the chapter. The explicit reconstruction of x_c from its samples, and its proof carried out in full *without*

distributions, are in Theorem 24.9. □

Remark 24.16: What we import in the comb Lemma and in this proof.

Lemma 24.2 and the proof of Theorem 24.8 carry out three formal steps with distributions, in which the Dirac comb is manipulated as if it were a function. They are not justified with the rigour of the rest of the chapter, and we isolate them here one by one, saying exactly what is missing.

- (P1) Fourier series expansion of the comb. One writes $\Delta_T(t) = \sum_n c_n e^{in\omega_c t}$ «like any periodic (distribution) function»: but Δ_T is not a function, and the L^2 convergence theorem for Fourier series (Theorem 17.1) does not cover it. One would need the notion of periodic tempered distribution and the convergence of the series in $\mathcal{S}'(\mathbb{R})$.
- (P2) Term-by-term transformation. The series $\frac{1}{T} \sum_n e^{in\omega_c t}$ converges at no point t , and it is transformed summand by summand. One would need the continuity of the Fourier transform on $\mathcal{S}'(\mathbb{R})$ and the identity $\hat{1} = 2\pi\delta$, itself defined only in $\mathcal{S}'(\mathbb{R})$.
- (P3) Product of a function by a distribution, and product/convolution duality. One forms $x_s = x_c \Delta_T$ and applies to it the duality $\widehat{fg} = \frac{1}{2\pi} \hat{f} * \hat{g}$: Lemma 24.1 proves it when the inverse transforms involved are summable functions, a hypothesis that Δ_T does not satisfy. One would need the product of a tempered distribution by a C^∞ function of slow growth and the convolution between distributions.

The three points are therefore **imported**. The theory that justifies them — tempered distributions, the delta never treated as a function (Proposition 19.8), the comb proved self-dual (Theorem 19.14) — is that of Section 19.12 of Chapter 19, with the sources already recorded there: NIST, Digital Library of Mathematical Functions, §1.16(v) «Tempered Distributions» and §1.16(vii) «Fourier Transforms of Tempered Distributions», and Bracewell, Chapter 10 «Sampling and Series» for the distributional apparatus of sampling. This is exactly what Remark 19.10 of that chapter states: the sampling theorem follows from the self-duality of the comb, «but its complete proof requires the product of a tempered distribution by a C^∞ function of slow growth and the convolution between distributions», which are not constructed there.

Consequently the route through the Dirac comb is to be read as a context remark — the quickest way of seeing why sampling periodizes the spectrum — and not as the formal proof of the result. The formal proof is that of Theorem 24.9, which uses distributions in no step and reconstructs the signal with an explicit formula; no later statement of the chapter depends on (P1)–(P3).

Remark 24.17: Practical relevance for control and automation. In choosing the scan time of a PLC, the theorem imposes an absolute upper limit (the sampling period cannot be slower than half the period of the fastest dynamics one wishes to observe or control), but in engineering practice one samples much faster than the theoretical minimum — typically 5–10 times the process bandwidth — both because an ideal low-pass filter does not exist (real anti-aliasing filters have a gradual transition, not a sharp cutoff) and to limit the delay introduced by sampling itself in the control loop. An analog sensor (temperature, pressure, level) should always be preceded by an analog anti-aliasing filter that attenuates components above $f_c/2$ before A/D conversion: otherwise high-frequency noise folds (aliasing) into the useful band and becomes indistinguishable from the signal.

Remark 24.18: Beyond Nyquist-Shannon: compressed sensing. Theorem 24.8 sets a sufficient limit on the sampling frequency for exact reconstruction of a generic bandlimited signal, but it is not necessary if the signal has richer structure than just a bandwidth limit: for example, that it is sparse in some basis (few non-zero components, perhaps not in the

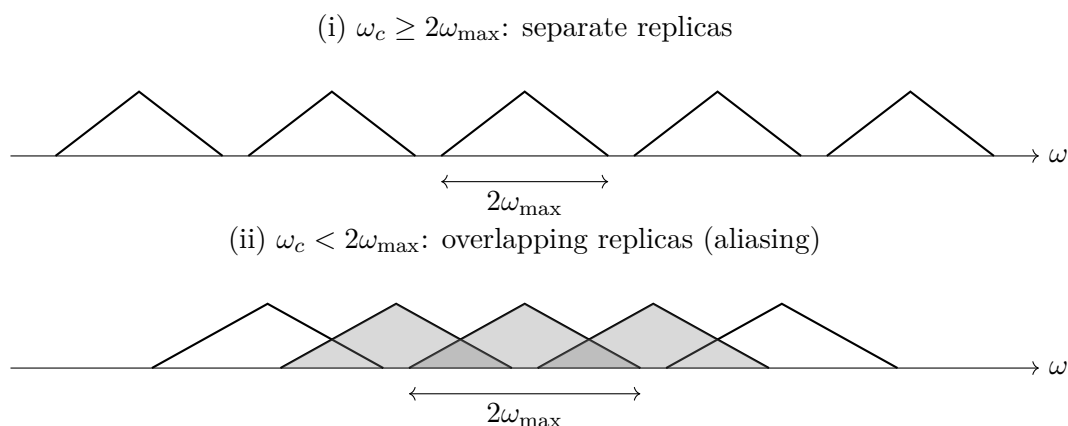


Figure 24.3: Sampling in time periodically replicates the spectrum with step $\omega_c = 2\pi/T$: if $\omega_c \geq 2\omega_{\max}$ the replicas remain separate (i) and the signal is exactly reconstructed; otherwise they overlap (ii, shaded areas) and lost information is no longer recoverable — exactly the two cases of the proof just seen.

time basis but in that of frequencies or wavelets). Compressed sensing (or compressive sampling), developed in the mid-2000s by Candès, Romberg and Tao (2006) and independently by Donoho (2006), shows that an S -sparse signal of length N can be exactly reconstructed from only $O(S \log N)$ appropriately chosen linear measurements (typically random), well below the N samples required by Theorem 24.8 in the generic case — provided one replaces linear reconstruction (the low-pass filter of the proof of Theorem 24.8) with a nonlinear optimization problem (ℓ_1 norm minimization). This result has had direct impacts in medical imaging (accelerated MRI), astronomy, and is today an active area of research also for compressed acquisition of industrial vibration signals, where battery-powered wireless sensors benefit from transmitting as few samples as possible. It is an instructive example of how a classical limit (Nyquist-Shannon, 1949) remains fully valid within its assumptions (generically bandlimited signal) while subsequent research explores its boundaries under more realistic additional assumptions.

24.9. The reconstruction formula with the sinc function

The proof of Theorem 24.8 establishes the structural fact — sampling periodicizes the spectrum, and if the replicas do not touch the information is fully preserved — by going through the Dirac comb, that is, through an object that is a distribution and not a function. This section completes the picture by proving the *explicit* reconstruction formula, and it does so without distributions: the only tool is the Fourier series expansion of the spectrum on the bandwidth interval, together with the inversion formula. The result is that the samples not only *determine* the signal, but reconstruct it with a formula that can be written down and evaluated.

Definition 24.10: Bandlimited and Nyquist frequency. A signal $x \in L^1(\mathbb{R})$ is **bandlimited** to $\omega_{\max} > 0$ if $\hat{x}(\omega) = 0$ for every $|\omega| > \omega_{\max}$. Given a sampling step $T > 0$, the **sampling angular frequency** is $\omega_c = 2\pi/T$ and the **sampling frequency** is $\nu_c = 1/T$; the **Nyquist frequency** is $\nu_c/2 = 1/(2T)$, i.e. half the sampling frequency. The condition of Theorem 24.8, $\omega_c \geq 2\omega_{\max}$, thus reads: the band of the signal must lie below the Nyquist frequency.

Theorem 24.9: Shannon's reconstruction formula. Let $x \in L^1(\mathbb{R})$ with $\hat{x}$ zero outside $[-\omega_{\max}, \omega_{\max}]$, and let $T > 0$ be such that

$$\Omega := \frac{\pi}{T} \geq \omega_{\max} \quad \left(\text{equivalently } \omega_c = \frac{2\pi}{T} \geq 2\omega_{\max} \right).$$

Then x coincides a.e. with a continuous function, still denoted x , and for every $t \in \mathbb{R}$

$$x(t) = \sum_{n \in \mathbb{Z}} x(nT) \operatorname{sinc}\left(\frac{t - nT}{T}\right),$$

where the series converges for every t in the sense of the symmetric sums $\sum_{|n| \leq N}$, $N \rightarrow \infty$.

Proof. Step 0 — x is continuous and the inversion formula applies. The function $\hat{x}$ is continuous (Theorem 19.1) and zero outside the compact set $[-\omega_{\max}, \omega_{\max}]$, hence bounded and with compact support: therefore $\hat{x} \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, with $\|\hat{x}\|_1 \leq 2\omega_{\max} \sup |\hat{x}| < \infty$ and similarly for the quadratic norm. Since $x \in L^1$ and $\hat{x} \in L^1$, Theorem 19.9 gives that x coincides a.e. with the continuous function

$$x(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{x}(\omega) e^{i\omega t} d\omega = \frac{1}{2\pi} \int_{-\Omega}^{\Omega} \hat{x}(\omega) e^{i\omega t} d\omega,$$

where in the second equality the domain was restricted because $\hat{x} = 0$ outside $[-\omega_{\max}, \omega_{\max}] \subseteq [-\Omega, \Omega]$, since $\Omega \geq \omega_{\max}$ by hypothesis. From here on x denotes this continuous representative.

Step 1 — the spectrum as a Fourier series on $[-\Omega, \Omega]$. Consider the restriction of $\hat{x}$ to $[-\Omega, \Omega]$ and transport it onto the circle with the linear change of variable

$$\omega = \frac{\Omega}{\pi} \theta, \quad \theta \in [-\pi, \pi],$$

setting $F(\theta) = \hat{x}(\Omega\theta/\pi)$. The function F is continuous on $[-\pi, \pi]$, hence $F \in L^2(\mathbb{T})$. By Theorem 17.1 of Chapter 17 its partial Fourier sums $S_N F = \sum_{|n| \leq N} \hat{F}(n) e_n$, with

$$\hat{F}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(\theta) e^{-in\theta} d\theta,$$

converge to F in the $L^2(\mathbb{T})$ norm. Returning to the variable ω (the change of variable is affine and bijective between $[-\pi, \pi]$ and $[-\Omega, \Omega]$, with $d\omega = \frac{\Omega}{\pi} d\theta$), this means: setting

$$c_n := \hat{F}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}\left(\frac{\Omega\theta}{\pi}\right) e^{-in\theta} d\theta = \frac{1}{2\Omega} \int_{-\Omega}^{\Omega} \hat{x}(\omega) e^{-in\pi\omega/\Omega} d\omega,$$

the functions $S_N(\omega) := \sum_{|n| \leq N} c_n e^{in\pi\omega/\Omega}$ converge to $\hat{x}$ in the $L^2([-\Omega, \Omega])$ norm.

Step 2 — the coefficients are the samples. Evaluate the inversion formula of Step 0 at the points $t = nT$. Since $T = \pi/\Omega$, one has $nT = n\pi/\Omega$ and

$$x(nT) = \frac{1}{2\pi} \int_{-\Omega}^{\Omega} \hat{x}(\omega) e^{in\pi\omega/\Omega} d\omega.$$

Comparing with the expression for c_{-n} of Step 1, which is $\frac{1}{2\Omega} \int_{-\Omega}^{\Omega} \hat{x}(\omega) e^{in\pi\omega/\Omega} d\omega$, one obtains

$$c_{-n} = \frac{2\pi}{2\Omega} x(nT) = \frac{\pi}{\Omega} x(nT) = T x(nT) \quad \text{for every } n \in \mathbb{Z}.$$

Step 3 — exchanging sum and integral. Fix $t \in \mathbb{R}$. By Step 0,

$$x(t) = \frac{1}{2\pi} \int_{-\Omega}^{\Omega} \hat{x}(\omega) e^{i\omega t} d\omega.$$

Replace $\hat{x}$ by S_N and estimate the error committed: by the Cauchy–Schwarz inequality on the bounded interval $[-\Omega, \Omega]$,

$$\left| \int_{-\Omega}^{\Omega} (\hat{x}(\omega) - S_N(\omega)) e^{i\omega t} d\omega \right| \leq \|\hat{x} - S_N\|_{L^2([-\Omega, \Omega])} \cdot \left(\int_{-\Omega}^{\Omega} |e^{i\omega t}|^2 d\omega \right)^{1/2} = \sqrt{2\Omega} \|\hat{x} - S_N\|_{L^2([-\Omega, \Omega])},$$

which tends to 0 as $N \rightarrow \infty$ by Step 1 (and the bound does not depend on t). Hence

$$x(t) = \lim_{N \rightarrow \infty} \frac{1}{2\pi} \int_{-\Omega}^{\Omega} S_N(\omega) e^{i\omega t} d\omega = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{c_n}{2\pi} \int_{-\Omega}^{\Omega} e^{i\omega(t+n\pi/\Omega)} d\omega,$$

where exchanging the *finite* sum and the integral is the linearity of the integral.

Step 4 — the elementary integral. For every $s \in \mathbb{R}$,

$$\int_{-\Omega}^{\Omega} e^{i\omega s} d\omega = \left[\frac{e^{i\omega s}}{is} \right]_{-\Omega}^{\Omega} = \frac{e^{i\Omega s} - e^{-i\Omega s}}{is} = \frac{2 \sin(\Omega s)}{s} \quad (s \neq 0),$$

and $\int_{-\Omega}^{\Omega} 1 d\omega = 2\Omega$ for $s = 0$, which is also the limit of the right-hand side. With $s = t + n\pi/\Omega = t + nT$ and $\Omega = \pi/T$:

$$\frac{1}{2\pi} \int_{-\Omega}^{\Omega} e^{i\omega(t+nT)} d\omega = \frac{1}{2\pi} \cdot \frac{2 \sin(\pi(t+nT)/T)}{t+nT} = \frac{1}{\pi} \cdot \frac{\sin(\pi(t+nT)/T)}{t+nT}.$$

Multiplying and dividing by T and using the definition $\text{sinc}(v) = \sin(\pi v)/(\pi v)$ with $v = (t+nT)/T$,

$$\frac{1}{\pi} \cdot \frac{\sin(\pi v)}{Tv} = \frac{1}{T} \cdot \frac{\sin(\pi v)}{\pi v} = \frac{1}{T} \text{sinc}\left(\frac{t+nT}{T}\right).$$

Step 5 — conclusion. Substituting into Step 3 and using Step 2 in the form $c_n = T x(-nT)$:

$$x(t) = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} T x(-nT) \cdot \frac{1}{T} \text{sinc}\left(\frac{t+nT}{T}\right) = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} x(-nT) \text{sinc}\left(\frac{t+nT}{T}\right).$$

With the change of index $m = -n$, which is a bijection of $\{-N, \dots, N\}$ onto itself,

$$x(t) = \lim_{N \rightarrow \infty} \sum_{|m| \leq N} x(mT) \text{sinc}\left(\frac{t-mT}{T}\right) = \sum_{m \in \mathbb{Z}} x(mT) \text{sinc}\left(\frac{t-mT}{T}\right),$$

which is the claim. □

Corollary 24.4: The sinc function is the impulse response of the reconstructor.

Under the hypotheses of Theorem 24.9, the reconstruction formula is the interpolation of the samples with the function

$$g(t) = \text{sinc}(t/T),$$

which has $g(0) = 1$ and $g(mT) = 0$ for every nonzero integer m . Consequently the formula returns exactly the sample at the sampling points: evaluating at $t = kT$, all the terms vanish except the one with $m = k$, which equals $x(kT) \cdot 1$.

Proof. By definition, $\text{sinc}(0) = 1$ and $\text{sinc}(v) = \sin(\pi v)/(\pi v) = 0$ exactly when $\sin(\pi v) = 0$ and $v \neq 0$, i.e. when v is a nonzero integer. Setting $v = (kT - mT)/T = k - m$, one has $v = 0$ if and only if $m = k$, and v a nonzero integer otherwise. Hence in the sum of Theorem 24.9 evaluated at $t = kT$ only the term $m = k$ survives, and $x(kT) = x(kT)$. □

Example 24.7: Reconstruction checked on a bandlimited signal. Take $x(t) = \text{sinc}(t)^2$. Its band is determined in two steps. First: by Proposition 19.2 and the inversion formula (Theorem 19.9), $\text{sinc}(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i\omega t} d\omega$, i.e. $\widehat{\text{sinc}} = \mathbf{1}_{[-\pi, \pi]}$ — the same computation as in Corollary 24.2 with $\Omega = \pi$. Second: by Lemma 24.1, applied to $F = G = \mathbf{1}_{[-\pi, \pi]} \in L^1(\mathbb{R})$ — whose inverse transforms are $f = g = \text{sinc}$, by the previous step, and whose product sinc^2 lies in $L^1(\mathbb{R})$ because it is continuous and $O(t^{-2})$ at infinity —,

$$\widehat{\text{sinc}^2} = \frac{1}{2\pi} (\mathbf{1}_{[-\pi, \pi]} * \mathbf{1}_{[-\pi, \pi]}),$$

and the convolution of two functions with support in $[-\pi, \pi]$ has support contained in $[-2\pi, 2\pi]$, because $|\omega| > 2\pi$ implies that for every ξ at least one of $\mathbf{1}_{[-\pi, \pi]}(\xi)$ and $\mathbf{1}_{[-\pi, \pi]}(\omega - \xi)$ vanishes. Hence $\omega_{\max} = 2\pi$, i.e. the signal has a 1 Hz band. The condition of Theorem 24.9 is $\Omega = \pi/T \geq 2\pi$, i.e. $T \leq 1/2$: one chooses $T = 1/2$, the critical step.

Check (Python script executed). Truncating the series to $|n| \leq 4000$:

t	$x(t)$ (closed form)	truncated series	gap
0.3	0.736 839 729 322 3	0.736 839 729 321 9	$3.83 \cdot 10^{-13}$
1.234	0.029 930 079 376 2	0.029 930 079 374 6	$1.65 \cdot 10^{-12}$
-0.75	0.090 063 274 348 7	0.090 063 274 349 8	$1.01 \cdot 10^{-12}$
2.5	0.016 211 389 382 8	0.016 211 389 382 8	0

✓The four gaps are of the order of the truncation error of the series, which for an x with t^{-2} decay and symmetric tails is of order N^{-1} but with strong cancellations; the value $t = 2.5 = 5T$ falls on a sampling point, where by Corollary 24.4 the series has only one nonzero term and the result is exact.

Counter-check: what happens below Nyquist. With the same x and $T = 1$ (i.e. $\nu_c = 1$ Hz, half the required minimum of 2 Hz), the same truncated series gives 0.858 393 691 3 at $t = 0.3$ against the true value 0.736 839 729 3: a gap of 0.121 553 962, i.e. over 16% of the value. And at $t = 1.234$ it gives -0.173 003 119 6 against 0.029 930 079 4: the reconstruction even changes sign. ✓The hypothesis $\Omega \geq \omega_{\max}$ is not a precaution, it is the validity condition.

24.10. Aliasing

Definition 24.11: Alias frequency. Let $\nu_c = 1/T$ be the sampling frequency and let $\nu_0 \geq 0$ be the frequency of a sinusoidal component of the signal. Its **alias frequency** is

$$\nu_a = |\nu_0 - k\nu_c|, \quad k = \text{an integer closest to } \nu_0/\nu_c,$$

(if there are two closest integers, i.e. if ν_0/ν_c is a half-integer, the choice does not matter because the two values of $|\nu_0 - k\nu_c|$ coincide and equal $\nu_c/2$), that is, the unique number in $[0, \nu_c/2]$ congruent to $\pm\nu_0$ modulo ν_c . If $\nu_0 \leq \nu_c/2$ one has $k = 0$ and $\nu_a = \nu_0$; if $\nu_0 > \nu_c/2$ one has $\nu_a < \nu_0$ and one speaks of **aliasing** or **folding**.

Theorem 24.10: Two sinusoids indistinguishable in the samples. Let $\nu_0 \geq 0$, $\nu_c > 0$, $A \in \mathbb{R}$, $\varphi_0 \in \mathbb{R}$, $T = 1/\nu_c$, and let ν_a be the alias frequency of Definition 24.11, with k the integer defined there. Set

$$x(t) = A \cos(2\pi\nu_0 t + \varphi_0), \quad \tilde{x}(t) = A \cos(2\pi\nu_a t + \varepsilon\varphi_0), \quad \varepsilon = \text{sgn}(\nu_0 - k\nu_c)$$

(with $\varepsilon = 1$ if $\nu_0 = k\nu_c$). Then

$$x(nT) = \tilde{x}(nT) \quad \text{for every } n \in \mathbb{Z} :$$

the two signals, of different frequencies, produce the same sample sequence, and no processor working only with the samples can distinguish them.

Proof. Step 1 — reduction modulo ν_c . Set $\mu = \nu_0 - k\nu_c$, so that $\nu_a = |\mu|$ and $\mu = \varepsilon\nu_a$ by the definition of ε . For every $n \in \mathbb{Z}$, since $T = 1/\nu_c$,

$$2\pi\nu_0 nT + \varphi_0 = 2\pi(\mu + k\nu_c)\frac{n}{\nu_c} + \varphi_0 = 2\pi\mu nT + 2\pi kn + \varphi_0.$$

Step 2 — periodicity of the cosine. Since $kn \in \mathbb{Z}$ and the cosine is 2π -periodic, $\cos(\alpha + 2\pi kn) = \cos \alpha$ for every α ; hence

$$x(nT) = A \cos(2\pi\mu nT + \varphi_0).$$

Step 3 — evenness of the cosine. If $\varepsilon = 1$, then $\mu = \nu_a$ and Step 2 already gives $x(nT) = A \cos(2\pi\nu_a nT + \varphi_0) = \tilde{x}(nT)$. If $\varepsilon = -1$, then $\mu = -\nu_a$ and, using $\cos(-\beta) = \cos \beta$ with $\beta = 2\pi\nu_a nT - \varphi_0$,

$$x(nT) = A \cos(-2\pi\nu_a nT + \varphi_0) = A \cos(2\pi\nu_a nT - \varphi_0) = A \cos(2\pi\nu_a nT + \varepsilon\varphi_0) = \tilde{x}(nT).$$

In both cases the equality holds for every $n \in \mathbb{Z}$. $\square$

Example 24.8: The numerical example, checked sample by sample. Sample at $\nu_c = 1\,000$ Hz, i.e. with $T = 1$ ms; the Nyquist frequency is 500 Hz.

(a) A component at $\nu_0 = 1\,300$ Hz. Here $\nu_0/\nu_c = 1.3$, hence $k = 1$ and $\mu = 1\,300 - 1\,000 = 300$ Hz, with $\varepsilon = +1$: the alias frequency is $\nu_a = 300$ Hz. The first six samples of $\cos(2\pi \cdot 1\,300 t)$, computed by hand from standard angles — $2\pi \cdot 1\,300 \cdot n/1\,000 = 2.6\pi n$, which modulo 2π equals $0.6\pi n$, i.e. $108^\circ \cdot n$ — are

$$1, \quad \cos 108^\circ, \quad \cos 216^\circ, \quad \cos 324^\circ, \quad \cos 432^\circ, \quad \cos 540^\circ = 1, \quad -0.309\,017, \quad -0.809\,017, \quad 0.809\,017, \quad 0.309\,017,$$

and are *exactly* the samples of $\cos(2\pi \cdot 300 t)$, for which $2\pi \cdot 300 \cdot n/1\,000 = 0.6\pi n$, i.e. the same angle. $\checkmark$

(b) A component at $\nu_0 = 700$ Hz. Here $\nu_0/\nu_c = 0.7$, hence $k = 1$ and $\mu = 700 - 1\,000 = -300$ Hz, with $\varepsilon = -1$: again $\nu_a = 300$ Hz, but with the phase sign changed. The two components at 700 and 1 300 Hz therefore fall *on the same* alias: a spectrum computed from the samples sums them without being able to separate them.

Independent check (Python script executed). Over 101 samples, from $n = -50$ to $n = 50$, the maximum gap between $\cos(2\pi \cdot 1\,300 nT)$ and $\cos(2\pi \cdot 300 nT)$ equals $5.64 \cdot 10^{-14}$; between $\cos(2\pi \cdot 700 nT)$ and $\cos(2\pi \cdot 300 nT)$ it equals $4.52 \cdot 10^{-14}$; and, with the phase, between $\cos(2\pi \cdot 700 nT + 0.7)$ and $\cos(2\pi \cdot 300 nT - 0.7)$ it equals $5.53 \cdot 10^{-14}$, also confirming the sign of ε in Theorem 24.10. $\checkmark$ All gaps are of the order of double-precision machine error.

(c) The industrial case. A sensor read by a PLC at $\nu_c = 50$ Hz and disturbed by the mains at $\nu_0 = 60$ Hz: $\nu_0/\nu_c = 1.2$, $k = 1$, $\nu_a = |60 - 50| = 10$ Hz. The mains disturbance appears in the data as a slow 10 Hz oscillation, indistinguishable from a real variation of the process. The maximum gap between the samples of $\cos(2\pi \cdot 60 t)$ and of $\cos(2\pi \cdot 10 t)$, over 101 samples, is $6.43 \cdot 10^{-14}$. $\checkmark$ No *digital* filter, however sophisticated, can remove it: the only defence is the *analog* anti-aliasing filter of Remark 24.17, which acts before conversion.

(d) A limiting case. If $\nu_0 = 2\,000$ Hz, then $k = 2$ and $\nu_a = 0$: the sampled signal is constant, and the sinusoid disappears entirely. This is the phenomenon that makes wagon wheels appear motionless in films shot at 24 frames per second.

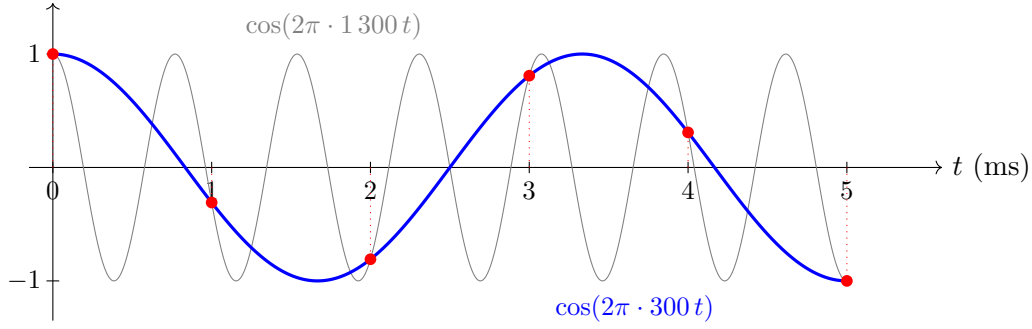


Figure 24.4: Aliasing, Example 24.8(a): sampling at $\nu_c = 1000$ Hz (step 1 ms) the 1300 Hz sinusoid (grey) and the 300 Hz one (blue) pass through the *same* six red points, whose values 1, -0.309017 , -0.809017 , 0.809017 , 0.309017 , -1 are the cosines of the angles 0° , 108° , 216° , 324° , 432° , 540° , computed by hand and confirmed by the script. From the samples alone the two sinusoids are indistinguishable, as Theorem 24.10 states.

Best practice. How to choose a sampling frequency. (1) Establish the band $\nu_{\max}$ of the *useful signal*, not of the signal one hopes to measure. (2) Insert an *analog* anti-aliasing filter that attenuates everything above $\nu_c/2$ by at least the resolution of the converter (for a 12-bit A/D, about 72 dB). (3) Choose ν_c equal to 5–10 times $\nu_{\max}$, not 2 times: the factor 2 is the mathematical limit of Theorem 24.8, not a design target, and the margin serves the non-ideal transition of the filter (Remark 24.17). (4) Verify the result by feeding the system a sinusoid of known frequency *above* $\nu_c/2$ and checking that its alias does not appear in the spectrum: if it appears, the filter is not attenuating enough. The computation of the expected alias is that of Definition 24.11, and must be done *before* the test, not after.

24.11. The analytic signal and the Hilbert transform

By Theorem 24.6(iii) the spectrum of a real signal is redundant: the negative half is the conjugate reflection of the positive half, and carries no new information. The **analytic signal** is the canonical way of eliminating the redundancy — the negative frequencies are suppressed and the positive ones are doubled — and the **Hilbert transform** is the operator that this suppression requires. This section defines both, proves the fact that justifies them, and states precisely the perimeter: what is proved here and what remains imported.

Definition 24.12: Hilbert transform on $L^2(\mathbb{R})$. For $x \in L^2(\mathbb{R})$, the **Hilbert transform** $\mathcal{H}x \in L^2(\mathbb{R})$ is defined as the unique function of $L^2(\mathbb{R})$ whose Fourier transform is

$$\widehat{\mathcal{H}x}(\omega) = -i \operatorname{sgn}(\omega) \hat{x}(\omega) \quad \text{for almost every } \omega \in \mathbb{R}.$$

$\mathcal{H}$ is also said to be the *filter* with frequency response $H_{\mathcal{H}}(\omega) = -i \operatorname{sgn}(\omega)$.

Proposition 24.4: The definition is well posed, and $\mathcal{H}$ is an isometry. Definition 24.12 determines a unique $\mathcal{H}x \in L^2(\mathbb{R})$ for every $x \in L^2(\mathbb{R})$, and the operator $\mathcal{H} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ so obtained is linear and satisfies

$$\|\mathcal{H}x\|_2 = \|x\|_2 \quad \text{and} \quad \mathcal{H}(\mathcal{H}x) = -x \quad \text{for every } x \in L^2(\mathbb{R}).$$

(The value of the multiplier at the single point $\omega = 0$ is irrelevant, because $\{0\}$ has measure zero and no L^2 norm is affected by it.)

Proof. Step 1 — existence and uniqueness. By Corollary 19.4 of Chapter 19 the Fourier transform is an isometric bijection (up to the factor $\sqrt{2\pi}$ of the convention) of $L^2(\mathbb{R})$ onto itself. The function $\omega \mapsto -i \operatorname{sgn}(\omega) \hat{x}(\omega)$ is measurable and satisfies $|-i \operatorname{sgn}(\omega) \hat{x}(\omega)| \leq |\hat{x}(\omega)|$ for every ω , hence it belongs to $L^2(\mathbb{R})$: it therefore has a unique inverse transform in $L^2(\mathbb{R})$, which is $\mathcal{H}x$. Uniqueness is that of the inverse transform, i.e. the injectivity of the transform on L^2 .

Step 2 — isometry. Since $|-i \operatorname{sgn}(\omega)| = 1$ for every $\omega \neq 0$, one has $|\widehat{\mathcal{H}x}(\omega)| = |\hat{x}(\omega)|$ for almost every ω (the only excluded point is $\omega = 0$, a set of measure zero). Hence $\|\widehat{\mathcal{H}x}\|_2 = \|\hat{x}\|_2$, and by Plancherel's isometry (Corollary 19.4) $\|\mathcal{H}x\|_2 = \|x\|_2$.

Step 3 — $\mathcal{H}^2 = -\operatorname{id}$. Applying the definition twice,

$$\widehat{\mathcal{H}(\mathcal{H}x)}(\omega) = -i \operatorname{sgn}(\omega) \widehat{\mathcal{H}x}(\omega) = (-i \operatorname{sgn}(\omega))^2 \hat{x}(\omega) = -\operatorname{sgn}(\omega)^2 \hat{x}(\omega) = -\hat{x}(\omega)$$

for almost every ω , because $\operatorname{sgn}(\omega)^2 = 1$ for $\omega \neq 0$ and $\{0\}$ has measure zero. By injectivity of the transform on L^2 , $\mathcal{H}(\mathcal{H}x) = -x$. $\square$

Definition 24.13: Analytic signal. Let $x \in L^2(\mathbb{R})$ be real-valued. Its **analytic signal** is

$$x_a = x + i \mathcal{H}x \in L^2(\mathbb{R}; \mathbb{C}).$$

The modulus $A(t) = |x_a(t)|$ is called the **envelope** of x , and, where x_a does not vanish, any continuous determination ψ of $\arg x_a$ is called the **instantaneous phase**; its derivative $\psi'(t)/(2\pi)$, where it exists, is the **instantaneous frequency**.

Theorem 24.11: The analytic signal suppresses the negative frequencies. Let $x \in L^2(\mathbb{R})$ and $x_a = x + i \mathcal{H}x$. Then, for almost every $\omega \in \mathbb{R}$,

$$\hat{x}_a(\omega) = (1 + \operatorname{sgn}(\omega)) \hat{x}(\omega) = \begin{cases} 2 \hat{x}(\omega), & \omega > 0, \\ \hat{x}(0), & \omega = 0, \\ 0, & \omega < 0. \end{cases}$$

In particular the spectrum of x_a is zero on the whole negative-frequency half-line, and on the positive one it is twice that of x .

Proof. Step 1 — linearity of the transform. Since $x \in L^2(\mathbb{R})$ and $\mathcal{H}x \in L^2(\mathbb{R})$ by Proposition 24.4, also $x_a = x + i \mathcal{H}x \in L^2(\mathbb{R}; \mathbb{C})$, and the Fourier transform on L^2 is linear (Corollary 19.4):

$$\hat{x}_a(\omega) = \hat{x}(\omega) + i \widehat{\mathcal{H}x}(\omega).$$

Step 2 — substituting the definition. By Definition 24.12, for almost every ω ,

$$\hat{x}_a(\omega) = \hat{x}(\omega) + i \cdot (-i \operatorname{sgn}(\omega)) \hat{x}(\omega) = \hat{x}(\omega) + (-i^2) \operatorname{sgn}(\omega) \hat{x}(\omega) = \hat{x}(\omega) + \operatorname{sgn}(\omega) \hat{x}(\omega) = (1 + \operatorname{sgn}(\omega)) \hat{x}(\omega),$$

having used $i \cdot (-i) = -i^2 = 1$.

Step 3 — the three cases. By definition of sgn : for $\omega > 0$ one has $1 + \operatorname{sgn}(\omega) = 1 + 1 = 2$; for $\omega < 0$ one has $1 + \operatorname{sgn}(\omega) = 1 - 1 = 0$; for $\omega = 0$ one has $1 + \operatorname{sgn}(0) = 1 + 0 = 1$. Substituting gives the three lines of the statement. $\square$

Corollary 24.5: Recovering the real signal and the envelope. With the notation of Definition 24.13, for x real one has

$$x = \operatorname{Re} x_a, \quad \mathcal{H}x = \operatorname{Im} x_a, \quad |x(t)| \leq A(t) = |x_a(t)| \text{ for every } t.$$

Proof. Step 1 — the spectrum of a real signal of L^2 is Hermitian. Let $x \in L^2(\mathbb{R})$ be real and let $x_N = x \mathbf{1}_{[-N, N]}$ be its truncations. By Cauchy–Schwarz $\int_{-N}^N |x| \leq \sqrt{2N} \|x\|_2 < \infty$, hence $x_N \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, and it is real; moreover $x_N \rightarrow x$ in $L^2(\mathbb{R})$ by dominated convergence. Theorem 24.6(iii), applied to $x_N \in L^1(\mathbb{R})$ real, gives $\hat{x}_N(-\omega) = \overline{\hat{x}_N(\omega)}$ for every ω . By Plancherel’s isometry (Corollary 19.4) one has $\hat{x}_N \rightarrow \hat{x}$ in $L^2(\mathbb{R})$, hence there is a subsequence along which $\hat{x}_N \rightarrow \hat{x}$ a.e.; the reflection $\omega \mapsto -\omega$ sends sets of measure zero to sets of measure zero, so along the same subsequence also $\hat{x}_N(-\cdot) \rightarrow \hat{x}(-\cdot)$ a.e. Passing to the limit on both sides (conjugation is continuous) gives $\hat{x}(-\omega) = \overline{\hat{x}(\omega)}$ for almost every ω .

Step 2 — $\mathcal{H}x$ is also real. For almost every ω ,

$$\widehat{\mathcal{H}x}(-\omega) = -i \operatorname{sgn}(-\omega) \hat{x}(-\omega) = i \operatorname{sgn}(\omega) \overline{\hat{x}(\omega)} = \overline{-i \operatorname{sgn}(\omega) \hat{x}(\omega)} = \overline{\widehat{\mathcal{H}x}(\omega)},$$

using $\operatorname{sgn}(-\omega) = -\operatorname{sgn}(\omega)$, Step 1 and $\overline{-iz} = iz$. A function of L^2 whose spectrum has Hermitian symmetry is real a.e.: indeed the function $\widehat{\mathcal{H}x}$ has as its transform $\omega \mapsto \widehat{\mathcal{H}x}(-\omega)$, which by what was just shown coincides a.e. with $\widehat{\mathcal{H}x}$; by injectivity of the transform on L^2 , $\widehat{\mathcal{H}x} = \mathcal{H}x$ a.e.

Step 3 — conclusion. Since x and $\mathcal{H}x$ are both real, the expression $x_a = x + i\mathcal{H}x$ is exactly the decomposition of x_a into real part and imaginary part: $x = \operatorname{Re} x_a$ and $\mathcal{H}x = \operatorname{Im} x_a$. Finally, for every $z = \alpha + i\beta$ with α, β real one has $|\alpha| \leq \sqrt{\alpha^2 + \beta^2} = |z|$: applied to $z = x_a(t)$ this gives $|x(t)| \leq A(t)$. $\square$

Definition 24.14: Extension to trigonometric polynomials. A sinusoid does not belong to $L^2(\mathbb{R})$ (Remark 24.1), and Definition 24.12 does not apply to it. For a **trigonometric polynomial** $p(t) = \sum_{j=1}^m c_j e^{i\omega_j t}$ (a *finite* sum, with $c_j \in \mathbb{C}$ and $\omega_j \in \mathbb{R}$) one then defines

$$\mathcal{H}p := \sum_{j=1}^m (-i \operatorname{sgn}(\omega_j)) c_j e^{i\omega_j t},$$

i.e. by applying the multiplier $-i \operatorname{sgn}$ to each exponential, as if it were an eigenfunction in the sense of Theorem 24.2.

Proposition 24.5: Hilbert sends the cosine to the sine. Let $\omega_0 > 0$. With Definition 24.14,

$$\mathcal{H}[\cos(\omega_0 t)] = \sin(\omega_0 t), \quad \mathcal{H}[\sin(\omega_0 t)] = -\cos(\omega_0 t),$$

and the analytic signal of the cosine is the complex exponential:

$$(\cos(\omega_0 \cdot))_a(t) = \cos(\omega_0 t) + i \sin(\omega_0 t) = e^{i\omega_0 t},$$

whose envelope is the constant $A \equiv 1$ and whose instantaneous frequency is the constant $\omega_0/(2\pi)$.

Proof. Step 1 — the cosine. By Euler’s formula, $\cos(\omega_0 t) = \frac{1}{2}e^{i\omega_0 t} + \frac{1}{2}e^{-i\omega_0 t}$: this is a trigonometric polynomial with $m = 2$, $c_1 = c_2 = \frac{1}{2}$, $\omega_1 = \omega_0 > 0$, $\omega_2 = -\omega_0 < 0$. By Definition 24.14,

$$\mathcal{H}[\cos(\omega_0 t)] = \frac{1}{2}(-i)(+1)e^{i\omega_0 t} + \frac{1}{2}(-i)(-1)e^{-i\omega_0 t} = -\frac{i}{2}(e^{i\omega_0 t} - e^{-i\omega_0 t}) = -\frac{i}{2} \cdot 2i \sin(\omega_0 t) = \sin(\omega_0 t),$$

using $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ and $-i \cdot i = 1$.

Step 2 — the sine. Similarly $\sin(\omega_0 t) = \frac{1}{2i}e^{i\omega_0 t} - \frac{1}{2i}e^{-i\omega_0 t}$, hence

$$\mathcal{H}[\sin(\omega_0 t)] = \frac{-i}{2i}e^{i\omega_0 t} - \frac{(-i)(-1)}{2i}e^{-i\omega_0 t} = -\frac{1}{2}e^{i\omega_0 t} - \frac{1}{2}e^{-i\omega_0 t} = -\cos(\omega_0 t),$$

having simplified $-i/(2i) = -1/2$ and $i/(2i) = 1/2$. Note that, consistently with Proposition 24.4, applying $\mathcal{H}$ twice to the cosine gives $\mathcal{H}[\sin] = -\cos$: i.e. $\mathcal{H}^2 = -\text{id}$ also on these signals.

Step 3 — the analytic signal. By Definition 24.13 and Step 1, $(\cos(\omega_0 \cdot))_a(t) = \cos(\omega_0 t) + i \sin(\omega_0 t)$, which is $e^{i\omega_0 t}$ by Euler's formula. Its modulus is $|e^{i\omega_0 t}| = 1$ for every t ; a continuous determination of the argument is $\psi(t) = \omega_0 t$, and $\psi'(t)/(2\pi) = \omega_0/(2\pi)$.

Independent numerical check. With $\omega_0 = 2\pi \cdot 5$ rad/s, at $t = 0.017$ s one has $\cos(\omega_0 t) + i \sin(\omega_0 t) = 0.860\,742\,027\,0 + 0.509\,041\,415\,8\,i$, which coincides with $e^{i\omega_0 t}$ computed independently within machine error, with modulus 1.000 000 0; the instantaneous frequency computed by centred differences with step 10^{-7} equals 4.999 999 999 89 Hz against the expected value of 5 Hz. $\checkmark$ $\square$

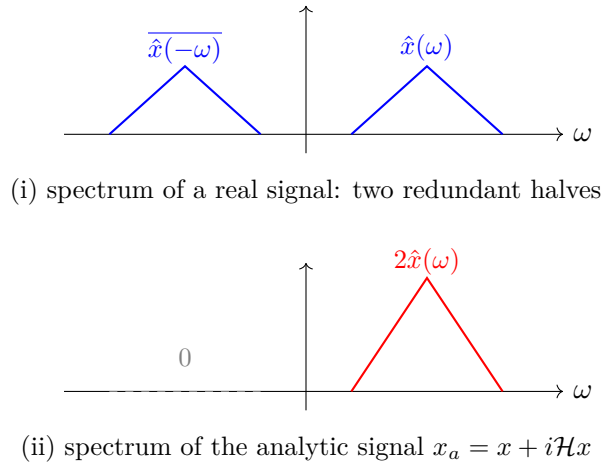


Figure 24.5: Theorem 24.11 in a picture. The spectrum of a real signal (i) has two halves linked by the Hermitian symmetry of Theorem 24.6(iii): the negative half carries no new information. The analytic signal (ii) eliminates it and doubles the positive half, so that the energy is conserved: indeed, by Theorem 24.11 and by Plancherel, $\|x_a\|_2^2 = \frac{1}{2\pi} \int_0^{+\infty} 4|\hat{x}|^2 = \frac{2}{2\pi} \int_{\mathbb{R}} |\hat{x}|^2 = 2\|x\|_2^2$, consistent with $\|x_a\|_2^2 = \|x\|_2^2 + \|\mathcal{H}x\|_2^2 = 2\|x\|_2^2$ of Proposition 24.4. The two vertices are at $(\pm 1.6, 0.9)$ in panel (i) and at $(1.6, 1.5)$ in panel (ii): the ratio $1.5/0.9 = 5/3$ is only graphical, the figure is schematic and not to scale in energy.

Remark 24.19: What we import about the Hilbert transform. Of this section, everything concerning the multiplier is proved in full: the well-posedness on L^2 (Proposition 24.4), the isometry, the identity $\mathcal{H}^2 = -\text{id}$, the suppression of the negative frequencies (Theorem 24.11) and the computation on cosine and sine (Proposition 24.5). What remains **imported**, that is, stated and used but not proved here, is the following.

(H1) The integral form. For $x \in L^2(\mathbb{R})$, for almost every t ,

$$(\mathcal{H}x)(t) = \frac{1}{\pi} \text{p.v.} \int_{\mathbb{R}} \frac{x(s)}{t-s} ds := \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0^+} \int_{|t-s| > \varepsilon} \frac{x(s)}{t-s} ds.$$

The equivalence between this formula and Definition 24.12 requires the theory of singular integrals (the kernel $1/(\pi t)$ is not integrable in any neighbourhood of the origin, and lies neither in L^1 nor in L^2 : the system does not fall under Definition 24.4). What is exactly missing: the proof that the limit exists a.e. and that the limit function has transform $-i \operatorname{sgn} \hat{x}$.

(H2) The M. Riesz theorem. $\mathcal{H}$ extends to a bounded operator on $L^p(\mathbb{R})$ for every $1 < p < \infty$, but not on L^1 nor on L^∞ . What is missing: the interpolation and the weak-type estimates on which the proof rests.

(H3) The Bedrosian theorem. If x is narrowband around ω_0 and a has band contained in $[-\omega_0, \omega_0]$, then $\mathcal{H}[a \cdot x] = a \cdot \mathcal{H}[x]$; this is the result that makes it legitimate to call the modulus $A(t)$ the “envelope” in a modulated signal $a(t) \cos(\omega_0 t)$. What is missing: the proof, which is a computation on frequency supports but requires (H1).

The source of (H1) and (H2) is King, *Hilbert Transforms*, entry [6] of the *Sources* environment; that of (H3), together with the engineering use of the envelope, is Oppenheim and Schaffer, entry [4]. None of these three results is used in any statement or proof of this chapter: they appear only in this remark, as **panorama** of what the theory contains beyond the perimeter covered here.

Remark 24.20: The envelope is not what it seems. Envelope and instantaneous frequency are defined for every real signal of $L^2(\mathbb{R})$, but they are useful only for narrowband signals. For an arbitrary signal the instantaneous frequency can turn out negative, or diverge where x_a vanishes, without this indicating anything physical: it is an artifact of the definition. The criterion that makes the concept reliable is the Bedrosian condition (H3) of Remark 24.19, and it must be checked before interpreting the results — for instance before reading the envelope of a vibration signal as the “instantaneous amplitude of the defect”. The case of Proposition 24.5 is the ideal case, and should be used as a benchmark: there the envelope is exactly constant and the instantaneous frequency exactly $\omega_0/(2\pi)$.

24.12. Exercises inherited from Volume II

The exercises below come from Volume II, Chapter 4, and concern the topics of this chapter. They are reproduced in full, together with their solutions, which follow in the same order as the exercises.

Exercises

Exercise 24.1. (Application to water treatment) — A turbidity sensor in a water treatment line has dynamics with significant frequency components up to 2 Hz. What is the minimum sampling frequency imposed by Nyquist? Why would an engineer in practice choose a higher PLC sampling frequency, for example 20 Hz?

Solutions

Solution. By Theorem 24.8, the minimum sampling frequency is $f_c \geq 2f_{\max} = 2 \times 2 \text{ Hz} = 4 \text{ Hz}$: this is the absolute theoretical limit below which information is irreversibly lost to aliasing.

In practice, however, an engineer would choose a higher frequency (for example 20 Hz, i.e. 5 times the bandwidth) for at least three reasons, already discussed in Remark 24.17: (a) real anti-aliasing filters do not have a sharp cutoff at $f_c/2$, so a margin is needed for the components just above 2 Hz to be sufficiently attenuated before sampling; (b) sampling exactly at the Nyquist limit introduces a phase lag and poor time resolution in the control loop (few samples per period of the fastest signal degrade the performance of the PID controller); (c) a margin also makes it possible to observe correctly any disturbances or noise at a frequency slightly above $f_{\max}$ without their aliasing into the useful band of the turbidity signal.

Inherited quiz

The items below come from the quiz of Volume II, Chapter 4, and are renumbered here.

1. The Nyquist-Shannon theorem requires a sampling frequency
 - A) equal to the bandwidth of the signal
 - B) at least twice the maximum frequency
 - C) arbitrary
 - D) lower than the bandwidth

BEST PRACTICE

Answer key.

1-B: Theorem 24.8.

24.13. Assigned exercises

These are *new* exercises, distinct both from the inherited ones of Section 24.12 and from the worked ones of Section 24.14. The answers are in the Teacher’s key (Section 24.16).

Exercise 24.2: A first-order filter, from beginning to end. *Goal:* to walk through the whole chain of the chapter on a single system — impulse response, stability, causality, frequency response, the two delays — and to know how to translate every number into a sentence about the physical behaviour. *Estimated time:* 45 minutes. *Type:* symbolic and numerical. *Level:* basic–intermediate. Consider the low-pass RC filter with $\tau = RC = 5$ ms.

- (a) Write h , verify $h \in L^1(\mathbb{R})$ by computing $\|h\|_1$ in closed form, and deduce BIBO stability and the gain from Theorem 24.4. Say why the system is causal by citing Theorem 24.5.
- (b) Compute $H(\omega)$ by direct integration, then $|H|$ and ϕ . Determine the -3 dB frequency in hertz and verify that at that frequency the gain equals $1/\sqrt{2}$ and the phase -45° .
- (c) Compute τ_p and τ_g in milliseconds at $\omega = 1/\tau$ and at $\omega = 3/\tau$, and the limiting value $\tau_g(0)$. Say which of the two delays is read as the “delay of the envelope” and why (Theorem 24.7).
- (d) Does the system satisfy the no-distortion condition of Corollary 24.3 on some band? Justify the answer *without* computing anything new, using only the form of $\tau_g(\omega)$.
- (e) Verify (b) and (c) with a script: H by numerical quadrature of the defining integral, τ_g by centred differences; report the gaps.

Exercise 24.3: Linear phase is not enough. *Goal:* to distinguish the *two* conditions of Corollary 24.3, which are often conflated into one. *Estimated time:* 40 minutes. *Type:* proof-based and numerical. *Level:* intermediate. Consider the moving average with window $T_0 = 10$ ms.

- (a) Derive $H(\omega)$ by retracing the computation of Example 24.5, and determine the frequencies in hertz of the first three zeros of the gain.
- (b) Show that, on every frequency interval where $\sin(\omega T_0/2) > 0$, the continuous phase is exactly $\phi(\omega) = -\omega T_0/2$ and hence $\tau_g \equiv T_0/2$, constant. What is its value in milliseconds?
- (c) Despite (b), the system *distorts* a signal with a band reaching 150 Hz. Explain why, indicating which of the two hypotheses of Corollary 24.3 fails, and compute $|H|$ at 25, 50, 100 and 150 Hz.

- (d) Determine the maximum band $\nu_{\max}$ within which the gain does not drop below 0.9, and verify that within that band the “no distortion” approximation is reasonable. (It suffices to solve $\text{sinc}(\nu T_0) = 0.9$ numerically by trial.)
- (e) What changes if the moving average is replaced with an RC filter of equal low-frequency group delay, i.e. with $\tau = T_0/2$? Compare the two $\tau_g(\omega)$ and comment.

Exercise 24.4: Sampling an industrial signal. *Goal:* to apply in sequence Theorem 24.8, Theorem 24.9 and Theorem 24.10 to a concrete case, and to know how to compute *before* the measurement what will appear in the spectrum. *Estimated time:* 50 minutes. *Type:* numerical and computational. *Level:* intermediate. An accelerometer on a gearbox produces a signal with significant components at 250, 1 200 and 3 000 Hz, plus a disturbance at 9 500 Hz and a resonance at 12 000 Hz. The converter operates at $\nu_c = 8\,000$ Hz.

- (a) Which components satisfy the condition of Theorem 24.8? What is the Nyquist frequency?
- (b) Compute, with Definition 24.11, the alias frequency of each of the two out-of-band components. In which zone of the measured spectrum will they appear, and with what risk of being mistaken for real phenomena?
- (c) Verify with a script that the samples of $\cos(2\pi \cdot 9\,500 t)$ at $\nu_c = 8\,000$ Hz coincide with those of the sinusoid at the alias frequency computed in (b), reporting the maximum gap over at least 200 samples; repeat with a nonzero initial phase and verify the sign of ε in Theorem 24.10.
- (d) Design the countermeasure: which sampling frequency would you choose, and which anti-aliasing filter? Apply the four points of the Good practice of Section 24.10 and justify each choice with a number.
- (e) Assuming the two out-of-band components have been eliminated, write explicitly the reconstruction formula of Theorem 24.9 for $\nu_c = 8\,000$ Hz and evaluate its first nonzero term at $t = 1/16\,000$ s.

24.14. Worked exercises

Exercise 24.5: Two tones through an RC : the waveform changes. *Type:* symbolic and numerical computation with cross-check. Consider the RC filter with $\tau = 1$ ms and the input

$$x(t) = \cos(\omega_1 t) + \cos(\omega_2 t), \quad \omega_1 = \frac{1}{\tau} = 1\,000, \quad \omega_2 = \frac{3}{\tau} = 3\,000 \text{ rad/s}.$$

Compute the output, show that it is not of the form $Ax(t - t_0)$ for any pair (A, t_0) , and quantify the distortion.

Solution. Strategy. The input is a *finite* sum of sinusoids: by Corollary 24.1 and by linearity it suffices to treat them one at a time, without ever computing a convolution integral. No distortion, by Corollary 24.3, requires the two tones to come out with the same gain *and* with the same delay: one checks that neither condition holds.

Step 1 — the two values of the frequency response. From the closed form $H(\omega) = 1/(1 + i\omega\tau)$ of Example 24.4, with $\omega_1\tau = 1$ and $\omega_2\tau = 3$:

$$H(\omega_1) = \frac{1}{1 + i} = \frac{1 - i}{2}, \quad H(\omega_2) = \frac{1}{1 + 3i} = \frac{1 - 3i}{10},$$

having rationalised with the conjugate: $\frac{1}{1+i} = \frac{1-i}{(1+i)(1-i)} = \frac{1-i}{2}$ and $\frac{1}{1+3i} = \frac{1-3i}{1+9} = \frac{1-3i}{10}$. Hence

$$|H(\omega_1)| = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} = 0.707\,106\,781\,2, \quad |H(\omega_2)| = \frac{\sqrt{1+9}}{10} = \frac{1}{\sqrt{10}} = 0.316\,227\,766\,0,$$

$$\phi_1 = -\arctan 1 = -\frac{\pi}{4} = -0.785\,398\,163\,4 \text{ rad} = -45^\circ, \quad \phi_2 = -\arctan 3 = -1.249\,045\,772\,4 \text{ rad} = -71.565^\circ$$

Step 2 — the output. By Corollary 24.1 applied to each term and by linearity (Proposition 24.2),

$$y(t) = 0.707\,106\,781\,2 \cos(1\,000t - 0.785\,398\,163\,4) + 0.316\,227\,766\,0 \cos(3\,000t - 1.249\,045\,772\,4).$$

Step 3 — the two phase delays do not coincide. For y to be $Ax(t - t_0)$ one would need $\phi_j = -\omega_j t_0$ with the *same* t_0 for $j = 1, 2$, i.e. $\tau_p(\omega_1) = \tau_p(\omega_2)$. Instead

$$\tau_p(\omega_1) = \frac{0.785\,398\,163\,4}{1\,000} = 0.785\,398\,163\,4 \text{ ms}, \quad \tau_p(\omega_2) = \frac{1.249\,045\,772\,4}{3\,000} = 0.416\,348\,590\,8 \text{ ms},$$

whose ratio equals 1.886 395 632 8: the first tone comes out delayed by almost twice the second. This alone excludes the form $Ax(t - t_0)$.

Step 4 — nor do the gains coincide. The ratio of the two gains is $|H(\omega_1)|/|H(\omega_2)| = \sqrt{10}/\sqrt{2} = \sqrt{5} = 2.236\,067\,977\,5$: the high tone comes out attenuated more than twice as much as the low one. So the first of the two conditions of Corollary 24.3 also fails.

Independent check 1: the two group delays. From the closed formula $\tau_g(\omega) = \tau/(1 + \omega^2\tau^2)$: $\tau_g(\omega_1) = \tau/2 = 0.5 \text{ ms}$ and $\tau_g(\omega_2) = \tau/10 = 0.1 \text{ ms}$, in an exact ratio of 5. The computation by centred differences with step 10^{-6} gives 0.499 999 985 9 ms and 0.100 000 008 3 ms, with gaps $1.41 \cdot 10^{-11}$ and $8.27 \cdot 10^{-12}$ ms. ✓

Independent check 2: the quadrature. Computing $H(\omega_1)$ and $H(\omega_2)$ by Simpson's quadrature of the integral $\int_0^{60\tau} h(t)e^{-i\omega t} dt$ with $6 \cdot 10^5$ subintervals gives $0.5 - 0.5i$ and $0.1 - 0.3i$ with gaps $2.14 \cdot 10^{-13}$ and $1.36 \cdot 10^{-14}$ from the closed values $\frac{1-i}{2}$ and $\frac{1-3i}{10}$. ✓

Interpretation. The filter does not “delay the signal”: it delays *each component by a different time* and changes their proportions. The result is a new waveform, not a shifted copy. If the tone at 3 000 rad/s were the diagnostically interesting part (for instance a defect harmonic), it would come out attenuated by a factor of $\sqrt{5}$ relative to the low tone: its relative amplitude would be reduced by $1 - 1/\sqrt{5} = 0.552\,786\,4$, i.e. by over 55%.

Typical errors. (1) Adding the magnitudes instead of treating the components separately: the gain is not a single number, it is a function of ω . (2) Confusing τ_p and τ_g : here $\tau_p(\omega_2) = 0.416 \text{ ms}$ and $\tau_g(\omega_2) = 0.100 \text{ ms}$, a factor-of-4 difference. (3) Concluding “there is no distortion because the frequency does not change”: Corollary 24.1 guarantees that frequencies *never* change in an LTI system; it is precisely for this reason that distortion, when present, concerns relative amplitudes and phases.

Extension. Repeat with the moving average of window $T_0 = 2 \text{ ms}$ (Example 24.5) and verify that the two phase delays coincide — both $T_0/2 = 1 \text{ ms}$ — while the gains do not: the residual distortion is only in amplitude, and can be corrected with an equaliser, while phase distortion cannot.

Exercise 24.6: The input that saturates the gain. *Type:* proof-based with numerical check. Let $h = \mathbf{1}_{[0,1]} - \mathbf{1}_{[1,2]}$. Compute $\|h\|_1$ and $\sup_\omega |H(\omega)|$, construct the critical input of Step 2 of the proof of Theorem 24.4 and verify that it actually attains the gain, while no sinusoid reaches it.

Solution. Strategy. Theorem 24.4 says the gain is $\|h\|_1$; Theorem 24.6(iv) says the maximum of $|H|$ can be strictly smaller. Here both numbers are computed and compared.

Step 1 — the L^1 norm. $|h| = \mathbf{1}_{[0,2]}$, hence $\|h\|_1 = \int_0^2 1 dt = 2$. By Theorem 24.4 the system is BIBO stable with gain exactly 2.

Step 2 — the frequency response. For $\omega \neq 0$,

$$H(\omega) = \int_0^1 e^{-i\omega t} dt - \int_1^2 e^{-i\omega t} dt = \frac{1 - e^{-i\omega}}{i\omega} - \frac{e^{-i\omega} - e^{-2i\omega}}{i\omega} = \frac{1 - 2e^{-i\omega} + e^{-2i\omega}}{i\omega} = \frac{(1 - e^{-i\omega})^2}{i\omega},$$

recognising the square of a binomial. The modulus is

$$|H(\omega)| = \frac{|1 - e^{-i\omega}|^2}{|\omega|} = \frac{2 - 2\cos\omega}{|\omega|} = \frac{4\sin^2(\omega/2)}{|\omega|},$$

and $H(0) = \int_0^1 1 dt - \int_1^2 1 dt = 0$.

Step 3 — the maximum of $|H|$. Setting the derivative of $4\sin^2(\omega/2)/\omega$ to zero for $\omega > 0$ gives $2\sin(\omega/2)\cos(\omega/2)\omega = 2\sin^2(\omega/2)$, i.e. $\omega\sin\omega = 2\sin^2(\omega/2) = 1 - \cos\omega$, which with $\sin\omega = 2\sin(\omega/2)\cos(\omega/2)$ and $1 - \cos\omega = 2\sin^2(\omega/2)$ reduces to $\tan(\omega/2) = \omega$. The smallest positive root, found numerically, is $\omega^* = 2.331\,122\,3$, where $|H| = 1.449\,222\,707\,6$. ✓ (Two independent computations agree: an exhaustive search with step 10^{-6} on $(0, 40)$ and a refining ternary search; and the check $\tan(\omega^*/2) = 2.331\,122\,3$ of the stationarity condition.) Hence

$$\sup_{\omega} |H(\omega)| = 1.449\,222\,707\,6 < 2 = \|h\|_1.$$

Step 4 — the critical input. Since h is real, Step 2 of the proof of Theorem 24.4 gives, for $N \geq 2$, $x_N(t) = \operatorname{sgn}(h(-t))$ on $[-N, N]$. Now $h(-t) \neq 0$ if and only if $-t \in [0, 2]$, i.e. $t \in [-2, 0]$; and it equals $+1$ for $t \in [-1, 0]$, -1 for $t \in [-2, -1]$. Hence

$$x_N = \mathbf{1}_{[-1,0]} - \mathbf{1}_{[-2,-1]} \quad (N \geq 2),$$

a square wave of amplitude 1 and duration 2.

Step 5 — the output at time 0. One has $x_N(-\tau) = +1$ for $\tau \in [0, 1]$ and -1 for $\tau \in [1, 2]$, hence

$$y(0) = \int_{\mathbb{R}} h(\tau)x_N(-\tau) d\tau = \int_0^1 (+1)(+1) d\tau + \int_1^2 (-1)(-1) d\tau = 1 + 1 = 2 = \|h\|_1. \quad \checkmark$$

The critical input therefore attains the gain *exactly* — here without even needing to pass to the limit, because h has compact support.

Independent check. $|H(1)| = 0.919\,395\,388\,3$ and $|H(4)| = 0.826\,821\,810\,4$, both well below 1.449 and even further below 2: the sinusoidal exploration never finds the value 2. ✓

Interpretation. This is Remark 24.8 put into numbers: the gain of a system (the worst amplification factor over all bounded inputs) is $\|h\|_1 = 2$, while the peak of the frequency response (the worst factor over sinusoidal inputs alone) is 1.449. The ratio $2/1.449\,222\,708 = 1.380$ is, literally, what a test with pure tones alone would not see: a margin of 38%.

Typical errors. (1) Believing that the gain is $\max |H|$: this is true only for systems with $h \geq 0$, such as the RC filter and the moving average, where $\max |H| = H(0) = \int h = \|h\|_1$. (2) Forgetting that x_N is defined with h *time-reversed*: using $\operatorname{sgn}(h(t))$ instead of $\operatorname{sgn}(h(-t))$ gives, in this example, $y(0) = \int_0^1 (+1)(-1) + \int_1^2 (-1)(+1) = -2$, i.e. the right value in magnitude but with the wrong sign — and in non-symmetric examples the wrong value.

Extension. Redo the computation with $h = \mathbf{1}_{[0,1]} + \mathbf{1}_{[1,2]} = \mathbf{1}_{[0,2]}$, where $h \geq 0$: here $\|h\|_1 = 2$, $H(0) = 2$ and $\max |H| = 2$, the two numbers coincide, and the critical input is the constant 1.

Exercise 24.7: Two components, a single alias. *Type:* numerical with sample-by-sample check. A signal $x(t) = 2 \cos(2\pi \cdot 200 t) + \cos(2\pi \cdot 900 t + 0.5)$ is sampled at $\nu_c = 1\,000$ Hz. Determine what signal is “seen” by someone who has only the samples, and verify the identity of the samples.

Solution. Strategy. Apply Definition 24.11 to each component and then Theorem 24.10, which also gives the sign of the phase.

Step 1 — the Nyquist frequency. $\nu_c/2 = 500$ Hz. The component at 200 Hz lies below it, the one at 900 Hz does not: the latter will be the one that aliases.

Step 2 — the first component. $\nu_0 = 200$, $\nu_0/\nu_c = 0.2$, the closest integer is $k = 0$, hence $\mu = 200 - 0 = 200$ and $\nu_a = 200$ Hz, with $\varepsilon = \text{sgn}(200) = +1$: the component is transferred intact, phase included. This is the case $\nu_0 \leq \nu_c/2$ of Definition 24.11.

Step 3 — the second component. $\nu_0 = 900$, $\nu_0/\nu_c = 0.9$, the closest integer is $k = 1$, hence

$$\mu = 900 - 1 \cdot 1\,000 = -100, \quad \nu_a = |-100| = 100 \text{ Hz}, \quad \varepsilon = \text{sgn}(-100) = -1.$$

By Theorem 24.10 the samples of $\cos(2\pi \cdot 900 t + 0.5)$ coincide with those of $\cos(2\pi \cdot 100 t - 0.5)$: frequency 100 Hz and phase *sign reversed*.

Step 4 — the apparent signal. Someone with only the samples measures

$$\tilde{x}(t) = 2 \cos(2\pi \cdot 200 t) + \cos(2\pi \cdot 100 t - 0.5) :$$

two components at 200 and 100 Hz, both *below* Nyquist, both perfectly plausible. Nothing in the data signals that the second is an artifact.

Independent check (Python script executed). Over 121 samples, from $n = -60$ to $n = 60$, the maximum gap between $x(nT)$ and $\tilde{x}(nT)$ equals $4.75 \cdot 10^{-14}$: the two sequences coincide within machine error. ✓

Interpretation. The error is not “small”, it is *invisible*: the two sequences do not differ at all. This is the fundamental difference between a measurement error — which can be estimated and reduced — and aliasing, which irreversibly destroys information even before the data exist. The operational consequence is in the Good practice of Section 24.10: the defence is analog and must be placed *before* the converter.

Typical errors. (1) Computing $\nu_a = \nu_c - \nu_0$ without the absolute value and without choosing the closest integer: for $\nu_0 = 1\,300$ and $\nu_c = 1\,000$ this would give -300 instead of 300. (2) Forgetting the sign change of the phase when $\varepsilon = -1$: the reconstruction would come out shifted in time. (3) Hoping that a digital filter, applied afterward, could separate the two components: by Theorem 24.10 they have the same samples, and no function of the samples can distinguish them.

Extension. Add a third component at 1 000 Hz and verify that its alias frequency is 0: in the samples it appears as a constant, i.e. as a drift of the mean value.

24.15. Self-assessment quiz

1. (Definition) A system $\mathcal{T}$ is *time-invariant* if
 - A) $\mathcal{T}(\alpha x) = \alpha \mathcal{T}x$ for every scalar α
 - B) $\mathcal{T}(\sigma_a x) = \sigma_a(\mathcal{T}x)$ for every x and every $a \in \mathbb{R}$
 - C) $\mathcal{T}x$ does not depend on t
 - D) the output to a periodic input is periodic of the same period
2. (Hypothesis) In Theorem 24.1 hypothesis (S3), i.e. $\|\mathcal{T}x\|_\infty \leq C\|x\|_1$, serves to

-
- A) guarantee that the system is causal B) make the functional $x \mapsto (\mathcal{T}x)(0)$ bounded on L^1 , so that Riesz's theorem gives its representative $g \in L^\infty$ C) ensure that $\mathcal{T}x$ is continuous D) make h belong to L^1
3. (Symbolic computation) For an LTI system with impulse response $h \in L^1(\mathbb{R})$ and frequency response $H = \hat{h}$, the output to the input $e^{i\omega_0 t}$ is
- A) $\hat{h}(t) e^{i\omega_0 t}$ B) $H(\omega_0) e^{i\omega_0 t}$ C) $(h * e^{i\omega_0 \cdot})(\omega_0)$, a number D) $|H(\omega_0)| e^{i\omega_0 t}$
4. (Numerical computation) For the RC filter with $\tau = 1$ ms, the group delay at angular frequency $\omega = 3/\tau$ equals
- A) 0.5 ms B) 0.416 348 6 ms C) 0.1 ms D) 1 ms
5. (Proof step) In the proof of Theorem 24.4, the signal $x_N(t) = \overline{h(-t)}/|h(-t)|$ of Step 2 serves to
- A) make h integrable B) eliminate every cancellation in the convolution integral at $t = 0$, so that the output equals $\int_{-N}^N |h|$ C) verify that the system is causal D) prove that $\sup_\omega |H(\omega)| = \|h\|_1$
6. (Error diagnosis) A test measures the gain of an LTI system by injecting unit-amplitude sinusoids at many frequencies and recording the maximum of the output. Which statement is correct?
- A) the test measures exactly the BIBO gain, because every bounded signal is a sum of sinusoids B) the test measures $\sup_\omega |H(\omega)|$, which can be strictly less than the BIBO gain $\|h\|_1$ C) the test overestimates the BIBO gain D) the test is equivalent to measuring $\|h\|_2$
7. (Transfer) A signal with components at 250, 1 200, 3 000 and 9 500 Hz is sampled at 8 000 Hz. In the spectrum computed from the samples the component at 9 500 Hz appears at
- A) 9 500 Hz B) 4 000 Hz C) 1 500 Hz D) it does not appear, because it is beyond Nyquist
8. (Source and metacognition) The article in which the sampling theorem is stated and proved in the form used here, with its exact location, is
- A) H. Nyquist, *Certain topics in telegraph transmission theory*, *Transactions of the AIEE* **47** (1928), pp. 617–644 B) C. E. Shannon, *Communication in the presence of noise*, *Proceedings of the IRE* **37** (1949), n. 1, pp. 10–21 C) D. Gabor, *Theory of communication*, *Journal of the IEE, Part III* **93** (1946), pp. 429–441 D) J. R. Higgins, *Five short stories about the cardinal series*, *Bulletin of the AMS* **12** (1985), pp. 45–89

24.16. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnosis (items 1–4).

1) **B.** *Rationale:* this is condition (TI) of Definition 24.2: the system commutes with translations. *Distractors:* A is homogeneity, i.e. half of linearity (L), not invariance; C confuses “the system does not change over time” with “the output is constant”; D is a true *consequence* for stable LTI systems, not the definition — and it also holds for non-LTI systems, for instance the squarer of Example 24.1(e), which sends T -periodic signals to T -periodic signals despite not being linear. *Diagnosed error:* confusing an observable

property with the definition that implies it. *Source*: Oppenheim and Willsky, entry [3] of the *Sources* environment. *Prerequisite*: Definition 24.2. *Remediation*: redo the counterexample of Example 24.1(f) with $a = \pi/(2\omega_0)$ instead of $a = \pi/\omega_0$.

2) B. Rationale: Step 1 of the proof uses (S3) precisely to conclude $|\Lambda(x)| \leq C\|x\|_1$, i.e. $\Lambda \in (L^1)^*$; without it Step 2 (Riesz) does not apply. *Distractors*: A is false, causality is independent and has its own theorem (Theorem 24.5); C is guaranteed by the hypothesis that $\mathcal{T}$ takes values in C_b , not by (S3); D is the wrong conclusion — the theorem produces $h \in L^\infty$, not $h \in L^1$, and indeed the ideal integrator of Example 24.2(d) has $h = u \in L^\infty \setminus L^1$. *Diagnosed error*: not knowing at which step a given hypothesis is needed. *Source*: Theorem 24.1 and Remark 24.3. *Prerequisite*: duality $(L^1)^* = L^\infty$. *Remediation*: rewrite Step 1 without (S3) and identify where the reasoning breaks down.

3) B. Rationale: this is Theorem 24.2: the exponential is an eigenfunction and the eigenvalue is $H(\omega_0) = \hat{h}(\omega_0)$, a complex number that multiplies the exponential. *Distractors*: A swaps the variable of the transform with time; C confuses the output (a function of t) with its value at a point; D throws away the phase, which is half of the information — and it is precisely the half that generates the delays of Section 24.7. *Diagnosed error*: treating the frequency response as a real gain. *Source*: Oppenheim and Willsky, entry [3] of the *Sources* environment; here Theorem 24.2. *Prerequisite*: Definition 19.1. *Remediation*: redo Step 2 of the proof by writing out $e^{i\omega(t-\tau)} = e^{i\omega t}e^{-i\omega\tau}$ explicitly.

4) C. Rationale: $\tau_g(\omega) = \tau/(1 + \omega^2\tau^2)$ (Example 24.6); with $\omega\tau = 3$ one has $\tau_g = \tau/10 = 0.1$ ms. *Distractors*: A is τ_g at $\omega = 1/\tau$, i.e. $\tau/2$; B is the *phase delay* at $\omega = 3/\tau$, $\arctan(3)/3000 = 0.416\,348\,6$ ms, and is the most insidious distractor because it is a correct number from the table, but from the other column; D is τ , i.e. τ_g at $\omega = 0$. *Diagnosed error*: confusing τ_p and τ_g , or reading the wrong row. *Source*: Definition 24.9 and the check by centred differences. *Prerequisite*: derivative of $\arctan$. *Remediation*: recompute all four columns of the table of Example 24.6 for $\omega\tau = 0.5$ (answers: $0.894\,427\,2$; $-26.565\,051^\circ$; $0.927\,295\,2$ ms; 0.8 ms).

BEST PRACTICE

Quiz — answers, rationales and diagnosis (items 5–8).

5) B. Rationale: choosing $x_N(-\tau) = \overline{h(\tau)}/|h(\tau)|$ gives $h(\tau)x_N(-\tau) = |h(\tau)| \geq 0$ for every τ : all contributions are aligned and the integral equals $\int_{-N}^N |h|$, as in Step 3. *Distractors*: A reverses cause and effect — the integrability of h is the *conclusion*, not a hypothesis; C confuses two distinct theorems, even though the construction of Theorem 24.5 is analogous; D is false, and Theorem 24.6(iv) shows that the two numbers can differ (in Worked exercise 24.6 they equal 1.449 and 2). *Diagnosed error*: not recognising the role of an ad hoc construction inside a proof. *Source*: Theorem 24.4, Steps 2–3; the discrete analogue is Theorem 21.10. *Prerequisite*: $z\bar{z} = |z|^2$. *Remediation*: carry out Steps 4–5 of Worked exercise 24.6 on $h = \mathbf{1}_{[0,1]} - \mathbf{1}_{[1,2]}$.

6) B. Rationale: a sinusoidal test explores $\sup_\omega |H(\omega)|$, which by Theorem 24.6(iv) is $\leq \|h\|_1$, with strict inequality in general; the BIBO gain is $\|h\|_1$ (Theorem 24.4). *Distractors*: A contains a widespread error: a bounded signal is *not* in general a sum of sinusoids with summable coefficients, and in any case the critical input of Step 2 is not one, being a square wave with infinitely many harmonics; C gets the direction of the inequality wrong; D confuses the L^2 norm of the impulse response (which governs energy, not amplitude) with the L^1 norm. *Diagnosed error*: identifying the worst case with the worst case *among those tested*. *Source*: Remark 24.8 and Worked exercise 24.6. *Prerequisite*: Theorem 24.4. *Remediation*: compute $\|h\|_1$ and $\max |H|$ for $h = \mathbf{1}_{[0,1]} - \mathbf{1}_{[1,2]}$ and compare (2 against $1.449\,222\,707\,6$).

7) C. Rationale: by Definition 24.11, $\nu_0/\nu_c = 9\,500/8\,000 = 1.1875$, the closest integer is

$k = 1$, hence $\nu_a = |9\,500 - 8\,000| = 1\,500$ Hz. *Distractors*: A ignores aliasing; B is the alias of the component at 12 000 Hz ($k = 2$, $|12\,000 - 16\,000| = 4\,000$ Hz), i.e. the right answer to the other question; D is the central conceptual error of Section 24.10 — the component does not disappear, it disguises itself. *Diagnosed error*: believing that sampling “cuts off” whatever is above Nyquist. *Source*: Theorem 24.10 and the sample-by-sample check of Example 24.8. *Prerequisite*: Definition 24.11. *Remediation*: redo the four cases of Exercise 24.4(b) and check them with the script of point (c).

8) B. *Rationale*: this is Shannon’s 1949 article, verified against the Crossref record of the DOI 10.1109/JRPROC.1949.232969: *Proceedings of the IRE*, volume 37, issue 1, pages 10–21. *Distractors*: A is Nyquist’s 1928 article, which studies the maximum signalling speed over a telegraph line and from which the name “Nyquist frequency” comes, but does not contain the reconstruction formula; C is Gabor’s article, the source of the *analytic signal* of Section 24.11 and not of sampling; D is a historical-mathematical survey article from 1985, valuable for the history of the cardinal series but not a primary source. *Diagnosed error*: attributing a result from memory to the name that appears in its label. *Source*: entries [1], [2], [5] and [7] of this chapter’s *Sources* environment. *Prerequisite*: none. *Remediation*: check the four references against a bibliographic archive and note, for each, what it contains and what it does not.

BEST PRACTICE

Assigned exercises — answers.

Exercise 24.2 ($\tau = 5$ ms). (a) $h(t) = \frac{1}{\tau}e^{-t/\tau}u(t)$; $\|h\|_1 = \frac{1}{\tau} \int_0^{+\infty} e^{-t/\tau} dt = 1$, hence $h \in L^1(\mathbb{R})$ and by Theorem 24.4 the system is BIBO stable with gain 1. It is causal because $h(t) = 0$ for $t < 0$ (Theorem 24.5). (b) $H(\omega) = 1/(1 + i\omega\tau)$, $|H| = (1 + \omega^2\tau^2)^{-1/2}$, $\phi = -\arctan(\omega\tau)$. The -3 dB frequency is $\nu_{3\text{dB}} = \frac{1}{2\pi\tau} = 31.830\,988\,6$ Hz, i.e. $\omega = 1/\tau = 200$ rad/s, where $|H| = 1/\sqrt{2} = 0.707\,106\,781\,2$ and $\phi = -45^\circ$. (c) $\tau_g(0) = \tau = 5$ ms; at $\omega = 1/\tau = 200$: $\tau_p = \frac{\pi}{4}\tau = 3.926\,990\,817$ ms and $\tau_g = \tau/2 = 2.5$ ms; at $\omega = 3/\tau = 600$: $\tau_p = \arctan(3)/600 = 2.081\,742\,954$ ms and $\tau_g = \tau/10 = 0.5$ ms. The delay of the envelope is τ_g , because Theorem 24.7 ties the shift of the waveform to the slope of the phase, not to its quotient. (d) No, on no non-degenerate band: $\tau_g(\omega) = \tau/(1 + \omega^2\tau^2)$ is strictly decreasing for $\omega > 0$, hence it is not constant on any interval, and the second condition of Corollary 24.3 always fails. (e) The typical gaps are $\sim 10^{-13}$ for H (Simpson with $6 \cdot 10^5$ subintervals on $[0, 60\tau]$) and $\sim 10^{-11}$ ms for τ_g (centred differences with step 10^{-6}).

Exercise 24.3 ($T_0 = 10$ ms). (a) $H(\omega) = e^{-i\omega T_0/2} \text{sinc}(\omega T_0/(2\pi))$; the zeros are at $\nu = k/T_0$, i.e. 100, 200, 300 Hz. (b) Where $\sin(\omega T_0/2) > 0$ the second factor is real and positive, so the phase is that of the exponential alone: $\phi(\omega) = -\omega T_0/2$, and $\tau_g = -\phi' = T_0/2 = 5$ ms, constant. (c) The *first* condition fails: the gain is not constant. Indeed $|H| = 0.900\,316\,316$ at 25 Hz, 0.636 619 772 at 50 Hz, 0 at 100 Hz and 0.212 206 591 at 150 Hz — and at 100 Hz the component is cancelled completely, which is the most radical distortion possible. (Note also that above 100 Hz the second factor becomes negative and the phase jumps by π : the phase remains piecewise linear, not linear over the whole band.) (d) $\text{sinc}(\nu T_0) = 0.9$ at $\nu \approx 25$ Hz: indeed at 25 Hz the gain equals 0.900 316 3, just above 0.9. Within $\nu_{\max} \approx 25$ Hz the gain varies by less than 10% and the phase is exactly linear, so the no-distortion approximation is reasonable. (e) With $\tau = T_0/2 = 5$ ms the RC has $\tau_g(0) = 5$ ms, equal to that of the moving average; but τ_g collapses immediately (2.5 ms already at $\omega = 1/\tau$), whereas that of the moving average stays 5 ms at all frequencies. The moving average costs more in-band attenuation but does not introduce phase dispersion.

Exercise 24.4 ($\nu_c = 8\,000$ Hz). (a) The Nyquist frequency is 4 000 Hz; the components at 250, 1 200 and 3 000 Hz satisfy the condition. (b) For 9 500 Hz: $9\,500/8\,000 = 1.1875$, $k = 1$, $\nu_a = |9\,500 - 8\,000| = 1\,500$ Hz. For 12 000 Hz: $12\,000/8\,000 = 1.5$, $k = 2$ (or

$k = 1$: the result does not change), $\nu_a = |12\,000 - 16\,000| = 4\,000$ Hz. The first alias falls at 1 500 Hz, i.e. between the useful components at 1 200 and 3 000 Hz: it is indistinguishable from a real gearbox harmonic. The second falls exactly on the Nyquist frequency. (c) The maximum gap is of order 10^{-13} ; with initial phase $\varphi_0 \neq 0$ the alias has phase $+\varphi_0$ because $\varepsilon = \text{sgn}(9\,500 - 8\,000) = +1$. (d) With the highest component at 12 000 Hz the theorem alone would require $\nu_c \geq 24\,000$ Hz; applying the factor of 5–10 of the Good practice to the *useful* band of 3 000 Hz one instead chooses $\nu_c = 20\,000$ – $30\,000$ Hz *with* an analog anti-aliasing filter that attenuates by at least 72 dB at $\nu_c/2$ (12-bit A/D): at that point the components at 9 500 and 12 000 Hz are suppressed before conversion. (e) With $T = 1/8\,000$ s the formula is $x(t) = \sum_n x(nT) \text{sinc}(8\,000t - n)$; at $t = 1/16\,000 = T/2$ the term $n = 0$ equals $x(0) \text{sinc}(1/2) = x(0) \cdot \frac{\sin(\pi/2)}{\pi/2} = x(0) \cdot \frac{2}{\pi} = 0.636\,619\,772\,4 x(0)$.

24.17. Chapter formulary

LTI systems and convolution. $(\sigma_a x)(t) = x(t - a)$; (L) linearity, (TI) $\mathcal{T}\sigma_a = \sigma_a \mathcal{T}$.

$$(h * x)(t) = \int_{\mathbb{R}} h(\tau)x(t - \tau) d\tau = \int_{\mathbb{R}} h(t - u)x(u) du = (x * h)(t),$$

$$\|h * x\|_{\infty} \leq \|h\|_1 \|x\|_{\infty} \quad (h \in L^1, x \in L^{\infty}), \quad \|h * x\|_{\infty} \leq \|h\|_{\infty} \|x\|_1 \quad (h \in L^{\infty}, x \in L^1).$$

Representation: $\mathcal{T} : L^1 \rightarrow C_b$ linear, TI and with $\|\mathcal{T}x\|_{\infty} \leq C\|x\|_1 \implies \mathcal{T}x = h * x$ with $h \in L^{\infty}$, $\|h\|_{\infty} \leq C$, unique a.e.

Eigenfunctions and frequency response. $H(\omega) = \hat{h}(\omega)$, $\hat{h}(\Omega) = \sum_k h[k]e^{-i\Omega k}$ in the discrete case.

$$\mathcal{T}_h e^{i\omega t} = H(\omega) e^{i\omega t}, \quad \mathcal{T}_h e^{i\Omega n} = \hat{h}(\Omega) e^{i\Omega n}.$$

$$A \cos(\omega_0 t + \varphi_0) \mapsto A |H(\omega_0)| \cos(\omega_0 t + \varphi_0 + \arg H(\omega_0)) \quad (h \text{ real}).$$

$|H(\omega)| \leq \|h\|_1$; H uniformly continuous; $H(-\omega) = \overline{H(\omega)}$ for h real; $|H(\omega)| \rightarrow 0$ as $|\omega| \rightarrow \infty$; $|H|_{\text{dB}} = 20 \log_{10} |H|$.

Stability and causality.

BIBO $\iff h \in L^1(\mathbb{R})$, gain $= \|h\|_1$ (critical input $x_N = \overline{h(-t)}/|h(-t)|$ on $[-N, N]$);

$$\text{causal} \iff h = 0 \text{ a.e. on } (-\infty, 0) \implies y(t) = \int_0^{+\infty} h(\tau)x(t - \tau) d\tau.$$

In the discrete case: BIBO $\iff h \in \ell^1(\mathbb{Z})$; for a causal rational system, poles inside the unit circle.

Delays and no distortion. With $H = |H|e^{i\phi}$:

$$\tau_p(\omega) = -\frac{\phi(\omega)}{\omega}, \quad \tau_g(\omega) = -\frac{d\phi}{d\omega}(\omega).$$

No distortion on $[-\Omega, \Omega]$ $\iff |H| \equiv A$ and $\tau_g \equiv t_0 \iff H(\omega) = Ae^{-i\omega t_0} \implies y(t) = Ax(t - t_0)$. Pure delay: $H(\omega) = e^{-i\omega t_0}$, $\tau_p = \tau_g = t_0$.

The two reference filters. *RC* filter, $h(t) = \frac{1}{\tau}e^{-t/\tau}u(t)$:

$$H(\omega) = \frac{1}{1 + i\omega\tau}, \quad |H| = \frac{1}{\sqrt{1 + \omega^2\tau^2}}, \quad \phi = -\arctan(\omega\tau), \quad \tau_p = \frac{\arctan(\omega\tau)}{\omega}, \quad \tau_g = \frac{\tau}{1 + \omega^2\tau^2}.$$

At $\omega = 1/\tau$: $|H| = 1/\sqrt{2}$, $-3.010\,299\,957$ dB, $\phi = -45^\circ$, $\tau_p = \frac{\pi}{4}\tau$, $\tau_g = \tau/2$; $\nu_{3\text{dB}} = 1/(2\pi\tau)$.
Moving average, $h = \frac{1}{T_0}\mathbf{1}_{[0, T_0]}$:

$$H(\omega) = e^{-i\omega T_0/2} \text{sinc}\left(\frac{\omega T_0}{2\pi}\right), \quad \text{zeros at } \nu = \frac{k}{T_0}, \quad k \neq 0, \quad \tau_g \equiv \frac{T_0}{2}.$$

Both have $\|h\|_1 = 1$: BIBO stable with gain 1, causal.

Sampling, reconstruction, aliasing. T step, $\omega_c = 2\pi/T$, $\nu_c = 1/T$, $\Omega = \pi/T$; Nyquist frequency $\nu_c/2$.

$$\hat{x}_s(\omega) = \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{x}_c(\omega - k\omega_c); \quad \omega_c \geq 2\omega_{\max} \implies x(t) = \sum_{n \in \mathbb{Z}} x(nT) \text{sinc}\left(\frac{t - nT}{T}\right),$$

with $\text{sinc}(v) = \sin(\pi v)/(\pi v)$, $\text{sinc}(0) = 1$, $\text{sinc}(m) = 0$ for $m \in \mathbb{Z} \setminus \{0\}$. Aliasing: $\nu_a = |\nu_0 - k\nu_c|$ with k the integer closest to ν_0/ν_c ; $\varepsilon = \text{sgn}(\nu_0 - k\nu_c)$ and $\cos(2\pi\nu_0 t + \varphi_0)$ has the same samples as $\cos(2\pi\nu_a t + \varepsilon\varphi_0)$. Examples: $\nu_c = 1000$: $1\,300 \rightarrow 300$, $700 \rightarrow 300$, $2\,000 \rightarrow 0$; $\nu_c = 50$: $60 \rightarrow 10$; $\nu_c = 8\,000$: $9\,500 \rightarrow 1\,500$, $12\,000 \rightarrow 4\,000$.

Hilbert and analytic signal. On $L^2(\mathbb{R})$:

$$\widehat{\mathcal{H}x}(\omega) = -i \text{sgn}(\omega) \hat{x}(\omega), \quad \|\mathcal{H}x\|_2 = \|x\|_2, \quad \mathcal{H}^2 = -\text{id};$$

$$x_a = x + i\mathcal{H}x, \quad \hat{x}_a(\omega) = (1 + \text{sgn} \omega) \hat{x}(\omega) = \begin{cases} 2\hat{x}(\omega), & \omega > 0 \\ 0, & \omega < 0 \end{cases}$$

$$\mathcal{H}[\cos \omega_0 t] = \sin \omega_0 t, \quad \mathcal{H}[\sin \omega_0 t] = -\cos \omega_0 t, \quad (\cos \omega_0 \cdot)_a = e^{i\omega_0 t}.$$

Envelope $A(t) = |x_a(t)| \geq |x(t)|$; instantaneous phase $\psi = \arg x_a$; instantaneous frequency $\psi'/(2\pi)$.

Sources and bibliography

- [1] C. E. Shannon, *Communication in the presence of noise*, *Proceedings of the IRE* **37** (1949), n. 1, pp. 10–21, Institute of Radio Engineers. DOI 10.1109/JRPROC.1949.232969. This is the primary source of the sampling theorem and of the reconstruction formula with the sinc function, restated here in Theorem 24.8 and proved in full in Theorem 24.9. Verified on 7 September 2026 (Crossref record of the DOI: author, title, journal, volume, issue, year and pages checked one by one).
- [2] H. Nyquist, *Certain topics in telegraph transmission theory*, *Transactions of the American Institute of Electrical Engineers* **47** (1928), n. 2, pp. 617–644. DOI 10.1109/T-AIEE.1928.5055024. Establishes the limit on the signalling speed of a line of given bandwidth, whence the name “Nyquist frequency” in Definition 24.10; it does *not* contain the reconstruction formula, which is Shannon’s. Verified on 7 September 2026 (Crossref record of the DOI: author, title, journal, volume, issue, year and pages).
- [3] A. V. Oppenheim, A. S. Willsky, with S. H. Nawab, *Signals and Systems*, 2nd edition,

Prentice Hall, ISBN 0-13-814757-4, 957 pages. This is the standard engineering exposition for the whole chain of this chapter: linearity and time-invariance (Definition 24.2, Example 24.1), the convolution representation in the continuous and discrete cases with the impulse response treated as a distribution — the framework cited as *panorama* in Remark 24.3 —, complex exponentials as eigenfunctions (Theorem 24.2) and sampling. *Verified on 7 September 2026* (Open Library catalog record of the ISBN: authors, publisher, page count, LCCN 96019945; the record gives the publication year as 1997. The edition’s table of contents was not consulted, so no chapter or page numbers are cited).

- [4] A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd edition, Pearson, 2010, ISBN 978-0-13-198842-2, 1108 pages. Source of the framework on minimum-phase systems and on the constraints causality imposes on the frequency response, declared *panorama* in Remarks 24.11 and 24.13, and of the engineering use of the analytic signal and of the discrete Hilbert transform recalled in Remark 24.19. *Verified on 7 September 2026* (Open Library catalog record of the ISBN: authors, publisher, year, page count, LCCN 2011292650. The edition’s table of contents was not consulted, so no chapter or page numbers are cited).
- [5] D. Gabor, *Theory of communication. Part 1: The analysis of information*, *Journal of the Institution of Electrical Engineers — Part III: Radio and Communication Engineering* **93** (1946), n. 26, pp. 429–441. DOI 10.1049/ji-3-2.1946.0074. This is the article in which the *analytic signal* of Definition 24.13 first appears, as a tool for defining the instantaneous amplitude and phase of a real signal. *Verified on 7 September 2026* (Crossref record of the DOI: author, full title including the “Part 1” indication, journal, volume, issue, year and pages).
- [6] F. W. King, *Hilbert Transforms*, Cambridge University Press, Cambridge, 2008, ISBN 978-0-521-88762-5, 800 pages (volume 1 of the two-volume work). This is the source of the *imported* facts (H1) and (H2) of Remark 24.19: the representation as a Cauchy principal value and its equivalence with the multiplier definition, and the M. Riesz theorem on boundedness on L^p for $1 < p < \infty$. *Verified on 7 September 2026* (Open Library catalog record of the ISBN: author, publisher, publication date, page count, LCCN 2008013534).
- [7] J. R. Higgins, *Five short stories about the cardinal series*, *Bulletin of the American Mathematical Society* **12** (1985), n. 1, pp. 45–89. DOI 10.1090/S0273-0979-1985-15293-0. A survey article on the cardinal series $\sum_n x(nT) \operatorname{sinc}((t - nT)/T)$, i.e. on the object of Theorem 24.9. It is cited as a bibliographic reference for the historical context and for variants of the statement, and not as the basis of any step of the proof, which is carried out here in full. *Verified on 7 September 2026* (Crossref record of the DOI: author, title, journal, volume, issue, year and pages).

The circle in control and transmission lines

In Chapter 3 we proved that Möbius transformations send circles and lines to circles and lines (Theorem, Volume II, Chapter 3), and in particular that $w = (z - 1)/(z + 1)$ sends the unit circle onto the imaginary axis (Corollary, Volume II, Chapter 3). In Chapter 4 we saw that the unit circle is the stability boundary of discrete-time systems (Theorem 21.12), and in Chapter 6 that the locus of a one-parameter admittance is a circle (Theorem, Volume II, Chapter 5).

This chapter gathers those three threads and shows they are one thread. The Nyquist criterion counts how many times a curve winds around a point; stability margins are read off the intersection with the unit circle; the constant-magnitude and constant-phase loci of the closed loop are circles; the root locus of a notable class of systems is a circle; and the Smith chart — the most used graphical tool in the history of microwave engineering — is literally the Möbius map of Chapter 3, drawn on paper.

INFO

The idea in one line. In feedback, the question “is the system stable?” becomes “how many times does a certain curve wind around the point -1 ?”; and “how far are we from instability?” becomes “how far is that curve from the unit circle?”. The circle is not an illustration: it is the object on which the answer is read.

Warning. Conventions of the chapter.

- **Laplace variable** $s = \sigma + i\omega$; **Z variable** z , as in Chapter 4. The half-plane $\text{Re } s > 0$ is called the *right half-plane* (RHP).
- **Unity negative feedback:** given the loop gain $L(s)$, the closed-loop transfer function is $T(s) = \frac{L(s)}{1 + L(s)}$.
- **Decibels:** $|H|_{\text{dB}} = 20 \log_{10} |H|$ for amplitude ratios.
- **Phase** in degrees on plots, in radians in computations.
- **Transmission lines:** characteristic impedance $Z_0 > 0$ real (lossless line), phase constant $\beta = 2\pi/\lambda$.

INFO

Prerequisites. Chapter 3 (complex numbers, Möbius maps, residues and the residue theorem); Chapter 4 (Laplace and Z transforms, BIBO stability); Chapter 6 (impedances, reflection coefficient of Exercise S1). Notions of linear system and feedback are recalled where needed.

25.1. Frequency response and transfer function

Definition 25.1: Transfer function and frequency response. For a causal linear time-invariant (LTI) system with impulse response h , the **transfer function** is $H(s) = \mathcal{L}\{h\}(s)$, and the **frequency response** is its restriction to the imaginary axis, $\omega \mapsto H(i\omega)$. The **Nyquist plot** is the image of the imaginary axis, i.e. the curve $\{H(i\omega) : \omega \in \mathbb{R}\} \subset \mathbb{C}$; the **Bode plots** are the two graphs $\omega \mapsto 20 \log_{10} |H(i\omega)|$ and $\omega \mapsto \arg H(i\omega)$ on a logarithmic frequency axis.

Theorem 25.1: Steady-state harmonic response. Let a causal LTI system have $h \in L^1(0, \infty)$ (hence BIBO stable, by the continuous-time version of Theorem 21.12). Then:

- (i) the input $u(t) = e^{i\omega t}$, defined for $t \in \mathbb{R}$, produces the output $y(t) = H(i\omega) e^{i\omega t}$, with $H(i\omega) = \int_0^\infty h(\tau) e^{-i\omega\tau} d\tau$;
- (ii) the real input $u(t) = A \cos(\omega t + \phi)$ produces $y(t) = A |H(i\omega)| \cos(\omega t + \phi + \arg H(i\omega))$;
- (iii) if the input is applied from $t = 0$ onwards, the output differs from that of (ii) by a transient tending to zero, provided all poles of H have negative real part.

Proof. (i) By the convolution representation of causal LTI systems,

$$y(t) = \int_0^\infty h(\tau) u(t - \tau) d\tau = e^{i\omega t} \int_0^\infty h(\tau) e^{-i\omega\tau} d\tau,$$

where pulling $e^{i\omega t}$ out is legitimate because it does not depend on τ , and the integral converges absolutely because $|h(\tau) e^{-i\omega\tau}| = |h(\tau)|$ with $h \in L^1$. The remaining integral is exactly $H(i\omega)$, the Laplace transform evaluated at $s = i\omega$ (Chapter 4).

(ii) By Exercise S3 of Chapter 6, $A \cos(\omega t + \phi) = \frac{1}{2} A e^{i\phi} e^{i\omega t} + \frac{1}{2} A e^{-i\phi} e^{-i\omega t}$. Applying (i) to each term and using linearity,

$$y(t) = \frac{1}{2} A e^{i\phi} H(i\omega) e^{i\omega t} + \frac{1}{2} A e^{-i\phi} H(-i\omega) e^{-i\omega t}.$$

Since h is real, $H(-i\omega) = \overline{H(i\omega)}$ (conjugate the definition of H), so the two terms are conjugate and their sum is $2 \operatorname{Re}$ of the first:

$$y(t) = A |H(i\omega)| \cos(\omega t + \phi + \arg H(i\omega)).$$

(iii) With the input applied from $t = 0$, the output is $y(t) = \int_0^t h(\tau) u(t - \tau) d\tau$, differing from (i) by $-\int_t^\infty h(\tau) u(t - \tau) d\tau$. Since $|u| \leq A$,

$$\left| \int_t^\infty h(\tau) u(t - \tau) d\tau \right| \leq A \int_t^\infty |h(\tau)| d\tau \xrightarrow{t \rightarrow \infty} 0,$$

being the tail of a convergent integral. If moreover all poles have negative real part, h is a combination of terms $t^k e^{-\alpha t}$ with $\alpha > 0$ and the convergence is exponential. $\square$

Remark 25.1: Why the frequency response is an image of the imaginary axis.

Theorem 25.1 says that $H(i\omega)$ is an eigenvalue: complex exponentials are eigenfunctions of every LTI system, and $H(i\omega)$ is the eigenvalue attached to $e^{i\omega t}$. This is the same property that in Chapter 4 made the characters e_n the right functions to diagonalise convolution (Observation, Volume II, Chapter 4): the Nyquist plot is nothing but the image of the imaginary axis — where the characters live — under the map H . In the discrete-time z -plane the locus of characters is the unit circle (Theorem, Volume II, Chapter 4), which is why there the frequency response is read on $\mathbb{T}$ and not on a line.

25.2. Bode plots

Theorem 25.2: Logarithmic additivity. If $H = H_1 H_2 \cdots H_m$, then for every ω with $H_k(i\omega) \neq 0$:

$$20 \log_{10} |H(i\omega)| = \sum_{k=1}^m 20 \log_{10} |H_k(i\omega)|, \quad \arg H(i\omega) = \sum_{k=1}^m \arg H_k(i\omega) \pmod{2\pi}.$$

Proof. By Theorem (Volume II, Chapter 3)(iii) the modulus is multiplicative, $|H| = \prod_k |H_k|$; applying $20 \log_{10}$ and the logarithm-of-a-product rule gives the first formula. For arguments, Theorem (Volume II, Chapter 3) gives $\arg(H_1 H_2) = \arg H_1 + \arg H_2$ as an equality of sets of values, i.e. modulo 2π ; induction on m concludes. $\square$

Remark 25.2: Why decibels and why a logarithmic frequency axis. Theorem 25.2 is the whole *raison d'être* of Bode plots: on a logarithmic scale the plot of a product is the sum of the plots, so it can be drawn by hand by superposing a few elementary bricks. The logarithmic abscissa serves the same purpose: it turns the slopes $\omega^{\pm k}$ into straight lines, as the next theorem shows.

Theorem 25.3: Asymptotic slopes of the elementary factors. Let $\omega_p > 0$. Then, with $\log_{10} \omega$ on the abscissa and decibels on the ordinate:

- (i) the factor s^n ($n \in \mathbb{Z}$) gives a straight line of slope $20n$ dB/decade and constant phase $n \cdot 90^\circ$;
- (ii) the factor $(1 + s/\omega_p)^{-1}$ (simple pole) has asymptotic magnitude 0 dB for $\omega \ll \omega_p$ and slope -20 dB/decade for $\omega \gg \omega_p$, with maximum error 3.01 dB at $\omega = \omega_p$; the phase goes from 0° to -90° , being exactly -45° at ω_p ;
- (iii) the factor $(1 + s/\omega_z)$ (simple zero) gives the same plots with the sign reversed.

Proof. (i) $|(i\omega)^n| = \omega^n$, so $20 \log_{10} \omega^n = 20n \log_{10} \omega$: with $x = \log_{10} \omega$ this is the line $y = 20nx$, of slope $20n$ dB per unit of x , i.e. per decade. The argument is $\arg(i^n) = n\pi/2$, constant.

(ii) Setting $u = \omega/\omega_p$,

$$\left| (1 + iu)^{-1} \right| = \frac{1}{\sqrt{1 + u^2}}, \quad \arg (1 + iu)^{-1} = -\arctan u.$$

For $u \ll 1$, $\sqrt{1 + u^2} \rightarrow 1$ and the magnitude in dB tends to 0; for $u \gg 1$, $\sqrt{1 + u^2} \sim u$ and $20 \log_{10}(1/u) = -20 \log_{10} u$, a line of slope -20 dB/decade. The two asymptotes meet at $u = 1$, i.e. $\omega = \omega_p$: the *break frequency*. The maximum gap between curve and asymptotes occurs exactly there, where the true value is $-20 \log_{10} \sqrt{2} = -3.0103$ dB against 0 dB for the asymptote. (That $u = 1$ is the point of maximum gap follows since the difference $-10 \log_{10}(1 + u^2) + 20 \log_{10} \max(1, u)$ is even in $\log u$ and has vanishing derivative only at $u = 1$.) Finally $\arctan 0 = 0$, $\arctan 1 = \pi/4$ and $\arctan u \rightarrow \pi/2$.

(iii) Since $(1 + s/\omega_z) = [(1 + s/\omega_z)^{-1}]^{-1}$, by Theorem 25.2 magnitude in dB and phase change sign. $\square$

Example 25.1: Bode plot of a second-order system. Sketch the asymptotic plots of

$$H(s) = \frac{100}{(s + 1)(s + 10)}.$$

Step 1 — normalised form. Factoring the constants out of the binomials:

$$H(s) = \frac{100}{1 \cdot 10} \cdot \frac{1}{(1+s)(1+s/10)} = \frac{10}{(1+s)(1+s/10)}.$$

The static gain is $H(0) = 10$, i.e. 20 dB.

Step 2 — elementary bricks. By Theorem 25.2 the plot is the sum of three contributions: the constant 10 (+20 dB, phase 0), the pole at $\omega_1 = 1$ and the pole at $\omega_2 = 10$.

Step 3 — asymptotic magnitude.

- $\omega < 1$: 20 dB, slope 0;
- $1 < \omega < 10$: slope -20 dB/dec; at $\omega = 10$ the asymptote equals $20 - 20 \log_{10} 10 = 0$ dB;
- $\omega > 10$: slope -40 dB/dec.

Step 4 — exact check values.

$$|H(i1)| = \frac{100}{\sqrt{2} \cdot \sqrt{101}} = \frac{100}{1.41421 \times 10.04988} = \frac{100}{14.21267} = 7.0359,$$

i.e. $20 \log_{10} 7.0359 = 16.947$ dB, against 20 dB for the asymptote: a gap of 3.05 dB, consistent with Theorem 25.3(ii) plus the small contribution of the second pole.

$$|H(i10)| = \frac{100}{\sqrt{101} \cdot \sqrt{200}} = \frac{100}{10.04988 \times 14.14214} = \frac{100}{142.127} = 0.70359,$$

i.e. -3.053 dB against 0 dB for the asymptote. ✓

Step 5 — phase.

$$\arg H(i\omega) = -\arctan \omega - \arctan \frac{\omega}{10}.$$

At $\omega = 1$: $-45^\circ - 5.71^\circ = -50.71^\circ$. At $\omega = \sqrt{10} = 3.162$ (geometric mean of the two poles): $-72.45^\circ - 17.55^\circ = -90.00^\circ$, exactly -90° by symmetry. At $\omega = 10$: $-84.29^\circ - 45^\circ = -129.29^\circ$. As $\omega \rightarrow \infty$: -180° .

Typical error. Reading the static gain straight off the numerator (100, i.e. 40 dB) without normalising the binomials. The form $\frac{100}{(s+1)(s+10)}$ has static gain $100/10 = 10$: the factor 10 in the denominator must be taken out.

Remark 25.3: Bode's gain-phase relation. In Example 25.1 the phase “follows” the slope of the magnitude: where the magnitude falls at 20 dB/dec the phase settles near -90° , where it falls at 40 dB/dec near -180° . This is no accident. Bode proved in 1945 that for minimum-phase systems (no poles or zeros in the right half-plane) the magnitude determines the phase, through the integral relation

$$\arg H(i\omega_0) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{d(\ln |H|)}{du} \ln \coth \frac{|u|}{2} du, \quad u = \ln \frac{\omega}{\omega_0}.$$

The kernel $\ln \coth(|u|/2)$ is strongly concentrated near $u = 0$, which explains the rule of thumb: the phase at ω_0 is determined mainly by the local slope of the magnitude, and a slope of $-20n$ dB/dec corresponds to roughly $-n \cdot 90^\circ$. Proving the integral relation requires the theory of analytic functions in a half-plane and the Hilbert relations, and goes beyond the level of this treatise: see Bode (1945), Chapter 14, cited in the bibliography. Below we use it only qualitatively, and never as a premise of a proof.

25.3. The argument principle and the Nyquist criterion

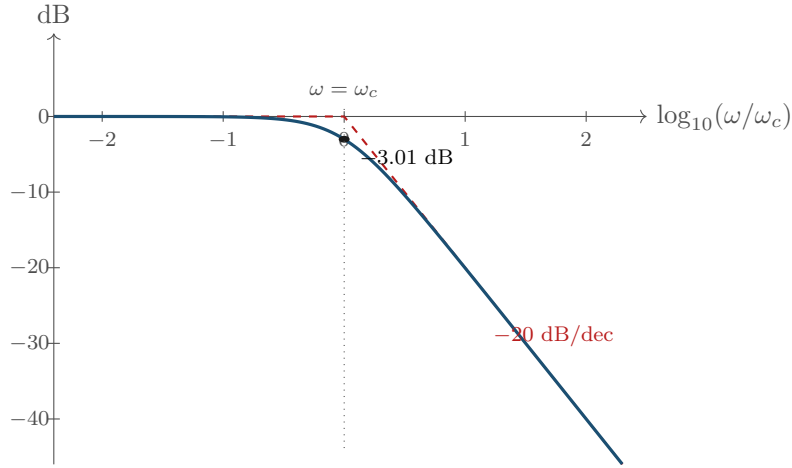


Figure 25.1: Bode magnitude plot of $G(s) = 1/(1 + s/\omega_c)$. Dashed in red are the two asymptotes (0 dB and -20 dB/decade) meeting at the corner frequency ω_c ; in blue the exact magnitude. The largest gap between curve and asymptotes occurs exactly at $\omega = \omega_c$ and equals $-20 \log_{10} \sqrt{2} = -3.01$ dB: this is why ω_c is also called the -3 dB frequency.

The Nyquist criterion is, in substance, a theorem of complex analysis applied to engineering: the *argument principle*, which counts the zeros and poles of a function inside a contour by looking at how many times the image of the contour winds around the origin. We prove it, because it is the logical heart of the whole chapter.

Theorem 25.4: Argument principle. Let F be meromorphic on an open set containing a simple closed contour γ and its interior, with no zeros or poles on γ . Writing Z and P for the sums of the multiplicities of the zeros and poles inside γ ,

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{F'(s)}{F(s)} ds = Z - P,$$

and the left-hand side is the **winding number** of the image curve $F \circ \gamma$ around the origin, counted with sign according to the direction of travel.

Proof. Step 1 — local structure of F'/F . Let s_0 be a zero of F of multiplicity $m \geq 1$: then $F(s) = (s - s_0)^m g(s)$ with g holomorphic and $g(s_0) \neq 0$ nearby. Differentiating,

$$F'(s) = m(s - s_0)^{m-1} g(s) + (s - s_0)^m g'(s),$$

hence

$$\frac{F'(s)}{F(s)} = \frac{m}{s - s_0} + \frac{g'(s)}{g(s)},$$

where the second term is holomorphic near s_0 (because $g(s_0) \neq 0$). So F'/F has a simple pole at s_0 with residue m . If instead s_0 is a pole of F of order p , write $F(s) = (s - s_0)^{-p} g(s)$ and the same computation gives residue $-p$.

Step 2 — residue theorem. The function F'/F is holomorphic on γ and inside it except at the zeros and poles of F , which are finitely many (otherwise they would accumulate, contradicting holomorphy). By Theorem 15.5 of Chapter 3,

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{F'}{F} ds = \sum_{\text{zeros}} m_k - \sum_{\text{poles}} p_j = Z - P.$$

Step 3 — interpretation as winding number. Parametrise γ by $s = \gamma(t)$, $t \in [0, 1]$, and set $w(t) = F(\gamma(t))$. Then

$$\oint_{\gamma} \frac{F'(s)}{F(s)} ds = \int_0^1 \frac{w'(t)}{w(t)} dt.$$

Since w never vanishes (no zeros on γ), the function $\Phi(t) = \int_0^t w'/w$ satisfies $\frac{d}{dt}(w(t)e^{-\Phi(t)}) = 0$, hence $w(t) = w(0)e^{\Phi(t)}$. Since $w(1) = w(0)$, we get $e^{\Phi(1)} = 1$, i.e. $\Phi(1) \in 2\pi i\mathbb{Z}$; and $\Phi(1)/(2\pi i)$ is by definition the winding number of w about the origin, since $\text{Im } \Phi$ is the continuous variation of the argument of w . $\square$

Theorem 25.5: Nyquist criterion. Let $L(s)$ be rational and proper, with no poles of L nor zeros of $1 + L$ on the imaginary axis, and let P be the number of poles of L in the right half-plane. Traverse the Nyquist contour γ_R — the imaginary axis from $-iR$ to $+iR$ followed by the right semicircle of radius R — clockwise, with R large enough, and let N be the number of times the curve $L(\gamma_R)$ winds around the point -1 , counted clockwise. Then the number Z of closed-loop poles in the right half-plane is

$$Z = N + P.$$

In particular the closed loop is stable if and only if $N = -P$, i.e. if and only if the Nyquist plot makes P counterclockwise encirclements of -1 .

Proof. Step 1 — the right function. The closed-loop poles are the zeros of $F(s) := 1 + L(s)$: indeed $T = L/(1 + L)$ has as poles the zeros of the denominator. On the other hand the poles of F coincide with those of L , since F and L differ by the constant 1.

Step 2 — applying the argument principle. Apply Theorem 25.4 to F on the contour γ_R , which for large R encloses all zeros and poles of F in the right half-plane (being F rational, they are finitely many and bounded). Since γ_R is traversed *clockwise*, i.e. opposite to the standard orientation, the winding number of $F(\gamma_R)$ about the origin, counted clockwise, equals $Z - P$.

Step 3 — translation. $F(\gamma_R) = 1 + L(\gamma_R)$: the curve $F(\gamma_R)$ is $L(\gamma_R)$ translated by $+1$. The encirclements of $F(\gamma_R)$ about 0 are therefore exactly those of $L(\gamma_R)$ about -1 , i.e. N . Equating, $N = Z - P$, hence $Z = N + P$.

Step 4 — the arc at infinity is irrelevant. Since L is proper, $L(s) \rightarrow L_{\infty}$ finite as $|s| \rightarrow \infty$ (with $L_{\infty} = 0$ if strictly proper): the image of the semicircle tends to a point and contributes no encirclements for large R . The Nyquist plot thus reduces to the image of the imaginary axis alone, closed through the point at infinity.

Step 5 — stability criterion. The closed loop is stable iff it has no RHP poles, i.e. $Z = 0$, i.e. $N = -P$: P counterclockwise encirclements. $\square$

Corollary 25.1: Open-loop stable case. If L has no RHP poles ($P = 0$), the closed loop is stable if and only if the Nyquist plot does not encircle the point -1 .

Proof. Immediate from Theorem 25.5 with $P = 0$: the condition $N = -P$ becomes $N = 0$. $\square$

Remark 25.4: Why the point -1 . The critical point is -1 and not something else because the feedback is negative and unity: the loop closes on $1 + L = 0$, i.e. $L = -1$. Physically, $L(i\omega) = -1$ means that a signal travelling round the loop returns to its starting point with the same amplitude ($|L| = 1$) and in antiphase ($\arg L = -180^\circ$): added to the input with the minus sign of the feedback, it reinforces it exactly. This is the condition for sustained oscillation, i.e. the boundary between stability and instability.

Definition 25.2: Stability margins. Let L have $P = 0$ and a Nyquist plot not encircling -1 .

- The **gain crossover frequency** ω_g is the one with $|L(i\omega_g)| = 1$, i.e. the intersection of the plot with the *unit circle*; the **phase margin** is

$$\text{PM} = 180^\circ + \arg L(i\omega_g).$$

- The **phase crossover frequency** ω_π is the one with $\arg L(i\omega_\pi) = -180^\circ$, i.e. the intersection with the negative real axis; the **gain margin** is

$$\text{GM} = \frac{1}{|L(i\omega_\pi)|}.$$

Theorem 25.6: Geometric meaning of the margins. With the previous notation:

- (i) GM is the factor by which the loop gain may be multiplied before the plot touches -1 ;
- (ii) PM is the angle by which the plot may be rotated clockwise before it touches -1 ;
equivalently, it is the angle $(-1)\widehat{OL}(i\omega_g)$ read on the unit circle;
- (iii) the maximum tolerable pure delay before instability is $\tau_{\max} = \text{PM} [\text{rad}] / \omega_g$.

Proof. (i) Multiplying L by $k > 0$ dilates the plot by k without rotating it (Theorem (Volume II, Chapter 3) of Chapter 3 with $\alpha = k$ real positive). The point $L(i\omega_\pi)$, lying on the negative real axis at distance $|L(i\omega_\pi)|$ from the origin, reaches -1 when $k|L(i\omega_\pi)| = 1$, i.e. $k = \text{GM}$.

(ii) Multiplying L by $e^{-i\theta}$ rotates the plot by $-\theta$ (Corollary, Volume II, Chapter 3) without changing moduli: in particular ω_g remains the point on the unit circle. That point has argument $\arg L(i\omega_g)$; it reaches -1 , of argument 180° , after a clockwise rotation of $\arg L(i\omega_g) - (-180^\circ) = \text{PM}$.

(iii) A pure delay τ has transfer function $e^{-s\tau}$, of modulus $|e^{-i\omega\tau}| = 1$ and argument $-\omega\tau$: it *does not change the magnitude* and subtracts $\omega\tau$ from the phase. At ω_g , where the magnitude is 1, the phase becomes $\arg L(i\omega_g) - \omega_g\tau$; the system reaches the critical point when this equals -180° , i.e. when $\omega_g\tau = \text{PM}$ in radians. $\square$

Example 25.2: Margins of a third-order loop. Let

$$L(s) = \frac{K}{s(s+1)(s+2)}, \quad K > 0.$$

Step 1 — phase crossover.

$$\arg L(i\omega) = -90^\circ - \arctan \omega - \arctan \frac{\omega}{2} = -180^\circ$$

requires $\arctan \omega + \arctan(\omega/2) = 90^\circ$, i.e. the two angles are complementary, i.e.

$$\omega = \frac{2}{\omega} \implies \omega_\pi = \sqrt{2}.$$

Step 2 — magnitude at ω_π .

$$|L(i\sqrt{2})| = \frac{K}{\sqrt{2} \cdot \sqrt{3} \cdot \sqrt{6}} = \frac{K}{6}.$$

Step 3 — critical gain and gain margin. The plot touches -1 when $K/6 = 1$, i.e. $K_{\text{crit}} = 6$: for $K > 6$ the closed loop is unstable. For $K = 3$,

$$\text{GM} = \frac{1}{3/6} = 2 = 6.02 \text{ dB}.$$

Step 4 — phase margin for $K = 3$. We seek ω_g with $|L| = 1$:

$$\omega \sqrt{1 + \omega^2} \sqrt{4 + \omega^2} = 3.$$

At $\omega = 0.95$ the left side is 2.9013; at $\omega = 0.97$ it is 3.0037. Interpolating, $\omega_g = 0.969$ rad/s. Then

$$\arg L(i\omega_g) = -90^\circ - \arctan(0.969) - \arctan(0.4845) = -90^\circ - 44.10^\circ - 25.85^\circ = -159.95^\circ,$$

so

$$\text{PM} = 180^\circ - 159.95^\circ = 20.05^\circ.$$

Step 5 — tolerable delay. By Theorem 25.6(iii),

$$\tau_{\max} = \frac{20.05^\circ \times \pi/180}{0.969} = \frac{0.34996}{0.969} = 0.361 \text{ s}.$$

A delay of just 361 ms in the feedback path — for instance the cycle time of a PLC sampling too slowly — suffices to destabilise this loop.

Step 6 — independent check of the critical gain. The characteristic polynomial is $s^3 + 3s^2 + 2s + K$. By the Routh–Hurwitz criterion for third order, stability requires $K > 0$ and $3 \cdot 2 > K$, i.e. $K < 6$. ✓The critical value matches the graphical one.

Typical error. Judging a large gain margin sufficient while ignoring the phase margin. Here GM = 6 dB looks reassuring, but PM = 20° is low: the step response will be strongly oscillatory. The two margins must be read *together*, since they measure the distance from -1 along two different directions.

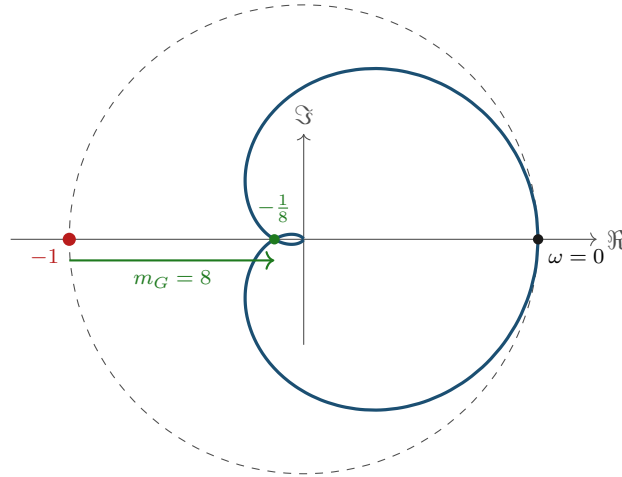


Figure 25.2: Nyquist plot of $L(s) = 1/(1+s)^3$ (the $\omega < 0$ branch is the conjugate of the $\omega > 0$ one). The curve does not encircle -1 : since L has no right half-plane poles, the Nyquist criterion gives $N = P = 0$ and the closed loop is stable. The negative real axis is crossed at $\omega = \sqrt{3}$, where $|L| = 1/8$: the gain margin is $m_G = 8$ (18.06 dB).

25.4. The M and N circles

The Nyquist plot shows the loop gain L ; but what matters is the *closed-loop* behaviour, i.e. $T = L/(1+L)$. The link between the two is a Möbius transformation, and by Theorem (Volume II, Chapter 3) of Chapter 3 the curves of constant $|T|$ and constant $\arg T$ in the L -plane can only be circles. This section computes them.

Theorem 25.7: Constant-magnitude circles (M circles). Let $T = L/(1 + L)$ and $M > 0$. In the complex plane of the variable $L = x + iy$, the locus $\{|T| = M\}$ is:

(i) for $M \neq 1$, the circle of centre and radius

$$c_M = -\frac{M^2}{M^2 - 1}, \quad r_M = \frac{M}{|M^2 - 1|},$$

both real (the centre lies on the real axis);

(ii) for $M = 1$, the vertical line $x = -\frac{1}{2}$.

Moreover the circles with $M > 1$ lie to the left of the line $x = -\frac{1}{2}$ and contain -1 ; those with $M < 1$ lie to the right and contain the origin.

Proof. Step 1 — translating the condition. $|T| = M$ is equivalent to $|L| = M|1 + L|$, i.e. upon squaring,

$$x^2 + y^2 = M^2[(1 + x)^2 + y^2].$$

Step 2 — rearranging. Expanding and collecting,

$$(1 - M^2)(x^2 + y^2) - 2M^2x - M^2 = 0. \quad (*)$$

Step 3 — case $M = 1$. The coefficient of $x^2 + y^2$ vanishes and $(*)$ becomes $-2x - 1 = 0$, i.e. $x = -\frac{1}{2}$: a vertical line.

Step 4 — case $M \neq 1$. Dividing by $1 - M^2 \neq 0$ and completing the square in x :

$$\left(x - \frac{M^2}{1 - M^2}\right)^2 + y^2 = \frac{M^4 + M^2(1 - M^2)}{(1 - M^2)^2} = \frac{M^2}{(1 - M^2)^2}.$$

Hence the centre is $M^2/(1 - M^2) = -M^2/(M^2 - 1)$ and the radius is $M/|1 - M^2|$, as claimed.

Step 5 — relative position. For $M > 1$ we have $M^2 - 1 > 0$, hence $c_M < 0$; and

$$c_M + r_M = \frac{-M^2 + M}{M^2 - 1} = -\frac{M}{M + 1} < -\frac{1}{2}$$

for $M > 1$: the whole circle lies left of $-\frac{1}{2}$. Moreover

$$c_M - r_M = -\frac{M}{M - 1} < -1 < -\frac{M}{M + 1} = c_M + r_M,$$

so -1 lies between the endpoints of the horizontal diameter: it is *inside* the circle. For $M < 1$ the signs reverse and an identical computation shows the circle contains the origin.

Remark on structure. The result was predictable without computation: the map $L \mapsto T = L/(1 + L)$ is a Möbius transformation (Definition (Volume II, Chapter 3) with $a = 1$, $b = 0$, $c = 1$, $d = 1$, determinant $1 \neq 0$), and the loci $|T| = M$ are circles in the T -plane; by Theorem (Volume II, Chapter 3) their preimages are circles or lines in the L -plane, and lines exactly when they pass through the pole of the inverse map. The computation only serves to locate centre and radius. $\square$

Theorem 25.8: Constant-phase circles (N circles). Let $T = L/(1 + L)$ and $\alpha \in (-\pi, \pi]$ with $\alpha \neq 0$, and set $N = \tan \alpha$. In the $L = x + iy$ plane, the locus $\{\arg T = \alpha\}$ is contained in the circle

$$\left(x + \frac{1}{2}\right)^2 + \left(y - \frac{1}{2N}\right)^2 = \frac{1}{4} + \frac{1}{4N^2},$$

of centre $(-\frac{1}{2}, \frac{1}{2N})$ and radius $\frac{1}{2}\sqrt{1+1/N^2}$. All these circles pass through the origin and through the point -1 .

Proof. Step 1 — expressing the phase. With $L = x + iy$, $\arg T = \arg L - \arg(1 + L)$ by Theorem (Volume II, Chapter 3). Taking tangents and using the subtraction formula,

$$N = \frac{\frac{y}{x} - \frac{y}{1+x}}{1 + \frac{y}{x} \cdot \frac{y}{1+x}} = \frac{y(1+x) - yx}{x(1+x) + y^2} = \frac{y}{x^2 + x + y^2},$$

the terms yx cancelling in the numerator.

Step 2 — equation of the locus. $N(x^2 + x + y^2) = y$ rewrites as $x^2 + x + y^2 - y/N = 0$, and completing both squares,

$$\left(x + \frac{1}{2}\right)^2 + \left(y - \frac{1}{2N}\right)^2 = \frac{1}{4} + \frac{1}{4N^2}.$$

Step 3 — the two common points. Substituting $(x, y) = (0, 0)$ into $x^2 + x + y^2 - y/N = 0$ gives $0 = 0$: the origin lies on every N circle. Substituting $(x, y) = (-1, 0)$: $1 - 1 = 0$: so does -1 . ✓

Step 4 — why “contained in” and not “equal to”. The equation obtained expresses $\tan(\arg T) = N$, which does not distinguish α from $\alpha + \pi$. Each circle therefore contains the points of phase α and those of phase $\alpha - \pi$: the locus of phase exactly α is one of the two arcs into which the points 0 and -1 divide the circle. □

Remark 25.5: What they were really for. Superposing the grid of M and N circles on the Nyquist plot lets one read, point by point, the magnitude and phase of the closed-loop response without computing it: the resonant peak M_r is the value of the smallest M circle tangent to the plot, and the corresponding frequency is that of the tangency point. Before computers this construction — known as the Hall chart — was the standard way to design a control loop; its close relative, the Nichols chart, is the same family of circles redrawn in (phase, gain in dB) coordinates. Today the computation is instant, but the geometric content remains: the distance of the plot from -1 controls the resonant peak, which is why margins are good indicators of damping.

25.5. The modulus margin and the sensitivity circle

Gain and phase margins measure the distance from the critical point -1 along two particular directions: the first along the real axis, the second along the unit circle. But a Nyquist plot may have both margins excellent and still pass very close to -1 in an oblique direction. The remedy is to measure the distance from -1 *in all directions at once*, that is to consider the largest circle centred at -1 that the plot does not touch. This is the *modulus margin*, and it is the criterion on which modern controller tuning, PID included, is based.

Definition 25.3: Sensitivity function and sensitivity peak. Given the loop gain $L(s)$, the *sensitivity function* is

$$S(s) = \frac{1}{1 + L(s)}$$

and the *sensitivity peak* is the number $M_s = \|S\|_\infty = \sup_{\omega \in \mathbb{R}} |S(j\omega)|$. The *modulus margin* is $m = 1/M_s$.

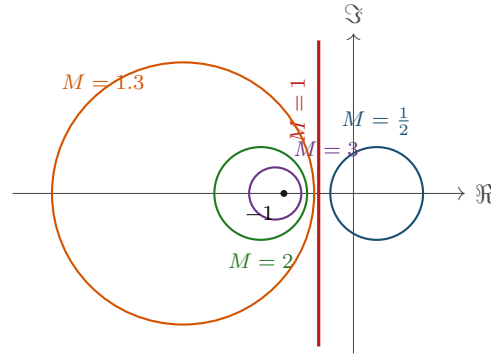


Figure 25.3: The family of M -circles: loci of points L in the complex plane where the closed-loop magnitude $|L/(1+L)|$ equals M . For $M \neq 1$ these are circles with centre $(-M^2/(M^2-1), 0)$ and radius $M/|M^2-1|$; for $M = 1$ the circle degenerates into the line $\Re L = -1/2$, the perpendicular bisector of the segment from the origin to -1 . Circles with $M > 1$ encircle the critical point -1 and shrink onto it as M grows: the resonant peak of the closed loop is read off as the smallest M whose circle touches the Nyquist plot.

Theorem 25.9: The modulus margin is a distance. Let $\Gamma = \{L(j\omega) : \omega \in \mathbb{R}\}$ be the Nyquist plot. Then

$$m = \frac{1}{M_s} = \min_{\omega \in \mathbb{R}} |1 + L(j\omega)| = \text{dist}(-1, \Gamma)$$

that is, m is the radius of the largest circle centred at -1 that does not meet the Nyquist plot. The locus $\{L : |1 + L| = m\}$ is that circle, and is called the sensitivity circle.

Proof. **Step 1 — from sup to min.** By definition

$$M_s = \sup_{\omega} \frac{1}{|1 + L(j\omega)|} = \frac{1}{\inf_{\omega} |1 + L(j\omega)|},$$

since $t \mapsto 1/t$ is decreasing on the positive reals and $|1 + L| > 0$ for every ω (if it vanished, the closed loop would have a pole on the imaginary axis). Hence $m = 1/M_s = \inf_{\omega} |1 + L(j\omega)|$, and the infimum is a minimum because $L(j\omega) \rightarrow L(\infty)$ as $\omega \rightarrow \pm\infty$ and the continuous function $\omega \mapsto |1 + L(j\omega)|$ attains its minimum on the compactification.

Step 2 — geometric reading. In the complex plane $|1 + L| = |L - (-1)|$ is the Euclidean distance between the point L and the point -1 . Hence $\min_{\omega} |1 + L(j\omega)|$ is the smallest distance between -1 and the points of the plot, that is $\text{dist}(-1, \Gamma)$.

Step 3 — the circle. The set $\{L \in \mathbb{C} : |L + 1| = m\}$ is by definition the circle of centre -1 and radius m . By Step 2 no point of Γ lies strictly inside that circle, and at least one point belongs to it (the one attaining the minimum): the sensitivity circle is therefore the largest circle centred at -1 disjoint from the interior of the plot, and it touches it.

Step 4 — why S . The sensitivity function is the transfer function from output disturbance to output, and also the one from reference to error. Its peak M_s is thus the largest amplification factor of disturbances: a large M_s means both “close to instability” and “amplifies disturbances”. The theorem says that the two are *the same thing*, measured by the same number.

□

Theorem 25.10: The modulus margin dominates the other two. If the Nyquist plot does not enter the sensitivity circle of radius $m < 1$, then the gain margin m_G and the phase

margin φ_m satisfy

$$m_G \geq \frac{1}{1-m} = \frac{M_s}{M_s-1}, \quad \varphi_m \geq 2 \arcsin \frac{m}{2} = 2 \arcsin \frac{1}{2M_s}$$

Proof. Step 1 — gain margin. Let ω_π be the frequency at which the plot crosses the negative real half-axis, and let $L(j\omega_\pi) = -1/m_G$ with $m_G > 1$ (this is the definition of gain margin). The distance of this point from -1 equals

$$\left| -\frac{1}{m_G} - (-1) \right| = 1 - \frac{1}{m_G},$$

positive because $m_G > 1$. Since the point belongs to Γ , by Theorem 25.9 that distance is $\geq m$:

$$1 - \frac{1}{m_G} \geq m \iff \frac{1}{m_G} \leq 1 - m \iff m_G \geq \frac{1}{1-m},$$

having used $m < 1$ in order to invert.

Step 2 — phase margin. Let ω_c be the gain crossover frequency, that is $|L(j\omega_c)| = 1$, and let φ_m be the phase margin, so that $L(j\omega_c) = e^{j(-\pi+\varphi_m)}$. The distance from $-1 = e^{j\pi}$ is computed by the chord formula:

$$|e^{j\theta} + 1| = |e^{j\theta/2}| \cdot |e^{j\theta/2} + e^{-j\theta/2}| = 2|\cos(\theta/2)|.$$

With $\theta = -\pi + \varphi_m$ one has $\cos(\theta/2) = \cos(-\frac{\pi}{2} + \frac{\varphi_m}{2}) = \sin(\frac{\varphi_m}{2})$, so the distance equals $2 \sin(\varphi_m/2)$.

Step 3 — conclusion. This point too belongs to Γ , hence

$$2 \sin \frac{\varphi_m}{2} \geq m \implies \sin \frac{\varphi_m}{2} \geq \frac{m}{2} \implies \varphi_m \geq 2 \arcsin \frac{m}{2},$$

the sine being increasing on $[0, \pi/2]$ and $\varphi_m/2 \in [0, \pi/2]$ in the cases of interest.

Step 4 — reading. A single number, m , simultaneously guarantees both classical margins. The converse is false: one may have large m_G and φ_m and small m , because the plot can approach -1 from a direction that is neither the real nor the tangential one. This is why M_s has superseded the pair (m_G, φ_m) as the single robustness indicator. $\square$

Example 25.3: The modulus margin of $L(s) = K/(1+s)^3$. **Step 1 — exact computation of $|1+L|^2$.** From $(1+j\omega)^3 = (1-3\omega^2) + j(3\omega-\omega^3)$ and $|(1+j\omega)^3|^2 = (1+\omega^2)^3$ it follows that

$$|1+L(j\omega)|^2 = 1 + \frac{2K(1-3\omega^2) + K^2}{(1+\omega^2)^3}.$$

Indeed, writing $L = K\overline{(1+j\omega)^3}/(1+\omega^2)^3$, the cross term equals $2K(1-3\omega^2)/(1+\omega^2)^3$ and the quadratic term $K^2(1+\omega^2)^3/(1+\omega^2)^6 = K^2/(1+\omega^2)^3$.

Step 2 — minimum. Setting $u = \omega^2$ and $f(u) = 1 + (2K(1-3u) + K^2)/(1+u)^3$, one has

$$f'(u) = \frac{3K(4u-4-K)}{(1+u)^4},$$

which vanishes at $u^* = 1 + K/4$ and is negative before, positive after: a minimum. Hence

$$m^2 = f(u^*) = 1 - \frac{4K + \frac{1}{2}K^2}{(2 + \frac{K}{4})^3}.$$

Step 3 — the case $K = 2$. One gets $u^* = 1.5$, that is $\omega^* = \sqrt{1.5} = 1.2247$, and

$$m^2 = 1 - \frac{8+2}{2.5^3} = 1 - \frac{10}{15.625} = 1 - 0.64 = 0.36,$$

hence $m = 0.600$ exactly and $M_s = 1/0.6 = 1.667$.

Step 4 — comparison with the classical margins. The crossing of the negative real axis occurs where $3 \arctan \omega = 180^\circ$, that is $\omega = \sqrt{3}$, where $|L| = 2/8 = 0.25$: hence $m_G = 4$. The unit-modulus crossing occurs where $(1 + \omega^2)^{3/2} = 2$, that is $\omega_c = 0.7664$, where the phase is $-3 \arctan(0.7664) = -112.36^\circ$: hence $\varphi_m = 67.6^\circ$. The estimates of Theorem 25.10 give

$$m_G \geq \frac{1}{1-0.6} = 2.5 \checkmark, \quad \varphi_m \geq 2 \arcsin(0.3) = 34.9^\circ \checkmark,$$

both verified and, as always, conservative.

Step 5 — tuning the gain. Current practice recommends $M_s \in [1.2, 2.0]$: below 1.2 the system is very sluggish, above 2.0 it is not robust. Here $M_s = 1.667$ is acceptable but close to the upper desirable limit. If one wants $M_s = 1.60$, that is $m = 0.625$, one solves Step 2 numerically for K : with $K = 1.85$ one gets $u^* = 1.4625$, $(2 + K/4)^3 = 2.4625^3 = 14.9324$, $4K + K^2/2 = 9.1113$, hence $m^2 = 1 - 0.61017 = 0.38983$ and $m = 0.6244$, that is $M_s = 1.602$. Reducing the gain by 7.5% is therefore enough.

Remark 25.6: What the three PID actions do to the sensitivity circle. A PID controller has loop gain $L(s) = K_p(1 + \frac{1}{T_i s} + T_d s)G(s)$, and each of the three actions moves the Nyquist plot in a characteristic way relative to the sensitivity circle:

- the proportional action K_p is a homothety centred at the origin: it dilates the plot without rotating it, and therefore brings it closer to -1 as K_p grows, reducing m ;
- the integral action adds -90° of phase at low frequencies and infinite gain at DC: it removes the steady-state error (because $S(0) = 0$), but rotates the initial part of the plot towards the left half-plane, that is towards -1 ;
- the derivative action adds up to $+90^\circ$ at high frequencies: it rotates the plot away from -1 precisely in the critical region, and this is why it increases the modulus margin — at the price of amplifying measurement noise, which is why in practice it is always used in the filtered form $T_d s/(1 + T_d s/N)$.

Tuning a PID therefore means choosing the three parameters so that the plot stays outside the prescribed sensitivity circle: this is the modern formulation of the problem, and it replaces the empirical rules of Ziegler and Nichols with a verifiable geometric condition. Note that the constraint is a constraint on a circle: the whole of classical control theory can be summarised by saying which circles the Nyquist plot must avoid.

Warning. The modulus margin cannot be made arbitrarily large. If L has relative degree at least 2 and no unstable poles, Bode's integral imposes $\int_0^\infty \ln |S(j\omega)| d\omega = 0$: the area where the sensitivity is smaller than 1 (on a logarithmic scale) exactly equals the area where it is larger than 1. Reducing $|S|$ at certain frequencies necessarily makes it grow at others — the so-called *waterbed effect*. Designing a controller is therefore not the minimisation of M_s , but the choice of *where* to pay the unavoidable amplification.

25.6. The circle criterion for nonlinear systems

The Nyquist criterion applies to linear systems. When the loop contains a static nonlinearity — a saturation, a dead zone, a relay — the criterion does not apply. There is however a substitute

that preserves the geometric reading, and that replaces the *point* -1 by a *circle*.

Definition 25.4: L^2 gain and sector. For a causal operator G on signals defined on $[0, \infty)$, the **gain** is

$$\|G\| = \sup \left\{ \|Gu\|_{2,T} : \|u\|_{2,T} \leq 1, T > 0 \right\},$$

where $\|\cdot\|_{2,T}$ is the L^2 norm of the truncation to $[0, T]$. A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ with $\varphi(0) = 0$ belongs to the **sector** $[k_1, k_2]$ if

$$k_1 e^2 \leq e \varphi(e) \leq k_2 e^2 \quad \text{for every } e \in \mathbb{R}.$$

Theorem 25.11: Small-gain theorem. Let G_1, G_2 be causal operators with finite gains γ_1, γ_2 , connected in feedback:

$$e_1 = u_1 - G_2 e_2, \quad e_2 = u_2 + G_1 e_1.$$

If $\gamma_1 \gamma_2 < 1$, then for every $T > 0$

$$\|e_1\|_{2,T} \leq \frac{\|u_1\|_{2,T} + \gamma_2 \|u_2\|_{2,T}}{1 - \gamma_1 \gamma_2},$$

and similarly for e_2 : the closed loop is input-output stable.

Proof. Step 1 — cross estimates. Applying the triangle inequality to the first equation, truncated to $[0, T]$, and the definition of gain:

$$\|e_1\|_{2,T} \leq \|u_1\|_{2,T} + \gamma_2 \|e_2\|_{2,T},$$

and similarly from the second,

$$\|e_2\|_{2,T} \leq \|u_2\|_{2,T} + \gamma_1 \|e_1\|_{2,T}.$$

(Causality is what makes truncation legitimate: the value of $G_k e_k$ on $[0, T]$ depends only on e_k on $[0, T]$.)

Step 2 — substitution. Inserting the second into the first,

$$\|e_1\|_{2,T} \leq \|u_1\|_{2,T} + \gamma_2 \|u_2\|_{2,T} + \gamma_1 \gamma_2 \|e_1\|_{2,T}.$$

Step 3 — solving. Since $\|e_1\|_{2,T}$ is finite for every T (the equations are well posed on every finite interval) and $1 - \gamma_1 \gamma_2 > 0$, we may move it to the left:

$$(1 - \gamma_1 \gamma_2) \|e_1\|_{2,T} \leq \|u_1\|_{2,T} + \gamma_2 \|u_2\|_{2,T},$$

which is the claim. The bound does not depend on T : if the inputs are in $L^2(0, \infty)$, so is e_1 . $\square$

Theorem 25.12: Circle criterion. Consider the loop formed by a stable LTI system of transfer function $G(s)$ and a static nonlinearity φ in the sector $[k_1, k_2]$ with $0 < k_1 < k_2$. If the Nyquist plot of G stays outside the closed disc $D(k_1, k_2)$ whose diameter is the segment of the real axis with endpoints $-1/k_1$ and $-1/k_2$, then the closed loop is input-output stable.

Proof. Step 1 — loop transformation. Set

$$k_m = \frac{k_1 + k_2}{2}, \quad r = \frac{k_2 - k_1}{2},$$

and define $\tilde{\varphi}(e) = \varphi(e) - k_m e$. The sector condition $k_1 e^2 \leq e\varphi(e) \leq k_2 e^2$ becomes $-r e^2 \leq e\tilde{\varphi}(e) \leq r e^2$, i.e. $|\tilde{\varphi}(e)| \leq r|e|$: the operator $\tilde{\varphi}$ has gain $\leq r$. At the same time, moving the linear term $k_m e$ into the linear block, the loop becomes the feedback between

$$\tilde{G}(s) = \frac{G(s)}{1 + k_m G(s)} \quad \text{and} \quad \tilde{\varphi},$$

as one checks by rewriting the loop equations and collecting.

Step 2 — applying small gain. By Theorem 25.11, the transformed loop is stable if $\|\tilde{G}\| \cdot r < 1$. For a stable LTI system the L^2 gain equals the infinity norm of the frequency response (Parseval's identity of Chapter 4 applied to the output), so the condition is

$$|\tilde{G}(i\omega)| < \frac{1}{r} \quad \text{for every } \omega.$$

Step 3 — geometric translation. The condition rewrites as $r|G| < |1 + k_m G|$, and squaring with $G = x + iy$:

$$(r^2 - k_m^2)(x^2 + y^2) - 2k_m x - 1 < 0.$$

Now

$$r^2 - k_m^2 = \frac{(k_2 - k_1)^2 - (k_1 + k_2)^2}{4} = -k_1 k_2,$$

and since $k_1 k_2 > 0$ we may divide by $-k_1 k_2 < 0$, reversing the inequality:

$$x^2 + y^2 + \frac{k_1 + k_2}{k_1 k_2} x + \frac{1}{k_1 k_2} > 0.$$

Step 4 — completing the square.

$$\left(x + \frac{k_1 + k_2}{2k_1 k_2}\right)^2 + y^2 > \frac{(k_1 + k_2)^2 - 4k_1 k_2}{4k_1^2 k_2^2} = \frac{(k_2 - k_1)^2}{4k_1^2 k_2^2}.$$

The centre is therefore

$$-\frac{k_1 + k_2}{2k_1 k_2} = -\frac{1}{2} \left(\frac{1}{k_1} + \frac{1}{k_2} \right),$$

the midpoint between $-1/k_1$ and $-1/k_2$, and the radius is

$$\frac{k_2 - k_1}{2k_1 k_2} = \frac{1}{2} \left(\frac{1}{k_1} - \frac{1}{k_2} \right),$$

half their distance. The strict inequality says $G(i\omega)$ lies *outside* that disc, which is exactly $D(k_1, k_2)$. $\square$

Remark 25.7: How the point becomes a circle. If the nonlinearity degenerates into a constant gain k , i.e. $k_1 = k_2 = k$, the disc $D(k, k)$ shrinks to the point $-1/k$, and “stay outside the disc” becomes “do not pass through $-1/k$ ”, i.e. the Nyquist condition for the gain kG . The circle criterion is thus the exact generalisation: uncertainty about the gain, expressed as the interval $[k_1, k_2]$, translates into inflating the critical point into a disc. The wider the interval, the larger the disc, the more stringent the condition — which is entirely reasonable: whoever knows less about their plant must stay further from the edge.

25.7. The root locus

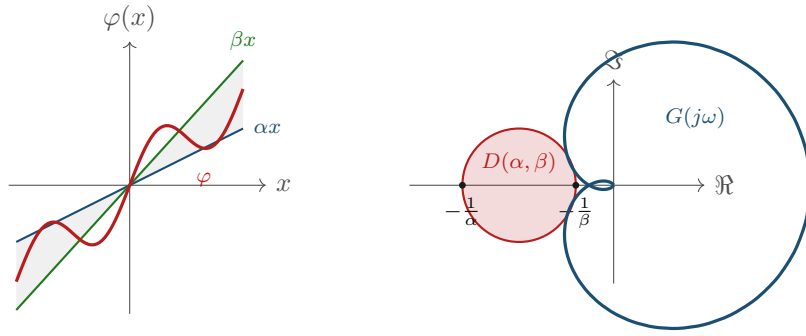


Figure 25.4: The circle criterion. Left: the nonlinearity φ lies in the sector $[\alpha, \beta]$, i.e. its graph stays confined between the lines $y = \alpha x$ and $y = \beta x$ (grey region). Right: the forbidden disc $D(\alpha, \beta)$, whose diameter is the real segment from $-1/\alpha$ to $-1/\beta$; if the Nyquist plot of the linear part G does not enter the disc (and encircles it as many times as G has unstable poles) the feedback system is absolutely stable. For $\alpha = \beta$ the disc collapses to the point $-1/\alpha$ and the Nyquist criterion is recovered.

Theorem 25.13: Magnitude and angle conditions. Let $L(s) = K \frac{N(s)}{D(s)}$ with N, D monic and $K > 0$. A point s_0 belongs to the **root locus** — i.e. is a closed-loop pole for some $K > 0$ — if and only if

$$\arg L(s_0) = \arg \frac{N(s_0)}{D(s_0)} \equiv 180^\circ \pmod{360^\circ},$$

and in that case the corresponding gain is

$$K = \frac{|D(s_0)|}{|N(s_0)|}.$$

Proof. The closed-loop poles are the zeros of $1 + L$, i.e. the solutions of $L(s) = -1$. By Theorem (Volume II, Chapter 3) two complex numbers coincide iff they have equal modulus and arguments congruent modulo 2π . Now -1 has modulus 1 and argument 180° , so $L(s_0) = -1$ is equivalent to

$$|L(s_0)| = 1, \quad \arg L(s_0) \equiv 180^\circ \pmod{360^\circ}.$$

Since $K > 0$ is real, it does not change the argument (Theorem, Volume II, Chapter 3) and the second condition involves only N/D : it is the angle condition, independent of K . Given s_0 satisfying it, the first condition determines $K = |D(s_0)|/|N(s_0)|$ uniquely, and it is positive. $\square$

Theorem 25.14: Asymptotes of the locus. With $n = \deg D$, $m = \deg N$ and $q := n - m \geq 1$, as $K \rightarrow +\infty$ exactly m branches of the locus tend to the zeros of N and q branches tend to infinity along asymptotes of directions

$$\theta_k = \frac{(2k+1)180^\circ}{q}, \quad k = 0, 1, \dots, q-1,$$

meeting at the real point (centroid)

$$\sigma_a = \frac{\sum_j p_j - \sum_i z_i}{q},$$

where p_j are the poles and z_i the zeros of L , counted with multiplicity.

Proof. Step 1 — behaviour at infinity. The characteristic equation is $D(s) + KN(s) = 0$. For $|s| \rightarrow \infty$, $D(s) \sim s^n$ and $N(s) \sim s^m$, hence $s^n + Ks^m \approx 0$, i.e. $s^q \approx -K$. The q solutions of $s^q = -K$ are, by Theorem (Volume II, Chapter 3) of Chapter 3,

$$s = K^{1/q} e^{i(180^\circ + k \cdot 360^\circ)/q}, \quad k = 0, \dots, q-1,$$

whose arguments are exactly the θ_k of the statement. The other m roots stay bounded and, dividing the characteristic equation by K and letting $K \rightarrow \infty$, satisfy $N(s) = 0$: they tend to the zeros.

Step 2 — the centroid. Write $D(s) = s^n + d_{n-1}s^{n-1} + \dots$ and $N(s) = s^m + e_{m-1}s^{m-1} + \dots$. By Vieta's formulas, $\sum_j p_j = -d_{n-1}$ and $\sum_i z_i = -e_{m-1}$.

The characteristic polynomial $D + KN$ has degree n and coefficient of s^{n-1} equal to d_{n-1} when $q \geq 2$ (then $m \leq n-2$ and KN does not contribute at that degree). For $q \geq 2$, the sum of *all* closed-loop roots equals $-d_{n-1} = \sum_j p_j$ for every K .

Step 3 — passing to the limit. For large K , m roots tend to the zeros, with sum $\rightarrow \sum_i z_i$; the remaining q behave as $\sigma_a + K^{1/q}e^{i\theta_k} + o(1)$ for a suitable real σ_a . But

$$\sum_{k=0}^{q-1} e^{i\theta_k} = 0,$$

because the θ_k are equally spaced by $360^\circ/q$ and their sum is that of the q -th roots of -1 , zero by Corollary (Volume II, Chapter 3) of Chapter 3 (after factoring out $e^{i180^\circ/q}$). Hence the sum of the q large roots tends to $q\sigma_a$. Equating the two expressions for the total sum,

$$\sum_j p_j = \sum_i z_i + q\sigma_a \implies \sigma_a = \frac{\sum_j p_j - \sum_i z_i}{q}.$$

(For $q = 1$ the single asymptote is the direction 180° and the result persists by continuity of the argument on the coefficient of s^{n-1} .) $\square$

Theorem 25.15: The circular root locus. Let

$$L(s) = K \frac{s+z}{(s+p_1)(s+p_2)}, \quad z, p_1, p_2 \in \mathbb{R} \text{ distinct},$$

with $(p_1 - z)(p_2 - z) > 0$. Then the non-real part of the root locus is contained in the circle of centre $-z$ (the zero) and radius

$$\rho = \sqrt{(p_1 - z)(p_2 - z)}.$$

Proof. Step 1 — eliminating K . The characteristic equation is

$$(s+p_1)(s+p_2) + K(s+z) = 0 \iff K = -\frac{(s+p_1)(s+p_2)}{s+z},$$

for $s \neq -z$. Since K must be real, the locus condition is

$$\operatorname{Im} \left[\frac{(s+p_1)(s+p_2)}{s+z} \right] = 0.$$

Step 2 — change of variable. Set $u = s + z$, $a = p_1 - z$, $b = p_2 - z$, so that $s + p_1 = u + a$ and $s + p_2 = u + b$, with a, b real. Then

$$\frac{(u+a)(u+b)}{u} = u + (a+b) + \frac{ab}{u}.$$

Step 3 — imaginary part. Writing $u = \xi + i\eta$ and using $\text{Im}(1/u) = -\eta/|u|^2$,

$$\text{Im} \left[u + (a+b) + \frac{ab}{u} \right] = \eta \left(1 - \frac{ab}{|u|^2} \right).$$

Step 4 — the two components of the locus. This vanishes iff $\eta = 0$ (the real part of the locus) or

$$|u|^2 = ab \iff |s+z|^2 = (p_1-z)(p_2-z),$$

i.e. s lies on the circle of centre $-z$ and radius $\rho = \sqrt{ab}$, real and positive by the hypothesis $ab > 0$. $\square$

Example 25.4: A circular root locus, with numerical check. Let

$$L(s) = K \frac{s+3}{(s+1)(s+2)}.$$

Step 1 — applying the theorem. Here $z = 3$, $p_1 = 1$, $p_2 = 2$, and $(1-3)(2-3) = 2 > 0$, so the complex locus is the circle of centre -3 and radius $\rho = \sqrt{2} = 1.41421$.

Step 2 — intersections with the real axis.

$$s = -3 \pm \sqrt{2} \implies s_1 = -1.58579, \quad s_2 = -4.41421.$$

These are the *breakaway* and *break-in* points.

Step 3 — corresponding gains. By Theorem 25.13,

$$K(s_1) = -\frac{(s_1+1)(s_1+2)}{s_1+3} = -\frac{(-0.58579)(0.41421)}{1.41421} = \frac{0.24264}{1.41421} = 0.17157,$$

$$K(s_2) = -\frac{(-3.41421)(-2.41421)}{-1.41421} = \frac{8.24264}{1.41421} = 5.82843.$$

Both positive $\checkmark$. One recognises the exact values $K_1 = 3 - 2\sqrt{2} = 0.171573$ and $K_2 = 3 + 2\sqrt{2} = 5.828427$, whose product is $9 - 8 = 1$ and whose sum is 6.

Step 4 — checking a complex point. Take $K = 1$. The characteristic equation is

$$s^2 + 4s + 5 = 0 \implies s = -2 \pm i.$$

Membership of the circle: $|(-2+i)+3| = |1+i| = \sqrt{2} = \rho \checkmark$

Step 5 — design reading. For K between 0.1716 and 5.8284 the closed-loop poles are complex conjugate and lie on the circle: is their modulus (i.e. the natural frequency) constant? No — the modulus is measured from the *origin*, not from the centre of the circle, and it varies. What stays constant is the distance from the *zero*: this is the property guaranteed by Theorem 25.15, and it makes drawing the locus immediate with ruler and compass.

25.8. Transmission lines and the reflection coefficient

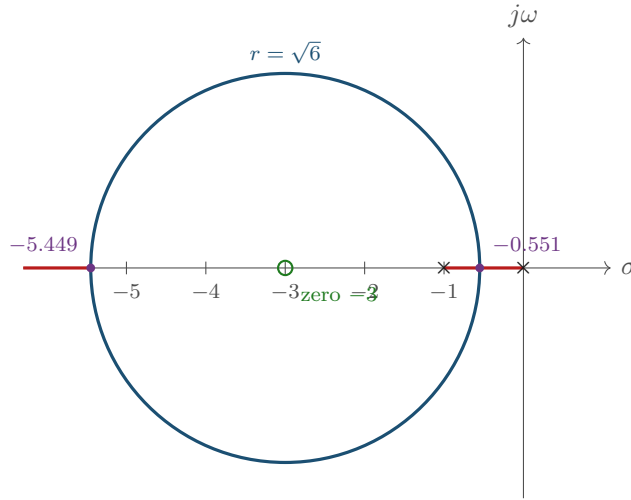


Figure 25.5: Root locus of $L(s) = K(s+3)/(s(s+1))$: poles at 0 and -1 (crosses), zero at -3 (small circle). The branches leaving the real axis trace exactly a circle centred at the zero -3 with radius $\sqrt{(3-0)(3-1)} = \sqrt{6} \simeq 2.449$. In red the real-axis portions belonging to the locus: the breakaway point is $-3 + \sqrt{6} = -0.551$, the break-in point $-3 - \sqrt{6} = -5.449$. Whenever exactly one finite zero is present, the complex locus is always a circle centred at that zero.

Definition 25.5: Lossless line and reflection coefficient. On a lossless transmission line of real characteristic impedance $Z_0 > 0$ and phase constant $\beta = 2\pi/\lambda$, voltage and current in the sinusoidal steady state are written, as functions of the distance d from the load towards the generator,

$$\underline{V}(d) = V^+ e^{i\beta d} + V^- e^{-i\beta d}, \quad \underline{I}(d) = \frac{1}{Z_0} (V^+ e^{i\beta d} - V^- e^{-i\beta d}),$$

where V^+ and V^- are the amplitudes of the forward and reflected waves. The **reflection coefficient** at d is

$$\Gamma(d) = \frac{V^- e^{-i\beta d}}{V^+ e^{i\beta d}} = \Gamma_L e^{-2i\beta d}, \quad \Gamma_L := \frac{V^-}{V^+}.$$

Theorem 25.16: The reflection coefficient as a Möbius map. Let $\underline{Z}_L$ be the load impedance and $z_L = \underline{Z}_L/Z_0$ its normalised form. Then:

- (i) $\Gamma_L = \frac{\underline{Z}_L - Z_0}{\underline{Z}_L + Z_0} = \frac{z_L - 1}{z_L + 1}$, and conversely $z_L = \frac{1 + \Gamma_L}{1 - \Gamma_L}$;
- (ii) the map $z \mapsto \Gamma$ is a Möbius transformation carrying the closed half-plane $\operatorname{Re} z \geq 0$ (passive loads) into the closed disc $|\Gamma| \leq 1$, and the imaginary axis $\operatorname{Re} z = 0$ (purely reactive loads) onto the unit circle $|\Gamma| = 1$;
- (iii) $\Gamma_L = 0$ if and only if $\underline{Z}_L = Z_0$ (matching); $\Gamma_L = +1$ for an open circuit, $\Gamma_L = -1$ for a short circuit.

Proof. (i) Evaluating Definition 25.5 at $d = 0$:

$$\underline{Z}_L = \frac{\underline{V}(0)}{\underline{I}(0)} = Z_0 \frac{1 + \Gamma_L}{1 - \Gamma_L},$$

after dividing numerator and denominator by $V^+ \neq 0$. Solving for Γ_L : $\Gamma_L(\underline{Z}_L + Z_0) = \underline{Z}_L - Z_0$, whence the formula. Dividing numerator and denominator by Z_0 gives the normalised form.

(ii) This is Exercise S1 of Chapter 6, proved there directly: with $\underline{Z} = R + iX$ and $Z_0 > 0$,

$$|\underline{Z} + Z_0|^2 - |\underline{Z} - Z_0|^2 = 4RZ_0,$$

which is ≥ 0 iff $R \geq 0$, i.e. $|\Gamma| \leq 1$ iff $\text{Re } \underline{Z} \geq 0$, with simultaneous equality. In normalised form ($Z_0 = 1$) the map is exactly $w = (z - 1)/(z + 1)$ of Corollary (Volume II, Chapter 3) of Chapter 3, read the other way round.

(iii) $\Gamma_L = 0 \iff \underline{Z}_L = Z_0$ from (i). For an open circuit, $\underline{Z}_L \rightarrow \infty$ and $\Gamma_L \rightarrow 1$; for a short circuit, $\underline{Z}_L = 0$ and $\Gamma_L = -1$. $\square$

Theorem 25.17: Propagation along the line: rotation on the circle. On a lossless line:

- (i) $|\Gamma(d)| = |\Gamma_L|$ for every d : moving along the line is a rotation on the circle of radius $|\Gamma_L|$, by the angle $-2\beta d$;
- (ii) Γ is periodic in d with period $\lambda/2$;
- (iii) the input impedance at distance d is

$$\underline{Z}_{\text{in}}(d) = Z_0 \frac{\underline{Z}_L + iZ_0 \tan \beta d}{Z_0 + i\underline{Z}_L \tan \beta d};$$

(iv) (quarter-wave transformer) for $d = \lambda/4$, $\underline{Z}_{\text{in}} = Z_0^2/\underline{Z}_L$.

Proof. (i) From Definition 25.5, $\Gamma(d) = \Gamma_L e^{-2i\beta d}$, and $|e^{-2i\beta d}| = 1$ (Corollary, Volume II, Chapter 3); hence the modulus is unchanged and the argument decreases by $2\beta d$: the rotation of Theorem (Volume II, Chapter 3).

(ii) $\Gamma(d + \Delta) = \Gamma(d)$ requires $2\beta\Delta \in 2\pi\mathbb{Z}$, i.e. $\Delta \in (\lambda/2)\mathbb{Z}$: the minimal period is $\lambda/2$, not λ . This is why the Smith chart carries a “wavelengths towards generator” scale graduated from 0 to 0.5.

(iii) By Theorem 25.16(i) applied at d ,

$$\underline{Z}_{\text{in}}(d) = Z_0 \frac{1 + \Gamma_L e^{-2i\beta d}}{1 - \Gamma_L e^{-2i\beta d}}.$$

Substituting $\Gamma_L = (\underline{Z}_L - Z_0)/(\underline{Z}_L + Z_0)$ and multiplying numerator and denominator by $(\underline{Z}_L + Z_0)e^{i\beta d}$, then collecting and using the inverse Euler formulas (Theorem, Volume II, Chapter 3),

$$\text{numerator} = 2\underline{Z}_L \cos \beta d + 2iZ_0 \sin \beta d, \quad \text{denominator} = 2Z_0 \cos \beta d + 2i\underline{Z}_L \sin \beta d.$$

Dividing both by $2 \cos \beta d$ gives the stated formula.

(iv) For $d = \lambda/4$ we have $\beta d = \pi/2$ and $\tan \beta d \rightarrow \infty$. Passing to the limit in (iii), the terms with $\tan$ dominate:

$$\underline{Z}_{\text{in}} \rightarrow \frac{Z_0^2}{\underline{Z}_L}.$$

(Equivalently, from (i): Γ rotates by $-\pi$, i.e. changes sign, and by (i) of Theorem 25.16 changing the sign of Γ amounts to replacing z by $1/z$.) $\square$

Theorem 25.18: Standing wave ratio. On a lossless line, the ratio of the maximum to the minimum of the voltage magnitude along the line is

$$\text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} \in [1, +\infty],$$

and it equals 1 if and only if the line is matched. Voltage maxima and minima are $\lambda/4$ apart.

Proof. Step 1 — voltage magnitude. From Definition 25.5,

$$\underline{V}(d) = V^+ e^{i\beta d} (1 + \Gamma(d)) \implies |\underline{V}(d)| = |V^+| \cdot |1 + \Gamma(d)|.$$

Step 2 — extremes of $|1 + \Gamma|$. By Theorem 25.17(i), $\Gamma(d)$ traverses the whole circle of centre the origin and radius $|\Gamma_L|$ as d varies. By the triangle inequality (Theorem (Volume II, Chapter 3)(iv),(v)),

$$1 - |\Gamma_L| \leq |1 + \Gamma(d)| \leq 1 + |\Gamma_L|,$$

with equality on the right when $\Gamma(d)$ is real positive and on the left when it is real negative. Both positions are attained because the argument of $\Gamma(d)$ takes all values.

Step 3 — ratio and spacing. The ratio of the two extremes is as stated. The two cases correspond to arguments of Γ differing by π ; since the argument is $\arg \Gamma_L - 2\beta d$, a difference of π corresponds to $2\beta \Delta d = \pi$, i.e. $\Delta d = \lambda/4$.

Step 4 — matched case. $\text{SWR} = 1 \iff |\Gamma_L| = 0 \iff \underline{Z}_L = Z_0$ by Theorem 25.16(iii). $\square$

25.9. The Smith chart

The Smith chart, published by Phillip Hagar Smith in 1939 and refined in 1944, is the graphical representation of the map $z \mapsto \Gamma$ of Theorem 25.16. On it one reads impedance, reflection coefficient and SWR simultaneously, and moves along the line simply by rotating. It is, to all effects, Theorem (Volume II, Chapter 3) of Chapter 3 printed on paper.

Theorem 25.19: Constant-resistance circles. Let $z = r + ix$ be the normalised impedance and $\Gamma = u + iv$. For each fixed $r \geq 0$, the locus $\{\text{Re } z = r\}$ is, in the Γ -plane, the circle

$$\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2,$$

of centre $(\frac{r}{1+r}, 0)$ and radius $\frac{1}{1+r}$. All these circles lie inside the unit disc, are mutually tangent at $\Gamma = 1$, and degenerate to the point $\Gamma = 1$ as $r \rightarrow \infty$.

Proof. Step 1 — expression for r . From $z = (1 + \Gamma)/(1 - \Gamma)$, multiplying numerator and denominator by the conjugate of the denominator:

$$z = \frac{(1 - u^2 - v^2) + 2iv}{(1 - u)^2 + v^2},$$

since $(1 + u + iv)(1 - u + iv) = 1 - u^2 - v^2 + 2iv$. Hence

$$r = \frac{1 - u^2 - v^2}{(1 - u)^2 + v^2}, \quad x = \frac{2v}{(1 - u)^2 + v^2}.$$

Step 2 — constant- r locus. The equation $r[(1 - u)^2 + v^2] = 1 - u^2 - v^2$ becomes

$$(1 + r)(u^2 + v^2) - 2ru + r - 1 = 0.$$

Step 3 — completing the square. Dividing by $1 + r > 0$ and completing,

$$\left(u - \frac{r}{1+r}\right)^2 + v^2 = \frac{r^2 - (r-1)(r+1)}{(1+r)^2} = \frac{1}{(1+r)^2},$$

giving the stated radius.

Step 4 — tangency and inclusion. The rightmost point of each circle is

$$\frac{r}{1+r} + \frac{1}{1+r} = 1,$$

independent of r : all pass through $\Gamma = 1$ and are mutually tangent there (they share a vertical tangent, the centre being on the real axis). The leftmost point is $\frac{r-1}{r+1} \geq -1$, with equality only for $r = 0$: the circles lie inside the unit disc, and the one with $r = 0$ is the unit circle, consistently with Theorem 25.16(ii). As $r \rightarrow \infty$ the radius tends to 0 and the centre to 1. $\square$

Theorem 25.20: Constant-reactance circles. For each fixed $x \in \mathbb{R} \setminus \{0\}$, the locus $\{\operatorname{Im} z = x\}$ is, in the Γ -plane, the arc inside the unit disc of the circle

$$(u-1)^2 + \left(v - \frac{1}{x}\right)^2 = \frac{1}{x^2},$$

of centre $(1, \frac{1}{x})$ and radius $\frac{1}{|x|}$. For $x = 0$ the locus is the real segment $[-1, 1]$. All these circles pass through $\Gamma = 1$.

Proof. **Step 1.** From Step 1 of the previous proof, $x[(1-u)^2 + v^2] = 2v$.

Step 2. For $x \neq 0$, dividing by x and rearranging: $u^2 - 2u + 1 + v^2 - \frac{2v}{x} = 0$.

Step 3 — completing the squares.

$$(u-1)^2 + \left(v - \frac{1}{x}\right)^2 = \frac{1}{x^2},$$

as claimed. Substituting $(u, v) = (1, 0)$ verifies that $\Gamma = 1$ lies on every circle.

Step 4 — the case $x = 0$. The equation $2v = 0$ gives $v = 0$: the locus is the real axis, of which only the segment $[-1, 1]$ is physically accessible because $|\Gamma| \leq 1$ for passive loads.

Step 5 — why only an arc. The circle of centre $(1, 1/x)$ and radius $1/|x|$ leaves the unit disc; only the portion with $|\Gamma| \leq 1$ corresponds to passive loads ($r \geq 0$), and that is the one drawn on the chart. $\square$

Remark 25.8: The Smith chart as an atlas of Möbius. The two families of circles of Theorems 25.19 and 25.20 are the images, under the Möbius map $\Gamma = (z-1)/(z+1)$, of the two families of lines $\{\operatorname{Re} z = r\}$ and $\{\operatorname{Im} z = x\}$, i.e. of the Cartesian grid of the impedance plane. That they are circles is Theorem (Volume II, Chapter 3); that they all pass through $\Gamma = 1$ is because all those lines pass through the point at infinity, which the map sends to 1 (Observation, Volume II, Chapter 3). That they meet at right angles — a property evident on the chart — follows from the conformality of Möbius maps, i.e. from their being holomorphic with non-vanishing derivative and hence angle-preserving: the Cartesian grid is orthogonal, and so is its image.

Example 25.5: Single-stub matching. A load $\underline{Z}_L = 100 + i50 \, \Omega$ is connected to a line with $Z_0 = 50 \, \Omega$. Match it with a shunt stub (a length of line terminated in a short or an open, connected in parallel).

Step 1 — normalisation and reflection coefficient.

$$z_L = 2 + i, \quad \Gamma_L = \frac{1+i}{3+i} = \frac{(1+i)(3-i)}{10} = 0.4 + i0.2.$$

Modulus and argument: $|\Gamma_L| = \sqrt{0.16 + 0.04} = \sqrt{0.2} = 0.44721$, $\arg \Gamma_L = \arctan(0.5) = 26.565^\circ$.

Step 2 — SWR. By Theorem 25.18,

$$\text{SWR} = \frac{1 + 0.44721}{1 - 0.44721} = \frac{1.44721}{0.55279} = 2.618.$$

Step 3 — normalised load admittance.

$$y_L = \frac{1}{z_L} = \frac{1}{2 + i} = \frac{2 - i}{5} = 0.4 - i 0.2.$$

Step 4 — distance at which the real part equals 1. By the admittance dual of Theorem 25.17(iii), with $t = \tan \beta d$:

$$y(d) = \frac{y_L + it}{1 + iy_L t} = \frac{0.4 + i(t - 0.2)}{(1 + 0.2t) + i 0.4t}.$$

Imposing $\text{Re } y = 1$ and rationalising, the numerator of the real part is $0.4 + 0.4t^2$ and the denominator $1 + 0.4t + 0.2t^2$; equality gives

$$0.4 + 0.4t^2 = 1 + 0.4t + 0.2t^2 \implies 0.2t^2 - 0.4t - 0.6 = 0 \implies t^2 - 2t - 3 = 0,$$

with roots $t = 3$ and $t = -1$.

Step 5 — first solution. With $t = 3$: $\beta d = \arctan 3 = 71.565^\circ$, i.e.

$$d = \frac{71.565}{360} \lambda = 0.1988 \lambda.$$

Substituting,

$$y = \frac{0.4 + i 2.8}{1.6 + i 1.2} = \frac{(0.4 + 2.8i)(1.6 - 1.2i)}{2.56 + 1.44} = \frac{4.0 + 4.0i}{4} = 1 + i. \checkmark$$

We therefore need a stub supplying normalised susceptance $-i$.

Step 6 — stub length. A *short-circuited* stub of length ℓ presents $y_{\text{stub}} = -i \cot \beta \ell$ (from Theorem 25.17(iii) with $\underline{Z}_L = 0$, then inverting). Imposing $-i \cot \beta \ell = -i$:

$$\cot \beta \ell = 1 \implies \beta \ell = 45^\circ \implies \ell = 0.125 \lambda.$$

With an *open* stub one would have $y = i \tan \beta \ell = -i$, i.e. $\beta \ell = 135^\circ$ and $\ell = 0.375 \lambda$.

Step 7 — final check. In parallel, admittances add (Theorem (Volume II, Chapter 5)(iv) of Chapter 6):

$$y_{\text{tot}} = (1 + i) + (-i) = 1 \implies \underline{Z} = 50 \Omega,$$

i.e. $\Gamma = 0$ and $\text{SWR} = 1$: matched line. $\checkmark$

Step 8 — second solution. With $t = -1$: $\beta d = 135^\circ$, $d = 0.375 \lambda$, and one finds $y = 1 - i$; a susceptance $+i$ is then needed, i.e. a shorted stub of 0.375λ or an open one of 0.125λ . Both solutions are valid; in practice one picks the shorter stub, which offers wider bandwidth.

Typical error. Working in *impedance* rather than *admittance* for a shunt stub. Since admittances add in parallel, the whole construction must be carried out on the Smith chart read as an admittance chart (which is the same chart rotated by 180° : indeed $y = 1/z$ corresponds to $\Gamma \mapsto -\Gamma$, by Theorem 25.17(iv)).

25.10. Back to the unit circle: discrete-time stability

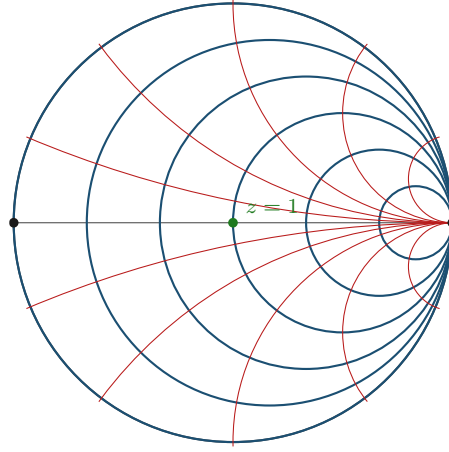


Figure 25.6: The skeleton of the Smith chart. In blue the constant normalised resistance circles r ($r = 0, 0.2, 0.5, 1, 2, 5$), with centre $(r/(1+r), 0)$ and radius $1/(1+r)$: they are all tangent to one another at $\Gamma = 1$. In red the constant normalised reactance arcs x ($x = \pm 0.2, \pm 0.5, \pm 1, \pm 2, \pm 5$), arcs of circles with centre $(1, 1/x)$ and radius $1/|x|$, again tangent at $\Gamma = 1$. The two families are orthogonal because they are the images, under the Möbius map $\Gamma = (z - 1)/(z + 1)$, of the orthogonal lines $\Re z = r$ and $\Im z = x$. The centre $\Gamma = 0$ is the matched condition $z = 1$; the rim $|\Gamma| = 1$ is the purely reactive load.

Chapter 4 established that a discrete-time system is stable if and only if all its poles lie inside the unit circle (Theorem 21.12). Checking that condition without computing the roots is the task of the Jury criterion; we prove the second-order case, the one met in practice when tuning digital controllers.

Theorem 25.21: Jury criterion, second order. Let $p(z) = z^2 + a_1 z + a_0$ with $a_0, a_1 \in \mathbb{R}$. Both roots have modulus < 1 if and only if

$$|a_0| < 1 \quad \text{and} \quad |a_1| < 1 + a_0.$$

Equivalently: $|a_0| < 1$, $p(1) > 0$ and $p(-1) > 0$.

Proof. Step 1 — equivalence of the two formulations. $p(1) = 1 + a_1 + a_0 > 0$ and $p(-1) = 1 - a_1 + a_0 > 0$ are equivalent to $-(1 + a_0) < a_1 < 1 + a_0$, i.e. $|a_1| < 1 + a_0$ (which implicitly requires $1 + a_0 > 0$, guaranteed by $|a_0| < 1$).

Step 2 — necessity. Let z_1, z_2 be the roots, with $|z_1|, |z_2| < 1$. By Vieta, $a_0 = z_1 z_2$, hence $|a_0| < 1$. Moreover $p(1) = (1 - z_1)(1 - z_2)$ and $p(-1) = (1 + z_1)(1 + z_2)$.

- If the roots are complex conjugate, $p(1) = |1 - z_1|^2 > 0$ and $p(-1) = |1 + z_1|^2 > 0$ (non-zero because $z_1 \neq \pm 1$, having modulus < 1).
- If they are real in $(-1, 1)$, then $1 - z_k > 0$ and $1 + z_k > 0$ for both, so both products are positive.

Step 3 — sufficiency, complex roots. If the discriminant is negative the roots are conjugate and $a_0 = |z_1|^2$. The hypothesis $|a_0| < 1$ then gives $|z_1| = |z_2| < 1$. ✓

Step 4 — sufficiency, real roots. Let $z_1 \leq z_2$ be real. From $p(1) = (1 - z_1)(1 - z_2) > 0$ the two factors have the same sign: either both roots are < 1 or both are > 1 . From $p(-1) = (1 + z_1)(1 + z_2) > 0$: either both > -1 or both < -1 .

If both were > 1 , then $a_0 = z_1 z_2 > 1$, contradicting $|a_0| < 1$. If both were < -1 , then again $z_1 z_2 > 1$, same contradiction. Only the compatible possibilities remain: both < 1 and both > -1 , i.e. both in $(-1, 1)$. ✓ □

Example 25.6: Checking discrete stability. Let $p(z) = z^2 - 1.2z + 0.5$.

Step 1 — criterion. $a_1 = -1.2$, $a_0 = 0.5$.

$$|a_0| = 0.5 < 1 \checkmark, \quad |a_1| = 1.2 < 1 + 0.5 = 1.5 \checkmark$$

The system is stable.

Step 2 — direct check.

$$z = \frac{1.2 \pm \sqrt{1.44 - 2}}{2} = \frac{1.2 \pm i\sqrt{0.56}}{2} = 0.6 \pm i 0.37417,$$

$$|z| = \sqrt{0.36 + 0.14} = \sqrt{0.5} = 0.70711 < 1 \checkmark$$

Note that $|z|^2 = 0.5 = a_0$: consistent with Step 3 of the proof.

Step 3 — limiting case. With $a_0 = 0.5$ fixed, stability is lost for $|a_1| \geq 1.5$. For $a_1 = -1.5$: $p(z) = z^2 - 1.5z + 0.5 = (z - 1)(z - 0.5)$, which has a root *on* the unit circle: exactly the boundary.

Control interpretation. In a digital controller a_1 and a_0 depend on the gain and the sampling time. The criterion says that raising the gain beyond a threshold moves the pair (a_1, a_0) outside the *stability triangle* bounded by the three lines $a_0 = 1$, $a_1 = 1 + a_0$ and $a_1 = -(1 + a_0)$: it is the discrete counterpart of the critical gain of Example 25.2.

25.11. Numerical exercises

All exercises are solved in Section 25.13, with an independent check and a typical error.

Exercise 25.1: N1 — Margins of a type-1 system. For $L(s) = \frac{20}{s(1 + s/5)}$ find the gain crossover frequency, the phase margin and the maximum tolerable delay. Explain why the gain margin is infinite.

Exercise 25.2: N2 — Critical gain analytically. For $L(s) = \frac{K}{(s + 1)(s + 2)(s + 5)}$ find the phase crossover frequency, the critical value of K and the gain margin for $K = 10$. Check the critical gain with the Routh–Hurwitz criterion.

Exercise 25.3: N3 — An M circle. Find centre, radius and the endpoints of the horizontal diameter of the M circle for $M = 1.4$, and verify that the point -1 is interior to it.

Exercise 25.4: N4 — A circular root locus. For $L(s) = K \frac{s + 4}{(s + 1)(s + 3)}$ find the centre and radius of the complex part of the locus, the breakpoints and the corresponding gains. Verify that the product of the two gains equals 4.

Exercise 25.5: N5 — Reflection coefficient and SWR. A load $\underline{Z}_L = 25 - i50 \, \Omega$ is connected to a line with $Z_0 = 50 \, \Omega$. Compute z_L , Γ_L in Cartesian and polar form, and the standing wave ratio.

Exercise 25.6: N6 — Quarter-wave transformer. Match a resistive load of $200\ \Omega$ to a line with $Z_0 = 50\ \Omega$ using a $\lambda/4$ section. Find the characteristic impedance of the section and check the result with Theorem 25.17(iv). State what happens at twice the frequency.

Exercise 25.7: N7 — Discrete stability. Apply the Jury criterion to $p(z) = z^2 + 0.4z - 0.3$ and verify the result by computing the roots.

Exercise 25.8: N8 — Circle criterion. A loop contains $G(s) = \frac{5}{(s+1)(s+2)}$ and a nonlinearity in the sector $[0.2, 2]$. Find the critical disc and verify, by computing the minimum distance of the Nyquist plot from the centre, that the circle criterion is satisfied.

Exercise 25.9: N9 — From phase margin to sampling time. A loop has $\omega_g = 5$ rad/s and $\text{PM} = 45^\circ$. (a) Compute the maximum tolerable delay. (b) Knowing that sampling with period T introduces an average delay $T/2$, find the largest admissible T if at least half the phase margin is to be preserved.

Exercise 25.10: N10 — Prewarping the bilinear transform. A digital filter with cutoff frequency 100 Hz is to be realised sampling at 1 kHz, using the bilinear transform $s = \frac{2}{T} \frac{z-1}{z+1}$. Compute the analogue frequency to be used in the design (*prewarping*) and the error incurred by ignoring the warping.

25.12. Symbolic exercises

Exercise 25.11: S1 — Symmetry of the Nyquist plot. Let L be rational with real coefficients and strictly proper. Prove that $L(-i\omega) = \overline{L(i\omega)}$, and deduce that the Nyquist plot is symmetric about the real axis and closes at the origin. Explain why it therefore suffices to draw it for $\omega \geq 0$.

Exercise 25.12: S2 — Rouché's theorem and continuity of the poles. Prove **Rouché's theorem**: if f, g are holomorphic on an open set containing a simple closed contour γ and its interior, and if $|f(s) - g(s)| < |g(s)|$ on γ , then f and g have the same number of zeros inside (with multiplicity). Deduce that the closed-loop poles depend continuously on the gain K .

Exercise 25.13: S3 — The modulus margin is a circle. Define the *sensitivity function* $S = 1/(1+L)$ and the *modulus margin* $s_m = \min_\omega |1+L(i\omega)|$. Prove that

$$s_m = \frac{1}{\|S\|_\infty},$$

and that s_m is the radius of the largest circle centred at -1 not meeting the Nyquist plot.

Exercise 25.14: S4 — Orthogonality of M and N circles. Prove that every M circle meets every N circle at right angles. (Hint: in the T -plane the two families are the circles centred at the origin and the rays from the origin; use the conformality of Möbius transformations.)

Exercise 25.15: S5 — The modulus margin dominates the other two. Prove that if the modulus margin is $s_m \in (0, 1)$, then

$$\text{PM} \geq 2 \arcsin \frac{s_m}{2}, \quad \text{GM} \geq \frac{1}{1 - s_m}.$$

Comment on why s_m is regarded as a more reliable robustness indicator than PM and GM taken separately.

Exercise 25.16: S6 — The sector $[0, k]$ and the half-plane. Prove that in the circle criterion, letting $k_1 \rightarrow 0^+$ with $k_2 = k$ fixed, the critical disc degenerates into the half-plane $\{\text{Re } G \leq -1/k\}$, and that the condition becomes: the Nyquist plot of G stays to the right of the vertical line $\text{Re } G = -1/k$. Recognise in this the simplified Popov criterion.

Exercise 25.17: S7 — The half-wave line. Prove that a lossless line section of length exactly $\lambda/2$ reproduces the load impedance at its input, whatever the load and whatever Z_0 . Deduce that Γ has period $\lambda/2$ and explain why the Smith chart scale runs to 0.5λ and not to λ .

Exercise 25.18: S8 — The stub is purely reactive. Prove that the input impedance of a lossless line section terminated in a short circuit is $Z_{\text{in}} = iZ_0 \tan \beta\ell$, and that of an open-circuited section is $-iZ_0 \cot \beta\ell$. Determine for which lengths the shorted stub behaves as an inductor and for which as a capacitor, and comment on the periodicity.

Exercise 25.19: S9 — The SWR circles. Prove that the constant- $|\Gamma|$ loci on the Smith chart are the circles centred at the origin, that each corresponds to a single value of SWR, and that moving towards the generator is a *clockwise* rotation on them. Find the relation between the SWR and the abscissa of the intersection of the circle with the positive real semiaxis.

Exercise 25.20: S10 — Jury via the bilinear transform. Prove Theorem 25.21 by an alternative route: substitute $z = \frac{1+w}{1-w}$ into $p(z)$, multiply by $(1-w)^2$ and apply the second-order Routh–Hurwitz criterion to the resulting polynomial in w . Check that exactly the same three conditions are obtained.

25.13. Solutions to the numerical exercises

Solution. N1. Step 1 — magnitude. $|L(i\omega)| = \frac{20}{\omega\sqrt{1+\omega^2/25}}.$

Step 2 — gain crossover. Imposing $|L| = 1$: $\omega\sqrt{1+\omega^2/25} = 20$. At $\omega = 9$ the left side is $9 \times \sqrt{1+3.24} = 9 \times 2.0591 = 18.53$; at $\omega = 9.4$ it is $9.4 \times \sqrt{1+3.5344} = 9.4 \times 2.1294 = 20.02$. Interpolating, $\omega_g = 9.39$ rad/s.

Step 3 — phase and margin.

$$\arg L(i\omega_g) = -90^\circ - \arctan \frac{9.39}{5} = -90^\circ - \arctan(1.878) = -90^\circ - 61.96^\circ = -151.96^\circ,$$

so $\text{PM} = 180^\circ - 151.96^\circ = 28.04^\circ$.

Step 4 — maximum delay. By Theorem 25.6(iii),

$$\tau_{\max} = \frac{28.04^\circ \times \pi/180}{9.39} = \frac{0.48939}{9.39} = 0.0521 \text{ s} = 52.1 \text{ ms}.$$

Step 5 — why $\text{GM} = \infty$. The phase is $-90^\circ - \arctan(\omega/5)$, which tends to -180° only as $\omega \rightarrow \infty$, never reaching it. There is thus no finite ω_π , the plot never crosses the negative real axis, and the gain margin is infinite: K can be increased without limit without destabilising. This is typical of systems with only two poles (including one at the origin).

Typical error. Concluding “infinite gain margin, hence a robust system”. The 28° phase margin says otherwise: the response will be oscillatory, and a 52 ms delay suffices to destabilise.

Solution. N2. *Step 1 — denominator in Cartesian form.*

$$(1 + i\omega)(2 + i\omega)(5 + i\omega) = (10 - 8\omega^2) + i(17\omega - \omega^3),$$

as one checks by expanding: real part $5(2 - \omega^2) - 3\omega^2 = 10 - 8\omega^2$; imaginary part $\omega(2 - \omega^2) + 15\omega = 17\omega - \omega^3$.

Step 2 — phase crossover. $L(i\omega)$ is real when the imaginary part of the denominator vanishes:

$$17\omega - \omega^3 = 0 \implies \omega_\pi = \sqrt{17} = 4.1231 \text{ rad/s}.$$

Step 3 — value at ω_π . The denominator equals $10 - 136 = -126$, so $L(i\omega_\pi) = -K/126$, real and negative ✓

Step 4 — critical gain. The plot touches -1 when $K/126 = 1$, i.e. $K_{\text{crit}} = 126$.

Step 5 — gain margin for $K = 10$. $\text{GM} = 126/10 = 12.6 = 22.0 \text{ dB}$.

Step 6 — Routh–Hurwitz check. The characteristic polynomial is $s^3 + 8s^2 + 17s + 10 + K$. For third order, stability requires all coefficients positive and $a_2a_1 > a_0$:

$$8 \times 17 > 10 + K \implies K < 126. \checkmark$$

Typical error. Looking for ω_π by solving $\arctan \omega + \arctan(\omega/2) + \arctan(\omega/5) = 180^\circ$ numerically. It works, but it is far longer and prone to quadrant errors: setting the imaginary part of the denominator to zero is the direct route.

Solution. N3. *Step 1 — centre and radius.* By Theorem 25.7 with $M = 1.4$: $M^2 = 1.96$, $M^2 - 1 = 0.96$, hence

$$c_M = -\frac{1.96}{0.96} = -2.04167, \quad r_M = \frac{1.4}{0.96} = 1.45833.$$

Step 2 — endpoints of the horizontal diameter.

$$c_M - r_M = -3.50000, \quad c_M + r_M = -0.58333,$$

(exact values $-7/2$ and $-7/12$).

Step 3 — the point -1 . Since $-3.5 < -1 < -0.58333$, the point -1 lies between the endpoints and is therefore interior. ✓ In agreement with Step 5 of the proof of Theorem 25.7: $c_M + r_M = -M/(M+1) = -1.4/2.4 = -0.58333$ and $c_M - r_M = -M/(M-1) = -1.4/0.4 = -3.5$ ✓

Step 4 — check on a point. The point $L = -0.58333$ must give $|T| = 1.4$:

$$T = \frac{-0.58333}{1 - 0.58333} = \frac{-0.58333}{0.41667} = -1.40000,$$

of modulus 1.4 ✓ And $L = -3.5$:

$$T = \frac{-3.5}{1 - 3.5} = \frac{-3.5}{-2.5} = 1.4 \checkmark$$

Reading. If the Nyquist plot is tangent to this circle, the closed-loop resonant peak is $M_r = 1.4$, i.e. +2.92 dB: a typical design value, associated with a damping $\zeta \approx 0.38$ and a step overshoot of about 27%.

Solution. N4. *Step 1 — applying the theorem.* Here $z = 4$, $p_1 = 1$, $p_2 = 3$, and $(1 - 4)(3 - 4) = 3 > 0$, so by Theorem 25.15 the complex part of the locus is the circle of centre -4 and radius $\rho = \sqrt{3} = 1.73205$.

Step 2 — breakpoints. $s_{1,2} = -4 \pm \sqrt{3} \implies s_1 = -2.26795$, $s_2 = -5.73205$.

Step 3 — corresponding gains. By Theorem 25.13, with $K = -\frac{(s+1)(s+3)}{s+4}$:

$$K_1 = -\frac{(-1.26795)(0.73205)}{1.73205} = \frac{0.92820}{1.73205} = 0.53590,$$

$$K_2 = -\frac{(-4.73205)(-2.73205)}{-1.73205} = \frac{12.92820}{1.73205} = 7.46410.$$

Exact values: $K_1 = 4 - 2\sqrt{3}$ and $K_2 = 4 + 2\sqrt{3}$.

Step 4 — product check. $K_1 K_2 = 16 - 12 = 4 \checkmark$ (and the sum is 8). Numerically $0.53590 \times 7.46410 = 4.00000 \checkmark$

Step 5 — checking a complex point. For $K = 2$ the characteristic equation is $s^2 + 6s + 11 = 0$, with roots $s = -3 \pm i\sqrt{2}$. Distance from the centre:

$$|(-3 + i\sqrt{2}) + 4| = \sqrt{1 + 2} = \sqrt{3} = \rho \checkmark$$

Typical error. Confusing the centre of the circle (the *zero*, at -4) with the asymptote centroid. Here $q = n - m = 1$: there is a single asymptote, the direction 180° , and no centroid to compute.

Solution. N5. *Step 1 — normalisation.* $z_L = 0.5 - i$.

Step 2 — reflection coefficient. By Theorem 25.16(i),

$$\Gamma_L = \frac{-0.5 - i}{1.5 - i} = \frac{(-0.5 - i)(1.5 + i)}{2.25 + 1}.$$

The numerator is $-0.75 - 0.5i - 1.5i - i^2 = -0.75 + 1 - 2i = 0.25 - 2i$. Hence

$$\Gamma_L = \frac{0.25 - 2i}{3.25} = 0.076923 - i 0.615385.$$

Step 3 — polar form.

$$|\Gamma_L| = \sqrt{0.0059172 + 0.3786982} = \sqrt{0.3846154} = 0.620174 \quad (= \sqrt{5/13}),$$

$$\arg \Gamma_L = \text{atan2}(-0.615385, 0.076923) = -82.87^\circ,$$

fourth quadrant: $\arctan(-8) = -82.87^\circ \checkmark$

Step 4 — SWR. By Theorem 25.18,

$$\text{SWR} = \frac{1 + 0.620174}{1 - 0.620174} = \frac{1.620174}{0.379826} = 4.2656.$$

Step 5 — consistency check. The load has positive real part ($R = 25 > 0$), so by Theorem 25.16(ii) we must have $|\Gamma_L| < 1$: and indeed $0.62 < 1 \checkmark$. An SWR of 4.27 means the voltage varies by a factor 4.27 between maximum and minimum along the line: a severe mismatch, which in transmission would mean a reflected power of $|\Gamma|^2 = 38.5\%$ of the incident one.

Solution. N6. *Step 1 — matching condition.* By Theorem 25.17(iv), a $\lambda/4$ section of characteristic impedance Z'_0 presents at its input $\underline{Z}_{\text{in}} = (Z'_0)^2/\underline{Z}_L$. We want 50Ω with $\underline{Z}_L = 200 \Omega$:

$$(Z'_0)^2 = 10\,000 \implies Z'_0 = 100 \Omega.$$

The characteristic impedance of the transformer is the *geometric mean* of the two impedances to be matched.

Step 2 — check. With $Z'_0 = 100$: $\underline{Z}_{\text{in}} = 50 \Omega \checkmark$, and the reflection coefficient seen from the main line is $\Gamma = 0 \checkmark$

Step 3 — at twice the frequency. At $2f$ the wavelength halves, so the section that was $\lambda/4$ at f becomes $\lambda'/2$ at $2f$. By Exercise S7, a $\lambda/2$ section reproduces the load impedance:

$$\underline{Z}_{\text{in}}(2f) = 200 \Omega,$$

and the reflection coefficient returns to $\Gamma = 150/250 = 0.6$, with $\text{SWR} = 4$. Quarter-wave matching is therefore *intrinsically narrowband*: it works perfectly at the design frequency and not at all at its second harmonic.

Typical error. Using the arithmetic mean $(50 + 200)/2 = 125 \Omega$ instead of the geometric one. With $Z'_0 = 125$ one would get $\underline{Z}_{\text{in}} = 78.1 \Omega$ and $\Gamma = 0.22$: matching missed.

Solution. N7. *Step 1 — criterion.* $p(z) = z^2 + 0.4z - 0.3$, so $a_1 = 0.4$ and $a_0 = -0.3$. By Theorem 25.21:

$$|a_0| = 0.3 < 1 \checkmark, \quad |a_1| = 0.4 < 1 + a_0 = 0.7 \checkmark$$

The system is stable.

Step 2 — equivalent form. $p(1) = 1 + 0.4 - 0.3 = 1.1 > 0 \checkmark$; $p(-1) = 1 - 0.4 - 0.3 = 0.3 > 0 \checkmark$

Step 3 — direct check.

$$z = \frac{-0.4 \pm \sqrt{0.16 + 1.2}}{2} = \frac{-0.4 \pm \sqrt{1.36}}{2} = \frac{-0.4 \pm 1.16619}{2} \implies z_1 = 0.38310, \quad z_2 = -0.78310,$$

both real with modulus $< 1 \checkmark$

Step 4 — Vieta check. $z_1 z_2 = -0.30000 = a_0 \checkmark$; $z_1 + z_2 = -0.40000 = -a_1 \checkmark$

Remark. Here $a_0 < 0$: the product of the roots is negative, so they have opposite signs. The condition $|a_1| < 1 + a_0$ tightens as a_0 becomes more negative; as $a_0 \rightarrow -1^+$ it forces $a_1 \rightarrow 0$: the stability triangle closes to a point.

Solution. N8. *Step 1 — the critical disc.* With $k_1 = 0.2$ and $k_2 = 2$:

$$-\frac{1}{k_1} = -5, \quad -\frac{1}{k_2} = -0.5,$$

so, by Theorem 25.12, the disc has

$$\text{centre} = \frac{-5 - 0.5}{2} = -2.75, \quad \text{radius} = \frac{5 - 0.5}{2} = 2.25.$$

Step 2 — Nyquist plot of G .

$$\operatorname{Re} G = \frac{5(2 - \omega^2)}{(2 - \omega^2)^2 + 9\omega^2}, \quad \operatorname{Im} G = \frac{-15\omega}{(2 - \omega^2)^2 + 9\omega^2}.$$

Step 3 — notable points. $\omega = 0$: $G = 2.5$; $\omega \rightarrow \infty$: $G \rightarrow 0$. Distances from -2.75 : 5.25 and 2.75, both > 2.25 .

Step 4 — searching for the closest point.

ω	$\operatorname{Re} G$	$\operatorname{Im} G$	distance from -2.75
2.0	-0.27778	-0.83333	2.6076
2.5	-0.28596	-0.50521	2.5153
2.8	-0.27897	-0.40126	2.5034
3.0	-0.26923	-0.34615	2.5048
3.5	-0.23803	-0.24384	2.5238

The minimum is about 2.503, near $\omega = 2.85$ rad/s.

Step 5 — conclusion. Since $2.503 > 2.25$, the plot stays strictly outside the critical disc: by Theorem 25.12 the loop is input-output stable *for every* nonlinearity in the sector $[0.2, 2]$, including saturations, dead zones and hysteresis-free relays.

Step 6 — how much margin is left. The ratio $2.503/2.25 = 1.11$ says the sector could be widened by about 11% in radius before losing the guarantee. Note the criterion is *sufficient*, not necessary: violating it does not imply instability, only loss of the guarantee.

Solution. N9. (a) *Maximum delay.* By Theorem 25.6(iii),

$$\tau_{\max} = \frac{45 \times \pi/180}{5} = \frac{0.785398}{5} = 0.157 \text{ s} = 157 \text{ ms.}$$

(b) *Sampling period.* Preserving at least half the margin means spending at most $\text{PM}/2 = 22.5^\circ = 0.392699$ rad on delay. The corresponding delay is

$$\tau = \frac{0.392699}{5} = 0.0785 \text{ s} = 78.5 \text{ ms.}$$

Since sampling introduces an average delay $T/2$,

$$\frac{T}{2} \leq 0.0785 \implies T \leq 0.157 \text{ s,}$$

i.e. a sampling frequency $f_s \geq 6.4$ Hz.

Step 3 — comparison with the rule of thumb. Industrial practice suggests f_s between 10 and 30 times the closed-loop bandwidth, here of order $\omega_g/(2\pi) = 0.80$ Hz: that would give f_s between 8 and 24 Hz, i.e. T between 42 and 125 ms. The exact computation (157 ms) is thus less demanding than the rule of thumb, which is reasonable: the rule also accounts for signal reconstruction quality, not just stability.

Typical error. Using T instead of $T/2$ as the equivalent delay. The average delay of a zero-order-hold sampler is $T/2$; using T would double the risk estimate and lead to needlessly oversizing the sampling rate.

Solution. N10. Step 1 — frequency warping. Substituting $z = e^{i\omega_d T}$ into the bilinear transform, factoring $e^{i\omega_d T/2}$ out of numerator and denominator and using the inverse Euler

formulas (Theorem, Volume II, Chapter 3):

$$s = \frac{2}{T} \cdot \frac{2i \sin(\omega_d T/2)}{2 \cos(\omega_d T/2)} = i \frac{2}{T} \tan \frac{\omega_d T}{2},$$

so the corresponding analogue frequency is $\omega_a = \frac{2}{T} \tan \frac{\omega_d T}{2}$. (One recognises Corollary (Volume II, Chapter 3) of Chapter 3: the unit circle of the z -plane goes to the imaginary axis of the s -plane, with $\tan$ as the correspondence law.)

Step 2 — numerical data. $f_s = 1$ kHz $\Rightarrow T = 1$ ms; desired cutoff $f_d = 100$ Hz $\Rightarrow \omega_d = 628.319$ rad/s.

Step 3 — prewarping.

$$\frac{\omega_d T}{2} = \frac{628.319 \times 0.001}{2} = 0.314159 \text{ rad} = 18.000^\circ,$$

$$\tan(0.314159) = 0.324920, \quad \omega_a = \frac{2}{0.001} \times 0.324920 = 649.84 \text{ rad/s}.$$

In frequency: $f_a = 649.84/(2\pi) = 103.42$ Hz.

Step 4 — error if warping is ignored. Designing the analogue filter directly at 628.319 rad/s and then discretising, the effective digital cutoff would solve $\frac{2}{T} \tan(\omega_d T/2) = 628.319$, i.e. $\tan(\omega_d T/2) = 0.314159$, i.e. $\omega_d T/2 = 0.304693$, hence $\omega_d = 609.39$ rad/s, i.e. 96.99 Hz instead of 100 Hz: a 3.0% error.

Step 5 — when warping matters. The relative error grows with $\omega_d T$: for $f_d/f_s = 0.1$ (our case) it is 3%; for $f_d/f_s = 0.25$ it becomes

$$\frac{\tan(\pi/4)}{\pi/4} = \frac{1}{0.785398} = 1.273,$$

i.e. 27%. Near the Nyquist frequency the warping diverges, consistently with $\tan$ having an asymptote at $\pi/2$: the whole semiaxis $[0, \infty)$ of analogue frequencies is compressed into the interval $[0, \pi/T)$ of digital ones.

25.14. Solutions to the symbolic exercises

Solution. S1. *Step 1 — Hermitian symmetry.* Let $L(s) = N(s)/D(s)$ with N, D having real coefficients. For a real-coefficient polynomial, $\overline{N(s)} = N(\bar{s})$: conjugating a sum of monomials $a_k s^k$ with $a_k \in \mathbb{R}$ gives $\sum a_k \bar{s}^k$. Hence

$$\overline{L(i\omega)} = \frac{N(-i\omega)}{D(-i\omega)} = L(-i\omega).$$

Step 2 — symmetry of the plot. The point $L(-i\omega)$ is thus the conjugate of $L(i\omega)$, i.e. its reflection in the real axis (Definition, Volume I, Chapter 7). The curve $\{L(i\omega) : \omega \in \mathbb{R}\}$ is therefore symmetric about the real axis, and the branch with $\omega < 0$ is determined by the one with $\omega > 0$: it suffices to draw the latter and mirror it.

Step 3 — closing at the origin. If L is strictly proper, $\deg N < \deg D$, hence

$$|L(i\omega)| \sim |\omega|^{\deg N - \deg D} \longrightarrow 0 \quad (\omega \rightarrow \pm\infty).$$

The curve starts and ends at the origin, and the semicircle at infinity of the Nyquist contour has as image the single point 0 (Step 4 of the proof of Theorem 25.5).

Step 4 — consequence for encirclements. Since the full plot is the union of two conjugate arcs, the winding number about -1 — which is real — receives symmetric contributions from the two arcs. This is why, in practice, one counts the crossings of the real axis to the left of -1 for $\omega > 0$ only and doubles them.

Solution. S2. *Step 1 — building the homotopy.* For $t \in [0, 1]$ set $h_t(s) = g(s) + t(f(s) - g(s))$, so $h_0 = g$ and $h_1 = f$. Each h_t is holomorphic on the given open set.

Step 2 — no zeros on the contour. For $s \in \gamma$ and $t \in [0, 1]$, by the reverse triangle inequality (Theorem (Volume II, Chapter 3)(v)),

$$|h_t(s)| \geq |g(s)| - |f(s) - g(s)| > 0$$

by hypothesis. So h_t has no zeros on γ , for any t .

Step 3 — the count is continuous and integer-valued. By Theorem 25.4, the number of zeros of h_t inside γ is

$$N(t) = \frac{1}{2\pi i} \oint_{\gamma} \frac{h'_t(s)}{h_t(s)} ds$$

(there are no poles, h_t being holomorphic). The integrand depends continuously on t uniformly on γ — because h_t and h'_t depend linearly on t and $|h_t|$ is bounded below by a positive constant, by Step 2 — so N is a continuous function of t . But $N(t)$ is an integer for every t : a continuous integer-valued function on an interval is constant.

Step 4 — conclusion. $N(1) = N(0)$, i.e. f and g have the same number of zeros inside. ✓

Step 5 — continuity of the closed-loop poles. Let $\chi_K(s) = D(s) + KN(s)$ be the characteristic polynomial. Fix K_0 and a root s_0 of χ_{K_0} of multiplicity m ; let γ be a small circle about s_0 containing no other root, and $\delta = \min_{s \in \gamma} |\chi_{K_0}(s)| > 0$. For $|K - K_0|$ small enough,

$$|\chi_K(s) - \chi_{K_0}(s)| = |K - K_0| \cdot |N(s)| < \delta \leq |\chi_{K_0}(s)| \quad \text{on } \gamma,$$

since $|N|$ is bounded on the compact γ . By Rouché, χ_K has exactly m roots inside γ : roots cannot “jump”, and as $K \rightarrow K_0$ they tend to s_0 . This is exactly what makes it meaningful to draw the root locus as a family of continuous *curves*.

Solution. S3. *Step 1 — identity between the two minima.* By definition $S(i\omega) = 1/(1 + L(i\omega))$, so $|S(i\omega)| = 1/|1 + L(i\omega)|$. Since $x \mapsto 1/x$ is decreasing on $(0, \infty)$, the maximum of $|S|$ occurs where $|1 + L|$ is minimal:

$$\|S\|_{\infty} = \frac{1}{\inf_{\omega} |1 + L(i\omega)|} = \frac{1}{s_m},$$

whence $s_m = 1/\|S\|_{\infty}$.

Step 2 — geometric interpretation. By Theorem (Volume II, Chapter 3) of Chapter 3, $|1 + L(i\omega)| = |L(i\omega) - (-1)|$ is the *distance* between the point $L(i\omega)$ and the point -1 . Hence s_m is the minimum distance between the Nyquist plot and the critical point.

Step 3 — the circle. The circle of centre -1 and radius r misses the plot iff $|L(i\omega) + 1| > r$ for every ω , i.e. iff $r < s_m$. The largest such circle (in the closed sense: tangent but not secant) therefore has radius exactly s_m . ✓

Step 4 — why it is the right robustness measure. $\|S\|_{\infty}$ is the maximum amplification factor of disturbances under feedback: a small s_m means a large $\|S\|_{\infty}$, i.e. amplified disturbances and poor robustness. Typical design values are $s_m \geq 0.5$, i.e. $\|S\|_{\infty} \leq 2$ (6 dB).

Solution. S4. *Step 1 — the two families in the T -plane.* In the T -plane, the locus $|T| = M$ is the circle of centre the origin and radius M ; the locus $\arg T = \alpha$ is the ray from the origin with inclination α . The two families are orthogonal: at every point $T_0 \neq 0$ the circle has tangent perpendicular to the radius, and the ray is the radius.

Step 2 — the map. The correspondence $L \mapsto T = L/(1 + L)$ is a Möbius transformation (Definition, Volume II, Chapter 3), hence holomorphic on its domain, with derivative

$$\frac{dT}{dL} = \frac{1}{(1 + L)^2},$$

never zero and defined everywhere except at $L = -1$.

Step 3 — conformality. A holomorphic function with non-zero derivative at a point preserves angles between curves there. Indeed, if two curves meet at L_0 with tangent vectors v_1, v_2 , the images have tangent vectors $T'(L_0)v_1$ and $T'(L_0)v_2$; the angle between them is

$$\arg \frac{T'(L_0)v_2}{T'(L_0)v_1} = \arg \frac{v_2}{v_1},$$

by Theorem (Volume II, Chapter 3): the common factor cancels and the angle is unchanged.

Step 4 — conclusion. The M and N circles are the preimages of the two orthogonal families of Step 1 under the inverse map $T \mapsto L = T/(1 - T)$, itself Möbius and hence conformal. Angles are preserved: the preimages meet at right angles. ✓

Remark. The same argument explains the orthogonality of the Smith chart grid (Observation 25.8): again, conformality of Möbius maps.

Solution. S5. *Step 1 — the constraint.* By Exercise S3, the plot stays outside the open circle of centre -1 and radius s_m : for every ω , $|L(i\omega) + 1| \geq s_m$.

Step 2 — phase margin estimate. At ω_g we have $|L(i\omega_g)| = 1$, hence $L(i\omega_g) = e^{i\theta}$ with $\theta = -180^\circ + \text{PM}$. The distance between two points of the unit circle with arguments θ_1, θ_2 is, by the chord formula (Theorem (Volume II, Chapter 3) of Chapter 3, or directly),

$$|e^{i\theta_1} - e^{i\theta_2}| = 2 \left| \sin \frac{\theta_1 - \theta_2}{2} \right|.$$

Here $\theta_1 - \theta_2 = \text{PM}$, so

$$|L(i\omega_g) + 1| = 2 \sin \frac{\text{PM}}{2} \geq s_m \implies \text{PM} \geq 2 \arcsin \frac{s_m}{2}.$$

Step 3 — gain margin estimate. At ω_π the point $L(i\omega_\pi)$ lies on the negative real axis at distance $1/\text{GM}$ from the origin. Its distance from -1 is

$$1 - \frac{1}{\text{GM}} \geq s_m,$$

(assuming $\text{GM} > 1$, i.e. a stable system), whence $\text{GM} \geq 1/(1 - s_m)$.

Step 4 — typical values. For $s_m = 0.5$: $\text{PM} \geq 2 \arcsin(0.25) = 28.96^\circ$ and $\text{GM} \geq 1/0.5 = 2$ (6 dB).

Step 5 — why s_m is more reliable. PM and GM measure the distance from -1 along *two particular directions*: along the unit circle and along the real axis respectively. One can construct systems with both margins excellent and yet a plot passing very close to -1 in an intermediate direction — hence fragile. The modulus margin, measuring the minimum distance *over all directions*, has no such defect: by Steps 2 and 3 it implies acceptable values of both others, while the converse is false.

Solution. S6. *Step 1 — endpoints of the diameter.* In Theorem 25.12 the disc has as diameter the segment with endpoints $-1/k_1$ and $-1/k_2$. With $k_2 = k$ fixed and $k_1 \rightarrow 0^+$, $-1/k_1 \rightarrow -\infty$ while $-1/k_2 = -1/k$ stays fixed.

Step 2 — limit of the disc. Centre and radius,

$$c = -\frac{1}{2} \left(\frac{1}{k_1} + \frac{1}{k} \right) \rightarrow -\infty, \quad r = \frac{1}{2} \left(\frac{1}{k_1} - \frac{1}{k} \right) \rightarrow +\infty,$$

both diverge, but the difference $c + r = -1/k$ stays constant: the right edge of the disc stays put at $-1/k$ while the rest expands indefinitely leftwards. The limit is thus the closed half-plane $\{\operatorname{Re} G \leq -1/k\}$.

Step 3 — form of the condition. “Stay outside the disc” becomes “stay to the right of the vertical line $\operatorname{Re} G = -1/k$ ”, i.e. $\operatorname{Re} G(i\omega) > -1/k$ for every ω .

Step 4 — direct check. The condition of Step 3 of the proof of Theorem 25.12 was $(r^2 - k_m^2)(x^2 + y^2) - 2k_mx - 1 < 0$ with $r^2 - k_m^2 = -k_1k_2$. As $k_1 \rightarrow 0$ the coefficient of $x^2 + y^2$ tends to 0 and $k_m \rightarrow k/2$, so the condition degenerates into $-kx - 1 < 0$, i.e. $x > -1/k$. ✓

Step 5 — connection with Popov. The condition obtained is the simplest version of the Popov criterion: for memoryless nonlinearities in the sector $[0, k]$, absolute stability is guaranteed if the Nyquist plot stays to the right of a vertical line. Popov (1962) showed that, also exploiting monotonicity of the nonlinearity, the vertical line can be replaced by a *slanted* line through $-1/k$, giving a less restrictive condition; the circle criterion is the vertical-line case, valid also for time-varying nonlinearities.

Solution. S7. *Step 1 — direct computation.* For $d = \lambda/2$ we have $\beta d = \pi$, hence $\tan \beta d = 0$. By Theorem 25.17(iii),

$$\underline{Z}_{\text{in}} = Z_0 \frac{\underline{Z}_L}{Z_0} = \underline{Z}_L,$$

independent of Z_0 . ✓

Step 2 — via the reflection coefficient. Alternatively, by Theorem 25.17(i),

$$\Gamma(\lambda/2) = \Gamma_L e^{-2\pi i} = \Gamma_L,$$

since $e^{-2\pi i} = 1$ (Theorem (Volume II, Chapter 3) of Chapter 3). Same Γ , hence same impedance by Theorem 25.16(i).

Step 3 — periodicity. From Step 2, $\Gamma(d + \lambda/2) = \Gamma(d)$ for every d : the period divides $\lambda/2$. That it is exactly $\lambda/2$ follows from $e^{-2i\beta\Delta} = 1$ requiring $\Delta \in (\lambda/2)\mathbb{Z}$.

Step 4 — the Smith chart scale. Since everything repeats every $\lambda/2$, a full 360° rotation on the chart corresponds to a displacement of $\lambda/2$ along the line: the perimeter scale is therefore graduated from 0 to 0.5 in wavelengths. It is the same “factor of two” as in Theorem 25.17(i): the rotation angle is $2\beta d$, not βd .

Step 5 — practical consequence. A cable section half a wavelength long is “invisible” as far as matching is concerned: it can be inserted or removed without changing anything, at the design frequency. At twice the frequency it becomes a full wavelength — still invisible; at $1.5f$ it becomes 0.75λ , i.e. equivalent to 0.25λ , and inverts the impedance. The property is thus strictly monochromatic.

Solution. S8. *Step 1 — short-circuited stub.* With $\underline{Z}_L = 0$ in Theorem 25.17(iii):

$$\underline{Z}_{\text{in}} = Z_0 \frac{iZ_0 \tan \beta \ell}{Z_0} = iZ_0 \tan \beta \ell,$$

purely imaginary.

Step 2 — open-circuited stub. With $\underline{Z}_L \rightarrow \infty$, dividing numerator and denominator by $\underline{Z}_L$ and passing to the limit:

$$\underline{Z}_{\text{in}} \longrightarrow \frac{Z_0}{i \tan \beta \ell} = -i Z_0 \cot \beta \ell,$$

also purely imaginary.

Step 3 — why this was predictable. A lossless line dissipates no energy, and neither does a short or an open circuit: the overall one-port cannot absorb active power, hence by Theorem (Volume II, Chapter 5)(ii) of Chapter 6 its real part vanishes. The conclusion is independent of the computation.

Step 4 — inductive or capacitive behaviour. For the shorted stub, $X = Z_0 \tan \beta \ell$, hence by Theorem (Volume II, Chapter 5) of Chapter 6:

- $X > 0$ (*inductive*) for $\tan \beta \ell > 0$, i.e. $0 < \ell < \lambda/4$ and more generally $\ell \in (\frac{n\lambda}{2}, \frac{n\lambda}{2} + \frac{\lambda}{4})$;
- $X < 0$ (*capacitive*) for $\lambda/4 < \ell < \lambda/2$ and translates by $\lambda/2$.

At $\ell = \lambda/4$, $X \rightarrow \infty$ (equivalent open circuit: it is the quarter-wave transformer applied to a short); at $\ell = \lambda/2$ one is back to the short circuit, as Exercise S7 predicts.

Step 5 — engineering reading. A stub is therefore a reactive component *tunable with scissors*: cut to the right length it gives any reactance, inductive or capacitive, without coils or capacitors — which at microwave frequencies would be unrealisable. The $\lambda/2$ periodicity guarantees that a short solution always exists.

Solution. S9. *Step 1 — constant- $|\Gamma|$ loci.* By definition, $\{|\Gamma| = \rho\}$ is the circle of centre the origin and radius ρ in the Γ -plane (Corollary (Volume II, Chapter 3) of Chapter 3). By Theorem 25.17(i), moving along a lossless line preserves $|\Gamma|$: the representative point stays on that circle.

Step 2 — correspondence with SWR. By Theorem 25.18, $\text{SWR} = (1 + \rho)/(1 - \rho)$, a strictly increasing function of ρ on $[0, 1)$ (its derivative is $2/(1 - \rho)^2 > 0$): each circle corresponds to exactly one SWR, and conversely. Inverting, $\rho = (\text{SWR} - 1)/(\text{SWR} + 1)$.

Step 3 — direction of rotation. By Theorem 25.17(i), $\Gamma(d) = \Gamma_L e^{-2i\beta d}$: as d grows (movement towards the generator) the argument *decreases*, i.e. the point rotates **clockwise**. This is why the Smith chart carries two perimeter scales, “towards generator” clockwise and “towards load” counterclockwise.

Step 4 — intersection with the positive real semiaxis. The point of the circle on the positive real semiaxis is $\Gamma = \rho$; by Theorem 25.16(i) the corresponding normalised impedance is

$$z = \frac{1 + \rho}{1 - \rho} = \text{SWR},$$

real and greater than 1. Hence *the abscissa of that intersection, read as a normalised resistance, is numerically equal to the SWR*: this is the classical way to read the SWR off the chart with no computation, and it corresponds physically to the point of the line where the voltage is maximal.

Step 5 — check. In Example 25.5 we had $\rho = 0.44721$ and $\text{SWR} = 2.618$; the intersection with the positive real semiaxis is $\Gamma = 0.44721$, corresponding to $z = 1.44721/0.55279 = 2.618$ ✓

Solution. S10. *Step 1 — the substitution.* Set $z = \frac{1 + w}{1 - w}$. By Corollary (Volume II, Chapter 3) of Chapter 3 (read backwards), this map carries the half-plane $\text{Re } w < 0$ into the disc $|z| < 1$: its inverse is $w = (z - 1)/(z + 1)$, which by Exercise S1 of Chapter 6 satisfies $|z| < 1 \iff \text{Re } w < 0$. Hence *the roots of p all lie in $|z| < 1$ if and only if the roots of the transformed polynomial all lie in $\text{Re } w < 0$.*

Step 2 — the transformed polynomial. Substituting into $p(z) = z^2 + a_1z + a_0$ and multiplying by $(1 - w)^2$ (which neither creates nor destroys roots, $w = 1$ being the image of $z = \infty$):

$$q(w) = (1 + w)^2 + a_1(1 + w)(1 - w) + a_0(1 - w)^2,$$

and expanding,

$$q(w) = (1 - a_1 + a_0)w^2 + 2(1 - a_0)w + (1 + a_1 + a_0).$$

One recognises $q_2 = p(-1)$, $q_0 = p(1)$ and $q_1 = 2(1 - a_0)$.

Step 3 — second-order Routh–Hurwitz. A real polynomial $q_2w^2 + q_1w + q_0$ has both roots with negative real part iff its three coefficients are *all of the same sign* (and non-zero). Proof: assuming $q_2 > 0$ and dividing, take $q(w) = w^2 + bw + c$; the roots have sum $-b$ and product c . If real, both negative $\iff$ sum negative and product positive, i.e. $b > 0$ and $c > 0$; if complex conjugate $\sigma \pm i\tau$, the real part is $-b/2$ and the product is $\sigma^2 + \tau^2 = c > 0$, so the condition is again $b > 0$.

Step 4 — translation. Normalising to $q_2 > 0$, the conditions are

$$1 - a_1 + a_0 > 0, \quad a_0 < 1, \quad 1 + a_1 + a_0 > 0.$$

Step 5 — comparison with Theorem 25.21. The first and third give $|a_1| < 1 + a_0$; added together they imply $2 + 2a_0 > 0$, i.e. $a_0 > -1$, which together with $a_0 < 1$ gives $|a_0| < 1$. Conversely the conditions of the theorem imply these three. The two formulations coincide.

✓

Remark. This proof has the merit of being *constructive and general*: the same procedure — bilinear transform followed by Routh–Hurwitz — gives the Jury criterion for polynomials of any degree, without repeating the analysis case by case. And once again it is the Möbius map of Chapter 3 that does all the work: turning the problem “inside the circle” into the problem “left of the line”, for which a criterion already exists.

25.15. Self-assessment quiz

- The Nyquist plot is the image, under L , of
 - the unit circle
 - the imaginary axis
 - the real axis
 - the right half-plane
- On a logarithmic scale, the magnitude plot of a product is
 - the product of the plots
 - the sum of the plots
 - the average
 - no simple relation
- Beyond the break frequency, a simple pole contributes a slope of
 - −10 dB/dec
 - −20 dB/dec
 - −40 dB/dec
 - −6 dB/dec
- At the break point, the gap between exact curve and asymptotes is
 - 0 dB
 - 3.01 dB
 - 6 dB
 - 20 dB
- The argument principle relates the winding number to
 - $Z + P$
 - $Z - P$
 - $Z \cdot P$
 - Z/P
- The critical point of the Nyquist criterion is
 - 0
 - −1
 - +1
 - ∞

-
7. With a stable open loop, the closed loop is stable if the plot
 - A) encircles -1 once
 - B) does not encircle -1
 - C) passes through -1
 - D) stays in the first quadrant
 8. The phase margin is read at the intersection of the plot with
 - A) the real axis
 - B) the unit circle
 - C) the imaginary axis
 - D) the point -1
 9. The maximum tolerable delay equals
 - A) $\text{PM} \cdot \omega_g$
 - B) PM/ω_g (in radians)
 - C) ω_g/PM
 - D) $1/\omega_g$
 10. The loci $|T| = M$ in the L -plane are
 - A) lines
 - B) circles (a line for $M = 1$)
 - C) ellipses
 - D) hyperbolas
 11. All N circles pass through
 - A) the origin and -1
 - B) only -1
 - C) only the origin
 - D) $+1$
 12. In the circle criterion, the critical point -1 is replaced by
 - A) a different point
 - B) a disc of diameter $[-1/k_1, -1/k_2]$
 - C) a line
 - D) the unit circle
 13. The small-gain theorem requires
 - A) $\gamma_1 + \gamma_2 < 1$
 - B) $\gamma_1 \gamma_2 < 1$
 - C) $\gamma_1 < \gamma_2$
 - D) $\gamma_1 \gamma_2 > 1$
 14. The membership condition for the root locus concerns the
 - A) magnitude
 - B) argument ($\equiv 180^\circ$)
 - C) derivative
 - D) residue
 15. For $L = K(s + z)/((s + p_1)(s + p_2))$ with real data, the complex part of the locus is
 - A) a vertical line
 - B) a circle centred at the zero
 - C) a hyperbola
 - D) a parabola
 16. The reflection coefficient $\Gamma = (z - 1)/(z + 1)$ maps passive loads
 - A) outside the unit disc
 - B) into the unit disc
 - C) onto the real axis
 - D) to infinity
 17. Moving along a lossless line corresponds, in the Γ -plane, to
 - A) a translation
 - B) a rotation at constant modulus
 - C) a dilation
 - D) a reflection
 18. The constant-resistance circles of the Smith chart all pass through
 - A) $\Gamma = 0$
 - B) $\Gamma = 1$
 - C) $\Gamma = -1$
 - D) infinity
 19. A $\lambda/4$ line section transforms $\underline{Z}_L$ into
 - A) $\underline{Z}_L$
 - B) $Z_0^2/\underline{Z}_L$
 - C) $-\underline{Z}_L$
 - D) Z_0
 20. The second-order Jury criterion requires
 - A) $|a_0| < 1$ and $|a_1| < 1 + a_0$
 - B) $a_0 > 1$
 - C) $a_1 = 0$
 - D) $|a_1| < 1$

BEST PRACTICE

Answer key.

1-B: Definition 25.1; in discrete time it would be the unit circle (Observation 25.1). **2-B:** Theorem 25.2. **3-B:** Theorem 25.3(ii). **4-B:** same theorem: $-20 \log_{10} \sqrt{2} = -3.0103$ dB. **5-B:** Theorem 25.4. **6-B:** Observation 25.4: it is the condition $1 + L = 0$. **7-B:** Corollary 25.1. **8-B:** Definition 25.2 and Theorem 25.6(ii). **9-B:** Theorem 25.6(iii).

10-B: Theorem 25.7. **11-A:** Theorem 25.8, Step 3. **12-B:** Theorem 25.12; for $k_1 = k_2$ the disc shrinks to the point (Observation 25.7). **13-B:** Theorem 25.11. **14-B:** Theorem 25.13: the magnitude condition only fixes K . **15-B:** Theorem 25.15; the radius is $\sqrt{(p_1 - z)(p_2 - z)}$. **16-B:** Theorem 25.16(ii), which is Exercise S1 of Chapter 6. **17-B:** Theorem 25.17(i). **18-B:** Theorem 25.19, Step 4: the rightmost point is always 1. **19-B:** Theorem 25.17(iv). **20-A:** Theorem 25.21.

25.16. Chapter formulary

Frequency response

$$u = e^{i\omega t} \implies y = H(i\omega)e^{i\omega t}, \quad H(-i\omega) = \overline{H(i\omega)} \text{ (} h \text{ real)}$$

Bode

$$20 \log_{10} |H_1 H_2| = 20 \log_{10} |H_1| + 20 \log_{10} |H_2|, \quad \arg(H_1 H_2) = \arg H_1 + \arg H_2$$

simple pole : -20 dB/dec, -45° at ω_p , gap 3.01 dB

Argument principle and Nyquist

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{F'}{F} ds = Z - P, \quad Z = N + P \text{ (clockwise encirclements of } -1)$$

$$\text{PM} = 180^\circ + \arg L(i\omega_g), \quad \text{GM} = \frac{1}{|L(i\omega_\pi)|}, \quad \tau_{\max} = \frac{\text{PM} [\text{rad}]}{\omega_g}$$

$$s_m = \min_{\omega} |1 + L(i\omega)| = \frac{1}{\|S\|_{\infty}}$$

M and N circles ($T = L/(1 + L)$)

$$|T| = M : \quad c_M = -\frac{M^2}{M^2 - 1}, \quad r_M = \frac{M}{|M^2 - 1|}$$

$$\arg T = \alpha, \quad N = \tan \alpha : \quad \text{centre } \left(-\frac{1}{2}, \frac{1}{2N}\right), \quad \text{radius } \frac{1}{2}\sqrt{1 + 1/N^2}$$

Circle criterion (sector $[k_1, k_2]$, $0 < k_1 < k_2$)

$$\text{disc of diameter } \left[-\frac{1}{k_1}, -\frac{1}{k_2}\right], \quad \gamma_1 \gamma_2 < 1 \text{ (small gain)}$$

Root locus

$$\arg \frac{N(s)}{D(s)} \equiv 180^\circ, \quad K = \frac{|D(s)|}{|N(s)|}, \quad \theta_k = \frac{(2k+1)180^\circ}{n-m}, \quad \sigma_a = \frac{\sum p_j - \sum z_i}{n-m}$$

$$L = K \frac{s+z}{(s+p_1)(s+p_2)} \implies \text{circle of centre } -z, \text{ radius } \sqrt{(p_1 - z)(p_2 - z)}$$

Transmission lines

$$\Gamma_L = \frac{z_L - 1}{z_L + 1}, \quad z_L = \frac{1 + \Gamma_L}{1 - \Gamma_L}, \quad \Gamma(d) = \Gamma_L e^{-2i\beta d}, \quad \beta = \frac{2\pi}{\lambda}$$

$$\underline{Z}_{\text{in}}(d) = Z_0 \frac{\underline{Z}_L + iZ_0 \tan \beta d}{Z_0 + i\underline{Z}_L \tan \beta d}, \quad \underline{Z}_{\text{in}}(\lambda/4) = \frac{Z_0^2}{\underline{Z}_L}$$

$$\text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}, \quad |\Gamma_L| = \frac{\text{SWR} - 1}{\text{SWR} + 1}$$

Smith chart

r constant : centre $\left(\frac{r}{1+r}, 0\right)$, radius $\frac{1}{1+r}$; x constant : centre $\left(1, \frac{1}{x}\right)$, radius $\frac{1}{|x|}$

Discrete time

$$s = \frac{2}{T} \cdot \frac{z-1}{z+1}, \quad \omega_a = \frac{2}{T} \tan \frac{\omega_d T}{2} \text{ (prewarping)}$$

$$z^2 + a_1 z + a_0 \text{ stable} \iff |a_0| < 1 \text{ and } |a_1| < 1 + a_0$$

Sources and bibliography

- [1] H. Nyquist, *Regeneration Theory*, Bell System Technical Journal, vol. 11, no. 1, 1932, pp. 126–147. Original memoir of the criterion bearing his name (Theorem 25.5).
- [2] H. W. Bode, *Network Analysis and Feedback Amplifier Design*, D. Van Nostrand, New York, 1945. Source of the logarithmic plots and of the gain–phase relation (Observation 25.3, Chapter 14 of the original).
- [3] P. H. Smith, *Transmission Line Calculator*, Electronics, vol. 12, no. 1, January 1939, pp. 29–31; and *An Improved Transmission Line Calculator*, Electronics, vol. 17, no. 1, January 1944, pp. 130–133 and 318–325. The two memoirs introducing the Smith chart (Section 25.9).
- [4] W. R. Evans, *Graphical Analysis of Control Systems*, Transactions of the AIEE, vol. 67, 1948, pp. 547–551. Introduction of the root locus (Section 25.7).
- [5] G. Zames, *On the input-output stability of time-varying nonlinear feedback systems*, IEEE Transactions on Automatic Control, vol. AC-11, 1966, parts I and II. Origin of the small-gain theorem and of the circle criterion in input–output form.
- [6] V. M. Popov, *Absolute stability of nonlinear systems of automatic control*, Automation and Remote Control, vol. 22, 1962, pp. 857–875. The Popov criterion, cited in Exercise S6.
- [7] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002. Modern rigorous treatment of the circle criterion and absolute stability.
- [8] K. J. Åström, R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, Princeton University Press, 2008 — freely available at <https://fbswiki.org/>. Stability margins, modulus margin and their geometric interpretation (Exercises S3 and S5).
- [9] K. Ogata, *Modern Control Engineering*, 5th ed., Pearson, 2010. Classic teaching reference for Bode, Nyquist, root locus and M – N circles.
- [10] G. F. Franklin, J. D. Powell, A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 8th ed., Pearson, 2019. Same function; consulted for margin conventions and the link between M_r and damping.
- [11] G. C. Goodwin, S. F. Graebe, M. E. Salgado, *Control System Design*, Prentice Hall, 2001 (University of Newcastle, Australia). Modern approach to frequency-domain design; Oceania-region reference.

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- [12] E. I. Jury, *Theory and Application of the z-Transform Method*, John Wiley & Sons, New York, 1964. Stability criterion for discrete-time systems (Theorem 25.21).
 - [13] D. M. Pozar, *Microwave Engineering*, 4th ed., John Wiley & Sons, 2011. Transmission lines, reflection coefficient, Smith chart and stub matching (Sections 25.8 and 25.9).
 - [14] R. E. Collin, *Foundations for Microwave Engineering*, 2nd ed., IEEE Press, 2001. Rigorous treatment of lines and their transformation properties.
 - [15] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw-Hill, 1979. Argument principle and Rouché's theorem (Chapter 4); basis of Theorem 25.4 and Exercise S2.
 - [16] *Encyclopedia of Mathematics*, entry “Argument, principle of the” — <https://encyclopediaofmath.org> Reference statement of the argument principle.
 - [17] MIT OpenCourseWare, *2.004 Dynamics and Control II* — <https://ocw.mit.edu/>. Teaching framework for frequency response and stability margins.
 - [18] University of Michigan, *Control Tutorials for MATLAB and Simulink* — <https://ctms.engin.umich.edu> Institutional teaching material on Bode, Nyquist and root locus; consulted for graphical conventions.
 - [19] NPTEL, Indian Institute of Technology, *Control Engineering*, online university course — <https://nptel.ac.in/>. Path through frequency analysis and stability criteria; Asian reference of the chapter.
 - [20] F. Nicolls, University of Cape Town, *EEE2047S Signals and Systems I* — <https://www.prasa.uct.ac.za> Frequency response and filtering from an engineering viewpoint.
 - [21] UNAM, Facultad de Ingeniería, *Matemáticas Avanzadas*, 2023-1 syllabus. Link between complex variables and systems theory in the Latin American curriculum.
 - [22] C. A. Barbieri, D. Miller-Cotto, S. N. Clerjuste, K. Chawla, *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*, Educational Psychology Review 35, art. 11, 2023 — <https://doi.org/10.1007/s10648-023-09745-1>.
 - [23] H. L. Roediger III, J. D. Karpicke, *Test-Enhanced Learning*, Psychological Science 17(3), 2006 — <https://doi.org/10.1111/j.1467-9280.2006.01693.x>.

25.17. Modulus margin of a third-order loop

Algorithm. For $L(s) = K/(1+s)^3$ one has $|1 + L(j\omega)|^2 = 1 + [2K(1 - 3\omega^2) + K^2]/(1 + \omega^2)^3$; tabulate and look for the minimum.

Datum. $K = 2$.

ω	$ 1 + L(j\omega) ^2$	$ 1 + L(j\omega) $
0	9.00000	3.00000
0.5	3.56000	1.88680
1.0	0.50000	0.70711
1.2247	0.36000	0.60000
1.5	0.44652	0.66822
2.0	0.68000	0.82462
3.0	0.90000	0.94868

The minimum falls at $\omega = \sqrt{3/2} = 1.2247$ and equals exactly 0.6: hence $m = 0.6$ and $M_s = 1/0.6 = 1.6667$.

Independent check (classical margins). The plot crosses the negative real axis at $\omega = \sqrt{3}$, where $|L| = 2/8 = 0.25$: gain margin $m_G = 4$. The estimate of Theorem 25.10 requires $m_G \geq 1/(1-m) = 2.5$ ✓. Unit modulus is crossed at $\omega_c = 0.7664$, where the phase is -112.36° : phase margin 67.6° , against the estimate $2 \arcsin(0.3) = 34.9^\circ$ ✓.

Telecommunications: complex baseband, IQ and modulations

A radio transmitter does not send the message into the ether: it sends a circle, and on the circle it writes the message. The sentence is less rhetorical than it sounds, and it is the technical content of this chapter. The signal that travels is a sinusoid at the carrier frequency f_c ; the information does not lie in the sinusoid, which is always the same, but in the pair (A, φ) of amplitude and phase with which it is emitted in each symbol interval, that is, in a *point of the complex plane* $Ae^{i\varphi}$. The set of admitted points is called a **constellation**, and the classical constellations are literally sets of points on one or more concentric circles: the M M -th roots of unity of Chapter 6 (Definition 6.2) rotated and scaled give the M -PSK modulation; two concentric circles with different populations give M -APSK; a square grid — which is still a union of circles, but with unequal populations — gives M -QAM.

The logical chain of the chapter is exactly that, and must be walked in order. First it is proved that a real narrowband signal around f_c *can be written in one and only one way* as the real part of a slow complex signal multiplied by $e^{i2\pi f_c t}$ (Theorem 26.1 and Theorem 26.2): this is the **complex envelope**, and its two real components are the **in-phase** and **quadrature** components. Then it is proved, with the integral computed in full and not with a wave of the hand, under exactly which conditions the two carriers $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ are orthogonal on the symbol interval, and what the error is worth when those conditions are not satisfied (Proposition 26.2). Everything rests on this orthogonality: it is what allows two independent streams to be transmitted on the same frequency and separated at reception.

Once the constellation is fixed, the receiver must decide. It is proved that, for additive white Gaussian noise and equiprobable symbols, the maximum-likelihood rule *coincides* with the minimum Euclidean-distance rule (Theorem 26.6), that the decision regions are the Voronoi cells of the constellation and that they are convex polyhedra (Theorem 26.7). Before deciding, however, one must measure: it is proved that among all linear filters the one that maximises the signal-to-noise ratio at the sampling instant is the **matched filter**, and that the maximum equals $2E/N_0$ (Theorem 26.8); the proof is an application of the Cauchy–Schwarz inequality, whose *equality case* is used in an essential way — and is therefore proved as well (Lemma 26.3).

Finally the numbers. The error probability of BPSK and QPSK is derived in full from the Q function (Theorem 26.9 and Theorem 26.10), with the relation between energy per bit and energy per symbol written explicitly at every step, and every numerical value computed twice by independent routes. The chapter closes with the minimum distance of constellations and its link with spectral efficiency (Section 26.8), and with the two tools that in the laboratory tell whether a link works: the **eye diagram** and the **error vector** (Section 26.9), defined precisely and not by analogy.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages at level **D**: the Nyquist criterion for the absence of intersymbol interference (Theorem 26.11), which uses the Poisson summation formula, and the optimal construction of the APSK constellation

(Theorem 26.5).

Internal prerequisites. Complex exponential, Euler's formula, M -th roots of unity and their arrangement at the vertices of a regular polygon (Chapter 6: Definition 6.2, Theorem 6.2, Theorem 6.3, Proposition 6.1); the Fourier transform on L^1 and on L^2 , the convolution theorem, the Plancherel–Parseval identity, the Poisson summation formula (Chapter 19: Definition 19.1, Theorem 19.3, Theorem 19.4, Theorem 19.10, Theorem 19.12, Proposition 19.2); the sampling theorem (Chapter 24, Theorem 24.8); frequency response and transfer function (Chapter 25, Definition 25.1).

External prerequisites, assumed and not reproved:

- elementary probability: random variable, density, independence, expected value, Bayes' formula;
- the real Gaussian density and the value of the Gauss integral, already proved in the treatise (Lemma 19.4);
- the existence of the *isonormal process* that models white noise: the probabilistic construction is stated in Definition 26.10 and *imported*, with a precise statement of what is missing, in Remark 26.8;
- the strong law of large numbers, used only once and declared *imported* in Remark 26.13;
- the central limit theorem and Shannon's channel capacity theorem, cited only in a panorama remark (Remark 26.11).

Everything else is proved here.

What is learned. Passing from the real signal travelling in the antenna to the complex number that represents it, and back; recognising that the orthogonality of the two carriers is a statement with precise hypotheses and not a dogma; reading a constellation as a set of points on circles and computing its average energy, minimum distance and peak-to-average power ratio; proving that "minimum distance" and "maximum likelihood" are the same rule under Gaussian hypotheses, and knowing what those hypotheses are; deriving the matched filter from Cauchy–Schwarz and understanding why the equality case is the essential part; deriving an error probability without copying it from a table; linking minimum distance, energy per bit and spectral efficiency; defining precisely the eye diagram and the error vector.

Warning. Conventions and notation of the chapter.

- The Fourier transform is taken in the *frequency* convention of Chapter 19:

$$\hat{x}(f) = \int_{-\infty}^{+\infty} x(t) e^{-i2\pi ft} dt,$$

with no 2π factors in the exponents. Time t is in seconds, frequency f in hertz.

- $\bar{z}$ is the conjugate of $z \in \mathbb{C}$, $\Re z$ and $\Im z$ the real part and the imaginary part, $|z|$ the modulus. The scalar product between finite-energy signals is

$$\langle u, v \rangle = \int_{-\infty}^{+\infty} u(t) \overline{v(t)} dt, \quad \|u\|^2 = \langle u, u \rangle = \int_{-\infty}^{+\infty} |u(t)|^2 dt.$$

For *real* signals conjugation is inert and $\langle u, v \rangle = \int uv$.

- $f_c > 0$ is the carrier frequency, $T > 0$ the symbol duration, $R_s = 1/T$ the symbol rate (in baud), M the cardinality of the constellation, $m = \log_2 M$ the number of bits per symbol. E_s is the average energy per symbol and E_b the average energy per bit; the relation between the two, proved in Proposition 26.3, is $E_s = mE_b$.
- N_0 is the *one-sided* power spectral density of the noise, in W/Hz. The two-sided density is therefore $N_0/2$. This is the dominant convention in the digital communications literature, and it is the one for which the error probability of BPSK is written

$Q(\sqrt{2E_b/N_0})$: Remark 26.6 warns about the formulas obtained with the other convention.

- The letter Q always denotes the tail function of the standard Gaussian (Definition 26.12); it must not be confused with the symbol $\mathbb{Q}$ of the rational numbers, which does not appear in this chapter, nor with the quadrature component, which is written s_Q with the subscript.
- Numerical values are rounded to the seventh decimal digit and computed *twice by independent routes*: a closed form (or an exact identity) and an actually executed Python script, which does not use the closed form but a different method — Simpson integration for the Q function, exhaustive enumeration of point pairs for minimum distances. The symbol $\checkmark$ signals that the comparison has been made, and the discrepancy is reported.
- All constellation figures are **redrawn from the formulas**: the coordinates printed in the TikZ sources are the values of $R\cos\theta$ and $R\sin\theta$ computed from the definitions of this chapter and re-checked with the same script. No figure is taken from a standardisation document.

26.1. The bandpass signal and its complex envelope

The signal leaving an antenna is *real*: it is a voltage, and voltages have no imaginary part. It is, however, a very particular real signal, because all its energy is concentrated in a narrow strip around a large frequency f_c . This section proves that such a signal can be written in *one and only one way* as the real part of the product of a complex carrier $e^{i2\pi f_c t}$ and a slow complex signal, and that therefore all the information lies in that slow signal: two real functions of bandwidth B instead of one real function of bandwidth $2B$ centred at f_c . This is the observation on which every digital radio is built.

Definition 26.1: Bandpass signal. Let $f_c > 0$ and $0 < B < f_c$. A signal $s: \mathbb{R} \rightarrow \mathbb{R}$ of class $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ is said to be **bandpass with carrier f_c and half-bandwidth B** if its Fourier transform satisfies

$$\hat{s}(f) = 0 \quad \text{for every } f \text{ such that } ||f| - f_c| > B.$$

In other words $\hat{s}$ vanishes outside the union of the two intervals $[-f_c - B, -f_c + B]$ and $[f_c - B, f_c + B]$. The **bandwidth** of the signal is $W = 2B$: it is the width of the interval occupied around $+f_c$.

Remark 26.1: Why $B < f_c$ is required. The condition $B < f_c$ is not a technical detail: it is what makes the two intervals $[-f_c - B, -f_c + B]$ and $[f_c - B, f_c + B]$ disjoint, because their distance equals $(f_c - B) - (-f_c + B) = 2(f_c - B) > 0$. All the proofs of this section consist in separating two contributions living on those two intervals, and without disjointness the separation is not possible: the real signal would no longer be reconstructible from the positive half of its spectrum alone, and the baseband representation would lose uniqueness (Theorem 26.2). A signal that also occupies the origin — for instance a baseband signal — is not bandpass in the sense of Definition 26.1, and for it the theory of this section does not apply: this is why, in practice, f_c is always chosen much larger than the bandwidth.

Definition 26.2: Analytic signal and complex envelope. Let s be bandpass with carrier f_c and half-bandwidth B . The **analytic signal** associated with s is the complex

signal s_+ defined by its Fourier transform

$$\widehat{s_+}(f) = 2\widehat{s}(f)\mathbf{1}_{\{f>0\}}(f) = \begin{cases} 2\widehat{s}(f), & f > 0, \\ 0, & f \leq 0, \end{cases}$$

and the **complex envelope** (or **complex baseband representation**) of s is

$$s_L(t) = s_+(t)e^{-i2\pi f_c t}.$$

The two real functions

$$s_I = \Re s_L, \quad s_Q = \Im s_L$$

are called the **in-phase component** and the **quadrature component** of s .

Theorem 26.1: Complex baseband representation. Let s be bandpass with carrier f_c and half-bandwidth $B < f_c$, and let s_+ , s_L , s_I , s_Q be as in Definition 26.2. Then:

- (i) $\widehat{s_L}(f) = 2\widehat{s}(f + f_c)$ for every f , and in particular $\widehat{s_L}(f) = 0$ for $|f| > B$: the complex envelope is a baseband signal of half-bandwidth B ;
- (ii) $s(t) = \Re s_+(t)$ for every t ;
- (iii) $s(t) = \Re\{s_L(t)e^{i2\pi f_c t}\}$ for every t ;
- (iv) $s(t) = s_I(t)\cos(2\pi f_c t) - s_Q(t)\sin(2\pi f_c t)$ for every t ;
- (v) s_I and s_Q are real and their transforms also vanish outside $[-B, B]$.

Proof. First observe that $\widehat{s}$ is well defined and continuous, because $s \in L^1(\mathbb{R})$ (Definition 19.1 and Proposition 19.1), and that $\widehat{s_+} \in L^1 \cap L^2$ because it is, up to the factor 2, the restriction of $\widehat{s}$ to a half-line: hence s_+ is well defined as an inverse transform.

Step 1 — Hermitian symmetry of the spectrum of a real signal. Since s is real-valued, $\overline{s(t)} = s(t)$, and therefore for every f

$$\overline{\widehat{s}(f)} = \overline{\int_{\mathbb{R}} s(t)e^{-i2\pi ft} dt} = \int_{\mathbb{R}} \overline{s(t)} e^{+i2\pi ft} dt = \int_{\mathbb{R}} s(t) e^{-i2\pi(-f)t} dt = \widehat{s}(-f).$$

Hence $\widehat{s}(-f) = \overline{\widehat{s}(f)}$ (Theorem 19.3). Moreover $\widehat{s}(0) = 0$, because 0 is at distance $f_c > B$ from $\pm f_c$, so $||0| - f_c| = f_c > B$.

Step 2 — the spectrum of $\overline{s_+}$. Let $v(t) = \overline{s_+(t)}$. Then

$$\widehat{v}(f) = \int_{\mathbb{R}} \overline{s_+(t)} e^{-i2\pi ft} dt = \overline{\int_{\mathbb{R}} s_+(t) e^{+i2\pi ft} dt} = \overline{\widehat{s_+}(-f)}.$$

By the definition of $\widehat{s_+}$, the number $\widehat{s_+}(-f)$ is zero if $-f \leq 0$, i.e. if $f \geq 0$, and equals $2\widehat{s}(-f)$ if $f < 0$. Using Step 1, $\overline{2\widehat{s}(-f)} = 2\widehat{s}(f)$. Hence

$$\widehat{v}(f) = \begin{cases} 2\widehat{s}(f), & f < 0, \\ 0, & f \geq 0. \end{cases}$$

Step 3 — proof of (ii). By linearity of the transform (Theorem 19.3), the signal $\frac{1}{2}(s_+ + v) = \Re s_+$ has transform

$$\frac{1}{2}(\widehat{s_+}(f) + \widehat{v}(f)) = \begin{cases} \frac{1}{2} \cdot 2\widehat{s}(f) = \widehat{s}(f), & f > 0, \\ \frac{1}{2} \cdot 0 = 0 = \widehat{s}(0), & f = 0, \\ \frac{1}{2} \cdot 2\widehat{s}(f) = \widehat{s}(f), & f < 0, \end{cases}$$

where in the case $f = 0$ we used $\widehat{s}(0) = 0$ from Step 1. The two functions $\Re s_+$ and s therefore have the same Fourier transform; both are continuous (the first because it is the inverse transform of an L^1 function, the second because s is, for the same reason), and by the L^1 inversion theorem (Theorem 19.9) they coincide everywhere. This is (ii).

Step 4 — proof of (i). Multiplication by $e^{-i2\pi f_c t}$ in time is a frequency shift (Theorem 19.3):

$$\widehat{s_L}(f) = \widehat{s_+}(f + f_c) = 2\widehat{s}(f + f_c) \mathbf{1}_{\{f+f_c>0\}}.$$

If $|f| > B$ then $f + f_c$ satisfies $||f + f_c| - f_c| > B$ or $f + f_c \leq 0$: indeed for $f > B$ we have $f + f_c > f_c + B > 0$ and $|f + f_c| - f_c = f > B$; for $f < -B$ either $f + f_c \leq 0$, and then the indicator kills everything, or $0 < f + f_c < f_c - B$, and then $f_c - |f + f_c| = -f > B$. In both cases $\widehat{s_L}(f) = 0$. For $|f| \leq B$ we have $f + f_c \geq f_c - B > 0$, the indicator equals 1 and we are left with $\widehat{s_L}(f) = 2\widehat{s}(f + f_c)$. This proves (i).

Step 5 — proof of (iii) and (iv). From $s_L(t) = s_+(t)e^{-i2\pi f_c t}$ it follows that $s_+(t) = s_L(t)e^{i2\pi f_c t}$, and by (ii)

$$s(t) = \Re s_+(t) = \Re\{s_L(t)e^{i2\pi f_c t}\},$$

which is (iii). Writing $s_L = s_I + is_Q$ with s_I, s_Q real and using Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$,

$$s_L(t)e^{i2\pi f_c t} = (s_I(t) + is_Q(t))(\cos(2\pi f_c t) + i \sin(2\pi f_c t)),$$

whose product has real part $s_I(t) \cos(2\pi f_c t) - s_Q(t) \sin(2\pi f_c t)$: this is (iv).

Step 6 — proof of (v). s_I and s_Q are real by construction. We have $s_I = \frac{1}{2}(s_L + \overline{s_L})$, and repeating the computation of Step 2 with s_L in place of s_+ gives that $\overline{s_L}$ has transform $f \mapsto \widehat{s_L}(-f)$, which vanishes for $|f| > B$ because $\widehat{s_L}$ vanishes for $|-f| = |f| > B$ by point (i). Hence $\widehat{s_I}(f) = \frac{1}{2}(\widehat{s_L}(f) + \widehat{s_L}(-f)) = 0$ for $|f| > B$. The same argument with $s_Q = \frac{1}{2i}(s_L - \overline{s_L})$ gives the conclusion for s_Q . $\square$

Theorem 26.2: Uniqueness of the representation. Let s be bandpass with carrier f_c and half-bandwidth $B < f_c$. If $u \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ is a complex-valued signal with $\widehat{u}(f) = 0$ for $|f| > B$ and

$$s(t) = \Re\{u(t)e^{i2\pi f_c t}\} \quad \text{for every } t,$$

then $u = s_L$. The complex envelope is therefore the unique signal of half-bandwidth B representing s in the form (iii) of Theorem 26.1.

Proof. Write $\Re z = \frac{1}{2}(z + \bar{z})$ with $z = u(t)e^{i2\pi f_c t}$:

$$s(t) = \frac{1}{2}u(t)e^{i2\pi f_c t} + \frac{1}{2}\overline{u(t)}e^{-i2\pi f_c t}.$$

We transform the two terms. The first, by the frequency-shift property (Theorem 19.3), has transform $\frac{1}{2}\widehat{u}(f - f_c)$, which vanishes outside $[f_c - B, f_c + B]$. The second: setting $v = \bar{u}$, the computation of Step 2 of the previous proof gives $\widehat{v}(f) = \widehat{\bar{u}}(-f)$, so $\bar{u}e^{-i2\pi f_c t}$ has transform $\widehat{v}(f + f_c) = \widehat{\bar{u}}(-f - f_c)$, which vanishes if $|-f - f_c| > B$, i.e. outside $[-f_c - B, -f_c + B]$. Hence

$$\widehat{s}(f) = \frac{1}{2}\widehat{u}(f - f_c) + \frac{1}{2}\overline{\widehat{u}(-f - f_c)},$$

a sum of two terms with *disjoint* supports: the first lives in $[f_c - B, f_c + B] \subset (0, +\infty)$ and the second in $[-f_c - B, -f_c + B] \subset (-\infty, 0)$, because $B < f_c$ (Remark 26.1). It follows that for every $f > 0$ the second term vanishes and

$$\widehat{s}(f) = \frac{1}{2}\widehat{u}(f - f_c), \quad \text{i.e.} \quad \widehat{u}(g) = 2\widehat{s}(g + f_c) \quad \text{for every } g > -f_c.$$

For $g \leq -f_c$ we have $|g| \geq f_c > B$, so $\widehat{u}(g) = 0$ by hypothesis; on the other hand $2\widehat{s}(g + f_c)$ also vanishes there by point (i) of Theorem 26.1. The equality $\widehat{u}(g) = 2\widehat{s}(g + f_c) = \widehat{s_L}(g)$ therefore holds for every $g \in \mathbb{R}$, and by the inversion theorem (Theorem 19.9) $u = s_L$. $\square$

Corollary 26.1: The IQ demodulator. Let s be bandpass with carrier f_c and half-bandwidth $B < f_c$, and let $\mathcal{L}_B$ be the ideal low-pass filtering operator of cutoff B , defined by $\widehat{\mathcal{L}_B x}(f) = \widehat{x}(f)\mathbf{1}_{[-B, B]}(f)$. Then

$$s_I = \mathcal{L}_B\{2s(t)\cos(2\pi f_c t)\}, \quad s_Q = \mathcal{L}_B\{-2s(t)\sin(2\pi f_c t)\},$$

with exact, not approximate, equality.

Proof. Set $\theta = \theta(t) = 2\pi f_c t$. By point (iv) of Theorem 26.1, $s = s_I \cos \theta - s_Q \sin \theta$, so

$$2s \cos \theta = 2s_I \cos^2 \theta - 2s_Q \sin \theta \cos \theta = s_I(1 + \cos 2\theta) - s_Q \sin 2\theta = s_I + \underbrace{(s_I \cos 2\theta - s_Q \sin 2\theta)}_{=:r},$$

where the identities $2\cos^2 \theta = 1 + \cos 2\theta$ and $2\sin \theta \cos \theta = \sin 2\theta$ were used. The term r is, by the same computation of Theorem 26.1(iv) read backwards with carrier $2f_c$, the real part of $s_L(t)e^{i4\pi f_c t}$; by Step 4 of that proof and by Step 2, its transform vanishes outside $[2f_c - B, 2f_c + B] \cup [-2f_c - B, -2f_c + B]$. Since $2f_c - B > B$ — indeed $2f_c - B > B$ is equivalent to $f_c > B$, which is the hypothesis — this set is disjoint from $[-B, B]$, and hence $\mathcal{L}_B r = 0$. On the other hand $\widehat{s}_I$ vanishes outside $[-B, B]$ by point (v), so $\mathcal{L}_B s_I = s_I$. By linearity $\mathcal{L}_B\{2s \cos \theta\} = s_I + 0 = s_I$.

Similarly, with $2\sin^2 \theta = 1 - \cos 2\theta$,

$$-2s \sin \theta = -2s_I \sin \theta \cos \theta + 2s_Q \sin^2 \theta = -s_I \sin 2\theta + s_Q(1 - \cos 2\theta) = s_Q - (s_I \sin 2\theta + s_Q \cos 2\theta),$$

and the term in parentheses is the *imaginary* part, with sign reversed, of $s_L e^{i4\pi f_c t}$, with the same spectral support around $\pm 2f_c$: the filter kills it and s_Q remains. $\square$

Proposition 26.1: Energy is halved when passing to baseband. Under the hypotheses of Theorem 26.1,

$$\int_{-\infty}^{+\infty} s(t)^2 dt = \frac{1}{2} \int_{-\infty}^{+\infty} |s_L(t)|^2 dt = \frac{1}{2} \int_{-\infty}^{+\infty} (s_I(t)^2 + s_Q(t)^2) dt.$$

Proof. As in the proof of Theorem 26.2 (with $u = s_L$),

$$\widehat{s}(f) = \frac{1}{2} \widehat{s}_L(f - f_c) + \frac{1}{2} \overline{\widehat{s}_L(-f - f_c)},$$

and the two terms have disjoint supports. For two functions a, b with disjoint supports, $|a + b|^2 = |a|^2 + |b|^2$ pointwise, because at every point at least one of the two vanishes. Hence, by the Plancherel–Parseval identity (Theorem 19.10),

$$\int_{\mathbb{R}} s^2 dt = \int_{\mathbb{R}} |\widehat{s}(f)|^2 df = \frac{1}{4} \int_{\mathbb{R}} |\widehat{s}_L(f - f_c)|^2 df + \frac{1}{4} \int_{\mathbb{R}} |\overline{\widehat{s}_L(-f - f_c)}|^2 df.$$

In the first integral the substitution $g = f - f_c$ gives $\frac{1}{4} \|\widehat{s}_L\|^2$; in the second the substitution $g = -f - f_c$, which is an affinity of modulus of determinant -1 , also gives $\frac{1}{4} \|\widehat{s}_L\|^2$ (the modulus of the conjugate is the modulus). In total $\|\widehat{s}\|^2 = \frac{1}{2} \|\widehat{s}_L\|^2$, and again by Plancherel–Parseval $\|s\|^2 = \frac{1}{2} \|s_L\|^2$. The last equality of the statement is $|s_L|^2 = s_I^2 + s_Q^2$, i.e. $|a + ib|^2 = a^2 + b^2$ for real a, b . $\square$

Remark 26.2: What the Hilbert transform is, and why it is not needed here. The literature usually defines the analytic signal as $s_+ = s + i\mathcal{H}s$, where $\mathcal{H}$ is the **Hilbert transform**, the operator that in frequency multiplies by $-i \operatorname{sgn}(f)$. The two definitions

coincide: if $\widehat{\mathcal{H}s}(f) = -i \operatorname{sgn}(f) \widehat{s}(f)$ then

$$\widehat{s + i\mathcal{H}s}(f) = (1 + \operatorname{sgn}(f))\widehat{s}(f) = \begin{cases} 2\widehat{s}(f), & f > 0, \\ \widehat{s}(0), & f = 0, \\ 0, & f < 0, \end{cases}$$

and $\widehat{s}(0) = 0$ by Step 1 of the proof of Theorem 26.1. We have preferred the spectral definition of Definition 26.2 because it requires neither constructing $\mathcal{H}$ nor discussing the principal-value integral by which it acts in time: the proof stays entirely within the tools of Chapter 19. Readers consulting the sources will find the other route; the two paths arrive at the same s_+ .

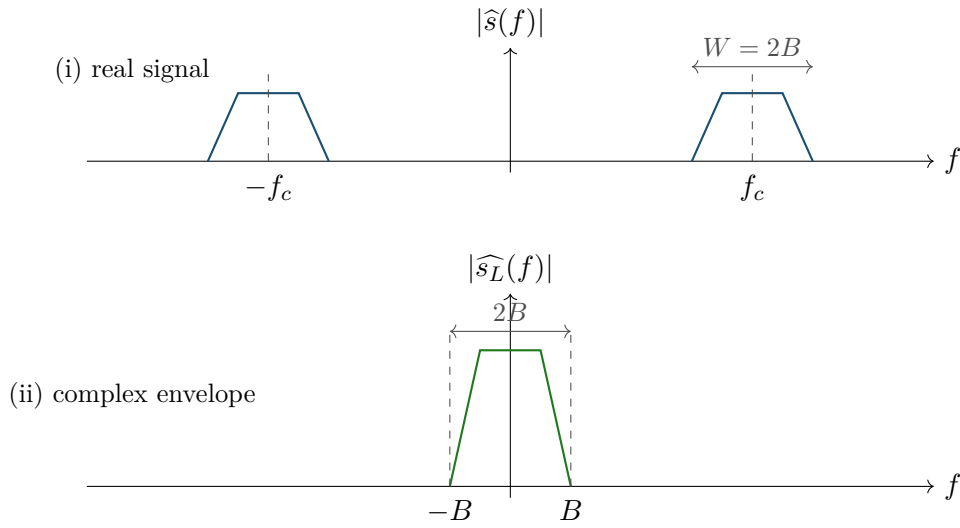


Figure 26.1: The shift of Definition 26.2, drawn with $f_c = 3.2$ and $B = 0.8$ in the units of the figure. In (i) the spectrum of a real bandpass signal: two humps symmetric about the origin, each of width $W = 2B$, linked by the Hermitian symmetry $\widehat{s}(-f) = \widehat{s}(f)$ (Step 1 of the proof of Theorem 26.1). In (ii) the spectrum of the complex envelope: only the positive hump is kept, it is doubled and shifted to the origin. The hump in (ii) is drawn twice as tall as the one in (i) precisely because $\widehat{s}_L(f) = 2\widehat{s}(f + f_c)$; it is not symmetric, and need not be, because s_L is not real. The heights and abscissas of the four vertices of each polyline are those written in the source and do not come from any measurement.

Example 26.1: A tone shifted in frequency. Let $s(t) = A \cos(2\pi(f_c + \nu)t + \varphi)$ with $A > 0$, $|\nu| < B < f_c$ and $\varphi \in \mathbb{R}$. Then

$$s(t) = \Re\left\{A e^{i\varphi} e^{i2\pi\nu t} e^{i2\pi f_c t}\right\},$$

and the factor $u(t) = A e^{i\varphi} e^{i2\pi\nu t}$ is a signal whose transform (in the sense of tempered distributions, Definition 19.7) is concentrated at the single point ν , with $|\nu| < B$. By Theorem 26.2 it follows that $s_L(t) = A e^{i\varphi} e^{i2\pi\nu t}$, and

$$s_I(t) = A \cos(2\pi\nu t + \varphi), \quad s_Q(t) = A \sin(2\pi\nu t + \varphi).$$

The tone, which in the real domain was a sinusoid at frequency $f_c + \nu$, in baseband is a point rotating on the circle of radius A with angular velocity $2\pi\nu$: a frequency offset is a *slow rotation* of the constellation, and this is exactly how a frequency error manifests itself in a real receiver.

Independent check. Applying Corollary 26.1 directly: $2s(t) \cos(2\pi f_c t) = A \cos(2\pi \nu t + \varphi) + A \cos(2\pi(2f_c + \nu)t + \varphi)$ by the product-to-sum formula $2 \cos \alpha \cos \beta = \cos(\alpha - \beta) + \cos(\alpha + \beta)$; the second term has frequency $2f_c + \nu > 2f_c - B > B$ and is removed by the filter, the first is s_I . ✓

26.2. Orthogonality of the two carriers

Corollary 26.1 recovers s_I and s_Q with an ideal low-pass filter, which is a mathematical object and not a circuit. The real receiver does something rougher and more useful: it multiplies by the two carriers and *integrates over the symbol interval*. For this to work, the two carriers must be orthogonal with respect to that scalar product. This section proves that they are — but not always: they are exactly orthogonal when $2f_c T$ is an integer, and otherwise they are orthogonal up to an error that can be computed and bounded. The difference between "orthogonal" and "orthogonal up to $1/(4\pi f_c)$ " is what separates a statement from a slogan.

Proposition 26.2: The three integrals of the carriers over a symbol interval. Let $f_c > 0$ and $T > 0$. Then

$$\int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt = \frac{\sin^2(2\pi f_c T)}{4\pi f_c},$$

$$\int_0^T \cos^2(2\pi f_c t) dt = \frac{T}{2} + \frac{\sin(4\pi f_c T)}{8\pi f_c}, \quad \int_0^T \sin^2(2\pi f_c t) dt = \frac{T}{2} - \frac{\sin(4\pi f_c T)}{8\pi f_c}.$$

In particular:

- (i) the three integrals equal exactly 0, $T/2$ and $T/2$ **if and only if** $2f_c T \in \mathbb{Z}$, i.e. if and only if the symbol duration is an integer multiple of the half-period of the carrier;
- (ii) in every case

$$0 \leq \int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt \leq \frac{1}{4\pi f_c}, \quad \left| \int_0^T \cos^2(2\pi f_c t) dt - \frac{T}{2} \right| \leq \frac{1}{8\pi f_c},$$

and hence the relative discrepancy from the value $T/2$ is bounded by $\frac{1}{2\pi f_c T}$, which tends to zero as the number $f_c T$ of carrier periods contained in a symbol grows.

Proof. Step 1 — the mixed product. From the duplication identity $2 \sin \alpha \cos \alpha = \sin 2\alpha$, with $\alpha = 2\pi f_c t$,

$$\int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt = \frac{1}{2} \int_0^T \sin(4\pi f_c t) dt = \frac{1}{2} \left[\frac{-\cos(4\pi f_c t)}{4\pi f_c} \right]_0^T = \frac{1 - \cos(4\pi f_c T)}{8\pi f_c}.$$

By the half-angle formula $1 - \cos x = 2 \sin^2(x/2)$, applied with $x = 4\pi f_c T$,

$$\frac{1 - \cos(4\pi f_c T)}{8\pi f_c} = \frac{2 \sin^2(2\pi f_c T)}{8\pi f_c} = \frac{\sin^2(2\pi f_c T)}{4\pi f_c},$$

which is the first formula.

Step 2 — the two squares. From $\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$,

$$\int_0^T \cos^2(2\pi f_c t) dt = \frac{1}{2} \int_0^T dt + \frac{1}{2} \int_0^T \cos(4\pi f_c t) dt = \frac{T}{2} + \frac{1}{2} \left[\frac{\sin(4\pi f_c t)}{4\pi f_c} \right]_0^T = \frac{T}{2} + \frac{\sin(4\pi f_c T)}{8\pi f_c}.$$

From $\sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha)$ one obtains the third formula with the opposite sign. Note, as a check, that the sum of the two integrals equals T , as it must by the identity $\cos^2 + \sin^2 = 1$. ✓

Step 3 — point (i). The first integral vanishes if and only if $\sin(2\pi f_c T) = 0$, i.e. if and only if $2\pi f_c T = k\pi$ for some $k \in \mathbb{Z}$, i.e. $2f_c T = k \in \mathbb{Z}$. If $2f_c T = k$ then $4\pi f_c T = 2k\pi$ and $\sin(4\pi f_c T) = 0$: the two squares also equal exactly $T/2$. Conversely, if the second integral equals $T/2$ then $\sin(4\pi f_c T) = 0$, i.e. $4f_c T \in \mathbb{Z}$: this is weaker than $2f_c T \in \mathbb{Z}$, and indeed for $2f_c T = k + \frac{1}{2}$ the two squares equal $T/2$ while the mixed product equals $\sin^2((k + \frac{1}{2})\pi)/(4\pi f_c) = 1/(4\pi f_c) \neq 0$: this is the worst case. The necessary and sufficient condition for *all three* equalities to hold is therefore that of the mixed product, $2f_c T \in \mathbb{Z}$.

Step 4 — point (ii). Since $0 \leq \sin^2 x \leq 1$ and $|\sin x| \leq 1$ for every real x , the three formulas immediately give $0 \leq \sin^2(2\pi f_c T)/(4\pi f_c) \leq 1/(4\pi f_c)$ and $|\sin(4\pi f_c T)/(8\pi f_c)| \leq 1/(8\pi f_c)$. Dividing the first bound by $T/2$:

$$\frac{1/(4\pi f_c)}{T/2} = \frac{1}{2\pi f_c T},$$

which is the announced relative estimate; for the second integral the relative discrepancy is half of this. $\square$

Example 26.2: Two numbers, one exact and one not. Take $f_c = 2,400$ Hz and $T = 1/600$ s. Then $2f_c T = 2 \cdot 2400/600 = 8 \in \mathbb{Z}$: we are in the exact case of Proposition 26.2(i), and

$$\int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt = 0, \quad \int_0^T \cos^2(2\pi f_c t) dt = \frac{T}{2} = 8.3333333 \cdot 10^{-4} \text{ s.}$$

Now take $f_c = 2,500$ Hz with the same $T = 1/600$ s. Now $2f_c T = 25/3 \notin \mathbb{Z}$, and $2\pi f_c T = 25\pi/3$; since $25\pi/3 - 8\pi = \pi/3$, we have $\sin(2\pi f_c T) = \sin(\pi/3) = \sqrt{3}/2$ and $\sin^2(2\pi f_c T) = 3/4$, hence

$$\int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt = \frac{3/4}{4\pi \cdot 2500} = \frac{3}{40000\pi} = 2.3873241 \cdot 10^{-5} \text{ s,}$$

i.e. 2.8647890% of $T/2$. Similarly, $4\pi f_c T = 50\pi/3$ and $50\pi/3 - 16\pi = 2\pi/3$, so $\sin(4\pi f_c T) = \sin(2\pi/3) = \sqrt{3}/2$ and

$$\int_0^T \cos^2(2\pi f_c t) dt = \frac{T}{2} + \frac{\sqrt{3}/2}{8\pi \cdot 2500} = 8.3333333 \cdot 10^{-4} + 1.3783222 \cdot 10^{-5} = 8.4711656 \cdot 10^{-4} \text{ s.}$$

The bound (ii) gives a relative discrepancy not exceeding $1/(2\pi f_c T) = 6/(50\pi) = 3.8197186\%$, consistent with the 2.8647890% found. $\checkmark$

Independent check. A Python script that numerically integrates the three integrands with Simpson's rule over 200,000 subintervals returns $9.6 \cdot 10^{-20}$ in the first case (i.e. zero up to machine error) and $2.3873241 \cdot 10^{-5}$ in the second, with a discrepancy from the closed form of $1.7 \cdot 10^{-20}$; the integral of the square gives $8.4711656 \cdot 10^{-4}$. $\checkmark$

Corollary 26.2: The two carriers as an orthonormal basis. If $2f_c T \in \mathbb{Z}$, the two functions

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t), \quad \phi_2(t) = -\sqrt{\frac{2}{T}} \sin(2\pi f_c t), \quad t \in [0, T],$$

form an orthonormal system in $L^2([0, T])$: $\langle \phi_1, \phi_1 \rangle = \langle \phi_2, \phi_2 \rangle = 1$ and $\langle \phi_1, \phi_2 \rangle = 0$. Consequently, if s_I and s_Q are constant on $[0, T]$ and equal $\Re a$ and $\Im a$ respectively for some

$a \in \mathbb{C}$, the signal $s(t) = \Re\{a e^{i2\pi f_c t}\}$ restricted to $[0, T]$ is written

$$s = \sqrt{\frac{T}{2}} \left((\Re a) \phi_1 + (\Im a) \phi_2 \right), \quad \int_0^T s(t)^2 dt = \frac{T}{2} |a|^2.$$

Proof. By Proposition 26.2 with $2f_c T \in \mathbb{Z}$: $\int_0^T \cos^2 = \int_0^T \sin^2 = T/2$ and $\int_0^T \cos \sin = 0$. Multiplying by the factor $2/T$ gives $\|\phi_1\|^2 = \|\phi_2\|^2 = (2/T)(T/2) = 1$ and $\langle \phi_1, \phi_2 \rangle = -(2/T) \cdot 0 = 0$.

For the second part, Theorem 26.1(iv) with $s_I \equiv \Re a$ and $s_Q \equiv \Im a$ gives $s(t) = (\Re a) \cos(2\pi f_c t) - (\Im a) \sin(2\pi f_c t)$, and substituting $\cos(2\pi f_c t) = \sqrt{T/2} \phi_1(t)$ and $-\sin(2\pi f_c t) = \sqrt{T/2} \phi_2(t)$ gives the formula. The energy is computed with the Pythagorean theorem in $L^2([0, T])$, valid because $\phi_1 \perp \phi_2$:

$$\|s\|^2 = \frac{T}{2} \left((\Re a)^2 \|\phi_1\|^2 + (\Im a)^2 \|\phi_2\|^2 \right) = \frac{T}{2} ((\Re a)^2 + (\Im a)^2) = \frac{T}{2} |a|^2.$$

□

Remark 26.3: Where the hypothesis $2f_c T \in \mathbb{Z}$ hides in real systems. No real system guarantees $2f_c T \in \mathbb{Z}$: the carrier comes from one oscillator and the symbol frequency from another, and the two are not in a controlled rational ratio. What is guaranteed is the inequality $f_c T \gg 1$ — typically f_c is of the order of gigahertz and $R_s = 1/T$ of the order of megabaud, so that $f_c T$ is of the order of 10^3 — and then Proposition 26.2(ii) says that the relative discrepancy from orthogonality is at most $1/(2\pi f_c T) \approx 1.6 \cdot 10^{-4}$, entirely negligible compared with every other imperfection. This is the honest way of saying "the two carriers are orthogonal": it is not an identity, it is an inequality with an explicit constant. It should be added that in practice one does not multiply by the carrier and integrate over $[0, T]$, but filters with the transmit pulse (Section 26.6), which does not change the substance of the argument because the pulse, too, is slow compared with f_c .

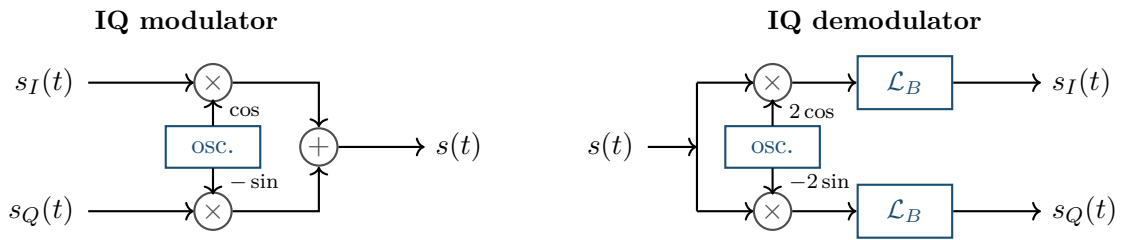


Figure 26.2: The quadrature modulator and demodulator. On the left, Theorem 26.1(iv), read as a block diagram: the two components multiply $\cos(2\pi f_c t)$ and $-\sin(2\pi f_c t)$ respectively and are summed. On the right, Corollary 26.1: one multiplies by $2\cos(2\pi f_c t)$ and by $-2\sin(2\pi f_c t)$ and removes with the low-pass filter $\mathcal{L}_B$ the term around $2f_c$ that this multiplication generates. The two branches do not disturb each other because the carriers are orthogonal (Proposition 26.2 and Corollary 26.2): this is the only reason two independent streams can occupy the same frequency.

26.3. Constellations: points on the circle

Once the way of transmitting a pair (s_I, s_Q) is fixed, it remains to decide *which* pairs to use. The set of admitted values is the constellation, and the choice of which finite set of points of the complex plane to use is the design core of digital modulation: one wants many points (to carry many bits per symbol) well separated from each other (to withstand noise) and of not too

different modulus (so as not to mistreat the power amplifier). The three wishes conflict, and this section and the next quantify them.

Definition 26.3: Constellation, energy, minimum distance. Let $M \geq 2$ be an integer. A **constellation** of order M is a finite set $\mathcal{A} = \{a_0, \dots, a_{M-1}\} \subset \mathbb{C}$ of M distinct points, called **symbols**. Assuming the symbols are **equiprobable**, one defines:

- the **average energy per symbol** in baseband $E_s = \frac{1}{M} \sum_{j=0}^{M-1} |a_j|^2$;
- the number of **bits per symbol** $m = \log_2 M$ (the constellations used here have M a power of two, so that $m \in \mathbb{N}$) and the **average energy per bit** $E_b = E_s/m$;
- the **minimum distance** $d_{\min}(\mathcal{A}) = \min_{j \neq l} |a_j - a_l|$;
- the **peak-to-average power ratio** $\text{PAPR}(\mathcal{A}) = \frac{\max_j |a_j|^2}{E_s}$.

A **mapping** is a bijection $\mu: \{0, 1\}^m \rightarrow \mathcal{A}$ that assigns each word of m bits to a symbol.

Proposition 26.3: Energy per bit and energy per symbol. Transmit a sequence of K symbols chosen independently and uniformly in $\mathcal{A}$. The overall average baseband energy is KE_s , the transmitted bits are Km , and hence the average energy spent per bit equals $E_s/m = E_b$, independently of K . If the symbol is transmitted in bandpass form over an interval of duration T as in Corollary 26.2, the energy of the real signal per symbol equals $\frac{T}{2}E_s$ and that per bit $\frac{T}{2}E_b$: the factor $T/2$ is common and does not affect any energy ratio.

Proof. Let $A_1, \dots, A_K$ be the transmitted symbols, independent and uniform on $\mathcal{A}$. The baseband energy of the sequence is $\sum_{k=1}^K |A_k|^2$, and by linearity of expectation

$$\mathbb{E} \left[\sum_{k=1}^K |A_k|^2 \right] = \sum_{k=1}^K \mathbb{E}[|A_k|^2] = \sum_{k=1}^K \frac{1}{M} \sum_{j=0}^{M-1} |a_j|^2 = K E_s,$$

where the definition of expected value for a variable uniform on a finite set was used. Each symbol carries m bits and there are K symbols: the bits are Km . The ratio between average energy and number of bits is $KE_s/(Km) = E_s/m$, which does not depend on K . The last statement is Corollary 26.2, according to which $\int_0^T s^2 = \frac{T}{2}|a|^2$: averaging over a uniform in $\mathcal{A}$ gives $\frac{T}{2}E_s$, and dividing by m gives $\frac{T}{2}E_b$. $\square$

Definition 26.4: M-PSK phase modulation. Let $M \geq 2$, $E_s > 0$ and $\varphi_0 \in \mathbb{R}$. The **M-PSK** constellation of energy E_s and initial phase φ_0 is

$$\mathcal{A}_{M\text{-PSK}} = \left\{ a_k = \sqrt{E_s} e^{i\left(\frac{2\pi k}{M} + \varphi_0\right)} : k = 0, 1, \dots, M-1 \right\}.$$

The particular cases have their own names: $M = 2$ gives **BPSK** (*binary phase shift keying*), which with $\varphi_0 = 0$ is $\{+\sqrt{E_s}, -\sqrt{E_s}\}$; $M = 4$ gives **QPSK** (*quaternary*), which with $\varphi_0 = \pi/4$ is $\{\sqrt{E_s}/2(\pm 1 \pm i)\}$; $M = 8$ gives **8-PSK**.

Remark 26.4: The roots of unity, rotated and dilated. Setting $\zeta_M = e^{i2\pi/M}$, Definition 26.4 is rewritten $a_k = \sqrt{E_s} e^{i\varphi_0} \zeta_M^k$: the M -PSK constellation is exactly the set of M -th roots of unity (Definition 6.2), which are M in number by Theorem 6.2, multiplied by the complex number $\sqrt{E_s} e^{i\varphi_0}$, i.e. rotated by φ_0 and dilated by $\sqrt{E_s}$. By Theorem 6.3 its points are the vertices of a regular M -gon inscribed in the circle of radius $\sqrt{E_s}$. Every geometric property of the regular polygon is therefore immediately a property of the modula-

tion: the rotational symmetry of order M says that all symbols are equivalent (and indeed the error probability does not depend on which symbol was transmitted, Corollary 26.4), and the formula for the side of the regular polygon is the minimum distance (Theorem 26.3). Even the elementary fact that $\sum_k \zeta_M^k = 0$ for $M \geq 2$ (Theorem 6.4) has a meaning: the centroid of the constellation is the origin, i.e. the modulation does not waste energy in a DC component.

Lemma 26.1: The chord of the unit circle. For every $\alpha, \beta \in \mathbb{R}$, $|e^{i\alpha} - e^{i\beta}| = 2 \left| \sin \frac{\alpha - \beta}{2} \right|$.

Proof. Factor out the average phase term:

$$e^{i\alpha} - e^{i\beta} = e^{i\frac{\alpha+\beta}{2}} \left(e^{i\frac{\alpha-\beta}{2}} - e^{-i\frac{\alpha-\beta}{2}} \right).$$

The equality is checked by multiplying out: the first product gives $e^{i(\alpha+\beta)/2+i(\alpha-\beta)/2} = e^{i\alpha}$ and the second $e^{i(\alpha+\beta)/2-i(\alpha-\beta)/2} = e^{i\beta}$. By Euler's formula, $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ with $\theta = (\alpha - \beta)/2$. Taking moduli and using $|e^{i\gamma}| = 1$ and $|2i \sin \theta| = 2|\sin \theta|$ concludes the proof. $\square$

Theorem 26.3: Minimum distance of M -PSK. For the M -PSK constellation of energy E_s (Definition 26.4),

$$d_{\min} = 2\sqrt{E_s} \sin \frac{\pi}{M},$$

for every $M \geq 2$ and every φ_0 , and the minimum is attained exactly by pairs of adjacent symbols, i.e. with indices k, l such that $k - l \equiv \pm 1 \pmod{M}$.

Proof. By Lemma 26.1, with $\alpha = \frac{2\pi k}{M} + \varphi_0$ and $\beta = \frac{2\pi l}{M} + \varphi_0$ (the phases φ_0 cancel in the difference),

$$|a_k - a_l| = \sqrt{E_s} |e^{i\alpha} - e^{i\beta}| = 2\sqrt{E_s} \left| \sin \frac{\pi(k-l)}{M} \right|.$$

Setting $j = \langle k - l \rangle_M \in \{1, \dots, M-1\}$ (the remainder of $k - l$ modulo M , nonzero because $k \neq l$), and observing that $\sin \frac{\pi(k-l)}{M}$ and $\sin \frac{\pi j}{M}$ differ at most by sign — indeed $k - l = j + qM$ and $\sin(\frac{\pi j}{M} + q\pi) = (-1)^q \sin \frac{\pi j}{M}$ — we get

$$d_{\min} = 2\sqrt{E_s} \cdot \min_{1 \leq j \leq M-1} \sin \frac{\pi j}{M},$$

where the absolute value has been dropped because for $1 \leq j \leq M-1$ the argument $\pi j/M$ lies in $(0, \pi)$ and there the sine is positive.

It remains to minimise $j \mapsto \sin(\pi j/M)$ over $\{1, \dots, M-1\}$. The identity $\sin \frac{\pi(M-j)}{M} = \sin(\pi - \frac{\pi j}{M}) = \sin \frac{\pi j}{M}$ holds: the function is symmetric under $j \mapsto M - j$, and it therefore suffices to minimise it for $1 \leq j \leq M/2$. On that interval $\pi j/M \in (0, \pi/2]$, where the sine is *strictly increasing* (the derivative $\cos$ is positive on $[0, \pi/2)$): the minimum is at $j = 1$ and equals $\sin(\pi/M)$. By symmetry the same value is attained at $j = M-1$, and at no other j if $M \geq 3$ (for $M = 2$ the two indices coincide). Hence $d_{\min} = 2\sqrt{E_s} \sin(\pi/M)$, attained only by pairs with $k - l \equiv \pm 1 \pmod{M}$. $\square$

Corollary 26.3: *BPSK and QPSK have the same minimum distance at equal energy per bit. With the notation of Definition 26.3:*

$$\text{BPSK } (m = 1, E_s = E_b) : \quad d_{\min} = 2\sqrt{E_s} = 2\sqrt{E_b};$$

$$\text{QPSK } (m = 2, E_s = 2E_b) : \quad d_{\min} = 2\sqrt{E_s} \sin \frac{\pi}{4} = \sqrt{2} \sqrt{E_s} = \sqrt{2} \sqrt{2E_b} = 2\sqrt{E_b}.$$

The two constellations therefore have the same minimum distance when the same energy per bit is spent, even though QPSK carries twice as many bits per symbol.

Proof. For $M = 2$, Theorem 26.3 gives $d_{\min} = 2\sqrt{E_s} \sin(\pi/2) = 2\sqrt{E_s}$, and $m = \log_2 2 = 1$ gives $E_b = E_s$ by Definition 26.3. For $M = 4$ we have $\sin(\pi/4) = \sqrt{2}/2$, hence $d_{\min} = 2\sqrt{E_s} \cdot \frac{\sqrt{2}}{2} = \sqrt{2}\sqrt{E_s}$; and $m = \log_2 4 = 2$ gives $E_s = 2E_b$, from which $\sqrt{2}\sqrt{E_s} = \sqrt{2} \cdot \sqrt{2E_b} = 2\sqrt{E_b}$. $\square$

Proposition 26.4: *QPSK is the superposition of two orthogonal BPSKs. Let $\mathcal{A}$ be the QPSK constellation of energy E_s with $\varphi_0 = \pi/4$, i.e.*

$$\mathcal{A} = \left\{ \sqrt{\frac{E_s}{2}} (\varepsilon_1 + i\varepsilon_2) : \varepsilon_1, \varepsilon_2 \in \{+1, -1\} \right\}.$$

Then the in-phase and quadrature components of a symbol each take only the two values $\pm\sqrt{E_s/2} = \pm\sqrt{E_b}$, i.e. they are two BPSKs of energy E_b , and the mapping $(\varepsilon_1, \varepsilon_2) \mapsto a$ makes them independent if the two bits are independent and uniform.

Proof. With $\varphi_0 = \pi/4$ the four phases are $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$, and $\cos$ and $\sin$ of each equal $\pm\frac{\sqrt{2}}{2}$: explicitly $\sqrt{E_s}e^{i\pi/4} = \sqrt{E_s}(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}) = \sqrt{E_s/2}(1+i)$, and similarly for the other three, which correspond to the four sign choices. Hence $\Re a = \varepsilon_1\sqrt{E_s/2}$ and $\Im a = \varepsilon_2\sqrt{E_s/2}$. By Definition 26.3 with $M = 4$ we have $m = 2$ and $E_b = E_s/2$, so $\sqrt{E_s/2} = \sqrt{E_b}$: each component is a BPSK of energy E_b , in agreement with Corollary 26.3. Independence is immediate: the pair $(\varepsilon_1, \varepsilon_2)$ determines a and conversely, so if the two bits are independent so are the two components. $\square$

Definition 26.5: Gray mapping. A mapping $\mu: \{0, 1\}^m \rightarrow \mathcal{A}$ is said to be **Gray** if every pair of symbols at distance $d_{\min}$ corresponds to a pair of binary words that differ in *exactly one bit*.

Proposition 26.5: *A Gray mapping exists for M-PSK. For every $M = 2^m$ the M-PSK constellation admits a Gray mapping.*

Proof. By Theorem 26.3 the pairs at distance $d_{\min}$ are exactly those with consecutive indices modulo M . It therefore suffices to exhibit a bijection $g: \{0, \dots, M-1\} \rightarrow \{0, 1\}^m$ such that $g(k)$ and $g(k+1)$ differ in exactly one bit for every k , and also $g(M-1)$ and $g(0)$. Take the **reflected Gray code**, defined for $k = \sum_{t=0}^{m-1} b_t 2^t$ by

$$g(k) = (c_{m-1}, \dots, c_0), \quad c_t = b_t \oplus b_{t+1} \quad (t < m-1), \quad c_{m-1} = b_{m-1},$$

where $\oplus$ is addition modulo 2. The map is bijective because it can be inverted explicitly: $b_{m-1} = c_{m-1}$ and $b_t = c_t \oplus b_{t+1}$ for decreasing t , which uniquely reconstructs k from $g(k)$. That $g(k)$ and $g(k+1)$ differ in exactly one bit is seen as follows. Let t_0 be the position of the least significant bit of k equal to 0; adding 1 to k flips that bit to 1 and all bits of position $< t_0$ to 0, leaving those of position $> t_0$ unchanged. So, writing b and b' for the bits of k and $k+1$: $b'_t = b_t$ for $t > t_0$, $b'_{t_0} = 1 \neq 0 = b_{t_0}$, and $b'_t = 0$, $b_t = 1$ for $t < t_0$. It follows, for $t > t_0$, that $c'_t = b'_t \oplus b'_{t+1} = b_t \oplus b_{t+1} = c_t$; for $t = t_0$, $c'_{t_0} = b'_{t_0} \oplus b'_{t_0+1} = 1 \oplus b_{t_0+1} \neq 0 \oplus b_{t_0+1} = c_{t_0}$;

for $t < t_0 - 1$, $c'_t = 0 \oplus 0 = 0$ and $c_t = 1 \oplus 1 = 0$, equal; for $t = t_0 - 1$ (if $t_0 \geq 1$), $c'_{t_0-1} = b'_{t_0-1} \oplus b'_{t_0} = 0 \oplus 1 = 1$ and $c_{t_0-1} = b_{t_0-1} \oplus b_{t_0} = 1 \oplus 0 = 1$, equal. In total only the bit of position t_0 changes.

There remains the cyclic pair $(M - 1, 0)$. For $k = M - 1 = 2^m - 1$ all the b_t equal 1, hence $c_{m-1} = 1$ and $c_t = 1 \oplus 1 = 0$ for $t < m - 1$: $g(M - 1) = (1, 0, \dots, 0)$. For $k = 0$ all the b_t equal 0 and $g(0) = (0, 0, \dots, 0)$. The two words differ in only the most significant bit. $\square$

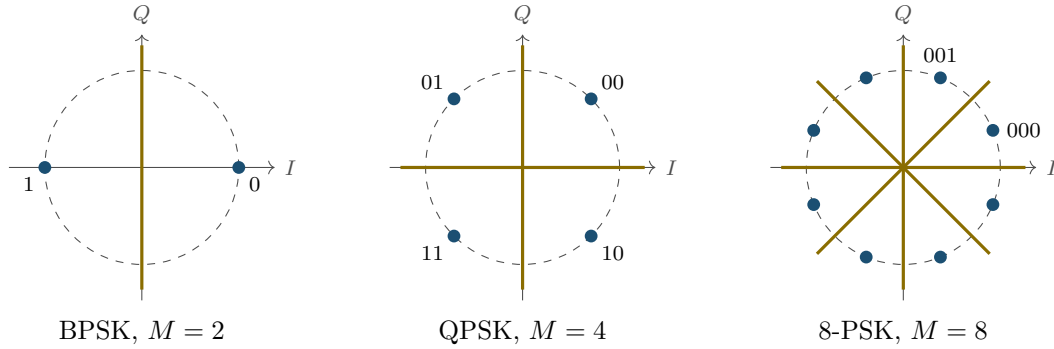


Figure 26.3: The three PSK constellations, drawn from Definition 26.4 with $\varphi_0 = 0$ for BPSK, $\varphi_0 = \pi/8$ for 8-PSK, and $\varphi_0 = \pi/4$ for QPSK. The radius of the dashed circle is $\sqrt{E_s}$, here 1.35 cm. The coordinates of the points are $R \cos \theta$ and $R \sin \theta$ with $R = 1.35$: for QPSK $1.35 \cdot \frac{\sqrt{2}}{2} = 0.9546$; for 8-PSK $1.35 \cos 22.5^\circ = 1.2473$ and $1.35 \sin 22.5^\circ = 0.5166$, and the boundaries between decision regions lie at $0^\circ, 45^\circ, \dots$ with $1.7 \cdot \frac{\sqrt{2}}{2} = 1.2021$. The thick lines are the boundaries of the minimum-distance decision regions (Theorem 26.7): for PSK these are the circular sectors bounded by the bisectors of the arcs between adjacent symbols. The binary labels follow a Gray mapping (Proposition 26.5).

Example 26.3: The numerical values of the minimum distance. With $E_s = 1$, Theorem 26.3 gives the following values.

M	$m = \log_2 M$	π/M	$d_{\min} = 2 \sin(\pi/M)$	$d_{\min}^2/(4E_b)$
2	1	$\pi/2$	2.0000000	1.0000000
4	2	$\pi/4$	1.4142136	1.0000000
8	3	$\pi/8$	0.7653669	0.4393398
16	4	$\pi/16$	0.3901806	0.1522409
32	5	$\pi/32$	0.1960343	0.0480368

The last column, computed with $E_b = E_s/m = 1/m$ and hence equal to $m d_{\min}^2/4$, is the ratio that will appear in the argument of the Q function (Theorem 26.9 and Remark 26.10). The first two rows are equal, which is the content of Corollary 26.3: QPSK doubles the bits per symbol *without* paying anything in minimum distance at equal energy per bit. From the third row on one pays, and dearly: going from 4 to 8 points the ratio drops by a factor $1/0.4393398 = 2.2761424$, and from 4 to 32 by a factor $1/0.0480368 = 20.8173$.

Independent check. A Python script that builds the M roots of unity in floating-point arithmetic and *enumerates all* $M(M - 1)$ ordered pairs, computing the modulus of the difference, returns, for $M = 2, 4, 8, 16, 32$, the same values with maximum discrepancy $6.9 \cdot 10^{-16}$ from the closed form. $\checkmark$

26.4. QAM and APSK: points on several circles

M -PSK spends all its energy on the same circle, and for large M the points crowd together: Theorem 26.3 says the minimum distance decreases like $2\pi\sqrt{E_s}/M$, because $\sin x \sim x$. To escape

this bottleneck one must also use the *modulus*, i.e. arrange the points on several concentric circles. The two classical families are QAM, which arranges the points on a square grid — and hence, read in polar coordinates, on circles with unequal populations — and APSK, which arranges them on a few circles with chosen populations.

Definition 26.6: Square M -QAM quadrature amplitude modulation. Let $M = L^2$ with $L \geq 2$ an even integer, and let $\Delta > 0$. The **square M -QAM** constellation of step Δ is

$$\mathcal{A}_{M\text{-QAM}} = \left\{ \Delta[(2u-1-L) + i(2v-1-L)] : u, v \in \{1, \dots, L\} \right\}.$$

The two coordinates independently range over the L levels $\Delta\{-(L-1), -(L-3), \dots, L-3, L-1\}$, symmetric about the origin and 2Δ apart from one another. For $L = 4$ the levels are $\Delta\{-3, -1, 1, 3\}$ and one obtains 16-QAM.

Lemma 26.2: Sum of squares of the levels. For every integer $L \geq 1$, $\sum_{u=1}^L (2u-1-L)^2 = \frac{L(L^2-1)}{3}$.

Proof. As u ranges from 1 to L the number $2u-1-L$ runs, in steps of two, through the values from $1-L$ to $L-1$: these are L integers, all of parity opposite to that of L , symmetric about 0.

Case L even. The values are $\pm 1, \pm 3, \dots, \pm(L-1)$, i.e. the first $L/2$ odd numbers taken with both signs. Setting $m = L/2$,

$$\sum_{u=1}^L (2u-1-L)^2 = 2 \sum_{j=1}^m (2j-1)^2.$$

Now

$$\sum_{j=1}^m (2j-1)^2 = \sum_{j=1}^m (4j^2 - 4j + 1) = 4 \cdot \frac{m(m+1)(2m+1)}{6} - 4 \cdot \frac{m(m+1)}{2} + m,$$

using the two known sums $\sum_{j \leq m} j^2 = \frac{m(m+1)(2m+1)}{6}$ and $\sum_{j \leq m} j = \frac{m(m+1)}{2}$. Reducing to the common denominator 3:

$$\frac{2m(m+1)(2m+1) - 6m(m+1) + 3m}{3} = \frac{m[2(m+1)(2m+1) - 6(m+1) + 3]}{3} = \frac{m[4m^2 - 1]}{3},$$

because $2(m+1)(2m+1) - 6(m+1) + 3 = (m+1)(4m+2-6) + 3 = (m+1)(4m-4) + 3 = 4m^2 - 1$. Hence

$$2 \sum_{j=1}^m (2j-1)^2 = \frac{2m(4m^2 - 1)}{3} = \frac{L(L^2 - 1)}{3},$$

substituting $m = L/2$, i.e. $2m = L$ and $4m^2 - 1 = L^2 - 1$.

Case L odd. The values are $0, \pm 2, \pm 4, \dots, \pm(L-1)$. Setting $m = (L-1)/2$,

$$\sum_{u=1}^L (2u-1-L)^2 = 2 \sum_{j=1}^m (2j)^2 = 8 \sum_{j=1}^m j^2 = 8 \cdot \frac{m(m+1)(2m+1)}{6} = \frac{4m(m+1)(2m+1)}{3}.$$

With $m = \frac{L-1}{2}$ we have $m+1 = \frac{L+1}{2}$ and $2m+1 = L$, hence

$$\frac{4}{3} \cdot \frac{L-1}{2} \cdot \frac{L+1}{2} \cdot L = \frac{L(L^2 - 1)}{3}.$$

The two cases exhaust all $L \geq 1$, and in both the same formula is obtained. $\square$

Theorem 26.4: Energy, minimum distance and PAPR of square QAM. For the square M -QAM constellation of step Δ , with $M = L^2$,

$$E_s = \frac{2(M-1)}{3} \Delta^2, \quad d_{\min} = 2\Delta = \sqrt{\frac{6E_s}{M-1}}, \quad \text{PAPR} = \frac{3(L-1)}{L+1} = \frac{3(\sqrt{M}-1)}{\sqrt{M}+1}.$$

Proof. Energy. By Definition 26.3 and Definition 26.6,

$$E_s = \frac{1}{L^2} \sum_{u=1}^L \sum_{v=1}^L \Delta^2 [(2u-1-L)^2 + (2v-1-L)^2].$$

The double sum splits: in the first part the index v is mute and contributes a factor L , in the second u is mute. Hence, by Lemma 26.2,

$$E_s = \frac{\Delta^2}{L^2} \left[L \sum_{u=1}^L (2u-1-L)^2 + L \sum_{v=1}^L (2v-1-L)^2 \right] = \frac{\Delta^2}{L^2} \cdot 2L \cdot \frac{L(L^2-1)}{3} = \frac{2(L^2-1)}{3} \Delta^2,$$

which is the first formula with $M = L^2$.

Minimum distance. Let $a = \Delta[(2u-1-L) + i(2v-1-L)]$ and $a' = \Delta[(2u'-1-L) + i(2v'-1-L)]$ be two distinct symbols. Then

$$|a - a'|^2 = 4\Delta^2 [(u-u')^2 + (v-v')^2],$$

because $(2u-1-L) - (2u'-1-L) = 2(u-u')$. Since $(u, v) \neq (u', v')$ and the differences are integers, the quantity $(u-u')^2 + (v-v')^2$ is an integer ≥ 1 , and equals exactly 1 for instance with $u' = u+1$, $v' = v$ (possible because $L \geq 2$). The minimum of $|a - a'|$ is therefore $\sqrt{4\Delta^2} = 2\Delta$. Substituting $\Delta^2 = \frac{3E_s}{2(M-1)}$ obtained from the first formula,

$$d_{\min} = 2\Delta = 2\sqrt{\frac{3E_s}{2(M-1)}} = \sqrt{\frac{4 \cdot 3E_s}{2(M-1)}} = \sqrt{\frac{6E_s}{M-1}}.$$

PAPR. The maximum modulus occurs at the four corner symbols, where both coordinates equal $\pm\Delta(L-1)$: there $|a|^2 = 2\Delta^2(L-1)^2$. That this is the maximum follows from the fact that $|a|^2$ is a sum of two squares and each coordinate has modulus at most $\Delta(L-1)$. Hence

$$\text{PAPR} = \frac{2\Delta^2(L-1)^2}{\frac{2}{3}(L^2-1)\Delta^2} = \frac{3(L-1)^2}{L^2-1} = \frac{3(L-1)^2}{(L-1)(L+1)} = \frac{3(L-1)}{L+1},$$

where $L-1 \neq 0$ was cancelled because $L \geq 2$. □

Example 26.4: The parameters of 16-QAM and its siblings. With $E_s = 1$, Theorem 26.4 gives:

M	$L = \sqrt{M}$	m	$d_{\min} = \sqrt{6/(M-1)}$	PAPR
4	2	2	1.4142136	1.0000000
16	4	4	0.6324555	1.8000000
64	8	6	0.3086067	2.3333333
256	16	8	0.1533930	2.6470588

The row $M = 4$ recovers QPSK: indeed $\sqrt{6/3} = \sqrt{2}$ and 4-QAM is the QPSK of Definition 26.4 with $\varphi_0 = \pi/4$ (its four points $\Delta(\pm 1 \pm i)$ all lie on the circle of radius $\Delta\sqrt{2}$), in agreement with Corollary 26.3. For $M = 16$ the comparison with 16-PSK is merciless:

0.6324555 against 0.3901806 (Example 26.3), i.e. a factor 1.6209302 in distance, which is $20 \log_{10} 1.6209302 = 4.1952855$ dB of gain.

Independent check. A Python script that builds the M -QAM constellations point by point from Definition 26.6, computes their average energy as the arithmetic mean of the $|a|^2$, normalises them to $E_s = 1$, and enumerates all $M(M-1)$ ordered pairs returns, for $M = 4, 16, 64, 256$, the same minimum distances with maximum discrepancy $2.3 \cdot 10^{-16}$, and energies in units of Δ^2 equal to 2, 10, 42, 170, which coincide with $\frac{2}{3}(M-1)$. ✓

Definition 26.7: APSK constellation with concentric circles. Let $r \geq 1$ be the number of circles, $M_1, \dots, M_r$ positive integers with $M = M_1 + \dots + M_r$, radii $0 < R_1 < \dots < R_r$ and phases $\varphi_1, \dots, \varphi_r \in \mathbb{R}$. The corresponding **APSK** (*amplitude and phase shift keying*) constellation is

$$\mathcal{A}_{\text{APSK}} = \bigcup_{\ell=1}^r \left\{ R_\ell e^{i\left(\frac{2\pi j}{M_\ell} + \varphi_\ell\right)} : j = 0, \dots, M_\ell - 1 \right\},$$

i.e. the union of r PSK constellations of different orders and radii. It is called $(M_1 + \dots + M_r)$ -**APSK**; for instance $4 + 12$ means $r = 2$, $M_1 = 4$, $M_2 = 12$, $M = 16$.

Theorem 26.5: The 4+12-APSK of maximum minimum distance. Consider the family of 4 + 12-APSK constellations with

$$\varphi_1 = \frac{\pi}{4}, \quad \varphi_2 = 0, \quad R_2 = \gamma R_1 \quad (\gamma \geq 1),$$

normalised to $E_s = 1$. Then:

- (i) normalisation forces $R_1 = \frac{2}{\sqrt{1+3\gamma^2}}$;
- (ii) the minimum distance equals

$$d_{\min}(\gamma) = \frac{2}{\sqrt{1+3\gamma^2}} \min \left\{ \sqrt{2}, \quad 2\gamma \sin \frac{\pi}{12}, \quad \sqrt{1+\gamma^2-2\gamma \cos \frac{\pi}{12}} \right\};$$

- (iii) $d_{\min}(\gamma)$ is maximal for $\gamma^* = 1 + \sqrt{3} = 2.7320508$, and the maximum value is

$$d_{\min}(\gamma^*) = \frac{2\sqrt{2}}{\sqrt{13+6\sqrt{3}}} = \sqrt{\frac{104-48\sqrt{3}}{61}} = 0.5848015;$$

- (iv) at γ^* the peak-to-average power ratio equals $\text{PAPR} = R_2^2 = \frac{64+8\sqrt{3}}{61} = 1.2763345$.

Proof. (i) Normalisation. By Definition 26.3 the 16 symbols have moduli R_1 (four times) and R_2 (twelve times), hence

$$E_s = \frac{4R_1^2 + 12R_2^2}{16} = \frac{R_1^2 + 3R_2^2}{4} = \frac{R_1^2(1+3\gamma^2)}{4}.$$

Imposing $E_s = 1$ gives $R_1^2 = 4/(1+3\gamma^2)$, i.e. the formula, having chosen the positive root because $R_1 > 0$.

(ii) The three families of pairs. Every pair of distinct symbols is of one of three types. *Both on the inner circle.* The four points $R_1 e^{i(\pi/4+k\pi/2)}$ form a 4-PSK of energy R_1^2 , so by Theorem 26.3 their minimum distance is $2R_1 \sin(\pi/4) = R_1 \sqrt{2}$.

Both on the outer circle. The twelve points $R_2 e^{i2\pi j/12}$ form a 12-PSK of energy R_2^2 , with minimum distance $2R_2 \sin(\pi/12) = 2\gamma R_1 \sin(\pi/12)$.

One per circle. For $a = R_1 e^{i\alpha}$ and $b = R_2 e^{i\beta}$,

$$\begin{aligned} |a - b|^2 &= (R_1 e^{i\alpha} - R_2 e^{i\beta}) \overline{(R_1 e^{i\alpha} - R_2 e^{i\beta})} = R_1^2 + R_2^2 - R_1 R_2 (e^{i(\alpha-\beta)} + e^{-i(\alpha-\beta)}) \\ &= R_1^2 + R_2^2 - 2R_1 R_2 \cos(\alpha - \beta), \end{aligned}$$

using $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ (Euler's formula). This quantity is smaller the larger $\cos(\alpha - \beta)$ is, i.e. the closer $\alpha - \beta$ is to a multiple of 2π . Now

$$\alpha - \beta = \frac{\pi}{4} + \frac{k\pi}{2} - \frac{j\pi}{6} = \frac{\pi}{12}(3 + 6k - 2j), \quad k \in \{0, 1, 2, 3\}, \quad j \in \{0, \dots, 11\},$$

and $3 + 6k - 2j$ is *odd*, because $6k - 2j$ is even. The minimum of $|3 + 6k - 2j|$ over a set of odd integers is therefore at least 1, and equals exactly 1 because $k = 0, j = 1$ gives $3 - 2 = 1$. Staying within $|\alpha - \beta| \leq \pi$ — which is legitimate because cosine is 2π -periodic and even — the minimum value of $|\alpha - \beta|$ is $\pi/12$, and the minimum distance between the two circles is $\sqrt{R_1^2 + R_2^2 - 2R_1 R_2 \cos(\pi/12)} = R_1 \sqrt{1 + \gamma^2 - 2\gamma \cos(\pi/12)}$.

Factoring out R_1 in the three values and taking the minimum gives formula (ii), with R_1 given by (i).

(iii) The optimum. Set $s = \sin \frac{\pi}{12}$, $c = \cos \frac{\pi}{12}$ and

$$h_1(\gamma) = \frac{2\sqrt{2}}{\sqrt{1+3\gamma^2}}, \quad h_2(\gamma) = \frac{4\gamma s}{\sqrt{1+3\gamma^2}}, \quad h_3(\gamma) = \frac{2\sqrt{1+\gamma^2-2\gamma c}}{\sqrt{1+3\gamma^2}},$$

so that $d_{\min} = \min\{h_1, h_2, h_3\}$.

The function h_1 is *strictly decreasing* on $[1, +\infty)$, because the numerator is constant and the denominator $\sqrt{1+3\gamma^2}$ is strictly increasing.

The function h_2 is *strictly increasing*: indeed $h_2(\gamma) = 4s/\sqrt{\gamma^{-2}+3}$ for $\gamma > 0$, and $\gamma^{-2}+3$ is strictly decreasing.

The two meet where $2\sqrt{2} = 4\gamma s$, i.e. $\gamma = \frac{\sqrt{2}}{2s} = \frac{1}{\sqrt{2}s}$. Since $s = \sin \frac{\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{4}$ — a value that follows from $\sin 15^\circ = \sin(45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}-\sqrt{2}}{4}$ — we have $\sqrt{2}s = \frac{\sqrt{12}-2}{4} = \frac{2\sqrt{3}-2}{4} = \frac{\sqrt{3}-1}{2}$, and

$$\gamma^* = \frac{2}{\sqrt{3}-1} = \frac{2(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{2(\sqrt{3}+1)}{2} = \sqrt{3}+1.$$

At γ^* we have $\gamma^{*2} = (\sqrt{3}+1)^2 = 4+2\sqrt{3}$ and hence $1+3\gamma^{*2} = 13+6\sqrt{3}$. Let us check that at γ^* the third term is not the minimum: with $c = \cos \frac{\pi}{12} = \frac{\sqrt{6}+\sqrt{2}}{4}$ (same computation as before with $\cos(45^\circ - 30^\circ)$),

$$1 + \gamma^{*2} - 2\gamma^* c = 1 + 4 + 2\sqrt{3} - 2(\sqrt{3}+1) \cdot \frac{\sqrt{6}+\sqrt{2}}{4} = 3.1861847 > 2,$$

so $h_3(\gamma^*) > h_1(\gamma^*) = h_2(\gamma^*)$ and $d_{\min}(\gamma^*) = h_1(\gamma^*) = \frac{2\sqrt{2}}{\sqrt{13+6\sqrt{3}}}$.

Now maximality. Let $\gamma < \gamma^*$: then $d_{\min}(\gamma) \leq h_2(\gamma) < h_2(\gamma^*) = d_{\min}(\gamma^*)$, because h_2 is strictly increasing. Let $\gamma > \gamma^*$: then $d_{\min}(\gamma) \leq h_1(\gamma) < h_1(\gamma^*) = d_{\min}(\gamma^*)$, because h_1 is strictly decreasing. In both cases the inequality is strict: the maximum is at γ^* and is unique. For the value, rationalising with $(13 - 6\sqrt{3})$ and using $13^2 - (6\sqrt{3})^2 = 169 - 108 = 61$:

$$d_{\min}(\gamma^*)^2 = \frac{8}{13+6\sqrt{3}} = \frac{8(13-6\sqrt{3})}{61} = \frac{104-48\sqrt{3}}{61} = 0.3419928,$$

whose square root is 0.5848015.

(iv) **PAPR.** The maximum modulus is R_2 , because $R_2 > R_1$ for $\gamma > 1$, and $E_s = 1$: hence $\text{PAPR} = R_2^2 = \gamma^{*2} R_1^2 = \frac{4\gamma^{*2}}{1+3\gamma^{*2}} = \frac{4(4+2\sqrt{3})}{13+6\sqrt{3}} = \frac{16+8\sqrt{3}}{13+6\sqrt{3}}$. Rationalising,

$$\frac{(16+8\sqrt{3})(13-6\sqrt{3})}{61} = \frac{208-96\sqrt{3}+104\sqrt{3}-48\cdot 3}{61} = \frac{208-144+8\sqrt{3}}{61} = \frac{64+8\sqrt{3}}{61} = 1.2763345.$$

As a consistency check, $E_s = \frac{R_1^2+3R_2^2}{4} = \frac{0.1709964+3\cdot 1.2763345}{4} = \frac{4.0000000}{4} = 1$. ✓

□

Remark 26.5: Why APSK exists, if QAM has larger minimum distance. The comparison at $M = 16$ and $E_s = 1$ is sharp and must be stated without sugar-coating:

	16-QAM	optimal 4 + 12-APSK
$d_{\min}$	0.6324555	0.5848015
$d_{\min}$ in relative dB	+0.6804302	0
PAPR	1.8000000	1.2763345
PAPR in dB	2.5527251	1.0596452

On a linear Gaussian channel QAM wins, by 0.6804302 dB in minimum distance. APSK is therefore not an absolutely better modulation: it is better when the channel is not linear. A power amplifier aboard a satellite operates near saturation so as not to waste energy, and in that regime its characteristic compresses large moduli and rotates their phase; a constellation with PAPR 1.2763345 (1.0596452 dB) passes through that nonlinearity much better than one with PAPR 1.8 (2.5527251 dB), and it also has only two distinct moduli instead of three, which makes compensation simple. The gain of 1.4930799 dB in PAPR is paid for with 0.6804302 dB of minimum distance: the balance is favourable exactly in the regime in which satellites operate. The constellation of Theorem 26.5 was built here from the optimisation of $d_{\min}$, not taken from any standardisation document; real systems use radius ratios chosen with different criteria (and generally dependent on the coding rate), and do not coincide with $\gamma^* = 1 + \sqrt{3}$.

26.5. Minimum-distance decision

The receiver observes a complex number and must guess which symbol was transmitted. The rule everyone uses — "choose the closest symbol" — is not a convention or a heuristic: under precise hypotheses, it is the rule that *minimises the error probability*. This section states those hypotheses and proves the claim.

Definition 26.8: Additive white Gaussian noise channel in baseband. Let $\mathcal{A}$ be a constellation of order M . The **additive white Gaussian noise channel** (AWGN) for one use, of one-sided spectral density $N_0 > 0$, is the model

$$r = a + n,$$

where a is the transmitted symbol and n is a complex random variable, independent of a , with *independent* real and imaginary parts, each Gaussian with zero mean and variance $N_0/2$. The joint density of n with respect to Lebesgue measure on $\mathbb{R}^2 \cong \mathbb{C}$ is therefore

$$p_n(z) = \frac{1}{2\pi \cdot \frac{N_0}{2}} \exp\left(-\frac{(\Re z)^2 + (\Im z)^2}{2 \cdot \frac{N_0}{2}}\right) = \frac{1}{\pi N_0} e^{-|z|^2/N_0}, \quad z \in \mathbb{C}.$$

Such a variable is called **circularly symmetric complex Gaussian** of variance N_0 , because $\mathbb{E}[|n|^2] = N_0$ and the density depends only on the modulus.

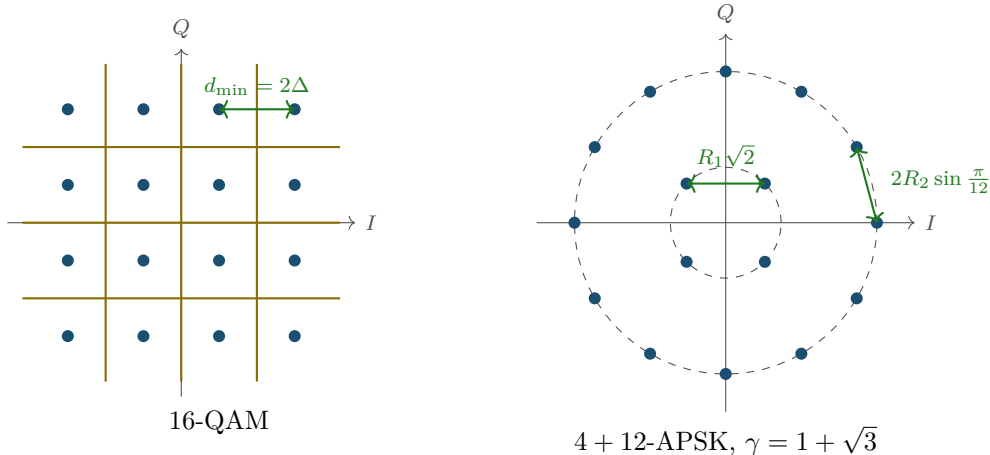


Figure 26.4: On the left, the 16-QAM of Definition 26.6 with $\Delta = 0.5$ cm: the sixteen points lie at $(\pm 0.5, \pm 0.5)$, $(\pm 0.5, \pm 1.5)$, $(\pm 1.5, \pm 0.5)$, $(\pm 1.5, \pm 1.5)$, and the minimum-distance decision lines are $I, Q \in \{-1, 0, 1\}$, i.e. the midpoints between adjacent levels (Theorem 26.7). On the right, the optimal 4 + 12-APSK of Theorem 26.5, drawn with $R_2 = 2$ cm and hence $R_1 = 2/\gamma^* = 2/(1 + \sqrt{3}) = 0.7321$ cm: the four inner points lie at $45^\circ + k \cdot 90^\circ$, with coordinates $0.7321 \cdot \frac{\sqrt{2}}{2} = 0.5176$, and the twelve outer ones at $j \cdot 30^\circ$, with coordinates $2 \cos 30^\circ = 1.7321$, $2 \cos 60^\circ = 1$, 2 and 0 . The two green segments are the two distances that at the optimum become equal: this is the coincidence $h_1 = h_2$ of the proof. All coordinates are computed from the formulas, not measured.

Remark 26.6: The two conventions on N_0 . Definition 26.8 assigns variance $N_0/2$ to each of the two real components and N_0 to the complex one. This is the digital communications convention, in which N_0 is the one-sided spectral density (all the noise power is counted on positive frequencies) and $N_0/2$ the two-sided one. Readers of signal theory texts often find the opposite convention, with σ^2 per component and no reference to N_0 : the formulas change accordingly, and the error probability of BPSK, which here is $Q(\sqrt{2E_b/N_0})$, then becomes $Q(\sqrt{E_b}/\sigma)$. The two are translated by $\sigma^2 = N_0/2$. Whenever a numerical result of this chapter is compared with a source, the first thing to check is which of the two conventions it uses: it is the most common source of 3 dB discrepancies.

Definition 26.9: Decision rule and decision regions. A **decision rule** is a measurable function $\delta: \mathbb{C} \rightarrow \mathcal{A}$. Its **decision regions** are the sets $\mathcal{R}_j = \delta^{-1}(\{a_j\})$, $j = 0, \dots, M - 1$, which form a partition of $\mathbb{C}$. The **error probability** of the rule is

$$P_e(\delta) = \Pr[\delta(r) \neq a] = \frac{1}{M} \sum_{j=0}^{M-1} \Pr[r \notin \mathcal{R}_j \mid a = a_j].$$

The rule is called **maximum likelihood** (ML) if $\delta(r) \in \arg \max_j p_n(r - a_j)$, **maximum a posteriori** (MAP) if $\delta(r) \in \arg \max_j \Pr[a = a_j] p_n(r - a_j)$, and **minimum distance** if $\delta(r) \in \arg \min_j |r - a_j|$.

Theorem 26.6: Minimum distance, maximum likelihood, minimum error probability. Consider the channel of Definition 26.8 with equiprobable symbols. Then:

- (i) the MAP rule coincides with the ML rule;

(ii) the ML rule coincides with the minimum-distance rule:

$$\arg \max_j p_n(r - a_j) = \arg \min_j |r - a_j| \quad \text{for every } r \in \mathbb{C};$$

(iii) every MAP rule minimises the error probability among all decision rules;

(iv) two minimum-distance rules that differ only in the choice made in case of a tie have the same error probability.

Proof. (i). By definition the MAP rule maximises $\Pr[a = a_j] p_n(r - a_j)$. If the symbols are equiprobable, $\Pr[a = a_j] = 1/M$ for every j , and multiplying all the quantities to be compared by the same positive constant $1/M$ does not change the set of maximisers. Hence the two sets of maximising arguments coincide.

(ii). From Definition 26.8,

$$p_n(r - a_j) = \frac{1}{\pi N_0} e^{-|r - a_j|^2 / N_0}.$$

The function $x \mapsto \frac{1}{\pi N_0} e^{-x/N_0}$ is strictly decreasing on $[0, +\infty)$, because $N_0 > 0$ and its derivative equals $-\frac{1}{\pi N_0^2} e^{-x/N_0} < 0$. A strictly decreasing function reverses order, hence

$$p_n(r - a_j) \geq p_n(r - a_l) \iff |r - a_j|^2 \leq |r - a_l|^2 \iff |r - a_j| \leq |r - a_l|,$$

the last step because $t \mapsto t^2$ is strictly increasing on $[0, +\infty)$. Maximising the likelihood is therefore equivalent to minimising the distance, and the two sets of optimal arguments coincide.

(iii). Let δ be any decision rule, with regions $\mathcal{R}_j$. The probability of correct decision is

$$P_c(\delta) = \sum_{j=0}^{M-1} \Pr[a = a_j] \int_{\mathcal{R}_j} p_n(r - a_j) dr = \int_{\mathbb{C}} \sum_{j=0}^{M-1} \mathbf{1}_{\mathcal{R}_j}(r) \Pr[a = a_j] p_n(r - a_j) dr,$$

where interchanging the finite sum and the integral is legitimate because the integrands are nonnegative. For each fixed r , the inner sum equals $\Pr[a = a_{j(r)}] p_n(r - a_{j(r)})$, where $j(r)$ is the unique index with $r \in \mathcal{R}_{j(r)}$ (the regions form a partition); hence

$$\sum_j \mathbf{1}_{\mathcal{R}_j}(r) \Pr[a = a_j] p_n(r - a_j) \leq \max_{0 \leq j \leq M-1} \Pr[a = a_j] p_n(r - a_j) =: F(r),$$

and integrating, $P_c(\delta) \leq \int_{\mathbb{C}} F(r) dr$. The bound is an equality if and only if for almost every r the index $j(r)$ attains the maximum, i.e. if and only if δ is (almost everywhere) a MAP rule. Since a MAP rule exists — the maximum over a finite set is always attained, and one can choose the smallest such index, which gives a measurable function because each choice set is a finite intersection of measurable sets — the value $\int F$ is attained, and $P_e = 1 - P_c$ is minimal exactly for MAP rules.

(iv). Two minimum-distance rules differ only on the set

$$\mathcal{T} = \bigcup_{j \neq l} \{r \in \mathbb{C} : |r - a_j| = |r - a_l|\}.$$

For every pair $j \neq l$, the set $\{|r - a_j| = |r - a_l|\}$ is the perpendicular bisector of the segment $a_j a_l$: expanding $|r - a_j|^2 = |r - a_l|^2$ gives, with $r = x + iy$ and $a_j = \alpha_j + i\beta_j$,

$$-2\alpha_j x - 2\beta_j y + |a_j|^2 = -2\alpha_l x - 2\beta_l y + |a_l|^2,$$

i.e. a linear equation in (x, y) not identically zero (the coefficients $2(\alpha_l - \alpha_j)$ and $2(\beta_l - \beta_j)$ are not both zero, because $a_j \neq a_l$): it is a line, of zero Lebesgue measure in $\mathbb{R}^2$. The finite union $\mathcal{T}$ therefore has zero measure, and since the law of r is absolutely continuous (it has density $p_n(\cdot - a_j)$ conditionally on $a = a_j$), $\Pr[r \in \mathcal{T}] = 0$. The two rules therefore coincide with probability 1 and have the same error probability. $\square$

Theorem 26.7: The decision regions are convex Voronoi cells. For the minimum-distance rule on a constellation $\mathcal{A}$, the region associated with a_j , taken in its closed form

$$\overline{\mathcal{R}}_j = \{r \in \mathbb{C} : |r - a_j| \leq |r - a_l| \quad \forall l \neq j\},$$

is the intersection of $M - 1$ closed half-planes:

$$\overline{\mathcal{R}}_j = \bigcap_{l \neq j} \left\{ r \in \mathbb{C} : 2\Re[r \overline{(a_l - a_j)}] \leq |a_l|^2 - |a_j|^2 \right\}.$$

In particular $\overline{\mathcal{R}}_j$ is closed, convex, and contains a_j ; it is therefore a convex polygon, possibly unbounded.

Proof. For $r, a \in \mathbb{C}$, $|r - a|^2 = (r - a)\overline{(r - a)} = |r|^2 - r\bar{a} - \bar{r}a + |a|^2$, and $r\bar{a} + \bar{r}a = 2\Re(r\bar{a})$. Hence

$$|r - a_j|^2 \leq |r - a_l|^2 \iff |r|^2 - 2\Re(r\bar{a}_j) + |a_j|^2 \leq |r|^2 - 2\Re(r\bar{a}_l) + |a_l|^2.$$

The term $|r|^2$ cancels; moving the terms in r to the left and the moduli to the right, and using the real linearity of $\Re$,

$$2\Re(r\bar{a}_l) - 2\Re(r\bar{a}_j) = 2\Re[r \overline{(a_l - a_j)}] \leq |a_l|^2 - |a_j|^2.$$

For each fixed l the condition has the form $\langle v, w_l \rangle_{\mathbb{R}^2} \leq c_l$ with w_l a nonzero real vector (corresponding to $2\overline{(a_l - a_j)} \neq 0$): it is a closed half-plane. As an intersection of closed half-planes, $\overline{\mathcal{R}}_j$ is closed and convex (the intersection of convex sets is convex, the intersection of closed sets is closed). It contains a_j because $|a_j - a_j| = 0 \leq |a_j - a_l|$ for every l . That it is a polygon, possibly unbounded, is precisely the definition of a finite intersection of half-planes. $\square$

Corollary 26.4: Symmetry of the error probability in M-PSK. For the M-PSK constellation (Definition 26.4) with minimum-distance decision on the AWGN channel, the conditional error probability $\Pr[\delta(r) \neq a_j \mid a = a_j]$ does not depend on j , and hence coincides with the symbol error probability P_s .

Proof. Let $\rho = e^{i2\pi/M}$. By Definition 26.4, $a_{k+1} = \rho a_k$ with indices modulo M , so multiplication by ρ is a bijection of $\mathcal{A}$ onto itself. By Theorem 26.7, $\overline{\mathcal{R}}_j$ is defined by conditions involving only the moduli $|r - a_l|$; since $|\rho z| = |z|$ for every z ,

$$|\rho r - \rho a_{l+1}| = |\rho r - \rho a_l| = |\rho| |r - a_l| = |r - a_l|,$$

and hence $r \in \overline{\mathcal{R}}_j \iff \rho r \in \overline{\mathcal{R}}_{j+1}$: the rotation by $2\pi/M$ carries the region of a_j to that of a_{j+1} . Now,

$$\Pr[r \notin \overline{\mathcal{R}}_j \mid a = a_j] = \int_{\mathbb{C} \setminus \overline{\mathcal{R}}_j} p_n(r - a_j) dr.$$

In the change of variable $r = \rho u$ — a rotation of the plane, whose Jacobian determinant has modulus 1 and carries $\mathbb{C} \setminus \overline{\mathcal{R}}_j$ to $\mathbb{C} \setminus \overline{\mathcal{R}}_{j-1}$ read backwards — one uses $p_n(\rho u - a_j) = \frac{1}{\pi N_0} e^{-|\rho u - \rho a_{j-1}|^2/N_0} = \frac{1}{\pi N_0} e^{-|u - a_{j-1}|^2/N_0} = p_n(u - a_{j-1})$, obtaining

$$\Pr[r \notin \overline{\mathcal{R}}_j \mid a = a_j] = \int_{\mathbb{C} \setminus \overline{\mathcal{R}}_{j-1}} p_n(u - a_{j-1}) du = \Pr[r \notin \overline{\mathcal{R}}_{j-1} \mid a = a_{j-1}].$$

Iterating, all the conditional probabilities are equal to one another; since $P_s = \frac{1}{M} \sum_j \Pr[\cdot \mid a = a_j]$ is their average, it coincides with the common value. $\square$

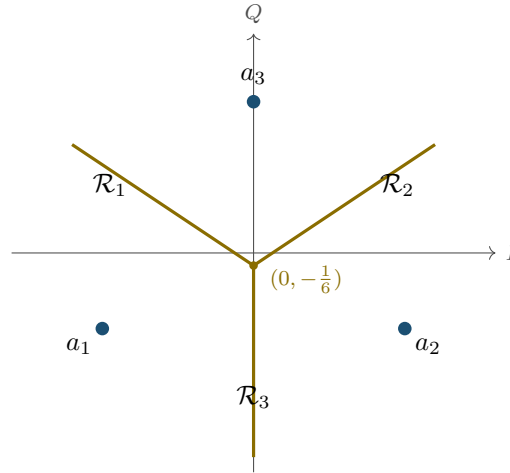


Figure 26.5: The decision regions of Theorem 26.7 for the three-point constellation $a_1 = -2 - i$, $a_2 = 2 - i$, $a_3 = 2i$ (units: centimetres). Each boundary is the perpendicular bisector of a segment, i.e. the edge of the half-plane in the statement; the three bisectors meet at the circumcentre of the triangle, which lies on $I = 0$ by symmetry and has ordinate determined by $4 + (y+1)^2 = (y-2)^2$, i.e. $6y = -1$, $y = -\frac{1}{6} = -0.1667$. The region $\mathcal{R}_3$ is the intersection of two half-planes and is an unbounded angular sector; the other two are obtained by reflection. No region is bounded: this is the general case for points lying on the convex hull of the constellation.

Remark 26.7: What happens if the symbols are not equiprobable. Point (i) of Theorem 26.6 is the only one using equiprobability, and it uses it essentially. If $\Pr[a = a_j] = \pi_j$ with the π_j not all equal, the optimal rule is still MAP, which maximises $\pi_j e^{-|r-a_j|^2/N_0}$; taking the (strictly increasing) logarithm and changing sign, it minimises

$$|r - a_j|^2 - N_0 \ln \pi_j,$$

i.e. a distance corrected by a bias term that favours the more probable symbols. The boundaries remain lines — the additional term is constant with respect to r — but they no longer pass through the midpoints: they shift towards the less probable symbol by an amount $\frac{N_0 \ln(\pi_j/\pi_l)}{2|a_j - a_l|}$. The regions are still Voronoi cells, but weighted. In classical modulations the symbols are made equiprobable by construction (by the source encoder and a scrambler), and this case does not arise; it does arise, instead, in probabilistic shaping techniques, in which the inequality between the probabilities is introduced on purpose.

26.6. The matched filter

Before deciding one must measure, and the way of measuring is not indifferent. The useful signal occupies an interval of time and a bandwidth; the noise occupies everything. A linear filter that collects the observation into a single number can be chosen in infinitely many ways, and the choice changes the signal-to-noise ratio of that number by several decibels. This section proves that an optimal choice exists, that it is unique up to a multiplicative constant, and that the signal-to-noise ratio it attains depends only on the *energy* of the signal — not on its shape.

Definition 26.10: White Gaussian noise as a family of integrals. Let $N_0 > 0$. A **white Gaussian noise** of two-sided spectral density $N_0/2$ is a family $\{\mathcal{N}(g)\}_{g \in L^2(\mathbb{R}, \mathbb{R})}$ of real random variables, defined on the same probability space, such that:

(N1) $g \mapsto \mathcal{N}(g)$ is linear: $\mathcal{N}(\lambda g + \mu h) = \lambda \mathcal{N}(g) + \mu \mathcal{N}(h)$ almost surely, for every $\lambda, \mu \in \mathbb{R}$;
(N2) each $\mathcal{N}(g)$ is Gaussian with zero mean;
(N3) $\mathbb{E}[\mathcal{N}(g)\mathcal{N}(h)] = \frac{N_0}{2} \langle g, h \rangle = \frac{N_0}{2} \int_{\mathbb{R}} g(t)h(t) dt$.
One suggestively writes $\mathcal{N}(g) = \int_{\mathbb{R}} g(t) w(t) dt$, but the notation is a *name*: no function w realises those integrals, and the well-defined object is the family, not its integrand.

Remark 26.8: What is imported. The **existence** of a family with properties (N1)–(N3) is **imported** and not proved in this treatise. What is missing, precisely, is the construction of a Gaussian measure on a probability space that simultaneously realises all the $\mathcal{N}(g)$ for $g \in L^2$: it is obtained with Kolmogorov’s extension theorem applied to the coherent family of multivariate Gaussians with covariance matrix $\frac{N_0}{2}(\langle g_i, g_j \rangle)_{i,j}$ (which is positive semidefinite because it is a Gram matrix), or by completing the isometry $g \mapsto \mathcal{N}(g)$ from a countable orthonormal system. Both routes require measure-theoretic tools that the treatise does not develop. It must be stressed that nothing in what follows uses anything other than (N1)–(N3): Theorem 26.8 is proved starting from those axioms, and is therefore valid for any realisation of them. The model and its consequences are found in Wozencraft and Jacobs and in Proakis and Salehi (Sources [4] and [1]).

Lemma 26.3: Cauchy–Schwarz inequality with the equality case. Let $u, v \in L^2(\mathbb{R}, \mathbb{R})$. Then

$$\left(\int_{\mathbb{R}} u(t)v(t) dt \right)^2 \leq \left(\int_{\mathbb{R}} u(t)^2 dt \right) \left(\int_{\mathbb{R}} v(t)^2 dt \right),$$

i.e. $\langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2$. Equality holds if and only if u and v are proportional, i.e. if and only if there exists $\lambda \in \mathbb{R}$ with $u = \lambda v$ almost everywhere, or $v = 0$ almost everywhere.

Proof. If $\|v\| = 0$ then $v = 0$ almost everywhere, both sides vanish and equality holds: this is the degenerate case allowed by the statement. Suppose therefore $\|v\| > 0$ and consider, for $\lambda \in \mathbb{R}$,

$$q(\lambda) = \|u - \lambda v\|^2 = \int_{\mathbb{R}} (u(t) - \lambda v(t))^2 dt = \|u\|^2 - 2\lambda \langle u, v \rangle + \lambda^2 \|v\|^2,$$

expanding the square and using linearity of the integral (all integrals are finite because $u, v \in L^2$ and, by the inequality $2|uv| \leq u^2 + v^2$, also $uv \in L^1$). The function q is a second-degree polynomial in λ with leading coefficient $\|v\|^2 > 0$, and $q(\lambda) \geq 0$ for every λ , because it is the integral of a square. A second-degree polynomial with positive leading coefficient is nonnegative on all of $\mathbb{R}$ if and only if its discriminant is ≤ 0 :

$$\Delta = 4\langle u, v \rangle^2 - 4\|u\|^2 \|v\|^2 \leq 0 \iff \langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2.$$

This is the inequality.

Equality case. $\langle u, v \rangle^2 = \|u\|^2 \|v\|^2$ holds if and only if $\Delta = 0$, i.e. if and only if the polynomial q has a real (double) root $\lambda_0 = \langle u, v \rangle / \|v\|^2$. But $q(\lambda_0) = \|u - \lambda_0 v\|^2 = 0$ means $\int (u - \lambda_0 v)^2 = 0$; the integrand is nonnegative, hence it vanishes almost everywhere (if a set of positive measure had $(u - \lambda_0 v)^2 > 0$, by continuity from above there would exist $\varepsilon > 0$ and a set of positive measure on which $(u - \lambda_0 v)^2 \geq \varepsilon$, and the integral would be positive). Hence $u = \lambda_0 v$ almost everywhere. Conversely, if $u = \lambda v$ almost everywhere then $\langle u, v \rangle = \lambda \|v\|^2$ and $\|u\|^2 = \lambda^2 \|v\|^2$, and both sides equal $\lambda^2 \|v\|^4$. $\square$

Definition 26.11: Linear filter, sampling and signal-to-noise ratio. Let $s \in L^2(\mathbb{R}, \mathbb{R})$ be a pulse of energy $E = \|s\|^2 > 0$, transmitted in the presence of white noise (Definition 26.10), and let $h \in L^2(\mathbb{R}, \mathbb{R})$ be the impulse response of a linear filter, with $g(\tau) = h(T - \tau)$ not identically zero. The **output sampled at instant T** is the random variable

$$y = \underbrace{\int_{\mathbb{R}} h(T - \tau) s(\tau) d\tau}_{y_s = \langle s, g \rangle} + \mathcal{N}(g),$$

and its **signal-to-noise ratio** is

$$\text{SNR}(h) = \frac{y_s^2}{\text{Var}[\mathcal{N}(g)]} = \frac{\langle s, g \rangle^2}{\frac{N_0}{2} \|g\|^2}.$$

The filter is called **matched** to s (at instant T) if $h(t) = c s(T - t)$ for some constant $c \neq 0$.

Theorem 26.8: The matched filter maximises the signal-to-noise ratio. With the notation of Definition 26.11, for every admissible filter

$$\text{SNR}(h) \leq \frac{2E}{N_0},$$

and equality holds **if and only if** h is matched to s , i.e. $h(t) = c s(T - t)$ almost everywhere for some $c \neq 0$. The maximum therefore depends only on the energy of the pulse and not on its shape.

Proof. Step 1 — the variance of the noise term. By property (N3) of Definition 26.10 with $h = g$, and since $\mathcal{N}(g)$ has zero mean by (N2),

$$\text{Var}[\mathcal{N}(g)] = \mathbb{E}[\mathcal{N}(g)^2] = \frac{N_0}{2} \|g\|^2 > 0,$$

positive because g is not identically zero. The ratio of Definition 26.11 is therefore well defined.

Step 2 — the bound. We apply Lemma 26.3 to the pair (s, g) :

$$\langle s, g \rangle^2 \leq \|s\|^2 \|g\|^2 = E \|g\|^2.$$

Dividing by the positive quantity $\frac{N_0}{2} \|g\|^2$,

$$\text{SNR}(h) = \frac{\langle s, g \rangle^2}{\frac{N_0}{2} \|g\|^2} \leq \frac{E \|g\|^2}{\frac{N_0}{2} \|g\|^2} = \frac{2E}{N_0},$$

where $\|g\|^2$ cancels because it is nonzero. Note that the bound is *independent of g* : whatever filter is chosen, the ratio does not exceed $2E/N_0$.

Step 3 — the equality case. The inequality of Step 2 is an equality if and only if the Cauchy–Schwarz inequality applied to (s, g) is. By the equality case of Lemma 26.3, and since $g \neq 0$ by hypothesis, this occurs if and only if there exists $c \in \mathbb{R}$ with

$$s = \frac{1}{c} g \quad \text{almost everywhere, i.e.} \quad g(\tau) = c s(\tau) \quad \text{almost everywhere,}$$

with $c \neq 0$ (if $c = 0$ then $g = 0$, excluded). Recalling that $g(\tau) = h(T - \tau)$, the condition is rewritten $h(T - \tau) = c s(\tau)$, i.e. setting $t = T - \tau$ and hence $\tau = T - t$,

$$h(t) = c s(T - t) \quad \text{almost everywhere,}$$

which is exactly the definition of a matched filter. Finally, by direct check, a matched filter attains the maximum: with $g = cs$, $\langle s, g \rangle = c\|s\|^2 = cE$ and $\|g\|^2 = c^2E$, hence

$$\text{SNR} = \frac{c^2 E^2}{\frac{N_0}{2} c^2 E} = \frac{2E}{N_0}. \quad \checkmark$$

□

Corollary 26.5: *The matched filter is a correlator, and its output is the AWGN channel.* Let $p \in L^2(\mathbb{R}, \mathbb{R})$ be a pulse of unit energy, $\|p\|^2 = 1$, and transmit in complex baseband the signal $ap(t)$ with $a \in \mathcal{A}$. Let the noise be the sum $w = w_I + iw_Q$ of two independent white noises of two-sided density $N_0/2$ each. If the receiver computes

$$z = \int_{\mathbb{R}} (ap(t) + w(t)) p(t) dt = a + (\mathcal{N}_I(p) + i\mathcal{N}_Q(p)),$$

then $z = a + n$ with n circularly symmetric complex Gaussian of variance N_0 : i.e. exactly the channel of Definition 26.8. Moreover the value z is obtained from the filter matched to p sampled at $t = 0$, and it attains for each component the ratio $2E/N_0$ of Theorem 26.8.

Proof. The deterministic part gives $\int ap(t)p(t) dt = a\|p\|^2 = a$. The two random parts are $\mathcal{N}_I(p)$ and $\mathcal{N}_Q(p)$: by (N2) they are Gaussian with zero mean, by (N3) they have variance $\frac{N_0}{2}\|p\|^2 = \frac{N_0}{2}$, and they are independent by hypothesis. The joint density of $n = \mathcal{N}_I(p) + i\mathcal{N}_Q(p)$ is therefore the product of the two Gaussian densities, i.e.

$$\frac{1}{\sqrt{2\pi\frac{N_0}{2}}} e^{-x^2/N_0} \cdot \frac{1}{\sqrt{2\pi\frac{N_0}{2}}} e^{-y^2/N_0} = \frac{1}{\pi N_0} e^{-(x^2+y^2)/N_0},$$

which is the density of Definition 26.8; and $\mathbb{E}[|n|^2] = \frac{N_0}{2} + \frac{N_0}{2} = N_0$. That this is the matched filter is Definition 26.11 with $T = 0$ and $c = 1$: there $h(t) = s(-t)$ and the sampled output is $\int h(-\tau)s(\tau)d\tau = \int s(\tau)^2 d\tau$; here p has energy 1 and the computation is the same, component by component. □

Example 26.5: Rectangular pulse: the integrator and the cost of the wrong filter.

Let $s(t) = A$ for $t \in [0, T]$ and $s(t) = 0$ elsewhere, with $A = 0.5$ V and $T = 2 \cdot 10^{-6}$ s. The energy is

$$E = \int_0^T A^2 dt = A^2 T = 0.25 \cdot 2 \cdot 10^{-6} = 5.0000000 \cdot 10^{-7} \text{ V}^2\text{s}.$$

The matched filter has $h(t) = s(T - t)$, which for a rectangle coincides with s itself: the filter matched to a rectangular pulse is the *integrator* over the symbol interval. With $N_0 = 5 \cdot 10^{-8}$ V²/Hz,

$$\text{SNR}_{\max} = \frac{2E}{N_0} = \frac{2 \cdot 5 \cdot 10^{-7}}{5 \cdot 10^{-8}} = 20.0000000, \quad 10 \log_{10} 20 = 13.0102999 \text{ dB}.$$

The cost of the wrong filter. Suppose one mistakenly integrates only over the first half of the interval, i.e. takes $g = \mathbf{1}_{[0, T/2]}$. Then

$$\langle s, g \rangle = A \cdot \frac{T}{2}, \quad \|g\|^2 = \frac{T}{2},$$

and by Definition 26.11

$$\text{SNR} = \frac{A^2 T^2 / 4}{\frac{N_0}{2} \cdot \frac{T}{2}} = \frac{A^2 T}{N_0} = \frac{E}{N_0} = 10.0000000,$$

exactly half of the maximum: one loses $10 \log_{10} 2 = 3.0102999$ dB. The loss can also be read directly from Lemma 26.3, whose ratio $\langle s, g \rangle^2 / (\|s\|^2 \|g\|^2)$ here equals

$$\frac{(AT/2)^2}{(A^2T)(T/2)} = \frac{A^2T^2/4}{A^2T^2/2} = \frac{1}{2},$$

i.e. the squared cosine of the angle between s and g in L^2 : g "sees" only half of the pulse, and collects half its energy.

Independent check. The two ratios equal $2E/N_0 = 20$ and $E/N_0 = 10$: their quotient is 2, and $10 \log_{10} 20 - 10 \log_{10} 10 = 13.0102999 - 10 = 3.0102999$, which coincides with $10 \log_{10} 2$.
✓

Remark 26.9: Why the maximum does not depend on the shape of the pulse. Theorem 26.8 says that $2E/N_0$ is attainable with any pulse of energy E : rectangular, Gaussian, raised-cosine, or of arbitrarily irregular shape. The shape of the pulse therefore does not serve to improve the signal-to-noise ratio, but to control two different things: the bandwidth occupation (Section 26.8) and the interference between consecutive symbols (Theorem 26.11). This is why no real system uses the rectangular pulse of Example 26.5, even though it is optimal from the point of view of the signal-to-noise ratio: its spectrum is a sinc function (Proposition 19.2), which decays only as $1/|f|$ and occupies bandwidth unacceptably.

26.7. The error probability of BPSK and QPSK

Now the errors are counted. The whole section rests on a single function, the tail of the standard Gaussian, and on two elementary geometries: a line separating two points (BPSK) and two orthogonal lines separating four points (QPSK). The final result — that BPSK and QPSK have the *same* error probability per bit — is the reason why, in real systems, QPSK is almost always preferred: it doubles the rate at no cost.

Definition 26.12: The Q function. The Q function is the right tail of the standard Gaussian density:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{+\infty} e^{-u^2/2} du, \quad x \in \mathbb{R}.$$

If Z is a Gaussian random variable with zero mean and variance 1, then $Q(x) = \Pr[Z > x]$.

Proposition 26.6: Elementary properties of the Q function. The following hold:

- (i) Q is well defined, $Q(-\infty) = 1$, $Q(+\infty) = 0$, and it is strictly decreasing on all of $\mathbb{R}$;
- (ii) $Q(0) = \frac{1}{2}$ and $Q(-x) = 1 - Q(x)$ for every x ;
- (iii) if X is Gaussian with mean μ and variance $\sigma^2 > 0$, then $\Pr[X > c] = Q\left(\frac{c-\mu}{\sigma}\right)$;
- (iv) $Q(x) = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$, where $\operatorname{erfc}(y) = \frac{2}{\sqrt{\pi}} \int_y^{+\infty} e^{-v^2} dv$.

Proof. (i). The integrand is positive and integrable on $\mathbb{R}$: for $|u| \geq 1$, $u^2 \geq |u|$ and hence $e^{-u^2/2} \leq e^{-|u|/2}$, which is integrable; on $[-1, 1]$ it is continuous. The function Q is therefore finite for every x , tends to $\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-u^2/2} du$ as $x \rightarrow -\infty$ and to 0 as $x \rightarrow +\infty$ (dominated convergence theorem, or simply additivity of the integral). By the fundamental theorem of calculus $Q'(x) = -\frac{1}{\sqrt{2\pi}} e^{-x^2/2} < 0$ for every x : Q is strictly decreasing.

That $Q(-\infty) = 1$ follows from the normalisation $\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-u^2/2} du = 1$, obtained from Lemma 19.4 — according to which $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$ — with the substitution $u = t\sqrt{2\pi}$,

$du = \sqrt{2\pi} dt$:

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-u^2/2} du = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\pi t^2} \sqrt{2\pi} dt = \int_{\mathbb{R}} e^{-\pi t^2} dt = 1,$$

having used $u^2/2 = (t\sqrt{2\pi})^2/2 = \pi t^2$.

(ii). The integrand $e^{-u^2/2}$ is even, hence $\int_{-\infty}^{-x} e^{-u^2/2} du = \int_x^{+\infty} e^{-u^2/2} du$ (substitution $u \mapsto -u$). Therefore

$$Q(-x) = \frac{1}{\sqrt{2\pi}} \int_{-x}^{+\infty} = \frac{1}{\sqrt{2\pi}} \left(\int_{\mathbb{R}} - \int_{-\infty}^{-x} \right) = 1 - \frac{1}{\sqrt{2\pi}} \int_x^{+\infty} = 1 - Q(x).$$

Setting $x = 0$: $Q(0) = 1 - Q(0)$, i.e. $Q(0) = \frac{1}{2}$.

(iii). If X is Gaussian (μ, σ^2) , then $Z = (X - \mu)/\sigma$ is standard Gaussian, and $X > c \iff Z > (c - \mu)/\sigma$; Definition 26.12 applies.

(iv). In the definition of Q set $u = v\sqrt{2}$, $du = \sqrt{2} dv$: when $u = x$, $v = x/\sqrt{2}$, and $u^2/2 = v^2$. Hence

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x/\sqrt{2}}^{+\infty} e^{-v^2} \sqrt{2} dv = \frac{1}{\sqrt{\pi}} \int_{x/\sqrt{2}}^{+\infty} e^{-v^2} dv = \frac{1}{2} \cdot \frac{2}{\sqrt{\pi}} \int_{x/\sqrt{2}}^{+\infty} e^{-v^2} dv = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right).$$

□

Theorem 26.9: Error probability of BPSK. Transmit a BPSK of energy per bit E_b (Definition 26.4 with $M = 2$, $\varphi_0 = 0$, $E_s = E_b$) over the AWGN channel of Definition 26.8, with minimum-distance decision and equiprobable symbols. Then the error probability per bit — which coincides with that per symbol, because $m = 1$ — equals exactly

$$P_b^{\text{BPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

Proof. Step 1 — the rule reduces to the sign of the real part. The two symbols are $a_0 = +\sqrt{E_b}$ and $a_1 = -\sqrt{E_b}$, both real. By Theorem 26.7 the boundary between the two regions is the locus $2\Re[r(a_1 - a_0)] = |a_1|^2 - |a_0|^2$, i.e. since $a_1 - a_0 = -2\sqrt{E_b}$ is real and $|a_1|^2 = |a_0|^2 = E_b$,

$$-4\sqrt{E_b} \Re r = 0 \iff \Re r = 0.$$

The region of a_0 is therefore the half-plane $\Re r > 0$ and that of a_1 the half-plane $\Re r < 0$: the decision is made by looking at the *sign of the real part* of r , and the imaginary part plays no role at all in the decision.

Step 2 — the law of the real part. Suppose $a_0 = +\sqrt{E_b}$ was transmitted. Then $r = \sqrt{E_b} + n$ and, by Definition 26.8,

$$\Re r = \sqrt{E_b} + \nu, \quad \nu = \Re n \text{ Gaussian with mean 0 and variance } \frac{N_0}{2}.$$

Hence $\Re r$ is Gaussian with mean $\sqrt{E_b}$ and variance $N_0/2$.

Step 3 — the computation. An error occurs if $\Re r < 0$. By Proposition 26.6(iii), applied to $-\Re r$, which is Gaussian with mean $-\sqrt{E_b}$ and variance $N_0/2$, and with threshold $c = 0$:

$$\Pr[\Re r < 0 \mid a = a_0] = \Pr[-\Re r > 0] = Q\left(\frac{0 - (-\sqrt{E_b})}{\sqrt{N_0/2}}\right) = Q\left(\frac{\sqrt{E_b}}{\sqrt{N_0/2}}\right).$$

The ratio under Q simplifies:

$$\frac{\sqrt{E_b}}{\sqrt{N_0/2}} = \sqrt{\frac{E_b}{N_0/2}} = \sqrt{\frac{2E_b}{N_0}}.$$

The tie event $\Re r = 0$ has probability zero (Theorem 26.6(iv)) and does not contribute.

Step 4 — the other symbol, and the conclusion. If $a_1 = -\sqrt{E_b}$ was transmitted, then $\Re r$ is Gaussian with mean $-\sqrt{E_b}$ and variance $N_0/2$, and an error occurs if $\Re r > 0$: by the same proposition the probability is $Q(\frac{0+\sqrt{E_b}}{\sqrt{N_0/2}})$, the same number. Averaging over the two equiprobable symbols gives the same value. Finally $m = \log_2 2 = 1$: each symbol is one bit, and the symbol error probability is the bit error probability. $\square$

Theorem 26.10: Error probability of QPSK. Transmit a QPSK of energy per symbol $E_s = 2E_b$ (Definition 26.4 with $M = 4$, $\varphi_0 = \pi/4$) over the AWGN channel, with minimum-distance decision, equiprobable symbols and a Gray mapping in which the first bit determines the sign of the in-phase component and the second that of the quadrature component. Setting for brevity

$$q = Q\left(\sqrt{\frac{2E_b}{N_0}}\right),$$

the following hold exactly:

$$P_s^{\text{QPSK}} = 1 - (1 - q)^2 = 2q - q^2, \quad P_b^{\text{QPSK}} = q = P_b^{\text{BPSK}}.$$

Proof. Step 1 — the regions are the four quadrants. By Proposition 26.4 the four symbols are $a = \sqrt{E_b}(\varepsilon_1 + i\varepsilon_2)$ with $\varepsilon_1, \varepsilon_2 \in \{\pm 1\}$, having used $\sqrt{E_s/2} = \sqrt{E_b}$. We apply Theorem 26.7 to the symbol $a_{00} = \sqrt{E_b}(1 + i)$. The three boundaries are obtained from the other three symbols; since all four have the same modulus $\sqrt{2E_b}$, the right-hand side $|a_l|^2 - |a_j|^2$ is always zero, and the conditions become $\Re[r(a_l - a_j)] \leq 0$. With $r = x + iy$:

- $a_l = \sqrt{E_b}(-1 + i)$: $a_l - a_j = -2\sqrt{E_b}$, condition $-2\sqrt{E_b}x \leq 0$, i.e. $x \geq 0$;
- $a_l = \sqrt{E_b}(1 - i)$: $a_l - a_j = -2i\sqrt{E_b}$, $\overline{a_l - a_j} = 2i\sqrt{E_b}$, and $\Re[(x + iy) \cdot 2i\sqrt{E_b}] = -2\sqrt{E_b}y \leq 0$, i.e. $y \geq 0$;
- $a_l = \sqrt{E_b}(-1 - i)$: $a_l - a_j = -2\sqrt{E_b}(1 + i)$, and $\Re[(x + iy) \cdot (-2\sqrt{E_b})(1 - i)] = -2\sqrt{E_b}(x + y) \leq 0$, i.e. $x + y \geq 0$, which is implied by the previous two.

The region of a_{00} is therefore the closed first quadrant $\{x \geq 0, y \geq 0\}$, and by symmetry the other three are the other three quadrants. The decision therefore reduces to *two independent sign decisions*: the sign of $\Re r$ and the sign of $\Im r$.

Step 2 — the two components are independent BPSKs. If $a = \sqrt{E_b}(\varepsilon_1 + i\varepsilon_2)$ was transmitted, then

$$\Re r = \varepsilon_1 \sqrt{E_b} + \Re n, \quad \Im r = \varepsilon_2 \sqrt{E_b} + \Im n,$$

and by Definition 26.8 $\Re n$ and $\Im n$ are *independent*, Gaussian with zero mean and variance $N_0/2$. Each of the two components is therefore, word for word, the situation of Step 2 of the proof of Theorem 26.9 with energy E_b : the probability that the sign of $\Re r$ is wrong equals $q = Q(\sqrt{2E_b/N_0})$, and the same for $\Im r$.

Step 3 — symbol error. The decision is correct if and only if *both* signs are correct. The two events are independent (because $\Re n$ and $\Im n$ are, and each event depends on only one of the two), hence

$$\Pr[\text{correct symbol}] = (1 - q)(1 - q) = (1 - q)^2,$$

and $P_s = 1 - (1 - q)^2 = 1 - 1 + 2q - q^2 = 2q - q^2$. By Corollary 26.4 the value does not depend on the transmitted symbol, and coincides with the symbol error probability.

Step 4 — bit error. With the mapping described, the first bit is determined by the sign of $\Re r$ alone and the second by the sign of $\Im r$ alone: this is a Gray mapping (Definition 26.5), because adjacent symbols — which by Theorem 26.3 are those differing in a single component — have labels differing in a single bit. Each bit is therefore wrong with probability q , and the

bit error probability, defined as the expected value of the fraction of wrong bits, equals

$$P_b = \frac{1}{2} \left(\Pr[\text{first bit wrong}] + \Pr[\text{second bit wrong}] \right) = \frac{q + q}{2} = q.$$

Comparing with Theorem 26.9, $P_b^{\text{QPSK}} = P_b^{\text{BPSK}}$: this is the probabilistic translation of Corollary 26.3, according to which the two constellations have the same minimum distance at equal energy per bit. $\square$

Example 26.6: The numbers, computed twice. The table gives $P_b = Q(\sqrt{2E_b/N_0})$ and $P_s^{\text{QPSK}} = 2P_b - P_b^2$ for a few values of E_b/N_0 . The second column is $10^{\text{dB}/10}$, the third the argument of the Q function.

E_b/N_0 (dB)	E_b/N_0	$\sqrt{2E_b/N_0}$	P_b (BPSK and QPSK)	P_s (QPSK)
0.0	1.0000000	1.4142136	$7.8649604 \cdot 10^{-2}$	$1.5111345 \cdot 10^{-1}$
2.0	1.5848932	1.7803894	$3.7506128 \cdot 10^{-2}$	$7.3605547 \cdot 10^{-2}$
4.0	2.5118864	2.2413774	$1.2500818 \cdot 10^{-2}$	$2.4845366 \cdot 10^{-2}$
6.0	3.9810717	2.8217270	$2.3882908 \cdot 10^{-3}$	$4.7708776 \cdot 10^{-3}$
8.0	6.3095734	3.5523439	$1.9090777 \cdot 10^{-4}$	$3.8177910 \cdot 10^{-4}$
9.6	9.1201084	4.2708567	$9.7361760 \cdot 10^{-6}$	$1.9472257 \cdot 10^{-5}$
10.0	10.0000000	4.4721360	$3.8721082 \cdot 10^{-6}$	$7.7442014 \cdot 10^{-6}$
12.0	15.8489319	5.6300856	$9.0060104 \cdot 10^{-9}$	$1.8012021 \cdot 10^{-8}$

First route: the closed form. The values of P_b are obtained from the relation $Q(x) = \frac{1}{2} \text{erfc}(x/\sqrt{2})$ (Proposition 26.6(iv)) with the `erfc` function of the standard library, which implements its rational approximation; for instance at 10 dB the argument is $\sqrt{20} = 4.4721360$ and $\frac{1}{2} \text{erfc}(4.4721360/\sqrt{2}) = \frac{1}{2} \text{erfc}(3.1622777) = 3.8721082 \cdot 10^{-6}$.

Second, independent route: numerical integration. A Python script directly integrates $\frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ from x to $x + 40$ with Simpson's rule over 200,000 subintervals, without using `erfc`. The relative discrepancies from the first route are at most $4.7 \cdot 10^{-14}$ over all eight values. $\checkmark$

Third, internal-consistency check. By Theorem 26.10 it must hold that $P_s = 1 - (1 - P_b)^2$: at 6 dB the direct computation gives $4.7708776 \cdot 10^{-3}$ and the form $2P_b - P_b^2$ gives the same number, with zero discrepancy to the first twelve digits; at 10 dB the discrepancy is $3.5 \cdot 10^{-20}$, i.e. pure machine rounding. $\checkmark$

Reading. The row at 9.6 dB is the one encountered most often in designs: $E_b/N_0 = 9.6$ dB is, to good approximation, the value that brings uncoded BPSK to an error probability of about 10^{-5} . Between 8 and 12 dB the probability drops by more than four orders of magnitude: the curve is extremely steep, and this is why in radio one speaks of a "threshold".

Remark 26.10: The union bound, and why it is not a theorem of this chapter.

For constellations other than BPSK and QPSK, the exact error probability generally has no elementary closed form, and one uses the **union bound**: since the event "error" is contained in the union of the events " r is closer to a_l than to a_j " as $l \neq j$ ranges, and each of these has probability $Q(|a_j - a_l|/\sqrt{2N_0})$ — this is the computation of Theorem 26.9 applied to the projection along the direction $a_l - a_j$ — subadditivity of probability gives

$$P_s \leq \frac{1}{M} \sum_j \sum_{l \neq j} Q\left(\frac{|a_j - a_l|}{\sqrt{2N_0}}\right) \leq (M-1) Q\left(\frac{d_{\min}}{\sqrt{2N_0}}\right),$$

and the estimate is better the smaller P_s is. This is a bound, not an equality, and this is why we report it in a remark and not in a theorem: its complete proof would require carefully treating the projection of complex Gaussian noise onto an arbitrary direction, which is not

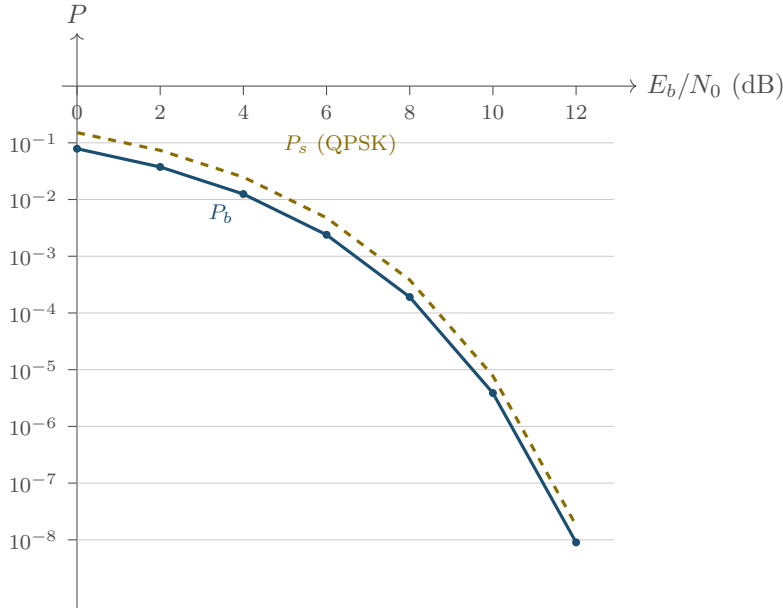


Figure 26.6: The two curves of Example 26.6, drawn from the tabulated values and not from a measurement. The horizontal scale is 0.55 cm per decibel and the vertical one 0.75 cm per decade: the ordinate of a point is $0.75 \log_{10} P$. For instance at 10 dB, $P_b = 3.8721082 \cdot 10^{-6}$, $\log_{10} P_b = -5.41205$ and ordinate -4.0590 ; at 12 dB, $\log_{10} P_b = -8.04547$ and ordinate -6.0341 . The solid curve holds both for BPSK and for the *bit* probability of QPSK (Theorem 26.10): the two coincide, and this is the graphical content of Corollary 26.3. The dashed curve is the *symbol* error probability of QPSK, which lies slightly above because $P_s = 2P_b - P_b^2 \approx 2P_b$ when P_b is small: a discrepancy of $10 \log_{10} 2 = 3.01$ dB in ordinate, not in abscissa.

needed here. For M -PSK, substituting $d_{\min} = 2\sqrt{E_s} \sin(\pi/M)$ from Theorem 26.3 and $E_s = mE_b$, the argument of Q becomes

$$\frac{2\sqrt{mE_b} \sin(\pi/M)}{\sqrt{2N_0}} = \sqrt{\frac{2mE_b}{N_0}} \sin \frac{\pi}{M},$$

a formula that reproduces exactly the results proved for $M = 2$ and $M = 4$ (where $m \sin^2(\pi/M) = 1$ in both cases, as shown by the last column of the table of Example 26.3), and which for $M \geq 8$ remains a bound.

Remark 26.11: Panorama: how far one can go. The curves of Figure 26.6 concern uncoded transmissions, in which each bit travels alone. Shannon's channel capacity theorem (1948) states that on the AWGN channel of bandwidth W and signal-to-noise ratio S/N there exists, for every transmission rate below $C = W \log_2(1 + S/N)$, a sequence of codes that makes the error probability arbitrarily small — and that no code can do so for higher rates. In the infinite-bandwidth limit this translates into the minimum value $E_b/N_0 = \ln 2 = 0.6931472$, i.e. -1.5917 dB, below which no reliable communication is possible. Between the 9.6 dB of uncoded BPSK at 10^{-5} and this limit there are about 11 dB: this is the space that error-correcting codes have filled in seventy years. Shannon's theorem is **not** proved in this treatise: its proof requires information theory (entropy, mutual information, the asymptotic equipartition property) and the central limit theorem, which are not developed here. We report it as a panorama, not as an available result: no proof in the chapter uses it.

26.8. Minimum distance, bandwidth and spectral efficiency

It remains to link the two quantities the designer cannot optimise simultaneously: how many bits per second pass through each occupied hertz, and how far apart the constellation points remain. The link runs through the waveform, and more precisely through the condition that the pulse not interfere with neighbouring symbols.

Definition 26.13: Transmit pulse and intersymbol interference. Transmit in complex baseband the signal $x(t) = \sum_{k \in \mathbb{Z}} a_k p(t - kT)$, with $a_k \in \mathcal{A}$ and p real continuous.

Sampling at instant $k_0 T$ gives

$$x(k_0 T) = a_{k_0} p(0) + \underbrace{\sum_{k \neq k_0} a_k p((k_0 - k)T)}_{\text{intersymbol interference}}.$$

The pulse p is called a **Nyquist pulse** for step T if $p(kT) = \delta_{k0}$ for every $k \in \mathbb{Z}$ (i.e. $p(0) = 1$ and $p(kT) = 0$ for $k \neq 0$): in that case the intersymbol interference is zero and $x(k_0 T) = a_{k_0}$.

Theorem 26.11: Nyquist criterion for the absence of intersymbol interference.

Let $p: \mathbb{R} \rightarrow \mathbb{R}$ be continuous, with transform $P = \hat{p}$, and suppose there exist $C > 0$ and $\varepsilon > 0$ such that

$$|p(t)| \leq \frac{C}{(1 + |t|)^{1+\varepsilon}} \quad \text{and} \quad |P(f)| \leq \frac{C}{(1 + |f|)^{1+\varepsilon}} \quad \text{for every } t, f \in \mathbb{R}.$$

Then, for every $T > 0$ and every $f \in \mathbb{R}$,

$$\sum_{k \in \mathbb{Z}} p(kT) e^{-i2\pi f k T} = \frac{1}{T} \sum_{n \in \mathbb{Z}} P\left(f + \frac{n}{T}\right),$$

and p is a Nyquist pulse for step T **if and only if**

$$\sum_{n \in \mathbb{Z}} P\left(f + \frac{n}{T}\right) = T \quad \text{for every } f \in \mathbb{R}.$$

Proof. Step 1 — reduction to the Poisson formula. Fix $f \in \mathbb{R}$ and set

$$h(u) = p(uT) e^{-i2\pi f u T}, \quad u \in \mathbb{R}.$$

The function h is continuous and satisfies hypothesis (H1) of Theorem 19.12: indeed $|h(u)| = |p(uT)| \leq C(1 + |uT|)^{-1-\varepsilon} \leq C'(1 + |u|)^{-1-\varepsilon}$ with $C' = C \max\{1, T^{-1-\varepsilon}\}$. Its transform is computed with the substitution $v = uT$, $du = dv/T$:

$$\hat{h}(\xi) = \int_{\mathbb{R}} p(uT) e^{-i2\pi f u T} e^{-i2\pi \xi u} du = \frac{1}{T} \int_{\mathbb{R}} p(v) e^{-i2\pi(f + \frac{\xi}{T})v} dv = \frac{1}{T} P\left(f + \frac{\xi}{T}\right),$$

and satisfies (H2) for the same reason, with constant $C/T \cdot \max\{1, T^{1+\varepsilon}\}$.

Step 2 — application. Theorem 19.12 with $t = 0$ gives $\sum_n h(n) = \sum_n \hat{h}(n)$, i.e.

$$\sum_{n \in \mathbb{Z}} p(nT) e^{-i2\pi f n T} = \frac{1}{T} \sum_{n \in \mathbb{Z}} P\left(f + \frac{n}{T}\right),$$

which is the first formula of the statement (with k in place of n); both series converge absolutely and uniformly in f .

Step 3 — the equivalence. The left-hand side, $S(f) = \sum_k p(kT)e^{-i2\pi f kT}$, is a continuous function periodic of period $1/T$ in f , and the numbers $p(kT)$ are *exactly* its Fourier coefficients: indeed, by uniform convergence one can integrate term by term and, by orthogonality of the exponentials over one period,

$$T \int_0^{1/T} S(f) e^{+i2\pi f k_0 T} df = T \sum_k p(kT) \int_0^{1/T} e^{-i2\pi f(k-k_0)T} df = p(k_0 T),$$

because the inner integral equals $1/T$ if $k = k_0$ and 0 otherwise.

If $p(kT) = \delta_{k0}$, then $S(f) = 1$ for every f , and by Step 2 $\sum_n P(f + n/T) = T$. Conversely, if $\sum_n P(f + n/T) = T$ for every f , then $S(f) = 1$ for every f ; the Fourier coefficients of S are therefore those of the constant function 1, i.e. $p(0) = 1$ and $p(kT) = 0$ for $k \neq 0$, by uniqueness of the Fourier coefficients of a continuous function (Corollary 17.2). This completes the equivalence. $\square$

Corollary 26.6: The minimum Nyquist bandwidth. If P vanishes outside $[-\frac{1}{2T}, \frac{1}{2T}]$ and p is a Nyquist pulse for step T , then necessarily $P = T \cdot \mathbf{1}_{[-\frac{1}{2T}, \frac{1}{2T}]}$ almost everywhere. The baseband half-bandwidth cannot therefore be smaller than $\frac{1}{2T}$, and the corresponding bandpass bandwidth (Definition 26.1) is $W = \frac{1}{T} = R_s$: one baud per hertz.

Proof. Let $|f| < \frac{1}{2T}$. For every $n \neq 0$, $|f + n/T| \geq \frac{|n|}{T} - |f| > \frac{1}{T} - \frac{1}{2T} = \frac{1}{2T}$, hence $P(f + n/T) = 0$: in the sum of Theorem 26.11 only the term $n = 0$ survives, and the condition becomes $P(f) = T$. Outside the interval P is zero by hypothesis. The value at the two endpoints does not matter, because two functions equal almost everywhere have the same inverse transform. For the last claim: the complex envelope has half-bandwidth $B = \frac{1}{2T}$ (Theorem 26.1(i)), and the bandwidth of the bandpass signal is $W = 2B = \frac{1}{T}$ (Definition 26.1). $\square$

Remark 26.12: The sinc function and the hypotheses of the theorem. The pulse achieving the limit of Corollary 26.6 is the inverse transform of $T \cdot \mathbf{1}_{[-\frac{1}{2T}, \frac{1}{2T}]}$, i.e.

$$p(t) = \text{sinc}\left(\frac{t}{T}\right) = \frac{\sin(\pi t/T)}{\pi t/T},$$

by Proposition 19.2. That it satisfies $p(kT) = \delta_{k0}$ is checked directly, without any theorem: for $k \neq 0$ the numerator $\sin(k\pi)$ vanishes and the denominator does not, and for $k = 0$ the standard limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ gives $p(0) = 1$. It must however be stated precisely that this p **does not** satisfy hypothesis (H1) of Theorem 26.11, because $|\text{sinc}(t/T)|$ decays only as $1/|t|$ and not as $(1 + |t|)^{-1-\varepsilon}$: the theorem does not apply to the sinc function, and there is no need for it to, because in that case the check is immediate. The fact is nonetheless a serious practical flaw: a tail decaying as $1/|t|$ makes the sum of intersymbol interference not absolutely convergent, and even a small error on the sampling instant produces unbounded interference. This is exactly why no real system uses the sinc pulse, accepting instead to occupy a little more bandwidth.

Proposition 26.7: *The raised-cosine pulse satisfies the Nyquist criterion. Let $T > 0$ and $\beta \in [0, 1]$ (roll-off factor). Define*

$$P_\beta(f) = \begin{cases} T, & |f| \leq \frac{1-\beta}{2T}, \\ \frac{T}{2} \left[1 + \cos\left(\frac{\pi T}{\beta} \left(|f| - \frac{1-\beta}{2T}\right)\right) \right], & \frac{1-\beta}{2T} \leq |f| \leq \frac{1+\beta}{2T}, \\ 0, & |f| \geq \frac{1+\beta}{2T} \end{cases}$$

(for $\beta = 0$ only the first case is meant). Then $\sum_{n \in \mathbb{Z}} P_\beta(f + \frac{n}{T}) = T$ for every f , and the baseband half-bandwidth occupied is $B = \frac{1+\beta}{2T}$.

Proof. Step 1 — the odd symmetry about $\frac{1}{2T}$. Let $0 \leq x \leq \frac{\beta}{2T}$. Then both $\frac{1}{2T} + x$ and $\frac{1}{2T} - x$ belong to the transition interval $[\frac{1-\beta}{2T}, \frac{1+\beta}{2T}]$. For the first,

$$\frac{\pi T}{\beta} \left(\frac{1}{2T} + x - \frac{1-\beta}{2T} \right) = \frac{\pi T}{\beta} \left(\frac{\beta}{2T} + x \right) = \frac{\pi}{2} + \frac{\pi T x}{\beta},$$

and since $\cos(\frac{\pi}{2} + \theta) = -\sin \theta$,

$$P_\beta\left(\frac{1}{2T} + x\right) = \frac{T}{2} \left[1 - \sin \frac{\pi T x}{\beta} \right].$$

For the second, with the same computation and $\cos(\frac{\pi}{2} - \theta) = \sin \theta$,

$$P_\beta\left(\frac{1}{2T} - x\right) = \frac{T}{2} \left[1 + \sin \frac{\pi T x}{\beta} \right].$$

Adding, the two sines cancel:

$$P_\beta\left(\frac{1}{2T} - x\right) + P_\beta\left(\frac{1}{2T} + x\right) = T.$$

Step 2 — the sum of the replicas. By periodicity of the sum in f of period $\frac{1}{T}$ it suffices to check the identity for $f \in [0, \frac{1}{2T}]$; write $f = \frac{1}{2T} - x$ with $x \in [0, \frac{1}{2T}]$. The terms with $|n| \geq 2$ vanish: indeed $|f + \frac{n}{T}| \geq \frac{2}{T} - \frac{1}{2T} = \frac{3}{2T} > \frac{1+\beta}{2T}$ because $\beta \leq 1 < 2$. What remains are $n = 0$ and $n = -1$, and

$$f - \frac{1}{T} = \frac{1}{2T} - x - \frac{1}{T} = -\left(\frac{1}{2T} + x\right),$$

whose modulus is $\frac{1}{2T} + x$. Since P_β is even,

$$\sum_n P_\beta\left(f + \frac{n}{T}\right) = P_\beta\left(\frac{1}{2T} - x\right) + P_\beta\left(\frac{1}{2T} + x\right).$$

If $x \leq \frac{\beta}{2T}$ the sum equals T by Step 1. If instead $x > \frac{\beta}{2T}$, then $\frac{1}{2T} + x > \frac{1+\beta}{2T}$ and the second term is zero, while $\frac{1}{2T} - x < \frac{1-\beta}{2T}$ and the first equals T : again the sum is T . In both cases the condition of Theorem 26.11 is satisfied.

Step 3 — the bandwidth. P_β is zero for $|f| > \frac{1+\beta}{2T}$ and nonzero just below (for $\beta < 1$, $T > 0$ holds on the flat stretch; for $\beta = 1$ the cosine equals $\frac{T}{2}(1 + \cos \theta) > 0$ for $\theta < \pi$): the half-bandwidth is therefore exactly $B = \frac{1+\beta}{2T}$. $\square$

Definition 26.14: Spectral efficiency. Let $R_b = mR_s = m/T$ be the bit rate in bit/s and W the bandwidth occupied by the bandpass signal in hertz. The **spectral efficiency** is $\eta = \frac{R_b}{W}$, measured in bits per second per hertz.

Corollary 26.7: Spectral efficiency of a modulation with a Nyquist pulse. With the raised-cosine pulse of roll-off β (Proposition 26.7) and a constellation of $M = 2^m$ points,

$$\eta = \frac{m}{1 + \beta} \quad \text{bit/s/Hz.}$$

In particular $\eta = m$ in the ideal limit $\beta = 0$, and $\eta = m/2$ for $\beta = 1$.

Proof. By Proposition 26.7 the baseband half-bandwidth is $B = \frac{1+\beta}{2T}$; by Definition 26.1 the bandwidth of the bandpass signal is $W = 2B = \frac{1+\beta}{T}$. The bit rate is $R_b = m/T$, hence

$$\eta = \frac{R_b}{W} = \frac{m/T}{(1+\beta)/T} = \frac{m}{1+\beta}.$$

□

Proposition 26.8: The price of spectral efficiency in minimum distance. At equal energy per bit E_b and equal roll-off β , square QAM of $M = 2^m$ points (m even) has efficiency $\eta = m/(1 + \beta)$ and

$$\frac{d_{\min}^2}{4E_b} = \frac{3m}{2(2^m - 1)},$$

a quantity that decreases exponentially in m : doubling the spectral efficiency requires, asymptotically, about 3 dB more energy per bit for each bit added to the symbol.

Proof. The efficiency is Corollary 26.7. By Theorem 26.4, $d_{\min}^2 = \frac{6E_s}{M-1}$, and by Definition 26.3 $E_s = mE_b$; hence

$$\frac{d_{\min}^2}{4E_b} = \frac{6mE_b}{4E_b(M-1)} = \frac{3m}{2(2^m - 1)}.$$

For the asymptotic behaviour, the ratio between two consecutive values (m and $m+1$) is

$$\frac{3(m+1)/(2(2^{m+1}-1))}{3m/(2(2^m-1))} = \frac{m+1}{m} \cdot \frac{2^m-1}{2^{m+1}-1} \xrightarrow{m \rightarrow \infty} 1 \cdot \frac{1}{2} = \frac{1}{2},$$

because $\frac{m+1}{m} \rightarrow 1$ and $\frac{2^m-1}{2^{m+1}-1} = \frac{1-2^{-m}}{2-2^{-m}} \rightarrow \frac{1}{2}$. A factor $\frac{1}{2}$ in the ratio $d_{\min}^2/(4E_b)$ must be compensated by a factor 2 in E_b to leave the argument of the Q function unchanged, i.e. by $10 \log_{10} 2 = 3.0102999$ dB. □

Example 26.7: The price, in numbers. With $\beta = 0$ (ideal limit) Proposition 26.8 gives:

M	$m = \eta$	$d_{\min}^2/(4E_b)$	in dB	discrepancy from previous row
4	2	1.0000000	0.0000000	—
16	4	0.4000000	−3.9794001	3.9794001
64	6	0.1428571	−8.4509804	4.4715803
256	8	0.0470588	−13.2735893	4.8226089

Each doubling of the points *per dimension* (i.e. each additional two bits per symbol) costs between 4 and 5 dB, and the cost tends to 6 dB, i.e. 3 dB per bit, as proved. Read the

table together with that of Example 26.6: at 10 dB of E_b/N_0 , QPSK sits at $P_b \approx 3.9 \cdot 10^{-6}$; 256-QAM, at the same E_b/N_0 , has the argument of the Q function reduced by the factor $\sqrt{0.0470588} = 0.2169$ and its error probability is of the order of 10^{-1} : four times the spectral efficiency is not free.

Independent check. The four values $\frac{3m}{2(2^m-1)}$ are recomputed by hand: $\frac{6}{2 \cdot 3} = 1$, $\frac{12}{2 \cdot 15} = \frac{2}{5} = 0.4$, $\frac{18}{2 \cdot 63} = \frac{1}{7} = 0.1428571$, $\frac{24}{2 \cdot 255} = \frac{4}{85} = 0.0470588$; the logarithms: $10 \log_{10}(2/5) = -3.9794001$, $10 \log_{10}(1/7) = -8.4509804$, $10 \log_{10}(4/85) = -13.2735893$. ✓

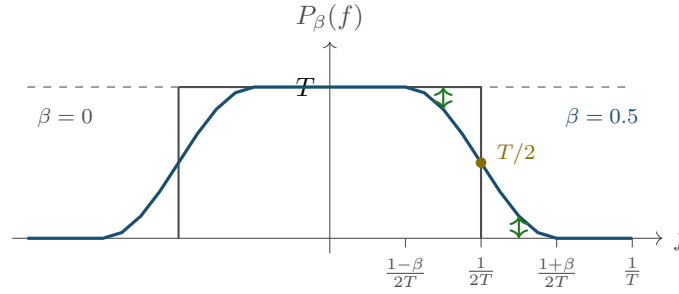


Figure 26.7: The raised-cosine spectrum of Proposition 26.7, drawn in units $1/(2T) = 2$ cm on the abscissa and $T = 2$ cm on the ordinate. In grey the ideal case $\beta = 0$, i.e. the rectangle of Corollary 26.6. In blue the case $\beta = 0.5$: the flat stretch reaches $\frac{1-\beta}{2T} = 0.25/T$ (abscissa 1 cm) and the transition ends at $\frac{1+\beta}{2T} = 0.75/T$ (abscissa 3 cm). The ordinates of the transition stretch are $2 \cdot \frac{1}{2} [1 + \cos(\pi(2Tf - 0.5))]$ computed at nine points: 2, 1.92388, 1.70711, 1.38268, 1, 0.61732, 0.29289, 0.07612, 0. The two green segments have the same length, 0.29289 cm: this is the odd symmetry about $(\frac{1}{2T}, \frac{T}{2})$ proved in Step 1, i.e. what makes the replicas sum to T .

26.9. Eye diagram and error vector

The two tools used in the laboratory to judge a digital link are the eye diagram, which looks at a real component in time, and the error vector, which looks at the complex symbol in the plane. Neither is an analogy: both have a precise definition and a provable relation to the quantities of this chapter.

Definition 26.15: Eye diagram. Let $x(t) = \sum_k a_k p(t - kT)$ as in Definition 26.13 and let $L \geq 1$ be an integer. The **eye diagram** of order L of the signal x is the superposition, on a single time axis $\tau \in [-\frac{LT}{2}, \frac{LT}{2}]$, of all the traces

$$\tau \mapsto x(\tau + k_0 T), \quad k_0 \in \mathbb{Z}.$$

The **vertical eye opening** at instant τ is

$$\mathcal{O}(\tau) = \inf \{x(\tau + k_0 T) : a_{k_0} = +1\} - \sup \{x(\tau + k_0 T) : a_{k_0} = -1\},$$

where the infimum and supremum are taken over all admissible sequences (a_k) . The eye is said to be **open** at τ if $\mathcal{O}(\tau) > 0$.

Proposition 26.9: Eye opening for an antipodal binary transmission. Let $a_k \in \{+1, -1\}$ for every k , and let p be such that $\sum_{k \in \mathbb{Z}} |p(\tau + kT)| < +\infty$ for every τ . Then,

setting

$$D(\tau) = p(\tau) - \sum_{k \neq 0} |p(\tau - kT)|,$$

we have $\mathcal{O}(\tau) = 2D(\tau)$, and the infimum is a minimum, attained by the sequence $a_k = -\text{sgn}(p(\tau - kT))$ for $k \neq 0$ (with $a_0 = +1$). In particular, if p is a Nyquist pulse (Definition 26.13), then $\mathcal{O}(0) = 2$: at the exact sampling instant the eye is completely open.

Proof. Set $k_0 = 0$ without loss of generality (the traces are the same up to renaming the indices). With $a_0 = +1$,

$$x(\tau) = \sum_{k \in \mathbb{Z}} a_k p(\tau - kT) = p(\tau) + \sum_{k \neq 0} a_k p(\tau - kT),$$

a series absolutely convergent by hypothesis (the coefficients a_k have modulus 1). By the triangle inequality, for every choice of signs,

$$\sum_{k \neq 0} a_k p(\tau - kT) \geq - \sum_{k \neq 0} |p(\tau - kT)|,$$

hence $x(\tau) \geq D(\tau)$. The inequality is an equality for the choice $a_k = -\text{sgn}(p(\tau - kT))$ (with $a_k = +1$ arbitrary where $p(\tau - kT) = 0$), because then $a_k p(\tau - kT) = -|p(\tau - kT)|$ for every $k \neq 0$. That choice is admissible, being a sequence of signs. Hence $\inf\{x(\tau) : a_0 = +1\} = D(\tau)$, and it is a minimum.

For $a_0 = -1$ one argues the same way with signs swapped: $x(\tau) \leq -D(\tau)$, with equality for the opposite sequence; hence $\sup\{x(\tau) : a_0 = -1\} = -D(\tau)$. Subtracting, $\mathcal{O}(\tau) = D(\tau) - (-D(\tau)) = 2D(\tau)$.

If p is Nyquist, at $\tau = 0$ we have $p(0) = 1$ and $p(-kT) = 0$ for $k \neq 0$, hence $D(0) = 1 - 0 = 1$ and $\mathcal{O}(0) = 2$. $\square$

Example 26.8: The eye closes at half a symbol. Take the raised-cosine pulse with $\beta = 0.5$ and $T = 1$, whose expression in time is

$$p(t) = \text{sinc}(t) \frac{\cos(\pi\beta t)}{1 - (2\beta t)^2} = \frac{\sin(\pi t)}{\pi t} \cdot \frac{\cos(\pi t/2)}{1 - t^2},$$

and consider only the three symbols a_{-1}, a_0, a_1 (the other contributions are smaller than $|p(2)|$ and are neglected in the computation, which is thus explicitly truncated). The values needed are

$$p(0) = 1, \quad p\left(\frac{1}{2}\right) = p\left(-\frac{1}{2}\right) = 0.6002109, \quad p\left(\frac{3}{2}\right) = -0.1200422.$$

Applying Proposition 26.9:

$$D(0) = p(0) - |p(-1)| - |p(1)| = 1 - 0 - 0 = 1, \quad \mathcal{O}(0) = 2,$$

because the raised cosine is Nyquist (Proposition 26.7 and Theorem 26.11); while at half a symbol

$$D\left(\frac{1}{2}\right) = p\left(\frac{1}{2}\right) - \left|p\left(-\frac{1}{2}\right)\right| - \left|p\left(\frac{3}{2}\right)\right| = 0.6002109 - 0.6002109 - 0.1200422 = -0.1200422 < 0.$$

The eye is **closed**: there exists a sequence of symbols — for instance $(a_{-1}, a_0, a_1) = (+1, +1, -1)$ — for which, even though $a_0 = +1$ was transmitted, the signal sampled at $t = T/2$ equals -0.1200422 , i.e. has the wrong sign *even in the absence of noise*. A timing

error of half a symbol destroys the link independently of the transmitted power: this is why timing recovery is, in a receiver, a subsystem as important as the demodulator.

Independent check. A Python script evaluates p from the closed form and builds the eight traces corresponding to the eight triples $(a_{-1}, a_0, a_1) \in \{\pm 1\}^3$ on a grid of step 0.1. At $t = 0$ all eight traces equal exactly ± 1 (the only two possible ordinates, as it must be for a Nyquist pulse); at $t = 0.5$ the values are ± 1.3205 , ± 0.1200422 : the trace $(+1, +1, -1)$ equals -0.1200422 , which coincides with $D(1/2)$ computed above. ✓

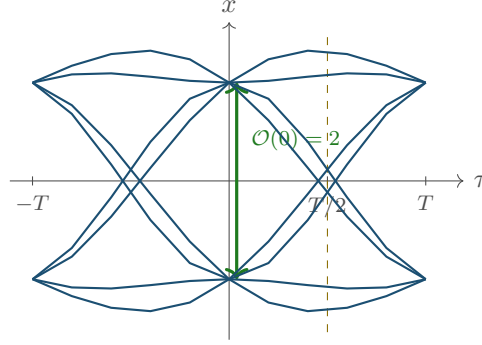


Figure 26.8: The order-2 eye diagram of Example 26.8, drawn from the values of the raised-cosine pulse with $\beta = 0.5$ and not from a measurement. Scale: $T = 2.6$ cm on the abscissa and one amplitude unit = 1.3 cm on the ordinate; each trace is the polyline on the eleven points $\tau = -1.0; -0.8; \dots; 1.0$ (in units of T). The eight traces correspond to the eight triples (a_{-1}, a_0, a_1) and are pairwise opposite, by linearity of the sum: the four with $a_{-1} = -1$ are the reflections of the four drawn first. At $\tau = 0$ all pass through ± 1 : the eye is open by $\mathcal{O}(0) = 2$ (green segment), which is the content of Proposition 26.9 for a Nyquist pulse. On the dashed vertical $\tau = T/2$ the traces cross through zero: there the eye is closed, $D(T/2) = -0.1200422$.

Definition 26.16: Error vector and EVM. Let $a_1, \dots, a_K \in \mathcal{A}$ be the transmitted symbols and $r_1, \dots, r_K \in \mathbb{C}$ the corresponding received symbols after the matched filter and synchronisation. The **error vector** of the k -th symbol is the complex number

$$e_k = r_k - a_k,$$

and the **error vector magnitude** (EVM), root-mean-square, is

$$\text{EVM}_K = \sqrt{\frac{\frac{1}{K} \sum_{k=1}^K |e_k|^2}{\frac{1}{K} \sum_{k=1}^K |a_k|^2}} = \sqrt{\frac{\sum_{k=1}^K |r_k - a_k|^2}{\sum_{k=1}^K |a_k|^2}},$$

usually expressed as a percentage. The two components of e_k in the basis $\{a_k/|a_k|, i a_k/|a_k|\}$ are called the **amplitude error** and the **phase error** of the symbol.

Proposition 26.10: EVM and signal-to-noise ratio. On the AWGN channel of Definition 26.8, with $r_k = a_k + n_k$ and n_k independent of each other and of the symbols,

$$\frac{\mathbb{E} \left[\sum_{k=1}^K |e_k|^2 \right]}{\mathbb{E} \left[\sum_{k=1}^K |a_k|^2 \right]} = \frac{N_0}{E_s} = \frac{1}{\text{SNR}_s}, \quad \text{i.e.} \quad \text{EVM} \approx \frac{1}{\sqrt{\text{SNR}_s}},$$

where $\text{SNR}_s = E_s/N_0$ is the signal-to-noise ratio per symbol. In particular $\text{SNR}_s = 20$ dB corresponds to $\text{EVM} = 10\%$.

Proof. By Definition 26.16, $e_k = r_k - a_k = n_k$. By Definition 26.8, $\mathbb{E}[|n_k|^2] = \mathbb{E}[(\Re n_k)^2] + \mathbb{E}[(\Im n_k)^2] = \frac{N_0}{2} + \frac{N_0}{2} = N_0$, hence, by linearity of expectation, $\mathbb{E}[\sum_k |e_k|^2] = KN_0$. By Proposition 26.3, $\mathbb{E}[\sum_k |a_k|^2] = KE_s$. The ratio is $KN_0/(KE_s) = N_0/E_s$, independent of K . For the last claim: $\text{SNR}_s = 20$ dB means $E_s/N_0 = 10^{20/10} = 100$, hence $1/\sqrt{\text{SNR}_s} = 1/10 = 0.1 = 10\%$. $\square$

Remark 26.13: Why " $\approx$ " and not "=", and what EVM really measures. Two clarifications, both important in practical use.

First. The quantity EVM_K of Definition 26.16 is the ratio of two sample averages, not of two expected values: Proposition 26.10 computes the ratio of the expected values. The two coincide in the limit $K \rightarrow \infty$ by the strong law of large numbers, which guarantees the almost-sure convergence of each of the two averages to its own expected value (the variables $|n_k|^2$ and $|a_k|^2$ are independent, identically distributed and integrable); the law of large numbers is a result of probability theory that is imported, not proved in this treatise. For finite K the measured EVM fluctuates, and laboratory measurements always state the number of symbols over which the average was taken.

Second. The equality $\text{EVM}^2 = N_0/E_s$ holds only if additive noise is the only impairment. In a real link the error vector also collects residual intersymbol interference (Definition 26.13), oscillator phase error, amplifier distortion, imbalance between the two IQ branches. This is exactly why the measurement is useful: an EVM much worse than $1/\sqrt{\text{SNR}_s}$ signals that the degradation is not noise, and the shape of the cloud of error vectors tells which cause is at work — a cloud elongated radially indicates amplitude compression, one elongated tangentially indicates phase noise, a cloud rotated as a block indicates a frequency error, i.e. the slow rotation of Example 26.1.

Example 26.9: From SNR to EVM and back. From Proposition 26.10:

SNR_s (dB)	SNR_s	$\text{EVM} = 1/\sqrt{\text{SNR}_s}$
10	10.0000000	31.6227766%
15	31.6227766	17.7827941%
20	100.0000000	10.0000000%
25	316.2277660	5.6234133%
30	1,000.0000000	3.1622777%

Note the practical rule that follows: every 20 dB of SNR corresponds to a factor 10 on EVM, an immediate consequence of $\text{EVM} = 10^{-\text{SNR}_{\text{dB}}/20}$.

Independent check. The two columns are obtained from each other in two different ways: $\text{SNR} = 10^{\text{dB}/10}$ followed by $\text{EVM} = 1/\sqrt{\text{SNR}}$, or directly $\text{EVM} = 10^{-\text{dB}/20}$. For 15 dB: the first route gives $10^{1.5} = 31.6227766$ and $1/\sqrt{31.6227766} = 0.1778279$; the second $10^{-0.75} = 0.1778279$. $\checkmark$

26.10. Assigned exercises

These are *new* exercises, distinct from the worked ones in Section 26.11. The answers are in the Teacher's key (Section 26.13).

Exercise 26.1: From the carrier to baseband, and back. *Objective:* to be able to compute the complex envelope of a given signal, verify uniqueness, and recognise where the hypothesis $B < f_c$ enters. *Estimated time:* 40 minutes. *Type:* symbolic computation and proof. *Level:* basic–intermediate.

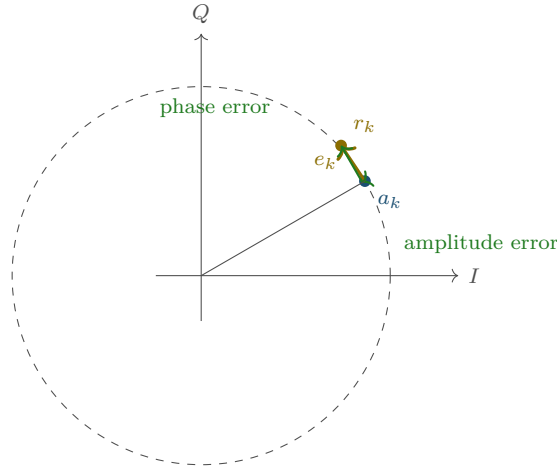


Figure 26.9: The error vector of Definition 26.16. The ideal symbol is $a_k = 2.5 e^{i\pi/6}$, i.e. $(2.5 \cos 30^\circ, 2.5 \sin 30^\circ) = (2.1651, 1.25)$ in centimetres; the received symbol is $r_k = (1.85, 1.72)$, chosen for the figure. The error vector is $e_k = r_k - a_k = (-0.3151, 0.4700)$, of modulus $\sqrt{0.3151^2 + 0.4700^2} = 0.5659$. Its decomposition in the basis $(\frac{a_k}{|a_k|}, i \frac{a_k}{|a_k|}) = ((0.8660, 0.5), (-0.5, 0.8660))$ gives the radial component $-0.3151 \cdot 0.8660 + 0.4700 \cdot 0.5 = -0.0379$ (amplitude error) and the tangential component $0.3151 \cdot 0.5 + 0.4700 \cdot 0.8660 = 0.5646$ (phase error). The square of the modulus is recovered from the two components: $0.0379^2 + 0.5646^2 = 0.3202 = 0.5659^2$. ✓

Consider the real signal

$$s(t) = A \cos(2\pi f_c t) + B \cos(2\pi(f_c + \nu)t + \varphi),$$

with $A = 2$, $B = 1$, $f_c = 1$ MHz, $\nu = 1$ kHz, $\varphi = \pi/3$.

- Write s_L , s_I and s_Q using Theorem 26.2, and compute their value at $t = 0$; say why the answer is unique and which hypothesis of Definition 26.1 guarantees it.
- Compute $|s_L(t)|^2$ in closed form and determine the maximum and minimum value of the modulus $|s_L|$, stating at which instants they are attained.
- Verify that s_L has half-bandwidth ν and that hence the ratio f_c/ν equals 1,000: how large must $f_c T$ be for the discrepancy from carrier orthogonality of Proposition 26.2(ii) to stay below 0.1%?
- If the demodulator mistakenly uses the carrier $\cos(2\pi f_c t + \theta)$ instead of $\cos(2\pi f_c t)$, show that the recovered components are $\tilde{s}_I = s_I \cos \theta + s_Q \sin \theta$ and $\tilde{s}_Q = -s_I \sin \theta + s_Q \cos \theta$, i.e. that the complex envelope is multiplied by $e^{-i\theta}$: the constellation rotates rigidly.

Exercise 26.2: Designing an optimal 8 + 8-APSK. *Objective:* to independently repeat the optimisation of Theorem 26.5 on a different geometry and to be able to interpret the result. *Estimated time:* 55 minutes. *Type:* proof-based and numerical. *Level:* intermediate-advanced.

Consider the family of 8 + 8-APSK constellations (Definition 26.7) with $M_1 = M_2 = 8$, phases $\varphi_1 = \pi/8$ and $\varphi_2 = 0$, $R_2 = \gamma R_1$ with $\gamma \geq 1$, normalised to $E_s = 1$.

- Prove that normalisation forces $R_1 = \sqrt{2/(1 + \gamma^2)}$.
- Prove that the minimum angular difference between a point of the inner circle and one of the outer circle equals $\pi/8$ (adapt the computation on odd integers of Theorem 26.5), and write $d_{\min}(\gamma)$ as the minimum of two quantities (not three: explain why one of the three is always redundant for $\gamma \geq 1$).

- (c) Determine γ^* by solving the quadratic equation $\gamma^2 - 2\gamma \cos \frac{\pi}{8} + (1 - 4 \sin^2 \frac{\pi}{8}) = 0$, showing that the constant term equals $\sqrt{2} - 1$ and that $2 \cos \frac{\pi}{8} = \sqrt{2 + \sqrt{2}}$; compute γ^* , R_1 , R_2 , $d_{\min}$ and the PAPR to seven decimal places.
- (d) Compare $d_{\min}$ and PAPR with those of 16-QAM (Example 26.4) and of 4 + 12-APSK (Theorem 26.5), and say which of the three one would choose for a satellite transponder near saturation and which for a linear fibre link, justifying with the numbers.

Exercise 26.3: A complete link, from decibels to the eye. *Objective:* to walk through the entire chain of the chapter in a single exercise on assigned numerical data. *Estimated time:* 50 minutes. *Type:* numerical with independent check. *Level:* intermediate.

A QPSK link uses $R_s = 2$ Mbaud, a raised-cosine pulse with $\beta = 0.35$, and is designed for $P_b = 10^{-6}$.

- (a) Compute the bit rate R_b , the occupied bandwidth W and the spectral efficiency η (Corollary 26.7).
- (b) Determine the value of E_b/N_0 (in dB) that gives $P_b = 10^{-6}$, by numerically solving $Q(x) = 10^{-6}$ by successive trials to at least three digits and obtaining $E_b/N_0 = x^2/2$ (Theorem 26.9 and Theorem 26.10); compare with the table of Example 26.6 and verify that the value falls between the rows at 10 and 12 dB.
- (c) Compute the corresponding *symbol* error probability $P_s = 2P_b - P_b^2$ and say by how many decibels E_b/N_0 would have to be raised to bring P_b to 10^{-9} .
- (d) If the instrument measures $\text{EVM} = 4\%$, what SNR_s does this imply (Proposition 26.10)? And what E_b/N_0 , recalling that for QPSK $E_s = 2E_b$? Does the link meet the specification of point (b)?
- (e) If in addition the sampling instant is wrong by $0.2T$, does the eye opening (Proposition 26.9) remain positive? Answer by computing $D(0.2)$ with only the three symbols a_{-1}, a_0, a_1 for the raised-cosine pulse with $\beta = 0.5$ of Example 26.8, and say what would change with $\beta = 0.35$.

26.11. Worked exercises

Exercise 26.4: What it costs to go from QPSK to 8-PSK. *Type:* numerical, with a proved upper and lower bound. Estimate the symbol error probability of 8-PSK at $E_b/N_0 = 10$ dB, bracketing it between two computed values, and compare it with QPSK at the same E_b/N_0 .

Solution. Strategy. For M -PSK the decision region is a circular sector (Figure 26.3), and its complement is the union of *two* half-planes: the one beyond the bisector toward the right neighbour and the one beyond the bisector toward the left neighbour. Hence a lower bound (a single half-plane) and an upper bound (the union of the two, by subadditivity). There is no need to compute the integral over the sector.

Step 1 — the distance from the boundary. Let $a_0 = \sqrt{E_s}$ be the transmitted symbol (the choice is irrelevant by Corollary 26.4). The boundary toward a_1 is the perpendicular bisector of the segment $a_0 a_1$ (Theorem 26.7), and the distance of a_0 from that bisector is half the length of the segment, i.e. $\frac{1}{2}d_{\min}$. By Theorem 26.3,

$$\frac{d_{\min}}{2} = \sqrt{E_s} \sin \frac{\pi}{M}.$$

Step 2 — the probability of crossing a boundary. The noise component orthogonal to the boundary is Gaussian with zero mean and variance $N_0/2$ — it is the same situation as Step 2 of the proof of Theorem 26.9, with the direction rotated, and the rotation does not change the law because the density of Definition 26.8 depends only on the modulus. Hence, by Proposition 26.6(iii),

$$p_1 := \Pr[\text{crossing a fixed boundary}] = Q\left(\frac{d_{\min}/2}{\sqrt{N_0/2}}\right) = Q\left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{M}\right).$$

Step 3 — the bracket. The event "error" contains the event "crossing the right boundary" and, by subadditivity of probability, is contained in the union of the two:

$$p_1 \leq P_s \leq 2p_1.$$

Step 4 — the numbers. For $M = 8$, $m = 3$ and $E_s = 3E_b$ (Definition 26.3), hence

$$\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{8} = \sqrt{\frac{6E_b}{N_0}} \sin \frac{\pi}{8}.$$

At $E_b/N_0 = 10$ dB, i.e. $E_b/N_0 = 10$:

$$\sqrt{60} = 7.7459667, \quad \sin \frac{\pi}{8} = 0.3826834, \quad \text{argument} = 2.9642531.$$

Hence $p_1 = Q(2.9642531) = 1.5170930 \cdot 10^{-3}$ and

$$1.5170930 \cdot 10^{-3} \leq P_s^{\text{8-PSK}} \leq 3.0341860 \cdot 10^{-3}.$$

Step 5 — the comparison. By Theorem 26.10, at the same $E_b/N_0 = 10$ dB, QPSK has $P_s = 7.7442014 \cdot 10^{-6}$. The ratio between the upper bound for 8-PSK and QPSK equals

$$\frac{3.0341860 \cdot 10^{-3}}{7.7442014 \cdot 10^{-6}} = 391.8010,$$

i.e. almost *four hundred times* more errors for having added a single bit per symbol. This is the probabilistic translation of the last column of the table of Example 26.3, where the ratio $d_{\min}^2/(4E_b)$ drops from 1 to 0.4393398.

Independent check. The argument of the Q function can be recomputed by a different route, starting from the minimum distance rather than from the angle: by the table of Example 26.3 with $E_s = 1$, $d_{\min} = 0.7653669\sqrt{E_s}$, so setting $E_b = 1$, $N_0 = 1/10$ and $E_s = 3$, $d_{\min} = 0.7653669\sqrt{3} = 1.3256543$, and

$$\frac{d_{\min}/2}{\sqrt{N_0/2}} = \frac{0.7653669\sqrt{3}/2}{\sqrt{1/20}} = \frac{0.7653669}{2} \cdot \sqrt{3} \cdot \sqrt{20} = \frac{0.7653669}{2} \cdot \sqrt{60} = 0.3826834 \cdot 7.7459667 = 2.9642531. \checkmark$$

The value of Q was computed in two ways (Example 26.6): with $\frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ and with Simpson integration, which agree to within $5 \cdot 10^{-14}$ in relative value. $\checkmark$

Interpretation. The bracket $p_1 \leq P_s \leq 2p_1$ is a factor of 2 wide, i.e. 3.01 dB in ordinate but very little in abscissa: on the extremely steep curve of Figure 26.6 a factor of 2 in probability corresponds to less than 0.3 dB of E_b/N_0 . For designing a link the bracket is therefore more than sufficient, which is why in practice the upper bound $2p_1$ is used without qualms.

Common mistakes. (1) Using $E_s = E_b$ instead of $E_s = 3E_b$: the argument is underestimated by a factor $\sqrt{3}$ and the error probability comes out enormously larger. (2) Confusing

symbol error and bit error: with Gray mapping and small probabilities, $P_b \approx P_s/m$, i.e. roughly a third, but it is not an equality. (3) Applying the union bound with $M - 1 = 7$ terms: this gives $7p_1 = 1.0619651 \cdot 10^{-2}$, correct but needlessly loose, because the five non-adjacent symbols contribute negligibly.

Extension. Repeat the computation for 16-PSK ($m = 4$, $E_s = 4E_b$, $\sin(\pi/16) = 0.1950903$) and verify that the argument at 10 dB equals $\sqrt{80} \cdot 0.1950903 = 1.7449409$, that $Q(1.7449409) = 4.0497580 \cdot 10^{-2}$ and that the bracket becomes $[4.0497580 \cdot 10^{-2}, 8.0995159 \cdot 10^{-2}]$: 16-PSK at 10 dB is unusable without coding.

Exercise 26.5: Filter matched to a triangular pulse. *Type:* computation of integrals with application of the equality case in Cauchy–Schwarz. Consider the triangular pulse $s(t) = A(1 - |t|/T)$ for $|t| \leq T$ and $s(t) = 0$ elsewhere. Determine the energy, the matched filter, the maximum signal-to-noise ratio, and the loss incurred by integrating instead over all of $[-T, T]$ with constant weight.

Solution. Strategy. Apply Theorem 26.8: the maximum equals $2E/N_0$ and is attained only for h proportional to the reflection of s . The loss of the wrong filter is read as the Cauchy–Schwarz ratio $\langle s, g \rangle^2 / (\|s\|^2 \|g\|^2)$, which is the squared cosine of the angle between the two signals in L^2 .

Step 1 — the energy. By evenness of the integrand,

$$E = \int_{-T}^T A^2 \left(1 - \frac{|t|}{T}\right)^2 dt = 2A^2 \int_0^T \left(1 - \frac{t}{T}\right)^2 dt.$$

With the substitution $u = 1 - t/T$, $du = -dt/T$, the endpoints $t = 0, T$ become $u = 1, 0$:

$$\int_0^T \left(1 - \frac{t}{T}\right)^2 dt = T \int_0^1 u^2 du = \frac{T}{3}.$$

Hence $E = \frac{2A^2T}{3}$.

Step 2 — the matched filter. By Definition 26.11 the filter matched at instant t_0 has $h(t) = c s(t_0 - t)$. Since s is *even*, $s(t_0 - t) = s(t - t_0)$: the matched filter is the pulse itself, shifted to t_0 . With $t_0 = 0$ and $c = 1$, $h = s$. The maximum signal-to-noise ratio is

$$\text{SNR}_{\max} = \frac{2E}{N_0} = \frac{4A^2T}{3N_0}.$$

Step 3 — the wrong filter. Take $g = \mathbf{1}_{[-T, T]}$, i.e. integrate the observation over the whole duration of the pulse with constant weight. Then

$$\langle s, g \rangle = \int_{-T}^T A \left(1 - \frac{|t|}{T}\right) dt = AT$$

— this is the area of the triangle of base $2T$ and height A , i.e. $\frac{1}{2} \cdot 2T \cdot A$ — and $\|g\|^2 = 2T$. By Definition 26.11,

$$\text{SNR} = \frac{(AT)^2}{\frac{N_0}{2} \cdot 2T} = \frac{A^2T}{N_0}.$$

Step 4 — the loss. The ratio between the two equals

$$\frac{\text{SNR}}{\text{SNR}_{\max}} = \frac{A^2T/N_0}{4A^2T/(3N_0)} = \frac{3}{4},$$

i.e. a loss of $10 \log_{10} \frac{4}{3} = 1.2493874$ dB.

Independent check. The same $3/4$ is obtained directly from Lemma 26.3 without going through N_0 :

$$\frac{\langle s, g \rangle^2}{\|s\|^2 \|g\|^2} = \frac{(AT)^2}{\frac{2A^2T}{3} \cdot 2T} = \frac{A^2T^2}{\frac{4}{3}A^2T^2} = \frac{3}{4}. \quad \checkmark$$

The value is strictly less than 1, as it must be by the equality case of the lemma: g is constant and s is not, so they are not proportional. $\checkmark$

Interpretation. The loss of 1.25 dB is the price of having given the same weight to samples taken when the signal is strong (near $t = 0$) and to those taken when it is weak (near $\pm T$). The matched filter "weighs" each instant proportionally to the signal found there: this is the engineering translation of the equality case of Cauchy–Schwarz.

Common mistakes. (1) Computing E as $A^2 \cdot 2T$, as if the pulse were rectangular: this overestimates by a factor 3. (2) Forgetting that the matched filter is s *reflected*: for an even pulse it makes no difference, for an asymmetric pulse it does, and the mistake is invisible until the waveform is changed. (3) Comparing the two ratios in different units (one linear, one in dB).

Extension. Repeat the computation with the half-cosine pulse $s(t) = A \cos \frac{\pi t}{2T}$ on $[-T, T]$: find $E = A^2T$, and verify that with $g = \mathbf{1}_{[-T, T]}$ the Cauchy–Schwarz ratio equals $8/\pi^2 = 0.8105695$, i.e. a loss of 0.9121 dB, smaller than that of the triangle because the half-cosine is "flatter."

Exercise 26.6: A constant phase error, in EVM and in decibels. *Type:* exact computation with metrological interpretation. A QPSK receiver has a constant phase error θ in carrier recovery (case (d) of Exercise 26.1). Express the resulting EVM as a function of θ , in *exact* form, and convert it into the equivalent signal-to-noise ratio.

Solution. Strategy. A phase error multiplies every received symbol by $e^{-i\theta}$ (Exercise 26.1(d)). The error vector is therefore $a(e^{-i\theta} - 1)$, and its modulus is computed with Lemma 26.1, with no small-angle approximation.

Step 1 — the error vector. With $r_k = a_k e^{-i\theta}$ (no additive noise, only the phase error),

$$e_k = r_k - a_k = a_k(e^{-i\theta} - 1), \quad |e_k| = |a_k| |e^{-i\theta} - e^{i0}| = |a_k| \cdot 2 \left| \sin \frac{-\theta - 0}{2} \right| = 2|a_k| \left| \sin \frac{\theta}{2} \right|,$$

having applied Lemma 26.1 with $\alpha = -\theta$ and $\beta = 0$.

Step 2 — the EVM. By Definition 26.16,

$$\text{EVM} = \sqrt{\frac{\sum_k |e_k|^2}{\sum_k |a_k|^2}} = \sqrt{\frac{4 \sin^2 \frac{\theta}{2} \sum_k |a_k|^2}{\sum_k |a_k|^2}} = 2 \left| \sin \frac{\theta}{2} \right|.$$

The result is *exact* and does not depend on the constellation: the factor $|a_k|^2$ cancels identically, so it holds for QPSK as much as for 16-QAM or any other.

Step 3 — the equivalent signal-to-noise ratio. By Proposition 26.10, a measured EVM is read as $\text{SNR}_s = 1/\text{EVM}^2$. Hence

$$\text{SNR}_{\text{eq}} = \frac{1}{4 \sin^2(\theta/2)}, \quad \text{SNR}_{\text{eq}}|_{\text{dB}} = -20 \log_{10} \left(2 \left| \sin \frac{\theta}{2} \right| \right).$$

Step 4 — the numbers.

θ	$\text{EVM} = 2 \sin(\theta/2)$	EVM in %	SNR_{eq} (dB)
1°	0.0174531	1.7453%	35.1625629
2°	0.0349048	3.4905%	29.1422937
5°	0.0872388	8.7239%	21.1858089
10°	0.1743115	17.4311%	15.1734799

Independent check. For small angles $2 \sin(\theta/2) \approx \theta$ in radians: for $\theta = 5^\circ = 0.0872665$ rad the exact formula gives 0.0872388, and the relative discrepancy is $3.2 \cdot 10^{-4}$, consistent with the expansion $2 \sin(x/2) = x - \frac{x^3}{24} + O(x^5)$, which for $x = 0.0872665$ predicts a relative discrepancy $x^2/24 = 3.17 \cdot 10^{-4}$. ✓ Moreover, SNR_{eq} at 5° equals $1/0.0872388^2 = 131.3956$, and $10 \log_{10} 131.3956 = 21.1858089$ dB, in agreement with the table. ✓

Interpretation. The practical reading is severe: a phase error of only 5° limits the EVM to 8.7%, i.e. it imposes a ceiling of 21.2 dB on the signal-to-noise ratio *whatever* the transmitted power. A 256-QAM, which by the table of Example 26.7 needs more than 13 dB more than QPSK, does not work with that ceiling: this is why higher-order modulations require a much tighter carrier-recovery loop than lower-order ones. The degradation cannot be fought with power, because it is not noise.

Common mistakes. (1) Using the approximation $\text{EVM} \approx \theta$ without stating it: for $\theta = 10^\circ$ the discrepancy is already 1.3 per mille, and for larger angles it grows as $\theta^2/24$. (2) Confusing degrees and radians in the sine. (3) Adding in decibels the contribution of the phase error and that of the noise: one must add the error *powers*, i.e. the squared EVMs, not the decibels.

Extension. Show that if the phase error and the additive noise are independent, then $\text{EVM}_{\text{tot}}^2 = \text{EVM}_\theta^2 + \text{EVM}_{\text{rum}}^2$, and compute the total EVM for $\theta = 2^\circ$ and $\text{SNR}_s = 25$ dB: one obtains $\sqrt{0.0349048^2 + 0.0562341^2} = 0.0661862$, i.e. 6.62%.

26.12. Self-assessment quiz

- (Definition) The *complex envelope* of a real bandpass signal s with carrier f_c and half-bandwidth $B < f_c$ is
 - a unique complex signal of half-bandwidth B such that $s(t) = \Re\{s_L(t)e^{i2\pi f_c t}\}$
 - the imaginary part of s
 - any complex signal whose real part is s
 - the Fourier transform of s restricted to positive frequencies
- (Hypothesis) In Theorem 26.2 the hypothesis $B < f_c$ serves to
 - guarantee that s is real-valued
 - make disjoint the supports of the two terms of $\hat{s}(f) = \frac{1}{2}\hat{u}(f - f_c) + \frac{1}{2}\hat{u}(-f - f_c)$
 - ensure the convergence of the integral defining $\hat{s}$
 - allow the Plancherel–Parseval identity to be applied
- (Symbolic computation) For the M -PSK constellation of energy per symbol E_s , the minimum distance equals
 - $\sqrt{E_s} \sin \frac{2\pi}{M}$
 - $2\sqrt{E_s} \sin \frac{\pi}{M}$
 - $\frac{2\sqrt{E_s}}{M}$
 - $\sqrt{\frac{6E_s}{M-1}}$
- (Numerical computation) At $E_b/N_0 = 10$ dB the *bit* error probability of QPSK equals
 - $3.8721082 \cdot 10^{-6}$
 - $7.7442014 \cdot 10^{-6}$
 - $1.9090777 \cdot 10^{-4}$
 - $2.3882908 \cdot 10^{-3}$
- (Proof step) In Theorem 26.8 the fact that the maximum of SNR is attained *only* by the matched filter follows

- A) from the positivity of N_0 B) from the equality case in the Cauchy–Schwarz inequality, i.e. from the proportionality of s and g C) from the linearity of the integral
D) from the fact that $\|g\|^2$ cancels in the ratio
6. (Error diagnosis) A student computes the error probability of QPSK as $Q(\sqrt{2E_s/N_0})$ instead of $Q(\sqrt{2E_b/N_0})$, and at $E_b/N_0 = 10$ dB obtains $1.3 \cdot 10^{-10}$ instead of $3.9 \cdot 10^{-6}$, i.e. a value thirty thousand times too optimistic. The mistake consists in having
- A) used the two-sided convention $N_0/2$ instead of the one-sided one B) forgotten that for QPSK $E_s = 2E_b$, i.e. having attributed to each of the two components the entire energy of the symbol C) confused Q with erfc D) used the minimum distance of BPSK
7. (Transfer) A 4 + 12-APSK and a 16-QAM are normalised to the same average energy per symbol. Compared with 16-QAM, the optimal APSK of Theorem 26.5 has
- A) larger minimum distance and larger PAPR B) smaller minimum distance (0.5848015 against 0.6324555) and smaller PAPR (1.2763345 against 1.8) C) both minimum distance and PAPR smaller, but by negligible amounts D) exactly the same parameters, because both have 16 points
8. (Source and metacognition) The reference text in which the baseband representation, the matched filter and the error probabilities of digital modulations are treated organically, in the edition verified here, is
- A) J. G. Proakis, M. Salehi, *Digital Communications*, 5th ed., McGraw-Hill, 2008, ISBN 978-0-07-295716-7 B) J. W. Cooley, J. W. Tukey, *An algorithm for the machine calculation of complex Fourier series*, 1965 C) E. T. Whittaker, *On the functions which are represented by the expansions of the interpolation theory*, 1915 D) C. Van Loan, *Computational Frameworks for the Fast Fourier Transform*, SIAM, 1992

26.13. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

- 1) **A.** *Rationale:* Theorem 26.1(iii) gives existence and Theorem 26.2 uniqueness, which is an essential part of the definition: without the half-bandwidth constraint B the representation would not be unique. *Distractors:* B confuses the complex envelope with the quadrature component; C is exactly what uniqueness excludes (there are infinitely many u with $\Re\{ue^{i2\pi f_c t}\} = s$ if the bandwidth constraint is dropped); D describes $\widehat{s}_+$, not s_L , and forgets the shift by f_c and the factor 2. *Diagnosed error:* not having understood that bandlimitness is a hypothesis, not an accessory. *Source:* Source [1], chapter on the representation of bandpass signals. *Prerequisite:* the Fourier transform and its injectivity (Theorem 19.9). *Remedial activity:* redo Step 3 of the proof of Theorem 26.1 and Example 26.1.
- 2) **B.** *Rationale:* this is literally where the proof of Theorem 26.2 uses the hypothesis: without disjointness of the supports one could not conclude $\widehat{s}(f) = \frac{1}{2}\widehat{u}(f - f_c)$ for $f > 0$. *Distractors:* A is false, s is real by an independent hypothesis; C is false, convergence follows from $s \in L^1$; D is irrelevant, Plancherel–Parseval is used elsewhere (Proposition 26.1) and does not require $B < f_c$. *Diagnosed error:* reading hypotheses as formalities. *Source:* Source [2], chapter on bandpass signals. *Prerequisite:* Remark 26.1. *Remedial activity:* construct a counterexample with $B > f_c$ and verify that the envelope is no longer unique.
- 3) **B.** *Rationale:* Theorem 26.3, proved from Lemma 26.1 and the monotonicity of the sine on $(0, \pi/2]$. *Distractors:* A uses the full angle between two symbols instead of half of it,

forgetting the factor $\frac{1}{2}$ of the chord formula; C is the approximation $\sin x \approx x$ without the π , i.e. an error by a factor π ; D is the formula for square QAM (Theorem 26.4), correct but for a different constellation. *Diagnosed error*: confusing the central angle with the half-angle in the chord formula. *Source*: Source [3], chapter on bandpass modulations. *Prerequisite*: roots of unity (Theorem 6.3). *Remedial activity*: derive $|e^{i\alpha} - e^{i\beta}|$ by hand and check it for $M = 4$, where it must give $\sqrt{2}\sqrt{E_s}$.

4) **A.** *Rationale*: Theorem 26.10, $P_b = q = Q(\sqrt{2E_b/N_0})$ with $\sqrt{20} = 4.4721360$: the value is in the table of Example 26.6. *Distractors*: B is the *symbol* error probability of QPSK, i.e. roughly double; C is the value at 8 dB; D is the value at 6 dB. *Diagnosed error*: confusing bit error and symbol error, or picking the wrong row in the table. *Source*: Source [1], chapter on optimal receivers for the AWGN channel. *Prerequisite*: Proposition 26.6. *Remedial activity*: recompute $Q(4.4721360)$ with $\frac{1}{2} \operatorname{erfc}(3.1622777)$ and compare with the table.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **B.** *Rationale*: Step 3 of the proof of Theorem 26.8 uses the equality case of Lemma 26.3, and it is the only point producing uniqueness of the optimal solution. *Distractors*: A is needed only for the ratio to be well defined; C is used in Step 1 but does not give uniqueness; D explains why the maximum does not depend on g , not why it is attained only by one g . *Diagnosed error*: using Cauchy–Schwarz as a pure bound, without ever asking when it is an equality. *Source*: Source [4], chapter on optimal reception. *Prerequisite*: the equality case proved in Lemma 26.3. *Remedial activity*: Exercise 26.5, where the ratio equals $3/4 < 1$ precisely because proportionality fails.

6) **B.** *Rationale*: $Q(\sqrt{2E_s/N_0}) = Q(\sqrt{4E_b/N_0}) = Q(\sqrt{40}) = Q(6.3245553) = 1.2698143 \cdot 10^{-10}$, against the correct value $Q(\sqrt{20}) = 3.8721082 \cdot 10^{-6}$: a ratio of 30,493.5. The point of the exercise is recognising that the error is *a factor 2 on energy*, i.e. 3 dB, and that on such a steep curve 3 dB are worth four orders of magnitude. By Theorem 26.10 each of the two components carries E_b , not E_s : half the symbol's energy goes into the in-phase branch and half into the quadrature branch. *Distractors*: A is a different error, which also produces 3 dB but on the noise rather than the signal (Remark 26.6); C would give a factor 2 on the result, not on the argument; D is false, the minimum distance of the two is the same (Corollary 26.3). *Diagnosed error*: not distinguishing E_s from E_b : this is the most frequent mistake in the whole chapter. *Source*: Source [3], chapter on error probabilities. *Prerequisite*: Proposition 26.3 and Proposition 26.4. *Remedial activity*: redo Step 2 of Theorem 26.10 writing explicitly $\sqrt{E_s}/2 = \sqrt{E_b}$.

7) **B.** *Rationale*: these are the numbers in the table of Remark 26.5, proved in Theorem 26.4 and in Theorem 26.5. *Distractors*: A reverses both inequalities; C is wrong in the quantification: 0.68 dB of minimum distance and 1.49 dB of PAPR are not negligible in a link budget; D confuses cardinality with geometry. *Diagnosed error*: believing there is an absolutely "better" constellation, rather than one better with respect to a stated constraint. *Source*: Source [1], chapter on digital modulations. *Prerequisite*: Definition 26.3 (PAPR). *Remedial activity*: Exercise 26.2, which repeats the optimisation on an $8 + 8$ geometry.

8) **A.** *Rationale*: this is Source [1] of the Sources environment, with the bibliographic data verified on 7 September 2026. *Distractors*: B is the source for Chapter 23 and concerns an algorithm, not modulations; C concerns the sampling theorem (Chapter 24); D is a text on the FFT. *Diagnosed error*: citing from memory without checking edition and year — the metacognitive point of the chapter. *Source*: the Sources environment of this chapter. *Prerequisite*: none. *Remedial activity*: check for oneself, in a library catalogue, the data of the four sources listed, and note that each of them exists in several editions with different contents and numbering.

BEST PRACTICE

Exercise 26.1 — answers.

(a) Factor out $e^{i2\pi f_c t}$: $s(t) = \Re\{(A + Be^{i(2\pi\nu t + \varphi)})e^{i2\pi f_c t}\}$, hence

$$s_L(t) = A + Be^{i(2\pi\nu t + \varphi)}, \quad s_I(t) = A + B \cos(2\pi\nu t + \varphi), \quad s_Q(t) = B \sin(2\pi\nu t + \varphi).$$

With $A = 2$, $B = 1$, $\varphi = \pi/3$: $s_L(0) = 2 + e^{i\pi/3} = 2.5 + 0.8660254i$ and $|s_L(0)| = \sqrt{2.5^2 + 0.8660254^2} = \sqrt{6.25 + 0.75} = \sqrt{7} = 2.6457513$. The answer is unique by Theorem 26.2, because s_L has half-bandwidth $\nu = 1$ kHz $< f_c = 1$ MHz: this is the hypothesis $B < f_c$ of Definition 26.1.

(b) $|s_L|^2 = (A + B \cos \psi)^2 + (B \sin \psi)^2 = A^2 + B^2 + 2AB \cos \psi$ with $\psi = 2\pi\nu t + \varphi$, i.e. $5 + 4 \cos \psi$. The maximum $|s_L| = A + B = 3$ occurs for $\cos \psi = 1$, i.e. $t = (k - \frac{1}{6})/\nu$ with $k \in \mathbb{Z}$; the minimum $|s_L| = |A - B| = 1$ for $\cos \psi = -1$, i.e. $t = (k + \frac{1}{3})/\nu$.

(c) $\widehat{s}_L$ is concentrated at $\{0, \nu\}$, hence the half-bandwidth is ν and $f_c/\nu = 10^6/10^3 = 1,000$. The relative discrepancy of Proposition 26.2(ii) is $\leq 1/(2\pi f_c T)$; imposing $1/(2\pi f_c T) < 10^{-3}$ gives $f_c T > 1/(2\pi \cdot 10^{-3}) = 159.1549$: 160 carrier periods per symbol suffice.

(d) With the offset carrier, $2s(t) \cos(2\pi f_c t + \theta) = 2\Re\{s_L e^{i2\pi f_c t}\} \cos(2\pi f_c t + \theta)$; writing $2\Re z \cos \alpha = \Re\{ze^{i\alpha}\} + \Re\{ze^{-i\alpha}\}$ with $z = s_L e^{i2\pi f_c t}$ and $\alpha = 2\pi f_c t + \theta$, the first term has frequency around $2f_c$ and is removed by the filter, the second equals $\Re\{s_L e^{-i\theta}\} = s_I \cos \theta + s_Q \sin \theta$. With the quadrature branch one gets $-\Im\{s_L e^{-i\theta}\} \cdot (-1) = -s_I \sin \theta + s_Q \cos \theta$. In complex form $\tilde{s}_L = s_L e^{-i\theta}$: the constellation rotates rigidly by $-\theta$, without deforming. The quantitative consequences are in Exercise 26.6.

BEST PRACTICE

Exercise 26.2 — answers.

(a) $E_s = \frac{8R_1^2 + 8R_2^2}{16} = \frac{R_1^2 + R_2^2}{2} = \frac{R_1^2(1+\gamma^2)}{2}$; imposing $E_s = 1$ gives $R_1 = \sqrt{2/(1+\gamma^2)}$.

(b) The phases are $\frac{\pi}{8} + \frac{k\pi}{4}$ and $\frac{j\pi}{4}$, and the difference equals $\frac{\pi}{8}(1 + 2(k-j))$: an *odd* multiple of $\pi/8$, hence of modulus at least $\pi/8$, a value attained for $k = j$. The three quantities are $2R_1 \sin \frac{\pi}{8}$, $2R_2 \sin \frac{\pi}{8}$ and $\sqrt{R_1^2 + R_2^2 - 2R_1 R_2 \cos \frac{\pi}{8}}$; since $R_2 = \gamma R_1 \geq R_1$, the second is always $\geq$ the first and can never be the minimum: two quantities remain.

(c) The equality $2R_1 \sin \frac{\pi}{8} = \sqrt{R_1^2 + R_2^2 - 2R_1 R_2 \cos \frac{\pi}{8}}$, divided by R_1^2 , gives $4 \sin^2 \frac{\pi}{8} = 1 + \gamma^2 - 2\gamma \cos \frac{\pi}{8}$, i.e. the equation of the statement. The constant term equals $1 - 4 \sin^2 \frac{\pi}{8} = 1 - 2(1 - \cos \frac{\pi}{4}) = 2 \cos \frac{\pi}{4} - 1 = \sqrt{2} - 1 = 0.4142136$, and $2 \cos \frac{\pi}{8} = \sqrt{2 + 2 \cos \frac{\pi}{4}} = \sqrt{2 + \sqrt{2}} = 1.8477590$. The root ≥ 1 is

$$\gamma^* = \cos \frac{\pi}{8} + \sqrt{\cos^2 \frac{\pi}{8} - \sqrt{2} + 1} = 0.9238795 + 0.6628272 = 1.5867067.$$

From this $R_1 = \sqrt{2/(1+\gamma^{*2})} = \sqrt{2/3.5176381} = 0.7540314$, $R_2 = \gamma^* R_1 = 1.1964266$, $d_{\min} = 2R_1 \sin \frac{\pi}{8} = 2 \cdot 0.7540314 \cdot 0.3826834 = 0.5771106$ and $\text{PAPR} = R_2^2 = 1.4314367$. Check: $E_s = (R_1^2 + R_2^2)/2 = (0.5685633 + 1.4314367)/2 = 1$. ✓

(d) Comparison at $M = 16$ and $E_s = 1$: 16-QAM $d_{\min} = 0.6324555$, PAPR 1.8; 4 + 12-APSK $d_{\min} = 0.5848015$, PAPR 1.2763345; 8 + 8-APSK $d_{\min} = 0.5771106$, PAPR 1.4314367. The 8 + 8 is dominated: it has both smaller minimum distance *and* larger PAPR than the 4 + 12, so it is never the best choice among the three. For the satellite transponder near saturation one chooses 4 + 12-APSK (minimum PAPR); for the linear fibre link, 16-QAM (maximum minimum distance, +0.6804302 dB relative to the 4 + 12).

BEST PRACTICE

Exercise 26.3 — answers.

(a) $m = 2$, hence $R_b = mR_s = 4$ Mbit/s; $W = R_s(1 + \beta) = 2 \cdot 1.35 = 2.7$ MHz; $\eta = R_b/W = 4/2.7 = 1.4814815$ bit/s/Hz, which coincides with $m/(1 + \beta) = 2/1.35$ (Corollary 26.7). ✓

(b) Solving $Q(x) = 10^{-6}$ by bisection gives $x = 4.7534243$, hence $E_b/N_0 = x^2/2 = 11.2975213$, i.e. 10.5298317 dB. The value falls between the row at 10 dB ($P_b = 3.87 \cdot 10^{-6}$) and that at 12 dB ($P_b = 9.01 \cdot 10^{-9}$) of the table of Example 26.6. ✓

(c) $P_s = 2 \cdot 10^{-6} - 10^{-12} = 1.999999 \cdot 10^{-6}$. For $P_b = 10^{-9}$ one needs $Q(y) = 10^{-9}$, i.e. $y = 5.9978070$ and $E_b/N_0 = y^2/2 = 17.9868445 = 12.5495498$ dB: an extra $12.5495498 - 10.5298317 = 2.0197181$ dB is needed. Three orders of magnitude on the probability therefore cost just over two decibels: this is the steepness of the curve.

(d) $\text{EVM} = 4\% = 0.04$ implies $\text{SNR}_s = 1/0.04^2 = 625 = 27.9588002$ dB (Proposition 26.10). Since for QPSK $E_s = 2E_b$, $E_b/N_0 = \text{SNR}_s/2 = 312.5 = 24.9485002$ dB, well above the 10.5298317 dB required: the specification is met with more than 14 dB of margin. *Mandatory caution:* the equality $\text{EVM}^2 = N_0/E_s$ holds only if the degradation is entirely additive noise (Remark 26.13); if part of the EVM comes from distortion or phase error, the real margin is smaller.

(e) For $\beta = 0.5$: $p(0.2) = 0.9267741$, $p(1.2) = -0.1095008$, $p(-0.8) = 0.2007514$, hence $D(0.2) = 0.9267741 - 0.1095008 - 0.2007514 = 0.6165219 > 0$: the eye remains open, with opening $2D = 1.2330438$ against the value 2 at exact sampling, i.e. a loss of 38.3478% in amplitude. For $\beta = 0.35$ one obtains $p(0.2) = 0.9312114$, $p(1.2) = -0.1317067$, $p(-0.8) = 0.2171851$ and $D(0.2) = 0.5823196$: the eye is *more* closed, because a smaller roll-off means slower tails and more intersymbol interference under mistimed sampling. This is the hidden price of the spectral efficiency of Corollary 26.7.

26.14. Chapter formulary

Complex baseband. For s real bandpass with carrier f_c and half-bandwidth $B < f_c$:

$$\widehat{s}_+(f) = 2\widehat{s}(f)\mathbf{1}_{\{f>0\}}, \quad s_L = s_+e^{-i2\pi f_c t}, \quad \widehat{s}_L(f) = 2\widehat{s}(f+f_c), \quad \widehat{s}_L(f) = 0 \text{ for } |f| > B.$$

$$s(t) = \Re\{s_L(t)e^{i2\pi f_c t}\} = s_I(t)\cos(2\pi f_c t) - s_Q(t)\sin(2\pi f_c t), \quad s_L = s_I + is_Q.$$

$$s_I = \mathcal{L}_B\{2s\cos(2\pi f_c t)\}, \quad s_Q = \mathcal{L}_B\{-2s\sin(2\pi f_c t)\}, \quad \int s^2 = \frac{1}{2} \int |s_L|^2.$$

Carrier phase error: $\tilde{s}_L = s_L e^{-i\theta}$ (rigid rotation).

Orthogonality of the carriers. For every $f_c > 0$, $T > 0$:

$$\int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt = \frac{\sin^2(2\pi f_c T)}{4\pi f_c}, \quad \int_0^T \cos^2(2\pi f_c t) dt = \frac{T}{2} + \frac{\sin(4\pi f_c T)}{8\pi f_c}.$$

Exact values $0, T/2, T/2 \iff 2f_c T \in \mathbb{Z}$; relative discrepancy always $\leq \frac{1}{2\pi f_c T}$. Orthonormal basis on $[0, T]$: $\phi_1 = \sqrt{2/T} \cos(2\pi f_c t)$, $\phi_2 = -\sqrt{2/T} \sin(2\pi f_c t)$; $\int_0^T s^2 = \frac{T}{2}|a|^2$.

Constellations. $E_s = \frac{1}{M} \sum_j |a_j|^2$, $m = \log_2 M$, $E_b = E_s/m$, $d_{\min} = \min_{j \neq l} |a_j - a_l|$, $\text{PAPR} = \max_j |a_j|^2/E_s$. Chord of the circle: $|e^{i\alpha} - e^{i\beta}| = 2|\sin \frac{\alpha-\beta}{2}|$.

$$M\text{-PSK} : a_k = \sqrt{E_s} e^{i(2\pi k/M + \varphi_0)}, \quad d_{\min} = 2\sqrt{E_s} \sin \frac{\pi}{M}.$$

$$\text{Square } M\text{-QAM: } E_s = \frac{2}{3}(M-1)\Delta^2, \quad d_{\min} = 2\Delta = \sqrt{\frac{6E_s}{M-1}}, \quad \text{PAPR} = \frac{3(\sqrt{M}-1)}{\sqrt{M}+1}.$$

With $E_s = 1$: $2 \sin \frac{\pi}{M} = 2.0000000$ ($M = 2$), 1.4142136 (4), 0.7653669 (8), 0.3901806 (16); $\sqrt{6/(M-1)} = 0.6324555$ (16-QAM), 0.3086067 (64), 0.1533930 (256); PAPR = 1.8, 2.3333333, 2.6470588.

Two-circle APSK, $E_s = 1$. With $M_1 + M_2$ points, $R_2 = \gamma R_1$:

$$R_1 = \sqrt{\frac{M}{M_1 + M_2 \gamma^2}}, \quad d_{\min} = R_1 \min \left\{ 2 \sin \frac{\pi}{M_1}, 2\gamma \sin \frac{\pi}{M_2}, \sqrt{1 + \gamma^2 - 2\gamma \cos \delta} \right\},$$

with δ the minimum angular difference between the two circles. Case 4 + 12 ($\varphi_1 = \pi/4$, $\varphi_2 = 0$, $\delta = \pi/12$):

$$\gamma^* = 1 + \sqrt{3} = 2.7320508, \quad d_{\min} = \frac{2\sqrt{2}}{\sqrt{13 + 6\sqrt{3}}} = 0.5848015, \quad \text{PAPR} = \frac{64 + 8\sqrt{3}}{61} = 1.2763345.$$

Case 8 + 8 ($\varphi_1 = \pi/8$, $\varphi_2 = 0$, $\delta = \pi/8$): $\gamma^* = \cos \frac{\pi}{8} + \sqrt{\cos^2 \frac{\pi}{8} - \sqrt{2} + 1} = 1.5867067$, $d_{\min} = 0.5771106$, PAPR = 1.4314367.

Decision and matched filter. AWGN channel: $p_n(z) = \frac{1}{\pi N_0} e^{-|z|^2/N_0}$, $\mathbb{E}|n|^2 = N_0$, variance $N_0/2$ per component. MAP = ML = minimum distance (equiprobable symbols). Regions:

$$\overline{\mathcal{R}}_j = \bigcap_{l \neq j} \left\{ r : 2\Re[r \overline{(a_l - a_j)}] \leq |a_l|^2 - |a_j|^2 \right\} \quad (\text{convex Voronoi cells}).$$

Matched filter: $\text{SNR}(h) = \frac{\langle s, g \rangle^2}{\frac{N_0}{2} \|g\|^2} \leq \frac{2E}{N_0}$, with equality $\iff h(t) = c s(T - t)$; Cauchy-Schwarz $\langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2$, equality $\iff u = \lambda v$.

Error probability. $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du = \frac{1}{2} \text{erfc}(x/\sqrt{2})$; $Q(0) = \frac{1}{2}$, $Q(-x) = 1 - Q(x)$.

$$P_b^{\text{BPSK}} = P_b^{\text{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right), \quad P_s^{\text{QPSK}} = 2P_b - P_b^2.$$

M -PSK: $Q(\sqrt{2mE_b/N_0} \sin \frac{\pi}{M}) \leq P_s \leq 2Q(\sqrt{2mE_b/N_0} \sin \frac{\pi}{M})$. Values of P_b : $7.8649604 \cdot 10^{-2}$ (0 dB), $1.2500818 \cdot 10^{-2}$ (4 dB), $2.3882908 \cdot 10^{-3}$ (6 dB), $1.9090777 \cdot 10^{-4}$ (8 dB), $9.7361760 \cdot 10^{-6}$ (9.6 dB), $3.8721082 \cdot 10^{-6}$ (10 dB), $9.0060104 \cdot 10^{-9}$ (12 dB). Thresholds: $Q(x) = 10^{-6}$ for $x = 4.7534243$ ($E_b/N_0 = 10.5298317$ dB); $Q(x) = 10^{-9}$ for $x = 5.9978070$ (12.5495498 dB).

Bandwidth, efficiency, eye, EVM. Nyquist criterion:

$$\sum_k p(kT) e^{-i2\pi f kT} = \frac{1}{T} \sum_n P\left(f + \frac{n}{T}\right); \quad p(kT) = \delta_{k0} \iff \sum_n P\left(f + \frac{n}{T}\right) = T.$$

Minimum bandwidth $B = \frac{1}{2T}$, $W = \frac{1}{T} = R_s$; raised cosine of roll-off β : $B = \frac{1+\beta}{2T}$, $W = \frac{1+\beta}{T}$,

$$\eta = \frac{R_b}{W} = \frac{m}{1+\beta} \text{ bit/s/Hz}, \quad \frac{d_{\min}^2}{4E_b} = \frac{3m}{2(2^m - 1)} \quad (\text{square QAM}).$$

Eye opening: $\mathcal{O}(\tau) = 2D(\tau)$, $D(\tau) = p(\tau) - \sum_{k \neq 0} |p(\tau - kT)|$; $\mathcal{O}(0) = 2$ for a Nyquist pulse.
 EVM: $\text{EVM} = \sqrt{\sum |r_k - a_k|^2 / \sum |a_k|^2}$, $\text{EVM} \approx 1/\sqrt{\text{SNR}_s} = 10^{-\text{SNR}_{\text{dB}}/20}$; pure phase error:
 $\text{EVM} = 2|\sin \frac{\theta}{2}|$.

Sources and bibliography

- [1] J. G. Proakis, M. Salehi, *Digital Communications*, 5th edition, McGraw-Hill Science/Engineering/Math, New York, 2008 (publication date declared by the publisher: 6 November 2007), 1168 pages, ISBN 978-0-07-295716-7 (ISBN-10 0-07-295716-6). It is the reference text for the complex baseband representation of bandpass signals, for the optimal receiver on the additive white Gaussian noise channel (matched filter, correlator, maximum-likelihood decision), and for the error probabilities of PSK and QAM constellations: these are the contents of Theorems 26.1, 26.6, 26.8, 26.9 and 26.10, here re-exposed and proved in full and independently. *Verified on 7 September 2026* (Open Library catalogue record for the ISBN-13: authors Proakis and Salehi, publisher, "5 edition" edition, date and page count checked one by one). Chapter or page numbers are *not* given, because internal location was not verified on a copy of the cited edition: earlier editions have different numbering.
- [2] S. Haykin, *Communication Systems*, 4th edition, John Wiley & Sons, New York, 2001, ISBN 0-471-17869-1 (ISBN-13 978-0-471-17869-9). It is the source for the orthogonality of the quadrature carriers and for the structure of the IQ modulator and demodulator (Proposition 26.2, Corollary 26.1, Figure 26.2), as well as for the treatment of the analytic signal and the Hilbert transform recalled in Remark 26.2. *Verified on 7 September 2026* (Open Library catalogue record: author Simon S. Haykin, publisher Wiley, year 2001, ISBN-10 and ISBN-13 checked). Here too no pages are cited, not having been verified on a copy.
- [3] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd edition, Prentice-Hall PTR, Upper Saddle River (New Jersey), 2001, 1079 pages, ISBN 0-13-084788-7. It is the source for the geometric presentation of constellations, for the Gray mapping, for the eye diagram, and for the link between minimum distance and spectral efficiency (Definition 26.5, Section 26.8, Section 26.9). *Verified on 7 September 2026* (Open Library catalogue record for the ISBN-10: title and subtitle, author Bernard Sklar, publisher Prentice-Hall PTR, "2nd ed.", year 2001, 1079 pages).
- [4] J. M. Wozencraft, I. M. Jacobs, *Principles of Communication Engineering*, John Wiley & Sons, New York, 1965, about 720 pages (reprint Waveland Press, 1990, ISBN 0-88133-554-1). It is the classic text in which the white Gaussian noise model and the theory of optimal decision are set up in the geometric form adopted here: signals as vectors of a space with a scalar product, noise described by its projections, decision as a partition of the space. It is the source of the model imported in Definition 26.10 and in Remark 26.8. *Verified on 7 September 2026* (Open Library catalogue record: authors John M. Wozencraft and Irwin Mark Jacobs, publishers Wiley and Waveland Press, years 1965 and 1990, ISBN of the reprint checked).
- [5] Sources internal to the treatise, used and not reproved: M -th roots of unity and regular polygons (Chapter 6); the Fourier transform, inversion, Plancherel–Parseval and the Poisson summation formula (Chapter 19); uniqueness of Fourier coefficients (Chapter 17); the sampling theorem (Chapter 24). Every cross-reference is made by label, not by a hand-written number.

OFDM: the circle that carries broadband

A telephone cable, a stretch of air between an antenna and a rooftop, a fibre: none of these channels carries a pulse without deforming it. The pulse leaves narrow and arrives spread out, because the signal reaches the receiver along several paths of different length and the replicas add up out of phase. If the symbols follow one another faster than that spreading lasts, each one invades the next: this is intersymbol interference, and fighting it in a channel with many taps requires an equalizer that, in the time domain, is an expensive and delicate machine.

Orthogonal frequency-division multiplexing — OFDM — solves the problem by changing its shape. Instead of one fast stream on a single carrier, N slow streams are sent on N nearby carriers; and the carriers are chosen so that, although their spectra overlap heavily, they are orthogonal over the interval of one symbol. At that point comes the discovery that made the idea practicable: the samples of the signal built this way are, literally, an inverse discrete Fourier transform of the data, and the receiver that analyses them performs a forward DFT. Transmitter and receiver are two FFTs, and nothing else.

One difficulty remains. The DFT turns *circular* convolution into a product (Theorem 22.5), but a physical channel performs a *linear* convolution: there is no reason for the time axis to close back on itself. The cyclic prefix is the device that forces it to do so. A copy of the block's tail is prepended to it; the channel, unable to tell that copy from the true signal, produces on the useful samples exactly the circular convolution; the channel matrix is then circulant, its diagonalisation is the one — unique, independent of the channel — of Theorem 22.8, and equalisation reduces to N divisions by a complex number. This is the chapter in which the time line is bent into a circle, and the circle settles the bill: the group $\mathbb{Z}/N\mathbb{Z}$ replaces $\mathbb{Z}$, and with it comes the whole algebra of Chapter 22.

The chapter follows this chain in order and in full. The orthogonality condition over the symbol time is proved, first with the integral (Theorem 27.1) and then in the sampled version (Theorem 27.2); writing the two computations side by side shows that the modulator *is* an IDFT and the demodulator a DFT (Theorems 27.3 and 27.4); it is proved, with the modular indices written out in full, that the cyclic prefix converts the channel's linear convolution into a circular one (Theorem 27.5) and that the condition $N_g \geq L - 1$ is not only sufficient but necessary (Theorem 27.6); from this follows the diagonality of the channel in frequency (Corollary 27.3) and single-coefficient equalisation, with the error in the presence of noise computed exactly (Theorem 27.8) and the equalizer optimal in mean square derived in full (Theorem 27.9). Next come the efficiency loss due to the prefix (Proposition 27.3), the carrier frequency offset with the exact formula for intercarrier interference and the signal-to-interference ratio (Lemma 27.3, Theorem 27.10, Corollary 27.6), and the ratio of peak power to average power, with the worst case proved (Theorem 27.11) and the statistical estimate stated under explicit hypotheses (Proposition 27.7) and declared imported for the part that is not closed here (Observation 27.13). The last two sections work out in full, in exact symbolic form and in numerical form, the examples $N = 4$ and $N = 8$, with the verification that the channel does become diagonal.

The results of Chapter 22 — the orthogonality of the exponential vectors (Theorem 22.1), the

inversion formula (Theorem 22.2), circular convolution (Definition 22.4 and Theorem 22.5), Parseval's identity (Theorem 22.6), circulant matrices (Definition 22.5, Proposition 22.5, Theorem 22.8) — and those of Chapter 23 on the cost of computation (Proposition 23.1, Theorem 23.5, Corollary 23.4, Theorem 23.10) are recalled and used here, and never reproved.

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages at level **D**: the equalizer optimal in mean square of Section 27.4 and the statistics of the peak ratio of Section 27.7, which require elementary notions of probability.

Internal prerequisites. N -th roots of unity and their arithmetic (Chapter 6); the discrete Fourier transform, orthogonality, inversion, circular convolution, Parseval, circulant matrices and their diagonalisation (Chapter 22: Definition 22.1, Definition 22.2, Definition 22.3, Definition 22.4, Definition 22.5, Theorem 22.1, Theorem 22.2, Theorem 22.5, Theorem 22.6, Theorem 22.8, Proposition 22.5, Proposition 22.4); the fast algorithm and its cost (Chapter 23: Proposition 23.1, Theorem 23.5, Corollary 23.4, Theorem 23.10); the Fourier transform on $\mathbb{R}$ and its inversion formula (Chapter 19).

External prerequisites, assumed and not reproved:

- the Riemann integral of a continuous complex-valued function on a compact interval, and its linearity;
- elementary linear algebra: matrix-vector product, determinant, eigenvalues and eigenvectors, unitary matrices;
- the Cauchy–Schwarz inequality in $\mathbb{C}^N$ and the triangle inequality;
- elementary notions of probability: expected value, variance, independent variables, the Gaussian distribution and change of variable in polar coordinates (used only in Section 27.7 and in Section 27.4, always under stated hypotheses).

Sign convention. As throughout the Volume, $\omega_N = e^{-2\pi i/N}$ and $\langle j \rangle_N$ is the unique integer in $\{0, \dots, N-1\}$ congruent to j modulo N (Definition 22.1 and Definition 22.4). The subcarriers therefore carry the *opposite* sign: $e^{+2\pi i k n/N} = \omega_N^{-kn}$. This is not an oversight, and it is the reason why the modulator turns out to be an *inverse* transform rather than a forward one.

27.1. The orthogonal subcarriers

The notation is fixed once and for all. The number of subcarriers is $N \geq 2$; the duration of the *useful* OFDM symbol is $T > 0$; the spacing between two consecutive subcarriers is $\Delta f > 0$; the sampling step is $T_s = T/N$ and the sampling instants are $t_n = nT_s = nT/N$ for $n = 0, \dots, N-1$. The data are N complex numbers $c = (c_0, \dots, c_{N-1}) \in \mathbb{C}^N$, the points of a constellation (BPSK, QPSK, QAM): what they are does not matter in this chapter, only that they are given complex numbers.

Definition 27.1: OFDM subcarriers. Given $T > 0$ and $\Delta f > 0$, the k -th subcarrier of duration T and spacing Δf is the function

$$\phi_k : [0, T] \longrightarrow \mathbb{C}, \quad \phi_k(t) = e^{2\pi i k \Delta f t}, \quad k = 0, 1, \dots, N-1.$$

On the space of continuous functions $[0, T] \rightarrow \mathbb{C}$ one considers the Hermitian product

$$\langle u, v \rangle_T = \int_0^T u(t) \overline{v(t)} dt.$$

The subcarriers are called **orthogonal** if $\langle \phi_k, \phi_\ell \rangle_T = 0$ for every pair $k \neq \ell$ of indices in $\{0, \dots, N-1\}$.

Note that $|\phi_k(t)| = 1$ for every t : each subcarrier turns on the unit circle at angular speed $2\pi k\Delta f$, completing $k\Delta f T$ turns over the interval $[0, T]$. The whole chapter depends on that number of turns being an integer.

Theorem 27.1: Orthogonality condition over the symbol time. Let $N \geq 2$, $T > 0$, $\Delta f > 0$ and let $\phi_0, \dots, \phi_{N-1}$ be the subcarriers of Definition 27.1. Then

$$\langle \phi_k, \phi_\ell \rangle_T = T \delta_{k\ell} \quad \text{for every } k, \ell \in \{0, \dots, N-1\}$$

if and only if the product $\Delta f T$ is a positive integer.

Proof. Step 1: the case $k = \ell$. For every k , $\phi_k(t)\overline{\phi_k(t)} = |\phi_k(t)|^2 = 1$, hence $\langle \phi_k, \phi_k \rangle_T = \int_0^T 1 dt = T$. This equality holds for every choice of $\Delta f > 0$ and is therefore not involved in the condition to be proved.

Step 2: the case $k \neq \ell$, computing the integral. Let $m = k - \ell$; since $k, \ell \in \{0, \dots, N-1\}$ and $k \neq \ell$, the integer m satisfies $1 \leq |m| \leq N-1$. Then

$$\langle \phi_k, \phi_\ell \rangle_T = \int_0^T e^{2\pi i k \Delta f t} \overline{e^{2\pi i \ell \Delta f t}} dt = \int_0^T e^{2\pi i (k-\ell) \Delta f t} dt = \int_0^T e^{2\pi i m \Delta f t} dt,$$

where $\overline{e^{i\theta}} = e^{-i\theta}$ for real θ was used. Since $m \neq 0$ and $\Delta f > 0$, the number $2\pi i m \Delta f$ is nonzero and the function $t \mapsto \frac{1}{2\pi i m \Delta f} e^{2\pi i m \Delta f t}$ is an antiderivative of the integrand; by the fundamental theorem of calculus

$$\langle \phi_k, \phi_\ell \rangle_T = \left[\frac{e^{2\pi i m \Delta f t}}{2\pi i m \Delta f} \right]_{t=0}^{t=T} = \frac{e^{2\pi i m \Delta f T} - 1}{2\pi i m \Delta f}. \quad (27.1)$$

The denominator is nonzero, so the left-hand side vanishes if and only if $e^{2\pi i m \Delta f T} = 1$, that is — since $e^{2\pi i x} = 1$ is equivalent to $x \in \mathbb{Z}$ for real x — if and only if $m \Delta f T \in \mathbb{Z}$.

Step 3: sufficiency. Suppose $\Delta f T = q$ with q an integer, $q \geq 1$. Then for every integer m one has $m \Delta f T = mq \in \mathbb{Z}$, and by Step 2 the integral (27.1) vanishes for every pair $k \neq \ell$. Together with Step 1 this gives $\langle \phi_k, \phi_\ell \rangle_T = T \delta_{k\ell}$.

Step 4: necessity. Suppose now that $\langle \phi_k, \phi_\ell \rangle_T = 0$ for every $k \neq \ell$. Since $N \geq 2$, the pair $k = 1, \ell = 0$ is admissible and gives $m = 1$; by Step 2 the vanishing of the integral forces $1 \cdot \Delta f T \in \mathbb{Z}$, that is $\Delta f T \in \mathbb{Z}$. On the other hand $\Delta f > 0$ and $T > 0$, hence $\Delta f T > 0$: a positive integer. This is what was to be shown. $\square$

Corollary 27.1: Minimum spacing and occupied bandwidth. Among all the admissible spacings of Theorem 27.1 the smallest is $\Delta f = 1/T$; with $\Delta f T = q$ a positive integer, the N subcarriers have frequencies $0, \Delta f, \dots, (N-1)\Delta f$ and occupy the frequency interval of width $B = (N-1)\Delta f$ between the first and the last, that is $B = q(N-1)/T$. The number of constellation symbols transmitted per unit time and per unit bandwidth is therefore

$$\frac{N}{BT} = \frac{N}{q(N-1)} \xrightarrow{N \rightarrow \infty} \frac{1}{q},$$

and it is maximal for $q = 1$.

Proof. By Theorem 27.1 the admissible spacings are exactly the numbers $\Delta f = q/T$ with q a positive integer; the smallest is obtained for $q = 1$ and equals $1/T$. The subcarrier frequencies are $k\Delta f$ for $k = 0, \dots, N-1$, so the distance between the first and the last is $(N-1)\Delta f = q(N-1)/T$. In a time interval T , N constellation symbols are transmitted, so

the ratio between their number and the product BT is

$$\frac{N}{BT} = \frac{N}{\frac{q(N-1)}{T} \cdot T} = \frac{N}{q(N-1)}.$$

The factor $\frac{N}{N-1} = 1 + \frac{1}{N-1}$ tends to 1 and is decreasing in N ; the factor $1/q$ is maximal for $q = 1$. This gives the last statement. $\square$

Remark 27.1: Why $q = 1$ is always chosen. Corollary 27.1 says that choosing $q = 2$ doubles the occupied bandwidth without transmitting one more symbol: the spectral efficiency is halved. The rest of the chapter therefore assumes

$$\Delta f = \frac{1}{T}, \quad \phi_k(t) = e^{2\pi i k t / T}.$$

With this choice the k -th subcarrier completes exactly k full turns on the unit circle over the interval $[0, T]$: the subcarriers are the characters of the group $\mathbb{R}/T\mathbb{Z}$, and the orthogonality of Theorem 27.1 is the orthogonality of characters already seen, in another guise, in Chapter 13.

Proposition 27.1: The spectrum of a truncated subcarrier. Let $\Delta f = 1/T$ and let $\psi_k = \phi_k \cdot \mathbf{1}_{[0, T]}$ be the k -th subcarrier extended by zero outside the symbol interval. Then its Fourier transform $\Psi_k(f) = \int_{\mathbb{R}} \psi_k(t) e^{-2\pi i f t} dt$ equals

$$\Psi_k(f) = T e^{-i\pi(fT-k)} \operatorname{sinc}(fT-k), \quad \operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \quad (x \neq 0), \quad \operatorname{sinc}(0) = 1.$$

In particular $|\Psi_k(f)| = T |\operatorname{sinc}(fT-k)|$, and

$$\Psi_k\left(\frac{\ell}{T}\right) = T \delta_{k\ell} \quad \text{for every } \ell \in \mathbb{Z} :$$

the spectrum of the k -th subcarrier vanishes exactly at the centres of all the others.

Proof. Since ψ_k vanishes outside $[0, T]$,

$$\Psi_k(f) = \int_0^T e^{2\pi i k t / T} e^{-2\pi i f t} dt = \int_0^T e^{-2\pi i a t} dt, \quad a := f - \frac{k}{T}.$$

If $a = 0$ the integrand equals 1 and the integral equals T , in agreement with the proposed formula, since in that case $fT - k = 0$, $\operatorname{sinc}(0) = 1$ and $e^0 = 1$. If $a \neq 0$, with the same antiderivative as in Step 2 of the proof of Theorem 27.1,

$$\Psi_k(f) = \frac{1 - e^{-2\pi i a T}}{2\pi i a}.$$

Let $\theta = 2\pi a T$ and factor out $e^{-i\theta/2}$:

$$1 - e^{-i\theta} = e^{-i\theta/2} (e^{i\theta/2} - e^{-i\theta/2}) = e^{-i\theta/2} \cdot 2i \sin \frac{\theta}{2},$$

where the last equality is Euler's formula for the sine. Substituting $\theta/2 = \pi a T$,

$$\Psi_k(f) = \frac{e^{-i\pi a T} 2i \sin(\pi a T)}{2\pi i a} = e^{-i\pi a T} \frac{\sin(\pi a T)}{\pi a} = T e^{-i\pi a T} \frac{\sin(\pi a T)}{\pi a T} = T e^{-i\pi a T} \operatorname{sinc}(aT).$$

Since $aT = fT - k$, the announced formula follows; the modulus follows from $|e^{-i\pi a T}| = 1$. For the last statement, let $f = \ell/T$ with $\ell \in \mathbb{Z}$: then $fT - k = \ell - k$ is an integer. If $\ell = k$ this is the case $a = 0$ and $\Psi_k(k/T) = T$. If $\ell \neq k$ then $\ell - k$ is a nonzero integer and $\sin(\pi(\ell - k)) = 0$, while $\pi(\ell - k) \neq 0$: hence $\operatorname{sinc}(\ell - k) = 0$ and $\Psi_k(\ell/T) = 0$. $\square$

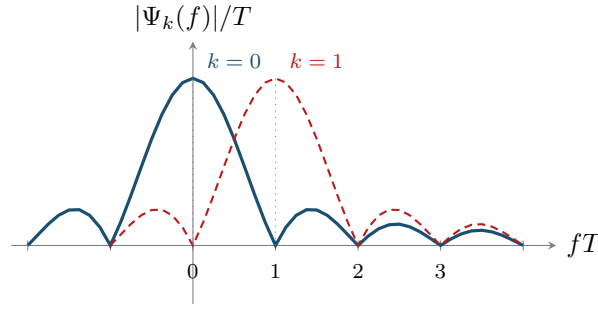


Figure 27.1: The modulus of the spectrum of two truncated subcarriers over the symbol interval (Proposition 27.1), plotted as $|\Psi_k(f)|/T$ against fT : solid blue for $k = 0$, dashed red for $k = 1$. The ordinates are the values of $|\text{sinc}(fT - k)|$ computed with step 0.125 (for instance $|\text{sinc}(\frac{1}{2})| = 2/\pi = 0.6366$ and $|\text{sinc}(\frac{3}{2})| = 2/(3\pi) = 0.2122$). The two spectra overlap over most of the main lobe, but each vanishes at the centre of the other: the zeros fall at $fT - k = \pm 1, \pm 2, \dots$, that is exactly at the centres $f = \ell/T$ of the other subcarriers. This is orthogonality read in the frequency domain.

Remark 27.2: Orthogonality is not band separation. Figure 27.1 shows the point that is most often misunderstood. The spectra of the subcarriers overlap heavily: the main lobe of Ψ_k is $2/T$ wide, that is twice the spacing, and halfway between two subcarriers ($fT - k = \frac{1}{2}$) the value $|\text{sinc}(1/2)| = 2/\pi = 0.6366$ is still 63.66% of the maximum. No filter would separate the subcarriers. What makes them distinguishable is not spectral separation but the choice of the reading point: the demodulator samples the spectrum exactly at the centres $f = \ell/T$, and there — by Proposition 27.1 — every subcarrier other than the ℓ -th one is zero. Shift the reading point, and orthogonality disappears: this is precisely the mechanism of the carrier frequency offset studied in Section 27.6.

Theorem 27.2: Orthogonality of the sampled subcarriers. Let $\Delta f = 1/T$ and let $t_n = nT/N$, $n = 0, \dots, N-1$, be the N sampling instants. Set $v_k = (\phi_k(t_n))_{n=0}^{N-1} \in \mathbb{C}^N$. Then

$$v_k = (\omega_N^{-kn})_{n=0}^{N-1} = e_k,$$

where e_k is the exponential vector of Theorem 22.1, and

$$\langle v_k, v_\ell \rangle = \sum_{n=0}^{N-1} \phi_k(t_n) \overline{\phi_\ell(t_n)} = N \delta_{k\ell}, \quad k, \ell \in \{0, \dots, N-1\}.$$

More precisely, for any integers k, ℓ the sum equals N if $k \equiv \ell \pmod{N}$ and 0 otherwise.

Proof. Step 1: identifying the vectors. By definition $\phi_k(t_n) = e^{2\pi i k t_n / T} = e^{2\pi i k (nT/N) / T} = e^{2\pi i k n / N}$. Since $\omega_N = e^{-2\pi i / N}$ one has $e^{2\pi i k n / N} = \omega_N^{-kn}$, which is by definition the n -th component of the vector e_k of Theorem 22.1. Hence $v_k = e_k$.

Step 2: orthogonality. For $k, \ell \in \{0, \dots, N-1\}$ Theorem 22.1 states $\langle e_k, e_\ell \rangle = N \delta_{k\ell}$, and Step 1 allows that statement to be read as the desired one. There is no need to repeat the computation.

Step 3: the case of arbitrary integer indices. If $k, \ell \in \mathbb{Z}$, one has

$$\langle v_k, v_\ell \rangle = \sum_{n=0}^{N-1} \omega_N^{-kn} \overline{\omega_N^{-\ell n}} = \sum_{n=0}^{N-1} \omega_N^{(\ell-k)n},$$

having used $|\omega_N| = 1$, that is $\overline{\omega_N^m} = \omega_N^{-m}$. Setting $m = \ell - k$, Lemma 22.1 gives $\sum_{n=0}^{N-1} \omega_N^{mn} = N$ if $N \mid m$ and 0 otherwise, which is exactly the statement. $\square$

Remark 27.3: The sampled subcarriers are counted modulo N . Step 3 of Theorem 27.2 contains a substantial difference from the continuous case. In the continuous case the subcarriers ϕ_k are pairwise orthogonal for every pair of distinct integers $k \neq \ell$; after sampling at N points, however, $v_k = v_{k+N}$, because $\omega_N^{-(k+N)n} = \omega_N^{-kn} \omega_N^{-Nn} = \omega_N^{-kn}$. The subcarriers of index k and $k + N$ become indistinguishable: this is aliasing, already seen in Chapter 21 and in Theorem 22.9. This is the reason why the number of usable subcarriers is exactly N , the number of samples: not one more, not one fewer.

27.2. The modulator is an IFFT, the demodulator is an FFT

Up to this point OFDM is only a sum of exponentials: one could build it with N oscillators and N multipliers, and that is exactly how the first paper on the subject, by Chang in 1966, described it. The step that turned it into a technology was the observation that the samples of that sum *already are* a discrete transform: no oscillator, no filter bank, a single call to the FFT. This section proves the claim by writing the two computations — the modulator's computation and the inverse transform's computation — side by side.

Definition 27.2: Baseband OFDM symbol and its samples. Let $c = (c_0, \dots, c_{N-1}) \in \mathbb{C}^N$ be the data vector and let $\Delta f = 1/T$. The **baseband OFDM symbol** associated with c is the function

$$s : [0, T) \longrightarrow \mathbb{C}, \quad s(t) = \sum_{k=0}^{N-1} c_k \phi_k(t) = \sum_{k=0}^{N-1} c_k e^{2\pi i k t / T}.$$

The vector of its samples at the instants $t_n = nT/N$ is denoted $s = (s[0], \dots, s[N-1]) \in \mathbb{C}^N$, where

$$s[n] := s(t_n) = \sum_{k=0}^{N-1} c_k e^{2\pi i k n / N} = \sum_{k=0}^{N-1} c_k \omega_N^{-kn}, \quad n = 0, \dots, N-1.$$

The map $c \mapsto s$ is called the **OFDM modulator**.

Theorem 27.3: The OFDM modulator is the inverse discrete transform. For every $c \in \mathbb{C}^N$ the vector s of Definition 27.2 satisfies

$$s = N \cdot \mathcal{F}_N^{-1} c, \quad \text{that is} \quad s[n] = N (\mathcal{F}_N^{-1} c)[n] \quad \text{for every } n,$$

where $\mathcal{F}_N^{-1}$ is the inverse discrete transform of Definition 22.2. Equivalently, in matrix form,

$$s = \overline{W_N} c = W_N^* c = \sqrt{N} F_N^* c,$$

with W_N and F_N the matrices of Definition 22.3.

Proof. Step 1: the two computations, side by side. Write out in full the definition of the modulator and the definition of the inverse transform, for the same index n :

$$\underbrace{s[n] = \sum_{k=0}^{N-1} c_k \omega_N^{-kn}}_{\text{Definition 27.2}} \quad \underbrace{(\mathcal{F}_N^{-1} c)[n] = \frac{1}{N} \sum_{k=0}^{N-1} c_k \omega_N^{-kn}}_{\text{Definition 22.2}}.$$

The two sums have the same summation index, the same bounds 0 and $N - 1$, and the same general term $c_k \omega_N^{-kn}$: they differ only by the factor $1/N$ in front of the second one. Hence, multiplying the second by N ,

$$N(\mathcal{F}_N^{-1}c)[n] = N \cdot \frac{1}{N} \sum_{k=0}^{N-1} c_k \omega_N^{-kn} = \sum_{k=0}^{N-1} c_k \omega_N^{-kn} = s[n],$$

for every $n = 0, \dots, N - 1$. Since two vectors of $\mathbb{C}^N$ with the same components are equal, $s = N\mathcal{F}_N^{-1}c$.

Step 2: the matrix form. By Definition 22.3 $(W_N)_{nk} = \omega_N^{nk}$; since $|\omega_N| = 1$ one has $\overline{\omega_N^{nk}} = \omega_N^{-nk}$, hence $(\overline{W_N})_{nk} = \omega_N^{-nk}$ and

$$(\overline{W_N}c)[n] = \sum_{k=0}^{N-1} (\overline{W_N})_{nk} c_k = \sum_{k=0}^{N-1} \omega_N^{-nk} c_k = s[n].$$

Moreover W_N is symmetric — $(W_N)_{nk} = \omega_N^{nk} = (W_N)_{kn}$ — so $W_N^* = \overline{W_N}^\top = \overline{W_N}$; and from $F_N = N^{-1/2}W_N$ it follows that $F_N^* = N^{-1/2}W_N^*$, that is $W_N^* = \sqrt{N}F_N^*$. $\square$

Theorem 27.4: The OFDM demodulator is the forward discrete transform. Let $s \in \mathbb{C}^N$ be the vector of samples of an OFDM symbol built with the data $c \in \mathbb{C}^N$. Then

$$c_k = \frac{1}{N} (\mathcal{F}_N s)[k] = \frac{1}{N} \sum_{n=0}^{N-1} s[n] \omega_N^{kn}, \quad k = 0, \dots, N - 1.$$

Recovering the data from the samples is therefore a forward DFT followed by division by N ; in particular the modulator is injective and the data are recovered exactly.

Proof. First route (by substitution and orthogonality). Substitute the definition of $s[n]$ into the sum on the right-hand side, calling j the internal dummy index so as not to confuse it with k :

$$\frac{1}{N} \sum_{n=0}^{N-1} s[n] \omega_N^{kn} = \frac{1}{N} \sum_{n=0}^{N-1} \left(\sum_{j=0}^{N-1} c_j \omega_N^{-jn} \right) \omega_N^{kn}.$$

Both sums are finite, so they may be exchanged without any convergence hypothesis:

$$= \frac{1}{N} \sum_{j=0}^{N-1} c_j \sum_{n=0}^{N-1} \omega_N^{(k-j)n}.$$

By Lemma 22.1 the inner sum equals N if $N \mid (k-j)$ and 0 otherwise; since $j, k \in \{0, \dots, N-1\}$ one has $|k-j| \leq N-1$, so $N \mid (k-j)$ occurs only for $j = k$. Only the term $j = k$ survives:

$$\frac{1}{N} \sum_{n=0}^{N-1} s[n] \omega_N^{kn} = \frac{1}{N} c_k \cdot N = c_k.$$

Second route (from Theorem 27.3). By Theorem 27.3, $s = N\mathcal{F}_N^{-1}c$; applying $\mathcal{F}_N$ to both sides and using Theorem 22.2, according to which $\mathcal{F}_N\mathcal{F}_N^{-1}$ is the identity of $\mathbb{C}^N$,

$$\mathcal{F}_N s = N \mathcal{F}_N \mathcal{F}_N^{-1} c = N c,$$

whence $c = \frac{1}{N} \mathcal{F}_N s$. The two routes give the same result: the first is the explicit computation, the second shows that this computation is nothing but the inversion formula of Chapter 22.

✓

Injectivity follows: if two data vectors c and c' produce the same s , the formula just proved gives $c_k = \frac{1}{N} (\mathcal{F}_N s)[k] = c'_k$ for every k . $\square$

Corollary 27.2: Cost of the modulator and the demodulator. Count, as in Chapter 23, one complex multiplication and one complex addition as elementary operations. Then:

- (i) computing s from Definition 27.2, sum by sum, costs N^2 complex multiplications and $N(N-1)$ complex additions;
- (ii) the same vector s , computed with the radix-2 algorithm for $N = 2^m$, costs $\frac{N}{2} \log_2 N$ complex multiplications and $N \log_2 N$ complex additions, of which only $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$ are nontrivial multiplications;
- (iii) the demodulator has exactly the same cost as the modulator, plus N divisions by the real number N .

Proof. (i) By Theorem 27.3, $s = \overline{W_N}c$: this is a matrix-vector product with an $N \times N$ matrix whose entries have modulus 1, the same structure as the product $X = W_N x$ counted in Proposition 23.1. The proof of that proposition counts N multiplications and $N-1$ additions for each of the N components, regardless of whether the entries are ω_N^{nk} or their conjugates ω_N^{-nk} ; the counts $N \cdot N = N^2$ and $N(N-1)$ follow.

(ii) Theorem 23.10 states that the inverse transform is obtained from the forward one by conjugation, without a second algorithm: precisely, $\mathcal{F}_N^{-1}X = \frac{1}{N} \overline{\mathcal{F}_N X}$. Applying it to $s = N \mathcal{F}_N^{-1}c$ gives $s = \overline{\mathcal{F}_N c}$: the modulator is realised by conjugating the data, performing *one* forward FFT, and conjugating the output. The two conjugations cost neither multiplications nor additions (they only change the sign of the imaginary part), and the factor $\frac{1}{N}$ of Theorem 23.10 cancels with the factor N of Theorem 27.3. What remains are the counts of Theorem 23.5 and Corollary 23.4, which are the ones stated.

(iii) By Theorem 27.4 the demodulator is $c = \frac{1}{N} \mathcal{F}_N s$: a forward FFT, with the same counts as in (i) and (ii) because it is the same matrix up to conjugation, followed by the N divisions by N . $\square$

Table 27.1: Cost of the OFDM modulator: direct evaluation of the sum of Definition 27.2 against radix-2 FFT (Corollary 27.2). All entries are exact numbers. The last column is the ratio between the multiplications of the two preceding columns; the values were recomputed with an independent script.

N	direct sum		radix-2 FFT		ratio
	mult. N^2	add. $N(N-1)$	mult. $\frac{N}{2} \log_2 N$	nontrivial	
4	16	12	4	0	4.000000
8	64	56	12	2	5.333333
64	4096	4032	192	98	21.333333
256	65536	65280	1024	642	64.000000
1024	1048576	1047552	5120	3586	204.800000
2048	4194304	4192256	11264	8194	372.363636
4096	16777216	16773120	24576	18434	682.666667

Example 27.1: The numbers for $N = 2048$, computed twice. Let $N = 2048 = 2^{11}$, hence $m = \log_2 N = 11$.

First route, closed formula. Direct sum: $N^2 = 2048^2 = 4194304$ multiplications. FFT: $\frac{N}{2} \log_2 N = 1024 \cdot 11 = 11264$ multiplications; nontrivial ones $11264 - \frac{3 \cdot 2048}{2} + 2 = 11264 - 3072 + 2 = 8194$. Ratio $\frac{4194304}{11264} = 372.363636\dots$, that is $\frac{4096}{11}$.

Second route, script. A script that recomputes N^2 , $\frac{N}{2}m$ and $\frac{N}{2}m - \frac{3N}{2} + 2$ for all powers of two from 2^2 to 2^{12} returns, for $N = 2048$, the values 4194304, 11264 and 8194, and the ratio 372.363636. $\checkmark$

Consistency check. The ratio is exactly $\frac{N^2}{2 \log_2 N} = \frac{2N}{\log_2 N} = \frac{4096}{11}$, and $4096 = 11 \cdot 372 + 4$, hence $\frac{4096}{11} = 372 + \frac{4}{11} = 372.\overline{36}$: the repeating decimal 36 confirms the reading of Table 27.1. ✓

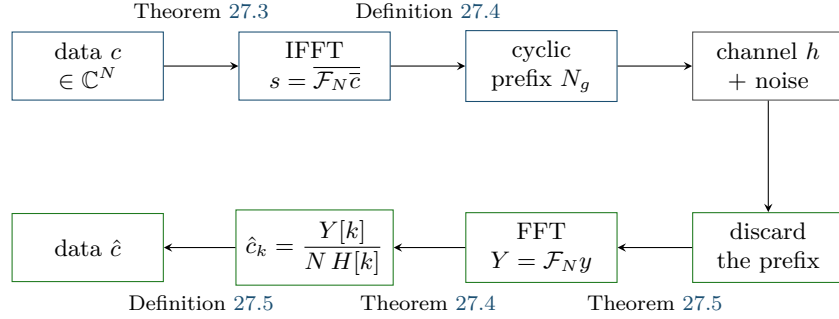


Figure 27.2: The complete OFDM chain. On top the transmitter, at the bottom the receiver read from right to left. Each block is a result of this chapter: the only costly operation is the FFT/IFFT pair (Corollary 27.2), and channel equalisation reduces to N complex divisions (Definition 27.5).

Remark 27.4: The three ideas, and who had them. The three ideas that make up OFDM arose separately, years apart. The orthogonality of subcarriers with overlapping spectra — Section 27.1 — is due to R. W. Chang, in a 1966 paper in the *Bell System Technical Journal* **45**, pp. 1775–1796. The realisation by DFT — Theorems 27.3 and 27.4 — is due to S. B. Weinstein and P. M. Ebert, *IEEE Transactions on Communication Technology* **19** (1971), pp. 628–634, who eliminated the bank of oscillators by replacing it with a transform. The cyclic prefix — Section 27.3 — is due to A. Peled and A. Ruiz, *ICASSP 1980*, vol. 5, pp. 964–967, who observed that filling the guard interval with a copy of the block’s tail, instead of with silence, turns the linear convolution into a circular one. All three sources are listed, with DOI and verification date, in the sources environment at the end of the chapter. Note that the historical order and the logical order coincide: without orthogonality there is nothing to transform, without the DFT the computation cannot be carried out, and without the cyclic prefix the DFT diagonalises the wrong matrix.

27.3. The channel, the cyclic prefix and circular convolution

Theorem 22.5 says that the DFT turns *circular* convolution into a product. A physical channel, however, performs a *linear* convolution: the echo of a sample arrives later, it does not reappear at the start of the block. Between the two worlds there is a discrepancy, and the cyclic prefix is what removes it. This section contains the heart of the chapter.

Definition 27.3: Finite impulse response channel. Let $L \geq 1$ be an integer and let $h = (h_0, h_1, \dots, h_{L-1}) \in \mathbb{C}^L$ with $h_{L-1} \neq 0$. The **channel** of impulse response h and length L is the map that associates to a sequence $x : \mathbb{Z} \rightarrow \mathbb{C}$ the sequence

$$r[n] = \sum_{l=0}^{L-1} h_l x[n-l], \quad n \in \mathbb{Z} \quad (\text{linear convolution}).$$

The integer $L - 1$ is called the **dispersion** (or *delay spread* in samples) of the channel: it is the maximum delay, measured in sampling steps, with which a replica of the signal reaches the receiver. In physical time the dispersion equals $\tau_{\max} = (L - 1) T_s$.

Definition 27.4: Cyclic prefix. Let $s \in \mathbb{C}^N$ be the samples of an OFDM symbol (Definition 27.2) and let $N_g \geq 0$ be an integer with $N_g \leq N$. The **block with cyclic prefix** of guard length N_g is the vector $\tilde{s} \in \mathbb{C}^{N+N_g}$ defined by

$$\tilde{s}[n] = s[\langle n - N_g \rangle_N], \quad n = 0, 1, \dots, N + N_g - 1.$$

Writing out the two parts in full,

$$\tilde{s}[n] = \begin{cases} s[N - N_g + n], & n = 0, 1, \dots, N_g - 1 \quad (\text{the prefix: the tail of } s), \\ s[n - N_g], & n = N_g, \dots, N + N_g - 1 \quad (\text{the useful block}). \end{cases}$$

The prefix is thus a literal copy of the last N_g samples of the block, prepended to the block itself.

Remark 27.5: The two forms of the definition are the same. It is worth checking the equivalence, because the whole section depends on it. For $n \in \{0, \dots, N_g - 1\}$ one has $-N_g \leq n - N_g \leq -1$, so $n - N_g$ is negative and greater than or equal to $-N$ (because $N_g \leq N$); the representative of $n - N_g$ in $\{0, \dots, N - 1\}$ is then $\langle n - N_g \rangle_N = n - N_g + N = N - N_g + n$. For $n \in \{N_g, \dots, N + N_g - 1\}$, on the other hand, $0 \leq n - N_g \leq N - 1$, and the representative is $n - N_g$ itself. ✓

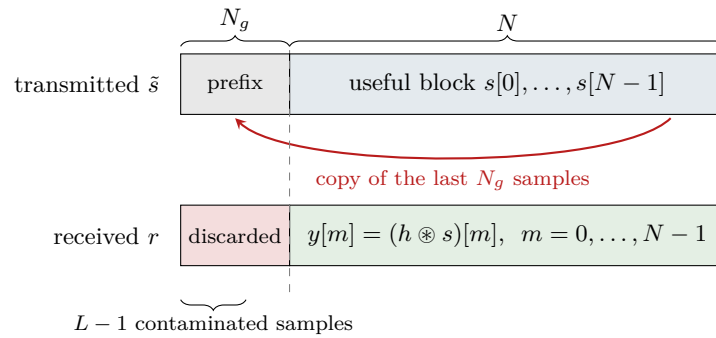


Figure 27.3: The cyclic prefix. On top the transmitted block: the last N_g samples of the symbol are copied to the front (red arrow). At the bottom the received block: the tail of the previous block contaminates only the first $L-1$ samples (Lemma 27.1), and if $N_g \geq L-1$ these all fall in the discarded region. On the remaining N samples the channel has produced exactly the circular convolution (Theorem 27.5). The figure is drawn for $N = 8$ and $N_g = 2$, the values of the example in Section 27.9.

Lemma 27.1: Interblock interference falls in the first $L-1$ samples. Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ be the transmitted sequence, with $x[n] = \tilde{s}[n]$ for $n \in \{0, \dots, N + N_g - 1\}$ and $x[n]$ arbitrary for $n < 0$ (the tail of the previous blocks), and let $r[n] = \sum_{l=0}^{L-1} h_l x[n-l]$ be the channel output. Then, for $n \geq L-1$, the value $r[n]$ does not depend on any $x[j]$ with $j < 0$; for $0 \leq n \leq L-2$ (possible only if $L \geq 2$) the value $r[n]$ depends on the $L-1-n$ past samples $x[-1], \dots, x[n-L+1]$.

Proof. In the sum $r[n] = \sum_{l=0}^{L-1} h_l x[n-l]$ exactly the indices $n-l$ with $l = 0, 1, \dots, L-1$ appear, that is, the integers from $n-L+1$ to n . One of these indices is negative if and only if $n-L+1 < 0$, that is $n < L-1$. Hence for $n \geq L-1$ all the indices $n-l$ are ≥ 0 and no past sample is involved. For $0 \leq n \leq L-2$ the negative indices are those with $n-l < 0$, that is $l > n$: these are $l = n+1, \dots, L-1$, in number $L-1-n$, and correspond to the samples

$x[-1], x[-2], \dots, x[n-L+1]$. The coefficient of $x[n-L+1]$ is $h_{L-1} \neq 0$ by hypothesis, so the dependence is effective. $\square$

Theorem 27.5: The cyclic prefix converts linear convolution into circular convolution. Let $N \geq L$, $N \geq N_g \geq L-1$, let $\tilde{s}$ be the block with cyclic prefix of Definition 27.4, let x be the transmitted sequence of Lemma 27.1 and let r be the channel output. Let the receiver discard the first N_g samples and set

$$y[m] := r[m + N_g], \quad m = 0, 1, \dots, N-1.$$

Finally let $h_p \in \mathbb{C}^N$ be the zero-padding of h , that is $h_p[l] = h_l$ for $0 \leq l \leq L-1$ and $h_p[l] = 0$ for $L \leq l \leq N-1$. Then

$$y = h_p \circledast s, \quad \text{that is} \quad y[m] = \sum_{l=0}^{N-1} h_p[l] s[\langle m-l \rangle_N], \quad m = 0, \dots, N-1,$$

with $\circledast$ the circular convolution of Definition 22.4.

Proof. Step 1: no past sample is involved. By Lemma 27.1 the value $r[n]$ is independent of the past as soon as $n \geq L-1$. The indices used are $n = m + N_g$ with $m \geq 0$, hence $n \geq N_g \geq L-1$ by hypothesis: the condition holds for every $m = 0, \dots, N-1$. From here on $x[j] = \tilde{s}[j]$ for every index that appears.

Step 2: the indices stay inside the block. For $m \in \{0, \dots, N-1\}$ and $l \in \{0, \dots, L-1\}$ set $n_l := m + N_g - l$. Then

$$n_l \geq 0 + N_g - (L-1) \geq 0 \quad (\text{because } N_g \geq L-1),$$

$$n_l \leq (N-1) + N_g - 0 = N + N_g - 1.$$

Hence $n_l \in \{0, \dots, N + N_g - 1\}$ and $\tilde{s}[n_l]$ is defined by Definition 27.4. Therefore

$$y[m] = r[m + N_g] = \sum_{l=0}^{L-1} h_l \tilde{s}[m + N_g - l].$$

Step 3: the modular index computation, written out in full. It is claimed that

$$\tilde{s}[m + N_g - l] = s[\langle m-l \rangle_N] \quad \text{for every } m \in \{0, \dots, N-1\}, l \in \{0, \dots, L-1\}. \quad (27.2)$$

Two cases are distinguished.

Case (a): $l \leq m$. Then $m-l \geq 0$ and $m-l \leq N-1$, so $\langle m-l \rangle_N = m-l$, that is, there is no wraparound. On the other hand $n_l = m + N_g - l \geq N_g$, so n_l falls in the *second* row of Definition 27.4 and

$$\tilde{s}[n_l] = s[n_l - N_g] = s[m + N_g - l - N_g] = s[m-l] = s[\langle m-l \rangle_N].$$

Case (b): $l > m$. Let $d := l - m$, so that $1 \leq d \leq L-1 \leq N_g$. Then

$$n_l = m + N_g - l = N_g - d, \quad 0 \leq N_g - d \leq N_g - 1,$$

so n_l falls in the *first* row of Definition 27.4, that is, inside the prefix, and

$$\tilde{s}[n_l] = s[N - N_g + n_l] = s[N - N_g + N_g - d] = s[N - d] = s[N + m - l].$$

On the other hand $m-l = -d$ with $1 \leq d \leq N_g \leq N$, that is $-N \leq m-l \leq -1$: if $d < N$ the representative is $\langle m-l \rangle_N = m-l + N = N-d$, and if $d = N$ (possible only for $N_g = N = d$) it is $\langle m-l \rangle_N = 0 = N-d$. In both sub-cases $s[\langle m-l \rangle_N] = s[N-d] = \tilde{s}[n_l]$.

The two cases exhaust the possibilities, so (27.2) is proved. It is here, and only here, that the cyclic prefix is needed: without it, in case (b) the sample $\tilde{s}[N_g - d]$ would belong to the previous block and would have no reason to equal $s[N - d]$.

Step 4: conclusion. Substituting (27.2) into Step 2,

$$y[m] = \sum_{l=0}^{L-1} h_l s[\langle m - l \rangle_N].$$

Since $L \leq N$, the sum can be extended up to $N - 1$ by adding zero terms, because $h_p[l] = 0$ for $l \geq L$:

$$y[m] = \sum_{l=0}^{N-1} h_p[l] s[\langle m - l \rangle_N] = (h_p \circledast s)[m],$$

which is exactly Definition 22.4 with $x = h_p$ and $y = s$. $\square$

Theorem 27.6: *The condition $N_g \geq L - 1$ is also necessary.* Let $L \geq 2$, $h_{L-1} \neq 0$ and $0 \leq N_g \leq L - 2$. Then there exist a data block $s \in \mathbb{C}^N$ and two transmitted sequences x and x' that agree on $\{0, \dots, N + N_g - 1\}$ — that is, that carry the same symbol with the same prefix — but produce different received vectors:

$$y'[0] - y[0] = h_{L-1} \neq 0,$$

while $h_p \circledast s$ is by construction the same in both cases. Consequently the identity $y = h_p \circledast s$ of Theorem 27.5 cannot hold for every transmission, and the condition $N_g \geq L - 1$ is necessary as well as sufficient.

Proof. Choose any $s \in \mathbb{C}^N$ and build $\tilde{s}$ as in Definition 27.4. Define two transmitted sequences $x, x' : \mathbb{Z} \rightarrow \mathbb{C}$ by setting, for both, $x[n] = x'[n] = \tilde{s}[n]$ on $\{0, \dots, N + N_g - 1\}$ and $x[n] = x'[n] = 0$ for every $n < 0$ except the index

$$j_0 := N_g - (L - 1),$$

where $x[j_0] = 0$ and $x'[j_0] = 1$ are set. The index j_0 is indeed negative: $j_0 = N_g - L + 1 \leq (L - 2) - L + 1 = -1$.

Now compute $y[0]$ and $y'[0]$. By definition $y[0] = r[N_g] = \sum_{l=0}^{L-1} h_l x[N_g - l]$, and likewise for x' . The two sequences differ only at the index $j_0 = N_g - (L - 1)$, which appears in that sum for $l = L - 1$ and for no other value of l (the map $l \mapsto N_g - l$ is injective). Hence, subtracting term by term,

$$y'[0] - y[0] = h_{L-1}(x'[j_0] - x[j_0]) = h_{L-1} \cdot (1 - 0) = h_{L-1} \neq 0.$$

The vector $h_p \circledast s$, instead, depends only on h and on s , which are the same in both cases. If the identity $y = h_p \circledast s$ held for both transmissions one would have $y[0] = (h_p \circledast s)[0] = y'[0]$, contradicting what was just computed. Hence it fails for at least one of the two, and no choice of $N_g \leq L - 2$ can guarantee it in general. $\square$

Corollary 27.3: *The channel becomes diagonal in the frequency domain.* Under the hypotheses of Theorem 27.5 set

$$H := \mathcal{F}_N h_p, \quad H[k] = \sum_{l=0}^{L-1} h_l \omega_N^{kl} = \sum_{l=0}^{L-1} h_l e^{-2\pi i k l / N}, \quad k = 0, \dots, N - 1.$$

Then, with $Y = \mathcal{F}_N y$,

$$Y[k] = H[k] S[k] = N H[k] c_k, \quad k = 0, \dots, N-1.$$

The channel, which in the time domain mixes all N samples together, in the frequency domain acts on each subcarrier separately, multiplying it by a single complex number.

Proof. By Theorem 27.5, $y = h_p \otimes s$; applying $\mathcal{F}_N$ and Theorem 22.5 (the discrete convolution theorem) gives $Y[k] = H[k] S[k]$ with $H = \mathcal{F}_N h_p$ and $S = \mathcal{F}_N s$. The explicit expression of $H[k]$ is Definition 22.1 applied to h_p , whose terms of index $\geq L$ vanish. Finally Theorem 27.4 gives $S = \mathcal{F}_N s = N c$, that is $S[k] = N c_k$. $\square$

Theorem 27.7: Matrix form: the channel is a circulant, and the basis that diagonalises it is the Fourier basis. Under the hypotheses of Theorem 27.5 let $C_h := \text{circ}(h_p)$ be the circulant matrix generated by h_p (Definition 22.5), that is $(C_h)_{mj} = h_p[\langle m - j \rangle_N]$. Then

- (i) $y = C_h s$;
- (ii) $F_N C_h F_N^* = \text{diag}(H[0], \dots, H[N-1])$, with F_N the unitary matrix of Definition 22.3;
- (iii) the eigenvalues of C_h are the numbers $H[k]$ and an eigenvector for $H[k]$ is the k -th sampled subcarrier $v_k = (\omega_N^{-kn})_{n=0}^{N-1}$ of Theorem 27.2;
- (iv) $\det C_h = \prod_{k=0}^{N-1} H[k]$, and C_h is invertible if and only if $H[k] \neq 0$ for every k .

Proof. (i) By Proposition 22.5, $\text{circ}(h_p) s = h_p \otimes s$, and Theorem 27.5 gives $h_p \otimes s = y$.
(ii) This is exactly Theorem 22.8 applied to the generator $c = h_p$: it states $F_N C F_N^* = \text{diag}(\lambda)$ with $\lambda = \mathcal{F}_N c$, and here $\mathcal{F}_N h_p = H$ by definition of H .
(iii) The same theorem identifies the eigenvector for $\lambda[k]$ as the vector $e_k = (\omega_N^{-kn})_n$; Theorem 27.2 identifies e_k with the k -th sampled subcarrier v_k . This is the central statement of the chapter, said in the language of linear algebra: *the subcarriers are the eigenvectors of the channel*, and they are so for *every* channel, because the basis that diagonalises every circulant of order N is a single one.
(iv) From (ii) it follows that $C_h = F_N^* \text{diag}(H) F_N$; the determinant is multiplicative and $\det(F_N^*) \det(F_N) = \det(F_N^* F_N) = \det I_N = 1$, hence $\det C_h = \det \text{diag}(H) = \prod_k H[k]$. A matrix is invertible if and only if its determinant is nonzero, and a product of complex numbers is nonzero if and only if every factor is. $\square$

Remark 27.6: What the prefix has really done: it has closed the time axis into a circle. It is worth saying in one sentence what has happened. A channel is invariant under translation on the time axis $\mathbb{Z}$: delaying the input delays the output. The linear maps of $\mathbb{C}^N$ into itself that are invariant under the circular translation $x[n] \mapsto x[\langle n-1 \rangle_N]$ are exactly the circulant matrices (Definition 22.5); and Theorem 22.8 says that all those matrices share the same eigenvectors, the characters of the cyclic group $\mathbb{Z}/N\mathbb{Z}$, that is, the sampled subcarriers. The cyclic prefix does nothing more than force the channel to behave as if time lived on $\mathbb{Z}/N\mathbb{Z}$ instead of on $\mathbb{Z}$: the tail of the block, copied to the front, makes the sample preceding $s[0]$ equal to $s[N-1]$, exactly as on the discrete circle. From there on, the whole algebra of Chapter 22 is available, and the diagonalisation comes for free. This is the reason this chapter belongs to a treatise on the circle: modern broadband rests on the idea of closing a line into a ring.

Remark 27.7: The prefix carries no information. The receiver discards the first N_g samples and loses nothing: by Definition 27.4 those samples are a copy of samples that will

arrive anyway, and by Lemma 27.1 they are also the only ones contaminated by the previous block. The prefix is thus pure redundancy: it costs transmission time and carries no bits. Exactly how much it costs is the subject of Section 27.5.

27.4. Single-coefficient equalisation per subcarrier

Corollary 27.3 reduces equalisation to a trivial operation: since $Y[k] = NH[k]c_k$, recovering c_k only takes a division. A channel that in the time domain would require an inverse filter of theoretically infinite length is equalised with N complex divisions. This section writes the formula, computes its error in the presence of noise, and shows where the procedure is fragile.

Definition 27.5: Zero-forcing equalizer (single-coefficient). Suppose $H[k] \neq 0$ for every k . The **zero-forcing equalizer** (ZF) is the map that associates to the received vector y

$$\hat{c}_k = \frac{Y[k]}{N H[k]}, \quad Y = \mathcal{F}_N y, \quad k = 0, \dots, N-1.$$

It is called *single-coefficient* because for each subcarrier the correction is a single complex number, $1/(NH[k])$: a gain and a rotation.

Proposition 27.2: In the absence of noise the ZF equalizer is exact. Under the hypotheses of Theorem 27.5, if $H[k] \neq 0$ for every k then $\hat{c}_k = c_k$ for every k : the data are reconstructed exactly.

Proof. By Corollary 27.3, $Y[k] = NH[k]c_k$. Since $N \neq 0$ and $H[k] \neq 0$ one may divide, obtaining $\hat{c}_k = \frac{NH[k]c_k}{NH[k]} = c_k$. $\square$

Noise changes the picture, and it changes it in a way that depends on k . The following model is adopted, made explicit once and for all.

Definition 27.6: Hypothesis (R) on the noise. The random vector $w = (w[0], \dots, w[N-1])$ with values in $\mathbb{C}^N$ is said to satisfy **hypothesis (R)** if

$$\mathbb{E}[w[n]] = 0, \quad \mathbb{E}[w[n]\overline{w[m]}] = \sigma^2 \delta_{nm}, \quad \mathbb{E}[w[n]w[m]] = 0 \quad \text{for every } n, m,$$

with $\sigma^2 > 0$, and if w is independent of the data vector c . The third condition is called *circular symmetry* and says that the real and imaginary parts have the same variance $\sigma^2/2$ and are uncorrelated. The received vector is then $y = h_p \otimes s + w$.

Lemma 27.2: The DFT preserves the whiteness of the noise. Under hypothesis (R), setting $W = \mathcal{F}_N w$, one has

$$\mathbb{E}[W[k]] = 0, \quad \mathbb{E}[W[k]\overline{W[k']}]= N\sigma^2 \delta_{kk'}, \quad \mathbb{E}[|W[k]|^2] = N\sigma^2.$$

Proof. The mean is zero by linearity of the expectation: $\mathbb{E}[W[k]] = \sum_n \omega_N^{kn} \mathbb{E}[w[n]] = 0$. For the covariance one uses the definition of the DFT (Definition 22.1) and again linearity:

$$\mathbb{E}[W[k]\overline{W[k']}]= \mathbb{E}\left[\sum_{n=0}^{N-1} w[n]\omega_N^{kn} \sum_{m=0}^{N-1} \overline{w[m]\omega_N^{k'm}}\right] = \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \mathbb{E}[w[n]\overline{w[m]}] \omega_N^{kn} \omega_N^{-k'm},$$

where $\overline{\omega_N^{k'm}} = \omega_N^{-k'm}$ was used (licit because $|\omega_N| = 1$). By hypothesis (R) the terms with

$n \neq m$ vanish and those with $n = m$ equal σ^2 :

$$\mathbb{E}[W[k]\overline{W[k']}] = \sigma^2 \sum_{n=0}^{N-1} \omega_N^{kn} \omega_N^{-k'n} = \sigma^2 \sum_{n=0}^{N-1} \omega_N^{(k-k')n}.$$

By Lemma 22.1 the last sum equals N if $N \mid (k-k')$ and 0 otherwise; since $k, k' \in \{0, \dots, N-1\}$, this happens only for $k = k'$. This gives $N\sigma^2\delta_{kk'}$, and for $k = k'$ the formula for the squared modulus. $\square$

Theorem 27.8: Error of the single-coefficient equalizer in the presence of noise.

Under hypothesis (R) and the hypotheses of Theorem 27.5, with $H[k] \neq 0$ for every k , the output of the ZF equalizer of Definition 27.5 is

$$\hat{c}_k = c_k + e_k, \quad e_k := \frac{W[k]}{N H[k]},$$

and the error satisfies

$$\mathbb{E}[e_k] = 0, \quad \mathbb{E}[|e_k|^2] = \frac{\sigma^2}{N |H[k]|^2}, \quad \mathbb{E}[e_k \overline{e_{k'}}] = 0 \quad (k \neq k').$$

In particular the equalizer is unbiased ($\mathbb{E}[\hat{c}_k] = \mathbb{E}[c_k]$) and the errors on the different subcarriers are uncorrelated, but the error power is inversely proportional to $|H[k]|^2$.

Proof. Step 1: shape of the error. The received vector is $y = h_p \otimes s + w$; by linearity of the DFT and by Corollary 27.3,

$$Y[k] = \mathcal{F}_N(h_p \otimes s)[k] + \mathcal{F}_N w[k] = N H[k] c_k + W[k].$$

Dividing by $NH[k]$, which is nonzero,

$$\hat{c}_k = \frac{NH[k]c_k + W[k]}{NH[k]} = c_k + \frac{W[k]}{NH[k]} = c_k + e_k.$$

Step 2: mean of the error. Since $NH[k]$ is a deterministic constant, $\mathbb{E}[e_k] = \frac{1}{NH[k]} \mathbb{E}[W[k]] = 0$ by Lemma 27.2. It follows that $\mathbb{E}[\hat{c}_k] = \mathbb{E}[c_k]$.

Step 3: power of the error. Again because $NH[k]$ is deterministic,

$$\mathbb{E}[|e_k|^2] = \mathbb{E}\left[\frac{|W[k]|^2}{|NH[k]|^2}\right] = \frac{\mathbb{E}[|W[k]|^2]}{N^2 |H[k]|^2} = \frac{N\sigma^2}{N^2 |H[k]|^2} = \frac{\sigma^2}{N |H[k]|^2},$$

where $\mathbb{E}[|W[k]|^2] = N\sigma^2$ from Lemma 27.2 was used.

Step 4: decorrelation. For $k \neq k'$,

$$\mathbb{E}[e_k \overline{e_{k'}}] = \frac{\mathbb{E}[W[k]\overline{W[k']}] }{N^2 H[k]\overline{H[k']}} = \frac{N\sigma^2 \delta_{kk'}}{N^2 H[k]\overline{H[k']}} = 0,$$

again by Lemma 27.2. $\square$

Corollary 27.4: Signal-to-noise ratio per subcarrier and noise amplification.

Suppose moreover that $\mathbb{E}[|c_k|^2] = P$ for every k . Then the ratio between the symbol power

and the error power on the k -th subcarrier is

$$\text{SNR}_k = \frac{\mathbb{E}[|c_k|^2]}{\mathbb{E}[|e_k|^2]} = \frac{N P |H[k]|^2}{\sigma^2},$$

and for two subcarriers k, k'

$$\frac{\text{SNR}_k}{\text{SNR}_{k'}} = \frac{|H[k]|^2}{|H[k']|^2}.$$

The ratio between the best and the worst subcarrier is therefore $\max_k |H[k]|^2 / \min_k |H[k]|^2$, and it diverges when a subcarrier falls into a deep fade, that is, when $H[k]$ approaches zero.

Proof. Immediate from Theorem 27.8: dividing P by $\sigma^2/(N|H[k]|^2)$ gives $NP|H[k]|^2/\sigma^2$; the ratio of two such values cancels N, P and σ^2 . $\square$

Example 27.2: The errors on the eight subcarriers of the example $N = 8$. Take $N = 8$ and the channel $h = (1, \frac{1}{2}, \frac{1}{4})$ of Section 27.9, with $\sigma^2 = 0.04$ and $P = 1$.

First route, closed formula. The squared moduli of the frequency response are (Section 27.9, where they are derived)

$$\begin{aligned} |H[0]|^2 &= \frac{49}{16} = 3.0625, & |H[1]|^2 &= |H[7]|^2 = \frac{21}{16} + \frac{5\sqrt{2}}{8} = 2.1963835, \\ |H[2]|^2 &= |H[6]|^2 = \frac{13}{16} = 0.8125, & |H[3]|^2 &= |H[5]|^2 = \frac{21}{16} - \frac{5\sqrt{2}}{8} = 0.4286165, \\ |H[4]|^2 &= \frac{9}{16} = 0.5625. \end{aligned}$$

By Theorem 27.8, $\mathbb{E}[|e_k|^2] = \frac{0.04}{8|H[k]|^2} = \frac{0.005}{|H[k]|^2}$, that is

(0.0016327, 0.0022765, 0.0061538, 0.0116654, 0.0088889, 0.0116654, 0.0061538, 0.0022765).

The worst subcarrier is the third one (and the fifth), with an error $\frac{0.0116654}{0.0016327} = 7.1450815$ times that of subcarrier 0: in decibels, $10 \log_{10} 7.1450815 = 8.5400719$ dB of disadvantage. The exact value of the ratio is

$$\frac{|H[0]|^2}{|H[3]|^2} = \frac{49/16}{\frac{21}{16} - \frac{5\sqrt{2}}{8}} = \frac{49}{21 - 10\sqrt{2}} = 7.1450815.$$

Second route, simulation. A script generated 200 000 independent realisations of circularly symmetric complex Gaussian noise with $\sigma^2 = 0.04$, computed their DFT, divided by $NH[k]$, and averaged $|e_k|^2$. The empirical values obtained are

(0.0016322, 0.0022752, 0.0061601, 0.0116692, 0.0089008, 0.0116804, 0.0061455, 0.0022778),

with empirical/theoretical ratios between 0.99864 and 1.00134: a discrepancy compatible with the statistical error of an average over 200 000 samples. $\checkmark$

Remark 27.8: Why ZF is not the optimal choice. Theorem 27.8 shows the flaw of the ZF equalizer: when $|H[k]|$ is small, the divisor $NH[k]$ is small and the noise is amplified exactly to the same extent that the signal is restored. In the limit $H[k] \rightarrow 0$ the error diverges, while common sense suggests that, if a subcarrier is completely cancelled by the channel, the sensible thing is to give it up rather than amplify noise until it swamps everything else. The

next theorem formalises this intuition and shows that the optimal equalizer, in the class of single-coefficient corrections, is not $1/H[k]$.

Theorem 27.9: The correction optimal in mean square. Fix k and assume hypothesis (R), together with $\mathbb{E}[c_k] = 0$ and $\mathbb{E}[|c_k|^2] = P > 0$. Consider the family of estimators

$$\hat{c}_k(g) = g \cdot \frac{Y[k]}{N}, \quad g \in \mathbb{C},$$

and set $J(g) = \mathbb{E}[|c_k - \hat{c}_k(g)|^2]$. Then J has a unique minimum point, given by

$$g^* = \frac{\overline{H[k]}}{|H[k]|^2 + \frac{\sigma^2}{NP}},$$

and the minimum value is

$$J(g^*) = \frac{P\sigma^2}{N|H[k]|^2 P + \sigma^2} < \frac{\sigma^2}{N|H[k]|^2} = \mathbb{E}[|e_k|^2].$$

Moreover $g^* \rightarrow 1/H[k]$ as $\sigma^2 \rightarrow 0$: in the noiseless limit the optimal equalizer coincides with the zero-forcing one.

Proof. Step 1: reduction to a scalar problem. Set $u := \frac{Y[k]}{N} = H[k]c_k + \frac{W[k]}{N}$ (Step 1 of the proof of Theorem 27.8). Write $H := H[k]$ for brevity. Since c_k and w are independent and $\mathbb{E}[W[k]] = \mathbb{E}[c_k] = 0$, one has $\mathbb{E}[c_k \overline{W[k]}] = \mathbb{E}[c_k] \mathbb{E}[\overline{W[k]}] = 0$; hence

$$\mathbb{E}[|u|^2] = |H|^2 \mathbb{E}[|c_k|^2] + \frac{\mathbb{E}[|W[k]|^2]}{N^2} = |H|^2 P + \frac{N\sigma^2}{N^2} = |H|^2 P + \frac{\sigma^2}{N} =: A > 0,$$

$$\mathbb{E}[u \overline{c_k}] = H \mathbb{E}[|c_k|^2] + \frac{1}{N} \mathbb{E}[W[k] \overline{c_k}] = HP.$$

Step 2: the function to minimise. Expanding the squared modulus,

$$J(g) = \mathbb{E}[(c_k - gu)(\overline{c_k - gu})] = \mathbb{E}[|c_k|^2] - \bar{g} \mathbb{E}[c_k \overline{u}] - g \mathbb{E}[\overline{c_k} u] + |g|^2 \mathbb{E}[|u|^2].$$

By Step 1, $\mathbb{E}[\overline{c_k} u] = HP$ and $\mathbb{E}[c_k \overline{u}] = \overline{HP} = \bar{H}P$ (because P is real). Hence

$$J(g) = P - \bar{g} \bar{H}P - gHP + |g|^2 A = P - 2\Re(gHP) + |g|^2 A.$$

Step 3: completing the square. Set $z := \frac{\bar{H}P}{A}$. Then

$$A|g - z|^2 = A|g|^2 - 2A\Re(g\bar{z}) + A|z|^2 = A|g|^2 - 2\Re(gHP) + \frac{|H|^2 P^2}{A},$$

having used $A\bar{z} = HP$ and $A|z|^2 = |H|^2 P^2 / A$. Comparing with Step 2,

$$J(g) = A|g - z|^2 + P - \frac{|H|^2 P^2}{A}.$$

Since $A > 0$, the first term is nonnegative and vanishes if and only if $g = z$: the minimum is unique and equals $g^* = z = \frac{\bar{H}P}{A}$. Dividing numerator and denominator by P ,

$$g^* = \frac{\bar{H}P}{|H|^2 P + \frac{\sigma^2}{N}} = \frac{\bar{H}}{|H|^2 + \frac{\sigma^2}{NP}},$$

which is the announced formula.

Step 4: the minimum value. Substituting,

$$\begin{aligned} J(g^*) &= P - \frac{|H|^2 P^2}{A} = \frac{PA - |H|^2 P^2}{A} = \frac{P(|H|^2 P + \frac{\sigma^2}{N}) - |H|^2 P^2}{|H|^2 P + \frac{\sigma^2}{N}} \\ &= \frac{\frac{P\sigma^2}{N}}{|H|^2 P + \frac{\sigma^2}{N}} = \frac{P\sigma^2}{N|H|^2 P + \sigma^2}, \end{aligned}$$

where in the last step numerator and denominator were multiplied by N .

Step 5: comparison with ZF. Since $\sigma^2 > 0$ one has $N|H|^2 P + \sigma^2 > N|H|^2 P > 0$, hence

$$J(g^*) = \frac{P\sigma^2}{N|H|^2 P + \sigma^2} < \frac{P\sigma^2}{N|H|^2 P} = \frac{\sigma^2}{N|H|^2} = \mathbb{E}[|e_k|^2].$$

The inequality is consistent with the fact that the ZF equalizer belongs to the family considered: it corresponds to $g = 1/H$, and the minimum cannot exceed the value at any given point.

Step 6: the noiseless limit. In the formula $g^* = \bar{H}/(|H|^2 + \frac{\sigma^2}{NP})$ the denominator tends to $|H|^2$ as $\sigma^2 \rightarrow 0$, and $|H|^2 \neq 0$; hence $g^* \rightarrow \bar{H}/|H|^2 = \bar{H}/(H\bar{H}) = 1/H$. $\square$

Remark 27.9: The price of optimality: bias. The equalizer of Theorem 27.9 is biased: one has

$$\mathbb{E}[\hat{c}_k(g^*) | c_k] = g^* H c_k = \frac{|H|^2}{|H|^2 + \frac{\sigma^2}{NP}} c_k \neq c_k,$$

that is, the estimated symbol is systematically shrunk towards the origin by a factor $\frac{|H|^2}{|H|^2 + \sigma^2/(NP)} < 1$, more pronounced the more attenuated the subcarrier is. ZF has no such flaw (Theorem 27.8, Step 2) but pays with a larger mean-square error. There is no absolute right choice: one chooses according to the criterion, and the two criteria — absence of bias and minimum mean-square error — are incompatible. In the numerical example of Example 27.2, with $\sigma^2 = 0.04$, $P = 1$ and $N = 8$, the MMSE gain over ZF ranges from a factor 0.99837 on the best subcarrier to 0.98847 on the worst: a small advantage, because the noise is weak, but systematic and growing as $|H[k]|$ decreases.

Remark 27.10: Deep fades and what is done about them in practice: panorama.

Corollary 27.4 describes a situation that no equalisation can solve: if $H[k]$ is nearly zero, the k -th subcarrier no longer carries usable information, and this is independent of the equalizer chosen — the ratio $NP|H[k]|^2/\sigma^2$ already says so, and it contains no reference to the receiver. The countermeasure is not in equalisation but in coding: the bits of each codeword are spread over distant subcarriers, so that the few cancelled ones can be recovered from the many good ones (frequency diversity), and the number of bits per subcarrier is adapted to its SNR_k (bit loading). Both techniques belong to coding theory and are neither treated nor proved here: only the panorama is given. The reader finds them presented in Y. S. Cho, J. Kim, W. Y. Yang, C. G. Kang, *MIMO-OFDM Wireless Communications with MATLAB*, Wiley, 2010, Chapter 4 “Introduction to OFDM”, pp. 111–151, and in J. G. Proakis, M. Salehi, *Digital Communications*, 5th edition, McGraw-Hill, 2008. This chapter stops at the point where the channel has been made diagonal: from there on the problem changes discipline.

27.5. How long the prefix must be, and what it costs

Theorem 27.5 and Theorem 27.6 together determine the length of the prefix: not less than $L - 1$, because otherwise the computation does not close, and not more, because every extra sample is transmission time thrown away. This section computes exactly how much that time costs.

Corollary 27.5: The condition on the prefix length. Let h be a channel of length L and dispersion $\tau_{\max} = (L - 1)T_s$ (Definition 27.3). Then the identity $y = h_p \circledast s$ holds for every transmission if and only if

$$N_g \geq L - 1, \quad \text{that is} \quad N_g T_s \geq \tau_{\max} :$$

the guard interval must last at least as long as the channel's dispersion. The minimal choice is $N_g = L - 1$.

Proof. Sufficiency is Theorem 27.5, necessity is Theorem 27.6. The rewriting in physical time is obtained by multiplying both sides of the inequality $N_g \geq L - 1$ by $T_s > 0$ and using $\tau_{\max} = (L - 1)T_s$. $\square$

Definition 27.7: Time efficiency. The transmitted block lasts $N + N_g$ samples, of which only N carry information (Observation 27.7). The ratio

$$\eta := \frac{N}{N + N_g} \in (0, 1],$$

is called the **time efficiency**, and the quantity $1 - \eta$ the **efficiency loss**. The loss expressed in decibels is $-10 \log_{10} \eta$.

Proposition 27.3: The loss due to the prefix. With the notation of Definition 27.7 and setting $\rho := N_g/N$:

- (i) $\eta = \frac{1}{1 + \rho}$ and $1 - \eta = \frac{\rho}{1 + \rho} = \frac{N_g}{N + N_g}$;
- (ii) the loss in decibels equals $-10 \log_{10} \eta = 10 \log_{10}(1 + \rho)$;
- (iii) if $N_g = L - 1$ is chosen minimal, then $\eta = \frac{N}{N + L - 1}$ is strictly increasing in N for $L \geq 2$, and $\eta \rightarrow 1$ as $N \rightarrow \infty$.

Proof. (i) Dividing numerator and denominator of $\frac{N}{N + N_g}$ by $N > 0$ gives $\frac{1}{1 + N_g/N} = \frac{1}{1 + \rho}$; and $1 - \frac{1}{1 + \rho} = \frac{1 + \rho - 1}{1 + \rho} = \frac{\rho}{1 + \rho}$, which, multiplying top and bottom by N , becomes $\frac{N_g}{N + N_g}$.

(ii) By (i), $-10 \log_{10} \eta = -10 \log_{10} \frac{1}{1 + \rho} = 10 \log_{10}(1 + \rho)$.

(iii) Consider $g(N) = \frac{N}{N + L - 1}$ as a function of the real variable $N > 0$, with $L \geq 2$ fixed. Then

$$g'(N) = \frac{(N + L - 1) \cdot 1 - N \cdot 1}{(N + L - 1)^2} = \frac{L - 1}{(N + L - 1)^2} > 0,$$

because $L - 1 \geq 1$: hence g is strictly increasing, and in particular so is the sequence of its values at the integers. Finally $g(N) = \frac{1}{1 + (L - 1)/N} \rightarrow \frac{1}{1} = 1$ as $N \rightarrow \infty$, because $(L - 1)/N \rightarrow 0$. $\square$

Example 27.3: A block of length 2048 with guard 144, computed twice. Take $N = 2048$ and $N_g = 144$, that is $\rho = \frac{144}{2048} = \frac{9}{128}$.

Table 27.2: Efficiency loss due to the cyclic prefix (Proposition 27.3). The values depend only on the ratio $\rho = N_g/N$: the first four rows collect the most commonly used fractions, the last three show that, at fixed dispersion, lengthening the block reduces the loss. All numbers were recomputed with an independent script.

N	N_g	$\rho = N_g/N$	η	loss	loss (dB)
4	1	1/4	0.8000000	20.000000%	0.969100
8	2	1/4	0.8000000	20.000000%	0.969100
64	8	1/8	0.8888889	11.111111%	0.511525
256	16	1/16	0.9411765	5.882353%	0.263289
1024	32	1/32	0.9696970	3.030303%	0.133640
256	16	1/16	0.9411765	5.882353%	0.263289
1024	16	1/64	0.9846154	1.538462%	0.067334
2048	16	1/128	0.9922481	0.775194%	0.033797

First route, closed formula. By Proposition 27.3(i),

$$\eta = \frac{2048}{2048 + 144} = \frac{2048}{2192} = \frac{128}{137} = 0.9343066\dots,$$

having simplified by 16 ($2048 = 16 \cdot 128$, $2192 = 16 \cdot 137$). The loss is $1 - \eta = \frac{9}{137} = 0.0656934\dots$, that is 6.569343%; in decibels, $10 \log_{10} \frac{137}{128} = 10 \log_{10} 1.0703125 = 0.2951058$ dB.

Second route, script. A script that computes η , $1 - \eta$ and $-10 \log_{10} \eta$ for a list of pairs (N, N_g) returns, for this pair, $\eta = 0.9343065693$, loss 0.0656934307, that is 6.569343%, and 0.295106 dB. ✓

Consistency check. The fraction $\frac{9}{137}$ has decimal expansion 0.06569343065693..., periodic with period 8; the first digits agree with those returned by the script. ✓

Remark 27.11: The trade-off, and why lengthening the block is not enough.

Proposition 27.3(iii) seems to suggest a simple recipe: at fixed dispersion, take N large and the loss vanishes. The suggestion is only half correct, because N appears elsewhere too. The symbol time is $T = NT_s$, so the subcarrier spacing is

$$\Delta f = \frac{1}{T} = \frac{1}{NT_s} :$$

doubling N halves Δf . An absolute frequency error δf (due to an imperfect oscillator or to Doppler effect) then corresponds to a normalised error $\varepsilon = \delta f / \Delta f$ that doubles, and Section 27.6 will show that the power of the intercarrier interference grows like ε^2 : it quadruples. Designing an OFDM system is exactly choosing a balance point between these two opposing pressures — N large to amortise the prefix, N small to tolerate the frequency error — with the prefix length bounded below by Corollary 27.5.

27.6. Carrier frequency offset and intercarrier interference

Observation 27.2 already anticipated the point: the orthogonality of the subcarriers depends on the demodulator reading the spectrum exactly at the points $f = \ell/T$. If the receiver's local oscillator is misaligned by δf hertz, all the reading points shift by the same amount, the zeros of Proposition 27.1 no longer fall where they are needed, and every subcarrier spills energy onto all the others. This section computes exactly how much.

Definition 27.8: Normalised frequency error. Suppose the receiver's oscillator differs from the transmitter's by δf hertz. The received baseband signal is then $e^{2\pi i \delta f t} s(t)$, and at the sampling instants $t_n = nT/N$

$$r[n] = e^{2\pi i \delta f nT/N} s[n] = e^{2\pi i \varepsilon n/N} s[n], \quad n = 0, \dots, N-1,$$

where

$$\varepsilon := \delta f \cdot T = \frac{\delta f}{\Delta f}$$

is called the **normalised frequency error** (*carrier frequency offset*, CFO): it is the error measured in units of subcarrier spacing. The energy that, due to ε , one subcarrier spills onto the others is called **intercarrier interference** (ICI). In this section the channel is assumed ideal, $h = (1)$, in order to isolate the effect of the frequency error alone.

Lemma 27.3: The kernel of the intercarrier interference. Let $\varepsilon \in \mathbb{R} \setminus \mathbb{Z}$ and $m \in \mathbb{Z}$, and set

$$I(m) := \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(m+\varepsilon)n/N}.$$

Then I is periodic of period N in m , and for every m with $|m + \varepsilon| < N$ the closed form holds

$$I(m) = \frac{\sin(\pi\varepsilon)}{N \sin(\pi(m+\varepsilon)/N)} e^{i\pi\varepsilon(1-\frac{1}{N})} e^{-i\pi m/N}$$

and in particular

$$|I(m)| = \frac{|\sin(\pi\varepsilon)|}{N |\sin(\pi(m+\varepsilon)/N)|}, \quad |I(0)| = \frac{|\sin(\pi\varepsilon)|}{N |\sin(\pi\varepsilon/N)|}.$$

Proof. Step 1: periodicity. Substituting $m + N$ for m ,

$$I(m+N) = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(m+N+\varepsilon)n/N} = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i(m+\varepsilon)n/N} e^{2\pi i n} = I(m),$$

because $e^{2\pi i n} = 1$ for every integer n .

Step 2: the geometric sum. Set $q := e^{2\pi i(m+\varepsilon)/N}$. One has $q = 1$ if and only if $(m+\varepsilon)/N \in \mathbb{Z}$; since $|m+\varepsilon| < N$ by hypothesis, this can happen only if $m+\varepsilon = 0$, that is $\varepsilon = -m \in \mathbb{Z}$, excluded by the hypothesis $\varepsilon \notin \mathbb{Z}$. Hence $q \neq 1$ and the geometric sum formula applies:

$$I(m) = \frac{1}{N} \sum_{n=0}^{N-1} q^n = \frac{1}{N} \cdot \frac{q^N - 1}{q - 1} = \frac{1}{N} \cdot \frac{e^{2\pi i(m+\varepsilon)} - 1}{e^{2\pi i(m+\varepsilon)/N} - 1}.$$

Step 3: the numerator. Since m is an integer, $e^{2\pi i m} = 1$ and hence $e^{2\pi i(m+\varepsilon)} = e^{2\pi i \varepsilon}$. With the identity $e^{i\theta} - 1 = 2i \sin(\theta/2) e^{i\theta/2}$ — verified by factoring out $e^{i\theta/2}$ and using Euler's formula for the sine, as in the corresponding step of the proof of Proposition 27.1 — with $\theta = 2\pi\varepsilon$:

$$e^{2\pi i \varepsilon} - 1 = 2i \sin(\pi\varepsilon) e^{i\pi\varepsilon}.$$

Step 4: the denominator. With the same identity and $\theta = 2\pi(m+\varepsilon)/N$:

$$e^{2\pi i(m+\varepsilon)/N} - 1 = 2i \sin\left(\frac{\pi(m+\varepsilon)}{N}\right) e^{i\pi(m+\varepsilon)/N}.$$

The sine in the denominator is nonzero: it would vanish only if $(m + \varepsilon)/N \in \mathbb{Z}$, already excluded in Step 2.

Step 5: the quotient. Dividing, the factors $2i$ cancel and what remains is

$$I(m) = \frac{1}{N} \cdot \frac{\sin(\pi\varepsilon)}{\sin(\pi(m + \varepsilon)/N)} \cdot e^{i\pi\varepsilon - i\pi(m + \varepsilon)/N}.$$

The exponent can be rewritten

$$\pi\varepsilon - \frac{\pi(m + \varepsilon)}{N} = \pi\varepsilon - \frac{\pi\varepsilon}{N} - \frac{\pi m}{N} = \pi\varepsilon\left(1 - \frac{1}{N}\right) - \frac{\pi m}{N},$$

giving the announced form. The modulus follows because both imaginary exponentials have modulus 1; for $m = 0$ the second formula is obtained. $\square$

Theorem 27.10: Decomposition of the demodulator output in the presence of a frequency error. Let $\varepsilon \in \mathbb{R} \setminus \mathbb{Z}$ and let $r[n] = e^{2\pi i \varepsilon n/N} s[n]$ as in Definition 27.8. Then the demodulator output $\hat{c}_k = \frac{1}{N} \sum_{n=0}^{N-1} r[n] \omega_N^{kn}$ equals

$$\hat{c}_k = \underbrace{I(0) c_k}_{\text{useful signal}} + \underbrace{\sum_{\substack{j=0 \\ j \neq k}}^{N-1} c_j I(j - k)}_{\text{intercarrier interference}}, \quad k = 0, \dots, N-1.$$

The factor $I(0)$ does not depend on k : the frequency error attenuates all the subcarriers by the same amount $|I(0)| \leq 1$ and rotates them all by the same angle $\pi\varepsilon(1 - 1/N)$.

Proof. Substitute Definition 27.2, $s[n] = \sum_{j=0}^{N-1} c_j e^{2\pi i j n/N}$, into the definition of $\hat{c}_k$, recalling that $\omega_N^{kn} = e^{-2\pi i k n/N}$:

$$\hat{c}_k = \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i \varepsilon n/N} \left(\sum_{j=0}^{N-1} c_j e^{2\pi i j n/N} \right) e^{-2\pi i k n/N}.$$

The sums are finite and may be exchanged; collecting the exponentials with the same base,

$$\hat{c}_k = \sum_{j=0}^{N-1} c_j \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i (j - k + \varepsilon) n/N} = \sum_{j=0}^{N-1} c_j I(j - k),$$

by the definition of I given in Lemma 27.3. Isolating the term $j = k$, which equals $c_k I(0)$, gives the desired decomposition.

For the last statement it suffices to observe that in the closed formula of Lemma 27.3 with $m = 0$ no k appears; and that $|I(0)| \leq 1$ because

$$|I(0)| = \left| \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i \varepsilon n/N} \right| \leq \frac{1}{N} \sum_{n=0}^{N-1} |e^{2\pi i \varepsilon n/N}| = \frac{1}{N} \cdot N = 1$$

by the triangle inequality. $\square$

Proposition 27.4: Power conservation: the kernel has norm one. For every $\varepsilon \in \mathbb{R} \setminus \mathbb{Z}$ the exact identity holds

$$\sum_{m=0}^{N-1} |I(m)|^2 = 1.$$

Proof. Set $u[n] := e^{2\pi i \varepsilon n/N}$, $n = 0, \dots, N-1$, and let $U = \mathcal{F}_N u$ be its DFT. By Definition 22.1,

$$U[k] = \sum_{n=0}^{N-1} u[n] \omega_N^{kn} = \sum_{n=0}^{N-1} e^{2\pi i \varepsilon n/N} e^{-2\pi i kn/N} = \sum_{n=0}^{N-1} e^{2\pi i (\varepsilon - k)n/N} = N I(-k),$$

by the definition of I (with $m = -k$). Hence $|I(-k)|^2 = \frac{1}{N^2} |U[k]|^2$. As k ranges over $\{0, \dots, N-1\}$ the index $-k$ runs through a complete system of residues modulo N , and I is N -periodic by Lemma 27.3: hence

$$\sum_{m=0}^{N-1} |I(m)|^2 = \sum_{k=0}^{N-1} |I(-k)|^2 = \frac{1}{N^2} \sum_{k=0}^{N-1} |U[k]|^2.$$

By the discrete Plancherel identity (Theorem 22.6), $\sum_k |U[k]|^2 = N \sum_n |u[n]|^2$; and $|u[n]| = 1$ for every n , so $\sum_n |u[n]|^2 = N$. In conclusion

$$\sum_{m=0}^{N-1} |I(m)|^2 = \frac{1}{N^2} \cdot N \cdot N = 1.$$

□

Corollary 27.6: Signal-to-interference ratio in closed form. Suppose the data $c_0, \dots, c_{N-1}$ have zero mean, are uncorrelated and have equal power, that is $\mathbb{E}[c_j] = 0$ and $\mathbb{E}[c_j \overline{c_{j'}}] = P \delta_{jj'}$. Then, for every k , the power of the useful term and that of the interference in Theorem 27.10 equal

$$\mathbb{E}[|I(0)c_k|^2] = |I(0)|^2 P, \quad \mathbb{E}\left[\left|\sum_{j \neq k} c_j I(j-k)\right|^2\right] = (1 - |I(0)|^2) P,$$

and hence the signal-to-interference ratio does not depend on k and equals

$$\text{SIR}(\varepsilon) = \frac{|I(0)|^2}{1 - |I(0)|^2}, \quad |I(0)|^2 = \left(\frac{\sin(\pi \varepsilon)}{N \sin(\pi \varepsilon/N)} \right)^2$$

Proof. Useful term. $\mathbb{E}[|I(0)c_k|^2] = |I(0)|^2 \mathbb{E}[|c_k|^2] = |I(0)|^2 P$, because $I(0)$ is a deterministic constant.

Interference term. Expanding the squared modulus of the sum,

$$\mathbb{E}\left[\left|\sum_{j \neq k} c_j I(j-k)\right|^2\right] = \sum_{j \neq k} \sum_{j' \neq k} \mathbb{E}[c_j \overline{c_{j'}}] I(j-k) \overline{I(j'-k)} = P \sum_{j \neq k} |I(j-k)|^2,$$

by the uncorrelation hypothesis, which annihilates all terms with $j \neq j'$.

Counting the indices. As j ranges over $\{0, \dots, N-1\}$ the difference $m = j-k$ ranges over the set $\{-k, -k+1, \dots, N-1-k\}$, which consists of N consecutive integers and is therefore a complete system of residues modulo N ; since I is N -periodic (Lemma 27.3), the sum $\sum_{j=0}^{N-1} |I(j-k)|^2$ coincides with $\sum_{m=0}^{N-1} |I(m)|^2$, which equals 1 by Proposition 27.4. Removing the term $j = k$,

which equals $|I(0)|^2$, leaves

$$\sum_{j \neq k} |I(j - k)|^2 = 1 - |I(0)|^2.$$

Conclusion. The ratio of the two powers is $\frac{|I(0)|^2 P}{(1 - |I(0)|^2)^P} = \frac{|I(0)|^2}{1 - |I(0)|^2}$, and does not depend on k ; the expression for $|I(0)|^2$ is that of Lemma 27.3. Note that $|I(0)| < 1$ for $\varepsilon \notin \mathbb{Z}$, otherwise Proposition 27.4 would force every other $|I(m)|$ to vanish, which is false: for instance $|I(1)| \neq 0$ because $\sin(\pi\varepsilon) \neq 0$. $\square$

Proposition 27.5: Behaviour for small frequency errors. *As $\varepsilon \rightarrow 0$ the expansions hold*

$$1 - |I(0)|^2 = \frac{\pi^2 \varepsilon^2}{3} \left(1 - \frac{1}{N^2}\right) + O(\varepsilon^4), \quad \text{SIR}(\varepsilon) \sim \frac{3}{\pi^2 \varepsilon^2 (1 - N^{-2})}.$$

In particular the interference power grows like the square of the frequency error, and for large N , $\text{SIR}(\varepsilon) \approx 3/(\pi\varepsilon)^2$, independent of N .

Proof. Set $a := \pi\varepsilon$ and $f(\varepsilon) := \frac{\sin(\pi\varepsilon)}{N \sin(\pi\varepsilon/N)}$, so that $|I(0)| = |f(\varepsilon)|$ and, for ε close to 0, $f(\varepsilon) > 0$.

Step 1: expansion of numerator and denominator. From the Taylor expansion $\sin x = x - \frac{x^3}{6} + O(x^5)$ one gets

$$\begin{aligned} \sin(\pi\varepsilon) &= a - \frac{a^3}{6} + O(a^5) = a \left(1 - \frac{a^2}{6} + O(a^4)\right), \\ N \sin\left(\frac{a}{N}\right) &= N \left(\frac{a}{N} - \frac{a^3}{6N^3} + O(a^5)\right) = a \left(1 - \frac{a^2}{6N^2} + O(a^4)\right). \end{aligned}$$

Step 2: the quotient. The factors a cancel (for $a \neq 0$) and, using $\frac{1}{1-u} = 1 + u + O(u^2)$ with $u = \frac{a^2}{6N^2} + O(a^4)$,

$$f = \left(1 - \frac{a^2}{6} + O(a^4)\right) \left(1 + \frac{a^2}{6N^2} + O(a^4)\right) = 1 - \frac{a^2}{6} \left(1 - \frac{1}{N^2}\right) + O(a^4).$$

Step 3: the square. Squaring and neglecting terms of order a^4 ,

$$f^2 = 1 - \frac{a^2}{3} \left(1 - \frac{1}{N^2}\right) + O(a^4), \quad 1 - f^2 = \frac{a^2}{3} \left(1 - \frac{1}{N^2}\right) + O(a^4).$$

Substituting $a = \pi\varepsilon$ gives the first formula.

Step 4: the asymptotic equivalence. By Corollary 27.6, $\text{SIR} = \frac{f^2}{1-f^2}$. Since $f^2 \rightarrow 1$ as $\varepsilon \rightarrow 0$ and $1 - f^2 \sim \frac{\pi^2 \varepsilon^2}{3} (1 - N^{-2})$,

$$\text{SIR}(\varepsilon) \cdot \frac{\pi^2 \varepsilon^2 (1 - N^{-2})}{3} = f^2 \cdot \frac{\pi^2 \varepsilon^2 (1 - N^{-2})/3}{1 - f^2} \rightarrow 1 \cdot 1 = 1,$$

which is the announced equivalence. Finally, for large N , $1 - N^{-2} \rightarrow 1$. $\square$

Remark 27.12: What these numbers say. Table 27.3 should be read together with Corollary 27.4. If thermal noise already produces an SNR of, say, 20 dB, then a frequency error of $\varepsilon = 0.05$ — that is, five per cent of the subcarrier spacing — adds an interference of the same order and, in practice, halves the useful ratio: the frequency error alone becomes the limiting factor. Two design consequences follow from this. The first is that

Table 27.3: Signal-to-interference ratio in the presence of a frequency error (Corollary 27.6), for $N = 64$, compared with the approximation of Proposition 27.5. The values in the middle column are the exact closed formula; those in the last column the asymptotic estimate. Both columns were recomputed with an independent script, which also verified $\sum_m |I(m)|^2 = 1$ up to 10^{-15} .

ε	$ I(0) ^2$	SIR	SIR (dB)	asympt. estimate (dB)
0.01	0.9996711	3 039.778	34.8284	34.8293
0.02	0.9986851	759.494	28.8052	28.8087
0.05	0.9918043	121.016	20.8284	20.8499
0.10	0.9675390	29.806	14.7431	14.8293
0.15	0.9281520	12.918	11.1120	11.3075
0.20	0.8751683	7.011	8.4577	8.8087
0.25	0.8106102	4.280	6.3146	6.8705

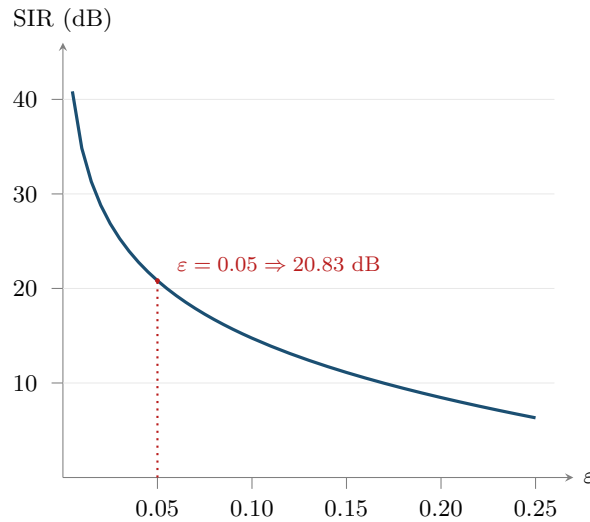


Figure 27.4: The signal-to-interference ratio as a function of the normalised frequency error, for $N = 64$ (Corollary 27.6). The ordinates are the exact values $10 \log_{10} \frac{|I(0)|^2}{1-|I(0)|^2}$ computed with step 0.005 up to $\varepsilon = 0.10$ and 0.01 beyond; the marked point corresponds to the third row of Table 27.3. The curve drops by about 6 dB for every doubling of ε , as predicted by Proposition 27.5: from 28.81 dB at $\varepsilon = 0.02$ to 22.77 dB at $\varepsilon = 0.04$, that is 6.0309 dB of difference.

every OFDM receiver contains an estimator of ε and a corrector that multiplies the received samples by $e^{-2\pi i \varepsilon n/N}$ before the FFT; the estimation algorithms are not treated here and are found in Cho, Kim, Yang and Kang, *MIMO-OFDM Wireless Communications with MATLAB*, Chapter 5 “Synchronization for OFDM”, pp. 153–185. The second is the one already anticipated in Observation 27.11: since $\varepsilon = \delta f N T_s$, for the same absolute error δf a block twice as long doubles ε and quadruples the interference power, that is, it loses about 6 dB of SIR.

27.7. Peak-to-average power ratio

The OFDM signal is a sum of N exponentials. At most instants the phases are scattered and the sum is modest; but there are instants at which they all line up, and then the amplitude is N times that of a single subcarrier while the average power is only N times that of one subcarrier: the peak-to-average ratio equals N . This is the structural flaw of OFDM, because a power amplifier that must stay linear up to the peak works, for almost all of the time, far below its own capacity.

Definition 27.9: Peak-to-average power ratio (PAPR). Let $s \in \mathbb{C}^N$ be nonzero. The peak-to-average power ratio (PAPR) of the sample sequence is

$$\text{PAPR}(s) = \frac{\max_{0 \leq n \leq N-1} |s[n]|^2}{\frac{1}{N} \sum_{n=0}^{N-1} |s[n]|^2},$$

and that of the continuous-time signal $s(t)$ of Definition 27.2 is

$$\text{PAPR}_c(s) = \frac{\sup_{t \in [0, T)} |s(t)|^2}{\frac{1}{T} \int_0^T |s(t)|^2 dt}.$$

Both are often expressed in decibels, as $10 \log_{10} \text{PAPR}$.

Lemma 27.4: The average power, in both versions, equals $\sum_k |c_k|^2$. Let s be the OFDM symbol associated with the data $c \in \mathbb{C}^N$. Then

$$\frac{1}{N} \sum_{n=0}^{N-1} |s[n]|^2 = \sum_{k=0}^{N-1} |c_k|^2 = \frac{1}{T} \int_0^T |s(t)|^2 dt.$$

The two average powers — the one computed on the samples and the one computed on the integral — coincide, and equal the sum of the data powers.

Proof. Discrete version. By Theorem 27.4, $S := \mathcal{F}_N s = Nc$, that is $|S[k]|^2 = N^2 |c_k|^2$. The discrete Plancherel identity (Theorem 22.6, second formula) gives

$$\sum_{n=0}^{N-1} |s[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |S[k]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} N^2 |c_k|^2 = N \sum_{k=0}^{N-1} |c_k|^2,$$

and dividing by N gives the first equality.

Continuous version. Expanding the squared modulus and exchanging finite sums with the integral,

$$\begin{aligned}\int_0^T |s(t)|^2 dt &= \int_0^T \left(\sum_k c_k \phi_k(t) \right) \overline{\left(\sum_\ell c_\ell \phi_\ell(t) \right)} dt \\ &= \sum_k \sum_\ell c_k \bar{c}_\ell \int_0^T \phi_k(t) \bar{\phi}_\ell(t) dt = \sum_k \sum_\ell c_k \bar{c}_\ell \langle \phi_k, \phi_\ell \rangle_T.\end{aligned}$$

By Theorem 27.1 (with $\Delta f = 1/T$, that is $\Delta f T = 1$ a positive integer), $\langle \phi_k, \phi_\ell \rangle_T = T \delta_{k\ell}$: only the terms with $\ell = k$ survive, and

$$\int_0^T |s(t)|^2 dt = T \sum_k c_k \bar{c}_k = T \sum_k |c_k|^2.$$

Dividing by T gives the second equality. $\square$

Theorem 27.11: The worst case: the PAPR does not exceed N , and N is attained. Let $c \in \mathbb{C}^N$ be nonzero and let s be the associated OFDM symbol. Then

$$1 \leq \text{PAPR}(s) \leq \frac{\left(\sum_{k=0}^{N-1} |c_k| \right)^2}{\sum_{k=0}^{N-1} |c_k|^2} \leq N,$$

and $\text{PAPR}(s) = N$ if and only if there exist $A > 0$, $\theta \in \mathbb{R}$ and $n_0 \in \{0, \dots, N-1\}$ such that

$$c_k = A e^{i\theta} \omega_N^{kn_0} = A e^{i\theta} e^{-2\pi i k n_0 / N} \quad \text{for every } k = 0, \dots, N-1.$$

In that case the peak is attained at the instant t_{n_0} and equals $|s[n_0]| = NA$.

Proof. Write $\mu := \frac{1}{N} \sum_n |s[n]|^2 = \sum_k |c_k|^2 > 0$, by Lemma 27.4 and because $c \neq 0$.

Step 1: the left-hand inequality. The maximum of a finite family of real numbers is never less than their arithmetic mean: $\max_n |s[n]|^2 \geq \frac{1}{N} \sum_n |s[n]|^2 = \mu$. Dividing by $\mu > 0$ gives $\text{PAPR}(s) \geq 1$.

Step 2: the numerator, via the triangle inequality. For every n ,

$$|s[n]| = \left| \sum_{k=0}^{N-1} c_k e^{2\pi i k n / N} \right| \leq \sum_{k=0}^{N-1} |c_k| |e^{2\pi i k n / N}| = \sum_{k=0}^{N-1} |c_k|,$$

since every imaginary exponential has modulus 1. Taking the maximum over n and squaring,

$$\max_n |s[n]|^2 \leq \left(\sum_k |c_k| \right)^2. \quad (27.3)$$

Dividing by $\mu = \sum_k |c_k|^2$ gives the middle inequality.

Step 3: Cauchy-Schwarz. Applying the Cauchy-Schwarz inequality in $\mathbb{R}^N$ to the vectors $(|c_0|, \dots, |c_{N-1}|)$ and $(1, 1, \dots, 1)$,

$$\sum_{k=0}^{N-1} |c_k| \cdot 1 \leq \sqrt{\sum_{k=0}^{N-1} |c_k|^2} \sqrt{\sum_{k=0}^{N-1} 1^2} = \sqrt{N} \sqrt{\sum_k |c_k|^2}, \quad (27.4)$$

whence, squaring, $\left(\sum_k |c_k| \right)^2 \leq N \sum_k |c_k|^2 = N\mu$. Dividing by μ gives the right-hand inequality, and with it $\text{PAPR}(s) \leq N$.

Step 4: the condition is sufficient. Suppose $c_k = Ae^{i\theta}\omega_N^{kn_0}$ with $A > 0$. Then, recalling $\omega_N^{kn_0} = e^{-2\pi i kn_0/N}$,

$$s[n_0] = \sum_{k=0}^{N-1} A e^{i\theta} e^{-2\pi i kn_0/N} e^{2\pi i kn_0/N} = A e^{i\theta} \sum_{k=0}^{N-1} 1 = N A e^{i\theta},$$

hence $|s[n_0]|^2 = N^2 A^2$; and $\mu = \sum_k |c_k|^2 = N A^2$. Therefore $\text{PAPR}(s) \geq \frac{N^2 A^2}{N A^2} = N$, and by Step 3 $\text{PAPR}(s) \leq N$: the two inequalities give equality.

Step 5: the condition is necessary. Suppose $\text{PAPR}(s) = N$. Then both inequalities (27.3) and (27.4) are equalities.

From (27.4): equality in the Cauchy–Schwarz inequality holds if and only if the two vectors are proportional; since the second is $(1, \dots, 1)$, there exists $A \geq 0$ with $|c_k| = A$ for every k , and $A > 0$ because $c \neq 0$.

From (27.3): let n_0 be an index at which the maximum is attained. Then

$$\left| \sum_{k=0}^{N-1} c_k e^{2\pi i kn_0/N} \right| = \sum_{k=0}^{N-1} |c_k| = \sum_{k=0}^{N-1} |c_k e^{2\pi i kn_0/N}|,$$

that is, there is equality in the triangle inequality for the family of nonzero complex numbers $z_k := c_k e^{2\pi i kn_0/N}$ (nonzero because $|z_k| = |c_k| = A > 0$). This occurs if and only if all the z_k have the same argument: there exists θ such that $z_k = |z_k| e^{i\theta} = A e^{i\theta}$ for every k . Solving for c_k ,

$$c_k = A e^{i\theta} e^{-2\pi i kn_0/N} = A e^{i\theta} \omega_N^{kn_0},$$

which is the announced form. The peak value is $|s[n_0]| = |N e^{i\theta} A| = N A$. $\square$

Corollary 27.7: The worst case in decibels. *The maximum PAPR, expressed in decibels, equals $10 \log_{10} N$: it grows by 3.0103 dB for every doubling of N . For $N = 4, 8, 64, 2048$ it equals, respectively, 6.0206, 9.0309, 18.0618 and 33.1133 dB.*

Proof. The maximum value is N by Theorem 27.11, and its expression in decibels is $10 \log_{10} N$ by definition. Doubling N gives $10 \log_{10}(2N) - 10 \log_{10} N = 10 \log_{10} 2 = 3.0103\dots$ The four numerical values are obtained with $\log_{10} 4 = 2 \log_{10} 2 = 0.60206$, $\log_{10} 8 = 3 \log_{10} 2 = 0.90309$, $\log_{10} 64 = 6 \log_{10} 2 = 1.80618$ and $\log_{10} 2048 = 11 \log_{10} 2 = 3.311330$, with $\log_{10} 2 = 0.3010300$. A script that computes $10 \log_{10} N$ for the same four values returns 6.020600, 9.030900, 18.061800 and 33.113300. $\checkmark$ $\square$

Example 27.4: The two extremes, realised. Both extremes of Theorem 27.11 are attained by QPSK data vectors, that is, with $c_k \in \{1, i, -1, -i\}$.

The worst case, $N = 4$. Let $c = (1, i, -1, -i)$, that is $c_k = i^k$. Since $i = e^{2\pi i/4} = \omega_4^{-1} = \omega_4^3$, one has $c_k = \omega_4^{3k}$, which is the form of Theorem 27.11 with $A = 1$, $\theta = 0$ and $n_0 = 3$. Hence the PAPR equals 4 and the peak falls at $n_0 = 3$. Indeed the direct computation (Section 27.8) gives $s = (0, 0, 0, 4)$: three zero samples and one equal to $NA = 4$. The PAPR is $\frac{16}{\frac{1}{4} \cdot 16} = \frac{16}{4} = 4 = N$, that is 6.0206 dB. $\checkmark$

The best case, $N = 8$. Let $c = (1, i, -1, 1, -1, i, 1, 1)$. The direct computation (Section 27.9) gives

$$s = (2 + 2i, 2 - 2i, -2 - 2i, 2 + 2i, -2 - 2i, 2 - 2i, 2 + 2i, 2 + 2i),$$

hence $|s[n]|^2 = 4 + 4 = 8$ for every n : the modulus is constant. The average power is 8, the peak is 8, and $\text{PAPR}(s) = 1$, that is 0 dB: the minimum possible by Step 1 of Theorem 27.11. $\checkmark$

What the comparison shows. The PAPR does not depend on the constellation but on the particular word transmitted: the same four QPSK symbols, arranged differently, give 6.02 dB in one case and 0 dB in the other. It is exactly this dependence on the data that PAPR-reduction techniques exploit.

Proposition 27.6: *The PAPR of the continuous signal is not smaller than that of the samples. For every $c \neq 0$, $\text{PAPR}_c(s) \geq \text{PAPR}(s)$.*

Proof. By Lemma 27.4 the two ratios have the same denominator, $\sum_k |c_k|^2$. As for the numerators, the sampling instants $t_0, \dots, t_{N-1}$ belong to the interval $[0, T)$, hence

$$\sup_{t \in [0, T)} |s(t)|^2 \geq \max_{0 \leq n \leq N-1} |s(t_n)|^2 = \max_{0 \leq n \leq N-1} |s[n]|^2,$$

because the supremum over a set is not less than the maximum over a subset of it. Dividing by the common, positive, denominator concludes the proof. $\square$

The question that really matters in practice remains: how *likely* is a high PAPR, if the data are random? The exact answer is not known in closed form; an estimate valid under a hypothesis that must be stated is available, and that is what is done here.

Proposition 27.7: *Distribution of the PAPR under the Gaussian hypothesis (G).*

Assume **hypothesis (G)**: the samples $s[0], \dots, s[N-1]$ are independent random variables, each a circularly symmetric complex Gaussian of zero mean and power $\mu > 0$, that is $s[n] = x_n + iy_n$ with x_n, y_n independent and Gaussian with zero mean and variance $\mu/2$. Set

$$\Gamma := \frac{\max_{0 \leq n \leq N-1} |s[n]|^2}{\mu}.$$

Then, for every $\gamma \geq 0$,

$$\Pr(\Gamma \leq \gamma) = (1 - e^{-\gamma})^N, \quad \Pr(\Gamma > \gamma) = 1 - (1 - e^{-\gamma})^N.$$

Proof. Step 1: the law of $|s[n]|^2$. The joint density of (x_n, y_n) , the product of two independent Gaussian densities of variance $\mu/2$, is

$$p(x, y) = \frac{1}{\sqrt{2\pi \cdot \frac{\mu}{2}}} e^{-x^2/\mu} \cdot \frac{1}{\sqrt{2\pi \cdot \frac{\mu}{2}}} e^{-y^2/\mu} = \frac{1}{\pi\mu} e^{-(x^2+y^2)/\mu}.$$

For $u \geq 0$, switching to polar coordinates $x = r \cos \vartheta$, $y = r \sin \vartheta$, with area element $r dr d\vartheta$,

$$\Pr(|s[n]|^2 > u) = \iint_{x^2+y^2 > u} p(x, y) dx dy = \int_0^{2\pi} \int_{\sqrt{u}}^{\infty} \frac{r}{\pi\mu} e^{-r^2/\mu} dr d\vartheta = \frac{2\pi}{\pi\mu} \int_{\sqrt{u}}^{\infty} r e^{-r^2/\mu} dr.$$

The antiderivative of $r e^{-r^2/\mu}$ is $-\frac{\mu}{2} e^{-r^2/\mu}$, hence

$$\Pr(|s[n]|^2 > u) = \frac{2}{\mu} \left[-\frac{\mu}{2} e^{-r^2/\mu} \right]_{\sqrt{u}}^{\infty} = \frac{2}{\mu} \cdot \frac{\mu}{2} e^{-u/\mu} = e^{-u/\mu}.$$

The squared modulus is thus an exponential random variable of mean μ , and $\Pr(|s[n]|^2 \leq u) = 1 - e^{-u/\mu}$.

Step 2: the maximum of independent variables. The event $\{\Gamma \leq \gamma\}$ coincides with $\bigcap_{n=0}^{N-1} \{|s[n]|^2 \leq \gamma\mu\}$: the maximum does not exceed $\gamma\mu$ if and only if none of the terms

exceeds it. By the independence postulated by hypothesis (G) the probability of the intersection is the product of the probabilities, and by Step 1 each factor equals $1 - e^{-\gamma\mu/\mu} = 1 - e^{-\gamma}$. Hence

$$\Pr(\Gamma \leq \gamma) = \prod_{n=0}^{N-1} (1 - e^{-\gamma}) = (1 - e^{-\gamma})^N,$$

and the second formula is the complement to 1 of the first. $\square$

Table 27.4: Probability that the PAPR exceeds a threshold γ , *under hypothesis (G)* of Proposition 27.7: $1 - (1 - e^{-\gamma})^N$ with $\gamma = 10^{\gamma_{\text{dB}}/10}$. The values were computed with a script. Compare them with the worst case of Corollary 27.7: for $N = 2048$ the theoretical maximum is 33.11 dB, but the threshold exceeded about once in four thousand is already at 12 dB. What in this table is proved and what is imported is stated in Observation 27.13.

N	Pr(PAPR > γ) under (G)		
	$\gamma = 8$ dB	$\gamma = 10$ dB	$\gamma = 12$ dB
64	$1.0998 \cdot 10^{-1}$	$2.9014 \cdot 10^{-3}$	$8.3767 \cdot 10^{-6}$
256	$3.7252 \cdot 10^{-1}$	$1.1555 \cdot 10^{-2}$	$3.3506 \cdot 10^{-5}$
2048	$9.7597 \cdot 10^{-1}$	$8.8789 \cdot 10^{-2}$	$2.6802 \cdot 10^{-4}$

Proposition 27.8: *The samples are uncorrelated, but not independent.* Let $N \geq 2$ and let $c_0, \dots, c_{N-1}$ have zero mean, be uncorrelated and have equal power P , that is $\mathbb{E}[c_k] = 0$ and $\mathbb{E}[c_k \bar{c}_\ell] = P\delta_{k\ell}$. Then

$$\mathbb{E}[s[n]] = 0, \quad \mathbb{E}[s[n] \overline{s[m]}] = NP\delta_{nm} :$$

the samples have zero mean, are uncorrelated and have equal power NP .

Suppose further that the c_k are independent and uniform on a constant-modulus constellation $\{Ae^{2\pi i j/q} : j = 0, \dots, q-1\}$ with $A > 0$ and $q \geq 2$ (for instance BPSK with $q = 2$ or QPSK with $q = 4$). Then the samples $s[0], \dots, s[N-1]$ are not independent.

Proof. Step 1: mean and covariance. The mean is zero by linearity: $\mathbb{E}[s[n]] = \sum_k \mathbb{E}[c_k] \omega_N^{-kn} = 0$. For the covariance,

$$\mathbb{E}[s[n] \overline{s[m]}] = \sum_{k=0}^{N-1} \sum_{\ell=0}^{N-1} \mathbb{E}[c_k \bar{c}_\ell] \omega_N^{-kn} \overline{\omega_N^{-\ell m}} = P \sum_{k=0}^{N-1} \omega_N^{-kn} \omega_N^{km} = P \sum_{k=0}^{N-1} \omega_N^{k(m-n)},$$

where the uncorrelation hypothesis was used, annihilating the terms with $\ell \neq k$, and $\overline{\omega_N^{-\ell m}} = \omega_N^{\ell m}$. By Lemma 22.1 the last sum equals N if $N \mid (m-n)$, that is — since $n, m \in \{0, \dots, N-1\}$ — if and only if $n = m$, and 0 otherwise. It follows that $\mathbb{E}[s[n] \overline{s[m]}] = NP\delta_{nm}$, and in particular $\mathbb{E}[|s[n]|^2] = NP = NA^2$, since for a constant-modulus constellation $P = \mathbb{E}[|c_k|^2] = A^2$.

Step 2: the sum of squared moduli is deterministic. By Lemma 27.4, for every realisation of the data,

$$\sum_{n=0}^{N-1} |s[n]|^2 = N \sum_{k=0}^{N-1} |c_k|^2 = N \cdot NA^2 = N^2 A^2,$$

because $|c_k| = A$ for every k and for every realisation. The random variable $\sum_n |s[n]|^2$ is thus constant, and its variance is zero.

Step 3: the contradiction. Suppose, for contradiction, that $s[0], \dots, s[N-1]$ are independent. Then so are the real variables $|s[0]|^2, \dots, |s[N-1]|^2$, being measurable functions of independent variables; and they have finite variance, because they take finitely many values (the

constellation is finite). By the formula for the variance of a sum of independent variables,

$$0 = \text{Var} \left(\sum_{n=0}^{N-1} |s[n]|^2 \right) = \sum_{n=0}^{N-1} \text{Var} (|s[n]|^2).$$

Being a sum of nonnegative terms equal to zero, every term is zero: hence each $|s[n]|^2$ is almost surely equal to its own mean, which by Step 1 equals NA^2 .

But this is false. The realisation $c_0 = c_1 = \dots = c_{N-1} = A$ has probability $q^{-N} > 0$, and for it

$$s[0] = \sum_{k=0}^{N-1} A \cdot 1 = NA, \quad |s[0]|^2 = N^2 A^2,$$

while $N^2 A^2 \neq NA^2$ because $N \geq 2$ and $A > 0$. The variable $|s[0]|^2$ thus takes, with positive probability, a value different from NA^2 , contradicting what was just deduced. The independence hypothesis is absurd. $\square$

Remark 27.13: What is imported on the statistics of the PAPR. Proposition 27.7 is proved in full, but under hypothesis (G); and Proposition 27.8 proves that hypothesis (G) is false for a real OFDM system: the samples are uncorrelated, not independent, and they are not Gaussian, because they take finitely many values. The formula $1 - (1 - e^{-\gamma})^N$ of Table 27.4 is therefore, in this treatise, **imported** as an approximation, not proved as a theorem. What is missing is stated precisely.

First, Gaussianity. For large N the sum $s[n] = \sum_k c_k e^{2\pi i k n / N}$ has many independent, bounded terms, and a central limit theorem makes it a complex Gaussian in the limit $N \rightarrow \infty$. But a limit theorem controls the convergence of the distribution in the bulk, not necessarily in the tail, and it is precisely the tail — probabilities of order 10^{-4} and smaller — that is needed in Table 27.4. An estimate with explicit control of the tail error requires concentration inequalities (for instance of Berry–Esseen type or large deviations) that are not developed here.

Second, independence. Proposition 27.8 shows that the samples are only uncorrelated. For jointly Gaussian variables, uncorrelation and independence coincide, but the samples are not jointly Gaussian: the implication does not apply, and the step from the maximum of dependent variables to the product of the marginal probabilities (Step 2 of Proposition 27.7) is not justified.

Third, the continuous signal. Proposition 27.6 proves only the inequality $\text{PAPR}_c \geq \text{PAPR}$. By how much it is larger — that is, how much oversampling is needed for the samples to capture the true peak, with the practical rule of thumb of a factor of four — is an experimental and asymptotic result that is neither proved nor used in any proof here.

Source of what is imported. Y. S. Cho, J. Kim, W. Y. Yang, C. G. Kang, *MIMO-OFDM Wireless Communications with MATLAB*, Wiley, 2010, Chapter 7 “PAPR Reduction”, pp. 209–250, where the tail formula, the Gaussian hypothesis and the practical oversampling rule are presented together. All the statements of this section that do not depend on (G) — Lemma 27.4, Theorem 27.11, Corollary 27.7, Propositions 27.6 and 27.8 — are instead proved here in full, and depend on nothing imported.

27.8. The example $N = 4$, worked out in full

The whole chain is now traced through on a case small enough to be worked by hand at every step, and rich enough to show every phenomenon. The data are

$$N = 4, \quad c = (1, i, -1, -i), \quad h = \left(1, \frac{1}{2}\right), \quad L = 2, \quad N_g = 1.$$

The choice $N_g = 1$ is the minimum allowed by Corollary 27.5, since $L - 1 = 1$.

27.8.1 The modulator, in exact symbolic form

Since $N = 4$ one has $e^{2\pi i k n / 4} = i^{kn}$ and, since $c_k = i^k$,

$$s[n] = \sum_{k=0}^3 c_k e^{2\pi i k n / 4} = \sum_{k=0}^3 i^k i^{kn} = \sum_{k=0}^3 (i^{n+1})^k.$$

The last expression is a geometric sum of ratio $q = i^{n+1}$, a fourth root of unity. By Lemma 22.1 it equals 4 if $q = 1$, that is if $4 \mid n + 1$, and 0 otherwise. For $n \in \{0, 1, 2, 3\}$ the only solution is $n = 3$, hence

$$s = (0, 0, 0, 4)$$

The computation can also be done by hand, term by term:

$$\begin{aligned} n = 0 : & \quad 1 + i + i^2 + i^3 = 1 + i - 1 - i = 0, \\ n = 1 : & \quad 1 + i^2 + i^4 + i^6 = 1 - 1 + 1 - 1 = 0, \\ n = 2 : & \quad 1 + i^3 + i^6 + i^9 = 1 - i - 1 + i = 0, \\ n = 3 : & \quad 1 + i^4 + i^8 + i^{12} = 1 + 1 + 1 + 1 = 4. \end{aligned}$$

✓ The two computations agree.

Check with the inverse transform. By Theorem 27.3 one must have $s = 4\mathcal{F}_4^{-1}c$. Applying Definition 22.2 with $\omega_4^{-1} = e^{2\pi i / 4} = i$:

$$4(\mathcal{F}_4^{-1}c)[3] = \sum_{k=0}^3 c_k i^{3k} = 1 + i \cdot (-i) + (-1) \cdot (-1) + (-i) \cdot i = 1 + 1 + 1 + 1 = 4. \quad \checkmark$$

The PAPR. By Lemma 27.4 the average power equals $\sum_k |c_k|^2 = 4$; the peak equals $|s[3]|^2 = 16$; hence $\text{PAPR}(s) = 16/4 = 4 = N$: this is the worst case, and the form $c_k = \omega_4^{3k}$ is exactly the one predicted by Theorem 27.11 with $A = 1$, $\theta = 0$, $n_0 = 3$ (Example 27.4).

27.8.2 The channel and its frequency response

The zero-padding is $h_p = (1, \frac{1}{2}, 0, 0)$, and by Corollary 27.3, with $\omega_4 = e^{-2\pi i / 4} = -i$,

$$H[k] = 1 + \frac{1}{2} \omega_4^k = 1 + \frac{1}{2} (-i)^k,$$

that is

$$H[0] = \frac{3}{2}, \quad H[1] = 1 - \frac{i}{2}, \quad H[2] = \frac{1}{2}, \quad H[3] = 1 + \frac{i}{2},$$

having used $(-i)^0 = 1$, $(-i)^1 = -i$, $(-i)^2 = -1$, $(-i)^3 = i$. The moduli are

$$|H[0]| = \frac{3}{2} = 1.5, \quad |H[1]| = |H[3]| = \sqrt{1 + \frac{1}{4}} = \frac{\sqrt{5}}{2} = 1.1180340, \quad |H[2]| = \frac{1}{2} = 0.5.$$

Note the Hermitian symmetry $H[3] = \overline{H[1]}$, a consequence of h being real (Theorem 22.7).

27.8.3 Cyclic prefix, channel, and the convolution that becomes circular

The transmitted block is, by Definition 27.4 with $N_g = 1$,

$$\tilde{s} = (s[3], s[0], s[1], s[2], s[3]) = (4, 0, 0, 0, 4) \in \mathbb{C}^5.$$

The channel output is $r[n] = \tilde{s}[n] + \frac{1}{2}\tilde{s}[n-1]$, where $\tilde{s}[-1] = p$ is the last sample of the previous block, arbitrary:

$$r = \left(4 + \frac{p}{2}, 0 + \frac{1}{2} \cdot 4, 0, 0, 4 + 0, 0 + \frac{1}{2} \cdot 4\right) = \left(4 + \frac{p}{2}, 2, 0, 0, 4, 2\right).$$

The receiver discards $r[0]$ — the only sample in which p appears, in agreement with Lemma 27.1 — and keeps

$$y = (r[1], r[2], r[3], r[4]) = (2, 0, 0, 4).$$

Note that y does not depend on p : the previous block has been eliminated.

Check: circular convolution. By Theorem 27.5 one must have $y = h_p \circledast s$. Writing the four rows of Definition 22.4 with $x = h_p = (1, \frac{1}{2}, 0, 0)$ and $s = (0, 0, 0, 4)$:

$$\begin{aligned} (h_p \circledast s)[0] &= h_p[0]s[0] + h_p[1]s[3] + h_p[2]s[2] + h_p[3]s[1] = 0 + \frac{1}{2} \cdot 4 + 0 + 0 = 2, \\ (h_p \circledast s)[1] &= h_p[0]s[1] + h_p[1]s[0] + h_p[2]s[3] + h_p[3]s[2] = 0 + 0 + 0 + 0 = 0, \\ (h_p \circledast s)[2] &= h_p[0]s[2] + h_p[1]s[1] + h_p[2]s[0] + h_p[3]s[3] = 0 + 0 + 0 + 0 = 0, \\ (h_p \circledast s)[3] &= h_p[0]s[3] + h_p[1]s[2] + h_p[2]s[1] + h_p[3]s[0] = 4 + 0 + 0 + 0 = 4. \end{aligned}$$

Hence $h_p \circledast s = (2, 0, 0, 4) = y$. ✓

Check: the matrix form. The circulant matrix of Theorem 27.7 is $(C_h)_{mj} = h_p[\langle m-j \rangle_4]$, that is

$$C_h = \begin{pmatrix} 1 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 1 & 0 & 0 \\ 0 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{1}{2} & 1 \end{pmatrix}, \quad C_h \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \cdot 4 \\ 0 \\ 0 \\ 1 \cdot 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \\ 4 \end{pmatrix} = y. \quad \checkmark$$

(The first row is obtained from $h_p[\langle 0-j \rangle_4]$ for $j = 0, 1, 2, 3$, that is $h_p[0], h_p[3], h_p[2], h_p[1]$, which equal $1, 0, 0, \frac{1}{2}$.)

27.8.4 Verifying that the channel has become diagonal

Compute $Y = \mathcal{F}_4 y$ with $y = (2, 0, 0, 4)$. By Definition 22.1, $Y[k] = 2\omega_4^0 + 4\omega_4^{3k} = 2 + 4i^k$, having used $\omega_4^3 = (-i)^3 = i$. Hence

$$Y = (6, 2 + 4i, -2, 2 - 4i).$$

On the other hand $S = \mathcal{F}_4 s = 4c = (4, 4i, -4, -4i)$ (Theorem 27.4), and Corollary 27.3 predicts $Y[k] = H[k]S[k]$. Check component by component:

$$\begin{aligned} k=0: & \quad H[0]S[0] = \frac{3}{2} \cdot 4 = 6 & = Y[0] & \checkmark \\ k=1: & \quad H[1]S[1] = (1 - \frac{i}{2})(4i) = 4i - 2i^2 = 2 + 4i & = Y[1] & \checkmark \\ k=2: & \quad H[2]S[2] = \frac{1}{2} \cdot (-4) = -2 & = Y[2] & \checkmark \\ k=3: & \quad H[3]S[3] = (1 + \frac{i}{2})(-4i) = -4i - 2i^2 = 2 - 4i = Y[3] & \checkmark \end{aligned}$$

All four equalities are exact: the channel, which in the time domain mixes the samples (the matrix C_h is not diagonal, it has two nonzero diagonals), has become, in the frequency domain, multiplication by four numbers.

The eigenvalues, and the determinant computed twice. By Theorem 27.7(iii) the eigenvalues of C_h are $\frac{3}{2}, 1 - \frac{i}{2}, \frac{1}{2}, 1 + \frac{i}{2}$. By part (iv),

$$\det C_h = \prod_{k=0}^3 H[k] = \frac{3}{2} \cdot \frac{1}{2} \cdot \left(1 - \frac{i}{2}\right) \left(1 + \frac{i}{2}\right) = \frac{3}{4} \cdot \left(1 + \frac{1}{4}\right) = \frac{3}{4} \cdot \frac{5}{4} = \frac{15}{16} = 0.9375.$$

Independent check, by an algebraic route. The $H[k]$ are the values of the polynomial $p(z) = 1 + \frac{z}{2}$ at the four fourth roots of unity. Hence

$$\prod_{k=0}^3 H[k] = \prod_{z^4=1} \left(1 + \frac{z}{2}\right) = \frac{1}{2^4} \prod_{z^4=1} (2 + z) = \frac{1}{16} \prod_{z^4=1} (-((-2) - z)) = \frac{(-1)^4}{16} \prod_{z^4=1} ((-2) - z).$$

Since $\prod_{z^4=1} (t - z) = t^4 - 1$ for every t (the fourth roots of unity are exactly the roots of $t^4 - 1$), with $t = -2$ one gets $(-2)^4 - 1 = 15$, and $\det C_h = \frac{15}{16} = 0.9375$. ✓ The two computations, by entirely independent routes, give the same number.

Third check, with a script. A numerical computation that builds C_h and takes its determinant with the linear algebra routine returns 0.9375, and the product of the numerical eigenvalues returns $0.9375 - 6.94 \cdot 10^{-17}i$; the discrepancy between Y and $H \cdot S$ is $1.78 \cdot 10^{-15}$, that is zero up to double-precision rounding. ✓

27.8.5 Equalisation

By Definition 27.5, $\hat{c}_k = \frac{Y[k]}{4H[k]}$:

$$\hat{c}_0 = \frac{6}{4 \cdot \frac{3}{2}} = \frac{6}{6} = 1 = c_0,$$

$$\hat{c}_1 = \frac{2 + 4i}{4(1 - \frac{i}{2})} = \frac{2 + 4i}{4 - 2i} = \frac{(2 + 4i)(4 + 2i)}{(4 - 2i)(4 + 2i)} = \frac{8 + 4i + 16i + 8i^2}{16 + 4} = \frac{20i}{20} = i = c_1,$$

$$\hat{c}_2 = \frac{-2}{4 \cdot \frac{1}{2}} = \frac{-2}{2} = -1 = c_2,$$

$$\hat{c}_3 = \frac{2 - 4i}{4(1 + \frac{i}{2})} = \frac{2 - 4i}{4 + 2i} = \frac{(2 - 4i)(4 - 2i)}{20} = \frac{8 - 4i - 16i + 8i^2}{20} = \frac{-20i}{20} = -i = c_3.$$

All four symbols are reconstructed *exactly*, as predicted by Proposition 27.2. ✓

27.8.6 What happens without the cyclic prefix

Repeat the transmission omitting the prefix: only the block $s = (0, 0, 0, 4)$ is sent, and the channel produces $r'[n] = s[n] + \frac{1}{2}s[n-1]$ with $s[-1] = p$ the last sample of the previous block. Then

$$y' = \left(\frac{p}{2}, 0, 0, 4\right).$$

Theorem 27.5 predicts, for the case with the prefix, $y[0] = 2$: the difference $y'[0] - y[0] = \frac{p}{2} - 2$ vanishes *if and only if* $p = 4 = s[3]$, that is if and only if the sample preceding the block is exactly its own tail — which is exactly what the cyclic prefix guarantees. Here is Theorem 27.5's statement seen up close: the prefix adds no information, it *creates the boundary condition* that makes the convolution circular.

Consider the consequences for $p = 0$ (previous block ending with a zero sample). Then $y' = (0, 0, 0, 4)$ and

$$Y'[k] = 4\omega_4^{3k} = 4i^k, \quad Y' = (4, 4i, -4, -4i),$$

whence

$$\hat{c}'_k = \frac{Y'[k]}{4H[k]} : \quad \hat{c}'_0 = \frac{4}{6} = \frac{2}{3} = 0.6666667, \quad \hat{c}'_1 = \frac{4i}{4 - 2i} = \frac{4i(4 + 2i)}{20} = -0.4 + 0.8i,$$

$$\hat{c}'_2 = \frac{-4}{2} = -2, \quad \hat{c}'_3 = \frac{-4i}{4 + 2i} = \frac{-4i(4 - 2i)}{20} = -0.4 - 0.8i.$$

None of the four coincides with the data: $\hat{c}' = (0.667, -0.4 + 0.8i, -2, -0.4 - 0.8i)$ against $c = (1, i, -1, -i)$. ✓ (Also verified with a script: the same four values, with maximum discrepancy $9.80 \cdot 10^{-16}$ against the values computed by hand.)

Warning. Note the most instructive point of the example: the error committed in the time domain concerns *only one* sample ($y'[0]$ instead of $y[0]$), but after the DFT it is spread over *all four* subcarriers. There is no subcarrier “spared” by the absence of the prefix: the transform of a concentrated error is a diffuse error. This is the reason why the guard interval cannot be shortened “just a little”.

Remark 27.14: The balance sheet of the example. Summarising the numbers: four QPSK symbols transmitted over five sampling intervals, that is an efficiency $\eta = \frac{4}{5} = 0.8$ and a loss of 20% (0.9691 dB, Table 27.2); a PAPR of 4, that is 6.0206 dB, which is the maximum possible (Corollary 27.7); a two-tap channel equalised with four complex divisions instead of an inverse filter; and an exact reconstruction. With $\varepsilon = 0.05$ of frequency error, the signal-to-interference ratio would be, by Corollary 27.6 with $N = 4$, $|I(0)|^2 = 0.9923123$ and SIR = 129.0783, that is 21.1085 dB. The next example repeats everything with $N = 8$ and a three-tap channel, where the computations are less trivial and the diagonality is more instructive.

27.9. The example $N = 8$, worked out in full

The whole path is repeated with eight subcarriers and a three-tap channel, where nothing simplifies by symmetry and the diagonality check is genuine. The data are

$$N = 8, \quad c = (1, i, -1, 1, -1, i, 1, 1), \quad h = \left(1, \frac{1}{2}, \frac{1}{4}\right), \quad L = 3, \quad N_g = 2,$$

and $N_g = 2 = L - 1$ is again the minimal choice of Corollary 27.5. Write $\beta := \frac{\sqrt{2}}{2} = 0.7071068$ and recall that $\zeta := e^{2\pi i/8} = \beta(1 + i)$ and $\omega_8 = e^{-2\pi i/8} = \beta(1 - i)$, with

$$\begin{aligned} \zeta^0 &= 1, & \zeta^1 &= \beta(1 + i), & \zeta^2 &= i, & \zeta^3 &= \beta(-1 + i), \\ \zeta^4 &= -1, & \zeta^5 &= \beta(-1 - i), & \zeta^6 &= -i, & \zeta^7 &= \beta(1 - i). \end{aligned}$$

27.9.1 The modulator, in exact symbolic form

By Definition 27.2, $s[n] = \sum_{k=0}^7 c_k \zeta^{kn}$. The first two samples are worked out in full.

Sample $n = 0$. All the ζ^0 equal 1, hence

$$s[0] = \sum_{k=0}^7 c_k = 1 + i - 1 + 1 - 1 + i + 1 + 1 = 2 + 2i.$$

Sample $n = 1$. Here $\zeta^{k \cdot 1} = \zeta^k$, and collecting the terms with β separately from those without:

$$s[1] = \underbrace{1 \cdot 1}_{k=0} + \underbrace{i \beta(1 + i)}_{k=1} + \underbrace{(-1) i}_{k=2} + \underbrace{1 \beta(-1 + i)}_{k=3} + \underbrace{(-1)(-1)}_{k=4} + \underbrace{i \beta(-1 - i)}_{k=5} + \underbrace{1(-i)}_{k=6} + \underbrace{1 \beta(1 - i)}_{k=7}.$$

The terms without β are $1 - i + 1 - i = 2 - 2i$. The terms with β have, for the sum of their coefficients,

$$i(1 + i) + (-1 + i) + i(-1 - i) + (1 - i) = (i - 1) + (-1 + i) + (-i + 1) + (1 - i) = 0,$$

having used $i \cdot i = -1$ and $i \cdot (-i) = 1$. Hence $s[1] = 2 - 2i$.

Proceeding in the same way for the other six samples gives

$$s = (2 + 2i, 2 - 2i, -2 - 2i, 2 + 2i, -2 - 2i, 2 - 2i, 2 + 2i, 2 + 2i)$$

All eight samples have the form $\pm 2 \pm 2i$: the terms with β always cancel. The result was rechecked with a script that computes both the defining sum and $8\mathcal{F}_8^{-1}c$, with maximum discrepancy $6.28 \cdot 10^{-16}$. ✓

The PAPR. One has $|s[n]|^2 = 2^2 + 2^2 = 8$ for every n : the modulus is constant. The average power is $\sum_k |c_k|^2 = 8$ (Lemma 27.4), the peak is 8, hence

$$\text{PAPR}(s) = \frac{8}{8} = 1 = 0 \text{ dB},$$

the minimum possible (Theorem 27.11, Step 1). The same QPSK alphabet that gave the worst case in the example $N = 4$ gives here the best case: the PAPR is a property of the word, not of the constellation (Example 27.4).

27.9.2 The frequency response of the channel

With $h_p = (1, \frac{1}{2}, \frac{1}{4}, 0, 0, 0, 0, 0)$ and $H[k] = 1 + \frac{1}{2}\omega_8^k + \frac{1}{4}\omega_8^{2k}$ (Corollary 27.3), using $\omega_8^2 = -i$, $\omega_8^3 = \beta(-1 - i)$, $\omega_8^4 = -1$, $\omega_8^6 = i$:

$$H[0] = 1 + \frac{1}{2} + \frac{1}{4} = \frac{7}{4} = 1.75,$$

$$H[1] = 1 + \frac{\beta}{2}(1 - i) + \frac{1}{4}(-i) = 1 + \frac{\sqrt{2}}{4} - i\left(\frac{\sqrt{2}}{4} + \frac{1}{4}\right) = 1.3535534 - 0.6035534i,$$

$$H[2] = 1 + \frac{1}{2}(-i) + \frac{1}{4}(-1) = \frac{3}{4} - \frac{1}{2}i = 0.75 - 0.5i,$$

$$H[3] = 1 + \frac{\beta}{2}(-1 - i) + \frac{1}{4}i = 1 - \frac{\sqrt{2}}{4} + i\left(\frac{1}{4} - \frac{\sqrt{2}}{4}\right) = 0.6464466 - 0.1035534i,$$

$$H[4] = 1 - \frac{1}{2} + \frac{1}{4} = \frac{3}{4} = 0.75,$$

and $H[5] = \overline{H[3]}$, $H[6] = \overline{H[2]}$, $H[7] = \overline{H[1]}$ by Hermitian symmetry (Theorem 22.7, applicable because h is real). The squared moduli, in exact form, are

$$|H[0]|^2 = \frac{49}{16}, \quad |H[1]|^2 = |H[7]|^2 = \frac{21}{16} + \frac{5\sqrt{2}}{8}, \quad |H[2]|^2 = |H[6]|^2 = \frac{13}{16},$$

$$|H[3]|^2 = |H[5]|^2 = \frac{21}{16} - \frac{5\sqrt{2}}{8}, \quad |H[4]|^2 = \frac{9}{16}.$$

For instance, for $k = 1$: $(1 + \frac{\sqrt{2}}{4})^2 + (\frac{1}{4} + \frac{\sqrt{2}}{4})^2 = 1 + \frac{\sqrt{2}}{2} + \frac{1}{8} + \frac{1}{16} + \frac{\sqrt{2}}{8} + \frac{1}{8} = \frac{21}{16} + \frac{5\sqrt{2}}{8}$, and for $k = 3$ the sign of $\sqrt{2}$ changes. In decimal form

$$|H[k]| = (1.75, 1.4820201, 0.9013878, 0.6546881, 0.75, 0.6546881, 0.9013878, 1.4820201).$$

The worst subcarrier is the third one (and the fifth): the gap with respect to the best one equals $\frac{49}{21-10\sqrt{2}} = 7.1450815$, that is 8.5400719 dB (Example 27.2).

27.9.3 Cyclic prefix and channel output

The transmitted block, by Definition 27.4 with $N_g = 2$, is $\tilde{s} = (s[6], s[7], s[0], \dots, s[7]) \in \mathbb{C}^{10}$, that is

$$\tilde{s} = (2 + 2i, 2 + 2i, \mid 2 + 2i, 2 - 2i, -2 - 2i, 2 + 2i, -2 - 2i, 2 - 2i, 2 + 2i, 2 + 2i),$$

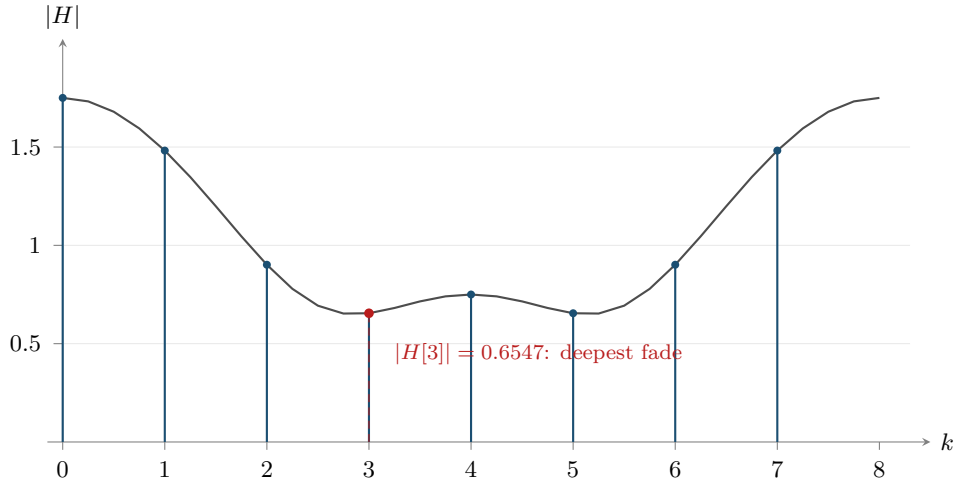


Figure 27.5: The frequency response of the channel $h = (1, \frac{1}{2}, \frac{1}{4})$ of the example $N = 8$. The grey curve is $|H(\theta)|$ as the frequency varies continuously, computed with step 0.25 in units of subcarrier spacing; the blue segments are the eight samples $|H[k]|$, that is, the moduli of the eigenvalues of the circulant matrix C_h (Theorem 27.7). The subcarrier $k = 3$ and its symmetric counterpart $k = 5$ fall in the minimum and suffer 8.54 dB of disadvantage relative to subcarrier $k = 0$ (Corollary 27.4 and Example 27.2). The value $|H[8]|$ does not exist: the plot is 8-periodic, and $k = 8$ coincides with $k = 0$ (Observation 27.3).

where the bar separates the prefix from the useful block. The channel output is $r[n] = \tilde{s}[n] + \frac{1}{2}\tilde{s}[n-1] + \frac{1}{4}\tilde{s}[n-2]$; assuming for simplicity that the two preceding samples are zero (Lemma 27.1 guarantees that their actual value does not affect the kept samples),

$$r = (2 + 2i, 3 + 3i, \mid 3.5 + 3.5i, 3.5 - 0.5i, -0.5 - 2.5i, 1.5 + 0.5i, \\ -1.5 - 1.5i, 1.5 - 2.5i, 2.5 + 0.5i, 3.5 + 2.5i, \dots).$$

For instance $r[2] = \tilde{s}[2] + \frac{1}{2}\tilde{s}[1] + \frac{1}{4}\tilde{s}[0] = (2+2i) + \frac{1}{2}(2+2i) + \frac{1}{4}(2+2i) = \frac{7}{4}(2+2i) = 3.5+3.5i$, and $r[3] = \tilde{s}[3] + \frac{1}{2}\tilde{s}[2] + \frac{1}{4}\tilde{s}[1] = (2-2i) + \frac{1}{2}(2+2i) + \frac{1}{4}(2+2i) = (2-2i) + (1+i) + (0.5+0.5i) = 3.5-0.5i$. Discarding the first two samples,

$$y = (3.5 + 3.5i, 3.5 - 0.5i, -0.5 - 2.5i, 1.5 + 0.5i, -1.5 - 1.5i, 1.5 - 2.5i, 2.5 + 0.5i, 3.5 + 2.5i).$$

Check: circular convolution. Theorem 27.5 predicts $y = h_p \circledast s$. Check component $m = 0$, the one in which modular wraparound occurs (case (b) of Step 3 of that proof):

$$\begin{aligned} (h_p \circledast s)[0] &= h_p[0]s[0] + h_p[1]s[\langle -1 \rangle_8] + h_p[2]s[\langle -2 \rangle_8] = 1 \cdot s[0] + \frac{1}{2}s[7] + \frac{1}{4}s[6], \\ &= (2 + 2i) + \frac{1}{2}(2 + 2i) + \frac{1}{4}(2 + 2i) = \frac{7}{4}(2 + 2i) = 3.5 + 3.5i = y[0]. \quad \checkmark \end{aligned}$$

And component $m = 2$, where there is no wraparound:

$$\begin{aligned} (h_p \circledast s)[2] &= s[2] + \frac{1}{2}s[1] + \frac{1}{4}s[0] = (-2 - 2i) + \frac{1}{2}(2 - 2i) + \frac{1}{4}(2 + 2i) \\ &= (-2 - 2i) + (1 - i) + (0.5 + 0.5i) = -0.5 - 2.5i = y[2]. \quad \checkmark \end{aligned}$$

A script verified all eight components: maximum discrepancy between y and $h_p \circledast s$ equal to $4.44 \cdot 10^{-16}$. $\checkmark$

27.9.4 Verifying that the channel has become diagonal

Compute $Y = \mathcal{F}_8 y$ and $S = \mathcal{F}_8 s = 8c$. The values of Y , in exact form, contain $2\sqrt{2} = 2.8284271$:

$$\begin{aligned} Y[0] &= 14, & Y[1] &= (2 + 2\sqrt{2}) + i(8 + 2\sqrt{2}), \\ Y[2] &= -6 + 4i, & Y[3] &= (8 - 2\sqrt{2}) + i(2 - 2\sqrt{2}), \\ Y[4] &= -6, & Y[5] &= (2 - 2\sqrt{2}) + i(8 - 2\sqrt{2}), \\ Y[6] &= 6 + 4i, & Y[7] &= (8 + 2\sqrt{2}) + i(2 + 2\sqrt{2}). \end{aligned}$$

The comparison with $H[k]S[k]$, predicted by Corollary 27.3, is in Table 27.5. Three rows are worked out by hand.

$$k = 0: \quad H[0]S[0] = \frac{7}{4} \cdot 8 = 14 = Y[0]. \quad \checkmark$$

$$\begin{aligned} k = 1: \quad H[1]S[1] &= \left(1 + \frac{\sqrt{2}}{4} - i\left(\frac{\sqrt{2}}{4} + \frac{1}{4}\right)\right) \cdot 8i = 8i\left(1 + \frac{\sqrt{2}}{4}\right) + 8\left(\frac{\sqrt{2}}{4} + \frac{1}{4}\right) \\ &= (2\sqrt{2} + 2) + i(8 + 2\sqrt{2}) = Y[1]. \quad \checkmark \end{aligned}$$

$$k = 2: \quad H[2]S[2] = \left(\frac{3}{4} - \frac{1}{2}i\right) \cdot (-8) = -6 + 4i = Y[2]. \quad \checkmark$$

In the computation for $k = 1$, $-i \cdot i = 1$ and $8 \cdot \frac{\sqrt{2}}{4} = 2\sqrt{2}$, $8 \cdot \frac{1}{4} = 2$ were used.

Table 27.5: The explicit check of diagonality for the example $N = 8$ (Corollary 27.3). The column $Y[k]$ is the DFT of the received vector y ; the column $H[k]S[k]$ is the product of the frequency response and the DFT of the transmitted symbol. The two columns agree in all eight rows. The last column is the reconstructed data $\hat{c}_k = Y[k]/(8H[k])$, which reproduces the transmitted data exactly. A script confirmed the two middle columns with maximum discrepancy $9.27 \cdot 10^{-15}$ and the reconstruction with maximum discrepancy $1.12 \cdot 10^{-15}$.

k	$Y[k]$	$H[k]S[k]$	$8H[k]$	$\hat{c}_k = Y[k]/(8H[k])$
0	14	14	14	1
1	$4.8284271 + 10.8284271i$	$4.8284271 + 10.8284271i$	$10.8284271 - 4.8284271i$	i
2	$-6 + 4i$	$-6 + 4i$	$6 - 4i$	-1
3	$5.1715729 - 0.8284271i$	$5.1715729 - 0.8284271i$	$5.1715729 - 0.8284271i$	1
4	-6	-6	6	-1
5	$-0.8284271 + 5.1715729i$	$-0.8284271 + 5.1715729i$	$5.1715729 + 0.8284271i$	i
6	$6 + 4i$	$6 + 4i$	$6 + 4i$	1
7	$10.8284271 + 4.8284271i$	$10.8284271 + 4.8284271i$	$10.8284271 + 4.8284271i$	1

Equalisation, row by row. Rows $k = 3$ and $k = 2$ of Table 27.5 are checked by hand. For $k = 3$, $8H[3] = Y[3]$, because $8H[3] = 8\left(1 - \frac{\sqrt{2}}{4}\right) + 8i\left(\frac{1}{4} - \frac{\sqrt{2}}{4}\right) = (8 - 2\sqrt{2}) + i(2 - 2\sqrt{2}) = Y[3]$; hence $\hat{c}_3 = 1 = c_3$. $\checkmark$ For $k = 2$, $8H[2] = 6 - 4i$ and

$$\hat{c}_2 = \frac{-6 + 4i}{6 - 4i} = \frac{-(6 - 4i)}{6 - 4i} = -1 = c_2. \quad \checkmark$$

The determinant, computed twice. By Theorem 27.7(iv), $\det C_h = \prod_{k=0}^7 H[k]$. The $H[k]$ are the values of the polynomial $p(z) = 1 + \frac{z}{2} + \frac{z^2}{4} = \frac{1}{4}(z^2 + 2z + 4)$ at the eight eighth roots of unity. The roots of $z^2 + 2z + 4$ are $r_{1,2} = -1 \pm i\sqrt{3} = 2e^{\pm 2\pi i/3}$, hence $p(z) = \frac{1}{4}(z - r_1)(z - r_2)$ and

$$\prod_{k=0}^7 H[k] = \prod_{z^8=1} p(z) = \frac{1}{4^8} \prod_{z^8=1} (z - r_1) \prod_{z^8=1} (z - r_2).$$

From $\prod_{z^8=1} (t - z) = t^8 - 1$ it follows that $\prod_{z^8=1} (z - r) = (-1)^8 \prod_{z^8=1} (r - z) = r^8 - 1$. Now $r_1^8 = 2^8 e^{16\pi i/3} = 256 e^{4\pi i/3} = 256\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = -128 - 128\sqrt{3}i$, having subtracted 4π from the

argument; hence $r_1^8 - 1 = -129 - 128\sqrt{3}i$ and, conjugating, $r_2^8 - 1 = -129 + 128\sqrt{3}i$. Their product is

$$(-129)^2 + (128\sqrt{3})^2 = 16\,641 + 49\,152 = 65\,793,$$

and $4^8 = 65\,536$, so that

$$\det C_h = \frac{65\,793}{65\,536} = 1.0039215.$$

Independent check. The numerical computation of the determinant of C_h with the linear algebra routine returns 1.0039215088, and the product of the eight numerical eigenvalues returns $1.0039215088 + 9.57 \cdot 10^{-17}i$: both agree with $\frac{65\,793}{65\,536} = 1.00392150878906 \dots \checkmark$

27.9.5 What happens without the cyclic prefix

Now only the block s is transmitted, without a prefix, and the previous block is assumed to end with the two samples $(3, -1 + 2i)$. The channel output changes on the first two samples:

$$y'[0] = s[0] + \frac{1}{2}(-1 + 2i) + \frac{1}{4} \cdot 3 = (2 + 2i) + (-0.5 + i) + 0.75 = 2.25 + 3i,$$

$$y'[1] = s[1] + \frac{1}{2}s[0] + \frac{1}{4}(-1 + 2i) = (2 - 2i) + (1 + i) + (-0.25 + 0.5i) = 2.75 - 0.5i,$$

while $y'[m] = y[m]$ for $m \geq 2$ (Lemma 27.1: only the first $L - 1 = 2$ samples are contaminated). Applying the demodulator and the equalizer anyway gives

$$\begin{aligned} \hat{c}' = & (0.857 - 0.036i, -0.138 + 0.941i, -1.163 - 0.067i, 0.863 - 0.016i, \\ & -1.083 - 0.083i, -0.167 + 0.827i, 0.760 - 0.048i, 0.827 - 0.018i), \end{aligned}$$

against $c = (1, i, -1, 1, -1, i, 1, 1)$. $\checkmark$ (Values obtained with the same script that produced Table 27.5.)

Warning. Two wrong samples out of eight in the time domain produce *eight* wrong symbols out of eight in the frequency domain. The discrepancies $|\hat{c}'_k - c_k|$ equal

$$(0.1473, 0.1502, 0.1768, 0.1375, 0.1179, 0.2400, 0.2451, 0.1735),$$

all of the same order: the largest, 0.2451, is on subcarrier $k = 6$, the smallest, 0.1179, on $k = 4$, and no subcarrier is spared. In terms of rotation, subcarrier $k = 5$ turns out to be rotated by 11.40 degrees and $k = 1$ by 8.35 degrees. This is the same phenomenon as in the example $N = 4$, and it is no accident: the DFT of an error concentrated in a few cells has modulus of the same order at every frequency.

Remark 27.15: The balance sheet of the example $N = 8$. Eight QPSK symbols over ten sampling intervals: $\eta = \frac{8}{10} = 0.8$, a loss of 20%, that is 0.9691 dB (Table 27.2); a PAPR of 1, that is 0 dB, the minimum possible; a three-tap channel reduced to eight complex multiplications; exact reconstruction of all eight symbols. The modulator cost for $N = 8$ is 12 complex multiplications against the 64 of the direct computation (Table 27.1), and with $\varepsilon = 0.05$ of frequency error the signal-to-interference ratio equals 122.9128, that is 20.8960 dB (Corollary 27.6). These are, on a miniature scale, exactly the four numbers that decide the design of a real system.

27.10. Assigned exercises

These are *new* exercises, distinct from the worked ones in Section 27.11. The answers are in the Teacher's Key (Section 27.13).

Exercise 27.1: Sizing the prefix and counting its price. *Goal:* to be able to pass, for a given channel, from the dispersion to the prefix length, and from there to the three numbers that describe the cost — efficiency, percentage loss, loss in decibels — and to the computational cost of the modulator. *Estimated time:* 40 minutes. *Type:* numerical and probative. *Level:* basic–intermediate.

- A channel has $L = 17$ taps. Determine, with Corollary 27.5, the minimal admissible prefix length; then compute η , the percentage loss and the loss in decibels for $N = 256$, using Proposition 27.3. Verify the three numbers with a script.
- Repeat part (a) for $N = 512$ and $N = 1024$, with the same channel. Prove *in general*, and not only for these three values, that doubling N reduces the loss $1 - \eta$ by a factor between 1 and 2, and determine that factor exactly as a function of $\rho = N_g/N$.
- For $N = 512$, compute, with Corollary 27.2, the complex multiplications of the modulator computed from the defining sum, those of the radix-2 FFT, the nontrivial ones, and the ratio between the first two. Compare with Table 27.1.
- With the same channel and the same T_s , going from $N = 256$ to $N = 1024$ gains efficiency; using Observation 27.11 and Proposition 27.5, state how many decibels are instead lost in signal-to-interference ratio for the same absolute frequency error δf , and justify the number.

Exercise 27.2: A four-subcarrier OFDM block, in full. *Goal:* to carry out on paper the whole chain of the chapter, with all the cross-checks called for. *Estimated time:* 45 minutes. *Type:* symbolic and numerical computation. *Level:* remedial–basic. Consider

$$N = 4, \quad c = (1, 1, -1, -1), \quad h = \left(1, -\frac{1}{2}\right), \quad N_g = 1.$$

- Compute s from Definition 27.2, in exact symbolic form, and verify it with Theorem 27.3.
- Compute the PAPR and compare it with the bound of Theorem 27.11; state why the maximum is not attained, identifying which of the two inequalities in the proof is strict.
- Compute $H[0], \dots, H[3]$ and verify the Hermitian symmetry.
- Write $\tilde{s}$, the channel output, the vector y after discarding the prefix, and verify the identity $y = h_p \otimes s$ by writing out the four rows of Definition 22.4 in full.
- Verify Corollary 27.3 on all four components, then equalise with Definition 27.5 and check that $\hat{c} = c$.
- Compute $\det C_h$ in two independent ways, as in Section 27.8.
- Repeat without the prefix, assuming the previous block ends with the sample p . Determine for which value of p the reconstruction remains exact, and compute $\hat{c}'$ for $p = 0$.

Exercise 27.3: Frequency error, and the PAPR of constant-modulus words. *Goal:* to use the closed form of the ICI kernel and the characterisation of the worst PAPR case as computational tools, not only as statements. *Estimated time:* 50 minutes. *Type:* probative and numerical. *Level:* intermediate–advanced.

- Prove, from Lemma 27.3, that $|I(0)|$ is an *even* function of ε , and that $|I(0)| \rightarrow 1$ and $|I(m)| \rightarrow 0$ for $m \neq 0$ as $\varepsilon \rightarrow 0$. Interpret the second limit in light of Theorem 27.10.
- For $N = 16$, compute $|I(0)|^2$, SIR and SIR in decibels for $\varepsilon = 0.02$ and $\varepsilon = 0.08$, with the exact formula of Corollary 27.6; compare with the asymptotic estimate of Proposition 27.5 and comment on which of the two cases the approximation is better

in, and why.

- (c) Let $N = 4$. Compute the PAPR of the four words $(1, 1, 1, 1)$, $(1, -1, 1, -1)$, $(1, i, -1, -i)$ and $(1, 1, 1, -1)$. All four have constant modulus: explain, using Steps 3 and 5 of the proof of Theorem 27.11, why three of them attain the maximum and one does not.
- (d) Prove that, for every N and every n_0 , the word $c_k = \omega_N^{kn_0}$ produces $s = (0, \dots, 0, N, 0, \dots, 0)$ with the only nonzero value in position n_0 ; and deduce that the worst PAPR case corresponds exactly to the OFDM symbols that concentrate all the energy in a single sample.

27.11. Worked exercises

Exercise 27.4: A prefix that is too short, measured. *Type:* numerical computation with cross-check. Take $N = 4$, $c = (1, i, -1, -i)$ and the channel $h = (1, \frac{1}{2}, \frac{1}{4})$, hence $L = 3$, but use a prefix $N_g = 1$, that is one shorter than the minimum required by Corollary 27.5. Compute the error committed, verify that it matches the prediction of Theorem 27.6, and measure the effect on the four reconstructed symbols.

Solution. Strategy. Theorem 27.6 does not merely say that with $N_g \leq L - 2$ the identity can fail: it says *by how much* it fails, namely by $h_{L-1} x[j_0]$ with $j_0 = N_g - (L - 1)$. Here $L - 1 = 2$, $N_g = 1$, hence $j_0 = -1$ and the missing term is $\frac{1}{4} x[-1]$: only component $m = 0$ is wrong, and the error is proportional to the last sample of the previous block. Everything will be computed twice, once by hand and once with a script.

Step 1 — the symbol and the correct circular convolution. As in Section 27.8, $s = (0, 0, 0, 4)$. With $h_p = (1, \frac{1}{2}, \frac{1}{4}, 0)$, the four rows of Definition 22.4 give

$$\begin{aligned} (h_p \otimes s)[0] &= 1 \cdot 0 + \frac{1}{2} \cdot 4 + \frac{1}{4} \cdot 0 + 0 \cdot 0 = 2, \\ (h_p \otimes s)[1] &= 1 \cdot 0 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 4 + 0 \cdot 0 = 1, \\ (h_p \otimes s)[2] &= 1 \cdot 0 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 0 + 0 \cdot 4 = 0, \\ (h_p \otimes s)[3] &= 1 \cdot 4 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 0 + 0 \cdot 0 = 4, \end{aligned} \quad h_p \otimes s = (2, 1, 0, 4).$$

Step 2 — what actually arrives. With $N_g = 1$ the transmitted block is $\tilde{s} = (4, 0, 0, 4)$ and the channel output, calling p_{-1} and p_{-2} the last two samples of the previous block, is

$$\begin{aligned} r[0] &= 4 + \frac{1}{2}p_{-1} + \frac{1}{4}p_{-2}, & r[1] &= 0 + \frac{1}{2} \cdot 4 + \frac{1}{4}p_{-1} = 2 + \frac{1}{4}p_{-1}, \\ r[2] &= 0 + 0 + \frac{1}{4} \cdot 4 = 1, & r[3] &= 0, \\ r[4] &= 4. \end{aligned}$$

Discarding only $r[0]$, what remains is $y = (2 + \frac{1}{4}p_{-1}, 1, 0, 4)$, while the circular convolution gives $(2, 1, 0, 4)$. The difference is

$$y - h_p \otimes s = \left(\frac{1}{4}p_{-1}, 0, 0, 0\right) = (h_{L-1} x[j_0], 0, 0, 0),$$

exactly as predicted by Theorem 27.6, with $h_{L-1} = h_2 = \frac{1}{4}$ and $j_0 = N_g - (L - 1) = -1$. ✓

Step 3 — the effect on the symbols, for $p_{-1} = 4$. Then $y = (3, 1, 0, 4)$. The frequency response is, with $\omega_4 = -i$,

$$H[k] = 1 + \frac{1}{2}(-i)^k + \frac{1}{4}(-1)^k, \quad H = \left(\frac{7}{4}, \frac{3}{4} - \frac{i}{2}, \frac{3}{4}, \frac{3}{4} + \frac{i}{2}\right),$$

and $Y[k] = 3 + (-i)^k + 4i^k$, that is $Y = (8, 3 + 3i, -2, 3 - 3i)$. Equalising:

$$\hat{c}_0 = \frac{8}{7} = 1.1428571, \quad \hat{c}_1 = \frac{3 + 3i}{3 - 2i} = \frac{(3 + 3i)(3 + 2i)}{13} = \frac{3 + 15i}{13} = 0.2307692 + 1.1538462i,$$

$$\hat{c}_2 = \frac{-2}{3} = -0.6666667, \quad \hat{c}_3 = \overline{\hat{c}_1} = 0.2307692 - 1.1538462i.$$

The discrepancies $|\hat{c}_k - c_k|$ equal

$$\left(\frac{1}{7}, \frac{1}{\sqrt{13}}, \frac{1}{3}, \frac{1}{\sqrt{13}}\right) = (0.1428571, 0.2773501, 0.3333333, 0.2773501).$$

(For instance $\hat{c}_1 - c_1 = \frac{3+15i}{13} - i = \frac{3+15i-13i}{13} = \frac{3+2i}{13}$, of modulus $\frac{\sqrt{13}}{13} = \frac{1}{\sqrt{13}}$.)

Independent check 1: the structure of the error. The error in time is the vector $(1, 0, 0, 0)$; its DFT is $(1, 1, 1, 1)$, that is, it has modulus 1 at *every* frequency. Dividing by $4H[k]$ gives errors $\frac{1}{4H[k]} = (\frac{1}{7}, \frac{1}{3-2i}, \frac{1}{3}, \frac{1}{3+2i})$, of moduli $(\frac{1}{7}, \frac{1}{\sqrt{13}}, \frac{1}{3}, \frac{1}{\sqrt{13}})$: the same values computed in Step 3. ✓

Independent check 2: script. A numerical computation that builds the transmitted sequence, convolves it with h , discards the first sample, transforms and equalises returns the same four values and the same four discrepancies, and confirms that for $p_{-1} = 0$ the reconstruction is instead *exact*. ✓

Interpretation. The last check is the most instructive. With $p_{-1} = 0$ everything works, even though the prefix is too short: the missing term is $\frac{1}{4} \cdot 0 = 0$. Theorem 27.6 is stated as an existence claim — *there exist* two transmissions that give different results — and not as a systematic failure, and this example shows why that formulation is necessary: a test run with the previous block set to zero would detect nothing. Note also that the error, concentrated in a single sample in time, hits all four subcarriers, and hits the attenuated ones harder: the worst is $k = 2$, where $|H[2]| = \frac{3}{4}$ is minimal (Corollary 27.4).

Typical mistakes. (1) Forgetting that the channel has *three* taps and writing $r[n] = \tilde{s}[n] + \frac{1}{2}\tilde{s}[n-1]$: the term $\frac{1}{4}$ is lost and with it the whole phenomenon. (2) Discarding $N_g = 2$ samples despite having transmitted only 1 of prefix: the received vector would then have length 3 and the DFT would no longer be a 4-point one. (3) Concluding, from the fact that for $p_{-1} = 0$ the computation works out, that the short prefix is acceptable: this is exactly the conclusion that Theorem 27.6 excludes.

Extension. Redo the computation with $N_g = 0$ (no prefix): the error becomes $(\frac{1}{2}p_{-1} + \frac{1}{4}p_{-2} - 2, \frac{1}{4}p_{-1} - 1, 0, 0)$, that is, two wrong components instead of one, in agreement with Lemma 27.1, which predicts $L - 1 - N_g = 2$.

Exercise 27.5: How much frequency error can be tolerated. *Type:* inverse problem, with a comparison between the exact formula and the asymptotic estimate. A system with $N = 1024$ subcarriers must guarantee a signal-to-interference ratio of at least 25 dB. Determine the maximum admissible normalised frequency error ε , first with the estimate of Proposition 27.5 and then with the exact formula of Corollary 27.6, and assess how conservative the estimate is.

Solution. Strategy. The exact formula $\text{SIR} = \frac{|I(0)|^2}{1 - |I(0)|^2}$ does not invert in closed form, because $|I(0)|$ contains two sines with different arguments. The asymptotic estimate, instead, inverts immediately, and provides a first-attempt value to be refined numerically.

Step 1 — the threshold. 25 dB corresponds to $\text{SIR} = 10^{25/10} = 10^{2.5} = 316.2277660$.

Step 2 — inverting the asymptotic estimate. From Proposition 27.5, $\text{SIR} \approx \frac{3}{\pi^2 \varepsilon^2 (1 - N^{-2})}$; equating with the threshold and solving for ε ,

$$\varepsilon \approx \sqrt{\frac{3}{\pi^2 \cdot 316.2277660 \cdot (1 - 1024^{-2})}}.$$

With $\pi^2 = 9.8696044$ and $1 - 1024^{-2} = 0.99999905$, the denominator equals $9.8696044 \cdot 316.2277660 \cdot 0.99999905 = 3121.0400$, hence

$$\varepsilon^2 \approx \frac{3}{3121.0400} = 9.6122 \cdot 10^{-4}, \quad \varepsilon \approx 0.0310035.$$

Step 3 — the exact formula. Compute the exact SIR at $\varepsilon = 0.0310035$:

$$|I(0)|^2 = \left(\frac{\sin(\pi \cdot 0.0310035)}{1024 \sin(\pi \cdot 0.0310035/1024)} \right)^2 = 0.9968417, \quad \text{SIR} = \frac{0.9968417}{0.0031583} = 315.6280,$$

that is 24.9918 dB: just *below* the threshold. The asymptotic estimate is thus slightly optimistic, as it must be: from the expansion in Step 3 of Proposition 27.5, the neglected term is of order ε^4 and has a sign that makes $1 - |I(0)|^2$ grow beyond the approximated value, hence makes the SIR drop.

Step 4 — the exact value. Solving numerically by bisection the equation $\text{SIR}(\varepsilon) = 316.2277660$ gives

$$\varepsilon_{\max} = 0.0309742,$$

with $\text{SIR}(\varepsilon_{\max}) = 316.2277660$, that is exactly 25.0000 dB. ✓

Independent check 1: consistency between the two values. The relative discrepancy between the estimate and the exact value is $\frac{0.0310035 - 0.0309742}{0.0309742} = 9.5 \cdot 10^{-4}$, that is less than one part per thousand: the asymptotic estimate is usable in design, provided one knows it is optimistic and leaves a margin. ✓

Independent check 2: Table 27.3. For $N = 64$ and $\varepsilon = 0.02$ that table gives 28.8052 dB exact against 28.8087 dB asymptotic: a discrepancy of 0.0035 dB in the same direction (optimistic estimate), consistent with what was found here. ✓

Interpretation. The number $\varepsilon_{\max} = 0.031$ should be read together with Observation 27.11: since $\varepsilon = \delta f N T_s$, with $N = 1024$ one is asking of the oscillator a precision relative to the subcarrier spacing of three per cent. If N were doubled to 2048 to halve the prefix loss, the same δf would double ε and the SIR would drop by about 6 dB, below 19 dB: the gain in efficiency would come at a high price.

Typical mistakes. (1) Inverting the exact formula as if it were $\text{SIR} = 3/(\pi\varepsilon)^2$, forgetting the factor $(1 - N^{-2})$: for $N = 1024$ the effect is negligible, but for $N = 4$ or $N = 8$ it is not at all. (2) Using $\text{SIR} = 10^{25}$ instead of $10^{2.5}$: confusing decibels and powers. (3) Forgetting that the estimate is optimistic and sizing the system exactly at $\varepsilon = 0.0310035$, obtaining 24.99 dB instead of the required 25.

Extension. Redo the computation for a threshold of 30 dB: the asymptotic estimate gives $\varepsilon \approx 0.0174346$, and verify that the exact value is again slightly lower, $\varepsilon_{\max} = 0.0174293$.

Exercise 27.6: Three words, the same alphabet, three different PAPRs. *Type:* exact symbolic computation with verification. For $N = 8$, compute the PAPR of the three words

$$\begin{aligned} c^{(1)} &= (1, 1, 1, 1, 1, 1, 1, 1), & c^{(2)} &= (1, i, -1, 1, -1, i, 1, 1), \\ c^{(3)} &= (1, 1, 1, 1, -1, -1, -1, -1), \end{aligned}$$

all of constant modulus $A = 1$, and explain the result in light of Theorem 27.11.

Solution. Strategy. The average power is the same for the three words, and equals $\sum_k |c_k|^2 = 8$ (Lemma 27.4); only the peak changes. For $c^{(1)}$ and $c^{(3)}$ it is convenient to sum the geometric series; for $c^{(2)}$ the computation has already been done in Section 27.9.

Step 1 — the constant word. With $c_k^{(1)} = 1$ for every k ,

$$s[n] = \sum_{k=0}^7 e^{2\pi i k n / 8} = \sum_{k=0}^7 (\zeta^n)^k,$$

a geometric sum of ratio ζ^n , an eighth root of unity. By Lemma 22.1 it equals 8 if $8 \mid n$ and 0 otherwise, hence $s = (8, 0, 0, 0, 0, 0, 0, 0)$. The peak is 64, the average 8, and $\text{PAPR} = \frac{64}{8} = 8 = N$: the worst case. Check consistency with Theorem 27.11: here $c_k = \omega_8^{k \cdot 0} = 1$, that is $A = 1$, $\theta = 0$, $n_0 = 0$, and indeed the peak falls at $n_0 = 0$. ✓

Step 2 — the word from the example. In Section 27.9, $|s[n]|^2 = 8$ was obtained for every n , hence peak 8, average 8 and $\text{PAPR} = 1$: the best case.

Step 3 — the “half and half” word. Here $c_k^{(3)} = 1$ for $k \leq 3$ and -1 for $k \geq 4$, hence

$$s[n] = \sum_{k=0}^3 \zeta^{kn} - \sum_{k=4}^7 \zeta^{kn} = \sum_{k=0}^3 \zeta^{kn} (1 - \zeta^{4n}) = (1 - (-1)^n) \sum_{k=0}^3 \zeta^{kn},$$

having factored out $\zeta^{4n} = (\zeta^4)^n = (-1)^n$ from the second sum (setting $k = k' + 4$). Hence $s[n] = 0$ for n even. For n odd the factor equals 2 and the geometric sum of ratio $\zeta^n \neq 1$ equals $\frac{\zeta^{4n}-1}{\zeta^n-1} = \frac{-2}{\zeta^n-1}$, so that

$$s[n] = \frac{-4}{\zeta^n - 1}, \quad n \text{ odd}.$$

With $\beta = \frac{\sqrt{2}}{2}$: for $n = 1$, $\zeta - 1 = (\beta - 1) + \beta i$ and

$$|\zeta - 1|^2 = (\beta - 1)^2 + \beta^2 = \beta^2 - 2\beta + 1 + \beta^2 = 2 - \sqrt{2} = 0.5857864,$$

having used $2\beta^2 = 1$ and $2\beta = \sqrt{2}$; hence $|s[1]|^2 = \frac{16}{2-\sqrt{2}}$. Rationalising,

$$|s[1]|^2 = \frac{16(2 + \sqrt{2})}{(2 - \sqrt{2})(2 + \sqrt{2})} = \frac{16(2 + \sqrt{2})}{2} = 8(2 + \sqrt{2}) = 16 + 8\sqrt{2} = 27.3137085.$$

For $n = 3$, $\zeta^3 - 1 = (-\beta - 1) + \beta i$ and $|\zeta^3 - 1|^2 = 2 + \sqrt{2}$, whence $|s[3]|^2 = \frac{16}{2+\sqrt{2}} = 8(2 - \sqrt{2}) = 4.6862915$. By symmetry $|s[7]| = |s[1]|$ and $|s[5]| = |s[3]|$.

The peak is thus $16 + 8\sqrt{2}$ and

$$\text{PAPR}(s^{(3)}) = \frac{16 + 8\sqrt{2}}{8} = 2 + \sqrt{2} = 3.4142136,$$

that is $10 \log_{10}(2 + \sqrt{2}) = 5.3329068$ dB.

Independent check 1: the total power. It must equal $\sum_n |s[n]|^2 = N \sum_k |c_k|^2 = 8 \cdot 8 = 64$ (Lemma 27.4). Indeed

$$2(16 + 8\sqrt{2}) + 2(16 - 8\sqrt{2}) = 32 + 32 = 64. \quad \checkmark$$

The terms in $\sqrt{2}$ cancel: this check immediately exposes a sign error in the rationalisation.

Independent check 2: script. The numerical computation of $8\mathcal{F}_8^{-1}c^{(3)}$ returns $s^{(3)} = (0, 2 + 4.8284271i, 0, 2 + 0.8284271i, 0, 2 - 0.8284271i, 0, 2 - 4.8284271i)$, with $|s[1]|^2 = 4 + 23.3137085 = 27.3137085$ and PAPR 3.4142136, that is 5.3329068 dB: the same values. ✓ (Note $4.8284271 = 2 + 2\sqrt{2}$.)

Interpretation. Three words with the same alphabet and the same power give 9.0309 dB, 0 dB and 5.3329 dB. Theorem 27.11 explains the first case (the word $c^{(1)}$ has the form $Ae^{i\theta}\omega_N^{kn_0}$ with $n_0 = 0$) and the third (neither inequality is saturated: the phases never align perfectly); the second is the opposite case, in which the energy is uniformly spread over all samples. It is on this dependence on the word that PAPR-reduction techniques are based, choosing among equivalent representations of the same data the one with the smaller peak (Observation 27.13 for the source).

Typical mistakes. (1) Computing the average power as $\frac{1}{N}\sum_k |c_k|^2$ instead of $\sum_k |c_k|^2$: the PAPR is wrong by a factor of N . (2) In Step 3, forgetting the factor $(1 - (-1)^n)$ and concluding that s has eight nonzero components. (3) Rationalising $\frac{16}{2-\sqrt{2}}$ by multiplying by $2 - \sqrt{2}$ instead of $2 + \sqrt{2}$.

Extension. Show that for $N = 8$ and BPSK words ($c_k \in \{\pm 1\}$) the maximum PAPR 8 is attained only by $c = (1, 1, \dots, 1)$ and by $c = (1, -1, 1, -1, 1, -1, 1, -1)$, and by their opposites. (Hint: Theorem 27.11 forces $c_k = e^{i\theta}\omega_8^{kn_0}$; for all the c_k to be real, $\omega_8^{n_0}$ must be real, that is $n_0 \in \{0, 4\}$, and $e^{i\theta} \in \{1, -1\}$.)

27.12. Self-assessment quiz

- (Definition) The *cyclic prefix* of length N_g is
 - an interval of N_g zero samples prepended to the block
 - a copy of the last N_g samples of the block, prepended to the block itself
 - a copy of the first N_g samples of the block, appended after the block
 - a known sequence of N_g samples used to estimate the channel
- (Hypothesis) In Theorem 27.5 the hypothesis $N_g \geq L - 1$ serves to
 - guarantee that h_p has length N
 - guarantee that all the indices $m + N_g - l$ are nonnegative, that is, that no sample of the previous block enters the kept samples
 - make the matrix C_h invertible
 - ensure that $H[k] \neq 0$ for every k
- (Symbolic computation) The modulus of the common attenuation factor produced by a normalised frequency error ε equals
 - $\frac{|\sin(\pi\varepsilon)|}{\pi\varepsilon}$
 - $\frac{|\sin(\pi\varepsilon)|}{N|\sin(\pi\varepsilon/N)|}$
 - $\frac{|\sin(\pi\varepsilon/N)|}{N|\sin(\pi\varepsilon)|}$
 - $\frac{1}{N}\left|\frac{\sin(\pi\varepsilon)}{\sin(\pi\varepsilon/N)}\right|$
- (Numerical computation) With $N = 2048$ and $N_g = 144$ the efficiency loss due to the cyclic prefix equals
 - 7.031250%
 - 6.569343%
 - 0.295106%
 - 20.000000%
- (Proof step) In Step 3 of Theorem 27.5, case (b) with $l > m$, the equality $\tilde{s}[N_g - d] = s[N - d]$ holds because
 - the index $N_g - d$ falls in the prefix, which by Definition 27.4 is a copy of the tail of s
 - circular convolution is commutative
 - h has finite length
 - $\langle m - l \rangle_N = m - l$ even when $m < l$
- (Error diagnosis) A designer shortens the prefix from $N_g = L - 1$ to $N_g = L - 2$ to gain efficiency, and tests the system on an ideal single-tap channel ($L = 1$, $h = (1)$). The test

passes. Why does it not detect the error?

- A) because with $L = 1$ the condition $N_g \geq L - 1$ is satisfied by any $N_g \geq 0$, and the critical case is never exercised
 - B) because the DFT is unitary and conserves energy
 - C) because the ZF equalizer automatically corrects interblock interference
 - D) because the PAPR does not depend on the prefix length
7. (Transfer) N is doubled while keeping fixed the sampling step T_s , the channel and the absolute frequency error δf . Then
- A) both the prefix loss and the intercarrier interference power decrease
 - B) the prefix loss roughly halves and the intercarrier interference power roughly quadruples
 - C) the prefix loss doubles and the interference halves
 - D) neither changes, because both depend only on the ratio N_g/N
8. (Source and metacognition) The paper that in 1971 replaced the bank of oscillators with the discrete Fourier transform, making OFDM realisable, is
- A) R. W. Chang, *Synthesis of band-limited orthogonal signals for multichannel data transmission*, *Bell System Technical Journal* **45** (1966), pp. 1775–1796
 - B) S. B. Weinstein, P. M. Ebert, *Data transmission by frequency-division multiplexing using the discrete Fourier transform*, *IEEE Transactions on Communication Technology* **19** (1971), pp. 628–634
 - C) A. Peled, A. Ruiz, *Frequency domain data transmission using reduced computational complexity algorithms*, ICASSP 1980, pp. 964–967
 - D) J. W. Cooley, J. W. Tukey, *An algorithm for the machine calculation of complex Fourier series*, *Mathematics of Computation* **19** (1965), pp. 297–301

27.13. Teacher’s key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

- 1) **B.** *Rationale:* this is literally Definition 27.4, $\tilde{s}[n] = s[\langle n - N_g \rangle_N]$, which for $n < N_g$ gives $s[N - N_g + n]$, that is, the last N_g samples. *Distractors:* A describes the empty guard interval, which eliminates interblock interference but does not produce circular convolution — this is exactly the difference between the pre-1980 solution and that of Peled and Ruiz; C reverses the order (the cyclic suffix exists, but it is a different technique and is not the one used here); D confuses the prefix with a pilot sequence. *Diagnosed error:* believing the prefix only serves to “leave room” for the echo. *Source:* Peled and Ruiz, ICASSP 1980, pp. 964–967. *Prerequisite:* Definition 22.4. *Remedy:* redo Section 27.8, the “without prefix” part, with $p = 0$ and with $p = 4$, and observe that only the second gives the circular result.
- 2) **B.** *Rationale:* Step 2 of that proof checks $n_l = m + N_g - l \geq N_g - (L - 1) \geq 0$; it is the only point where the hypothesis is used, and it serves to exclude negative indices, that is, samples of the previous block (Lemma 27.1). *Distractors:* A confuses the hypothesis $N_g \geq L - 1$ with the hypothesis $L \leq N$, used in Step 4 for the zero-padding; C and D are independent consequences or conditions (the invertibility of C_h and the non-vanishing of the $H[k]$ are the same thing by Theorem 27.7(iv), and have nothing to do with N_g). *Diagnosed error:* not being able to identify at which step a given hypothesis is used. *Source:* here, Theorem 27.5, Step 2. *Prerequisite:* Lemma 27.1. *Remedy:* redo Step 2 with $N_g = L - 2$ and identify for which pair (m, l) the index becomes negative.
- 3) **B.** *Rationale:* this is $|I(0)|$ of Lemma 27.3, with $m = 0$. *Distractors:* A is the limit as $N \rightarrow \infty$ of the correct answer, that is, the sinc function: it is the formula for the continuous-time case, not the sampled one, and it is already wrong for $N = 4$; C swaps numerator and

denominator, and would give an *amplification* greater than 1; D is identically $1/N$ and does not depend on ε , so it cannot describe an attenuation that grows with the error. *Diagnosed error*: substituting the discrete Dirichlet kernel with the continuous sinc function. *Source*: here, Lemma 27.3; cf. Lemma 22.3 for the same structure in Chapter 22. *Prerequisite*: geometric sum, Lemma 22.1. *Remedy*: compute $|I(0)|$ for $N = 4$ and $\varepsilon = 0.25$ with both formulas: the correct answer gives $\frac{\sin(\pi/4)}{4\sin(\pi/16)} = \frac{0.7071068}{0.7803612} = 0.9061274$, distractor A gives $\text{sinc}(0.25) = \frac{0.7071068}{0.7853982} = 0.9003163$; the difference, 0.0057962, is small but nonzero, and grows as N decreases.

4) **B. Rationale**: by Proposition 27.3(i) the loss is $\frac{N_g}{N+N_g} = \frac{144}{2192} = \frac{9}{137} = 0.0656934\dots$, that is 6.569343% (Example 27.3). *Distractors*: A is $\rho = \frac{144}{2048} = \frac{9}{128} = 7.03125\%$, that is the ratio N_g/N and not $N_g/(N+N_g)$: this is the most frequent mistake; C is the value of the loss *in decibels* (0.295106), read as a percentage; D is the loss for $\rho = \frac{1}{4}$, that is, another row of Table 27.2. *Diagnosed error*: confusing ρ with $1-\eta$, and decibels with percentages. *Source*: Proposition 27.3 and verification with a script. *Prerequisite*: $\frac{1}{1+\rho}$ against $\frac{\rho}{1+\rho}$. *Remedy*: redo the computation for $\rho = \frac{1}{8}$ (answer: 11.11111% and 0.511525 dB) and for $\rho = \frac{1}{16}$ (5.882353% and 0.263289 dB), checking against Table 27.2.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **A. Rationale**: in case (b), $n_l = N_g - d$ with $1 \leq d \leq N_g$, hence $0 \leq n_l \leq N_g - 1$: the index falls in the first row of Definition 27.4, where $\tilde{s}[n] = s[N - N_g + n]$, and substituting $n = N_g - d$ gives $s[N - d]$. *Distractors*: B is true but unused (commutativity of circular convolution does not appear in that proof); C is used elsewhere, to guarantee that the sum over l is finite; D is *false*: when $m < l$ the representative is $m - l + N$, and this is exactly the point of the proof. *Diagnosed error*: skipping the computation of the modular representative and relying on intuition. *Source*: here, Theorem 27.5, Step 3. *Prerequisite*: Observation 27.5. *Remedy*: rewrite the table of $\tilde{s}[n]$ for $N = 8$, $N_g = 2$ and check the equality by hand for all pairs (m, l) with $l > m$.

6) **A. Rationale**: the condition of Corollary 27.5 is $N_g \geq L - 1$; with $L = 1$ it becomes $N_g \geq 0$, satisfied by any prefix, even an empty one. The test never exercises the critical case, which requires $L \geq 2$ and $h_{L-1} \neq 0$: these are exactly the hypotheses of Theorem 27.6. *Distractors*: B is true but irrelevant — and worse, it is a check that does *not* detect the error, as seen in Worked Exercise 27.4; C is false: the ZF equalizer assumes diagonality, it does not restore it; D is true but beside the point. *Diagnosed error*: testing outside the regime in which the hypothesis is binding. *Source*: here, Theorem 27.6 and Worked Exercise 27.4. *Prerequisite*: Corollary 27.5. *Remedy*: redo Worked Exercise 27.4 first with $p_{-1} = 0$ (the test passes) and then with $p_{-1} = 4$ (the test fails).

7) **B. Rationale**: at fixed N_g , $1 - \eta = \frac{N_g}{N+N_g}$ roughly halves when N doubles (Proposition 27.3(iii)); and since $\varepsilon = \delta f N T_s$, doubling N doubles ε , and by Proposition 27.5 the interference power grows like ε^2 , hence quadruples (−6.03 dB of SIR; Figure 27.4). *Distractors*: A and C reverse one of the two effects; D forgets that ε depends on N and not only on ρ . *Diagnosed error*: optimising a parameter while looking at only one constraint. *Source*: here, Observation 27.11. *Prerequisite*: Definition 27.8. *Remedy*: work through Worked Exercise 27.5 and then redo it with $N = 2048$.

8) **B. Rationale**: this is the paper by Weinstein and Ebert, which introduced the DFT-based realisation (here, Theorems 27.3 and 27.4). *Distractors*: A is Chang, 1966, who introduced the orthogonality of the subcarriers but not the transform-based realisation; C is Peled and Ruiz, 1980, to whom the cyclic prefix is due; D is Cooley and Tukey, 1965, who provide the *algorithm* used to compute the transform but which has nothing to do

with telecommunications and is the source of Chapter 23. *Diagnosed error*: attributing a composite idea to a single paper, or confusing the algorithm with the application. *Source*: the four citations are verified and reported in the sources environment of this chapter and of Chapter 23. *Prerequisite*: Observation 27.4. *Remedy*: reread Observation 27.4 and match each of the three ideas to its own paper and to its own section of the chapter.

BEST PRACTICE

Assigned exercises — answers.

Exercise 27.1. (a) $N_g = L - 1 = 16$; for $N = 256$, $\eta = \frac{256}{272} = \frac{16}{17} = 0.9411765$, loss $\frac{1}{17} = 5.882353\%$, that is 0.263289 dB. (b) $N = 512$: $\eta = \frac{32}{33} = 0.9696970$, loss 3.030303%, 0.133640 dB. $N = 1024$: $\eta = \frac{64}{65} = 0.9846154$, loss 1.538462%, 0.067334 dB. In general, going from N to $2N$ at fixed N_g , the loss goes from $\frac{\rho}{1+\rho}$ to $\frac{\rho/2}{1+\rho/2} = \frac{\rho}{2+\rho}$, hence it is reduced by the factor $\frac{2+\rho}{1+\rho}$, which for $\rho > 0$ is strictly between 1 and 2 and tends to 2 as $\rho \rightarrow 0$. (For $\rho = \frac{1}{16}$ the factor equals $\frac{33}{17} = 1.9412$.) (c) $N = 512 = 2^9$: direct sum $512^2 = 262\,144$ multiplications; FFT $\frac{512}{2} \cdot 9 = 2\,304$; nontrivial $2\,304 - 768 + 2 = 1\,538$; ratio $\frac{262\,144}{2\,304} = 113.777778$. (d) ε quadruples (from N to $4N$), the ICI power grows like ε^2 hence by a factor of 16: $10 \log_{10} 16 = 12.0412$ dB of signal-to-interference ratio are lost.

Exercise 27.2. (a) $s[n] = 1 + i^n - i^{2n} - i^{3n}$, hence $s = (0, 2 + 2i, 0, 2 - 2i)$. (b) Average power $\sum_k |c_k|^2 = 4$; peak $|s[1]|^2 = 8$; PAPR = 2, that is 3.0103 dB, against the maximum 4. The Cauchy–Schwarz inequality is saturated (all $|c_k|$ equal 1), the triangle one is not: at no instant do the four phases align. (c) $H[k] = 1 - \frac{1}{2}(-i)^k$, that is $H = (\frac{1}{2}, 1 + \frac{i}{2}, \frac{3}{2}, 1 - \frac{i}{2})$; $H[3] = \overline{H[1]}$. (d) $\tilde{s} = (2 - 2i, 0, 2 + 2i, 0, 2 - 2i)$; $r = (2 - 2i, -1 + i, 2 + 2i, -1 - i, 2 - 2i, -1 + i)$; $y = (-1 + i, 2 + 2i, -1 - i, 2 - 2i) = h_p \circledast s$. (e) $S = 4c = (4, 4, -4, -4)$, $Y = (2, 4 + 2i, -6, -4 + 2i)$, and $Y[k] = H[k]S[k]$ for all four; $\hat{c} = (1, 1, -1, -1) = c$. (f) $\det C_h = \frac{1}{2} \cdot \frac{3}{2} \cdot |1 + \frac{i}{2}|^2 = \frac{3}{4} \cdot \frac{5}{4} = \frac{15}{16} = 0.9375$; by an algebraic route $\prod_{z^4=1} (1 - \frac{z}{2}) = \frac{1}{16} \prod_{z^4=1} (2 - z) = \frac{2^4 - 1}{16} = \frac{15}{16}$. (g) The reconstruction remains exact if and only if $p = s[3] = 2 - 2i$. For $p = 0$ one gets $y' = (0, 2 + 2i, -1 - i, 2 - 2i)$ and $\hat{c}' = (1.5 - 0.5i, 1.1 - 0.3i, -0.8333333 - 0.1666667i, -0.7 - 0.1i)$: all four symbols are wrong.

Exercise 27.3. (a) In the formula of Lemma 27.3 with $m = 0$, substituting $-\varepsilon$ for ε changes the sign of both $\sin(\pi\varepsilon)$ and $\sin(\pi\varepsilon/N)$, and the quotient is unchanged: $|I(0)|$ is even. As $\varepsilon \rightarrow 0$, $|I(0)| \rightarrow 1$ by Proposition 27.5; and for $m \not\equiv 0 \pmod{N}$ the numerator $\sin(\pi\varepsilon) \rightarrow 0$ while the denominator $N \sin(\pi m/N) \neq 0$ stays bounded away from zero, hence $|I(m)| \rightarrow 0$. In Theorem 27.10 this means that without a frequency error only the term $j = k$ remains: orthogonality is recovered. (b) $N = 16$, $\varepsilon = 0.02$: $|I(0)|^2 = 0.9986899$, SIR = 762.2882, that is 28.8212 dB, against the asymptotic estimate 762.8889, that is 28.8246 dB. $\varepsilon = 0.08$: $|I(0)|^2 = 0.9792019$, SIR = 47.0813, that is 16.7285 dB, against 47.6806, that is 16.7834 dB. The approximation is better for small ε , because the neglected term is of order ε^4 : the discrepancy goes from 0.0034 dB to 0.0549 dB, that is, it grows by a factor of 16 when ε quadruples, as predicted. (c) $(1, 1, 1, 1)$, $(1, -1, 1, -1)$ and $(1, i, -1, -i)$ have PAPR 4, that is 6.0206 dB: they are of the form $\omega_4^{kn_0}$ with $n_0 = 0$, $n_0 = 2$ and $n_0 = 3$ respectively, and give $s = (4, 0, 0, 0)$, $s = (0, 0, 4, 0)$, $s = (0, 0, 0, 4)$. The word $(1, 1, 1, -1)$, instead, has $s = (2, 2i, 2, -2i)$, all of modulus 2, hence PAPR 1: the Cauchy–Schwarz inequality is saturated (all moduli equal) but the triangle one never is, because $(1, 1, 1, -1)$ is not of the form $Ae^{i\theta}\omega_4^{kn_0}$ for any n_0 . (d) With $c_k = \omega_N^{kn_0}$ one has $s[n] = \sum_k \omega_N^{kn_0} \omega_N^{-kn} = \sum_k \omega_N^{k(n_0-n)}$, which by Lemma 22.1 equals N if $n = n_0$ and 0 otherwise. The average power is $\sum_k |c_k|^2 = N$ and the peak N^2 : PAPR = N . Conversely Step 5 of Theorem 27.11 shows that these are the only words with maximum PAPR, up to the factor $Ae^{i\theta}$.

27.14. Chapter formulary

Subcarriers and orthogonality. $\Delta f = 1/T$, $T_s = T/N$, $t_n = nT/N$.

$$\begin{aligned}\phi_k(t) &= e^{2\pi i k \Delta f t}, & \langle \phi_k, \phi_\ell \rangle_T &= T \delta_{k\ell} \iff \Delta f T \in \{1, 2, 3, \dots\}, \\ \Psi_k(f) &= T e^{-i\pi(fT-k)} \text{sinc}(fT-k), & \Psi_k(\ell/T) &= T \delta_{k\ell}, & \phi_k(t_n) &= \omega_N^{-kn}.\end{aligned}$$

Modulator and demodulator. With $\omega_N = e^{-2\pi i/N}$:

$$s[n] = \sum_{k=0}^{N-1} c_k \omega_N^{-kn} = N(\mathcal{F}_N^{-1}c)[n] = (\overline{\mathcal{F}_N c})[n], \quad c_k = \frac{1}{N}(\mathcal{F}_N s)[k].$$

Cost: N^2 multiplications from the definition; $\frac{N}{2} \log_2 N$ with radix-2 FFT, of which $\frac{N}{2} \log_2 N - \frac{3N}{2} + 2$ nontrivial. For $N = 2048$: 4 194 304 against 11 264 (ratio $\frac{4096}{11} = 372.363636$).

Cyclic prefix and diagonalisation. Channel $h \in \mathbb{C}^L$, h_p its zero-padding to length N .

$$\tilde{s}[n] = s[\langle n - N_g \rangle_N], \quad n = 0, \dots, N + N_g - 1; \quad y[m] = r[m + N_g].$$

If $N_g \geq L - 1$ (necessary and sufficient condition) then

$$\begin{aligned}y &= h_p \otimes s = C_h s, & C_h &= \text{circ}(h_p), & F_N C_h F_N^* &= \text{diag}(H), \\ H[k] &= \sum_{l=0}^{L-1} h_l \omega_N^{kl}, & Y[k] &= H[k] S[k] = N H[k] c_k, & \det C_h &= \prod_{k=0}^{N-1} H[k].\end{aligned}$$

In physical time: $N_g T_s \geq \tau_{\max} = (L - 1) T_s$.

Single-coefficient equalisation. With noise under hypothesis (R), $\mathbb{E}|W[k]|^2 = N\sigma^2$:

$$\begin{aligned}\text{ZF: } \hat{c}_k &= \frac{Y[k]}{N H[k]} = c_k + \frac{W[k]}{N H[k]}, & \mathbb{E}[|e_k|^2] &= \frac{\sigma^2}{N |H[k]|^2}, & \text{SNR}_k &= \frac{NP |H[k]|^2}{\sigma^2}. \\ \text{MMSE: } g^* &= \frac{\overline{H[k]}}{|H[k]|^2 + \frac{\sigma^2}{NP}}, & J(g^*) &= \frac{P\sigma^2}{N |H[k]|^2 P + \sigma^2} < \frac{\sigma^2}{N |H[k]|^2}.\end{aligned}$$

Efficiency. With $\rho = N_g/N$:

$$\eta = \frac{N}{N + N_g} = \frac{1}{1 + \rho}, \quad 1 - \eta = \frac{\rho}{1 + \rho}, \quad \text{loss in dB} = 10 \log_{10}(1 + \rho).$$

Values: $\rho = \frac{1}{4} \Rightarrow 20\%$ and 0.969100 dB; $\rho = \frac{1}{8} \Rightarrow 11.111111\%$ and 0.511525 dB; $\rho = \frac{1}{16} \Rightarrow 5.882353\%$ and 0.263289 dB; $\rho = \frac{9}{128} \Rightarrow 6.569343\%$ and 0.295106 dB.

Frequency error. $\varepsilon = \delta f / \Delta f = \delta f N T_s$, $r[n] = e^{2\pi i \varepsilon n / N} s[n]$.

$$I(m) = \frac{\sin(\pi \varepsilon)}{N \sin(\pi(m + \varepsilon)/N)} e^{i\pi \varepsilon(1 - \frac{1}{N})} e^{-i\pi m / N}, \quad I(m + N) = I(m), \quad \sum_{m=0}^{N-1} |I(m)|^2 = 1.$$

$$\hat{c}_k = I(0)c_k + \sum_{j \neq k} c_j I(j-k), \quad \text{SIR}(\varepsilon) = \frac{|I(0)|^2}{1 - |I(0)|^2} \sim \frac{3}{\pi^2 \varepsilon^2 (1 - N^{-2})}.$$

For $N = 64$: $\varepsilon = 0.05 \Rightarrow 20.8284$ dB; $\varepsilon = 0.10 \Rightarrow 14.7431$ dB.

PAPR.

$$\text{PAPR}(s) = \frac{\max_n |s[n]|^2}{\frac{1}{N} \sum_n |s[n]|^2}, \quad \frac{1}{N} \sum_n |s[n]|^2 = \sum_k |c_k|^2 = \frac{1}{T} \int_0^T |s(t)|^2 dt.$$

$$1 \leq \text{PAPR}(s) \leq \frac{(\sum_k |c_k|)^2}{\sum_k |c_k|^2} \leq N, \quad \text{PAPR} = N \iff c_k = A e^{i\theta} \omega_N^{kn_0}.$$

In decibels the maximum is $10 \log_{10} N$: 6.0206 dB for $N = 4$, 9.0309 dB for $N = 8$, 33.1133 dB for $N = 2048$. Under the Gaussian hypothesis (G), *imported*: $\Pr(\text{PAPR} > \gamma) = 1 - (1 - e^{-\gamma})^N$.

The numbers of the examples. $N = 4$, $c = (1, i, -1, -i)$, $h = (1, \frac{1}{2})$, $N_g = 1$:

$$s = (0, 0, 0, 4), \quad H = \left(\frac{3}{2}, 1 - \frac{i}{2}, \frac{1}{2}, 1 + \frac{i}{2}\right), \quad y = (2, 0, 0, 4), \quad Y = (6, 2 + 4i, -2, 2 - 4i),$$

$\det C_h = \frac{15}{16}$, $\text{PAPR} = 4$. $N = 8$, $c = (1, i, -1, 1, -1, i, 1, 1)$, $h = (1, \frac{1}{2}, \frac{1}{4})$, $N_g = 2$:

$$s = (2 + 2i, 2 - 2i, -2 - 2i, 2 + 2i, -2 - 2i, 2 - 2i, 2 + 2i, 2 + 2i), \quad \text{PAPR} = 1,$$

$$H[0] = \frac{7}{4}, \quad H[2] = \frac{3}{4} - \frac{i}{2}, \quad H[4] = \frac{3}{4}, \quad |H[1]|^2 = \frac{21}{16} + \frac{5\sqrt{2}}{8}, \quad |H[3]|^2 = \frac{21}{16} - \frac{5\sqrt{2}}{8},$$

$$y = (3.5 + 3.5i, 3.5 - 0.5i, -0.5 - 2.5i, 1.5 + 0.5i, -1.5 - 1.5i, 1.5 - 2.5i, 2.5 + 0.5i, 3.5 + 2.5i),$$

$$\det C_h = \frac{65793}{65536} = 1.0039215.$$

Sources and bibliography

- [1] R. W. Chang, *Synthesis of Band-Limited Orthogonal Signals for Multichannel Data Transmission*, *Bell System Technical Journal* **45** (1966), n. 10, pp. 1775–1796. DOI 10.1002/j.1538-7305.1966.tb02435.x, ISSN 0005-8580. This is the paper in which the idea of subcarriers with overlapping yet orthogonal spectra first appears: it is the historical source of Section 27.1 (Theorem 27.1 and Proposition 27.1), here re-presented and proved in full. *Verified on 7 September 2026* (Crossref DOI record: author, title, journal, volume, issue, year and page range checked one by one).
- [2] S. B. Weinstein, P. M. Ebert, *Data Transmission by Frequency-Division Multiplexing Using the Discrete Fourier Transform*, *IEEE Transactions on Communication Technology* **19** (1971), n. 5, pp. 628–634. DOI 10.1109/TCOM.1971.1090705, ISSN 0018-9332. This is the paper that replaces the bank of oscillators with a discrete transform: historical source of Theorems 27.3 and 27.4 and of Corollary 27.2. *Verified on 7 September 2026* (Crossref DOI record: authors, title, journal, volume, issue, year and page range checked one by one).
- [3] A. Peled, A. Ruiz, *Frequency Domain Data Transmission Using Reduced Computational Complexity Algorithms*, in *ICASSP '80. IEEE International Conference on Acoustics, Speech, and Signal Processing*, vol. 5 (1980), pp. 964–967. DOI 10.1109/ICASSP.1980.1171076. This is the paper in which the empty guard interval is replaced by a copy of the

block's tail: historical source of Section 27.3 (Definition 27.4 and Theorem 27.5), here proved in full with the modular indices written out in full. *Verified on 7 September 2026* (Crossref DOI record: authors, title, proceedings, volume, year and page range checked one by one).

- [4] Y. S. Cho, J. Kim, W. Y. Yang, C. G. Kang, *MIMO-OFDM Wireless Communications with MATLAB*, John Wiley & Sons, Singapore, 2010, ISBN 978-0-470-82561-7 (print), 978-0-470-82563-1 (electronic), DOI 10.1002/9780470825631. Chapter 4 “Introduction to OFDM”, pp. 111–151 (general picture of the OFDM chain, frequency diversity and bit loading, recalled in Observation 27.10); Chapter 5 “Synchronization for OFDM”, pp. 153–185 (estimation and correction of the frequency error, Observation 27.12); Chapter 7 “PAPR Reduction”, pp. 209–250 (source of what is **imported** in Observation 27.13: the Gaussian hypothesis, the tail formula $1 - (1 - e^{-\gamma})^N$ and the practical rule of thumb on oversampling). *Verified on 7 September 2026* (Crossref DOI record of the volume and of the DOIs of the individual chapters ...ch4, ...ch5 and ...ch7, with titles and respective page ranges).
- [5] J. G. Proakis, M. Salehi, *Digital Communications*, 5th edition, McGraw-Hill, New York, 2008, ISBN 978-0-07-295716-7, 1150 pages, LCCN 2007036509 (bibliography on pp. 1109–1141). General reference text for digital communications and for multicarrier systems; it is the second source of the overall picture in Observation 27.10. No result of this chapter depends on it: every statement is proved here, except for what is explicitly declared imported in Observation 27.13. *Verified on 7 September 2026* (Open Library catalogue record 0L21874530M: title, authors, publisher, year, edition, ISBN-10 and ISBN-13, number of pages and LCCN; internal chapter or page numbers are not cited, as they were not verified).

Part E

Applications to automation

Rotations of space, SO(3) and quaternions

Chapter 11 established *what* the rotations of space are: by Euler's theorem (Theorem 11.8) each has an axis and an angle; by Theorem 11.10 the unit quaternions realise all of them, each twice. This chapter takes up the next question, the engineer's rather than the geometer's: *how one writes* a rotation, how its representations are converted into one another, and what breaks when the conversion passes near a singularity.

Three representations are in use. The **rotation matrix** is the most direct but uses nine numbers for three degrees of freedom, and keeping it orthogonal in finite-precision arithmetic costs effort. **Euler angles** (in the *roll-pitch-yaw* variant used by robots) use three parameters, the minimum possible, and precisely for that reason carry an unavoidable singularity: *gimbal lock*. The **unit quaternion** uses four parameters tied by a single quadratic constraint, has no singularity, composes with sixteen multiplications instead of twenty-seven, and renormalises by dividing by a norm. Between matrix and axis-angle the bridge is Rodrigues' formula, from 1840; between quaternion and matrix the bridge is Hamilton's calculus, from 1843. The two bridges are the same formula written twice, and this coincidence — proved in full in Section 28.5 — is the technical heart of the chapter.

The thread that runs through it all is still the circle. Every rotation of space is a plane rotation in the plane orthogonal to its own axis: Rodrigues' formula does nothing more than write $\cos \theta$ and $\sin \theta$ in that plane and leave the axis fixed. And the unit quaternions are the sphere S^3 , which covers $SO(3)$ twice exactly as, one dimension down, S^1 covers $SO(2)$ once (Theorem 11.7).

INFO

Level and prerequisites. Level C (undergraduate), with Section 28.6 at level C–D.

Internal prerequisites. The orthogonal group and the rotation group (Definition 11.2); Euler's theorem on the axis of rotation (Theorem 11.8); the Cartan–Dieudonné theorem (Theorem 11.9); the algebra of quaternions (Definition 11.3) and the action $\Phi_q(v) = qv\bar{q}$ together with its character as a double covering (Theorem 11.10); the Hopf fibration (Theorem 11.11); $SO(2) \cong S^1$ (Theorem 11.7); Lagrange's identity $\|\mathbf{a} \times \mathbf{b}\|^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \langle \mathbf{a}, \mathbf{b} \rangle^2$ (Lemma 10.1) and the invariance of the cross product under $SO(3)$ (Proposition 3.4, Step 1); trigonometric functions and the addition and duplication formulas (Volume II, Chapter 1).

External prerequisites, assumed and not reproved (a first course in linear algebra and analysis):

- the matrix exponential series $e^M = \sum_{n \geq 0} M^n/n!$ converges in norm for every real 3×3 matrix M , because $\|M^n\| \leq \|M\|^n$ by submultiplicativity of the operator norm and $\sum \|M\|^n/n!$ converges;
- an absolutely convergent series may be reordered and regrouped;
- the rank of a matrix is the maximum number of independent columns, and a square matrix is invertible if and only if its determinant is nonzero;
- the inverse function theorem in $\mathbb{R}^3$.

Everything else is proved here.

What is learned. Proving Rodrigues' formula in matrix form and recognising it as the exponential of a skew-symmetric matrix; multiplying quaternions in scalar-vector form and proving that the norm is multiplicative; converting in both directions among matrix, quaternion and axis-angle, treating $\theta = 0$ and $\theta = \pi$ explicitly; proving *gimbal lock* as a rank drop of the Jacobian; choosing the sign of the quaternion before interpolating, and knowing why.

Warning. Conventions of the chapter.

- Vectors of $\mathbb{R}^3$ are columns; $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is the canonical basis; $\langle \cdot, \cdot \rangle$ is the Euclidean scalar product and $\times$ the cross product.
- I is the 3×3 identity matrix and $M^\top$ the transpose. Matrices act on the left: $R\mathbf{v}$.
- An **active** rotation moves the vector while leaving the axes fixed. All formulas of the chapter are active; for the passive reading (axes rotating, vector fixed) one substitutes $R^\top$ for R .
- The orientation is right-handed: $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$, and a rotation through a positive angle about $\mathbf{u}$ is counterclockwise as seen from the tip of $\mathbf{u}$.
- A quaternion is written $q = (w, \mathbf{v})$ with $w \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}^3$, that is $q = w + v_1i + v_2j + v_3k$: this is Definition 11.3 with the real part placed first. The conjugate is $\bar{q} = (w, -\mathbf{v})$ and the norm $\|q\|^2 = w^2 + \|\mathbf{v}\|^2$.
- θ is the angle of rotation; the half-angle $\theta/2$ appears throughout. Angles are written in degrees in numerical calculations and in radians in formulas.

28.1. The rotation group and the elementary rotations

Recall Definition 11.2: the *rotation group* of space is

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} : R^\top R = I, \det R = +1\},$$

a group under matrix multiplication. Its three *elementary rotations* are the rotations about the coordinate axes.

Definition 28.1: Elementary rotations. For $\alpha \in \mathbb{R}$ one sets

$$R_x(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix}, \quad R_y(\alpha) = \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix},$$

$$R_z(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The sign of R_y is opposite to the other two because the cyclic triple is (x, y, z) : in the rotation about y the rotated plane is (z, x) and not (x, z) .

Proposition 28.1: The elementary rotations are rotations. For every α one has $R_x(\alpha), R_y(\alpha), R_z(\alpha) \in SO(3)$, and the one-parameter group identities hold

$$R_*(\alpha)R_*(\beta) = R_*(\alpha + \beta), \quad R_*(\alpha)^{-1} = R_*(-\alpha) = R_*(\alpha)^\top, \quad * \in \{x, y, z\}.$$

Proof. Step 1 — orthogonality. The computation is carried out for R_z ; the other two cases are identical after permuting the indices. The columns of $R_z(\alpha)$ are $c_1 = (\cos \alpha, \sin \alpha, 0)^\top$, $c_2 = (-\sin \alpha, \cos \alpha, 0)^\top$, $c_3 = (0, 0, 1)^\top$. Then $\langle c_1, c_1 \rangle = \cos^2 \alpha + \sin^2 \alpha = 1$, $\langle c_2, c_2 \rangle = \sin^2 \alpha + \cos^2 \alpha = 1$, and $\langle c_1, c_2 \rangle = 0$. Similarly, $\langle c_1, c_3 \rangle = \langle c_2, c_3 \rangle = 0$.

$\cos^2 \alpha = 1$, $\langle c_3, c_3 \rangle = 1$, $\langle c_1, c_2 \rangle = -\cos \alpha \sin \alpha + \sin \alpha \cos \alpha = 0$ and $\langle c_1, c_3 \rangle = \langle c_2, c_3 \rangle = 0$. The columns therefore form an orthonormal basis, that is $R_z^\top R_z = I$.

Step 2 — determinant. Expanding along the third row, $\det R_z(\alpha) = 1 \cdot (\cos^2 \alpha + \sin^2 \alpha) = 1$. Likewise $\det R_x(\alpha) = \cos^2 \alpha + \sin^2 \alpha = 1$ expanding along the first row, and $\det R_y(\alpha) = \cos^2 \alpha - (\sin \alpha)(-\sin \alpha) = 1$ expanding along the second row. Hence all three belong to $SO(3)$.

Step 3 — one-parameter group. Again for R_z :

$$R_z(\alpha)R_z(\beta) = \begin{pmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -\cos \alpha \sin \beta - \sin \alpha \cos \beta & 0 \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and the addition formulas (Volume II, Chapter 1) recognise in this matrix $R_z(\alpha + \beta)$. Setting $\beta = -\alpha$ gives $R_z(\alpha)R_z(-\alpha) = R_z(0) = I$, that is $R_z(\alpha)^{-1} = R_z(-\alpha)$; and $R_z(-\alpha) = R_z(\alpha)^\top$ by direct inspection, since $\cos$ is even and $\sin$ is odd. The other two cases are identical. $\square$

The first structural fact is that, unlike $SO(2)$, the group $SO(3)$ is not commutative. Theorem 11.7 had already announced this; here an explicit counterexample is given, which will be used repeatedly in what follows.

Theorem 28.1: *$SO(3)$ is not commutative. Set $A = R_z(90^\circ)R_x(90^\circ)$ and $B = R_x(90^\circ)R_z(90^\circ)$. Then*

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix},$$

and in particular $Ae_1 = e_2$ while $Be_1 = e_3$. Hence $A \neq B$ and $SO(3)$ is not abelian.

Proof. Step 1 — the two elementary matrices. By Definition 28.1, with $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$,

$$R_z(90^\circ) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_x(90^\circ) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Step 2 — the product $A = R_z(90^\circ)R_x(90^\circ)$. This is computed row by row. The first row of $R_z(90^\circ)$ is $(0, -1, 0)$; multiplied by the columns $(1, 0, 0)^\top$, $(0, 0, 1)^\top$, $(0, -1, 0)^\top$ of $R_x(90^\circ)$ this gives $0 \cdot 1 + (-1) \cdot 0 + 0 \cdot 0 = 0$, then $0 \cdot 0 + (-1) \cdot 0 + 0 \cdot 1 = 0$, then $0 \cdot 0 + (-1) \cdot (-1) + 0 \cdot 0 = 1$: the first row of A is $(0, 0, 1)$. The second row of $R_z(90^\circ)$ is $(1, 0, 0)$ and reproduces the first row of $R_x(90^\circ)$, namely $(1, 0, 0)$. The third row of $R_z(90^\circ)$ is $(0, 0, 1)$ and reproduces the third row of $R_x(90^\circ)$, namely $(0, 1, 0)$. Hence A is the matrix of the statement.

Step 3 — the product $B = R_x(90^\circ)R_z(90^\circ)$. The rows of $R_x(90^\circ)$ are $(1, 0, 0)$, $(0, 0, -1)$, $(0, 1, 0)$: the first reproduces the first row of $R_z(90^\circ)$, namely $(0, -1, 0)$; the second gives the opposite of the third row of $R_z(90^\circ)$, namely $(0, 0, -1)$; the third gives the second row of $R_z(90^\circ)$, namely $(1, 0, 0)$. Hence B is the matrix of the statement.

Step 4 — comparison. The first column of A is $(0, 1, 0)^\top = e_2$, that of B is $(0, 0, 1)^\top = e_3$. Since $e_2 \neq e_3$, the two matrices already differ in the first column: $A \neq B$. Both belong to $SO(3)$ because they are products of elements of $SO(3)$ (Proposition 28.1 and closure of the group). Hence there exist in $SO(3)$ two elements that do not commute.

Step 5 — geometric reading. The computation says something that can be checked by hand with a book: rotating it a quarter turn about the vertical axis and then a quarter turn about the front axis does not take it where the reversed sequence does. The difference is measurable: it will be seen in Example 28.2 that A and B have the *same* angle, 120° , but

different axes.

Step 6 — independent check. The symbolic computation was repeated in `numpy` by building $R_z(\pi/2)$ and $R_x(\pi/2)$ from the trigonometric functions and multiplying them in the two orders: one obtains exactly the two matrices of the statement, with $A\mathbf{e}_1 = (0, 1, 0)^\top$ and $B\mathbf{e}_1 = (0, 0, 1)^\top$. ✓ $\square$

***Remark 28.1: What is lost and what is gained.** Non-commutativity has two immediate practical consequences. First: the order in which a controller applies rotations is part of the specification, not a detail; the ZYX , ZYZ , XYZ conventions of manufacturers describe different orders and produce different matrices from the same three numbers. Second: there is no “sum of orientations”. Interpolating two triples of Euler angles component by component does not in general produce the shortest path between the two orientations — see Proposition 28.6. In $SO(2)$, where the group is commutative, the average of two angles is still a sensible angle: this is why the plane case does not prepare one for the pitfalls of the spatial case.*

A fact already proved elsewhere in the treatise and used constantly in what follows is that rotations “commute with the cross product”: for $A \in SO(3)$ and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ one has

$$A(\mathbf{u} \times \mathbf{v}) = (A\mathbf{u}) \times (A\mathbf{v}), \quad (28.1)$$

while for A orthogonal with $\det A = -1$ the right-hand side changes sign. This is Step 1 of the proof of Proposition 3.4, obtained from the characterisation $\langle \mathbf{u} \times \mathbf{v}, \mathbf{w} \rangle = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ and the multiplicativity of the determinant. Section 28.2 draws from it the matrix form needed here (Lemma 28.3).

28.2. The skew-symmetric matrix of the axis

The entire matrix treatment of rotations rests on a single matrix: the one representing the cross product with a fixed vector.

Definition 28.2: Skew-symmetric matrix associated with a vector. To $\mathbf{u} = (u_1, u_2, u_3)^\top \in \mathbb{R}^3$ one associates the matrix

$$[\mathbf{u}]_\times = \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3},$$

called the **skew-symmetric matrix of the axis** (or *hat map*). When $\mathbf{u}$ is a unit vector one writes $K = [\mathbf{u}]_\times$ for brevity.

Lemma 28.1: Elementary properties of the hat map. For every $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$:

- (i) $[\mathbf{u}]_\times \mathbf{v} = \mathbf{u} \times \mathbf{v}$;
- (ii) $[\mathbf{u}]_\times^\top = -[\mathbf{u}]_\times$, and conversely every real 3×3 skew-symmetric matrix is $[\mathbf{u}]_\times$ for a unique $\mathbf{u}$;
- (iii) $[\mathbf{u}]_\times \mathbf{u} = \mathbf{0}$;
- (iv) $[\mathbf{u}]_\times^2 = \mathbf{u}\mathbf{u}^\top - \|\mathbf{u}\|^2 I$.

Proof. Step 1 — (i). The cross product, component by component, is $\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)^\top$. On the other hand

$$[\mathbf{u}]_\times \mathbf{v} = \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} -u_3v_2 + u_2v_3 \\ u_3v_1 - u_1v_3 \\ -u_2v_1 + u_1v_2 \end{pmatrix},$$

which is the same vector.

Step 2 — (ii). Transposition swaps the entry at position $(1, 2)$, namely $-u_3$, with the one at position $(2, 1)$, namely u_3 , and likewise for the other two off-diagonal pairs; the diagonal is zero. Hence $[\mathbf{u}]_\times^\top = -[\mathbf{u}]_\times$. Conversely, let S be skew-symmetric: $S^\top = -S$ forces $s_{ii} = -s_{ii}$, that is $s_{ii} = 0$, and $s_{ji} = -s_{ij}$. Setting $u_1 = s_{32}$, $u_2 = s_{13}$, $u_3 = s_{21}$ gives $S = [\mathbf{u}]_\times$; and this choice is forced by comparing entries, hence $\mathbf{u}$ is unique.

Step 3 — (iii). By (i), $[\mathbf{u}]_\times \mathbf{u} = \mathbf{u} \times \mathbf{u} = \mathbf{0}$, because the cross product is alternating.

Step 4 — (iv). By (i), $[\mathbf{u}]_\times^2 \mathbf{v} = \mathbf{u} \times (\mathbf{u} \times \mathbf{v})$. The double cross product identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$ (also used in Chapter 3) gives

$$\mathbf{u} \times (\mathbf{u} \times \mathbf{v}) = \mathbf{u}(\mathbf{u} \cdot \mathbf{v}) - \mathbf{v}(\mathbf{u} \cdot \mathbf{u}) = (\mathbf{u}\mathbf{u}^\top) \mathbf{v} - \|\mathbf{u}\|^2 \mathbf{v},$$

having observed that $\mathbf{u}(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u}(\mathbf{u}^\top \mathbf{v}) = (\mathbf{u}\mathbf{u}^\top) \mathbf{v}$ by associativity of matrix multiplication. Since $\mathbf{v}$ is arbitrary, the two matrices coincide. $\square$

Lemma 28.2: Recurrence of the powers. Let $\mathbf{u}$ be a unit vector and $K = [\mathbf{u}]_\times$. Then

$$K^3 = -K,$$

and more generally, for every integer $m \geq 0$,

$$K^{2m+1} = (-1)^m K, \quad K^{2m+2} = (-1)^m K^2.$$

Proof. Step 1 — the cube. By Lemma 28.1(iv) with $\|\mathbf{u}\| = 1$ one has $K^2 = \mathbf{u}\mathbf{u}^\top - I$. Hence

$$K^3 = K \cdot K^2 = K(\mathbf{u}\mathbf{u}^\top - I) = (K\mathbf{u})\mathbf{u}^\top - K = \mathbf{0} \cdot \mathbf{u}^\top - K = -K,$$

where $K\mathbf{u} = \mathbf{0}$ is Lemma 28.1(iii).

Step 2 — induction on the odd powers. For $m = 0$ the claim $K^1 = (-1)^0 K$ is trivial. Assuming $K^{2m+1} = (-1)^m K$, one multiplies by K^2 :

$$K^{2m+3} = K^{2m+1} K^2 = (-1)^m K K^2 = (-1)^m K^3 = (-1)^m (-K) = (-1)^{m+1} K,$$

using Step 1. The induction principle concludes.

Step 3 — even powers. Multiplying the identity of Step 2 by K : $K^{2m+2} = K^{2m+1} K = (-1)^m K \cdot K = (-1)^m K^2$.

Step 4 — reading. The recurrence says that the space spanned by I , K , K^2 is closed under multiplication by K : any power series in K reduces to a combination of these three matrices. This is exactly what makes Rodrigues' formula *closed*, that is free of residual series. $\square$

Lemma 28.3: The hat map under conjugation. For every $A \in SO(3)$ and every $\mathbf{b} \in \mathbb{R}^3$ one has

$$A[\mathbf{b}]_\times A^\top = [A\mathbf{b}]_\times.$$

Proof. Step 1. Let $\mathbf{x} \in \mathbb{R}^3$ be arbitrary. By Lemma 28.1(i),

$$A[\mathbf{b}]_\times A^\top \mathbf{x} = A(\mathbf{b} \times (A^\top \mathbf{x})).$$

Step 2. By invariance (28.1) of the cross product under $SO(3)$ (Proposition 3.4, Step 1),

$$A(\mathbf{b} \times (A^\top \mathbf{x})) = (A\mathbf{b}) \times (AA^\top \mathbf{x}) = (A\mathbf{b}) \times \mathbf{x} = [A\mathbf{b}]_\times \mathbf{x},$$

since $AA^\top = I$.

Step 3. The two matrices $A[\mathbf{b}]_\times A^\top$ and $[A\mathbf{b}]_\times$ act the same way on every $\mathbf{x} \in \mathbb{R}^3$, hence they coincide. $\square$

Proposition 28.2: The hat map is a linear isomorphism. The map $\mathbf{u} \mapsto [\mathbf{u}]_\times$ is a linear isomorphism from $\mathbb{R}^3$ onto the space $\mathfrak{so}(3)$ of real 3×3 skew-symmetric matrices, which therefore has dimension 3.

Proof. **Step 1 — linearity.** Every entry of $[\mathbf{u}]_\times$ is 0 or $\pm u_i$: it is a linear function of the components of $\mathbf{u}$. Hence $[\alpha\mathbf{u} + \beta\mathbf{v}]_\times = \alpha[\mathbf{u}]_\times + \beta[\mathbf{v}]_\times$.

Step 2 — injectivity. If $[\mathbf{u}]_\times = 0$, then the entries at positions (3, 2), (1, 3), (2, 1) give $u_1 = u_2 = u_3 = 0$.

Step 3 — surjectivity onto $\mathfrak{so}(3)$. This is the second half of Lemma 28.1(ii): every skew-symmetric matrix is $[\mathbf{u}]_\times$ for a suitable $\mathbf{u}$. Hence the map is an isomorphism and $\dim \mathfrak{so}(3) = \dim \mathbb{R}^3 = 3$.

Step 4 — reading. The dimension 3 is the number of degrees of freedom of the orientation of a rigid body: three, not four nor nine. The space $\mathfrak{so}(3)$ is the Lie algebra of the group $SO(3)$, the subject of Chapter 29; here it is needed only as the space of “generators” of rotations. $\square$

28.3. Rodrigues' formula

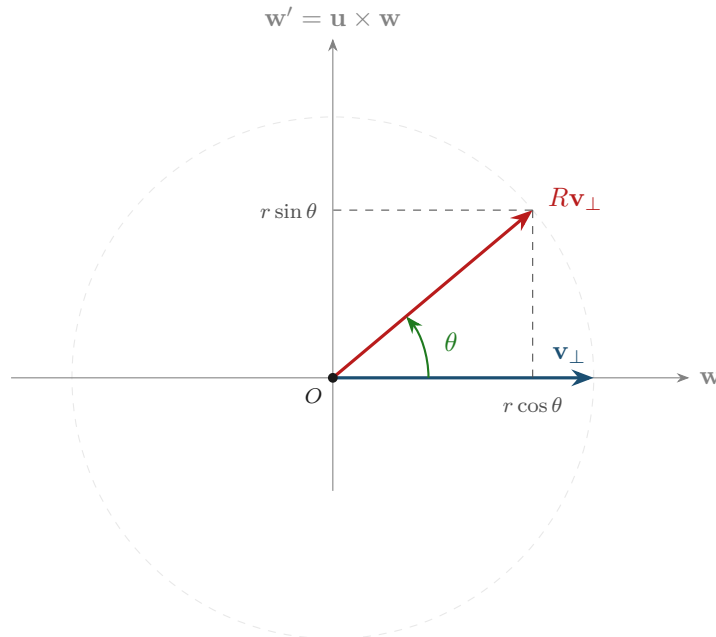


Figure 28.1: Rodrigues' formula seen in the plane orthogonal to the axis. If $\mathbf{u}$ is the unit vector of the axis and $\mathbf{v}_\perp \neq \mathbf{0}$ is the component of $\mathbf{v}$ orthogonal to $\mathbf{u}$, the pair $\mathbf{w} = \mathbf{v}_\perp / r$, $\mathbf{w}' = \mathbf{u} \times \mathbf{w}$ (with $r = \|\mathbf{v}_\perp\|$) is an orthonormal basis of $\mathbf{u}^\perp$ positively oriented as seen from the tip of $\mathbf{u}$. The rotation sends $\mathbf{v}_\perp$ to $r \cos \theta \mathbf{w} + r \sin \theta \mathbf{w}'$: it is a plane rotation, and that is all that Rodrigues' formula states. The coordinates drawn are $r = 3$ and $\theta = 40^\circ$, so that $r \cos \theta = 3 \times 0.766044 = 2.298133$ and $r \sin \theta = 3 \times 0.642788 = 1.928363$.

Figure 28.1 anticipates the geometric content of what follows: a rotation of space is a *plane*

rotation in the plane orthogonal to the axis, and the whole of Rodrigues' formula consists in writing that plane in an orthonormal basis built from the vector to be rotated.

In Chapter 11 Rodrigues' formula appears in its *vector* form (Exercise S5 of that chapter), deduced from the quaternion calculus. Here it is proved in *matrix* form, independently and without quaternions, and identified with the exponential of a skew-symmetric matrix: this is the form algorithms use, because it produces directly the nine entries of R .

Theorem 28.2: Rodrigues' formula. Let $\mathbf{u} \in \mathbb{R}^3$ be a unit vector, $K = [\mathbf{u}]_{\times}$ and $\theta \in \mathbb{R}$. Set

$$R(\mathbf{u}, \theta) = I + \sin \theta K + (1 - \cos \theta) K^2.$$

Then:

- (i) $R(\mathbf{u}, \theta) = e^{\theta K} = \sum_{n \geq 0} \frac{\theta^n K^n}{n!}$;
- (ii) $R(\mathbf{u}, \theta) \in SO(3)$, fixes $\mathbf{u}$ and is the rotation through angle θ about $\mathbf{u}$, that is in the positive orthonormal basis $\{\mathbf{u}, \mathbf{w}, \mathbf{u} \times \mathbf{w}\}$ (with $\mathbf{w}$ a unit vector orthogonal to $\mathbf{u}$) it has

$$\text{matrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix};$$

- (iii) in vector form, for every $\mathbf{v} \in \mathbb{R}^3$,

$$R(\mathbf{u}, \theta)\mathbf{v} = \cos \theta \mathbf{v} + \sin \theta (\mathbf{u} \times \mathbf{v}) + (1 - \cos \theta) \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{u}.$$

Proof. Step 1 — the exponential series converges and can be reordered. The operator norm is submultiplicative, hence $\|\theta^n K^n / n!\| \leq (|\theta| \|K\|)^n / n!$ and the series $\sum_n \theta^n K^n / n!$ converges absolutely (entry by entry) for every θ . An absolutely convergent series can be reordered and regrouped at will: this is what will be done in Step 3.

Step 2 — separating the powers. One isolates the term $n = 0$ and then the terms of odd and even index, writing $n = 2m + 1$ and $n = 2m + 2$ with $m \geq 0$:

$$e^{\theta K} = I + \sum_{m \geq 0} \frac{\theta^{2m+1}}{(2m+1)!} K^{2m+1} + \sum_{m \geq 0} \frac{\theta^{2m+2}}{(2m+2)!} K^{2m+2}.$$

Every integer $n \geq 1$ is of exactly one of the two forms, so no term is counted twice or omitted.

Step 3 — using the recurrence. By Lemma 28.2, $K^{2m+1} = (-1)^m K$ and $K^{2m+2} = (-1)^m K^2$. The two sums become

$$\left(\sum_{m \geq 0} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} \right) K + \left(\sum_{m \geq 0} \frac{(-1)^m \theta^{2m+2}}{(2m+2)!} \right) K^2.$$

The first series is the expansion of $\sin \theta$. The second is

$$\sum_{m \geq 0} \frac{(-1)^m \theta^{2m+2}}{(2m+2)!} = - \sum_{m \geq 0} \frac{(-1)^{m+1} \theta^{2(m+1)}}{(2(m+1))!} = - \sum_{p \geq 1} \frac{(-1)^p \theta^{2p}}{(2p)!} = -(\cos \theta - 1) = 1 - \cos \theta,$$

having set $p = m + 1$ and recognised in $\sum_{p \geq 0} (-1)^p \theta^{2p} / (2p)!$ the expansion of $\cos \theta$, from which the term $p = 0$, namely 1, has been removed. Hence $e^{\theta K} = I + \sin \theta K + (1 - \cos \theta) K^2$: this is (i).

Step 4 — the axis is fixed. By Lemma 28.1(iii), $K\mathbf{u} = \mathbf{0}$, and hence also $K^2\mathbf{u} = K(K\mathbf{u}) = \mathbf{0}$. Therefore $R\mathbf{u} = \mathbf{u} + \sin \theta \cdot \mathbf{0} + (1 - \cos \theta) \cdot \mathbf{0} = \mathbf{u}$.

Step 5 — action on the orthogonal plane. Let $\mathbf{w}$ be a unit vector orthogonal to $\mathbf{u}$ and $\mathbf{w}' = \mathbf{u} \times \mathbf{w}$. Then $\mathbf{w}'$ is orthogonal to both $\mathbf{u}$ and $\mathbf{w}$, and $\|\mathbf{w}'\| = \|\mathbf{u}\| \|\mathbf{w}\| \sin 90^\circ = 1$ by Lagrange's identity (Lemma 10.1) with $\langle \mathbf{u}, \mathbf{w} \rangle = 0$. Hence $\{\mathbf{u}, \mathbf{w}, \mathbf{w}'\}$ is orthonormal;

and it is positively oriented because $\det[\mathbf{u} \ \mathbf{w} \ \mathbf{w}'] = \langle \mathbf{u} \times \mathbf{w}, \mathbf{w}' \rangle = \langle \mathbf{w}', \mathbf{w}' \rangle = 1 > 0$. Now $K\mathbf{w} = \mathbf{u} \times \mathbf{w} = \mathbf{w}'$ and, by Lemma 28.1(iv) with $\|\mathbf{u}\| = 1$,

$$K^2\mathbf{w} = (\mathbf{u}\mathbf{u}^\top - I)\mathbf{w} = \mathbf{u} \langle \mathbf{u}, \mathbf{w} \rangle - \mathbf{w} = -\mathbf{w},$$

and likewise $K^2\mathbf{w}' = -\mathbf{w}'$, whence $K\mathbf{w}' = K(K\mathbf{w}) = K^2\mathbf{w} = -\mathbf{w}$. Substituting:

$$R\mathbf{w} = \mathbf{w} + \sin \theta \mathbf{w}' + (1 - \cos \theta)(-\mathbf{w}) = \cos \theta \mathbf{w} + \sin \theta \mathbf{w}',$$

$$R\mathbf{w}' = \mathbf{w}' + \sin \theta (-\mathbf{w}) + (1 - \cos \theta)(-\mathbf{w}') = -\sin \theta \mathbf{w} + \cos \theta \mathbf{w}'.$$

Step 6 — conclusion of (ii). Steps 4 and 5 say that, in the positive orthonormal basis $\{\mathbf{u}, \mathbf{w}, \mathbf{w}'\}$, the matrix of R is the one in the statement. That matrix is orthogonal with determinant $1 \cdot (\cos^2 \theta + \sin^2 \theta) = 1$; since the change of basis is orthogonal and direct, R is orthogonal with determinant $+1$, that is $R \in SO(3)$. And by Step 4 of the proof of Theorem 11.8, a matrix that in a positive orthonormal basis $\{\mathbf{u}, \mathbf{w}, \mathbf{w}'\}$ has that form is precisely the rotation through angle θ about $\mathbf{u}$: in the plane $\mathbf{u}^\perp$ it sends $\mathbf{w}$ to $\cos \theta \mathbf{w} + \sin \theta \mathbf{w}'$, that is it rotates by θ counterclockwise as seen from the tip of $\mathbf{u}$.

Step 7 — the vector form (iii). For every $\mathbf{v}$, using again $K^2 = \mathbf{u}\mathbf{u}^\top - I$ and $K\mathbf{v} = \mathbf{u} \times \mathbf{v}$:

$$R\mathbf{v} = \mathbf{v} + \sin \theta (\mathbf{u} \times \mathbf{v}) + (1 - \cos \theta)(\langle \mathbf{u}, \mathbf{v} \rangle \mathbf{u} - \mathbf{v}) = \cos \theta \mathbf{v} + \sin \theta (\mathbf{u} \times \mathbf{v}) + (1 - \cos \theta) \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{u}.$$

This is the vector form obtained by another route in Exercise S5 of Chapter 11: the two proofs are independent and check each other.

Step 8 — one-parameter group. From (i) and the fact that θK and $\theta' K$ commute it follows that $R(\mathbf{u}, \theta)R(\mathbf{u}, \theta') = e^{\theta K} e^{\theta' K} = e^{(\theta+\theta')K} = R(\mathbf{u}, \theta + \theta')$: the rotations about a fixed axis form a commutative group isomorphic to $SO(2)$, that is to the circle (Theorem 11.7). The non-commutativity of Theorem 28.1 arises only from comparing different axes. $\square$

A particular case, used constantly in what follows, is that of the coordinate axes. For $\mathbf{u} = \mathbf{e}_3$ one has $K = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $K^2 = \mathbf{e}_3 \mathbf{e}_3^\top - I = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, hence

$$R(\mathbf{e}_3, \alpha) = I + \sin \alpha K + (1 - \cos \alpha)K^2 = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} = R_z(\alpha),$$

and by the same computation $R(\mathbf{e}_1, \alpha) = R_x(\alpha)$ and $R(\mathbf{e}_2, \alpha) = R_y(\alpha)$: the elementary rotations of Definition 28.1 are the special cases of Rodrigues' formula with a coordinate axis, and in particular $R_*(\alpha) = e^{\alpha[\mathbf{e}_*] \times}$.

Corollary 28.1: Trace and skew-symmetric part. Let $R = R(\mathbf{u}, \theta)$ as in Theorem 28.2. Then

$$\operatorname{tr} R = 1 + 2 \cos \theta, \quad R - R^\top = 2 \sin \theta K, \quad R + R^\top = 2I + 2(1 - \cos \theta)K^2.$$

Proof. **Step 1 — the trace.** The trace is linear. One has $\operatorname{tr} I = 3$; $\operatorname{tr} K = 0$ because the diagonal of $[\mathbf{u}]_\times$ is zero; and, by Lemma 28.1(iv) with $\|\mathbf{u}\| = 1$, $\operatorname{tr} K^2 = \operatorname{tr}(\mathbf{u}\mathbf{u}^\top) - \operatorname{tr} I = \|\mathbf{u}\|^2 - 3 = 1 - 3 = -2$. Hence

$$\operatorname{tr} R = 3 + \sin \theta \cdot 0 + (1 - \cos \theta)(-2) = 3 - 2 + 2 \cos \theta = 1 + 2 \cos \theta,$$

in agreement with Theorem 11.8. $\checkmark$

Step 2 — skew-symmetric and symmetric part. By Lemma 28.1(ii), $K^\top = -K$; consequently $(K^2)^\top = (K^\top)^2 = (-K)^2 = K^2$, that is K^2 is symmetric. Transposing Rodrigues'

formula,

$$R^\top = I - \sin \theta K + (1 - \cos \theta)K^2,$$

and subtracting or adding gives the two identities of the statement. $\square$

Corollary 28.2: Conjugation of a rotation. For every $A \in SO(3)$, every unit vector $\mathbf{u}$ and every θ ,

$$A R(\mathbf{u}, \theta) A^\top = R(A\mathbf{u}, \theta).$$

In words: conjugating a rotation by A means rotating its axis by A , leaving the angle unchanged.

Proof. Step 1. Let $K = [\mathbf{u}]_\times$. Since $A^\top A = I$, $AK^2A^\top = AK A^\top A K A^\top = (AK A^\top)^2$; and by Lemma 28.3, $AK A^\top = [A\mathbf{u}]_\times$.

Step 2. Conjugating Rodrigues' formula, and using $AIA^\top = I$ and linearity of conjugation:

$$A R(\mathbf{u}, \theta) A^\top = I + \sin \theta (AK A^\top) + (1 - \cos \theta)(AK A^\top)^2 = I + \sin \theta [A\mathbf{u}]_\times + (1 - \cos \theta)[A\mathbf{u}]_\times^2.$$

Since $A\mathbf{u}$ is still a unit vector (A is orthogonal), the right-hand side is $R(A\mathbf{u}, \theta)$ by Theorem 28.2. $\square$

The following corollary is the inverse conversion — from matrix to axis and angle — and it is here that the two degenerate cases that will accompany the whole chapter first appear.

Corollary 28.3: From matrix to axis–angle, with the degenerate cases. Let $R \in SO(3)$. Set $\theta = \arccos((\operatorname{tr} R - 1)/2) \in [0, \pi]$. Then:

(i) if $0 < \theta < \pi$, the axis is uniquely determined by

$$\mathbf{u} = \frac{1}{2 \sin \theta} \begin{pmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{pmatrix},$$

and $R = R(\mathbf{u}, \theta)$;

(ii) if $\theta = 0$, then $R = I$ and every unit vector $\mathbf{u}$ satisfies $R = R(\mathbf{u}, 0)$: the axis is undetermined;

(iii) if $\theta = \pi$, then $R = R^\top$ and $\mathbf{u}\mathbf{u}^\top = (R + I)/2$; choosing the index i that maximises R_{ii} one has $u_i = \sqrt{(R_{ii} + 1)/2} > 0$ and $u_j = R_{ij}/(2u_i)$ for $j \neq i$. The axis is determined up to sign, and $R(\mathbf{u}, \pi) = R(-\mathbf{u}, \pi)$.

Proof. Step 0 — every rotation is an $R(\mathbf{n}, \vartheta)$. If $R = I$ take $\vartheta = 0$ and $\mathbf{n}$ arbitrary. If $R \neq I$, by Theorem 11.8 there is a unit vector $\mathbf{n}$ with $R\mathbf{n} = \mathbf{n}$ and, choosing an orthonormal basis $\{\mathbf{n}, \mathbf{e}'_1, \mathbf{e}'_2\}$ with $\mathbf{e}'_1, \mathbf{e}'_2$ a basis of $\mathbf{n}^\perp$, the matrix of R in that basis is the block form of Step 4 of that proof, with some angle ϑ . The basis may be assumed *positive*: if $\det[\mathbf{n} \ \mathbf{e}'_1 \ \mathbf{e}'_2] = -1$ it suffices to swap $\mathbf{e}'_1$ with $\mathbf{e}'_2$, which changes ϑ into $-\vartheta$ — immaterial, because Step 5 shows that the sign of the angle is absorbed by changing the sign of the axis. With the positive basis, $\mathbf{e}'_2 = \mathbf{n} \times \mathbf{e}'_1$ and by Theorem 28.2(ii) also $R(\mathbf{n}, \vartheta)$ has, in the *same* basis, the *same* matrix; two endomorphisms with the same matrix in a basis coincide, hence $R = R(\mathbf{n}, \vartheta)$.

Step 1 — θ is well defined and equals $\theta = \pm\vartheta$ modulo 2π . By Corollary 28.1, $\operatorname{tr} R = 1 + 2 \cos \vartheta$, hence $(\operatorname{tr} R - 1)/2 = \cos \vartheta \in [-1, 1]$ and the arccosine is defined; moreover $\cos \theta = \cos \vartheta$ with $\theta \in [0, \pi]$. By Step 5, replacing $\mathbf{n}$ with $-\mathbf{n}$ if necessary one may always assume $\vartheta = \theta \in [0, \pi]$; this is assumed in what follows and one writes $\mathbf{u} = \mathbf{n}$.

Step 2 — case $0 < \theta < \pi$. Then $\sin \theta > 0$. By Corollary 28.1 applied to $R = R(\mathbf{u}, \theta)$ one has $R - R^\top = 2 \sin \theta [\mathbf{u}]_\times$, that is $[\mathbf{u}]_\times = (R - R^\top)/(2 \sin \theta)$. Reading the entries of $[\mathbf{u}]_\times$ at

positions $(3, 2)$, $(1, 3)$, $(2, 1)$ — which equal respectively u_1 , u_2 , u_3 by Definition 28.2 — gives the formula of the statement, and $\mathbf{u}$ is a unit vector, being the axis supplied by Theorem 11.8. Uniqueness of the axis for $R \neq I$ is Step 5 of that proof; and with $\theta \in (0, \pi)$ fixed the sign is no longer free, because $R(-\mathbf{u}, \theta) \neq R(\mathbf{u}, \theta)$ (their skew-symmetric parts are opposite and nonzero).

Step 3 — case $\theta = 0$. Then $\text{tr } R = 3$; but $\text{tr } R = 1 + 2 \cos \vartheta$ forces $\cos \vartheta = 1$, that is $\vartheta \in 2\pi\mathbb{Z}$, and Rodrigues' formula gives $R = I + 0 \cdot K + 0 \cdot K^2 = I$ for any axis whatsoever. Consistently, $R - R^\top = 0$ carries no information about the axis: the axis-angle parametrisation is singular at the identity, as already observed for the equivalent-axis representation.

Step 4 — case $\theta = \pi$. Then $\sin \theta = 0$ and $R - R^\top = 0$: the matrix is symmetric and Step 2 does not apply. But $\cos \theta = -1$, so Rodrigues' formula gives

$$R = I + 0 \cdot K + 2K^2 = I + 2(\mathbf{u}\mathbf{u}^\top - I) = 2\mathbf{u}\mathbf{u}^\top - I,$$

whence $\mathbf{u}\mathbf{u}^\top = (R + I)/2$. Comparing entries: $u_i u_j = (R_{ij} + \delta_{ij})/2$, and in particular $u_i^2 = (R_{ii} + 1)/2$. Since $\text{tr } R = 1 + 2 \cos \pi = -1$, the diagonal has sum -1 and its maximum satisfies $R_{ii} \geq -1/3$, hence $u_i^2 \geq 1/3 > 0$: the square root can be extracted without division by zero, and this is the numerically most stable choice. Once the sign $u_i > 0$ is fixed, the other components follow from $u_i u_j = R_{ij}/2$. The substitution $\mathbf{u} \mapsto -\mathbf{u}$ leaves $\mathbf{u}\mathbf{u}^\top$, hence R , unchanged: the axis is determined only up to sign.

Step 5 — why $\theta \in [0, \pi]$ suffices. If $\vartheta \in (\pi, 2\pi)$, note that $R(\mathbf{u}, \vartheta) = R(-\mathbf{u}, 2\pi - \vartheta)$: indeed replacing $\mathbf{u}$ with $-\mathbf{u}$ changes K into $-K$ (linearity of the hat map, Proposition 28.2) while K^2 stays unchanged, and $\sin(2\pi - \vartheta) = -\sin \vartheta$ while $\cos(2\pi - \vartheta) = \cos \vartheta$; the two terms whose sign is changed compensate one another. Hence every rotation can be written with $\theta \in [0, \pi]$, and this is the normalisation adopted in the chapter. $\square$

Example 28.1: An exact case: 120° about the diagonal of the cube. Rodrigues' formula is applied with $\mathbf{u} = (1, 1, 1)^\top / \sqrt{3}$ and $\theta = 120^\circ$. The result is an integer matrix, and this makes the case ideal for a hand check.

Step 1 — the skew-symmetric matrix. By Definition 28.2,

$$K = [\mathbf{u}]_\times = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Step 2 — the square. By Lemma 28.1(iv), since $\mathbf{u}\mathbf{u}^\top = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$,

$$K^2 = \mathbf{u}\mathbf{u}^\top - I = \frac{1}{3} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

Step 3 — the coefficients. $\sin 120^\circ = \frac{\sqrt{3}}{2}$ and $\cos 120^\circ = -\frac{1}{2}$, hence $1 - \cos \theta = \frac{3}{2}$. Thus

$$\sin \theta K = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix},$$

$$(1 - \cos \theta)K^2 = \frac{3}{2} \cdot \frac{1}{3} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

Note that the factor $\sqrt{3}$ cancels: this is why the result is an integer.

Step 4 — the sum.

$$R = I + \frac{1}{2} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} = I + \frac{1}{2} \begin{pmatrix} -2 & 0 & 2 \\ 2 & -2 & 0 \\ 0 & 2 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Step 5 — reading. The resulting matrix sends $\mathbf{e}_1 \mapsto \mathbf{e}_2$, $\mathbf{e}_2 \mapsto \mathbf{e}_3$, $\mathbf{e}_3 \mapsto \mathbf{e}_1$: it is the cyclic permutation of the axes. That this is a rotation of 120° about the diagonal $(1, 1, 1)$ of the cube is also seen geometrically: the diagonal is the only axis that cyclically permutes the three edges leaving the vertex, and a third of a turn suffices to do it.

Step 6 — three independent checks. (a) $\text{tr } R = 0 + 0 + 0 = 0$, while $1 + 2 \cos 120^\circ = 1 - 1 = 0$: consistent with Corollary 28.1. ✓ (b) $R^3 = I$, as it must be because $3 \times 120^\circ = 360^\circ$; and indeed applying the cyclic permutation three times each axis returns to its place. ✓ (c) $R\mathbf{u} = \mathbf{u}$: adding the columns, $R(1, 1, 1)^\top = (1, 1, 1)^\top$. ✓ The computation was repeated in `numpy`, building K , K^2 and R from the numerical values of $\sin 120^\circ$ and $\cos 120^\circ$: one obtains the integer matrix with maximum deviation 1.1×10^{-16} . ✓

Example 28.2: The axes of the non-commutativity counterexample. $A = R_z(90^\circ)R_x(90^\circ)$ and $B = R_x(90^\circ)R_z(90^\circ)$ of Theorem 28.1 are taken up again and their axis and angle computed with Corollary 28.3.

Step 1 — the angles. $\text{tr } A = 0 + 0 + 0 = 0$ and $\text{tr } B = 0 + 0 + 0 = 0$. Hence for both $\cos \theta = (0 - 1)/2 = -\frac{1}{2}$ and $\theta = 120^\circ$, with $\sin \theta = \frac{\sqrt{3}}{2}$ and $2 \sin \theta = \sqrt{3}$.

Step 2 — the axis of A . From $A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ one reads $A_{32} - A_{23} = 1 - 0 = 1$, $A_{13} - A_{31} = 1 - 0 = 1$, $A_{21} - A_{12} = 1 - 0 = 1$, hence

$$\mathbf{u}_A = \frac{1}{\sqrt{3}}(1, 1, 1)^\top.$$

This is the diagonal of the cube of Example 28.1: indeed the two matrices coincide. ✓

Step 3 — the axis of B . From $B = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}$ one reads $B_{32} - B_{23} = 0 - (-1) = 1$, $B_{13} - B_{31} = 0 - 1 = -1$, $B_{21} - B_{12} = 0 - (-1) = 1$, hence

$$\mathbf{u}_B = \frac{1}{\sqrt{3}}(1, -1, 1)^\top.$$

Step 4 — reading. Reversing the order of two quarter-turn rotations does not change the resulting *angle*, which remains 120° , but changes its *axis*: from one diagonal of the cube to another. The non-commutativity of $SO(3)$ is entirely this, and it is not a defect of matrices: it is a property of space.

Step 5 — independent check. Reconstructing with Rodrigues' formula $R(\mathbf{u}_A, 120^\circ)$ and $R(\mathbf{u}_B, 120^\circ)$ in `numpy` recovers A and B with maximum deviation 1.1×10^{-15} ; and the angles computed from $\arccos((\text{tr} - 1)/2)$ both equal 120.000000° . ✓

Example 28.3: A complete numerical case and the inverse conversion. Take $\mathbf{u} = \frac{1}{3}(2, -1, 2)^\top$ (a unit vector, because $2^2 + 1^2 + 2^2 = 9$) and $\theta = 40^\circ$. *Convention on digits:* all values are computed in double precision and rounded to the seventh decimal digit only for printing; adding the printed values may therefore give a difference of one unit in the last digit.

Step 1 — the matrices.

$$K = \frac{1}{3} \begin{pmatrix} 0 & -2 & -1 \\ 2 & 0 & -2 \\ 1 & 2 & 0 \end{pmatrix}, \quad K^2 = \mathbf{u}\mathbf{u}^\top - I = \frac{1}{9} \begin{pmatrix} -5 & -2 & 4 \\ -2 & -8 & -2 \\ 4 & -2 & -5 \end{pmatrix}.$$

Step 2 — the coefficients. $\sin 40^\circ = 0.6427876$, $\cos 40^\circ = 0.7660444$, $1 - \cos 40^\circ = 0.2339556$.

Step 3 — the rotation matrix. Summing $I + 0.6427876 K + 0.2339556 K^2$ entry by entry:

$$R = \begin{pmatrix} 0.8700247 & -0.4805152 & -0.1102823 \\ 0.3765349 & 0.7920395 & -0.4805152 \\ 0.3182428 & 0.3765349 & 0.8700247 \end{pmatrix}.$$

For example, $R_{11} = 1 + 0 - 0.2339556 \times \frac{5}{9} = 1 - 0.1299753 = 0.8700247$ and $R_{12} = -0.6427876 \times \frac{2}{3} - 0.2339556 \times \frac{2}{9} = -0.4285251 - 0.0519901 = -0.4805152$.

Step 4 — independent check (trace). $\text{tr } R = 0.8700247 + 0.7920395 + 0.8700247 = 2.5320889$, while $1 + 2 \cos 40^\circ = 1 + 1.5320889 = 2.5320889$: they agree to seven decimals (Corollary 28.1). ✓

Step 5 — independent check (axis). $R\mathbf{u} = \mathbf{u}$: the first component is $0.8700247 \times \frac{2}{3} - 0.4805152 \times (-\frac{1}{3}) - 0.1102823 \times \frac{2}{3} = 0.5800165 + 0.1601717 - 0.0735215 = 0.6666667 = \frac{2}{3}$. ✓

Step 6 — the inverse conversion. With Corollary 28.3(i): $\theta = \arccos(0.7660444) = 40.000000^\circ$; $2 \sin \theta = 1.2855752$; and

$$R_{32} - R_{23} = 0.3765349 + 0.4805152 = 0.8570501,$$

$$R_{13} - R_{31} = -0.1102823 - 0.3182428 = -0.4285251,$$

$$R_{21} - R_{12} = 0.3765349 + 0.4805152 = 0.8570501,$$

whence

$$\mathbf{u} = \frac{(0.8570501, -0.4285251, 0.8570501)^\top}{1.2855752} = (0.6666667, -0.3333333, 0.6666667)^\top,$$

that is $\mathbf{u} = \frac{1}{3}(2, -1, 2)^\top$. ✓

Step 7 — check with the exponential series. Summing in `numpy` the first 60 terms of $\sum_n \theta^n K^n / n!$ gives the same matrix with maximum deviation 1.7×10^{-16} : (i) of Theorem 28.2 is verified numerically as well, and the same check repeated on five different pairs $(\mathbf{u}, \theta)$ (including $\theta = 180^\circ$ and negative θ) gives deviations always below 2.3×10^{-16} . ✓

Step 8 — orthogonality. $R^\top R - I$ has maximum norm 2.3×10^{-16} and $\det R = 1$ up to 10^{-16} : Rodrigues' formula produces orthogonal matrices *by construction*, with no need for re-orthogonalisation. This is why, in an implementation, it is preferable to update the axis and angle and rebuild R , instead of accumulating products of matrices. ✓

***Remark 28.2: Why the formula closes, and what happens in dimension 4.** The reason why the exponential series collapses into only three terms is Lemma 28.2: the minimal polynomial of K divides $\lambda^3 + \lambda = \lambda(\lambda^2 + 1)$, hence I, K, K^2 generate all the powers. In dimension n a skew-symmetric matrix has minimal polynomial of degree up to n , and the exponential can no longer be written with three terms: already in $\mathbb{R}^4$ one needs, in general, the two “rotation angles” of the two invariant planes (see Step 7 of the proof of Theorem 11.8). Rodrigues' formula is thus a privilege of dimension three — the same dimension in which*

the cross product exists, and this is no coincidence: the two things are the same thing seen twice (Proposition 28.2).

28.4. Quaternions: Hamilton's rule

The algebra $\mathbb{H}$ of quaternions is Definition 11.3: $\mathbb{R}^4$ with basis $\{1, i, j, k\}$ and associative product determined by the *fundamental formula*

$$i^2 = j^2 = k^2 = ijk = -1,$$

whence $ij = k$, $jk = i$, $ki = j$ and $ji = -k$, $kj = -i$, $ik = -j$. Here the notation $q = (w, \mathbf{v})$ is adopted, with $w = q_0 \in \mathbb{R}$ and $\mathbf{v} = (q_1, q_2, q_3)^\top \in \mathbb{R}^3$: this is the form in which the formulas become readable, because the product is expressed through the scalar product and the cross product of $\mathbb{R}^3$.

Proposition 28.3: Hamilton's rule in scalar-vector form. Let $p = (a, \mathbf{u})$ and $q = (b, \mathbf{v})$ be quaternions. Then

$$pq = (ab - \langle \mathbf{u}, \mathbf{v} \rangle, \quad a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v}).$$

Proof. Step 1 — bilinearity. The quaternion product is bilinear by definition (it extends by distributivity from the products of the generators). Writing $p = a + u_1i + u_2j + u_3k$ and $q = b + v_1i + v_2j + v_3k$, the product is the sum of 16 terms, which group into four families.

Step 2 — the terms with at least one real part. The term $a \cdot b$ gives the real ab . The three terms $a \cdot v_\ell e_\ell$ (where $e_1 = i$, $e_2 = j$, $e_3 = k$) give the vector $a\mathbf{v}$, because the real commutes with i, j, k . The three terms $u_\ell e_\ell \cdot b$ give the vector $b\mathbf{u}$.

Step 3 — the three diagonal terms. These are $u_\ell v_\ell e_\ell^2$ with $\ell = 1, 2, 3$, and $e_\ell^2 = -1$: they give the real $-(u_1v_1 + u_2v_2 + u_3v_3) = -\langle \mathbf{u}, \mathbf{v} \rangle$.

Step 4 — the six mixed terms. These are grouped in pairs. For the pair (i, j) : $u_1v_2ij + u_2v_1ji = u_1v_2k - u_2v_1k = (u_1v_2 - u_2v_1)k$, and $u_1v_2 - u_2v_1$ is the third component of $\mathbf{u} \times \mathbf{v}$. For the pair (j, k) : $u_2v_3jk + u_3v_2kj = (u_2v_3 - u_3v_2)i$, the first component of $\mathbf{u} \times \mathbf{v}$. For the pair (k, i) : $u_3v_1ki + u_1v_3ik = (u_3v_1 - u_1v_3)j$, the second component. In total the six terms give the vector $\mathbf{u} \times \mathbf{v}$.

Step 5 — sum. Real part: $ab - \langle \mathbf{u}, \mathbf{v} \rangle$; vector part: $a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v}$. This is the claim.

Step 6 — reading. The non-commutativity of $\mathbb{H}$ is concentrated in the single term $\mathbf{u} \times \mathbf{v}$, which is antisymmetric: qp has the same real part as pq and vector part $a\mathbf{v} + b\mathbf{u} - \mathbf{u} \times \mathbf{v}$. Hence $pq = qp$ if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$, that is if and only if the vector parts are parallel: two rotations with the same axis commute, and this is exactly Step 8 of the proof of Theorem 28.2.

□

Corollary 28.4: The norm is multiplicative. For all $p, q \in \mathbb{H}$ one has $\|pq\| = \|p\| \|q\|$. In particular the unit quaternions form a multiplicative group, the sphere S^3 .

Proof. Step 1 — expansion. With $p = (a, \mathbf{u})$ and $q = (b, \mathbf{v})$, Proposition 28.3 gives

$$\|pq\|^2 = (ab - \langle \mathbf{u}, \mathbf{v} \rangle)^2 + \|a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v}\|^2.$$

Step 2 — the first term. $(ab - \langle \mathbf{u}, \mathbf{v} \rangle)^2 = a^2b^2 - 2ab\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle^2$.

Step 3 — the second term. Expanding the squared norm of a sum of three vectors:

$$a^2\|\mathbf{v}\|^2 + b^2\|\mathbf{u}\|^2 + \|\mathbf{u} \times \mathbf{v}\|^2 + 2ab\langle \mathbf{v}, \mathbf{u} \rangle + 2a\langle \mathbf{v}, \mathbf{u} \times \mathbf{v} \rangle + 2b\langle \mathbf{u}, \mathbf{u} \times \mathbf{v} \rangle.$$

The last two terms are zero, because $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

Step 4 — simplification. Adding Steps 2 and 3, the terms $\mp 2ab\langle \mathbf{u}, \mathbf{v} \rangle$ cancel and what remains is

$$\|pq\|^2 = a^2b^2 + a^2\|\mathbf{v}\|^2 + b^2\|\mathbf{u}\|^2 + (\langle \mathbf{u}, \mathbf{v} \rangle^2 + \|\mathbf{u} \times \mathbf{v}\|^2).$$

Step 5 — Lagrange's identity. By Lemma 10.1, $\|\mathbf{u} \times \mathbf{v}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2$. Hence

$$\|pq\|^2 = a^2b^2 + a^2\|\mathbf{v}\|^2 + b^2\|\mathbf{u}\|^2 + \|\mathbf{u}\|^2\|\mathbf{v}\|^2 = (a^2 + \|\mathbf{u}\|^2)(b^2 + \|\mathbf{v}\|^2) = \|p\|^2\|q\|^2.$$

Taking the square root gives the claim. Written out in coordinates, this identity is Euler's *four-square identity*, invoked without proof in Step 1 of the proof of Theorem 11.10: here it is proved.

Step 6 — the group. If $\|p\| = \|q\| = 1$ then $\|pq\| = 1$: the product of unit quaternions is unitary. The unit $(1, \mathbf{0})$ is unitary, and the inverse of a unitary q is $\bar{q}$ (Proposition 28.4), which is also unitary. Hence $S^3 = \{q : \|q\| = 1\}$ is a multiplicative subgroup of $\mathbb{H} \setminus \{0\}$. $\square$

Proposition 28.4: Conjugate and inverse. For $q = (w, \mathbf{v})$ set $\bar{q} = (w, -\mathbf{v})$. Then, for all $p, q \in \mathbb{H}$:

$$\overline{pq} = \bar{q}\bar{p}, \quad q\bar{q} = \bar{q}q = \|q\|^2, \quad q^{-1} = \frac{\bar{q}}{\|q\|^2} \quad (q \neq 0),$$

and in particular $q^{-1} = \bar{q}$ if $\|q\| = 1$.

Proof. **Step 1 — the anti-automorphism.** With $p = (a, \mathbf{u})$, $q = (b, \mathbf{v})$, Proposition 28.3 gives $\overline{pq} = (ab - \langle \mathbf{u}, \mathbf{v} \rangle, -a\mathbf{v} - b\mathbf{u} - \mathbf{u} \times \mathbf{v})$. On the other hand

$$\bar{q}\bar{p} = (b, -\mathbf{v})(a, -\mathbf{u}) = (ba - \langle -\mathbf{v}, -\mathbf{u} \rangle, b(-\mathbf{u}) + a(-\mathbf{v}) + (-\mathbf{v}) \times (-\mathbf{u})),$$

and since $(-\mathbf{v}) \times (-\mathbf{u}) = \mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}$, the right-hand side equals

$$(ab - \langle \mathbf{u}, \mathbf{v} \rangle, -b\mathbf{u} - a\mathbf{v} - \mathbf{u} \times \mathbf{v}).$$

The two results coincide.

Step 2 — the norm as a product. Again by Proposition 28.3, with $p = q = (w, \mathbf{v})$ and conjugating the second factor:

$$q\bar{q} = (w \cdot w - \langle \mathbf{v}, -\mathbf{v} \rangle, w(-\mathbf{v}) + w\mathbf{v} + \mathbf{v} \times (-\mathbf{v})) = (w^2 + \|\mathbf{v}\|^2, \mathbf{0}) = \|q\|^2,$$

because $\mathbf{v} \times \mathbf{v} = \mathbf{0}$. Exchanging the roles gives $\bar{q}q = \|q\|^2$.

Step 3 — the inverse. If $q \neq 0$ then $\|q\|^2 > 0$ and Step 2 gives $q \cdot \bar{q}/\|q\|^2 = 1 = \bar{q}/\|q\|^2 \cdot q$: this is the definition of inverse. For $\|q\| = 1$ this reduces to $q^{-1} = \bar{q}$. $\square$

Remark 28.3: Why controllers distribute quaternions. Three reasons, all quantitative. (1) Memory: four numbers against nine. (2) Cost of composition: the product of two quaternions has four components, each a sum of four real products: $4 \times 4 = 16$ multiplications. The product of two 3×3 matrices requires $3 \times 3 \times 3 = 27$. (3) Stability: rounding error carries q outside S^3 , and the remedy is dividing by $\|q\|$ — one square root and four divisions. It carries R outside $SO(3)$ instead, and the remedy requires re-orthogonalisation (Gram–Schmidt or polar factorisation), which is far more costly and not unique. The constraint $\|q\| = 1$ is one equation on four parameters; the constraint $R^\top R = I$ is six equations on nine. This is why inertial navigation packages, graphics engines and robotic middleware publish orientation as a unit quaternion and convert to a matrix only at the moment of use.

28.5. Euler parameters and conversions between representations

Definition 28.3: Euler parameters. Given a unit vector $\mathbf{u} \in \mathbb{R}^3$ and $\theta \in \mathbb{R}$, the **Euler parameters** (or *Euler–Rodrigues parameters*) of the rotation $R(\mathbf{u}, \theta)$ are the four numbers

$$e_0 = \cos \frac{\theta}{2}, \quad (e_1, e_2, e_3)^\top = \sin \frac{\theta}{2} \mathbf{u},$$

that is the components of the quaternion $q = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \mathbf{u})$. They satisfy the **normalisation constraint**

$$e_0^2 + e_1^2 + e_2^2 + e_3^2 = 1.$$

The constraint is immediate: $e_0^2 + \|(e_1, e_2, e_3)\|^2 = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \|\mathbf{u}\|^2 = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} = 1$. Four parameters tied by one equation describe a manifold of dimension $4 - 1 = 3$, exactly the degrees of freedom of orientation (Proposition 28.2, Step 4): the count works out, and this is the structural reason why quaternions can be global where three free parameters cannot (Remark 28.7).

Theorem 28.3: From quaternion to rotation matrix. Let $q = (w, \mathbf{v})$ with $\|q\| = 1$ and $\mathbf{v} = (x, y, z)^\top$, and let $\Phi_q(\mathbf{p}) = q \mathbf{p} \bar{q}$ be the map of Theorem 11.10. Then the matrix of Φ_q in the canonical basis is

$$R(q) = (w^2 - \|\mathbf{v}\|^2)I + 2\mathbf{v}\mathbf{v}^\top + 2w[\mathbf{v}]_\times = \begin{pmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{pmatrix}. \quad (28.2)$$

If moreover q is the quaternion of the Euler parameters of $(\mathbf{u}, \theta)$ (Definition 28.3), then

$$R(q) = R(\mathbf{u}, \theta) = I + \sin \theta [\mathbf{u}]_\times + (1 - \cos \theta) [\mathbf{u}]_\times^2 :$$

the quaternionic construction and Rodrigues' formula are the same matrix.

Proof. Step 1 — first product. Let $\mathbf{x} \in \mathbb{R}^3$ and let $p = (0, \mathbf{x})$ be the corresponding pure quaternion. By Proposition 28.3,

$$qp = (w, \mathbf{v})(0, \mathbf{x}) = (-\langle \mathbf{v}, \mathbf{x} \rangle, w\mathbf{x} + \mathbf{v} \times \mathbf{x}).$$

Step 2 — second product, real part. Again by the same proposition, with $\bar{q} = (w, -\mathbf{v})$,

$$(qp)\bar{q} = \left(-\langle \mathbf{v}, \mathbf{x} \rangle w - \langle w\mathbf{x} + \mathbf{v} \times \mathbf{x}, -\mathbf{v} \rangle, \cdot \right).$$

The real part equals $-w\langle \mathbf{v}, \mathbf{x} \rangle + w\langle \mathbf{x}, \mathbf{v} \rangle + \langle \mathbf{v} \times \mathbf{x}, \mathbf{v} \rangle = 0$, because $\mathbf{v} \times \mathbf{x} \perp \mathbf{v}$. The result is therefore pure: this recovers Step 2 of the proof of Theorem 11.10, here by a direct route.

Step 3 — second product, vector part.

$$(-\langle \mathbf{v}, \mathbf{x} \rangle)(-\mathbf{v}) + w(w\mathbf{x} + \mathbf{v} \times \mathbf{x}) + (w\mathbf{x} + \mathbf{v} \times \mathbf{x}) \times (-\mathbf{v}).$$

The first term is $\langle \mathbf{v}, \mathbf{x} \rangle \mathbf{v}$; the second is $w^2 \mathbf{x} + w(\mathbf{v} \times \mathbf{x})$; the third, using $\mathbf{a} \times (-\mathbf{b}) = -\mathbf{a} \times \mathbf{b}$ and $\mathbf{x} \times \mathbf{v} = -\mathbf{v} \times \mathbf{x}$, equals $w(\mathbf{v} \times \mathbf{x}) - (\mathbf{v} \times \mathbf{x}) \times \mathbf{v}$.

Step 4 — the double cross product. By the identity already used in Lemma 28.1(iv),

$$(\mathbf{v} \times \mathbf{x}) \times \mathbf{v} = -\mathbf{v} \times (\mathbf{v} \times \mathbf{x}) = -(\mathbf{v}\langle \mathbf{v}, \mathbf{x} \rangle - \mathbf{x}\|\mathbf{v}\|^2) = -\langle \mathbf{v}, \mathbf{x} \rangle \mathbf{v} + \|\mathbf{v}\|^2 \mathbf{x}.$$

Step 5 — sum. Putting together Steps 3 and 4:

$$\Phi_q(\mathbf{x}) = \langle \mathbf{v}, \mathbf{x} \rangle \mathbf{v} + w^2 \mathbf{x} + 2w(\mathbf{v} \times \mathbf{x}) + \langle \mathbf{v}, \mathbf{x} \rangle \mathbf{v} - \|\mathbf{v}\|^2 \mathbf{x} = (w^2 - \|\mathbf{v}\|^2) \mathbf{x} + 2\langle \mathbf{v}, \mathbf{x} \rangle \mathbf{v} + 2w(\mathbf{v} \times \mathbf{x}).$$

Since $\langle \mathbf{v}, \mathbf{x} \rangle \mathbf{v} = (\mathbf{v} \mathbf{v}^\top) \mathbf{x}$ and $\mathbf{v} \times \mathbf{x} = [\mathbf{v}]_\times \mathbf{x}$ (Lemma 28.1(i)), and since $\mathbf{x}$ is arbitrary, the first equality of (28.2) follows.

Step 6 — the entries. On the diagonal, for example, $R_{11} = (w^2 - x^2 - y^2 - z^2) + 2x^2 = w^2 + x^2 - y^2 - z^2$; and using $w^2 + x^2 + y^2 + z^2 = 1$ one has $w^2 + x^2 = 1 - y^2 - z^2$, that is $R_{11} = 1 - 2(y^2 + z^2)$. The other two diagonal entries are identical up to permuting the names.

Off diagonal, $2\mathbf{v}\mathbf{v}^\top$ contributes $2xy$, $2xz$, $2yz$ symmetrically, while $2w[\mathbf{v}]_\times = 2w \begin{pmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{pmatrix}$ contributes antisymmetrically: this gives the six entries written in (28.2).

Step 7 — identification with Rodrigues. Substitute $w = \cos \frac{\theta}{2}$, $\mathbf{v} = \sin \frac{\theta}{2} \mathbf{u}$ with $\|\mathbf{u}\| = 1$, and write $K = [\mathbf{u}]_\times$. By the duplication formulas (Volume II, Chapter 1):

$$w^2 - \|\mathbf{v}\|^2 = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta,$$

$$2\mathbf{v}\mathbf{v}^\top = 2\sin^2 \frac{\theta}{2} \mathbf{u}\mathbf{u}^\top = (1 - \cos \theta) \mathbf{u}\mathbf{u}^\top, \quad 2w[\mathbf{v}]_\times = 2\cos \frac{\theta}{2} \sin \frac{\theta}{2} K = \sin \theta K.$$

Hence $R(q) = \cos \theta I + (1 - \cos \theta) \mathbf{u}\mathbf{u}^\top + \sin \theta K$. Now $\mathbf{u}\mathbf{u}^\top = K^2 + I$ by Lemma 28.1(iv), so that

$$R(q) = \cos \theta I + (1 - \cos \theta) K^2 + (1 - \cos \theta) I + \sin \theta K = I + \sin \theta K + (1 - \cos \theta) K^2,$$

which is exactly $R(\mathbf{u}, \theta)$ of Theorem 28.2.

Step 8 — reading. The half-angle is not an arbitrary convention: it appears because the quaternion acts *twice* on $\mathbf{x}$, once from the left and once from the right, and each application contributes half the angle. This is the same reason why the covering is $2:1$ (Section 28.6). $\square$

Theorem 28.4: From matrix to quaternion: the four branches. Let $R \in SO(3)$ and set

$$c_0 = 1 + \operatorname{tr} R, \quad c_1 = 1 + 2R_{11} - \operatorname{tr} R, \quad c_2 = 1 + 2R_{22} - \operatorname{tr} R, \quad c_3 = 1 + 2R_{33} - \operatorname{tr} R.$$

Then $c_0 + c_1 + c_2 + c_3 = 4$, all the c_m are nonnegative, and if $q = (w, x, y, z)$ is a unit quaternion with $R(q) = R$ one has

$$c_0 = 4w^2, \quad c_1 = 4x^2, \quad c_2 = 4y^2, \quad c_3 = 4z^2.$$

In particular $\max_m c_m \geq 1$; letting m^* be an index of the maximum and $s = \frac{1}{2}\sqrt{c_{m^*}} \geq \frac{1}{2}$, the remaining components are obtained by dividing by $4s$ according to the scheme

$$\begin{aligned} m^* = 0: & \quad w = s, \quad x = \frac{R_{32} - R_{23}}{4s}, \quad y = \frac{R_{13} - R_{31}}{4s}, \quad z = \frac{R_{21} - R_{12}}{4s}; \\ m^* = 1: & \quad x = s, \quad w = \frac{R_{32} - R_{23}}{4s}, \quad y = \frac{R_{12} + R_{21}}{4s}, \quad z = \frac{R_{13} + R_{31}}{4s}; \\ m^* = 2: & \quad y = s, \quad w = \frac{R_{13} - R_{31}}{4s}, \quad x = \frac{R_{12} + R_{21}}{4s}, \quad z = \frac{R_{23} + R_{32}}{4s}; \\ m^* = 3: & \quad z = s, \quad w = \frac{R_{21} - R_{12}}{4s}, \quad x = \frac{R_{13} + R_{31}}{4s}, \quad y = \frac{R_{23} + R_{32}}{4s}. \end{aligned}$$

The only solutions are q and $-q$.

Proof. Step 1 — existence of q . By Theorem 11.10(iv) the map $S^3 \rightarrow SO(3)$, $q \mapsto \Phi_q$, is surjective: there is a unit q with $\Phi_q = R$, and by Theorem 28.3 its matrix is (28.2). In what follows one works with that $q = (w, x, y, z)$.

Step 2 — the trace. Summing the diagonal entries of (28.2) in the first form, $\operatorname{tr} R = 3(w^2 - \|\mathbf{v}\|^2) + 2\|\mathbf{v}\|^2 = 3w^2 - \|\mathbf{v}\|^2$; with $\|\mathbf{v}\|^2 = 1 - w^2$ one gets $\operatorname{tr} R = 4w^2 - 1$, that is

$$c_0 = 4w^2.$$

Step 3 — the other three. From $R_{11} = w^2 + x^2 - y^2 - z^2$ and $\text{tr } R = 4w^2 - 1$:

$$c_1 = 1 + 2(w^2 + x^2 - y^2 - z^2) - (4w^2 - 1) = 2 - 2w^2 + 2x^2 - 2y^2 - 2z^2.$$

Substituting $1 - w^2 = x^2 + y^2 + z^2$ gives $2 - 2w^2 = 2(x^2 + y^2 + z^2)$, hence $c_1 = 4x^2$. Likewise $c_2 = 4y^2$ and $c_3 = 4z^2$.

Step 4 — sum and lower bound. $c_0 + c_1 + c_2 + c_3 = 4(w^2 + x^2 + y^2 + z^2) = 4$. Being four nonnegative numbers summing to 4, the maximum is at least 1; hence $s = \frac{1}{2}\sqrt{c_{m^*}} \geq \frac{1}{2}$ and dividing by $4s \geq 2$ does not degrade precision.

Step 5 — the off-diagonal combinations. From (28.2):

$$R_{32} - R_{23} = (2yz + 2wx) - (2yz - 2wx) = 4wx, \quad R_{13} - R_{31} = 4wy, \quad R_{21} - R_{12} = 4wz,$$

$$R_{12} + R_{21} = 4xy, \quad R_{13} + R_{31} = 4xz, \quad R_{23} + R_{32} = 4yz.$$

Step 6 — the four branches. Suppose $m^* = 0$. Then $|w| = s$ and, choosing the representative with $w = +s$ (the other choice gives $-q$), Step 5 supplies $x = (R_{32} - R_{23})/(4w)$, $y = (R_{13} - R_{31})/(4w)$, $z = (R_{21} - R_{12})/(4w)$: this is the first row of the scheme. If $m^* = 1$, then $|x| = s$ and, setting $x = +s$, Step 5 gives $w = (R_{32} - R_{23})/(4x)$, $y = (R_{12} + R_{21})/(4x)$, $z = (R_{13} + R_{31})/(4x)$. The cases $m^* = 2, 3$ are identical with the roles permuted.

Step 7 — uniqueness up to sign. This is Theorem 11.10(iv): $\Phi_q = \Phi_{q'}$ if and only if $q' = \pm q$.

Step 8 — the two degenerate cases are automatic. If $R = I$ then $\text{tr } R = 3$, $c_0 = 4$ and $c_1 = c_2 = c_3 = 0$: the branch chosen is $m^* = 0$ and one obtains $q = (1, 0, 0, 0)$. If instead the angle is π , then $\text{tr } R = -1$ and $c_0 = 0$: the branch $m^* = 0$ is *never* chosen, and the extracted component is one of the three vector ones. Neither case requires separate treatment: this is the merit of the four-branch scheme over the naive single-branch formula, which divides by w and loses precision when $w \rightarrow 0$ (Remark 28.4). $\square$

Corollary 28.5: Between quaternion and axis-angle, both ways. Let $q = (w, \mathbf{v})$ be unitary. Normalise the sign by setting $q \mapsto -q$ if $w < 0$, so that $w \geq 0$. Then:

- (i) $\theta = 2 \text{atan2}(\|\mathbf{v}\|, w) \in [0, \pi]$ is the angle of the rotation Φ_q ;
- (ii) if $\mathbf{v} \neq \mathbf{0}$, the axis is $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$, and $q = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \mathbf{u})$;
- (iii) if $\mathbf{v} = \mathbf{0}$, then $q = (1, \mathbf{0})$, $\Phi_q = I$, $\theta = 0$ and the axis is undetermined;
- (iv) if $w = 0$, then $\theta = \pi$, $\|\mathbf{v}\| = 1$ and $\mathbf{u} = \mathbf{v}$; in this case $q = (0, \mathbf{u})$ and $-q = (0, -\mathbf{u})$ represent the same rotation, so the axis is determined only up to sign.

In the opposite direction, given a unit vector $\mathbf{u}$ and $\theta \in [0, \pi]$, the quaternion is the one of Definition 28.3.

Proof. Step 1 — existence of the axis-angle form. By Theorem 11.10(iii) and by Corollary 28.3 (Step 0 and Step 5), there exist a unit vector $\mathbf{u}$ and an angle $\theta \in [0, \pi]$ with $\Phi_q = R(\mathbf{u}, \theta)$; by Theorem 11.10(iv) the only quaternions realising that rotation are $\pm(\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \mathbf{u})$. Since $\theta/2 \in [0, \pi/2]$ one has $\cos \frac{\theta}{2} \geq 0$: the representative with $w \geq 0$ is $q = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \mathbf{u})$, except for the case $\cos \frac{\theta}{2} = 0$ treated in (iv).

Step 2 — (i) and (ii). With $w = \cos \frac{\theta}{2}$ and $\mathbf{v} = \sin \frac{\theta}{2} \mathbf{u}$ one has $\|\mathbf{v}\| = |\sin \frac{\theta}{2}| = \sin \frac{\theta}{2} \geq 0$, since $\frac{\theta}{2} \in [0, \frac{\pi}{2}]$. Hence the point $(w, \|\mathbf{v}\|)$ of the plane lies in the first quadrant (boundary included) and

$$\text{atan2}(\|\mathbf{v}\|, w) = \text{atan2}\left(\sin \frac{\theta}{2}, \cos \frac{\theta}{2}\right) = \frac{\theta}{2},$$

because $\text{atan2}(\sin \alpha, \cos \alpha) = \alpha$ for $\alpha \in (-\pi, \pi]$. Multiplying by 2 gives (i). If $\mathbf{v} \neq \mathbf{0}$ then $\sin \frac{\theta}{2} > 0$ and $\mathbf{v}/\|\mathbf{v}\| = \mathbf{u}$: this is (ii).

Step 3 — (iii). If $\mathbf{v} = \mathbf{0}$ and $\|q\| = 1$ with $w \geq 0$, then $w = 1$ and $q = (1, \mathbf{0})$; $\Phi_q(\mathbf{x}) = 1 \cdot \mathbf{x} \cdot 1 = \mathbf{x}$, hence $\Phi_q = I$. Every unit vector $\mathbf{u}$ satisfies $R(\mathbf{u}, 0) = I$: the axis is not defined, in agreement with Corollary 28.3(ii).

Step 4 — (iv). If $w = 0$ then $\cos \frac{\theta}{2} = 0$ with $\frac{\theta}{2} \in [0, \frac{\pi}{2}]$, that is $\theta = \pi$; and $\|\mathbf{v}\| = 1$ by normalisation. Part (i) gives $\theta = 2 \operatorname{atan2}(1, 0) = 2 \cdot \frac{\pi}{2} = \pi$. ✓ Here the normalisation $w \geq 0$ no longer selects a unique representative, because $-q = (0, -\mathbf{v})$ still has zero real part: this is the same sign indeterminacy of the axis found in Corollary 28.3(iii), seen from the side of quaternions.

Step 5 — opposite direction. This is Definition 28.3 together with Theorem 11.10(iii). □

Example 28.4: The three representations of the same orientation. The rotation of Example 28.3 is taken up again: $\mathbf{u} = \frac{1}{3}(2, -1, 2)^\top$, $\theta = 40^\circ$, and the cycle axis-angle $\rightarrow$ quaternion $\rightarrow$ matrix $\rightarrow$ quaternion $\rightarrow$ axis-angle is traced out. The convention on digits declared in that example holds, here with eight decimals.

Step 1 — axis-angle $\rightarrow$ quaternion. By Definition 28.3, with $\cos 20^\circ = 0.93969262$ and $\sin 20^\circ = 0.34202014$:

$$q = (0.93969262, 0.22801343, -0.11400671, 0.22801343),$$

because $0.34202014 \times \frac{2}{3} = 0.22801343$ and $0.34202014 \times (-\frac{1}{3}) = -0.11400671$. *Constraint check:* the four squares equal 0.88302222, 0.05199012, 0.01299753, 0.05199012 and sum to 0.99999999: this is 1, and the deficit of 10^{-8} is the rounding of the four printed digits (in double precision the sum equals 1 up to 10^{-16}). ✓

Step 2 — quaternion $\rightarrow$ matrix. With (28.2):

$$R_{11} = 1 - 2(y^2 + z^2) = 1 - 2 \times 0.06498765 = 1 - 0.12997530 = 0.87002470,$$

$$R_{12} = 2xy - 2wz = -0.05199012 - 0.42852507 = -0.48051519,$$

$$R_{13} = 2xz + 2wy = 0.10398025 - 0.21426254 = -0.11028229,$$

and continuing one obtains the matrix of Example 28.3, which to seven decimals is $R_{11} = 0.8700247$, $R_{12} = -0.4805152$, $R_{13} = -0.1102823$. ✓ The two routes — Rodrigues and quaternion — are independent and give the same result: this is Step 7 of the proof of Theorem 28.3 checked on the numbers.

Step 3 — matrix $\rightarrow$ quaternion, with the four branches. With $\operatorname{tr} R = 2.53208889$, $2R_{11} = 2R_{33} = 1.74004938$ and $2R_{22} = 1.58407901$:

$$c_0 = 1 + 2.53208889 = 3.53208889, \quad c_1 = 1 + 1.74004938 - 2.53208889 = 0.20796049,$$

$$c_2 = 1 + 1.58407901 - 2.53208889 = 0.05199012, \quad c_3 = c_1 = 0.20796049.$$

Check: $3.53208889 + 0.20796049 + 0.05199012 + 0.20796049 = 3.99999999$, that is 4 up to rounding, as prescribed by Step 4 of the proof of Theorem 28.4. ✓ The maximum is c_0 , hence $m^* = 0$ and $s = \frac{1}{2}\sqrt{3.53208889} = \frac{1}{2} \times 1.87938524 = 0.93969262 = w$. ✓ With $4s = 3.75877048$:

$$x = \frac{R_{32} - R_{23}}{4s} = \frac{0.85705015}{3.75877048} = 0.22801343,$$

$$y = \frac{R_{13} - R_{31}}{4s} = \frac{-0.42852507}{3.75877048} = -0.11400671,$$

and $z = x = 0.22801343$: the quaternion of Step 1 is recovered. ✓

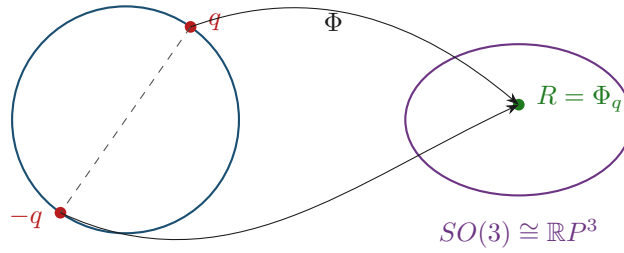
Step 4 — quaternion $\rightarrow$ axis-angle. $\|\mathbf{v}\| = \sqrt{x^2 + y^2 + z^2} = 0.34202014$, which is $\sin 20^\circ$. ✓ Hence $\theta = 2 \operatorname{atan2}(0.34202014, 0.93969262) = 2 \times 20^\circ = 40^\circ$ and $\mathbf{u} = \mathbf{v}/0.34202014 = (0.66666667, -0.33333333, 0.66666667)^\top = \frac{1}{3}(2, -1, 2)^\top$. ✓

Step 5 — independent check in numpy. The same cycle was run in double-precision arithmetic, comparing the matrix obtained from the closed formula (28.2) with the one obtained by computing column by column $q \mathbf{e}_\ell \bar{q}$ with the quaternion product: the maximum deviation is 1.1×10^{-16} . Repeating the four-branch extraction on 20 000 random rotations, the maximum deviation between R and $R(q_{\text{extracted}})$ is 1.3×10^{-15} . ✓

Remark 28.4: Two numerical traps, with the numbers. First trap: $\theta = 2 \arccos w$. The derivative of $\arccos$ diverges at $w = 1$; since $w = \cos \frac{\theta}{2} \approx 1 - \theta^2/8$ for small θ , the information about the angle is compressed near 1, where double-precision rounding ($\varepsilon \approx 2.2 \times 10^{-16}$) is amplified. The formula $\theta = 2 \operatorname{atan2}(\|\mathbf{v}\|, w)$ does not have this defect, because $\|\mathbf{v}\| = \sin \frac{\theta}{2}$ is of order θ . Measured: for $\theta = 10^{-7}$ the error of $2 \arccos w$ is 1.2×10^{-9} , that is over 1% relative; that of $2 \operatorname{atan2}(\|\mathbf{v}\|, w)$ is zero to machine precision. For $\theta = 10^{-9}$ the error of $2 \arccos w$ is 1.0×10^{-9} , that is 100%: the computed angle is zero.

Second trap: a single branch in the extraction. The naive formula $w = \frac{1}{2} \sqrt{1 + \operatorname{tr} R}$ followed by $x = (R_{32} - R_{23})/(4w)$ divides by a number that tends to zero as $\theta \rightarrow \pi$. Measured: for $\theta = 179.9999^\circ$ the single-branch reconstruction is wrong by 1.9×10^{-6} , while the four-branch scheme of Theorem 28.4 is wrong by 2.2×10^{-16} : ten orders of magnitude. Step 4 of that proof explains why — the branch chosen always has $4s^2 \geq 1$ — and this is why serious libraries implement all four branches.

28.6. The double covering and the choice of sign



S^3 : the unit quaternions

Figure 28.2: The double covering. The homomorphism $\Phi : S^3 \rightarrow SO(3)$, $q \mapsto \Phi_q$, is surjective and has kernel $\{\pm 1\}$ (Theorem 11.10(iv)): every rotation has exactly two quaternions, antipodal on the sphere. The quotient of S^3 by antipodal identification is real projective space $\mathbb{R}P^3$, and it is the rotation group. The two points drawn are diametrically opposite: on the circle of radius 1.5 centred at $(-2.6, 0)$ they correspond to the angles 55° and 235° , that is to the points $(-2.6 + 1.5 \cos 55^\circ, 1.5 \sin 55^\circ) = (-1.7396, 1.2287)$ and its reflection through the centre.

Theorem 11.10(iv) establishes that $\Phi : S^3 \rightarrow SO(3)$ is a surjective homomorphism with kernel $\{\pm 1\}$; Figure 28.2 gives its picture. Here the consequences are drawn out, and there are three: one topological, one algebraic and one operational.

Corollary 28.6: The kernel, the fibre and $\mathbb{R}P^3$. Let $\Phi : S^3 \rightarrow SO(3)$, $\Phi(q) = \Phi_q$. Then:

- (i) $\ker \Phi = \{+1, -1\}$, a group of two elements;
- (ii) for every $R \in SO(3)$ the fibre $\Phi^{-1}(R)$ has exactly two elements, antipodal: $\Phi^{-1}(R) = \{q, -q\}$;

- (iii) $SO(3) \cong S^3/\{\pm 1\} = \mathbb{R}P^3$ as groups;
- (iv) the parametrisation with $w \geq 0$ of Corollary 28.5 chooses a representative for each fibre, except on the equator $w = 0$ (rotations by π), where the two representatives both have $w = 0$ and remain indistinguishable.

Proof. Step 1 — (i). This is the first half of Theorem 11.10(iv): $\Phi_q = \text{id}$ if and only if $q = \pm 1$. The set $\{\pm 1\}$ is closed under product and inverse, hence is a subgroup of order 2; and it is normal, being the kernel of a homomorphism.

Step 2 — (ii). Let $R \in SO(3)$. By surjectivity there is q with $\Phi_q = R$. If also $\Phi_{q'} = R$, then $\Phi_{q'q^{-1}} = \Phi_{q'}\Phi_q^{-1} = \text{id}$ by the homomorphism property (Theorem 11.10(ii)), hence $q'q^{-1} \in \ker \Phi = \{\pm 1\}$ and $q' = \pm q$. Conversely $\Phi_{-q}(\mathbf{p}) = (-q)\mathbf{p}(-q) = q\mathbf{p}q = \Phi_q(\mathbf{p})$ because the two signs cancel. Hence the fibre is exactly $\{q, -q\}$, which has two distinct elements because $q \neq -q$ (since $\|q\| = 1 \neq 0$).

Step 3 — (iii). Φ is a surjective group homomorphism with kernel $\{\pm 1\}$: by the first isomorphism theorem, $S^3/\{\pm 1\} \cong SO(3)$. The quotient of S^n by the antipodal identification $q \sim -q$ is by definition real projective space $\mathbb{R}P^n$; with $n = 3$ this gives the claim.

Step 4 — (iv). The normalisation $w \geq 0$ discards, from each pair $\{q, -q\}$, the one with negative real part: the choice is unique as long as $w \neq 0$. If $w = 0$, both representatives have zero real part and the rule does not decide. By Corollary 28.5(iv) this happens exactly for the rotations by angle π , which form the equator $S^2 \subset S^3$: these are the same rotations for which Corollary 28.3(iii) leaves the axis undetermined in sign. The two indeterminacies are the same thing.

Step 5 — topological reading. That $SO(3)$ is $\mathbb{R}P^3$ and not S^3 is not a detail: $\mathbb{R}P^3$ is not simply connected (its fundamental group has two elements), while S^3 is. This is why there exists a non-contractible closed path of rotations — the “plate trick” of Remark 28.5 — and why there is no continuous, global choice of the sign of the quaternion: if there were, Φ would have a continuous section and $SO(3)$ would be homeomorphic to half of S^3 , contradicting the connectedness of the antipodal fibre. $\square$

Proposition 28.5: S^3 as a matrix group: $SU(2)$. Identify $q = w + xi + yj + zk$ with the pair of complex numbers $a = w + xi$, $b = y + zi$, and set

$$M(q) = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \in \mathbb{C}^{2 \times 2}.$$

Then $M(pq) = M(p)M(q)$, $\det M(q) = \|q\|^2$, and M is a group isomorphism between S^3 and

$$SU(2) = \{M \in \mathbb{C}^{2 \times 2} : M^*M = I, \det M = 1\}.$$

Proof. Step 1 — the writing $q = a + bj$. With $k = ij$ one has $q = w + xi + yj + zij = (w + xi) + (y + zi)j = a + bj$, where a, b lie in the subalgebra $\mathbb{C} = \{u + vi\}$. Moreover, by Hamilton’s rules, $ji = -ij$, hence for every $c = u + vi \in \mathbb{C}$ one has $jc = (u - vi)j = \bar{c}j$.

Step 2 — the product. Let $p = a + bj$ and $q = c + dj$. Then

$$pq = (a + bj)(c + dj) = ac + adj + bjc + bjdj = ac + adj + b\bar{c}j + b\bar{d}jj = (ac - b\bar{d}) + (ad + b\bar{c})j,$$

having used Step 1 twice and $j^2 = -1$.

Step 3 — multiplicativity of M . By Step 2, $M(pq) = \begin{pmatrix} ac - b\bar{d} & ad + b\bar{c} \\ -(\bar{a}d + b\bar{c}) & \bar{a}\bar{c} - b\bar{d} \end{pmatrix}$. On the other hand

$$M(p)M(q) = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \begin{pmatrix} c & d \\ -\bar{d} & \bar{c} \end{pmatrix} = \begin{pmatrix} ac - b\bar{d} & ad + b\bar{c} \\ -\bar{b}c - \bar{a}\bar{d} & -\bar{b}d + \bar{a}\bar{c} \end{pmatrix}.$$

Now $-\overline{(ad + bc)} = -\bar{a}\bar{d} - \bar{b}\bar{c}$ and $\overline{(ac - bd)} = \bar{a}\bar{c} - \bar{b}\bar{d}$: the two matrices coincide.

Step 4 — determinant. $\det M(q) = a\bar{a} - b(-\bar{b}) = |a|^2 + |b|^2 = w^2 + x^2 + y^2 + z^2 = \|q\|^2$. Hence $q \in S^3$ implies $\det M(q) = 1$, and then

$$M(q)^{-1} = \begin{pmatrix} \bar{a} & -b \\ \bar{b} & a \end{pmatrix} = \begin{pmatrix} \bar{a} & \overline{-b} \\ \bar{b} & \bar{a} \end{pmatrix} = M(q)^*,$$

that is $M(q) \in SU(2)$.

Step 5 — surjectivity onto $SU(2)$. Let $M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SU(2)$. From $\det M = 1$ it follows that $M^{-1} = \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$; from $M^*M = I$ it follows that $M^{-1} = M^* = \begin{pmatrix} \bar{\alpha} & \bar{\gamma} \\ \bar{\beta} & \bar{\delta} \end{pmatrix}$. Comparing: $\delta = \bar{\alpha}$ and $\gamma = -\bar{\beta}$. Hence $M = M(q)$ with $a = \alpha$, $b = \beta$, and $\det M = |\alpha|^2 + |\beta|^2 = 1$ says that the corresponding q is unitary.

Step 6 — injectivity. $M(q) = M(q')$ forces $a = a'$ and $b = b'$, that is $q = q'$. Together with Steps 3 and 5, $M : S^3 \rightarrow SU(2)$ is a group isomorphism. Composing it with Φ gives the classical double covering $SU(2) \rightarrow SO(3)$. $\square$

We now turn to the operational consequence, which is the reason why every orientation interpolation library begins with a sign check.

Proposition 28.6: The angle of the relative rotation and the choice of sign. Let $q_0, q_1 \in S^3$ and let $\Omega = \arccos \langle q_0, q_1 \rangle \in [0, \pi]$ be the angle between them in $\mathbb{R}^4$. Then:

- (i) the real part of $\bar{q}_0 q_1$ equals $\langle q_0, q_1 \rangle = \cos \Omega$;
- (ii) the angle of the relative rotation $\Phi_{q_0}^{-1} \Phi_{q_1}$, normalised in $[0, \pi]$, equals 2Ω if $\Omega \leq \frac{\pi}{2}$ and $2(\pi - \Omega)$ if $\Omega > \frac{\pi}{2}$;
- (iii) replacing q_1 with $-q_1$ replaces Ω with $\pi - \Omega$, without changing the rotation Φ_{q_1} ;
- (iv) consequently, the choice of sign that makes $\langle q_0, q_1 \rangle \geq 0$ is the one — and the only one — for which the great-circle arc of S^3 joining q_0 to q_1 has length $\Omega \leq \frac{\pi}{2}$ and describes the relative rotation of minimal angle $2\Omega \leq \pi$.

Proof. **Step 1 — (i).** With $q_0 = (w_0, \mathbf{v}_0)$ and $q_1 = (w_1, \mathbf{v}_1)$, Proposition 28.3 applied to $\bar{q}_0 = (w_0, -\mathbf{v}_0)$ gives as the real part of $\bar{q}_0 q_1$

$$w_0 w_1 - \langle -\mathbf{v}_0, \mathbf{v}_1 \rangle = w_0 w_1 + \langle \mathbf{v}_0, \mathbf{v}_1 \rangle = \langle q_0, q_1 \rangle,$$

where the last scalar product is the Euclidean one on $\mathbb{R}^4$. Since q_0, q_1 are unitary, $\langle q_0, q_1 \rangle = \cos \Omega$ by definition of the angle between two unit vectors.

Step 2 — the relative rotation. By Theorem 11.10(ii) the map Φ is a homomorphism, and by Proposition 28.4 $q_0^{-1} = \bar{q}_0$. Hence $\Phi_{q_0}^{-1} \Phi_{q_1} = \Phi_{\bar{q}_0 q_1}$. Set $p = \bar{q}_0 q_1$, unitary by Corollary 28.4.

Step 3 — (ii). By Corollary 28.5(i), the angle of Φ_p normalised in $[0, \pi]$ is obtained after making the real part nonnegative, and equals $2 \arccos |\operatorname{Re} p| = 2 \arccos |\cos \Omega|$. If $\Omega \leq \frac{\pi}{2}$, then $|\cos \Omega| = \cos \Omega$ and the angle is 2Ω ; if $\Omega > \frac{\pi}{2}$, then $|\cos \Omega| = -\cos \Omega = \cos(\pi - \Omega)$ with $\pi - \Omega \in [0, \frac{\pi}{2}]$, and the angle is $2(\pi - \Omega)$.

Step 4 — (iii). $\langle q_0, -q_1 \rangle = -\cos \Omega = \cos(\pi - \Omega)$, hence the angle between q_0 and $-q_1$ is $\pi - \Omega$. And $\Phi_{-q_1} = \Phi_{q_1}$ by Corollary 28.6(ii): the rotation does not change.

Step 5 — (iv). Of the two choices $\pm q_1$, exactly one has scalar product with q_0 nonnegative (the other has it nonpositive); for that one, $\Omega \leq \frac{\pi}{2}$ and Step 3 gives the angle $2\Omega \leq \pi$. On the unit sphere S^3 the geodesic distance between two points is the angle between them, hence the great-circle arc from q_0 to q_1 has length Ω : the choice of the nearer sign is also the one giving the shortest arc. If the two choices give $\Omega = \frac{\pi}{2}$ (that is $\langle q_0, q_1 \rangle = 0$) they are equivalent, and the relative rotation is by π : this is the only ambiguous configuration, the same as the equator of Corollary 28.6(iv).

Step 6 — operational reading. A planner that interpolates between q_0 and q_1 without the

sign check may traverse the long arc: the resulting rotation is the same at the endpoints, but the path is the one “on the wrong side”, with angle up to nearly 2π . On robots this shows up as a full turn of the last axis halfway through the trajectory. The fix is one line of code: `if dot(q0,q1) < 0: q1 = -q1`. $\square$

Example 28.5: Ten degrees or three hundred and fifty. Interpolation is performed between the initial orientation $q_0 = (1, 0, 0, 0)$ (identity) and the final orientation obtained by rotating by 350° about z .

Step 1 — the final quaternion. By Definition 28.3, with $\theta/2 = 175^\circ$: $\cos 175^\circ = -0.99619470$ and $\sin 175^\circ = 0.08715574$, hence

$$q_1 = (-0.99619470, 0, 0, 0.08715574).$$

Note the *negative* real part: this is the “far” sign.

Step 2 — the scalar product. $\langle q_0, q_1 \rangle = 1 \times (-0.99619470) = -0.99619470$, hence $\Omega = \arccos(-0.99619470) = 175^\circ$.

Step 3 — without the choice of sign. The great-circle arc from q_0 to q_1 has length 175° and describes a rotation of $2 \times 175^\circ = 350^\circ$: the robot makes almost a full turn.

Step 4 — with the choice of sign. Since $\langle q_0, q_1 \rangle < 0$, q_1 is replaced with $-q_1 = (0.99619470, 0, 0, -0.08715574)$, which represents the same rotation (Corollary 28.6(ii)). Now $\langle q_0, -q_1 \rangle = 0.99619470$ and $\Omega' = \arccos(0.99619470) = 5^\circ$: the rotation traversed is $2 \times 5^\circ = 10^\circ$.

Step 5 — independent verification. The rotation of 350° about $\mathbf{e}_3$ coincides with the one of 10° about $-\mathbf{e}_3$, because $350^\circ = 360^\circ - 10^\circ$: this is Step 5 of the proof of Corollary 28.3. Hence 10° is indeed the minimal path, and 350° its complement. $\checkmark$

Step 6 — numerical check. The computation performed in `numpy` gives $2 \arccos \langle q_0, q_1 \rangle = 350.0000^\circ$ without the choice of sign and $2 \arccos |\langle q_0, q_1 \rangle| = 10.0000^\circ$ with the choice, with a deviation from the expected value below 10^{-9} degrees. Moreover, over 5 000 random pairs of unit quaternions, the real part of $\bar{q}_0 q_1$ coincides with $\langle q_0, q_1 \rangle$ up to 0 units in the last place. $\checkmark$

Remark 28.5: The “plate trick”, spin and the Hopf fibration. The two-point fibre of Corollary 28.6 has a physical manifestation. Follow the quaternion continuously along a rotation of 2π about a fixed axis: $q(t) = (\cos \frac{t}{2}, \sin \frac{t}{2} \mathbf{u})$ for $t \in [0, 2\pi]$ gives $q(0) = (1, \mathbf{0})$ and $q(2\pi) = (-1, \mathbf{0})$. The rotation has returned to the identity, the quaternion has not: $t = 4\pi$ is needed. The phenomenon can be exhibited by hand, holding a glass on one’s palm and rotating it through two full turns without releasing the grip — the plate trick — and it is the geometric origin of spin $\frac{1}{2}$: the representations of $SU(2)$ that do not descend to $SO(3)$ are those of half-integer spin. In the plane nothing of the kind happens, because $S^1 \rightarrow SO(2)$ is an isomorphism and not a double covering (Theorem 11.7).

The fine structure of S^3 is described by the Hopf fibration (Theorem 11.11), which decomposes S^3 into disjoint circles parametrised by S^2 ; in the quaternionic reading of Remark 11.5 the projection $q \mapsto q i \bar{q}$ is precisely the axis of the rotation Φ_q . The fibres are thus the sets of rotations with the same axis: a circle, as it must be by Step 8 of the proof of Theorem 28.2.

Remark 28.6: Interpolation: what is proved here and what in Chapter 30. Spherical linear interpolation (SLERP)

$$\text{slerp}(q_0, q_1; t) = \frac{\sin((1-t)\Omega)}{\sin \Omega} q_0 + \frac{\sin(t\Omega)}{\sin \Omega} q_1, \quad \cos \Omega = \langle q_0, q_1 \rangle, \quad t \in [0, 1],$$

traces the great-circle arc of S^3 between q_0 and q_1 at constant angular speed; the precise

statement and proof are in Chapter 30 (Theorem 30.2), where SLERP is placed alongside splines and jerk-limited profiles. What belongs to this chapter, and what has been proved above, is the premise: without the choice of sign of Proposition 28.6 the formula is still correct as a formula, but describes the wrong path. This is why the reference implementation of SLERP — Shoemake's, from 1985, cited among the sources — begins with the comparison $\langle q_0, q_1 \rangle < 0$.

28.7. Euler angles, RPY and gimbal lock

Definition 28.4: Roll, pitch and yaw angles. In the **RPY** convention (roll–pitch–yaw, that is ZYX with fixed axes) the orientation is

$$R(\phi, \theta, \psi) = R_z(\psi) R_y(\theta) R_x(\phi),$$

with ϕ the **roll** (about x), θ the **pitch** (about y) and ψ the **yaw** (about z). The three matrices are those of Definition 28.1. The same convention is called A, B, C by KUKA and, with the order reversed, W, P, R by FANUC.

Proposition 28.7: The RPY matrix in full. Setting $c_\alpha = \cos \alpha$ and $s_\alpha = \sin \alpha$,

$$R(\phi, \theta, \psi) = \begin{pmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{pmatrix}.$$

Proof. Step 1 — the product $R_y(\theta)R_x(\phi)$. Multiplying row by column,

$$R_y(\theta)R_x(\phi) = \begin{pmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{pmatrix} = \begin{pmatrix} c_\theta & s_\theta s_\phi & s_\theta c_\phi \\ 0 & c_\phi & -s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{pmatrix}.$$

For example the first row: $(c_\theta, 0, s_\theta)$ against the three columns $(1, 0, 0)^\top$, $(0, c_\phi, s_\phi)^\top$, $(0, -s_\phi, c_\phi)^\top$ gives c_θ , $s_\theta s_\phi$, $s_\theta c_\phi$.

Step 2 — left multiplication by $R_z(\psi)$. The first row of the result is $c_\psi \cdot$ (first row of Step 1) $- s_\psi \cdot$ (second row), that is $(c_\psi c_\theta, c_\psi s_\theta s_\phi - s_\psi c_\phi, c_\psi s_\theta c_\phi + s_\psi s_\phi)$. The second row is $s_\psi \cdot$ (first) $+ c_\psi \cdot$ (second), that is $(s_\psi c_\theta, s_\psi s_\theta s_\phi + c_\psi c_\phi, s_\psi s_\theta c_\phi - c_\psi s_\phi)$. The third row of $R_z(\psi)$ is $(0, 0, 1)$ and reproduces the third row of Step 1.

Step 3 — independent check. The formula was compared in `numpy` with the product of the three matrices on 4000 random triples of angles in $[-\pi, \pi]$: maximum deviation 1.1×10^{-16} . ✓ A second check: the third row $(-s_\theta, c_\theta s_\phi, c_\theta c_\phi)$ has norm $s_\theta^2 + c_\theta^2(s_\phi^2 + c_\phi^2) = s_\theta^2 + c_\theta^2 = 1$, as it must be for an orthogonal matrix. ✓ □

To prove *gimbal lock* one more tool is needed: the relation between the velocities of the three angles and the angular velocity of the body. This is the content of the next lemma.

Lemma 28.4: The Jacobian of the RPY angles. Let ϕ, θ, ψ be differentiable functions of time and $R = R(\phi, \theta, \psi)$. Then $\dot{R}R^\top = [\omega]_\times$ with

$$\omega = J(\theta, \psi) \begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix}, \quad J(\theta, \psi) = \left[R_z(\psi)R_y(\theta)\mathbf{e}_1 \mid R_z(\psi)\mathbf{e}_2 \mid \mathbf{e}_3 \right] = \begin{pmatrix} c_\psi c_\theta & -s_\psi & 0 \\ s_\psi c_\theta & c_\psi & 0 \\ -s_\theta & 0 & 1 \end{pmatrix}.$$

The vector $\boldsymbol{\omega}$ is the (spatial) angular velocity of the body.

Proof. Step 1 — derivative of an elementary rotation. By Theorem 28.2(i) and by the identification $R_*(\alpha) = R(\mathbf{e}_*, \alpha)$ established in Section 28.3, one has $R_z(\alpha) = e^{\alpha[\mathbf{e}_3]_\times}$; differentiating the series term by term (legitimate by absolute and uniform convergence on compact sets) gives $\frac{d}{d\alpha}R_z(\alpha) = [\mathbf{e}_3]_\times e^{\alpha[\mathbf{e}_3]_\times} = [\mathbf{e}_3]_\times R_z(\alpha)$, and likewise for R_x with $\mathbf{e}_1$ and for R_y with $\mathbf{e}_2$. This can also be checked directly: differentiating the entries of $R_z(\alpha)$ gives $\begin{pmatrix} -s_\alpha & -c_\alpha & 0 \\ c_\alpha & -s_\alpha & 0 \\ 0 & 0 & 0 \end{pmatrix}$, which is $[\mathbf{e}_3]_\times R_z(\alpha)$.

Step 2 — derivative of the product. With $R = R_z(\psi)R_y(\theta)R_x(\phi)$, the Leibniz rule gives

$$\dot{R} = \dot{\psi}[\mathbf{e}_3]_\times R_z R_y R_x + \dot{\theta} R_z[\mathbf{e}_2]_\times R_y R_x + \dot{\phi} R_z R_y[\mathbf{e}_1]_\times R_x.$$

Step 3 — multiplication by $R^\top$. Since $R^\top = R_x^\top R_y^\top R_z^\top$, everything simplifies in the first term and $\dot{\psi}[\mathbf{e}_3]_\times$ remains; in the second $R_z[\mathbf{e}_2]_\times R_z^\top$ remains; in the third $R_z R_y[\mathbf{e}_1]_\times (R_z R_y)^\top$. Hence

$$\dot{R}R^\top = \dot{\psi}[\mathbf{e}_3]_\times + \dot{\theta} R_z[\mathbf{e}_2]_\times R_z^\top + \dot{\phi} (R_z R_y)[\mathbf{e}_1]_\times (R_z R_y)^\top.$$

Step 4 — the conjugation lemma. By Lemma 28.3, $R_z[\mathbf{e}_2]_\times R_z^\top = [R_z \mathbf{e}_2]_\times$ and $(R_z R_y)[\mathbf{e}_1]_\times (R_z R_y)^\top = [R_z R_y \mathbf{e}_1]_\times$. By linearity of the hat map (Proposition 28.2),

$$\dot{R}R^\top = [\dot{\phi} R_z R_y \mathbf{e}_1 + \dot{\theta} R_z \mathbf{e}_2 + \dot{\psi} \mathbf{e}_3]_\times,$$

which is the claim with $\boldsymbol{\omega}$ given by the statement.

Step 5 — the columns in full. $R_y(\theta)\mathbf{e}_1 = (c_\theta, 0, -s_\theta)^\top$ (first column of R_y), and applying $R_z(\psi)$ gives $(c_\psi c_\theta, s_\psi c_\theta, -s_\theta)^\top$. Then $R_z(\psi)\mathbf{e}_2 = (-s_\psi, c_\psi, 0)^\top$ (second column of R_z), and $\mathbf{e}_3 = (0, 0, 1)^\top$. These are the three columns of J .

Step 6 — numerical check. The three columns were recomputed by finite differences, evaluating $(R(\cdot + h\mathbf{e}_\ell) - R)R^\top/h$ with $h = 10^{-6}$ and extracting the associated vector, on 2000 random triples: maximum deviation 3.2×10^{-10} , of the order of the truncation error of finite differences. $\checkmark$

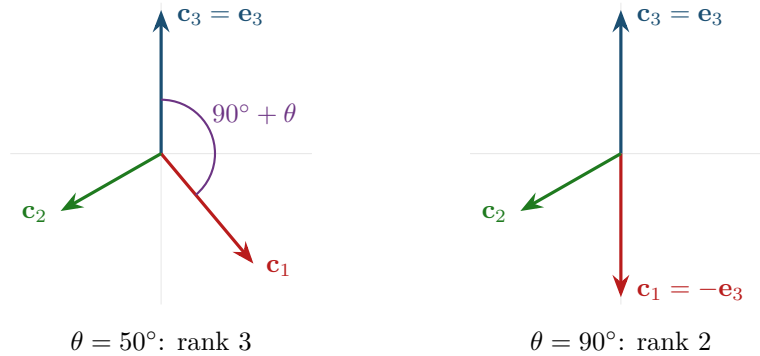


Figure 28.3: *Gimbal lock* as the collapse of two columns of the Jacobian. With $\psi = 0$ the columns of J are $\mathbf{c}_1 = (\cos \theta, 0, -\sin \theta)^\top$, $\mathbf{c}_2 = (0, 1, 0)^\top$ and $\mathbf{c}_3 = (0, 0, 1)^\top$: the first and the third lie in the plane xz (here the plane of the page, with z upward), the second comes out of the page and is drawn slanted. The angle between $\mathbf{c}_1$ and $\mathbf{c}_3$ equals $90^\circ + \theta$: at $\theta = 50^\circ$ it is 140° and the three columns are independent; at $\theta = 90^\circ$ it is 180° , that is $\mathbf{c}_1 = -\mathbf{c}_3$, and the rank drops to 2. The coordinates drawn at $\theta = 50^\circ$ are $2 \cos 50^\circ = 1.2856$ and $-2 \sin 50^\circ = -1.5321$.

Theorem 28.5: Gimbal lock: rank drop at $\theta = \pm\pi/2$. Let $J(\theta, \psi)$ be the Jacobian of Lemma 28.4. Then:

- (i) $\det J(\theta, \psi) = \cos \theta$, for every ψ ;
(ii) J is invertible if and only if $\theta \neq \pm \frac{\pi}{2}$ (modulo 2π);
(iii) for $\theta = \pm \frac{\pi}{2}$ the rank of J is exactly 2, and the kernel is generated by $(1, 0, 1)^\top$ if $\theta = +\frac{\pi}{2}$, by $(1, 0, -1)^\top$ if $\theta = -\frac{\pi}{2}$;
(iv) consistently, for $\theta = +\frac{\pi}{2}$ one has $R(\phi, \frac{\pi}{2}, \psi) = R_z(\psi - \phi)R_y(\frac{\pi}{2})$, which depends on the two angles only through $\psi - \phi$; and for $\theta = -\frac{\pi}{2}$ one has $R(\phi, -\frac{\pi}{2}, \psi) = R_z(\psi + \phi)R_y(-\frac{\pi}{2})$, which depends only on $\psi + \phi$. In both cases one degree of freedom is lost.

Proof. Step 1 — the determinant. One expands along the third column of $J = \begin{pmatrix} c_\psi c_\theta & -s_\psi & 0 \\ s_\psi c_\theta & c_\psi & 0 \\ -s_\theta & 0 & 1 \end{pmatrix}$, which has two zeros:

$$\det J = 1 \cdot \det \begin{pmatrix} c_\psi c_\theta & -s_\psi \\ s_\psi c_\theta & c_\psi \end{pmatrix} = c_\psi^2 c_\theta + s_\psi^2 c_\theta = c_\theta (c_\psi^2 + s_\psi^2) = \cos \theta.$$

Step 2 — (ii). A square matrix is invertible if and only if the determinant is nonzero: here $\cos \theta \neq 0$, that is $\theta \neq \pm \frac{\pi}{2}$ modulo 2π .

Step 3 — rank exactly 2 at $\theta = +\frac{\pi}{2}$ (Figure 28.3). Then $c_\theta = 0$ and $s_\theta = 1$, hence

$$\mathbf{c}_1 = R_z(\psi)R_y(\frac{\pi}{2})\mathbf{e}_1 = (0, 0, -1)^\top = -\mathbf{e}_3 = -\mathbf{c}_3, \quad \mathbf{c}_2 = (-s_\psi, c_\psi, 0)^\top, \quad \mathbf{c}_3 = (0, 0, 1)^\top.$$

The columns $\mathbf{c}_1$ and $\mathbf{c}_3$ are proportional, hence the rank is at most 2; and it is at least 2 because $\mathbf{c}_2$ and $\mathbf{c}_3$ are independent: $\mathbf{c}_2$ has zero third component and first two components $(-s_\psi, c_\psi) \neq (0, 0)$ (since $s_\psi^2 + c_\psi^2 = 1$), while $\mathbf{c}_3 = \mathbf{e}_3$. Hence the rank is 2.

Step 4 — the kernel at $\theta = +\frac{\pi}{2}$. By the rank theorem the kernel has dimension $3 - 2 = 1$. Now

$$J \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \mathbf{c}_1 + \mathbf{c}_3 = -\mathbf{e}_3 + \mathbf{e}_3 = \mathbf{0},$$

and $(1, 0, 1)^\top \neq \mathbf{0}$: hence the kernel is generated by $(1, 0, 1)^\top$. The meaning is that the combination $\dot{\phi} = \dot{\psi}$, $\dot{\theta} = 0$ produces zero angular velocity: roll and yaw cancel each other out.

Step 5 — the case $\theta = -\frac{\pi}{2}$. Now $s_\theta = -1$ and $\mathbf{c}_1 = (0, 0, 1)^\top = \mathbf{c}_3$; with the same argument the rank is 2 and $J(1, 0, -1)^\top = \mathbf{c}_1 - \mathbf{c}_3 = \mathbf{0}$: the kernel is generated by $(1, 0, -1)^\top$.

Step 6 — (iv), closed form at $\theta = +\frac{\pi}{2}$. By Corollary 28.2 with $A = R_y(\frac{\pi}{2})$ and $R_x(\phi) = R(\mathbf{e}_1, \phi)$,

$$R_y(\frac{\pi}{2})R_x(\phi)R_y(\frac{\pi}{2})^\top = R(R_y(\frac{\pi}{2})\mathbf{e}_1, \phi).$$

But $R_y(\frac{\pi}{2})\mathbf{e}_1 = (\cos \frac{\pi}{2}, 0, -\sin \frac{\pi}{2})^\top = (0, 0, -1)^\top = -\mathbf{e}_3$. Moreover $R(-\mathbf{e}_3, \phi) = R(\mathbf{e}_3, -\phi) = R_z(-\phi)$: indeed, setting $K = [\mathbf{e}_3]_\times$, one has $[-\mathbf{e}_3]_\times = -K$ and $(-K)^2 = K^2$, hence

$$R(-\mathbf{e}_3, \phi) = I + \sin \phi (-K) + (1 - \cos \phi) K^2 = I + \sin(-\phi) K + (1 - \cos(-\phi)) K^2 = R(\mathbf{e}_3, -\phi),$$

and $R(\mathbf{e}_3, \alpha) = R_z(\alpha)$ by the identification established in Section 28.3. Hence $R_y(\frac{\pi}{2})R_x(\phi) = R_z(-\phi)R_y(\frac{\pi}{2})$ and

$$R(\phi, \frac{\pi}{2}, \psi) = R_z(\psi)R_y(\frac{\pi}{2})R_x(\phi) = R_z(\psi)R_z(-\phi)R_y(\frac{\pi}{2}) = R_z(\psi - \phi)R_y(\frac{\pi}{2}),$$

using Proposition 28.1. The right-hand side depends on ϕ and ψ only through $\psi - \phi$, hence moving ϕ and ψ by the same amount leaves the orientation unchanged: this is the “integrated” translation of the kernel of Step 4.

Step 7 — the case $\theta = -\frac{\pi}{2}$. $R_y(-\frac{\pi}{2})\mathbf{e}_1 = (\cos(-\frac{\pi}{2}), 0, -\sin(-\frac{\pi}{2}))^\top = (0, 0, 1)^\top = \mathbf{e}_3$, hence $R_y(-\frac{\pi}{2})R_x(\phi) = R_z(\phi)R_y(-\frac{\pi}{2})$ and $R(\phi, -\frac{\pi}{2}, \psi) = R_z(\psi + \phi)R_y(-\frac{\pi}{2})$.

Step 8 — independent numerical checks. (a) $\det J = \cos \theta$ on 4000 random triples: maximum deviation 3.3×10^{-16} . ✓ (b) The numerical rank of J at $\theta = \pm \frac{\pi}{2}$, computed with `numpy.linalg.matrix_rank` for three values of ψ each, equals 2 in all six cases, with $\det J \approx 6.1 \times 10^{-17}$. ✓ (c) Identity (iv) was verified on 3000 random triples comparing $R(\phi + t, \pm \frac{\pi}{2}, \psi \pm t)$ with $R(\phi, \pm \frac{\pi}{2}, \psi)$: maximum deviation 1.0×10^{-15} . ✓ □

Example 28.6: The velocities that blow up. Gimbal lock is not a problem only *at* the singular point: it is a problem throughout a whole neighbourhood, because J^{-1} has a norm that diverges. Let the body be asked for a constant angular velocity $\boldsymbol{\omega} = (1, 0, 0)^\top$ rad/s, with $\psi = 0$, and solve $J\boldsymbol{\xi} = \boldsymbol{\omega}$ for $\boldsymbol{\xi} = (\dot{\phi}, \dot{\theta}, \dot{\psi})^\top$.

Step 1 — closed-form solution. With $\psi = 0$, $J = \begin{pmatrix} c_\theta & 0 & 0 \\ 0 & 1 & 0 \\ -s_\theta & 0 & 1 \end{pmatrix}$, hence $J\boldsymbol{\xi} = (\dot{\phi}c_\theta, \dot{\theta}, -\dot{\phi}s_\theta + \dot{\psi})^\top$. Setting this equal to $(1, 0, 0)^\top$ gives

$$\dot{\phi} = \frac{1}{\cos \theta} = \sec \theta, \quad \dot{\theta} = 0, \quad \dot{\psi} = \dot{\phi} \sin \theta = \tan \theta.$$

Step 2 — the values.

θ	$\det J = \cos \theta$	$\dot{\phi} = \sec \theta$	$\dot{\psi} = \tan \theta$
0°	1.000000	1.0000	0.0000
45°	0.707107	1.4142	1.0000
80°	0.173648	5.7588	5.6713
89°	0.017452	57.2987	57.2900
89.9°	0.001745	572.9581	572.9572
89.99°	0.000175	5729.5780	5729.5779

Step 3 — reading. At a tenth of a degree from the singularity, to rotate the body at 1 rad/s (about 57°/s) the joints would have to move at 573 rad/s, that is over 5000 revolutions per minute: no actuator does that, and the controller either saturates or stops. This is the **wrist singularity** of six-axis robots, which occurs when axis J5 passes through 0° and axes J4 and J6 become parallel.

Step 4 — independent check. The same values were obtained in `numpy` by solving the linear system $J\boldsymbol{\xi} = \boldsymbol{\omega}$ with `numpy.linalg.solve`, without using the closed form of Step 1: they agree to four decimals. ✓ A consistency check: for $\boldsymbol{\omega} = (0, 0, 1)^\top$ (rotation about z) the solution is $\boldsymbol{\xi} = (0, 0, 1)^\top$ for *every* θ , and indeed yaw is always available: the singularity does not prevent every motion, only those in the lost direction. ✓

Corollary 28.7: Extracting the RPY angles from a matrix. Let $R \in SO(3)$ with $R_{31} \neq \pm 1$. Then there is a unique triple (ϕ, θ, ψ) with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\phi, \psi \in (-\pi, \pi]$ such that $R = R(\phi, \theta, \psi)$, and it is

$$\theta = \text{atan2}\left(-R_{31}, \sqrt{R_{11}^2 + R_{21}^2}\right), \quad \psi = \text{atan2}(R_{21}, R_{11}), \quad \phi = \text{atan2}(R_{32}, R_{33}).$$

If instead $R_{31} = \mp 1$, then $\theta = \pm \frac{\pi}{2}$ and only $\psi \mp \phi$ is determined; a conventional choice is $\phi = 0$.

Proof. Step 1 — the pitch. From Proposition 28.7, $R_{31} = -s_\theta$, $R_{11} = c_\psi c_\theta$, $R_{21} = s_\psi c_\theta$. Hence $R_{11}^2 + R_{21}^2 = c_\theta^2$ and $\sqrt{R_{11}^2 + R_{21}^2} = |c_\theta| = c_\theta$ for $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Therefore $\text{atan2}(-R_{31}, \sqrt{R_{11}^2 + R_{21}^2}) = \text{atan2}(s_\theta, c_\theta) = \theta$.

Step 2 — yaw and roll. Since $c_\theta > 0$, the pair $(R_{11}, R_{21}) = c_\theta(c_\psi, s_\psi)$ is a positive multiple of (c_ψ, s_ψ) , and atan2 is invariant under multiplying both arguments by a positive scalar: hence

$\text{atan2}(R_{21}, R_{11}) = \psi$. Likewise $(R_{33}, R_{32}) = c_\theta(c_\phi, s_\phi)$ gives $\text{atan2}(R_{32}, R_{33}) = \phi$.

Step 3 — uniqueness. The three formulas determine the triple; and every triple with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ that produces R must satisfy them, by Steps 1 and 2.

Step 4 — the degenerate case. If $R_{31} = -1$ then $s_\theta = 1$, that is $\theta = \frac{\pi}{2}$ (in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$), and $c_\theta = 0$: the first column becomes $(0, 0, -1)^\top$ and the formulas of Steps 1–2 give $\text{atan2}(0, 0)$, undefined. By Theorem 28.5(iv) the matrix depends only on $\psi - \phi$, which is read off from the identity $R = R_z(\psi - \phi)R_y(\frac{\pi}{2})$. Carrying out the product,

$$R_z(\gamma)R_y(\frac{\pi}{2}) = \begin{pmatrix} 0 & -s_\gamma & c_\gamma \\ 0 & c_\gamma & s_\gamma \\ -1 & 0 & 0 \end{pmatrix}, \quad \gamma = \psi - \phi,$$

hence $\psi - \phi = \text{atan2}(-R_{12}, R_{13})$. For $R_{31} = +1$ instead one has $\theta = -\frac{\pi}{2}$ and

$$R_z(\delta)R_y(-\frac{\pi}{2}) = \begin{pmatrix} 0 & -s_\delta & -c_\delta \\ 0 & c_\delta & -s_\delta \\ 1 & 0 & 0 \end{pmatrix}, \quad \delta = \psi + \phi,$$

whence $\psi + \phi = \text{atan2}(-R_{12}, -R_{13})$. Fixing $\phi = 0$ is a convention, not a result: the lost information cannot be recovered.

Step 5 — numerical check. The formulas of Steps 1–2 were applied to 5000 matrices $R(\phi, \theta, \psi)$ with random angles and $\theta \in (-\frac{\pi}{2} + 0.05, \frac{\pi}{2} - 0.05)$, then reconstructing R from the extracted triple: maximum deviation 3.1×10^{-16} . $\checkmark$ $\square$

Proposition 28.8: No three-parameter chart covers all of $SO(3)$. No open set $U \subseteq \mathbb{R}^3$ is homeomorphic to $SO(3)$. In particular there is no three-parameter parametrisation, continuous with continuous inverse, that covers the entire rotation group.

Proof. **Step 1 — $SO(3)$ is compact.** Consider $SO(3)$ as a subset of $\mathbb{R}^{3 \times 3} \cong \mathbb{R}^9$. It is *closed*: the map $F(M) = (M^\top M - I, \det M - 1)$ is continuous (the entries are polynomials in the entries of M) and $SO(3) = F^{-1}(\{0\})$ is the preimage of a closed set. It is *bounded*: if $M \in SO(3)$, each column is a unit vector, hence $\|M\|_F^2 = \sum_j \|M\mathbf{e}_j\|^2 = 3$. By the Heine–Borel theorem a closed and bounded subset of $\mathbb{R}^9$ is compact.

Step 2 — the absurd hypothesis. Suppose there is a nonempty open set $U \subseteq \mathbb{R}^3$ and a homeomorphism $h : U \rightarrow SO(3)$. The continuous image of a compact set is compact and a homeomorphism has continuous inverse, hence $U = h^{-1}(SO(3))$ is compact.

Step 3 — contradiction. A compact subset of $\mathbb{R}^3$ is closed and bounded (again Heine–Borel). Hence U is at once open, closed and nonempty in $\mathbb{R}^3$; but $\mathbb{R}^3$ is connected, so $U = \mathbb{R}^3$. Then $\mathbb{R}^3$ would be bounded, which is false (it contains the points $(n, 0, 0)$ for every n). The hypothesis is absurd.

Step 4 — reading. Gimbal lock is not a defect of the RPY convention: it is the local manifestation of a global obstruction. Changing convention $(ZYZ, XYZ, ZXZ, \dots)$ moves the singularity, it does not remove it — this can also be seen directly, because the computation of Theorem 28.5 always produces a determinant that is a sine or cosine of the second angle, and this vanishes somewhere. $\square$

Remark 28.7: What remains imported. Proposition 28.8 proves that no homeomorphic chart covers $SO(3)$, and this suffices to explain why every triple of angles has singular points. What remains *imported*, that is assumed and not proved here, is Stuelpnagel’s stronger statement: every global parametrisation of $SO(3)$ with three parameters, even a non-injective one, has points where the differential is singular, and the smallest dimension

in which a non-singular global parametrisation exists is four (J. Stuelpnagel, *SIAM Review* 6 (1964), pp. 422–430, §§2–3). What is missing to prove it here: covering-space theory applied to $\mathbb{R}P^3$, that is the fact that the universal covering of $SO(3)$ is S^3 and that $\pi_1(SO(3))$ has two elements. The tools are those of Chapter 12 (lifting of paths and homotopies), but applied in dimension three rather than one; the treatise does not develop them at that level. The fact is nonetheless consistent with what is proved here: Corollary 28.6 exhibits $S^3 \rightarrow SO(3)$ as a two-sheeted covering, and Definition 28.3 exhibits the non-singular global four-parameter parametrisation whose minimality Stuelpnagel asserts.

28.8. Industrial conventions and implementation

Remark 28.8: From ZYX angles to the quaternion. In the ZYX convention — the angles A (about z), B (about y), C (about x) of KUKA controllers — the orientation is $R = R_z(A)R_y(B)R_x(C)$, that is Definition 28.4 with $\psi = A$, $\theta = B$, $\phi = C$. The corresponding quaternion is the product $q = q_z(A)q_y(B)q_x(C)$, with $q_z(A) = (\cos \frac{A}{2}, 0, 0, \sin \frac{A}{2})$ and similarly for the others. Carrying out the product with Proposition 28.3 and setting $c_A = \cos \frac{A}{2}$, $s_A = \sin \frac{A}{2}$ and so on:

$$\begin{aligned} w &= c_A c_B c_C + s_A s_B s_C, & x &= c_A c_B s_C - s_A s_B c_C, \\ y &= c_A s_B c_C + s_A c_B s_C, & z &= s_A c_B c_C - c_A s_B s_C. \end{aligned}$$

The opposite direction goes through Theorem 28.4 and Corollary 28.7: one builds R from the quaternion and then extracts the angles from it, remembering that the triple is undefined when $B = \pm 90^\circ$. Check: the two routes (quaternion product and closed form) were compared in `numpy` on 5 000 random angle triples, and $R(q)$ was compared with $R_z(A)R_y(B)R_x(C)$: maximum deviation 1.2×10^{-15} in both comparisons. ✓

Example 28.7: A KUKA target: $A = 30^\circ$, $B = 60^\circ$, $C = 45^\circ$. Convert the triple $(A, B, C) = (30^\circ, 60^\circ, 45^\circ)$ into the other two representations.

Step 1 — the matrix, in exact form. From Proposition 28.7 with $\psi = 30^\circ$, $\theta = 60^\circ$, $\phi = 45^\circ$, and using $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 30^\circ = \frac{1}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 45^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2}$:

$$R = \begin{pmatrix} \frac{\sqrt{3}}{4} & \frac{\sqrt{2}}{8} & \frac{5\sqrt{2}}{8} \\ \frac{1}{4} & \frac{3\sqrt{6}}{8} & -\frac{\sqrt{6}}{8} \\ -\frac{\sqrt{3}}{2} & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}}{4} \end{pmatrix} = \begin{pmatrix} 0.4330127 & 0.1767767 & 0.8838835 \\ 0.2500000 & 0.9185587 & -0.3061862 \\ -0.8660254 & 0.3535534 & 0.3535534 \end{pmatrix}.$$

For instance $R_{12} = c_\psi s_\theta s_\phi - s_\psi c_\phi = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} - \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{8} - \frac{2\sqrt{2}}{8} = \frac{\sqrt{2}}{8}$, and $R_{13} = \frac{3\sqrt{2}}{8} + \frac{2\sqrt{2}}{8} = \frac{5\sqrt{2}}{8}$.

Step 2 — the quaternion. With $c_A = \cos 15^\circ = 0.96592583$, $s_A = \sin 15^\circ = 0.25881905$, $c_B = \cos 30^\circ = 0.86602540$, $s_B = \sin 30^\circ = 0.5$, $c_C = \cos 22.5^\circ = 0.92387953$, $s_C = \sin 22.5^\circ = 0.38268343$, the formulas of Remark 28.8 give

$$q = (0.82236317, 0.20056212, 0.53197570, 0.02226003).$$

Constraint check: the sum of the squares equals 1 to within 10^{-16} . ✓

Step 3 — axis and angle. $\text{tr } R = \frac{\sqrt{3}}{4} + \frac{3\sqrt{6}}{8} + \frac{\sqrt{2}}{4} = 1.70512475$, hence $\cos \theta = (1.70512475 - 1)/2 = 0.35256237$ and $\theta = 69.355878^\circ$. By Corollary 28.5, $\theta = 2 \text{atan2}(\|\mathbf{v}\|, w)$ with $\|\mathbf{v}\| = 0.56896293$: one obtains 69.355878° again. ✓ The axis is $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\| = (0.35250473, 0.93499184, 0.03912386)^\top$, of norm 1 to within 10^{-12} ; the same vector is

obtained from Corollary 28.3(i) with $2 \sin \theta = 1.87157663$: $\mathbf{u} = (R_{32} - R_{23}, R_{13} - R_{31}, R_{21} - R_{12})^\top / 1.87157663$. ✓

Step 4 — reading. Three numbers readable on the *teach pendant* (30° , 60° , 45°) correspond to a single rotation of nearly 70° about an axis that is none of the coordinate axes. This is why reading Euler angles gives no geometric intuition: none of the three numbers is the rotation angle, and none indicates the axis.

A compact implementation covering all the conversions of the chapter:

Listing 28.1: Conversions between matrix, quaternion and axis-angle

```
import numpy as np

def qmul(a, b):
    w1, x1, y1, z1 = a
    w2, x2, y2, z2 = b
    return np.array([w1*w2 - x1*x2 - y1*y2 - z1*z2,
                    w1*x2 + x1*w2 + y1*z2 - z1*y2,
                    w1*y2 - x1*z2 + y1*w2 + z1*x2,
                    w1*z2 + x1*y2 - y1*x2 + z1*w2])

def q_da_asse_angolo(u, theta):
    u = np.asarray(u, float); u = u/np.linalg.norm(u)
    return np.concatenate([[np.cos(theta/2)], np.sin(theta/2)*u])

def asse_angolo_da_q(q):
    if q[0] < 0: q = -q # sign normalisation
    nv = np.linalg.norm(q[1:])
    theta = 2*np.arctan2(nv, q[0]) # NOT 2*arccos(q[0])
    u = q[1:]/nv if nv > 1e-12 else np.array([0., 0., 1.])
    return u, theta

def R_da_q(q):
    w, x, y, z = q
    return np.array([[1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
                    [2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
                    [2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]])

def q_da_R(R): # four branches (Shepperd)
    t = np.trace(R)
    c = np.array([1+t, 1+2*R[0,0]-t, 1+2*R[1,1]-t, 1+2*R[2,2]-t])
    m = int(np.argmax(c)); s = np.sqrt(c[m])/2; f = 4*s
    if m == 0: return np.array([s, (R[2,1]-R[1,2])/f, (R[0,2]-R[2,0])/f,
                                (R[1,0]-R[0,1])/f])
    if m == 1: return np.array([(R[2,1]-R[1,2])/f, s, (R[0,1]+R[1,0])/f,
                                (R[0,2]+R[2,0])/f])
    if m == 2: return np.array([(R[0,2]-R[2,0])/f, (R[0,1]+R[1,0])/f, s,
                                (R[1,2]+R[2,1])/f])
    return np.array([(R[1,0]-R[0,1])/f, (R[0,2]+R[2,0])/f, (R[1,2]+R[2,1])/f, s])

def segno_vicino(q0, q1): # to use BEFORE
    interpolating
    return -q1 if np.dot(q0, q1) < 0.0 else q1

qz = q_da_asse_angolo([0, 0, 1], np.pi/2)
qx = q_da_asse_angolo([1, 0, 0], np.pi/2)
print(np.round(qmul(qz, qx), 6)) # -> [0.5 0.5 0.5 0.5] (120
```

```
degrees about (1,1,1))
print(np.round(qmul(qx, qz), 6))      # -> [0.5 0.5 -0.5 0.5] (120
degrees about (1,-1,1))
```

Best practice. Conventions by brand, and what to check. Every manufacturer exposes orientation in a different way, but all of them describe the same element of $SO(3)$: **KUKA** uses the angles A, B, C (ZYX : A about z , B about y , C about x); **FANUC** uses W, P, R (XYZ); **ABB** stores the unit quaternion directly in the `rot` field of the `robtarg` of the RAPID language. Before exchanging poses between two systems, three things must be checked, in this order: (1) the order of the axes (Definition 28.4 is ZYX : changing it changes the matrix, by Theorem 28.1); (2) whether the axes are fixed or attached to the body (a passive reading requires $R^\top$); (3) the position of the real component of the quaternion, which some libraries place last (x, y, z, w) rather than first. An error on any one of the three produces a plausible and wrong pose. The singularity of Theorem 28.5 shows up on the robot as a **wrist singularity**, when the J5 axis crosses zero and J4 and J6 become parallel: controllers handle it with dedicated functions or with a damped least-squares inversion, that is by replacing J^{-1} with $J^\top(JJ^\top + \lambda^2 I)^{-1}$, which stays bounded even where $\det J = 0$.

28.9. Assigned exercises

The answers are in the Teacher's key (Section 28.12).

Exercise 28.1: A half-turn rotation. *Aim:* to handle the degenerate case $\theta = \pi$ in the two conversion directions. *Estimated time:* 30 minutes. *Type:* symbolic with numerical check.

Let $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1, 0)^\top$ and $\theta = \pi$.

- Compute $K = [\mathbf{u}]_\times$, K^2 and $R = R(\mathbf{u}, \pi)$ with Theorem 28.2, and check that R has integer entries.
- Check $R^2 = I$ and $\text{tr } R = -1$, and explain why both facts are forced.
- Compute the four numbers $c_0, \dots, c_3$ of Theorem 28.4, show that $c_0 = 0$ and deduce which branch is chosen. What happens if there are two equal maxima?
- Extract the quaternion and check $R(q) = R$.
- Show that the naive formula $w = \frac{1}{2}\sqrt{1 + \text{tr } R}$ followed by division by $4w$ fails here, and say exactly where.

Exercise 28.2: Gimbal lock in the ZYZ convention. *Aim:* to redo the gimbal-lock proof for a different convention and to observe that the singularity moves but does not disappear. *Estimated time:* 40 minutes. *Type:* proof-based.

Consider the ZYZ convention: $R(\alpha, \beta, \gamma) = R_z(\alpha)R_y(\beta)R_z(\gamma)$.

- Repeat Lemma 28.4 for this convention and write the three columns of the Jacobian J in the order $(\dot{\alpha}, \dot{\beta}, \dot{\gamma})$.
- Compute $\det J$ and determine the singular configurations.
- For each singular configuration, show that the rank of J is exactly 2 and compute the kernel.
- Deduce, with Corollary 28.2, the closed form of R at the singular configurations and say on which combination of α and γ it depends.
- Compare with Theorem 28.5: is the singularity in the same place? Is it of the same type? What does Proposition 28.8 say about this?

Exercise 28.3: Interpolating without taking the long way round. *Aim:* to measure the effect of the choice of sign and the cost of being close to a singularity. *Estimated time:* 45 minutes. *Type:* numerical/computational.

- Let q_0 be the quaternion of the 60° rotation about z and q_1 that of the 300° rotation about z . Compute $\langle q_0, q_1 \rangle$ and the angle of the relative rotation with and without the choice of sign of Proposition 28.6.
- Sample $t \in \{0, \frac{1}{10}, \dots, 1\}$ along the great-circle arc in the two cases, convert each sample into a matrix and plot the cumulative angle: check that in the correct case it grows from 0° to 120° and in the other from 0° to 240° .
- Generate 10 000 random pairs of unit quaternions and numerically check point (i) of Proposition 28.6.
- For the RPY convention, tabulate $\|J^{-1}\|_2$ for $\theta = 80^\circ, 89^\circ, 89.9^\circ, 89.99^\circ$ with $\psi = 0$, and compare with $1/|\cos \theta|$.
- Repeat (d) with the damped inversion $J^\top(JJ^\top + \lambda^2 I)^{-1}$ for $\lambda = 10^{-2}$ and comment on what is gained and what is lost.

28.10. Worked exercises

Exercise 28.4: Three quarter-turns that make one. *Type:* symbolic/proof-based. Compute $R = R_z(90^\circ)R_y(90^\circ)R_x(90^\circ)$ in two independent ways — with matrices and with quaternions — and determine its axis and angle.

Solution. Strategy. One uses Proposition 28.7 with $\psi = \theta = \phi = 90^\circ$ for the matrix route, and Remark 28.8 with $A = B = C = 90^\circ$ for the quaternion route; then the axis is extracted with Corollary 28.3. The comparison between the two routes is the required independent check.

Step 1 — the matrix route. With $c_\psi = c_\theta = c_\phi = 0$ and $s_\psi = s_\theta = s_\phi = 1$, the matrix of Proposition 28.7 becomes

$$R = \begin{pmatrix} 0 \cdot 0 & 0 \cdot 1 \cdot 1 - 1 \cdot 0 & 0 \cdot 1 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 0 & 1 \cdot 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 \cdot 0 - 0 \cdot 1 \\ -1 & 0 \cdot 1 & 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Step 2 — recognition. The resulting matrix is exactly $R_y(90^\circ)$ of Definition 28.1: composing three quarter-turns about z, y, x is equivalent to a single quarter-turn about y .

Step 3 — axis and angle from the matrix. $\text{tr } R = 0 + 1 + 0 = 1$, hence $\cos \theta = (1 - 1)/2 = 0$ and $\theta = 90^\circ$, with $\sin \theta = 1$ and $2 \sin \theta = 2$. Then $R_{32} - R_{23} = 0 - 0 = 0$, $R_{13} - R_{31} = 1 - (-1) = 2$, $R_{21} - R_{12} = 0 - 0 = 0$, hence $\mathbf{u} = \frac{1}{2}(0, 2, 0)^\top = (0, 1, 0)^\top = \mathbf{e}_2$: the axis is y , as expected from Step 2. ✓

Step 4 — the quaternion route. With $A = B = C = 90^\circ$ one has $c_A = c_B = c_C = s_A = s_B = s_C = \cos 45^\circ = \frac{\sqrt{2}}{2} = 0.70710678$, and $(\frac{\sqrt{2}}{2})^3 = \frac{\sqrt{2}}{4} = 0.35355339$. The formulas of Remark 28.8 give

$$w = \frac{\sqrt{2}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{2}}{2}, \quad x = \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4} = 0, \quad y = \frac{\sqrt{2}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{2}}{2}, \quad z = \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4} = 0,$$

that is $q = (\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}, 0)$. By Definition 28.3 this is the quaternion of axis $\mathbf{e}_2$ and angle θ with $\cos \frac{\theta}{2} = \frac{\sqrt{2}}{2}$, that is $\frac{\theta}{2} = 45^\circ$ and $\theta = 90^\circ$: Step 3 is recovered by an entirely different route. ✓

Independent check. The two routes were recomputed in `numpy`: $R_z(\pi/2)R_y(\pi/2)R_x(\pi/2) - R_y(\pi/2)$ has maximum norm 1.1×10^{-16} , and the prod-

uct $q_z q_y q_x$ equals $(0.70710678, 0, 0.70710678, 0)$ with deviation 1.1×10^{-16} from the exact value. ✓

Interpretation. The result is not a coincidence: it is gimbal lock. With $\theta = 90^\circ$, Theorem 28.5(iv) gives $R(\phi, \frac{\pi}{2}, \psi) = R_z(\psi - \phi)R_y(\frac{\pi}{2})$, and here $\psi = \phi = 90^\circ$, hence $\psi - \phi = 0$ and $R = R_z(0)R_y(90^\circ) = R_y(90^\circ)$. The cancellation of the x and z components of the quaternion, $c_{ACBSC} - s_{ASBCC} = 0$ and $s_{ACBCC} - c_{ASBSC} = 0$, is the same fact seen from the quaternion side.

Typical mistakes. (1) Writing $R = R_x(90^\circ)R_y(90^\circ)R_z(90^\circ)$: the ZYX convention of Definition 28.4 places R_z on the left, and the order matters (Theorem 28.1). The result of the reversed order is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, which has trace -1 : it is a 180° rotation about $\frac{1}{\sqrt{2}}(1, 0, 1)^\top$, an entirely different object. (2) Extracting the axis from the transposed matrix, obtaining $-\mathbf{e}_2$. (3) Applying $\theta = 2\arccos w$ without normalising the sign of q : here $w > 0$ and no harm follows, but in general it would.

Extension. Check that $R_z(90^\circ)R_y(90^\circ)R_x(90^\circ) = R_y(90^\circ)$ is a particular case of the identity $R_z(\alpha)R_y(90^\circ)R_x(\alpha) = R_y(90^\circ)$ for every α : it is exactly the gimbal lock of Theorem 28.5(iv) with $\psi = \phi = \alpha$, that is $\psi - \phi = 0$.

Exercise 28.5: The case $\theta = \pi$ solved with the four branches. *Type:* exact symbolic. Let $\mathbf{u} = \frac{1}{7}(2, 3, 6)^\top$ (a unit vector, since $2^2 + 3^2 + 6^2 = 49$) and $R = R(\mathbf{u}, \pi)$. Compute R in exact form, extract the quaternion with Theorem 28.4 and explain why the single-branch formula does not work.

Solution. Strategy. For $\theta = \pi$ Rodrigues' formula reduces to $R = 2\mathbf{u}\mathbf{u}^\top - I$ (Corollary 28.3, Step 4), which is computed in exact fractions; then the four branches are applied.

Step 1 — the matrix. $\mathbf{u}\mathbf{u}^\top = \frac{1}{49} \begin{pmatrix} 4 & 6 & 12 \\ 6 & 9 & 18 \\ 12 & 18 & 36 \end{pmatrix}$, hence

$$R = 2\mathbf{u}\mathbf{u}^\top - I = \frac{1}{49} \begin{pmatrix} 8-49 & 12 & 24 \\ 12 & 18-49 & 36 \\ 24 & 36 & 72-49 \end{pmatrix} = \frac{1}{49} \begin{pmatrix} -41 & 12 & 24 \\ 12 & -31 & 36 \\ 24 & 36 & 23 \end{pmatrix}.$$

Step 2 — immediate checks. $\text{tr } R = \frac{-41-31+23}{49} = \frac{-49}{49} = -1$, consistent with $1 + 2\cos\pi = -1$ (Corollary 28.1). ✓ Also R is symmetric, as it must be because $\sin\pi = 0$ annihilates the antisymmetric part. ✓ And $R\mathbf{u} = 2\mathbf{u}\langle\mathbf{u}, \mathbf{u}\rangle - \mathbf{u} = 2\mathbf{u} - \mathbf{u} = \mathbf{u}$. ✓

Step 3 — the four numbers.

$$c_0 = 1 + \text{tr } R = 1 - 1 = 0, \quad c_1 = 1 + \frac{2(-41)}{49} + 1 = 2 - \frac{82}{49} = \frac{98-82}{49} = \frac{16}{49},$$

$$c_2 = 2 - \frac{62}{49} = \frac{36}{49}, \quad c_3 = 2 + \frac{46}{49} = \frac{144}{49}.$$

Check: $0 + \frac{16}{49} + \frac{36}{49} + \frac{144}{49} = \frac{196}{49} = 4$, as prescribed by Step 4 of the proof of Theorem 28.4. ✓

Step 4 — the branch and the components. The maximum is $c_3 = \frac{144}{49}$, hence $m^* = 3$ and $s = \frac{1}{2}\sqrt{\frac{144}{49}} = \frac{1}{2} \cdot \frac{12}{7} = \frac{6}{7}$, with $4s = \frac{24}{7}$. Note that $s = \frac{6}{7} > \frac{1}{2}$, as guaranteed. Then

$$w = \frac{R_{21} - R_{12}}{4s} = \frac{\frac{12}{49} - \frac{12}{49}}{24/7} = 0, \quad x = \frac{R_{13} + R_{31}}{4s} = \frac{48/49}{24/7} = \frac{48}{49} \cdot \frac{7}{24} = \frac{2}{7},$$

$$y = \frac{R_{23} + R_{32}}{4s} = \frac{72/49}{24/7} = \frac{3}{7}, \quad z = s = \frac{6}{7}.$$

Hence $q = (0, \frac{2}{7}, \frac{3}{7}, \frac{6}{7})$: zero real part and vector part equal to $\mathbf{u}$, exactly as predicted by Corollary 28.5(iv). ✓

Step 5 — why the naive branch fails. The single-branch formula sets $w = \frac{1}{2}\sqrt{1 + \text{tr } R} = \frac{1}{2}\sqrt{0} = 0$ and then divides by $4w = 0$: it is a division by zero. This is not an inaccuracy but an undefined formula. Step 4 of the proof of Theorem 28.4 instead guarantees that at least one of the four c_m equals at least 1: here it is $c_3 = \frac{144}{49} = 2.938776$.

Independent check. The computation was repeated in exact arithmetic with Python's fractions (`fractions.Fraction`), obtaining the same fractions; and in double precision it was checked that $R(q) = R$ with deviation 0 and $R^2 = I$ with deviation 1.1×10^{-16} . ✓

Interpretation. The half-turn rotation is the only one whose quaternion is *pure*. The set of these rotations is the equator $w = 0$ of S^3 , an S^2 : it is the same S^2 that appears as the base of the Hopf fibration (Theorem 11.11), and this is not a coincidence, because both parametrise axes.

Typical mistakes. (1) Concluding from $c_0 = 0$ that “the quaternion does not exist”: it exists, it simply has zero real part. (2) Choosing the branch with the smallest index among the nonzero ones instead of the maximum: it works but loses precision, which is exactly what the theorem avoids. (3) Forgetting that q and $-q$ are both valid, and looking for a criterion for “the” correct sign when $w = 0$: none exists (Corollary 28.6(iv)).

Extension. Repeat with $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1, 0)^\top$: now $c_1 = c_2 = 2$ and $c_3 = 0$, and the two maxima are equal. The theorem remains valid — either of the two maximal indices may be chosen — and one obtains $q = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$ from both.

Exercise 28.6: The cost of approaching a singularity. *Type:* numerical/computational with a proof-based part. Prove that, in the RPY convention with $\psi = 0$, the roll rate needed to produce a unit angular velocity about x equals $\sec \theta$, and determine at what distance from the singularity a joint with limit $\dot{\phi}_{\max} = 3$ rad/s saturates.

Solution. Strategy. One writes the Jacobian of Lemma 28.4 with $\psi = 0$, solves the linear system in closed form and imposes the limit.

Step 1 — the Jacobian. By Lemma 28.4 with $\psi = 0$ ($c_\psi = 1$, $s_\psi = 0$),

$$J = \begin{pmatrix} c_\theta & 0 & 0 \\ 0 & 1 & 0 \\ -s_\theta & 0 & 1 \end{pmatrix}, \quad J \begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} \dot{\phi} c_\theta \\ \dot{\theta} \\ -\dot{\phi} s_\theta + \dot{\psi} \end{pmatrix}.$$

Step 2 — the solution. Imposing $J\xi = \omega = (1, 0, 0)^\top$ one obtains, from the first row, $\dot{\phi} = 1/\cos \theta = \sec \theta$ (defined because $\cos \theta \neq 0$ away from the singularity); from the second, $\dot{\theta} = 0$; from the third, $\dot{\psi} = \dot{\phi} \sin \theta = \tan \theta$. This is the claim.

Step 3 — the saturation threshold. Imposing $\sec \theta \leq 3$, that is $\cos \theta \geq \frac{1}{3}$, that is $|\theta| \leq \arccos \frac{1}{3}$. Now $\arccos \frac{1}{3} = 70.528779^\circ$: the joint saturates as soon as $|\theta|$ exceeds 70.53° , that is at barely *twenty degrees* from the singularity. One need not get very close to run into trouble.

Step 4 — the overall cost. $\|\xi\| = \sqrt{\sec^2 \theta + \tan^2 \theta} = \sqrt{1 + 2 \tan^2 \theta}$, using $\sec^2 = 1 + \tan^2$. At $\theta = 80^\circ$: $\tan 80^\circ = 5.671282$, hence $\|\xi\| = \sqrt{1 + 2 \times 32.163437} = \sqrt{65.326875} = 8.082504$.

Independent check. The same system was solved in `numpy` with `numpy.linalg.solve`, without using the closed form: at $\theta = 80^\circ$ one obtains $\xi = (5.7588, 0, 5.6713)^\top$ with $\|\xi\| = 8.0825$, which agrees with $\sec 80^\circ = 5.758770$, $\tan 80^\circ = 5.671282$ and with Step 4 to four decimal places. ✓ A second check: $\det J = \cos \theta$ (Theorem 28.5(i)), and at $\theta = 80^\circ$ it equals 0.173648 , whose reciprocal 5.758770 is exactly $\dot{\phi}$: consistent, since the first column of J^{-1} has norm of order $1/\det J$. ✓

Interpretation. The singularity is not an isolated point to be avoided but a *cone* of degraded configurations. Industrial planners know this and keep a safety distance from the vertical pitch, or resort to damped inversion, which replaces J^{-1} with $J^{\top}(JJ^{\top} + \lambda^2 I)^{-1}$: the velocities remain bounded, at the cost of a tracking error in the lost direction.

Typical mistakes. (1) Concluding that “it is enough not to pass exactly through $\theta = 90^\circ$ ”: Step 3 shows that at a distance of 20° one is already saturated. (2) Confusing the singularity of the *parametrisation* with a singularity of the *robot*: here the body can rotate perfectly well, it is the three angles that cannot describe it. (3) Using $\|J^{-1}\| = 1/\det J$: this is false in general; the correct relation goes through the singular values.

Extension. Repeat the computation with $\boldsymbol{\omega} = (0, 0, 1)^\top$: the solution is $\boldsymbol{\xi} = (0, 0, 1)^\top$ for every θ , and no velocity blows up. The singularity only prevents motions in the lost direction, that is the one orthogonal to the image of J — here the direction $\mathbf{e}_1$ when $\theta = \pm 90^\circ$, as read off from the columns computed in Step 3 of the proof of Theorem 28.5.

28.11. Self-assessment quiz

- (Definition) For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, the matrix $[\mathbf{u}]_{\times}$ of Definition 28.2 satisfies
 - $[\mathbf{u}]_{\times} \mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{u}$
 - $[\mathbf{u}]_{\times} \mathbf{v} = \mathbf{u} \times \mathbf{v}$
 - $[\mathbf{u}]_{\times} \mathbf{v} = \mathbf{v} \times \mathbf{u}$
 - $[\mathbf{u}]_{\times} \mathbf{v} = (\mathbf{u}\mathbf{u}^\top) \mathbf{v}$
- (Hypothesis) In Lemma 28.2 the identity $K^3 = -K$ holds thanks to a hypothesis that cannot be dropped:
 - K is skew-symmetric
 - $\|\mathbf{u}\| = 1$
 - $\theta \in [0, \pi]$
 - $\det K = 1$
- (Symbolic computation) For every unit vector $\mathbf{u}$ and every θ , the trace of $R(\mathbf{u}, \theta)$ equals
 - $3 \cos \theta$
 - $1 + 2 \cos \theta$
 - $2 + \cos \theta$
 - $1 + 2 \sin \theta$
- (Numerical computation) The unit quaternion of the 120° rotation about $\frac{1}{\sqrt{3}}(1, 1, 1)^\top$ is
 - $\frac{1}{2}(1, 1, 1, 1)$
 - $(\frac{\sqrt{3}}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
 - $\frac{1}{2}(1, 1, 1, -1)$
 - $(0, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$
- (Proof step) In the proof of Theorem 28.2, the recurrence $K^{2m+1} = (-1)^m K$, $K^{2m+2} = (-1)^m K^2$ is used to
 - prove that R is orthogonal
 - reduce all powers of K to I , K and K^2 and close the exponential series in three terms
 - compute $\det R$
 - prove the uniqueness of the axis
- (Error diagnosis) A student extracts the axis of $R \in SO(3)$ with $\mathbf{u} = (R_{32} - R_{23}, R_{13} - R_{31}, R_{21} - R_{12})^\top / (2 \sin \theta)$, finds $\text{tr } R = -1$ and gets $\frac{0}{0}$; the student concludes that “ R has no axis”. The mistake is
 - $\text{tr } R = -1$ is impossible in $SO(3)$
 - for $\theta = \pi$ the antisymmetric part is zero and the axis must be extracted from $\mathbf{u}\mathbf{u}^\top = (R + I)/2$
 - the student forgot to normalise $\mathbf{u}$
 - the denominator should be $2 \cos \theta$
- (Transfer) Before interpolating between q_0 and q_1 the sign is checked because
 - otherwise the interpolation formula is undefined
 - q_1 and $-q_1$ represent different rotations
 - q_1 and $-q_1$ represent the *same* rotation, but are at distance Ω and $\pi - \Omega$ from q_0 respectively: without the check one may traverse the long arc
 - the quaternion would lose its unit norm
- (Source and metacognition) Rodrigues’ formula in matrix form, together with the recurrence for the powers of the skew-symmetric matrix, appears with this numbering in

- A) Murray, Li, Sastry, *A Mathematical Introduction to Robotic Manipulation*, chap. 2, Lemma 2.3 and equation (2.14), p. 28 B) Kuipers, *Quaternions and Rotation Sequences*, chap. 1 C) Diebel, *Representing Attitude*, §8 D) Hamilton, *Elements of Quaternions*, Book I

28.12. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (1–4).

1) **B.** *Rationale:* this is Lemma 28.1(i), checked component by component in Step 1 of its proof. *Distractors:* A and D describe the projection onto $\mathbf{u}$, which is $\mathbf{u}\mathbf{u}^\top$ and appears in K^2 , not in K ; C has the wrong sign, because the cross product is antisymmetric. *Diagnosed error:* confusing the symmetric and antisymmetric parts of the Rodrigues decomposition. *Source:* Murray–Li–Sastry, chap. 2, §2.2, p. 27. *Prerequisite:* the cross product in coordinates. *Remediation:* compute $[\mathbf{e}_3]_\times \mathbf{e}_1$ and compare it with $\mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2$.

2) **B.** *Rationale:* Step 1 of the proof of Lemma 28.2 uses $K^2 = \mathbf{u}\mathbf{u}^\top - I$, which by Lemma 28.1(iv) requires $\|\mathbf{u}\| = 1$; without it, $K^2 = \mathbf{u}\mathbf{u}^\top - \|\mathbf{u}\|^2 I$ and $K^3 = -\|\mathbf{u}\|^2 K$. *Distractors:* A is true but not sufficient (it holds for every $\mathbf{u}$); C concerns the normalisation of the angle, which does not enter the lemma; D is false, because every 3×3 skew-symmetric matrix has zero determinant. *Diagnosed error:* applying $K^3 = -K$ to a non-normalised axis — the most common mistake in implementations. *Source:* Murray–Li–Sastry, chap. 2, Lemma 2.3, eq. (2.12)–(2.13), pp. 27–28, which explicitly states $\hat{a}^3 = -\|a\|^2 \hat{a}$. *Prerequisite:* norm and normalisation. *Remediation:* compute K^3 for $\mathbf{u} = (2, 0, 0)$ and check that it equals $-4K$.

3) **B.** *Rationale:* this is Corollary 28.1, Step 1, with $\text{tr } K = 0$ and $\text{tr } K^2 = -2$; it coincides with Theorem 11.8. *Distractors:* A confuses it with the trace of $\cos \theta I$; C adds the contributions wrongly; D swaps sine and cosine, that is the antisymmetric part with the symmetric one. *Diagnosed error:* forgetting that the trace of a skew-symmetric matrix is zero. *Source:* Corollary 28.1; Diebel, §6.5. *Prerequisite:* linearity of the trace. *Remediation:* check the formula on the matrix of Example 28.1, where $\text{tr } R = 0$ and $\theta = 120^\circ$.

4) **A.** *Rationale:* by Definition 28.3, $w = \cos 60^\circ = \frac{1}{2}$ and the vector part is $\sin 60^\circ \cdot \frac{1}{\sqrt{3}}(1, 1, 1)^\top = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}}(1, 1, 1)^\top = \frac{1}{2}(1, 1, 1)^\top$. *Distractors:* B uses the full angle instead of the half-angle ($\cos 120^\circ \neq \frac{\sqrt{3}}{2}$, and in any case the norm would not be 1); C gets the sign of one component wrong, obtaining the axis $(1, 1, -1)$; D is the quaternion of a 180° rotation, not 120° . *Diagnosed error:* using θ instead of $\theta/2$: this is the single most frequent quaternion mistake. *Source:* Example 28.1 and Example 28.2; Murray–Li–Sastry, p. 34. *Prerequisite:* double-angle formulas. *Remediation:* check that quaternion A, cubed, gives -1 , consistent with $3 \times 120^\circ = 360^\circ$ and with the double covering.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (5–8).

5) **B.** *Rationale:* Step 3 of the proof of Theorem 28.2 substitutes the powers and recognises the series for sin and cos; without the recurrence the series would not close. *Distractors:* A instead follows from Step 6 (block matrix in an orthonormal basis); C follows from A; D is Theorem 11.8, proved elsewhere and by other means. *Diagnosed error:* attributing to one lemma the role of a different step. *Source:* Murray–Li–Sastry, chap. 2, eq. (2.14), p. 28. *Prerequisite:* the exponential series of matrices. *Remediation:* redo Step 3 writing out explicitly the first six terms of the series.

6) **B.** *Rationale:* this is Corollary 28.3(iii): for $\theta = \pi$ one has $R = R^\top$, the antisymmetric

part is zero and the axis is read from $(R + I)/2 = \mathbf{u}\mathbf{u}^\top$. *Distractors*: A is false, $\text{tr } R = -1$ corresponds to $\theta = \pi$ and is very common; C is irrelevant, the vector obtained is already zero; D makes no sense, because Corollary 28.1 gives $R - R^\top = 2\sin\theta K$. *Diagnosed error*: not recognising the degenerate case and concluding that the object does not exist. *Source*: Corollary 28.3; Theorem 28.4, Step 8; Diebel, §5.8. *Prerequisite*: the symmetric and antisymmetric parts of a matrix. *Remediation*: redo Worked exercise 28.5.

7) C. *Rationale*: this is Proposition 28.6(iii)–(iv) together with Example 28.5: ten degrees against three hundred and fifty. *Distractors*: A is false, the formula is defined for every pair with $\sin\Omega \neq 0$; B contradicts Corollary 28.6(ii); D is false, $-q_1$ is as unit as q_1 . *Diagnosed error*: believing that “same rotation” implies “same path”. *Source*: Proposition 28.6; Shoemake, SIGGRAPH 1985, pp. 245–254. *Prerequisite*: the double covering (Theorem 11.10(iv)). *Remediation*: compute $\langle q_0, q_1 \rangle$ for the two pairs of Assigned exercise 28.3(a).

8) A. *Rationale*: Murray–Li–Sastry’s Lemma 2.3 gives the recurrence (equations 2.12–2.13) and equation (2.14) is Rodrigues’ formula, both on p. 28. *Distractors*: B, Kuipers’s text is an introduction to rotation sequences and does not use this numbering; C, §8 of Diebel is the catalogue of the twelve Euler parametrisations, not Rodrigues; D, Book I of Hamilton’s *Elements* treats vectors “without reference to angles or rotations” — the title of the book itself says so. *Diagnosed error*: citing from memory without checking the exact location. *Source*: list at the end of the chapter, entries [1] and [3]. *Prerequisite*: none. *Remediation*: open the three sources and locate in each of them where, if at all, the recurrence and the formula appear.

BEST PRACTICE

Assigned exercises — answers.

Exercise 28.1. (a) $K = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$, $K^2 = \mathbf{u}\mathbf{u}^\top - I = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & -1 \end{pmatrix}$, and with $\sin\pi = 0$, $1 - \cos\pi = 2$: $R = I + 2K^2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$, with integer entries. (b) $R^2 = I$ because two half-turns make a full turn (Theorem 28.2, Step 8: $R(\mathbf{u}, \pi)^2 = R(\mathbf{u}, 2\pi) = I$); $\text{tr } R = 0 + 0 - 1 = -1 = 1 + 2\cos\pi$ (Corollary 28.1). (c) $c_0 = 1 - 1 = 0$, $c_1 = 1 + 0 + 1 = 2$, $c_2 = 1 + 0 + 1 = 2$, $c_3 = 1 - 2 + 1 = 0$; sum 4. ✓ There are *two* maxima, $c_1 = c_2 = 2$: Theorem 28.4 calls for a maximal index, not a unique maximum, and the two branches give the same quaternion (up to the global sign). (d) With $m^* = 1$: $s = \frac{\sqrt{2}}{2}$, $4s = 2\sqrt{2} = 2.82842712$, $x = s$, $w = (R_{32} - R_{23})/(4s) = 0$, $y = (R_{12} + R_{21})/(4s) = 2/(2\sqrt{2}) = \frac{1}{\sqrt{2}}$, $z = (R_{13} + R_{31})/(4s) = 0$; hence $q = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0) = (0, \mathbf{u})$, and $R(q) = R$ by Theorem 28.3. With $m^* = 2$ one obtains the same q . (e) The single-branch formula sets $w = \frac{1}{2}\sqrt{1 + \text{tr } R} = 0$ and then divides by $4w = 0$: it fails at the division, not at the extraction of the root.

Exercise 28.2. (a) With the same scheme as Lemma 28.4 (derivative, multiplication by $R^\top$, Lemma 28.3) one obtains $\boldsymbol{\omega} = \dot{\alpha} \mathbf{e}_3 + \dot{\beta} R_z(\alpha) \mathbf{e}_2 + \dot{\gamma} R_z(\alpha) R_y(\beta) \mathbf{e}_3$, that is

$$J = \begin{pmatrix} 0 & -s_\alpha & c_\alpha s_\beta \\ 0 & c_\alpha & s_\alpha s_\beta \\ 1 & 0 & c_\beta \end{pmatrix}.$$

(b) Expanding along the first column, $\det J = 1 \cdot \det \begin{pmatrix} -s_\alpha & c_\alpha s_\beta \\ c_\alpha & s_\alpha s_\beta \end{pmatrix} = -s_\alpha^2 s_\beta - c_\alpha^2 s_\beta = -\sin\beta$. The singular configurations are $\beta = 0$ and $\beta = \pi$. (c) For $\beta = 0$: $R_y(0) = I$, hence the third column is $R_z(\alpha) \mathbf{e}_3 = \mathbf{e}_3$, equal to the first: rank 2 (the second column $(-s_\alpha, c_\alpha, 0)^\top$ is independent of $\mathbf{e}_3$), kernel spanned by $(1, 0, -1)^\top$. For $\beta = \pi$: $R_y(\pi) \mathbf{e}_3 = (\sin\pi, 0, \cos\pi)^\top = -\mathbf{e}_3$, hence the third column is $-\mathbf{e}_3$: rank 2, kernel spanned by $(1, 0, 1)^\top$. (d) For $\beta = 0$,

$R = R_z(\alpha)R_z(\gamma) = R_z(\alpha + \gamma)$ (Proposition 28.1): it depends only on $\alpha + \gamma$. For $\beta = \pi$, Corollary 28.2 gives $R_y(\pi)R_z(\gamma)R_y(\pi)^\top = R(R_y(\pi)\mathbf{e}_3, \gamma) = R(-\mathbf{e}_3, \gamma) = R_z(-\gamma)$, hence $R = R_z(\alpha)R_z(-\gamma)R_y(\pi) = R_z(\alpha - \gamma)R_y(\pi)$: it depends only on $\alpha - \gamma$. (e) The singularity has moved from $\theta = \pm \frac{\pi}{2}$ to $\beta \in \{0, \pi\}$ and is of the same type (rank drops to 2, loss of one degree of freedom); the fact that it cannot be removed by changing convention is Proposition 28.8. Note the practical difference: in ZYZ the singularity lies at the “rest” configuration $\beta = 0$, which makes this convention worse than RPY for a manipulator working about the vertical. *Numerical check:* $\det J + \sin \beta$ on 4 000 random pairs has maximum deviation 3.3×10^{-16} ; the columns of J recomputed by finite differences differ by at most 3.3×10^{-10} ; the identities of (d) are checked on 3 000 triples with maximum deviation 1.0×10^{-15} ; the rank at $\beta = 0$ and $\beta = \pi$ equals 2 in every case tested. ✓

Exercise 28.3. (a) $q_0 = (\cos 30^\circ, 0, 0, \sin 30^\circ) = (0.8660254, 0, 0, 0.5)$ and $q_1 = (\cos 150^\circ, 0, 0, \sin 150^\circ) = (-0.8660254, 0, 0, 0.5)$, hence $\langle q_0, q_1 \rangle = -0.75 + 0.25 = -0.5$ and $\Omega = 120^\circ$: without the choice of sign the rotation traversed is $2\Omega = 240^\circ$; with the choice ($q_1 \mapsto -q_1$) one has $\Omega = 60^\circ$ and the rotation is 120° . (b) The samples with the correct sign give cumulative angles $0^\circ, 12^\circ, 24^\circ, \dots, 120^\circ$; without it, $0^\circ, 24^\circ, 48^\circ, \dots, 240^\circ$. In both cases the endpoints are the same orientations. (c) The real part of $\bar{q}_0 q_1$ coincides with $\langle q_0, q_1 \rangle$ up to machine error (measured: maximum deviation 0 on 5 000 pairs). (d) $\|J^{-1}\|_2$ equals 8.1131, 81.0295, 810.2848, 8102.8469 for $\theta = 80^\circ, 89^\circ, 89.9^\circ, 89.99^\circ$, against $1/|\cos \theta| = 5.7588, 57.2987, 572.9581, 5729.5780$: the ratio tends to $\sqrt{2}$, hence $\|J^{-1}\|_2$ diverges like $1/|\cos \theta|$, the reciprocal of the determinant (Theorem 28.5(i)). (e) With $\lambda = 10^{-2}$, the norm of the damped inverse $J^\top(JJ^\top + \lambda^2 I)^{-1}$ equals 8.0601, 48.9138, 12.1562, 1.2339 at the same four points: it grows up to a maximum around $\theta = 89^\circ$ (where the smallest singular value is comparable to λ) and then *decreases*, because the deeper one goes into the singularity the more the term $\lambda^2 I$ dominates. It diverges nowhere: this is exactly the property sought. One gains boundedness of the joint velocities; one loses tracking accuracy, because in the lost direction the angular velocity produced is smaller than the one requested by a factor $\sigma^2/(\sigma^2 + \lambda^2)$, with σ the singular value tending to zero.

28.13. Chapter formulary

Rotation group and elementary rotations.

$$SO(3) = \{R : R^\top R = I, \det R = +1\}, \quad R_*(\alpha)R_*(\beta) = R_*(\alpha + \beta),$$

$$R_*(\alpha)^{-1} = R_*(-\alpha) = R_*(\alpha)^\top.$$

$$R_x(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\alpha & -s_\alpha \\ 0 & s_\alpha & c_\alpha \end{pmatrix}, \quad R_y(\alpha) = \begin{pmatrix} c_\alpha & 0 & s_\alpha \\ 0 & 1 & 0 \\ -s_\alpha & 0 & c_\alpha \end{pmatrix}, \quad R_z(\alpha) = \begin{pmatrix} c_\alpha & -s_\alpha & 0 \\ s_\alpha & c_\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Non-commutativity: $R_z(90^\circ)R_x(90^\circ) = R(\frac{1}{\sqrt{3}}(1, 1, 1), 120^\circ)$ and $R_x(90^\circ)R_z(90^\circ) = R(\frac{1}{\sqrt{3}}(1, -1, 1), 120^\circ)$.

Hat map and Rodrigues' formula. For $\|\mathbf{u}\| = 1$ and $K = [\mathbf{u}]_\times$:

$$K^\top = -K, \quad K\mathbf{u} = \mathbf{0}, \quad K^2 = \mathbf{u}\mathbf{u}^\top - I, \quad K^3 = -K,$$

$$K^{2m+1} = (-1)^m K, \quad K^{2m+2} = (-1)^m K^2.$$

$$R(\mathbf{u}, \theta) = e^{\theta K} = I + \sin \theta K + (1 - \cos \theta)K^2,$$

$$R(\mathbf{u}, \theta)\mathbf{v} = \cos \theta \mathbf{v} + \sin \theta (\mathbf{u} \times \mathbf{v}) + (1 - \cos \theta)\langle \mathbf{u}, \mathbf{v} \rangle \mathbf{u},$$

$$\operatorname{tr} R = 1 + 2 \cos \theta, \quad R - R^\top = 2 \sin \theta K, \quad A R(\mathbf{u}, \theta) A^\top = R(A\mathbf{u}, \theta) \quad (A \in SO(3)).$$

Inverse (logarithm). $\theta = \arccos \frac{\operatorname{tr} R - 1}{2} \in [0, \pi]$; if $0 < \theta < \pi$, $\mathbf{u} = \frac{1}{2 \sin \theta} (R_{32} - R_{23}, R_{13} - R_{31}, R_{21} - R_{12})^\top$; if $\theta = 0$, $R = I$ and the axis is undetermined; if $\theta = \pi$, $\mathbf{u}\mathbf{u}^\top = (R + I)/2$ and $\mathbf{u}$ is determined up to sign.

Quaternions. With $p = (a, \mathbf{u})$, $q = (b, \mathbf{v})$:

$$pq = (ab - \langle \mathbf{u}, \mathbf{v} \rangle, a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v}), \quad \overline{pq} = \bar{q}\bar{p}, \quad q\bar{q} = \|q\|^2, \quad q^{-1} = \frac{\bar{q}}{\|q\|^2},$$

$$\|pq\| = \|p\| \|q\| \quad (\text{four-square identity}).$$

Euler parameters. $q = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \mathbf{u})$, with $e_0^2 + e_1^2 + e_2^2 + e_3^2 = 1$; four parameters, one constraint, three degrees of freedom.

Conversions. From $q = (w, x, y, z)$ unitary:

$$R(q) = (w^2 - \|\mathbf{v}\|^2)I + 2\mathbf{v}\mathbf{v}^\top + 2w[\mathbf{v}]_\times = \begin{pmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{pmatrix}.$$

From R to q (four branches): $c_0 = 1 + \operatorname{tr} R$, $c_m = 1 + 2R_{mm} - \operatorname{tr} R$; $c_0 + c_1 + c_2 + c_3 = 4$, $c_0 = 4w^2$, $c_m = 4(m\text{-th component})^2$; the maximal m^* is chosen, $s = \frac{1}{2}\sqrt{c_{m^*}} \geq \frac{1}{2}$, and one divides by $4s$ the combinations

$$R_{32} - R_{23} = 4wx, \quad R_{13} - R_{31} = 4wy, \quad R_{21} - R_{12} = 4wz,$$

$$R_{12} + R_{21} = 4xy, \quad R_{13} + R_{31} = 4xz, \quad R_{23} + R_{32} = 4yz.$$

From q to axis-angle (after making $w \geq 0$): $\theta = 2 \operatorname{atan2}(\|\mathbf{v}\|, w)$ and $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$; $\mathbf{v} = \mathbf{0}$ gives $R = I$, $w = 0$ gives $\theta = \pi$ with the axis determined up to sign.

Double covering and interpolation.

$$\Phi : S^3 \rightarrow SO(3), \quad \ker \Phi = \{\pm 1\}, \quad \Phi^{-1}(R) = \{q, -q\},$$

$$SO(3) \cong S^3 / \{\pm 1\} = \mathbb{R}P^3 \cong SU(2) / \{\pm I\}.$$

$$\operatorname{Re}(\bar{q}_0 q_1) = \langle q_0, q_1 \rangle = \cos \Omega; \quad \text{relative angle} = 2\Omega \quad (\Omega \leq \frac{\pi}{2}), \quad 2(\pi - \Omega) \quad (\Omega > \frac{\pi}{2}).$$

Operating rule: if $\langle q_0, q_1 \rangle < 0$, replace q_1 with $-q_1$ before interpolating.

RPY angles (ZYX) and gimbal lock. $R = R_z(\psi)R_y(\theta)R_x(\phi)$:

$$R = \begin{pmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{pmatrix}, \quad J = \begin{pmatrix} c_\psi c_\theta & -s_\psi & 0 \\ s_\psi c_\theta & c_\psi & 0 \\ -s_\theta & 0 & 1 \end{pmatrix},$$

$$\dot{R}R^\top = [\boldsymbol{\omega}]_\times, \quad \boldsymbol{\omega} = J(\dot{\phi}, \dot{\theta}, \dot{\psi})^\top, \quad \det J = \cos \theta.$$

Gimbal lock: at $\theta = \pm \frac{\pi}{2}$ the rank of J is 2, the kernel is $(1, 0, \pm 1)^\top$, and $R(\phi, \pm \frac{\pi}{2}, \psi) = R_z(\psi \mp \phi)R_y(\pm \frac{\pi}{2})$. Extraction (for $\cos \theta \neq 0$): $\theta = \operatorname{atan2}(-R_{31}, \sqrt{R_{11}^2 + R_{21}^2})$, $\psi = \operatorname{atan2}(R_{21}, R_{11})$, $\phi = \operatorname{atan2}(R_{32}, R_{33})$. From ZYX angles to quaternion, with $c_A = \cos \frac{A}{2}$

and so on:

$$\begin{aligned} w &= c_A c_B c_C + s_A s_B s_C, & x &= c_A c_B s_C - s_A s_B c_C, \\ y &= c_A s_B c_C + s_A c_B s_C, & z &= s_A c_B c_C - c_A s_B s_C. \end{aligned}$$

Sources and bibliography

- [1] O. Rodrigues, *Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace, et de la variation des coordonnées provenant de ces déplacements considérés indépendamment des causes qui peuvent les produire*, «Journal de Mathématiques Pures et Appliquées», 1st series, vol. 5 (1840), pp. 380–440 (publisher Gauthier-Villars). Original memoir in which the half-angle parameters known today as the Euler–Rodrigues parameters, and the composition of rotations, first appear. Full reproduction in the NUMDAM archive, reference JMPA_1840_1_5__380_0; English translation with commentary by R. Friedberg, arXiv:2211.07787 — verified on 6 September 2026.
- [2] W. R. Hamilton, *Elements of Quaternions*, Longmans, Green & Co., London, 1866 (posthumous edition edited by his son W. E. Hamilton; vi + lix + 762 pp.). Book II, Chap. I, Sect. 10, «On a System of Three Right Versors, in Three Rectangular Planes; and on the Laws of the Symbols i, j, k », pp. 157–162: the *fundamental formula* $i^2 = j^2 = k^2 = ijk = -1$ and the quadrinomial form $q = w + ix + jy + kz$, recalled in Section 28.4. Book III, Art. 297, item (15.), p. 333: the operator $q(\)q^{-1}$ is described as a *conical rotation* of the operand about the axis of q through an angle equal to twice the angle of q — the historical statement of the half-angle of Theorem 28.3, with a reference to Art. 179 (1.) and 191 (5.). Full digital copy: Internet Archive, identifier `elementsofquater00hamirich` — verified on 6 September 2026.
- [3] R. M. Murray, Z. Li, S. S. Sastry, *A Mathematical Introduction to Robotic Manipulation*, CRC Press, Boca Raton, 1994 (ISBN 978-0-8493-7981-9). Chapter 2 «Rigid Body Motion», §2 «Rotational Motion in $\mathbb{R}^3$ »: §2.1 «Properties of rotation matrices», p. 23; §2.2 «Exponential coordinates for rotation», pp. 27–30, with Lemma 2.3 (recurrence $\hat{a}^3 = -\|a\|^2 \hat{a}$, eqq. 2.12–2.13) and Rodrigues' formula, eq. (2.14), p. 28; Proposition 2.4 (the exponential of a skew-symmetric matrix lies in $SO(3)$), p. 28; Proposition 2.5 (surjectivity of the exponential) and Theorem 2.6 (Euler), pp. 29–30. §2.3 «Other representations», pp. 31–34: Euler angles ZYZ (pp. 31–32), ZYX «yaw, pitch, roll» angles with the singularity stated at $\theta = -\pi/2$ (p. 32), quaternions (pp. 32–34) with the product in scalar–vector form $Q \cdot P = (q_0 p_0 - \vec{q} \cdot \vec{p}, q_0 \vec{p} + p_0 \vec{q} + \vec{q} \times \vec{p})$ on p. 33 and the correspondence $Q = (\cos(\theta/2), \omega \sin(\theta/2))$ with the extraction $\theta = 2 \cos^{-1} q_0$ on p. 34. Full text in PDF distributed by the authors and hosted among the materials of the robotics course at Lehigh University — verified on 6 September 2026.
- [4] J. B. Kuipers, *Quaternions and Rotation Sequences: A Primer with Applications to Orbits, Aerospace and Virtual Reality*, Princeton University Press, Princeton (New Jersey), 1999, 372 pp. (ISBN 0-691-05872-5; paperback reprint 2002, ISBN 978-0-691-10298-6). Elementary and complete treatment of the algebra of quaternions and of rotation sequences, with the aerospace conventions of *roll*, *pitch* and *yaw* used in Definition 28.4. Bibliographic data verified against the Princeton University Press catalogue and the review that appeared in «The Aeronautical Journal» (Cambridge Core) — verified on 6 September 2026.
- [5] J. Diebel, *Representing Attitude: Euler Angles, Unit Quaternions, and Rotation Vectors*, technical report, Stanford University, Stanford (California), 20 October 2006, 35 pp. §5 «Euler Angles», pp. 7–13, with §5.8 «Singularities», p. 13; §6 «Quaternions», pp. 14–

18, with §6.2 «Quaternion Multiplication» (p. 14), §6.4 «Unit Quaternion $\Rightarrow$ Rotation Matrix» and §6.5 the reverse direction (p. 15), §§6.12–6.13 the conversions with the axis–angle representation (pp. 17–18); §8 «A Catalog of Euler Angle Parameterizations», pp. 22–34, which lists all twelve sequences with their respective singularities (the sequence $(3, 2, 1)$, that is the RPY of this chapter, is on p. 33). It is the reference source for the conversion formulas and for their consistency across different conventions — verified on 6 September 2026.

- [6] J. Stuelpnagel, *On the Parametrization of the Three-Dimensional Rotation Group*, «SIAM Review» 6 (1964), no. 4, pp. 422–430 (DOI 10.1137/1006093). Contains the statement declared *imported* in Remark 28.7: every global parametrisation of $SO(3)$ with three parameters is singular at some point, and four is the smallest dimension in which a non-singular global parametrisation exists — verified on 6 September 2026.
- [7] S. W. Shepperd (Charles Stark Draper Laboratory, Cambridge, Massachusetts), *Quaternion from rotation matrix*, «Journal of Guidance and Control», vol. 1, June 1978. It is the origin of the four-branch method of Theorem 28.4: the abstract states that «the four components of the quaternion are treated on an equal footing», which «leads to a unified, compact and singularity-free procedure for determining the quaternion given the matrix». Author, title, venue and year verified on the record of the NASA Technical Reports Server, identifier 19780048191 (issue number and pages are not reported, as they are not confirmed against the primary source) — verified on 6 September 2026.
- [8] K. Shoemake, *Animating rotation with quaternion curves*, in «Computer Graphics» (SIGGRAPH '85 Proceedings) 19 (1985), no. 3, pp. 245–254 (DOI 10.1145/325334.325242). Introduction of SLERP and of the sign-choice rule proved in Proposition 28.6; the complete treatment of interpolation is in Chapter 30 — verified on 6 September 2026.

Lie groups and kinematics: from $SE(3)$ to the product of exponentials

Rodrigues' formula and the unit quaternions of Chapter 28 describe the *orientation* of a rigid body. A robot, however, does not only orient itself: it moves. The complete pose of the tool is a pair (rotation, position), and the set of all poses is a group, $SE(3)$, which is neither a direct product of $SO(3)$ by $\mathbb{R}^3$ nor something more exotic: it is a *semidirect product*, and the twist that the semidirect product introduces is exactly what makes the kinematics of manipulators a nontrivial problem.

The thread running through the chapter is a single one, and it is the same one as for the circle: the exponential map. On the circle $t \mapsto e^{it}$ unrolled S^1 onto the line (Chapter 12); on $SO(3)$ the exponential of a skew-symmetric matrix gave Rodrigues' formula; on $SE(3)$ the exponential of a *twist* gives a **screw motion**, that is, a rotation about an axis composed with a translation along that same axis. Chasles' theorem (1830) states that *every* rigid motion of space is of this form: the exponential is surjective. From there the forward kinematics of an open kinematic chain is written as a product of exponentials, the geometric Jacobian is obtained by differentiating that product instead of postulating it, and singularities become a rank condition on a $6 \times n$ matrix whose columns are twists. The gimbal lock of Chapter 28 reappears here in its mechanical incarnation: wrist singularity.

INFO

Level and prerequisites. Level C–D (undergraduate/graduate).

Internal prerequisites. Rotations in space, the skew-symmetric matrix associated with a vector and Rodrigues' formula (Chapter 28); groups of isometries and the classification of $O(2)$, $O(3)$ (Chapter 5); the exponential as a homomorphism of one-parameter groups (Chapter 13); the notions of rank, determinant and singular value decomposition of a real matrix (basic linear algebra, Volume II, Chapter 1).

External prerequisites, assumed and not reproved:

- the matrix exponential series $e^A = \sum_{k \geq 0} A^k/k!$ converges absolutely in every submultiplicative norm, can be differentiated term by term, and satisfies $\frac{d}{dt}e^{tA} = Ae^{tA} = e^{tA}A$ and $\det e^A = e^{\text{tr} A}$;
- if $AB = BA$ then $e^{A+B} = e^Ae^B$; in general not;
- the double cross product identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$;
- the existence of the singular value decomposition $A = U\Sigma V^T$ and the fact that the number of nonzero singular values is the rank of A .

Everything else — including the closed form of the exponential of a twist, the inverse of the matrix $G(\theta)$, Chasles' theorem with explicit axis and pitch, the product-of-exponentials formula and the two Jacobians — is proved here.

What one learns. Recognising $SE(3)$ as a matrix Lie group and as a semidirect product; computing the exponential of a twist in closed form and its logarithm; decomposing a given rigid motion into axis, angle and pitch; writing the forward kinematics of an open chain with no intermediate frames; deriving (not postulating) the spatial and the body Jacobian;

identifying singular configurations and verifying them with singular values.

Warning. Conventions of this chapter.

- Vectors of $\mathbb{R}^3$ are columns; $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is the canonical basis; I is the identity of size inferred from context.
- A rigid motion is written either as a pair $(R, \mathbf{p})$ or as a 4×4 homogeneous matrix; the two languages are used interchangeably.
- The coordinates of a twist are ordered **angular-first**: $\xi = (\omega, \mathbf{v}) \in \mathbb{R}^6$. This is the Lynch–Park convention; Murray–Li–Sastry uses the opposite order $\xi = (\mathbf{v}, \omega)$, and with it the adjoint matrix Ad_g appears block-transposed. When comparing formulas with the literature, the first thing to check is the order.
- θ always denotes the parameter of the screw motion (angle if $\omega \neq 0$, length if $\omega = 0$); θ_i is the variable of joint i .
- “Axis” means an oriented line in $\mathbb{R}^3$; its direction is that of ω .
- Numerical values are rounded to six decimal places, with the decimal point as separator.

29.1. The group of rigid motions $SE(3)$

Definition 29.1: Rigid motion and the group $SE(3)$. A map $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a **direct rigid motion** if there exist $R \in SO(3)$ and $\mathbf{p} \in \mathbb{R}^3$ such that

$$f(\mathbf{x}) = R\mathbf{x} + \mathbf{p} \quad \text{for every } \mathbf{x} \in \mathbb{R}^3.$$

The set of such maps is denoted $SE(3)$ (*special Euclidean group*). To $f = (R, \mathbf{p})$ one associates the **homogeneous matrix**

$$T = \begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 4},$$

which acts on points through *homogeneous coordinates* $\tilde{\mathbf{x}} = \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix}$: $T\tilde{\mathbf{x}} = \begin{pmatrix} R\mathbf{x} + \mathbf{p} \\ 1 \end{pmatrix}$.

The correspondence $f \leftrightarrow T$ is a bijection: R and $\mathbf{p}$ are read off from T , and conversely. From here on, $SE(3)$ denotes indifferently the set of maps or the set of homogeneous matrices.

Proposition 29.1: $SE(3)$ is a group. $SE(3)$, with matrix multiplication, is a group. Explicitly:

$$(R_1, \mathbf{p}_1) \cdot (R_2, \mathbf{p}_2) = (R_1 R_2, R_1 \mathbf{p}_2 + \mathbf{p}_1), \quad (R, \mathbf{p})^{-1} = (R^\top, -R^\top \mathbf{p}),$$

and the identity element is $(I, \mathbf{0})$.

Proof. Step 1 — closure. Multiply the homogeneous matrices in block form:

$$\begin{pmatrix} R_1 & \mathbf{p}_1 \\ \mathbf{0}^\top & 1 \end{pmatrix} \begin{pmatrix} R_2 & \mathbf{p}_2 \\ \mathbf{0}^\top & 1 \end{pmatrix} = \begin{pmatrix} R_1 R_2 & R_1 \mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0}^\top & 1 \end{pmatrix}.$$

The bottom-left block is $\mathbf{0}^\top R_2 + 1 \cdot \mathbf{0}^\top = \mathbf{0}^\top$ and the bottom-right one is $\mathbf{0}^\top \mathbf{p}_2 + 1 = 1$: the homogeneous form is preserved. It remains to see that $R_1 R_2 \in SO(3)$: indeed $(R_1 R_2)^\top (R_1 R_2) = R_2^\top R_1^\top R_1 R_2 = R_2^\top R_2 = I$ and $\det(R_1 R_2) = \det R_1 \det R_2 = 1$. Hence the product lies in

$SE(3)$. (In terms of maps: $f_1(f_2(\mathbf{x})) = R_1(R_2\mathbf{x} + \mathbf{p}_2) + \mathbf{p}_1 = R_1R_2\mathbf{x} + R_1\mathbf{p}_2 + \mathbf{p}_1$, the same formula.)

Step 2 — associativity. This is the associativity of 4×4 matrix multiplication, which holds in $\mathbb{R}^{4 \times 4}$ and is inherited by any subset closed under the product.

Step 3 — identity. $(I, \mathbf{0})$ has homogeneous matrix I_4 and satisfies $(I, \mathbf{0})(R, \mathbf{p}) = (R, \mathbf{p}) = (R, \mathbf{p})(I, \mathbf{0})$ by the formula of Step 1.

Step 4 — inverse. Set $S = (R^\top, -R^\top \mathbf{p})$; since $R^\top = R^{-1} \in SO(3)$ (the inverse of a special orthogonal matrix is special orthogonal: $(R^\top)^\top R^\top = RR^\top = I$ and $\det R^\top = \det R = 1$), $S \in SE(3)$. Applying Step 1 twice:

$$(R, \mathbf{p})S = (RR^\top, R(-R^\top \mathbf{p}) + \mathbf{p}) = (I, -\mathbf{p} + \mathbf{p}) = (I, \mathbf{0}),$$

$$S(R, \mathbf{p}) = (R^\top R, R^\top \mathbf{p} - R^\top \mathbf{p}) = (I, \mathbf{0}).$$

Thus S is the two-sided inverse. All group axioms are verified. $\square$

Before speaking of a Lie algebra we need a definition of Lie group that can be used without building differentiable manifolds. We adopt the *matrix* definition, standard in introductory texts (Hall, *An Elementary Introduction to Groups and Representations*, Definition 2.2): a matrix Lie group is a subgroup of $GL(n, \mathbb{R})$ that is *closed* in $GL(n, \mathbb{R})$.

Definition 29.2: Matrix Lie group. Let $GL(n, \mathbb{R}) = \{A \in \mathbb{R}^{n \times n} : \det A \neq 0\}$, with the topology induced from $\mathbb{R}^{n \times n}$. A subgroup $G \subseteq GL(n, \mathbb{R})$ is a **matrix Lie group** if it is closed in $GL(n, \mathbb{R})$: every sequence $(A_k) \subset G$ that converges to some $A \in GL(n, \mathbb{R})$ has limit $A \in G$.

Theorem 29.1: $SE(3)$ is a matrix Lie group. $SE(3)$, viewed as a set of 4×4 matrices, is a closed subgroup of $GL(4, \mathbb{R})$; in particular it is a matrix Lie group.

Proof. **Step 1 — $SE(3) \subseteq GL(4, \mathbb{R})$.** Expanding the determinant along the last row of the homogeneous matrix gives $\det T = 1 \cdot \det R = 1 \neq 0$.

Step 2 — it is a subgroup. This is Proposition 29.1: $SE(3)$ is closed under the product and the inverse and contains I_4 .

Step 3 — it is closed. Let $T_k = \begin{pmatrix} R_k & \mathbf{p}_k \\ \mathbf{0}^\top & 1 \end{pmatrix} \in SE(3)$ with $T_k \rightarrow T \in GL(4, \mathbb{R})$. Convergence in $\mathbb{R}^{4 \times 4}$ is entrywise convergence; hence the last row of T is the limit of the last rows, that is $(0, 0, 0, 1)$, and T has the form $\begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix}$ with $R_k \rightarrow R$ and $\mathbf{p}_k \rightarrow \mathbf{p}$. The maps $A \mapsto A^\top A$ and $A \mapsto \det A$ are continuous (they are polynomial in the entries), so

$$R^\top R = \lim_k R_k^\top R_k = I, \quad \det R = \lim_k \det R_k = 1,$$

that is $R \in SO(3)$ and $T \in SE(3)$. Closedness is proved, and with it the statement. $\square$

Theorem 29.2: $SE(3)$ as a semidirect product. Let $N = \{(I, \mathbf{p}) : \mathbf{p} \in \mathbb{R}^3\}$ be the subset of translations and $H = \{(R, \mathbf{0}) : R \in SO(3)\}$ that of rotations about the origin. Then:

- (i) N is a normal subgroup of $SE(3)$, isomorphic to the additive group $(\mathbb{R}^3, +)$;
- (ii) H is a subgroup isomorphic to $SO(3)$;
- (iii) $N \cap H = \{(I, \mathbf{0})\}$ and $NH = SE(3)$;
- (iv) every $g \in SE(3)$ can be written uniquely as $g = (I, \mathbf{p})(R, \mathbf{0})$, and the product is

computed by the rule

$$(\mathbf{p}_1, R_1) \cdot (\mathbf{p}_2, R_2) = (\mathbf{p}_1 + R_1 \mathbf{p}_2, R_1 R_2).$$

Hence $SE(3) = \mathbb{R}^3 \rtimes SO(3)$, a semidirect product with respect to the natural action of $SO(3)$ on $\mathbb{R}^3$.

Proof. (i). That N is a subgroup follows from Proposition 29.1: $(I, \mathbf{p}_1)(I, \mathbf{p}_2) = (I, \mathbf{p}_1 + \mathbf{p}_2)$ and $(I, \mathbf{p})^{-1} = (I, -\mathbf{p})$. The same formula shows that $\mathbf{p} \mapsto (I, \mathbf{p})$ is an isomorphism of $(\mathbb{R}^3, +)$ onto N (it is a bijection and turns the sum into the product). For normality we compute the conjugate:

$$(R, \mathbf{q})(I, \mathbf{p})(R, \mathbf{q})^{-1} = (R, R\mathbf{p} + \mathbf{q})(R^\top, -R^\top \mathbf{q}) = (RR^\top, -RR^\top \mathbf{q} + R\mathbf{p} + \mathbf{q}) = (I, R\mathbf{p}),$$

which belongs to N . Hence $gNg^{-1} \subseteq N$ for every g , and applying the same inclusion to g^{-1} gives $gNg^{-1} = N$: N is normal. Note already the key formula

$$(R, \mathbf{q})(I, \mathbf{p})(R, \mathbf{q})^{-1} = (I, R\mathbf{p}), \quad (29.1)$$

which is exactly the action of $SO(3)$ on $\mathbb{R}^3$ by rotation.

(ii). $(R_1, \mathbf{0})(R_2, \mathbf{0}) = (R_1 R_2, \mathbf{0})$ and $(R, \mathbf{0})^{-1} = (R^\top, \mathbf{0})$: H is a subgroup and $R \mapsto (R, \mathbf{0})$ is an isomorphism onto $SO(3)$.

(iii). If $(I, \mathbf{p}) = (R, \mathbf{0})$ then $R = I$ and $\mathbf{p} = \mathbf{0}$: the intersection is trivial. Moreover, for every $(R, \mathbf{p})$,

$$(I, \mathbf{p})(R, \mathbf{0}) = (R, \mathbf{p}),$$

hence $NH = SE(3)$.

(iv). Existence is the formula just written. For uniqueness, if $(I, \mathbf{p})(R, \mathbf{0}) = (I, \mathbf{p}')(R', \mathbf{0})$ then $(R, \mathbf{p}) = (R', \mathbf{p}')$, that is $R = R'$ and $\mathbf{p} = \mathbf{p}'$. Identifying g with the pair $(\mathbf{p}, R)$ and again using Proposition 29.1:

$$(\mathbf{p}_1, R_1)(\mathbf{p}_2, R_2) = (R_1, \mathbf{p}_1)(R_2, \mathbf{p}_2) = (R_1 R_2, R_1 \mathbf{p}_2 + \mathbf{p}_1) \longleftrightarrow (\mathbf{p}_1 + R_1 \mathbf{p}_2, R_1 R_2).$$

This is, by definition, the multiplication of the semidirect product $\mathbb{R}^3 \rtimes SO(3)$ associated with the homomorphism $SO(3) \rightarrow \text{Aut}(\mathbb{R}^3)$, $R \mapsto (\mathbf{p} \mapsto R\mathbf{p})$, which is a homomorphism because $(R_1 R_2)\mathbf{p} = R_1(R_2 \mathbf{p})$. $\square$

Remark 29.1: Why “semidirect” and not “direct”. If the product were direct one would have $(\mathbf{p}_1, R_1)(\mathbf{p}_2, R_2) = (\mathbf{p}_1 + \mathbf{p}_2, R_1 R_2)$: translations and rotations would not see each other. Instead $R_1 \mathbf{p}_2$ appears: the translation of the second factor gets rotated by the first. It is the same asymmetry by which “advance one metre, then turn 90° ” is not the same command as “turn 90° , then advance one metre”; and it is the reason a robotic arm cannot be programmed by separately summing rotations and translations. Formally: H is not normal in $SE(3)$, because for $\mathbf{q} \neq \mathbf{0}$ and $R \neq I$

$$(I, \mathbf{q})(R, \mathbf{0})(I, -\mathbf{q}) = (R, \mathbf{q} - R\mathbf{q}) \notin H \quad \text{if } R\mathbf{q} \neq \mathbf{q}.$$

Remark 29.2: What we import: from the closed subgroup to the manifold. With Definition 29.2, Theorem 29.1 is complete: everything needed in the rest of the chapter (Lie algebra, exponential, one-parameter subgroups) is built from closedness alone. What remains, however, is an **imported** statement, used only to connect the matrix language with that of differentiable manifolds:

Closed subgroup theorem (Cartan). *Every closed subgroup of a Lie group is a closed embedded submanifold, hence a Lie group in the abstract sense, and its Lie algebra coincides with $\{X : e^{tX} \in G \ \forall t \in \mathbb{R}\}$.*

What is missing to prove it here: *the construction of local charts by means of the exponential and the technical lemma that rules out “spurious directions” of the group (in Lee’s proof: that a sequence $h_i \rightarrow e$ in H produces, after reparametrisation, a vector of the algebra), which require the inverse function theorem on manifolds — tools outside the scope of this volume.*
Source: J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer (GTM 218), New York 2013, Theorem 20.12, p. 522 (see the “Sources” section). In what follows the imported result is never used: when we write “ $\mathfrak{se}(3)$ has dimension 6” we mean the dimension as a real vector space, which is computed explicitly in Theorem 29.3.

29.2. The Lie algebra $\mathfrak{se}(3)$ and twists

Definition 29.3: Hat and vee maps on $\mathbb{R}^3$. For $\omega = (\omega_1, \omega_2, \omega_3)^\top \in \mathbb{R}^3$ set

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3},$$

the unique matrix such that $[\omega]_\times \mathbf{x} = \omega \times \mathbf{x}$ for every $\mathbf{x}$. The map $\omega \mapsto [\omega]_\times$ is a linear isomorphism of $\mathbb{R}^3$ onto the space $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3 \times 3} : A^\top = -A\}$ of skew-symmetric matrices; the inverse map is denoted $A \mapsto A^\vee$.

That it is an isomorphism is seen at once: the map is linear in the entries, it is injective (if $[\omega]_\times = 0$ then $\omega = 0$, reading off the entries), and $\mathfrak{so}(3)$ has dimension 3 (a 3×3 skew-symmetric matrix has zero diagonal and three free entries above the diagonal), equal to that of $\mathbb{R}^3$.

Lemma 29.1: Powers of $[\omega]_\times$. For every $\omega \in \mathbb{R}^3$, setting $K = [\omega]_\times$:

$$(i) \ K^2 = \omega \omega^\top - \|\omega\|^2 I, \quad (ii) \ K^3 = -\|\omega\|^2 K, \quad (iii) \ K \omega = 0.$$

In particular, if $\|\omega\| = 1$ then $K^3 = -K$, $K^4 = -K^2$, $K^5 = K$, and in general the powers of K reduce cyclically to K and K^2 .

Proof. (iii) $K \omega = \omega \times \omega = 0$.

(i) For every $\mathbf{x} \in \mathbb{R}^3$,

$$K^2 \mathbf{x} = \omega \times (\omega \times \mathbf{x}) = \omega(\omega \cdot \mathbf{x}) - \mathbf{x}(\omega \cdot \omega) = (\omega \omega^\top - \|\omega\|^2 I) \mathbf{x},$$

where the double cross product identity (prerequisite) was used with $\mathbf{a} = \mathbf{b} = \omega$, $\mathbf{c} = \mathbf{x}$. Two matrices agreeing on every $\mathbf{x}$ are equal.

(ii) Multiplying (i) on the left by K :

$$K^3 = K \omega \omega^\top - \|\omega\|^2 K = (K \omega) \omega^\top - \|\omega\|^2 K = -\|\omega\|^2 K,$$

because $K \omega = 0$ by (iii).

For $\|\omega\| = 1$: $K^3 = -K$; hence $K^4 = K \cdot K^3 = -K^2$, $K^5 = K \cdot K^4 = -K^3 = K$, and the sequence of powers is periodic with period 4 starting from K . $\square$

Lemma 29.2: *The hat map turns the cross product into the commutator. For every $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$,*

$$[[\mathbf{a}]_{\times}, [\mathbf{b}]_{\times}] := [\mathbf{a}]_{\times}[\mathbf{b}]_{\times} - [\mathbf{b}]_{\times}[\mathbf{a}]_{\times} = [\mathbf{a} \times \mathbf{b}]_{\times}.$$

Hence $\mathfrak{so}(3)$, with the commutator, is a Lie algebra isomorphic to $(\mathbb{R}^3, \times)$.

Proof. Apply both sides to a generic $\mathbf{x} \in \mathbb{R}^3$ and use the double cross product identity twice:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{x}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{x}) - \mathbf{x}(\mathbf{a} \cdot \mathbf{b}), \quad \mathbf{b} \times (\mathbf{a} \times \mathbf{x}) = \mathbf{a}(\mathbf{b} \cdot \mathbf{x}) - \mathbf{x}(\mathbf{a} \cdot \mathbf{b}).$$

Subtracting, the terms in $\mathbf{x}(\mathbf{a} \cdot \mathbf{b})$ cancel:

$$[[\mathbf{a}]_{\times}, [\mathbf{b}]_{\times}]\mathbf{x} = \mathbf{b}(\mathbf{a} \cdot \mathbf{x}) - \mathbf{a}(\mathbf{b} \cdot \mathbf{x}).$$

On the other hand, again by the same identity applied to $\mathbf{x} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\mathbf{x} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{x} \cdot \mathbf{a})$, one has

$$[\mathbf{a} \times \mathbf{b}]_{\times}\mathbf{x} = (\mathbf{a} \times \mathbf{b}) \times \mathbf{x} = -\mathbf{x} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{x}) - \mathbf{a}(\mathbf{b} \cdot \mathbf{x}).$$

The two results agree for every $\mathbf{x}$, hence the matrices are equal. Since $\omega \mapsto [\omega]_{\times}$ is a linear isomorphism (Definition 29.3) that turns $\times$ into $[\cdot, \cdot]$, it is an isomorphism of Lie algebras. $\square$

Example 29.1: The commutator of the canonical generators. With $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ the canonical basis:

$$[\mathbf{e}_1]_{\times} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad [\mathbf{e}_2]_{\times} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad [\mathbf{e}_3]_{\times} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The two products, computed row by column, are

$$[\mathbf{e}_1]_{\times}[\mathbf{e}_2]_{\times} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$[\mathbf{e}_2]_{\times}[\mathbf{e}_1]_{\times} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

(for instance, entry (1, 2) of the second product is $0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 = 1$). The difference is

$$[[\mathbf{e}_1]_{\times}, [\mathbf{e}_2]_{\times}] = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = [\mathbf{e}_3]_{\times} = [\mathbf{e}_1 \times \mathbf{e}_2]_{\times},$$

in agreement with Lemma 29.2. $\checkmark$

Structural check. The commutator of two skew-symmetric matrices is skew-symmetric: $[A, B]^{\top} = B^{\top}A^{\top} - A^{\top}B^{\top} = BA - AB = -[A, B]$. If the result of such a computation comes out symmetric, the computation is wrong — and it is the most frequent sign error, because $[\mathbf{e}_2]_{\times}[\mathbf{e}_1]_{\times}$ has its single 1 *above* the diagonal, not below. The same three matrices satisfy, by the same lemma, $[[\mathbf{e}_2]_{\times}, [\mathbf{e}_3]_{\times}] = [\mathbf{e}_1]_{\times}$ and $[[\mathbf{e}_3]_{\times}, [\mathbf{e}_1]_{\times}] = [\mathbf{e}_2]_{\times}$: these are the structure relations of $\mathfrak{so}(3)$. Independent check in double precision on 2 000 randomly drawn pairs $(\mathbf{a}, \mathbf{b})$: maximum discrepancy $|[[\mathbf{a}]_{\times}, [\mathbf{b}]_{\times}] - [\mathbf{a} \times \mathbf{b}]_{\times}| = 0$ (the entries are sums and differences of exact floating-point products). $\checkmark$

Definition 29.4: The Lie algebra of a matrix Lie group. Let $G \subseteq GL(n, \mathbb{R})$ be a matrix Lie group. Its **Lie algebra** is

$$\mathfrak{g} = \{X \in \mathbb{R}^{n \times n} : e^{tX} \in G \text{ for every } t \in \mathbb{R}\}.$$

For $G = SE(3)$ one writes $\mathfrak{se}(3)$; for $G = SO(3)$ one writes $\mathfrak{so}(3)$ (the notation is consistent, as shown by Theorem 29.3).

Theorem 29.3: Structure of $\mathfrak{se}(3)$: twists. With $SE(3)$ as in Definition 29.1 and $\mathfrak{se}(3)$ as in Definition 29.4,

$$\mathfrak{se}(3) = \left\{ [\xi] := \begin{pmatrix} [\omega]_{\times} & \mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix} : \omega, \mathbf{v} \in \mathbb{R}^3 \right\},$$

and likewise $\mathfrak{so}(3) = \{[\omega]_{\times} : \omega \in \mathbb{R}^3\}$ is the space of skew-symmetric matrices. In particular:

- (i) $\mathfrak{se}(3)$ is a real vector subspace of $\mathbb{R}^{4 \times 4}$ of dimension 6; the map $\xi = (\omega, \mathbf{v}) \mapsto [\xi]$ is a linear isomorphism $\mathbb{R}^6 \rightarrow \mathfrak{se}(3)$, and ξ is called a **twist** (or twist coordinate vector);
- (ii) $\mathfrak{se}(3)$ is closed under the commutator, and

$$[[\xi_1], [\xi_2]] = [\xi_3], \quad \xi_3 = (\omega_1 \times \omega_2, \omega_1 \times \mathbf{v}_2 - \omega_2 \times \mathbf{v}_1);$$

- (iii) $\{[\xi] : \omega = 0\}$ is an abelian ideal of $\mathfrak{se}(3)$ isomorphic to $\mathbb{R}^3$: it is the infinitesimal counterpart of the normal subgroup of translations of Theorem 29.2.

Proof. Step 1 — structure of the powers. Let $X = \begin{pmatrix} A & \mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix}$ with $A \in \mathbb{R}^{3 \times 3}$. By induction on $k \geq 1$,

$$X^k = \begin{pmatrix} A^k & A^{k-1}\mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix}. \quad (29.2)$$

The case $k = 1$ is the definition (with $A^0 = I$). If it holds for k , then

$$X^{k+1} = X^k X = \begin{pmatrix} A^k & A^{k-1}\mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix} \begin{pmatrix} A & \mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix} = \begin{pmatrix} A^{k+1} & A^k\mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix}.$$

Summing the exponential series gives, for every $t \in \mathbb{R}$,

$$e^{tX} = I_4 + \sum_{k \geq 1} \frac{t^k}{k!} X^k = \begin{pmatrix} e^{tA} & \left(\sum_{k \geq 1} \frac{t^k}{k!} A^{k-1} \right) \mathbf{v} \\ \mathbf{0}^{\top} & 1 \end{pmatrix}, \quad (29.3)$$

where the block series converges because the series for e^{tX} converges and the blocks are continuous projections.

Step 2 — inclusion $\supseteq$. Let $A = [\omega]_{\times}$, so $A^{\top} = -A$. Then

$$(e^{tA})^{\top} e^{tA} = e^{tA^{\top}} e^{tA} = e^{-tA} e^{tA} = e^{-tA+tA} = I,$$

where the penultimate step uses that $-tA$ and tA commute, and

$$\det e^{tA} = e^{t \operatorname{tr} A} = e^0 = 1$$

because a skew-symmetric matrix has zero diagonal. Hence $e^{tA} \in SO(3)$ and, by (29.3), $e^{tX} \in SE(3)$ for every t : $X \in \mathfrak{se}(3)$.

Step 3 — inclusion $\subseteq$. Let $X \in \mathbb{R}^{4 \times 4}$ with $e^{tX} \in SE(3)$ for every t . Differentiating at $t = 0$ (the exponential series can be differentiated term by term, a prerequisite) gives $X = \frac{d}{dt}\big|_{t=0} e^{tX}$. The last row of e^{tX} is constantly $(0, 0, 0, 1)$, hence its derivative vanishes and the last row of X is $(0, 0, 0, 0)$. Therefore $X = \begin{pmatrix} A & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix}$, and by (29.3) the top-left 3×3 block of e^{tX} is e^{tA} , which must therefore lie in $SO(3)$ for every t . From $(e^{tA})^\top e^{tA} = I$, differentiating at $t = 0$ with the product rule,

$$0 = \frac{d}{dt}\bigg|_{t=0} (e^{tA^\top} e^{tA}) = A^\top + A,$$

so A is skew-symmetric, that is $A = [\omega]_\times$ for a unique ω (Definition 29.3). The two inclusions give equality. The same argument, applied in $\mathbb{R}^{3 \times 3}$, gives $\mathfrak{so}(3) = \{[\omega]_\times\}$.

(i). The set described is the image of the injective linear map $(\omega, \mathbf{v}) \mapsto [\xi]$, hence it is a subspace of dimension $3 + 3 = 6$; injectivity follows from that of the hat map.

(ii). With $X_i = \begin{pmatrix} [\omega_i]_\times & \mathbf{v}_i \\ \mathbf{0}^\top & 0 \end{pmatrix}$ one computes

$$X_1 X_2 = \begin{pmatrix} [\omega_1]_\times [\omega_2]_\times & [\omega_1]_\times \mathbf{v}_2 \\ \mathbf{0}^\top & 0 \end{pmatrix}, \quad X_2 X_1 = \begin{pmatrix} [\omega_2]_\times [\omega_1]_\times & [\omega_2]_\times \mathbf{v}_1 \\ \mathbf{0}^\top & 0 \end{pmatrix},$$

hence

$$[X_1, X_2] = \begin{pmatrix} [[\omega_1]_\times, [\omega_2]_\times] & \omega_1 \times \mathbf{v}_2 - \omega_2 \times \mathbf{v}_1 \\ \mathbf{0}^\top & 0 \end{pmatrix} = \begin{pmatrix} [\omega_1 \times \omega_2]_\times & \omega_1 \times \mathbf{v}_2 - \omega_2 \times \mathbf{v}_1 \\ \mathbf{0}^\top & 0 \end{pmatrix},$$

using Lemma 29.2. The result has the required form, hence $\mathfrak{se}(3)$ is closed under the commutator.

(iii). Let $\mathfrak{n} = \{[\xi] : \omega = 0\}$. It is a subspace of dimension 3. By (ii), with ω_1 arbitrary and $\omega_2 = 0$,

$$[[\xi_1], [\xi_2]] = (0, \omega_1 \times \mathbf{v}_2)^\wedge \in \mathfrak{n},$$

hence $[\mathfrak{se}(3), \mathfrak{n}] \subseteq \mathfrak{n}$: $\mathfrak{n}$ is an ideal. Taking also $\omega_1 = 0$ gives a zero commutator: $\mathfrak{n}$ is abelian. The isomorphism with $\mathbb{R}^3$ is $\mathbf{v} \mapsto [(0, \mathbf{v})]$. $\square$

Remark 29.3: What a twist “is”. If a rigid body moves according to $T(t) \in SE(3)$, the matrix $[\mathcal{V}] = \dot{T}T^{-1}$ belongs to $\mathfrak{se}(3)$: indeed the curve $s \mapsto T(t+s)T(t)^{-1}$ passes through the identity at $s = 0$ and has derivative $\dot{T}T^{-1}$, and one checks directly that $\dot{T}T^{-1} = \begin{pmatrix} \dot{R}R^\top & \dot{\mathbf{p}} - \dot{R}R^\top \mathbf{p} \\ \mathbf{0}^\top & 0 \end{pmatrix}$ with $\dot{R}R^\top$ skew-symmetric (differentiating $RR^\top = I$). Its coordinates $\mathcal{V} = (\omega, \mathbf{v})$ are the angular velocity of the body and the velocity of the (possibly imaginary) body point that, at that instant, passes through the origin. The component $\mathbf{v}$ is not the velocity of the centre of the tool: it is the velocity of the body-fixed point that occupies the origin of the fixed frame, and this is the first stumbling block for anyone reading a Jacobian table without knowing what it contains.

Lemma 29.3: Conjugation of a skew-symmetric matrix. For every $R \in SO(3)$ and $\omega \in \mathbb{R}^3$,

$$R[\omega]_\times R^\top = [R\omega]_\times.$$

Proof. **Step 1 — equivariance of the cross product.** For $R \in SO(3)$ and $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$, the scalar triple product is a determinant:

$$((R\mathbf{a}) \times (R\mathbf{b})) \cdot R\mathbf{c} = \det \begin{bmatrix} R\mathbf{a} & R\mathbf{b} & R\mathbf{c} \end{bmatrix} = \det (R[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]) = \det R \cdot \det[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c},$$

since $\det R = 1$. On the other hand $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (R(\mathbf{a} \times \mathbf{b})) \cdot R\mathbf{c}$, because R preserves the scalar product. Hence $[(R\mathbf{a}) \times (R\mathbf{b}) - R(\mathbf{a} \times \mathbf{b})] \cdot R\mathbf{c} = 0$ for every $\mathbf{c}$; since R is invertible, $R\mathbf{c}$ ranges over all of $\mathbb{R}^3$ and the vector in square brackets vanishes:

$$(R\mathbf{a}) \times (R\mathbf{b}) = R(\mathbf{a} \times \mathbf{b}). \quad (29.4)$$

Step 2 — conclusion. For every $\mathbf{x}$, using $R^\top = R^{-1}$ and (29.4) with $\mathbf{a} = \boldsymbol{\omega}$, $\mathbf{b} = R^\top \mathbf{x}$:

$$R[\boldsymbol{\omega}]_\times R^\top \mathbf{x} = R(\boldsymbol{\omega} \times R^\top \mathbf{x}) = (R\boldsymbol{\omega}) \times (RR^\top \mathbf{x}) = (R\boldsymbol{\omega}) \times \mathbf{x} = [R\boldsymbol{\omega}]_\times \mathbf{x}.$$

□

Lemma 29.4: Exponential of a conjugate. If $S \in GL(n, \mathbb{R})$ and $X \in \mathbb{R}^{n \times n}$, then $e^{SXS^{-1}} = S e^X S^{-1}$.

Proof. By induction, $(SXS^{-1})^k = SX^k S^{-1}$: true for $k = 0, 1$, and $(SXS^{-1})^{k+1} = SX^k S^{-1} \cdot SXS^{-1} = SX^{k+1} S^{-1}$. Since the exponential series converges absolutely and $Y \mapsto SY S^{-1}$ is linear and continuous, it can be exchanged with the sum:

$$e^{SXS^{-1}} = \sum_{k \geq 0} \frac{1}{k!} SX^k S^{-1} = S \left(\sum_{k \geq 0} \frac{X^k}{k!} \right) S^{-1} = S e^X S^{-1}.$$

□

Proposition 29.2: The adjoint Ad_g . For $g = (R, \mathbf{p}) \in SE(3)$ and $\xi = (\boldsymbol{\omega}, \mathbf{v}) \in \mathbb{R}^6$,

$$g[\xi]g^{-1} = [\text{Ad}_g \xi], \quad \text{Ad}_g = \begin{pmatrix} R & 0 \\ [\mathbf{p}]_\times R & R \end{pmatrix} \in \mathbb{R}^{6 \times 6}.$$

Moreover $\text{Ad}_{g_1 g_2} = \text{Ad}_{g_1} \text{Ad}_{g_2}$, $\text{Ad}_I = I_6$ and Ad_g is invertible with $(\text{Ad}_g)^{-1} = \text{Ad}_{g^{-1}}$. Finally

$$g e^{[\xi]\theta} g^{-1} = e^{[\text{Ad}_g \xi]\theta} \quad \text{for every } \theta \in \mathbb{R}.$$

Proof. Step 1 — block computation. With $g^{-1} = (R^\top, -R^\top \mathbf{p})$ (Proposition 29.1):

$$g[\xi] = \begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix} \begin{pmatrix} [\boldsymbol{\omega}]_\times & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix} = \begin{pmatrix} R[\boldsymbol{\omega}]_\times & R\mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix},$$

$$g[\xi]g^{-1} = \begin{pmatrix} R[\boldsymbol{\omega}]_\times & R\mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix} \begin{pmatrix} R^\top & -R^\top \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix} = \begin{pmatrix} R[\boldsymbol{\omega}]_\times R^\top & -R[\boldsymbol{\omega}]_\times R^\top \mathbf{p} + R\mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix}.$$

By Lemma 29.3, $R[\boldsymbol{\omega}]_\times R^\top = [R\boldsymbol{\omega}]_\times$; hence the result is an element of $\mathfrak{se}(3)$ with angular part $R\boldsymbol{\omega}$ and linear part

$$-[R\boldsymbol{\omega}]_\times \mathbf{p} + R\mathbf{v} = -(R\boldsymbol{\omega}) \times \mathbf{p} + R\mathbf{v} = \mathbf{p} \times (R\boldsymbol{\omega}) + R\mathbf{v} = [\mathbf{p}]_\times R\boldsymbol{\omega} + R\mathbf{v},$$

where the antisymmetry of the cross product was used. In coordinates: $\text{Ad}_g \xi = (R\boldsymbol{\omega}, [\mathbf{p}]_\times R\boldsymbol{\omega} + R\mathbf{v})$, that is the block matrix of the statement.

Step 2 — homomorphism. For every ξ ,

$$[\text{Ad}_{g_1 g_2} \xi] = (g_1 g_2)[\xi](g_1 g_2)^{-1} = g_1(g_2[\xi]g_2^{-1})g_1^{-1} = g_1[\text{Ad}_{g_2} \xi]g_1^{-1} = [\text{Ad}_{g_1} \text{Ad}_{g_2} \xi],$$

and since $\eta \mapsto [\eta]$ is injective, one concludes $\text{Ad}_{g_1 g_2} = \text{Ad}_{g_1} \text{Ad}_{g_2}$. With $g_1 = g$, $g_2 = g^{-1}$ and $\text{Ad}_I = I_6$ (evident from the formula, with $R = I$, $\mathbf{p} = \mathbf{0}$), invertibility follows.

Step 3 — exponential. By Lemma 29.4 with $S = g$ and $X = [\xi]\theta$:

$$g e^{[\xi]\theta} g^{-1} = e^{g[\xi]\theta g^{-1}} = e^{[\text{Ad}_g \xi]\theta}.$$

Independent numerical check on 3000 randomly drawn pairs of elements of $SE(3)$:
 $\max |\text{Ad}_{g_1 g_2} - \text{Ad}_{g_1} \text{Ad}_{g_2}| = 5.3 \cdot 10^{-15}$ and $\max |g[\xi]g^{-1} - [\text{Ad}_g \xi]| = 5.6 \cdot 10^{-16}$. ✓ $\square$

29.3. The exponential of a twist in closed form

The series $e^{[\xi]\theta} = \sum_k \frac{\theta^k}{k!} [\xi]^k$ cannot be evaluated by hand term by term, but the powers of $[\omega]_\times$ repeat cyclically (Lemma 29.1), and the series collapses into just three terms. The result is the $SE(3)$ counterpart of Rodrigues' formula.

Theorem 29.4: Exponential of a twist. Let $\xi = (\omega, \mathbf{v}) \in \mathbb{R}^6$ and $\theta \in \mathbb{R}$.

(a) (Purely translational case.) If $\omega = \mathbf{0}$, then

$$e^{[\xi]\theta} = \begin{pmatrix} I & \mathbf{v}\theta \\ \mathbf{0}^\top & 1 \end{pmatrix}.$$

(b) (General case, $\|\omega\| = 1$.) If $\|\omega\| = 1$, then

$$e^{[\xi]\theta} = \begin{pmatrix} e^{[\omega]_\times \theta} & G(\theta)\mathbf{v} \\ \mathbf{0}^\top & 1 \end{pmatrix},$$

where

$$e^{[\omega]_\times \theta} = I + \sin \theta [\omega]_\times + (1 - \cos \theta) [\omega]_\times^2 \quad (\text{Rodrigues}),$$

$$G(\theta) = I\theta + (1 - \cos \theta) [\omega]_\times + (\theta - \sin \theta) [\omega]_\times^2.$$

(c) (Reduction.) If $\omega \neq \mathbf{0}$, setting $\hat{\omega} = \omega/\|\omega\|$, $\hat{\mathbf{v}} = \mathbf{v}/\|\omega\|$ and $\hat{\theta} = \|\omega\|\theta$, one has
 $[\xi]\theta = [\hat{\xi}]\hat{\theta}$ with $\hat{\xi} = (\hat{\omega}, \hat{\mathbf{v}})$, which reduces to (b).

In every case $e^{[\xi]\theta} \in SE(3)$.

Proof. (a). If $\omega = \mathbf{0}$ then $[\xi] = \begin{pmatrix} 0 & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix}$ and by (29.2) (with $A = 0$) $[\xi]^2 = 0$, hence $[\xi]^k = 0$ for $k \geq 2$. The series stops at first order: $e^{[\xi]\theta} = I_4 + \theta[\xi]$, which is the matrix of the statement.
(b). Start from (29.3) with $A = [\omega]_\times$ and $t = \theta$:

$$e^{[\xi]\theta} = \begin{pmatrix} e^{[\omega]_\times \theta} & G(\theta)\mathbf{v} \\ \mathbf{0}^\top & 1 \end{pmatrix}, \quad G(\theta) = \sum_{k \geq 1} \frac{\theta^k}{k!} [\omega]_\times^{k-1}.$$

The angular block is Rodrigues' formula, already proved in Chapter 28 (Theorem 28.2); it is rederived here anyway, for completeness, with the same grouping that follows.

Write $K = [\omega]_\times$ and split the terms according to the residue of $k - 1$ modulo 4, using Lemma 29.1 with $\|\omega\| = 1$:

$$K^0 = I, \quad K^1 = K, \quad K^2, \quad K^3 = -K, \quad K^4 = -K^2, \quad K^5 = K, \dots$$

Hence

$$G(\theta) = \underbrace{\theta I}_{k=1} + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \right)}_{k=2,4,6,\dots} K + \underbrace{\left(\frac{\theta^3}{3!} - \frac{\theta^5}{5!} + \frac{\theta^7}{7!} - \dots \right)}_{k=3,5,7,\dots} K^2.$$

The two coefficients are well-known power series:

$$\sum_{m \geq 1} (-1)^{m+1} \frac{\theta^{2m}}{(2m)!} = 1 - \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) = 1 - \cos \theta,$$

$$\sum_{m \geq 1} (-1)^{m+1} \frac{\theta^{2m+1}}{(2m+1)!} = \theta - \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) = \theta - \sin \theta.$$

Substituting gives exactly $G(\theta) = I\theta + (1 - \cos \theta)K + (\theta - \sin \theta)K^2$. The rearrangement is legitimate because the exponential series converges absolutely.

With the same grouping applied to $e^{K\theta} = \sum_{k \geq 0} \frac{\theta^k}{k!} K^k$ one finds

$$e^{K\theta} = I + \left(\theta - \frac{\theta^3}{3!} + \cdots\right)K + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \cdots\right)K^2 = I + \sin \theta K + (1 - \cos \theta)K^2.$$

(c). $[\xi]\theta$ is a matrix, and one checks directly that

$$[\hat{\xi}]\hat{\theta} = \begin{pmatrix} [\omega/\|\omega\|]_{\times} & \mathbf{v}/\|\omega\| \\ \mathbf{0}^{\top} & 0 \end{pmatrix} \|\omega\|\theta = \begin{pmatrix} [\omega]_{\times}\theta & \mathbf{v}\theta \\ \mathbf{0}^{\top} & 0 \end{pmatrix} = [\xi]\theta,$$

using the linearity of the hat map. The exponentials agree because the matrices agree.

Membership in $SE(3)$. This is Step 2 of the proof of Theorem 29.3. $\square$

Example 29.2: Numerical verification of the closed form. Take

$$\omega = \frac{1}{3}(1, 2, 2)^{\top} \ (\|\omega\| = 1), \quad \mathbf{v} = (1, 0, -1)^{\top}, \quad \theta = 1.2.$$

The closed form of Theorem 29.4(b) gives

$$e^{[\xi]\theta} = \begin{pmatrix} 0.433207 & -0.479661 & 0.763058 & 0.477171 \\ 0.763058 & 0.645754 & -0.027283 & 0.578095 \\ -0.479661 & 0.594076 & 0.645754 & -1.416681 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

First check (independent). The same matrix, computed from the exponential series to 40 decimal digits (`mpmath`), agrees with the above with maximum discrepancy $5.6 \cdot 10^{-17}$: the closed form and the sum of the series give the same result. $\checkmark$

Second check (structural). The 3×3 block must lie in $SO(3)$: $\max |R^{\top}R - I| = 1.1 \cdot 10^{-16}$ and $\det R = 1.000000$. $\checkmark$

Case $\omega = \mathbf{0}$. With $\mathbf{v} = (1, 0, -1)/\sqrt{2}$ and $\theta = 2.5$, part (a) predicts the translation $\mathbf{v}\theta = (1.767767, 0, -1.767767)$; the series to 40 digits gives the same vector with discrepancy 0 (the series has only two terms). $\checkmark$

The matrix $G(\theta)$ is the delicate point of the inverse passage, that is, of the *logarithm*: given a rigid motion, reconstructing the twist that generates it requires inverting $G(\theta)$. The next result states exactly when this is possible and with which formula.

Proposition 29.3: Invertibility of $G(\theta)$ and explicit formula. Let $\|\omega\| = 1$, $K = [\omega]_{\times}$ and $G(\theta) = I\theta + (1 - \cos \theta)K + (\theta - \sin \theta)K^2$. Then

$$\det G(\theta) = 4\theta \sin^2 \frac{\theta}{2},$$

so $G(\theta)$ is invertible if and only if $\theta \neq 0$ and $\theta \notin 2\pi\mathbb{Z}$; in particular for $\theta \in (0, 2\pi)$. In that

case

$$G(\theta)^{-1} = \frac{1}{\theta}I - \frac{1}{2}K + \left(\frac{1}{\theta} - \frac{1}{2}\cot\frac{\theta}{2}\right)K^2.$$

Proof. Step 1 — orthogonal decomposition. Since $\|\omega\| = 1$, Lemma 29.1 gives $K\omega = 0$ and $K^2 = \omega\omega^\top - I$. Decompose $\mathbb{R}^3 = \langle\omega\rangle \oplus \omega^\perp$ (orthogonal sum). On the two summands:

- on $\langle\omega\rangle$: $K\omega = 0$ and $K^2\omega = 0$, hence $G(\theta)\omega = \theta\omega$: G acts as $\theta \text{ id}$;
- on $\omega^\perp$: if $\omega \cdot \mathbf{x} = 0$ then $K^2\mathbf{x} = \omega(\omega \cdot \mathbf{x}) - \mathbf{x} = -\mathbf{x}$, hence $K^2 = -I$ on $\omega^\perp$; moreover K maps $\omega^\perp$ into itself (because $\omega \cdot (\omega \times \mathbf{x}) = 0$). Hence

$$G(\theta)|_{\omega^\perp} = \theta I + (1 - \cos\theta)K - (\theta - \sin\theta)I = \sin\theta I + (1 - \cos\theta)K.$$

Both subspaces are invariant under $G(\theta)$, so the determinant is the product of the determinants of the restrictions.

Step 2 — the determinant. On $\langle\omega\rangle$ (dimension 1) the determinant is θ . On $\omega^\perp$ (dimension 2), $J := K|_{\omega^\perp}$ satisfies $J^2 = -I$: fixing an orthonormal basis $(\mathbf{u}, K\mathbf{u})$ of $\omega^\perp$ — orthonormal because $\|K\mathbf{u}\| = \|\omega \times \mathbf{u}\| = \|\mathbf{u}\|$ and $\mathbf{u} \cdot K\mathbf{u} = 0$ — the matrix of J is $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and

$$\det(\sin\theta I + (1 - \cos\theta)J) = \det\begin{pmatrix} \sin\theta & -(1 - \cos\theta) \\ 1 - \cos\theta & \sin\theta \end{pmatrix},$$

$$\det(\sin\theta I + (1 - \cos\theta)J) = \sin^2\theta + (1 - \cos\theta)^2 = 2 - 2\cos\theta = 4\sin^2\frac{\theta}{2}.$$

Hence $\det G(\theta) = \theta \cdot 4\sin^2(\theta/2)$, which vanishes if and only if $\theta = 0$ or $\sin(\theta/2) = 0$, that is $\theta \in 2\pi\mathbb{Z}$.

Step 3 — the inverse. We look for G^{-1} in the form $aI + bK + cK^2$ (which is legitimate because the space spanned by I, K, K^2 is closed under the product, by Lemma 29.1, and contains the inverse when it exists, as verified a posteriori). We require $aI + bK + cK^2$ to invert G on each of the two subspaces.

- On $\langle\omega\rangle$: $aI + bK + cK^2$ acts as $a \text{ id}$ and G as $\theta \text{ id}$: we need $a\theta = 1$, that is $a = 1/\theta$.
- On $\omega^\perp$: $K^2 = -I$, hence $aI + bK + cK^2$ acts as $(a - c)I + bK$, and we require

$$((a - c)I + bJ)(\sin\theta I + (1 - \cos\theta)J) = I.$$

Using $J^2 = -I$ and expanding:

$$[(a - c)\sin\theta - b(1 - \cos\theta)]I + [(a - c)(1 - \cos\theta) + b\sin\theta]J = I,$$

that is, the system

$$(a - c)\sin\theta - b(1 - \cos\theta) = 1, \quad (a - c)(1 - \cos\theta) + b\sin\theta = 0.$$

The determinant of the system in the unknowns $(a - c, b)$ is $\sin^2\theta + (1 - \cos\theta)^2 = 4\sin^2(\theta/2) \neq 0$; the solution is

$$a - c = \frac{\sin\theta}{4\sin^2(\theta/2)}, \quad b = -\frac{1 - \cos\theta}{4\sin^2(\theta/2)}.$$

With the half-angle formulas $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$:

$$a - c = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{4\sin^2\frac{\theta}{2}} = \frac{1}{2}\cot\frac{\theta}{2}, \quad b = -\frac{2\sin^2\frac{\theta}{2}}{4\sin^2\frac{\theta}{2}} = -\frac{1}{2}.$$

From $a = 1/\theta$ it follows that $c = a - \frac{1}{2}\cot\frac{\theta}{2} = \frac{1}{\theta} - \frac{1}{2}\cot\frac{\theta}{2}$. Substituting gives the formula of the statement. Since $aI + bK + cK^2$ and $G(\theta)$ agree, being mutual inverses on two complementary invariant subspaces, the product is the identity on all of $\mathbb{R}^3$.

Independent numerical check. With $\omega = \frac{1}{3}(1, 2, 2)^\top$ we computed $\max |G(\theta)G(\theta)^{-1} - I|$ for $\theta \in \{0.3; 1; \pi/2; 2; 3; 6\}$: the values lie between $7 \cdot 10^{-17}$ and $1.6 \cdot 10^{-15}$. In the same table $\det G(\theta)$ computed by `numpy` agrees with $4\theta \sin^2(\theta/2)$ to ten decimal digits; for instance $\theta = \pi/2$ gives 3.1415926536 both ways (that is π : indeed $4 \cdot \frac{\pi}{2} \sin^2 \frac{\pi}{4} = 2\pi \cdot \frac{1}{2} = \pi$). ✓ $\square$

Remark 29.4: Why $\theta = 2\pi$ is excluded, and why this is not a defect. For $\theta = 2\pi$ the rotation returns to the identity, but the translation accumulated along the axis is not zero: $G(2\pi)\mathbf{v} = 2\pi(\omega \cdot \mathbf{v})\omega$. The matrix $G(2\pi)$ has rank 1 and the equation $G(\theta)\mathbf{v} = \mathbf{p}$ is no longer solvable for arbitrary $\mathbf{p}$: the pair (ω, θ) is no longer sufficient to describe the motion. This is the same phenomenon as the complex logarithm, which is not defined continuously on all of $\mathbb{C}^*$ (Chapter 6); and the construction of Chasles' theorem avoids the problem by always choosing $\theta \in (0, \pi]$, as we shall see in the next section.

29.4. Chasles' theorem

Definition 29.5: Screw motion, axis and pitch. Let $\mathcal{L} = \{\mathbf{q} + \lambda\omega : \lambda \in \mathbb{R}\}$ be an oriented line of $\mathbb{R}^3$ (with $\|\omega\| = 1$), $\theta \in \mathbb{R}$ and $h \in \mathbb{R}$. The **screw motion** of axis $\mathcal{L}$, angle θ and **pitch** h is the map

$$\text{Scr}_{\mathcal{L}, h, \theta}(\mathbf{x}) = \mathbf{q} + e^{[\omega] \times \theta}(\mathbf{x} - \mathbf{q}) + h\theta\omega,$$

that is, the rotation of angle θ about $\mathcal{L}$ composed with the translation of $h\theta$ along $\mathcal{L}$. The case $h = 0$ is *pure rotation* about $\mathcal{L}$; the case $\theta = 0$, with $h\theta$ finite, is understood as *pure translation* (infinite pitch).

Note that the definition does not depend on the point $\mathbf{q}$ chosen on the axis: if $\mathbf{q}' = \mathbf{q} + \mu\omega$ then, since $e^{[\omega] \times \theta}\omega = \omega$ (because $[\omega] \times \omega = 0$, Lemma 29.1),

$$\mathbf{q}' + e^{[\omega] \times \theta}(\mathbf{x} - \mathbf{q}') = \mathbf{q} + \mu\omega + e^{[\omega] \times \theta}(\mathbf{x} - \mathbf{q}) - \mu\omega = \mathbf{q} + e^{[\omega] \times \theta}(\mathbf{x} - \mathbf{q}).$$

Lemma 29.5: Axis and pitch of a twist. Let $\xi = (\omega, \mathbf{v})$ with $\|\omega\| = 1$. Then there exist a unique $h \in \mathbb{R}$ and a unique $\mathbf{q} \in \mathbb{R}^3$ with $\mathbf{q} \perp \omega$ such that

$$\mathbf{v} = h\omega - \omega \times \mathbf{q},$$

and they are given by

$$h = \omega^\top \mathbf{v}, \quad \mathbf{q} = \omega \times \mathbf{v}.$$

The line $\mathcal{L} = \{\mathbf{q} + \lambda\omega\}$ is called the **axis of the twist** and h its **pitch**.

Proof. Existence and formulas. Decompose $\mathbf{v} = \mathbf{v}_\parallel + \mathbf{v}_\perp$ with $\mathbf{v}_\parallel = (\omega^\top \mathbf{v})\omega$ and $\mathbf{v}_\perp = \mathbf{v} - \mathbf{v}_\parallel \perp \omega$. Set $h = \omega^\top \mathbf{v}$ and seek $\mathbf{q} \perp \omega$ with $-\omega \times \mathbf{q} = \mathbf{v}_\perp$. Applying $\omega \times (\cdot)$ to both sides and using Lemma 29.1(i) with $\|\omega\| = 1$:

$$\omega \times \mathbf{v} = \omega \times \mathbf{v}_\perp = -\omega \times (\omega \times \mathbf{q}) = -[\omega]_\times^2 \mathbf{q} = -(\omega\omega^\top - I)\mathbf{q} = \mathbf{q}$$

(the last step because $\omega^\top \mathbf{q} = 0$). Hence necessarily $\mathbf{q} = \omega \times \mathbf{v}$. Conversely, with this choice, $\mathbf{q} \perp \omega$ (because $\omega \cdot (\omega \times \mathbf{v}) = 0$) and

$$-\omega \times \mathbf{q} = -\omega \times (\omega \times \mathbf{v}) = -(\omega(\omega \cdot \mathbf{v}) - \mathbf{v}) = \mathbf{v} - h\omega = \mathbf{v}_\perp,$$

that is $\mathbf{v} = h\omega - \omega \times \mathbf{q}$, as required.

Uniqueness. If $h\omega - \omega \times \mathbf{q} = h'\omega - \omega \times \mathbf{q}'$ with $\mathbf{q}, \mathbf{q}' \perp \omega$, projecting onto ω (the term $\omega \times \mathbf{q}$ is orthogonal to ω) gives $h = h'$; what remains is $\omega \times (\mathbf{q} - \mathbf{q}') = \mathbf{0}$, that is $\mathbf{q} - \mathbf{q}' \parallel \omega$, and since both are orthogonal to ω , $\mathbf{q} = \mathbf{q}'$. $\square$

Theorem 29.5: The exponential of a twist is a screw motion. Let $\xi = (\omega, \mathbf{v})$ with $\|\omega\| = 1$, and let $h = \omega^\top \mathbf{v}$, $\mathcal{L} = \{\mathbf{q} + \lambda\omega\}$ as in Lemma 29.5. Then for every $\theta \in \mathbb{R}$ and every $\mathbf{x} \in \mathbb{R}^3$

$$e^{[\xi]\theta} \mathbf{x} = \mathbf{q} + e^{[\omega] \times \theta} (\mathbf{x} - \mathbf{q}) + h\theta \omega = \text{Scr}_{\mathcal{L}, h, \theta}(\mathbf{x}),$$

that is, $e^{[\xi]\theta}$ is the screw motion of axis $\mathcal{L}$, angle θ and pitch h . In particular the points of $\mathcal{L}$ translate rigidly by $h\theta\omega$, and $\mathcal{L}$ is invariant as a set.

Proof. Write $R = e^{[\omega] \times \theta}$ and $K = [\omega]_\times$. By Theorem 29.4(b), $e^{[\xi]\theta} \mathbf{x} = R\mathbf{x} + G(\theta)\mathbf{v}$; it therefore suffices to prove

$$G(\theta)\mathbf{v} = \mathbf{q} - R\mathbf{q} + h\theta\omega. \quad (29.5)$$

Step 1 — action of G and of R on $\langle \omega \rangle$ and on $\omega^\perp$. As in the proof of Proposition 29.3: $G(\theta)\omega = \theta\omega$, and on $\omega^\perp$ one has $K^2 = -I$, hence $G(\theta)\omega = \theta\omega$, and on $\omega^\perp$ one has $K^2 = -I$, hence

$$G(\theta)|_{\omega^\perp} = \sin \theta I + (1 - \cos \theta)K, \quad R|_{\omega^\perp} = I + \sin \theta K + (1 - \cos \theta)K^2 = \cos \theta I + \sin \theta K.$$

Step 2 — computing $G(\theta)\mathbf{v}$. By Lemma 29.5, $\mathbf{v} = h\omega - K\mathbf{q}$ with $\mathbf{q} \perp \omega$. By linearity,

$$G(\theta)\mathbf{v} = hG(\theta)\omega - G(\theta)K\mathbf{q} = h\theta\omega - G(\theta)K\mathbf{q}.$$

The vector $K\mathbf{q} = \omega \times \mathbf{q}$ belongs to $\omega^\perp$, so by Step 1

$$G(\theta)K\mathbf{q} = \sin \theta K\mathbf{q} + (1 - \cos \theta)K^2\mathbf{q} = \sin \theta K\mathbf{q} - (1 - \cos \theta)\mathbf{q},$$

having used $K^2\mathbf{q} = -\mathbf{q}$ ($\mathbf{q} \in \omega^\perp$). Hence

$$G(\theta)\mathbf{v} = h\theta\omega - \sin \theta K\mathbf{q} + (1 - \cos \theta)\mathbf{q}.$$

Step 3 — computing $\mathbf{q} - R\mathbf{q}$. Again by Step 1, since $\mathbf{q} \in \omega^\perp$,

$$R\mathbf{q} = \cos \theta \mathbf{q} + \sin \theta K\mathbf{q} \implies \mathbf{q} - R\mathbf{q} = (1 - \cos \theta)\mathbf{q} - \sin \theta K\mathbf{q}.$$

Step 4 — comparison. Steps 2 and 3 give

$$G(\theta)\mathbf{v} = h\theta\omega + [(1 - \cos \theta)\mathbf{q} - \sin \theta K\mathbf{q}] = h\theta\omega + \mathbf{q} - R\mathbf{q},$$

which is (29.5). Substituting:

$$e^{[\xi]\theta} \mathbf{x} = R\mathbf{x} + \mathbf{q} - R\mathbf{q} + h\theta\omega = \mathbf{q} + R(\mathbf{x} - \mathbf{q}) + h\theta\omega.$$

Step 5 — points of the axis. If $\mathbf{x} = \mathbf{q} + \lambda\omega$, then $R(\mathbf{x} - \mathbf{q}) = \lambda R\omega = \lambda\omega$ (because $K\omega = 0$ implies $R\omega = \omega$), hence

$$e^{[\xi]\theta} \mathbf{x} = \mathbf{q} + \lambda\omega + h\theta\omega = \mathbf{x} + h\theta\omega :$$

every point of the axis translates by $h\theta\omega$, and the axis maps to itself. $\square$

Theorem 29.6: Chasles' theorem. Let $g = (R, \mathbf{p}) \in SE(3)$. Then there exist $\xi \in \mathbb{R}^6$ and $\theta \in \mathbb{R}$ with $g = e^{[\xi]\theta}$: the exponential map $\exp : \mathfrak{se}(3) \rightarrow SE(3)$ is **surjective**. More precisely, every direct rigid motion of space is a screw motion, and the three alternatives are:

- (i) Identity. If $R = I$ and $\mathbf{p} = \mathbf{0}$, then $g = e^{[\xi]\cdot 0}$ for every ξ : $\theta = 0$, and there is no axis.
- (ii) Pure translation. If $R = I$ and $\mathbf{p} \neq \mathbf{0}$, set $\omega = \mathbf{0}$, $\mathbf{v} = \mathbf{p}/\|\mathbf{p}\|$, $\theta = \|\mathbf{p}\|$; then $g = e^{[\xi]\theta}$. The motion is a translation: infinite pitch, undetermined axis (every line parallel to $\mathbf{p}$ has points that translate along themselves).
- (iii) Generic case $R \neq I$. By Euler's theorem there exist $\theta \in (0, \pi]$ and ω with $\|\omega\| = 1$ such that $R = e^{[\omega]\times\theta}$ (ω is unique if $\theta < \pi$, unique up to sign if $\theta = \pi$). Setting

$$\mathbf{v} = G(\theta)^{-1}\mathbf{p}, \quad h = \omega^\top \mathbf{v}, \quad \mathbf{q} = \omega \times \mathbf{v},$$

one has $g = e^{[\xi]\theta}$ with $\xi = (\omega, \mathbf{v})$, and g is the screw motion of axis $\mathcal{L} = \{\mathbf{q} + \lambda\omega\}$, angle θ and pitch h . If $h = 0$ the motion is a pure rotation about $\mathcal{L}$.

Proof. (i). $e^0 = I_4$.

(ii). This is Theorem 29.4(a): with $\omega = \mathbf{0}$, $e^{[\xi]\theta} = \begin{pmatrix} I & \mathbf{v}\theta \\ \mathbf{0}^\top & 1 \end{pmatrix}$, and $\mathbf{v}\theta = \frac{\mathbf{p}}{\|\mathbf{p}\|}\|\mathbf{p}\| = \mathbf{p}$.

(iii). *Step 1 — existence of (ω, θ) .* By Euler's rotation theorem (Theorem 5.9) there exist a unit vector ω and an angle θ such that R is, in an orthonormal basis $\{\omega, w_1, w_2\}$, the rotation by θ in the plane $\omega^\perp$; replacing, if needed, ω by $-\omega$ and θ by $-\theta$, and then θ by $\theta + 2k\pi$, we can reduce to $\theta \in [0, \pi]$, with $\theta \neq 0$ because $R \neq I$. Rodrigues' formula (Chapter 28, Theorem 28.2) says that this rotation is $e^{[\omega]\times\theta}$, hence $R = e^{[\omega]\times\theta}$. Operationally:

$$\theta = \arccos \frac{\text{tr } R - 1}{2} \in (0, \pi], \quad [\omega]_\times = \frac{R - R^\top}{2 \sin \theta} \text{ if } \theta \in (0, \pi),$$

while for $\theta = \pi$ the axis is read off as the eigenvector of R for the eigenvalue $+1$ (the previous formula degenerates because $\sin \pi = 0$).

Step 2 — existence of $\mathbf{v}$. Since $\theta \in (0, \pi] \subset (0, 2\pi)$, Proposition 29.3 guarantees that $G(\theta)$ is invertible; set $\mathbf{v} = G(\theta)^{-1}\mathbf{p}$, that is, the unique solution of $G(\theta)\mathbf{v} = \mathbf{p}$.

Step 3 — conclusion. By Theorem 29.4(b),

$$e^{[\xi]\theta} = \begin{pmatrix} e^{[\omega]\times\theta} & G(\theta)\mathbf{v} \\ \mathbf{0}^\top & 1 \end{pmatrix} = \begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix} = g.$$

By Theorem 29.5, $e^{[\xi]\theta}$ is the screw motion of axis $\mathcal{L}$, angle θ and pitch h . If $h = \omega^\top \mathbf{v} = 0$, the screw motion formula reduces to $\mathbf{x} \mapsto \mathbf{q} + R(\mathbf{x} - \mathbf{q})$: a pure rotation about $\mathcal{L}$, which has $\mathcal{L}$ as its set of fixed points.

The three cases exhaust $SE(3)$: either $R = I$ (and then (i) or (ii) according to whether $\mathbf{p}$ is zero or not), or $R \neq I$ and (iii) holds. $\square$

Proposition 29.4: Uniqueness of the axis and variational characterisation. Let $g = (R, \mathbf{p})$ with $R \neq I$, and let $\omega, \theta, h, \mathcal{L}$ be as in Theorem 29.6(iii). Then, for every $\mathbf{x} \in \mathbb{R}^3$,

(i) $\omega^\top(g(\mathbf{x}) - \mathbf{x}) = h\theta$: the component of the displacement along the axis is the same for all points (in particular $h\theta = \omega^\top \mathbf{p}$);

(ii) $\|g(\mathbf{x}) - \mathbf{x}\|^2 = (h\theta)^2 + \|(R - I)(\mathbf{x} - \mathbf{q})\|^2 \geq (h\theta)^2$;

(iii) equality holds if and only if $\mathbf{x} \in \mathcal{L}$.

Consequently $\mathcal{L}$ is the set of points of minimal displacement, and depends only on g : the screw axis is unique, and with it the translation vector $h\theta\omega$.

Proof. (i). By Theorem 29.5, $g(\mathbf{x}) - \mathbf{x} = (R - I)(\mathbf{x} - \mathbf{q}) + h\theta\omega$. Since $R\omega = \omega$ and $R^\top = R^{-1}$, one has $R^\top\omega = \omega$, hence for every $\mathbf{y}$

$$\omega^\top(R - I)\mathbf{y} = (R^\top\omega - \omega)^\top\mathbf{y} = 0.$$

Hence $\omega^\top(g(\mathbf{x}) - \mathbf{x}) = h\theta\omega^\top\omega = h\theta$. Setting $\mathbf{x} = \mathbf{0}$ gives $h\theta = \omega^\top\mathbf{p}$.

(ii). The identity just proved shows that $(R - I)(\mathbf{x} - \mathbf{q})$ is orthogonal to ω , so the two terms of $g(\mathbf{x}) - \mathbf{x} = (R - I)(\mathbf{x} - \mathbf{q}) + h\theta\omega$ are orthogonal and the Pythagorean theorem applies.

(iii). Equality in (ii) holds if and only if $(R - I)(\mathbf{x} - \mathbf{q}) = \mathbf{0}$, that is $\mathbf{x} - \mathbf{q} \in \ker(R - I)$. Since $R \neq I$ is a rotation by angle $\theta \in (0, \pi]$ about ω , its eigenspace for eigenvalue 1 is exactly $\langle\omega\rangle$: indeed R acts on $\omega^\perp$ as $\cos\theta I + \sin\theta K$ (Step 1 of the proof of Theorem 29.5), and on a nonzero vector $\mathbf{y} \in \omega^\perp$, $R\mathbf{y} = \mathbf{y}$ would force $(\cos\theta - 1)\mathbf{y} + \sin\theta K\mathbf{y} = \mathbf{0}$; since $\mathbf{y}$ and $K\mathbf{y}$ are orthogonal and nonzero, this would give $\cos\theta = 1$ and $\sin\theta = 0$, that is $\theta \in 2\pi\mathbb{Z}$, which is excluded. Hence $\mathbf{x} \in \mathbf{q} + \langle\omega\rangle = \mathcal{L}$.

Uniqueness. The set $\{\mathbf{x} : \|g(\mathbf{x}) - \mathbf{x}\| \text{ minimal}\}$ is defined from g alone; by (ii)–(iii) it coincides with $\mathcal{L}$. The translation vector is $h\theta\omega = g(\mathbf{x}) - \mathbf{x}$ for $\mathbf{x} \in \mathcal{L}$, likewise a function of g alone. $\square$

Remark 29.5: What remains non-unique. Unique are the axis $\mathcal{L}$ as a line and the vector $h\theta\omega$. Not unique are the pair (ω, θ) , and hence h : the opposite choice $(-\omega, 2\pi - \theta)$ generates the same R , the same line $\mathcal{L}$ and the same total translation, but pitch $h' = -h\theta/(2\pi - \theta)$. The convention $\theta \in (0, \pi]$ of Theorem 29.6 removes the ambiguity except for $\theta = \pi$, where the two possibilities $\pm\omega$ remain (and then h changes sign together with ω , so that $h\theta\omega$ stays the same). There is nothing pathological about this: it is the same non-uniqueness of the logarithm on S^1 (Chapter 12), lifted to $SE(3)$.

Example 29.3: Screw decomposition of a given motion. Consider the rigid motion $g = (R, \mathbf{p})$ with

$$R = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (\text{rotation by } 90^\circ \text{ about } z), \quad \mathbf{p} = (1, 2, 3)^\top.$$

Axis and angle of the rotation. $\text{tr } R = 1$, hence $\theta = \arccos \frac{1-1}{2} = \frac{\pi}{2}$, and $\omega = \mathbf{e}_3 = (0, 0, 1)^\top$.

The twist. With $K = [\mathbf{e}_3]_\times$ one has $K^2 = \text{diag}(-1, -1, 0)$ and $\cot \frac{\pi}{4} = 1$, so by Proposition 29.3

$$G\left(\frac{\pi}{2}\right)^{-1} = \frac{2}{\pi}I - \frac{1}{2}K + \left(\frac{2}{\pi} - \frac{1}{2}\right)K^2 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{2}{\pi} \end{pmatrix},$$

(the diagonal entries (1, 1) and (2, 2) equal $\frac{2}{\pi} - (\frac{2}{\pi} - \frac{1}{2}) = \frac{1}{2}$), and hence

$$\mathbf{v} = G\left(\frac{\pi}{2}\right)^{-1}\mathbf{p} = \left(\frac{3}{2}, \frac{1}{2}, \frac{6}{\pi}\right)^\top = (1.500000, 0.500000, 1.909859)^\top.$$

Pitch and axis.

$$h = \omega^\top\mathbf{v} = \frac{6}{\pi} = 1.909859, \quad \mathbf{q} = \omega \times \mathbf{v} = \left(-\frac{1}{2}, \frac{3}{2}, 0\right)^\top,$$

hence the axis is the vertical line $\mathcal{L} = \{(-\frac{1}{2}, \frac{3}{2}, \lambda)\}$ and the translation along it equals $h\theta = \frac{6}{\pi} \cdot \frac{\pi}{2} = 3$.

First independent check (geometric). The motion leaves the height unchanged and in the xy plane acts as $\mathbf{x} \mapsto R_2\mathbf{x} + (1, 2)$ with $R_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$: its unique fixed point is

$$(I - R_2)^{-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 3/2 \end{pmatrix},$$

which is exactly the projection of $\mathcal{L}$ onto the xy plane. ✓

Second independent check (algebraic). $h\theta$ must equal $\omega^\top \mathbf{p} = 3$ by Proposition 29.4(i): it coincides. ✓

Third check (numerical). Reconstructing $e^{[\xi]\theta}$ from the closed form and, independently, summing the exponential series to 40 digits, one obtains g with maximum discrepancy $1.1 \cdot 10^{-16}$ and $2.2 \cdot 10^{-16}$ respectively. Moreover the minimal displacement, sought over 200 000 randomly drawn points in a cube, equals 3.000098, against the exact value $|h\theta| = 3$ predicted by Proposition 29.4 (the excess is because no sample point falls exactly on the axis). ✓

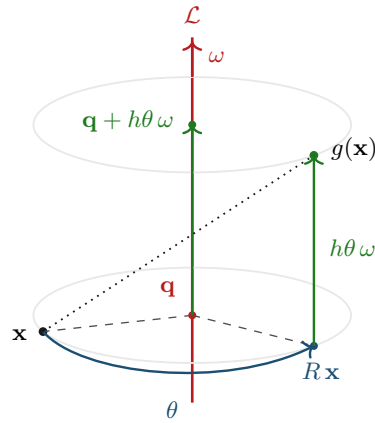


Figure 29.1: The screw motion of Theorem 29.5. The point $\mathbf{x}$ rotates by θ about the axis $\mathcal{L}$ (blue arrow, along the ellipse that is the circle of rotation seen in perspective) and then translates by $h\theta\omega$ along the axis (green arrow). The composition, shown as a dotted line, is the total displacement $g(\mathbf{x}) - \mathbf{x}$. Points of the axis undergo translation alone, which is why they are the points of minimal displacement (Proposition 29.4). The ellipses have semi-axes 2 and 0.6; the two points on the lower circle correspond to parameters 200° and 320° , that is $(-1.8794, -0.2052)$ and $(1.5321, -0.3857)$.

Remark 29.6: Chasles, Mozzi and screw theory. The statement was published by Michel Chasles in 1830 in *Férussac's Bulletin des Sciences Mathématiques*, in a six-page note on the “déplacement fini, ou infiniment petit, d’un corps solide libre.” Priority is however disputed: Giulio Mozzi had already described the screw axis (the spontaneous axis of rotation) in 1763, and the historical reconstruction by M. Ceccarelli (*Mechanism and Machine Theory* 35, 2000) documents the Mozzi–Cauchy–Poinsot–Chasles chain; for this reason modern English-language literature often writes “Chasles–Mozzi theorem.” The dual of the theorem, due to Poinsot, states that every system of forces on a rigid body reduces to a force along a line plus a couple about the same line: this is the notion of wrench, which stands to twists as forces stand to velocities. On these two foundations Robert S. Ball built, in 1900, screw theory, of which the present chapter is the modern matrix version.

29.5. The product-of-exponentials formula

Definition 29.6: Open kinematic chain. An **open kinematic chain** with n joints is a sequence of rigid bodies (*links*) $L_0, L_1, \dots, L_n$ such that L_{i-1} and L_i are connected by **joint** i , with L_0 fixed (*base*). Joint i is:

- **revolute** if it allows only rotation of L_i relative to L_{i-1} about a line fixed in L_{i-1} ;
- **prismatic** if it allows only translation of L_i relative to L_{i-1} along a direction fixed in L_{i-1} .

We fix a **base** frame S attached to L_0 and a **tool** frame T attached to L_n ; the **forward kinematics** is the map $\theta = (\theta_1, \dots, \theta_n) \mapsto T(\theta) \in SE(3)$ giving the pose of the tool frame relative to the base frame. The **reference configuration** is $\theta = 0$, and we set $M = T(0)$.

Definition 29.7: Twist of a joint. In the reference configuration, joint i has an axis that, read in the base frame, is the line $\{\mathbf{q}_i + \lambda\omega_i\}$ with $\|\omega_i\| = 1$ (revolute joint), or a direction $\mathbf{v}_i$ with $\|\mathbf{v}_i\| = 1$ (prismatic joint). The **twist of joint** i is

$$\xi_i = (\omega_i, -\omega_i \times \mathbf{q}_i) \quad (\text{revolute}), \quad \xi_i = (\mathbf{0}, \mathbf{v}_i) \quad (\text{prismatic}).$$

The choice of linear part $-\omega_i \times \mathbf{q}_i$ is not arbitrary: by Lemma 29.5 it makes the twist ξ_i have pitch $h = 0$ and axis exactly the joint's line. Indeed $h = \omega_i^\top(-\omega_i \times \mathbf{q}_i) = 0$ and, by the computation in Lemma 29.5, $\omega_i \times (-\omega_i \times \mathbf{q}_i)$ is the component of $\mathbf{q}_i$ orthogonal to ω_i , that is the point of the axis closest to the origin. By Theorem 29.5, $e^{[\xi_i]\theta_i}$ is therefore the *pure rotation* by angle θ_i about the axis of joint i , and for a prismatic joint $e^{[\xi_i]\theta_i}$ is the translation by θ_i along $\mathbf{v}_i$ (Theorem 29.4(a)).

Lemma 29.6: How a pose transforms. Let a rigid body have body-fixed frame B at pose $T \in SE(3)$ relative to the fixed frame S (that is, the S -coordinates of a point with B -coordinates $\mathbf{y}$ are $T\tilde{\mathbf{y}}$). If the rigid displacement $g \in SE(3)$, read in S (that is, every point of space with coordinates $\mathbf{x}$ goes to $g\mathbf{x}$), is applied to the body, the new pose is gT .

Proof. Let P be a point of the body, with coordinates $\mathbf{y}$ in B : before the displacement its coordinates in S are $T\tilde{\mathbf{y}}$. After the displacement, its coordinates in S are $gT\tilde{\mathbf{y}}$. Since the body is rigid, the coordinates of P in the body-fixed frame B do not change, remaining $\mathbf{y}$. The new pose T' is by definition the matrix sending the B -coordinates to the new S -coordinates: $T'\tilde{\mathbf{y}} = gT\tilde{\mathbf{y}}$ for every $\mathbf{y}$, hence $T' = gT$ (two matrices agreeing on all vectors with last component 1 agree, because such vectors span $\mathbb{R}^4$). $\square$

Theorem 29.7: Product-of-exponentials formula. For an open kinematic chain with n joints, with twists $\xi_1, \dots, \xi_n$ read in the reference configuration and $M = T(0)$, the forward kinematics is

$$T(\theta_1, \dots, \theta_n) = e^{[\xi_1]\theta_1} e^{[\xi_2]\theta_2} \dots e^{[\xi_n]\theta_n} M$$

Proof. We prove this by downward induction on the index of the first joint moved. For $k = n+1, n, \dots, 1$ let

$$\mathcal{P}(k) : \quad T(0, \dots, 0, \theta_k, \dots, \theta_n) = e^{[\xi_k]\theta_k} \dots e^{[\xi_n]\theta_n} M \quad \text{for every } \theta_k, \dots, \theta_n.$$

Base case ($k = n+1$). All joints are at zero: $T(0, \dots, 0) = M$, which is the empty product times M . True by the definition of M .

Inductive step ($k+1 \Rightarrow k$). Assume $\mathcal{P}(k+1)$ and fix $\theta_k, \dots, \theta_n$. Consider the subchain $C_k = L_k \cup L_{k+1} \cup \dots \cup L_n$, together with the joints $k+1, \dots, n$ that articulate it. Moving

joint k by θ_k , while keeping $\theta_1 = \dots = \theta_{k-1} = 0$ and the values $\theta_{k+1}, \dots, \theta_n$ fixed, moves C_k *rigidly* (the relative positions inside C_k depend only on $\theta_{k+1}, \dots, \theta_n$, which do not change), and it moves it by the screw motion about the axis of joint k .

That axis, since $\theta_1 = \dots = \theta_{k-1} = 0$, is still in the reference position: joint k is fixed in link L_{k-1} , and L_{k-1} is at the reference pose because all joints preceding it are at zero. Hence the motion is $g = e^{[\xi_k]\theta_k}$, with ξ_k the twist of the joint read in the reference configuration (Definition 29.7); here Theorem 29.5 is used to identify the exponential with the rotation about that axis (or, for a prismatic joint, Theorem 29.4(a) for the translation).

The tool frame T is fixed in $L_n \subset C_k$, so by Lemma 29.6 its pose changes from $T(0, \dots, 0, \theta_{k+1}, \dots, \theta_n)$ to

$$T(0, \dots, 0, \theta_k, \theta_{k+1}, \dots, \theta_n) = e^{[\xi_k]\theta_k} T(0, \dots, 0, \theta_{k+1}, \dots, \theta_n) = e^{[\xi_k]\theta_k} e^{[\xi_{k+1}]\theta_{k+1}} \dots e^{[\xi_n]\theta_n} M,$$

where the last equality is $\mathcal{P}(k+1)$. Hence $\mathcal{P}(k)$ holds.

Conclusion. By induction $\mathcal{P}(1)$ is true, which is the statement. $\square$

The formula, in the form proved here, is due to Roger W. Brockett, who introduced it in 1984 as a systematic tool for the kinematics of open-chain robots; see the source cited at the end of the chapter.

Remark 29.7: Where the difficulty lies. The proof is short, but its delicate point is a single one: the joints are moved from the last to the first, because moving joint k when the preceding ones are at zero leaves its axis in the reference position. If one attempted the induction in the opposite order (first joint 1, then joint 2, ...) the axis of joint 2 would already be displaced by θ_1 , and its twist would no longer be ξ_2 but $\text{Ad}_{\exp([\xi_1]\theta_1)}\xi_2$: one would obtain

$$T(\theta) = e^{[\text{Ad}_{e^{[\xi_1]\theta_1}}\xi_2]\theta_2} e^{[\xi_1]\theta_1} M = e^{[\xi_1]\theta_1} e^{[\xi_2]\theta_2} M$$

for $n = 2$, using Proposition 29.2 — the same formula, but by a longer route. This is exactly why the literature numbers twists “from the base toward the tool” and applies them in that order.

Proposition 29.5: Independence from configuration and frames. Let $T(\theta) = e^{[\xi_1]\theta_1} \dots e^{[\xi_n]\theta_n} M$ be the forward kinematics of an open chain. Then:

- (i) (Change of base frame.) If S is replaced by a new frame S' such that $T'(\theta) = PT(\theta)$ with $P \in SE(3)$ constant, then

$$T'(\theta) = e^{[\xi'_1]\theta_1} \dots e^{[\xi'_n]\theta_n} M', \quad \xi'_i = \text{Ad}_P \xi_i, \quad M' = PM :$$

the formula has the same structure, with the twists carried by the adjoint.

- (ii) (Change of tool frame.) If $T'(\theta) = T(\theta)Q$ with $Q \in SE(3)$ constant, then $T'(\theta) = e^{[\xi_1]\theta_1} \dots e^{[\xi_n]\theta_n} M'$ with $M' = MQ$ and the same twists ξ_i .

- (iii) (Change of reference configuration.) If one chooses as reference configuration $\theta = \bar{\theta}$ instead of $\theta = 0$, setting $\phi_i = \theta_i - \bar{\theta}_i$, one still has $T = e^{[\bar{\xi}_1]\phi_1} \dots e^{[\bar{\xi}_n]\phi_n} \bar{M}$ with $\bar{M} = T(\bar{\theta})$ and $\bar{\xi}_i = \text{Ad}_{e^{[\xi_1]\bar{\theta}_1} \dots e^{[\xi_{i-1}]\bar{\theta}_{i-1}}} \xi_i$.

Proof. (i). By Proposition 29.2, $P e^{[\xi_i]\theta_i} P^{-1} = e^{[\text{Ad}_P \xi_i]\theta_i}$. Inserting $P^{-1}P$ between consecutive factors:

$$P e^{[\xi_1]\theta_1} \dots e^{[\xi_n]\theta_n} M = (P e^{[\xi_1]\theta_1} P^{-1}) \dots (P e^{[\xi_n]\theta_n} P^{-1}) PM = e^{[\xi'_1]\theta_1} \dots e^{[\xi'_n]\theta_n} PM.$$

(ii). Immediate: $T(\theta)Q = e^{[\xi_1]\theta_1} \dots e^{[\xi_n]\theta_n} (MQ)$.

(iii). Write, for every i , $\theta_i = \bar{\theta}_i + \phi_i$; since $[\xi_i]\bar{\theta}_i$ and $[\xi_i]\phi_i$ commute (they are multiples of the

same matrix),

$$e^{[\xi_i]\theta_i} = e^{[\xi_i]\bar{\theta}_i} e^{[\xi_i]\phi_i}.$$

Set $A_i = e^{[\xi_1]\bar{\theta}_1} \dots e^{[\xi_i]\bar{\theta}_i}$ (with $A_0 = I$), so that $A_i = A_{i-1} e^{[\xi_i]\bar{\theta}_i}$, and prove by induction on $m \leq n$ that

$$\prod_{i=1}^m \left(e^{[\xi_i]\bar{\theta}_i} e^{[\xi_i]\phi_i} \right) = \left[\prod_{i=1}^m A_{i-1} e^{[\xi_i]\phi_i} A_{i-1}^{-1} \right] A_m, \quad (29.6)$$

where the products are ordered by increasing index from the left. For $m = 0$ both sides equal I . Assuming it holds for $m - 1$, multiply on the right by $e^{[\xi_m]\bar{\theta}_m} e^{[\xi_m]\phi_m}$ and use the commutativity of the two factors:

$$\Pi_{m-1} A_{m-1} e^{[\xi_m]\phi_m} e^{[\xi_m]\bar{\theta}_m} = \Pi_{m-1} (A_{m-1} e^{[\xi_m]\phi_m} A_{m-1}^{-1}) A_{m-1} e^{[\xi_m]\bar{\theta}_m},$$

having set $\Pi_{m-1} = \prod_{i=1}^{m-1} A_{i-1} e^{[\xi_i]\phi_i} A_{i-1}^{-1}$; and $A_{m-1} e^{[\xi_m]\bar{\theta}_m} = A_m$: this gives (29.6) for m . Applying Proposition 29.2 again, $A_{i-1} e^{[\xi_i]\phi_i} A_{i-1}^{-1} = e^{[\text{Ad}_{A_{i-1}} \xi_i] \phi_i}$, and $A_n M = T(\bar{\theta}) = \bar{M}$: this gives the desired formula. $\square$

Remark 29.8: What “with no intermediate frames” means. The merit of the formula is not brevity but the absence of choices. To apply it one needs only: the tool pose M in some configuration, and for every joint a line (or a direction) in space, read in that same configuration. No intermediate frame needs to be assigned, and Proposition 29.5 says that moving the base, the tool, or the reference configuration breaks nothing: only the data change, not the structure. This is why modern kinematics toolkits adopt this description: modelling errors reduce to errors on measurable lines, not on conventions.

Definition 29.8: Denavit–Hartenberg parameters. In the Denavit–Hartenberg convention, each link is assigned a frame following precise rules (axis z_i along the axis of joint $i + 1$, axis x_i along the common normal between z_{i-1} and z_i), and the transformation between consecutive frames depends on only four parameters $(\theta_i, d_i, a_i, \alpha_i)$:

$$T_{i-1,i} = \text{Rot}_z(\theta_i) \text{Trans}_z(d_i) \text{Trans}_x(a_i) \text{Rot}_x(\alpha_i) \in SE(3),$$

and the forward kinematics is $T_{0,n} = T_{0,1} T_{1,2} \dots T_{n-1,n}$.

Remark 29.9: Denavit–Hartenberg versus product of exponentials. The two descriptions give the same map $T(\theta)$ and are both correct; they differ in modelling cost and in robustness.

- **DH** uses four numbers per joint instead of the twist’s six, but pays for it with the obligation to assign frames according to rigid rules, and with degenerate cases (consecutive parallel or intersecting axes) in which the common normal is not unique and the choice of parameters becomes arbitrary.
- **PoE** uses six numbers per joint (of which one is redundant for a revolute joint, because $\|\omega_i\| = 1$, and three for a prismatic one), but the data are directly measurable physical quantities — lines in space — and there are no degenerate cases.
- Numerically, a calibration error on a line propagates continuously in the PoE; in DH an axis nearly parallel to the previous one makes the parameters a_i, α_i jump even though the geometry changes little. This is the main reason modern kinematic calibration methods are formulated on $\mathfrak{se}(3)$.

Every industrial controller — **KUKA, ABB, FANUC, Yaskawa** — nonetheless computes the tool pose as a product of 4×4 homogeneous transformations; what changes from brand

to brand is the convention by which the orientation is then exposed to the operator (A, B, C angles for KUKA, W, P, R for FANUC, a quaternion in ABB's RAPID *robtarg*et), not the underlying mathematics (Chapter 28).

Example 29.4: Product of exponentials for the planar 3R arm. Consider a planar arm with three revolute joints with axes parallel to $\mathbf{e}_3$ and links of length $\ell_1 = 2$, $\ell_2 = 1.5$, $\ell_3 = 1$. In the reference configuration the arm is stretched out along the x axis, hence

$$\mathbf{q}_1 = (0, 0, 0)^\top, \quad \mathbf{q}_2 = (2, 0, 0)^\top, \quad \mathbf{q}_3 = (3.5, 0, 0)^\top, \quad \omega_i = \mathbf{e}_3,$$

$$M = \begin{pmatrix} I & (4.5, 0, 0)^\top \\ \mathbf{0}^\top & 1 \end{pmatrix}.$$

The twists are $\xi_i = (\mathbf{e}_3, -\mathbf{e}_3 \times \mathbf{q}_i)$, that is

$$\xi_1 = (0, 0, 1, 0, 0, 0), \quad \xi_2 = (0, 0, 1, 0, -2, 0), \quad \xi_3 = (0, 0, 1, 0, -3.5, 0).$$

Comparison with the closed form. Elementary trigonometry gives, for the tool position,

$$x = \ell_1 \cos \theta_1 + \ell_2 \cos(\theta_1 + \theta_2) + \ell_3 \cos(\theta_1 + \theta_2 + \theta_3),$$

$$y = \ell_1 \sin \theta_1 + \ell_2 \sin(\theta_1 + \theta_2) + \ell_3 \sin(\theta_1 + \theta_2 + \theta_3), \quad \varphi = \theta_1 + \theta_2 + \theta_3.$$

Evaluating both expressions at three configurations:

$(\theta_1, \theta_2, \theta_3)$	(x, y) from PoE	(x, y) from closed form
$(30^\circ, 60^\circ, -45^\circ)$	(2.4391576; 3.2071068)	(2.4391576; 3.2071068)
$(30^\circ, 0^\circ, 0^\circ)$	(3.8971143; 2.2500000)	(3.8971143; 2.2500000)
$(0.4; -0.9; 1.3) \text{ rad}$	(3.8552025; 0.7770545)	(3.8552025; 0.7770545)

The rotation block of the PoE agrees with $e^{[\mathbf{e}_3] \times \varphi}$ with maximum discrepancy $1.1 \cdot 10^{-16}$. ✓

Reading. The fact that in the second row $\|(x, y)\| = 4.5 = \ell_1 + \ell_2 + \ell_3$ signals that the arm is fully outstretched: this is the configuration that will turn out to be singular in Section 29.7.

29.6. Geometric Jacobian: spatial and body

Definition 29.9: Spatial twist and body twist. Let $t \mapsto T(t) \in SE(3)$ be a differentiable curve. Set

$$[\mathcal{V}_s] = \dot{T}T^{-1} \in \mathfrak{se}(3) \quad (\text{spatial twist}), \quad [\mathcal{V}_b] = T^{-1}\dot{T} \in \mathfrak{se}(3) \quad (\text{body twist}),$$

and $\mathcal{V}_s, \mathcal{V}_b \in \mathbb{R}^6$ are the corresponding coordinate vectors.

That $\dot{T}T^{-1}$ and $T^{-1}\dot{T}$ lie in $\mathfrak{se}(3)$ is checked at once: if $T = \begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix}$, then $\dot{T} = \begin{pmatrix} \dot{R} & \dot{\mathbf{p}} \\ \mathbf{0}^\top & 0 \end{pmatrix}$ and

$$\dot{T}T^{-1} = \begin{pmatrix} \dot{R} & \dot{\mathbf{p}} \\ \mathbf{0}^\top & 0 \end{pmatrix} \begin{pmatrix} R^\top & -R^\top \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix} = \begin{pmatrix} \dot{R}R^\top & \dot{\mathbf{p}} - \dot{R}R^\top \mathbf{p} \\ \mathbf{0}^\top & 0 \end{pmatrix},$$

and $\dot{R}R^\top$ is skew-symmetric because differentiating $RR^\top = I$ gives $\dot{R}R^\top + R\dot{R}^\top = 0$, that is $\dot{R}R^\top = -(\dot{R}R^\top)^\top$. Similarly for $T^{-1}\dot{T}$, with $R^\top \dot{R}$ skew-symmetric.

Proposition 29.6: Relation between the two twists. $\mathcal{V}_s = \text{Ad}_T \mathcal{V}_b$ and $\mathcal{V}_b = \text{Ad}_{T^{-1}} \mathcal{V}_s$.

Proof. $[\mathcal{V}_s] = \dot{T}T^{-1} = T(T^{-1}\dot{T})T^{-1} = T[\mathcal{V}_b]T^{-1} = [\text{Ad}_T \mathcal{V}_b]$ by Proposition 29.2; injectivity of the map $\xi \mapsto [\xi]$ gives the first equality, and the second is obtained by applying $\text{Ad}_{T^{-1}} = (\text{Ad}_T)^{-1}$. $\square$

Theorem 29.8: Spatial Jacobian. Let $T(\theta) = e^{[\xi_1]\theta_1} \dots e^{[\xi_n]\theta_n} M$ be the forward kinematics of an open chain (Theorem 29.7) and let $t \mapsto \theta(t)$ be differentiable. Then

$$\mathcal{V}_s = J_s(\theta) \dot{\theta}, \quad J_s(\theta) = [J_{s,1} \ J_{s,2} \ \dots \ J_{s,n}] \in \mathbb{R}^{6 \times n},$$

with

$$J_{s,1} = \xi_1, \quad J_{s,i} = \text{Ad}_{e^{[\xi_1]\theta_1} \dots e^{[\xi_{i-1}]\theta_{i-1}}} \xi_i \quad (i \geq 2).$$

Column i is therefore the twist of joint i carried into the current configuration by the joints preceding it.

Proof. Step 1 — derivative of the product. By the product rule applied to $T(\theta(t))$,

$$\dot{T} = \sum_{i=1}^n \frac{\partial T}{\partial \theta_i} \dot{\theta}_i, \quad \text{hence} \quad [\mathcal{V}_s] = \dot{T}T^{-1} = \sum_{i=1}^n \left(\frac{\partial T}{\partial \theta_i} T^{-1} \right) \dot{\theta}_i.$$

Since $\xi \mapsto [\xi]$ is linear and injective, it suffices to identify $\frac{\partial T}{\partial \theta_i} T^{-1}$ as an element of $\mathfrak{se}(3)$.

Step 2 — the partial derivative. Only the i -th factor depends on θ_i , and $\frac{\partial}{\partial \theta_i} e^{[\xi_i]\theta_i} = [\xi_i] e^{[\xi_i]\theta_i}$ (a prerequisite about the matrix exponential). Hence, setting $A_{i-1} = e^{[\xi_1]\theta_1} \dots e^{[\xi_{i-1}]\theta_{i-1}}$ and $B_i = e^{[\xi_i]\theta_i} \dots e^{[\xi_n]\theta_n} M$ (so that $T = A_{i-1} B_i$),

$$\frac{\partial T}{\partial \theta_i} = A_{i-1} [\xi_i] B_i.$$

Step 3 — simplification. With $T^{-1} = B_i^{-1} A_{i-1}^{-1}$,

$$\frac{\partial T}{\partial \theta_i} T^{-1} = A_{i-1} [\xi_i] B_i B_i^{-1} A_{i-1}^{-1} = A_{i-1} [\xi_i] A_{i-1}^{-1} = [\text{Ad}_{A_{i-1}} \xi_i],$$

by Proposition 29.2. For $i = 1$ one has $A_0 = I$ and the column is ξ_1 .

Step 4 — conclusion. Substituting into Step 1 and passing to coordinates:

$$\mathcal{V}_s = \sum_{i=1}^n (\text{Ad}_{A_{i-1}} \xi_i) \dot{\theta}_i = J_s(\theta) \dot{\theta}.$$

Note that $J_{s,i}$ does not depend on $\theta_i, \dots, \theta_n$: column i is determined by only the upstream joints, which makes the computation recursive ($A_i = A_{i-1} e^{[\xi_i]\theta_i}$). $\square$

Theorem 29.9: Body Jacobian. With the same notation,

$$\mathcal{V}_b = J_b(\theta) \dot{\theta}, \quad J_{b,n} = \text{Ad}_{M^{-1}} \xi_n, \quad J_{b,i} = \text{Ad}_{(e^{[\xi_{i+1}]\theta_{i+1}} \dots e^{[\xi_n]\theta_n} M)^{-1}} \xi_i \quad (i < n),$$

and moreover

$$J_s(\theta) = \text{Ad}_{T(\theta)} J_b(\theta), \quad J_b(\theta) = \text{Ad}_{T(\theta)^{-1}} J_s(\theta).$$

Proof. Step 1. With $B_{i+1} = e^{[\xi_{i+1}]\theta_{i+1}} \dots e^{[\xi_n]\theta_n} M$ one has $T = A_{i-1} e^{[\xi_i]\theta_i} B_{i+1}$, and by Step 2 of the previous proof $\frac{\partial T}{\partial \theta_i} = A_{i-1} [\xi_i] e^{[\xi_i]\theta_i} B_{i+1}$. Since $[\xi_i]$ and $e^{[\xi_i]\theta_i}$ commute, $\frac{\partial T}{\partial \theta_i} =$

$A_{i-1}e^{[\xi_i]\theta_i}[\xi_i]B_{i+1}$, hence

$$T^{-1}\frac{\partial T}{\partial \theta_i} = B_{i+1}^{-1}e^{-[\xi_i]\theta_i}A_{i-1}^{-1} \cdot A_{i-1}e^{[\xi_i]\theta_i}[\xi_i]B_{i+1} = B_{i+1}^{-1}[\xi_i]B_{i+1} = [\text{Ad}_{B_{i+1}^{-1}} \xi_i].$$

For $i = n$ one has $B_{n+1} = M$ and the column is $\text{Ad}_{M^{-1}}\xi_n$.

Step 2. As in the previous theorem, $[\mathcal{V}_b] = T^{-1}\dot{T} = \sum_i (T^{-1}\frac{\partial T}{\partial \theta_i})\dot{\theta}_i$, whence $\mathcal{V}_b = J_b(\theta)\dot{\theta}$.

Step 3 — relation between the two. For every $\dot{\theta}$,

$$J_s\dot{\theta} = \mathcal{V}_s = \text{Ad}_T\mathcal{V}_b = \text{Ad}_T J_b\dot{\theta}$$

by Proposition 29.6; since $\dot{\theta}$ is arbitrary, the matrices agree. The second formula follows by multiplying by $\text{Ad}_{T^{-1}} = (\text{Ad}_T)^{-1}$ (Proposition 29.2). $\square$

Remark 29.10: Which of the two to use. The columns of J_s are twists read in the base frame; those of J_b in the tool frame. For force control and for tracking trajectories expressed relative to the tool, J_b is convenient, because the body twist does not depend on the choice of base frame (Theorem 29.10(i) proves this for the rank, and the same computation proves it for the matrix); for statics and for obstacles described in the world, J_s is convenient. The two descriptions are related by an invertible matrix: any property that depends only on the rank, such as singularity, does not distinguish between them.

Example 29.5: Spatial Jacobian of the planar 3R arm. With the data of Example 29.4 and $\theta = (0.4; -0.9; 1.3)$ radians, the formula of Theorem 29.8 gives

$$J_s(\theta) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0.778837 & 0.059698 \\ 0 & -1.842122 & -3.158496 \\ 0 & 0 & 0 \end{pmatrix}.$$

Structure. The first three rows (angular part) are $\mathbf{e}_3$ for every column: all axes remain parallel to z , as must hold for a planar arm. The sixth row is zero: no joint produces velocity along z .

Independent check. Approximating $\dot{T}T^{-1}$ with centred finite differences of step 10^{-6} on the forward kinematics computed numerically gives the same matrix with maximum discrepancy $4.2 \cdot 10^{-10}$, of the order of the truncation error expected for a centred difference ($O(\delta^2) = 10^{-12}$ multiplied by the scale of the third derivatives, plus $O(\varepsilon/\delta) \approx 10^{-10}$ of rounding error). $\checkmark$

Rank. $\text{rank } J_s = 3$ and $\text{rank } J_b = 3$: the configuration is not singular, consistently with the fact that $\sin \theta_2 = \sin(-0.9) \neq 0$ (Example 29.6). $\checkmark$

29.7. Kinematic singularities

Definition 29.10: Singular configuration. Let $J(\theta) \in \mathbb{R}^{6 \times n}$ be the Jacobian (spatial or body, per Definition 29.9) of an open chain with n joints (Definition 29.6). The configuration θ is called **singular** if

$$\text{rank } J(\theta) < \min(6, n),$$

that is, if $J(\theta)$ does not have full rank. If the assigned task is not the full pose but only part

of it (for instance, position alone, or motion in a plane), J is replaced with the corresponding **task Jacobian matrix** and the same definition is applied.

Proposition 29.7: *The rank does not depend on the choice of Jacobian. For every θ , $\text{rank } J_s(\theta) = \text{rank } J_b(\theta)$.*

Proof. By Theorem 29.9, $J_s = \text{Ad}_T J_b$; by Proposition 29.2 the matrix $\text{Ad}_T \in \mathbb{R}^{6 \times 6}$ is invertible. Multiplying on the left by an invertible matrix does not change the rank: $\text{rank}(\text{Ad}_T J_b) = \text{rank } J_b$, because Ad_T is an isomorphism of $\mathbb{R}^6$ and sends the image of J_b to a subspace of the same dimension. $\square$

Theorem 29.10: *Singularities do not depend on the frames. Let $T(\theta)$ be the forward kinematics of an open chain.*

- (i) *If the base frame is changed, that is T is replaced with $T'(\theta) = P T(\theta)$ with $P \in SE(3)$ constant, then $J'_b(\theta) = J_b(\theta)$.*
- (ii) *If the tool frame is changed, that is $T'(\theta) = T(\theta)Q$ with $Q \in SE(3)$ constant, then $J'_s(\theta) = J_s(\theta)$.*

In both cases the set of singular configurations remains the same.

Proof. (i). $T' = PT$ with P constant gives $\dot{T}' = P\dot{T}$, hence

$$(T')^{-1}\dot{T}' = T^{-1}P^{-1}P\dot{T} = T^{-1}\dot{T},$$

that is $\mathcal{V}'_b = \mathcal{V}_b$ for every $\dot{\theta}$; since $\mathcal{V}_b = J_b\dot{\theta}$ and $\mathcal{V}'_b = J'_b\dot{\theta}$ for every $\dot{\theta}$, one concludes $J'_b = J_b$.

(ii). $T' = TQ$ with Q constant gives $\dot{T}' = \dot{T}Q$, hence

$$\dot{T}'(T')^{-1} = \dot{T}Q Q^{-1}T^{-1} = \dot{T}T^{-1},$$

that is $J'_s = J_s$ by the same argument.

Conclusion. In (i) the rank of J_b is unchanged and, by Proposition 29.7, so is that of J_s ; in (ii) that of J_s is unchanged, and hence so is that of J_b . The condition in Definition 29.10 does not change. $\square$

Remark 29.11: *Why singularity is a property of the mechanism. Theorem 29.10 says that moving the robot within the cell (change of base) or mounting a different tool (change of tool frame) neither creates nor removes singularities. This is a fact that is not intuitive to programmers: the temptation to “escape a singularity by redefining the TCP” is understandable and wrong. What can be done is to change the task — for instance, giving up control of orientation about one axis — and then the task Jacobian matrix is a different matrix, with different singularities.*

Theorem 29.11: *Two collinear revolute axes give a singularity. Let there be an open chain with $n \leq 6$ joints, of which two — joint i and joint j , $i \neq j$ — are revolute, and suppose that in configuration θ their axes, read in the base frame, are the same line (same direction up to sign and same support). Then θ is singular.*

Proof. Step 1 — the columns are the twists of the current axes. By Theorem 29.8, $J_{s,k} = \text{Ad}_{A_{k-1}} \xi_k$. By Proposition 29.2, $\text{Ad}_{A_{k-1}}$ sends the twist of a line to the twist of the transformed line: if $\xi_k = (\omega_k, -\omega_k \times \mathbf{q}_k)$ and $A_{k-1} = (R, \mathbf{p})$, then

$$\text{Ad}_{A_{k-1}} \xi_k = (R\omega_k, [\mathbf{p}]_{\times} R\omega_k + R(-\omega_k \times \mathbf{q}_k)) = (\tilde{\omega}_k, -\tilde{\omega}_k \times \tilde{\mathbf{q}}_k),$$

with $\tilde{\omega}_k = R\omega_k$ and $\tilde{\mathbf{q}}_k = R\mathbf{q}_k + \mathbf{p}$. Checking the last equality is a direct computation:

$$-\tilde{\omega}_k \times \tilde{\mathbf{q}}_k = -(R\omega_k) \times (R\mathbf{q}_k + \mathbf{p}) = -R(\omega_k \times \mathbf{q}_k) - (R\omega_k) \times \mathbf{p} = R(-\omega_k \times \mathbf{q}_k) + [\mathbf{p}]_{\times} R\omega_k,$$

where the equivariance (29.4) and the antisymmetry of the cross product were used. Hence column k is the twist of the axis of joint k in the current position, with zero pitch.

Step 2 — collinearity. Let $\tilde{\omega}_i, \tilde{\mathbf{q}}_i$ and $\tilde{\omega}_j, \tilde{\mathbf{q}}_j$ be the data of the two current axes. By hypothesis the two lines coincide, hence $\tilde{\omega}_j = \varepsilon \tilde{\omega}_i$ with $\varepsilon \in \{+1, -1\}$ and $\tilde{\mathbf{q}}_j = \tilde{\mathbf{q}}_i + \mu \tilde{\omega}_i$ for some $\mu \in \mathbb{R}$. Then

$$J_{s,j} = (\varepsilon \tilde{\omega}_i, -\varepsilon \tilde{\omega}_i \times (\tilde{\mathbf{q}}_i + \mu \tilde{\omega}_i)) = \varepsilon (\tilde{\omega}_i, -\tilde{\omega}_i \times \tilde{\mathbf{q}}_i) = \varepsilon J_{s,i},$$

because $\tilde{\omega}_i \times \tilde{\omega}_i = \mathbf{0}$.

Step 3 — conclusion. Two columns of J_s are proportional, hence linearly dependent, and $\text{rank } J_s \leq n - 1$. Since $n \leq 6$, $\min(6, n) = n$, and hence $\text{rank } J_s < \min(6, n)$: the configuration is singular by Definition 29.10. $\square$

***Remark 29.12: Why the hypothesis $n \leq 6$.** The hypothesis is not a technicality. In a redundant robot ($n > 6$) the loss of one independent column does not necessarily bring the rank below 6: two aligned axes remove one internal degree of freedom without preventing any motion of the tool, and the loss shows up in the null space of J , not in its rank. For industrial six-axis manipulators, which are by far the most frequent case, J is square and the condition $\det J(\theta) = 0$ is equivalent to Definition 29.10: this is what controllers monitor.*

First worked example: the three-joint planar arm

Example 29.6: Singularity of the planar 3R arm. We return to the arm of Example 29.4: three revolute joints parallel to $\mathbf{e}_3$, links $\ell_1 = 2$, $\ell_2 = 1.5$, $\ell_3 = 1$. The task is the pose planar pose (x, y, φ) , so the task Jacobian matrix is 3×3 .

Step 1 — the matrix. Differentiating the formulas of Example 29.4, with $s_1 = \sin \theta_1$, $s_{12} = \sin(\theta_1 + \theta_2)$, $s_{123} = \sin(\theta_1 + \theta_2 + \theta_3)$ and analogously for the cosines:

$$J = \begin{pmatrix} -\ell_1 s_1 - \ell_2 s_{12} - \ell_3 s_{123} & -\ell_2 s_{12} - \ell_3 s_{123} & -\ell_3 s_{123} \\ \ell_1 c_1 + \ell_2 c_{12} + \ell_3 c_{123} & \ell_2 c_{12} + \ell_3 c_{123} & \ell_3 c_{123} \\ 1 & 1 & 1 \end{pmatrix}.$$

Step 2 — the determinant. Subtract the second column from the first and the third from the second (operations that do not change the determinant):

$$\det J = \det \begin{pmatrix} -\ell_1 s_1 & -\ell_2 s_{12} & -\ell_3 s_{123} \\ \ell_1 c_1 & \ell_2 c_{12} & \ell_3 c_{123} \\ 0 & 0 & 1 \end{pmatrix} = \det \begin{pmatrix} -\ell_1 s_1 & -\ell_2 s_{12} \\ \ell_1 c_1 & \ell_2 c_{12} \end{pmatrix} = \ell_1 \ell_2 (c_1 s_{12} - s_1 c_{12}),$$

having expanded along the last row. By the subtraction formula, $c_1 s_{12} - s_1 c_{12} = \sin((\theta_1 + \theta_2) - \theta_1) = \sin \theta_2$, hence

$$\boxed{\det J = \ell_1 \ell_2 \sin \theta_2}$$

Step 3 — the singular set. $\det J = 0$ if and only if $\sin \theta_2 = 0$, that is $\theta_2 \in \{0, \pi\}$ (modulo 2π): a fully outstretched arm ($\theta_2 = 0$, *elbow singularity*) or a fully folded one ($\theta_2 = \pi$). The result depends neither on θ_1 nor on θ_3 nor on ℓ_3 : the last is a structural fact, because the last link contributes to the position but its contribution is identical in the three columns and disappears in the subtractions of Step 2.

Step 4 — what is lost. In the outstretched configuration $\theta = (30^\circ, 0, 0)$ the tool is located at $(3.8971143; 2.2500000)$, at distance $\|P\| = 4.5 = \ell_1 + \ell_2 + \ell_3$ from the origin — the maximum reachable. The left singular vector associated with the zero singular value, computed with the singular value decomposition, is

$$\mathbf{u} = (0.8660254; 0.5000000; 0) = (\cos 30^\circ, \sin 30^\circ, 0),$$

that is, the direction of the outstretched arm: no combination of joint velocities can produce a tool velocity with a component along the arm (at constant orientation φ). This is geometrically obvious in hindsight — the distance from the origin is maximal, hence its derivative is zero — but it is the rank computation that produces it without need of intuition.

Step 5 — independent numerical check. For four configurations we computed the determinant with `numpy` and the singular values of J ; the product of the singular values must equal $|\det J|$.

$(\theta_1, \theta_2, \theta_3)$	$\det J$	$\ell_1 \ell_2 \sin \theta_2$	singular values
$(30^\circ, 60^\circ, -45^\circ)$	2.5980762	2.5980762	4.952994; 0.807799; 0.649353
$(30^\circ, 5.73^\circ, -45^\circ)$	0.2995002	0.2995002	5.181025; 0.981248; 0.058912
$(30^\circ, 0^\circ, 0^\circ)$	0	0	5.466340; 0.786845; 0
$(30^\circ, 180^\circ, 22.9^\circ)$	0	0	3.091766; 0.884932; 0

(the second row corresponds to $\theta_2 = 0.1$ rad, the fourth to $\theta_3 = 0.4$ rad). In the first two rows the product of the singular values reproduces $|\det J|$ to ten digits ($4.952994 \cdot 0.807799 \cdot 0.649353 = 2.5980762114$ and $5.181025 \cdot 0.981248 \cdot 0.058912 = 0.2995002499$); in the last two the rank computed with tolerance 10^{-9} equals 2. ✓ Moreover J obtained by centred finite differences ($\delta = 10^{-6}$) from the forward kinematics agrees with the analytic one to within $2.9 \cdot 10^{-10}$. ✓

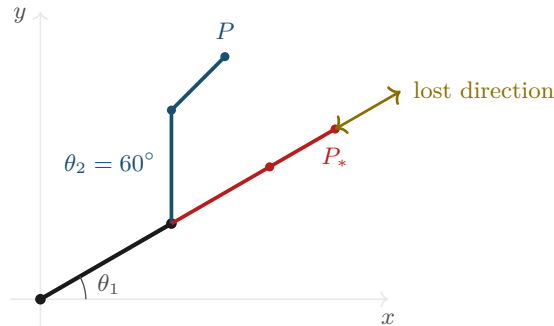


Figure 29.2: The planar 3R arm with lengths $\ell_1 = 2$, $\ell_2 = 1.5$, $\ell_3 = 1$. The first link (dark grey) is common to both configurations, which share $\theta_1 = 30^\circ$. In blue the configuration $(30^\circ, 60^\circ, -45^\circ)$, with vertices $(0, 0)$, $(1.7321; 1)$, $(1.7321; 2.5)$, $(2.4392; 3.2071)$ and $\det J = 2.598076$. In red the outstretched configuration $(30^\circ, 0, 0)$, with vertices $(0, 0)$, $(1.7321; 1)$, $(3.0311; 1.75)$, $(3.8971; 2.25)$: here $\det J = \ell_1 \ell_2 \sin 0 = 0$ and the tool, located at the maximum distance 4.5 from the origin, cannot move instantaneously along the direction $(\cos 30^\circ, \sin 30^\circ)$ of the outstretched arm (double orange arrow).

Second worked example: the spherical wrist

Example 29.7: Singularity of the spherical wrist. A **spherical wrist** is formed by three revolute joints whose axes meet at a point $\mathbf{c}$ (the *wrist centre*). Consider the reference configuration in which the axes are, in order, $\mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_3$ (a “ZYZ” wrist, by far the most common form in six-axis industrial robots, where the three joints are J4, J5, J6).

Step 1 — reduction to 3×3 . Take $\mathbf{c}$ as the origin of the base frame. Then every joint twist has the form $\xi_k = (\omega_k, -\omega_k \times \mathbf{0}) = (\omega_k, \mathbf{0})$. It follows that every $e^{[\xi_k]\theta_k}$ is a rotation about a line through the origin, that is, of the form $(R, \mathbf{0})$ (Theorem 29.5 with $h = 0$ and $\mathbf{q} = \mathbf{0}$); hence also $A_{k-1} = e^{[\xi_1]\theta_1} \dots e^{[\xi_{k-1}]\theta_{k-1}}$ has the form $(R_{k-1}, \mathbf{0})$, and by Proposition 29.2

$$J_{s,k} = \text{Ad}_{A_{k-1}} \xi_k = (R_{k-1}\omega_k, [\mathbf{0}]_{\times} R_{k-1}\omega_k + R_{k-1}\mathbf{0}) = (R_{k-1}\omega_k, \mathbf{0}).$$

All columns have zero linear part: geometrically, no wrist joint moves the centre $\mathbf{c}$. Hence

$$J_s(\theta) = \begin{pmatrix} J_\omega(\theta) \\ 0 \end{pmatrix}, \quad \text{rank } J_s = \text{rank } J_\omega, \quad J_\omega = \begin{bmatrix} \omega_1 & R_1\omega_2 & R_1R_2\omega_3 \end{bmatrix} \in \mathbb{R}^{3 \times 3},$$

with $R_1 = e^{[\mathbf{e}_3] \times \theta_1}$ and $R_2 = e^{[\mathbf{e}_2] \times \theta_2}$.

Step 2 — the three columns. With $c_k = \cos \theta_k$, $s_k = \sin \theta_k$:

$$\mathbf{c}_1 = \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{c}_2 = R_1\mathbf{e}_2 = \begin{pmatrix} -s_1 \\ c_1 \\ 0 \end{pmatrix}, \quad \mathbf{c}_3 = R_1R_2\mathbf{e}_3 = R_1 \begin{pmatrix} s_2 \\ 0 \\ c_2 \end{pmatrix} = \begin{pmatrix} c_1s_2 \\ s_1s_2 \\ c_2 \end{pmatrix}.$$

Step 3 — the determinant. Expanding along the first column (whose only nonzero entry is in position $(3, 1)$):

$$\det J_\omega = (+1) \cdot \det \begin{pmatrix} -s_1 & c_1s_2 \\ c_1 & s_1s_2 \end{pmatrix} = -s_1^2s_2 - c_1^2s_2 = -s_2,$$

hence

$$\boxed{\det J_\omega = -\sin \theta_2}$$

and the wrist is singular if and only if $\theta_2 \in \{0, \pi\}$: the fifth axis (J5) at 0° or 180° , that is, axes J4 and J6 aligned. This is the **wrist singularity**.

Step 4 — singular values in closed form. The three columns are unit vectors, and

$$\mathbf{c}_1 \cdot \mathbf{c}_2 = 0, \quad \mathbf{c}_2 \cdot \mathbf{c}_3 = -s_1c_1s_2 + c_1s_1s_2 = 0, \quad \mathbf{c}_1 \cdot \mathbf{c}_3 = c_2.$$

The Gram matrix is therefore

$$J_\omega^\top J_\omega = \begin{pmatrix} 1 & 0 & c_2 \\ 0 & 1 & 0 \\ c_2 & 0 & 1 \end{pmatrix},$$

whose eigenvalues are read off by inspection: 1 (eigenvector $(0, 1, 0)$) and $1 \pm c_2$ (eigenvectors $(1, 0, \pm 1)$). The singular values of J_ω are the square roots:

$$\sigma = \left\{ \sqrt{1 + \cos \theta_2}, 1, \sqrt{1 - \cos \theta_2} \right\} = \left\{ \sqrt{2} \left| \cos \frac{\theta_2}{2} \right|, 1, \sqrt{2} \left| \sin \frac{\theta_2}{2} \right| \right\}.$$

The product equals $\sqrt{(1 + c_2)(1 - c_2)} = \sqrt{1 - c_2^2} = |\sin \theta_2| = |\det J_\omega|$: consistent with Step 3. ✓

Step 5 — independent numerical check. The singular value decomposition of J_ω computed with `numpy` (with $\theta_1 = 0.7$ and $\theta_3 = -1.1$, which do not matter) gives:

θ_2 (rad)	$\det J_\omega$	$-\sin \theta_2$	singular values (numerical)
$\pi/3$	-0.8660254	-0.8660254	1.2247449; 1; 0.7071068
0.3	-0.2955202	-0.2955202	1.3983335; 1; 0.2113374
0.01	-0.0099998	-0.0099998	1.4141959; 1; 0.0070710
0.001	-0.0010000	-0.0010000	1.4142134; 1; 0.0007071
0	0	0	1.4142136; 1; 0
π	0	0	1.4142136; 1; 0

The maximum discrepancy between the numerical singular values and those of Step 4 is $5.5 \cdot 10^{-15}$. ✓ Note the behaviour $\sigma_{\min} \approx |\theta_2|/\sqrt{2}$ as $\theta_2 \rightarrow 0$, which is also read off from the closed form: $\sqrt{2} |\sin(\theta_2/2)| \approx |\theta_2|/\sqrt{2}$.

Step 6 — what is lost. At $\theta_2 = 0$ one has $\mathbf{c}_3 = \mathbf{c}_1 = \mathbf{e}_3$, hence the image of J_ω is $\langle \mathbf{e}_3, R_1 \mathbf{e}_2 \rangle$ and the lost direction is $R_1 \mathbf{e}_1 = (\cos \theta_1, \sin \theta_1, 0)$. For $\theta_1 = 0.7$ the singular value decomposition gives, as the left singular vector of the zero singular value, $\pm(0.7648422; 0.6442177; 0)$, and indeed $(\cos 0.7; \sin 0.7; 0) = (0.7648422; 0.6442177; 0)$. ✓ The wrist cannot rotate instantaneously about the direction orthogonal to the plane of the first two axes: this is the *gimbal lock* of Chapter 28 (Theorem 28.5) in its mechanical incarnation, with the — essential — difference that here it is not an artefact of the parametrisation but a property of the mechanism: two physical axes are truly aligned.

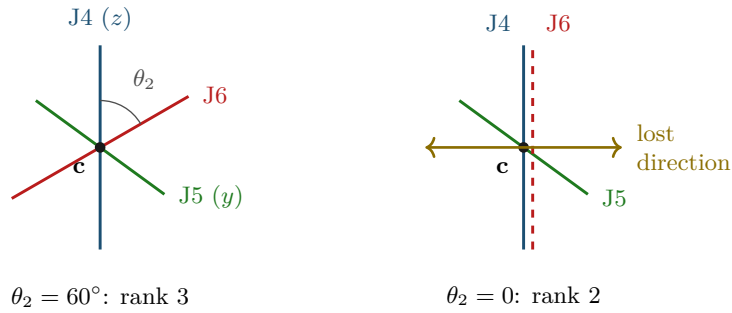


Figure 29.3: The ZYZ spherical wrist, seen in the xz plane of the page with $\theta_1 = 0$; axis J5, which points along y , that is out of the page, is drawn obliquely. On the left the regular configuration $\theta_2 = 60^\circ$: the three axes meet at the centre $\mathbf{c}$ and are linearly independent; J6 makes the angle θ_2 with J4 and is directed along $(\sin 60^\circ, \cos 60^\circ) = (0.8660; 0.5000)$. On the right the singular configuration $\theta_2 = 0$: J6 returns to coincide with J4 (dashed red line, shifted slightly for legibility), the independent columns drop to two and the wrist cannot rotate about the direction $R_1 \mathbf{e}_1$, orthogonal to the plane of J4 and J5 (double orange arrow). This is the wrist singularity, the one controllers flag when the fifth axis crosses zero.

Remark 29.13: What controllers do: damped least squares. Near a singularity the exact solution $\dot{\theta} = J^{-1}\mathcal{V}$ blows up like $1/\sigma_{\min}$. The standard countermeasure is the damped least squares (DLS) inverse

$$\dot{\theta}_\lambda = J^\top (JJ^\top + \lambda^2 I)^{-1} \mathcal{V},$$

which, in terms of singular values, replaces the factor $1/\sigma_i$ with $\sigma_i/(\sigma_i^2 + \lambda^2)$: the latter is bounded by $1/(2\lambda)$ and tends to 0 as $\sigma_i \rightarrow 0$, instead of diverging. On the planar 3R arm of Example 29.6, requesting the twist $\mathcal{V} = (\cos 30^\circ, \sin 30^\circ, 0)$ — precisely the direction that is lost — with $\lambda = 0.05$:

θ_2 (rad)	$\ J^{-1}\mathcal{V}\ $	$\ \dot{\theta}_\lambda\ $	error $\ J\dot{\theta}_\lambda - \mathcal{V}\ $
0.2	7.1584	6.3164	0.11531
0.05	28.6718	9.3688	0.67239
0.01	143.3716	2.7364	0.98086
0.001	1433.7208	0.2789	0.99980

One reads the trade-off: the joint velocity stays bounded, but the tracking error grows until the requested motion is given up entirely. Industrial controllers implement this strategy or variants (**FANUC** Singular Avoidance, **ABB** SingArea, **KUKA** \$SINGULARITY_HANDLING); none of them “eliminates” the singularity, which by Theorem 29.10 is a property of the mechanism, but all of them make passing through it numerically harmless.

Remark 29.14: Panorama: the Baker–Campbell–Hausdorff formula. The chapter has repeatedly used the fact that $e^A e^B \neq e^{A+B}$ when A and B do not commute: this is why the order of the factors in the product of exponentials matters, and why there are no “coordinates” in which composing two rigid motions reduces to adding. The quantitative measure of the discrepancy is the **Baker–Campbell–Hausdorff** formula, reported here as a panorama, without proof:

$$e^A e^B = \exp \left(A + B + \frac{1}{2}[A, B] + \frac{1}{12}([A, [A, B]] - [B, [A, B]]) + \cdots \right),$$

where the subsequent terms are rational combinations of nested commutators and the series converges for $\|A\|, \|B\|$ sufficiently small. The first correction term $\frac{1}{2}[A, B]$ is, for two twists, the commutator computed in Theorem 29.3(ii): to first order, reversing the order of two infinitesimal screw motions produces a net screw motion with twist $[\xi_1, \xi_2]$. This is the principle behind nonholonomic manoeuvring — think of parallel parking, where alternating steering and forward motion generates a lateral displacement that neither command possesses on its own.

What is missing to prove it here: the construction of the Dynkin series and the convergence estimate, which require the free Lie algebra and the Poincaré–Birkhoff–Witt theorem. Sources: Murray, Li, Sastry, *A Mathematical Introduction to Robotic Manipulation*, Theorem 8.5, p. 381 (stated in the form reported here); B. C. Hall, *An Elementary Introduction to Groups and Representations*, arXiv:math-ph/0005032, Chapter 4 (in particular §4.3, series form), for the complete proof.

29.8. Assigned exercises

Answers are in the Teacher’s key (Section 29.11).

Exercise 29.1: Axis and pitch, and how they are transported. *Objective:* handle the correspondence between twist and screw (axis + pitch) and understand how the adjoint transports it. *Estimated time:* 30 minutes. *Type:* symbolic/proof-based.

Let $\xi = (\omega, \mathbf{v})$ with $\|\omega\| = 1$, and let $h = \omega^\top \mathbf{v}$ and $\mathcal{L} = \{\omega \times \mathbf{v} + \lambda \omega\}$ be its pitch and its axis.

- Compute the pitch and axis of $\xi = (\mathbf{e}_3, (1, 0, 2)^\top)$ and describe in words the motion $e^{[\xi]\theta}$.
- Prove that for every $g \in SE(3)$ the twist $\text{Ad}_g \xi$ has the same pitch h and axis the line $g(\mathcal{L})$.
- Deduce that the pitch is an invariant of the screw motion under changes of frame, and relate the result to Theorem 29.10.
- Verify (b) numerically with $g = (e^{[\mathbf{e}_2] \times 0.8}, (1, -2, \frac{1}{2})^\top)$ and the twist of (a): compute $\text{Ad}_g \xi$, its pitch and axis point, and check that the latter differs from $g(\omega \times \mathbf{v})$ by a multiple of the new ω .

Exercise 29.2: Screw decomposition of a given motion. *Objective:* carry out the full algorithm of Theorem 29.6 and check it by two independent routes. *Estimated time:* 40 minutes. *Type:* numerical/computational.

Consider $g = (R, \mathbf{p})$ with

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{p} = (2, 1, -1)^\top.$$

- (a) Determine θ and ω .
- (b) Write $G(\theta)^{-1}$ explicitly and compute $\mathbf{v} = G(\theta)^{-1}\mathbf{p}$.
- (c) Compute the pitch h , the axis point $\mathbf{q}$ and the translation along the axis $h\theta$; verify that $h\theta = \omega^\top \mathbf{p}$.
- (d) *Independent geometric check:* the motion acts on the yz plane as a planar rotation composed with a translation; compute its fixed point and compare it with $\mathbf{q}$.
- (e) *Numerical check:* verify $e^{[\xi]\theta} = g$ and estimate $\min_{\mathbf{x}} \|g(\mathbf{x}) - \mathbf{x}\|$ on a random sample, comparing it with $|h\theta|$.

Exercise 29.3: The RPY wrist and gimbal lock. *Objective:* transfer the method of Example 29.7 to a different wrist architecture and recognise that the singularity changes position but not nature. *Estimated time:* 40 minutes. *Type:* computational with a proof-based part.

Consider a spherical wrist whose three axes, in the reference configuration, are $\omega_1 = \mathbf{e}_3$, $\omega_2 = \mathbf{e}_2$, $\omega_3 = \mathbf{e}_1$ (a “ZYX” wrist, that is, the *roll-pitch-yaw* convention).

- (a) Write the three columns of $J_\omega(\theta)$.
- (b) Compute $\det J_\omega$ and determine the singular set.
- (c) Compute the Gram matrix $J_\omega^\top J_\omega$ and derive from it the singular values in closed form; verify that their product equals $|\det J_\omega|$.
- (d) Identify the lost rotation direction at the singular configuration and verify it numerically with the singular value decomposition for $\theta_1 = 0.9$.
- (e) Compare the result with Theorem 28.5 of Chapter 28 and explain in what sense the singularity here is *not* an artefact of the parametrisation.

29.9. Worked exercises

Exercise 29.4: Composing two rotations: intersecting axes and skew axes. *Type:* symbolic/geometric with numerical check. Let

- g_1 be the rotation by 90° about the z axis (line $\{(0, 0, \lambda)\}$);
- g_2 be the rotation by 90° about the line $\{(t, 0, 1)\}$, parallel to the x axis and passing through $(0, 0, 1)$.

Determine the screw axis and pitch of $g = g_2 \circ g_1$. Repeat replacing g_2 with the rotation by 90° about the line $\{(t, 1, 0)\}$, which is *skew* to the z axis, and comment on the difference.

Solution. Strategy. We write g_1 and g_2 as pairs $(R, \mathbf{p})$ using the form $\mathbf{x} \mapsto \mathbf{c} + R(\mathbf{x} - \mathbf{c})$ for a rotation about a line through $\mathbf{c}$; compose with Proposition 29.1; apply the algorithm of Theorem 29.6(iii).

Step 1 — the two matrices. $g_1 = (R_1, \mathbf{0})$ with $R_1 = e^{[\mathbf{e}_3] \times \pi/2} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (the axis passes through the origin, so $\mathbf{p}_1 = \mathbf{0}$). For g_2 , with $\mathbf{c} = (0, 0, 1)^\top$ and $R_2 = e^{[\mathbf{e}_1] \times \pi/2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$, one has $\mathbf{x} \mapsto R_2\mathbf{x} + (\mathbf{c} - R_2\mathbf{c})$, that is

$$\mathbf{p}_2 = \mathbf{c} - R_2\mathbf{c} = (0, 0, 1)^\top - (0, -1, 0)^\top = (0, 1, 1)^\top.$$

Step 2 — the composition. By Proposition 29.1, $g = g_2 g_1 = (R_2 R_1, R_2 \mathbf{0} + \mathbf{p}_2) = (R, \mathbf{p})$ with

$$R = R_2 R_1 = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{p} = (0, 1, 1)^\top.$$

(The matrix R cyclically permutes the axes: $\mathbf{e}_1 \mapsto \mathbf{e}_3 \mapsto -\mathbf{e}_2 \mapsto \dots$; read off column by column.)

Step 3 — angle and axis of the rotation. $\text{tr } R = 0$, hence

$$\theta = \arccos \frac{0 - 1}{2} = \frac{2\pi}{3} = 120^\circ.$$

Since $\sin \theta = \frac{\sqrt{3}}{2}$,

$$[\omega]_\times = \frac{R - R^\top}{2 \sin \theta} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \implies \omega = \frac{1}{\sqrt{3}}(1, -1, 1)^\top.$$

Check: $R(1, -1, 1)^\top = (1, -1, 1)^\top$, so ω is indeed the axis. ✓

Step 4 — pitch. By Proposition 29.4(i),

$$h\theta = \omega^\top \mathbf{p} = \frac{1}{\sqrt{3}}(1 \cdot 0 + (-1) \cdot 1 + 1 \cdot 1) = 0,$$

so $h = 0$: the composition is a *pure rotation* by 120° .

Step 5 — the axis. With $\theta = \frac{2\pi}{3}$ one has $\cot \frac{\theta}{2} = \cot \frac{\pi}{3} = \frac{1}{\sqrt{3}}$, and Proposition 29.3 gives $\mathbf{v} = G(\theta)^{-1} \mathbf{p} = \frac{1}{\sqrt{3}}(1, 1, 0)^\top = (0.577350; 0.577350; 0)$. Hence

$$\mathbf{q} = \omega \times \mathbf{v} = \left(-\frac{1}{3}, \frac{1}{3}, \frac{2}{3}\right)^\top.$$

Independent check (geometric). The two lines of rotation meet at the point $\mathbf{c} = (0, 0, 1)$, common to $\{(0, 0, \lambda)\}$ and $\{(t, 0, 1)\}$. Since $\mathbf{c}$ lies on the axis of g_1 , $g_1(\mathbf{c}) = \mathbf{c}$, and since it lies on the axis of g_2 , $g_2(\mathbf{c}) = \mathbf{c}$: hence $g(\mathbf{c}) = \mathbf{c}$. A rigid motion with a fixed point and $R \neq I$ is a pure rotation whose axis passes through that point — which independently confirms $h = 0$ and tells us where the axis is. Finally, one checks that $\mathbf{q}$ lies on the line through $\mathbf{c}$ with direction ω : the point of that line closest to the origin is

$$\mathbf{c} - (\mathbf{c}^\top \omega) \omega = (0, 0, 1)^\top - \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}(1, -1, 1)^\top = \left(-\frac{1}{3}, \frac{1}{3}, \frac{2}{3}\right)^\top = \mathbf{q}. \quad \checkmark$$

The numerical computation confirms $(\mathbf{q} - \mathbf{c}) \times \omega = \mathbf{0}$ up to machine zero.

Numerical check. $\|e^{[\xi]\theta} - g\|_{\max} = 2.2 \cdot 10^{-16}$ with the closed form and $3.1 \cdot 10^{-16}$ with the series to 40 digits. ✓

Second case: skew axes. With $\mathbf{c}' = (0, 1, 0)^\top$ one has $\mathbf{p}'_2 = \mathbf{c}' - R_2 \mathbf{c}' = (0, 1, 0)^\top - (0, 0, 1)^\top = (0, 1, -1)^\top$, hence $g' = (R, (0, 1, -1)^\top)$ with the same R and hence the same $\theta = \frac{2\pi}{3}$ and the same ω . Now, however,

$$h\theta = \omega^\top \mathbf{p}' = \frac{1}{\sqrt{3}}(0 - 1 - 1) = -\frac{2}{\sqrt{3}} = -1.154701,$$

$$h = \frac{h\theta}{\theta} = -\frac{3}{\sqrt{3}\pi} = -\frac{\sqrt{3}}{\pi} = -0.551329.$$

Computing $\mathbf{v} = G(\theta)^{-1}\mathbf{p}'$ gives $\mathbf{v} = (-0.125860; 0.125860; -0.703210)$ and $\mathbf{q} = \omega \times \mathbf{v} = \left(\frac{1}{3}, \frac{1}{3}, 0\right)^\top$: a genuine screw motion, with nonzero pitch.

Independent check of the second case. By Proposition 29.4, $\min_{\mathbf{x}} \|g'(\mathbf{x}) - \mathbf{x}\| = |h\theta| = 1.154701$; searching for the minimum over 400 000 randomly drawn points gives 1.154724, which exceeds the exact value by $2.3 \cdot 10^{-5}$ because no sample falls exactly on the axis. ✓

Interpretation. The composition of two rotations with *intersecting* axes is a rotation (about an axis through the common point); with *skew* axes it is a screw motion with nonzero pitch. The rotational part is the same in both cases — it depends only on R_1, R_2 — and only the translation changes. This is why the singularities of a spherical wrist (concurrent axes) are separate from those of the arm: in the wrist there is no translation to compose.

Typical errors. (1) Writing $g_2 = (R_2, \mathbf{c})$ instead of $(R_2, \mathbf{c} - R_2\mathbf{c})$: the translation of a rotation about a line not through the origin is not the point of the line. (2) Composing in the wrong order: $g_1g_2 \neq g_2g_1$, and the two products have different axes (the same θ , because $\text{tr}(R_1R_2) = \text{tr}(R_2R_1)$, but different axis and pitch). (3) Concluding “ $h\theta = 0$ hence $\theta = 0$ ”: zero pitch means pure rotation, not absence of motion.

Extension. Prove, with the same method, that the composition of two rotations about *parallel* and distinct axes, of angles α and $-\alpha$, is a pure translation (case (ii) of Theorem 29.6); and that if the angles are α and β with $\alpha + \beta \notin 2\pi\mathbb{Z}$ one obtains a rotation about a third axis parallel to the first two.

Exercise 29.5: The spatial elbow arm: two families of singularities. *Type:* computational with a symbolic part. Consider the anthropomorphic arm with three revolute joints with: $\omega_1 = \mathbf{e}_3$ (vertical axis through the origin, *shoulder rotation*), $\omega_2 = \mathbf{e}_2$ (horizontal axis through the origin, *shoulder elevation*), $\omega_3 = \mathbf{e}_2$ (horizontal axis through $(a_2, 0, 0)$, *elbow*), with the wrist at $(a_2 + a_3, 0, 0)$ in the reference configuration; $a_2 = 1.2$, $a_3 = 1$.

- Write the position of the wrist centre as a function of $(\theta_1, \theta_2, \theta_3)$.
- Compute the determinant of the position Jacobian matrix.
- Describe the singular configurations.
- Verify everything numerically with the singular values.

Solution. Strategy. We observe that joints 2 and 3 move the arm in a vertical plane, that joint 1 rotates that plane about the z axis, and we write the columns of the Jacobian in an adapted basis — $\mathbf{u}$ radial, $\mathbf{w}$ tangential, $\mathbf{e}_3$ vertical — in which the determinant factorises.

Step 1 — forward kinematics. A rotation by θ about $\mathbf{e}_2$ sends $\mathbf{e}_1$ to $(\cos \theta, 0, -\sin \theta)$. Hence, in the vertical plane $\theta_1 = 0$, the wrist centre is at

$$r = a_2 \cos \theta_2 + a_3 \cos(\theta_2 + \theta_3), \quad z = -(a_2 \sin \theta_2 + a_3 \sin(\theta_2 + \theta_3)),$$

and rotating by θ_1 about z :

$$\mathbf{P}(\theta) = (r \cos \theta_1, r \sin \theta_1, z).$$

Here r is the (signed) distance from the vertical axis and z the height.

Step 2 — the columns of the Jacobian. Set $\mathbf{u} = (\cos \theta_1, \sin \theta_1, 0)$ and $\mathbf{w} = (-\sin \theta_1, \cos \theta_1, 0)$, so that $(\mathbf{u}, \mathbf{w}, \mathbf{e}_3)$ is a right-handed orthonormal basis. Differentiating:

$$\frac{\partial \mathbf{P}}{\partial \theta_1} = r \mathbf{w}, \quad \frac{\partial \mathbf{P}}{\partial \theta_2} = r' \mathbf{u} + z' \mathbf{e}_3, \quad \frac{\partial \mathbf{P}}{\partial \theta_3} = r'' \mathbf{u} + z'' \mathbf{e}_3,$$

with

$$r' = -(a_2 \sin \theta_2 + a_3 \sin(\theta_2 + \theta_3)) = z, \quad z' = -r,$$

$$r'' = -a_3 \sin(\theta_2 + \theta_3), \quad z'' = -a_3 \cos(\theta_2 + \theta_3).$$

Step 3 — the determinant. In the coordinates of the basis $(\mathbf{u}, \mathbf{w}, \mathbf{e}_3)$ the three columns are $(0, r, 0)$, $(r', 0, z')$, $(r'', 0, z'')$. Expanding along the second row (the only nonzero entry is in position $(2, 1)$, with sign $(-1)^{2+1} = -1$):

$$\det J = -r (r' z'' - r'' z').$$

One computes, with $s_2 = \sin \theta_2$, $c_2 = \cos \theta_2$, $s_{23} = \sin(\theta_2 + \theta_3)$, $c_{23} = \cos(\theta_2 + \theta_3)$:

$$\begin{aligned} r' z'' - r'' z' &= [-(a_2 s_2 + a_3 s_{23})](-a_3 c_{23}) - (-a_3 s_{23})[-(a_2 c_2 + a_3 c_{23})] \\ &= a_2 a_3 (c_{23} s_2 - s_{23} c_2) + a_3^2 (c_{23} s_{23} - s_{23} c_{23}) = a_2 a_3 \sin(\theta_2 - (\theta_2 + \theta_3)) = -a_2 a_3 \sin \theta_3. \end{aligned}$$

Hence

$$\det J = a_2 a_3 \sin \theta_3 (a_2 \cos \theta_2 + a_3 \cos(\theta_2 + \theta_3)) = a_2 a_3 r \sin \theta_3$$

Step 4 — the two families. $\det J = 0$ if and only if

- $\sin \theta_3 = 0$: **elbow singularity**, arm outstretched ($\theta_3 = 0$) or folded ($\theta_3 = \pi$); this is the spatial analogue of that of the planar arm (Example 29.6);
- $r = 0$: **shoulder singularity**, the wrist centre falls on the vertical axis of the first joint. At that position rotating θ_1 does not move the wrist, and the first column of the Jacobian vanishes identically.

Step 5 — independent numerical check. With $a_2 = 1.2$, $a_3 = 1$:

$(\theta_1, \theta_2, \theta_3)$ rad	$\det J$ (numpy)	$a_2 a_3 r \sin \theta_3$	singular values
(0.5; 0.6; 0.9)	0.9974634	0.9974634	2.178457; 1.061140; 0.431494
(0.3; 1.1; 1.6)	-0.4315241	-0.4315241	1.693609; 0.708244; 0.359757
(0.5; 0.6; 0)	0	0	2.416609; 1.815738; 0
(0.5; 0.6; π)	0	0	1.019804; 0.165067; 0
(0.5; 0.6; 2.4029373)	0	0	1.017082; 0.794383; 0

The last row is the shoulder singularity: θ_3 is the solution of $a_2 \cos \theta_2 + a_3 \cos(\theta_2 + \theta_3) = 0$ for $\theta_2 = 0.6$, and the wrist centre comes out at $(0; 0; -0.8157825)$, that is, exactly on the vertical axis. ✓ In the first two rows the product of the singular values reproduces $|\det J|$ ($2.178457 \cdot 1.061140 \cdot 0.431494 = 0.9974633634$). ✓ The analytic Jacobian agrees with the one from centred finite differences ($\delta = 10^{-6}$) to within $1.6 \cdot 10^{-10}$ in every row. ✓

Structural remark. In the first row the intermediate singular value, 1.061140, equals r exactly. This is no coincidence: the first column is $r\mathbf{w}$ and the other two lie in the plane $\langle \mathbf{u}, \mathbf{e}_3 \rangle$, orthogonal to $\mathbf{w}$; hence $\mathbf{w}$ is a left singular vector with singular value $|r|$. The shoulder singularity is precisely the vanishing of this singular value, the elbow one the vanishing of the determinant of the residual 2×2 block.

Typical errors. (1) Forgetting the sign in $R_y(\theta)\mathbf{e}_1 = (\cos \theta, 0, -\sin \theta)$ and obtaining z with the wrong sign: the determinant changes sign but not the singular set, so the error remains invisible until a velocity is computed. (2) Concluding that the shoulder singularity requires particular θ_2 and θ_3 : it requires only $r = 0$, which is a curve in the (θ_2, θ_3) plane. (3) Looking for the shoulder singularity at “small” r : $\det J$ vanishes only for exactly $r = 0$, but $\sigma_{\min} \leq |r|$, so conditioning already worsens in a neighbourhood.

Extension. Adding a spherical wrist (Example 29.7) gives the six-axis manipulator of industrial use. For a robot with a spherical wrist, the 6×6 Jacobian can be written in block-triangular form by choosing the origin at the wrist centre, and then $\det J_{6 \times 6} = \det J_{\text{arm}} \cdot \det J_{\omega, \text{wrist}}$: the three families of singularities (shoulder, elbow, wrist) are exactly the zeros of the three factors.

Exercise 29.6: Changing reference configuration without changing the robot.
Type: numerical/proof-based. For the planar 3R arm of Example 29.4, adopt as reference configuration $\bar{\theta} = (30^\circ, 60^\circ, -45^\circ)$ instead of $\bar{\theta} = 0$. Compute the new twists $\bar{\xi}_i$ and the new $\bar{M}$, verify that their axes are the current positions of the joints, and check numerically that the two product-of-exponentials formulas give the same kinematics.

Solution. Strategy. We apply Proposition 29.5(iii): the new twists are $\bar{\xi}_i = \text{Ad}_{A_{i-1}} \xi_i$ with $A_{i-1} = e^{[\xi_1]\bar{\theta}_1} \dots e^{[\xi_{i-1}]\bar{\theta}_{i-1}}$, and the new reference pose is $\bar{M} = T(\bar{\theta})$. We then check that the result matches that of Step 1 of the proof of Theorem 29.11: $\bar{\xi}_i$ must be the twist of the axis of joint i in the position it occupies when $\theta = \bar{\theta}$.

Step 1 — the new reference pose. From Example 29.4, $T(\bar{\theta})$ has position (2.4391576; 3.2071068; 0) and orientation $\varphi = 30^\circ + 60^\circ - 45^\circ = 45^\circ$:

$$\bar{M} = \begin{pmatrix} 0.7071068 & -0.7071068 & 0 & 2.4391576 \\ 0.7071068 & 0.7071068 & 0 & 3.2071068 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Step 2 — the new twists. Computing $\text{Ad}_{A_{i-1}} \xi_i$ gives

$$\bar{\xi}_1 = (0, 0, 1, 0, 0, 0), \quad \bar{\xi}_2 = (0, 0, 1, 1, -1.7320508, 0), \quad \bar{\xi}_3 = (0, 0, 1, 2.5, -1.7320508, 0).$$

Step 3 — geometric reading. By Lemma 29.5, the axis of $\bar{\xi}_i$ passes through $\bar{\mathbf{q}}_i = \omega \times \bar{\mathbf{v}}_i$ and has pitch $\omega^\top \bar{\mathbf{v}}_i$. This gives

$$\bar{\mathbf{q}}_1 = (0, 0, 0), \quad \bar{\mathbf{q}}_2 = (1.7320508; 1; 0), \quad \bar{\mathbf{q}}_3 = (1.7320508; 2.5; 0),$$

with zero pitch in all three cases (computed values: exactly 0 in double precision). Comparing with the joint positions in configuration $(30^\circ, 60^\circ, -45^\circ)$, computed in Figure 29.2 — (0, 0), (1.7321; 1), (1.7321; 2.5) — one sees that the new axes are exactly the axes of the joints *in that configuration*. ✓ This is the geometric content of Proposition 29.5(iii): changing the reference configuration means reading the axes where they currently are.

Step 4 — numerical verification. For three values of θ we compared $T(\theta) = e^{[\xi_1]\theta_1} e^{[\xi_2]\theta_2} e^{[\xi_3]\theta_3} M$ with $\bar{T}(\theta) = e^{[\bar{\xi}_1]\phi_1} e^{[\bar{\xi}_2]\phi_2} e^{[\bar{\xi}_3]\phi_3} \bar{M}$, $\phi_i = \theta_i - \bar{\theta}_i$:

θ (rad)	$\max T - \bar{T} $	tool position
(0.4; -0.9; 1.3)	$1.3 \cdot 10^{-15}$	(3.8552025; 0.7770545)
(1; 0.2; -0.3)	$4.4 \cdot 10^{-16}$	(2.2457512; 3.8643275)
$\bar{\theta}$	0	(2.4391576; 3.2071068)

The two descriptions agree within machine error, and at $\theta = \bar{\theta}$ (where $\phi = 0$) they agree exactly because both reduce to $\bar{M}$. ✓

Interpretation. The result is what makes the product of exponentials usable on the shop floor: there is no need to move the robot to a “nominal” configuration to disassemble it. It suffices to measure the axes where they are and record the tool pose in that same configuration; the formula has the same structure, with different data.

Typical errors. (1) Changing the twists but forgetting to change M : one obtains kinematics that differ by a right multiplicative constant — a systematic pose error that is not visible when comparing *motions*, only absolute positions. (2) Using θ_i instead of $\phi_i = \theta_i - \bar{\theta}_i$: the robot ends up rotated by $\bar{\theta}$ in every configuration. (3) Transporting the twists with Ad_{A_i} instead of $\text{Ad}_{A_{i-1}}$: this includes joint i in the transport, and for joint i this does not change its own axis (the axis is invariant under rotation about itself), but it shifts all the following ones.

Extension. Repeat the exercise with a change of base frame $P \in SE(3)$ (for instance the translation by $(0, 0, 1)$ composed with a rotation by 20° about z) and verify part (i) of Proposition 29.5: the new twists are $\text{Ad}_P \xi_i$ and the new reference pose is PM ; the set of singular configurations remains the same (Theorem 29.10).

29.10. Self-assessment quiz

1. (Definition) The Lie algebra $\mathfrak{se}(3)$ consists of the 4×4 matrices of the form

A) $\begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix}$ with $R \in SO(3)$
symmetric

B) $\begin{pmatrix} [\omega]_\times & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix}$ with $\omega, \mathbf{v} \in \mathbb{R}^3$

C) $\begin{pmatrix} S & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix}$ with S

D) all traceless matrices
2. (Hypothesis) In Theorem 29.6, the hypothesis that guarantees solvability of $G(\theta)\mathbf{v} = \mathbf{p}$ is

A) $\|\omega\| = 1$

B) $\theta \in (0, 2\pi)$

C) $\mathbf{p} \neq \mathbf{0}$

D) R symmetric
3. (Symbolic computation) For the planar 3R arm with links ℓ_1, ℓ_2, ℓ_3 the determinant of the task Jacobian matrix (x, y, φ) equals

A) $\ell_1 \ell_2 \ell_3 \sin \theta_2$

B) $\ell_1 \ell_2 \sin \theta_2$

C) $\ell_2 \ell_3 \sin \theta_3$

D) $\ell_1 \ell_2 \sin(\theta_1 + \theta_2)$
4. (Numerical computation) For $g = (R, \mathbf{p})$ with R the rotation by 90° about z and $\mathbf{p} = (1, 2, 3)^\top$, the pitch h of the screw motion equals

A) 3

B) $6/\pi \approx 1.909859$

C) 0

D) $\sqrt{14} \approx 3.741657$
5. (Proof step) In the proof of Theorem 29.5, the fact that the points of the axis only translate follows from

A) $[\omega]_\times \omega = \mathbf{0}$, whence $e^{[\omega]_\times \theta} \omega = \omega$

B) the invertibility of $G(\theta)$

C) $\det R = 1$

D) the Baker–Campbell–Hausdorff formula
6. (Error diagnosis) A student claims: “The robot is at a wrist singularity; I redefine the TCP by shifting it 50 mm and the singularity disappears.” The error is

A) the TCP cannot be redefined

B) changing the tool frame leaves J_s unchanged, hence it does not change the rank

C) the singularity depends on the controller brand

D) one must move the base, not the tool
7. (Transfer) A ZYZ spherical wrist with axes $\mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_3$ in the reference configuration is singular if and only if

A) $\theta_1 = 0$

B) $\theta_2 \in \{0, \pi\}$

C) $\theta_3 = \pi/2$

D) $\theta_1 + \theta_3 = 0$
8. (Source and metacognition) Chasles’ theorem, in the form “every rigid motion can be realised by a rotation about an axis composed with a translation parallel to that axis,” is found stated in

A) Murray–Li–Sastry, Theorem 2.11, p. 49, and Lynch–Park, Proposition 3.25, p. 105

B) Hatcher, *Algebraic Topology*, §1.1

C) Murray–Li–Sastry, Theorem 8.5, p. 381

D) Lee, *Introduction to Smooth Manifolds*, Theorem 20.12

29.11. Teacher’s key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (1–4).

1) **B.** *Rationale:* this is Theorem 29.3: the algebra is the space of twists, with *skew-symmetric* angular block and zero last row. *Distractors:* A describes the *group* $SE(3)$, not the algebra (confusing e^X with X); C replaces “skew-symmetric” with “symmetric”, and then $e^{tS} \notin SO(3)$; D is the algebra $\mathfrak{sl}(4, \mathbb{R})$, much larger (dimension 15 versus 6). *Diagnosed error:* confusing group and algebra. *Source:* Theorem 29.3; Lynch–Park, *Modern Robotics*, §3.3.2. *Prerequisite:* the definition of $\mathfrak{g}$ as $\{X : e^{tX} \in G \ \forall t\}$. *Remediation:* redo Step 3 of the proof of Theorem 29.3, where $(e^{tA})^\top e^{tA} = I$ is differentiated at $t = 0$.

2) **B.** *Rationale:* Proposition 29.3 gives $\det G(\theta) = 4\theta \sin^2(\theta/2)$, which is nonzero exactly for $\theta \notin 2\pi\mathbb{Z}$ and $\theta \neq 0$; the choice $\theta \in (0, \pi]$ of the theorem guarantees the condition. *Distractors:* A is a normalisation, necessary for the formulas to have the stated form but not sufficient (with $\theta = 2\pi$ and $\|\omega\| = 1$ the matrix G is singular); C plays no role (if $\mathbf{p} = \mathbf{0}$ the system is still solvable, with $\mathbf{v} = \mathbf{0}$); D is false for a generic rotation. *Diagnosed error:* confusing a normalisation with a hypothesis of invertibility. *Source:* Proposition 29.3; Remark 29.4. *Prerequisite:* determinant of a block-diagonal operator. *Remediation:* compute $G(2\pi)$ and verify its rank.

3) **B.** *Rationale:* Example 29.6, Step 2: after the column subtractions the determinant reduces to a 2×2 one and equals $\ell_1 \ell_2 \sin \theta_2$. *Distractors:* A adds ℓ_3 , which instead vanishes; C shifts the index to the third joint, which does not produce a singularity; D confuses $\sin \theta_2$ with $\sin(\theta_1 + \theta_2)$, that is, forgets the subtraction formula $c_1 s_{12} - s_1 c_{12} = \sin \theta_2$. *Diagnosed error:* not recognising that the determinant is invariant under elementary column operations. *Source:* Example 29.6. *Prerequisite:* properties of the determinant. *Remediation:* redo the computation with $\ell_3 = 0$ and verify that the result does not change.

4) **B.** *Rationale:* Example 29.3: $\theta = \pi/2$ and $h\theta = \omega^\top \mathbf{p} = 3$, hence $h = 3/(\pi/2) = 6/\pi = 1.909859$. *Distractors:* A confuses the pitch h (translation *per radian*) with the total translation $h\theta = 3$; C would be the pure rotation case, excluded because $\omega^\top \mathbf{p} = 3 \neq 0$; D is $\|\mathbf{p}\|$, which is not the translation along the axis. *Diagnosed error:* confusing pitch and total translation. *Source:* Definition 29.5; Proposition 29.4(i). *Prerequisite:* orthogonal projection onto a unit vector. *Remediation:* compute h and $h\theta$ for the same $\mathbf{p}$ but with $\theta = \pi$ and compare.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (5–8).

5) **A.** *Rationale:* in Step 5 of the proof one uses $R\omega = \omega$, a consequence of $[\omega]_\times \omega = \mathbf{0}$ (Lemma 29.1(iii)) and of the exponential series; this is what removes the rotational part on the points of the axis. *Distractors:* B is needed for the *logarithm*, that is, for the inverse passage; C holds for every rotation and does not single out the axis; D does not appear in that proof (and is declared a panorama in Remark 29.14). *Diagnosed error:* attributing to a hypothesis the wrong part of the proof. *Source:* Theorem 29.5, Step 5. *Prerequisite:* $Ke^{K\theta} = e^{K\theta}K$. *Remediation:* verify by hand that $e^{[\omega]_\times \theta} \omega = \omega$ by summing the series.

6) **B.** *Rationale:* by Theorem 29.10(ii), $T' = TQ$ with Q constant gives $\dot{T}'(T')^{-1} = \dot{T}T^{-1}$, that is $J'_s = J_s$: the rank, and hence the singularity, does not change. *Distractors:* A is false (the TCP can always be redefined); C confuses the mathematics with the implementation (controllers differ in how they *handle* the singularity, not in whether they have it); D is equally false by part (i) of the same theorem. *Diagnosed error:* believing that singularities depend on conventions rather than on the mechanism. *Source:* Theorem 29.10; Lynch–Park, *Modern Robotics*, §5.3, pp. 192–193. *Prerequisite:* invariance of rank under multiplication by invertible matrices. *Remediation:* redo the two two-line computations of the theorem

and discuss Remark 29.11.

7) B. Rationale: Example 29.7, Step 3: $\det J_\omega = -\sin \theta_2$, which vanishes only for $\theta_2 \in \{0, \pi\}$; at those configurations the first and third axes become collinear (Theorem 29.11). *Distractors:* A and D involve θ_1 and θ_3 , which do not appear in the determinant — and must not, because rotating the first or third joint does not alter the angles between the axes; C confuses the ZYZ wrist with the ZYX one of Exercise 29.3, where the determinant is $-\cos \theta_2$ and the singularity is at $\theta_2 = \pm\pi/2$. *Diagnosed error:* applying the condition of one architecture to another without redoing the computation. *Source:* Example 29.7; Murray–Li–Sastry, ch. 3, §4.3 “Singularities,” p. 125. *Prerequisite:* expansion of a 3×3 determinant. *Remediation:* redo Exercise 29.3 and compare the two answers.

8) A. Rationale: Theorem 2.11 of Murray–Li–Sastry, p. 49, is literally the statement quoted. The same claim, in constructive form, is Proposition 3.25 of Lynch–Park, p. 105, which is its constructive form (the exponential of a twist in closed form), followed on p. 106 by the statement that this derivation “is a constructive proof of the Chasles–Mozzi theorem.” *Distractors:* B is the source of the fundamental group of the circle (Chapter 12); C is the Baker–Campbell–Hausdorff formula; D is the closed subgroup theorem, quoted in Remark 29.2. *Diagnosed error:* citing a book instead of a statement. *Source:* see the “Sources” section. *Prerequisite:* knowing how to read a reference with theorem number and page. *Remediation:* locate the two statements in the two books and compare the conventions on the order of the twist coordinates.

BEST PRACTICE

Assigned exercise 29.1 — answers. (a) $h = \omega^\top \mathbf{v} = (0, 0, 1) \cdot (1, 0, 2) = 2$ and $\mathbf{q} = \omega \times \mathbf{v} = (0, 1, 0)$: the axis is the vertical line through $(0, 1, 0)$ and the motion $e^{[\xi]\theta}$ rotates by θ about it while rising by 2θ along z . (b) Let $\eta = \text{Ad}_g \xi = (R\omega, [\mathbf{p}]_\times R\omega + R\mathbf{v})$ with $g = (R, \mathbf{p})$. The pitch is

$$(R\omega)^\top ([\mathbf{p}]_\times R\omega + R\mathbf{v}) = (R\omega)^\top (\mathbf{p} \times R\omega) + (R\omega)^\top R\mathbf{v} = 0 + \omega^\top R^\top R\mathbf{v} = \omega^\top \mathbf{v} = h,$$

because $\mathbf{p} \times R\omega \perp R\omega$ and $R^\top R = I$. For the axis: Theorem 29.5 and Proposition 29.2 give $e^{[\eta]\theta} = g e^{[\xi]\theta} g^{-1}$; the “minimal-translation” fixed points of $e^{[\eta]\theta}$ are therefore the images under g of those of $e^{[\xi]\theta}$ (Proposition 29.4), that is, $g(\mathcal{L})$. (c) A change of frame acts on twists by the adjoint; since pitch and axis transform covariantly and the pitch stays the same number, the pitch is an invariant. The same structure — conjugation by a fixed element of $SE(3)$ — is what leaves the rank of J unchanged in Theorem 29.10. (d) With the given g one obtains $\text{Ad}_g \xi = (0.717356; 0; 0.696707; 0.738005; -0.338029; 2.110770)$; the pitch equals 2.000000, unchanged ✓; the axis point is $(0.235507; -1; -0.242487)$ while $g(\mathbf{q}) = (1; -1; 0.5)$: the difference is $(0.764493; 0; 0.742487)$, parallel to $R\omega$ to within 10^{-16} . ✓

BEST PRACTICE

Assigned exercise 29.2 — answers. (a) $\text{tr } R = 1$, hence $\theta = \arccos 0 = \pi/2$; the axis is $\omega = \mathbf{e}_1$ (read off from $R\mathbf{e}_1 = \mathbf{e}_1$). (b) With $K = [\mathbf{e}_1]_\times$, $K^2 = \text{diag}(0, -1, -1)$ and $\cot(\pi/4) = 1$:

$$G(\pi/2)^{-1} = \frac{2}{\pi}I - \frac{1}{2}K + \left(\frac{2}{\pi} - \frac{1}{2}\right)K^2 = \begin{pmatrix} 2/\pi & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & -1/2 & 1/2 \end{pmatrix}, \quad \mathbf{v} = \left(\frac{4}{\pi}, 0, -1\right)^\top.$$

Numerically $\mathbf{v} = (1.273240; 0; -1)$. (c) $h = \omega^\top \mathbf{v} = 4/\pi = 1.273240$; $\mathbf{q} = \omega \times \mathbf{v} = (0, 1, 0)$; $h\theta = \frac{4}{\pi} \cdot \frac{\pi}{2} = 2$, and $\omega^\top \mathbf{p} = 2$: they agree. ✓ (d) In the yz plane the motion is

$\begin{pmatrix} y \\ z \end{pmatrix} \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, whose fixed point is $\frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, that is $(y, z) = (1, 0)$: exactly the projection of the axis $\{(\lambda, 1, 0)\}$. ✓(e) $\|e^{[\xi]\theta} - g\|_{\max} = 1.1 \cdot 10^{-16}$ both with the closed form and with the series to 40 digits; the minimum of $\|g(\mathbf{x}) - \mathbf{x}\|$ over 300 000 random points equals 2.000014, against the exact value $|h\theta| = 2$. ✓

BEST PRACTICE

Assigned exercise 29.3 — answers. (a) With $R_1 = e^{[\mathbf{e}_3] \times \theta_1}$, $R_2 = e^{[\mathbf{e}_2] \times \theta_2}$:

$$\mathbf{c}_1 = (0, 0, 1)^\top, \quad \mathbf{c}_2 = (-s_1, c_1, 0)^\top, \quad \mathbf{c}_3 = R_1 R_2 \mathbf{e}_1 = (c_1 c_2, s_1 c_2, -s_2)^\top.$$

(b) Expanding along the third row, whose entries are $1, 0, -s_2$:

$$\det J_\omega = 1 \cdot \det \begin{pmatrix} -s_1 & c_1 c_2 \\ c_1 & s_1 c_2 \end{pmatrix} + (-s_2) \cdot \det \begin{pmatrix} 0 & -s_1 \\ 0 & c_1 \end{pmatrix} = -c_2 + 0 = -\cos \theta_2.$$

The singular set is $\theta_2 = \pm\pi/2$ (modulo 2π). (c) $\mathbf{c}_1 \cdot \mathbf{c}_2 = 0$, $\mathbf{c}_2 \cdot \mathbf{c}_3 = -s_1 c_1 c_2 + c_1 s_1 c_2 = 0$, $\mathbf{c}_1 \cdot \mathbf{c}_3 = -s_2$, hence

$$J_\omega^\top J_\omega = \begin{pmatrix} 1 & 0 & -s_2 \\ 0 & 1 & 0 \\ -s_2 & 0 & 1 \end{pmatrix}, \quad \sigma = \left\{ \sqrt{1 + |\sin \theta_2|}, 1, \sqrt{1 - |\sin \theta_2|} \right\},$$

whose product is $\sqrt{1 - \sin^2 \theta_2} = |\cos \theta_2| = |\det J_\omega|$. ✓(d) At $\theta_2 = \pi/2$ one has $\mathbf{c}_3 = (0, 0, -1)^\top = -\mathbf{c}_1$: the independent columns are two, the image is $\langle \mathbf{e}_3, R_1 \mathbf{e}_2 \rangle$ and the lost direction is $R_1 \mathbf{e}_1 = (\cos \theta_1, \sin \theta_1, 0)$. For $\theta_1 = 0.9$ the singular value decomposition gives $\pm(0.621610; 0.783327; 0)$, and $(\cos 0.9; \sin 0.9; 0) = (0.621610; 0.783327; 0)$. ✓Check table (numerical singular values against closed form; maximum discrepancy $6.3 \cdot 10^{-16}$):

θ_2 (rad)	$\det J_\omega$	singular values
0	-1	1; 1; 1
0.4	-0.9210610	1.178736; 1; 0.781397
1	-0.5403023	1.357008; 1; 0.398157
1.5207963	-0.0499792	1.413772; 1; 0.035352
$\pi/2$	0	1.414214; 1; 0

(e) Theorem 28.5 says that the RPY *parametrisation* of $SO(3)$ degenerates at $\theta = \pm\pi/2$: there the Jacobian of the map $(\phi, \theta, \psi) \mapsto R$ loses rank. Here the matrix is the same, but the three parameters are not coordinates on a manifold: they are the positions of three physical axes. The degeneracy is therefore mechanical — two axes really do align — and cannot be removed by changing chart, as is instead done by passing to quaternions (Remark 28.7). It is the same formula with two different meanings, and the difference lies entirely in the fact that here θ_2 is a quantity measured by an encoder.

29.12. Formulary of the chapter

Group and algebra.

$$SE(3) = \left\{ \begin{pmatrix} R & \mathbf{p} \\ \mathbf{0}^\top & 1 \end{pmatrix} : R \in SO(3), \mathbf{p} \in \mathbb{R}^3 \right\}, \quad (R, \mathbf{p})^{-1} = (R^\top, -R^\top \mathbf{p}),$$

$$(R_1, \mathbf{p}_1)(R_2, \mathbf{p}_2) = (R_1 R_2, R_1 \mathbf{p}_2 + \mathbf{p}_1), \quad SE(3) = \mathbb{R}^3 \rtimes SO(3).$$

$$\mathfrak{se}(3) = \left\{ [\xi] = \begin{pmatrix} [\omega]_{\times} & \mathbf{v} \\ \mathbf{0}^{\top} & 0 \end{pmatrix}, \quad [[\xi_1], [\xi_2]] = (\omega_1 \times \omega_2, \omega_1 \times \mathbf{v}_2 - \omega_2 \times \mathbf{v}_1)^{\wedge} \right\}.$$

Hat identities: $[\omega]_{\times}^2 = \omega\omega^{\top} - \|\omega\|^2 I$, $[\omega]_{\times}^3 = -\|\omega\|^2 [\omega]_{\times}$, $[[\mathbf{a}]_{\times}, [\mathbf{b}]_{\times}] = [\mathbf{a} \times \mathbf{b}]_{\times}$, $R[\omega]_{\times} R^{\top} = [R\omega]_{\times}$.

Exponential and logarithm. For $\|\omega\| = 1$:

$$e^{[\omega]_{\times}\theta} = I + \sin\theta [\omega]_{\times} + (1 - \cos\theta)[\omega]_{\times}^2,$$

$$e^{[\xi]\theta} = \begin{pmatrix} e^{[\omega]_{\times}\theta} & G(\theta)\mathbf{v} \\ \mathbf{0}^{\top} & 1 \end{pmatrix}, \quad G(\theta) = I\theta + (1 - \cos\theta)[\omega]_{\times} + (\theta - \sin\theta)[\omega]_{\times}^2.$$

For $\omega = \mathbf{0}$: $e^{[\xi]\theta} = \begin{pmatrix} I & \mathbf{v}\theta \\ \mathbf{0}^{\top} & 1 \end{pmatrix}.$

$$\det G(\theta) = 4\theta \sin^2 \frac{\theta}{2}, \quad G(\theta)^{-1} = \frac{1}{\theta}I - \frac{1}{2}[\omega]_{\times} + \left(\frac{1}{\theta} - \frac{1}{2} \cot \frac{\theta}{2}\right)[\omega]_{\times}^2.$$

Logarithm: $\theta = \arccos \frac{\text{tr } R - 1}{2}$, $[\omega]_{\times} = \frac{R - R^{\top}}{2 \sin \theta}$ for $\theta \in (0, \pi)$, $\mathbf{v} = G(\theta)^{-1} \mathbf{p}$.

Chasles: axis and pitch. For $\xi = (\omega, \mathbf{v})$ with $\|\omega\| = 1$:

$$h = \omega^{\top} \mathbf{v}, \quad \mathbf{q} = \omega \times \mathbf{v}, \quad \mathcal{L} = \{\mathbf{q} + \lambda\omega\},$$

$$e^{[\xi]\theta} \mathbf{x} = \mathbf{q} + e^{[\omega]_{\times}\theta}(\mathbf{x} - \mathbf{q}) + h\theta\omega, \quad \omega^{\top}(g(\mathbf{x}) - \mathbf{x}) = h\theta = \omega^{\top} \mathbf{p},$$

$$\|g(\mathbf{x}) - \mathbf{x}\|^2 = (h\theta)^2 + \|(R - I)(\mathbf{x} - \mathbf{q})\|^2,$$

with equality if and only if $\mathbf{x} \in \mathcal{L}$ (the axis is the locus of minimal displacement).

Adjoint, product of exponentials, Jacobians.

$$\text{Ad}_g = \begin{pmatrix} R & 0 \\ [\mathbf{p}]_{\times} R & R \end{pmatrix}, \quad g[\xi]g^{-1} = [\text{Ad}_g \xi], \quad g e^{[\xi]\theta} g^{-1} = e^{[\text{Ad}_g \xi]\theta}.$$

Joint twists: $\xi_i = (\omega_i, -\omega_i \times \mathbf{q}_i)$ (revolute), $\xi_i = (\mathbf{0}, \mathbf{v}_i)$ (prismatic).

$$T(\theta) = e^{[\xi_1]\theta_1} e^{[\xi_2]\theta_2} \dots e^{[\xi_n]\theta_n} M, \quad M = T(0).$$

$$[\mathcal{V}_s] = \dot{T}T^{-1}, \quad [\mathcal{V}_b] = T^{-1}\dot{T}, \quad \mathcal{V}_s = \text{Ad}_T \mathcal{V}_b.$$

$$J_{s,i} = \text{Ad}_{e^{[\xi_1]\theta_1} \dots e^{[\xi_{i-1}]\theta_{i-1}}} \xi_i, \quad J_{b,i} = \text{Ad}_{(e^{[\xi_{i+1}]\theta_{i+1}} \dots e^{[\xi_n]\theta_n} M)^{-1}} \xi_i, \quad J_s = \text{Ad}_T J_b.$$

Singularities. θ singular $\iff \text{rank } J(\theta) < \min(6, n)$; $\text{rank } J_s = \text{rank } J_b$; changes of base and tool frame do not alter the singular set. Two collinear revolute axes $\Rightarrow$ singularity (for $n \leq 6$). Notable cases:

$$\text{planar 3R: } \det J = \ell_1 \ell_2 \sin \theta_2; \quad \text{ZZY wrist: } \det J_{\omega} = -\sin \theta_2;$$

$$\text{ZYZ wrist: } \det J_{\omega} = -\cos \theta_2;$$

$$\text{elbow arm: } \det J = a_2 a_3 \sin \theta_3 (a_2 \cos \theta_2 + a_3 \cos(\theta_2 + \theta_3)).$$

Damped least squares: $\dot{\theta}_{\lambda} = J^{\top}(JJ^{\top} + \lambda^2 I)^{-1} \mathcal{V}$, which replaces $1/\sigma_i$ with $\sigma_i/(\sigma_i^2 + \lambda^2) \leq 1/(2\lambda)$.

Sources and bibliography

- [1] R. W. Brockett, *Robotic manipulators and the product of exponentials formula*, in *Mathematical Theory of Networks and Systems* (Proc. International Symposium MTNS, Beer Sheva, Israel, 1983), Lecture Notes in Control and Information Sciences vol. 58, Springer-Verlag, Berlin, 1984, pp. 120–129, DOI 10.1007/BFb0031048. First introduces the product-of-exponentials formula for the forward kinematics of open-chain robots (Theorem 29.7 of this chapter); bibliographic data verified on Springer Link and Semantic Scholar on 16 September 2026.
- [2] M. Chasles, *Note sur les propriétés générales du système de deux corps semblables entr’eux et placés d’une manière quelconque dans l’espace; et sur le déplacement fini, ou infiniment petit, d’un corps solide libre*, Bulletin des Sciences Mathématiques, Astronomiques, Physiques et Chimiques (de Férussac), vol. 14, Paris, November 1830, pp. 321–326. This is the original publication of the statement of Theorem 29.6. It was not possible to consult the original issue directly online; the bibliographic data (volume, year, pages) were checked against the historical reconstruction by Ceccarelli cited in the next entry, which reports them identically — verified on 6 September 2026.
- [3] M. Ceccarelli, *Screw axis defined by Giulio Mozzi in 1763 and early studies on helical motion*, Mechanism and Machine Theory 35 (2000), pp. 761–770, DOI 10.1016/S0094-114X(99)00046-4. Historical reconstruction of Mozzi’s priority (1763) on the screw axis and of the Mozzi–Cauchy–Poinsot–Chasles chain; it is the source of Remark 29.6 — verified on 6 September 2026.
- [4] R. M. Murray, Z. Li, S. S. Sastry, *A Mathematical Introduction to Robotic Manipulation*, CRC Press, Boca Raton, 1994. Chapter 2 “Rigid Body Motion”: Rodrigues’ formula p. 28; Proposition 2.8 (exponential from $\mathfrak{se}(3)$ to $SE(3)$, with the closed form, equations (2.32) and (2.36)) pp. 41–42; Proposition 2.9 (surjectivity of the exponential onto $SE(3)$, constructive proof) pp. 42–43; §3.3 “Screws: a geometric description of twists” pp. 45–50, with Theorem 2.11 (Chasles) on p. 49. Chapter 3 “Manipulator Kinematics” (sections are numbered per chapter): §2.2 “The product of exponentials formula,” equation (3.3), pp. 85–87; §2.3 “Parameterization of manipulators via twists,” p. 91, with the subsection on Denavit–Hartenberg parameters on p. 93; §4.1 “End-effector velocity,” spatial Jacobian, equations (3.53)–(3.54), p. 116, and body Jacobian, equations (3.55)–(3.56), p. 117; §4.3 “Singularities,” pp. 123–127, with the example of collinear revolute axes and the wrist singularity on p. 125; §4.4 “Manipulability,” p. 127, with singular values on p. 128. Chapter 8: Theorem 8.5 (Campbell–Baker–Hausdorff formula) p. 381. Full text freely available from the author’s page at Caltech and from copies deposited at universities (the full 1994 PDF copy was consulted) — verified on 6 September 2026.
- [5] K. M. Lynch, F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Cambridge University Press, Cambridge, 2017 (May 2017 preprint version, distributed by the authors at modernrobotics.org). Chapter 3 “Rigid-Body Motions,” §3.3.3 (exponential coordinates for rigid motions): Chasles–Mozzi theorem and Proposition 3.25 (exponential of a twist in closed form, equations (3.86)–(3.89)) pp. 105–106, with the logarithm algorithm and the formula for $G^{-1}(\theta)$, equation (3.92), p. 106. Chapter 4 “Forward Kinematics”: §4.1 “Product of Exponentials Formula,” pp. 140–151, in particular §4.1.1 “First Formulation: Screw Axes in the Base Frame,” p. 141. Chapter 5 “Velocity Kinematics and Statics”: §5.1.1 “Space Jacobian” p. 178, §5.1.2 “Body Jacobian” p. 183, §5.1.4 “Relationship between the Space and Body Jacobian” p. 187, §5.3 “Singularity Analysis” pp. 191–195, with the invariance of singularities under change of frame (pp. 192–193) and the “two collinear revolute axes” case (p. 193). Appendix C

“Denavit–Hartenberg Parameters,” pp. 585 onward — verified on 6 September 2026.

- [6] J. M. Selig, *Geometric Fundamentals of Robotics*, 2nd ed. (revision of *Geometrical Methods in Robotics*, 1996), Springer, New York, 2005, xv+398 pp., ISBN 0-387-20874-7, Monographs in Computer Science series. Chapter 2 “Lie Groups,” §2.5 “The Proper Euclidean Group,” subsection 2.5.2 “Chasles’s Theorem”: treatment of the theorem in the language of Lie groups, with $SE(3)$ as a semidirect product — verified on 6 September 2026.
- [7] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer (Graduate Texts in Mathematics 218), New York, 2013, ISBN 978-1-4419-9981-8. Chapter 20 “The Exponential Map,” section “The Closed Subgroup Theorem,” Theorem 20.12, p. 522: every closed subgroup of a Lie group is an embedded Lie subgroup. This is the statement declared *imported* in Remark 29.2 — verified on 6 September 2026.
- [8] B. C. Hall, *An Elementary Introduction to Groups and Representations*, arXiv:math-ph/0005032 (2000). Definition 2.2 (§2.1): a matrix Lie group is a closed subgroup of $GL(n, \mathbb{C})$ — this is the convention adopted in Definition 29.2, here restricted to $GL(n, \mathbb{R})$. Chapter 4 “The Baker–Campbell–Hausdorff Formula,” §4.3 “The Series Form”: proof of the formula reported as a panorama in Remark 29.14. The text is the freely available preliminary version of the volume *Lie Groups, Lie Algebras, and Representations*, 2nd ed., Springer (GTM 222), 2015 — verified on 6 September 2026.
- [9] E. Minguzzi, *A geometrical introduction to screw theory*, European Journal of Physics 34 (2013), pp. 613–632 (preprint arXiv:1201.4497). Unified presentation of twists and wrenches as dual elements of screw theory, useful for the Poinot part recalled in Remark 29.6 — verified on 6 September 2026.
- [10] J. Denavit, R. S. Hartenberg, *A kinematic notation for lower-pair mechanisms based on matrices*, ASME Journal of Applied Mechanics, June 1955, pp. 215–221. This is the original publication of the convention recalled in Definition 29.8. *Caveat*: the two secondary sources used here disagree on the volume number — Lynch–Park (bibliography, entry [34]) write “23:215–221, 1955,” Murray–Li–Sastry (entry [25]) omit the volume and report only “pages 215–221, June 1955”; not having been able to consult the original volume, the volume number is not given here. The comparison between the two descriptions (Remark 29.9) follows Murray–Li–Sastry, §2.3 “Parameterization of manipulators via twists,” subsection “Relationship with Denavit–Hartenberg parameters,” p. 93, and Lynch–Park, Appendix C — verified on 6 September 2026.

Trajectories and interpolation: SLERP and vibration reduction

Planning a motion means constructing a smooth curve that passes through (or near) assigned points, with controlled velocity and acceleration profiles. This chapter collects the standard tools: Bézier and B-spline curves for position, SLERP and dual quaternions for orientation, jerk-limited profiles for timing — all in use in robot controllers and motion PLCs.

The thread that ties them together is the circle. Position is interpolated with polynomials, but an arc of circle is *not* a polynomial curve (Proposition 30.4): representing it exactly requires a fraction of polynomials, and this is why CAD/CAM uses NURBS; Theorem 30.1 exhibits the arc as a rational quadratic Bézier and computes its time law in closed form. Orientation is interpolated on S^3 , and the SLERP formula is not postulated here but *derived* (Theorem 30.2) from the single requirement of traversing at constant angular velocity the great-circle arc joining two unit quaternions; Theorem 30.3 shows that the curve so obtained is, in space, a fixed-axis rotation at constant speed, and from this the “short path” rule (Corollary 30.3) follows in full, being a consequence of the double cover $S^3 \rightarrow SO(3)$ and not a convention. Timing, finally, is governed with trapezoidal and S-curve profiles: matching conditions and minimum times are derived (Theorems 30.4 and 30.5), and it is proved that Singer and Seering’s ZV shaper cancels *exactly* the residual vibration of a second-order elastic mode (Theorem 30.7), with its robust extension ZVD (Theorem 30.8) and a quantification of the robustness (Proposition 30.9).

Two surprises link the two halves of the chapter. The first: the residual vibration of a train of impulses is the modulus of its Laplace transform evaluated at the system’s pole (Theorem 30.6), so that designing a shaper means placing zeros — the same trade as Chapter 21. The second: the time law of the circular arc as rational Bézier is *exactly* that of normalized linear interpolation between quaternions (Remark 30.3), the very one that Proposition 30.7 condemns for its non-constant speed.

INFO

Level and prerequisites. Level **C** (undergraduate, advanced), with two points at level **D**: the equality case in Lemma 30.1 and the lower bound on jerk-limited times (Proposition 30.8).

Internal prerequisites. Great circles of the sphere and their minimality (Chapter 7, Theorem 7.7); the structure of S^3 , quaternion algebra, and the double cover $S^3 \rightarrow SO(3)$ (Chapter 11, Definition 11.3 and Theorem 11.10); geodesics and the calculus of variations (Chapter 9); rotations, Euler angles, and gimbal lock (Chapter 28); rigid-body kinematics and $\mathfrak{se}(3)$ (Chapter 29); transfer function and frequency response (Chapter 25, Definition 25.1); zeros of a transform and their effect (Chapter 21). From Volume I we recall the rational parametrization of the circle (Volume I, Chapter 3); from Volume II the computation of arc lengths (Volume II, Chapter 2).

External prerequisites, assumed and not reproved:

- single-variable differential and integral calculus: the fundamental theorem, Taylor’s formula with Lagrange remainder, the Cauchy–Schwarz inequality in $\mathbb{R}^n$;

- absolutely continuous functions: $f(b) - f(a) = \int_a^b f'$, and $|f'| \leq g$ almost everywhere implies $|f(b) - f(a)| \leq \int_a^b g$;
- uniqueness of the solution of a linear constant-coefficient Cauchy problem (existence, uniqueness, and the form of the general integral at second order);
- $\mathbb{C}[t]$ is an integral domain and its invertible elements are the nonzero constants.

Everything else is proved here.

What is learned. Deriving an interpolation formula from a kinematic requirement instead of postulating it; recognizing when a rational curve is indispensable and when a polynomial one suffices; reading the choice of a sign as a consequence of a topological property (the double cover); deriving minimum times from integral inequalities rather than from recipes; designing a two- or three-impulse filter by imposing the vanishing of a transform at an assigned point, and quantifying its robustness.

Warning. Conventions of the chapter.

- $\langle \cdot, \cdot \rangle$ is the Euclidean scalar product of $\mathbb{R}^4$, which on quaternions coincides with $\langle p, q \rangle = p_0q_0 + p_1q_1 + p_2q_2 + p_3q_3$; $\|q\|$ is the corresponding norm and $S^3 = \{q \in \mathbb{H} : \|q\| = 1\}$ (Definition 11.3).
- For $q_0, q_1 \in S^3$ we always set $\Omega = \arccos \langle q_0, q_1 \rangle \in [0, \pi]$, called the **angle between the two quaternions**; the angle of the corresponding rotation in $SO(3)$ is instead 2Ω (Theorem 30.3).
- $\Phi_q \in SO(3)$ is the rotation $v \mapsto qv\bar{q}$ associated with $q \in S^3$ (Theorem 11.10).
- In motion profiles s is the position, $v = \dot{s}$ the velocity, $a = \dot{v}$ the acceleration, $j = \dot{a}$ the jerk; $L, V, A, J > 0$ are respectively the travel and the limits on velocity, acceleration, and jerk.
- An *elastic mode* is a second-order system of natural frequency $\omega_n > 0$ and damping $\zeta \in [0, 1]$; we set $\beta = \sqrt{1 - \zeta^2}$, $\omega_d = \omega_n\beta$ (damped frequency), $T_d = 2\pi/\omega_d$ (damped period), and $K = e^{-\zeta\pi/\beta} \in (0, 1]$.
- Units are SI; numerical values are rounded to the digit shown, and every rounding is flagged with $\approx$. Checks marked $\checkmark$ were performed twice by independent routes.

30.1. Bézier curves

Definition 30.1: Bézier curve. Given $n + 1$ control points $P_0, \dots, P_n$ of $\mathbb{R}^d$, the **Bézier curve** of degree n is

$$B(t) = \sum_{i=0}^n b_{i,n}(t) P_i, \quad t \in [0, 1], \quad b_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i},$$

a combination of the points with the **Bernstein polynomials** $b_{i,n}$. The polygonal path $P_0P_1 \cdots P_n$ is called the **control polygon**.

Proposition 30.1: Elementary properties of the Bézier curve. Let B be as in Definition 30.1, with $n \geq 1$. Then:

- (i) $\sum_{i=0}^n b_{i,n}(t) = 1$ and $b_{i,n}(t) \geq 0$ for $t \in [0, 1]$;
- (ii) $B(0) = P_0$ and $B(1) = P_n$;
- (iii) $B'(t) = n \sum_{i=0}^{n-1} b_{i,n-1}(t) (P_{i+1} - P_i)$; in particular $B'(0) = n(P_1 - P_0)$ and $B'(1) = n(P_n - P_{n-1})$;

- (iv) $B(t)$ belongs to the convex hull of $\{P_0, \dots, P_n\}$ for every $t \in [0, 1]$;
(v) for every affine map T one has $T(B(t)) = \sum_i b_{i,n}(t)T(P_i)$: transforming the control points is the same as transforming the curve.

Proof. Step 1 — (i). The binomial theorem applied to $(t + (1-t))^n$ gives $\sum_{i=0}^n \binom{n}{i} t^i (1-t)^{n-i} = (t + (1-t))^n = 1^n = 1$. Nonnegativity on $[0, 1]$ is immediate because $t \geq 0$, $1-t \geq 0$, and $\binom{n}{i} > 0$.

Step 2 — (ii). At $t = 0$ the factor t^i kills every term with $i \geq 1$, while $b_{0,n}(0) = \binom{n}{0} 0^0 1^n = 1$; hence $B(0) = P_0$. At $t = 1$ the factor $(1-t)^{n-i}$ kills every term with $i \leq n-1$ and $b_{n,n}(1) = 1$: $B(1) = P_n$.

Step 3 — the derivative of the Bernstein polynomials. For $1 \leq i \leq n-1$,

$$b'_{i,n}(t) = \binom{n}{i} \left[i t^{i-1} (1-t)^{n-i} - (n-i) t^i (1-t)^{n-i-1} \right].$$

Now $\binom{n}{i} i = \frac{n!}{(i-1)!(n-i)!} = n \binom{n-1}{i-1}$ and $\binom{n}{i} (n-i) = \frac{n!}{i!(n-i-1)!} = n \binom{n-1}{i}$, hence

$$b'_{i,n}(t) = n [b_{i-1,n-1}(t) - b_{i,n-1}(t)],$$

a formula that also holds for $i = 0$ and $i = n$ with the convention $b_{-1,n-1} \equiv b_{n,n-1} \equiv 0$ (for $i = 0$ the first summand is absent and the computation gives $-n(1-t)^{n-1} = -n b_{0,n-1}$; for $i = n$ the second is absent and gives $n t^{n-1} = n b_{n-1,n-1}$).

Step 4 — (iii). Summing,

$$B'(t) = \sum_{i=0}^n b'_{i,n}(t) P_i = n \sum_{i=0}^n b_{i-1,n-1}(t) P_i - n \sum_{i=0}^n b_{i,n-1}(t) P_i.$$

In the first sum the nonzero terms have $1 \leq i \leq n$; setting $j = i - 1$ it becomes $n \sum_{j=0}^{n-1} b_{j,n-1}(t) P_{j+1}$. In the second the nonzero terms have $0 \leq i \leq n-1$. Subtracting term by term gives $B'(t) = n \sum_{j=0}^{n-1} b_{j,n-1}(t) (P_{j+1} - P_j)$. Evaluating at $t = 0$ and at $t = 1$ with Step 2 applied to degree $n-1$ gives $B'(0) = n(P_1 - P_0)$ and $B'(1) = n(P_n - P_{n-1})$: the curve is tangent to the two extreme sides of the control polygon.

Step 5 — (iv). By Step 1 the coefficients $b_{i,n}(t)$ are nonnegative and sum to 1: $B(t)$ is a convex combination of the P_i , hence belongs to their convex hull, which by definition is the set of all such combinations.

Step 6 — (v). Let $T(x) = Mx + c$ with M linear. Then

$$T(B(t)) = M \left(\sum_i b_{i,n}(t) P_i \right) + c = \sum_i b_{i,n}(t) M P_i + \left(\sum_i b_{i,n}(t) \right) c = \sum_i b_{i,n}(t) T(P_i),$$

where in the second step $\sum_i b_{i,n}(t) = 1$ from Step 1 was used to distribute the translation c inside the sum. $\square$

Proposition 30.2: De Casteljau's algorithm. Fix $t \in [0, 1]$, and set $P_i^0 = P_i$ for $i = 0, \dots, n$ and, for $r = 1, \dots, n$,

$$P_i^r(t) = (1-t) P_i^{r-1}(t) + t P_{i+1}^{r-1}(t), \quad i = 0, \dots, n-r.$$

Then $P_0^n(t) = B(t)$. In particular the value of the curve is obtained with $\frac{n(n+1)}{2}$ linear interpolations, without computing powers or binomial coefficients.

Proof. **Step 1 — the statement to be proved by induction.** We show that

$$P_i^r(t) = \sum_{j=0}^r b_{j,r}(t) P_{i+j} \quad \text{for every } r = 0, \dots, n \text{ and every } i = 0, \dots, n-r.$$

For $r = n$ and $i = 0$ this is exactly $P_0^n(t) = \sum_j b_{j,n}(t) P_j = B(t)$.

Step 2 — base case. For $r = 0$ the sum has only the term $j = 0$ with $b_{0,0}(t) = \binom{0}{0} t^0 (1-t)^0 = 1$, so it reduces to $P_i^0 = P_i$: true by definition.

Step 3 — inductive step. Assuming the formula holds for $r-1$,

$$P_i^r = (1-t) \sum_{j=0}^{r-1} b_{j,r-1} P_{i+j} + t \sum_{j=0}^{r-1} b_{j,r-1} P_{i+1+j} = \sum_{j=0}^r \left[(1-t)b_{j,r-1} + t b_{j-1,r-1} \right] P_{i+j},$$

having set $j \rightarrow j+1$ in the second sum and used the convention $b_{-1,r-1} = b_{r,r-1} = 0$.

Step 4 — the coefficient is $b_{j,r}$. For $1 \leq j \leq r-1$,

$$(1-t)b_{j,r-1} + t b_{j-1,r-1} = \binom{r-1}{j} t^j (1-t)^{r-j} + \binom{r-1}{j-1} t^j (1-t)^{r-j} = \binom{r}{j} t^j (1-t)^{r-j} = b_{j,r}(t)$$

by Pascal's rule $\binom{r-1}{j} + \binom{r-1}{j-1} = \binom{r}{j}$. For $j = 0$ this leaves $(1-t)b_{0,r-1} = (1-t)^r = b_{0,r}$ and for $j = r$ it leaves $t b_{r-1,r-1} = t^r = b_{r,r}$. The formula of Step 1 is proved, and with it the statement.

Step 5 — the operation count. At level r one computes $n-r+1$ points, hence in all $\sum_{r=1}^n (n-r+1) = \sum_{k=1}^n k = \frac{n(n+1)}{2}$ linear interpolations. $\square$

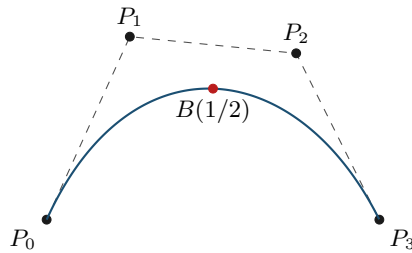


Figure 30.1: Cubic Bézier curve (in blue) and its control polygon (dashed): the curve starts at P_0 , ends at P_3 , is tangent to the extreme sides of the polygon (Proposition 30.1(iii)) and stays within its convex hull. The midpoint, computed with de Casteljau (Proposition 30.2), is $B(\frac{1}{2}) = \frac{1}{8}(P_0 + 3P_1 + 3P_2 + P_3) = \frac{1}{8}(0 + 3 + 9 + 4; 0 + 6.6 + 6 + 0) = \frac{1}{8}(16; 12.6) = (2; 1.575)$. Moving a control point reshapes the entire curve — unlike B-splines, which have local control.

Exercise 30.1. Compute the point at $t = 1/2$ of the quadratic Bézier curve with $P_0 = (0, 0)$, $P_1 = (1, 2)$, $P_2 = (2, 0)$ using the de Casteljau algorithm.

Solution. De Casteljau (Proposition 30.2) interpolates in pairs: first level $P_0^1 = \frac{1}{2}(P_0 + P_1) = (0.5; 1)$, $P_1^1 = \frac{1}{2}(P_1 + P_2) = (1.5; 1)$; second level $B(\frac{1}{2}) = \frac{1}{2}(P_0^1 + P_1^1) = (1; 1)$. The midpoint of the curve is $(1; 1)$: under the apex of the control “triangle”, at half its height, as expected. *Independent check* with the closed formula of Definition 30.1: $B(\frac{1}{2}) = \frac{1}{4}P_0 + \frac{1}{2}P_1 + \frac{1}{4}P_2 = (0 + 0.5 + 0.5; 0 + 1 + 0) = (1; 1)$. $\checkmark$

30.2. B-splines, NURBS, and the exact circular arc

Definition 30.2: B-spline. Given a nondecreasing sequence of **knots** $t_0 \leq t_1 \leq \dots \leq t_m$, the basis functions $N_{i,p}$ of degree p are defined by recurrence (Cox–de Boor):

$$N_{i,0}(t) = \begin{cases} 1 & t \in [t_i, t_{i+1}), \\ 0 & \text{otherwise,} \end{cases} \quad N_{i,p}(t) = \frac{t - t_i}{t_{i+p} - t_i} N_{i,p-1}(t) + \frac{t_{i+p+1} - t}{t_{i+p+1} - t_{i+1}} N_{i+1,p-1}(t),$$

with the convention that a term with vanishing denominator equals 0. A **B-spline** of degree p with control points P_i is $C(t) = \sum_i N_{i,p}(t) P_i$.

Proposition 30.3: Local support of the B-spline basis. For every i and every $p \geq 0$, $N_{i,p}(t) = 0$ for $t \notin [t_i, t_{i+p+1})$. Consequently, moving only the control point P_i modifies $C(t)$ only for $t \in [t_i, t_{i+p+1})$, i.e. on at most $p + 1$ consecutive spans.

Proof. **Step 1 — induction on p .** For $p = 0$ the statement is the definition itself: $N_{i,0}$ is the characteristic function of $[t_i, t_{i+1}) = [t_i, t_{i+0+1})$.

Step 2 — inductive step. Assume the statement true for $p - 1$: then $N_{i,p-1}$ vanishes outside $[t_i, t_{i+p})$ and $N_{i+1,p-1}$ outside $[t_{i+1}, t_{i+p+1})$. In the recurrence of Definition 30.2, $N_{i,p}$ is a combination (with coefficients that are functions of t , but finite on every compact set where the denominators do not vanish) of these two functions; hence it vanishes where both vanish, that is, outside $[t_i, t_{i+p}) \cup [t_{i+1}, t_{i+p+1}) \subseteq [t_i, t_{i+p+1})$.

Step 3 — consequence for the curve. If P_i is replaced by $P_i + \delta$, the curve changes by $N_{i,p}(t) \delta$, which vanishes outside $[t_i, t_{i+p+1})$: the part of the curve outside that knot interval is unchanged. Since $[t_i, t_{i+p+1})$ is the union of the spans $[t_i, t_{i+1}), \dots, [t_{i+p}, t_{i+p+1})$, at most $p + 1$ spans are involved. $\square$

Remark 30.1: What we import about B-splines. Of the theory of B-splines this chapter proves only local support (Proposition 30.3), which is what is needed for the comparison with Bézier curves. Three facts remain imported, that is, assumed without proof and cited to their source (Piegl and Tiller, *The NURBS Book*, Chapter 2): (a) the partition of unity $\sum_i N_{i,p}(t) = 1$ on $[t_p, t_{m-p}]$, which gives the convex hull property; (b) the regularity C^{p-k} of the basis across a knot of multiplicity k , whence C^{p-1} for simple knots; (c) the linear independence of the $N_{i,p}$, which makes the control points unique. None of the three is used in the proofs that follow: where needed, they are recalled explicitly as hypotheses. The verification of (a) and (b) is done by induction on the degree with the same technique as Step 2 above, but requires the full discussion of knot multiplicities, which lies outside the scope of the chapter.

Definition 30.3: NURBS. A **NURBS** (Non-Uniform Rational B-Spline) adds to a B-spline weights $w_i > 0$:

$$C(t) = \frac{\sum_i N_{i,p}(t) w_i P_i}{\sum_i N_{i,p}(t) w_i}.$$

In the special case where the basis is the Bernstein basis ($N_{i,p} = b_{i,n}$, Definition 30.1), one speaks of a **rational Bézier curve** of degree n :

$$C(t) = \frac{\sum_{i=0}^n b_{i,n}(t) w_i P_i}{\sum_{i=0}^n b_{i,n}(t) w_i}, \quad t \in [0, 1].$$

The two propositions that follow explain why CAD/CAM uses *rational* curves and not plain polynomial ones: the circle, the most used primitive after the line, is not polynomial but is

rational, and already at degree 2.

Proposition 30.4: No arc of circle is a polynomial curve. Let $x, y \in \mathbb{R}[t]$ be polynomials such that $x(t)^2 + y(t)^2 = 1$ for every t in an interval $I \subseteq \mathbb{R}$ not reduced to a point. Then x and y are constant. In particular no polynomial Bézier curve (Definition 30.1) traces a nondegenerate arc of circle.

Proof. Step 1 — from the interval to the whole line. The polynomial $x^2 + y^2 - 1$ vanishes on I , which contains infinitely many points; a nonzero polynomial has finitely many roots, hence $x^2 + y^2 - 1$ is the zero polynomial and $x(t)^2 + y(t)^2 = 1$ holds *identically* in $\mathbb{R}[t]$, and therefore also in $\mathbb{C}[t]$.

Step 2 — factorization in $\mathbb{C}[t]$. In $\mathbb{C}[t]$ one factors

$$(x + iy)(x - iy) = x^2 + y^2 = 1.$$

Hence $x + iy$ is invertible in $\mathbb{C}[t]$.

Step 3 — the invertible elements of $\mathbb{C}[t]$ are the constants. If $fg = 1$ with $f, g \in \mathbb{C}[t]$, then $\deg f + \deg g = \deg 1 = 0$ (degree is additive because $\mathbb{C}$ is a field and hence $\mathbb{C}[t]$ is an integral domain), whence $\deg f = \deg g = 0$: f is a nonzero constant.

Step 4 — conclusion. There is therefore $\lambda \in \mathbb{C} \setminus \{0\}$ with $x + iy = \lambda$ and $x - iy = \lambda^{-1}$. Adding and subtracting,

$$x = \frac{\lambda + \lambda^{-1}}{2}, \quad y = \frac{\lambda - \lambda^{-1}}{2i},$$

which are constants. If moreover x, y are real, λ has modulus 1 and is written $\lambda = e^{i\varphi}$: then $x = \cos \varphi$ and $y = \sin \varphi$, i.e. the “curve” is a single point of the circle.

Step 5 — the Bézier case. The components of a Bézier curve are polynomials in t (Definition 30.1); if the curve lay on a circle of center Z and radius $r > 0$, the components of $(B - Z)/r$ would satisfy the hypothesis on $I = [0, 1]$ and would be constant: the curve would reduce to a point, i.e. the arc would be degenerate. $\square$

Theorem 30.1: The circular arc as a rational quadratic Bézier curve. Let $0 < \theta < \pi$ and set $c = \cos \frac{\theta}{2}$, $s = \sin \frac{\theta}{2}$. In the plane identified with $\mathbb{C}$, consider the control points and weights

$$P_0 = 1, \quad P_1 = 1 + i \tan \frac{\theta}{2}, \quad P_2 = e^{i\theta}; \quad w_0 = 1, \quad w_1 = c, \quad w_2 = 1.$$

Then:

- (i) P_1 is the intersection of the tangents to the unit circle at P_0 and at P_2 ;
- (ii) setting $w(t) = (1 - t) + t e^{i\theta/2}$, the rational quadratic Bézier curve of Definition 30.3 equals

$$C(t) = \frac{w(t)}{\overline{w(t)}},$$

and in particular $|C(t)| = 1$ for every $t \in [0, 1]$;

- (iii) $C(t) = e^{i\psi(t)}$ with

$$\psi(t) = 2 \arctan \frac{ts}{(1-t) + tc}, \quad \psi'(t) = \frac{2s}{1 - 2t(1-t)(1-c)} > 0,$$

and ψ increases strictly from $\psi(0) = 0$ to $\psi(1) = \theta$: the curve traces exactly once the arc $\{e^{i\psi} : 0 \leq \psi \leq \theta\}$.

Proof. Step 1 — (i), the intersection of the tangents. The tangent to the unit circle at the point $P_0 = 1$ is the line $\{z : \operatorname{Re} z = 1\}$, because it is orthogonal to the radius OP_0 (directed as 1) and passes through P_0 . Likewise the tangent at $P_2 = e^{i\theta}$ is $\{z : \operatorname{Re}(ze^{-i\theta}) = 1\}$. Seeking the intersection in the form $z = 1 + iy$ with $y \in \mathbb{R}$:

$$\operatorname{Re}((1 + iy)(\cos \theta - i \sin \theta)) = \cos \theta + y \sin \theta = 1,$$

whence, since $\sin \theta \neq 0$ for $0 < \theta < \pi$,

$$y = \frac{1 - \cos \theta}{\sin \theta} = \frac{2s^2}{2sc} = \frac{s}{c} = \tan \frac{\theta}{2},$$

having used $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$ and $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$, and $c \neq 0$ because $0 < \frac{\theta}{2} < \frac{\pi}{2}$. Hence $z = P_1$.

Step 2 — the weighted control point is $e^{i\theta/2}$. Setting $z = e^{i\theta/2} = c + is$,

$$w_1 P_1 = c \left(1 + i \frac{s}{c}\right) = c + is = z,$$

while $w_0 P_0 = 1$ and $w_2 P_2 = e^{i\theta} = z^2$.

Step 3 — the numerator is a perfect square. With $b_{i,2}(t)$ from Definition 30.1,

$$N(t) = \sum_{i=0}^2 b_{i,2}(t) w_i P_i = (1-t)^2 \cdot 1 + 2t(1-t)z + t^2 z^2 = ((1-t) + tz)^2 = w(t)^2.$$

Step 4 — the denominator is $|w(t)|^2$. Since $|z| = 1$ and $\operatorname{Re} z = c$,

$$|w(t)|^2 = ((1-t) + tz) \overline{((1-t) + tz)} = (1-t)^2 + t^2 |z|^2 + 2t(1-t) \operatorname{Re} z = (1-t)^2 + t^2 + 2t(1-t)c,$$

which is exactly $D(t) = \sum_i b_{i,2}(t) w_i = (1-t)^2 + 2t(1-t)c + t^2$.

Step 5 — D does not vanish. From $(1-t)^2 + t^2 = 1 - 2t(1-t)$ it follows that

$$D(t) = 1 - 2t(1-t)(1-c).$$

On $[0, 1]$ one has $0 \leq 2t(1-t) \leq \frac{1}{2}$ (maximum at $t = \frac{1}{2}$) and $0 < 1 - c < 1$, hence $D(t) \geq 1 - \frac{1-c}{2} = \frac{1+c}{2} > 0$.

Step 6 — (ii). By Steps 3–5,

$$C(t) = \frac{N(t)}{D(t)} = \frac{w(t)^2}{|w(t)|^2} = \frac{w(t)^2}{w(t)\overline{w(t)}} = \frac{w(t)}{\overline{w(t)}},$$

which is legitimate because $w(t) \neq 0$ (its squared modulus is $D(t) > 0$). The modulus of the ratio of a number to its conjugate is 1: $|C(t)| = |w|/|\overline{w}| = 1$.

Step 7 — (iii), the argument. Writing $w = \rho e^{i\alpha}$ gives $C = w/\overline{w} = e^{2i\alpha}$, hence $\psi = 2\alpha = 2 \arg w$. Now

$$\operatorname{Re} w(t) = (1-t) + tc \geq \min\{1, c\} > 0, \quad \operatorname{Im} w(t) = ts \geq 0,$$

(the first because $(1-t) + tc$ is a convex combination of 1 and c , both positive), so $w(t)$ lies in the closed right half-plane and its principal argument is $\arg w = \arctan \frac{ts}{(1-t) + tc}$. The formula

for ψ follows; in particular $\psi(0) = 2 \arctan 0 = 0$ and $\psi(1) = 2 \arctan \frac{s}{c} = 2 \arctan \tan \frac{\theta}{2} = \theta$, since $\frac{\theta}{2} \in (0, \frac{\pi}{2})$.

Step 8 — monotonicity. Setting $\mathcal{A}(t) = (1-t) + tc$ and $\mathcal{B}(t) = ts$, one has $\mathcal{A}' = c - 1$, $\mathcal{B}' = s$, and $\mathcal{A}^2 + \mathcal{B}^2 = |w|^2 = D$, hence

$$\frac{d}{dt} \arctan \frac{\mathcal{B}}{\mathcal{A}} = \frac{\mathcal{B}'\mathcal{A} - \mathcal{A}'\mathcal{B}}{\mathcal{A}^2 + \mathcal{B}^2} = \frac{s[(1-t) + tc] - (c-1)ts}{D(t)} = \frac{s(1-t) + stc - stc + st}{D(t)} = \frac{s}{D(t)}.$$

Hence $\psi'(t) = 2s/D(t) > 0$ by Step 5 and $s > 0$: ψ is strictly increasing from 0 to θ , and by the intermediate value theorem it takes every value in $[0, \theta]$ exactly once.

Step 9 — numerical check (quarter circle). For $\theta = \pi/2$ one has $c = s = \frac{\sqrt{2}}{2}$, $\tan \frac{\theta}{2} = 1$, hence $P_0 = (1; 0)$, $P_1 = (1; 1)$, $P_2 = (0; 1)$ and $w_1 = \frac{\sqrt{2}}{2} \approx 0.7071068$. At $t = \frac{1}{2}$:

$$N = \frac{1}{4}(1; 0) + \frac{1}{2} \cdot \frac{\sqrt{2}}{2}(1; 1) + \frac{1}{4}(0; 1) = \left(\frac{1}{4} + \frac{\sqrt{2}}{4}; \frac{\sqrt{2}}{4} + \frac{1}{4}\right), \quad D = \frac{1}{4} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} + \frac{1}{4} = \frac{1}{2} + \frac{\sqrt{2}}{4},$$

and the ratio is $(\frac{1+\sqrt{2}}{2+\sqrt{2}}, \frac{1+\sqrt{2}}{2+\sqrt{2}})$. Now $\frac{1+\sqrt{2}}{2+\sqrt{2}} = \frac{(1+\sqrt{2})(2-\sqrt{2})}{(2+\sqrt{2})(2-\sqrt{2})} = \frac{2-\sqrt{2}+2\sqrt{2}-2}{2} = \frac{\sqrt{2}}{2}$, hence $C(\frac{1}{2}) = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$, which has modulus 1 and argument $45^\circ = \theta/2$, as predicted by Step 7 by symmetry. ✓

Step 10 — independent check, by direct algebra. Without using Step 3, expand $|N|^2$ with $\alpha = (1-t)^2$, $\gamma = t^2$, $\beta = 2t(1-t)$, recalling $\operatorname{Re} z = c$ and $\operatorname{Re} z^2 = \cos \theta = 2c^2 - 1$:

$$|N|^2 = \alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta c + 2\beta\gamma c + 2\alpha\gamma \cos \theta, \quad D^2 = \alpha^2 + \beta^2 c^2 + \gamma^2 + 2\alpha\beta c + 2\beta\gamma c + 2\alpha\gamma,$$

whence

$$|N|^2 - D^2 = \beta^2(1 - c^2) + 2\alpha\gamma(\cos \theta - 1) = s^2(\beta^2 - 4\alpha\gamma) = s^2(4t^2(1-t)^2 - 4(1-t)^2t^2) = 0,$$

having used $1 - c^2 = s^2$ and $1 - \cos \theta = 2s^2$. ✓

Step 11 — numerical check. A Python script evaluates $||C(t)| - 1|$ at 1001 equally spaced values of $t \in [0, 1]$ for $\theta = \pi/2$: the maximum is $2.2 \cdot 10^{-16}$, i.e. the rounding error in double precision. ✓ □

Corollary 30.1: The quarter circle in NURBS form. *The quarter of unit circle from the point (1; 0) to the point (0; 1) is exactly the rational quadratic Bézier curve with control points (1; 0), (1; 1), (0; 1) and weights 1, $\frac{\sqrt{2}}{2}$, 1. Four arcs of this type, joined along the axes, give the whole circle as a degree-2 NURBS with nine control points (the four vertices of the circumscribed square, the four points of contact, and the repetition of the first).*

Proof. The first statement is Theorem 30.1 with $\theta = \pi/2$, made explicit in Step 9 of its proof. For the second: let R be the rotation by 90° about the origin, which is linear. Replacing P_i by RP_i and leaving the weights unchanged, the numerator becomes $\sum_i b_{i,2} w_i RP_i = R(\sum_i b_{i,2} w_i P_i)$ by linearity, while the denominator $\sum_i b_{i,2} w_i$ does not depend on the points and stays the same: the new curve is $R \circ C$, i.e. the rotated arc. Iterating with R, R^2, R^3 gives the three rational Bézier curves of the other three quarters; the end point of each arc coincides with the starting point of the next, and the four arcs cover the circle. The overall control points, listed in order and without repeating the joining points except for the first, are (1; 0), (1; 1), (0; 1), (−1; 1), (−1; 0), (−1; −1), (0; −1), (1; −1), (1; 0): nine points, four of which are at the vertices of the square $[-1, 1]^2$ and four at the points of contact. □

Remark 30.2: Why NURBS for the circle. *Proposition 30.4 and Theorem 30.1 are two faces of the same answer: an arc of circle is not polynomial, but is rational, and already at degree 2. The classical rational parametrization $(\frac{1-u^2}{1+u^2}, \frac{2u}{1+u^2})$ (Volume I, Chapter 3) is the same curve reparametrized: setting $u = \tan \frac{\psi}{2}$ recovers $(\cos \psi, \sin \psi)$. This is why the CAD/CAM standard and advanced controllers (NURBS interpolators on **Siemens SINUMERIK**, **FANUC**) use NURBS: a single primitive exactly represents lines, arcs, conics, and free-form curves, without approximating arcs with chains of polynomial Bézier curves — an approximation that, by Proposition 30.4, can never be exact.*

Exercise 30.2. Why, in a B-spline, does moving one control point not alter the whole curve?

Solution. Because the basis $N_{i,p}$ has compact support: by Proposition 30.3 it is nonzero only on $[t_i, t_{i+p+1})$. Moving P_i changes $C(t)$ only where $N_{i,p}(t) \neq 0$, i.e. on at most $p+1$ consecutive spans. Outside that range the curve is unchanged: this is the *local control* property, absent from Bézier curves, where by Definition 30.1 every $b_{i,n}$ with $1 \leq i \leq n-1$ is strictly positive on the whole of $(0, 1)$, so every control point influences every point of the curve. Local control is what makes the interactive editing of a long trajectory usable.

30.3. Interpolating orientation: SLERP

Interpolating positions means moving in $\mathbb{R}^3$, where the segment is the shortest path. Interpolating *orientations* means moving in $SO(3)$, which is not a vector space: the average of two rotation matrices is not a rotation, and the average of two triples of Euler angles has no geometric meaning at all (besides running into the gimbal-lock singularities, Chapter 28). The correct route passes through unit quaternions, that is, through the sphere S^3 (Definition 11.3), where the shortest path is a great-circle arc. This section derives the SLERP formula from that single requirement.

30.3.1 The shortest path on a sphere of any dimension

Theorem 7.7 of Chapter 7 establishes that on the sphere $S^2 \subset \mathbb{R}^3$ the minimizing paths are the arcs of great circle. Here the same statement is needed on $S^3 \subset \mathbb{R}^4$. The proof of Chapter 7 is for the most part independent of the dimension, but its Step 9 uses the cross product $A \times B$, which exists only in $\mathbb{R}^3$; the lemma that follows redoes that step in a way valid in every dimension, and for completeness retraces the entire chain of inequalities.

Lemma 30.1: Minimal arc on S^{n-1} . Let $n \geq 2$ and $A, B \in S^{n-1} \subset \mathbb{R}^n$ with $\Omega = \arccos\langle A, B \rangle \in (0, \pi)$. For every curve $\gamma : [0, 1] \rightarrow S^{n-1}$ continuous and piecewise C^1 with $\gamma(0) = A$ and $\gamma(1) = B$,

$$L(\gamma) = \int_0^1 \|\gamma'(t)\| dt \geq \Omega,$$

and the value Ω is attained. Equality holds if and only if γ traverses monotonically the minor arc of the great circle $S^{n-1} \cap \text{span}\{A, B\}$, possibly pausing at the endpoints.

Proof. Step 1 — the angle function. Set $u(t) = \langle A, \gamma(t) \rangle \in [-1, 1]$ and, on the stretches where $|u(t)| < 1$, $g(t) = \arccos u(t) \in (0, \pi)$: this is the angle at the center between A and $\gamma(t)$.

Step 2 — the pointwise inequality $|g'| \leq \|\gamma'\|$. From $\|\gamma\|^2 \equiv 1$ it follows, by differentiating, that $\langle \gamma, \gamma' \rangle = 0$. Decompose $A = u\gamma + A_\perp$ with $\langle A_\perp, \gamma \rangle = 0$: then $\|A_\perp\|^2 = \|A\|^2 - u^2 = 1 - u^2$ and

$$u' = \langle A, \gamma' \rangle = \langle A_\perp, \gamma' \rangle, \quad \text{hence} \quad |u'| \leq \|A_\perp\| \|\gamma'\| = \sqrt{1 - u^2} \|\gamma'\|$$

by the Cauchy–Schwarz inequality (using $\langle u\gamma, \gamma' \rangle = u\langle \gamma, \gamma' \rangle = 0$). Since where $|u| < 1$ one has $g' = -u'/\sqrt{1 - u^2}$, we conclude $|g'(t)| \leq \|\gamma'(t)\|$. None of these steps uses the dimension.

Step 3 — the last passage through A . The set $\{t \in [0, 1] : \gamma(t) = A\}$ is closed by continuity, contains 0, and does not contain 1 (because $\Omega > 0$ implies $B \neq A$): let $c < 1$ be its maximum. Then $\gamma(c) = A$, $u(c) = 1$, and $u(t) < 1$ for $t \in (c, 1]$.

Step 4 — the case where γ avoids $-A$ after c . If $\gamma(t) \neq -A$ for every $t \in [c, 1]$, then $|u| < 1$ on $(c, 1]$, so g is defined and piecewise C^1 there, and continuous up to c with

$g(c) = \arccos 1 = 0$. For every $\varepsilon \in (0, 1 - c)$, using Step 2,

$$L(\gamma) \geq \int_{c+\varepsilon}^1 \|\gamma'\| \geq \int_{c+\varepsilon}^1 |g'| \geq \left| \int_{c+\varepsilon}^1 g' \right| = g(1) - g(c + \varepsilon),$$

and as $\varepsilon \rightarrow 0^+$, by continuity of g at c , $L(\gamma) \geq g(1) - g(c) = \arccos \langle A, B \rangle = \Omega$.

Step 5 — the case where γ touches $-A$. If there is $t \in [c, 1]$ with $\gamma(t) = -A$, let τ be the minimum of such t . On (c, τ) one has $|u| < 1$, and Step 4 is repeated on the interval $[c, \tau]$, giving $L(\gamma) \geq g(\tau) - g(c) = \arccos(-1) = \pi > \Omega$. The inequality therefore always holds.

Step 6 — the minimum is attained. Since $\Omega < \pi$, the vectors A and B are linearly independent; setting

$$E = \frac{B - \langle A, B \rangle A}{\|B - \langle A, B \rangle A\|} = \frac{B - \cos \Omega A}{\sin \Omega},$$

(the denominator equals $\sin \Omega \neq 0$ because $\|B - \cos \Omega A\|^2 = 1 - 2 \cos^2 \Omega + \cos^2 \Omega = \sin^2 \Omega$), the vector E is a unit vector orthogonal to A , and the curve

$$\gamma_0(t) = \cos(\Omega t) A + \sin(\Omega t) E, \quad t \in [0, 1],$$

satisfies $\|\gamma_0\| \equiv 1$, $\gamma_0(0) = A$, $\gamma_0(1) = \cos \Omega A + \sin \Omega E = B$, and $\|\gamma'_0\| \equiv \Omega$, hence $L(\gamma_0) = \Omega$.

Step 7 — the equality case: the curve stays put, then rises. Let $L(\gamma) = \Omega < \pi$; by Step 5 we are in the case of Step 4, and all the inequalities of that step are equalities. In particular:

- (a) $\int_0^c \|\gamma'\| = 0$, hence $\gamma \equiv A$ on $[0, c]$;
- (b) $\|\gamma'\| = |g'|$ almost everywhere on $(c, 1)$, i.e. equality holds in Cauchy–Schwarz in Step 2: $\gamma'(t)$ is parallel to $A_\perp(t) = A - u(t)\gamma(t)$, and hence $\gamma' \in \text{span}\{A, \gamma\}$;
- (c) $g' \geq 0$ almost everywhere, i.e. g is nondecreasing.

Step 8 — from (b) it follows that γ lies in the plane $\Pi = \text{span}\{A, B\}$ (the step valid in every dimension). Let $w \in \mathbb{R}^n$ be any vector orthogonal to both A and B , and set $\psi(t) = \langle \gamma(t), w \rangle$. By (b) we write $\gamma' = \alpha A + \beta \gamma$ with α, β piecewise continuous functions, hence

$$\psi' = \alpha \langle A, w \rangle + \beta \langle \gamma, w \rangle = \beta \psi,$$

because $\langle A, w \rangle = 0$. Let $\mathcal{B}$ be an antiderivative of β : then $(\psi e^{-\mathcal{B}})' = e^{-\mathcal{B}}(\psi' - \beta \psi) = 0$, i.e. $\psi e^{-\mathcal{B}}$ is constant; since $\psi(1) = \langle B, w \rangle = 0$, the constant is zero and hence $\psi \equiv 0$. So $\gamma(t)$ is orthogonal to every vector of $\Pi^\perp$, that is, $\gamma(t) \in \Pi$ for every t : the curve lies on $S^{n-1} \cap \Pi$, which is the great circle through A and B . (In $\mathbb{R}^3$ this argument reduces to that of Step 9 of Theorem 7.7 with $w = A \times B$; in any dimension it applies to all $w \in \Pi^\perp$ simultaneously.)

Step 9 — it is the minor arc, traversed monotonically. By (c) the function g increases from 0 to $\Omega < \pi$ without ever exceeding Ω : the curve never reaches $-A$, hence it traverses the arc of the great circle that does not contain it, i.e. the minor arc, and traverses it monotonically. Conversely, every monotone parametrization of that arc has length Ω by Step 6: the equality case is exactly this.

Step 10 — numerical check. On S^3 , $A = (1, 0, 0, 0)$ and $B = (\frac{\sqrt{2}}{2}, 0, 0, \frac{\sqrt{2}}{2})$ give $\langle A, B \rangle = \frac{\sqrt{2}}{2}$ and $\Omega = \frac{\pi}{4} = 0.7853982$. A competing path passing through $(0, 0, 0, 1)$ measures $\arccos 0 + \arccos \frac{\sqrt{2}}{2} = \frac{\pi}{2} + \frac{\pi}{4} = 2.3561945$, i.e. three times as much. $\checkmark$ $\square$

Lemma 30.2: Orthonormal frame of the plane of two quaternions. Let $q_0, q_1 \in S^3$ with $\Omega = \arccos \langle q_0, q_1 \rangle \in (0, \pi)$ and $\Pi = \text{span}\{q_0, q_1\}$. Setting

$$E = \frac{q_1 - \cos \Omega q_0}{\sin \Omega},$$

the pair $\{q_0, E\}$ is an orthonormal basis of Π , one has $q_1 = \cos \Omega q_0 + \sin \Omega E$, and every point of $S^3 \cap \Pi$ is written uniquely as $\cos \varphi q_0 + \sin \varphi E$ with $\varphi \in (-\pi, \pi]$.

Proof. Step 1 — E is well defined. From $\Omega \in (0, \pi)$ it follows that $\sin \Omega > 0$.

Step 2 — E is orthogonal to q_0 . $\langle q_1 - \cos \Omega q_0, q_0 \rangle = \langle q_1, q_0 \rangle - \cos \Omega \|q_0\|^2 = \cos \Omega - \cos \Omega = 0$.

Step 3 — E is a unit vector. By the Pythagorean theorem applied to the orthogonal decomposition $q_1 = \cos \Omega q_0 + (q_1 - \cos \Omega q_0)$,

$$\|q_1 - \cos \Omega q_0\|^2 = \|q_1\|^2 - \cos^2 \Omega = 1 - \cos^2 \Omega = \sin^2 \Omega,$$

hence $\|E\| = 1$.

Step 4 — basis of Π . $\{q_0, E\}$ is orthonormal, hence independent; both vectors belong to Π (E is a combination of q_0 and q_1) and $\dim \Pi = 2$ because q_0, q_1 are independent ($\Omega \neq 0, \pi$ excludes $q_1 = \pm q_0$). From the definition of E one reads off directly $q_1 = \cos \Omega q_0 + \sin \Omega E$.

Step 5 — parametrization of the great circle. A point of Π is written uniquely as $x = \lambda q_0 + \mu E$ and, by orthonormality, $\|x\|^2 = \lambda^2 + \mu^2$. Hence $x \in S^3 \cap \Pi$ if and only if $\lambda^2 + \mu^2 = 1$, i.e. if and only if there is a unique $\varphi \in (-\pi, \pi]$ with $\lambda = \cos \varphi$ and $\mu = \sin \varphi$. $\square$

30.3.2 The SLERP formula, derived

Theorem 30.2: SLERP. Let $q_0, q_1 \in S^3$ with $\Omega = \arccos \langle q_0, q_1 \rangle \in (0, \pi)$, and let Π and E be as in Lemma 30.2. Call a curve of constant angular velocity on the great circle $S^3 \cap \Pi$ any curve of the form $\gamma(t) = \cos \varphi(t) q_0 + \sin \varphi(t) E$ with φ an affine function of t . Then:

(i) the curves of constant angular velocity on $S^3 \cap \Pi$ with $\gamma(0) = q_0$ and $\gamma(1) = q_1$ are exactly the

$$\gamma_k(t) = \cos((\Omega + 2k\pi)t) q_0 + \sin((\Omega + 2k\pi)t) E, \quad k \in \mathbb{Z};$$

(ii) $L(\gamma_k) = |\Omega + 2k\pi|$, and the minimum is attained only by γ_0 , of length Ω ;

(iii) γ_0 is written without reference to E in the form

$$\text{slerp}(q_0, q_1; t) = \frac{\sin((1-t)\Omega)}{\sin \Omega} q_0 + \frac{\sin(t\Omega)}{\sin \Omega} q_1;$$

(iv) γ_0 is the unique path of minimal length from q_0 to q_1 on S^3 , up to monotone reparametrizations.

Proof. Step 1 — (i), the form of admissible curves. By Lemma 30.2 every curve on $S^3 \cap \Pi$ is written $\gamma(t) = \cos \varphi(t) q_0 + \sin \varphi(t) E$; by hypothesis $\varphi(t) = \omega t + \varphi_0$ with ω, φ_0 constants. The condition $\gamma(0) = q_0$ forces $\cos \varphi_0 = 1$ and $\sin \varphi_0 = 0$, i.e. $\varphi_0 \in 2\pi\mathbb{Z}$; since φ_0 appears only through sine and cosine, we may choose $\varphi_0 = 0$ without loss of generality. The condition $\gamma(1) = q_1 = \cos \Omega q_0 + \sin \Omega E$ and the uniqueness of components in the orthonormal basis $\{q_0, E\}$ force $\cos \omega = \cos \Omega$ and $\sin \omega = \sin \Omega$, i.e. $\omega = \Omega + 2k\pi$ with $k \in \mathbb{Z}$. Conversely each of these γ_k satisfies the requirements.

Step 2 — (ii), the lengths. Differentiating,

$$\gamma'_k(t) = -\omega \sin(\omega t) q_0 + \omega \cos(\omega t) E, \quad \omega = \Omega + 2k\pi,$$

and by orthonormality of $\{q_0, E\}$ $\|\gamma'_k(t)\|^2 = \omega^2(\sin^2(\omega t) + \cos^2(\omega t)) = \omega^2$, i.e. $\|\gamma'_k\| \equiv |\omega|$ and $L(\gamma_k) = |\omega| = |\Omega + 2k\pi|$. Since $0 < \Omega < \pi$, for $k \geq 1$ one has $|\Omega + 2k\pi| \geq \Omega + 2\pi > \Omega$ and for $k \leq -1$ one has $|\Omega + 2k\pi| = 2|k|\pi - \Omega \geq 2\pi - \Omega > \pi > \Omega$: the strict minimum is at $k = 0$.

Step 3 — (iii), eliminating E . Substitute $E = (q_1 - \cos \Omega q_0)/\sin \Omega$ from Lemma 30.2 into $\gamma_0(t) = \cos(t\Omega)q_0 + \sin(t\Omega)E$:

$$\gamma_0(t) = \cos(t\Omega)q_0 + \frac{\sin(t\Omega)}{\sin \Omega}(q_1 - \cos \Omega q_0) = \frac{\cos(t\Omega)\sin \Omega - \sin(t\Omega)\cos \Omega}{\sin \Omega}q_0 + \frac{\sin(t\Omega)}{\sin \Omega}q_1.$$

The numerator of the first coefficient is, by the sine subtraction formula, $\sin(\Omega - t\Omega) = \sin((1-t)\Omega)$. This gives the formula of the statement.

Step 4 — (iv), global minimality. By Lemma 30.1 applied with $n = 4$, $A = q_0$, $B = q_1$, every path from q_0 to q_1 on S^3 has length $\geq \Omega$, with equality only for the monotone parametrizations of the minor arc of the great circle $S^3 \cap \Pi$; and γ_0 is precisely one of them, because $t \mapsto t\Omega$ is strictly increasing from 0 to $\Omega < \pi$. Hence SLERP is the minimizing geodesic, and it is the only curve that is at once a minimizing geodesic and of constant angular velocity.

Step 5 — numerical check. With $q_0 = 1$ and $q_1 = \cos 45^\circ + \sin 45^\circ k$, one has $\langle q_0, q_1 \rangle = \cos 45^\circ$ and $\Omega = 45^\circ$. The formula gives, at $t = \frac{1}{2}$, equal coefficients $\sin 22.5^\circ / \sin 45^\circ = 0.3826834/0.7071068 = 0.5411961$, and

$$\text{slerp}(q_0, q_1; \tfrac{1}{2}) = 0.5411961(1) + 0.5411961(0.7071068 + 0.7071068k) = 0.9238795 + 0.3826834k,$$

i.e. exactly $\cos 22.5^\circ + \sin 22.5^\circ k$: the rotation halfway there. ✓ *Independent check:* a Python script evaluates the formula at 1001 values of t and finds $|\|\text{slerp}\| - 1| \leq 4.4 \cdot 10^{-16}$ and $|\text{angular velocity} - \Omega| \leq 7.6 \cdot 10^{-8}$ (the second using difference quotients of step 10^{-4} , whose expected error is of order 10^{-8}). ✓ □

Corollary 30.2: SLERP is unit-norm for every t . Under the hypotheses of Theorem 30.2, $\|\text{slerp}(q_0, q_1; t)\| = 1$ for every $t \in \mathbb{R}$: the formula requires no renormalization.

Proof. First proof (geometric). By Step 3 of the proof of Theorem 30.2, SLERP coincides with $\cos(t\Omega)q_0 + \sin(t\Omega)E$, and $\{q_0, E\}$ is orthonormal (Lemma 30.2): hence $\|\text{slerp}\|^2 = \cos^2(t\Omega) + \sin^2(t\Omega) = 1$.

Second proof (direct check on the formula, independent of the first). Setting $\sigma_0 = \sin((1-t)\Omega)$ and $\sigma_1 = \sin(t\Omega)$,

$$\|\sigma_0 q_0 + \sigma_1 q_1\|^2 = \sigma_0^2 + \sigma_1^2 + 2\sigma_0\sigma_1 \cos \Omega.$$

With the product-to-sum formulas $2 \sin X \sin Y = \cos(X-Y) - \cos(X+Y)$ and $\cos X + \cos Y = 2 \cos \frac{X+Y}{2} \cos \frac{X-Y}{2}$:

$$\sigma_0^2 + \sigma_1^2 = \frac{1 - \cos(2(1-t)\Omega)}{2} + \frac{1 - \cos(2t\Omega)}{2} = 1 - \cos \Omega \cos((1-2t)\Omega),$$

$$2\sigma_0\sigma_1 \cos \Omega = [\cos((1-2t)\Omega) - \cos \Omega] \cos \Omega.$$

Adding, the two terms in $\cos \Omega \cos((1-2t)\Omega)$ cancel and there remains $1 - \cos^2 \Omega = \sin^2 \Omega$. Dividing by $\sin^2 \Omega$ gives 1. ✓ □

Proposition 30.5: Constant angular velocity. Let $\gamma(t) = \text{slerp}(q_0, q_1; t)$ under the hypotheses of Theorem 30.2. Then, for every $s, t \in [0, 1]$:

$$\langle \gamma(t), \gamma(s) \rangle = \cos((t-s)\Omega), \quad \|\gamma'(t)\| \equiv \Omega, \quad L(\gamma|_{[s,t]}) = |t-s|\Omega.$$

In particular the angle between q_0 and $\gamma(t)$ equals $t\Omega$: orientations produced at equally spaced instants are angularly equally spaced.

Proof. Step 1 — scalar product. With $\gamma(t) = \cos(t\Omega)q_0 + \sin(t\Omega)E$ and $\{q_0, E\}$ orthonormal (Lemma 30.2),

$$\langle \gamma(t), \gamma(s) \rangle = \cos(t\Omega) \cos(s\Omega) + \sin(t\Omega) \sin(s\Omega) = \cos((t-s)\Omega)$$

by the cosine subtraction formula.

Step 2 — modulus of the derivative. This is Step 2 of the proof of Theorem 30.2 with $k = 0$: $\|\gamma'\| \equiv \Omega$.

Step 3 — length of the stretches. $L(\gamma|_{[s,t]}) = \int_s^t \|\gamma'\| = |t-s|\Omega$.

Step 4 — the statement about angles. With $s = 0$, Step 1 gives $\langle q_0, \gamma(t) \rangle = \cos(t\Omega)$ and, since $t\Omega \in [0, \Omega] \subset [0, \pi)$, the angle between q_0 and $\gamma(t)$ is exactly $t\Omega$: two instants at distance Δt produce orientations at angular distance $\Delta t \Omega$, independent of where one is along the path.

□

30.3.3 What SLERP does in space: fixed axis and short path

The formula of Theorem 30.2 lives on S^3 . The theorem that follows translates it into $SO(3)$ and shows that the resulting motion is the simplest imaginable: a rotation about a fixed axis, at an angle that grows linearly in time.

Theorem 30.3: SLERP is a fixed-axis, constant-speed rotation. Let $q_0, q_1 \in S^3$ with $\Omega = \arccos \langle q_0, q_1 \rangle \in (0, \pi)$ and let $r = \bar{q}_0 q_1$. Then:

- (i) $r \in S^3$, its scalar part equals $\cos \Omega$, and it is written uniquely as $r = \cos \Omega + \sin \Omega u$ with u a unit pure quaternion;
- (ii) setting $r^t := \cos(t\Omega) + \sin(t\Omega)u$, one has $\text{slerp}(q_0, q_1; t) = q_0 r^t$ for every $t \in \mathbb{R}$;
- (iii) denoting by $R_{w,\alpha} \in SO(3)$ the rotation by angle α about the unit axis w ,

$$\Phi_{\text{slerp}(q_0, q_1; t)} = \Phi_{q_0} \circ R_{u, 2t\Omega} = R_{n, 2t\Omega} \circ \Phi_{q_0}, \quad n = \Phi_{q_0}(u);$$

- (iv) the body therefore rotates about the fixed spatial axis n with constant angular velocity 2Ω , and the total rotation swept from $t = 0$ to $t = 1$ equals 2Ω .

Proof. Step 1 — r is a unit quaternion. The quaternion norm is multiplicative ($\|pq\| = \|p\|\|q\|$: this is Step 1 of the proof of Theorem 11.10) and $\|\bar{q}\| = \|q\|$ by definition of the conjugate; hence $\|r\| = \|\bar{q}_0\|\|q_1\| = 1$.

Step 2 — the scalar part of $\bar{p}q$ is $\langle p, q \rangle$. For every $p, q \in \mathbb{H}$ one has, on one hand, $\|p - q\|^2 = \|p\|^2 + \|q\|^2 - 2\langle p, q \rangle$ (Euclidean norm on $\mathbb{R}^4$) and, on the other,

$$\|p - q\|^2 = (p - q)(\overline{p - q}) = p\bar{p} - p\bar{q} - q\bar{p} + q\bar{q} = \|p\|^2 + \|q\|^2 - (p\bar{q} + \overline{p\bar{q}}) = \|p\|^2 + \|q\|^2 - 2\text{Re}(p\bar{q}),$$

because $\overline{p\bar{q}} = q\bar{p}$ and $x + \bar{x} = 2\text{Re } x$. Comparing, $\text{Re}(p\bar{q}) = \langle p, q \rangle$. Applying this identity to the pair $(\bar{q}_0, \bar{q}_1)$ and observing that $\bar{\bar{q}}_1 = q_1$ and that the scalar product is invariant under simultaneous conjugation ($\langle \bar{p}, \bar{q} \rangle = p_0q_0 + (-p_1)(-q_1) + (-p_2)(-q_2) + (-p_3)(-q_3) = \langle p, q \rangle$), one obtains

$$\text{Re}(\bar{q}_0 q_1) = \langle \bar{q}_0, \bar{q}_1 \rangle = \langle q_0, q_1 \rangle = \cos \Omega.$$

Step 3 — (i), the decomposition. Write $r = \cos \Omega + R$ with R pure (Step 2). The scalar part and the pure part are orthogonal in $\mathbb{R}^4$, hence $\|R\|^2 = \|r\|^2 - \cos^2 \Omega = 1 - \cos^2 \Omega = \sin^2 \Omega$; since $\sin \Omega > 0$, the quaternion $u = R/\sin \Omega$ is pure and unit, and is the only one with $r = \cos \Omega + \sin \Omega u$.

Step 4 — the vector E is $q_0 u$. From $\bar{q}_0 q_0 = \|q_0\|^2 = 1$ it follows that $q_0 r = q_0 \bar{q}_0 q_1 = q_1$. Hence, by linearity of the quaternion product,

$$q_0 u = q_0 \frac{r - \cos \Omega}{\sin \Omega} = \frac{q_0 r - \cos \Omega q_0}{\sin \Omega} = \frac{q_1 - \cos \Omega q_0}{\sin \Omega} = E,$$

with E as in Lemma 30.2.

Step 5 — (ii). Using Step 4 and again linearity,

$$q_0 r^t = q_0 (\cos(t\Omega) + \sin(t\Omega)u) = \cos(t\Omega) q_0 + \sin(t\Omega) q_0 u = \cos(t\Omega) q_0 + \sin(t\Omega) E,$$

which is $\gamma_0(t) = \text{slerp}(q_0, q_1; t)$ by Step 3 of the proof of Theorem 30.2.

Step 6 — the rotation associated with r^t . The quaternion r^t is unit ($\cos^2 + \sin^2 = 1$ and u a unit pure quaternion) and is of the form $\cos \frac{\vartheta}{2} + \sin \frac{\vartheta}{2} u$ with $\vartheta = 2t\Omega$: by Theorem 11.10(iii), $\Phi_{r^t} = R_{u, 2t\Omega}$.

Step 7 — (iii), first form. By Theorem 11.10(ii) the map $q \mapsto \Phi_q$ is a homomorphism, hence $\Phi_{\text{slerp}(t)} = \Phi_{q_0 r^t} = \Phi_{q_0} \circ \Phi_{r^t} = \Phi_{q_0} \circ R_{u, 2t\Omega}$.

Step 8 — (iii), second form (spatial axis). Compute the quaternion conjugate $q_0 r^t \bar{q}_0$:

$$q_0 (\cos(t\Omega) + \sin(t\Omega)u) \bar{q}_0 = \cos(t\Omega) q_0 \bar{q}_0 + \sin(t\Omega) q_0 u \bar{q}_0 = \cos(t\Omega) + \sin(t\Omega) \Phi_{q_0}(u),$$

where $q_0 \bar{q}_0 = 1$ and the definition $\Phi_{q_0}(u) = q_0 u \bar{q}_0$ were used, which is legitimate because u is pure and $\Phi_{q_0}(u)$ remains pure and unit (Theorem 11.10(i)). By Step 6 applied to this quaternion, $\Phi_{q_0 r^t \bar{q}_0} = R_{n, 2t\Omega}$ with $n = \Phi_{q_0}(u)$; and by the homomorphism property, $\Phi_{q_0 r^t \bar{q}_0} = \Phi_{q_0} \circ \Phi_{r^t} \circ \Phi_{q_0}^{-1}$ (note $\Phi_{\bar{q}_0} = \Phi_{q_0}^{-1}$ because $q_0 \bar{q}_0 = 1$). Composing on the right with Φ_{q_0} gives $\Phi_{q_0} \circ \Phi_{r^t} = R_{n, 2t\Omega} \circ \Phi_{q_0}$.

Step 9 — (iv). Neither u nor n depends on t : the axis is fixed. The angle $2t\Omega$ is linear in t , hence the angular velocity $\frac{d}{dt}(2t\Omega) = 2\Omega$ is constant, and the rotation swept from $t = 0$ to $t = 1$ is 2Ω . Note the factor 2: the angle *between the quaternions* is Ω , the angle *of the rotation* is twice that.

Step 10 — numerical check. With $q_0 = 1$ and $q_1 = \frac{1}{2}(1 + i + j + k)$ one has $\|q_1\|^2 = 4 \cdot \frac{1}{4} = 1$ and $\langle q_0, q_1 \rangle = \frac{1}{2}$, hence $\Omega = 60^\circ$. Here $r = \bar{q}_0 q_1 = q_1$, whose scalar part is $\frac{1}{2} = \cos 60^\circ \checkmark$ and whose pure part has norm $\sqrt{3}/2 = 0.8660254 = \sin 60^\circ \checkmark$; the axis is $u = (i + j + k)/\sqrt{3}$. The rotation swept is $2\Omega = 120^\circ$ about the diagonal $(1, 1, 1)$, which is precisely the rotation that cyclically permutes the axes $i \mapsto j \mapsto k \mapsto i$: this is checked with Theorem 11.10(iii), since $q_1 = \cos 60^\circ + \sin 60^\circ u$ with $\vartheta/2 = 60^\circ$. *Independent check in Python:* $\arccos\langle q_0, q_1 \rangle = 1.0471976$ and $\pi/3 = 1.0471976$; $\|u\| = 1.0000000$ and $1/\sqrt{3} = 0.5773503$ on each component. $\checkmark$ □

Corollary 30.3: Choice of sign: the short path. Let $q_0, q_1 \in S^3$ with $q_1 \neq \pm q_0$ and $\Omega = \arccos\langle q_0, q_1 \rangle$. Then:

- (i) $\text{slerp}(q_0, q_1; \cdot)$ and $\text{slerp}(q_0, -q_1; \cdot)$ join in $SO(3)$ the same pair of rotations Φ_{q_0} and Φ_{q_1} ;
- (ii) the angle between q_0 and $-q_1$ equals $\pi - \Omega$, and the rotations swept overall by the two paths are 2Ω and $2\pi - 2\Omega$, summing to 2π ;
- (iii) if $\langle q_0, q_1 \rangle > 0$ the short path is the one through q_1 ; if $\langle q_0, q_1 \rangle < 0$ it is the one through $-q_1$; if $\langle q_0, q_1 \rangle = 0$ the two paths both sweep π and the choice is genuinely ambiguous. The operational rule “if $\langle q_0, q_1 \rangle < 0$ replace q_1 with $-q_1$ ” is therefore not a convention, but the consequence of the fact that $S^3 \rightarrow SO(3)$ is a two-to-one cover.

Proof. **Step 1 — (i).** By Theorem 11.10(iv), $\Phi_q = \Phi_{q'}$ if and only if $q' = \pm q$; in particular $\Phi_{-q_1} = \Phi_{q_1}$. Both SLERPs start at q_0 and arrive respectively at q_1 and at $-q_1$, which project to the same rotation: in $SO(3)$ the two paths have the same endpoints.

Step 2 — (ii), the angle. $\langle q_0, -q_1 \rangle = -\langle q_0, q_1 \rangle = -\cos \Omega = \cos(\pi - \Omega)$, and $\pi - \Omega \in (0, \pi)$ because $\Omega \in (0, \pi)$ (the hypothesis $q_1 \neq \pm q_0$ excludes $\Omega \in \{0, \pi\}$): hence the angle is $\pi - \Omega$. By Theorem 30.3(iv) the rotations swept are 2Ω and $2(\pi - \Omega) = 2\pi - 2\Omega$, whose sum is 2π .

Step 3 — (iii). From $2\Omega + (2\pi - 2\Omega) = 2\pi$ it follows that one of the two values is less than π and the other greater, except when they are both equal to π . Precisely $2\Omega < \pi \iff \Omega <$

$\frac{\pi}{2} \iff \cos \Omega > 0 \iff \langle q_0, q_1 \rangle > 0$; $2\Omega > \pi \iff \langle q_0, q_1 \rangle < 0$; and $2\Omega = \pi$ if and only if $\langle q_0, q_1 \rangle = 0$, in which case the other path also sweeps $2\pi - \pi = \pi$.

Step 4 — why the double cover is needed. If the map $S^3 \rightarrow SO(3)$ were injective, each pair of rotations would correspond to a single pair of quaternions and there would be nothing to choose. It is the “two-to-one” property of Theorem 11.10(iv) that produces two distinct paths on S^3 with the same endpoints in $SO(3)$; and they are distinct *even as paths in $SO(3)$* , because by Theorem 30.3(iv) they sweep different rotations (2Ω versus $2\pi - 2\Omega$) about opposite axes. The sign rule chooses between the two the one of smaller rotation.

Step 5 — numerical check. With $q_0 = 1$ and $q_1 = \frac{1}{2}(1+i+j+k)$ one has $\langle q_0, q_1 \rangle = 0.5 > 0$, $\Omega = 60^\circ$, and rotation swept 120° ; with $-q_1$ one would have $\langle q_0, -q_1 \rangle = -0.5$, $\Omega' = 120^\circ$, and rotation swept 240° : both bring the body to the same final orientation, but the second makes it turn two-thirds of a revolution more. $120^\circ + 240^\circ = 360^\circ \checkmark$, as predicted by Step 2. $\square$

30.3.4 The degenerate case and comparison with linear interpolation

Proposition 30.6: Limit as $\Omega \rightarrow 0$: SLERP degenerates to the linear one. For

$a \in [0, 1]$ and $\Omega \in (0, \pi)$ set $F(a, \Omega) = \frac{\sin(a\Omega)}{\sin \Omega}$. Then:

- (i) $\lim_{\Omega \rightarrow 0^+} F(a, \Omega) = a$ for every $a \in [0, 1]$;
- (ii) for $0 < \Omega \leq 1$ the quantitative estimate holds

$$\left| F(a, \Omega) - a - \frac{a(1-a^2)}{6} \Omega^2 \right| \leq \frac{\Omega^4}{15};$$

- (iii) consequently $\text{slerp}(q_0, q_1; t) \rightarrow (1-t)q_0 + tq_1$ as $\Omega \rightarrow 0^+$, uniformly in $t \in [0, 1]$: in the limit of nearby orientations SLERP is linear interpolation, and the discrepancy is $O(\Omega^2)$.

Proof. **Step 1 — (i).** If $a = 0$ the statement is trivial. If $a > 0$,

$$F(a, \Omega) = \frac{\sin(a\Omega)}{a\Omega} \cdot \frac{\Omega}{\sin \Omega} \cdot a \xrightarrow{\Omega \rightarrow 0^+} 1 \cdot 1 \cdot a = a,$$

by the standard limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ applied twice.

Step 2 — Taylor’s formula with Lagrange remainder. For every $x \in \mathbb{R}$, the degree-4 Taylor polynomial of $\sin$ at 0 is $x - \frac{x^3}{6}$ (the degree-4 term vanishes) and the Lagrange remainder is $\frac{\cos \xi}{120} x^5$ for some ξ between 0 and x ; hence

$$\left| \sin x - \left(x - \frac{x^3}{6} \right) \right| \leq \frac{|x|^5}{120}, \quad \text{and for } 0 < x \leq 1 \text{ also } \sin x \geq x - \frac{x^3}{6} \geq \frac{5}{6}x,$$

the second because for $0 < x \leq 1 < \frac{\pi}{2}$ one has $\xi \in (0, x)$ and $\cos \xi > 0$, and because $x - \frac{x^3}{6} = x(1 - \frac{x^2}{6}) \geq x(1 - \frac{1}{6})$.

Step 3 — the numerator of the difference. For $0 < \Omega \leq 1$ and $a \in [0, 1]$, with $r_1 = \sin(a\Omega) - (a\Omega - \frac{a^3\Omega^3}{6})$ and $r_2 = \sin \Omega - (\Omega - \frac{\Omega^3}{6})$, one has $|r_1| \leq \frac{a^5\Omega^5}{120} \leq \frac{\Omega^5}{120}$ and $|r_2| \leq \frac{\Omega^5}{120}$, and

$$\sin(a\Omega) - a \sin \Omega = \left(a\Omega - \frac{a^3\Omega^3}{6} + r_1 \right) - a \left(\Omega - \frac{\Omega^3}{6} + r_2 \right) = \frac{a(1-a^2)}{6} \Omega^3 + \varepsilon,$$

with $\varepsilon = r_1 - ar_2$ and $|\varepsilon| \leq \frac{2\Omega^5}{120} = \frac{\Omega^5}{60}$.

Step 4 — dividing by $\sin \Omega$. Hence

$$F(a, \Omega) - a = \frac{a(1-a^2)}{6} \frac{\Omega^3}{\sin \Omega} + \frac{\varepsilon}{\sin \Omega} = \frac{a(1-a^2)}{6} \Omega^2 \cdot \frac{\Omega}{\sin \Omega} + \frac{\varepsilon}{\sin \Omega}.$$

By Step 2, $\sin \Omega \geq \Omega(1 - \frac{\Omega^2}{6}) > 0$, hence

$$1 \leq \frac{\Omega}{\sin \Omega} \leq \frac{1}{1 - \frac{\Omega^2}{6}} = 1 + \frac{\Omega^2/6}{1 - \Omega^2/6} \leq 1 + \frac{\Omega^2}{5}$$

(the last step because $\Omega \leq 1$ gives $1 - \frac{\Omega^2}{6} \geq \frac{5}{6}$).

Step 5 — (ii), the estimate. Subtracting the leading term,

$$\left| F(a, \Omega) - a - \frac{a(1-a^2)}{6} \Omega^2 \right| \leq \frac{a(1-a^2)}{6} \Omega^2 \cdot \frac{\Omega^2}{5} + \frac{\Omega^5/60}{\frac{5}{6}\Omega} \leq \frac{\Omega^4}{30} + \frac{\Omega^4}{50} = \frac{8\Omega^4}{150} < \frac{\Omega^4}{15},$$

having bounded $a(1-a^2) \leq 1$ on $[0, 1]$.

Step 6 — (iii). Applying (i) with $a = 1 - t$ and with $a = t$ to the two coefficients of the formula of Theorem 30.2 gives pointwise convergence to $(1-t)q_0 + tq_1$; estimate (ii), valid uniformly in $a \in [0, 1]$, makes it uniform in t , with error

$$\left\| \text{slerp}(q_0, q_1; t) - [(1-t)q_0 + tq_1] \right\| \leq \frac{\Omega^2}{6} \left[(1-t)t(2-t) + t(1-t)(1+t) \right] + \frac{2\Omega^4}{15} = \frac{t(1-t)}{2} \Omega^2 + \frac{2\Omega^4}{15},$$

where $\|q_0\| = \|q_1\| = 1$, the triangle inequality, and $(2-t) + (1+t) = 3$ were used.

Step 7 — numerical check. For $\Omega = 0.1$ and $t = 0.3$ (i.e. $a = 0.7$ for the first coefficient) the exact formula gives $F = 0.700595548933$, the linear one gives 0.7 , and the predicted correction term is $\frac{0.7 \cdot 0.51}{6} \cdot 0.01 = 5.950 \cdot 10^{-4}$, against an actual discrepancy of $5.955 \cdot 10^{-4}$: a difference of $5 \cdot 10^{-7}$, well within the bound $\Omega^4/15 = 6.7 \cdot 10^{-6}$. ✓ For $\Omega = 10^{-2}$ the actual discrepancy is $5.950 \cdot 10^{-6}$ and the prediction $5.950 \cdot 10^{-6}$; for $\Omega = 10^{-3}$ both are $5.950 \cdot 10^{-8}$: the ratio scales as Ω^2 , with a factor 100 per decade. ✓ □

Best practice. Implementing SLERP without losing digits. As $\Omega \rightarrow 0$ the formula of Theorem 30.2 divides two quantities that both tend to zero, and in finite arithmetic loses significant digits. Proposition 30.6 says exactly when it is safe to switch: if $\Omega \leq \Omega_*$ the discrepancy between SLERP and linear interpolation is at most $\frac{t(1-t)}{2} \Omega^2 + \frac{2}{15} \Omega^4 \leq \frac{\Omega_*^2}{8} + \frac{2\Omega_*^4}{15}$. For $\Omega_* = 10^{-3}$ rad the bound is $1.3 \cdot 10^{-7}$, below the resolution of a 22-bit encoder over one turn ($2\pi/2^{22} = 1.5 \cdot 10^{-6}$ rad); one then uses the linear formula, followed by a normalization. Above that threshold one uses the exact formula. The same Proposition 30.6 supplies the criterion: not “when Ω is small”, but “when $\Omega^2/8$ is below the required tolerance”.

Definition 30.4: Normalized linear interpolation (NLERP). For $q_0, q_1 \in S^3$ with $q_1 \neq -q_0$, the **NLERP** is the curve

$$\text{nlerp}(q_0, q_1; t) = \frac{(1-t)q_0 + tq_1}{\|(1-t)q_0 + tq_1\|}, \quad t \in [0, 1].$$

Proposition 30.7: NLERP traces the same arc, but not at constant speed. Let $q_0, q_1 \in S^3$ with $\Omega = \arccos \langle q_0, q_1 \rangle \in (0, \pi)$, and let E be as in Lemma 30.2. Then:

- (i) $\text{nlerp}(q_0, q_1; t) = \cos \phi(t) q_0 + \sin \phi(t) E$, where $\phi(t) \in [0, \Omega]$ is the argument of the complex number $(1-t) + t e^{i\Omega}$; when $(1-t) + t \cos \Omega > 0$ — in particular always, if $\Omega \leq \frac{\pi}{2}$ — this argument is written

$$\phi(t) = \arctan \frac{t \sin \Omega}{(1-t) + t \cos \Omega}.$$

In particular NLERP traces the same minor arc as SLERP, with the same endpoints;

(ii) its angular velocity equals

$$\phi'(t) = \frac{\sin \Omega}{D(t)}, \quad D(t) = 1 - 2t(1-t)(1 - \cos \Omega) \in \left[\cos^2 \frac{\Omega}{2}, 1 \right];$$

(iii) ϕ' attains its minimum $\sin \Omega$ at $t = 0$ and $t = 1$ and its maximum $2 \tan \frac{\Omega}{2}$ at $t = \frac{1}{2}$, with

$$\frac{\phi'(1/2)}{\phi'(0)} = \frac{1}{\cos^2 \frac{\Omega}{2}} > 1 \quad \text{for every } \Omega \in (0, \pi);$$

hence ϕ' is constant if and only if $\Omega = 0$: NLERP never has constant angular velocity when the two orientations are distinct;

(iv) the ratio between maximum speed and mean speed equals $\frac{2 \tan \frac{\Omega}{2}}{\Omega}$, which tends to 1 as $\Omega \rightarrow 0^+$ and to $+\infty$ as $\Omega \rightarrow \pi^-$.

Proof. Step 1 — components in the basis $\{q_0, E\}$. By Lemma 30.2, $q_1 = \cos \Omega q_0 + \sin \Omega E$, hence

$$v(t) := (1-t)q_0 + tq_1 = \underbrace{[(1-t) + t \cos \Omega]}_{\mathcal{A}(t)} q_0 + \underbrace{t \sin \Omega}_{\mathcal{B}(t)} E.$$

Step 2 — the norm. By orthonormality,

$$\|v(t)\|^2 = \mathcal{A}^2 + \mathcal{B}^2 = (1-t)^2 + 2t(1-t) \cos \Omega + t^2 \cos^2 \Omega + t^2 \sin^2 \Omega = (1-t)^2 + t^2 + 2t(1-t) \cos \Omega,$$

and since $(1-t)^2 + t^2 = 1 - 2t(1-t)$,

$$\|v(t)\|^2 = 1 - 2t(1-t)(1 - \cos \Omega) = D(t).$$

On $[0, 1]$ the function $2t(1-t)$ ranges between 0 (endpoints) and $\frac{1}{2}$ (at $t = \frac{1}{2}$), hence D ranges between $1 - \frac{1 - \cos \Omega}{2} = \frac{1 + \cos \Omega}{2} = \cos^2 \frac{\Omega}{2}$ and 1; in particular $D > 0$ for $\Omega < \pi$ and NLERP is well defined.

Step 3 — (i). Dividing by $\|v\| = \sqrt{D}$ gives a unit vector of Π , which by Lemma 30.2 is written $\cos \phi q_0 + \sin \phi E$ with $\cos \phi = \mathcal{A}/\sqrt{D}$ and $\sin \phi = \mathcal{B}/\sqrt{D}$; that is, ϕ is the argument of the complex number $\mathcal{A} + i\mathcal{B} = (1-t) + t e^{i\Omega}$. Since $(\mathcal{A}, \mathcal{B})$ is a combination with nonnegative coefficients of $(1, 0)$ (the value at $t = 0$) and of $(\cos \Omega, \sin \Omega)$ (the value at $t = 1$), the point $\mathcal{A} + i\mathcal{B}$ lies in the convex sector of opening $\Omega < \pi$ bounded by those two rays, and is never zero because $\mathcal{A}^2 + \mathcal{B}^2 = D > 0$: hence $\phi(t) \in [0, \Omega]$, with $\phi(0) = 0$ and $\phi(1) = \Omega$. If moreover $\mathcal{A} > 0$ — which holds for every $t \in [0, 1]$ when $\cos \Omega \geq 0$, since $\mathcal{A}$ is a convex combination of 1 and $\cos \Omega$ — the argument in the right half-plane is $\arctan(\mathcal{B}/\mathcal{A})$. The curve therefore traces the same minor arc as SLERP.

Step 4 — (ii), the derivative. The argument of a nowhere-vanishing differentiable curve $t \mapsto \mathcal{A}(t) + i\mathcal{B}(t)$ has derivative $\frac{\mathcal{B}'\mathcal{A} - \mathcal{A}'\mathcal{B}}{\mathcal{A}^2 + \mathcal{B}^2}$ (this is checked locally by writing the argument as $\arctan(\mathcal{B}/\mathcal{A})$ where $\mathcal{A} \neq 0$ and as $-\arctan(\mathcal{A}/\mathcal{B}) + \text{const}$ where $\mathcal{B} \neq 0$; the two cases cover the whole curve and the expressions agree where both hold). With $\mathcal{A}' = \cos \Omega - 1$, $\mathcal{B}' = \sin \Omega$, and $\mathcal{A}^2 + \mathcal{B}^2 = D$,

$$\phi' = \frac{\sin \Omega [(1-t) + t \cos \Omega] - (\cos \Omega - 1) t \sin \Omega}{D(t)}.$$

The numerator equals $\sin \Omega [(1-t) + t \cos \Omega - t \cos \Omega + t] = \sin \Omega \cdot 1 = \sin \Omega$: the terms in $t \cos \Omega$ cancel. Hence $\phi' = \sin \Omega / D$.

Step 5 — (iii). By Step 2, D has maximum 1 at the endpoints and minimum $\cos^2 \frac{\Omega}{2}$ at $t = \frac{1}{2}$; since $\phi' = \sin \Omega / D$ with $\sin \Omega > 0$, the angular velocity has minimum $\sin \Omega$ at $t \in \{0, 1\}$ and maximum

$$\frac{\sin \Omega}{\cos^2 \frac{\Omega}{2}} = \frac{2 \sin \frac{\Omega}{2} \cos \frac{\Omega}{2}}{\cos^2 \frac{\Omega}{2}} = 2 \tan \frac{\Omega}{2}$$

at $t = \frac{1}{2}$. The ratio is $\frac{2 \tan \frac{\Omega}{2}}{\sin \Omega} = \frac{2 \tan \frac{\Omega}{2}}{2 \sin \frac{\Omega}{2} \cos \frac{\Omega}{2}} = \frac{1}{\cos^2 \frac{\Omega}{2}}$, which is > 1 as soon as $\cos \frac{\Omega}{2} < 1$, i.e. $\Omega > 0$. If instead ϕ' were constant, the ratio would be 1 and one would have $\Omega = 0$.

Step 6 — (iv). The mean speed is $\frac{\phi(1) - \phi(0)}{1 - 0} = \Omega$; the ratio to the maximum is $2 \tan \frac{\Omega}{2} / \Omega$. As $\Omega \rightarrow 0^+$ one has $\tan x \sim x$ and the ratio tends to 1; as $\Omega \rightarrow \pi^-$ one has $\tan \frac{\Omega}{2} \rightarrow +\infty$ and the ratio diverges.

Step 7 — numerical check ($\Omega = 90^\circ$). One has $\sin \Omega = 1$, $\cos \Omega = 0$, so $D(t) = 1 - 2t(1 - t)$ and $\phi'(t) = 1/(1 - 2t + 2t^2)$. Then $\phi'(0) = 1$, $\phi'(\frac{1}{4}) = 1/(1 - \frac{1}{2} + \frac{1}{8}) = 1/0.625 = 1.6$, $\phi'(\frac{1}{2}) = 1/0.5 = 2$: the angular velocity *doubles* at midpoint, in agreement with $1/\cos^2 45^\circ = 2$. ✓ Compared to the mean $\Omega = \pi/2 = 1.5707963$, the peak equals $2/1.5707963 = 1.2732395$: a 27.3% overspeed, which matches $2 \tan 45^\circ / (\pi/2) = 1.2732395$. ✓ *Independent check:* difference quotients of step 10^{-6} on $\phi(t) = \arctan \frac{t}{1-t}$ give $\phi'(0) = 1.000001$, $\phi'(0.25) = 1.600000$, $\phi'(0.5) = 2.000000$. ✓

Step 8 — the angular discrepancy. Still for $\Omega = 90^\circ$, the maximum of $|\phi(t) - t\Omega|$ occurs at $t = 0.239$ and equals 4.0746° out of 90° : NLERP *lags* in the first half and catches up in the second. In terms of rotation of the body the error doubles (Theorem 30.3(iv)): 8.1° out of 180° . □

Remark 30.3: The same time law as the circular NURBS. Theorem 30.1(iii) gives for the circular arc in rational Bézier form the time law $\psi(t) = 2 \arctan \frac{t \sin \frac{\theta}{2}}{(1-t) + t \cos \frac{\theta}{2}}$, and Proposition 30.7(i) gives for NLERP $\phi(t) = \arctan \frac{t \sin \Omega}{(1-t) + t \cos \Omega}$: they are the same function, with $\Omega = \frac{\theta}{2}$ and a factor 2. This is no coincidence: both arise from normalizing (dividing respectively by $\bar{w}$ and by $\|v\|$) a linear combination of the two endpoints. The practical consequence is that a NURBS interpolator that traverses an arc at equally spaced t does not traverse it at constant speed — the very defect that Proposition 30.7 charges to NLERP, and for the same reason. This is why CNC controllers reparametrize arcs by arc length before applying the velocity profile of Section 30.4: the rational parametrization is exact as a set of points, not as a time law.

Remark 30.4: Squad, dual quaternions, and ScLERP: what remains imported.

To concatenate more than two orientations with C^1 continuity one uses *squad* (spherical spline quaternion), a quaternion spline construction for interpolating more than two orientations, attributed to Shoemake; to interpolate position and orientation simultaneously (that is, in $SE(3)$) one uses **dual quaternions**, which encode a screw motion (Chasles's theorem, Chapter 29) in a single algebraic object, and the corresponding ScLERP (screw linear interpolation). Of these two tools the chapter gives only the operational definition: their properties — that *squad* is indeed C^1 at the knots and that ScLERP is a geodesic of $SE(3)$ for a suitable metric — remain imported, that is, assumed without proof. What is missing, precisely, is: (a) the computation of the derivative of *squad* at the knots, which requires the differentiation formula for q^t with respect to t and the choice of intermediate control quaternions; (b) the construction of the metric on $SE(3)$ with respect to which ScLERP minimizes — and it must be noted that on $SE(3)$ there is no bi-invariant metric, so the statement “ScLERP is the geodesic of $SE(3)$ ” is meaningless until the metric is fixed. Nothing proved in this chapter depends on (a) or (b). The source for both is Shoemake (1985) for *squad*

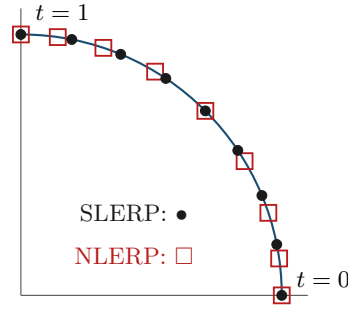


Figure 30.2: SLERP versus NLERP on the arc between two orientations at $\Omega = 90^\circ$ (a two-dimensional cross-section of S^3 seen as a circle; drawing radius 3). Nine equally spaced instants $t = 0, \frac{1}{8}, \dots, 1$. The black points (SLERP) fall at the angles $0^\circ, 11.25^\circ, \dots, 90^\circ$, exactly equally spaced (Proposition 30.5). The red squares (NLERP) fall at the angles $\arctan \frac{t}{1-t}$, i.e. $0^\circ, \arctan \frac{1}{7} = 8.130^\circ, \arctan \frac{1}{3} = 18.435^\circ, \arctan \frac{2}{5} = 30.964^\circ, 45^\circ$, and symmetrically: they are *crowded at the endpoints* and sparse at the center, because by Proposition 30.7(iii) the angular velocity of NLERP is minimal at the endpoints and double at midpoint. Hence the tool's non-uniform motion.

and the literature on rigid-body kinematics (Lynch and Park, *Modern Robotics, Chapters 3 and 9*) for $SE(3)$.

Exercise 30.3. Compute slerp at $t = 1/2$ between $q_0 = 1$ and $q_1 = \cos 45^\circ + \sin 45^\circ k$, and the total rotation swept.

Solution. $\cos \Omega = \langle q_0, q_1 \rangle = \cos 45^\circ = 1/\sqrt{2}$, hence $\Omega = 45^\circ$. At $t = \frac{1}{2}$ the two coefficients of the formula of Theorem 30.2 are equal and equal $\sin 22.5^\circ / \sin 45^\circ = 0.5411961$, and the result is $q_{1/2} = \cos 22.5^\circ + \sin 22.5^\circ k$: by Corollary 30.2 it is already unit, with no renormalization. The total rotation swept is $2\Omega = 90^\circ$ about z (Theorem 30.3(iv)), and at midpoint the body has rotated by 45° : the exact half, at constant angular velocity (Proposition 30.5). *Independent check:* $\|q_{1/2}\|^2 = \cos^2 22.5^\circ + \sin^2 22.5^\circ = 1 \checkmark$; and $\langle q_0, q_{1/2} \rangle = \cos 22.5^\circ$, i.e. angle $22.5^\circ = \Omega/2 \checkmark$.

30.4. Motion profiles: trapezoidal and jerk-limited

Once the geometric curve has been chosen, it remains to decide *when* to traverse each of its points: this is the problem of the time law. The two standard laws are the trapezoidal velocity profile (piecewise-constant acceleration) and the S-curve profile (bounded jerk). This section derives in full the matching conditions and the times involved.

Definition 30.5: Trapezoidal profile and S-curve profile. Given $L, V, A > 0$, a **trapezoidal profile** for a travel of length L is a motion $s : [0, T] \rightarrow \mathbb{R}$ with $s(0) = 0, s(T) = L, v(0) = v(T) = 0$, whose velocity $v = \dot{s}$ is continuous, piecewise linear, with $|v| \leq V$ and $|\dot{v}| \leq A$: the acceleration is piecewise constant, hence discontinuous, and the **jerk** $j = \dot{a}$ is impulsive. Given in addition $J > 0$, an **S-curve profile** is a motion with $|v| \leq V, |a| \leq A$, and $|j| \leq J$ almost everywhere, with a continuous: the acceleration is continuous and the jerk is bounded.

Theorem 30.4: Minimum time with velocity and acceleration limits. Let $L, V, A > 0$ and, for $T > 0$, let $\mathcal{C}(T)$ be the set of motions $s : [0, T] \rightarrow \mathbb{R}$ with $v = \dot{s}$ absolutely continuous, $s(0) = 0$, $s(T) = L$, $v(0) = v(T) = 0$, $|v| \leq V$ on $[0, T]$, and $|\dot{v}| \leq A$ almost everywhere. Set

$$T^* = \begin{cases} \frac{L}{V} + \frac{V}{A} & \text{if } L \geq \frac{V^2}{A} \quad (\text{trapezoidal case}), \\ 2\sqrt{\frac{L}{A}} & \text{if } L < \frac{V^2}{A} \quad (\text{triangular case}). \end{cases}$$

Then $\mathcal{C}(T) \neq \emptyset$ if and only if $T \geq T^*$, and $\mathcal{C}(T^*)$ contains exactly one element, that of velocity

$$v^*(t) = \min\{V, At, A(T^* - t)\}.$$

In the trapezoidal case v^* rises linearly to V at $t_a = V/A$, stays at V for $t_v = \frac{L}{V} - \frac{V}{A}$, and falls in t_a ; in the triangular case the peak velocity is $v_p = \sqrt{AL} < V$, reached at $t_a = \sqrt{L/A}$.

Proof. Step 1 — three bounds on the velocity. Let $s \in \mathcal{C}(T)$. From $v(0) = 0$ and $|\dot{v}| \leq A$ it follows, for every t , that $v(t) = \int_0^t \dot{v} \leq At$; from $v(T) = 0$ it follows that $v(t) = -\int_t^T \dot{v} \leq A(T - t)$; and by hypothesis $v(t) \leq V$. Hence

$$v(t) \leq m_T(t) := \min\{V, At, A(T - t)\} \quad \text{for every } t \in [0, T].$$

Step 2 — the function F . Integrating, $L = \int_0^T v \leq \int_0^T m_T =: F(T)$. F is computed by distinguishing two cases. (a) $T \geq 2V/A$. Then $At \leq V$ for $t \leq V/A \leq T/2$, and by symmetry $A(T - t) \leq V$ for $t \geq T - V/A$; in between the minimum is V . Hence

$$F(T) = 2 \int_0^{V/A} At \, dt + V \left(T - \frac{2V}{A} \right) = \frac{V^2}{A} + VT - \frac{2V^2}{A} = VT - \frac{V^2}{A}.$$

(b) $T < 2V/A$. Then $At < V$ for every $t < T/2$ and the minimum is At on $[0, T/2]$, $A(T - t)$ on $[T/2, T]$:

$$F(T) = 2 \int_0^{T/2} At \, dt = \frac{AT^2}{4}.$$

The two expressions agree at $T = 2V/A$ (both equal V^2/A), so F is continuous; it is strictly increasing because for $T' > T$ one has $m_{T'} \geq m_T$ on $[0, T]$ and $m_{T'} > 0$ on $(T, T']$; and it is surjective onto $(0, +\infty)$ because $F(0^+) = 0$ and $F(+\infty) = +\infty$.

Step 3 — T^* is the solution of $F(T) = L$. In the case $L \geq V^2/A$ the solution is sought in branch (a): $VT - \frac{V^2}{A} = L$ gives $T = \frac{L}{V} + \frac{V}{A}$, and this is $\geq \frac{2V}{A}$ if and only if $\frac{L}{V} \geq \frac{V}{A}$, that is $L \geq \frac{V^2}{A}$: consistent. In the case $L < V^2/A$ the solution is sought in branch (b): $\frac{AT^2}{4} = L$ gives $T = 2\sqrt{L/A}$, which is $< \frac{2V}{A}$ if and only if $\sqrt{L/A} < V/A$, that is $L < \frac{V^2}{A}$: consistent. In both cases $F(T^*) = L$.

Step 4 — necessity. If $s \in \mathcal{C}(T)$ then $L \leq F(T)$ by Step 2; since F is strictly increasing and $F(T^*) = L$, it follows that $T \geq T^*$.

Step 5 — sufficiency and construction. Define $v^* = m_{T^*}$ and $s^*(t) = \int_0^t v^*$. Then v^* is continuous, piecewise linear, with $v^*(0) = v^*(T^*) = 0$ (because $m_{T^*}(0) = \min\{V, 0, AT^*\} = 0$ and likewise at T^*), $0 \leq v^* \leq V$, and $|\dot{v}^*| \leq A$ almost everywhere (the slopes are $A, 0, -A$). Finally $s^*(T^*) = F(T^*) = L$ by Step 3: hence $s^* \in \mathcal{C}(T^*)$. For $T > T^*$ an element of $\mathcal{C}(T)$ is constructed by lengthening the constant-velocity segment (trapezoidal case) or by lowering the peak until $\int v = L$ holds; alternatively, $F(T) > L$ and the function $\lambda \mapsto \int_0^T \min\{\lambda, At, A(T - t)\} \, dt$ is continuous and increasing from 0 to $F(T)$, so it takes the value L for some $\lambda \in (0, V]$, which gives an element of $\mathcal{C}(T)$.

Step 6 — uniqueness at T^* . If $s \in \mathcal{C}(T^*)$ then $\int_0^{T^*} (m_{T^*} - v) = F(T^*) - L = 0$ with continuous, non-negative integrand (Step 1): hence $v \equiv m_{T^*} = v^*$ and, integrating with $s(0) = 0$, $s \equiv s^*$.

Step 7 — reading the two cases. In the trapezoidal case $t_a = V/A$ is the instant at which At reaches V ; the constant-velocity segment lasts $T^* - 2t_a = \frac{L}{V} + \frac{V}{A} - \frac{2V}{A} = \frac{L}{V} - \frac{V}{A} = t_v$. In the triangular case the peak occurs at $t = T^*/2 = \sqrt{L/A} = t_a$ and equals $v_p = A\sqrt{L/A} = \sqrt{AL}$, which is $< V$ because $L < V^2/A$.

Step 8 — numerical check. With $L = 1$ m, $V = 1$ m/s, $A = 2$ m/s² one has $V^2/A = 0.5 \leq 1 = L$: trapezoidal case, $t_a = 0.5$ s, $t_v = 1 - 0.5 = 0.5$ s, $T^* = 1 + 0.5 = 1.5$ s. *Independent check:* the distance travelled, computed by numerical integration of the profile with $2 \cdot 10^6$ steps, equals 0.99999999996 against $L = 1$. ✓ With $L = 0.1$ m and the same V, A one has $V^2/A = 0.5 > 0.1$: triangular case, $t_a = \sqrt{0.05} = 0.2236068$ s, $T^* = 0.4472136$ s, $v_p = \sqrt{0.2} = 0.4472136$ m/s. *Check:* the distance is $2 \cdot \frac{1}{2} At_a^2 = At_a^2 = 2 \cdot 0.05 = 0.1 = L$ ✓, whereas the trapezoidal formula would erroneously give $0.1 + 0.5 = 0.6$ s, a value *larger* than the true one: applying the wrong formula does not make the motion slower, it makes it impossible (it would require $v > V$ or a longer travel). □

30.4.1 The seven-segment S-curve profile

Bounding the jerk by a maximum J means making the acceleration continuous. The resulting profile has up to **seven segments**: positive jerk, constant acceleration, negative jerk, constant velocity, negative jerk, constant deceleration, positive jerk. Position, velocity, and acceleration remain continuous; only the jerk is discontinuous (in finite steps), no longer impulsive as in the trapezoidal profile. The lemma that follows is the key to the computation: it says that the acceleration phase, however complicated, covers exactly the distance it would cover at the average velocity $V/2$.

Lemma 30.3: Antisymmetry of the acceleration phase. Let $T_a > 0$ and let $j : [0, T_a] \rightarrow \mathbb{R}$ be bounded and measurable with $j(T_a - \sigma) = -j(\sigma)$ for almost every σ . Setting $a(t) = \int_0^t j$ and $v(t) = \int_0^t a$, one has:

$$a(T_a) = 0, \quad a(T_a - \sigma) = a(\sigma), \quad v(\sigma) + v(T_a - \sigma) = v(T_a), \quad \int_0^{T_a} v = \frac{T_a}{2} v(T_a).$$

Proof. Step 1 — $a(T_a) = 0$. With the substitution $\sigma = T_a - r$,

$$a(T_a) = \int_0^{T_a} j(\sigma) d\sigma = \int_0^{T_a} j(T_a - r) dr = - \int_0^{T_a} j(r) dr = -a(T_a),$$

hence $a(T_a) = 0$.

Step 2 — symmetry of a . For every $\sigma \in [0, T_a]$,

$$a(T_a - \sigma) = a(T_a) - \int_{T_a - \sigma}^{T_a} j = 0 - \int_0^{\sigma} j(T_a - r) dr = \int_0^{\sigma} j(r) dr = a(\sigma).$$

Step 3 — antisymmetry of v . Likewise, using Step 2,

$$v(T_a - \sigma) = v(T_a) - \int_{T_a - \sigma}^{T_a} a = v(T_a) - \int_0^{\sigma} a(T_a - r) dr = v(T_a) - \int_0^{\sigma} a(r) dr = v(T_a) - v(\sigma).$$

Step 4 — the integral. With the substitution $\sigma \rightarrow T_a - \sigma$ and Step 3,

$$\int_0^{T_a} v(\sigma) d\sigma = \int_0^{T_a} v(T_a - \sigma) d\sigma = \int_0^{T_a} [v(T_a) - v(\sigma)] d\sigma = T_a v(T_a) - \int_0^{T_a} v,$$

whence $2 \int_0^{T_a} v = T_a v(T_a)$. □

Theorem 30.5: The S-curve profile: matching, durations, and total time. Let $L, V, A, J > 0$.

(a) Seven-segment case. If $V \geq \frac{A^2}{J}$ and $L \geq V\left(\frac{V}{A} + \frac{A}{J}\right)$, set

$$t_j = \frac{A}{J}, \quad t_a = \frac{V}{A} - \frac{A}{J}, \quad T_a = 2t_j + t_a = \frac{V}{A} + \frac{A}{J}, \quad t_v = \frac{L}{V} - T_a,$$

and let j be the piecewise-constant function equal to $+J, 0, -J, 0, -J, 0, +J$ on seven consecutive intervals of durations $t_j, t_a, t_j, t_v, t_j, t_a, t_j$, with $a(0) = v(0) = s(0) = 0$. Then $t_j, t_a, t_v \geq 0$ and the resulting motion satisfies: a continuous, v of class C^1 , s of class C^2 ; $\max |a| = A$, $\max v = V$; $a(T) = v(T) = 0$ and $s(T) = L$; and

$$T = \frac{L}{V} + \frac{V}{A} + \frac{A}{J}.$$

(b) Four-segment case (the acceleration never reaches A). If $V < \frac{A^2}{J}$ and $L \geq 2V\sqrt{V/J}$, the same statement holds with $t_j = \sqrt{V/J}$, $t_a = 0$, $T_a = 2\sqrt{V/J}$, $t_v = \frac{L}{V} - T_a$, $\max |a| = \sqrt{VJ} < A$, and

$$T = \frac{L}{V} + 2\sqrt{\frac{V}{J}}.$$

Proof. Step 1 — sign of the durations. In case (a), $t_j > 0$ and $t_a = \frac{V}{A} - \frac{A}{J} \geq 0$ is equivalent to $VJ \geq A^2$, that is to the hypothesis $V \geq A^2/J$; $t_v = \frac{L}{V} - \frac{V}{A} - \frac{A}{J} \geq 0$ is equivalent to $L \geq V\left(\frac{V}{A} + \frac{A}{J}\right)$, the second hypothesis. In case (b), $t_j = \sqrt{V/J} > 0$ and $t_v \geq 0$ is equivalent to $L \geq 2V\sqrt{V/J}$.

Step 2 — regularity and matching. The jerk j is piecewise constant and bounded, so $a(t) = \int_0^t j$ is continuous and piecewise linear, $v = \int_0^t a$ is of class C^1 (continuous derivative a), and $s = \int_0^t v$ is of class C^2 . The matching conditions — continuity of s, v, a at every junction — are automatic because s, v, a are defined as integrals of bounded functions; what is *not* automatic, and what determines the durations, are the three terminal conditions $a(T) = 0$, $v(T) = 0$, $s(T) = L$, verified in the steps that follow.

Step 3 — the acceleration phase, case (a). On $[0, T_a]$ the jerk equals $+J, 0, -J$ on intervals of durations t_j, t_a, t_j : it is antisymmetric about the midpoint $T_a/2$, because $j(T_a - \sigma) = -J = -j(\sigma)$ for $\sigma \in (0, t_j)$ and $j = 0$ on the central segment. Integrating explicitly:

$$a(t_j) = Jt_j = A, \quad v(t_j) = \frac{1}{2}Jt_j^2 = \frac{A^2}{2J},$$

$$a(t_j + t_a) = A, \quad v(t_j + t_a) = \frac{A^2}{2J} + At_a = \frac{A^2}{2J} + V - \frac{A^2}{J} = V - \frac{A^2}{2J},$$

$$a(T_a) = A - Jt_j = 0, \quad v(T_a) = V - \frac{A^2}{2J} + \left(At_j - \frac{1}{2}Jt_j^2\right) = V - \frac{A^2}{2J} + \frac{A^2}{2J} = V.$$

Hence at the end of the acceleration phase the velocity is exactly V and the acceleration has returned to zero; the maximum of $|a|$ is A (reached on the central segment) and v is increasing, so $\max v = V$. By Lemma 30.3 the distance covered in this phase is

$$\int_0^{T_a} v = \frac{T_a}{2} V.$$

Step 4 — the constant-velocity phase and the deceleration. On $[T_a, T_a + t_v]$ the jerk is zero and $a(T_a) = 0$: hence $a \equiv 0$ and $v \equiv V$, with distance Vt_v . On $[T_a + t_v, 2T_a + t_v]$ the

jerk is $-J, 0, +J$ over durations t_j, t_a, t_j , that is, the opposite of that of the acceleration phase: by linearity of integration the acceleration is the opposite of that of the acceleration phase, $a(T_a + t_v + \sigma) = -a(\sigma)$, and hence the velocity is $v(T_a + t_v + \sigma) = V - v(\sigma)$ for $\sigma \in [0, T_a]$; in particular it falls from V to $V - V = 0$ and the acceleration returns to 0. The distance covered in deceleration is therefore

$$\int_0^{T_a} (V - v(\sigma)) d\sigma = V T_a - \frac{T_a}{2} V = \frac{T_a}{2} V,$$

again by Lemma 30.3: equal to that of the acceleration phase, as symmetry requires.

Step 5 — the total distance and time. Summing the three contributions,

$$s(T) = \frac{T_a}{2} V + V t_v + \frac{T_a}{2} V = V(T_a + t_v) = V \left(\frac{V}{A} + \frac{A}{J} + \frac{L}{V} - \frac{V}{A} - \frac{A}{J} \right) = L,$$

and the total duration is

$$T = 2T_a + t_v = 2 \left(\frac{V}{A} + \frac{A}{J} \right) + \frac{L}{V} - \frac{V}{A} - \frac{A}{J} = \frac{L}{V} + \frac{V}{A} + \frac{A}{J}.$$

Step 6 — case (b). If $V < A^2/J$ there is no constant-acceleration segment: the jerk equals $+J, -J$ on two intervals of duration t_j . Then $a(t_j) = Jt_j$, $v(t_j) = \frac{1}{2}Jt_j^2$, and at the end $a(2t_j) = 0$, $v(2t_j) = \frac{1}{2}Jt_j^2 + (Jt_j \cdot t_j - \frac{1}{2}Jt_j^2) = Jt_j^2$. Imposing $v(2t_j) = V$ gives $t_j = \sqrt{V/J}$ and $\max |a| = Jt_j = \sqrt{VJ}$, which is $< A$ because $V < A^2/J$: the acceleration constraint is not active. The jerk remains antisymmetric about $T_a/2 = t_j$, so Lemma 30.3 again gives distance $\frac{T_a}{2} V$, and Steps 4–5 repeat identically with $T_a = 2\sqrt{V/J}$, giving $T = \frac{L}{V} + 2\sqrt{V/J}$.

Step 7 — numerical check. With $L = 1$ m, $V = 1$ m/s, $A = 2$ m/s², $J = 10$ m/s³: $A^2/J = 0.4 \leq 1 = V$ (case (a)); $t_j = 0.2$ s, $t_a = 0.5 - 0.2 = 0.3$ s, $T_a = 0.7$ s, $V T_a = 0.7 \leq 1 = L$ ✓, $t_v = 1 - 0.7 = 0.3$ s. The durations sum to $0.2 + 0.3 + 0.2 + 0.3 + 0.2 + 0.3 + 0.2 = 1.7$ s, which agrees with $\frac{L}{V} + \frac{V}{A} + \frac{A}{J} = 1 + 0.5 + 0.2 = 1.7$ s. ✓ *Independent check:* integrating exactly segment by segment (constant jerk, formulas $s = v\delta + \frac{1}{2}a\delta^2 + \frac{1}{6}j\delta^3$) gives $s(T) = 1.000000000$, $v(T) = -5.1 \cdot 10^{-17}$, $a(T) = 1.3 \cdot 10^{-16}$, $\max v = 1.000000000$, $\max |a| = 2.000000000$. ✓ The S-curve profile costs $1.7 - 1.5 = 0.2$ s more than the trapezoidal one (Theorem 30.4), that is 13.3%; and the additional term is exactly A/J . □

Proposition 30.8: How far the S-curve profile is from the absolute minimum time. Let $L, V, A, J > 0$ with $V \geq \frac{A^2}{J}$ and $L \geq V \left(\frac{V}{A} + \frac{A}{J} \right)$, and let $\mathcal{S}(T)$ be the set of motions $s : [0, T] \rightarrow \mathbb{R}$ with $a = \ddot{s}$ absolutely continuous, $s(0) = 0$, $s(T) = L$, $v(0) = v(T) = 0$, $a(0) = a(T) = 0$, $|v| \leq V$, $|a| \leq A$, and $|\dot{a}| \leq J$ almost everywhere. Then every T with $\mathcal{S}(T) \neq \emptyset$ satisfies

$$T > T_{\text{inf}} := \frac{L}{V} + \frac{V}{A} + \frac{A}{J} - \frac{A^3}{12J^2V} = T_S - \frac{A^3}{12J^2V},$$

where T_S is the time of the seven-segment profile (Theorem 30.5). The seven-segment profile is thus optimal up to $\frac{A^3}{12J^2V}$ in absolute value, that is, up to a fraction $\frac{A^3}{12J^2VT_S}$ of the total time.

Proof. Step 1 — bounding the acceleration. Let $s \in \mathcal{S}(T)$. From $a(0) = 0$ and $|\dot{a}| \leq J$ it follows that $a(\sigma) \leq J\sigma$; from $a(T) = 0$ it follows that $-a(\sigma) \leq J(T - \sigma)$. With $|a| \leq A$:

$$a(\sigma) \leq \min\{A, J\sigma\}, \quad -a(\sigma) \leq \min\{A, J(T - \sigma)\}.$$

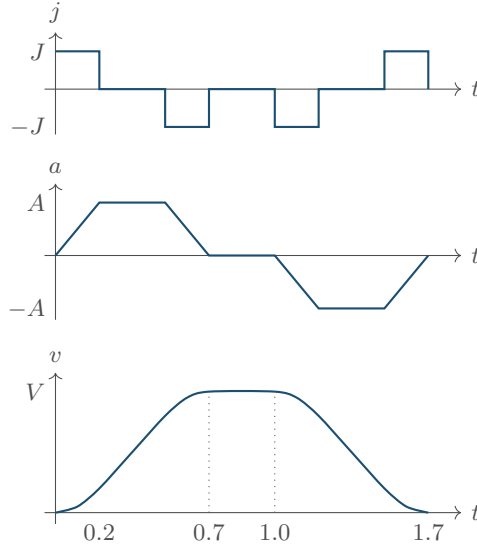


Figure 30.3: Seven-segment S-curve profile for $L = 1$ m, $V = 1$ m/s, $A = 2$ m/s², $J = 10$ m/s³ (Theorem 30.5, case (a)). Matching instants, in seconds: 0, 0.2, 0.5, 0.7, 1.0, 1.2, 1.5, 1.7; at those instants the acceleration equals 0, 2, 2, 0, 0, -2, -2, 0 (m/s²), the velocity 0; 0.2; 0.8; 1; 1; 0.8; 0.2; 0 (m/s), and the position 0; 0.0133; 0.1633; 0.35; 0.65; 0.8367; 0.9867; 1 (m). The jerk is piecewise constant, the acceleration is continuous (trapezoidal), the velocity is C^1 , and the position C^2 . The ordinates of the drawing are to scale: $J \rightarrow 0.5$, $A \rightarrow 0.7$, $V \rightarrow 1.6$ units.

Step 2 — bounding the velocity. Setting $G(t) = \int_0^t \min\{A, Jr\} dr$, that is,

$$G(t) = \begin{cases} \frac{Jt^2}{2} & t \leq \frac{A}{J}, \\ \frac{A^2}{2J} + A\left(t - \frac{A}{J}\right) & t \geq \frac{A}{J}, \end{cases}$$

one has $v(t) = \int_0^t a \leq G(t)$ and $v(t) = -\int_t^T a \leq \int_t^T \min\{A, J(T - \sigma)\} d\sigma = G(T - t)$. With $v \leq V$:

$$v(t) \leq \mu_T(t) := \min\{V, G(t), G(T - t)\}.$$

Step 3 — computing $F_J(T) = \int_0^T \mu_T$. Let t_1 be the instant at which G reaches V : from $\frac{A^2}{2J} + A(t_1 - \frac{A}{J}) = V$ one obtains $t_1 = \frac{V}{A} + \frac{A}{2J}$, and $t_1 \geq \frac{A}{J}$ because $V \geq \frac{A^2}{J} \geq \frac{A^2}{2J}$. Assuming $T \geq 2t_1$ (verified a posteriori in Step 5),

$$F_J(T) = 2 \int_0^{t_1} G + V(T - 2t_1).$$

Now, setting $\tau = \frac{A}{J}$,

$$\int_0^\tau G = \int_0^\tau \frac{Jr^2}{2} dr = \frac{J\tau^3}{6} = \frac{A\tau^2}{6},$$

$$\int_\tau^{t_1} G = \int_\tau^{t_1} \left(Ar - \frac{A\tau}{2}\right) dr = \frac{A}{2}(t_1^2 - \tau^2) - \frac{A\tau}{2}(t_1 - \tau) = \frac{A}{2}(t_1 - \tau)(t_1 + \tau - \tau) = \frac{A}{2}t_1(t_1 - \tau),$$

having used $\frac{A^2}{2J} - \frac{A^2}{J} = -\frac{A^2}{2J} = -\frac{A\tau}{2}$. Hence

$$F_J(T) = VT - 2Vt_1 + \frac{A\tau^2}{3} + At_1^2 - At_1\tau.$$

Substituting $t_1 = \frac{V}{A} + \frac{\tau}{2}$:

$$2Vt_1 = \frac{2V^2}{A} + V\tau, \quad At_1^2 = \frac{V^2}{A} + V\tau + \frac{A\tau^2}{4}, \quad At_1\tau = V\tau + \frac{A\tau^2}{2},$$

and summing algebraically

$$F_J(T) = VT - \frac{V^2}{A} - V\tau + A\tau^2\left(\frac{1}{3} + \frac{1}{4} - \frac{1}{2}\right) = VT - \frac{V^2}{A} - \frac{VA}{J} + \frac{A^3}{12J^2},$$

since $\frac{1}{3} + \frac{1}{4} - \frac{1}{2} = \frac{1}{12}$ and $A\tau^2 = \frac{A^3}{J^2}$.

Step 4 — the inequality. By Step 2, $L = \int_0^T v \leq F_J(T)$, and F_J is strictly increasing in T (the bound μ_T grows with T pointwise and the domain widens). Solving $F_J(T) = L$ gives

$$T = \frac{L}{V} + \frac{V}{A} + \frac{A}{J} - \frac{A^3}{12J^2V} = T_{\text{inf}},$$

hence $\mathcal{S}(T) \neq \emptyset$ implies $T \geq T_{\text{inf}}$.

Step 5 — consistency of the hypothesis $T \geq 2t_1$. One has $2t_1 = \frac{2V}{A} + \frac{A}{J}$ and $T_{\text{inf}} \geq \frac{L}{V} + \frac{V}{A} + \frac{A}{J} - \frac{A^3}{12J^2V}$; since $L \geq V(\frac{V}{A} + \frac{A}{J})$ gives $\frac{L}{V} \geq \frac{V}{A} + \frac{A}{J}$, one obtains $T_{\text{inf}} \geq \frac{2V}{A} + \frac{2A}{J} - \frac{A^3}{12J^2V}$. It thus suffices to check $\frac{A}{J} \geq \frac{A^3}{12J^2V}$, that is $12JV \geq A^2$, true because $JV \geq A^2$ by hypothesis.

Step 6 — the inequality is strict. If $T = T_{\text{inf}}$ held for some $s \in \mathcal{S}(T)$, Step 4 would give $\int_0^T (\mu_T - v) = 0$ with continuous, non-negative integrand, hence $v \equiv \mu_T$. But μ_T is not differentiable with J -Lipschitz derivative: at $t = t_1$ the left derivative is $G'(t_1) = A$ and the right one is 0 (because there the minimum switches from $G(t)$ to V), and $t_1 < T - t_1$ by Step 5, so the jump is effective. The acceleration $a = \dot{v}$ would then jump by A at an instant, against $|\dot{a}| \leq J$. Hence no element of $\mathcal{S}(T_{\text{inf}})$ exists and the inequality is strict.

Step 7 — numerical check. With $L = 1$, $V = 1$, $A = 2$, $J = 10$ (the units of Theorem 30.5): $\frac{A^3}{12J^2V} = \frac{8}{1200} = 0.0066667$ s, hence $T_{\text{inf}} = 1.7 - 0.0066667 = 1.6933333$ s against $T_S = 1.7$ s: the seven-segment profile wastes at most 0.392% of the time. *Independent check:* $t_1 = \frac{V}{A} + \frac{A}{2J} = 0.5 + 0.1 = 0.6$ s and $2t_1 = 1.2 \leq 1.6933$ ✓; and re-evaluating the formula of Step 3 at $T = 1.6933333$, with $\frac{V^2}{A} = 0.5$, $\frac{VA}{J} = 0.2$, and $\frac{A^3}{12J^2} = 0.0066667$,

$$F_J(T) = 1.6933333 - 0.5 - 0.2 + 0.0066667 = 1.0000000 = L. \quad \checkmark$$

□

Remark 30.5: Why bound the jerk. Jerk is the number-one enemy of motion quality: an abrupt change in acceleration excites the elastic modes of the mechanics (gearboxes, belts, long shafts, flexible arms), producing residual vibrations that degrade precision and finish and fatigue the actuators. A real mechanical system is not a rigid body: it is a collection of masses and springs with natural frequencies ω_n , and a “hard” command drives them into resonance. Quantitatively: in the trapezoidal profile the acceleration jumps by A at an instant, that is, the jerk is a Dirac distribution of intensity A , whose spectrum is flat and therefore excites all frequencies equally; in the S-curve profile the jerk is bounded by J and its spectrum decays, the more rapidly the longer the ramp $t_j = A/J$. The price, by Theorem 30.5, is exactly A/J seconds more: the very same t_j that softens the command. Section 30.5 tackles the same problem from the other side — not softening the command, but cancelling the vibration with a second command in phase opposition — and achieves a delay of $T_d/2$, that is, half the period of the mode to be suppressed.

Exercise 30.4. An axis must travel $L = 1$ m with maximum velocity $V = 1$ m/s and acceleration $A = 2$ m/s² (trapezoidal profile). How long does the motion last? And with $J = 10$ m/s³?

Solution. Since $V^2/A = 0.5 \leq 1 = L$, this is the trapezoidal case of Theorem 30.4: acceleration time $t_a = V/A = 0.5$ s, distance in acceleration $\frac{1}{2}At_a^2 = 0.25$ m (equal in deceleration), total 0.5 m in the ramp phases; the remaining 0.5 m at constant velocity, in 0.5 s. Total time $T = t_a + t_v + t_a = 0.5 + 0.5 + 0.5 = 1.5$ s, in agreement with $T^* = \frac{L}{V} + \frac{V}{A} = 1 + 0.5$. With the jerk limit, since $A^2/J = 0.4 \leq V$ and $V(\frac{V}{A} + \frac{A}{J}) = 0.7 \leq L$, Theorem 30.5 gives $T = \frac{L}{V} + \frac{V}{A} + \frac{A}{J} = 1 + 0.5 + 0.2 = 1.7$ s: 0.2 s more, that is, 13.3%. *Independent check:* the seven durations $0.2 + 0.3 + 0.2 + 0.3 + 0.2 + 0.3 + 0.2$ sum to 1.7 s ✓.

30.5. Residual vibration and shaped command

A jerk-limited profile attenuates the excitation of the elastic modes, but does not cancel it. The technique of *input shaping* does something different and more radical: it replaces the command with its convolution with a short train of impulses, chosen so that the vibrations produced by the individual impulses cancel *exactly*. This section derives in full the two classical shapers, ZV and ZVD.

Definition 30.6: Elastic mode and residual vibration. An **elastic mode** of natural frequency $\omega_n > 0$ and damping $\zeta \in [0, 1)$ is the second-order linear system

$$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 u, \quad \text{that is} \quad H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

(Definition 25.1), with $\beta = \sqrt{1 - \zeta^2}$, $\omega_d = \omega_n\beta$, $T_d = 2\pi/\omega_d$, and $K = e^{-\zeta\pi/\beta}$. If the command u is zero (or constant) from some instant t_N onward, the amplitude of the oscillation of y for $t \geq t_N$, that is, the coefficient C in the expression $y(t) = y_\infty + C e^{-\zeta\omega_n t} \sin(\omega_d t + \vartheta)$, is called the **residual vibration**.

Lemma 30.4: Impulse and step response of an elastic mode. With the notation of Definition 30.6:

(i) the **impulse response**, that is, the solution of $\ddot{h} + 2\zeta\omega_n\dot{h} + \omega_n^2 h = 0$ with $h(0) = 0$ and $\dot{h}(0) = \omega_n^2$, is

$$h(t) = \frac{\omega_n^2}{\omega_d} e^{-\zeta\omega_n t} \sin(\omega_d t), \quad t \geq 0;$$

(ii) the unit **step response**, that is, the solution of $\ddot{g} + 2\zeta\omega_n\dot{g} + \omega_n^2 g = \omega_n^2$ with $g(0) = \dot{g}(0) = 0$, is

$$g(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\beta} \sin \omega_d t \right) = 1 - \frac{1}{\beta} e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi),$$

where $\varphi \in (0, \pi)$ is defined by $\cos \varphi = \zeta$, $\sin \varphi = \beta$;

(iii) $\dot{g}(t) = \frac{\omega_n}{\beta} e^{-\zeta\omega_n t} \sin(\omega_d t)$, hence g is increasing on $[0, \pi/\omega_d]$ and its first maximum equals $g(\pi/\omega_d) = 1 + K$.

Proof. Step 1 — roots of the characteristic equation. The polynomial $s^2 + 2\zeta\omega_n s + \omega_n^2$ has discriminant $4\zeta^2\omega_n^2 - 4\omega_n^2 = -4\omega_n^2\beta^2 < 0$ for $\zeta < 1$: the roots are $s_\pm = -\zeta\omega_n \pm i\omega_d$,

complex conjugates and distinct. The general integral of the homogeneous equation is thus $e^{-\zeta\omega_n t}(C_1 \cos \omega_d t + C_2 \sin \omega_d t)$.

Step 2 — (i). Imposing $h(0) = C_1 = 0$ and $\dot{h}(0) = -\zeta\omega_n C_1 + \omega_d C_2 = \omega_d C_2 = \omega_n^2$ gives $C_2 = \omega_n^2/\omega_d$, that is, the formula of the statement. *Independent check:* from the table of transforms (Volume III, Appendix C), $\mathcal{L}\{e^{-at} \sin(bt)\} = \frac{b}{(s+a)^2 + b^2}$, hence

$$\mathcal{L}\{h\} = \frac{\omega_n^2}{\omega_d} \cdot \frac{\omega_d}{(s + \zeta\omega_n)^2 + \omega_d^2} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \zeta^2\omega_n^2 + \omega_n^2 - \zeta^2\omega_n^2} = H(s). \quad \checkmark$$

Step 3 — (ii). The constant function 1 is a particular solution of the equation with forcing term ω_n^2 ; the general integral is thus $g = 1 + e^{-\zeta\omega_n t}(C_1 \cos \omega_d t + C_2 \sin \omega_d t)$. From $g(0) = 1 + C_1 = 0$ it follows that $C_1 = -1$; from $\dot{g}(0) = -\zeta\omega_n C_1 + \omega_d C_2 = \zeta\omega_n + \omega_d C_2 = 0$ it follows that $C_2 = -\zeta\omega_n/\omega_d = -\zeta/\beta$. For the compact form:

$$\cos \omega_d t + \frac{\zeta}{\beta} \sin \omega_d t = \frac{1}{\beta} (\beta \cos \omega_d t + \zeta \sin \omega_d t) = \frac{1}{\beta} \sin(\omega_d t + \varphi)$$

with $\sin \varphi = \beta$ and $\cos \varphi = \zeta$, which is legitimate because $\zeta^2 + \beta^2 = 1$; and $\varphi \in (0, \pi)$ because $\beta > 0$.

Step 4 — (iii). Differentiating the first form of (ii),

$$\dot{g} = \zeta\omega_n e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\beta} \sin \omega_d t \right) - e^{-\zeta\omega_n t} \left(-\omega_d \sin \omega_d t + \frac{\zeta\omega_d}{\beta} \cos \omega_d t \right).$$

The cos terms cancel because $\frac{\zeta\omega_d}{\beta} = \zeta\omega_n$; the sin terms give the coefficient $\frac{\zeta^2\omega_n}{\beta} + \omega_d = \frac{\zeta^2\omega_n + \omega_n\beta^2}{\beta} = \frac{\omega_n}{\beta}$. Hence $\dot{g} = \frac{\omega_n}{\beta} e^{-\zeta\omega_n t} \sin(\omega_d t)$, which is ≥ 0 for $\omega_d t \in [0, \pi]$: g increases on $[0, \pi/\omega_d]$. Its value at the first endpoint is

$$g\left(\frac{\pi}{\omega_d}\right) = 1 - e^{-\zeta\omega_n\pi/\omega_d} (\cos \pi + 0) = 1 + e^{-\zeta\pi/\beta} = 1 + K.$$

□

Theorem 30.6: *The residual vibration is the modulus of the shaper's transform at the pole. Let $A_1, \dots, A_N \in \mathbb{R}$ and $0 = t_1 < t_2 < \dots < t_N$, and let*

$$\widehat{A}(s) = \sum_{i=1}^N A_i e^{-st_i}$$

be the Laplace transform of the impulse train $\sum_i A_i \delta(t - t_i)$. Let $s_0 = -\zeta\omega_n + i\omega_d$ be the pole of the elastic mode. Then, for every $t \geq t_N$:

(i) *the response to the impulse train equals*

$$y(t) = \frac{\omega_n^2}{\omega_d} |\widehat{A}(s_0)| e^{-\zeta\omega_n t} \sin(\omega_d t + \arg \widehat{A}(s_0));$$

(ii) *the response to the shaped step $u(t) = \sum_i A_i \mathbf{1}_{[t_i, +\infty)}(t)$ equals*

$$y(t) = \sum_{i=1}^N A_i - \frac{1}{\beta} |\widehat{A}(s_0)| e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi + \arg \widehat{A}(s_0));$$

(iii) *in both cases the residual vibration is zero if and only if $\widehat{A}(s_0) = 0$.*

Proof. Step 1 — superposition. The system is linear and time-invariant, hence for $t \geq t_N$ (that is, once all the impulses have already acted) $y(t) = \sum_i A_i h(t - t_i)$ with h from Lemma 30.4(i).

Step 2 — sum of damped sinusoids. Using $\sin x = \operatorname{Im} e^{ix}$,

$$y(t) = \frac{\omega_n^2}{\omega_d} \sum_i A_i e^{-\zeta\omega_n(t-t_i)} \operatorname{Im} e^{i\omega_d(t-t_i)} = \frac{\omega_n^2}{\omega_d} e^{-\zeta\omega_n t} \operatorname{Im} \left[e^{i\omega_d t} \sum_i A_i e^{(\zeta\omega_n - i\omega_d)t_i} \right],$$

having pulled out of the sum the factors that do not depend on i and used the linearity of Im with respect to real coefficients.

Step 3 — recognizing the transform. Since $-s_0 = \zeta\omega_n - i\omega_d$, the sum is $\sum_i A_i e^{-s_0 t_i} = \hat{A}(s_0)$. Writing $\hat{A}(s_0) = |\hat{A}(s_0)| e^{i \arg \hat{A}(s_0)}$ gives

$$\operatorname{Im} [e^{i\omega_d t} \hat{A}(s_0)] = |\hat{A}(s_0)| \sin(\omega_d t + \arg \hat{A}(s_0)),$$

that is, (i).

Step 4 — (ii). For $t \geq t_N$ each step $\mathbf{1}_{[t_i, \infty)}$ is already active, hence $y(t) = \sum_i A_i g(t - t_i)$ with g from Lemma 30.4(ii). Separating the constant part,

$$y(t) = \sum_i A_i - \frac{1}{\beta} \sum_i A_i e^{-\zeta\omega_n(t-t_i)} \sin(\omega_d(t-t_i) + \varphi),$$

and repeating Steps 2–3 with $e^{i(\omega_d t + \varphi)}$ in place of $e^{i\omega_d t}$ gives the formula of the statement.

Step 5 — (iii). If $\hat{A}(s_0) = 0$ the oscillating term is identically zero. Conversely, if the oscillating term is zero for every $t \geq t_N$, then $|\hat{A}(s_0)| e^{-\zeta\omega_n t} \sin(\omega_d t + \vartheta) \equiv 0$; since $e^{-\zeta\omega_n t} > 0$ and $\sin(\omega_d t + \vartheta)$ is not identically zero on any interval (it has isolated zeros), it must be that $|\hat{A}(s_0)| = 0$.

Step 6 — reading. Designing a shaper thus means *placing a zero of its transform exactly at the pole of the system*: this is the same craft as the pole-zero cancellation of Chapter 25 and the placement of zeros in the FIR filters of Chapter 21, with the difference that here the zero does not cancel a pole of the loop, but the *free mode* that pole generates. $\square$

Theorem 30.7: Zero-vibration ZV shaper. *Seek two impulses of amplitudes $A_1, A_2 > 0$ at instants $t_1 = 0$ and $t_2 = \Delta > 0$, with $A_1 + A_2 = 1$ (unit gain), such that the residual vibration of the elastic mode vanishes. Then the conditions determine uniquely, up to the choice of the minimal Δ :*

$$\Delta = \frac{T_d}{2} = \frac{\pi}{\omega_d}, \quad A_1 = \frac{1}{1+K}, \quad A_2 = \frac{K}{1+K}, \quad K = e^{-\zeta\pi/\sqrt{1-\zeta^2}}.$$

More precisely, the solutions are all and only those with $\Delta = mT_d/2$ for m a positive odd integer and the same amplitudes with K replaced by K^m ; the minimal delay is at $m = 1$.

Proof. Step 1 — the condition. By Theorem 30.6(iii) the requirement is $\hat{A}(s_0) = 0$, with $\hat{A}(s) = A_1 + A_2 e^{-s\Delta}$ and $s_0 = -\zeta\omega_n + i\omega_d$. Now

$$e^{-s_0 \Delta} = e^{\zeta\omega_n \Delta} e^{-i\omega_d \Delta} = e^{\zeta\omega_n \Delta} (\cos \omega_d \Delta - i \sin \omega_d \Delta),$$

hence the equation $\hat{A}(s_0) = 0$ splits into the two real conditions

$$(\text{real part}) \quad A_1 + A_2 e^{\zeta\omega_n \Delta} \cos(\omega_d \Delta) = 0, \quad (\text{imaginary part}) \quad -A_2 e^{\zeta\omega_n \Delta} \sin(\omega_d \Delta) = 0.$$

Step 2 — the delay. Since $A_2 > 0$ and $e^{\zeta\omega_n \Delta} > 0$, the second condition gives $\sin(\omega_d \Delta) = 0$, that is, $\omega_d \Delta = m\pi$ with m a positive integer, that is, $\Delta = m \frac{\pi}{\omega_d} = m \frac{T_d}{2}$.

Step 3 — the parity of m . With $\omega_d \Delta = m\pi$ one has $\cos(\omega_d \Delta) = (-1)^m$ and the first condition becomes $A_1 = -(-1)^m A_2 e^{\zeta \omega_n \Delta}$. The left-hand side is positive and $A_2 e^{\zeta \omega_n \Delta} > 0$; hence $(-1)^m = -1$, that is, m is *odd*. (With m even the two impulses would arrive in phase and would add up instead of cancelling.)

Step 4 — the amplitudes. With m odd,

$$e^{\zeta \omega_n \Delta} = e^{\zeta \omega_n m\pi/\omega_d} = e^{m\zeta\pi/\beta} = K^{-m},$$

and the first condition gives $A_1 = A_2 K^{-m}$, that is, $A_2 = K^m A_1$. With the constraint $A_1 + A_2 = 1$:

$$A_1(1 + K^m) = 1 \implies A_1 = \frac{1}{1 + K^m}, \quad A_2 = \frac{K^m}{1 + K^m}.$$

For $m = 1$ one obtains the formulas of the statement; the delay grows with m , so the minimum is at $m = 1$. Note that $A_1, A_2 \in (0, 1)$ and $A_1 > A_2$ because $K \in (0, 1]$: the first impulse is always the larger one, and for $\zeta = 0$ (no damping) the two impulses are equal to $\frac{1}{2}$.

Step 5 — numerical check. For $f_n = 2$ Hz and $\zeta = 0.05$: $\omega_n = 2\pi f_n = 12.5663706$ rad/s, $\beta = 0.9987492$, $\omega_d = 12.5506528$ rad/s, $T_d = 0.5006262$ s, $\Delta = T_d/2 = 0.2503131$ s, $K = e^{-0.05\pi/0.9987492} = e^{-0.1572764} = 0.8544679$. Hence $A_1 = 1/1.8544679 = 0.5392382$ and $A_2 = 0.8544679/1.8544679 = 0.4607618$, summing to 1.0000000. ✓ *Independent check:* evaluating $|\hat{A}(s_0)| = |A_1 + A_2 e^{-s_0 T_d/2}|$ in double precision gives $3.1 \cdot 10^{-16}$, that is, zero up to rounding. ✓

Corollary 30.4: The ZV shaped step settles exactly at $T_d/2$. With the ZV shaper of Theorem 30.7, the response to the shaped step $u = A_1 \mathbf{1}_{[0,\infty)} + A_2 \mathbf{1}_{[T_d/2,\infty)}$ satisfies

$$y(t) = A_1 g(t) \text{ increasing on } \left[0, \frac{T_d}{2}\right], \quad y(t) = 1 \text{ for every } t \geq \frac{T_d}{2}.$$

The motion thus ends exactly at the instant $T_d/2$, with no overshoot and no oscillating tail.

Proof. Step 1 — before the second impulse. For $t < T_d/2$ only the first step acts, hence $y = A_1 g$, which is increasing by Lemma 30.4(iii), since $T_d/2 = \pi/\omega_d$.

Step 2 — after the second impulse. For $t \geq T_d/2$ Theorem 30.6(ii) gives $y(t) = \sum_i A_i - \frac{1}{\beta} |\hat{A}(s_0)| e^{-\zeta \omega_n t} \sin(\dots)$; since $\sum_i A_i = 1$ and $\hat{A}(s_0) = 0$ by Theorem 30.7, it remains that $y \equiv 1$.

Step 3 — continuity and absence of overshoot. At $t = T_d/2$ the value from the left is $A_1 g(T_d/2) = \frac{1+K}{1+K} = 1$ by Lemma 30.4(iii): it coincides with the value from the right. On the whole $[0, T_d/2]$ one then has $y = A_1 g \leq A_1 g(T_d/2) = 1$, hence $\max y = 1$: no overshoot.

Step 4 — elementary and independent check. Setting $E(t) = e^{-\zeta \omega_n t} (\cos \omega_d t + \frac{\zeta}{\beta} \sin \omega_d t)$, that is, $g = 1 - E$, and $\tau = T_d/2 = \pi/\omega_d$, one has for every $t \geq \tau$

$$E(t - \tau) = e^{-\zeta \omega_n (t - \tau)} \left(\cos(\omega_d t - \pi) + \frac{\zeta}{\beta} \sin(\omega_d t - \pi) \right) = -e^{\zeta \omega_n \tau} E(t) = -\frac{1}{K} E(t),$$

using $\cos(x - \pi) = -\cos x$, $\sin(x - \pi) = -\sin x$ and $e^{\zeta \omega_n \tau} = e^{\zeta \pi/\beta} = 1/K$. Therefore, for $t \geq \tau$,

$$y(t) = A_1(1 - E(t)) + A_2(1 - E(t - \tau)) = 1 - \left(A_1 - \frac{A_2}{K}\right) E(t) = 1 - \left(\frac{1}{1 + K} - \frac{1}{1 + K}\right) E(t) = 1. \quad \checkmark$$

Step 5 — numerical check. For $f_n = 2$ Hz, $\zeta = 0.05$: the peak-to-peak excursion of the response on the window $[2T_d, 3T_d]$ equals 0.9885584 without the shaper and $1.1 \cdot 10^{-15}$ with the ZV shaper (closed-form evaluation), that is, zero up to rounding; the absolute maximum of the response goes from $1.8544679 = 1 + K$ (an overshoot of 85.4%) to 1.0000000 (no

overshoot). ✓ *Independent check* by numerical integration of the differential equation with fourth-order Runge–Kutta, step $2 \cdot 10^{-6}$ s: peak-to-peak excursion 0.9885584 without the shaper and $3.6 \cdot 10^{-6}$ with the shaper, that is, 3.6 parts per million — the residual is the integration error, not the vibration. ✓ $\square$

Theorem 30.8: ZVD shaper: double zero at the pole. *Seek three impulses of amplitudes A_1, A_2, A_3 at instants $0, \tau = T_d/2, 2\tau = T_d$, with $A_1 + A_2 + A_3 = 1$, such that s_0 is a double zero of $\hat{A}$, that is, $\hat{A}(s_0) = \hat{A}'(s_0) = 0$. Then the solution exists, is unique, and equals*

$$A_1 = \frac{1}{(1+K)^2}, \quad A_2 = \frac{2K}{(1+K)^2}, \quad A_3 = \frac{K^2}{(1+K)^2},$$

that is, amplitudes proportional to $1, 2K, K^2$. In particular $|\hat{A}(s)| = O(|s - s_0|^2)$ as $s \rightarrow s_0$: the residual vibration vanishes to second order with respect to an error on the pole, and not only to first order.

Proof. Step 1 — reduction to a polynomial. Setting $z = e^{-s\tau}$, one has $\hat{A}(s) = A_1 + A_2 z + A_3 z^2 =: P(z)$, with $\frac{dz}{ds} = -\tau z$. Since $z_0 := e^{-s_0\tau} \neq 0$ and $\tau > 0$, the derivative $\frac{dz}{ds}$ does not vanish at s_0 ; hence, by the chain rule $\hat{A}'(s) = P'(z) \frac{dz}{ds}$, the two conditions $\hat{A}(s_0) = \hat{A}'(s_0) = 0$ are equivalent to $P(z_0) = P'(z_0) = 0$, that is, to saying that z_0 is a double root of P .

Step 2 — the value of z_0 . As in Step 4 of Theorem 30.7,

$$z_0 = e^{-s_0\tau} = e^{\zeta\omega_n\tau} e^{-i\omega_d\tau} = \frac{1}{K} e^{-i\pi} = -\frac{1}{K}.$$

Step 3 — the form of P . If $A_3 = 0$, then P has degree ≤ 1 and cannot have a double root unless it is the zero polynomial, which would contradict $P(1) = A_1 + A_2 + A_3 = 1$ (note $z = 1$ for $s = 0$). Hence $A_3 \neq 0$ and

$$P(z) = A_3 (z - z_0)^2 = A_3 \left(z + \frac{1}{K}\right)^2 = A_3 \left(z^2 + \frac{2}{K}z + \frac{1}{K^2}\right),$$

whence, comparing coefficients, $A_2 = \frac{2A_3}{K}$ and $A_1 = \frac{A_3}{K^2}$.

Step 4 — normalization. From $A_1 + A_2 + A_3 = 1$:

$$A_3 \left(\frac{1}{K^2} + \frac{2}{K} + 1\right) = A_3 \frac{1 + 2K + K^2}{K^2} = A_3 \frac{(1+K)^2}{K^2} = 1 \implies A_3 = \frac{K^2}{(1+K)^2},$$

and hence $A_2 = \frac{2K}{(1+K)^2}$ and $A_1 = \frac{1}{(1+K)^2}$: existence and uniqueness. All three are positive.

Step 5 — the order of the zero. From $P(z) = A_3(z - z_0)^2$ and the differentiability of $s \mapsto z(s)$ it follows that $\hat{A}(s) = A_3(z(s) - z(s_0))^2$ and $z(s) - z(s_0) = z'(s_0)(s - s_0) + O(|s - s_0|^2)$, hence $\hat{A}(s) = A_3 \tau^2 z_0^2 (s - s_0)^2 + O(|s - s_0|^3)$: the zero is exactly double, because $A_3 \tau^2 z_0^2 \neq 0$.

Step 6 — numerical check. For $f_n = 2$ Hz, $\zeta = 0.05$ ($K = 0.8544679$, $(1+K)^2 = 3.4390512$):

$$A_1 = 0.2907779, \quad A_2 = 0.4969207, \quad A_3 = 0.2123014,$$

summing to 1.0000000 ✓. *Independent check:* $|\hat{A}(s_0)| = 5.6 \cdot 10^{-17}$ and $|\hat{A}'(s_0)| = 8.2 \cdot 10^{-17}$ in double precision, that is, both zero ✓; and the peak-to-peak excursion of the response to the shaped step on $[2T_d, 3T_d]$ equals $8.9 \cdot 10^{-16}$ in closed form and $3.3 \cdot 10^{-6}$ with Runge–Kutta at step $2 \cdot 10^{-6}$ s. ✓ $\square$

30.5.1 Robustness to error in the frequency estimate

In practice ω_n is not known exactly: it is estimated from an impact test or an identifier, and it varies with the load and the configuration. One therefore needs to know how much vibration survives when the shaper is designed for ω_n but the system oscillates at $\omega \neq \omega_n$.

Definition 30.7: Normalized residual vibration and sensitivity curve. Given a shaper $(A_i, t_i)_{i=1}^N$ with $\sum_i A_i = 1$, designed for (ω_n, ζ) , and given a real system of natural frequency $\omega > 0$ and damping ζ , the **normalized residual vibration** is

$$V(\omega) = |\hat{A}(s(\omega))|, \quad s(\omega) = -\zeta\omega + i\omega\sqrt{1-\zeta^2},$$

and the **sensitivity curve** is the function $r \mapsto V(r\omega_n)$ of the frequency ratio $r = \omega/\omega_n$.

Proposition 30.9: Robustness of ZV and ZVD. The quantity $V(\omega)$ of Definition 30.7 is exactly the ratio between the amplitude of the residual vibration produced by the shaped command and that produced by the same command without the shaper. Moreover, in the undamped case $\zeta = 0$ and with $r = \omega/\omega_n$:

- (i) $V_{ZV}(r) = |\cos \frac{\pi r}{2}|$ and $V_{ZVD}(r) = \cos^2 \frac{\pi r}{2} = V_{ZV}(r)^2$;
- (ii) setting $\delta = r - 1$, one has $V_{ZV} = |\sin \frac{\pi \delta}{2}|$ and $V_{ZVD} = \sin^2 \frac{\pi \delta}{2}$; hence the band in which the residual vibration stays below 5% is

$$|\delta| \leq \frac{2}{\pi} \arcsin \frac{1}{20} = 0.0318443 \quad (\pm 3.18\%) \quad \text{for ZV,}$$

$$|\delta| \leq \frac{2}{\pi} \arcsin \frac{1}{\sqrt{20}} = 0.1435663 \quad (\pm 14.36\%) \quad \text{for ZVD,}$$

that is, ZVD is 4.508 times wider;

- (iii) for small errors $V_{ZV} \leq \frac{\pi}{2}|\delta|$ (linear dependence) and $V_{ZVD} \leq \frac{\pi^2}{4}\delta^2$ (quadratic dependence): doubling the precision of the estimate of ω_n halves the residual vibration with ZV and reduces it to a quarter with ZVD.

Proof. Step 1 — V is a ratio of amplitudes. Apply Theorem 30.6(i) to the real system, of frequency ω and pole $s(\omega)$: the residual vibration has amplitude $\frac{\omega^2}{\omega\sqrt{1-\zeta^2}} |\hat{A}(s(\omega))|$ multiplied by the envelope $e^{-\zeta\omega t}$. For the unshaped command (a single unit impulse at $t = 0$) one has $\hat{A} \equiv 1$ and the same amplitude with the factor 1 in place of $|\hat{A}(s(\omega))|$; the envelope and the prefactor are identical because they depend only on the system. The ratio is thus $|\hat{A}(s(\omega))| = V(\omega)$.

Step 2 — (i), the case $\zeta = 0$ for ZV. For $\zeta = 0$ one has $\beta = 1$, $K = e^0 = 1$, $\omega_d = \omega_n$, $\tau = T_d/2 = \pi/\omega_n$, and by Theorem 30.7 $A_1 = A_2 = \frac{1}{2}$; moreover $s(\omega) = i\omega$. Hence

$$\hat{A}(i\omega) = \frac{1}{2}(1 + e^{-i\omega\tau}).$$

For every $\theta \in \mathbb{R}$ one has $|1 + e^{-i\theta}| = |e^{-i\theta/2}| \cdot |e^{i\theta/2} + e^{-i\theta/2}| = 2|\cos \frac{\theta}{2}|$, hence

$$V_{ZV} = \frac{1}{2} \cdot 2 \left| \cos \frac{\omega\tau}{2} \right| = \left| \cos \frac{\pi\omega}{2\omega_n} \right| = \left| \cos \frac{\pi r}{2} \right|.$$

Step 3 — (i), ZVD. For $\zeta = 0$ Theorem 30.8 gives $A = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ at instants $0, \tau, 2\tau$, hence

$$\hat{A}(i\omega) = \frac{1}{4}(1 + 2e^{-i\omega\tau} + e^{-2i\omega\tau}) = \frac{1}{4}(1 + e^{-i\omega\tau})^2,$$

recognizing the square of a binomial. Taking the modulus and using Step 2, $V_{ZVD} = \frac{1}{4} \cdot 4 \cos^2 \frac{\omega\tau}{2} = \cos^2 \frac{\pi r}{2} = V_{ZV}^2$.

Step 4 — (ii), the shift. With $r = 1 + \delta$,

$$\cos \frac{\pi(1+\delta)}{2} = \cos \left(\frac{\pi}{2} + \frac{\pi\delta}{2} \right) = -\sin \frac{\pi\delta}{2},$$

whence $V_{ZV} = |\sin \frac{\pi\delta}{2}|$ and $V_{ZVD} = \sin^2 \frac{\pi\delta}{2}$. The condition $|\sin \frac{\pi\delta}{2}| \leq \frac{1}{20}$ is equivalent, for $|\frac{\pi\delta}{2}| \leq \frac{\pi}{2}$ (that is, $|\delta| \leq 1$, always satisfied in the band sought) and by the monotonicity of $\sin$ on $[0, \frac{\pi}{2}]$, to $\frac{\pi}{2}|\delta| \leq \arcsin \frac{1}{20}$, that is, $|\delta| \leq \frac{2}{\pi} \arcsin \frac{1}{20}$. Likewise $\sin^2 \frac{\pi\delta}{2} \leq \frac{1}{20}$ is equivalent to $|\sin \frac{\pi\delta}{2}| \leq \frac{1}{\sqrt{20}}$ and hence to $|\delta| \leq \frac{2}{\pi} \arcsin \frac{1}{\sqrt{20}}$.

Step 5 — the numerical values. $\arcsin 0.05 = 0.0500209$, hence $\frac{2}{\pi} \cdot 0.0500209 = 0.0318443$; $\sqrt{0.05} = 0.2236068$, $\arcsin 0.2236068 = 0.2255124$, hence $\frac{2}{\pi} \cdot 0.2255124 = 0.1435663$. The ratio of the two widths is $0.1435663/0.0318443 = 4.5084$. *Independent check:* evaluating the formulas of (i) at the points found gives $|\cos(\pi \cdot 1.0318443/2)| = 0.0500000$ and $\cos^2(\pi \cdot 1.1435663/2) = 0.0500000$. ✓

Step 6 — (iii). From $|\sin x| \leq |x|$ it follows that $V_{ZV} \leq \frac{\pi}{2}|\delta|$ and $V_{ZVD} = V_{ZV}^2 \leq \frac{\pi^2}{4}\delta^2$. The quadratic character is the quantitative translation of the double zero of Theorem 30.8: a zero of order k at the pole produces a residual vibration $O(|\delta|^k)$.

Step 7 — the damped case, numerically. With $\zeta = 0.05$ the closed formulas of (i) no longer hold exactly and the band is computed numerically from $V(\omega) = |\hat{A}(s(\omega))|$. The result, for $f_n = 2$ Hz:

	band at 5% (in r)	V at $r = 1.10$	V at $r = 1.20$
ZV	[0.97044; 1.02942]	0.17026	0.33899
ZVD	[0.86578; 1.13140]	0.02899	0.11491

The widths are 0.05898 and 0.26562, in ratio 4.503: damping slightly narrows both bands (from $\pm 3.18\%$ to about $\pm 2.95\%$ for ZV) but does not change the ratio between the two, which stays close to 4.5. One also reads the quadratic signature: at $r = 1.10$ ZV leaves 17% and ZVD 2.9%, that is, approximately the square ($0.17026^2 = 0.02899$). ✓ □

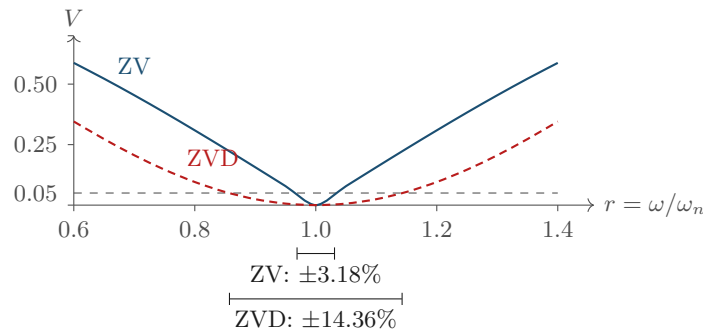


Figure 30.4: Sensitivity curves of ZV (solid) and ZVD (dashed) in the undamped case, $V_{ZV}(r) = |\cos \frac{\pi r}{2}|$ and $V_{ZVD}(r) = \cos^2 \frac{\pi r}{2}$ (Proposition 30.9). Ordinates of the drawing: 1 unit = 0.25 of residual vibration; the horizontal dashed line is the 5% threshold. Exact values used: $V_{ZV}(0.9) = 0.156434$, $V_{ZVD}(0.9) = 0.024472$; $V_{ZV}(0.8) = 0.309017$, $V_{ZVD}(0.8) = 0.095492$. The two bars below the axis measure the 5% bands: $\pm 3.18\%$ for ZV and $\pm 14.36\%$ for ZVD, in ratio 4.51.

Remark 30.6: Beyond ZVD: what remains imported. The construction of Theorem 30.8 iterates: imposing a zero of order k at the pole gives the shaper ZVD^{k-1} , with $k+1$ impulses of amplitudes proportional to the binomial coefficients $\binom{k}{j}K^j$ and total delay $kT_d/2$ — this is proved in exactly the same way as Theorem 30.8, replacing $(z - z_0)^2$ with

$(z - z_0)^k$ and normalizing; the residual vibration becomes V_{ZV}^k in the undamped case. The EI family (Extra-Insensitive) instead follows a different criterion: it does not cancel the vibration at ω_n , but keeps it below a threshold V_{tol} over an assigned frequency interval, which leads to a Chebyshev approximation problem. The explicit formulas for the EI shapers and the proof of their optimality remain imported here: what is missing is the formulation of the minimax problem and the equioscillation theorem that characterizes its solution. The source is Singhose, *Command shaping for flexible systems: a review of the first 50 years (2009)*, which collects and compares the ZV, ZVD, ZVD^k, EI families and their two-mode variants. Nothing proved in this chapter depends on the EI shapers.

30.5.2 Notch filters

Definition 30.8: Notch filter. A **notch filter** (band-elimination) of center frequency ω_n is

$$H_N(s) = \frac{s^2 + 2\zeta_z\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\omega_n s + \omega_n^2}, \quad 0 < \zeta_z < \zeta_p < 1.$$

Inserted in the velocity loop of the servo drive, it selectively attenuates a narrow band around the mechanical resonance, allowing high gains to be maintained.

Proposition 30.10: Depth and selectivity of the notch. For the filter of Definition 30.8:

- (i) $|H_N(0)| = 1$ and $\lim_{\omega \rightarrow +\infty} |H_N(i\omega)| = 1$: the filter alters neither DC nor high frequencies;
- (ii) $|H_N(i\omega)| < 1$ for every $\omega > 0$;
- (iii) the minimum of $|H_N(i\omega)|$ occurs exactly at $\omega = \omega_n$ and equals $\frac{\zeta_z}{\zeta_p}$, that is, an attenuation of $20 \log_{10} \frac{\zeta_p}{\zeta_z}$ decibels.

Proof. **Step 1 — the squared modulus.** Substituting $s = i\omega$,

$$H_N(i\omega) = \frac{(\omega_n^2 - \omega^2) + 2i\zeta_z\omega_n\omega}{(\omega_n^2 - \omega^2) + 2i\zeta_p\omega_n\omega}, \quad |H_N(i\omega)|^2 = \frac{X + 4\zeta_z^2\omega_n^2\omega^2}{X + 4\zeta_p^2\omega_n^2\omega^2}, \quad X = (\omega_n^2 - \omega^2)^2.$$

Step 2 — (i). For $\omega = 0$ numerator and denominator both equal ω_n^4 , hence $|H_N| = 1$. As $\omega \rightarrow +\infty$ the term $X \sim \omega^4$ dominates the terms in ω^2 both above and below, and the ratio tends to 1.

Step 3 — (ii). The difference between numerator and denominator is $4\omega_n^2\omega^2(\zeta_z^2 - \zeta_p^2) < 0$ for $\omega > 0$, because $\zeta_z < \zeta_p$: hence the ratio is < 1 , and since both are positive so is $|H_N(i\omega)| < 1$.

Step 4 — (iii), where the minimum lies. Rewrite

$$|H_N(i\omega)|^2 = 1 - \frac{4\omega_n^2\omega^2(\zeta_p^2 - \zeta_z^2)}{X + 4\zeta_p^2\omega_n^2\omega^2} = 1 - \frac{4\omega_n^2(\zeta_p^2 - \zeta_z^2)}{\frac{X}{\omega^2} + 4\zeta_p^2\omega_n^2}.$$

Minimizing $|H_N|$ is equivalent to minimizing X/ω^2 , and

$$\frac{X}{\omega^2} = \frac{(\omega_n^2 - \omega^2)^2}{\omega^2} = \left(\frac{\omega_n^2}{\omega} - \omega\right)^2 \geq 0,$$

with equality if and only if $\omega^2 = \omega_n^2$, that is, $\omega = \omega_n$.

Step 5 — the value of the minimum. At $\omega = \omega_n$ one has $X = 0$ and

$$|H_N(i\omega_n)|^2 = \frac{4\zeta_z^2\omega_n^4}{4\zeta_p^2\omega_n^4} = \frac{\zeta_z^2}{\zeta_p^2} \implies |H_N(i\omega_n)| = \frac{\zeta_z}{\zeta_p}.$$

The attenuation in decibels is $-20 \log_{10} \frac{\zeta_z}{\zeta_p} = 20 \log_{10} \frac{\zeta_p}{\zeta_z}$.

Step 6 — numerical check. With $\zeta_z = 0.02$ and $\zeta_p = 0.4$ one has $|H_N(i\omega_n)| = 0.05$, that is, -26.02 dB. *Independent check:* evaluating numerically $|H_N(i\omega)|$ for $\omega_n = 12.5663706$ gives $|H_N(i\omega_n)| = 0.0500000$, $|H_N(i \cdot 10^{-9})| = 1.0000000$, and $|H_N(100 i\omega_n)| = 0.9999681$. ✓ □

Anti-vibration strategies, in summary. (1) *S-curve profile / bounded jerk* (Theorem 30.5): acts upstream, softening the command and reducing the spectral content of the jerk; costs A/J seconds and does not require knowledge of ω_n . (2) *Input shaping ZV/ZVD* (Theorems 30.7 and 30.8): cancels the vibration *exactly* at a known frequency, in feedforward; costs $T_d/2$ (or T_d for ZVD) and requires an estimate of ω_n , with robustness quantified by Proposition 30.9. (3) *Notch filter* (Proposition 30.10): removes the resonance from inside the control loop, with *adaptive* versions that track ω_n as the load varies; costs no time but introduces phase lag, which erodes the phase margin (Definition 25.2). Commercial products combine all three: jerk limiting and filters on **Siemens SINAMICS/SIMOTION**, **Rockwell**, **B&R**, **Beckhoff**; vibration suppression *TrueMove/QuickMove* on **ABB**; *Vibration Control* on **FANUC** and **Yaskawa**. On PLCs the PLCopen functions `MC_MoveAbsolute`/`MC_MoveVelocity` accept velocity, acceleration, and jerk parameters; on robots (**ABB**, **KUKA**, **FANUC**, **Yaskawa**) the planner generates S-curve profiles for position and SLERP for orientation. Curvature continuity (Chapter 2), orientation continuity (Theorem 30.2), and jerk continuity are the three levels of “smoothness” of industrial motion.

Generating an S-curve profile and designing a ZV or ZVD input shaper in Python is straightforward:

Listing 30.1: S-curve profile and ZV/ZVD input shaper in Python

```
import numpy as np

def profilo_s(L, V, A, J):
    # Seven-segment case: V >= A^2/J and L >= V*(V/A + A/J)
    tj, ta = A/J, V/A - A/J
    tv = L/V - (V/A + A/J)
    assert ta >= 0 and tv >= 0, "use a degenerate case"
    durate = [tj, ta, tj, tv, tj, ta, tj]
    jerk = [J, 0., -J, 0., -J, 0., J]
    return durate, jerk, L/V + V/A + A/J      # -> ..., T

def shaper(wn, zeta, tipo="ZV"):
    # Amplitudes and instants of the shaper for frequency wn and damping
    # zeta
    beta = np.sqrt(1.0 - zeta**2)
    K = np.exp(-zeta*np.pi/beta)
    Td = 2*np.pi/(wn*beta)
    if tipo == "ZV":
        return np.array([1.0, K])/(1.0 + K), np.array([0.0, Td/2])
    if tipo == "ZVD":
        return np.array([1.0, 2*K, K*K])/(1.0 + K)**2, np.array([0.0, Td/2, Td])
    raise ValueError(tipo)

def vibrazione_residua(amp, tempi, w, zeta):
    # |A(s)| at the pole of the real system of frequency w: this is the
```

```

ratio
# between the residual vibration with the shaper and without it (
  Prop. robustness)
s = -zeta*wn + 1j*wn*np.sqrt(1.0 - zeta**2)
return abs(np.sum(amp*np.exp(-s*tempi)))

wn, zeta = 2*np.pi*2.0, 0.05
a, t = shaper(wn, zeta, "ZV")
print(a, t) # -> [0.5392 0.4608] [0.
0.2503]
print(vibrazione_residua(a, t, wn, zeta)) # -> 3.1e-16 (zero)
print(vibrazione_residua(a, t, 1.10*wn, zeta)) # -> 0.1703
b, u = shaper(wn, zeta, "ZVD")
print(vibrazione_residua(b, u, 1.10*wn, zeta)) # -> 0.0290

```

Exercise 30.5. An arm has a mode at $f_n = 2$ Hz with damping $\zeta = 0.05$. Compute the times and amplitudes of the ZV shaper, and the residual vibration if the true frequency is 2.2 Hz.

Solution. $\omega_n = 2\pi f_n = 12.5663706$ rad/s; $\beta = \sqrt{1 - 0.05^2} = 0.9987492$, hence $\omega_d = 12.5506528$ rad/s, $T_d = 2\pi/\omega_d = 0.5006262$ s, and the spacing is $T_d/2 = 0.2503131$ s. $K = e^{-0.05\pi/0.9987492} = 0.8544679$. By Theorem 30.7 the amplitudes are $A_1 = 1/(1 + K) = 0.5392382$ at $t = 0$ and $A_2 = K/(1 + K) = 0.4607618$ at $t = 0.2503131$ s, summing to 1. The command convolved with these two impulses does not excite the 2 Hz mode: by Corollary 30.4 the response to the shaped step equals exactly 1 from $t = 0.2503$ s onward. If the true frequency is 2.2 Hz, that is, $r = 1.10$, Proposition 30.9 gives, in the undamped case, the estimate $V = |\sin(\pi \cdot 0.10/2)| = \sin(9^\circ) = 0.1564$; the exact computation with $\zeta = 0.05$ gives $V = 0.1703$. About 17% of the vibration thus remains: outside the band, because the 5% band of ZV is about $\pm 3\%$. With ZVD it would drop to 0.0290, that is, 2.9%, at the cost of a delay of $T_d = 0.5006$ s instead of 0.2503 s. *Independent check:* $0.1703^2 = 0.0290$, as predicted by the relation $V_{ZVD} = V_{ZV}^2$ of Step 3 of Proposition 30.9 (exact for $\zeta = 0$ and an excellent approximation for $\zeta = 0.05$). ✓

30.6. Assigned exercises

The answers are in the Teacher's Key (Section 30.9).

Exercise 30.6: SLERP between two orientations at 60° . *Objective:* use the derived formula, verify its unit norm, and read off the rotation swept. *Estimated time:* 35 minutes. *Type:* symbolic/numerical. Let $q_0 = 1$ and $q_1 = \frac{1}{2}(1 + i + j + k)$.

- Verify that $q_1 \in S^3$ and compute Ω .
- Compute $\text{slerp}(q_0, q_1; t)$ for $t = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$ and verify in each case that the norm equals 1 (Corollary 30.2).
- Determine the axis u and the total rotation swept (Theorem 30.3), and verify that Φ_{q_1} cyclically permutes i, j, k .
- Repeat with $-q_1$ in place of q_1 : what is Ω now? And the rotation swept? Which of the two paths does the rule of Corollary 30.3 choose?

Exercise 30.7: Trapezoidal, S-curve, and minimum time. *Objective:* apply the two theorems on times and quantify the price of the jerk limit. *Estimated time:* 30 minutes.

Type: applied numerical. A linear axis must travel $L = 0.4$ m with $V = 0.8$ m/s, $A = 4$ m/s², and $J = 60$ m/s³.

- Verify that this is the trapezoidal case and compute t_a , t_v , and T^* (Theorem 30.4).
- Verify the two hypotheses of the seven-segment case and compute t_j , t_a , t_v , and T (Theorem 30.5).
- Compute the lower bound $T_{\inf}$ of Proposition 30.8 and the relative gap of the seven-segment profile.
- Reduce the travel to $L = 0.1$ m keeping V and A : is this still the trapezoidal case? Compute T^* and the peak velocity.

Exercise 30.8: Designing a shaper for a 5 Hz mode. *Objective:* design ZV and ZVD and quantify their robustness. *Estimated time:* 40 minutes. *Type:* proof-based/numerical. A gantry has an elastic mode at $f_n = 5$ Hz with $\zeta = 0.02$.

- Compute ω_n , β , ω_d , T_d , and K .
- Derive the amplitudes and instants of the ZV shaper and of the ZVD one.
- Compute the normalized residual vibration of both for frequency estimation errors of +5%, +10%, and +20%, and verify the relation $V_{ZVD} \approx V_{ZV}^2$.
- If the specification requires “residual vibration below 5% with an uncertainty of $\pm 10\%$ on the frequency,” which of the two shapers satisfies it? And how much more time does it cost?

30.7. Worked exercises

Exercise 30.9: How wrong is NLERP on a 120° rotation. *Type:* numerical/proof-based. Two orientations are $\Omega = 60^\circ$ apart on the sphere of quaternions, that is, 120° of rotation. By how much does NLERP depart from SLERP: in peak angular velocity, in maximum angular deviation, and what does the tool mounted on the wrist see?

Solution. Strategy. Use the closed formulas of Proposition 30.7 with $\Omega = \pi/3$ and translate the results into $SO(3)$ by doubling the angles (Theorem 30.3(iv)).

Step 1 — the angular velocity at the endpoints. By Proposition 30.7(iii), $\phi'(0) = \phi'(1) = \sin 60^\circ = \frac{\sqrt{3}}{2} = 0.8660254$ rad per unit of t .

Step 2 — the angular velocity at the midpoint. $\phi'(\frac{1}{2}) = 2 \tan 30^\circ = \frac{2}{\sqrt{3}} = 1.1547005$. The ratio is

$$\frac{\phi'(1/2)}{\phi'(0)} = \frac{2/\sqrt{3}}{\sqrt{3}/2} = \frac{4}{3} = 1.3333333 = \frac{1}{\cos^2 30^\circ} = \frac{1}{3/4}. \quad \checkmark$$

Step 3 — comparison with SLERP. SLERP has constant angular velocity $\Omega = \pi/3 = 1.0471976$ (Proposition 30.5). NLERP thus runs at $\frac{0.8660254}{1.0471976} = 82.7\%$ of the average velocity at the endpoints and at $\frac{1.1547005}{1.0471976} = 110.3\%$ at the midpoint: a velocity modulation from -17.3% to $+10.3\%$, requested by no one.

Step 4 — the angular deviation. The maximum of $|\phi(t) - t\Omega|$, computed over 10^4 equally spaced values of t , occurs at $t = 0.2225$ and equals 1.116986° out of 60° overall, that is, 1.86% of the angular travel.

Step 5 — what the tool sees. In $SO(3)$ all angles double: the total rotation is 120° , the maximum orientation deviation is $2 \cdot 1.116986 = 2.233973^\circ$, and the rotation speed oscillates between $2 \cdot 0.8660254 = 1.7320508$ and $2 \cdot 1.1547005 = 2.3094011$ rad per unit of t . A tool 200 mm from the wrist traces, for an orientation error of 2.234° , a position deviation of $200 \cdot \tan(2.233973^\circ) = 200 \cdot 0.0390 = 7.80$ mm: well beyond the typical tolerances of a

finishing operation.

Independent check. The values of ϕ' obtained by incremental ratio (step 10^{-6}) agree with the formula to the sixth digit: $\phi'(0) = 0.866026$, $\phi'(0.5) = 1.154701$. ✓ And the ratio $2 \tan \frac{\Omega}{2} / \Omega = 1.1026578$ agrees with $1.1547005 / 1.0471976 = 1.1026578$. ✓

Interpretation. NLERP does not “get the path wrong”: by Proposition 30.7(i) it traces exactly the same arc. It gets the *timing* wrong. This is why the error shows up in surface finish and vibrations, not in the geometry of the part: the trajectory is right, the time law is not.

Typical errors. (1) Confusing Ω (angle between quaternions) with the rotation angle, which is double: here 60° versus 120° . (2) Believing that, since $\phi(\frac{1}{2}) = \frac{\Omega}{2}$ (NLERP passes exactly through the midpoint of the arc, as verified from the symmetry of the formula), the error is zero: the error is maximal elsewhere, at $t \approx 0.22$ and $t \approx 0.78$. (3) Applying NLERP without normalizing, obtaining a quaternion of norm $\sqrt{D(t)} < 1$ and hence a matrix that is not a rotation.

Extension. For $\Omega = 90^\circ$ the ratio becomes $1/\cos^2 45^\circ = 2$ and the maximum deviation 4.07° (Proposition 30.7, Steps 7–8); as $\Omega \rightarrow 180^\circ$ the ratio diverges. This is why, when NLERP is used, the interval is subdivided into subintervals with small Ω : for $\Omega \leq 10^\circ$ the ratio is $1/\cos^2 5^\circ = 1.0077$, that is, a velocity deviation below 1%.

Exercise 30.10: When the trapezoid degenerates into a triangle. *Type:* conceptual with numerical computation. An axis has $V = 1 \text{ m/s}$ and $A = 2 \text{ m/s}^2$. For which travels is the maximum velocity actually reached? Compute the time for $L = 0.1 \text{ m}$ and compare it with what the trapezoidal-case formula would give, explaining why the latter is meaningless.

Solution. Strategy. Apply Theorem 30.4, which distinguishes the two cases by the single inequality $L \geq V^2/A$, and interpret the meaning of the threshold.

Step 1 — the threshold. The maximum velocity is reached if and only if $L \geq V^2/A = \frac{1}{2} = 0.5 \text{ m}$. The meaning of V^2/A is immediate: it is the distance covered accelerating from 0 to V ($\frac{V^2}{2A}$) plus that for decelerating from V to 0 (again $\frac{V^2}{2A}$). If the travel is shorter, there is no room to reach V .

Step 2 — the case $L = 0.1 \text{ m}$. Since $0.1 < 0.5$, this is the triangular case:

$$t_a = \sqrt{\frac{L}{A}} = \sqrt{0.05} = 0.2236068 \text{ s}, \quad T^* = 2t_a = 0.4472136 \text{ s},$$

$$v_p = \sqrt{AL} = \sqrt{0.2} = 0.4472136 \text{ m/s}.$$

The peak velocity is less than half of V : the velocity limit never comes into play.

Step 3 — what the wrong formula would give. $\frac{L}{V} + \frac{V}{A} = 0.1 + 0.5 = 0.6 \text{ s}$. This is a value *larger* than the true time 0.4472 s , which at first sight seems merely conservative; but it is not, because the corresponding motion does not exist: it would require $t_v = \frac{L}{V} - \frac{V}{A} = 0.1 - 0.5 = -0.4 \text{ s}$, that is, a constant-velocity phase of negative duration. Whoever implements it obtains a profile with negative distance in the central segment, or — in implementations that “clip” to zero — a profile that travels $2 \cdot \frac{V^2}{2A} = 0.5 \text{ m}$ instead of 0.1 : five times too much.

Step 4 — independent check. The distance of the triangular profile equals $2 \cdot \frac{1}{2} A t_a^2 = A t_a^2 = 2 \cdot 0.05 = 0.1 \text{ m} = L$ ✓; and the peak velocity $A t_a = 2 \cdot 0.2236068 = 0.4472136 \text{ m/s}$ agrees with $\sqrt{AL} = \sqrt{0.2}$ ✓. Moreover the formula of branch (b) of Step 2 of the proof of Theorem 30.4 gives $F(T^*) = \frac{A(T^*)^2}{4} = \frac{2 \cdot 0.2}{4} = 0.1 = L$ ✓.

Step 5 — continuity at the threshold. At $L = 0.5 \text{ m}$ the two formulas agree: $\frac{L}{V} + \frac{V}{A} = 0.5 + 0.5 = 1.0$ and $2\sqrt{L/A} = 2\sqrt{0.25} = 1.0$. ✓ This is the continuity of F proved in Step 2 of Theorem 30.4.

Interpretation. The threshold V^2/A separates two physically different regimes: above it, the motion is limited by *velocity* and the time grows linearly with L ($T = \frac{L}{V} + \text{const}$); below it, it is limited by *acceleration* and the time grows like $\sqrt{L}$. In a machine that makes many short moves, increasing V is useless: one must increase A .

Typical errors. (1) Applying the trapezoidal formula without checking $L \geq V^2/A$. (2) Believing that $v_p = V/2$ in the triangular case: $v_p = \sqrt{AL}$, which here equals 0.447 and not 0.5; they coincide only for $L = V^2/(4A)$. (3) Forgetting that, with the jerk limit, the threshold changes: the seven-segment case requires $L \geq V(\frac{V}{A} + \frac{A}{J})$, more restrictive than $L \geq \frac{V^2}{A}$.

Extension. With $J = 10 \text{ m/s}^3$, the threshold for the S-curve profile becomes $L \geq V(\frac{V}{A} + \frac{A}{J}) = 1 \cdot (0.5 + 0.2) = 0.7 \text{ m}$, versus 0.5 m for the trapezoid: between 0.5 and 0.7 m the trapezoid reaches V but the S-curve does not, and the formulas of the degenerate cases are needed.

Exercise 30.11: A crane at 0.5 Hz: ZV or ZVD? *Type:* design with numerical computation. The load hanging from a crane oscillates like a pendulum at $f_n = 0.5 \text{ Hz}$ with $\zeta = 0.02$. The cable length varies during the shift, and with it the frequency, by about $\pm 10\%$. Design the shaper and decide between ZV and ZVD.

Solution. Strategy. Compute the two shapers with Theorems 30.7 and 30.8, then evaluate the residual vibration at the extreme frequencies with Proposition 30.9 and compare the time cost.

Step 1 — the mode parameters. $\omega_n = 2\pi \cdot 0.5 = 3.1415927 \text{ rad/s}$; $\beta = \sqrt{1 - 0.02^2} = 0.9998000$; $\omega_d = 3.1409643 \text{ rad/s}$; $T_d = 2\pi/\omega_d = 2.0004001 \text{ s}$; $K = e^{-0.02\pi/0.9998} = e^{-0.0628444} = 0.9390896$.

Step 2 — the ZV shaper. Delay $T_d/2 = 1.0002001 \text{ s}$; amplitudes

$$A_1 = \frac{1}{1+K} = \frac{1}{1.9390896} = 0.5157059, \quad A_2 = \frac{K}{1+K} = 0.4842941,$$

summing to 1.0000000 ✓. With ζ so small the two impulses are nearly equal, as predicted by Step 4 of Theorem 30.7 ($K \rightarrow 1$ as $\zeta \rightarrow 0$).

Step 3 — the ZVD shaper. $(1+K)^2 = 3.7600684$, hence

$$A_1 = \frac{1}{3.7600684} = 0.2659526, \quad A_2 = \frac{2K}{3.7600684} = 0.4995066,$$

$$A_3 = \frac{K^2}{3.7600684} = 0.2345407,$$

at instants 0, 1.0002001, 2.0004001 s; the sum is 0.9999999 (rounding at the seventh digit). ✓

Step 4 — the residual vibration at the extreme frequencies. Evaluating $V = |\hat{A}(s(\omega))|$ (Definition 30.7):

$r = \omega/\omega_n$	0.90	1.00	1.10
ZV	0.160875	0.000000	0.161889
ZVD	0.025881	0.000000	0.026208

Independent check: the closed formula for $\zeta = 0$ of Proposition 30.9(ii) gives $V_{ZV} = |\sin(\pi \cdot 0.1/2)| = \sin 9^\circ = 0.156434$ and $V_{ZVD} = \sin^2 9^\circ = 0.024472$: the values with $\zeta = 0.02$ differ from these by 3.5% (ZV) and 7.1% (ZVD): damping shifts the sensitivity curve only slightly, and for ZVD the *relative* discrepancy exactly doubles because V_{ZVD} is the square of V_{ZV} . ✓ And $0.161889^2 = 0.026208$, exactly the tabulated ZVD value. ✓

Step 5 — the decision. With ZV, at $\pm 10\%$ error 16% of the vibration remains: on a load that swings by 200 mm without a shaper, 32 mm remain, which for a crane is unacceptable. With ZVD 2.6% remains, that is, 5 mm. The price is the delay: 1.00 s for ZV versus 2.00 s for ZVD. Since the frequency *certainly varies* by $\pm 10\%$ by construction of the problem, and since the 5% band of ZV is barely $\pm 3\%$ (Proposition 30.9(ii)), the choice is ZVD.

Interpretation. The resulting rule of thumb: choose ZV when the frequency is known better than 3% (for example a rigid structure, measured once and stable), ZVD when the uncertainty reaches 14%, and the EI families (Remark 30.6) beyond that. Each order of robustness costs half a period $T_d/2$ of delay: robustness is bought in seconds.

Typical errors. (1) Using $T_d = 1/f_n$ instead of $2\pi/\omega_d$: with $\zeta = 0.02$ the difference is only 0.02%, but with $\zeta = 0.3$ it becomes 4.8%, enough to push the shaper out of band. (2) Normalizing the amplitudes to something other than 1: if $\sum_i A_i \neq 1$ the command arrives scaled and the axis stops in the wrong place, as read from Theorem 30.6(ii), where the final value is $\sum_i A_i$. (3) Applying the shaper *after* the position loop instead of at the reference: the loop reacts to the discontinuity of the shaped command and the cancellation breaks down.

Extension. If the load has two modes (for example the cable pendulum and the arm's bending), the two shapers are convolved: the amplitudes multiply and the delays add, because the transform of a convolution product is the product of the transforms, and a zero at each of the two poles is preserved (Theorem 30.6). The total delay is the sum of the two, that is, $T_d^{(1)}/2 + T_d^{(2)}/2$ for two ZVs.

30.8. Self-assessment quiz

1. (Definition) For $q_0, q_1 \in S^3$ with $\Omega = \arccos\langle q_0, q_1 \rangle \in (0, \pi)$, SLERP equals

A) $(1-t)q_0 + tq_1$ B) $\frac{\sin((1-t)\Omega)q_0 + \sin(t\Omega)q_1}{\sin \Omega}$ C) $\frac{\cos((1-t)\Omega)q_0 + \cos(t\Omega)q_1}{\cos \Omega}$
 D) $\frac{(1-t)q_0 + tq_1}{\|(1-t)q_0 + tq_1\|}$

2. (Hypothesis) In Theorem 30.2 the hypothesis $\Omega \in (0, \pi)$ serves to

A) guarantee that q_0 and q_1 are unit B) exclude $q_1 = \pm q_0$, that is, guarantee $\sin \Omega \neq 0$: otherwise E is not defined and $\text{span}\{q_0, q_1\}$ is not a plane C) guarantee that the rotation swept is less than 180° D) guarantee that SLERP has norm 1

3. (Symbolic computation) For $\Omega \in (0, \pi)$, $\text{slerp}(q_0, q_1; \frac{1}{2})$ equals

A) $\frac{q_0 + q_1}{2}$ B) $\frac{q_0 + q_1}{\|q_0 + q_1\|}$ C) $\frac{q_0 + q_1}{2 \cos \Omega}$ D) none of the above

4. (Numerical computation) For a mode at $f_n = 2$ Hz with $\zeta = 0.05$, the delay of the second impulse of the ZV shaper equals, to four decimal places,

A) 0.2500 s B) 0.2503 s C) 0.5006 s D) 0.1250 s

5. (Proof step) In the derivation of the ZV shaper, the condition $\sin(\omega_d \Delta) = 0$ arises

A) from the vanishing of the imaginary part of $\hat{A}(s_0)$, combined with $A_2 > 0$ B) from the unit-gain constraint $A_1 + A_2 = 1$ C) from the definition of T_d D) from the continuity of the step response

6. (Error diagnosis) A student writes: “NLERP is perfectly fine: the result is normalized, so it stays on S^3 , and it passes through the same endpoints.” The error is

- A) NLERP does not stay on S^3 B) NLERP traces an arc different from that of SLERP
 C) NLERP traces the *same* arc, but with non-constant angular velocity — minimal at the endpoints and double at the midpoint for $\Omega = 90^\circ$ D) NLERP is not defined for $\Omega < \pi/2$
7. (Transfer) If at a certain frequency estimation error a ZV shaper leaves 16% residual vibration, at the same frequency a ZVD shaper leaves approximately
 A) 16% B) 8% C) 2.6% D) 0%
8. (Source and metacognition) The SLERP formula and the ZV shaper were published, respectively, in
 A) Shoemake, SIGGRAPH '85, Computer Graphics 19 (1985) / Singer and Seering, ASME J. Dyn. Sys. Meas. Control 112 (1990) B) Piegl and Tiller, *The NURBS Book* (1997) / Singhose (2009) C) Hamilton (1843) / Bode (1945) D) Farin (2002) / Biagiotti and Melchiorri (2008)

30.9. Teacher's key

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 1–4).

1) **B.** *Rationale:* this is the formula derived in Step 3 of Theorem 30.2, not postulated but deduced from the requirement of traversing the great circle at constant angular velocity. *Distractors:* A is LERP, which does not lie on S^3 ; C makes no dimensional sense and at $t = 0$ would give $q_0 + \frac{q_1}{\cos \Omega}$, not q_0 ; D is NLERP (Definition 30.4), which lies on S^3 and traces the same arc but not at constant velocity. *Diagnosed error:* confusing the formula with any weighted average. *Source:* Shoemake (1985), §3. *Prerequisite:* sine addition formulas. *Remediation:* verify that formula B gives q_0 at $t = 0$ and q_1 at $t = 1$, and that A and C do not.

2) **B.** *Rationale:* the entire construction rests on $E = (q_1 - \cos \Omega q_0) / \sin \Omega$ (Lemma 30.2), which requires $\sin \Omega \neq 0$; and if $q_1 = \pm q_0$ there is no plane $\text{span}\{q_0, q_1\}$ nor a unique great circle (for $q_1 = -q_0$ there are infinitely many, as in Step 11 of Theorem 7.7). *Distractors:* A is a separate hypothesis ($q_0, q_1 \in S^3$); C is *false*, because the rotation swept is 2Ω and reaches up to 360° ; D is a *consequence* (Corollary 30.2), not the purpose of the hypothesis. *Diagnosed error:* mistaking a consequence for the hypothesis that makes it possible. *Source:* Lemma 30.2, Step 1. *Prerequisite:* orthogonal decomposition in a Euclidean space. *Remediation:* try applying the formula with $q_1 = -q_0$ and observe where it breaks down.

3) **B.** *Rationale:* at $t = \frac{1}{2}$ the two coefficients are equal to $\frac{\sin(\Omega/2)}{\sin \Omega} = \frac{1}{2 \cos(\Omega/2)}$, hence $\text{slerp}(\frac{1}{2}) = \frac{q_0 + q_1}{2 \cos(\Omega/2)}$; and $\|q_0 + q_1\|^2 = 2 + 2 \cos \Omega = 4 \cos^2 \frac{\Omega}{2}$, that is, $\|q_0 + q_1\| = 2 \cos \frac{\Omega}{2}$: the two expressions coincide. *Distractors:* A is not unit (it has norm $\cos \frac{\Omega}{2} < 1$); C uses $\cos \Omega$ instead of $\cos \frac{\Omega}{2}$ and is already wrong for $\Omega = \pi/2$, where it would give division by zero; D is ruled out by B. *Diagnosed error:* forgetting the half-angle. *Source:* Theorem 30.2 and Corollary 30.2. *Prerequisite:* sine double-angle formula. *Remediation:* numerically verify for $\Omega = 90^\circ$ that both give $(q_0 + q_1) / \sqrt{2}$.

4) **B.** *Rationale:* $\omega_d = \omega_n \sqrt{1 - \zeta^2} = 12.5506528$, $T_d = 2\pi / \omega_d = 0.5006262$, delay $T_d/2 = 0.2503131 \approx 0.2503$ s (Theorem 30.7). *Distractors:* A is $\frac{1}{2f_n} = 0.2500$, that is, the *undamped* half-period, obtained by ignoring the factor $\beta = 0.9987$; C is $T_d = 0.5006$, the delay of ZVD; D is $\frac{1}{4f_n} = 0.1250$, a quarter period, which cancels nothing because it would leave the two impulses in quadrature instead of in opposition. *Diagnosed error:* using f_n in place of f_d . *Source:* Theorem 30.7, Step 5. *Prerequisite:* relation among ω_n , ζ , and ω_d . *Remediation:*

redo the computation with $\zeta = 0.3$ and observe that the error rises to 4.8%.

BEST PRACTICE

Quiz — answers, rationale, and diagnosis (items 5–8).

5) A. *Rationale:* this is Step 2 of the proof of Theorem 30.7: the imaginary part of $\hat{A}(s_0) = A_1 + A_2 e^{\zeta \omega_n \Delta} (\cos \omega_d \Delta - i \sin \omega_d \Delta)$ is $-A_2 e^{\zeta \omega_n \Delta} \sin(\omega_d \Delta)$, and it vanishes only if $\sin(\omega_d \Delta) = 0$, because $A_2 > 0$ and the exponential is positive. *Distractors:* B fixes the scale, not the delay (and by itself allows any Δ); C is circular, because T_d is defined earlier and independently; D is true but does not enter the computation. *Diagnosed error:* not separating the real and imaginary parts of a complex equation. *Source:* Singer and Seering (1990), §2. *Prerequisite:* a complex number is zero if and only if both its parts are. *Remediation:* solve the same system with $A_2 < 0$ and see that the parity of m changes.

6) C. *Rationale:* Proposition 30.7(i) proves that NLERP stays on S^3 and traces *exactly* the same minor arc with the same endpoints — the student is right about everything stated; the point is that not enough was said. Part (iii) shows that the angular velocity equals $\sin \Omega$ at the endpoints and $2 \tan \frac{\Omega}{2}$ at the midpoint, in ratio $1/\cos^2 \frac{\Omega}{2}$, which for $\Omega = 90^\circ$ equals 2. *Distractors:* A and B are false (they are disproved by Proposition 30.7(i)); D is false, NLERP is defined for every $q_1 \neq -q_0$. *Diagnosed error:* mistaking the correctness of the *image* of a curve for the correctness of its *parametrization*. *Source:* Proposition 30.7, Steps 5–8. *Prerequisite:* distinction between the trace and the parametrization of a curve. *Remediation:* draw Figure 30.2 for $\Omega = 60^\circ$ and measure the spacings.

7) C. *Rationale:* by Proposition 30.9(i), $V_{ZVD} = V_{ZV}^2$ in the undamped case (and to excellent approximation for small ζ): $0.16^2 = 0.0256 \approx 2.6\%$. *Distractors:* A ignores the robustness gain; B corresponds to believing that ZVD halves it (linear rather than quadratic dependence); D is true only exactly at $\omega = \omega_n$, not at an off-frequency. *Diagnosed error:* confusing “also cancels the derivative” with “always cancels.” *Source:* Proposition 30.9, Step 3. *Prerequisite:* multiple zeros and order of vanishing. *Remediation:* verify in the table that $0.161889^2 = 0.026208$ (Worked exercise 30.11).

8) A. *Rationale:* Shoemake introduced SLERP at SIGGRAPH in 1985; Singer and Seering published the ZV/ZVD shaper in the ASME Journal of Dynamic Systems, Measurement and Control in 1990. *Distractors:* B cites two real but wrong sources — Piegl and Tiller is the reference for NURBS (Theorem 30.1) and Singhose (2009) is the later *survey*, not the original article; C confuses the invention of quaternions with that of SLERP, and Bode plots with input shaping; D cites two reference texts on curves and trajectories, not the primary sources of the two results. *Diagnosed error:* attributing a result to the text where one studied it rather than to the original publication. *Source:* see the “Sources” section. *Prerequisite:* distinguishing an original article, a survey, and a textbook. *Remediation:* compile a “result / author / year / venue” table for the five main results of the chapter.

BEST PRACTICE

Assigned exercise 30.6 — answers. (a) $\|q_1\|^2 = 4 \cdot \frac{1}{4} = 1$, hence $q_1 \in S^3$; $\langle q_0, q_1 \rangle = \frac{1}{2}$ and $\Omega = \arccos \frac{1}{2} = 60^\circ = 1.0471976$ rad. (b) The coefficients are $\sin((1-t)60^\circ)/\sin 60^\circ$ and $\sin(t \cdot 60^\circ)/\sin 60^\circ$; the result is always $\cos(60^\circ t) + \sin(60^\circ t) u$ with $u = (i+j+k)/\sqrt{3}$, that is,

$$t = \frac{1}{3} : \cos 20^\circ + \sin 20^\circ u = 0.9396926 + 0.1974654(i+j+k),$$

$$t = \frac{1}{2} : \cos 30^\circ + \sin 30^\circ u = 0.8660254 + 0.2886751(i+j+k),$$

$$t = \frac{2}{3} : \cos 40^\circ + \sin 40^\circ u = 0.7660444 + 0.3711136(i+j+k).$$

Norms: $0.9396926^2 + 3 \cdot 0.1974654^2 = 0.8830222 + 0.1169778 = 1.0000000$; $0.75 + 3 \cdot$

$0.0833333 = 1; 0.5868241 + 3 \cdot 0.1377253 = 1.0000000$. ✓(c) $r = \bar{q}_0 q_1 = q_1$ has scalar part $\frac{1}{2} = \cos 60^\circ$ and pure part of norm $\frac{\sqrt{3}}{2} = \sin 60^\circ$, hence $u = (i + j + k)/\sqrt{3}$; the rotation swept is $2\Omega = 120^\circ$ (Theorem 30.3(iv)). By Theorem 11.10(iii) Φ_{q_1} is the rotation by 120° about the diagonal $(1, 1, 1)$, which cyclically permutes the axes: this is verified directly, $q_1 i \bar{q}_1 = j$, $q_1 j \bar{q}_1 = k$, $q_1 k \bar{q}_1 = i$. (d) $\langle q_0, -q_1 \rangle = -\frac{1}{2}$, hence $\Omega' = 120^\circ$ and the rotation swept is 240° ; $120^\circ + 240^\circ = 360^\circ$ (Corollary 30.3(ii)). ✓ Since $\langle q_0, q_1 \rangle = +\frac{1}{2} > 0$, the rule chooses q_1 and the 120° path.

BEST PRACTICE

Assigned exercise 30.7 — answers. (a) $V^2/A = 0.64/4 = 0.16 \leq 0.4 = L$: trapezoidal case. $t_a = V/A = 0.2$ s, $t_v = \frac{L}{V} - \frac{V}{A} = 0.5 - 0.2 = 0.3$ s, $T^* = 0.5 + 0.2 = 0.7$ s. (b) $A^2/J = 16/60 = 0.2666667 \leq 0.8 = V$ ✓; $V(\frac{V}{A} + \frac{A}{J}) = 0.8 \cdot (0.2 + 0.0666667) = 0.2133333 \leq 0.4 = L$ ✓. Hence $t_j = A/J = 0.0666667$ s, $t_a = 0.2 - 0.0666667 = 0.1333333$ s, $T_a = 0.2666667$ s, $t_v = 0.5 - 0.2666667 = 0.2333333$ s, and $T = 0.5 + 0.2 + 0.0666667 = 0.7666667$ s: 0.0666667 s more than the trapezoid, that is, exactly A/J . (c) $\frac{A^3}{12J^2V} = \frac{64}{12 \cdot 3600 \cdot 0.8} = \frac{64}{34560} = 0.0018519$ s, hence $T_{\text{inf}} = 0.7648148$ s and the relative gap of the seven-segment profile is $0.0018519/0.7666667 = 0.2415\%$: practically optimal. *Independent check*: integrating the seven segments exactly gives $s(T) = 0.400000000000$, $v(T) = -2.8 \cdot 10^{-17}$, $a(T) = 0$. ✓(d) With $L = 0.1$: $V^2/A = 0.16 > 0.1$, hence the *triangular* case; $T^* = 2\sqrt{L/A} = 2\sqrt{0.025} = 0.3162278$ s and $v_p = \sqrt{AL} = \sqrt{0.4} = 0.6324555$ m/s $< 0.8 = V$. ✓

BEST PRACTICE

Assigned exercise 30.8 — answers. (a) $\omega_n = 2\pi \cdot 5 = 31.4159265$ rad/s; $\beta = \sqrt{1 - 0.02^2} = 0.9998000$; $\omega_d = 31.4096427$ rad/s; $T_d = 2\pi/\omega_d = 0.2000400$ s; $K = e^{-0.02\pi/0.9998} = 0.9390896$. (b) ZV: $A_1 = 1/(1 + K) = 0.5157059$ at $t = 0$ and $A_2 = K/(1 + K) = 0.4842941$ at $t = T_d/2 = 0.1000200$ s. ZVD: $(1 + K)^2 = 3.7600684$ and $A = (0.2659526; 0.4995066; 0.2345407)$ at instants $0, 0.1000200, 0.2000400$ s. (c) Normalized residual vibration (Definition 30.7):

error	+5%	+10%	+20%
ZV	0.081067	0.161889	0.320799
ZVD	0.006572	0.026208	0.102912

and indeed $0.081067^2 = 0.006572$, $0.161889^2 = 0.026208$, $0.320799^2 = 0.102912$: the relation $V_{\text{ZVD}} = V_{\text{ZV}}^2$ of Proposition 30.9(i) is verified to the sixth digit even with $\zeta = 0.02$. ✓(d) Only ZVD: at +10% it leaves 2.6%, below the 5% threshold, while ZV leaves 16.2%. The additional cost is $T_d - T_d/2 = T_d/2 = 0.1000200$ s, that is, the delay goes from 0.1000 to 0.2000 s: a tenth of a second more for a reduction of the vibration by a factor of 6.

30.10. Chapter formulary

Bézier, de Casteljau, NURBS.

$$B(t) = \sum_{i=0}^n b_{i,n}(t) P_i, \quad b_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}; \quad B'(t) = n \sum_{i=0}^{n-1} b_{i,n-1}(t) (P_{i+1} - P_i).$$

De Casteljau: $P_i^r = (1-t)P_i^{r-1} + tP_{i+1}^{r-1}$, $B(t) = P_0^n(t)$. Circular arc ($0 < \theta < \pi$): points $1, 1 + i \tan \frac{\theta}{2}, e^{i\theta}$; weights $1, \cos \frac{\theta}{2}, 1$;

$$C(t) = \frac{w(t)}{w(t)}, \quad w(t) = (1-t) + t e^{i\theta/2}, \quad \psi(t) = 2 \arg w(t),$$

$$\psi'(t) = \frac{2 \sin \frac{\theta}{2}}{1 - 2t(1-t)(1 - \cos \frac{\theta}{2})}.$$

No arc of circle is polynomial: $x^2 + y^2 = 1$ in $\mathbb{R}[t]$ forces x, y constant.

SLERP. For $q_0, q_1 \in S^3$, $\Omega = \arccos \langle q_0, q_1 \rangle \in (0, \pi)$, $E = \frac{q_1 - \cos \Omega q_0}{\sin \Omega}$:

$$\text{slerp}(q_0, q_1; t) = \frac{\sin((1-t)\Omega)}{\sin \Omega} q_0 + \frac{\sin(t\Omega)}{\sin \Omega} q_1 = \cos(t\Omega) q_0 + \sin(t\Omega) E = q_0 (\bar{q}_0 q_1)^t.$$

$$\|\text{slerp}\| \equiv 1; \quad \langle \gamma(t), \gamma(s) \rangle = \cos((t-s)\Omega); \quad \|\gamma'\| \equiv \Omega; \quad \Phi_{\gamma(t)} = \Phi_{q_0} \circ R_{u, 2t\Omega}.$$

Rotation swept = 2Ω . Short path: if $\langle q_0, q_1 \rangle < 0$ use $-q_1$ (then $\Omega \rightarrow \pi - \Omega$ and $2\Omega \rightarrow 2\pi - 2\Omega$). Degenerate case: $\frac{\sin(a\Omega)}{\sin \Omega} = a + \frac{a(1-a^2)}{6}\Omega^2 + \rho$, $|\rho| \leq \frac{\Omega^4}{15}$ for $0 < \Omega \leq 1$.

NLERP. $\text{nlerp} = \frac{(1-t)q_0 + tq_1}{\|(1-t)q_0 + tq_1\|} = \cos \phi q_0 + \sin \phi E$, with

$$\phi(t) = \arg((1-t) + te^{i\Omega}), \quad \phi'(t) = \frac{\sin \Omega}{D(t)}, \quad D(t) = 1 - 2t(1-t)(1 - \cos \Omega).$$

$$\min \phi' = \phi'(0) = \sin \Omega, \quad \max \phi' = \phi'(\frac{1}{2}) = 2 \tan \frac{\Omega}{2}, \quad \frac{\max}{\min} = \frac{1}{\cos^2 \frac{\Omega}{2}} > 1.$$

Motion profiles. Trapezoidal/triangular (limits V, A ; travel L):

$$T^* = \frac{L}{V} + \frac{V}{A} \text{ if } L \geq \frac{V^2}{A}; \quad T^* = 2\sqrt{\frac{L}{A}}, \quad v_p = \sqrt{AL} \text{ if } L < \frac{V^2}{A}.$$

Seven-segment (in addition $|j| \leq J$; requires $V \geq \frac{A^2}{J}$ and $L \geq V(\frac{V}{A} + \frac{A}{J})$):

$$t_j = \frac{A}{J}, \quad t_a = \frac{V}{A} - \frac{A}{J}, \quad t_v = \frac{L}{V} - \frac{V}{A} - \frac{A}{J}, \quad T = \frac{L}{V} + \frac{V}{A} + \frac{A}{J};$$

lower bound: $T > \frac{L}{V} + \frac{V}{A} + \frac{A}{J} - \frac{A^3}{12J^2V}$. If $V < \frac{A^2}{J}$: $t_j = \sqrt{V/J}$, $T = \frac{L}{V} + 2\sqrt{V/J}$.

Elastic mode and input shaping. $\beta = \sqrt{1 - \zeta^2}$, $\omega_d = \omega_n \beta$, $T_d = \frac{2\pi}{\omega_d}$, $K = e^{-\zeta\pi/\beta}$, $s_0 = -\zeta\omega_n + i\omega_d$.

$$h(t) = \frac{\omega_n^2}{\omega_d} e^{-\zeta\omega_n t} \sin \omega_d t, \quad g(t) = 1 - \frac{1}{\beta} e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi), \quad \cos \varphi = \zeta.$$

Residual vibration $\propto |\hat{A}(s_0)|$ with $\hat{A}(s) = \sum_i A_i e^{-st_i}$; zero $\iff \hat{A}(s_0) = 0$.

$$\text{ZV: } A = \frac{(1, K)}{1 + K} \text{ at } \left(0, \frac{T_d}{2}\right); \quad \text{ZVD: } A = \frac{(1, 2K, K^2)}{(1 + K)^2} \text{ at } \left(0, \frac{T_d}{2}, T_d\right).$$

Sensitivity ($\zeta = 0$, $r = \omega/\omega_n$, $\delta = r - 1$): $V_{\text{ZV}} = |\cos \frac{\pi r}{2}| = |\sin \frac{\pi \delta}{2}|$, $V_{\text{ZVD}} = V_{\text{ZV}}^2$; 5% band: $\pm 3.18\%$ (ZV), $\pm 14.36\%$ (ZVD). Notch: $|H_N(i\omega_n)| = \frac{\zeta_z}{\zeta_p}$, minimum exactly at ω_n .

Sources and bibliography

- [1] K. Shoemake, *Animating rotation with quaternion curves*, in *Proceedings of the 12th annual conference on Computer graphics and interactive techniques* (SIGGRAPH '85), ACM SIGGRAPH Computer Graphics 19 (1985), n. 3, pp. 245–254, DOI 10.1145/325165.325242. Introduction of SLERP (the formula of Theorem 30.2), of the sign choice for the short path, and of *squad*; here the formula is *derived* rather than cited, while *squad* remains imported (Remark 30.4) — verified on 6 September 2026.
- [2] N. C. Singer, W. P. Seering, *Preshaping Command Inputs to Reduce System Vibration*, ASME Journal of Dynamic Systems, Measurement, and Control 112 (1990), n. 1, pp. 76–82, DOI 10.1115/1.2894142. Original article on input shaping: ZV and ZVD shapers, conditions for zero vibration and zero derivative, experimental results on the Space Shuttle robotic arm simulator. This is the source of Theorems 30.7 and 30.8, proved here in full — verified on 6 September 2026.
- [3] W. Singhose, *Command shaping for flexible systems: a review of the first 50 years*, International Journal of Precision Engineering and Manufacturing 10 (2009), n. 4, pp. 153–168, DOI 10.1007/s12541-009-0084-2. Survey of the shaper families (ZV, ZVD, ZVD^k, EI, multi-mode) and of their industrial applications; it is the source for the EI shapers, which remain *imported* (Remark 30.6) — verified on 6 September 2026.
- [4] L. Biagiotti, C. Melchiorri, *Trajectory Planning for Automatic Machines and Robots*, Springer, Berlin–Heidelberg, 2008, ISBN 978-3-540-85628-3 (eBook DOI 10.1007/978-3-540-85629-0). Systematic treatment of motion profiles: trapezoidal, seven-segment double-S, matching conditions and degenerate cases (Chapter 3), and multi-point trajectories. It is the reference for Theorems 30.4 and 30.5, proved here with the integral inequalities of their respective Steps 1–2 — verified on 6 September 2026.
- [5] L. Piegl, W. Tiller, *The NURBS Book*, 2nd edition, Springer, Berlin–Heidelberg, 1997. Chapter 1 (rational Bézier curves) and Chapter 7 (*Conics and Circles*, with the rational quadratic arc and the weights $1, \cos \frac{\theta}{2}, 1$): source of Theorem 30.1 and of the three facts about B-splines declared imported (Remark 30.1) — verified on 6 September 2026.
- [6] G. Farin, *Curves and Surfaces for CAGD: A Practical Guide*, 5th edition, Morgan Kaufmann, San Francisco, 2002, ISBN 978-1-55860-737-8. De Casteljau’s algorithm (Proposition 30.2), Bernstein polynomials and their derivatives, and an elementary treatment of rational conics — verified on 6 September 2026.
- [7] K. M. Lynch, F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Cambridge University Press, Cambridge, 2017, ISBN 978-1-107-15630-2. Chapter 9, *Trajectory Generation* (time laws, trapezoidal and S-curve profiles, via-point trajectories) and Chapter 3 for $SE(3)$ and screw motions, referenced in Remark 30.4 — verified on 6 September 2026.

Part F

Numerics and research

Estimating the circle from data: geometric and algebraic fitting

Up to now the circle has been a *given* object: assigned by a center and a radius, or by an equation, or by a geometric condition. In this chapter the circle is instead an object *to be estimated*. One has n points of the plane — readings of a probe on a ground shaft, echoes of a laser scanner on the rim of a tank, contour pixels extracted from an image — that are assumed to lie *approximately* on a circle, and one asks which circle describes them best. The question has no unique answer until one says what "best" means: the chapter takes the most natural definition from the geometric point of view, the sum of the squares of the distances of the points from the circle, and follows its consequences all the way through.

The consequences are not trivial. The resulting objective function is invariant under isometries and equivariant under similarities (Section 31.2), but it is *not* a polynomial in the parameters, because it contains n square roots: its normal equations are nonlinear (Section 31.4). Worse, the minimum may fail to exist: if the points are collinear the minimizing sequence escapes to infinity and the circle degenerates into a line (Theorem 31.3 and Corollary 31.4). Where it exists, it is reached by a Gauss–Newton iteration, whose Jacobian has a remarkably simple form — the rows are the radial unit vectors with sign reversed (Proposition 31.9) — and whose direction is one of descent as soon as the Jacobian has full rank (Theorem 31.5); when the rank is not full or the step is too long, Levenberg–Marquardt damping repairs both defects and interpolates between Gauss–Newton and the gradient (Theorem 31.7).

The delicate point is conditioning. On a short arc three very different parameter triples produce almost the same arc, and the problem becomes ill-conditioned in a *quantifiable* way: the condition number of the Jacobian grows like the inverse square of the arc's half-aperture, $\kappa_2(J) \sim \sqrt{180} \alpha^{-2}$ (Theorem 31.8 and Corollary 31.6), an estimate we verify numerically on generated data. This is the reason a roundness measurement on a 30° sector is a much harder operation than the same measurement on a full circle.

Kåsa's algebraic method and the Hough transform were already built in Volume I, Chapter 10, and are not repeated here: we use them as *providers of the starting point* for the iteration (Section 31.8). We do prove, however, the link between the algebraic minimum and the geometric one — the factorization $\rho^2 - r^2 = d(2r + d)$ of Lemma 31.6, the sandwich estimate of Proposition 31.13 and the exact inequality $r_{\text{alg}} \geq r_{\text{geo}}$ at fixed center of Proposition 31.14 — and we state explicitly, in an observation, which part of the bias analysis remains *imported* and from where (Observation 31.13).

INFO

Level and prerequisites. Level **C** (undergraduate), with two passages at level **D**: the proof of the behavior at infinity of the objective function (Theorem 31.2) and the asymptotic expansion of the condition number on a short arc (Corollary 31.6).

Internal prerequisites. Isometries of the plane, the structure $f(x) = Ax + b$ and the preservation of circles (Chapter 5, Definition 5.1, Theorem 5.1, Theorem 5.2); orthogonal matrices and their characterizations (Chapter 5, Proposition 5.1); parametrization of a circle and arc

length (Chapter 2). Kåsa's algebraic method, the Hough transform and the circle through three points are in Volume I, Chapter 10 and Chapter 2: they are recalled by textual cross-reference and never reproduced.

External prerequisites, assumed and not reproved:

- linear algebra: the Euclidean scalar product, symmetric matrices, the spectral theorem in finite dimension, the Rayleigh quotient, the singular value decomposition of a real $n \times k$ matrix;
- multivariable calculus: differential, gradient, Hessian matrix, Weierstrass's theorem on compact sets, the first-order necessary condition and the second-order sufficient condition for a local minimum;
- Taylor's formula with Lagrange remainder for a real function of one variable;
- linear least squares: if A has full column rank, the minimum of $\|A\beta - z\|_2$ is attained at the unique point solving $A^\top A \beta = A^\top z$ (used only to recall the method of Volume I).

Everything else is proved here.

What you will learn. To write the distance of a point from a circle and recognize when the foot of the projection is not unique; to formulate the geometric fit as a nonlinear least-squares problem; to exploit invariance and equivariance to normalize the data; to decide whether the minimum exists by looking at the smallest eigenvalue of the scatter matrix; to derive the normal equations and the Jacobian; to build a Gauss–Newton step and prove its descent property; to damp it à la Levenberg–Marquardt and read the two limits of the damping; to estimate the condition number on an arc and understand why short arcs are a hard problem; to use an algebraic fit as a starting point and to know how biased it is.

Warning. Conventions of the chapter.

- The data are n points $p_i = (x_i, y_i) \in \mathbb{R}^2$, $i = 1, \dots, n$, not necessarily distinct. $\bar{p} = \frac{1}{n} \sum_i p_i$ is their **centroid** and $w_i = p_i - \bar{p}$ the deviation from the centroid.
- The circle's parameters are $\theta = (a, b, r)$ with $c = (a, b)$ the center and $r \geq 0$ the radius; $\mathcal{C}(c, r) = \{q \in \mathbb{R}^2 : \|q - c\| = r\}$ is the corresponding circle. We write $\rho_i(c) = \|p_i - c\|$ and $d_i(\theta) = \rho_i(c) - r$.
- $\|\cdot\|$ is always the Euclidean norm, $\langle \cdot, \cdot \rangle$ the Euclidean scalar product; for a matrix, $\|\cdot\|_2$ is the spectral norm.
- I is the identity matrix (of order 2 or 3 according to context); $A^\top$ is the transpose. Vectors of $\mathbb{R}^2$ and $\mathbb{R}^3$ are columns.
- "Geometric fit" always means minimization of the sum of squares of the *distances*; "algebraic fit" means minimization of the sum of squares of the residuals $\rho_i^2 - r^2$, i.e. the problem of Volume I, Chapter 10.
- Numerical values are rounded to the seventh decimal place and recomputed twice by independent routes; the sign $\checkmark$ marks the second computation.

31.1. The problem: distance from a circle and the objective function

Before minimizing a sum of distances one needs a formula for the distance of a point from a circle. It is an elementary fact, but it must be proved with care because the degenerate case — the point coinciding with the center — is exactly the one that undermines every derivative formula for the rest of the chapter.

Definition 31.1: Sample, candidate circle, parameters. A **sample** is a finite list $P = (p_1, \dots, p_n)$ of points of $\mathbb{R}^2$, with $n \geq 1$. A **candidate circle** is a circle $\mathcal{C}(c, r) = \{q \in \mathbb{R}^2 : \|q - c\| = r\}$ with $c = (a, b) \in \mathbb{R}^2$ and $r \geq 0$; the triple $\theta = (a, b, r) \in \mathbb{R}^2 \times [0, +\infty)$ is the **parameter vector**. For $r = 0$ the circle reduces to the single point c and is called **degenerate**.

Lemma 31.1: Distance of a point from a circle. Let $c \in \mathbb{R}^2$, $r \geq 0$ and $p \in \mathbb{R}^2$. Setting $\rho = \|p - c\|$, one has

$$\text{dist}(p, \mathcal{C}(c, r)) := \min_{q \in \mathcal{C}(c, r)} \|p - q\| = |\rho - r|.$$

Moreover:

- (i) if $p \neq c$, the minimum is attained at the unique point $q^* = c + r \frac{p - c}{\rho}$, which is the intersection of the ray from c through p with $\mathcal{C}(c, r)$;
- (ii) if $p = c$ and $r > 0$, the distance equals r and the minimum is attained at every point of $\mathcal{C}(c, r)$;
- (iii) if $r = 0$ the distance equals ρ and the minimizing point is $q^* = c$.

Proof. Step 1 — the minimum exists. $\mathcal{C}(c, r)$ is closed and bounded in $\mathbb{R}^2$, hence compact, and $q \mapsto \|p - q\|$ is continuous: by Weierstrass's theorem the minimum is attained.

Step 2 — the lower bound. Let $q \in \mathcal{C}(c, r)$, i.e. $\|q - c\| = r$. By the triangle inequality applied to $p - c = (p - q) + (q - c)$ we get $\rho = \|p - c\| \leq \|p - q\| + r$, i.e. $\|p - q\| \geq \rho - r$; and applying it to $q - c = (q - p) + (p - c)$ we get $r \leq \|p - q\| + \rho$, i.e. $\|p - q\| \geq r - \rho$. The two inequalities together give

$$\|p - q\| \geq \max\{\rho - r, r - \rho\} = |\rho - r| \quad \text{for every } q \in \mathcal{C}(c, r). \quad (31.1)$$

Step 3 — the upper bound when $p \neq c$. Let $u = (p - c)/\rho$, which is well defined because $\rho > 0$, and is a unit vector. The point $q^* = c + ru$ satisfies $\|q^* - c\| = r\|u\| = r$, hence $q^* \in \mathcal{C}(c, r)$, and

$$p - q^* = (p - c) - ru = \rho u - ru = (\rho - r)u, \quad \text{so} \quad \|p - q^*\| = |\rho - r|.$$

With (31.1) this proves that the minimum equals $|\rho - r|$ and is attained at q^* .

Step 4 — uniqueness when $p \neq c$. Suppose $\|p - q\| = |\rho - r|$ with $\|q - c\| = r$. If $\rho > r$, the equality $\rho = \|p - q\| + \|q - c\|$ is the equality case in the triangle inequality for $p - c = (p - q) + (q - c)$: the two vectors $p - q$ and $q - c$ are then parallel and same-direction, i.e. $q - c = \lambda(p - q)$ with $\lambda \geq 0$ (or one of the two is zero, but $\|q - c\| = r$ and $\|p - q\| = \rho - r$ are both positive if $0 < r < \rho$; if $r = 0$ this falls back into case (iii)). From $q - c = \lambda(p - q)$ it follows that q lies on the line through c and p , on the side of p ; the only such point at distance r from c is $c + ru = q^*$. If instead $\rho < r$, the equality $r = \|q - p\| + \|p - c\|$ says that p lies on the segment between q and c , so again q is on the ray from c through p and $q = q^*$. If $\rho = r$ then $|\rho - r| = 0$ and $q = p = q^*$.

Step 5 — the case $p = c$, $r > 0$. Here $\rho = 0$ and $|\rho - r| = r$; for every $q \in \mathcal{C}(c, r)$ one has $\|p - q\| = \|c - q\| = r$. Hence the function $q \mapsto \|p - q\|$ is constant on $\mathcal{C}(c, r)$ and every point is a minimizer: the foot of the projection is not unique.

Step 6 — the case $r = 0$. $\mathcal{C}(c, 0) = \{c\}$, hence the minimum is $\|p - c\| = \rho = |\rho - 0|$, attained at $q^* = c$. $\square$

Remark 31.1: The center is the only singularity. Lemma 31.1 shows that the function $p \mapsto \text{dist}(p, \mathcal{C}(c, r))$ is smooth everywhere except on the circle itself (where the absolute value has a corner) and at the center (where the foot of the projection jumps). In the rest of the chapter we will instead minimize the signed quantity $\rho - r$, which removes the corner on the circle; the singularity at the center remains, i.e. at the points θ where c coincides with one of the data points. All the derivative formulas that follow require $\rho_i(c) > 0$ for every i , a hypothesis that will always be written explicitly.

Definition 31.2: Geometric residual and objective function. Given a sample P and a parameter vector $\theta = (a, b, r)$ with $c = (a, b)$, set

$$\rho_i(c) = \|p_i - c\| = \sqrt{(x_i - a)^2 + (y_i - b)^2}, \quad d_i(\theta) = \rho_i(c) - r,$$

and call the **objective function of the geometric fit** the function

$$F(\theta) = F(a, b, r) = \sum_{i=1}^n d_i(\theta)^2 = \sum_{i=1}^n \left(\sqrt{(x_i - a)^2 + (y_i - b)^2} - r \right)^2.$$

The number $d_i(\theta)$ is the **signed geometric residual** of the point p_i : positive if p_i is outside $\mathcal{C}(c, r)$, negative if inside, zero if it lies on the circle. A **geometric least-squares circle** of the sample is a θ^* that minimizes F on $\mathbb{R}^2 \times [0, +\infty)$.

Proposition 31.1: The objective function is the sum of squared distances. For every $\theta = (a, b, r)$ with $r \geq 0$ one has

$$F(\theta) = \sum_{i=1}^n \text{dist}(p_i, \mathcal{C}(c, r))^2.$$

Proof. By Lemma 31.1, $\text{dist}(p_i, \mathcal{C}(c, r)) = |\rho_i(c) - r| = |d_i(\theta)|$ for every i (the statement of the lemma covers all cases, including $p_i = c$ and $r = 0$). Squaring, $\text{dist}(p_i, \mathcal{C}(c, r))^2 = d_i(\theta)^2$, and summing over i gives the claim. $\square$

Remark 31.2: Why the absolute value disappears. Proposition 31.1 is the technical reason why one minimizes the sum of squares and not, for instance, the sum of distances: squaring makes the absolute value disappear and the function d_i becomes differentiable wherever $\rho_i > 0$, while $|d_i|$ is not differentiable on the circle. The price is the well-known sensitivity of least squares to outliers, which in Volume I, Chapter 10 is exactly the reason why the Hough transform is credited with a robustness that least squares does not have.

Proposition 31.2: Optimal radius at a fixed center. Let $c \in \mathbb{R}^2$ be fixed and set $\bar{\rho}(c) = \frac{1}{n} \sum_{i=1}^n \rho_i(c)$. Then the function $r \mapsto F(a, b, r)$ is a second-degree polynomial with leading coefficient $n > 0$, has a unique minimizer on $\mathbb{R}$ at $r = \bar{\rho}(c)$, and

$$\min_{r \in \mathbb{R}} F(a, b, r) = G(c) := \sum_{i=1}^n (\rho_i(c) - \bar{\rho}(c))^2.$$

Since $\bar{\rho}(c) \geq 0$, the minimum over $r \geq 0$ coincides with the minimum over $\mathbb{R}$; in particular

$$\inf_{\theta \in \mathbb{R}^2 \times [0, +\infty)} F(\theta) = \inf_{c \in \mathbb{R}^2} G(c).$$

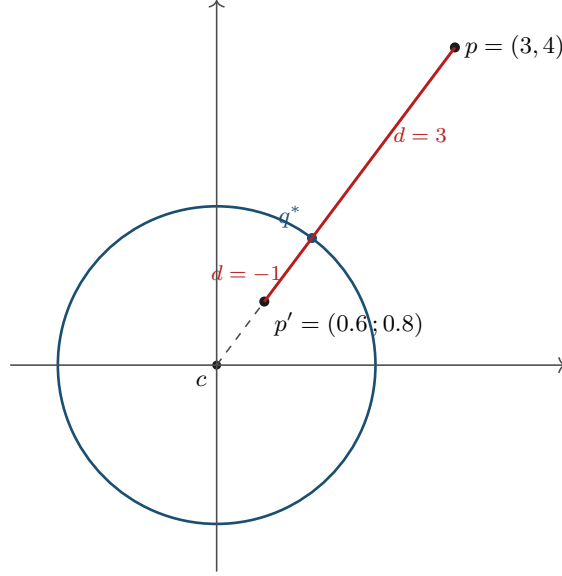


Figure 31.1: Lemma 31.1 on an example with exact coordinates: circle of center $c = (0,0)$ and radius $r = 2$. For $p = (3,4)$ we have $\rho = \sqrt{9+16} = 5$, residual $d = 5 - 2 = 3$ and foot $q^* = c + 2(3,4)/5 = (1.2; 1.6)$. For $p' = (0.6; 0.8)$ we have $\rho' = \sqrt{0.36+0.64} = 1$, residual $d' = 1 - 2 = -1$ and *the same* foot $q^* = (1.2; 1.6)$, because p and p' lie on the same ray from c . Check: $\|q^*\| = \sqrt{1.44 + 2.56} = \sqrt{4} = 2 = r$. ✓

Proof. Step 1 — quadratic form in r . Expanding the square,

$$F(a, b, r) = \sum_{i=1}^n (\rho_i(c) - r)^2 = nr^2 - 2r \sum_{i=1}^n \rho_i(c) + \sum_{i=1}^n \rho_i(c)^2,$$

which is a degree-2 polynomial in r with leading coefficient $n > 0$: hence an upward-opening parabola, which has a unique minimum at its vertex.

Step 2 — the vertex. The derivative with respect to r is $\partial_r F = 2nr - 2 \sum_i \rho_i(c)$, which vanishes if and only if $r = \frac{1}{n} \sum_i \rho_i(c) = \bar{\rho}(c)$.

Step 3 — the minimum value. Substituting $r = \bar{\rho}(c)$ in the definition,

$$F(a, b, \bar{\rho}(c)) = \sum_{i=1}^n (\rho_i(c) - \bar{\rho}(c))^2 = G(c).$$

Step 4 — the constraint $r \geq 0$ does not bind. Every $\rho_i(c) = \|p_i - c\|$ is ≥ 0 , hence $\bar{\rho}(c) \geq 0$: the unconstrained minimizer already lies in the admissible half-plane and the two minima coincide.

Step 5 — the infimum. For every c one has $\inf_r F(a, b, r) = G(c)$ by the previous steps; taking the infimum over c of both sides, and using that the infimum of a function of two groups of variables can be computed in two stages, one obtains $\inf_\theta F = \inf_c \inf_r F(a, b, r) = \inf_c G(c)$.
□

***Remark 31.3: The problem really has only two unknowns.** Proposition 31.2 eliminates exactly one parameter: the geometric fit is the minimization of the function $G : \mathbb{R}^2 \rightarrow [0, +\infty)$, which measures the dispersion of the distances from the candidate center. This reduction will be the main tool of Section 31.3, because G is defined on $\mathbb{R}^2$ rather than on a half-space of $\mathbb{R}^3$, and because its interpretation — "how far the points are from being equidistant from c " — makes the behavior at infinity readable. In the numerical iteration,*

however, it is convenient to keep all three parameters: the Jacobian with respect to (a, b, r) has a much simpler form than that of G .

Example 31.1: Three non-collinear points: the fit is exact. Let P be the sample $p_1 = (0, 0)$, $p_2 = (4, 0)$, $p_3 = (0, 3)$. The three points are not collinear (the vectors $p_2 - p_1 = (4, 0)$ and $p_3 - p_1 = (0, 3)$ have determinant $4 \cdot 3 - 0 \cdot 0 = 12 \neq 0$), so by Volume I, Chapter 2 the circumscribed circle of the triangle exists. Its center is the intersection of the perpendicular bisectors: the bisector of $p_1 p_2$ is $x = 2$, the bisector of $p_1 p_3$ is $y = \frac{3}{2}$, so $c^* = (2; 1.5)$ and

$$r^* = \|p_1 - c^*\| = \sqrt{4 + 2.25} = \sqrt{6.25} = 2.5.$$

Check on the other two points: $\|p_2 - c^*\| = \sqrt{(4 - 2)^2 + 1.5^2} = \sqrt{6.25} = 2.5$ and $\|p_3 - c^*\| = \sqrt{2^2 + (3 - 1.5)^2} = \sqrt{6.25} = 2.5$. ✓ Hence $d_1 = d_2 = d_3 = 0$ and $F(\theta^*) = 0$: since $F \geq 0$ by definition, θ^* is a *global* minimum. *Independent check:* the algebraic method of Volume I, Chapter 10, applied to these three points, returns $a = 2.0000000$, $b = 1.5000000$, $r = 2.5000000$ and zero algebraic residual — as it must, because on exactly concyclic points both functionals vanish (Lemma 31.6). ✓

31.2. Invariance under isometries and equivariance under similarities

A criterion of fit between data and circle that depended on the reference frame would be useless: rotating the workpiece on the measuring machine would change the "best" circle. The objective function of Definition 31.2 does not have this defect, and the proof is a direct application of the structure of isometries from Chapter 5.

Definition 31.3: Action of a similarity on the parameters. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be of the form $T(x) = \lambda Ax + v$ with $\lambda > 0$, $A \in O(2)$ and $v \in \mathbb{R}^2$ (a **similarity** of ratio λ ; for $\lambda = 1$ it is an isometry, by Theorem 5.1). For a parameter vector $\theta = (c, r)$ set

$$T_*\theta := (T(c), \lambda r).$$

For a sample $P = (p_1, \dots, p_n)$ set $T(P) = (T(p_1), \dots, T(p_n))$, and write F_P for the objective function of the sample P .

Proposition 31.3: Similarities carry circles to circles. With the notation of Definition 31.3, $T(\mathcal{C}(c, r)) = \mathcal{C}(T(c), \lambda r)$ for every $c \in \mathbb{R}^2$ and $r \geq 0$.

Proof. For every $q \in \mathbb{R}^2$,

$$\|T(q) - T(c)\| = \|\lambda Aq + v - \lambda Ac - v\| = \lambda \|A(q - c)\| = \lambda \|q - c\|,$$

where the last equality is the implication (i) $\Rightarrow$ (ii) of Proposition 5.1. Hence $\|q - c\| = r$ if and only if $\|T(q) - T(c)\| = \lambda r$. Since T is bijective (it is the composition of the linear bijection λA , invertible because $\lambda > 0$ and $\det A = \pm 1 \neq 0$ by Corollary 5.1, with a translation), the image set of $\mathcal{C}(c, r)$ is exactly $\{q' : \|q' - T(c)\| = \lambda r\}$. For $\lambda = 1$ this is the special case already proved in Theorem 5.2. $\square$

Theorem 31.1: Invariance under isometries and equivariance under similarities.

Let $T(x) = \lambda Ax + v$ be a similarity of ratio $\lambda > 0$. Then for every sample P and every $\theta = (c, r)$ with $r \geq 0$

$$F_{T(P)}(T_*\theta) = \lambda^2 F_P(\theta).$$

In particular, if T is an isometry ($\lambda = 1$) then $F_{T(P)}(T_*\theta) = F_P(\theta)$: the objective function is **invariant** under isometries and **equivariant of weight 2** under similarities.

Proof. Step 1 — the radial distances rescale. As in the proof of Proposition 31.3, $\|T(p_i) - T(c)\| = \lambda\|p_i - c\|$, i.e. $\rho_i^T(T(c)) = \lambda\rho_i(c)$, where ρ^T denotes the distances computed in the transformed sample.

Step 2 — the residuals rescale. The second component of $T_*\theta$ is λr , so the i -th residual of the transformed sample equals

$$d_i^T(T_*\theta) = \rho_i^T(T(c)) - \lambda r = \lambda\rho_i(c) - \lambda r = \lambda d_i(\theta).$$

Step 3 — sum of squares. Squaring and summing,

$$F_{T(P)}(T_*\theta) = \sum_{i=1}^n \left(\lambda d_i(\theta) \right)^2 = \lambda^2 \sum_{i=1}^n d_i(\theta)^2 = \lambda^2 F_P(\theta).$$

For $\lambda = 1$ the factor disappears. □

Corollary 31.1: The minimizers transport. With the notation of Theorem 31.1: θ^* is a minimum point (global, or local) of F_P if and only if $T_*\theta^*$ is a minimum point (global, respectively local) of $F_{T(P)}$; and the minimum values are in ratio λ^2 . In particular F_P admits a minimum if and only if $F_{T(P)}$ does.

Proof. Step 1 — T_* is a bijection of the parameter space. The map $\theta = (c, r) \mapsto (T(c), \lambda r)$ is a bijection of $\mathbb{R}^2 \times [0, +\infty)$ onto itself, with inverse $\theta' = (c', r') \mapsto (T^{-1}(c'), \lambda^{-1}r')$: indeed T is bijective (see the final step of the proof of Proposition 31.3) and $\lambda > 0$. It is moreover a homeomorphism, because T and T^{-1} are affine and hence continuous, and multiplication by $\lambda > 0$ is as well.

Step 2 — comparison of values. By Theorem 31.1, $F_{T(P)}(T_*\theta) = \lambda^2 F_P(\theta)$ for every θ . If θ^* is a global minimizer for F_P , then for every θ' in the parameter space there is $\theta = T_*^{-1}\theta'$ and $F_{T(P)}(\theta') = \lambda^2 F_P(\theta) \geq \lambda^2 F_P(\theta^*) = F_{T(P)}(T_*\theta^*)$: hence $T_*\theta^*$ is a global minimizer. The converse follows by applying the same argument to T^{-1} , which is again a similarity, of ratio λ^{-1} .

Step 3 — local minimizers. If θ^* is a local minimizer, there is a neighborhood U of θ^* on which $F_P \geq F_P(\theta^*)$; since T_* is a homeomorphism (Step 1), $T_*(U)$ is a neighborhood of $T_*\theta^*$, and on it $F_{T(P)} = \lambda^2 F_P \circ T_*^{-1} \geq \lambda^2 F_P(\theta^*)$. □

Corollary 31.2: Normalization of the sample. Let P be a sample with at least two distinct points, $\bar{p}$ its centroid and $M = \max_i \|p_i - \bar{p}\| > 0$. Setting $T(x) = M^{-1}(x - \bar{p})$, the sample $T(P)$ has centroid at the origin and $\max_i \|T(p_i)\| = 1$, and the fitting problems for P and for $T(P)$ are equivalent in the sense of Corollary 31.1.

Proof. T is the similarity of ratio $\lambda = M^{-1} > 0$ with $A = I \in O(2)$ and $v = -M^{-1}\bar{p}$. The centroid of $T(P)$ is $\frac{1}{n} \sum_i M^{-1}(p_i - \bar{p}) = M^{-1}(\frac{1}{n} \sum_i p_i - \bar{p}) = M^{-1} \cdot 0 = 0$, by definition of $\bar{p}$. Moreover $\max_i \|T(p_i) - 0\| = \max_i M^{-1}\|p_i - \bar{p}\| = M^{-1}M = 1$. The equivalence of the two problems is Corollary 31.1. □

Remark 31.4: The algebraic functional is also invariant, but with a different weight. The algebraic residual $\rho_i^2 - r^2$ of Volume I, Chapter 10 transforms, under the similarity T , into $\lambda^2(\rho_i^2 - r^2)$: the sum of its squares is therefore equivariant of weight 4, not weight 2. For isometries ($\lambda = 1$) the two functionals are both invariant, and this is what makes it legitimate to compare them; but the different weight already signals that the algebraic functional is not homogeneous of the same power of lengths, and that its minimum cannot coincide with the geometric one except in cases where both vanish. The difference is quantified in Section 31.8.

31.3. Existence of the minimum, and the cases where it fails

The minimum of F may fail to exist. The reason is geometric and can be seen by eye: if the center is moved off to infinity along a direction u while the radius is kept equal to the mean distance, the arcs of circle that pass near the data tend to a line, and the objective function tends to the sum of squared distances of the points from that line. If a line describes the data better than any circle, the infimum is attained only "at infinity" and there is no optimal circle. This section makes the argument precise.

Definition 31.4: Scatter matrix of the sample. Given a sample P with centroid $\bar{p}$, set $w_i = p_i - \bar{p}$ and

$$S = \sum_{i=1}^n w_i w_i^\top \in \mathbb{R}^{2 \times 2},$$

the **scatter matrix** of the sample. Denote by $\lambda_{\min}(S)$ the smaller of its two eigenvalues and set $M = \max_i \|w_i\|$.

Lemma 31.2: Rayleigh quotient for the scatter matrix. S is symmetric and positive semidefinite, and

$$\min_{\|u\|=1} u^\top S u = \lambda_{\min}(S),$$

with the minimum attained at a unit eigenvector of S corresponding to $\lambda_{\min}(S)$. Moreover $u^\top S u = \sum_{i=1}^n \langle u, w_i \rangle^2$ for every u .

Proof. Step 1 — symmetry and quadratic form. Each $w_i w_i^\top$ is symmetric, hence so is S . For every $u \in \mathbb{R}^2$,

$$u^\top S u = \sum_{i=1}^n u^\top w_i w_i^\top u = \sum_{i=1}^n (w_i^\top u)^2 = \sum_{i=1}^n \langle u, w_i \rangle^2 \geq 0,$$

which proves both the last claim and positive semidefiniteness.

Step 2 — the minimum on the unit circle. By the spectral theorem there is an orthonormal basis (e_1, e_2) of $\mathbb{R}^2$ with $S e_j = \mu_j e_j$ and $\mu_1 = \lambda_{\min}(S) \leq \mu_2$. Writing $u = \xi_1 e_1 + \xi_2 e_2$ with $\xi_1^2 + \xi_2^2 = 1$, we get $u^\top S u = \mu_1 \xi_1^2 + \mu_2 \xi_2^2 \geq \mu_1 (\xi_1^2 + \xi_2^2) = \mu_1$, with equality for $u = e_1$ (i.e. $\xi_1 = 1, \xi_2 = 0$), or for every unit u if $\mu_1 = \mu_2$. $\square$

Lemma 31.3: Expansion of the radial distance for distant centers. Let P be a sample with $M = \max_i \|w_i\| > 0$, let $u \in \mathbb{R}^2$ be a unit vector and let $c = \bar{p} + t u$ with $t \geq 8M$. Then, for every i ,

$$\rho_i(c) = t - \langle u, w_i \rangle + \eta_i, \quad |\eta_i| \leq \frac{4M^2}{t}.$$

Proof. Step 1 — the square of the radial distance. Since $p_i - c = w_i - tu$ and $\|u\| = 1$,

$$\rho_i(c)^2 = \|w_i - tu\|^2 = \|w_i\|^2 - 2t\langle u, w_i \rangle + t^2 = t^2(1 + x_i), \quad x_i = -\frac{2\langle u, w_i \rangle}{t} + \frac{\|w_i\|^2}{t^2}.$$

Step 2 — x_i is small. By Cauchy–Schwarz $|\langle u, w_i \rangle| \leq \|w_i\| \leq M$, hence

$$|x_i| \leq \frac{2M}{t} + \frac{M^2}{t^2} = \frac{2M}{t} \left(1 + \frac{M}{2t}\right) \leq \frac{2M}{t} \cdot \frac{17}{16} \leq \frac{17}{64} < \frac{1}{2},$$

using $t \geq 8M$, i.e. $M/t \leq 1/8$, both in the parenthesis ($M/(2t) \leq 1/16$) and in the factor ($2M/t \leq 1/4$).

Step 3 — Taylor with Lagrange remainder. Let $f(x) = \sqrt{1+x}$. Then $f(0) = 1$, $f'(0) = \frac{1}{2}$ and $f''(x) = -\frac{1}{4}(1+x)^{-3/2}$. For $|x| \leq \frac{1}{2}$ the intermediate point ξ of Taylor's formula satisfies $1 + \xi \geq \frac{1}{2}$, hence $|f''(\xi)| \leq \frac{1}{4} \cdot 2^{3/2} = \frac{\sqrt{2}}{2} \leq 1$ and

$$\left| \sqrt{1+x} - 1 - \frac{x}{2} \right| = \left| \frac{f''(\xi)}{2} x^2 \right| \leq \frac{x^2}{2}.$$

Step 4 — reassembly. From $\rho_i(c) = t\sqrt{1+x_i}$ (the positive root, because $t > 0$) and from Step 3,

$$\rho_i(c) = t \left(1 + \frac{x_i}{2} + e_i\right), \quad |e_i| \leq \frac{x_i^2}{2},$$

and, substituting x_i ,

$$t \cdot \frac{x_i}{2} = -\langle u, w_i \rangle + \frac{\|w_i\|^2}{2t}.$$

Hence $\rho_i(c) = t - \langle u, w_i \rangle + \eta_i$ with $\eta_i = \frac{\|w_i\|^2}{2t} + t e_i$.

Step 5 — estimate of the remainder. From Step 2, $|x_i| \leq \frac{17}{16} \cdot \frac{2M}{t} = \frac{17M}{8t}$, so

$$t|e_i| \leq \frac{t x_i^2}{2} \leq \frac{t}{2} \cdot \frac{289M^2}{64t^2} = \frac{289}{128} \cdot \frac{M^2}{t} \leq \frac{5M^2}{2t},$$

while $\frac{\|w_i\|^2}{2t} \leq \frac{M^2}{2t}$. Adding, $|\eta_i| \leq \frac{M^2}{2t} + \frac{5M^2}{2t} = \frac{3M^2}{t} \leq \frac{4M^2}{t}$. □

Theorem 31.2: Behavior of the objective function at infinity. Let P be a sample with $M = \max_i \|w_i\| > 0$ and let G be the function of Proposition 31.2. Then for every unit vector u and every $t \geq 8M$

$$\left| G(\bar{p} + tu) - u^\top S u \right| \leq \frac{24nM^3}{t}.$$

In particular $G(\bar{p} + tu) \rightarrow u^\top S u$ as $t \rightarrow +\infty$, uniformly in u , and

$$\liminf_{\|c\| \rightarrow +\infty} G(c) \geq \lambda_{\min}(S).$$

Proof. Step 1 — the residuals relative to the mean. Let $c = \bar{p} + tu$ with $t \geq 8M$. By Lemma 31.3, $\rho_i(c) = t - \langle u, w_i \rangle + \eta_i$ with $|\eta_i| \leq 4M^2/t$. Averaging over i and using $\sum_i w_i = \sum_i (p_i - \bar{p}) = n\bar{p} - n\bar{p} = 0$, hence $\frac{1}{n} \sum_i \langle u, w_i \rangle = \langle u, \frac{1}{n} \sum_i w_i \rangle = 0$, we get $\bar{\rho}(c) = t + \bar{\eta}$ with $\bar{\eta} = \frac{1}{n} \sum_i \eta_i$. Therefore

$$\rho_i(c) - \bar{\rho}(c) = -\langle u, w_i \rangle + B_i, \quad B_i = \eta_i - \bar{\eta}, \quad |B_i| \leq |\eta_i| + |\bar{\eta}| \leq \frac{8M^2}{t}.$$

Step 2 — comparison of the sums of squares. Setting $A_i = \langle u, w_i \rangle$, we have $(\rho_i - \bar{\rho})^2 = (B_i - A_i)^2 = A_i^2 - 2A_i B_i + B_i^2$, so

$$\left| G(c) - \sum_i A_i^2 \right| = \left| \sum_i (-2A_i B_i + B_i^2) \right| \leq 2 \sum_i |A_i| |B_i| + \sum_i B_i^2.$$

By Cauchy–Schwarz $|A_i| \leq \|w_i\| \leq M$, and by Step 1 $|B_i| \leq 8M^2/t$; hence

$$\left| G(c) - \sum_i A_i^2 \right| \leq 2nM \cdot \frac{8M^2}{t} + n \cdot \frac{64M^4}{t^2} = \frac{16nM^3}{t} + \frac{64nM^4}{t^2}.$$

Since $t \geq 8M$ implies $M/t \leq \frac{1}{8}$, the second term is $\frac{64nM^3}{t} \cdot \frac{M}{t} \leq \frac{8nM^3}{t}$, and the sum is at most $\frac{24nM^3}{t}$.

Step 3 — identification of the limit. By Lemma 31.2, $\sum_i A_i^2 = \sum_i \langle u, w_i \rangle^2 = u^\top S u$. This proves the inequality of the statement; the right-hand side does not depend on u and tends to 0 as $t \rightarrow +\infty$, so the convergence is uniform in u .

Step 4 — the liminf at infinity. Let c have $\|c - \bar{p}\| = t \geq 8M$ and let $u = (c - \bar{p})/t$, a unit vector. By the previous steps and by Lemma 31.2,

$$G(c) \geq u^\top S u - \frac{24nM^3}{t} \geq \lambda_{\min}(S) - \frac{24nM^3}{t}.$$

Since $\|c\| \rightarrow +\infty$ is equivalent to $t = \|c - \bar{p}\| \rightarrow +\infty$, passing to the liminf gives $\liminf_{\|c\| \rightarrow \infty} G(c) \geq \lambda_{\min}(S)$. $\square$

***Remark 31.5: What $u^\top S u$ is: the limiting line.** The quantity $u^\top S u = \sum_i \langle u, w_i \rangle^2$ is the sum of squared distances of the points p_i from the line $\ell_u = \{q : \langle u, q - \bar{p} \rangle = 0\}$, i.e. from the line through the centroid orthogonal to u : indeed the distance of p_i from ℓ_u equals $|\langle u, p_i - \bar{p} \rangle| = |\langle u, w_i \rangle|$. Theorem 31.2 therefore says that, as the center moves off in the direction u , the optimal circle degenerates into the line through the centroid perpendicular to u , and the objective function tends to the residual of the orthogonal linear fit on that line. By Lemma 31.2 the best of these residuals is $\lambda_{\min}(S)$, obtained by choosing u in the direction of minimum dispersion, i.e. by taking as the limiting line the one of maximum dispersion.*

***Corollary 31.3: The infimum does not exceed the residual of the optimal line.** If $M > 0$, then $\inf_{c \in \mathbb{R}^2} G(c) \leq \lambda_{\min}(S)$, and hence $\inf_{\theta} F(\theta) \leq \lambda_{\min}(S)$.*

Proof. Let u^* be a unit eigenvector of S corresponding to $\lambda_{\min}(S)$, which exists by Lemma 31.2. By Theorem 31.2, for every $t \geq 8M$,

$$G(\bar{p} + t u^*) \leq (u^*)^\top S u^* + \frac{24nM^3}{t} = \lambda_{\min}(S) + \frac{24nM^3}{t}.$$

Hence $\inf_c G(c) \leq \lambda_{\min}(S) + \frac{24nM^3}{t}$ for every $t \geq 8M$; letting $t \rightarrow +\infty$ gives $\inf_c G \leq \lambda_{\min}(S)$. The second claim follows from Proposition 31.2, by which $\inf_{\theta} F = \inf_c G$. $\square$

***Theorem 31.3: Existence of the minimum.** Let P be a sample with $M > 0$. If there exists $c_0 \in \mathbb{R}^2$ with*

$$G(c_0) < \lambda_{\min}(S),$$

then G attains a minimum on $\mathbb{R}^2$ and F attains a minimum on $\mathbb{R}^2 \times [0, +\infty)$; every minimizer of G belongs to the closed disc of center $\bar{p}$ and radius

$$t_0 = \max \left\{ 8M, \frac{24nM^3}{\lambda_{\min}(S) - G(c_0)} \right\}.$$

Proof. Step 1 — G is continuous. Each $\rho_i(c) = \|p_i - c\|$ is continuous in c (the norm is 1-Lipschitz), hence so are $\bar{\rho}$ and, by composition of continuous functions, $G(c) = \sum_i (\rho_i(c) - \bar{\rho}(c))^2$.

Step 2 — outside a disc G is too large. Let $m = G(c_0)$ and $\varepsilon = \lambda_{\min}(S) - m > 0$. Let c have $t = \|c - \bar{p}\| > t_0$. Then $t \geq 8M$ and $\frac{24nM^3}{t} < \frac{24nM^3}{t_0} \leq \varepsilon$, so by Step 4 of the proof of Theorem 31.2

$$G(c) \geq \lambda_{\min}(S) - \frac{24nM^3}{t} > \lambda_{\min}(S) - \varepsilon = m.$$

The inequality $\frac{24nM^3}{t} < \varepsilon$ holds because $t > t_0 \geq \frac{24nM^3}{\varepsilon}$; the condition $t \geq 8M$ holds because $t > t_0 \geq 8M$.

Step 3 — Weierstrass on the disc. Let $K = \{c : \|c - \bar{p}\| \leq t_0\}$, closed and bounded, hence compact; it is nonempty and, by Step 2, contains c_0 (because $G(c_0) = m$ and outside K one has $G > m$). The function G , continuous by Step 1, attains on K a minimum point c^* , with $G(c^*) \leq G(c_0) = m$.

Step 4 — the minimum on K is global. For every $c \notin K$ one has $G(c) > m \geq G(c^*)$ by Step 2; for every $c \in K$ one has $G(c) \geq G(c^*)$ by Step 3. Hence c^* is a global minimizer of G on $\mathbb{R}^2$, and every other global minimizer lies in K for the same reason.

Step 5 — from G to F . Setting $r^* = \bar{\rho}(c^*) \geq 0$ and $\theta^* = (c^*, r^*)$, Proposition 31.2 gives $F(\theta^*) = G(c^*) = \inf_c G = \inf_\theta F$: hence θ^* is a global minimizer of F . $\square$

Corollary 31.4: Collinear points: the minimum does not exist. Let P be a sample whose points all lie on a line ℓ and among which at least three are distinct. Then $\lambda_{\min}(S) = 0$, $\inf_c G(c) = 0$ and $G(c) > 0$ for every $c \in \mathbb{R}^2$: the objective function F has no minimum.

Proof. Step 1 — $\lambda_{\min}(S) = 0$. Let v be a direction unit vector of ℓ and let u be a unit vector orthogonal to v . All the points lie on ℓ , so the centroid $\bar{p} = \frac{1}{n} \sum p_i$ also belongs to it (it is a convex combination of points of a line), and each $w_i = p_i - \bar{p}$ is parallel to v , i.e. $\langle u, w_i \rangle = 0$. By Lemma 31.2, $u^\top S u = \sum_i \langle u, w_i \rangle^2 = 0$; since S is positive semidefinite, $\lambda_{\min}(S) = 0$. Moreover $M > 0$, because at least two points are distinct.

Step 2 — $\inf G = 0$. By Corollary 31.3, $\inf_c G \leq \lambda_{\min}(S) = 0$; and $G \geq 0$ by definition. Hence $\inf_c G = 0$.

Step 3 — G never vanishes. Suppose $G(c) = 0$ for some c . Then all the terms $(\rho_i(c) - \bar{\rho}(c))^2$ vanish, i.e. $\rho_1(c) = \dots = \rho_n(c) =: \varrho$: all the points are at distance ϱ from c . Parametrize ℓ as $q + sv$, $s \in \mathbb{R}$, and let s_1, s_2, s_3 be the parameters of three distinct points of the sample. The function

$$\psi(s) = \|q + sv - c\|^2 = s^2 \|v\|^2 + 2s \langle v, q - c \rangle + \|q - c\|^2 = s^2 + 2s \langle v, q - c \rangle + \|q - c\|^2$$

is a polynomial of degree exactly 2 in s (leading coefficient $\|v\|^2 = 1$). The equation $\psi(s) = \varrho^2$ is thus a quadratic equation, and has at most two solutions; but it is satisfied by s_1, s_2, s_3 , which are three distinct values (distinct points of a line have distinct parameters). Contradiction: hence $G(c) > 0$ for every c .

Step 4 — conclusion. The infimum of G equals 0 (Step 2) but is not attained at any point (Step 3): G has no minimum. By Proposition 31.2, $\inf_\theta F = \inf_c G = 0$, and if F had a minimum at $\theta^* = (c^*, r^*)$ we would have $G(c^*) \leq F(\theta^*) = 0$, absurd. Hence F has no minimum either. $\square$

Example 31.2: Five collinear points: the escape to infinity, in numbers. Let P be the sample $(0, 0), (1, 0), (2, 0), (3, 0), (4, 0)$. The centroid is $\bar{p} = (2, 0)$, the deviations are $w_i = (s_i, 0)$ with $s_i \in \{-2, -1, 0, 1, 2\}$, and

$$S = \sum_i w_i w_i^\top = \begin{pmatrix} \sum_i s_i^2 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 10 & 0 \\ 0 & 0 \end{pmatrix}, \quad \lambda_{\min}(S) = 0.$$

Take $c = (2, t)$ with $t > 0$. Then $\rho_i = \sqrt{s_i^2 + t^2}$ and, expanding as in Lemma 31.3 but one order further, $\rho_i = t + \frac{s_i^2}{2t} - \frac{s_i^4}{8t^3} + O(t^{-5})$, whence $\rho_i - \bar{\rho} = \frac{s_i^2 - m_2}{2t} + O(t^{-3})$ with $m_2 = \frac{1}{5} \sum_i s_i^2 = 2$. Therefore

$$\begin{aligned} G(2, t) &= \frac{1}{4t^2} \sum_i (s_i^2 - 2)^2 + O(t^{-4}) = \frac{2^2 + 1^2 + 2^2 + 1^2 + 2^2}{4t^2} + O(t^{-4}) \\ &= \frac{14}{4t^2} + O(t^{-4}) = \frac{3.5}{t^2} + O(t^{-4}). \end{aligned}$$

Independent check (direct evaluation of G in double precision): $G(2, 10) = 3.4244729 \cdot 10^{-2}$ against $3.5/10^2 = 3.5 \cdot 10^{-2}$; $G(2, 100) = 3.4992252 \cdot 10^{-4}$ against $3.5 \cdot 10^{-4}$; $G(2, 1000) = 3.4999922 \cdot 10^{-6}$ against $3.5 \cdot 10^{-6}$. ✓ The ratio of the computed value to the estimate tends to 1 as $1 + O(t^{-2})$, exactly as predicted. The infimum 0 is never attained: the optimal radius $\bar{\rho}(2, t) \approx t + 1/t$ diverges, and the circle degenerates into the line $y = 0$, which is the line of Observation 31.5 for $u = (0, 1)$.

Proposition 31.4: Samples with few points: the minimum exists but is not unique.

- (i) If $n = 1$, then $F(\theta) = 0$ for every $c \in \mathbb{R}^2$ and $r = \|p_1 - c\|$: the set of global minimizers is a two-dimensional surface.
- (ii) If $n = 2$ and $p_1 \neq p_2$, then $F(\theta) = 0$ if and only if c belongs to the perpendicular bisector of the segment $p_1 p_2$ and $r = \|p_1 - c\|$: the set of global minimizers is a curve.
- (iii) If all the points of the sample coincide ($M = 0$), then $F(\theta) = 0$ if and only if $r = \|p_1 - c\|$, for every c .

In all three cases the minimum exists, equals 0, and is not unique.

Proof. (i) For $n = 1$, $F(\theta) = (\rho_1(c) - r)^2$, which is ≥ 0 and vanishes exactly for $r = \rho_1(c) = \|p_1 - c\|$. Since c ranges freely over $\mathbb{R}^2$, the set of minimizers is $\{(c, \|p_1 - c\|) : c \in \mathbb{R}^2\}$, the graph of a continuous function on $\mathbb{R}^2$.

(ii) For $n = 2$, $F(\theta) = (\rho_1 - r)^2 + (\rho_2 - r)^2 \geq 0$ vanishes if and only if $\rho_1(c) = \rho_2(c) = r$. The condition $\|p_1 - c\| = \|p_2 - c\|$ defines, by definition, the perpendicular bisector of the segment $p_1 p_2$, which is a line because $p_1 \neq p_2$; on it $r = \|p_1 - c\|$ is determined by c . The set of minimizers is thus the graph of a continuous function on a line.

(iii) If $p_1 = \dots = p_n =: p$, then $\rho_i(c) = \|p - c\|$ for every i and $F(\theta) = n(\|p - c\| - r)^2$, which vanishes if and only if $r = \|p - c\|$.

In all cases $F \geq 0$ by definition and the value 0 is attained: these are therefore global minima, and the sets described have more than one element. □

Remark 31.6: The existence criterion is verifiable. Theorem 31.3 can be used in practice: one only needs to compute $\lambda_{\min}(S)$ (two eigenvalues of a 2×2 matrix: a closed-form operation) and compare it with the value of G at the center proposed by an algebraic method. If $G(c_0) < \lambda_{\min}(S)$ the minimum exists, lies in the explicit disc of the statement,

and any descent method starting from c_0 that decreases G stays in a compact set. On the two numerical samples of the chapter the margin is enormous: in sample A (Solved exercise 31.4) one has $\lambda_{\min}(S) = 75.7684689$ against $G(c^*) = 0.3463347$; in sample B, which is a short arc (Solved exercise 31.5), one has $\lambda_{\min}(S) = 0.2201842$ against $G(c^*) = 0.0017718$: the ratio is still two orders of magnitude, but one already sees that shortening the arc brings the two numbers closer. In the limit of collinear points they coincide and the minimum disappears (Corollary 31.4).

31.4. The normal equations and their nonlinearity

Proposition 31.5: Regularity of the objective function. Let $\Omega = \{\theta = (c, r) : \rho_i(c) > 0 \text{ for every } i\}$, i.e. the set of parameters whose center does not coincide with any data point. Then Ω is an open subset of $\mathbb{R}^3$ and F is of class C^∞ on Ω , with

$$\frac{\partial \rho_i}{\partial c} = -u_i^\top, \quad u_i = u_i(c) = \frac{p_i - c}{\rho_i(c)} \quad (\|u_i\| = 1).$$

Proof. Step 1 — Ω is open. The function $c \mapsto \min_i \rho_i(c)$ is continuous (finite minimum of continuous functions) and Ω is the preimage of the open set $(0, +\infty)$ multiplied by $\mathbb{R}$ in the variable r : it is therefore open in $\mathbb{R}^3$.

Step 2 — derivative of ρ_i . On Ω one has $\rho_i(c) = \sqrt{g_i(c)}$ with $g_i(c) = (x_i - a)^2 + (y_i - b)^2 > 0$; g_i is a polynomial and $s \mapsto \sqrt{s}$ is C^∞ on $(0, +\infty)$, hence $\rho_i \in C^\infty(\Omega)$. Differentiating,

$$\frac{\partial \rho_i}{\partial a} = \frac{-2(x_i - a)}{2\sqrt{g_i}} = -\frac{x_i - a}{\rho_i}, \quad \frac{\partial \rho_i}{\partial b} = -\frac{y_i - b}{\rho_i},$$

i.e. $\partial \rho_i / \partial c = -u_i^\top$ with $u_i = (p_i - c) / \rho_i$. That u_i is a unit vector follows from $\|p_i - c\| = \rho_i$.

Step 3 — F . $d_i(\theta) = \rho_i(c) - r$ is C^∞ on Ω and $F = \sum_i d_i^2$ is a sum of squares of C^∞ functions: it is C^∞ . $\square$

Proposition 31.6: Gradient of the objective function. On Ω one has

$$\frac{\partial F}{\partial c} = -2 \sum_{i=1}^n d_i u_i^\top, \quad \frac{\partial F}{\partial r} = -2 \sum_{i=1}^n d_i,$$

$$\text{i.e. } \nabla F(\theta) = -2 \sum_{i=1}^n d_i \begin{pmatrix} u_i \\ 1 \end{pmatrix}.$$

Proof. By Proposition 31.5, $F = \sum_i d_i^2$ is differentiable on Ω and $\partial F / \partial \theta = 2 \sum_i d_i \partial d_i / \partial \theta$. Now $\partial d_i / \partial c = \partial \rho_i / \partial c = -u_i^\top$ (the term $-r$ does not depend on c) and $\partial d_i / \partial r = -1$ (the term ρ_i does not depend on r). Substituting gives the two formulas, which collected into a single column vector give the final expression. $\square$

Theorem 31.4: Normal equations of the geometric fit. Let $\theta^* = (c^*, r^*) \in \Omega$ with $r^* > 0$ be a local minimizer of F . Then the three equations

$$\sum_{i=1}^n d_i = 0, \quad \sum_{i=1}^n d_i u_i = 0 \in \mathbb{R}^2, \quad (31.2)$$

hold, equivalent to the system

$$r^* = \bar{\rho}(c^*) = \frac{1}{n} \sum_{i=1}^n \rho_i(c^*), \quad c^* = \bar{p} - \frac{r^*}{n} \sum_{i=1}^n u_i(c^*). \quad (31.3)$$

Conversely, every $\theta \in \Omega$ satisfying (31.2) is a stationary point of F .

Proof. Step 1 — first-order necessary condition. θ^* is interior to the domain $\mathbb{R}^2 \times (0, +\infty)$ (because $r^* > 0$) and F is differentiable in a neighborhood (Proposition 31.5); hence $\nabla F(\theta^*) = 0$. By Proposition 31.6 this is equivalent to $\sum_i d_i = 0$ and $\sum_i d_i u_i = 0$, i.e. to (31.2).

Step 2 — the first equation gives the radius. $\sum_i d_i = \sum_i \rho_i - nr = 0$ is equivalent to $r = \frac{1}{n} \sum_i \rho_i = \bar{\rho}(c)$: this is the first equation of (31.3), and coincides with Proposition 31.2.

Step 3 — the second equation gives the center. Since $\rho_i u_i = p_i - c$ by definition of u_i , we have

$$0 = \sum_{i=1}^n d_i u_i = \sum_{i=1}^n (\rho_i - r) u_i = \sum_{i=1}^n \rho_i u_i - r \sum_{i=1}^n u_i = \sum_{i=1}^n (p_i - c) - r \sum_{i=1}^n u_i = n(\bar{p} - c) - r \sum_{i=1}^n u_i.$$

Dividing by n and isolating c gives $c = \bar{p} - \frac{r}{n} \sum_i u_i$, i.e. the second equation of (31.3). The steps are reversible, so the two formulations are equivalent.

Step 4 — converse. If $\theta \in \Omega$ satisfies (31.2), then $\nabla F(\theta) = 0$ by Proposition 31.6: θ is stationary. $\square$

Remark 31.7: A fixed-point system, not a linear system. Form (31.3) is instructive: the optimal center is the centroid of the data corrected by a vector proportional to the mean of the radial unit vectors. If the data are distributed symmetrically around the center, that mean vanishes and the optimal center coincides with the centroid; on a short arc, however, all the unit vectors u_i point roughly in the same direction, the mean has norm close to 1 and the correction is nearly r : the optimal center is very far from the centroid. The unknowns, however, also appear on the right-hand side, inside $u_i(c)$ and $\bar{\rho}(c)$: (31.3) is a fixed-point equation, not a linear system, and this is why an iteration is needed.

Proposition 31.7: Hessian matrix. On Ω the Hessian matrix of F , written in blocks with respect to the decomposition $\theta = (c, r)$, is

$$\nabla^2 F(\theta) = 2 \begin{pmatrix} \sum_i u_i u_i^\top & \sum_i u_i \\ \sum_i u_i^\top & n \end{pmatrix} + 2 \begin{pmatrix} \sum_i \frac{d_i}{\rho_i} (I - u_i u_i^\top) & 0 \\ 0 & 0 \end{pmatrix}.$$

Proof. Step 1 — derivative of the radial unit vector. On Ω , $u_i = (p_i - c)/\rho_i$. Differentiating the product and using $\partial \rho_i / \partial c = -u_i^\top$ (Proposition 31.5),

$$\frac{\partial u_i}{\partial c} = \frac{\partial(p_i - c)}{\partial c} \cdot \frac{1}{\rho_i} + (p_i - c) \frac{\partial}{\partial c} \left(\frac{1}{\rho_i} \right) = -\frac{I}{\rho_i} + (p_i - c) \cdot \frac{u_i^\top}{\rho_i^2} = \frac{-I + u_i u_i^\top}{\rho_i},$$

where we used $\partial(1/\rho_i)/\partial c = -\rho_i^{-2} \partial \rho_i / \partial c = \rho_i^{-2} u_i^\top$ and $p_i - c = \rho_i u_i$.

Step 2 — the cc block. Differentiating $\partial F / \partial c = -2 \sum_i d_i u_i^\top$ (Proposition 31.6) with respect to c , with the product rule and with $\partial d_i / \partial c = -u_i^\top$,

$$\frac{\partial^2 F}{\partial c^2} = -2 \sum_i \left[u_i \frac{\partial d_i}{\partial c} + d_i \frac{\partial u_i}{\partial c} \right] = -2 \sum_i \left[-u_i u_i^\top + d_i \frac{-I + u_i u_i^\top}{\rho_i} \right]$$

$$= 2 \sum_i \left[u_i u_i^\top + \frac{d_i}{\rho_i} (I - u_i u_i^\top) \right].$$

Step 3 — the mixed block and the rr block. Differentiating $\partial F / \partial c = -2 \sum_i d_i u_i^\top$ with respect to r , and recalling that u_i does not depend on r while $\partial d_i / \partial r = -1$, we get $\partial^2 F / \partial c \partial r = -2 \sum_i (-1) u_i = 2 \sum_i u_i$. Differentiating $\partial F / \partial r = -2 \sum_i d_i$ with respect to r gives $\partial^2 F / \partial r^2 = -2 \sum_i (-1) = 2n$.

Step 4 — reassembly. Putting together the three blocks and separating the term with d_i / ρ_i from the rest gives the formula of the statement. $\square$

Proposition 31.8: The objective function is not quadratic. *There is a sample P for which $\nabla^2 F$ is not constant on Ω ; in particular F is not a second-degree polynomial in (a, b, r) and the normal equations (31.2) are not a linear system.*

Proof. Take the sample with a single point $p_1 = (1, 0)$ (or, if one prefers a sample with $n \geq 3$, the three points $(1, 0), (1, 0), (1, 0)$, for which everything is multiplied by 3). For $c = (a, 0)$ with $a < 1$ one has $\rho_1 = 1 - a$, $u_1 = (1, 0)$, and by Proposition 31.7 the cc block of the Hessian equals

$$2 \left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{(1-a)-r}{1-a} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right] = 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{r}{1-a} \end{pmatrix}.$$

The $(2, 2)$ entry equals $2(1 - \frac{r}{1-a})$, which depends on a : for $r = 1$ it equals 0 at $a = 0$ and -2 at $a = \frac{1}{2}$. Hence $\nabla^2 F$ is not constant.

If F were a polynomial of degree ≤ 2 in the three variables, its Hessian would be the constant matrix of the second-degree coefficients; and the gradient of such a polynomial is affine, so that $\nabla F = 0$ would be a linear system. Since the Hessian is not constant, F is not quadratic and $\nabla F = 0$ is not a linear system. $\square$

Remark 31.8: Where the nonlinearity exactly lies. *The comparison with Volume I, Chapter 10 is sharp. There one minimizes $\sum_i (\rho_i^2 - r^2)^2$: setting $u = 2a$, $v = 2b$, $w = r^2 - a^2 - b^2$ and $z_i = x_i^2 + y_i^2$, the residual becomes $z_i - ux_i - vy_i - w$, i.e. an affine function of the new parameters, and the minimization is a linear least-squares problem with a closed-form solution. Here instead the residual is $\sqrt{(x_i - a)^2 + (y_i - b)^2} - r$, and the square root cannot be absorbed by any change of variables of the three parameters: the term $\frac{d_i}{\rho_i} (I - u_i u_i^\top)$ of the Hessian — which is exactly the piece depending on the curvature of the residual — is not constant. This is the technical reason why the geometric fit requires an iteration and the algebraic fit does not; and it is also the reason why the algebraic fit does not solve the same problem (Section 31.8).*

Example 31.3: A symmetric sample with an exact solution. Let P be the sample $p_1 = (1, 0)$, $p_2 = (-1, 0)$, $p_3 = (0, 2)$, $p_4 = (0, -2)$, and let $\theta^* = (0, 0, \frac{3}{2})$.

The normal equations. One has $\rho_1 = \rho_2 = 1$, $\rho_3 = \rho_4 = 2$, so $\bar{\rho} = \frac{1+1+2+2}{4} = \frac{3}{2} = r^*$: the first normal equation is satisfied, and the residuals are $d_1 = d_2 = -\frac{1}{2}$, $d_3 = d_4 = +\frac{1}{2}$. The radial unit vectors are $u_1 = (1, 0)$, $u_2 = (-1, 0)$, $u_3 = (0, 1)$, $u_4 = (0, -1)$, and

$$\sum_i d_i u_i = -\frac{1}{2}(1, 0) - \frac{1}{2}(-1, 0) + \frac{1}{2}(0, 1) + \frac{1}{2}(0, -1) = (0, 0) :$$

the second equation is satisfied as well. By Theorem 31.4, θ^* is stationary, and

$$F(\theta^*) = 4 \cdot \left(\frac{1}{2}\right)^2 = 1.$$

It is a minimum. By Proposition 31.7, since $\sum_i u_i u_i^\top = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ and $\sum_i u_i = 0$,

$$\sum_i \frac{d_i}{\rho_i} (I - u_i u_i^\top) = 2 \cdot \left(-\frac{1}{2}\right) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + 2 \cdot \left(\frac{1}{4}\right) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -1 \end{pmatrix},$$

whence

$$\nabla^2 F(\theta^*) = 2 \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} + 2 \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

The eigenvalues are 2, 5, 8, all positive: by the second-order sufficient condition θ^* is a strict local minimum.

Independent check 1 (grid). Evaluating G of Proposition 31.2 on a 401×401 grid of the square $[-1, 1]^2$ one finds the minimum 1.0000000 at $(0.00000; 0.00000)$ with $\bar{\rho} = 1.500000$. ✓

Independent check 2 (direct method). A Nelder–Mead iteration on the variable c alone, started at $(0.3; 0.4)$, converges in 57 steps to $c = (6.0 \cdot 10^{-8}, 6.3 \cdot 10^{-8})$ with $G = 1.0000000$ and $\bar{\rho} = 1.5000000$. ✓

Remark. The algebraic fit of the same sample gives the same center but radius $\sqrt{\frac{1+1+4+4}{4}} = \sqrt{2.5} = 1.5811388$, larger than 1.5: this is the inequality of Proposition 31.14, verified here in exact form, with gap $\sqrt{2.5} - \frac{3}{2} = 0.0811388$ and $\frac{\text{Var}(\rho)}{\sqrt{2.5} + \frac{3}{2}} = \frac{0.25}{3.0811388} = 0.0811388$. ✓

31.5. Gauss–Newton: Jacobian, step, descent direction

Definition 31.5: Residual vector and Jacobian. For $\theta \in \Omega$ set

$$d(\theta) = (d_1(\theta), \dots, d_n(\theta))^\top \in \mathbb{R}^n, \quad J(\theta) = \frac{\partial d}{\partial \theta} \in \mathbb{R}^{n \times 3}.$$

Then $F(\theta) = \|d(\theta)\|^2 = d^\top d$.

Proposition 31.9: The rows of the Jacobian are the radial unit vectors. For $\theta \in \Omega$, the i -th row of $J(\theta)$ is

$$J_{i\bullet} = -\begin{pmatrix} u_i^\top & 1 \end{pmatrix} = \begin{pmatrix} -\frac{x_i - a}{\rho_i} & -\frac{y_i - b}{\rho_i} & -1 \end{pmatrix},$$

and

$$\nabla F(\theta) = 2J^\top d, \quad J^\top J = \begin{pmatrix} \sum_i u_i u_i^\top & \sum_i u_i \\ \sum_i u_i^\top & n \end{pmatrix}.$$

In particular every entry of J has modulus ≤ 1 and J is dimensionless.

Proof. Step 1 — the rows. $d_i(\theta) = \rho_i(c) - r$, so by Proposition 31.5 $\partial d_i / \partial c = -u_i^\top$ and $\partial d_i / \partial r = -1$: the i -th row of J is $(-u_i^\top, -1)$.

Step 2 — the gradient. Differentiating $F = \sum_i d_i^2$, $\nabla F = 2 \sum_i d_i \nabla d_i = 2J^\top d$, where we used that ∇d_i is the transpose of the i -th row of J . The result coincides with Proposition 31.6, since $2J^\top d = 2 \sum_i d_i \cdot (- (u_i^\top, 1))^\top = -2 \sum_i d_i (u_i, 1)^\top$.

Step 3 — $J^\top J$. By definition of matrix product, $J^\top J = \sum_i J_{i\bullet}^\top J_{i\bullet} = \sum_i \begin{pmatrix} u_i \\ 1 \end{pmatrix} \begin{pmatrix} u_i^\top & 1 \end{pmatrix}$ (the two minus signs cancel), i.e. the block matrix of the statement.

Step 4 — **boundedness**. The components of u_i have modulus $\leq \|u_i\| = 1$, and the third column is the constant -1 . Since every entry is either a component of a unit vector or -1 , J has no physical dimensions: the three parameters a, b, r are all lengths and so are the residuals, hence the derivatives are pure numbers. This is what makes it meaningful to speak of the condition number of J without having to rescale the columns. $\square$

Lemma 31.4: Full rank of the Jacobian. Let $\theta \in \Omega$. Then $\text{rank } J(\theta) = 3$ if and only if the unit vectors $u_1, \dots, u_n$ take at least three distinct values. In particular $n \geq 3$ is necessary, and the rank is not full exactly when all the data lie on at most two rays issuing from the center c .

Proof. **Step 1** — **reformulation**. Set $v_i = (u_i^\top, 1)^\top \in \mathbb{R}^3$; the rows of J are $-v_i^\top$, hence $\text{rank } J = \dim \text{span}\{v_1, \dots, v_n\}$, and $\text{rank } J < 3$ if and only if there is a nonzero $(z, \gamma) \in \mathbb{R}^2 \times \mathbb{R}$ with $\langle z, u_i \rangle + \gamma = 0$ for every i .

Step 2 — **necessarily** $z \neq 0$. If $z = 0$, the condition would give $\gamma = 0$ and hence $(z, \gamma) = 0$, excluded. Hence $z \neq 0$ and the condition says that all the u_i belong to the line $L = \{u \in \mathbb{R}^2 : \langle z, u \rangle = -\gamma\}$.

Step 3 — **a line meets the unit circle in at most two points**. Parametrizing L as $q + s \hat{z}^\perp$ with $\hat{z}^\perp$ a unit vector orthogonal to z and $q \in L$, the condition $\|q + s \hat{z}^\perp\| = 1$ becomes $s^2 + 2s\langle q, \hat{z}^\perp \rangle + \|q\|^2 - 1 = 0$, a quadratic equation in s with leading coefficient $1 \neq 0$: it has at most two solutions. Since $\|u_i\| = 1$ for every i , the u_i take at most two distinct values.

Step 4 — **converse**. Suppose the u_i take at most two distinct values, say u and u' (possibly equal). If $u \neq u'$, let $z \neq 0$ be a vector orthogonal to $u - u'$ and let $\gamma = -\langle z, u \rangle$: then $\langle z, u \rangle + \gamma = 0$ and $\langle z, u' \rangle + \gamma = \langle z, u' - u \rangle = 0$. If $u = u'$, take $z = u \neq 0$ and $\gamma = -1$: $\langle u, u \rangle - 1 = 0$. In both cases $(z, \gamma) \neq 0$ satisfies the condition of Step 1, hence $\text{rank } J < 3$.

Step 5 — **geometric reading**. $u_i = (p_i - c)/\rho_i$ is the direction of the ray from c toward p_i ; two points have the same unit vector if and only if they lie on the same ray issuing from c . Hence "the u_i take at most two values" means "the data lie on at most two rays issuing from c ". Finally three distinct values require at least three points. $\square$

Definition 31.6: Gauss–Newton step. Let $\theta \in \Omega$ with $J = J(\theta)$ of rank 3. The **Gauss–Newton step** at θ is the vector $\delta_{\text{GN}} \in \mathbb{R}^3$ defined by

$$J^\top J \delta_{\text{GN}} = -J^\top d, \quad \text{i.e.} \quad \delta_{\text{GN}} = -(J^\top J)^{-1} J^\top d,$$

and the **Gauss–Newton iteration** is $\theta^{(k+1)} = \theta^{(k)} + \delta_{\text{GN}}(\theta^{(k)})$.

Lemma 31.5: The step minimizes the linear model of the residual. Let J have rank 3. Then δ_{GN} is the unique minimizer of the function $\delta \mapsto \|d + J\delta\|^2$, and is well defined because $J^\top J$ is invertible.

Proof. **Step 1** — **$J^\top J$ is positive definite**. For every $\delta \in \mathbb{R}^3$, $\delta^\top J^\top J \delta = \|J\delta\|^2 \geq 0$, with equality if and only if $J\delta = 0$; since $\text{rank } J = 3$ and J has three columns, the kernel of J is $\{0\}$, hence $\delta^\top J^\top J \delta > 0$ for $\delta \neq 0$. A symmetric positive definite matrix is invertible.

Step 2 — **minimum of the model**. The function $q(\delta) = \|d + J\delta\|^2 = d^\top d + 2d^\top J\delta + \delta^\top J^\top J \delta$ is a second-degree polynomial in δ whose quadratic part is positive definite (Step 1): it is thus strictly convex and has a unique minimizer, which is its only stationary point. From $\nabla q(\delta) = 2J^\top d + 2J^\top J \delta = 0$ it follows that $\delta = -(J^\top J)^{-1} J^\top d = \delta_{\text{GN}}$. $\square$

Theorem 31.5: The Gauss–Newton step is a descent direction. Let $\theta \in \Omega$ with $J = J(\theta)$ of rank 3 and $J^\top d \neq 0$. Then

$$\langle \nabla F(\theta), \delta_{\text{GN}} \rangle = -2 (J^\top d)^\top (J^\top J)^{-1} (J^\top d) < 0,$$

and consequently there exists $\tau_0 > 0$ such that $F(\theta + \tau \delta_{\text{GN}}) < F(\theta)$ for every $\tau \in (0, \tau_0)$. If instead $J^\top d = 0$, then $\delta_{\text{GN}} = 0$ and θ is a stationary point of F .

Proof. Step 1 — the scalar product. By Proposition 31.9, $\nabla F = 2J^\top d$. Setting $g = J^\top d \in \mathbb{R}^3$ and $A = J^\top J$, we have $\delta_{\text{GN}} = -A^{-1}g$ and

$$\langle \nabla F, \delta_{\text{GN}} \rangle = \langle 2g, -A^{-1}g \rangle = -2g^\top A^{-1}g.$$

Step 2 — the sign. A is symmetric positive definite (Step 1 of Lemma 31.5); hence so is A^{-1} : if $Av = \mu v$ with $\mu > 0$ and $v \neq 0$, then $A^{-1}v = \mu^{-1}v$, and the eigenvalues of A^{-1} are the reciprocals of those of A , all positive. Hence $g^\top A^{-1}g > 0$ for $g \neq 0$, and $\langle \nabla F, \delta_{\text{GN}} \rangle < 0$.

Step 3 — actual decrease. F is differentiable at $\theta \in \Omega$ (Proposition 31.5) and Ω is open, so the function $\varphi(\tau) = F(\theta + \tau \delta_{\text{GN}})$ is defined and differentiable in a neighborhood of $\tau = 0$, with $\varphi'(0) = \langle \nabla F(\theta), \delta_{\text{GN}} \rangle < 0$. By definition of derivative there exists $\tau_0 > 0$ such that $\frac{\varphi(\tau) - \varphi(0)}{\tau} < 0$ for $0 < \tau < \tau_0$, i.e. $\varphi(\tau) < \varphi(0) = F(\theta)$.

Step 4 — the case $J^\top d = 0$. Then $\delta_{\text{GN}} = -A^{-1} \cdot 0 = 0$ and $\nabla F = 2J^\top d = 0$: θ is stationary, and the iteration stops. $\square$

Corollary 31.5: Fixed points of the iteration and stationary points. If $J(\theta)$ has rank 3, then θ is a fixed point of the Gauss–Newton iteration if and only if θ satisfies the normal equations (31.2).

Proof. θ is a fixed point if and only if $\delta_{\text{GN}}(\theta) = 0$, i.e., since $J^\top J$ is invertible (Lemma 31.5), if and only if $J^\top d = 0$, i.e. $\nabla F(\theta) = 0$ (Proposition 31.9), i.e., by Theorem 31.4, if and only if (31.2) holds. $\square$

Remark 31.9: What Gauss–Newton discards. Comparing Proposition 31.7 with Proposition 31.9 one sees that

$$\nabla^2 F(\theta) = 2J^\top J + 2 \begin{pmatrix} \sum_i \frac{d_i}{\rho_i} (I - u_i u_i^\top) & 0 \\ 0 & 0 \end{pmatrix} :$$

the Gauss–Newton method replaces the exact Hessian with only the part $2J^\top J$, discarding the term proportional to the residuals. The substitution is the more harmless the smaller the residuals d_i are compared with the radial distances ρ_i : on the data of Solved exercise 31.4 the ratio $\max_i |d_i| / \min_i \rho_i$ equals $0.2875391 / 4.8294136 = 0.0595391$, and indeed the iteration converges in three steps. On very noisy data, or on a fit that starts far from the minimum, the discarded term is not small and Gauss–Newton may diverge: this is the problem that the damping of Section 31.6 solves.

Remark 31.10: What we import about convergence. The convergence rate of the Gauss–Newton iteration — local quadratic convergence in the case of zero residual, linear with a factor proportional to the norm of the discarded term in the case of nonzero residual, and possible divergence without a line search — is a result of numerical analysis that here is **imported** and not proved: see Nocedal–Wright, *Numerical Optimization*, 2nd ed.,

Chapter 10 "Least-Squares Problems", pp. 245–269. What exactly is missing: the estimate $\|\theta^{(k+1)} - \theta^*\| \leq C\|\theta^{(k)} - \theta^*\|^2 + L\|\theta^{(k)} - \theta^*\|$ with L tied to the term $\sum_i \frac{d_i}{\rho_i}(I - u_i u_i^\top)$ of Proposition 31.7, and the Lipschitz regularity hypotheses on the Jacobian in a neighborhood of the minimum that support it. In this chapter we prove only that the step is one of descent (Theorem 31.5) and that the fixed points are exactly the stationary points (Corollary 31.5): these are the two facts needed to guarantee the correctness of the algorithm with a line search, not its speed.

31.6. Levenberg–Marquardt: damping and its two limits

The Gauss–Newton step has two twin defects: it is undefined when the Jacobian loses rank (Lemma 31.4), and when the rank is nearly lost the step is enormous, so much so that it leaves the region where the linear model of the residual is credible. Levenberg–Marquardt damping repairs both with a single parameter.

Definition 31.7: Damped Levenberg–Marquardt step. Let $\theta \in \Omega$, $J = J(\theta)$, $d = d(\theta)$ and $\lambda > 0$ the **damping parameter**. The **Levenberg–Marquardt step** is the vector $\delta(\lambda) \in \mathbb{R}^3$ defined by

$$(J^\top J + \lambda I) \delta(\lambda) = -J^\top d.$$

Theorem 31.6: The damped step is always well defined and is one of descent. Let $\theta \in \Omega$ and $\lambda > 0$. Then:

- (i) $J^\top J + \lambda I$ is symmetric positive definite, whatever the rank of J ; in particular it is invertible and $\delta(\lambda)$ exists and is unique;
- (ii) $\delta(\lambda)$ is the unique minimizer of the function $\delta \mapsto \|d + J\delta\|^2 + \lambda\|\delta\|^2$;
- (iii) if $J^\top d \neq 0$, then $\langle \nabla F(\theta), \delta(\lambda) \rangle < 0$, and hence there exists $\tau_0 > 0$ with $F(\theta + \tau\delta(\lambda)) < F(\theta)$ for every $\tau \in (0, \tau_0)$.

Proof. (i) The matrix $J^\top J$ is symmetric and positive semidefinite, because $\delta^\top J^\top J \delta = \|J\delta\|^2 \geq 0$. Hence for $\delta \neq 0$

$$\delta^\top (J^\top J + \lambda I) \delta = \|J\delta\|^2 + \lambda\|\delta\|^2 \geq \lambda\|\delta\|^2 > 0,$$

and the matrix is positive definite; it is symmetric because it is a sum of symmetric matrices. A positive definite matrix has trivial kernel, hence it is invertible.

(ii) The function $q_\lambda(\delta) = \|d + J\delta\|^2 + \lambda\|\delta\|^2 = d^\top d + 2d^\top J\delta + \delta^\top (J^\top J + \lambda I)\delta$ is a second-degree polynomial whose quadratic part is positive definite by (i): it is thus strictly convex and has a unique minimum, its only stationary point. From $\nabla q_\lambda = 2J^\top d + 2(J^\top J + \lambda I)\delta = 0$ it follows that $\delta = \delta(\lambda)$.

(iii) Setting $g = J^\top d$ and $A_\lambda = J^\top J + \lambda I$, we have $\nabla F = 2g$ (Proposition 31.9) and $\delta(\lambda) = -A_\lambda^{-1}g$, hence

$$\langle \nabla F, \delta(\lambda) \rangle = -2g^\top A_\lambda^{-1}g < 0,$$

because A_λ^{-1} is positive definite (the inverse of a symmetric positive definite matrix has as eigenvalues the reciprocals, all positive) and $g \neq 0$. The decrease for small steps follows as in Step 3 of the proof of Theorem 31.5. $\square$

Theorem 31.7: The two limits of the damping. Let $\theta \in \Omega$ and $g = J^\top d$.

- (i) If $\text{rank } J = 3$, then $\lim_{\lambda \rightarrow 0^+} \delta(\lambda) = \delta_{\text{GN}}$.
- (ii) $\lim_{\lambda \rightarrow +\infty} \lambda \delta(\lambda) = -g = -\frac{1}{2} \nabla F(\theta)$. In particular, if $g \neq 0$, the unit vector of the step

tends to the unit vector of the steepest-descent direction:

$$\lim_{\lambda \rightarrow +\infty} \frac{\delta(\lambda)}{\|\delta(\lambda)\|} = -\frac{\nabla F(\theta)}{\|\nabla F(\theta)\|}.$$

Proof. (i) If $\text{rank } J = 3$ then $J^\top J$ is invertible (Step 1 of Lemma 31.5). The map $A \mapsto A^{-1}$ is continuous on the open set of invertible matrices (the entries of the inverse are, by Cramer's rule, rational functions of the entries with denominator $\det A \neq 0$), and $\lambda \mapsto J^\top J + \lambda I$ is continuous with value $J^\top J$ at $\lambda = 0$; for λ small enough the matrix remains invertible. Hence

$$\lim_{\lambda \rightarrow 0^+} \delta(\lambda) = \lim_{\lambda \rightarrow 0^+} -(J^\top J + \lambda I)^{-1} g = -(J^\top J)^{-1} g = \delta_{\text{GN}}.$$

(ii) For $\lambda > 0$,

$$\lambda \delta(\lambda) = -\lambda (J^\top J + \lambda I)^{-1} g = -\left(\frac{J^\top J}{\lambda} + I\right)^{-1} g,$$

where we used $\lambda(B + \lambda I)^{-1} = (\lambda^{-1}B + I)^{-1}$, valid because $\lambda(B + \lambda I)^{-1} \cdot (\lambda^{-1}B + I) = \lambda(B + \lambda I)^{-1} \cdot \lambda^{-1}(B + \lambda I) = I$. Now $\lambda^{-1}J^\top J \rightarrow 0$ as $\lambda \rightarrow +\infty$, hence $\lambda^{-1}J^\top J + I \rightarrow I$, which is invertible; by the continuity of inversion (Part (i)) we get $\lambda \delta(\lambda) \rightarrow -I^{-1}g = -g$. Since $\nabla F = 2g$, the limit is $-\frac{1}{2}\nabla F$.

Unit vectors. If $g \neq 0$, by what we just showed $\|\lambda \delta(\lambda)\| \rightarrow \|g\| > 0$, hence $\lambda \delta(\lambda) \neq 0$ for large λ and

$$\frac{\delta(\lambda)}{\|\delta(\lambda)\|} = \frac{\lambda \delta(\lambda)}{\|\lambda \delta(\lambda)\|} \rightarrow \frac{-g}{\|g\|} = -\frac{\nabla F}{\|\nabla F\|},$$

by continuity of the map $v \mapsto v/\|v\|$ away from the origin. $\square$

Proposition 31.10: The step length decreases with the damping. Let $\theta \in \Omega$ with $g = J^\top d \neq 0$. Then the function $\lambda \mapsto \|\delta(\lambda)\|$ is strictly decreasing on $(0, +\infty)$, with $\lim_{\lambda \rightarrow +\infty} \|\delta(\lambda)\| = 0$; if moreover $\text{rank } J = 3$, then $\lim_{\lambda \rightarrow 0^+} \|\delta(\lambda)\| = \|\delta_{\text{GN}}\|$ and $\|\delta(\lambda)\| < \|\delta_{\text{GN}}\|$ for every $\lambda > 0$.

Proof. **Step 1 — singular value decomposition.** Let $J = U\Sigma V^\top$ be a reduced singular value decomposition, with $U \in \mathbb{R}^{n \times 3}$ having orthonormal columns, $V \in \mathbb{R}^{3 \times 3}$ orthogonal and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$, $\sigma_j \geq 0$. Then $J^\top J = V\Sigma^2 V^\top$ and

$$J^\top J + \lambda I = V(\Sigma^2 + \lambda I)V^\top, \quad g = J^\top d = V\Sigma U^\top d.$$

Step 2 — explicit form of the step. Setting $h = U^\top d \in \mathbb{R}^3$ with components h_1, h_2, h_3 ,

$$\delta(\lambda) = -V(\Sigma^2 + \lambda I)^{-1}\Sigma h, \quad \text{i.e.} \quad V^\top \delta(\lambda) = \left(-\frac{\sigma_1 h_1}{\sigma_1^2 + \lambda}, -\frac{\sigma_2 h_2}{\sigma_2^2 + \lambda}, -\frac{\sigma_3 h_3}{\sigma_3^2 + \lambda}\right)^\top.$$

Since V is orthogonal, $\|\delta(\lambda)\| = \|V^\top \delta(\lambda)\|$ and

$$\|\delta(\lambda)\|^2 = \sum_{j=1}^3 \frac{\sigma_j^2 h_j^2}{(\sigma_j^2 + \lambda)^2}.$$

Step 3 — monotonicity. Each term is a nonincreasing function of $\lambda > 0$, because the numerator is constant and the denominator grows; and it is *strictly* decreasing if $\sigma_j h_j \neq 0$. Since $g = V\Sigma h \neq 0$ and V is invertible, $\Sigma h \neq 0$, i.e. at least one $\sigma_j h_j \neq 0$: the sum is strictly decreasing.

Step 4 — the limits. Each term tends to 0 as $\lambda \rightarrow +\infty$, hence $\|\delta(\lambda)\| \rightarrow 0$. If $\text{rank } J = 3$ all the σ_j are positive and each term tends to h_j^2/σ_j^2 as $\lambda \rightarrow 0^+$; the sum is $\|\Sigma^{-1}h\|^2 = \|\delta_{\text{GN}}\|^2$,

consistent with Theorem 31.7(i). The strict inequality for $\lambda > 0$ follows from the strict monotonicity of Step 3. $\square$

Remark 31.11: Damping as a trust region. Proposition 31.10 says that λ controls the length of the step monotonically: increasing λ shortens the step and rotates it toward the gradient. The fact that $\delta(\lambda)$ is exactly the solution of the constrained problem

$$\min_{\|\delta\| \leq \Delta} \|d + J\delta\|^2$$

for a suitable $\Delta = \Delta(\lambda)$ — i.e. that Levenberg–Marquardt is a trust-region method — is a result that here is **imported** and not proved: see Nocedal–Wright, *Numerical Optimization*, 2nd ed., Chapter 4 "Trust-Region Methods", pp. 66–100, and Chapter 10, pp. 245–269. What is missing: the first-order optimality conditions for the constrained problem with Lagrange multiplier λ and the discussion of the so-called hard case, in which the minimum of the model falls on the boundary of the region without the multiplier being determined by the equation $\|\delta(\lambda)\| = \Delta$ alone. Here we prove only well-posedness (Theorem 31.6), the two limits (Theorem 31.7) and the monotonicity of the length (Proposition 31.10), which are the facts on which the practical update rule for λ rests.

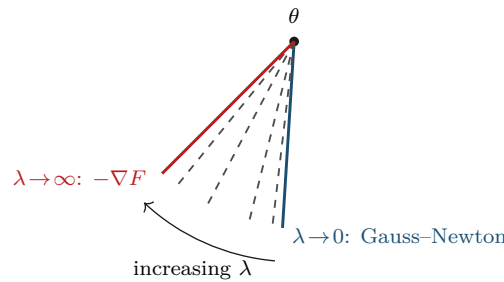


Figure 31.2: The Levenberg–Marquardt "fan" in the model case $J^\top J = \text{diag}(4; 0.25)$, $J^\top d = (1; 1)$, for which $\delta(\lambda) = -(\frac{1}{4+\lambda}; \frac{1}{0.25+\lambda})$. The directions drawn correspond to $\lambda = 0$ (-93.5763°), 0.25 (-96.7098°), 1 (-104.0362°), 4 (-117.9795°), 16 (-129.0939°) and $\lambda \rightarrow \infty$ (-135° , i.e. the direction of $-\nabla F = -2(1; 1)$). Check of the limiting case: as $\lambda \rightarrow \infty$ the two components of $\lambda\delta(\lambda)$ both tend to -1 , hence the angle tends to $\arctan(-1/-1)$ in the third quadrant, i.e. -135° , as in Theorem 31.7(ii). $\checkmark$

Best practice. Damping update rule. At each iteration one computes $\delta(\lambda)$ and compares $F(\theta + \delta(\lambda))$ with $F(\theta)$:

- if F has decreased, the step is accepted and λ is *reduced* (typically $\lambda \leftarrow \lambda/10$), moving closer to Gauss–Newton;
- if F has not decreased, the step is rejected and λ is *increased* (typically $\lambda \leftarrow 10\lambda$), shortening the step and rotating it toward the gradient.

Theorem 31.6(iii) guarantees that, for λ large enough, an acceptable step always exists, provided $\nabla F(\theta) \neq 0$: indeed $\delta(\lambda)$ is one of descent and its length tends to 0 (Proposition 31.10), so it eventually falls in the interval $(0, \tau_0)$ of the theorem. The loop can therefore never get stuck at a nonstationary point.

31.7. Conditioning, short arcs and stopping criteria

Definition 31.8: Condition number of the Jacobian. Let $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$ be the singular values of $J(\theta)$. If $\sigma_3 > 0$ set

$$\kappa_2(J) = \frac{\sigma_1}{\sigma_3} \geq 1,$$

the **spectral condition number** of J ; if $\sigma_3 = 0$ set $\kappa_2(J) = +\infty$. Since the eigenvalues of $J^\top J$ are the σ_j^2 , one has $\kappa_2(J)^2 = \kappa_2(J^\top J)$.

Proposition 31.11: Sensitivity of the step to the data. Let J have rank 3 and let δ_{GN} and $\tilde{\delta}_{\text{GN}}$ be the Gauss–Newton steps computed with the residual vectors d and $d + e$ respectively (same J). Then

$$\|\tilde{\delta}_{\text{GN}} - \delta_{\text{GN}}\| \leq \frac{\|e\|}{\sigma_3}, \quad \text{and, if } \delta_{\text{GN}} \neq 0, \quad \frac{\|\tilde{\delta}_{\text{GN}} - \delta_{\text{GN}}\|}{\|\delta_{\text{GN}}\|} \leq \kappa_2(J) \frac{\|e\|}{\|J\delta_{\text{GN}}\|}.$$

Proof. Step 1 — the difference. By linearity, $\tilde{\delta}_{\text{GN}} - \delta_{\text{GN}} = -(J^\top J)^{-1} J^\top e$.

Step 2 — norm of the pseudoinverse. With $J = U\Sigma V^\top$ as in Step 1 of the proof of Proposition 31.10, and Σ invertible because the rank is 3,

$$(J^\top J)^{-1} J^\top = (V\Sigma^2 V^\top)^{-1} V\Sigma U^\top = V\Sigma^{-2} \Sigma U^\top = V\Sigma^{-1} U^\top.$$

For every $e \in \mathbb{R}^n$, since U has orthonormal columns and V is orthogonal, $\|V\Sigma^{-1} U^\top e\| = \|\Sigma^{-1} U^\top e\| \leq \sigma_3^{-1} \|U^\top e\| \leq \sigma_3^{-1} \|e\|$. This proves the first estimate.

Step 3 — the relative form. Since $\|J\delta_{\text{GN}}\| \leq \sigma_1 \|\delta_{\text{GN}}\|$ (the largest singular value is the spectral norm), one has $\|\delta_{\text{GN}}\| \geq \|J\delta_{\text{GN}}\|/\sigma_1$ and hence

$$\frac{\|\tilde{\delta}_{\text{GN}} - \delta_{\text{GN}}\|}{\|\delta_{\text{GN}}\|} \leq \frac{\|e\|/\sigma_3}{\|J\delta_{\text{GN}}\|/\sigma_1} = \frac{\sigma_1}{\sigma_3} \cdot \frac{\|e\|}{\|J\delta_{\text{GN}}\|} = \kappa_2(J) \frac{\|e\|}{\|J\delta_{\text{GN}}\|}.$$

□

Theorem 31.8: Conditioning on a symmetric arc. Let $c \in \mathbb{R}^2$, $r > 0$ and $\varphi_1, \dots, \varphi_n$ be angles such that the list $(\varphi_1, \dots, \varphi_n)$ is symmetric about 0 (i.e. counted with multiplicity, $-\varphi$ occurs as many times as φ). Let $p_i = c + r(\cos \varphi_i, \sin \varphi_i)$ and $\theta = (c, r)$. Setting

$$C_1 = \frac{1}{n} \sum_{i=1}^n \cos \varphi_i, \quad C_2 = \frac{1}{n} \sum_{i=1}^n \cos^2 \varphi_i,$$

one has

$$J(\theta)^\top J(\theta) = n M, \quad M = \begin{pmatrix} C_2 & 0 & C_1 \\ 0 & 1 - C_2 & 0 \\ C_1 & 0 & 1 \end{pmatrix},$$

and the eigenvalues of M are

$$\mu_0 = 1 - C_2, \quad \mu_{\pm} = \frac{(1 + C_2) \pm \sqrt{(1 - C_2)^2 + 4C_1^2}}{2},$$

all nonnegative. Consequently $\kappa_2(J(\theta))^2 = \frac{\max\{\mu_0, \mu_+, \mu_-\}}{\min\{\mu_0, \mu_+, \mu_-\}}$.

Proof. Step 1 — the radial unit vectors. By construction $p_i - c = r(\cos \varphi_i, \sin \varphi_i)$ and $\rho_i = \|p_i - c\| = r$, hence $u_i = (\cos \varphi_i, \sin \varphi_i)$ and $\theta \in \Omega$ because $r > 0$.

Step 2 — the sums. By Proposition 31.9,

$$J^\top J = \begin{pmatrix} \sum_i \cos^2 \varphi_i & \sum_i \cos \varphi_i \sin \varphi_i & \sum_i \cos \varphi_i \\ \sum_i \cos \varphi_i \sin \varphi_i & \sum_i \sin^2 \varphi_i & \sum_i \sin \varphi_i \\ \sum_i \cos \varphi_i & \sum_i \sin \varphi_i & n \end{pmatrix}.$$

The function $\varphi \mapsto \sin \varphi$ is odd, and the list of angles is symmetric: pairing φ with $-\varphi$ gives $\sum_i \sin \varphi_i = 0$. Also $\varphi \mapsto \cos \varphi \sin \varphi = \frac{1}{2} \sin 2\varphi$ is odd, hence $\sum_i \cos \varphi_i \sin \varphi_i = 0$. Finally $\sum_i \cos^2 \varphi_i = nC_2$, $\sum_i \sin^2 \varphi_i = \sum_i (1 - \cos^2 \varphi_i) = n(1 - C_2)$ and $\sum_i \cos \varphi_i = nC_1$. Substituting gives $J^\top J = nM$.

Step 3 — the eigenvalues. The vector $e_2 = (0, 1, 0)^\top$ satisfies $Me_2 = (1 - C_2)e_2$, hence $\mu_0 = 1 - C_2$ is an eigenvalue. The orthogonal complement $\text{span}\{e_1, e_3\}$ is invariant under M (the entries $(1, 2)$, $(2, 1)$, $(2, 3)$, $(3, 2)$ are zero), and the restriction has matrix $N = \begin{pmatrix} C_2 & C_1 \\ C_1 & 1 \end{pmatrix}$, with trace $1 + C_2$ and determinant $C_2 - C_1^2$. The eigenvalues of a 2×2 matrix are the roots of $\mu^2 - (\text{tr } N)\mu + \det N = 0$, i.e.

$$\mu_{\pm} = \frac{(1 + C_2) \pm \sqrt{(1 + C_2)^2 - 4(C_2 - C_1^2)}}{2} = \frac{(1 + C_2) \pm \sqrt{(1 - C_2)^2 + 4C_1^2}}{2},$$

having used $(1 + C_2)^2 - 4C_2 = (1 - C_2)^2$.

Step 4 — nonnegativity. $C_2 \leq 1$ because $\cos^2 \leq 1$, hence $\mu_0 \geq 0$. By the Cauchy–Schwarz inequality applied to the vectors $(\cos \varphi_1, \dots, \cos \varphi_n)$ and $(1, \dots, 1)$, $(\sum_i \cos \varphi_i)^2 \leq n \sum_i \cos^2 \varphi_i$, i.e. $C_1^2 \leq C_2$: hence $\det N = C_2 - C_1^2 \geq 0$ and, since also $\text{tr } N = 1 + C_2 \geq 0$, both eigenvalues $\mu_{\pm}$ are ≥ 0 .

Step 5 — the condition number. The eigenvalues of $J^\top J = nM$ are $n\mu_0$, $n\mu_{\pm}$, and coincide with the squares of the singular values of J . The ratio between the largest and the smallest is thus $\kappa_2(J)^2$, and the factor n cancels. $\square$

Corollary 31.6: Degradation on short arcs. Replace in Theorem 31.8 the sample averages with the continuous averages on $[-\alpha, \alpha]$, i.e.

$$C_1 = \frac{1}{2\alpha} \int_{-\alpha}^{\alpha} \cos \varphi \, d\varphi = \frac{\sin \alpha}{\alpha}, \quad C_2 = \frac{1}{2\alpha} \int_{-\alpha}^{\alpha} \cos^2 \varphi \, d\varphi = \frac{1}{2} + \frac{\sin 2\alpha}{4\alpha}.$$

Then, as $\alpha \rightarrow 0^+$,

$$\mu_+ = 2 - \frac{\alpha^2}{3} + O(\alpha^4), \quad \mu_0 = \frac{\alpha^2}{3} + O(\alpha^4), \quad \mu_- = \frac{\alpha^4}{90}(1 + O(\alpha^2)),$$

and hence

$$\kappa_2(J) = \frac{\sqrt{180}}{\alpha^2}(1 + O(\alpha^2)) = \frac{13.4164079}{\alpha^2}(1 + O(\alpha^2)).$$

Proof. Step 1 — the two integrals. $\int_{-\alpha}^{\alpha} \cos \varphi \, d\varphi = 2 \sin \alpha$, whence $C_1 = \sin \alpha / \alpha$. With $\cos^2 \varphi = \frac{1 + \cos 2\varphi}{2}$, $\int_{-\alpha}^{\alpha} \cos^2 \varphi \, d\varphi = \alpha + \frac{\sin 2\alpha}{2}$, whence $C_2 = \frac{1}{2} + \frac{\sin 2\alpha}{4\alpha}$.

Step 2 — Taylor expansions. From $\sin s = s - \frac{s^3}{6} + \frac{s^5}{120} + O(s^7)$ it follows that $\frac{\sin s}{s} = 1 - \frac{s^2}{6} + \frac{s^4}{120} + O(s^6)$. Hence

$$C_1 = 1 - \frac{\alpha^2}{6} + \frac{\alpha^4}{120} + O(\alpha^6),$$

and, with $s = 2\alpha$,

$$C_2 = \frac{1}{2} + \frac{1}{2} \cdot \frac{\sin 2\alpha}{2\alpha} = \frac{1}{2} + \frac{1}{2} \left(1 - \frac{4\alpha^2}{6} + \frac{16\alpha^4}{120} \right) + O(\alpha^6) = 1 - \frac{\alpha^2}{3} + \frac{\alpha^4}{15} + O(\alpha^6).$$

Step 3 — trace and determinant. It follows immediately that $\mu_0 = 1 - C_2 = \frac{\alpha^2}{3} - \frac{\alpha^4}{15} + O(\alpha^6)$ and $\text{tr } N = 1 + C_2 = 2 - \frac{\alpha^2}{3} + O(\alpha^4)$. For the determinant,

$$C_1^2 = \left(1 - \frac{\alpha^2}{6} + \frac{\alpha^4}{120} \right)^2 + O(\alpha^6) = 1 - \frac{\alpha^2}{3} + \left(\frac{1}{60} + \frac{1}{36} \right) \alpha^4 + O(\alpha^6) = 1 - \frac{\alpha^2}{3} + \frac{2\alpha^4}{45} + O(\alpha^6),$$

where $\frac{1}{60} + \frac{1}{36} = \frac{3+5}{180} = \frac{2}{45}$; hence

$$\det N = C_2 - C_1^2 = \frac{\alpha^4}{15} - \frac{2\alpha^4}{45} + O(\alpha^6) = \frac{\alpha^4}{45} + O(\alpha^6).$$

Step 4 — the two eigenvalues of the block. Let $T = \text{tr } N$ and $D = \det N$. Since $T \rightarrow 2$ and $D = O(\alpha^4)$, for small α $4D/T^2 = O(\alpha^4)$ and

$$\sqrt{T^2 - 4D} = T \sqrt{1 - \frac{4D}{T^2}} = T \left(1 - \frac{2D}{T^2} + O(\alpha^8) \right) = T - \frac{2D}{T} + O(\alpha^8).$$

Hence

$$\begin{aligned} \mu_+ &= \frac{T + \sqrt{T^2 - 4D}}{2} = T - \frac{D}{T} + O(\alpha^8) = 2 - \frac{\alpha^2}{3} + O(\alpha^4), \\ \mu_- &= \frac{T - \sqrt{T^2 - 4D}}{2} = \frac{D}{T} + O(\alpha^8). \end{aligned}$$

Substituting $D = \frac{\alpha^4}{45}(1 + O(\alpha^2))$ and $T = 2(1 + O(\alpha^2))$ gives $\mu_- = \frac{\alpha^4}{90}(1 + O(\alpha^2))$.

Step 5 — ordering and condition number. For α small enough $\mu_+ \approx 2$, $\mu_0 \approx \alpha^2/3$ and $\mu_- \approx \alpha^4/90$ are in strictly decreasing order; hence by Theorem 31.8

$$\kappa_2(J)^2 = \frac{\mu_+}{\mu_-} = \frac{2(1 + O(\alpha^2))}{\frac{\alpha^4}{90}(1 + O(\alpha^2))} = \frac{180}{\alpha^4}(1 + O(\alpha^2)),$$

and taking the square root (which preserves expansions of the type $1 + O(\alpha^2)$) $\kappa_2(J) = \frac{\sqrt{180}}{\alpha^2}(1 + O(\alpha^2))$. Finally $\sqrt{180} = 6\sqrt{5} = 13.4164079$. $\square$

Example 31.4: Numerical check of the estimate $\kappa \sim \sqrt{180}\alpha^{-2}$. Three quantities are compared: the *exact* condition number computed with the singular value decomposition of J for $n = 201$ equally spaced points in $[-\alpha, \alpha]$ on a unit circle; the condition number obtained from the formula of Theorem 31.8 with the *continuous* moments of Corollary 31.6; and the asymptotic estimate $\sqrt{180}/\alpha^2$.

α	$\kappa_2(J), n = 201$	$\kappa_2, \text{continuous moments}$	$\sqrt{180}/\alpha^2$	ratio
1.00	12.2382	12.3685	13.4164	0.92189
0.50	51.9205	52.4494	53.6656	0.97734
0.30	146.3455	147.8159	149.0712	0.99158
0.20	330.8339	334.1425	335.4102	0.99622
0.10	1327.1312	1340.3655	1341.6408	0.99905
0.05	5312.3489	5365.2860	5366.5631	0.99976

The last column is the ratio between the third and the fourth: it tends to 1 as $1 + O(\alpha^2)$,

exactly as predicted by Corollary 31.6 (the gap from unity shrinks by a factor of four every time α is halved, from 0.2 downward: 0.00378, 0.00095, 0.00024, in ratio 3.98 and 3.96, i.e. ≈ 4 , as it must be for a term of order α^2). ✓

Independent check. The eigenvalues were computed twice: once with the singular value decomposition routine applied directly to J , the other with the closed-form formulas $\mu_0 = 1 - C_2$ and $\mu_{\pm}$ of Theorem 31.8. For $\alpha = 0.30$ and $n = 201$ the two computations give $\mu_+ = 1.9701541$, $\mu_0 = 0.0297539$, $\mu_- = 9.1990 \cdot 10^{-5}$ with agreement down to the last printed digit; and $\det N = 1.812347 \cdot 10^{-4}$ against $\alpha^4/45 = 1.8 \cdot 10^{-4}$. ✓

Remark 31.12: What $\kappa \approx 1000$ means. For an arc of half-aperture $\alpha = 0.1$ radians (a little less than 6°) the condition number is about $1.3 \cdot 10^3$: by Proposition 31.11, a relative error of 10^{-3} on the residuals can produce a relative error of order unity on the parameters. In double precision this is not yet a problem of arithmetic, but it is a statistical problem: measurement noise, not rounding, is amplified. On sample B of Solved exercise 31.5, with measured half-aperture 0.2828 radians, one has $\kappa_2(J) = 134.9473$, and shifting the ordinate of the endpoint by 0.010 changes the estimated radius by 0.0367: an amplification factor of 3.7 on the radius alone, which becomes much larger on the coordinates of the center.

Proposition 31.12: Stopping criteria: from the gradient to the residual step. Let $\theta \in \Omega$ with $J = J(\theta)$ of rank 3 and singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$. Then

$$\|\delta_{\text{GN}}\| \leq \frac{\|J^\top d\|}{\sigma_3^2} = \frac{\|\nabla F(\theta)\|}{2\sigma_3^2}, \quad \|\nabla F(\theta)\| \leq 2\sigma_1 \|d\|.$$

In particular a stopping criterion on the gradient controls the length of the step still to be taken, and the conversion constant is σ_3^{-2} , i.e. it worsens as the square of the condition number for fixed σ_1 .

Proof. Step 1 — first estimate. With the singular value decomposition of Step 1 of the proof of Proposition 31.10, $(J^\top J)^{-1} = V\Sigma^{-2}V^\top$, whose spectral norm is σ_3^{-2} (the eigenvalues are the σ_j^{-2} and the maximum occurs for the smallest σ_j). Hence $\|\delta_{\text{GN}}\| = \|(J^\top J)^{-1} J^\top d\| \leq \sigma_3^{-2} \|J^\top d\|$. The equality with $\|\nabla F\|/(2\sigma_3^2)$ follows from $\nabla F = 2J^\top d$ (Proposition 31.9).

Step 2 — second estimate. $\|J^\top d\| \leq \|J^\top\|_2 \|d\| = \sigma_1 \|d\|$, because the spectral norm of $J^\top$ is its largest singular value, which coincides with that of J . Multiplying by 2 gives the second estimate. □

Best practice. The four stopping criteria, and why all are needed. The iteration is stopped when *at least one* of the following holds:

1. *small gradient:* $\|J^\top d\| \leq \varepsilon_g (1 + \|d\|)$ — this is the first-order necessary condition (Theorem 31.4), made dimensionless by the factor $1 + \|d\|$;
2. *small step:* $\|\delta\| \leq \varepsilon_\theta (1 + \|\theta\|)$ — useful because, by Proposition 31.12, a small gradient alone does not guarantee a small step if σ_3 is tiny; the converse holds instead, with constant σ_1^2 ;
3. *small relative decrease:* $F(\theta^{(k)}) - F(\theta^{(k+1)}) \leq \varepsilon_F F(\theta^{(k)})$ — it recognizes stalling on ill-conditioned problems, where the step remains large but useless;
4. *maximum number of iterations* — the only guarantee of termination, since we have proved nothing about global convergence (Observation 31.10).

Typical values in double precision: $\varepsilon_g = 10^{-10}$, $\varepsilon_\theta = 10^{-12}$, $\varepsilon_F = 10^{-14}$, 50 iterations. On both numerical samples of the chapter criterion (1) triggers at the third iteration.

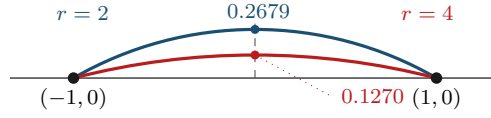


Figure 31.3: Why a short arc is ill-conditioned. The two arcs pass through the same endpoints $(\pm 1, 0)$ and have radii $r = 2$ and $r = 4$, i.e. centers $(0; -\sqrt{3}) = (0; -1.7320508)$ and $(0; -\sqrt{15}) = (0; -3.8729833)$: the sagittas (maximum distance from the chord) equal $2 - \sqrt{3} = 0.2679492$ and $4 - \sqrt{15} = 0.1270167$, and differ by 0.1409325. Doubling the radius moves the arc by a little more than a tenth of the chord's unit length: a perturbation of the data of that order completely changes the estimated parameters. The half-aperture equals $\arcsin(1/2) = 0.5235988$ for the first arc and $\arcsin(1/4) = 0.2526803$ for the second; the estimate of Corollary 31.6 gives respectively $\kappa \approx 48.9$ and $\kappa \approx 210.1$. ✓

31.8. The algebraic fit as a starting point, and how biased it is

Kåsa's method and the Hough transform are already built and proved in Volume I, Chapter 10: there one finds the linearization that turns the algebraic normal equations into a 3×3 linear system in the auxiliary unknowns $u = 2a$, $v = 2b$, $w = r^2 - a^2 - b^2$, the computational cost of the Hough transform and its robustness to outliers, with numerical examples. We repeat none of this. Here we use them for one thing only — to supply the starting point $\theta^{(0)}$ of the iteration of Sections 31.5 and 31.6 — and we prove exactly what separates their minimum from ours.

Definition 31.9: Algebraic residual and algebraic objective function. For a sample P and $\theta = (c, r)$ set

$$g_i(\theta) = \rho_i(c)^2 - r^2, \quad F_{\text{alg}}(\theta) = \sum_{i=1}^n g_i(\theta)^2.$$

F_{alg} is the objective function of the algebraic fit of Volume I, Chapter 10; F of Definition 31.2 is that of the geometric fit. To distinguish them one sometimes writes $F_{\text{geo}} = F$.

Lemma 31.6: Factorization of the algebraic residual. For every $\theta = (c, r)$ with $r \geq 0$ and every i ,

$$g_i(\theta) = d_i(\theta) (\rho_i(c) + r) = d_i(\theta) (2r + d_i(\theta)).$$

In particular, if $r > 0$, then $g_i(\theta) = 0$ if and only if $d_i(\theta) = 0$; and $F_{\text{alg}}(\theta) = 0$ if and only if $F(\theta) = 0$.

Proof. Step 1 — the factorization. By the difference of squares, $\rho_i^2 - r^2 = (\rho_i - r)(\rho_i + r) = d_i(\rho_i + r)$; and $\rho_i + r = (d_i + r) + r = 2r + d_i$.

Step 2 — vanishing. The factor $\rho_i + r$ is a sum of two nonnegative numbers, hence it vanishes if and only if $\rho_i = r = 0$. If $r > 0$ it is therefore $\geq r > 0$, and $g_i = 0$ is equivalent to $d_i = 0$.

Step 3 — the sums. $F_{\text{alg}} = \sum_i g_i^2$ and $F = \sum_i d_i^2$ are sums of squares: they vanish if and only if every term vanishes. By Step 2 (with $r > 0$) the two conditions "all g_i zero" and "all d_i zero" coincide. □

Proposition 31.13: Sandwich estimate between the two functionals. Let $\theta = (c, r)$ with $r > 0$ and suppose there is $\varepsilon \in [0, 2)$ with $|d_i(\theta)| \leq \varepsilon r$ for every i . Then

$$4r^2 \left(1 - \frac{\varepsilon}{2}\right)^2 F(\theta) \leq F_{\text{alg}}(\theta) \leq 4r^2 \left(1 + \frac{\varepsilon}{2}\right)^2 F(\theta).$$

Proof. Step 1 — control of the factor. By Lemma 31.6, $g_i = d_i(2r + d_i)$. From the hypothesis $|d_i| \leq \varepsilon r$ it follows that

$$2r - \varepsilon r \leq 2r + d_i \leq 2r + \varepsilon r, \quad \text{i.e.} \quad 2r \left(1 - \frac{\varepsilon}{2}\right) \leq 2r + d_i \leq 2r \left(1 + \frac{\varepsilon}{2}\right).$$

Since $\varepsilon < 2$, the left-hand side is positive: all three terms are positive and can be squared while preserving the inequalities,

$$4r^2 \left(1 - \frac{\varepsilon}{2}\right)^2 \leq (2r + d_i)^2 \leq 4r^2 \left(1 + \frac{\varepsilon}{2}\right)^2.$$

Step 2 — summing. Multiplying by $d_i^2 \geq 0$ and summing over i ,

$$4r^2 \left(1 - \frac{\varepsilon}{2}\right)^2 \sum_i d_i^2 \leq \sum_i d_i^2 (2r + d_i)^2 = \sum_i g_i^2 = F_{\text{alg}} \leq 4r^2 \left(1 + \frac{\varepsilon}{2}\right)^2 \sum_i d_i^2,$$

which is the claim because $\sum_i d_i^2 = F(\theta)$. $\square$

Corollary 31.7: The algebraic minimum is nearly optimal in the geometric sense too. Let $\theta_A = (c_A, r_A)$ be a minimizer of F_{alg} and $\theta_G = (c_G, r_G)$ a minimizer of F , both with positive radius. If there is $\varepsilon \in [0, 2)$ such that $|d_i(\theta_A)| \leq \varepsilon r_A$ and $|d_i(\theta_G)| \leq \varepsilon r_G$ for every i , then

$$F(\theta_A) \leq \frac{\left(1 + \frac{\varepsilon}{2}\right)^2}{\left(1 - \frac{\varepsilon}{2}\right)^2} \cdot \frac{r_G^2}{r_A^2} F(\theta_G).$$

Proof. Applying Proposition 31.13 at θ_A (left inequality) and at θ_G (right inequality), and using in between the minimality of θ_A for F_{alg} ,

$$4r_A^2 \left(1 - \frac{\varepsilon}{2}\right)^2 F(\theta_A) \leq F_{\text{alg}}(\theta_A) \leq F_{\text{alg}}(\theta_G) \leq 4r_G^2 \left(1 + \frac{\varepsilon}{2}\right)^2 F(\theta_G).$$

Dividing by $4r_A^2(1 - \frac{\varepsilon}{2})^2 > 0$ gives the claim. $\square$

Proposition 31.14: At a fixed center the algebraic radius is never smaller than the geometric one. Let $c \in \mathbb{R}^2$ be fixed, and let

$$r_{\text{geo}}(c) = \frac{1}{n} \sum_{i=1}^n \rho_i(c) = \bar{\rho}(c), \quad r_{\text{alg}}(c) = \sqrt{\frac{1}{n} \sum_{i=1}^n \rho_i(c)^2}$$

be the radii minimizing, respectively, $r \mapsto F(c, r)$ and $r \mapsto F_{\text{alg}}(c, r)$ on $[0, +\infty)$. Then

$$r_{\text{alg}}(c) \geq r_{\text{geo}}(c), \quad r_{\text{alg}}(c)^2 - r_{\text{geo}}(c)^2 = \text{Var}(\rho) := \frac{1}{n} \sum_{i=1}^n (\rho_i(c) - \bar{\rho}(c))^2 = \frac{G(c)}{n},$$

and, if $r_{\text{alg}}(c) + r_{\text{geo}}(c) > 0$,

$$r_{\text{alg}}(c) - r_{\text{geo}}(c) = \frac{\text{Var}(\rho)}{r_{\text{alg}}(c) + r_{\text{geo}}(c)}.$$

Equality $r_{\text{alg}} = r_{\text{geo}}$ holds if and only if all the $\rho_i(c)$ are equal, i.e. if and only if the points are concyclic with center c .

Proof. Step 1 — the algebraic minimum in r . Setting $s = r^2 \geq 0$, $F_{\text{alg}}(c, r) = \sum_i (\rho_i^2 - s)^2$ is a second-degree polynomial in s with leading coefficient $n > 0$; its unique minimum on $\mathbb{R}$ is at the vertex $s^* = \frac{1}{n} \sum_i \rho_i^2$, which is ≥ 0 as a mean of nonnegative quantities. Hence the minimum over $r \geq 0$ is attained at $r = \sqrt{s^*} = r_{\text{alg}}(c)$. That $r_{\text{geo}}(c) = \bar{\rho}(c)$ is the minimum of $F(c, \cdot)$ is Proposition 31.2.

Step 2 — the difference of squares. By the variance identity,

$$r_{\text{alg}}^2 - r_{\text{geo}}^2 = \frac{1}{n} \sum_i \rho_i^2 - \bar{\rho}^2 = \frac{1}{n} \sum_i (\rho_i - \bar{\rho})^2 = \text{Var}(\rho),$$

verifiable by expanding the square: $\frac{1}{n} \sum_i (\rho_i - \bar{\rho})^2 = \frac{1}{n} \sum_i \rho_i^2 - 2\bar{\rho} \cdot \frac{1}{n} \sum_i \rho_i + \bar{\rho}^2 = \frac{1}{n} \sum_i \rho_i^2 - \bar{\rho}^2$. The right-hand side is $\frac{1}{n} G(c)$ by the definition of G (Proposition 31.2).

Step 3 — the sign. $\text{Var}(\rho) \geq 0$ as a sum of squares, hence $r_{\text{alg}}^2 \geq r_{\text{geo}}^2$; being both nonnegative, $r_{\text{alg}} \geq r_{\text{geo}}$.

Step 4 — the difference. If $r_{\text{alg}} + r_{\text{geo}} > 0$,

$$r_{\text{alg}} - r_{\text{geo}} = \frac{r_{\text{alg}}^2 - r_{\text{geo}}^2}{r_{\text{alg}} + r_{\text{geo}}} = \frac{\text{Var}(\rho)}{r_{\text{alg}} + r_{\text{geo}}}.$$

Step 5 — the equality case. $\text{Var}(\rho) = 0$ if and only if $\rho_i = \bar{\rho}$ for every i , i.e. if and only if all the points are equidistant from c : they are concyclic on the circle of center c and radius $\bar{\rho}$. $\square$

Remark 31.13: Two different statements about bias, and what is imported.

Proposition 31.14 is a statement at fixed center: if c is kept fixed and only the radius is optimized, the algebraic radius is always $\geq$ the geometric one. The fact well known in the literature — that Kåsa's algebraic fit systematically underestimates the radius on short arcs, as already observed in Volume I, Chapter 10 — concerns instead the joint minimization in (c, r) : there the center moves, and the shift of the center dominates the effect on the radius. The two statements are not in contradiction, because they speak of two different problems; Example 31.5 shows both on the same data.

The second fact is here **imported** and not proved. What exactly is missing: a statistical noise model (points $p_i = p_i^* + \sigma \xi_i$ with ξ_i independent normal on the true circle), the asymptotic expansion of Kåsa's estimator in powers of σ and the computation of the order- σ^2 term of the expected value of r_{alg} , which is what produces the sign of the bias and its dependence on α^{-4} for arcs of half-aperture α . The complete treatment is in N. Chernov, *Circular and Linear Regression*, CRC Press 2010, chapters "Algebraic circle fits" (pp. 129–166) and "Statistical analysis of circle fits" (pp. 201–236); the experimental comparison between algebraic and geometric fits, with the same conclusion, is in W. Gander, G. H. Golub, R. Strebler, *Least-squares fitting of circles and ellipses*, BIT **34** (1994), pp. 558–578. In this chapter we prove only the three deterministic facts of the section: the factorization (Lemma 31.6), the sandwich (Proposition 31.13) and the fixed-center inequality (Proposition 31.14), and we verify the rest numerically.

Example 31.5: Kåsa's bias in a numerical experiment. We generate $K = 20000$ independent samples of $n = 9$ points on a symmetric arc of half-aperture α of the circle centered at the origin with radius 10, with equally spaced angles and Gaussian noise of standard deviation $\sigma = 0.02$ added independently to each coordinate. For each sample the

radius of the algebraic fit (method of Volume I, Chapter 10) and that of the geometric fit (Gauss–Newton started from the previous one, four iterations) are computed. The averages over the 20 000 samples are:

α (rad)	mean of r_{alg}	mean of r_{geo}	true radius
0.30	9.91880	10.01461	10
0.60	9.99598	10.00172	10
1.00	9.99926	9.99990	10
π (full circle)	10.00004	10.00003	10

On an arc of half-aperture 0.30 (about 17°) the algebraic fit underestimates the radius by 0.81%, while the geometric one overestimates it by 0.15%: the algebraic distortion is more than five times larger. As the arc grows both disappear, and on the full circle the two methods are indistinguishable to the fourth decimal place. ✓

Consistency with Proposition 31.14. The fixed-center inequality holds at any center, even an absurd one. On sample B of Solved exercise 31.5, fixing the center at the centroid $\bar{p}$ (which lies *inside* the arc and is not at all a good center), one gets $r_{\text{alg}}(\bar{p}) = 1.9297218$ and $r_{\text{geo}}(\bar{p}) = 1.6792973$, with difference 0.2504245; and the right-hand side of the formula of Step 4, $\text{Var}(\rho)/(r_{\text{alg}} + r_{\text{geo}})$, equals 0.2504245. ✓ The algebraic radius is thus *larger* at fixed center, and *smaller* when the center is also allowed to move (10.6002914 against 10.6836501, see Solved exercise 31.5): this is exactly the distinction of Observation 31.13.

Proposition 31.15: The algebraic fit gives an admissible starting point. Let P be a sample with $M > 0$ and let $\theta^{(0)} = (c^{(0)}, r^{(0)})$ be any parameter vector with

$$F(\theta^{(0)}) < \lambda_{\min}(S).$$

Then the sublevel set $\Sigma_0 = \{\theta \in \mathbb{R}^2 \times [0, +\infty) : F(\theta) \leq F(\theta^{(0)})\}$ is compact and nonempty, contains a global minimizer of F , and every sequence $\theta^{(k)}$ with $F(\theta^{(k+1)}) \leq F(\theta^{(k)})$ stays in Σ_0 .

Proof. Step 1 — nonempty and closed. $\theta^{(0)} \in \Sigma_0$. Moreover F is continuous on all of $\mathbb{R}^2 \times [0, +\infty)$ (composition of continuous functions: the square root is continuous even at 0), hence Σ_0 is closed in $\mathbb{R}^2 \times [0, +\infty)$, which is itself closed in $\mathbb{R}^3$: Σ_0 is closed in $\mathbb{R}^3$.

Step 2 — the centers are bounded. Let $\theta = (c, r) \in \Sigma_0$. By Proposition 31.2, $G(c) = \min_{r'} F(c, r') \leq F(\theta) \leq F(\theta^{(0)}) < \lambda_{\min}(S)$. By Step 2 of the proof of Theorem 31.3 (applied with $c_0 = c^{(0)}$ and $m = F(\theta^{(0)})$), every c with $G(c) \leq m$ satisfies $\|c - \bar{p}\| \leq t_0$, with t_0 given by the same theorem.

Step 3 — the radii are bounded. Let $\theta = (c, r) \in \Sigma_0$ and let $m = F(\theta^{(0)})$. From $\sum_i (\rho_i(c) - r)^2 \leq m$ it follows in particular that $|\rho_1(c) - r| \leq \sqrt{m}$, hence

$$0 \leq r \leq \rho_1(c) + \sqrt{m} = \|p_1 - c\| + \sqrt{m} \leq \|p_1 - \bar{p}\| + \|c - \bar{p}\| + \sqrt{m} \leq M + t_0 + \sqrt{m},$$

using Step 2 and the triangle inequality.

Step 4 — compactness and conclusion. By Steps 2 and 3, Σ_0 is bounded; by Step 1 it is closed: it is compact. The continuous function F attains a minimum on it, at a point θ^* ; and for every $\theta \notin \Sigma_0$ one has $F(\theta) > F(\theta^{(0)}) \geq F(\theta^*)$, hence θ^* is a global minimizer. Finally, if $F(\theta^{(k+1)}) \leq F(\theta^{(k)})$ for every k , by induction $F(\theta^{(k)}) \leq F(\theta^{(0)})$, i.e. $\theta^{(k)} \in \Sigma_0$. □

Remark 31.14: The hypothesis is checked in a single stroke. Proposition 31.15 turns the algebraic fit into a guarantee: one computes $\theta^{(0)}$ with the method of Volume I, Chapter 10 (a 3×3 linear solution), evaluates $F(\theta^{(0)})$ and compares it with $\lambda_{\min}(S)$ (two eigenvalues of a

2×2 matrix). If the inequality holds, the minimum exists, lies in an explicit compact set, and any monotone method — Gauss–Newton with a line search, or Levenberg–Marquardt with the update rule of Section 31.6 — stays confined to it. On the two samples of the chapter: $F(\theta^{(0)}) = 0.3466706$ against $\lambda_{\min}(S) = 75.7684689$ for sample A, and $F(\theta^{(0)}) = 0.0017857$ against $\lambda_{\min}(S) = 0.2201842$ for sample B. ✓ When instead the points are nearly collinear the two numbers come close, and the check becomes informative rather than a formality.

31.9. Assigned exercises

The three exercises that follow are distinct from the solved ones of Section 31.10. The complete answers are in the Teacher’s key (Section 31.12).

Exercise 31.1: Distance, invariance and degenerate cases. *Goal:* to handle Lemma 31.1 and Theorem 31.1 in both directions, theoretical and numerical. *Estimated time:* 35 minutes. *Type:* proof/numerical.

- Prove that if $p = c$ and $r > 0$ the foot of the projection of p onto $\mathcal{C}(c, r)$ is not unique, and deduce that the function $c \mapsto \rho_i(c)$ is not differentiable at $c = p_i$. Why does this not prevent F from being continuous on all of $\mathbb{R}^3$?
- Consider the sample $P = \{(1, 0), (-1, 0), (0, 2), (0, -2)\}$ of Example 31.3 and the isometry $T(x) = R_{\pi/3}x + (2, -1)$, where $R_{\pi/3}$ is the rotation by $\frac{\pi}{3}$. Compute $T(P)$ to seven decimal places, compute $F_{T(P)}(T_*\theta^*)$ with $\theta^* = (0, 0, \frac{3}{2})$ and verify that it equals 1.
- Repeat with the similarity $T'(x) = 3R_{\pi/3}x + (2, -1)$ and verify that the objective function becomes 9, as predicted by Theorem 31.1. What is the optimal radius of the transformed sample?
- Prove that $\Phi(\theta) = \sum_i |d_i(\theta)|$ is also invariant under isometries, and explain in one line why it is not used in place of F . What would happen if one used $\sum_i d_i(\theta)$ without absolute value or square?

Exercise 31.2: A stationary point that is not the global minimum. *Goal:* to see the normal equations, the degenerate Hessian and the existence criterion at work together. *Estimated time:* 45 minutes. *Type:* proof/numerical. Consider the sample

$$P_C = \{(3, 0), (-3, 0), (0, 1), (0, -1)\}$$

and the parameter vector $\theta_0 = (0, 0, 2)$.

- Verify by hand that θ_0 satisfies the normal equations (31.2) and compute $F(\theta_0)$.
- Compute $\nabla^2 F(\theta_0)$ with Proposition 31.7 and show that it is singular. What is the degenerate direction?
- Verify numerically that $G(a, 0) - G(0, 0)$ behaves like $\frac{3}{4}a^4$ for small a (use $a = 0.05, 0.1, 0.2$), and deduce that θ_0 is a *local* minimum.
- Compute $\lambda_{\min}(S)$ and compare it with $G(0, 0)$. Is the hypothesis of Theorem 31.3 satisfied? Compute $G(0, t)$ for $t = 10, 50, 200, 1000$ and deduce that θ_0 is *not* a global minimum. What is the limiting line of Observation 31.5?

Exercise 31.3: Conditioning of an arc, by hand and with the asymptotic formula.

Goal: to use Theorem 31.8 in exact form and understand when the asymptotic estimate of Corollary 31.6 is reliable. *Estimated time:* 40 minutes. *Type:* numerical computation. Take $n = 7$ points on the unit circle at equally spaced angles in $[-0.4; 0.4]$, i.e. $\varphi_k = -0.4 + \frac{0.8k}{6}$, $k = 0, \dots, 6$.

- (a) Compute C_1 and C_2 to seven decimal places, and verify that the list of angles is symmetric about 0 as required by the theorem.
- (b) Compute μ_0 , μ_+ , μ_- and $\kappa_2(J)$ with the closed-form formulas, and check the result with a singular value decomposition of J .
- (c) Compare with the estimate $\sqrt{180}/\alpha^2$ and with the value obtained from the continuous moments $C_1 = \sin \alpha / \alpha$, $C_2 = \frac{1}{2} + \frac{\sin 2\alpha}{4\alpha}$. Why does the sample with $n = 7$ give a *smaller* condition number than the continuous value? (Hint: compare the two values of $C_2 - C_1^2$.)
- (d) Using Proposition 31.11, estimate by how much the Gauss–Newton step can change if the residuals are perturbed by 10^{-3} in norm, assuming $\|J\delta_{\text{GN}}\| = 10^{-2}$.

31.10. Solved exercises

Exercise 31.4: From the algebraic fit to the geometric fit on six points. *Type:* complete numerical computation with three independent checks. Consider the sample

$$P_A = \{(8; 2), (5; 6.9), (-0.1; 5.4), (-1.9; 0.1), (1.2; -2.7), (6.7; -2.1)\}.$$

Compute the geometric least-squares circle, starting from the algebraic fit of Volume I, Chapter 10, and verify the result.

Solution. Strategy. The algebraic fit costs a single 3×3 linear solve and supplies a starting point; then the Gauss–Newton iteration (Definition 31.6) refines it, and the normal equations (31.2) give the final check. First of all, though, we check with Theorem 31.3 that a minimum exists.

Step 1 — centroid and scatter matrix. $\sum_i x_i = 18.9$ and $\sum_i y_i = 9.6$, so $\bar{p} = (3.15; 1.6)$. The deviations $w_i = p_i - \bar{p}$ give

$$S = \begin{pmatrix} 79.415 & 2.22 \\ 2.22 & 77.12 \end{pmatrix},$$

with $\text{tr } S = 156.535$ and $\det S = 79.415 \cdot 77.12 - 2.22^2 = 6124.4848 - 4.9284 = 6119.5564$. The eigenvalues are the roots of $\mu^2 - 156.535\mu + 6119.5564 = 0$:

$$\mu = \frac{156.535 \pm \sqrt{156.535^2 - 4 \cdot 6119.5564}}{2} = \frac{156.535 \pm \sqrt{24.980625}}{2} = \frac{156.535 \pm 4.9980621}{2},$$

i.e. $\lambda_{\min}(S) = 75.7684689$ and $\lambda_{\max}(S) = 80.7665311$ (sum 156.535 ✓, product 6119.5564 ✓).

Step 2 — the algebraic fit. With $z_i = x_i^2 + y_i^2$ one has $z = (68; 72.61; 29.17; 3.62; 8.73; 49.3)$ and

$$A^T A = \begin{pmatrix} 138.95 & 32.46 & 18.9 \\ 32.46 & 92.48 & 9.6 \\ 18.9 & 9.6 & 6 \end{pmatrix}, \quad A^T z = \begin{pmatrix} 1238.041 \\ 667.788 \\ 231.43 \end{pmatrix},$$

whence $(u, v, w) = (6.3070658; 3.6760673; 12.8227016)$ and

$$a^{(0)} = \frac{u}{2} = 3.1535329, \quad b^{(0)} = \frac{v}{2} = 1.8380337,$$

$$r^{(0)} = \sqrt{w + a^2 + b^2} = \sqrt{26.1458396} = 5.1133002.$$

The value of the *geometric* objective function at this point is $F(\theta^{(0)}) = 0.3466706$.

Step 3 — the minimum exists. $F(\theta^{(0)}) = 0.3466706 < 75.7684689 = \lambda_{\min}(S)$: by Theorem 31.3 and Proposition 31.15 the minimum exists, lies in a compact set and the monotone iteration stays in it.

Step 4 — the Gauss–Newton iterations.

k	$a^{(k)}$	$b^{(k)}$	$r^{(k)}$	$F(\theta^{(k)})$	$\ \nabla F\ $
0	3.1535329	1.8380337	5.1133002	0.346670569	$7.76 \cdot 10^{-2}$
1	3.1535637	1.8310288	5.1073286	0.346334745	$5.03 \cdot 10^{-4}$
2	3.1535025	1.8309733	5.1073286	0.346334724	$6.17 \cdot 10^{-6}$
3	3.1535022	1.8309723	5.1073285	0.346334724	$7.93 \cdot 10^{-8}$
4	3.1535022	1.8309722	5.1073285	0.346334724	$1.01 \cdot 10^{-9}$

The stopping criterion on the gradient triggers at the third iteration. The solution is

$$a^* = 3.1535022, \quad b^* = 1.8309722, \quad r^* = 5.1073285$$

with $F(\theta^*) = 0.3463347$ and root-mean-square residual $\sqrt{F/n} = 0.2402550$.

Independent check 1: the normal equations. The final residuals are

$$d = (-0.2578841; 0.2875391; -0.2779149; 0.2344080; -0.1731741; 0.1870260),$$

with $\sum_i d_i = 2.7 \cdot 10^{-15}$ and $\sum_i d_i u_i = (-3.6 \cdot 10^{-12}; -5.5 \cdot 10^{-12})$: the three equations (31.2) are satisfied up to machine error. ✓ Moreover $r^* - \bar{\rho}(c^*) = 8.9 \cdot 10^{-16}$ and $c^* - (\bar{p} - \frac{r^*}{n} \sum_i u_i) = (6.7 \cdot 10^{-15}; 1.2 \cdot 10^{-14})$, i.e. the fixed-point form (31.3). ✓

Independent check 2: a derivative-free method. Minimizing only the function G of Proposition 31.2 with the Nelder–Mead method (which uses neither gradient nor Jacobian), starting from (3; 2), one obtains in 57 steps $c = (3.1535022; 1.8309723)$, $G = 0.346334724$ and $\bar{\rho} = 5.1073285$: the same seven decimal places. ✓

Independent check 3: the conditioning. The singular values of $J(\theta^*)$ are 2.4543455, 1.7538045, 1.7030437, hence $\kappa_2(J) = 1.4412$: the problem is *well* conditioned, as it must be for points distributed over the whole circle. Consistently, the convergence is immediate.

Interpretation. The algebraic fit misses the center by $\|c^{(0)} - c^*\| = 0.0070616$ and the radius by 0.0059717, i.e. about 0.12% of the radius; the geometric fit reduces the sum of squares from 0.3466706 to 0.3463347, i.e. by 0.097%. This is the order of magnitude predicted by Corollary 31.7: with $\varepsilon = \max_i |d_i|/r = 0.0565353$ the upper bound is 1.1171, and the true ratio is 1.0009697.

Typical errors. (1) Confusing F and F_{alg} : the minimizer of F_{alg} is $\theta^{(0)}$, not θ^* , and indeed $F_{\text{alg}}(\theta^{(0)}) = 36.1107472 < 36.1461246 = F_{\text{alg}}(\theta^*)$; the two rankings are reversed with respect to F , as they must be. (2) Forgetting the minus sign in the rows of the Jacobian (Proposition 31.9): the step changes direction and the iteration diverges. (3) Stopping at the algebraic fit when the required tolerance is tighter than the bias: here the bias on the radius is $6 \cdot 10^{-3}$, and if the tolerance were 10^{-3} the algebraic fit alone would not suffice.

Extension. Applying to the sample the similarity of ratio $\lambda = 10$ and then redoing the fit, by Theorem 31.1 one must find $F = 100 \cdot 0.3463347 = 34.63347$ and parameters multiplied by 10: it is a convenient way to check that an implementation contains no hidden dimensional constants.

Exercise 31.5: A short arc, Levenberg–Marquardt and the price of ill-conditioning. *Type:* numerical computation with damping and sensitivity analysis. Con-

sider the sample of nine points on a short arc

$$P_B = \left\{ \begin{array}{ccc} (9.563; -2.963) & (9.764; -2.249) & (9.887; -1.477) \\ (9.992; -0.721) & (9.997; 0.010) & (9.945; 0.748) \\ (9.876; 1.477) & (9.738; 2.238) & (9.574; 3.001) \end{array} \right\}.$$

Compute the geometric least-squares circle with Levenberg–Marquardt starting from the *deliberately wrong* point $\theta^{(0)} = (0, 0, 10)$, measure its conditioning and discuss its sensitivity.

Solution. Strategy. On a short arc the pure Gauss–Newton step, starting far away, is enormous; the damping of Definition 31.7 shortens it as long as the objective function decreases, and then lets it lengthen again (Proposition 31.10).

Step 1 — existence. The centroid is $\bar{p} = (9.8151111; 0.0071111)$ and the scatter matrix

$$S = \begin{pmatrix} 0.2203729 & -0.0790031 \\ -0.0790031 & 33.2940629 \end{pmatrix}$$

has eigenvalues 0.2201842 and 33.2942516 (sum 33.5144358 = $\text{tr } S$ ✓): hence $\lambda_{\min}(S) = 0.2201842$. The algebraic fit of Volume I, Chapter 10, gives

$$\theta_{\text{alg}}^{(0)} = (-0.6080292; -0.0151995; 10.6002914), \quad F = 0.0017857,$$

and $0.0017857 < 0.2201842$: by Proposition 31.15 the minimum exists.

Step 2 — Levenberg–Marquardt from $\theta^{(0)} = (0, 0, 10)$, $\lambda_0 = 1$. Rule: step accepted $\Rightarrow \lambda \leftarrow \lambda/10$; step rejected $\Rightarrow \lambda \leftarrow 10\lambda$.

k	λ	$a^{(k)}$	$b^{(k)}$	$r^{(k)}$	$F(\theta^{(k)})$
0	10^0	0.000000	0.000000	10.000000	0.0029633619
1	10^{-1}	0.000556	-0.003469	10.002209	0.0028567346
2	10^{-2}	-0.007478	-0.011510	10.010258	0.0027961493
3	10^{-3}	-0.079458	-0.014013	10.080908	0.0025793176
4	10^{-4}	-0.398338	-0.014807	10.393951	0.0019472930
5	10^{-5}	-0.660562	-0.015374	10.651829	0.0017738768
6	10^{-6}	-0.692368	-0.015426	10.683221	0.0017717622
7	10^{-7}	-0.692801	-0.015424	10.683649	0.0017717618

No step is rejected: λ decreases by a factor of 10 at each iteration, i.e. the method progressively passes from the gradient to Gauss–Newton, exactly as described by Theorem 31.7. The solution is

$$a^* = -0.6928022, \quad b^* = -0.0154242, \quad r^* = 10.6836501$$

with $F(\theta^*) = 0.0017717618$ and root-mean-square residual 0.0140308.

Independent check 1: Gauss–Newton from the algebraic fit. Starting instead from $\theta_{\text{alg}}^{(0)}$ and using the undamped step one gets

$$\begin{array}{ll} k = 0 : & (-0.608029; -0.015199; 10.600291) \quad F = 0.0017857210 \\ k = 1 : & (-0.692104; -0.015428; 10.682952) \quad F = 0.0017717638 \\ k = 2 : & (-0.692802; -0.015424; 10.683650) \quad F = 0.0017717618 \end{array}$$

with $\|\nabla F\| = 1.97 \cdot 10^{-11}$ at the third iteration: the same seven decimal places as the Levenberg–Marquardt table. ✓

Independent check 2: Nelder–Mead on G . Starting from $(0; 0)$, the derivative-free method converges in 104 steps to $c = (-0.692802; -0.015424)$ with $G = 0.0017717618$ and $\bar{\rho} = 10.683650$. ✓

Step 3 — the conditioning. The singular values of $J(\theta^*)$ are 4.2080080, 0.5400890, 0.0311826, hence

$$\kappa_2(J) = \frac{4.2080080}{0.0311826} = 134.9473.$$

The angles of the nine points as seen from the estimated center range from -0.27986 to 0.28576 radians, i.e. a half-aperture $\alpha = 0.28281$. Theorem 31.8 applied to nine equally spaced points on an arc of that half-aperture gives $\kappa_2 = 135.4666$: it differs by 0.4% from the observed value, a difference due to the fact that the real points are noisy and not exactly equally spaced. ✓ The asymptotic estimate of Corollary 31.6, $\sqrt{180}/\alpha^2 = 167.7$, overestimates by 24%: with only nine points the asymptotic regime is not yet reached (see also Assigned exercise 31.3).

Step 4 — sensitivity. Shifting the ordinate of a single point by $+0.010$ and redoing the fit:

shifted point	a	b	r
none	−0.692802	−0.015424	10.683650
the first, (9.563; −2.963)	−0.661303	−0.012763	10.652358
the fifth, (9.997; 0.010)	−0.692560	−0.015404	10.683415
the ninth, (9.574; 3.001)	−0.729840	−0.012745	10.720388

A perturbation of 0.010 on an endpoint changes the radius by 0.0367: a factor of 3.7. The same perturbation on the *central* point changes it by 0.00024, i.e. about one hundred fifty times less: it is the points at the ends of the arc that determine the curvature, and this is why on a short arc the problem is ill-conditioned (Proposition 31.11).

Interpretation. The data were generated on a circle of radius 10 with noise of standard deviation 0.02 per coordinate — a relative noise of 0.2% — and the fit returns $r^* = 10.68$, i.e. an error of 6.8%: an amplification of about 34. This is not a defect of the algorithm, it is the conditioning of the problem: any method, applied to these nine points, must give the same result, because it is the minimum of the objective function. To tighten the radius one must lengthen the arc or increase the number of points, not change the optimizer.

Typical errors. (1) Interpreting the difference between 10.68 and 10 as a programming error. (2) Increasing the number of iterations hoping for a better result: the gradient is already 10^{-11} . (3) Using the algebraic fit and believing that its smaller radius (10.6003) is more accurate: this is the distortion of Observation 31.13, which here by chance moves closer to the true value but for the wrong reason, and which over the set of replicates of Example 31.5 turns out to be systematic and larger than the geometric one.

Extension. Repeating with the same angles but doubling the half-aperture ($\alpha = 0.6$) and the same noise, the condition number drops by about a factor of four and the mean radius bias goes from -0.081 to -0.004 for the algebraic fit (Example 31.5): doubling the arc is worth more than quadrupling the number of points.

Exercise 31.6: Does the minimum exist? Five nearly collinear points. *Type:* application of the existence criterion, with numerical check. Consider

$$P_Q = \{(0; 0), (1; 0.05), (2; 0.02), (3; 0.09), (4; 0)\}.$$

Determine whether the geometric fit admits a minimum and, if so, compute it.

Solution. Strategy. The points are nearly collinear: by Corollary 31.4, if they were exactly collinear the minimum would not exist. The decisive criterion is the comparison between $\lambda_{\min}(S)$ and a value of G actually attained (Theorem 31.3).

Step 1 — the scatter matrix. The centroid is $\bar{p} = (2; 0.032)$, and

$$S = \begin{pmatrix} 10 & 0.04 \\ 0.04 & 0.00588 \end{pmatrix}.$$

The characteristic polynomial is $\mu^2 - 10.00588\mu + (0.0588 - 0.0016) = 0$, i.e. $\mu^2 - 10.00588\mu + 0.0572 = 0$, with roots

$$\lambda_{\min}(S) = 0.0057199, \quad \lambda_{\max}(S) = 10.0001601.$$

Check: sum $10.00588 = \text{tr } S$ ✓, product $0.0572000 = \det S$ ✓. Moreover $M = \max_i \|w_i\| = 2.0002560$.

Step 2 — the algebraic starting point. The method of Volume I, Chapter 10 gives

$$\theta^{(0)} = (2.0630466; -15.6916434; 15.7872769), \quad F(\theta^{(0)}) = 0.0083910.$$

Warning: $0.0083910 > 0.0057199 = \lambda_{\min}(S)$. The criterion of Theorem 31.3 is *not* yet satisfied at this point: nothing can be concluded, either way.

Step 3 — Gauss–Newton.

k	$a^{(k)}$	$b^{(k)}$	$r^{(k)}$	$F(\theta^{(k)})$
0	2.063047	-15.691643	15.787277	0.008390970
1	2.100540	-25.101450	25.159347	0.005084032
2	2.136008	-33.991682	34.049679	0.003517934
3	2.152962	-38.234580	38.292610	0.003405368
4	2.155353	-38.830564	38.888597	0.003403967
5	2.155393	-38.840256	38.898289	0.003403966

At the fifth iteration $\|\nabla F\| = 2.26 \cdot 10^{-8}$ and at the sixth $2.7 \cdot 10^{-13}$. The solution is

$$a^* = 2.1553928, \quad b^* = -38.8402627, \quad r^* = 38.8982963, \quad F(\theta^*) = 0.0034040.$$

Step 4 — now the criterion applies. $G(c^*) = F(\theta^*) = 0.0034040$ (the radius is already optimal, by Step 2 of Theorem 31.4), and

$$0.0034040 < 0.0057199 = \lambda_{\min}(S).$$

By Theorem 31.3 the minimum *exists* and every minimizer lies in the disc of center $\bar{p}$ and radius $t_0 = \max\{8M, 24nM^3/(\lambda_{\min} - G(c^*))\} = 414677.3$. The center found is 38.8726 from the centroid: the bound t_0 is true but very loose, as is typical of bounds obtained from an asymptotic expansion.

Independent check 1: behavior at infinity. Theorem 31.2 predicts $G(\bar{p} + tu) \rightarrow u^\top Su$. Taking $u = (0, \pm 1)$ (nearly the eigenvector of $\lambda_{\min}$, because S is nearly diagonal):

$$G(2; 10^4) = 0.0058980, \quad G(2; -10^4) = 0.0058620,$$

both close to $\lambda_{\min}(S) = 0.0057199$ and *larger* than $G(c^*) = 0.0034040$. ✓ The minimum thus does not escape to infinity.

Independent check 2: the residuals. The final residuals are

$$d = (0.0017259; 0.0091255; -0.0377228; 0.0411274; -0.0142560),$$

with $\sum_i d_i = 1.5 \cdot 10^{-16}$ and $\sum_i d_i^2 = 0.0034040 = F(\theta^*)$. ✓

Step 5 — the conditioning is terrible. The singular values of $J(\theta^*)$ are 3.1612324, 0.0812967, 0.0008757, hence $\kappa_2(J) = 3610.0$. The optimal circle has radius almost 39 for

a sample of width 4: it is an arc of half-aperture $\approx 4/(2 \cdot 38.9) = 0.051$ radians, and the estimate $\sqrt{180}/\alpha^2 \approx 5100$ is of the same order of magnitude as the true value 3610. ✓

Interpretation. The sample is a hair's breadth from the pathological case: it suffices to bring the ordinates to zero for the minimum to disappear (Corollary 31.4). The margin $\lambda_{\min}(S) - G(c^*) = 0.0023159$ is the exact measure of how much the sample is "more circular than straight"; as it tends to zero, the optimal radius diverges and the condition number explodes.

Typical errors. (1) Concluding "the points are nearly collinear, hence the minimum does not exist": near-collinearity is not enough, the numerical comparison is needed. (2) Stopping at Step 2 and declaring that the criterion fails: the criterion must be applied to the *best* point available, not the first one. (3) Trusting the digits of the center: with $\kappa = 3610$ the last digits of b^* are not significant relative to a perturbation of the data at the third decimal place.

Extension. Replacing the ordinate 0.09 of the fourth point with 0.045 gives a sample closer to alignment; recomputing, $\lambda_{\min}(S)$ decreases and the margin narrows. The critical value of the ordinate at which the margin vanishes is the boundary between samples that admit an optimal circle and those that do not.

31.11. Self-assessment quiz

- (Definition) The *signed geometric residual* of the point p_i with respect to the circle $\mathcal{C}(c, r)$ is
 A) $\rho_i^2 - r^2$ B) $\rho_i - r$ C) $|\rho_i - r|$ D) $(\rho_i - r)^2$
- (Hypothesis) In Theorem 31.5 the hypothesis rank $J = 3$ is needed to
 A) guarantee that the points are not collinear B) make $J^\top J$ positive definite, so that δ_{GN} exists and $g^\top (J^\top J)^{-1} g > 0$ C) ensure quadratic convergence D) guarantee the existence of the minimum of F
- (Symbolic computation) The i -th row of the Jacobian $J(\theta) = \partial d / \partial \theta$ equals
 A) $(u_i^\top \ 1)$ B) $-(u_i^\top \ 1)$ C) $(2(x_i - a) \ 2(y_i - b) \ -2r)$ D) $(u_i^\top \ 0)$
- (Numerical computation) For a symmetric arc of half-aperture $\alpha = 0.20$, with the continuous moments of Corollary 31.6, the condition number $\kappa_2(J)$ equals about
 A) 33.5 B) 83.9 C) 334.1 D) 1341.6
- (Proof step) In the proof of Theorem 31.2 the term $u^\top S u$ appears because
 A) it is the trace of the scatter matrix B) the expansion gives $\rho_i - \bar{\rho} = -\langle u, w_i \rangle + O(t^{-1})$ and $\sum_i \langle u, w_i \rangle^2 = u^\top S u$ C) the optimal radius tends to t D) S is positive semidefinite
- (Error diagnosis) On a short arc a student finds radius 10.60 with the algebraic fit and 10.68 with the geometric one; knowing that the true radius is 10, they conclude that the algebraic method is more accurate. The error is
 A) the geometric radius is computed wrongly B) drawing from a single sample a conclusion about average behavior: the algebraic distortion is systematic and, over the replicates of Example 31.5, five times larger than the geometric one C) the two methods minimize the same function, hence the radii should coincide D) the true radius is not defined for noisy data

7. (Transfer) For five points that are *exactly* collinear and pairwise distinct, the geometric fit
- A) has a minimum with $r = 0$ B) has a unique minimum C) has no minimum:
 $\inf F = 0$ is not attained D) has infinitely many minima
8. (Source and metacognition) The statistical analysis of the bias of algebraic fits, which this chapter declares imported, is found in
- A) N. Chernov, *Circular and Linear Regression*, chapter "Statistical analysis of circle fits", pp. 201–236 B) J. Nocedal, S. J. Wright, *Numerical Optimization*, Chapter 3 "Line Search Methods", pp. 30–65 C) I. Kåsa, IEEE Trans. Instrum. Meas. **IM-25** (1976), pp. 8–14 D) W. Gander, G. H. Golub, R. Strebel, BIT **34** (1994), pp. 558–578

31.12. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Definition 31.2: the *signed* residual is $d_i = \rho_i - r$, positive outside and negative inside. *Distractors:* A is the *algebraic* residual (Definition 31.9); C is the distance, i.e. $|d_i|$ (Lemma 31.1); D is its square, the term of F . *Diagnosed error:* confusing the residual, the distance and the term of the objective function. *Source:* Chernov, *Circular and Linear Regression*, chapter "Fitting circles: Theory", pp. 77–98. *Prerequisite:* point-to-circle distance. *Remediation:* redo Figure 31.1 with an interior and an exterior point and write down the four numbers ρ , r , d , $|d|$.

2) **B.** *Rationale:* Step 1 of Lemma 31.5 uses the full rank *only* to conclude that $\ker J = \{0\}$, and hence that $J^\top J$ is positive definite and invertible; this is what makes δ_{GN} well defined and the scalar product of Theorem 31.5 strictly negative. *Distractors:* A is false, because the rank depends on the *current* center and not only on the data (Lemma 31.4); C concerns a result that is not proved here (Observation 31.10); D is Theorem 31.3, which does not use the rank of the Jacobian. *Diagnosed error:* attributing a global role to a technical hypothesis. *Source:* Nocedal–Wright, Chapter 10, pp. 245–269. *Prerequisite:* positive definite matrices. *Remediation:* build a sample with all points on only two rays issuing from c and verify that $\det J^\top J = 0$.

3) **B.** *Rationale:* Proposition 31.9: from $d_i = \rho_i - r$ and $\partial \rho_i / \partial c = -u_i^\top$ it follows that $J_{i\bullet} = (-u_i^\top, -1)$. *Distractors:* A forgets the signs (and produces an ascent step); C is the gradient of the *algebraic* residual $\rho_i^2 - r^2$; D forgets the derivative with respect to r , i.e. treats the radius as constant. *Diagnosed error:* differentiating a square root without the sign of the denominator. *Source:* Gander–Golub–Strebel, BIT **34** (1994), pp. 558–578, §2. *Prerequisite:* derivative of the Euclidean norm. *Remediation:* redo Step 2 of Proposition 31.5 in coordinates.

4) **C.** *Rationale:* the table of Example 31.4 gives 334.1425 for $\alpha = 0.20$; the asymptotic estimate $\sqrt{180}/\alpha^2 = 335.4102$ confirms it to about 1%. *Distractors:* A is the value for an arc with $\alpha \approx 0.63$; B is the value for $\alpha = 0.40$; D is the value for $\alpha = 0.10$, i.e. one has made a factor-of-two error on the aperture, which becomes a factor of four on the conditioning. *Diagnosed error:* not remembering that the dependence is in α^{-2} . *Source:* Corollary 31.6. *Prerequisite:* Taylor expansions of $\sin s/s$. *Remediation:* recompute the table for $\alpha = 0.25$ and check that the value falls between those for 0.20 and 0.30.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **B.** *Rationale:* in Step 1 of the proof of Theorem 31.2 the mean $\bar{\rho}$ absorbs the term t and, since $\sum_i w_i = 0$, what remains is $\rho_i - \bar{\rho} = -\langle u, w_i \rangle + B_i$ with $B_i = O(t^{-1})$; Step 2 squares and sums, and Lemma 31.2 recognizes $\sum_i \langle u, w_i \rangle^2$ as $u^\top S u$. *Distractors:* A confuses the quadratic form with the trace; C is true but does not explain the appearance of S ; D is a property of S , not a step of the proof. *Diagnosed error:* skipping the point where the mean removes the dominant term. *Source:* Chernov, chapter "Fitting circles: Theory", pp. 77–98. *Prerequisite:* Rayleigh quotient. *Remediation:* redo the computation with $n = 3$ explicit points and $t = 100$.

6) **B.** *Rationale:* a single sample says nothing about the bias, which is a property of the *mean* over the space of samples; Example 31.5 shows over 20 000 replicates that the algebraic fit underestimates on average by 0.81% and the geometric one overestimates by 0.15%. *Distractors:* A is ruled out by the three independent checks of Solved exercise 31.5; C is false, the objective functions are different (Observation 31.4); D is false, the data are generated from a known circle. *Diagnosed error:* confusing the error on one sample with the bias. *Source:* Chernov, chapters "Algebraic circle fits", pp. 129–166, and "Statistical analysis of circle fits", pp. 201–236, declared imported in Observation 31.13. *Prerequisite:* notion of a biased estimator. *Remediation:* rerun the experiment with $\alpha = 1$ and observe that the bias disappears.

7) **C.** *Rationale:* this is Corollary 31.4: with at least three distinct collinear points $\lambda_{\min}(S) = 0$, hence $\inf G = 0$ by Corollary 31.3, but $G(c) > 0$ for every c because a line meets a circle in at most two points. *Distractors:* A would give $F = \sum_i \rho_i^2 > 0$, which is not the infimum; B and D presuppose that the minimum exists. *Diagnosed error:* assuming that a continuous nonnegative function always attains its infimum on $\mathbb{R}^2$. *Source:* Chernov, chapter "Fitting circles: Theory", pp. 77–98. *Prerequisite:* Weierstrass's theorem and its need for compactness. *Remediation:* redo Example 31.2 with seven collinear points.

8) **A.** *Rationale:* this is exactly the source cited in Observation 31.13. *Distractors:* B is the source for the line search, not for the bias; C is the original paper of the algebraic method, which contains an error analysis but not the σ^2 asymptotics of the bias; D is the experimental comparison between methods, cited as confirmation but not as proof. *Diagnosed error:* not distinguishing the source of a method from the source of its statistical analysis. *Source:* see the *Sources* environment at the end of the chapter. *Prerequisite:* critical reading of a bibliography. *Remediation:* list which statements of the chapter are proved and which are imported.

BEST PRACTICE

Assigned exercise 31.1 — answers.

(a) This is Step 5 of Lemma 31.1: for $p = c$ the function $q \mapsto \|p - q\|$ is constant $= r$ on $\mathcal{C}(c, r)$, so every point is a minimizer. The non-differentiability of $c \mapsto \rho_i(c) = \|p_i - c\|$ at $c = p_i$ is seen at once: along the line $c = p_i + s v$ with $\|v\| = 1$ one has $\rho_i = |s|$, which at $s = 0$ has right derivative $+1$ and left derivative -1 . Nonetheless F remains *continuous* on all of $\mathbb{R}^3$, because it is a composition of continuous functions: the norm and the square root are continuous even where they are not differentiable. This is why Theorem 31.3 and Proposition 31.15 can use Weierstrass without regularity hypotheses, while the normal equations require $\theta \in \Omega$.

(b) With $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} = 0.8660254$:

$$T(P) = \{(2.5; -0.1339746), (1.5; -1.8660254), (0.2679492; 0), (3.7320508; -2)\},$$

and $T_*\theta^* = ((2; -1), \frac{3}{2})$. The four distances from the new center are $\|(0.5; 0.8660254)\| = 1$,

$\|(-0.5; -0.8660254)\| = 1$, $\|(-1.7320508; 1)\| = 2$, $\|(1.7320508; -1)\| = 2$: the same as before, hence $d = (-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $F_{T(P)}(T_*\theta^*) = 4 \cdot \frac{1}{4} = 1$. ✓

(c) With $\lambda = 3$:

$$T'(P) = \{(3.5; 1.5980762), (0.5; -3.5980762), (-3.1961524; 2), (7.1961524; -4)\},$$

center $(2; -1)$, optimal radius $3 \cdot \frac{3}{2} = \frac{9}{2} = 4.5$. The distances are 3, 3, 6, 6, the residuals $\pm \frac{3}{2}$ and $F = 4 \cdot \frac{9}{4} = 9 = 3^2 \cdot 1$. ✓ The optimal radius of the transformed sample is 4.5, i.e. λ times the original one, as in Definition 31.3.

(d) Φ is invariant because each $|d_i|$ is: in Step 2 of the proof of Theorem 31.1 one has $d_i^T = \lambda d_i$, hence $|d_i^T| = \lambda |d_i|$ and for $\lambda = 1$ nothing changes. It is not used because $|d_i|$ is not differentiable where $d_i = 0$, i.e. exactly at the points that lie on the candidate circle: there are no normal equations and Gauss–Newton cannot be applied (the criterion Φ is the fit in ℓ^1 norm, robust but not smooth). Using instead $\sum_i d_i$ without absolute value would be senseless: keeping c fixed and sending $r \rightarrow +\infty$ one has $\sum_i d_i = \sum_i \rho_i - nr \rightarrow -\infty$, so the function is unbounded below and has no minimum.

BEST PRACTICE

Assigned exercise 31.2 — answers.

(a) $\rho = (3, 3, 1, 1)$, $\bar{\rho} = \frac{3+3+1+1}{4} = 2 = r$: the first normal equation is satisfied, and $d = (1, 1, -1, -1)$, $\sum_i d_i = 0$. The unit vectors are $u = (1, 0), (-1, 0), (0, 1), (0, -1)$ and $\sum_i d_i u_i = (1, 0) + (-1, 0) - (0, 1) - (0, -1) = (0, 0)$: the second holds too. Hence θ_0 is stationary and $F(\theta_0) = 1 + 1 + 1 + 1 = 4$.

(b) $\sum_i u_i u_i^\top = \text{diag}(2, 2)$ and $\sum_i u_i = 0$; moreover

$$\sum_i \frac{d_i}{\rho_i} (I - u_i u_i^\top) = 2 \cdot \frac{1}{3} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + 2 \cdot (-1) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & \frac{2}{3} \end{pmatrix},$$

whence, by Proposition 31.7, $\nabla^2 F(\theta_0) = \text{diag}(0, \frac{16}{3}, 8)$. It is singular, and the degenerate direction is that of the first coordinate of the center, i.e. a : moving the center along the x -axis leaves the objective function unchanged to second order.

(c) $G(a, 0) - G(0, 0)$ equals $4.6816498 \cdot 10^{-6}$ for $a = 0.05$, $7.4627327 \cdot 10^{-5}$ for $a = 0.1$ and $1.1765837 \cdot 10^{-3}$ for $a = 0.2$, against $\frac{3}{4}a^4 = 4.6875 \cdot 10^{-6}$, $7.5 \cdot 10^{-5}$ and $1.2 \cdot 10^{-3}$: the agreement is to 0.1%, 0.5% and 2%, consistent with a next-order term in a^6 . ✓ Since the difference is positive in every direction tried (and $G(0, t) > 4$ for small t), θ_0 is a *local* minimum, but the growth is fourth order, not second: this is the signature of the singular Hessian of part (b).

(d) $\bar{p} = (0, 0)$ and $S = \text{diag}(18, 2)$, hence $\lambda_{\min}(S) = 2$. Since $G(0, 0) = 4 > 2 = \lambda_{\min}(S)$, the hypothesis of Theorem 31.3 is *not* satisfied and the theorem says nothing. Indeed

$$G(0; 10) = 2.1938698, \quad G(0; 50) = 2.0080855,$$

$$G(0; 200) = 2.0005062, \quad G(0; 1000) = 2.0000203,$$

all smaller than 4: hence θ_0 is **not** a global minimum, and $\inf F \leq 2$. By Corollary 31.3 one also has $\inf F \leq 2$, and the tabulated values show that the infimum $2 = \lambda_{\min}(S)$ is only approached by sending the center to infinity along $u = (0, 1)$. The limiting line of Observation 31.5 is the x -axis, and the sum of squared distances of the four points from it equals $0 + 0 + 1 + 1 = 2$, i.e. exactly $\lambda_{\min}(S)$. ✓ Moral: the criterion of Theorem 31.3 is sufficient but not necessary, and a stationary point is not necessarily a global minimum.

BEST PRACTICE

Assigned exercise 31.3 — answers.

(a) $\varphi = (-0.4; -0.2666667; -0.1333333; 0; 0.1333333; 0.2666667; 0.4)$: the list is symmetric about 0 because $\varphi_{6-k} = -\varphi_k$. One gets $C_1 = 0.9648114$ and $C_2 = 0.9317826$.

(b) $\mu_0 = 1 - C_2 = 0.0682174$; $\text{tr } N = 1 + C_2 = 1.9317826$ and $\det N = C_2 - C_1^2 = 0.0009216$, whence

$$\mu_+ = 1.9313055, \quad \mu_- = 0.0004772, \quad \kappa_2(J) = \sqrt{\frac{1.9313055}{0.0004772}} = 63.6179.$$

Check with the singular value decomposition: $\sigma(J) = (3.6768381; 0.6910293; 0.0577957)$ and $3.6768381/0.0577957 = 63.6179$. ✓ Note also $\sigma_j^2 = n\mu_j$: $7 \cdot 1.9313055 = 13.5191385$ and $\sqrt{13.5191385} = 3.6768381$. ✓

(c) $\sqrt{180}/0.4^2 = 83.8525$; with the continuous moments $C_1 = \sin(0.4)/0.4 = 0.9735459$ and $C_2 = \frac{1}{2} + \frac{\sin 0.8}{1.6} = 0.9483476$, whence $\kappa_2 = 82.6145$. The discrete sample gives a *smaller* value because its $\det N$ is larger: 0.0009216 against 0.0005560 for the continuous case (and $\alpha^4/45 = 0.0005689$). The reason is that the seven equally spaced angles *with the endpoints included* weight the tails of the interval more heavily than the uniform distribution, increasing the variance of $\cos \varphi$, which is precisely $C_2 - C_1^2$; and $\mu_- \approx \det N / \text{tr } N$ grows with it, lowering κ .

(d) By Proposition 31.11, with $\sigma_3 = 0.0577957$,

$$\|\tilde{\delta}_{\text{GN}} - \delta_{\text{GN}}\| \leq \frac{10^{-3}}{0.0577957} = 1.73 \cdot 10^{-2},$$

and in relative form, with $\|J\delta_{\text{GN}}\| = 10^{-2}$,

$$\frac{\|\tilde{\delta}_{\text{GN}} - \delta_{\text{GN}}\|}{\|\delta_{\text{GN}}\|} \leq 63.6179 \cdot \frac{10^{-3}}{10^{-2}} = 6.36,$$

i.e. the step can change by 636%: on an arc of 0.4 radians a perturbation of one thousandth on the residuals renders the Gauss–Newton step meaningless, and this is exactly the situation in which the damping of Section 31.6 becomes indispensable.

31.13. Formulary of the chapter

Distance, residuals, objective function.

$$\text{dist}(p, \mathcal{C}(c, r)) = |\|p - c\| - r|, \quad q^* = c + r \frac{p - c}{\|p - c\|} \quad (p \neq c).$$

$$\rho_i(c) = \|p_i - c\|, \quad d_i = \rho_i - r, \quad F(\theta) = \sum_{i=1}^n d_i^2, \quad u_i = \frac{p_i - c}{\rho_i}.$$

$$\min_r F(c, r) = G(c) = \sum_{i=1}^n (\rho_i(c) - \bar{\rho}(c))^2, \quad r_{\text{opt}} = \bar{\rho}(c) = \frac{1}{n} \sum_i \rho_i(c).$$

Invariance, existence, degenerations. For $T(x) = \lambda Ax + v$ with $A \in O(2)$, $\lambda > 0$: $F_{T(P)}(T_*\theta) = \lambda^2 F_P(\theta)$; for $\lambda = 1$, invariance.

$$w_i = p_i - \bar{p}, \quad S = \sum_{i=1}^n w_i w_i^\top, \quad u^\top S u = \sum_i \langle u, w_i \rangle^2, \quad \min_{\|u\|=1} u^\top S u = \lambda_{\min}(S).$$

$$|G(\bar{p} + tu) - u^\top Su| \leq \frac{24nM^3}{t} \quad (t \geq 8M), \quad \inf_c G \leq \lambda_{\min}(S).$$

If there exists c_0 with $G(c_0) < \lambda_{\min}(S)$, the minimum exists; collinear points (≥ 3 distinct) $\Rightarrow$ no minimum.

Normal equations, Jacobian, Hessian.

$$\sum_{i=1}^n d_i = 0, \quad \sum_{i=1}^n d_i u_i = 0 \quad \Longleftrightarrow \quad r = \bar{\rho}(c), \quad c = \bar{p} - \frac{r}{n} \sum_{i=1}^n u_i.$$

$$J_{i\bullet} = -(u_i^\top \quad 1), \quad \nabla F = 2J^\top d, \quad J^\top J = \begin{pmatrix} \sum_i u_i u_i^\top & \sum_i u_i \\ \sum_i u_i & n \end{pmatrix}.$$

$$\nabla^2 F = 2J^\top J + 2 \begin{pmatrix} \sum_i \frac{d_i}{\rho_i} (I - u_i u_i^\top) & 0 \\ 0 & 0 \end{pmatrix}.$$

rank $J = 3$ if and only if the unit vectors u_i take at least three distinct values.

Gauss–Newton, Levenberg–Marquardt, conditioning.

$$J^\top J \delta_{\text{GN}} = -J^\top d, \quad \langle \nabla F, \delta_{\text{GN}} \rangle = -2 g^\top (J^\top J)^{-1} g < 0, \quad g = J^\top d.$$

$$(J^\top J + \lambda I) \delta(\lambda) = -J^\top d; \quad \delta(\lambda) \xrightarrow{\lambda \rightarrow 0^+} \delta_{\text{GN}}, \quad \lambda \delta(\lambda) \xrightarrow{\lambda \rightarrow \infty} -\frac{1}{2} \nabla F.$$

$$\|\delta(\lambda)\|^2 = \sum_{j=1}^3 \frac{\sigma_j^2 h_j^2}{(\sigma_j^2 + \lambda)^2}, \quad h = U^\top d; \quad \kappa_2(J) = \frac{\sigma_1}{\sigma_3}, \quad \|\delta_{\text{GN}}\| \leq \frac{\|J^\top d\|}{\sigma_3^2}.$$

Symmetric arc, $C_1 = \frac{1}{n} \sum \cos \varphi_i$, $C_2 = \frac{1}{n} \sum \cos^2 \varphi_i$:

$$\mu_0 = 1 - C_2, \quad \mu_{\pm} = \frac{(1 + C_2) \pm \sqrt{(1 - C_2)^2 + 4C_1^2}}{2}, \quad \kappa_2(J) = \frac{\sqrt{180}}{\alpha^2} (1 + O(\alpha^2)).$$

Algebraic versus geometric.

$$g_i = \rho_i^2 - r^2 = d_i(\rho_i + r) = d_i(2r + d_i), \quad F_{\text{alg}} = \sum_i g_i^2.$$

$$|d_i| \leq \varepsilon r \quad (\varepsilon < 2) \implies 4r^2 \left(1 - \frac{\varepsilon}{2}\right)^2 F \leq F_{\text{alg}} \leq 4r^2 \left(1 + \frac{\varepsilon}{2}\right)^2 F.$$

At a fixed center:

$$r_{\text{geo}} = \bar{\rho}, \quad r_{\text{alg}} = \sqrt{\frac{1}{n} \sum_i \rho_i^2}, \quad r_{\text{alg}} - r_{\text{geo}} = \frac{\text{Var}(\rho)}{r_{\text{alg}} + r_{\text{geo}}} \geq 0.$$

The numerical values of the chapter. Sample A (six points): $c^* = (3.1535022; 1.8309722)$, $r^* = 5.1073285$, $F = 0.3463347$, $\kappa_2(J) = 1.4412$, $\lambda_{\min}(S) = 75.7684689$.

Sample B (short arc): $c^* = (-0.6928022; -0.0154242)$, $r^* = 10.6836501$, $F = 0.0017717618$, $\kappa_2(J) = 134.9473$, $\lambda_{\min}(S) = 0.2201842$.

Sample Q (nearly collinear): $c^* = (2.1553928; -38.8402627)$, $r^* = 38.8982963$, $F = 0.0034040$, $\kappa_2(J) = 3610.0$, $\lambda_{\min}(S) = 0.0057199$.

Conditioning constant: $\sqrt{180} = 6\sqrt{5} = 13.4164079$.

Sources and bibliography

- [1] N. Chernov, *Circular and Linear Regression: Fitting Circles and Lines by Least Squares*, Chapman and Hall/CRC, Boca Raton, 2010 (CRC Press; 286 pages; ISBN 978-1-4398-3591-3; DOI 10.1201/EBK1439835906). Chapters used: "Fitting circles: Theory", pp. 77–98 (existence and nonexistence of the minimum, degeneration into a line: reference source for Theorem 31.3 and Corollary 31.4); "Geometric circle fits", pp. 99–128 (normal equations, Gauss–Newton, Levenberg–Marquardt, fixed-point method on the center: reference source for Theorem 31.4 and Section 31.5); "Algebraic circle fits", pp. 129–166, and "Statistical analysis of circle fits", pp. 201–236 (bias analysis of algebraic fits: this is the *imported* part, declared in Observation 31.13). Bibliographic data, DOI and the chapter index with pagination verified against the Crossref records of the volume and its chapters — verified on 6 September 2026.
- [2] W. Gander, G. H. Golub, R. Strebler, *Least-squares fitting of circles and ellipses*, BIT Numerical Mathematics **34** (1994), issue 4, pp. 558–578 (DOI 10.1007/BF01934268). This is the classical reference for the systematic comparison between algebraic and geometric fits of circles and ellipses, for the formulation of the geometric problem as nonlinear least squares and for the use of Gauss–Newton and Levenberg–Marquardt started from an algebraic fit: it supports the strategy of Section 31.8 and the choice of Jacobian of Proposition 31.9. Title, authors, volume, issue, pages and DOI verified against the Crossref record — verified on 6 September 2026.
- [3] I. Kåsa, *A circle fitting procedure and its error analysis*, IEEE Transactions on Instrumentation and Measurement **IM-25** (1976), issue 1, pp. 8–14 (DOI 10.1109/TIM.1976.6312298). This is the original paper of the algebraic method, recalled by cross-reference (Volume I, Chapter 10) and used here only as a supplier of the starting point (Proposition 31.15). Author, title, journal, volume, issue, pages and DOI verified against the Crossref record — verified on 6 September 2026.
- [4] J. Nocedal, S. J. Wright, *Numerical Optimization*, 2nd edition, Springer Series in Operations Research and Financial Engineering, Springer, New York, 2006 (ISBN 978-0-387-30303-1; DOI 10.1007/978-0-387-40065-5). Chapters used: Chapter 4 "Trust-Region Methods", pp. 66–100, and Chapter 10 "Least-Squares Problems", pp. 245–269, which are the sources of the two results here *imported* and not proved: the convergence rate of Gauss–Newton (Observation 31.10) and the equivalence between Levenberg–Marquardt and trust-region methods (Observation 31.11); Chapter 3 "Line Search Methods", pp. 30–65, for the line search recalled in Theorem 31.5. Chapter index with pagination, series, ISBN and DOI verified against the Crossref records of the volume and its chapters — verified on 6 September 2026.
- [5] For the results of the treatise recalled with \ref: the structure of the isometries of the plane, the characterization of orthogonal matrices and the preservation of circles are proved in Chapter 5; the parametrization of plane curves in Chapter 2. Kåsa's algebraic method, the Hough transform and the circle through three points are in Volume I, Chapter 10 and Chapter 2, and are recalled by textual cross-reference without being reproduced, as prescribed by the Executive plan, §3.4, line 31.
- [6] All the numerical values of the chapter were computed twice by independent routes with Python scripts run on 6 September 2026: the geometric fit with the Gauss–Newton iteration and, as a check, with the derivative-free Nelder–Mead method applied to the function G alone; the eigenvalues of $J^\top J$ with the singular value decomposition and, as a check, with the closed-form formulas of Theorem 31.8; the bias experiment (Example 31.5) over

20 000 independent replicates.

Open problems and research literacy

Thirty-one chapters have shown the circle as a finite object: it is defined, measured, transformed, and every statement closes with a proof. This chapter shows the other half of the trade. Around the circle there are questions phrased in one line, comprehensible to anyone who has read Chapter 2, that no one can solve; there are others that have been solved, but in the opposite sense to what everyone expected; and there is at least one whose status changed while this volume was being written.

The didactic risk of a chapter like this is precise and must be named at once: presenting a conjecture with the same typography as a theorem. In the rest of the treatise every **theorem** is followed by a complete **proof**; here most of the material is *not*, and never will be in these pages. So a rigid typographic rule holds, declared in Section 32.1 and respected line by line: *in this chapter no open or unproved statement appears in a **teorema** environment*. The environments **teorema**, **proposizione**, **lemma** and **corollario** are reserved for the results that are proved in full in this chapter — and there are some, because every great open problem has an elementary core that is solved completely: centrally symmetric convex bodies inscribe squares (Theorem 32.1) and indeed rectangles of every shape (Theorem 32.2); the kissing number of the plane equals 6 (Theorem 32.4); there exists on the circle a dense set of points with all mutual distances rational (Theorem 32.5); the curvatures of an integral Apollonian packing remain integral forever (Theorem 32.3 and Corollary 32.2). Everything else — Toeplitz, Greene and Lobb, the *local-global* conjecture, kissing numbers in high dimension, Erdős and Ulam — sits in **osservazione** or **infobox**, as panorama, with the date.

The last section (Section 32.8) teaches the craft needed not to get hurt by this material: reading a preprint, distinguishing a proved result from an announced one, understanding what "under review" means, citing so that the reader can check. It is not a rhetorical appendix. While this chapter was being written, the identifier **arXiv:2604.17116**, indicated at the planning stage as "the result of Greene and Lobb on aspect ratios", was verified: it exists, it is by Greene and Lobb, but it is *not* the article published in the *Annals* in 2021 — it is a preprint of 18 April 2026, with a different and weaker statement, not yet submitted to any journal at the time of verification. Confusing the two would have produced a chapter that attributes to a published article a theorem that article does not contain. Section 32.8 explains how this error is avoided, and Section 32.3 keeps the two works rigorously distinct.

INFO

Level and prerequisites. Level **B–C**: the chapter's proofs use nothing beyond single-variable analysis and elementary plane geometry; the level rises only in reading the sources, which are research articles.

Internal prerequisites. Plane curves and Jordan curves (Chapter 2); global theorems on closed curves (Chapter 4); tangencies between circles, Ford circles and optimality of packings (Chapter 14); hyperspheres in any dimension and the volume of the n -dimensional ball (Chapter 11); rational parametrization of the circle (Chapter 6).

Prerequisites in other volumes, recalled only in text. Descartes' circle theorem and the construction of the Apollonian packing are in Volume I, Chapter 5; points with rational

coordinates on the circle and Pythagorean triples are in Volume I, Chapter 9. Here they are used as acquired results and not reproof: Observation 32.8 states exactly what is imported. *External prerequisites, assumed and not reproof:*

- the intermediate value theorem and Weierstrass's theorem for continuous functions on a compact set;
- elementary notions on plane convex sets: convexity, interior, boundary;
- arithmetic of the rationals and the notion of a dense subset of $\mathbb{R}$;
- the volume of the n -dimensional ball and the notion of a lattice, for Observation 32.11 alone (the data on the lattices E_8 and Leech are *imported* and attributed to their source).

Everything stated in a **teorema**, **proposizione**, **lemma** or **corollario** environment is proved here in full.

What is learned. Recognizing the difference between an open problem, a refuted conjecture and a theorem; isolating the special case of an open problem that *can* be solved, and proving it; reading the header of a preprint and deriving its editorial status; building a verifiable citation; distinguishing "announced", "submitted", "accepted", "published"; dating a statement about the state of research.

Warning. Conventions of the chapter.

- Every statement about the *status* of a problem is accompanied by the date of verification. All the checks in this chapter were made on **7 September 2026**; Table 32.2 collects the status of every statement cited as of that date. A status can change: it is the one thing that distinguishes this chapter from the others.
- A **Jordan curve** is, as in Chapter 2, the image of a continuous injective map $\gamma: S^1 \rightarrow \mathbb{R}^2$; "smooth" means of class C^∞ and with derivative never zero; "rectifiable" means of finite length.
- The four points of an *inscribed* square or rectangle must be distinct; degenerate configurations (four points coinciding pairwise, or collinear) do not count.
- The curvatures of a circle are the reciprocals of the radii, with a negative sign for the circle that contains the others: the convention is that of Volume I, Chapter 5.
- Decimals are written with a point, 0.5, and thousands with a comma in braces, 196,560.

32.1. What "open" means

Definition 32.1: Conjecture, open problem, theorem. Let P be a precise mathematical statement, that is, a statement for which the setting, the hypotheses and the conclusion are fixed without ambiguity.

- P is a **theorem** if there exists a proof of P accepted by the mathematical community, that is, written, published and checked.
- P is **refuted** if the negation of P is a theorem: someone has exhibited a counterexample, or proved that P is false.
- P is an **open problem** if it falls under neither case (i) nor case (ii). A **conjecture** is an open problem for which someone has publicly declared they expect an affirmative answer, usually on the basis of proved special cases or computer verifications.

Remark 32.1: Why typography matters. In earlier chapters the sequence "teorema $\rightarrow$ dimostrazione" is a contract: the reader knows that every line of the statement has been paid for. If, in this chapter, an open problem appeared in a **teorema** environment, that

contract would be broken, and the reader would no longer be able to tell, while skimming, what is established from what is merely hoped for.

Hence: the environments `teorema`, `proposizione`, `lemma` and `corollario` are used in this chapter only for the eleven results that are proved in full here (*Lemma 32.1*, *Theorem 32.1*, *Proposition 32.1*, *Theorem 32.2*, *Corollary 32.1*, *Proposition 32.2*, *Theorem 32.3*, *Corollary 32.2*, *Theorem 32.4*, *Lemma 32.2*, *Theorem 32.5*). Every other statement — whether open, or proved elsewhere by means out of reach here — sits in `osservazione` or `infobox`, and is declared as such on its first line.

32.2. Toeplitz's inscribed square problem

Definition 32.2: Square inscribed in a curve. Let $\gamma \subset \mathbb{R}^2$ be a Jordan curve. We say that γ **inscribes a square** if there exist four distinct points $A, B, C, D \in \gamma$ that, in this order, are the vertices of a non-degenerate square, that is

$$|AB| = |BC| = |CD| = |DA| > 0, \quad |AC| = |BD|.$$

It is not required that the four points follow one another in the order in which the curve traces them out, nor that the square lie inside the region enclosed by γ .

INFO

Panorama — the inscribed square problem (Toeplitz, 1911). Open as of 7 September 2026.

Problem. Is it true that every Jordan curve $\gamma \subset \mathbb{R}^2$ inscribes a square?

Otto Toeplitz posed the question in a communication to the Swiss Society of Natural Sciences in 1911; the exact reference is in the list of sources at the end of the chapter. The statement carries no regularity hypothesis: the curve is merely continuous, simple and closed. A hundred and fifteen years later the answer is **not known**. No counterexample has been exhibited, and no general proof has been accepted: the problem is open in the sense of Definition 32.1(iii).

What is known is a long list of *classes* of curves for which the answer is yes: Table 32.1 summarizes it, with authors and years. All these classes share some form of regularity — convexity, analyticity, differentiability, finite total curvature, local monotonicity — and the open problem is exactly what remains when every regularity is removed.

The elementary core of the problem, instead, is solved completely, and it is proved here. The simplest class for which the answer is yes is that of *centrally symmetric* convex bodies: for these no topology is needed, the intermediate value theorem applied to the radial function is enough. The preparatory work is all in the next lemma.

Definition 32.3: Convex body and radial function. A **convex body** of the plane is a set $K \subset \mathbb{R}^2$ that is compact, convex and has non-empty interior. If 0 is an interior point of K , the **radial function** of K is

$$\rho: S^1 \rightarrow (0, +\infty), \quad \rho(u) = \max\{t \geq 0 : tu \in K\},$$

where $S^1 = \{u \in \mathbb{R}^2 : |u| = 1\}$. One also writes $\rho(\varphi)$ for $\rho(u_\varphi)$, with $u_\varphi = (\cos \varphi, \sin \varphi)$; in this notation ρ is a 2π -periodic function of $\varphi \in \mathbb{R}$. The body K is **centrally symmetric about 0** if $K = -K$.

Table 32.1: Classes of Jordan curves for which the inscribed square problem has an affirmative answer. List drawn from the historical survey of B. Matschke and from the bibliographies of the two Greene and Lobb papers, all listed among the sources at the end of the chapter and verified on 7 September 2026. The last row is the open problem.

Class of curves	Author	Year
Piecewise analytic curves, with finitely many singularities	A. Emch	1913–1916
Convex curves	Emch; Zindler; Christensen	1913 onward
Smooth curves (Jordan, of class C^∞)	L. Šnirel'man	1929
Analytic curves	R. P. Jerrard	1961
Locally monotone curves	W. Stromquist	1989
Polygonal curves (discrete case)	I. Pak	2008
C^1 curves and curves of finite total curvature without cusps	Cantarella, Denne, McCleary	2022
Curves that are the union of two Lipschitz graphs of slope < 1	T. Tao	2017
Arbitrary Jordan curves (merely continuous)	—	open

Lemma 32.1: *The radial function is well defined and Lipschitz. Let $K \subset \mathbb{R}^2$ be a convex body with 0 interior, and let $0 < r \leq R$ be such that*

$$rB \subseteq K \subseteq RB, \quad B = \{x \in \mathbb{R}^2 : |x| \leq 1\}.$$

Then:

- (i) $\rho(u)$ exists and is finite for every $u \in S^1$, and $r \leq \rho(u) \leq R$;
- (ii) for every $x \in K$ and every $s \in (0, 1)$, the closed disc of centre $(1-s)x$ and radius sr is contained in K ;
- (iii) $|\rho(u) - \rho(v)| \leq \frac{R^2}{r} |u - v|$ for every $u, v \in S^1$; in particular ρ is continuous;
- (iv) $\partial K = \{\rho(u)u : u \in S^1\}$;
- (v) if $K = -K$, then $\rho(-u) = \rho(u)$ for every u , that is, $\rho(\varphi + \pi) = \rho(\varphi)$: the function ρ is π -periodic.

Proof. (i). Fix $u \in S^1$, and let $T_u = \{t \geq 0 : tu \in K\}$. The set T_u is non-empty because $0 \in K$; it is bounded because $K \subseteq RB$ implies $|tu| = t \leq R$ for every $t \in T_u$; it is closed because K is closed and $t \mapsto tu$ is continuous. A non-empty, closed and bounded subset of $\mathbb{R}$ has a maximum: hence $\rho(u) = \max T_u$ exists and is $\leq R$. Moreover $ru \in rB \subseteq K$, that is $r \in T_u$, so $\rho(u) \geq r > 0$.

(ii). Let $x \in K$, $s \in (0, 1)$, and let y be a point of the closed disc of centre $(1-s)x$ and radius sr . Then $y - (1-s)x$ has norm $\leq sr$, that is, one can write $y = (1-s)x + sb$ with $b = \frac{1}{s}(y - (1-s)x)$ and $|b| \leq r$. Hence $b \in rB \subseteq K$. Since $x \in K$, $b \in K$, K is convex and $(1-s)x + sb = y$ with $0 < s < 1$, we get $y \in K$.

(iii). Let $u, v \in S^1$ and set

$$\delta = |u - v|, \quad s = \frac{R\delta}{r + R\delta} \in [0, 1).$$

If $\delta = 0$ there is nothing to prove; so let $\delta > 0$, so that $s \in (0, 1)$. Note that

$$1 - s = \frac{r}{r + R\delta}, \quad \text{hence} \quad (1-s)R\delta = \frac{rR\delta}{r + R\delta} = sr. \quad (32.1)$$

Let $x = \rho(u)u$; by definition of ρ , $x \in K$. The point $y = (1-s)\rho(u)v$ is at distance from

$(1-s)x = (1-s)\rho(u)u$ exactly

$$|(1-s)\rho(u)v - (1-s)\rho(u)u| = (1-s)\rho(u)|v-u| \leq (1-s)R\delta \stackrel{(32.1)}{=} sr,$$

where $\rho(u) \leq R$ from point (i) was used. Hence y belongs to the closed disc of centre $(1-s)x$ and radius sr , which by point (ii) is contained in K . Thus $(1-s)\rho(u) \in T_v$, that is

$$\rho(v) \geq (1-s)\rho(u).$$

It follows that

$$\rho(u) - \rho(v) \leq s\rho(u) \leq sR = \frac{R^2\delta}{r+R\delta} \leq \frac{R^2}{r}\delta,$$

having used $\rho(u) \leq R$ and $r+R\delta \geq r$. Exchanging the roles of u and v gives the same bound for $\rho(v) - \rho(u)$, hence $|\rho(u) - \rho(v)| \leq \frac{R^2}{r}\delta$. A Lipschitz function is continuous.

(iv). Let $u \in S^1$ and $x = \rho(u)u$. For every $t > \rho(u)$ we have $tu \notin K$ (otherwise $t \in T_u$ would contradict the maximality of $\rho(u)$); since $t \mapsto tu$ is continuous and $x \in K$, every neighbourhood of x contains points of K (the point x itself) and points not in K (the points tu with $t > \rho(u)$ close enough to $\rho(u)$). Hence $x \in \partial K$.

Conversely let $y \in \partial K$. Since 0 is interior to K and y is a boundary point, $y \neq 0$; write $y = |y|u$ with $u = y/|y| \in S^1$. From $y \in K$ it follows that $|y| \in T_u$, that is $|y| \leq \rho(u)$. If $|y| < \rho(u)$ held, set $s = 1 - \frac{|y|}{\rho(u)} \in (0, 1)$ and $x = \rho(u)u \in K$: then $(1-s)x = |y|u = y$, and by point (ii) the closed disc of centre y and radius $sr > 0$ is contained in K , that is, y would be interior to K , against $y \in \partial K$. Hence $|y| = \rho(u)$ and $y = \rho(u)u$.

(v). If $K = -K$, then $tu \in K \iff -tu \in K$, that is $T_u = T_{-u}$, hence $\rho(u) = \rho(-u)$. In terms of φ : $u_{\varphi+\pi} = -u_\varphi$, hence $\rho(\varphi+\pi) = \rho(\varphi)$. $\square$

Theorem 32.1: Centrally symmetric convex bodies inscribe a square. Let $K \subset \mathbb{R}^2$ be a convex body, centrally symmetric about one of its interior points, which may be assumed to be the origin, and let $\gamma = \partial K$. Then γ inscribes a square, and the centre of that square is the origin.

Proof. Apply Lemma 32.1: the radial function $\rho: \mathbb{R} \rightarrow [r, R]$, $\varphi \mapsto \rho(u_\varphi)$, is continuous (point (iii)) and π -periodic (point (v)), and $\rho(\varphi)u_\varphi \in \gamma$ for every φ (point (iv)).

Step 1 — a function that must vanish. Set

$$h(\varphi) = \rho(\varphi) - \rho(\varphi + \frac{\pi}{2}), \quad \varphi \in \mathbb{R}.$$

The function h is continuous, being the difference of two continuous functions. Moreover, using the π -periodicity of ρ ,

$$h(\varphi + \frac{\pi}{2}) = \rho(\varphi + \frac{\pi}{2}) - \rho(\varphi + \pi) = \rho(\varphi + \frac{\pi}{2}) - \rho(\varphi) = -h(\varphi). \quad (32.2)$$

In particular $h(0)$ and $h(\pi/2) = -h(0)$ have opposite signs (or are both zero).

Step 2 — the intermediate value theorem. If $h(0) = 0$, take $\varphi_0 = 0$. Otherwise $h(0)$ and $h(\pi/2)$ are non-zero and of opposite sign by (32.2); since h is continuous on the interval $[0, \pi/2]$, the intermediate value theorem gives $\varphi_0 \in (0, \pi/2)$ with $h(\varphi_0) = 0$, that is

$$\rho(\varphi_0) = \rho(\varphi_0 + \frac{\pi}{2}) =: \varrho > 0.$$

Step 3 — the four points. Set

$$A = \varrho u_{\varphi_0}, \quad B = \varrho u_{\varphi_0+\pi/2}, \quad C = -A, \quad D = -B.$$

All four lie on γ : for A and B this is point (iv) of Lemma 32.1 with $\rho(\varphi_0) = \rho(\varphi_0 + \pi/2) = \varrho$; for $C = \varrho u_{\varphi_0 + \pi}$ and $D = \varrho u_{\varphi_0 + 3\pi/2}$ the same point (iv) is used together with the π -periodicity (point (v)), which gives $\rho(\varphi_0 + \pi) = \rho(\varphi_0) = \varrho$ and $\rho(\varphi_0 + 3\pi/2) = \rho(\varphi_0 + \pi/2) = \varrho$.

The four points are distinct: $A \neq C$ and $B \neq D$ because $\varrho > 0$; $A \neq \pm B$ because u_{φ_0} and $u_{\varphi_0 + \pi/2}$ are orthogonal and non-zero.

Step 4 — $ABCD$ is a square. Since $C = -A$ and $D = -B$, the segments AC and BD both have the origin as midpoint, hence $ABCD$ is a parallelogram. The sides:

$$\vec{AB} = B - A, \quad \vec{BC} = C - B = -A - B.$$

Their dot product equals

$$\langle B - A, -A - B \rangle = -\langle B, A \rangle - \langle B, B \rangle + \langle A, A \rangle + \langle A, B \rangle = |A|^2 - |B|^2 = \varrho^2 - \varrho^2 = 0,$$

so consecutive sides are orthogonal: $ABCD$ is a rectangle. The lengths: since $\langle A, B \rangle = \varrho^2 \langle u_{\varphi_0}, u_{\varphi_0 + \pi/2} \rangle = 0$,

$$|AB|^2 = |B - A|^2 = |A|^2 - 2\langle A, B \rangle + |B|^2 = 2\varrho^2,$$

$$|BC|^2 = |A + B|^2 = |A|^2 + 2\langle A, B \rangle + |B|^2 = 2\varrho^2.$$

Hence $|AB| = |BC| = \varrho\sqrt{2} > 0$, and by the parallelogram property also $|CD| = |AB|$ and $|DA| = |BC|$: the four sides are equal and the angles are right angles. Finally $|AC| = |2A| = 2\varrho = |2B| = |BD|$. The quadrilateral $ABCD$ is therefore a non-degenerate square inscribed in γ , centred at the origin. $\square$

Example 32.1: The square inscribed in the ellipse $x^2/4 + y^2 = 1$. The ellipse $E = \{(x, y) : x^2/4 + y^2 = 1\}$ is the boundary of a convex body centrally symmetric about the origin, so Theorem 32.1 applies. Here the square can be found in closed form.

The radial function is obtained by imposing $\rho^2 \cos^2 \varphi/4 + \rho^2 \sin^2 \varphi = 1$:

$$\rho(\varphi) = \left(\frac{\cos^2 \varphi}{4} + \sin^2 \varphi \right)^{-1/2}.$$

The condition $\rho(\varphi) = \rho(\varphi + \pi/2)$ of Step 2 becomes

$$\frac{\cos^2 \varphi}{4} + \sin^2 \varphi = \frac{\sin^2 \varphi}{4} + \cos^2 \varphi \iff \left(\frac{1}{4} - 1 \right) (\cos^2 \varphi - \sin^2 \varphi) = 0 \iff \cos 2\varphi = 0,$$

that is $\varphi_0 = \pi/4$. Then

$$\varrho = \rho(\pi/4) = \left(\frac{1}{8} + \frac{1}{2} \right)^{-1/2} = \left(\frac{5}{8} \right)^{-1/2} = \sqrt{\frac{8}{5}} = \frac{2\sqrt{2}}{\sqrt{5}},$$

and the four vertices are

$$\left(\pm \frac{2}{\sqrt{5}}, \pm \frac{2}{\sqrt{5}} \right) = (\pm 0.8944272, \pm 0.8944272).$$

Independent check (direct substitution into the equation of the ellipse): with $x = y = 2/\sqrt{5}$ one has $x^2 = y^2 = 4/5$ and

$$\frac{x^2}{4} + y^2 = \frac{4/5}{4} + \frac{4}{5} = \frac{1}{5} + \frac{4}{5} = 1. \quad \checkmark$$

The side of the square equals $2 \cdot 2/\sqrt{5} = 4/\sqrt{5} = 1.7888544$, and the diagonal $2\varrho = 2\sqrt{8/5} = 2.5298221$; the check side $\cdot \sqrt{2} = 1.7888544 \cdot 1.4142136 = 2.5298221$ works out. $\checkmark$

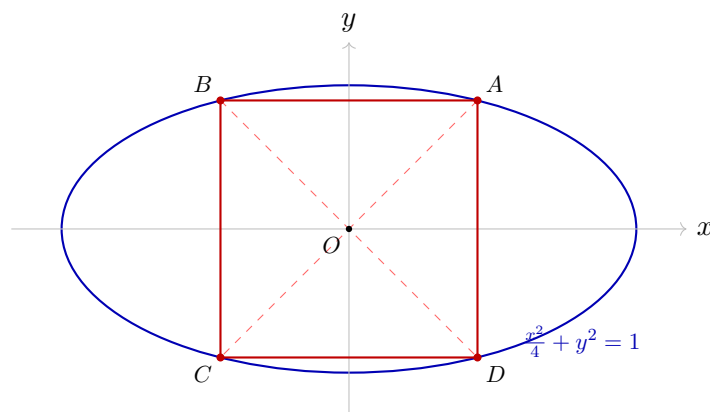


Figure 32.1: The square inscribed in the ellipse $x^2/4 + y^2 = 1$ constructed in Example 32.1: the vertices are $(\pm 2/\sqrt{5}, \pm 2/\sqrt{5}) = (\pm 0.8944272, \pm 0.8944272)$, the centre is the origin, the side equals $4/\sqrt{5} = 1.7888544$. The coordinates are computed by hand and verified by substitution into the equation of the ellipse.

Remark 32.2: What we import: the history of the Toeplitz problem. Of the picture summarized in Table 32.1, only one row is proved in full in this chapter — and not even one of those: Theorem 32.1 concerns centrally symmetric convex bodies, which are a proper subset of the convex bodies of the second row. Every other row is **imported**: the statements are reported with author and year, the proofs are not here.

What exactly is missing:

- Convex curves in general. The classical proof does not use central symmetry but a continuity argument on the family of circumscribed squares, or an approximation argument; it cannot be reconstructed here in a few lines with the certainty the treatise requires.
- Smooth curves. Šnirel'man's argument is topological: one counts modulo 2 the number of inscribed squares of a regular curve, shows the count is invariant under deformation and equals 1 for the circle. It requires intersection theory and transversality.
- C^1 curves and curves of finite total curvature. The argument of Cantarella, Denne and McCleary uses multijet transversality in configuration spaces.
- Lipschitz curves. Tao's argument replaces the topological count with an integral identity.

None of these tools is developed in the treatise, and none of them solves the general problem: the bold row of Table 32.1 remains empty. Status verified: 7 September 2026.

32.3. Inscribed rectangles: Greene and Lobb

If the square is replaced by a rectangle of assigned shape, the problem changes nature: from a question about a single shape it becomes a question about a one-parameter family. It is convenient to parametrize the family by the angle between the diagonals, not by the ratio of the sides: the angle varies in a bounded interval, the ratio does not, and it is the angle that appears in the statements found in the literature.

Definition 32.4: θ -rectangle and aspect ratio. Let $\theta \in (0, \pi)$. A **θ -rectangle** is a non-degenerate rectangle of the plane whose diagonals meet at angle θ (the unoriented angle between two rays from the point of intersection, measured in $(0, \pi)$). The **aspect ratio** of a rectangle with sides $p \leq q$ is the number $p/q \in (0, 1]$.

A Jordan curve γ **inscribes** a θ -rectangle if there exist four distinct points of γ that are its vertices.

Proposition 32.1: *The angle of the diagonals determines the aspect ratio.* Let $\theta \in (0, \pi)$ and let $\mathcal{R}$ be a θ -rectangle of diagonal $d > 0$. Then the sides of $\mathcal{R}$ measure

$$d \sin\left(\frac{\theta}{2}\right) \quad \text{and} \quad d \cos\left(\frac{\theta}{2}\right),$$

and the aspect ratio of $\mathcal{R}$ equals

$$\min \left\{ \tan\left(\frac{\theta}{2}\right), \cot\left(\frac{\theta}{2}\right) \right\}.$$

In particular:

- (i) $\mathcal{R}$ is a square if and only if $\theta = \pi/2$;
- (ii) the function $\theta \mapsto \tan(\theta/2)$ is a strictly increasing bijection from $(0, \pi/2]$ onto $(0, 1]$; consequently, for every aspect ratio $\lambda \in (0, 1]$ there is exactly one $\theta \in (0, \pi/2]$ such that θ -rectangles have aspect ratio λ , and it is $\theta = 2 \arctan \lambda$;
- (iii) a θ -rectangle and a $(\pi - \theta)$ -rectangle have the same aspect ratio.

Proof. Step 1 — normal form. The diagonals of a rectangle are equal and bisect each other: letting O be their meeting point, the four vertices lie on the circle centred at O with radius $d/2$, and are pairwise opposite with respect to O . By an isometry of the plane (Chapter 5) one can move O to the origin and arrange the vertices symmetrically about the x -axis: let

$$A = \frac{d}{2} \left(\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \right), \quad B = \frac{d}{2} \left(\cos \frac{\theta}{2}, -\sin \frac{\theta}{2} \right), \quad C = -A, \quad D = -B.$$

The angle between the rays OA and OB is θ by construction, and the diagonals are AC and BD : so this configuration, and only this one up to isometries, is a θ -rectangle of diagonal d . That $ABCD$ is indeed a rectangle is checked as in Step 4 of the proof of Theorem 32.1: $|A| = |B| = d/2$ implies $\langle B - A, -A - B \rangle = |A|^2 - |B|^2 = 0$.

Step 2 — the sides. A direct computation gives

$$A - B = \left(0, d \sin \frac{\theta}{2} \right) \implies |AB| = d \sin \frac{\theta}{2},$$

$$B - C = B + A = \left(d \cos \frac{\theta}{2}, 0 \right) \implies |BC| = d \cos \frac{\theta}{2}.$$

Both are positive because $\theta/2 \in (0, \pi/2)$. Since in a rectangle opposite sides are equal, these two measures exhaust the sides.

Step 3 — the ratio. The aspect ratio is the quotient of the smaller side by the larger one, that is

$$\frac{\min\{|AB|, |BC|\}}{\max\{|AB|, |BC|\}} = \min \left\{ \frac{\sin(\theta/2)}{\cos(\theta/2)}, \frac{\cos(\theta/2)}{\sin(\theta/2)} \right\} = \min \left\{ \tan \frac{\theta}{2}, \cot \frac{\theta}{2} \right\}.$$

(i). $\mathcal{R}$ is a square $\iff |AB| = |BC| \iff \sin(\theta/2) = \cos(\theta/2) \iff \tan(\theta/2) = 1$. Since $\theta/2 \in (0, \pi/2)$ and the tangent is strictly increasing and injective on $(0, \pi/2)$, this happens only for $\theta/2 = \pi/4$, that is $\theta = \pi/2$.

(ii). On $(0, \pi/2]$ one has $\theta/2 \in (0, \pi/4]$, where $\tan(\theta/2) \in (0, 1]$ and $\cot(\theta/2) \geq 1$: hence the minimum of Step 3 is $\tan(\theta/2)$. The map $\theta \mapsto \tan(\theta/2)$ is the composition of $\theta \mapsto \theta/2$, increasing, with the tangent, strictly increasing on $(0, \pi/4]$; it is thus strictly increasing, hence injective. It is surjective onto $(0, 1]$ because it is continuous, with $\tan(\theta/2) \rightarrow 0^+$ as $\theta \rightarrow 0^+$

and $\tan(\pi/4) = 1$: the intermediate value theorem covers the whole of $(0, 1]$. The inverse is $\lambda \mapsto 2 \arctan \lambda$.

(iii). Substituting θ with $\pi - \theta$ gives $\sin \frac{\pi-\theta}{2} = \cos \frac{\theta}{2}$ and $\cos \frac{\pi-\theta}{2} = \sin \frac{\theta}{2}$: the two side measures swap, and the minimum of Step 3 does not change. $\square$

Example 32.2: Three angles and three shapes. By Proposition 32.1, with $d = 2$:

θ	sides	aspect ratio	shape
$\pi/2 = 1.5707963$	1.4142136; 1.4142136	1	square
$\pi/3 = 1.0471976$	1; 1.7320508	0.5773503	$1 : \sqrt{3}$
$2 \arctan \frac{1}{2} = 0.9272952$	0.8944272; 1.7888544	0.5	$1 : 2$

Independent check (closed form against direct computation of the distances between the vertices, run with a Python script): for $\theta = 2 \arctan \frac{1}{2}$ one has $\theta/2 = \arctan \frac{1}{2}$, so $\tan(\theta/2) = 0.5$ exactly, and $\sin(\theta/2) = 1/\sqrt{5} = 0.4472136$, $\cos(\theta/2) = 2/\sqrt{5} = 0.8944272$; the sides equal $2 \cdot 0.4472136 = 0.8944272$ and $2 \cdot 0.8944272 = 1.7888544$, whose ratio is 0.5. The script, which builds the four vertices and measures the four sides and the angle between the diagonals, returns sides 0.8944272 and 1.7888544, ratio 2.0000000 (larger side over smaller) and angle between the diagonals 0.9272952: the three numbers agree with those from the formula. $\checkmark$

Note $2 \arctan \frac{1}{2} = 0.9272952 < \pi/2 = 1.5707963$: the $1 : 2$ rectangle corresponds to a diagonal angle *smaller* than that of the square. The more elongated the rectangle, the smaller the angle $\theta \in (0, \pi/2]$.

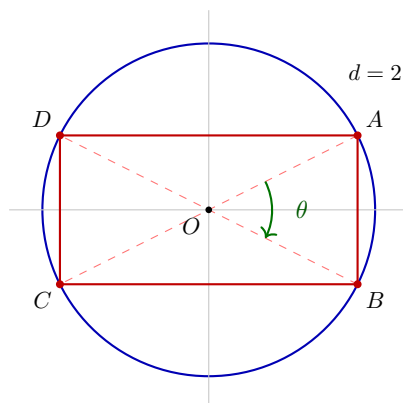


Figure 32.2: A θ -rectangle inscribed in the unit circle, with $\theta = 2 \arctan \frac{1}{2} = 0.9272952$ (that is 53.1301024°) and diagonal $d = 2$. The vertices are $(\pm 0.8944272, \pm 0.4472136)$, that is $(\pm 2/\sqrt{5}, \pm 1/\sqrt{5})$; the sides measure $2/\sqrt{5} = 0.8944272$ and $4/\sqrt{5} = 1.7888544$, in ratio $1 : 2$ (Proposition 32.1). Check: $(2/\sqrt{5})^2 + (1/\sqrt{5})^2 = \frac{4}{5} + \frac{1}{5} = 1$, the four vertices lie on the unit circle. $\checkmark$

Theorem 32.2: Centrally symmetric convex bodies inscribe every shape of rectangle. Let $K \subset \mathbb{R}^2$ be a convex body, centrally symmetric about one of its interior points, taken to be the origin, and let $\gamma = \partial K$. Then for every $\theta \in (0, \pi)$ the curve γ inscribes a θ -rectangle centred at the origin. Equivalently (Proposition 32.1(ii)), γ inscribes a rectangle of every aspect ratio $\lambda \in (0, 1]$.

Proof. Let ρ be the radial function of K : by Lemma 32.1 it is continuous, π -periodic and takes values in $[r, R]$ with $0 < r \leq R$, and $\rho(\varphi)u_\varphi \in \gamma$ for every φ .

Step 1 — a continuous function on the circle. Fix $\theta \in (0, \pi)$, and set

$$h_\theta(\varphi) = \rho(\varphi) - \rho(\varphi + \theta), \quad \varphi \in \mathbb{R}.$$

It is continuous and π -periodic (because ρ is).

Step 2 — h_θ changes sign (or vanishes). The function ρ is continuous on the compact set $[0, \pi]$, hence by Weierstrass's theorem it attains a maximum and a minimum: let $\varphi_M, \varphi_m \in [0, \pi]$ be such that

$$\rho(\varphi_M) = \max_{[0, \pi]} \rho, \quad \rho(\varphi_m) = \min_{[0, \pi]} \rho.$$

By π -periodicity these are the maximum and minimum of ρ over all of $\mathbb{R}$. Then

$$h_\theta(\varphi_M) = \rho(\varphi_M) - \rho(\varphi_M + \theta) \geq 0, \quad h_\theta(\varphi_m) = \rho(\varphi_m) - \rho(\varphi_m + \theta) \leq 0,$$

because $\rho(\varphi_M)$ is the maximum and $\rho(\varphi_m)$ the minimum of ρ on $\mathbb{R}$.

Step 3 — the intermediate value. The function h_θ is continuous on the closed interval with endpoints φ_m and φ_M (one of the two intervals $[\varphi_m, \varphi_M]$ or $[\varphi_M, \varphi_m]$, depending on the order), where it takes a value ≤ 0 and a value ≥ 0 . By the intermediate value theorem there exists φ_0 in that interval with $h_\theta(\varphi_0) = 0$, that is

$$\rho(\varphi_0) = \rho(\varphi_0 + \theta) =: \varrho \in [r, R], \quad \varrho > 0.$$

Step 4 — the four points and their membership in γ . Set

$$A = \varrho u_{\varphi_0}, \quad B = \varrho u_{\varphi_0 + \theta}, \quad C = -A = \varrho u_{\varphi_0 + \pi}, \quad D = -B = \varrho u_{\varphi_0 + \theta + \pi}.$$

By Lemma 32.1(iv) and π -periodicity (point (v)) all four belong to γ : indeed $\rho(\varphi_0 + \pi) = \rho(\varphi_0) = \varrho$ and $\rho(\varphi_0 + \theta + \pi) = \rho(\varphi_0 + \theta) = \varrho$.

The four points are distinct: $A \neq C$ and $B \neq D$ because $\varrho > 0$; $A \neq B$ and $A \neq D$ because u_{φ_0} , $u_{\varphi_0 + \theta}$ and $u_{\varphi_0 + \theta + \pi}$ are pairwise distinct unit vectors, since $\theta \in (0, \pi)$.

Step 5 — $ABCD$ is a θ -rectangle. The diagonals AC and BD both have the origin as midpoint, so $ABCD$ is a parallelogram; and

$$\langle B - A, -A - B \rangle = |A|^2 - |B|^2 = \varrho^2 - \varrho^2 = 0,$$

so consecutive sides are orthogonal and $ABCD$ is a rectangle. Its diagonals are the segments AC and BD , which lie on the lines through the origin with directions u_{φ_0} and $u_{\varphi_0 + \theta}$: the angle between the rays OA and OB is θ , hence $ABCD$ is a θ -rectangle. The diagonal measures $2\varrho > 0$, and the sides $2\varrho \sin(\theta/2) > 0$ and $2\varrho \cos(\theta/2) > 0$ by Proposition 32.1: the rectangle is non-degenerate.

Step 6 — every aspect ratio. Let $\lambda \in (0, 1]$. By Proposition 32.1(ii) there exists $\theta = 2 \arctan \lambda \in (0, \pi/2]$ whose θ -rectangles have aspect ratio λ ; Steps 1–5 provide a θ -rectangle inscribed in γ . $\square$

Corollary 32.1: *The square as the case $\theta = \pi/2$. Theorem 32.1 is the case $\theta = \pi/2$ of Theorem 32.2.*

Proof. Apply Theorem 32.2 with $\theta = \pi/2$: one obtains a $\pi/2$ -rectangle inscribed in γ and centred at the origin. By Proposition 32.1(i) a θ -rectangle is a square if and only if $\theta = \pi/2$; hence the rectangle obtained is a square, and this is exactly the conclusion of Theorem 32.1. $\square$

Remark 32.3: *Why central symmetry is a genuine hypothesis and not a convenience. In the proof, central symmetry is used twice, and in two different ways. In Step 4*

it guarantees that the points $-A$ and $-B$ still lie on γ : without it the four points would not all be on the curve. In Steps 1–3 it makes ρ π -periodic, which is what allows $\rho(\varphi)$ to be compared with $\rho(\varphi + \theta)$ on a compact domain and the argument to be concluded with the intermediate value theorem.

Remove the hypothesis, and the argument collapses entirely: nothing forces the four points to be pairwise opposite about a centre, and the proof provides no substitute. It is exactly here that the problem becomes hard: for curves without symmetry the inscribed rectangle has no privileged centre, and finding the four points becomes an intersection problem in a four-parameter configuration space. This is the leap that separates this page from the Greene and Lobb theorem of Observation 32.4.

Remark 32.4: Panorama — the Greene and Lobb theorem (2021). The result that follows **is not proved in this chapter**: it is reported as panorama, with the verified source, and for this reason it appears in a remark and not in a **teorema** environment.

Statement (J. E. Greene, A. Lobb, *The rectangular peg problem*, *Annals of Mathematics* **194** (2021), no. 2, pp. 509–517; verified on 7 September 2026). For every smooth Jordan curve γ and every rectangle R of the Euclidean plane, there exists a rectangle similar to R whose vertices lie on γ .

In the language of Definition 32.4: every smooth Jordan curve inscribes a θ -rectangle for every $\theta \in (0, \pi)$, that is, a rectangle of every aspect ratio. Theorem 32.2 is the same conclusion under the much stronger hypothesis that γ is the boundary of a centrally symmetric convex body — a hypothesis that allows the elementary proof given above.

What is left out. The hypothesis of smoothness cannot be removed with Greene and Lobb’s proof: for a merely continuous Jordan curve the statement is, as of 7 September 2026, **open** — just as the special case $\theta = \pi/2$ is open, namely the Toeplitz problem of Section 32.2.

Remark 32.5: What we import from the Greene and Lobb theorem. The theorem of Observation 32.4 is **imported**: its statement and exact location are reported, not its proof. What is missing, precisely: the translation of the problem into a question of symplectic geometry in $\mathbb{C}^2$; the construction, starting from a smooth curve with no inscribed θ -rectangles, of a smoothly immersed Lagrangian Klein bottle in $\mathbb{C}^2$; and the non-immersion theorem of V. Shevchishin (2009), cited by the authors, which asserts that no such immersion exists. None of these three ingredients is developed in the treatise, and they cannot be reconstructed here with the certainty required: for this reason the statement remains in a remark. The article’s bibliography, verified on 7 September 2026, is the source of this description.

Definition 32.5: The isodiametric constant $B(\gamma)$. Let $\gamma \subset \mathbb{R}^2$ be a Jordan curve, let $A(\gamma)$ be the area of the bounded region it encloses, and let

$$R(\gamma) = \frac{1}{2} \text{diam}(\gamma) = \frac{1}{2} \sup_{P, Q \in \gamma} |PQ|$$

be its **radius**, that is, half the diameter. One sets

$$B(\gamma) = \frac{A(\gamma)}{R(\gamma)^2}.$$

Remark 32.6: Panorama — a 2026 preprint, and how to read it. On 18 April 2026 Greene and Lobb deposited on arXiv the work *Jordan curves inscribe a positive measure*

of rectangles ([arXiv:2604.17116](#)). As of 7 September 2026 the deposit record shows a single version (*v1*, 18 April 2026), no journal reference and no editorial DOI: it is a **preprint** in the sense of Definition 32.9, not a published article. The two statements it contains are reported here as *panorama*.

Statement A. If $\gamma \subset \mathbb{C}$ is a Jordan curve, there exists a subset $\mathcal{S}(\gamma) \subset (0, \pi)$ of measure at least $B(\gamma)$ such that γ inscribes a θ -rectangle for every $\theta \in \mathcal{S}(\gamma)$.

Statement B. If $\gamma \subset \mathbb{C}$ is a rectifiable Jordan curve, then γ inscribes a θ -rectangle for every $\theta \in (0, B(\gamma))$.

These are statements about all Jordan curves, with no smoothness hypothesis — and this is where their interest lies with respect to the 2021 theorem — but they do not solve the problem: they give a set of angles of positive measure, not all angles. The authors explicitly declare open both the inscribed square problem and the rectangle problem for continuous curves.

How this material should be used. Until a journal reference appears, a work of this kind is cited as a preprint, with identifier, version and date, and its correctness has not yet been certified by any peer review (see Section 32.8). It must not be confused with the 2021 article of Observation 32.4, which is published, has different hypotheses (smooth curves) and a stronger conclusion (all angles).

Example 32.3: What Statement B says for concrete curves. $B(\gamma)$ is computed for five curves, using Definition 32.5. For a polygon, $R(\gamma)$ is half the maximum distance between two vertices.

γ	$A(\gamma)$	$R(\gamma)$	$B(\gamma)$	$B(\gamma)/\pi$
circle of radius 1	π	1	$\pi = 3.1415927$	1
regular hexagon of side 1	$\frac{3\sqrt{3}}{2}$	1	$\frac{3\sqrt{3}}{2} = 2.5980762$	0.8269933
square of side 1	1	$\frac{\sqrt{2}}{2}$	2	0.6366198
equilateral triangle of side 1	$\frac{\sqrt{3}}{4}$	$\frac{1}{2}$	$\sqrt{3} = 1.7320508$	0.5513289
ellipse $a = 2$, $b = 1$	2π	2	$\frac{\pi}{2} = 1.5707963$	0.5

Independent check (closed form against numerical computation): the area of the three polygons was recomputed with the Gauss ("shoelace") formula $A = \frac{1}{2} |\sum_i (x_i y_{i+1} - x_{i+1} y_i)|$ on the vertices, and the diameter as the maximum of the pairwise distances between vertices, using a Python script. The values obtained are: square, area 1, diameter 1.4142136, $B = 1.9999999\dots$ (floating-point rounding of 2); triangle, area 0.4330127 = $\sqrt{3}/4$, diameter 1, $B = 1.7320508$; hexagon, area 2.5980762, diameter 2, $B = 2.5980762$. These agree with the $B(\gamma)$ column. ✓

Reading. All five curves are rectifiable, so Statement B applies: for the square it guarantees inscribed θ -rectangles for every $\theta \in (0, 2)$, that is for 63.66% of the possible angles; for the equilateral triangle for every $\theta \in (0; 1.7320508)$, that is 55.13%; for the circle for every $\theta \in (0, \pi)$, that is all of them — and this is consistent, because the circle is the boundary of a centrally symmetric convex body and Theorem 32.2 gives the same conclusion by elementary means. The square, hexagon and ellipse are also centrally symmetric, and for them Theorem 32.2 gives *more* than Statement B; the equilateral triangle, which is not, is the case where Statement B says something new.

Proposition 32.2: From a threshold of angles to every aspect ratio. Let γ be a Jordan curve and let $b \in (0, \pi]$ be a number such that γ inscribes a θ -rectangle for every $\theta \in (0, b)$. If $b > \pi/2$, then γ inscribes a rectangle of every aspect ratio $\lambda \in (0, 1]$, including the square. If $b = \pi/2$, then γ inscribes a rectangle of every aspect ratio $\lambda \in (0, 1)$, but this hypothesis alone implies nothing about the square.

Proof. Let $\lambda \in (0, 1]$. By Proposition 32.1(ii) the rectangles of aspect ratio λ are exactly the θ_λ -rectangles with

$$\theta_\lambda = 2 \arctan \lambda \in (0, \pi/2],$$

and $\lambda \mapsto \theta_\lambda$ is strictly increasing with $\theta_1 = 2 \arctan 1 = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}$.

If $b > \pi/2$, then for every $\lambda \in (0, 1]$ one has $\theta_\lambda \leq \pi/2 < b$, that is $\theta_\lambda \in (0, b)$: by hypothesis γ inscribes a θ_λ -rectangle, which has aspect ratio λ . In particular for $\lambda = 1$ one obtains a square. If $b = \pi/2$, then for $\lambda \in (0, 1)$ one has $\theta_\lambda = 2 \arctan \lambda < 2 \arctan 1 = \pi/2 = b$, hence $\theta_\lambda \in (0, b)$ and the conclusion holds as above; but for $\lambda = 1$, $\theta_1 = \pi/2 = b \notin (0, b)$, and the hypothesis says nothing. $\square$

Remark 32.7: The computation that really matters. Combining Proposition 32.2 with Statement B of Observation 32.6 gives a criterion that can be checked by hand: a rectifiable Jordan curve with

$$B(\gamma) = \frac{A(\gamma)}{R(\gamma)^2} > \frac{\pi}{2} = 1.5707963,$$

that is, enclosing more than half the area of the disc of the same diameter, inscribes rectangles of every aspect ratio, including the square. In the table of Example 32.3, the circle (π), the hexagon (2.5980762), the square (2) and the equilateral triangle (1.7320508) exceed the threshold; the ellipse with semi-axes 2 and 1 reaches it exactly ($\pi/2$) and stays in the second case of the proposition. Note that this criterion depends on an imported statement not yet subjected to review: it is a good example of a conditional conclusion, and must be cited as such.

32.4. Apollonian packings and the local–global conjecture

Descartes' circle theorem and the construction of the Apollonian packing are developed in Volume I, Chapter 5, and tangencies between circles are taken up again in this volume in Chapter 14. Here only Descartes' statement is used, and from it the arithmetic fact that opens the problem is derived in full: *if four starting curvatures are integers, all curvatures of the packing are integers.*

Remark 32.8: What we import: Descartes' circle theorem. The following result is **imported** from Volume I, Chapter 5, where it is proved: if four circles of the plane are pairwise tangent at four distinct points, and a, b, c, d are their signed curvatures (the reciprocal of the radius, negative for the circle that contains the other three), then

$$(a + b + c + d)^2 = 2(a^2 + b^2 + c^2 + d^2). \quad (32.3)$$

What is missing here: the geometric proof of (32.3) and the discussion of degenerate cases (lines as circles of zero curvature, external and internal tangencies). Nothing that follows reproves (32.3): it is used as the starting algebraic identity.

Definition 32.6: Descartes quadruple, integral Apollonian packing. A **Descartes quadruple** is an ordered quadruple (a, b, c, d) of real numbers satisfying (32.3). The **Apollonian packing** generated by four mutually tangent circles is the set of circles obtained by repeatedly filling every curvilinear gap between three mutually tangent circles with the circle tangent to all three. The packing is called **integral** if all the curvatures of its circles are integers.

Theorem 32.3: Vieta's reflection. Let a, b, c, d be real numbers satisfying (32.3), and set $s = a + b + c$. Then:

(i) d is a root of the quadratic equation

$$X^2 - 2sX + (2(a^2 + b^2 + c^2) - s^2) = 0;$$

(ii) the other root is

$$d' = 2(a + b + c) - d,$$

and (a, b, c, d') also satisfies (32.3);

(iii) if a, b, c, d are integers, so is d' .

Proof. (i). Set $q = a^2 + b^2 + c^2$. Identity (32.3) can be rewritten

$$(s + d)^2 = 2(q + d^2),$$

that is $s^2 + 2sd + d^2 = 2q + 2d^2$, that is

$$d^2 - 2sd + (2q - s^2) = 0.$$

This is exactly the equation $X^2 - 2sX + (2q - s^2) = 0$ evaluated at $X = d$: hence d is a root of it.

(ii). A monic quadratic polynomial $X^2 - \sigma X + \pi_0$ has the sum of its roots equal to σ (Vieta's formulas). Here $\sigma = 2s$: if d is a root, the other is

$$d' = 2s - d = 2(a + b + c) - d.$$

Since d' is a root of the same equation, $d'^2 - 2sd' + (2q - s^2) = 0$, and retracing the steps of point (i) with d' in place of d :

$$d'^2 - 2sd' + 2q - s^2 = 0 \iff s^2 + 2sd' + d'^2 = 2q + 2d'^2 \iff (s + d')^2 = 2(q + d'^2),$$

that is $(a + b + c + d')^2 = 2(a^2 + b^2 + c^2 + d'^2)$: the quadruple (a, b, c, d') satisfies (32.3).

(iii). $d' = 2(a + b + c) - d$ is obtained from a, b, c, d using only sums, products by 2 and a difference: if $a, b, c, d \in \mathbb{Z}$, then $d' \in \mathbb{Z}$, because $\mathbb{Z}$ is closed under these operations. $\square$

Corollary 32.2: Hereditary integrality of Apollonian packings. Let $\mathcal{A}$ be an Apollonian packing generated by four mutually tangent circles whose curvatures a, b, c, d are integers. Then all the curvatures of the circles of $\mathcal{A}$ are integers: $\mathcal{A}$ is integral.

Proof. Every circle of $\mathcal{A}$ is obtained from the initial quadruple by a finite sequence of steps, in each of which one of the four circles of a mutually tangent quadruple is replaced by the other circle tangent to the remaining three. Let n be the number of steps; the proof proceeds by induction on n .

Base $n = 0$: the quadruple is (a, b, c, d) , with integer curvatures by hypothesis.

Inductive step: suppose that after n steps a quadruple of mutually tangent circles with integer curvatures (a_n, b_n, c_n, d_n) has been obtained. The four curvatures satisfy (32.3) (Observation 32.8). At step $n + 1$ one of the four circles, say the fourth, is replaced by the other circle tangent to the first three: its curvature d_{n+1} is the other root of the equation in Theorem 32.3(i), that is

$$d_{n+1} = 2(a_n + b_n + c_n) - d_n,$$

which is an integer by Theorem 32.3(iii). The same holds if the circle replaced is the first, second or third, by symmetry of the roles in (32.3). Hence the quadruple after $n + 1$ steps also

has integer curvatures.

By induction, every circle reachable in finitely many steps has integer curvature; since $\mathcal{A}$ is by construction the set of circles reachable in this way, all its curvatures are integers. $\square$

Example 32.4: The packing $(-1, 2, 2, 3)$, with the computations. Start from the circles: Γ_0 of centre $(0, 0)$ and radius 1, which contains the others (curvature -1); Γ_1, Γ_2 of centres $(0, \pm \frac{1}{2})$ and radius $\frac{1}{2}$ (curvature 2); Γ_3 of centre $(\frac{2}{3}, 0)$ and radius $\frac{1}{3}$ (curvature 3).

Checking the tangencies (by hand, distance between centres against sum or difference of the radii):

$$\begin{array}{ll} \Gamma_1 \text{ and } \Gamma_0 : & |(0, \frac{1}{2})| = \frac{1}{2}, & 1 - \frac{1}{2} = \frac{1}{2} \quad \checkmark \text{ (internal tangency)} \\ \Gamma_1 \text{ and } \Gamma_2 : & |(0, 1)| = 1, & \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark \\ \Gamma_3 \text{ and } \Gamma_0 : & |(\frac{2}{3}, 0)| = \frac{2}{3}, & 1 - \frac{1}{3} = \frac{2}{3} \quad \checkmark \\ \Gamma_3 \text{ and } \Gamma_1 : & |(\frac{2}{3}, -\frac{1}{2})| = \sqrt{\frac{4}{9} + \frac{1}{4}} = \sqrt{\frac{25}{36}} = \frac{5}{6}, & \frac{1}{3} + \frac{1}{2} = \frac{5}{6} \quad \checkmark \end{array}$$

Checking Descartes' identity for $(-1, 2, 2, 3)$:

$$(-1 + 2 + 2 + 3)^2 = 6^2 = 36, \quad 2((-1)^2 + 2^2 + 2^2 + 3^2) = 2(1 + 4 + 4 + 9) = 2 \cdot 18 = 36. \quad \checkmark$$

Vieta's reflection. Replacing the curvature -1 with the other root:

$$d' = 2(2 + 2 + 3) - (-1) = 14 + 1 = 15.$$

Check on $(15, 2, 2, 3)$: $(15 + 2 + 2 + 3)^2 = 22^2 = 484$ and $2(225 + 4 + 4 + 9) = 2 \cdot 242 = 484$.

$\checkmark$ The corresponding circle has radius $\frac{1}{15}$; solving the system of the three tangencies gives centre $(\frac{4}{15}, 0)$, and indeed $|(\frac{4}{15}, 0) - (0, \frac{1}{2})| = \sqrt{\frac{16}{225} + \frac{1}{4}} = \sqrt{\frac{289}{900}} = \frac{17}{30} = \frac{1}{2} + \frac{1}{15}$. $\checkmark$

Replacing instead one of the curvatures 2:

$$d' = 2(-1 + 2 + 3) - 2 = 8 - 2 = 6,$$

quadruple $(-1, 2, 3, 6)$: $(-1 + 2 + 3 + 6)^2 = 100$ and $2(1 + 4 + 9 + 36) = 100$. $\checkmark$ The circle has radius $\frac{1}{6}$ and centre $(\frac{2}{3}, \frac{1}{2})$; checks: $|(\frac{2}{3}, \frac{1}{2})| = \frac{5}{6} = 1 - \frac{1}{6}$ (internal tangency to Γ_0), $|(\frac{2}{3}, 0)| = \frac{1}{2} = \frac{1}{3} + \frac{1}{6}$ (tangency to Γ_3), and the distance from $(0, \frac{1}{2})$ equals $\frac{2}{3} = \frac{1}{2} + \frac{1}{6}$ (tangency to Γ_1). $\checkmark$

Independent check. A Python script recomputed $(\sum_i k_i)^2$ and $2 \sum_i k_i^2$ for the quadruples $(-1, 2, 2, 3)$, $(15, 2, 2, 3)$ and $(-1, 2, 3, 6)$, obtaining in all three cases two equal numbers (36 and 36; 484 and 484; 100 and 100), and verified that the control quadruple $(-1, 3, 6, 7)$ — *not* a Descartes quadruple — instead gives $225 \neq 190$. $\checkmark$

INFO

Panorama — the *local-global* conjecture: posed in 2003, refuted in 2024. None of the statements in this box is proved here.

Let $\mathcal{A}$ be an integral, primitive Apollonian packing (that is, with curvatures sharing no common factor). A classical fact, due to Graham, Lagarias, Mallows, Wilks and Yan (*Journal of Number Theory* **100** (2003), pp. 1–45), is that the set of curvatures of $\mathcal{A}$ occupies only certain residue classes modulo 24: these are the *local conditions*. A positive integer that satisfies the local conditions and *does not* occur among the curvatures is called **missing**.

Local-global conjecture (Graham–Lagarias–Mallows–Wilks–Yan, 2003; Fuchs–Sandén, 2011). *For every primitive integral Apollonian packing the number of missing curvatures is finite:* that is, every sufficiently large integer that is admissible modulo 24 actually occurs as a curvature.

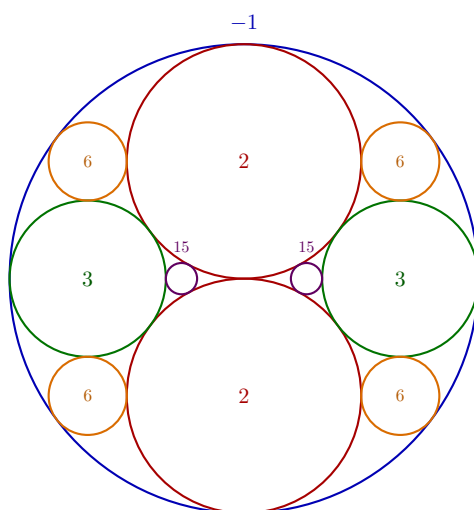


Figure 32.3: The first generations of the Apollonian packing with root quadruple $(-1, 2, 2, 3)$, with the curvatures shown. Centres and radii, all computed by hand and verified in Example 32.4: curvature -1 , centre $(0, 0)$, radius 1 ; curvature 2 , centres $(0, \pm\frac{1}{2})$, radius $\frac{1}{2}$; curvature 3 , centres $(\pm\frac{2}{3}, 0)$, radius $\frac{1}{3}$; curvature 6 , centres $(\pm\frac{2}{3}, \pm\frac{1}{2})$, radius $\frac{1}{6}$; curvature 15 , centres $(\pm\frac{4}{15}, 0)$, radius $\frac{1}{15}$. All curvatures are integers, as required by Corollary 32.2.

In support of the conjecture, Bourgain and Kontorovich proved in 2014 (*Inventiones mathematicae* **196**, pp. 589–650) that the set of missing curvatures has density zero. The conjecture has, however, been **refuted**:

Theorem (S. Haag, C. Kertzer, J. Rickards, K. E. Stange, *The local-global conjecture for Apollonian circle packings is false*, *Annals of Mathematics* **200** (2024), no. 2, pp. 749–770; verified on 7 September 2026). *There exist infinitely many primitive integral Apollonian packings that have at least $c\sqrt{N}$ missing curvatures up to N ; in particular the number of missing curvatures is infinite and the conjecture is false.*

The obstruction is not congruential in nature but one of *reciprocity*: the authors exhibit quadratic and quartic families of integers that are admissible modulo 24 and yet never occur. An explicit example given in the article: in the packing with root quadruple $(-23, 48, 49, 52)$ no integer of the form n^4 , $4n^4$, $9n^4$, or $36n^4$ occurs among the curvatures (type $(6, 1, 1, -1)$ in the article’s obstruction table), even though infinitely many of these are admissible modulo 24: the conjecture is false in every admissible residue class for this packing.

The result was deposited as a preprint ([arXiv:2307.02749](https://arxiv.org/abs/2307.02749)) on 6 July 2023, with a final revision on 5 September 2024, and appeared in the *Annals* in the September 2024 issue: it is the case, described in Section 32.8, in which the preprint and the published article coincide in content and the correct citation is that of the journal.

Example 32.5: The counterexample quadruple. The quadruple $(-23, 48, 49, 52)$ cited in the box is indeed a Descartes quadruple. Verification by hand:

$$-23 + 48 + 49 + 52 = 126, \quad 126^2 = 15,876;$$

$$23^2 = 529, \quad 48^2 = 2,304, \quad 49^2 = 2,401, \quad 52^2 = 2,704,$$

$$2(529 + 2,304 + 2,401 + 2,704) = 2 \cdot 7,938 = 15,876. \quad \checkmark$$

Independent check: the same computation, run with a Python script, returns 15,876 for both sides. $\checkmark$

Remark 32.9: What remains open after the refutation. Refuting a conjecture does not close a field: it reopens it in a more precise form. As of 7 September 2026 the situation is this.

- Haag, Kertzer, Rickards and Stange propose, in the same article, a **refined conjecture**: calling sporadic a missing integer that does not belong to the quadratic or quartic families they identify, the set of sporadic integers of every packing is finite. This conjecture is **open**.
- The phenomenon is not limited to Apollonian circles. Shi, Shi, Whitehead, Williams-Tracy and Zhang extended it to four other families of circle packings — octahedral, cubic, square and triangular — in a work deposited as a preprint ([arXiv:2510.21702](#)) on 24 October 2025, revised on 25 March 2026 and published in the *Journal of Number Theory* **288** (2026), pp. 103–157. There too the obstruction comes from quadratic reciprocity.
- A historical reconstruction of the whole affair, from formulation to refutation, is in the account by E. Fuchs in the *Bulletin of the American Mathematical Society* **63** (2025), no. 1, pp. 59–77.

What is **imported** in this section: everything in the preceding box and in this remark. What is missing here is the theory of thin groups, the circle method applied to the orbits of the Apollonian group, and the theory of reciprocity obstructions: these are the tools of the cited works and are not developed in the treatise. What is instead proved in full, here, is only the elementary part: the hereditary integrality of Corollary 32.2.

32.5. Kissing numbers and packings in dimension 8 and 24

Chapter 14 treats tangencies between circles and the optimality of the hexagonal packing of the plane; Chapter 11 introduces hyperspheres and the volume of the n -dimensional ball. Here we take the simplest question one can ask about a sphere — *how many other equal spheres can it touch?* — and observe that the answer is known in only five dimensions.

Definition 32.7: Kissing number. The **kissing number** τ_n is the maximum number of unit balls of $\mathbb{R}^n$, with pairwise disjoint interiors, that can all be tangent to a fixed unit ball.

Theorem 32.4: The kissing number of the plane equals 6. $\tau_2 = 6$.

Proof. Fix the unit ball centred at the origin. A unit ball of centre C is tangent to it if and only if $|C| = 2$, and two unit balls of centres C_i, C_j have disjoint interiors if and only if $|C_i - C_j| \geq 2$.

Step 1 — translation into angles. Let $C_1, \dots, C_m$ be the centres of m unit balls, all tangent to the central one and with pairwise disjoint interiors. From $|C_i| = 2$ one writes $C_i = 2u_{\alpha_i}$ with $u_{\alpha} = (\cos \alpha, \sin \alpha)$. Let $\delta_{ij} \in [0, \pi]$ be the angle between C_i and C_j ; then

$$|C_i - C_j|^2 = |C_i|^2 + |C_j|^2 - 2\langle C_i, C_j \rangle = 4 + 4 - 8 \cos \delta_{ij} = 8(1 - \cos \delta_{ij}) = 16 \sin^2 \frac{\delta_{ij}}{2},$$

having used $1 - \cos x = 2 \sin^2(x/2)$. Hence $|C_i - C_j| = 4 \sin \frac{\delta_{ij}}{2}$, and the condition $|C_i - C_j| \geq 2$ is equivalent to

$$\sin \frac{\delta_{ij}}{2} \geq \frac{1}{2}.$$

Since $\delta_{ij}/2 \in [0, \pi/2]$, where sine is increasing, this is equivalent to $\delta_{ij}/2 \geq \pi/6$, that is

$$\delta_{ij} \geq \frac{\pi}{3} \quad \text{for every } i \neq j. \quad (32.4)$$

Step 2 — $m \leq 6$. Choose representatives $\alpha_1 < \alpha_2 < \dots < \alpha_m$ in $[0, 2\pi)$ (they are pairwise distinct by (32.4), which forces $\delta_{ij} > 0$). Set the *gaps*

$$g_i = \alpha_{i+1} - \alpha_i \quad (1 \leq i \leq m-1), \quad g_m = 2\pi - \alpha_m + \alpha_1,$$

which are positive and satisfy $\sum_{i=1}^m g_i = 2\pi$. For each i , the angle between C_i and C_{i+1} (indices modulo m) is $\delta = \min\{g_i, 2\pi - g_i\}$. If $g_i \leq \pi$ then $\delta = g_i$, and (32.4) gives $g_i \geq \pi/3$. If instead $g_i > \pi$, then a fortiori $g_i > \pi \geq \pi/3$. In every case $g_i \geq \pi/3$, and hence

$$2\pi = \sum_{i=1}^m g_i \geq m \cdot \frac{\pi}{3} \quad \implies \quad m \leq 6.$$

Step 3 — $m = 6$ is achieved. Take $C_k = 2u_{k\pi/3}$, $k = 0, 1, \dots, 5$. Each C_k has norm 2, so the corresponding ball is tangent to the central one. For $k \neq l$ the angle between C_k and C_l equals $\delta = \frac{\pi}{3}|k-l|$ or $2\pi - \frac{\pi}{3}|k-l|$, and in either case $\delta \geq \pi/3$; by Step 1 this is equivalent to $|C_k - C_l| \geq 2$, that is disjoint interiors. For the closest pair, $\delta = \pi/3$, one has exactly $|C_k - C_l| = 4 \sin \frac{\pi}{6} = 4 \cdot \frac{1}{2} = 2$: the balls touch without overlapping. From Steps 2 and 3, $\tau_2 = 6$. □

Remark 32.10: Numerical check of Theorem 32.4. A Python script built the six centres $C_k = 2u_{k\pi/3}$ and computed the minimum distance between any two of them, obtaining 1.9999999999999996, that is 2 up to floating-point rounding: the configuration of Step 3 is admissible and saturates the inequality. The closed-form check is the computation of Step 3: $4 \sin(\pi/6) = 4 \cdot \frac{1}{2} = 2$. ✓

INFO

Panorama — what is known about kissing numbers. Verified on 7 September 2026. None of the values that follow, except τ_1 and τ_2 , is proved here.

n	τ_n	who, when
1	2	elementary
2	6	elementary (Theorem 32.4)
3	12	K. Schütte, B. L. van der Waerden, 1953
4	24	O. R. Musin, 2008
8	240	V. I. Levenshtein; A. M. Odlyzko, N. J. A. Sloane, 1979
24	196,560	Levenshtein; Odlyzko–Sloane, 1979

For **every other dimension** n the exact value of τ_n is **unknown**: only a lower bound (given by an explicit configuration) and an upper bound (given by a linear programming method) are known, and the two do not coincide. Already for $n = 5$ the problem is open. The phenomenon of dimensions 8 and 24 does not concern kissing alone but also the density of the packing:

Theorem (M. S. Viazovska, *The sphere packing problem in dimension 8*, *Annals of Mathematics* **185** (2017), no. 3, pp. 991–1015). *No packing of unit balls of $\mathbb{R}^8$ has density greater than that of the lattice packing E_8 .*

Theorem (H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska, *The sphere packing problem in dimension 24*, *Annals of Mathematics* **185** (2017), no. 3, pp. 1017–1033). *The Leech lattice gives the densest packing of congruent balls in $\mathbb{R}^{24}$, and it is the unique optimal periodic packing.*

With these two results the sphere packing problem is solved in dimensions 1, 2, 3, 8, 24 and in no other: for $n = 4$, for $n = 5$, and in general for every other n , it is **open**.

Remark 32.11: The two densities, computed twice. Only two pieces of information about the lattices are **imported**, both classical and attributed to their source: the lattice $E_8 \subset \mathbb{R}^8$ and the Leech lattice $\Lambda_{24} \subset \mathbb{R}^{24}$ are even and unimodular (determinant 1, hence covolume 1) and have minimum norm 2 and 4 respectively, that is minimum distance between distinct points $\sqrt{2}$ and 2. Neither their existence nor these properties is proved here. From this data the density can be computed. For a lattice Λ of covolume V and minimum distance $2r$, the balls of radius r centred at the points of Λ have disjoint interiors and the density of the packing is the ratio of the volume of one ball to the covolume:

$$\Delta = \frac{\text{vol } B_n(r)}{V}, \quad \text{vol } B_n(r) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} r^n$$

(the volume formula is that of Chapter 11).

- E_8 : $n = 8$, $V = 1$, minimum distance $\sqrt{2}$, hence $r = \frac{\sqrt{2}}{2}$ and $r^8 = (\frac{\sqrt{2}}{2})^8 = \frac{2^4}{2^8} = \frac{1}{16}$; moreover $\Gamma(5) = 4! = 24$. Hence

$$\Delta_{E_8} = \frac{\pi^4}{24} \cdot \frac{1}{16} = \frac{\pi^4}{384}.$$

- Leech: $n = 24$, $V = 1$, minimum distance 2, hence $r = 1$ and $r^{24} = 1$; moreover $\Gamma(13) = 12! = 479,001,600$. Hence

$$\Delta_{\Lambda_{24}} = \frac{\pi^{12}}{12!}.$$

Numerical values and double check. The closed form gives

$$\frac{\pi^4}{384} = 0.2536695, \quad \frac{\pi^{12}}{12!} = 0.0019295743.$$

A Python script that evaluates $\text{vol } B_n(r)$ using the Γ function (hence by a route different from the hand simplification) returns 0.2536695079010481 for E_8 and 0.001929574309403922 for Leech; the same digits are obtained by evaluating $\pi^4/384$ and $\pi^{12}/12!$ directly. ✓ The two values agree with those reported in H. Cohn's account in the Notices of the AMS **64** (2017), no. 2, pp. 102–115 (verified on 7 September 2026), which writes them as $\pi^4/384 = 0.2536\dots$ and $\pi^{12}/12! = 0.001929\dots$ ✓

32.6. The Erdős–Ulam problem

Points with rational coordinates on the circle and their parametrization by Pythagorean triples are treated in Volume I, Chapter 9. From that parametrization one derives, with a complete proof, a surprising fact: *on the circle there exists a dense set of points whose mutual distances are all rational*. This is why the analogous question in the plane — the Erdős–Ulam problem — is hard: the obvious candidate for a counterexample is not there.

Definition 32.8: Rational-distance set. A set $E \subset \mathbb{R}^2$ is said to be **of rational distances** if $|PQ| \in \mathbb{Q}$ for every $P, Q \in E$. It is said to be **dense** in a set X if every point of X is a limit of points of E .

Lemma 32.2: *The chord between two rational points of the circle. For $t \in \mathbb{R}$ set*

$$P(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right).$$

Then:

- (i) $P(t)$ belongs to the unit circle S^1 , and precisely $P(t) = u_{2\arctan t}$ with $u_\alpha = (\cos \alpha, \sin \alpha)$;
- (ii) $P: \mathbb{R} \rightarrow S^1 \setminus \{(-1, 0)\}$ is a homeomorphism;
- (iii) for every $s, t \in \mathbb{R}$

$$|P(t) - P(s)| = \frac{2|t-s|}{\sqrt{1+t^2}\sqrt{1+s^2}}.$$

Proof. (i). Let $\varphi = \arctan t \in (-\pi/2, \pi/2)$, so that $\tan \varphi = t$. From the double-angle formulas, dividing numerator and denominator by $\cos^2 \varphi \neq 0$,

$$\cos 2\varphi = \cos^2 \varphi - \sin^2 \varphi = \frac{1 - \tan^2 \varphi}{1 + \tan^2 \varphi} = \frac{1 - t^2}{1 + t^2},$$

$$\sin 2\varphi = 2 \sin \varphi \cos \varphi = \frac{2 \tan \varphi}{1 + \tan^2 \varphi} = \frac{2t}{1 + t^2},$$

where $1 = \cos^2 \varphi + \sin^2 \varphi$ was used, that is $1/\cos^2 \varphi = 1 + \tan^2 \varphi$. Hence $P(t) = (\cos 2\varphi, \sin 2\varphi) = u_{2\arctan t}$, which has norm 1.

(ii). The map $t \mapsto 2\arctan t$ is an increasing homeomorphism of $\mathbb{R}$ onto $(-\pi, \pi)$, and $\alpha \mapsto u_\alpha$ is a homeomorphism of $(-\pi, \pi)$ onto $S^1 \setminus \{(-1, 0)\}$ (it is continuous, injective on an interval of width 2π with one endpoint removed, and its inverse is the continuous determination of the argument in $(-\pi, \pi)$). The composition is a homeomorphism.

(iii). Let $\alpha = 2\arctan t$ and $\beta = 2\arctan s$. By point (i),

$$|P(t) - P(s)|^2 = |u_\alpha - u_\beta|^2 = 2 - 2\cos(\alpha - \beta) = 4\sin^2 \frac{\alpha - \beta}{2},$$

so $|P(t) - P(s)| = 2|\sin \frac{\alpha - \beta}{2}| = 2|\sin(\arctan t - \arctan s)|$. Setting $\varphi = \arctan t$, $\psi = \arctan s$,

$$\cos \varphi = \frac{1}{\sqrt{1+t^2}}, \quad \sin \varphi = \frac{t}{\sqrt{1+t^2}}, \quad \cos \psi = \frac{1}{\sqrt{1+s^2}}, \quad \sin \psi = \frac{s}{\sqrt{1+s^2}}$$

(the cosines are positive because $\varphi, \psi \in (-\pi/2, \pi/2)$). By the subtraction formula,

$$\sin(\varphi - \psi) = \sin \varphi \cos \psi - \cos \varphi \sin \psi = \frac{t}{\sqrt{1+t^2}\sqrt{1+s^2}} - \frac{s}{\sqrt{1+t^2}\sqrt{1+s^2}} = \frac{t-s}{\sqrt{1+t^2}\sqrt{1+s^2}},$$

whence the claim follows by taking absolute values and multiplying by 2. $\square$

Theorem 32.5: *A dense set of rational distances exists on the circle. Set*

$$T = \{t \in \mathbb{Q} : 1+t^2 \text{ is the square of a rational}\}, \quad E = \{P(t) : t \in T\} \subset S^1.$$

Then:

- (i) E is a set of rational distances;
- (ii) every point of E has both coordinates rational;
- (iii) E is dense in S^1 .

In particular there exist infinite sets, dense in the unit circle, with all mutual distances rational.

Proof. (i). Let $t, s \in T$ and let $p, q \in \mathbb{Q}$, $p, q > 0$, with $p^2 = 1 + t^2$ and $q^2 = 1 + s^2$; then $\sqrt{1 + t^2} = p$ and $\sqrt{1 + s^2} = q$. By Lemma 32.2(iii),

$$|P(t) - P(s)| = \frac{2|t - s|}{pq} \in \mathbb{Q},$$

because the numerator and denominator are rational and $pq \neq 0$.

(ii). If $t \in \mathbb{Q}$ then $1 + t^2 \in \mathbb{Q}$ and $1 + t^2 \geq 1 > 0$, so $\frac{1-t^2}{1+t^2}$ and $\frac{2t}{1+t^2}$ are rational.

(iii). Consider

$$\phi: (0, +\infty) \rightarrow \mathbb{R}, \quad \phi(\sigma) = \frac{1 - \sigma^2}{2\sigma} = \frac{1}{2\sigma} - \frac{\sigma}{2}.$$

First: ϕ sends rationals to elements of T . If $\sigma \in \mathbb{Q}$, $\sigma > 0$, then $\phi(\sigma) \in \mathbb{Q}$, and

$$1 + \phi(\sigma)^2 = 1 + \frac{(1 - \sigma^2)^2}{4\sigma^2} = \frac{4\sigma^2 + 1 - 2\sigma^2 + \sigma^4}{4\sigma^2} = \frac{1 + 2\sigma^2 + \sigma^4}{4\sigma^2} = \left(\frac{1 + \sigma^2}{2\sigma} \right)^2,$$

the square of a positive rational. Hence $\phi(\mathbb{Q} \cap (0, +\infty)) \subseteq T$.

Second: ϕ is a homeomorphism of $(0, +\infty)$ onto $\mathbb{R}$. It is differentiable with

$$\phi'(\sigma) = -\frac{1}{2\sigma^2} - \frac{1}{2} < 0 \quad \text{for every } \sigma > 0,$$

hence strictly decreasing, therefore injective. Moreover $\lim_{\sigma \rightarrow 0+} \phi(\sigma) = +\infty$ and $\lim_{\sigma \rightarrow +\infty} \phi(\sigma) = -\infty$; being continuous, by the intermediate value theorem ϕ takes every real value: it is surjective onto $\mathbb{R}$. A continuous, strictly monotone bijection between intervals has a continuous inverse: ϕ is a homeomorphism.

Third: T is dense in $\mathbb{R}$. The set $\mathbb{Q} \cap (0, +\infty)$ is dense in $(0, +\infty)$; a homeomorphism sends dense sets to dense sets, hence $\phi(\mathbb{Q} \cap (0, +\infty))$ is dense in $\mathbb{R}$. Being contained in T , a fortiori T is dense in $\mathbb{R}$.

Fourth: E is dense in S^1 . By Lemma 32.2(ii) the map P is a homeomorphism of $\mathbb{R}$ onto $S^1 \setminus \{(-1, 0)\}$; it therefore sends the dense set T to a subset E dense in $S^1 \setminus \{(-1, 0)\}$. Finally $S^1 \setminus \{(-1, 0)\}$ is dense in S^1 , because $(-1, 0)$ is the limit of points u_α with $\alpha \rightarrow \pi^-$, which belong to it. The closure of E therefore contains S^1 , that is, E is dense in S^1 .

Conclusion. E is infinite (it contains the image of a dense set, hence is infinite) and by (i) and (iii) it is a set dense in the circle with all mutual distances rational. $\square$

Example 32.6: Six explicit points, with the distance matrix. Six parameters of T and the corresponding points are chosen (all with rational coordinates, as Theorem 32.5(ii) requires):

t	$\sqrt{1 + t^2}$	$P(t)$
0	1	(1, 0)
3/4	5/4	$(\frac{7}{25}, \frac{24}{25})$
4/3	5/3	$(-\frac{7}{25}, \frac{24}{25})$
5/12	13/12	$(\frac{119}{169}, \frac{120}{169})$
12/5	13/5	$(-\frac{119}{169}, \frac{120}{169})$
8/15	17/15	$(\frac{161}{289}, \frac{240}{289})$

The six values of $\sqrt{1 + t^2}$ are rational because the chosen fractions come from the Pythagorean triples (3, 4, 5), (5, 12, 13) and (8, 15, 17): for example $1 + (\frac{3}{4})^2 = \frac{25}{16} = (\frac{5}{4})^2$.

The fifteen distances, computed with Lemma 32.2(iii):

	0	3/4	4/3	5/12	12/5	8/15
0	0	$\frac{6}{5}$	$\frac{8}{5}$	$\frac{10}{13}$	$\frac{24}{13}$	$\frac{16}{17}$
3/4	$\frac{6}{5}$	0	$\frac{14}{25}$	$\frac{65}{32}$	$\frac{65}{66}$	$\frac{85}{26}$
4/3	$\frac{8}{5}$	$\frac{14}{25}$	0	$\frac{65}{66}$	$\frac{32}{65}$	$\frac{72}{85}$
5/12	$\frac{10}{13}$	$\frac{65}{32}$	$\frac{65}{66}$	0	$\frac{238}{169}$	$\frac{42}{221}$
12/5	$\frac{24}{13}$	$\frac{65}{66}$	$\frac{32}{65}$	$\frac{238}{169}$	0	$\frac{280}{221}$
8/15	$\frac{16}{17}$	$\frac{26}{85}$	$\frac{72}{85}$	$\frac{42}{221}$	$\frac{280}{221}$	0

All rational. **Independent check:** the same fifteen distances were recomputed with a Python script *without* using the lemma's formula, that is, from the exact coordinates of the points with fraction arithmetic, extracting the exact square root of $|P - Q|^2$; the fifteen values agree one by one with those in the table. ✓ For example, between $(1, 0)$ and $(\frac{7}{25}, \frac{24}{25})$:

$$\left(1 - \frac{7}{25}\right)^2 + \left(\frac{24}{25}\right)^2 = \left(\frac{18}{25}\right)^2 + \left(\frac{24}{25}\right)^2 = \frac{324 + 576}{625} = \frac{900}{625}, \quad \sqrt{\frac{900}{625}} = \frac{30}{25} = \frac{6}{5}. \quad \checkmark$$

INFO

Panorama — the Erdős–Ulam problem. Open as of 7 September 2026. Nothing that follows is proved here.

Problem (Ulam, 1945). Does there exist a set $E \subset \mathbb{R}^2$ dense in the plane and of rational distances?

Theorem 32.5 shows that the answer is *yes* if the plane is replaced by the circle; the same holds, trivially, for a line (the points with rational coordinate). Erdős conjectured that lines and circles were the only exceptions, and that no dense rational-distance set exists in the plane. As of 7 September 2026 the problem is **open**. What is known:

- J. Solymosi and F. de Zeeuw (*Discrete & Computational Geometry* **43** (2010), no. 2, pp. 393–401) proved Erdős's conjecture for algebraic curves: *no irreducible algebraic curve other than a line or a circle contains an infinite rational-distance set*. Theorem 32.5 is thus an extremal case: the circle is one of only two curves that can host infinite sets of that kind.
- J. Shaffaf (*Discrete & Computational Geometry* **60** (2018), no. 2, pp. 283–293) and, independently, K. Ascher, L. Braune and A. Turchet (*Bulletin of the London Mathematical Society* **52** (2020), no. 6, pp. 1053–1063) proved that, *assuming* the Bombieri–Lang conjecture, no dense rational-distance set exists in the plane.

The second item is the chapter's most instructive example of what a *conditional* result is: it is a theorem in full standing, but its conclusion depends on a conjecture that is itself open. Citing it as "proved that it does not exist" would be false.

32.7. Gauss's circle problem: what is counted and what remains open

INFO

Level B for counting and the elementary estimate; level D for reading research. Bibliographic verification: 10 September 2026. This section distinguishes proofs developed here, external results and conjectures. The supplied structure document places Gauss's circle problem in printed Chapter 12 of Volume I, but that volume was not supplied: its proof is not assumed. The elementary estimate is proved here to make this addition readable, without attributing

it to an unconsulted page.

32.7.1 Conventions and a first proved bound

Count integer lattice points in the closed disk of radius $r \geq 0$, including the origin and the boundary:

$$N(r) = \#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq r^2\}, \quad E(r) = N(r) - \pi r^2.$$

Area suggests $N(r) \approx \pi r^2$, but does not specify the error. For a paper using $X = r^2$, write $R(X) = N(\sqrt{X}) - \pi X$: then $E(r) = R(r^2)$. An exponent θ in the variable X becomes 2θ in the variable r .

Proposition 32.3: Estimate using unit squares. For every $r \geq 0$, setting $c = \sqrt{2}/2$ gives

$$\pi \max(r - c, 0)^2 \leq N(r) \leq \pi(r + c)^2, \quad |E(r)| \leq \sqrt{2}\pi r + \frac{\pi}{2}.$$

In particular, $E(r) = O(r)$ as $r \rightarrow \infty$, and $|E(r)| \leq 2\sqrt{2}\pi r$ for $r \geq 1$.

Proof. Associate each lattice point p in the disk with the square $Q_p = p + [-1/2, 1/2]^2$. Their interiors are disjoint and each has area one, so the finite union U_r has area $N(r)$; overlaps along edges have area zero. Every point of Q_p lies at distance at most c from p , so U_r is contained in the disk of radius $r + c$, giving the upper bound.

If $r \geq c$ and $|x| \leq r - c$, round both coordinates of x to their nearest integers, obtaining p with $|p - x| \leq c$. Then $|p| \leq |x| + c \leq r$ and $x \in Q_p \subset U_r$. Thus the disk of radius $r - c$ is contained in the union. For $r < c$, the zero lower bound is immediate. Subtracting πr^2 from the upper estimate gives $E(r) \leq 2\pi cr + \pi c^2$. If $r \geq c$, the lower estimate gives $E(r) \geq -2\pi cr + \pi c^2 \geq -2\pi cr - \pi c^2$. If $r < c$, use $E(r) \geq -\pi r^2 \geq -\pi c^2$. This proves the absolute bound. Finally, for $r \geq 1$, $\pi/2 \leq \sqrt{2}\pi r$, giving the coarser estimate. $\square$

The small-radius restriction matters: $N(0) = 1$ and $E(r) \rightarrow 1$ as $r \downarrow 0$. No estimate $|E(r)| \leq Cr$ with finite C can hold for every positive radius. The asymptotic statement concerns $r \rightarrow \infty$.

Proposition 32.4: Exact counting by vertical slices.

$$N(r) = \sum_{m=-\lfloor r \rfloor}^{\lfloor r \rfloor} \left(2 \left\lfloor \sqrt{r^2 - m^2} \right\rfloor + 1 \right).$$

Proof. For integer abscissa m with $|m| \leq r$, admissibility is

$$-\sqrt{r^2 - m^2} \leq n \leq \sqrt{r^2 - m^2}.$$

Admissible integers run from $-\lfloor \sqrt{r^2 - m^2} \rfloor$ to its opposite, including both endpoints, giving exactly $2\lfloor \sqrt{r^2 - m^2} \rfloor + 1$ points. Columns are disjoint and the sum exhausts them. $\square$

For integer radius R , the code uses the exact integer square root $\text{isqrt}(R^2 - m^2)$ and compares it with independent enumeration of points in $[-R, R]^2$. Floating-point square roots are not used to decide boundary membership. Radii 0, 1, 5, 10, 100, 1000 and their counts are recorded in the JSON. This checks the algorithm and examples, not an asymptotic exponent.

32.7.2 Research overview: external results, not proofs in this section

Remark 32.12: Lower bounds and the conjecture. The introduction of [14] recalls Hardy’s historical result (1915): in the variable X , $R(X) = \Omega((X \log X)^{1/4})$. This is reported as an external result, not given a fictitious proof. In this section’s convention, it implies that $E(r) = O(r^\alpha)$ is impossible for $\alpha < 1/2$. Indeed, along an unbounded sequence, the error magnitude is at least a positive constant times $r^{1/2}(\log(r^2))^{1/4}$, whose ratio to r^α tends to infinity.

The usual conjecture is that for every $\varepsilon > 0$ there exists C_ε such that $|E(r)| \leq C_\varepsilon r^{1/2+\varepsilon}$ for every sufficiently large radius. One cannot remove ε merely by taking a limit: the constant depends on it. The cited lower bound even rules out uniform $O(r^{1/2})$, but is compatible with every exponent $1/2 + \varepsilon$.

Remark 32.13: A dated comparison of sources. The supplied plan calls Huxley’s bound current and dates it to 2000. The verified publisher record for Exponential Sums and Lattice Points III instead gives 2003: [13], volume 87, issue 3, pp. 591–609. Its radius exponent is $131/208 \simeq 0.6298076923$, with logarithmic factors; this implies the weaker form $E(r) = O_\varepsilon(r^{131/208+\varepsilon})$. The lengthy analytic-number-theory proof is not reproduced here.

A later improvement is stated by Xiaochun Li and Xuerui Yang in preprint [14], version 2 dated 14 September 2023, Theorem 1.2: $R(X) = O_\varepsilon(X^{\theta_*+\varepsilon})$, where $\theta_* = 0.3144831759741\dots$. In the radius variable this becomes

$$E(r) = O_\varepsilon(r^{2\theta_*+\varepsilon}), \quad 2\theta_* = 0.6289663519482\dots$$

Replacing 2ε by ε is valid because the result holds for every positive number: apply the X estimate with $\varepsilon/2$. This is a consulted preprint, not a journal publication whose acceptance has been verified. The authors’ exact definition is that $-\theta_*$ is the unique solution $x \in [-0.35, -0.3]$ of

$$-\frac{8}{25}x - \frac{1}{200} \left(\sqrt{2(1-14x)} - 5\sqrt{-1-8x} \right)^2 + \frac{51}{200} = -x.$$

The formula identifies the number without treating its decimal approximation as exact; uniqueness here is part of the cited result, not a new proof.

The Bourgain–Watt preprint [15], arXiv:1709.04340, is marked withdrawn on 16 July 2023: the notice reports gaps in the proofs of Theorems 1–3. The numerical bound often associated with that text must therefore not be listed as a verified theorem without another valid source. This comparison corrects the plan using the consulted documents; it does not claim an exhaustive census of every manuscript available by the verification date.

32.7.3 Guided reading with answers

Read the introduction and Theorem 1.2 of [14], comparing the publisher record of [13] and the withdrawal notice of [15]. The survey [16] is a bibliographic suggestion for further reading, not a source for a record updated to 2026.

1. Why can $131/416$ and $131/208$ describe the same bound? **Answer:** the first is an exponent of $X = r^2$; the second is the radius exponent.
2. Does an upper bound $O(r^\alpha)$ prove that the error has exact order r^α ? **Answer:** no; an upper bound allows much smaller errors and supplies no matching lower bound.
3. Does Ω notation require large error at every radius? **Answer:** under the convention used

here, an unbounded sequence of radii satisfying the lower bound suffices; not every radius.

4. Can a million counts prove the conjecture? **Answer:** no: a finite check cannot control every radius beyond every threshold. It can find counterexamples to a specified numerical inequality or test code.
5. What separates this overview from a solution of the problem? **Answer:** a proof of the conjectured exponent, or an asymptotic counterexample ruling it out. A new decimal place in an upper bound is not by itself a solution of the conjecture.

32.8. Research literacy

The preceding sections have cited fourteen works, of which two are unpublished preprints, one was published after being a preprint for a year, one refutes a twenty-one-year-old conjecture, and two prove a statement *under the hypothesis of another conjecture*. Holding this material together without confusion is not a matter of memory but of method, and the method can be learned. This section sets it out.

32.8.1 Preprint, submission, review, publication

Definition 32.9: The editorial states of a mathematical work. A mathematical work normally passes through these states.

- (a) **Preprint** (or *e-print*): the text is deposited in a public archive — for mathematics, almost always **arXiv** — and anyone can read it. The deposit involves a relevance check, *not* a correctness check. A preprint may contain errors, even fatal ones, and remain visible for years.
- (b) **Submitted**: the authors have sent the work to a journal. No one, externally, has yet judged anything.
- (c) **Under review** (peer review): the editorial office has entrusted the text to one or more anonymous experts, who check its correctness, originality and interest, and recommend acceptance, revision or rejection. This is the only stage at which someone is *required* to check the proofs.
- (d) **Accepted** (*to appear*): the journal has decided to publish it; the final text exists, but has no volume, issue or pages yet.
- (e) **Published**: the work has a stable location — journal, volume, issue, year, pages — and usually a DOI. This is the state in which the citation becomes definitive.

The same work can be in (a) and in (e) at the same time: the preprint stays online even after publication, often with the journal indication added in the *Journal reference* field.

Best practice. How to read the header of an arXiv preprint. Before the text, the deposit record already says almost everything needed to decide how much weight to give it.

- **Identifier** AAMM.NNNNN: the first two digits are the year, the next two the month of the *first* deposit. Thus 2604.17116 was deposited in April 2026, and 2307.02749 in July 2023. This is the first consistency check to make: an identifier that does not follow this scheme, or that points to a future month, is wrong.
- **Version** v1, v2, ... and *submission history*: the list of deposit dates. A single version can mean "work just released" or "work never touched again"; several versions mean the authors corrected or expanded it, often after referee comments.
- **Journal reference** and **DOI**: if present, the work is published and *the journal is cited*, not the archive. If absent, it is a preprint.

- **Comments:** a free-text field, where authors write the number of pages, whether the work has been submitted, accepted, or which earlier version it replaces.
- **Subjects** (MSC classification): indicates the area, and hence which expertise is needed to check the work.

Example 32.7: Three real cases, read line by line. The following three records were consulted on 7 September 2026.

- (a) **arXiv:2005.09193**, Greene and Lobb, *The rectangular peg problem*, deposited on 19 May 2020, 6 pages. The record shows the journal reference: *Annals of Mathematics* **194** (2021), no. 2, pp. 509–517. *Status: published*. The journal is cited.
- (b) **arXiv:2307.02749**, Haag, Kertzer, Rickards and Stange, *The local–global conjecture for Apollonian circle packings is false*, deposited on 6 July 2023, last revised on 5 September 2024. Journal reference: *Annals of Mathematics* **200** (2024), no. 2, pp. 749–770. *Status: published*, after fourteen months and at least one revision. The journal is cited.
- (c) **arXiv:2604.17116**, Greene and Lobb, *Jordan curves inscribe a positive measure of rectangles*, deposited on 18 April 2026, a single version, no journal reference, no editorial DOI. *Status: preprint*. The archive is cited, with identifier, version and date, and it is stated that the work has not yet been submitted to review — or that this is not on record.

Cases (a) and (c) have the same authors and closely related topics: this is exactly the situation in which one goes wrong. Work (a) proves the *smooth* case for *all* aspect ratios; work (c) proves, for *arbitrary* curves, a statement about a *set of positive measure* of ratios. Attributing to the first the statement of the second, or vice versa, produces a false citation even if the authors' names are correct.

Warning. Five mistakes that cost a false citation.

1. *Citing an arXiv identifier without opening it.* A number reported by a third party can be wrong, or correspond to a work different from the one believed. The check costs thirty seconds.
2. *Confusing "announced" with "proved".* An announcement — in a seminar, in an abstract, in a message — is not a checkable proof. Until the text is public and verifiable, it is not cited as a theorem.
3. *Treating a conditional result as unconditional.* "Assuming the Bombieri–Lang conjecture, there is no dense rational-distance set" is not "there is no dense rational-distance set".
4. *Copying pages and DOIs from someone else's bibliography.* Bibliographic errors propagate; the page number must be read from the source, or from an editorial registry (Crossref, the journal's website).
5. *Not dating statements about the status of a problem.* "It is open" is a statement with an expiry date. It must be written "open as of ...", with the date of verification.

32.8.2 How to read a research article

Best practice. A five-pass reading. A research article is not read from the first line to the last, like a textbook chapter. It is read like this.

1. *Header* (two minutes). Authors, date, editorial status, length, classification. What is being read, and who has already checked it.
2. *Main statement* (ten minutes). One looks for the first numbered theorem and tran-

scribes it by hand, separating *hypotheses* and *conclusion*. Every technical word whose exact definition is unknown is underlined.

3. *Special cases* (half an hour). The general object is replaced with the simplest possible example and one checks that the statement says something sensible. For the Greene and Lobb theorem: the curve is a circle, and then the conclusion is obvious; it is the boundary of an ellipse, and then one recovers Theorem 32.2.
4. *Structure of the proof* (one hour). Not the details: the list of steps and, for each, which hypothesis it consumes. One marks where external results, cited and not proved, enter: these are the points where the reader will have to trust or go and check.
5. *Details* (days). Only if needed, and only on the steps identified as critical.

Remark 32.14: How to tell a proved result from an announced one. *The operative criterion is a single one: does a public text exist that contains the proof, and has someone checked it? From here one descends through five rungs, from the most solid to the most fragile:*

- (1) *published in a peer-reviewed journal, and subsequently cited and used by other works — the case of Greene–Lobb 2021, of Haag–Kertzer–Rickards–Stange 2024, of Viazovska 2017;*
- (2) *published, but recent and still little used;*
- (3) *accepted, not yet typeset;*
- (4) *public preprint, with a complete proof in the text, not submitted or pending — the case of arXiv:2604.17116 as of 7 September 2026;*
- (5) *announcement with no public text: a seminar title, an abstract, a communication. It is not cited as a result.*

*In the treatise, levels (1)–(3) can appear in a **teorema** environment if the proof is reported, and in **osservazione** or **infobox** as panorama if it is not; levels (4) and (5) appear only as panorama, with the status declared.*

32.8.3 How to cite

Best practice. What a citation must contain for the reader to be able to check it.

- *Published article:* authors (initial and surname, all of them), exact title, journal, volume in bold, year in parentheses, issue number, page range, DOI. Example:
J. E. Greene, A. Lobb, *The rectangular peg problem*, *Annals of Mathematics* **194** (2021), no. 2, pp. 509–517, DOI 10.4007/annals.2021.194.2.4.
- *Preprint:* authors, title, archive and identifier with version, deposit date, and the explicit word "preprint". Example:
J. E. Greene, A. Lobb, *Jordan curves inscribe a positive measure of rectangles*, preprint arXiv:2604.17116v1, 18 April 2026.
- *Book:* authors, title, edition, publisher, city, year, ISBN, and the chapter or page range actually used.
- *Internal cross-reference in the text:* the precise point, not the work as a whole — "Theorem 1.3", "Section 4", "pp. 752–755". A citation pointing to seventeen pages is not verifiable in practice.
- *Verification date:* in a work that states something about the *status* of research, every entry carries the date on which it was checked. This is the convention followed in the **fonti** environment of this chapter.

WARNING

What is not a source. A collaborative encyclopedia, a blog, a forum, an answer generated by an automated system, and a seminar slide can be excellent *clues*: they say where to look. They are not final authorities, and must never be cited in place of the original work. The rule, in this treatise, is that every entry of the **fonti** environment has been opened and read in its editorial location, and that the date of that check is written down.

Table 32.2: Status of every statement cited in the chapter, as of 7 September 2026. "Proved here" means the complete proof is in this chapter; "panorama" means the statement is reported with its source and the proof is elsewhere; "open" means neither the statement nor its negation is known.

Statement	Status	Where, in this chapter
Every centrally symmetric convex body inscribes a square and every θ -rectangle	proved here	Theorems 32.1 and 32.2
Aspect ratio of a θ -rectangle $= \min\{\tan(\theta/2), \cot(\theta/2)\}$	proved here	Proposition 32.1
Every Jordan curve inscribes a square (Toeplitz, 1911)	open	Section 32.2
Every <i>smooth</i> Jordan curve inscribes every rectangle (Greene–Lobb, 2021)	theorem, published	Observation 32.4
Every Jordan curve inscribes θ -rectangles for a set of angles of measure $\geq B(\gamma)$ (Greene–Lobb, 2026)	preprint	Observation 32.6
The curvatures of an integral Apollonian packing remain integral	proved here	Corollary 32.2
Apollonian <i>local–global</i> conjecture (2003)	refuted in 2024	Section 32.4
Refined conjecture on sporadic integers (2024)	open	Observation 32.9
$\tau_2 = 6$	proved here	Theorem 32.4
$\tau_3 = 12, \tau_4 = 24, \tau_8 = 240, \tau_{24} = 196,560$	theorems, published	Section 32.5
τ_n for $n \notin \{1, 2, 3, 4, 8, 24\}$	open	Section 32.5
Optimal packing in dimension 8 and 24 (2017)	theorems, published	Section 32.5
Optimal packing in dimension $n \notin \{1, 2, 3, 8, 24\}$	open	Section 32.5
A dense rational-distance set exists on the circle	proved here	Theorem 32.5
A dense rational-distance set exists in the <i>plane</i> (Erdős–Ulam)	open	Section 32.6
It does not exist, assuming Bombieri–Lang (2018, 2020)	<i>conditional</i> theorem	Section 32.6

32.9. Assigned exercises

These are *new* exercises, distinct from those worked out in Section 32.10. The answers are in the Teacher’s Key (Section 32.12).

Warning. None of the exercises in this chapter asks to prove a conjecture. The problems of Sections 32.2, 32.4, 32.5 and 32.6 are open: asking for a proof of them in an exercise would be dishonest toward the reader. The first two exercises are *guided readings* of two real articles, with comprehension questions; the third is computational, on the results proved in the chapter.

Exercise 32.1: Guided reading 1 — Greene and Lobb, *The rectangular peg problem*. *Goal:* to read a six-page research article and extract its statement, hypotheses, proof structure and what remains outside it, without attempting to verify the technical details. *Estimated time:* 90 minutes. *Type:* reading and comprehension. *Level:* intermediate.

Text. J. E. Greene, A. Lobb, *The rectangular peg problem*, *Annals of Mathematics* **194** (2021), no. 2, pp. 509–517; freely accessible version: preprint [arXiv:2005.09193](#), deposited on 19 May 2020.

- (a) *Header.* Open the preprint record. Report: identifier, date of first deposit, number of versions, number of pages, subject classification, *Journal reference* field. Deduce the editorial status according to Definition 32.9 and say which of the two citations — archive or journal — should be used, and why.
- (b) *Statement.* Transcribe the main statement, separating the hypotheses and the conclusion into two columns. Indicate which words are technical terms requiring a precise definition.
- (c) *Translation into the chapter’s language.* Rewrite the statement using Definition 32.4 and Proposition 32.1, that is, in terms of θ -rectangles instead of similarities. Check that the two formulations say the same thing.
- (d) *Special cases.* Apply the statement to: (1) the unit circle; (2) the boundary of an ellipse; (3) the boundary of a square. For which of the three does Theorem 32.2 of this chapter already give the same conclusion? For which, instead, does the article’s statement not apply at all, and why?
- (e) *Structure.* List the steps of the proof without going into detail, and identify the external result the authors declare they need. Who proved it, and in what year? What would happen to the article if that result were false?
- (f) *What remains open.* Indicate which hypothesis of the statement cannot be removed with this proof, and which classical problem is obtained by removing it.
- (g) *Citation.* Write the complete citation of the article according to the scheme of Section 32.8, and the complete citation of the preprint. State in which situation the second would be used instead of the first.

Exercise 32.2: Guided reading 2 — Haag, Kertzer, Rickards and Stange, *The local-global conjecture ... is false*. *Goal:* to read an article that *refutes* a conjecture, to distinguish the refuted statement, the counterexample and the replacement statement, and to reconstruct the editorial chronology. *Estimated time:* 100 minutes. *Type:* reading and comprehension. *Level:* intermediate-advanced.

Text. S. Haag, C. Kertzer, J. Rickards, K. E. Stange, *The local-global conjecture for Apollonian circle packings is false*, *Annals of Mathematics* **200** (2024), no. 2, pp. 749–770; preprint [arXiv:2307.02749](#).

- (a) *Chronology.* From the preprint record, derive: date of first deposit, date of last revision, number of versions, journal reference. Compute how many months elapsed between first deposit and publication. What does this interval say about the review process?
- (b) *The refuted statement.* Transcribe, as formulated in the article, the statement of the *local-global* conjecture, with the attribution and year the authors assign to it. What

exactly is a "missing" curvature?

- (c) *The local conditions.* Explain what role the number 24 plays in the statement. Why is a condition modulo a fixed integer called "local"?
- (d) *The main theorem.* Transcribe the statement of the theorem that refutes the conjecture, with its numbering in the article, and say why a bound of the type $\Omega(\sqrt{N})$ on missing curvatures up to N is sufficient to falsify the statement of point (b).
- (e) *The explicit counterexample.* Identify in the article a concrete packing and the families of integers missing from it. Verify by hand, using identity (32.3), that the cited quadruple is indeed a Descartes quadruple.
- (f) *The replacement conjecture.* Transcribe the refined conjecture proposed by the authors, with its numbering. What changes with respect to the refuted one? Is it proved?
- (g) *Metacognition.* In Table 32.2 two rows concern this article, with different statuses. Explain why, and what would be wrong with writing "the *local-global* problem is solved."

Exercise 32.3: Computations on the proved results. *Goal:* to use the theorems proved in the chapter, and to distinguish what they give from what an imported statement gives. *Estimated time:* 60 minutes. *Type:* symbolic and numerical computation. *Level:* basic–intermediate.

- (a) Let γ be the boundary of a rectangle of sides p and q . Derive in closed form

$$B(\gamma) = \frac{4pq}{p^2 + q^2}$$

and compute its value for $(p, q) = (1, 1), (1, 2), (1, 3), (2, 3)$, to seven decimal places. For which of these rectangles is $B(\gamma) > \pi/2$? What can be concluded with Proposition 32.2, and what does Theorem 32.2 give instead, for the same rectangles? Which of the two is stronger, and why?

- (b) For the 1×3 rectangle of point (a), determine the set of aspect ratios λ for which Statement B of Observation 32.6 guarantees an inscribed rectangle. (Hint: $\lambda = \tan(\theta/2)$, and θ ranges over $(0, B(\gamma))$.)
- (c) In the Apollonian packing with root quadruple $(-1, 2, 2, 3)$, start from the quadruple $(-1, 2, 3, 6)$ and apply Theorem 32.3 by replacing the curvature -1 ; then start from $(2, 2, 3, 15)$ and replace the curvature 3. Write the two new quadruples and check identity (32.3) for each, by hand and with a script. Finally list the curvatures found so far and their residues modulo 24.
- (d) From the Pythagorean triple $(20, 21, 29)$ derive a value $t \in T$ (with T as in Theorem 32.5) and the corresponding point $P(t)$. Compute its distances from the points $P(0) = (1, 0)$ and $P(3/4) = (\frac{7}{25}, \frac{24}{25})$ in two ways: with the formula of Lemma 32.2(iii) and directly from the coordinates, showing that both give the same rational number.

32.10. Worked exercises

Exercise 32.4: Squares and rectangles inscribed in the rhombus $|x| + |y| \leq 1$. *Type:* direct application of Theorems 32.1 and 32.2, with exact computation. Let $K = \{(x, y) : |x| + |y| \leq 1\}$ and $\gamma = \partial K$. Determine (i) all the squares inscribed in γ produced by the construction of Theorem 32.1; (ii) an inscribed $\frac{\pi}{3}$ -rectangle, with exact vertices, sides and aspect ratio.

Solution. Strategy. K is compact, convex, with non-empty interior and $K = -K$: both theorems apply. Everything reduces to studying the radial function.

Step 1 — the radial function. A point ϱu_φ lies on γ if and only if $|\varrho \cos \varphi| + |\varrho \sin \varphi| = 1$, that is, since $\varrho > 0$,

$$\rho(\varphi) = \frac{1}{|\cos \varphi| + |\sin \varphi|}.$$

Note that $r = \frac{\sqrt{2}}{2}$ and $R = 1$ work in Lemma 32.1, because $1 \leq |\cos \varphi| + |\sin \varphi| \leq \sqrt{2}$.

Step 2 — ρ has period $\pi/2$. Since $|\cos(\varphi + \frac{\pi}{2})| = |\sin \varphi|$ and $|\sin(\varphi + \frac{\pi}{2})| = |\cos \varphi|$, the denominator does not change: $\rho(\varphi + \frac{\pi}{2}) = \rho(\varphi)$ for every φ .

Step 3 — (i) all the squares. The function h of Step 1 of the proof of Theorem 32.1 is $h(\varphi) = \rho(\varphi) - \rho(\varphi + \frac{\pi}{2}) \equiv 0$: every φ_0 gives an inscribed square. The rhombus therefore has infinitely many inscribed squares, a one-parameter family. Two specimens:

$$\varphi_0 = 0 : \quad \varrho = \rho(0) = 1, \quad \text{vertices } (\pm 1, 0), (0, \pm 1), \quad \text{side } \sqrt{2};$$

$$\varphi_0 = \frac{\pi}{4} : \quad \varrho = \rho\left(\frac{\pi}{4}\right) = \frac{1}{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}} = \frac{1}{\sqrt{2}} = 0.7071068, \quad \text{vertices } \left(\pm \frac{1}{2}, \pm \frac{1}{2}\right), \quad \text{side } 1.$$

Membership check: $|1| + |0| = 1 \checkmark$ and $\left|\frac{1}{2}\right| + \left|\frac{1}{2}\right| = 1 \checkmark$.

Step 4 — (ii) the $\frac{\pi}{3}$ -rectangle. We need φ_0 with $\rho(\varphi_0) = \rho(\varphi_0 + \frac{\pi}{3})$. Set $f(\varphi) = |\cos \varphi| + |\sin \varphi|$, of period $\frac{\pi}{2}$ by Step 2, and observe that on $[0, \frac{\pi}{2}]$ one has $f(\varphi) = f(\frac{\pi}{2} - \varphi)$ (because there $f(\varphi) = \cos \varphi + \sin \varphi$, symmetric about $\frac{\pi}{4}$). For $\varphi \in [\frac{\pi}{6}, \frac{\pi}{2}]$ one has $\varphi + \frac{\pi}{3} \in [\frac{\pi}{2}, \frac{5\pi}{6}]$ and hence, by the $\frac{\pi}{2}$ periodicity,

$$f\left(\varphi + \frac{\pi}{3}\right) = f\left(\varphi - \frac{\pi}{6}\right).$$

The condition $f(\varphi) = f(\varphi - \frac{\pi}{6})$ holds if the two arguments are symmetric about $\frac{\pi}{4}$, that is if

$$\varphi - \frac{\pi}{6} = \frac{\pi}{2} - \varphi \implies 2\varphi = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3} \implies \varphi_0 = \frac{\pi}{3}.$$

Direct check: $f(\frac{\pi}{3}) = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1+\sqrt{3}}{2}$ and $f(\frac{\pi}{6}) = \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{1+\sqrt{3}}{2}$: equal. $\checkmark$ Hence

$$\varrho = \rho\left(\frac{\pi}{3}\right) = \frac{2}{1 + \sqrt{3}} = \frac{2(\sqrt{3} - 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = \frac{2(\sqrt{3} - 1)}{2} = \sqrt{3} - 1 = 0.7320508.$$

Step 5 — the four vertices. With $\varphi_0 = \frac{\pi}{3}$ and $\varphi_0 + \theta = \frac{2\pi}{3}$:

$$A = (\sqrt{3} - 1) \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = \left(\frac{\sqrt{3}-1}{2}, \frac{3-\sqrt{3}}{2}\right) = (0.3660254, 0.6339746),$$

$$B = (\sqrt{3} - 1) \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = (-0.3660254, 0.6339746), \quad C = -A, \quad D = -B.$$

Membership check for γ :

$$|x_A| + |y_A| = \frac{\sqrt{3}-1}{2} + \frac{3-\sqrt{3}}{2} = \frac{2}{2} = 1. \quad \checkmark$$

The same for B , and for C, D by symmetry.

Step 6 — sides and aspect ratio. By Proposition 32.1 with diagonal $d = 2\varrho = 2(\sqrt{3} - 1)$ and $\theta = \frac{\pi}{3}$:

$$\text{short side} = d \sin \frac{\pi}{6} = 2(\sqrt{3} - 1) \cdot \frac{1}{2} = \sqrt{3} - 1 = 0.7320508,$$

$$\text{long side} = d \cos \frac{\pi}{6} = 2(\sqrt{3} - 1) \cdot \frac{\sqrt{3}}{2} = \sqrt{3}(\sqrt{3} - 1) = 3 - \sqrt{3} = 1.2679492,$$

$$\text{ratio} = \frac{\sqrt{3}-1}{3-\sqrt{3}} = \frac{\sqrt{3}-1}{\sqrt{3}(\sqrt{3}-1)} = \frac{1}{\sqrt{3}} = 0.5773503 = \tan \frac{\pi}{6}. \quad \checkmark$$

Independent check. A Python script that builds the four vertices from the numerical coordinates and measures the four sides, the two diagonals and the angle between them returns: sides 0.7320508 and 1.2679492 (alternating), diagonals both $1.4641016 = 2(\sqrt{3}-1)$, aspect ratio 0.5773503, angle between the diagonals $1.0471976 = \pi/3$. All the numbers agree with the closed-form ones. $\checkmark$

Interpretation. The rhombus is the case in which the construction of Theorem 32.1 produces not one but infinitely many squares: the function h is identically zero, not merely equipped with one zero. It is the opposite extreme of a "generic" curve, where the zeros of h are isolated.

Typical mistakes. (1) Believing the inscribed square is unique: here there is a one-parameter family. (2) Taking the square with vertices $(\pm 1, 0), (0, \pm 1)$ for the rhombus and concluding that the "inscribed square" coincides with the curve: this is true only for $\varphi_0 \equiv 0 \pmod{\pi/2}$. (3) Forgetting that $\theta = \frac{\pi}{3}$ gives ratio $\tan \frac{\pi}{6}$ and not $\tan \frac{\pi}{3}$.

Extension. Repeat for $\theta = \frac{\pi}{4}$: one finds $\varphi_0 = \frac{3\pi}{8}$ by the same symmetry argument, and aspect ratio $\tan \frac{\pi}{8} = \sqrt{2} - 1 = 0.4142136$.

Exercise 32.5: The half-disc: a case of exact boundary. *Type:* computation of $B(\gamma)$ and correct reading of an imported statement. Let γ be the boundary of the half-disc $D = \{(x, y) : x^2 + y^2 \leq 1, y \geq 0\}$, that is the upper semicircle joined to the diameter $[-1, 1] \times \{0\}$. Compute $B(\gamma)$ and state exactly what Statement B of Observation 32.6 guarantees.

Solution. Strategy. The enclosed area and the diameter are needed; then apply Definition 32.5 and Proposition 32.2.

Step 1 — the area. D is half the unit disc, hence $A(\gamma) = \frac{\pi}{2} = 1.5707963$.

Step 2 — the diameter. Let $P, Q \in \gamma$. Both belong to the closed unit disc, so $|PQ| \leq 2$; and the two points $(-1, 0)$ and $(1, 0)$ lie on γ (they are the endpoints of the diameter) and are at distance exactly 2. Hence $\text{diam}(\gamma) = 2$ and $R(\gamma) = 1$.

Step 3 — the constant.

$$B(\gamma) = \frac{A(\gamma)}{R(\gamma)^2} = \frac{\pi/2}{1} = \frac{\pi}{2} = 1.5707963.$$

Step 4 — what follows. The curve γ is rectifiable (length $\pi + 2 = 5.1415927$), so Statement B applies and guarantees an inscribed θ -rectangle for every $\theta \in (0, \pi/2)$. We are exactly in the second case of Proposition 32.2, with $b = \pi/2$: it follows that γ inscribes a rectangle of every aspect ratio $\lambda \in (0, 1)$, but *nothing* follows about the square ($\lambda = 1$, that is $\theta = \pi/2$, which is excluded from the open interval).

Independent check. A Python script sampled 802 points of γ (401 on the semicircle and 401 on the diameter) and computed the maximum of the pairwise distances, obtaining 2.0000000: the diameter of Step 2 is confirmed. The area $\pi/2 = 1.5707963$ is that of the half-disc, and was recovered as half of $\pi \cdot 1^2$. $\checkmark$

Interpretation. The half-disc is the boundary case: it reaches the threshold $\pi/2$ of Proposition 32.2 without exceeding it. A reading lesson follows: a statement with an *open* interval $(0, B(\gamma))$ says nothing about the endpoint, and here the difference between " $<$ " and " $\leq$ " separates "all rectangles except the square" from "all rectangles". Does the half-disc inscribe a square or not? This does not follow from Statement B: does it follow, instead, from the 2021 theorem of Greene and Lobb? No: the boundary of the half-disc has two corners,

hence is not smooth, and that theorem does not apply either. As of 7 September 2026 the question, for this curve, is not covered by either of the two statements cited in the chapter.

Typical mistakes. (1) Taking $R(\gamma)$ to be the radius of the circle, that is 1, for the wrong reason: here it coincides, but only because the diameter of the figure is 2. For a circular sector of angular width $\frac{\pi}{3}$ and radius 1 the diameter is 1, not 2, and $R = \frac{1}{2}$. (2) Concluding "the half-disc inscribes a square" from $B(\gamma) = \pi/2$: this is exactly the case the statement does not cover. (3) Using Statement B without checking rectifiability.

Extension. Compute $B(\gamma)$ for the circular sector of radius 1 and angular width $\alpha \in (0, \pi]$: one finds $A(\gamma) = \alpha/2$, while the diameter equals $\max\{1, 2\sin(\alpha/2)\}$, that is 1 (the radius) for $\alpha \leq \frac{\pi}{3}$ and the chord $2\sin(\alpha/2)$ for $\alpha \geq \frac{\pi}{3}$. Discuss the two regimes and say for which α the threshold $\pi/2$ of Proposition 32.2 is exceeded.

Exercise 32.6: A fourth point at rational distances. *Type:* exact computation with fractions, double check. Using the Pythagorean triple $(20, 21, 29)$, produce an element $t \in T$ and the point $P(t)$ of the unit circle, and compute its distances from $P(0) = (1, 0)$ and from $P(3/4) = (\frac{7}{25}, \frac{24}{25})$ in two independent ways.

Solution. Strategy. An element of T is a rational t such that $1 + t^2$ is the square of a rational. A Pythagorean triple (a, b, c) with $a^2 + b^2 = c^2$ gives $t = a/b$, because then $1 + t^2 = \frac{b^2 + a^2}{b^2} = (\frac{c}{b})^2$.

Step 1 — the parameter. From the triple $(20, 21, 29)$, with $20^2 + 21^2 = 400 + 441 = 841 = 29^2$, take

$$t = \frac{20}{21}, \quad 1 + t^2 = \frac{441 + 400}{441} = \frac{841}{441} = \left(\frac{29}{21}\right)^2, \quad \sqrt{1 + t^2} = \frac{29}{21} \in \mathbb{Q}.$$

Hence $t \in T$.

Step 2 — the point. By Lemma 32.2,

$$P(t) = \left(\frac{1 - \frac{400}{441}}{\frac{841}{441}}, \frac{2 \cdot \frac{20}{21}}{\frac{841}{441}} \right) = \left(\frac{441 - 400}{841}, \frac{2 \cdot 20 \cdot 21}{841} \right) = \left(\frac{41}{841}, \frac{840}{841} \right).$$

Check: $41^2 + 840^2 = 1,681 + 705,600 = 707,281 = 841^2$. ✓ The point lies on the unit circle and has rational coordinates.

Step 3 — distance from $P(0) = (1, 0)$, first route (formula). With $s = 0$, $\sqrt{1 + s^2} = 1$:

$$|P(t) - P(0)| = \frac{2 \left| \frac{20}{21} - 0 \right|}{\frac{29}{21} \cdot 1} = \frac{\frac{40}{21}}{\frac{29}{21}} = \frac{40}{29} = 1.3793103.$$

Step 4 — same distance, second route (coordinates).

$$\left(1 - \frac{41}{841}\right)^2 + \left(\frac{840}{841}\right)^2 = \left(\frac{800}{841}\right)^2 + \left(\frac{840}{841}\right)^2 = \frac{640,000 + 705,600}{707,281} = \frac{1,345,600}{707,281}.$$

Now $1,345,600 = 1,600 \cdot 841$ and $707,281 = 841^2$, so the fraction equals $\frac{1,600}{841}$, whose square root is $\frac{40}{29}$. ✓ The two routes agree.

Step 5 — distance from $P(3/4)$, two routes. With $s = \frac{3}{4}$ and $\sqrt{1 + s^2} = \frac{5}{4}$:

$$\left| \frac{20}{21} - \frac{3}{4} \right| = \frac{80 - 63}{84} = \frac{17}{84}, \quad 2 \cdot \frac{17}{84} = \frac{17}{42}, \quad \frac{29}{21} \cdot \frac{5}{4} = \frac{145}{84},$$

$$|P(t) - P(s)| = \frac{17/42}{145/84} = \frac{17}{42} \cdot \frac{84}{145} = \frac{17 \cdot 2}{145} = \frac{34}{145} = 0.2344828.$$

By coordinates: $\frac{41}{841} - \frac{7}{25} = \frac{1,025-5,887}{21,025} = -\frac{4,862}{21,025}$ and $\frac{840}{841} - \frac{24}{25} = \frac{21,000-20,184}{21,025} = \frac{816}{21,025}$; the sum of the squares equals $\frac{4,862^2+816^2}{21,025^2} = \frac{23,639,044+665,856}{21,025^2} = \frac{24,304,900}{21,025^2}$, and since $24,304,900 = 1,156 \cdot 21,025$ the fraction is $\frac{1,156}{21,025}$, whose square root is $\frac{34}{145}$. ✓

Independent check. A Python script with exact fraction arithmetic returns $|P(t) - P(0)|^2 = \frac{1,600}{841}$ and $|P(t) - P(0)| = \frac{40}{29}$, $|P(t) - P(3/4)|^2 = \frac{1,156}{21,025}$ and $|P(t) - P(3/4)| = \frac{34}{145}$, and confirms that the formula of Lemma 32.2(iii) gives the same two rationals. ✓

Interpretation. Every Pythagorean triple provides a new point to add to the set E of Theorem 32.5 without spoiling the rationality of the distances: this is what makes E infinite, and density then comes from the parameter $\phi(\sigma) = \frac{1-\sigma^2}{2\sigma}$, which produces as many triples as wanted, arbitrarily close to any direction.

Typical mistakes. (1) Taking any rational t : for $t = 1$ one has $1 + t^2 = 2$, which is not a square in $\mathbb{Q}$, and indeed $|P(1) - P(0)| = |(0, 1) - (1, 0)| = \sqrt{2}$ is irrational. (2) Confusing "rational coordinates" with "rational distances": all points $P(t)$ with $t \in \mathbb{Q}$ have rational coordinates, but only those with $t \in T$ have rational mutual distances. (3) Getting the sign of $\frac{1-t^2}{1+t^2}$ wrong when $t > 1$.

Extension. Using the triples $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(20, 21, 29)$ and $(7, 24, 25)$, construct eight points (also counting the reciprocals $t \mapsto 1/t$, which still lie in T) and write the 8×8 matrix of distances, checking that they are all rational.

32.11. Self-assessment quiz

- (Definition) A precise statement P is called an *open problem* when
 - no one has worked on it yet
 - neither P nor its negation has been proved
 - it has been proved only in special cases
 - it has been announced but not yet published
- (Hypothesis) In Theorem 32.2 the hypothesis $K = -K$ is used
 - only to guarantee that ρ is continuous
 - only to guarantee that the four points are distinct
 - twice: to make ρ π -periodic and to guarantee that $-A$ and $-B$ lie on γ
 - to guarantee that the origin is interior to K
- (Symbolic computation) The aspect ratio of a θ -rectangle, for $\theta \in (0, \pi/2]$, equals
 - $\sin \theta$
 - $\tan \theta$
 - $\tan(\theta/2)$
 - $\cos(\theta/2)/2$
- (Numerical computation) For the boundary γ of the equilateral triangle of side 1, $B(\gamma) =$
 - 0.4330127
 - 0.5513289
 - 1.7320508
 - 3.4641016
- (Proof step) In the proof of Theorem 32.4, the step from $|C_i - C_j| \geq 2$ to $\delta_{ij} \geq \pi/3$ is justified
 - by the triangle inequality
 - by the identity $|C_i - C_j| = 4 \sin(\delta_{ij}/2)$ and the monotonicity of sine on $[0, \pi/2]$
 - by observing that six angles of 60° make a full turn
 - by the intermediate value theorem
- (Error diagnosis) A student writes: "Ascher, Braune and Turchet proved in 2020 that no dense rational-distance set exists in the plane: the Erdős–Ulam problem is solved." The error is
 - the date, which is 2018
 - the attribution: the result is by Solymosi and de Zeeuw
 - the omission of the hypothesis: the result holds *assuming* the Bombieri–Lang conjecture, which is itself open
 - none: the sentence is correct

7. (Transfer) Let γ be the boundary of the square of side 1. What does Statement B of Observation 32.6 guarantee?
- A) nothing, because γ is not smooth B) an inscribed θ -rectangle for every $\theta \in (0, 2)$, hence rectangles of every aspect ratio, including the square C) an inscribed θ -rectangle for every $\theta \in (0, \pi)$ D) an inscribed θ -rectangle only for $\theta = \pi/2$
8. (Source and metacognition) As of 7 September 2026, the correct way to cite the work arXiv:2604.17116 is
- A) J. E. Greene, A. Lobb, *The rectangular peg problem*, *Annals of Mathematics* **194** (2021), pp. 509–517 B) J. E. Greene, A. Lobb, *Jordan curves inscribe a positive measure of rectangles*, preprint arXiv:2604.17116v1, 18 April 2026 C) J. E. Greene, A. Lobb, *Jordan curves inscribe a positive measure of rectangles*, *Annals of Mathematics* **200** (2026) D) Greene and Lobb, personal communication, 2026

32.12. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

1) **B.** *Rationale:* this is Definition 32.1(iii): open means that neither P nor its negation has been proved. *Distractors:* A confuses "open" with "neglected" — the Toeplitz problem has a hundred and fifteen years of literature and remains open; C describes the typical situation of an open problem, but is not the definition (even a theorem can have been first proved in special cases); D describes an *editorial* status (Definition 32.9), not a logical one. *Diagnosed error:* confusing the logical status of a statement with its history or its editorial location. *Source:* Section 32.1 and Table 32.2. *Prerequisite:* none. *Remediation:* classify, with Table 32.2 covered, the sixteen statements of the chapter into the three categories.

2) **C.** *Rationale:* in the proof of Theorem 32.2 the hypothesis $K = -K$ enters in Steps 1–3 through Lemma 32.1(v), which gives the π -periodicity of ρ and allows the use of a maximum and minimum over a period, and in Step 4, where it guarantees that $-A$ and $-B$ still lie on γ . This is exactly what is discussed in Observation 32.3. *Distractors:* A is false, the continuity of ρ comes from convexity and compactness (Lemma 32.1(iii)), not from symmetry; B is false, the distinctness of the points follows from $\varrho > 0$ and $\theta \in (0, \pi)$; D is false, that the origin is interior is a separate hypothesis. *Diagnosed error:* not being able to identify where, in a proof, a given hypothesis is consumed. *Source:* Theorem 32.2, Steps 1–4. *Prerequisite:* Lemma 32.1. *Remediation:* rewrite Steps 1–5 assuming K convex but not symmetric and mark the first line that fails.

3) **C.** *Rationale:* this is Proposition 32.1: the sides equal $d \sin(\theta/2)$ and $d \cos(\theta/2)$, and for $\theta \in (0, \pi/2]$ the ratio of the smaller to the larger is $\tan(\theta/2)$. *Distractors:* A and B come from forgetting the *half*-angle (the diagonals bisect each other at the centre and each side subtends $\theta/2$); D is dimensionally wrong and does not equal 1 for $\theta = \pi/2$. *Diagnosed error:* confusing the angle between the diagonals with the angle between a diagonal and a side. *Source:* Proposition 32.1, Steps 2–3. *Prerequisite:* double-angle formulas. *Remediation:* redo Example 32.2 for $\theta = \pi/2$ and check that the ratio equals 1.

4) **C.** *Rationale:* for the equilateral triangle of side 1 one has $A = \frac{\sqrt{3}}{4}$ and $\text{diam} = 1$, hence $R = \frac{1}{2}$ and

$$B = \frac{\sqrt{3}/4}{1/4} = \sqrt{3} = 1.7320508.$$

Distractors: A is the area $\sqrt{3}/4 = 0.4330127$, that is, forgetting the division by R^2 ; B is $B/\pi = 0.5513289$, that is, the *fraction* of covered angles, not the constant; D is $2\sqrt{3}$, that

is, using $R = \text{diam}$ instead of $R = \text{diam}/2$ in one single factor. *Diagnosed error*: confusing radius and diameter in Definition 32.5. *Source*: Example 32.3. *Prerequisite*: area of the equilateral triangle. *Remediation*: redo the table of Example 32.3 row by row, marking each time what R is.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) B. *Rationale*: this is Step 1 of the proof of Theorem 32.4: from $|C_i| = |C_j| = 2$ it follows that $|C_i - C_j| = 4 \sin(\delta_{ij}/2)$, and since $\delta_{ij}/2 \in [0, \pi/2]$, where sine is increasing, $|C_i - C_j| \geq 2$ is equivalent to $\sin(\delta_{ij}/2) \geq \frac{1}{2}$, that is $\delta_{ij} \geq \pi/3$. *Distractors*: A produces no angular bound; C is the conclusion of Step 2, not its justification, and is circular reasoning if used here; D is the tool of other proofs in the chapter (Theorems 32.1 and 32.2), not of this one. *Diagnosed error*: skipping the metric-to-angular step and assuming as given what must be proved. *Source*: Theorem 32.4, Step 1. *Prerequisite*: $1 - \cos x = 2 \sin^2(x/2)$. *Remediation*: redo the computation with centres at distance 3 from the origin instead of 2 and see how the minimum angle changes.

6) C. *Rationale*: the result of Ascher, Braune and Turchet (and, independently, that of Shaffaf) is *conditional* on the Bombieri–Lang conjecture, which as of 7 September 2026 is itself open: the conclusion is not unconditional and the Erdős–Ulam problem remains open (Table 32.2). *Distractors*: A and B shift attention to bibliographic data that are correct in the sentence (Shaffaf is from 2018, Ascher–Braune–Turchet from 2020, and Solymosi–de Zeeuw proved a different thing: no irreducible algebraic curve other than a line or a circle contains an infinite rational-distance set); D accepts the error. *Diagnosed error*: removing the hypothesis of a conditional theorem — error no. 3 of the list in Section 32.8. *Source*: final box of Section 32.6. *Prerequisite*: none. *Remediation*: rewrite the sentence correctly, in one line, and compare it with the corresponding row of Table 32.2.

7) B. *Rationale*: for the square of side 1 one has $A = 1$, $\text{diam} = \sqrt{2}$, $R = \frac{\sqrt{2}}{2}$ and $B(\gamma) = 2$ (Example 32.3); the curve is rectifiable, so Statement B gives θ -rectangles for every $\theta \in (0, 2)$, and since $2 > \pi/2 = 1.5707963$ Proposition 32.2 gives every aspect ratio, including the square. *Distractors*: A confuses Statement B (no smoothness hypothesis) with the 2021 theorem (smooth curves); C gives the right interval only if $B(\gamma) = \pi$, that is only for the circle; D confuses the conclusion with the single case $\theta = \pi/2$. *Diagnosed error*: applying to a curve the wrong statement among two statements by the same authors. *Source*: Observation 32.6 and Example 32.3. *Prerequisite*: Proposition 32.2. *Remediation*: repeat the reasoning for the 1×3 rectangle, where $B(\gamma) = 1.2 < \pi/2$ and the conclusion changes.

8) B. *Rationale*: as of 7 September 2026 the record of arXiv:2604.17116 shows no journal reference and no editorial DOI: the work must be cited as a preprint, with identifier, version and date (Section 32.8). *Distractors*: A is the correct citation of *another* work by the same authors (arXiv:2005.09193), with a different statement; C invents an editorial location that does not exist — it is the most serious error, because a reader looking for that volume would find nothing; D downgrades to a private communication a text that is publicly available. *Diagnosed error*: attributing a publication to a preprint, or swapping two works by the same authors. *Source*: Example 32.7, case (c). *Prerequisite*: Definition 32.9. *Remediation*: rewrite the three citations of Example 32.7 from memory and compare them with the records.

BEST PRACTICE

Assigned exercises — answers and grading criteria.

Exercise 32.1 (guided reading 1). (a) Identifier arXiv:2005.09193, first deposit 19 May 2020, six pages, areas *Geometric Topology*, *Metric Geometry*, *Symplectic Geometry*; the

Journal reference field shows *Annals of Mathematics* **194** (2021), no. 2, pp. 509–517. Status: *published*; the journal is cited, because it has a stable location and definitive pagination; the preprint is added, if desired, as a freely accessible version. (b) Hypotheses: γ a *smooth* Jordan curve; R any rectangle of the plane. Conclusion: there exists a rectangle *similar* to R with all four vertices on γ . Technical terms to define: "Jordan curve", "smooth", "similar". (c) In the chapter's language: for every $\theta \in (0, \pi)$ the curve γ inscribes a θ -rectangle. The equivalence is Proposition 32.1(ii): two rectangles are similar if and only if they have the same aspect ratio, and aspect ratios are in bijection with angles $\theta \in (0, \pi/2]$. (d) (1) and (2) are also covered by Theorem 32.2 of this chapter, because the circle and the ellipse are boundaries of centrally symmetric convex bodies; (3) the boundary of the square is *not* smooth (it has four corners), so the article's statement does not apply — while Theorem 32.2, which requires no regularity, does. (e) The external result is V. Shevchishin's theorem (2009) on the non-existence of smoothly immersed Lagrangian Klein bottles in $\mathbb{C}^2$; if it were false, the article's proof would collapse (see Observation 32.5). (f) The hypothesis that cannot be removed is smoothness; removing it gives the inscribed rectangle problem for continuous curves, of which the case $\theta = \pi/2$ is the Toeplitz problem — both open as of 7 September 2026. (g) The two citations are those of the examples in Section 32.8; the preprint citation is used when one wants to indicate the freely accessible version, or when citing an earlier version different from the published one. *Grading criterion*: evaluate (b), (d) and (f); a paper that does not distinguish hypotheses and conclusion in (b) does not go beyond a passing grade.

Exercise 32.2 (guided reading 2). (a) First deposit 6 July 2023; last revision 5 September 2024; publication: *Annals of Mathematics* **200** (2024), no. 2, pp. 749–770, September 2024 issue. About fourteen months elapse between first deposit and publication: this is the typical time for a peer review at a journal of this level, and the September 2024 revision is almost certainly the final accepted version. (b) The conjecture states that the number of *missing* curvatures of a primitive integral Apollonian packing is finite; a missing integer is a positive integer that satisfies the local conditions modulo 24 and does not occur among the curvatures. Attribution: Graham–Lagarias–Mallows–Wilks–Yan (2003) and Fuchs–Sandén (2011). (c) The 24 is the modulus with respect to which the curvatures of a given packing occupy only certain residue classes; "local" because the condition is checked one congruence at a time, that is, in each of the reductions modulo prime powers, while "global" is the statement about the actual set of curvatures. (d) The main theorem (Theorem 1.3 of the article) states that there exist infinitely many primitive integral packings with $\Omega(\sqrt{N})$ missing curvatures up to N . This is sufficient because $\Omega(\sqrt{N}) \rightarrow \infty$: the number of missing curvatures is infinite, and the conjecture required it to be finite. (e) An explicit example given is the packing with root quadruple $(-23, 48, 49, 52)$, in which no integers of the form n^4 , $4n^4$, $9n^4$, or $36n^4$ occur (type $(6, 1, 1, -1)$). The check of identity (32.3) is in Example 32.5: $126^2 = 15,876 = 2 \cdot 7,938$. (f) The refined conjecture (Conjecture 1.5 of the article) states that the set of *sporadic* missing integers — those outside the identified quadratic and quartic families — is finite. It is not proved: it is open. (g) The two rows of Table 32.2 are "local–global conjecture: refuted" and "refined conjecture: open". Writing "the local–global problem is solved" would be ambiguous and misleading: the question posed in 2003 is solved (with a negative answer), but the complete description of missing curvatures remains unknown. *Grading criterion*: point (g) is the one that distinguishes a passive reading from a critical one.

Exercise 32.3 (computations). (a) For the rectangle of sides p, q : $A = pq$, $\text{diam} =$

$\sqrt{p^2 + q^2}$ (the diagonal), $R = \frac{1}{2}\sqrt{p^2 + q^2}$, hence $B = \frac{pq}{\frac{1}{4}(p^2 + q^2)} = \frac{4pq}{p^2 + q^2}$. Values:

(p, q)	$B(\gamma)$	$B(\gamma) > \pi/2 = 1.5707963?$
(1, 1)	2.0000000	yes
(1, 2)	1.6000000	yes
(1, 3)	1.2000000	no
(2, 3)	1.8461538	yes

For the three that exceed the threshold, Proposition 32.2 (combined with Statement B) gives inscribed rectangles of every aspect ratio. But the boundary of a rectangle is centrally symmetric, so Theorem 32.2 gives the same conclusion *for all four*, including the 1×3 one, and without depending on an imported statement: here the elementary result proved in the chapter is stronger than the imported one. (b) For the 1×3 : $\theta \in (0; 1.2)$, hence $\lambda = \tan(\theta/2) \in (0; \tan(0.6)) = (0; 0.6841368)$. (c) From $(-1, 2, 3, 6)$ replacing -1 : $d' = 2(2+3+6)+1 = 23$, quadruple $(23, 2, 3, 6)$, with $34^2 = 1,156$ and $2(529+4+9+36) = 1,156$. ✓ From $(2, 2, 3, 15)$ replacing 3 : $d' = 2(2+2+15) - 3 = 35$, quadruple $(2, 2, 15, 35)$, with $54^2 = 2,916$ and $2(4+4+225+1,225) = 2,916$. ✓ Curvatures found: $-1, 2, 3, 6, 15, 23, 35$; residues modulo 24: $23, 2, 3, 6, 15, 23, 11$, that is the set $\{2, 3, 6, 11, 15, 23\}$. (d) $t = \frac{20}{21}$, $\sqrt{1+t^2} = \frac{29}{21}$, $P(t) = (\frac{41}{841}, \frac{840}{841})$; distances $\frac{40}{29} = 1.3793103$ from $(1, 0)$ and $\frac{34}{145} = 0.2344828$ from $(\frac{7}{25}, \frac{24}{25})$. The computations by the direct route are in Steps 4 and 5 of Exercise 32.6. *Grading criterion:* in (a) evaluate the comparison between the two statements; this is the point that reveals whether the student distinguishes what is proved from what is imported.

32.13. Chapter formulary

Radial function of a convex body. With $rB \subseteq K \subseteq RB$ and 0 interior:

$$\rho(u) = \max\{t \geq 0 : tu \in K\} \in [r, R], \quad |\rho(u) - \rho(v)| \leq \frac{R^2}{r} |u - v|,$$

$$\partial K = \{\rho(u)u : u \in S^1\}, \quad K = -K \implies \rho(\varphi + \pi) = \rho(\varphi).$$

Construction of the four points: find φ_0 with $\rho(\varphi_0) = \rho(\varphi_0 + \theta) = \varrho$ and set $A = \varrho u_{\varphi_0}$, $B = \varrho u_{\varphi_0 + \theta}$, $C = -A$, $D = -B$.

θ -rectangles. Diagonal d , angle $\theta \in (0, \pi)$ between the diagonals:

$$\text{sides } d \sin \frac{\theta}{2} \text{ and } d \cos \frac{\theta}{2}, \quad \text{aspect ratio } \min \left\{ \tan \frac{\theta}{2}, \cot \frac{\theta}{2} \right\},$$

$$\theta \in (0, \frac{\pi}{2}] : \lambda = \tan \frac{\theta}{2}, \quad \theta = 2 \arctan \lambda, \quad \lambda = 1 \iff \theta = \frac{\pi}{2} \text{ (square)}.$$

Values: $\theta = \frac{\pi}{2} \Rightarrow \lambda = 1$; $\theta = \frac{\pi}{3} \Rightarrow \lambda = \tan \frac{\pi}{6} = 0.5773503$; $\lambda = \frac{1}{2} \Rightarrow \theta = 2 \arctan \frac{1}{2} = 0.9272952$; $\lambda = \sqrt{2} - 1 = 0.4142136 \Rightarrow \theta = \frac{\pi}{4}$.

Isodiametric constant. $B(\gamma) = A(\gamma)/R(\gamma)^2$ with $R(\gamma) = \frac{1}{2} \text{diam}(\gamma)$.

γ	$B(\gamma)$	$B(\gamma)/\pi$
circle	$\pi = 3.1415927$	1
regular hexagon	$\frac{3\sqrt{3}}{2} = 2.5980762$	0.8269933
square	2	0.6366198
rectangle $p \times q$	$\frac{4pq}{p^2+q^2}$	
equilateral triangle	$\sqrt{3} = 1.7320508$	0.5513289
ellipse $a \geq b$	$\frac{\pi b}{a}$	$\frac{b}{a}$
half-disc	$\frac{\pi}{2} = 1.5707963$	0.5

Threshold: $B(\gamma) > \pi/2$ is the condition of Proposition 32.2.

Descartes and Vieta's reflection.

$$(a + b + c + d)^2 = 2(a^2 + b^2 + c^2 + d^2), \quad d' = 2(a + b + c) - d,$$

$$d, d' \text{ roots of } X^2 - 2(a + b + c)X + (2(a^2 + b^2 + c^2) - (a + b + c)^2) = 0.$$

Quadruples checked in the chapter: $(-1, 2, 2, 3)$, $(-1, 2, 3, 6)$, $(15, 2, 2, 3)$, $(23, 2, 3, 6)$, $(2, 2, 15, 35)$, $(-23, 48, 49, 52)$.

Kissing and density. Centres at distance 2 from the origin:

$$|C_i - C_j| = 4 \sin \frac{\delta_{ij}}{2}, \quad |C_i - C_j| \geq 2 \iff \delta_{ij} \geq \frac{\pi}{3}, \quad \tau_2 = 6.$$

$$\tau_1 = 2, \tau_2 = 6, \tau_3 = 12, \tau_4 = 24, \tau_8 = 240, \tau_{24} = 196,560.$$

Lattice densities, with covolume 1 and radius $r = \frac{1}{2} \times \text{minimum distance}$:

$$\Delta = \frac{\pi^{n/2} r^n}{\Gamma(\frac{n}{2} + 1)}, \quad \Delta_{E_8} = \frac{\pi^4}{384} = 0.2536695, \quad \Delta_{\Lambda_{24}} = \frac{\pi^{12}}{12!} = 0.0019295743.$$

Rational-distance points of the circle.

$$P(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right) = u_{2 \arctan t}, \quad |P(t) - P(s)| = \frac{2|t-s|}{\sqrt{1+t^2}\sqrt{1+s^2}},$$

$$T = \{t \in \mathbb{Q} : 1+t^2 \in (\mathbb{Q})^2\}, \quad \phi(\sigma) = \frac{1-\sigma^2}{2\sigma} \implies 1+\phi(\sigma)^2 = \left(\frac{1+\sigma^2}{2\sigma} \right)^2.$$

From a Pythagorean triple (a, b, c) : $t = a/b \in T$ and $\sqrt{1+t^2} = c/b$. Examples: $t = \frac{3}{4}$ (5/4), $t = \frac{5}{12}$ (13/12), $t = \frac{8}{15}$ (17/15), $t = \frac{20}{21}$ (29/21).

Reading an arXiv deposit. Identifier AAMM.NNNNN: year and month of the first deposit. Fields to read: *version* (vN) and *submission history*; *Journal reference* and *DOI* (if present: work published, the journal is cited; if absent: preprint); *Comments*; *Subjects*. States: preprint $\rightarrow$ submitted $\rightarrow$ under review $\rightarrow$ accepted $\rightarrow$ published.

Sources and bibliography

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- [1] O. Toeplitz, *Ueber einige Aufgaben der Analysis situs*, *Verhandlungen der Schweizerischen Naturforschenden Gesellschaft* (1911), no. 4, p. 197. This is the communication in which the inscribed square problem was posed (Section 32.2). *Verified on 7 September 2026* in the form reported by the authors who cite it: the entry was read in the bibliography of the preprint [arXiv:2604.17116v1](#) by Greene and Lobb (entry [14]). The 1911 volume was not consulted directly, and for this reason the entry is reported without adding data not appearing in that bibliography.
- [2] B. Matschke, *A survey on the square peg problem*, *Notices of the American Mathematical Society* **61** (2014), no. 4, DOI [10.1090/noti1100](#). Historical picture of the classes of curves for which the problem is solved (Table 32.1). *Verified on 7 September 2026* (Crossref and OpenAlex records for the DOI: author, title, journal, volume, issue, year and starting page 346). *Note:* the page range was not confirmed — the editorial registries report only the starting page and a third-party bibliographic index gives 346–355; the exact range is therefore not stated here.
- [3] B. Matschke, *On the square peg problem and its relatives*, *Transactions of the American Mathematical Society* **375** (2022), no. 9, pp. 6255–6280, DOI [10.1090/tran/8649](#). Expanded, peer-reviewed version of the 2014 survey material. *Verified on 7 September 2026* (Crossref record for the DOI: author, title, journal, volume, issue, year and pages).
- [4] J. E. Greene, A. Lobb, *The rectangular peg problem*, *Annals of Mathematics* **194** (2021), no. 2, pp. 509–517, DOI [10.4007/annals.2021.194.2.4](#); freely accessible version: preprint [arXiv:2005.09193](#), deposited on 19 May 2020, six pages. Source of Observation 32.4 and of the guided reading of Exercise 32.1. *Verified on 7 September 2026* (Crossref record for the DOI; page range read on the article’s page on the *Annals* website, issue 194-2; main statement and bibliography read in the preprint).
- [5] J. E. Greene, A. Lobb, *Jordan curves inscribe a positive measure of rectangles*, **preprint** [arXiv:2604.17116v1](#), deposited on 18 April 2026. Source of Statements A and B of Observation 32.6 and of the definition of $B(\gamma)$. *Verified on 7 September 2026* (deposit record: identifier, authors, title, single version v1 of 18 April 2026, areas math.GT/math.CO/math.SG; no journal reference and no editorial DOI — the only DOI present is the deposit DOI [10.48550/arXiv.2604.17116](#); statements and bibliography read in the text).
- [6] J. E. Greene, A. Lobb, *Cyclic quadrilaterals and smooth Jordan curves*, *Inventiones mathematicae* **234** (2023), no. 3, pp. 931–935, DOI [10.1007/s00222-023-01212-6](#). Follow-up to the 2021 work, cited in the bibliography of the 2026 preprint. *Verified on 7 September 2026* (Crossref record for the DOI: authors, title, journal, volume, issue, year and pages).
- [7] T. Tao, *An integration approach to the Toeplitz square peg problem*, *Forum of Mathematics, Sigma* **5** (2017), article e30, DOI [10.1017/fms.2017.23](#). Source of the row “curves that are the union of two Lipschitz graphs of slope less than 1” of Table 32.1. *Verified on 7 September 2026* (Crossref record for the DOI: author, title, journal, volume, article number, year; content of the abstract).
- [8] W. Stromquist, *Inscribed squares and square-like quadrilaterals in closed curves*, *Mathematika* **36** (1989), no. 2, pp. 187–197, DOI [10.1112/S0025579300013061](#). Locally monotone curves. *Verified on 7 September 2026* (Crossref record for the DOI).
- [9] R. P. Jerrard, *Inscribed squares in plane curves*, *Transactions of the American Mathematical Society* **98** (1961), no. 2, pp. 234–241, DOI [10.1090/S0002-9947-1961-0120604-3](#). Analytic curves. *Verified on 7 September 2026* (Crossref record for the DOI).
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- [10] A. Emch, *Some properties of closed convex curves in a plane*, *American Journal of Mathematics* **35** (1913), no. 4, starting at p. 407, DOI 10.2307/2370404; and *On some properties of the medians of closed continuous curves formed by analytic arcs*, *ibid.* **38** (1916), no. 1, starting at p. 6, DOI 10.2307/2370541. Verified on 7 September 2026 (Crossref record for both DOIs: author, title, journal, volume, issue, year and starting page; the page range is not recorded and is not stated).
- [11] J. Cantarella, E. Denne, J. McCleary, *Configuration spaces, multijet transversality, and the square-peg problem*, *Illinois Journal of Mathematics* **66** (2022), no. 3, DOI 10.1215/00192082-10120454. C^1 curves and curves of finite total curvature without cusps. Verified on 7 September 2026 (Crossref record for the DOI: authors, title, journal, volume, issue, year).
- [12] R. L. Graham, J. C. Lagarias, C. L. Mallows, A. R. Wilks, C. H. Yan, *Apollonian circle packings: number theory*, *Journal of Number Theory* **100** (2003), no. 1, pp. 1–45, DOI 10.1016/S0022-314X(03)00015-5. Local conditions modulo 24 and original formulation of the conjecture. Verified on 7 September 2026 (Crossref record for the DOI: authors, title, journal, volume, issue, year and pages).
- [13] J. Bourgain, A. Kontorovich, *On the local-global conjecture for integral Apollonian gaskets*, *Inventiones mathematicae* **196** (2014), no. 3, pp. 589–650, DOI 10.1007/s00222-013-0475-y. Zero density of the set of missing curvatures. Verified on 7 September 2026 (Crossref record for the DOI: authors, title, journal, volume, issue, pages; print date June 2014, first published online 10 July 2013).
- [14] S. Haag, C. Kertzer, J. Rickards, K. E. Stange, *The local-global conjecture for Apollonian circle packings is false*, *Annals of Mathematics* **200** (2024), no. 2, pp. 749–770, DOI 10.4007/annals.2024.200.2.6; preprint [arXiv:2307.02749](#), deposited on 6 July 2023, last revised on 5 September 2024. Source of the refutation, of the example $(-23, 48, 49, 52)$ with the families n^2 and $6n^2$, and of the refined conjecture. Verified on 7 September 2026 (Crossref record for the DOI; page range read on the article’s page on the *Annals* website, issue 200-2; statements and chronology read in the preprint).
- [15] H. Shi, W. Shi, I. Whitehead, H. Williams-Tracy, J. Z. Zhang, *The Local-Global Conjecture is false for generalized circle packings*, *Journal of Number Theory* **288** (2026), pp. 103–157, DOI 10.1016/j.jnt.2026.03.006; preprint [arXiv:2510.21702](#), deposited on 24 October 2025, revised on 25 March 2026. Extension of the phenomenon to the octahedral, cubic, square and triangular packings. Verified on 7 September 2026 (Crossref record for the DOI and the preprint deposit record).
- [16] E. Fuchs, *Apollonian packings: the rise and fall of the local to global conjecture*, *Bulletin of the American Mathematical Society* **63** (2025), no. 1, pp. 59–77, DOI 10.1090/bull/1874. Reconstruction of the affair from formulation to refutation. Verified on 7 September 2026 (Crossref record for the DOI: author, title, journal, volume, issue, year and pages).
- [17] M. S. Viazovska, *The sphere packing problem in dimension 8*, *Annals of Mathematics* **185** (2017), no. 3, pp. 991–1015, DOI 10.4007/annals.2017.185.3.7. Verified on 7 September 2026 (Crossref record for the DOI; page range read on the article’s page on the *Annals* website, issue 185-3; abstract read on the preprint [arXiv:1603.04246](#)).
- [18] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska, *The sphere packing problem in dimension 24*, *Annals of Mathematics* **185** (2017), no. 3, pp. 1017–1033, DOI 10.4007/annals.2017.185.3.8. Verified on 7 September 2026 (Crossref record for the

DOI; page range read on the article's page on the *Annals* website, issue 185-3; abstract read on the preprint [arXiv:1603.06518](#)).

- [19] H. Cohn, *A conceptual breakthrough in sphere packing*, *Notices of the American Mathematical Society* **64** (2017), no. 2, pp. 102–115, DOI [10.1090/noti1474](#). Source of the values $\pi^4/384$ and $\pi^{12}/12!$ for the densities of E_8 and of the Leech lattice (Observation [32.11](#)) and of the list of solved dimensions (1, 2, 3, 8, 24). *Verified on 7 September 2026* (Crossref record for the DOI for the bibliographic data; density values and list of dimensions read in the text of the preprint [arXiv:1611.01685](#)).
- [20] A. M. Odlyzko, N. J. A. Sloane, *New bounds on the number of unit spheres that can touch a unit sphere in n dimensions*, *Journal of Combinatorial Theory, Series A* **26** (1979), no. 2, pp. 210–214, DOI [10.1016/0097-3165\(79\)90074-8](#). Determination of $\tau_8 = 240$ and $\tau_{24} = 196,560$, obtained independently also by V. I. Levenshtein in the same year. *Verified on 7 September 2026* (Crossref record for the DOI for the bibliographic data; the attribution of the two values to Levenshtein and to Odlyzko–Sloane in 1979, and that of $\tau_3 = 12$ to Schütte and van der Waerden in 1953, were read in the introduction of the following source).
- [21] O. R. Musin, *The kissing number in four dimensions*, *Annals of Mathematics* **168** (2008), no. 1, pp. 1–32, DOI [10.4007/annals.2008.168.1](#); preprint [arXiv:math/0309430](#). Determination of $\tau_4 = 24$ and overview of the known values (Section [32.5](#)). *Verified on 7 September 2026* (Crossref record for the DOI: author, title, journal, volume, issue, year and pages; introduction read on the preprint).
- [22] J. H. Conway, R. A. Parker, N. J. A. Sloane, *The covering radius of the Leech lattice*, *Proceedings of the Royal Society of London, Series A* **380** (1982), pp. 261–290. Source of the data imported in Observation [32.11](#) about the Leech lattice: an even unimodular lattice of minimum norm 4, discovered by Leech in 1965. *Verified on 7 September 2026* (first page of the article read directly: header with journal, volume, pages and year, and the statement "an even unimodular lattice with minimum norm 4").
- [23] J. Solymosi, F. de Zeeuw, *On a question of Erdős and Ulam*, *Discrete & Computational Geometry* **43** (2010), no. 2, pp. 393–401, DOI [10.1007/s00454-009-9179-x](#). No irreducible algebraic curve other than a line or a circle contains an infinite rational-distance set. *Verified on 7 September 2026* (Crossref record for the DOI: authors, title, journal, volume, issue, pages, print March 2010, online 7 May 2009; statement read in the abstract of the preprint).
- [24] J. Shaffaf, *A solution of the Erdős–Ulam problem on rational distance sets assuming the Bombieri–Lang conjecture*, *Discrete & Computational Geometry* **60** (2018), no. 2, pp. 283–293, DOI [10.1007/s00454-018-0003-3](#). *Verified on 7 September 2026* (Crossref record for the DOI).
- [25] K. Ascher, L. Braune, A. Turchet, *The Erdős–Ulam problem, Lang’s conjecture and uniformity*, *Bulletin of the London Mathematical Society* **52** (2020), no. 6, pp. 1053–1063, DOI [10.1112/blms.12381](#). *Verified on 7 September 2026* (Crossref record for the DOI: authors, title, journal, volume, issue, year and pages).

Epilogue: the map of the three volumes

We started from a circle and arrived at the topology of spheres in any dimension, the quaternions that orient robots, the optimal packing of cables. It is worth, at the end, retracing the thread to see it whole.

This chapter adds nothing: it contains no theorems, no new definitions, it introduces no notation. It does one thing only, and does it fully: **it shows how the results chain together**. A treatise in three volumes runs one precise risk, that of becoming a collection of topics that touch only by title; the antidote is to exhibit the chains, that is, the sequences of statements in which each uses the previous one and no link is skipped. The chains are set out in Section 33.1: there are five, each running through all three volumes, and each is written with explicit cross-references, without the vague “as we have seen”.

The chapter then does three more things, all of them a form of accounting. It states the **double register** in which the treatise is written — the demonstrative one and the applicative one — and the rules by which the two are never mixed (Section 33.2); it draws up the **balance sheet** between what is proved here and what is imported from outside, in three distinct categories and with cross-references to the observations that declare it (Section 33.3); it indicates **what remains open** (Section 33.5), referring to Chapter 32 for the research problems and to Volume IV for the developments already planned. A section of numerical checks (Section 33.4) and a note on the working method (Section 33.6) close the account.

INFO

Level and prerequisites. Level **A** (reading), with three numerical checks at level **B** in Section 33.4. This is a synthesis chapter: it is read in full without paper and pencil, and reread with paper and pencil only if one wants to redo the three calculations.

Internal prerequisites. None in the strict sense: every result cited is recalled with its cross-reference, and anyone who does not remember it will find it at the label given. In a substantive sense the chapter presupposes having read *at least one* of the six Parts of the volume, because a map is useful to someone who has already walked part of the road. The chains of Section 33.1 use, as anchor points, Chapters 2, 4, 5, 6, 11, 12, 13, 14, 17, 19, 20, 21, 22, 23, 25, 28, 29 and 30.

External prerequisites. None. The three numerical checks of Section 33.4 use only elementary trigonometry and the definition of the DFT (Chapter 22).

What one learns. How to recognize a chain of results and how to write it out in full, link by link; how to distinguish, when reading a text of applied mathematics, what is proved from what is assumed and from what is merely described; where to look for a result in a work of thirty-three chapters; how to read the balance sheet of a text’s imports the way one reads a company’s balance sheet, that is, to understand what that text is accountable for and what it is not.

Warning. Conventions of this chapter.

- Cross-references *inside* Volume III are always a `\ref` to a label: a chapter number written by hand grows stale at the first reorganization, a label does not.
- Cross-references to the *other* volumes are given in plain text, in the form «Volume I, Chapter 5» or «Volume II, Chapter 4», because L^AT_EX’s cross-references do not cross document boundaries. The numbering cited is that of the two volumes in their current edition: Volume I in twelve chapters, Volume II in eight.
- The three tables of Sections 33.1 and 33.2 are not summaries: they are indexes. Each row points to the place where the thing is *done*, and no row claims to replace it.
- The numerical values of Section 33.4 are rounded to the seventh decimal digit and computed twice by independent routes; the symbol ✓ signals that the comparison has been made and reports the discrepancy.

33.1. How the three volumes chain together

33.1.1 The faces of the circle

The circle has appeared, chapter after chapter, in ever new but recognizable forms. The table that follows lists its faces and, for each, the place where that face is *constructed*: not a summary, an index.

Face of the circle	Where
Best local approximant (osculating circle)	Ch. 2
Only plane curve of constant curvature	Ch. 2
Case $\tau = 0$ of the circular helix	Ch. 3
Isoperimetric maximum at fixed perimeter	Ch. 4
Only $O(2)$ -invariant conic; $\mathbb{T} \cong SO(2)$	Chs. 5, 6
Roots of unity, regular polygons, Möbius	Ch. 6
Plane section of the sphere and the cone	Ch. 7
Great circles as the “lines” of the sphere	Chs. 8, 9
Case $n = 2$ of the sphere S^{n-1}	Ch. 11
Fiber S^1 of the Hopf fibration	Chs. 11, 12
$\pi_1(S^1) \cong \mathbb{Z}$: the hole that classifies loops	Ch. 12
Compact abelian group and its characters	Ch. 13
Optimal packing of the plane	Ch. 14
Boundary of the disc, where one reads what happens inside	Chs. 15, 16
Domain of Fourier series	Chs. 17, 18
Set of DFT nodes (the N roots of unity)	Chs. 22, 23
Stability boundary: $ z = 1$	Chs. 21, 25
Constellation and carrier in the complex plane	Chs. 26, 27
Base of the double cover $S^3 \rightarrow SO(3)$	Ch. 28
Constant-curvature arc (G02/G03 interpolation)	Chs. 2, 30
Three-parameter model to be estimated from data	Ch. 31

Twenty rows, and none of them is a metaphor: in each the circle is *the same set* $\{(x, y) : x^2 + y^2 = r^2\}$, viewed with the tools of a different theory. The treatise is, literally, the story of this set told twenty times.

33.1.2 From Volume I to Volume II: from figure to function

Volume I treats the circle as a *figure*: a locus of points in the Euclidean plane, with its chords, its tangents, its angles, its measure. The tools are those of synthetic geometry and elementary analytic geometry; the results are equalities among segments, angles and areas.

Volume II takes a single step, but a decisive one: it replaces the figure with a *function*. The circle stops being a set and becomes $\gamma(t) = (r \cos t, r \sin t)$, or $z(t) = re^{it}$, that is, an object on which one can differentiate and integrate. Three results of Volume I are transformed, in this passage, into three results of analysis:

- the arc length $\ell = r\theta$ (Volume I, Chapter 6), defined there as a proportion between arcs and angles, becomes in Volume II the integral $\int_0^\theta |\gamma'(t)| dt$, and the fact that the two values coincide is what justifies the elementary definition;
- the area πr^2 (Volume I, Chapter 6), obtained there by exhaustion with polygons, becomes a double integral in polar coordinates (Volume II, Chapter 2), and Archimedes' formula becomes a case of Green's formula;
- the trigonometry of the unit circle (Volume I, Chapter 8) becomes the correspondence $\theta \mapsto e^{i\theta}$ (Volume II, Chapter 3), which turns the addition formulas into a single identity, $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$.

Volume II then adds four applicative registers — basic harmonic analysis, electrical engineering in sinusoidal steady state, directional statistics, discrete circle — and closes with a synthesis chapter (Volume II, Chapter 8) that is the direct predecessor of this one: there the map is drawn from the point of view of the *introduction*, here from the point of view of the whole work.

33.1.3 From Volume II to Volume III: from function to structure

Volume III takes the second step: it replaces the function with the *structure*. It no longer asks what can be computed on the circle, but *of what the circle is an example*, and which properties survive when it is deformed, is generalized, or is replaced.

Structure	What the circle becomes, and where
Regular curve	a case of constant curvature, and the only one (Theorem 2.5); Part A
Manifold	the fiber of the Hopf fibration (Theorem 11.11); Ch. 11
Topological space	the only space with $\pi_1 \cong \mathbb{Z}$ among those considered (Theorem 12.7); Ch. 12
Group	the connected compact abelian group of dimension 1 (Theorem 13.2); Ch. 13
Measure domain	the support of the unique invariant measure (Theorem 13.4); Ch. 13
Boundary	the frontier on which the interior value is read (Theorem 16.6); Ch. 16
Set of nodes	the N roots of unity, basis of the DFT (Theorem 22.1); Ch. 22
Stability locus	the circle $ z = 1$ (Theorem 21.6, Theorem 25.5); Chs. 21, 25
Subgroup	the orbit of a fixed axis in $SO(3)$ and in $SE(3)$ (Theorem 29.6); Ch. 29

The change of viewpoint has a cost and a gain. The cost is that almost none of these statements can be verified with a drawing: they must be proved. The gain is that each of them *transports*: the characterization by curvature holds for every plane curve, the classification of characters holds for every compact abelian group, the diagonalization of circulants holds for every cyclic convolution. Volume III is long because every transport must be justified.

33.1.4 Five chains that run through the three volumes

A *chain* is a sequence of results in which each uses the previous one and no link is skipped. The five that follow are the load-bearing chains of the treatise: every chapter of Volume III rests on at least one of them. They are written out in full, with pointed cross-references, precisely so that the reader can check that there are no gaps.

Chain A — Measure: from $\ell = r\theta$ to concentration in high dimension.

1. Volume I, Chapter 6: length $\ell = r\theta$ and area πr^2 , by exhaustion with inscribed and circumscribed polygons.
2. Volume II, Chapter 2: the same two quantities as integrals, with Green's formula linking area and circulation on the boundary.
3. Chapter 4, Theorem 4.5: among all simple closed curves of given perimeter, the circle is the only one that maximizes the area. It is the first statement in the chain that cannot be checked with a drawing.
4. Chapter 4, Theorem 4.3: at a given constant width the perimeter is the same for all of them, circle included; the area, on the other hand, is not (Theorem 4.4). Perimeter and area, which in Volume I seemed like two faces of the same thing, here come apart.
5. Chapter 11, Theorem 11.1: the same measure in $\mathbb{R}^n$, with the ratio between the volume of the ball and the area of the sphere.
6. Chapter 11, Theorem 11.2 and Theorem 11.4: in high dimension the unit ball has vanishing volume and almost all the measure of the sphere lies near any given equator. The intuition built in Volume I about the circle, applied to S^{n-1} , gives wrong answers: it is the point at which the chain *corrects* its own beginning.
7. Chapter 14, Theorem 14.4: the same measure, counted over infinitely many circles, gives the density $\pi/\sqrt{12}$ of the triangular packing, checked numerically in Example 33.1.

Chain B — The group: from $\cos(\alpha + \beta)$ to Chasles' theorem.

1. Volume I, Chapter 8: the addition formulas of trigonometry, proved with the figure.
2. Volume II, Chapter 3: $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$; the addition formulas become a single multiplication.
3. Chapter 6, Theorem 6.1: $\mathbb{T}$ is a group, isomorphic to $SO(2)$ and to $\mathbb{R}/2\pi\mathbb{Z}$; the n -th roots form the finite subgroup μ_n (Theorem 6.2), whose elements are the vertices of a regular polygon (Theorem 6.3).
4. Chapter 5, Theorem 5.4 and Theorem 5.6: $O(2)$ is classified in full, and its connected component *is* the circle.
5. Chapter 13, Theorem 13.5: the characters of S^1 are exactly $\chi_n(z) = z^n$, $n \in \mathbb{Z}$. It is the link that connects this chain to Chain D: the characters are the exponentials of Fourier.
6. Chapter 11, Theorem 11.8 and Theorem 11.10: in dimension three every rotation has an axis, and the unit quaternions — that is, S^3 — realize $SO(3)$.
7. Chapter 28, Theorem 28.1 and Theorem 28.2: $SO(3)$ is not commutative, and Rodrigues' formula tells how a rotation about a given axis composes.

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8. Chapter 29, Theorem 29.1, Theorem 29.4 and Theorem 29.6: adding translations one arrives at $SE(3)$, where every rigid displacement is a screw motion. The last link is the kinematics of an industrial arm.

Remark 33.1: Where Chain B loses a property, and what it costs. *Between point 5 and point 7 of Chain B the most important thing in the treatise happens: the group stops being commutative. As long as the group is S^1 , its characters are complex-valued functions (Theorem 13.5) and harmonic analysis is a sequence of numbers; this one-dimensionality is the reason why the Fourier series, the transform, the DFT and the FFT exist in the form used throughout Part C. In $SO(3)$ the irreducible representations have dimension $2\ell + 1$, the “coefficients” become matrices, and multiplication no longer commutes (Theorem 28.1). The consequence is not abstract: it is the wrist singularity of manipulators, that is Theorem 28.5, and it is the reason why a robot cannot cross certain configurations at bounded speed. The fact that S^1 is abelian, compact and connected is the ultimate reason why it admits a complete theory, and hence the reason why this treatise can exist.*

Chain C — Curvature: from the radius to trajectory control.

1. Volume I, Chapters 2 and 6: the radius as the only quantity that determines the circle up to isometries.
2. Volume II, Chapter 1: arc-length parametrization and the first and second derivatives.
3. Chapter 2, Theorem 2.1 and Theorem 2.2: the Frenet formulas in the plane and the value $\kappa \equiv 1/r$ for the circle.
4. Chapter 2, Theorem 2.5: the property becomes a *characterization*, that is, constant nonzero κ implies that the curve is an arc of a circle; and Theorem 2.4: the osculating circle is the only circle with second-order contact.
5. Chapter 3, Theorem 3.3 and Theorem 3.2: adding a third dimension torsion appears, and the pair (κ, τ) determines the curve up to rigid motions; the circle is the case $\tau \equiv 0$.
6. Chapter 4, Theorem 4.1 and Theorem 4.6: curvature, integrated over the whole closed curve, gives an integer; and every simple closed curve has at least four vertices, while the circle has infinitely many. Constant curvature is not a local property as it seemed at point 3: it is an extremely rigid global condition.
7. Chapter 10, Theorem 10.2: the curvature of a surface is intrinsic, that is, it does not depend on how the surface is embedded.
8. Chapter 30, Theorem 30.4 and Theorem 30.5: on the constant-curvature arc one builds the jerk-limited motion profile, that is, the G02/G03 circular interpolation of numerically controlled machines. The “constant κ ” constraint of point 4 becomes here the command sent to a tool.

Chain D — Fourier: from the orthogonality of angles to OFDM. This is the longest chain in the treatise: eleven links, six chapters of theory and four of application.

1. Volume I, Chapter 8: cos and sin on the trigonometric circle, and the first integral identities between them.
2. Volume II, Chapter 4: Fourier coefficients, orthogonality of the trigonometric system, first partial sums.

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3. Chapter 6, Theorem 6.5: orthogonality is a fact about roots of unity, that is, a geometric sum equal to 0 or to n .
 4. Chapter 13, Theorem 13.4, Theorem 13.5 and Theorem 13.6: the measure with respect to which one integrates is unique, the characters are exactly the z^n , and their orthogonality follows from invariance. Here orthogonality stops being a computation and becomes a theorem of structure.
 5. Chapter 17, Theorem 17.1: the system is *complete*, hence the series converges in L^2 for every square-summable function; Theorem 17.4 for pointwise convergence and Theorem 17.6 for the Cesàro sum, which converges where the ordinary series does not converge.
 6. Chapter 16, Theorem 16.6: the same series, read on the disc, solves the Dirichlet problem; the Poisson kernel is the approximate identity that closes the argument.
 7. Chapter 18, Theorem 18.3 and Theorem 18.5: the same series solves Laplace's equation and the heat equation, and produces the Bessel functions when the domain is the circular membrane.
 8. Chapter 19, Theorem 19.4 and Theorem 19.5: passing from the circle to the line the series becomes an integral, and convolution becomes a product.
 9. Chapter 20, Theorem 20.2, and Chapter 21, Theorem 21.6: adding a damping factor the transform extends to a half-plane (continuous time) or to an annulus (discrete time), and the DTFT is the restriction of the Z-transform to $|z| = 1$. The circle reappears as a *locus*: it is where the two-sided transform coincides with frequency analysis.
 10. Chapter 22, Theorem 22.1, Theorem 22.2, Theorem 22.5 and Theorem 22.6: sampling the circle at N points gives the DFT, with the same orthogonality as point 3 and the same identities as points 5 and 8; Chapter 23, Theorem 23.3 and Theorem 23.5: it is computed in $\frac{N}{2} \log_2 N$ multiplications instead of N^2 , without changing a single digit of the result.
 11. Chapters 24, 26 and 27: sampling, complex baseband and orthogonal subcarriers. The last link is the way a modern radio connection transmits: N carriers that do not interfere for the same reason why, at point 3, a sum of roots of unity is zero.

Chain E — Stability: from the phasor to the gain margin.

1. Volume II, Chapter 5: the phasor, that is, the complex number of constant modulus that rotates; impedances, power, power-factor correction.
2. Chapter 15, Theorem 15.3 and Theorem 15.5: what happens inside a circle is read on the circle; the residue is the computation that makes this possible.
3. Chapter 20, Theorem 20.2 and Theorem 20.3: the region of convergence is a half-plane, and the position of the poles relative to the imaginary axis decides stability in continuous time.
4. Chapter 21, Theorem 21.1 and Theorem 21.6: in discrete time the half-plane becomes an annulus, and the imaginary axis becomes the circle $|z| = 1$.
5. Chapter 25, Theorem 25.4 and Theorem 25.5: the Nyquist criterion counts the turns of the diagram around a point, and is the argument principle applied to a transfer function; Theorem 25.6: the gain and phase margins are distances measured on a circle.

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6. The winding number that the criterion counts is the same object as Chapter 12, Theorem 12.5 and Theorem 12.6: a homotopy invariant with integer values. The stability of a control loop is, literally, a topological property.

Remark 33.2: How to read a chain, and what it is for. The five chains are not an alternative index: they are a control tool. A student can use them in three ways. First, to check one's own understanding: one covers the column of cross-references and tries to say, for each link, which hypothesis of the previous one is being used; the point at which one cannot answer is the point to reread. Second, to orient oneself: a result one needs is looked up on the chain to which it belongs, not in the analytic index, because the chain also says what must be read first. Third, to recognize a missing link: if in one's own reasoning one jumps from one link to the third one after it, one has almost always skipped a hypothesis. Exercise 33.1 asks for exactly this kind of work on a chain not listed here.

33.2. The double register: mathematics and workshop

Every strand of this thread has had a concrete counterpart. Continuous curvature has become the steering constraint of mobile robots and the clothoid transitions of CNC machines; the isometries of the circle have extended to $SO(3)$ and to the quaternions that ABB, KUKA, FANUC and Yaskawa use to orient the tool; the geodesics of the sphere are the orthodromic routes; the non-commutativity of $SO(3)$ is what becomes wrist singularity; limited jerk and *input shaping* are the workshop's translation of the demand for "smoothness" of motion. The same mathematics that describes the circle describes the way in which machines move.

This coexistence has a name and rules. The treatise is written in two distinct registers, never mixed.

- The **demonstrative register** is that of the environments *definizione*, *teorema*, *lemma*, *proposizione*, *corollario*, each followed by a complete *dimostrazione*. Here nothing is written that is not proved or explicitly declared as imported. The criterion of truth is internal: a statement is true if and only if its proof closes.
- The **applicative register** is that of the environments *esempio*, *osservazione*, *buonapratica*, *attenzione*, *infobox*, of the numerical tables and of the worked exercises. Here it is stated what the result produces when it is used: a value, a tolerance, a machine command, a fault diagnosis. The criterion of truth is external: a statement is true if the computation checks out, if the measurement confirms it, if the script reproduces it.

The two registers do not contradict each other and do not replace each other. An observation never proves a theorem, and a theorem never by itself justifies a design choice: between the statement and the choice there is always at least one modeling hypothesis, which the applicative register must declare. The table that follows sets in columns about ten pairs: on the left the result with its cross-reference, on the right the workshop consequence.

Demonstrative register	Applicative register
constant κ characterizes the circle (Theorem 2.5)	G02/G03 circular interpolation: an arc is defined by center, radius and direction, not by a table of points (Ch. 30)
the rotations of $\mathbb{R}^3$ do not commute (Theorem 28.1)	the order of orientation commands changes the result; hence wrist singularity (Theorem 28.5)
$S^3 \rightarrow SO(3)$ is a double cover (Theorem 11.10)	q and $-q$ are the same pose: SLERP interpolation must be done on the right branch (Theorem 30.2)
residual vibration is the modulus of a transform (Theorem 30.6)	ZV <i>input shaping</i> cancels vibration by placing a zero at the natural frequency (Ch. 30)
the DFT is computed in $\frac{N}{2} \log_2 N$ multiplications (Theorem 23.5)	online spectral analysis becomes possible on modest hardware; the count is however not the execution time (Ch. 23)
the Nyquist criterion counts turns (Theorem 25.5)	gain and phase margin as distances on the polar diagram (Theorem 25.6)
the ROC of the Z-transform is an annulus (Theorem 21.1)	BIBO stability of a digital filter: poles inside $ z = 1$ (Ch. 21)
the triangular packing has density $\pi/\sqrt{12}$ (Theorem 14.4)	number of cables in a conduit, of tubes on a pallet (Theorem 14.5)
the geodesics of S^2 are the great circles (Theorem 9.3)	orthodromic route and spherical triangulation (Ch. 8)
the geometric fit has explicit normal equations (Theorem 31.4)	measuring the radius of a hole from a cloud of points, with its conditioning (Ch. 31)

Remark 33.3: The three rules of the double register. *The rules followed throughout the treatise are three, and can be checked line by line.*

First rule: the modeling hypothesis is always written down. *No passage from the demonstrative register to the applicative one happens silently. When Theorem 30.6 becomes a tuning rule, the single-degree-of-freedom model with viscous damping is stated in full; when the count of Theorem 23.5 becomes a time prediction, it is declared that the cost model ignores memory accesses.*

Second rule: applicative numbers are checked twice. *Every value that appears in an example or in a table is obtained by two independent routes, typically a closed form and an executed script, and the discrepancy is reported. Section 33.4 redoes three of these checks.*

Third rule: the applicative register never promotes. *An observation, an infobox or a good practice cannot turn into a theorem something that is not proved. The great results that the treatise uses without proving them stand in declared observations, never in teorema environments: this is what Section 33.3 is about.*

Remark 33.4: The error that the double register prevents. *The error the double register is built against is tacit promotion: using as a proved premise something that was read in an example. Two forms of it can be seen. The first is citing as a theorem of the treatise a statement that the treatise exposes only as panorama — for instance the Kepler conjecture in the form proved by Hales, which here is in Remark 14.5 and in no teorema environment. The second is carrying a numerical value outside the hypotheses under which it was computed: the density $\pi/\sqrt{12}$ holds for the infinite packing of the plane, and applying it to a conduit of twenty cables gives an optimistic estimate, because the boundary effect is not negligible at that number of elements. In both cases the remedy is the same: go back to*

the label and read the hypotheses.

33.3. What the treatise proves and what it imports

A text of applied mathematics is also judged by how honest it is about its own debts. This treatise declares three types, kept separate and marked differently: the *prerequisites*, which are assumed known; the *panoramas*, which are exposed without proof; the *imports*, that is, the individual lemmas used inside an otherwise complete proof. The rule that governs them is a single one, repeated in every chapter: if a proof does not close with certainty, it is not written; it is reduced to lemmas named with their source, and it is declared, in an observation, what is left out.

33.3.1 First type: the prerequisites

These are the results of Volumes I and II, and the basic notions of linear algebra, differential and integral calculus, and elementary measure theory, which the volume uses without reproving them. They are collected in **Appendix A**, “Prerequisites recalled”, which lists them as a reminder and not as a treatment: dot product and cross product, the spectral theorem for symmetric matrices, Euler’s formula, the plane Frenet frame, the Gaussian integral and the Gamma function. The infobox “Level and prerequisites” at the head of each chapter adds, chapter by chapter, the specific prerequisites and declares which ones are assumed and not reproved: see for example the bulleted list of Chapter 23 (induction, linear algebra, modular arithmetic, binary representation, the floating-point arithmetic model) and that of Chapter 31.

A prerequisite is not a hidden debt: it is a choice of perimeter, declared before beginning. The criterion adopted is that a prerequisite must be *standard*, that is, found in any basic university course, and *independent of the circle*, that is, it must not be a part of the theory that the treatise has the task of building.

33.3.2 Second type: the panoramas

These are the great theorems that the treatise *cites* because they are needed to situate what it proves, but whose proof is beyond the reach of a didactic work of this level. They all stand in *osservazione* or *infobox* environments, never in *teorema* environments: this is the third rule of the double register (Remark 33.3). Here they are, with the place where they are exposed.

- **Packings.** The theorem of Thue and Fejes Tóth on the optimality of the hexagonal packing in the plane (Remark 14.4); the Kepler conjecture proved by Hales and the exceptional cases in dimensions 8 and 24 (Remark 14.5); the circle packing theorem of Koebe, Andreev and Thurston (Remark 14.6).
- **Topology.** The computation of $\pi_1(S^n)$ for $n \geq 2$ (Remark 12.5); the hairy ball theorem (Remark 12.6); the non-triviality of the Hopf fibration (Remark 12.7, and Remark 11.6 for what is left out of Chapter 11).
- **Groups.** The general theory of Lie groups (Remark 13.2); the existence and uniqueness of the Haar measure on any compact group (Remark 13.4); Pontryagin duality (Remark 13.6); Weyl’s equidistribution in dimension n (Remark 13.8); the Baker–Campbell–Hausdorff formula (Remark 29.14).
- **Complex and harmonic analysis.** Picard’s great theorem (Remark 15.18); the Poincaré metric and the Schwarz–Pick lemma (Remark 15.17); the two extremes of the theory

of pointwise convergence, that is, Kolmogorov’s example and Carleson’s theorem (Remark 17.9); the completeness of the Bessel–Fourier system (Remark 18.16).

- **Constructibility.** The Gauss–Wantzel theorem on polygons constructible with straight-edge and compass (Remark 6.4).

A panorama is not a gap: it is a boundary. The reader who crosses it knows exactly at what moment they are leaving what the treatise has built, and the source given tells them where to continue.

33.3.3 Third type: the imports

These are the most delicate cases, because they stand *inside* a proof. A chain of steps is complete except for one link, that link is a known and citable result, and so it is isolated, given a name, its source is indicated, and it is declared in an observation that this step is imported. The advantage of this form over keeping silent is that the reader sees the structure of the proof in full and knows exactly which single brick they must trust.

Where	What is imported
Ch. 3	completeness of $\mathbb{R}$ in the form used for the fundamental theorem of curves (Rem. 3.6)
Ch. 4	mixed areas and Pál’s theorem for Blaschke–Lebesgue (Rem. 4.2); the supporting osculating circle theorem for the strong form of the four-vertex theorem (Rem. 4.6); global Gauss–Bonnet (Rem. 4.9)
Ch. 6	identification of the rotations of the sphere with a subgroup of Möbius transformations (Rem. 6.6)
Ch. 9	local existence and uniqueness for differential equations (Picard–Lindelöf, Rem. 9.1); Jacobi fields and conjugate points (Rem. 9.2)
Ch. 10	Brioschi’s formula (Rem. 10.1); the characterization of surfaces with $K \equiv 0$ (Rem. 10.3)
Ch. 15	the notion of simple connectedness and what is assumed about it (Rem. 15.2)
Ch. 16	what fails in dimension greater than two (Rem. 16.2)
Ch. 17	the uniform part of the Dirichlet–Jordan criterion (Rem. 17.12) and the chapter’s balance sheet (Rem. 17.15)
Ch. 18	the tabulated values of the zeros of Bessel functions (Rem. 18.12) and McMahon’s asymptotic expansion (Rem. 18.13)
Ch. 19	the Plancherel extension to L^2 (Rem. 19.8); the limits of what can be said about sampling (Rem. 19.10)
Ch. 20	the Bromwich inversion in the general case (Rem. 20.6); the residue of BIBO stability (Rem. 20.9)
Ch. 21	Parseval’s theorem for finite-energy signals (Rem. 21.7)
Ch. 22	the multiplicities of the four eigenvalues of the DFT matrix (Rem. 22.3); the values of the side lobes (Rem. 22.13)
Ch. 23	the existence of a primitive root modulo p , used once in Rader’s algorithm (Rem. 23.11); the standard model of floating-point arithmetic (Rem. 23.12)
Ch. 28	the nonexistence of a global three-parameter chart for $SO(3)$ (Rem. 28.7)
Ch. 29	the closed subgroup theorem (Rem. 29.2)
Ch. 30	the properties of B-splines (Rem. 30.1); Squad, dual quaternions and ScLERP (Rem. 30.4); the robustness of shapers beyond ZVD (Rem. 30.6)
Ch. 31	the convergence of Gauss–Newton (Rem. 31.10); damping as a trust region (Rem. 31.11); the two statements about bias (Rem. 31.13)

Remark 33.5: What remains imported in Volume III, and how to recheck it.

The preceding table and the list of panoramas are the complete balance sheet of what this volume uses without proving it. It is worth stating precisely what this does not mean. It does not mean that those results are doubtful: they are all established theorems, with their source indicated and verified in the Fonti environment of the chapter that uses them. It means that, for those steps and only those, the reader who wants a closed deductive chain must leave here and go to the source.

Three remarks on the balance sheet. First, the imports concentrate where one would expect: in chapters that touch theories much wider than the circle (differential equations in Chapter 9, measure theory in Chapter 19, nonlinear optimization in Chapter 31). Second, no import is used to prove a result about the circle: the central statements of the treatise — Theorem 2.5, Theorem 4.5, Theorem 13.5, Theorem 22.1, Theorem 23.3 — have complete proofs. Third, the balance sheet is a living thing: it is derived mechanically from the sources with the script `registro_importati.py`, which lists every block containing a declaration of an import or of a panorama. Anyone reading an edition later than this one must rerun it: the applicative chapters 24, 26, 27 and Chapter 32, in particular, add their own imports, and the table above must be read as the snapshot of one edition, not as a law.

33.4. Three numbers, three registers

A synthesis chapter introduces no new computations, but it can redo the old ones. The three that follow are chosen because each belongs to a different register — elementary geometry, classical analysis, numerical computation — and because each is, at the same time, a result of Volume I reread with the tools of Volume III. All three are computed twice by independent routes, as prescribed by the second rule of the double register (Remark 33.3).

Example 33.1: The density $\pi/\sqrt{12}$, two ways. Theorem 14.4 states that the triangular packing of equal circles in the plane has density $\pi/\sqrt{12}$.

First way (closed form). The value is

$$\frac{\pi}{\sqrt{12}} = \frac{\pi}{2\sqrt{3}} = 0.9068997\dots$$

Second way (ratio of areas, with only the tools of Volume I). In the triangular lattice each circle of radius r has, as its cell, the regular hexagon of apothem r . That hexagon has side $L = 2r/\sqrt{3}$, because its circumscribed radius R satisfies $r = R \cos 30^\circ = R\sqrt{3}/2$ and in the regular hexagon $L = R$; it is made of six triangles of base L and height r , hence

$$A_{\text{hexagon}} = 6 \cdot \frac{Lr}{2} = 3 \frac{2r}{\sqrt{3}} r = 2\sqrt{3} r^2.$$

The density is the ratio between the area of the circle and that of its cell:

$$\frac{A_{\text{circle}}}{A_{\text{hexagon}}} = \frac{\pi r^2}{2\sqrt{3} r^2} = \frac{\pi}{2\sqrt{3}} = \frac{\pi}{\sqrt{12}}.$$

The two ways give the same number, and the second explains *why*: the optimal density is the ratio between a circle and the hexagon surrounding it, that is, an area computation from Volume I, Chapter 6.

Numerical check. The executed script gives 0.9068996821 for the first way and 0.9068996821 for the second, with discrepancy $2.220 \cdot 10^{-16}$, that is, one unit in the last digit of double precision. ✓

What the number does not say. The density is a limit over the whole plane: in a finite region the boundary effect lowers it. A count on a square of side $800r$ gives 0.9040, that is, 0.3% less, and on a conduit of twenty cables the discrepancy is much larger. This is the error described in Remark 33.4.

Example 33.2: Archimedes' polygons, two ways. Volume I, Chapter 1, reports Archimedes' estimate $\frac{223}{71} < \pi < \frac{22}{7}$, obtained with the regular inscribed and circumscribed polygons of 96 sides. With the tools of Volume II those semiperimeters have a closed form. *First way (closed form).* The regular polygon of n sides inscribed in the unit circle has semiperimeter $n \sin(\pi/n)$, the circumscribed one $n \tan(\pi/n)$. For $n = 96$:

$$96 \sin \frac{\pi}{96} = 3.1410320, \quad 96 \tan \frac{\pi}{96} = 3.1427146.$$

Second way (doubling the sides, as Archimedes proceeded). If ℓ_n is the side of the polygon of n sides inscribed in the unit circle, doubling gives

$$\ell_{2n} = \sqrt{2 - \sqrt{4 - \ell_n^2}},$$

and one starts from $\ell_6 = 1$, the side of the hexagon, which equals the radius. Four doublings take one from 6 to 96 sides; the resulting semiperimeter is $96 \ell_{96}/2 = 3.1410320$, with discrepancy $2.0 \cdot 10^{-14}$ from the closed form. ✓

Comparison with the ancient estimate. $\frac{223}{71} = 3.1408451 < 3.1410320$ and $3.1427146 < 3.1428571 = \frac{22}{7}$: Archimedes' fractions are correct and, as is well known, slightly wider than the exact values of the two polygons, because obtained by rounding down and up at each step. The true value $\pi = 3.1415927$ lies between the two semiperimeters, as it must.

Example 33.3: The same Parseval identity in three chapters. The identity linking the energy of a signal to that of its spectrum appears in the treatise three times: as Parseval's identity for Fourier series (Chapter 17, with the completeness of the trigonometric system, Theorem 17.1), as Plancherel's theorem for the transform (Chapter 19, for the imported part see Remark 19.8) and as a discrete identity (Chapter 22, Theorem 22.6). In its third form it says

$$\sum_{k=0}^{N-1} |X[k]|^2 = N \sum_{n=0}^{N-1} |x[n]|^2.$$

Let us check it on the ramp $x = (0, 1, 2, \dots, 7)$, $N = 8$, the same one from the worked example of Chapter 23.

Left side, by closed form. For that ramp $X[0] = 28$ and $X[k] = 8/(\omega_8^k - 1)$ for $k \neq 0$, with $\omega_8 = e^{-2\pi i/8}$; hence $|X[k]|^2 = 64/|\omega_8^k - 1|^2$ and $|\omega_8^k - 1|^2 = 2 - 2\cos(\pi k/4)$. The seven values are

$$\begin{aligned} \frac{64}{2 - \sqrt{2}} &= 64 + 32\sqrt{2} \quad (k = 1, 7), & 32 \quad (k = 2, 6), \\ \frac{64}{2 + \sqrt{2}} &= 64 - 32\sqrt{2} \quad (k = 3, 5), & 16 \quad (k = 4), \end{aligned}$$

and their sum is exactly $2(64 + 32\sqrt{2}) + 2 \cdot 32 + 2(64 - 32\sqrt{2}) + 16 = 128 + 64 + 128 + 16 = 336$, where the terms in $\sqrt{2}$ cancel. Adding $|X[0]|^2 = 784$ gives $784 + 336 = 1,120$.

Right side, by direct computation. $\sum_{n=0}^7 n^2 = \frac{7 \cdot 8 \cdot 15}{6} = 140$, and $N \cdot 140 = 8 \cdot 140 = 1,120$. The two sides coincide *exactly*, not up to rounding: both equal 1,120. The script that computes the DFT from the definition and sums the squared moduli gives 1,120.0000000

with discrepancy $2.0 \cdot 10^{-12}$ from the exact value, which is the rounding error expected in double precision for a sum of eight complex numbers of order 10^2 . ✓

Why this is a good synthesis example. The left side was computed with a closed form coming from complex analysis (sum of a truncated geometric series, Chapter 6); the right side with a formula for the sum of squares learned in high school; and the equality between the two is a theorem of linear algebra on the unitarity of the DFT matrix (Theorem 22.3). Three registers, one number.

Remark 33.6: Why two ways and not three. *The treatise’s rule requires two independent ways, not two calculations. Two ways are independent when an error made in the first cannot be repeated identically in the second: a closed form and a script that evaluates it are not independent, because the script copies the formula. In the preceding examples independence is guaranteed as follows: in Example 33.1 the second way reconstructs the value from a ratio of areas, without ever writing $\pi/\sqrt{12}$; in Example 33.2 the second way is a recursion that uses neither $\sin$ nor π ; in Example 33.3 the two sides are computed with different theories and the comparison is the statement itself. An executed script serves, in all three cases, as a third check, and it is the one that catches transcription errors.*

33.5. The roads that remain open

A treatise closes, the subject does not. The open roads are of three kinds, and it is worth not confusing them.

First kind: genuine open problems, that is, questions to which no one today knows the answer. Chapter 32 collects a few that closely concern the circle — Toeplitz’s inscribed square problem, inscribed rectangles, Apollonian packings and the local–global conjecture — and exposes them with the state of research and the date to which that state refers. That chapter also contains the part on research literacy: how to read a preprint, how one verifies a source, why an announced proof is not a proof.

Second kind: the panoramas of this volume, listed in Section 33.3.2. They are not open for mathematics — they are proved theorems — but they are open *for the reader*, in the sense that they remain to be studied. The difference is important and is easily forgotten: Carleson’s theorem (Remark 17.9) is not an unsolved problem, it is a result from 1966 that is not proved here.

Third kind: the directions already planned, that is, what the treatise will address in the next volume. These are the threads that this volume has deliberately left hanging: the Hopf fibration as the first example of a non-trivial fiber bundle (Remark 12.7), the general theory of Lie groups (Remark 13.2), Riemannian geometry beyond the case of the surfaces of Chapter 10.

Remark 33.7: Toward Volume IV. *Threads remain to be pulled, and they are the ones left hanging on purpose. **Fiber bundle theory**, of which the Hopf fibration of Theorem 11.11 is the first non-trivial example and of which only the structure of the fibers has been proved here (Proposition 12.5), while non-triviality remains a panorama (Remark 12.7). **Differentiable manifolds** and **Riemannian geometry** in general, of which Chapter 10 treats the case of surfaces in $\mathbb{R}^3$ and Chapter 9 the case of geodesics, with Jacobi fields left out (Remark 9.2). **Lie groups** such as $SU(2)$ and $SE(3)$ in their full structure, of which the properties needed for kinematics have been seen here (Theorem 29.1, Theorem 29.4) while the general theory, the Baker–Campbell–Hausdorff formula and the classification remain imported (Remark 13.2, Remark 29.14). And **geometric control** of robots, that is, what*

is obtained when the trajectory is no longer designed in joint space but on the manifold of configurations, of which Chapters 29 and 30 constitute the prerequisites.

The circle, once again, will be the starting point: the simplest of the non-trivial examples, the one on which everything else is understood. All four directions listed are, literally, the question “what remains true when S^1 is replaced by something larger?”, and it is the same question with which Part B of this volume opened.

33.6. Note on the working method

The rules with which this volume was written are few and can be checked from the outside. It is worth stating them, because a reader who knows them can use the text in a stricter way, and because anyone who wants to continue the work knows what to hold to.

Best practice. The five drafting rules of the treatise.

1. **No proof left half-done.** Every *teorema*, *lemma*, *proposizione* and *corollario* is followed by a complete proof, in which every step is justified and every result used is recalled with its label. The phrases “it is evident” and “one easily sees” do not appear. If a proof does not close with certainty, it is not written: it is reduced to named lemmas and the import is declared (Section 33.3). The check is automatic: the script `audit_dimostrazioni.py` lists statements without a proof, and the acceptance criterion for the volume is zero.
2. **Every value computed twice.** No number appears in the text unless it has been obtained by two independent routes, typically a closed form and an actually executed script; the discrepancy between the two is reported, and the symbol $\checkmark$ signals that the comparison has been made. On the independence of the two routes see Remark 33.6.
3. **Every source verified.** In the *Fonti* environment of each chapter every entry carries authors, title, venue, year, pages or section, and the date on which it was verified. No page number and no DOI is written from memory. The sources collected are poured into a single register, one line per central statement.
4. **Every figure with computed coordinates.** The drawings are built from coordinates obtained by computation and checked numerically, not traced by eye; a figure that “looks right” is not a verification.
5. **Three languages in parallel.** Every chapter exists in Italian, English and Ukrainian with the same labels, the same sequence of statements and the same numerical values; the only differences allowed are the text, the decimal separator and the groups of the analytic index. The comparison among the three editions is automatic and the allowed discrepancy is zero.

These rules have a cost, and it is high: they are the reason the volume is long. But they also have a precise consequence, which is the real reason they were adopted: a reader who finds a statement here can always establish, in finite time, whether it is proved, imported or merely described — and, in the first two cases, where to go to check.

33.7. Assigned exercises

Exercise 33.1: A sixth chain, written out in full. *Goal:* to be able to build a chain of results oneself, with cross-references, without gaps. *Estimated time:* 45 minutes.

The five chains of Section 33.1.4 do not exhaust the treatise. At least one is missing, which starts from the circular inversion of Volume I, Chapter 5, and reaches the Ford circles of Chapter 14.

(a) Write it in the same form as the others: from four to six numbered links, each with the cross-reference to the precise statement (not to the chapter).

(b) For each link, say in one line which hypothesis of the previous link is used.

(c) Indicate at which link the chain leaves the Euclidean plane, and what the substitution entails.

(d) Say whether the chain contains an import or a panorama, and which one.

Exercise 33.2: Proved, imported, or panorama?. *Goal:* to be able to place a statement in the volume's balance sheet. *Estimated time:* 25 minutes.

For each of the six statements below, say whether in Volume III it is (i) proved in full, (ii) imported inside an otherwise complete proof, or (iii) exposed as a panorama; and indicate the exact label of the place where it is found.

(a) Among simple closed curves of given perimeter, the circle maximizes the area. (b) The densest packing of equal spheres in $\mathbb{R}^3$ is the face-centered one. (c) A Cauchy problem with a locally Lipschitz right-hand side has one and only one local solution. (d) The radix-2 decimation-in-time FFT algorithm computes the DFT exactly. (e) The Fourier series of a square-summable function converges almost everywhere. (f) There exists a primitive root modulo every prime p .

For cases (ii) and (iii), add a line stating *what* exactly is missing for the proof to close inside the volume.

Exercise 33.3: Redoing the three checks with other data. *Goal:* to apply the rule of the two independent ways. *Estimated time:* 40 minutes.

(a) Repeat Example 33.3 with the ramp $x = (0, 1, \dots, 15)$, $N = 16$: compute $\sum_n |x[n]|^2$ with the formula for the sum of squares, compute $\sum_k |X[k]|^2$ with the closed form $|X[k]|^2 = N^2/(2 - 2\cos(2\pi k/N))$ for $k \neq 0$ and $X[0] = 120$, and compare.

(b) Repeat Example 33.2 with $n = 192$ sides, for both ways (closed form and doubling recursion), and say by how much the approximation of π improves relative to $n = 96$.

(c) Compute, with the same reasoning as Example 33.1, the density of the packing on the *square* lattice, and say by how much it is less dense than the triangular one. Verify that the ratio between the two densities does not depend on the radius.

33.8. Worked exercises

Exercise 33.4: What survives passing from S^1 to S^2 . For each of the following five properties of S^1 , say whether it still holds for S^2 and with which statement of the volume it is decided: (i) it is compact and connected; (ii) it has fundamental group $\mathbb{Z}$; (iii) it is a group; (iv) it admits a nowhere-vanishing tangent vector field; (v) it is the boundary of a domain on which the Dirichlet problem is solvable.

Solution. *Strategy.* Each item is decided with a precise statement, not with intuition: the sphere “resembles” the circle in four cases out of five and differs precisely where the analogy

is most tempting.

(i) *Yes.* S^1 is compact and connected by Theorem 12.1; for S^2 the same holds, and it is a case of the definition of n -dimensional sphere from Chapter 11 (Definition 11.1): the sphere is closed and bounded in $\mathbb{R}^n$, hence compact, and it is path-connected for $n \geq 2$.

(ii) *No.* $\pi_1(S^1) \cong \mathbb{Z}$ is Theorem 12.7; for S^n with $n \geq 2$ the fundamental group is trivial, and it is one of the declared panoramas (Remark 12.5). Watch the step: the proof of Theorem 12.7 uses the covering $E : \mathbb{R} \rightarrow S^1$ (Theorem 12.4) and path lifting; for S^2 that covering does not exist, so it is not that the proof “gives another result”: it simply does not get started.

(iii) *No.* S^1 is a group (Theorem 13.1), S^3 also is — these are the unit quaternions, Theorem 11.10 — but S^2 is not. The most elementary reason available here is (iv): a Lie group has nowhere-vanishing invariant vector fields, and S^2 admits none.

(iv) *No.* S^1 can be combed (Proposition 12.4); S^2 cannot, by the hairy ball theorem, which is a declared panorama (Remark 12.6). It is the only one of the five answers that cannot be proved here: it is taken from the source indicated in that observation.

(v) *Yes.* For the disc the result is Theorem 16.6, which uses the Poisson kernel; for the ball in higher dimension the analogous statement holds, but the construction must be redone, and Remark 16.2 says exactly what of the two-dimensional case does *not* carry over (in particular the connection with holomorphic functions, which is specific to the plane).

Independent check. The five answers can also be derived by another route: the Euler characteristic. $\chi(S^1) = 0$ and $\chi(S^2) = 2$; the Poincaré–Hopf theorem links χ to the sum of the indices of a vector field, and a nowhere-vanishing field forces $\chi = 0$. This explains (iii) and (iv) together, and confirms (ii), because a non-trivial covering of S^2 would contradict its simple connectedness. This route is consistent with the answers given, but uses the Poincaré–Hopf theorem, which this volume does not prove: it serves as a check, not as a proof.

Typical errors. The first is answering “yes” to (iii) by thinking of spherical coordinates: two angles do not make a group, because the composition of rotations is not the sum of coordinates (Theorem 28.1). The second is answering “no” to (i) by confusing compactness and boundedness in infinite dimension, which is not relevant here.

Extension. Redo the five questions for the torus $S^1 \times S^1$: one finds yes, no (the fundamental group is $\mathbb{Z} \times \mathbb{Z}$, Corollary 12.7), yes (it is a group, a product of two copies of S^1), yes (it can be combed, because it is a Lie group), yes. The torus, from this point of view, is much more similar to the circle than the sphere is.

Exercise 33.5: Where to look: a workshop question and its path. A technician must estimate the radius of a hole from 200 points measured on the boundary with a probe, and must state how reliable the estimate is. Trace the reading path in the treatise: which chapters, in which order, and which result answers which part of the question.

Solution. *Strategy.* The question contains three different questions — how one estimates, how stable the estimate is, how biased it is — and each has its place. The path is built starting from the last one and working backward.

Step 1: the model. The circle as a three-parameter model (center and radius) is Chapter 31. The first thing to read is the distinction between a *geometric* fit (minimizing radial distances) and an *algebraic* one; the second is treated in Volume I, Chapter 11, with the methods of Kåsa and the Hough transform, and is not repeated here.

Step 2: the estimate. Theorem 31.4 gives the normal equations of the geometric fit, Theorem 31.3 guarantees that the minimum exists, and Theorem 31.1 ensures that the estimate does not depend on where the origin of the measurement system was placed: a property

far from obvious, and indispensable for the stated number to be reproducible on another machine.

Step 3: the algorithm. Theorem 31.5 and Theorem 31.6: the Gauss–Newton step is a descent step, and the damped Levenberg–Marquardt version is always well defined. Here one must also read Remark 31.10, because global convergence is not proved in the volume: it is an import, and the technician must know this before writing “the method converges” in an acceptance-test report.

Step 4: reliability. Remark 31.13 distinguishes two different statements about the bias of the estimate and says which of the two is imported. This is the point at which the answer “how reliable is it” stops being a property of the algorithm and becomes a property of the noise model: without a hypothesis on the noise there is no reliability to declare.

Independent check. The path can be validated backward: take the final question (“what uncertainty do I state?”), look for the last result of the path, and check that all its prerequisites are in earlier steps. Here this is the case: Remark 31.13 requires the model of Step 1 and the estimate of Step 2, both already acquired.

Interpretation. Of the four stages, three lie in the demonstrative register and one — Step 4 — straddles the two: it is exactly the point at which the modeling hypothesis must be declared, as prescribed by the first rule of Remark 33.3.

Typical errors. The most common is skipping Step 2 and directly using a minimizer: one obtains a number, but does not know whether it is invariant under a change of origin, and two operators with two different reference frames obtain two different radii. The second is confusing the residual of the minimization with the uncertainty of the estimate: the first measures how much the points deviate from the circle found, the second how much the circle found deviates from the true one.

Extension. Redo the path for the question “with how many points is it best to measure”, which adds to the path the conditioning of the normal equations of Chapter 31 and, for the choice of the distribution of points on the arc, the concentration of measure of Chapter 11.

Exercise 33.6: The balance sheet of a chapter. Draw up the balance sheet of Chapter 14 in the three categories of Section 33.3, and say why that chapter has, proportionally, more panoramas than any other.

Solution. *Strategy.* One scans the statements of the chapter, separates those followed by a proof from those that stand in an observation, and looks for the structural reason for the proportion.

Proved. The chapter proves in full the results on Ford circles and Farey fractions (Theorem 14.1 and Theorem 14.2), the kissing number of the plane (Theorem 14.3), the density of the triangular packing (Theorem 14.4) and the applicative count (Theorem 14.5). These are five statements, all closed.

Imported. No imports inside the proofs: the chapter does not need external lemmas to close what it states.

Panorama. Three: the optimality of the hexagonal packing among *all* packings, not only the lattice ones, that is, the theorem of Thue and Fejes Tóth (Remark 14.4); the Kepler conjecture in Hales’s proof and the cases of dimensions 8 and 24 (Remark 14.5); the circle packing theorem (Remark 14.6).

Why that proportion. The reason lies in the nature of the statements. The Theorem 14.4 computes the density of a *given* packing, and is an area computation; the statement that no other packing does better is instead a statement about an infinite and unparametrized family of configurations, and its proof requires techniques — Delaunay decompositions, local estimates on cells, and in Hales’s case a computer-assisted verification — that are out of

scale for a chapter. The treatise’s rule then requires one to state, not to prove: and to say so.

Independent check. The same balance sheet is obtained mechanically by running the script `registro_importati.py` on the volume and reading the lines relating to that chapter: the result is the three observations listed, and no others. The automatic check and the manual count agree.

Typical errors. Counting among the panoramas results that are merely *mentioned* in a sentence of context: a panorama is an explicit declaration of what is not proved, not a mention. And counting among the proved results the applicative count of the tubes without noting that it holds under the declared hypothesis that the pallet be much larger than the radius.

Extension. Repeat the balance sheet for Chapter 13, which has the opposite proportion: many statements proved about the circle and four panoramas, all relating to generalizations (Remarks 13.2, 13.4, 13.6, 13.8).

33.9. Self-assessment quiz

- (Definition) In this treatise a *panorama* is
 - a result whose proof is deferred to an appendix
 - a theorem exposed in an observation or in an infobox, not proved here, with the source indicated
 - an exercise left to the reader
 - a synthesis chapter like this one
- (Hypothesis) Which property of S^1 is the ultimate reason why Part C can develop the Fourier series, the transform, the DFT and the FFT in the form used?
 - compactness, alone
 - being an *abelian* group, as well as compact and connected
 - being a differentiable manifold of dimension 1
 - having fundamental group $\mathbb{Z}$
- (Symbolic computation) Derived as the ratio between the area of the circle of radius r and that of the regular hexagon of apothem r , the density of the triangular packing equals
 - $\frac{\pi}{2\sqrt{3}}$
 - $\frac{\pi}{4}$
 - $\frac{\pi}{4\sqrt{3}}$
 - $\frac{2\pi}{3\sqrt{3}}$
- (Numerical computation) For the ramp $x = (0, 1, \dots, 15)$ with $N = 16$, the sum $\sum_k |X[k]|^2$ equals
 - 1,240
 - 19,840
 - 1,120
 - 240
- (Proof step) In Example 33.3 the sum of the seven terms $|X[k]|^2$ with $k \neq 0$ equals 336 because
 - the seven terms are all equal
 - the pairs $k = 1, 7$ and $k = 3, 5$ give $2(64 + 32\sqrt{2}) + 2(64 - 32\sqrt{2}) = 256$, in which the terms in $\sqrt{2}$ cancel, and $32 + 32 + 16 = 80$ remain
 - $336 = 8 \cdot 42$
 - Parseval’s identity forces that value
- (Error diagnosis) A report states that, “as is proved in Volume III, the densest way to arrange equal spheres in space is the face-centered one.” The error is that
 - there is no error
 - the volume exposes it as a panorama, without proving it
 - the statement is false
 - the statement concerns only $\mathbb{R}^2$
- (Transfer) To establish whether a digital filter is stable, and then whether it remains so once a feedback loop is closed, the correct reading path is

- A) Chapter 31, then Chapter 11 B) Chapter 21 (region of convergence and poles inside $|z| = 1$), then Chapter 25 (argument principle and Nyquist criterion) C) Chapter 23, then Chapter 22 D) Chapter 12, then Chapter 13
8. (Source and metacognition) The *up-to-date* list of what Volume III imports and of what it exposes as panorama is obtained
- A) by reading the analytic index B) by running on the sources the script that lists the blocks in which an import or a panorama is declared C) from the volume's general bibliography D) from Appendix A of the prerequisites

33.10. Teacher's key

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 1–4).

1) **B.** *Rationale:* this is the third rule of the double register (Remark 33.3): results not proved here stand in *osservazione* or *infobox* environments, never in *teorema* environments, and carry the source. The complete list is in Section 33.3.2. *Distractors:* A confuses the panorama with an internal cross-reference, and Appendix A contains prerequisites, not panoramas; C confuses what the text does not prove with what it asks the reader to do; D confuses a category of statements with a type of chapter. *Diagnosed error:* not distinguishing the three forms of the text's debt (prerequisite, panorama, import). *Source:* Section 33.3. *Prerequisite:* none. *Recovery:* open three observations from the list in Section 33.3.2 and check that each names the result, says that it does not prove it, and indicates where to find it.

2) **B.** *Rationale:* commutativity is what makes the characters complex-valued functions (Theorem 13.5), hence the coefficients numbers and not matrices; this is the content of Remark 33.1. *Distractors:* A is necessary — it is needed to have a finite Haar measure, Theorem 13.4 — but not sufficient, as $SO(3)$ shows, which is compact and non-abelian; C is also true for the line, which has no Fourier series; D is a topological property that plays no role in the construction of harmonic analysis. *Diagnosed error:* mistaking a necessary condition for the decisive one. *Source:* Chapter 13. *Prerequisite:* Theorem 13.5. *Recovery:* reread Remark 33.1 and say which step of Chain D would break if the group were not abelian.

3) **A.** *Rationale:* the hexagon of apothem r has side $L = 2r/\sqrt{3}$ and area $6 \cdot \frac{Lr}{2} = 2\sqrt{3}r^2$; the ratio with πr^2 is $\pi/(2\sqrt{3}) = \pi/\sqrt{12} = 0.9068997$ (Example 33.1, Theorem 14.4). *Distractors:* B is the density of the *square* lattice, that is, the ratio between the circle and the circumscribed square, and equals 0.7853982; C equals 0.4534498 and is obtained by forgetting the factor $\frac{1}{2}$ in the area of the six triangles, that is, writing $A_{\text{hexagon}} = 6Lr = 4\sqrt{3}r^2$; D equals 1.2091996 and arises from swapping apothem and side (hexagon of side r , of area $\frac{3\sqrt{3}}{2}r^2$), and must be discarded without computation because a density cannot exceed 1. *Diagnosed error:* getting the lattice cell wrong, confusing apothem and side, or getting the area of the regular polygon wrong. *Source:* Chapter 14. *Prerequisite:* area of the regular hexagon (Volume I, Chapter 6). *Recovery:* redo the computation drawing the cell and marking which segment equals r .

4) **B.** *Rationale:* $\sum_{n=0}^{15} n^2 = \frac{15 \cdot 16 \cdot 31}{6} = 1,240$ and Theorem 22.6 gives $\sum_k |X[k]|^2 = N \sum_n |x[n]|^2 = 16 \cdot 1,240 = 19,840$; this is item (a) of Exercise 33.3. *Distractors:* A is $\sum_n |x[n]|^2$, that is, the *right* side without the factor N ; C is the value for $N = 8$ from Example 33.3, reported without redoing the computation; D is $2 \sum_n x[n] = 2 \cdot 120 = 240$, that is, twice $X[0]$: a number that appears in the computation but is neither of the two

sides. *Diagnosed error*: forgetting the factor N in the discrete form of the identity. *Source*: Theorem 22.6. *Prerequisite*: sum of the first squares. *Recovery*: check the case $N = 8$ of Example 33.3, where $140 \cdot 8 = 1,120$, and then redo $N = 16$.

BEST PRACTICE

Quiz — answers, rationale and diagnosis (items 5–8).

5) **B**. *Rationale*: the four terms with $k \in \{1, 3, 5, 7\}$ are, in pairs, $64 + 32\sqrt{2}$ and $64 - 32\sqrt{2}$, and add up to 256 because the irrational parts cancel; the three remaining give $32 + 32 + 16 = 80$; in total 336 (Example 33.3). *Distractors*: A is false, the terms equal 109.2548, 32, 18.7452 and 16; C is a true arithmetic remark but with no content, which does not explain where 336 comes from; D is the circular answer, that is, using the thesis as the reason: it is exactly what verification by two independent ways is meant to avoid (Remark 33.6). *Diagnosed error*: accepting as a check a piece of reasoning that presupposes the result to be checked. *Source*: Example 33.3. *Prerequisite*: rationalization of $64/(2 \pm \sqrt{2})$. *Recovery*: redo the computation writing out the seven values in full and adding them in the order $k = 1, \dots, 7$.

6) **B**. *Rationale*: the result is true and proved — by Hales — but not in this volume, which exposes it as a panorama in Remark 14.5; writing “as proved in Volume III” attributes to the text a proof that it declares it does not give. This is the *tacit promotion* of Remark 33.4. *Distractors*: A is the answer of someone who does not distinguish the registers; C is false, the Kepler conjecture is a theorem; D is false, the volume also treats the three-dimensional case, but precisely as a panorama. *Diagnosed error*: citing as proved what was read in an observation. *Source*: Chapter 14. *Prerequisite*: Section 33.3.2. *Recovery*: rewrite the sentence in correct form, with the primary source in place of the volume.

7) **B**. *Rationale*: the stability of the isolated filter is read on the region of convergence (Theorem 21.1) and on the position of the poles relative to $|z| = 1$; the stability of the closed loop is the Nyquist criterion (Theorem 25.5), which is the argument principle (Theorem 25.4) applied to the transfer function. This is Chain E of Section 33.1.4. *Distractors*: A concerns the estimation of parameters from data, not stability; C concerns the *computation* of the transform, not its region of convergence — a very frequent confusion; D contains the conceptual tool (the winding number) but not the operational criterion, and by itself does not answer. *Diagnosed error*: looking for the answer in the chapter that contains the right word rather than in the one that contains the right statement. *Source*: Chapters 21 and 25. *Prerequisite*: Theorem 21.6. *Recovery*: go through Chain E link by link, saying, at each step, which question that link solves.

8) **B**. *Rationale*: the balance sheet is generated from the sources, not transcribed by hand; this is stated explicitly in Remark 33.5, which also warns that the printed table is the snapshot of one edition. *Distractors*: A gives the occurrences of terms, not the status of what is proved; C lists the works cited, not what the volume assumes; D contains the *prerequisites*, which are the first of the three categories and not the other two. *Diagnosed error*: confusing the bibliographic apparatus with the epistemic balance sheet of the text. *Source*: Remark 33.5. *Prerequisite*: none. *Recovery*: run the script and compare its result with the table of Section 33.3.3, noting the differences.

BEST PRACTICE

Assigned exercises — answers.

Exercise 33.1. A correct answer has five links. (1) Volume I, Chapter 5: circular inversion, which sends circles and lines to circles and lines. (2) Volume II, Chapter 3, and Chapter 6, Theorem 6.6: the Möbius transformations send generalized circles to generalized circles; the link uses from the previous one the fact that a line and a circle must be treated as a single

object. (3) Chapter 7, Theorem 7.6: the stereographic projection is conformal and sends circles to circles or lines; it uses from the previous one the notion of generalized circle, and realizes that notion as a “circle on the sphere.” Here the chain leaves the Euclidean plane, item (c): the line stops being a degenerate case and becomes the circle passing through the pole. (4) Chapter 14, Theorem 14.2: two Ford circles are tangent or disjoint; tangency is a contact condition preserved by the transformations of item (2). (5) Theorem 14.1: tangent Ford circles correspond to consecutive fractions in the Farey sequence, that is, to $|ad - bc| = 1$. For item (d): the chain contains a panorama, the circle packing theorem of Remark 14.6, and an import concerning the rotations of the sphere as Möbius transformations (Remark 6.6).

Exercise 33.2. (a) Proved: Theorem 4.5. (b) Panorama: Remark 14.5; the entire proof by Hales is missing, in particular the computer-assisted part. (c) Imported: Remark 9.1; the proof of the Picard–Lindelöf theorem is missing, used for the local existence and uniqueness of geodesics. (d) Proved: Theorem 23.3, by complete induction. (e) Panorama: Remark 17.9; the proof of Carleson’s theorem is missing, and the chapter instead proves convergence in L^2 (Theorem 17.1) and pointwise convergence under the hypothesis of bounded variation (Theorem 17.4). (f) Imported: Remark 23.11; the proof of the cyclicity of the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$ is missing, used once in Rader’s algorithm.

Exercise 33.3. (a) $\sum_{n=0}^{15} n^2 = \frac{15 \cdot 16 \cdot 31}{6} = 1,240$, hence the right side equals $16 \cdot 1,240 = 19,840$. The left side, with $X[0] = 120$ and $|X[k]|^2 = 256/(2 - 2\cos(\pi k/8))$, gives 14,400 from the term $k = 0$ and 5,440 from the other fifteen, in total 19,840. The script confirms 19,840.000000 with discrepancy $2.9 \cdot 10^{-11}$. ✓ (b) $192 \sin(\pi/192) = 3.1414525$ and $192 \tan(\pi/192) = 3.1418730$; the doubling recursion, starting from $\ell_6 = 1$ and doubling five times, gives the same 3.1414525. The gap between the two semiperimeters goes from $3.1427146 - 3.1410320 = 1.68 \cdot 10^{-3}$ for $n = 96$ to $3.1418730 - 3.1414525 = 4.21 \cdot 10^{-4}$ for $n = 192$: it shrinks by a factor 4.0008, that is, 4 to within one part in a thousand, as it must, because the width of the gap is $O(n^{-2})$ and the number of sides doubles. ✓ (c) In the square lattice the cell of each circle of radius r is the square of side $2r$, of area $4r^2$; the density is $\pi r^2/(4r^2) = \pi/4 = 0.7853982$. The ratio between the two densities is $\frac{\pi/\sqrt{12}}{\pi/4} = \frac{4}{\sqrt{12}} = \frac{2}{\sqrt{3}} = 1.1547005$: the triangular lattice is denser by 15.47%. The ratio does not depend on r because both densities are ratios between areas that scale as r^2 , and the factor r^2 cancels: it is the same invariance under similarity that Volume I attributes to π .

33.11. Chapter formulary

The five chains, one line each.

A — Measure $\ell = r\theta$, $A = \pi r^2 \rightarrow L^2 \geq 4\pi A \rightarrow \text{vol}(B^n) \rightarrow 0 \rightarrow \pi/\sqrt{12}$.

B — Group $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta} \rightarrow \mathbb{T} \cong SO(2) \rightarrow \chi_n(z) = z^n \rightarrow S^3 \rightarrow SO(3) \rightarrow SE(3)$, screw motion.

C — Curvature $\kappa \equiv 1/r \rightarrow \kappa \text{ constant} \Rightarrow \text{circle} \rightarrow (\kappa, \tau) \text{ determines the curve} \rightarrow \int \kappa ds \in 2\pi\mathbb{Z} \rightarrow G02/G03$.

D — Fourier orthogonality of the roots $\rightarrow$ characters of $S^1 \rightarrow$ completeness in $L^2 \rightarrow$ Poisson and Dirichlet $\rightarrow$ transform $\rightarrow$ DFT $\rightarrow$ FFT $\rightarrow$ OFDM.

E — Stability phasor $\rightarrow$ residues $\rightarrow$ half-plane $\text{Re } s < 0 \rightarrow$ annulus $|z| < 1 \rightarrow$ turns of the Nyquist diagram = winding number.

The chapter’s three check numbers.

$$\frac{\pi}{\sqrt{12}} = \frac{\pi}{2\sqrt{3}} = \frac{A_{\text{circle}}}{A_{\text{hexagon of apothem } r}} = 0.9068997, \quad \frac{\pi}{4} = 0.7853982 \text{ (square lattice),}$$

$$n \sin \frac{\pi}{n} < \pi < n \tan \frac{\pi}{n}, \quad \ell_{2n} = \sqrt{2 - \sqrt{4 - \ell_n^2}}, \quad \ell_6 = 1;$$

$$96 \sin \frac{\pi}{96} = 3.1410320, \quad 96 \tan \frac{\pi}{96} = 3.1427146, \quad \frac{223}{71} = 3.1408451, \quad \frac{22}{7} = 3.1428571;$$

$$\sum_{k=0}^{N-1} |X[k]|^2 = N \sum_{n=0}^{N-1} |x[n]|^2 : \quad N = 8 : 1,120 = 8 \cdot 140, \quad N = 16 : 19,840 = 16 \cdot 1,240.$$

For the ramp $x[n] = n$: $X[0] = \frac{N(N-1)}{2}$ and $|X[k]|^2 = \frac{N^2}{2 - 2 \cos(2\pi k/N)}$ for $k \neq 0$.

The balance sheet in three categories.

Category	Where it stands	What it guarantees
Prerequisite	Appendix A, infobox	standard, independent of the circle
Panorama	<i>osservazione</i> , <i>infobox</i>	statement with source, not proved
Import	<i>osservazione</i> “imported”	proof closed except for one link

Operational rule: *no teorema environment without a complete proof*; the check is `audit_dimostrazioni.py`, the census of imports is `registro_importati.py`.

The five drafting rules. (1) No proof left half-done; what does not close is declared imported. (2) Every value by two independent ways, with the discrepancy reported and the sign ✓. (3) Every source with authors, title, venue, year, pages and date of verification. (4) Every figure with computed and checked coordinates. (5) Three languages with the same labels, the same statements and the same numbers.

Sources and bibliography

[1] The statements of this chapter are statements *about the treatise*: each is verifiable by opening the place indicated by the label. The external sources of the results cited stand in the *Fonti* environments of the chapters that prove or declare them, and are not repeated here; the complete list, one line per central statement, is in the source register of Volume III, which collects author, title, venue, year, pages and date of verification for every entry cited in the volume. *Verified on 7 September 2026*.

[2] *Master Plan — The Circle: complete guidelines*, a project document of the treatise. It is the source of the perimeter of the work and of the correspondence between its chapters and the three volumes, used in Section 33.1. *Verified on 7 September 2026* (chapter-by-chapter correspondence rechecked against the volume’s list of sources).

[3] *Executive Plan of Volumes II and III*, a project document of the treatise, §3.4 (content and origin of each chapter of Volume III) and §6.1 (apparatuses of the volume, including this epilogue with the map of the three volumes). It is the source of the structure in six Parts and of the division of contents between Volume II and Volume III described in Sections 33.1.2 and 33.1.3. *Verified on 7 September 2026*.

[4] *Structure and compilation of the treatise*, a maintenance document of the repository: writing conventions (decimal separator, identical labels across the three languages), the list of the model’s environments and of the checking tools (`audit_dimostrazioni.py`, `registro_importati.py`, `parita_lingue.py`, `verifica_coerenza.py`). It is the source of the five drafting rules of Section 33.6. *Verified on 7 September 2026*.

[5] The three numerical checks of Section 33.4 and the answers to Exercise 33.3 were carried

out with a Python script in double precision, which computes each value by the two ways described in the text and prints the discrepancy; the discrepancies reported ($2.2 \cdot 10^{-16}$, $2.0 \cdot 10^{-14}$, $2.0 \cdot 10^{-12}$ and $2.9 \cdot 10^{-11}$) are those actually obtained. *Executed and verified on 7 September 2026.*

A further chain: from motion to control

Rigid-body kinematics describes motion; the mass matrix in Proposition 36.1 measures its energy. Theorem 36.3 connects forces and coordinates, while Theorem 36.4 reduces the ideal error to a linear system. Kalman's rank criterion, Theorem 34.1, decides whether its state can be controlled. The recursion in Theorem 35.1 reconstructs the state from noisy data under its stated assumptions. Finally, Theorem 37.1 gives the sampled model and Theorem 37.3 provides synthesis in the unit disk. The disk as a universal conformal domain is now Theorem 15.21; lattice points remain a boundary between proved results and open research, not a promise of a solution.

Part G

State, estimation, dynamics and digital control

State-space control

INFO

Level C–D. Prerequisites: real and complex linear algebra, Cayley–Hamilton, quadratic forms, matrix integrals and series. Linear dynamics connects to the rotation exponential in Chapter 28; discrete synthesis connects to the Z transform in Chapter 21, not the DFT in Chapter 22. The physical model is assumed; reachability, estimation and optimality consequences are proved. References for comparison: [2, 3].

34.1. The model and its solution

Definition 34.1: Continuous-time linear system. The state $x(t) \in \mathbb{R}^n$, input $u(t) \in \mathbb{R}^m$ and output $y(t) \in \mathbb{R}^p$ satisfy

$$\dot{x} = Ax + Bu, \quad y = Cx + Du.$$

The matrices are constant and real. On finite intervals inputs belong to L^2 , hence also to L^1 . State equations hold almost everywhere for absolutely continuous functions.

Proposition 34.1: Variation of constants.

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-s)}Bu(s)ds.$$

This is the unique solution with $x(0) = x_0$.

Proof. In a submultiplicative norm, $e^{At} = \sum_{j \geq 0} A^j t^j / j!$ converges uniformly on bounded intervals, dominated by $e^{\|A\||t|}$. The differentiated series also converges uniformly and gives $(e^{At})' = Ae^{At}$. The product $e^{-At}e^{At}$ has derivative zero and initial value I , so it equals I . Multiply the state equation by e^{-At} and differentiate the product:

$$\frac{d}{dt}(e^{-At}x(t)) = e^{-At}Bu(t).$$

Integrate and multiply by e^{At} to obtain the formula. Conversely, differentiation verifies the equation and initial condition. For two solutions with the same data, e^{-At} times their difference has derivative zero and initial value zero: the difference vanishes. $\square$

34.2. Controllability and observability

Theorem 34.1: Kalman rank criterion. Fix any $T > 0$. Every final state is reachable from zero at time T if and only if

$$\text{rank } \mathcal{C} = n, \quad \mathcal{C} = [B \ AB \ \cdots \ A^{n-1}B].$$

In this case every transfer between two states is possible in the same time.

Proof. 1. Gramian. Set

$$W_T = \int_0^T e^{As} B B^\top e^{A^\top s} ds, \quad v^\top W_T v = \int_0^T \|B^\top e^{A^\top s} v\|^2 ds.$$

Thus W_T is symmetric positive semidefinite. If the last quantity vanishes, its continuous nonnegative integrand vanishes for every s . Differentiation at $s = 0$ gives $v^\top A^j B = 0$ for all $j \geq 0$. Conversely, if these equalities hold for $0 \leq j < n$, Cayley–Hamilton extends them to every power: A^n is a combination of preceding powers, and induction applies. The exponential series then gives $B^\top e^{A^\top s} v = 0$. Consequently

$$\ker W_T = \ker \mathcal{C}^\top, \quad W_T > 0 \iff \text{rank } \mathcal{C} = n.$$

2. Necessity. If the rank is less than n , take $v \neq 0$ in $\ker \mathcal{C}^\top$. For any input, the solution formula gives $v^\top x(T) = 0$ when $x_0 = 0$. The state v cannot be reached because $v^\top v > 0$.

3. Constructive sufficiency. If $W_T > 0$, for target x_f and initial state x_0 set $d = x_f - e^{AT} x_0$ and choose

$$u_*(s) = B^\top e^{A^\top(T-s)} W_T^{-1} d.$$

Substitution in the solution, with $r = T - s$, makes the integral equal $W_T W_T^{-1} d = d$. Hence $x(T) = x_f$. The input is continuous and belongs to $L^2[0, T]$. $\square$

Corollary 34.1: Minimum transfer energy. Among all inputs realizing that transfer, u_* uniquely minimizes $\int_0^T \|u\|^2 ds$, and the minimum is $d^\top W_T^{-1} d$.

Proof. Write $u = u_* + v$. The difference leaves the final state unchanged, so $\int_0^T e^{A(T-s)} B v(s) ds = 0$. Hence

$$\int_0^T u_*^\top v ds = d^\top W_T^{-1} \int_0^T e^{A(T-s)} B v(s) ds = 0.$$

Expanding the square gives $\|u\|_{L^2}^2 = \|u_*\|_{L^2}^2 + \|v\|_{L^2}^2$. Substituting u_* makes the first term $d^\top W_T^{-1} d$; equality requires $v = 0$ almost everywhere. $\square$

Theorem 34.2: Observability and duality. With known input, the initial state is uniquely determined by the output on $[0, T]$, $T > 0$, if and only if

$$\text{rank } \mathcal{O} = n, \quad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}.$$

This is exactly controllability of the dual pair $(A^\top, C^\top)$.

Proof. Subtract from y the known-input response, including Du , to obtain $\tilde{y}(t) = Ce^{At}x_0$. Define

$$W_o = \int_0^T e^{A^\top t} C^\top C e^{At} dt.$$

Theorem 34.1, applied to $(A^\top, C^\top)$, gives $W_o > 0$ exactly when $\mathcal{O}$ has full rank. If $W_o > 0$,

$$x_0 = W_o^{-1} \int_0^T e^{A^\top t} C^\top \tilde{y}(t) dt.$$

If W_o is singular, a vector $v \neq 0$ in its kernel satisfies $Ce^{At}v = 0$ for every t , by the same nonnegative integral identity. The states x_0 and $x_0 + v$ produce identical outputs and cannot be distinguished. $\square$

34.3. Pole placement: a construction, not a promise

Theorem 34.3: Ackermann's formula for one input. If (A, b) is controllable and $p(\lambda) = \lambda^n + \sum_{j=0}^{n-1} d_j \lambda^j$ has real coefficients, the feedback $u = -Kx$ with

$$K = e_n^\top C^{-1} p(A)$$

makes p the characteristic polynomial of $A - bK$.

Proof. Let $r = e_n^\top C^{-1}$. Then $rA^j b = 0$ for $j < n-1$ and $rA^{n-1}b = 1$. Form S with rows $r, rA, \dots, rA^{n-1}$. The product SC has zeros before the antidiagonal and ones on it, so it is invertible, as is S . In coordinates $z = Sx$, $Sb = e_n$. The first $n-1$ rows of SAS^{-1} shift a coordinate to the next one. If $\det(\lambda I - A) = \lambda^n + \sum a_j \lambda^j$, Cayley–Hamilton gives last row $(-a_0, \dots, -a_{n-1})$.

$$SAS^{-1} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}.$$

Feedback in the new basis, $u = -\sum (d_j - a_j)z_{j+1}$, replaces the last row by $(-d_0, \dots, -d_{n-1})$. Expanding $\det(\lambda I - A_c)$ recursively along the first column of the companion matrix gives $p(\lambda)$. Returning to x ,

$$K = \sum_{j=0}^{n-1} (d_j - a_j) r A^j = r \left(A^n + \sum_{j=0}^{n-1} d_j A^j \right) = r p(A).$$

The last equality again uses Cayley–Hamilton. A desired multiset of roots closed under conjugation has a real monic polynomial; conjugate complex poles and repeated poles are both included. $\square$

Theorem 34.4: Extension to multiple inputs. If (A, B) is controllable, every real monic polynomial of degree n is the characteristic polynomial of $A - BK$ for some real matrix K .

Proof. Choose $b = B\beta \neq 0$ and build a basis $v_1 = b, v_2, \dots, v_n$ with $v_{j+1} = Av_j + Bu_j$. Suppose $k < n$ independent vectors have been constructed and let $V_k = \text{span}(v_1, \dots, v_k)$. If no choice of u_k produces a vector outside V_k , then $Av_k \in V_k$ and $\text{im } B \subset V_k$. The preceding relations also give $Av_j = v_{j+1} - Bu_j \in V_k$ for $j < k$. Thus V_k is A -invariant and contains $\text{im } B$, hence every column of $\mathcal{C}$, contradicting rank n . The basis can therefore be completed.

Define F by $Fv_j = u_j$ for $j < n$ and choose any value on v_n . Then $(A + BF)^{j-1}b = v_j$ for $1 \leq j \leq n$, so $(A + BF, b)$ is single-input controllable. Theorem 34.3 gives a row k assigning the polynomial to $A + BF - bk$. Since $b = B\beta$, set $K = \beta k - F$. $\square$

Corollary 34.2: Observer and separation principle. *If (A, C) is observable, the poles of $A - LC$ can be assigned arbitrarily. With*

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x} - Du), \quad u = -K\hat{x},$$

the combined poles are those of $A - BK$ and $A - LC$, counted with multiplicity.

Proof. Apply pole placement to the dual pair $(A^\top, C^\top)$ and transpose the gain. With $e = x - \hat{x}$, subtracting the equations gives $\dot{e} = (A - LC)e$ and

$$\frac{d}{dt} \begin{bmatrix} x \\ e \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix}.$$

The determinant of a block triangular matrix is the product of its diagonal-block determinants. $\square$

34.4. Finite-horizon LQR

Lemma 34.1: Local existence for the Riccati equation. *A matrix equation $\dot{P} = F(P)$ with polynomial F has a unique local solution for every initial value. A solution bounded on a finite interval extends to its endpoint.*

Proof. On a closed ball around P_0 , F is bounded by M and Lipschitz with constant L , because its derivative is continuous and bounded. Choose h with hM smaller than the radius and $hL < 1$. The operator $\mathcal{T}P(t) = P_0 + \int_0^t F(P(s))ds$ maps continuous functions in the ball to themselves and contracts the uniform distance by hL . Differences of successive iterates are dominated by a geometric series, so the iterates converge uniformly. The limit solves the integral equation and hence the differential equation. The same contraction gives uniqueness. If P stays bounded, so does $F(P)$; thus P is uniformly Lipschitz in time and has a limit at the endpoint. Restart the local construction from that limit. $\square$

Theorem 34.5: Riccati and completion of the square. *Let $Q = Q^\top \geq 0$, $R = R^\top > 0$, $S = S^\top \geq 0$ and $T < \infty$. For the state system, the cost*

$$J_T(u) = x(T)^\top Sx(T) + \int_0^T (x^\top Qx + u^\top Ru) dt$$

has minimum $x_0^\top P(0)x_0$, uniquely attained by $u_ = -R^{-1}B^\top P(t)x$, where*

$$-\dot{P} = A^\top P + PA - PBR^{-1}B^\top P + Q, \quad P(T) = S.$$

This equation has a symmetric positive semidefinite solution on all of $[0, T]$.

Proof. 1. Identity. Begin with the local solution of Lemma 34.1, integrated backwards. Its transpose solves the same equation with the same data, so $P = P^\top$. Set $K = R^{-1}B^\top P$.

Differentiate $x^\top Px$ and substitute the two equations:

$$\begin{aligned}\frac{d}{dt}(x^\top Px) &= x^\top (\dot{P} + A^\top P + PA)x + 2x^\top PBu \\ &= -x^\top Qx + x^\top K^\top RKx + 2x^\top K^\top Ru \\ &= -x^\top Qx - u^\top Ru + (u + Kx)^\top R(u + Kx).\end{aligned}$$

Integrating from t to T ,

$$J_{[t,T]}(u) = x(t)^\top P(t)x(t) + \int_t^T (u + Kx)^\top R(u + Kx) ds.$$

The choice $u = -Kx$ makes the integral zero; every other choice makes it positive unless equality holds almost everywhere. The closed linear system has a unique solution, so the optimal input is unique.

2. No Riccati blow-up. The cost is nonnegative, so the optimal choice shows $P(t) \geq 0$ wherever the solution exists. The cost of $u = 0$ gives, for every initial vector ξ ,

$$0 \leq \xi^\top P(t)\xi \leq \xi^\top \left[e^{A^\top(T-t)} S e^{A(T-t)} + \int_0^{T-t} e^{A^\top s} Q e^{As} ds \right] \xi.$$

The bracketed matrix is uniformly bounded for $0 \leq t \leq T$. For symmetric positive semidefinite matrices, this also bounds the norm of P . The continuation lemma extends the solution to zero. The identity in step 1 therefore holds over the entire interval. $\square$

34.5. Infinite horizon: hypotheses and existence proved

Definition 34.2: Stabilizability and detectability. A matrix is Hurwitz when every eigenvalue has negative real part. The pair (A, B) is stabilizable if some F makes $A - BF$ Hurwitz. The pair (H, A) is detectable if some L makes $A - LH$ Hurwitz. Here $H = Q^{1/2}$, the symmetric positive semidefinite square root of Q .

Theorem 34.6: Stabilizing solution of the algebraic Riccati equation. Under stabilizability and detectability, with $Q \geq 0$ and $R > 0$, there is a symmetric solution $P \geq 0$ of

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0.$$

The matrix $A - BR^{-1}B^\top P$ is Hurwitz. Feedback $u_* = -R^{-1}B^\top Px$ minimizes $J_\infty = \int_0^\infty (x^\top Qx + u^\top Ru)dt$, with value $x_0^\top Px_0$. The stabilizing solution is unique. Positive definiteness $P > 0$ is not asserted without an additional hypothesis.

Proof. 1. Monotone limit. Let P_T be the initial value matrix of the finite problem with terminal weight $S = 0$. Extending the horizon adds nonnegative cost, so $x_0^\top P_T x_0$ increases with T for every x_0 . Stabilizing feedback $u = -Fx$ provides the upper bound

$$P_T \leq \bar{P} = \int_0^\infty e^{(A-BF)^\top t} (Q + F^\top RF) e^{(A-BF)t} dt.$$

The integral converges: Jordan form shows that the exponential of a Hurwitz matrix is bounded by $Me^{-\alpha t}$, absorbing polynomial block factors into a slightly slower exponential. Bounded monotone quadratic forms have limits. Applying this to basis vectors and their sums yields, by polarization, a matrix limit $P \geq 0$.

2. Algebraic equation. By time invariance, P_T , regarded as a function of T , satisfies

$$\frac{dP_T}{dT} = A^\top P_T + P_T A - P_T B R^{-1} B^\top P_T + Q, \quad P_0 = 0.$$

The right-hand side converges to the same expression in P . If any limiting component were nonzero, the corresponding derivative would eventually have fixed sign and modulus bounded away from zero, preventing that component of P_T from converging. The limiting right-hand side is therefore zero, giving the algebraic equation.

3. Stability. Set $K = R^{-1} B^\top P$ and $A_c = A - BK$. Riccati implies

$$A_c^\top P + P A_c = -Q - K^\top R K.$$

If $A_c v = \lambda v$ for a nonzero complex vector,

$$2 \operatorname{Re} \lambda v^* P v = -\|Hv\|^2 - \|R^{1/2} K v\|^2.$$

For $\operatorname{Re} \lambda \geq 0$, the left side is nonnegative and the right side nonpositive, so $Hv = Kv = 0$. Then $Av = \lambda v$ and $(A - LH)v = \lambda v$, contradicting detectability. Thus A_c is Hurwitz.

4. The boundary term cannot be ignored. For an input of finite cost, u and Hx belong to $L^2[0, \infty)$. Write

$$\dot{x} = (A - LH)x + v(t), \quad v = Bu + LHx \in L^2.$$

Variation of constants gives $x(t) \rightarrow 0$. Specifically, split the convolution into $[0, S]$ and $[S, t]$. The first integral tends to zero by exponential kernel decay and integrability of v on $[0, S]$. Cauchy–Schwarz bounds the second by $(\int_0^\infty M^2 e^{-2\alpha s} ds)^{1/2} \|v\|_{L^2(S, \infty)}$. First let $t \rightarrow \infty$, then $S \rightarrow \infty$. Thus $x(t)^\top P x(t) \rightarrow 0$. Integrating the completed-square identity with constant P gives

$$J_\infty(u) = x_0^\top P x_0 + \int_0^\infty (u + Kx)^\top R(u + Kx) dt.$$

The input $u_* = -Kx$ has finite cost since A_c is Hurwitz, and makes the integral zero. Infinite-cost inputs cannot improve it.

5. Stabilizing uniqueness. For two symmetric stabilizing solutions P_1, P_2 , set $G = BR^{-1}B^\top$ and $D = P_1 - P_2$. Subtracting their Riccati equations gives

$$(A - GP_1)^\top D + D(A - GP_2) = 0.$$

The derivative of $e^{(A-GP_1)^\top t} D e^{(A-GP_2)t}$ is zero, so this product always equals D . Stability makes it tend to zero; hence $D = 0$. $\square$

Remark 34.1: Semidefinite does not mean definite. For $A = -1$, $B = 1$, $Q = 0$, $R = 1$, the system is stabilizable and detectable, but $P = 0$ and the minimum cost is zero with $u = 0$. Thus the plan's assertion $P > 0$ needs correction. If $(Q^{1/2}, A)$ is observable instead, $x_0^\top P x_0 = 0$ forces, under the optimum, $u = 0$ and $Q^{1/2} e^{At} x_0 = 0$ for every t . Observability forces $x_0 = 0$, hence $P > 0$.

34.6. Discrete time and a checkable example

Proposition 34.2: Discrete reachability and observability. For $x_{k+1} = Ax_k + Bu_k$, $y_k = Cx_k + Du_k$, we have

$$x_N = A^N x_0 + \sum_{j=0}^{N-1} A^{N-1-j} Bu_j.$$

Reachability from zero in n steps is equivalent to full rank of $\mathcal{C}$; observability from $y_0, \dots, y_{n-1}$ is equivalent to full rank of $\mathcal{O}$. Pole-placement theorems remain unchanged because they are algebraic.

Proof. The formula for x_N follows by induction: multiply the formula for N by A and add Bu_N . For $N = n$, the final vector from zero is $\mathcal{C}(u_{n-1}^\top, \dots, u_0^\top)^\top$, so it is arbitrary exactly when $\mathcal{C}$ is onto. For $N > n$, Cayley–Hamilton prevents growth of the reachable space. Subtract known-input contributions from the outputs; the stacked vector is $\mathcal{O}x_0$, which identifies x_0 exactly when $\ker \mathcal{O} = \{0\}$. $\square$

In discrete time, driving every state to zero is not synonymous with reachability from zero: with $A = 0$, $B = 0$, every state disappears in one step, but no nonzero state is reachable from zero. Asymptotic stability requires $|\lambda(A)| < 1$; Jordan form proves this because $k^j |\lambda|^k \rightarrow 0$, while an eigenvector with $|\lambda| \geq 1$ prevents decay.

Example 34.1: An oscillator, two pole checks. Consider $\ddot{q} + \dot{q} + 4q = u$, with $x = (q, \dot{q})^\top$ and output q . Parameters are $\omega_n = 2$ and $\zeta = 1/4$, using the elastic-mode conventions of Chapter 30.

$$A = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathcal{C} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathcal{O} = I.$$

The determinant of $\mathcal{C}$ is -1 . For poles $-3, -4$, the desired polynomial is $s^2 + 7s + 12$. With $K = (k_1, k_2)$, the closed polynomial is $s^2 + (1 + k_2)s + 4 + k_1$, hence $K = (8, 6)$. Ackermann's formula, calculated separately, gives the same result. Finally

$$A - BK = \begin{bmatrix} 0 & 1 \\ -12 & -7 \end{bmatrix}, \quad \det(sI - A + BK) = (s + 3)(s + 4).$$

The code also computes the numerical eigenvalues directly, rather than inferring them only from the coefficients.

34.7. Assigned and worked exercises

Exercise 34.1: Complete scalar LQR. For $\dot{x} = u$, $Q = R = 1$, $S = 0$, derive $P(t)$, $x_*(t)$ and the cost.

Solution. Riccati is $-\dot{P} = 1 - P^2$, with $P(T) = 0$. Separating variables gives $\operatorname{artanh} P = T - t$, hence $P(t) = \tanh(T - t)$. Integrating $\dot{x} = -P(t)x$ gives

$$x_*(t) = x_0 \frac{\cosh(T - t)}{\cosh T}, \quad u_*(t) = -x_0 \frac{\sinh(T - t)}{\cosh T}, \quad J_T^* = x_0^2 \tanh T.$$

The cost is also checked by directly integrating $x_*^2 + u_*^2 = x_0^2 \cosh(2(T - t)) / \cosh^2 T$. As

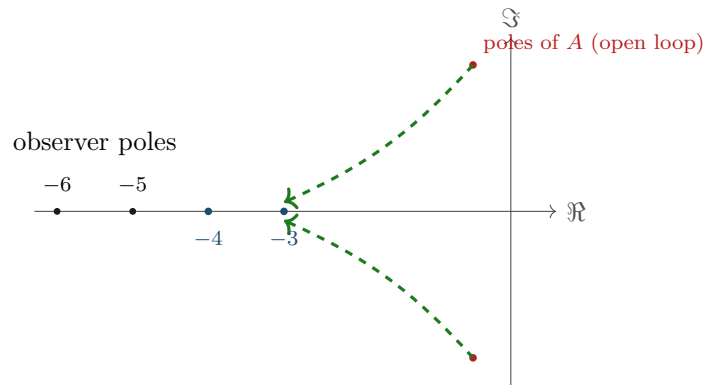


Figure 34.1: The oscillator example at a glance: the open-loop poles (red, $-0.5 \pm 1.9365j$, roots of $s^2 + s + 4$) are moved by state feedback $K = (8, 6)$ to the desired poles $-3, -4$ (blue, roots of $s^2 + 7s + 12$, Theorem 34.3); the observer poles $-5, -6$ (black, gain $L = (10, 16)$, Exercise 34.3) remain distinct, as required by the separation principle.

$T \rightarrow \infty$, $P \rightarrow 1$, $u = -x$ and $J_\infty^* = x_0^2$.

Exercise 34.2: An invisible mode. For $A = \text{diag}(-1, 2)$, $B = (1, 0)^\top$, $C = (1, 0)$, determine controllability, observability and stabilizability.

Solution. Both rank matrices have rank one. State x_2 is absent from the output and always satisfies $\dot{x}_2 = 2x_2$, independently of u . No feedback stabilizes it. Replacing 2 by -2 leaves the system uncontrollable and unobservable, but makes it stabilizable and detectable: the hidden mode already decays on its own.

Exercise 34.3: Design the oscillator observer. Assign poles $-5, -6$ to $A - LC$ in the example.

Solution. For $L = (\ell_1, \ell_2)^\top$,

$$\det(sI - A + LC) = (s + \ell_1)(s + 1) + 4 + \ell_2.$$

Comparison with $s^2 + 11s + 30$ gives $\ell_1 = 10$, $\ell_2 = 16$. With $K = (8, 6)$ the four observer-controller poles are $-3, -4, -5, -6$ by separation.

34.8. Self-assessment: eight statements

Mark each statement true or false before consulting the key.

1. Continuous-time controllability depends on the particular $T > 0$.
2. $W_T > 0$ is equivalent to full rank of $\mathcal{C}$.
3. Observability uses known input, not necessarily zero input.
4. Real feedback can assign a lone nonreal pole without its conjugate.
5. Finite Riccati requires controllability to have a solution on a finite interval.
6. Under the infinite-horizon theorem, P is always positive definite.

-
7. Separation combines the two pole multisets, with multiplicity.
 8. In discrete time, dying out in one step implies reachability from zero.

Teacher's key

1 False: the rank criterion is independent of $T > 0$. 2 True: the kernels coincide. 3 True: subtract the known response. 4 False: a real characteristic polynomial has conjugate roots. 5 False: the finite proof only uses $Q, S \geq 0$ and $R > 0$. 6 False: $Q = 0, A = -1$ gives $P = 0$. 7 True: the combined matrix is block triangular. 8 False: $A = B = 0$ is a counterexample.

34.9. Formulary and sources

$$\mathcal{C} = [B \ AB \ \cdots \ A^{n-1}B], \quad \mathcal{O}^\top = [C^\top \ A^\top C^\top \ \cdots \ (A^\top)^{n-1}C^\top].$$

$$K_{\text{SISO}} = e_n^\top \mathcal{C}^{-1} p(A), \quad K_{\text{LQR}} = R^{-1} B^\top P.$$

For numerical use, explicitly forming $\mathcal{C}^{-1}$ in an ill-conditioned system is not recommended: the formula proves existence and correctness, while a linear solver avoids an unnecessary inversion. The supplied checks retain both exact calculations and floating-point comparisons. References [2, 3] are recorded with URL, section and consultation date in the bibliography and source register. The multiple-input proof and infinite-horizon limit proof are developed in the text rather than deferred to imported results.

State estimation: Kalman and quaternion attitude

INFO

Level C–D. Prerequisites: means, covariances and Gaussian densities; quaternion algebra from Chapter 28; the state model of Chapter 34. An exact derivation for linear systems is distinguished from a local linearization on $SO(3)$. The latter does not automatically inherit the former’s optimality. Sources: Kalman [4], Lall [5], Markley–Bauer [6] and Solà [7].

35.1. Which error is minimized

For a random vector X with mean μ , $\text{Cov } X = \mathbb{E}[(X - \mu)(X - \mu)^\top]$. The quadratic loss is $\mathbb{E}\|X - \hat{X}\|^2$. If the error has zero mean, this loss is the trace of its covariance. Here unbiased means $\mathbb{E}(X - \hat{X}) = 0$ under the stated probability model; it does not mean unbiasedness for every possible unknown deterministic state.

Lemma 35.1: Linear projection and the orthogonality principle. *Let X, Y be random vectors with finite second moments and $S = \text{Cov } Y > 0$. Among affine estimators $a + KY$, the unique quadratic-loss minimizer is*

$$\hat{X} = \mathbb{E}X + \text{Cov}(X, Y)S^{-1}(Y - \mathbb{E}Y).$$

Its error is orthogonal in expectation to every component of $Y - \mathbb{E}Y$.

Proof. For fixed K , separating the mean from the centred error shows that the loss contains $\|\mathbb{E}X - a - K\mathbb{E}Y\|^2$, minimized by $a = \mathbb{E}X - K\mathbb{E}Y$. Centre X, Y , set $P = \text{Cov } X$, $V = \text{Cov}(X, Y)$ and expand

$$\begin{aligned} \text{Cov}(X - KY) &= P - KV^\top - VK^\top + KSK^\top \\ &= P - VS^{-1}V^\top + (K - VS^{-1})S(K - VS^{-1})^\top. \end{aligned}$$

The last term is positive semidefinite, and its trace vanishes only for $K = VS^{-1}$ since $S > 0$. Moreover, $\mathbb{E}[(X - KY)Y^\top] = V - KS = 0$ for that value. This is also the projection criterion: any other estimator differs by a combination of Y , orthogonal to the residual, and the loss decomposes into a sum of two squares. $\square$

35.2. The linear filter, fully derived

Theorem 35.1: Discrete Kalman recursion. *Consider deterministic matrices, possibly depending on k , and known deterministic inputs:*

$$x_{k+1} = A_k x_k + B_k u_k + w_k, \quad z_k = H_k x_k + v_k.$$

The initial error, all w_k and all v_k have zero means and finite second moments and are mutually uncorrelated; $\text{Cov } w_k = Q_k \geq 0$ and $\text{Cov } v_k = R_k > 0$. Initialized with the correct mean and covariance, the following formulas give the best affine estimator in all available observations:

$$\hat{x}_k^- = A_{k-1}\hat{x}_{k-1}^+ + B_{k-1}u_{k-1}, \quad P_k^- = A_{k-1}P_{k-1}^+A_{k-1}^\top + Q_{k-1}, \quad (35.1)$$

$$r_k = z_k - H_k\hat{x}_k^-, \quad S_k = H_kP_k^-H_k^\top + R_k, \quad (35.2)$$

$$K_k = P_k^-H_k^\top S_k^{-1}, \quad \hat{x}_k^+ = \hat{x}_k^- + K_k r_k, \quad (35.3)$$

$$P_k^+ = (I - K_k H_k)P_k^-(I - K_k H_k)^\top + K_k R_k K_k^\top.$$

Initially use $\hat{x}_0^- = \mathbb{E}x_0$, $P_0^- = \text{Cov } x_0$ before the first measurement.

Proof. 1. Prediction. If $e_{k-1}^+ = x_{k-1} - \hat{x}_{k-1}^+$, then $e_k^- = A_{k-1}e_{k-1}^+ + w_{k-1}$. Its mean is zero, and uncorrelatedness removes the two cross terms in the covariance, yielding (35.1).

2. Update. Suppress index k . The innovation is $r = He^- + v$, so $\text{Cov } r = S = HP^-H^\top + R > 0$ and $\text{Cov}(e^-, r) = P^-H^\top$. After correction by Kr , the error is $e^+ = (I - KH)e^- - Kv$. Expanding its covariance gives the Joseph form (35.3). Lemma 35.1 gives

$$P^+(K) = P^- - P^-H^\top S^{-1}HP^- + (K - K_*)S(K - K_*)^\top, \quad K_* = P^-H^\top S^{-1}.$$

This does more than minimize the trace: no other gain improves the covariance in the positive-semidefinite order. Explicitly, the directional derivative of its trace in direction E is $2\text{tr}((KS - P^-H^\top)E^\top)$, vanishing for every E exactly when $K = K_*$. The second variation is $2\text{tr}(ESE^\top) > 0$ for $E \neq 0$.

3. Why the latest innovation is enough. Inductively, the prediction error is orthogonal to all centred past observations. Initially this is true because there are no observations. The innovation $r_k = H_k e_k^- + v_k$ is also orthogonal to the past. After updating, e_k^+ remains orthogonal to the past and is orthogonal to the new innovation because

$$\mathbb{E}[e_k^+ r_k^\top] = P_k^- H_k^\top - K_k S_k = 0.$$

The new affine data space is the previous space plus the components of r_k , since $z_k = r_k + H_k \hat{x}_k^-$. Orthogonality therefore characterizes the optimum with respect to *all* observations. Propagation by $A_k e_k^+ + w_k$ preserves orthogonality, completing the induction. Error means remain zero at every step. $\square$

Corollary 35.1: When the filter is optimal among all estimators. If the initial vector and all noises are jointly Gaussian, the filter is the conditional mean and minimizes quadratic error even among nonlinear estimators with finite second moment. Without Gaussianity, only affine optimality has been proved.

Proof. At a fixed time, the state and stacked observations Y are affine transformations of a Gaussian vector, hence jointly Gaussian. The residual $e = x - \hat{x}$ is jointly Gaussian with Y and has zero cross-covariance by the theorem. Two uncorrelated Gaussian blocks are independent: their joint characteristic function is

$$\exp\left(-\frac{1}{2}t^\top \Sigma_e t - \frac{1}{2}s^\top \Sigma_Y s\right) = \exp\left(-\frac{1}{2}t^\top \Sigma_e t\right) \exp\left(-\frac{1}{2}s^\top \Sigma_Y s\right),$$

after centring. Equivalently, diagonalizing each block covariance makes the Gaussian density on its support a product of the two block densities; zero-variance coordinates are constant. Thus $\mathbb{E}(e | Y) = 0$. For any $g(Y)$ with finite second moment,

$$\mathbb{E}\|x - g(Y)\|^2 = \mathbb{E}\|e\|^2 + \mathbb{E}\|\hat{x} - g(Y)\|^2,$$

because conditioning on Y makes the cross term zero. The minimum is therefore $g = \hat{x}$ almost surely. $\square$

Example 35.1: Two measurements of a static state. Take initial mean zero, initial variance 1, $H = 1$, $Q = 0$, $R = 1/4$, and measurements $z_1 = 6/5$, $z_2 = 11/10$. The first update gives

$$K_1 = \frac{4}{5}, \quad \hat{x}_1 = \frac{24}{25}, \quad P_1 = \frac{1}{5}.$$

At the second step,

$$K_2 = \frac{1/5}{1/5 + 1/4} = \frac{4}{9}, \quad \hat{x}_2 = \frac{24}{25} + \frac{4}{9} \left(\frac{11}{10} - \frac{24}{25} \right) = \frac{46}{45}, \quad P_2 = \frac{1}{9}.$$

Independent check: in the Gaussian case the posterior density is proportional to $\exp[-(x^2 + 4(z_1 - x)^2 + 4(z_2 - x)^2)/2]$. The coefficient of x^2 is 9, and the linear coefficient is $-8(z_1 + z_2)$. Completing the square gives mean $4(z_1 + z_2)/9 = 46/45$ and variance $1/9$, without reusing the recursion.

35.3. Kinematics and multiplicative error

We adopt the volume's convention: scalar-first quaternion $q = (q_0, q_v)$, Hamilton product and active rotation from body to fixed frame. Write $[a]_{\times} b = a \times b$, $R(q_1 \otimes q_2) = R(q_1)R(q_2)$ and

$$\text{Exp}_q(\phi) = \left(\cos \frac{\|\phi\|}{2}, \frac{\sin(\|\phi\|/2)}{\|\phi\|} \phi \right), \quad \text{Exp}(\phi) = e^{[\phi]_{\times}}.$$

At zero use the limiting ratio. The rotation logarithm Log returns a vector of norm less than π on the local branch considered. Near angle π , the branch becomes ambiguous: the following linearization is not a global description.

Proposition 35.1: Linear dynamics of local error. Angular velocity is expressed in the body frame. The gyroscope measures $\omega_m = \omega + b + n_g$, with bias b and noise n_g . For nominal estimates $\hat{q}, \hat{b}$, define

$$q = \hat{q} \otimes \delta q, \quad \delta q = \text{Exp}_q(\delta \theta), \quad \delta b = b - \hat{b}, \quad \hat{\omega} = \omega_m - \hat{b}.$$

To first order in $\delta \theta, \delta b, n_g$, between samples with constant bias,

$$\frac{d}{dt} \begin{bmatrix} \delta \theta \\ \delta b \end{bmatrix} = F \begin{bmatrix} \delta \theta \\ \delta b \end{bmatrix} + \begin{bmatrix} -n_g \\ 0 \end{bmatrix}, \quad F = \begin{bmatrix} -[\hat{\omega}]_{\times} & -I \\ 0 & 0 \end{bmatrix}.$$

Proof. 1. Kinematics. Over dt , the body-frame increment is $\text{Exp}_q(\omega dt) = (1, \omega dt/2) + O(dt^2)$. Right composition gives $q(t + dt) = q(t) \otimes \text{Exp}_q(\omega dt) + O(dt^2)$, hence

$$\dot{q} = \frac{1}{2} q \otimes (0, \omega), \quad \dot{\hat{q}} = \frac{1}{2} \hat{q} \otimes (0, \hat{\omega}).$$

2. Exact error. Differentiate $q = \hat{q} \otimes \delta q$ and multiply on the left by $\hat{q}^{-1}$:

$$\dot{\delta q} = \frac{1}{2} [\delta q \otimes (0, \hat{\omega} - \delta b - n_g) - (0, \hat{\omega}) \otimes \delta q].$$

3. Vector part. Use $\delta q = (1, \delta \theta/2) + O(\|\delta \theta\|^2)$ and Hamilton's rule $(a, u) \otimes (c, v) = (ac - u^{\top} v, av + cu + u \times v)$. To first order, the vector part of the bracket is

$$-\delta b - n_g + \frac{1}{2} \delta \theta \times \hat{\omega} - \frac{1}{2} \hat{\omega} \times \delta \theta = -\delta b - n_g - [\hat{\omega}]_{\times} \delta \theta.$$

The derivative of the vector part of δq is $\delta\theta/2$ to first order, so the factor $1/2$ cancels on both sides. The scalar part is at least second order. Both biases are constant within the interval, so $\dot{\delta b} = 0$. The cross-product sign is fixed by the choice of *right* error; reversing that convention changes the formulas. $\square$

35.4. A consistent noise discretization

Proposition 35.2: Error transition and process covariance. Assume $\hat{\omega}$ and n_g constant during an interval of duration T . The sample n_g has covariance Σ_g . At the interval end, the true bias receives an independent increment η_b of covariance Σ_b . With $\phi = T\hat{\omega}$, the linearized transition is

$$\begin{aligned} \Phi = e^{FT} &= \begin{bmatrix} \text{Exp}(-\phi) & -TJ_r(\phi) \\ 0 & I \end{bmatrix}, & \Gamma_g &= \begin{bmatrix} -TJ_r(\phi) \\ 0 \end{bmatrix}, & \Gamma_b &= \begin{bmatrix} 0 \\ I \end{bmatrix}, \\ Q_d &= \Gamma_g \Sigma_g \Gamma_g^\top + \Gamma_b \Sigma_b \Gamma_b^\top, & P^- &= \Phi P^+ \Phi^\top + Q_d, \end{aligned}$$

where the right Jacobian is

$$J_r(\phi) = I - \frac{1 - \cos \theta}{\theta^2} [\phi]_\times + \frac{\theta - \sin \theta}{\theta^3} [\phi]_\times^2, \quad \theta = \|\phi\|.$$

Proof. The second error component is constant until the final increment. Variation of constants for the first gives

$$\delta\theta(T) = e^{-T[\hat{\omega}]_\times} \delta\theta(0) - \left(\int_0^T e^{-s[\hat{\omega}]_\times} ds \right) (\delta b + n_g).$$

Rodrigues' formula from Chapter 28, integrated term by term, gives

$$\int_0^1 e^{-s[\phi]_\times} ds = J_r(\phi).$$

This gives Φ and the two noise-input matrices. Independence makes the covariance of the sum equal the sum of transformed covariances. At zero the ratios tend to $1/2$ and $1/6$, so $J_r(\phi) = I - [\phi]_\times/2 + [\phi]_\times^2/6 + O(\|\phi\|^3)$. $\square$

This is a *sampled-noise* convention: Σ_g is angular-rate variance per sample, not a continuous white-noise spectral density. The two cannot be interchanged without changing the time scaling. Nominal propagation is $\hat{q}_{k+1}^- = \hat{q}_k^+ \otimes \text{Exp}_q(T\hat{\omega}_k)$, $\hat{b}_{k+1}^- = \hat{b}_k^+$.

35.5. Measurement, correction and reset

The numerical example assumes a *full* attitude measurement $R_m = R(q) \text{Exp}(\nu)$, with small zero-mean error ν and covariance Σ_a . This can model an optical system or an absolute estimate built from multiple directions. A single accelerometer is not credited with three observable attitude degrees of freedom.

$$r = \text{Log}(R(\hat{q}^-)^\top R_m) = \delta\theta + \nu + O((\|\delta\theta\| + \|\nu\|)^2), \quad H = [I \ 0].$$

The first-order equality follows by expanding $\text{Exp}(\delta\theta) \text{Exp}(\nu) = I + [\delta\theta + \nu]_\times + O(2)$ and the logarithm near I . Apply the linear Kalman equations to this local model with $R = \Sigma_a$. If $a = Kr = (a_\theta, a_b)$, correct by

$$\hat{q}^+ = \hat{q}^- \otimes \text{Exp}_q(a_\theta), \quad \hat{b}^+ = \hat{b}^- + a_b.$$

A product of unit quaternions preserves unit norm in exact arithmetic. Renormalization in code only corrects roundoff; it does not replace geometric treatment of the covariance.

Proposition 35.3: Correct reset Jacobian. *After nominal correction, the covariance P_J obtained in Joseph form must be expressed in the new error frame. To first order in the residual fluctuation about a_θ ,*

$$P^+ = G_r P_J G_r^\top, \quad G_r = \begin{bmatrix} J_r(a_\theta) & 0 \\ 0 & I \end{bmatrix}.$$

Proof. Before correction, write the angular error as $a_\theta + \varepsilon$. The new error satisfies

$$\text{Exp}(\delta\theta^+) = \text{Exp}(-a_\theta) \text{Exp}(a_\theta + \varepsilon).$$

We derive this expression rather than postulating the reset sign. For $A = [a_\theta]_\times$, $E = [\varepsilon]_\times$, differentiate the differential equation for $e^{t(A+hE)}$ with respect to h at $h = 0$. Variation of constants gives

$$\left. \frac{d}{dh} e^{A+hE} \right|_{h=0} = \int_0^1 e^{(1-s)A} E e^{sA} ds.$$

Multiplying on the left by e^{-A} and using $R[v]_\times R^\top = [Rv]_\times$,

$$e^{-A} \left. \frac{d}{dh} e^{A+hE} \right|_{h=0} = \left[\int_0^1 e^{-sA} \varepsilon ds \right]_\times = [J_r(a_\theta) \varepsilon]_\times.$$

The derivative of the logarithm at I is the identity, from $\log(I + hE) = hE + O(h^2)$. Thus $\delta\theta^+ = J_r(a_\theta) \varepsilon + O(\|\varepsilon\|^2)$. For the bias, $\delta b^+ = \delta b - a_b$, so its centred fluctuation does not rotate. This gives G_r , and covariance transforms by left and right multiplication. In particular, the leading reset term is $I - [a_\theta]_\times/2$, not $I + [a_\theta]_\times/2$ under this convention. $\square$

Proposition 35.4: One direction does not observe full attitude. *For a nonzero fixed direction v observed in the body, $y = R(q)^\top v + n$, the local angular-error measurement matrix has rank two. Two nonparallel directions remove this instantaneous angular ambiguity.*

Proof. Set $\hat{v} = R(\hat{q})^\top v$. Since $R(q) = R(\hat{q}) \text{Exp}(\delta\theta)$,

$$R(q)^\top v = (I - [\delta\theta]_\times) \hat{v} + O(\|\delta\theta\|^2) = \hat{v} + [\hat{v}]_\times \delta\theta + O(\|\delta\theta\|^2).$$

The matrix is $[\hat{v}]_\times$, whose kernel is the line spanned by $\hat{v}$: the cross product vanishes exactly for parallel vectors. Its rank is two. Stacking two such matrices makes the kernel the intersection of two lines, which is zero for nonparallel directions. This alone does not establish temporal observability of the bias. $\square$

35.6. Reproducible experiment and limits of the conclusion

The code uses true rotation $R(t) = \text{Exp}(t\omega)$ with $\omega = (0.2, -0.1, 0.4)$ rad/s and true bias $b = (0.01, -0.02, 0.005)$ rad/s. The step is $T = 0.02$ s and duration 12 s; a full attitude measurement arrives every five steps. Per-sample gyro noise has standard deviation 0.01 rad/s and angular measurement noise 0.02 rad. The filter allows a bias increment with standard deviation 10^{-5} rad/s per step, while the test trajectory has constant bias: this is a stated conservative model choice, not a hidden equality.

The initial nominal error is a rotation vector $(0.15, -0.10, 0.08)$ rad; the initial bias estimate is

zero. Initial variances are 0.2^2 for the three angular components and 0.05^2 for the three bias components. Checks cover unit norm, covariance symmetry and nonnegativity, final angular error below 0.05 rad, and comparison with uncorrected gyro integration.

The cross-check does not call the same function twice: one route uses an explicitly implemented quaternion product and analytic Jacobians; the other uses rotation matrices and Jacobians obtained by centred differences of the nonlinear map, including reset. The linear check is separate: a Kalman recursion is compared with batch Gaussian conditioning on all observations. Actual numerical outcomes, seed and versions are in the report and JSON. Figure F.1 in the expansion apparatus (at the end of the volume) shows, on a logarithmic scale, the convergence of the MEKF orientation error compared with gyroscope-only integration.

***Remark 35.1: What a simulation does not prove.** The MEKF neglects quadratic terms, uses a local logarithm branch and approximates the error distribution. Global optimality, convergence from every initialization and universal statistical consistency of its covariance have not been proved. A small final error for the stated seed is a reproducible experimental outcome, not any of these theorems. Agreement between two implementations is not an independent mathematical review of the entire chapter either.*

35.7. Assigned and worked exercises

Exercise 35.1: How much a noisy measurement weighs. For scalar $H = 1$, $P^- = p > 0$, $R = r > 0$, find the gain and variance limits as $r \rightarrow 0$ and $r \rightarrow \infty$.

Solution.

$$K = \frac{p}{p+r}, \quad P^+ = p - \frac{p^2}{p+r} = \frac{pr}{p+r}.$$

As $r \rightarrow 0$, $K \rightarrow 1$ and $P^+ \rightarrow 0$: a perfect measurement is trusted. As $r \rightarrow \infty$, $K \rightarrow 0$ and $P^+ \rightarrow p$: the measurement adds no useful information. These are limits of the correct model, not a reason to artificially declare $R = 0$.

Exercise 35.2: Bias drift. On a fixed axis, a noiseless gyro has constant bias β . If integrated without correction from the correct initial angle, what is the error after t ?

Solution. The integrated rates differ by β , so the unwrapped error is βt . For $\beta = 0.01$ rad/s and $t = 10$ s it is 0.1 rad. Constant bias uncertainty therefore creates growing angular error, not merely constant measurement variance. This comparison is local before wrapping modulo 2π .

Exercise 35.3: The reset sign. For a small correction $a_\theta = (0, 0, \alpha)$, compute the first-order reset on $\varepsilon = (1, 0, 0)^\top$.

Solution.

$$J_r(a_\theta)\varepsilon = \left(I - \frac{1}{2}[a_\theta]_\times\right)\varepsilon + O(\alpha^2) = (1, -\alpha/2, 0)^\top + O(\alpha^2).$$

The transverse component is negative. A finite-difference check of

$$\frac{\text{Log}(\text{Exp}(-a_\theta) \text{Exp}(a_\theta + h\varepsilon))}{h}$$

must converge to the exact Jacobian vector as $h \rightarrow 0$ with fixed α . The term $O(\alpha^2)$ instead concerns the approximation of the Jacobian itself.

35.8. Self-assessment quiz

1. The linear derivation requires Gaussianity to minimize over affine estimators.
2. For the optimal gain, updated error is uncorrelated with the innovation.
3. Joseph form is a sum of positive semidefinite matrices.
4. Under Gaussianity, the conditional mean is optimal even among nonlinear estimators.
5. Normalizing the quaternion makes covariance reset unnecessary.
6. One measured vector instantaneously observes all three rotations.
7. In the sampled model, gyro noise enters the angle with a factor proportional to T .
8. A successful simulation proves global convergence of the MEKF.

Teacher's key

1 False: second moments and uncorrelatedness suffice. 2 True: $P^-H^\top - KS = 0$. 3 True: each term has form $AMA^\top$ with $M \geq 0$. 4 True: the Gaussian residual is independent of the data. 5 False: norm and error coordinates are different issues. 6 False: the measured direction is the kernel. 7 True: Γ_g contains $-TJ_r$. 8 False: linearization is local and the test is finite.

35.9. Formulary and sources

$$K = P^-H^\top(HP^-H^\top + R)^{-1}, \quad \hat{x}^+ = \hat{x}^- + K(z - H\hat{x}^-).$$

$$\hat{q}^+ = \hat{q}^- \otimes \text{Exp}_q(a_\theta), \quad P^+ = \text{diag}(J_r(a_\theta), I)P_J \text{diag}(J_r(a_\theta), I)^\top.$$

For [4], volume, issue, pages and DOI are verified. For [6], the 2003 publication is cited; its copy lists only F. Landis Markley. The NASA preprint record and its scan disagree on the author list, a discrepancy recorded rather than resolved by guessing. Conventions and local calculations are also compared with the primary copy of [7], §§4.3.3, 5.3 and 6.3. Markley's quaternion conventions differ from ours: formulas are not transferred without rederiving the signs.

Manipulator dynamics: from energy to torque

INFO

Level C–D. Prerequisites: kinematics and Jacobians from Chapter 29, profiles from Chapter 30, and control from Chapter 34. In this chapter q denotes real joint coordinates, not a quaternion. Rigid bodies, ideal joints, positive inertias and uniform gravity are assumed. Newton’s laws and virtual work are physical-model assumptions; the resulting equations and identities are derived. Sources: [8, 9].

36.1. The Jacobian enters the energy

For body i , express angular velocity ω_i and translational velocity v_i in the same body-fixed centre-of-mass frame. Preserve Chapter 29’s order: angular first, linear second.

$$V_i = \begin{bmatrix} \omega_i \\ v_i \end{bmatrix} = J_i(q)\dot{q}, \quad \mathcal{I}_i = \begin{bmatrix} I_i & 0 \\ 0 & m_i I_3 \end{bmatrix}.$$

Here I_i is the inertia tensor about the centre of mass. The twist’s linear part must refer to that point; using a Jacobian at another origin without transport gives the wrong energy. Transport follows by differentiating the centre-of-mass position or using $v_C = v_O + \omega \times r_{OC}$ from rigid-body kinematics.

Proposition 36.1: Mass matrix and nondegeneracy.

$$T(q, \dot{q}) = \frac{1}{2} \sum_i V_i^\top \mathcal{I}_i V_i = \frac{1}{2} \dot{q}^\top M(q) \dot{q}, \quad M(q) = \sum_i J_i(q)^\top \mathcal{I}_i J_i(q).$$

If $m_i > 0$, $I_i > 0$, and the matrix obtained by stacking the J_i has trivial kernel, then $M = M^\top > 0$.

Proof. Translational energy is $m_i \|v_i\|^2/2$ and rotational energy is $\omega_i^\top I_i \omega_i/2$; summation gives the first equality. Substituting $V_i = J_i \dot{q}$ gives the second. Every term in M is symmetric. For $z \neq 0$,

$$z^\top M z = \sum_i (J_i z)^\top \mathcal{I}_i (J_i z) \geq 0.$$

Equality would force $J_i z = 0$ for every body, contradicting the stacked matrix’s trivial kernel. Thus the form is positive definite. The condition concerns all bodies, not only the tool Jacobian: a tool kinematic singularity does not imply a singular mass matrix. $\square$

36.2. Lagrange and energy structure

Theorem 36.1: Equations of motion in joint coordinates. Let $L = T - V$, $T = \dot{q}^\top M(q) \dot{q}/2$ and V be potential energy. The action principle with virtual work yields

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau, \quad g_i = \partial_i V,$$

$$C_{ij} = \sum_k c_{ijk} \dot{q}_k, \quad c_{ijk} = \frac{1}{2}(\partial_k M_{ij} + \partial_j M_{ik} - \partial_i M_{jk}).$$

For this Christoffel choice, $\dot{M} - 2C$ is skew-symmetric and

$$\frac{d}{dt}(T + V) = \dot{q}^\top \tau.$$

Proof. 1. Variation. Let η be a smooth variation vanishing at the endpoints. The physical action condition with virtual work is

$$0 = \int (\partial_q L \cdot \eta + \partial_{\dot{q}} L \cdot \dot{\eta} + \tau \cdot \eta) dt = \int \left(\partial_q L - \frac{d}{dt} \partial_{\dot{q}} L + \tau \right) \cdot \eta dt.$$

The last equality is integration by parts, with zero boundary term. Each continuous component in parentheses must vanish: if positive or negative at a point, it retains that sign nearby, and a nonnegative variation supported there produces a nonzero integral. Thus $\frac{d}{dt} \partial_{\dot{q}_i} L - \partial_i L = \tau_i$.

2. Expansion. By symmetry of M ,

$$\begin{aligned} \frac{d}{dt} \partial_{\dot{q}_i} L &= \sum_j M_{ij} \ddot{q}_j + \sum_{j,k} \partial_k M_{ij} \dot{q}_j \dot{q}_k, \\ \partial_i L &= \frac{1}{2} \sum_{j,k} \partial_i M_{jk} \dot{q}_j \dot{q}_k - \partial_i V. \end{aligned}$$

In the double product $\dot{q}_j \dot{q}_k$ we may interchange j and k and average. The resulting coefficient is precisely c_{ijk} , giving $M\ddot{q} + C\dot{q} + g = \tau$.

3. Skew-symmetry. Add the expressions with i, j interchanged; the noncommon terms cancel:

$$C_{ij} + C_{ji} = \sum_k \partial_k M_{ij} \dot{q}_k = \dot{M}_{ij}.$$

Hence $(\dot{M} - 2C)^\top = \dot{M} - 2C^\top = -(\dot{M} - 2C)$. For every vector v , $v^\top (\dot{M} - 2C)v = 0$, since transposing this scalar makes it its own negative.

4. Balance. Differentiate kinetic energy and substitute the equation of motion:

$$\begin{aligned} \dot{T} &= \dot{q}^\top M \ddot{q} + \frac{1}{2} \dot{q}^\top \dot{M} \dot{q} \\ &= \dot{q}^\top (\tau - g) + \frac{1}{2} \dot{q}^\top (\dot{M} - 2C) \dot{q} = \dot{q}^\top \tau - \dot{q}^\top g. \end{aligned}$$

Since $\dot{V} = \nabla V^\top \dot{q} = g^\top \dot{q}$, the conclusion follows. $\square$

The matrix C is not unique if only its product $C\dot{q}$ is prescribed. The matrix identity just proved refers to the explicit Christoffel choice, not every arbitrary matrix representing that product.

36.3. Planar two-link arm: every coefficient

The first angle q_1 is measured from the positive horizontal axis; q_2 is relative to the first link. Lengths are l_1, l_2 , centre-of-mass distances from the joints are c_1, c_2 , masses are m_1, m_2 , and

scalar centre-of-mass inertias are $I_1, I_2 > 0$. Gravity is $(0, -g_0)$, with $g_0 > 0$. Introduce

$$e(\theta) = (\cos \theta, \sin \theta)^\top, \quad t(\theta) = (-\sin \theta, \cos \theta)^\top, \quad q_{12} = q_1 + q_2.$$

$$p_1 = c_1 e(q_1), \quad p_2 = l_1 e(q_1) + c_2 e(q_{12}).$$

Theorem 36.2: Explicit dynamics of the 2R arm. Set

$$a = I_1 + I_2 + m_1 c_1^2 + m_2 (l_1^2 + c_2^2), \quad b = m_2 l_1 c_2, \quad d = I_2 + m_2 c_2^2, \quad h = b \sin q_2.$$

The motion matrices are

$$M = \begin{bmatrix} a + 2b \cos q_2 & d + b \cos q_2 \\ d + b \cos q_2 & d \end{bmatrix}, \quad (36.1)$$

$$C = \begin{bmatrix} -h\dot{q}_2 & -h(\dot{q}_1 + \dot{q}_2) \\ h\dot{q}_1 & 0 \end{bmatrix}, \quad (36.2)$$

$$g(q) = g_0 \begin{bmatrix} (m_1 c_1 + m_2 l_1) \cos q_1 + m_2 c_2 \cos q_{12} \\ m_2 c_2 \cos q_{12} \end{bmatrix}. \quad (36.3)$$

Proof. 1. Velocities. Differentiate the positions:

$$\dot{p}_1 = c_1 t(q_1) \dot{q}_1, \quad \dot{p}_2 = l_1 t(q_1) \dot{q}_1 + c_2 t(q_{12}) (\dot{q}_1 + \dot{q}_2).$$

Since $t(q_1)^\top t(q_{12}) = \cos q_2$,

$$\|\dot{p}_2\|^2 = l_1^2 \dot{q}_1^2 + c_2^2 (\dot{q}_1 + \dot{q}_2)^2 + 2l_1 c_2 \cos q_2 \dot{q}_1 (\dot{q}_1 + \dot{q}_2).$$

Angular velocities are $\dot{q}_1$ and $\dot{q}_1 + \dot{q}_2$. Adding the two translational and two rotational energies gives

$$2T = (a + 2b \cos q_2) \dot{q}_1^2 + 2(d + b \cos q_2) \dot{q}_1 \dot{q}_2 + d \dot{q}_2^2,$$

which identifies M .

2. Quadratic terms. The only nonzero derivatives of M are $\partial_2 M_{11} = -2h$ and $\partial_2 M_{12} = \partial_2 M_{21} = -h$. In the Christoffel formula:

$$c_{112} = c_{121} = -h, \quad c_{122} = -h, \quad c_{211} = h,$$

all other coefficients being zero. Thus $C_{11} = -h\dot{q}_2$, $C_{12} = -h\dot{q}_1 - h\dot{q}_2$, $C_{21} = h\dot{q}_1$, $C_{22} = 0$, as claimed.

3. Gravity. Potential energy is mass times g_0 times height:

$$V = g_0 [(m_1 c_1 + m_2 l_1) \sin q_1 + m_2 c_2 \sin q_{12}].$$

Its two partial derivatives give exactly (36.3). The cosine rather than sine results from measuring angles from the horizontal. $\square$

Corollary 36.1: Direct check of structural properties. For the 2R arm, $M > 0$ in every configuration and

$$\dot{M} - 2C = \begin{bmatrix} 0 & h(2\dot{q}_1 + \dot{q}_2) \\ -h(2\dot{q}_1 + \dot{q}_2) & 0 \end{bmatrix}.$$

Proof. The lower principal entry $M_{22} = d$ is positive. The determinant is

$$\begin{aligned}\det M &= d(a - d) - b^2 \cos^2 q_2 \\ &= d(I_1 + m_1 c_1^2) + I_2 m_2 l_1^2 + m_2^2 l_1^2 c_2^2 \sin^2 q_2 > 0.\end{aligned}$$

The principal-minor criterion for a symmetric 2×2 matrix gives $M > 0$. Equivalently, rotational energy contains $I_1 \dot{q}_1^2/2 + I_2(\dot{q}_1 + \dot{q}_2)^2/2$, vanishing only if both rates vanish. Also

$$\dot{M} = \begin{bmatrix} -2h\dot{q}_2 & -h\dot{q}_2 \\ -h\dot{q}_2 & 0 \end{bmatrix}.$$

Subtracting $2C$ gives the claimed matrix, verifying skew-symmetry without appealing to the general case. $\square$

36.4. Recursive Newton–Euler

For a serial chain of revolute joints, now express *all* vectors in fixed coordinates. Let o_i be joint i , z_i its unit axis, c_i the vector from that joint to body i 's centre of mass, and $r_i = o_i - o_{i-1}$. Fixed-frame inertia is $I_i^w = R_i I_i R_i^\top$. The base is stationary and there are no external terminal loads.

Proposition 36.2: The two recursions. *With initial values $\omega_0 = \alpha_0 = a_{o_0} = 0$, forward propagation is*

$$\begin{aligned}\omega_i &= \omega_{i-1} + z_i \dot{q}_i, \\ \alpha_i &= \alpha_{i-1} + z_i \ddot{q}_i + \omega_{i-1} \times (z_i \dot{q}_i), \\ a_{o_i} &= a_{o_{i-1}} + \alpha_{i-1} \times r_i + \omega_{i-1} \times (\omega_{i-1} \times r_i), \\ a_{C_i} &= a_{o_i} + \alpha_i \times c_i + \omega_i \times (\omega_i \times c_i), \\ f_i &= m_i(a_{C_i} - g_w), \quad n_i = I_i^w \alpha_i + \omega_i \times (I_i^w \omega_i).\end{aligned}$$

With $F_{n+1} = N_{n+1} = 0$, backward propagation is

$$F_i = f_i + F_{i+1}, \quad N_i = n_i + c_i \times f_i + N_{i+1} + (o_{i+1} - o_i) \times F_{i+1}, \quad \tau_i = z_i^\top N_i.$$

For $i = n$, the last lever arm is irrelevant because $F_{n+1} = 0$.

Proof. 1. Kinematics. Angular velocities add. Since the joint axis is fixed in the preceding body, $\dot{z}_i = \omega_{i-1} \times z_i$; differentiating the first equation gives the second. For a body-fixed vector r , $\dot{r} = \omega \times r$ and $\ddot{r} = \alpha \times r + \omega \times (\omega \times r)$. Apply this first to the next origin and then to the centre of mass to obtain both accelerations.

2. Own force and moment. Newton gives the required nongravitational force $f_i = m_i(a_{C_i} - g_w)$. Angular momentum about the centre of mass is $I_i^w \omega_i$. Since $\dot{I}_i^w = [\omega_i]_\times I_i^w - I_i^w [\omega_i]_\times$, its derivative is $I_i^w \alpha_i + \omega_i \times (I_i^w \omega_i)$; the term $I_i^w (\omega_i \times \omega_i)$ is zero.

3. Subtree balance. Joint i supplies body i 's force plus that required by all later bodies: $F_i = f_i + F_{i+1}$. Transporting moments to o_i adds $c_i \times f_i$ for its own body and $(o_{i+1} - o_i) \times F_{i+1}$ for the remainder. The moment component along the joint's free axis is motor torque τ_i ; the other components are ideal constraint reactions. This proves the recursion from force and moment balance. $\square$

36.4.1 Explicit equivalence of both routes for the 2R arm

Theorem 36.3: Newton–Euler returns exactly Lagrange. Applied to the 2R arm, the recursions above produce matrices (36.1)–(36.3), with identical signs and angles.

Proof. Write $e_1 = e(q_1)$, $e_{12} = e(q_{12})$, and similarly t_1, t_{12} . The planar cross product is the scalar $u \times v = u_x v_y - u_y v_x$. Differentiate positions twice:

$$\begin{aligned} a_1 &= c_1(t_1 \ddot{q}_1 - e_1 \dot{q}_1^2), \\ a_2 &= l_1(t_1 \ddot{q}_1 - e_1 \dot{q}_1^2) + c_2(t_{12}(\ddot{q}_1 + \ddot{q}_2) - e_{12}(\dot{q}_1 + \dot{q}_2)^2), \\ f_1 &= m_1(a_1 + (0, g_0)^\top), \quad f_2 = m_2(a_2 + (0, g_0)^\top). \end{aligned}$$

1. Distal joint. The required moment is

$$\tau_2 = I_2(\ddot{q}_1 + \ddot{q}_2) + c_2 e_{12} \times f_2.$$

Use $e_{12} \times t_1 = \cos q_2$, $e_{12} \times e_1 = -\sin q_2$, $e_{12} \times t_{12} = 1$ and $e_{12} \times e_{12} = 0$. Each term becomes

$$\begin{aligned} \tau_2 &= I_2(\ddot{q}_1 + \ddot{q}_2) + m_2 l_1 c_2 \cos q_2 \ddot{q}_1 + m_2 l_1 c_2 \sin q_2 \dot{q}_1^2 \\ &\quad + m_2 c_2^2 (\ddot{q}_1 + \ddot{q}_2) + m_2 g_0 c_2 \cos q_{12} \\ &= (d + b \cos q_2) \ddot{q}_1 + d \ddot{q}_2 + h \dot{q}_1^2 + g_2(q). \end{aligned}$$

This is the second row of Lagrange's equation.

2. Proximal joint. Transport the second joint's moment and force to the first:

$$\tau_1 = \tau_2 + I_1 \ddot{q}_1 + c_1 e_1 \times f_1 + l_1 e_1 \times f_2.$$

The two products are

$$\begin{aligned} c_1 e_1 \times f_1 &= m_1 c_1^2 \ddot{q}_1 + m_1 g_0 c_1 \cos q_1, \\ l_1 e_1 \times f_2 &= m_2 l_1^2 \ddot{q}_1 + b \cos q_2 (\ddot{q}_1 + \ddot{q}_2) \\ &\quad - h(\dot{q}_1 + \dot{q}_2)^2 + m_2 g_0 l_1 \cos q_1. \end{aligned}$$

Adding the distal result makes the acceleration coefficients $a + 2b \cos q_2$ and $d + b \cos q_2$. Velocity terms reduce to

$$h \dot{q}_1^2 - h(\dot{q}_1 + \dot{q}_2)^2 = -h(2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2).$$

Gravity terms sum to $g_1(q)$. Hence

$$\tau_1 = (a + 2b \cos q_2) \ddot{q}_1 + (d + b \cos q_2) \ddot{q}_2 - h(2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2) + g_1(q),$$

the required first row. The two derivations do not share their coefficient calculation: one differentiates energy, the other sums forces and moments. $\square$

36.5. Computed-torque control

Theorem 36.4: Exact error linearization. Let $q_d(t)$ be a twice continuously differentiable trajectory. With an exact model, available state, no saturation and $e = q - q_d$, choose

$$\tau = M(q)(\ddot{q}_d - K_d \dot{e} - K_p e) + C(q, \dot{q})\dot{q} + g(q).$$

Then $\ddot{e} + K_d \dot{e} + K_p e = 0$. If K_p, K_d are symmetric positive definite, the error converges exponentially to zero.

Proof. Substitute τ into the model: $C\dot{q} + g$ cancels and leaves $M(q)(\ddot{q} - \ddot{q}_d + K_d \dot{e} + K_p e) = 0$. Invertibility of M gives the error equation. To prove stability even when K_p and K_d do not commute, consider an eigenvalue λ of the error state matrix. There is $v \neq 0$ such that

$$(\lambda^2 I + \lambda K_d + K_p)v = 0, \quad \|v\|^2 \lambda^2 + (v^* K_d v) \lambda + v^* K_p v = 0.$$

The last quadratic has three positive real coefficients. Real roots have negative sum and positive product, so both are negative. Conjugate roots have real part $-(v^* K_d v)/(2\|v\|^2) < 0$. Thus all eigenvalues lie in the left half-plane. The exponential estimate for Hurwitz matrices from Chapter 34 completes the proof. $\square$

The proof does not authorize cancellation of unknown dynamics. Wrong inertia, friction, varying loads, estimation noise or saturation add terms to the error equation. The law is a synthesis and simulation model here, not a certified procedure for operating a real robot.

36.6. Numerical example with three comparisons

Choose SI units and illustrative values $l_1 = l_2 = 1$, $c_1 = c_2 = 1/2$, $m_1 = m_2 = 1$, $I_1 = I_2 = 1/12$, $g_0 = 9.81$. Then $a = 5/3$, $b = 1/2$, $d = 1/3$. At

$$q = (0, \pi/2)^\top, \quad \dot{q} = (0.4, -0.2)^\top, \quad \ddot{q} = (0.3, -0.1)^\top,$$

without approximating $\cos(\pi/2)$ in the exact calculation, we obtain

$$M = \begin{bmatrix} 5/3 & 1/3 \\ 1/3 & 1/3 \end{bmatrix}, \quad \det M = 4/9, \quad C\dot{q} = \begin{bmatrix} 0.06 \\ 0.08 \end{bmatrix}, \quad g = \begin{bmatrix} 14.715 \\ 0 \end{bmatrix}.$$

$$\tau = \begin{bmatrix} 15.241666 \dots \\ 0.146666 \dots \end{bmatrix} \text{ N m.}$$

The first check compares these torques with a direct Newton–Euler moment sum. The second uses symbolic identities for M , C , skew-symmetry and equality of the two dynamics. The third integrates a trajectory under computed-torque control.

For the trajectory, take $q_A = (0, 0)^\top$, $q_B = (0.8, -0.6)^\top$, motion duration $T_f = 3$ s, and

$$q_d(t) = q_A + (q_B - q_A)s(t/T_f), \quad s(\xi) = 10\xi^3 - 15\xi^4 + 6\xi^5 \quad (0 \leq \xi \leq 1).$$

Before and after motion the profile is constant. Choose $K_p = \text{diag}(25, 36)$, $K_d = \text{diag}(10, 12)$, $e(0) = (0.1, -0.08)^\top$, $\dot{e}(0) = 0$. Each error component has polynomial $(\lambda + \omega_i)^2$, with $\omega_1 = 5$, $\omega_2 = 6$, so

$$e_i(t) = e_i(0)(1 + \omega_i t)e^{-\omega_i t}.$$

Simulation on $[0, 6]$ s compares a DOP853 integrator using Lagrangian dynamics with a Radau integrator using independently reconstructed Newton–Euler forces and moments, then compares both with the analytic error. Actual discrepancies are in the JSON; the required comparison error is below 10^{-7} rad. Stability is proved above, not by a plot or numerical tolerance. Figure F.2 in the expansion apparatus (at the end of the volume) shows the two joint positions q_1, q_2 together with the quintic references $q_{d,1}, q_{d,2}$.

36.7. Assigned and worked exercises

Exercise 36.1: Derive the quintic profile. Find the polynomial of degree at most five with $s(0) = 0$, $s(1) = 1$ and zero velocity and acceleration at both endpoints.

Solution. The three conditions at zero eliminate the constant, linear and quadratic coefficients. Write $s = a\xi^3 + b\xi^4 + c\xi^5$. At one:

$$a + b + c = 1, \quad 3a + 4b + 5c = 0, \quad 6a + 12b + 20c = 0.$$

Subtracting three times the first equation from the second gives $b + 2c = -3$; subtracting six times the first from the third gives $6b + 14c = -6$. Eliminate b to obtain $2c = 12$, hence $c = 6$, $b = -15$, $a = 10$. Also $s' = 30\xi^2(1 - \xi)^2 \geq 0$, so motion does not reverse within the interval.

Exercise 36.2: Horizontal equilibrium. For the numerical parameters, calculate the torques needed to hold $q = (0, 0)^\top$ at rest.

Solution. With zero velocity and acceleration, $\tau = g(q)$. Both cosines equal one, so $\tau_1 = 9.81(1/2 + 1 + 1/2) = 19.62$ N m and $\tau_2 = 9.81/2 = 4.905$ N m. Newton–Euler confirms: at the first joint, weight lever arms are $1/2$ and $3/2$; at the second, the second weight has lever arm $1/2$.

Exercise 36.3: Gravity does not create energy. With $\tau = 0$ and no friction, is kinetic energy constant?

Solution. Not necessarily: $\dot{T} = -\dot{q}^\top g = -\dot{V}$. The sum $T + V$ is constant, not each term separately. If $g = 0$ as well, the energy identity gives $\dot{T} = 0$. The Coriolis matrix does not create or destroy energy: its contribution compensates that of $\dot{M}/2$.

36.8. Self-assessment quiz

1. In the angular-first convention, centre-of-mass inertia is $\text{diag}(I, mI_3)$.
2. Every tool-Jacobian singularity makes M singular.
3. The Christoffel choice satisfies $C + C^\top = \dot{M}$.
4. Angles measured from the horizontal produce cosines in g .
5. Newton–Euler first propagates accelerations and then forces backwards.
6. Computed torque guarantees the same error equation even with saturation.
7. For $\tau = 0$, $T + V$ is conserved in the ideal model.
8. With $K_p = \omega^2 I$, $K_d = 2\omega I$, an error with zero initial velocity is $e(0)(1 + \omega t)e^{-\omega t}$.

Teacher's key

1 True: the rotational block comes first. 2 False: the stacked Jacobian of all bodies matters. 3 True: this is the energy-proof calculation. 4 True: differentiation of a sine-containing potential

gives cosines. 5 True: these are the two recursions. 6 False: cancellation requires the torque actually applied. 7 True: external power is zero. 8 True: this is the critically damped solution with those initial data.

36.9. Formulary and sources

$$M = \sum_i J_i^\top \mathcal{I}_i J_i, \quad g = \nabla V, \quad M\ddot{q} + C\dot{q} + g = \tau, \quad \dot{T} + \dot{V} = \dot{q}^\top \tau.$$

Source [8] treats dynamics in Chapter 4; its 2R example and coefficients appear on printed pp. 164–165, while gravity and the full Newton–Euler comparison are added here. Source [9] covers dynamics in Chapter 8 and computed torque in §11.4. For the connection to kinematics, [8] attributes the product-of-exponentials formula to R. W. Brockett; the original 1984 reference is listed as historical attribution, distinct from sources actually consulted for the proofs.

Digital control: from the half-plane to the unit disk

INFO

Level C–D. Prerequisites: the Laplace transform (Chapter 20), the Z-transform (Chapter 21), and feedback and controllability from Chapter 34. Three operations are distinguished: exact sampling of a held-input plant, approximation of a continuous controller, and direct design of a discrete controller. They are not interchangeable. All equations describe ideal models without computation delays, saturation or quantization unless explicitly stated.

37.1. Zero-order hold and the exact model

Theorem 37.1: Exact state discretization. Let $T > 0$ be the period and $u(t) = u_k$ on $[kT, (k+1)T)$. For $\dot{x} = Ax + Bu$ and $x_k = x(kT)$,

$$x_{k+1} = A_d x_k + B_d u_k, \quad A_d = e^{AT}, \quad B_d = \int_0^T e^{A\tau} B d\tau.$$

The formula does not require invertibility of A . If A is invertible, $B_d = A^{-1}(A_d - I)B$.

Proof. Apply variation of constants on the sampling interval:

$$x((k+1)T) = e^{AT} x(kT) + \int_{kT}^{(k+1)T} e^{A((k+1)T-s)} B u_k ds.$$

The substitution $\tau = (k+1)T - s$ gives the formula. Integrating $\frac{d}{d\tau} e^{A\tau} = A e^{A\tau}$ yields $AB_d = (A_d - I)B$, hence the invertible case. A method avoiding this inverse is

$$\exp\left(T \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix}.$$

Indeed, for $j \geq 1$ the upper blocks of the j th power are A^j and $A^{j-1}B$. Summing the exponential series gives the upper-right block $\sum_{j \geq 1} T^j A^{j-1} B / j! = \int_0^T e^{A\tau} B d\tau$. $\square$

Proposition 37.1: The ZOH formula through the step response. For a causal strictly proper system with transfer function $G(s)$, zero initial state and step response $s_c(t) = \mathcal{L}^{-1}\{G(s)/s\}$, the transfer function from u_k to $y(kT)$ is

$$G_d(z) = (1 - z^{-1}) \sum_{k=0}^{\infty} s_c(kT) z^{-k}.$$

Proof. The discrete impulse $u_0 = 1$, $u_k = 0$ for $k \neq 0$ becomes the rectangle $1_{[0,T)}(t)$ after the hold: the difference of a step and that step delayed by T . Linearity and time invariance give the output $s_c(t) - s_c(t - T)$, with $s_c(t) = 0$ for $t < 0$. Sampling yields $h_k = s_c(kT) - s_c((k-1)T)$. Shifting the index in the second Z-transform sum gives the factor z^{-1} . Strict properness excludes instantaneous-feedthrough ambiguities at the rectangle boundary; a direct term D must be added separately. $\square$

Example 37.1: A first-order system, two derivations. For $\dot{x} = -ax + bu$, $y = x$, $a > 0$, the step response is $s_c(t) = b(1 - e^{-at})/a$. Set $\alpha = e^{-aT}$; the preceding formula gives

$$\begin{aligned} G_d(z) &= \frac{b}{a}(1 - z^{-1}) \left(\frac{1}{1 - z^{-1}} - \frac{1}{1 - \alpha z^{-1}} \right) \\ &= \frac{\beta z^{-1}}{1 - \alpha z^{-1}}, \quad \beta = \frac{b}{a}(1 - \alpha). \end{aligned}$$

Independently, state integration gives $x_{k+1} = \alpha x_k + \beta u_k$ and therefore the same transfer function. The factor z^{-1} records that u_k affects x_{k+1} , not x_k . As $a \rightarrow 0$, $(1 - e^{-aT})/a \rightarrow T$, recovering the integrator with $\beta = bT$.

37.2. Tustin: a Möbius transformation

Theorem 37.2: From trapezoids to the stability disk. The trapezoidal rule applied to $\dot{v} = u$ produces the substitution

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}, \quad z = \frac{1 + sT/2}{1 - sT/2}.$$

It maps $\operatorname{Re} s < 0$ bijectively onto $|z| < 1$. On the imaginary axis, continuous angular frequency Ω and digital frequency $\omega \in (-\pi, \pi)$ satisfy

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}, \quad \omega = 2 \arctan \frac{\Omega T}{2}.$$

Proof. The trapezoidal rule gives $v_k - v_{k-1} = T(u_k + u_{k-1})/2$. With zero initial conditions, transformation yields $V/U = (T/2)(1 + z^{-1})/(1 - z^{-1})$, to be compared with $1/s$. Solving for z gives the second formula. Moreover,

$$|1 + sT/2|^2 - |1 - sT/2|^2 = 2T \operatorname{Re} s.$$

The denominator does not vanish in the left half-plane, so $|z| < 1$ is equivalent to $\operatorname{Re} s < 0$. The inverse exists in the disk because $z + 1 \neq 0$. Finally, set $z = e^{i\omega}$ and multiply numerator and denominator of $(1 - e^{-i\omega})/(1 + e^{-i\omega})$ by $e^{i\omega/2}$ to obtain $i \tan(\omega/2)$. The tangent increases strictly from $-\infty$ to $+\infty$ on $(-\pi/2, \pi/2)$, giving the full correspondence. $\square$

Stability is preserved, but the linear frequency scale is not. With $T = 0.1$ s and $\Omega = 10$ rad/s, $\omega = 2 \arctan(1/2) \simeq 0.927295218$ rad/sample: the digital physical frequency ω/T is about 9.27295218 rad/s, not 10. To match a chosen physical frequency Ω_c , with $0 < \Omega_c T < \pi$, use $s = \kappa(1 - z^{-1})/(1 + z^{-1})$ with

$$\kappa = \frac{\Omega_c}{\tan(\Omega_c T/2)}.$$

Substitution of $z = e^{i\Omega_c T}$ gives exactly $s = i\Omega_c$: this is frequency prewarping. It does not simultaneously correct every frequency.

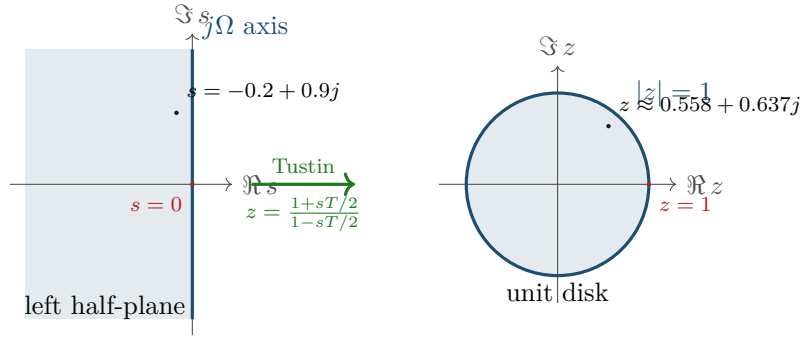


Figure 37.1: The Tustin transformation maps the left half-plane of the s -plane (shaded) to the unit disk of the z -plane (shaded), and the imaginary axis $j\Omega$ to the circle $|z| = 1$: exactly Theorem 37.2. Verified point with $T = 1$: $s = -0.2 + 0.9j$ gives $1 + sT/2 = 0.9 + 0.45j$, $1 - sT/2 = 1.1 - 0.45j$, hence $z = (0.9 + 0.45j)/(1.1 - 0.45j) = 0.5575 + 0.6372j$, with $|z| \approx 0.847 < 1$; independent check via the identity from the proof: $|1 + sT/2|^2 - |1 - sT/2|^2 = 1.0125 - 1.4125 = -0.4 = 2T \operatorname{Re} s$.

37.3. Pole placement and deadbeat control

Theorem 37.3: Finite-step regulation. If (A_d, B_d) is controllable, one can choose K with all eigenvalues of $F = A_d - B_d K$ zero; then $x_n = 0$ for every x_0 under $u_k = -Kx_k$. In the single-input case, the nilpotency index of F is exactly n : among these constant state feedbacks, n is the minimum settling time uniform over all initial states.

Proof. Pole placement in Chapter 34 is algebraic in $A - BK$, so applies unchanged in discrete time. Choosing characteristic polynomial λ^n , Cayley–Hamilton gives $F^n = 0$. Induction gives $x_k = F^k x_0$, hence $x_n = 0$. For SISO minimality, observe that $F^j B_d$ equals $A_d^j B_d$ plus a combination of $B_d, \dots, A_d^{j-1} B_d$: this follows inductively by multiplying by $A_d - B_d K$. The two controllability matrices therefore differ by a triangular matrix with unit diagonal and have the same rank n . If $F^m = 0$ for $m < n$, all vectors $F^j B_d$ with $j \geq m$ would vanish, giving rank at most m , a contradiction. Finally, $F^{n-1} \neq 0$ implies the existence of x_0 with $F^{n-1} x_0 \neq 0$. $\square$

This does not say that every state needs n steps, or that n is minimal for multi-input systems. With $B_d = I$, choosing $K = A_d$ gives one-step settling. Constant-reference tracking first requires an admissible equilibrium $(x_\star, u_\star)$ satisfying $x_\star = A_d x_\star + B_d u_\star$; use $u_k = u_\star - K(x_k - x_\star)$. Pole placement cannot make an output reference compatible with an equilibrium when it is not.

Example 37.2: Double integrator: a complete symbolic design. For $\ddot{p} = u$, the ZOH-sampled state $x = (p, \dot{p})^\top$ satisfies

$$A_d = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}, \quad \det[B_d \ A_d B_d] = -T^3 \neq 0.$$

Writing $K = (k_1, k_2)$, the closed-loop trace and determinant are

$$\operatorname{tr} F = 2 - \frac{T^2}{2} k_1 - T k_2, \quad \det F = 1 - T k_2 + \frac{T^2}{2} k_1.$$

Set the trace to $\rho_1 + \rho_2$ and determinant to $\rho_1 \rho_2$. Subtracting the equations gives $T^2 k_1 = (1 - \rho_1)(1 - \rho_2)$; substitution into the second gives

$$k_1 = \frac{(1 - \rho_1)(1 - \rho_2)}{T^2}, \quad k_2 = \frac{3 - \rho_1 - \rho_2 - \rho_1 \rho_2}{2T}.$$

For $\rho_1 = \rho_2 = 0$, $K = (T^{-2}, 3/(2T))$ and

$$F = \begin{bmatrix} 1/2 & T/4 \\ -1/T & -1/2 \end{bmatrix}, \quad F^2 = \begin{bmatrix} 1/4 - 1/4 & T/8 - T/8 \\ -1/(2T) + 1/(2T) & -1/4 + 1/4 \end{bmatrix} = 0.$$

This is the matrix to use. With $T = 0.1$ and $x_0 = (1, 0)^\top$, $u_0 = -100$, $x_1 = (1/2, -10)^\top$, $u_1 = 100$, $x_2 = 0$. Large accelerations show that algebraic deadbeat regulation is not torque-constrained or comfort-constrained design; reducing T makes the position gain grow as T^{-2} .

37.4. Finite-horizon discrete LQR

Theorem 37.4: Riccati by completing the square. For $x_{k+1} = Ax_k + Bu_k$, $Q, S \succeq 0$, $R \succ 0$, consider

$$J = \sum_{k=0}^{N-1} (x_k^\top Q x_k + u_k^\top R u_k) + x_N^\top S x_N.$$

Set $P_N = S$. The recursion

$$\begin{aligned} U_k &= R + B^\top P_{k+1} B, & K_k &= U_k^{-1} B^\top P_{k+1} A, \\ P_k &= Q + A^\top P_{k+1} A - A^\top P_{k+1} B U_k^{-1} B^\top P_{k+1} A \end{aligned}$$

is well defined, gives $P_k \succeq 0$, and produces the unique optimal control $u_k = -K_k x_k$, with cost $x_0^\top P_0 x_0$.

Proof. Work backwards. The terminal cost has the stated form. If the minimum cost from step $k+1$ is $x^\top P_{k+1} x$, the cost at step k for input u is

$$\begin{aligned} & x^\top Q x + u^\top R u + (Ax + Bu)^\top P_{k+1} (Ax + Bu) \\ &= x^\top (Q + A^\top P_{k+1} A) x + 2u^\top B^\top P_{k+1} A x + u^\top U_k u \\ &= x^\top P_k x + (u + K_k x)^\top U_k (u + K_k x). \end{aligned}$$

Since $P_{k+1} \succeq 0$, $U_k \succ 0$. The square has unique minimum zero at $u = -K_k x$, and the minimum cost is nonnegative because every original term is nonnegative, hence $P_k \succeq 0$. Induction justifies every step. Summation and cancellation of intermediate state costs also give

$$J = x_0^\top P_0 x_0 + \sum_{k=0}^{N-1} (u_k + K_k x_k)^\top U_k (u_k + K_k x_k).$$

The identity directly establishes optimality and uniqueness of the whole sequence. No external dynamic-programming theorem is needed. $\square$

37.5. Digital PID: comparing three discretizations

An ideal differentiator amplifies high frequencies without bound. We therefore use the proper model

$$C(s) = K_p + \frac{K_i}{s} + K_d \frac{s}{1 + \tau_f s}, \quad \tau_f > 0.$$

Let I be the integral before multiplication by K_i , and d the filtered derivative before multiplication by K_d ; then $u_k = K_p e_k + K_i I_k + K_d d_k$.

Proposition 37.2: Explicit recurrences. The three substitutions and corresponding recurrences are

$$\begin{aligned} s_{\text{FE}} &= (z - 1)/T, & I_k^{\text{FE}} &= I_{k-1}^{\text{FE}} + Te_{k-1}, \\ s_{\text{BE}} &= (1 - z^{-1})/T, & I_k^{\text{BE}} &= I_{k-1}^{\text{BE}} + Te_k, \\ s_{\text{TU}} &= \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}, & I_k^{\text{TU}} &= I_{k-1}^{\text{TU}} + \frac{T}{2}(e_k + e_{k-1}), \end{aligned}$$

$$\begin{aligned} d_k^{\text{FE}} &= \left(1 - \frac{T}{\tau_f}\right) d_{k-1}^{\text{FE}} + \frac{e_k - e_{k-1}}{\tau_f}, \\ d_k^{\text{BE}} &= \frac{\tau_f}{\tau_f + T} d_{k-1}^{\text{BE}} + \frac{e_k - e_{k-1}}{\tau_f + T}, \\ d_k^{\text{TU}} &= \frac{2\tau_f - T}{2\tau_f + T} d_{k-1}^{\text{TU}} + \frac{2(e_k - e_{k-1})}{2\tau_f + T}. \end{aligned}$$

Proof. For the integrator, left-rectangle, right-rectangle and trapezoidal quadrature of $\int_{(k-1)T}^{kT} e(t)dt$ directly give the three rows. For the differentiator, write $D/E = s/(1 + \tau_f s)$ and substitute. In the FE case,

$$\frac{D}{E} = \frac{1 - z^{-1}}{\tau_f + (T - \tau_f)z^{-1}}.$$

Multiplication by the denominator and interpretation of z^{-1} as a delay give the first recurrence. For BE the denominator is $T + \tau_f - \tau_f z^{-1}$ with numerator $1 - z^{-1}$; for TU it is $T + 2\tau_f + (T - 2\tau_f)z^{-1}$ with numerator $2(1 - z^{-1})$. Isolating d_k gives the other two formulas. $\square$

Corollary 37.1: Stability of the derivative filter alone. The FE filter is asymptotically stable if and only if $0 < T < 2\tau_f$; BE and TU are asymptotically stable for every $T > 0$.

Proof. The free responses are powers of the three coefficients multiplying d_{k-1} . Stability requires modulus strictly less than one. For FE, $|1 - T/\tau_f| < 1$ is equivalent to the stated inequality; for BE, $0 < \tau_f/(\tau_f + T) < 1$; for TU, $|2\tau_f - T| < 2\tau_f + T$. $\square$

This does not imply stability of the whole feedback loop: the PID contains an integrator with a pole at $z = 1$, and the plant enters the characteristic equation. One must not simply set $\tau_f = 0$ in every recurrence: the ideal FE differentiator is anticipatory, while TU introduces a pole at -1 . A real controller also needs saturation handling and anti-windup, which this linear derivation does not certify.

37.5.1 Reproducible numerical comparison

The comparison uses $K_p = 2$, $K_i = 1$, $K_d = 0.1$, $\tau_f = 0.05$ s, $T = 0.01$ s, $\Omega = 10$ rad/s and $z = e^{i\Omega T}$. The first method evaluates $C(s_{\text{FE}})$, $C(s_{\text{BE}})$, $C(s_{\text{TU}})$ as complex numbers. The second runs the recurrences on $e_k = \cos(\Omega kT)$ and estimates amplitude and phase by regression on cosine, sine and a constant after the transient. The constant is needed because the integrator retains its initial offset. The JSON reports magnitudes, phases and discrepancies, with comparison tolerance 10^{-8} . The continuous reference is $C(10i)$, not $C(i\omega)$: ω is in radians per sample, Ω in radians per second.

37.6. Assigned and worked exercises

Exercise 37.1: Nonzero poles. For the double integrator with $T = 0.1$, place the poles at $1/2$ and $3/5$.

Solution. The formula gives $k_1 = (1/2)(2/5)/(1/100) = 20$ and $k_2 = (3 - 1/2 - 3/5 - 3/10)/(1/5) = 8$. The closed-loop matrix is

$$F = \begin{bmatrix} 9/10 & 3/50 \\ -2 & 1/5 \end{bmatrix}, \quad \det(\lambda I - F) = \lambda^2 - \frac{11}{10}\lambda + \frac{3}{10} = (\lambda - 1/2)(\lambda - 3/5).$$

Both poles lie inside the unit circle; uniform finite-step settling does not occur because F is invertible.

Exercise 37.2: One Riccati step. For $x_1 = x_0 + u_0$, $N = 1$, $Q = R = S = 1$, find the optimal control.

Solution. $U_0 = 2$, $K_0 = 1/2$, $P_0 = 1 + 1 - 1/2 = 3/2$. Directly, $J = x_0^2 + u_0^2 + (x_0 + u_0)^2 = (3/2)x_0^2 + 2(u_0 + x_0/2)^2$, with unique minimum at $u_0 = -x_0/2$.

Exercise 37.3: Sampling too slowly. With $\tau_f = 0.02$ s and $T = 0.05$ s, compare the three derivative-filter poles.

Solution. FE: $1 - T/\tau_f = -3/2$, unstable. BE: $\tau_f/(\tau_f + T) = 2/7$, stable. TU: $(2\tau_f - T)/(2\tau_f + T) = -1/9$, stable with alternating free response. Stability of two filters does not guarantee that the sampling period is adequate for the desired loop bandwidth.

37.7. Self-assessment quiz

1. ZOH requires invertibility of A .
2. Tustin maps the left half-plane into the unit disk.
3. Under Tustin, digital frequency always equals $\omega = \Omega T$.
4. Controllable SISO deadbeat feedback has nilpotency index n .
5. Finite discrete LQR requires an imported solution of an algebraic Riccati equation.
6. The FE derivative filter is stable for every $T > 0$.
7. The factor z^{-1} can represent the step from u_k to x_{k+1} .
8. Two-step settling permits ignoring actuator limits.

Teacher's key

1 False: the exponential integral suffices. 2 True: the difference of squared moduli proves it. 3 False: a tangent is involved. 4 True: controllability excludes a smaller index. 5 False: finite recursion and completing the square suffice. 6 False: $T < 2\tau_f$ is required. 7 True: this follows from the stated timing convention. 8 False: the required torques may be unrealizable.

37.8. Formulary and sources

$$(A_d, B_d) = \left(e^{AT}, \int_0^T e^{A\tau} B d\tau \right), \quad z = \frac{1 + sT/2}{1 - sT/2}, \quad u_k = -K_k x_k.$$

For PID context and modifications required for implementation, see [12], §10.5 of the stated electronic version. For discrete LQR, see [11], PDF pp. 20–23. The recurrences, the deadbeat minimality proof within the stated scope, and the two-way checks are developed explicitly here.

Final assessment of the Volume

INFO

How to use this assessment. Fifty multiple-choice questions spanning the whole volume, from the circle as a generator (Chapter 1) to the epilogue (Chapter 33), along the six Parts: differential and global geometry, complex and harmonic analysis on the disc, transforms, signals and telecommunications, automation, numerics and research. As in the end-of-chapter quizzes, but here at the scale of the whole volume, the **answer is given immediately after each question**, together with a pointer to the chapter and the result that justify it: it is designed for revision, not for grading. Anyone wishing to use it as a closed test should cover the boldface line with a sheet of paper.

Indicative scoring: ≥ 45 excellent; 38–44 good; 30–37 sufficient, with targeted revision of the Parts missed; < 30 it is worth rereading the volume following the logical chain that leads from Chapter 1 to Chapter 33.

Part A — Differential and global geometry (Chapters 1–14)

- Chapter 1 recounts that Lindemann proved the transcendence of π in 1882. In Volume III this result
 - is proved in full in the prologue
 - is cited as an imported historical fact, not proved in the treatise
 - is refuted by Archimedes' estimates
 - is equivalent to the four-vertex theorem

Answer: B. Chapter 1 cites it as a historical fact (Lindemann, 1882): consistently with the rule that great theorems outside the perimeter of the circle remain panorama, it is not proved in the volume.

- The fundamental theorem of plane curves (Chapter 2) guarantees, starting from a continuous curvature function k , an initial point and an initial direction,
 - the existence of infinitely many non-congruent curves with that curvature
 - the existence and uniqueness of a curve with that curvature, unique up to a direct rigid motion
 - existence only, without uniqueness
 - existence only if k is constant

Answer: B. This is the content of Theorem 2.9: existence, uniqueness and rigidity (Chapter 2).

- The Frenet–Serret formulas (Chapter 3) express the derivatives of T, N, B along a biregular curve through a matrix that is
 - symmetric
 - antisymmetric, with curvature κ and torsion τ

- C) diagonal
- D) zero whenever $\tau = 0$

Answer: B. Theorem 3.1 (Chapter 3) writes $\frac{d}{ds}(T, N, B)^\top$ as the product with the antisymmetric matrix carrying $\pm\kappa, \pm\tau$; even with $\tau = 0$ one still has $T' = \kappa N \neq 0$ and $N' = -\kappa T$.

4. The four-vertex theorem, in its general version (Chapter 4), applies to

- A) convex curves only
- B) every closed, simple, regular plane curve of class C^2 , convex or not
- C) ellipses only
- D) curves of constant curvature only

Answer: B. Theorem 4.7 (Chapter 4) extends the convex case to every closed, simple, regular C^2 curve, relying only on the Jordan curve theorem as an imported topological prerequisite.

5. For a closed, regular, simple plane curve, the rotation index theorem (Umlaufsatz, Chapter 4) states that the total curvature integral $\int_0^L k(s) ds$ equals

- A) π
- B) 0
- C) 2π in absolute value, with sign given by the orientation
- D) a value depending on the shape of the curve

Answer: C. Theorem 4.1 (Chapter 4): $\text{ind}(\gamma) = \pm 1$, hence $\int_0^L k ds = 2\pi \text{ind}(\gamma)$.

6. The global Gauss–Bonnet theorem (Chapter 4), for a compact, orientable, boundaryless surface of Gaussian curvature K , states that $\iint_M K dA$ equals

- A) always 4π
- B) $2\pi \chi(M)$, where $\chi(M)$ is the Euler characteristic of a triangulation
- C) always 0
- D) the area of M

Answer: B. Theorem 4.8 (Chapter 4). The combinatorial step $2\pi V - \pi F = 2\pi \chi$ is proved in full in the volume; three supporting facts for general surfaces remain imported (Observation 4.9).

7. By the classification of $O(2)$ (Chapter 5), every 2×2 orthogonal matrix is

- A) always a rotation
- B) either a rotation R_θ (determinant $+1$) or a reflection S_θ (determinant -1)
- C) always a reflection
- D) neither a rotation nor a reflection in general

Answer: B. Theorem 5.4 (Chapter 5): $O(2) = SO(2) \cup \{S_\theta\}$.

8. Euler's rotation theorem (Chapter 5) guarantees that every $A \in SO(3)$

- A) has no fixed axis

- B) fixes a unit vector u (the axis) and rotates the plane $u^\perp$ by an angle θ
- C) is always the identity
- D) has three distinct fixed axes

Answer: B. Theorem 5.9 (Chapter 5): every rotation of space is a rotation about an axis.

9. The theorem of Chapter 6 on Möbius transformations states that the image of a generalised circle (a circle or a line) under a Möbius transformation is
- A) always a line
 - B) always a circle, never a line
 - C) always a generalised circle (a circle or a line)
 - D) in general neither a circle nor a line

Answer: C. Theorem 6.6 (Chapter 6), proved via decomposition into translations, similarities and inversion.

10. Stereographic projection σ from the sphere (Chapter 7) has the properties:
- A) it is conformal but does not map circles to circles
 - B) it maps circles to circles or lines, but is not conformal
 - C) it is conformal, and maps circles through the pole to lines, other circles to circles
 - D) it is neither conformal nor circle-preserving

Answer: C. Theorem 7.6 (Chapter 7).

11. The spherical law of cosines (Chapter 8), for a spherical triangle with sides a, b, c and angle A opposite a , states that
- A) $\cos a = \cos b \cos c + \sin b \sin c \cos A$
 - B) $a^2 = b^2 + c^2 - 2bc \cos A$, as in the plane
 - C) $\cos a = \cos b + \cos c$
 - D) $\sin a = \sin b \sin c \cos A$

Answer: A. Theorem 8.1 (Chapter 8). Answer B is the *planar* law of cosines, not valid on the sphere: it is the most common error in transferring plane trigonometry to the sphere.

12. Girard's theorem (Chapter 8) relates the area of a spherical triangle to its excess $E = A + B + C - \pi$. In the chapter's Rome–New York example, with angles $A \approx 59.00^\circ$, $B \approx 57.33^\circ$, $C \approx 86.49^\circ$, the excess is approximately
- A) 0° (the triangle would be flat)
 - B) 22.82° , a value confirmed independently with L'Huilier's formula
 - C) 180°
 - D) 360°

Answer: B. Theorem 8.6 (Chapter 8): $\text{Area}(T) = Er^2$, with E in radians in the area formula and reported here in degrees for readability.

13. Chapter 9 proves that the geodesics of the sphere S^2 are

- A) the curves of zero curvature, which do not exist on the sphere
- B) the arcs of great circles, traversed at constant speed
- C) the parallels of any colatitude
- D) logarithmic spirals

Answer: B. Theorem 9.3 (Chapter 9). Answer C is false in general: among the parallels, only the equator is a great circle and hence a geodesic.

14. Clairaut's relation (Chapter 9), for a geodesic on a surface of revolution, states that along the path the following stays constant:
- A) the height
 - B) the product $r \sin \psi$ of the radius of the parallel and the sine of the angle with the meridian
 - C) the length travelled so far
 - D) the Gaussian curvature

Answer: B. Theorem 9.4 (Chapter 9).

15. Gauss's Theorema Egregium (Chapter 10) states that the Gaussian curvature K
- A) depends on how the surface is embedded in $\mathbb{R}^3$
 - B) is intrinsic: it is computed from the coefficients E, F, G alone and is preserved under local isometries
 - C) is always zero for curved surfaces
 - D) always coincides with the mean curvature H

Answer: B. Theorem 10.2 (Chapter 10). Answer D is false: the plane and the cylinder are locally isometric yet have different mean curvatures, because H is not intrinsic.

16. For the volume of the n -dimensional ball (Chapter 11), $V_n = \pi^{n/2} / \Gamma(\frac{n}{2} + 1)$. Using the recursion $V_n = \frac{2\pi}{n} V_{n-2}$ with $V_1 = 2$, $V_2 = \pi$, the value of V_5 is
- A) π^2
 - B) $\frac{8\pi^2}{15} \approx 5.26$
 - C) $\frac{4}{3}\pi$
 - D) $2\pi^2$

Answer: B. Theorem 11.1 (Chapter 11): $V_3 = \frac{4}{3}\pi$, $V_5 = \frac{2\pi}{5} V_3 = \frac{8\pi^2}{15}$.

17. The map $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it}$ (Chapter 12), is a covering map because
- A) it is injective
 - B) every point of S^1 has a neighbourhood whose preimage is a disjoint union of copies homeomorphic to intervals of $\mathbb{R}$, translated by $2\pi\mathbb{Z}$
 - C) it is surjective but not continuous
 - D) $\mathbb{R}$ is compact

Answer: B. Theorem 12.4 (Chapter 12); E is not injective (the fibre above every point is $t_0 + 2\pi\mathbb{Z}$), and it is precisely this that makes it a covering map rather than a homeomorphism.

18. The winding number $w(\gamma)$ of a closed path $\gamma : I \rightarrow S^1$ (Chapter 12)

- A) can be any real number
- B) is always an integer, invariant under homotopy, and equals n for $\omega_n(s) = e^{2\pi i n s}$
- C) depends on the lift chosen
- D) is defined only if γ is injective

Answer: B. Theorem 12.5 (Chapter 12).

19. Chapter 12 proves that the fundamental group $\pi_1(S^1, 1)$ is isomorphic to

- A) $\mathbb{Z}/2\mathbb{Z}$
- B) $\mathbb{R}$
- C) $\mathbb{Z}$, infinite cyclic, generated by $[\omega_1]$
- D) the trivial group, because S^1 would be simply connected

Answer: C. Theorem 12.7 (Chapter 12). Answer D is a widespread error: S^1 is not simply connected, and it is exactly $\pi_1(S^1) \cong \mathbb{Z} \neq \{1\}$ that proves it.

20. The orthogonality relations for the characters $\chi_n(z) = z^n$ of S^1 (Chapter 13) state that $\int_{S^1} \chi_n \overline{\chi_m} dm$ equals

- A) always 1
- B) δ_{nm} : 1 if $n = m$, 0 otherwise
- C) $n - m$
- D) 2π for every n, m

Answer: B. Theorem 13.6 (Chapter 13).

21. The kissing number of the plane (Chapter 14), that is, the maximum number of circles of radius r externally tangent to a fixed circle of the same radius without overlapping, equals

- A) 4
- B) 5
- C) 6, as in the hexagonal packing with centres at angular distance $\pi/3$
- D) 12

Answer: C. Theorem 14.3 (Chapter 14). Answer D is the kissing number of three-dimensional *space*, not of the plane.

Part B — Complex and harmonic analysis on the disc (Chapters 15–18)

22. The Schwarz lemma (Chapter 15) applies to a holomorphic function $f : \mathbb{D} \rightarrow \mathbb{D}$ under the additional hypothesis

- A) f bijective

- B) $f(0) = 0$
- C) f even
- D) $|f'(0)| = 0$

Answer: B. Lemma 15.9 (Chapter 15): under $f(0) = 0$ one has $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality in one non-trivial case if and only if $f(z) = \lambda z$ for some λ of modulus 1.

23. The maximum modulus principle (Chapter 15) states that if $|f|$ has a local maximum at a point of a connected open set on which f is holomorphic, then
- A) f has a zero at that point
 - B) f is constant on the whole connected domain
 - C) f is bounded, but not necessarily constant
 - D) f has a pole at that point

Answer: B. Theorem 15.11 (Chapter 15).

24. For continuous $f : \mathbb{T} \rightarrow \mathbb{R}$, the Dirichlet problem on the disc (Chapter 16) admits
- A) infinitely many harmonic solutions with boundary value f
 - B) no solution in general
 - C) exactly one harmonic solution, continuous up to the boundary, given by the Poisson formula
 - D) a solution only if f is constant

Answer: C. Theorem 16.6 (Chapter 16).

25. The Dirichlet–Jordan theorem (Chapter 17) guarantees that, at a point with finite one-sided limits and derivatives, the Fourier partial sum converges to
- A) always $f(\theta_0)$, even at jumps
 - B) the average of the one-sided limits $\frac{f(\theta_0^+) + f(\theta_0^-)}{2}$
 - C) 0 at jump points
 - D) it never converges at jump points

Answer: B. Theorem 17.4 (Chapter 17).

26. The Gibbs phenomenon (Chapter 17), for the square wave, produces at the first maximum an overshoot that, as N grows, tends to
- A) 0: adding more terms makes the overshoot disappear
 - B) about 17.9% of the half-amplitude of the jump, and it does NOT tend to zero
 - C) exactly 50%
 - D) a value depending only on the fundamental frequency

Answer: B. Theorem 17.9 (Chapter 17): the limit is $\frac{2}{\pi} \text{Si}(\pi) = 1.178979744\dots$, that is, a persistent overshoot of 17.898% of the half-amplitude. Answer A is the most common misconception about this phenomenon.

27. In Chapter 18 the first positive zero $j_{0,1}$ of the Bessel function J_0 is enclosed, with a sign change verified in exact rational arithmetic, in the interval

- A) (1.0, 1.5)
- B) (2.40, 2.41)
- C) (3.0, 3.5)
- D) (0.5, 1.0)

Answer: B. Proposition 18.13 (Chapter 18).

Part C — Transforms (Chapters 19–23)

28. The convolution theorem for the continuous Fourier transform (Chapter 19), proved in full in the chapter, requires as a hypothesis that

- A) f, g both be real
- B) $f, g \in L^1(\mathbb{R})$
- C) f, g be periodic
- D) $f = g$

Answer: B. Theorem 19.4 (Chapter 19): under that hypothesis $f * g \in L^1(\mathbb{R})$ and $\widehat{f * g} = \hat{f} \hat{g}$.

29. The initial value theorem for the Laplace transform (Chapter 20) states that, under suitable regularity hypotheses,

- A) $\lim_{s \rightarrow 0} s F(s) = f(0^+)$
- B) $\lim_{\operatorname{Re}(s) \rightarrow +\infty} s F(s) = f(0^+)$
- C) $\lim_{s \rightarrow \infty} F(s) = f(0^+)$
- D) $s F(s) = f(0^+)$ for every s

Answer: B. Theorem 20.12 (Chapter 20); not to be confused with the final value theorem, which instead uses $s \rightarrow 0^+$.

30. For a causal discrete LTI system with transfer function $H(z)$, Chapter 21 proves that BIBO stability is equivalent to having

- A) all poles with $|z| > 1$
- B) all poles with $|z| < 1$, strictly inside the unit circle
- C) poles on the unit circle
- D) no poles

Answer: B. Theorem 21.12 (Chapter 21); a pole *on* the unit circle gives only marginal stability.

31. The discrete exponential vectors $e_k = (\omega_N^{-kn})_{n=0}^{N-1}$ (Chapter 22) satisfy $\langle e_k, e_\ell \rangle$

- A) $= 1$ always
- B) $= N \delta_{k\ell}$: they are orthogonal and have norm $\sqrt{N}$
- C) $= 0$ always, even for $k = \ell$
- D) depends on the sign of $k - \ell$

Answer: B. Theorem 22.1 (Chapter 22).

32. The DFT inversion formula (Chapter 22) reconstructs $x[n]$ from $X[k]$ as

A) $x[n] = \sum_k X[k] \omega_N^{-kn}$, without the factor $1/N$

B) $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \omega_N^{-kn}$

C) $x[n] = \frac{1}{N} \sum_k X[k]$

D) $x[n] = X[n]$

Answer: B. Theorem 22.2 (Chapter 22). Forgetting the factor $1/N$ (answer A) is the most frequent error when applying the inversion.

33. The discrete convolution theorem (Chapter 22) states that the DFT transforms circular convolution $x \otimes y$ into

A) the sum $X + Y$

B) the componentwise product $X[k] Y[k]$

C) the linear convolution of X and Y

D) the ratio $X[k]/Y[k]$

Answer: B. Theorem 22.5 (Chapter 22).

34. The cost recursion of the Cooley–Tukey algorithm, $T(1) = c_0$, $T(N) = 2T(N/2) + cN$ (Chapter 23), has solution

A) $T(N) = cN^2$

B) $T(N) = cN \log_2 N + c_0 N$, that is $T(N) = \Theta(N \log_2 N)$

C) $T(N) = cN$

D) $T(N) = c_0 \log_2 N$

Answer: B. Theorem 23.6 (Chapter 23).

35. For the ramp signal $x = (0, 1, 2, \dots, 7)$, its 8-point DFT (Chapter 23) has $X[0] = \sum_n x[n] = 28$; using the closed form $X[k] = N/(\omega_N^k - 1)$ for $k \neq 0$, the value of $X[4]$ is exactly

A) 0

B) -4

C) 28

D) $4i$

Answer: B. Lemma 23.5 (Chapter 23): since $\omega_8^4 = -1$, $X[4] = 8/(-1 - 1) = -4$, a real value as required by the conjugate symmetry of a real signal at $k = N/2$.

Part D — Signals, control, telecommunications (Chapters 24–27)

36. The Nyquist–Shannon sampling theorem (Chapter 24) requires, to exactly reconstruct a band-limited signal $|\omega| \leq \omega_{\max}$, a sampling frequency f_c such that

A) $f_c \geq f_{\max}$

- B) $f_c \geq 2f_{\max}$
- C) $f_c \leq 2f_{\max}$
- D) any positive f_c

Answer: B. Theorem 24.8 (Chapter 24). Answer A forgets the factor of 2: it is the most common error about the sampling theorem.

37. For a continuous LTI system with impulse response h , Chapter 24 proves that BIBO stability is equivalent to
- A) h bounded
 - B) $h \in L^1(\mathbb{R})$, with exact gain $\|h\|_1$
 - C) h periodic
 - D) $h(0) = 0$

Answer: B. Theorem 24.4 (Chapter 24).

38. The Nyquist criterion for closed-loop stability (Chapter 25) relates the number Z of unstable closed-loop poles to the number N of clockwise encirclements of -1 by the diagram of L and to the number P of unstable poles of L through
- A) $Z = N - P$
 - B) $Z = N + P$
 - C) $Z = P/N$
 - D) $Z = 0$ always, if $P = 0$

Answer: B. Theorem 25.5 (Chapter 25): the closed loop is stable if and only if $N = -P$, that is, with P counter-clockwise encirclements.

39. The phase margin (Chapter 25) is defined, at the gain crossover frequency ω_g (where $|L(i\omega_g)| = 1$), as
- A) $\text{PM} = 180^\circ - \arg L(i\omega_g)$
 - B) $\text{PM} = 180^\circ + \arg L(i\omega_g)$
 - C) $\text{PM} = \arg L(i\omega_g)$
 - D) $\text{PM} = 1/|L(i\omega_g)|$

Answer: B. Definition 25.2 (Chapter 25); answer D is instead the *gain* margin, defined at the phase crossover frequency.

40. The bit error probability of BPSK over the AWGN channel (Chapter 26) equals exactly $P_b = Q(\sqrt{2E_b/N_0})$; at $E_b/N_0 = 10$ dB, with independent double verification in the text, this gives
- A) $P_b \approx 0.5$
 - B) $P_b \approx 3.87 \times 10^{-6}$
 - C) $P_b = 0$
 - D) $P_b \approx 1$

Answer: B. Theorem 26.9 (Chapter 26).

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41. The cyclic prefix in OFDM (Chapter 27), if at least as long as $L - 1$ samples of the channel response, turns the effect of the channel on the transmitted block into a
- A) linear convolution
 - B) circular convolution, diagonalised by the DFT
 - C) direct sum, with no convolution
 - D) multiplication in the time domain

Answer: B. Theorem 27.5 (Chapter 27).

Part E — Applications to automation (Chapters 28–30)

42. Rodrigues' formula (Chapter 28), $R(\mathbf{u}, \theta) = I + \sin \theta K + (1 - \cos \theta)K^2$ with $K = [\mathbf{u}]_{\times}$, applied to a 120° rotation about $\mathbf{u} = (1, 1, 1)/\sqrt{3}$ gives a matrix with trace
- A) 3
 - B) 0
 - C) -1
 - D) 1

Answer: B. Theorem 28.2 (Chapter 28): $\text{tr } R = 1 + 2 \cos \theta = 1 + 2 \cos 120^\circ = 1 - 1 = 0$, also verified numerically on the integer matrix $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.

43. Chapter 28 proves that the map $\Phi : S^3 \rightarrow SO(3)$ has, for every $R \in SO(3)$, a fibre $\Phi^{-1}(R)$ consisting of
- A) a single quaternion
 - B) exactly two antipodal quaternions $\{q, -q\}$
 - C) infinitely many quaternions
 - D) no quaternion, in general

Answer: B. Corollary 28.6 (Chapter 28): $SO(3) \cong S^3/\{\pm 1\} = \mathbb{R}P^3$, the double cover by the unit quaternions.

44. Chasles' theorem (Chapter 29) states that every direct rigid motion of space is
- A) always a pure translation
 - B) always a pure rotation about a point
 - C) a screw motion: a rotation about an axis combined with a translation along the same axis
 - D) impossible to represent by means of an axis

Answer: C. Theorem 29.6 (Chapter 29); the identity and pure translation cases are degenerate cases of the screw motion.

45. The SLERP formula between two unit quaternions q_0, q_1 , with angle $\Omega = \arccos \langle q_0, q_1 \rangle$ (Chapter 30), is
- A) $\text{slerp}(q_0, q_1; t) = (1 - t)q_0 + tq_1$, linear interpolation

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- B) $\text{slerp}(q_0, q_1; t) = \frac{\sin((1-t)\Omega)}{\sin \Omega} q_0 + \frac{\sin(t\Omega)}{\sin \Omega} q_1$, the unique shortest path on S^3 at constant angular velocity
- C) $\text{slerp}(q_0, q_1; t) = q_0^{1-t} q_1^t$
- D) $\text{slerp}(q_0, q_1; t) = q_0 + t(q_1 - q_0)$, then renormalised

Answer: B. Theorem 30.2 (Chapter 30). Answers A and D describe lerp (linear interpolation, possibly renormalised): it does not have constant angular velocity and is not the shortest path.

46. The zero-vibration (ZV) shaper (Chapter 30), for two impulses at times $t_1 = 0$, $t_2 = \Delta$ with $A_1 + A_2 = 1$, cancels the residual vibration by choosing
- A) arbitrary Δ and $A_1 = A_2 = \frac{1}{2}$
- B) $\Delta = T_d/2 = \pi/\omega_d$ and $A_1 = 1/(1+K)$, $A_2 = K/(1+K)$, with $K = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$
- C) $\Delta = T_d$ and $A_1 = 1$, $A_2 = 0$
- D) $\Delta = 0$

Answer: B. Theorem 30.7 (Chapter 30).

Part F — Numerics and research (Chapters 31–33)

47. Least-squares geometric circle fitting (Chapter 31) may fail to have a minimum when
- A) the sampled points are in general position
- B) the sampled points (at least three distinct ones) are all collinear
- C) there are more than ten points
- D) the points lie exactly on a circle

Answer: B. Corollary 31.4 (Chapter 31): if the points are collinear then $\inf_c G(c) = 0$ but $G(c) > 0$ for every c , so F has no minimum; Theorem 31.3 instead gives sufficient conditions for existence.

48. Chapter 32 proves in full that a curve $\gamma = \partial K$ inscribes a square when
- A) γ is any Jordan curve (the general case of Toeplitz’s problem, 1911)
- B) K is a convex body centrally symmetric about an interior point of its own
- C) γ is a polygon with an even number of sides
- D) γ has rational length

Answer: B. Theorem 32.1 (Chapter 32). Answer A is instead the general case of Toeplitz’s problem (Definition 32.2), which after one hundred and fifteen years remains open.

49. The *local–global* conjecture for Apollonian circle packings (2003), according to Chapter 32, as of the verification date reported in the text is classified as
- A) still an open problem, with no counterexamples
- B) a theorem, proved true in 2014 by Bourgain and Kontorovich

-
- C) refuted: Haag, Kertzer, Rickards and Stange (2024) exhibited infinitely many packings with at least $c\sqrt{N}$ missing curvatures up to N
- D) never rigorously formulated

Answer: C. By Definition 32.1(ii) (Chapter 32): a conjecture is refuted when its negation is a theorem. Bourgain and Kontorovich (2014) had proved only that the missing curvatures have density zero, a result in the conjecture’s favour but not equivalent to it; the explicit counterexample is verified by hand in Example 32.5.

50. By the third rule of the double register stated in the epilogue (Chapter 33), an observation, an infobox or a good-practice note
- A) may promote an unproved statement to a theorem if it is very plausible
- B) never promote an unproved statement to the rank of theorem: the great results used without proof remain in an observation or infobox, with the source declared
- C) replace proofs whenever the computation is too long
- D) have no value and may be ignored

Answer: B. Observation 33.3 (Chapter 33): it is the same rule — “never invent anything” — that runs through the whole treatise, from the prologue of Chapter 1 to here.

INFO

After the assessment. The six Parts follow the order of the volume’s six families of ideas: from the curvature of a curve to the curvature of a surface, from the sphere to the topology of the circle as a group, from tangencies between circles to the transforms that carry their arithmetic into signals, all the way to robot arms that rotate according to the same laws by which a circle wraps around itself. A reader who answered questions 1–27 (geometry and analysis) well but the later ones poorly has the geometric intuition and must practise the calculus of transforms and systems; one who did the opposite can compute but would do well to return to curvature and the sphere, because it is from there that the transforms inherit their symmetries. The last question is no accident: the double register that distinguishes what is proved from what is merely plausible is the same rule by which this volume — and the whole treatise — was written.

CHAPTER A

Recalled Prerequisites

This appendix recalls, without proof, the results from Volumes I and II used in the text. It serves as a quick reference, not a treatment.

A.1. Linear algebra and complex numbers

Dot product $\langle v, w \rangle = \sum v_i w_i$; norm $|v| = \sqrt{\langle v, v \rangle}$; cross product in $\mathbb{R}^3$ with $|v \times w| = |v| |w| \sin \theta$. Eigenvalues and eigenvectors: $Av = \lambda v$. Skew-symmetric matrix $A^\top = -A$; symmetric $A^\top = A$ (diagonalizable on an orthonormal basis, spectral theorem).

Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$; multiplication by $e^{i\theta}$ = rotation by θ in the complex plane; $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ is the unit circle, multiplicative group.

A.2. Calculus and plane curves (plane Frenet)

Arc length $s(t) = \int |\gamma'| dt$; arc-length parametrization $|\gamma'(s)| = 1$. Plane Frenet frame $\{T, N\}$ with $T' = \kappa N$, $N' = -\kappa T$. The circle has constant curvature $\kappa \equiv 1/r$ (Volume II).

A.3. Notable integrals

Gaussian integral $\int_{\mathbb{R}} e^{-t^2} dt = \sqrt{\pi}$ (Volume II). Gamma function $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ with $\Gamma(s+1) = s\Gamma(s)$, $\Gamma(n+1) = n!$, $\Gamma(1/2) = \sqrt{\pi}$. Arc length on circle $l = r\theta$ (Volume I).

A.4. Imported results, chapter by chapter

This section is an index, not a treatise within the treatise: it lists, chapter by chapter, the results that Volume III *uses* but does *not* prove, with a concise statement, the source, and a reference to the chapter where they occur. Results proved elsewhere in the same treatise (Volume I or II, or an earlier chapter of this volume) and simply recalled by reference are excluded: those are not imported, they are only not repeated. The list below was drawn from the “what we import” observations and the panoramas scattered through the 33 chapters.

Chapter 3 — Space curves: Frenet–Serret and torsion

- **Completeness of the space of continuous functions.** The space $(C([\alpha, \beta], \mathbb{R}^n), \|\cdot\|_\infty)$ is complete: a Cauchy sequence of continuous functions on a compact set converges uniformly to a continuous function. Source: classical result of functional analysis; see K. Wang, *Fundamental Theorem of the Local Theory of Curves*, §3.

Chapter 4 — Global theorems on plane curves

- **Jordan curve theorem.** If $J \subset \mathbb{R}^2$ is a Jordan curve (a homeomorphic image of a circle), then $\mathbb{R}^2 \setminus J$ has exactly two connected components, one bounded and one unbounded, each

with boundary J . Source: J. R. Munkres, *Topology*, 2nd ed., §63; C. Thomassen, *The Jordan–Schönflies theorem and the classification of surfaces*, Amer. Math. Monthly 99 (1992).

- **Two facts about the Minkowski mixed area.** For two planar convex bodies K, L and $\lambda \geq 0$: (M1) the area of $K + \lambda L$ is a quadratic polynomial in λ whose coefficients (the mixed areas) are symmetric and monotone with respect to inclusion; (M2) the mixed area is bilinear with respect to Minkowski addition. Source: R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., §5.1.
- **Three prerequisites for the global Gauss–Bonnet theorem.** (P1) The local form of Gauss–Bonnet on a geodesic triangle; (P2) existence of a geodesic triangulation of any compact oriented regular surface; (P3) topological invariance of the Euler characteristic χ . Source: M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, §4-5; T. Radó, *Über den Begriff der Riemannschen Fläche*, 1926.

Chapter 9 — Geodesics and the calculus of variations

- **Picard–Lindelöf theorem.** A first-order system of differential equations with C^1 right-hand side has, locally, a unique solution for given initial data, depending continuously on the data. Source: T. Shifrin, *Differential Geometry: A First Course in Curves and Surfaces*, Appendix §3.
- **Jacobi fields and conjugate points in general.** On any surface (not only the sphere, treated in full in the chapter), Jacobi fields exist along every geodesic, and the resulting theory of conjugate points is general. Source: M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Ch. 5.

Chapter 10 — Local theory of surfaces

- **Brioschi’s formula.** The Gaussian curvature K can be written as a ratio of two determinants in the coefficients E, F, G of the first fundamental form and their derivatives alone, even when $F \neq 0$. Source: Lee-Peng Teo, *The Brioschi Formula for the Gaussian Curvature*, arXiv:2404.00835, 2024.
- **Minding’s theorem (converse direction).** Every surface with identically zero Gaussian curvature is locally isometric to the Euclidean plane (the chapter proves only the direction “isometric to the plane $\Rightarrow K \equiv 0$ ”). Source: M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Ch. 4.

Chapter 11 — The circle in any dimension

- **Classification of group-spheres.** S^0 , S^1 and S^3 are the only spheres carrying a Lie group structure; this is a classification, not a list of examples, and requires algebraic-topology tools not developed in the chapter (stable homotopy theory; Bott, Milnor and Adams, 1958–1960, as stated in Remark 11.6). J. Stillwell, *Naive Lie Theory*, Springer, 2008, gives an elementary treatment of the context (low-rank Lie groups), not a proof of the classification.

Chapter 12 — Topology of the circle

- $\pi_1(S^n) = 0$ for $n \geq 2$. Every loop on a sphere of dimension at least two contracts to a point. Source: A. Hatcher, *Algebraic Topology*, Proposition 1.14.
- **Hairy ball theorem.** On S^2 there is no continuous nowhere-zero tangent vector field; more generally S^n admits one if and only if n is odd. Source: A. Hatcher, *Algebraic Topology*, Theorem 2.28.

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- **Non-triviality of the Hopf fibration.** The Hopf map $h : S^3 \rightarrow S^2$ is not a trivial fibration ($S^3 \not\cong S^2 \times S^1$), and $\pi_3(S^2) \cong \mathbb{Z}$ is generated by the class of h . Source: A. Hatcher, *Algebraic Topology*, Examples 4.45 and 4.51.

Chapter 13 — The circle as a group

- **Compact abelian Lie groups.** Every connected, compact, abelian Lie group is isomorphic to a torus $(S^1)^n$. Source: T. Bröcker, T. tom Dieck, *Representations of Compact Lie Groups*; F. W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*.
- **Haar's theorem.** On every locally compact Hausdorff topological group there exists, unique up to a positive factor, a nonzero left-translation-invariant Radon measure. Source: W. Rudin, *Fourier Analysis on Groups*; G. B. Folland, *A Course in Abstract Harmonic Analysis*.
- **Pontryagin duality.** For every locally compact abelian group G , the dual $\hat{G}$ is itself locally compact abelian, and $\hat{\hat{G}} \cong G$. Source: W. Rudin, *Fourier Analysis on Groups*.
- **Weyl's equidistribution criterion in dimension n .** If the averages of $e^{2\pi i k \cdot w^n}$ tend to zero for every $k \neq 0$, the sequence (w^n) is not merely dense but equidistributed. Source: H. Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Math. Ann. 77 (1916); E. M. Stein, R. Shakarchi, *Fourier Analysis: An Introduction*.

Chapter 14 — Tangencies and packings

- **Thue–Fejes Tóth theorem.** The maximum density of a packing of equal circles in the plane is $\pi/\sqrt{12} = 0.9069\dots$. Source: A. Thue, 1910; L. Fejes Tóth, *Lagerungen in der Ebene, auf der Kugel und im Raum*, 1953; H.-C. Chang, L.-C. Wang, arXiv:1009.4322, 2010 (elementary proof).
- **Kepler conjecture and optimal packings in dimension 8 and 24.** The face-centered cubic arrangement is densest in space; the E_8 and Leech lattices are optimal in their respective dimensions. Source: T. C. Hales, *A proof of the Kepler conjecture*, 2005; M. S. Viazovska, *The sphere packing problem in dimension 8*, 2017; H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. S. Viazovska, *The sphere packing problem in dimension 24*, 2017.
- **Circle packing theorem (Koebe–Andreev–Thurston).** Every finite simple planar graph is the tangency graph of some circle packing in the plane. Source: K. Stephenson, *Introduction to Circle Packing*, 2005.

Chapter 15 — Complex analysis on the disk

- **General (homological) form of Cauchy's theorem.** On any open set, $\oint_\gamma f = 0$ for every cycle γ that is homologically zero in the open set. Source: L. V. Ahlfors, *Complex Analysis*, §4.
- **Poincaré metric and the Schwarz–Pick lemma.** The unit disk carries a metric invariant under automorphisms, with respect to which every automorphism is an isometry. Source: J. B. Conway, *Functions of One Complex Variable I*, Ch. VI.
- **Great Picard theorem.** In every punctured neighborhood of an essential singularity, a holomorphic function takes every complex value, with at most one exception, infinitely often. Source: J. B. Conway, *Functions of One Complex Variable I*, Ch. XII.

Chapter 17 — Convergence of Fourier series

- **Real-analysis prerequisites.** Density of step functions in L^1 of a bounded interval and of $C(\mathbb{T})$ in $L^2(\mathbb{T})$; dominated convergence; Lebesgue differentiation; integration by parts for absolutely continuous functions; the uniform boundedness principle. Source: W. Rudin, *Real and Complex Analysis*, 3rd ed., Ch. 3 and 5.
- **Kolmogorov's and Carleson's theorems.** Kolmogorov (1926): there exists $f \in L^1(\mathbb{T})$ whose Fourier series diverges at every point. Carleson (1966): for every $f \in L^2(\mathbb{T})$ the Fourier series converges almost everywhere. Source: L. Carleson, *On convergence and growth of partial sums of Fourier series*, Acta Math. 116 (1966); A. Zygmund, *Trigonometric Series*, Ch. VIII (for Kolmogorov's example).

Chapter 18 — Partial differential equations on the circle and in the disc

- **McMahon's asymptotic expansion.** For fixed $\nu \geq 0$ and $m \rightarrow \infty$, the m -th positive zero $j_{\nu,m}$ of the Bessel function J_ν has an explicit asymptotic expansion in $a = (m + \nu/2 - 1/4)\pi$. Source: NIST, *Digital Library of Mathematical Functions*, §10.21(vi), eq. 10.21.19.
- **Completeness of the Bessel–Fourier system.** Every reasonable function on $[0, 1]$ expands in a series of functions $J_\nu(j_{\nu,m}t)$. Source: NIST, *Digital Library of Mathematical Functions*, §10.23(iii); G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Ch. 18.

Chapter 19 — The Fourier transform

- **Plancherel extension to L^2 .** The chapter proves that $\mathcal{F}$ preserves the L^2 norm on the Schwartz class $\mathcal{S}(\mathbb{R})$; that a unique surjective isometric extension $\mathcal{F}_2 : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ exists remains **imported**, and reduces to a single fact not proved here: the density of $\mathcal{S}(\mathbb{R})$ in $L^2(\mathbb{R})$ (the rest of the construction, via Cauchy sequences, is mechanical and is indicated for completeness). Source: W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw-Hill, 1987, Ch. 9 "Fourier Transforms", section "The Plancherel theorem".
- **Full rigour of the sampling theorem.** The Whittaker–Kotel'nikov–Shannon theorem follows from the chapter's periodisation theorem, but its complete proof requires the product of a tempered distribution by a slowly growing C^∞ function and the convolution of distributions, tools beyond this chapter's scope: it remains **imported**. Source: R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd ed., McGraw-Hill, 1999.

Chapter 20 — The Laplace transform

- **Bromwich's inversion formula in the general case.** Even when F is not integrable on the vertical line $\operatorname{Re}(s) = c$, the inversion formula remains true when read as a principal value. Source: G. Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Ch. 24–26.
- **Necessity of $h \in L^1$ for BIBO stability in the general case.** Beyond proper rational transfer functions (treated in full in the chapter), the condition $h \in L^1$ is also necessary for bounded-input/bounded-output stability. Source: A. V. Oppenheim, A. S. Willsky, S. H. Nawab, *Signals and Systems*, 2nd ed., Ch. 9.

Chapter 21 — The z -transform and the DTFT

- **Riesz–Fischer theorem.** The space $L^2(\mathbb{T})$ is complete; this extends Parseval's identity from $\ell^1(\mathbb{Z})$ to $\ell^2(\mathbb{Z})$. Source: W. Rudin, *Real and Complex Analysis*, 3rd ed., Ch. 3.

Chapter 22 — The discrete Fourier transform

- **Multiplicities of the DFT eigenvalues.** The DFT matrix F_N has exactly four distinct eigenvalues $1, -i, -1, i$, with multiplicities known in closed form. Source: J. H. McClellan, T. W. Parks, *Eigenvalue and eigenvector decomposition of the discrete Fourier transform*, 1972.
- **Asymptotic decay of sidelobes.** The usual spectral windows have known asymptotic decay laws for their sidelobes as the window length grows. Source: F. J. Harris, *On the use of windows for harmonic analysis with the discrete Fourier transform*, Proc. IEEE 66 (1978).

Chapter 23 — The fast Fourier transform

- **Existence of a primitive root modulo a prime.** For every prime p the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic, i.e. it admits a primitive root; this is the hypothesis on which Rader's algorithm rests. Source: G. H. Hardy, E. M. Wright, *An Introduction to the Theory of Numbers*.
- **Standard model of floating-point error.** $\text{fl}(a * b) = (a * b)(1 + \delta)$ with $|\delta| \leq u$, and its propagation in the error analysis of the FFT. Source: N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., Ch. 2 and 24.

Chapter 24 — Signals and linear systems

- **Riesz representation theorem.** For μ σ -finite, $(L^1(\mu))^* \cong L^\infty(\mu)$. Source: classical result of measure theory, used in representing LTI systems; cf. A. V. Oppenheim, A. S. Willsky, S. H. Nawab, *Signals and Systems*.
- **Paley–Wiener constraints between causality and frequency response.** A causal, stable system admits a holomorphic extension of its Laplace transform to the right half-plane, which constrains the magnitude and phase of the frequency response. Source: A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed.
- **Minimum-phase systems.** Among all causal, stable systems with the same gain $|H(\omega)|$, there is a unique one that is minimum-phase, with no zeros in the right half-plane. Source: A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed.
- **Hilbert transform.** Its representation as a Cauchy principal value, its equivalence with the multiplier definition, and M. Riesz's boundedness theorem in L^p for $1 < p < \infty$. Source: F. W. King, *Hilbert Transforms*, vol. 1, 2008.

Chapter 26 — Telecommunications: complex baseband, IQ and modulations

- **Existence of Gaussian white noise.** There exists a Gaussian measure on a probability space that simultaneously realizes the projections of white noise onto every function in L^2 . Source: J. M. Wozencraft, I. M. Jacobs, *Principles of Communication Engineering*, 1965.
- **Channel capacity theorem (Shannon, 1948).** On the AWGN channel of bandwidth W and signal-to-noise ratio S/N , every transmission rate below $C = W \log_2(1 + S/N)$ is achievable with arbitrarily small error probability. Source: C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27** (1948), pp. 379–423 and 623–656 (it does not appear as a standalone bibliographic entry in the volume's source register: exact theorem and part number to be checked against a primary copy).

Chapter 27 — OFDM

- **Frequency diversity against deep fades.** No equalization can recover a nearly-zero sub-carrier: channel coding with frequency diversity is required. Source: Y. S. Cho, J. Kim, W. Y. Yang, C. G. Kang, *MIMO-OFDM Wireless Communications with MATLAB*, Ch. 4.
- **Gaussian approximation of PAPR statistics.** The hypothesis (G) that the samples are independent Gaussian is convenient for computing the PAPR's CCDF, but is false for a real OFDM system, whose samples are uncorrelated yet not independent. Source: Y. S. Cho, J. Kim, W. Y. Yang, C. G. Kang, *MIMO-OFDM Wireless Communications with MATLAB*, Ch. 7.

Chapter 28 — Rotations of space, $SO(3)$ and quaternions

- **Stuelpnagel's theorem.** No global parametrization of $SO(3)$ with three parameters is free of singularities; four is the minimum number of parameters for a non-singular global parametrization. Source: J. Stuelpnagel, *On the Parametrization of the Three-Dimensional Rotation Group*, SIAM Review 6 (1964).

Chapter 29 — Lie groups and kinematics

- **Closed subgroup theorem (Cartan).** Every closed subgroup of a Lie group is, with the induced topology, an embedded Lie subgroup. Source: J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Theorem 20.12.
- **Baker–Campbell–Hausdorff formula.** The product $e^A e^B$ of two matrix exponentials that do not commute equals $e^{C(A,B)}$ for an explicit series $C(A,B)$ in the iterated commutators of A and B . Source: B. C. Hall, *An Elementary Introduction to Groups and Representations*, arXiv:math-ph/0005032, §4.3.

Chapter 30 — Trajectories and interpolation

- **Three properties of B-splines.** Beyond local support (proved in the chapter): partition of unity, non-negativity of the basis functions, and continuity of the join between segments. Source: L. Piegl, W. Tiller, *The NURBS Book*, 2nd ed., Ch. 2.
- **Squad and dual quaternions.** Squad (*spherical spline quaternion*) extends SLERP to more than two orientations with C^1 continuity; dual quaternions interpolate position and orientation simultaneously in $SE(3)$. Source: K. Shoemake, *Animating rotation with quaternion curves*, SIGGRAPH '85, 1985.
- **Robust shaper families beyond ZVD.** EI shapers damp residual vibration while remaining robust to an error in the estimated natural frequency, unlike ZVD. Source: W. Singhose, *Command shaping for flexible systems: a review of the first 50 years*, 2009.

Chapter 31 — Estimating the circle from data

- **Convergence of Gauss–Newton.** The iteration converges quadratically when the residual at the minimum is zero, linearly otherwise (with a factor proportional to the norm of the neglected term), and may diverge without a line search. Source: J. Nocedal, S. J. Wright, *Numerical Optimization*, 2nd ed., Ch. 10.
- **Levenberg–Marquardt as a trust-region method.** The damped Levenberg–Marquardt step exactly solves a minimization problem constrained to a ball of suitable radius. Source: J. Nocedal, S. J. Wright, *Numerical Optimization*, 2nd ed., Ch. 4.

-
- **Bias of Kåsa’s algebraic fit.** The radius estimated by Kåsa’s algebraic method is systematically greater than or equal to that of the geometric fit, for the same data. Source: N. Chernov, *Circular and Linear Regression: Fitting Circles and Lines by Least Squares*, 2010.

Chapter 32 — Open problems

- **History of Toeplitz’s inscribed square problem (1911).** A survey of the classes of Jordan curves for which it is now proved that every curve in the class inscribes a square. Source: B. Matschke, *A survey on the square peg problem*, 2014; *On the square peg problem and its relatives*, 2022.
- **Inscribed rectangle theorem (Greene–Lobb) and the 2026 preprint.** Every smooth Jordan curve inscribes a rectangle; a 2026 preprint gives a positive measure of them under additional hypotheses. Source: J. E. Greene, A. Lobb, *The rectangular peg problem*, Ann. of Math. 194 (2021); *Jordan curves inscribe a positive measure of rectangles*, arXiv:2604.17116, 2026.

Dependencies of the additions dated 10 September 2026

This update does not reclassify the appendix’s historical entries. The new sections prove the holomorphic logarithm, Montel and Riemann; Kalman’s rank criterion, multi-input pole placement and finite/infinite-horizon Riccati; linear filtering with precise optimality conditions; 2R dynamics and digital control. The prerequisites remain those stated in the chapters. The MEKF is local and does not inherit the linear model’s Gaussian optimality. Advanced Gauss circle results are reported as external, distinguishing articles, preprints and withdrawal; see [13, 14, 15]. Montel is not registered as imported because its proof has been added.

CHAPTER B

Tables for Automation and Practical References

This appendix collects the conventions of the main platforms as an operational reference to connect the mathematics of the text to real programming.

B.1. Orientation conventions by brand

Brand	Representation	Convention
KUKA	A,B,C angles	Euler ZYX ($A=R_z$, $B=R_y$, $C=R_x$)
FANUC	W,P,R angles	Euler XYZ (fixed)
ABB	quaternion (q_1, q_2, q_3, q_4)	unit, in <code>robtarg</code> (RAPID)
Yaskawa	Rx,Ry,Rz angles	proprietary format, convertible to quaternion

All controllers internally use a 4×4 homogeneous matrix with 3×3 rotation block in $SO(3)$: the representations above are equivalent interfaces toward the operator. The conversion always passes through the rotation matrix (Chapter 4).

B.2. Circular motion commands and interpolation

Platform	Circular motion	Notes
Siemens SINUMERIK (CNC)	G02/G03	plane G17/G18/G19, center I, J, K
PLCopen (Siemens, Schneider, Rockwell, Delta, Elco)	MC_MoveCircularAbsolute	vendor- neutral stan- dard, param. veloc- ity/accel./jerk
ABB (RAPID)	MoveC	intermediate point + final point
KUKA (KRL)	CIRC	auxiliary point + final point
FANUC (TP)	CIRC / Circular	circular motion
Yaskawa (INFORM)	MOVc	circular motion

Mathematical correspondences: an arc is the locus of constant curvature $\kappa = 1/r$ (Chapter 1); helical interpolation is the primitive of the helix with constant κ, τ (Chapter 2); orientation along the path is interpolated with SLERP on quaternions (Chapter 11); wrist singularities are the mechanical gimbal lock (Chapters 4 and 10).

B.3. Synthesis exercises (with solution)

Exercise B.1. A KUKA robot has orientation A=90, B=0, C=0. What is the rotation matrix?

Solution. *ZYX* convention with only A= 90° about *z*: $R = R_z(90^\circ) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (Chapter 4). B and C zero do not contribute. The equivalent quaternion (ABB format) is $q = \cos 45^\circ + \sin 45^\circ \mathbf{k} = \frac{1}{\sqrt{2}}(1, 0, 0, 1)$. $\square$

Exercise B.2. In SINUMERIK G-code, how do you program a counterclockwise arc of radius 50 from the current point to (100,0) in the *xy* plane?

Solution. Select the plane with G17, then G03 X100 Y0 CR=50 (counterclockwise arc, radius 50), or specify the center with I, J. Mathematically one traverses a segment with constant curvature $\kappa = 1/50 = 0.02$; a continuous transition to a straight line would require a variable-curvature transition (Chapter 1). $\square$

CHAPTER C

Tables of transforms and identities

C.1. Notable Fourier Transforms

Signal $f(t)$	Transform $\hat{f}(\omega)$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$\Pi(t)$ (rectangle, width 1)	$\text{sinc}(\omega/2) = \frac{\sin(\omega/2)}{\omega/2}$
$e^{-a t }$, $a > 0$	$\frac{2a}{a^2 + \omega^2}$
$e^{-at}u(t)$, $a > 0$	$\frac{1}{a + i\omega}$
e^{-at^2} , $a > 0$ (gaussian)	$\sqrt{\pi/a} e^{-\omega^2/(4a)}$

Convention used in this treatise: $\hat{f}(\omega) = \int_{\mathbb{R}} f(t)e^{-i\omega t}dt$, $f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega t}d\omega$ (Definition 4.3, Chapter 4).

C.2. Notable Laplace Transforms (causal signals)

Signal $f(t)u(t)$	Transform $F(s)$
$\delta(t)$	1
$u(t)$ (step)	$1/s$
t (ramp)	$1/s^2$
t^n , $n \in \mathbb{N}$	$n!/s^{n+1}$
e^{-at}	$1/(s + a)$
te^{-at}	$1/(s + a)^2$
$\cos(\omega t)$	$s/(s^2 + \omega^2)$
$\sin(\omega t)$	$\omega/(s^2 + \omega^2)$
$e^{-at}\cos(\omega t)$	$(s + a)/((s + a)^2 + \omega^2)$
$e^{-at}\sin(\omega t)$	$\omega/((s + a)^2 + \omega^2)$

C.3. Notable Z-Transforms (causal sequences)

Sequence $x[n]u[n]$	Transform $X(z)$	ROC
$\delta[n]$	1	$\mathbb{C}$ entire
$u[n]$	$z/(z - 1)$	$ z > 1$
a^n	$z/(z - a)$	$ z > a $
na^n	$az/(z - a)^2$	$ z > a $
$\cos(\Omega_0 n)$	$\frac{z^2 - z \cos \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}$	$ z > 1$
$\sin(\Omega_0 n)$	$\frac{z \sin \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}$	$ z > 1$

C.4. DFT and FFT: formulas and complexity

Quantity	Formula
Direct DFT (N points)	$X[k] = \sum_{n=0}^{N-1} x[n]e^{-2\pi i kn/N}$
Inverse DFT	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{2\pi i kn/N}$
Relation with Z-Transform	$X[k] = X(\zeta_N^k)$, $\zeta_N = e^{2\pi i/N}$ (Theorem 4.11, Chapter 4)
Periodicity	$X[k + N] = X[k]$ for every k
Circular convolution	$\text{DFT}\{x *_N y\}[k] = X[k] \cdot Y[k]$
Discrete Parseval	$\sum_{n=0}^{N-1} x[n] ^2 = \frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$
Radix-2 FFT butterfly	$X[k] = E[k] + \zeta_N^k O[k]$, $X[k + N/2] = E[k] - \zeta_N^k O[k]$ (Theorem 4.12)

N	Direct DFT $O(N^2)$	FFT $O(N \log_2 N)$	Factor
8	64	24	$\approx 2.7\times$
64	4 096	384	$\approx 10.7\times$
1 024	1 048 576	10 240	$\approx 102\times$
4 096	16 777 216	49 152	$\approx 341\times$

Note: the count is in complex arithmetic operations; see Remark 4.15 (Chapter 4) for the derivation of the recurrence $T(N) = 2T(N/2) + cN$.

CHAPTER D

Symbols and Notation

List of the main symbols used in the volume, with their meaning and reference chapter.

Symbol	Meaning	Ch.
$\gamma(t), \gamma(s)$	parametric curve / arc-length parametrization	2
κ	curvature (signed in the plane)	2
τ	torsion	3
$\rho = 1/ \kappa $	radius of curvature	2
T, N, B	Frenet frame vectors (tangent, normal, binormal)	3
$O(n), SO(n)$	orthogonal group, special orthogonal group	4
$\mathbb{T}$	unit circle $\{ z = 1\} \cong SO(2)$	5
R_θ, S_θ	plane rotation, reflection	4
$[\mathbf{u}]_\times$	skew-symmetric matrix of $\mathbf{u}$	6
$\mathbb{H}, SU(2)$	quaternions, special unitary group	6
$\bar{q}, q $	quaternion conjugate and norm	6
I, II	first and second fundamental form	7
K, H	Gaussian and mean curvature	7
S^{n-1}, B^n	n -dimensional sphere and ball	12
$\Gamma(s)$	Euler's Gamma function	12
$\pi_1(X)$	fundamental group	13
$\chi(M)$	Euler characteristic	16
$\mathfrak{so}(3), \mathfrak{se}(3)$	Lie algebras of $SO(3), SE(3)$	14
$SE(3)$	rigid motion group	14
$J(\theta)$	geometric Jacobian	14
$E = A + B + C - \pi$	spherical excess	10
δ	packing density	17

CHAPTER E

Numerical laboratory

Across the 33 chapters of this volume every numerical check appears at least twice: a few dozen scripts, of varying length and style, are scattered through the proofs, the examples and the solved exercises. This appendix does not collect all of them — that would only repeat the volume itself — but selects five, one per family of numerical technique, as a representative sample and as a starting point for anyone who wants to reproduce the checks outside the text. None of the scripts below is new: each one is already written, and executed, in the chapter indicated.

E.1. The convention followed by the treatise

Every numerical value appearing in the text is computed **twice, by independent routes**: once with the closed formula or the algorithm proved on the preceding pages, once with a Python script that was actually run (never merely imagined). Agreement between the two routes is marked with the symbol ✓. The scripts appear in two forms, depending on what is being checked:

- as a complete `lstlisting` block, when the *code itself* is the object under discussion — for instance the recursion of the radix-2 FFT, or the four conversions between matrix, quaternion and axis-angle;
- described in prose, numbered step by numbered step, inside a proof or a solved exercise, when it is the *numerical result* — not the code — that matters: this is the case for the circle fit, the Bessel zeros, and the SLERP check below.

In neither case does the treatise declare the exact interpreter version, a seed, or a line-by-line numerical tolerance: the convention actually followed is simpler and verifiable by eye, the double computation marked with ✓, and this appendix documents it as it really is, without promising a finer reproducibility than the text actually offers.

E.2. How to reproduce the checks

- **Interpreter and libraries.** All scripts in the volume are written for CPython 3 with NumPy (linear algebra, FFT). A few need more: Chapter 31 uses the derivative-free Nelder–Mead method as an independent check; Chapter 18 uses an arbitrary-precision arithmetic library for the Bessel zeros. Wherever an extra tool is needed, the point is stated in the text cited below.
- **Determinism.** The five checks in this appendix are deterministic: same input data, same output, no random generator. The one exception in the whole volume is the Monte Carlo experiment on the bias of the algebraic fit (Chapter 31, Example 31.5, 20,000 replicas): the chapter does not declare a seed, so individual replicas change from one run to the next, but the averages reported over 20,000 trials stay stable to the digits quoted.
- **To reproduce a check:** copy a `lstlisting` block into a `.py` file and run it with a Python 3 interpreter that has NumPy installed; for the checks described in prose (circle fit, Bessel zeros, SLERP) the numbered steps of the text are enough on their own to

rebuild the script, because every intermediate step is already spelled out in full in the chapter cited.

E.3. Five representative checks

E.3.1 Eight-point fast Fourier transform

From Chapter 22: the decimation-in-time radix-2 FFT, written recursively and checked on the eight alternating samples $+1, -1$ against `numpy.fft.fft` — the same numerical example on which Appendix C tabulates the comparison between $O(N^2)$ and $O(N \log_2 N)$.

Listing E.1: Recursive radix-2 FFT in Python (Chapter 22; for teaching purposes: for real use, use `numpy.fft.fft`, which is optimized and not restricted to N a power of 2).

```
import numpy as np

def fft_radix2(x):
    """Recursive FFT, decimation in time. len(x) must be a power of 2."""
    N = len(x)
    if N == 1:
        return x
    if N % 2 != 0:
        raise ValueError("N must be a power of 2")
    E = fft_radix2(x[0::2]) # sub-problem on even indices
    O = fft_radix2(x[1::2]) # sub-problem on odd indices
    twiddles = np.exp(-2j * np.pi * np.arange(N // 2) / N)
    butterfly = twiddles * O
    return np.concatenate([E + butterfly, E - butterfly])

# Check against numpy.fft.fft
x = np.array([1, -1, 1, -1, 1, -1, 1, -1], dtype=complex)
X_mine = fft_radix2(x)
X_numpy = np.fft.fft(x)
print(np.allclose(X_mine, X_numpy)) # True
print(np.round(X_mine, 6))
# [ 0.+0.j  0.+0.j  0.+0.j  0.+0.j  8.-0.j  0.+0.j  0.+0.j  0.+0.j ]
```

E.3.2 Estimating a circle from data: the Gauss–Newton iteration

From Chapter 31: the geometric fit of a circle to six sample points, started from the linear algebraic fit and refined with the Gauss–Newton step (Definition 31.6) until the stopping criterion on the gradient fires.

k	$a^{(k)}$	$b^{(k)}$	$r^{(k)}$	$F(\theta^{(k)})$	$\ \nabla F\ $
0	3.1535329	1.8380337	5.1133002	0.346670569	$7.76 \cdot 10^{-2}$
1	3.1535637	1.8310288	5.1073286	0.346334745	$5.03 \cdot 10^{-4}$
2	3.1535025	1.8309733	5.1073286	0.346334724	$6.17 \cdot 10^{-6}$
3	3.1535022	1.8309723	5.1073285	0.346334724	$7.93 \cdot 10^{-8}$
4	3.1535022	1.8309722	5.1073285	0.346334724	$1.01 \cdot 10^{-9}$

The stopping criterion on the gradient fires at the third iteration: $a^* = 3.1535022$, $b^* = 1.8309722$, $r^* = 5.1073285$. *Independent check:* minimizing only the function G of Proposition 31.2 with the Nelder–Mead method (which uses neither gradient nor Jacobian), starting from (3; 2), one obtains in 57 steps $c = (3.1535022; 1.8309723)$ and $\bar{\rho} = 5.1073285$: the same seven decimal places. ✓

E.3.3 Zeros of the Bessel function J_n

From Chapter 18: the first eight positive zeros of J_0, J_1, J_2, J_3 , needed for the eigenfrequencies of a circular membrane, computed by three independent routes:

- **A** — a sign scan with step 10^{-3} , 200 bisections and 60 Newton iterations, in double precision;
- **B** — a multiple-precision arithmetic library with 30 significant digits, using a completely different algorithm;
- **C** — McMahon’s asymptotic expansion (Observation 18.13), truncated after the fourth term.

zero	A	B	C (McMahon)
$j_{0,1}$	2.4048255577	2.4048255577	2.4064624434
$j_{1,1}$	3.8317059702	3.8317059702	3.8316382972
$j_{2,1}$	5.1356223018	5.1356223018	5.1358891046
$j_{0,2}$	5.5200781103	5.5200781103	5.5200867370
$j_{3,1}$	6.3801618959	6.3801618959	6.3828008747
$j_{1,2}$	7.0155866698	7.0155866698	7.0155851349
$j_{2,2}$	8.4172441404	8.4172441404	8.4172441404
$j_{0,3}$	8.6537279129	8.6537279129	8.6537283466

Columns A and B agree to every digit reported (largest discrepancy $5.6 \cdot 10^{-13}$ on $j_{2,3}$); column C, being only asymptotic, deviates more the smaller k and the larger n . ✓

E.3.4 Quaternions, axis–angle, and Rodrigues’ formula

From Chapter 28: the conversions between rotation matrix, unit quaternion, and axis–angle pair, including the sign choice (`segno_vicino`) to apply *before* any interpolation between orientations.

Listing E.2: Conversions between matrix, quaternion and axis–angle (Chapter 28).

```
import numpy as np

def qmul(a, b):
    w1, x1, y1, z1 = a
    w2, x2, y2, z2 = b
    return np.array([w1*w2 - x1*x2 - y1*y2 - z1*z2,
                    w1*x2 + x1*w2 + y1*z2 - z1*y2,
                    w1*y2 - x1*z2 + y1*w2 + z1*x2,
                    w1*z2 + x1*y2 - y1*x2 + z1*w2])

def q_da_asse_angolo(u, theta):
    u = np.asarray(u, float); u = u/np.linalg.norm(u)
    return np.concatenate([[np.cos(theta/2)], np.sin(theta/2)*u])

def asse_angolo_da_q(q):
    if q[0] < 0: q = -q # sign normalisation
    nv = np.linalg.norm(q[1:])
    theta = 2*np.arctan2(nv, q[0]) # NOT 2*arccos(q[0])
    u = q[1:]/nv if nv > 1e-12 else np.array([0., 0., 1.])
    return u, theta

def R_da_q(q):
    w, x, y, z = q
    return np.array([[1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
                    [2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
```

```

[2*(x*z-w*y),    2*(y*z+w*x),    1-2*(x*x+y*y)]]

def q_da_R(R):
    # four branches (Shepperd)
    t = np.trace(R)
    c = np.array([1+t, 1+2*R[0,0]-t, 1+2*R[1,1]-t, 1+2*R[2,2]-t])
    m = int(np.argmax(c)); s = np.sqrt(c[m])/2; f = 4*s
    if m == 0: return np.array([s, (R[2,1]-R[1,2])/f, (R[0,2]-R[2,0])/f,
                                (R[1,0]-R[0,1])/f])
    if m == 1: return np.array([(R[2,1]-R[1,2])/f, s, (R[0,1]+R[1,0])/f,
                                (R[0,2]+R[2,0])/f])
    if m == 2: return np.array([(R[0,2]-R[2,0])/f, (R[0,1]+R[1,0])/f, s,
                                (R[1,2]+R[2,1])/f])
    return np.array([(R[1,0]-R[0,1])/f, (R[0,2]+R[2,0])/f, (R[1,2]+R[2,1])/f, s])

def segno_vicino(q0, q1):
    # to use BEFORE
    interpolating
    return -q1 if np.dot(q0, q1) < 0.0 else q1

qz = q_da_asse_angolo([0, 0, 1], np.pi/2)
qx = q_da_asse_angolo([1, 0, 0], np.pi/2)
print(np.round(qmul(qz, qx), 6))      # -> [0.5 0.5 0.5 0.5] (120
    degrees about (1,1,1))
print(np.round(qmul(qx, qz), 6))      # -> [0.5 0.5 -0.5 0.5] (120
    degrees about (1,-1,1))

```

The same chapter checks Rodrigues' formula (Theorem 28.2) by an independent route: summing in `numpy` the first 60 terms of the exponential series $\sum_n \theta^n K^n / n!$ recovers the matrix produced by the closed formula, with maximum deviation 1.7×10^{-16} over five different axis-angle pairs, including $\theta = 180^\circ$ (Example 28.3, Step 7). ✓

E.3.5 SLERP: checking constant angular velocity

From Chapter 30: the SLERP formula (Theorem 30.2) applied to $q_0 = 1$ and $q_1 = \cos 45^\circ + \sin 45^\circ k$ ($\Omega = 45^\circ$). At $t = \frac{1}{2}$ the two coefficients are equal, $\sin 22.5^\circ / \sin 45^\circ = 0.5411961$, and

$$\text{slerp}(q_0, q_1; \tfrac{1}{2}) = 0.5411961(1) + 0.5411961(0.7071068 + 0.7071068k) = 0.9238795 + 0.3826834k,$$

that is, exactly $\cos 22.5^\circ + \sin 22.5^\circ k$: the rotation halfway there. ✓ *Independent check:* a Python script evaluates the formula at 1001 values of t and finds $|\|\text{slerp}\| - 1| \leq 4.4 \cdot 10^{-16}$ and $|\text{angular velocity} - \Omega| \leq 7.6 \cdot 10^{-8}$ (the latter using difference quotients of step 10^{-4}). ✓

CHAPTER F

Additional exercises

Collection of step-by-step solved exercises, organized by topic, to consolidate the concepts of the volume. Each exercise is independent; the solutions recall the theorems of the corresponding chapters.

F.1. Plane curves and curvature

Exercise F.1. Calculate the curvature of the curve $y = \sin x$ at the maximum point $x = \pi/2$.

Solution. With $\gamma(t) = (t, \sin t)$: $\gamma' = (1, \cos t)$, $\gamma'' = (0, -\sin t)$. At $t = \pi/2$: $\gamma' = (1, 0)$, $\gamma'' = (0, -1)$. Curvature $\kappa = \frac{1 \cdot (-1) - 0 \cdot 0}{(1 + 0)^{3/2}} = -1$. The negative sign indicates that the curve bends downward; the osculating radius is $\rho = 1$, center at $(\pi/2, 0)$. $\square$

Exercise F.2. Determine the minimum radius of curvature of the ellipse $x^2/9 + y^2/4 = 1$.

Solution. Here $a = 3$, $b = 2$. The minimum radius is at the vertices of the major axis: $\rho_{\min} = b^2/a = 4/3$. The maximum, at the vertices of the minor axis, $\rho_{\max} = a^2/b = 9/2$. $\square$

Exercise F.3. Verify that the logarithmic spiral $\gamma(t) = e^t(\cos t, \sin t)$ intersects every ray emanating from the origin with the same angle.

Solution. The position vector is γ and the tangent $\gamma' = e^t(\cos t - \sin t, \sin t + \cos t)$. The angle ψ between γ and γ' satisfies $\cos \psi = \frac{\gamma \cdot \gamma'}{|\gamma||\gamma'|}$. Now $\gamma \cdot \gamma' = e^{2t}[(\cos t)(\cos t - \sin t) + (\sin t)(\sin t + \cos t)] = e^{2t}$, and $|\gamma| = e^t$, $|\gamma'| = \sqrt{2}e^t$, therefore $\cos \psi = 1/\sqrt{2}$, i.e., $\psi = 45^\circ$ constant: the logarithmic spiral is *equiangular* (hence its historical name). $\square$

F.2. Space curves

Exercise F.4. For the helix $\gamma(t) = (3 \cos t, 3 \sin t, 4t)$, calculate curvature, torsion and pitch.

Solution. With $a = 3$, $b = 4$: $a^2 + b^2 = 25$. Curvature $\kappa = a/(a^2 + b^2) = 3/25 = 0.12$; torsion $\tau = b/(a^2 + b^2) = 4/25 = 0.16$. The pitch (advance per turn) is $p = 2\pi b = 8\pi \approx 25.1$. The speed is $|\gamma'| = \sqrt{a^2 + b^2} = 5$. $\square$

Exercise F.5. Show that a curve with non-zero constant torsion and constant curvature is necessarily a helix.

Solution. By the fundamental theorem of space curves (Chapter 3), the functions κ, τ determine the curve up to rigid motion. If both are constant, the solution is unique and is realized by the helix with $a = \kappa/(\kappa^2 + \tau^2)$, $b = \tau/(\kappa^2 + \tau^2)$; therefore every curve with those data is that helix (up to rototranslation). $\square$

F.3. Rotations, quaternions, kinematics

Exercise F.6. Find the axis and angle of the rotation R that cyclically permutes the axes $\mathbf{e}_1 \rightarrow \mathbf{e}_2 \rightarrow \mathbf{e}_3 \rightarrow \mathbf{e}_1$.

Solution. $R = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ has trace 0, therefore from $\text{tr } R = 1 + 2 \cos \theta = 0$ we have $\cos \theta = -1/2$, $\theta = 120^\circ$. The axis is the eigenvector with eigenvalue +1: solving $R\mathbf{u} = \mathbf{u}$ we find $\mathbf{u} = (1, 1, 1)/\sqrt{3}$. The rotation is by 120° about the main diagonal. $\square$

Exercise F.7. Compose via quaternions a rotation of 180° about x followed by one of 180° about y . What rotation is obtained?

Solution. $q_x = \cos 90^\circ + \sin 90^\circ \mathbf{i} = \mathbf{i}$; $q_y = \mathbf{j}$. Composition (first x , then y): $q = q_y q_x = \mathbf{j}\mathbf{i} = -\mathbf{k}$. Since $-\mathbf{k}$ and $\mathbf{k}$ give the same rotation (2 : 1 coverage), q represents the rotation of 180° about z . Two rotations of π about orthogonal axes compose into a rotation of π about the third axis. $\square$

Exercise F.8. A planar 2R arm has $\ell_1 = \ell_2 = 1$. At which point in the plane is the TCP for $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$?

Solution. $x = \cos 30^\circ + \cos 90^\circ = 0.866 + 0 = 0.866$; $y = \sin 30^\circ + \sin 90^\circ = 0.5 + 1 = 1.5$. The TCP is at (0.866, 1.5), with orientation $\phi = \theta_1 + \theta_2 = 90^\circ$. $\square$

F.4. Sphere, geodesics, spherical trigonometry

Exercise F.9. What is the geodesic distance (on Earth, $R = 6371$ km) between two points on the same meridian at latitudes 30°N and 60°N ?

Solution. On the same meridian the central angle is the latitude difference, $\Delta = 30^\circ = \pi/6$. Distance $d = R\Delta = 6371 \cdot 0.5236 \approx 3336$ km. The meridians are great circles, therefore geodesics. $\square$

Exercise F.10. A spherical triangle has angles $A = B = C = 80^\circ$. What is its area on a sphere of radius $r = 2$?

Solution. Excess $E = (3 \cdot 80^\circ) - 180^\circ = 240^\circ - 180^\circ = 60^\circ = \pi/3$. Area $= Er^2 = \frac{\pi}{3} \cdot 4 = \frac{4\pi}{3} \approx 4.19$. Since the angles are $< 90^\circ$ the excess is modest: “almost Euclidean” triangle. $\square$

Exercise F.11. Verify the spherical cosine theorem for a triangle with $b = c = 90^\circ$ and $A = 50^\circ$.

Solution. $\cos a = \cos b \cos c + \sin b \sin c \cos A = \cos 90^\circ \cos 90^\circ + \sin 90^\circ \sin 90^\circ \cos 50^\circ = 0 + 1 \cdot 1 \cdot \cos 50^\circ = \cos 50^\circ$, therefore $a = 50^\circ = A$. When two sides are quadrants (90°), the third side equals the opposite angle: a notable property of birectangular triangles. $\square$

F.5. Dimension n and topology

Exercise F.12. Calculate the volume and area of the unit sphere in dimension $n = 6$ ($\Gamma(4) = 6$, $\Gamma(3) = 2$).

Solution. $\text{Vol}(B^6, 1) = \pi^3/\Gamma(4) = \pi^3/6 \approx 5.17$; $\text{Area}(S^5, 1) = 2\pi^3/\Gamma(3) = \pi^3/2 \approx 31.0$. One verifies $\text{Area} = \frac{d}{dr} \text{Vol} \Big|_{r=1} = 6 \cdot \text{Vol} = \pi^3$. $\square$

Exercise F.13. In $B^{50}(O, 1)$, what percentage of the volume is closer to the boundary than to the center (beyond $r = 0.5$)?

Solution. The fraction inside $r = 0.5$ is $0.5^{50} = 2^{-50} \approx 8.9 \times 10^{-16}$: practically nothing. Over 99.99999% of the volume lies beyond half the radius: in dimension 50 the “center” is empty, a manifestation of measure concentration. $\square$

Exercise F.14. What is the fundamental group of the torus T^2 and what does it imply for a robot with two revolute joints?

Solution. $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$: loops are classified by two integers (windings around the two “holes”). For a robot with two independent revolute joints, every periodic trajectory is characterized by how many times each joint completes full turns — information relevant for cable twist accumulation. $\square$

F.6. Interpolation and motion

Exercise F.15. Design a ZV shaper for a mode at $f_n = 5$ Hz, $\zeta = 0.1$: calculate K , T_d and the amplitudes.

Solution. $\omega_n = 2\pi \cdot 5 = 31.42$ rad/s; $\sqrt{1 - \zeta^2} = \sqrt{0.99} = 0.995$. $T_d = 2\pi/(\omega_n \cdot 0.995) = 0.201$ s, spacing $T_d/2 = 0.100$ s. $K = e^{-0.1\pi/0.995} = e^{-0.3157} = 0.729$. Amplitudes $A_1 = 1/(1 + K) = 0.578$ at $t = 0$, $A_2 = K/(1 + K) = 0.422$ at $t = 0.100$ s. $\square$

Exercise F.16. An axis performs a point-to-point motion of 200 mm with trapezoidal profile, $v_{\max} = 100$ mm/s, $a = 500$ mm/s². Calculate the total time.

Solution. $t_a = v/a = 0.2$ s, acceleration distance $\frac{1}{2}at_a^2 = 10$ mm (same in deceleration, 20 mm total). Constant-speed segment: $200 - 20 = 180$ mm in $180/100 = 1.8$ s. Total time $t = 0.2 + 1.8 + 0.2 = 2.2$ s. With an S-profile the time increases slightly but the jerk remains finite, reducing vibrations. $\square$

Exercise F.17. To represent exactly a quarter circle with a quadratic NURBS, what weight does the central control point have?

Solution. A circle arc is represented with a rational quadratic NURBS with three control points, where the central point (at the intersection of the tangents at the ends) has weight $w = \cos(\alpha/2)$, with α the angle subtended by the arc. For a quarter circle $\alpha = 90^\circ$, therefore $w = \cos 45^\circ = 1/\sqrt{2} \approx 0.707$. It is this “non-polynomial” weight that allows the exact representation of the circle. $\square$

F.7. Deepening and synthesis

Exercise F.18. A clothoid segment connects a straight line to an arc of radius $r = 30$ m on a road. If the clothoid parameter is $A = 60$ m (with $A^2 = rL$), how long is the segment L ?

Solution. From the relation $A^2 = rL$ we obtain $L = A^2/r = 3600/30 = 120$ m. The curvature grows linearly from 0 to $1/30 \approx 0.033 \text{ m}^{-1}$ over the 120 m: smooth transition, without curvature jumps. $\square$

Exercise F.19. Verify that the area of the unit sphere in $\mathbb{R}^3$ is four times the area of its great circle.

Solution. Sphere area $= 4\pi r^2$; great circle area $= \pi r^2$. The ratio is 4: the famous result of Archimedes, who wanted it engraved on his tomb together with the circumscribed cylinder (volume ratio 3 : 2). $\square$

Exercise F.20. A 6-axis robot must rotate the tool from an identity orientation to a rotation of 120° about $(1, 1, 1)/\sqrt{3}$. Write the target quaternion.

Solution. $q = \cos 60^\circ + \sin 60^\circ \frac{1}{\sqrt{3}}(1, 1, 1) = \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k}) = \frac{1}{2} + \frac{1}{2}(\mathbf{i} + \mathbf{j} + \mathbf{k}) = \frac{1}{2}(1, 1, 1, 1)$. It is a unit quaternion ($|q|^2 = 4 \cdot \frac{1}{4} = 1$); the SLERP from identity to q gives a smooth rotation at constant angular velocity. $\square$

Exercise F.21. On a machine tool, the axis resonance is at 30 Hz. At what frequency should the notch filter be centered and why?

Solution. The notch filter must be centered exactly on the resonance frequency, $\omega_n = 2\pi \cdot 30 \approx 188 \text{ rad/s}$ (30 Hz): it selectively attenuates that band in the velocity loop, suppressing the vibration without reducing gains at other frequencies. If the load varies and the resonance shifts, an *adaptive* notch that tracks ω_n is used. $\square$

Exercise F.22. Prove that, among all rectangles of fixed perimeter, the square has maximum area, and compare with the circle's optimality.

Solution. Rectangle with sides x , $p/2 - x$ (perimeter p): area $S(x) = x(p/2 - x)$, maximum for $S'(x) = p/2 - 2x = 0$, i.e., $x = p/4$: the square. It is the discrete analog of isoperimetry; among *all* figures (not just rectangles) of perimeter p , however, the absolute maximum is the circle (Chapter 16), with area $p^2/4\pi$ against $p^2/16$ for the square. $\square$

Exercise F.23. Calculate the Gaussian curvature of a cylinder of radius 5 and explain why a sheet metal can be wrapped into a cylinder but not onto a sphere.

Solution. The cylinder has Gaussian curvature $K = 0$ (one principal curvature is $1/5$, the other 0; the product is zero). By the Theorema Egregium (Chapter 7) K is preserved under isometry: plane and cylinder both have $K = 0$, so the sheet wraps without stretching. The sphere has $K = 1/r^2 \neq 0$: flattening it would require deforming lengths, impossible without stretching. $\square$

Exercise F.24. Do a great circle and a parallel at 60° latitude have the same length? On a spherical Earth of radius R , calculate both.

Solution. The great circle (e.g., the equator) has length $2\pi R$. The parallel at 60° has radius $R \cos 60^\circ = R/2$, therefore length $2\pi(R/2) = \pi R$: half. Parallels other than the equator are small circles, shorter, and are not geodesics (Chapter 11). $\square$

Exercise F.25. Verify numerically that composing two rotations of 90° (about x and about y) is not commutative.

Solution. With quaternions $q_x = \frac{1}{\sqrt{2}}(1 + \mathbf{i})$, $q_y = \frac{1}{\sqrt{2}}(1 + \mathbf{j})$: $q_x q_y = \frac{1}{2}(1 + \mathbf{i})(1 + \mathbf{j}) = \frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{ij}) = \frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{k})$, while $q_y q_x = \frac{1}{2}(1 + \mathbf{j})(1 + \mathbf{i}) = \frac{1}{2}(1 + \mathbf{j} + \mathbf{i} + \mathbf{ji}) = \frac{1}{2}(1 + \mathbf{i} + \mathbf{j} - \mathbf{k})$. The two $\mathbf{k}$ parts have opposite signs: $q_x q_y \neq q_y q_x$, i.e., $SO(3)$ is not commutative — the mathematical root of the importance of axis order in RPY programming. $\square$

Exercise F.26. What is the spherical excess of a triangle covering an entire hemisphere (three vertices on the equator at 120°)?

Solution. Such a “degenerate” triangle has its three sides on the equator and area equal to half the sphere $2\pi r^2$. From $E = \text{Area}/r^2$ we get $E = 2\pi$: the sum of angles exceeds π by 2π , i.e., equals 3π (three straight angles). It is the limiting maximal case: the excess of a spherical triangle is always $0 < E < 2\pi$. $\square$

F.8. Further solved problems

Exercise F.27. A robotic arm follows a helical trajectory around a tube (radius $a = 100$ mm, pitch $p = 50$ mm). Calculate curvature and torsion, and the helix inclination angle.

Solution. $b = p/(2\pi) = 50/(2\pi) \approx 7.96 \text{ mm}$; $a^2 + b^2 \approx 10063 \text{ mm}^2$. Curvature $\kappa = a/(a^2 + b^2) \approx 9.94 \times 10^{-3} \text{ mm}^{-1}$, torsion $\tau = b/(a^2 + b^2) \approx 7.9 \times 10^{-4} \text{ mm}^{-1}$. The inclination angle (between tangent and horizontal plane) is $\arctan(b/a) = \arctan(0.0796) \approx 4.6^\circ$: helix slightly inclined, almost circular. $\square$

Exercise F.28. Prove that the composition of two reflections with respect to lines forming an angle α is a rotation by 2α .

Solution. From Chapter 4, $S_\beta S_\gamma = R_{\beta-\gamma}$. If the two lines form an angle α , their axes differ by α and the product is $R_{2\alpha}$ (the rotation angle is twice the angle between the axes). It follows that every rotation factors into two reflections, with infinitely many ways to choose the pair: it is the germ of the Cartan–Dieudonné theorem. $\square$

Exercise F.29. A solar panel must point at the Sun: given the hour angle H and declination δ , set up the elevation h calculation using spherical trigonometry (latitude ϕ).

Solution. In the spherical triangle pole–zenith–Sun, applying the cosine theorem for sides one obtains $\sin h = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H$. From h the azimuth is obtained with the sine theorem. It is the same formula for antenna and telescope pointing: spherical trigonometry (Chapter 10) is the common tool for tracking on spherical surfaces. $\square$

Exercise F.30. Why is a trajectory with infinite jerk (ideal trapezoidal profile) harmful for a robot with elastic joints?

Solution. An acceleration jump (impulsive jerk) contains spectral components at all frequencies, including those of the elastic joints and transmissions resonances: it excites them, generating residual vibration that degrades precision and service life. An S-profile limits jerk and concentrates the command energy at low frequencies, while input shaping (Chapter 15) cancels the components at the resonance frequency. They are the two faces of motion “smoothness”: upstream (profile) and targeted (shaper). $\square$

Exercise F.31. Calculate how many unit circles are needed, at minimum, to completely surround an equal one in the plane, and verify with the kissing number.

Solution. Each external unit circle, touching the central one, subtends an angle of 60° from the center (the triangle of the three centers is equilateral, side 2). To cover 360° one needs $360/60 = 6$ circles: exactly the kissing number in the plane (Chapter 17). Six circles surround the seventh perfectly, in the hexagonal arrangement. $\square$

Exercise F.32. Find the radius of curvature of the parabola $y = x^2/(2p)$ (parameter p) at the vertex, and interpret p .

Solution. $y' = x/p$, $y'' = 1/p$. At the vertex $x = 0$: $\kappa = y''/(1 + y'^2)^{3/2} = 1/p$, therefore $\rho = p$. The parameter p of the parabola is exactly the radius of the osculating circle at the vertex: the parabola “starts” as a circle of radius p . $\square$

Exercise F.33. A geostationary satellite orbits at $R = 42164$ km from the Earth's center. What is its angular and linear velocity?

Solution. Geostationary means period $T = 86164$ s (sidereal day), therefore angular velocity $\omega = 2\pi/T \approx 7.29 \times 10^{-5}$ rad/s. Linear velocity $v = \omega R \approx 7.29 \times 10^{-5} \cdot 42164 \approx 3.07$ km/s. The motion is uniform circular: constant curvature $1/R$, the astronomical example of the circle as trajectory. $\square$

Exercise F.34. Prove that the principal curvatures of the torus vary in sign, and identify where $K > 0$, $K < 0$, $K = 0$.

Solution. On the torus (tube radius r , major radius R), the outer part is convex in both directions ($K > 0$), the inner part is saddle-shaped ($K < 0$), and the two top and bottom circles (where the tangent plane is “vertical”) have $K = 0$. The integral $\iint_M K dA = 0$ (Gauss–Bonnet, $\chi(\text{torus}) = 0$): the positive and negative regions cancel exactly. $\square$

Exercise F.35. A rotating axis must perform 270° without twisting the cable, which tolerates at most $\pm 360^\circ$. From which initial position is it convenient to start?

Solution. The joint's C-space is S^1 and cable configurations are classified by winding number ($\pi_1(S^1) = \mathbb{Z}$, Chapter 13). To stay within $[-360^\circ, +360^\circ]$ during a rotation of 270° , it is convenient to start “pre-loaded” in the opposite direction, e.g., from -135° , so as to reach $+135^\circ$ without exceeding the limits: the management of accumulated turns is a topological problem, not just a mechanical one. $\square$

Exercise F.36. Verify that the general formula for the volume of B^n gives, for $n = 2$ and $n = 3$, the known values.

Solution. $n = 2$: $\Gamma(2) = 1$, $\text{Vol} = \pi^1/1 \cdot r^2 = \pi r^2$ (circle area). $n = 3$: $\Gamma(5/2) = \frac{3}{4}\sqrt{\pi}$, $\text{Vol} = \pi^{3/2}/(\frac{3}{4}\sqrt{\pi}) r^3 = \frac{4}{3}\pi r^3$ (ball volume). The formula unifies circle and sphere as cases $n = 2, 3$. $\square$

Exercise F.37. Two successive reflections with respect to planes forming a dihedral of 30° : what rotation do they produce in $\mathbb{R}^3$?

Solution. In $\mathbb{R}^3$ the composition of two reflections with respect to planes intersecting along a line ℓ , forming a dihedral α , is a rotation about ℓ of angle 2α . With $\alpha = 30^\circ$ one obtains a rotation of 60° about the line of intersection: the spatial analog of the planar result (Cartan–Dieudonné). $\square$

Glossary

Abscissa of convergence

Infimum σ_c of the $\text{Re}(s)$ for which the Laplace integral converges; the abscissa of absolute convergence $\sigma_a \geq \sigma_c$ bounds absolute convergence instead (Definition 20.4, Chapter 20).

Aliasing

Overlap, after sampling, of components at different frequencies that become indistinguishable modulo the sampling frequency; it occurs when the signal exceeds the Nyquist frequency (Definition 24.11, Chapter 24).

Analytic signal (and Hilbert transform)

$x_a = x + i\mathcal{H}x$, where $\mathcal{H}$ is the filter with frequency response $-i \text{sgn}(\omega)$; the modulus of x_a is the envelope and the derivative of its phase is the instantaneous frequency (Definitions 24.12 and 24.13, Chapter 24).

Apollonian packing (Descartes quadruple)

Family of circles generated by recursively filling every curvilinear gap between three mutually tangent circles; the curvatures of four mutually tangent circles satisfy Descartes' circle theorem (Definition 32.6, Chapter 32).

Arc length

Parameter s such that $|\gamma'(s)| = 1$; measures the length traversed along the curve.

AWGN channel

Model $r = a + n$ with complex zero-mean Gaussian noise n , used to analyze the error probability of a constellation (Definition 26.8, Chapter 26).

Bessel's equation (and Bessel function)

$x^2 y'' + xy' + (x^2 - n^2)y = 0$; the solution bounded at the origin is $J_n(x) = \sum_k \frac{(-1)^k}{k!(n+k)!} (x/2)^{n+2k}$ (Definitions 18.6 and 18.7, Chapter 18).

Bézier curve

Curve $\sum_i b_{i,n}(t)P_i$ defined by Bernstein polynomials; passes through the first and last control point (Definition 30.1, Chapter 30).

BIBO (stability)

“Bounded Input, Bounded Output”: a system is BIBO stable if every bounded input produces a bounded output; equivalent to poles in $\text{Re}(s) < 0$ (continuous) or $|z| < 1$ (discrete) (Definitions 24.6, 20.2 and 21.6, Chapters 24, 20 and 21).

Bilinear transform (generalized)

Family of Möbius transformations ($z \mapsto \frac{z-1}{z+1}$ and variants) that discretize a continuous system by mapping the imaginary axis to the unit circle $\mathbb{T}$ (Definition 21.7, Chapter 21).

Bit reversal

Permutation that writes the index in binary on m digits and reads it backwards; it reorders the input (or output) of the radix-2 FFT computed in place (Definition 23.5, Chapter 23).

B-spline / NURBS

Locally controlled curves defined by basis functions on a knot vector; NURBS (rational) represent circle arcs and conics exactly (Definitions 30.2 and 30.3, Chapter 30).

Butterfly

Elementary operation $(a, b) \mapsto (a + wb, a - wb)$ on a pair of complex numbers with twiddle factor w ; it is the building block of every step of the FFT algorithm (Definition 23.3, Chapter 23).

Butterworth (filter)

Family of IIR filters characterized by a *maximally flat* magnitude response in the passband (no ripple, unlike Chebyshev or elliptic filters); in the digital domain they are typically designed in the continuous domain and discretized with the bilinear transform (Section 21.7, Chapter 21).

Cauchy integral formula

Representation of a holomorphic function via an integral over the circular boundary of a disk.

Characters of a group

Continuous homomorphisms from a topological group to $\mathbb{T}$; for $\mathbb{T}$ itself, the characters are $\chi_n(z) = z^n$, $n \in \mathbb{Z}$.

Circle fitting (geometric and algebraic)

Search for the circle that best approximates a sample of points: the geometric fit minimizes the sum of squares of signed distances from the circle, the algebraic fit the sum of squares of $\rho_i(c)^2 - r^2$, simpler to solve but less geometrically faithful (Definitions 31.2 and 31.9, Chapter 31).

Circular convolution

Product $(x \circledast y)[n] = \sum_m x[m] y[\langle n - m \rangle_N]$ between vectors of $\mathbb{C}^N$, with indices taken modulo N ; the DFT turns it into the pointwise product of the transforms (Definition 22.4, Chapter 22).

Circular membrane

Problem $u_{tt} = c^2 \Delta u$ in the disk with fixed boundary; the separation of time and space leads to Helmholtz's equation and to the normal modes of the drum, whose eigenvalues are linked to the zeros of the Bessel functions (Definition 18.5, Chapter 18).

Clothoid

Curve with linear curvature $\kappa(s) = \kappa_0 + cs$; Euler spiral, used in continuous-curvature transitions.

Compressed sensing

Technique that reconstructs sparse signals from a number of samples below the Nyquist limit, exploiting the signal's sparsity in a known domain (Remark 24.18, Chapter 24).

Constellation

Finite set of points of the complex plane used as symbols of a modulation (for example M -PSK or M -QAM), characterized by average energy per symbol and minimum distance between symbols (Definition 26.3, Chapter 26).

Curvature (signed)

$\kappa = \varphi'(s)$, rate of rotation of the tangent angle; for 3D curves $\kappa = |T'|$.

Cyclic prefix

Copy of the last N_g samples of the OFDM block prepended to the block itself; it turns linear convolution with the channel into circular convolution, removable with a one-tap equalizer per subcarrier (Definition 27.4, Chapter 27).

Darboux vector

$\omega = \tau T + \kappa B$; angular velocity of the Frenet frame.

Denavit–Hartenberg

Four-parameter convention (θ, d, a, α) for describing manipulator kinematics via homogeneous transformations in $SE(3)$ (Definition 29.8, Chapter 29).

DFT (Discrete Fourier Transform)

Finite N -sample version of the Fourier series, which uses the N -th roots of unity as its basis (Definition 22.1, Chapter 22).

Dirac comb

$\sum_{n \in \mathbb{Z}} \delta_n$; its Fourier transform is again a Dirac comb, a fact underlying sampling and periodization (Definition 19.10, Chapter 19).

Dirac delta

Functional $\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0)$ on a test function; it is the identity element of convolution among tempered distributions (Definition 19.8, Chapter 19).

Dirichlet kernel

$D_N(\theta) = \sum_{n=-N}^N e^{in\theta}$; the Fourier partial sum is the convolution of f with D_N .

Dirichlet problem (in the disc)

Find a harmonic function in the disk D_a with prescribed boundary data $u|_{\partial D_a} = f$; solved by series through separation of variables (Definition 18.2, Chapter 18).

Discrete Dirichlet kernel

N -point analog of the continuous Dirichlet kernel; the circular convolution of a signal with it is linked to its truncated DFT (Definition 22.6, Chapter 22).

Discrete-time Fourier transform (DTFT)

$\hat{x}(\Omega) = \sum_n x[n]e^{-in\Omega}$; it is the restriction of the Z-transform to the unit circle $\mathbb{T}$, when this is contained in the region of convergence (Definition 21.5, Chapter 21).

Double cover

$2 : 1$ homomorphism $SU(2) \rightarrow SO(3)$: q and $-q$ give the same rotation.

Eigenvalue

Scalar λ with $Av = \lambda v$, $v \neq 0$; for non-trivial plane rotations they are $e^{\pm i\theta}$.

Erdős–Ulam conjecture

Question (Ulam, 1945) whether there exists a set dense in the plane whose pairwise distances are all rational; on the circle such a dense set does exist (with a complete proof), which makes the planar case harder: there is no obvious candidate counterexample (Section 32.6, Definition 32.8, Chapter 32).

Euclidean group $SE(3)$ (rigid motion)

Maps $\mathbf{x} \mapsto R\mathbf{x} + \mathbf{p}$ with $R \in SO(3)$; represented by 4×4 homogeneous matrices, they form a matrix Lie group whose algebra is $\mathfrak{se}(3)$ (Definition 29.1, Chapter 29).

Euler's rotation theorem

Every element of $SO(3)$ is a rotation about an axis (eigenvector with eigenvalue $+1$).

Evolute

Locus of curvature centers of a curve; the normals of the curve are its tangents.

Eye diagram

Superposition of the traces of the received signal over intervals lasting one symbol period; its vertical opening visually measures the margin against intersymbol interference (Definition 26.15, Chapter 26).

FFT (Fast Fourier Transform)

Algorithm (Cooley–Tukey) that computes the DFT in $O(N \log N)$ operations instead of $O(N^2)$, by decimation in time or in frequency (Definition 23.4, Chapter 23).

First fundamental form

Metric $E du^2 + 2F du dv + G dv^2$; measures lengths and angles on the surface.

Fourier transform (on $\mathbb{R}$)

$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$, with inversion formula $f(t) = \frac{1}{2\pi} \int \hat{f}(\omega)e^{i\omega t} d\omega$ (Definition 19.2, Chapter 19).

Frenet–Serret

Moving frame $\{T, N, B\}$ and formulas $T' = \kappa N$, $N' = -\kappa T + \tau B$, $B' = -\tau N$.

Fundamental group π_1

Homotopy classes of loops; $\pi_1(S^1) = \mathbb{Z}$, $\pi_1(S^n) = 0$ for $n \geq 2$.

Gamma function

$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt$; generalizes the factorial, appears in the volume of B^n .

Gauss–Bonnet

Relates integrated curvature and angles: $\alpha + \beta + \gamma - \pi = K \text{Area}(T)$ on the sphere.

Gauss–Newton method

Iteration that, at each step, solves $J^\top J \delta = -J^\top d$ to minimize the linear model of the residual; converges quickly near the minimum but requires J of full rank (Definition 31.6, Chapter 31).

Gaussian curvature

$K = \kappa_1 \kappa_2$, product of principal curvatures; intrinsic (Theorema Egregium). Constant $1/r^2$ on the sphere.

Geodesic

Locally shortest path; purely normal acceleration; on the sphere it is an arc of a great circle.

Gibbs phenomenon

Persistent overshoot (approximately 9%) of the Fourier partial sum in a neighborhood of a jump discontinuity, which does not vanish as N grows.

Gimbal lock

Singularity of 3-angle charts (Euler/RPY) where one degree of freedom is lost.

Great circle

Intersection of a sphere with a plane through the center; they are the geodesics of S^2 .

Heat equation (on the circle)

$u_t = \kappa u_{\theta\theta}$ on $\mathbb{T}$; the solution with initial data f is the convolution of f with the heat kernel $H_t(\theta) = \frac{1}{2\pi} \sum_n e^{-\kappa n^2 t} e^{in\theta}$ (Definition 18.3, Chapter 18).

Hopf fibration

Non-trivial fibration $h : S^3 \rightarrow S^2$ with fiber S^1 , constructed with quaternions.

Input shaping

Feedforward technique that convolves the command with impulses to cancel residual vibration at a known frequency (ZV, ZVD shaper) (Chapter 30; see also Residual vibration).

Inscribed square problem (Toeplitz)

Conjecture (1911), still open, that every Jordan curve in the plane inscribes a square, that is, has four points that are its vertices (Definition 32.2, Chapter 32).

IQ signal (complex envelope)

Representation $s(t) = \text{Re}\{\tilde{s}(t)e^{i2\pi f_c t}\}$ of a bandpass signal via its complex envelope $\tilde{s}$, whose real and imaginary parts are the in-phase (I) and quadrature (Q) components (Definitions 26.1 and 26.2, Chapter 26).

Jerk

Derivative of acceleration; its limitation (S-curve profile) reduces vibrations and mechanical stresses (Definition 30.5, Chapter 30).

Kissing number

Maximum number of pairwise disjoint unit balls that can all be tangent to one fixed unit ball; in the plane it equals 6 (proved in this chapter) and in dimension 3 it equals 12 (Schütte–van der Waerden, 1953), but for most dimensions, starting at $n = 5$, the exact value remains an open problem (Definition 32.7, Theorem 32.4, Chapter 32).

Laplace operator

$\Delta u = \partial^2 u / \partial x^2 + \partial^2 u / \partial y^2$; u is called harmonic if $\Delta u = 0$ (Definition 18.1, Chapter 18).

Laplace transform

$F(s) = \int_0^\infty f(t)e^{-st}dt$; generalizes Fourier to non-integrable causal signals, basis of stability analysis in the continuous domain (Definition 20.1, Chapter 20).

Levenberg–Marquardt method

Damped variant of Gauss–Newton, $(J^\top J + \lambda I)\delta = -J^\top d$; always well defined and a descent direction, whatever the rank of J (Definition 31.7, Chapter 31).

LTI system (continuous-time)

Map that is linear and invariant under time translations between signal spaces; it is always of the form of a convolution with an impulse response (Definition 24.2, Chapter 24).

Matrix Lie group

Closed subgroup of $GL(n, \mathbb{R})$; it has an associated matrix Lie algebra via the exponential map (Definition 29.2, Chapter 29).

NLERP (normalized linear interpolation)

$\text{nlerp}(q_0, q_1; t) = \frac{(1-t)q_0 + tq_1}{\|(1-t)q_0 + tq_1\|}$; traces the same arc as SLERP but not at constant angular velocity (Definition 30.4, Chapter 30).

Notch filter

Band-stop filter $H_N(s) = \frac{s^2 + 2\zeta_z\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\omega_n s + \omega_n^2}$ that selectively attenuates the mechanical resonance of an axis without altering the other frequencies (Definition 30.8, Chapter 30).

OFDM (orthogonal frequency-division multiplexing)

Transmission over N subcarriers $\phi_k(t) = e^{2\pi i k \Delta f t}$ mutually orthogonal over the symbol time; the symbol is generated and demodulated with an IDFT/DFT (Definitions 27.1 and 27.2, Chapter 27).

Orthogonal group $O(n)$

Matrices with $A^\top A = I$; $SO(n)$ the subgroup with $\det = +1$ (rotations).

Osculating circle

Circle of radius $1/|\kappa|$ with second-order contact with the curve at a point.

PAPR (peak-to-average power ratio)

$\max_n |s[n]|^2$ divided by the average power of the samples; typically high in OFDM because of the sum of many independent subcarriers, it limits amplifier efficiency (Definition 27.9, Chapter 27).

Quaternion (unit)

$q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ with $|q| = 1$; the sphere S^3 , rotation group via $qp\bar{q}$.

Radius of curvature

$\rho = 1/|\kappa|$, radius of the osculating circle.

Residual vibration

Amplitude of the oscillation of a second-order elastic mode that persists after the end of the command; input shaping cancels it by convolving the command with a sequence of impulses (Definition 30.6, Chapter 30).

Riemann sphere

The sphere $S^2 \cong \mathbb{C} \cup \{\infty\}$ obtained by compactifying the complex plane with a point at infinity.

ROC (region of convergence)

Set of $z \in \mathbb{C}$ for which the Laurent series of the Z-transform converges absolutely; for a causal signal it is the exterior of a circle, for an anticausal one the interior (Definition 21.4, Chapter 21).

Rodrigues' formula

Axis-angle rotation: with $K = [\mathbf{u}]_{\times}$, $R = I + \sin \theta K + (1 - \cos \theta)K^2$.

Sampling theorem

(Nyquist-Shannon) A bandlimited signal $\omega_{\max}$ is exactly reconstructible if sampled at frequency $f_c \geq 2f_{\max}$ (Definition 24.10 and Theorem 24.8, Chapter 24).

Schwartz class

C^∞ functions all of whose derivatives decay faster than every power; it is the space of test functions on which tempered distributions are defined (Definition 19.5, Chapter 19).

Screw motion

Rotation by angle θ about an axis composed with a translation $h\theta$ along the same axis, of pitch h ; every rigid motion is a screw motion (Definition 29.5, Chapter 29).

Second fundamental form

$e du^2 + 2f du dv + g dv^2$; measures the curvature of the surface in space.

Singular configuration

Configuration of a kinematic chain in which the Jacobian (spatial or body) does not have full rank; at such a configuration at least one degree of freedom of the assigned task is locally lost (Definition 29.10, Chapter 29).

SLERP (spherical linear interpolation)

Path of constant angular velocity and minimal length between two unit quaternions, $\text{slerp}(q_0, q_1; t) = \frac{\sin((1-t)\Omega)}{\sin \Omega} q_0 + \frac{\sin(t\Omega)}{\sin \Omega} q_1$; it is the geodesic of S^3 between q_0 and q_1 (Theorem 30.2, Chapter 30).

Spectral window (and leakage)

Function $w[n]$ with which samples are weighted before the DFT to attenuate discontinuities at the edges of the observation window; without windowing (equivalent to a rectangular window) the energy of a tone not aligned to a bin spreads to neighboring bins, a phenomenon called *leakage* (Definition 22.7 and Remark 22.11, Chapter 22).

Sphere S^{n-1}

$\{x \in \mathbb{R}^n : |x| = r\}$, of intrinsic dimension $n - 1$.

Spherical excess

$E = \alpha + \beta + \gamma - \pi > 0$ for a spherical triangle, equal to $\text{Area}(T)/r^2$.

Stereographic projection

Conformal bijection between the sphere minus a pole and the plane; maps circles to circles or lines.

Tempered distribution

Continuous linear functional on the Schwartz class $\mathcal{S}(\mathbb{R})$; generalizes slowly growing functions and includes the Dirac delta and comb (Definition 19.7, Chapter 19).

Theorema Egregium

Gaussian curvature depends only on the first fundamental form.

Torsion

τ defined by $B' = -\tau N$; measures how much a 3D curve deviates from being planar.

Twist / $\mathfrak{se}(3)$

Element of the Lie algebra of rigid motions, pair (angular velocity, linear velocity); generates helical motions via exponential (Theorem 29.3, Chapter 29).

Wave equation (on the circle)

$u_{tt} = c^2 u_{\theta\theta}$ on $\mathbb{T}$, with initial position and velocity data; solved by factorizing the wave operator (Definition 18.4, Chapter 18).

Z-transform

$X(z) = \sum_n x[n]z^{-n}$; the discrete analog of Laplace, with $\mathbb{T}$ as the critical stability locus (Definition 21.3, Chapter 21).

Formula Cheat Sheet

Plane curves (Ch. 2)

- $\kappa = \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}}$; Frenet: $T' = \kappa N$, $N' = -\kappa T$
- Osculating circle: $\rho = 1/|\kappa|$, $c = \gamma + \frac{1}{\kappa}N$
- Fund. theorem: $\kappa(s) \Rightarrow$ curve (up to isometry); clothoid $\kappa(s) = \kappa_0 + cs$

Space curves (Ch. 3)

- $T' = \kappa N$, $N' = -\kappa T + \tau B$, $B' = -\tau N$
- $\kappa = \frac{|\gamma' \times \gamma''|}{|\gamma'|^3}$, $\tau = \frac{\det(\gamma', \gamma'', \gamma''')}{|\gamma' \times \gamma''|^2}$

- Helix: $\kappa = \frac{a}{a^2 + b^2}$, $\tau = \frac{b}{a^2 + b^2}$

Global theorems (Ch. 4)

- Umlaufsatz: $\oint \kappa ds = 2\pi \text{ind}(\gamma)$
- Isoperimetric: $L^2 \geq 4\pi A$ (equality: circle)
- Global Gauss–Bonnet: $\iint_M K dA = 2\pi\chi(M)$, $\chi = 2 - 2g$

Isometries and the complex circle (Ch. 5–6)

- $O(2)$: $A^\top A = I$; R_θ ($\det +1$), S_θ ($\det -1$); $SO(2) \cong \mathbb{T}$
- Roots of unity: $\zeta_k = e^{2\pi i k/n}$, $\sum \zeta_k = 0$
- Möbius: $z \mapsto \frac{az+b}{cz+d}$; inversion $z \mapsto r^2/\bar{z}$

Sphere (Ch. 7)

- Spherical distance: $d(A, B) = r\theta = r \arccos \frac{\langle A, B \rangle}{r^2}$
- Girard: $\text{Area}(T) = r^2(A + B + C - \pi)$
- Conformal stereographic map: $E = G$, $F = 0$; sends circles to circles or lines

Spherical trigonometry (Ch. 8)

- Cosine law (sides): $\cos a = \cos b \cos c + \sin b \sin c \cos A$
- Sines: $\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c}$
- L'Huilier: $\tan \frac{E}{4} = \sqrt{\tan \frac{s}{2} \tan \frac{s-a}{2} \tan \frac{s-b}{2} \tan \frac{s-c}{2}}$

Geodesics and surfaces (Ch. 9–10)

- Euler–Lagrange: $\frac{d}{ds} \frac{\partial L}{\partial q'} - \frac{\partial L}{\partial q} = 0$
- Geodesics: $\ddot{u}^k + \Gamma_{ij}^k \dot{u}^i \dot{u}^j = 0$; Clairaut $r \sin \psi = \text{const}$
- $K = \frac{eg - f^2}{EG - F^2}$; Theorema Egregium: K is intrinsic

Dimension n and topology (Ch. 11–13)

- $\text{Vol}(B^n, r) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)} r^n$
- $\pi_1(S^1) \cong \mathbb{Z}$; winding number $w(\gamma) = \frac{1}{2\pi i} \oint_\gamma \frac{dz}{z}$
- Characters $\chi_n(z) = z^n$; $\int_{S^1} \chi_n \overline{\chi_m} dm = \delta_{nm}$

Circle packing (Ch. 14)

- Descartes: $(k_1 + k_2 + k_3 + k_4)^2 = 2 \sum k_i^2$
- Kissing number $\tau_2 = 6$; density $\delta_\Delta = \pi/(2\sqrt{3})$

Complex and harmonic analysis on the disc (Ch. 15–18)

- Cauchy: $f(z) = \frac{1}{2\pi i} \oint \frac{f(\zeta)}{\zeta - z} d\zeta$
- Schwarz: $f(0) = 0$, $|f| \leq 1 \Rightarrow |f(z)| \leq |z|$
- Poisson: $P_r(t) = \frac{1 - r^2}{1 - 2r \cos t + r^2}$; mean value $u(w) = \frac{1}{2\pi} \int u$

Transforms (Ch. 19–21)

- Fourier: $\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} dt$
- Laplace: $F(s) = \int_0^\infty f(t) e^{-st} dt$, ROC: half-plane $\text{Re } s > \sigma_c$
- Zeta: $X(z) = \sum_n x[n] z^{-n}$, ROC: annulus; DTFT = $X|_{|z|=1}$; BIBO $\iff$ poles in $|z| < 1$

DFT and FFT (Ch. 22–23)

- $X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}$, $\omega_N = e^{-2\pi i/N}$
- Orthogonality: $\sum_n \omega_N^{jn} = N\delta(N \mid j)$; circular convolution $\leftrightarrow$ product; Parseval $\sum |x|^2 = \frac{1}{N} \sum |X|^2$
- Butterfly: $X[k] = E[k] + \omega_N^k O[k]$, $X[k + \frac{N}{2}] = E[k] - \omega_N^k O[k]$; cost $\frac{N}{2} \log_2 N$

Signals (Ch. 24)

- Nyquist–Shannon: $f_c \geq 2f_{\max}$, $x(t) = \sum_n x(nT) \text{sinc}(\frac{t-nT}{T})$
- LTI: $y = h * x$; BIBO $\iff h \in L^1$, gain = $\|h\|_1$

SO(3), Lie theory, trajectories (Ch. 28–30)

- Rodrigues: $R = I + \sin \theta K + (1 - \cos \theta) K^2$, $K = [\mathbf{u}]_\times$
- Quaternion: $q = \cos \frac{\theta}{2} + \mathbf{u} \sin \frac{\theta}{2}$; rotation $p \mapsto qp\bar{q}$
- SLERP: $\text{slerp} = q_0(\bar{q}_0 q_1)^t$; exp: $\mathfrak{se}(3) \rightarrow SE(3)$

Fitting (Ch. 31)

- Normal equations: $\sum_i d_i = 0$, $\sum_i d_i \mathbf{u}_i = 0 \iff c = \bar{p} - \frac{\tau}{n} \sum_i \mathbf{u}_i$
- Gauss–Newton: $J^\top J \delta = -J^\top d$

Supplementary apparatus

Register of dependencies and scope

The proofs reconstruct the new results from the stated prerequisites. They do not rebuild the real numbers, integral calculus, linear algebra or Gaussian distributions axiomatically. Labels containing **ext**- identify additions only: they do not certify results in the original text.

Area	Dependency or limitation
Conformal mapping	Cauchy, Rouché and Schwarz are prerequisites from Chapter 15. Holomorphic roots, Montel normality, limits of injective maps and the extremal argument are proved in the additions. A reference to Montel does not replace its proof.
Linear control	Cayley–Hamilton, Jordan form and square roots of positive symmetric matrices are algebraic prerequisites. Rank criteria, multi-input placement, separation and finite/infinite-horizon Riccati results are proved.
Kalman / MEKF	The linear result requires finite second moments; optimality among all estimators requires Gaussianity. The MEKF uses local linearization and a stated sampled-noise model: no global convergence or nonlinear optimality is proved.
Mechanics and actuation	Rigid bodies, positive inertias, ideal constraints and virtual work are model assumptions. Lagrange and Newton–Euler are derived for that model; computed-torque cancellation is exact only with an exact model, available state and feasible torques.
Gauss circle problem	Exact counting and the elementary bound are proved. Advanced results attributed to Hardy, Huxley and Li–Yang are external, with source status stated explicitly. The conjecture is not presented as a theorem.

Actual verification results

The run with seed 20260910 passed all 40 checks recorded in `verifiche/risultati_numerici.json`. Versions, tolerances and unrounded results are in that file. The tables give comparison errors, not universal rigorous bounds on rounding error. None of these checks replaces a proof or validation on real machinery.

Comparison	Error	Threshold
Strip: two formulas and inverses	$1.69 \cdot 10^{-15}$	$1.00 \cdot 10^{-12}$
Poles: Ackermann / numerical algorithm	$1.78 \cdot 10^{-15}$	$1.00 \cdot 10^{-12}$
Gramian: quadrature / exponential	$5.69 \cdot 10^{-16}$	$1.00 \cdot 10^{-12}$
Kalman: recursion / joint Gaussian	$3.33 \cdot 10^{-14}$	$1.00 \cdot 10^{-10}$
MEKF: transition Jacobian	$2.73 \cdot 10^{-12}$	$1.00 \cdot 10^{-7}$
MEKF: reset Jacobian	$1.95 \cdot 10^{-11}$	$1.00 \cdot 10^{-7}$
Lagrange / Newton–Euler: 100 cases	$5.33 \cdot 10^{-15}$	$1.00 \cdot 10^{-11}$
Dynamics: DOP853 / Radau	$6.90 \cdot 10^{-12}$	$1.00 \cdot 10^{-7}$
Discrete LQR: recursion / batch minimum	$7.77 \cdot 10^{-16}$	$1.00 \cdot 10^{-11}$
Tustin PID: formula / recurrence	$4.58 \cdot 10^{-16}$	$1.00 \cdot 10^{-8}$

The symbolic equalities between Lagrangian and recursive dynamics, and the Christoffel identity, have exactly zero residual in SymPy. The square of the deadbeat matrix is also symbolically zero; in floating-point arithmetic the state residual after two steps is about $1.11 \cdot 10^{-16}$.

Attitude estimation

In the experiment described in Chapter 35, the final MEKF error is 0.00735314 rad, against 0.338431 rad for gyroscope-only integration. The root-mean-square error over the last 300 samples is respectively 0.00543608 and 0.316953 rad. These are values from this single realization, not probabilistic guarantees. The maximum

discrepancy between the quaternion implementation with analytic Jacobians and the matrix implementation with finite differences is $5.24 \cdot 10^{-13}$ rad.

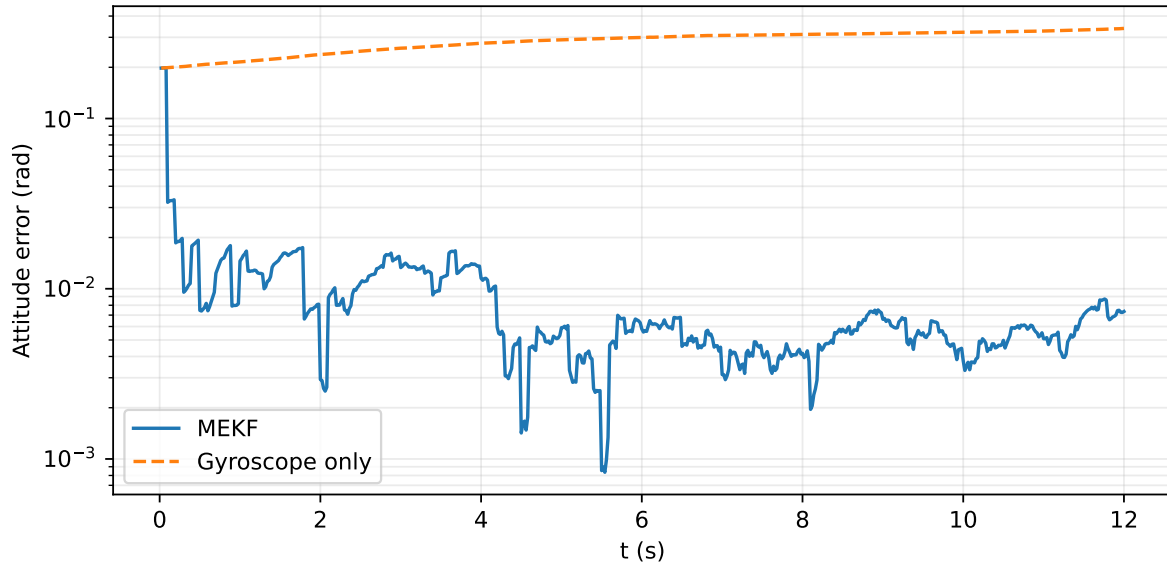


Figure F.1: Geodesic attitude error in radians. The vertical scale is logarithmic; data are in `mekf_traccia.csv`.

2R-arm tracking

The comparison between DOP853 on the Lagrangian formulation and Radau on dynamics reconstructed from Newton–Euler gives a maximum position discrepancy of $6.90 \cdot 10^{-12}$ rad. Comparison with the analytic error solution gives $6.88 \cdot 10^{-12}$ rad. The stated threshold is 10^{-7} rad, not a precision guarantee for every other problem.

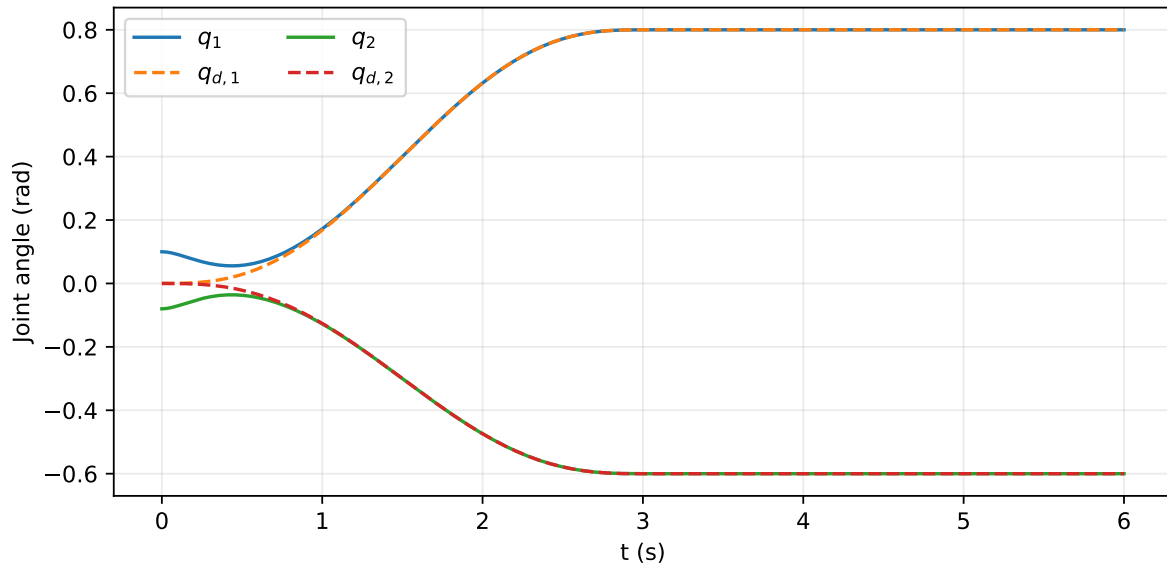


Figure F.2: Joint positions and quintic references from Chapter 36. The two integrations are indistinguishable at the scale of the plot; their separate data remain in `dinamica_traccia.csv`.

PID: comparison on the same signal

For the parameters of Chapter 37, the continuous controller has $C(i\Omega) = 2.4 + 0.7i$. The recurrence response is estimated from 9000 samples after discarding the transient and compared with rational substitution at $z = e^{i\Omega T}$. Phase is in radians.

	$ C_d $	$\arg C_d$	$ C_d - C(i\Omega) $
FE	2.48026224	0.29965243	0.04414666
BE	2.51807581	0.26857060	0.04225694
TU	2.50064796	0.28391994	0.00072030

FE denotes forward Euler, BE backward Euler, and TU Tustin. In this example Tustin approximates the continuous response more closely: this does not imply a universal ranking of control performance.

Exact lattice counts

r	$N(r)$	$E(r)$
0	1	1.00000000
1	5	1.85840735
5	81	2.46018366
10	317	2.84073464
100	31417	1.07346410
1000	3141549	-43.65358979

Row counting with an integer square root and direct disk-membership counting agree exactly. The digits of $E(r)$, by contrast, depend on the numerical approximation of π .

Trilingual glossary and index of new topics

Page references belong to this supplement. In the complete integration the same labels are recalculated by LaTeX; the original chapter numbers stay unchanged.

Term / equivalents in the other languages	Page
Conformal mapping / Rappresentazione conforme	458
Normal family / Famiglia normale	457
Lattice points / Punti reticolari	1347
Controllability / Controllabilità	1274
Observability / Osservabilità	1274
Pole placement / Assegnazione dei poli	1275
Detectability / Rilevabilità	1277
Linear quadratic regulator / Regolatore lineare quadratico	1276
Innovation / Innovazione	1282
Multiplicative error / Errore moltiplicativo	1284
Right Jacobian / Jacobiano destro	1286
Mass matrix / Matrice di massa	1289
Computed torque / Coppia calcolata	1293
Zero-order hold / Tenuta di ordine zero	1297
Deadbeat control / Azzeramento in tempo finito	1299
Frequency warping / Distorsione in frequenza	1298

In English, *circle* denotes the curve, *disk* the region, and *circumference* its length. The spelling *disc*, when it appears in an original source or received chapter title, does not change the mathematical meaning.

Bibliography

General bibliography of Volume III, gathering in a single list, by topic and without duplicates, the main sources of the 33 chapters (each chapter still carries, at its end, its own *Sources* environment with pointwise references to theorems and sections). International sources (United States, Europe, China, India, South America) collected during writing; the complete list, chapter by chapter, is in the volume's source register.

General reference texts

- M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, 2nd ed., Dover, 2016 (1st ed. Prentice-Hall, 1976).
- D. J. Struik, *Lectures on Classical Differential Geometry*, 2nd ed., Dover, 1988.
- A. Pressley, *Elementary Differential Geometry*, 2nd ed., Springer Undergraduate Mathematics Series, 2010.
- A. Gray, E. Abbena, S. Salamon, *Modern Differential Geometry of Curves and Surfaces with Mathematica*, 3rd ed., Chapman & Hall/CRC, 2006.
- W. Kühnel, *Differential Geometry: Curves — Surfaces — Manifolds*, 3rd ed., Student Mathematical Library 77, American Mathematical Society, 2015.
- T. Shifrin, *Differential Geometry: A First Course in Curves and Surfaces*, University of Georgia, 2021.
- J. Gallier, *Geometric Methods and Applications for Computer Science and Engineering*, 2nd ed., Springer, Texts in Applied Mathematics 38, ch. 19.
- I. M. Gelfand, S. V. Fomin, *Calculus of Variations*, Dover, 2000 (1st ed. Prentice-Hall, 1963).
- A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.
- J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000.
- R. Murray, Z. Li, S. Sastry, *A Mathematical Introduction to Robotic Manipulation*, CRC Press, 1994.
- L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw-Hill, 1979.
- W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw-Hill, 1987.
- W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976.

Primary historical sources

- Archimedes, *On the Measurement of a Circle; On the Sphere and Cylinder*, 3rd century BC.
- Apollonius of Perga, *Conics*, books I–IV, 3rd century BC.
- Liu Hui, Zu Gengzhi, commentary on the *Nine Chapters on the Mathematical Art* and *Zhui Shu*, 3rd–6th century AD.
- Bhaskara II, *Siddhanta Shiromani*, 1150.
- Nasir al-Din al-Tusi, *al-Tadhkira fi 'ilm al-hay'a*, ca. 1261.
- G. Cardano, *De subtilitate libri XXI*, Nuremberg, 1550.
- A. Girard, *Invention nouvelle en l'algèbre*, Amsterdam, 1629.
- J. Wallis, *Tractatus duo. Prior, de cycloide...*, Oxford, 1659.
- G. P. Dandelin, *Mémoire sur quelques propriétés remarquables de la focale parabolique*, Nouveaux Mémoires de l'Académie Royale de Bruxelles, vol. 2, 1822, pp. 171–202.
- C. F. Gauss, *Disquisitiones generales circa superficies curvas*, Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores, vol. VI, Göttingen, 1828.
- L. Euler, *Formulae generales pro translatione quacunque corporum rigidorum*, Novi Commentarii academiae scientiarum Petropolitanae 20 (1776), pp. 189–207.
- O. Rodrigues, *Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace...*, Journal de Mathématiques Pures et Appliquées 5 (1840), pp. 380–440.
- W. R. Hamilton, *Lectures on Quaternions*, Hodges and Smith, 1853; *Elements of Quaternions*, Longmans, Green & Co., 1866.

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- M. Chasles, *Note sur les propriétés générales du système de deux corps semblables entr'eux...*, Bulletin des Sciences Mathématiques, Astronomiques, Physiques et Chimiques 14, 1830, pp. 321–326.
 - F. Lindemann, transcendence of π , 1882.
 - H. Hopf, *Über die Drehung der Tangenten und Sehnen ebener Kurven*, Compositio Mathematica 2 (1935), pp. 50–62; *Über die Abbildungen der dreidimensionalen Sphäre auf die Kugelfläche*, Mathematische Annalen 104 (1931), pp. 637–665.
 - A. Hurwitz, *Sur quelques applications géométriques des séries de Fourier*, Annales scientifiques de l'École Normale Supérieure, s. 3, 19 (1902), pp. 357–408.
 - E. Barbier, *Note sur le problème de l'aiguille et le jeu du joint couvert*, Journal de mathématiques pures et appliquées, s. 2, 5 (1860), pp. 273–286.
 - W. Blaschke, *Konvexe Bereiche gegebener konstanter Breite und kleinsten Inhalts*, Mathematische Annalen 76 (1915), pp. 504–513.
 - H. Lebesgue, *Sur le problème des isopérimètres et sur les domaines de largeur constante*, Bulletin de la Société Mathématique de France, Comptes Rendus 7 (1914), pp. 72–76.
 - S. Mukhopadhyaya, *New methods in the geometry of a plane arc*, Bulletin of the Calcutta Mathematical Society 1 (1909), pp. 31–37.
 - A. Kneser, *Bemerkungen über die Anzahl der Extreme der Krümmung auf geschlossenen Kurven...*, Festschrift Heinrich Weber, Teubner, 1912, pp. 170–180.
 - J. Pál, *Ein Minimumproblem für Ovale*, Mathematische Annalen 83 (1921), pp. 311–319.
 - T. Bonnesen, W. Fenchel, *Theorie der konvexen Körper*, Springer, 1934.
 - T. Radó, *Über den Begriff der Riemannschen Fläche*, Acta Litterarum ac Scientiarum Regiae Universitatis Hungaricae Francisco-Josephinae 2 (1926), pp. 101–121.
 - Á. Császár, *A polyhedron without diagonals*, Acta Scientiarum Mathematicarum (Szeged) 13 (1949–1950), pp. 140–142.
 - S.-S. Chern, *A Simple Intrinsic Proof of the Gauss–Bonnet Formula for Closed Riemannian Manifolds*, Annals of Mathematics 45 (1944), pp. 747–752.
 - H. Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Mathematische Annalen 77 (1916), pp. 313–352.
 - A. Thue, *Über die dichteste Zusammenstellung von kongruenten Kreisen in einer Ebene*, Skrifter Videnskabs-Selskabet i Christiania, 1910, no. 1.
 - L. Fejes Tóth, *Lagerungen in der Ebene, auf der Kugel und im Raum*, Grundlehren der mathematischen Wissenschaften 65, Springer, 1953.
 - L. R. Ford, *Fractions*, The American Mathematical Monthly 45 (1938), pp. 586–601.
 - G. H. Hardy, E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
 - C. E. Shannon, *Communication in the presence of noise*, Proceedings of the IRE 37 (1949), pp. 10–21.
 - H. Nyquist, *Certain topics in telegraph transmission theory*, Transactions of the AIEE 47 (1928), pp. 617–644; *Regeneration Theory*, Bell System Technical Journal 11 (1932), pp. 126–147.
 - D. Gabor, *Theory of communication. Part 1: The analysis of information*, Journal of the Institution of Electrical Engineers, Part III, 93 (1946), pp. 429–441.
 - H. W. Bode, *Network Analysis and Feedback Amplifier Design*, D. Van Nostrand, 1945.
 - P. H. Smith, *Transmission Line Calculator*, Electronics 12 (1939), pp. 29–31; *An Improved Transmission Line Calculator*, Electronics 17 (1944), pp. 130–133 and 318–325.
 - W. R. Evans, *Graphical Analysis of Control Systems*, Transactions of the AIEE 67 (1948), pp. 547–551.
 - G. Zames, *On the input-output stability of time-varying nonlinear feedback systems*, IEEE Transactions on Automatic Control AC-11 (1966).
 - V. M. Popov, *Absolute stability of nonlinear systems of automatic control*, Automation and Remote Control 22 (1962), pp. 857–875.
 - E. I. Jury, *Theory and Application of the z-Transform Method*, John Wiley & Sons, 1964.
 - J. W. Cooley, J. W. Tukey, *An algorithm for the machine calculation of complex Fourier series*, Mathematics of Computation 19 (1965), pp. 297–301.
 - J. Stuelpnagel, *On the Parametrization of the Three-Dimensional Rotation Group*, SIAM Review 6 (1964), pp. 422–430.
 - S. W. Shepperd, *Quaternion from rotation matrix*, Journal of Guidance and Control 1 (1978).
 - K. Shoemake, *Animating rotation with quaternion curves*, Computer Graphics (SIGGRAPH '85 Proceedings) 19 (1985), pp. 245–254.
-

-
- N. C. Singer, W. P. Seering, *Preshaping Command Inputs to Reduce System Vibration*, ASME Journal of Dynamic Systems, Measurement, and Control 112 (1990), pp. 76–82.
 - J. Denavit, R. S. Hartenberg, *A kinematic notation for lower-pair mechanisms based on matrices*, ASME Journal of Applied Mechanics, 1955, pp. 215–221.
 - O. Toeplitz, *Ueber einige Aufgaben der Analysis situs*, Verhandlungen der Schweizerischen Naturforschenden Gesellschaft, 1911, no. 4, p. 197.
 - I. Kása, *A circle fitting procedure and its error analysis*, IEEE Transactions on Instrumentation and Measurement IM-25 (1976), pp. 8–14.
 - R. Osserman, *The Four-or-More Vertex Theorem*, The American Mathematical Monthly 92 (1985), pp. 332–337.

Differential geometry: curves, surfaces and global theorems

- M. Powell, *M435: Introduction to Differential Geometry*, course notes, University of Glasgow.
- K. Wang, *Fundamental Theorem of the Local Theory of Curves*, University of Chicago REU, 2018.
- L. C. B. da Silva, *Differential Geometry of Rotation Minimizing Frames, Spherical Curves, and Quantum Mechanics of a Constrained Particle*, thesis, arXiv:1806.08830, 2018.
- E. Ghys, S. Tabachnikov, V. Timorin, *Osculating curves: around the Tait–Kneser theorem*, The Mathematical Intelligencer 35 (2013), pp. 61–66.
- Y.-B. Jia, *Geodesics*, Com S 477/577 Notes, Iowa State University, 2015.
- A. C. Clairaut, memoirs to the Académie Royale des Sciences, Paris, 1733 and 1739.
- Lee-Peng Teo, *The Brioschi Formula for the Gaussian Curvature*, arXiv:2404.00835, 2024.
- R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge University Press, 2014.
- I. M. Yaglom, V. G. Boltyanskii, *Convex Figures*, Holt, Rinehart and Winston, 1961.
- H. Rademacher, O. Toeplitz, *The Enjoyment of Mathematics*, Princeton University Press, 1957.
- G. D. Chakerian, *Sets of Constant Width*, Pacific Journal of Mathematics 19 (1966), pp. 13–21.
- E. Harrell, *A direct proof of a theorem of Blaschke and Lebesgue*, Journal of Geometric Analysis 12 (2002), pp. 81–88.
- A. Petrunin, S. Zamora Barrera, *Moon in a puddle and the four-vertex theorem*, arXiv:2107.08455, 2021; The American Mathematical Monthly 129 (2022), pp. 475–479.
- D. DeTurck, H. Gluck, D. Pomerleano, D. S. Vela-Vick, *The four vertex theorem and its converse*, Notices of the American Mathematical Society 54 (2007), pp. 192–207.
- C. Thomassen, *The Jordan–Schönflies theorem and the classification of surfaces*, The American Mathematical Monthly 99 (1992), pp. 116–130.
- P. J. Federico, *Descartes on Polyhedra*, Sources in the History of Mathematics and Physical Sciences 4, Springer-Verlag, 1982.
- ISO 1101, *Geometrical product specifications (GPS) — Geometrical tolerancing*, International Organization for Standardization.
- ISO 53:1998, *Cylindrical gears for general and heavy engineering — Standard basic rack tooth profile*.
- S. Oh, Q. Chen, H. E. Tseng, G. Pandey, G. Orosz, *Sharable Clothoid-based Continuous Motion Planning for Connected Automated Vehicles*, arXiv:2312.10880, 2023.
- A. Shaju, S. Southward, M. Ahmadian, *Enhancing Autonomous Vehicle Navigation with a Clothoid-Based Lateral Controller*, Applied Sciences 14 (2024), no. 5, art. 1817.

Isometries, the complex circle and Möbius transformations

- D. Klain, *Lecture notes on orthogonal matrices (with exercises)*, University of Massachusetts Lowell, 2004.
- RES.18-011 *Algebra I Student Notes*, MIT OpenCourseWare, 2021 (edited by M. Artin).
- A. Felikson, *Geometry III/IV — Circles in Euclidean geometry: some facts and proofs*, Durham University.
- T. Needham, *Visual Complex Analysis*, Clarendon Press–Oxford University Press, 1997.
- J. Orloff, *18.04 Complex Variables with Applications*, MIT OpenCourseWare, spring 2018.
- T. W. Judson, *Abstract Algebra: Theory and Applications*, Stephen F. Austin State University.
- J. B. Conway, *Functions of One Complex Variable I*, 2nd ed., Graduate Texts in Mathematics 11, Springer-Verlag, 1978.

The sphere and spherical trigonometry

- D. Hilbert, S. Cohn-Vossen, *Anschauliche Geometrie*, Springer, 1932 (English trans. *Geometry and the Imagination*, Chelsea, 1952).
- M. Berger, *Géométrie*, vols. 1–5, Cedic/Fernand Nathan, 1977 (English trans. *Geometry I–II*, Springer, 1987).
- H. S. M. Coxeter, *Introduction to Geometry*, 2nd ed., John Wiley & Sons, 1969; *The problem of Apollonius*, The American Mathematical Monthly 75 (1968), pp. 5–15.
- I. Todhunter, *Spherical Trigonometry, For the Use of Colleges and Schools*, 5th ed., Macmillan and Co., 1886.
- National Geospatial-Intelligence Agency, *Department of Defense World Geodetic System 1984*, NGA.STND.0036_1.0.0_WGS84, 2014.
- E. Y.-M. Chiang, course notes for MATH 5030, Hong Kong University of Science and Technology.
- J. Shurman, *Stereographic projection is conformal*, course notes for Math 311, Reed College.
- University of California, Riverside, *Length minimizing property of great circles*, Math 138A, 2006.
- OpenStreetMap, Nominatim geocoding service.

Dimension n , Lie groups and the topology of the circle

- J. Stillwell, *Naive Lie Theory*, Springer, 2008.
- K. Ball, *An Elementary Introduction to Modern Convex Geometry*, in *Flavors of Geometry*, MSRI Publications 31, Cambridge University Press, 1997, pp. 1–58.
- M. Ledoux, *The Concentration of Measure Phenomenon*, Mathematical Surveys and Monographs 89, American Mathematical Society, 2001.
- V. D. Milman, G. Schechtman, *Asymptotic Theory of Finite Dimensional Normed Spaces*, Lecture Notes in Mathematics 1200, Springer, 1986.
- A. N. Gorban, I. Y. Tyukin, *Blessing of dimensionality: mathematical foundations of the statistical physics of data*, Philosophical Transactions of the Royal Society A 376 (2018), art. 20170237; arXiv:1801.03421.
- R. Bellman, *Dynamic Programming*, Princeton University Press, 1957.
- M. Lackenby, *Topology and Groups*, course notes, University of Oxford, 2017.
- W. Rudin, *Fourier Analysis on Groups*, Interscience Tracts in Pure and Applied Mathematics 12, Interscience/Wiley, 1962.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed., CRC Press/Chapman & Hall, 2016.
- E. M. Stein, R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, 2003.
- Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, 2004.
- T. Bröcker, T. tom Dieck, *Representations of Compact Lie Groups*, Graduate Texts in Mathematics 98, Springer, 1985.
- F. W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Graduate Texts in Mathematics 94, Springer, 1983.

Tangency and packing of circles

- H.-C. Chang, L.-C. Wang, *A Simple Proof of Thue’s Theorem on Circle Packing*, arXiv:1009.4322, 2010.
- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der mathematischen Wissenschaften 290, Springer, 1999.
- T. C. Hales, *A proof of the Kepler conjecture*, Annals of Mathematics 162 (2005), pp. 1065–1185; *A formal proof of the Kepler conjecture*, Forum of Mathematics, Pi 5 (2017), e2.
- M. S. Viazovska, *The sphere packing problem in dimension 8*, Annals of Mathematics 185 (2017), pp. 991–1015.
- H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. S. Viazovska, *The sphere packing problem in dimension 24*, Annals of Mathematics 185 (2017), pp. 1017–1033.
- K. Stephenson, *Introduction to Circle Packing: The Theory of Discrete Analytic Functions*, Cambridge University Press, 2005.

Complex analysis, Fourier series and Bessel functions

- S. Axler, P. Bourdon, W. Ramey, *Harmonic Function Theory*, 2nd ed., Graduate Texts in Mathematics, Springer, 2001.
- A. Zygmund, *Trigonometric Series*, 3rd ed., Cambridge Mathematical Library, Cambridge University Press, 2002.
- H. Dym, H. P. McKean, *Fourier Series and Integrals*, Probability and Mathematical Statistics 14, Academic Press, 1972.
- L. Carleson, *On convergence and growth of partial sums of Fourier series*, Acta Mathematica 116 (1966), pp. 135–157.
- R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd ed., McGraw-Hill, 2000.
- NIST, *Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>.
- G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge University Press, 1944.
- W. A. Strauss, *Partial Differential Equations: An Introduction*, 2nd ed., John Wiley & Sons, 2008.
- L. C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics, American Mathematical Society, 2010.

Laplace transform, z -transform, DFT and FFT

- G. Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer-Verlag, 1974.
- A. V. Oppenheim, A. S. Willsky, S. H. Nawab, *Signals and Systems*, 2nd ed., Prentice Hall, 1997.
- A. V. Oppenheim, R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed., Pearson, 2010.
- S. K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., McGraw-Hill, 2011.
- W. L. Briggs, V. E. Henson, *The DFT: An Owner's Manual for the Discrete Fourier Transform*, SIAM, 1995.
- C. Van Loan, *Computational Frameworks for the Fast Fourier Transform*, Frontiers in Applied Mathematics 10, SIAM, 1992.
- F. J. Harris, *On the use of windows for harmonic analysis with the discrete Fourier transform*, Proceedings of the IEEE 66 (1978), pp. 51–83.
- J. H. McClellan, T. W. Parks, *Eigenvalue and eigenvector decomposition of the discrete Fourier transform*, IEEE Transactions on Audio and Electroacoustics 20 (1972), pp. 66–74.
- M. T. Heideman, D. H. Johnson, C. S. Burrus, *Gauss and the history of the fast Fourier transform*, IEEE ASSP Magazine 1 (1984), pp. 14–21.
- N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, 2002.
- M. Frigo, S. G. Johnson, *The Design and Implementation of FFTW3*, Proceedings of the IEEE 93 (2005), pp. 216–231.

Signals, automatic control and telecommunications

- J. R. Higgins, *Five short stories about the cardinal series*, Bulletin of the American Mathematical Society 12 (1985), pp. 45–89.
- F. W. King, *Hilbert Transforms*, vol. 1, Cambridge University Press, 2008.
- H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002.
- K. J. Åström, R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, Princeton University Press, 2008.
- K. Ogata, *Modern Control Engineering*, 5th ed., Pearson, 2010.
- G. F. Franklin, J. D. Powell, A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 8th ed., Pearson, 2019.
- G. C. Goodwin, S. F. Graebe, M. E. Salgado, *Control System Design*, Prentice Hall, 2001.
- D. M. Pozar, *Microwave Engineering*, 4th ed., John Wiley & Sons, 2011.
- R. E. Collin, *Foundations for Microwave Engineering*, 2nd ed., IEEE Press, 2001.
- Encyclopedia of Mathematics, entry “Argument, principle of the”.
- J. G. Proakis, M. Salehi, *Digital Communications*, 5th ed., McGraw-Hill, 2008.
- S. Haykin, *Communication Systems*, 4th ed., John Wiley & Sons, 2001.
- B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed., Prentice-Hall PTR, 2001.

-
- J. M. Wozencraft, I. M. Jacobs, *Principles of Communication Engineering*, John Wiley & Sons, 1965.
 - R. W. Chang, *Synthesis of Band-Limited Orthogonal Signals for Multichannel Data Transmission*, Bell System Technical Journal 45 (1966), pp. 1775–1796.
 - S. B. Weinstein, P. M. Ebert, *Data Transmission by Frequency-Division Multiplexing Using the Discrete Fourier Transform*, IEEE Transactions on Communication Technology 19 (1971), pp. 628–634.
 - A. Peled, A. Ruiz, *Frequency Domain Data Transmission Using Reduced Computational Complexity Algorithms*, ICASSP '80, vol. 5, pp. 964–967.
 - Y. S. Cho, J. Kim, W. Y. Yang, C. G. Kang, *MIMO-OFDM Wireless Communications with MATLAB*, John Wiley & Sons, 2010.

Rigid-body kinematics, quaternions and trajectories

- M. Ceccarelli, *Screw axis defined by Giulio Mozzi in 1763 and early studies on helicoidal motion*, Mechanism and Machine Theory 35 (2000), pp. 761–770.
- J. B. Kuipers, *Quaternions and Rotation Sequences*, Princeton University Press, 1999.
- J. Diebel, *Representing Attitude: Euler Angles, Unit Quaternions, and Rotation Vectors*, technical report, Stanford University, 2006.
- K. M. Lynch, F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Cambridge University Press, 2017.
- J. M. Selig, *Geometric Fundamentals of Robotics*, 2nd ed., Monographs in Computer Science, Springer, 2005.
- J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Graduate Texts in Mathematics 218, Springer, 2013.
- B. C. Hall, *An Elementary Introduction to Groups and Representations*, arXiv:math-ph/0005032, 2000.
- E. Minguzzi, *A geometrical introduction to screw theory*, European Journal of Physics 34 (2013), pp. 613–632.
- W. Singhose, *Command shaping for flexible systems: a review of the first 50 years*, International Journal of Precision Engineering and Manufacturing 10 (2009), pp. 153–168.
- L. Biagiotti, C. Melchiorri, *Trajectory Planning for Automatic Machines and Robots*, Springer, 2008.
- L. Piegl, W. Tiller, *The NURBS Book*, 2nd ed., Springer, 1997.
- G. Farin, *Curves and Surfaces for CAD: A Practical Guide*, 5th ed., Morgan Kaufmann, 2002.

Fitting, numerical optimization and open problems

- N. Chernov, *Circular and Linear Regression: Fitting Circles and Lines by Least Squares*, Chapman and Hall/CRC, 2010.
- W. Gander, G. H. Golub, R. Strebler, *Least-squares fitting of circles and ellipses*, BIT Numerical Mathematics 34 (1994), pp. 558–578.
- J. Nocedal, S. J. Wright, *Numerical Optimization*, 2nd ed., Springer Series in Operations Research and Financial Engineering, Springer, 2006.
- B. Matschke, *A survey on the square peg problem*, Notices of the American Mathematical Society 61 (2014); *On the square peg problem and its relatives*, Transactions of the American Mathematical Society 375 (2022), pp. 6255–6280.
- J. E. Greene, A. Lobb, *The rectangular peg problem*, Annals of Mathematics 194 (2021), pp. 509–517; *Cyclic quadrilaterals and smooth Jordan curves*, Inventiones mathematicae 234 (2023), pp. 931–935; *Jordan curves inscribe a positive measure of rectangles*, arXiv:2604.17116, 2026.
- T. Tao, *An integration approach to the Toeplitz square peg problem*, Forum of Mathematics, Sigma 5 (2017), art. e30.
- W. Stromquist, *Inscribed squares and square-like quadrilaterals in closed curves*, Mathematika 36 (1989), pp. 187–197.
- R. P. Jerrard, *Inscribed squares in plane curves*, Transactions of the American Mathematical Society 98 (1961), pp. 234–241.
- A. Emch, *Some properties of closed convex curves in a plane*, American Journal of Mathematics 35 (1913); *On some properties of the medians of closed continuous curves formed by analytic arcs*, ibid. 38 (1916).
- J. Cantarella, E. Denne, J. McCleary, *Configuration spaces, multijet transversality, and the square-peg problem*, Illinois Journal of Mathematics 66 (2022).
- R. L. Graham, J. C. Lagarias, C. L. Mallows, A. R. Wilks, C. H. Yan, *Apollonian circle packings: number theory*, Journal of Number Theory 100 (2003), pp. 1–45.

-
- J. Bourgain, A. Kontorovich, *On the local-global conjecture for integral Apollonian gaskets*, *Inventiones mathematicae* 196 (2014), pp. 589–650.
 - S. Haag, C. Kertzer, J. Rickards, K. E. Stange, *The local-global conjecture for Apollonian circle packings is false*, *Annals of Mathematics* 200 (2024), pp. 749–770.
 - H. Shi, W. Shi, I. Whitehead, H. Williams-Tracy, J. Z. Zhang, *The Local-Global Conjecture is false for generalized circle packings*, *Journal of Number Theory* 288 (2026), pp. 103–157.
 - E. Fuchs, *Apollonian packings: the rise and fall of the local to global conjecture*, *Bulletin of the American Mathematical Society* 63 (2025), pp. 59–77.
 - A. M. Odlyzko, N. J. A. Sloane, *New bounds on the number of unit spheres that can touch a unit sphere in n dimensions*, *Journal of Combinatorial Theory, Series A* 26 (1979), pp. 210–214.
 - O. R. Musin, *The kissing number in four dimensions*, *Annals of Mathematics* 168 (2008), pp. 1–32.
 - J. H. Conway, R. A. Parker, N. J. A. Sloane, *The covering radius of the Leech lattice*, *Proceedings of the Royal Society of London A* 380 (1982), pp. 261–290.
 - H. Cohn, *A conceptual breakthrough in sphere packing*, *Notices of the American Mathematical Society* 64 (2017), pp. 102–115.
 - J. Solymosi, F. de Zeeuw, *On a question of Erdős and Ulam*, *Discrete & Computational Geometry* 43 (2010), pp. 393–401.
 - J. Shaffaf, *A solution of the Erdős–Ulam problem on rational distance sets assuming the Bombieri–Lang conjecture*, *Discrete & Computational Geometry* 60 (2018), pp. 283–293.
 - K. Ascher, L. Braune, A. Turchet, *The Erdős–Ulam problem, Lang’s conjecture and uniformity*, *Bulletin of the London Mathematical Society* 52 (2020), pp. 1053–1063.

Technical standards and international teaching materials

- MIT OpenCourseWare, *18.950 Differential Geometry; 18.02 Multivariable Calculus; 18.700 Linear Algebra; 18.900 Geometry and Topology in the Plane; 12.215 Modern Navigation; 2.004 Dynamics and Control II; 18.03SC Differential Equations; 6.341 Discrete-Time Signal Processing* — <https://ocw.mit.edu/>.
- NPTEL, Indian Institute of Technology, *Differential Geometry of Curves and Surfaces; Multivariable Calculus and Classical Mechanics; Control Engineering* — <https://nptel.ac.in/>.
- UNAM, Facultad de Ciencias / Facultad de Ingeniería, curricula in *Geometría Diferencial, Geometría Analítica e Álgebra Lineal, Matemáticas Avanzadas*.
- AIMS South Africa, African Institute for Mathematical Sciences, teaching materials — <https://aims.ac.za/>.
- University of Michigan, *Control Tutorials for MATLAB and Simulink* — <https://ctms.engin.umich.edu/>.
- F. Nicolls, University of Cape Town, *EEE2047S Signals and Systems I*.

Teaching methodology

- C. A. Barbieri, D. Miller-Cotto, S. N. Clerjuste, K. Chawla, *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*, *Educational Psychology Review* 35 (2023), art. 11.
- H. L. Roediger III, J. D. Karpicke, *Test-Enhanced Learning*, *Psychological Science* 17 (2006).

Supplement bibliography

- [1] Y.-T. Siu, *Riemann Mapping Theorem*, Harvard Math 113, Spring 2024, 4 pp.
https://people.math.harvard.edu/~siu/math113/15_riemann_mapping_theorem.pdf Primary copy consulted; comparison with the extremal proof.
- [2] K. J. Åström, R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, Princeton University Press, 2008, Chapter 6, pp. 167–199; Chapter 7, pp. 201–227. ISBN 978-0-691-13576-2.
<https://authors.library.caltech.edu/records/yzs24-xsx88> Contents and metadata verified in the institutional repository; indexed text of the state-feedback chapter consulted. Full-copy downloads returned access errors.
- [3] Stanford University, EE363, Winter 2008–09, Lecture 4, *Continuous time linear quadratic regulator*, 28 pp.
<https://ee363.stanford.edu/archive/lectures/clqr.pdf> Primary course copy consulted; continuous Riccati and the quadratic problem.
- [4] R. E. Kalman, “A New Approach to Linear Filtering and Prediction Problems”, *Journal of Basic Engineering*, 82(1), 1960, pp. 35–45. DOI: 10.1115/1.3662552.
https://www.cs.cmu.edu/~motionplanning/papers/sbp_papers/k/Kalman1960.pdf Scan of the original article consulted.
- [5] S. Lall, *The Kalman filter*, EE363, Stanford University, 28 pp., no publication date stated on the consulted title page.
https://ee363.stanford.edu/lectures/021_kf.pdf Primary copy consulted, particularly pp. 13–17.
- [6] F. Landis Markley, “Attitude Error Representations for Kalman Filtering”, *Journal of Guidance, Control, and Dynamics*, 26(2), 2003, pp. 311–317.
https://www.researchgate.net/profile/Landis-Markley/publication/245432681_Attitude_Error_Representations_for_Kalman_Filtering/links/557c5d4108aeea18b776a727/Attitude-Error-Representations-for-Kalman-Filtering.pdf Article copy on the author’s profile, verified against title page and text. The paper’s passive, scalar-last conventions are not confused with the active, scalar-first conventions of this supplement.
- [7] J. Solà, *Quaternion kinematics for the error-state Kalman filter*, arXiv:1711.02508v1, 2017, 95 pp., §§4.3.3, 5.3, 6.3.
<https://arxiv.org/abs/1711.02508> Primary preprint consulted; right Jacobian, local error and reset.
- [8] R. M. Murray, Z. Li, S. S. Sastry, *A Mathematical Introduction to Robotic Manipulation*, CRC Press, 1994, Chapter 4; 2R example, printed pp. 164–165.
<https://atc.home.ece.ust.hk/files/mls94-complete.pdf> Primary copy on a coauthor’s website consulted.
- [9] K. M. Lynch, F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Cambridge University Press, 2017, Chapter 8 and §11.4.
<https://modernrobotics.northwestern.edu/chapters/chapter8/>
<https://modernrobotics.northwestern.edu/nu-gm-book-resource/11-4-motion-control-with-torque-or-force-inputs-part-3-of-3/> Authors’ official contents and transcripts consulted; no unviewed printed page numbers are attributed.
- [10] R. W. Brockett, “Robotic manipulators and the product of exponentials formula”, in *Mathematical Theory of Networks and Systems*, Springer, 1984, pp. 120–129. Historical attribution checked in the bibliography of [8]; the original article was not directly consulted.

-
- [11] Stanford University, EE363, Winter 2008–09, Lecture 1, *Linear quadratic regulator: Discrete-time finite horizon*, 32 pp., pp. 20–23.
<https://ee363.stanford.edu/archive/lectures/dlqr.pdf> Primary copy consulted.
- [12] K. J. Åström, R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, Princeton University Press, 2008, Chapter 10, pp. 293–314; §10.5, pp. 308–312.
<https://authors.library.caltech.edu/records/yzs24-xsx88> Contents and publication data verified in the institutional repository; also compared with indexed text of the 24 July 2020 electronic revision, where PID is Chapter 11. Direct download failed. The present recurrences are explicitly derived, not attributed to unread pages.
- [13] M. N. Huxley, “Exponential Sums and Lattice Points III”, *Proceedings of the London Mathematical Society*, 87(3), 2003, pp. 591–609. DOI: 10.1112/S0024611503014485.
<https://doi.org/10.1112/S0024611503014485> Publisher record and abstract verified; the bound is reported as an external result, not proved here.
- [14] X. Li, X. Yang, *An improvement on Gauss’s Circle Problem and Dirichlet’s Divisor Problem*, arXiv:2308.14859v2, 14 September 2023, 32 pp., Theorem 1.2.
<https://arxiv.org/abs/2308.14859> Primary preprint consulted. It also supplies the attribution of the lower bound to Hardy (1915), not a direct verification of Hardy’s article.
- [15] J. Bourgain, N. Watt, *Mean square of zeta function, circle problem and divisor problem revisited*, arXiv:1709.04340, withdrawn 16 July 2023.
<https://arxiv.org/abs/1709.04340> Withdrawal notice consulted; not used as a source of a valid theorem.
- [16] B. C. Berndt, S. Kim, A. Zaharescu, “The Circle Problem of Gauss and the Divisor Problem of Dirichlet—Still Unsolved”, *The American Mathematical Monthly*, 125(2), 2018, pp. 99–114. DOI: 10.1080/00029890.2018.1401853.
<https://doi.org/10.1080/00029890.2018.1401853> Metadata checked in the authors’ institutional profile; suggested reading, not the primary source of updated bounds.

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